SCALAR CONTINUUM LIMITS OF THREE-DIMENSIONAL LATTICE YANG–MILLS THEORY

Abstract. We study the action density of the three-dimensional SU(2) Wilson measure at lattice spacing L^{-K} and inverse coupling $L^K\mathsf{b}$, for $\mathsf{b} > 0$. A local probability kernel draws the coarse links. Contraction and source estimates for the resulting renormalisation give full-sequence convergence of separated cumulants of order at least two on fixed tori and subsequential limits when the side tends to infinity. For translation-invariant Gibbs states the limits have full Euclidean symmetry and reflection positivity. Their canonical clustering components reconstruct local scalar Wightman fields with unique invariant vacua. A reflected block-field test gives nontriviality and a two-point lower bound at short distances in each component.

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Introduction

Let U have the SU(2) Wilson law on a three-dimensional periodic lattice of spacing $a_K = L^{-K}$. We study the action density smeared against f,

$$\mathcal{O}_K(f) = a_K^{-1} \sum_p f(x(p)) \left(1 - \tfrac{1}{2} \operatorname{tr} \partial U(p)\right).$$

Here $x(p)$ is the plaquette barycentre and $\partial U(p)$ is the plaquette holonomy. We take the inverse lattice coupling to be $\beta = L^K\mathsf{b}$, with positive label b. Subtracting the mean gives the centered observable. Cumulants of order at least two are independent of this subtraction.

Theorem 0. Fix a sufficiently large admissible odd blocking factor L. For each positive label b, the action-density cumulants of order at least two converge on real continuous tests with pairwise disjoint supports as $K \to \infty$ on a fixed torus of side $2L^m$. On $\mathbb{R}^3$, taking the cutoff and volume limits in either order or jointly gives subsequential limits with a common subsequence for every separated tuple. These limits have cubic symmetry and coordinate reflection positivity.

For sequences of translation-invariant infinite-lattice Gibbs states whose labels tend to b, the continuum limits are invariant under all Euclidean isometries and reflection positive at every plane. Their canonical clustering components yield local scalar Wightman fields with unique invariant vacua. The connected two-point kernel in each component is bounded below by a positive multiple of $\mathsf{b}^{-2}r^{-6}$ at sufficiently small distances r. ■□

The precise statement, including the conditions on the blocking factor, the three limiting procedures and the test spaces, is Thm. A.

The construction is in the style of OpenAI's nonlinear sigma-model paper *The canonical massive continuum limit of the two-dimensional* O(3) *model*, from result family #215 [1]. Given the fine links U, draw each coarse link independently from a normalized SU(2) density around a gauge-covariant local average $\mathsf{M}(U)(c)$. Here x is the reference inverse coupling at the fine scale and $a > 0$ is the smoothing parameter of the kernel. If ρ is the fine density, the coarse density is

$$\rho'(V') = \int \rho(U) \prod_c k_{ax}(V'(c), \mathsf{M}(U)(c))\, dU.$$

The kernel k_{ax} has total mass one in its first argument. This preserves the partition integral at every step and lets us compare cutoffs by comparing effective interactions in a common class of densities. The Gaussian rows describe the reference quadratic form. After matching the rows, a weighted contraction controls the difference of the interactions. Its complex-source version controls separated cumulants by differentiation. A chessboard estimate moves the volume factor into a bound per occupied cell. This makes the continuum estimates uniform in the side.

The construction and its Gaussian estimates occupy §2–§5. The source analysis and the limits are in §6–§8. Growth, symmetry and nontriviality are proved next. In §12 we pass the lower bound to each clustering component and apply the scalar reconstruction theorem of Appendix F.

1. The setting and the main theorem

1.1. **Wilson measures, sources and dilation.** Write $G = \mathrm{SU}(2)$. On the torus $T_\ell = \mathbb{R}^3/\ell\mathbb{Z}^3$, with $\ell = 2L^m$ and $a_K = L^{-K}$, take the lattice $\Lambda_{K,\ell} = a_K\mathbb{Z}^3/\ell\mathbb{Z}^3$. We allow integers K, m with $K + m \geq 1$ in this convention. Each positively oriented coordinate bond has one independent variable in G, with the inverse variable on the reversed bond. The plaquettes are counted once, in the three positive coordinate planes, and the trace of the ordered plaquette product is independent of the choice of its base point and orientation. Write dU for the product of normalized Haar probabilities, $a_p(U) = 1 - \frac{1}{2}\operatorname{tr}\partial U(p)$ and $A(U) = \sum_p a_p(U)$. The Wilson probability is $Z_{K,\ell,\beta}(0)^{-1}e^{-\beta A(U)}dU$. The signed permutations W_3 and lattice translations act on bonds and plaquettes. They preserve this probability, since they permute bond variables with possible inversions and preserve A.

Definition 1.1 (Action-density sources)**.** For a bounded f on T_ℓ define

$$\mathcal{O}_K(f) = a_K^{-1} \sum_p f(x(p))a_p(U),$$

where $x(p)$ is the plaquette barycentre. For a tuple $f = (f_1, \ldots, f_n)$ put

$$Z_{K,\ell,\beta}(z) = \int \exp\left\{-\beta A(U) + \sum_i z_i \mathcal{O}_K(f_i)\right\} dU.$$

The joint cumulant $S^T_{n;K,\ell,\beta}(f)$ is the derivative $\partial_{z_1}\cdots\partial_{z_n}\log(Z(z)/Z(0))|_{z=0}$. The ratio is entire by boundedness of the observables. It equals 1 at zero, so the logarithm is holomorphic on a neighborhood of zero. ■

Lemma 1.2 (Centered moments and cumulants)**.** Let X_i be bounded random variables and $Y_i = X_i - \mathbb{E}X_i$. Then

$$\mathbb{E}\prod_{i=1}^{n} Y_i = \sum_{\pi\in\mathrm{Part}_{\geq 2}(n)}\prod_{B\in\pi}\kappa(X_i : i\in B). \tag{1.i}$$

■

Proof. In the commutative algebra generated by z_i subject to $z_i^2 = 0$, write $z_B = \prod_{i\in B} z_i$. The coefficient of z_B in $\log\mathbb{E}\prod_i(1+z_iX_i)$ is $\kappa(X_i : i\in B)$. Centering subtracts $\sum_i z_i\mathbb{E}X_i$ from this logarithm, so its coefficients in degrees at least two are unchanged and its linear coefficients vanish. Exponentiating gives the partition sum. For each partition the factorial in the exponential cancels the number of orders of its blocks. The empty partition has weight 1. □

Lemma 1.3 (Exact change of length units)**.** Let $j\in\mathbb{Z}$ and $f^{[j]}(x) = f(L^{-j}x)$. For each tuple f of length $n\geq 1$ one has

$$S^T_{n;K,\ell,\beta}(f) = L^{jn}S^T_{n;K-j,L^j\ell,\beta}(f^{[j]}). \tag{1.ii}$$

The full-cutoff limits at (ℓ,b) and $(L^j\ell, L^j\mathsf{b})$ exist together. When they exist,

$$S^{(\ell,\mathsf{b})}_n(f) = L^{jn}S^{(L^j\ell,L^j\mathsf{b})}_n(f^{[j]}). \tag{1.iii}$$

■

Proof. Dilation maps the lattice bijectively onto the lattice with spacing L^{j-K}. It preserves the graph, plaquette products, Haar law and β. The observable on the rescaled lattice is $\mathcal{O}_{K-j}(f^{[j]}) = L^{-j}\mathcal{O}_K(f)$. Multilinearity of the joint cumulant gives the finite-cutoff identity. For the limit use $\beta = L^K\mathsf{b} = L^{K-j}(L^j\mathsf{b})$. □

1.2. **Parameters and limits.** The parameters below ensure that the coarse interaction stays in the class on which the step and the comparison estimates are proved.

Definition 1.4 ((S1): the parameter table)**.** The gauge group is $G = \mathrm{SU}(2)$, and the dimension is 3. We work with one table, (O0) to (O13), with all its floors: the table of item (1) of §6.1, repeated in item 1 of §8.1. The number L is odd and is the L of the table, so that $L\geq L_*^{\mathrm{su}}\geq 10^{2048.9}$ (see the remark following Thm. 8.2), and it satisfies (L3su) and (L5*). Further, $M = L^{m'}$ with $m'\geq 1$, subject to (M2). We fix $\kappa\geq 6$ and put $\kappa_1 = \kappa+1$.

We set $p_0(x) = A_0(\log x)^{\mathfrak{p}_0}$ with $\mathfrak{p}_0 > 2$, and $\varepsilon(x) = p_0(x)x^{-1/2}$; further $c_a(x) = E_a p_0(x)^3 x^{-1/2}$ and $c_h(x) = e^{-c_h p_0(x)^2}$, where c_h without an argument, as in the exponent, is the number of (O11), while $c_h(x)$, with an argument, is the function defined here; and $r(x) = \lceil 20c_R\log x/M\rceil$, $s = 8r$, $\check{r} = r+3$, $\rho(x) = \lceil c_R\log x\rceil$, which is the radius denoted $\varrho(x)$ in §2.1 and in item 1 of §8.1. The number $x_s\geq e^2$ is the last of the floors.

The constants are fixed in this order: first the table, then the label, then the numbers of (O14). Every number of item 2 but one belongs to (O14), so no floor depends on it; the exception is the threshold $N^*(K,\mathsf{b}')$ of Thm. 10.4 (4.9) (i). ■

Definition 1.5 ((S2): the data of a torus)**.** The data are those of Thm. 8.2, which agree with those of item 1 of §7.3. They consist of a label $\mathsf{b}'\geq x_s$; of a side $N = 2L^{m_l}$ with $m_l\geq m'$, subject to (Tor), that is, $N\geq 2Ms(L\mathsf{b}')$; and of a cutoff $K\geq 1$, with $a_K = L^{-K}$ and $\beta = L^K\mathsf{b}'$. By $\mu_{K,N,\beta}$ we denote Wilson's measure on $a_K\mathbb{Z}^3/N\mathbb{Z}^3$. The label, the side and the cutoff, with a_K and β, are what we call the data. ■

Definition 1.6 ((S3): the field, cumulants, the centred functional, distances)**.** For a function f we put $\mathcal{O}_K f = a_K^{-1}\sum_p f(x(p))a_p(U)$, the sum running over the plaquettes p, where $a_p = 1 - \frac{1}{2}\operatorname{tr}\partial U(p) \in [0,2]$ and $x(p)$ is the barycentre of p. On a torus we write S_n^T, and in a state ρ we write $S_n^T[\rho]$, for the joint cumulants of the $\mathcal{O}_K f_i$; and we put $\sigma_{K,N}(f_1,\dots,f_n) = \langle\prod(\mathcal{O}_K f_i - \langle\mathcal{O}_K f_i\rangle)\rangle$.

For a family $S = (S_n)_{n\ge 2}$, the centred functional σ_S is that of eq. 1.i:

$$\sigma_S(X) = \sum\prod_J S_{|J|}(X_J),$$

where the sum runs over the partitions of X into blocks J of size ≥ 2, and $\sigma_S(\emptyset) = 1$ for the empty tuple $\emptyset$.

Distances are sup distances unless we say "Euclidean"; a tuple whose Euclidean type is $(d;R)$ has the sup type $(d/\sqrt{3};R)$ (item 1 of §8.1). We put $\check{\delta}_n := \min\{d_{\sup}, 1/(4n)\}$, where $d_{\sup}$ is a lower bound for separation in the sup distance; we take $d_{\sup} = d$ in Sections 8.3 and 9, and $d_{\sup} = d/\sqrt{3}$ in Sections 10 and 12. Further, $\|f\|_\sharp = \max\{\|f\|_\infty, 40M\operatorname{Lip} f\}$, where $\operatorname{Lip} f$ is taken with respect to the Euclidean distance, as in item 1 of §8.1; $f^{[z]}(x) = f(L^{-z}x)$, and W_3 denotes the cubic group.

In the limit statements, a separated tuple consists of real continuous functions with compact, pairwise disjoint supports; on a fixed torus the functions are continuous on that torus. For a volume limit, a tuple on $\mathbb{R}^3$ is periodised once the side exceeds twice the diameter of the union of its supports. The definition of Lim_G in (S6) uses the C_c^1 subspace. Write $\mathfrak{S}(O)$ for the separated C_c^1 tuples in O, including the empty tuple, and $\mathbb{C}\mathfrak{S}(O)$ for their finite complex linear combinations. The finite-cutoff estimates also allow the bounded sources specified in their statements. For a tuple of distinct points, write $d(x) = \min_{i\ne j}|x_i - x_j|_2$, with $d(x) = \infty$ when $n = 1$. ■

Definition 1.7 ((S4): the drawn block fields and the subtractions)**.** The fields $(V_0 = U, V_1, \dots, V_K)$ are distributed by the joint law of the K drawn steps (see item 2 of §5.5, under "The step" and "Runs", and Step 11 (0) of the proof of Thm. 8.2). The level k is the lattice $L^{-(K-k)}\mathbb{Z}^3/N\mathbb{Z}^3$, whose depth is $K-k$, and $x_k = L^{K-k}\mathsf{b}'$. We write ρ_k^z for the k-fold drawn density of Wilson's weight multiplied by $\exp\{\sum_t z_t\mathcal{O}_K g_t\}$. For $0 \le s \le K-1$ the subtractions are $c_K^{(s)}(f) := \frac{d}{dz}\log\rho_{K-s}^{[zf]}(1)$ at $z = 0$, where $\rho_k^{[zf]}$ is the density ρ_k^z above for the single source f with the variable z, that is, the k-fold drawn density of Wilson's weight multiplied by $\exp\{z\mathcal{O}_K f\}$, and the argument 1 is the trivial field of the level k; and $\hat{c}_K := c_K^{(0)}$. These numbers depend on K and N. ■

Definition 1.8 ((S5): sources, the setting of Def. 8.6)**.** The number $Q \ge 1$ is arbitrary, and $\check{\delta}, \check{\rho} > 0$. The sources $g_1,\dots,g_Q$ are bounded functions on $\mathbb{R}^3/N\mathbb{Z}^3$, complex in §8.2 and real from §8.3 on. The sets $\{g_t \ne 0\}$ are pairwise $\ge \check{\delta}$ apart, and each of them lies within $\check{\rho}$ of a single point. ■

Definition 1.9 ((S6): states and limits)**.** For any $\mathsf{b}' > 0$ we let $\mathbb{G}_K(\mathsf{b}')$ consist of those Gibbs states ν of Wilson's action at $\beta = L^K\mathsf{b}'$ on $a_K\mathbb{Z}^3$ (B.1 (2)) for which $(\tau_v)_*\nu = \nu$ holds for every $v \in a_K\mathbb{Z}^3$. This set lies in $\mathcal{G}^{\mathrm{per}}$ (B.1 (3)) and contains $\mathcal{G}^{\mathrm{tor}}$, the set of all limit points of tori (B.1 (4), B.3 (c)).

For $\mathsf{b} > 0$, $\mathrm{Lim}_G(\mathsf{b})$ is the set of the families $S = (S_n)_{n\ge 2}$ for which $S_n(X) = \lim_j S_n^T[\nu_j](X)$ holds on all separated C_c^1 tuples over $\mathbb{R}^3$, for some $K_j \to \infty$, $\nu_j \in \mathbb{G}_{K_j}(\mathsf{b}_j)$ and $\mathsf{b}_j \to \mathsf{b}$. The subset $\mathrm{Lim}_{\mathrm{tor}}(\mathsf{b})$ consists of the families obtained in this way with $\nu_j \in \mathcal{G}^{\mathrm{tor}}$.

By the limits of (R5) we mean the limits of Thm. 8.17 (R5) in any of its three orders: (a) the continuum first, (b) the volume first, (c) both jointly. Every limit of (R5) (b) belongs to $\mathrm{Lim}_{\mathrm{tor}}(\mathsf{b})$, by item 1 of §10. We call the σ_ξ of a canonical decomposition $(\mathsf{X}, \lambda, (\sigma_\xi))$ of σ_S (E.12) the components of S. ■

Definition 1.10 ((S7): labels; $z_0(\mathsf{b})$, $z(\mathsf{b})$, $U(\mathsf{b})$; normal form)**.** We use $\mathsf{b} > 0$ for the label of a limit and b' for that of an approximant, that is, of a torus or of a state. In a dilation, the letter z

written without an argument, as opposed to $z_0(\mathsf{b})$ and $z(\mathsf{b})$ below, is a bound variable: it stands for any integer that the clause admits.

We put $z_0(\mathsf{b}) :=$ the least integer $z \geq 0$ with $L^z\mathsf{b} \geq x_s$; it brings one label into the data. We put $z(\mathsf{b}) :=$ the least integer $z \geq 0$ with $L^z\mathsf{b} > x_s$; it brings a whole neighbourhood of the label into the data. Thus $z(\mathsf{b}) = z_0(\mathsf{b})$, except when $L^{z_0(\mathsf{b})}\mathsf{b} = x_s$, in which case $z(\mathsf{b}) = z_0(\mathsf{b}) + 1$.

Let $U(\mathsf{b}) := [\mathsf{b}/2, 2\mathsf{b}] \cap [L^{-z(\mathsf{b})}x_s, \infty)$. This is a compact neighbourhood of b, determined by b alone; it satisfies $L^{z(\mathsf{b})}U(\mathsf{b}) = U(L^{z(\mathsf{b})}\mathsf{b}) \subset [x_s, \infty)$; and for a sequence $\mathsf{b}_j \to \mathsf{b}$, at most finitely many of the labels lie outside $U(\mathsf{b})$. We leave out of the sequence the j with $\mathsf{b}_j \notin U(\mathsf{b})$. This changes no convergence and no limit.

The label of a limit is in normal form if $\mathsf{b} > x_s$; then $z(\mathsf{b}) = 0$ and $U(\mathsf{b}) = [\mathsf{b}/2, 2\mathsf{b}] \cap [x_s, \infty)$. Any other $\mathsf{b} > 0$ is brought to normal form by the dilation $L^{z(\mathsf{b})}$. ■

Definition 1.11 ((S8): numbers of (O14) at a label)**.** Let X be any number of (O14), regarded as a function of the label. The suprema below are taken over its original formula. (n1) Except in (n2), if the label belongs to the data (a torus, a state at $\mathsf{b}' \geq x_s$, or a limit at a fixed label $\mathsf{b} \geq x_s$), then $X(\mathsf{b}')$ is simply the value of its formula.

(n2) In Sections 10 and 12, in a clause on $\mathrm{Lim}_G(\mathsf{b})$ or on a frame, and with the label in normal form, we put $X(\mathsf{b}) := \sup_{\mathsf{b}' \in U(\mathsf{b})} X(\mathsf{b}')$ and $j^\circ_U(\mathsf{b}, \check{\delta}) := \sup_{U(\mathsf{b})} j^\circ(\cdot, \check{\delta})$. These suprema are finite, by (Lab1) in item 1 of §9 and by item 1 of §10. For $\mathsf{b} \leq x_s$, a number taken "at $L^z\mathsf{b}$", with $z = z(\mathsf{b})$, means its supremum over $L^zU(\mathsf{b})$.

(n3) In §9, at a limit of (R5) whose labels move, that is, a limit taken along approximants with labels $\mathsf{b}_j \to \mathsf{b}$ rather than at the fixed label of (n1), the number is that of (Lab3), namely $\limsup_j X(\mathsf{b}_j)$ along any subsequence; for a label in normal form, this is $\leq$ the number of (n2). ■

Remark 1.12 (The one label $\mathsf{b} = x_s$). Let $\mathsf{b} = x_s$. For every sequence, the numbers that enter Thm. 10.4 (4.6) and (4.9) (ii) are then given by (4.8) at $z = 1$; and when $\mathsf{b}_j \geq x_s$ along the sequence, the suprema over $[x_s, 2x_s]$ may be used as well, as the remark in item 1 of §10 explains.

Definition 1.13 ((S9): boards)**.** We use a single family of boards $B_i(o)$, for $-m_l \leq i \leq K-1$ and $o \in a_K\mathbb{Z}^3$: the walls of $B_i(o)$ are $o_\mu + L^{-i}(\mathbb{Z} + \frac{1}{2})$, and its cells have side L^{-i}. We write $B_i := B_i(0)$; its cell 0 is centred at 0. In the notation of A.4 (2) the board $B_i(o)$ has the parameters $(a_K, \lambda = L^{K-i}, \mathsf{n}_\mu = 2L^{m_l+i}, o)$, and in that of B.10 the parameters (i, o).

Without arguments, B_0 is the unit board. As to offsets, §8.3 takes $o \in \mathbb{Z}^3$ or $o \in L^{-s}\mathbb{Z}^3$, where s, with $0 \leq s \leq K-1$, is a depth as in (S4); §10 takes any $o \in a_K\mathbb{Z}^3$ and $i \geq 0$; and in Sections 9, 12 and 11 the boards carry no offset. ■

Definition 1.14 (Item 2: constants in the main theorem)**.** The constants are those of the results cited below, evaluated at the label according to (S8). Except for $N^*(K, \mathsf{b}')$, they belong to (O14) and are fixed after the table and the label; no floor depends on them.

The source constants j°, r°, e_h and $G_1(\mathsf{b}, \check{\delta}, \check{\rho})$ are defined in Def. 8.8; in particular, $\mathrm{sep}^\circ = 8\log(8L^{1/2})$. Their depth-$s$ versions r°_s, $e_h^{(s)}$ and $G_{1,s}$ are defined in item 2 of §8.3. That item also sets

$$c_{\mathrm{vol}}(\mathsf{b}) = 225/M^3 + \tfrac{3}{2}\mathsf{N}_Z p_0(\mathsf{b})^2 + 3\log(384/\varepsilon(\mathsf{b})^3),$$
$$1 + 0.182C_{\mathrm{vol}}(N, \mathsf{b}) \leq e^{c_{\mathrm{vol}}(\mathsf{b})N^3}.$$

With A_4 from item 1 of §9, put $A_5 = 64(4e)^5A_4$, $A_6 = 3^{5/2}A_5$ and $C_E = (10^{53})^6M\,8A_6$. For $0 < \mathsf{b} < x_s$ use the dilated constants of Thm. 9.4 (G2)(iv) and (G4)(e). The insertion constant $\check{\Gamma}_*(\mathsf{b}, \eta)$ and norm $\|\!|h|\!\|_{\eta,y}$ are those of Thm. 9.4 (G6) and (G6+). Set

$$C_n^\infty(d; R) = 1.45^{n+1}n!\,A_6^n(n!)^5\max\{1, d^{-5n}\}(1+R)^{3n},$$
$$C_1(R) = A_6(1+R)^3, \quad C^{\mathsf{b}} = (C_1, (C_n^\infty)).$$

The constants $\hat G$, $\Gamma(\mathsf{b},\mathsf{l})$, C_φ and C_n^σ are those of item 2 of §10 and Thm. 10.4 (4.6); in particular, $(c_{\rm I}, C_{\rm I}, \mathfrak{p}_{\rm I}) = (1/(48L), 24Le^{c_{\rm vol}}\hat G, 1)$. The threshold $N^*(K,\mathsf{b}')$ is the one in Thm. 10.4 (4.9)(i). The definition "New numbers" of §12 gives $\bar c_* = 3A_6/(4\pi)$ and $C_{\rm pt}$; set $C_\varphi = 32C_{\rm pt}$, $\varphi = 5$, $c_0 = 1/2$ and $C_* = C_\varphi + \bar c_*$.

For the lower bounds, Thm. 11.1 defines $J(\mathsf{b})$, $x^\#$, $x^{\#\#}$ and the pure constants $D_0 = 10^{140}$, $c_F \in [5.4\cdot 10^{-5}, 4.0\cdot 10^{-4}]D_0^{-5}$ and $c_* = c_F^2/7 \ge 10^{-1410}$. For $\mathsf{b} \ge x_s$, let

$$y(\mathsf{b}) = \mathsf{b}\max\{c_{\rm vol}(\mathsf{b})^{1/3}, L^{1-m'}\},$$
$$j_*(\mathsf{b}) = \min\{j \ge 1 : L^j\mathsf{b} \ge \max\{x^\#, y(\mathsf{b})^{4/3}\}\},$$

and let $j^+(\mathsf{b})$ be the supremum of j_* over $[\mathsf{b}/2, 2\mathsf{b}] \cap [x_s, \infty)$. For $0 < \mathsf{b} < x_s$, set $j_*(\mathsf{b}) = z_0(\mathsf{b}) + j_*(L^{z_0(\mathsf{b})}\mathsf{b})$. Finally, $c_{**} = (4/L)^6 c_*$ and, with $z = z(\mathsf{b})$, $r_*(\mathsf{b}) = L^{-z}D_0L^{-j^+(L^z\mathsf{b})}$, as in Thm. 12.6 (C4). ■

Theorem A (Bounds and continuum limits)**.** Assume (S1). Then (Y0) to (Y7), (Y7′), (Y8) and (Y9) below hold, and so does the corollary with which they close. ■□

(Y0) *Subtraction.* For real f the number $c_K^{(s)}(f)$ is real, and it is linear in f. One has $c_K^{(s)}(f\circ R) = c_K^{(s)}(f)$ for the reflections $R : y_\mu \mapsto c - y_\mu$ with any $c \in L^{-s}\mathbb{Z}$, and for the translations by $L^{-s}\mathbb{Z}^3$; for $\hat c_K$ every integer c is admitted, and the reflections with odd c are the mirrors in the walls of B_0. Moreover, for given data and $0 \le s \le K-1$,

$$c_K^{(s)}(f) = \frac{d}{dz}\Big|_{z=0} \sum_{1\le j\le K-s} V_j(z) + \sum_X \delta^h[f](K-s)(X,1),$$

where X runs over $\mathcal{D}$ of the level $K-s$ and the $V_j(z)$ are given by Thm. 7.1 (h) along the run whose single source is f, with the variable z; with the argument z, the $V_j(z)$ are numbers, not the fields V_j of (S4). If $\{f \ne 0\}$ lies within $\check\rho$ of a single point, as each source of (S5) does, then the last sum has modulus $\le e_h^{(s)}\|f\|_\infty/r_s^\circ$, at any $\check\delta > 0$.

Proof. Clause (Y0) is Thm. 8.17 (R0), with Lems. 8.20 and 8.21. At $s = 0$ the formula for $c_K^{(0)}(f)$ and the bound on its last sum come from Prop. 8.11 (J7°) (vii) and (J6°) (e). □

(Y1) *Per unit of volume.* In (Y1), in the bound on cells and in lines 1 and 2 of (Y2), and in (Y7) (D), (N0) to (N3) and the first assertion of (N4), b is the label of the data, written b' in (S2), so that $\mathsf{b} \ge x_s$; in the rest of (N4) and in (N5) it is the label of the limit, that is, a fixed label or the limit of the sequence of labels. Throughout (Y7), except in (N6′), we assume $\mathsf{b} \ge x_s$, and for the parts (N3), (N4) and (N5) of (Y7), every other $\mathsf{b} > 0$ is treated in (Y8). In (N6′) the label is any $\mathsf{b} > 0$. For given data, with sources as in (S5), and for all $K \ge 1$ and all $Q \ge 1$,

$$|\langle\prod_t(\mathcal{O}_K g_t - \hat c_K g_t)\rangle| \le G_1(\mathsf{b},\check\delta,\check\rho)^Q(1 + 0.182C_{\rm vol}(N,\mathsf{b}))\prod\|g_t\|_\infty \le G_1^Q e^{c_{\rm vol}(\mathsf{b})N^3}\prod\|g_t\|_\infty.$$

Here $C_{\rm vol}$ enters to the first power only. For every bounded Borel function F of V_K, with the same K, Q and sources, one has $|\langle F(V_K)\prod_t(\mathcal{O}_K g_t - \hat c_K g_t)\rangle| \le \sup|F|\,G_1^Q(1+0.182C_{\rm vol}(N,\mathsf{b}))\prod\|g_t\|_\infty \le \sup|F|\,G_1^Q e^{c_{\rm vol}(\mathsf{b})N^3}\prod\|g_t\|_\infty$.

Proof. The proof is that of Cor. 8.14 (a), (b), which Proof 11 of §8.2 gives. At a depth $s \ge 0$ the bound is Thm. 8.17 (R8) (d) (i), with Lems. 8.21 to 8.25. □

(Y2) *Bounds free of the side, of the cutoff and of the places. Cells.* Take the board $B_i(o)$ with $o \in \mathbb{Z}^3$ and $-m_l \le i \le 0$, and let $h = L^{-i}$. If every support lies in one open cell, at distance $\ge \check\delta/2$ from the walls, and q cells are occupied, then the moment of (Y1) is $\le G_1^Q e^{qc_{\rm vol}(\mathsf{b})h^3}\prod\|g_t\|_\infty$.

Lines 1, 2. Let $f = (f_1, \dots, f_n)$ be a tuple of real bounded functions on $\mathbb{R}^3/N\mathbb{Z}^3$ whose sets $\{f_i \neq 0\}$ are pairwise $\geq d$ apart, each lying in some sup-ball of radius R_f. Wherever these balls lie, for all $K \geq 1$, all N of the data, all $n \geq 2$ and all $d, R_f > 0$,

$$|\sigma_{K,N}(f)| \leq A_5(\mathsf{b})^n (n!)^5 \max\{1, d^{-5n}\}(1 + R_f)^{3n} \prod \|f_i\|_\infty,$$

and $|S_n^T(f)| \leq 1.45^{n+1} n!$ times the right side of this bound. The limits of Thm. 8.2 (T3) obey these two bounds too.

Line 3. For $\mathsf{b}' \geq x_s$, all $\rho, \nu \in \mathbb{G}_K(\mathsf{b}')$, all $K \geq 1$ and $n \geq 1$, and tuples of the Euclidean type $(d; R)$,

$$|\rho(\prod(\mathcal{O}_K f_i - \nu(\mathcal{O}_K f_i)))| \leq A_6(\mathsf{b}')^n (n!)^5 \max\{1, d^{-5n}\}(1 + R)^{3n} \prod \|f_i\|_\infty;$$

and for $n \geq 2$, $|S_n^T[\rho]| \leq C_n^\infty(d; R) \prod \|f_i\|_\infty$. In line 3 the label is b'.

Line 4. Let $K_j \to \infty$ and $\mathsf{b}_j \to \mathsf{b} > x_s$. After discarding the finitely many labels outside $U(\mathsf{b})$, the family $(K_j, \mathbb{G}_{K_j}(\mathsf{b}_j))$ is a class system that obeys C^{b}, in the sense of (Cls1), (Cls2), (Bd1) and (Bd2) of E.8.

Proof. The bound on cells is Thm. 8.17 (R2), with Lem. 8.30. Thm. 9.4 (G3), with Lem. 9.7, is a bound of the same kind. It covers more than (R2) in that each occupied cell t has data $(\check{\delta}_t, \check{\rho}_t)$ of its own, and less in that (G3) is stated on the board B_0 alone, for $\check{\delta}_t \in (0, 1]$ and $K \geq \max_t j^\circ(\mathsf{b}, \check{\delta}_t)$. Bounds of the same kind, as seminorms with free centres, are Thm. 10.4 (4.1), (4.2), by Steps 1 and 2 of its proof.

Lines 1 and 2 are Thm. 9.4 (G2) (i), (ii), proved in Lem. 9.9; a form with $C_n^{(2)}$ is Thm. 8.17 (R3), with Lem. 8.33. For the limits of Thm. 8.2 (T3), Thm. 8.17 (R3) gives the bounds with $C_n^{(2)}$; with A_5 they follow from Thm. 9.4 (G2) (i), (ii) at fixed N, as $K \to \infty$.

Line 3 is proved in Thm. 10.4 (4.3) (a), (b), by Step 3 of its proof, and also in Thm. 12.4 (E0), by Step 1 of §12. Line 4 is Thm. 12.4 (E0). □

(Y3) *Limits, for every* $\mathsf{b} > 0$. (a) Continuum first: for each side, the limit as $K \to \infty$ exists along the whole sequence of cutoffs, without passing to a subsequence; every sequence of sides $l_j = 2L^{m_j}$ with $l_j \to \infty$ has a subsequence converging on all tuples; and the same holds for $\mathsf{b}_j \to \mathsf{b}$. (b) Volume first: along one subsequence of sides $l_j \to \infty$, common to every fixed K, the measures $\mu_{K,l_j,L^K\mathsf{b}}$ tend weakly to a state $\nu_K \in \mathbb{G}_K(\mathsf{b})$ that is invariant under W_3 and reflection positive at planes of sites and at mid planes; after this one passes to a subsequence of K. (c) Jointly: every sequence (K_j, l_j) with $K_j \to \infty$ and $l_j \to \infty$ has a convergent subsequence, and so does every such sequence at labels $\mathsf{b}_j \to \mathsf{b}$. (d) If $K_j/l_j^3 \to \infty$, then the limit points in (c) coincide with those in (a) along the same sides and labels.

Every sequence $(K_j, \rho_j, \mathsf{b}_j)$ with $\rho_j \in \mathbb{G}_{K_j}(\mathsf{b}_j)$, $K_j \to \infty$ and $\mathsf{b}_j \to \mathsf{b}$ has a convergent subsequence; in particular $\mathrm{Lim}_{\mathrm{tor}}(\mathsf{b})$ and $\mathrm{Lim}_G(\mathsf{b})$ are not empty. For any $\mathsf{b} > 0$, the limit points of joint sequences (K_j, N_j), in which the N_j are the sides themselves, at labels $\mathsf{b}_j \to \mathsf{b}$ with $N_j \geq N^*(K_j, \mathsf{b}_j)$ are functionals of members of $\mathrm{Lim}_{\mathrm{tor}}(\mathsf{b})$.

Proof. The limits are Thm. 8.17 (R5), with Lem. 8.36, together with Thm. 10.4 (4.3) (c), (4.8) and (4.9) (i). The assertion on joint sequences with $N_j \geq N^*(K_j, \mathsf{b}_j)$ is Thm. 10.4 (4.9) (i), proved in Step 8 of its proof; Thm. 12.4 (E6) is the same statement. □

(Y4) *Every limit* S *of* (R5). Each S_n is real, symmetric and n-linear. The cumulants S_n and centered moments σ_S obey the bounds of lines 1 and 2 of (Y2), with the label conventions of (S8). The family S is invariant under all translations and under W_3, and σ_S is reflection positive at every coordinate mirror. On the set of points of $(\mathbb{R}^3)^n$ whose entries are pairwise different, there is exactly one distribution that takes the value $S_n(f_1, \dots, f_n)$ at $f_1 \otimes \cdots \otimes f_n$ for every smooth separated tuple $(f_1, \dots, f_n)$, and its order on compact subsets of that set is $\leq 3n + 1$. For every integer z, one has $S_n^{\mathsf{b}}(f) = L^{zn} S_n^{L^z\mathsf{b}}(f^{[z]})$ along matched subsequences: each side and cutoff (l, K) is changed to

$(L^z l, K - z)$, with the same order of limits. For moving labels, the matching is $\mathsf{b}_j \mapsto L^z \mathsf{b}_j$ at each member of the sequence. The symmetry group of S is W_3 or $\mathrm{O}(3)$.

The family S determines $\mathcal{H}(S)$, Ω, e^{-tH} with $H \geq 0$, the momenta, W_2, and the class $[X]$ of continuous entries. Its centered functional σ_S obeys the bounds of (Y2), with A_6 for the Euclidean type, and the growth condition Def. C.17 with $(13, 1, C_E(\mathsf{b}), 13)$. The extensions of its moments to $\mathcal{S}_{\neq}$, the Schwartz tests flat on the diagonals, satisfy

$$|S_n^{\sigma_S}(\varphi)| \leq (10^{36} A_5)^n (n!)^5 |\varphi|_{12n+1}.$$

For the extensions of the cumulants S_n, the same estimate has the further factor $1.45^{n+1} n!$. At finite (K, N) of the data, the centered lattice distributions obey the displayed bound for smooth tests flat on the diagonals and supported in $(-N/4+1, N/4-1)^{3n}$, as in Thm. 9.4 (G4)(d). The extensions of σ_S are invariant under translations and W_3, satisfy the positivity eq. C.xxx at coordinate mirrors, and are symmetric.

For $\eta \in (0, 1]$, $F, F' \in \mathbb{C}\mathfrak{S}(\{x_3 > \eta\})$, real $h \in C_c^1(\{|x_3| < \eta/4\})$ and $y \in \mathbb{R}^3$, a field between two vectors obeys a bound free of n,

$$|\sigma_S(\theta \bar{F} \cdot (h) \cdot F')| \leq \check{\Gamma}_*(\mathsf{b}, \eta) \, \|\!|h|\!\|_{\eta, y} \, \|[\tau_{-\eta e_3} F]\| \, \|[\tau_{-\eta e_3} F']\|,$$

in which θ is the map $x_3 \mapsto -x_3$; the bound holds in the restricted form, that is, at a fixed label $\mathsf{b} \geq x_s$, or with $\mathsf{b}_j \geq x_s$ for infinitely many j. At every $\mathsf{b} > 0$ a bound of the same form holds on a thinner slab, with its numbers at the dilated data, by the last bullet of Thm. 9.4 (G6). On the whole slab, (G6+) gives bounds in two cases: in (m), for labels of the restricted form and any $\eta > 0$, a bound of the same form with the numbers of (G6+) (m); in (n), for every $\mathsf{b} > 0$ and $\eta \in (0, 1]$, a bound with its numbers at the dilated data.

The centered functional σ_S has (R0) to (R3), (R5) and (R7) of Def. D.10. For S the following three are equivalent: the symmetry group $\mathrm{O}(3)$; reflection positivity at the one mirror $((e_1 + e_3)/\sqrt{2}, 0)$; and (R4) together with (R6) of D.10.

Proof. The assertions of (Y4) from reality up to the symmetry group are Thm. 8.17 (R6) (i) to (vii), with Lem. 8.37. The objects $\mathcal{H}(S)$, Ω, e^{-tH}, the momenta, W_2 and $[X]$ come from Thm. 8.17 (R7), with Lem. 8.38. The bounds with A_6, the growth condition, the sharper bound and the properties on $\mathcal{S}_{\neq}$ are Thm. 9.4 (G2) (iii), (G4), (G5), proved in Lems. 9.9 to 9.12. The bound on a field between two vectors is Thm. 9.4 (G6), (G6+), with Lem. 9.14: the main clause of (G6) gives it in the restricted form, the last bullet of (G6) gives its form at the dilated data for every $\mathsf{b} > 0$, and (G6+) gives it on the whole slab. The properties of Def. D.10 and the equivalence of the three conditions are Thm. 10.4 (4.9) (ii), proved in Step 8 of its proof. □

(Y5) *Every* $S \in \mathrm{Lim}_G(\mathsf{b})$. In (Y5) the label is in normal form, $\mathsf{b} > x_s$, and every other $\mathsf{b} > 0$ is treated in (Y8). On the lattice the insertion bound holds with $2\,\Gamma(\mathsf{b}', \mathsf{l}) \|f\|_\infty$, and the plates are $\leq C_\varphi t^{-7/2}$. The functional σ_S has (R0) to (R7) of Def. D.10, with C_n^σ, $(c_{\mathrm{I}}, C_{\mathrm{I}}, \mathsf{p}_{\mathrm{I}})$ and $\mathsf{a}_\Gamma = 7/2 < 4$. The family S is invariant under $\mathbb{R}^3 \rtimes \mathrm{O}(3)$, and σ_S is reflection positive at every plane. Moreover $H \geq |P_k|$ for $k = 1, 2$, the joint spectrum lies in $\{E \geq |q|\}$, and $\mathrm{O}(2)$ acts on $\mathcal{H}(\sigma_S)$.

Proof. Clause (Y5) is Thm. 10.4 (4.4), (4.5), (4.6), (4.7), proved in Steps 4 to 7 of its proof. □

(Y6) *Components of* $S \in \mathrm{Lim}_G(\mathsf{b})$. In (Y6) the label is in normal form, $\mathsf{b} > x_s$, and every other $\mathsf{b} > 0$ is treated in (Y8). (E1) Every sequence $(K_j, \nu_j, \mathsf{b}_j)$ with $K_j \to \infty$, $\nu_j \in \mathbb{G}_{K_j}(\mathsf{b}_j)$ and $\mathsf{b}_j \to \mathsf{b}$ has a convergent subsequence. A member S of $\mathrm{Lim}_G(\mathsf{b})$ is obtained along such a sequence if and only if (K_j, ν_j), which by Def. E.9 is a frame in the class system $(K_j, \mathbb{G}_{K_j}(\mathsf{b}_j))$, converges in the sense of eq. E.xxv. (E2) The canonical decomposition exists and is unique up to one isomorphism; $\ker H = L^2(\mathsf{X}, \lambda)$ and $\#\mathsf{X} = \dim \ker H$ when X is finite, and $\ker H$ is infinite-dimensional when X is infinite.

(E3) Every limit σ' in the frame, hence every σ_ξ, coincides, up to the shift sh_c, with the centred functional of some member of $\mathrm{Lim}_G(\mathsf{b})$. It is invariant under all isometries and reflection positive at every mirror, it has order $\le 4n+1$, its first member satisfies $S_1 = c\int f$ with $|c| \le \bar{c}_*$, it obeys the bound of line 3 of (Y2), and it satisfies the growth condition C.17, with $(13, 1, C_E(\mathsf{b}), 13)$ as its quadruple. Moreover $\ker H_\xi = \mathbb{C}\,\Omega$, and σ_ξ clusters.

(E4) Every σ_ξ satisfies (E0′), (E1) for $\mathbb{R}^3 \rtimes \mathrm{O}(3)$, (E2), (E3) and (E4) of F.17, as well as (PtB) and (Rec1) to (Rec5). Its point functions are real-analytic, with $|\mathsf{S}^\xi_n(x)| \le [C_* \max\{1, (2n^2/d(x))^5\}]^n$. It yields a Wightman theory of a single Hermitian scalar field on $\mathbb{R}^{1+2}$.

Proof. Clause (Y6) is Thm. 12.4 (E1) to (E4) of §12. □

(Y7) *Non-triviality on tori and for every limit of* (R5). (D) The number $x^\#$ is finite, $[j_*(\mathsf{b}), \infty)$ lies in $J(\mathsf{b})$, and j_* is bounded on compact sets. The clauses (N0) to (N3) hold for every $j \in J(\mathsf{b})$, hence for all $j \ge j_*(\mathsf{b})$, for all $K \ge j+1$, and for all N of the data.

(N0) $\mathrm{Cov}(G_-, \mathcal{O}_K f)^2 \le \mathrm{Var}(G_-)\, S_2^T(\theta f, f)$ for each real bounded f with $f(x(p)) = 0$ whenever some site of the plaquette p lies outside the region $\frac{1}{2} < y_1 < \frac{1}{2} + NL^j/2$; here the unit of y is that of the level $k = K - j$, namely L^{-j}, and y_1 is represented in the fundamental domain $\frac{1}{2} - NL^j/2 < y_1 \le \frac{1}{2} + NL^j/2$, the unit and the domain being as in "Geometry" and (N0) of Thm. 11.1. Throughout (Y7), θ is the reflection $y_1 \to 1 - y_1$ in units of the level k, and G_- and f_* are those of "Geometry" of Thm. 11.1. (N1) $\mathsf{b}\,\mathrm{Cov}(G_-, \mathcal{O}_K f_*) \ge c_F$. (N2) $\mathrm{Var}(G_-) \le 7$. (N3) $\mathsf{b}^2 S^T_{2;K,N,L^K\mathsf{b}}(\theta f_*, f_*) \ge c_*$, where c_* is a pure number.

(N4) The same bound holds for $S_2^{(N,\mathsf{b})}$ of Thm. 8.2 (T3); it holds for each limit of (R5) (a), (b), (c) taken at the fixed label b; and it holds for (R5) (a), (c) along labels $\mathsf{b}_n \to \mathsf{b}$ with $\mathsf{b}_n \ge x_s$, at every $j \ge j^+(\mathsf{b})$.

(N5) Let τ_s be the translation by the distance s perpendicular to the mirror of θ, towards the side of that mirror on which f_* is supported. Every limit S of (R5) satisfies $S_2(\theta\tau_s f_*, \tau_s f_*) > 0$ for all $s \ge 0$ and has $\dim \mathcal{H}(S) \ge 2$; for a limit along labels $\mathsf{b}_n \to \mathsf{b}$, in place of a fixed label, this holds at the j of (N4) or at the i of (N6′). (N6′) Let f_0 and the mirror θ_i be those of Thm. 11.1. For every $\mathsf{b} > 0$ and every integer i with $L^i\mathsf{b} \ge x^{\#\#}$, the inequality $\mathsf{b}^2 S_2(\theta_i f_0(L^i\cdot), f_0(L^i\cdot)) \ge c_*$ holds for each limit of (R5) taken at the fixed label b; on a torus at the label b it holds provided that $K \ge i+1$ and that, for some integer z with $L^z\mathsf{b} \in [x_s, L^2 x_s]$, the side of the torus meets the conditions that Thm. 8.17 (R4) imposes on a side at that z. With $Lx^{\#\#}$ in place of $x^{\#\#}$, limits along labels $\mathsf{b}_n \to \mathsf{b}$ satisfy it as well.

Proof. Clause (Y7) is Thm. 11.1, proved with Lems. 11.3 to 11.22; its part (D) is Lem. 11.22. □

(Y7′) *Every periodic state, every component.* Throughout (Y7′), g_f, V, P_ρ, θ_*, G_-, f_* and s_ξ are those of Thms. 12.4, 12.5 and 12.6; P_0 is the projection on $\ker H$; and θ is the map $x_3 \mapsto -x_3$, as in (Y4), not the θ of (Y7). Thm. 12.4 (E7) gives, for every $f \in C^1_c(\{x_3 > 0\})$, the identity $S_2(\theta f, f) = \int g_f\, d\lambda + V(\int f)^2$, in which V is a number. For a given ξ, it also states four conditions on the component σ_ξ that are equivalent to one another. A component is trivial if it satisfies them, and non-trivial if it does not.

Let $\mathsf{b}' \ge x_s$, $j \ge j_*(\mathsf{b}')$ and $K \ge j+1$; these are the data of Thm. 12.5. For every $\rho \in \mathbb{G}_K(\mathsf{b}')$ that theorem gives the following.

(P0) In product form, $\mathrm{Cov}_{P_\rho}(G_-, \mathcal{O}_K f)^2 \le \mathrm{Var}_{P_\rho}(G_-)\, S_2^T[\rho](\theta_* f, f)$ for every real bounded local f carried by the open right half.
(P1) $\mathsf{b}'\,\mathrm{Cov}_{P_\rho}(G_-, \mathcal{O}_K f_*) \ge c_F$.
(P2) $\mathrm{Var}_{P_\rho}(G_-) \le 7$.
(P3) $\mathsf{b}'^2 S_2^T[\rho](\theta_* f_*, f_*) \ge c_*$.
(P4) Conditionally on the invariant sets, the same holds.

In normal form, $\mathsf{b} > x_s$, Thm. 12.6 gives the following for every ξ. At every other $\mathsf{b} > 0$, the same conclusions hold with the test and radius of its clause (C4), as stated in (Y8).

(C1) For every $j \geq j^+(\mathsf{b})$, with the mirror carried to $(e_3, 0)$, we have $\mathsf{b}^2 g_{f_*}(\xi) \geq c_*$.
(C2) On $0 < r \leq r_*(\mathsf{b})$ we have $s_\xi(r) \geq c_{**}\mathsf{b}^{-2}r^{-6}$; moreover H_ξ is unbounded, and $\dim \mathcal{H}(\sigma_\xi)$ is infinite.
(C3) $\|(1 - P_0)[(f_*)]\|^2 \geq c_*/\mathsf{b}^2$.

Proof. Clause (Y7′) follows from Thm. 12.4 (E7), from Thms. 12.5 and 12.6, and from Steps 9 to 13 of §12. □

(Y8) *Every* $\mathsf{b} > 0$ *and every side* $2L^m$. By the dilation L^z, the following hold, in their dilated forms, at every $\mathsf{b} > 0$ and every side $2L^m$: the two bounds of lines 1 and 2 of (Y2), on $\sigma_{K,N}(f)$ and on $S_n^T(f)$; lines 3 and 4 of (Y2); the clauses (Y4), (Y5) and (Y6); the parts (N3), (N4) and (N5) of (Y7); and clause (Y7′). The exact finite-torus dilation allows every integer z satisfying the four conditions of Thm. 8.17 (R4), including $K - z \geq 1$. The simplified constants in Thm. 9.4 (G2)(iv) use $z \geq 0$. For finite-cutoff Gibbs-state bounds, take $K \geq z+1$ as well. For the clauses on $\mathrm{Lim}_G(\mathsf{b})$, frames and components, take $z = z(\mathsf{b})$, so that $L^z\mathsf{b} > x_s$, and use the constants of Thm. 12.4 (E5) and the tests and radius of Thm. 12.6 (C4). These choices apply to fixed and convergent labels, including $\mathsf{b} = x_s$, and give (Y5), (Y6) and (Y7′) at every $\mathsf{b} > 0$.

Proof. Clause (Y8) follows from Thm. 8.17 (R4) and (R6) (vi); from Thm. 9.4 (G2) (iv) and (G4) (e), from the last bullet of (G6) and from (G6+) (n); from Thm. 10.4 (4.8) together with Lem. 10.5; from Thm. 12.4 (E5) and Thm. 12.6 (C4); and from Thm. 11.1 (N6) and (N6′). □

(Y9) *Block fields at every depth.* Let $s \geq 0$ be any depth and let $K \geq s + 1$. The block fields at depth s have the following properties.

(a) Locality, conditional independence and covariance hold.
(b) Reflection positivity holds without a collar at every $c \in L^{-s}(\mathbb{Z} + \frac{1}{2})$.
(c) First half: along a further subsequence of sides, the laws of Cor. 5.65 (b) converge locally in total variation; this half uses Thm. 8.2 (T7), which supplies these laws at each side. Second half: every sequence (K_j, N_j) with $N_j \to \infty$ has a subsequence along which the same holds for all $s \leq \liminf K_j - 1$.
(d) Mixed bounds hold, both without cells and with cells.

Proof. Clause (Y9) follows from Thm. 8.17 (R8) and from Lems. 8.25, 8.27, 8.30 and 8.39. □

The closing corollary is Cor. 12.7, and §12 proves it. It states that $\mathrm{Lim}_G(\mathsf{b})$ is not empty for every $\mathsf{b} > 0$, and that every component σ_ξ of every member of $\mathrm{Lim}_G(\mathsf{b})$, and hence of every limit taken volume first, satisfies (E0′) to (E4) of F.17 with full Euclidean invariance and has $S_2^{\sigma_\xi} \neq 0$.

Remark 1.15 (Two orders of distributions). The bound $\leq 3n + 1$ of (Y4), on the order of the distribution whose value at $f_1 \otimes \cdots \otimes f_n$ is $S_n(f_1, \ldots, f_n)$, holds for the limits of Thm. 8.17 (R5); it is Thm. 8.17 (R6) (v). The bound $\leq 4n+1$ of (E3) in (Y6), on the order of the limit σ' in the frame, holds for every limit in a frame of $\mathrm{Lim}_G(\mathsf{b})$; it is Thm. 12.4 (E3) (i), which comes from E.18 (e). These are two upper bounds, on two objects and for two classes.

A limit taken volume first lies in both classes, and so the bound $\leq 3n+1$ holds for the distribution of its S_n.

2. Blocking setup and free kernels

2.1. **Blocking geometry and scales.** The step takes a density at coupling x on a fine torus to a density at coupling x/L on a coarse torus. We specify the class of densities, the geometry of a step and the order of the constants here. Thm. 4.39 shows that one step preserves the class, and Thm. 5.7 iterates this result from Wilson's weight.

The bounds on the free kernels in §2.2 determine the radii used to localize the nonlinear estimates. The parameters in (O0) to (O13) are fixed before the lower bounds on x in (A0) to (A11). We write c_h for the constant in the exponential decay and $c_h(x)$ for the resulting bound at x. In a tower the coupling is $x_k = L^{-k}\beta$. A subscript on a fixed threshold, as in x_9 of Thm. 5.15, is a label for that threshold.

Item 1. Setting. Throughout, $G = \mathrm{SU}(2)$ and $d = 3$.

Definition 2.1 (Lattice)**.** The full definitions are those of items 3.1, 3.3 and 3.5 of §4.2. The radii are those of item 2 below.

The number L is odd, and $\mathsf{h}_L = (L-1)/2$; further $M = L^{m'}$ with $m' \geq 1$; and N is even, with $M \mid N$.

The fine torus is $\mathbb{T} = \mathbb{Z}^3/NL\mathbb{Z}^3$, and the coarse torus is $\mathbb{T}' = L\mathbb{Z}^3/NL\mathbb{Z}^3$.

Fine bonds and fine plaquettes are denoted b and p; coarse ones are denoted c and P.

For an object i, $\pi(i)$ is its base site, and $y(i)$ is the coarse site whose block $B(y)$ contains $\pi(i)$; the distance $\mathrm{dist}(i,j) = L^{-1}|y(i) - y(j)|_\infty$ is measured in coarse units.

The combs are those of item 3.1 of §4.2. I° is the set of the fine bonds off the combs, of which a block has at most $3L^3$, and $\mathfrak{L}_\mathfrak{g}$ is the space of $\mathfrak{g}$-valued functions on I°.

We also use d, Q_1, E, $T(y,y'')$, K^V, ℓ^V, $\mathsf{B}^V = 1 + \ell^{V*}\ell^V$, $q = q_\ell^V$ and $\mathsf{g}^V = \Pi_\mathfrak{L}\mathrm{d}^{V*}\mathsf{B}^V E$. Here $\Pi_\mathfrak{L}$ is the orthogonal projection from Ω^1 onto the $\mathfrak{L}$ of Def. 2.21.

Norms: $|\cdot|$ is the operator norm of a 2×2 matrix, $\|Y\|^2 = \frac{1}{2}\operatorname{tr} Y^*Y$ and $Y \cdot Y' = \frac{1}{2}\operatorname{tr} YY'$. Then $\|Y\| \leq |Y| \leq \sqrt{2}\|Y\|$, and $|Y| = \|Y\|$ on $\mathfrak{g}$. ■

Definition 2.2 (Cubes)**.** At each level the cubes consist of M^3 sites of a fixed grid, and a coarse cube is called a cell $\square$. For a cell, $\hat{\square}$ collects the fine sites that lie in the blocks of the sites of $\square$; for a set A of cells, $\hat{A}$ collects the fine objects based there.

The grids fit: if the grid of cells has the origin o', then the grid of fine cubes has the origin $o = Lo' - \mathsf{h}_L \bmod M$.

Then $\hat{\square}$ is the box of side LM whose lower corner is $L(o' + Mn) - \mathsf{h}_L = o + LMn$, where n is a point of $\mathbb{Z}^3$; it is made up of exactly L^3 fine cubes.

An object belongs to the cube of its base site, and to the cell of $y(i)$.

Distances between cubes, and between cells, are sup distances, measured in cubes and in cells respectively.

$\mathcal{D}$ is the family of the non-empty sets of cubes that are connected through faces, edges or corners.

Count 1. In a graph of degree at most 26, at most $(26e)^{n-1}$ connected sets of n vertices contain a prescribed vertex. This applies to fine cubes and to coarse cells with the same adjacency.

Proof of Count 1. Order the neighbours of each vertex. A breadth-first tree from the prescribed vertex encodes a connected set as a rooted subtree of the tree with 26 ordered children per vertex. This encoding is injective, since following the recorded neighbour indices recovers the set. If $T(z)$ counts these subtrees by size, then $T = z(1+T)^{26}$. For completeness, substituting $z = u/(1+u)^{26}$ in the formal residue for $[z^n]T$ gives

$$[z^n]T = [u^{n-1}](1+u)^{26n-1}(1-25u) = \frac{1}{n}\binom{26n}{n-1}.$$

The last number is at most $26^{n-1}n^{n-1}/n! \leq (26e)^{n-1}$, using $\log n! \geq \int_1^n \log t\,\mathrm{d}t$. ■

$\mathcal{P}$ is the family of the non-empty finite sets of cubes.

For $X \in \mathcal{P}$, $\mathrm{hull}(X)$ is a member of $\mathcal{D}$ that contains X with the least number of cubes, chosen by a fixed order. We put $|X|_c = \#\mathrm{hull}(X)$; for $X \in \mathcal{D}$, $\mathrm{hull}(X) = X$.

Count 2. Given a cube, at most $(52e)^n$ members of $\mathcal{P}$ with $|X|_c = n$ contain it.

Proof of Count 2. Such an X is a subset of $\mathrm{hull}(X)$, and by Count 1 the hull is one of at most $(26e)^{n-1}$ members of $\mathcal{D}$ with n cubes that contain the given cube; as a set of n cubes has 2^n subsets, multiplying the two numbers gives the bound. ∎

For a set X of cubes, X^{+j} is the set of the cubes at distance at most j from X, and for a set X of cells it is the set of the cells at distance at most j from X, the distances being the sup distances set above; and we put $X^{\sim} = X^{+1}$.

For a set A of fine cubes, $\mathrm{cl}(A)$ is the set of the cells $\square$ with $\hat{\square} \cap A \neq \emptyset$.

The bonds of A are the bonds with both ends in A, and the paths and the plaquettes of A are those with all their sites in A. This is the convention of item 3.3 of §4.2 and of item C.II 1.4 of §3, and the definitions of $\mathcal{U}_X(x)$, (ca), (ch), (cΞ), as well as Thm. 5.18, are understood in this sense.

A plaquette p is in A when its base site lies in A; this is the sense of "in" in 1_A and in $D(V) \cap X$, and of the words "holds" and "based in".

We say that a field given on the bonds of A is tame there when $\phi_p \leq \varepsilon$ holds at every plaquette of A.

The convention on "of A" and "in A" is in force throughout Sections 2.1, 4.4, 5.2. ∎

Definition 2.3 (The cells $Y_0(X)$)**.** For a set X of fine cubes let $n = \#X$ if $X \in \mathcal{D}$, and $n = |X|_c$ if $X \in \mathcal{P}$. For $X \in \mathcal{D}$ both choices give the same n, since then $\mathrm{hull}(X) = X$. We call X small if $n \leq L/2 - 2$. The cubes and the cells are ordered once and for all, and "first" refers to these orders.

If X is small, $Y_0(X)$ is the one cell that contains the first cube of X, and $\mathrm{hull}(X)^{\sim}$ lies within $M/2$ coarse units of it.

If X is not small, $Y_0(X)$ is the set of the cells that meet $\mathrm{hull}(X)^{\sim}$; this set is connected, and $\#Y_0 \leq 8 + 432n/L$.

In both cases $\mathrm{cl}(X) \subset Y_0^{+1}$.

Proof for small X. The set $\mathrm{hull}(X)$ is connected, has n cubes and contains the first cube of X; hence every cube of $\mathrm{hull}(X)^{\sim}$ is within $n \leq L/2 - 2$ cubes of that first cube. Put $\square := Y_0(X)$. Then the sites of $\mathrm{hull}(X)^{\sim}$ are within $(L/2-2)M$ fine sites of $\hat{\square}$, and their blocks lie at most $(L/2-2)M/L + 1 \leq M/2$ coarse units away from the sites of $\square$, as $M \geq L$. Since $M/2 < M$, the cells that meet $\mathrm{hull}(X)^{\sim}$, those of $\mathrm{cl}(X)$ among them, lie in $\square^{+1}$. ∎

Proof for X not small. The set $\mathrm{hull}(X)^{\sim}$ is connected and has $m \leq 27n$ cubes; two neighbouring cubes lie in one cell or in two neighbouring cells, so Y_0 is connected. Walking twice along a spanning tree of $\mathrm{hull}(X)^{\sim}$ we obtain a closed walk through all of its cubes with at most $2m$ steps, which we cut into at most $2m/L + 1$ pieces of at most L steps each. Each piece lies in a box of side $L + 1$ cubes, while a cell has side L cubes because the grids fit; hence a box meets at most 2 cells along each coordinate direction, so at most 8 cells altogether, and $\#Y_0 \leq 8(2m/L + 1) \leq 8 + 432n/L$. Finally $\mathrm{cl}(X) \subset Y_0$, since every cell of $\mathrm{cl}(X)$ meets $X \subset \mathrm{hull}(X)^{\sim}$. ∎

∎

Definition 2.4 (At x)**.** The coupling of the step is x, with $x' = x/L$, and a prime marks a quantity evaluated at x'.

We set $p_0 = A_0(\log x)^{\mathfrak{p}_0}$ and $\varepsilon = p_0 x^{-1/2}$; then $\varepsilon' \leq L^{1/2}\varepsilon$.

We set $R = \varrho(x) = \lceil c_R \log x \rceil$, $r(x) = \lceil 20 c_R \log x / M \rceil$, $s(x) = 8r(x)$ and $\ell_w(x) = 6Mr(x) + 12$.

For sets of fine cubes, $X^{\approx} = X^{+r(x)}$ and $X^{\approx\approx} = X^{+s(x)}$; we write $r' = r(x')$ and $s' = s(x')$; for sets of cells, $Y^{\approx} = Y^{+r'}$; and $v = (LM)^3$.

The caps are $c_a(x) = E_a p_0^3 x^{-1/2}$ and $c_h(x) = e^{-c_h p_0^2}$. ∎

Definition 2.5 (Fields, step, pair, energy)**.** *Fields.* We set $\phi_p = \mathrm{ang}\, \partial U(p)$, $F_p = \mathsf{L}(\partial U(p))$ and $D(U) = \{\phi_p > \varepsilon\}$, and on the coarse lattice we use the angles $\phi'_P = \mathrm{ang}\, \partial V'(P)$ and $D' = \{\phi'_P > \varepsilon'\}$.

Here ang and L, with $\mathsf{L}(-1)=0$, are those of item C.I 4 of §3, and F_p, ϕ_p, $D(U)$ are as in item 3.5 of §4.2.

Step. The step is given by

$$\rho'(V') = \int \rho(U) \prod_c k_{ax}(V'(c), \mathsf{M}(U)(c))\, dU.$$

Here k_{ax} is the kernel of the step and $\mathsf{M}(U)$ is the average, both as in item 3.4 of §4.2.

Pair. The second entry of a pair is $e \in \mathfrak{E} = \prod_P S^2$, where S^2 is the unit sphere of $(\mathfrak{g}, \|\cdot\|)$, one copy for each coarse plaquette P; see also item 3.7 of §4.2 and E2 of §4.1.

On each of these spheres we take the probability measure invariant under rotations, and de denotes the product of these measures over all P.

Property 1: $\int_{\mathfrak{E}} de = 1$. Property 2: under de the e_P are independent.

Property 3: for every G-valued u, the map $e \mapsto \mathrm{Ad}_u\, e$, $(\mathrm{Ad}_u\, e)_P = \mathrm{Ad}_{u(y_P)}\, e_P$, where y_P is the base site of P, preserves de, since every $\mathrm{Ad}_{u(y_P)}$ is a rotation of the three-dimensional Euclidean space $(\mathfrak{g}, \|\cdot\|)$.

The mean over e is the integral against de.

We put $X'_P(V', e) = \mathsf{L}(\partial V'(P)) + \pi e_P 1\{\partial V'(P) = -1\}$. At every P one has $|X'_P| = \phi'_P$, and e_P enters only at $P \in D'$.

We write $1_A(V) = 1\{\phi_p \le \varepsilon \text{ for all } p \text{ in } A\}$, and V is tame on A when this indicator is 1.

Energy. Following E2 of §4.3,

$$\mathrm{En}_\ell(U) = \sum_p S_\varepsilon(\phi_p) + \frac{1}{2}\sum_y m_y(U) |\ell_y^U \zeta|^2.$$

The energy $\mathrm{En}'(V')$ is the same expression at x', defined with the rows ℓ'^{c}, the cut of ℓ' at $\varrho(x')$. ■

Definition 2.6 (Domains)**.** We let $\mathcal{U}_X(x)$ be the set of the $\mathrm{SL}(2,\mathbb{C})$ fields V on the bonds of $X^{\approx}$, with $V(\bar b) = V(b)^{-1}$, that satisfy (u1), (u2) and (u3):

(u1) $|V(b)| < 1 + \vartheta(x)$, where $\vartheta(x) = \vartheta_0^\flat(1 - 1/\log x)$;
(u2) $|V(\pi)| < 2 - 1/\log x$ for the paths π of $X^{\approx}$ with at most $\ell_w(x)$ bonds;
(u3) $|\partial V(p) - 1| < 2\varepsilon$ at every plaquette p of $X^{\approx}$.

The domain $U^\tau(B; x')$ is that of item 3.3 of §4.2, given by (e1), (e2), (e3).

$\mathfrak{V}$ is the set of the $\mathrm{SL}(2,\mathbb{C})$ fields on $\mathbb{T}'$ with $|V'(c)| \le 1 + \vartheta_0^\flat$, and $\mathfrak{V}^\circ$ is its interior.

We extend any field that is given on a set of bonds by setting it equal to 1 on all other bonds.

Convention. Let a complex field be given on the bonds of a set B, and let it satisfy (u3) or (e3). We then put $X'_P := -i \log \partial V'(P)$ for those P whose four sites all lie in B, and $X'_P := 0$ for every other P; this is the convention of item 3.7 of §4.2.

The convention does no harm. Indeed, $a_{0,b}$ depends on X' within R_3 of $y(b)$, by the last clause of (Γ3) of Step 0 of the proof of Thm. 4.39; and a letter with $J_\alpha \subset Y$ depends on the field in $Y^{+(r'-2)}$, by item 5 of Step 7 of that proof. Hence none of the values in use depends on the convention. ■

Definition 2.7 (Members)**.** For $A \ge 1$, $M_A^{\mathrm{su}}(x)$ is the set of the triples $u = (E^a, E^h, w)$ that satisfy (ca_A), (ch_A) and $(\mathrm{c}\Xi_A)$:

(ca_A) for $X \in \mathcal{D}$, the function $E^a(X, \cdot)$ is holomorphic on $\mathcal{U}_X(x)$, is invariant under G-valued gauge transformations, and vanishes at the pure gauges, by which we mean here and below the fields 1^u with u G-valued; and $\sup |E^a(X,\cdot)| \le A c_a(x) e^{-\kappa(\#X - 1)}$, the supremum being taken over $\mathcal{U}_X(x)$;

(ch_A) $E^h(X,\cdot)$ is a complex function, Borel and gauge invariant, of those G-valued fields on the bonds of $X^{\approx}$ which are tame there, and $\sup |E^h(X,\cdot)| \le A c_h(x) e^{-\kappa(\#X-1)}$, where the supremum runs over these fields;

(cΞ_A) for $X \in \mathcal{P}$, $w(X,\cdot)$ is a complex function of V on the bonds of $X^{\approx\approx}$, Borel and gauge invariant, and $|w(X,V)| \le A e^{-\kappa(|X|_c - 1)}$ at every V; this bound is the weak cap.

The strong cap is the bound

$$|w(X,V)| \le \exp\{-c_\Xi x\varepsilon \sum_{D(V)\cap X} \phi_p - \kappa(|X|_c - 1)\}.$$

We call u real if E^a is real at G-valued fields and E^h, w are real.

The density of u with the rows ℓ is

$$\rho[u;\ell] = \exp\{-x\mathrm{En}_\ell + \sum_{X\in\mathcal{D}} 1_{X^\approx}[E^a + E^h](X,\cdot)\}\,\Xi_w,$$

where $\Xi_w(V) = \sum_{\{X_i\}} \prod_i w(X_i,V)$; the sum runs over the finite families $\{X_i\}$ of sets in $\mathcal{P}$ at pairwise distance $> s(x)$ such that each X_i holds a plaquette of $D(V)$ and $D(V)$ lies inside their union. The empty family, with the product 1, is admitted if and only if $D(V) = \emptyset$. ■

Definition 2.8 (The class)**.** The class of densities at x with the rows ℓ is

$$C_\flat^{\mathrm{su}}(x;\ell) = \{e^{\mathsf{V}}\rho[u;\ell] > 0 :\ \mathsf{V} \text{ real},\ u \in M_1^{\mathrm{su}}(x) \text{ real, strong cap}\}.$$

The clause "V real" is called (c0); see item 6 of Step 13 of the proof of Thm. 4.39. ■

Definition 2.9 (The data (SD))**.** The data (SD) of a step at x consist of the side N, the grids and the rows ℓ, subject to the following conditions. First, N is even, $M \mid N$, (Tor) $N \ge 2Ms(x)$ holds, and the grids fit.

The rows of (SD) are $\ell \in \mathcal{R}(b_*, K_*, \lambda_*, \frac{1}{9})$ on $\mathbb{T}$, of range $\le \varrho(x)$.

Empty rows are admitted. The family $\ell = \emptyset$, with $\mathsf{H}_y = \{0\}$ for all y, lies in $\mathcal{R}(1,0,\lambda_*,0) \subset \mathcal{R}(b_*, K_*, \lambda_*, \frac{1}{9})$, by Def. 2.48, with its display and its last sentence; its range is 0.

At $\ell = \emptyset$ we have $\mathrm{En}_\ell = \sum_p S_\varepsilon(\phi_p)$, $\mathsf{B}^V = 1$ and $q = |\mathrm{d}^V \cdot|^2 + a|Q_1^V \cdot|^2$; and the numbers b, N_B, γ, ϱ_L and all the radii keep the values that belong to $(K_*, \lambda_*, \varrho(x))$.

(SD) involves no level and no depth.

The last clause of (SD) is a condition on the rows: the pair (ℓ, ℓ'), where ℓ' denotes the rows built from ℓ by Prop. 2.28 (with $\mathsf{B}' = 1 + \ell'^{\mathsf{T}}\ell'$, tensored with $1_\mathfrak{g}$; at $\ell = \emptyset$ the group (5) of the list of that proposition is void), satisfies the four conclusions of Thm. 2.64 listed below.

(1) (Dec)(C_D, θ_D);
(2) (N′) with eq. 2.vii at every cut;
(3) $\mathsf{t}_\rho \le \frac{1}{8}(\frac{1}{2} - d_0 - c_\Xi)$;
(4) (L1), (L2), taken as (L1^{su}), (L2^{su}).

Here (C_D, θ_D) are those of (O6).

The number

$$\mathsf{t}_\rho := \|\ell'^{\mathrm{c}}\|_{\mathrm{r}} \sup_{P_0} \sum_{y' :\, \mathrm{dist}(y', P_0) > \rho_0} |\ell'^{\mathrm{c}}_{y'}(P_0)|,$$

as in E5 of §4.3, is a quantity of the rows.

The part "Conclusion" of Thm. 2.65 contains (Fr0), (Fr2), (Fr3′) and (Fr4). At the (C_D, θ_D) of (O6) and for the pair (ℓ, ℓ') of that theorem, the conclusions (1), (2), (3) are (Fr2) for the sixteen kernels, (Fr3′) and (Fr4), respectively, and (Fr0) gives the two conditions of (SD) on the rows, that $\ell \in \mathcal{R}(b_*, K_*, \lambda_*, \frac{1}{9})$ and that ℓ is of range $\le \varrho(x)$.

The four conclusions of (SD) imply (Fr2), (Fr3) of item C.II 2.2 of §3, and (Fr1) is Lem. 2.38 (c), which is algebra and holds for all rows. ■

Item 2. The radii.

Definition 2.10 (The radii)**.** The radii set here are not letters of the class $C_\flat^{\mathrm{su}}$. In coarse units we set

$$\varrho_L = \lceil R/L \rceil, \quad r_w = \lfloor R/2 \rfloor, \quad r_G = r_w + \varrho_L + 1, \quad R_2 = 2R + 2, \quad R_3 = 2R + \varrho_L + 5,$$

$$R_\chi = R + \varrho_L + 2, \quad R_v = 2R + 2\varrho_L + 7, \quad R_6 = 4R + 6\varrho_L + 11,$$

$$D_w = \lceil \ell_w(x)/L \rceil + 1, \quad \omega = M + \lceil Mr(x)/L \rceil + D_w + R_3 + 2.$$

In cells we set $r_l = \lceil (\varrho_L + 2)/M \rceil$, $r_1 = \lceil R_2/M \rceil + 3r_l + 5$ (the r_1 of E8 of §4.3), $s_0 = 2r'$ and $\varrho_* = 2r_1 + \lceil s(x)/L \rceil + 2$.

Identically, $r_1 \geq 9$ and $r_1 M \geq 2R + 3\varrho_L + 8 + 5M$; hence $r_1 M \geq R_v + 1$, $r_1 M \geq R_\chi$ and $r_1 M + 1 > 2(R_\chi + 1)$; and $\lceil R_2/M \rceil \leq r_1 - 8$.

Proof. Since $\lceil R_2/M \rceil \geq 1$ and $r_l \geq 1$, we find $r_1 \geq 9$ and $\lceil R_2/M \rceil = r_1 - 3r_l - 5 \leq r_1 - 8$. Since $M\lceil R_2/M \rceil \geq R_2$ and $Mr_l \geq \varrho_L + 2$, we find $r_1 M \geq R_2 + 3(\varrho_L + 2) + 5M = 2R + 3\varrho_L + 8 + 5M$. Finally $R_v + 1 = 2R + 2\varrho_L + 8$ and $2(R_\chi + 1) = 2R + 2\varrho_L + 6$, and $M \geq 1$; together with the last bound these give the three inequalities for $r_1 M$. ■

■

Item 3. The letters, in the order of the table.

Definition 2.11 (The table)**.** We choose the constants of Sections 2.2 to 5.5 in the order in which they are listed below. Wherever one constant is subject to two conditions, we impose the stricter of them: at (O11) it is the smaller bound, at (O12) and (O13) the larger value.

The entries are as follows.

(O0) The pure numbers $b_*, \gamma_*, \lambda_*, K_*, L_A, L_B, c_1^*, c_2^*, \lambda_2^*, \theta_J$, and $\mathsf{g}_* = \mathsf{g}(\lambda_*)$. All of them but L_B are those of §2.2; L_B is a pure number set in §5.4, (O0), and it enters here only through $L \geq L_B$ in (O4).

(O1) $d_0 \in (0, \frac{1}{2})$.

(O2) $0 < c_\Xi \leq \frac{1}{2}(\frac{1}{2} - d_0)$.

(O3) The integer ρ_0, fixed by eq. 2.vi of §2.2.

Then the following functions of L: $\mathsf{I}_L = 1 + \log L$, $\mathsf{N}_Z = \mathsf{N}_{\mathsf{B}} = 1 + 3K_*^2\mathsf{g}_*^2$, $\mathsf{P}_\rho = 1 + 3(2 + 3(2\rho_0 + 1)^3)K_*^2\mathsf{g}_*^2$ (this is eq. 2.vii of §2.2), and

$$\mathsf{N}_\sharp = c_1^*\mathsf{I}_L\mathsf{N}_Z\mathsf{P}_\rho, \quad \mathsf{N}_\Xi = c_2^*\mathsf{N}_Z, \quad \mathsf{N}_\nu = \mathsf{N}_\sharp + (3 + 4c_2^*L)\mathsf{N}_Z, \quad \mathsf{N}_\sigma = 3K_*\mathsf{g}_*c_1^*\mathsf{I}_L\mathsf{N}_Z,$$

$$\mathsf{N}_\nu^\flat = \mathsf{N}_Z + 2L\mathsf{N}_\Xi + (3 + 3(2\rho_0 + 1)^3)K_*\mathsf{g}_*\mathsf{N}_\sigma, \qquad \mathsf{N}_{\mathrm{c}} = 17(\mathsf{N}_\sharp + 1).$$

(O4) L is odd and $L \geq L_*^{\mathrm{su}}$, where L_*^{su} is the least integer $\geq \max\{L_A, L_B\}$ such that every $L \geq L_*^{\mathrm{su}}$ satisfies the three conditions below:

$$[\mathsf{N}_\nu/L + 2200\mathsf{N}_Z + 40\mathsf{N}_\Xi]L^{-1/2} \leq d_0/4, \tag{L1^{su}}$$

$$[\mathsf{N}_\nu^\flat/L + 2200\mathsf{N}_Z + 40\mathsf{N}_\Xi]L^{-1/2} \leq \frac{3}{16}d_0, \tag{L1$^\flat$}$$

$$L^{1/2} \geq 1.2 \cdot 10^4\mathsf{N}_{\mathrm{c}}^2. \tag{L2^{su}}$$

Further, $c_L = (L^2 - 1)/(6L^2)$ and $a \geq 3/(c_L L)$, so that $aL \geq 18$.

Of these three conditions, the proof of Thm. 4.39 itself depends on (L2^{su}) alone. It gives $L \geq 1.2 \cdot 10^{13}$, as $\mathsf{N}_{\mathrm{c}} \geq 17$, and $T := L^{3/2}/(19\mathsf{N}_{\mathrm{c}}) > 1$.

Indeed, $T \geq (1.2 \cdot 10^4\mathsf{N}_{\mathrm{c}}^2)^3/(19\mathsf{N}_{\mathrm{c}}) \geq 1.7 \cdot 10^{12} \cdot 17^5/19 > 10^{17}$, so that $T \geq 2$ as well.

(O5) $\kappa \geq 6$, and $\kappa^\sharp = 10(\kappa + 4)$.

(O6) Formed from (L, a, K_*, λ_*) as in item C.I 1.2 of §3: $\gamma = (73L^4)^{-1}$, $b = 16\mathsf{N_B} + a/L$, $A^\flat$, $\theta^\flat$, $\theta_{\mathbb{C}} = \theta^\flat/6$, $\vartheta_0^\flat = \theta^\flat/24$, θ_2, $\Lambda = 2 + A^\flat/b$, $\mu = \gamma\theta_2/(4b\log\Lambda)$, and $\mathsf{G}_0 \le C_\varphi^\flat$.

Next, $\theta_D = \frac{1}{2}\min\{\lambda_2^*, \theta_{\mathbb{C}}, \theta_J\}$.

The number C_D is at least the expression that Thm. 2.65 gives for C_D in item (6) of its part "The order of the constants", evaluated at this θ_D; that expression is $\ge A_q \ge 32e^2$, so $C_D \ge 4$.

The number C_D belongs to the table and is fixed here, before any rows are chosen: for every C at least that expression and every $\ell \in \mathcal{R}(b_*, K_*, \lambda_*, \frac{1}{9})$, (Dec)$(C, \theta_D)$ holds, by Step H13 of the proof of Thm. 2.65, which is Prop. 2.62 with $\theta_{\mathbb{C}}$ in place of θ_q. Every later constant rises with C_D, by (Fr2) of item C.II 2.2 of §3.

Then $c_R = 64/\theta_D$, so that $c_R\theta_{\mathbb{C}} \ge 128$.

The condition (R$^\flat$) $c_R\mu \ge 1.35$ follows, by a chain of four links. First, $\lambda_* L \ge 12$ by $L \ge L_A$, see item C.II 5 of §3, under "Conditions on the letters"; hence $\theta_2 = \frac{1}{2}$ and $\theta^\flat = \gamma/(2A^\flat)$.

Second, $c_R \ge 768/\theta^\flat = 1536A^\flat/\gamma$, and $c_R\mu \ge 192\, t/\log(2+t)$ with $t = A^\flat/b \ge 14$.

Third, $A^\flat \ge 14b$, by the "Conditions on the letters" in item C.II 5 of §3: there $2e^2 \ge 14$ and $384e^8\mathsf{g}_2(\lambda/2)^2 \ge 14 \cdot 48\mathsf{g}(\lambda)^2$, and a quotient of two sums is at least the least quotient of their terms.

Fourth, $t/\log(2+t)$ increases with t; hence $c_R\mu \ge 969$.

After these come $\mathsf{K}_{\mathrm{c}}^0$ and a_{c}, from item C.II 4 of §3, under "Constants".

(O7) $M = L^{m'}$ with (M2) $M \ge 1920(\kappa+4)c_R$ and (M4$^\flat$) $M \ge (54/\mu)(\kappa^\sharp + 3)$. The condition (M4$^\sharp$) is a consequence of (M4$^\flat$). Further (M5r) $M \ge \rho_0$, for Thms. 5.48 and 5.63.

After these come $\kappa_+ = M/(40c_R)$ and $\mathsf{K}_1^\flat$, the number of item C.II 5 of §3.

(O8) $\mathfrak{p}_0 > 2$, strictly, and $A_0 > 0$.

(O9) $c_\xi = (6.5C_\varphi^\flat)^{-1}$, and $\mathsf{n} = 60$.

(O10) First, the numbers that come from the cited theorems. $\mathsf{K}_{\mathrm{c}} := \sqrt{2}\mathsf{K}_{\mathrm{c}}^0$ is used for (Cen) alone, that is Cor. 3.38 (a); there it is the corollary's own number, for each $0 < \hat{\varepsilon} \le \frac{1}{40}$.

We take the numbers $\mathsf{K}_1^\flat$ and $1.5\mathsf{K}_1^\flat$ in the two bounds on $\tilde{\mathsf{D}} = -\mathsf{D}_k/2$, of Cor. 3.43; further $K_{\mathrm{q}} := K_{\mathrm{q},k}$, of Thm. 4.2, and $K_v := K_v^\flat$, of Thm. 4.5.

From item F of §4.3: h_*, u_b, u_t, u_1, $\delta_3 = \min\{d_0/1536,\ 3c_\xi^2/(256h_*^2)\}$, $e_1^\flat$, Res and e^o, with the primed u_t', u_1', $\delta_3' = \min\{d_0/2304,\ 3c_\xi^2/(256h_*^2)\}$ and $e^{o\prime}$.

Our own numbers are $a_\infty := 4\pi\mathsf{N_B}(C_\varphi^\flat)^2$ and $n_a := (3v/2)a_\infty^2$.

As to dependence: first, $\mathsf{K}_{\mathrm{c}}, \mathsf{K}_1^\flat, K_{\mathrm{q}}, K_v, h_*, \delta_3, \delta_3', a_\infty, n_a$ are numbers; none of them varies with c_h, E_a, x, N or a member, and of ℓ only the pure numbers b_*, K_*, λ_* and the range enter them.

Second, $u_b, u_t, u_1, u_t', u_1', e_1^\flat$ are functions of x built from those numbers and $(A_0, \mathfrak{p}_0)$, and u_t, u_t' depend in addition on $r_1(x)$.

Third, Res, e^o, $e^{o\prime}$ depend in addition on the ϕ_P' of D' and on the t_ρ of the rows; no floor contains them.

Fourth, not one of all these quantities depends on c_h, E_a, N or a member.

(O11) $0 < c_h \le \delta_3/2$. The table takes $c_h \le \delta_3'/2$: Thms. 5.3 and 4.39 require $c_h \le \delta_3/2$, and $\delta_3' \le \delta_3$ by (O10).

(O12)
$$E_a \ge E_a^0 := \max\{32L^{-1/2}K_v,\ 32K_{\mathrm{q}}\}$$
. The table takes the larger value $\max\{32K_{\mathrm{q}},\ 1500L^{-1/2}K_v\}$; Thm. 5.3 requires $E_a \ge E_a^0$ alone.

(O13) Last: $x_\flat \ge x_\flat^0 = x_\flat^0(c_h, E_a) := 1 + \inf S(c_h, E_a)$, where $S(c_h, E_a)$ is the set of those $y \ge e^2$ for which (A0) to (A11) hold at all $x \ge Ly$.

The set S is an upper half line, and it is not empty, by Step 14 of the proof of Thm. 4.39.

The set S need not contain its infimum. The floors contain ceilings, namely r', r_1, $\lceil s(x)/L \rceil$, R, ϱ_L, which are continuous from the left and jump upwards on the right; so a least such number need not exist.

But $1 + \inf S \in S$, and it depends on what S depends on.

The table takes $x_\flat \geq \max\{x_\flat^{(1)}, x_\flat^0, x_9/L\}$, together with (X12).

Here $x_\flat^{(1)}$ is the number of Thm. 5.3, namely $x_\flat^0$ evaluated at $(c_h, E_a) = (\delta_3/2, E_a^0)$, and hence computed from (O0) to (O10) alone.

And x_9 is that of (O13) of §5.3, determined by (O0) to (O12) alone; so there is no circle.

§5.5 adds (X13) to (X18) under its (O13).

■

Item 4. The floors.

Definition 2.12 (The numbers of the floors)**.** We set $r_\eta := (0.22 + L^{1/2}\mathsf{a_c})\varepsilon$, $r_T := \sqrt{2}C_\varphi^\flat c_\xi \varepsilon + 17T\mathsf{a_c}\varepsilon'$, $\varepsilon'' := x^{-1/4}$ and $\upsilon := e^{-c_\Xi p_0^2}$.

We set $r_\Xi := \frac{1}{2} - d_0 - \mathsf{t}_\rho - 3e_1^\flat - c_\Xi$; it depends on the rows through t_ρ.

We set $r_\Xi^\flat := \frac{7}{8}(\frac{1}{2} - d_0 - c_\Xi) - 3e_1^\flat$; it depends on x and the letters alone.

By (SD), $\mathsf{t}_\rho \leq \frac{1}{8}(\frac{1}{2} - d_0 - c_\Xi)$; hence $r_\Xi \geq r_\Xi^\flat$. ■

Definition 2.13 (The floors (A0) to (A11))**.** The floors are the following conditions on x.

(A0) *Part 1.* The floor $x \geq x_{\mathrm{fl}}$ of item G of §4.3, which contains (W); ($\mathrm{F_{law}}$) of Thm. 3.6 (k); $x \geq L^2$, $R \geq 6$, (P1) and (P3) of item C.II 4 of §3; and (S1), (S2) of item C.II 5 of §3, with $r_u = \lceil R_2/M \rceil$.

Part 2. The floors of Thm. 4.2, that is ($\mathrm{G1}_k$), ($\mathrm{F8}_k$), ($\mathrm{F9}_k$), and those of Thm. 4.5, that is (W0) to (W5).

On (P0). Of (P0), which (W0) names, only the two clauses $x \geq L^2$ and $R \geq 6$ are floors. In item C.II 4 of §3, under "Floors", (P0) carries a third clause, (Tor). It involves N: for fixed N it is violated by every large enough x. So (Tor) is a datum of (SD) and no floor, and the same is true in the other sections from which (A0) takes its floors, for instance in Step 14 of the proof of Thm. 4.5, in item G of §4.3, and in item G of §4.1. Where a statement taken from another section and listed in Step 0 of the proof of Thm. 4.39 uses (Tor), (SD) supplies it, as for the statements (Γ3) and (Γ4) of that step. Hence none of the conditions that define $S(c_h, E_a)$ involves N.

Part 3. $3D_w \leq \ell_w(x')$; $17T\varepsilon' \leq \frac{1}{40}$; and (D2), (D3) of Thm. 5.18 at $(\tau, r) = (1, r_\eta)$ and at $(17T, r_T)$.

Here (D2)(τ, r) reads $\ell_w(x)(7.1r + 18D_w\tau\varepsilon') \leq \mathsf{g}_x/2$, and (D3)$(\tau, r)$ reads $(4.1r + 2.1\tau\varepsilon')^2 \leq 0.1\varepsilon$.

Part 4. (C0), (C1), (C2) of Thm. 5.22, at this ω and this T.

These are: (C0) $18\omega\varepsilon' \leq \frac{1}{10}$ and $160T\omega\varepsilon' \leq 1$; (C1) $2640\,\ell_w(x')(T+1)\omega\varepsilon' \leq \mathsf{g}_x$; (C2) $102400\,T\omega^2\varepsilon' \leq 1$.

Part 5. $\mathsf{K_c}(17T\varepsilon')^2 \leq 0.12\varepsilon$.

Notes. Here $\mathsf{g}_x = \vartheta_0^\flat \log L/(4(\log x)^2)$. The condition (D0)$(\tau)$ is: $2D_w + 1 < N$, $3D_w \leq \ell_w(x')$, $\tau\varepsilon' \leq \frac{1}{40}$; its first clause is no floor, since it follows from (Tor), see Step 0 of the proof of Thm. 4.39, at the clauses on N.

(A1) $\varepsilon \leq \frac{1}{40}$; $\rho_0 \leq \varrho(x')$; $\varrho_L + 2 \leq \lceil R/2 \rceil$; and $0.72(4.1r_\eta + 2.1\varepsilon')^2 + \mathsf{K_c}\varepsilon'^2 \leq 0.1\varepsilon$.

(A2) $5r_1 + 2\lceil s(x)/L \rceil + 6 \leq r'$, $\omega \leq Mr'$ and $R_6 + 1 \leq Mr'$.

(A3) $50x^{-1/8}(2(s_0 + \varrho_* + 3r_1 + 2) + 1)^3 \leq 1$.

(A4) $(c_a + c_h)(x) \leq \frac{1}{2}$.

(A5)
$$x^{1/4}(2s'+1)^3\cdot 50e^{(\kappa+1)s'}\left[5e^{\kappa+3}L^3(c_a+c_h)(x)+2e^{\kappa+3}K_v p_0^3 x^{-1/2}+x^{-20}+e^{\kappa+3}u_b+u_1\right]\le x^{-1/8}.$$

(A6)
$$\left[u_t e^{\kappa+5}+220(2r_1+1)^3L^3e^{\kappa+5}\upsilon+6((2r_1+1)L+2r(x))^3e^{\kappa+5}(c_a+c_h)(x)\right]e^{(\kappa+4)s'}\le\frac{1}{2}\varepsilon''.$$

(A7) $3e_1^\flat\le\frac{1}{8}(\frac{1}{2}-d_0-c_\Xi)$; and, with
$$\upsilon_f:=\exp\{-r_\Xi^\flat p_0'^2+1.01\mathsf{K}_1^\flat(2r'+1)^3+4+(\kappa+4)(1+2s')\},$$
defined with $r_\Xi^\flat$ and not with r_Ξ so that this floor depends on no rows, $e^{(\kappa+1)s'}\upsilon_f^{1/2}\le\frac{1}{4}\varepsilon''$.

(A8) $(p_0'/p_0)^4\ge\frac{1}{2}$; $25.5\mathsf{K}_1^\flat\le E_a p_0'^2$; $111\cdot 4.86L^3x^{-1/4}\le 0.0026L^{1/2}$; $111x^{-1/4}(2K_v/E_a+1)\le 0.003L^{1/2}$; $x^{-20}\le c_a(x)$; and $2x^{-20}\le 0.001c_a(x')$.

(A9) $10^3L^3e^{-c_h(p_0^2-p_0'^2)}\le\frac{1}{2}$; $e^{\delta_3 p_0^2/2}\ge 200$; and $250v(c_\xi p_0+1/(c_\xi p_0))e^{-0.49c_\xi^2p_0^2}\le\frac{1}{4}$.

(A10) $4.04x^{-20}\le(\gamma/2)x^{-2}$; $2.02x^{-19}n_a+38.8v\gamma^{-1}x^{-20}(2r_1+1)^3\le x^{-1/8}$; and $\varsigma':=4.04x^{-20}\cdot 3L^3(R+1)^3(C_\varphi^\flat)^2\le\frac{1}{2}$ together with $9v\varsigma'(2r_1+1)^3+2.02x^{-19}n_a\le x^{-1/8}$.

(A11) $\mathsf{K}_1^\flat e^{-\kappa^\sharp r'}\le 1$, and $(2r'+1)^3\cdot 2.01\mathsf{K}_1^\flat e^{\kappa+1}e^{-(\kappa+26.74)r'}\le 1$.

■

Remark 2.14 (Independence of the floors). No floor depends on N, since (Tor) is a datum of (SD) and no floor, by (A0); and no floor depends on ℓ beyond the pure numbers b_*, K_*, λ_* and the range, nor on a member, since t_ρ enters none of them, (A7) being defined with $r_\Xi^\flat$.

Hence one and the same number $x_\flat^0$ applies to every choice of rows in (SD), the empty rows included, every N and every level k.

2.2. **Free kernels and row estimates.** The quadratic part of one blocking step is a constrained minimization. The fine quadratic form is $1+\ell^{\mathsf{T}}\ell$, with rows ℓ indexed by sites. We first identify the minimizer and show that its coarse form again has a sum-of-squares representation.

To iterate the construction, we need estimates on the new rows that are uniform in the number of steps. The exponential bounds below give this control and quantify the error from cutting off distant entries. The sharper sup bounds control the centre of the Gaussian integration. Thm. 2.65 assembles these estimates at the parameters used in the nonlinear argument.

Remark 2.15 (Letters with several senses). The letter K is the bound of the rows in $\mathcal{R}(b,K,\lambda,\tau)$, the number of steps in L^{K-k}, and a matrix in $\|K\|_\theta$.

On the base line of a formula, T is the kernel of Def. 2.35, while a raised ^{T} is the transpose.

The letter N, when it denotes the side of a torus, has two senses, and both occur in Thm. 2.57: where a single step is treated, as in part (a) of that theorem, N is the side of the coarse torus of that step; where a tower is treated, as in part (b), N is the side of the last torus $\mathbb{T}_K$, so that step k has a coarse torus of side $N'=NL^{K-k-1}$.

Remark 2.16 (Standing conditions with $a>0$). In the next five subsections, $L\ge 3$ is odd, $N\ge 2$ is even and $a>0$. We state the stronger bounds on a where they first enter. The row iteration will use $aL\ge 3/c_L$.

2.2.1. *Lattices, forms, averages and distances.*

Definition 2.17 (Lattices, forms and the four averages)**.** Let $L\ge 3$ be odd, $\mathsf{h}_L=\frac{1}{2}(L-1)$ and $c_L=\frac{L^2-1}{6L^2}<\frac{1}{6}$. Let $N\ge 2$ be even. The fine and the coarse torus are $\mathbb{T}=\mathbb{Z}^3/NL\mathbb{Z}^3$ and $\mathbb{T}'=L\mathbb{Z}^3/NL\mathbb{Z}^3$, and Ω^i,Ω'^i denote the real i-cochains on them. We write d and $\delta=\mathrm{d}^*$ for the fine coboundary and its adjoint, and d', δ' for the coarse ones; on 2-forms, d'^* denotes the adjoint of d' on 1-forms, whereas δ' on 3-forms is the adjoint of d' on 2-forms. Fine bonds, plaquettes and

cubes are denoted b, p, q, and coarse ones c, P, q'. The block $B(y)$ is centred at y; W_c is the double block of c, and W_P is made of the four blocks at the corners of P.

The four averages are $(Q_0\lambda)(y) = L^{-3}\sum_{x\in B(y)}\lambda(x)$ and $(Q_1 A)(c) = L^{-3}\sum_{x\in B(c_-)} A([x,x'])$, where, for $c=(y,\mu)$, $c_- = y$, $x' = x + Le_\mu$ and $[x,x']$ is the straight path from x to x'; then $W = Q_2$, given by

$$(WF)(y;\mu\nu) = L^{-3}\sum_{x\in B(y)}\sum_{0\le i,j<L} F(x+ie_\mu+je_\nu;\mu\nu),$$

and Q_3, given by

$$(Q_3\Theta)(y) = L^{-3}\sum_{x\in B(y)}\sum_{0\le i,j,k<L}\Theta(x+ie_1+je_2+ke_3).$$

■

Lemma 2.18 (Facts on the averages)**.** Let $t(s) = L - |s - \mathsf{h}_L|$ for $-\mathsf{h}_L \le s \le 3\mathsf{h}_L$, and $t(s) = 0$ for all other s. In the kernel formulas below, t and the indicator of $[-\mathsf{h}_L, \mathsf{h}_L]$ are periodized modulo NL. Thus, for example, $t(z_\mu - y_\mu)$ in a kernel means $\sum_{m\in\mathbb{Z}} t(z_\mu - y_\mu + mNL)$. For $p = (z;\mu\nu)$ and $P = (y;\mu\nu)$ we have, with λ the third direction, normal to p,

$$W(P,p) = L^{-3}t(z_\mu - y_\mu)t(z_\nu - y_\nu)1\{|z_\lambda - y_\lambda| \le \mathsf{h}_L\},$$

and, for the coarse cube q' at y and the fine cube q at z,

$$Q_3(q',q) = L^{-3}\prod_\mu t(z_\mu - y_\mu) =: \psi_{q'}(q).$$

The function t satisfies $\sum_{y\in L\mathbb{Z}} t(s-y) = L$, $\sum_s t(s) = L^2$ and $t \le L$.

The averages intertwine the coboundaries: $\mathrm{d}'Q_i = Q_{i+1}\mathrm{d}$. For $i=1$ this follows from $(\mathrm{d}'Q_1A)(P) = L^{-3}\sum_x A(\partial S_x)$, where S_x is the $L\times L$ square at x. For $i = 0, 2$ the corresponding identities are that $(\mathrm{d}'Q_0\lambda)(c)$ equals L^{-3} times the sum over $x \in B(c_-)$ of λ on the boundary of the path $[x,x']$, and that $(\mathrm{d}'WF)(q')$ equals L^{-3} times the sum over x of F on the boundary of the $L\times L\times L$ cube at x.

The kernel of W satisfies $W(P,p) \ge 0$, $\sum_P W(P,p) = L^{-1}$, $\sum_p W(P,p) = L^2$ and $W(P,p) \le L^{-1}$, and $W(P,p) \ne 0$ only for $p \subset W_P$. Further, $0 \le \psi_{q'} \le 1$, $\sum_{q'}\psi_{q'} = 1$ and $\sum_q \psi_{q'}(q) = L^3$. By Schur's test, $\|Q_1\|^2 \le L^{-1}$, $\|W\|^2 \le L$ and $\|Q_3\|^2 \le L^3$. Moreover, $\sum_s t(s)^2 = \frac{1}{3}L(2L^2+1)$ and $\sum_s t(s)t(s-L) = \frac{1}{6}L(L^2-1)$.

Let $\Delta'_\mu = 2 - \mathsf{S}_\mu - \mathsf{S}_\mu^*$, where S_μ is the coarse shift. Then $Q_3Q_3^* = L^3\hat{\mathsf{J}}$, with $\hat{\mathsf{J}} = \prod_\mu(1 - c_L\Delta'_\mu) \ge (1-4c_L)^3 \ge \frac{1}{27}$, and on the component $\mu\nu$, $L^{-1}WW^* = (1-c_L\Delta'_\mu)(1-c_L\Delta'_\nu)$. Furthermore, $\mathsf{P} := 1 - L^{-1}WW^* - (aL)^{-1}\mathrm{d}'\mathrm{d}'^* \ge 0$ if $aL \ge 3/c_L$. Indeed, $\mathrm{d}'\mathrm{d}'^* \le 2\,\mathrm{diag}(\Delta'_\mu + \Delta'_\nu)$; on the component $\mu\nu$ the operator $1 - L^{-1}WW^*$ equals $c_L(\Delta'_\mu + \Delta'_\nu) - c_L^2\Delta'_\mu\Delta'_\nu$, and as $\Delta'_\mu\Delta'_\nu \le 2(\Delta'_\mu + \Delta'_\nu)$,

$$c_L(\Delta'_\mu + \Delta'_\nu) - c_L^2\Delta'_\mu\Delta'_\nu \ge (c_L - 2c_L^2)(\Delta'_\mu + \Delta'_\nu) \ge \frac{2}{3}c_L(\Delta'_\mu + \Delta'_\nu);$$

lastly, $\frac{2}{aL} \le \frac{2}{3}c_L$. The condition $aL \ge 3/c_L$ also gives $aL \ge 18$. ■

Definition 2.19 (Base sites and the two distances)**.** Every cell i has a base site $\pi(i) \in \mathbb{T}$, which for a coarse cell is a point of $\mathbb{T}' \subset \mathbb{T}$. The fine distance, in block units, is $\mathsf{d}(i,j) := L^{-1}|\pi(i) - \pi(j)|_\infty$; it is a pseudo-metric on all cells of the step. Let $y(i)$ be the coarse site whose block contains $\pi(i)$; the coarse distance is $\mathrm{dist}(i,j) := L^{-1}|y(i) - y(j)|_\infty$. Since $|\pi(i) - y(i)|_\infty \le \mathsf{h}_L$, we have $|\mathsf{d} - \mathrm{dist}| < 1$, and $\mathsf{d} = \mathrm{dist}$ for two coarse cells. Finally, $\mathsf{g}(\lambda) := \sum_{n\in\mathbb{Z}^3} e^{-\lambda|n|_\infty}$. ■

Definition 2.20 (The weighted norm of a matrix between cells)**.** For $\theta \ge 0$ and a matrix K between two kinds of cells we set

$$\|K\|_\theta := \max\{\sup_i \sum_j |K(i,j)|e^{\theta\mathrm{dist}(i,j)},\ \sup_j\sum_i |K(i,j)|e^{\theta\mathrm{dist}(i,j)}\}.$$

The triangle inequality gives $\|K_1K_2\|_\theta \le \|K_1\|_\theta\|K_2\|_\theta$. By Schur's test, the norm $\|K\|_\infty$ of K as a map $\ell^\infty \to \ell^\infty$ and its operator norm $\|K\|$ on ℓ^2 are both $\le \|K\|_0$. Moreover, $\sup_i \sum_{\mathrm{dist}(i,j)>R} |K(i,j)| \le e^{-\theta R}\|K\|_\theta$. ■

2.2.2. *Edge lifts and the slice.*

Definition 2.21 (Edge lifts, combs and the slice)**.** The crossing bonds of a coarse bond $c = (y, \mu)$ are the fine bonds $b = (z, \mu)$ with $z_\mu = y_\mu + \mathsf{h}_L$ and $|z_\nu - y_\nu| \le \mathsf{h}_L$ for $\nu \ne \mu$; among them, b_c is the one with $z = y + \mathsf{h}_L e_\mu$. The edge plaquettes under a coarse plaquette $P = (y; \mu\nu)$ are the plaquettes $p = (z; \mu\nu)$ with $z_\mu = y_\mu + \mathsf{h}_L$, $z_\nu = y_\nu + \mathsf{h}_L$ and $|z_\lambda - y_\lambda| \le \mathsf{h}_L$, where λ is the third direction. There are L of them. Under the coarse cube at y lies its corner cube, the fine cube at $z = y + \mathsf{h}_L(1,1,1)$. The lifts are given by $(\mathsf{J}A')_b = A'_c$ on the crossing bonds of c, with $\mathsf{J}A' = 0$ on the bonds inside blocks; by $(EX')_p = X'_P$ on the edge plaquettes under P, with $EX' = 0$ elsewhere; and by $(E_3\Theta)_q = \Theta_{q'}$ on the corner cube under q', with $E_3\Theta = 0$ elsewhere.

For $x \in B(y)$, the comb $\Gamma(y, x)$ is the path from y to x that runs first along e_1, then along e_2 and then along e_3; the bonds of all combs form a set $\mathcal{T}$, which is a forest with one tree to a block. The slice is $\mathfrak{L} = \{\varphi \in \Omega^1 : \varphi = 0 \text{ on } \mathcal{T}\}$. ■

2.2.3. *Rows and the mask.*

Definition 2.22 (Rows, their norms, and the mask)**.** Rows of a level are given by a Euclidean space H_y of finite dimension for each site y, together with a linear map $\ell_y : \Omega^2 \to \mathsf{H}_y$ whose columns are $\ell_y(p) \in \mathsf{H}_y$; we write $\ell = (\ell_y)_y : \Omega^2 \to \mathsf{H} = \bigoplus \mathsf{H}_y$. The norm $|\cdot|$ on H_y is the Euclidean one, and for $\sigma \in \mathsf{H}$ we set $|\sigma|_{2,\infty} = \sup_y |\sigma_y|$. In $\|\cdot\|_\theta$, the entry of ℓ at (y, p) is taken to be $|\ell_y(p)|$. We put

$$\|\ell\|_{\mathrm{r}} = \sup_y \sum_p |\ell_y(p)|, \qquad \|\ell\|_{\mathrm{c}} = \sup_p \sum_y |\ell_y(p)|, \qquad \mathsf{h}_\ell^2 = \sup_y \sum_p |\ell_y(p)|^2 \ (\le \|\ell\|_{\mathrm{r}}^2).$$

Nothing that follows depends on the dimension of H_y; all of it is invariant under orthogonal maps of each H_y. The rows have range ϱ if $\ell_y(p) = 0$ whenever, in units of that level, the distance between y and p is $> \varrho$. We set $\mathsf{B} = 1 + \ell^{\mathsf{T}}\ell$, $\mathsf{R} = \mathsf{B}^{-1} \le 1$ and $b_{\mathsf{B}} = \|\mathsf{B}\| \le 1 + \|\ell\|_0^2$. At depth 0, $\mathsf{H} = 0$.

The mask is $m_y(V) = 1\{\text{no plaquette of } D(V) \text{ within the distance } \rho_0 \text{ of } y\}$, the distance being measured in units of the level of V and y; a single plaquette of $D(V)$ switches off at most $(2\rho_0+1)^3$ sites. The mask of the class has a second clause, in item 3.5 of §4.2: in addition, no p with $\phi_p > 1$ lies within $\varrho(x)+1$ of y. The arguments of §2.2 do not depend on the clauses of a mask: Prop. 2.40, (N′) and (Fr3′) are stated for *every* map m from the sites to $\{0, 1\}$. ■

2.2.4. *The lifts, the form on the slice, and an inverse.*

Lemma 2.23 (The lifts invert the averages and intertwine the coboundaries)**.** We have $Q_1\mathsf{J} = 1$, $\mathrm{d}\mathsf{J} = E\mathrm{d}'$, $WE = 1$, $\mathrm{d}E = E_3\mathrm{d}'$ and $Q_3E_3 = 1$. ■

Proof. For $x \in B(c_-)$, the path $[x, x']$ contains exactly one crossing bond, and this bond belongs to c; hence $Q_1\mathsf{J} = 1$.

Let $p = (z; \mu\nu)$, and say that z_μ is at the top of its block when $z_\mu = y_\mu + \mathsf{h}_L$. If neither z_μ nor z_ν is at the top, the four bonds of p lie in one block. If exactly one of them is at the top, say z_μ, then (z, μ) and $(z + e_\nu, \mu)$ are crossing bonds of one and the same c and cancel, while the bonds of p in the direction ν lie in blocks. If both are at the top, the four bonds of p are crossing bonds for the four bonds of P, with the signs of d'. So $\mathrm{d}\mathsf{J} = E\mathrm{d}'$.

Since $t(\mathsf{h}_L) = L$ and $t(\mathsf{h}_L \pm L) = 0$, we have $W(P, p) = L^{-1}$ at the L edge plaquettes under P and $W(P, p) = 0$ at those under any other P'; hence $WE = 1$. The same two values of t, inserted in $\psi_{q'}(q) = L^{-3}\prod_\mu t(z_\mu - y_\mu)$, give $\psi_{q'} = 1$ at the corner cube under q' and $\psi_{q'} = 0$ at the corner cubes under the other coarse cubes; hence $Q_3E_3 = 1$.

Let q be the cube at z. If z_μ, z_ν are at the top and z_λ is not, the faces $(z; \mu\nu)$ and $(z + e_\lambda; \mu\nu)$ lie under one P and cancel, and no other face is an edge plaquette. The six faces of a corner cube lie under the six faces of the coarse cube. Hence $\mathrm{d}E = E_3\mathrm{d}'$. □

For $\varphi \in \Omega^1$ let $f := \mathrm{d}\varphi$, and let f° be f with the edge plaquettes of Subsection 2.2.2 left out. On $\mathfrak{L}$ we use the form $q(\varphi) := \langle \mathrm{d}\varphi, \mathsf{B}\mathrm{d}\varphi\rangle + a|Q_1\varphi|^2$, that is, $q = \Pi_{\mathfrak{L}}(\mathrm{d}^*\mathsf{B}\mathrm{d} + aQ_1^*Q_1)\Pi_{\mathfrak{L}}$, as in the proof of Prop. 2.62.

Lemma 2.24 (Bounds on the form on the slice)**.** Let φ vanish on the combs. The local estimates are valid for cochains with values in a real Hilbert space. For a block $B(y)$ and a coarse bond c they read, with $f = \mathrm{d}\varphi$,

$$\sum_{b\subset B(y)} |\varphi_b|^2 \le \tfrac{1}{2}L^3 \sum_{p\subset B(y)} |f_p|^2, \tag{2.i}$$

$$\sum_{b \text{ crossing } c} |\varphi_b|^2 \le 6L^2|(Q_1\varphi)_c|^2 + 12L^4 \sum_{p\subset W_c} |f_p|^2. \tag{2.ii}$$

Summing gives $|\varphi|^2 \le 73L^4|f^\circ|^2 + 6L^2|Q_1\varphi|^2$. Hence $\gamma_q|\varphi|^2 \le |f^\circ|^2 + a|Q_1\varphi|^2 \le q(\varphi) \le b_q|\varphi|^2$, the first inequality for $a \ge 6/(73L^2)$ and the other two for every $a > 0$, where $\gamma_q := (73L^4)^{-1}$ and $b_q := 16b_{\mathsf{B}} + a/L$. If $aL \ge 3/c_L$, then $a \ge 6/(73L^2)$. ∎

Proof. Step 1: bonds inside a block. Let $b = (z,\nu) \subset B(y)$. If $\nu = 3$, then $b \in \mathcal{T}$. If $\nu = 2$, the closed path $\Gamma(y,z) + b - \Gamma(y,z+e_2)$ bounds a ladder of $|z_3 - y_3|$ plaquettes $\{23\}$. If $\nu = 1$, the closed path $\Gamma(y,z) + b - \Gamma(y,z+e_1)$ bounds a ladder of $|z_3 - y_3|$ plaquettes $\{13\}$ together with a ladder of $|z_2 - y_2|$ plaquettes $\{12\}$ at the height y_3. As φ vanishes on the combs, in each case $\varphi_b = \sum_{p\in\sigma_b} \pm f_p$ with $\#\sigma_b < L$ and $\sigma_b \subset B(y)$, and a plaquette belongs to at most $\mathsf{h}_L L \le \frac{1}{2}L^2$ of the sets σ_b. Hence, by the Cauchy–Schwarz inequality,

$$\sum_{b\subset B(y)} |\varphi_b|^2 \le \tfrac{1}{2}L^3 \sum_{p\subset B(y)} |f_p|^2$$

in every block $B(y)$.

Step 2: crossing bonds. On the combs, $\varphi = 0$. For a coarse bond $c = (y,y')$ of direction μ, for $x \in B(y)$ and $x' = x + Le_\mu$, let $\gamma_{c,x} = \Gamma(y,x) + [x,x'] - \Gamma(y',x')$. Then $\gamma_{c,x} - \gamma_{c,y} = \partial n_{c,x}$, where $n_{c,x}$ consists of fewer than L^2 plaquettes of W_c, each occurring once. To see this, we move the bonds of $\Gamma(y,x)$ that are not parallel to e_μ by L unit steps in the direction e_μ; they sweep out at most $2\mathsf{h}_L$ strips of L plaquettes each, these strips are disjoint, and their boundaries telescope. On every torus with $N \ge 2$ even, the identity is the image of the identity of chains on $\mathbb{Z}^3$, and still each plaquette occurs once: the L steps of a strip are pairwise distinct mod NL, and the other two coordinates of a strip stay within one block. So $\varphi([x,x']) - \varphi([y,y']) = \sum_{n_{c,x}} \pm f_p$.

Let $\mathsf{F}_c = \sum_{p\subset W_c} |f_p|^2$. Averaging over x, we get $|\varphi([y,y'])|^2 \le 2|(Q_1\varphi)_c|^2 + 2L^2\mathsf{F}_c$. For a crossing bond $b = (z,\mu)$ of c we take $x = z$; then $\varphi_b = \varphi([y,y']) + \sum_{n_{c,x}} \pm f_p - \beta_b$, where β_b is the sum of φ over the $L-1$ other bonds of $[x,x']$, which lie inside $B(y')$. The paths of different b are disjoint, so the Cauchy–Schwarz inequality and Step 1, applied in $B(y')$, give $\sum_b |\beta_b|^2 \le L \cdot L^3\mathsf{F}_c$. Therefore

$$\sum_{b \text{ of } c} |\varphi_b|^2 \le 3L^2[2|(Q_1\varphi)_c|^2 + 3L^2\mathsf{F}_c] + 3L^4\mathsf{F}_c = 6L^2|(Q_1\varphi)_c|^2 + 12L^4\mathsf{F}_c$$

for every coarse bond c.

A plaquette lies in at most six of the sets W_c, and an edge plaquette in none. Summing over the blocks and the coarse bonds, we obtain $|\varphi|^2 \le (L^3 + 72L^4)|f^\circ|^2 + 6L^2|Q_1\varphi|^2$. Since $L^3 \le L^4$, this proves the summed estimate. If $a \ge 6/(73L^2)$, then $6L^2 \le 73L^4 a$, giving the lower bound with γ_q. The second inequality, $|f^\circ|^2 + a|Q_1\varphi|^2 \le q(\varphi)$, holds because $\mathsf{B} \ge 1$ and $|f^\circ| \le |f|$. The last inequality follows from $\langle f, \mathsf{B}f\rangle \le b_{\mathsf{B}}|f|^2$, from $|\mathrm{d}\varphi|^2 \le 16|\varphi|^2$ and from $\|Q_1\|^2 \le L^{-1}$ of Lem. 2.18. Since $c_L < \frac{1}{6}$, the condition $aL \ge 3/c_L$ gives $aL > 18$, and so $a > 18/L > 6/(73L^2)$. □

Lemma 2.25 (Decay of an inverse; a ratio of determinants)**.** Let I be a finite set with a pseudo-metric dist, let A be real symmetric on $\ell^2(I)$ with $0 < \gamma \le \mathsf{A}$, and let $\varpi_{\mathsf{A}}(\lambda) := \sup_i \sum_j |\mathsf{A}(i,j)|(e^{\lambda \mathrm{dist}(i,j)} - 1)$.

(a) If $\varpi_{\mathsf{A}}(\lambda) \le \frac{\gamma}{2}$, then $|\mathsf{A}^{-1}(i,j)| \le \frac{2}{\gamma} e^{-\lambda \mathrm{dist}(i,j)}$.
(b) If $\mathsf{A}_1, \mathsf{A}_2 > 0$, $\operatorname{rank}(\mathsf{A}_1 - \mathsf{A}_2) \le r$ and $\|\mathsf{A}_1\|\|\mathsf{A}_2^{-1}\| \ge 1$, then $\det \mathsf{A}_1 / \det \mathsf{A}_2 \le (\|\mathsf{A}_1\|\|\mathsf{A}_2^{-1}\|)^r$. ■

Proof. Step (a): the decay of the inverse. For $\lambda \le 0$, the bound $\|\mathsf{A}^{-1}\| \le \gamma^{-1}$ proves the assertion directly. Hence assume $\lambda > 0$. Fix j_0, and let $\rho = \mathrm{dist}(\cdot, j_0)$ and $\mathsf{A}_\lambda = e^{\lambda\rho} \mathsf{A} e^{-\lambda\rho}$. By the triangle inequality for dist, $|\mathsf{A}_\lambda(i,j) - \mathsf{A}(i,j)| \le |\mathsf{A}(i,j)|(e^{\lambda \mathrm{dist}(i,j)} - 1)$, and Schur's test gives $\|\mathsf{A}_\lambda - \mathsf{A}\| \le \varpi_{\mathsf{A}}(\lambda)$. So $\|\mathsf{A}_\lambda u\| \ge \frac{\gamma}{2}\|u\|$, and $\mathsf{A}^{-1}(i,j_0) = e^{-\lambda\rho(i)} \mathsf{A}_\lambda^{-1}(i,j_0)$; since $|\mathsf{A}_\lambda^{-1}(i,j_0)| \le \frac{2}{\gamma}$, this proves (a).

Step (b): the ratio of determinants. Write $\mathsf{A}_2^{-1/2} \mathsf{A}_1 \mathsf{A}_2^{-1/2} = 1 + K$ with $\operatorname{rank} K \le r$. All but r eigenvalues of $1 + K$ are 1, and the others are $\le \|\mathsf{A}_1\|\|\mathsf{A}_2^{-1}\|$. As $\det \mathsf{A}_1 / \det \mathsf{A}_2 = \det(1+K)$ and $\|\mathsf{A}_1\|\|\mathsf{A}_2^{-1}\| \ge 1$, this proves (b). □

2.2.5. *Pairs and the minimiser of one step.*

Definition 2.26 (Pairs, the move, and $R_\sharp$)**.** Let $\mathcal{H} = \Omega^2 \oplus \Omega'^1$, with $\langle (F, \mathsf{n}), (F', \mathsf{n}') \rangle_{\mathcal{H}} = \langle F, \mathsf{B} F' \rangle + a \langle \mathsf{n}, \mathsf{n}' \rangle$ and the norm $\|\cdot\|_{\mathcal{H}}$. For $Z \in \Omega'^2$ and $\Xi \in \Omega^3$ let $Y = \mathbb{Y}(Z, \Xi) := W^* Z + \delta \Xi$, $\mathrm{pair}(Z, \Xi) := (\nu, \omega) = (\mathsf{R} Y, a^{-1} \mathrm{d}'^* Z)$ and $\sigma = \ell \nu$. Then $\nu + \ell^{\mathsf{T}} \sigma = Y$, and the identity

$$\|\mathrm{pair}(Z, \Xi)\|_{\mathcal{H}}^2 = \langle Y, \mathsf{R} Y \rangle + a^{-1} |\mathrm{d}'^* Z|^2 = |\nu|^2 + |\sigma|^2 + a|\omega|^2$$

holds.

We set $\mathcal{A} := \{\mathrm{pair}(Z, \Xi) : Z \in \Omega'^2,\ \Xi \in \Omega^3\}$ and call its elements admissible; further, $\mathcal{K} = \{(\mathrm{d}\varphi, -Q_1 \varphi) : \varphi \in \Omega^1\}$. For all φ, by $\mathrm{d}' Q_1 = W \mathrm{d}$ of Lem. 2.18 and $\mathrm{dd} = 0$,

$$\langle \mathrm{pair}(Z, \Xi), (\mathrm{d}\varphi, -Q_1 \varphi) \rangle_{\mathcal{H}} = \langle Z, W \mathrm{d}\varphi - \mathrm{d}' Q_1 \varphi \rangle + \langle \Xi, \mathrm{dd}\varphi \rangle = 0 : \qquad \mathcal{A} \perp \mathcal{K}. \tag{2.iii}$$

The adjoint of $Q_3 \mathrm{d} = \mathrm{d}' W$ is $\delta Q_3^* = W^* \delta'$; together with $\mathrm{d}'^* \delta' = 0$ it gives $\mathrm{pair}(-\delta' \Theta, Q_3^* \Theta) = 0$ for all $\Theta \in \Omega'^3$. This identity is called the move. For $X' \in \Omega'^2$ we set $g(Z, \Xi; X') = \langle Z, X' \rangle - \frac{1}{2}\|\mathrm{pair}(Z, \Xi)\|_{\mathcal{H}}^2$.

Let $\Omega_0^3 = \ker Q_3$, and let $\Xi_Z \in \Omega_0^3$ be the minimiser of $\langle Y, \mathsf{R} Y \rangle$ over Ω_0^3. It is unique and linear in Z: for $\Xi \in \Omega_0^3$, $\delta \Xi = 0$ makes Ξ a constant κ, and $Q_3 \kappa = L^3 \kappa$ then forces $\kappa = 0$. We set $\mathsf{T}_\sharp Z = W^* Z + \delta \Xi_Z$ and $R_\sharp = \mathsf{T}_\sharp^* \mathsf{R} \mathsf{T}_\sharp + a^{-1} \mathrm{d}' \mathrm{d}'^*$; thus $\langle Z, R_\sharp Z \rangle = \min_{\Xi \in \Omega_0^3} \|\mathrm{pair}(Z, \Xi)\|_{\mathcal{H}}^2$. ■

Proposition 2.27 (The minimiser of one step)**.** (a) $R_\sharp > 0$. Let $\mathsf{B}' := L R_\sharp^{-1}$, $\hat{R} = \mathsf{B}'^{-1}$, $H^\sharp := L^{-1} \mathsf{R} \mathsf{T}_\sharp \mathsf{B}'$ and $\mathsf{N}^\sharp := (aL)^{-1} \mathrm{d}'^* \mathsf{B}'$. For every $X' \in \Omega'^2$, the element $(\bar{F}, \bar{\mathsf{n}}) = (H^\sharp X', \mathsf{N}^\sharp X') = \mathrm{pair}(\bar{Z}, \Xi_{\bar{Z}})$, where $\bar{Z} = L^{-1} \mathsf{B}' X'$, satisfies $W \bar{F} + \mathrm{d}' \bar{\mathsf{n}} = X'$ and $\|(\bar{F}, \bar{\mathsf{n}})\|_{\mathcal{H}}^2 = L^{-1} \langle X', \mathsf{B}' X' \rangle$.
(b) $\hat{R} \le 1 - \mathsf{P} \le 1$; the first inequality holds for every $a > 0$, and the second, which is the inequality $\mathsf{P} \ge 0$ of Lem. 2.18, holds if $aL \ge 3/c_L$.
(c) $\mathcal{A} = \mathcal{K}^\perp$, and $\mathcal{K} = \{(\mathrm{d}\varphi, -Q_1 \varphi) : \varphi \in \mathfrak{L}\}$. Let $\mathsf{g} = \Pi_{\mathfrak{L}} \mathrm{d}^* \mathsf{B} E$, $H_\circ := E - \mathrm{d} q^{-1} \mathsf{g}$ and $\mathsf{N}_\circ := Q_1 q^{-1} \mathsf{g}$. Here $q > 0$ on $\mathfrak{L}$ for every $a > 0$, by the summed estimate and the bound $|f^\circ|^2 + a|Q_1 \varphi|^2 \le q(\varphi)$ in Lem. 2.24, which give $|\varphi|^2 \le \max\{73 L^4, 6L^2/a\}\, q(\varphi)$ for $\varphi \in \mathfrak{L}$. For every $X' \in \Omega'^2$, $(H_\circ X', \mathsf{N}_\circ X')$ is the orthogonal projection of $(EX', 0)$ on $\mathcal{A}$. ■

Proof. Step (a): positivity, and the minimiser. To prove $R_\sharp > 0$, let $\mathsf{T}_\sharp Z = 0$ and $\mathrm{d}'^* Z = 0$. We pair $\mathsf{T}_\sharp Z$ with EX' and use Lem. 2.23: $0 = \langle Z, W E X' \rangle + \langle \Xi_Z, \mathrm{d} E X' \rangle = \langle Z + \delta' E_3^{\mathsf{T}} \Xi_Z, X' \rangle$ for all X', so $Z = \delta' \Theta$ with $\Theta = -E_3^{\mathsf{T}} \Xi_Z$. Then, by $\delta Q_3^* = W^* \delta'$, $0 = W^* \delta' \Theta + \delta \Xi_Z = \delta(Q_3^* \Theta + \Xi_Z)$; hence $Q_3^* \Theta + \Xi_Z = \kappa$ is a constant, and $\kappa = Q_3^* \kappa$, where on the right κ is the coarse cochain with the same constant value. Thus $\Xi_Z = Q_3^*(\kappa - \Theta)$, and $\Xi_Z \in \ker Q_3 \cap \operatorname{ran} Q_3^* = 0$; so $Q_3 Q_3^* (\Theta - \kappa) = 0$, which gives $\Theta = \kappa$ because $Q_3 Q_3^* = L^3 \hat{\mathsf{J}}$ is invertible, and $Z = 0$ because $\delta' \kappa = 0$.

Next, $\sup_{\Xi\in\Omega^3_0} g = \langle Z, X'\rangle - \frac{1}{2}\langle Z, R_\sharp Z\rangle$, which is maximal at $\bar Z = R_\sharp^{-1}X' = L^{-1}\mathsf{B}'X'$. As the maximum over both variables is attained at $(\bar Z, \Xi_{\bar Z})$, the function g is stationary in Z at fixed $\Xi = \Xi_{\bar Z}$; this gives $W\mathsf{R}\bar Y + a^{-1}\mathrm{d}'\mathrm{d}'^*\bar Z = X'$, with $\bar Y = \mathbb{Y}(\bar Z, \Xi_{\bar Z})$, which is $W\bar F + \mathrm{d}'\bar{\mathsf{n}} = X'$. Finally, $\|(\bar F, \bar{\mathsf{n}})\|^2_{\mathcal{H}} = \langle \bar Z, R_\sharp \bar Z\rangle = L^{-1}\langle X', \mathsf{B}'X'\rangle$.

Step (b): the bound on $\hat R$. With the competitor $\Xi = 0$ and $\mathsf{R} \le 1$, $\langle Z, R_\sharp Z\rangle \le |W^*Z|^2 + a^{-1}|\mathrm{d}'^*Z|^2 = L\langle Z, (1-\mathsf{P})Z\rangle$; since $\hat R = L^{-1}R_\sharp$, this is the first inequality of (b).

Step (c): admissible pairs, and the projection. The relation $\mathcal{A} \perp \mathcal{K}$ in the display of Def. 2.26 gives $\mathcal{A} \subset \mathcal{K}^\perp$. For the converse inclusion $\mathcal{K}^\perp \subset \mathcal{A}$, an element of $\mathcal{K}^\perp$ has the form $(\mathsf{R}G, \mathsf{n})$ with $\langle G, \mathrm{d}\varphi\rangle = a\langle \mathsf{n}, Q_1\varphi\rangle$ for all $\varphi \in \Omega^1$, and we show that $(\mathsf{R}G, \mathsf{n}) \in \mathcal{A}$. The range of the map $(Z, \Xi) \mapsto (W^*Z + \delta\Xi, \mathrm{d}'^*Z)$ is the orthogonal complement of $\{(F, n) : WF + \mathrm{d}'n = 0,\ \mathrm{d}F = 0\}$. For such (F, n) we have $F = \mathrm{d}\varphi + h$ with h constant. Indeed, on a nonzero Fourier mode the coboundary is exterior multiplication by a nonzero vector v. Contraction by $\bar v/|v|^2$ gives a contracting homotopy, since its anticommutator with exterior multiplication by v is the identity. Thus closed forms are exact on nonzero modes. The zero mode is constant, and an exact form has zero constant mode. Since $WF = \mathrm{d}'Q_1\varphi + L^2 h$ and $WF + \mathrm{d}'n = 0$, the constant mode gives $h = 0$. Then $\mathrm{d}'(n + Q_1\varphi) = 0$, so the same argument on the coarse torus gives $n = -Q_1\varphi + \mathrm{d}'\lambda' + h'$ with h' constant. Then $\langle G, F\rangle + a\langle \mathsf{n}, n\rangle = a\langle \mathsf{n}, \mathrm{d}'\lambda' + h'\rangle = 0$, the last equality by the tests $\varphi = \mathsf{J}\mathrm{d}'\lambda'$ and $\varphi = \mathsf{J}h'$, for which $\mathrm{d}\varphi = E\mathrm{d}'(\cdot) = 0$, while $Q_1\varphi = \mathrm{d}'\lambda'$ for the first test and $Q_1\varphi = h'$ for the second. Hence $(G, a\mathsf{n})$ lies in that range, and $(\mathsf{R}G, \mathsf{n}) \in \mathcal{A}$.

For the slice, every $\varphi \in \Omega^1$ is of the form $\varphi = \varphi_{\mathfrak{L}} + \mathrm{d}\lambda$, with $\lambda(x) = \varphi(\Gamma(y, x)) + \kappa_y$ on $B(y)$: a constant κ_y on a block alters $\mathrm{d}\lambda$ on crossing bonds only, so $\varphi - \mathrm{d}\lambda \in \mathfrak{L}$ for every choice of the κ_y. We choose them with $Q_0\lambda = 0$; then $Q_1\mathrm{d}\lambda = \mathrm{d}'Q_0\lambda = 0$ and $\mathrm{d}\mathrm{d}\lambda = 0$, so that $(\mathrm{d}\varphi, -Q_1\varphi) = (\mathrm{d}\varphi_{\mathfrak{L}}, -Q_1\varphi_{\mathfrak{L}})$, and $\varphi \in \mathfrak{L}$ suffices to describe $\mathcal{K}$.

For the projection, $\bar\eta = -q^{-1}\mathsf{g}X'$ minimises

$$\frac{1}{2}\|(EX' + \mathrm{d}\eta, -Q_1\eta)\|^2_{\mathcal{H}} = \frac{1}{2}\langle EX', \mathsf{B}EX'\rangle + \langle \mathsf{g}X', \eta\rangle + \frac{1}{2}q(\eta)$$

over $\eta \in \mathfrak{L}$; hence the orthogonal projection of $(EX', 0)$ on $\mathcal{K}^\perp = \mathcal{A}$ is $(EX' + \mathrm{d}\bar\eta, -Q_1\bar\eta) = (H_\circ X', \mathsf{N}_\circ X')$. □

Proposition 2.28 (B' as one plus a sum of squares of the next rows)**.** Suppose that $aL \ge 3/c_L$, and put $\alpha = c_L - \frac{2}{aL}$, $\beta = \frac{1}{2}c_L^2$ and $u, v = \frac{1}{2}(\sqrt{\alpha} \pm \sqrt{\alpha - 4\beta})$. Then $\alpha > 4\beta > 0$, and therefore u and v are real. Moreover $\mathsf{B}' = 1 + \ell'^{\mathsf{T}}\ell'$, where $\ell'_{y'}\zeta \in \mathsf{H}'_{y'}$ is the list whose entries are those of (1) to (5):

(1) the six numbers $((u + v\mathsf{S}_{\rho'})\partial'_\rho\zeta_{\mu\nu})(y')$, for $\mu < \nu$ and $(\rho, \rho') \in \{(\mu, \nu), (\nu, \mu)\}$;
(2) the three numbers $(aL)^{-1/2}(\mathrm{d}^\dagger\zeta)(y', \cdot)$;
(3) the numbers $L^{1/2}((H^\sharp - L^{-1}W^*)\zeta)_p$, one for each p based in $B(y')$;
(4) the numbers $(aL)^{1/2}((\mathsf{N}^\sharp - (aL)^{-1}\mathrm{d}'^*)\zeta)_c$, one for each c based at y';
(5) the entries $L^{1/2}\ell_y H^\sharp\zeta$, one for each $y \in B(y')$.

Here $\mathrm{d}^\dagger$ is obtained from d'^* by reversing the sign of the second of its two terms, and $(\partial'_\rho\zeta_{\mu\nu})(y') = \zeta_{\mu\nu}(y' + Le_\rho) - \zeta_{\mu\nu}(y')$; the same formula defines ∂'_ρ on every function on $\mathbb{T}'$, and $\partial'^*_\rho\partial'_\rho = \partial'_\rho\partial'^*_\rho = \Delta'_\rho$. ■

Proof. Let $(\bar F, \bar{\mathsf{n}})$ be the pair of Prop. 2.27 (a) at $X' = \zeta$. Expanding the squares, we find

$$L|\bar F - L^{-1}W^*\zeta|^2 + aL|\bar{\mathsf{n}} - (aL)^{-1}\mathrm{d}'^*\zeta|^2 = L|\bar F|^2 + aL|\bar{\mathsf{n}}|^2 - 2\langle \zeta, W\bar F + \mathrm{d}'\bar{\mathsf{n}}\rangle + \langle \zeta, (1-\mathsf{P})\zeta\rangle.$$

To both members of this identity we add $|\zeta|^2 + \langle \zeta, \mathsf{P}\zeta\rangle + L|\ell\bar F|^2$; the resulting sum is $L\|(\bar F, \bar{\mathsf{n}})\|^2_{\mathcal{H}} = \langle \zeta, \mathsf{B}'\zeta\rangle$.

We now turn to P, for which we use the identity

$$1-(1-c_L\Delta'_\mu)(1-c_L\Delta'_\nu)=c_L\Delta'_\mu(1-\frac{c_L}{2}\Delta'_\nu)+c_L\Delta'_\nu(1-\frac{c_L}{2}\Delta'_\mu).$$

Every component of $\mathrm{d}'^*\zeta$ has the form s_1A+s_2B, where $s_1,s_2\in\{+1,-1\}$ and A,B are two of the six functions $\partial'^*_\rho\zeta_{\mu\nu}$, and each of the six appears in exactly one component. From $(s_1A+s_2B)^2+(s_1A-s_2B)^2=2A^2+2B^2$ we get

$$|\mathrm{d}'^*\zeta|^2=2\sum\langle\zeta_{\mu\nu},(\Delta'_\mu+\Delta'_\nu)\zeta_{\mu\nu}\rangle-|\mathrm{d}^\dagger\zeta|^2,$$

and consequently

$$\mathsf{P}=\sum\partial'^*_\rho(\alpha-\beta\Delta'_{\rho'})\partial'_\rho+\frac{1}{aL}\mathrm{d}^{\dagger\mathsf{T}}\mathrm{d}^\dagger.$$

Moreover, $(u+v\mathsf{S})^*(u+v\mathsf{S})=u^2+v^2+uv(\mathsf{S}+\mathsf{S}^*)=(u+v)^2-uv\Delta'=\alpha-\beta\Delta'$.

Finally, $\alpha>4\beta$ holds because $\frac{2}{aL}\le\frac{2}{3}c_L<c_L(1-2c_L)$. □

Definition 2.29 (The map to the next form)**.** The form B' of Prop. 2.27 depends on the rows ℓ only through B; we write $\Phi(\mathsf{B}):=\mathsf{B}'$. ■

Definition 2.30 (The auxiliary form)**.** We put $\mathsf{S}_0:=L^{-5}$. On $\Omega'^2\oplus\Omega^3$ we define $\mathfrak{R}(Z,\Xi)=\|\mathrm{pair}(Z,\Xi)\|^2_{\mathcal{H}}+\mathsf{S}_0|Q_3\Xi|^2$, and we write $\iota X'=(X',0)$. Lems. 2.31 and 2.38 are valid for every $\mathsf{S}_0>0$, with S_0 kept in the formulas, except for the last member $\mathsf{B}'+\delta'\hat{\mathsf{J}}^{-2}\mathrm{d}'$ in the formula for $L\iota^{\mathsf{T}}\mathfrak{R}^{-1}\iota$, which is valid at $\mathsf{S}_0=L^{-5}$. The kernels $H^\sharp$ and B', as well as the kernels T, X, $\mathfrak{G}$, K_1, K_2 introduced below, are independent of S_0; by contrast, G_Z and G_Ξ depend on it, through the move. ■

Lemma 2.31 (The penalty does not move the pair, at every depth)**.** For $X'\in\Omega'^2$ we have

$$\mathfrak{R}^{-1}\iota X'=(\bar Z-\delta'\Theta,\ \Xi_{\bar Z}+Q_3^*\Theta),\quad \Theta=-\mathsf{S}_0^{-1}(Q_3Q_3^*)^{-2}\mathrm{d}'X';\qquad H^\sharp=\mathsf{R}\mathbb{Y}\mathfrak{R}^{-1}\iota,\quad H_\circ=\mathsf{R}\mathbb{Y}\mathfrak{R}^{-1}(\iota+(0,E_3\mathrm{d}')); \tag{2.iv}$$

moreover $L\iota^{\mathsf{T}}\mathfrak{R}^{-1}\iota=\mathsf{B}'+L\mathsf{S}_0^{-1}\delta'(Q_3Q_3^*)^{-2}\mathrm{d}'=\mathsf{B}'+\delta'\hat{\mathsf{J}}^{-2}\mathrm{d}'$, and the pair of $\mathfrak{R}^{-1}(X',E_3\mathrm{d}'X')$ is $(H_\circ X',\mathsf{N}_\circ X')$. ■

Proof. Given (Z,Ξ), we write $\Xi=\Xi_0+Q_3^*\Theta$ with $\Xi_0\in\Omega^3_0$, the two summands being orthogonal, and we put $\tilde Z=Z+\delta'\Theta$. The move gives $\mathrm{pair}(Z,\Xi)=\mathrm{pair}(\tilde Z,\Xi_0)$, while $Q_3\Xi=Q_3Q_3^*\Theta$. Therefore

$$\langle Z,X'\rangle-\frac{1}{2}\mathfrak{R}(Z,\Xi)=g(\tilde Z,\Xi_0;X')-\langle\Theta,\mathrm{d}'X'\rangle-\frac{\mathsf{S}_0}{2}|Q_3Q_3^*\Theta|^2.$$

The maximiser is $\tilde Z=\bar Z$, $\Xi_0=\Xi_{\bar Z}$, with Θ as in the statement, and $\mathfrak{R}>0$ follows.

For the source $(X',E_3\mathrm{d}'X')$ there is one more term, $\langle\Xi,E_3\mathrm{d}'X'\rangle=\langle\Xi_0,E_3\mathrm{d}'X'\rangle+\langle\Theta,\mathrm{d}'X'\rangle$, because $Q_3E_3=1$; as a result, the maximiser has $\Theta=0$. Then $(\tilde Z,\Xi_0)$ maximises

$$\langle\tilde Z,X'\rangle+\langle\Xi_0,E_3\mathrm{d}'X'\rangle-\frac{1}{2}\|\mathrm{pair}\|^2=\langle\mathrm{pair}(\tilde Z,\Xi_0),(EX',0)\rangle_{\mathcal{H}}-\frac{1}{2}\|\mathrm{pair}\|^2,$$

the equality being a consequence of $\langle\mathsf{R}Y,\mathsf{B}EX'\rangle=\langle Z,WEX'\rangle+\langle\Xi,\mathrm{d}EX'\rangle$; this identity holds by $\mathsf{RB}=1$, $Y=W^*Z+\delta\Xi$ and adjointness, and by $WE=1$ and $\mathrm{d}E=E_3\mathrm{d}'$ of Lem. 2.23 its right side is $\langle Z,X'\rangle+\langle\Xi,E_3\mathrm{d}'X'\rangle$. As every pair is of the form $\mathrm{pair}(\tilde Z,\Xi_0)$, the maximiser is the projection of $(EX',0)$ on $\mathcal{A}$, as in Prop. 2.27 (c). □

Definition 2.32 (The numbers w_*, w_1, $\mathsf{r}_\sharp$, γ_1, λ_1; the natural norm)**.** We set $w_*:=3^{-6}\le w_1:=(1-4c_L)^6$, and further $\mathsf{r}_\sharp:=w_*/(1.3\cdot10^7)$, $\gamma_1:=\mathsf{r}_\sharp/3074$ and $\lambda_1:=\frac{1}{40}\gamma_1^{1/2}$. The numbers w_*, $\mathsf{r}_\sharp$, γ_1 and λ_1 are free of L, a and N, while w_1 depends on L through c_L. The natural norm is given by $\mathsf{D}=\mathrm{diag}(L\text{ on }\Omega'^2,\ L^{-2}\text{ on }\Omega^3)$, with $N(Z,\Xi)=L|Z|^2+L^{-2}|\Xi|^2$. ■

Proposition 2.33 (Poincaré bounds for empty rows and a lower bound for the auxiliary form)**.** Let $\ell=\emptyset$, let $L\ge3$ be odd and $N\ge2$ even, and let $a>0$. Then the following hold.

(a) For $u\in\Omega^3$, with $\bar u:=L^{-3}Q_3u$, we have $|u|^2\le768L^2|\delta u|^2+2L^{-3}|Q_3u|^2$.

(b) $|\delta'\bar{u}|^2 \le 1.4\cdot 10^5 L^{-1}|\delta u|^2$.
(c) For $Z\in\Omega'^2$ and $\Xi_0\in\Omega_0^3$ we have $|W^*Z+\delta\Xi_0|^2 \ge \mathsf{r}_\sharp L|Z|^2$; consequently $\hat{R}\ge \mathsf{r}_\sharp$ and $\mathsf{B}'\le \mathsf{r}_\sharp^{-1}$.
(d) $\mathfrak{R}\ge \gamma_1 \mathsf{D}$.

■

Proof. We prove (a). First,

$$|u|^2=\sum_{q'}\sum_{q}\psi_{q'}(q)u_q^2 \le 2\sum_{q'}\sum_{q}\psi_{q'}(u_q-\bar{u}_{q'})^2+2L^3|\bar{u}|^2,$$

and for every q'

$$\sum_q \psi_{q'}(u_q-\bar{u}_{q'})^2=\frac{1}{2L^3}\sum_{q,\tilde{q}}\psi_{q'}(q)\psi_{q'}(\tilde{q})(u_q-u_{\tilde{q}})^2.$$

Inside the box of side $<2L$ we join q to $\tilde{q}$ by the coordinate path that moves first in the direction e_1, then in the direction e_2 and last in the direction e_3; the number of its steps is $<6L$. A given dual bond is crossed by at most $(2L)^4$ of these paths. Indeed, a bond along e_1 fixes q_2,q_3, and the free data are q_1 and $\tilde{q}$; a bond along e_2 fixes $\tilde{q}_1$ and q_3, and leaves $q_1,q_2,\tilde{q}_2,\tilde{q}_3$ free; a bond along e_3 fixes $\tilde{q}_1$ and $\tilde{q}_2$, and leaves q and $\tilde{q}_3$ free. The double sum is therefore $\le \frac{1}{2L^3}6L(2L)^4\sum_{\mathrm{box}}(\delta u)^2 = 48L^2\sum_{\mathrm{box}}(\delta u)^2$, and each dual bond belongs to at most 8 boxes.

We prove (b). For neighbouring q',q'', Jensen's inequality gives

$$(\bar{u}_{q'}-\bar{u}_{q''})^2 \le L^{-6}\sum_{q,\tilde{q}}\psi_{q'}(q)\psi_{q''}(\tilde{q})(u_q-u_{\tilde{q}})^2.$$

For $N\ge 4$, the two supports lie in a box of side less than $3L$ that embeds in the fine torus. The coordinate paths within the box have length at most $9L$, and a dual bond belongs to at most $(3L)^4$ paths, by the count in part (a). Thus the last sum with its factor L^{-6} is bounded by $729L^{-1}$ times the sum of $(\delta u)^2$ in the box. Each dual bond belongs to at most $3\cdot 4^3=192$ such boxes. Thus the total coefficient is at most $729\cdot 192L^{-1}\le 1.4\cdot 10^5 L^{-1}$.

For $N=2$, use the Jensen sum over the fine torus of side $2L$. Join its two endpoints by positive coordinate steps in the order $1,2,3$. The paths have length less than $6L$. A given dual bond belongs to at most $(2L)^4$ paths, by the same count of free coordinates as in part (a). Since $\psi_{q'}\le 1$, each coarse plaquette contributes at most $6L(2L)^4L^{-6}|\delta u|^2 = 96L^{-1}|\delta u|^2$. There are 24 coarse plaquettes, so $2304L^{-1}$ suffices in this case. Part (a) applies unchanged, because its boxes have side $2L-1$.

We prove (c). We decompose $Z=Z_{\mathrm{cl}}+\delta'\Theta_1$ into two orthogonal parts, where $\mathrm{d}'Z_{\mathrm{cl}}=0$; then $Y=W^*Z+\delta\Xi_0=W^*Z_{\mathrm{cl}}+\delta u$, where $u=Q_3^*\Theta_1+\Xi_0$.

The closed part is bounded by means of a test field. For $z=y+r$ with $y\in L\mathbb{Z}^3$ and $0\le r_\rho<L$ we put

$$(\mathsf{W}_2 Z)(z;\mu\nu)=L^{-2}[(1-\frac{r_\lambda}{L})Z(y;\mu\nu)+\frac{r_\lambda}{L}Z(y+Le_\lambda;\mu\nu)]$$

and $(\mathsf{W}_3\Theta)(z)=L^{-3}\Theta(y)$. Then $\mathrm{d}\mathsf{W}_2=\mathsf{W}_3\mathrm{d}'$, because the difference along e_λ equals $L^{-3}\partial'_\lambda Z_{\mu\nu}(y)$ for each value of r_λ, the value $r_\lambda=L-1$ included. By convexity, $|\mathsf{W}_2 Z|^2\le L^{-1}|Z|^2$, the weight being $L^{-4}\cdot L^2\sum_{r_\lambda}((1-\frac{r_\lambda}{L})+\frac{r_\lambda}{L})$.

Next, $W\mathsf{W}_2=\prod_\rho(1-c''\Delta'_\rho)=:\mathsf{K}$, with $c''=\frac{L^2-1}{8L^2}$. We look at one coordinate at a time. Along μ, for a function g on the coarse sites and with $\lfloor z\rfloor$ denoting the coarse site y' for which $z=y'+r$ and $0\le r<L$ in the coordinate at hand, the sum $\sum_z L^{-1}t(z-y)L^{-1}g(\lfloor z\rfloor)$ assigns to $g(y\pm L)$ the weight $L^{-2}(1+\cdots+\mathsf{h}_L)=\frac{\mathsf{h}_L(\mathsf{h}_L+1)}{2L^2}=c''$. Along λ, the average $L^{-1}\sum_{|s|\le \mathsf{h}_L}$ of the linear interpolation assigns to $g(y\pm L)$ the weight $L^{-1}\sum_{s=1}^{\mathsf{h}_L}\frac{s}{L}=c''$. The weights sum to 1. We have $\mathsf{K}\ge(1-4c'')^3\ge\frac{1}{8}$; moreover K acts in the same manner on every component and commutes with translations, hence with d' and δ'.

Hence $\langle Y, \mathsf{W}_2 Z_{\mathrm{cl}}\rangle = \langle Z, \mathsf{K} Z_{\mathrm{cl}}\rangle + \langle \Xi_0, \mathsf{W}_3 \mathrm{d}' Z_{\mathrm{cl}}\rangle = \langle Z_{\mathrm{cl}}, \mathsf{K} Z_{\mathrm{cl}}\rangle \geq \frac{1}{8}|Z_{\mathrm{cl}}|^2$, and therefore $|Y|^2 \geq \frac{L}{64}|Z_{\mathrm{cl}}|^2$.

For the co-exact part, $|\delta u| \leq |Y| + L^{1/2}|Z_{\mathrm{cl}}| \leq 9|Y|$. Further, $\bar{u} = L^{-3} Q_3 Q_3^* \Theta_1 = \hat{\mathsf{J}}\Theta_1$, and $\hat{\mathsf{J}} \geq w_1^{1/2}$ commutes with δ'. Using (b), we get $w_1 |\delta' \Theta_1|^2 \leq |\delta' \bar{u}|^2 \leq 1.4 \cdot 10^5 L^{-1} \cdot 81 |Y|^2$.

Adding the bounds for the two parts, we obtain $|Z|^2 \leq (64 + 1.134 \cdot 10^7 / w_*) L^{-1} |Y|^2 \leq \mathsf{r}_\sharp^{-1} L^{-1} |Y|^2$. For the consequences concerning $\hat{R}$ and B' we note that $\langle Z, R_\sharp Z\rangle \geq |\mathsf{T}_\sharp Z|^2$.

We prove (d). Let $\Xi = \Xi_0 + Q_3^* \Theta$, $\tilde{Z} = Z + \delta' \Theta$ and $Y = W^* \tilde{Z} + \delta \Xi_0$, as in the proof of Lem. 2.31; then $L|\tilde{Z}|^2 \leq \mathsf{r}_\sharp^{-1} |Y|^2$. Further, $|\delta \Xi_0| \leq |Y| + L^{1/2}|\tilde{Z}| \leq 2\mathsf{r}_\sharp^{-1/2}|Y|$, so that (a) gives $|\Xi_0|^2 \leq 3072 L^2 \mathsf{r}_\sharp^{-1} |Y|^2$. As $\|\delta'\|^2 \leq 12$, we have $L|Z|^2 \leq 2L|\tilde{Z}|^2 + 24L|\Theta|^2$; in addition $|\Xi|^2 \leq |\Xi_0|^2 + L^3 |\Theta|^2$ and $|\Theta|^2 \leq |Q_3 \Xi|^2 / (L^6 w_1)$. Altogether,

$$N(Z, \Xi) \leq 3074\, \mathsf{r}_\sharp^{-1} |Y|^2 + 25L \cdot L^{-6} w_*^{-1} |Q_3 \Xi|^2 = 3074\, \mathsf{r}_\sharp^{-1} |Y|^2 + 25 w_*^{-1} \mathsf{S}_0 |Q_3 \Xi|^2 \leq \gamma_1^{-1} \mathfrak{R}(Z, \Xi).$$

□

Remark 2.34 (The torus of side two). Prop. 2.33 includes $N = 2$. Its proof uses the whole fine torus for the path estimate in part (b) at that side.

Definition 2.35 (The kernels X and T)**.** For $v \in \Omega'^2$ we set $\mathsf{X} v := \Xi_{L^{-1} v}$ and $\mathsf{T} v := \mathsf{R}\mathsf{T}_\sharp L^{-1} v$. Thus $\mathrm{pair}(L^{-1} v, \mathsf{X} v) = (\mathsf{T} v, (aL)^{-1} \mathrm{d}'^* v)$, and $H^\sharp = \mathsf{T}\mathsf{B}'$. ■

Remark 2.36 (Decay of $\hat{\mathsf{J}}^{-2}$; the rate θ_J). We have $\hat{\mathsf{J}}^2 \geq w_*$; moreover $\hat{\mathsf{J}}^2$ has range 2, and its absolute row sums are $\leq (1 + 4c_L)^6 \leq 25$. Applying Lem. 2.25 (a) to $\mathsf{A} = \hat{\mathsf{J}}^2$, for which $\varpi_{\hat{\mathsf{J}}^2}(\lambda) \leq 25(e^{2\lambda} - 1)$, we obtain $|\hat{\mathsf{J}}^{-2}(i, j)| \leq \frac{2}{w_*} e^{-\theta_J \mathrm{dist}}$, where $\theta_J := \frac{1}{2} \log(1 + \frac{w_*}{50})$ is chosen so that $25(e^{2\theta_J} - 1) = \frac{1}{2} w_*$.

Definition 2.37 (The kernels with a source on coarse cubes)**.** For $\theta \in \Omega'^3$ we define $(\mathsf{G}_Z \theta, \mathsf{G}_\Xi \theta) := \mathfrak{R}^{-1}(0, E_3 \theta)$ and $\mathfrak{G}\theta := \mathrm{pair}(\mathsf{G}_Z \theta, \mathsf{G}_\Xi \theta) \in \mathcal{A}$. We denote by $\mathsf{G}\theta$ the first component of $\mathfrak{G}\theta$; its second component is given by $\mathsf{G}_\mathsf{n} := a^{-1} \mathrm{d}'^* \mathsf{G}_Z$. ■

Lemma 2.38 (The projected lift: a minimiser plus a term in $\mathrm{d}' X'$)**.**

(a) For all (Z, Ξ) and all θ, $\langle \mathrm{pair}(Z, \Xi), \mathfrak{G}\theta\rangle_{\mathcal{H}} = \langle E_3^\mathsf{T} \Xi, \theta\rangle - \langle Q_3 \Xi, (Q_3 Q_3^*)^{-1} \theta\rangle$.

(b) $\|\mathfrak{G}\theta\|_{\mathcal{H}}^2 = \langle \theta, \mathsf{K}_2 \theta\rangle$, where $\mathsf{K}_2 := E_3^\mathsf{T} \mathsf{G}_\Xi - \mathsf{S}_0^{-1} (Q_3 Q_3^*)^{-2}$.

(c) For every $X' \in \Omega'^2$, $(H_\circ X', \mathsf{N}_\circ X') = (H^\sharp X', \mathsf{N}^\sharp X') + \mathfrak{G}\mathrm{d}' X'$; in particular $H_\circ = H^\sharp + \mathsf{G}\mathrm{d}'$. Moreover, with $\mathsf{m}_\circ(X') := \frac{1}{2}\|(H_\circ X', \mathsf{N}_\circ X')\|_{\mathcal{H}}^2 = \frac{1}{2}\langle E X', \mathsf{B} E X'\rangle - \frac{1}{2}\langle \mathsf{g} X', q^{-1} \mathsf{g} X'\rangle$ and $\mathsf{K}_1 := E_3^\mathsf{T} \mathsf{X} \mathsf{B}'$, we have

$$\mathsf{m}_\circ(X') = \tfrac{1}{2L}\langle X', \mathsf{B}' X'\rangle + \langle \mathsf{K}_1 X', \mathrm{d}' X'\rangle + \tfrac{1}{2}\langle \mathrm{d}' X', \mathsf{K}_2 \mathrm{d}' X'\rangle.$$

(d) For $v \in \Omega'^2$, $\|\mathrm{pair}(L^{-1} v, \mathsf{X} v)\|_{\mathcal{H}}^2 = L^{-1}\langle v, \hat{R} v\rangle$.

■

Proof. We prove (a). The vector $\mathfrak{R}^{-1}(0, E_3\theta)$ is the stationary point of $\langle \Xi, E_3 \theta\rangle - \frac{1}{2}\mathfrak{R}(Z, \Xi)$; this means that for all (Z, Ξ)

$$\langle \Xi, E_3 \theta\rangle = \langle \mathrm{pair}(Z, \Xi), \mathfrak{G}\theta\rangle_{\mathcal{H}} + \mathsf{S}_0 \langle Q_3 \Xi, Q_3 \mathsf{G}_\Xi \theta\rangle.$$

At $(Z, \Xi) = (-\delta' \Theta, Q_3^* \Theta)$ the pair vanishes, and $Q_3 E_3 = 1$; hence $\langle \Theta, \theta\rangle = \mathsf{S}_0 \langle Q_3 Q_3^* \Theta, Q_3 \mathsf{G}_\Xi \theta\rangle$ for every Θ, which gives $Q_3 \mathsf{G}_\Xi \theta = \mathsf{S}_0^{-1} (Q_3 Q_3^*)^{-1} \theta$.

Part (b) is (a) taken at $(Z, \Xi) = (\mathsf{G}_Z \theta, \mathsf{G}_\Xi \theta)$.

In (c), the first claim follows from Lem. 2.31, since $\mathfrak{R}^{-1}$ and pair are linear. To obtain the formula for $\mathsf{m}_\circ$ we expand the square: on the one hand $\|(H^\sharp X', \mathsf{N}^\sharp X')\|^2 = L^{-1}\langle X', \mathsf{B}' X'\rangle$; on the other hand the cross term is given by (a) at $(\bar{Z}, \Xi_{\bar{Z}})$, where $Q_3 \Xi_{\bar{Z}} = 0$ and $\Xi_{\bar{Z}} = \mathsf{X}\mathsf{B}' X'$. The second expression for $\mathsf{m}_\circ$ is the minimum in the proof of Prop. 2.27 (c).

Part (d) is $\langle Z, R_\sharp Z\rangle$ evaluated at $Z = L^{-1} v$. □

Definition 2.39 (Cut rows, masks, the number t_ρ)**.** The cut rows are $\ell'^{\mathrm{c}}_{y'}(P) := \ell'_{y'}(P)1\{\mathrm{dist}(y', P) \le \varrho(x')\}$, and we put $\ell'^{>} = \ell' - \ell'^{\mathrm{c}}$. Further, $\varsigma_> := (\|\ell'^{>}\|_{\mathrm{r}}\|\ell'^{>}\|_{\mathrm{c}})^{1/2} \ge \|\ell'^{>}\|$ and $\delta_{\mathrm{c}} := 3\varsigma_>$. A mask m is a map from the coarse sites to $\{0,1\}$; for a mask we set $\mathsf{B}'^{\mathrm{c}}_m := 1 + \ell'^{\mathrm{c}\mathsf{T}} m\, \ell'^{\mathrm{c}}$, and $\mathsf{B}'^{\mathrm{c}} = \mathsf{B}'^{\mathrm{c}}_1$. Finally,

$$\mathsf{t}_\rho := \|\ell'^{\mathrm{c}}\|_{\mathrm{r}} \sup_{P_0} \sum_{y' :\, \mathrm{dist}(y', P_0) > \rho_0} |\ell'^{\mathrm{c}}_{y'}(P_0)|.$$

■

Proposition 2.40 (Cutting and masking cost a factor $1+\delta_{\mathrm{c}}$)**.** Suppose that $\varsigma_> \le 1$. For every mask, $\mathsf{B}'^{\mathrm{c}}_m \le (1+\delta_{\mathrm{c}})\mathsf{B}'$; consequently $\hat{R} \le (1+\delta_{\mathrm{c}})(\mathsf{B}'^{\mathrm{c}}_m)^{-1}$. ■

Proof. If $\varsigma_> = 0$, the claim reduces to $m \le 1$. Otherwise, for $\delta > 0$ we have $|m^{1/2}\ell'^{\mathrm{c}}\zeta|^2 \le (1+\delta)|m^{1/2}\ell'\zeta|^2 + (1+\delta^{-1})|\ell'^{>}\zeta|^2$. The choice $\delta = \varsigma_>$ yields $\mathsf{B}'^{\mathrm{c}}_m \le 1 + (1+\varsigma_>)\ell'^{\mathsf{T}}\ell' + (\varsigma_>^2 + \varsigma_>) \le (1+3\varsigma_>)\mathsf{B}'$. Passing to inverses reverses the order, which gives the bound on $\hat{R}$. □

Definition 2.41 (Uniform exponential decay of the sixteen kernels)**.** Condition (Dec)(C_D, θ_D) holds if $C_D \ge 4$, $\theta_D \in (0,1]$, $\varpi_q(\theta_D) \le \frac{1}{2}\gamma_q$, and each of the kernels

$$q,\ \mathsf{g},\ \ell,\ \mathsf{T},\ \ell\mathsf{T},\ \mathsf{X},\ \mathsf{G},\ \ell\mathsf{G},\ \mathsf{G}_Z,\ \mathsf{G}_\Xi,\ H^\sharp,\ \mathsf{N}^\sharp,\ \mathsf{B}',\ \ell',\ \mathsf{K}_1,\ \mathsf{K}_2$$

satisfies $\|\cdot\|_{\theta_D} \le C_D$ on every torus of Def. 2.17. In that case we set $c_R := 64/\theta_D$. ■

Definition 2.42 (Sup bounds on the masked minimiser; two conditions on L)**.** We are given rows ℓ at some level, the next rows ℓ'^{c} and a radius ρ_0; the mask m ranges over all masks, and ζ' over the elements of Ω'^2 with $|\zeta'|_\infty \le 1$. We put $Z = L^{-1}\mathsf{B}'^{\mathrm{c}}_m\zeta'$, $\Xi = \Xi_Z$, $(\nu,\omega) = \mathrm{pair}(Z,\Xi)$ and $\sigma = \ell\nu$. Condition (N′) holds if there exist numbers $\mathsf{N}_\sharp, \mathsf{N}_Z, \mathsf{N}_\Xi$, free of N and x, such that

$$L^2\big[|\nu|_\infty + \big(2\|\ell\|_{\mathrm{c}} + (2\rho_0+1)^3\|\ell\|_{\mathrm{r}}\big)|\sigma|_{2,\infty}\big] \le \mathsf{N}_\sharp, \qquad L|Z|_\infty \le \mathsf{N}_Z, \qquad L|\Xi|_\infty \le \mathsf{N}_\Xi.$$

The two conditions on L read

$$\textbf{(L1)}\quad \mathsf{c}_{\mathrm{loss}} := \big(\mathsf{N}_\sharp L^{-1} + 60\,\mathsf{N}_Z + 4\,\mathsf{N}_\Xi\big)L^{-1/2} \le \tfrac{1}{4}d_0; \qquad \textbf{(L2)}\quad L \ge 115\,\mathsf{N}_\sharp^2.$$

Since $\mathsf{N}_Z \ge 1$, as the mask $m \equiv 0$ gives $LZ = \zeta'$, condition (L1) requires $L > (240/d_0)^2 > 2.3\cdot 10^5$. For the mask $m \equiv 1$ one has $\nu = H^\sharp\zeta' - \mathsf{T}(\mathsf{B}' - \mathsf{B}'^{\mathrm{c}})\zeta'$. ■

Definition 2.43 (Distances in the units of a level)**.** For a site y and plaquettes p, p' of one and the same level we put $|y-p| := |y - \pi(p)|_\infty$ and $|p-p'| := |\pi(p) - \pi(p')|_\infty$; these distances are periodic and measured in the units of that level. We write $\beta(p) \in \{1,2,3\}$ for the direction normal to p. For $F \in \Omega^2$, $F_\beta(z)$ denotes the value of F at the plaquette based at z with normal e_β. ■

Definition 2.44 (The sums g_1 and g_2)**.** We put

$$\mathsf{g}_1(\lambda) := \sum_{u\in\mathbb{Z}^3} |u|_\infty e^{-\lambda|u|_\infty}, \qquad \mathsf{g}_2(\lambda) := \sum_{u\in\mathbb{Z}^3} (1+|u|_\infty)^2 e^{-\lambda|u|_\infty}.$$

For $r \ge 1$ there are $24r^2 + 2$ points $u \in \mathbb{Z}^3$ with $|u|_\infty = r$. For $0 < \lambda \le 1$ we have $\mathsf{g}(\lambda) \le 51\lambda^{-3}$ and $\mathsf{g}_1(\lambda) \le 182\lambda^{-4}$. Moreover, $\mathsf{g}(\lambda) \le \mathsf{g}_2(\lambda)$ and $\mathsf{g}_1(\lambda) \le \mathsf{g}_2(\lambda)$. ■

Proof of the bounds. Put $t = \lambda/2 \le \frac{1}{2}$; all sums below run over $r \ge 1$. Since $e^\lambda - 1 \ge \lambda$ and $\sinh t \ge t$, we have $\sum e^{-\lambda r} = (e^\lambda - 1)^{-1} \le \lambda^{-1}$ and $\sum r e^{-\lambda r} = (4\sinh^2 t)^{-1} \le \lambda^{-2}$. Next, $\sum r^2 e^{-\lambda r} = \cosh t/(4\sinh^3 t) \le (4t^3)^{-1} = 2\lambda^{-3}$, where the inequality is equivalent to $\cosh t \le (\sinh t/t)^3$. This holds because

$$\cosh t \le 1 + \frac{t^2}{2} + \frac{t^4}{23} \le 1 + \frac{t^2}{2} + \frac{13t^4}{120} \le (1 + \frac{t^2}{6} + \frac{t^4}{120})^3 \le (\sinh t/t)^3,$$

the first and the last step by the power series of $\cosh t$ and $\sinh t$ for $t \le \frac{1}{2}$, the third by expanding the cube.

Further,

$$\sum r^3 e^{-\lambda r} = (2\cosh 2t + 4)/(16\sinh^4 t) \le [6(1+\frac{2t^2}{3}) + 1.5t^4]/[16t^4(1+\frac{2t^2}{3})] \le 6\lambda^{-4} + 0.1,$$

where the middle step bounds $2\cosh 2t + 4$ from above and $\sinh^4 t$ from below by power series, and the last step uses $16t^4 = \lambda^4$. By the count of points, and since $\lambda \le 1$,

$$\mathsf{g}(\lambda) = 1 + 24\sum r^2 e^{-\lambda r} + 2\sum e^{-\lambda r} \le 1 + 48\lambda^{-3} + 2\lambda^{-1} \le 51\lambda^{-3}$$

and

$$\mathsf{g}_1(\lambda) = 24\sum r^3 e^{-\lambda r} + 2\sum r e^{-\lambda r} \le 144\lambda^{-4} + 2.4 + 2\lambda^{-2} \le 182\lambda^{-4}.$$

□

Definition 2.45 (Differences, gradient and Laplacian)**.** For a cochain F of any degree and a direction α we put $(\partial_\alpha F)_p = F_{p+e_\alpha} - F_p$, where e_α is the unit step of the level of F, and $|\nabla F|^2 = \sum_\alpha |\partial_\alpha F|^2$. The operator $\nabla^*\nabla = \sum_\alpha \partial_\alpha^*\partial_\alpha$ acts on each component as the scalar Laplacian, whose symbol is $\hat{\Delta}(\kappa) = \sum_\alpha |e^{i\kappa_\alpha} - 1|^2 \le 12$. We write ∇' for the same objects on $\mathbb{T}'$. ■

Definition 2.46 (Translation covariant rows)**.** Rows ℓ are called translation covariant if for each translation t of the lattice and each site y there is an orthogonal map $U_{\mathsf{t},y} : \mathsf{H}_y \to \mathsf{H}_{y+\mathsf{t}}$ such that $\ell_{y+\mathsf{t}}(p+\mathsf{t}) = U_{\mathsf{t},y}\ell_y(p)$ for all p. For such rows B and R commute with the translations, $|\ell_y(p)|$ does not change when y and p are moved by the same translation, and $|\ell_y h|$, for a constant form h, does not change when y is moved by a translation. ■

Definition 2.47 (The defect on constant forms)**.** The defect of rows ℓ on constant forms is

$$\tau(\ell) := \sup_y \sup\{|\ell_y h| : h \in \Omega^2 \text{ constant}, |h|_\infty \le 1\}.$$

■

Definition 2.48 (The sets of rows)**.** Let $b \ge 1$, $K \ge 0$, $0 < \lambda \le 1$ and $\tau \ge 0$. We denote by $\mathcal{R}(b, K, \lambda, \tau)$ the set of translation covariant rows ℓ on a torus with

$$b_{\mathsf{B}} \le b, \qquad |\ell_y(p)| \le K e^{-\lambda|y-p|} \text{ for all } y, p, \qquad \tau(\ell) \le \tau.$$

Rows in this set satisfy $\|\ell\|_{\mathrm{r}} \le 3K\mathsf{g}(\lambda)$, $\|\ell\|_{\mathrm{c}} \le K\mathsf{g}(\lambda)$ and $|(\mathsf{B}-1)(p,p')| \le K^2\mathsf{g}(\frac{\lambda}{2})e^{-\frac{\lambda}{2}|p-p'|}$. For the third bound we write $(\mathsf{B}-1)(p,p') = \sum_y \langle \ell_y(p), \ell_y(p')\rangle$ and use $\sum_y e^{-\lambda(|y-p|+|y-p'|)} \le e^{-\frac{\lambda}{2}|p-p'|}\sum_y e^{-\frac{\lambda}{2}|y-p|}$, which follows from the triangle inequality. The empty rows belong to the family: $\emptyset \in \mathcal{R}(1, 0, \lambda, 0)$. ■

Definition 2.49 (Towers of tori and chains of rows)**.** Let N be even and $K \ge 1$. The tower of tori consists of $\mathbb{T}_k := \mathbb{Z}^3/NL^{K-k}\mathbb{Z}^3$, $0 \le k \le K$, each in its own units, and step k of the tower is the block step with $\mathbb{T} = \mathbb{T}_k$ and $\mathbb{T}' \cong \mathbb{T}_{k+1}$. Given radii $\varrho_1, \dots, \varrho_K \in \mathbb{N} \cup \{\infty\}$, the chain of rows starts with $\ell_0 = \emptyset$ on $\mathbb{T}_0$ and continues with $\ell_{k+1} = ((\ell_k)')^{\mathrm{c}}$ for $0 \le k < K$, that is, with the rows of Prop. 2.28 built from ℓ_k and cut at ϱ_{k+1} as in Def. 2.39; its forms are $\mathsf{B}_k = 1 + \ell_k^{\mathsf{T}}\ell_k$. The chain is uncut when $\varrho_k = \infty$ for all k. ■

Remark 2.50 (Standing conditions). Throughout the part of the section that runs from the subsection "Four tools", which opens with Lem. 2.51, to the end of the proof of Thm. 2.64, we assume that $L \ge 3$ is odd, $aL \ge 3/c_L$, $N \ge 2$ is even, and ℓ are rows on $\mathbb{T}$. Lem. 2.58 has hypotheses of its own, and they take precedence over the standing conditions: in that lemma the side of the torus is an integer ≥ 2, L is a real number with $1 \le L \le$ half that side, and R is not tied to rows.

2.2.6. *Four tools.*

Lemma 2.51 (Weighted decay of a column of an inverse)**.** Let I and J be finite sets, each point of which is placed in a pseudo-metric space with distance d. Let $\mathbb{A}$ be a real $J \times I$ matrix and $\mathfrak{R} = \mathbb{A}^{\mathsf{T}}\mathbb{A}$; let D be a positive diagonal matrix on I, with $N(\phi) = \langle \phi, \mathsf{D}\phi\rangle$; and let $\gamma > 0$ be such that $\mathfrak{R} \geq \gamma \mathsf{D}$. Let $\lambda > 0$ and $0 \leq \varkappa \leq \frac{1}{4}\gamma^{1/2}$ be such that

$$\sum_{r\in J}\Big(\sum_{i\in I} |\mathbb{A}(r,i)| \sinh\big(\lambda \mathsf{d}(r,i)\big)|\phi_i|\Big)^2 \leq \varkappa^2 N(\phi) \quad \text{for all } \phi. \tag{sinh}$$

Then for every $j \in I$ the vector $\Phi := \mathfrak{R}^{-1}\delta_j$ satisfies

$$\text{(a)}\ \sum_{i\in I} \mathsf{D}_i e^{2\lambda \mathsf{d}(i,j)}|\Phi_i|^2 \leq 4\gamma^{-2}\mathsf{D}_j^{-1}; \qquad \text{(b)}\ \sum_{r\in J} e^{2\lambda \mathsf{d}(r,j)}|(\mathbb{A}\Phi)_r|^2 \leq 4\gamma^{-1}\mathsf{D}_j^{-1}.$$

If (sinh) holds at λ, it holds with the same $\varkappa$ at every smaller λ. ■

Proof. Put $w := \lambda \mathsf{d}(\cdot, j)$ on $I \cup J$. By the triangle inequality, $|w_i - w_r| \leq \lambda \mathsf{d}(r,i)$ for $i \in I$ and $r \in J$. We let $e^{\pm w}$ act on I as diagonal matrices and put $\mathfrak{R}_w := e^w \mathfrak{R} e^{-w}$.

For real ϕ put $\mathsf{C}_r = \sum_i \mathbb{A}(r,i)\cosh(w_i - w_r)\phi_i$ and $\mathsf{S}_r = \sum_i \mathbb{A}(r,i)\sinh(w_i - w_r)\phi_i$, and let C and S be the vectors with these entries. Writing $e^{\pm(w_i - w_r)} = \cosh(w_i - w_r) \pm \sinh(w_i - w_r)$, we find $e^{-w_r}(\mathbb{A}e^w\phi)_r = \mathsf{C}_r + \mathsf{S}_r$ and $e^{w_r}(\mathbb{A}e^{-w}\phi)_r = \mathsf{C}_r - \mathsf{S}_r$. Consequently $\langle \phi, \mathfrak{R}_w\phi\rangle = \langle \mathbb{A}e^w\phi, \mathbb{A}e^{-w}\phi\rangle = |\mathsf{C}|^2 - |\mathsf{S}|^2$.

Since $|w_i - w_r| \leq \lambda \mathsf{d}(r,i)$ and $\cosh t - 1 \leq \sinh|t|$, both $|\sinh(w_i - w_r)|$ and $\cosh(w_i - w_r) - 1$ are at most $\sinh(\lambda \mathsf{d}(r,i))$; so (sinh) yields $|\mathsf{S}|, |\mathsf{C} - \mathbb{A}\phi| \leq \varkappa N(\phi)^{1/2}$. From $\mathfrak{R} \geq \gamma\mathsf{D}$ and $\varkappa \leq \frac{1}{4}\gamma^{1/2}$ we get $|\mathbb{A}\phi| \geq \gamma^{1/2}N(\phi)^{1/2} \geq 4\varkappa N(\phi)^{1/2}$. Therefore

$$\langle \phi, \mathfrak{R}_w\phi\rangle \geq (|\mathbb{A}\phi| - \varkappa N(\phi)^{1/2})^2 - \varkappa^2 N(\phi) \geq \frac{1}{2}|\mathbb{A}\phi|^2 \geq \frac{\gamma}{2}N(\phi),$$

where the first step uses $|\mathsf{C}| \geq |\mathbb{A}\phi| - \varkappa N(\phi)^{1/2} \geq 0$ and the second uses $\varkappa N(\phi)^{1/2} \leq \frac{1}{4}|\mathbb{A}\phi|$.

We now choose $\phi = \mathfrak{R}_w^{-1}\delta_j$. Since $w_j = 0$, we have $e^{-w}\delta_j = \delta_j$, and therefore $\phi = e^w\mathfrak{R}^{-1}\delta_j = e^w\Phi$.

For (a): as $\mathfrak{R}_w\phi = \delta_j$, the lower bound just proved gives $\frac{\gamma}{2}N(\phi) \leq \langle \phi, \delta_j\rangle \leq N(\phi)^{1/2}\mathsf{D}_j^{-1/2}$, the second step because $\mathsf{D}_j\phi_j^2 \leq N(\phi)$; hence $N(\phi) \leq 4\gamma^{-2}\mathsf{D}_j^{-1}$. Since $\phi_i = e^{\lambda \mathsf{d}(i,j)}\Phi_i$, the number $N(\phi)$ is the left side of (a).

For (b): now $e^{w_r}(\mathbb{A}\Phi)_r = \mathsf{C}_r - \mathsf{S}_r$, so the left side of (b) equals $|\mathsf{C} - \mathsf{S}|^2 \leq (|\mathsf{C}| + |\mathsf{S}|)^2$. By (a) and $\varkappa^2 \leq \gamma/16$,

$$|\mathsf{S}|^2 \leq \varkappa^2 N(\phi) \leq \frac{\gamma}{16}\cdot\frac{4}{\gamma^2}\mathsf{D}_j^{-1} = \frac{1}{4\gamma}\mathsf{D}_j^{-1}.$$

Moreover $|\mathsf{C}|^2 - |\mathsf{S}|^2 = \langle \phi, \mathfrak{R}_w\phi\rangle = \langle \phi, \delta_j\rangle$, and $\langle \phi, \delta_j\rangle \leq N(\phi)^{1/2}\mathsf{D}_j^{-1/2} \leq \frac{2}{\gamma}\mathsf{D}_j^{-1}$ by the bounds for (a); so

$$|\mathsf{C}|^2 = \langle \phi, \delta_j\rangle + |\mathsf{S}|^2 \leq (\frac{2}{\gamma} + \frac{1}{4\gamma})\mathsf{D}_j^{-1}.$$

Hence $|\mathsf{C}| + |\mathsf{S}| \leq (\frac{3}{2} + \frac{1}{2})\gamma^{-1/2}\mathsf{D}_j^{-1/2}$, and squaring gives (b). □

In the rest of this section, $\mathfrak{R}$ denotes the form of Def. 2.30 on $\Omega'^2 \oplus \Omega^3$, and D the matrix of Def. 2.32, outside Lem. 2.51, where both matrices are general.

Lemma 2.52 (Unfolding R by one more variable)**.** On the space $\Omega'^2 \oplus \Omega^3 \oplus \mathsf{H}$ we define

$$\mathbb{A}^{\mathrm{e}}(Z, \Xi, \sigma) := (W^*Z + \delta\Xi - \ell^{\mathsf{T}}\sigma,\ \ \sigma,\ \ a^{-1/2}\mathrm{d}'^*Z,\ \ \mathsf{S}_0^{1/2}Q_3\Xi), \qquad \mathfrak{R}^{\mathrm{e}} := \mathbb{A}^{\mathrm{e}\mathsf{T}}\mathbb{A}^{\mathrm{e}}, \qquad \mathsf{D}^{\mathrm{e}} := \mathrm{diag}(L, L^{-2}, 1).$$

The rows of $\mathbb{A}^{\mathrm{e}}$ are indexed by the plaquettes p, by the components of the spaces H_y, each component being placed at its site y, by the coarse bonds c, and by the coarse cubes q'.

(a) Let $\psi \in \Omega'^2 \oplus \Omega^3$, and let $(Z,\Xi) = \mathfrak{R}^{-1}\psi$ and $\nu = \mathsf{R}\mathbb{Y}(Z,\Xi)$. Then $(\mathfrak{R}^{\mathrm{e}})^{-1}(\psi,0) = (Z,\Xi,\ell\nu)$ and $\mathbb{A}^{\mathrm{e}}(Z,\Xi,\ell\nu) = (\nu,\ell\nu,a^{-1/2}\mathrm{d}'^{*}Z,\mathsf{S}_0^{1/2}Q_3\Xi)$. In the case $\psi = \iota\delta_P$ one has $\nu = H^{\sharp}\delta_P$ and $a^{-1}\mathrm{d}'^{*}Z = \mathsf{N}^{\sharp}\delta_P$.

(b) The form $\mathfrak{R}^{\mathrm{e}}$ satisfies $\mathfrak{R}^{\mathrm{e}} \geq \gamma_{\mathrm{e}}\mathsf{D}^{\mathrm{e}}$, where $\gamma_{\mathrm{e}} := \gamma_1/2b_{\mathsf{B}}$; this stays true if any $b \geq b_{\mathsf{B}}$ takes the place of b_{B}.

(c) Let $|\ell_y(p)| \leq Ke^{-\lambda|y-p|}$ and $b_{\mathsf{B}} \leq b$, let $\gamma_{\mathrm{e}} = \gamma_1/2b$, and put

$$\lambda' := \min\Big\{\frac{\gamma_{\mathrm{e}}^{1/2}}{44},\ \frac{\lambda L}{2},\ \frac{\lambda L\,\gamma_{\mathrm{e}}^{1/2}}{12K\mathsf{g}(\lambda/2)}\Big\} \quad \text{(the first alone if } K=0\text{)}.$$

Then (sinh) holds for $\mathbb{A}^{\mathrm{e}}$ and D^{e}, with the distance d of Def. 2.19, at the rate λ' and with a constant $\varkappa \leq \frac{1}{4}\gamma_{\mathrm{e}}^{1/2}$.

■

Proof. (a) Fix Y. As a function of $\sigma \in \mathsf{H}$, the quantity $|Y - \ell^{\mathsf{T}}\sigma|^2 + |\sigma|^2$ takes its least value at the point where $\sigma = \ell(Y - \ell^{\mathsf{T}}\sigma)$. We write $\nu := Y - \ell^{\mathsf{T}}\sigma$ at this point. Then $\sigma = \ell\nu$, hence $\nu + \ell^{\mathsf{T}}\ell\nu = Y$, which means $\nu = \mathsf{R}Y$; and the least value is $|\nu|^2 + |\ell\nu|^2 = \langle\nu, \mathsf{B}\nu\rangle = \langle Y, \mathsf{R}Y\rangle$.

Now let $Y = \mathbb{Y}(Z,\Xi)$. By the definition of $\mathbb{A}^{\mathrm{e}}$, the value $\mathfrak{R}^{\mathrm{e}}(Z,\Xi,\sigma)$ is the sum of $|Y-\ell^{\mathsf{T}}\sigma|^2+|\sigma|^2$ and of $a^{-1}|\mathrm{d}'^{*}Z|^2 + \mathsf{S}_0|Q_3\Xi|^2$, which does not involve σ; by Def. 2.26 of pair and Def. 2.30 of $\mathfrak{R}$, the value $\mathfrak{R}(Z,\Xi)$ is the sum of $\langle Y, \mathsf{R}Y\rangle$ and of the same two terms. Therefore $\min_\sigma \mathfrak{R}^{\mathrm{e}}(Z,\Xi,\sigma) = \mathfrak{R}(Z,\Xi)$. The vector $(\mathfrak{R}^{\mathrm{e}})^{-1}(\psi,0)$ maximises $\langle\psi,(Z,\Xi)\rangle - \frac{1}{2}\mathfrak{R}^{\mathrm{e}}$. We may maximise first over σ, which yields $\sigma = \ell\nu$, and then maximise $\langle\psi,(Z,\Xi)\rangle - \frac{1}{2}\mathfrak{R}(Z,\Xi)$, which yields $(Z,\Xi) = \mathfrak{R}^{-1}\psi$; so the maximiser is $(Z,\Xi,\ell\nu)$ as in (a). The first component of $\mathbb{A}^{\mathrm{e}}(Z,\Xi,\ell\nu)$ is $Y - \ell^{\mathsf{T}}\ell\nu = \nu$, and its other components are read off from the definition.

For the case $\psi = \iota\delta_P$, Lem. 2.31 gives $H^{\sharp} = \mathsf{R}\mathbb{Y}\mathfrak{R}^{-1}\iota$, so that $\nu = \mathsf{R}\mathbb{Y}\mathfrak{R}^{-1}\iota\delta_P = H^{\sharp}\delta_P$; the same lemma gives $Z = \bar{Z} - \delta'\Theta$, with $\bar{Z}$ and Θ evaluated at $X' = \delta_P$. As $\mathrm{d}'^{*}\delta' = 0$ and $\bar{Z} = L^{-1}\mathsf{B}'\delta_P$, we obtain $a^{-1}\mathrm{d}'^{*}Z = a^{-1}\mathrm{d}'^{*}\bar{Z} = \mathsf{N}^{\sharp}\delta_P$.

(b) The minimum over σ found above shows that $\mathfrak{R}^{\mathrm{e}} \geq \mathfrak{R}$ as forms in (Z,Ξ,σ). Let $\mathfrak{R}_{\ell=\emptyset}$ be the form $\mathfrak{R}$ at $\ell = \emptyset$, where $\mathsf{R} = 1$. We apply the bound $\mathsf{R} \geq b_{\mathsf{B}}^{-1}$ to the term $\langle Y, \mathsf{R}Y\rangle$, and $b_{\mathsf{B}} \geq 1$ to the two remaining terms; together with Prop. 2.33 (d) this gives

$$\mathfrak{R}^{\mathrm{e}} \geq \mathfrak{R} \geq b_{\mathsf{B}}^{-1}\mathfrak{R}_{\ell=\emptyset} \geq b_{\mathsf{B}}^{-1}\gamma_1\mathsf{D}.$$

Here γ_1 and D are those of Def. 2.32, with $\mathsf{D} = \mathrm{diag}(L \text{ on } \Omega'^2,\ L^{-2} \text{ on } \Omega^3)$. On the other hand $\mathfrak{R}^{\mathrm{e}} \geq |\sigma|^2$, since σ is the second component of $\mathbb{A}^{\mathrm{e}}(Z,\Xi,\sigma)$. Half the sum of these two lower bounds is at least $\gamma_{\mathrm{e}}\mathsf{D}^{\mathrm{e}}$, because $\gamma_1/b_{\mathsf{B}} \leq 1$.

(c) We begin by bounding, in each row of $\mathbb{A}^{\mathrm{e}}$, the distance from the row to every entry that occurs in it; the facts needed come from Lem. 2.18 and Def. 2.19. Write $p = (z;\mu\nu)$ and $P = (y;\mu\nu)$, and let λ denote the direction normal to p. For the kernel supports, take the representatives of the periodic displacements from Lem. 2.18. We have $W(P,p) \neq 0$ only when both $z_\mu - y_\mu$ and $z_\nu - y_\nu$ belong to $[-\mathsf{h}_L, 3\mathsf{h}_L]$ while $|z_\lambda - y_\lambda| \leq \mathsf{h}_L$; moreover $3\mathsf{h}_L/L < \frac{3}{2}$. The two cubes of p have the base points z and $z - e_\lambda$. Every row (y,α) has a single entry, namely $\sigma_{y,\alpha}$, which is located at y; in these rows of σ, and in $\mathsf{d}(p,y)$ below, y is a fine site. The four $P \ni c$ have their base points at c_- or at $c_- - Le_\nu$. Lastly, if a fine cube q has its base point at z and a coarse cube q' at y, as in Lem. 2.18, then $\psi_{q'}(q) \neq 0$ only if $z - y \in [-\mathsf{h}_L, 3\mathsf{h}_L]^3$. It follows that $\mathsf{d}(p,P) \leq \frac{3}{2}$ whenever $W(P,p) \neq 0$, that the distance between p and each of its two cubes is $\leq \frac{1}{L}$, that $\mathsf{d}(p,y) = |y-p|/L$, that in the rows of σ the distance is 0, that the distance between c and each of the four $P \ni c$ is ≤ 1, and that the distance is $\leq \frac{3}{2}$ whenever $\psi_{q'}(q) \neq 0$.

Put $\mathsf{s} := \sinh(\frac{3}{2}\lambda')$; then $\mathsf{s} \leq 1.8\lambda'$, because $\lambda' \leq \gamma_{\mathrm{e}}^{1/2}/44 \leq \frac{1}{44}$. Since sinh is convex on $[0,\infty)$ and vanishes at 0, we have $\sinh(\theta t) \leq \theta \sinh t$ for $0 \leq \theta \leq 1$ and $t \geq 0$. With $\theta = 2/(3L)$ and $t = \frac{3}{2}\lambda'$, this bound gives $\sinh(\lambda'/L) \leq \frac{2}{3L}\mathsf{s} \leq \mathsf{s}/L$. We put $\theta_L := 2\lambda'/\lambda L$ for every K. If $K > 0$,

then $\theta_L \le 1$ by the second entry of the minimum that defines λ', and the bound $\sinh(\theta t) \le \theta \sinh t$ at $\theta = \theta_L$ gives $\sinh(\lambda' n/L) \le \theta_L \sinh(\frac{\lambda}{2} n) \le \frac{1}{2}\theta_L e^{\lambda n/2}$ for $n \ge 0$. If $K = 0$, then $\ell = 0$, the rows p contain no entry of σ, and this bound is not used.

Consider now the row of a plaquette p. The components of σ_y occur in it with the coefficients of $\ell_y(p)$, and they are controlled through $\sum_\alpha |\ell_y(p)_\alpha||\sigma_{y,\alpha}| \le |\ell_y(p)||\sigma_y|$. Let x_1, x_2 and x_3 be the parts of the sum of this row in (sinh) that come from the entries of Z, of Ξ and of σ; we use $(x_1 + x_2 + x_3)^2 \le 3\sum x_i^2$. Since $\mathsf{d}(p, P) \le \frac{3}{2}$ where $W(P, p) \ne 0$, and since $\sum_P W(P, p) = L^{-1}$, the Cauchy–Schwarz inequality gives $x_1^2 \le \mathsf{s}^2 L^{-1} \sum_P W(P, p) Z_P^2$. The entries of Ξ in this row belong to the two cubes q and $\tilde{q}$ of p, carry the coefficients ± 1 and lie at distance at most $\frac{1}{L}$, whence $x_2^2 \le 2\mathsf{s}^2 L^{-2}(\Xi_q^2 + \Xi_{\tilde{q}}^2)$. Finally, $x_3 \le \sum_y |\ell_y(p)| \sinh(\lambda' |y - p|/L)|\sigma_y|$, by the Cauchy–Schwarz bound over the components α and because $\mathsf{d}(p, y) = |y - p|/L$.

We now sum over p. For the first term we use $\sum_p W(P, p) = L^2$, and for the second the fact that every cube has six faces. For the third we apply Schur's test: by the hypothesis on ℓ and the bound obtained from convexity, the sums of its kernel are $\le \frac{1}{2}\theta_L K \mathsf{g}(\frac{\lambda}{2})$ over y and $\le \frac{3}{2}\theta_L K \mathsf{g}(\frac{\lambda}{2})$ over p. Altogether the rows p contribute at most

$$3\mathsf{s}^2 L |Z|^2 + 36\mathsf{s}^2 L^{-2} |\Xi|^2 + \frac{9}{4}\theta_L^2 K^2 \mathsf{g}(\frac{\lambda}{2})^2 |\sigma|^2.$$

The row of a coarse bond c contains $a^{-1/2}$ times the entries $\pm Z_P$ of the four $P \ni c$, at distance at most 1. By the Cauchy–Schwarz inequality, and as every P has four bonds, the rows c contribute at most $16 a^{-1} \mathsf{s}^2 |Z|^2 \le \mathsf{s}^2 L |Z|^2$; the last step holds because $aL \ge 3/c_L > 18 \ge 16$ by the standing conditions of Rem. 2.50. The row of a coarse cube q' contains the entries $\mathsf{S}_0^{1/2} \psi_{q'}(q) \Xi_q$, at distance at most $\frac{3}{2}$; by the Cauchy–Schwarz inequality and $\sum_q \psi_{q'}(q) = L^3$, the square of its sum is at most $\mathsf{S}_0 \mathsf{s}^2 L^3 \sum \psi_{q'} \Xi^2$, and the total over all q', by $\sum_{q'} \psi_{q'} = 1$ and $\mathsf{S}_0 = L^{-5}$, is $\mathsf{s}^2 L^{-2} |\Xi|^2$. The rows of σ contribute nothing, as their distances vanish.

Adding the contributions of all rows and comparing with $\mathsf{D}^{\mathrm{e}} = \mathrm{diag}(L, L^{-2}, 1)$, we find that (sinh) holds with $\varkappa = \max\{\sqrt{37}\,\mathsf{s}, \frac{3}{2}\theta_L K \mathsf{g}(\frac{\lambda}{2})\}$. For the first member, $\sqrt{37}\,\mathsf{s} \le \sqrt{37} \cdot 1.8\lambda' \le 10.95\lambda' \le \frac{1}{4}\gamma_{\mathrm{e}}^{1/2}$, by the first entry of the minimum that defines λ'. For the second member, $\frac{3}{2}\theta_L K \mathsf{g}(\frac{\lambda}{2}) = 3\lambda' K \mathsf{g}(\frac{\lambda}{2})/\lambda L \le \frac{1}{4}\gamma_{\mathrm{e}}^{1/2}$; for $K > 0$ this follows from the third entry of that minimum, and for $K = 0$ the second member is 0, so that $\varkappa = \sqrt{37}\,\mathsf{s}$. □

Lemma 2.53 (Admitted pairs and the minimiser)**.** Let $X' \in \Omega'^2$. We call a pair $(F, \mathsf{n}) \in \Omega^2 \oplus \Omega'^1$ *admitted at* X' when $WF + \mathrm{d}'\mathsf{n} = X'$ and, for every $\Xi \in \ker Q_3$, $\langle \Xi, \mathrm{d}F \rangle = 0$. The notion is independent of ℓ.

(a) The pair $(\bar{F}, \bar{\mathsf{n}}) = (H^\sharp X', \mathsf{N}^\sharp X')$ is admitted at X', and $L^{-1}\langle X', \mathsf{B}' X' \rangle = \|(\bar{F}, \bar{\mathsf{n}})\|_{\mathcal{H}}^2 \le \|(F, \mathsf{n})\|_{\mathcal{H}}^2$ holds for every pair (F, n) admitted at X'. Every such pair satisfies, moreover, $\|(F, \mathsf{n}) - (\bar{F}, \bar{\mathsf{n}})\|_{\mathcal{H}}^2 = \|(F, \mathsf{n})\|_{\mathcal{H}}^2 - \|(\bar{F}, \bar{\mathsf{n}})\|_{\mathcal{H}}^2$.

(b) If $\mathsf{B}_1 \le \mathsf{B}_2$, then $\Phi(\mathsf{B}_1) \le \Phi(\mathsf{B}_2)$.

(c) Let $\mathsf{B}_1, \mathsf{B}_2 \ge 1$, put $\mathsf{B}_s = \mathsf{B}_1 + s(\mathsf{B}_2 - \mathsf{B}_1)$ for $0 \le s \le 1$, and let $H_s^\sharp$ be defined with B_s. Then

$$\Phi(\mathsf{B}_2) - \Phi(\mathsf{B}_1) = L \int_0^1 H_s^{\sharp\mathsf{T}} (\mathsf{B}_2 - \mathsf{B}_1) H_s^\sharp \, ds.$$

■

Proof. (a) By Prop. 2.27 (a), $W\bar{F} + \mathrm{d}'\bar{\mathsf{n}} = X'$. Since $\Xi_{\bar{Z}}$ minimises $\langle Y, \mathsf{R}Y \rangle$ over $\ker Q_3$, the first variation at $\Xi_{\bar{Z}}$ vanishes: $\langle \delta\Xi, \mathsf{R}\bar{Y} \rangle = 0$ for $\Xi \in \ker Q_3$. As $\mathsf{R}\bar{Y} = \bar{F}$ and $\delta = \mathrm{d}^*$, this is the condition $\langle \Xi, \mathrm{d}\bar{F} \rangle = 0$ for $\Xi \in \ker Q_3$, and so $(\bar{F}, \bar{\mathsf{n}})$ is admitted at X'. Now let (F, n) be any pair admitted at X'. Using first $\bar{F} = \mathsf{R}\bar{Y}$ and $a\bar{\mathsf{n}} = \mathrm{d}'^* \bar{Z}$, then $\bar{Y} = W^* \bar{Z} + \delta \Xi_{\bar{Z}}$, and finally the two conditions of admission together with $\Xi_{\bar{Z}} \in \ker Q_3$, we get

$$\langle (\bar{F}, \bar{\mathsf{n}}), (F, \mathsf{n}) \rangle_{\mathcal{H}} = \langle \bar{Y}, F \rangle + \langle \mathrm{d}'^* \bar{Z}, \mathsf{n} \rangle = \langle \bar{Z}, WF + \mathrm{d}'\mathsf{n} \rangle + \langle \Xi_{\bar{Z}}, \mathrm{d}F \rangle = \langle \bar{Z}, X' \rangle.$$

This number is the same for all pairs admitted at X'. Taking $(F,\mathsf{n}) = (\bar F,\bar{\mathsf{n}})$ we see that it equals $\|(\bar F,\bar{\mathsf{n}})\|^2_{\mathcal H}$, and since $\bar Z = L^{-1}\mathsf{B}'X'$ it also equals $L^{-1}\langle X',\mathsf{B}'X'\rangle$. Expanding the square in $\|(F,\mathsf{n})-(\bar F,\bar{\mathsf{n}})\|^2_{\mathcal H}$ yields the last identity of (a); as its left side is ≥ 0, the inequality follows too.

(b) Let $(\bar F_2,\bar{\mathsf{n}}_2)$ be the minimiser for B_2. By (a) it is admitted at X', and admission is independent of ℓ. Hence (a) applied with B_1, then $\mathsf{B}_1\leq\mathsf{B}_2$, and (a) applied with B_2 give

$$L^{-1}\langle X',\Phi(\mathsf{B}_1)X'\rangle \leq \langle \bar F_2,\mathsf{B}_1\bar F_2\rangle + a|\bar{\mathsf{n}}_2|^2 \leq L^{-1}\langle X',\Phi(\mathsf{B}_2)X'\rangle$$

for every X', which is (b).

(c) Put $m(s) := L^{-1}\langle X',\Phi(\mathsf{B}_s)X'\rangle$, $D=\mathsf{B}_2-\mathsf{B}_1$ and $\bar F_s = H^\sharp_s X'$, and let $s,s'\in[0,1]$. The minimiser for B_s is admitted at X', and the square of its norm in the space $\mathcal H$ defined with $\mathsf{B}_{s'} = \mathsf{B}_s + (s'-s)D$ equals $m(s)+(s'-s)\langle \bar F_s, D\bar F_s\rangle$; so (a) applied with $\mathsf{B}_{s'}$ gives $m(s')\leq m(s)+(s'-s)\langle \bar F_s,D\bar F_s\rangle$. Using this as it stands and with s and s' exchanged, we obtain for $s'>s$

$$\langle \bar F_{s'},D\bar F_{s'}\rangle \leq \frac{m(s')-m(s)}{s'-s} \leq \langle \bar F_s,D\bar F_s\rangle.$$

The operator $H^\sharp_s$ depends continuously on s: it is a rational function of the entries of B_s that has no poles for $0\leq s\leq 1$, because $R_\sharp>0$. Letting s' tend to s from above in this display, and from below in the display with s and s' exchanged, we conclude that $m'(s)=\langle \bar F_s,D\bar F_s\rangle$. Integrating from 0 to 1 we obtain the identity of (c) between the quadratic forms of its two sides at X', and polarisation gives (c). □

Lemma 2.54 (The gradient of the minimiser)**.** Let ℓ be translation covariant, and let $b_{\mathsf{B}}\leq b$ and $\|\mathsf{B}'\|\leq b'$. Then for every $X'\in\Omega'^2$

$$|\nabla H^\sharp X'| \leq 23\, b\, b'\, L^{-1}|X'|.$$

■

The natural size of $|\nabla H^\sharp X'|/|X'|$ is $L^{-3/2}$; the bound above is weaker because W^*Z jumps across the seams of the blocks.

Proof. Step (i): the gradient of W^*Z. We show that $|\nabla W^*Z|^2\leq(4+\frac{48}{L})|Z|^2\leq 20|Z|^2$ for $Z\in\Omega'^2$. Take the component $\mu\nu$ and a direction $\alpha\in\{\mu,\nu\}$. We use the product form of $W(P,p)$ in Lem. 2.18, with the tent function t. When p moves by e_α, one tent factor of $W(P,p)$ changes by at most 1 while the other one stays at most L, so $|W(P,p+e_\alpha)-W(P,p)|\leq L^{-3}\cdot 1\cdot L$. For a given p this difference is non-zero for at most $3\cdot 2\cdot 1=6$ plaquettes P, and for a given P for at most $2L\cdot 2L\cdot L$ plaquettes p. By Schur's test, $|\partial_\alpha W^*Z|^2\leq 6L^{-2}\cdot 4L\,|Z|^2$, and there are two such directions.

In the normal direction $\alpha=\lambda$, the number $W(P,p)$ depends on z_λ only through the layer of blocks in which z_λ lies. Hence $\partial_\lambda W^*Z$ is zero unless z_λ lies at the top of its block, and at such p it equals $\sum_P W(P,p)(Z_{P+Le_\lambda}-Z_P)$. By the Cauchy–Schwarz inequality and $\sum_P W(P,p)=L^{-1}$, its square is at most $L^{-1}\sum_P W(P,p)(Z_{P+Le_\lambda}-Z_P)^2$; moreover $\sum_{p:\,z_\lambda\text{ fixed}} W(P,p)=L$. Therefore $|\partial_\lambda W^*Z|^2\leq\sum_P(Z_{P+Le_\lambda}-Z_P)^2\leq 4|Z|^2$. Adding the two tangential directions and the normal one over all components gives the claim of Step (i); its last inequality holds as $L\geq 3$.

Step (ii): the bound on $\nabla H^\sharp X'$. Put $\nu := H^\sharp X' = \mathsf{R}Y$, where $Y = W^*\bar Z+\delta\Xi$ and $\Xi=\Xi_{\bar Z}$. Since $\bar Z = L^{-1}\mathsf{B}'X'$, we have $|\bar Z|\leq L^{-1}b'|X'|$. As R commutes with translations and $\|\mathsf{R}\|\leq 1$, we get $|\nabla\nu|\leq|\nabla Y|\leq|\nabla W^*\bar Z|+|\nabla\delta\Xi|$. Since the difference operators commute,

$$|\nabla\delta\Xi|^2=\sum_{\alpha,\beta}|\partial_\alpha\partial^*_\beta\Xi|^2=\langle\Xi,(\sum_\alpha\partial^*_\alpha\partial_\alpha)(\sum_\beta\partial_\beta\partial^*_\beta)\Xi\rangle=|\mathrm{d}\delta\Xi|^2.$$

We now use Fourier series on $\mathbb{T}$. Let $\hat u(\kappa)\in\mathbb{C}^3$ be the symbol of δ on Ω^3, whose entries are $\pm(1-e^{-i\kappa_\beta})$, and let $\hat{\mathsf{R}}(\kappa)$ be the 3×3 symbol of R. Then $\langle\hat u,\hat{\mathsf{R}}\hat u\rangle$ is the symbol of the scalar operator $\mathrm{d}\mathsf{R}\delta$, and $\langle\hat u,\hat{\mathsf{R}}\hat u\rangle\geq b^{-1}|\hat u|^2$ as $\mathsf{R}\geq b^{-1}$, where $|\hat u|^2=\hat\Delta$ is the symbol of $\mathrm{d}\delta$. Since $\mathsf{R}\delta\Xi=\nu-\mathsf{R}W^*\bar Z$,

this gives $|\mathrm{d}\delta\Xi| \le b|\mathrm{d}\mathsf{R}\delta\Xi| \le b(|\mathrm{d}\nu| + |\mathrm{d}\mathsf{R}W^*\bar{Z}|)$. For $F \in \Omega^2$ one has $(\mathrm{d}\mathsf{R}F)(z) = \sum_\alpha \pm(\mathsf{R}\partial_\alpha F)_\alpha(z)$, hence $|\mathrm{d}\mathsf{R}F|^2 \le 3|\nabla F|^2$, again by $\|\mathsf{R}\| \le 1$.

By Lem. 2.53 (a) the pair $(\nu, \bar{\mathsf{n}})$ is admitted at X'. So $\mathrm{d}\nu$ is orthogonal to $\ker Q_3$, that is $\mathrm{d}\nu \in \operatorname{ran} Q_3^*$, and $Q_3\mathrm{d}\nu = \mathrm{d}'W\nu = \mathrm{d}'(X' - \mathrm{d}'\bar{\mathsf{n}}) = \mathrm{d}'X'$, the first equality by $Q_3\mathrm{d} = \mathrm{d}'W$ from Lem. 2.18. With $Q_3Q_3^* = L^3\hat{\mathsf{J}}$, $\hat{\mathsf{J}} \ge \frac{1}{27}$, again from Lem. 2.18, and with $|\mathrm{d}'X'|^2 \le 12|X'|^2$, it follows that $|\mathrm{d}\nu|^2 = \langle \mathrm{d}'X', (Q_3Q_3^*)^{-1}\mathrm{d}'X'\rangle \le 27L^{-3}\cdot 12|X'|^2$. We collect the bounds, using Step (i) for $\bar{Z}$:

$$|\nabla\nu| \le \sqrt{20}(1+\sqrt{3}b)b'L^{-1}|X'| + 18bL^{-3/2}|X'| \le (4.48 + 7.75 + 10.4)\, bb'L^{-1}|X'|,$$

where the second inequality uses $b, b' \ge 1$ and $L \ge 3$. As $4.48 + 7.75 + 10.4 < 23$, the lemma follows. □

2.2.7. *One step on rows.*

Proposition 2.55 (One step on a set of rows)**.** Let $\ell \in \mathcal{R}(b, K, \lambda, \tau)$ on $\mathbb{T}$, let $\gamma_\mathrm{e} = \gamma_1/2b$, let λ' be the number of Lem. 2.52 (c), and put $K_\natural := 3\gamma_\mathrm{e}^{-1/2}$.

(a) The rows ℓ' of Prop. 2.28 are translation covariant on $\mathbb{T}'$, and $|\ell'_{y'}(P)| \le K_\natural e^{-\lambda'\mathrm{dist}(y',P)}$ for all y', P. Furthermore, for all cells i, j of $\Omega'^2 \oplus \Omega^3$ and every $0 < \lambda'' \le \lambda'$ we have

$$|\mathfrak{R}^{-1}(i,j)| \le \frac{2}{\gamma_\mathrm{e}}(\mathsf{D}_i\mathsf{D}_j)^{-1/2}e^{-\lambda''\mathsf{d}(i,j)},$$

and, for $(Z, \Xi) = \mathfrak{R}^{-1}\iota\delta_P$ and $\nu = H^\sharp\delta_P$,

$$\sum_q e^{2\lambda''\mathsf{d}(q,P)}\Xi_q^2 \le 4\gamma_\mathrm{e}^{-2}L, \qquad \sum_p e^{2\lambda''\mathsf{d}(p,P)}\nu_p^2 \le 4\gamma_\mathrm{e}^{-1}L^{-1}.$$

(b) The form B' of the rows ℓ' satisfies

$$1 \le \mathsf{B}' \le \mathsf{r}_\sharp^{-1} + 9522\,\mathsf{r}_\sharp^{-2}K^2\mathsf{g}_1(\lambda)^2L^{-1} + 2\tau^2\mathsf{r}_\sharp^{-1} =: b'.$$

(c) We have $\tau(\ell') \le \tau$. If $\tau = 0$, then $H^\sharp h' = L^{-2}h$ for every constant form h' on $\mathbb{T}'$, where h denotes the constant form on $\mathbb{T}$ with the components of h'.

(d) Let ϱ be a cut radius with $\mathsf{t} := 3K_\natural\mathsf{g}(\frac{\lambda'}{2})e^{-\lambda'\varrho/2} \le \frac{1}{3}$. Then $\|\ell'^{>}\|_{\mathrm{r},\varsigma_>} \le \mathsf{t}$, $\delta_\mathrm{c} \le 3\mathsf{t}$, and $\ell'^\mathrm{c} \in \mathcal{R}\big((1+3\mathsf{t})b', K_\natural, \lambda', \tau + \mathsf{t}\big)$.

■

Proof. (a) We begin with the covariance. Let $\mathsf{t}' \in L\mathbb{Z}^3$. The maps $W, Q_3, \mathrm{d}, \delta, \mathrm{d}'$ commute with the shift by t', and so does R. Hence the shift maps $\ker Q_3$ onto itself and carries Ξ_Z to the minimiser that belongs to the shifted Z, and it commutes with $R_\sharp$, B', $H^\sharp$ and $\mathsf{N}^\sharp$. It follows that the groups (1) to (4) of $\ell'_{y'+\mathsf{t}'}$, evaluated at the shifted form, consist of the numbers of $\ell'_{y'}$ with shifted indices. For the group (5) one has $\ell_{y+\mathsf{t}'}H^\sharp(\text{shifted } \zeta) = U_{\mathsf{t}',y}\ell_y H^\sharp\zeta$. Thus ℓ' is translation covariant on $\mathbb{T}'$.

Next we prove the bound. Fix P, put $\Phi = (\mathfrak{R}^\mathrm{e})^{-1}(\iota\delta_P, 0)$, and note that $\mathsf{D}_P^\mathrm{e} = L$. By Lem. 2.52 (a), $\Phi = (Z, \Xi, \ell\nu)$ with $(Z, \Xi) = \mathfrak{R}^{-1}\iota\delta_P$ and $\nu = H^\sharp\delta_P$, and the column $\ell'_{y'}(P)$ splits into a sum of two vectors. The first of them carries the numbers $L^{1/2}\nu_p$ for the p based in $B(y')$, the numbers $L^{1/2}a^{-1/2}(\mathrm{d}^*Z)_c = (aL)^{1/2}\mathsf{N}^\sharp(c, P)$ for the c based at y', the vectors $L^{1/2}(\ell_y\nu)$ for $y \in B(y')$, and zeros in the groups (1) and (2). Apart from these zeros, the entries of the first vector are the entries $(\mathbb{A}^\mathrm{e}\Phi)_r$ multiplied by $L^{1/2}$, for the rows r of three kinds: the p based in $B(y')$, the c based at y', and the (y, α) with $y \in B(y')$. For each of these rows, $\mathrm{dist}(r, P) = \mathrm{dist}(y', P)$; hence $\mathsf{d}(r, P) > \mathrm{dist}(y', P) - 1$, because $|\mathsf{d} - \mathrm{dist}| < 1$ by Def. 2.19. By Lem. 2.52 (b) and (c) and by Lem. 2.51 (b), the norm of the first vector is therefore at most $L^{1/2}\cdot 2\gamma_\mathrm{e}^{-1/2}L^{-1/2}e^{\lambda'}e^{-\lambda'\mathrm{dist}(y',P)}$.

The second vector is explicit, and it vanishes for $\mathrm{dist}(y', P) > 1$. In the group (1) of Prop. 2.28, where $\zeta = \delta_P$ with $P = (y; \mu\nu)$, the four of its six numbers that involve the two other components of ζ vanish, so that at most the two that involve $\zeta_{\mu\nu}$ differ from 0. Each of these two is the

value of $u(\partial'_\rho \delta_P)$ at y' plus the value of $v(\partial'_\rho \delta_P)$ at the neighbour of y' along ρ'. Four sites enter here: y', the neighbours of y' along ρ and along ρ', and the site reached from y' by one step along ρ and one along ρ'; they are distinct, even for $N = 2$. Hence each of the two numbers has at most one non-zero term and lies in $\{\pm u, \pm v, 0\}$. So this group has norm at most $\sqrt{2} \cdot 0.41$, since $0 < v \le u \le \sqrt{\alpha} < \sqrt{c_L} < 6^{-1/2} < 0.41$ for the number α of Prop. 2.28. The group (2) has norm at most $18^{-1/2}\sqrt{3}$. The norm of the group (3) is $(\sum_p L^{-1} W(P,p)^2)^{1/2} \le (L^{-2} \sum_p W(P,p))^{1/2} = 1$, since $W(P,p) \le L^{-1}$ and $\sum_p W(P,p) = L^2$. The group (4) has norm at most $18^{-1/2}\sqrt{3}$. Altogether the second vector has norm at most 2.4. Where it does not vanish, $e^{\lambda'} e^{-\lambda' \mathrm{dist}(y',P)} \ge 1$; so the two bounds add up to $|\ell'_{y'}(P)| \le (2\gamma_{\mathrm{e}}^{-1/2} + 2.4) e^{\lambda'} e^{-\lambda' \mathrm{dist}(y',P)}$ for all y', P, and $(2\gamma_{\mathrm{e}}^{-1/2} + 2.4) e^{\lambda'} \le 3\gamma_{\mathrm{e}}^{-1/2}$ because $\lambda' \le \frac{1}{44}$ and $\gamma_{\mathrm{e}} = \gamma_1 / 2b \le \gamma_1 / 2 \le 1.72 \cdot 10^{-14} \le 10^{-13}$. This estimate of γ_{e} rests on $b \ge 1$ and on the numbers of Def. 2.32.

The remaining claims of (a) follow from Lem. 2.51 (a) and (b), applied with the rate λ'', because by Lem. 2.52 (a) the block of $(\mathfrak{R}^{\mathrm{e}})^{-1}$ in the variables (Z, Ξ) is $\mathfrak{R}^{-1}$.

(b) The lower bound $\mathsf{B}' \ge 1$ is Prop. 2.27 (b). For the upper bound, let $X' \in \Omega'^2$, and let $(\bar{F}_0, \bar{\mathsf{n}}_0)$ denote the minimiser for $\ell = \emptyset$ on the same two lattices. Lem. 2.53 (a) shows that this pair is admitted at X', and admission is independent of ℓ; so Lem. 2.53 (a) for the rows ℓ, together with Prop. 2.33 (c), gives

$$L^{-1} \langle X', \mathsf{B}' X' \rangle \le |\bar{F}_0|^2 + a |\bar{\mathsf{n}}_0|^2 + |\ell \bar{F}_0|^2 \le (L \mathsf{r}_\sharp)^{-1} |X'|^2 + |\ell \bar{F}_0|^2 .$$

We claim that every $F \in \Omega^2$ satisfies $|\ell F| \le 3K \mathsf{g}_1(\lambda) |\nabla F| + \tau |F|$. For the proof, let $h^{(y)}$ be the constant form whose components are the numbers $F_\beta(y)$. Splitting $\ell_y F = \ell_y (F - h^{(y)}) + \ell_y h^{(y)}$ and using the decay of the entries $\ell_y(p)$, we get $|\ell_y F| \le \sum_{\beta,u} K e^{-\lambda |u|_\infty} |F_\beta(y+u) - F_\beta(y)| + |\ell_y h^{(y)}|$. We take the norm in $\ell^2(y)$ and use Minkowski's inequality. The norm in $\ell^2(y)$ of the function $y \mapsto F_\beta(y+u) - F_\beta(y)$ is at most $\sum_i |u_i| |\partial_i F_\beta|$, and $\sum_{\beta,i} |\partial_i F_\beta| \le 3 |\nabla F|$; hence the first sum contributes at most $3K \mathsf{g}_1(\lambda) |\nabla F|$. For the last term, $|\ell_y h^{(y)}| \le \tau |h^{(y)}|_\infty \le \tau (\sum_\beta F_\beta(y)^2)^{1/2}$, and the norm of the right side in $\ell^2(y)$ is $\tau |F|$. This proves the claim.

Lem. 2.54 applied with $\ell = \emptyset$, where $b = 1$ and $b' = \mathsf{r}_\sharp^{-1}$, gives $|\nabla \bar{F}_0| \le 23 \mathsf{r}_\sharp^{-1} L^{-1} |X'|$, and the display above gives $|\bar{F}_0| \le (L \mathsf{r}_\sharp)^{-1/2} |X'|$. By the claim, $|\ell \bar{F}_0|$ is at most $|X'|$ times $69K \mathsf{g}_1(\lambda) \mathsf{r}_\sharp^{-1} L^{-1} + \tau (L \mathsf{r}_\sharp)^{-1/2}$; since the square of a sum of two terms is at most twice the sum of their squares,

$$L (69K \mathsf{g}_1(\lambda) \mathsf{r}_\sharp^{-1} L^{-1} + \tau (L \mathsf{r}_\sharp)^{-1/2})^2 \le 9522 K^2 \mathsf{g}_1(\lambda)^2 \mathsf{r}_\sharp^{-2} L^{-1} + 2\tau^2 \mathsf{r}_\sharp^{-1} .$$

Multiplying the first display of the proof of (b) by L and inserting the bound just found for $L |\ell \bar{F}_0|^2 / |X'|^2$, we obtain (b).

(c) Let h' be a constant form on $\mathbb{T}'$, and let h be the constant form on $\mathbb{T}$ with the components of h'. The pair $(L^{-2} h, 0)$ is admitted at h', because $\sum_p W(P,p) = L^2$ and $dh = 0$. Lem. 2.53 (a) therefore gives $L^{-1} \langle h', \mathsf{B}' h' \rangle \le L^{-4} (|h|^2 + |\ell h|^2)$, where $|h|^2 = L^3 |h'|^2$ and $|\ell h|^2 \le (NL)^3 \tau^2 |h|_\infty^2$. Hence

$$\sum_{y'} |\ell'_{y'} h'|^2 = \langle h', (\mathsf{B}' - 1) h' \rangle \le N^3 \tau^2 |h|_\infty^2 ,$$

and the N^3 terms on the left are equal by covariance; this gives $\tau(\ell') \le \tau$. If $\tau = 0$, the bound reads $L^{-1} \langle h', \mathsf{B}' h' \rangle \le L^{-1} |h'|^2$; as $\mathsf{B}' \ge 1$, the trial pair $(L^{-2} h, 0)$ attains the minimum, and the last identity of Lem. 2.53 (a) then shows that it is the minimiser $(H^\sharp h', \mathsf{N}^\sharp h')$.

(d) By (a), $\sum_{\mathrm{dist} > \varrho} |\ell'_{y'}(P)| \le 3 K_\natural e^{-\lambda' \varrho / 2} \mathsf{g}(\frac{\lambda'}{2})$ for the sum over P at fixed y', and the sum over y' at fixed P is at most a third of this bound. Hence $\|\ell'^{>}\|_{\mathrm{r}} \le \mathsf{t}$ and $\varsigma_> \le \mathsf{t}$, so that $\delta_{\mathrm{c}} \le 3\mathsf{t}$. The cut is decided by $\mathrm{dist}(y', P)$, which is invariant under a common translation of y' and P, so the cut rows ℓ'^{c} inherit the translation covariance of ℓ'. For the form of the cut rows we apply Prop. 2.40

with $m \equiv 1$. Finally, for a constant form h' we have $|\ell'^{\mathrm{c}}_{y'} h'| \le |\ell'_{y'} h'| + \|\ell'^{>}\|_{\mathrm{r}} |h'|_\infty$, which bounds the defect of ℓ'^{c} by $\tau + \mathsf{t}$, by (c). □

2.2.8. *The absorbing set of rows.*

Definition 2.56 (Pure numbers of the absorbing set)**.** A number is called *pure* if it depends neither on L, a, N, k, nor on the cuts, nor on x. We put

$$b_* := \frac{2}{\mathsf{r}_\sharp}, \qquad \gamma_* := \frac{\gamma_1}{2b_*} = \frac{\mathsf{r}_\sharp^2}{12296}, \qquad \lambda_* := \frac{\gamma_*^{1/2}}{44}, \qquad K_* := 3\gamma_*^{-1/2}, \qquad \mathsf{t}_*(\varrho) := 3K_* \mathsf{g}(\tfrac{\lambda_*}{2}) e^{-\lambda_* \varrho/2},$$

and

$$L_A := \max\Big\{ 12 K_* \mathsf{g}(\tfrac{\lambda_*}{2}) \gamma_*^{-1/2},\ 38088\, K_*^2 \mathsf{g}_1(\lambda_*)^2 \mathsf{r}_\sharp^{-1} \Big\}.$$

■

Theorem 2.57 (One step, and the whole chain)**.** Let $L \ge L_A$ be odd, let $aL \ge 3/c_L$, and let $N \ge 2$ be even.

(a) *One step.* Let $\ell \in \mathcal{R}(b_*, K, \lambda_*, \tau)$, where $\tau \le \frac{1}{9}$ and where K satisfies $12K\mathsf{g}(\frac{\lambda_*}{2})\gamma_*^{-1/2} \le L$ and $38088 K^2 \mathsf{g}_1(\lambda_*)^2 \mathsf{r}_\sharp^{-1} \le L$. Then $|\ell'_{y'}(P)| \le K_* e^{-\lambda_* \mathrm{dist}(y', P)}$, $1 \le \mathsf{B}' \le \frac{3}{2}\mathsf{r}_\sharp^{-1}$ and $\tau(\ell') \le \tau$. Moreover, $\ell'^{\mathrm{c}} \in \mathcal{R}(b_*, K_*, \lambda_*, \tau + \mathsf{t}_*(\varrho))$ for every ϱ with $\mathsf{t}_*(\varrho) \le \frac{1}{9}$.

(b) *The chain.* The condition on the radii of the cuts is

$$\sum_{k=1}^{K} \mathsf{t}_*(\varrho_k) \le \frac{1}{9}. \tag{Cut}$$

Let the radii satisfy (Cut). Then, for every tower, $\ell_k \in \mathcal{R}(b_*, K_*, \lambda_*, \frac{1}{9})$ for $0 \le k \le K$, and $(\ell_k)'$ obeys the bounds of (a) for $0 \le k < K$. For the uncut chain, $\tau(\ell_k) = 0$.

■□

Proof. (a) We apply Prop. 2.55 with $b = b_*$ and $\lambda = \lambda_*$, so that $\gamma_{\mathrm{e}} = \gamma_*$. As $L \ge 3$, and by the first condition on K, we get $\lambda' = \min\{\lambda_*, \frac{1}{2}\lambda_* L, \lambda_* \cdot L\gamma_*^{1/2}/12K\mathsf{g}(\frac{\lambda_*}{2})\} = \lambda_*$; moreover $K_\natural = K_*$. Prop. 2.55 (b) gives $b' \le \mathsf{r}_\sharp^{-1}(1 + \frac{1}{4} + \frac{2}{81}) \le \frac{3}{2}\mathsf{r}_\sharp^{-1}$, where the term $\frac{1}{4}$ comes from the second condition on K and the term $\frac{2}{81}$ from $\tau \le \frac{1}{9}$. With Prop. 2.55 (a) and (c), this proves the bounds on ℓ', B' and $\tau(\ell')$ in (a). If $\mathsf{t}_*(\varrho) \le \frac{1}{9}$, then Prop. 2.55 (d) applies with $\mathsf{t} = \mathsf{t}_*(\varrho)$, and $(1+3\mathsf{t}_*(\varrho))b' \le \frac{4}{3} \cdot \frac{3}{2}\mathsf{r}_\sharp^{-1} = b_*$; this proves the claim on ℓ'^{c}.

(b) We argue by induction on k, starting from $\emptyset \in \mathcal{R}(1, 0, \lambda_*, 0)$ and carrying the bound $\tau(\ell_k) \le \sum_{j \le k} \mathsf{t}_*(\varrho_j)$, which is at most $\frac{1}{9}$ by (Cut). The sets $\mathcal{R}(\cdot)$ become larger when b, K or τ is raised and λ is held fixed, so that $\mathcal{R}(1, 0, \lambda_*, 0) \subset \mathcal{R}(b_*, K_*, \lambda_*, 0)$. At every step K_* satisfies the two conditions on K in (a), by the definition of L_A. At step k the two tori are $\mathbb{T}_k$ and $\mathbb{T}_{k+1}$, and the side NL^{K-k-1} of the coarse one is even. We apply (a) to ℓ_k, with K_* as the bound K of the rows in (a), with $\tau = \tau(\ell_k)$ and with $\varrho = \varrho_{k+1}$; this choice of ϱ is allowed in (a), since (Cut) also bounds each single term, $\mathsf{t}_*(\varrho_{k+1}) \le \frac{1}{9}$. It yields the bounds on $(\ell_k)'$ and shows that ℓ_{k+1} lies in $\mathcal{R}(b_*, K_*, \lambda_*, \tau(\ell_k) + \mathsf{t}_*(\varrho_{k+1}))$. For the uncut chain, $\varrho_j = \infty$ gives $\mathsf{t}_*(\varrho_j) = 0$, and the carried bound yields $\tau(\ell_k) = 0$. □

2.2.9. *Kernels on cubes and their decay.*

Lemma 2.58 (The kernel of $(\mathrm{d}\mathsf{R}\delta + L^{-2})^{-1}$ on cubes)**.** Let $\mathcal{N} \ge 2$ be an integer, and let R be a real symmetric operator on $\Omega^2(\mathbb{Z}^3/\mathcal{N}\mathbb{Z}^3)$, commuting with translations, with $b^{-1} \le \mathsf{R} \le 1$ and $|\mathsf{R}(p, p')| \le Ce^{-\theta|p-p'|}$, where $b \ge 1$, $C \ge 1$ and $0 < \theta \le 1$. Put

$$\eta_0 := \min\Big\{\tfrac{1}{2}, \tfrac{\theta}{2}, \frac{1}{24bC\mathsf{g}_1(\theta/2)}\Big\}, \quad \omega := \min\{(156b)^{-1/2}, \tfrac{1}{4}\eta_0\}, \quad n_0 := 18\pi bC\mathsf{g}_1(\theta), \quad c^{\mathbb{Z}} := 4b\,(12\eta_0^{-2} + 900b^2),$$

$$\omega_{\mathcal{G}} := \tfrac{1}{2}\omega, \qquad c_{\mathcal{G}} := \max\{c^{\mathbb{Z}}\mathsf{g}(\tfrac{\omega}{2}),\ n_0^2(1+n_0)^2 e^{n_0}\}.$$

Then, for each real L in the range $1 \le L \le \frac{1}{2}\mathcal{N}$, the kernel $\mathcal{G} = (\mathrm{d}\mathsf{R}\delta + L^{-2})^{-1}$ on Ω^3 obeys, with the periodic distance $r = |x-y|_\infty$ and with x^+ a neighbour of x,

$$|\mathcal{G}(x,y)| \le c_{\mathcal{G}}(1+r)^{-1}e^{-\omega_{\mathcal{G}} r/L}, \qquad |\mathcal{G}(x^+,y) - \mathcal{G}(x,y)| \le c_{\mathcal{G}}(1+r)^{-2}e^{-\omega_{\mathcal{G}} r/L}. \tag{G_{R}}$$

If $\mathcal{N} \ge n_0$, these bounds hold with the first member of the maximum in place of $c_{\mathcal{G}}$, and $c^{\mathbb{Z}}\mathsf{g}(\frac{\omega}{2}) \le 2.2\cdot 10^{30}\, b^6 C^5 \theta^{-20}$. ∎

Proof. We write $n := \mathcal{N}$ and $m^2 := L^{-2}$. Because R commutes with translations, the entry of R that links the plaquettes with normals e_α, e_β based at z, z' is a function of $z - z'$ alone, which we denote by $\mathsf{R}_{\alpha\beta}(z-z')$. The symmetry of R gives $\mathsf{R}_{\alpha\beta}(u) = \mathsf{R}_{\beta\alpha}(-u)$.

Step (1): a lift to $\mathbb{Z}^3$. For $u \in \mathbb{Z}^3$ let $[u]$ be its class modulo n, and put $\mathsf{R}^{\mathbb{Z}}(u) := 2^{-\#\{i:|u_i|=n/2\}}\mathsf{R}([u])$ if $|u|_\infty \le \frac{n}{2}$, and $\mathsf{R}^{\mathbb{Z}}(u) := 0$ otherwise. Then $\sum_{j\in\mathbb{Z}^3}\mathsf{R}^{\mathbb{Z}}(u+nj) = \mathsf{R}([u])$, $\mathsf{R}^{\mathbb{Z}}_{\alpha\beta}(u) = \mathsf{R}^{\mathbb{Z}}_{\beta\alpha}(-u)$ and $|\mathsf{R}^{\mathbb{Z}}_{\alpha\beta}(u)| \le Ce^{-\theta|u|_\infty}$. We set $\tilde{\mathsf{R}}(k) := \sum_u \mathsf{R}^{\mathbb{Z}}(u)e^{-ik\cdot u}$. This is a 3×3 matrix of trigonometric polynomials, Hermitian for real k; at the points $k \in \frac{2\pi}{n}\mathbb{Z}^3$ it coincides with the symbol of R, whose spectrum lies in $[b^{-1},1]$.

Step (2): real k. From this step to the end of Step (6) we assume $n \ge n_0$; the case $n < n_0$ is treated in Step (7). For real k, let k_0 be a point of the grid $\frac{2\pi}{n}\mathbb{Z}^3$ with $|k-k_0|_\infty \le \frac{\pi}{n}$. The norm of a 3×3 matrix is at most $3\max|\text{entry}|$, and $|e^{-i(k-k_0)u}-1| \le \frac{\pi}{n}3|u|_\infty$. Hence $\|\tilde{\mathsf{R}}(k) - \tilde{\mathsf{R}}(k_0)\| \le \frac{9\pi C}{n}\mathsf{g}_1(\theta) \le \frac{1}{2b}$, the second bound by $n \ge n_0$. Since the spectrum of $\tilde{\mathsf{R}}(k_0)$ lies in $[b^{-1},1]$ and $b \ge 1$, we conclude that $\frac{1}{2b} \le \tilde{\mathsf{R}}(k) \le \frac{3}{2}$.

Step (3): the strip. We have $(\delta\Xi)_\beta(z) = s_\beta(\Xi(z) - \Xi(z-e_\beta))$ with signs $s_\beta = \pm 1$, so δ and d have the symbols $u_\beta(k) = s_\beta(1-e^{-ik_\beta})$ and $\tilde{u}_\beta(k) = s_\beta(1-e^{ik_\beta})$. The function $E(k) := \sum_{\alpha\beta}\tilde{u}_\alpha(k)\tilde{\mathsf{R}}_{\alpha\beta}(k)u_\beta(k)$ is entire and 2π-periodic. Let $k = \kappa + i\eta e_i$ with κ real and $|\eta| \le \eta_0$, and put $v := u(\kappa)$, so that $|v|^2 = \hat{\Delta}(\kappa) = \sum_\beta 2(1-\cos\kappa_\beta)$. We write $u(k) = v + \mathsf{a}$ and $\tilde{u}(k) = \bar{v} + \mathsf{a}'$; as $|\eta| \le \frac{1}{2}$, we have $|\mathsf{a}|, |\mathsf{a}'| \le e^{|\eta|} - 1 \le 1.3|\eta|$. Further, we write $\tilde{\mathsf{R}}(k) = \tilde{\mathsf{R}}(\kappa) + \varrho$; by the definition of η_0, $\|\varrho\| \le 3C\sum_u e^{-\theta|u|_\infty}(e^{|\eta||u|_\infty}-1) \le 3C|\eta|\mathsf{g}_1(\frac{\theta}{2}) \le \frac{1}{8b}$.

We expand $E(k)$ in these terms. The main term, with $\tilde{\mathsf{R}} = \tilde{\mathsf{R}}(\kappa)$, satisfies $\bar{v}^{\mathsf{T}}\tilde{\mathsf{R}}v \ge \frac{1}{2b}|v|^2$ by Step (2). Next, $|\bar{v}^{\mathsf{T}}\varrho v| \le \frac{1}{8b}|v|^2$. The four terms with one factor a or a' are together at most $2(\frac{3}{2}+\frac{1}{8})1.3|\eta||v| = 4.225|\eta||v| \le \frac{1}{8b}|v|^2 + 35.8b\eta^2$, and the two terms with both a' and a are at most $(\frac{3}{2}+\frac{1}{8})1.69\eta^2 \le 2.8\eta^2$. As $b \ge 1$, it follows that

$$\operatorname{Re} E(\kappa + i\eta e_i) \ \ge\ \tfrac{1}{4b}\hat{\Delta}(\kappa) - 39b\,\eta^2. \tag{2.v}$$

Step (4): the heat kernel on $\mathbb{Z}^3$. For $t \ge 0$ and $x \in \mathbb{Z}^3$ we define

$$P_t(x) := (2\pi)^{-3}\int_{[-\pi,\pi]^3} e^{ik\cdot x - tE(k)}dk.$$

Because $E \ge 0$ at real k, the kernel $\mathcal{G}^{\mathbb{Z}}$ is an integral of P_t over t:

$$\mathcal{G}^{\mathbb{Z}}(x) := (2\pi)^{-3}\int \frac{e^{ik\cdot x}}{E(k)+m^2}dk = \int_0^\infty e^{-tm^2}P_t(x)dt.$$

Let $r = |x|_\infty = |x_i|$, where i is a direction in which the maximum is attained. As a function of k_i the integrand of $P_t(x)$ is entire and periodic, so the shift $k_i \to k_i + i\eta\operatorname{sgn}x_i$, $0 \le \eta \le \eta_0$, of the path of integration leaves $P_t(x)$ unchanged. We write $s := t/4b$ and $A := 156b^2$.

We use eq. 2.v and the inequality $1-\cos\theta \ge 2\theta^2/\pi^2$, valid for $|\theta| \le \pi$. Since $\hat{\Delta}(\kappa) = \sum_\alpha 2(1-\cos\kappa_\alpha)$, the factor e^{-tE} of the shifted integrand has modulus at most $e^{A\eta^2 s}\prod_\alpha e^{-4s\kappa_\alpha^2/\pi^2}$, and the factor $e^{ik\cdot x}$ has modulus $e^{-\eta r}$. We put $\mathsf{a}_s := \min\{1, 0.45s^{-1/2}\} \ge \frac{1}{2\pi}\int e^{-4s\theta^2/\pi^2}d\theta$ and $\mathsf{b}_s := \min\{2, 0.4s^{-1}\} \ge \frac{1}{2\pi}\int|\theta|e^{-4s\theta^2/\pi^2}d\theta$, with integrals over $[-\pi,\pi]$; the bounds $0.45s^{-1/2}$ and $0.4s^{-1}$

follow by integrating on $\mathbb{R}$, where the exact coefficients are $\sqrt{\pi}/4$ and $\pi/8$. The bounds 1 and 2 follow by dropping the exponential on $[-\pi,\pi]$. In a difference $P_t(x\pm e_j)-P_t(x)$ the integrand has the additional factor $e^{\pm ik_j}-1$, and after the shift $|e^{\pm i(\kappa_j+i\eta_j)}-1|\le|\kappa_j|+1.3\eta$, where η_j, of modulus at most η, is the imaginary part of the shifted coordinate k_j. Integrating in each coordinate separately, we obtain

$$|P_t(x)|\le e^{-\eta r+A\eta^2 s}\mathsf{a}_s^3,\qquad |P_t(x\pm e_j)-P_t(x)|\le e^{-\eta r+A\eta^2 s}(\mathsf{b}_s+1.3\eta\mathsf{a}_s)\mathsf{a}_s^2.$$

Here only the bound splits into a product over the three coordinates; P_t itself does not.

We choose $\eta=\min\{\eta_0, r/2As\}$. In region II, that is, for $s\ge r/2A\eta_0$, the exponent $-\eta r+A\eta^2 s$ then equals $-r^2/4As$; in region I, the smaller values of s, we have $\eta=\eta_0$ and the exponent is $\le-\frac12\eta_0 r$.

Step (5): the integrals over s. The substitution $t=4bs$ turns the formula of Step (4) into $\mathcal{G}^{\mathbb{Z}}=4b\int_0^\infty e^{-\mu s}P_{4bs}ds$, with $\mu=4b/L^2$. We estimate the integral $\int_0^\infty e^{-\mu s}|P_{4bs}(x)|\,ds$, and the same integral with $P_{4bs}(x\pm e_j)-P_{4bs}(x)$ in place of $P_{4bs}(x)$, separately over the two regions; the factor $4b$ is restored at the end. Splitting the integrals at the breakpoints 0.2 and 0.2025 gives $\int_0^\infty \mathsf{a}_s^3\,\mathrm{d}s=0.6075$ and $\int_0^\infty \mathsf{b}_s\mathsf{a}_s^2\,\mathrm{d}s=0.8+0.4\log(0.2025/0.2)<0.805$.

Consider first $\mathcal{G}^{\mathbb{Z}}$ with $r\ge1$. Region I contributes at most $0.61e^{-\eta_0 r/2}$, and since $(1+r)e^{-\eta_0 r/4}\le\frac{4}{\eta_0}e^{-7/8}$, this is at most $\frac{1.02}{\eta_0}(1+r)^{-1}e^{-\eta_0 r/4}$. In region II, $\mathsf{a}_s^3\le0.0912s^{-3/2}$, and we write $e^{-r^2/4As}$ as the product of two factors $e^{-r^2/8As}$. One of them, together with $e^{-\mu s}$, is at most $e^{-r/(L\sqrt{78b})}$, because $\frac{r^2}{8As}+\mu s\ge r(\frac{\mu}{2A})^{1/2}=\frac{r}{L\sqrt{78b}}$; the other is integrated by means of $\int_0^\infty s^{-3/2}e^{-r^2/8As}ds=\sqrt{8\pi A}/r\le 62.7b/r$. Hence region II contributes at most $5.72\,b\,r^{-1}e^{-r/(L\sqrt{78b})}\le11.5b(1+r)^{-1}e^{-\omega r/L}$.

Consider next the differences, again with $r\ge1$. In region I we have $1.3\eta=1.3\eta_0\le0.65$, so this region contributes at most $(0.805{+}0.65{\cdot}0.6075)e^{-\eta_0 r/2}\le1.2e^{-\eta_0 r/2}$; since $(1{+}r)^2e^{-\eta_0 r/4}\le\frac{64}{\eta_0^2}e^{-15/8}$, this is at most $\frac{11.8}{\eta_0^2}(1+r)^{-2}e^{-\eta_0 r/4}$. In region II,

$$1.3\eta\mathsf{a}_s e^{-r^2/8As}\le1.3\cdot0.45\cdot\frac{1}{2\sqrt{A}}\sup_y(ye^{-y^2/8})s^{-1}\le0.0285s^{-1},$$

where $y=r/\sqrt{As}$; the supremum equals $2e^{-1/2}$, and $\sqrt{A}=\sqrt{156}\,b\ge12.48$, which yields the value 0.02844. With $\mathsf{b}_s\le0.4s^{-1}$ and $\mathsf{a}_s^2\le0.2025s^{-1}$, the integrand is therefore at most $0.4285\cdot 0.2025\,s^{-2}e^{-r^2/8As-\mu s}$. Splitting $e^{-r^2/8As}$ once more into two equal factors, we use $\frac{r^2}{16As}+\mu s\ge\frac{r}{L\sqrt{156b}}$ and $\int_0^\infty s^{-2}e^{-r^2/16As}ds=16A/r^2$; so region II contributes at most $216.6\,b^2r^{-2}e^{-r/(L\sqrt{156b})}\le867b^2(1+r)^{-2}e^{-\omega r/L}$.

For $r=0$ we take $\eta=0$, and the two integrals are at most 0.61 and 0.81. As $L\ge1$ and $\omega\le\frac14\eta_0$, we have $e^{-\eta_0 r/4}\le e^{-\omega r/L}$. We add the contributions of the two regions and multiply by $4b$. Since $\eta_0\le\frac12$ and $b\ge1$, the constants so obtained are at most $4b(12\eta_0^{-2}+900b^2)=c^{\mathbb{Z}}$, and for all $x\in\mathbb{Z}^3$

$$|\mathcal{G}^{\mathbb{Z}}(x)|\le c^{\mathbb{Z}}(1+r)^{-1}e^{-\omega r/L},\qquad |\mathcal{G}^{\mathbb{Z}}(x\pm e_j)-\mathcal{G}^{\mathbb{Z}}(x)|\le c^{\mathbb{Z}}(1+r)^{-2}e^{-\omega r/L}.$$

Step (6): the torus. The function $f(k)=(E(k)+m^2)^{-1}$ has the Fourier coefficients $\mathcal{G}^{\mathbb{Z}}$, which by Step (5) are absolutely summable. Consequently, for $x\in\mathbb{Z}^3$,

$$n^{-3}\sum_{k\in\frac{2\pi}{n}\mathbb{Z}^3_n}f(k)e^{ik\cdot x}=\sum_y\mathcal{G}^{\mathbb{Z}}(y)1\{y\equiv x\},$$

where $y\equiv x$ is congruence modulo n. On the grid $\frac{2\pi}{n}\mathbb{Z}^3$ the function E is the symbol of $\mathrm{dR}\delta$, so the left side is $\mathcal{G}(x,0)$, and $\mathcal{G}(x,0)=\sum_j\mathcal{G}^{\mathbb{Z}}(x+nj)$, where j runs over $\mathbb{Z}^3$. Let $|x|_\infty=r\le\frac{n}{2}$. For $j\ne0$ we have $|x+nj|_\infty\ge\max\{r,\frac{n}{2}|j|_\infty\}\ge\frac{r}{2}+\frac{n}{4}|j|_\infty$, and since $n\ge2L$, the weight satisfies $e^{-\omega|\cdot|_\infty/L}\le e^{-\omega r/2L}e^{-\omega|j|_\infty/2}$ at the point $x+nj$. As moreover $(1+|x+nj|_\infty)^{-1}\le(1+r)^{-1}$, the

bounds of Step (5), summed over j, give the bounds of Lem. 2.58 for $n \geq n_0$, with the rate $\omega_{\mathcal{G}} = \frac{1}{2}\omega$ and the constant $c^{\mathbb{Z}}\mathsf{g}(\frac{\omega}{2}) \leq c_{\mathcal{G}}$.

Step (7): the case $n < n_0$. Since $L \leq \frac{1}{2}n$, we have $0 < \mathcal{G} \leq L^2 \leq \frac{1}{4}n_0^2$ as an operator; so the entries of $\mathcal{G}$ and the differences of its entries are bounded by $\frac{1}{2}n_0^2$ in absolute value. As $r \leq \frac{1}{2}n_0$, $L \geq 1$ and $\omega_{\mathcal{G}} \leq 1$, we have $(1+r)^{-2}e^{-\omega_{\mathcal{G}} r/L} \geq (1+n_0)^{-2}e^{-n_0}$, and since $c_{\mathcal{G}} \geq n_0^2(1+n_0)^2e^{n_0}$, the bounds of Lem. 2.58 hold in this case too.

Step (8): bounds by powers of b, C and θ^{-1}. By the bounds $\mathsf{g}_1(\lambda) \leq 182\lambda^{-4}$ and $\mathsf{g}(\lambda) \leq 51\lambda^{-3}$ of Def. 2.44, valid for $0 < \lambda \leq 1$, the first applied to $\mathsf{g}_1(\frac{\theta}{2})$, we have $\eta_0^{-1} \leq 24 \cdot 182 \cdot 16\, bC\theta^{-4} \leq 7 \cdot 10^4 bC\theta^{-4}$. It follows that $c^{\mathbb{Z}} \leq 2.4 \cdot 10^{11} b^3C^2\theta^{-8}$ and $\omega^{-1} \leq 2.8 \cdot 10^5 bC\theta^{-4}$. The second of the two bounds of Def. 2.44 used above, $\mathsf{g}(\lambda) \leq 51\lambda^{-3}$, applied with $\lambda = \frac{\omega}{2}$, gives $\mathsf{g}(\frac{\omega}{2}) \leq 408\omega^{-3}$. We insert the bound on ω^{-1} into $\mathsf{g}(\frac{\omega}{2}) \leq 408\omega^{-3}$ and multiply the result by the bound on $c^{\mathbb{Z}}$. In the product, the powers are b^{3+3}, C^{2+3} and θ^{-8-12}, and the numerical factor is $2.4 \cdot 10^{11} \cdot 408 \cdot (2.8 \cdot 10^5)^3 \leq 2.2 \cdot 10^{30}$; hence $c^{\mathbb{Z}}\mathsf{g}(\frac{\omega}{2}) \leq 2.2 \cdot 10^{30}\, b^6C^5\theta^{-20}$, which is the last bound of Lem. 2.58. □

Lemma 2.59 (Sums over all cubes and over seam cubes)**.** Let $0 < \varpi \leq 1$ and $L \geq 3$, let $K_i(r) := (1+r)^{-i}e^{-\varpi r/L}$ for $i = 1, 2$ and $r \geq 0$, and let $\mathsf{l}_L := 1 + \log L$. A seam cube of direction β is a cube z whose coordinate z_β lies at the top of its block. Then, for every cube x of $\mathbb{T}$, with $r = |x-y|_\infty$ and with sums over the cubes y of $\mathbb{T}$,

$$\sum_y K_1 \leq \frac{25L^2}{\varpi^2}, \quad \sum_y K_2 \leq \frac{25L}{\varpi}, \quad \sum_\beta \sum_{y \text{ seam of } \beta} K_1 \leq \frac{216L}{\varpi^2}, \quad \sum_\beta \sum_{y \text{ seam of } \beta} K_2 \leq \frac{144}{\varpi}(1 + \log \tfrac{3}{\varpi})\mathsf{l}_L.$$

■

Proof. Let $\alpha := \varpi/L$; then $\alpha \leq \frac{1}{3}$. For $r \geq 1$ we have $\frac{24r^2+2}{1+r} \leq 24r$ and $\frac{24r^2+2}{(1+r)^2} \leq 24$, and also $\sum_{r\geq 1} re^{-\alpha r} \leq \alpha^{-2}$ and $\sum_{r\geq 1} e^{-\alpha r} \leq \alpha^{-1}$. At most $24r^2 + 2$ cubes y lie at distance $r \geq 1$ from x, and the cube $y = x$ contributes 1; as $1 \leq \alpha^{-1} \leq \alpha^{-2}$, this gives $\sum_y K_1 \leq 1 + 24\alpha^{-2} \leq 25\alpha^{-2}$ and $\sum_y K_2 \leq 1 + 24\alpha^{-1} \leq 25\alpha^{-1}$, which are the first two bounds.

We estimate the two sums over seam cubes one plane at a time. In one plane $y_\beta = s$ let $\mathsf{a} = |s - x_\beta|$, and let τ be the transverse sup distance from y to x; the plane contains at most 8τ cubes with transverse distance $\tau \geq 1$ and one with $\tau = 0$, and $r = \max\{\mathsf{a}, \tau\}$. For K_2, the cubes with $\tau \leq \mathsf{a}$, of which there are at most $(2\mathsf{a}+1)^2 \leq 4(1+\mathsf{a})^2$, contribute at most $4e^{-\alpha\mathsf{a}}$; the cubes with $\tau > \mathsf{a}$ contribute at most $8e^{-\alpha\mathsf{a}/2}\sum_{\tau\geq 1}\frac{e^{-\alpha\tau/2}}{1+\tau}$. Put $\beta_0 = \frac{\alpha}{2} \leq \frac{1}{6}$. In the last sum, the terms with $\tau \leq 1/\beta_0$ add up to at most $\sum_{\tau\leq 1/\beta_0}\frac{1}{1+\tau} \leq \log(1+\frac{2L}{\varpi}) \leq \log L + \log\frac{3}{\varpi}$, and the terms with $\tau > 1/\beta_0$ to at most $\beta_0\frac{e^{-1}}{1-e^{-\beta_0}} \leq 0.4$. So for K_2 the plane contributes at most $e^{-\alpha\mathsf{a}/2}[4 + 8(\mathsf{l}_L + \log\frac{3}{\varpi} - 0.6)] \leq 8e^{-\alpha\mathsf{a}/2}(\mathsf{l}_L + \log\frac{3}{\varpi})$.

For K_1 the plane contributes at most $4(1+\mathsf{a})e^{-\alpha\mathsf{a}} + \frac{8}{\alpha}e^{-\alpha\mathsf{a}}$; since $(1+\mathsf{a})e^{-\alpha\mathsf{a}/2} \leq \frac{2}{\alpha}e^{-5/6}$, this is at most $\frac{12}{\alpha}e^{-\alpha\mathsf{a}/2}$.

The seam cubes of direction β fill the planes $y_\beta = s$ with s at the top of a block. For each integer $j \geq 0$ at most two of these planes have $\mathsf{a} \in [jL, (j+1)L)$, so that over the seam planes of one direction $\sum e^{-\alpha\mathsf{a}/2} \leq \frac{2}{1-e^{-\varpi/2}} \leq \frac{6}{\varpi}$. There are three directions β; hence the sum over seam cubes is at most $3 \cdot \frac{12}{\alpha} \cdot \frac{6}{\varpi} = \frac{216L}{\varpi^2}$ for K_1, and at most $3 \cdot 8 \cdot \frac{6}{\varpi}(\mathsf{l}_L + \log\frac{3}{\varpi}) \leq \frac{144}{\varpi}(1 + \log\frac{3}{\varpi})\mathsf{l}_L$ for K_2, the last step because $\mathsf{l}_L \geq 1$. □

Theorem 2.60 (Decay of the kernels of one step)**.** Let $\ell \in \mathcal{R}(b, K, \lambda, \tau)$ be rows on $\mathbb{T}$, and let L, a and N satisfy the standing conditions of Rem. 2.50; none of the constants and bounds below depends on τ. Let $\gamma_{\mathrm{e}} = \gamma_1/2b$, let λ' be the rate of Lem. 2.52 (c), let $K_\natural = 3\gamma_{\mathrm{e}}^{-1/2}$, and let $0 < \lambda^\circ \leq \lambda'$. We define the following constants, in the order in which they are listed. First, $\theta_R := \min\{\frac{\lambda}{4}, [6K^2\mathsf{g}(\frac{\lambda}{2})\mathsf{g}_1(\frac{\lambda}{4})]^{-1}\}$, read as $\frac{\lambda}{4}$ if $K = 0$. Next, $(c_{\mathcal{G}}, \omega_{\mathcal{G}})$ are the constants of Lem. 2.58

at $(b, C, \theta) = (b, 2, \theta_R)$; then $\varpi := \frac{1}{2}\omega_{\mathcal{G}}$ and $\lambda_B := \min\{\lambda^\circ, \varpi\}$. Further, $\theta_J := \frac{1}{2}\log(1 + \frac{w_*}{50})$, a pure number; it is the rate in the bound $|\hat{\mathsf{J}}^{-2}(i,j)| \le \frac{2}{w_*} e^{-\theta_J \mathrm{dist}}$ of Rem. 2.36. Finally,

$$c_\Xi^\natural := \frac{6000\, c_{\mathcal{G}}\, \mathsf{g}_2(\theta_R/4)}{\gamma_{\mathrm{e}} \varpi^2}, \qquad c_H^\natural := \frac{180\, \mathsf{g}(\theta_R/2)\, c_{\mathcal{G}}\, c_\Xi^\natural}{\varpi}, \qquad \lambda_2^\natural := \min\Big\{\frac{\lambda^\circ}{4}, \frac{1}{6K_\natural^2 \mathsf{g}(\lambda^\circ/2)\mathsf{g}_1(\lambda^\circ/4)}\Big\},$$

and

$$c_1^\natural := 19\, c_H^\natural\, \mathsf{g}(\lambda_B)\mathsf{g}(\lambda_2^\natural), \qquad c_2^\natural := 19\, c_\Xi^\natural\, \mathsf{g}(\lambda_B)\mathsf{g}(\lambda_2^\natural) + 72\, w_*^{-1}\mathsf{g}(\theta_J)\mathsf{g}(\lambda_2^\natural).$$

We denote by pr_Ξ the projection of $\Omega'^2 \oplus \Omega^3$ onto Ω^3, that is, $\mathrm{pr}_\Xi(Z, \Xi) = \Xi$. Then the following bounds hold for all cells.

(a) $|\mathsf{R}(p, p')| \le 2e^{-\theta_R |p - p'|}$.

(b) $|(\mathrm{pr}_\Xi \mathfrak{R}^{-1} \iota)(q, P)| \le c_\Xi^\natural L^{-1} e^{-\lambda_B \mathsf{d}(q,P)}$, $\quad |H^\sharp(p, P)| \le c_H^\natural\, \mathsf{l}_L\, L^{-2} e^{-\lambda_B \mathsf{d}(p,P)}$.

(c) $|(\mathsf{B}' - 1)(P, P')| \le K_\natural^2 \mathsf{g}(\frac{\lambda^\circ}{2}) e^{-\frac{1}{2}\lambda^\circ \mathrm{dist}(P,P')}$, $\quad |\hat{R}(P, P')| \le 2e^{-\lambda_2^\natural \mathrm{dist}(P,P')}$.

(d) $L^2 \|\mathsf{T}\|_\infty \le c_1^\natural \mathsf{l}_L$, $\quad L\|\mathsf{X}\|_\infty \le c_2^\natural$, $\quad L^2 \sup_{|v|_\infty \le 1} |\ell \mathsf{T} v|_{2,\infty} \le 3K \mathsf{g}(\lambda) c_1^\natural \mathsf{l}_L$.

Each of these constants is a function of $(b, K, \lambda, \lambda^\circ)$ alone; none of them depends on L, a or N. ■□

Proof. Step (a): the kernel R. We have $\mathsf{B} \ge 1$. For $\theta \le \frac{\lambda}{4}$, the third bound in the second paragraph of Def. 2.48, $|(\mathsf{B} - 1)(p, p')| \le K^2 \mathsf{g}(\frac{\lambda}{2}) e^{-\frac{\lambda}{2}|p - p'|}$, and the inequality $e^t - 1 \le te^t$ give

$$\varpi_{\mathsf{B}}(\theta) \le 3K^2 \mathsf{g}(\frac{\lambda}{2}) \sum_u e^{-\frac{\lambda}{2}|u|_\infty} (e^{\theta|u|_\infty} - 1) \le 3K^2 \mathsf{g}(\frac{\lambda}{2}) \theta \mathsf{g}_1(\frac{\lambda}{4}).$$

At $\theta = \theta_R$ the right side is at most $\frac{1}{2}$, and Lem. 2.25 (a), with the distance $|p - p'|$, gives the bound (a).

Step (b): the rates, and the equation for Ξ. By the definitions of ω and η_0 in Lem. 2.58, at $\theta = \theta_R$, we have $\omega_{\mathcal{G}} = \frac{1}{2}\omega \le \frac{1}{8}\eta_0 \le \frac{1}{16}\theta_R$. Hence $\lambda_B \le \varpi \le \frac{1}{32}\theta_R$; moreover $\varpi \le \frac{1}{128}$, because $\theta_R \le \frac{\lambda}{4} \le \frac{1}{4}$, and $\lambda_B + \varpi \le \omega_{\mathcal{G}}$.

We fix a coarse plaquette P and put $(Z, \Xi) = \mathfrak{R}^{-1} \iota \delta_P$. Prop. 2.55 (a), applied with $\lambda'' = \lambda_B$, gives the two bounds (e1) $|Z_{P'}| \le \frac{2}{\gamma_{\mathrm{e}}} L^{-1} e^{-\lambda_B \mathsf{d}(P', P)}$ for all P', and (e2) $\sum_q e^{2\lambda_B \mathsf{d}(q,P)} \Xi_q^2 \le 4\gamma_{\mathrm{e}}^{-2} L$. The Ξ-component of the equation $\mathfrak{R}(Z, \Xi) = \iota \delta_P$ reads $\mathrm{d}\mathsf{R}(W^* Z + \delta\Xi) + \mathsf{S}_0 Q_3^* Q_3 \Xi = 0$; in other words,

$$(\mathrm{d}\mathsf{R}\delta + L^{-2})\Xi = s_1 + s_2 + s_3, \qquad s_1 = -\mathrm{d}\mathsf{R} W^* Z, \quad s_2 = -\mathsf{S}_0 Q_3^* Q_3 \Xi, \quad s_3 = L^{-2}\Xi,$$

and $\mathcal{G} = (\mathrm{d}\mathsf{R}\delta + L^{-2})^{-1}$ is the kernel of Lem. 2.58 with $\mathcal{N} = NL \ge 2L$. We put $\mathsf{z} := \frac{2.1}{\gamma_{\mathrm{e}}} L^{-1}$; the numeral 2.1 rests on $2e^{3\lambda_B} \le 2e^{3/128} \le 2.0475 \le 2.1$.

Step (b): the source s_1. Let $Y = W^* Z$. Then $(\mathrm{d}\mathsf{R} Y)(z) = \sum_\alpha \pm (\mathsf{R} \partial_\alpha Y)_\alpha(z)$. By Step (i) of the proof of Lem. 2.54, for $\alpha \ne \beta$ we have $|(\partial_\alpha Y)_\beta(z')| \le 6L^{-2} \max |Z_{P'}|$. For $\alpha = \beta$ the value $(\partial_\beta Y)_\beta(z')$ is zero unless z', as a base site, is a seam cube of β, and there it is at most $2L^{-1} \max |Z_{P'}|$. In both bounds the maximum is taken over the P' with $\mathsf{d}(P', z') \le \frac{5}{2}$, and by (e1) it is therefore at most $\mathsf{z} e^{-\lambda_B \mathsf{d}(z', P)}$. Using (a) and $e^{\lambda_B \mathsf{d}(z,P)} \le e^{\lambda_B \mathsf{d}(z', P)} e^{\frac{1}{2}\theta_R |z - z'|}$, we arrive at

$$e^{\lambda_B \mathsf{d}(z,P)} |s_1(z)| \le \mathsf{z}\Big[72\, \mathsf{g}(\tfrac{\theta_R}{2}) L^{-2} + 4L^{-1} \mathcal{S}(z)\Big], \qquad \mathcal{S}(z) := \sum_\beta \sum_{z' \text{ seam of } \beta} e^{-\frac{1}{2}\theta_R |z - z'|}$$

in which $72 = 6 \cdot 2 \cdot 6$ comes from the six pairs $\alpha \ne \beta$ and $4 = 2 \cdot 2$ from the three pairs $\alpha = \beta$.

Step (b): the source s_2. By the Cauchy–Schwarz inequality and (e2),

$$|(Q_3 \Xi)_{q'}| \le (\sum \psi_{q'}^2)^{1/2} (\sum_{\psi_{q'} \ne 0} \Xi^2)^{1/2} \le L^{3/2} \frac{2}{\gamma_{\mathrm{e}}} L^{1/2} e^{-\lambda_B (\mathsf{d}(q', P) - \frac{3}{2})}.$$

Since $\sum_{q'} \psi_{q'} = 1$, it follows that $e^{\lambda_B \mathsf{d}(q,P)} |s_2(q)| \le \mathsf{z} L^{-2}$.

Step (b): the bound on Ξ. We have $e^{\lambda_B \mathsf{d}(x,P)} \le e^{\lambda_B \mathsf{d}(y,P)} e^{\lambda_B r/L}$, with $r = |x-y|_\infty$, and $\omega_{\mathcal{G}} - \lambda_B \ge \varpi$; so the kernels that take the weighted sources to the weighted field are $c_{\mathcal{G}} K_i$, with the functions K_i of Lem. 2.59 at the present ϖ. For the seam term of s_1 we use a smearing, which compares the kernel K_i at the cube z with K_i at the seam cube z': $K_i(|x-z|) \le K_i(|x-z'|)(1+|z-z'|)^i e^{\varpi|z-z'|/L}$, and therefore $\sum_z K_i(|x-z|) e^{-\frac12\theta_R|z-z'|} \le \mathsf{g}_2(\frac{\theta_R}{4}) K_i(|x-z'|)$. For s_3 we apply the Cauchy–Schwarz inequality together with (e2), and $\sum_y K_1(r)^2 \le \sum_y K_2 \le 25L/\varpi$. With Lem. 2.59 this gives

$$e^{\lambda_B \mathsf{d}(x,P)} |\Xi_x| \le c_{\mathcal{G}} \mathsf{z} [72\mathsf{g} \cdot \frac{25}{\varpi^2} + 4\mathsf{g}_2 \cdot \frac{216}{\varpi^2} + \frac{25}{\varpi^2}] + L^{-2} c_{\mathcal{G}} (\frac{25L}{\varpi})^{1/2} \frac{2}{\gamma_{\mathrm{e}}} L^{1/2}$$

$$\le \frac{c_{\mathcal{G}}}{\gamma_{\mathrm{e}} \varpi^2} L^{-1} [2.1(1800 + 864 + 25) + 10] \mathsf{g}_2(\tfrac{\theta_R}{4}) \le c_\Xi^\natural L^{-1},$$

where g and g_2 stand for $\mathsf{g}(\frac{\theta_R}{2})$ and $\mathsf{g}_2(\frac{\theta_R}{4})$; in the second step we apply $\mathsf{g}(\frac{\theta_R}{2}) \le \mathsf{g}_2(\frac{\theta_R}{4})$, $1 \le \mathsf{g}_2$ and $\varpi^{-1/2} \le \varpi^{-2}$. Since $\Xi = \mathrm{pr}_\Xi \mathfrak{R}^{-1} \iota \delta_P$, this is the first bound in (b).

Step (b): the bounds on $\delta\Xi$ *and on* $H^\sharp$. Let the plaquette p lie between the cubes x and x^+. We bound $(\delta\Xi)_p$ through the same three sources, now with the differences of $\mathcal{G}$: that is, with $c_{\mathcal{G}} K_2$ in place of $c_{\mathcal{G}} K_1$ and with the second and fourth sums of Lem. 2.59 in place of the first and third; and s_3 is now bounded by the bound on Ξ just proved, $|s_3| \le L^{-3} c_\Xi^\natural e^{-\lambda_B \mathsf{d}}$. We have $(\delta\Xi)_p = \pm(\Xi_{x^+} - \Xi_x) = \pm \sum_y [\mathcal{G}(x^+, y) - \mathcal{G}(x,y)](s_1 + s_2 + s_3)(y)$. The base sites of p and of x are at most one fine step apart, so that $\mathsf{d}(p,P) \le \mathsf{d}(x,P) + \frac{1}{L}$; as $\lambda_B \le \frac{1}{128}$ and $L \ge 3$, we have $e^{\lambda_B/L} \le e^{1/384} \le 1.1$, which is the factor 1.1 of the following bound. With g and g_2 abbreviated as in the bound on Ξ, we obtain

$$e^{\lambda_B \mathsf{d}(p,P)} |(\delta\Xi)_p| \le 1.1 c_{\mathcal{G}} \{\mathsf{z}[72\mathsf{g} L^{-2} \frac{25L}{\varpi} + 4L^{-1} \mathsf{g}_2 \frac{144}{\varpi}(1 + \log\frac{3}{\varpi}) \mathsf{l}_L + L^{-2} \frac{25L}{\varpi}] + L^{-3} c_\Xi^\natural \frac{25L}{\varpi}\}.$$

By $\log\frac{3}{\varpi} \le \frac{3}{\varpi}$ and $\varpi \le \frac12$,

$$2.1[1800 + 576(1 + \frac{3}{\varpi}) + 25] \le \frac{6150}{\varpi} \le \frac{6000}{\varpi^2}.$$

We use $c_{\mathcal{G}} \ge 1$, which follows from $c_{\mathcal{G}} \ge c^{\mathbb{Z}} \ge 4(48 + 900)$, as well as $\mathsf{l}_L \ge 1$ and $\mathsf{g}(\frac{\theta_R}{2}) \le \mathsf{g}_2(\frac{\theta_R}{4})$. So the expression in braces in the bound on $e^{\lambda_B \mathsf{d}(p,P)} |(\delta\Xi)_p|$ is at most $L^{-2} \mathsf{l}_L \frac{26}{\varpi} c_\Xi^\natural$, and

$$|(\delta\Xi)_p| \le \frac{29}{\varpi} c_{\mathcal{G}} c_\Xi^\natural \mathsf{l}_L L^{-2} e^{-\lambda_B \mathsf{d}(p,P)}.$$

Moreover $|(W^* Z)_p| \le L^{-1} \max|Z| \le \mathsf{z} L^{-1} e^{-\lambda_B \mathsf{d}(p,P)}$ and $\frac{2.1}{\gamma_{\mathrm{e}}} \le c_\Xi^\natural$, so that $|(W^* Z + \delta\Xi)_p| \le \frac{30}{\varpi} c_{\mathcal{G}} c_\Xi^\natural \mathsf{l}_L L^{-2} e^{-\lambda_B \mathsf{d}(p,P)}$. By Lem. 2.31, $H^\sharp \delta_P = \mathsf{R}(W^* Z + \delta\Xi)$. We apply (a) with $e^{\lambda_B \mathsf{d}(p,P)} \le e^{\lambda_B \mathsf{d}(p',P)} e^{\frac12 \theta_R |p-p'|}$; as $\sum_{p'} 2e^{-\theta_R|p-p'|} e^{\frac12\theta_R|p-p'|} \le 6\mathsf{g}(\frac{\theta_R}{2})$ and $6 \cdot 30 = 180$, we obtain the bound on $H^\sharp$ in (b).

Step (c): the kernels B' *and* $\hat{R}$. We have $\mathsf{B}' - 1 = \ell'^{\mathsf{T}} \ell'$. Prop. 2.55 (a), whose bound holds with the rate λ° in place of λ' because $\lambda^\circ \le \lambda'$, bounds the rows ℓ' on $\mathbb{T}'$ with the constant $K_\natural$ at the rate λ°; the third bound in the second paragraph of Def. 2.48, applied to the rows ℓ' on $\mathbb{T}'$ with $K_\natural$, λ° and $\mathrm{dist}(P,P')$ in place of K, λ and $|p-p'|$, then gives the first bound in (c). For the second bound we carry out Step (a) on $\mathbb{T}'$, with B', $K_\natural$, λ°, $\lambda_2^\natural$, $\mathrm{dist}(P,P')$ and $\hat{R}$ in place of B, K, λ, θ_R, $|p-p'|$ and R: we have $\mathsf{B}' \ge 1$, by the identity just stated; for $\theta \le \frac{\lambda^\circ}{4}$, the same bound of Def. 2.48 for the rows ℓ' and $e^t - 1 \le te^t$ give $\varpi_{\mathsf{B}'}(\theta) \le 3K_\natural^2 \mathsf{g}(\frac{\lambda^\circ}{2}) \theta \mathsf{g}_1(\frac{\lambda^\circ}{4})$, which is at most $\frac12$ at $\theta = \lambda_2^\natural$; and Lem. 2.25 (a), with the distance $\mathrm{dist}(P,P')$, gives the bound on $\hat{R}$.

Step (d): the norms of T, X *and* $\ell\mathsf{T}$. We have $\sum_P e^{-\lambda_B \mathsf{d}(p,P)} \le 3e^{\lambda_B} \mathsf{g}(\lambda_B) \le 3.1\mathsf{g}(\lambda_B)$, and by (c), $\|\hat{R}\|_\infty \le 6\mathsf{g}(\lambda_2^\natural)$. Since $\mathsf{T} = H^\sharp \hat{R}$, the bound on $H^\sharp$ in (b), together with $3.1 \cdot 6 \le 19$, gives the

first bound in (d). By Lem. 2.31, applied with $X' = \hat{R}v$, where $\bar{Z} = L^{-1}v$,

$$\mathsf{X} = \mathrm{pr}_\Xi \mathfrak{R}^{-1} \iota \hat{R} + \mathsf{S}_0^{-1} Q_3^* (Q_3 Q_3^*)^{-2} \mathrm{d}' \hat{R} = \mathrm{pr}_\Xi \mathfrak{R}^{-1} \iota \hat{R} + L^{-1} Q_3^* \hat{\mathsf{J}}^{-2} \mathrm{d}' \hat{R}.$$

Here $\|Q_3^*\|_\infty = 1$, $\|\hat{\mathsf{J}}^{-2}\|_\infty \le \frac{2}{w_*}\mathsf{g}(\theta_J)$ and $\|\mathrm{d}'\|_\infty = 6$; with the first bound in (b) for the first term, this gives the second bound in (d). Finally $|\ell_y \mathsf{T} v| \le \|\ell\|_{\mathrm{r}} |\mathsf{T} v|_\infty$, and $\|\ell\|_{\mathrm{r}} \le 3K\mathsf{g}(\lambda)$ by the first bound in the second paragraph of Def. 2.48; with the first bound in (d) this gives the third. □

Definition 2.61 (Constants at the pure numbers)**.** We denote by $\theta_R^*, c_{\mathcal{G}}^*, \varpi^*, \lambda_B^*, c_\Xi^*, c_H^*, \lambda_2^*, c_1^*, c_2^*$ the constants of Thm. 2.60 at $(b, K, \lambda, \lambda^\circ) = (b_*, K_*, \lambda_*, \lambda_*)$. They are pure numbers. ■

At these values $K_\natural = K_*$, since then $\gamma_{\mathrm{e}} = \gamma_*$; hence the formula for θ_R and the formula for $\lambda_2^\natural$ in Thm. 2.60 coincide, and $\lambda_2^* = \theta_R^*$.

2.2.10. *(Dec) on the absorbing set.*

Proposition 2.62 (The condition (Dec) on the absorbing set)**.** Let $L \ge L_A$ be odd, $aL \ge 3/c_L$, $N \ge 2$ even, and $\ell \in \mathcal{R}(b_*, K_*, \lambda_*, \frac{1}{9})$. With $\gamma_q = (73L^4)^{-1}$, let

$$A_q := 2e^2(16+a) + 48e^2 K_*^2 \mathsf{g}(\tfrac{\lambda_*}{2})\mathsf{g}_2(\tfrac{\lambda_*}{4}), \quad \theta_q := \min\Big\{\frac{\lambda_*}{4}, \frac{\gamma_q}{2A_q}\Big\}, \quad \theta_D := \tfrac{1}{2}\min\{\lambda_2^*, \theta_q, \theta_J\},$$

$$\mathsf{C}_0 := 4L^3 \mathsf{g}(\theta_D)\frac{2.1L^2}{\gamma_*}, \quad \mathsf{C}_R := 6e\,\mathsf{g}(\tfrac{1}{2}\theta_R^*), \quad \mathsf{C}_\ell := 3eK_*\mathsf{g}(\tfrac{\lambda_*}{2}), \quad C_D := \max\{(4L^3\mathsf{C}_0)^4 \mathsf{C}_R \mathsf{C}_\ell,\ (4L^3)^2(1+\mathsf{C}_\ell^2),\ A_q\}.$$

Then (Dec)(C_D, θ_D) holds for the pair (ℓ, ℓ'). The numbers C_D and θ_D depend on L, a and pure numbers alone, and on none of ℓ, N, the cuts, x and k. Here θ_D is defined with θ_q; in the number θ_D of Thm. 2.65 the number $\theta_{\mathbb{C}}$ takes the place of θ_q, as in Steps H13 and H16 (ii) of its proof. ■

Proof. The form q. For cells of the fine level, dist $< |\cdot|/L + 1$. We have $q = \Pi_{\mathfrak{L}}(\mathrm{d}^*\mathrm{d} + aQ_1^*Q_1 + \mathrm{d}^*(\mathsf{B}-1)\mathrm{d})\Pi_{\mathfrak{L}}$, where, throughout this section, $\Pi_{\mathfrak{L}}$ denotes the orthogonal projection of Ω^1 onto $\mathfrak{L}$, with $\mathfrak{L}$ and $\mathcal{T}$ as in Def. 2.21: for $\varphi \in \Omega^1$, $(\Pi_{\mathfrak{L}}\varphi)_b = \varphi_b$ if $b \notin \mathcal{T}$ and $(\Pi_{\mathfrak{L}}\varphi)_b = 0$ if $b \in \mathcal{T}$. The first two terms of q have coarse range at most 2 and absolute row sums at most $16 + a$, for the following reasons. First, $(\mathrm{d}^*\mathrm{d})(b, b') = \sum_p \mathrm{d}(p, b)\mathrm{d}(p, b')$, where a bond lies in four plaquettes p and a plaquette has four bonds b', which gives 16; and any two bonds of one plaquette are based within one fine step, at coarse distance at most 1. Second, $(Q_1^*Q_1)(b, b') = \sum_c Q_1(c, b)Q_1(c, b')$ with $Q_1(c, b) = L^{-3}\#\{x \in B(c_-) : b \in [x, x']\} \ge 0$; here $\sum_c Q_1(c, b) = L^{-2}$, since there are L points x, each in one block, and $\sum_{b'} Q_1(c, b') = L$, so that the row sums are $a/L \le a$; and b, b' are based in W_c, at coarse distance at most 1. Third, the factors $\Pi_{\mathfrak{L}}$ on the two sides set every row and every column with an index $b \in \mathcal{T}$ to zero and leave the other entries unchanged. These facts show that the contribution of the first two terms of q to $\varpi_q(\theta)$ is at most $(16+a)(e^{2\theta} - 1) \le 2e^2(16+a)\theta$ when $\theta \le 1$.

For the third term, $|(\cdot)(b, b')| \le 16K_*^2 \mathsf{g}(\frac{\lambda_*}{2}) e^{\lambda_*} e^{-\frac{1}{2}\lambda_* n}$, where n is the distance of the base sites of b and b': each bond lies in four plaquettes, and the plaquettes at b lie at distance $\ge n - 2$ from those at b'. Since $e^{\theta \mathrm{dist}} - 1 \le \theta(1+n)e^{\theta(1+n)}$ and three bonds are based at each site, the third term contributes, for $\theta \le \frac{\lambda_*}{4}$, at most $48e^{\lambda_* + \theta} K_*^2 \mathsf{g}(\frac{\lambda_*}{2})\mathsf{g}_2(\frac{\lambda_*}{4})\theta$. So $\varpi_q(\theta) \le A_q\theta$, and $\varpi_q(\theta_D) \le \frac{1}{2}\gamma_q$.

Further, $\|q\|_{\theta_D} \le \|q\|_0 + \varpi_q(\theta_D) \le \frac{1}{2}A_q + 1 \le A_q$. The first inequality rests on $e^{\theta \mathrm{dist}} = 1 + (e^{\theta \mathrm{dist}} - 1)$. For $\|q\|_0 \le \frac{1}{2}A_q$, note that the first two terms of q have absolute row sums $\le 16 + a \le e^2(16+a)$, while the third term, by the bound on its entries and because three bonds are based at each site, has absolute row sums

$$\le 48e^{\lambda_*} K_*^2 \mathsf{g}(\frac{\lambda_*}{2}) \sum_{u \in \mathbb{Z}^3} e^{-\frac{1}{2}\lambda_* |u|_\infty} = 48e^{\lambda_*} K_*^2 \mathsf{g}(\frac{\lambda_*}{2})^2 \le 24e^2 K_*^2 \mathsf{g}(\frac{\lambda_*}{2})\mathsf{g}_2(\frac{\lambda_*}{4}),$$

because $2e^{\lambda_*} \le e^2$ and $\mathsf{g}(\frac{\lambda_*}{2}) \le \mathsf{g}(\frac{\lambda_*}{4}) \le \mathsf{g}_2(\frac{\lambda_*}{4})$ by Def. 2.44, whose bound $\mathsf{g}(\lambda) \le \mathsf{g}_2(\lambda)$ we apply at $\lambda = \frac{\lambda_*}{4}$; and q is symmetric, so its column sums equal its row sums. Finally $\varpi_q(\theta_D) \le \frac{1}{2}\gamma_q \le 1 \le \frac{1}{2}A_q$, because $A_q \ge 32e^2$.

Basic kernels; the kernel R*, the rows* ℓ*, the local maps.* The basic kernels $\mathfrak{R}^{-1}, \mathsf{B}', \hat{R}, \hat{\mathsf{J}}^{-2}$ are bounded entrywise by $\frac{2.1L^2}{\gamma_*} e^{-2\theta_D \mathrm{dist}}$. Since $\ell \in \mathcal{R}(b_*, K_*, \lambda_*, \frac{1}{9})$ and $L \ge L_A$, we have $\lambda' = \lambda_*$ by part (a) of the proof of Thm. 2.57. Hence the bound of Prop. 2.55 (a) holds with $\lambda'' = \lambda_*$, and Thm. 2.60 applies with $\lambda^\circ = \lambda_*$. For $\mathfrak{R}^{-1}$ the entrywise bound follows from Prop. 2.55 (a) with $\lambda'' = \lambda_*$, together with $\mathsf{D}_i \ge L^{-2}$, $\mathsf{d} > \mathrm{dist} - 1$ and $2e^{\lambda_*} \le 2.1$. For B' and $\hat{R}$ it follows from Thm. 2.60 (c), with $1 + K_*^2 \mathsf{g}(\frac{\lambda_*}{2}) = 1 + 9\mathsf{g}(\frac{\lambda_*}{2})/\gamma_* \le L^2/\gamma_*$, by $L \ge L_A \ge 36\mathsf{g}(\frac{\lambda_*}{2})/\gamma_*$. For $\hat{\mathsf{J}}^{-2}$ it follows from the bound on $\hat{\mathsf{J}}^{-2}$ of Rem. 2.36. In each of the four bounds just cited the rate is at least $2\theta_D$, since $2\theta_D \le \lambda_2^* \le \frac{1}{4}\lambda_*$ and $2\theta_D \le \theta_J$. A block holds at most $4L^3$ cells of one kind, so that $\|\cdot\|_{\theta_D} \le \mathsf{C}_0$ for each basic kernel.

For R let $n = |p - p'|$, so that $\mathrm{dist} < n/L + 1$. Since three plaquettes are based at each site, we have $\sum_{p'} 2e^{-\theta_R^* n} e^{\theta_D(n/L+1)} \le 2 \cdot 3 \cdot e\, \mathsf{g}(\theta_R^* - \theta_D) \le 6e\, \mathsf{g}(\frac{1}{2}\theta_R^*) = \mathsf{C}_R$. In the first inequality we also use that $\theta_D \le 1$ and $L \ge 1$ give $e^{\theta_D(n/L+1)} \le e \cdot e^{\theta_D n}$, and that the sum over the torus is at most the sum over $\mathbb{Z}^3$; the second inequality follows from $\theta_D \le \frac{1}{2}\lambda_2^* = \frac{1}{2}\theta_R^*$. For ℓ we have $\|\ell\|_{\theta_D} \le \mathsf{C}_\ell$, and $\|\ell K\|_\theta \le \|\ell\|_\theta \|K\|_\theta$ for every kernel K whose first index is a plaquette, because $|(\ell K)_y(j)| \le \sum_p |\ell_y(p)||K(p,j)|$. The local maps $W, W^*, \delta, \mathrm{d}, \mathrm{d}^*, \mathrm{d}', \delta', \mathrm{d}'^*, Q_3, Q_3^*, E, E_3, \iota$ have coarse range at most 1 and row and column sums at most L^3, so $\|\cdot\|_{\theta_D} \le 4L^3$ for each of them.

Composites. The composite kernels are $H^\sharp = \mathsf{R}\mathbb{Y}\mathfrak{R}^{-1}\iota$, $\mathsf{T} = H^\sharp \hat{R}$ and $\ell\mathsf{T}$; $\mathsf{X} = \mathrm{pr}_\Xi \mathfrak{R}^{-1}\iota\hat{R} + L^{-1}Q_3^*\hat{\mathsf{J}}^{-2}\mathrm{d}'\hat{R}$; $\mathsf{N}^\sharp = (aL)^{-1}\mathrm{d}'^*\mathsf{B}'$; $(\mathsf{G}_Z, \mathsf{G}_\Xi) = \mathfrak{R}^{-1}(0, E_3\cdot)$, $\mathsf{G} = \mathsf{R}\mathbb{Y}\mathfrak{R}^{-1}(0, E_3\cdot)$ and $\ell\mathsf{G}$; and $\mathsf{K}_1 = E_3^\mathsf{T}\mathsf{X}\mathsf{B}'$, $\mathsf{K}_2 = E_3^\mathsf{T}\mathsf{G}_\Xi - L^{-1}\hat{\mathsf{J}}^{-2}$. Each of them is a sum of at most two products, each of at most three basic kernels, four local maps, one R and one ℓ; hence each has norm $\|\cdot\|_{\theta_D} \le 2\mathsf{C}_0^3(4L^3)^4\mathsf{C}_R\mathsf{C}_\ell$. The kernel $\mathsf{g} = \Pi_\mathfrak{L}\mathrm{d}^*\mathsf{B}E$ has norm at most $(4L^3)^2(1 + \mathsf{C}_\ell^2)$. For the rows ℓ', $\|\ell'\|_{\theta_D} \le 3K_*\mathsf{g}(\frac{\lambda_*}{2}) \le \mathsf{C}_\ell$ by Thm. 2.57 (a). Since $\mathsf{C}_0 \ge 2$ and $\mathsf{C}_0, \mathsf{C}_R, \mathsf{C}_\ell, 4L^3 \ge 1$, we have $2\mathsf{C}_0^3(4L^3)^4\mathsf{C}_R\mathsf{C}_\ell \le (4L^3\mathsf{C}_0)^4\mathsf{C}_R\mathsf{C}_\ell \le C_D$, and the same holds for products with fewer factors. Moreover $\|\mathsf{B}'\|_{\theta_D} \le \mathsf{C}_0 \le C_D$, $\|\ell\|_{\theta_D}, \|\ell'\|_{\theta_D} \le \mathsf{C}_\ell \le C_D$, $\|q\|_{\theta_D} \le A_q \le C_D$ and $\|\mathsf{g}\|_{\theta_D} \le (4L^3)^2(1+\mathsf{C}_\ell^2) \le C_D$. The composite kernels, B', ℓ, ℓ', q and g are the sixteen kernels of Def. 2.41, so each of them has norm $\|\cdot\|_{\theta_D}$ at most C_D. The bound $\varpi_q(\theta_D) \le \frac{1}{2}\gamma_q$ was shown above, and we also have $C_D \ge 4$ and $\theta_D \le 1$. □

2.2.11. *(S-chain); K steps.*

Definition 2.63 (The order of the letters; the choice of ρ_0, L_* and $x_\flat$)**.** The letters are chosen in this order: d_0; c_Ξ; ρ_0; L, a; κ; (C_D, θ_D), which fix $c_R = 64/\theta_D$; M; $\mathfrak{p}_0$, A_0; c_ξ; c_h; E_a; and, last of all, $x_\flat$.

Given $0 < d_0 < \frac{1}{2}$ and $0 < c_\Xi \le \frac{1}{2}(\frac{1}{2} - d_0)$, we define

$$\rho_0 := \text{the least integer} \ge 0 \text{ with } \quad 3K_*^2\, \mathsf{g}(\lambda_*)\, \mathsf{g}(\tfrac{\lambda_*}{2})\, e^{-\lambda_*\rho_0/2} \le \tfrac{1}{8}(\tfrac{1}{2} - d_0 - c_\Xi); \tag{2.vi}$$

and, with this ρ_0, the four numbers

$$\mathsf{N}_Z := 1 + 3K_*^2\mathsf{g}(\lambda_*)^2, \quad \mathsf{P}_\rho := 1 + 3(2 + 3(2\rho_0+1)^3)K_*^2\mathsf{g}(\lambda_*)^2, \quad \mathsf{N}_\sharp := c_1^*\mathsf{I}_L\mathsf{N}_Z\mathsf{P}_\rho, \quad \mathsf{N}_\Xi := c_2^*\mathsf{N}_Z. \tag{2.vii}$$

The number $L_*(d_0, c_\Xi)$, which is defined next, and the conditions (L1), (L2) are those of this section. In item 3 of §2.1, their place is taken by L_*^{su}, (L1^{su}) and (L2^{su}) of (O4). As to the size of L, Thm. 2.65 requires $L \ge L_A$ alone.

We let $L_*(d_0, c_\Xi)$ be the least integer $\ge L_A$ for which the two inequalities

$$(c_1^*\mathsf{I}_L\mathsf{P}_\rho L^{-1} + 60 + 4c_2^*)\mathsf{N}_Z \le \tfrac{1}{4}d_0 L^{1/2}, \qquad 115\,(c_1^*\mathsf{I}_L\mathsf{N}_Z\mathsf{P}_\rho)^2 \le L$$

hold for every real $L \geq L_*$; such an integer exists, because $\mathsf{l}_L^2/L \to 0$. The floor condition on $x_\flat$ is (X12): $3K_*\mathsf{g}(\frac{\lambda_*}{2})\, x_\flat^{-e_R}/(1-L^{-e_R}) \leq \frac{1}{9}$, where $e_R := \frac{1}{2}\lambda_* c_R$. ■

Theorem 2.64 (Decay and bounds along the cut chain)**.** Let $0 < d_0 < \frac{1}{2}$ and $0 < c_\Xi \leq \frac{1}{2}(\frac{1}{2} - d_0)$, let ρ_0 be given by eq. 2.vi, let $L \geq L_*(d_0, c_\Xi)$ be odd and $a \geq 3/(c_L L)$, let (C_D, θ_D) be the constants of Prop. 2.62, $c_R = 64/\theta_D$ and $\varrho(x) = \lceil c_R \log x \rceil$, and let $x_\flat > 1$ satisfy (X12). For every x_0, consider the chain $x_k = L^{-k}x_0$, $\ell_0 = \emptyset$, $\ell_{k+1} = ((\ell_k)')^{\mathrm{c}}$, in which the cut is taken at $\varrho(x_{k+1})$. Then, for every k with $x_{k+1} \geq x_\flat$ and every tower of $K \geq k+1$ steps with $N \geq 2$ even, the pair $(\ell_k, (\ell_k)')$ satisfies (Dec)(C_D, θ_D), and it satisfies (N′) with the numbers eq. 2.vii and with this ρ_0, whatever the radius at which $(\ell_k)'$ is cut; moreover $\mathsf{t}_\rho \leq \frac{1}{8}(\frac{1}{2} - d_0 - c_\Xi)$, and (L1), (L2) hold; we call the assertion of this theorem, as a whole, (S-chain). ■□

Proof. Step (Cut): the condition (Cut) *along the chain.* Since $\varrho(x) \geq c_R \log x$ and $e_R = \frac{1}{2}\lambda_* c_R$, we have $\mathsf{t}_*(\varrho(x)) \leq 3K_*\mathsf{g}(\frac{\lambda_*}{2})x^{-e_R}$. The inequality $x_j \geq L^{k+1-j}x_\flat$ holds for every $j \leq k+1$, as $x_{k+1} \geq x_\flat$; summing the resulting geometric series in j, we find that $\sum_{j\leq k+1} \mathsf{t}_*(\varrho(x_j))$ is at most the left side of (X12), hence at most $\frac{1}{9}$. This choice of the radii is not circular: the number L_A, Thm. 2.57 and Prop. 2.62 depend on no cut radius, the number c_R is defined after them, and the floor (X12) is chosen last. We may therefore apply Thm. 2.57 (b) to the tower of $K' = k+1$ steps and side $N' = NL^{K-k-1}$, an even number. That tower has the same tori $\mathbb{T}_0$ to $\mathbb{T}_{k+1}$ and the same rows ℓ_0 to ℓ_{k+1}, because these rows are determined by the radii $\varrho(x_1)$ to $\varrho(x_{k+1})$ alone; and its condition (Cut), which reads $\sum_{j\leq k+1} \mathsf{t}_*(\varrho(x_j)) \leq \frac{1}{9}$, is the bound just proved. The argument is independent of the radii $\varrho(x_j)$, $j > k+1$, of a tower with $K > k+1$; for such j, x_j can be smaller than $x_\flat$. Thm. 2.57 (b) thus yields $\ell_k \in \mathcal{R}(b_*, K_*, \lambda_*, \frac{1}{9})$ and $|(\ell_k)'_{y'}(P)| \leq K_* e^{-\lambda_* \mathrm{dist}(y', P)}$.

The condition (Dec) *and the bound on* t_ρ. Prop. 2.62 gives (Dec)(C_D, θ_D). Next, the chain of inequalities

$$\mathsf{t}_\rho \leq 3K_*\mathsf{g}(\lambda_*) \cdot K_* \sum_{|u|_\infty > \rho_0} e^{-\lambda_*|u|_\infty} \leq 3K_*^2 \mathsf{g}(\lambda_*)\mathsf{g}(\frac{\lambda_*}{2}) e^{-\lambda_*\rho_0/2}$$

holds, the second inequality because $e^{-\lambda_*|u|_\infty} \leq e^{-\lambda_*\rho_0/2} e^{-\lambda_*|u|_\infty/2}$ for $|u|_\infty > \rho_0$; by eq. 2.vi, its right side is at most $\frac{1}{8}(\frac{1}{2} - d_0 - c_\Xi)$.

The conditions (N′)*,* (L1) *and* (L2). For a mask m and $|\zeta'|_\infty \leq 1$ we have $|LZ|_\infty = |\mathsf{B}'^{\mathrm{c}}_m \zeta'|_\infty \leq 1 + \|\ell'^{\mathrm{c}}\|_{\mathrm{c}} \|\ell'^{\mathrm{c}}\|_{\mathrm{r}} \leq \mathsf{N}_Z$. The two inequalities hold because the bound $|(\ell_k)'_{y'}(P)| \leq K_* e^{-\lambda_* \mathrm{dist}(y', P)}$ at the end of Step (Cut) gives $\|\ell'^{\mathrm{c}}\|_{\mathrm{r}} \leq 3K_*\mathsf{g}(\lambda_*)$ and $\|\ell'^{\mathrm{c}}\|_{\mathrm{c}} \leq K_*\mathsf{g}(\lambda_*)$ for every radius of the cut, and because a mask raises neither norm. Here $Z = L^{-1}\mathsf{B}'^{\mathrm{c}}_m \zeta'$, Ξ, $(\nu, \omega) = \mathrm{pair}(Z, \Xi)$ and σ are the objects of (N′) in Def. 2.42, with $\ell = \ell_k$; by that definition and by Def. 2.35, taken at $v = LZ$, they satisfy the identities $\nu = \mathsf{RT}_\sharp Z = \mathsf{T}(LZ)$, $\Xi = \Xi_Z = \mathsf{X}(LZ)$ and $\sigma = \ell_k \nu$. The first two bounds in the second paragraph of Def. 2.48 are $\|\ell\|_{\mathrm{r}} \leq 3K\mathsf{g}(\lambda)$ and $\|\ell\|_{\mathrm{c}} \leq K\mathsf{g}(\lambda)$. They hold for the rows ℓ_k of $\mathcal{R}(b_*, K_*, \lambda_*, \frac{1}{9})$ and give $|\sigma|_{2,\infty} \leq 3K_*\mathsf{g}(\lambda_*)|\nu|_\infty$ and $2\|\ell_k\|_{\mathrm{c}} + (2\rho_0+1)^3\|\ell_k\|_{\mathrm{r}} \leq (2+3(2\rho_0+1)^3)K_*\mathsf{g}(\lambda_*)$. Thm. 2.60 (d), applied, as in Def. 2.61, with $(b, K, \lambda, \lambda^\circ) = (b_*, K_*, \lambda_*, \lambda_*)$, where $\lambda' = \lambda_*$ because $L \geq L_A$, now gives (N′) with the numbers eq. 2.vii. Finally, (L1) and (L2) follow from the definition of L_* in Def. 2.63. □

Theorem 2.65 (Uniform bounds in the cut tower)**.** Use the one-step geometry of Def. 2.17 on the tower of Def. 2.49. In that geometry the coarse side at step k is $N' = NL^{K-k-1}$. The kernels are the scalar kernels defined above, and $\mathsf{l}_L = 1 + \log L$.

The chain. We put $x_j = L^{-j}x_0$; ℓ_0 is empty, and $\ell_{j+1} = ((\ell_j)')^{\mathrm{c}}$, with the cut taken at $\varrho(x_{j+1})$, where $\varrho(x) = \lceil c_R \log x \rceil$.

The order of the constants. The constants are fixed in the order (0) to (8).

(0) The pure numbers are defined exactly by

$$w_* = 3^{-6}, \quad \mathsf{r}_\sharp = \frac{w_*}{1.3\cdot 10^7}, \quad \gamma_1 = \frac{\mathsf{r}_\sharp}{3074}, \quad b_* = \frac{2}{\mathsf{r}_\sharp} = 1.8954\cdot 10^{10}, \quad \gamma_* = \frac{\gamma_1}{2b_*} = \frac{\mathsf{r}_\sharp^2}{12296},$$

$$\lambda_* = \frac{\gamma_*^{1/2}}{44}, \qquad K_* = 3\gamma_*^{-1/2}, \qquad \theta_J = \tfrac{1}{2}\log(1+w_*/50).$$

The following outward intervals, rather than rounded equalities, may be used in numerical bounds:

$$\mathsf{r}_\sharp \in [1.05518, 1.05519]\cdot 10^{-10}, \quad \gamma_1 \in [3.43261, 3.43262]\cdot 10^{-14},$$
$$\gamma_* \in [9.05512, 9.05513]\cdot 10^{-25}, \quad \lambda_* \in [2.16269, 2.16270]\cdot 10^{-14},$$
$$K_* \in [3.15263, 3.15264]\cdot 10^{12}, \quad \theta_J \in [1.37172, 1.37173]\cdot 10^{-5}.$$

The number $L_A = \max\{12K_*\mathsf{g}(\lambda_*/2)\gamma_*^{-1/2},\ 38088K_*^2\mathsf{g}_1(\lambda_*)^2/\mathsf{r}_\sharp\}$ is at most $2.5\cdot 10^{153}$, and $\lambda_2^* = \theta_R^* = \min\{\lambda_*/4,\ [6K_*^2\mathsf{g}(\lambda_*/2)\mathsf{g}_1(\lambda_*/4)]^{-1}\}$ is at least $1.9\cdot 10^{-129}$. The pair $(c_{\mathcal{G}}^*, \omega_{\mathcal{G}}^*)$ is the one given by Lem. 2.58 at $(b, C, \theta) = (b_*, 2, \theta_R^*)$; we put $\varpi^* = \omega_{\mathcal{G}}^*/2$ and $\lambda_B^* = \min\{\lambda_*, \varpi^*\}$. Then $c_\Xi^* = 6000c_{\mathcal{G}}^*\mathsf{g}_2(\theta_R^*/4)/(\gamma_*\varpi^{*2})$, $c_H^* = 180\mathsf{g}(\theta_R^*/2)c_{\mathcal{G}}^*c_\Xi^*/\varpi^*$ and $c_1^* = 19c_H^*\mathsf{g}(\lambda_B^*)\mathsf{g}(\lambda_2^*)$; moreover $c_2^* = 19c_\Xi^*\mathsf{g}(\lambda_B^*)\mathsf{g}(\lambda_2^*) + 72w_*^{-1}\mathsf{g}(\theta_J)\mathsf{g}(\lambda_2^*)$, which is the $c_2^\natural$ of Thm. 2.60, taken at the starred values, and enters (Fr3′). Further, $\mathsf{N}_Z = 1 + 3K_*^2\mathsf{g}(\lambda_*)^2$.

(1) $d_0 \in (0, \frac{1}{2})$.

(2) $0 < c_\Xi \le \frac{1}{2}(\frac{1}{2} - d_0)$.

(3) The integer ρ_0 is the least integer ≥ 0 with $3K_*^2\mathsf{g}(\lambda_*)\mathsf{g}(\lambda_*/2)e^{-\lambda_*\rho_0/2} \le \frac{1}{8}(\frac{1}{2} - d_0 - c_\Xi)$, and $\mathsf{P}_\rho = 1 + 3(2 + 3(2\rho_0+1)^3)K_*^2\mathsf{g}(\lambda_*)^2$.

(4) L is odd, $L \ge L_A$, and $a \ge 3/(c_L L)$; we put $\mathsf{N}_\sharp = c_1^*\mathsf{I}_L\mathsf{N}_Z\mathsf{P}_\rho$. Where the later sections apply the theorem, at the values of item 3 of §2.1, (O4) of that item gives $L \ge L_*^{\mathrm{su}}(d_0, c_\Xi) \ge L_A$; here L_*^{su} is the number of (O4), built from the N_ν of (O3) in that item. This bound is not among the hypotheses of the theorem: the lower bound on L that this theorem uses is $L \ge L_A$.

(5) The letter κ.

(6) The rate and the bound, defined in the following order: $\gamma = (73L^4)^{-1}$; $A^\flat = 2e^2(16+a) + 384e^8K_*^2\mathsf{g}_2(\lambda_*/2)^2$; $\theta_1 = \min\{1, \lambda_*L/12\}$; $\theta_{\mathbb{C}} = \frac{1}{6}\min\{\theta_1, \gamma/2A^\flat\}$; $\theta_D = \frac{1}{2}\min\{\lambda_2^*, \theta_{\mathbb{C}}, \theta_J\}$. After θ_D come $\mathsf{C}_0 = 4L^3\mathsf{g}(\theta_D)\,2.1L^2/\gamma_*$, $\mathsf{C}_R = 6e\,\mathsf{g}(\theta_R^*/2)$, $\mathsf{C}_\ell = 3eK_*\mathsf{g}(\lambda_*/2)$ and $A_q = 2e^2(16+a) + 48e^2K_*^2\mathsf{g}(\lambda_*/2)\mathsf{g}_2(\lambda_*/4)$; then $C_D = \max\{(4L^3\mathsf{C}_0)^4\mathsf{C}_R\mathsf{C}_\ell,\ (4L^3)^2(1+\mathsf{C}_\ell^2),\ A_q\}$, and $c_R = 64/\theta_D$.

(7) The letters M; $\mathfrak{p}_0$, A_0; c_ξ; c_h; E_a.

(8) Last, a number $x_\flat > 1$ satisfying (X12), that is, $3K_*\mathsf{g}(\lambda_*/2)x_\flat^{-e_R}/(1 - L^{-e_R}) \le \frac{1}{9}$, where $e_R = \lambda_* c_R/2$.

This order is not circular: L_A, Thm. 2.57 and Prop. 2.62 are independent of the cut; c_R is defined after them; and (X12) is a lower bound on $x_\flat$.

Conclusion. Let x_0 be arbitrary, let $k \ge 0$ satisfy $x_{k+1} \ge x_\flat$, take any tower with $N \ge 2$ even, and let $(\ell, \ell') = (\ell_k, (\ell_k)')$. Then the following hold.

(Fr0) $\ell_k \in \mathcal{R}(b_*, K_*, \lambda_*, \frac{1}{9})$, of range $\le \varrho(x_k)$; here ℓ_0 is empty, and for $k \ge 1$ the range bound comes from the cut. Moreover $|(\ell_k)'_{y'}(P)| \le K_* e^{-\lambda_*\mathrm{dist}(y', P)}$ and $1 \le \mathsf{B}' \le \frac{3}{2}/\mathsf{r}_\sharp$.

(Fr2) $\|K\|_{\theta_D} \le C_D$ for $K \in \{\mathsf{G}, \mathsf{G}_Z, H^\sharp, \mathsf{N}^\sharp, \mathsf{T}, \ell'\}$. The bound $\|\cdot\|_{\theta_D} \le C_D$ also holds for each of the sixteen kernels of (Dec), and $\varpi_q(\theta_D) \le \gamma/24 \le \gamma_q/2$.

(Fr3) At every cut radius, $L^2\|H^\sharp - \mathsf{T}(\mathsf{B}' - \mathsf{B}'^{\mathrm{c}})\|_\infty = L^2\|\mathsf{T}\mathsf{B}'^{\mathrm{c}}\|_\infty \le c_1^*\mathsf{I}_L\mathsf{N}_Z \le \mathsf{N}_\sharp$, and $\|\mathsf{B}'^{\mathrm{c}}\|_\infty \le \mathsf{N}_Z$.

(Fr3′) Condition (N′) of Def. 2.42 holds with the numbers eq. 2.vii. Explicitly, let a cut radius, a mask m and ζ' with $|\zeta'|_\infty \le 1$ be arbitrary, and let $\mathsf{B}'^{\mathrm{c}}_m = 1 + \ell'^{\mathrm{c}\mathsf{T}} m\, \ell'^{\mathrm{c}}$, $Z = L^{-1}\mathsf{B}'^{\mathrm{c}}_m\zeta'$,

$\Xi = \Xi_Z$, $(\nu,\omega) = \mathrm{pair}(Z,\Xi)$ and $\sigma = \ell\nu$; then $L^2[|\nu|_\infty + (2\|\ell\|_{\mathrm{c}} + (2\rho_0+1)^3\|\ell\|_{\mathrm{r}})|\sigma|_{2,\infty}] \le \mathsf{N}_\sharp$, $L|Z|_\infty \le \mathsf{N}_Z$ and $L|\Xi|_\infty \le \mathsf{N}_\Xi := c_2^*\mathsf{N}_Z$; here N_Ξ is a pure number.

(Fr4) At every cut radius, $\mathsf{t}_\rho \le \frac{1}{8}(\frac{1}{2} - d_0 - c_\Xi)$, with t_ρ the number of Def. 2.39.

Concerning dependence: C_D and θ_D are determined by (L,a) together with pure numbers; N_Z is a pure number; and $\mathsf{N}_\sharp$ depends on L only by way of I_L, and on (d_0, c_Ξ) only by way of ρ_0. Each of these numbers is independent of k, N, K, x_0, of the cuts and of the rows. ■□

Remark 2.66 (The four hypotheses). The proof of Thm. 2.65 rests on four hypotheses: $L \ge L_A$ odd; $aL \ge 3/c_L$, which is stronger than $aL \ge 18$; N even; and $x_{k+1} \ge x_\flat$, with $x_\flat$ satisfying (X12). In §2.1, item 1 requires N to be even; (O4) of item 3 requires L to be odd, with $L \ge L_*^{\mathrm{su}} \ge L_A$, and $a \ge 3/(c_L L)$; and (O13) of item 3 requires (X12). By Step H16 (v) of the proof of Thm. 2.65, condition (X12) holds automatically for every $x_\flat \ge 2$. Part (Fr0) is proved in Step H9, from Thm. 2.57 (a) and (b); parts (Fr3′) and (Fr4) are proved in Step H14.

Proof of Thm. 2.65. We apply the one-step estimates to the cut tower and check the chosen rate for the Gaussian window. The estimates for a single pair of rows are proved in Props. 2.55, 2.62 and Thm. 2.60.

Step H9. The cut tower. For $j \le k+1$, we have $x_j \ge L^{k+1-j}x_\flat$. The definition of ϱ and (X12) give

$$\sum_{j=1}^{k+1} \mathsf{t}_*(\varrho(x_j)) \le \frac{3K_*\mathsf{g}(\lambda_*/2)x_\flat^{-e_R}}{1 - L^{-e_R}} \le \frac{1}{9}.$$

Thus Thm. 2.57(b) applies to the first $k+1$ steps, viewed as a tower with final side $N' = NL^{K-k-1}$. It gives (Fr0). Only radii through step $k+1$ enter this argument. The numerical bounds on L_A and λ_2^* in item (0) follow by inserting $\mathsf{g}(\lambda) \le 51\lambda^{-3}$ and $\mathsf{g}_1(\lambda) \le 182\lambda^{-4}$ in their defining formulas.

Step H13. The decay rate for the window. The proof of Prop. 2.62 applies at any rate θ_D with $\theta_D \le 1$, $2\theta_D \le \min\{\lambda_2^*, \theta_J\}$ and $\varpi_q(\theta_D) \le \gamma_q/2$. Its constants are evaluated at that rate. The choice in item (6) satisfies the first two conditions and gives $\theta_D \le \lambda_*/4$, since $\lambda_2^* \le \lambda_*/4$. The bound $\varpi_q(\theta) \le A_q\theta$ in that proof and $A_q \le A^\flat$ from Step H16(ii) give $\varpi_q(\theta_D) \le A_q\gamma/(24A^\flat) \le \gamma/24$. By Step H9 the rows are in the required class, so these estimates give (Fr2).

As to the group, tensoring a scalar kernel with $1_{\mathfrak{g}}$ preserves the norms in Def. 2.20, since the norm of each matrix entry becomes the operator norm of that scalar times the identity.

Step H14: Thm. 2.64; (Cut), (N′), and hence (Fr3). For every k with $x_{k+1} \ge x_\flat$, the chain cut at $\varrho(x) = \lceil c_R \log x \rceil$ satisfies: $\ell_k \in \mathcal{R}(b_*, K_*, \lambda_*, \frac{1}{9})$; (Dec) holds; (N′) holds with eq. 2.vii, at every cut; and $\mathsf{t}_\rho \le \frac{1}{8}(\frac{1}{2} - d_0 - c_\Xi)$.

Condition (Cut) holds by Step H9, and Thm. 2.57 (b) then applies; by Step H13, (Dec)(C_D, θ_D), with C_D and θ_D as in item (6) of the theorem, holds for $(\ell_k, (\ell_k)')$.

For (N′), we have $|\mathsf{B}_m'^{\mathrm{c}}\zeta'|_\infty \le 1 + \|\ell'^{\mathrm{c}}\|_{\mathrm{c}}\|\ell'^{\mathrm{c}}\|_{\mathrm{r}} \le 1 + K_*\mathsf{g}_* \cdot 3K_*\mathsf{g}_* = \mathsf{N}_Z$, where $\mathsf{g}_* = \mathsf{g}(\lambda_*)$; here cutting and masking lower both norms. As $\nu = \mathsf{T}(LZ)$, Thm. 2.60 (d) at the starred values of Def. 2.61 gives $L^2|\nu|_\infty \le c_1^*\mathsf{I}_L\mathsf{N}_Z$. Further, $|\sigma|_{2,\infty} \le 3K_*\mathsf{g}_*|\nu|_\infty$ and $2\|\ell\|_{\mathrm{c}} + (2\rho_0+1)^3\|\ell\|_{\mathrm{r}} \le (2 + 3(2\rho_0+1)^3)K_*\mathsf{g}_*$, and together these give the factor P_ρ. As $\Xi = \mathsf{X}(LZ)$, Thm. 2.60 (d) at the starred values gives $L|\Xi|_\infty \le c_2^*\mathsf{N}_Z = \mathsf{N}_\Xi$. This is (Fr3′) at every mask and at every cut. The bound on t_ρ at the end of this step, with the choice eq. 2.vi of ρ_0, is (Fr4); it holds at every cut, because cutting lowers both factors of t_ρ. The step rests on $L \ge L_A$, $aL \ge 3/c_L$, the evenness of N and $x_{k+1} \ge x_\flat$, and it depends on no other condition on L.

For (Fr3), the identity $H^\sharp = \mathsf{T}\mathsf{B}'$ gives $H^\sharp - \mathsf{T}(\mathsf{B}' - \mathsf{B}'^{\mathrm{c}}) = \mathsf{T}\mathsf{B}'^{\mathrm{c}}$ exactly. Hence $L^2\|\cdot\|_\infty \le L^2\|\mathsf{T}\|_\infty\|\mathsf{B}'^{\mathrm{c}}\|_\infty \le c_1^*\mathsf{I}_L\mathsf{N}_Z \le \mathsf{N}_\sharp$, since $\mathsf{P}_\rho \ge 1$; this bound does not depend on the cut radius. Finally $\mathsf{t}_\rho \le 3K_*\mathsf{g}(\lambda_*)K_*\sum_{|u|>\rho_0} e^{-\lambda_*|u|} \le 3K_*^2\mathsf{g}(\lambda_*)\mathsf{g}(\lambda_*/2)e^{-\lambda_*\rho_0/2}$, and by eq. 2.vi the last member is $\le \frac{1}{8}(\frac{1}{2} - d_0 - c_\Xi)$.

Step H16: the passage to $c_R = 64/\theta_D$ with $\theta_D \le \theta_{\mathbb{C}}/2$, and (X12). We prove five assertions. (i) For $\theta' \le \theta$, (Dec)(C,θ) implies (Dec)(C,θ'). (ii) $A_q \le A^\flat$, and $\theta_{\mathbb{C}} \le \theta_q/6$, with θ_q that of Prop. 2.62;

hence, writing θ_D^{sc} for the θ_D given by the display of Prop. 2.62, the smaller of θ_D^{sc} and $\theta_{\mathbb{C}}/2$ is the θ_D of Thm. 2.65. (iii) When the chain is cut with the larger c_R, it remains in the absorbing set $\mathcal{R}(b_*, K_*, \lambda_*, \frac{1}{9})$. (iv) (N′) holds at every cut. (v) (X12) holds. In Thm. 2.65 and in its proof, θ_D, C_D and c_R are those of its item (6), and $e_R = \lambda_* c_R/2$ is that of its item (8), whereas the display of Prop. 2.62 yields θ_D^{sc} together with the C_D evaluated at θ_D^{sc}.

For (i): $e^{\theta \mathrm{dist}}$ and $e^{\theta \mathrm{dist}} - 1$ increase with θ.

For (ii), we first treat $A_q \le A^\flat$, for which it is enough that $\mathsf{g}(\mu)\mathsf{g}_2(\mu/2) \le 8e^6 \mathsf{g}_2(\mu)^2$ at $\mu = \lambda_*/2$. To begin with, $\mathsf{g}(\mu) \le \mathsf{g}_2(\mu)$. Writing $u = 2v + \varepsilon$ with $\varepsilon \in \{0,1\}^3$, we get $(1+|u|)^2 \le 4(1+|v|)^2$ and $e^{-\mu|u|/2} \le e^{\mu/2} e^{-\mu|v|}$. Since $\mu = \lambda_*/2$ with $\lambda_* = \gamma_*^{1/2}/44 \le \frac{1}{44}$, we have $\mu \le \frac{1}{2}$ and hence $e^{\mu/2} \le e^{1/4}$. So $\mathsf{g}_2(\mu/2) \le 32 e^{1/4} \mathsf{g}_2(\mu) \le 41.1\, \mathsf{g}_2(\mu) \le 3227\, \mathsf{g}_2(\mu)$, where $32 = 8 \cdot 4$ accounts for the eight values of ε and the factor 4. Since $8e^6 = 3227.4\ldots$, we have $3227 \le 8e^6$, and together with $\mathsf{g}(\mu) \le \mathsf{g}_2(\mu)$ this gives $\mathsf{g}(\mu)\mathsf{g}_2(\mu/2) \le 8e^6 \mathsf{g}_2(\mu)^2$.

Hence $\theta_{\mathbb{C}} \le \gamma/(12A^\flat) \le \frac{1}{6}\gamma_q/(2A_q)$. Also $\theta_{\mathbb{C}} \le 1/(12 \cdot 384 e^8 \cdot 9/\gamma_*) \le \gamma_*/(1.236 \cdot 10^8) \le \gamma_*^{1/2}/1056 = \lambda_*/24$, since $\gamma_*^{1/2} \le 1$ and $44 \cdot 24 = 1056$. Combining this with the preceding bound, $\theta_{\mathbb{C}} \le \frac{1}{6}\min\{\lambda_*/4, \gamma_q/(2A_q)\} = \theta_q/6$. Consequently $\min\{\theta_D^{\mathrm{sc}}, \theta_{\mathbb{C}}/2\} = \frac{1}{2}\min\{\lambda_2^*, \theta_{\mathbb{C}}, \theta_J\}$. We define C_D with $\mathsf{g}(\theta_D)$ at this θ_D, following item (6) of Thm. 2.65; by (i), the C_D evaluated at θ_D^{sc}, which is the smaller one, would suffice equally well.

For (iii): Thm. 2.57 (a) and Prop. 2.62 hold for every ℓ in the set, and neither of these depends on a radius. The radii enter through (Cut) alone, and $\mathsf{t}_*(\varrho)$ decreases as ϱ increases. For (iv): by Step H14, $\|\ell'^{\mathrm{c}}\|_{\mathrm{r,c}} \le \|\ell'\|_{\mathrm{r,c}}$ at every radius.

For (v): $x^{-e_R}/(1 - L^{-e_R})$ decreases as $e_R = \lambda_* c_R/2$ increases, so a larger c_R helps. Moreover $\theta_D \le \lambda_2^*/2 \le \lambda_*/8$ gives $e_R = 32\lambda_*/\theta_D \ge 256$, and

$$27 K_* \mathsf{g}(\lambda_*/2) \le 27 \cdot 3.153 \cdot 10^{12} \cdot 4.034 \cdot 10^{43} \le 3.44 \cdot 10^{57} < 1.157 \cdot 10^{77} \le 2^{256}(1 - 3^{-256}).$$

Here the bound $\mathsf{g}(\lambda) \le 51\lambda^{-3}$ of Def. 2.44 gives $\mathsf{g}(\lambda_*/2) \le 51(\lambda_*/2)^{-3} \le 4.034 \cdot 10^{43}$. So (X12) holds for every $x_\flat \ge 2$ with $c_R = 64/\theta_D$, the pair (θ_D, c_R) being that of item (6) of Thm. 2.65, and also with $c_R = 64/\theta_D^{\mathrm{sc}}$: both θ_D and θ_D^{sc} are $\le \lambda_2^*/2$, so that $e_R \ge 256$ in both cases.

In addition, $e_R \ge 64\lambda_*/\lambda_2^*$, because $\theta_D \le \lambda_2^*/2$. The step depends on e_R only through $e_R \ge 256$. The requirement $x_\flat \ge e^2$ is part of (O13), in item 3 of §2.1 and in item 1 of §5.5; so (X12) imposes no lower bound of its own on $x_\flat$. □

3. Local Gaussian integration

In this section a block field V' is given. On the fine bonds off the combs we study a quadratic form q_ℓ^V, built with rows and dressed with transports of V', and from it we construct a Gaussian law of finite range. Part C.I shows that $\gamma \le q \le b$ at every V' and proves a weighted bound; it then forms a matrix G row by row from windows, establishes the properties (a) to (k) of the law $\varphi = G^V \xi$, whose range is at most R, and gives an identity that writes the step as a mean over the white noise ξ. Part C.II proves, for this law, the bounds on the centre, (S-cen$_k$), and the locality in V' of $\log\det q_\ell^V$ and of the remainder, (S-loc$_k$). Item C.II 4, in which (S-cen$_k$) is established, uses the free scalar model (Fr1) to (Fr3), which is stated in item C.II 2.2 and proved in §2.2, whereas (S-loc$_k$) is independent of (Fr1) to (Fr3).

Remark 3.1 (The level and the rows). The level is of arbitrary depth. The rows are those of item C.I 1.1 and of item C.II 1.2, with $K \ge 0$; the empty rows, for which $\mathsf{H}_y = \{0\}$ for every y, are therefore admitted.

Part C.I. (Law$_k$): the law of windows.

C.I 1. Objects.

Remark 3.2 (Notation). In Part C.I some letters have two senses, and the context makes clear which one is meant: b is the number of item C.I 1.2 or a bond; G is the group or the matrix of the law of item C.I 3; Y is a subset of I° or a matrix in $\mathfrak{g}_{\mathbb{C}}$; x is a fine site or the parameter of the law; $\mathfrak{g}$ denotes either the sums $\mathfrak{g}(\lambda)$ and $\mathfrak{g}_2(\lambda)$ or the kernel $\mathfrak{g}^V$.

C.I 1.1. Group, lattices, transports, form. Let $G = \mathrm{SU}(2)$, let $\mathfrak{g}$ be the space of hermitian traceless 2×2 matrices, and write $\mathrm{Ad}_g Y = gYg^{-1}$. A field V is a map from the oriented bonds to G, or to $\mathrm{SL}(2,\mathbb{C})$, such that $V(\bar b) = V(b)^{-1}$, where $\bar b$ is the bond b with the opposite orientation. For a path π from π_- to π_+, $V(\pi)$ is the product of the values of V on the bonds of π, taken in the order in which π runs through them. For a gauge u we set $V^u(b) = u(b_-)V(b)u(b_+)^{-1}$, where b_- and b_+ are the initial and the final site of b. For a plaquette $p = (z;\mu\nu)$ with $\mu<\nu$ we put

$$\partial V(p) = V(z,\mu)V(z+e_\mu,\nu)V(z+e_\nu,\mu)^{-1}V(z,\nu)^{-1}.$$

The letters V' and V' denote fields on the coarse lattice, and $\mathcal{V}'$ is the set of those with values in G.

The integer L is odd, and $\mathsf{h}_L = \frac12(L-1)$. The integer N is even; it is the coarse side of the present step, that is, the number $N' = NL^{K-k-1}$ of §2.2, in which N denotes the side of the last torus. We set $\mathbb{T} = \mathbb{Z}^3/NL\mathbb{Z}^3$ and $\mathbb{T}' = L\mathbb{Z}^3/NL\mathbb{Z}^3$. Fine bonds and fine plaquettes are written b and p; coarse bonds, coarse plaquettes and coarse cubes are written c, P and q'. Each cell i has a base site $\pi(i)$, namely its least corner, and $y(i)$ is the coarse site whose block $B(y) = \{x : |x-y|_\infty \le \mathsf{h}_L\}$ contains $\pi(i)$. The distance of two cells is $\mathrm{dist}(i,j) = L^{-1}|y(i)-y(j)|_\infty$, taken periodically. For $x\in B(y)$, the comb $\Gamma(y,x)$ is the path from y to x that runs first along e_1, then along e_2, then along e_3; the set of the bonds of all combs is $\mathcal{T}$.

The set I°. We denote by I° the set of the fine bonds off $\mathcal{T}$, that is, of the fine bonds that lie on no comb; at most $3L^3$ bonds of I° are based in a block.

We write $\mathfrak{L}_{\mathfrak{g}}$ for the space of the $\mathfrak{g}$-valued functions on I°. A coarse bond $c = (y,y')$ of direction μ has $c_- = y$ and $c_+ = y' = y + Le_\mu$, and we set $W_c = B(y)\cup B(y')$. For $x\in B(y)$ we put $x' = x + Le_\mu$, which is a fine site. We write $[x,x']$ for the straight path from x to x', and $A([x,x'])$ for the sum of A over its L bonds. A crossing bond of c is a bond (z,μ) with $z\in B(y)$ and $z+e_\mu\in B(y')$. The coboundary d acts by $(\mathrm{d}\eta)_p = \sum_b \mathrm{d}(p,b)\eta_b$, and the kernel Q_1 by

$$(Q_1 A)(c) = L^{-3}\sum_{x\in B(c_-)} A([x,x']).$$

The edge lift E is given by $(EX')_p = X'_P$ when p is one of the L edge plaquettes under P, and by 0 at every other plaquette. The edge plaquettes under $P = (y;\mu\nu)$ are the plaquettes $p = (z;\mu\nu)$ with $z_\mu = y_\mu + \mathsf{h}_L$, $z_\nu = y_\nu + \mathsf{h}_L$ and $|z_\lambda - y_\lambda| \le \mathsf{h}_L$ for the direction λ normal to P, the one other than μ and ν. They are based in $B(y_P)$, where $y_P = \pi(P)$.

For coarse sites y and y'' at sup distance less than half the side of the torus, $\Gamma(y,y'')$ is the coordinate path from y to y'', along e_1, then e_2, then e_3, each time in the shorter direction, and the transport between them is $T(y,y'') := \mathrm{Ad}_{V'(\Gamma(y,y''))}$. A scalar kernel K is dressed with transports by $K^V(i,j) := K(i,j)T(y(i),y(j))$ and $(K^V)^*(j,i) := K(i,j)T(y(i),y(j))^{-1}$. The rows act on a $\mathfrak{g}$-valued function F on the fine plaquettes by $(\ell^V F)_y = \sum_p \ell_y(p)\otimes T(y(y),y(p))F_p$. In particular,

$$(\mathrm{d}^V\eta)_p = \sum_b \mathrm{d}(p,b)T(y(p),y(b))\eta_b, \qquad (Q_1^V\eta)_c = \sum_b Q_1(c,b)T(c_-,y(b))\eta_b.$$

The displayed adjoints are transposes for the bilinear form on complex cochains, with inverse transports. With $\mathsf{B}^V = 1 + \ell^{V*}\ell^V$ we define the form and the kernel

$$q_\ell^V(\eta) = \langle \mathrm{d}^V\eta, \mathsf{B}^V\mathrm{d}^V\eta\rangle + a\langle Q_1^V\eta, Q_1^V\eta\rangle, \qquad \mathsf{g}^V = \Pi_{\mathfrak{L}}\mathrm{d}^{V*}\mathsf{B}^V E,$$

where $a>0$ is the parameter of the form and $\Pi_{\mathfrak{L}}$ is the restriction to I°.

Norms. The operator norm of a 2×2 matrix is written $|\cdot|$. On $\mathfrak{g}_{\mathbb{C}}$, the traceless complex 2×2 matrices, we use $\|Y\|^2 = \frac12 \mathrm{tr}\, Y^*Y$ and $Y\cdot Y' = \frac12 \mathrm{tr}\, YY'$; then $\|Y\| \le |Y| \le \sqrt{2}\|Y\|$ and

$\|AYB\| \le |A|\|Y\||B|$. For $g \in \mathrm{SL}(2,\mathbb{C})$ we have $|g^{-1}| = |g| \ge 1$ and $\|\operatorname{Ad}_g\| \le |g|^2$. In the real basis $\sigma_1,\sigma_2,\sigma_3$ of $\mathfrak{g}$, the form $Y\cdot Y'$ reads $\sum y_i y_i'$ and the square of the norm reads $\sum |y_i|^2$. The transpose ^{T} is taken with respect to $Y\cdot Y'$; it preserves operator norms, and $\operatorname{Ad}_g^{\mathsf{T}} = \operatorname{Ad}_g^{-1}$. The blocks of a matrix on $\ell^2(I^\circ;\mathfrak{g}_{\mathbb{C}})$ are measured in the operator norm for $\|\cdot\|$, the norm $\|\cdot\|$ of the matrix itself is its operator norm on ℓ^2, and $\|A\|_\infty := \sup_b \sum_{b'} \|A(b,b')\|$.

For matrices Y_k with $S = \sum |Y_k| \le 1$ we have

$$|e^{iY_1}\cdots e^{iY_n} - \mathbf{1} - i\sum Y_k| \le e^S - 1 - S \le 0.72 S^2. \tag{E}$$

Proof of (E). We expand every factor in its power series and multiply out: the product $e^{iY_1}\cdots e^{iY_n}$ becomes an absolutely convergent sum, over all $(m_1,\dots,m_n)$ with integers $m_k \ge 0$, of the products $\prod_k (iY_k)^{m_k}/m_k!$, whose factors keep their order. The terms with $\sum m_k = 0$ add up to $\mathbf{1}$, and those with $\sum m_k = 1$ add up to $i\sum Y_k$. By the triangle inequality and the submultiplicativity of the operator norm, the norm of the remaining terms is therefore at most the sum, over the (m_k) with $\sum m_k \ge 2$, of the numbers $\prod_k |Y_k|^{m_k}/m_k!$. This sum equals $e^S - 1 - S$: summed over all (m_k), these numbers give $\prod_k e^{|Y_k|} = e^S$, and those with $\sum m_k \le 1$ give $1+S$. The first inequality of (E) thus holds for every $S \ge 0$; this is its form in item C.II 1.1, and it is applied with $S \le 1.236$ in Lem. 3.27 (d). For the second inequality we observe that $(e^S - 1 - S)/S^2 = \sum_{j\ge 2} S^{j-2}/j!$ increases with S and takes the value $e - 2 < 0.7183$ at $S = 1$; hence this quotient is at most 0.72 whenever $S \le 1$. □

We also write

$$\mathsf{g}(\lambda) = \sum_{n\in\mathbb{Z}^3} e^{-\lambda|n|_\infty}, \qquad \mathsf{g}_2(\lambda) = \sum_n (1+|n|_\infty)^2 e^{-\lambda|n|_\infty}.$$

Rows. The rows form a family $\ell = (\ell_y)$, in which y runs over the sites of $\mathbb{T}$ and each $\ell_y(p)$ is a vector of a Euclidean space H_y with inner product $\langle\cdot,\cdot\rangle$. They satisfy $|\ell_y(p)| \le K e^{-\lambda|y-p|}$, where $|y-p| = |y-\pi(p)|_\infty$ is measured in fine units, $K \ge 0$ and $0 < \lambda \le 1$; and their range is at most ϱ, with $\frac{\varrho}{L} + 2 < \frac{N}{2}$. It follows that $\|\ell\|_{\mathrm{r}} \le 3K\mathsf{g}(\lambda)$ and $\|\ell\|_{\mathrm{c}} \le K\mathsf{g}(\lambda)$, where $\|\ell\|_{\mathrm{r}} = \sup_y \sum_p |\ell_y(p)|$ and $\|\ell\|_{\mathrm{c}} = \sup_p \sum_y |\ell_y(p)|$; the factor 3 comes from the three plaquettes based at each site.

Condition ($\mathrm{u1}^\flat$). A field V' satisfies ($\mathrm{u1}^\flat$) if it is an $\mathrm{SL}(2,\mathbb{C})$ field with $\mathsf{V}'(\bar c) = \mathsf{V}'(c)^{-1}$ and $|\mathsf{V}'(c)| \le 1 + \vartheta_0^\flat$ at every bond c. A field that is given only on a set of bonds is extended by $\mathbf{1}$ to the remaining bonds. At a field V' of this kind, T, d^{V}, ℓ^{V}, q_ℓ^{V} and g^{V} are given by the same formulas as above, with * defined as for $(K^V)^*$; consequently q_ℓ^{V} is symmetric for $Y\cdot Y'$ rather than hermitian.

C.I 1.2. Numbers. We set $\gamma = (73L^4)^{-1}$ and

$$b = 16(1+3K^2\mathsf{g}(\lambda)^2) + \tfrac{a}{L}, \quad A^\flat = 2e^2(16+a) + 384e^8K^2\mathsf{g}_2(\tfrac{\lambda}{2})^2, \quad \theta_1 = \min\{1, \tfrac{\lambda L}{12}\}, \quad \theta^\flat = \min\{\theta_1, \tfrac{\gamma}{2A^\flat}\}, \quad \theta_{\mathbb{C}} = \tfrac{1}{6}\theta^\flat, \quad \vartheta_0^\flat = \tfrac{1}{24}\theta^\flat,$$

$$\varkappa = 1 - \frac{\gamma}{2b}, \qquad C_\varphi^\flat = \frac{2b^{1/2}}{\gamma}\,\frac{3L^3\mathsf{g}(\theta_{\mathbb{C}})}{1-e^{-\theta_{\mathbb{C}}}}, \qquad c_k = 4^{-k}\tbinom{2k}{k}.$$

These numbers satisfy $\vartheta_0^\flat \le \frac{\gamma}{48A^\flat} \le 10^{-7}$; the second of the two bounds rests on $L \ge 3$, a hypothesis of every statement, which gives $\gamma \le \frac{1}{5913}$ and $A^\flat \ge 32e^2$.

In the chain these numbers are evaluated at $(K,\lambda) = (K_*,\lambda_*)$ at every level, the level of the empty rows included (item C.II 1.3; item 3, (O6), of §2.1); they thus depend on the letters of the bound and are independent of the rows.

We further put $\theta_2 = \frac{1}{2}\theta_1$ and $\Lambda = 2 + A^\flat/b$, and, for $R \in \mathbb{N}$,

$$r_w := \lfloor R/2\rfloor, \quad \varrho_L := \lceil \varrho/L\rceil, \quad r_q := 2\varrho_L + 2, \quad \mathsf{R}_{\mathsf{E}} := 2r_w + 3\varrho_L + 3,$$

$$d_R(\theta) := 9L^3\, b\, \gamma^{-3/2}(R+3)^3\, \mathsf{g}(\theta)\, e^{-\theta R/2}, \quad d_R := d_R(\theta_{\mathbb{C}}),$$

$$\mathsf{G}_0 := (\tfrac{2}{\gamma})^{1/2} 3L^3\mathsf{g}(\theta_{\mathbb{C}})\ (\le C_\varphi^\flat), \qquad \mathsf{K}_L := \frac{153\, A^\flat}{\gamma^{3/2}\theta^\flat}.$$

In what follows we use the identities $\sum_k c_k\kappa^k = (1-\kappa)^{-1/2}$ and $\sum_k kc_k\kappa^{k-1} = \frac{1}{2}(1-\kappa)^{-3/2}$; at $\kappa = \varkappa$ they give $b^{-1/2}\sum_k c_k\varkappa^k = (\frac{2}{\gamma})^{1/2}$ and $b^{-1/2}\sum_k kc_k\varkappa^{k-1} = \sqrt{2}\,b\gamma^{-3/2}$. We also use $\sum_k |\binom{1/2}{k}| = 2$ and the inequality $(1+t)e^{-\delta t} \le \frac{1}{\delta}$ for $t \ge 0$ and $0 < \delta \le 1$.

C.I 1.3. Steps, sub-forms. The matrix of the form is a sum over steps,

$$q_\ell^V(b,b') = \sum_{\mathsf{s}:b\to b'} q_\mathsf{s}\mathsf{T}_\mathsf{s},$$

in which a step s from b to b' is of one of the three kinds (P), (Q), (R), with the following weight q_s and transport T_s.

(P) A step $\mathsf{s} = (p)$ with $p \ni b, b'$ has $q_\mathsf{s} = \mathrm{d}(p,b)\mathrm{d}(p,b')$ and $\mathsf{T}_\mathsf{s} = T(y(p),y(b))^{-1}T(y(p),y(b'))$.
(Q) A step $\mathsf{s} = (c)$ has $q_\mathsf{s} = aQ_1(c,b)Q_1(c,b')$ and $\mathsf{T}_\mathsf{s} = T(c_-,y(b))^{-1}T(c_-,y(b'))$.
(R) A step $\mathsf{s} = (y,p,p')$ with $p \ni b$ and $p' \ni b'$ has $q_\mathsf{s} = \mathrm{d}(p,b)\mathrm{d}(p',b')\langle \ell_y(p), \ell_y(p')\rangle$ and

$$\mathsf{T}_\mathsf{s} = T(y(p),y(b))^{-1}T(y(y),y(p))^{-1}T(y(y),y(p'))T(y(p'),y(b')).$$

For each kind, $\mathsf{T}_\mathsf{s} = \mathrm{Ad}_{V'(\pi_\mathsf{s})}$, where π_s is the coarse path from $y(b)$ to $y(b')$ that the product of transports traces, taken as it stands. We write $|\mathsf{s}|$ for the number of its bonds, backtracks included, and $\bar{\mathsf{s}} : b' \to b$ for the reversed step. The weight q_s is independent of the field. For the steps (y,p,p') of kind (R) only the sites y with $|y-p| \le \varrho$ and $|y-p'| \le \varrho$ are admitted: for every other y the range of the rows gives $q_\mathsf{s} = 0$, and so the transport $T(y(y),y(p))$ is required only for admitted y. If $\mathsf{H}_y = \{0\}$, every step (y,p,p') of kind (R) has $q_\mathsf{s} = 0$. Every sum over steps below carries the factor q_s; hence the steps with $q_\mathsf{s} = 0$ change no sum and no bound.

Sub-forms. Let S be a set of coarse sites, and let $I_\mathsf{S} := \{b \in I^\circ : y(b) \in \mathsf{S}\}$. A step $\mathsf{s} : b \to b'$ lies *in* S if $b, b' \in I_\mathsf{S}$ and if, for a step of kind (R), all sites of $\Gamma(y(y),y(p))$ and of $\Gamma(y(y),y(p'))$ belong to S. For $\zeta \in \mathbb{C}$ and $Y \subset I_\mathsf{S}$ we define

$$q_\mathsf{S}(\zeta) := \sum_{\mathsf{s}\text{ in }\mathsf{S}} \zeta^{|\mathsf{s}|} q_\mathsf{s}\mathsf{T}_\mathsf{s}, \qquad q_\mathsf{S} := q_\mathsf{S}(1), \qquad K_{\mathsf{S},Y}(\zeta) := \mathbf{1}_Y - b^{-1}\mathbf{1}_Y q_\mathsf{S}(\zeta)\mathbf{1}_Y \ \text{ on } \ell^2(Y;\mathfrak{g}_\mathbb{C}),$$

and wherever a product with another matrix is taken, $K_{\mathsf{S},Y}(\zeta)$ is extended by 0 to $\ell^2(I^\circ)$. We also write $K_{\mathsf{S},Y} := K_{\mathsf{S},Y}(1)$ and $K_\mathsf{S} := K_{\mathsf{S},I_\mathsf{S}}$. When S is the set of all sites, the index S is omitted: then $q = q_\ell^V$, $K = 1 - q/b$, and K_Y stands for $K_{\mathsf{S},Y}$. Finally, for a polynomial $A(\zeta)$, $[A(\zeta)]_j$ denotes its coefficient of ζ^j.

C.I 2. The form; one weighted bound.

Lemma 3.3 (The form and its sub-forms)**.** In the setting of item C.I 1.1, with rows as there, let $L \ge 3$ be odd and let $aL \ge 18$. Then the following hold.

(a) At every $V' \in \mathcal{V}'$, for every S and every $\eta \in \mathfrak{L}_\mathfrak{g}$ supported in I_S, we have $\gamma|\eta|^2 \le \langle \eta, q_\mathsf{S}\eta\rangle \le b|\eta|^2$. In consequence, $0 \le K_{\mathsf{S},Y} \le 1 - \frac{\gamma}{b}$ for all $Y \subset I_\mathsf{S}$.
(b) Let $0 \le \beta \le \theta_1$, and write $\sum^{(b)} := \sup_b \sum_{b'} \sum_{\mathsf{s}:b\to b'}$. Then we have the first bound $\sum^{(b)} |q_\mathsf{s}| \le b$, the second bound $\sum^{(b)} |q_\mathsf{s}|(e^{\beta|\mathsf{s}|} - 1) \le A^\flat\beta$, and the third bound $\sum^{(b)} |q_\mathsf{s}|\,|\mathsf{s}|(1+|\mathsf{s}|)e^{\beta|\mathsf{s}|} \le 9A^\flat$. These three bounds also hold for the sums over b at fixed b'.
(c) For every step $\mathsf{s} : b \to b'$ we have $\mathrm{dist}(b,b') \le \min\{|\mathsf{s}|, r_q\}$, and each site of π_s lies within $\varrho_L + 1$ of $y(b)$ or of $y(b')$.
(d) For every step s and the reversed step $\bar{\mathsf{s}}$ we have $\mathsf{T}_{\bar{\mathsf{s}}} = \mathsf{T}_\mathsf{s}^{-1} = \mathsf{T}_\mathsf{s}^\mathsf{T}$, $q_{\bar{\mathsf{s}}} = q_\mathsf{s}$ and $|\bar{\mathsf{s}}| = |\mathsf{s}|$. Moreover, for each map u from the coarse sites to $\mathrm{SL}(2,\mathbb{C})$ and each step $\mathsf{s} : b \to b'$, $\mathsf{T}_\mathsf{s}[\mathsf{V}'^u] = \mathrm{Ad}_{u(y(b))}\,\mathsf{T}_\mathsf{s}[\mathsf{V}']\,\mathrm{Ad}^{-1}_{u(y(b'))}$.
(e) The entries of g^V are $\mathsf{g}^V(b,P) = \sum_{\mathsf{t}:b\to P} g_\mathsf{t}\,\mathrm{Ad}_{V'(\pi_\mathsf{t})}$, the sum running over steps of two kinds:
(E) $\mathsf{t} = (p)$, where $p \ni b$ is an edge plaquette under P, with $g_\mathsf{t} = \mathrm{d}(p,b)$;

(RE) $\mathsf{t} = (y, p, p')$, where $p \ni b$ and p' is an edge plaquette under P, with $g_{\mathsf{t}} = \mathrm{d}(p,b)\langle \ell_y(p), \ell_y(p')\rangle$; as in item C.I 1.3, the site y is restricted to $|y-p| \le \varrho$ and $|y-p'| \le \varrho$, and $g_{\mathsf{t}} = 0$ for every other y.

For each of these steps, π_{t} is a coarse path from $y(b)$ to y_P, all of whose sites lie within $2\varrho_L + 1$ of $y(b)$. We write $|\mathsf{t}|$ for the number of bonds of π_{t}, backtracks included, just as $|\mathsf{s}|$ is defined in item C.I 1.3. Moreover $\sup_b \sum_P \sum_{\mathsf{t}} |g_{\mathsf{t}}| \le \frac{b}{4}$ and $\sup_b \sum_P \sum_{\mathsf{t}} |g_{\mathsf{t}}| e^{2\vartheta_0^\flat |\mathsf{t}|} \le b$, and for $\beta \le \theta_1$ also $\sup_b \sum_P \sum_{\mathsf{t}} |g_{\mathsf{t}}||\mathsf{t}|(1+|\mathsf{t}|)e^{\beta|\mathsf{t}|} \le 9A^\flat$.

■

Proof. Step (a): the lower bound for the form without rows. Let $\eta \in \mathfrak{L}_{\mathfrak{g}}$, and put $f = \mathrm{d}^V \eta$. We define the form without rows, which is the form q_ℓ^V at the empty rows, by $q_\emptyset^V(\eta) := |\mathrm{d}^V\eta|^2 + a|Q_1^V\eta|^2$, that is, $q_\emptyset^V(\eta) = |f|^2 + a|Q_1^V\eta|^2$; by the display that defines q_ℓ^V in item C.I 1.1, $q_\ell^V(\eta) = q_\emptyset^V(\eta) + |\ell^V \mathrm{d}^V\eta|^2$.

In each block all transports are the identity, so eq. 2.i applies. For a double block W_c of $c = (y, y')$, keep the values of η based in $B(y)$ and apply $\mathrm{Ad}_{V'(c)}$ to those based in $B(y')$. On a plaquette of W_c the ordinary coboundary of the transformed cochain is $\mathrm{d}^V\eta$ expressed in the frame at y. Across the seam this follows from $T(y,y') = \mathrm{Ad}_{V'(c)}$, and inside either block it follows because all its bonds are in the same frame. The transformed cochain vanishes on the combs and its average on c is $(Q_1^V\eta)_c$. The transport is an isometry, so eq. 2.ii gives the same inequality with $f = \mathrm{d}^V\eta$ and $Q_1^V\eta$. Summing, as in Lem. 2.24, gives $|\eta|^2 \le 73L^4|f|^2 + 6L^2|Q_1^V\eta|^2 \le 73L^4 q_\emptyset^V(\eta)$. The last inequality uses $aL \ge 18$.

Step (a): the sub-forms. We put $\ell_y^{[\mathsf{S}]}(p) := \ell_y(p)1\{\text{all sites of } \Gamma(y(y), y(p)) \text{ lie in } \mathsf{S}\}$. Then $q_{\mathsf{S}} = \mathbf{1}_{I_{\mathsf{S}}} q_\emptyset^V \mathbf{1}_{I_{\mathsf{S}}} + (\ell^{[\mathsf{S}]V}\mathrm{d}^V\mathbf{1}_{I_{\mathsf{S}}})^*(\ell^{[\mathsf{S}]V}\mathrm{d}^V\mathbf{1}_{I_{\mathsf{S}}})$: when the right side is multiplied out, its first term yields the steps (P) and (Q) between bonds of I_{S}, and its second term the steps (R) in S. Thus q_{S} is the sum of a compression of $q_\emptyset^V$ and of a square, where $q_\emptyset^V \ge \gamma$ by the local estimates just proved; this gives the lower bound of (a).

Step (a): the upper bound. The upper bound of (a) follows from Schur's test. Indeed, every transport has $\|\mathsf{T}_{\mathsf{s}}\| = 1$, and the sums over the steps that make up q_{S} are sub-sums of the sums in the first bound of (b), for rows and for columns.

Step (b): the three bounds. For the first bound, the three kinds of steps contribute as follows.

(P) There are four plaquettes $p \ni b$, each with four bonds b', which gives 16; as $\Gamma(y(p), y(b))$ is empty or consists of one bond, $|\mathsf{s}| \le 2$.

(Q) As $\sum_c Q_1(c,b) = L^{-2}$ and $\sum_{b'} Q_1(c,b') = L$, the contribution is $\frac{a}{L} \le a$; as $y(b) \in \{c_-, c_+\}$, again $|\mathsf{s}| \le 2$.

(R) The contribution is $\le 4 \cdot 4\, \|\ell\|_{\mathrm{c}} \|\ell\|_{\mathrm{r}} \le 48K^2 \mathsf{g}(\lambda)^2$.

These three contributions add up to at most b, which proves the first bound. For the other two bounds we consider a step of kind (R) with the site y and the plaquettes p and p', and we put $n_1 = |y-p|$ and $n_2 = |y-p'|$. Then $L^{-1}|y(y) - y(p)|_\infty < \frac{n_1}{L} + 1$, and so $\Gamma(y(y), y(p))$ has at most $3(\frac{n_1}{L} + 1)$ bonds; the path from $y(y)$ to $y(p')$ obeys the same bound with n_2 in place of n_1. Consequently $m := |\mathsf{s}| \le 8 + \frac{3}{L}(n_1 + n_2) \le 8(1+n_1)(1+n_2)$, hence $1 + m \le 9(1+n_1)(1+n_2)$, and also $e^{\beta m} \le e^8 e^{\frac{\lambda}{4}(n_1+n_2)}$. In addition we use $e^{\beta m} - 1 \le \beta m e^{\beta m}$. The contributions to the second and to the third sum are then bounded as follows.

(P), (Q) Since $|\mathsf{s}| \le 2$, these two kinds contribute at most $(16+a)2\beta e^2$ to the second sum and at most $(16+a)6e^2$ to the third.

(R) As there are three plaquettes to each site, the contribution to the second sum is at most $16 \cdot 8e^8 K^2 \beta \cdot 3[\sum_u (1+|u|)e^{-\frac{3}{4}\lambda|u|}]^2 \le 384 e^8 K^2 \mathsf{g}_2(\frac{\lambda}{2})^2 \beta$, where u runs over $\mathbb{Z}^3$ and $|u| = |u|_\infty$, here and in the next sum over u, and the contribution to the third sum is at most $16 \cdot 72 e^8 K^2 \cdot 3[\sum_u (1+|u|)^2 e^{-\frac{3}{4}\lambda|u|}]^2 \le 3456 e^8 K^2 \mathsf{g}_2(\frac{\lambda}{2})^2$.

The contributions to the second sum add up to at most $(16+a)2\beta e^2 + 384e^8K^2\mathsf{g}_2(\frac{\lambda}{2})^2\beta = A^\flat\beta$, and those to the third sum add up to at most $(16+a)6e^2 + 3456e^8K^2\mathsf{g}_2(\frac{\lambda}{2})^2 \le 9A^\flat$; in both cases we use the definition $A^\flat = 2e^2(16+a)+384e^8K^2\mathsf{g}_2(\frac{\lambda}{2})^2$ in item C.I 1.2. This proves the second and the third bound of (b). For the columns, that is, for the sums over b at fixed b', the same bounds follow from (d).

Step (c): distances and paths. The path π_s joins $y(b)$ to $y(b')$. For the plaquette $p \ni b$ of a step, $y(p)$ is within 1 of $y(b)$. For a step through a site y we have $L^{-1}|y(y)-y(p)|_\infty < \frac{\varrho}{L}+1$, and as this number is an integer, it is at most ϱ_L. Finally, a coordinate path does not leave the box spanned by its two ends; this proves (c).

Step (d): reversal and gauge transformations. The reversed step runs along the reversed path, and $\mathrm{Ad}_g^\mathsf{T} = \mathrm{Ad}_g^{-1}$; this gives the first part of (d). The second part follows from $\mathsf{V}'^u(\pi) = u(\pi_-)\mathsf{V}'(\pi)u(\pi_+)^{-1}$, where π_- and π_+ are the initial and the final site of the path π.

Step (e): the matrix g^V. The two kinds of steps arise on multiplying out $\Pi_\mathfrak{L}\mathrm{d}^{V*}(1+\ell^{V*}\ell^V)E$; here an edge plaquette under P is based in $B(y_P)$. The sums in (e) are sub-sums of those in (b), without the last factor 4, because no $b' \in \partial p'$ occurs, and with $|\mathsf{t}| \le 7 + \frac{3}{L}(n_1+n_2)$; in particular, the third bound of (e) follows from the third bound of (b). The first sum in (e) is therefore at most $4 + 4\cdot 3K^2\mathsf{g}(\lambda)^2 \le \frac{b}{4}$, and, by the second bound of (b), the second sum is at most $\frac{b}{4} + 2A^\flat\vartheta_0^\flat \le b$. □

Lemma 3.4 (One weighted bound for all sub-forms)**.** Under the hypotheses of Lem. 3.3, that is, with rows as in item C.I 1.1, with $L \ge 3$ odd and with $aL \ge 18$, consider a family of retained steps on a set $Y \subset I^\circ$, closed under reversal. Write $q_{\mathrm{ret}}(\zeta)$ for its step sum as in item C.I 1.3 and $K_{\mathrm{ret}}(\zeta) = 1 - b^{-1}q_{\mathrm{ret}}(\zeta)$. When ζ is omitted, it is set to 1. Assume that $q_{\mathrm{ret}} \ge \gamma$ at every G-valued field. The families defining $K_{\mathsf{S},Y}$ satisfy this assumption by Lem. 3.3(a). Let w be real on Y with $|w(b_1) - w(b_2)| \le \theta\,\mathrm{dist}(b_1,b_2)$. Then the following bounds hold for this family.

(r) At every $V' \in \mathcal{V}'$, if $\theta \le \theta^\flat$, then $\|e^w K_{\mathrm{ret}} e^{-w}\| \le \varkappa$.

(c) At every V' satisfying (u1$^\flat$), if $\theta \le \theta_\mathbb{C}$, then $\|e^w K_{\mathrm{ret}} e^{-w}\| \le 1 - \frac{5\gamma}{8b} \le \varkappa$.

(z) At every V' satisfying (u1$^\flat$), $\|K_{\mathrm{ret}}(\zeta)\| \le \Lambda$ for $|\zeta| \le e^{\theta_2}$; consequently $\|[K_{\mathrm{ret}}(\zeta)^k]_j\| \le \Lambda^k e^{-\theta_2 j}$.

■

Proof. At a real field the lower bound on the form is the assumption. Its upper bound by b follows from Schur's test and the first bound of Lem. 3.3(b), since the transports are orthogonal. Thus $0 \le K_{\mathrm{ret}} \le 1 - \gamma/b$.

Step (r): real fields. The entry (b_1,b_2) of $e^w K_{\mathrm{ret}} e^{-w} - K_{\mathrm{ret}}$ equals

$$-b^{-1}\sum_{\mathsf{s}} q_\mathsf{s}\mathsf{T}_\mathsf{s}(e^{w(b_1)-w(b_2)} - 1)$$

. For real t one has $|e^t - 1| \le e^{|t|} - 1$, and $|w(b_1) - w(b_2)| \le \theta|\mathsf{s}|$ by Lem. 3.3 (c). Schur's test and Lem. 3.3 (b) thus bound the norm of the difference by $b^{-1}A^\flat\theta^\flat \le \frac{\gamma}{2b}$. Combined with $0 \le K_{\mathrm{ret}} \le 1 - \gamma/b$ at a real field, this proves (r).

Step (c): fields satisfying (u1$^\flat$). We decompose every $\mathsf{V}'(c)$ in polar form, $\mathsf{V}'(c) = \mathsf{p}_c k_c$, where $k_c \in G$ and $\mathsf{p}_c > 0$ has the eigenvalues $s^{\pm 1}$ with $1 \le s \le 1+\vartheta_0^\flat$; thus $|\mathsf{p}_c^{\pm 1} - \mathbf{1}| \le \vartheta_0^\flat$. Then $V_0 := (k_c)$ is a field with values in G. Consider a path of m bonds, and let $g = \mathsf{V}'(\pi)$ and $g_0 = V_0(\pi)$. Every factor of g is the sum of a unitary matrix and a matrix of norm $\le \vartheta_0^\flat$, so that $|g| \le (1+\vartheta_0^\flat)^m$ and $|g - g_0| \le (1+\vartheta_0^\flat)^m - 1$. From the identity $gYg^{-1} - g_0Yg_0^{-1} = (g-g_0)Yg^{-1} + g_0Yg^{-1}(g_0-g)g_0^{-1}$ we deduce $\|\mathrm{Ad}_g - \mathrm{Ad}_{g_0}\| \le 2|g||g-g_0| \le 2(e^{2\vartheta_0^\flat m} - 1)$. The entries of $e^w q_{\mathrm{ret}}^{\mathsf{V}} e^{-w} - q_{\mathrm{ret}}^{V_0}$ are therefore, in norm, $\le \sum |q_\mathsf{s}|[(e^{\theta m} - 1) + 2e^{\theta m}(e^{2\vartheta_0^\flat m} - 1)] \le 3\sum |q_\mathsf{s}|(e^{(\theta+2\vartheta_0^\flat)m} - 1)$, by the elementary inequality $2e^{A+B} - e^A - 1 \le 3e^{A+B} - 3$ for $A, B \ge 0$. Here $\theta_\mathbb{C} + 2\vartheta_0^\flat = \frac{1}{4}\theta^\flat$. The row sums and the column sums share one majorant, and Schur's test, with the second bound of Lem. 3.3 (b), bounds

the norm of this difference by $\frac{3}{4}A^\flat\theta^\flat \le \frac{3}{8}\gamma$. Using $0 \le K_{\mathrm{ret}}[V_0] \le 1-\gamma/b$, we conclude that $\|e^w K^{\mathsf V}_{\mathrm{ret}} e^{-w}\| \le 1-\frac{\gamma}{b}+\frac{3\gamma}{8b} \le \varkappa$. The coercivity assumption is used at the polar field V_0.

Step (z): the variable ζ. We have $\|\mathsf T_{\mathsf s}\| \le |\mathsf V'(\pi_{\mathsf s})|^2 \le e^{2\vartheta_0^\flat|\mathsf s|}$ and $\theta_2 + 2\vartheta_0^\flat \le \theta_1$; Schur's test and Lem. 3.3 (b) give $\|q_{\mathrm{ret}}(\zeta)\| \le b + A^\flat\theta_1$, and from this the first assertion of (z) follows. The second assertion is Cauchy's estimate for the polynomial $K(\zeta)^k$ on the circle $|\zeta| = e^{\theta_2}$.

□

C.I 3. (Law$_k$).

Definition 3.5 (Windows, the matrix of the law, the law)**.** Let $R \in \mathbb{N}$. For $b \in I^\circ$, the window of b is $Y_b := \{b' \in I^\circ : \mathrm{dist}(b,b') \le r_w\}$; we let $K_b := K_{Y_b}$, taken as in item C.I 1.3 with $\mathsf S$ the set of all sites, and we write $q_b = \mathbf{1}_{Y_b} q \mathbf{1}_{Y_b} = b(\mathbf{1}_{Y_b} - K_b)$. The matrix of the law, $G^{\mathsf V}$, and the matrices C, $\mathsf E$, F and Δ are defined by

$$G^{\mathsf V}(b,b') := b^{-1/2}\sum_{k\ge 0} c_k\,(K_b)^k(b,b') \quad (b' \in Y_b), \quad 0 \text{ for all other } b' \in I^\circ;$$

$$C := GG^{\mathsf T}, \quad \mathsf E := G^{\mathsf T} q G - 1, \quad F := b^{-1/2}\sum_k c_k K^k, \quad \Delta := F - G.$$

Let $x > 1$ be the parameter of the law, and let $\xi \sim \mathcal N(0, x^{-1}\mathbf 1)$ on $\mathfrak L_{\mathfrak g}$; this is a product measure, and it depends on no field. The law is the image of $\mathcal N(0,x^{-1}\mathbf 1)$ under the map $\xi \mapsto \varphi := G^V\xi$. Under the hypotheses of Thm. 3.6, at every $V' \in \mathcal V'$ the matrix G^V is real, as Thm. 3.6 (a) asserts, and the law at such a field is a probability measure on $\mathfrak L_{\mathfrak g}$.

■

Theorem 3.6 (The law of windows, (Law$_k$))**.** In the setting of item C.I 1.1, with rows as there, let $L \ge 3$ be odd, let $aL \ge 18$ and let $R \in \mathbb N$, and let G, C, $\mathsf E$, F and Δ be as in Def. 3.5. Each clause below holds at every $V' \in \mathcal V'$ with $\theta = \theta^\flat$, and at every $\mathsf V'$ with (u1$^\flat$) with $\theta = \theta_{\mathbb C}$.

(a) *Well defined.* The series converge in norm, $F^2 q = 1$, and $F^{\mathsf T} = F$. At $V' \in \mathcal V'$, moreover, $F = (q^V_\ell)^{-1/2}$, $G^V(b,\cdot)$ is the row b of $q_b^{-1/2}$, and G^V is real.

(b) *Range and dependence.* We have $G(b,b') = 0$ unless $\mathrm{dist}(b,b') \le r_w \le \frac12 R$. The matrix C is symmetric, and $C(b,b') = 0$ unless $\mathrm{dist}(b,b') \le 2r_w \le R$. The component $(G\xi)_b$ is a function of the values of ξ on Y_b. The row $G(b,\cdot)$ depends only on $\mathrm{Ad}\,\mathsf V'$ on the coarse bonds whose two ends lie within $r_w + \varrho_L + 1$ of $y(b)$, and the row $C(b,\cdot)$ only on $\mathrm{Ad}\,\mathsf V'$ on those whose two ends lie within $3r_w + \varrho_L + 1$ of $y(b)$.

(c) *Covariance.* For G-valued u, $G[\mathsf V'^u] = \mathrm{Ad}_u\, G[\mathsf V']\,\mathrm{Ad}_u^{-1}$, where $(\mathrm{Ad}_u\,\eta)_b = \mathrm{Ad}_{u(y(b))}\,\eta_b$; the same holds for $C, \mathsf E, F$.

(c′) The identities of (c) hold also for $\mathrm{SL}(2,\mathbb C)$-valued u, provided both $\mathsf V'$ and $\mathsf V'^u$ satisfy (u1$^\flat$). For a single row b it suffices to impose the bound (u1$^\flat$) and the gauge relation on the union of the paths of the steps between bonds of Y_b. These paths are contained in the region of part (b).

(d) *Decay in the sup norm.* We have $\|G(b,b')\|, \|F(b,b')\| \le (\frac{2}{\gamma})^{1/2} e^{-\theta\,\mathrm{dist}(b,b')}$. Consequently $\|G\|_\infty, \|G^{\mathsf T}\|_\infty \le \mathsf G_0$ and $\|(G\xi)_b\| \le \mathsf G_0 \sup_{Y_b}\|\xi_{b'}\| \le C^\flat_\varphi |\xi|_\infty$, where $|\xi|_\infty := \sup_{b'\in I^\circ}\|\xi_{b'}\|$.

(e) *Closeness.* We have $\|\Delta(b,b')\| \le \sqrt{2}\,b\,\gamma^{-3/2}\,e^{-\theta\max\{r_w+1,\ \mathrm{dist}(b,b')\}}$; consequently $\|\Delta\|, \|\Delta\|_\infty, \|\Delta^{\mathsf T}\|_\infty \le d_R(\theta)$.

(f) *Inverse.* If $4b^{1/2}d_R(\theta) \le 1$, then G is invertible and $\|G^{-1}\| \le 4b^{1/2}$; under the same condition, $S := C^{-1} - q$ is symmetric, $S = -G^{-\mathsf T}\mathsf E G^{-1}$, and $\|S\| \le 48\,b^{3/2}d_R(\theta)$. Whatever the value of $d_R(\theta)$, $\|q^{-1} - C\|_\infty \le 2\mathsf G_0 d_R(\theta) + d_R(\theta)^2$.

(g) *The ratio in white noise.* The matrix $\mathsf E$ is symmetric, and $\mathsf E(b,b') = 0$ unless $\mathrm{dist}(b,b') \le 2r_w + r_q \le \mathsf R_{\mathsf E}$; the row $\mathsf E(b,\cdot)$ depends only on $\mathrm{Ad}\,\mathsf V'$ on the bonds within $\mathsf R_{\mathsf E}$ of $y(b)$.

Further, $\mathsf{E} = -F^{-1}\Delta - \Delta^{\mathsf{T}}F^{-1} + \Delta^{\mathsf{T}}q\Delta$, and if $4b^{1/2}d_R(\theta) \le 1$, then $\|\mathsf{E}\| \le 5\,b^{1/2}d_R(\theta)$ and $\|\mathsf{E}\|_\infty \le 3L^3(2\mathsf{R}_\mathsf{E}+1)^3\|\mathsf{E}\|$.

(h) *Determinant.* We have $(\det G)^2 \det q = \det(1+\mathsf{E})$. If $4b^{1/2}d_R(\theta) \le 1$, then at $V' \in \mathcal{V}'$ we have $\det G^V > 0$ and $\log\det G^V = -\frac{1}{2}\log\det q_\ell^V + \frac{1}{2}\operatorname{tr}\log(1+\mathsf{E}^V)$; the logarithm here is that of the real symmetric matrix $1+\mathsf{E}^V = (1-\mathsf{X})^{\mathsf{T}}(1-\mathsf{X})$, where $\mathsf{X} := F^{-1}\Delta$, and the spectrum of this matrix lies in $[\frac{1}{4},\frac{9}{4}]$. If $5b^{1/2}d_R(\theta) < 1$, this logarithm is the sum of its power series.

(i) *Holomorphy.* On the set of the fields satisfying (u1$^\flat$), each entry of G, C, E, F is the uniform limit of polynomials in the entries of those matrices $\mathsf{V}'(c)$ on which the entry depends; hence it is continuous on this set and holomorphic in its interior. This holds independently of any bound on products along paths and of any bound on the curvature.

(j) *Lipschitz bound in the field, for differences growing linearly with the distance.* Let $\mathsf{V}_1, \mathsf{V}_2$ be fields satisfying (u1$^\flat$), let $b_0 \in I^\circ$ and $\epsilon_1 \ge 0$, and let $\|\operatorname{Ad}^{\pm1}_{\mathsf{V}_1(c)} - \operatorname{Ad}^{\pm1}_{\mathsf{V}_2(c)}\| \le \epsilon_1(1+L^{-1}|w - y(b_0)|_\infty)$ for each coarse bond $c = (w,\mu)$ on a transport path of a step between bonds of Y_{b_0}. Then for all b'

$$\|(G^{\mathsf{V}_1} - G^{\mathsf{V}_2})(b_0,b')\| \le \mathsf{K}_L\,\epsilon_1\,e^{-\frac{1}{2}\theta_{\mathbb{C}}\operatorname{dist}(b_0,b')}.$$

The constant K_L depends neither on R nor on ϱ nor on N.

(k) *Numbers.* Let $x > 1$ and $R = \lceil c_R \log x\rceil$, where $c_R\theta_{\mathbb{C}} \ge 128$, and let the floor condition

$$432\,L^3 b^{5/2}\gamma^{-3/2}\mathsf{g}(\theta_{\mathbb{C}})(R+3)^3 \le x^4 \qquad (\mathrm{F_{law}})$$

be satisfied. Then $4b^{1/2}d_R \le 1$, $\|\mathsf{E}\| \le x^{-60}$ and $\|S\| \le x^{-60}$.

■□

Proof. Preliminary step: the bound (W). Let w be a real function on I° with $|w(b_1) - w(b_2)| \le \theta\operatorname{dist}(b_1,b_2)$, and let Y be arbitrary. By Lem. 3.4 (r) at the fields of $\mathcal{V}'$, and by Lem. 3.4 (c) at the fields satisfying (u1$^\flat$),

$$\|e^w K_Y e^{-w}\| \le \varkappa. \qquad (\mathrm{W})$$

For $w = 0$ this gives $\|K_Y\| \le \varkappa$.

Step (a): the series. The coefficients satisfy $c_k \ge 0$ and $\sum_k c_k\varkappa^k < \infty$, so by (W) at $w = 0$ all the series converge in norm. As power series, $(\sum_k c_k\kappa^k)^2(1-\kappa) = 1$, and the series converge absolutely at $\varkappa$; hence $F^2 b(1-K) = 1$, which is the identity $F^2 q = 1$. By Lem. 3.3 (d), $K^{\mathsf{T}} = K$, and so $F^{\mathsf{T}} = F$. At real fields the matrices K and K_b are real and symmetric with $0 \le K, K_b \le 1 - \frac{\gamma}{b}$, and the series are the functions $(b(1-\kappa))^{-1/2}$ of these matrices; this gives the remaining assertions of (a).

Step (b): range and dependence. The first assertion of (b) holds by definition, and so does the assertion on $(G\xi)_b$. The formula $C(b,b') = \sum_{b''} G(b,b'')G(b',b'')^{\mathsf{T}}$ gives the assertions on C. The row $G(b,\cdot)$ is a function of the entries $q(b_1,b_2)$ with $b_i \in Y_b$, and Lem. 3.3 (c) bounds the region on which these entries depend. The row $C(b,\cdot)$ is computed from the rows $G(b',\cdot)$ with $\operatorname{dist}(b,b') \le 2r_w$.

Steps (c) and (c′): covariance. Let $\mathsf{U} := \operatorname{diag}(\operatorname{Ad}_{u(y(b))})$. This matrix is bounded and has a bounded inverse, and it commutes with every $\mathbf{1}_Y$. By Lem. 3.3 (d), $K_Y[\mathsf{V}'^u] = \mathsf{U}K_Y[\mathsf{V}']\mathsf{U}^{-1}$, and therefore $\sum_k c_k K_Y[\mathsf{V}'^u]^k = \mathsf{U}(\sum_k c_k K_Y[\mathsf{V}']^k)\mathsf{U}^{-1}$. As $\mathsf{U}^{\mathsf{T}} = \mathsf{U}^{-1}$, the same follows for G^{T}, C and E. For a single row, extend each field by the identity off the transport paths of the window. The entries of K_b are related by the same similarity, since only the products along those paths enter them.

Step (d): decay. Let $w := \theta\operatorname{dist}(\cdot,b')$. Since $w(b') = 0$, we have

$$(K_b^k)(b,b') = e^{-w(b)}(e^w K_b e^{-w})^k(b,b')$$

. By (W), and because $b^{-1/2}\sum_k c_k \varkappa^k = (\frac{2}{\gamma})^{1/2}$, this gives the bound on $\|G(b,b')\|$. The same computation with K in place of K_b gives the bound on $\|F(b,b')\|$. There are at most $3L^3$ bonds of I° to a block, and $(\frac{2}{\gamma})^{1/2} \le \frac{2b^{1/2}}{\gamma}$; with these two facts, summing the bounds on the entries yields the remaining assertions of (d).

Step (e): closeness. Fix b, let $Y = Y_b$, and let $\hat K := K_Y$, extended by 0, with the convention $\hat K^0 := 1$. As $b \in Y$, we have $G(b,\cdot) = b^{-1/2}\sum_k c_k \delta_b^{\mathsf T} \hat K^k$. Moreover $K^k - \hat K^k = \sum_{j<k} \hat K^j (K - \hat K) K^{k-1-j}$, and every row vector $u^{\mathsf T}$ supported in Y satisfies $u^{\mathsf T}(K-\hat K) = u^{\mathsf T} K \mathbf{1}_{Y^c}$. It follows that $\Delta(b,b') = b^{-1/2}\sum_k c_k \sum_{j<k} \delta_b^{\mathsf T} \mathsf A\, \mathbf{1}_{Y^c}\, \mathsf B\, \delta_{b'}$, with $\mathsf A = \hat K^j K$ and $\mathsf B = K^{k-1-j}$. Let $w_1 = \theta\,\mathrm{dist}(b,\cdot)$ and $w_2 = \theta\,\mathrm{dist}(\cdot,b')$. Then $\mathsf A(b,b_2) = \tilde{\mathsf A}(b,b_2) e^{-w_1(b_2)}$, where $\tilde{\mathsf A} = e^{-w_1}\mathsf A e^{w_1}$ satisfies $\|\tilde{\mathsf A}\| \le \varkappa^{j+1}$, and $\mathsf B(b_2,b') = e^{-w_2(b_2)}\tilde{\mathsf B}(b_2,b')$, where $\tilde{\mathsf B} = e^{w_2}\mathsf B e^{-w_2}$ satisfies $\|\tilde{\mathsf B}\| \le \varkappa^{k-1-j}$; both norm bounds come from (W). Hence the term with the indices k and j equals $(\delta_b^{\mathsf T}\tilde{\mathsf A})\,\mathsf M\,(\tilde{\mathsf B}\delta_{b'})$, where $\mathsf M$ is the diagonal matrix $\mathsf M(b_2) = \mathbf{1}_{Y^c}(b_2) e^{-\theta(\mathrm{dist}(b,b_2)+\mathrm{dist}(b_2,b'))}$. Off Y, $\mathrm{dist}(b,b_2) \ge r_w + 1$; together with the triangle inequality this yields $\|\mathsf M\| \le e^{-\theta\max\{r_w+1, \mathrm{dist}(b,b')\}}$. As $b^{-1/2}\sum_k k c_k \varkappa^k \le \sqrt{2} b \gamma^{-3/2}$, the bound on $\|\Delta(b,b')\|$ follows.

Step (e), continued: the sums. The bound on $\|\Delta(b,b')\|$ is symmetric in (b,b'). Put $r = r_w + 1$. At most $3L^3(2r+1)^3$ bonds b' of I° satisfy $\mathrm{dist}(b,b') \le r$, for a given bond b. Moreover $\sum_{|n|_\infty > r} e^{-\theta|n|_\infty} = \sum_{m\ge 1}(24(m+r)^2+2) e^{-\theta(m+r)} \le (1+r)^2 e^{-\theta r}\mathsf g(\theta)$, since $(m+r)^2 \le m^2(1+r)^2$. So the sums of $\|\Delta(b,b')\|$ over a row and over a column are $\le \sqrt{2} b\gamma^{-3/2} 3L^3[(2r+1)^3 + (1+r)^2\mathsf g(\theta)] e^{-\theta r} \le \sqrt{2}\cdot 3\cdot 2\, L^3 b\gamma^{-3/2}(R+3)^3 \mathsf g(\theta) e^{-\theta R/2}$, because $2r+1 \le R+3$ and $r \ge \frac12 R$. The bound on $\|\Delta\|$ follows by Schur's test.

Step (f): the inverse. Let $F^{-1} := b^{1/2}\sum_k \binom{1/2}{k}(-K)^k$. Then $FF^{-1} = 1$, $F^{-2} = q$, and $\|F^{-1}\| \le 2b^{1/2}$. We write $G = F(1 - F^{-1}\Delta)$; under the condition of (f), $\|F^{-1}\Delta\| \le \frac12$, so that $\|G^{-1}\| \le 4b^{1/2}$. Next, $G^{-1} - F^{-1} = G^{-1}\Delta F^{-1}$, and this difference has norm $\le 8b\, d_R(\theta)$; further, $C^{-1} - F^{-2} = G^{-\mathsf T}(G^{-1} - F^{-1}) + (G^{-1} - F^{-1})^{\mathsf T} F^{-1}$, of norm $\le 6b^{1/2}\cdot 8b\, d_R(\theta)$, and since $F^{-2} = q$ this is the bound on $\|S\|$. The identity $G^{-\mathsf T}(1+\mathsf E)G^{-1} = q$ gives the formula for S. Finally, $q^{-1} - C = F^2 - (F-\Delta)(F-\Delta^{\mathsf T}) = F\Delta^{\mathsf T} + \Delta F - \Delta\Delta^{\mathsf T}$, and (d) and (e) yield the last bound of (f). In this identity no inverse of G occurs, and $\|\cdot\|_\infty$ is submultiplicative; so the last bound holds whatever the value of $d_R(\theta)$.

Step (g): the ratio. We have $\mathsf E(b,b') + \delta_{bb'} = \sum_{b_1,b_2} G(b_1,b)^{\mathsf T} q(b_1,b_2) G(b_2,b')$, and only terms with $\mathrm{dist}(b,b_1) \le r_w$, $\mathrm{dist}(b_1,b_2) \le r_q$ and $\mathrm{dist}(b_2,b') \le r_w$ contribute; this gives the range of $\mathsf E$. As to the dependence, measured from $y(b)$: the row $G(b_1,\cdot)$ depends on bonds within $2r_w + \varrho_L + 1$, the entry $q(b_1,b_2)$ on bonds within $r_w + r_q + \varrho_L + 1$, and the row $G(b_2,\cdot)$ on bonds within $(r_w + r_q) + (r_w + \varrho_L + 1) = \mathsf R_{\mathsf E}$. The formula for $\mathsf E$ results from expanding $(F-\Delta^{\mathsf T})q(F-\Delta)$ and using $FqF = 1$ and $Fq = qF = F^{-1}$. Since $\|q\| \le b(1+\varkappa) \le 2b$, under the condition of (g) we obtain $\|\mathsf E\| \le 4b^{1/2} d_R(\theta) + 2b\, d_R(\theta)^2 \le 4.5\, b^{1/2} d_R(\theta)$. Lastly, each entry of $\mathsf E$ has norm at most $\|\mathsf E\|$, and the bound on $\|\mathsf E\|_\infty$ follows.

Step (h): the determinant. The first identity follows from $\det G^{\mathsf T} = \det G$. At real fields $\det F > 0$; moreover, the eigenvalues of the real matrix $1 - F^{-1}\Delta$ lie in the disc $|z-1| \le \frac12$, and those which are not real come in conjugate pairs, so that its determinant is > 0 and $\det G > 0$. With $\mathsf X = F^{-1}\Delta$ and $\|\mathsf X\| \le \frac12$, the relations $G = F(1-\mathsf X)$, $F^{\mathsf T} = F$ and $FqF = 1$ give $1 + \mathsf E = G^{\mathsf T} q G = (1-\mathsf X)^{\mathsf T}(1-\mathsf X)$. At real fields this matrix is real and symmetric, while the singular values of $1 - \mathsf X$ lie in $[\frac12, \frac32]$. By the spectral theorem, then, $\log(1+\mathsf E)$ exists, and $\det(1+\mathsf E) = \exp \mathrm{tr}\log(1+\mathsf E)$. As $\det G > 0$ and $\det q > 0$, we may take logarithms in the first identity. If $\|\mathsf E\| < 1$, the power series of the logarithm converges, and its sum is the logarithm just described; by (g), $\|\mathsf E\| < 1$ whenever $5b^{1/2} d_R(\theta) < 1$.

Step (i): holomorphy. We have $\mathrm{Ad}_g Y = gYg^{-1}$, and g^{-1} is the adjugate of g because $\det g = 1$; thus the entries of $\mathrm{Ad}_g^{\pm 1}$ are polynomials in the entries of g. The partial sums of the series are therefore polynomials, and the bounds $\varkappa^k$ make the convergence uniform.

Step (j): the Lipschitz bound. Let $Y = Y_{b_0}$, $K_i := K_Y[\mathsf{V}_i]$, $D := K_1 - K_2$, $t(b) := \operatorname{dist}(b_0, b)$, $w = \theta_{\mathbb{C}} t$ and $w' = \frac{1}{2}\theta_{\mathbb{C}} t$. For a step $\mathsf{s} : b_1 \to b_2$ whose path is made of the m bonds $c_1, \dots, c_m$,

$$\operatorname{Ad}_{\mathsf{V}_1(\pi)} - \operatorname{Ad}_{\mathsf{V}_2(\pi)} = \sum_i \operatorname{Ad}_{\mathsf{V}_1(c_1 \cdots c_{i-1})} [\operatorname{Ad}_{\mathsf{V}_1(c_i)} - \operatorname{Ad}_{\mathsf{V}_2(c_i)}] \operatorname{Ad}_{\mathsf{V}_2(c_{i+1} \cdots c_m)} .$$

The base of c_i lies within $t(b_1)+m$ of $y(b_0)$, and $\| \operatorname{Ad}^{\pm 1}_{\mathsf{V}(c)} \| \le e^{2\vartheta_0^\flat}$ for both fields. By the hypothesis of (j), therefore, $\|\mathsf{T}_{\mathsf{s}}^{(1)} - \mathsf{T}_{\mathsf{s}}^{(2)}\| \le m e^{2\vartheta_0^\flat m} \epsilon_1 (1+t(b_1)+m) \le \epsilon_1 (1+t(b_1))\, m(1+m) e^{2\vartheta_0^\flat m}$. In $e^{-w} D e^{w'}$ the entry (b_1, b_2) carries the factor $e^{-\theta_{\mathbb{C}} t(b_1) + \frac{1}{2}\theta_{\mathbb{C}} t(b_2)} \le e^{-\frac{1}{2}\theta_{\mathbb{C}} t(b_1)} e^{\frac{1}{2}\theta_{\mathbb{C}} m}$. Also $(1+t) e^{-\frac{1}{2}\theta_{\mathbb{C}} t} \le \frac{2}{\theta_{\mathbb{C}}} = \frac{12}{\theta^\flat}$ and $2\vartheta_0^\flat + \frac{1}{2}\theta_{\mathbb{C}} = \frac{1}{6}\theta^\flat \le \theta_1$. Schur's test, with the third bound of Lem. 3.3 (b) for rows and columns alike, now gives $\|e^{-w} D e^{w'}\| \le \frac{108 A^\flat}{b\,\theta^\flat} \epsilon_1$. Because $w(b_0) = w'(b_0) = 0$, we have $\delta_{b_0}^{\mathsf{T}} K_1^l D K_2^m e^{w'} = \delta_{b_0}^{\mathsf{T}} (e^{-w} K_1 e^{w})^l (e^{-w} D e^{w'}) (e^{-w'} K_2 e^{w'})^m$, and by (W) the norm of this row is at most $\varkappa^{l+m} \frac{108 A^\flat}{b\theta^\flat} \epsilon_1$. Further, $K_1^k - K_2^k = \sum_{l+m=k-1} K_1^l D K_2^m$. Finally, $b^{-1/2} \sum_k k c_k \varkappa^{k-1} = \sqrt{2} b \gamma^{-3/2}$ and $108\sqrt{2} \le 153$; summing first over $l+m = k-1$ and then over k, we obtain the bound of (j).

Step (k): the bound on d_R. By the choice of R and of c_R we have $e^{-\theta_{\mathbb{C}} R/2} \le x^{-64}$. With the definition of d_R and the floor ($\mathrm{F}_{\mathrm{law}}$) this gives $48 b^{3/2} d_R \le 432 L^3 b^{5/2} \gamma^{-3/2} \mathsf{g}(\theta_{\mathbb{C}}) (R+3)^3 x^{-64} \le x^{-60}$. Since $b \ge 16$, we obtain $4 b^{1/2} d_R \le 5 b^{1/2} d_R \le 48 b^{3/2} d_R \le x^{-60} \le 1$. By clause (f) we have $\|S\| \le 48 b^{3/2} d_R$, and by clause (g) we have $\|\mathsf{E}\| \le 5 b^{1/2} d_R \le 48 b^{3/2} d_R$. Hence the three bounds of clause (k) hold: $4 b^{1/2} d_R \le 1$, $\|\mathsf{E}\| \le x^{-60}$ and $\|S\| \le x^{-60}$. □

Remark 3.7 (On Thm. 3.6). (1) Neither Thm. 3.6 nor its proof depends on a symmetry of G. The covariance of $G\xi$ is $x^{-1} G G^{\mathsf{T}} = x^{-1} C$.

(2) The theorem is free of M; of N it uses only $\frac{\varrho}{L} + 2 < \frac{N}{2}$; and its clauses (a) to (j) inclusive are free of x. Along the chain, the condition $c_R \theta_{\mathbb{C}} \ge 128$ comes from $c_R = 64/\theta_D$ together with $\theta_D \le \frac{1}{2}\theta_{\mathbb{C}}$, as in (Fr2) of item C.II 2.2.

(3) If $\varrho \le R$, then $3 r_w + \varrho_L + 1 \le 2R + 1$. Indeed, $\varrho_L \le \lceil R/3 \rceil$, and $\frac{3}{2}R + \frac{1}{3}(R+2) + 1 \le 2R + 1$ for $R \ge 4$; for $R = 0, 1, 2, 3$ the quantity $3 r_w + \varrho_L + 1$ is at most $1, 2, 5, 5$, respectively, while $2R + 1$ takes the values $1, 3, 5, 7$.

(4) Under the hypotheses of clause (k), $\|C\| \le \frac{2}{\gamma}$ at G-valued fields. Indeed, there $q \ge \gamma$ and S is real symmetric, so that $C^{-1} = q + S \ge \gamma - x^{-60} \ge \frac{\gamma}{2}$, since ($\mathrm{F}_{\mathrm{law}}$) gives $x^{60} \ge x^4 \ge 432 \gamma^{-3/2} \ge \frac{2}{\gamma}$. At the complex fields on which (u1$^\flat$) holds, Lem. 3.26 (a) gives instead $\|C\| \le \frac{4}{\gamma}$. The hypotheses stated in that lemma are the clauses ($\lambda 0$) and ($\rho 5$) of item C.II 2.1, the bound $\frac{2}{\gamma} x^{-60} \le \frac{1}{2}$, and (u1$^\flat$). The proof of $\|C\| \le \frac{4}{\gamma}$ in Lem. 3.26 (a) uses, besides these hypotheses, one fact from the setting of the lemma, namely $\|q^{-1}\| \le \frac{2}{\gamma}$, which holds for the rows of item C.I 1.1 by Lem. 3.4 (c) at $w = 0$. With exponent 60, ($\lambda 0$) and ($\rho 5$) follow from Thm. 3.6 (a), (f), (k), and $x^{60} \ge x^4 \ge 432 \gamma^{-3/2} \ge \frac{4}{\gamma}$ gives $\frac{2}{\gamma} x^{-60} \le \frac{1}{2}$.

(5) The floor ($\mathrm{F}_{\mathrm{law}}$) holds on a half line of values of x. Indeed, $R + 3 \le c_R \log x + 4$, and the function $(c_R \log x + 4)^3 x^{-4}$ decreases for $\log x \ge \frac{3}{4}$, because there $3 c_R < 4 c_R \log x + 16$.

C.I 4. White noise: the exact formula and the letters of the ratio.

Definition 3.8 (The step, the chart)**.** We put $\operatorname{ang}(g) := \arccos \frac{1}{2} \operatorname{tr} g \in [0, \pi]$. For $g \ne -\mathbf{1}$ we let $\mathsf{L}(g) \in \mathfrak{g}$ be the unique Y with $e^{iY} = g$ and $|Y| < \pi$, and we set $\mathsf{L}(-\mathbf{1}) := 0$.

For a G-valued fine field U we write $\mathsf{t}_U(x) := U(\Gamma(y(x), x))$. For $c = (y, y')$ and $x \in B(y)$ we put $h_{c,x} := \mathsf{t}_U(x)\, U([x, x'])\, \mathsf{t}_U(x')^{-1}$, then $\bar\tau_c := L^{-3} \sum_{x \in B(y)} \mathsf{L}(h_{c,x} h_{c,y}^{-1})$ and $\mathsf{M}(U)(c) := e^{i\bar\tau_c} h_{c,y}$. Consequently $\mathsf{M}(U^u)(c) = u(y) \mathsf{M}(U)(c) u(y')^{-1}$.

Let $z(s) := \int_G e^{-s(1-\cos \operatorname{ang} g)} dg$. We put $k_s(g,h) = z(s)^{-1} e^{-s(1-\cos \operatorname{ang}(gh^{-1}))}$ with $s = ax$. The step is

$$\rho'(V') = \int \rho(U) \prod_c k_s(V'(c), \mathsf{M}(U)(c))\, dU,$$

where dU denotes the Haar measure. The chart is the field $\mathsf{U}(\eta; V')$ defined by $\mathsf{U}(\eta; V')_b := \mathbf{1}$ on $\mathcal{T}$, by the value $e^{i\eta_b}$ at the other bonds inside blocks, and by the value $e^{i\eta_b} V'(c)$ at a crossing bond of c.

Further, $\mathfrak{D} = \{|\eta_b| < \pi \ \forall b\}$, $\mathsf{j}(\eta_b) = (\sin|\eta_b|/|\eta_b|)^2$, $\mathsf{n}_{\mathfrak{L}} = \#I^\circ$ and $\mathsf{c}_0 = (2\pi^2)^{-\mathsf{n}_{\mathfrak{L}}}$. Finally, $\mathsf{n}_c = \operatorname{ang}(V'(c)\mathsf{M}(U)(c)^{-1})$. ∎

Lemma 3.9 (The step in the coordinates of the chart)**.** For every bounded, Borel, gauge invariant ρ,

$$\rho'(V') = \mathsf{c}_0 \int_{\mathfrak{D}} \rho(\mathsf{U}(\eta; V')) \prod_c k_s(V'(c), \mathsf{M}(\mathsf{U}(\eta; V'))(c)) \prod_{b \in I^\circ} \mathsf{j}(\eta_b)\, d\eta.$$

∎

Proof. Fix the variables on $\mathcal{T}$. For each block let $u(y) = \mathbf{1}$ at its centre and set $u(x) = U(\Gamma(y,x))$ at its other sites. The transformation $U \mapsto U^u$ sends the fixed tree variables to the identity. On every bond outside the forest it is left and right multiplication by fixed elements of G, and therefore preserves the conditional product Haar measure. The integrand is invariant under this transformation, since u is the identity at the centres. Consequently the conditional integral is the same for every choice of tree variables. Their product Haar probability integrates to 1.

On crossing bonds use right invariance of Haar measure to pass from U_b to $U_b V'(c)^{-1}$. In Pauli coordinates, SU(2) is the unit sphere S^3 with normalized surface measure, since multiplication preserves the Euclidean norm of the four real coordinates. The exponential of iY, for $Y = r\omega \cdot \sigma$ and $|\omega| = 1$, has coordinates $(\cos r, \sin r\, \omega)$. Thus $dg = (2\pi^2)^{-1} \sin^2 r \, \mathrm{d}r \, \mathrm{d}\omega = (2\pi^2)^{-1} \mathsf{j}(Y)\, \mathrm{d}^3 Y$ for $r < \pi$. The omitted element $-\mathbf{1}$ has Haar measure zero. □

Proposition 3.10 (The step in white noise)**.** In the setting of Thm. 3.6 and under its hypotheses, let $4b^{1/2} d_R \le 1$, and let V' satisfy (u1$^\flat$). Let X' be an arbitrary $\mathfrak{g}_{\mathbb{C}}$-valued function on the coarse plaquettes, let ξ be a function on I°, and put

$$h := G^{\mathsf{T}} \mathsf{g} X', \quad a_0 := Gh = C\mathsf{g}X', \quad \eta := G\xi - a_0, \quad F^\flat := EX' + \mathrm{d}^{\mathsf{V}}\eta, \quad \bar{F} := EX' - \mathrm{d}^{\mathsf{V}} a_0, \quad \mathcal{Q}(\eta) := \tfrac{1}{2} F^\flat \cdot \mathsf{B} F^\flat + \tfrac{a}{2} (Q_1^{\mathsf{V}} \eta) \cdot (Q_1^{\mathsf{V}} \eta).$$

(a) *Algebra.* The function $\mathcal{Q}$ satisfies

$$\mathcal{Q}(\eta) = \mathcal{Q}(-a_0) + \tfrac{1}{2}\, \xi \cdot \xi + \tfrac{1}{2}\, \xi \cdot \mathsf{E} \xi - \xi \cdot \mathsf{E} h, \qquad \mathcal{Q}(-a_0) = \tfrac{1}{2} \bar{F} \cdot \mathsf{B} \bar{F} + \tfrac{a}{2} (Q_1^{\mathsf{V}} a_0) \cdot (Q_1^{\mathsf{V}} a_0) = \mathsf{m}_C + \tfrac{1}{2} h \cdot \mathsf{E} h,$$

where $\mathsf{m}_C = \frac{1}{2} EX' \cdot \mathsf{B} EX' - \frac{1}{2} \mathsf{g} X' \cdot C \mathsf{g} X'$; moreover $\eta \cdot S\eta = -(\xi - h) \cdot \mathsf{E}(\xi - h)$.

(b) *Exact formula, at every $V' \in \mathcal{V}'$.* Let ρ be bounded, Borel and gauge invariant; let $n^\flat = -Q_1^V \eta$ and $U = \mathsf{U}(\eta; V')$; and let $X' = \mathsf{L}(\partial V')$, or else any other $\mathfrak{g}$-valued X' on the coarse plaquettes that does not depend on η, whatever its dependence on V' and on other variables; in particular X' can be the $X'(V', e)$ of the pairs of item E2 of Thm. 4.2. Let E_ξ denote the mean over $\xi \sim \mathcal{N}(0, x^{-1}\mathbf{1})$ on $\mathfrak{L}_{\mathfrak{g}}$. Then

$$\rho'(V') = \mathsf{c}_0\, z(s)^{-3\#\mathbb{T}'} \left(\tfrac{2\pi}{x}\right)^{\frac{3}{2}\mathsf{n}_{\mathfrak{L}}} \det G^V \ e^{-x\mathcal{Q}(-a_0)} \ E_\xi \Big[e^{-\frac{x}{2}\langle \xi, \mathsf{E}\xi\rangle + x\langle \xi, \mathsf{E}h\rangle} \ 1_{\mathfrak{D}}(\eta) \prod_b \mathsf{j}(\eta_b) \ \rho(U)\, e^{\frac{x}{2}\langle F^\flat, \mathsf{B}^V F^\flat\rangle} \prod_c e^{\frac{s}{2}|n_c^\flat|^2 - s(1-\cos \mathsf{n}_c)} \Big].$$

(c) *The letters of the ratio.* Let $M \in \mathbb{N}$ with $M \mid N$. The cells are the cubes of M^3 coarse sites that tile $\mathbb{T}'$, and $b \in \hat{\square}$ means $y(b) \in \square$. For cells $\square, \square'$ we put

$$T_{\square\square'}(\xi, \mathsf{V}') := -\frac{x}{2} \sum_{b \in \hat{\square},\, b' \in \hat{\square}'} \xi_b \cdot \mathsf{E}(b, b') \xi_{b'} + \delta_{\square\square'}\, x \sum_{b \in \hat{\square}} \xi_b \cdot (\mathsf{E}h)_b.$$

Then the ratio, by which we mean the factor $e^{-\frac{x}{2}\langle \xi, \mathsf{E}\xi\rangle + x\langle \xi, \mathsf{E}h\rangle}$ under E_ξ in the formula of clause (b), has $\sum_{\square,\square'} T_{\square\square'}$ as its exponent, and $T_{\square\square'} = 0$ unless the coarse distance from $\square$

to $\square'$ is at most R_{E}. In the following three assertions, on the dependence, the holomorphy and the covariance of $T_{\square\square'}$, we take X' to be given by $X'_P = -i \log \partial \mathsf{V}'(P)$, at fields with $|\partial \mathsf{V}'(P) - \mathbf{1}| < 1$. The function $T_{\square\square'}$ depends only on the values of ξ on $\hat{\square} \cup \hat{\square}'$ and on those of V' on the bonds within distance $R_t := 4r_w + 5\varrho_L + 5$ of $\square$. It is holomorphic in V' on the interior of the region where (u1$^\flat$) holds and $|\partial \mathsf{V}'(P) - \mathbf{1}| < 1$. For G-valued u we have $T_{\square\square'}(\mathrm{Ad}_u\, \xi, \mathsf{V}'^u) = T_{\square\square'}(\xi, \mathsf{V}')$. Finally, let ξ be real and put $|\xi_{\hat{\square}}|^2 := \sum_{b \in \hat{\square}} |\xi_b|^2$. Then, with the supremum taken over the P on which $(\mathsf{E}h)_b$, $b \in \hat{\square}$, depends,

$$|T_{\square\square'}| \le \tfrac{x}{4} \|\mathsf{E}\| (|\xi_{\hat{\square}}|^2 + |\xi_{\hat{\square}'}|^2) + \delta_{\square\square'}\, x\, \|\mathsf{E}\|_\infty\, \mathsf{G}_0\, b \sup \|X'_P\| \sum_{b \in \hat{\square}} |\xi_b|.$$

■

Proof. Step (a): the algebra. Since B and q are symmetric for $Y \cdot Y'$, we have $\mathcal{Q}(\eta) = \frac{1}{2} EX' \cdot \mathsf{B} EX' + \mathsf{g} X' \cdot \eta + \frac{1}{2} \eta \cdot q\eta$. As $\eta = G(\xi - h)$, it follows that $\eta \cdot q\eta = (\xi - h) \cdot (1 + \mathsf{E})(\xi - h)$ and $\mathsf{g}X' \cdot \eta = h \cdot (\xi - h)$. Next, $\frac{1}{2}(\xi - h) \cdot (\xi - h) + h \cdot (\xi - h) = \frac{1}{2} \xi \cdot \xi - \frac{1}{2} h \cdot h$, where $h \cdot h = \mathsf{g}X' \cdot C \mathsf{g} X'$. Therefore $\mathcal{Q}(\eta) = \mathsf{m}_C + \frac{1}{2} \xi \cdot \xi + \frac{1}{2}(\xi - h) \cdot \mathsf{E}(\xi - h)$, and at $\xi = 0$ we have $\eta = -a_0$. Finally, $S = -G^{-\mathsf{T}} \mathsf{E} G^{-1}$.

Step (b): the exact formula. In the formula of Lem. 3.9 we multiply and divide by $e^{-x\mathcal{Q}(\eta)}$, noting that $x \frac{a}{2} |Q_1^V \eta|^2 = \frac{s}{2} |n^\flat|^2$. We then substitute $\eta = G\xi - a_0$, which is a bijection of $\mathfrak{L}_{\mathfrak{g}}$ with Jacobian $\det G^V > 0$ (Thm. 3.6 (f), (h)), and we use $\int e^{-\frac{x}{2}|\xi|^2} d\xi = (2\pi/x)^{\frac{3}{2}\mathsf{n}_{\mathfrak{L}}}$.

The argument depends on X' through two facts alone. First, $a_0 = C\mathsf{g}X'$ is a fixed vector of $\mathfrak{L}_{\mathfrak{g}}$, because at $V' \in \mathcal{V}'$ the matrices C and g^V are real (Thm. 3.6 (a)) while X' is $\mathfrak{g}$-valued and free of η; hence $\xi \mapsto G\xi - a_0$ is the bijection just used. Second, the argument uses clause (a), which is stated for every X'.

Step (c): the letters of the ratio. The range of E, and the variables on which E depends, are given by Thm. 3.6 (g). We have $h_b = \sum_{b_1} G(b_1, b)^{\mathsf{T}} (\mathsf{g}X')_{b_1}$ with $\mathrm{dist}(b, b_1) \le r_w$. Here $G(b_1, \cdot)$ depends on the field within $2r_w + \varrho_L + 1$ of $y(b)$, and $(\mathsf{g}X')_{b_1}$ depends on the field within $r_w + 2\varrho_L + 2$ of $y(b)$, by Lem. 3.3 (e) and since $X'_P = -i \log \partial \mathsf{V}'(P)$ depends on V' through the bonds of P. Hence $(\mathsf{E}h)_b$ depends on the field within $\mathsf{R}_{\mathsf{E}} + 2r_w + 2\varrho_L + 2 = R_t$ of $y(b)$. Holomorphy and covariance follow from Thm. 3.6 (i), (c). The bound follows from the Cauchy–Schwarz inequality; in it, $\|h_b\| \le \mathsf{G}_0 \sup \|(\mathsf{g}X')_{b_1}\| \le \mathsf{G}_0 b \sup \|X'_P\|$, by Lem. 3.3 (e) and $\|\mathrm{Ad}_{\mathsf{V}'(\pi)}\| \le e^{2\vartheta_0^\flat |\pi|}$. □

Part C.II. (S-cen$_k$) and (S-loc$_k$) at the law of windows.

C.II 1. Objects.

Remark 3.11. The k of $\mathsf{D}_k(Y, \cdot)$, $S_k(Y; \cdot)$, (S-loc$_k$), (S-cen$_k$), (Law$_k$) is part of the name of each of them and is not the number of a level. In Part C.II the letter b denotes either a bond or the number given in item C.II 1.3, the letter G denotes either the matrix of the law or the group, and the context shows which of the two senses is meant.

Definition 3.12 (Item C.II 1.1: lattices, transports and norms)**.** The objects of this item are those of item C.I 1.1. Throughout Part C.II, $L \ge 3$ is odd, the number h_L is computed from it as in item C.I 1.1, and $aL \ge 18$. We fix $M = L^{m'}$ with $m' \ge 1$; then $M \ge L \ge 3$, and M is odd, so that every cell has a centre site. The number N is even, and $M \mid N$.

The letter N stands for the coarse side of the present step, and the condition (Tor) below bears on this number. In §2.2 the same number is written $N' = NL^{K-k-1}$, and there N denotes the side of the last torus.

We use the tori, base sites, distance, combs, transport convention and norms of item C.I 1.1. The coarse coboundary is d'. With $\mathsf{g}_I(\lambda) := 3L^3 \mathsf{g}(\lambda)$, the count of bonds per block gives

$$\sum_{j \in I^\circ} \mathrm{dist}(i, j) e^{-\lambda \mathrm{dist}(i,j)} \le \lambda^{-1} \mathsf{g}_I(\lambda/2),$$

since $te^{-\lambda t/2} \le 1/\lambda$ for $t \ge 0$.

We also use $\|\operatorname{Ad}_g^{\pm 1} - 1\| \le 2|g||g-\mathbf{1}|$ for $g \in \mathrm{SL}(2,\mathbb{C})$. For the positive sign, write $gYg^{-1} - Y = [(g-\mathbf{1})Y - Y(g-\mathbf{1})]g^{-1}$. For the negative sign, write $g^{-1}Yg - Y = g^{-1}[Y(g-\mathbf{1}) - (g-\mathbf{1})Y]$. The norm bound follows from $|g^{-1}| = |g|$. The map Ad_g preserves trace and the bilinear form $Y \cdot Y'$.

The first bound in (E) of item C.I 1.1 holds for every $S \ge 0$. For $S \le 1$ we also use its bound $0.72S^2$.

From now on $x > Le$ is a parameter and x' denotes the number $x' = x/L$; the fine site $x' = x + Le_\mu$ of item C.I 1.1 appears from here on only within the symbol $[x, x']$. We put $p_0(x) = A_0(\log x)^{\mathfrak{p}_0}$, where $A_0 > 0$ and $\mathfrak{p}_0 > 2$, and $\varepsilon = p_0(x)x^{-1/2}$; further $p_0' = p_0(x')$ and $\varepsilon' = p_0' x'^{-1/2}$.

We also set $R = R(x) = \varrho(x) = \lceil c_R \log x \rceil$, $r(x) = \lceil \frac{20 c_R}{M} \log x \rceil$ with $r' = r(x')$, $\ell_w(x) = 6Mr(x) + 12$, and $\mathsf{b}(x) = 2 - \frac{1}{\log x}$.

A cell $\square$ consists of M^3 coarse sites, and $\hat{\square}$ is the set of fine sites in the blocks of its sites; we assign each object to the cell of its base site. For a set Y of cells, Y^{+j} is the set of those cells whose sup distance to some cell of Y, counted in cells, is $\le j$, and we write $Y^\sim = Y^{+1}$ and $Y^\approx = Y^{+r'}$.

The condition (Tor) is $N \ge 16Mr(x)$. The number κ satisfies $\kappa \ge 6$, and $\kappa^\sharp = 10(\kappa + 4)$.

We write $\mathcal{V}'$ for the set of G-valued fields on $\mathbb{T}'$. For V' in this set we put $\phi_P' = \operatorname{ang} \partial V'(P)$, where $\operatorname{ang} g = \arccos \frac{1}{2} \operatorname{tr} g$, and we say that V' is *tame on* Y if $\phi_P' \le \varepsilon'$ for the plaquettes P of Y.

For a set Y of cells, $\mathcal{U}_Y(x')$ consists of the maps V that send the oriented coarse bonds of $Y^\approx$ into $\mathrm{SL}(2,\mathbb{C})$, satisfy $\mathsf{V}(\bar{c}) = \mathsf{V}(c)^{-1}$ and obey (u1), (u2) and (u3) below; a bond, a path or a plaquette of $Y^\approx$ is one whose sites all lie in $Y^\approx$.

(u1) $|\mathsf{V}(c)| < 1 + \vartheta(x')$ at every bond c, where ϑ denotes the function of item C.II 1.3, built with $\vartheta_0^\flat$.

(u2) $|\mathsf{V}(\pi)| < \mathsf{b}(x')$ for each path π in $Y^\approx$ with at most $\ell_w(x')$ bonds.

(u3) $|\partial \mathsf{V}(P) - \mathbf{1}| < 2\varepsilon'$ at each plaquette of $Y^\approx$.

Finally we set

$$\varrho_L := \lceil \varrho(x)/L \rceil, \qquad R_2 = 2R + 2, \qquad R_3 = R_2 + \varrho_L + 3, \qquad r_u := \lceil R_2/M \rceil.$$

■

We use the graph of cells with adjacency at sup distance one. Count 1 in §2.1 bounds the number of connected sets of size n through a cell by $(26e)^{n-1}$.

Definition 3.13 (Item C.II 1.2: rows, the form and its steps)**.** The rows, the form and its steps are those of items C.I 1.1 and C.I 1.3, where H_y, $\langle \cdot, \cdot \rangle$, $|y - p|$ and $\Pi_{\mathfrak{L}}$ are defined. As in Def. 2.22, a row ℓ is of range $\le \varrho$ if $\ell_y(p) = 0$ whenever $|y - p| > \varrho$. Rows ℓ on $\mathbb{T}$ obey $|\ell_y(p)| \le Ke^{-\lambda|y-p|}$, where $0 < \lambda \le 1$ and $K \ge 0$, and they are of range $\le \varrho(x)$; moreover $\varrho_L + 2 < \frac{N}{2}$.

We use the form and the two step expansions of item C.I 1.3 and Lem. 3.3(e). The transport of a step s from b stays within $r_q = 2\varrho_L + 2$ of $y(b)$: its sites lie within $\varrho_L + 1$ of one of the endpoints, and for a step of kind (R) the second leg from $y(y)$ to $y(p')$ lies within ϱ_L of $y(y)$, which is within $\varrho_L + 1$ of $y(b)$. The last leg has length at most one. Thus the row $q(b, \cdot)$ depends on the coarse bonds in that neighbourhood. The paths of the g-steps from b stay within $2\varrho_L + 1$ of $y(b)$ by Lem. 3.3(e). ■

Definition 3.14 (Item C.II 1.3: the numbers)**.** Use the numbers of item C.I 1.2. In addition, set

$$\mathsf{N}_{\mathsf{B}} = 1 + 3K^2 \mathsf{g}(\lambda)^2, \qquad \mu = \frac{\gamma \theta_2}{4b \log \Lambda}, \qquad \kappa_+ = \frac{M}{40 c_R}.$$

We use $\vartheta(x) = \vartheta_0^\flat(1 - 1/\log x)$ and $\mathsf{g}_x = \vartheta_0^\flat \log L/[4(\log x)^2]$.

For ℓ in the set $\mathcal{R}(b_*, K_*, \lambda_*, \frac{1}{9})$, the empty rows included, all these numbers are evaluated at (K_*, λ_*), that is, with $K = K_*$ and $\lambda = \lambda_*$, and not with the constants of the bound that the rows at hand obey.

We have $\vartheta_0^\flat \le 10^{-7}$, as in item C.I 1.2, because $\vartheta_0^\flat \le \frac{\gamma}{48A^\flat}$, $\gamma \le \frac{1}{5913}$ and $A^\flat \ge 32e^2$; a fortiori $\vartheta_0^\flat \le 10^{-4}$. Moreover $\theta_{\mathbb{C}} \le \frac{1}{6}\theta_1 \le \frac{1}{6}$.

Throughout, we require $c_R\theta_{\mathbb{C}} \ge 128$. If the numbers of this item are evaluated at $K = K_*$ and $\lambda = \lambda_*$, this inequality follows from the relations $\theta_D \le \frac{1}{2}\theta_{\mathbb{C}}$ and $c_R = 64/\theta_D$ in (Fr2) of item C.II 2.2. For rows of item C.II 1.2 with other values of K and λ, it is a requirement of this item and not a consequence of these relations. As $R \ge c_R \log x$, it follows that $e^{-\theta_{\mathbb{C}} R} \le x^{-128}$. ■

Definition 3.15 (Item C.II 1.4: fields on a box)**.** The data are the following: a box B of coarse sites, that is, a product of three intervals embedded in the torus, whose bonds, paths, plaquettes and cubes are those with all their sites in B; a site $o \in \mathsf{B}$; a number $0 < \hat\varepsilon \le \frac{1}{40}$; and an $\mathrm{SL}(2,\mathbb{C})$ field V' on the bonds of B.

We say that V' has (h) at $\hat\varepsilon$ if it satisfies (h1), (h2) and (h3).

(h1) $|\mathsf{V}'(c)| \le 1 + \vartheta_0^\flat$ at every bond c of B.
(h2) $|\mathsf{V}'(\pi)| \le 2$ for each path π in B with at most $\ell_w(x')$ bonds.
(h3) $|\partial\mathsf{V}'(P) - \mathbf{1}| \le 2\hat\varepsilon$ at every plaquette P of B.

We also use the two strict bounds below.

(e1) $|\mathsf{V}'(c)| < 1 + \vartheta(x') + \mathsf{g}_x$ at every bond c of B.
(e2) $|\mathsf{V}'(\pi)| < \mathsf{b}(x') + \mathsf{g}_x$ for each path π in B with at most $\ell_w(x')$ bonds.

These bounds give (h1) and (h2), with the room $\vartheta_0^\flat - \vartheta(x') - \mathsf{g}_x \ge \frac{3\vartheta_0^\flat}{4\log x}$ and $2 - \mathsf{b}(x') - \mathsf{g}_x \ge \frac{3}{4\log x}$. The room follows from the definitions of ϑ and b, since $\mathsf{g}_x \le \frac{\vartheta_0^\flat}{4\log x}$ and $\log x' \le \log x$.

We put $X'_P := -i\log\partial\mathsf{V}'(P)$, defined by the power series; under (h3) it satisfies $|X'_P| \le 2.06\hat\varepsilon$. With $\mathsf{k}(y) := \mathsf{V}'(\Gamma(o,y))$ we define $\mathsf{W}(c) := \mathsf{k}(c_-)\mathsf{V}'(c)\mathsf{k}(c_+)^{-1}$ and $\tilde X_P := \mathrm{Ad}_{\mathsf{k}(y_P)} X'_P = T(o,y_P)X'_P$. If π is a path in B from y to y'', then $\mathsf{W}(\pi) = \mathsf{k}(y)\mathsf{V}'(\pi)\mathsf{k}(y'')^{-1}$.

Every field on B is extended by $\mathbf{1}$ on the remaining bonds of $\mathbb{T}'$, and we set $X'_P := 0$ for $P \not\subset \mathsf{B}$. ■

Definition 3.16 (Item C.II 1.5: sub-forms)**.** A *family* $\mathfrak{S}$ consists of a set $I_{\mathfrak{S}} \subset I^\circ$ and of steps joining bonds of $I_{\mathfrak{S}}$, the set of steps being closed under reversal. On $\ell^2(I_{\mathfrak{S}};\mathbb{C}^3)$ we set $q^{\mathsf{V}}_{\mathfrak{S}} := \sum_{\mathsf{s}\in\mathfrak{S}} q_{\mathsf{s}}\mathsf{T}_{\mathsf{s}}$, and we put $K^{\mathsf{V}}_{\mathfrak{S}}(\zeta) := 1 - b^{-1}\sum_{\mathsf{s}\in\mathfrak{S}} \zeta^{|\mathsf{s}|} q_{\mathsf{s}}\mathsf{T}_{\mathsf{s}}$ and $K^{\mathsf{V}}_{\mathfrak{S}} = K^{\mathsf{V}}_{\mathfrak{S}}(1)$. We call $\mathfrak{S}$ *coercive* when $\gamma \le q^{V_0}_{\mathfrak{S}}$ for all G-valued V_0.

Families of two kinds occur, (w) and (c).

(w) Windows: $I_{\mathfrak{S}} = \mathsf{Y} \subset I^\circ$, with all the steps between bonds of Y; for them $q_{\mathfrak{S}} = \mathbf{1}_{\mathsf{Y}} q_\ell \mathbf{1}_{\mathsf{Y}} =: q_{\mathsf{Y}}$.
(c) Sets of cells: given a set Y of cells, we let I_{Y} consist of the bonds of I° based in $\hat{\mathsf{Y}}$. To a step s of kind (P) or (Q) we attach $\mathsf{c}(\mathsf{s}) := \{$the cells of $y(b)$, $y(b')\}$, and to a step of kind (R) we attach these cells together with the cells met by $\Gamma(y(y), y(p))$ and by $\Gamma(y(y), y(p'))$. The family consists of the steps s with $\mathsf{c}(\mathsf{s}) \subset \mathsf{Y}$, and its form is

$$q_{[\mathsf{Y}]} := q_{\mathfrak{S}} = \mathbf{1}_{\mathsf{Y}} q_\emptyset \mathbf{1}_{\mathsf{Y}} + \mathbf{1}_{\mathsf{Y}} \mathrm{d}^{V*} \ell^{[\mathsf{Y}]V*} \ell^{[\mathsf{Y}]V} \mathrm{d}^V \mathbf{1}_{\mathsf{Y}},$$

where $\ell_y^{[\mathsf{Y}]}(p) = \ell_y(p) 1\{$the cells met by $\Gamma(y(y), y(p))$ lie in $\mathsf{Y}\}$.

Both kinds are coercive by Lem. 3.24: a window yields a compression of a form $\ge \gamma$, and a set of cells yields a compression plus a square.

For every step, $\mathsf{c}(\mathsf{s}) \subset \{$cells of the sites of $\pi_{\mathsf{s}}\}$, and $\mathsf{c}(\mathsf{s})$ is connected, since two coarse sites at sup distance ≤ 1 always lie in one cell or in two neighbouring cells, whatever $M \ge 1$ is. ■

C.II 2.1. The law: what Part C.II uses of (Law_k).

Definition 3.17 (The law and the sets of fields)**.** In Part C.II, $(G^{\mathsf{V}}, C^{\mathsf{V}})$ is the law of windows of Part C.I, given by Def. 3.5, taken at $R = R(x)$ and for rows of range $\le \varrho(x) = R$, in the setting where $c_R\theta_{\mathbb{C}} \ge 128$ and the floor $(\mathrm{F}_{\mathrm{law}})$ of Thm. 3.6 (k) hold.

We write $\mathfrak{V}$ for the set of $\mathrm{SL}(2,\mathbb{C})$ fields on $\mathbb{T}'$ that satisfy (u1$^\flat$) of item C.I 1.1, namely $|\mathsf{V}'(c)| \le 1 + \vartheta_0^\flat$ for all bonds c, and $\mathfrak{V}^\circ$ for the set given by the same bounds with strict inequalities. Then $\mathcal{V}' \subset \mathfrak{V}$.

In Part C.II, ϱ_L stands for $\lceil \varrho(x)/L \rceil$; it is at least the ϱ_L of Part C.I, and every radius below increases with it.

The proofs of Part C.II rest on the law through the clauses listed in the proposition below and through nothing else. Each of these clauses holds at every field of $\mathfrak{V}$, complex fields included, by the clause of Thm. 3.6 named at its end, applied with $\theta = \theta_{\mathbb{C}}$. ∎

Proposition 3.18 (Item C.II 2.1: What Part C.II uses of the law)**.** The law has the following properties.

- (λ0) G^{V} and C^{V} are defined at every field of $\mathfrak{V}$ (by the part of Thm. 3.6 that precedes its clauses, and by its clause (a)).
- (λ1) $G^{\mathsf{V}}(b,b') = 0$ whenever $\mathrm{dist}(b,b') > \frac{1}{2}R$, and $(G\xi)_b$ depends only on the values of ξ on Y_b (Thm. 3.6 (b)).
- (λ2) $G^{\mathsf{V}}(b,\cdot)$ is determined by the restriction of V' to the coarse bonds within $r_w + \varrho_L + 1$ of $y(b)$, and $r_w + \varrho_L + 1 \le R_3$ (Thm. 3.6 (b)).
- (λ3) $G[\mathsf{V}'^u] = \mathrm{Ad}_u\, G[\mathsf{V}']\, \mathrm{Ad}_u^{-1}$ for $u : \mathbb{T}' \to G$ (Thm. 3.6 (c)).
- (λ4) G^{V} has entries that are holomorphic on $\mathfrak{V}^\circ$ and real at G-valued fields (Thm. 3.6 (i) and (a)).
- (λ5) $\|(G^{\mathsf{V}}\xi)_b\| \le \mathsf{G}_0 \sup_{Y_b} \|\xi_{b'}\| \le C_\varphi^\flat |\xi|_\infty$ (Thm. 3.6 (d)).
- (λ6) = (ρ5) C^{V} is invertible, and $\|(C^{\mathsf{V}})^{-1} - q_\ell^{\mathsf{V}}\| \le x^{-\mathsf{n}}$ in the operator norm on $\ell^2(I^\circ;\mathbb{C}^3)$, with $\mathsf{n} = 60$ (Thm. 3.6 (f) and (k)).
- (λ7) $C = GG^{\mathsf{T}}$ (by Def. 3.5).
- (ρ1) $C^{\mathsf{V}}(b,b') = 0$ whenever $\mathrm{dist}(b,b') > R$ (Thm. 3.6 (b), since $2r_w \le R$).
- (ρ2) $C^{\mathsf{V}}(b,\cdot)$ is determined by the restriction of V' to the coarse bonds within $R_2 - 1$ of $y(b)$ (Thm. 3.6 (b), since $3r_w + \varrho_L + 1 \le 2R + 1$ by Remark (3) after Thm. 3.6).
- (ρ3) $C[\mathsf{V}'^u] = \mathrm{Ad}_u\, C[\mathsf{V}']\, \mathrm{Ad}_u^{-1}$ for G-valued u (Thm. 3.6 (c)).
- (ρ4) C^{V} has entries that are holomorphic on $\mathfrak{V}^\circ$ (Thm. 3.6 (i)).
- (ρ6) $C^{\mathsf{T}} = C$, and at G-valued fields the matrix C is real with $\det C = (\det G)^2 > 0$ (Thm. 3.6 (b), (a) and (f)).

∎

Remark 3.19 (Item C.II 2.1: dependence on n, and notation)*.* In each statement below, the exponent n of (ρ5) is a letter, and a law is any map $\mathsf{V}' \mapsto (G^{\mathsf{V}}, C^{\mathsf{V}})$ on $\mathfrak{V}$ that satisfies the clauses on which the statement rests. These statements hold for every such law whose exponent lies in the range $2 \le \mathsf{n} \le 127$ and, in a statement that assumes (S2), also obeys $\mathsf{n} \ge 4$. The law of windows fixed at the start of this item is among these laws, with $\mathsf{n} = 60$. The proofs from Lem. 3.26 to Cor. 3.38 depend on n only through this range. The passage from the law to a window, in Lem. 3.26 (c), needs $\mathsf{n} \le 128$: there the term of the window carries the factor $e^{-\theta_{\mathbb{C}} R} \le x^{-128}$ of item C.II 1.3, and $x^{-128} \le x^{-\mathsf{n}}$. This condition follows from the range. The proofs of Thm. 3.42 and Cor. 3.43 need, in addition, $\mathsf{n} \ge 4$, which holds in the setting (S2) of item C.II 5, under "Floors". Indeed, the geometric sum that occurs in Thm. 3.42 (c) is controlled through $x^{-(\mathsf{n}-2)} \le \frac{1}{2}$, and Cor. 3.43 rests on $x^{-(\mathsf{n}-3)} \le \varepsilon'$.

We write $S^{\mathsf{V}} := (C^{\mathsf{V}})^{-1} - q_\ell^{\mathsf{V}}$, $a_0 := C^{\mathsf{V}} \mathsf{g}^{\mathsf{V}} X'$ and $\mathsf{u}^{\mathsf{V}} := 1 - C^{\mathsf{V}} q_\ell^{\mathsf{V}}$.

C.II 2.2. (Fr)*: the free scalar model at a level with rows.*

Proposition 3.20 (The free scalar model at a level with rows)**.** All kernels here are real and scalar, and none of them depends on a field. We write $q = q_\ell^{\mathbf{1}}$, $\mathsf{g} = \Pi_{\mathfrak{L}} \mathrm{d}^* \mathsf{B} E$, $H_\circ = E - \mathrm{d} q^{-1} \mathsf{g}$ and $\mathsf{N}_\circ = Q_1 q^{-1} \mathsf{g}$, as in Prop. 2.27 (c). The kernels $\mathsf{G}, \mathsf{G}_Z, \mathsf{G}_{\mathsf{n}}, H^\sharp, \mathsf{N}^\sharp, \mathsf{T}$, the next rows ℓ' with B'

and the cut rows ℓ'^{c} with B'^{c} are those of Props. 2.27 and 2.28, Defs. 2.35 and 2.37, Lem. 2.38 and Def. 2.39, the weighted norm $\|\cdot\|_\theta$ is that of Def. 2.20, and the constants N_Z, $\mathsf{N}_\sharp$ are those defined in the remark after this proposition. We consider the following hypotheses.

(H1) $L \geq L_A$, and L is odd.
(H2) $aL \geq 3/c_L$, where $c_L = \frac{L^2-1}{6L^2}$; in particular $aL > 18$.
(H3) N is even.
(H4) $x' = x_{k+1} \geq x_\flat$. Together with (X12) of §2.2, this keeps the rows $\ell_{k+1} = ((\ell_k)')^{\mathrm{c}}$ of the chain, cut at $\varrho(x_{k+1})$, in $\mathcal{R}(b_*, K_*, \lambda_*, \frac{1}{9})$.

Clause (Fr1) below holds for all rows ℓ. Under (H1), (H2) and (H3), clauses (Fr2) and (Fr3) below hold for every $\ell \in \mathcal{R}(b_*, K_*, \lambda_*, \frac{1}{9})$ and at every cut; under the same three hypotheses, (H4) and (X12) keep the rows of the chain in this set, and (H4) is used for that alone. The relations $C_D \geq 4$, $0 < \theta_D \leq \frac{1}{2}\theta_{\mathbb{C}}$ and $c_R = 64/\theta_D$ in (Fr2) are fixed by the choice of the letters at (O6) in item 3 of §2.1; they are relations between letters of (O6), with $\theta_{\mathbb{C}}$ taken at (K_*, λ_*), and they involve no rows, no torus and none of (H1) to (H3).

(Fr1) $H_\circ = H^\sharp + \mathsf{G}\mathrm{d}'$ and $\mathsf{N}_\circ = \mathsf{N}^\sharp + \mathsf{G}_{\mathsf{n}}\mathrm{d}'$, with $\mathsf{G}_{\mathsf{n}} = a^{-1}\mathrm{d}'^*\mathsf{G}_Z$. This is Lem. 2.38 (c), which is formulated and proved there for rows ℓ.
(Fr2) $C_D \geq 4$, $0 < \theta_D \leq \frac{1}{2}\theta_{\mathbb{C}}$ and $c_R = 64/\theta_D$, and each of $\mathsf{G}, \mathsf{G}_Z, H^\sharp, \mathsf{N}^\sharp, \mathsf{T}, \ell'$ satisfies $\|\cdot\|_{\theta_D} \leq C_D$ on every torus. This comes from (Dec)(C_D, θ_D) of Def. 2.41, applied to the pair (ℓ, ℓ'), that is, from Prop. 2.62 with $\theta_D := \frac{1}{2}\min\{\lambda_2^*, \theta_{\mathbb{C}}, \theta_J\}$, which amounts to putting $\theta_{\mathbb{C}}$ in the place of θ_q. The constant C_D is the letter C_D of (O6) in item 3 of §2.1; it is at least the value at this θ_D of the expression that Prop. 2.62 displays for its own C_D, a maximum built from A_q, C_0, C_R and C_ℓ, and in Thm. 2.65 that expression is item (6) of the order of the constants. The bound (Dec)(C, θ_D) holds for every C at least that expression, and every constant below increases with C_D. Since $\mathsf{G}_{\mathsf{n}} = a^{-1}\mathrm{d}'^*\mathsf{G}_Z$ and $aL \geq 18$, it follows that $\|\mathsf{G}_{\mathsf{n}}\|_{\theta_D} \leq \frac{L}{18} \cdot 4e\, C_D \leq LC_D$; moreover $e^{-\theta_D R(x)} \leq x^{-64}$.
(Fr3) $L^2\|H^\sharp - \mathsf{T}(\mathsf{B}' - \mathsf{B}'^{\mathrm{c}})\|_\infty \leq \mathsf{N}_\sharp$ and $\|\mathsf{B}'^{\mathrm{c}}\|_\infty \leq \mathsf{N}_Z$, where $\mathsf{N}^\sharp = (aL)^{-1}\mathrm{d}'^*\mathsf{B}'$. This rests on (N′) of Def. 2.42, taken at $m \equiv 1$ and with the objects m, ζ', Z, Ξ, ν, ω, σ that are defined there just before (N′).

■

Remark 3.21 (Source and hypotheses). The objects of this item include the next rows ℓ' and $\mathsf{B}' = 1 + \ell'^{\mathsf{T}}\ell'$, and the cut rows ℓ'^{c}, namely ℓ' cut at $\varrho(x')$, together with $\ell'^{>} := \ell' - \ell'^{\mathrm{c}}$ and $\mathsf{B}'^{\mathrm{c}} = 1 + \ell'^{\mathrm{c}\mathsf{T}}\ell'^{\mathrm{c}}$. The statements and definitions named in the proposition build on Defs. 2.26 and 2.30 and on Lem. 2.31.

The weighted norm of Def. 2.20 is

$$\|K\|_\theta = \max\{\sup_i \sum_j |K(i,j)|e^{\theta \mathrm{dist}(i,j)},\ \sup_j \sum_i |K(i,j)|e^{\theta \mathrm{dist}(i,j)}\}$$

and $\|K\|_0$ denotes the same expression at $\theta = 0$. The sets $\mathcal{R}(b, K, \lambda, \tau)$ are those of §2.2, from Def. 2.46 to Def. 2.49. On $\mathfrak{g}$-valued cochains a scalar kernel K operates as $K \otimes 1$ and keeps its norms, so these norms are free of the group.

Clause (Fr1) is algebra. The proofs of (Fr2) and (Fr3) for $\ell \in \mathcal{R}(b_*, K_*, \lambda_*, \frac{1}{9})$ are given in §2.2, in Lems. 2.51 to 2.54, Prop. 2.55, Thm. 2.57, Lems. 2.58 and 2.59, Thm. 2.60, Prop. 2.62 and Thm. 2.64, and in Props. 2.27, 2.28, 2.33 and 2.40.

In Thm. 2.65 the conclusion is stated for the pairs $(\ell_k, (\ell_k)')$ of the chain. The two clauses hold, however, for every ℓ of the set, as follows. Clause (Fr2) is obtained in Step H13 of the proof of Thm. 2.65, namely from Prop. 2.62 and its proof, taken with $\theta_{\mathbb{C}}$ in place of θ_q, together with Step H16 (i), (ii) of that proof. Clause (Fr3) follows from the argument of Step H14 of the same proof; this argument uses of ℓ only that it belongs to the set, and it goes as follows.

Thm. 2.57 (a) applied with $K = K_*$ and $\tau \le \frac{1}{9}$, which $L \ge L_A$ allows, yields $|\ell'_{y'}(P)| \le K_* e^{-\lambda_* \mathrm{dist}(y',P)}$; consequently, at every cut, $\|\mathsf{B}'^{\mathrm{c}}\|_\infty \le 1 + K_* \mathsf{g}(\lambda_*) \cdot 3K_* \mathsf{g}(\lambda_*) = \mathsf{N}_Z$. Next, Thm. 2.60 (d) with $(b, K, \lambda, \lambda^\circ) = (b_*, K_*, \lambda_*, \lambda_*)$, as in Def. 2.61, yields $L^2 \|\mathsf{T}\|_\infty \le c_1^* \mathsf{I}_L$; here $\lambda' = \lambda_*$, by part (a) of the proof of Thm. 2.57. Finally, the identity $H^\sharp = \mathsf{T}\mathsf{B}'$ gives $H^\sharp - \mathsf{T}(\mathsf{B}' - \mathsf{B}'^{\mathrm{c}}) = \mathsf{T}\mathsf{B}'^{\mathrm{c}}$, and the two bounds just found give, for this kernel, $L^2 \| \cdot \|_\infty \le c_1^* \mathsf{I}_L \mathsf{N}_Z \le \mathsf{N}_\sharp$, where $\mathsf{N}_\sharp = c_1^* \mathsf{I}_L \mathsf{N}_Z \mathsf{P}_\rho$, with $\mathsf{P}_\rho \ge 1$, is the constant that Thm. 2.65 fixes in items (3), (4) of the order of the constants.

Each statement of Part C.II that rests on (Fr2) or on (Fr3) carries with it the hypotheses (H1) to (H4).

Where $\ell \in \mathcal{R}(b_*, K_*, \lambda_*, \frac{1}{9})$ is part of the data, as in (SD), the hypotheses (H1) to (H3) suffice: Prop. 2.62 requires nothing more, and the argument for (Fr3) given above rests on membership alone. For the empty rows membership holds: $\emptyset \in \mathcal{R}(1, 0, \lambda, 0)$, by Def. 2.48.

The constants $C_D, \theta_D, \mathsf{N}_\sharp, \mathsf{N}_Z$ depend neither on k nor on N nor on x. Moreover $\theta_D \le \frac{1}{2}\theta_{\mathbb{C}} \le \frac{1}{8}\lambda_*$, a bound on which Thm. 5.63 rests; it follows from $\theta_{\mathbb{C}} \le \frac{\gamma}{12A^\flat}$, $A^\flat \ge 384 e^8 K_*^2$, $K_*^2 = 9/\gamma_*$, $\lambda_* = \gamma_*^{1/2}/44$ and $\gamma, \gamma_* \le 1$.

The data (SD) of item 1 of §2.1, namely the four conclusions that Thm. 2.64 draws for the pair (ℓ, ℓ'), imply (Fr1) to (Fr3), but they form a different statement: (Fr2) is (Dec) for six of the sixteen kernels of (SD), (Fr3) is (N′) at the mask $m \equiv 1$, and (Fr1) is Lem. 2.38 (c).

Thms. 3.32 and 3.34 rest on (Fr2) for G, G_Z alone, and Thms. 3.40, 3.42 and Cor. 3.43 are free of (Fr1) to (Fr3).

Definition 3.22 (The matrices of (S-cen$_k$))**.** The matrices of (S-cen$_k$) are built from the matrix $C^{\mathbf{1}}$ of the law, that is, from C^{V} at the trivial field. We put $H^{(x)} := E - \mathrm{d}C^{\mathbf{1}}\mathsf{g}$ and $\mathsf{N}^{(x)} := Q_1 C^{\mathbf{1}}\mathsf{g}$. For a fine plaquette $i = p$ and for a coarse bond $i = c$, let $\mathcal{Q}_i$ denote the set of those coarse cubes each face of which lies within R_2 of i. With these sets we define

$$H^\natural(p,P) := H^{(x)}(p,P) - \sum_{q' \in \mathcal{Q}_p} \mathsf{G}(p,q')\mathrm{d}'(q',P), \qquad \mathsf{N}^\natural(c,P) := \mathsf{N}^{(x)}(c,P) - \sum_{q' \in \mathcal{Q}_c} \mathsf{G}_{\mathsf{n}}(c,q')\mathrm{d}'(q',P).$$

■

Remark 3.23. A priori the entries of these matrices are 3×3 blocks. For a constant G-valued u one has $\mathbf{1}^u = \mathbf{1}$; so ($\rho$3), taken at this field, shows that each block $C^{\mathbf{1}}(b,b')$ commutes with all Ad_u; by (ρ6) each such block is also real, since the field $\mathbf{1}$ is G-valued, and therefore $C^{\mathbf{1}}(b,b')$ is a real number times $1_{\mathfrak{g}}$.

C.II 3. Tools.

Lemma 3.24 (Bounds on the quadratic form q_ℓ^V and sums over steps)**.** In the setting of item C.II 1.2, with rows as there, let $L \ge 3$ be odd and $aL \ge 18$. Then the following hold.

(a) At every $V' \in \mathcal{V}'$ and for every η, $\gamma|\eta|^2 \le q_\emptyset^V(\eta) \le q_\ell^V(\eta) \le b|\eta|^2$.
(b) $\sup_b \sum_{\mathsf{s}\ \text{from}\ b} |q_{\mathsf{s}}| \le b$; and $\varpi^\flat(\theta) := \sup_b \sum_{\mathsf{s}\ \text{from}\ b} |q_{\mathsf{s}}|(e^{\theta|\mathsf{s}|} - 1) \le A^\flat\theta$ whenever $0 \le \theta \le \theta_1$.
(c) $\sup_b \sum_{\mathsf{s}} |q_{\mathsf{s}}||\mathsf{s}| \le A^\flat$, $\sup_b \sum_{\mathsf{s}} |q_{\mathsf{s}}||\mathsf{s}|^2 \le 2A^\flat/\theta_1$ and $\sup_b \sum_{\mathsf{s}} |q_{\mathsf{s}}||\mathsf{s}| e^{\theta_2|\mathsf{s}|} \le 2A^\flat$.
(d) $\sup_b \sum_{\mathsf{t}\ \text{from}\ b} |g_{\mathsf{t}}| \le 4\mathsf{N}_{\mathsf{B}}$; moreover, for each bond b and each increasing function $f \ge 0$, $\sum_{\mathsf{t}\ \text{from}\ b} |g_{\mathsf{t}}| f(|\mathsf{t}|) \le \sum_{\mathsf{s}\ \text{from}\ b} |q_{\mathsf{s}}| f(|\mathsf{s}|)$.

■

Proof. Part (a) is Lem. 3.3(a) on the full bond set, applied both to the empty rows and to the given rows. Adding rows adds a nonnegative square. Part (b) is the first and second bounds of Lem. 3.3(b).

Step (c): moments of the length. For $m \ge 0$ we have $e^{\theta m} - 1 \ge \theta m$ and $e^{\theta m} - 1 \ge \frac{1}{2}\theta^2 m^2$. With $\theta = \theta_1$, these inequalities and (b) give the first two bounds of (c). For the third we apply

$ue^u \le e^{2u}-1$, valid for $u \ge 0$, at $u = \theta_2 m$; as $\theta_2 = \frac{1}{2}\theta_1$, this gives $\sum |q_{\mathsf{s}}||\mathsf{s}|e^{\theta_2|\mathsf{s}|} \le \theta_2^{-1}\varpi^\flat(\theta_1)$, and by (b) the right side is at most $2A^\flat$.

Step (d): the g*-steps.* A bond b lies in four plaquettes $p \ni b$; for each of them, the sum of $|g_{\mathsf{t}}|$ over the g-steps t from b through p runs once over the g-step (p) and once over each g-step (y,p,p''), because each plaquette p'' lies under a single P; this sum is at most $\sum_{p''}[\delta_{pp''} + \sum_y |\langle \ell_y(p), \ell_y(p'')\rangle|]$, which by the two row bounds in Def. 2.48 is $\le 1 + \sum_y |\ell_y(p)| \sum_{p''} |\ell_y(p'')| \le 1 + K\mathsf{g}(\lambda)\cdot 3K\mathsf{g}(\lambda) = \mathsf{N}_{\mathsf{B}}$. This gives the first assertion of (d). For the second we map the g-steps from b into the steps from b. A g-step (p) is sent to the step $(p) : b \to b$ of kind (P); then $|\mathsf{s}| = 2|\mathsf{t}|$. A g-step (y,p,p'') is sent to the step $(y,p,p'') : b \to b''$ of kind (R), where b'' is the bond of p'' based at $\pi(p'')$ that points in the first direction of p''. This b'' is a crossing bond and therefore lies in I°; moreover $y(b'') = y(p'')$, so that $|\mathsf{s}| = |\mathsf{t}|$. The map so defined is injective, with $|q_{\mathsf{s}}| = |g_{\mathsf{t}}|$ and $|\mathsf{s}| \ge |\mathsf{t}|$; as $f \ge 0$ is increasing, the second assertion follows. □

Lemma 3.25 (A weighted bound for families and a crude bound on the circle)**.** Let $\mathfrak{S}$ be a coercive family, let V' be an $\mathrm{SL}(2,\mathbb{C})$ field such that $|\mathsf{V}'(c)| \le 1 + \vartheta_0^\flat$ for every bond c on which $\mathfrak{S}$ depends, and let w be a real function on $I_{\mathfrak{S}}$ with $|w(b) - w(b')| \le \theta_{\mathbb{C}}\mathrm{dist}(b,b')$.

(a) $\|e^w K^{\mathsf{V}}_{\mathfrak{S}} e^{-w}\| \le \varkappa$; in fact this norm is at most $1 - \frac{5}{8}\frac{\gamma}{b}$, so that $\|(q^{\mathsf{V}}_{\mathfrak{S}})^{-1}\| \le \frac{8}{5\gamma}$. Consequently $q^{\mathsf{V}}_{\mathfrak{S}}$ is invertible, with $\|e^w (q^{\mathsf{V}}_{\mathfrak{S}})^{-1} e^{-w}\| \le \frac{2}{\gamma}$, $\|e^w q^{\mathsf{V}}_{\mathfrak{S}} e^{-w}\| \le 2b$ and $\|(q^{\mathsf{V}}_{\mathfrak{S}})^{-1}(b,b')\| \le \frac{2}{\gamma} e^{-\theta_{\mathbb{C}}\mathrm{dist}(b,b')}$. At G-valued fields, $0 \le K_{\mathfrak{S}} \le 1 - \frac{\gamma}{b}$.

(b) For $|\zeta| \le e^{\theta_2}$ we have $\|K^{\mathsf{V}}_{\mathfrak{S}}(\zeta)\| \le \Lambda$; hence, in the polynomial $K^{\mathsf{V}}_{\mathfrak{S}}(\zeta)^k$, the coefficient of ζ^j has norm at most $\Lambda^k e^{-\theta_2 j}$.

■

Proof. The family satisfies the hypotheses of Lem. 3.4, which supplies the real bounds, the sharp weighted contraction and the coefficient estimate of part (b). Only the bonds in the transport paths matter, so the field can be extended by the identity off them.

Write $q = q^{\mathsf{V}}_{\mathfrak{S}}$ and $K = K^{\mathsf{V}}_{\mathfrak{S}}$. Then $q = b(1-K)$, and the Neumann series $q^{-1} = b^{-1}\sum_k K^k$ converges after conjugation with e^w, with $\|e^w q^{-1} e^{-w}\| \le b^{-1}(1-\varkappa)^{-1} = \frac{2}{\gamma}$; at $w = 0$ the sharper bound, used in place of $\varkappa$, gives $\|q^{-1}\| \le \frac{8}{5\gamma}$. From $q = b(1-K)$ we also obtain $\|e^w q e^{-w}\| \le 2b$. For the entries we take $w = \theta_{\mathbb{C}}\mathrm{dist}(\cdot, b')$, which satisfies the hypothesis on w by the triangle inequality and vanishes at b'; the bound on $e^w q^{-1} e^{-w}$ then gives the bound on the entry at (b,b').

□

Lemma 3.26 (From the law to a window)**.** Let a law satisfy (λ0) and (ρ5), and let $\frac{2}{\gamma}x^{-\mathsf{n}} \le \frac{1}{2}$ and $\mathsf{V}' \in \mathfrak{V}$. Clause (c) rests in addition on the standing conditions $2 \le \mathsf{n} \le 127$ (item C.II 2.1) and $c_R\theta_{\mathbb{C}} \ge 128$ (item C.II 1.3).

(a) $\|C^{\mathsf{V}}\| \le \frac{4}{\gamma}$ and $\|C^{\mathsf{V}} - (q^{\mathsf{V}}_\ell)^{-1}\| \le 8\gamma^{-2}x^{-\mathsf{n}}$. Moreover $\mathsf{u}^{\mathsf{V}} = C^{\mathsf{V}}S^{\mathsf{V}}$, with $\|\mathsf{u}^{\mathsf{V}}\| \le \frac{4}{\gamma}x^{-\mathsf{n}}$ and $\|\mathsf{u}^{\mathsf{V}}\| \le \frac{2}{3}$, and $S^{\mathsf{V}} = q^{\mathsf{V}}_\ell \sum_{m\ge 1}(\mathsf{u}^{\mathsf{V}})^m$.

(b) Let $\mathsf{Y} \subset I^\circ$ and $b \in \mathsf{Y}$ be such that every bond of $I^\circ \setminus \mathsf{Y}$ lies at distance at least ρ from b. Then, for all $b' \in \mathsf{Y}$, $\|(q^{\mathsf{V}}_{\mathsf{Y}})^{-1}(b,b') - (q^{\mathsf{V}}_\ell)^{-1}(b,b')\| \le 8b\gamma^{-2}e^{-\theta_{\mathbb{C}}\rho}$.

(c) Let o be a coarse site, and let Y_o be the set of those bonds of I° whose base lies in a block of the box of radius $R+1$ centred at o. For a bond b with $\mathrm{dist}(b,o) \le 1$ and for $b' \in \mathsf{Y}_o$,

$$\|C^{\mathsf{V}}(b,b') - (q^{\mathsf{V}}_{\mathsf{Y}_o})^{-1}(b,b')\| \le \tau_1 := 8\gamma^{-2}(1+b)\,x^{-\mathsf{n}}.$$

If the law also satisfies (ρ1), then $C^{\mathsf{V}}(b,b') = 0$ for $b' \notin \mathsf{Y}_o$. Finally, $q^{\mathsf{V}}_{\mathsf{Y}_o}$ depends on V' only through its values on the bonds of the box of radius $R + \varrho_L + 2$ centred at o, other bonds entering at most through products $T^{-1}T = 1$.

■

Proof. Step (a). We write $q = q_\ell^{\mathsf{V}}$, $C = C^{\mathsf{V}}$, $S = S^{\mathsf{V}}$ and $\mathsf{u} = \mathsf{u}^{\mathsf{V}}$. The family of all steps is coercive by Lem. 3.24 (a), and $\mathsf{V}' \in \mathfrak{V}$; so Lem. 3.25 (a), applied to this family with $w = 0$, gives $\|q^{-1}\| \le \frac{2}{\gamma}$. By ($\rho$5) we have $C^{-1} = q(1+\mathsf{x})$ with $\mathsf{x} = q^{-1}S$, and $\|\mathsf{x}\| \le \frac{1}{2}$ by the hypothesis $\frac{2}{\gamma}x^{-\mathsf{n}} \le \frac{1}{2}$. Hence $C = (1+\mathsf{x})^{-1}q^{-1}$ and $\|C\| \le \frac{4}{\gamma}$. Further, $q^{-1} - C = \mathsf{x}(1+\mathsf{x})^{-1}q^{-1}$, whose norm is at most $2 \cdot \frac{2}{\gamma}x^{-\mathsf{n}} \cdot \frac{2}{\gamma}$; this is the second bound.

From $C(q+S) = 1$ we get $1 - Cq = CS$, that is, $\mathsf{u} = CS$, and $\|C\| \le \frac{4}{\gamma}$ gives $\|\mathsf{u}\| \le \frac{4}{\gamma}x^{-\mathsf{n}}$. Next, $Cq = 1 - \mathsf{u}$ yields $C^{-1} = q(1-\mathsf{u})^{-1}$; as $S = C^{-1} - q$, this gives the series for S^{V} in (a), provided that series converges. It does: the sharper bound of Lem. 3.25 (a) gives $\|K\| \le 1 - \frac{5}{8}\frac{\gamma}{b}$, hence $\|q^{-1}\| \le \frac{8}{5\gamma}$ and $\|\mathsf{x}\| \le \frac{8}{5\gamma}x^{-\mathsf{n}} \le \frac{2}{5}$. Since $\mathsf{u} = CS = (1+\mathsf{x})^{-1}q^{-1}S = (1+\mathsf{x})^{-1}\mathsf{x}$, we obtain $\|\mathsf{u}\| \le \frac{2/5}{3/5} = \frac{2}{3}$.

Step (b). Multiplying the identity $\mathbf{1}_{\mathsf{Y}}q(\mathbf{1}_{\mathsf{Y}} + \mathbf{1}_{\mathsf{Y}^c})q^{-1}\mathbf{1}_{\mathsf{Y}} = \mathbf{1}_{\mathsf{Y}}$ from the left by q_{Y}^{-1}, we obtain $q_{\mathsf{Y}}^{-1} - \mathbf{1}_{\mathsf{Y}}q^{-1}\mathbf{1}_{\mathsf{Y}} = \mathsf{M}\,\mathbf{1}_{\mathsf{Y}^c}q^{-1}\mathbf{1}_{\mathsf{Y}}$, where $\mathsf{M} := q_{\mathsf{Y}}^{-1}\mathbf{1}_{\mathsf{Y}}q$. Let $w = \theta_{\mathbb{C}}\mathrm{dist}(b,\cdot)$. Then $\mathsf{M}(b,b_2) = e^{-w(b_2)}(e^{-w}\mathsf{M}e^{w})(b,b_2)$, and Lem. 3.25 (a), used twice with the weight $-w$, once for the window Y and once for all steps, gives $\|e^{-w}\mathsf{M}e^{w}\| \le \frac{2}{\gamma} \cdot 2b$. Off Y we have $w \ge \theta_{\mathbb{C}}\rho$, so the row $\delta_b^{\mathsf{T}}\mathsf{M}\mathbf{1}_{\mathsf{Y}^c}$ has ℓ^2 norm at most $\frac{4b}{\gamma}e^{-\theta_{\mathbb{C}}\rho}$. Multiplied by $\|q^{-1}\| \le \frac{2}{\gamma}$, this gives (b).

Step (c). A bond off Y_o lies at distance at least $R+2$ from o, hence at least $R+1$ from b. As $e^{-\theta_{\mathbb{C}}(R+1)} \le x^{-128} \le x^{-\mathsf{n}}$, the bounds of (a) and of (b), the latter taken with $\mathsf{Y} = \mathsf{Y}_o$ and $\rho = R+1$, add up to τ_1. For $b' \notin \mathsf{Y}_o$ we get $\mathrm{dist}(b,b') > R$, and (ρ1) gives $C^{\mathsf{V}}(b,b') = 0$.

The dependence on V' comes from the steps as described in item C.II 1.2. Let $\mathsf{s} : b_1 \to b_2$ be a step between bonds of Y_o. Such a step of kind (P) or (Q) with $y(b_1) = y(b_2)$ has $\mathsf{T}_{\mathsf{s}} = 1$. Indeed, by item C.II 1.2, for these two kinds T_{s} is the inverse of the transport from one site to $y(b_1)$ times the transport from the same site to $y(b_2)$, and at $y(b_1) = y(b_2)$ the two transports are equal. These $\mathsf{T}_{\mathsf{s}} = 1$ are the products $T^{-1}T = 1$ in the last assertion of (c). In every case the sites of π_{s} are within $\varrho_L + 1$ of an end of the step, by item C.II 1.2, and so within $R + \varrho_L + 2$ of o. □

Lemma 3.27 (The comb gauge of a box, paths, and the Bianchi identity)**.** Let B be a box and $o \in \mathsf{B}$ a site such that every site of B lies within ρ_{B} of o, where $3\rho_{\mathsf{B}} + 1 \le \ell_w(x')$, and let V' be a field on B that satisfies (h2) and (h3) at $\hat{\varepsilon} \le \frac{1}{40}$. In this lemma and in its proof, coordinates of coarse sites, $|\cdot|_\infty$ and e_λ are given in coarse units.

(a) Let $c = (w,\rho)$ be a bond of B, and let $t = |w - o|_\infty$. Then $|\mathsf{W}(c) - \mathbf{1}| \le e^{18t\hat{\varepsilon}} - 1$ and $|\mathsf{W}(c)| \le e^{18t\hat{\varepsilon}}$. Moreover $|\mathsf{k}(y)| \le 2$ and $\|\mathrm{Ad}_{\mathsf{k}(y)}^{\pm 1}\| \le 4$ for the sites y of B.

(b) For a path π in B of $m \le \ell_w(x')$ bonds whose sites all lie within τ of o, $\|\mathrm{Ad}_{\mathsf{W}(\pi)}^{\pm 1} - 1\| \le 2340\, m\, \tau\, \hat{\varepsilon}$.

(c) For every plaquette P of B, $|\tilde{X}_P| \le 8.24\,\hat{\varepsilon}$ and $e^{i\tilde{X}_P} = \partial\mathsf{W}(P)$.

(d) Let $q' \subset \mathsf{B}$ be a cube at z, let $t = |z - o|_\infty$, and let $(\mathrm{d}'\tilde{X})_{q'} = \sum_P \mathrm{d}'(q',P)\tilde{X}_P$. Then $\|(\mathrm{d}'\tilde{X})_{q'}\| \le 6 \cdot 10^4\,(1+t)\,\hat{\varepsilon}^2$.

None of these bounds requires $\rho_{\mathsf{B}}\hat{\varepsilon}$ to be small. Of (h2), clause (a) needs only the bounds for the combs $\Gamma(o,y)$, $y \in \mathsf{B}$; since a comb has at most $3\rho_{\mathsf{B}}$ bonds, (a) is valid already under $3\rho_{\mathsf{B}} \le \ell_w(x')$, and the additional bond is used in (d). ■

Proof. Step (a). By definition, $\mathsf{W}(c) = \mathsf{V}'(\Gamma(o,c_-)\,c\,\Gamma(o,c_+)^{-1})$. If $\rho = 3$, the two combs differ by the bond c alone, and $\mathsf{W}(c) = \mathbf{1}$. If $\rho = 2$, the loop is a single ladder of $|w_3 - o_3|$ plaquettes; if $\rho = 1$, it consists of two ladders, with $|w_3 - o_3| + |w_2 - o_2| \le 2t$ plaquettes in all.

A ladder is described by the following data: a path σ from o; steps $u_0, \dots, u_{n-1}$, each equal to $\pm e_\lambda$, that start from the end of σ; the paths $\tau_j = u_0 \cdots u_{j-1}$; the bond h_j of direction ρ at the end of $\sigma\tau_j$; the step v_j, which is u_j shifted by e_ρ; and the loop $\mathsf{p}_j = u_j h_{j+1} v_j^{-1} h_j^{-1}$ along the boundary

of one plaquette. For these data

$$\tau_n h_n v_{n-1}^{-1} \cdots v_0^{-1} h_0^{-1} = \prod_{j=n-1}^{0} \tau_j \mathsf{p}_j \tau_j^{-1}.$$

Indeed, $\tau_j \mathsf{p}_j \tau_j^{-1} = \tau_{j+1} h_{j+1} v_j^{-1} h_j^{-1} \tau_j^{-1}$, and the product telescopes.

The identity holds between words and requires of the u_j, h_j, v_j nothing but invertibility. Hence it holds in any group, and also when paths of a fine field take the place of the h_j; in this form it enters Step 3, (3), of the proof of Thm. 4.19.

Here $\sigma = \Gamma(o,(w_1,w_2,o_3))$ and $\lambda = 3$; for $\rho = 1$ the second ladder has $\sigma = \Gamma(o,(w_1,o_2,o_3))$ and $\lambda = 2$. So every $\sigma\tau_j$ is a comb $\Gamma(o,\cdot)$. Thus each plaquette of a ladder contributes the factor $\mathsf{k}(y)\,\partial\mathsf{V}'(P)^{\pm1}\mathsf{k}(y)^{-1}$, in which $\mathsf{k}(y)$ is a comb. When the step is $-e_\lambda$, the plaquette P is based at the endpoint of that step, and the comb leading to this endpoint is the conjugating one.

By (h2), $|\mathsf{k}| \le 2$; hence $\|\operatorname{Ad}_{\mathsf{k}(y)}^{\pm1}\| \le 4$ by item C.II 1.1. For a plaquette factor $\partial = \partial\mathsf{V}'(P)$ we have $|\partial^{-1} - \mathbf{1}| \le 1.05 \cdot 2\hat\varepsilon$: this follows from $|g^{-1} - \mathbf{1}| \le |g^{-1}||g - \mathbf{1}|$ and $|g^{-1}| = |g| \le 1 + 2\hat\varepsilon \le 1.05$ at $g = \partial$, valid on $\mathrm{SL}(2,\mathbb{C})$ by item C.II 1.1 and (h3). With $|\mathsf{k}| \le 2$, each factor is therefore of the form $\mathbf{1} + \delta$ with $|\delta| \le 8.4\hat\varepsilon$, and $(1 + 8.4\hat\varepsilon)^{2t} \le e^{16.8t\hat\varepsilon}$; since $16.8 \le 18$, the two bounds on $\mathsf{W}(c)$ follow.

Step (b). By (a), $|\mathsf{W}(c)^{-1} - \mathbf{1}| \le |\mathsf{W}(c)||\mathsf{W}(c) - \mathbf{1}| \le e^{36t\hat\varepsilon} - 1$. Thus $g = \mathsf{W}(\pi)$ is a product of m factors $\mathbf{1} + h_i$ with $1 + |h_i| \le e^{36\tau\hat\varepsilon}$, and with $\varsigma := 36m\tau\hat\varepsilon$ we get $|g| \le e^\varsigma$ and $|g - \mathbf{1}| \le e^\varsigma - 1$. On the other hand, if π runs from y to y'', then $|g| = |\mathsf{k}(y)\mathsf{V}'(\pi)\mathsf{k}(y'')^{-1}| \le 8$ by (h2). With the bounds on Ad_g of item C.II 1.1 we conclude that $\|\operatorname{Ad}_g^{\pm1} - 1\| \le \min\{65,\ 2e^\varsigma(e^\varsigma - 1)\} \le 65\varsigma$. For the last inequality, take first $\varsigma \le 1$: the function $2e^\varsigma(e^\varsigma - 1)$ is convex, is 0 at 0 and is below 9.35 at 1, hence it is at most 9.35ς; for $\varsigma > 1$ we have $65 \le 65\varsigma$. Since $65 \cdot 36 = 2340$, this is (b).

Step (c). Since $|\mathsf{k}| \le 2$ and $|X'_P| \le 2.06\hat\varepsilon$, we have $|\mathsf{k}X'\mathsf{k}^{-1}| \le 4 \cdot 2.06\hat\varepsilon$, which is the bound on $\tilde X_P$. The exponential commutes with conjugation, $e^{iX'_P} = \partial\mathsf{V}'(P)$, and $\partial\mathsf{W}(P) = \mathsf{k}(y_P)\partial\mathsf{V}'(P)\mathsf{k}(y_P)^{-1}$; hence $e^{i\tilde X_P} = \partial\mathsf{W}(P)$.

Step (d). We use Bianchi's identity, which is an identity of invertible matrices. Take the six monotone paths from z to $z + e_1 + e_2 + e_3$ in the cyclic order $123, 132, 312, 321, 231, 213$; any two neighbours in this order differ by one exchange. The quotients of neighbouring paths are then the six matrices $\kappa_1\partial(z + e_1; 23)\kappa_1^{-1}$, $\partial(z; 13)$, $\kappa_3\partial(z + e_3; 12)\kappa_3^{-1}$, $\partial(z; 23)^{-1}$, $\kappa_2\partial(z + e_2; 13)^{-1}\kappa_2^{-1}$ and $\partial(z; 12)^{-1}$, where κ_λ denotes the bond (z, λ); the product of these six is $\mathbf{1}$.

For the field W, by (c), this means $\prod_{k=1}^6 e^{iY_k} = \mathbf{1}$, where $Y_k = \epsilon_k \operatorname{Ad}_{\kappa_k} \tilde X_{P_k}$ and $\epsilon_k = \mathrm{d}'(q', P_k)$, with $\kappa_k = \mathbf{1}$ for the faces based at z and $\kappa_k = \mathsf{W}(z,\lambda)$ for the face based at $z + e_\lambda$. Now $\operatorname{Ad}_{\mathsf{W}(z,\lambda)} \tilde X_P = \operatorname{Ad}_{\mathsf{k}(z)\mathsf{V}'(z,\lambda)} X'_P$, and $|\mathsf{k}(z)\mathsf{V}'(z,\lambda)| \le 2$ by (h2), applied to the comb $\Gamma(o,z)$ followed by the bond (z,λ), a path of at most $3\rho_{\mathsf{B}} + 1$ bonds. Therefore $|Y_k| \le 8.24\hat\varepsilon$ for every k, and the sum S of (E) satisfies $S \le 49.44\hat\varepsilon \le 1.236$.

As the product is $\mathbf{1}$, (E) gives $|\sum Y_k| \le e^S - 1 - S \le 0.7894\, S^2 \le 1929.6\,\hat\varepsilon^2 \le 1931\,\hat\varepsilon^2$. Here we used that $(e^S - 1 - S)/S^2$ increases with S and is ≤ 0.78931 at $S = 1.236$, and that $0.7894 \cdot 49.44^2 \le 1929.55$. Finally, $(\mathrm{d}'\tilde X)_{q'} - \sum Y_k = \sum_\lambda \pm(1 - \operatorname{Ad}_{\mathsf{W}(z,\lambda)})\tilde X_{P_\lambda}$; by (b), with $m = 1$ and $\tau = t + 1$, and by (c), its norm is at most $3 \cdot 2340(t+1)\hat\varepsilon \cdot 8.24\hat\varepsilon \le 57845(t+1)\hat\varepsilon^2$. As $1931 + 57845 \le 6 \cdot 10^4$, the two bounds together give (d). □

C.II 4. (S-cen$_k$).

Definition 3.28 (Constants of the centre)**.** Throughout item C.II 4 we write $\theta = \theta_{\mathbb{C}}$. We define

$$\mathsf{K}_A := 1.24 \cdot 10^6\, \frac{A^\flat}{\gamma\,\theta}\, \mathsf{g}_I(\tfrac{\theta}{2})\Big[\frac{8\mathsf{N}_{\mathsf{B}}}{\gamma}\mathsf{g}_I(\theta) + 1\Big], \qquad \mathsf{K}_{\mathrm{c}}^0 := (L+1)\mathsf{K}_A + 6 \cdot 10^4\, L\, C_D\, \theta_D^{-1}, \qquad \mathsf{K}_{\mathrm{c}}^1 := 289\,\mathsf{K}_{\mathrm{c}}^0 + 17,$$

together with

$$\mathsf{a}_{\mathrm{c}} := \frac{100}{\gamma}\mathsf{N}_{\mathsf{B}}\, \mathsf{g}_I(\theta), \qquad \tau_{\mathrm{c}} := 165\,(L+1)\,\mathsf{N}_{\mathsf{B}}\cdot 3L^3(2R+3)^3\,\tau_1.$$

Each of these constants except τ_{c} is computed from $(L, a, K, \lambda, C_D, \theta_D)$; as K_*, λ_* are pure numbers, in the case $K = K_*$ and $\lambda = \lambda_*$ each such constant depends on (L, a, C_D, θ_D) and on pure numbers alone. ■

Definition 3.29 (Floors on x and on the torus)**.** The hypotheses (P0) to (P2) separate the coupling bounds from the torus-size requirements.

(P0) $x \geq L^2$, $R \geq 6$, and (Tor).
The condition (Tor) of item C.II 1.1 is a datum of (SD) (item 1 of §2.1). It is no floor on x, since for fixed N it is violated once x is large.
From (P0) we obtain $3R_3 + 1 \leq 7R + 19 \leq 14 c_R \log x' + 26 \leq \ell_w(x')$, $|\mathsf{s}|, |\mathsf{t}| \leq 8 + 6\varrho/L \leq \ell_w(x')$, $r_q \leq R + 2$ and $2R_3 + 1 < N$.

(P1) $\frac{4}{\gamma} x^{-\mathsf{n}} \leq 1$, and $165(L+1)\mathsf{N}_{\mathsf{B}}\, 3L^3(2R+3)^3\, 8\gamma^{-2}(1+b) \leq x$; by the latter, $\tau_{\mathrm{c}} \leq x^{-(\mathsf{n}-1)}$.

(P2) $10^4\, \ell_w(x')\, R_3^2\, \tau_{\mathrm{c}} \log x \leq \vartheta_0^\flat$, and $2(R_3 + \lceil 1/\vartheta_0^\flat \rceil + 1) + 1 < N$.

With the letters fixed, every clause that is free of N holds once x is large enough. Only (Tor) and the second clause of (P2) involve N; at the given x they are conditions on N, and (Tor) entails $N \geq 320 c_R \log x$. ■

Definition 3.30 (The remainders Λ_i)**.** Let $i = p$ be a fine plaquette or $i = c$ a coarse bond; put $o := y(p)$, resp. $o := c_-$, and let $\mathsf{B}(i)$ be the box of radius R_3 centred at o. The remainders are

$$\Lambda_p(\mathsf{V}') := \big(EX' - \mathrm{d}^{\mathsf{V}} C^{\mathsf{V}} \mathsf{g}^{\mathsf{V}} X'\big)_p - \sum_P H^\natural(p,P) T(y(p), y_P) X'_P, \qquad \Lambda_c(\mathsf{V}') := (Q_1^{\mathsf{V}} C^{\mathsf{V}} \mathsf{g}^{\mathsf{V}} X')_c - \sum_P \mathsf{N}^\natural(c,P) T(c_-, y_P) X'_P.$$

■

Remark 3.31. Each coarse site o is the end c_- of some coarse bond c; hence Thm. 3.32, applied with $i = c$ for each coarse bond c, covers all coarse sites o, and in this form Sections 4.2 and 4.4 use its clauses (a) and (c).

Theorem 3.32 (The remainders: a quadratic bound with a small linear term)**.** Let $L \geq 3$ be odd with $aL \geq 18$, let the rows be as in item C.II 1.2, let a law with $(\lambda 0)$, $(\rho 1)$, $(\rho 2)$ and $(\rho 5)$ be given, let G, G_Z satisfy (Fr2), and let (P0) and (P1) hold. Then, for every i, every $0 < \hat{\varepsilon} \leq \frac{1}{40}$ and every $\mathrm{SL}(2,\mathbb{C})$ field V' on $\mathsf{B}(i)$ with (h) at $\hat{\varepsilon}$, the following hold.

(a) $\Lambda_i(\mathsf{V}')$, and $a_{0,b}$ at the bonds with $\mathrm{dist}(b, o) \leq 1$, depend on V' only through its values on the bonds of $\mathsf{B}(i)$, and on X' only through its values at the plaquettes of $\mathsf{B}(i)$. Moreover, independently of (h2), (h3) and (Fr2): whenever two pairs, each consisting of a field of $\mathfrak{V}$ and a $\mathfrak{g}_{\mathbb{C}}$-valued function X' on the plaquettes of $\mathbb{T}'$, coincide on the bonds and on the plaquettes of $\mathsf{B}(i)$, the formulas of Λ_i and of $a_{0,b} = (C^{\mathsf{V}} \mathsf{g}^{\mathsf{V}} X')_b$ take the same values at both pairs. Consequently, when the field is given on a larger set and X' is defined on the plaquettes of that set, the values obtained are those computed on $\mathsf{B}(i)$.

(b) $\|\Lambda_i(\mathsf{V}')\| \leq \mathsf{K}_{\mathrm{c}}^0\, \hat{\varepsilon}^2 + \tau_{\mathrm{c}}\, \hat{\varepsilon}$.

(c) $|a_{0,b}| \leq \sqrt{2}\|a_{0,b}\| \leq \mathsf{a}_{\mathrm{c}} \hat{\varepsilon}$ at the bonds with $\mathrm{dist}(b, o) \leq 1$.

(By item C.II 1.4, every field with (e1), (e2) and $|\partial \mathsf{V}'(P) - \mathbf{1}| \leq 2\hat{\varepsilon}$ has (h) at $\hat{\varepsilon}$. A G-valued field always satisfies (h1) and (h2), and it satisfies (h3) if and only if $\sin \frac{1}{2}\phi'_P \leq \hat{\varepsilon}$ at the plaquettes of $\mathsf{B}(i)$.) ■□

Proof. We take $i = p$ and write $\mathsf{B} = \mathsf{B}(p)$. We extend V' by $\mathbf{1}$; by (h1), the extended field belongs to $\mathfrak{V}$. Let $\mathsf{Y} := \mathsf{Y}_o$ be the set of Lem. 3.26 (c); its size is $\#\mathsf{Y} \leq 3L^3(2R+3)^3$. We put $t_j := \mathrm{dist}(j, o)$.

Step (0): on what the objects depend. Let b be a bond of p, or of a path $[x, x']$ with $x \in B(c_-)$. Then $y(b) \in \{o\} \cup \{o + Le_\mu\}$; therefore $\mathrm{dist}(b, o) \leq 1$, and $T(o, y(b)) = \mathrm{Ad}_{\mathsf{k}(y(b))}$, because the path

$\Gamma(o,y(b))$ is either empty or a single bond. By $(\rho1)$, $a_{0,b}=\sum_{j\in\mathsf{Y}}C^{\mathsf{V}}(b,j)\mathsf{g}_j$, where $\mathsf{g}_j:=(\mathsf{g}^{\mathsf{V}}X')_j$. By $(\rho2)$, $C^{\mathsf{V}}(b,\cdot)$ depends on V' only within $R_2\le R_3$ of o. For $j\in\mathsf{Y}$, the term g_j depends only on paths π_{t} and plaquettes P all of whose sites lie within $(R+1)+2\varrho_L+2\le R_3$ of o; this inequality holds as $\varrho_L\le R+2$.

By Lem. 3.37 (a), $H^{\natural}(p,P)\neq0$ only for $\operatorname{dist}(p,P)\le R_2$. The proof of clause (a) of that lemma rests on $(\rho1)$, $R\ge6$ and $L\ge3$ alone and is independent of the present theorem. All four sites of such a plaquette P lie within $R_2+1\le R_3$ of o, and the path $\Gamma(o,y_P)$ is contained in B; further, $(EX')_p$ equals X'_P for a plaquette P with $y_P=o$, and 0 otherwise. This proves (a): none of these objects sees the extension of V'.

Step (0) also proves the second sentence of (a), because it is independent of the sizes of $\mathsf{V}'(\pi)$, of $\partial\mathsf{V}'(P)$ and of X', and of the formula that defines X': the properties $(\lambda0)$, $(\rho1)$, $(\rho2)$ hold at every field of $\mathfrak{V}$, and the ranges of d, Q_1, g, $H^{\natural}$, $\mathsf{N}^{\natural}$ are the same for every field.

Step (1): sizes in the given gauge. For $j\in\mathsf{Y}$ we have $\|\mathsf{T}_{\mathsf{t}}\|\le|\mathsf{V}'(\pi_{\mathsf{t}})|^2\le4$ by (h2); with Lem. 3.24 (d) this gives $\|\mathsf{g}_j\|\le4\mathsf{N}_{\mathsf{B}}\cdot4\cdot2.06\hat\varepsilon\le33\mathsf{N}_{\mathsf{B}}\hat\varepsilon$.

Step (2): from the law to the window. Let $\hat a:=(q^{\mathsf{V}}_{\mathsf{Y}})^{-1}\mathbf{1}_{\mathsf{Y}}\mathsf{g}$, where g denotes the vector $(\mathsf{g}_j)_j$ of Step (0). Lem. 3.26 (c) yields $\|a_{0,b}-\hat a_b\|\le\tau_1\#\mathsf{Y}\cdot33\mathsf{N}_{\mathsf{B}}\hat\varepsilon$, and together with Lem. 3.25 (a) we obtain

$$\|a_{0,b}\|\le[\frac{2}{\gamma}\mathsf{g}_I(\theta)+\tau_1\#\mathsf{Y}]33\mathsf{N}_{\mathsf{B}}\hat\varepsilon\le\frac{70}{\gamma}\mathsf{N}_{\mathsf{B}}\mathsf{g}_I(\theta)\hat\varepsilon,$$

the second inequality by (P1). Since $\sqrt{2}\cdot70\le100$, this is (c).

Step (3): the gauge. Let $\hat{\mathsf{G}}:=\operatorname{diag}(\operatorname{Ad}_{\mathsf{k}(y(j))})_{j\in\mathsf{Y}}$. For a path π in B from y to y'', the relation $\operatorname{Ad}_{\mathsf{k}(y)}\operatorname{Ad}_{\mathsf{V}'(\pi)}\operatorname{Ad}^{-1}_{\mathsf{k}(y'')}=\operatorname{Ad}_{\mathsf{W}(\pi)}$ is an identity of matrices. Hence $q^{\mathsf{W}}:=\hat{\mathsf{G}}q^{\mathsf{V}}_{\mathsf{Y}}\hat{\mathsf{G}}^{-1}$ has the entries $\sum q_{\mathsf{s}}\operatorname{Ad}_{\mathsf{W}(\pi_{\mathsf{s}})}$, where the steps with $\mathsf{T}_{\mathsf{s}}=1$ and $y(b)=y(b')$ keep the factor 1. Also $\hat{\mathsf{G}}\mathbf{1}_{\mathsf{Y}}\mathsf{g}=\mathsf{g}^{\mathsf{W}}\tilde X$, with $\mathsf{g}^{\mathsf{W}}(j,P):=\sum g_{\mathsf{t}}\operatorname{Ad}_{\mathsf{W}(\pi_{\mathsf{t}})}$. Put $\mathsf{a}^{\mathsf{W}}:=(q^{\mathsf{W}})^{-1}\mathsf{g}^{\mathsf{W}}\tilde X=\hat{\mathsf{G}}\hat a$; then Step (0) gives

$$(\mathrm{d}^{\mathsf{V}}\hat a)_p=\sum_b\mathrm{d}(p,b)\operatorname{Ad}_{\mathsf{k}(y(b))}\hat a_b=(d\,\mathsf{a}^{\mathsf{W}})_p.$$

Since $\mathsf{k}(o)=\mathbf{1}$ and an edge plaquette p sits under some P with $y_P=o$, we have $(EX')_p=(E\tilde X)_p$. From now on, g once more denotes the kernel of item C.II 2.2, that is, g^{V} at $\mathsf{V}'=\mathbf{1}$. By the definitions of $\tilde X$ and of $H^{\natural}$,

$$\sum_P H^{\natural}(p,P)T(o,y_P)X'_P=(H^{\natural}\tilde X)_p=(E\tilde X)_p-(\mathrm{d}C^{\mathbf{1}}\mathsf{g}\tilde X)_p-\sum_{\mathcal{Q}_p}\mathsf{G}(p,q')(\mathrm{d}'\tilde X)_{q'}.$$

Let $q^{\mathbf{1}}:=q^{\mathbf{1}}_{\mathsf{Y}}$ and $\mathsf{a}^{\mathbf{1}}:=(q^{\mathbf{1}})^{-1}\mathbf{1}_{\mathsf{Y}}\mathsf{g}\tilde X$. Lem. 3.26 (c), applied with $\mathbf{1}$ as its field V' and used with $\|(\mathsf{g}\tilde X)_j\|\le4\mathsf{N}_{\mathsf{B}}\cdot8.24\hat\varepsilon\le33\mathsf{N}_{\mathsf{B}}\hat\varepsilon$, gives $\|(C^{\mathbf{1}}\mathsf{g}\tilde X)_b-\mathsf{a}^{\mathbf{1}}_b\|\le\tau_1\#\mathsf{Y}\,33\mathsf{N}_{\mathsf{B}}\hat\varepsilon$.

Inserting all this into the definition of Λ_p, we find

$$\Lambda_p=-\big(d[\mathsf{a}^{\mathsf{W}}-\mathsf{a}^{\mathbf{1}}]\big)_p+\sum_{q'\in\mathcal{Q}_p}\mathsf{G}(p,q')(\mathrm{d}'\tilde X)_{q'}+\mathsf{ti},\qquad\|\mathsf{ti}\|\le4\,(4+1)\,33\,\mathsf{N}_{\mathsf{B}}\tau_1\#\mathsf{Y}\,\hat\varepsilon.$$

Here the factor 4 inside $(4+1)$ enters because the error of Step (2) is transported by $\operatorname{Ad}_{\mathsf{k}}$, whose norm is at most 4. When $i=c$, the row of Q_1 sums to L; in either case $\|\mathsf{ti}\|\le\tau_{\mathrm{c}}\hat\varepsilon$.

Step (4): the resolvent identity, row by row. We write $\mathsf{a}^{\mathsf{W}}-\mathsf{a}^{\mathbf{1}}=\mathrm{A}+\mathrm{B}$, where $\mathrm{A}:=-(q^{\mathsf{W}})^{-1}\delta\,(q^{\mathbf{1}})^{-1}\mathsf{g}\tilde X$ with $\delta:=q^{\mathsf{W}}-q^{\mathbf{1}}$, and $\mathrm{B}:=(q^{\mathsf{W}})^{-1}(\mathsf{g}^{\mathsf{W}}-\mathsf{g})\tilde X$. The entries obey $\|(q^{\mathsf{W}})^{-1}(b,j)\|\le16\cdot\frac{2}{\gamma}e^{-\theta\operatorname{dist}(b,j)}$; this comes from Lem. 3.25 (a) for $q^{\mathsf{V}}_{\mathsf{Y}}$, which applies by (h1) and independently of any closeness to $q^{\mathbf{1}}$, while the conjugation costs the factor $4\cdot4$. Moreover $|(q^{\mathbf{1}})^{-1}(j,j')|\le\frac{2}{\gamma}e^{-\theta\mathrm{dist}}$.

All sites of a step from j lie within $t_j+|\mathsf{s}|$ of o. Therefore Lem. 3.27 (b) and Lem. 3.24 (c) give

$$\sum_{j'}\|\delta(j,j')\|\le\sum_{\mathsf{s}\text{ from }j}|q_{\mathsf{s}}|\,2340\,|\mathsf{s}|(t_j+|\mathsf{s}|)\hat\varepsilon\le2340A^{\flat}(t_j+\tfrac{2}{\theta_1})\hat\varepsilon.$$

Lem. 3.24 (d) gives the same bound for $\sum_P \|(\mathsf{g}^{\mathsf{W}}-\mathsf{g})(j,P)\|$. If $\mathrm{dist}(b,o)\le 1$, then $t_j \le 1+\mathrm{dist}(b,j)$, and so $\sum_j e^{-\theta \mathrm{dist}(b,j)}(t_j+\frac{2}{\theta_1}) \le \mathsf{g}_I(\frac{\theta}{2})[1+\frac{2}{\theta_1}+\frac{1}{\theta}] \le \frac{2}{\theta}\mathsf{g}_I(\frac{\theta}{2})$. For the last inequality we use $\theta \le \frac{1}{6}\theta_1 \le \frac{1}{6}$, by which the bracket is at most $\frac{1}{\theta}$ times $\frac{1}{6}+\frac{1}{3}+1$.

Consequently

$$\|\mathrm{A}_b\| \le \tfrac{32}{\gamma}\cdot 2340 A^\flat \hat\varepsilon\, \tfrac{2}{\theta}\mathsf{g}_I(\tfrac{\theta}{2})\cdot \tfrac{2}{\gamma}\mathsf{g}_I(\theta)\cdot 4\mathsf{N}_{\mathsf{B}}\, 8.24\hat\varepsilon, \qquad \|\mathrm{B}_b\| \le \tfrac{32}{\gamma}\cdot 2340 A^\flat \hat\varepsilon\, \tfrac{2}{\theta}\mathsf{g}_I(\tfrac{\theta}{2})\cdot 8.24\hat\varepsilon.$$

Since $32\cdot 2340\cdot 2\cdot 8.24 = 1234022.4$, it follows that $\|\mathrm{A}_b\|+\|\mathrm{B}_b\| \le \mathsf{K}_A\hat\varepsilon^2$. For the first term of Λ_i we use that the row of d has four entries, that the row of Q_1 sums to L, and that $4 \le L+1$.

Step (5): the monopole sum. The corners of a cube of $\mathcal{Q}_p$ lie within $R_2+1 \le R_3$ of o. Lem. 3.27 (d), together with $\mathrm{dist}(q',o) = \mathrm{dist}(p,q')$, $1+t \le \theta_D^{-1}e^{\theta_D t}$ and (Fr2), bounds the norm of the monopole sum of Step (3) by $\sum_{q'} |\mathsf{G}(p,q')|\, 6\cdot 10^4(1+\mathrm{dist}(p,q'))\hat\varepsilon^2 \le 6\cdot 10^4 C_D\theta_D^{-1}\hat\varepsilon^2$. For G_{n}, which takes the place of G when $i=c$, the factor C_D is replaced by LC_D. Together with Steps (3) and (4), this proves (b). □

Lemma 3.33 (Covariance of a holomorphic law at complex pure gauges)**.** Let a law satisfy (λ0), (ρ2), (ρ3) and (ρ4), and let (P0) and the second clause of (P2) hold. Let B be the box of radius R_3 about o, let $u : \mathsf{B} \to \mathrm{SL}(2,\mathbb{C})$ satisfy $|u(y)| \le 2$, and $|u(c_-)u(c_+)^{-1}| < 1+\vartheta_0^\flat$ on the bonds of B, and let $\mathsf{V}_0(c) := u(c_-)u(c_+)^{-1}$ on B, extended by $\mathbf{1}$. Then, for $\mathrm{dist}(b,o)\le 1$ and for all b' based in the blocks of B,

$$C^{\mathsf{V}_0}(b,b') = \mathrm{Ad}_{u(y(b))}\, C^{\mathbf{1}}(b,b')\, \mathrm{Ad}^{-1}_{u(y(b'))}\,.$$

■

Proof. Step (1): an extension of the gauge. We write u in polar form, $u(y) = k_0(y)e^{h(y)}$ with $k_0 \in G$ and h hermitian traceless; then $|e^{h(y)}| \le 2$. Let $D := \lceil 1/\vartheta_0^\flat\rceil + 1$; let B^+ be the box of radius R_3+D about o, which embeds by (P2); let $\varpi : \mathsf{B}^+ \to \mathsf{B}$ be the map that, in each direction μ, sends the coordinate y_μ of a site y to the point of the interval of B nearest to y_μ, so that ϖ fixes the sites of B; and let $\chi(y) := 1-\mathrm{dist}(y,\mathsf{B})/D$. For $0\le s\le 1$ we define $u_s(y) := k_0(\varpi y)e^{s\chi(y)h(\varpi y)}$ on B^+, and $u_s(y) := \mathbf{1}$ off it.

We claim that $\mathbf{1}^{u_s} \in \mathfrak{V}^\circ$. Take a lattice step in B^+ in the direction μ, from y to y'. Either the step leaves ϖy unchanged, or it moves ϖy along a bond c of B and leaves $\mathrm{dist}(y,\mathsf{B})$ unchanged. The former happens when y_μ remains outside the interval of B, when it arrives from outside at an end of this interval, or when it leaves the interval from one of its ends; then χ changes by at most $\frac{1}{D}$, and $|u_s(y)u_s(y')^{-1}| = |e^{s(\chi-\chi')h}| \le |e^h|^{1/D} \le 2^{1/D} < 1+\vartheta_0^\flat$, by $\frac{\log 2}{D} < 0.7\vartheta_0^\flat < \log(1+\vartheta_0^\flat)$. In the latter case, with h_1 and h_2 the values of h at c_- and at c_+, Cordes' inequality bounds the norm of $u_s(y)u_s(y')^{-1}$: $|e^{s\chi h_1}e^{-s\chi h_2}| \le |e^{h_1}e^{-h_2}|^{s\chi} = |u(c_-)u(c_+)^{-1}|^{s\chi} < 1+\vartheta_0^\flat$.

Cordes' inequality says that $|A^\alpha B^\alpha| \le |AB|^\alpha$ for positive matrices A,B and $0\le\alpha\le 1$. For the proof, consider the set of those α for which $|A^\alpha B^\alpha| \le |AB|^\alpha$. It is closed, it contains 0 and 1, and together with two points α,β it contains their midpoint $m = \frac{\alpha+\beta}{2}$, because, with r the spectral radius,

$$|A^m B^m|^2 = \mathrm{r}(B^m A^{2m} B^m) = \mathrm{r}(A^{\alpha+\beta}B^{\alpha+\beta}) = \mathrm{r}(B^\alpha A^\alpha A^\beta B^\beta) \le |A^\alpha B^\alpha||A^\beta B^\beta|.$$

Hence this set is the whole interval $[0,1]$.

At the rim, where $\chi = 0$, the gauge u_s is unitary; so, on every bond from y to y' that leaves B^+, the value $u_s(y)u_s(y')^{-1}$ of $\mathbf{1}^{u_s}$ has norm 1. This proves the claim.

Step (2): the derivative vanishes. For $\mathsf{Z} \in \mathfrak{V}^\circ$ and $\lambda : \mathbb{T}' \to \mathfrak{g}_{\mathbb{C}}$ we set

$$\mathfrak{d}(\mathsf{Z};\lambda) := \frac{d}{d\sigma}\big[\mathrm{Ad}^{-1}_{v_\sigma(y(b))}\, C^{\mathsf{Z}^{v_\sigma}}(b,b')\, \mathrm{Ad}_{v_\sigma(y(b'))}\big]_{\sigma=0},$$

where $v_\sigma = e^{i\sigma\lambda}$. This derivative is $\mathbb{C}$-linear in λ, since (ρ4) makes the bracket a holomorphic function of v near $\mathbf{1}$ and the tangent vector of the curve v_σ is $\mathbb{C}$-linear in λ. If λ is $\mathfrak{g}$-valued, the bracket is constant by (ρ3). Every λ can be written as $\lambda_1 + i\lambda_2$ with $\mathfrak{g}$-valued λ_1,λ_2; hence $\mathfrak{d} \equiv 0$.

Let $\Psi(s) := \mathrm{Ad}^{-1}_{u_s(y(b))}\, C^{\mathbf{1}^{u_s}}(b,b')\,\mathrm{Ad}_{u_s(y(b'))}$. We have $\mathbf{1}^{u_{s+\epsilon}} = (\mathbf{1}^{u_s})^{u_{s+\epsilon}u_s^{-1}}$ and $u_{s+\epsilon}u_s^{-1} = \exp(\epsilon \dot u_s u_s^{-1} + o(\epsilon))$, where $\dot u_s u_s^{-1}$ is traceless. As $\mathbf{1}^{u_s} \in \mathfrak{V}^\circ$ by Step (1), this gives $\Psi'(s) = \mathrm{Ad}^{-1}_{u_s}\,\mathfrak{d}(\mathbf{1}^{u_s}; -i\dot u_s u_s^{-1})\,\mathrm{Ad}_{u_s} = 0$. Moreover $\Psi(0) = C^{\mathbf{1}}(b,b')$ by $(\rho 3)$, because u_0 is G-valued.

Step (3): comparison on the box. Since $u_1 = u$ on B, we have $\mathbf{1}^{u_1} = \mathsf{V}_0$ on the bonds of B. By $(\rho 2)$, $C(b,\cdot)$ depends on the field only within $1 + (R_2 - 1) \le R_3$ of o; hence $C^{\mathsf{V}_0}(b,b') = C^{\mathbf{1}^{u_1}}(b,b')$. As Ψ is constant by Step (2), this is the assertion. □

Theorem 3.34 (The remainders without the linear term, (S-cen$_k$))**.** Let $L \ge 3$ be odd with $aL \ge 18$, let the rows be as in item C.II 1.2, in particular $\ell \in \mathcal{R}(b_*, K_*, \lambda_*, \frac{1}{9})$ of range $\le \varrho(x)$, let a law with $(\lambda 0)$, $(\rho 1)$ to $(\rho 5)$ be given, let G, G_Z satisfy (Fr2), and let (P0), (P1) and (P2) hold. Then, for every fine plaquette p and every coarse bond c, for every $0 < \hat\varepsilon \le \frac{1}{40}$, and for every $\mathrm{SL}(2,\mathbb{C})$ field V' on the box of radius R_3 about $y(p)$, resp. c_-, that satisfies (e1), (e2) and $|\partial \mathsf{V}'(P) - \mathbf{1}| \le 2\hat\varepsilon$,

$$\|\Lambda_p(\mathsf{V}')\| \le \mathsf{K}_{\mathrm{c}}^1 \hat\varepsilon^2, \qquad \|\Lambda_c(\mathsf{V}')\| \le \mathsf{K}_{\mathrm{c}}^1 \hat\varepsilon^2, \qquad \|(C^{\mathsf{V}} \mathsf{g}^{\mathsf{V}} X')_b\| \le \mathsf{a}_{\mathrm{c}} \hat\varepsilon \quad (\mathrm{dist}(b,o) \le 1).$$

■□

Proof. The third bound is Thm. 3.32 (c). If $\hat\varepsilon \ge \tau_{\mathrm{c}}$, the first two bounds follow from Thm. 3.32 (b), because then $\tau_{\mathrm{c}}\hat\varepsilon \le \hat\varepsilon^2$. Now let $\hat\varepsilon < \tau_{\mathrm{c}}$, and let $T := \tau_{\mathrm{c}}/\hat\varepsilon > 1$. We take $i = p$; for $i = c$ the proof is the same, with Λ_c in place of Λ_p and the box about c_- in place of the box about $y(p)$.

We first collect the numerical facts that Step (1) uses. The first clause of (P2), together with $\vartheta_0^\flat \le 10^{-4}$, $\ell_w(x') \ge 12$ and $\log x \ge 1$, gives $R_3^2\tau_{\mathrm{c}} \le 10^{-9}$; besides, we use $R_3 \ge 17$. With $\hat\varepsilon < \tau_{\mathrm{c}}$ and $T\hat\varepsilon = \tau_{\mathrm{c}}$, and with the number a of Step (1), these facts give: $18R_3\hat\varepsilon \le \frac{1}{10}$, so that $e^{18R_3\hat\varepsilon} - 1 \le 19R_3\hat\varepsilon \le 0.106$; $(T+1)\mathsf{a} < 42R_3\tau_{\mathrm{c}} \le 1$, so that $4(e^{(T+1)\mathsf{a}} - 1) \le 8(T+1)\mathsf{a}$; $4T\mathsf{a} = 84R_3\tau_{\mathrm{c}} \le 1$; $56448\, R_3^2\tau_{\mathrm{c}} \le 2$; and $17\tau_{\mathrm{c}} \le \frac{1}{40}$.

Step (1): the curve, with the room of item C.II 1.4. By Lem. 3.27 (a), $|\mathsf{W}(c) - \mathbf{1}| \le e^{18R_3\hat\varepsilon} - 1 \le 19R_3\hat\varepsilon \le 0.106$. Hence $A'_c := -i \log \mathsf{W}(c)$ is traceless, with $|A'_c| \le \mathsf{a} := 21R_3\hat\varepsilon$. We set $\mathsf{V}'_t(c) := \mathsf{k}(c_-)^{-1} e^{itA'_c} \mathsf{k}(c_+)$; this is entire in t, and $\mathsf{V}'_1 = \mathsf{V}'$. For $|t| < T$ we have $(T+1)\mathsf{a} < 42R_3\tau_{\mathrm{c}}$, and $\mathsf{V}'_t(c) = \mathsf{V}'(c)(\mathbf{1} + \delta_c)$ with $\mathbf{1} + \delta_c = \mathsf{k}(c_+)^{-1} e^{i(t-1)A'_c} \mathsf{k}(c_+)$, where $|\delta_c|, |(\mathbf{1}+\delta_c)^{-1} - \mathbf{1}| \le 4(e^{(T+1)\mathsf{a}} - 1) \le \delta := 336 R_3\tau_{\mathrm{c}}$.

We now check (h) for V'_t with $|t| < T$. First, (h1) holds strictly: $|\mathsf{V}'_t(c)| < (1+\delta)(1 + \vartheta(x') + \mathsf{g}_x) \le 1 + \vartheta_0^\flat$, since $1.01\delta \le \frac{3\vartheta_0^\flat}{4 \log x}$ by (P2). Next, (h2): in a path of $n \le \ell_w(x')$ bonds we carry the factors $(\mathbf{1}+\delta_c)^{\pm 1}$ to the left through the partial products, whose norms are < 2; we obtain $|\mathsf{V}'_t(\pi)| \le (1+4\delta)^n |\mathsf{V}'(\pi)| < e^{\mathsf{y}}(2 - \mathsf{r})$ with $\mathsf{y} = 4n\delta$ and $\mathsf{r} = \frac{3}{4 \log x}$. Now $(2-\mathsf{r})(1+2\mathsf{y}) \le 2$ as soon as $4\mathsf{y} \le \mathsf{r}$, and the latter holds because $5376\, \ell_w(x') R_3 \tau_{\mathrm{c}} \le \frac{3}{4 \log x}$ by (P2).

Finally, (h3): $\partial \mathsf{V}'_t(P) = \mathsf{k}(y_P)^{-1} \Pi_t \mathsf{k}(y_P)$ with $\Pi_t = e^{itA'_1} e^{itA'_2} e^{-itA'_3} e^{-itA'_4}$, and (E) gives $|\Pi_t - \mathbf{1} - it(\mathrm{d}'A')_P| \le (4|t|\mathsf{a})^2$, as $4T\mathsf{a} \le 1$. At $t = 1$ this yields $|(\mathrm{d}'A')_P| \le 8\hat\varepsilon + 16\mathsf{a}^2$. Therefore $|\partial \mathsf{V}'_t(P) - \mathbf{1}| \le 32T\hat\varepsilon + 128T^2\mathsf{a}^2 \le 34T\hat\varepsilon$, by $128 \cdot 441\, T R_3^2 \hat\varepsilon = 56448\, R_3^2 \tau_{\mathrm{c}} \le 2$. Thus V'_t has (h) at $\hat\varepsilon_T := 17T\hat\varepsilon = 17\tau_{\mathrm{c}} \le \frac{1}{40}$.

Step (2): holomorphy and a first bound. The function $f(t) := \Lambda_p(\mathsf{V}'_t)$ is holomorphic on $|t| < T$. This follows from $(\rho 4)$, since $t \mapsto \mathsf{V}'_t$, extended by $\mathbf{1}$, is a holomorphic curve in $\mathfrak{V}^\circ$, since X' is a power series, and since the transports are polynomial. By Thm. 3.32 (b), $\|f(t)\| \le \mathsf{K}_{\mathrm{c}}^0 (17T\hat\varepsilon)^2 + \tau_{\mathrm{c}} 17 T\hat\varepsilon$.

Step (3): the value and the derivative at $t = 0$. We have $f(0) = 0$, because $\partial \mathsf{V}'_0 = \mathbf{1}$ and hence $X' = 0$. We also have $f'(0) = 0$. Indeed, only X' is differentiated, and $\dot X_P = \mathrm{Ad}^{-1}_{\mathsf{k}(y_P)}(\mathrm{d}'A')_P$. On the box $\mathsf{B} = \mathsf{B}(p)$ of radius R_3 about $y(p)$, $\mathsf{V}'_0 = \mathbf{1}^u$ with $u = \mathsf{k}^{-1}$; here $|u| \le 2$, and $|u(c_-)u(c_+)^{-1}| = |\mathsf{V}'_0(c)| < 1 + \vartheta_0^\flat$ by Step (1). The transports at V'_0 are $T_0(y,y'') = \mathrm{Ad}^{-1}_{\mathsf{k}(y)}\,\mathrm{Ad}_{\mathsf{k}(y'')}$; therefore $(\mathsf{g}^{\mathsf{V}_0}\dot X)_j = \mathrm{Ad}^{-1}_{\mathsf{k}(y(j))}(\mathsf{g}\mathrm{d}'A')_j$. Lem. 3.33 gives $C^{\mathsf{V}_0}(b,j) = \mathrm{Ad}^{-1}_{\mathsf{k}(y(b))}\, C^{\mathbf{1}}(b,j)\,\mathrm{Ad}_{\mathsf{k}(y(j))}$. In addition,

$T_0(o, y(b)) = \mathrm{Ad}_{\mathsf{k}(y(b))}$ and $T_0(o, y_P)\dot{X}_P = (\mathrm{d}'A')_P$. Hence

$$f'(0) = \big((E - \mathrm{d}C^{\mathbf{1}}\mathsf{g} - H^\natural)\mathrm{d}'A'\big)_p = \sum_{q' \in \mathcal{Q}_p} \mathsf{G}(p, q')(\mathrm{d}'\mathrm{d}'A')_{q'} = 0.$$

Step (4): the maximum principle. The function $f(t)/t^2$ is holomorphic on $|t| < T$, and the maximum principle for its norm, at $t = 1$, gives $\|f(1)\| \le T^{-2} \sup \|f\| \le 289\mathsf{K}_{\mathrm{c}}^0 \hat{\varepsilon}^2 + 17\tau_{\mathrm{c}}\hat{\varepsilon}/T = (289\mathsf{K}_{\mathrm{c}}^0 + 17)\hat{\varepsilon}^2$. □

Remark 3.35 (The three constants). (1) Three constants occur: $\mathsf{K}_{\mathrm{c}}^0$, which appears together with the linear term $\tau_{\mathrm{c}}\hat{\varepsilon}$ in Thm. 3.32 (b); $\mathsf{K}_{\mathrm{c}}^1 = 289\mathsf{K}_{\mathrm{c}}^0 + 17$, in Thm. 3.34, which assumes (P2); and $\sqrt{2}\mathsf{K}_{\mathrm{c}}^0$, in (Cen), that is, in Cor. 3.38 (a). Each of the three bounds holds for every $0 < \hat{\varepsilon} \le \frac{1}{40}$.

(2) For $\tau_{\mathrm{c}} \le \hat{\varepsilon} \le \frac{1}{40}$, in the setting of Thm. 3.32 and with its hypotheses alone, $\|\Lambda_i(\mathsf{V}')\| \le (\mathsf{K}_{\mathrm{c}}^0 + 1)\hat{\varepsilon}^2 \le \mathsf{K}_{\mathrm{c}}\hat{\varepsilon}^2$, where $\mathsf{K}_{\mathrm{c}} := \sqrt{2}\mathsf{K}_{\mathrm{c}}^0$. Indeed, $\tau_{\mathrm{c}}\hat{\varepsilon} \le \hat{\varepsilon}^2$, and $\mathsf{K}_{\mathrm{c}}^0 \ge 6 \cdot 10^4 L C_D \theta_D^{-1} \ge 7.2 \cdot 10^5 > (\sqrt{2} - 1)^{-1}$. This K_{c} is the letter K_{c} of the table of letters, entered at (O10) in item 3 of §2.1.

(3) If (P1) holds at $\mathsf{n} = 60$, $p_0' \ge 1$ and $\hat{\varepsilon} = \tau\varepsilon'$ with $\tau \ge 1$, then $\hat{\varepsilon} \ge \tau_{\mathrm{c}}$, which is the lower bound on $\hat{\varepsilon}$ in (2). Indeed, $\varepsilon' \ge x'^{-1/2} > x^{-59} \ge \tau_{\mathrm{c}}$, by (P1) applied with the value 60 of its letter n.

Definition 3.36 (Two numbers and the floor (P3))**.** We set

$$\tau_2 := 24L^3(2R+1)^3\gamma^{-2}x^{-\mathsf{n}} + \tfrac{2}{\gamma}\mathsf{g}_I(\tfrac{\theta}{2})x^{-64}, \qquad \mathsf{ti}_{\mathrm{c}} := (16 + 4L)\,\mathsf{N}_{\mathsf{B}}\,\tau_2 + 7e\,(L+1)\,C_D\,x^{-128}.$$

The floor (P3) is the condition

$$\textbf{(P3)} \quad L^2(1+a)(\mathsf{ti}_{\mathrm{c}} + 8C_D^3 x'^{-64}) \le 1.$$

■

Lemma 3.37 (Range, closeness to the free matrices, and size of $H^\natural$ and $\mathsf{N}^\natural$)**.** Let a law satisfy (λ0), (ρ1) and (ρ5), and let (Fr1) to (Fr3), (P0) and (P1) hold. Then the following hold.

(a) $H^\natural(p, P) = 0 = \mathsf{N}^\natural(c, P)$ for dist $> R_2$. Of the properties of the law, (ρ1) alone suffices for this clause, together with $R \ge 6$ and $L \ge 3$; (Fr1) to (Fr3) are not needed for it.

(b) $\|H^\natural - H^\sharp\|_\infty + \|\mathsf{N}^\natural - \mathsf{N}^\sharp\|_\infty \le \mathsf{ti}_{\mathrm{c}}$.

(c) If, in addition, (P3) holds, then $L^2\|H^\natural\|_\infty \le \mathsf{N}_\sharp + 1$ and $aL\|\mathsf{N}^\natural\|_\infty \le 4\mathsf{N}_Z + 1$.

■

Proof. Step (a): the ranges. By (ρ1), $C^{\mathbf{1}}$ has range $\le R$; the matrices d and Q_1 have range ≤ 1, and g has range $\le 2\varrho_L + 1$. Moreover $1 + R + 2\varrho_L + 1 \le R_2$, by $R \ge 6$ and $L \ge 3$, through $2\varrho_L \le \frac{2}{3}R + 2$. No cube of $\mathcal{Q}_i$ has a face beyond R_2. This gives (a).

Step (b): closeness to the free matrices. By (Fr1), $H^{(x)} = H_\circ + d(q^{-1} - C^{\mathbf{1}})\mathsf{g} = H^\sharp + \mathsf{G}\mathrm{d}' + \mathsf{Ti}$, the last term being $\mathsf{Ti} = d(q^{-1} - C^{\mathbf{1}})\mathsf{g}$. We bound the absolute row sums of $q^{-1} - C^{\mathbf{1}}$. Its entries at (b, b') with $\mathrm{dist}(b, b') \le R$ are $\le 8\gamma^{-2}x^{-\mathsf{n}}$ by Lem. 3.26 (a), and a row has $3L^3(2R+1)^3$ of them. For the entries with $\mathrm{dist}(b, b') > R$, we have $C^{\mathbf{1}} = 0$ and $|q^{-1}(b, b')| \le \frac{2}{\gamma}e^{-\theta\mathrm{dist}}$, and they contribute at most $\frac{2}{\gamma}e^{-\theta R/2}\mathsf{g}_I(\frac{\theta}{2})$ to a row sum. Altogether, the row sums are $\le \tau_2$, and so $\|\mathsf{Ti}\|_\infty \le 4 \cdot \tau_2 \cdot 4\mathsf{N}_{\mathsf{B}}$.

For $\mathrm{dist}(p, P) \le R_2$ one has $(H^\natural - H^\sharp)(p, P) = \mathsf{Ti}(p, P) + \sum_{q' \notin \mathcal{Q}_p} \mathsf{G}(p, q')\mathrm{d}'(q', P)$, and beyond R_2 the difference is $-H^\sharp(p, P)$. A cube off $\mathcal{Q}_p$ lies at distance $\ge R_2 - 1$ and has six faces, so that such cubes add at most $6C_D e^{-\theta_D(R_2-1)}$. The tail of $H^\sharp$ contributes $C_D e^{-\theta_D R_2}$. Here $e^{-\theta_D R_2} \le x^{-128}$ and $e^{\theta_D} \le e$. For $\mathsf{N}^\natural$, the matrix Q_1, whose rows sum to L, replaces d, which gives $L\tau_2 4\mathsf{N}_{\mathsf{B}}$, and LC_D replaces C_D. Adding these contributions, we obtain (b).

Step (c): the sizes. By (Fr3), $\|H^\sharp\|_\infty \le L^{-2}\mathsf{N}_\sharp + \|\mathsf{T}\|_0\|\mathsf{B}' - \mathsf{B}'^{\mathrm{c}}\|_0$. As $\mathsf{B}' - \mathsf{B}'^{\mathrm{c}} = \ell'^{>\mathsf{T}}\ell' + \ell'^{\mathrm{c}\mathsf{T}}\ell'^{>}$, we have $\|\mathsf{B}' - \mathsf{B}'^{\mathrm{c}}\|_0 \le 2C_D\|\ell'^{>}\|_0 \le 2C_D^2 e^{-\theta_D\varrho(x')} \le 2C_D^2 x'^{-64}$. Further, $aL\|\mathsf{N}^\sharp\|_\infty \le 4\|\mathsf{B}'\|_\infty \le 4\mathsf{N}_Z + 8C_D^2 x'^{-64}$. With (b) and (P3), these bounds give (c). □

Corollary 3.38 (The centre bound and the shrinking of the analytic terms given at x)**.** Let the hypotheses of Thm. 3.32 hold, together with (Fr1) to (Fr3) and (P3). Then $\tau_{\mathrm{c}} \le L^{-2}$, since $\tau_{\mathrm{c}} \le x^{-(\mathsf{n}-1)} \le x^{-1} \le L^{-2}$ by (P1) and (P0), and the two assertions below are true.

(a) Condition (Cen)$(\mathsf{N}_{\mathrm{c}}, \sqrt{2}\mathsf{K}_{\mathrm{c}}^0, \mathsf{a}_{\mathrm{c}})$ holds with $\mathsf{N}_{\mathrm{c}} = 17(\mathsf{N}_\sharp + 1)$. Spelled out, this means the following. Take any fine plaquette p, any bond b of p, any number $\hat\varepsilon$ in the range $0 < \hat\varepsilon \le \frac{1}{40}$, and any field defined on the box of radius R_3 about $y(p)$ that satisfies (e1), (e2) and $|\partial\mathsf{V}'(P) - \mathbf{1}| \le 2\hat\varepsilon$ at its plaquettes P. At such a field the first bound $|a_{0,b}| \le \mathsf{a}_{\mathrm{c}}\hat\varepsilon$ holds, and so does the second bound

$$|(EX' - \mathrm{d}^{\mathsf{V}}a_0)_p| \le \mathsf{N}_{\mathrm{c}}L^{-2}\hat\varepsilon + \sqrt{2}\mathsf{K}_{\mathrm{c}}^0\hat\varepsilon^2.$$

(b) *At the law of windows, the analytic terms given at x shrink.* We begin with the objects that enter this clause. The fine cubes are the cubes with M^3 fine sites, taken from the grid in which each $\hat\square$ is a union of L^3 such cubes (see the definition "Cubes" in item 1 of §2.1); the distance between two fine cubes is measured in units of cubes. For a set X of fine cubes we set $X^{\approx} := X^{+r(x)}$, and by a bond, a path or a plaquette of $X^{\approx}$ we mean one whose sites all belong to $X^{\approx}$.

We write $\mathcal{U}_X(x)$ for the set of those $\mathrm{SL}(2,\mathbb{C})$ fields U on the bonds of $X^{\approx}$ that satisfy the three conditions below.

(u1) $|\mathsf{U}(b)| < 1 + \vartheta(x)$ at each bond b of $X^{\approx}$.
(u2) $|\mathsf{U}(\pi)| < \mathsf{b}(x)$ for each path π in $X^{\approx}$ that has at most $\ell_w(x)$ bonds.
(u3) $|\partial\mathsf{U}(p) - \mathbf{1}| < 2\varepsilon$ at each plaquette p of $X^{\approx}$.

For a set B of coarse sites and a number $\tau \ge 1$, the set $U^\tau(\mathsf{B}; x')$ consists of the $\mathrm{SL}(2,\mathbb{C})$ fields on the bonds of B that obey the three conditions below.

(e1) Condition (e1) of item C.II 1.4.
(e2) Condition (e2) of item C.II 1.4, for those paths in B that have at most $\ell_w(x')$ bonds.
(e3) $|\partial\mathsf{V}'(P) - \mathbf{1}| < 2\tau\varepsilon'$ at each plaquette P of B.

We put $c_a(x) := E_a p_0(x)^3 x^{-1/2}$ with a number $E_a > 0$, and we let $T := L^{3/2}/(19\mathsf{N}_{\mathrm{c}})$, $c_\xi := (6.5C_\varphi^\flat)^{-1}$ and $\mathsf{K}_{\mathrm{c}} = \sqrt{2}\mathsf{K}_{\mathrm{c}}^0$. With these numbers we define $r_T := \sqrt{2}C_\varphi^\flat c_\xi\varepsilon + 17T\mathsf{a}_{\mathrm{c}}\varepsilon'$. Further, $D_w := \lceil \ell_w(x)/L\rceil + 1$ and $\omega := M + \lceil Mr(x)/L\rceil + D_w + R_3 + 2$; these are the numbers D_w, ω of item 2 of §2.1.

In what follows, X is a connected set of at most $\frac{L}{2} - 2$ fine cubes (such a set is called small) whose first cube, with respect to a fixed ordering of all cubes, lies in $\hat\square_0$. Let o be the centre site of the cell $\square_0$, and let B_ω be the box of radius ω about o. We define $\chi_X(\xi) := 1\{|\xi_b| \le c_\xi\varepsilon$ for all bonds b of I° that lie within $R(x)$ of some block meeting $X^{\approx}\}$.

Let $E^a(X,\cdot)$ be holomorphic on $\mathcal{U}_X(x)$ and invariant under the gauge transformations with values in G, and let $\sup|E^a(X,\cdot)| \le \mathsf{A}c_a(x)e^{-\kappa(\#X-1)} \le \frac{1}{2}$ hold for a number $\mathsf{A} \ge 1$; for the class of item 1 of §2.1 one has $\mathsf{A} = 1$. We put

$$\mathcal{G}_X(\mathsf{V}') := E_\xi[\chi_X(\xi)(e^{E^a(X,\mathsf{U})} - 1)]$$

with $\xi \sim \mathcal{N}(0, x^{-1}\mathbf{1})$, where $\mathsf{U} = \mathsf{U}(G^{\mathsf{V}}\xi - a_0; \mathsf{V}')$ is the chart of item C.I 4, defined at a complex field by the same expression. Here $E^a(X,\mathsf{U})$ stands for the value of $E^a(X,\cdot)$ at the restriction of U to the bonds of $X^{\approx}$. Wherever $\chi_X(\xi) = 0$, the integrand is taken to be 0: for such ξ the restriction need not belong to $\mathcal{U}_X(x)$, and $E^a(X,\mathsf{U})$ need not be defined.

The hypotheses of clause (b) are those of clause (a), a law with (λ0) to (λ7), the condition (Tor), and nine floors. Four of these floors have no name: $T > 1$, $17T\varepsilon' \le \frac{1}{40}$, $\mathsf{K}_{\mathrm{c}}(17T\varepsilon')^2 \le 0.12\varepsilon$ and $\omega \le Mr'$. The remaining five, (D2), (D3), (C0), (C1) and (C2), are as follows.

(D2) $\ell_w(x)(7.1r_T + 18D_w \cdot 17T\varepsilon') \le \frac{1}{2}\mathsf{g}_x$.
(D3) $(4.1r_T + 2.1 \cdot 17T\varepsilon')^2 \le 0.1\varepsilon$.
(C0) $18\omega\varepsilon' \le \frac{1}{10}$ and $160T\omega\varepsilon' \le 1$.
(C1) $2640\,\ell_w(x')(T+1)\omega\varepsilon' \le \mathsf{g}_x$.
(C2) $102400\,T\omega^2\varepsilon' \le 1$.

Each of the nine floors except $T > 1$ appears among (A0), (A2) of item 4 of §2.1, and $T > 1$ follows from (L2^{su}), which is the condition $L^{1/2} \ge 1.2 \cdot 10^4 \mathsf{N}_{\mathrm{c}}^2$ of claim (3) below. When the letters have the values fixed in item 3 of §2.1, all these floors are satisfied for every sufficiently large x; this is shown in Step 14 of the proof of Thm. 4.39 and in Step 0 of the proof of Thm. 5.15. Under the hypotheses of clause (b), the following three claims hold.

(1) On $U^{17T}(\mathsf{B}_\omega; x')$ the function $\mathcal{G}_X$ is defined, holomorphic and invariant under the gauge transformations with values in G, and there $|\mathcal{G}_X| \le 2\mathsf{A}c_a(x)e^{-\kappa(\#X-1)}$.
(2) The bound
$$|\mathcal{G}_X(\mathsf{V}') - \mathcal{G}_X(\mathbf{1})| \le 4T^{-2}\mathsf{A}c_a(x)e^{-\kappa(\#X-1)}$$
holds at every field V' on B_ω that satisfies (u1), (u2), (u3) at x', and hence at the restriction to B_ω of each member of $\mathcal{U}_{\{\square_0\}}(x')$.
(3) If, in addition, $(p_0'/p_0)^4 \ge \frac{1}{2}$ and (L2^{su}) $L^{1/2} \ge 1.2 \cdot 10^4 \mathsf{N}_{\mathrm{c}}^2$ hold, then the right sides of (2), summed over all small X whose first cube lies in $\hat{\square}_0$, total at most $\frac{1}{4}\mathsf{A}c_a(x')$.

■□

Proof of Cor. 3.38 (a). By the parenthetical remark at the end of Thm. 3.32, the conditions (e1), (e2) and $|\partial\mathsf{V}'(P) - \mathbf{1}| \le 2\hat{\varepsilon}$ on the field give (h) at $\hat{\varepsilon}$. By Step (0) of the proof of Thm. 3.32, every bond b of p has $\mathrm{dist}(b, o) \le 1$ for $o = y(p)$; hence Thm. 3.32 (c), applied with $i = p$ and $o = y(p)$, gives the first bound $|a_{0,b}| \le \mathsf{a}_{\mathrm{c}}\hat{\varepsilon}$. The sum $\sum_P H^\natural(p, P)\tilde{X}_P$ obeys $\|\sum_P H^\natural(p, P)\tilde{X}_P\| \le \|H^\natural\|_\infty 8.24\hat{\varepsilon}$, since $\|\tilde{X}_P\| \le |\tilde{X}_P| \le 8.24\hat{\varepsilon}$ at every plaquette P of the box of radius R_3 about $y(p)$: the first inequality is the bound $\|Y\| \le |Y|$ of item C.II 1.1, and the second is Lem. 3.27 (c), applied to that box with $o = y(p)$ and $\rho_{\mathsf{B}} = R_3$, which is allowed because $3R_3 + 1 \le \ell_w(x')$ by (P0). Combined with Lem. 3.37 (c) and Thm. 3.32 (b), this gives the estimate $\|(EX' - \mathrm{d}^{\mathsf{V}}a_0)_p\| \le [8.24(\mathsf{N}_\sharp + 1) + 1]L^{-2}\hat{\varepsilon} + \mathsf{K}_{\mathrm{c}}^0\hat{\varepsilon}^2$, where the $+1$ in the bracket accounts for the term $\tau_{\mathrm{c}}\hat{\varepsilon}$ of Thm. 3.32 (b), since $\tau_{\mathrm{c}} \le L^{-2}$ gives $\tau_{\mathrm{c}}\hat{\varepsilon} \le L^{-2}\hat{\varepsilon}$. The second bound of clause (a) now follows from $|\cdot| \le \sqrt{2}\|\cdot\|$ and $\sqrt{2} \cdot 9.24 \le 17$. □

Proof of Cor. 3.38 (b). Step (o): dependence on the data, and sizes. We put $\hat{\varepsilon} := 17T\varepsilon'$. Let $\mathsf{V}' \in U^{17T}(\mathsf{B}_\omega; x')$, let ξ be such that $\chi_X(\xi) = 1$, and set $\eta := G^{\mathsf{V}}\xi - a_0$ and $F^\flat := EX' + \mathrm{d}^{\mathsf{V}}\eta$. Because $\omega \le Mr'$, we have $3D_w + 1 \le 3\omega + 1 \le \ell_w(x')$. Next, $r_T \ge 0.2175\varepsilon$; indeed $r_T \ge \sqrt{2}C_\varphi^\flat c_\xi\varepsilon = \frac{\sqrt{2}}{6.5}\varepsilon$, and $\frac{\sqrt{2}}{6.5} = 0.21757\ldots$. As $(4.1 \cdot 0.2175)^2 \ge 0.7952$, the floor (D3) gives $0.79\varepsilon^2 \le 0.1\varepsilon$, whence $\varepsilon \le 0.13$.

We turn to the geometry. Each site z of $X^\approx$ lies at sup distance $< M(\frac{L}{2} - 2 + r(x))$ from some site z_0 of the first cube of X. Consequently $L^{-1}|y(z) - y(z_0)|_\infty < \frac{M}{2} - \frac{2M}{L} + \frac{Mr(x)}{L} + 1 \le \frac{M}{2} + \frac{Mr(x)}{L} - 1$, where the second inequality rests on $M \ge L$, which holds as $m' \ge 1$ (item C.II 1.1). The coarse site $y(z_0) \in \square_0$ is within $\frac{1}{2}(M-1)$ of o. It follows that the blocks meeting $X^\approx$ lie within $M + \lceil Mr(x)/L \rceil$ of o, and that B_ω contains the box of radius D_w and the box of radius R_3 about the coarse site of each of these blocks. Every bond and every plaquette of $X^\approx$ is based in a block of this kind.

The field given on B_ω is extended by $\mathbf{1}$ to the remaining coarse bonds; (λ0) of the proposition in item C.II 2.1 applies to the extended field, because (e1) gives (h1). By (λ1) and (λ2), the variable $(G^{\mathsf{V}}\xi)_b$ depends on ξ through Y_b alone, hence within $\frac{1}{2}R$, and on the field within $r_w + \varrho_L + 1 \le R_3$. By the second and the third sentence of Thm. 3.32 (a), a_0 depends on the field only within R_3. In detail: the field is given on B_ω, and X' is defined there; hence $a_{0,b}$ and $(EX' - \mathrm{d}^{\mathsf{V}}a_0)_p$ take the values that are computed on the box of radius R_3 about $y(b)$ and about $y(p)$, respectively, and it is

to these values that we apply Thm. 3.32 (c) and clause (a) of the corollary, in (η1) and (η2) of this step. It follows from these dependences and the geometry above that none of the quantities used in this proof depends on the extension of the field by $\mathbf{1}$.

The data (η1), (η2) of Step (ii), with r_T in the role of r, are obtained as follows.

(η1) By (λ5), $|(G^{\mathsf{V}}\xi)_b| \le \sqrt{2}C_\varphi^\flat c_\xi \varepsilon$: the supremum in ($\lambda$5) runs over Y_b, and there $\chi_X(\xi) = 1$ bounds ξ. Thm. 3.32 (c), applied with $i = c$ and $c_- = y(b)$, gives $|a_{0,b}| \le \mathsf{a}_\mathrm{c}\hat\varepsilon$. Hence $|\eta_b| \le r_T$ at every bond of $X^{\approx}$.

(η2) We use clause (a), applied with the value $\hat\varepsilon = 17T\varepsilon'$ of this step; the inequality $\varepsilon' \le L^{1/2}\varepsilon$, which holds because $p_0' \le p_0$; the third floor $\mathsf{K}_\mathrm{c}(17T\varepsilon')^2 \le 0.12\varepsilon$; and $|\operatorname{Ad}_g Y| \le |g|^2|Y| \le 1.01|Y|$. Combined, they yield

$$|F_p^\flat| \le \mathsf{N}_\mathrm{c}L^{-2}\hat\varepsilon + \mathsf{K}_\mathrm{c}\hat\varepsilon^2 + 4\cdot 1.01\sqrt{2}C_\varphi^\flat c_\xi\varepsilon \le 1.9\varepsilon$$

at each plaquette of $X^{\approx}$. As to the numerals, $17T\mathsf{N}_\mathrm{c}L^{-3/2} = \frac{17}{19}$ and $4.04\sqrt{2}/6.5 \le 0.879$, and with 0.12 the total is at most 1.894, hence ≤ 1.9.

In the invariance argument of Step (iii), a gauge transformation u is extended by $\mathbf{1}$ outside B_ω. The extended field, once transformed, and the extension of V'^u disagree only at the bonds leaving B_ω; at these bonds the transformed field takes unitary values, so that it lies in $\mathfrak{V}$. For a bond b and a plaquette p of $X^{\approx}$, the dependences and the geometry described earlier in this step show that the row $G^{\mathsf{V}}(b,\cdot)$ and the values $a_{0,b}$ and $(EX' - \mathrm{d}^{\mathsf{V}}a_0)_p$ do not involve the field at the bonds leaving B_ω. Finally, B_ω is embedded: by (Tor) we have $2\omega + 1 < N$, as $2\omega + 1 \le 2Mr' + 1 < 16Mr(x)$.

Step (i): an expansion of the curvature of the chart. The data are: numbers $\hat\varepsilon \le \frac{1}{40}$ and $r > 0$ such that $4.1r + 2.1\hat\varepsilon \le 1$; a fine plaquette $p = (z;\mu\nu)$, with $y = y(p)$; a field V' that satisfies (h1) at the coarse bonds and (h3) with parameter $\hat\varepsilon$ at the coarse plaquette that occurs below; and η with $|\eta_b| \le r$ at the bonds of p, where $\eta := 0$ on $\mathcal{T}$. We claim the bound

$$|\partial\mathsf{U}(p) - \mathbf{1} - iF_p^\flat| \le 0.72(4.1r + 2.1\hat\varepsilon)^2$$

for $\mathsf{U} = \mathsf{U}(\eta;\mathsf{V}')$ and $F^\flat = EX' + \mathrm{d}^{\mathsf{V}}\eta$.

For the proof, let $\eta_1,\dots,\eta_4$ denote the values of η at (z,μ), $(z+e_\mu,\nu)$, $(z+e_\nu,\mu)$, (z,ν); we say that z_μ is at the top if $z_\mu = y_\mu + \mathsf{h}_L$, and the same definition, with ν in place of μ, applies to z_ν. If neither z_μ nor z_ν is at the top, then $\partial\mathsf{U}(p) = e^{i\eta_1}e^{i\eta_2}e^{-i\eta_3}e^{-i\eta_4}$. If only z_μ is at the top, let $c = (y,\mu)$: the first and the third bond of p cross c, while the second is based in $B(c_+)$, so that

$$\partial\mathsf{U}(p) = e^{i\eta_1}\mathsf{V}'(c)e^{i\eta_2}\mathsf{V}'(c)^{-1}e^{-i\eta_3}e^{-i\eta_4} = e^{i\eta_1}e^{i\operatorname{Ad}_{\mathsf{V}'(c)}\eta_2}e^{-i\eta_3}e^{-i\eta_4}.$$

If only z_ν is at the top, let $c = (y,\nu)$: now the second and the fourth bond cross c, while the third is based in $B(c_+)$, and $\partial\mathsf{U}(p)$ equals $e^{i\eta_1}e^{i\eta_2}e^{-i\operatorname{Ad}_{\mathsf{V}'(c)}\eta_3}e^{-i\eta_4}$.

If both z_μ and z_ν are at the top, then p is an edge plaquette under $P = (y;\mu\nu)$, whose curvature, written through its four coarse bonds, is $\partial\mathsf{V}'(P) = \mathsf{V}'(c_1)\mathsf{V}'(c_2)\mathsf{V}'(c_3)^{-1}\mathsf{V}'(c_4)^{-1}$, and

$$\partial\mathsf{U}(p) = e^{i\eta_1}\mathsf{V}'(c_1)e^{i\eta_2}\mathsf{V}'(c_2)\mathsf{V}'(c_3)^{-1}e^{-i\eta_3}\mathsf{V}'(c_4)^{-1}e^{-i\eta_4} = e^{i\eta_1}e^{i\operatorname{Ad}_{\mathsf{V}'(c_1)}\eta_2}e^{iX'_P}e^{-i\operatorname{Ad}_{\mathsf{V}'(c_4)}\eta_3}e^{-i\eta_4}.$$

In all four cases the transports that occur are the $T(y(p),y(b))$ of d^{V}; hence the sum of the exponents is $iF_p^\flat$. The norms of the exponents add up to at most $(2 + 2(1+\vartheta_0^\flat)^2)r + 2.06\hat\varepsilon \le 4.1r + 2.1\hat\varepsilon$, and (E) of item C.I 1.1 gives the claim.

Step (ii): the fine field lies in $\mathcal{U}_X(x)$. The data are: a number τ with $1 \le \tau \le 17T$ and $\hat\varepsilon = \tau\varepsilon' \le \frac{1}{40}$; a number $r > 0$ such that (D2), (D3) hold at (τ, r), that is, $\ell_w(x)(7.1r + 18D_w\hat\varepsilon) \le \frac{1}{2}\mathsf{g}_x$ and $(4.1r + 2.1\hat\varepsilon)^2 \le 0.1\varepsilon \le 1$; the inequality $3D_w \le \ell_w(x')$; an arbitrary non-empty set X of fine cubes; a set B of coarse sites containing the box of radius D_w about each site whose block meets $X^{\approx}$; a field $\mathsf{V}' \in U^\tau(\mathsf{B};x')$; and η with the two properties below.

(η1) $|\eta_b| \le r$ at the bonds of $X^{\approx}$.

(η2) $|F_p^\flat| \le 1.9\varepsilon$ at the plaquettes of $X^{\approx}$.

We claim that $\mathsf{U}(\eta;\mathsf{V}')$, restricted to the bonds of $X^{\approx}$, belongs to $\mathcal{U}_X(x)$. We prove the three conditions one by one.

(u3) The coarse bonds and the coarse plaquette that enter Step (i) lie within 1 of $y(p)$, hence in B, where (e1) and (e3) give (h1) and (h3). Step (i) therefore yields $|\partial\mathsf{U}(p)-\mathbf{1}| \le 1.9\varepsilon + 0.072\varepsilon < 2\varepsilon$.

(u1) We have $|\mathsf{U}(b)| \le e^r(1+\vartheta(x')+\mathsf{g}_x) < 1+\vartheta(x')+4\mathsf{g}_x \le 1+\vartheta(x)$; here $r \le \frac{1}{14}\mathsf{g}_x$ by (D2), and $\vartheta(x)-\vartheta(x') = \vartheta_0^\flat \frac{\log L}{\log x \log x'} \ge 4\mathsf{g}_x$.

(u2) Let π be a path in $X^{\approx}$ with $n \le \ell_w(x)$ bonds, say $b_1,\dots,b_n$, which passes through the sites $z_0,\dots,z_n$, and write $y_i := y(z_i)$ and $o_1 := y_0$. Then $L^{-1}|y_i - o_1|_\infty < \frac{n}{L}+1$, so every y_i is within D_w of o_1, and the box of radius D_w about o_1 is contained in B. Since by its definition in item C.II 1.4 a box is embedded, the hypothesis on B includes $2D_w+1<N$; in Step (iii), that inequality holds because $D_w<\omega$ and $2\omega+1<N$ by Step (o).

Inside this box, $\mathsf{k}(y) := \mathsf{V}'(\Gamma(o_1,y))$ obeys $|\mathsf{k}(y)| < \mathsf{b}(x') + \mathsf{g}_x < 2$ by (e2), since every comb there has at most $3D_w$ bonds. With $g_i := \mathsf{k}(y_{i-1})\mathsf{U}(b_i)^{\pm1}\mathsf{k}(y_i)^{-1}$ we have $\mathsf{U}(\pi) = g_1\cdots g_n\mathsf{k}(y_n)$. For a bond of π that lies inside a block, $g_i = \mathsf{k}e^{\pm i\eta}\mathsf{k}^{-1}$, and $|g_i| \le 1+4(e^r-1) \le e^{7.1r}$. For a crossing bond of c, the factor g_i is one of this kind multiplied by $\mathsf{W}(c)^{\pm1}$, and $|\mathsf{W}(c)^{\pm1}| = |\mathsf{W}(c)| \le e^{18D_w\hat\varepsilon}$ by Lem. 3.27 (a), applied in this box; the proof of that clause invokes (h2) only along combs, so that $3D_w \le \ell_w(x')$ suffices for it.

Therefore

$$|\mathsf{U}(\pi)| < e^{\ell_w(x)(7.1r+18D_w\hat\varepsilon)}(\mathsf{b}(x')+\mathsf{g}_x) \le e^{\mathsf{g}_x/2}(\mathsf{b}(x')+\mathsf{g}_x) < \mathsf{b}(x')+4\mathsf{g}_x \le \mathsf{b}(x),$$

where for the last inequality we use $\mathsf{b}(x)-\mathsf{b}(x') = \frac{\log L}{\log x\log x'} \ge 4\mathsf{g}_x$.

Step (iii): proof of claim (1). By Step (o), Step (ii) applies with $(\tau,r)=(17T,r_T)$ and $\mathsf{B}=\mathsf{B}_\omega$; thus, wherever $\chi_X(\xi)=1$, the field U on the bonds of $X^{\approx}$ belongs to $\mathcal{U}_X(x)$. Since $|e^w-1| \le 2|w|$ for $|w| \le \frac12$, the bound on $E^a(X,\cdot)$ gives the bound of claim (1). As to holomorphy: by (λ4) and (λ7) the entries of G^{V} and C^{V} are holomorphic at the extended field, which belongs to $\mathfrak{V}^\circ$ by (e1); by (e3), X' is given by a power series; the chart is a polynomial in V' and an entire function of η; and the integrand is bounded and depends on finitely many ξ_b only. As to invariance: (λ3) and the covariance of g^{V} and of X' give $\eta[\mathsf{V}'^u, \mathrm{Ad}_u\,\xi] = \mathrm{Ad}_u\,\eta[\mathsf{V}',\xi]$; moreover $\mathsf{U}(\mathrm{Ad}_u\,\eta;\mathsf{V}'^u) = \mathsf{U}(\eta;\mathsf{V}')^{\tilde u}$ with the fine gauge transformation $\tilde u(z) := u(y(z))$; and E^a, χ_X and the law of ξ are invariant. This proves claim (1).

Step (iv): the change of a bounded holomorphic invariant function. The data are: an embedded box B all of whose sites lie within ω of a site $o\in\mathsf{B}$; the inequalities $3\omega+1\le\ell_w(x')$, $T>1$ and $17T\varepsilon' \le \frac{1}{40}$, together with (C0), (C1), (C2); the set $\Omega := U^{17T}(\mathsf{B};x')$; and a function Φ on Ω that is holomorphic, invariant under the gauge transformations with values in G, and bounded by $|\Phi|\le\mathsf{S}$. The claim is that $|\Phi(\mathsf{V}')-\Phi(\mathbf{1})| \le 2\mathsf{S}T^{-2}$ at every field V' on B that satisfies (u1), (u2), (u3) at x'.

We first build a curve of fields. The field V' has (h2) and (h3) at ε'. By Lem. 3.27 (a) and (C0), $|\mathsf{W}(c)-\mathbf{1}| \le e^{18\omega\varepsilon'}-1 \le 19\omega\varepsilon' \le 0.106$, and $A'_c := -i\log\mathsf{W}(c)$ is a traceless matrix with $|A'_c| \le \mathsf{a} := 21\omega\varepsilon'$. For complex t we put $\mathsf{V}'_t(c) := \mathsf{k}(c_-)^{-1}e^{itA'_c}\mathsf{k}(c_+) = \mathsf{V}'(c)(\mathbf{1}+\delta_c)$, where $\mathbf{1}+\delta_c = \mathsf{k}(c_+)^{-1}e^{i(t-1)A'_c}\mathsf{k}(c_+)$; this field is entire in t, and $\mathsf{V}'_1 = \mathsf{V}'$. Now let $|t|<T$. Then $(T+1)\mathsf{a}\le 1$ by (C1), and therefore $|\delta_c|, |(\mathbf{1}+\delta_c)^{-1}-\mathbf{1}| \le 4(e^{(T+1)\mathsf{a}}-1) \le \delta := 168(T+1)\omega\varepsilon'$. We show that V'_t has the three properties that define Ω.

(e1) $|\mathsf{V}'_t(c)| < (1+\delta)(1+\vartheta(x')) \le 1+\vartheta(x')+1.01\delta \le 1+\vartheta(x')+\mathsf{g}_x$, where the last inequality comes from (C1).

(e2) Take a path of $n\le\ell_w(x')$ bonds, and shift each factor $(\mathbf{1}+\delta_c)^{\pm1}$ to the left past the partial products of V', whose norms are <2 by (u2). This gives

$$|\mathsf{V}'_t(\pi)| \le (1+4\delta)^n|\mathsf{V}'(\pi)| < e^{4n\delta}\mathsf{b}(x') \le \mathsf{b}(x')+2(e^{4n\delta}-1) < \mathsf{b}(x')+\mathsf{g}_x$$

with $4n\delta \le \frac{672}{2640}\mathsf{g}_x$ by (C1).

(e3) The computation of Step (1) of the proof of Thm. 3.34 gives $\partial \mathsf{V}'_t(P) = \mathsf{k}(y_P)^{-1}\Pi_t \mathsf{k}(y_P)$ with $\Pi_t = e^{itA'_1}e^{itA'_2}e^{-itA'_3}e^{-itA'_4}$. By (E), $|\Pi_t - \mathbf{1} - it(\mathrm{d}'A')_P| \le (4|t|\mathsf{a})^2$, where $4T\mathsf{a} = 84T\omega\varepsilon' \le 1$ by (C0). At $t = 1$, (u3) gives $|\Pi_1 - \mathbf{1}| < 8\varepsilon'$, and therefore $|(\mathrm{d}'A')_P| < 8\varepsilon' + 16\mathsf{a}^2$. It follows that $|\partial \mathsf{V}'_t(P) - \mathbf{1}| < 32T\varepsilon' + 128T^2\mathsf{a}^2 \le 34T\varepsilon'$; for the last inequality note that $128 \cdot 441\, T\omega^2\varepsilon' = 56448\, T\omega^2\varepsilon' \le 2$ by (C2).

Thus $\mathsf{V}'_t \in \Omega$ for $|t| < T$; on this disc the function $f(t) := \Phi(\mathsf{V}'_t)$ is holomorphic, and $|f| \le \mathsf{S}$.

We now bring in complex gauge transformations. The set Ω is open. Let $\mathsf{Z} \in \Omega$ and $\lambda : \mathsf{B} \to \mathfrak{g}_{\mathbb{C}}$. The derivative of $\Phi(\mathsf{Z}^{\exp(i\sigma\lambda)})$ at $\sigma = 0$ is $\mathbb{C}$-linear in λ, as Φ is holomorphic, and it vanishes whenever λ is $\mathfrak{g}$-valued, as Φ is invariant; since every λ can be written as $\lambda_1 + i\lambda_2$ with $\mathfrak{g}$-valued summands, the derivative vanishes for every λ. By the argument of Step (2) of the proof of Lem. 3.33, it follows that $\Phi(\mathsf{Z}^{u_s})$ is constant along each C^1 curve of gauge transformations $u_s : \mathsf{B} \to \mathrm{SL}(2,\mathbb{C})$ for which $\mathsf{Z}^{u_s} \in \Omega$ throughout.

Write $\mathsf{k}(y)^{-1} = k_0(y)e^{h(y)}$ in polar form, with $k_0 \in G$ and h hermitian and traceless, and set $u_s := k_0 e^{sh}$ for $0 \le s \le 1$. For a path π from y to y'', Cordes' inequality, in the form used in Step (1) of the proof of Lem. 3.33, gives

$$|\mathbf{1}^{u_s}(\pi)| = |e^{sh(y)}e^{-sh(y'')}| \le |e^{h(y)}e^{-h(y'')}|^s = |\mathsf{V}'_0(\pi)|^s \le |\mathsf{V}'_0(\pi)|.$$

Since $\mathsf{V}'_0 = \mathbf{1}^{u_1}$ lies in Ω, each $\mathbf{1}^{u_s}$ therefore has (e1) and (e2); its curvature is $\mathbf{1}$; and so $\mathbf{1}^{u_s} \in \Omega$. Consequently $f(0) = \Phi(\mathbf{1}^{u_1}) = \Phi(\mathbf{1}^{u_0}) = \Phi(\mathbf{1})$, because u_0 takes values in G.

Next we show $f'(0) = 0$. The set $\{\mathbf{1}^{u_s}\}$ is compact and contained in the open set Ω; hence some $t_0 > 0$ has the property that $\mathsf{Z}_t^{u_s} \in \Omega$ whenever $|t| < t_0$, for every s, where $\mathsf{Z}_t := (e^{itA'_c})$. As $\mathsf{V}'_t = \mathsf{Z}_t^{u_1}$, we obtain $f(t) = \Phi(\mathsf{Z}_t^{u_0}) = \Phi(\mathsf{Z}_t)$ for these t. The derivative of $\Phi(\mathsf{Z}_t)$ at $t = 0$ is a $\mathbb{C}$-linear function of A', and it is unchanged under $A' \mapsto \mathrm{Ad}_u A'$ for every constant $u \in G$, because $\mathsf{Z}_t^u = (e^{it\,\mathrm{Ad}_u A'_c})$. It therefore takes the same value at $\int_G \mathrm{Ad}_u A'\, du$. Now $\int_G \mathrm{Ad}_u Y\, du$ commutes with G and is traceless, so it equals 0, and the derivative vanishes.

To finish, $(f(t) - f(0))/t^2$ is holomorphic on $|t| < T$, and $|f - f(0)| \le 2\mathsf{S}$ there. The maximum principle on the circles $|t| = \rho$ with $1 < \rho < T$, followed by the limit $\rho \to T$, gives $|f(1) - f(0)| \le 2\mathsf{S}T^{-2}$, and this is the claim.

Step (v): proof of claims (2) *and* (3). Claim (2) is the case $\mathsf{B} = \mathsf{B}_\omega$, $\Phi = \mathcal{G}_X$, $\mathsf{S} = 2\mathsf{A}c_a(x)e^{-\kappa(\#X-1)}$ of Step (iv), the bound S being supplied by claim (1); moreover $\mathsf{B}_\omega \subset \square_0^{\approx}$, since $\omega \le Mr'$. For claim (3), observe that $\hat{\square}_0$ contains L^3 cubes that can serve as first cubes. The bound of item C.II 1.1 on the number of connected sets gives, for any given cube, $\sum_{X \ni \text{the given cube}} e^{-\kappa(\#X-1)} \le \sum_{n\ge1}(26e^{1-\kappa})^{n-1} \le (1 - 26e^{-5})^{-1} \le 1.2124$. Moreover $L^3T^{-2} = 361\mathsf{N}_{\mathrm{c}}^2$, and $c_a(x') = L^{1/2}(p'_0/p_0)^3 c_a(x) \ge 2^{-3/4}L^{1/2}c_a(x)$. Hence the sum in claim (3) is $\le 4 \cdot 1.2124 \cdot 361\, \mathsf{N}_{\mathrm{c}}^2\, 2^{3/4}L^{-1/2}\mathsf{A}c_a(x') \le \frac{1}{4}\mathsf{A}c_a(x')$, where the last inequality follows from ($\mathrm{L2}^{\mathrm{su}}$) because $4 \cdot 1.2124 \cdot 361/(0.25 \cdot 2^{-3/4}) \le 11777.3 \le 1.2 \cdot 10^4$. □

C.II 5. (S-loc$_k$).

Definition 3.39 (Walks and their traces sorted by sets of cells)**.** Let a family of steps be given. A walk ω of k steps in this family consists of bonds $b_0, \dots, b_k$ together with, for each i, either a step $\mathsf{s}_i : b_{i-1} \to b_i$ of the family or the trivial step; the trivial step has $b_i = b_{i-1}$, weight 1 and length 0, and its set of cells is $\mathsf{c} = \{\text{the cell of } b_i\}$. The length of ω is $|\omega| := \sum |\mathsf{s}_i|$, and its set of cells, $\mathsf{c}(\omega) := \bigcup \mathsf{c}(\mathsf{s}_i)$, is a connected set of cells. Whatever the number k of its steps,

$$\#\mathsf{c}(\omega) \le 27(\frac{|\omega|}{M} + 1). \tag{Cnt}$$

Indeed, $\pi_\omega = \pi_{\mathsf{s}_1} \cdots \pi_{\mathsf{s}_k}$ is a coarse lattice path of $|\omega|$ bonds, and $\mathsf{c}(\omega) \subset \{\text{cells of its sites}\}$. Along M consecutive sites this path moves by at most $M - 1$, so such sites lie in the cube of 27 cells centred at the cell of the first of them; and the sites of the path, in the order in which it passes through

them, split into $\lceil \frac{|\omega|+1}{M} \rceil \le \frac{|\omega|}{M}+1$ runs of at most M consecutive sites, each of which therefore meets at most 27 cells. This proves (Cnt).

For a set Y of cells, let $\mathfrak{S}$ be the family of Y (item C.II 1.5 (c)), and let $K^{\mathsf{V}}_{[\mathsf{Y}]}(\zeta) := K^{\mathsf{V}}_{\mathfrak{S}}(\zeta)$, where $K^{\mathsf{V}}_{\mathfrak{S}}(\zeta)$ is the operator of item C.II 1.5. We write $K_{[\mathsf{Y}]} = K_{[\mathsf{Y}]}(1)$, and $K_{[\mathsf{Y}]}[\mathsf{V}']$ for $K_{[\mathsf{Y}]}$ at the field V'. For a non-empty finite set Y of cells, for $k \ge 1$ and for $j \ge 0$, we define the traces sorted by sets of cells,

$$T_k(Y;\mathsf{V}') := \sum_{\mathsf{Y}\subset Y} (-1)^{\#Y\setminus\mathsf{Y}} \operatorname{tr}(K^{\mathsf{V}}_{[\mathsf{Y}]})^k = \sum_j T_{k,j}(Y;\mathsf{V}'), \qquad T_{k,j} := \sum_{\mathsf{Y}\subset Y} (-1)^{\#Y\setminus\mathsf{Y}} \big[\operatorname{tr} K^{\mathsf{V}}_{[\mathsf{Y}]}(\zeta)^k\big]_j,$$

where $[\,\cdot\,]_j$ stands for the coefficient of ζ^j in the polynomial $\operatorname{tr} K^{\mathsf{V}}_{[\mathsf{Y}]}(\zeta)^k$, and we put

$$\mathsf{D}^q(Y,\mathsf{V}') := \sum_{k\ge 1} \tfrac{1}{k}\big[T_k(Y;\mathsf{V}') - T_k(Y;\mathbf{1})\big].$$

■

Conditions on the letters. We use the following conditions.

(M4$^\flat$) $M \ge \frac{54}{\mu}(\kappa^\sharp+3) = 216\,\frac{b}{\gamma}\,\frac{\log\Lambda}{\theta_2}\,(\kappa^\sharp+3)$.

(R$^\flat$) $c_R\,\mu \ge 1.35$.

The condition (M4$^\sharp$), that is $M \ge 216\frac{b}{\gamma}(\kappa^\sharp+3)$, is a consequence of (M4$^\flat$). Indeed, the right side of (M4$^\flat$) is that of (M4$^\sharp$) multiplied by the factor $\frac{\log\Lambda}{\theta_2} \ge 2\log 16 \ge 5.5$; this lower bound holds since $\theta_2 \le \frac12$ and $A^\flat \ge 14b$, and the latter inequality holds since $2e^2 \ge 14$ and $384e^8\mathsf{g}_2(\frac{\lambda}{2})^2 \ge 14\cdot 48\,\mathsf{g}(\lambda)^2$. The numbers Λ, θ_2, μ are computed from (L, a, K, λ) before M is chosen, and they do not depend on M.

If $\lambda L \ge 12$, then (R$^\flat$) is a consequence of $c_R\theta_{\mathbb{C}} \ge 128$. In that case $\theta_2 = \frac12$, and with $t = A^\flat/b$ we get $c_R \ge 768/\theta^\flat \ge 1536\,t\frac{b}{\gamma} \ge 10.8\log(2+t)\frac{b}{\gamma}$, which is the lower bound on c_R to which (R$^\flat$) reduces when $\theta_2 = \frac12$. At the starred values the inequality $\lambda_* L \ge 12$ holds, because $L \ge L_A \ge 36\,\mathsf{g}(\frac{\lambda_*}{2})/\gamma_*$, $\lambda_* = \gamma_*^{1/2}/44$ and $\gamma_* \le 1$, and because $\mathsf{g}(\lambda) \ge 1 + 26e^{-1} + 98e^{-2} > 16$ for $0 < \lambda \le 1$, so that $\mathsf{g}(\frac{\lambda_*}{2}) \ge 16$. The starred values and L_A are those of §2.2.

Further, we use the following condition.

(M2) $M \ge 1920(\kappa+4)c_R = 192\kappa^\sharp c_R$.

We define the number

$$\mathsf{K}_1^\flat := 18\,(LM)^3\Big[\frac{\Lambda}{1-e^{-\theta_2}} + \frac{2b}{\gamma}\Big]e^{27(\kappa^\sharp+3+\kappa_+)}.$$

Floors. We use the following floors (S1) and (S2) on x.

(S1) The floor (S1) consists of four clauses, in this order: $p_0' \ge 1$; $r' \ge 1$; $18M(r'+1)\varepsilon' \le \frac{1}{10}$; $4\big(170\,M(r'+1)\log x'/\vartheta_0^\flat\big)^2\varepsilon' \le 1$.

(S2) The floor (S2) has six clauses. The first four are $3L^3(2R_2+1)^3\frac{4}{\gamma} \le x$; $(\kappa^\sharp+1)(3r_u+3) \le \log x$; $18(LM)^3 \le x$; $4(b+A^\flat)e^{27\kappa^\sharp} \le x$. The last two are $r_u \le r'$ and $x^{-(\mathsf{n}-3)} \le \varepsilon'$, where $\mathsf{n} \ge 4$.

Under (M2) we have $3r_u + 3 \le \frac{6c_R\log x+12}{M} + 6$ and $(\kappa^\sharp+1)\frac{6c_R}{M} \le 0.04$, so that the second clause of (S2) holds as soon as $\log x \ge 11(\kappa^\sharp+1)$. Thus, if (M2) is assumed, every clause of (S1) and of (S2) holds for all large x.

The next theorem shows that $\log\det q_\ell^V$ is local in V', at a level whose rows may have any depth: the rows are those of item C.II 1.2 with $K \ge 0$, so that $\mathsf{H}_y = \{0\}$ is admitted.

Theorem 3.40 (The determinant of the form is local in the coarse field, at a level with rows)**.** Let $L \ge 3$ be odd with $aL \ge 18$, let the rows be as in item C.II 1.2, and let $\kappa \ge 6$; we assume (M4$^\flat$), (R$^\flat$) and (Tor). Then the following hold.

(a) At every $V' \in \mathcal{V}'$ we have $\log\det q_\ell^{\mathbf{1}} - \log\det q_\ell^{V} = \sum_Y \mathsf{D}^q(Y, V')$. Moreover, $T_{k,j}(Y;\cdot) = 0$ unless Y is connected and $\#Y \le 27(\frac{j}{M}+1)$; in particular $\mathsf{D}^q(Y,\cdot) = 0$ if Y is not connected.
(b) Every gauge transformation u with values in G leaves $\mathsf{D}^q(Y,\cdot)$ unchanged; so does every u taking its values in $\mathrm{SL}(2,\mathbb{C})$, provided that V' and V'^u both belong to the domain defined at the end of this clause. The function $\mathsf{D}^q(Y,\cdot)$ is real at G-valued fields, and it depends on the field only through $\mathrm{Ad}_{\mathsf{V}'(c)}$ for the bonds c of $Y^\sim$. It is holomorphic on the domain $\{|\mathsf{V}'(c)| < 1 + \vartheta_0^\flat$ on the bonds of $Y^\sim\}$, and it vanishes at those pure gauges $\mathbf{1}^u$, with u taking values in G or in $\mathrm{SL}(2,\mathbb{C})$, that belong to this domain.
(c) At every $V' \in \mathcal{V}'$, and at every field of this domain, $|\mathsf{D}^q(Y,\mathsf{V}')| \le \mathsf{K}_1^\flat\, e^{-(\kappa^\sharp+\kappa_+)\#Y}$.
(d) If (S1) holds, then on $\mathcal{U}_Y(x')$, and hence at every G-valued field that is tame on $Y^{\approx}$, we have $|\mathsf{D}^q(Y,\mathsf{V}')| \le \mathsf{K}_1^\flat\, \varepsilon'\, e^{-\kappa^\sharp(\#Y-1)}$.

■□

Proof. Throughout the proof $n := \#Y$ and $j_0 := \max\{0, M(\frac{n}{27} - 1)\}$, and we put $\mathsf{d}_Y := 9(LM)^3 n$; for every $\mathsf{Y} \subset Y$ this number bounds the dimension of $\ell^2(I_{\mathsf{Y}};\mathbb{C}^3)$.

Step (a): sums over sets of cells. By Lem. 3.25 (a), $\|K_{[\mathsf{Y}]}\| \le \varkappa < 1$; consequently $-\log\det(q_{[\mathsf{Y}]}/b) = \sum_k \frac{1}{k}\operatorname{tr} K_{[\mathsf{Y}]}^k$. For the set $\mathsf{Y} = \mathfrak{c}$ of all cells, $q_{[\mathfrak{c}]} = q_\ell$. Möbius inversion states that $\sum_{Y\subset\mathfrak{c}}\sum_{\mathsf{Y}\subset Y}(-1)^{\#Y\setminus\mathsf{Y}} f(\mathsf{Y}) = f(\mathfrak{c})$, and we use it for functions with $f(\emptyset) = 0$: applied to $-\log\det(q_{[\mathsf{Y}]}/b)$, at V' and at $\mathbf{1}$, it yields, through the definition of D^q, the identity of (a). Next we expand the k-th power and write $\operatorname{tr} K_{[\mathsf{Y}]}(\zeta)^k$ as a sum indexed by the closed walks ω of k steps with $\mathsf{c}(\omega) \subset \mathsf{Y}$, the term of ω being $\zeta^{|\omega|}$ multiplied by a number and by $\operatorname{tr}_{\mathfrak{g}}\mathrm{Ad}_{\mathsf{V}'(\pi_\omega)}$. This rests on the fact that a step lies in the family of Y precisely when $\mathsf{c}(\mathsf{s}) \subset \mathsf{Y}$ (item C.II 1.5 (c)). For a fixed walk, $\sum_{\mathsf{c}(\omega)\subset\mathsf{Y}\subset Y}(-1)^{\#Y\setminus\mathsf{Y}} = 1\{\mathsf{c}(\omega) = Y\}$. Therefore only the walks with $\mathsf{c}(\omega) = Y$ and $|\omega| = j$ contribute to $T_{k,j}(Y;\cdot)$; for these, Y is connected, and (Cnt) gives $\#Y \le 27(\frac{j}{M}+1)$.

Step (b): invariance, dependence and holomorphy. Let u be a gauge transformation with values in $\mathrm{SL}(2,\mathbb{C})$. For a step $\mathsf{s} : b \to b'$ we have the identity of matrices $\mathsf{T}_{\mathsf{s}}[\mathsf{V}'^u] = \mathrm{Ad}_{u(y(b))}\,\mathsf{T}_{\mathsf{s}}\,\mathrm{Ad}^{-1}_{u(y(b'))}$, so that $K_{[\mathsf{Y}]}[\mathsf{V}'^u] = \mathrm{Ad}_u\, K_{[\mathsf{Y}]}[\mathsf{V}']\,\mathrm{Ad}_u^{-1}$, and the traces at the two fields coincide for every k and every $\mathsf{Y} \subset Y$. By Lem. 3.25 (a), the series defining D^q converges at each of the two fields V', V'^u as soon as that field belongs to the domain; this proves the invariance. For the unit field the same identity reads $T_k(Y;\mathbf{1}^u) = T_k(Y;\mathbf{1})$, and so $\mathsf{D}^q(Y,\cdot)$ vanishes at the pure gauges of the domain. At G-valued fields the function $\mathsf{D}^q(Y,\cdot)$ is real: for $g \in G$ the map Ad_g is real on $\mathfrak{g}$, being real orthogonal in the real basis $\sigma_1,\sigma_2,\sigma_3$, and q_{s} is real, so that $K_{[\mathsf{Y}]}$ and every trace $\operatorname{tr} K_{[\mathsf{Y}]}^k$ are real. As to the dependence on the field, a step of kind (R) depends only on bonds with both ends in cells of Y, a step of kind (P) on at most two bonds from $y(p)$, each within 1 of $y(b)$, and a step of kind (Q) on one bond from c_- within 1 of $y(b)$; this gives the dependence on the bonds of $Y^\sim$ alone. Finally, again by Lem. 3.25 (a), the series converges uniformly on the domain, and its terms are polynomials in the entries of the matrices $\mathrm{Ad}^{\pm1}_{\mathsf{V}'(c)}$; hence $\mathsf{D}^q(Y,\cdot)$ is holomorphic there.

Step (c): the bound on the domain. By (a), $T_k = \sum_{j\ge j_0} T_{k,j}$, and we bound T_k in two ways at every field of the domain. The first bound, $|T_k(Y)| \le 2^n \mathsf{d}_Y \varkappa^k$, comes from Lem. 3.25 (a). For the second, Lem. 3.25 (b) gives $|T_{k,j}(Y)| \le 2^n \mathsf{d}_Y \Lambda^k e^{-\theta_2 j}$, and summation over $j \ge j_0$ yields $|T_k(Y)| \le 2^n \mathsf{d}_Y \Lambda^k \frac{e^{-\theta_2 j_0}}{1-e^{-\theta_2}}$. Let $k_1 := \lfloor \frac{\theta_2 j_0}{2\log\Lambda} \rfloor$. For $k \le k_1$ we use the second bound, together with $\sum_{k\le k_1}\frac{1}{k}\Lambda^k \le \Lambda^{k_1+1} \le \Lambda e^{\theta_2 j_0/2}$, which holds as $\Lambda \ge 2$. For $k > k_1$ we use the first, together with $\sum_{k>k_1}\frac{1}{k}\varkappa^k \le \frac{2b}{\gamma}\varkappa^{k_1+1} \le \frac{2b}{\gamma}e^{-\frac{\gamma}{2b}\frac{\theta_2 j_0}{2\log\Lambda}} = \frac{2b}{\gamma}e^{-\mu j_0}$. Since $\mu \le \frac{1}{2}\theta_2$, we add the bounds for $k \le k_1$ and for $k > k_1$ and obtain $\sum_k \frac{1}{k}|T_k(Y)| \le 2^n \mathsf{d}_Y [\frac{\Lambda}{1-e^{-\theta_2}} + \frac{2b}{\gamma}] e^{-\mu j_0}$.

Now ($\mathrm{R}^\flat$) gives $\kappa_+ = \frac{M}{40 c_R} \le \frac{\mu M}{54}$, and (M4$^\flat$) gives $\frac{\mu M}{54} \ge \kappa^\sharp + 3$; therefore $\frac{\mu M}{27} \ge \kappa^\sharp + 3 + \kappa_+$. If $n \le 27$, then $j_0 = 0$, so $e^{-\mu j_0} = 1 \le e^{(\kappa^\sharp+3+\kappa_+)(27-n)}$. If $n > 27$, then $j_0 = \frac{M}{27}(n-27)$, so

$e^{-\mu j_0} = e^{-\frac{\mu M}{27}(n-27)} \le e^{-(\kappa^\sharp+3+\kappa_+)(n-27)}$. In both cases $e^{-\mu j_0} \le e^{27(\kappa^\sharp+3+\kappa_+)} e^{-(\kappa^\sharp+3+\kappa_+)n}$. As $\mathsf{d}_Y = 9(LM)^3 n$ and $n 2^n e^{-3n} \le 1$, the sum $\sum_k \frac{1}{k}|T_k(Y)|$ is at most one half of the bound in (c). Since D^q is defined in terms of the values of T_k at the two fields V' and $\mathbf{1}$, both of which belong to the domain, (c) follows.

Step (d): the bound on $\mathcal{U}_Y(x')$. Suppose first that $n > r'$. Then $e^{-\kappa_+ n} \le e^{-\kappa_+ r'} \le e^{-\frac{1}{2}\log x'} \le \varepsilon'$, where we use $r' \ge \frac{20 c_R}{M} \log x'$ and $p_0' \ge 1$. A member of $\mathcal{U}_Y(x')$ satisfies (u1), and $\vartheta(x') < \vartheta_0^\flat$; hence it belongs to the domain of (b), and (c) yields the bound.

Suppose next that $n \le r'$. We may take Y connected, since otherwise $\mathsf{D}^q(Y,\cdot) = 0$ by (a). Fix a cell $\square \in Y$ and a site o of $\square$. Then $Y^\sim \subset \square^{+n} \subset \mathsf{B} :=$ the sites of $\square^{+r'} \subset Y^\approx$; here B is a box, embedded by (Tor), all of whose sites lie within $\omega_\mathsf{B} := M(r'+1)$ of o, and $3\omega_\mathsf{B} + 1 \le \ell_w(x')$. Let $\Omega := \{$fields on the bonds of B with $|\mathsf{V}(c)| < 1 + \vartheta_0^\flat\}$. By (b) and (c), the function $\Phi := \mathsf{D}^q(Y,\cdot)$ is holomorphic on Ω, and $|\Phi| \le \mathsf{S} := \mathsf{K}_1^\flat e^{-\kappa^\sharp n}$.

Let $\mathsf{V}' \in \mathcal{U}_Y(x')$. By (u2) and (u3), it satisfies (h2) and (h3) on B, with $\hat\varepsilon = \varepsilon'$. The hypothesis $\hat\varepsilon \le \frac{1}{40}$ of Lem. 3.27 reads $\varepsilon' \le \frac{1}{40}$ here; this follows from the third clause of (S1), which together with $M \ge 3$ and $r' \ge 1$ even gives $\varepsilon' \le \frac{1}{1080}$. Lem. 3.27 (a) and (S1) then give $|\mathsf{W}(c) - \mathbf{1}| \le 19\omega_\mathsf{B}\varepsilon'$. We set $A_c' = -i \log \mathsf{W}(c)$. This matrix satisfies $|A_c'| \le \mathsf{a} := 21\omega_\mathsf{B}\varepsilon'$, and it is traceless because $\det \mathsf{W}(c) = 1$ and $|\mathsf{W}(c) - \mathbf{1}| < 1$. For complex t we consider the field $\mathsf{V}_t'(c) = \mathsf{k}(c_-)^{-1} e^{itA_c'} \mathsf{k}(c_+) = \mathsf{V}'(c)(\mathbf{1} + \delta_c)$, where $\mathbf{1} + \delta_c = \mathsf{k}(c_+)^{-1} e^{i(t-1)A_c'} \mathsf{k}(c_+)$. By Lem. 3.27 (a) we have $|\mathsf{k}| \le 2$; hence, if $(T+1)\mathsf{a} \le 1$, then $|\delta_c| \le 4(e^{(T+1)\mathsf{a}} - 1) \le 4(e-1)(T+1)\mathsf{a} \le 8(T+1)\mathsf{a}$ for $|t| < T$. We choose $T + 1 := \vartheta_0^\flat / (170\, \omega_\mathsf{B} \varepsilon' \log x')$. Then $(T+1)\mathsf{a} = \frac{21}{170} \frac{\vartheta_0^\flat}{\log x'} \le 1$, since $\vartheta_0^\flat \le 10^{-4}$ (item C.II 1.3) and $\log x' > 1$, the latter because $x > Le$ and $x' = x/L$ (item C.II 1.1); so the bound on $|\delta_c|$ holds for $|t| < T$, and $1.01 \cdot 8(T+1)\mathsf{a} \le \frac{169.7}{170} \frac{\vartheta_0^\flat}{\log x'}$. Hence, by (u1), $|\mathsf{V}_t'(c)| < (1 + |\delta_c|)(1 + \vartheta_0^\flat - \frac{\vartheta_0^\flat}{\log x'}) \le 1 + \vartheta_0^\flat$, that is, $\mathsf{V}_t' \in \Omega$. Membership in Ω depends neither on the curvature of V_t' nor on its values along paths.

The function $f(t) := \Phi(\mathsf{V}_t')$ is therefore holomorphic on the disc $|t| < T$, and $|f| \le \mathsf{S}$ there. By (b), $f(0) = 0$, because $\mathsf{V}_0' = \mathbf{1}^{\mathsf{k}^{-1}}$. For small $|t|$ we also have $\mathsf{Z}_t := (e^{itA_c'}) \in \Omega$, and $f(t) = \Phi(\mathsf{Z}_t^{\mathsf{k}^{-1}}) = \Phi(\mathsf{Z}_t)$ by (b): the gauge k^{-1} takes its values in $\mathrm{SL}(2,\mathbb{C})$, and Z_t and $\mathsf{Z}_t^{\mathsf{k}^{-1}} = \mathsf{V}_t'$ both lie in Ω, hence in the domain of (b). Moreover $\frac{d}{dt}\Phi(\mathsf{Z}_t)|_0 = 0$. Indeed, this derivative, which we denote by $\Lambda(A')$, is $\mathbb{C}$-linear in A', and, by (b), $\Lambda(\mathrm{Ad}_u A') = \Lambda(A')$ for every constant $u \in G$; hence $\Lambda(A') = \Lambda(\int_G \mathrm{Ad}_u A'\, du)$, whereas the matrix $\int_G \mathrm{Ad}_u A_c'\, du$, for each bond c, commutes with G and is traceless, so that it is 0.

Consequently $f(t)/t^2$ is holomorphic on the disc, and the maximum principle gives $|f(1)| \le \mathsf{S} T^{-2}$, once $t = 1$ is known to lie in the disc. By the last clause of (S1), $(T+1)^{-2} \le \frac{1}{4}\varepsilon'$, and the third clause of (S1) gives $\varepsilon' < 1$, so that $T > 1$ and $t = 1$ lies in the open disc $|t| < T$. Then $T \ge \frac{1}{2}(T+1)$, and so $T^{-2} \le \varepsilon'$. Since V_1' coincides with V' on the bonds of B, the value $f(1)$ equals $\mathsf{D}^q(Y, \mathsf{V}')$, and the bound of (d) follows. □

Definition 3.41 (The u-steps and the terms D^u, S°, S_k, D_k)**.** Let $\mathsf{u} = \mathsf{u}^\mathsf{V} = 1 - C^\mathsf{V} q_\ell^\mathsf{V}$. If ($\rho$1), ($\rho$2) and (P0) hold, then $\mathsf{u}(b,b') = 0$ for $\mathrm{dist}(b,b') > R + r_q$, and $R + r_q \le R_2$. Moreover, $\mathsf{u}(b,\cdot)$ depends on V' only through its values on the coarse bonds within R_2 of $y(b)$. Indeed, $C(b,\cdot)$ has this property with $R_2 - 1$ in place of R_2, while for $\mathrm{dist}(b,b'') \le R$ the row $q(b'',\cdot)$ has it with r_q and $y(b'')$ in place of R_2 and $y(b)$, by the bound of item C.II 1.2 that is measured from $y(b'')$ alone: every site of π_s, for a step s from b'', lies within r_q of $y(b'')$.

A u-step $b \to b'$ carries the set $\mathsf{c}(b,b') :=$ the cells met by $\Gamma(y(b), y(b'))$; this set is connected and has at most $\mathsf{n}_u := 3\lceil R_2/M \rceil + 3 = 3r_u + 3$ members, since the coordinate path is made of three straight stretches of at most R_2 bonds each. Under the hypotheses of Thm. 3.42 and at every $\mathsf{V}' \in \mathfrak{V}$, each row sum and each column sum of $\|\mathsf{u}(b,b')\|$ is at most $\upsilon_1 := 3L^3(2R_2+1)^3 \frac{4}{\gamma} x^{-\mathsf{n}}$. This

follows from Lem. 3.26 (a): the lemma assumes ($\lambda 0$), ($\rho 5$), $\frac{2}{\gamma}x^{-\mathsf{n}} \le \frac{1}{2}$ and $\mathsf{V}' \in \mathfrak{V}$, and (P1) gives $\frac{4}{\gamma}x^{-\mathsf{n}} \le 1$.

Let $\mathsf{V}' \in \mathfrak{V}$, and let Y be a connected set of cells. Under the hypotheses of Thm. 3.42 we have $\upsilon_1 \le \frac{1}{2}$ by (S2), so that the series in the following definitions converge absolutely:

$$\mathsf{D}^u(Y,\mathsf{V}') := -\sum_{m\ge 1}\frac{1}{m}\sum_{\substack{b_0,\dots,b_{m-1};\ b_m=b_0\\ \bigcup \mathsf{c}(b_i,b_{i+1})=Y}} \operatorname{tr}_{\mathfrak{g}} \prod_{i=0}^{m-1}\mathsf{u}(b_i,b_{i+1}),$$

$$S^\circ(Y;\mathsf{V}')(b,b') := \sum_{m\ge 1}\sum_{\substack{\mathsf{s}:b\to b_1;\ b_2,\dots,b_{m+1}=b'\\ \mathsf{c}(\mathsf{s})\cup\bigcup \mathsf{c}(b_i,b_{i+1})=Y}} q_{\mathsf{s}}\mathsf{T}_{\mathsf{s}}\prod_{i=1}^{m}\mathsf{u}(b_i,b_{i+1}),$$

and

$$S_k(Y;\mathsf{V}') := \tfrac{1}{2}(S^\circ(Y;\mathsf{V}') + S^\circ(Y;\mathsf{V}')^{\mathsf{T}}), \qquad \mathsf{D}_k(Y,\mathsf{V}') := -\mathsf{D}^q(Y,\mathsf{V}') - [\mathsf{D}^u(Y,\mathsf{V}') - \mathsf{D}^u(Y,\mathbf{1})].$$

■

Theorem 3.42 (The terms $\mathsf{D}^u(Y,\cdot)$ and $S_k(Y;\cdot)$ are local in the coarse field)**.** Let $L \ge 3$ be odd with $aL \ge 18$, let the rows be as in item C.II 1.2, and let $\kappa \ge 6$; we take a law with ($\lambda 0$), ($\rho 1$) to ($\rho 6$), and we assume (M4$^\flat$), (P0), (P1) and (S2). Then the following hold at every $\mathsf{V}' \in \mathfrak{V}$, and so in particular at every $V' \in \mathcal{V}'$, with Y ranging over the connected sets of cells.

(a) At $V' \in \mathcal{V}'$, $\log\det C^V = -\log\det q_\ell^V + \sum_Y \mathsf{D}^u(Y,V')$; moreover $S^{\mathsf{V}} = \sum_Y S_k(Y;\mathsf{V}')$; both sums converge absolutely.

(b) The functions $\mathsf{D}^u(Y,\cdot)$ and $S_k(Y;\cdot)$ depend on V' only through its values on the coarse bonds within R_2 of the sites of Y, and hence only through its values on bonds of Y^{+r_u}. They are holomorphic on $\mathfrak{V}^\circ$ and real at G-valued fields. For G-valued u, $\mathsf{D}^u(Y,\mathsf{V}'^u) = \mathsf{D}^u(Y,\mathsf{V}')$ and $S_k(Y;\mathsf{V}'^u) = \operatorname{Ad}_u S_k(Y;\mathsf{V}')\operatorname{Ad}_u^{-1}$. Finally, $S_k(Y)^{\mathsf{T}} = S_k(Y)$, and $S_k(Y)(b,b') = 0$ unless $b,b' \in \hat{Y}$.

(c) $|\mathsf{D}^u(Y,\mathsf{V}')| \le x^{-(\mathsf{n}-3)}e^{-\kappa^\sharp \#Y}$ and $\|S_k(Y;\mathsf{V}')\| \le x^{-(\mathsf{n}-3)}e^{-\kappa^\sharp\#Y}$.

■□

Proof. By (S2), $\upsilon_1 \le x^{-(\mathsf{n}-1)} \le \frac{1}{2}$ and $e^{(\kappa^\sharp+1)\mathsf{n}_u} \le x$.

Step (a): the two expansions. We have $\det C \det q = \det(1-\mathsf{u})$ with $\|\mathsf{u}\| < 1$, so that $\log\det(1-\mathsf{u}) = -\sum \frac{1}{m}\operatorname{tr}\mathsf{u}^m$, a real number by ($\rho 6$) and Lem. 3.24 (a). We write each trace as a sum over closed walks of u-steps and sort these walks by the union $\bigcup \mathsf{c}$ of their sets of cells; this gives the first identity. Likewise $S = q\sum_{m\ge1}\mathsf{u}^m$ by Lem. 3.26 (a); we sort the walks in the same manner, and since $S^{\mathsf{T}} = S$ by ($\rho 6$), the second identity follows.

Step (b): dependence, holomorphy and gauge behaviour. The dependence on the field is read off from the walks and from the clause ($\rho 2$) of the law. By ($\rho 3$), $\mathsf{u}[\mathsf{V}'^u] = \operatorname{Ad}_u \mathsf{u}\operatorname{Ad}_u^{-1}$, while the sets $\mathsf{c}(\cdot)$ depend on no field; this gives the behaviour under gauge transformations. Holomorphy follows from ($\rho 4$) and the uniform bounds of (c). At G-valued fields the matrix C is real by ($\rho 6$), and q_ℓ is real, since q_{s} is real and Ad is real on $\mathfrak{g}$; hence u is real, and so are $\mathsf{D}^u(Y,\cdot)$, $S^\circ(Y;\cdot)$ and $S_k(Y;\cdot)$. The definition of S_k gives $S_k(Y)^{\mathsf{T}} = S_k(Y)$. Finally, in a walk that contributes to $S^\circ(Y)(b,b')$, the cells of b and b' lie in $\mathsf{c}(\mathsf{s})$ and in $\mathsf{c}(b_m,b')$, respectively, and so in Y.

Step (c): the bounds. Let $n = \#Y$. A closed walk of m u-steps whose cells make up Y has $n \le m\mathsf{n}_u$. Its first bond satisfies $b_0 \in \hat{Y}$, which leaves at most $3(LM)^3 n$ choices; moreover $|\operatorname{tr}_{\mathfrak{g}}| \le 3\|\cdot\|$, and the sum over the remaining bonds is controlled by the row sums. Hence $|\mathsf{D}^u(Y)| \le 9(LM)^3 n\sum_{m\ge m_0}\upsilon_1^m \le 18(LM)^3 n\,\upsilon_1^{m_0}$, where $m_0 = \max\{1,\lceil n/\mathsf{n}_u\rceil\}$. Here $\upsilon_1^{m_0} \le x^{-(\mathsf{n}-2)}x^{-m_0}$ and $x^{-m_0} \le e^{-(\kappa^\sharp+1)\mathsf{n}_u m_0} \le e^{-(\kappa^\sharp+1)n}$; together with $18(LM)^3 n e^{-n} \le x$ this gives the bound on $|\mathsf{D}^u(Y,\mathsf{V}')|$ stated in (c).

For S°, the set Y of a walk obeys $\#Y \le \#\mathsf{c}(\mathsf{s}) + m\mathsf{n}_u$, where $\#\mathsf{c}(\mathsf{s}) \le 27(\frac{|\mathsf{s}|}{M}+1)$ by (Cnt); moreover $\|\mathsf{T}_\mathsf{s}\| \le e^{2\vartheta_0^\flat|\mathsf{s}|}$. By (M4$^\flat$), $\frac{27\kappa^\sharp}{M} \le \frac{1}{8}\theta_2$, and so $2\vartheta_0^\flat + \frac{27\kappa^\sharp}{M} \le \theta_1$; hence Lem. 3.24 (b) yields $\sum_{\mathsf{s}\text{ from }b} |q_\mathsf{s}| \|\mathsf{T}_\mathsf{s}\| e^{\kappa^\sharp \#\mathsf{c}(\mathsf{s})} \le e^{27\kappa^\sharp}(b + A^\flat)$. Also $\sum_{m\ge1}(\upsilon_1 e^{\kappa^\sharp \mathsf{n}_u})^m \le 2x^{-(\mathsf{n}-2)}$. Combining these, $e^{\kappa^\sharp \#Y} \sum_{b'} \|S^\circ(Y)(b,b')\| \le 2(b+A^\flat)e^{27\kappa^\sharp}x^{-(\mathsf{n}-2)} \le \frac{1}{2}x^{-(\mathsf{n}-3)}$. The column sums obey the same bound; for them one reverses the steps and uses the column sums of u. The Schur test now bounds the norm of $S^\circ(Y)$, the norm of its transpose $S^\circ(Y)^\mathsf{T}$ obeys the same bound, and the bound on $\|S_k(Y;\mathsf{V}')\|$ stated in (c) follows. □

Corollary 3.43 ((S-loc$_k$): the determinant and S^V as sums of local terms)**.** Let the hypotheses of Thms. 3.40 and 3.42 hold, assume (S1) in addition, and consider the families $\mathsf{D}_k(Y,\cdot)$ and $S_k(Y;\cdot)$, where Y ranges over the connected sets of cells. Then, at every $V' \in \mathcal{V}'$,

$$\log\det C^{\mathbf{1}} - \log\det C^V = \sum_Y \mathsf{D}_k(Y,V'), \qquad S^V = \sum_Y S_k(Y;V').$$

Each $\mathsf{D}_k(Y,\cdot)$ is real and gauge invariant, takes the value 0 at pure gauges, and depends on V' only through the bonds of $Y^{+r_u} \subset Y^{\approx}$. At every V' it obeys $|\mathsf{D}_k(Y,V')| \le 2\mathsf{K}_1^\flat e^{-\kappa^\sharp(\#Y-1)}$, and it is continuous on $\mathcal{V}'$. It continues to a holomorphic function on $\mathcal{U}_Y(x')$, where $|\mathsf{D}_k| \le 3\mathsf{K}_1^\flat\,\varepsilon'\,e^{-\kappa^\sharp(\#Y-1)}$, and this continuation is invariant under the G-valued gauge transformations on the sites of $Y^{\approx}$. Finally, $\|S_k(Y;V')\| \le x^{-(\mathsf{n}-3)}e^{-\kappa^\sharp\#Y}$, and $S_k(Y;\cdot)$ has the properties of Thm. 3.42 (b). ■□

Proof. The two identities follow from Thm. 3.40 (a) and Thm. 3.42 (a), applied at V' and at $\mathbf{1}$. At a pure gauge $\mathbf{1}^u$ we have $\mathsf{D}^q = 0$ and $\mathsf{D}^u(Y,\mathbf{1}^u) = \mathsf{D}^u(Y,\mathbf{1})$, so that $\mathsf{D}_k(Y,\cdot)$ vanishes there. A member of $\mathcal{U}_Y(x')$, extended by $\mathbf{1}$ off $Y^{\approx}$, is a member of $\mathfrak{V}^\circ$ by (u1), and by (S2) the bonds on whose values $\mathsf{D}^q(Y,\cdot)$ and $\mathsf{D}^u(Y,\cdot)$ depend lie in $Y^{+r_u} \subset Y^{\approx}$; this yields the continuation to $\mathcal{U}_Y(x')$. As $2x^{-(\mathsf{n}-3)} \le 2\varepsilon' \le \mathsf{K}_1^\flat\varepsilon'$, the two bounds on $|\mathsf{D}_k|$ stated in the corollary follow from Thm. 3.40 (c), (d) and Thm. 3.42 (c).

Continuity on $\mathcal{V}'$ holds because $\mathsf{D}^q(Y,\cdot)$ is holomorphic on the open set of Thm. 3.40 (b) and $\mathsf{D}^u(Y,\cdot)$ is holomorphic on $\mathfrak{V}^\circ$ by Thm. 3.42 (b); both sets contain every G-valued field, and $\mathsf{D}^u(Y,\mathbf{1})$ is a number.

For the invariance on $\mathcal{U}_Y(x')$, let $\mathsf{V} \in \mathcal{U}_Y(x')$ and let u be G-valued on the sites of $Y^{\approx}$. Then $\mathsf{V}^u \in \mathcal{U}_Y(x')$, since $|u_1 g u_2| = |g|$ and $|u_1 g u_1^{-1} - \mathbf{1}| = |g - \mathbf{1}|$ for unitary u_1, u_2. We extend u by $\mathbf{1}$. On the bonds that leave $Y^{\approx}$ the transform of the extended field then takes unitary values, which places it in $\mathfrak{V}^\circ$. Only on these bonds can it differ from the extension of V^u; none of them is a bond of Y^{+r_u}, and the terms do not depend on them. Thm. 3.40 (b) and Thm. 3.42 (b), both applied with G-valued u, now give the invariance. □

Remark 3.44 (The special case of the law of windows). For the law of windows, where $\mathsf{n} = 60$, we have $\|S_k(Y;\mathsf{V}')\| \le x^{-57}e^{-\kappa^\sharp\#Y}$ and $|\mathsf{D}^u(Y,\mathsf{V}')| \le x^{-57}e^{-\kappa^\sharp\#Y}$ on $\mathfrak{V}$. Let $\tilde{\mathsf{D}}(Y,V') := -\frac{1}{2}\mathsf{D}_k(Y,V')$. Then $\det G^V = (\det C^V)^{1/2} = (\det C^{\mathbf{1}})^{1/2}\exp\sum_Y \tilde{\mathsf{D}}(Y,V')$, where $\det G^V > 0$ by Thm. 3.6 (h); moreover $|\tilde{\mathsf{D}}| \le \mathsf{K}_1^\flat e^{-\kappa^\sharp(\#Y-1)}$ at every V', and $|\tilde{\mathsf{D}}| \le 1.5\,\mathsf{K}_1^\flat\varepsilon' e^{-\kappa^\sharp(\#Y-1)}$ on $\mathcal{U}_Y(x')$.

The conditions, whole. The conditions of Thms. 3.40, 3.42 and Cor. 3.43 are $\kappa \ge 6$, (M4$^\flat$), (R$^\flat$), (M2), (Tor) and the floors (F$_{\text{law}}$), (P0), (P1), (S1), (S2). The condition (M2) gives $6c_R(\kappa^\sharp+1)/M \le 0.04$. Since $2c_R\log x/M \le r_u \le (2c_R\log x + 4)/M + 1$, the second clause of (S2) holds for all large x if $M > 6c_R(\kappa^\sharp+1)$, and for no x if $M \le 6c_R(\kappa^\sharp+1)$.

The ratio in the formula of Prop. 3.10 (b) is the same function whether we write it in terms of φ or of ξ; here $\varphi = G\xi$, $\eta = \varphi - a_0$ and $h = G^\mathsf{T}\mathsf{g}X'$. For the ratio we use the letters $T_Y := \frac{x}{2}\langle\varphi, S_k(Y)\varphi\rangle - x\langle\varphi, S_k(Y)a_0\rangle$. Then $\sum_Y T_Y = \frac{x}{2}\langle\eta, S\eta\rangle - \frac{x}{2}\langle a_0, Sa_0\rangle = -\frac{x}{2}\langle\xi, \mathsf{E}\xi\rangle + x\langle\xi, \mathsf{E}h\rangle$, which is the exponent of Prop. 3.10 (b); the reason is Prop. 3.10 (a), by which $\eta\cdot S\eta = -(\xi-h)\cdot\mathsf{E}(\xi-h)$ and $a_0\cdot Sa_0 = -h\cdot\mathsf{E}h$.

4. The nonlinear step

4.1. **The local prefactor.** The Gaussian integral in Prop. 3.10 has prefactor $e^{-x\mathcal{Q}(-a_0)}$. We separate its exponent into the next-scale energy and local terms,

$$x\mathcal{Q}(-a_0) = x'\mathrm{En}'(V') + \sum_{\square} T_{\square}.$$

The identity holds for every real coarse field. In tame windows the $T_{\square}$ are holomorphic and small enough for the analytic cap. Near rough plaquettes the same terms enter the weights of the corresponding clusters.

Remark 4.1 (Notation from earlier sections; letters and tags with more than one sense). The letters G, G_Z and $\mathsf{G}_{\mathsf{n}} = a^{-1}\mathrm{d}'^{*}\mathsf{G}_Z$ are the free kernels of §2.2, introduced in Def. 2.37.

The letter b is the number of (O6) or a fine bond (item C1); e is the second member of a pair (V', e) or the number $2.718\ldots$, and e_μ is a unit vector; q is either a matrix, namely the form q_ℓ^V of item C4, or a fine cube (item C1), and q' is a coarse cube. In $M = L^{m'}$ of (O7), m' is the exponent, whereas $m'_{y'}$ and the m' in $\ell'^{cV*}m'\ell'^{cV}$ refer to the mask of item E2. The tag (h), said of a field at a number $\hat{\varepsilon}$ or ε', denotes the conditions (h1) to (h3) of (C3) in item D, and otherwise the monotonicity clause (h) of item H. In the names $K_{\mathrm{q},k}$, (G1$_k$), (F8$_k$) and (F9$_k$), the subscript k is a fixed label, not an index.

The sign $\partial V'(P)$ denotes the plaquette product $\partial V(p)$ of item C.I 1.1 of §3, evaluated at the coarse field. For $P = (y'; \mu\nu)$ with $\mu < \nu$ and the bonds $c_1 = (y', \mu)$, $c_2 = (y' + Le_\mu, \nu)$, $c_3 = (y' + Le_\nu, \mu)$, $c_4 = (y', \nu)$, it is

$$\partial V'(P) = V'(c_1)V'(c_2)V'(c_3)^{-1}V'(c_4)^{-1}.$$

With no argument (Step 6), $\partial V'$ is the family of these products.

We write $V'(\pi)$ for the ordered product along a path π, and $V'^{u}(c) = u(c_-)V'(c)u(c_+)^{-1}$; also $\operatorname{ang} g = \arccos(\frac{1}{2}\operatorname{tr} g) \in [0, \pi]$ (items C.I 1.1, C.I 4 of §3).

Theorem 4.2 (The prefactor at a level with rows)**.** The rows may be of any depth, and the empty rows are admitted (item B and Remark (4) of item I). The group is SU(2), and $d = 3$. The step is the drawn step of item C.I 4 of §3, set out there under "The step, the chart"; it coincides with the step of item 1 of §2.1, which is given there under "Fields" and "Step".

For the table of the letters, with (O0) to (O13) and L_*^{su}, we refer to item 3 of §2.1, and for (SD) to item 1 of §2.1.

A *Letters.* The letters follow the order (O0) to (O13) of item 3 of §2.1.

(O0) The pure numbers are b_*, K_*, λ_*, L_A, λ_2^*, θ_J, as in clause (0) of the order of the constants in Thm. 2.65. Further, $\mathsf{g}(t) = \sum_{n\in\mathbb{Z}^3} e^{-t|n|_\infty}$, $\mathsf{g}_2(t) = \sum_n (1 + |n|_\infty)^2 e^{-t|n|_\infty}$ and $\mathsf{g}_I(t) = 3L^3\mathsf{g}(t)$.

(O1) The number $d_0 \in (0, \frac{1}{2})$ enters in two ways: through $S_{\varepsilon'}$, and through the number ρ_0 of the display eq. 2.vi of §2.2.

(O3) The display eq. 2.vi of §2.2 gives ρ_0; moreover $\mathsf{N}_{\mathsf{B}} = 1 + 3K_*^2\mathsf{g}(\lambda_*)^2$.

(O4) The number L is odd, with $L \geq L_*^{\mathrm{su}}$ of item 3 of §2.1; of this the theorem uses only $L \geq 3$ odd and $L \geq L_A$. We set $c_L = (L^2 - 1)/(6L^2)$. Further, $a \geq 3/(c_L L)$, so that $aL \geq 18$.

(O5) The letter $\kappa \geq 6$ enters the theorem only through the bound of clause (ca).

(O6) These are the numbers of item C.II 1.3 of §3, evaluated at K_*, λ_*: $\gamma = (73L^4)^{-1}$, $b = 16\mathsf{N}_{\mathsf{B}} + a/L$, $A^\flat = 2e^2(16 + a) + 384e^8K_*^2\mathsf{g}_2(\lambda_*/2)^2$, $\theta_1 = \min\{1, \lambda_* L/12\}$, $\theta^\flat = \min\{\theta_1, \gamma/(2A^\flat)\}$, $\theta_{\mathbb{C}} = \theta^\flat/6$ and $\vartheta_0^\flat = \theta^\flat/24$ ($\leq 10^{-4}$). In addition $\theta_D = \frac{1}{2}\min\{\lambda_2^*, \theta_{\mathbb{C}}, \theta_J\}$ (≤ 1).
The number $C_D \geq 4$ is that of (O6) in item 3 of §2.1. It is not smaller than the maximum by which Prop. 2.62 defines its own C_D, taken at this θ_D; this number also

occurs in clause (6) of the order of the constants in Thm. 2.65. Moreover (Dec)(C, θ_D) holds for every C at least as large as that maximum. Each K defined below grows with C_D.

Finally $c_R = 64/\theta_D$, so that $c_R \theta_{\mathbb{C}} \geq 128$.

(O7) Here $M = L^{m'}$ with $m' \geq 1$; it serves as the side of a cell.

(O8) The letters are $\mathfrak{p}_0 > 2$ and $A_0 > 0$, and $p_0(x) = A_0(\log x)^{\mathfrak{p}_0}$, $\varepsilon = p_0 x^{-1/2}$, $x' = x/L$; a prime on a letter means that it is taken at x'. So $x\varepsilon'^2 = L p_0'^2$.

(O9) By Thm. 3.6 (k), $\mathsf{n} = 60$.

(O10) We take from item C.II 4 ("Constants") of §3 the numbers

$$\mathsf{K}_A = 1.24 \cdot 10^6 \, \frac{A^\flat}{\gamma \theta_{\mathbb{C}}} \, \mathsf{g}_I(\theta_{\mathbb{C}}/2) \big[\frac{8 \mathsf{N}_{\mathsf{B}}}{\gamma} \mathsf{g}_I(\theta_{\mathbb{C}}) + 1 \big]$$

and $\mathsf{K}_{\mathrm{c}}^0 = (L+1)\mathsf{K}_A + 6 \cdot 10^4 L C_D/\theta_D$, $\mathsf{a}_{\mathrm{c}} = \frac{100}{\gamma} \mathsf{N}_{\mathsf{B}} \mathsf{g}_I(\theta_{\mathbb{C}})$.

We put $\mathsf{K}_{\mathrm{c}} := \sqrt{2} \mathsf{K}_{\mathrm{c}}^0$. As $\mathsf{K}_{\mathrm{c}}^0 \geq 6 \cdot 10^4 \cdot 3 \cdot 4 = 7.2 \cdot 10^5 > 2.42$, it follows that $\mathsf{K}_{\mathrm{c}} \geq \mathsf{K}_{\mathrm{c}}^0 + 1$.

The numbers of item F come next in the order.

(O12) A single condition is imposed on E_a, namely $E_a \geq 32 K_{\mathrm{q},k}$, and it enters only the last inequality of (d). Since $c_a(x) = E_a p_0^3 x^{-1/2}$, we have $c_a(x') = E_a p_0'^2 \varepsilon'$.

(O13) The letter $x_\flat \geq e^2$ is the last one, and the floors of item G hold for every $x \geq L x_\flat$.

B *Data.* These are the data (SD) of item 1 of §2.1. We take $x \geq L x_\flat$. The side N is even, $M \mid N$, and (Tor) holds: $N \geq 2Ms(x) = 16Mr(x)$.

Here N is the coarse side of the present step: it is the $N' = NL^{K-k-1}$ of §2.2, where the N of that formula is the side of the last torus. The fine torus is $\mathbb{Z}^3/NL\mathbb{Z}^3$, and the coarse torus is $L\mathbb{Z}^3/NL\mathbb{Z}^3$.

The rows $\ell \in \mathcal{R}(b_*, K_*, \lambda_*, \frac{1}{9})$ live on the fine torus and have range $\leq \varrho(x)$. The empty rows are admitted: $\ell = \emptyset$, with $\mathsf{H}_y = \{0\}$ for all y, belongs to $\mathcal{R}(1, 0, \lambda_*, 0)$, a subset of that set (see Def. 2.48), and has range 0. Neither a level nor a depth is among the data; see Remark (4) of item I.

In the sense of Def. 2.22, rows attach a Euclidean space H_y to each fine site y, together with columns $\ell_y(p) \in \mathsf{H}_y$. Since $\ell \in \mathcal{R}(b_*, K_*, \lambda_*, \frac{1}{9})$, the second condition in the display of Def. 2.48, taken with the distance $|y - p| := |y - \pi(p)|_\infty$ of Def. 2.43, gives $|\ell_y(p)| \leq K_* e^{-\lambda_* |y - \pi(p)|_\infty}$. The rows are such that the pair (ℓ, ℓ') satisfies the four conclusions of Thm. 2.64. Of these data on the rows the theorem uses only $\ell \in \mathcal{R}(b_*, K_*, \lambda_*, \frac{1}{9})$, the range, and the bounds that (Rows) of item D asserts.

Finally, (V', e) is any pair (item E2 below).

C *First objects: lattice, norms, fields, form, radii, domains.*

C1 *Lattice, blocks, distances, averages, lifts.* The objects come from Subsections 2.2.1 and 2.2.2, including Lem. 2.18 and Defs. 2.19, 2.20 and 2.21, and from item C.II 1.1 of §3. Let $\mathsf{h}_L = (L-1)/2$. Fine bonds, plaquettes, cubes and sites are denoted b, p, q, y, and the coarse ones c, P, q', y'.

Every object i has a base site $\pi(i)$, its least corner; $B(y') = y' + [-\mathsf{h}_L, \mathsf{h}_L]^3$; and $y(i)$ denotes the coarse site whose block contains $\pi(i)$. We write $y_P = \pi(P)$, $y_{q'} = \pi(q')$, $y(c) = c_-$. Distances are $\mathrm{dist}(i, j) = L^{-1} |y(i) - y(j)|_\infty$, measured in coarse units, and "within r of o" means dist $\leq r$. A quantity depends on V' within r of o when it is a function of V' restricted to the coarse bonds whose two ends both lie within r of o.

Concerning the combs, I° is the set of the fine bonds off the combs, at most $3L^3$ of which are based in a block; $\mathfrak{L}$ is the space of functions on I° with values in $\mathfrak{g}$ or $\mathfrak{g}_{\mathbb{C}}$; and $\Pi_{\mathfrak{L}}$ is the restriction to I°. The maps d and d′ are coboundaries. On cubes, $\delta = \mathrm{d}^*$.

For $c = (c_-, \mu)$ we put

$$(Q_1 A)(c) = L^{-3} \sum_{x \in B(c_-)} A([x, x + Le_\mu])$$

and, for a function Θ on the fine cubes (Subsection 2.2.1),

$$(Q_3\Theta)(y') = L^{-3} \sum_{x \in B(y')} \sum_{0 \le i,j,k < L} \Theta(x + ie_1 + je_2 + ke_3).$$

The kernel $W = Q_2$ (Subsection 2.2.1 and Lem. 2.18) satisfies $W \ge 0$, $\sum_P W(P,p) = L^{-1}$ and $\sum_p W(P,p) = L^2$, and $W(P,p) = 0$ unless $y(p)$ is a corner of P.
The edge plaquettes under $P = (y'; \mu\nu)$ are the $p = (z; \mu\nu)$ with $z_\mu = y'_\mu + \mathsf{h}_L$, $z_\nu = y'_\nu + \mathsf{h}_L$ and $|z_\lambda - y'_\lambda| \le \mathsf{h}_L$; there are L of them, all based in $B(y_P)$. The corner cube under q' is the cube at $y_{q'} + \mathsf{h}_L(1,1,1)$. The lift E is given by $(EX')_p = X'_P$ on the edge plaquettes under P and by 0 on all other plaquettes, and the lift E_3 by $(E_3\Theta)_q = \Theta_{q'}$ on the corner cube under q' and by 0 on all other cubes. Hence $(E^*F)_P$ is the sum of F_p over the L edge plaquettes under P, and $(E_3^*\Xi)_{q'}$ is the value of Ξ at the corner cube under q'.

C2 *Norms.* We follow item C.II 1.1 of §3 and Def. 2.20. Here $\mathfrak{g}$ is the space of hermitian traceless 2×2 matrices, and $\mathfrak{g}_\mathbb{C}$ is its complexification. We use the bilinear form $Y \cdot Y' = \frac{1}{2} \operatorname{tr} YY'$, the pairing $\langle F, F' \rangle = \sum_i F_i \cdot F'_i$ and the norm $\|Y\|^2 = \frac{1}{2} \operatorname{tr} Y^*Y$, and $|\cdot|$ is the operator norm of a 2×2 matrix. Then $\|Y\| \le |Y| \le \sqrt{2}\|Y\|$, where the first two agree on $\mathfrak{g}$, and $|Y \cdot Y'| \le \|Y\|\,\|Y'\|$.
For $g \in \mathrm{SL}(2,\mathbb{C})$ one has $|g^{-1}| = |g|$ and $\|\operatorname{Ad}_g^{\pm 1}\| \le |g|^2$, and Ad_g^{-1} is the transpose of Ad_g with respect to $Y \cdot Y'$. By $\|A\|$ we mean the operator norm on $\ell^2(I^\circ; \mathfrak{g}_\mathbb{C})$, and $\|K\|_\infty = \sup_i \sum_j \|K(i,j)\|$. The norm $\|K\|_\theta$ is the larger of $\sup_i \sum_j |K(i,j)| e^{\theta \operatorname{dist}(i,j)}$ and $\sup_j \sum_i |K(i,j)| e^{\theta \operatorname{dist}(i,j)}$; for rows, the entry at (y,p) is $|\ell_y(p)|$. Then $\|K\|_\infty \le \|K\|_0 \le \|K\|_\theta$, and $\sup_i \sum_{\mathrm{dist} > r} |K(i,j)|$, as well as the corresponding sums over columns, is $\le e^{-\theta r} \|K\|_\theta$.

C3 *Fields, transports, dressed kernels.* We follow items C.II 1.1, C.II 1.2 of §3. We let $\mathcal{V}'$ be the set of $\mathrm{SU}(2)$-valued fields on the coarse bonds, with $V'(\bar{c}) = V'(c)^{-1}$, and $\mathfrak{V}$ the set of $\mathrm{SL}(2,\mathbb{C})$ fields with $|V'(c)| \le 1 + \vartheta_0^\flat$ at every bond; $\mathfrak{V}^\circ$ is the set of those with $|V'(c)| < 1 + \vartheta_0^\flat$ at every bond. We extend by 1 every field that is given on a set of bonds only.
The coordinate path $\Gamma(y', y'')$ runs first in the direction e_1, then in e_2, then in e_3, each time the short way. It has no more than $3 \operatorname{dist}(y', y'')$ bonds, and each of its sites lies within $\operatorname{dist}(y', y'')$ both of y' and of y'', since every coordinate of any site of the path lies on the short arc joining the matching coordinates of y' and y''. We set $T(y', y'') = \operatorname{Ad}_{V'(\Gamma(y', y''))}$.
For a free kernel K we set $K^V(i,j) = K(i,j)T(y(i), y(j))$; then $(K^V)^*(j,i) = K(i,j)T(y(i), y(j))^{-1}$, and $(K^V)^*$ is the transpose of K^V for $\langle , \rangle$. So $(\ell^V F)_y = \sum_p \ell_y(p) \otimes T(y(y), y(p)) F_p$ and $(\mathrm{d}'^V X')_{q'} = \sum_P \mathrm{d}'(q', P) T(y_{q'}, y_P) X'_P$. The lifts E and E_3 join objects i, j with $y(i) = y(j)$ and carry no transport.

C4 *The form and the kernel* g^V. We follow item C.II 1.2 of §3. With $\mathsf{B}^V = 1 + \ell^{V*}\ell^V$ we set

$$q = q_\ell^V = \Pi_\mathfrak{L} [\mathrm{d}^{V*} \mathsf{B}^V \mathrm{d}^V + a Q_1^{V*} Q_1^V] \Pi_\mathfrak{L}$$

and $\mathsf{g}^V = \Pi_\mathfrak{L} \mathrm{d}^{V*} \mathsf{B}^V E$. Here $\ell_y(p) = 0$ unless $\operatorname{dist}(y,p) \le \varrho_L$. The form satisfies $q(b,b') = 0$ beyond the distance $2\varrho_L + 2$, and $q(b,\cdot)$ depends on V' within $2\varrho_L + 2$ of $y(b)$.

The reason is the following. By item C3, a coordinate path keeps each of its sites, as seen from either end, within the distance between its two ends. Starting from $y(b)$ and following a step (y, p, p') from b to b', we find $y(p)$ within 1, the path $\Gamma(y(y), y(p))$ within $\varrho_L + 1$, the path $\Gamma(y(y), y(p'))$ within $2\varrho_L + 1$, and $y(b')$ within $2\varrho_L + 2$. Item C.II 1.2 of §3 makes the same estimate.

Moreover $\mathsf{g}^V(b, P) = 0$ unless y_P lies within $2\varrho_L + 1$ of $y(b)$, and $\mathsf{g}^V(b, P) = \sum_{\mathsf{t}} g_{\mathsf{t}} \operatorname{Ad}_{V'(\pi_{\mathsf{t}})}$, the sum running over the g-steps t of item C.II 1.2 of §3, where π_{t} is a single coarse path from $y(b)$ to y_P with at most $6\varrho_L + 8$ bonds, all of whose sites lie within $2\varrho_L + 1$ of $y(b)$.

C5 *Functions of x, radii, cells, the window.* At x we set $R = R(x) = \varrho(x) = \lceil c_R \log x \rceil$, $r(x) = \lceil 20 c_R \log x / M \rceil$, $s(x) = 8r(x)$, $r' = r(x')$, $\ell_w(x') = 6Mr' + 12$, $\mathsf{b}(x') = 2 - 1/\log x'$ and $\vartheta(x') = \vartheta_0^{\flat}(1 - 1/\log x')$. In coarse units the radii are $\varrho_L = \lceil R/L \rceil$, $r_w = \lfloor R/2 \rfloor$, $R_2 = 2R + 2$, $R_3 = 2R + \varrho_L + 5$ and $R_T := 4R + 2\varrho_L + 6$. In item E8 of §4.3, R_5 stands for $4R + 4\varrho_L + 8$ and R_6 for $4R + 6\varrho_L + 11$; see (h).

The cells $\square$ are the cubes of the fixed grid that contain M^3 coarse sites each. Every object i, fine or coarse, is assigned to the cell containing $y(i)$. This assignment defines the relations $P \in \square$, $q' \in \square$ and $b \in \hat{\square}$. The distance from one cell to another is their sup distance counted in cells; Y^{+j} is the set of the cells within distance j of Y, and $\square^{\approx} = \square^{+r'}$. The window $W_{\square}$ consists of the coarse sites within R_T of some site of $\square$; it is a box of side $M + 2R_T$.

C6 *Domains of complex fields; tame fields.* Let A be a set of coarse sites; by the bonds, paths and plaquettes of A we mean those whose sites all lie in A. Let a field with values in $\mathrm{SL}(2, \mathbb{C})$ be given on the bonds of A. It has (u) on A if the following hold.

(u1) $|V'(c)| < 1 + \vartheta(x')$.

(u2) $|V'(\pi)| < \mathsf{b}(x')$ for every path in A with at most $\ell_w(x')$ bonds.

(u3) $|\partial V'(P) - 1| < 2\varepsilon'$.

We let $\mathcal{U}_{\{\square\}}(x')$ be the set of fields on the bonds of $\square^{\approx}$ with (u) on $\square^{\approx}$. This set is the domain of clause (ca) of the class $C_{\flat}^{\mathrm{su}}$ at x', with $X = \{\square\}$: in item 1 of §2.1, under "Domains" and "Members", these are $\mathcal{U}_X(x)$ and (ca_A), taken at x' with $X = \{\square\}$ and $A = 1$. In the same way $U_{\square}^T(x')$ consists of the fields on the bonds of $W_{\square}$ that have (u) on $W_{\square}$.

For a field with (u3) we put $X'_P := -i \log \partial V'(P)$, the logarithm being its power series, and then $|X'_P| \le 2.06\varepsilon'$ by item C.II 1.4 of §3. When a field is given on the bonds of A only and is extended by 1, this definition is used at the plaquettes of A, while $X'_P := 0$ at every plaquette which is not a plaquette of A; this is the rule of item C.II 1.4 of §3. For the extended field, $\partial V'(P)$ at such a plaquette is a product of no more than three bond matrices, and there the power series may fail to converge. By this rule X'_P has a value at every coarse plaquette of the torus, and so X', $\mathsf{g}^V X'$, $a_0 = C\mathsf{g}^V X'$, v_* and Ξ_* are defined at each of their indices.

The rule $X'_P := 0$ at a plaquette which is not a plaquette of A enters neither a value nor a bound of $T_{\square}$, $P_{\square}^T$, (f) or (f′) (see Step 2 of the proof for the first two, and Steps 4 and 8 with (0.1) for the last two).

We call a field of $\mathcal{V}'$ tame on A if $\phi'_P \le \varepsilon'$ holds at every plaquette of A.

D *Inputs.* The following inputs are statements quoted from earlier sections, and none of them is a hypothesis of the theorem.

(C2) *The law.* We use Thm. 3.6 (a), (b), (c), (f), (i), (k), and only through $C = C^V = G^V (G^V)^{\mathsf{T}}$, in the following form.

(C2.0) The matrix C is defined at all fields of $\mathfrak{V}$.

(C2.1) By Thm. 3.6 (b), $C(b,b')=0$ unless $\mathrm{dist}(b,b')\le 2r_w$ ($\le R$), and $C(b,\cdot)$ depends on V' within $3r_w+\varrho_L+1$ ($\le R_2-1$) of $y(b)$.
(C2.2) By Thm. 3.6 (c), $C[V'^u]=\mathrm{Ad}_u\,C[V']\,\mathrm{Ad}_u^{-1}$ for SU(2)-valued u, where $(\mathrm{Ad}_u\,\eta)_b=\mathrm{Ad}_{u(y(b))}\,\eta_b$.
(C2.3) By Thm. 3.6 (i), (a), each entry of C is continuous on $\mathfrak{V}$, holomorphic on $\mathfrak{V}^\circ$ and real at SU(2)-valued fields.
(C2.4) By Thm. 3.6 (f), (k), under ($\mathrm{F_{law}}$) and $c_R\theta_{\mathbb{C}}\ge 128$, the matrix C is invertible on all of $\mathfrak{V}$, and $S:=C^{-1}-q$ has $\|S\|\le x^{-60}$.
(C2.5) From (C2.4) and (P1), Lem. 3.26 (a) gives $\|C\|\le 4/\gamma$ and $\|C-q^{-1}\|\le 8\gamma^{-2}x^{-60}$ on $\mathfrak{V}$.

(C3) *The centre.* We use Thm. 3.32 (a), (b), (c) together with Lem. 3.37 (a), (b), each applied with the law C of (C2) in the role of the law in its hypotheses. For the matrices we follow item C.II 2.2 of §3 ("The matrices"): q^1, g^1, C^1 are q, g^V, C at $V'=1$, $H^{(x)}=E-\mathrm{d}C^1\mathsf{g}^1$, $\mathsf{N}^{(x)}=Q_1C^1\mathsf{g}^1$, and $\mathcal{Q}_i$ is the set of those coarse cubes all of whose faces lie within R_2 of i. We set

$$H^\natural(p,P)=H^{(x)}(p,P)-\sum_{q'\in\mathcal{Q}_p}\mathsf{G}(p,q')\mathrm{d}'(q',P)$$

and

$$\mathsf{N}^\natural(c,P)=\mathsf{N}^{(x)}(c,P)-\sum_{q'\in\mathcal{Q}_c}\mathsf{G}_{\mathsf{n}}(c,q')\mathrm{d}'(q',P).$$

The numbers are $\tau_1=8\gamma^{-2}(1+b)x^{-60}$, $\tau_{\mathrm{c}}=165(L+1)\mathsf{N}_{\mathsf{B}}\,3L^3(2R+3)^3\tau_1$,

$$\tau_2=24L^3(2R+1)^3\gamma^{-2}x^{-60}+(2/\gamma)\mathsf{g}_I(\theta_{\mathbb{C}}/2)x^{-64}$$

and $\mathsf{ti}_{\mathrm{c}}=(16+4L)\mathsf{N}_{\mathsf{B}}\tau_2+7e(L+1)C_Dx^{-128}$.
Throughout (C3), either $i=p$ and $o=y(p)$, or $i=c$ and $o=c_-$, and $\mathsf{B}(i)$ denotes the box of radius R_3 about o. A field on $\mathsf{B}(i)$ is said to have (h) at $\hat\varepsilon$ if the following hold.
(h1) $|V'(c)|\le 1+\vartheta_0^\flat$.
(h2) $|V'(\pi)|\le 2$ for every path in the box with at most $\ell_w(x')$ bonds.
(h3) $|\partial V'(P)-1|\le 2\hat\varepsilon$.
We set

$$\Lambda_p=(EX'-\mathrm{d}^VC\mathsf{g}^VX')_p-\sum_P H^\natural(p,P)T(y(p),y_P)X'_P$$

and

$$\Lambda_c=(Q_1^VC\mathsf{g}^VX')_c-\sum_P\mathsf{N}^\natural(c,P)T(c_-,y_P)X'_P.$$

Under (P0) and (P1), for every i, every $0<\hat\varepsilon\le\frac{1}{40}$ and every field on $\mathsf{B}(i)$ with (h) at $\hat\varepsilon$, the following hold.
(C3.1) The quantities Λ_i and $a_{0,b}=(C\mathsf{g}^VX')_b$, the latter at the bonds with $\mathrm{dist}(b,o)\le 1$, depend on V' only through the restriction of V' to the bonds of $\mathsf{B}(i)$, and on X' only at the plaquettes of $\mathsf{B}(i)$; Thm. 3.32 (a) contains both statements.
By the part of Thm. 3.32 (a) that does not depend on (h2), (h3) or (Fr2), this clause holds with no bound on the size of the field: if a field of $\mathfrak{V}$ is given on a bigger set and X' is defined on all plaquettes of that set (item C6), then Λ_i and $a_{0,b}$ keep the values that they have when computed on $\mathsf{B}(i)$.
(C3.2) $\|\Lambda_i\|\le\mathsf{K}_{\mathrm{c}}^0\hat\varepsilon^2+\tau_{\mathrm{c}}\hat\varepsilon$; so $\|\Lambda_i\|\le\mathsf{K}_{\mathrm{c}}\hat\varepsilon^2$ when $\tau_{\mathrm{c}}\le\hat\varepsilon\le\frac{1}{40}$.
(C3.3) $\|a_{0,b}\|\le\mathsf{a}_{\mathrm{c}}\hat\varepsilon$ at those bonds.
(C3.4) $H^\natural(p,P)=0=\mathsf{N}^\natural(c,P)$ for $\mathrm{dist}>R_2$.
(C3.5) $\|H^\natural-H^\sharp\|_\infty+\|\mathsf{N}^\natural-\mathsf{N}^\sharp\|_\infty\le\mathsf{ti}_{\mathrm{c}}$.

(Rows) *Decay of the free kernels.* For the pair (ℓ,ℓ'), (Dec)(C_D,θ_D) of Def. 2.41 holds: $\|\cdot\|_{\theta_D}\le C_D$ for ℓ, X, $H^\sharp$, B', ℓ', as well as for T, $\mathsf{N}^\sharp$ in clause (f′) of item H, and for G, G_Z inside (C3). None of ℓ, X, B' belongs to the six kernels of (Fr2) in item C.II 2.2 of §3, but every kernel named here is one of the sixteen kernels of (Dec).
This comes from Prop. 2.62, for every $\ell\in\mathcal{R}(b_*,K_*,\lambda_*,\frac{1}{9})$, $L\ge L_A$ odd and $aL\ge 3/c_L$, taken with $\theta_{\mathbb{C}}$ for its θ_q; the proof of Thm. 2.65 obtains it in Step H13 with the help of Step H16 (i), (ii). Thus (Rows) holds for every ℓ of that set; it is not a hypothesis on ℓ, and it adds no condition to item B.

(Tools) Lem. 3.24 (d) gives $\sup_b\sum_{\mathsf{t}\text{ from }b}|g_{\mathsf{t}}|\le 4\mathsf{N}_{\mathsf{B}}$, and Lem. 3.25 (a) gives $\|q^{-1}(b,b')\|\le(2/\gamma)e^{-\theta_{\mathbb{C}}\mathrm{dist}(b,b')}$ on $\mathfrak{V}$. We invoke Lem. 3.27 (b), (c) and Bianchi's identity, which the proof of Lem. 3.27 (d) establishes as an identity of invertible matrices, together with its six quotients. We use Lem. 2.23 as well.
Finally, (E) of item C.I 1.1 of §3, where it is proved, states that

$$|e^{iY_1}\cdots e^{iY_n}-1-i\sum Y_k|\le e^S-1-S\le 0.72S^2$$

whenever the matrices Y_k satisfy $S=\sum|Y_k|\le 1$.

E *Second objects.*

E1 *The free kernels and their cuts.* These kernels are taken from §2.2, namely from Def. 2.26, from Prop. 2.27 and its proof, from Prop. 2.28 and from Def. 2.35; each of them is real and depends on no field. We set $\mathsf{B}=1+\ell^{\mathsf{T}}\ell$, and, given Ξ on the fine cubes and Z on the coarse plaquettes, $Y=W^*Z+\delta\Xi$.
Let $\Xi_Z\in\ker Q_3$ be the minimiser of $\langle Y,\mathsf{B}^{-1}Y\rangle$; then $\mathsf{T}_\sharp Z=W^*Z+\delta\Xi_Z$ and $R_\sharp=\mathsf{T}_\sharp^*\mathsf{B}^{-1}\mathsf{T}_\sharp+a^{-1}\mathrm{d}'\mathrm{d}'^*>0$. We put $\mathsf{B}'=LR_\sharp^{-1}$, $H^\sharp=L^{-1}\mathsf{B}^{-1}\mathsf{T}_\sharp\mathsf{B}'$ and $\mathsf{N}^\sharp=(aL)^{-1}\mathrm{d}'^*\mathsf{B}'$ (Prop. 2.27 (a)), as well as $\mathsf{X}v=\Xi_{v/L}$ and $\mathsf{T}v=\mathsf{B}^{-1}\mathsf{T}_\sharp v/L$. The rows ℓ' are those of Prop. 2.28, with $\mathsf{B}'=1+\ell'^{\mathsf{T}}\ell'$. These definitions give the identities
(I1) $\mathsf{B}\mathsf{T}=L^{-1}W^*+\delta\mathsf{X}$;
(I2) $H^\sharp=\mathsf{T}\mathsf{B}'$, so $\mathsf{B}H^\sharp=(L^{-1}W^*+\delta\mathsf{X})\mathsf{B}'$.
The cuts are $\ell'^{\mathrm{c}}=\ell'\,1\{\mathrm{dist}\le\varrho(x')\}$, the cuts $\mathsf{X}^<$, $\mathsf{T}^<$ at $R(x')$ and the cuts $H^{\sharp<}$, $\mathsf{N}^{\sharp<}$ at R_2; we also set $K^>=K-K^<$ and $\mathsf{B}'^{\mathrm{c}}=1+\ell'^{\mathrm{c}\mathsf{T}}\ell'^{\mathrm{c}}$.

E2 *Pairs, clipped curvature, mask, energy.* The objects are those of items E2, E7 of §4.3 and of item 1 of §2.1, under "Pair" and "Energy". Here $\mathfrak{E}=\prod_P S^2$, a product of unit spheres of $\mathfrak{g}$.
We set $\phi'_P=\mathrm{ang}\,\partial V'(P)$ and

$$X'_P(V',e)=\mathsf{L}(\partial V'(P))+\pi e_P 1\{\partial V'(P)=-1\},$$

where L is the logarithm with $|\mathsf{L}|<\pi$ and $\mathsf{L}(-1)=0$. So $|X'_P|=\phi'_P$, and X' is a Borel function of (V',e); moreover $X'_P(V'^u,\mathrm{Ad}_u\,e)=\mathrm{Ad}_{u(y_P)}X'_P(V',e)$, where $(\mathrm{Ad}_u\,e)_P=\mathrm{Ad}_{u(y_P)}\,e_P$, and e_P plays a role only where $\phi'_P=\pi$.
Let $D'=\{P:\phi'_P>\varepsilon'\}$, $\zeta'_P=X'_P\min\{1,\varepsilon'/\phi'_P\}$, and $S_{\varepsilon'}(t)=t^2/2$ $(t\le\varepsilon')$, $d_0\varepsilon' t$ $(t>\varepsilon')$. Here and below a clipping multiplier $\min\{1,\alpha/t\}$, with $\alpha>0$ and $t\ge 0$, has value 1 at $t=0$. The mask is $m'_{y'}=1$ when no P of D' has $\mathrm{dist}(y',P)\le\rho_0$ and no P with $\phi'_P>1$ has $\mathrm{dist}(y',P)\le\varrho(x')+1$, and 0 in every other case. The energy

$$\mathrm{En}'(V')=\sum_P S_{\varepsilon'}(\phi'_P)+\frac{1}{2}\sum_{y'}m'_{y'}|(\ell'^{\mathrm{c}V}\zeta')_{y'}|^2$$

does not depend on e.

E3 *The centre.* For a field of $\mathfrak{V}$ and an arbitrary $\mathfrak{g}_{\mathbb{C}}$-valued X' we set $a_0=C\mathsf{g}^VX'$, $S=C^{-1}-q$ and $u^T:=1-qC$ $(=SC)$. Because $C^{\mathsf{T}}=C$ (Thm. 3.6 (b)), u^T equals the transpose of the u that item C.II 2.1 of §3 defines. Further, $\bar F=EX'-\mathrm{d}^Va_0$ and

$\bar n = Q_1^V a_0$, and

$$\mathsf{m}_C := \frac{1}{2}\langle EX', \mathsf{B}^V EX'\rangle - \frac{1}{2}\langle \mathsf{g}^V X', C\mathsf{g}^V X'\rangle$$

is the m_C of Prop. 3.10 (a).

E4 *The multipliers.*
We set $v_* = (1+\ell'^{cV*} m' \ell'^{cV})\zeta'$, $Z_* = v_*/L$, $\Xi_* = (\mathsf{X}^<)^V v_*$, $Y_* = W^{V*} Z_* + \mathrm{d}^{V*}\Xi_*$ and $\Delta_Y = \mathsf{B}^V \bar F - Y_*$. For clauses (f′) and (g) of item H we set $\nu^{\mathrm{ref}} = (\mathsf{T}^<)^V v_*$, $\sigma_* = \ell^V \nu^{\mathrm{ref}}$, $\nu_* = Y_* - \ell^{V*}\sigma_*$, $\omega_* = (aL)^{-1}\mathrm{d}'^{V*} v_*$, $\Delta^{\mathrm{ref}} = \bar F - \nu^{\mathrm{ref}}$, $\Delta_\sigma = \ell^V \bar F - \sigma_*$, $\Delta_F = \bar F - \nu_*$ and $\Delta_n = \bar n - \omega_*$.
At a field with (u3), the tame formulas are $X'_P = -i \log \partial V'(P)$ at every plaquette of the set on which the field is given, with $X'_P := 0$, by convention, at all other plaquettes (item C6), and $v_* = (1+\ell'^{cV*}\ell'^{cV})X'$; every other object is computed from these by the same formulas.

E5 *The local term $T_\square$ and its holomorphic form.* With $\mathrm{En}'_P = S_{\varepsilon'}(\phi'_P) + \frac{1}{2}\langle \zeta'_P, (\ell'^{cV*} m' \ell'^{cV}\zeta')_P\rangle$ we set

$$T_\square := \sum_{P\in\square}\big[\frac{x'}{2}\langle X'_P, v_{*P}\rangle - x'\mathrm{En}'_P + \frac{x}{2}\langle X'_P, (E^*\Delta_Y)_P\rangle\big] + \frac{x}{2}\sum_{q'\in\square}\langle (\mathrm{d}'^V X')_{q'}, (E_3^*\Xi_*)_{q'}\rangle - \frac{x}{2}\sum_{b\in\hat\square}\langle a_{0,b}, (u^T \mathsf{g}^V X')_b\rangle.$$

For $V' \in U_\square^T(x')$, extended by 1, we define $P_\square^T(V')$ as the sum of the three sums

$$\frac{x}{2}\sum_{P\in\square}\langle X'_P, (E^*\Delta_Y)_P\rangle + \frac{x}{2}\sum_{q'\in\square}\langle (\mathrm{d}'^V X')_{q'}, (E_3^*\Xi_*)_{q'}\rangle - \frac{x}{2}\sum_{b\in\hat\square}\langle a_{0,b}, (u^T\mathsf{g}^V X')_b\rangle,$$

in which all objects are given by the tame formulas of item E4. For $V' \in \mathcal{U}_{\{\square\}}(x')$ we set $P(\{\square\}, V') := P_\square^T(V'|W_\square)$, and $P(Y,\cdot) := 0$ for $\#Y \ge 2$.

E6 *Marked cells, class partitions, the weights w_j.* We let $\mathrm{mk}' = \mathrm{mk}'(V')$ be the set of the cells that hold a plaquette of D'.
By a class partition we mean a partition $\{\mathrm{mk}'_j\}$ of mk' into sets whose pairwise distance is $> 2r'$ cells. We call classes of the chain the classes into which mk' falls under the relation generated by "distance $\le 2s_0 + 4r_1 + 2$", with $s_0 = 2r'$, $r_l = \lceil(\varrho_L + 2)/M\rceil$ and $r_1 = \lceil R_2/M\rceil + 3r_l + 5$; since their pairwise distance is $> 4r'$, they form a class partition.
We set $w_j := \sum_{\square\in \mathrm{mk}'^{+r'}_j} T_\square$ and $1_{\square^\approx}(V') = 1\{\phi'_P \le \varepsilon' \text{ for all } P \in \square^\approx\}$.
In (e) and in Step 7, for a field V' of $\mathcal{V}'$, $P(\{\square\}, V')$ stands for $P(\{\square\}, V'|\square^\approx)$, where $V'|\square^\approx$ is the restriction to the bonds of $\square^\approx$. By convention, the product $1_{\square^\approx}(V')P(\{\square\}, V')$ has the value 0 wherever $1_{\square^\approx}(V') = 0$, whether or not $P(\{\square\}, V')$ is defined there. At such fields the restriction $V'|\square^\approx$ may lie outside $\mathcal{U}_{\{\square\}}(x')$, and then P may be undefined.
Where $1_{\square^\approx}(V') = 1$, we have $V'|\square^\approx \in \mathcal{U}_{\{\square\}}(x')$ by (0.6).

F *Constants.* In the order of the letters, these numbers come after M and before c_h, E_a. We set

$$K_Y := 4\mathsf{N}_{\mathsf{B}}\mathsf{K}_{\mathrm{c}} + 10^6\theta_D^{-2}(1+C_D^2)(C_D + eL + 6eC_D) + 1$$

and

$$K_{\mathrm{q},k} := M^3[3.09L^2K_Y + 520LC_D(1+4C_D^2) + 1].$$

For clause (f′) of item H we set $K_{\mathrm{ref}} := \mathsf{K}_{\mathrm{c}} + 10^6\theta_D^{-2}C_D(1+C_D^2) + 1$, $K_\sigma := 4C_D K_{\mathrm{ref}}$, $K_F := K_Y + 16C_D^2 K_{\mathrm{ref}}$ and $K_n := \mathsf{K}_{\mathrm{c}} + 10^6\theta_D^{-2}(1+C_D^2) + 1$. These numbers depend on $(L, a, K_*, \lambda_*, C_D, \theta_D)$, and $K_{\mathrm{q},k}$ on M as well. None of them depends on x, N, c_h, E_a, a member of a class, or on more of ℓ than its starred constants and its range.

G *Floors.* Each floor is a condition on x and on the letters (O0) to (O10) alone; every one of them holds for all large x, and all of them enter the choice of $x_\flat$.

($\mathrm{F_{law}}$) $432L^3b^{5/2}\gamma^{-3/2}\mathsf{g}(\theta_{\mathbb{C}})(R+3)^3 \le x^4$.

(P0) $x \ge L^2$; $R \ge 6$.
The (P0) of item C.II 4 of §3 is this condition with (Tor) as a third clause. Since (Tor) depends on N, it is a datum here (item B) and not a floor. So the conditions (P0), (P1) as (C3) assumes them, together with item B, make up the whole hypothesis (P0), (P1) of Thm. 3.32 and of Lem. 3.37.

(P1) $(4/\gamma)x^{-60} \le 1$; $165(L+1)\mathsf{N_B}\,3L^3(2R+3)^3\,8\gamma^{-2}(1+b) \le x$.

($\mathrm{G1}_k$) $Mr' \ge R_T + 1$; $\rho_0 \le R(x')$.
This floor is a consequence of ($\mathrm{G1}^\flat$) of item G of §4.3, which states $Mr' \ge 5R+4\varrho_L+12$ and $\rho_0 \le R(x')$ and, besides, $R \ge \max\{M, 6\}$.

($\mathrm{F8}_k$) $\varepsilon' \le \frac{1}{40}$; $35100R_T^2\varepsilon' \le 1$; $p_0' \ge 1$.

($\mathrm{F9}_k$) $33\mathsf{N_B}(1+C_D)\mathsf{ti_c} + 100\mathsf{N_B}C_D^4x'^{-64} \le \varepsilon'$; $198L^3\mathsf{a_c}\gamma^{-1}\mathsf{N_B}(3L^3)^{1/2}(2R+4\varrho_L+5)^{3/2} \le x^{59}$.

H *Conclusions.* In this setting the following hold.

(a) At every (V', e),

$$x\,\mathsf{m}_C - \frac{x}{2}\langle a_0, Sa_0\rangle = \frac{x}{2}\big[\langle \bar F, \mathsf{B}^V \bar F\rangle + a|\bar n|^2\big] = x'\mathrm{En}'(V') + \sum_{\square} T_{\square}.$$

(b) The term $T_\square$ is real and jointly Borel in (V', e), and $T_\square(V'^u, \mathrm{Ad}_u\, e) = T_\square(V', e)$ for every SU(2)-valued u. It depends on V' only through the restriction of V' to the bonds of $W_\square$, hence within $R_T \le Mr' - 1$ of $\square$, and on e_P only at those plaquettes P of $W_\square$ where $\phi'_P = \pi$.

(c) Let V' be tame on $W_\square$; this is the case in particular when $\phi'_P \le \varepsilon'$ at each P within $R_T + 1$ of $\square$. Then $T_\square(V', e) = P_\square^T(V'|W_\square)$ and $|T_\square(V', e)| \le K_{\mathrm{q},k}p_0'^2\varepsilon'$.

(d) $P_\square^T$ is holomorphic on $U_\square^T(x')$, with $\sup|P_\square^T| \le K_{\mathrm{q},k}p_0'^2\varepsilon'$.
It follows that $P(\{\square\}, \cdot)$ is holomorphic on $\mathcal{U}_{\{\square\}}(x')$, is invariant under SU(2)-valued gauge transformations, is real at SU(2)-valued fields, vanishes at the pure gauges, and has $\sup|P(\{\square\}, \cdot)| \le K_{\mathrm{q},k}p_0'^2\varepsilon' \le \frac{1}{32}c_a(x')$.
Therefore the family P satisfies clause (ca) at x' with $\frac{1}{32}$ of the cap.

(e) At every pair (V', e) and for every class partition $\{\mathrm{mk}'_j\}$, so in particular for the classes of the chain,

$$x\,\mathsf{m}_C - \frac{x}{2}\langle a_0, Sa_0\rangle = x'\mathrm{En}'(V') + \sum_j w_j + \sum_{\square} 1_{\square\approx}(V')\,P(\{\square\}, V').$$

Moreover, each w_j depends on V' within $2r' \le s(x')$ cells of mk'_j.

(f) Let p be a fine plaquette, let $o = y(p)$, and let a field with values in $\mathrm{SL}(2,\mathbb{C})$ satisfy (u) on the box of radius R_T about o; the field is extended by $\mathbf{1}$, and the objects are defined by the tame formulas. Then $\|\Delta_{Y,p}\| \le K_Y\varepsilon'^2$.

(f′) Let i be a fine plaquette p, a fine site y or a coarse bond c, put $o = y(i)$, and let the field satisfy the hypothesis of (f) about this o. Then, according to the kind of i, $\|\Delta_p^{\mathrm{ref}}\| \le K_{\mathrm{ref}}\varepsilon'^2$, $\|\Delta_{\sigma,y}\| \le K_\sigma\varepsilon'^2$, $\|\Delta_{F,p}\| \le K_F\varepsilon'^2$ and $\|\Delta_{n,c}\| \le K_n\varepsilon'^2$.

(g) For every field of $\mathfrak{V}$, every choice of v_* and every $\psi \in \mathfrak{L}$,

$$\langle \nu_*, \mathrm{d}^V\psi\rangle + \langle \sigma_*, \ell^V\mathrm{d}^V\psi\rangle - a\langle \omega_*, Q_1^V\psi\rangle = \langle Z_*, (W^V\mathrm{d}^V - \mathrm{d}'^V Q_1^V)\psi\rangle + \langle \Xi_*, \mathrm{d}^V\mathrm{d}^V\psi\rangle,$$

and this identity is exact: it has no remainder. Moreover, at every pair (V', e) and for every $b \in I^\circ$,

$$|a_{0,b}| \le (8.4/\gamma)\mathsf{N_B}\mathsf{g}_I(\theta_{\mathbb{C}})\sup|X'_P|,$$

where the supremum runs over the P whose site y_P lies within $R + 2\varrho_L + 1$ of $y(b)$.

(h) Clauses (b) and (c) are monotone in the radius: they remain true when R_T is replaced by any integer R' with $R_T \le R' \le Mr' - 1$. Here the letter R' denotes a free integer; it is not the radius $R(x')$. For (b) this means that $T_\square$ is a function of V' on the bonds, and of the e_P at the plaquettes, of the box consisting of the sites within R' of $\square$. For (c) it means that the hypothesis becomes tameness on that box, whereas the value is still $P^T_\square(V'|W_\square)$, with the window $W_\square$ defined with R_T; no function $P^T_\square$ is defined with R'. In the setting (G1$^\flat$) of Thm. 4.19 (item G of §4.3), the admissible values of R' include $4R + 4\varrho_L + 8$, which is the radius R_5 of Thm. 4.19, and $4R + 6\varrho_L + 11$, which is the radius R_6 of Thms. 4.19 and 4.39; both radii R_5 and R_6 are those of item E8 of §4.3.

I Remarks.

(1) The middle member of (a) is the quantity $x\mathcal{Q}(-a_0)$ of Prop. 3.10 (a), and $e^{-x\mathcal{Q}(-a_0)}$ is the prefactor in the exact formula of Prop. 3.10 (b). That proposition is stated for every $\mathfrak{g}$-valued X', and so it holds at $X' = X'(V', e)$.

(2) Clauses (a), (b) and (e) hold also when e is set to 0, that is, with $X'_P = \mathsf{L}(\partial V'(P))$, and with $X'_P = 0$ where $\partial V'(P) = -\mathbf{1}$; in this form they hold at every V', and e does not enter them. We keep the pairs (V', e) because Thm. 4.19 uses $|X'_P| = \phi'_P$.

(3) No clause of the theorem rests on Thm. 5.3.

(4) We consider the case $\ell = \emptyset$. Then $\mathsf{H}_y = \{0\}$, and $\emptyset \in \mathcal{R}(b_*, K_*, \lambda_*, \frac{1}{9})$; this follows from Def. 2.48 of the sets of rows $\mathcal{R}(b, K, \lambda, \tau)$, according to which $\emptyset \in \mathcal{R}(1, 0, \lambda, 0)$. The conclusion of Thm. 2.65 is stated for every $k \ge 0$ with ℓ_0 empty, and item B admits this case. Here $\mathsf{B}^V = 1$, $\Delta_Y = \bar{F} - Y_*$ and $\sigma_* = 0$, and the steps (y, p, p') of item C4 carry the weight 0. Every clause holds with the same letters, since the constants are computed from (K_*, λ_*) and not from the rows.

■□

Proof. Step 0: what the data and the floors give. We collect the following facts.

(0.1) For $L \ge 3$ and $R \ge 6$ we have $2r_w \le R$, $3r_w + \varrho_L + 1 \le R_2 - 1$ and $\max\{R_2, R + 2\varrho_L + 3\} \le R_3$; each of the numbers $R_3 + 2\varrho_L$, $3R + 3$, $2R + 2\varrho_L + 3$ and $R + 4\varrho_L + 4$ is $\le R_T$; and $2\varrho_L + R_2 \le R_T - 1$, $3R + 1 \le R_T - 1$, $2\varrho_L + \max\{R_3, 3R + 1\} \le R_T$, $R(x') \le R$.

(0.2) We have $\ell_w(x') \ge 6R_T + 18$. Hence $3R_T + 1 \le \ell_w(x')$, and a path of at most $6\varrho_L + 8$, $6\varrho(x')$, $3R_2$ or $3R_T$ bonds has at most $\ell_w(x')$ bonds.

(0.3) We have $r' \ge 1$ and $N \ge 16(R_T + 1)$. Hence every box of radius $\le R_T$, every window $W_\square$ and every set $\square^\approx$ embeds in the coarse torus, and $4R_T + 2 \le N$.

(0.4) We have $W_\square \subset \square^\approx$, and no bond of $W_\square$ has an end outside $\square^\approx$ or in the outermost layer of sites of $\square^\approx$.

(0.5) We have $\tau_{\mathrm{c}} \le \varepsilon'$. Hence (C3.2) holds at $\hat{\varepsilon} = \varepsilon'$, in the form $\|\Lambda_i\| \le \mathsf{K}_{\mathrm{c}}\varepsilon'^2$.

(0.6) A field with (u) on a box satisfies, on that box, the conditions (h) at $\hat{\varepsilon} = \varepsilon'$. A field of $\mathcal{V}'$ that is tame on a set A satisfies (u) on A, and on A we have $X'_P(V', e) = -i \log \partial V'(P)$.

(0.7) On a set A where (u2) holds, the operators $\mathrm{Ad}^{\pm 1}$ of V' along a single path in A of at most $\ell_w(x')$ bonds have norm ≤ 4.

(0.8) We have $e^{-\theta_D R_2} \le x^{-128}$, $e^{-\theta_D R(x')} \le x'^{-64}$ and $\|\mathsf{B}' - \mathsf{B}'^{\mathrm{c}}\|_0 \le 2C_D^2 x'^{-64}$.

We prove these facts in turn.

(0.1) For $L \ge 3$ we have $\varrho_L \le (R+2)/3$; moreover $r_w \le R/2$, which is the first inequality. Hence $3r_w + \varrho_L + 1 \le 11R/6 + 5/3 \le 2R + 1$ for $R \ge 4$, and $2R + 1 = R_2 - 1$. The inequality $R + 2\varrho_L + 3 \le 2R + \varrho_L + 5$, that is $R + 2\varrho_L + 3 \le R_3$, holds if and only if $\varrho_L \le R + 2$, which is the case. The inequality $2R + 3\varrho_L + 5 \le R_T$, that is $R_3 + 2\varrho_L \le R_T$, holds if and only if $\varrho_L \le 2R + 1$, and $R + 4\varrho_L + 4 \le R_T$ holds if and only if $2\varrho_L \le 3R + 2$; both conditions are met. The remaining inequalities are read off the definitions of the radii.

(0.2) By (G1$_k$), $\ell_w(x') = 6Mr' + 12 \geq 6(R_T+1)+12 = 6R_T+18$, so that $3R_T+1 \leq \ell_w(x')$. The numbers $6\varrho_L+8$, $6R$, $3R_2 = 6R+6$ and $3R_T$ are all $\leq 6R_T+18$, and $6\varrho(x') \leq 6R$ by (0.1).

(0.3) By (G1$_k$), $Mr' \geq R_T+1 > 0$, so that $r' \geq 1$. By (Tor), $N \geq 16Mr(x) \geq 16Mr' \geq 16(R_T+1)$, which gives $4R_T+2 \leq N$. The sides of $W_\square$ and of $\square^{\approx}$ are $M+2R_T$ and $M(2r'+1)$, and both are $\leq 3Mr' < N$.

(0.4) A site within $R_T \leq Mr'-1$ of a site of $\square$ lies in a cell within r' of $\square$, and its distance from $\square$ is $\leq Mr'-1$, whereas the sites of the outermost layer of $\square^{\approx}$ are at distance Mr' from $\square$.

(0.5) By (P1), $\tau_{\mathrm{c}} \leq x \cdot x^{-60} = x^{-59}$, while $\varepsilon' = p_0' x'^{-1/2} \geq x'^{-1/2} \geq x^{-59}$, because $p_0' \geq 1$ and $x \geq x' \geq 1$. Hence $\tau_{\mathrm{c}} \leq \varepsilon'$, and therefore $\mathsf{K}_{\mathrm{c}}^0 \varepsilon'^2 + \tau_{\mathrm{c}}\varepsilon' \leq (\mathsf{K}_{\mathrm{c}}^0+1)\varepsilon'^2 \leq \sqrt{2}\mathsf{K}_{\mathrm{c}}^0 \varepsilon'^2$, where the last step rests on $\mathsf{K}_{\mathrm{c}}^0 \geq 2.42$.

(0.6) We have $\vartheta(x') < \vartheta_0^\flat$ and $\mathsf{b}(x') < 2$, and (u3) is (h3) at ε'; this gives the first assertion. For a real field, $|V'(c)| = |V'(\pi)| = 1$, and $1 < 1+\vartheta(x')$, $1 < \mathsf{b}(x')$, because $\log x' \geq \log x_\flat \geq 2$. At a tame plaquette, $|\partial V'(P) - \mathbf{1}| = 2\sin(\phi_P'/2) \leq \varepsilon' < 2\varepsilon'$; and for $\phi_P' \leq \varepsilon' < \pi$ the logarithm L is given by its power series, so that no e_P enters. This gives the second assertion.

(0.7) Let g be the matrix of $\mathrm{SL}(2,\mathbb{C})$ that V' assigns to the path. Then $\| \mathrm{Ad}_g^{\pm 1} \| \leq |g|^2 < \mathsf{b}(x')^2 < 4$.

(0.8) We have $\theta_D c_R = 64$, $R_2 \geq 2c_R \log x$ and $R(x') \geq c_R \log x'$, which give the first two bounds. For the third, $\mathsf{B}' - \mathsf{B}'^{\mathrm{c}} = \ell'^{>\mathsf{T}} \ell' + \ell'^{\mathrm{c}\mathsf{T}} \ell'^{>}$, and $\|\ell'^{>}\|_0 \leq e^{-\theta_D \varrho(x')} \|\ell'\|_{\theta_D} \leq C_D x'^{-64}$.

Step 1: the identity, and clause (a). Let a field of $\mathfrak{V}$ be given, together with an arbitrary $\mathfrak{g}_{\mathbb{C}}$-valued X', an arbitrary Z on the coarse plaquettes, an arbitrary Ξ on the fine cubes and an arbitrary symmetric invertible C, and put $\Delta := \mathsf{B}^V \bar{F} - W^{V*} Z - \mathrm{d}^{V*}\Xi$. We claim that

$$2\mathsf{m}_C - \langle a_0, S a_0\rangle = \langle \bar{F}, \mathsf{B}^V \bar{F}\rangle + a\langle \bar{n}, \bar{n}\rangle = \langle X', Z\rangle + \langle \mathrm{d}'^V X', E_3^* \Xi\rangle + \langle EX', \Delta\rangle - \langle a_0, u^T \mathsf{g}^V X'\rangle,$$

and that clause (a) follows from this identity.

Throughout, the brackets denote the bilinear form; B^V and q are symmetric for it, and $a_0 \in \mathfrak{L}$.

(i) The first equality. Since $a_0 = C\mathsf{g}^V X'$, we have $\langle a_0, C^{-1} a_0\rangle = \langle a_0, \mathsf{g}^V X'\rangle$, and with the definitions of m_C and of $S = C^{-1} - q$ this gives

$$2\mathsf{m}_C - \langle a_0, S a_0\rangle = \langle EX', \mathsf{B}^V EX'\rangle - 2\langle \mathsf{g}^V X', a_0\rangle + \langle a_0, q a_0\rangle.$$

For $\psi \in \mathfrak{L}$ we have $\langle \mathsf{g}^V X', \psi\rangle = \langle \mathsf{B}^V EX', \mathrm{d}^V \psi\rangle$; moreover $\langle a_0, q a_0\rangle = \langle \mathrm{d}^V a_0, \mathsf{B}^V \mathrm{d}^V a_0\rangle + a\langle Q_1^V a_0, Q_1^V a_0\rangle$. Inserting these, the first with $\psi = a_0$, we find that the sum of the three terms is $\langle \bar{F}, \mathsf{B}^V \bar{F}\rangle + a\langle \bar{n}, \bar{n}\rangle$.

(ii) For $\psi \in \mathfrak{L}$,

$$\langle \bar{F}, \mathsf{B}^V \mathrm{d}^V \psi\rangle - a\langle \bar{n}, Q_1^V \psi\rangle = \langle \mathsf{g}^V X', \psi\rangle - \langle q a_0, \psi\rangle = \langle u^T \mathsf{g}^V X', \psi\rangle;$$

the first equality follows from the definitions of $\bar{F}$ and $\bar{n}$ and from the identities in (i), the second from $u^T = 1 - qC$ and $a_0 = C\mathsf{g}^V X'$.

(iii) Since $\bar{F} = EX' - \mathrm{d}^V a_0$, we have $\langle \bar{F}, \mathsf{B}^V \bar{F}\rangle = \langle \mathsf{B}^V \bar{F}, EX'\rangle - \langle \bar{F}, \mathsf{B}^V \mathrm{d}^V a_0\rangle$. Taking $\psi = a_0$ in (ii) and using $\bar{n} = Q_1^V a_0$, we obtain

$$\langle \bar{F}, \mathsf{B}^V \bar{F}\rangle + a\langle \bar{n}, \bar{n}\rangle = \langle \mathsf{B}^V \bar{F}, EX'\rangle - \langle u^T \mathsf{g}^V X', a_0\rangle.$$

(iv) We show that $W^V E = 1$ and $\mathrm{d}^V E = E_3 \mathrm{d}'^V$. By the proof of Lem. 2.23, $W(P', p) = L^{-1}\delta_{PP'}$ at the L edge plaquettes p under P; these are based in $B(y_P)$, so that $T(y_P, y(p)) = 1$, and hence $(W^V EX')_P = X'_P$. For the second identity, take a fine cube at z and two directions μ, ν, and let λ be the third direction. The face $(z; \mu\nu)$ of the cube is an edge plaquette if and only if z_μ and z_ν are at the top of their block, and the opposite face $(z + e_\lambda; \mu\nu)$ is an edge plaquette under the same condition. If z_λ is not at the top of its block, both faces lie under one P, in the block of z, with transport 1 and with opposite signs, so that their contributions cancel; and no other face of the cube is an edge plaquette, as that would

require z_λ to be at the top. If z_μ, z_ν and z_λ are all at the top, the cube is the corner cube under q'. Its face at z lies under $(y_{q'};\mu\nu)$, with transport 1; its face at $z+e_\lambda$ is based in $B(y_{q'}+Le_\lambda)$ and lies under $(y_{q'}+Le_\lambda;\mu\nu)$, with transport $T(y_{q'},y_{q'}+Le_\lambda)=T(y_{q'},y_P)$; and the signs are those of d'. Therefore $(\mathrm{d}^V EX')_q=(\mathrm{d}'^V X')_{q'}$ at the corner cube, and $(\mathrm{d}^V EX')_q=0$ at all other cubes.

(v) By the definition of Δ and by (iv),

$$\langle \mathsf{B}^V\bar F, EX'\rangle=\langle Z, W^V EX'\rangle+\langle \Xi,\mathrm{d}^V EX'\rangle+\langle\Delta,EX'\rangle=\langle X',Z\rangle+\langle E_3^*\Xi,\mathrm{d}'^V X'\rangle+\langle EX',\Delta\rangle.$$

Together with (iii), this proves the claimed identity.

We turn to clause (a). At the pair (V',e) we take $(Z,\Xi)=(Z_*,\Xi_*)$; then $\Delta=\Delta_Y$. We multiply the identity by $x/2$ and use $x\langle X',Z_*\rangle=x'\langle X',v_*\rangle$. Further, $\langle EX',\Delta_Y\rangle=\sum_P\langle X'_P,(E^*\Delta_Y)_P\rangle$. Each P, each q' and each $b\in I^\circ$ lies in exactly one cell, because $M\mid N$, so that the sums over these objects split into sums over the cells. Finally we add and subtract $x'\sum_P \mathrm{En}'_P$; this sum equals $x'\mathrm{En}'(V')$, since

$$\sum_P\langle\zeta'_P,(\ell'^{cV*}m'\ell'^{cV}\zeta')_P\rangle=\langle\ell'^{cV}\zeta',m'\ell'^{cV}\zeta'\rangle=\sum_{y'}m'_{y'}|(\ell'^{cV}\zeta')_{y'}|^2.$$

Step 2: the sets on which the objects depend. We claim that every quantity entering $T_\square$ or $P^T_\square$, taken at an object i with $o=y(i)$ in $\square$, depends on V' within R_T of o, hence only on the bonds of $W_\square$; and that it depends on e only through the e_P at those plaquettes of $W_\square$ where $\phi'_P=\pi$.

We give the radii of dependence about $o=y(i)$, counting all four sites of a plaquette. Recall that the distance of every site of a coordinate path from the start of the path is at most the distance between its two ends (item C3). The radius is 1 for X'_P, ζ'_P and ϕ'_P, and $\varrho(x')+1$ for $(\ell'^{cV}\zeta')_{y'}$. For $m'_{y'}$ it is $\max\{\rho_0,\varrho(x')+1\}+1\le R(x')+2$, as $\rho_0\le R(x')$. For v_{*P} and En'_P, which involve the y' within $\varrho(x')$ of y_P, it is $\varrho(x')+R(x')+2\le 2R+2$. For Ξ_{*q}, which involves the P within $R(x')$ of q, it is $3R+2$, and for Y_{*p}, which involves the P and the q within 1 of p, it is $3R+3$. For $(\mathsf{g}^VX')_b$ it is $2\varrho_L+2$ (item C4), and for $a_{0,b}$ it is $\max\{3r_w+\varrho_L+1,\ R+2\varrho_L+2\}\le R_3-1$, by (C2.1). For $\bar F_p$ and $\bar n_c$, which involve the bonds within 1 of o, it is $\le R_3$; for $(\mathsf{B}^V\bar F)_p$, which involves the p' within $2\varrho_L$ of p and paths within $2\varrho_L$, it is $R_3+2\varrho_L$; and for $(\mathrm{d}'^VX')_{q'}$ it is 1.

Finally,

$$(u^T\mathsf{g}^VX')_b=(\mathsf{g}^VX')_b-\sum_{b''}q(b,b'')\sum_{b'}C(b'',b')(\mathsf{g}^VX')_{b'}.$$

Here $q(b,\cdot)$ has radius $2\varrho_L+2$; the factor $C(b'',\cdot)$ adds R_2-1, which leads to $2R+2\varrho_L+3$; and $(\mathsf{g}^VX')_{b'}$, with b' within $R+2\varrho_L+2$ of b, adds $2\varrho_L+2$, which leads to $R+4\varrho_L+4$. By (0.1), all these radii are $\le R_T$.

The edge plaquettes under P and the corner cube under q' are based in $B(y_P)$ and in $B(y_{q'})$ respectively; hence $y(p)=y_P$ and $y(q)=y_{q'}$ for them. The second member e of the pair enters through X' alone. If a field is given on $W_\square$ and extended by $\mathbf{1}$, nothing depends on the continuation, and X' enters at no plaquette other than the plaquettes of $W_\square$ (item C6).

Step 3: telescoping in the comb gauge of a box. We prove the following lemma.

Lemma 4.3 (The difference between dressed and free products of kernels in a box)**.** Let Bx be the box of radius ρ_B about a coarse site o, where $3\rho_B+1\le\ell_w(x')$ and $4\rho_B+2\le N$. Let V' be an $\mathrm{SL}(2,\mathbb{C})$ field on the bonds of Bx that satisfies (h2) and (h3) at some $\hat\varepsilon\le 1/40$ with $35100\rho_B^2\hat\varepsilon\le 1$. Let $K_1,\dots,K_n$, where $1\le n\le 5$, be free kernels of finite coarse ranges whose sum is $\le\rho_B-1$; each of them is scalar or consists of rows, and rows occur either as a neighbouring pair made up of an adjoint and a row, or as the kernel K_1 $(k=1)$. Let A_k be K_k^V or $(K_k^V)^*$, and let A_k^0 be the same object at $V'=\mathbf{1}$. Let i_0 be an index with $y(i_0)=o$, and let the last index run over the coarse plaquettes. Put $\tilde X_P:=T(o,y_P)X'_P$ at the plaquettes of Bx and $\tilde X_P=0$ at all other plaquettes,

where $X'_P = -i \log \partial V'(P)$ at the plaquettes of Bx (item C.II 1.4 of §3), and let $0 < \theta \le 1$. Then

$$\|(A_1 \cdots A_n X')_{i_0} - (A_1^0 \cdots A_n^0 \tilde X)_{i_0}\| \le 7020 \cdot 8 \cdot 1.2^{n-1} \theta^{-2} \prod_k \|K_k\|_\theta \, \hat\varepsilon \sup_P \|\tilde X_P\| \le 10^6 \theta^{-2} \prod_k \|K_k\|_\theta \, \hat\varepsilon^2,$$

and the first inequality holds for any $\mathfrak{g}_{\mathbb{C}}$-valued X'. If $n = 1$ and $A_1 = K_1^V$, the left side vanishes. ∎

Proof. We use the following facts from item C.II 1.4 of §3 and from Lem. 3.27; they apply because $3\rho_B + 1 \le \ell_w(x')$ and (h2), (h3) hold. The comb is $\mathsf{k}(y') = V'(\Gamma(o, y'))$, and for a path π in Bx from y' to y'' the field in the comb gauge is $\mathsf{W}(\pi) = \mathsf{k}(y') V'(\pi) \mathsf{k}(y'')^{-1}$. By Lem. 3.27 (b), if π has $m \le \ell_w(x')$ bonds and all its sites lie within τ of o, then $\| \operatorname{Ad}^{\pm 1}_{\mathsf{W}(\pi)} - 1\| \le 2340\, m\, \tau\, \hat\varepsilon$. By Lem. 3.27 (c), $|\tilde X_P| \le 8.24 \hat\varepsilon$.

We pass to the comb gauge. We have $T(y', y'') = \operatorname{Ad}^{-1}_{\mathsf{k}(y')} \operatorname{Ad}_{\mathsf{W}(\Gamma(y',y''))} \operatorname{Ad}_{\mathsf{k}(y'')}$, and an adjoint carries the inverse of this operator. Along a chain of indices $i_0, \dots, i_n$ the factors $\operatorname{Ad}_{\mathsf{k}}$ therefore cancel; moreover $\mathsf{k}(o) = \mathbf{1}$ and $\operatorname{Ad}_{\mathsf{k}(y_P)} X'_P = \tilde X_P$. Consequently $(A_1 \cdots A_n X')_{i_0} = (A_1^{\mathsf{W}} \cdots A_n^{\mathsf{W}} \tilde X)_{i_0}$, where A_k^{W} arises from A_k when $\operatorname{Ad}^{\pm 1}_{\mathsf{W}(\Gamma)}$ is put in the place of $T^{\pm 1}$. This is an identity of matrices as soon as all paths lie in Bx, which we now verify. Put $d_k = \operatorname{dist}(i_{k-1}, i_k)$ and $\ell = \sum_k d_k \le \rho_B - 1$. A site of the path between $y(i_{k-1})$ and $y(i_k)$ lies within d_k of each of the two ends, whichever of them is the start (item C3; the short way is unique, since $d_k \le \rho_B - 1 < N/4$); in an adjoint the path is $\Gamma(y(i_k), y(i_{k-1}))$. As the end $y(i_{k-1})$ lies within $d_1 + \cdots + d_{k-1}$ of o, the site lies within $d_1 + \cdots + d_k \le \ell$ of o. For the last plaquette, y_P lies within $\rho_B - 1$ of o, and so all its sites lie in Bx. Since $4\rho_B + 2 \le N$, the short way is unique and Bx is embedded. When $n = 1$ and $A_1 = K_1^V$, the transport is $T(o, y_P)$ itself, and the left side of the claim is 0.

Now consider one chain. Its k-th path has at most $3d_k \le \ell_w(x')$ bonds and lies within ℓ of o, so Lem. 3.27 (b) gives $\| \operatorname{Ad}^{\pm 1}_{\mathsf{W}} - 1\| \le 2340 \cdot 3 d_k \cdot \ell \cdot \hat\varepsilon =: \delta_k$, and $\delta_k \le 7020 \rho_B^2 \hat\varepsilon \le 0.2$. Writing a_k for the factor of the k-th path, we telescope: $\prod a_k - \prod 1 = \sum_k a_1 \cdots a_{k-1}(a_k - 1)$. Since $\|a_j\| \le 1.2$, this difference has norm $\le 1.2^{n-1} \sum_k \delta_k = 1.2^{n-1}\, 7020\, \ell^2 \hat\varepsilon$; the sum over k produces ℓ^2, and not $n\ell^2$, because $\sum_k d_k = \ell$.

To sum over the chains, we use $u^2 \le 2e^u$ (indeed $\sup u^2 e^{-u} \le 0.5414$) at $u = \theta\ell/2$, which gives $\ell^2 \le (8/\theta^2) e^{\theta\ell/2}$. The sum, over the chains that start at i_0, of $\prod_k |K_k(i_{k-1}, i_k)| e^{\theta d_k}$ is $\le \prod_k \|K_k\|_\theta$; as this norm is the larger of the weighted row sums and the weighted column sums, adjoints are covered too. For a pair of rows we use $|\langle \ell_y(p), \ell_y(p')\rangle| \le |\ell_y(p)||\ell_y(p')|$, and for rows at the place $k = 1$ we use $\| \sum_p \ell_y(p) \otimes w_p\| \le \sum_p |\ell_y(p)|\, \|w_p\|$. As to the numerals, $3 \cdot 2340 = 7020$, $5 \cdot 7020 = 35100$ and $7020 \cdot 8 \cdot 1.2^4 = 116453.376$, and $116453.376 \cdot 8.24 = 959575.81824 \le 10^6$; with $|\tilde X_P| \le 8.24\hat\varepsilon$, this yields the second inequality. □

We also note three norms, for $\theta \le 1$ and range 1. First, $\|L^{-1} W\|_\theta \le eL$, the sums that enter being $L^{-1} L^2$ and $L^{-1} L^{-1}$. Second, $\|d\|_\theta \le 6e$ between cubes and plaquettes, since a cube has six faces and a plaquette lies in two cubes. Third, $\|\mathrm{d}'\|_\theta \le 4e$ between coarse plaquettes and bonds.

Step 4: clause (f). Let p be a fine plaquette and $o = y(p)$, and let an $\mathrm{SL}(2, \mathbb{C})$ field satisfy (u) on the box Bx of radius R_T about o; the field is extended by $\mathbf{1}$, and the objects are defined by the tame formulas. We claim that $\|(\mathsf{B}^V \bar F)_p - Y_{*p}\| \le K_Y \varepsilon'^2$; this is clause (f).

The extended field lies in $\mathfrak{V}^\circ$. By (0.2), (0.3), (0.6) and (F8$_k$), Lem. 4.3 applies with $\rho_B = R_T$ and $\hat\varepsilon = \varepsilon'$. We define $\tilde X$ at o. Then $\sup \|\tilde X_P\| \le 8.24 \varepsilon'$.

(1) The neighbours. We write $(\mathsf{B}^V \bar F)_p = \sum_{p'} \mathsf{B}^V(p, p') \bar F_{p'}$, where

$$\mathsf{B}^V(p, p') = \delta_{pp'} + \sum_y \langle \ell_y(p), \ell_y(p')\rangle T(y(y), y(p))^{-1} T(y(y), y(p')),$$

and only p' with $y(p')$ within $2\varrho_L$ of o occur. The transport in this formula is Ad of V' along a single path of at most $6\varrho_L$ bonds in Bx, so its norm is ≤ 4 by (0.7). For the dressed

entries we use the absolute row bound

$$\sum_{p'}\Big(\delta_{pp'}+\sum_{y}|\ell_y(p)|\,|\ell_y(p')|\Big)\le 1+3K_*^2\mathsf{g}(\lambda_*)^2=\mathsf{N}_\mathsf{B}.$$

The box of radius R_3 about $y(p')$ lies in Bx by (0.1), and by (0.6) the field satisfies the conditions (h) at ε' there. By (C3.1), $\bar F_{p'}$ coincides with the one computed in that box, and by (0.5),

$$\bar F_{p'}=\sum_P H^\natural(p',P)T(y(p'),y_P)X'_P+\Lambda_{p'},$$

with $\|\Lambda_{p'}\|\le \mathsf{K}_\mathrm{c}\varepsilon'^2$. Hence $\|(\mathsf{B}^V\Lambda)_p\|\le 4\mathsf{N}_\mathsf{B}\mathsf{K}_\mathrm{c}\varepsilon'^2$.

(2) The split. Put $dH:=H^\natural-H^{\sharp<}$. By (C3.4) this kernel vanishes beyond R_2, and within R_2 it equals $H^\natural-H^\sharp$; so by (C3.5) its absolute row sums are $\le \mathsf{ti}_\mathrm{c}$. The transport $T(y(p'),y_P)$ runs along at most $3R_2$ bonds in Bx and so has norm ≤ 4. Thus

$$\|(\mathsf{B}^V(dH)^VX')_p\|\le 4\mathsf{N}_\mathsf{B}\cdot\mathsf{ti}_\mathrm{c}\cdot 4\cdot 2.06\varepsilon'=32.96\mathsf{N}_\mathsf{B}\mathsf{ti}_\mathrm{c}\varepsilon'.$$

This bound does not depend on a weighted norm of dH; indeed ti_c is of the order of $(\log x)^3x^{-60}$, whereas $e^{\theta_D R_2/2}$ is of the order of x^{64}.

(3) We apply Lem. 4.3 to the products $(H^{\sharp<})$ and $(\ell^*,\ell,H^{\sharp<})$. Their ranges are $\le 2\varrho_L+R_2\le R_T-1$, and $\|\ell\|_{\theta_D},\|H^\sharp\|_{\theta_D}\le C_D$; this gives

$$\|(\mathsf{B}^V(H^{\sharp<})^VX')_p-(\mathsf{B}H^{\sharp<}\tilde X)_p\|\le 10^6\theta_D^{-2}(1+C_D^2)C_D\varepsilon'^2.$$

(4) We apply Lem. 4.3 to $Y_*=(L^{-1}W^{V*}+\mathrm{d}^{V*}(\mathsf{X}^<)^V)(1+\ell'^{\mathrm{c}V*}\ell'^{\mathrm{c}V})X'$, which is made of four products of 1 to 4 kernels, with ranges $\le 1+R(x')+2\varrho(x')\le 3R+1\le R_T-1$. With the norms noted at the end of Step 3, this gives

$$\|Y_{*p}-((L^{-1}W^*+\delta\mathsf{X}^<)\mathsf{B}'^{\mathrm{c}}\tilde X)_p\|\le 10^6\theta_D^{-2}(eL+6eC_D)(1+C_D^2)\varepsilon'^2.$$

(5) The free identity. By (I2), as an identity of matrices and hence also on $\tilde X$ extended by 0,

$$\mathsf{B}H^{\sharp<}-(L^{-1}W^*+\delta\mathsf{X}^<)\mathsf{B}'^{\mathrm{c}}=-\mathsf{B}H^{\sharp>}+\delta\mathsf{X}^>\mathsf{B}'+(L^{-1}W^*+\delta\mathsf{X}^<)(\mathsf{B}'-\mathsf{B}'^{\mathrm{c}}).$$

By (0.8), the three terms on the right have row sums at most $\mathsf{N}_\mathsf{B}C_Dx^{-128}$, $2C_Dx'^{-64}\cdot C_D$ and $(L^{-2}+2C_D)\cdot 2C_D^2x'^{-64}$. Their sum is $\le(\mathsf{N}_\mathsf{B}C_D+8C_D^3)x'^{-64}$, because $2C_D^2+(2/9)C_D^2+4C_D^3\le 8C_D^3$ for $C_D\ge 4$; and it is to be multiplied by $8.24\varepsilon'$.

(6) The sum. The terms of (2) and (5) are linear in ε', and together they are at most

$$[33\mathsf{N}_\mathsf{B}\mathsf{ti}_\mathrm{c}+8.24(\mathsf{N}_\mathsf{B}C_D+8C_D^3)x'^{-64}]\varepsilon'\le[33\mathsf{N}_\mathsf{B}(1+C_D)\mathsf{ti}_\mathrm{c}+100\mathsf{N}_\mathsf{B}C_D^4x'^{-64}]\varepsilon'\le\varepsilon'^2,$$

the last inequality by (F9$_k$). Adding the bounds of (1), (3) and (4), we arrive at the total $4\mathsf{N}_\mathsf{B}\mathsf{K}_\mathrm{c}+10^6\theta_D^{-2}(1+C_D^2)(C_D+eL+6eC_D)+1=K_Y$.

Step 5: the sizes of the three sums on $U^T_\square(x')$, which give the sup bound of (d) *and the bound of* (c). Let $V'\in U^T_\square(x')$, extended by $\mathbf{1}$. We claim that $|P^T_\square(V')|\le K_{\mathrm{q},k}p_0'^2\varepsilon'$.

The extended field lies in $\mathfrak{V}^\circ$. For o in $\square$, the box of radius R_T about o lies in $W_\square$, so that (u) holds on it. We note that $x\varepsilon'^2=Lp_0'^2$.

(1) The first sum. There are $3M^3$ plaquettes P in $\square$, and for each of the L edge plaquettes p under such a P we have $y(p)=y_P$ in $\square$. Clause (f) is stated for a field that is given on the box of radius R_T about $o=y(p)$ and extended by $\mathbf{1}$, while here the field is given on $W_\square$. By Step 2, where the radii are $R_3+2\varrho_L$ for $(\mathsf{B}^V\bar F)_p$ and $3R+3$ for Y_{*p}, both $\le R_T$ about o when the four sites of a plaquette are counted, and by item C6, the quantity $\Delta_{Y,p}$ is a function of V' on the bonds of that box and of X' at its plaquettes, and of nothing else; hence it takes the same value at the two fields. The restriction to the box satisfies (u) there,

because the paths and plaquettes of the box are paths and plaquettes of $W_\square$. Therefore (f) gives

$$|\frac{x}{2}\langle X'_P, (E^*\Delta_Y)_P\rangle| \le \frac{x}{2}\cdot 2.06\varepsilon'\cdot L\cdot K_Y\varepsilon'^2 = 1.03L^2K_Y p_0'^2\varepsilon',$$

and the first sum is at most $3.09M^3L^2K_Y p_0'^2\varepsilon'$.

(2) The second sum. We use the Bianchi identity, which the proof of Lem. 3.27 (d) establishes as an identity between invertible matrices. For a coarse cube q' at z it states that $\prod_{k=1}^6 \kappa_k\,\partial V'(P_k)^{\varepsilon_k}\kappa_k^{-1} = \mathbf{1}$, where $\varepsilon_k = \mathrm{d}'(q', P_k)$, $\kappa_k = \mathbf{1}$ at the faces based at z, and $\kappa_k = V'(z,\lambda)$ at the face based at $z + Le_\lambda$. With $Y_k = \varepsilon_k \operatorname{Ad}_{\kappa_k} X'_{P_k}$ we get $\prod e^{iY_k} = \mathbf{1}$ and $\sum_k Y_k = (\mathrm{d}'^V X')_{q'}$, the latter because $T(z, z+Le_\lambda) = \operatorname{Ad}_{V'(z,\lambda)}$. Each $|Y_k| \le (1+\vartheta_0^\flat)^2\cdot 2.06\varepsilon'$, so that $S = \sum|Y_k| \le 12.3625\varepsilon' \le 0.31$. By (E) of item C.I 1.1 of §3, which item D lists under (Tools),

$$|\sum Y_k| = |\prod e^{iY_k} - \mathbf{1} - i\sum Y_k| \le e^S - 1 - S \le 0.72S^2,$$

where the ratio of $e^S - 1 - S$ to S^2 is ≤ 0.556 at $S = 0.31$; and this is $\le 110.04\varepsilon'^2 \le 126\varepsilon'^2$. Next we bound the multipliers. We have $v_* = X' + \ell'^{\mathrm{c}V*}\ell'^{\mathrm{c}V}X'$, where the transports run along a single path of at most $6\varrho(x')$ bonds and so have norm ≤ 4; since $\|\ell'\|_0 \le C_D$, this gives $\|v_{*P}\| \le (1 + 4C_D^2)\cdot 2.06\varepsilon'$. With paths of at most $3R(x')$ bonds, $\|\Xi_{*q}\| \le C_D\cdot 4\cdot\|v_*\|_{\sup} = 8.24C_D(1+4C_D^2)\varepsilon'$. Summing over the M^3 cubes, we find that the second sum is at most

$$M^3\frac{x}{2}\cdot 126\varepsilon'^2\cdot 8.24C_D(1+4C_D^2)\varepsilon' = 519.12M^3LC_D(1+4C_D^2)p_0'^2\varepsilon' \le 520M^3LC_D(1+4C_D^2)p_0'^2\varepsilon'.$$

(3) The third sum. There are at most $3L^3M^3$ bonds b in $\hat\square$. For each of them $\|a_{0,b}\| \le \mathsf{a_c}\varepsilon'$, by (C3.3) in the box of radius R_3 about $o = y(b)$: we take $i = c$, the coarse bond $(y(b), e_1)$, so that $o = c_- = y(b)$ and $\operatorname{dist}(b, o) = 0 \le 1$. Since $u^T = SC$, we have $\|u^T\| \le x^{-60}\cdot 4/\gamma$ at the extended field, by (C2.4) and (C2.5). The row b of $u^T = 1 - qC$ vanishes beyond $R + 2\varrho_L + 2$, so that at most $3L^3(2R + 4\varrho_L + 5)^3$ bonds b' occur. At these $\|(\mathsf{g}^V X')_{b'}\| \le 4\mathsf{N_B}\cdot 4\cdot 2.06\varepsilon' \le 33\mathsf{N_B}\varepsilon'$, by Lem. 3.24 (d) and by (0.7) for one path π_t of at most $6\varrho_L + 8$ bonds. Consequently

$$\|(u^T\mathsf{g}^V X')_b\| \le \|u^T\|\,(\text{number of } b')^{1/2}\sup\|(\mathsf{g}^V X')_{b'}\| \le (4/\gamma)x^{-60}(3L^3)^{1/2}(2R + 4\varrho_L + 5)^{3/2}\cdot 33\mathsf{N_B}\varepsilon'.$$

With the numeral $3\cdot\frac12\cdot 4\cdot 33 = 198$, the third sum is at most

$$198L^3M^3\mathsf{a_c}\gamma^{-1}\mathsf{N_B}(3L^3)^{1/2}(2R + 4\varrho_L + 5)^{3/2}x^{-59}\varepsilon'^2 \le M^3\varepsilon'^2,$$

the last inequality by (F9$_k$); and $M^3\varepsilon'^2 \le M^3 p_0'^2\varepsilon'$, since $\varepsilon' \le 1/40 \le 1 \le p_0'^2$ by (F8$_k$).

The sum of the bounds in (1) to (3) is $K_{\mathrm{q},k}p_0'^2\varepsilon'$.

Step 6: clauses (c), (d) *and* (b). We show that $T_\square = P_\square^T$ on tame windows; that $P_\square^T$ is holomorphic; that $P(\{\square\},\cdot)$ is holomorphic, invariant and real, vanishes at pure gauges and satisfies (ca); and that $T_\square$ is real, Borel and invariant.

For (c), let $V' \in \mathcal{V}'$ be tame on $W_\square$. A plaquette of $W_\square$ has its base within R_T of $\square$; hence the hypothesis that $\phi'_P \le \varepsilon'$ within $R_T + 1$ of $\square$ implies tameness on $W_\square$. By (0.6), $V'|W_\square \in U_\square^T(x')$, and $X'(V', e) = -i\log\partial V'$ on $W_\square$. By Step 2, all that $T_\square$ depends on lies in $W_\square$. There $\zeta' = X'$, and we meet neither a plaquette of D' nor one with $\phi'_P > 1$; so $m' = 1$ at every y' that enters, and the objects are those of the tame formulas. Since $S_{\varepsilon'}(\phi'_P) = \frac12|X'_P|^2$, we obtain, row by row,

$$x'\mathrm{En}'_P = \frac{x'}{2}\langle X'_P, X'_P + (\ell'^{\mathrm{c}V*}\ell'^{\mathrm{c}V}X')_P\rangle = \frac{x'}{2}\langle X'_P, v_{*P}\rangle.$$

What is left of $T_\square$ is therefore $P_\square^T(V'|W_\square)$. The operators C, q and u^T are evaluated at V' itself, but only entries depending on V' in $W_\square$ occur (Step 2), and these are the entries of $V'|W_\square$ extended by $\mathbf{1}$. The bound in (c) is that of Step 5.

For (d), we prove holomorphy. The function $P^T_\square$ is a finite sum of products of factors of three kinds. The factors of the first kind are the entries of $\mathrm{Ad}^{\pm1}_{V'(c)}$; they are polynomials in the entries of $V'(c)$, because $V'(c)^{-1}$ is the adjugate. Those of the second kind are the X'_P, which are power series in $\partial V'(P)-\mathbf{1}$, where $|\partial V'(P)-\mathbf{1}|<2\varepsilon'<1$. Those of the third kind are the entries of C^V; by (C2.3) they are holomorphic on $\mathfrak{V}^\circ$, and they are composed with the extension by $\mathbf{1}$, which maps $U^T_\square(x')$ into $\mathfrak{V}^\circ$ because $|V'(c)|<1+\vartheta(x')<1+\vartheta_0^\flat$. The restriction $\mathcal{U}_{\{\square\}}(x')\to U^T_\square(x')$ is well defined by (0.4), and it is a projection of coordinates.

Next, invariance. Let u be SU(2)-valued on $\square^\approx$ and extended by $\mathbf{1}$. Then $V'^u\in\mathcal{U}_{\{\square\}}(x')$, since the norms are unchanged. Now $P(\{\square\},V')=P^T_\square(V'|W_\square)$ is evaluated at $V'|W_\square$ extended by $\mathbf{1}$ off $W_\square$. By Step 2 and item C6, the three sums are functions of V' on the bonds of $W_\square$ and of X' at the plaquettes of $W_\square$, and of nothing else; they therefore have the same value at every field of $\mathfrak{V}$ which coincides with V' on the bonds of $W_\square$. One such field is V' extended by $\mathbf{1}$ off $\square^\approx$, and we work with it. This extension of V', transformed by u, and the extension of V'^u by $\mathbf{1}$ off $\square^\approx$ both belong to $\mathfrak{V}$, and they differ only on bonds with one end off $\square^\approx$; by (0.4), no such bond is a bond of $W_\square$. By (C2.2), by the covariance of all dressed kernels and of X', and by the orthogonality of Ad_u for $Y\cdot Y'$, each bracket is unchanged.

Reality at SU(2)-valued fields holds because there the transports are real on $\mathfrak{g}$, X' takes values in $\mathfrak{g}$, and C is real by (C2.3). At a pure gauge, given by a u with values in SU(2) or in $\mathrm{SL}(2,\mathbb{C})$ such that the field lies in the domain, we have $\partial V'=\mathbf{1}$ and hence $X'=0$; as each of the three sums is bilinear in X', $P(\{\square\},\cdot)$ vanishes there.

The bound on the supremum follows from Step 5 and from $K_{\mathrm{q},k}p_0'^2\varepsilon'\le\frac{1}{32}E_a p_0'^2\varepsilon'=\frac{1}{32}c_a(x')$; for $\#Y\ge2$ we have $P(Y,\cdot)=0$. The properties obtained are the five clauses of (ca), which item 1 of §2.1 states for members as (ca_A) at $A=1$ together with reality: holomorphy on $\mathcal{U}_Y(x')$; invariance under SU(2)-valued gauge transformations; reality at SU(2)-valued fields; vanishing at the pure gauges; and $\sup\le c_a(x')e^{-\kappa(\#Y-1)}$.

For (b): the dependence is that of Step 2. Reality follows by the argument given for $P(\{\square\},\cdot)$: at SU(2)-valued fields the transports are real on $\mathfrak{g}$, X' is $\mathfrak{g}$-valued, and C is real. The function $T_\square$ is Borel, since C is continuous on $\mathfrak{V}$ by (C2.3), X' is Borel in (V',e), and ϕ', ζ', m' are Borel. As to invariance under $(V',e)\to(V'^u,\mathrm{Ad}_u\,e)$: X' and ζ' are covariant, ϕ', D' and m' are invariant, and C and the dressed kernels are covariant.

Step 7: clauses (e) *and* (h)*, and remark* (2) *of item I.* We prove the decomposition over an arbitrary class partition, together with the set on which w_j depends; the monotonicity in the radius; and the case $e=0$.

For (e): the set $\square^\approx$ holds a plaquette of D' if and only if a mark lies within r' cells of $\square$. Hence the cells with $1_{\square^\approx}(V')=0$ form exactly the union of the sets $\mathrm{mk}_j'^{+r'}$. These sets are disjoint, as a common cell would put two classes within $2r'$ of each other. In a cell with $1_{\square^\approx}(V')=1$, every plaquette based in $\square^\approx$ is tame; so $V'|\square^\approx\in\mathcal{U}_{\{\square\}}(x')$ by (0.6), V' is tame on $W_\square$ by (0.4), and (c) yields $T_\square=P^T_\square(V'|W_\square)=P(\{\square\},V')$. Summing (a) over the cells now gives the identity of (e). As to the dependence, a cell of $\mathrm{mk}_j'^{+r'}$ lies within r' of mk'_j, and $T_\square$ depends on V' within $R_T\le Mr'$ coarse units, that is, within r' cells of $\square$; the two distances add up to $2r'\le8r'=s(x')$. For the classes of the chain, any two classes are farther apart than $2s_0+4r_1+2>4r'$, for every $r_1\ge0$.

For (h): a function of V' on the bonds within R_T of $\square$ is a function of V' on the bonds within R', and tameness within R' implies tameness on $W_\square$. Further, $4R+6\varrho_L+11\le Mr'-1$ follows from $Mr'\ge5R+4\varrho_L+12$ together with $2\varrho_L\le R$. The former is part of $(\mathrm{G1}^\flat)$, the setting in which (h) admits the radii R_5 and R_6; as to the latter, the proof of (0.1) gives $\varrho_L\le(R+2)/3$, so that $2\varrho_L\le(2R+4)/3\le R$ for $R\ge4$. Thus $4R+6\varrho_L+11\le5R+4\varrho_L+11\le Mr'-1$, and $R_T\le R_5\le R_6$.

For remark (2): the identity of Step 1 holds for any X'. With $X'_P=\mathsf{L}(\partial V'(P))$ and $\zeta'_P=X'_P\min\{1,\varepsilon'/\phi'_P\}$, the sum of the En'_P is still $\mathrm{En}'(V')$: since a row y' with $m'_{y'}=1$ meets no P

with $\phi'_P > 1$, the number En' is the same. The proofs of (b), (c) and (e) depend on $|X'_P| = \phi'_P$ at tame plaquettes only.

Step 8: clauses (f′) *and* (g)*, for the triple* $\boldsymbol{\nu} = (\nu_*, \sigma_*, \omega_*)$ *of item E6 of §4.3.* We show the following: for a field with (u) on the box of radius R_T about $o = y(i)$, as in (f′), the objects being defined by the tame formulas, $\|\Delta_p^{\mathrm{ref}}\| \le K_{\mathrm{ref}}\varepsilon'^2$, $\|\Delta_{\sigma,y}\| \le K_\sigma\varepsilon'^2$, $\|\Delta_{F,p}\| \le K_F\varepsilon'^2$ and $\|\Delta_{n,c}\| \le K_n\varepsilon'^2$; the identity of (g); and the sup bound on a_0 at every (V', e). For three of these bounds a smaller box suffices: for Δ_p^{ref}, the box of radius R_{ref} about $y(p)$, by (1) below; for $\Delta_{\sigma,y}$, the box of radius $\varrho_L + R_{\mathrm{ref}}$ about $y(y)$, by (2); and for $\Delta_{n,c}$, the box of radius R_3 about c_-, by (3). Here R_{ref} is the radius that (1) sets. The bound on $\Delta_{F,p}$ uses the bound on $\Delta_{\sigma,y}$ at the y within ϱ_L of p, with the smaller box of (2); for $\Delta_{F,p}$ itself the box remains the one of radius R_T about $y(p)$, because of (f).

(1) The bound on Δ_p^{ref}. Let $o = y(p)$, and let (u) hold on the box of radius $R_{\mathrm{ref}} := \max\{R_3, 3R+1\}$ about o. By (C3.1) and (0.5), and since $T(o, y_P)X'_P = \tilde{X}_P$, we have the exact equality $\bar{F}_p = (H^\natural \tilde{X})_p + \Lambda_p$, with $\|\Lambda_p\| \le \mathsf{K}_{\mathrm{c}}\varepsilon'^2$. We split $(H^\natural\tilde{X})_p = (H^{\sharp<}\tilde{X})_p + (dH\,\tilde{X})_p$, where $\|(dH\,\tilde{X})_p\| \le 8.24\mathsf{ti}_{\mathrm{c}}\varepsilon'$. Lem. 4.3, with $\rho_B = R_{\mathrm{ref}} \le R_T$, applied to $(\mathsf{T}^<)$ and to $(\mathsf{T}^<, \ell'^{c*}, \ell'^c)$, whose ranges are $\le 3R$, gives $\|\nu_p^{\mathrm{ref}} - (\mathsf{T}^<\mathsf{B}'^c\tilde{X})_p\| \le 10^6\theta_D^{-2}C_D(1+C_D^2)\varepsilon'^2$. On the free side, (I2) gives

$$H^{\sharp<} - \mathsf{T}^<\mathsf{B}'^c = -H^{\sharp>} + \mathsf{T}^>\mathsf{B}' + \mathsf{T}^<(\mathsf{B}' - \mathsf{B}'^c),$$

with row sums $\le (C_D + C_D^2 + 2C_D^3)x'^{-64}$, which are to be multiplied by $8.24\varepsilon'$. The terms linear in ε' add up to at most $[8.24\mathsf{ti}_{\mathrm{c}} + 8.24(C_D + C_D^2 + 2C_D^3)x'^{-64}]\varepsilon' \le \varepsilon'^2$, by $(\mathrm{F9}_k)$. The sum of all contributions is K_{ref}.

(2) The bounds on Δ_σ and Δ_F. We have $\Delta_\sigma = \ell^V(\bar{F} - \nu^{\mathrm{ref}}) = \ell^V\Delta^{\mathrm{ref}}$. The p with $\ell_y(p) \ne 0$ lie within ϱ_L of y, and their boxes of radius R_{ref} lie within $\varrho_L + R_{\mathrm{ref}} \le R_T$ of $y(y)$; the transports have norm ≤ 4, and $\sum_p |\ell_y(p)| \le C_D$; hence $\|\Delta_{\sigma,y}\| \le 4C_DK_{\mathrm{ref}}\varepsilon'^2$. Next, $\Delta_F = \bar{F} - Y_* + \ell^{V*}\sigma_* = \Delta_Y - \ell^{V*}\Delta_\sigma$ identically, because $\mathsf{B}^V = 1 + \ell^{V*}\ell^V$. Only the y within ϱ_L of p occur, and $2\varrho_L + R_{\mathrm{ref}} \le R_T$ by (0.1); so $\|(\ell^{V*}\Delta_\sigma)_p\| \le C_D \cdot 4 \cdot K_\sigma\varepsilon'^2 = 16C_D^2K_{\mathrm{ref}}\varepsilon'^2$. Combined with (f), this gives the constant K_F.

(3) The bound on $\Delta_{n,c}$. Let $o = c_-$, and consider the box of radius R_3 about o. Here $\bar{n}_c = (\mathsf{N}^\natural\tilde{X})_c + \Lambda_c$ exactly. The kernel $dN := \mathsf{N}^\natural - \mathsf{N}^{\sharp<}$ vanishes beyond R_2 and has row sums $\le \mathsf{ti}_{\mathrm{c}}$, so that $\|(dN\,\tilde{X})_c\| \le 8.24\mathsf{ti}_{\mathrm{c}}\varepsilon'$. We apply Lem. 4.3 to $((aL)^{-1}\mathrm{d}'^*)$ and to $((aL)^{-1}\mathrm{d}'^*, \ell'^{c*}, \ell'^c)$, with ranges $\le 1 + 2\varrho(x')$; the norms contribute $(aL)^{-1}4e(1+C_D^2) \le (1+C_D^2)$, since $4e/18 \le 0.605$, and therefore $\|\omega_{*c} - ((aL)^{-1}\mathrm{d}'^*\mathsf{B}'^c\tilde{X})_c\| \le 10^6\theta_D^{-2}(1+C_D^2)\varepsilon'^2$. On the free side,

$$\mathsf{N}^{\sharp<} - (aL)^{-1}\mathrm{d}'^*\mathsf{B}'^c = -\mathsf{N}^{\sharp>} + (aL)^{-1}\mathrm{d}'^*(\mathsf{B}' - \mathsf{B}'^c),$$

with row sums $C_Dx^{-128} + (4/18)2C_D^2x'^{-64}$, to be multiplied by $8.24\varepsilon'$. The terms linear in ε' are $\le \varepsilon'^2$, by $(\mathrm{F9}_k)$. The sum of all contributions is K_n.

(4) The identity (g). For $\psi \in \mathfrak{L}$,

$$\langle\nu_*, \mathrm{d}^V\psi\rangle + \langle\sigma_*, \ell^V\mathrm{d}^V\psi\rangle = \langle\nu_* + \ell^{V*}\sigma_*, \mathrm{d}^V\psi\rangle = \langle Y_*, \mathrm{d}^V\psi\rangle = \langle Z_*, W^V\mathrm{d}^V\psi\rangle + \langle\Xi_*, \mathrm{d}^V\mathrm{d}^V\psi\rangle,$$

and

$$a\langle\omega_*, Q_1^V\psi\rangle = L^{-1}\langle\mathrm{d}'^{V*}v_*, Q_1^V\psi\rangle = \langle Z_*, \mathrm{d}'^VQ_1^V\psi\rangle.$$

Subtracting the second identity from the first, we obtain (g).

It remains to prove the sup bound on a_0; let $V' \in \mathcal{V}'$. By (C2.1), Lem. 3.25 (a) and (C2.5),

$$\sum_{b'}\|C(b,b')\| \le \sum_{\mathrm{dist}\le R}[\|q^{-1}(b,b')\| + \|(C-q^{-1})(b,b')\|] \le (2/\gamma)\mathsf{g}_I(\theta_{\mathbb{C}}) + 8\gamma^{-2}x^{-60}\,3L^3(2R+1)^3.$$

By (P1), the second term is $\le x^{-59}/(165(L+1)\mathsf{N}_{\mathsf{B}}(1+b)) \le 1/(165\cdot 4\cdot 17) < 10^{-4}$, whereas $(0.1/\gamma)\mathsf{g}_I(\theta_{\mathbb{C}}) \ge 0.1\cdot 5913\cdot 81$. Hence the sum is $\le (2.1/\gamma)\mathsf{g}_I(\theta_{\mathbb{C}})$. Moreover $|(\mathsf{g}^VX')_{b'}| \le$

$4\mathsf{N}_\mathsf{B}\sup|X'_P|$, with the supremum over the P with y_P within $2\varrho_L+1$ of $y(b')$, since the transports are isometries (Lem. 3.24 (d)). As $2.1\cdot 4=8.4$, the bound of (g) follows.

Step 9: the law, the order of the constants, and the floors. We show that the law of windows has (C2.0) to (C2.5); that the hypotheses of Thm. 3.32 and of Lem. 3.37 (a), (b) are met; that the letters are fixed without a circle; and that each floor of item G holds for all large x and depends neither on N, ℓ, c_h or E_a nor on a member.

The law of windows is that of item C.I 3 of §3, Thm. 3.6. The head of Thm. 3.6, before its clauses, states that they hold at every field with (u1$^\flat$), that is, on $\mathfrak{V}$; this is (C2.0). Lem. 3.3 and Thm. 3.6 require $L\geq 3$ odd, $aL\geq 18$ and rows as in item C.I 1.1 of §3; Lem. 3.24 and Thm. 3.32 require rows as in item C.II 1.2 of §3; and Lems. 3.25, 3.26, 3.27 and 3.37 are stated under the same requirements. These requirements are met. Indeed, $L\geq 3$ odd and $aL\geq 18$ is (O4), and N is even by item B. Item B supplies rows of the kind described in item C.I 1.1 of §3, and equally of the kind described in item C.II 1.2 of §3, at $(K,\lambda,\varrho)=(K_*,\lambda_*,\varrho(x))$, since it gives the bound $|\ell_y(p)|\leq K_* e^{-\lambda_*|y-\pi(p)|_\infty}$ and the range $\leq\varrho(x)$. Here $K_*\geq 0$ and $0<\lambda_*\leq 1$, because these are the pure numbers of (O0), which Def. 2.56 gives as $K_*=3\gamma_*^{-1/2}$ and $\lambda_*=\gamma_*^{1/2}/44$, and because $\gamma_*\leq 1$, as clause (0) of the order of the constants in Thm. 2.65 gives $\gamma_*=\mathsf{r}_\sharp^2/12296\leq 9.05513\cdot 10^{-25}$. Further, $\varrho(x)/L+2\leq\varrho_L+2<N/2$, since $N\geq 16(R_T+1)$ by (0.3) and $R_T+1=4R+2\varrho_L+7>\varrho_L+2$. At $\ell=\emptyset$ the same requirements read $0\leq K_* e^{-\lambda_*|y-\pi(p)|_\infty}$ and range $0\leq\varrho(x)$.

Thm. 3.6 (b) gives (C2.1): the range is $2r_w$ and the radius of dependence is $3r_w+\varrho_L+1$, and by (0.1) these are $\leq R$ and $\leq R_2-1$, which are (ρ1) and (ρ2) of the proposition "What Part C.II uses of the law" in item C.II 2.1 of §3. Thm. 3.6 (c) gives (C2.2). Thm. 3.6 (i), together with Thm. 3.6 (a), by which G is real at $\mathcal{V}'$ and so $C=GG^{\mathsf{T}}$ is real, gives (C2.3). Thm. 3.6 (f), (k), which apply since ($\mathrm{F}_{\mathrm{law}}$) and $c_R\theta_{\mathbb{C}}\geq 128$ hold, give (C2.4) in the norm of $\ell^2(I^\circ;\mathfrak{g}_{\mathbb{C}})$, that is, ($\rho$5) of the same proposition with $\mathsf{n}=60$. Finally (C2.5) is Lem. 3.26 (a): with $\mathsf{w}:=q^{-1}S$ we have $\|\mathsf{w}\|\leq(2/\gamma)x^{-60}\leq 1/2$ by (P1), hence $C=(1+\mathsf{w})^{-1}q^{-1}$ and $q^{-1}-C=\mathsf{w}(1+\mathsf{w})^{-1}q^{-1}$.

For the input (C3): Thm. 3.32 requires (λ0), (ρ1), (ρ2), (ρ5), (Dec) for G,G_Z, (P0) and (P1). Lem. 3.37 (a) assumes (ρ1) and $R\geq 6$. Lem. 3.37 (b) requires (Fr1), (Dec) for $\mathsf{G},\mathsf{G}_Z,H^\sharp,\mathsf{N}^\sharp$, and ($\lambda$0), ($\rho$1), ($\rho$5), (P0), (P1); here (Fr1) is Lem. 2.38 (c), and (λ0) and (ρ1) are (C2.0) and (C2.1). All clauses of Lem. 3.37 are stated under (Fr1) to (Fr3). Of these, (Fr1) is algebra and holds for all rows, and (Fr2) is (Rows). Condition (Fr3) holds as well, although the proof of Lem. 3.37 (b) does not depend on it: by the remark "Source and hypotheses" in item C.II 2.2 of §3, it holds for every ℓ of $\mathcal{R}(b_*,K_*,\lambda_*,\frac{1}{9})$ in the setting (H1) $L\geq L_A$ odd, (H2) $aL\geq 3/c_L$, (H3) N even, which is (O4) together with item B. Hence the hypotheses of the lemma are met. Finally, (Tor), which Part C.II of §3 lists in (P0), belongs to the data. The hypotheses of Thm. 3.32 are met as well: the preceding paragraph gives (ρ2) and (ρ5), (Dec) for $\mathsf{G},\mathsf{G}_Z,H^\sharp,\mathsf{N}^\sharp$ is (Rows) of item D, and (P0) and (P1) are floors of item G.

The letters are fixed in the following order. First come $\gamma,\mathsf{N}_\mathsf{B},b,A^\flat,\theta_{\mathbb{C}},\vartheta_0^\flat$, which depend on (L,a,K_*,λ_*); then θ_D,C_D,c_R; then $\mathsf{K}_A,\mathsf{K}_{\mathrm{c}}^0,\mathsf{a}_{\mathrm{c}},\mathsf{K}_{\mathrm{c}}$; then K_Y, together with $K_{\mathrm{ref}},K_\sigma,K_F,K_n$. After M comes $K_{\mathrm{q},k}$, and then c_h, on which the proof does not depend. Then comes the condition $E_a\geq 32K_{\mathrm{q},k}$, which is a lower bound on E_a. It can be met, because Thm. 5.3 is stated for all E_a above its own two floors, namely for all $E_a\geq E_a^0$, where $E_a^0=\max\{32L^{-1/2}K_v,32K_{\mathrm{q}}\}$ is the number of (O12) in item 3 of §2.1 and $K_{\mathrm{q}}=K_{\mathrm{q},k}$. No clause of Thm. 4.2 uses this fact about Thm. 5.3. The condition $E_a\geq 32K_{\mathrm{q},k}$ itself enters only the last inequality of (d) and, through that inequality, (ca) with $\frac{1}{32}$ of the cap. Last comes $x_\flat$. The constant a_{c} enters none of the constants K, only (F9$_k$), and nothing earlier in the order depends on E_a; hence there is no circle.

Finally, each floor holds for all large x. In ($\mathrm{F}_{\mathrm{law}}$), in (P1) and in the second half of (F9$_k$), powers of $\log x$ are compensated by powers of x. In (P0) we need $x\geq L^2$, and $R\geq 6$ holds once $c_R\log x>5$. For (G1$_k$), $Mr'\geq 20c_R\log x'$, while

$$R_T+1\leq 4(c_R\log x+1)+2((c_R\log x+1)/3+1)+7=(14/3)c_R\log x+41/3;$$

since $\log x = \log x' + \log L$, the floor holds as soon as $(46/3)c_R \log x' \geq (14/3)c_R \log L + 41/3$; and $\rho_0 \leq R(x')$ holds for large x. In (F8$_k$), $\varepsilon' = A_0(\log x')^{\mathfrak{p}_0} x'^{-1/2}$ is compared with R_T^{-2}, and $p_0' \geq 1$ for large x. In the first half of (F9$_k$), $\mathsf{ti}_c \leq \mathrm{const}(\log x)^3 x^{-60} + \mathrm{const}\, x^{-64}$ and x'^{-64} are measured relative to $\varepsilon' \geq x'^{-1/2}$. None of the floors depends on N, c_h, E_a or a member, nor on more of ℓ than $(K_*, \lambda_*, \mathrm{range})$. □

4.2. **Local vertices.** We compare the exact integrand in the chart $\eta = G\xi - a_0$ with its quadratic approximation. The cell remainder $V_\square$ collects the differences in the plaquette action, the row energy and the step kernel, as well as the logarithm of the Haar density. The corresponding real factor $A_\square$ is defined on the entire chart. The key estimate is a local holomorphic bound of order $p_0^3 x^{-1/2}$ for $V_\square$ when the test $\chi_\square$ passes. In that region $A_\square = e^{V_\square}$.

Remark 4.4 (Letters with more than one meaning). The letter b is a fine bond where it is an index or an argument, and the number $b = 16\mathsf{N}_\mathsf{B} + a/L$ of (O6) where it is the base of a power (b^{-1}, $b^{-1/2}$, $b^{1/2}$); the letter d is the coboundary, but in $b \in dp$ the set dp consists of the four bonds of p. The letter y is a coarse site in $B(y)$ and in $T(y, y'')$; as the index of a row ℓ_y it is a fine site, and then $y(y)$ is its coarse site. The letter u, where it acts on a field or on a vector, is a gauge transformation; in $\rho[u; \ell]$ it is the triple (E^a, E^h, w) of members, and both meanings occur in Step 1 (3) of the proof of Thm. 4.5.

Theorem 4.5 (The vertex letter at a level of any depth)**.** The setting is the group $G = \mathrm{SU}(2)$ in dimension $d = 3$, with the drawn step of item C.I 4 ("The step, the chart") of §3.

Item 1: the letters. The letters follow the order (O0) to (O13) of item 3 of §2.1. The whole list is that of item 3 of §2.1, and the functions of L are written out in item A of §4.3. Some signs are only named below and are set elsewhere: I_L and P_ρ in item A of §4.3, under (O3); $A^\flat$ in item C.I 1.2 of §3; c_1^*, λ_2^*, θ_J and C_D in the statement of Thm. 2.65, where c_1^* is set in the order of the constants (0); (M2), (M4$^\flat$) and (M5r) in item 3 of §2.1, under (O7); K_c^0 in item C.II 4 of §3, under "Constants"; δ_3' in item F of §4.3; $K_{\mathrm{q},k}$ in item F of §4.1; (F_law) in Thm. 3.6 (k); (P0) and (P1) in item C.II 4 of §3, under "Floors", and (P3) there, before Lem. 3.37; $\mathfrak{E}$ in item E2 of §4.1; and $\mathcal{R}(b_*, K_*, \lambda_*, \frac{1}{9})$ in Defs. 2.46, 2.47 and 2.48.

(O0) The pure numbers of §2.2, of which the theorem uses K_* and λ_*, where $0 < \lambda_* \leq 1$. With them come the function $\mathsf{g}(t) = \sum_{n \in \mathbb{Z}^3} e^{-t|n|_\infty}$ of Def. 2.19 and the function $\mathsf{g}_1(t) = \sum_n |n|_\infty e^{-t|n|_\infty}$ of the display of g_1, g_2 in Def. 2.44; we put $\mathsf{g}_* = \mathsf{g}(\lambda_*)$, $\mathsf{g}_1 = \mathsf{g}_1(\lambda_*)$ and $\mathsf{g}_I(t) = 3L^3\mathsf{g}(t)$.

(O1) A number $d_0 \in (0, \frac{1}{2})$, which enters through the function S_ε and, by the display eq. 2.vi of §2.2, through ρ_0.

(O3) The radius ρ_0 of the display eq. 2.vi of §2.2, which enters the mask and (W0). Further, $\mathsf{N}_\mathsf{B} = \mathsf{N}_Z = 1 + 3K_*^2\mathsf{g}_*^2$, $\mathsf{N}_\sharp = c_1^*\mathsf{I}_L\mathsf{N}_Z\mathsf{P}_\rho$ and $\mathsf{N}_\mathrm{c} = 17(\mathsf{N}_\sharp + 1)$; the last two enter (d) alone.

(O4) An odd $L \geq L_*^{\mathrm{su}}$, with the floor L_*^{su} of item 3 of §2.1, and $a \geq 3/(c_L L)$. Of this, the theorem uses that $L \geq 3$ is odd and that $aL \geq 18$. Clause (d) depends on the condition ($\mathrm{L2}^{\mathrm{su}}$), $L^{1/2} \geq 1.2 \cdot 10^4 \mathsf{N}_\mathrm{c}^2$, only through $\mathsf{N}_\mathrm{c} L^{-3/2} \leq 0.2$.

(O6) From item C.I 1.2 of §3, taken at (K_*, λ_*): $\gamma = (73L^4)^{-1}$, $b = 16\mathsf{N}_\mathsf{B} + a/L$, the number $A^\flat$, and $\theta^\flat = \min\{1, \lambda_* L/12, \gamma/(2A^\flat)\}$. Then $\theta_\mathbb{C} = \theta^\flat/6$ and $\vartheta_0^\flat = \theta^\flat/24$, which is at most 10^{-4}. Further, $\theta_D = \frac{1}{2}\min\{\lambda_2^*, \theta_\mathbb{C}, \theta_J\}$, $C_D \geq 4$ and $c_R = 64/\theta_D$; hence $c_R\theta_\mathbb{C} \geq 128$ and $c_R \geq 768$.

(O7) The cell side $M = L^{m'}$, $m' \geq 1$, with (M2), (M4$^\flat$) and (M5r). The theorem depends on no condition on M.

(O8) Numbers $\mathfrak{p}_0 > 2$ and $A_0 > 0$. We put $p_0(x) = A_0(\log x)^{\mathfrak{p}_0}$, $\varepsilon = p_0 x^{-1/2}$ and $x' = x/L$, and a prime means "at x'". Then $\varepsilon' \leq L^{1/2}\varepsilon$; indeed $\varepsilon' = A_0(\log x')^{\mathfrak{p}_0} L^{1/2} x^{-1/2}$, and $0 < \log x' \leq \log x$ because $x' = x/L \geq L > 1$.

(O9) The numbers $\mathsf{G}_0 = (2/\gamma)^{1/2}\mathsf{g}_I(\theta_{\mathbb{C}}) \le C_\varphi^\flat = (2b^{1/2}/\gamma)\,\mathsf{g}_I(\theta_{\mathbb{C}})/(1-e^{-\theta_{\mathbb{C}}})$; further $\mathsf{n} = 60$ and $c_\xi = (6.5\,C_\varphi^\flat)^{-1}$.

(O10) From item C.II 4 of §3, under "Constants", the number $\mathsf{a}_{\mathrm{c}} = (100/\gamma)\mathsf{N}_{\mathsf{B}}\mathsf{g}_I(\theta_{\mathbb{C}})$, which exceeds 7, and the number $\mathsf{K}_{\mathrm{c}}^0$ of that item; we put $\mathsf{K}_{\mathrm{c}} := \sqrt{2}\mathsf{K}_{\mathrm{c}}^0$, which is the number of (Cen). The theorem has three numbers of its own. The first is $C_\eta^\flat := \frac{1}{6.5} + \max\{\mathsf{a}_{\mathrm{c}}, 7\}L^{1/2}$, and for $\tau \ge 1$ we set $C_\eta^\flat(\tau) := \frac{1}{6.5} + \max\{\mathsf{a}_{\mathrm{c}}, 7\}\,\tau L^{1/2}$. The second is $K_\ell := 291K_*^2\mathsf{g}_*\,[1320\mathsf{g}_1 + 216\mathsf{g}_*]$. The third is $K_v^\flat := (LM)^3(1157 + 474a + K_\ell)(C_\eta^\flat)^3$, and $K_v^\flat(\tau)$ is given by the same formula with $C_\eta^\flat(\tau)$ in place of $C_\eta^\flat$. These three numbers are computed from $(L, M, a, K_*, \lambda_*)$ alone, and from τ where it occurs.

(O11) A number c_h with $0 < c_h \le \delta_3'/2$, on which the theorem does not depend.

(O12) A number $E_a \ge \max\{32K_{\mathrm{q},k},\, 1500K_v^\flat L^{-1/2}\}$, whose two entries are those of item 3 of §2.1 under (O12). Although the condition (O12) involves $K_v^\flat$, the order of the constants is not circular, because no statement and no proof of this section depends on (O12).

(O13) The letter $x_\flat$ comes last; it is taken so that (W0) to (W5) of item 6 hold for all $x \ge Lx_\flat$.

Item 2: the data. The data are the data (SD) of item 1 of §2.1. They consist of a coupling $x \ge L^2$; of an even coarse side N with $M \mid N$ and (Tor) $N \ge 2Ms(x) = 16Mr(x)$, together with the tori $\mathbb{T} = \mathbb{Z}^3/NL\mathbb{Z}^3$ and $\mathbb{T}' = L\mathbb{Z}^3/NL\mathbb{Z}^3$; and of rows $\ell \in \mathcal{R}(b_*, K_*, \lambda_*, \frac{1}{9})$ of range at most $\varrho(x)$ such that the pair (ℓ, ℓ') satisfies the four conclusions of Thm. 2.64.

The empty rows are admitted. Indeed, $\ell = \emptyset$, with $\mathsf{H}_y = \{0\}$ for all y, has range 0 and lies in $\mathcal{R}(1, 0, \lambda_*, 0)$, which is contained in $\mathcal{R}(b_*, K_*, \lambda_*, \frac{1}{9})$ by the display and the last sentence of Def. 2.48. The data refer to no level and to no depth. At $\ell = \emptyset$, the second sum in $V_\square$, the second product in $A_\square$ and the mask are void, and K_ℓ is a bound for 0; all numbers and radii remain those evaluated at $(K_*, \lambda_*, \varrho(x))$.

The four conclusions are stated in full in item 1 of §2.1, under (SD). The fourth, (L1), (L2), is taken in the form (L1^{su}), (L2^{su}) of item 3 of §2.1, under (O4), as (SD) prescribes. The first, (Dec)(C_D, θ_D), is taken at the θ_D of (O6), which is defined with $\theta_{\mathbb{C}}$; in this form, and for all sixteen kernels of (Dec), it is the conclusion of Thm. 2.65, whereas Prop. 2.62 has θ_q in the place of $\theta_{\mathbb{C}}$.

The four conclusions imply (Fr2) and (Fr3) of item C.II 2.2 of §3, and (Fr1) is Lem. 2.38 (c), which is algebra and holds for all rows. Of the rows ℓ, the proofs of (a), (b), (c), (e), (f), (g) and (h) depend only on the bound $|\ell_y(p)| \le K_* e^{-\lambda_*|y-p|}$ and on the range, while (d) uses the rest, through Cor. 3.38 (a).

Item 3: the objects.

Definition 4.6 (Item 3.1: the lattice)**.** We use the objects of item C.I 1.1 of §3. There are the blocks $B(y)$; the base site $\pi(i)$ of an object i, which is its least corner; the coarse site $y(i)$ whose block contains $\pi(i)$; the site $y_P = \pi(P)$; and the distance $\mathrm{dist}(i, j) = L^{-1}|y(i) - y(j)|_\infty$. The combs are $\Gamma(y, x)$, $\mathcal{T}$ is the set of their bonds, I° is the set of fine bonds off $\mathcal{T}$, and $\mathfrak{L}_{\mathfrak{g}}$ is the space of $\mathfrak{g}$-valued functions on I°. Each coarse bond c has its crossing bonds, and each coarse plaquette P has the edge plaquettes under it. Besides the coboundary d, we use $(Q_1A)(c) = L^{-3}\sum_{x \in B(c_-)} A([x, x'])$ and the edge lift, $(EX')_p = X'_P$ on the L edge plaquettes under P and 0 elsewhere. In fine units, $|y - p| = |y - \pi(p)|_\infty$. At a level, $\Gamma(z, z')$ is the coordinate path that runs along e_1, e_2, e_3 in this order, each time the short way.

A cell $\square$ is a cube of M^3 coarse sites of a fixed grid, and $\hat{\square}$ is the set of the fine sites of the blocks of $\square$. We write $p, b, y \in \hat{\square}$ for a fine plaquette, bond or site that is based there, and $c \in \square$ for a coarse bond with $c_- \in \square$. Let $\mathrm{dist}(i, \square) = \min_{y \in \square} L^{-1}|y(i) - y|_\infty$; let $\square_{[j]}$ be the box of the coarse sites within j of $\square$; and let $\square^{+k}$ be $\square$ enlarged by k layers of cells; its sites are $\square_{[kM]}$. Finally $r(x) = \lceil 20c_R \log x/M \rceil$, $r' = r(x')$, $s(x) = 8r(x)$ and $\square^{\approx} = \square^{+r'}$. ∎

Definition 4.7 (Item 3.2: the group and the norms)**.** Here $G = \mathrm{SU}(2)$, and $\mathfrak{g}$ is the space of hermitian traceless 2×2 matrices; we use the angle ang and the logarithm $\mathsf{L}(g)$ on G, with $\mathsf{L}(-1) = 0$. The operator norm is written $|\cdot|$. On $\mathfrak{g}_{\mathbb{C}} = \mathfrak{sl}_2(\mathbb{C})$ we use the bilinear form $Y \cdot Y' = \frac{1}{2}\operatorname{tr} YY'$ and the norm given by $\|Y\|^2 = \frac{1}{2}\operatorname{tr} Y^*Y$; on $\mathfrak{g}$, $\|Y\| = |Y|$. For $|g-1| < 1$ we put $L_{\mathbb{C}}(g) = -i\sum_{k\geq1}\frac{(-1)^{k-1}}{k}(g-1)^k$. Further, $\mathsf{j}(Y) = s(Y\cdot Y)^2$ with $s(t) = \sum_k (-t)^k/(2k+1)!$; on $\mathfrak{g}$ this equals $(\sin|Y|/|Y|)^2$. The real Euclidean space of the row y is H_y. On $\mathsf{H}_y \otimes \mathfrak{g}_{\mathbb{C}}$, in an orthonormal basis e_i, we write $A = \sum e_i \otimes A_i$ and set $A \cdot A' = \sum A_i \cdot A_i'$ and $\|A\|^2 = \sum \|A_i\|^2$. ■

Lemma 4.8 (Complex 2×2 matrices: inverse, logarithm, products of exponentials, trace and Haar density)**.** In this lemma g, u and Y_k are complex 2×2 matrices.

(a) For $g \in \mathrm{SL}(2,\mathbb{C})$ we have $|g^{-1}| = |g| \geq 1$, $|g^{-1}-1| = |g-1|$, $\|\operatorname{Ad}_g\| \leq |g|^2$ and $\|\operatorname{Ad}_g - 1\| \leq 2|g||g-1|$; moreover $\operatorname{tr} Y$ and $Y\cdot Y'$ are invariant under Ad_g.

(b) Let $|g-1| = t < 1$. Then the series $L_{\mathbb{C}}(g)$ converges, $|L_{\mathbb{C}}(g)| \leq t + t^2/(2(1-t))$, $e^{iL_{\mathbb{C}}(g)} = g$. For every invertible u, the same logarithm series converges at ugu^{-1} and equals $uL_{\mathbb{C}}(g)u^{-1}$. If $\det g = 1$, then $\operatorname{tr} L_{\mathbb{C}}(g) = 0$; if $\det g = 1$ and $t < \frac{1}{2}$, then $L_{\mathbb{C}}(g^{-1}) = -L_{\mathbb{C}}(g)$; and if $g \in \mathrm{SU}(2)$, then $L_{\mathbb{C}}(g) = \mathsf{L}(g)$.

(c) Let $S = \sum|Y_k| \leq \frac{1}{4}$ and $g = \prod e^{iY_k}$. Then $|g-1| \leq e^S - 1$ and $|L_{\mathbb{C}}(g) - \sum Y_k| \leq \frac{3}{2}S^2$.

(d) Let $Y \in \mathfrak{sl}_2(\mathbb{C})$ and $t = Y\cdot Y$ with $|t| \leq 1$. Then $|1 - \frac{1}{2}\operatorname{tr} e^{iY} - \frac{t}{2}| \leq |t|^2/23$ and $|\log \mathsf{j}(Y)| \leq |t|/2$, where $\log \mathsf{j}(Y) := 2\log s(t)$ with the principal value of the logarithm. For all Y and Y' in $\mathfrak{g}_{\mathbb{C}}$, $|Y\cdot Y'| \leq \|Y\|\,\|Y'\|$.

■

Proof. Step (a): the inverse and the adjoint action. Let $g \in \mathrm{SL}(2,\mathbb{C})$. Then g^{-1} is the adjugate of g. For 2×2 matrices the adjugate is linear in the matrix: $\operatorname{adj} A = JA^{\mathsf{T}}J^{-1}$ with the unitary $J = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$. Since transposition and conjugation by a unitary matrix leave the operator norm unchanged, we have $|\operatorname{adj} A| = |A|$; this gives $|g^{-1}| = |g|$ and, by linearity, $|g^{-1}-1| = |\operatorname{adj}(g-1)| = |g-1|$. Moreover $|g|^2 \geq |\det g| = 1$. For the two bounds on Ad_g we refer to the paragraph on norms in item C.II 1.1 of §3. Finally, the trace of a conjugate equals the trace of the matrix, so $\operatorname{tr}(gYg^{-1}) = \operatorname{tr} Y$ and $\operatorname{tr}(gYg^{-1}gY'g^{-1}) = \operatorname{tr} YY'$; these two identities prove the last assertion of (a), that $\operatorname{tr} Y$ and $Y\cdot Y'$ are invariant under Ad_g.

Step (b): the logarithm. Let $|g-1| = t < 1$. The k-th term of the series $L_{\mathbb{C}}(g)$ has norm at most t^k/k, and $\sum_{k\geq1} t^k/k \leq t + (t^2/2)\sum_{k\geq0} t^k$, which is the bound stated in (b); so the series converges and obeys that bound. With $A = g-1$, the identity $e^{iL_{\mathbb{C}}(g)} = g$ reads $\exp(\log(1+A)) = 1+A$, an identity of power series that converge absolutely for $|A| < 1$. At ugu^{-1} each series term is the conjugate of the corresponding term at g. Its norm is bounded by $|u|\,|u^{-1}|t^k/k$, which proves absolute convergence and the covariance formula.

Now let $\det g = 1$, and let z and $1/z$ be the eigenvalues of g. They are eigenvalues of $1+(g-1)$ and of $1+(g^{-1}-1)$ respectively, and $|g^{-1}-1| = |g-1| = t$ by (a); hence $|z-1| \leq t$ and $|1/z - 1| \leq t$. So both lie in the region $\operatorname{Re} w > 0$, in which the series is the principal logarithm and $\log(1/z) = -\log z$. In a basis in which g is triangular, the diagonal of $L_{\mathbb{C}}(g)$ consists of $-i\log z$ and $-i\log(1/z)$; hence the trace of $L_{\mathbb{C}}(g)$ is 0.

If moreover $t < \frac{1}{2}$, then $|L_{\mathbb{C}}(g)| \leq -\log(1-t) < \log 2$. The identity $\log(\exp X) = X$ holds as an identity of power series that converge absolutely for $|X| < \log 2$. At $X = -iL_{\mathbb{C}}(g)$ we have $\exp X = g^{-1}$, and there the identity says $L_{\mathbb{C}}(g^{-1}) = -L_{\mathbb{C}}(g)$. Finally, let $g \in \mathrm{SU}(2)$ have the eigenvalues $e^{\pm i\vartheta}$ with $0 \leq \vartheta \leq \pi$. Then $|g-1| = 2\sin(\vartheta/2) < 1$ gives $\vartheta < \pi/3$, and the series produces the matrix that has the eigenvectors of g, carries the eigenvalues $\pm\vartheta$ on them and is hermitian and traceless; this matrix is $\mathsf{L}(g)$.

Step (c): products of exponentials. By (E) of item C.I 1.1 of §3, $|g - 1 - i\sum Y_k| \leq e^S - 1 - S$, so $|g-1| \leq e^S - 1 =: w \leq 0.2841$. The terms of the series with $k \geq 2$ give $|L_{\mathbb{C}}(g) + i(g-1)| \leq w^2/(2(1-w))$. We split $L_{\mathbb{C}}(g) - \sum Y_k = [L_{\mathbb{C}}(g) + i(g-1)] - i[g - 1 - i\sum Y_k]$, so $|L_{\mathbb{C}}(g) - \sum Y_k|$

is at most the sum of the two bounds found. Both bounds, e^S-1-S and $w^2/(2(1-w))$, divided by S^2, increase with S; at $S=\frac14$ they are at most 0.5445 and 0.9014, and their sum is at most $1.4459\le\frac32$. This proves $|L_{\mathbb C}(g)-\sum Y_k|\le\frac32 S^2$. Since $S\mapsto e^S-1$ and $S\mapsto\frac32S^2$ are increasing functions of S, both estimates of (c) hold when any bound $\le\frac14$ for $\sum|Y_k|$ replaces S.

Step (d): trace and Haar density. Since $\operatorname{tr}Y=0$ and $\operatorname{tr}Y^2=2t$, the Cayley–Hamilton theorem gives $Y^2=t1$. Hence $\frac12\operatorname{tr}e^{iY}=\sum_k(-t)^k/(2k)!$. In $1-\frac12\operatorname{tr}e^{iY}-\frac t2$ the terms with $k\le1$ cancel, and the terms with $k\ge2$ have modulus at most $|t|^2(\cosh1-\frac32)\le0.04309|t|^2\le|t|^2/23$; this is the first bound. Next, $|s(t)-1|\le(\sinh1-1)|t|=:u\le0.17521$, so $|\log\mathsf j(Y)|=2|\log s(t)|\le 2u/(1-u)\le0.4249|t|$, which gives the second bound. Finally, the Cauchy–Schwarz inequality gives $|\operatorname{tr}YY'|\le(\operatorname{tr}Y^*Y)^{1/2}(\operatorname{tr}Y'^*Y')^{1/2}$, and after division by 2 this is $|Y\cdot Y'|\le\|Y\|\,\|Y'\|$. ■□

Definition 4.9 (Item 3.3: fields and domains)**.** The set of G-valued coarse fields is $\mathcal V'$. We put

$$\mathfrak V=\{\mathrm{SL}(2,\mathbb C)\text{ fields on }\mathbb T',\ V'(\bar c)=V'(c)^{-1},\ |V'(c)|\le1+\vartheta_0^\flat\},$$

and $\mathfrak V^\circ$ is defined in the same way with $<$. Further, $\vartheta(x)=\vartheta_0^\flat(1-1/\log x)$, $\mathsf b(x)=2-1/\log x$, the room $\mathsf g_x=\vartheta_0^\flat\log L/(4(\log x)^2)$, and $\ell_w(x)=6Mr(x)+12$.

Given a set B of coarse sites, the bonds, the paths and the plaquettes of B are those all of whose sites belong to B. For $\tau\ge1$, the domain $U^\tau(B;x')$ is the set of $\mathrm{SL}(2,\mathbb C)$ fields on the bonds of B that satisfy three clauses:

(e1) $|V'(c)|<1+\vartheta(x')+\mathsf g_x$;
(e2) $|V'(\pi)|<\mathsf b(x')+\mathsf g_x$ for paths in B of at most $\ell_w(x')$ bonds;
(e3) $|\partial V'(P)-1|<2\tau\varepsilon'$.

The domain $\mathcal U_{\{\square\}}(x')$ is the set of $\mathrm{SL}(2,\mathbb C)$ fields on the bonds of $\square^\approx$ that satisfy

(u1) $|V'(c)|<1+\vartheta(x')$;
(u2) $|V'(\pi)|<\mathsf b(x')$ for paths in $\square^\approx$ of at most $\ell_w(x')$ bonds;
(u3) $|\partial V'(P)-1|<2\varepsilon'$.

Hence $\mathcal U_{\{\square\}}(x')\subset U^1(\text{sites of }\square^\approx;x')$. We write $\mathrm{cont}(V')$ for V' extended by 1 to the other bonds of $\mathbb T'$. Finally, $T(y,y'')=\mathrm{Ad}_{V'(\Gamma(y,y''))}$, and for a scalar kernel K we put $K^V(i,j)=K(i,j)T(y(i),y(j))$; in an adjoint, the transport appears as T^{-1}. ■

Definition 4.10 (Item 3.4: the step and the chart)**.** The chart is the fine field $\mathsf U(\eta;V')$ with $\mathsf U(\eta;V')_b=1$ on $\mathcal T$, $e^{i\eta_b}$ at the other bonds inside blocks, and $e^{i\eta_b}V'(c)$ at a crossing bond of c. At any G-valued fine field U, let $h_{c,x}=U(\Gamma(y,x))\,U([x,x'])\,U(\Gamma(y',x'))^{-1}$ for $c=(y,y')$ and $x\in B(y)$. With it, the three formulas below define $\tau_{c,x}$, $\bar\tau_c$ and $\mathsf M(U)(c)$ at every such U, as in item C.I 4 of §3, and $\mathsf M$ is a Borel map of U computed from U, L and the blocks alone. The formulas are $\tau_{c,x}=\mathsf L(h_{c,x}h_{c,y}^{-1})$, with $L_{\mathbb C}$ in place of $\mathsf L$ at complex fields, $\bar\tau_c=L^{-3}\sum_x\tau_{c,x}$ and $\mathsf M(U)(c)=e^{i\bar\tau_c}h_{c,y}$. In the chart, $h_{c,x}=U([x,x'])$.

With $s=ax$ and with the number $z(s)$ of item C.I 4 of §3, the kernel of the step is $k_s(g,h)=z(s)^{-1}e^{-s(1-\cos\mathrm{ang}(gh^{-1}))}$, and $\mathsf n_c=\mathrm{ang}(V'(c)\mathsf M(U)(c)^{-1})$. Finally, $\mathfrak D=\{|\eta_b|<\pi\text{ for all }b\}$ and $\mathsf c_0=(2\pi^2)^{-\#I^\circ}$. ■

Definition 4.11 (Item 3.5: rows and energy)**.** The dressed rows act by $(\ell^VF)_y=\sum_p\ell_y(p)\otimes T(y(y),y(p))F_p$, and $\mathsf B^V=1+\ell^{V*}\ell^V$. The quadratic form is $q(\eta)=q_\ell^V(\eta)=\langle\mathrm d^V\eta,\mathsf B^V\mathrm d^V\eta\rangle+a\langle Q_1^V\eta,Q_1^V\eta\rangle$, with the bilinear pairing of item 3.2; at real V' and η it equals $|\mathrm d^V\eta|^2+|\ell^V\mathrm d^V\eta|^2+a|Q_1^V\eta|^2$. The kernel is $\mathsf g^V=\Pi_{\mathfrak L}\mathrm d^{V*}\mathsf B^VE$, where $\Pi_{\mathfrak L}$ is the restriction to I°. At a fine field U we set $\ell_y^UF=\sum_p\ell_y(p)\otimes\mathrm{Ad}_{U(\Gamma(y,\pi(p)))}F_p$, where Γ is now the fine coordinate path.

At a real field U we put $F_p=\mathsf L(\partial U(p))$, $\phi_p=\mathrm{ang}\,\partial U(p)$, $D(U)=\{\phi_p>\varepsilon\}$ and $\zeta_p=F_p\min\{1,\varepsilon/\phi_p\}$, with multiplier 1 at $\phi_p=0$. Thus $|F_p|\le\phi_p$, with equality unless $\partial U(p)=-1$; at that point $F_p=\zeta_p=0$ and $\phi_p=\pi$, so the plaquette remains rough. The function S_ε is $S_\varepsilon(t)=t^2/2$

for $t \le \varepsilon$, and $d_0 \varepsilon t$ for $t > \varepsilon$. The mask is given by $m_y(U) = 1$ if and only if no p of $D(U)$ has $|y - p| \le \rho_0$ and no p with $\phi_p > 1$ has $|y - p| \le \varrho(x) + 1$; otherwise it is 0.

The class $C^{\mathrm{su}}_{\flat}(x; \ell)$ uses exactly this mask, with both of its radii. In §2.2, the mask of Def. 2.22 keeps only the first of the two clauses; the difference is harmless, because (Fr3′), in the conclusion of Thm. 2.65, is stated for every mask m. The energy is $\mathrm{En}_\ell(U) = \sum_p S_\varepsilon(\phi_p) + \frac{1}{2}\sum_y m_y(U) |\ell^U_y \zeta|^2$. ■

Definition 4.12 (Item 3.6: the law)**.** The law is that of Def. 3.5, in item C.I 3 of §3, taken at $R = R(x) = \varrho(x) = \lceil c_R \log x \rceil$. Let $r_w = \lfloor R/2 \rfloor$ and $Y_b = \{b' \in I^\circ : \mathrm{dist}(b, b') \le r_w\}$. Put $K_b = 1_{Y_b} - b^{-1} 1_{Y_b} q 1_{Y_b}$ and $c_k = 4^{-k}\binom{2k}{k}$. The matrix of the law is $G(b, b') = b^{-1/2} \sum_k c_k (K_b)^k (b, b')$ on Y_b, and 0 elsewhere. Further, $C = GG^{\mathsf{T}}$, $\mathsf{E} = G^{\mathsf{T}} q G - 1$ and $S = C^{-1} - q$, and the field is drawn as $\xi \sim \mathcal{N}(0, x^{-1} 1)$ on $\mathfrak{L}_{\mathfrak{g}}$. ■

Definition 4.13 (Item 3.7: the curvature, the centre and the field)**.** At a pair $(V', e) \in \mathcal{V}' \times \mathfrak{E}$, as in item E2 of §4.1, we put $X'_P = X'_P(V', e) = \mathsf{L}(\partial V'(P)) + \pi e_P 1\{\partial V'(P) = -1\}$. At a field V' that lives on the bonds of a set B and satisfies (e3), the field in use is $\mathrm{cont}(V')$, and we put $X'_P := L_{\mathbb{C}}(\partial V'(P))$ for each P with all four sites in B and $X'_P := 0$ for every other P; this is the convention of item C.II 1.4 of §3.

In both cases, for real $\xi \in \mathfrak{L}_{\mathfrak{g}}$, we set $h = G^{\mathsf{T}} \mathsf{g}^V X'$, $a_0 = Gh = C \mathsf{g}^V X'$, $\eta = G\xi - a_0$, $U = \mathsf{U}(\eta; V')$, $F^\flat = EX' + \mathrm{d}^V \eta$, $\bar{F} = EX' - \mathrm{d}^V a_0$ and $n^\flat = -Q_1^V \eta$. Finally $\mathcal{Q}(\eta) = \frac{1}{2} F^\flat \cdot \mathsf{B}^V F^\flat + \frac{a}{2} (Q_1^V \eta) \cdot (Q_1^V \eta)$ and $\mathsf{m}_C = \frac{1}{2} EX' \cdot \mathsf{B}^V EX' - \frac{1}{2} \mathsf{g}^V X' \cdot C \mathsf{g}^V X'$. ■

Definition 4.14 (Item 3.8: the vertex term)**.** With $F_p := L_{\mathbb{C}}(\partial U(p))$, the vertex term is

$$\begin{aligned} V_\square(\xi, V') := {} & \frac{x}{2} \sum_{p \in \hat{\square}} [F^\flat_p \cdot F^\flat_p - F_p \cdot F_p] \\ & + \frac{x}{2} \sum_{y \in \hat{\square}} [(\ell^V F^\flat)_y \cdot (\ell^V F^\flat)_y - (\ell^U_y F) \cdot (\ell^U_y F)] \\ & + \sum_{c \in \square} [\tfrac{s}{2} n^\flat_c \cdot n^\flat_c - s(1 - \tfrac{1}{2} \operatorname{tr} V'(c) \mathsf{M}(U)(c)^{-1})] \\ & + \sum_{b \in I^\circ \cap \hat{\square}} \log \mathsf{j}(\eta_b). \end{aligned}$$

■

Definition 4.15 (Item 3.9: the vertex letter)**.** At every real ξ and every pair (V', e), the vertex letter is

$$\begin{aligned} A_\square(\xi, V', e) := {} & \prod_{p \in \hat{\square}} \exp\{\tfrac{x}{2} |F^\flat_p|^2 - x S_\varepsilon(\phi_p)\} \\ & \cdot \prod_{y \in \hat{\square}} \exp\{\tfrac{x}{2} |(\ell^V F^\flat)_y|^2 - \tfrac{x}{2} m_y(U) |\ell^U_y \zeta|^2\} \\ & \cdot \prod_{c \in \square} \exp\{\tfrac{s}{2} |n^\flat_c|^2 - s(1 - \cos \mathsf{n}_c)\} \\ & \cdot \prod_{b \in I^\circ \cap \hat{\square}} 1\{|\eta_b| < \pi\} \mathsf{j}(\eta_b) \ \ge 0. \end{aligned}$$

■

Definition 4.16 (Item 3.10: the tests)**.** Let $R' \ge R^{\min}_\chi$ be an integer. The test is given by $\chi_\square(\xi) = 1$ if and only if $|\xi_b| \le c_\xi \varepsilon$ at every $b \in I^\circ$ with $\mathrm{dist}(b, \square) \le R'$; otherwise it is 0. The radius in force is $R' = R_\chi$. Further, $b_\square(\xi) = 1$ if and only if $|\xi_b| > c_\xi \varepsilon$ at a bond of I° in $\hat{\square}$, as in item E10 of §4.3. ■

Item 4: the radii. The radii form the one set of item 2 of §2.1; they are not letters of the class, and those from ϱ_L to r_1 are computed from x and the letters L, M and c_R. They are $\varrho_L = \lceil R/L \rceil$, $r_w = \lfloor R/2 \rfloor$, $R_2 = 2R + 2$, $R_3 = 2R + \varrho_L + 5$ and $R_v = \varrho_L + 2 + R_3 = 2R + 2\varrho_L + 7$; then $R_\chi^{\min} = r_w + \varrho_L + 2$ and $R_\chi = R + \varrho_L + 2$; and, as in item E8 of §4.3, $r_l = \lceil (\varrho_L + 2)/M \rceil$ and $r_1 = \lceil R_2/M \rceil + 3r_l + 5$. We also put $r := C_\eta^\flat \varepsilon$ and $r_\tau := C_\eta^\flat(\tau)\varepsilon$.

Item 5: the inputs. The first group concerns the law and comes from Part C.I of §3. Of Thm. 3.6 it contains the following clauses.

(a) The matrix G is real at $\mathcal{V}'$.
(b) The ranges, and what each object depends on.
(c) Covariance.
(d) The bound $\|(G\xi)_b\| \le \mathsf{G}_0 \sup_{Y_b} \|\xi_{b'}\|$.
(i) Holomorphy on $\mathfrak{V}^\circ$ and continuity on $\mathfrak{V}$.

This group also contains Thm. 3.6 (f), (h), (k): under ($\mathrm{F}_{\mathrm{law}}$), and with $c_R\theta_{\mathbb{C}} \ge 128$, the matrix G is invertible, $\det G^V > 0$ at $\mathcal{V}'$, and $\|S\| \le x^{-60}$. It contains further Lem. 3.3 (c), (d), (e) and Prop. 3.10 (a), which is algebra and holds for any X'.

The second group concerns the centre and comes from Part C.II of §3. It contains Thm. 3.32 (a), (c): one has $|a_{0,b}| \le \sqrt{2}\|a_{0,b}\| \le \mathsf{a}_{\mathrm{c}}\hat{\varepsilon}$ at $\mathrm{dist}(b, o) \le 1$ for every coarse site o, every $0 < \hat{\varepsilon} \le \frac{1}{40}$ and every $\mathrm{SL}(2, \mathbb{C})$ field on the box of radius R_3 about o that has the three properties

(h1) $|V'(c)| \le 1 + \vartheta_0^\flat$;
(h2) $|V'(\pi)| \le 2$ on paths in the box of at most $\ell_w(x')$ bonds;
(h3) $|\partial V'(P) - 1| \le 2\hat{\varepsilon}$.

Item C.II 4 of §3 states Thm. 3.32 for every i, with $i = p$ or $i = c$ and with $o = y(p)$ or $o = c_-$ respectively; since every coarse site o is the c_- of the bond $c = (o, 1)$, the form given here, at a coarse site o, is that theorem taken with $i = c$. The proof of Thm. 3.32 obtains (a) for a_0 and (c) in its Steps (0) to (2), and these steps rest on Thm. 3.6 (b), (f), (k), on (P0) and on (P1), and on no condition (Fr). Among its hypotheses, Thm. 3.32 lists (Fr2) for G and G_Z, the two free kernels of item C.II 2.2 of §3; in the setting (SD) this hypothesis holds, by §2.2, so Thm. 3.32 applies here as it stands.

For (d) the second group contains Cor. 3.38 (a), which is (Cen)$(\mathsf{N}_{\mathrm{c}}, \mathsf{K}_{\mathrm{c}}, \mathsf{a}_{\mathrm{c}})$: under (Fr1) to (Fr3), (P0), (P1), (P3), for every p, every $0 < \hat{\varepsilon} \le \frac{1}{40}$ and every field on the box of radius R_3 about $y(p)$ that satisfies (e1), (e2) and $|\partial V'(P) - 1| \le 2\hat{\varepsilon}$, we have $|(EX' - \mathrm{d}^V a_0)_p| \le \mathsf{N}_{\mathrm{c}} L^{-2}\hat{\varepsilon} + \mathsf{K}_{\mathrm{c}}\hat{\varepsilon}^2$.

The third group consists of Lem. 4.8 of item 3.2 and of the formula of the chart for bounded, Borel, gauge invariant ρ, together with its justification, from Lem. 3.9.

The fourth group concerns the other sections. Of §4.3, the definitions of its items E8, E10 and E11 enter. Of §4.1, item E2 (the set $\mathfrak{E}$, and $X'(V', e)$ with the facts that X' is Borel in (V', e) and that $X'(V'^u, \mathrm{Ad}_u\, e) = \mathrm{Ad}_u\, X'(V', e)$) enters. Nothing of Thm. 5.3 is used by any clause of the theorem or by any step that proves one.

Item 6: the floors. Each floor below holds for all large x when the letters and τ are fixed.

(W0) $x \ge L^2$; $R \ge 12$; $\rho_0 \le \varrho(x) + 1$; the floor ($\mathrm{F}_{\mathrm{law}}$) of Thm. 3.6 (k); and, from item C.II 4 of §3, the clauses $x \ge L^2$ and $R \ge 6$ of (P0), together with (P1). The third clause of (P0) in item C.II 4 of §3, which is (Tor), is not a floor, since it involves N and, for each fixed N, fails once x is large; it is a datum of item 2. Hence no clause of (W0) to (W5) depends on N.
($\mathrm{W1}_\tau$) $22LC_\eta^\flat(\tau)\varepsilon \le 1$.
($\mathrm{W2}_\tau$) $17.1\, C_\eta^\flat(\tau)\, \varepsilon\, \varrho(x) \le 0.1$.
(W3) $p_0(x) \ge 1$.
($\mathrm{W4}_\tau$) $K_v^\flat(\tau) p_0^3 x^{-1/2} \le 1$.
(W5) For (d) alone: (P3) of §3, $\mathsf{N}_{\mathrm{c}} L^{-3/2} \le 0.2$, and $(\mathsf{K}_{\mathrm{c}} L + 54(C_\eta^\flat)^2)\varepsilon \le 0.18$.

We write (W1), (W2) and (W4) for the cases $\tau = 1$.

Item 7: the conclusions. Assume (W0) to (W4), and let $\square$ be a cell. In (a), (b) and (c), V' is any field in $\mathcal{U}_{\{\square\}}(x')$ and ξ is any real field with $\chi_\square(\xi) = 1$.

(a) Every series in $V_\square$ converges. As a function of ξ, $V_\square$ depends only on the values at the bonds of I° within $r_w + \varrho_L + 1$ of $\square$. As a function of V', it depends only on the coarse bonds with both ends in $\square_{[R_v]}$; since $R_v \le Mr'$, these are bonds of $\square^\approx$. It depends on x, on ℓ and on the letters of the law, and on no member E^a, E^h, w, on no e, and on neither c_h nor E_a. The map $V' \mapsto V_\square(\xi, V')$ is holomorphic on $\mathcal{U}_{\{\square\}}(x')$.

(b) For every G-valued u on the sites of $\square^\approx$, extended by 1, and with $(\mathrm{Ad}_u \xi)_b = \mathrm{Ad}_{u(y(b))}\, \xi_b$, one has $V_\square(\mathrm{Ad}_u \xi, V'^u) = V_\square(\xi, V')$ and $\chi_\square(\mathrm{Ad}_u \xi) = \chi_\square(\xi)$.

(c) One has $\|\eta_b\| \le r$ at every b with $\mathrm{dist}(b, \square) \le \varrho_L + 2$, and $|X'_P| \le 2.06\varepsilon'$. At every fine plaquette p with $\mathrm{dist}(p, \square) \le \varrho_L + 1$, $|F^\flat_p| \le 6r$, $|F_p| \le 8.25r$ and $|F_p - F^\flat_p| \le 54r^2$. For $y \in \hat\square$ and $|y - p| \le \varrho(x)$, $\|\mathrm{Ad}_{U(\Gamma(y,\pi(p)))} - T(y(y), y(p))\| \le 160\, r\, |y - p|$. Finally $|V_\square| \le K^\flat_v p_0^3 x^{-1/2}$ and $|e^{V_\square} - 1| \le 2K^\flat_v p_0^3 x^{-1/2}$.

(d) Assume (W5) as well, let (V', e) be any pair on $\mathbb{T}'$ such that the restriction of V' to the bonds of $\square^\approx$ lies in $\mathcal{U}_{\{\square\}}(x')$, and let $\chi_\square(\xi) = 1$. This condition on V' holds in particular when $\phi'_P \le \varepsilon'$ on $\square^\approx$; that case rests on $\log x' > 1$, which is true because $x' \ge L \ge 3$. Then $V_\square$ is real, and $|\eta_b| < \pi$ at every b with $\mathrm{dist}(b, \square) \le \varrho_L + 2$; next, $\phi_p \le \varepsilon$ at every fine plaquette p with $|y - p| \le \varrho(x) + 1$ for some $y \in \hat\square$; hence $\zeta_p = F_p$ at these plaquettes, $m_y(U) = 1$ for $y \in \hat\square$, and $S_\varepsilon(\phi_p) = \frac{1}{2}|F_p|^2$; and finally $A_\square(\xi, V', e) = e^{V_\square(\xi, V')}$ for every e.

(e) Here ξ is any real field and (V', e) any pair; (e.i) and (e.ii) hold for every $\mathfrak{g}$-valued X' put in place of $X'(V', e)$.

Of the floors, (e), (g) and (h) use the following. The letter $A_\square$, and the objects of items 3.1 to 3.7 that enter it, are defined at every real ξ and every pair once $L \ge 3$ is odd, $aL \ge 18$, and the inequalities on N, namely $M + 2R_v < N$, $\varrho(x) + 1 < NL/2 - L$ and $\varrho_L + 2 < N/2$, hold; here Thm. 3.6 (a) holds with no condition on the size of R. These inequalities on N rest on $x \ge L^2$, on $R \ge 6$ and on (Tor): the first and the third are data of item 2, and $R \ge 6$ follows from $x \ge L^2$, because $c_R \ge 768$ and $\log x \ge 2 \log 3$. Beyond this, (e.i), (e.iv), (e.v) and (g) depend on no floor. The clauses (e.iii) and (h) use, of (W0), only $\rho_0 \le \varrho(x) + 1$, for the mask. The clause (e.ii) uses only ($\mathrm{F}_{\mathrm{law}}$), with $c_R \theta_{\mathbb{C}} \ge 128$ of (O6). The floor (P1) and the floors (W1) to (W4) enter (a) to (d) and (f) alone. So (e.i), (e.iv), (e.v) and (g) hold in the setting of the data of item 2 alone.

(e.i) The product over all cells is

$$\prod_\square A_\square = 1_{\mathfrak{D}}(\eta) \prod_{b \in I^\circ} \mathrm{j}(\eta_b)\, e^{-x \mathrm{En}_\ell(U)}\, e^{\frac{x}{2}\langle F^\flat, \mathsf{B}^V F^\flat\rangle} \prod_c e^{\frac{s}{2}|n^\flat_c|^2 - s(1 - \cos \mathsf{n}_c)}.$$

For every set $\mathfrak{Z}$ of cells, $\prod_{\square \in \mathfrak{Z}} A_\square = [A]^\ell_{\mathfrak{Z}}$, as in item E11 of §4.3. These identities do not depend on the law.

(e.ii) Let $\rho = e^{\mathsf{V}} \rho[u; \ell]$ be a density of the class $C^{\mathrm{su}}_\flat(x; \ell)$, or any $\rho = e^{\mathsf{V}} \rho[u; \ell]$ with $\mathsf{V} \in \mathbb{C}$ and $u = (E^a, E^h, w) \in M^{\mathrm{su}}_A(x)$, $A \ge 1$, real or complex. The signs $\rho[u; \ell]$, Ξ_w, E^a, E^h, w, $X^\approx$, $M^{\mathrm{su}}_A(x)$ and $C^{\mathrm{su}}_\flat(x; \ell)$ are those of item 1 of §2.1, under "At x", "Members" and "The class"; the sum over X below runs over the connected sets of fine cubes. The

step is linear in ρ, while nothing is linear in u. Then

$$\rho'(V') = \mathsf{c}_0\, z(s)^{-3\#\mathbb{T}'} \Big(\frac{2\pi}{x}\Big)^{\frac{3}{2}\#I^\circ} \det G^V\ e^{\mathsf{V}-x\mathcal{Q}(-a_0)}$$
$$E_\xi\Big[e^{-\frac{x}{2}\langle\xi,\mathsf{E}\xi\rangle+x\langle\xi,\mathsf{E}h\rangle}$$
$$\prod_{\square} A_\square$$
$$\exp\Big\{\sum_X 1_{X\approx}(U)(E^a+E^h)(X,U)\Big\}\,\Xi_w(U)\Big],$$

where $\mathcal{Q}(-a_0) = \mathsf{m}_C - \frac{1}{2}\langle a_0, Sa_0\rangle$ and E_ξ is the mean over $\xi \sim \mathcal{N}(0, x^{-1}1)$ on $\mathfrak{L}_\mathfrak{g}$ of item 3.6. This formula rests on Thm. 3.6 (f), (h), (k).

(e.iii) The letter $A_\square$ depends on ξ only through the bonds of I° within $R_\chi^{\min}$ of $\square$, and on (V', e) only through bonds and plaquettes in $\square_{[R_v]}$.

(e.iv) One has $A_\square(\mathrm{Ad}_u\,\xi, V'^u, \mathrm{Ad}_u\, e) = A_\square(\xi, V', e)$ for every $u : \mathbb{T}' \to G$.

(e.v) The letter $A_\square$ is a Borel function of (ξ, V', e).

(f) Let $\tau \geq 1$, and let B be a set of coarse sites with $B \supset \square_{[R_v]}$. Suppose that (W0), (W1$_\tau$), (W2$_\tau$), (W3) and (W4$_\tau$) hold. These conditions by themselves give $\tau\varepsilon' \leq 1/40$, and the clauses (a), (b) and (c) hold for every $V' \in U^\tau(B; x')$, where B takes the place of $\square^\approx$, r_τ that of r, $\tau\varepsilon'$ that of ε' in the bound on X', and $K_v^\flat(\tau)$ that of $K_v^\flat$.

(g) The functions $V_\square$ and $A_\square$ are built from the cell $\square$ (and thus from M and the grid of cells) on the torus (and thus from N), from x, ℓ, L, a and $R(x)$ (and thus from c_R), and from the number b of (O6) (and thus from K_* and λ_*). In $A_\square$ there occur, in addition, d_0, ρ_0, A_0 and $\mathfrak{p}_0$. Neither function is built from anything else. Each of these data is a letter of the table or a datum of (SD), and none of them is a member, c_h or E_a. Consequently, any two densities belonging to a single class $C_\flat^{\mathrm{su}}(x;\ell)$ on a single torus share their vertex letters. This clause is independent of the reality and of the positivity of the members.

(h) The radii obey the inequalities

$$R_\chi^{\min} \leq R_\chi, \qquad 2\lceil R_\chi/M\rceil \leq r_1 - 5, \qquad \lceil R_v/M\rceil \leq r_1 - 5.$$

Consequently, if $b_{\square'}(\xi) = 0$ for every cell $\square'$ of $\square^{+r_1}$, then $\chi_\square(\xi) = 1$. Moreover, $A_\square$ depends on ξ only under $\square^{+r_1}$ and on (V', e) only in $\square^{+r_1}$. Finally, if two cells $\square$ and $\tilde\square$ are more than $2\lceil R_\chi^{\min}/M\rceil$ cells apart, then the letters $A_\square$ and $A_{\tilde\square}$ depend on disjoint sets of the variables ξ.

Item 8: what is used. The proofs of (a), (b), (c) and (f) rest on the following results: Thm. 3.6 (a) to (d), (f), (i) and (k); Lem. 3.3 (c), (d) and (e); Thm. 3.32 (c); Lem. 4.8.

The proof of (d) rests, in addition, on Cor. 3.38 (a), and hence, for the pair (ℓ, ℓ') of the data, on (Fr1) to (Fr3) of item C.II 2.2 of §3, which hold in the setting (SD) of item 2 of this section. ■□

Proof of Thm. 4.5. Step 1: the letters as factors of the integrand; clauses (e.i) *and* (e.ii). Let ξ be real and $V' \in \mathcal{V}'$, and let X' be any $\mathfrak{g}$-valued function on the coarse plaquettes, for instance $X'(V', e)$. We claim three things. First,

$$\prod_\square A_\square = 1_\mathfrak{D}(\eta) \prod_{b\in I^\circ} \mathsf{j}(\eta_b)\, e^{-x\mathrm{En}_\ell(U)}\, e^{(x/2)\langle F^\flat, \mathsf{B}^V F^\flat\rangle} \prod_c e^{(s/2)|n_c^\flat|^2 - s(1-\cos \mathsf{n}_c)}.$$

Second, $\prod_{\square\in\mathfrak{Z}} A_\square = [A]_\mathfrak{Z}^\ell$. Third, if ($\mathrm{F}_{\mathrm{law}}$) holds, then the display of (e.ii) holds.

(1) *Splitting.* For real V' the transports satisfy $T^{-1} = T^\mathsf{T}$, so $(\ell^V)^*$ is the adjoint of ℓ^V, and therefore

$$\langle F^\flat, \mathsf{B}^V F^\flat\rangle = \sum_p |F_p^\flat|^2 + \sum_y |(\ell^V F^\flat)_y|^2.$$

By item 3.5, $\mathrm{En}_\ell(U) = \sum_p S_\varepsilon(\phi_p) + \frac12 \sum_y m_y(U) |\ell_y^U \zeta|^2$; moreover $1_{\mathfrak{D}}(\eta) = \prod_b 1\{|\eta_b| < \pi\}$. Each fine plaquette, each fine site and each bond of I° is based in exactly one $\hat\square$, and for each coarse bond c the site c_- lies in exactly one $\square$. So the right side of the first claim is the product, over the cells, of the four products of item 3.9, which proves the first claim; no estimate enters here, and this part is independent of G and of X'. Put $\hat{\mathfrak{Z}} := \bigcup_{\square\in\mathfrak{Z}} \hat\square$. Taken over the set $\hat{\mathfrak{Z}}$, the four products agree, letter by letter, with those of item E11 of §4.3, because $ax = s$; this proves the second claim.

(2) *Algebra.* This part is Prop. 3.10 (a), which is stated for every X'; we give its derivation. We have $\mathcal{Q}(\eta) = \frac12 EX' \cdot \mathsf{B} EX' + \mathsf{g}^V X' \cdot \eta + \frac12 \eta \cdot q\eta$, where q and B are symmetric. With $\eta = G(\xi - h)$ and $h = G^{\mathsf{T}} \mathsf{g}^V X'$ we get $\mathsf{g}^V X' \cdot \eta = h \cdot (\xi - h)$ and $\eta \cdot q\eta = (\xi - h) \cdot (1 + \mathsf{E})(\xi - h)$, and hence

$$\mathcal{Q}(\eta) = [\frac12 EX' \cdot \mathsf{B} EX' - \frac12 h \cdot h + \frac12 h \cdot \mathsf{E} h] + \frac12 \xi \cdot \xi + \frac12 \xi \cdot \mathsf{E}\xi - \xi \cdot \mathsf{E} h.$$

Since $h \cdot h = \mathsf{g}^V X' \cdot C \mathsf{g}^V X'$, the bracket equals $\mathcal{Q}(-a_0) = \mathsf{m}_C + \frac12 h \cdot \mathsf{E} h$. With $S = -G^{-\mathsf{T}} \mathsf{E} G^{-1}$, as given by Thm. 3.6 (f), we obtain $h \cdot \mathsf{E} h = -a_0 \cdot S a_0$.

(3) *Change of variables.* The formula of the chart, Lem. 3.9, states that for every bounded, Borel, gauge invariant ρ

$$\rho'(V') = \mathsf{c}_0 \int_{\mathfrak{D}} \rho(\mathsf{U}(\eta; V')) \prod_c k_s(V'(c), \mathsf{M}(U)(c)) \prod_{b \in I^\circ} \mathsf{j}(\eta_b)\, d\eta.$$

As there are $3\#\mathbb{T}'$ coarse bonds, the kernels contribute the factor $z(s)^{-3\#\mathbb{T}'} \prod_c e^{-s(1-\cos \mathsf{n}_c)}$. We multiply and divide the integrand by $e^{-x\mathcal{Q}(\eta)}$, where $x\mathcal{Q}(\eta) = (x/2)\langle F^\flat, \mathsf{B}^V F^\flat\rangle + (s/2)\sum_c |n_c^\flat|^2$. If $(\mathrm{F}_{\mathrm{law}})$ holds, then, as $c_R \theta_{\mathbb{C}} \geq 128$, Thm. 3.6 (k) gives $4b^{1/2} d_R \leq 1$, and by Thm. 3.6 (a), (f) and (h) the matrix G^V is real and invertible with $\det G^V > 0$. Therefore $\eta = G\xi - a_0$ is a bijection of $\mathfrak{L}_{\mathfrak{g}}$, with $d\eta = \det G^V\, d\xi$, and a_0 does not depend on ξ. We also have $\int e^{-(x/2)|\xi|^2} d\xi = (2\pi/x)^{(3/2)\#I^\circ}$. Now, in the exponent $-x\mathcal{Q}(\eta)$, we replace $\mathcal{Q}(\eta)$ by the expression for it found in part (2), and for $\rho(U)$ we put the formula of the class from item 1 of §2.1, where the members are defined,

$$\rho[u; \ell] = \exp\{-x\mathrm{En}_\ell + \sum_{X \in \mathcal{D}} 1_{X^\approx}[E^a + E^h](X, \cdot)\} \Xi_w;$$

using in addition the splitting of part (1), we arrive at the display of (e.ii).

It remains to see that $\rho = e^{\mathsf{V}} \rho[u; \ell]$, with $u \in M_A^{\mathrm{su}}(x)$ real or complex, is admissible in this formula: it is gauge invariant, bounded and Borel on the G-valued fine fields of a finite torus. As to invariance, ϕ_p, $D(U)$, $m_y(U)$, $1_{X^\approx}$, $S_\varepsilon(\phi_p)$ and $|\ell_y^U \zeta|$ are invariant, and so is En_ℓ. Indeed, $\partial U^u(p) = u(\pi(p)) \partial U(p) u(\pi(p))^{-1}$, so that $\zeta_p \to \mathrm{Ad}_{u(\pi(p))}\, \zeta_p$, and this holds also at $\partial U(p) = -1$, where both sides are 0; and the transports are conjugated. The functions E^a, E^h and w are invariant by their clauses (ca_A), (ch_A) and $(\mathrm{c}\Xi_A)$.

As to boundedness, $\mathrm{En}_\ell \geq 0$ gives $e^{-x\mathrm{En}_\ell} \leq 1$, and there are only finitely many sets X. A G-valued field that is tame on $X^\approx$ lies in $\mathcal{U}_X(x)$ (item 1 of §2.1, where the domains are defined): we have $|U(b)| = 1 < 1 + \vartheta(x)$ and $|U(\pi)| = 1 < \mathsf{b}(x)$, both because $\log x > 1$, which holds since $x \geq L^2 \geq 9$; and $|\partial U(p) - 1| = 2\sin(\phi_p/2) \leq \varepsilon < 2\varepsilon$. Hence $|1_{X^\approx}(E^a + E^h)(X, \cdot)| \leq A(c_a(x) + c_h(x))$ and $|w| \leq A$. As to measurability, ϕ_p is continuous; L, the clip, the masks and the indicators are Borel; E^a is continuous on $\mathcal{U}_X(x)$; E^h and w are Borel by their clauses; and all sums and products are finite.

The left side does not depend on e, so the display holds for every e. The display is linear in ρ; applying it to the real and to the imaginary part of ρ, we find that it holds for complex bounded Borel invariant $\rho[u; \ell]$.

(4) *The ratio factor through the $S_k(Y)$.* Put $\varphi = G\xi$ and $T_Y = (x/2)\langle \varphi, S_k(Y)\varphi\rangle - x\langle \varphi, S_k(Y) a_0\rangle$, where $S = \sum_Y S_k(Y)$ by Thm. 3.42 (a). Here Thm. 3.42 (a) is taken under its own hypotheses, among them $\kappa \geq 6$, $(\mathrm{M4}^\flat)$ and its floor (S2); Thms. 4.39 and 5.15 suppose these, Thm. 4.5 does

not, and no clause of Thm. 4.5 depends on part (4). Then

$$\sum_Y T_Y = (x/2)\eta\cdot S\eta - (x/2)a_0\cdot Sa_0 = -(x/2)\xi\cdot \mathsf{E}\xi + x\,\xi\cdot \mathsf{E}h,$$

where we have used $\eta\cdot S\eta = -(\xi-h)\cdot\mathsf{E}(\xi-h)$. Hence the ratio factor defined with the T_Y is that of (e.ii).

Step 2: on what the letters depend, and $R_v \le Mr'$; geometry only, with no field. We claim the following.

(i) $V_\square$ is a function of three things: the η_b with $b\in I^\circ$ and $\operatorname{dist}(b,\square)\le \varrho_L+1$; the values of V' on bonds with both ends in $\square_{[\varrho_L+1]}$; and the X'_P with P in $\square_{[\varrho_L+1]}$. For $A_\square$, and for the angles ϕ_p of (d), the same holds with ϱ_L+2 in place of ϱ_L+1.

(ii) Let o be a coarse site and let $\operatorname{dist}(b,o)\le 1$. Then $(G\xi)_b$ depends on ξ on Y_b; and the row $G(b,\cdot)$ and $a_{0,b}$ depend only on the field on those bonds, and on X' on those plaquettes, all of whose sites lie within R_3 of o.

(iii) Hence two fields of $\mathfrak{V}$ which, together with their X', agree on the box of radius R_3 about o have the same $G(b,\cdot)$ and the same $a_{0,b}$. Statements (ii) and (iii) hold at every field of $\mathfrak{V}$ and for every X' on the coarse plaquettes, and therefore at every pair (V',e): their proof uses the ranges alone and is independent of the size of the curvature.

(iv) $R_v\le 20c_R\log x'\le Mr'$ and $3\varrho_L\le \ell_w(x')$; the box $\square_{[R_v]}$ has side $M+2R_v<N$, so that this box and every box inside it embed in $\mathbb{T}'$; and $\varrho(x)+1<NL/2-L$.

Proof of (i). In the first and in the fourth of the four sums of item 3.8, p and b lie in $\hat\square$. The four corners of a fine plaquette p lie in blocks within 1 of $y(p)$, so that its bonds have $\operatorname{dist}(b,y(p))\le 1$. The letters $V'(c)$ in $\partial U(p)$, and the transports along one bond in $(\mathrm{d}^V\eta)_p$, have both ends within 1 of $y(p)$. An edge plaquette p lies under a P with $y_P=y(p)$, and the sites of P are within 1 of $y(p)$. In the third sum, the bonds of $[x,x']$ with $x\in B(c_-)$ are based in $B(c_-)$ or in $B(c_+)$, and $V'(c)$ enters; this gives distance at most 1. In the second sum, $y\in\hat\square$ and $|y-p|$ is at most the range of the rows, which is at most $\varrho(x)=R$; so $L^{-1}|y(y)-y(p)|_\infty<R/L+1$, and since the left side is an integer, it is at most ϱ_L (Lem. 3.3 (c)). The bonds of the fine path $\Gamma(y,\pi(p))$ are based in the box spanned by y and $\pi(p)$, and are therefore at distance at most ϱ_L. The path $\Gamma(y(y),y(p))$ stays in the box of its ends, inside $\square_{[\varrho_L]}$, and the plaquette p adds 1. By (W0) we have $\rho_0\le\varrho(x)+1$, so the mask $m_y(U)$ uses the plaquettes p with $|y-p|\le\max\{\rho_0,\varrho(x)+1\}=R+1$; clause (d) concerns those with $|y-p|\le\varrho(x)+1=R+1$. From $|y-p|\le R+1$ we get $L^{-1}|y(y)-y(p)|_\infty<(R+1)/L+1\le\varrho_L+1+1/L$; as the left side is an integer, it is at most ϱ_L+1, and once more the plaquette adds 1.

Proof of (ii). By Thm. 3.6 (b), $(G\xi)_b$ depends on ξ on Y_b; $G(b,\cdot)$ depends on the bonds with both ends within $r_w+\varrho_L+1$ of $y(b)$; $C(b,b')=0$ unless $\operatorname{dist}(b,b')\le 2r_w$; and $C(b,\cdot)$ depends on bonds within $3r_w+\varrho_L+1$ of $y(b)$. By Lem. 3.3 (e), $(\mathsf{g}^VX')_{b'}$ depends on paths that end at y_P and all of whose sites lie within $2\varrho_L+1$ of $y(b')$, and on X'_P. The sites of P are then within $2\varrho_L+2$ of $y(b')$, hence within $2r_w+2\varrho_L+2$ of $y(b)$. We add 1 for $\operatorname{dist}(b,o)\le 1$. For the rows of G and C this gives $3r_w+\varrho_L+2\le\frac32R+\varrho_L+2\le R_3$; for X' it gives $2r_w+2\varrho_L+3\le R+2\varrho_L+3\le 2R+\varrho_L+5=R_3$, since $\varrho_L\le R$. Here the ϱ_L of Part C.I of §3 (item 1.2 there) is defined with the range, which is at most R, so it is not larger than the ϱ_L of item 4 of this section.

Proof of (iii). Statement (iii) follows from (ii), applied to each of the two fields.

Proof of (iv). We have $\varrho_L\le R/3+1$ and $R\le c_R\log x+1$, and $\log x\le 2\log x'$ because $x\ge L^2$. Therefore

$$R_v = 2R+2\varrho_L+7\le\frac83R+9\le\frac{16}{3}c_R\log x'+11.67\le 20c_R\log x',$$

where the last inequality holds because $\frac{44}{3}c_R\log x'\ge\frac{44}{3}\cdot 768\cdot\log 3>11.67$. Next, $Mr'\ge 20c_R\log x'$, and so $3\varrho_L\le 3R_v\le 3Mr'\le\ell_w(x')$. By (Tor), $N\ge 16Mr(x)\ge 320c_R\log x$. So the side $M+2R_v\le M(2r'+1)$ of $\square_{[R_v]}$ is less than N, and $NL/2-L\ge L(160c_R\log x-1)>R+1$. Also $N/2\ge 160c_R\log x\ge 160(R-1)>R/3+3\ge\varrho_L+2\ge R/L+2$, since $R\ge 6$ by (P0) and $L\ge 3$.

Rows of range at most $\varrho(x)$ are thus rows as in item C.I 1.1 of §3, where $\varrho/L+2<N/2$ is required, and rows as in item C.II 1.2 of §3, where $\varrho_L+2<N/2$ is required. This is the hypothesis on N under which we apply, in this section, Lem. 3.3, Thm. 3.6, Prop. 3.10, Thm. 3.32 and Cor. 3.38.

By (i) and (ii), $V_\square$ and $A_\square$ depend on the field and on X' only through the bonds and plaquettes in $\square_{[\varrho_L+2+R_3]}=\square_{[R_v]}$. Only the $\xi_{b'}$ with $b'\in I^\circ$ and $\operatorname{dist}(b',\square)\le r_w+\varrho_L+1$ occur in $V_\square$, and only those with $\operatorname{dist}(b',\square)\le r_w+\varrho_L+2=R_\chi^{\min}$ occur in $A_\square$ and in the angles ϕ_p of (d).

Step 3: the size of η and of X'. From this step up to Step 10 we keep the following convention: $\tau\ge 1$, $B\supset\square_{[R_v]}$, $V'\in U^\tau(B;x')$ and $\chi_\square(\xi)=1$; we put $\hat\varepsilon:=\tau\varepsilon'$; and r stands for r_τ and C_η for $C_\eta^\flat(\tau)$. In the case $\tau=1$, with B the set of sites of $\square^\approx$, the set $U^1(B;x')$ contains $\mathcal{U}_{\{\square\}}(x')$. We claim: $\hat\varepsilon\le 1/462$; $\operatorname{cont}(V')\in\mathfrak{V}^\circ$; $|V'(\pi)|<2$ on paths in B of at most $\ell_w(x')$ bonds; $|X'_P|\le 2.06\hat\varepsilon$ for P in B; and $\|\eta_b\|\le r$ at every $b\in I^\circ$ with $\operatorname{dist}(b,\square)\le\varrho_L+2$.

By $(\mathrm{W1}_\tau)$, together with $C_\eta\ge 7\tau L^{1/2}$, we have $154\tau L^{3/2}\varepsilon\le 1$, and hence $\hat\varepsilon\le\tau L^{1/2}\varepsilon\le 1/(154L)\le 1/462$.

Room. We argue as in item C.II 1.4 of §3 and give the derivation. Since $\log L\le\log x$, we have $\mathsf{g}_x\le\vartheta_0^\flat/(4\log x)$; also $\log x'\le\log x$. So $\vartheta_0^\flat-\vartheta(x')-\mathsf{g}_x\ge\vartheta_0^\flat/\log x-\vartheta_0^\flat/(4\log x)>0$ and $2-\mathsf{b}(x')-\mathsf{g}_x\ge 1/\log x-1/(4\log x)>0$. Hence (e1) gives $|V'(c)|<1+\vartheta_0^\flat$, and as the continuation has norm 1, $\operatorname{cont}(V')\in\mathfrak{V}^\circ$; and (e2) gives $|V'(\pi)|<2$.

The field X'. Let $t=|\partial V'(P)-1|$. Then $t<2\hat\varepsilon\le 0.05$, and Lem. 4.8 (b) gives $|X'_P|\le t+t^2/(2(1-t))\le 1.0264t\le 2.06\hat\varepsilon$ and $e^{iX'_P}=\partial V'(P)$.

Noise. Let $\operatorname{dist}(b,\square)\le\varrho_L+2$. The bonds of Y_b lie within $r_w+\varrho_L+2=R_\chi^{\min}\le R'$ of $\square$, so that $|\xi_{b'}|\le c_\xi\varepsilon$ there. By Thm. 3.6 (d), $\|(G\xi)_b\|\le\mathsf{G}_0\sup_{Y_b}\|\xi_{b'}\|\le C_\varphi^\flat c_\xi\varepsilon=\varepsilon/6.5$.

Centre. Let $o=y(b)$. The box of radius R_3 about o lies in $\square_{[R_v]}$, and so in B. The restriction of V' to this box satisfies the hypotheses of Thm. 3.32 on the field, (h1), (h2) and (h3). Indeed, (h1) follows from (e1) and the bound $|V'(c)|<1+\vartheta_0^\flat$ of the paragraph "Room", and (h2) from (e2) and the bound $|V'(\pi)|<2$ on paths in B of at most $\ell_w(x')$ bonds, which that paragraph also gives; and (h3), with $\hat\varepsilon=\tau\varepsilon'$, is (e3). The parameter $\hat\varepsilon$ lies in the range $0<\hat\varepsilon\le\frac{1}{40}$ of that theorem, because $\hat\varepsilon\le 1/462$. By Step 2 (iii), the $a_{0,b}$ evaluated at $\operatorname{cont}(V')$ with the X' of item 3.7 equals the $a_{0,b}$ evaluated at the restriction extended by 1, with $X'=0$ off the box. We apply Thm. 3.32 (c). Its hypotheses (P0) and (P1) are contained in (W0), with one exception: the third clause of (P0) is (Tor), which is a datum of item 2 and is not contained in (W0). Those on the law, (λ0), (ρ1), (ρ2) and (ρ5), hold for the law of windows by item C.II 2.1 of §3, which derives them from Thm. 3.6 (a), (b), (f), (k) under ($\mathrm{F}_{\mathrm{law}}$) of (W0) and $c_R\theta_{\mathbb{C}}\ge 128$ of (O6), namely (λ0) from (a), (ρ1) and (ρ2) from (b), and (ρ5) from (f) and (k). The hypothesis (Fr2) of Thm. 3.32 holds in the setting (SD). Thm. 3.32 (c) therefore gives $\|a_{0,b}\|\le\mathsf{a}_{\mathrm{c}}\hat\varepsilon/\sqrt{2}\le\mathsf{a}_{\mathrm{c}}\tau L^{1/2}\varepsilon$.

Sum. Hence $\|\eta_b\|\le(1/6.5+\mathsf{a}_{\mathrm{c}}\tau L^{1/2})\varepsilon\le r$.

Step 4: four or five exponentials; the first sum. We claim that at every fine plaquette p with $\operatorname{dist}(p,\square)\le\varrho_L+1$ the following holds: $\partial U(p)=\prod_k e^{iY_k}$ with 4 or 5 factors, where $\sum Y_k=F_p^\flat$ and $\sum|Y_k|\le S:=6r\le 1/11$; further, $|F_p^\flat|\le 6r$, $|F_p-F_p^\flat|\le 54r^2$ and $|F_p|\le 8.25r$; and finally $(x/2)|F_p^\flat\cdot F_p^\flat-F_p\cdot F_p|\le 384.75\,xr^3$.

The argument rests on the identity of matrices

$$\partial U(p)=e^{i\eta_1}e^{iT_2\eta_2}[\partial V'(P)]e^{-iT_3\eta_3}e^{-i\eta_4},\qquad T_j=T(y(p),y(b_j))\in\{1,\mathrm{Ad}_{V'(c_1)},\mathrm{Ad}_{V'(c_4)}\},$$

in which the bracket is present exactly when p is an edge plaquette under $P=(y(p);\mu\nu)$; in that case $\partial V'(P)=e^{iX'_P}$ by Step 3.

To prove the identity we fix notation: $p=(z;\mu\nu)$, $b_1=(z,\mu)$, $b_2=(z+e_\mu,\nu)$, $b_3=(z+e_\nu,\mu)$ and $b_4=(z,\nu)$, so that $\partial U(p)=U_{b_1}U_{b_2}U_{b_3}^{-1}U_{b_4}^{-1}$; further $y=y(z)$, $c_1=(y,\mu)$, $c_4=(y,\nu)$, $c_2=(y+Le_\mu,\nu)$ and $c_3=(y+Le_\nu,\mu)$; and $\eta_j=\eta_{b_j}$, where we put 0 on a comb bond. Let $\alpha=1$ if $z+e_\mu\notin B(y)$ and $\alpha=0$ otherwise, and let $\beta=1$ if $z+e_\nu\notin B(y)$ and $\beta=0$ otherwise. If

$\alpha=\beta=0$, there are four exponentials. If $\alpha=1$ and $\beta=0$, the bonds b_1 and b_3 cross c_1 and b_2 lies in $B(y+Le_\mu)$, so that

$$\partial U(p)=e^{i\eta_1}V'(c_1)e^{i\eta_2}V'(c_1)^{-1}e^{-i\eta_3}e^{-i\eta_4}.$$

If $\alpha=0$ and $\beta=1$, the bonds b_2 and b_4 cross c_4 and b_3 lies in $B(y+Le_\nu)$, so that

$$\partial U(p)=e^{i\eta_1}e^{i\eta_2}V'(c_4)e^{-i\eta_3}V'(c_4)^{-1}e^{-i\eta_4}.$$

If $\alpha=\beta=1$, which is the case of the L edge plaquettes under $P=(y;\mu\nu)$, then

$$\partial U(p)=e^{i\eta_1}V'(c_1)e^{i\eta_2}V'(c_2)V'(c_3)^{-1}e^{-i\eta_3}V'(c_4)^{-1}e^{-i\eta_4}=e^{i\eta_1}e^{i\operatorname{Ad}_{V'(c_1)}\eta_2}\,\partial V'(P)\,e^{-i\operatorname{Ad}_{V'(c_4)}\eta_3}e^{-i\eta_4}.$$

Each time we use $ge^{iY}g^{-1}=e^{i\operatorname{Ad}_g Y}$, and $\Gamma(y,y(b_j))$ is empty or is the one bond named. The exponents add up to $(\mathrm{d}^V\eta)_p+(EX')_p=F^\flat_p$.

The bonds of p have $\operatorname{dist}(b,\square)\le\varrho_L+2$, so by Step 3 we have $|\eta_b|\le\sqrt{2}\|\eta_b\|\le\sqrt{2}r$ and $|\operatorname{Ad}_{V'(c)}\eta_b|\le|V'(c)|^2\sqrt{2}r\le(1+10^{-4})^2\sqrt{2}r\le1.42r$. Hence $\sum|Y_k|\le(2\sqrt{2}+2.84)r+2.06\hat\varepsilon\le 5.6685r+(2.06/7)r\le 6r$, because $r\ge 7\tau L^{1/2}\varepsilon\ge 7\hat\varepsilon$. By $(\mathrm{W1}_\tau)$, $r\le 1/(22L)\le 1/66$, so $S\le 1/11\le 1/4$. Then $|\partial U(p)-1|\le e^S-1<1$, and the series of F_p converges. Lem. 4.8 (c) gives $|F_p-F^\flat_p|\le\frac32 S^2\le 54r^2$, and hence $|F_p|\le S+\frac32 S^2\le 1.375S\le 8.25r$. Finally, $F\cdot F-F^\flat\cdot F^\flat=(F-F^\flat)\cdot(F+F^\flat)$ and $|Y\cdot Y'|\le\|Y\|\|Y'\|\le|Y||Y'|$, so the difference is at most $\frac32 S^2\cdot 2.375S=3.5625S^3$; multiplied by $x/2$, this is $1.78125\cdot 216\,xr^3=384.75\,xr^3$.

Step 5: the fine transport along a coordinate path compared with the coarse transport, at complex fields. Let $y\in\hat\square$, let p be a fine plaquette with $n:=|y-p|\le\varrho(x)$, and put $z=y$ and $z'=\pi(p)$. We claim that $U(\Gamma(z,z'))=\Pi\,V'(\Gamma(y(z),y(z')))$ with $\Pi\in\mathrm{SL}(2,\mathbb{C})$, $|\Pi-1|\le 18rn$ and $|\Pi|\le 1.106$, and that

$$\|\operatorname{Ad}_{U(\Gamma(z,z'))}-T(y(z),y(z'))\|\le 160rn,\qquad \|\operatorname{Ad}_{U(\Gamma(z,z'))}\|\le 4.9,\qquad \|T(y(z),y(z'))\|<4.$$

(1) *Geometry, without the field.* Only seams of the direction μ are crossed by a fine stretch along $\pm e_\mu$, and they are crossed one after the other. The stretch along e_1 goes from the block of $y(z)$ to the block whose first coordinate is $y(z')_1$, and so on for the other directions. In each direction the coarse displacement is less than $n+L<NL/2$ (Step 2 (iv)), and it has the sign of the fine displacement or is 0; so the crossings take the short way round $\mathbb{T}'$, as $\Gamma(y(z),y(z'))$ does by its definition. Hence the path of the crossings of $\Gamma(z,z')$ is $\pi':=\Gamma(y(z),y(z'))$.

(2) *An identity.* A factor of $U(\Gamma)$ is 1, $e^{\pm i\eta_b}$, $e^{i\eta_b}V'(c)$ or $V'(c)^{-1}e^{-i\eta_b}$. We move every exponential to the left past the letters $V'^{\pm1}$ and obtain $U(\Gamma)=[\prod_i g_i e^{\pm i\eta_{b_i}}g_i^{-1}]V'(\pi')$ with $g_i=V'(\pi'_{j(i)})$; here π'_j consists of the first j bonds of π', and $j(i)$ is the number of letters to the left of the factor in question.

(3) *The factors.* The path π'_j lies in $\square_{[\varrho_L]}$, hence in B, and it has at most $3\varrho_L\le\ell_w(x')$ bonds (Step 2). So $|g_i|=|g_i^{-1}|<2$ (Step 3 and Lem. 4.8 (a)). The bonds b of Γ have $\operatorname{dist}(b,\square)\le\varrho_L$, so $|\eta_b|\le\sqrt{2}r$. By $(\mathrm{W2}_\tau)$, together with $\varrho(x)\ge1$, we have $\sqrt{2}r\le\sqrt{2}\cdot 0.1/17.1\le 0.0083\le 0.01$. So each bracketed factor is $1+\delta_i$ with $|\delta_i|\le 4(e^{\sqrt{2}r}-1)\le 4\cdot 1.00502\cdot 1.41422\,r\le 5.7r$.

(4) *The product.* Since Γ has at most $3n$ bonds, $|\Pi-1|\le(1+5.7r)^{3n}-1\le e^{17.1rn}-1$. By $(\mathrm{W2}_\tau)$, $17.1rn\le 0.1$, and then the right side is at most $1.05171\cdot 17.1rn\le 18rn$. Also $|\Pi|\le e^{0.1}\le 1.106$; and $\det\Pi=1$, because the η_b are traceless, the blocks of G acting on $\mathfrak{g}_\mathbb{C}$.

(5) *The adjoint action.* Lem. 4.8 (a) gives $\|\operatorname{Ad}_\Pi-1\|\le 2|\Pi||\Pi-1|\le 2\cdot 1.106\cdot 18rn\le 39.82rn$, and $\|T\|\le|V'(\pi')|^2<4$. As $\operatorname{Ad}_{U(\Gamma)}=\operatorname{Ad}_\Pi T$, the difference is at most $4\cdot 39.82rn\le 159.3rn\le 160rn$, and the norm is at most $e^{0.2}\cdot 4\le 4.9$.

Step 6: the second sum, over $y\in\hat\square$, with the rows. We claim that for every $y\in\hat\square$

$$(x/2)|(\ell^V F^\flat)_y\cdot(\ell^V F^\flat)_y-(\ell^U_y F)\cdot(\ell^U_y F)|\le K_\ell\,xr^3,$$

where K_ℓ does not depend on $\varrho(x)$.

On $\mathsf{H}_y \otimes \mathfrak{g}_{\mathbb{C}}$ we have $|A \cdot A'| \le \|A\|\|A'\|$, by the Cauchy–Schwarz inequality in $\mathbb{C}^3$ and in i, and $\|\sum_p \ell_y(p) \otimes Y_p\| \le \sum_p |\ell_y(p)|\|Y_p\|$. Put $A = \ell_y^U F$, $B = (\ell^V F^\flat)_y$ and $n_p = |y - p| \le \varrho(x)$. In the chart $U = 1$ on the combs, so the frame of a fine site is that of its block; by Step 4, $F_p^\flat$ is the sum of the exponents of $\partial U(p)$ in the frame of $\pi(p)$. Therefore

$$A - B = \sum_p \ell_y(p) \otimes \{[\mathrm{Ad}_{U(\Gamma(y,\pi(p)))} - T(y(y), y(p))]F_p + T(y(y), y(p))(F_p - F_p^\flat)\}.$$

By Steps 4 and 5, and since $\|\cdot\| \le |\cdot|$, we get $\|A - B\| \le \sum_p |\ell_y(p)|[160 r n_p \cdot 8.25 r + 4 \cdot 54 r^2]$. There are three plaquettes to a site, and the range embeds in $\mathbb{Z}^3$ (Step 2 (iv)); so $\sum_p |\ell_y(p)| n_p \le 3K_* \mathsf{g}_1$ and $\sum_p |\ell_y(p)| \le 3K_* \mathsf{g}_*$, and hence $\|A - B\| \le 3K_*[1320\mathsf{g}_1 + 216\mathsf{g}_*] r^2$. Further, $\|A\| + \|B\| \le \sum_p |\ell_y(p)|(4.9 \cdot 8.25 + 4 \cdot 6) r \le 64.425 r \cdot 3K_* \mathsf{g}_* \le 194 K_* \mathsf{g}_* r$. Now $A \cdot A - B \cdot B = (A - B) \cdot (A + B)$, and $(x/2) \cdot 3 \cdot 194 = 291x$; this gives the bound $291 K_*^2 \mathsf{g}_*[1320\mathsf{g}_1 + 216\mathsf{g}_*]\, x r^3$. The decay $e^{-\lambda_* n_p}$ of $|\ell_y(p)|$ outweighs the weight n_p from Step 5, and $\varrho(x)$ does not occur in the inequality $\sum_p |\ell_y(p)| n_p \le 3K_* \mathsf{g}_1$; so no logarithm appears.

Step 7: the kernel and the Jacobian. We claim: $\varsigma := 1.42 L r \le 1/15$; for $c \in \square$ the series of $\tau_{c,x}$, of $\mathsf{M}(U)(c)$ and of $n_c := L_{\mathbb{C}}(V'(c)\mathsf{M}(U)(c)^{-1})$ converge; $|n_c - n_c^\flat| \le 28\varsigma^2$;

$$|\frac{1}{2} n_c^\flat \cdot n_c^\flat - (1 - \frac{1}{2} \operatorname{tr} V'(c)\mathsf{M}(U)(c)^{-1})| \le 55\varsigma^3;$$

and $|\log \mathsf{j}(\eta_b)| \le r^2/2$ for $b \in \hat{\square}$.

By $(\mathrm{W1}_\tau)$, $\varsigma \le 1.42/22 \le 0.0646 \le 1/15$. Put $\tilde\eta_b = \eta_b$ for b based in $B(c_-)$ and $\tilde\eta_b = \mathrm{Ad}_{V'(c)}\, \eta_b$ for b based in $B(c_+)$; by Step 4, $|\tilde\eta_b| \le 1.42 r$. The crossing bond of $[x, x']$ is based in $B(c_-)$ and carries $e^{i\eta_b} V'(c)$. Hence $g_x := h_{c,x} V'(c)^{-1} = \prod_{b \in [x,x']} e^{i\tilde\eta_b}$, a product of L factors. In this step, g_x and g_y denote these matrices, x and y being sites of $B(c_-)$. Let $\Sigma_x := \sum \tilde\eta_b$. Then $\sum |\tilde\eta_b| \le \varsigma$, and the mean of Σ_x over $x \in B(c_-)$ is $(Q_1^V \eta)_c = -n_c^\flat$.

We have $h_{c,x} h_{c,y}^{-1} = g_x g_y^{-1}$, and for this product the sum of the absolute values of the exponents is the S of Lem. 4.8 (c), not the $S = 6r$ of Step 4, and satisfies $S \le 2\varsigma \le 1/4$. So Lem. 4.8 (c) gives $|\tau_{c,x} - \Sigma_x + \Sigma_y| \le \frac{3}{2}(2\varsigma)^2 = 6\varsigma^2$, and therefore $|\bar\tau_c| \le 2\varsigma + 6\varsigma^2 \le 2.4\varsigma$. Next, $\mathsf{M}(U)(c) V'(c)^{-1} = e^{i\bar\tau_c} g_y$, and for this product $S \le 3.4\varsigma \le 3.75\varsigma \le 1/4$; its distance from 1 is at most $e^{1/4} - 1 \le 0.2841 < 1/2$. So $L_{\mathbb{C}}$ of its inverse is minus its $L_{\mathbb{C}}$ (Lem. 4.8 (b)). The hypothesis $\det(e^{i\bar\tau_c} g_y) = 1$ of that clause is met: the $\tilde\eta_b$ are traceless, so $\det g_x = 1$; then $\operatorname{tr} \tau_{c,x} = 0$ by the clause "$\det g = 1 \Rightarrow \operatorname{tr} L_{\mathbb{C}}(g) = 0$" of Lem. 4.8 (b), and so $\operatorname{tr} \bar\tau_c = 0$; and the same clause gives $\operatorname{tr} n_c = 0$, which Lem. 4.8 (d) requires below.

Applying Lem. 4.8 (c) to $e^{i\bar\tau_c} g_y$, we obtain $|-n_c - \bar\tau_c - \Sigma_y| \le \frac{3}{2}(3.75\varsigma)^2 \le 21.1\varsigma^2$. Together with the bound on $\tau_{c,x}$, averaged over x, this gives $|n_c + (Q_1^V \eta)_c| \le 27.1\varsigma^2 \le 28\varsigma^2$. Further, $|n_c^\flat| \le \varsigma$ and $|n_c| \le \varsigma + 28\varsigma^2 \le 2.867\varsigma \le 2.9\varsigma$, so that $\frac{1}{2}|n \cdot n - n^\flat \cdot n^\flat| \le \frac{1}{2} \cdot 28\varsigma^2 \cdot 3.9\varsigma = 54.6\varsigma^3$. Moreover $|n \cdot n| \le 8.5\varsigma^2 \le 1$ and $e^{in_c} = V'(c)\mathsf{M}(U)(c)^{-1}$, and Lem. 4.8 (d) gives $|1 - \frac{1}{2}\operatorname{tr} e^{in} - \frac{1}{2} n \cdot n| \le 72.25\varsigma^4/23 \le 0.21\varsigma^3$. As $54.6 + 0.21 \le 55$, the bound on the kernel follows. Finally, $|\eta_b \cdot \eta_b| \le r^2 \le 1$, and Lem. 4.8 (d) gives $|\log \mathsf{j}(\eta_b)| \le r^2/2$.

Step 8: the sum; clause (c)*, also within* (f). We claim that $|V_\square| \le K_v^\flat(\tau) p_0^3 x^{-1/2}$ and $|e^{V_\square} - 1| \le 2K_v^\flat(\tau) p_0^3 x^{-1/2}$.

The set $\hat\square$ consists of the $(LM)^3$ fine sites of the blocks of $\square$. Its $3(LM)^3$ plaquettes give at most $3 \cdot 384.75 = 1154.25$ times $(LM)^3 x r^3$, by Step 4. Its $(LM)^3$ sites give at most K_ℓ times $(LM)^3 x r^3$, by Step 6. Each of the $3M^3$ coarse bonds gives at most $55 s \varsigma^3 = 55a(1.42)^3 L^3\, x r^3 \le 158 a L^3\, x r^3$, by Step 7, which makes $474 a (LM)^3\, x r^3$ in all. The at most $3(LM)^3$ bonds of I° give $\frac{3}{2} r^2 (LM)^3 \le \frac{3}{2}(LM)^3\, x r^3$, because $x r = C_\eta p_0 x^{1/2} \ge 1$ by (W3), where we also use $C_\eta^\flat \ge 7 L^{1/2} \ge 1$ and $x \ge 1$. Adding up, we obtain $(LM)^3(1155.75 + 474a + K_\ell)\, x r^3 \le (LM)^3(1157 + 474a + K_\ell) C_\eta^3 p_0^3 x^{-1/2}$. For

$|z| \le 1$ we have $|e^z - 1| \le (e-1)|z| \le 2|z|$, and we apply $(\mathrm{W4}_\tau)$. The other bounds of (c) come from Steps 3, 4 and 5.

Step 9: clause (a)*: convergence, dependence, holomorphy.* We claim that $V' \mapsto V_\square(\xi, V')$ is holomorphic on $U^\tau(B; x')$, hence on $\mathcal{U}_{\{\square\}}(x')$, and that its dependence is the one stated in (a).

The dependence is that of Step 2. The formula of item 3.8 contains no member, no e, no c_h and no E_a. On the domain we have $\partial V'(P) \neq -1$ at every P in B, while e_P occurs in X'_P only where $\partial V'(P) = -1$; hence there is no dependence on e. The convergence follows from Steps 3, 4 and 7.

As to holomorphy, $U^\tau(B; x')$ is open in $\mathrm{SL}(2,\mathbb{C})^{\text{bonds of } B}$, and $V' \mapsto \mathrm{cont}(V')$ maps it holomorphically into $\mathfrak{V}^\circ$ (Step 3). By Thm. 3.6 (i), the entries of G and of C are holomorphic on $\mathfrak{V}^\circ$. The entries of $\mathrm{Ad}^{\pm 1}_{V'(c)}$ are polynomials in those of $V'(c)$, the inverse being the adjugate; so d^V, Q_1^V, ℓ^V and g^V are polynomial. Each X'_P is either a power series in $\partial V'(P) - 1$, which has norm $< 2\hat{\varepsilon} < 1$, or 0. So η, $F^\flat$ and $n^\flat$ are holomorphic. The functions $e^{i\eta_b}$, $\partial U(p)$, $U(\Gamma)$ and $h_{c,x}$ are entire in (η, V'), and $\mathrm{Ad}_{U(\Gamma)}$ is polynomial in $U(\Gamma)$, because $\det = 1$. The functions F_p, $\tau_{c,x}$, $\mathsf{M}(U)(c)$ and $\log \mathsf{j}(\eta_b)$ are power series whose arguments stay inside the discs of convergence on the whole domain, since there $|\partial U(p) - 1| \le e^{1/11} - 1$, $|g_x g_y^{-1} - 1| \le e^{1/4} - 1$ and $|\eta \cdot \eta| \le 1$ (Steps 4 and 7, whose bounds are uniform on the domain). All sums are finite. Finally, $\mathcal{U}_{\{\square\}}(x')$ is an open subset of $U^1(\text{sites of } \square^{\approx}; x')$.

Step 10: clause (b)*: invariance, with the extension by* 1. We claim that for every G-valued u on the sites of B we have $V_\square(\mathrm{Ad}_u \xi, V'^u) = V_\square(\xi, V')$, and that the domain and $\chi_\square$ are invariant.

(1) Norms do not change under unitary factors; moreover $V'^u(\pi) = uV'(\pi)u^{-1}$ and $\partial(V'^u)(P) - 1 = u(\partial V'(P) - 1)u^{-1}$. So (e1) to (e3), and likewise (u1) to (u3), are invariant; and $|\mathrm{Ad}_u \xi_b| = |\xi_b|$.

(2) Let $W = \mathrm{cont}(V')$, let $\tilde{u} = u$ on B and $\tilde{u} = 1$ elsewhere, and let $W_2 = \mathrm{cont}(V'^u)$. Then $W^{\tilde{u}} \in \mathfrak{V}^\circ$, and $W^{\tilde{u}}$ agrees with W_2 on every bond with both ends in B, while the two may differ on bonds that leave B. For P in B we have $X'[W_2]_P = \mathrm{Ad}_{u(y_P)} X'[W]_P$ (Lem. 4.8 (b)), and elsewhere both vanish; so $X'[W_2] = \mathrm{Ad}_{\tilde{u}} X'[W]$.

(3) Thm. 3.6 (c), applied to fields of $\mathfrak{V}$ and to the G-valued $\tilde{u}$ on all of $\mathbb{T}'$, gives $G[W^{\tilde{u}}] = \mathrm{Ad}_{\tilde{u}} G[W] \mathrm{Ad}_{\tilde{u}}^{-1}$, and the same for C. Lem. 3.3 (d) and (e) give $\mathsf{g}^{W^{\tilde{u}}} = \mathrm{Ad}_{\tilde{u}} \mathsf{g}^W \mathrm{Ad}_{\tilde{u}}^{-1}$. Hence at the field $W^{\tilde{u}}$, with $\mathrm{Ad}_{\tilde{u}} X'$ and $\mathrm{Ad}_{\tilde{u}} \xi$, we have $\eta = \mathrm{Ad}_{\tilde{u}} \eta[W]$.

(4) By Step 2 (ii) and (iii), the rows $G(b,\cdot)$ and the $a_{0,b}$ at $\mathrm{dist}(b, \square) \le \varrho_L + 2$ depend only on the field and on X' in $\square_{[R_v]}$, which lies inside B; and there $W^{\tilde{u}} = W_2$. The b' that enter have $y(b') \in \square_{[R_v]}$, where $\tilde{u} = u$. So $\eta[W_2, \mathrm{Ad}_u \xi]_b = \mathrm{Ad}_{u(y(b))} \eta[W, \xi]_b$ at these bonds.

(5) *The rest.* We have $T^u(y, y'') = \mathrm{Ad}_{u(y)} T \mathrm{Ad}^{-1}_{u(y'')}$. Hence $F_p^\flat \to \mathrm{Ad}_{u(y(p))} F_p^\flat$, $n_c^\flat \to \mathrm{Ad}_{u(c_-)} n_c^\flat$ and $(\ell^V F^\flat)_y \to (1 \otimes \mathrm{Ad}_{u(y(y))})(\ell^V F^\flat)_y$. Further, $\mathsf{U}(\mathrm{Ad}_u \eta; V'^u) = \mathsf{U}(\eta; V')^{\hat{u}}$ with $\hat{u}(x) = u(y(x))$: inside a block $e^{i \mathrm{Ad}_u \eta} = u e^{i\eta} u^{-1}$, and at a crossing bond $e^{i \mathrm{Ad}_{u(c_-)} \eta} u(c_-) V'(c) u(c_+)^{-1} = u(c_-) e^{i\eta} V'(c) u(c_+)^{-1}$. So $F_p \to \mathrm{Ad}_{\hat{u}(\pi(p))} F_p$, $\mathrm{Ad}_{U(\Gamma(y, \pi(p)))} \to \mathrm{Ad}_{\hat{u}(y)} \mathrm{Ad}_{U(\Gamma)} \mathrm{Ad}^{-1}_{\hat{u}(\pi(p))}$ and $\ell^U_y F \to (1 \otimes \mathrm{Ad}_{\hat{u}(y)}) \ell^U_y F$. Also $h_{c,x} \to u(c_-) h_{c,x} u(c_+)^{-1}$, and $\mathsf{M}(U)(c)$ transforms in the same way. Traces and the products $Y \cdot Y'$ are Ad-invariant (Lem. 4.8 (a)).

Step 11: clause (d). Let (V', e) be given on $\mathbb{T}'$, with the restriction of V' to $\square^{\approx}$ in $\mathcal{U}_{\{\square\}}(x')$; let $\chi_\square(\xi) = 1$, and let (W5) hold. We claim that $\phi_p \le \varepsilon$ at every fine plaquette p with $|y - p| \le \varrho(x) + 1$ for some $y \in \hat{\square}$; that $V_\square$ is real; that $|\eta_b| < \pi$; and that $A_\square(\xi, V', e) = e^{V_\square(\xi, V')}$.

(1) *One η.* On $\square^{\approx}$ we have $|\partial V'(P) - 1| < 2\varepsilon' < 1$. Hence $\partial V'(P) \neq -1$, and $X'_P(V', e) = \mathsf{L}(\partial V'(P)) = L_{\mathbb{C}}(\partial V'(P))$ (Lem. 4.8 (b)). The global pair $(V', X'(V', e))$ agrees on $\square_{[R_v]}$ with the pair consisting of $\mathrm{cont}(V'|\square^{\approx})$ and the X' of item 3.7. By Step 2, the η_b at $\mathrm{dist}(b, \square) \le \varrho_L + 2$, and everything on which $A_\square$ depends, are the same for the two; and e does not enter.

(2) *Reality.* The matrix G^V is real (Thm. 3.6 (a)), and X' is $\mathfrak{g}$-valued. So $\eta_b \in \mathfrak{g}$ and $|\eta_b| = \|\eta_b\| \le r < \pi$. On $\mathfrak{g}$ we have $Y \cdot Y = |Y|^2$, and $L_{\mathbb{C}} = \mathsf{L}$ where $|g - 1| < 1$. Therefore $F_p = \mathsf{L}(\partial U(p))$, $|F_p| = \phi_p$, $\mathsf{M}(U)$ is the one of item 3.4, and $1 - \frac{1}{2} \mathrm{tr}\, g = 1 - \cos \mathrm{ang}\, g$. So every bracket of $V_\square$ is real.

(3) *The angle.* Let p be as in the claim. Then $\operatorname{dist}(p,\square) \le \varrho_L+1$, its bonds are at distance at most ϱ_L+2, and the box of radius R_3 about $y(p)$ lies in $\square_{[R_v]}$ (Step 2). We have $F^\flat_p = \bar F_p + (\mathrm{d}^V G\xi)_p$. The restriction of V' to that box satisfies (e1) and (e2), which follow from (u1) and (u2), and $|\partial V'(P)-1| \le 2\varepsilon'$; and its $a_{0,b}$, $b \in dp$, are the ones at hand (Step 2 (iii)). Cor. 3.38 (a), applied with the value $\hat\varepsilon = \varepsilon'$ of its parameter, gives $|\bar F_p| \le \mathsf{N_c} L^{-2}\varepsilon' + \mathsf{K_c}\varepsilon'^2 \le (\mathsf{N_c}L^{-3/2} + \mathsf{K_c}L\varepsilon)\varepsilon$. At real fields T is an isometry, so $|(\mathrm{d}^V G\xi)_p| \le 4\sup|(G\xi)_b| \le (4/6.5)\varepsilon \le 0.61539\varepsilon$ (Step 3). By Step 4, $|F_p - F^\flat_p| \le 54(C^\flat_\eta)^2\varepsilon^2$. Adding the three bounds, we get from (W5) that $\phi_p \le [0.61539 + 0.2 + 0.18]\varepsilon = 0.99539\varepsilon < \varepsilon$. We note that (L2^{su}) gives $\mathsf{N_c}L^{-3/2} \le (1.2\cdot 10^4 \mathsf{N_c}L)^{-1} \le 10^{-4}$.

(4) *Consequences.* Let $y \in \hat\square$. At every p in the range of ℓ_y we have $\zeta_p = F_p$. No plaquette of $D(U)$ lies within $\rho_0 \le \varrho(x)+1$ of y (by (W0)); and no plaquette with $\phi_p > 1 \ge \varepsilon$ lies within $\varrho(x)+1$ of y, where $\varepsilon \le 1$ by (W1), because $C^\flat_\eta \ge 1$. So $m_y(U) = 1$. For $p \in \hat\square$ we have $S_\varepsilon(\phi_p) = \frac12|F_p|^2$. Moreover $1\{|\eta_b| < \pi\} = 1$ and $\mathsf{j}(\eta_b) = e^{\log \mathsf{j}(\eta_b)} > 0$. Therefore the four products of item 3.9 are the exponentials of the four sums of item 3.8.

Remark 4.17 (On Step 11). If the tests were taken at ε instead of at $c_\xi\varepsilon$, the second term would be bounded only by $4C^\flat_\varphi\varepsilon$, and the step would fail.

Step 12: clauses (e.iii), (e.iv), (e.v) *and* (g)*: $A_\square$ at every real point.* We claim that $A_\square(\xi, V', e)$ has the dependence, the covariance and the measurability that these clauses state, and that no member enters it.

Proof of (e.iii). We use Step 2 (i) with $\varrho_L + 2$, and Step 2 (ii); no field is estimated there. The value e_P enters only through X'_P, hence only at P in $\square_{[R_v]}$, and only where $\partial V'(P) = -1$.

Proof of (e.iv). Here $u : \mathbb{T}' \to G$ is global, and no continuation is constructed. We have $X'(V'^u, \mathrm{Ad}_u\, e) = \mathrm{Ad}_u\, X'(V', e)$ (item E2 of §4.1). By Thm. 3.6 (c) and Lem. 3.3 (d), (e), $\eta \to \mathrm{Ad}_u\,\eta$. As in Step 10 (5), $U \to U^{\hat u}$. The quantities ϕ_p, $D(U)$, $m_y(U)$ and $|\ell^U_y\zeta|$ are invariant under fine gauge transformations, because $\zeta_p \to \mathrm{Ad}_{\hat u(\pi(p))}\,\zeta_p$ and the transports are conjugated. Moreover $\mathsf{L}(ugu^{-1}) = \mathrm{Ad}_u\,\mathsf{L}(g)$, also at $g = -1$; this gives $\tau_{c,x} \to \mathrm{Ad}_{u(c_-)}\,\tau_{c,x}$ and $\mathsf{M}(U)(c) \to u(c_-)\mathsf{M}(U)(c)u(c_+)^{-1}$, so that n_c is invariant. Finally, $|F^\flat_p|$, $|(\ell^V F^\flat)_y|$, $|n^\flat_c|$ and $|\eta_b|$ are invariant, since Ad_u is orthogonal.

Proof of (e.v). The matrices G^V and C^V are continuous on $\mathfrak{V}$ (Thm. 3.6 (i)), and g^V, d^V, Q_1^V, ℓ^V are polynomial. By item E2 of §4.1, X' is Borel in (V', e); so η is Borel in (ξ, V', e). The field U is continuous in (η, V'), ang is continuous and L is Borel; and m_y, the clip and S_ε are Borel functions of these.

Proof of (g). The formulas of items 3.6 to 3.9 contain x, ℓ, L, a and $R(x)$; the number b of (O6), in K_b and in G (item 3.6); the sets $\hat\square$ and $\square$, and with them M and the grid; and the lattice operations of item 3.1, and with them N. Besides these, $A_\square$ contains ε, and with it A_0 and $\mathfrak{p}_0$, as well as d_0 and ρ_0. Nothing else occurs.

Step 13: clause (f)*, with* (a) *to* (c) *as its case* $\tau = 1$. We claim that (a), (b) and (c) hold on $U^\tau(B; x')$ with r_τ and $K^\flat_v(\tau)$, and on $\mathcal{U}_{\{\square\}}(x')$ with r and $K^\flat_v$.

Steps 3 to 10 are carried out for $\tau \ge 1$, $B \supset \square_{[R_v]}$ and $V' \in U^\tau(B; x')$. They depend on the field only through $\mathrm{cont}(V') \in \mathfrak{V}^\circ$ (from (e1)), through $|V'(\pi)| < 2$ on paths in B of at most $\ell_w(x')$ bonds (from (e2)), and through $|\partial V'(P) - 1| < 2\tau\varepsilon'$, which is (e3). Of the floors, they depend only on (W0), (W1$_\tau$), (W2$_\tau$), (W3) and (W4$_\tau$). The inequality $\tau\varepsilon' \le 1/40$ is a consequence of these, by Step 3. At $\tau = 1$, with B the set of sites of $\square^\approx$, which contains $\square_{[R_v]}$ by Step 2 (iv), the conditions (u1), (u2) and (u3) imply (e1), (e2) and (e3). Hence $\mathcal{U}_{\{\square\}}(x') \subset U^1(B; x')$, with the constants $C^\flat_\eta$ and $K^\flat_v$.

Step 14: the order of the letters, and the floors; no circle. We claim that $C^\flat_\eta$, K_ℓ and $K^\flat_v$ depend only on earlier entries of the table, and that (W0) to (W5), together with (W1$_\tau$), (W2$_\tau$) and (W4$_\tau$)

for every $\tau \geq 1$, are conditions on nothing but x, the letters (O0) to (O10) and τ, and that each of them holds for all large x once the letters and τ are fixed.

Order. The numbers γ, N_B, b, $\theta_\mathbb{C}$, $\vartheta_0^\flat$, G_0, $C_\varphi^\flat$, c_ξ and a_c depend on (L, a, K_*, λ_*), and K_c^0 depends, besides, on C_D and θ_D. The constant $C_\eta^\flat$ comes after a_c; K_ℓ is computed from K_* and λ_*; $K_v^\flat$ comes after M; $E_a \geq 1500 K_v^\flat L^{-1/2}$ comes after $K_v^\flat$; and $x_\flat$ comes last. The number N_c is a function of L through I_L, and as such it occurs in ($\mathrm{L2}^{\mathrm{su}}$), a condition to which L is subject. No number of Thm. 4.5 enters a condition on L, M or c_R. As $\mathsf{a}_\mathrm{c} \geq 100 \cdot 73 \cdot 3^4 > 7$, the maximum in $C_\eta^\flat$ is a_c.

Floors. The number $\varepsilon = A_0 (\log x)^{\mathfrak{p}_0} x^{-1/2}$ tends to 0, and on the range $\log x > 2\mathfrak{p}_0$ it is a decreasing function of x. In ($\mathrm{W1}_\tau$) and in the last clause of (W5) a number times ε is bounded; in ($\mathrm{W2}_\tau$), a number times $\varepsilon(c_R \log x + 1)$; and in ($\mathrm{W4}_\tau$), a number times $(\log x)^{3\mathfrak{p}_0} x^{-1/2}$. Condition (W3) amounts to $\log x \geq A_0^{-1/\mathfrak{p}_0}$. Condition (W0) rests on $R \to \infty$. Condition ($\mathrm{F}_{\mathrm{law}}$) compares $(R+3)^3$ with x^4; (P1) compares $(2R+3)^3$ with x, and contains x^{-60}; and (P3) contains negative powers of x and x'. The second clause of (W5), $\mathsf{N}_\mathrm{c} L^{-3/2} \leq 0.2$, involves no x: it is a condition on the letters alone, and Step 11 (3) shows that $\mathsf{N}_\mathrm{c} L^{-3/2} \leq 10^{-4}$. The clauses of (P0) on N are (Tor), which belongs to the data. None of these conditions depends on c_h, E_a, N, a member, or on more of ℓ than K_*, λ_* and the range. All the conditions are one-sided: a larger K_c or a_c only raises the floors and E_a.

Step 15: clause (h)*, the inequalities that compare the radii, counted in cells, with* r_1. We claim that $R_\chi^{\min} \leq R_\chi$, $2\lceil R_\chi/M \rceil \leq r_1 - 5$ and $\lceil R_v/M \rceil \leq r_1 - 5$, together with the three consequences stated in (h).

Since $r_w \leq R$, we have $R_\chi^{\min} \leq R_\chi$. Next, $\lceil (R+\varrho_L+2)/M \rceil \leq \lceil R/M \rceil + r_l$; moreover $2\lceil t \rceil \leq \lceil 2t \rceil + 1$, which gives $2\lceil R/M \rceil \leq \lceil R_2/M \rceil + 1$. Hence $2\lceil R_\chi/M \rceil \leq \lceil R_2/M \rceil + 1 + 2r_l = r_1 - r_l - 4 \leq r_1 - 5$, because $r_l \geq 1$. Further, $R_v = R_2 + 2(\varrho_L + 2) + 1 \leq M\lceil R_2/M \rceil + 2Mr_l + M$, and therefore $\lceil R_v/M \rceil \leq \lceil R_2/M \rceil + 2r_l + 1 = r_1 - r_l - 4 \leq r_1 - 5$.

Consequences. A bond of I° with $\mathrm{dist}(b, \square) \leq R_\chi$ is based under $\square^{+\lceil R_\chi/M \rceil}$, which lies inside $\square^{+r_1}$; so, if the cell tests there are passed, then $\chi_\square = 1$. Clause (e.iii) places the ξ on which $A_\square$ depends under $\square^{+\lceil R_\chi^{\min}/M \rceil}$, and the (V', e) on which it depends in $\square^{+\lceil R_v/M \rceil}$. Two cells that are more than $2\lceil R_\chi^{\min}/M \rceil$ cells apart have disjoint sets $\square_{[R_\chi^{\min}]}$. □

4.3. **Gaussian integration on zones.** We estimate the cell factors after integrating the white noise. Compatible sets have disjoint noise supports, so their integrals factor. The main bound suppresses a failed test by charging either a large fluctuation in a tame block or a rough plaquette. It follows from a lower bound for the fine energy plus the noise, after subtracting the coarse energy and controlling the terms at the boundary of the region.

Remark 4.18 (Notation and letters with more than one meaning). The lift E_3 maps a function $\Theta = (\Theta_{q'})$ on the coarse cubes q' to the function $E_3\Theta = ((E_3\Theta)_q)$ on the fine cubes q: $(E_3\Theta)_q = \Theta_{q'}$ when q is the corner cube under q', that is, the fine cube at $y_{q'} + \mathsf{h}_L(1,1,1)$ with $\mathsf{h}_L = (L-1)/2$, and $(E_3\Theta)_q = 0$ at the other fine cubes (§4.1, C1); the identity $\mathrm{d}^V E = E_3 \mathrm{d}'^V$ is Step 1 (iv) of the proof of Thm. 4.2. We write E_ξ for the mean over ξ. Moreover, $\mathrm{e} = \exp(1)$, and $\mathfrak{g}$ and $\mathfrak{L}_\mathfrak{g}$ are those of item C.I 1.1 of §3.

The exponent in $M = L^{m'}$ of (O7) is m', whereas $m'_{y'}$, and every other m' of this section, denote the mask at x'. The letter G denotes the matrix of the law and also the group SU(2), as in "G-valued" and "$g_i \in G$". The letters k_F, k_n, k_p are the components of $\mathfrak{k}$, while k_p in Step 3 (3) is a matrix.

The coboundaries are d, d', while in "$b \in dp$" and "$c \in dP$", dp and dP are the sets of the four bonds.

Theorem 4.19 (Means of zone and test letters at any depth)**.** *Item A: letters.* The letters are chosen in the order below. This order, from (O0) to (O13), is that of the table of the letters in

item 3 of §2.1. After each tag we state what this theorem uses of the entry of the table under that tag.

(O0) The pure numbers $b_*, K_*, \lambda_*, L_A, c_1^*, c_2^*, \lambda_2^*, \theta_J$ of Thm. 2.65, the order of the constants, (0). We put $K = K_*$ and $\lambda = \lambda_*$, and we set $\mathsf{g}(s) = \sum_{n\in\mathbb{Z}^3} e^{-s|n|_\infty}$, $\mathsf{g}_* = \mathsf{g}(\lambda)$, $\mathsf{g}_1 = \sum_{n\in\mathbb{Z}^3} |n|_\infty e^{-\lambda|n|_\infty}$, $l_r := 3K\mathsf{g}_*$ and $l_c := K\mathsf{g}_*$.

(O1) A number $d_0 \in (0, 1/2)$.

(O2) A number c_Ξ with $0 < c_\Xi \le \frac{1}{2}(\frac{1}{2} - d_0)$.

(O3) The number ρ_0 is the least integer ≥ 0 for which $3K^2\mathsf{g}(\lambda)\mathsf{g}(\lambda/2)e^{-\lambda\rho_0/2} \le \frac{1}{8}(\frac{1}{2} - d_0 - c_\Xi)$; this is display eq. 2.vi of §2.2. The functions of L are $\mathsf{l}_L = 1 + \log L$ and the four numbers of display eq. 2.vii of §2.2, namely $\mathsf{N}_Z = \mathsf{N}_\mathsf{B} = 1 + 3K^2\mathsf{g}_*^2$, $\mathsf{P}_\rho = 1 + 3(2 + 3(2\rho_0+1)^3)K^2\mathsf{g}_*^2$, $\mathsf{N}_\sharp = c_1^*\mathsf{l}_L\mathsf{N}_Z\mathsf{P}_\rho$ and $\mathsf{N}_\Xi = c_2^*\mathsf{N}_Z$. Moreover $\mathsf{N}_\sigma = 3K\mathsf{g}_*c_1^*\mathsf{l}_L\mathsf{N}_Z$, $N_1 = \mathsf{N}_Z + 2L\mathsf{N}_\Xi + l_c\mathsf{N}_\sigma$, and

$$\mathsf{N}_\nu^\flat = N_1 + (2l_c + (2\rho_0+1)^3 l_r)\mathsf{N}_\sigma = \mathsf{N}_Z + 2L\mathsf{N}_\Xi + (3 + 3(2\rho_0+1)^3)K\mathsf{g}_*\mathsf{N}_\sigma.$$

(O4) An odd integer L with the following three properties.
(i) $L \ge L_A$.
(ii) (L1$^\flat$) $[\mathsf{N}_\nu^\flat L^{-1} + 2200\mathsf{N}_Z + 40\mathsf{N}_\Xi]L^{-1/2} \le \frac{3}{16}d_0$.
(iii) $(\mathsf{N}_\sharp + 2)L^{-3/2} \le 0.08$.

Properties (ii) and (iii) hold for all large L, because $\mathsf{N}_Z, \mathsf{N}_\Xi$ are free of L, while each of $\mathsf{N}_\sharp$, N_σ, $\mathsf{N}_\nu^\flat/L$ is at most a number free of L times $(1 + \log L)$. We set $c_L = (L^2-1)/(6L^2)$, and the letter a satisfies $a \ge 3/(c_L L)$, so that $aL \ge 18$. The condition $L \ge L_*^{\mathrm{su}}$ of item 3 of §2.1, (O4), implies (i), (ii) and (iii): it includes $L \ge L_A$ and (L1$^\flat$), and its (L2^{su}), which reads $L^{1/2} \ge 1.2 \cdot 10^4 \mathsf{N}_\mathrm{c}^2$ with $\mathsf{N}_\mathrm{c} = 17(\mathsf{N}_\sharp + 1)$, gives $L^{3/2} \ge 10^{12}\mathsf{N}_\mathrm{c}^6 \ge 12.5(\mathsf{N}_\sharp + 2)$.

(O5) A number $\kappa \ge 6$.

(O6) The numbers of item C.I 1.2 of §3, evaluated at (K_*, λ_*): $\gamma = (73L^4)^{-1}$, $b_h = 16(1 + 3K^2\mathsf{g}_*^2) + a/L$,

$$A^\flat = 2\mathrm{e}^2(16 + a) + 384\mathrm{e}^8 K^2 \mathsf{g}_2(\lambda/2)^2$$

with $\mathsf{g}_2(s) = \sum_{n\in\mathbb{Z}^3}(1 + |n|_\infty)^2 e^{-s|n|_\infty}$, then $\theta_1 = \min\{1, \lambda L/12\}$, $\theta^\flat = \min\{\theta_1, \gamma/(2A^\flat)\}$, $\theta_\mathbb{C} = \theta^\flat/6$ and $\vartheta_0^\flat = \theta^\flat/24$. We also set $\theta_D = \frac{1}{2}\min\{\lambda_2^*, \theta_\mathbb{C}, \theta_J\}$. The number $C_D \ge 4$ is fixed at this stage, before any rows are chosen: it is the number given, at this θ_D, by the expression for C_D in Thm. 2.65, the order of the constants, (6), or else any number above that one, and that expression is $\ge A_q \ge 32\mathrm{e}^2 > 4$. That (Dec) holds at this C_D for the pair (ℓ, ℓ') is itself one of the data of item C. A number C_D with (Dec)(C_D, θ_D) comes from Prop. 2.62, taken with $\theta_\mathbb{C}$ in the place of θ_q; with this change, Step H13 of the proof of Thm. 2.65 states the assertion of that proposition for every $\ell \in \mathcal{R}(b_*, K_*, \lambda_*, 1/9)$, and clause (Fr2) of the part "Conclusion" of that theorem states it for all sixteen kernels of (Dec) at the pairs $(\ell_k, (\ell_k)')$ of the chain. Finally $c_R = 64/\theta_D$, so that $c_R\theta_\mathbb{C} \ge 128$.

(O7) The side $M = L^{m'}$ with $m' \ge 1$. We put $v := (LM)^3$.

(O8) Numbers $\mathfrak{p}_0 > 2$ and $A_0 > 0$. We put $p_0 = p_0(x) = A_0(\log x)^{\mathfrak{p}_0}$, $\varepsilon = p_0 x^{-1/2}$ and $x' = x/L$, and a prime on a quantity means that it is taken at x'. Then $\varepsilon' \le L^{1/2}\varepsilon$, since $\varepsilon' = A_0(\log x')^{\mathfrak{p}_0}L^{1/2}x^{-1/2}$ and $0 < \log x' \le \log x$, the latter because $x' = x/L \ge L > 1$.

(O9) The numbers

$$\mathsf{G}_0 = (2/\gamma)^{1/2}3L^3\mathsf{g}(\theta_\mathbb{C}) \le C_\varphi^\flat = (2b_h^{1/2}/\gamma)3L^3\mathsf{g}(\theta_\mathbb{C})/(1 - e^{-\theta_\mathbb{C}})$$

and $c_\xi = (6.5C_\varphi^\flat)^{-1}$. We use §3, Part C.II, with the value $\mathsf{n} = 60$ of its exponent.

(O10) The number K_c^0 of item C.II 4 of §3, "Constants"; here $\mathsf{K}_\mathrm{c}^0 \ge 6 \cdot 10^4$. We also define $\mathsf{a}_\mathrm{c} = (100/\gamma)\mathsf{N}_\mathsf{B}3L^3\mathsf{g}(\theta_\mathbb{C})$ and $\mathsf{K}_\mathrm{c} := \sqrt{2}\mathsf{K}_\mathrm{c}^0 \ge \mathsf{K}_\mathrm{c}^0 + 1$. The constants of item F, δ_3 and δ_3' among them, are defined in that item, and they come at (O10) in the order (O0) to (O13) of the letters.

(O11), (O12) The letters c_h, E_a play no role in this section.

(O13) The number x_{fl}, which is defined in item G. To apply this theorem for every $x \geq Lx_\flat$, one needs $Lx_\flat \geq x_{\text{fl}}$.

Item C: data. The coupling satisfies $x \geq x_{\text{fl}}$. The coarse side N is even, $M \mid N$, and the condition (Tor) $N \geq 2Ms(x)$ holds. The fine torus is $\mathbb{Z}^3/NL\mathbb{Z}^3$ and the coarse torus is $L\mathbb{Z}^3/NL\mathbb{Z}^3$, with N^3 sites; the grids of cubes are fitted as in item 1 of §2.1, Cubes, so that a cell fills exactly L^3 fine cubes.

The rows are $\ell \in \mathcal{R}(b_*, K_*, \lambda_*, 1/9)$ of range $\leq \varrho(x)$, the sets of rows being those of §2.2, from "Translation covariant rows" to "The sets of rows". The empty rows are admitted: $\ell = \emptyset$, with $\mathsf{H}_y = \{0\}$ for all y, lies in $\mathcal{R}(1, 0, \lambda_*, 0)$, which is a part of that set (Def. 2.48, with its display and its last sentence), and its range is 0. At $\ell = \emptyset$ we have $\mathsf{B} = 1$ and $\sigma_* = 0$, and the entries at the site rows of a triple of Hil lie in the spaces $\mathsf{H}_y = \{0\}$, so that they vanish and (Ta), (Tb), (Tc), Rim_3 and the "Second" determinant of Step 8 are void. There, every quantity that $l_r, l_c, K\mathsf{g}_1, c_{\text{tr}}$ bound is 0, and all numbers and radii are those evaluated at $(K_*, \lambda_*, \varrho(x))$.

We denote by ℓ' the family of rows obtained from ℓ by Prop. 2.28, and we set $\mathsf{B}' = 1 + \ell'^{\mathsf{T}}\ell'$. The pair (ℓ, ℓ') satisfies the four conclusions of Thm. 2.64: (Dec)(C_D, θ_D); (N′) with eq. 2.vii at every cut; $\mathsf{t}_\rho \leq \frac{1}{8}(\frac{1}{2} - d_0 - c_\Xi)$; and (L1), (L2). Here (L1), (L2) are taken as (L1^{su}), (L2^{su}) of item 3 of §2.1, (O4), as in item 1 of §2.1, (SD).

The pair (V', e) is any pair of §4.1, E2. The system $w(X, \cdot)$, $X \in \mathcal{P}$, is any system of complex Borel functions, each of which is a function of U on the bonds of $X^{+s(x)}$. A further datum is a number $A \geq 1$, the constant in the two caps that follow. The system obeys either the strong cap

$$|w(X, U)| \leq A \exp\{-c_\Xi x\varepsilon \sum_{D(U)\cap X} \phi_p - \kappa(|X|_c - 1)\}$$

or the weak cap

$$|w(X, U)| \leq Ae^{-\kappa(|X|_c - 1)}.$$

Item D: inputs.

(C2) We take Thm. 3.6 (a), (b), (d), (f), (k), together with Lem. 3.3 (a), (b), (c), (e), for real V'; they give the following. The matrix G is real, and $G(b, b') = 0$ beyond the distance r_w; $(G\xi)_b$ depends on ξ only through its values on $Y_b = \{b' \in I^\circ : \text{dist}(b, b') \leq r_w\}$; $G(b, \cdot)$ depends on V' only within $r_w + \varrho_L + 1$ of $y(b)$, and $C(b, \cdot)$ only within $3r_w + \varrho_L + 1$. Next, $C = GG^{\mathsf{T}}$, $C(b, b') = 0$ beyond the distance $2r_w \leq R$, and $\|G\|_\infty, \|G^{\mathsf{T}}\|_\infty \leq \mathsf{G}_0$. In the setting ($\text{F}_{\text{law}}$), with $c_R\theta_{\mathbb{C}} \geq 128$, the matrix G is invertible and $S = C^{-1} - q$ is symmetric with $\|S\| \leq x^{-60}$. Further, $\gamma \leq q_0^V := |\mathrm{d}^V \cdot|^2 + a|Q_1^V \cdot|^2 \leq q_\ell^V \leq b_h$; the first inequality is Lem. 3.3 (a) applied to the empty rows $\ell = \emptyset$, which item C.I 1.1 of §3 admits, with $K = 0$, and γ depends on no row. The absolute row sums of q_ℓ^V are at most b_h, $q_\ell^V(b, b') = 0$ beyond the distance $r_q = 2\varrho_L + 2$, and $|\mathsf{g}^V X'|_\infty \leq (b_h/4) \sup |X'|$. As a consequence, at real (G-valued) fields, $\|C\| \leq 2/\gamma$, because $C^{-1} = q + S \geq \gamma - x^{-60} \geq \gamma/2$, where $\gamma^{-2} \leq x^8$ by (F9$^\flat$).

(C3) We take Thm. 3.32 (a), (b), (c) for real fields and with $\hat{\varepsilon} = \varepsilon'$, as in §3, Part C.II, Remarks (2), (3) after Thm. 3.34; the hypothesis $\tau_{\text{c}} \leq \varepsilon'$ of these remarks holds by (P1) and by $x^{-59} \leq \varepsilon'$ of (FA). For $i = p$ or c, with $o = y(p)$ or $o = c_-$ respectively, they give $|\Lambda_i| \leq \mathsf{K}_{\text{c}}^0\varepsilon'^2 + \tau_{\text{c}}\varepsilon' \leq \mathsf{K}_{\text{c}}\varepsilon'^2$, where Λ_i is the remainder given by the display before Thm. 3.32 in item C.II 4 of §3, and differs from the Λ_i of item H (a); they also give $|a_{0,b}| \leq \mathsf{a}_{\text{c}}\varepsilon'$ whenever $\text{dist}(b, o) \leq 1$. Both quantities depend on V' only through its values on the bonds in the box of radius R_3 about o, at fields V' tame there; a G-valued field with $\phi'_P \leq \varepsilon'$ has (h) at ε', since $\sin(\phi'/2) \leq \varepsilon'$ (the passage in parentheses after clause (c) of Thm. 3.32). The law of (C2) has the clauses (λ0), (ρ1), (ρ2), (ρ5) that Thm. 3.32 assumes of a law:

(λ0), (ρ1) Clause (λ0) follows from clause (a) of Thm. 3.6 together with the part of that theorem preceding its clauses, and clause (ρ1) holds by Thm. 3.6 (b).

(ρ2) It holds because $3r_w + \varrho_L + 1 \le R_2 - 1$.

(ρ5) It holds with $\mathsf{n} = 60$ by Thm. 3.6 (f), (k).

Lem. 3.37 (b), (c) are inputs as well, with the matrices $H^\natural, \mathsf{N}^\natural$ of item C.II 2.2 of §3, "The matrices of (S-cen$_k$)". At $\mathsf{n} = 60$ we have $\tau_1 = 8\gamma^{-2}(1 + b_h)x^{-60}$ (Lem. 3.26 (c)) and $\tau_c = 165(L+1)\mathsf{N_B}3L^3(2R+3)^3\tau_1$ (item C.II 4 of §3, "Constants"), and $\tau_c \le x^{-59}$ by (P1). The display before Lem. 3.37 gives

$$\tau_2 = 24L^3(2R+1)^3\gamma^{-2}x^{-60} + (2/\gamma)3L^3\mathsf{g}(\theta_{\mathbb{C}}/2)x^{-64}$$

and

$$\mathsf{ti_c} = (16 + 4L)\mathsf{N_B}\tau_2 + 7\mathrm{e}(L+1)C_D x^{-128}.$$

These inputs rest on the following three conditions (Fr1), (Fr2), (Fr3) of item C.II 2.2 of §3, all of them implied by the data.

(Fr1) This is Lem. 2.38 (c).

(Fr2) This is (Dec), also for the free kernels G, G_Z, which belong to the list of (Dec) in §2.2.

(Fr3) This is (N′) at $m = 1$ (Def. 2.42).

(Rows) On the rows we take Thm. 2.64, Thm. 2.60 (d), Thm. 2.57 (a) and Prop. 2.62, together with Def. 2.26, Def. 2.35, Prop. 2.27 (a) and (Dec).

(C5) The present theorem takes from §4.1 the lift E_3 of item C1 there, the pairs (V', e) and X' of item E2 there, and the definitions of $T_\square$ and w_j in items E5 and E6 there, which are also those of items E7 and E9 of this section. It also uses Step 1 (iv) (the identities $W^V E = 1$ and $\mathrm{d}^V E = E_3 \mathrm{d}'^V$) and Step 5 (Bianchi's identity) of the proof of Thm. 4.2.

(C6) The present theorem depends on five passages of §4.2. The first is the identity of matrices in Step 4 of the proof of Thm. 4.5, which writes out $\partial U(p)$ in the chart case by case. The second is Step 5 (1), (2) of that proof, a piece of geometry and an identity. The third is the sentence of item 3.4 of §4.2 stating that $\mathsf{M}(U)(c)$ is a Borel map of U. The fourth is the passage "As to invariance" of Step 1 (3) of that proof. The fifth consists of the inequalities $\varrho_L + 2 < N/2$, $\varrho(x)/L + 2 < N/2$ and $\varrho(x) + L < NL/2$ of Step 2 of that proof: the first two appear in the proof of statement (iv) of that step, and the third follows from the stronger inequality $\varrho(x) + 1 < NL/2 - L$ of statement (iv) itself.

The first two passages are valid at every field, for every fine site y, every $\eta \in \mathfrak{L}_{\mathfrak{g}}$ and every p with $|y - p| \le \varrho(x)$; at a pair (V', e) we have $\partial V'(P) = e^{iX'_P}$ also where $\partial V'(P) = -1$, because $e^{i\pi e_P} = -1$. The third and the fourth are statements about every G-valued fine field U. The fifth rests on $x \ge L^2$, $R \ge 6$, $L \ge 3$ and (Tor) alone, which hold here by item G, (O4) and item C. None of the five depends on any of the floors (W1) to (W5) of Thm. 4.5, on a condition that bounds $|\eta_b|$ from above, on $\chi_\square = 1$, on a standing hypothesis of Steps 3 to 10 of its proof, or on anything of the present theorem.

We also take results from earlier sections. Bianchi's identity appears in the proof of Lem. 3.27 (d), and the decay of an inverse and the quotient of two determinants are Lem. 2.25 (a) and (b). The formula of the chart is given in Lem. 3.9.

Lemma 4.20 (Angles and logarithms in SU(2))**.** On $G = \mathrm{SU}(2)$, the angle of an element g is fixed by $\cos \mathrm{ang}\, g = \frac{1}{2}\operatorname{tr} g$ and $\mathrm{ang}\, g \in [0, \pi]$, and $|\mathsf{L}\, g| = \mathrm{ang}\, g$ for every g with the one exception $\mathsf{L}(-1) = 0$. For finitely many $Y_k \in \mathfrak{g}$ let $S = \sum |Y_k|$.

(a) For $g, h, u \in G$ and $Y \in \mathfrak{g}$, we have $\mathrm{ang}(gh) \le \mathrm{ang}\, g + \mathrm{ang}\, h$, also $\mathrm{ang}(g^{-1}) = \mathrm{ang}(ugu^{-1}) = \mathrm{ang}\, g$, and $\mathrm{ang}\, e^{iY} \le |Y|$.

(b) For $g \in G$, $\|\mathrm{Ad}_g - 1\| \le \min\{2, 2\,\mathrm{ang}\, g\}$.

(c) If $S < \pi$, then $|\mathsf{L}(\prod e^{iY_k}) - \sum Y_k| \le S^2$.

(c′) If $\prod e^{iY_k} = 1$, then $|\sum Y_k| \le \frac{1}{2}S^2$, whatever the value of S.

■

Proof. Step (a)*: the arc distance.* We view SU(2) as the unit sphere S^3. For $g, h \in G$, the number $\operatorname{ang}(g^{-1}h)$ is the arc distance between g and h on $S^3 = \mathrm{SU}(2)$, and this metric is invariant under left and under right translations. Thus $\operatorname{ang} g$ is the distance from 1 to g; the triangle inequality and the left invariance give $\operatorname{ang}(gh) \le \operatorname{ang} g + \operatorname{ang} h$, and the invariance on both sides gives $\operatorname{ang}(g^{-1}) = \operatorname{ang}(ugu^{-1}) = \operatorname{ang} g$. The curve $t \mapsto e^{itY}$, $0 \le t \le 1$, leads from 1 to e^{iY} with the speed $|Y|$; its length is therefore $|Y|$, and $\operatorname{ang} e^{iY} \le |Y|$.

Step (b)*: the adjoint action.* The map Ad_g is the rotation of $\mathfrak{g} = \mathbb{R}^3$ about the axis of g by the angle $2 \operatorname{ang} g$. A rotation by an angle α moves each unit vector by at most $2|\sin(\alpha/2)|$, with equality for the unit vectors orthogonal to its axis. Hence $\|\mathrm{Ad}_g - 1\| = 2\sin(\operatorname{ang} g)$, and this number is at most 2 and at most $2 \operatorname{ang} g$.

Step (c)*: the logarithm of a product.* Write g for $\prod e^{iY_k}$. If $S \ge 2$, then $|\mathsf{L}\, g| \le S$ by (a), as $|\mathsf{L}\, g|$ never exceeds $\operatorname{ang} g$; moreover $|\sum Y_k| \le S$, and the triangle inequality together with $2S \le S^2$ gives the claim. Now let $S < 2$. We reach the product $\prod e^{iY_k}$ factor by factor, along a curve $g(s)$ of speed 1 whose parameter s is the arc length: the interval $[0, S]$ is cut into consecutive subintervals, one of length $|Y_k|$ for each nonzero Y_k in order. On that subinterval put $Y(s) = Y_k/|Y_k|$. The path $g(s)$ is the solution of $g(0) = 1$ and $g' = g\, iY(s)$ on $0 \le s \le S$. So $g(S) = \prod e^{iY_k}$ and $\int_0^S Y = \sum Y_k$. Put $Z = \mathsf{L}\, g(s)$ and $\check\theta = |Z| = \operatorname{ang} g(s) \le s$; indeed $g(s)$ is a product of elements e^{iY} whose $|Y|$ add up to s, so that (a) bounds its angle by s, and $s < 2 < \pi$ excludes $g(s) = -1$.

The identity $e^{-A}(e^A)' = ((1 - e^{-\operatorname{ad} A})/\operatorname{ad} A)A'$ at $A = iZ$ gives $Z' = f(\mathrm{ad}_{iZ})Y$ with $f(z) = z/(1 - e^{-z})$. The operator ad_{iZ} is skew on $\mathfrak{g}$ and has the eigenvalues $0, \pm 2i\check\theta$, and $f(0) = 1$, $f(\pm 2i\check\theta) = \check\theta \cot\check\theta \pm i\check\theta$. Since $|Y(s)| = 1$, it follows that $|Z' - Y| \le m(\check\theta) := |\check\theta \cot\check\theta - 1 + i\check\theta|$. The function $1/t - \cot t$ is increasing on $(0, 2]$, since its derivative is $1/\sin^2 t - 1/t^2 \ge 0$. At $t = 2$ it is less than $0.958 < \sqrt{3}$. Since $m(\check\theta)^2 = (1 - \check\theta \cot\check\theta)^2 + \check\theta^2$, we obtain $m(\check\theta) \le 2\check\theta$ for $0 < \check\theta \le 2$. Since $Z(0) = 0$ and $\check\theta \le s < 2$, integration over $[0, S]$ yields $|Z(S) - \int_0^S Y| \le \int_0^S 2s\, ds = S^2$. This is (c), because $Z(S) = \mathsf{L}(\prod e^{iY_k})$ and $\int_0^S Y = \sum Y_k$.

Step (c′)*: a product equal to* 1. If $S \ge 2$, then $|\sum Y_k| \le S \le S^2/2$. Let $S < 2$, and define $g(s)$, Z and $\check\theta$ as in Step (c). Here $g(S) = 1$; so $g(s)^{-1} = g(s)^{-1} g(S)$ is a product of elements e^{iY} whose $|Y|$ add up to $S - s$, and (a) gives $\check\theta(s) \le \min\{s, S - s\}$. Also $Z(S) = 0$, and $Z(0) = 0$. The bound $|Z' - Y| \le m(\check\theta) \le 2\check\theta$, proved in Step (c) for $\check\theta \le 2$, now gives $|\sum Y_k| = |Z(S) - \int_0^S Y| \le \int_0^S 2\min\{s, S - s\} ds = S^2/2$. ■□

Lemma 4.21 (A comb gauge on a box)**.** Let Bx be an embedded box with a site o, let V be a G-valued field on the bonds of Bx, and let $\phi_p = \operatorname{ang} \partial V(p)$. Let Γ be the coordinate path, taken in the order e_1, e_2, e_3; put $\check\gamma(z) = V(\Gamma(o, z))$ and $V^{\check\gamma}(b) = \check\gamma(b_-) V(b) \check\gamma(b_+)^{-1}$.

(1) Let $b = (z, \cdot)$ be a bond. Then $\operatorname{ang} V^{\check\gamma}(b) \le \sum_p \phi_p$, where p ranges over the plaquettes of the ladders of b, which are as follows: for the direction 3 there is none; for the direction 2 there is one ladder of $|z_3 - o_3|$ plaquettes $\{23\}$; for the direction 1 there are two, one ladder of $|z_3 - o_3|$ plaquettes $\{13\}$ and one of $|z_2 - o_2|$ plaquettes $\{12\}$ at the height o_3.
(2) Distinct bonds of one straight stretch have disjoint ladders. Hence, along a straight stretch, $\operatorname{ang} V^{\check\gamma} \le \sum_{p \in \mathrm{Bx}} \phi_p$.
(3) Along a closed path in Bx made of k straight stretches, $\operatorname{ang} V \le k \sum_{p \in \mathrm{Bx}} \phi_p$.

■

Proof. Step (1)*: the closed path of a bond.* The path $\Gamma(o, b_-) + b - \Gamma(o, b_+)$ is closed, and it bounds the ladders of b named in (1); see Step 1 of the proof of Lem. 2.24. By the definition of $\check\gamma$, the element $V^{\check\gamma}(b)$ is the product of V along this closed path. Along a closed path that bounds ladders, the product of V is an ordered product of conjugates of the $\partial V(p)^{\pm 1}$, in which every plaquette of

the ladders enters once; see the proof of Lem. 3.27 (a), the passage "a ladder". So Lem. 4.20 (a) bounds $\operatorname{ang} V^{\check\gamma}(b)$ by the sum of the ϕ_p over these plaquettes.

Step (2)*: disjoint ladders.* Two bonds of a straight stretch have different values of the coordinate in the direction of the stretch, and the plaquettes of their ladders differ in that coordinate too; moreover a plaquette $\{13\}$ is never a plaquette $\{12\}$. So the ladders of distinct bonds of the stretch are disjoint. The product of $V^{\check\gamma}$ along the stretch is the product of the $V^{\check\gamma}(b)$ over its bonds; Lem. 4.20 (a) bounds its angle by the sum of their angles, and (1) with the disjointness bounds that sum by $\sum_{p\in \mathrm{Bx}}\phi_p$.

Step (3)*: a closed path.* Let π be a closed path in Bx made of k straight stretches, so that $\pi_- = \pi_+$. The gauge factors cancel in pairs along π, and $V(\pi) = \check\gamma(\pi_-)^{-1}V^{\check\gamma}(\pi)\check\gamma(\pi_+)$; as the path is closed, this is a conjugation. Here $V^{\check\gamma}(\pi)$ is the product of the $V^{\check\gamma}$ of the k stretches, and Lem. 4.20 (a) together with (2) gives $\operatorname{ang} V(\pi) \le k\sum_{p\in\mathrm{Bx}}\phi_p$. ■□

Remark 4.22. In the chart, $U = 1$ on the combs. Therefore, for the field U of the chart, the box $\mathrm{Bx} = B(y)$ and $o = y$, one has $\check\gamma = 1$.

Lemma 4.23 (Sizes of the vectors $\tau_{c,x}$ and $\bar\tau_c$ in the chart)**.** In the chart, let $c = (y, y')$ be of direction μ, and let $x \in B(y)$. Here $h_{c,x}$ is that of item C.I 4 of §3, and, as in item 3.4 of §4.2, $\tau_{c,x} = \mathsf{L}(h_{c,x}h_{c,y}^{-1})$ and $\bar\tau_c = L^{-3}\sum_{z\in B(y)}\tau_{c,z}$. Then $|\tau_{c,x}| \le \operatorname{ang}(h_{c,x}h_{c,y}^{-1}) \le s_c$; consequently $|\bar\tau_c| \le s_c$. ■

Proof. Let $v_0 = y, \dots, v_m = x$ be the sites of the comb $\Gamma(y, x)$, in their order along it; for a site v write $v' = v + Le_\mu$, and put

$$a_k = U(\Gamma(y, v_k))U([v_k, v'_k])U(\Gamma(y', v'_k))^{-1}.$$

Then $a_0 = h_{c,y}$ and $a_m = h_{c,x}$. If the bond (v_k, v_{k+1}) is parallel to e_μ, then this bond followed by $[v_{k+1}, v'_{k+1}]$, and $[v_k, v'_k]$ followed by the bond (v'_k, v'_{k+1}), are two paths from v_k to v'_{k+1} on one straight line of direction e_μ; since steps forth and back along this line cancel, the products of U along the two paths agree, and $a_{k+1} = a_k$. If it is not, then $a_{k+1}a_k^{-1}$ is conjugate to the product of U along the boundary of a strip of L plaquettes, and that product is an ordered product of L conjugates of $\partial U(p)^{\pm1}$. No two of these strips meet, and all of them lie in W_c; see Step 2 of the proof of Lem. 2.24. As $h_{c,x}h_{c,y}^{-1} = a_m a_0^{-1}$ is the ordered product of the $a_{k+1}a_k^{-1}$, Lem. 4.20 (a) bounds $\operatorname{ang}(h_{c,x}h_{c,y}^{-1})$ by the sum of the $\operatorname{ang}\partial U(p)$ over $p \in W_c$, which is s_c. Finally, $|\mathsf{L}\, g| \le \operatorname{ang} g$ holds at every g, also at $g = -1$; this gives $|\tau_{c,x}| \le \operatorname{ang}(h_{c,x}h_{c,y}^{-1})$, and with it the bound for $|\bar\tau_c|$. ■□

Item E: objects.

E1 *Lattice, radii at x, norms, cell and fine cube of an object.* We take from item C.I 1.1 of §3: the blocks $B(y)$; $\pi(i)$ and $y(i)$; the distance $\operatorname{dist}(i, j) = L^{-1}|y(i) - y(j)|_\infty$, in coarse units; the combs; the set I° of the fine bonds off the combs, of which at most $3L^3$ are based in one block; d, Q_1 and E; the coarse path $\Gamma(y, y'')$ and $T(y, y'')$; K^V and $(K^V)^*$; ℓ^V; $\mathsf{B}^V = 1 + \ell^{V*}\ell^V$; $q = q_\ell^V$; $\mathsf{g}^V = \Pi_{\mathfrak{Q}}\mathrm{d}^{V*}\mathsf{B}^V E$; and $|y - p| = |y - \pi(p)|_\infty$, in fine units.

From Def. 2.17 and Lem. 2.18 we take $W = Q_2$ with the weights $W(P, p)$; the double block W_c of c; the union W_P of the four blocks at the corners of P, which is a box $2L\times 2L\times L$ holding at most $12L^3$ plaquettes; and d'. The cubes, the cells $\square$, $\hat\square$, $\mathcal{P}$, $|X|_c$ and X^{+j} are those of the definition "Cubes" of item 1 of §2.1.

The radii at x are $R = R(x) = \varrho(x) = \lceil c_R \log x\rceil$, $r(x) = \lceil 20c_R \log x/M\rceil$ and $s(x) = 8r(x)$. We let $S_\varepsilon(t) = t^2/2$ for $t \le \varepsilon$, and $d_0\varepsilon t$ for $t > \varepsilon$. The norms $\|K\|_\theta, \|K\|_0, \|K\|_\infty$ are those of Def. 2.20.

An object i belongs to the cell containing $y(i)$ and to the fine cube in which its base site lies. For a set A of cells, $\hat A$ denotes the set of the fine objects that belong to a cell of A (the objects i with $y(i)$ in a cell of A), and $\langle , \rangle_A$, $\|\cdot\|_A$ are defined by summing over the

objects (rows) of A. The quantity $|\xi_A|^2$ is the sum of the $|\xi_b|^2$ over the bonds b of I° in $\hat{A}$, and $|\varphi_A|^2$ is the same sum with $(G\xi)_b$ in place of ξ_b. The distance between two cells, or between two cubes, is the sup distance measured in cells, or in cubes. We say that V' is tame on the box of radius ρ about i if $\phi'_P \le \varepsilon'$ for all P with $\operatorname{dist}(P,i) \le \rho$.

E2 *Fields, masks, energy, noise, defects.* As in E2 of §4.1, $X'_P = X'_P(V',e) = \mathsf{L}(\partial V'(P)) + \pi e_P 1\{\partial V'(P) = -1\}$ and $\phi'_P = |X'_P| = \operatorname{ang} \partial V'(P)$. Let $D' = \{\phi'_P > \varepsilon'\}$ and $\zeta'_P = X'_P \min\{1, \varepsilon'/\phi'_P\}$, and let ℓ'^{c} denote the cut of ℓ' at $\varrho(x')$. We put $m'_{y'} = 1$ exactly when no plaquette of D' is within ρ_0 of y' and none with $\phi'_P > 1$ is within $\varrho(x')+1$ of it, and $m'_{y'} = 0$ otherwise.

The chart $\mathsf{U}(\eta; V')$ is that of item C.I 4 of §3, and there $\mathsf{j}(\eta_b) \le 1$. For every G-valued fine field U we let $F_p = \mathsf{L}(\partial U(p))$, $\phi_p = \operatorname{ang} \partial U(p)$, $D = D(U) = \{\phi_p > \varepsilon\}$ and $\zeta_p = F_p \min\{1, \varepsilon/\phi_p\}$, with multiplier 1 at $\phi_p = 0$. Below, we also define $\ell^U_y \zeta$, $m_y(U)$, $\mathrm{En}_\ell(U)$ and the noise n_c, n_c at every such U. Where η is given, namely in the defects of E2, in E4, E11 and E12, and in (a) to (e) of item H, all these functions of U, when written without argument, are taken at $U = \mathsf{U}(\eta; V')$ for the η of E3. As in Step 5 (1) of the proof of Thm. 4.5, for fine sites z, z' the coordinate path $\Gamma(z,z')$ runs first along e_1, then along e_2 and last along e_3, always by the shorter way. We set $\ell^U_y \zeta = \sum_p \ell_y(p) \otimes \mathrm{Ad}_{U(\Gamma(y,\pi(p)))}\, \zeta_p$, and $m_y(U) = 1$ exactly when there is no p of D with $|y-p| \le \rho_0$ and no p with $\phi_p > 1$ and $|y-p| \le \varrho(x)+1$, while $m_y(U) = 0$ otherwise. The energy is

$$\mathrm{En}_\ell(U) := \sum_p S_\varepsilon(\phi_p) + \frac{1}{2} \sum_y m_y(U) |\ell^U_y \zeta|^2.$$

The noise is given by $\mathsf{n}_c = \operatorname{ang}(V'(c) \mathsf{M}(U)(c)^{-1}) \in [0,\pi]$ and $n_c = \mathsf{L}(V'(c)\mathsf{M}(U)(c)^{-1}) \in \mathfrak{g}$, where $|n_c| = \mathsf{n}_c$ if $\mathsf{n}_c < \pi$. The defects are $\mathrm{de}_P = X'_P - (W^V F)_P - (\mathrm{d}'^V n)_P$ and $\mathrm{db}_q = (\mathrm{d}^V F)_q$. We put $\Sigma_P = \sum_{p \in W_P} \phi_p$, $\mathrm{na}_P = \sum_{c \in dP} \mathsf{n}_c$, $s_c = \sum_{p \in W_c} \phi_p$ and $s_B = \sum_{p \in B} \phi_p$; a superscript D on Σ_P, s_c or s_B restricts the sum to the rough p, and a superscript t restricts it to the tame p, where a fine plaquette p is rough if $p \in D(U)$ and tame if $p \notin D(U)$.

E3 *White noise and centre.* Let $\xi \sim \mathcal{N}(0, x^{-1}1)$ on $\mathfrak{L}_\mathfrak{g}$, and put $\varphi = G^V \xi$, $a_0 = C\mathfrak{g}^V X'$ and $\eta = \varphi - a_0$; further, $u^T = 1 - qC$, $F^\flat = EX' + \mathrm{d}^V \eta$ and $n^\flat = -Q_1^V \eta$.

E4 *The space of triples.* The space Hil consists of the triples (F, Σ, n) with F on the fine plaquettes, Σ on the fine sites and n on the coarse bonds, which together are its rows; it carries $\langle , \rangle_{\mathrm{Hil}} = \langle F, F' \rangle + \langle \Sigma, \Sigma' \rangle + a \langle n, n' \rangle$. Let $\iota(F,n) = (F, \ell^V F, n)$ and $k\psi = \iota(\mathrm{d}^V \psi, -Q_1^V \psi)$; thus $q_\ell^V = \|k \cdot\|^2$, and k_i denotes the row i of k. We put $\bar{F} = \iota(EX', 0) - ka_0$, and $\mathfrak{k} = \iota(k_F, k_n)$ with $k_F = F^\flat - F$ and $k_n = n^\flat - n$. We set $d_1 = W^V \mathrm{d}^V - \mathrm{d}'^V Q_1^V$ and $d_2 = \mathrm{d}^V \mathrm{d}^V$.

E5 *Free kernels and their cuts.* The kernels T, X, $\mathsf{B} = 1 + \ell^{\mathsf{T}} \ell$, B', $H^\sharp$, $\mathsf{N}^\sharp$ are those of Def. 2.26, of Def. 2.35, and of Prop. 2.27 (a). We write $\mathsf{T}^<, \mathsf{X}^<$ for the cuts of T and X at $R(x')$, and $\mathsf{B}'^{\mathrm{c}} = 1 + \ell'^{\mathrm{c}\mathsf{T}} \ell'^{\mathrm{c}}$. With $\|\cdot\|_{\mathrm{r}}$ the largest absolute row sum,

$$\mathsf{t}_\rho := \|\ell'^{\mathrm{c}}\|_{\mathrm{r}} \sup_{P_0} \sum_{y' : \operatorname{dist}(y', P_0) > \rho_0} |\ell'^{\mathrm{c}}_{y'}(P_0)|,$$

as in Def. 2.39.

E6 *Multipliers and the pair map.* We put $v_* = (1 + \ell'^{\mathrm{c}V*} m' \ell'^{\mathrm{c}V}) \zeta'$, $Z_* = v_*/L$, $\Xi_* = (\mathsf{X}^<)^V v_*$ and $Y_* = W^{V*} Z_* + \mathrm{d}^{V*} \Xi_*$; further, $\nu^{\mathrm{ref}} = (\mathsf{T}^<)^V v_*$, $\sigma_* = \ell^V \nu^{\mathrm{ref}}$, $\nu_* = Y_* - \ell^{V*} \sigma_*$ and $\omega_* = (aL)^{-1} \mathrm{d}'^{V*} v_*$. The pair map is

$$\mathrm{pm}(v) = (W^{V*} v/L + \mathrm{d}^{V*} (\mathsf{X}^<)^V v - \ell^{V*} \ell^V (\mathsf{T}^<)^V v,\ \ell^V (\mathsf{T}^<)^V v,\ (aL)^{-1} \mathrm{d}'^{V*} v),$$

and we put $\boldsymbol{\nu} = \mathrm{pm}(v_*) = (\nu_*, \sigma_*, \omega_*)$ and $\hat{R}^V = L\, \mathrm{pm}^* \mathrm{pm}$. Let $\Delta = \bar{F} - \boldsymbol{\nu}$, with the components $\Delta_p, \Delta_y, \Delta_c$, and let $\Delta_Y = \mathsf{B}^V \bar{F}_F - Y_*$, where $\bar{F}_F$ is the plaquette component of $\bar{F}$. With $r_* = \mathsf{B}^V \nu^{\mathrm{ref}} - Y_*$ we then have $\nu_* = \nu^{\mathrm{ref}} - r_*$.

E7 *Coarse energy and the prefactor of a cell.* For a coarse plaquette P let $\mathrm{En}'_P = S_{\varepsilon'}(\phi'_P) + \frac{1}{2}\langle \zeta'_P, (\ell'^{\mathrm{cV}*} m' \ell'^{\mathrm{cV}} \zeta')_P\rangle$. The coarse energy is

$$\mathrm{En}'(V') := \sum_P \mathrm{En}'_P = \sum_P S_{\varepsilon'}(\phi'_P) + \frac{1}{2}\sum_{y'} m'_{y'} |\ell'^{\mathrm{cV}}_{y'} \zeta'|^2,$$

and for a cell $\square$ we put

$$\begin{aligned} T_\square := &\sum_{P\in\square} [(x'/2)\langle X'_P, v_{*P}\rangle - x'\mathrm{En}'_P + (x/2)\langle X'_P, (E^*\Delta_Y)_P\rangle] \\ &+ (x/2)\sum_{q'\in\square} \langle (\mathrm{d}'^V X')_{q'}, (E_3^*\Xi_*)_{q'}\rangle \\ &- (x/2)\sum_{b\in\hat\square} \langle a_{0,b}, (u^T\, \mathsf{g}^V X')_b\rangle. \end{aligned}$$

This $T_\square$ coincides, term by term, with the $T_\square$ of E5 of §4.1, where the $\bar F$ of that section is the $\bar F_F$ of E6, the plaquette component of the triple $\bar F$ of E4. The ν^{ref} and ν_* of E4 of §4.1 are the same objects as in E6, given by the same formulas $\nu^{\mathrm{ref}} = (\mathsf{T}^<)^V v_*$ and $\nu_* = Y_* - \ell^{V*}\sigma_*$; hence the $\Delta_F = \bar F - \nu_*$ of §4.1 is the Δ_p of E6. By E6, $\mathsf{B}^V(\bar F_F - \nu^{\mathrm{ref}}) + r_* = \Delta_Y$; expressed in the components of Δ from E6, $\Delta_Y = (\Delta_p)_p + \ell^{V*}(\Delta_y)_y$.

E8 *Radii.* In coarse units, $\varrho_L = \lceil R/L\rceil$, $r_w = \lfloor R/2\rfloor$, $r_q = 2\varrho_L + 2$, $R_2 = 2R + 2$, $R_3 = 2R + \varrho_L + 5$, $R_5 = 4R + 4\varrho_L + 8$ and $R_6 = R_5 + 2\varrho_L + 3$. In cells, $r_l = \lceil (\varrho_L + 2)/M\rceil$, which is ≥ 1, $r_1 = \lceil R_2/M\rceil + 3r_l + 5$ and $s_0 = 2r(x')$.

E9 *Marks, forced sets, classes.* Let mk' be the set of the cells that hold a plaquette of D', and $f = \mathrm{mk}'^{+s_0}$. The classes mk'_j are those of mk' under the relation generated by distance $\leq 2s_0 + 4r_1 + 2$. We put $f_j = \mathrm{mk}'^{+s_0}_j$; any two of these sets are at a distance $> 4r_1 + 2$. The set D'_j consists of the plaquettes of D' in mk'_j, and f_J, D'_J are the unions over $j \in J$. Further, $w_j = \sum_{\square\in \mathrm{mk}'^{+r(x')}_j} T_\square$.

E10 *Tests.* For a cell $\square$, $b_\square(\xi) = 1$ if $|\xi_b| > c_\xi \varepsilon$ at some bond b of I° in $\hat\square$, and $b_\square(\xi) = 0$ otherwise.

E11 *The factor of a set of cells.* For a set $\mathfrak{Z}$ of cells, $[A]^\ell_{\mathfrak{Z}}$ is the product of $\exp\{(x/2)|F^\flat_p|^2 - xS_\varepsilon(\phi_p)\}$ over $p \in \hat{\mathfrak{Z}}$, of $\exp\{(x/2)|(\ell^V F^\flat)_y|^2 - (x/2)m_y|\ell^U_y\zeta|^2\}$ over $y \in \hat{\mathfrak{Z}}$, of $\exp\{(ax/2)|n^\flat_c|^2 - ax(1-\cos \mathsf{n}_c)\}$ over $c \in \mathfrak{Z}$, and of $1\{|\eta_b| < \pi\}\mathsf{j}(\eta_b)$ over $b \in I^\circ \cap \hat{\mathfrak{Z}}$. We put $A_\square := [A]^\ell_{\{\square\}}$; since every object lies in one cell, $[A]^\ell_{\mathfrak{Z}} = \prod_{\square\in\mathfrak{Z}} A_\square$.

E12 *Cell data and letters at V'.* A cell datum (J, t) consists of a set J of classes and a finite set t of cells off f_J, such that $\mathfrak{b} := f_J \cup t$ is non-empty; here $\mathfrak{b}$ is any such set, connected or disconnected. The sets of the datum are $\mathfrak{Z} = \mathfrak{b}^{+r_1}$, $\mathfrak{Z}' = \mathfrak{b}^{+r_1-r_l}$, $\mathfrak{S} = \mathfrak{b}^{+r_1-r_l-1}$, $\mathfrak{S}^- = \mathfrak{b}^{+r_1-r_l-2}$, $\mathfrak{Z}_2 = \mathfrak{b}^{+2r_1}$ and the rim $\mathrm{Lr} = \mathfrak{b}^{+r_1+r_l+1} \setminus \mathfrak{b}^{+r_1-2r_l-3}$. We write χ for the restriction to the objects of $\mathfrak{Z}$, and χ_S for the restriction to the bonds based in $\hat{\mathfrak{S}}$.

We put

$$\mathcal{G}_{\mathfrak{Z}} = \sum_{P\in\mathfrak{Z}} [\langle v_*, X'\rangle_P - \frac{1}{2}\langle v_*, \hat R^V v_*\rangle_P]$$

and

$$\hat\lambda_{(J,t)} := e^{-\sum_J w_j} \prod_t b_\square \prod_{\mathfrak{Z}_2\setminus\mathfrak{b}} (1 - b_\square)[A]^\ell_{\mathfrak{Z}} \geq 0.$$

A cell datum is admitted if (H17) holds: each plaquette of D' that lies in a cell at distance at most $4r_1$ from $\mathfrak{b}$ lies in D'_j for some $j \in J$. Two cell data are compatible if each cell of the one set $\mathfrak{b}$ is at distance more than $4r_1 + 2$ from each cell of the other.

In the rest of E12, an object is a forced class f_j, whose cells are those of f_j, or a cell, or a polymer $X \in \mathcal{P}$. Letters are made of these objects, which are distinct from the objects i of E1. Two objects with cells are adjacent if they have cells at distance $\le 4r_1 + 2$; a polymer X and an object with a cell $\square$ are adjacent if and only if X contains a cube of $\widehat{\square^{+r_1}}$. A letter $\check\zeta = (J, t, \mathfrak{X})$ is a set of objects that is connected under adjacency and in which (J, t) is a cell datum, each X is adjacent to some object with cells, and any two X have distance $> s(x)$. The letter is bare if $\mathfrak{X}$ is empty.

Two letters are compatible if each object of the first is different from, and not adjacent to, every object of the second, and if any polymer of the first and any polymer of the second have distance $> s(x)$; the cell data of two compatible letters are then compatible. A letter is admitted if its cell datum is. We put $1_{\mathrm{ind}} = 1\{D(U) \cap \hat{\mathfrak{Z}}$ is contained in the union of $\mathfrak{X}$, and every X of $\mathfrak{X}$ holds a plaquette of $D(U) \cap \hat{\mathfrak{Z}}\}$, as well as $\Lambda_{\check\zeta} = \hat\lambda_{(J,t)} 1_{\mathrm{ind}}$ and $\lambda^o_{\check\zeta} = \Lambda_{\check\zeta} \prod_{\mathfrak{X}} w(X, U)$. We set $I(\check\zeta) = \mathfrak{Z}_2 \cup \{$cells that meet some $X^{+s(x)}\}$.

Item F: constants. We set $a^\flat_\infty = \mathsf{G}_0^2 b_h/4$ and $c^\flat_q = \pi b_h \mathsf{G}_0(1 + a^\flat_\infty)$. Further, we put $K_D = \mathsf{K}_{\mathrm{c}} + 1 + 6700\theta_D^{-2} C_D^3$, $K_r = 1 + 23000\theta_D^{-2}(C_D^5 + LC_D^2)$ and $K^\flat_\Delta = (1 + l_r)(K_D + K_r)$, and $K^\flat_{\mathrm{q}} = M^3[1.5L^2(\mathsf{N}_{\mathsf{B}} K_D + K_r) + 9\mathsf{N}_\Xi + 1]$. Next, $a_\nu = \mathrm{e}L + 6\mathrm{e}C_D + C_D^3$ and $C^\flat_{\mathrm{hol}} = 162\mathsf{g}(\theta_D)(1 + C_D^2)[La_\nu^2 + LC_D^4 + 7]$; then $c_B = 6\mathsf{g}(\theta_D/(4C_D^2))$ and $e^\flat_1 = \frac12 \mathsf{N}_Z^2 c_B C^\flat_{\mathrm{hol}} \varepsilon'$. We also set $\mathsf{ti}^\flat_0 := 8C_D^2 L a_\nu^2 x'^{-64}$ and $c^\flat_\eta = 1 + L^{1/2}(1 + \mathsf{a}_{\mathrm{c}})$.

With these,

$$K^\flat_s = 200vL^3(1 + a)(1 + K^\flat_\Delta)(\mathsf{N}_Z + \mathsf{N}_\Xi + \mathsf{N}^\flat_\nu + \mathsf{N}_\sigma)(1 + l_r)(1 + K\mathsf{g}_1)(1 + c^\flat_\eta)^2$$

and

$$\sigma^\flat_{\mathrm{tot}} = K^\flat_s p_0^3 x^{-1/2} + K^\flat_{\mathrm{q}} p_0'^2 \varepsilon' + (9e^\flat_1 + \frac92 \mathsf{N}_Z^2 c_B \mathsf{ti}^\flat_0) M^3 p_0'^2.$$

Let $t_0 = \varepsilon^{1/2}$ and, in this section alone, $\theta_2 = \varepsilon/12$; this θ_2 differs from the $\theta_2 = \theta_1/2$ of item 3 of §2.1. We put

$$\vartheta^\flat_F = [(86400\mathsf{N}_Z + 110\mathsf{N}_\Xi)L^{5/2} + 2949K\mathsf{g}_1\mathsf{N}_\sigma L^{9/2}]\varepsilon$$

and

$$\vartheta^\flat_n = [(192\mathsf{N}_Z + \mathsf{N}_\Xi)L^{-1/2} + 3K\mathsf{g}_1\mathsf{N}_\sigma L^{1/2}]\varepsilon,$$

and further, $z_b = 21\mathsf{N}_Z L^{-1/2}\varepsilon$, $r_o = x^{-3/8}$ and $\check\theta_x = x^{-1/4}$.

Let $\mathrm{py} = \min\{d_0 p_0^2/2, 0.05 a x^{1/4}\}$ and $\kappa_1 = \mathrm{py}/(240\pi^2 L^3 x)$; $h_* = C^\flat_\varphi(4 + 12K\mathsf{g}_* + a^{1/2}L)$, $t_Q = 1/(12h_*^2)$ and $\rho_Q = 3c_\xi^2 p_0^2/(128h_*^2)$; $c_{\mathrm{tr}} = (4/\gamma)144K^2\mathsf{g}(2\lambda)$ and $c_\tau = \max\{6.1, c_{\mathrm{tr}}\}/12$. Moreover

$$Q^\flat_\theta = [1.3 \cdot 10^5 L^3 + 4704aL^4 + (8.4 \cdot 10^4 l_r^2 + 1296K^2\mathsf{g}_1^2)L^3]M^3$$

and $u_b = 2.5v(c_\xi p_0 + (c_\xi p_0)^{-1})e^{-c_\xi^2 p_0^2/2}$.

We put $u_t = 3v \max\{e^{c_\tau - \rho_Q/2}, e^{-\mathrm{py}/(768(2r_1+1)^3)}\}$, $u_1 = \max\{e^{-\mathrm{py}/768}, e^{-\rho_Q/2}\}$ and $\delta_3 = \min\{d_0/1536, 3c_\xi^2/(256h_*^2)\}$. The constants u_t', u_1', δ_3' are given by the same formulas, with 1152 in place of 768 and 2304 in place of 1536, and in them, as in $e^{\circ\prime}$ and $e^{o\prime}$ below, the prime is not that of (O8), and p_0, py, ρ_Q, r_1 are those at x.

Then

$$e^\flat_2 = 9e^\flat_1 + \frac92 \mathsf{N}_Z^2 c_B \mathsf{ti}^\flat_0 + 6.3(\mathsf{N}^\flat_\nu)^2 \vartheta^\flat_F + 3\mathsf{N}_Z^2(\theta_2 + L\vartheta^\flat_n)$$

and

$$\mathrm{Res}_{(J,t)} = \exp\{-(\frac12 - d_0 - \mathsf{t}_\rho - 3e^\flat_1)x'\varepsilon' \sum_J \sum_{D'_j} \phi'_P\}.$$

For a cell datum (J,t) let $e^{\circ}_{(J,t)} = \operatorname{Res} u_t^{\#t} u_1^{1[t\neq\emptyset]}$ and $e^{\circ\prime}_{(J,t)} = \operatorname{Res} u_t'^{\#t} u_1'^{1[t\neq\emptyset]}$; for a letter $\check\zeta$ let

$$e^{\circ}_{\check\zeta} = e^{\circ}_{(J,t)} \prod_{\mathfrak{X}} A e^{-c_\Xi p_0^2 - \kappa(|X|_c - 1)}$$

and

$$e^{\circ\prime}_{\check\zeta} = e^{\circ\prime}_{(J,t)} \prod_{\mathfrak{X}} A e^{-\mathrm{py}/1152 - \kappa(|X|_c - 1)}.$$

Item G: floors. Let S_{fl} denote the set of all $y \geq L^2\mathrm{e}$ with the property that every floor in the list below holds at each $x \geq y$. If S_{fl} contains a number y, then it contains every number larger than y, so it is an upper half line; by Step 13 of the proof of Thm. 4.19 it is not empty.

The set S_{fl} need not contain its infimum. Indeed, R, ϱ_L, $r(x')$ and r_1 are values that the ceiling function $\lceil\cdot\rceil$ takes; as functions of x they are continuous from the left, and their jumps are all upward. A floor with an inequality in which R, ϱ_L, $r(x')$ or r_1 appears on the side asserted to be the larger, as in (G1$^\flat$), can therefore fail at a point x_0 and yet hold at every $x > x_0$.

We put $x_{\mathrm{fl}} := 1 + \inf S_{\mathrm{fl}}$. As S_{fl} is an upper half line and x_{fl} lies above its infimum, $x_{\mathrm{fl}} \in S_{\mathrm{fl}}$; the number x_{fl} depends only on what S_{fl} depends on. Since $x_{\mathrm{fl}} \in S_{\mathrm{fl}}$, the defining property of S_{fl} shows that all the floors hold at every $x \geq x_{\mathrm{fl}}$. This definition of x_{fl} has the same form as the definition of $x_\flat^0$ in Thm. 5.3. Each floor is a condition on x and on the letters of (O0) to (O10) alone. The floors are the following.

(G1$^\flat$) $R \geq \max\{M, 6\}$, $Mr(x') \geq 5R + 4\varrho_L + 12$ and $\rho_0 \leq R(x')$.

($\mathrm{F_{law}}$) $432 L^3 b_h^{5/2} \gamma^{-3/2} \mathsf{g}(\theta_{\mathbb{C}})(R+3)^3 \leq x^4$ (see Thm. 3.6 (k)).

(P1) $(4/\gamma)x^{-60} \leq 1$ and $165(L+1)\mathsf{N_B}\, 3L^3(2R+3)^3\, 8\gamma^{-2}(1+b_h) \leq x$ (see the floors in item C.II 4 of §3).

(P3) $L^2(1+a)(\mathsf{ti_c} + 8C_D^3 x'^{-64}) \leq 1$ (see the display before Lem. 3.37 in §3, C.II).

Among the floors in item C.II 4 of §3, the floor (P0) consists of $x \geq L^2$, $R \geq 6$ and (Tor). The first two hold by what precedes, because S_{fl} lies above $L^2\mathrm{e}$ and $R \geq 6$ is part of (G1$^\flat$); the third is a datum. The inequality $2R_3 + 1 < N$, which that section derives from (P0), follows as well: $2R_3 + 1 = 4R + 2\varrho_L + 11$, and

$$N \geq 16 Mr(x) \geq 320 c_R \log x > 320(R-1) > 4R + 2\varrho_L + 11,$$

where we use (Tor) with $s(x) = 8r(x)$, the definition of $r(x)$, the inequality $R < c_R \log x + 1$, and $R \geq 6$ together with $\varrho_L \leq R$. Since $N > 4R + 2\varrho_L + 11$ and $\varrho(x) = R$, also $\varrho_L + 2 < N/2$ and $\varrho(x)/L + 2 < N/2$ (see Step 2 (iv) of the proof of Thm. 4.5); hence every row of item C is a row as in 1.1 of §3, C.I, and also a row as in 1.2 of C.II there.

(FA) $\mathsf{K_c} L^2 \varepsilon' \leq 1$, $\varepsilon' \leq 1/40$, $\varepsilon \leq \varepsilon'$, $x^{-59} \leq \varepsilon'$ and $\mathsf{ti_c} + [3C_D^3 + (\mathsf{N_B}+6)C_D \mathsf{N}_Z] x'^{-64} \leq \varepsilon'$.

(F0$^\flat$) $\mathsf{ti}_0^\flat \leq x^{-5}$.

(F1) $12L^3\varepsilon \leq \pi/24$, $90L^3\varepsilon \leq 1$ and $2L^2 c_2^* \varepsilon \leq 1$.

(F3$^\flat$) $4 \cdot 10^5 \mathsf{N}_Z^2 \varepsilon \leq d_0$ and $3.2 \cdot 10^4 \mathsf{N}_Z^2 L \varepsilon^2 \leq 1$.

(F4$^\flat$) $\vartheta_F^\flat \leq 1/16$ and $\vartheta_n^\flat \leq a/20$.

(F5) $\mathsf{N}_Z L^{1/2} \varepsilon^{1/2} \leq 0.01$.

(Fm) $l_r \mathsf{N}_\sigma L^{-3/2}(2\varrho(x)+3)^3 \varepsilon \leq d_0/16$.

(Fc) $127 K \mathsf{g}_1 \mathsf{N}_\sigma L^{3/2} \varepsilon \leq d_0/16$.

(F8) $8400 Mr(x')\varepsilon' \leq \gamma/b_h$, $800(Mr(x'))^2 \varepsilon' \leq 1$, $\varepsilon' \leq 0.04$ and $x^{-4} \leq \varepsilon'$.

(F9$^\flat$) One has

$$10^6 v^2 b_h^3 \gamma^{-2}(1+a)(1+l_r)^2(1+La_\infty^\flat)^2(1+\mathsf{G}_0)^2(2R+4\varrho_L+5)^3 \leq x^8,$$

and $\mathrm{py} \geq 1$.

(U0$^\flat$) $60L^3\varepsilon \leq r_o \leq t_0$, $(2/9)\mathsf{N}_Z L^{1/2}\varepsilon \leq 0.09 r_o$, $4\vartheta_F^\flat \leq \check\theta_x/2$, $\theta_2 + 4\vartheta_n^\flat/a \leq \check\theta_x/2$, $6Lr_o < \pi$, $\kappa_1 \leq \gamma/2$, $\check\theta_x \leq 1/6$ and $p_0^2 x^{-1/4} \leq 1$.

(U1$^\flat$) $19L^3\log(2(b_h+1)/\kappa_1)+(c_{\mathrm{tr}}/2)L^3(2\varrho_L+5)^3\le \mathrm{py}/192$.

(Z0$^\flat$) $c_\eta^\flat\varepsilon\le r_o$, $25(c_\eta^\flat)^2\varepsilon\le 0.3$, $h_*(1+a^{1/2})[K_\Delta^\flat\varepsilon'^2+8\theta_2\mathsf{N}_Z(aL)^{-1}\varepsilon']\le c_\xi\varepsilon/4$ and $c_\xi p_0\ge 1$.

(Z1$^\flat$) One has

$$\begin{aligned}&\sigma_{\mathrm{tot}}^\flat+5K_s^\flat p_0^3x^{-1/2}\\&\quad+[6.3(\mathsf{N}_\nu^\flat)^2\vartheta_F^\flat+3\mathsf{N}_Z^2(\theta_2+L\vartheta_n^\flat)]M^3p_0'^2\\&\quad+(Q_\theta^\flat+60c_\tau v)x^{-1/4}\\&\quad+27M^3e^{-\mathrm{py}/384}+3v((c_q^\flat)^2+c_\xi^2p_0^2+2)x^{-1}+x^{-30}\\&\quad\le x^{-1/8}.\end{aligned}$$

(W) $0.05ax^{1/4}\ge d_0p_0^2/2$.

Item H: conclusions. Let $n\ge 0$. Let (J_i,t_i), $i=1,\dots,n$, be pairwise compatible admitted cell data, or, in the clauses on letters, let $\check\zeta_i$, $i=1,\dots,n$, be pairwise compatible admitted letters; $\mathfrak{b}_i$ denotes the set $\mathfrak{b}$ of (J_i,t_i), or of $\check\zeta_i$. Let J_0 be a finite set of cells off $\bigcup_i\mathfrak{b}_i^{+2r_1}$, let Y be a finite set of cells, and let $0\le\varsigma_1\le x^{-2}$ and $0\le\varsigma\le(\gamma/2)x^{-2}$. Then the following hold.

(a) For each i, the functions $\hat\lambda_i$ and $\Lambda_i=\hat\lambda_i1_{\mathrm{ind},i}$ depend only on the values of ξ at the bonds of I° under $\mathfrak{b}_i^{+2r_1}$; here the product $[A]_{\mathfrak{Z}}^\ell1_{\mathrm{ind}}$ depends on η only at bonds within the coarse distance ϱ_L+2 of $\hat{\mathfrak{Z}}$. The sets of bonds obtained in this way for the different i, and the bonds of $\hat J_0$, form pairwise disjoint sets. With $\mathsf{Y}_i:=\mathsf{Y}\cap\mathfrak{b}_i^{+2r_1}$ and $\mathsf{Y}_r:=\mathsf{Y}\setminus(\bigcup_i\mathfrak{b}_i^{+2r_1}\cup J_0)$, the following identity holds exactly, where for each i the bracket $(1_{\mathrm{ind},i})$ is a factor that is present on both sides or absent from both:

$$E_\xi[\prod_i\hat\lambda_i(1_{\mathrm{ind},i})\prod_{J_0}b_\square\,e^{(x/2)\varsigma_1|\xi_{\mathsf{Y}}|^2}]=\prod_iE_\xi[\hat\lambda_i(1_{\mathrm{ind},i})e^{(x/2)\varsigma_1|\xi_{\mathsf{Y}_i}|^2}]\cdot\prod_{\square\in J_0}E_\xi[b_\square e^{(x/2)\varsigma_1|\xi_{\mathsf{Y}\cap\{\square\}}|^2}]\cdot E_\xi[e^{(x/2)\varsigma_1|\xi_{\mathsf{Y}_r}|^2}].$$

Pointwise $|\lambda^o|\le\Lambda\prod_{\mathfrak{X}}(\mathrm{cap})$, and wherever $1_{\mathrm{ind}}=1$ the strong cap is at most the constant $Ae^{-c_\Xi p_0^2-\kappa(|X|_c-1)}$.

(a′) The function $\lambda^o_{\check\zeta}$ depends only on the values of ξ under $\mathfrak{Z}_2$ and within r_w of the bonds based in the cells of $I(\check\zeta)$. If the sets $I(\check\zeta)$ of some letters lie pairwise more than r_1 cells apart, then the functions $\lambda^o_{\check\zeta}$ of these letters are independent, and so are the products $\lambda^o_{\check\zeta}e^{(x/2)\varsigma_1|\xi_{\mathsf{Y}(\check\zeta)}|^2}$ with $\mathsf{Y}(\check\zeta)\subset I(\check\zeta)$.

(b) For every cell $\square$, $E_\xi[b_\square e^{(x/2)\varsigma_1|\xi_{\{\square\}}|^2}]\le e^{x^{-1/8}}u_b$; moreover $E_\xi[e^{(x/2)\varsigma_1|\xi_{\mathsf{Y}}|^2}]\le e^{4.8v\varsigma_1\#\mathsf{Y}}$.

(c°) In the bare case, for every admitted cell datum (J,t) and every $\mathsf{Y}\subset\mathfrak{Z}_2$,

$$E_\xi[\hat\lambda_{(J,t)}e^{(x/2)\varsigma_1|\xi_{\mathsf{Y}}|^2}]\le e^{x^{-1/8}\#\mathfrak{Z}_2}e^\circ_{(J,t)}.$$

The mean on the left contains no indicator, no polymer and no member.

(c) With the strong cap, for an admitted letter $\check\zeta$ and every $\mathsf{Y}\subset\mathfrak{Z}_2$,

$$E_\xi[|\lambda^o_{\check\zeta}|e^{(x/2)\varsigma_1|\xi_{\mathsf{Y}}|^2}]\le e^{x^{-1/8}\#\mathfrak{Z}_2}e^o_{\check\zeta}.$$

(c′) With the weak cap, the bound of (c) holds with $e^{o\prime}_{\check\zeta}$ in the place of $e^o_{\check\zeta}$. Even the mean of $\Lambda_{\check\zeta}$ satisfies

$$E_\xi[\Lambda_{\check\zeta}e^{(x/2)\varsigma_1|\xi_{\mathsf{Y}}|^2}]\le e^{x^{-1/8}\#\mathfrak{Z}_2}e^{\circ\prime}_{(J,t)}e^{-(\mathrm{py}/1152)\#\mathfrak{X}}.$$

(c″) The bounds (c) and (c′) remain true when 1_{ind} is replaced by any Borel function $0\le\mathsf{th}(\xi)\le 1$ such that, on its support, every X of $\mathfrak{X}$ holds a plaquette of $D(U)\cap\hat{\mathfrak{Z}}$. Since $\hat{\mathfrak{b}}\subset\hat{\mathfrak{Z}}$, they remain true in particular with the indicator of the same kind built with $D(U)\cap\hat{\mathfrak{b}}$ instead of $D(U)\cap\hat{\mathfrak{Z}}$.

(d°) For the cell data (J_i, t_i) above,

$$E_\xi[\prod_i \hat\lambda_i \prod_{J_0} b_\square\, e^{(x/2)\varsigma|\varphi_\Upsilon|^2}] \le \exp\{4.8v(2/\gamma)\varsigma(2r_1{+}1)^3 \#\mathsf{Y} + x^{-1/8}(\sum_i \#\mathfrak{b}_i^{+2r_1} + \#J_0)\}\, u_b^{\#J_0} \prod_i e^\circ_{(J_i,t_i)}.$$

(d) For the letters $\check\zeta_i$ above, the bound of (d°) holds with $|\lambda_i^o|$ in the place of $\hat\lambda_i$, and with $e^o_{\check\zeta_i}$ for the strong cap, or $e^{o\prime}_{\check\zeta_i}$ for the weak cap, in the place of $e^\circ_{(J_i,t_i)}$.

(d′) The bound of (d°) also holds with Λ_i in the place of $\hat\lambda_i$ and with $e^{\circ\prime}_{(J_i,t_i)} e^{-(\mathrm{py}/1152)\#\mathfrak{X}_i}$ in the place of $e^\circ_{(J_i,t_i)}$.

(e) One has $u_1 = e^{-\delta_3 p_0^2}$, $u_1' = e^{-\delta_3' p_0^2}$ and $e^{-\mathrm{py}/1152} = e^{-d_0 p_0^2/2304}$, where δ_3 and δ_3' are computed from $(d_0, L, a, K_*, \lambda_*)$.

(f) For all U and V',

$$L[\mathrm{En}_\ell(U) + a\sum_c (1-\cos \mathsf{n}_c)] \ge \mathrm{En}'(V') + (\frac12 - d_0 - \mathsf{t}_\rho - 3e_1^\flat)\varepsilon' \sum_{D'} \phi'_P + \frac12 L d_0 \varepsilon \sum_D \phi_p - e_2^\flat \varepsilon'^2 N^3.$$

(g*1) Let O be a finite set of objects of E12, consisting of classes, which form a set J, of cells off f_J, which form a set t such that $\mathfrak{b}$ is non-empty, and of polymers, which form a family $\mathfrak{X}$; any two of these polymers are farther apart than $s(x)$, and each of them is adjacent to an object with cells. The set O need not be connected. Let (H17) hold for (J,t) at V'; this is so, for instance, when (J,t) is the cell datum of a zone letter that is actual at V' (see (Act2) in Step 6 of the proof of Thm. 4.39). Then the adjacency of E12 splits O into components $\check\zeta^{(m)} = (J_m, t_m, \mathfrak{X}_m)$, which are admitted letters, any two of them compatible. Moreover $\mathfrak{Z}$ is the disjoint union of the $\mathfrak{Z}_m$, and $\mathfrak{Z}_2$ that of the $\mathfrak{Z}_{2,m}$; $1_{\mathrm{ind}} = \prod_m 1_{\mathrm{ind},m}$; $\hat\lambda_{(J,t)} 1_{\mathrm{ind}} \prod_{\mathfrak{X}} w = \prod_m \lambda^o_{\check\zeta^{(m)}}$; the factors Res and $u_t^{\#t}$ split into products over the components; and $\prod_m u_1^{1[t_m \ne \emptyset]} \le u_1^{1[t\ne\emptyset]}$. Hence (d) applies to the product $\prod_m \lambda^o_{\check\zeta^{(m)}}$.

(g*2) Let the same hypothesis hold for (J,t). Then the components (J_k, t_k) of (J,t) under the adjacency of cells alone are pairwise compatible admitted bare letters, $\hat\lambda_{(J,t)} = \prod_k \hat\lambda_{(J_k,t_k)}$, and Res and $u_t^{\#t}$ split into products over these components.

No conclusion of this theorem depends on the reality or the positivity of a member. ■□

Proof. Step 1: the free kernels and the multipliers.

Lemma 4.24 (Norms, identities and tails of the free kernels)**.** Let (ℓ, ℓ') be a pair of item C of Thm. 4.19. Then the following hold.

(i) Each of $\ell, \mathsf{T}, \mathsf{X}, \ell', \mathsf{B}', H^\sharp, \mathsf{N}^\sharp$, and each of the free kernels G, G_Z, has $\|\cdot\|_{\theta_D} \le C_D$.

(ii) $\mathsf{BT} = L^{-1}W^* + \mathrm{d}^*\mathsf{X}$, $H^\sharp = \mathsf{T}\mathsf{B}'$, $\mathsf{N}^\sharp = (aL)^{-1}\mathrm{d}'^*\mathsf{B}'$ and $\mathsf{B}'(L\mathsf{T}^*\mathsf{B}\mathsf{T} + (aL)^{-1}\mathrm{d}'\mathrm{d}'^*) = 1$.

(iii) $L^2\|\mathsf{T}^<\|_\infty \le c_1^* \mathsf{I}_L$, $L\|\mathsf{X}^<\|_\infty \le c_2^*$, $\|\mathsf{B}'^{\mathrm{c}}\|_\infty \le \mathsf{N}_Z$ and $\mathsf{t}_\rho \le \frac18(\frac12 - d_0 - c_\Xi)$.

(iv) For each of the kernels of (i), the tail that the cut at $R(x')$ removes has $\|\cdot\|_0 \le C_D x'^{-64}$; and $\|\mathsf{B}' - \mathsf{B}'^{\mathrm{c}}\|_0 \le 2C_D^2 x'^{-64}$.

■

Lemma 4.25 (Sizes of the multipliers)**.** At every pair (V', e) one has $|v_*| \le \mathsf{N}_Z\varepsilon'$, $|Z_*| \le \mathsf{N}_Z L^{-1}\varepsilon'$, $|\Xi_*| \le \mathsf{N}_\Xi L^{-1}\varepsilon'$, $|\sigma_*| \le \mathsf{N}_\sigma L^{-2}\varepsilon'$, $|\ell^{V*}\sigma_*| \le l_c \mathsf{N}_\sigma L^{-2}\varepsilon'$, $|\nu_*| \le N_1 L^{-2}\varepsilon'$ and $|\omega_*| \le 4\mathsf{N}_Z(aL)^{-1}\varepsilon'$. The components of $\mathrm{pm}(\chi v_*)$ and of $\mathrm{pm}((1-\chi)v_*)$ obey the same bounds. ■

Proof of Lem. 4.24. (i) The kernels in (i) are among those that the condition $(\mathrm{Dec})(C_D, \theta_D)$ of Def. 2.41 lists, and Thm. 2.64 gives this condition as its first conclusion.

(ii) By Def. 2.26, $\nu = \mathsf{R}Y$ satisfies $\nu + \ell^\mathsf{T}\sigma = Y$ with $\sigma = \ell\nu$; as $\mathsf{B} = 1 + \ell^\mathsf{T}\ell$, this says $\mathsf{B}\nu = Y$. By Def. 2.35, $\mathrm{pair}(v/L, \mathsf{X}v) = (\mathsf{T}v, (aL)^{-1}\mathrm{d}'^*v)$. For this pair $\nu = \mathsf{T}v$ and $Y = L^{-1}W^*v + \mathrm{d}^*\mathsf{X}v$, so the identity $\mathsf{B}\nu = Y$ yields $\mathsf{BT} = L^{-1}W^* + \mathrm{d}^*\mathsf{X}$. Def. 2.35 also

gives $H^\sharp = \mathsf{T}\mathsf{B}'$, and the identity for $\mathsf{N}^\sharp$ is its definition $\mathsf{N}^\sharp := (aL)^{-1}\mathrm{d}'^*\mathsf{B}'$ in Prop. 2.27 (a). Next, Prop. 2.27 (a), applied with $v = \mathsf{B}'X'$, gives $\|\mathrm{pair}(\mathsf{B}'X'/L, \mathsf{X}\mathsf{B}'X')\|^2 = L^{-1}\langle X', \mathsf{B}'X'\rangle$. Written with the components $\mathsf{T}v$ and $(aL)^{-1}\mathrm{d}'^*v$ of the pair, the left side is $\langle \mathsf{B}'X', (\mathsf{T}^*\mathsf{B}\mathsf{T} + a(aL)^{-2}\mathrm{d}'\mathrm{d}'^*)\mathsf{B}'X'\rangle$. Both B' and $\mathsf{T}^*\mathsf{B}\mathsf{T} + a(aL)^{-2}\mathrm{d}'\mathrm{d}'^*$ are symmetric, $\mathsf{B}' = 1 + \ell'^{\mathsf{T}}\ell'$ is invertible, and the identity holds for every X'; hence $\mathsf{B}'^{-1} = L\mathsf{T}^*\mathsf{B}\mathsf{T} + (aL)^{-1}\mathrm{d}'\mathrm{d}'^*$, because $La(aL)^{-2} = (aL)^{-1}$.

(iii) Thm. 2.60 (d), taken at the starred parameters, gives the first two bounds, since a cut can only lower absolute row sums. Thm. 2.57 (a) applies, as $L \geq L_A$ and $\ell \in \mathcal{R}(b_*, K_*, \lambda_*, \frac{1}{9})$, and gives $|\ell'_{y'}(P)| \leq K_* e^{-\lambda_* \mathrm{dist}}$; hence the absolute row sums of ℓ'^{c} are at most l_r, its absolute column sums are at most l_c, and $\|\mathsf{B}'^{\mathrm{c}}\|_\infty \leq 1 + l_c l_r = \mathsf{N}_Z$. The bound on t_ρ is the third conclusion of Thm. 2.64.

(iv) The tail bound in Def. 2.20, combined with (i), bounds the norm $\|\cdot\|_0$ of each such tail by $C_D e^{-\theta_D R(x')}$; and $e^{-\theta_D R(x')} \leq x'^{-64}$, since $R(x') \geq c_R \log x'$ and $c_R\theta_D = 64$. Finally $\mathsf{B}' - \mathsf{B}'^{\mathrm{c}} = (\ell' - \ell'^{\mathrm{c}})^{\mathsf{T}}\ell' + \ell'^{\mathrm{c}\mathsf{T}}(\ell' - \ell'^{\mathrm{c}})$. Each of the two terms is a tail of ℓ', or its transpose, times a kernel of norm at most C_D, and so $\|\mathsf{B}' - \mathsf{B}'^{\mathrm{c}}\|_0 \leq 2C_D^2 x'^{-64}$.

□

Proof of Lem. 4.25. At real V' every transport $T(y, y'')$ is an isometry; so the absolute row sums obey the same bounds as in the free model, and a mask can only lower them. Since $|\zeta'| \leq \varepsilon'$, we get $|v_*| \leq \|\mathsf{B}'^{\mathrm{c}}\|_\infty \varepsilon' \leq \mathsf{N}_Z\varepsilon'$ by Lem. 4.24 (iii), and $|Z_*| = |v_*|/L \leq \mathsf{N}_Z L^{-1}\varepsilon'$. Next, $|\Xi_*| \leq \|\mathsf{X}^<\|_\infty |v_*|$, and by Lem. 4.24 (iii) this is at most $c_2^* L^{-1}\mathsf{N}_Z\varepsilon' = \mathsf{N}_\Xi L^{-1}\varepsilon'$. As $\nu^{\mathrm{ref}} = (\mathsf{T}^<)^V v_*$, the bound on $L^2\|\mathsf{T}^<\|_\infty$ in Lem. 4.24 (iii) gives $|\nu^{\mathrm{ref}}| \leq c_1^* \mathsf{I}_L L^{-2}\mathsf{N}_Z\varepsilon'$; so $|\sigma_*| \leq l_r|\nu^{\mathrm{ref}}| \leq l_r c_1^* \mathsf{I}_L L^{-2}\mathsf{N}_Z\varepsilon' = \mathsf{N}_\sigma L^{-2}\varepsilon'$.

Since $\sum_P W(P,p) \leq 1/L$ and a fine plaquette lies in two fine cubes, $|Y_*| \leq (\mathsf{N}_Z + 2L\mathsf{N}_\Xi)L^{-2}\varepsilon'$. The absolute column sums of ℓ are at most l_c, so $|\ell^{V*}\sigma_*| \leq l_c\mathsf{N}_\sigma L^{-2}\varepsilon'$, and $|\nu_*| \leq |Y_*| + l_c|\sigma_*| \leq N_1 L^{-2}\varepsilon'$ by the definition of N_1. A coarse bond lies in four coarse plaquettes P, and this gives $|\omega_*| \leq 4\mathsf{N}_Z(aL)^{-1}\varepsilon'$. In these arguments v_* enters only through the bound $|v|_\infty \leq \mathsf{N}_Z\varepsilon'$, which χv_* and $(1-\chi)v_*$ satisfy as well; so the components of $\mathrm{pm}(\chi v_*)$ and $\mathrm{pm}((1-\chi)v_*)$ obey the same bounds. □

Step 2: the geometry of an admitted cell datum.

Lemma 4.26 (Distances of rough plaquettes, tame boxes, where objects are based, and inequalities between the radii)**.** Let (J,t) be an admitted cell datum, and let x satisfy (G1$^\flat$). Then the following hold.

(i) A plaquette of D' outside D'_J is at coarse distance $> R_6$ from $\mathfrak{Z}_2$.

(ii) Let an object be based in $\mathfrak{Z}_2$, and let it either be based off f_J or lie within $r_w + \varrho_L + 2$ of a site off f_J. Then V' is tame on the box of radius R_6 about this object.

(iii) Every plaquette of D' within $4R + 4\varrho_L + 6$ of $\mathfrak{Z}$ belongs to D'_J, and $D' \cap \mathfrak{Z} = D'_J$.

(iv) An object within $R(x') + 2\varrho_L + 3$ both of $\hat{\mathfrak{Z}}$ and of its complement is based in $\mathfrak{Z}_2 \setminus \mathfrak{b}$.

(v) Every site within $4R + 4\varrho_L + 5$ of D'_J lies in a cell of f_J; so does every object based at such a site, and every field supported on such objects is supported in f_J.

(vi) The bonds based in the rim Lr are farther than r_w both from $\hat{\mathfrak{b}}$ and from the complement of $\hat{\mathfrak{Z}}_2$.

(vii) $r_w + 2\varrho_L + 5 \leq r_1 M$ and $3r_w + \varrho_L + 1 \leq R_2 - 1$.

(viii) For the terms Rim_3, rim_0, bd_1, bd_2 and bd_3 of Steps 3 and 7 of the proof of Thm. 4.19, the following bonds are based in Lr: for each row $y \in \hat{\mathfrak{Z}} \setminus \hat{\mathfrak{Z}}'$ of Rim_3, the bonds of every plaquette p with $|y - p| \leq \varrho(x) + 1$ and those of the paths $\Gamma(y, \pi(p))$; for each row p at which k_p enters rim_0 or bd_1, all four bonds of p; for each row c at which k_n enters rim_0 or bd_1, every bond of W_c; and the bonds at which φ and η enter bd_2 or bd_3, those of bd_2 being the bonds off $\hat{\mathfrak{S}}$ within r_l cells of a row of $\mathfrak{Z}$.

(ix) $R_6 \le Mr(x')$ and $r_1 - r_l - 1 - \lceil R/M \rceil \ge 1$.

■

Proof of Lem. 4.26. We use five facts.

(A) Sites of two cells at cell distance $d \ge 1$ are at coarse distance $\ge (d-1)M+1$; and if a site is at coarse distance at most D from a cell, then the cell that contains the site is within $\lceil D/M \rceil$ cells of that cell.
(B) $r_l \ge 1$ and $r_l M \ge \varrho_L + 2$, by the definition $r_l = \lceil (\varrho_L+2)/M \rceil$.
(C) $r_1 M \ge R_2 + 3(\varrho_L+2) + 5M = 2R + 3\varrho_L + 8 + 5M$, by the definition of r_1 and (B).
(D) $s_0 M = 2Mr(x') \ge 10R + 8\varrho_L + 24$, by (G1$^\flat$).
(E) $2\varrho_L \le R$ and $R(x') \le R$. Indeed $L \ge 3$ gives $\varrho_L \le \lceil R/3 \rceil$, and $2\lceil R/3 \rceil \le 2(R+2)/3 \le R$ for $R \ge 4$, which holds as $R \ge 6$; and $R(x') \le R$ because $x' \le x$.

With these facts we prove the clauses in turn.

(i) By (H17) the cell of such a plaquette is farther than $4r_1$ from $\mathfrak{b}$, so it is at cell distance $\ge 2r_1+1$ from $\mathfrak{Z}_2 = \mathfrak{b}^{+2r_1}$. By (A) and (C) its coarse distance from $\mathfrak{Z}_2$ is then $\ge 2r_1 M + 1 \ge 4R + 6\varrho_L + 17 + 10M$, which exceeds $R_6 = 4R + 6\varrho_L + 11$.
(ii) Let $j \in J$. A site off f_j is at cell distance $\ge s_0 + 1$ from mk'_j, hence, by (A), at coarse distance $\ge s_0 M + 1$ from D'_j. An object within $r_w + \varrho_L + 2$ of such a site is therefore at coarse distance at least $s_0 M + 1 - (r_w + \varrho_L + 2) \ge 9.5R + 7\varrho_L + 23 > R_6$ from D'_j, by (D) and $r_w \le R/2$. A site off f_J is off every f_j with $j \in J$, so the box of radius R_6 about the object holds no plaquette of D'_J. The other rough plaquettes are farther than R_6 from $\mathfrak{Z}_2$ by (i), and the object is based in $\mathfrak{Z}_2$; hence V' is tame on that box.
(iii) Let a plaquette of D' lie within $4R + 4\varrho_L + 6$ of $\mathfrak{Z}$. If it were not in D'_J, then by (H17) its cell would be farther than $4r_1$ from $\mathfrak{b}$, hence at cell distance $\ge 3r_1 + 1$ from $\mathfrak{Z} = \mathfrak{b}^{+r_1}$, and so, by (A) and (C), at coarse distance $\ge 3r_1 M + 1 > 4R + 4\varrho_L + 6$ from $\mathfrak{Z}$. A plaquette of D' in $\mathfrak{Z}$ lies within $r_1 \le 4r_1$ of $\mathfrak{b}$, hence in D'_J by (H17); conversely $D'_J \subset \mathrm{mk}'_J \subset f_J \subset \mathfrak{Z}$. Thus $D' \cap \mathfrak{Z} = D'_J$.
(iv) By (E), such an object lies within $R + 2\varrho_L + 3$ of $\hat{\mathfrak{Z}}$ and of its complement. The complement of $\mathfrak{Z}$ is at cell distance $\ge r_1 + 1$ from $\mathfrak{b}$, and so, by (A) and (C), at coarse distance $\ge r_1 M + 1 > R + 2\varrho_L + 3$ from $\hat{\mathfrak{b}}$; hence the object is not based in $\mathfrak{b}$. Since $\lceil (R + 2\varrho_L + 3)/M \rceil \le r_1$ by (C), the object lies, by (A), in a cell within r_1 of $\mathfrak{Z}$, that is, in $\mathfrak{Z}_2$.
(v) By (D), $4R + 4\varrho_L + 5 \le s_0 M$. So, by (A), a site within $4R + 4\varrho_L + 5$ of $D'_j \subset \mathrm{mk}'_j$ lies in a cell within s_0 of mk'_j, that is, in f_j.
(vi) A cell of Lr is at cell distance $\ge r_1 - 2r_l - 2$ from $\mathfrak{b}$. By (A), the bonds based in it are at coarse distance $\ge (r_1 - 2r_l - 3)M + 1$ from $\hat{\mathfrak{b}}$, and by the definition of r_1 this is $\ge R_2 + (r_l+2)M + 1 > r_w$. Such a cell is also at cell distance $\ge (2r_1+1) - (r_1 + r_l + 1) = r_1 - r_l$ from the complement of $\mathfrak{Z}_2$, and $(r_1 - r_l - 1)M + 1 > r_w$.
(vii) As $r_w \le R/2$, the first inequality follows from (C). For the second, $3r_w \le 1.5R$, and $1.5R + \varrho_L + 1 \le 2R + 1 = R_2 - 1$ by (E).
(viii) For $\mathrm{Rim}_{\mathfrak{Z}}$, the summand at a site $y \in \hat{\mathfrak{Z}} \setminus \hat{\mathfrak{Z}}'$ depends on the plaquettes p with $|y - p| \le \varrho(x) + 1$ and on the paths $\Gamma(y, \pi(p))$; these are within the coarse distance $\varrho_L + 2 \le r_l M$ of y, hence, by (A) and (B), within r_l cells of y, and their bonds lie one cell further out; all of these lie in $\mathfrak{b}^{+r_1+r_l+1} \setminus \mathfrak{b}^{+r_1-2r_l-1}$. For rim_0, the bonds of the plaquettes $p \in \widehat{\mathfrak{Z}^{+r_l}} \setminus \hat{\mathfrak{Z}}'$ are based in $\mathfrak{b}^{+r_1+r_l+1} \setminus \mathfrak{b}^{+r_1-r_l-1}$. The term rim_0 also lives on the bonds $c \in \mathfrak{Z} \setminus \mathfrak{Z}'$, and since such a c is based in a cell of $\mathfrak{b}^{+r_1} \setminus \mathfrak{b}^{+r_1-r_l}$ and $W_c = B(c_-) \cup B(c_+)$ lies under that cell or a neighbouring one, every bond of W_c is based in $\mathfrak{b}^{+r_1+1} \setminus \mathfrak{b}^{+r_1-r_l-1}$, inside the set obtained above for rim_0. For bd_1, the contributions at the P and q in $\mathfrak{Z}'^{+1} \setminus \mathfrak{S}$ depend on W_P, and the bonds lie one cell further out; so these bonds are based in $\mathfrak{b}^{+r_1-r_l+3} \setminus \mathfrak{b}^{+r_1-r_l-3}$, which lies inside Lr because $r_l \ge 1$. For bd_2, the bonds off $\hat{\mathfrak{S}}$ that are within reach of a row of $\mathfrak{Z}$,

that is, within r_l cells of it, lie in $\mathfrak{b}^{+r_1+r_l} \setminus \mathfrak{S}$. For bd_3, the P and q lie in $\mathfrak{S}^{+1} \setminus \mathfrak{S}^-$, and their bonds one cell further out. All these sets lie in $\mathrm{Lr} = \mathfrak{b}^{+r_1+r_l+1} \setminus \mathfrak{b}^{+r_1-2r_l-3}$.

(ix) By (E), $R_6 = 4R + 6\varrho_L + 11 \le 5R + 4\varrho_L + 12$, and $5R + 4\varrho_L + 12 \le Mr(x')$ by (G1$^\flat$). Moreover $r_1 \ge \lceil R/M \rceil + 3r_l + 5$, as $R_2 \ge R$; hence $r_1 - r_l - 1 - \lceil R/M \rceil \ge 2r_l + 4 \ge 1$.

□

Step 3: a local lower bound for the fine energy.

Lemma 4.27 (The energy of a set of cells bounded below by a linear term, defects and squares, minus volume and rim terms)**.** Let (L1$^\flat$), (F1), (F3$^\flat$), (F4$^\flat$), (F5), (Fm), (Fc), $\rho_0 \le \varrho(x)$ and (Tor) hold. Let V' be any coarse field, let η be any element of $\mathfrak{L}_\mathfrak{g}$, and let $U = \mathsf{U}(\eta; V')$. Let $\mathfrak{Z}, \mathfrak{Z}', \mathfrak{S}$ be any sets of cells with $\mathfrak{Z}'^{+r_l} \subset \mathfrak{Z}$ and $\mathfrak{S}^{+1} \subset \mathfrak{Z}'$. Let $(\nu, \sigma, \omega, \mathsf{zf}, \Xi)$ be any fields on fine plaquettes, fine sites, coarse bonds, coarse plaquettes and fine cubes, in this order, that satisfy

$$|\nu| \le N_1 L^{-3/2}\varepsilon, \quad |\sigma| \le \mathsf{N}_\sigma L^{-3/2}\varepsilon, \quad |\mathsf{zf}| \le \mathsf{N}_Z L^{-1/2}\varepsilon, \quad |\Xi| \le \mathsf{N}_\Xi L^{-1/2}\varepsilon, \quad |\omega| \le (2/9)\mathsf{N}_Z L^{1/2}\varepsilon, \tag{Sup}$$

and write $\boldsymbol{\nu} = (\nu, \sigma, \omega)$. With

$$\mathcal{A}_\mathfrak{Z} := \sum_{\hat{\mathfrak{Z}}} S_\varepsilon(\phi_p) + \frac{1}{2}\sum_{y \in \hat{\mathfrak{Z}}} m_y |\ell_y^U \zeta|^2 + a \sum_{c \in \mathfrak{Z}} (1 - \cos \mathsf{n}_c)$$

one then has

$$\mathcal{A}_\mathfrak{Z} \ge \langle \boldsymbol{\nu}, \iota(F, n)\rangle_\mathfrak{Z} - \frac{1}{2}\|\boldsymbol{\nu}\|_\mathfrak{Z}^2 + |\langle \mathsf{zf}, \mathrm{de}\rangle_\mathfrak{S}| + |\langle \Xi, \mathrm{db}\rangle_\mathfrak{S}| + \mathrm{Sq}_\mathfrak{Z} + \mathrm{Pay}_\mathfrak{Z} - \mathrm{vol}_\mathfrak{Z} - \mathrm{Rim}_\mathfrak{Z},$$

where the terms on the right are defined as follows.

The squares are

$$\mathrm{Sq}_\mathfrak{Z} = \frac{1}{2}(1 - 4\vartheta_F^\flat) \sum_{\hat{\mathfrak{Z}} \setminus D} |F_p - \nu_p|^2 + \frac{1}{2} \sum_{y \in \hat{\mathfrak{Z}}, m_y = 1} |\ell_y^U \zeta - \sigma_y|^2 + \sum_{c \in \mathfrak{Z}, \mathsf{n}_c \le t_0} \mathsf{kc}_c |n_c - \tilde{\omega}_c|^2,$$

where $\tilde{\omega} = \omega/(1 - \theta_2)$; here $\mathsf{kc}_c = (a/2)(1 - \theta_2) - 2\vartheta_n^\flat$ off Cb and $\mathsf{kc}_c = a/4$ on Cb, and Cb is the set of the bonds of those P in $\mathfrak{S}$ that have $\Sigma_P \ge \pi/12$ or $\mathrm{na}_P \ge \pi/3$. The payments are

$$\mathrm{Pay}_\mathfrak{Z} = (a/20) \sum_{c \in \mathfrak{Z}, \mathsf{n}_c > t_0} \mathsf{n}_c^2 + \frac{1}{2} d_0 \varepsilon \sum_{D \cap \hat{\mathfrak{Z}}} \phi_p.$$

The volume term is

$$\mathrm{vol}_\mathfrak{Z} = 2\vartheta_F^\flat |\nu|_\infty^2 \, 3L^3 M^3 \#\mathfrak{Z} + (a\theta_2 + 2.1\vartheta_n^\flat) |\omega|_\infty^2 \, 3M^3 \#\mathfrak{Z},$$

and the rim term is

$$\mathrm{Rim}_\mathfrak{Z} = \sum_{y \in \hat{\mathfrak{Z}} \setminus \hat{\mathfrak{Z}}'} |\sigma_y| [1\{m_y = 1\}(|\mathsf{dl}_y| + |(\ell^V(F - \zeta))_y|) + 1\{m_y = 0\}|(\ell^V F)_y|],$$

with $\mathsf{dl}_y = \ell_y^U \zeta - (\ell^V \zeta)_y$.

For the starred fields $(\nu_*, \sigma_*, \omega_*, Z_*, \Xi_*)$, condition (Sup) holds by Lem. 4.25, by $\varepsilon' \le L^{1/2}\varepsilon$ and by $aL \ge 18$; and for these fields

$$L \, \mathrm{vol}_\mathfrak{Z} \le [6N_1^2 \vartheta_F^\flat + 3\mathsf{N}_Z^2(\theta_2 + L\vartheta_n^\flat)]\varepsilon'^2 M^3 \#\mathfrak{Z}.$$

■

Proof of Lem. 4.27. As $t_0 = \varepsilon^{1/2}$, the bound on $|\omega|$ in (Sup) together with (F5) gives $|\omega| \le 0.0023 t_0$. By (F1), $90L^3\varepsilon \le 1$, and $L \ge 3$; hence $\varepsilon \le 1/2430$ and $\theta_2 = \varepsilon/12 \le 3.5 \cdot 10^{-5}$. It follows that $1/(1 - \theta_2) \le 1.01$ and $2/(1 - \theta_2)^2 \le 2.1$.

(1) *One row.* We treat one row at a time. At a plaquette $p \notin D$ the term $S_\varepsilon(\phi_p)$ equals $\frac{1}{2}\phi_p^2$, and completing the square gives $\frac{1}{2}\phi_p^2 = \langle \nu_p, F_p\rangle - \frac{1}{2}|\nu_p|^2 + \frac{1}{2}|F_p - \nu_p|^2$. At a plaquette $p \in D$ the term

equals $d_0\varepsilon\phi_p$; since $|F_p| \le \phi_p$, we have $\langle \nu_p, F_p\rangle \le |\nu|_\infty \phi_p$, and hence $d_0\varepsilon\phi_p \ge \langle \nu_p, F_p\rangle - \frac{1}{2}|\nu_p|^2 + (d_0\varepsilon - |\nu|_\infty)\phi_p$.

At a fine site y with $m_y = 1$ we set $a = \ell^U_y \zeta = (\ell^V F)_y + \mathsf{dl}_y - (\ell^V(F-\zeta))_y$; in this case only, a denotes a vector and not the letter a of (O4). Then $\frac{1}{2}|a|^2 = \langle \sigma_y, a\rangle - \frac{1}{2}|\sigma_y|^2 + \frac{1}{2}|a-\sigma_y|^2$. At a fine site y with $m_y = 0$, the inequality $\langle \sigma_y, (\ell^V F)_y\rangle \le |\sigma_y||(\ell^V F)_y|$ gives $0 \ge \langle \sigma_y, (\ell^V F)_y\rangle - \frac{1}{2}|\sigma_y|^2 - |\sigma_y||(\ell^V F)_y|$. At a coarse bond c we define et_c by $a(1-\cos \mathsf{n}_c) = a\langle \omega_c, n_c\rangle - (a/2)|\omega_c|^2 + \mathsf{et}_c$.

We sum these relations over the rows of $\mathfrak{Z}$. The site terms of no definite sign at the sites $y \in \hat{\mathfrak{Z}} \setminus \hat{\mathfrak{Z}}'$ make up $\mathrm{Rim}_{\mathfrak{Z}}$. Those at the sites $y \in \hat{\mathfrak{Z}}'$ involve the plaquettes within $\varrho(x)+1$ of y, the paths $\Gamma(y, \pi(p))$, and their blocks and double blocks; all of these lie within the coarse distance $\varrho_L + 2 \le r_l M$ of y, hence, as $\mathfrak{Z}'^{+r_l} \subset \mathfrak{Z}$, in $\hat{\mathfrak{Z}}$ and in $\mathfrak{Z}$.

(2) *Noise.* Let c be a coarse bond. If $\mathsf{n}_c \le t_0$, note that $t^2/12 \le \varepsilon/12 = \theta_2$ for $0 \le t \le t_0$, so that $1-\cos t \ge t^2/2 - t^4/24$ gives $1-\cos t \ge (1-\theta_2)t^2/2$. With $|n_c| \le \mathsf{n}_c$ and $\tilde{\omega}_c = \omega_c/(1-\theta_2)$, completing the square in the definition of et_c then yields $\mathsf{et}_c \ge (a/2)(1-\theta_2)|n_c - \tilde{\omega}_c|^2 - a\theta_2|\omega_c|^2$. If $\mathsf{n}_c > t_0$, then $1-\cos \mathsf{n}_c \ge 2\mathsf{n}_c^2/\pi^2$ and $|\langle \omega_c, n_c\rangle| \le |\omega|\mathsf{n}_c \le (|\omega|/t_0)\mathsf{n}_c^2$; so $\mathsf{et}_c \ge a\,\mathsf{n}_c^2(2/\pi^2 - |\omega|/t_0) \ge 0.2a\,\mathsf{n}_c^2$, by the bound $|\omega| \le 0.0023 t_0$.

(3) *The sum of* $|\mathrm{de}_P|$ *over* $\mathfrak{S}$. We start from a bound on the plaquette defect at a single coarse plaquette.

Lemma 4.28 (Sizes of the plaquette defect and the cube defect)**.** Let $\eta \in \mathfrak{L}_{\mathfrak{g}}$, let $U = \mathsf{U}(\eta; V')$, and let (V', e) be any pair. Then (a) and (b) hold for every coarse plaquette P; a third clause (c), on the cube defect db_q at every fine cube q and for the same η, U and (V', e), is stated and proved in (4) below.

(a) $|\mathrm{de}_P| \le \pi + L^{-1}\Sigma_P + \mathrm{na}_P$.
(b) If $5\Sigma_P + \mathrm{na}_P < 3\pi/4$, then $|\mathrm{de}_P| \le 6(5\Sigma_P + \mathrm{na}_P)^2$.

■

Proof of Lem. 4.28 (a), (b). (a) In $\mathrm{de}_P = X'_P - (W^V F)_P - (\mathrm{d}'^V n)_P$ we have $|X'_P| \le \pi$. By Lem. 2.18, $0 \le W(P,p) \le L^{-1}$, and $W(P,p) = 0$ unless p lies in W_P; moreover $|F_p| \le \phi_p$, $|n_c| \le \mathsf{n}_c$, and the transports T are isometries. Hence $|(W^V F)_P| \le L^{-1}\Sigma_P$ and $|(\mathrm{d}'^V n)_P| \le \mathrm{na}_P$, which proves (a).

(b) We write $\mathsf{Sg} = \Sigma_P$ and $\mathrm{na} = \mathrm{na}_P$, and $P = (y; \mu\nu)$, with $y_1 = y + Le_\mu$, $y_2 = y + Le_\nu$, $y_3 = y_1 + Le_\nu$ and with the bonds $c_1 = (y, \mu)$, $c_2 = (y_1, \nu)$, $c_3 = (y_2, \mu)$, $c_4 = (y, \nu)$ of P; the letters $n_j, \bar{\tau}_j, \tau_{j,x}$ belong to c_j.

1. The gauge. We transform (U, V', e) by the u that is constant on blocks and has $u(y) = 1$, $u(y_1) = V'(c_1)$, $u(y_2) = V'(c_4)$, $u(y_3) = V'(c_1)V'(c_2)$. The combs keep the value 1, the values $V'(c_1), V'(c_2), V'(c_4)$ become 1, and $V'(c_3)$ becomes $\partial V'(P)^{-1}$. For $p \in W_P$ the value F_p becomes $T(y, y(p))F_p$, and n_c becomes $T(y, c_-)n_c$, while $X'_P, \phi_p, \mathsf{n}_c, s_c, s_B, \mathrm{ang}(e^{i\eta_b})$ keep their values. So de_P keeps its value, and in the new field $\mathrm{de}_P = X'_P - \sum_p W(P,p)F_p - (n_1 + n_2 - n_3 - n_4)$.

2. The letters. By Lem. 4.23, $\mathrm{ang}(h_{c,x}h^{-1}_{c,c_-}) \le s_c \le \mathsf{Sg} < \pi$, the last inequality by the hypothesis of (b), which also gives $\mathsf{n}_c < \pi$. Combining these gives $h_{c,x} = e^{i\tau_{c,x}}e^{-i\bar{\tau}_c}\mathsf{M}(U)(c)$ and $\mathsf{M}(U)(c) = e^{-in_c}V'(c)$.

3. One square. For $x \in B(y)$ let $x_1 = x + Le_\mu$ and $x_2 = x + Le_\nu$, and put

$$H_x := h_{c_1,x}h_{c_2,x_1}h^{-1}_{c_3,x_2}h^{-1}_{c_4,x};$$

this is the product of U along the boundary of the square S_x of $L \times L$ fine plaquettes whose least corner is x. Solving for $\partial V'(P)$ with the help of items 1 and 2, we find

$$\partial V'(P) = e^{in_2}e^{i\bar{\tau}_2}e^{-i\tau_{2,x_1}}e^{in_1}e^{i\bar{\tau}_1}e^{-i\tau_{1,x}}H_x e^{i\tau_{4,x}}e^{-i\bar{\tau}_4}e^{-in_4}e^{i\tau_{3,x_2}}e^{-i\bar{\tau}_3}e^{-in_3}.$$

4. The square as a product. In a fixed order, column by column and each column as a ladder,

$$H_x = \prod_{p \in S_x} e^{i \operatorname{Ad}_{k_p} F_p}.$$

We derive it from the ladder identity that appears in the proof of Lem. 3.27 (a). That identity is a telescoping product, valid in an arbitrary group, so it remains true when its h_j are paths. In a first application, u_j and v_j are the lowest and the highest bond of column number j, and h_j is the straight path of L bonds that starts at $x + je_\mu$ and runs along e_ν; then H_x is the ordered product of the loops along the boundaries of the columns, each conjugated by U along e_μ from x to $x + je_\mu$. A second application, inside each column, concerns the inverse of the loop of that column; now the steps go along e_ν and the rungs point in the direction μ, and the loops that appear are the $\partial U(p)^{-1}$, each conjugated by U along e_ν from $x + je_\mu$ to $\pi(p)$. Thus k_p is U along the path from x along e_μ, then along e_ν, to $\pi(p)$; and $e^{iF_p} = \partial U(p)$ because $\phi_p \le \mathsf{Sg} < \pi$.

5. Size. The sets W_{c_1} and W_{c_3} are disjoint parts of W_P, and so are W_{c_2} and W_{c_4}; hence $\sum_j s_{c_j} \le 2\mathsf{Sg}$. By Lem. 4.23, the norms of all the exponents in the formula of item 3, with H_x written as in item 4, add up to at most $\mathrm{na} + 2\sum_j s_{c_j} + \mathsf{Sg} \le 5\mathsf{Sg} + \mathrm{na} < 3\pi/4$. Therefore $\partial V'(P) \neq -1$ and $X'_P = \mathsf{L}(\partial V'(P))$, and Lem. 4.20 (c) gives

$$X'_P = (n_1 + n_2 - n_3 - n_4) + [\bar\tau_1 - \tau_{1,x} + \bar\tau_2 - \tau_{2,x_1} - \bar\tau_3 + \tau_{3,x_2} - \bar\tau_4 + \tau_{4,x}] + \sum_{p \in S_x} \operatorname{Ad}_{k_p} F_p + \mathrm{err}_x,$$

with $|\mathrm{err}_x| \le (5\mathsf{Sg} + \mathrm{na})^2$.

6. Mean over $x \in B(y)$. As x runs through $B(y)$, the points x_1 and x_2 run through $B(y_1)$ and $B(y_2)$; so the bracket has mean 0. Moreover $L^{-3}\sum_x \sum_{p \in S_x} F_p = \sum_p W(P,p) F_p$, by the formula that gives $W = Q_2$ in Def. 2.17.

7. Tails. The path of k_p runs inside blocks along at most four stretches, and it uses at most two crossing bonds: one of them belongs to c_1, the other to c_2 or to c_4, and none belongs to c_3. By Lem. 4.21 (2), applied with $\mathrm{Bx} = B$ and o the centre of B, each stretch inside a block B has $\mathrm{ang} \le s_B$, and together the stretches contribute at most $2\mathsf{Sg}$. On a crossing bond $V'(c) = 1$ now, so $\mathrm{ang}\, U_b = \mathrm{ang}(e^{i\eta_b}) \le 9s_c + \mathsf{n}_c$ by (Tc) (i) in (4′) below; the proof of that bound rests on Lems. 4.20, 4.23 and 4.21 alone. As $s_c \le \mathsf{Sg}$ for each bond c of P, we obtain $\mathrm{ang}\, k_p \le 20\mathsf{Sg} + \mathrm{na}$, and Lem. 4.20 (b) gives $\sum_{p \in S_x} \| \operatorname{Ad}_{k_p} - 1 \| \phi_p \le 2(20\mathsf{Sg} + \mathrm{na})\mathsf{Sg}$.

8. Conclusion. By items 1 and 6, averaging the identity of item 5 over $x \in B(y)$ shows that de_P is the mean over x of $\sum_{p \in S_x} (\operatorname{Ad}_{k_p} - 1) F_p + \mathrm{err}_x$. With $|F_p| \le \phi_p$ and the bounds of items 5 and 7 we obtain

$$|\mathrm{de}_P| \le (5\mathsf{Sg} + \mathrm{na})^2 + 40\mathsf{Sg}^2 + 2\mathsf{Sg}\,\mathrm{na} \le 2.6(5\mathsf{Sg} + \mathrm{na})^2 \le 6(5\mathsf{Sg} + \mathrm{na})^2.$$

□

Let $\mathrm{Pl}_1 = \{\Sigma_P \ge \pi/12\}$. For P in Pl_1, the tame part of the sum over W_P is at most $12L^3\varepsilon \le \pi/24$; hence $\Sigma^D \ge \pi/24$, and Lem. 4.28 (a), combined with the inequalities $\pi \le 24\Sigma^D$ and $\Sigma_P \le 2\Sigma^D$, yields $|\mathrm{de}_P| \le (24 + 2/L)\Sigma^D + \mathrm{na}_P \le 25\Sigma^D + \mathrm{na}_P$. Let Pl_2 be the set of those P outside Pl_1 with $\mathrm{na}_P \ge \pi/3$. On Pl_2 we combine Lem. 4.28 (a) with $\Sigma_P < \pi/12$, $L \ge 3$ and $\pi \le 3\mathrm{na}_P$ and obtain $|\mathrm{de}_P| \le \pi(1 + 1/36) + \mathrm{na}_P \le 5\mathrm{na}_P$. Every remaining P has $5\Sigma + \mathrm{na} < 3\pi/4$, because $\Sigma_P < \pi/12$ and $\mathrm{na}_P < \pi/3$; this is the hypothesis of Lem. 4.28 (b), and that clause gives $|\mathrm{de}_P| \le 6(5\Sigma + \mathrm{na})^2 \le 300\Sigma^2 + 12\mathrm{na}^2$. Here $\Sigma^2 \le (\pi/6)\Sigma^D + 24L^3 \sum_{\mathrm{tame}} \phi^2$, with the tame sum taken over W_P, and $\mathrm{na}_P^2 \le 4\sum \mathsf{n}_c^2$, with the sum taken over the four bonds c of P.

A fine plaquette lies in twelve boxes W_P, and a bond lies in four plaquettes P. Summing the three cases over $\mathfrak{S}$, we arrive at

$$\sum_{\mathfrak{S}} |\mathrm{de}_P| \le 2200 \sum_D \phi + 86400 L^3 \sum_{\text{tame}} \phi^2 + 192 \sum_{\mathsf{n}_c < \pi/3} \mathsf{n}_c^2 + 20 \sum_{\mathsf{Cb}} \mathsf{n}_c .$$

The four coefficients come from the bound $300 + 600\pi \le 2184.96 \le 2200$ and from the numbers $300 \cdot 24 \cdot 12$, $12 \cdot 4 \cdot 4$ and $5 \cdot 4$.

(4) *The sum of* $|\mathrm{db}_q|$ *over* $\mathfrak{S}$. We show that Lem. 4.28 (c) gives

$$\sum_{\mathfrak{S}} |\mathrm{db}_q| \le 40 \sum_D \phi + 110 L^3 \sum_{\text{tame}} \phi^2 + \sum_{\mathsf{n}_c < \pi/3} \mathsf{n}_c^2 + 2L^2 \varepsilon \sum_{\mathsf{n}_c \ge \pi/3} \mathsf{n}_c .$$

In this bound the rough plaquettes carry at most the factor $6\pi + 2 + 1 \le 22$, and the tame ones the factor $6 + 99L^3 + L^2 \le 110 L^3$.

The size of the cube defect is the subject of clause (c) of Lem. 4.28, which reads as follows.

(c) For every $\eta \in \mathfrak{L}_{\mathfrak{g}}$, every pair (V', e) and every fine cube q, with $U = \mathsf{U}(\eta; V')$,

$$|\mathrm{db}_q| \le \frac{1}{2} S_q^2 + \sum_{\lambda=1,2,3} \phi_{p_\lambda} \min\{2, 2(9A + \mathrm{na})\},$$

where $S_q = \sum_{p \in dq} \phi_p$ and p_λ is the face of q based at $\pi(q) + e_\lambda$; here $(A, \mathrm{na}) = (s_B, 0)$, with $B = B(y(q))$, if $\pi(q) + e_\lambda$ lies in the block of $\pi(q)$, and otherwise $(A, \mathrm{na}) = (s_c, \mathsf{n}_c)$, with $c = (y(q), \lambda)$.

Proof of Lem. 4.28 (c). The transports T are isometries and $|F_p| \le \phi_p$, so $|\mathrm{db}_q| \le S_q$; if $S_q \ge 2$, it follows that $|\mathrm{db}_q| \le S_q \le \frac{1}{2} S_q^2$.

Now let $S_q < 2$. Then $e^{iF_p} = \partial U(p)$ at each of the six faces p of q.

We apply Bianchi's identity to U at the fine cube q, with $z = \pi(q)$; as in the proof of Lem. 3.27 (d), it is an identity of invertible matrices. It states that $\prod_{k=1,\dots,6} e^{iY_k} = 1$, where $Y_k = d(q, p_k)\, \mathrm{Ad}_{\kappa_k} F_{p_k}$, with $\kappa_k = 1$ at the faces based at z and $\kappa_k = U_b$, $b = (z, \lambda)$, at the face p_λ.

By Lem. 4.20 (c′), $|\sum_k Y_k| \le \frac{1}{2} S_q^2$. Moreover

$$\mathrm{db}_q - \sum_k Y_k = \sum_\lambda d(q, p_\lambda) [T(y(q), y(p_\lambda)) - \mathrm{Ad}_{U_b}] F_{p_\lambda} .$$

If b lies in the block B, then $T = 1$ and $\mathrm{ang}\, U_b \le s_B$ (on a comb, $U_b = 1$). If b crosses c, then $T = \mathrm{Ad}_{V'(c)}$ and $U_b = e^{i\eta_b} V'(c)$, and $\mathrm{ang}(e^{i\eta_b}) \le 9 s_c + \mathsf{n}_c$.

These two angle bounds are proved in the cases (o) and (i) of the family (Tc) in (4′) below. Given them, Lem. 4.20 (b) bounds the norm of the bracket by $\min\{2, 2(9A + \mathrm{na})\}$, and clause (c) follows. □

We now sum over the cubes q of $\mathfrak{S}$. A cube has six faces and a plaquette is a face of two cubes; therefore

$$\sum_q \frac{1}{2} S_q^2 \le 3 \sum_q \sum_{dq} \phi_p^2 = 6 \sum_p \phi_p^2 \le 6\pi \sum_D \phi + 6 \sum_{\text{tame}} \phi^2 .$$

A plaquette is the face p_λ of one cube only. If p_λ is rough, its term is at most $2\phi_{p_\lambda}$; if it is tame, its term is at most $\phi_{p_\lambda}(18A + \min\{2, 2\mathrm{na}\})$.

The region of A is the set of those fine plaquettes p for which ϕ_p is a term of A: the block B if $A = s_B$, and W_c if $A = s_c$. It holds at most $6L^3$ plaquettes, and a given plaquette lies in the region of at most $3L^3 + 6L^2 \le 5L^3$ faces p_λ, namely the faces of its block and L^2 seam faces for each of six W_c.

So the part with A^D is at most $18\varepsilon\, 5L^3 \sum_D \phi \le \sum_D \phi$, by (F1), and the part with A^t is at most $18 \cdot \frac{1}{2}(6 + 5) L^3 \sum_{\text{tame}} \phi^2 = 99 L^3 \sum_{\text{tame}} \phi^2$.

In the part with n_c, each bond c has L^2 seam faces. If $\mathsf{n}_c < \pi/3$, then $2\mathsf{n}_c \sum_{\text{seam}} \phi \le \mathsf{n}_c^2 + L^2 \sum_{\text{seam}} \phi^2$. Otherwise the part is at most $2L^2\varepsilon \le 2L^2\varepsilon\, \mathsf{n}_c$, as $\pi/3 > 1$.

For P and q in $\mathfrak{S}$, everything that enters these bounds lies in $\mathfrak{S}^{+1} \subset \mathfrak{Z}$, and restricting P and q to $\mathfrak{S}$ does not increase the multiplicities counted in (3) and (4).

(4′) *The three families of site terms of no definite sign at $y \in \hat{\mathfrak{Z}}'$.*

(Ta) Where $m_y = 1$, we have $|F_p - \zeta_p| \le \phi_p 1_D$, and therefore $\sum_y |\sigma_y||(\ell^V(F-\zeta))_y| \le |\sigma|_\infty l_c \sum_D \phi$.

(Tb) Where $m_y = 0$, the rough part of $(\ell^V F)_y$ contributes at most $|\sigma|_\infty l_c \sum_D \phi$, while the tame part is at most $l_r\varepsilon$ at each site. The number of these sites obeys $\#\{m_y = 0\} \le (2\rho_0+1)^3 \#D + (2\varrho(x)+3)^3 \#\{\phi_p > 1\}$, and $\varepsilon \#D \le \sum_D \phi$, $\varepsilon \#\{\phi_p > 1\} \le \varepsilon \sum_D \phi$; so the tame part is in all at most $|\sigma|_\infty l_r[(2\rho_0+1)^3 + (2\varrho(x)+3)^3\varepsilon] \sum_D \phi$.

(Tc) We put $\text{ang}_b := \min\{1, \text{ang}(e^{i\eta_b})\}$ and $z_p := |\zeta_p| \le \min\{\phi_p, \varepsilon\}$.

Let $\Gamma = \Gamma(y, \pi(p))$. The crossings of Γ form the path $\Gamma' = \Gamma(y(y), y(p))$. We claim that, at every η,

$$\text{ang}(U(\Gamma)V'(\Gamma')^{-1}) \le \sum \text{ang}(e^{i\eta_b}),$$

the sum running over the bonds b of Γ in I°.

To see this, we use parts (1) and (2) of Step 5 of the proof of Thm. 4.5, which give

$$U(\Gamma) = [\prod_i g_i e^{\pm i\eta_{b_i}} g_i^{-1}] V'(\Gamma'),$$

with $g_i \in G$ and b_i the bonds of Γ in I°. Part (1) is a statement on the paths alone and does not involve the field, and part (2) is an identity between products of group elements. They do not depend on the size of η; they apply because $\varrho(x) + L < NL/2$ by (Tor).

Since ang is subadditive and invariant under conjugation and under passage to the inverse (Lem. 4.20 (a)), the claim follows. Lem. 4.20 (b) then gives $\| \text{Ad}_U - T\| \le \min\{2, 2\sum \text{ang}\} \le 2\sum \text{ang}_b$, and hence

$$|\mathsf{dl}_y| \le 2 \sum_p |\ell_y(p)| z_p \sum_{b \in \Gamma(y,\pi(p))} \text{ang}_b \,.$$

We now bound $\text{ang}(e^{i\eta_b})$ at every η; in the chart, $U = 1$ on the combs.

(o) Let b lie inside $B = B(y)$. Then $e^{i\eta_b} = U_b$, and U_b is the variable of the bond b when the box B is put in its comb gauge with origin y; so its angle is at most s_B, by Lem. 4.21.

(i) Let $b = (z, \lambda)$ be a crossing bond of c, and put $z' = z + Le_\lambda$ and $h_{c,z} = U([z, z'])$. Then

$$e^{i\eta_b} = U_b V'(c)^{-1} = [U_b h_{c,z}^{-1}][h_{c,z} \mathsf{M}(U)(c)^{-1}][\mathsf{M}(U)(c) V'(c)^{-1}].$$

The first bracket is conjugate to the closed path that leads from $z + e_\lambda$ back along its comb to c_+, then along the comb to z', and then straight back. It consists of $3 + 3 + 1 = 7$ straight segments in $B(c_+)$, so the angle of the first bracket is at most $7s_c$ by Lem. 4.21.

The second bracket equals $h_{c,z} h_{c,y}^{-1} e^{-i\bar{\tau}_c}$ with $y = c_-$. Its angle is at most $s_c + |\bar{\tau}_c| \le 2s_c$, because $|\tau_{c,x}| \le \text{ang}(h_{c,x} h_{c,y}^{-1}) \le s_c$ for every $x \in B(y)$, by Lem. 4.23 and the convention $\mathsf{L}(-1) = 0$.

The angle of the third bracket is n_c. Hence $\text{ang}(e^{i\eta_b}) \le 9s_c + \mathsf{n}_c$, and $\min\{1, 9s_c + \mathsf{n}_c\} \le 9s_c + \min\{1, \mathsf{n}_c\}$.

As a result, $\text{ang}_b \le s_B^D + s_B^t$ inside a block B, and $\text{ang}_b \le 9s_c^D + 9s_c^t + \min\{1, \mathsf{n}_c\}$ at a crossing bond of c.

For a bond b of direction μ and a displacement u, at most $|u_\mu|$ sites y have $b \in \Gamma(y, y+u)$. With three plaquettes to a site, this gives

$$\mu_\ell := \sup_b \sum_{(y,p): b \in \Gamma} |\ell_y(p)| \le 3K\mathsf{g}_1,$$

and likewise $\sum_y |\ell_y(p)||\Gamma| \le K \sum_u |u|_1 e^{-\lambda_*|u|_\infty} \le 3K\mathsf{g}_1$.

The part with D is at most $2\varepsilon\mu_\ell(3L^3 + 9\cdot 6L^2)\sum_D \phi \le K\mathsf{g}_1(18L^3+324L^2)\varepsilon\sum_D\phi$, by $\mu_\ell \le 3K\mathsf{g}_1$.

For the tame part we use $2z_p a \le z_p^2 + a^2$, which holds for any number $a \ge 0$; we put $a = \mathrm{ang}_b^t$, the tame term in the bound on ang_b, that is, $\mathrm{ang}_b^t := s_B^t$ at a bond b inside a block B and $\mathrm{ang}_b^t := 9s_c^t$ at a crossing bond b of c. We obtain $\sum|\ell||\Gamma|z_p^2 \le 3K\mathsf{g}_1[\sum_{\mathrm{tame}}\phi^2 + \varepsilon\sum_D\phi]$ and

$$\mu_\ell \sum_b (\mathrm{ang}_b^t)^2 \le 3K\mathsf{g}_1[3L^3\cdot 3L^3 + 81\cdot 6L^3\cdot L^2\cdot 6]\sum_{\mathrm{tame}}\phi^2.$$

For the noise at bonds with $\mathsf{n}_c < \pi/3$, the inequality $2z_p\min\{1,\mathsf{n}_c\} \le z_p^2 + \mathsf{n}_c^2$ gives once more $3K\mathsf{g}_1\sum z_p^2$, together with $3K\mathsf{g}_1L^2\mathsf{n}_c^2$ for each c.

For the noise at bonds with $\mathsf{n}_c \ge \pi/3$, where $\mathsf{n}_c > 1$, the term is at most $2\varepsilon\,\mathsf{n}_c^2$, and we obtain $6K\mathsf{g}_1L^2\varepsilon\,\mathsf{n}_c^2 \le 3K\mathsf{g}_1L^2\mathsf{n}_c^2$.

Adding all parts, we find

$$\sum_y |\mathsf{dl}_y| \le K\mathsf{g}_1(18L^3+324L^2+6)\varepsilon\sum_D\phi + 3K\mathsf{g}_1(2+9L^6+2916L^5)\sum_{\mathrm{tame}}\phi^2 + 3K\mathsf{g}_1L^2\sum_{c\in\mathfrak{Z}}\mathsf{n}_c^2,$$

and the right side is at most

$$127K\mathsf{g}_1L^3\varepsilon\sum_D\phi + 2949K\mathsf{g}_1L^6\sum_{\mathrm{tame}}\phi^2 + 3K\mathsf{g}_1L^2\sum\mathsf{n}_c^2.$$

The last step holds for $L \ge 3$, where $18L^3+324L^2+6 \le 126.23L^3$ and $2+9L^6+2916L^5 \le 981.01L^6$; moreover $2949 = 3\cdot 983$.

(5) *Collecting the bounds.* A bond of $\mathfrak{S}$ with $\mathsf{n}_c \ge \pi/3$ lies in Cb; hence the sum over these bonds in (4) runs over bonds of Cb, and we add it to the sum over Cb in (3). By (Sup), (3), (4) and (4′), the sum of the term $|\nu|_\infty\sum_{D\cap\hat{\mathfrak{Z}}}\phi$ from (1), of the site terms of (Ta), (Tb) and (Tc) at $y \in \hat{\mathfrak{Z}}'$ and of $|\langle \mathsf{zf}, \mathrm{de}\rangle_{\mathfrak{S}}| + |\langle \Xi, \mathrm{db}\rangle_{\mathfrak{S}}|$ is at most

$$c_{\mathrm{loss}}\varepsilon\sum_{D\cap\hat{\mathfrak{Z}}}\phi + \vartheta_F^\flat\sum_{\mathrm{tame}}\phi^2 + \vartheta_n^\flat\sum_{c\in\mathfrak{Z}}\mathsf{n}_c^2 + z_b\sum_{\mathsf{Cb}}\mathsf{n}_c,$$

with $(20\mathsf{N}_Z + 2L^2\varepsilon\mathsf{N}_\Xi)L^{-1/2}\varepsilon \le z_b$ by (F1), whose third clause, together with $\mathsf{N}_\Xi = c_2^*\mathsf{N}_Z$ from (O3), gives $2L^2\varepsilon\mathsf{N}_\Xi \le \mathsf{N}_Z$, and with

$$c_{\mathrm{loss}} = [\mathsf{N}_\nu^\flat L^{-1} + 2200\mathsf{N}_Z + 40\mathsf{N}_\Xi]L^{-1/2} + l_r\mathsf{N}_\sigma L^{-3/2}(2\varrho(x)+3)^3\varepsilon + 127K\mathsf{g}_1\mathsf{N}_\sigma L^{3/2}\varepsilon \le (\frac{3}{16}+\frac{1}{16}+\frac{1}{16})d_0.$$

The three terms of c_{loss} are at most $\frac{3}{16}d_0$, $d_0/16$ and $d_0/16$, by (L1$^\flat$), (Fm) and (Fc) in this order. Here $\mathsf{N}_\nu^\flat$ collects N_1, which comes from $|\nu|_\infty$, and $(2l_c + (2\rho_0+1)^3 l_r)\mathsf{N}_\sigma$.

On the tame plaquettes, $\phi_p^2 \le 2|F_p - \nu_p|^2 + 2|\nu_p|^2$. On the bonds with $\mathsf{n}_c \le t_0$, $\mathsf{n}_c^2 \le 2|n_c - \tilde\omega_c|^2 + 2.1|\omega_c|^2$. On the bonds $c \in \mathsf{Cb}$ with $\mathsf{n}_c \le t_0$ we have in addition $z_b\mathsf{n}_c \le (a/8)|n_c - \tilde\omega_c|^2 + 2z_b^2/a + 1.01z_b|\omega_c|$. This follows from $z_b\mathsf{n}_c \le z_b|n_c - \tilde\omega_c| + z_b|\tilde\omega_c|$, which uses $|n_c| = \mathsf{n}_c$, valid when $\mathsf{n}_c \le t_0$ because $t_0 = \varepsilon^{1/2} < \pi$, and from $|\tilde\omega_c| = |\omega_c|/(1-\theta_2) \le 1.01|\omega_c|$, as at the start of the proof. At such a bond, the coefficient of $|n_c - \tilde\omega_c|^2$ is $(a/2)(1-\theta_2)$ in (2), and what remains of it after we subtract $2\vartheta_n^\flat$ and $a/8$ is $(a/2)(1-\theta_2) - 2\vartheta_n^\flat - a/8 \ge a/4$, by (F4$^\flat$) and $\theta_2 \le 3.5\cdot 10^{-5}$. On the bonds with $\mathsf{n}_c > t_0$, $\vartheta_n^\flat\mathsf{n}_c^2 + z_b\mathsf{n}_c \le 0.05a\,\mathsf{n}_c^2 + 0.05a\,\mathsf{n}_c^2 + 5z_b^2/a$. The cost of a bond of Cb consists of the terms without n_c on the right side of the bound above on $z_b\mathsf{n}_c$ that applies to it: of $2z_b^2/a + 1.01z_b|\omega_c|$ if $\mathsf{n}_c \le t_0$, and of $5z_b^2/a$ if $\mathsf{n}_c > t_0$. Since $z_b = 21\mathsf{N}_ZL^{-1/2}\varepsilon$ by item F of §4.3, and $aL \ge 18$, we have $5z_b^2/a \le (5\cdot 441/18)\mathsf{N}_Z^2\varepsilon^2 = 122.5\mathsf{N}_Z^2\varepsilon^2$ and $2z_b^2/a \le 49\mathsf{N}_Z^2\varepsilon^2$, and (Sup) gives $1.01z_b|\omega_c| \le 4.72\mathsf{N}_Z^2\varepsilon^2$. So in both cases the cost is at most $123\mathsf{N}_Z^2\varepsilon^2$.

The bonds of Pl_1 number at most $4\#\mathrm{Pl}_1 \le (4\cdot 12\cdot 24/\pi)\sum_D\phi \le 366.694\sum_D\phi$. Their cost is thus at most $45104\mathsf{N}_Z^2\varepsilon\cdot\varepsilon\sum_D\phi \le \frac{1}{8}d_0\varepsilon\sum_D\phi$, by $8\cdot 45104 \le 4\cdot 10^5$ and (F3$^\flat$). For $P \in \mathrm{Pl}_2$ we

have $\mathrm{na}_P \geq \pi/3$, and na_P is the sum of n_c over the four bonds c of P; hence $\mathsf{n}_c \geq \pi/12$ at one of these bonds, and $\pi/12 > t_0$. At that bond, $\mathsf{et}_c \geq 0.2a\,\mathsf{n}_c^2$ by (2); when we subtract from it the terms $0.05a\,\mathsf{n}_c^2 + 0.05a\,\mathsf{n}_c^2$ of the bound above on the bonds with $\mathsf{n}_c > t_0$, the term $0.1a\,\mathsf{n}_c^2$ is left, and $0.1a\,\mathsf{n}_c^2 \geq 0.1233/L$ as $aL \geq 18$. From this term we subtract the costs of the bonds of every $P \in \mathrm{Pl}_2$ of which c is a bond. A bond lies in four plaquettes P, and each of them has four bonds, so these are at most sixteen bonds, and as $1968 = 16 \cdot 123$, their cost is at most $1968\mathsf{N}_Z^2\varepsilon^2 \leq 0.0615/L$ by (F3$^\flat$). This cost is at most $0.05a\,\mathsf{n}_c^2$, because $0.05a\,\mathsf{n}_c^2 \geq 0.06168/L$, and so the term $0.05a\,\mathsf{n}_c^2$ is left at that bond. The coefficient of $\varepsilon\sum_{D\cap\hat{\mathfrak{Z}}}\phi$ that is left, once the three terms of c_{loss} and the $\frac{1}{8}d_0$ for the bonds of Pl_1 are subtracted from the d_0 of (1), is at least $d_0(1-\frac{3}{16}-\frac{1}{16}-\frac{1}{16}-\frac{1}{8}) = \frac{9}{16}d_0 \geq \frac{1}{2}d_0$, and $\frac{1}{2}d_0$ is its coefficient in $\mathrm{Pay}_{\mathfrak{Z}}$. At the sites with $m_y = 1$, the squares $\frac{1}{2}|\ell_y^U\zeta - \sigma_y|^2$ enter $\mathrm{Sq}_{\mathfrak{Z}}$ with the full coefficient $\frac{1}{2}$; none of the bounds in (3) to (5) uses any part of them.

For the starred fields, the contributions are $L\cdot 2\cdot 3L^3N_{\mathrm{I}}^2L^{-4} = 6N_{\mathrm{I}}^2$ and

$$L\cdot 3\cdot 16\mathsf{N}_Z^2(aL)^{-2}(a\theta_2 + 2.1\vartheta_n^\flat) \leq 48\mathsf{N}_Z^2(\theta_2/18 + 2.1L\vartheta_n^\flat/324) \leq 3\mathsf{N}_Z^2(\theta_2 + L\vartheta_n^\flat).$$

□

Step 4: tame windows at real fields.

Lemma 4.29 (Tame windows at real fields: centre, remainder terms and prefactor)**.** In the setting of Thm. 4.19, the following hold.

(a) At every (V', e), $|a_{0,b}| \leq a_\infty^\flat \sup|X'| \leq \pi a_\infty^\flat$; the matrix $u^T = SC$ has range at most $R + 2\varrho_L + 2$; and $|(u^T\mathsf{g}^V X')_b| \leq x^{-50}$.

(b) If V' is tame on the box of radius R_5 about i, then $|\bar{F}_p - \nu_p^{\mathrm{ref}}| \leq K_D\varepsilon'^2$, $|r_{*p}| \leq K_r\varepsilon'^2$, $|\Delta_p| \leq (K_D + K_r)\varepsilon'^2$, $|\Delta_c| \leq K_D\varepsilon'^2$, $|a_{0,b}| \leq \mathsf{a}_{\mathrm{c}}\varepsilon'$ and $|\bar{F}_p| \leq (\mathsf{N}_\sharp + 2)L^{-2}\varepsilon'$. If the first of these bounds, $|\bar{F}_p - \nu_p^{\mathrm{ref}}| \leq K_D\varepsilon'^2$, holds at all p within $\varrho(x)$ of y, then $|\Delta_y| \leq l_r K_D\varepsilon'^2$. Consequently, if V' is tame on the box of radius R_6 about i, then $|\Delta_i| \leq K_\Delta^\flat\varepsilon'^2$ for rows i of all three kinds.

(c) If all P within R_6 of $\square$ are tame, then $|T_\square| \leq K_{\mathrm{q}}^\flat p_0'^2\varepsilon'$.

■

Proof of Lem. 4.29. We begin with (a). Since $\|C\|_\infty \leq \|G\|_\infty\|G^{\mathsf{T}}\|_\infty \leq \mathsf{G}_0^2$ and $|\mathsf{g}^V X'| \leq (b_h/4)\sup|X'|$, the centre $a_0 = C\mathsf{g}^V X'$ obeys the first bound of (a). Next, $u^T = 1 - qC = SC$, and $1 - qC$ has range at most $2r_w + r_q \leq R + 2\varrho_L + 2$. A row of u^T contains at most $3L^3(2R + 4\varrho_L + 5)^3$ blocks, each of norm at most $\|S\|\|C\| \leq (2/\gamma)x^{-60}$; hence

$$|(u^T\mathsf{g}^V X')_b| \leq 3L^3(2R + 4\varrho_L + 5)^3(2/\gamma)(b_h/4)\pi x^{-60} \leq x^{-52}$$

by (F9$^\flat$).

For (b) we work in the comb gauge at $o = y(i)$. Let $\check{\gamma}(y) = V'(\Gamma(o, y))$ and $\tilde{X}_P := \mathrm{Ad}_{\check{\gamma}(y_P)}X'_P = T(o, y_P)X'_P$; then $|\tilde{X}_P| = |X'_P| \leq \varepsilon'$.

The radius ρ of the gauge enters, at real fields, only through three facts:

(i) the embedding of the box of radius ρ about o in the torus, which Lem. 4.21 requires;

(ii) the bound $\|T^{\check{\gamma}}(y, y'')^{\pm 1} - 1\| \leq 130rt\varepsilon' \leq 0.2$ for y, y'' within $t \leq \rho$ of o at distance r, so that $r \leq 2t$, where $T^{\check{\gamma}}$ are the transports of $V'^{\check{\gamma}}(c) = \check{\gamma}(c_-)V'(c)\check{\gamma}(c_+)^{-1}$;

(iii) in Lem. 4.30 below, the bound ρ on the sum of the ranges.

We take $\rho = R_5$; by the hypothesis of (b), V' is tame on the box of this radius about i, and $R_5 \leq Mr(x')$ by (G1$^\flat$). At this radius the three facts hold, as follows.

(i) $2R_5 + 1 \leq 2Mr(x') + 1 < 16Mr(x) \leq N$, by (Tor).

(ii) If both ends of a bond lie within t of o, Lem. 4.21 with origin o, applied to two ladders of at most t plaquettes, gives $\mathrm{ang}\,V'^{\check{\gamma}}(c) \leq 2t\varepsilon'$; so $\|\mathrm{Ad}_{V'^{\check{\gamma}}(c)}^{\pm 1} - 1\| \leq 4t\varepsilon'$ by Lem. 4.20 (b).

The path $\Gamma(y,y'')$ has at most $3r$ bonds, all within t of o; hence the left side of the bound in (ii) is at most $(1+4t\varepsilon')^{3r}-1 \le e^{12rt\varepsilon'}-1$. By the second clause of (F8), $12rt\varepsilon' \le 24(Mr(x'))^2\varepsilon' \le 0.03$, so the left side is at most $12.2rt\varepsilon' \le 130rt\varepsilon'$ and at most $0.031 \le 0.2$.

(iii) The sums of the ranges that occur below are at most $4R+2\varrho_L+3 \le R_5 \le Mr(x')$, by (G1$^\flat$).

Moreover $T^{\check\gamma}(o,y)=1$, and every dressed kernel satisfies $K^{V^{\check\gamma}}(i,j) = \mathrm{Ad}_{\check\gamma(y(i))}\, K^V(i,j)\, \mathrm{Ad}^{-1}_{\check\gamma(y(j))}$. Indeed, products along paths telescope, $V'^{\check\gamma}(\pi) = \check\gamma(\pi_-)V'(\pi)\check\gamma(\pi_+)^{-1}$, by the argument in the proof of Lem. 3.3 (d). At $\pi=\Gamma(o,y)$ this equals $\check\gamma(o)\check\gamma(y)\check\gamma(y)^{-1}=1$, because $\check\gamma(o)=1$; at $\pi=\Gamma(y(i),y(j))$ we take Ad and multiply by $K(i,j)$.

The norms are $\|\cdot\|_{\theta_D}$. We have $\|\mathsf T^<\|, \|\mathsf X^<\|, \|\ell\|, \|\ell'^{\mathrm c}\| \le C_D$; further $\|L^{-1}W^*\| \le \mathrm e L$, because $\sum_p W(P,p) = L^2$, $\sum_P W(P,p) = L^{-1}$ and the range is 1; and $\|\mathrm d^*\| \le 6\mathrm e$, $\|\mathrm d'^*\| \le 4\mathrm e$.

For each coarse plaquette P and each coarse site y' occurring in the quantities bounded in (b), we have $\zeta'_P = X'_P$ and $m'_{y'}=1$, so that $v_* = (1+\ell'^{\mathrm cV*}\ell'^{\mathrm cV})X'$.

Item 1. By Thm. 3.32 (b), $|\bar F_p - (H^\natural \tilde X)_p| = |\Lambda_p| \le (\mathsf K_{\mathrm c}^0+1)\varepsilon'^2$, since $\tau_{\mathrm c} \le x^{-59} \le \varepsilon'$.

Item 2. By Lem. 3.37 (b) and Lem. 4.24, $\|H^\natural - \mathsf T^<\mathsf B'^{\mathrm c}\|_\infty \le \mathsf{ti}_{\mathrm c} + (C_D^2+2C_D^3)x'^{-64} \le \varepsilon'$, where the last inequality is (FA).

Item 3. By Lem. 4.30, which we state and prove now, replacing a dressed product of n kernels by the free one costs at most $1040\theta_D^{-2}\varepsilon' n \prod 1.2\|K_k\|$, times $\sup|\tilde X|$.

Lemma 4.30 (Dressed and free products of kernels in the comb gauge)**.** In the setting of Thm. 4.19 and in the comb gauge at o of radius ρ as above, let $K_1,\dots,K_n$ be kernels of the free model, each of finite range, and let the sum of their ranges be at most ρ. We assume that the entries of these kernels are scalars, or vectors of $\mathsf H_y$ or $\mathsf H'_{y'}$, and that the vectors occur in the pairs $\ell^*\ell$, $\ell'^{\mathrm c*}\ell'^{\mathrm c}$. An entry contributes its modulus $|K_k(i,j)|$ to the norms $\|K_k\|_\theta$. Let A_k be K_k or its adjoint, dressed with $V'^{\check\gamma}$, put $A_k^0 = K_k\otimes 1$ in the first case and $A_k^0 = K_k^*\otimes 1$ in the second, let $y(i_0)=o$, and let $0<\theta\le 1$. Then $\sum_j \|(A_1\cdots A_n - A_1^0\cdots A_n^0)(i_0,j)\| \le 130\varepsilon' n(8/\theta^2)\prod_k 1.2\|K_k\|_\theta$. ∎

Proof of Lem. 4.30. We write $\prod A - \prod A^0 = \sum_k A_1\cdots A_{k-1}(A_k-A_k^0)A_{k+1}^0\cdots A_n^0$ and expand each term along chains of indices. For a chain $i_0,\dots,i_n$ of length $\mathsf{le} = \sum_k \mathrm{dist}(i_{k-1},i_k) \le \rho$, each $y(i_k)$ is within le of o. By (ii), therefore, $\|A_k(i,j)\| \le 1.2|K_k(i,j)|$ and $\|(A_k-A_k^0)(i_{k-1},i_k)\| \le |K_k(i_{k-1},i_k)|130\mathsf{le}^2\varepsilon'$. From $u^2 \le 2e^u$ we get $\mathsf{le}^2 \le (8/\theta^2)e^{\theta\mathsf{le}/2}$, and the sum over the chains of $\prod|K_k|e^{\theta\mathsf{le}/2}$ is bounded by $\prod\|K_k\|_\theta$. Here, whenever A_k is the dressed adjoint, $|K_k(i,j)|$ denotes, in the two bounds derived from (ii) and in this sum, the modulus of the entry $K_k^*(i,j)$ of the adjoint kernel. Since $\|\cdot\|_\theta$ is the larger of the weighted row and column sums (Def. 2.20), we have $\|K_k^*\|_\theta = \|K_k\|_\theta$. The n terms of the sum over k give the factor n. □

We return to item 3. By covariance and $\check\gamma(o)=1$, we have $(K^V X')_{i_0} = (K^{V^{\check\gamma}}\tilde X)_{i_0}$. For $(\mathsf T^<)^V$ and $(\mathsf T^<\ell'^{\mathrm c*}\ell'^{\mathrm c})^V$ the lemma gives $1040\theta_D^{-2}(1.2C_D+5.184C_D^3)\varepsilon'^2 \le 5470\theta_D^{-2}C_D^3\varepsilon'^2$, for $C_D\ge 4$. Adding the bounds of items 1 to 3 gives the constant K_D.

Item 4. We have $r_* = [(1+\ell^*\ell)\mathsf T^< - L^{-1}W^* - \mathrm d^*\mathsf X^<]^V(1+\ell'^{\mathrm c*}\ell'^{\mathrm c})^V X'$. For the products that occur, the numbers $n\prod 1.2\|K_k\|$ are $1.2C_D$ for $\mathsf T^<$, $3\cdot 1.728C_D^3$ for $\ell^*\ell\mathsf T^<$, $3\cdot 1.728C_D^3$ for $\mathsf T^<\ell'^{\mathrm c*}\ell'^{\mathrm c}$, $5\cdot 2.48832C_D^5$ for $\ell^*\ell\mathsf T^<\ell'^{\mathrm c*}\ell'^{\mathrm c}$, $2\cdot 1.44\cdot 6\mathrm e C_D$ for $\mathrm d^*\mathsf X^<$, $4\cdot 2.0736\cdot 6\mathrm e C_D^3$ for $\mathrm d^*\mathsf X^<\ell'^{\mathrm c*}\ell'^{\mathrm c}$, $1.2\mathrm e L$ for $L^{-1}W^*$, and $3\cdot 1.728\mathrm e LC_D^2$ for $L^{-1}W^*\ell'^{\mathrm c*}\ell'^{\mathrm c}$.

Their sum is $r(C_D)C_D^5 + \mathrm e(1.2C_D^{-2}+5.184)LC_D^2$, where

$$r(C) = [1.2C + 10.368C^3 + 12.4416C^5 + 6\mathrm e(2.88C + 8.2944C^3)]C^{-5}$$

is a decreasing function with $r(4) \le 21.733$, and where $\mathrm e(1.2/16+5.184) \le 14.296$. Multiplied by 1040, the sum is at most $22603C_D^5 + 14868LC_D^2 \le 23000(C_D^5+LC_D^2)$.

By Lem. 4.24 (ii), the free product is $[\mathsf B(\mathsf T^<-\mathsf T) - \mathrm d^*(\mathsf X^<-\mathsf X)]\mathsf B'^{\mathrm c}$. Its sup norm is at most $(\mathsf N_{\mathsf B}+2)C_D x'^{-64}\mathsf N_Z$, and multiplied by ε' it is at most ε'^2, by (FA). This yields K_r. Every coarse

plaquette P on which r_{*p} depends, by way of X'_P, ζ'_P or m', lies within $2(\varrho_L{+}1){+}3R(x'){+}R{+}1 \le R_5$ of i, with i as in the hypothesis of (b).

Item 5. The relation $\nu_* = \nu^{\mathrm{ref}} - r_*$ gives the bound on $|\Delta_p|$. Next, $\Delta_c = (Q_1 a_0)_c - \omega_{*c}$, and Thm. 3.32 (b) for Λ_c gives

$$|(Q_1 a_0)_c - \sum_P \mathsf{N}^\natural(c,P) T(c_-, y_P) X'_P| \le (\mathsf{K}_{\mathrm{c}}^0 + 1)\varepsilon'^2.$$

Further, $\|\mathsf{N}^\natural - \mathsf{N}^\sharp\|_\infty \le \mathsf{ti}_{\mathrm{c}}$, and $\|\mathsf{N}^\sharp - (aL)^{-1}\mathrm{d}'^* \mathsf{B}'^{\mathrm{c}}\|_0 \le (4/18) 2 C_D^2 x'^{-64}$; multiplied by ε', the two are together at most ε'^2, by (FA). Lem. 4.30 for $(aL)^{-1}\mathrm{d}'^{V*}(1{+}\ell'^{\mathrm{c}V*}\ell'^{\mathrm{c}V})$ gives $1040\theta_D^{-2}(4\mathrm{e}/18)(1.2{+}5.184C_D^2)\varepsilon'^2 \le 3304\theta_D^{-2}C_D^2\varepsilon'^2 \le 6700\theta_D^{-2}C_D^3\varepsilon'^2$. The sum of these bounds is at most $K_D\varepsilon'^2$.

The site term $\Delta_y = (\ell^V(\bar{F}_F - \nu^{\mathrm{ref}}))_y$ is bounded by means of the row sums of ℓ^V. The bound on $|a_{0,b}|$ is given by Thm. 3.32 (c). Finally, $|\bar{F}_p| \le \|H^\natural\|_\infty \varepsilon' + \mathsf{K}_{\mathrm{c}}\varepsilon'^2 \le (\mathsf{N}_\sharp + 1)L^{-2}\varepsilon' + L^{-2}\varepsilon'$, by Lem. 3.37 (c) and (FA). The site rows use the p within $\varrho_L + 1$, and $R_5 + \varrho_L + 1 \le R_6$.

We turn to (c). At a tame P with $m' = 1$ on its reach, $(x'/2)\langle X', v_*\rangle_P - x'\mathrm{En}'_P = 0$. Moreover

$$|\Delta_Y| = |\mathsf{B}^V(\bar{F}_F - \nu^{\mathrm{ref}}) + r_*| \le (\mathsf{N}_{\mathsf{B}} K_D + K_r)\varepsilon'^2,$$

where the windows at the p' within $2(\varrho_L + 1)$ enter; this accounts for the radius R_6. The operator E^* sums L edge plaquettes, a cell holds $3M^3$ plaquettes P, and $x\varepsilon'^2 = Lp_0'^2$; so this part contributes $1.5M^3L^2(\mathsf{N}_{\mathsf{B}}K_D + K_r)p_0'^2\varepsilon'$.

For the corner cubes we use Bianchi's identity in the real case, as in the proof of Lem. 3.27 (d) and in Step 5 of the proof of Thm. 4.2: $\prod_{k=1,\dots,6} e^{iY_k} = 1$ with $Y_k = \pm \mathrm{Ad}_{\kappa_k} X'_{P_k}$. Here $\kappa_k = 1$ at the faces based at $z = \pi(q')$ and $\kappa_k = V'(z,\lambda)$ at the face based at $z + Le_\lambda$, and the signs are $\mathrm{d}'(q', P_k)$.

Thus $\sum_k Y_k = (\mathrm{d}'^V X')_{q'}$ with $|Y_k| \le \varepsilon'$, and Lem. 4.20 (c′), applied with $S \le 6\varepsilon'$, gives $|\sum_k Y_k| \le \frac{1}{2}S^2$, which is at most $18\varepsilon'^2$. A q' has one corner cube, so this part contributes $M^3(x/2)18\varepsilon'^2\mathsf{N}_\Xi L^{-1}\varepsilon' = 9M^3\mathsf{N}_\Xi p_0'^2\varepsilon'$.

The last sum is at most $(x/2)3v\mathsf{a}_{\mathrm{c}}\varepsilon' x^{-50}$, and $v\mathsf{a}_{\mathrm{c}} \le 4.42 v\gamma^{-1/2} b_h \mathsf{G}_0 \le x^8$ by (F9$^\flat$). As $\varepsilon \le \varepsilon'$, $c_\xi p_0 \ge 1$, $c_\xi \le 1/6.5$ and $x \ge L^2$, we have $p_0'^2 \ge p_0^2/L \ge 42x^{-1/2}$; hence the last sum is at most $1.5x^{-41}\varepsilon' \le M^3 p_0'^2 \varepsilon'$. □

Step 5: a lower bound for $\mathcal{G}_{\mathfrak{Z}}$.

Lemma 4.31 (Lower bound for $\mathcal{G}_{\mathfrak{Z}}$ in terms of the coarse energy and the rough coarse plaquettes)**.** In the setting of Thm. 4.19, let (J,t) be an admitted cell datum. Then

$$\mathcal{G}_{\mathfrak{Z}} \ge \sum_{P\in\mathfrak{Z}} \mathrm{En}'_P + (\frac{1}{2} - d_0 - \mathsf{t}_\rho - 3e_1^\flat)\varepsilon' \sum_{D'_J} \phi'_P - (9e_1^\flat + \frac{9}{2}\mathsf{N}_Z^2 c_B \mathsf{ti}_0^\flat)\varepsilon'^2 M^3 \#\mathfrak{Z}.$$

■

Proof of Lem. 4.31. In this step alone, ℓ stands for $\ell'^{\mathrm{c}V}$, except in $\nu^<$, in $\sigma^<$ and in the uncut ν and σ of (4) below, where ℓ has the fine rows. We put $B_1^V := 1 + \ell^*\ell$, $\mathsf{Hm} := B_1^V \hat{R}^V - 1$ and $\hat{R}^< := L(\mathrm{pm}^<)^*(\mathrm{pm}^<)$, where $\mathrm{pm}^<$ is the pair map at $V' = 1$; its three components are $\nu^< = L^{-1}W^* + \mathrm{d}^*\mathsf{X}^< - \ell^*\ell\mathsf{T}^<$, $\sigma^< = \ell\mathsf{T}^<$ and $(aL)^{-1}\mathrm{d}'^*$, and in the first two of them ℓ has the fine rows.

The operator $\hat{R}^V$ is symmetric, since $\hat{R}^V = L\,\mathrm{pm}^*\mathrm{pm}$. The ranges are at most $R(x') + 2\varrho_L + 2$ for pm, $2R(x') + 4\varrho_L + 4$ for $\hat{R}^V$, and $4R(x') + 4\varrho_L + 4$ for Hm.

(1) *The quadratic term.* Let $\mathsf{ww} := (1 - m')\ell\zeta'$ and $\mathsf{uu} := \ell^*\mathsf{ww}$. Then $v_* = B_1^V\zeta' - \mathsf{uu}$, and $\hat{R}^V v_* = \zeta' + \mathsf{Hm}^*\zeta' - \hat{R}^V\mathsf{uu}$.

A site with $m' = 0$ lies within $R(x') + 1$ of D'. Hence uu lives within $R(x')$ of such sites, and $\hat{R}^V\mathsf{uu}$ within $4R(x') + 4\varrho_L + 5$ of D'.

We split $\mathsf{uu} = \mathsf{uu}^J + \mathsf{uu}^o$ as follows: $\mathsf{ww}^J := \mathsf{ww}$ at those sites with $m' = 0$ that lie within $R(x')+1$ of D'_J, and 0 at every other site; $\mathsf{ww}^o := \mathsf{ww} - \mathsf{ww}^J$; $\mathsf{uu}^J := \ell^*\mathsf{ww}^J$ and $\mathsf{uu}^o := \ell^*\mathsf{ww}^o$. A site at which ww^o does not vanish has $m' = 0$ and is not within $R(x') + 1$ of D'_J. Since every site with $m' = 0$ is within $R(x') + 1$ of D', such a site is within $R(x') + 1$ of $D' \setminus D'_J$. No site lies within $R(x') + 1$ of D'_J and of $D' \setminus D'_J$ at once: a site within $R(x') + 1$ of both D'_J and $D' \setminus D'_J$ would place the latter within $2R(x') + 2$ of $\mathfrak{Z}_2$, in contradiction with Lem. 4.26 (i). By Lem. 4.26 (iii) and (v), $\hat{R}^V \mathsf{uu}^o = 0$ on $\mathfrak{Z}$, while uu^J and $\hat{R}^V \mathsf{uu}^J$ live in $f_J \subset \mathfrak{Z}$.

It follows that

$$\langle v_*, \hat{R}^V \mathsf{uu}\rangle_{\mathfrak{Z}} = \langle \hat{R}^V v_*, \mathsf{uu}^J\rangle = \langle \zeta' + \mathsf{Hm}^*\zeta' - \hat{R}^V \mathsf{uu}^J, \mathsf{uu}^J\rangle,$$

and therefore

$$\langle v_*, \hat{R}^V v_*\rangle_{\mathfrak{Z}} = \langle v_*, \zeta'\rangle_{\mathfrak{Z}} - \langle \zeta', \mathsf{uu}^J\rangle + \langle \mathsf{uu}^J, \hat{R}^V \mathsf{uu}^J\rangle + \langle v_*, \mathsf{Hm}^*\zeta'\rangle_{\mathfrak{Z}} - \langle \mathsf{Hm}^*\zeta', \mathsf{uu}^J\rangle.$$

Now $\hat{R}^V = (B_1^V)^{-1}(1 + \mathsf{Hm})$, and $\langle \ell^*\mathsf{ww}^J, (1 + \ell^*\ell)^{-1}\ell^*\mathsf{ww}^J\rangle \le |\mathsf{ww}^J|^2 = \langle \ell\zeta', \mathsf{ww}^J\rangle = \langle \zeta', \mathsf{uu}^J\rangle$, since $(1 - m')^2 = 1 - m'$. Hence $\langle v_*, \hat{R}^V v_*\rangle_{\mathfrak{Z}} \le \langle v_*, \zeta'\rangle_{\mathfrak{Z}} + \mathrm{Hol}$, with the number

$$\mathrm{Hol} = |\langle v_*, \mathsf{Hm}^*\zeta'\rangle_{\mathfrak{Z}}| + |\langle \mathsf{Hm}^*\zeta', \mathsf{uu}^J\rangle| + |\langle (B_1^V)^{-1}\mathsf{uu}^J, \mathsf{Hm}\,\mathsf{uu}^J\rangle|.$$

(2) *The inverse of B_1^V, and a bound for* Hol *through the row and column sums of* Hm. The fields v_* and uu obey $|v_*|,\ |\mathsf{uu}| \le \mathsf{N}_Z\varepsilon'$. We show that $\|(B_1^V)^{-1}\|_\infty \le c_B$. To invert B_1^V we apply Lem. 2.25 (a) with $\gamma = 1$, in the form of Lem. 4.32, which we state first.

Lemma 4.32 (Two bounds for symmetric matrices of 3×3 blocks)**.** The assertions of Lem. 2.25 (a) and (b) remain valid for symmetric matrices whose entries are real 3×3 blocks, where $|\cdot|$ now denotes the operator norm of a block. ■

Proof. The proofs of Lem. 2.25 (a) and (b), given after that lemma in §2.2, rest on Schur's test and on eigenvalues alone. Both arguments go through unchanged for symmetric matrices with real 3×3 blocks as entries, once the modulus of each entry is replaced by the operator norm of the block. □

The matrix B_1^V is of this kind, and $B_1^V \ge 1$. For real V' all transports T are isometries; hence, with $|\cdot|$ the operator norm of a block,

$$|(B_1^V - 1)(P, P')| \le \sum_{y'} |\ell_{y'}(P)|\,|\ell_{y'}(P')|.$$

For $0 < s \le \theta_D$ and $d \ge 0$, convexity gives $e^{sd} - 1 \le (s/\theta_D)(e^{\theta_D d} - 1)$. Since $\|\ell'^{\mathrm{c}}\|_{\theta_D} \le C_D$, it follows that, for $s = \theta_D/(4C_D^2)$,

$$\sup_P \sum_{P'} |(B_1^V - 1)(P, P')|\,(e^{s\mathrm{dist}(P,P')} - 1) \le (s/\theta_D)C_D^2 = \frac{1}{4}.$$

Because $\frac{1}{4} \le \frac{1}{2} = \gamma/2$ at $\gamma = 1$, the hypothesis $\varpi_{\mathsf{A}}(\lambda) \le \frac{\gamma}{2}$ of Lem. 2.25 (a) holds. Lem. 4.32 now yields $|(B_1^V)^{-1}(P, P')| \le 2e^{-s\mathrm{dist}(P,P')}$, and as there are three plaquettes P to a site, the bound $\|(B_1^V)^{-1}\|_\infty \le c_B$ follows. Consequently

$$\mathrm{Hol} \le 3\mathsf{N}_Z^2 c_B \varepsilon'^2 \max\{\sum_{P\in\mathfrak{Z}}\sum_{P'} |\mathsf{Hm}(P, P')|,\ \sum_{P\in\mathfrak{Z}}\sum_{P'} |\mathsf{Hm}(P', P)|\}.$$

(3) *Row and column sums of* Hm. We split

$$\mathsf{Hm}(P, P') = \mathrm{Hol}(P, P') + (\mathsf{B}'^{\mathrm{c}}\hat{R}^< - 1)(P, P')\,T(y_P, y_{P'}).$$

Within (3), $\mathrm{Hol}(P, P')$ and $\mathrm{Hol}(i_0, i_n)$ denote kernels, whereas outside (3) Hol is the number defined in (1). We bound the kernel $\mathrm{Hol}(P, P')$ by the following lemma, used at fixed i_0 for the row sums and at fixed i_n for the column sums.

Lemma 4.33 (The deviation of a product of dressed kernels from the bare product times one transport)**.** Let V' be any G-valued field and let $\theta \geq 0$. Let $K_1, \dots, K_n$ be kernels of finite range whose entries are scalars, or vectors of H_y and $\mathsf{H}'_{y'}$ that occur in pairs $\ell^*\ell$; only the moduli of the entries enter. For each k, choose $\tilde{K}_k = K_k^V$ or $\tilde{K}_k = (K_k^V)^*$ and let $A_k^0 = K_k$ or K_k^*, respectively. Assume the chosen factors are composable. Define

$$\mathrm{Hol}(i_0, i_n) := (\tilde{K}_1 \cdots \tilde{K}_n)(i_0, i_n) - (A_1^0 \cdots A_n^0)(i_0, i_n)\, T(y(i_0), y(i_n)).$$

Assume that the sum of the ranges of $K_1, \dots, K_n$ is $< N/2 - 1$. Then

$$\sum_{i_n} |\mathrm{Hol}(i_0, i_n)| \leq 6(n+1) \prod_k \|K_k\|_\theta \sum_{P''} e^{-\theta \mathrm{dist}(i_0, P'')} \phi'_{P''},$$

and the column sums $\sum_{i_0} |\mathrm{Hol}(i_0, i_n)|$ obey the same bound with $\mathrm{dist}(i_n, P'')$ in place of $\mathrm{dist}(i_0, P'')$. In the bound on the row sums, P'' runs through the plaquettes with $\mathrm{dist}(i_0, P'')$ at most the sum of the ranges of $K_1, \dots, K_n$, and in the bound on the column sums through those with $\mathrm{dist}(i_n, P'')$ at most the same sum. ∎

Proof. By the hypothesis on the ranges, the box spanned by the sites $y(i_k)$ embeds into the coarse torus. Expanding both products we obtain

$$\mathrm{Hol} = \sum_{i_1, \dots, i_{n-1}} \prod_k A_k^0(i_{k-1}, i_k)\, [\mathrm{Ad}_{V'(\mathrm{chain})} - \mathrm{Ad}_{V'(\Gamma(y(i_0), y(i_n)))}].$$

Here the chain is the path through $y(i_0), \dots, y(i_n)$ made of the coordinate paths $\Gamma(y(i_{k-1}), y(i_k))$, except that, where the factor $\tilde{K}_k$ is the adjoint of a dressed kernel, the path $\Gamma(y(i_k), y(i_{k-1}))$, run backwards, takes the place of $\Gamma(y(i_{k-1}), y(i_k))$. Each coordinate path has at most 3 straight stretches, so that the chain has at most $3n$ of them and $\Gamma(y(i_0), y(i_n))$ at most 3 more. Together with $\Gamma(y(i_0), y(i_n))$, the chain makes up a closed path in that box, with at most $3(n+1)$ straight stretches. By Lem. 4.21 and Lem. 4.20 (b), the bracket is therefore at most $6(n+1) \sum_{P'' \in \mathrm{box}} \phi'_{P''}$. For every plaquette P'' of the box, both $\mathrm{dist}(i_0, P'')$ and $\mathrm{dist}(i_n, P'')$ are bounded by $\mathsf{le} := \sum_k \mathrm{dist}(i_{k-1}, i_k)$. Hence, for fixed i_0 and P'',

$$\sum_{i_1, \dots, i_n} \prod_k |A_k^0|\, \mathbf{1}[\mathsf{le} \geq \mathrm{dist}(i_0, P'')] \leq e^{-\theta \mathrm{dist}(i_0, P'')} \prod_k \|K_k\|_\theta,$$

Here $\|A_k^0\|_\theta = \|K_k\|_\theta$, since the norm controls both row and column sums. Summing over P'' gives the row bound. For fixed i_n and P'' the same estimate holds with column sums and with $\mathrm{dist}(i_n, P'')$, because $\|\cdot\|_\theta$ controls the column sums as well (Def. 2.20); this gives the bound on the column sums. □

Measured in $\|\cdot\|_{\theta_D}$, the ν-map has norm at most $\mathrm{e}L + 6\mathrm{e}C_D + C_D^3 = a_\nu$ and the σ-map has norm at most C_D^2, while $(aL)^{-1}\|\mathrm{d}'\|^2 \leq 16\mathrm{e}^2/18 \leq 7$. Every product in $B_1^V \hat{R}^V$ has $n \leq 2+3+3 = 8$ factors, so that $6(n+1) \leq 54$, and the products of the norms add up to at most $(1 + C_D^2)[La_\nu^2 + LC_D^4 + 7]$. For each of these products, the sum of the ranges of the factors is $\leq 4R(x') + 4\varrho_L + 4 < N/2 - 1$ by (Tor), since $N \geq 16Mr(x) \geq 320 c_R \log x$. With $54 \cdot 3\mathsf{g}(\theta_D) = 162\mathsf{g}(\theta_D)$ and the constant $C_{\mathrm{hol}}^\flat$ of item F of §4.3, Lem. 4.33 gives

$$\sum_{P'} |\mathsf{Hm}(P, P')| \leq \frac{C_{\mathrm{hol}}^\flat}{3\mathsf{g}(\theta_D)} \sum_{P''} e^{-\theta_D \mathrm{dist}(P, P'')} \phi'_{P''} + \|\mathsf{B}'^{\mathrm{c}} \hat{R}^< - 1\|_0.$$

For the column sums, the bound of Lem. 4.33 with $\mathrm{dist}(i_n, P'')$ gives the same right-hand side,

$$\sum_{P'} |\mathsf{Hm}(P', P)| \leq \frac{C_{\mathrm{hol}}^\flat}{3\mathsf{g}(\theta_D)} \sum_{P''} e^{-\theta_D \mathrm{dist}(P, P'')} \phi'_{P''} + \|\mathsf{B}'^{\mathrm{c}} \hat{R}^< - 1\|_0,$$

because $\|\cdot\|_0$ is the larger of the largest row sum and the largest column sum (Def. 2.20).

(4) *Cuts.* In (4) alone we write $t := x'^{-64}$. Without the cuts, ν and σ denote the components of pm and $\hat{R}$ denotes $L\,\mathrm{pm}^*\mathrm{pm}$, at $V' = 1$ and with T, X in place of $\mathsf{T}^<, \mathsf{X}^<$. In (4), ℓ denotes the fine rows at $V' = 1$, as in $\sigma^<$, and not ℓ'^{cV}. By definition, the second component of the uncut pm is $\sigma = \ell\mathsf{T}$. With $\mathsf{B} = 1 + \ell^{\mathsf{T}}\ell$, the first identity of Lem. 4.24 (ii), $\mathsf{B}\mathsf{T} = L^{-1}W^* + \mathrm{d}^*\mathsf{X}$, gives $\nu = L^{-1}W^* + \mathrm{d}^*\mathsf{X} - \ell^{\mathsf{T}}\ell\mathsf{T} = \mathsf{B}\mathsf{T} - \ell^{\mathsf{T}}\ell\mathsf{T} = \mathsf{T}$. Written out, $\hat{R} = L(\mathsf{T}^*\mathsf{T} + \mathsf{T}^*\ell^{\mathsf{T}}\ell\mathsf{T}) + La(aL)^{-2}\mathrm{d}'\mathrm{d}'^* = L\mathsf{T}^*\mathsf{B}\mathsf{T} + (aL)^{-1}\mathrm{d}'\mathrm{d}'^*$, and the last identity of Lem. 4.24 (ii) yields $\mathsf{B}'\hat{R} = 1$. For the cut maps, $\|\nu^< - \mathsf{T}\|_0 \le (6C_D + C_D^3)t \le a_\nu t$ and $\|\sigma^< - \ell\mathsf{T}\|_0 \le C_D^2 t$. Hence $\|\hat{R}^< - \hat{R}\|_0 \le L[2a_\nu^2 + 2C_D^4]t \le 2.125\,La_\nu^2 t$, since $a_\nu^2 \ge 16C_D^4$. Further, $\|\hat{R}^<\|_0 \le La_\nu^2 + LC_D^4 + 7 \le 1.07\,La_\nu^2$, $\|\mathsf{B}'^c - \mathsf{B}'\|_0 \le 2C_D^2 t$ and $\|\mathsf{B}'\|_0 \le C_D$. From $\mathsf{B}'^c\hat{R}^< - 1 = (\mathsf{B}'^c - \mathsf{B}')\hat{R}^< + \mathsf{B}'(\hat{R}^< - \hat{R})$ we conclude

$$\|\mathsf{B}'^c\hat{R}^< - 1\|_0 \le (2.14C_D^2 + 2.125C_D)La_\nu^2 t \le 2.7\,C_D^2 La_\nu^2 t \le \mathsf{ti}_0^\flat.$$

(5) *The sum over $P \in \mathfrak{Z}$.* By Lem. 4.26 (iii), every plaquette P'' that enters either lies in D'_J or is tame. Moreover, $\sum_P e^{-\theta_D \mathrm{dist}(P,P'')} \le 3\mathsf{g}(\theta_D)$. Summing the bound of (3) over $P \in \mathfrak{Z}$ and inserting (4), we find

$$\sum_{P\in\mathfrak{Z}}\sum_{P'} |\mathsf{Hm}(P,P')| \le C_{\mathrm{hol}}^\flat[\sum_{D'_J}\phi' + 3M^3\#\mathfrak{Z}\,\varepsilon'] + 3M^3\#\mathfrak{Z}\,\mathsf{ti}_0^\flat.$$

Summed in the same way, the bound of (3) on the column sums gives the same bound for $\sum_{P\in\mathfrak{Z}}\sum_{P'}|\mathsf{Hm}(P',P)|$. Hence, by (2),

$$\frac{1}{2}\mathrm{Hol} \le 3e_1^\flat\varepsilon'\sum_{D'_J}\phi' + (9e_1^\flat + \frac{9}{2}\mathsf{N}_Z^2 c_B \mathsf{ti}_0^\flat)\varepsilon'^2 M^3\#\mathfrak{Z}.$$

(6) *The main part, row by row.* Let $\mathsf{et}' = X' - \zeta'$; this field is parallel to ζ'. At each plaquette P,

$$\langle v_{*P}, X'_P\rangle - \frac{1}{2}\langle v_{*P}, \zeta'_P\rangle = \frac{1}{2}|\zeta'_P|^2 + \frac{1}{2}\langle\zeta'_P, (\ell^* m'\ell\zeta')_P\rangle + \varepsilon'|\mathsf{et}'_P| + \langle(\ell^* m'\ell\zeta')_P, \mathsf{et}'_P\rangle.$$

For tame P the right-hand side equals En'_P. Now let $P \in D'$; for P in $\mathfrak{Z}$ this means $P \in D'_J$. The mask m' vanishes at all sites within ρ_0 of P, and so $|(\ell^* m'\ell\zeta')_P| \le \mathsf{t}_\rho\varepsilon'$. At such P we also have $|\mathsf{et}'_P| = \phi'_P - \varepsilon'$, and

$$\frac{1}{2}\varepsilon'^2 + \varepsilon'(\phi' - \varepsilon') = d_0\varepsilon'\phi' + (\frac{1}{2} - d_0)\varepsilon'\phi' + \frac{1}{2}\varepsilon'(\phi' - \varepsilon').$$

Summing over $P \in \mathfrak{Z}$ and combining the result with (1) and (5) proves Lem. 4.31. □

Step 6: the balance and its three rim brackets.

Lemma 4.34 (The balance, with three rim brackets)**.** Let (J, t) be an admitted cell datum, and set

$$\mathrm{Bal} := \frac{x}{2}\|\Delta\|_3^2 + x\langle\Xi_*, \mathrm{d}^V EX'\rangle_{\mathfrak{S}} - x\langle Z_*, d_1 a_0\rangle_{\mathfrak{S}^-} - x\langle\Xi_*, d_2 a_0\rangle_{\mathfrak{S}^-} - \sum_J w_j.$$

Then

$$\mathrm{Bal} \le -(\frac{1}{2} - d_0 - \mathsf{t}_\rho - 3e_1^\flat)x'\varepsilon'\sum_{D'_J}\phi'_P + \sigma_{\mathrm{tot}}^\flat\#\mathfrak{Z}_2.$$

■

Proof. Identities. What follows holds at every V' and for all $\mathsf{zf}, \Xi, \sigma, F, n, \psi$. Put

$$\mathsf{pair}(\mathsf{zf}, \Xi; \sigma) := (W^{V*}\mathsf{zf} + \mathrm{d}^{V*}\Xi - \ell^{V*}\sigma,\ \sigma,\ a^{-1}\mathrm{d}'^{V*}\mathsf{zf}).$$

Then

$$\langle\mathsf{pair}(\mathsf{zf}, \Xi; \sigma), \iota(F, n)\rangle_{\mathrm{Hil}} = \langle\mathsf{zf}, W^V F + \mathrm{d}'^V n\rangle + \langle\Xi, \mathrm{d}^V F\rangle, \tag{E1$_k$}$$

and at $\iota(F, n) = k\psi$ the right-hand side of (E1$_k$) becomes $\langle\mathsf{zf}, d_1\psi\rangle + \langle\Xi, d_2\psi\rangle$. Moreover,

$$\langle\bar{F}, k\psi\rangle_{\mathrm{Hil}} = \langle EX', \mathsf{B}^V\mathrm{d}^V\psi\rangle - \langle a_0, q\psi\rangle = \langle u^T\,\mathsf{g}^V X', \psi\rangle. \tag{E3$_k$}$$

We have $\bar F = \iota(EX' - \mathrm{d}^V a_0,\ Q_1^V a_0)$, and we write $\bar n = Q_1^V a_0$ for its bond part. Because $W^V E = 1$ (Step 1 (iv) of the proof of Thm. 4.2), it follows that $W^V \bar F_F + \mathrm{d}'^V \bar n = X' - d_1 a_0$; also $\mathrm{d}^V \bar F_F = \mathrm{d}^V E X' - d_2 a_0$. Finally, $\mathrm{d}^V E = E_3 \mathrm{d}'^V$, by Step 1 (iv) of the proof of Thm. 4.2.

(1) *Cut potentials.* Put

$$\Delta^\chi := \iota(E\chi X', 0) - k(\chi a_0) - \mathrm{pm}(\chi v_*),$$

and $\Xi^\chi = (\mathsf{X}^<)^V \chi v_*$. Let ν^χ be the plaquette component of $\mathrm{pm}(\chi v_*)$, the first of its three components. We bound the components of $\mathbf{1}_{\mathfrak{Z}}\Delta - \Delta^\chi$, beginning with the plaquette rows, where three terms occur. The first term vanishes, as $E\chi X' = \mathbf{1}_{\mathfrak{Z}} E X'$. The second, $\mathbf{1}_{\mathfrak{Z}}\mathrm{d}^V a_0 - \mathrm{d}^V \chi a_0$, is at most $4 \sup|a_0|$ and is nonzero only at those plaquette rows p that have a bond b with $p \in \hat{\mathfrak{Z}}$ and $b \notin \hat{\mathfrak{Z}}$, or with $p \notin \hat{\mathfrak{Z}}$ and $b \in \hat{\mathfrak{Z}}$. The third, $\mathbf{1}_{\mathfrak{Z}}\nu_* - \nu^\chi$, is the ν-component of $\mathrm{pm}((1-\chi)v_*)$ inside $\mathfrak{Z}$ and that of $-\mathrm{pm}(\chi v_*)$ outside; it is at most $N_1 L^{-2}\varepsilon'$ and is nonzero only at rows of $\hat{\mathfrak{Z}}$ that lie within $R(x') + 2\varrho_L + 3$ of the complement of $\hat{\mathfrak{Z}}$ and at rows off $\hat{\mathfrak{Z}}$ that lie within $R(x') + 2\varrho_L + 3$ of $\hat{\mathfrak{Z}}$. At the site rows there are again three terms. The first is $(\ell^V E(1-\chi)X')_y$ at $y \in \hat{\mathfrak{Z}}$ and $-(\ell^V E\chi X')_y$ off it, and is at most $l_r\varepsilon'$. The second, $\mathbf{1}_{\mathfrak{Z}}(y)(\ell^V \mathrm{d}^V a_0)_y - (\ell^V \mathrm{d}^V \chi a_0)_y$, equals $(\ell^V \mathrm{d}^V (1-\chi) a_0)_y$ at $y \in \hat{\mathfrak{Z}}$ and $-(\ell^V \mathrm{d}^V \chi a_0)_y$ off it, and is at most $4 l_r \sup|a_0|$: the absolute row sum of $\ell^V \mathrm{d}^V$ is at most $4l_r$, and a bond b contributes only if it lies within $\varrho_L + 2$ of y, with $b \notin \hat{\mathfrak{Z}}$ when $y \in \hat{\mathfrak{Z}}$ and $b \in \hat{\mathfrak{Z}}$ when $y \notin \hat{\mathfrak{Z}}$. The third, the σ-component, is at most $\mathsf{N}_\sigma L^{-2}\varepsilon'$. At the bond rows, which carry the weight a, the component is at most $L \sup|a_0| + 4\mathsf{N}_Z (aL)^{-1}\varepsilon'$. All these rows satisfy the hypothesis of Lem. 4.26 (iv); hence they are based in $\mathfrak{Z}_2 \setminus \mathfrak{b}$, and since $f_J \subset \mathfrak{b}$ by item E12 of §4.3, they lie off f_J. By Lem. 4.26 (ii) and Lem. 4.29 (b), at these rows $|\Delta_i| \le K_\Delta^\flat \varepsilon'^2$ and $|a_0| \le \mathsf{a}_\mathrm{c}\varepsilon'$. The rows that belong to one cell are those of $3v$ fine plaquettes, v fine sites and $3M^3$ coarse bonds. Hence

$$\frac{x}{2}\big|\,\|\Delta\|_{\mathfrak{Z}}^2 - \langle \Delta, \Delta^\chi\rangle_{\mathrm{Hil}}\big| \le \frac{x}{2} K_\Delta^\flat \varepsilon'^3 [3v(4\mathsf{a}_\mathrm{c} + N_1) + v(l_r(4\mathsf{a}_\mathrm{c}+1) + \mathsf{N}_\sigma) + 3M^3(aL\mathsf{a}_\mathrm{c} + 4\mathsf{N}_Z)]\#\mathfrak{Z}_2.$$

To simplify the bracket, let $c = c_\eta^\flat \ge \mathsf{a}_\mathrm{c}$ and $\mathsf{Nn} = \mathsf{N}_Z + \mathsf{N}_\nu^\flat + \mathsf{N}_\sigma \ge 1$, and note that $M^3 L \le v/9$. Then $12c + 3\mathsf{Nn} \le 12(1+c)\mathsf{Nn}$, $4l_r(1+c) + \mathsf{Nn} \le 4(1+l_r)(1+c)\mathsf{Nn}$ and $\frac{1}{3}ac + \frac{4}{9}\mathsf{Nn} \le \frac{4}{3}(1+a)(1+c)\mathsf{Nn}$. The bracket is therefore at most $17.34\, v(1+a)(1+l_r)(1+c)\mathsf{Nn}$, and the part of the bound above that comes from the objects based in one cell is at most $8.67\, L^{3/2} v K_\Delta^\flat (1+a)(1+l_r)(1+c)\mathsf{Nn}\, p_0^3 x^{-1/2}$.

(2) *Six pairings.* With the identities above, the six pairings are as follows. First, $\langle \bar F, \iota(E\chi X', 0)\rangle = \langle \mathsf{B}^V \bar F_F, E X'\rangle_{\mathfrak{Z}}$. Second, $\langle \bar F, k\chi a_0\rangle = \langle u^T\, \mathsf{g}^V X', \chi a_0\rangle$. Third,

$$\langle \bar F, \mathrm{pm}(\chi v_*)\rangle = \langle \chi Z_*, X' - d_1 a_0\rangle + \langle \Xi^\chi, \mathrm{d}^V E X' - d_2 a_0\rangle.$$

Fourth, $\langle \boldsymbol{\nu}, \iota(E\chi X', 0)\rangle = \langle Y_*, EX'\rangle_{\mathfrak{Z}}$. Fifth, $\langle \boldsymbol{\nu}, k\chi a_0\rangle = \langle Z_*, d_1 \chi a_0\rangle + \langle \Xi_*, d_2 \chi a_0\rangle$. Sixth, $\langle \boldsymbol{\nu}, \mathrm{pm}(\chi v_*)\rangle = L^{-1}\langle v_*, \hat R^V v_*\rangle_{\mathfrak{Z}}$. Combining the six pairings, we obtain

$$\begin{aligned}\langle \Delta, \Delta^\chi\rangle = {} & \langle \Delta_Y, EX'\rangle_{\mathfrak{Z}} - \langle u^T\, \mathsf{g}^V X', \chi a_0\rangle \\ & - \langle X', Z_*\rangle_{\mathfrak{Z}} - \langle \mathrm{d}^V E X', \Xi^\chi\rangle \\ & + \langle \chi Z_*, d_1 a_0\rangle + \langle Z_*, d_1 \chi a_0\rangle \\ & + \langle \Xi^\chi, d_2 a_0\rangle + \langle \Xi_*, d_2 \chi a_0\rangle \\ & + L^{-1}\langle v_*, \hat R^V v_*\rangle_{\mathfrak{Z}}.\end{aligned}$$

(3) *The prefactor.* We have $\mathrm{mk}_j'^{+r(x')} \subset f_j \subset \mathfrak{Z}$. A cell of $\mathfrak{Z}$ off $\mathrm{mk}_J'^{+r(x')}$ is farther than $Mr(x') \ge R_6$ from D'_J, and by Lem. 4.26 (i) it is also farther than R_6 from the other rough plaquettes. Lem. 4.29 (c) therefore gives

$$\Big|\sum_J w_j - \sum_{\square \in \mathfrak{Z}} T_\square\Big| \le K_\mathrm{q}^\flat p_0'^2 \varepsilon' \#\mathfrak{Z}.$$

By item E7 of §4.3,

$$\sum_{\mathfrak{Z}} T_\square = \frac{x}{2}\langle X', Z_*\rangle_{\mathfrak{Z}} - x'\sum_{\mathfrak{Z}} \mathrm{En}'_P + \frac{x}{2}\langle EX', \Delta_Y\rangle_{\mathfrak{Z}} + \frac{x}{2}\langle \mathrm{d}^V EX', \Xi_*\rangle_{\mathfrak{Z}} - \frac{x}{2}\langle a_0, u^T\, \mathrm{g}^V X'\rangle_{\mathfrak{Z}}.$$

(4) *The sum; support and sizes of the rim.* We consider the expression

$$\frac{x}{2}\langle \Delta, \Delta^\chi\rangle + x\langle \Xi_*, \mathrm{d}^V EX'\rangle_{\mathfrak{S}} - x\langle Z_*, d_1 a_0\rangle_{\mathfrak{S}^-} - x\langle \Xi_*, d_2 a_0\rangle_{\mathfrak{S}^-} - \sum_{\mathfrak{Z}} T_\square.$$

Once (2) and (3) are inserted, the terms with Δ_Y and those with u^T cancel. The term $-\frac{x}{2}\langle X', Z_*\rangle_{\mathfrak{Z}}$ appears twice and gives $-x'\langle v_*, X'\rangle_{\mathfrak{Z}}$, and the terms $+\frac{x'}{2}\langle v_*, \hat{R}^V v_*\rangle_{\mathfrak{Z}} + x'\sum \mathrm{En}'$ remain. So the expression equals $-x'\mathcal{G}_{\mathfrak{Z}} + x'\sum_{\mathfrak{Z}} \mathrm{En}' + \mathrm{rim}$, where

$$\begin{aligned}\mathrm{rim} = {} & x[\langle \mathrm{d}^V EX', \Xi_*\rangle_{\mathfrak{S}} - \frac{1}{2}\langle \mathrm{d}^V EX', \Xi^\chi\rangle - \frac{1}{2}\langle \mathrm{d}^V EX', \Xi_*\rangle_{\mathfrak{Z}}] \\ & + x[\frac{1}{2}\langle \chi Z_*, d_1 a_0\rangle + \frac{1}{2}\langle Z_*, d_1 \chi a_0\rangle - \langle Z_*, d_1 a_0\rangle_{\mathfrak{S}^-}] \\ & + x[\frac{1}{2}\langle \Xi^\chi, d_2 a_0\rangle + \frac{1}{2}\langle \Xi_*, d_2 \chi a_0\rangle - \langle \Xi_*, d_2 a_0\rangle_{\mathfrak{S}^-}].\end{aligned}$$

Support. The first bracket is a sum over corner cubes q. At $q \in \mathfrak{S}$ farther than $R(x')$ from the complement of $\mathfrak{Z}$ we have $\Xi^\chi = \Xi_*$, and the coefficients give $1 - \frac{1}{2} - \frac{1}{2} = 0$; outside $\mathfrak{Z}$ enlarged by $R(x')$ the first bracket vanishes. In the second bracket, $(d_1\psi)_P$ depends only on the bonds based in $B(y_3(P))$, one block away; this is Prop. 4.35, which we now state and prove.

Proposition 4.35 (d_1 and d_2 see the curvature of a single coarse plaquette)**.** Let V' be any G-valued field, let $\psi \in \mathfrak{L}_{\mathfrak{g}}$ and $P = (y; \mu\nu)$ with $\mu < \nu$, and let $y_1, y_2, y_3 = y_3(P)$ and c_1 to c_4 be as in Step 3 (3) of the proof of Thm. 4.19. Put $T_a := \mathrm{Ad}_{V'(c_1)V'(c_2)} = T(y, y_3)$ and $T_b := \mathrm{Ad}_{V'(c_4)V'(c_3)}$. Then the following hold.

(1) There are numbers be_b with $\sum_b |\mathsf{be}_b| \le L - 1$ such that $(d_1\psi)_P = \sum_{b \text{ based in } B(y_3)} \mathsf{be}_b\, (T_b - T_a)\psi_b$. Hence $|(d_1\psi)_P| \le 2\phi'_P(L-1)|\psi|_\infty \le 2\phi'_P\, 2L|\psi|_\infty$.

(2) Let q be a cube and $z = \pi(q)$. Then $(d_2\psi)_q$ is a sum over those pairs $\mu < \nu$ for which z_μ and z_ν lie at the top of the block of z. The term of such a pair is $\pm(T_a - T_b)\psi_b$, where $b = (z + e_\mu + e_\nu, \lambda)$ with λ the third direction, and T_a, T_b are those of $P = (y(q); \mu\nu)$. There are L such cubes to a P, and the share of P in a sum $\sum_q |\Xi_q|\,|(d_2\psi)_q|$ is at most $|\Xi|_\infty 2\phi'_P L|\psi|_\infty$.

■

Proof. We have $T_a T_b^{-1} = \mathrm{Ad}_{\partial V'(P)}$, so that $\|T_b - T_a\| \le 2\phi'_P$ by Lem. 4.20 (b); with this, the bounds in (1) and (2) follow from the formulas there, which we now prove.

(1). As scalar kernels, $W\mathrm{d} = \mathrm{d}'Q_1$ (Lem. 2.18). In $W^V\mathrm{d}^V$ the bond b carries the transport $T(y, y(p))T(y(p), y(b))$, and in $\mathrm{d}'^V Q_1^V$ it carries $T(y, c_-)T(c_-, y(b))$. Each of these is Ad of V' along a monotone path from y to $y(b)$ whose sites are corners of P. To y, y_1 and y_2 there is just one such path; to y_3 there are two, and they give T_a and T_b. If T_a is put in place of T_b, the sum becomes $\sum_b (W\mathrm{d} - \mathrm{d}'Q_1)(P, b)\, T(y, y(b))\psi_b = 0$. Hence only the terms carrying T_b remain, with $T_b - T_a$ in place of T_b, and these terms are of two kinds. The terms of the first kind are those of the plaquettes $p = (z; \mu\nu)$ based in $B(y_2)$ with $z + e_\mu \in B(y_3)$; a plaquette based in $B(y)$ has no bond based in $B(y_3)$. For these p we have $W(P, p) = L^{-3}L\, t(z_\nu - y_\nu)$, where $t(\cdot)$ denotes the tent function of the weights $W(P, p)$; its values run through $1, \dots, L-1$, at each of L heights, and these weights add up to $(L-1)/2$. The terms of the second kind are those of the coarse bond c_3, for which $\sum_{b \text{ based in } B(y_3)} Q_1(c_3, b) = L^{-3}L^2 \sum_{k=0}^{L-1} k = (L-1)/2$. The two kinds together give $\sum_b |\mathsf{be}_b| \le L - 1$.

(2). As scalars, $dd = 0$: every bond b of q lies in two faces of q, with opposite signs. These faces are based at z or at some $z + e_\rho$, and their bonds lie at most one step further. The two transports from $y(q)$ to $y(b)$ can therefore differ only if $b = (z + e_\mu + e_\nu, \lambda)$ and both steps cross a seam. In that case the two faces are $(z + e_\mu; \nu\lambda)$ and $(z + e_\nu; \mu\lambda)$, and the transports are T_a and T_b. □

We now return to the second bracket of rim. For $P \in \mathfrak{S}^-$, the χ in χZ_* is evaluated at P and, by Prop. 4.35, the χ in $d_1\chi a_0$ at the bonds based in $B(y_3(P))$, one block away. Since $\mathfrak{S}^- = \mathfrak{b}^{+r_1-r_l-2}$ and $\mathfrak{Z} = \mathfrak{b}^{+r_1}$, we have $P \in \mathfrak{Z}$, and these bonds lie in $\hat{\mathfrak{Z}}$. Hence both cuts of this bracket change nothing at P, and the coefficients give $\frac{1}{2} + \frac{1}{2} - 1 = 0$; off $\mathfrak{Z}^{+1}$ the second bracket is 0. The third bracket is treated as the first, with $\mathfrak{S}^-$ in place of $\mathfrak{S}$: at $q \in \mathfrak{S}^-$ farther than $R(x')$ from the complement of $\mathfrak{Z}$ we have $\Xi^\chi = \Xi_*$, and, since by Prop. 4.35 the χ in $d_2\chi a_0$ is evaluated at the bonds $(z + e_\mu + e_\nu, \lambda)$ with $z = \pi(q)$, which lie in $\hat{\mathfrak{Z}}$, this χ changes nothing at q, so that the coefficients give $\frac{1}{2} + \frac{1}{2} - 1 = 0$; outside $\mathfrak{Z}$ enlarged by $R(x')$ the third bracket vanishes. What remains is based in $\mathfrak{b}^{+r_1+\lceil R/M\rceil+1} \setminus \mathfrak{b}^{+r_1-r_l-2-\lceil R/M\rceil}$, which by Lem. 4.26 (ix) lies inside $\mathfrak{Z}_2 \setminus \mathfrak{b}$ and off f_J; its boxes are tame by Lem. 4.26 (ii).

Sizes. There $|(\mathrm{d}^V EX')_q| = |(\mathrm{d}'^V X')_{q'}| \le 18\varepsilon'^2$, with M^3 cubes q' to a cell, and $|\Xi_*|$, $|\Xi^\chi| \le \mathsf{N}_\Xi L^{-1}\varepsilon'$, $|Z_*| \le \mathsf{N}_Z L^{-1}\varepsilon'$. Prop. 4.35 with $\phi'_P \le \varepsilon'$ gives $|(d_1\psi)_P| \le 2\varepsilon' \cdot 2L|\psi|_\infty$ and, for the sum over the L cubes q that go with one P, $\sum_q |\Xi_q|\,|(d_2\psi)_q| \le |\Xi|_\infty 2\varepsilon' L|\psi|_\infty$. Here $|\psi|_\infty \le \mathsf{a}_\mathrm{c}\varepsilon'$, also at the bonds based in $B(y_3(P))$, one block further. This rests on the following two facts, which together are the two hypotheses of the second alternative of Lem. 4.26 (ii). First, these bonds are based in $\mathfrak{b}^{+r_1+\lceil R/M\rceil+2}$, which is part of $\mathfrak{Z}_2$ because $r_1 \ge \lceil R_2/M\rceil + 8$. Second, they lie within $1 \le r_w + \varrho_L + 2$ of the site y_P, which is off f_J; so the second alternative of Lem. 4.26 (ii) holds, giving a tame box of radius R_6, and Lem. 4.29 (b) yields the bound $\mathsf{a}_\mathrm{c}\varepsilon'$. There are $3M^3$ plaquettes P to a cell, and in each bracket the moduli of the coefficients add up to at most 2. The parts of the three brackets that come from the objects based in one cell are therefore at most $36\mathsf{N}_\Xi$, $24\mathsf{N}_Z\mathsf{a}_\mathrm{c}$ and $12\mathsf{N}_\Xi\mathsf{a}_\mathrm{c}$, each times $x\varepsilon'^3 M^3$, and

$$|\mathrm{rim}| \le 36L^{3/2}v(\mathsf{N}_Z + \mathsf{N}_\Xi)(1 + c_\eta^\flat)p_0^3 x^{-1/2}\#\mathfrak{Z}_2.$$

(5) *Conclusion.* We have

$$\mathrm{Bal} = \frac{x}{2}[\|\Delta\|_3^2 - \langle\Delta, \Delta^\chi\rangle_{\mathrm{Hil}}] + [-x'\mathcal{G}_\mathfrak{Z} + x'\sum_\mathfrak{Z} \mathrm{En}' + \mathrm{rim}] + [\sum_{\square\in\mathfrak{Z}} T_\square - \sum_J w_j],$$

where the three summands are, in this order, the error of (1), the expression of (4) and the error of (3). We treat the third summand below. By (1) and (4), the parts of the first summand and of the term rim of the second summand that come from the objects based in one cell of $\mathfrak{Z}_2$ are together at most

$$\begin{aligned} &L^{3/2}v(1 + c_\eta^\flat)\,[\,8.67\,K_\Delta^\flat(1 + a)(1 + l_r)\mathsf{N}\mathsf{n} + 36(\mathsf{N}_Z + \mathsf{N}_\Xi)\,]\,p_0^3 x^{-1/2} \\ &\quad \le 44.67\,L^{3/2}v(1 + a)(1 + K_\Delta^\flat)(1 + l_r)(1 + c_\eta^\flat)(\mathsf{N}_Z + \mathsf{N}_\Xi + \mathsf{N}_\nu^\flat + \mathsf{N}_\sigma)p_0^3 x^{-1/2} \\ &\quad \le K_s^\flat p_0^3 x^{-1/2}. \end{aligned}$$

Indeed, the product $K_\Delta^\flat(1 + a)(1 + l_r)(1 + c_\eta^\flat)\mathsf{N}\mathsf{n}$ from (1) and the product $(\mathsf{N}_Z + \mathsf{N}_\Xi)(1 + c_\eta^\flat)$ from (4) are both at most $(1 + a)(1 + K_\Delta^\flat)(\mathsf{N}_Z + \mathsf{N}_\Xi + \mathsf{N}_\nu^\flat + \mathsf{N}_\sigma)(1 + l_r)(1 + c_\eta^\flat)^2$, while $44.67L^{3/2} \le 200L^3$. Besides the two errors of (1) and (4), there is a third, the error $K_\mathrm{q}^\flat p_0'^2\varepsilon'\#\mathfrak{Z}$ of (3), which yields the second term of $\sigma_{\mathrm{tot}}^\flat$ because $\#\mathfrak{Z} \le \#\mathfrak{Z}_2$. The third term, $(9e_1^\flat + \frac{9}{2}\mathsf{N}_Z^2 c_B \mathsf{ti}_0^\flat)M^3 p_0'^2$, is what the last term of Lem. 4.31 becomes after multiplication by x', with $x'\varepsilon'^2 = p_0'^2$ and again $\#\mathfrak{Z} \le \#\mathfrak{Z}_2$. Together with the bound $K_s^\flat p_0^3 x^{-1/2}$ above, these are the three terms of $\sigma_{\mathrm{tot}}^\flat$; both $K_s^\flat$ and $\sigma_{\mathrm{tot}}^\flat$ are defined in item F of §4.3. The inequality of Lem. 4.31, multiplied by x', gives the main term $-(\frac{1}{2} - d_0 - \mathsf{t}_\rho - 3e_1^\flat)x'\varepsilon'\sum_{D'_j}\phi'_P$. This proves the claim. □

Step 7: the Gaussian weight row by row, the rim, and a summation by parts. Let (J,t) be an admitted cell datum. We call the set $\mathfrak{Z}_2 \setminus \mathfrak{b}$ the collar. By item E10 of §4.3, $b_\square(\xi)=0$ for a cell $\square$ exactly when $|\xi_b| \le c_\xi \varepsilon$ at every bond b of I° in $\hat{\square}$. We claim that, on the event that $b_\square(\xi)=0$ for every cell $\square$ of the collar and that $|\eta_b|<\pi$ at each bond b of $I^\circ \cap \hat{\mathfrak{Z}}$,

$$\begin{aligned} &-\sum_J w_j - \frac{x}{2}\|k\varphi\|_{\mathfrak{Z}}^2 + \frac{x}{2}\|\iota(F^\flat,n^\flat)\|_{\mathfrak{Z}}^2 - x\mathcal{A}_{\mathfrak{Z}} \\ &\le -(\frac{1}{2} - d_0 - \mathsf{t}_\rho - 3e_1^\flat)x'\varepsilon' \sum_{D'_J} \phi'_P \\ &\quad - x\,\mathrm{Sq}_{\mathfrak{Z}} - x\,\mathrm{Pay}_{\mathfrak{Z}} \\ &\quad + [\sigma_{\mathrm{tot}}^\flat + 5K_s^\flat p_0^3 x^{-1/2} + x^{-40} \\ &\quad + (6.3(\mathsf{N}_\nu^\flat)^2\vartheta_F^\flat + 3\mathsf{N}_Z^2(\theta_2 + L\vartheta_n^\flat))M^3 p_0'^2]\#\mathfrak{Z}_2. \end{aligned}$$

No term linear in φ is left on the right-hand side.

Substep 1: the exponent of the Gaussian weight cancelled row by row. As G is invertible and $C = GG^{\mathsf{T}}$, we have $|\xi|^2 = \langle\varphi, C^{-1}\varphi\rangle = \|k\varphi\|_{\mathrm{Hil}}^2 + \langle\varphi, S\varphi\rangle$, and

$$E_\xi[f] = (x/2\pi)^{\frac{3}{2}\#I^\circ}(\det C)^{-1/2}\int f\, e^{-\frac{x}{2}\langle\varphi,C^{-1}\varphi\rangle}d\varphi.$$

Moreover,

$$[A]_{\mathfrak{Z}}^\ell = \exp\{\frac{x}{2}\|\iota(F^\flat,n^\flat)\|_{\mathfrak{Z}}^2 - x\mathcal{A}_{\mathfrak{Z}}\}\prod \mathbf{1}\{|\eta_b|<\pi\}\mathsf{j}(\eta_b),$$

with $\mathsf{j}\le 1$. Since $\iota(F^\flat,n^\flat) = \bar{F} + k\varphi$ and $\iota(F,n) = \bar{F} + k\varphi - \mathfrak{k}$, we have, row by row on $\mathfrak{Z}$,

$$-\frac{1}{2}\|k\varphi\|^2 + \frac{1}{2}\|\bar{F}+k\varphi\|^2 - \langle\boldsymbol{\nu},\iota(F,n)\rangle + \frac{1}{2}\|\boldsymbol{\nu}\|^2 = \frac{1}{2}\|\Delta\|^2 + \langle\Delta,k\varphi\rangle + \langle\boldsymbol{\nu},\mathfrak{k}\rangle.$$

With Lem. 4.27, applied to the starred fields and the sets of item E12 of §4.3, the left-hand side of the claim without $-\sum_J w_j$ is at most

$$\frac{x}{2}\|\Delta\|_{\mathfrak{Z}}^2 + x\langle\Delta,k\varphi\rangle_{\mathfrak{Z}} + x\langle\boldsymbol{\nu},\mathfrak{k}\rangle_{\mathfrak{Z}} - x|\langle Z_*,\mathrm{d}e\rangle_{\mathfrak{S}}| - x|\langle\Xi_*,\mathrm{d}b\rangle_{\mathfrak{S}}| - x\,\mathrm{Sq} - x\,\mathrm{Pay} + x\,\mathrm{vol} + x\,\mathrm{Rim}_{\mathfrak{Z}}.$$

Substep 2: the rim, and the one use of the collar tests. By Lem. 4.26 (vi), and as $G(b,b')=0$ beyond the distance r_w by (C2) of item D of §4.3, at a bond b based in Lr the value $(G\xi)_b$ depends only on the values of ξ at the bonds b' of I° in the cells of $\mathfrak{Z}_2\setminus\mathfrak{b}$. At these bonds $|\xi_{b'}|\le c_\xi\varepsilon$, so that $|\varphi_b| \le \mathsf{G}_0 c_\xi\varepsilon \le \varepsilon/6.5$. By Lem. 4.26 (ii) and (C3) of item D of §4.3, $|a_{0,b}| \le \mathsf{a}_{\mathrm{c}}\varepsilon'$. Consequently $|\eta_b| \le r_\eta := c_\eta^\flat\varepsilon \le r_o$. We now apply the following lemma.

Lemma 4.36 (The deviations of curvature and noise from their linear parts, in local form)**.** For the chart $U = \mathsf{U}(\eta;V')$ the following hold; each assertion involves only the η_b that it names.

(i) Let $\sigma_p := \sum_{b\in dp}|\eta_b|$. If p is not an edge plaquette and $\sigma_p<\pi$, then $|F_p - (\mathrm{d}^V\eta)_p| \le \sigma_p^2$. If p is an edge plaquette under P and $\sigma_p + \phi'_P < \pi$, then $|F_p - X'_P - (\mathrm{d}^V\eta)_p| \le (\sigma_p+\phi'_P)^2$; and always $\phi'_P \le \phi_p + \sigma_p$.

(ii) For each $r>0$ with $3Lr<\pi$, if $|\eta_b|\le r$ on the bonds of W_c, then $|n_c + (Q_1^V\eta)_c| \le 14L^2r^2$. ∎

Proof. (i). By Step 4 of the proof of Thm. 4.5,

$$\partial U(p) = e^{i\eta_1}e^{iT_2\eta_2}[\partial V'(P)]e^{-iT_3\eta_3}e^{-i\eta_4},$$

where the factor in brackets is present only for an edge plaquette under P. Here $\partial V'(P) = e^{iX'_P}$, and the T_j are isometries. The exponents add up to $(\mathrm{d}^V\eta)_p$ if p is not an edge plaquette, and to $X'_P + (\mathrm{d}^V\eta)_p$ if it is one. Lem. 4.20 (c) now gives the two bounds of (i), and Lem. 4.20 (a) gives its last assertion.

(ii). Let $c = (y, y')$ have direction μ, so that $y = c_-$, and let $x \in B(y)$ and $x' = x + Le_\mu$; here $h_{c,x}$, $\bar\tau_c$ and $\mathsf{M}(U)(c)$ are as in item C.I 4 of §3, and $\tau_{c,x} = \mathsf{L}(h_{c,x}h_{c,y}^{-1})$. We write $h_{c,x}V'(c)^{-1} = \prod_{b\in[x,x']} e^{i\tilde\eta_b} =: e^{i\mathsf{sg}_x}$, where $\tilde\eta = \mathrm{Ad}_{V'(c)}\,\eta$ in $B(c_+)$ and $\tilde\eta = \eta$ elsewhere. Then $|\mathsf{sg}_x| \le Lr$ and $|\mathsf{sg}_x - \sum\tilde\eta_b| \le L^2r^2$. Next, $\tau_{c,x} = \mathsf{L}(e^{i\mathsf{sg}_x}e^{-i\mathsf{sg}_y})$, whence $|\tau_{c,x} - \mathsf{sg}_x + \mathsf{sg}_y| \le 4L^2r^2$ and $|\bar\tau_c| \le 2Lr$. Further, $-n_c = \mathsf{L}(e^{i\bar\tau_c}e^{i\mathsf{sg}_y})$, whence $|n_c + \bar\tau_c + \mathsf{sg}_y| \le 9L^2r^2$. The mean over x of $\sum\tilde\eta_b$ equals $(Q_1^V\eta)_c$. Adding the three errors, with $1+4+9=14$, we obtain (ii). □

The hypotheses of Lem. 4.36 are met, since $6Lr_o < \pi$ and $\phi'_P \le \varepsilon'$. Therefore $|k_p| \le (4r_\eta + \varepsilon')^2 \le 25(c_\eta^\flat)^2\varepsilon^2$, because $L^{1/2} \le c_\eta^\flat$, and $|k_n| \le 14L^2(c_\eta^\flat)^2\varepsilon^2$. On these rows $\partial U(p) \ne -1$: the identity of Step 4 of the proof of Thm. 4.5 and Lem. 4.20 (a) give $\operatorname{ang}\partial U(p) \le \sigma_p + \phi'_P \le 4r_o + \varepsilon' < \pi$, since $6Lr_o < \pi$ by (U0$^\flat$) and $\varepsilon' \le 1/40$ by (FA). So L is the genuine logarithm there, and $\phi_p = |F_p|$. Here $F_p = \bar F_p + (\mathrm{d}^V\varphi)_p - k_p$, and, as Lem. 4.26 (ii) gives a tame box, Lem. 4.29 (b) yields $|\bar F_p| \le (\mathsf{N}_\sharp + 2)L^{-2}\varepsilon'$. With $\varepsilon' \le L^{1/2}\varepsilon$, (O4) (iii) and (Z0$^\flat$), it follows that

$$\phi_p \le |\bar F_p| + 4\varepsilon/6.5 + |k_p| \le [(\mathsf{N}_\sharp + 2)L^{-3/2} + 0.6154 + 0.3]\varepsilon \le 0.9954\,\varepsilon.$$

As a result, no rough plaquette lies on the rows named in Lem. 4.26 (viii), and on these rows $\zeta = F$ and $m_y = 1$.

Substep 3: summation by parts, and five rim terms. (i) We have

$$\langle\boldsymbol\nu, \mathfrak k\rangle_{\mathfrak Z} = \sum_{\hat{\mathfrak Z}}\langle\nu_{*p}, k_p\rangle + \sum_p\langle(\ell^{V*}\mathbf 1_{\mathfrak Z}\sigma_*)_p, k_p\rangle + a\langle\omega_*, k_n\rangle_{\mathfrak Z}.$$

For $p \in \hat{\mathfrak Z}'$, every y with $\ell_y(p) \ne 0$ lies in $\hat{\mathfrak Z}$, and so $\nu_{*p} + (\ell^{V*}\mathbf 1_{\mathfrak Z}\sigma_*)_p = Y_{*p}$. It follows that $\langle\boldsymbol\nu, \mathfrak k\rangle_{\mathfrak Z} = \langle Y_*, k_F\rangle_{\mathfrak Z'} + a\langle\omega_*, k_n\rangle_{\mathfrak Z'} + \mathrm{rim}_0$, where rim_0 lives on the plaquettes $p \in \widehat{\mathfrak Z^{+r_l}} \setminus \hat{\mathfrak Z}'$ and on the bonds $c \in \mathfrak Z \setminus \mathfrak Z'$. For the part of rim_0 that comes from the objects based in one cell,

$$x|\mathrm{rim}_0| \le [75v(N_1 + l_c\mathsf{N}_\sigma)L^{-2} + 168M^3L\mathsf{N}_Z]L^{1/2}(c_\eta^\flat)^2p_0^3x^{-1/2},$$

the numerals arising as $3v\cdot 25$ and $3M^3\cdot a\cdot 4\mathsf{N}_Z(aL)^{-1}\cdot 14L^2$. Using $Y_* = W^{V*}Z_* + \mathrm{d}^{V*}\Xi_*$ and $a\omega_* = \mathrm{d}'^{V*}Z_*$, we write the main part as

$$\langle Z_*, W^V\mathbf 1_{\mathfrak Z'}k_F + \mathrm{d}'^V\mathbf 1_{\mathfrak Z'}k_n\rangle + \langle\Xi_*, \mathrm{d}^V\mathbf 1_{\mathfrak Z'}k_F\rangle.$$

At $P, q \in \mathfrak S$ the cut by $\mathbf 1_{\mathfrak Z'}$ changes nothing. Indeed, $(W^V\mathbf 1_{\mathfrak Z'}k_F + \mathrm{d}'^V\mathbf 1_{\mathfrak Z'}k_n)_P$ involves the values of k_F at the p with $W(P,p) \ne 0$ and those of k_n at the four bonds of P, all of them within one block of y_P, and $(\mathrm{d}^V\mathbf 1_{\mathfrak Z'}k_F)_q$ involves the values of k_F at the six faces of q. As $\mathfrak S = \mathfrak b^{+r_1-r_l-1}$ and $\mathfrak Z' = \mathfrak S^{+1}$, these plaquettes, bonds and faces all lie in $\mathfrak Z'$. Since $W^VF^\flat + \mathrm{d}'^Vn^\flat = X' + d_1\eta$ and $\mathrm{d}^VF^\flat = \mathrm{d}^VEX' + d_2\eta$, we obtain there $(W^Vk_F + \mathrm{d}'^Vk_n)_P = \mathrm{d}e_P + (d_1\eta)_P$ and $(\mathrm{d}^Vk_F)_q = (\mathrm{d}^VEX')_q + (d_2\eta)_q - \mathrm{d}b_q$. The remaining P and q lie in $\mathfrak Z'^{+1} \setminus \mathfrak S$, and the term bd_1 is x times their contribution to the main part of $\langle\boldsymbol\nu, \mathfrak k\rangle_{\mathfrak Z}$. At such a P the contribution is at most

$$\mathsf{N}_ZL^{-1}\varepsilon'(L^{-1}12L^3\cdot 25 + 4\cdot 14L^2)(c_\eta^\flat)^2\varepsilon^2 = \mathsf{N}_ZL^{-1}\varepsilon'\,356L^2(c_\eta^\flat)^2\varepsilon^2,$$

with $3M^3$ plaquettes P to a cell; at such a q it is at most $\mathsf{N}_\Xi L^{-1}\varepsilon'\cdot 6\cdot 25(c_\eta^\flat)^2\varepsilon^2$, with v cubes to a cell. As $1068/L^2 \le 118.7$, this yields, for the part of bd_1 that comes from the objects based in one cell,

$$|\mathrm{bd}_1| \le x\varepsilon'\varepsilon^2(c_\eta^\flat)^2[1068\mathsf{N}_ZLM^3 + 150\mathsf{N}_\Xi v/L] \le 150L^{1/2}v(\mathsf{N}_Z + \mathsf{N}_\Xi)(1 + c_\eta^\flat)^2p_0^3x^{-1/2}.$$

(ii) The field $k(\chi_S\varphi)$ lives on rows within $\varrho_L + 2$ of $\hat{\mathfrak S}$, all of which belong to $\mathfrak Z$; hence $x\langle\Delta, k\varphi\rangle_{\mathfrak Z} = x\langle\Delta, k\chi_S\varphi\rangle_{\mathrm{Hil}} + \mathrm{bd}_2$. Moreover, $\langle\Delta, k\psi\rangle_{\mathrm{Hil}} = \langle u^T\,\mathsf g^VX', \psi\rangle - \langle Z_*, d_1\psi\rangle - \langle\Xi_*, d_2\psi\rangle$. Here $\mathrm{bd}_2 = x\langle\Delta, k(1-\chi_S)\varphi\rangle_{\mathfrak Z}$; the absolute row sums of k are at most 4, $4l_r$, and L with the weight a, while

$|\Delta_i| \le K_\Delta^\flat \varepsilon'^2$ and $|\varphi_b| \le \varepsilon/6.5$. Thus, for the part of bd_2 that comes from the objects based in one cell,

$$|\mathrm{bd}_2| \le K_\Delta^\flat L(12v + 4l_r v + 3aLM^3)/6.5\, p_0^3 x^{-1/2}.$$

On $\hat{\mathfrak{Z}}$ we have $|\varphi_b| < \pi(1 + a_\infty^\flat)$, and therefore

$$x|\langle u^T\, \mathrm{g}^V X', \chi_S \varphi\rangle| \le x\, x^{-50}\pi(1 + a_\infty^\flat)3v\#\mathfrak{Z} \le x^{-40}\#\mathfrak{Z}.$$

(iii) For $P, q \in \mathfrak{S}^-$, the bonds on which d_1 and d_2 depend lie in $\hat{\mathfrak{S}}$, and $\eta - \varphi = -a_0$. Hence

$$x\langle Z_*, d_1\eta\rangle_{\mathfrak{S}} - x\langle Z_*, d_1\chi_S\varphi\rangle + x\langle \Xi_*, d_2\eta\rangle_{\mathfrak{S}} - x\langle \Xi_*, d_2\chi_S\varphi\rangle = -x\langle Z_*, d_1 a_0\rangle_{\mathfrak{S}^-} - x\langle \Xi_*, d_2 a_0\rangle_{\mathfrak{S}^-} + \mathrm{bd}_3.$$

The term bd_3 lives at $P, q \in \mathfrak{S}^{+1} \setminus \mathfrak{S}^-$; at a P it is at most $\mathsf{N}_Z L^{-1}\varepsilon'\, 2\varepsilon'\, 2L \cdot 2c_\eta^\flat\varepsilon + \mathsf{N}_\Xi L^{-1}\varepsilon'\, 2\varepsilon' L \cdot 2c_\eta^\flat\varepsilon$. Therefore, as there are $3M^3$ plaquettes P to a cell, we obtain, for the part of bd_3 that the objects based in one cell contribute,

$$|\mathrm{bd}_3| \le x\varepsilon'^2\, 2c_\eta^\flat\varepsilon(4\mathsf{N}_Z + 2\mathsf{N}_\Xi)3M^3 \le 24Lv(\mathsf{N}_Z + \mathsf{N}_\Xi)c_\eta^\flat p_0^3 x^{-1/2}.$$

(iv) The two moduli of Substep 1 absorb the terms

$$x\langle Z_*, \mathrm{de}\rangle_{\mathfrak{S}} - x\langle \Xi_*, \mathrm{db}\rangle_{\mathfrak{S}}$$

. The term $x\,\mathrm{Rim}_{\mathfrak{Z}}$ lives in the rim, where $\mathrm{Rim}_{\mathfrak{Z}} = \sum |\sigma_{*y}|\,|\mathsf{dl}_y|$ and

$$|\mathsf{dl}_y| \le 2\varepsilon r_\eta \sum_p |\ell_y(p)|\,|\Gamma(y, \pi(p))| \le 2\varepsilon r_\eta\, 9K\mathsf{g}_1;$$

hence, for the part that comes from the objects based in one cell,

$$x\,\mathrm{Rim}_{\mathfrak{Z}} \le v\mathsf{N}_\sigma L^{-3/2} 18K\mathsf{g}_1 c_\eta^\flat p_0^3 x^{-1/2}$$

. For each of the five terms $x\,\mathrm{rim}_0$, bd_1, bd_2, bd_3 and $x\,\mathrm{Rim}_{\mathfrak{Z}}$, the share of one cell is at most $K_s^\flat p_0^3 x^{-1/2}$, as a comparison factor by factor shows, with $243 \le 200L^3$. What is left is Bal.

Substep 4: the balance and the volume term. We bound Bal by Lem. 4.34. Moreover, $x\,\mathrm{vol}_{\mathfrak{Z}} = x'L\,\mathrm{vol}_{\mathfrak{Z}}$ and $6N_1^2 \le 6.3(\mathsf{N}_\nu^\flat)^2$. This proves the claim of Step 7.

Step 8: a bound for the contribution of one loud set $\mathfrak{N}$ and one map β.

Proposition 4.37 (A bound for one loud set and one choice of tested bonds)**.** Let (J, t) be an admitted cell datum, let $\mathsf{Y} \subset \mathfrak{Z}_2$ and $0 \le \varsigma_1 \le x^{-2}$, let th be a Borel function of ξ with $0 \le \mathsf{th}(\xi) \le 1$, let $\mathfrak{N}$ be a set of blocks, and let β be a map that assigns to each $\square \in t$ a bond of I° in $\hat{\square}$. In the mean $E_\xi[\hat{\lambda}_{(J,t)}\, \mathsf{th}\, e^{\frac{x}{2}\varsigma_1|\xi_{\mathsf{Y}}|^2}]$, bound the product of the tests by $\prod_t b_\square \le \sum_\beta \prod 1\{|\xi_{\beta(\square)}| > c_\xi\varepsilon\}$. Then the contribution of the event $\{\text{loud set} = \mathfrak{N}\}$ and of the map β to this mean is

$$\le \mathrm{Res}\; e^{\sigma'\#\mathfrak{Z}_2}\, e^{-(\mathrm{py}/192)\#\mathfrak{N}}\, (e^{c_\tau - \rho_Q})^{\#\beta_Q},$$

where σ' is the left side of $(\mathrm{Z1}^\flat)$ without the term $27M^3 e^{-\mathrm{py}/384}$. Moreover, $\#\{\text{type L}\} \le (2r_1 + 1)^3\#\mathfrak{N}$. ∎

The loud set is introduced in Substep 5 of the proof, and the cells of type L and of type Q, together with the set β_Q, in Substep 6.

Proof. Since $\mathsf{th} \le 1$ and the integrand is nonnegative, we may drop th.

Substep 5: quiet and loud blocks; the splitting; the payments. A coarse plaquette P is quiet if $W_P \cap D \cap \hat{\mathfrak{Z}}$ is empty and if $\mathsf{n}_c \le r_o$ holds at each of its bonds c that lie in $\mathfrak{Z}$. A block is quiet if all twelve of its coarse plaquettes P are quiet, and loud otherwise. The set of the loud blocks, the loud set, lies inside $\mathfrak{Z}^{+1}$; on the event of the proposition it is $\mathfrak{N}$.

The splitting over $\mathfrak{N}$ and β. We fix the pair (V', e). Then $\eta = G\xi - a_0$ is affine in ξ, the chart $\mathsf{U}(\eta; V')$ is continuous in η, and $\phi_p = \mathrm{ang}\,\partial U(p)$ is continuous in U. By item 3.4 of §4.2, $\mathsf{M}(U)(c)$ is a Borel map of U; hence n_c is Borel in ξ. Consequently, for each set $\mathfrak{N}$ of blocks of $\mathfrak{Z}^{+1}$ the

set $\{\xi : \text{loud set} = \mathfrak{N}\}$ is Borel; there are finitely many such sets, they are pairwise disjoint, and together they cover the whole space.

The integrand is nonnegative and Borel. Indeed, at the fixed pair (V', e) the vectors $F^\flat$, $n^\flat$ and η are affine in ξ. The chart and all path products are continuous, the logarithm with value 0 at -1 is Borel, and the masks, cutoff factors and tests are indicators of finite combinations of open or closed sets. The block map M is Borel by §4.2, item 3.4, so its noise angle is Borel as well. The members and th are Borel by hypothesis. Every sum and product defining $\hat{\lambda}$ is finite. Thus splitting the integral over the finitely many loud sets and bounding the tests by the sum over β gives the contributions in the proposition.

Let Pq be the set of the plaquettes of $\hat{\mathfrak{Z}}$ whose corners all lie in quiet blocks, and Cq the set of the bonds of $\mathfrak{Z}$ whose two ends are quiet. A site row y of $\hat{\mathfrak{Z}}$ is quiet if every block within $\varrho_L + 2$ of $y(y)$ is quiet; Yq is the set of the quiet site rows, and qu is the union of Pq, Cq and Yq. We let $\mathrm{Bd}(\mathfrak{N})$ be the set of the bonds of I° that lie inside a loud block, or that cross between two blocks at least one of which is loud; a loud block has $\le 3L^3 + 6L^2 \le 5L^3$ of them. The rows are normalised by $k_c \to a^{1/2}k_c$. We set $\mathsf{sh}_p := -\Delta_p$, $\mathsf{sh}_y := -\Delta_y$, $\mathsf{sh}_c := a^{1/2}(-\Delta_c + \theta_2(1-\theta_2)^{-1}\omega_{*c})$, and $\mathsf{dv}_i := k_i\varphi - \mathsf{sh}_i$ for every row i.

(1) On the bonds based in $\mathfrak{Z}^{+r_l}$ off $\mathrm{Bd}(\mathfrak{N})$ we have $|\eta_b| \le 2r_o$. For a bond based in $\mathfrak{S}$, the block of the bond when the bond lies inside a block, or W_c when the bond crosses along c, lies in $\hat{\mathfrak{Z}}$ and is tame, and (Tc) (o), (i) in Step 3 of the proof of Thm. 4.19 bounds $\mathrm{ang}(e^{i\eta_b})$ at such a bond b. If b lies inside its block B, then (o) gives $\mathrm{ang}(e^{i\eta_b}) \le s_B \le 3L^3\varepsilon$; if b is a crossing bond of c, then (i) gives $\mathrm{ang}(e^{i\eta_b}) \le 9s_c + \mathsf{n}_c$ with $s_c \le 6L^3\varepsilon$ and $\mathsf{n}_c \le r_o$, the latter because c lies in $\mathfrak{Z}$ and is a bond of a quiet P. In both cases $\mathrm{ang}(e^{i\eta_b}) \le 54L^3\varepsilon + r_o$. To pass from the angle to the modulus, note that $b \in I^\circ \cap \hat{\mathfrak{Z}}$; hence $|\eta_b| < \pi$ by the indicator $1\{|\eta_b| < \pi\}$ in $[A]^\ell_{\mathfrak{Z}}$ (item E11 of §4.3), so that $\mathsf{L}(e^{i\eta_b}) = \eta_b$ by the definition of L in item C.I 4 of §3, and $|\eta_b| = \mathrm{ang}(e^{i\eta_b})$ by Lem. 4.20. Finally, $54L^3\varepsilon + r_o \le 2r_o$, since $60L^3\varepsilon \le r_o$ by $(\mathrm{U0}^\flat)$. For a bond based in $\mathfrak{Z}^{+r_l} \setminus \mathfrak{S}$, which lies inside Lr, the bound is that of Substep 2 of Step 7 of the proof of Thm. 4.19.

Consequently every $p \in \mathsf{Pq}$ has $p \notin D$ and $|k_p| \le (17r_o)^2$, the latter because $\phi'_P \le \phi_p + \sigma_p \le 9r_o$, where $\sigma_p = \sum_{b \in dp} |\eta_b| \le 8r_o$ is the number that enters Lem. 4.36 (i). Every $c \in \mathsf{Cq}$ has $c \notin \mathsf{Cb}$, $\mathsf{n}_c \le r_o \le t_0$ and $|k_n| \le 56L^2r_o^2$. Every $y \in \mathsf{Yq}$ has $m_y = 1$ and $\zeta = F$ on its reach; moreover $\ell^U_y\zeta - \sigma_{*y} = (k\varphi)_y + \Delta_y - (\ell^V k_F)_y + \mathsf{dl}_y$, where the sum of the last two terms of the right side, the rest, satisfies $|\mathrm{rest}| \le 289 l_r r_o^2 + 36K\mathsf{g}_1\varepsilon r_o$.

The reach of a quiet site row, including the part outside $\hat{\mathfrak{Z}}$. Take $y \in \mathsf{Yq}$ and a plaquette p with $|y - p| \le \varrho(x) + 1$. The corners of p and the sites of $\Gamma(y, \pi(p))$ lie within $\varrho(x) + 2$ of y, and so, as $L \ge 3$, in blocks within $\varrho_L + 1$ of $y(y)$. These blocks are quiet, and they lie in $\mathfrak{Z}^{+r_l}$, since $\varrho_L + 2 \le r_l M$. Therefore the bonds of p and of $\Gamma(y, \pi(p))$ are based in $\mathfrak{Z}^{+r_l}$ and lie off $\mathrm{Bd}(\mathfrak{N})$, and the first assertion of (1) gives $|\eta_b| \le 2r_o$ on them.

If $y \in \hat{\mathfrak{Z}}'$, then $p \in \hat{\mathfrak{Z}}$ by item (1) of Step 3 of the proof of Thm. 4.19, and $p = (z; \mu\nu) \in W_P$ for $P = (y(z); \mu\nu)$, which is one of the twelve P of the quiet block $B(y(z))$; hence $p \notin D$. If $y \in \hat{\mathfrak{Z}} \setminus \hat{\mathfrak{Z}}'$, then y is a row of $\mathrm{Rim}_{\mathfrak{Z}}$; on the support of $\hat{\lambda}$ the tests of $\mathfrak{Z}_2 \setminus \mathfrak{b}$ pass, and by Lem. 4.26 (viii) and the last assertion of Substep 2 of Step 7 of the proof of Thm. 4.19 no rough plaquette lies within $\varrho(x) + 1$ of y. In either case $\zeta_p = F_p$; and $m_y = 1$, because $\rho_0 \le \varrho(x)$.

Further, $\sigma_p \le 8r_o$ and $\phi'_P \le \phi_p + \sigma_p \le 9r_o$. Lem. 4.36 (i), which applies because $17r_o < \pi$ by $(\mathrm{U0}^\flat)$, then gives $|k_p| \le (17r_o)^2$, also at the plaquettes p based off $\hat{\mathfrak{Z}}$; hence $|(\ell^V k_F)_y| \le 289 l_r r_o^2$. The estimate of item (iv) of Substep 3 of Step 7 of the proof of Thm. 4.19, here with $|\eta_b| \le 2r_o$ on the bonds of the paths, gives $|\mathsf{dl}_y| \le 2\varepsilon\, 2r_o \sum_p |\ell_y(p)|\, |\Gamma(y, \pi(p))| \le 36K\mathsf{g}_1\varepsilon r_o$. When the rows are empty, Yq is empty.

(2) We have $F_p - \nu_{*p} = \mathsf{dv}_p - k_p$ and $n_c - \tilde\omega_c = a^{-1/2}\mathsf{dv}_c - k_n$. For vectors u, r one has $|u+r|^2 \ge (1-\check\theta_x)|u|^2 - \check\theta_x^{-1}|r|^2$; this and (U0$^\flat$) give

$$x\,\mathrm{Sq}_{\mathfrak Z} \ge \frac{x}{2}(1-2\check\theta_x)\sum_{\mathsf{qu}}|\mathsf{dv}_i|^2 + [\text{the squares of the bonds with } r_o < \mathsf{n}_c \le t_0] - r_\theta,$$

with $r_\theta \le Q^\flat_\theta x^{-1/4}\#\mathfrak Z$. The numerals behind this bound are these: for one block, the plaquette rows give $\frac{x}{2}x^{1/4}17^4x^{-3/2}3L^3 \le 1.3\cdot10^5L^3x^{-1/4}$, the bond rows $4704aL^4x^{-1/4}$, and the site rows $L^3(83521l_r^2 + 1296K^2\mathsf{g}_1^2p_0^2x^{-1/4})x^{-1/4}$. A cell has M^3 blocks, and $p_0^2x^{-1/4}\le 1$ by (U0$^\flat$); hence the sum for one cell is at most $Q^\flat_\theta x^{-1/4}$, where $Q^\flat_\theta$ is the constant of item F of §4.3.

(3) Every rough plaquette of $\hat{\mathfrak Z}$ pays $\ge d_0p_0^2/2$ out of $x\,\mathrm{Pay}$. A bond of $\mathfrak Z$ with $\mathsf{n}_c > r_o$ has loud ends, does not belong to Cq, and keeps $\ge 0.05ax\mathsf{n}_c^2 \ge 0.05ax^{1/4}$: if $\mathsf{n}_c > t_0$, this is taken from Pay, and otherwise $\mathsf{kc}_c|n_c-\tilde\omega_c|^2 \ge \frac{a}{4}(0.9\mathsf{n}_c)^2$. One object makes at most 48 blocks loud (12 boxes with 4 corners each); so the payments are $\ge (\mathrm{py}/48)\#\mathfrak N$.

(4) For $b\in\mathrm{Bd}(\mathfrak N)$ we have $|\eta_b|<\pi$, by the indicator or by Substep 2 of Step 7 of the proof of Thm. 4.19, because a bond of $\mathrm{Bd}(\mathfrak N)$ that is not based in $\hat{\mathfrak Z}$ is based in the rim Lr. Indeed, the distance in blocks from any loud block to $\mathfrak Z$ is ≤ 1; hence the base of a crossing bond with an end in a loud block has distance in blocks $\le 2 \le M-1$ to $\mathfrak Z$, and so it lies in $\mathfrak Z^{+1}$; and $\mathfrak Z^{+1}\setminus\mathfrak Z = \mathfrak b^{+r_1+1}\setminus\mathfrak b^{+r_1}\subset \mathrm{Lr}$ because $r_l\ge 1$. Hence $1 \le e^{\frac{x}{2}\kappa_1\pi^2}e^{-\frac{x}{2}\kappa_1|\varphi_b-a_{0,b}|^2}$ on these bonds, at the cost $\frac{x}{2}\kappa_1\pi^2\,5L^3\#\mathfrak N = (\mathrm{py}/96)\#\mathfrak N$.

Substep 6: the tested bonds; type L and type Q; the tilt. As $\xi = G^{-1}\varphi = G^{\mathsf T}(q+S)\varphi$, we have $\xi_b = \sum_i h(b,i)k_i\varphi + (G^{\mathsf T}S\varphi)_b$ with $h(b,i) = (G^{\mathsf T}k_i^*)(b)$, and $h(b,i)$ is 0 beyond $r_w+\varrho_L+2$. The row sums of h are $\le \mathsf G_0(4+4l_c+a^{1/2}L^{-2})$ and its column sums are $\le \mathsf G_0\max\{4,4l_r,a^{1/2}L\}$; both bounds are $\le h_*$.

A cell $\square\in t$ is of type L if a loud block lies within $r_w+2\varrho_L+5$ ($\le r_1M$) of $\beta(\square)$; this gives $\#\{\text{type L}\}\le(2r_1+1)^3\#\mathfrak N$, the last assertion of the proposition. All other cells of t are of type Q. We write β_Q for the bonds of type Q and Π_β for the projection on them. For a bond b of type Q, every row i with $h\ne0$ is a row of $\mathfrak Z$; such a row is quiet and is based in $\mathfrak Z_2$, within $r_w+\varrho_L+2$ of a site off f_J. By Lem. 4.26 (ii), Lems. 4.29 and 4.25, and (Z0$^\flat$), $\sum_i|h(b,i)|\,|\mathsf{sh}_i|\le c_\xi\varepsilon/4$.

Therefore $|\xi_b|>c_\xi\varepsilon$ gives $2|\sum h\,\mathsf{dv}_i|^2 + 2|(G^{\mathsf T}S\varphi)_b|^2 > \frac{9}{16}c_\xi^2\varepsilon^2$, and $|\sum h\,\mathsf{dv}_i|^2 \le h_*\sum|h|\,|\mathsf{dv}_i|^2$. Chebyshev's inequality with the parameter $\frac{x}{2}t_Q$ yields

$$\prod_{\beta_Q}1\{\dots\} \le \exp\{\frac{x}{2}\sum_{\mathsf{qu}}\tau_i|\mathsf{dv}_i|^2 + \frac{x}{2}2t_Q\langle\varphi, SG\Pi_\beta G^{\mathsf T}S\varphi\rangle - \rho_Q\#\beta_Q\},$$

where $\tau_i := 2t_Qh_*\sum_{b\in\beta_Q}|h(b,i)| \le \frac16$ for $i\in\mathsf{qu}$, while τ_i is 0 for all other rows; moreover $\sum\tau_i \le \#\beta_Q/6$ and $\frac{9}{32}t_Qc_\xi^2p_0^2 = \rho_Q$.

The tilt. In the collar, $|\xi_b|\le c_\xi\varepsilon$, which gives one factor $e^{\frac32 v\varsigma_1c_\xi^2p_0^2}$ for every cell of $\mathsf Y$ that lies in the collar. For $b\in\hat{\mathfrak b}$, $(G^{\mathsf T}q\varphi)_b$ depends on φ within $r_w+r_q\le r_1M$, hence in $\hat{\mathfrak Z}$, where $|\varphi|<\pi(1+a^\flat_\infty)$; so $|(G^{\mathsf T}q\varphi)_b|\le c_q^\flat$ and $|\xi_b|^2\le 2(c_q^\flat)^2 + 2|(G^{\mathsf T}S\varphi)_b|^2$, whence one factor $e^{3vx\varsigma_1(c_q^\flat)^2}$ for each cell of $\mathsf Y$ in $\mathfrak b$. We put $S' := 2t_QSG\Pi_\beta G^{\mathsf T}S + 2\varsigma_1SG\mathbf 1_{\hat{\mathfrak b}}G^{\mathsf T}S \ge 0$; then $\|S'\|\le(2t_Q+2\varsigma_1)(2/\gamma)x^{-120}\le x^{-110}$.

Substep 7: the Gaussian integral; three quotients of determinants. Let

$$\sigma'' := \sigma^\flat_{\mathrm{tot}} + 5K_s^\flat p_0^3x^{-1/2} + x^{-40} + (6.3(\mathsf N_\nu^\flat)^2\vartheta_F^\flat + 3\mathsf N_Z^2(\theta_2+L\vartheta_n^\flat))M^3p_0'^2 + Q^\flat_\theta x^{-1/4} + \frac32 vc_\xi^2p_0^2x^{-2} + 3v(c_q^\flat)^2x^{-1}.$$

The terms up to $M^3p_0'^2$ form the bracket of Step 7 of the proof of Thm. 4.19. The term $Q^\flat_\theta x^{-1/4}$ takes care of r_θ of item (2) of Substep 5, as $\#\mathfrak Z\le\#\mathfrak Z_2$. The last two terms are the tilt of Substep 6 for $\varsigma_1\le x^{-2}$: a cell of $\mathsf Y$ lies in the collar or in $\mathfrak b$, and the sum covers both cases. The number σ'' depends only on x and on the letters (O0) to (O10).

The exponent $-(\mathrm{py}/96)\#\mathfrak{N}$ below combines $-(\mathrm{py}/48)\#\mathfrak{N}$ of item (3) of Substep 5 with $+(\mathrm{py}/96)\#\mathfrak{N}$ of item (4) of Substep 5. The integrand is at most

$$\mathrm{Res}\,\exp\{\sigma''\#\mathfrak{Z}_2 - (\mathrm{py}/96)\#\mathfrak{N} - \rho_Q\#\beta_Q - \frac{x}{2}\Phi(\varphi)\},$$

where $\Phi = \langle\varphi,(S-S')\varphi\rangle + \Psi$; here $\mathsf{co}_i := 1 - 2\check{\theta}_x - \tau_i \geq \frac{1}{2}$ and

$$\Psi(\varphi) := \|k\varphi\|^2 \text{ over the rows off } \mathfrak{Z} + \sum_{\mathsf{qu}} \mathsf{co}_i|k_i\varphi - \mathsf{sh}_i|^2 + \kappa_1 \sum_{\mathrm{Bd}(\mathfrak{N})} |\varphi_b - a_{0,b}|^2,$$

and Ψ_2 denotes the quadratic part of Ψ. We also set $\mathsf{Ez} := \sum_{\mathsf{qu}} \mathsf{co}_i|\mathsf{sh}_i|^2 + \kappa_1 \sum_{\mathrm{Bd}(\mathfrak{N})} |a_{0,b}|^2$ and $\mathbb{A}_2 := (S-S') + \Psi_2$.

Coercivity. In Ψ_2, each plaquette row whose corners all lie in quiet blocks, and each bond row whose two ends are quiet, appears with a coefficient $\geq \frac{1}{2}$, anywhere on the torus. These rows depend on the bonds off $\mathrm{Bd}(\mathfrak{N})$ alone, and on those bonds their sum is $\geq \gamma$, by Lem. 4.38, which we state and prove now.

Lemma 4.38 (A lower bound for the form away from the loud blocks)**.** For every $\varphi \in \mathfrak{L}_{\mathfrak{g}}$,

$$\sum_{b\in I^\circ \text{ off } \mathrm{Bd}(\mathfrak{N})} |\varphi_b|^2 \leq 73L^4\Big[\sum_{p \text{ with all corners in quiet blocks}} |(\mathrm{d}^V\varphi)_p|^2 + a \sum_{c \text{ with quiet ends}} |(Q_1^V\varphi)_c|^2\Big].$$

■

Proof. Apply eq. 2.i to each quiet block and eq. 2.ii to each pair c with quiet ends, after the blockwise unitary transports used in the proof of Lem. 3.3(a). These estimates give, respectively,

$$\sum_{b \text{ inside } B(y)} |\varphi_b|^2 \leq \tfrac{1}{2}L^3 \sum_{p \text{ inside } B(y)} |(\mathrm{d}^V\varphi)_p|^2$$

and

$$\sum_{b \text{ crossing } c} |\varphi_b|^2 \leq 6L^2|(Q_1^V\varphi)_c|^2 + 12L^4 \sum_{p \text{ inside } W_c} |(\mathrm{d}^V\varphi)_p|^2.$$

Both estimates use only the bonds inside the indicated block or pair of blocks. A bond of I° off $\mathrm{Bd}(\mathfrak{N})$ lies in a quiet block, or it is a crossing bond of some c with two quiet ends. We sum the first bound over the quiet blocks and the second over these c; a plaquette lies in at most six of the W_c. Finally, $L^3 + 72L^4 \leq 73L^4$ and $6L^2 \leq 73L^4 a$. □

Hence $\Psi_2 \geq \min\{\gamma/2, \kappa_1\} = \kappa_1$. From $|u-e|^2 \geq \frac{1}{2}|u|^2 - |e|^2$ for vectors u, e we obtain $\Psi \geq \frac{1}{2}\langle\varphi,\Psi_2\varphi\rangle - \mathsf{Ez}$, and so $|\varphi|^2 \leq (2/\kappa_1)(\Psi + \mathsf{Ez})$. We have $s_S = \|S\| + \|S'\| \leq 2x^{-60} \leq \kappa_1/2$. Indeed, $\kappa_1 = \mathrm{py}/(240\pi^2L^3x) \geq (2369L^3x)^{-1}$ because $\mathrm{py} \geq 1$ (F9$^\flat$), and $4\cdot 2369L^3x \leq x^{60}$ because $10^6v^2 \leq x^8$ (F9$^\flat$). It follows that $\Phi \geq \Psi(1 - 2s_S/\kappa_1) - 2s_S\mathsf{Ez}/\kappa_1 \geq -2s_S\mathsf{Ez}/\kappa_1$.

At every V' we have

$$|\mathsf{sh}_i| \leq (1+a^{1/2})(1+l_r)(\pi(1+(4+L)a_\infty^\flat)+1).$$

For this, note that $|\bar{F}_p| \leq \pi + 4\pi a_\infty^\flat$, $|(\ell^V\bar{F}_F)_y| \leq l_r(\pi + 4\pi a_\infty^\flat)$ and $|(Q_1^V a_0)_c| \leq L\pi a_\infty^\flat$ (Lem. 4.29 (a)), and that $\mathsf{sh}_c = a^{1/2}(-(Q_1^V a_0)_c + \omega_{*c}/(1-\theta_2))$. The summand 1 covers ν_*, σ_* and ω_*, by Lem. 4.25 and $\varepsilon' \leq 1/40$ (FA): $|\nu_*| \leq N_1L^{-2}\varepsilon' \leq \mathsf{N}_\nu^\flat L^{-3/2} \leq 1$ by (L1$^\flat$); $|\sigma_*| \leq \mathsf{N}_\sigma L^{-2}\varepsilon' \leq \mathsf{N}_\sharp L^{-2} \leq 1$, because $3K\mathsf{g}_* \leq 1 + 15K^2\mathsf{g}_*^2 \leq \mathsf{P}_\rho$, and by (O4) (iii); and $|\omega_*|/(1-\theta_2) \leq 1.03\cdot\frac{2}{9}\mathsf{N}_Z L^{1/2}\varepsilon \leq 1$ (F5).

With $(\dots) = \pi(1+(4+L)a_\infty^\flat)+1$, these components give $|\mathsf{sh}_p| \leq (\dots)$, $|\mathsf{sh}_y| \leq (1+l_r)(\dots)$ and $|\mathsf{sh}_c| \leq a^{1/2}(\dots)$: neither sh_p nor sh_y has a factor $a^{1/2}$, which only sh_c carries. So $|\mathsf{sh}_i|^2 \leq \max\{1,a\}(1+l_r)^2(\dots)^2 \leq (1+a)(1+l_r)^2(\dots)^2$. A cell carries at most $4.12v$ rows, and the second sum of Ez is at most $\kappa_1\, 5L^3\, 27M^3(\pi a_\infty^\flat)^2$ per cell; this number is at most $(5v - 4.12v)(1+a)(1+l_r)^2(\dots)^2$, which is what these rows leave of the bound $5v(1+a)(1+l_r)^2(\dots)^2$ for one cell. For the

last inequality, note that $5L^3 \cdot 27M^3 = 135v$, that $(\dots) \geq \pi(4+L)a_\infty^\flat \geq 7\pi a_\infty^\flat$ as $L \geq 3$, and that $\kappa_1 \leq \gamma/2$ by (U0$^\flat$), where $\gamma = (73L^4)^{-1}$. Hence $135\kappa_1(\pi a_\infty^\flat)^2 \leq 0.88 \cdot 49(\pi a_\infty^\flat)^2$, and therefore

$$\mathsf{Ez} \leq 5v\#\mathfrak{Z}\,(1+a)(1+l_r)^2(\pi(1+(4+L)a_\infty^\flat)+1)^2,$$

and, by (F9$^\flat$),

$$\frac{x}{2}\,2s_S\mathsf{Ez}/\kappa_1 \leq 2.2\cdot 10^6 L^3 v(1+a)(1+l_r)^2(1+La_\infty^\flat)^2 x^{-58}\#\mathfrak{Z} \leq x^{-40}\#\mathfrak{Z}_2.$$

The numerals: $s_S \leq 2x^{-60}$; $1/\kappa_1 \leq 2369L^3x$; $4+L \leq \frac{7}{3}L$ gives $\pi(1+(4+L)a_\infty^\flat)+1 \leq 7.34(1+La_\infty^\flat)$; and $2\cdot 2369\cdot 5\cdot 53.9 \leq 1.28\cdot 10^6$.

About its minimiser, $\Phi = \min\Phi + \langle\cdot, \mathbb{A}_2\cdot\rangle$, where $\mathbb{A}_2 \geq \kappa_1 - s_S \geq \kappa_1/2 > 0$; hence the integral is $\leq e^{x^{-40}\#\mathfrak{Z}_2}(\det C^{-1}/\det\mathbb{A}_2)^{1/2}$.

The determinants. Let $\mathbb{A}_1 := (S-S') + \sum_{i \text{ off } \mathfrak{Z}} k_i^* k_i + \sum_{i\in\mathfrak{Z}} \mathsf{co}_i k_i^* k_i$, and let $\mathbb{A}_{3/2} := \mathbb{A}_1 - \sum \mathsf{co}_y k_y^* k_y$, where the sum runs over the site rows y of $\hat{\mathfrak{Z}}$ that are not quiet. Then

$$\det C^{-1}/\det\mathbb{A}_2 = (\det C^{-1}/\det\mathbb{A}_1)(\det\mathbb{A}_1/\det\mathbb{A}_{3/2})(\det\mathbb{A}_{3/2}/\det\mathbb{A}_2),$$

and we bound the three quotients one after the other.

First. We have $C^{-1} - \mathbb{A}_1 = S' + \sum_{\mathfrak{Z}}(1-\mathsf{co}_i)k_i^*k_i \geq 0$, as S cancels; so the first quotient is $\leq \exp\operatorname{tr}[\mathbb{A}_1^{-1}(C^{-1}-\mathbb{A}_1)]$. This rests on the following fact: for symmetric $0 < A_2 \leq A_1$, $\det A_1/\det A_2 = \det(1+X) \leq e^{\operatorname{tr}X}$, where $X = A_2^{-1/2}(A_1-A_2)A_2^{-1/2} \geq 0$ and $\operatorname{tr}X = \operatorname{tr}[A_2^{-1}(A_1-A_2)]$; "Second" uses it as well. Next, $\mathbb{A}_1 \geq (1-2s_S/\gamma)\sum \mathsf{co}_j k_j^* k_j \geq \gamma/4$, and for the plaquette and bond rows $k_i\mathbb{A}_1^{-1}k_i^* \leq 2/(1-2s_S/\gamma)$ on $\mathfrak{g}$. The lower bound for $\mathbb{A}_1$ holds because $\mathsf{co}_j = 1$ off $\mathfrak{Z}$ makes $\sum \mathsf{co}_j k_j^* k_j \geq q/2 \geq \gamma/2$, while $S - S' \geq -s_S$. The upper bound for the rows holds because $A > 0$, $A \geq c\,k^*k$ and $c > 0$ imply $kA^{-1}k^* \leq 1/c$: for $B = kA^{-1/2}$ we have $cB^*B \leq 1$, hence $cBB^* \leq 1$; here $c = (1-2s_S/\gamma)\mathsf{co}_i \geq (1-2s_S/\gamma)/2$.

The trace of a plaquette or bond row is ≤ 6.1: there are three components, and $6/(1-2s_S/\gamma) \leq 6.1$ requires $2s_S/\gamma \leq 1/61$. This holds: $s_S \leq 2x^{-60}$ and $\gamma^{-1} \leq x^4$ by (F9$^\flat$), so that $2s_S/\gamma \leq 4x^{-56} \leq 1/61$, because $x \geq L^2\mathrm{e} \geq 24$. The same inequality yields $\mathbb{A}_1 \geq \gamma/4$ and $\mathbb{A}_{3/2} \geq \gamma/4$. For the site rows we use a bound free of the dimension of H_y: $\operatorname{tr} \leq (4/\gamma)\operatorname{tr} k_y k_y^* \leq (4/\gamma)\,3{\cdot}4{\cdot}4\sum_p |\ell_y(p)|^2 \leq c_{\mathrm{tr}}$. Both trace bounds are $\leq 12c_\tau$, and a cell carries at most $5v$ rows; so the square root of the first quotient is $\leq e^{c_\tau\#\beta_Q + 60c_\tau v\check{\theta}_x\#\mathfrak{Z} + x^{-40}\#\mathfrak{Z}_2}$.

The term $x^{-40}\#\mathfrak{Z}_2$ arises as follows: $S' \geq 0$ has rank $\leq 3\#\beta_Q + 9v\#\mathfrak{b} \leq 12v\#\mathfrak{Z}_2$ and norm $\leq x^{-110}$ (Substep 6), and $\mathbb{A}_1^{-1} \leq 4/\gamma$; hence $\frac{1}{2}\operatorname{tr}[\mathbb{A}_1^{-1}S'] \leq (2/\gamma)12vx^{-110}\#\mathfrak{Z}_2 \leq x^{-40}\#\mathfrak{Z}_2$, because $24v/\gamma \leq x^8$ (F9$^\flat$). A cell thus receives three terms x^{-40}, namely the one of Step 7 of the proof of Thm. 4.19, the one with Ez, and this one; and $3x^{-40} \leq x^{-30}$, the last term of (Z1$^\flat$).

Second. Here $\mathbb{A}_1 - \mathbb{A}_{3/2} \geq 0$, and each plaquette row and each bond row of the torus enters $\mathbb{A}_{3/2}$ with a coefficient $\geq \frac{1}{2}$; thus $\mathbb{A}_{3/2} \geq \frac{1}{2}q_0^V - s_S \geq \gamma/4$. The second quotient is $\leq \exp\{(4/\gamma)\sum \operatorname{tr} k_y k_y^*\} \leq \exp\{c_{\mathrm{tr}}\#(\text{rows})\}$, with at most $L^3(2\varrho_L+5)^3\#\mathfrak{N}$ rows; so its square root is $\leq e^{(c_{\mathrm{tr}}/2)L^3(2\varrho_L+5)^3\#\mathfrak{N}}$.

Third. Here $\mathbb{A}_{3/2} - \mathbb{A}_2 = \sum \mathsf{co}_i k_i^* k_i - \kappa_1 \mathbf{1}_{\mathrm{Bd}(\mathfrak{N})}$, the sum running over the plaquette and bond rows of $\mathfrak{Z}$ off qu, and the rank of this difference is $\leq 3[7.2L^3+6+5L^3]\#\mathfrak{N} \leq 38L^3\#\mathfrak{N}$. Indeed, a plaquette row off Pq has a corner in a loud block, so its base site lies in that block enlarged by one layer below: $3(L+1)^3 \leq 7.2L^3$ for $L \geq 3$, since $3(4/3)^3 \leq 7.12$. A bond row off Cq has a loud end, and there are six of these to a block. Moreover $\#\mathrm{Bd}(\mathfrak{N}) \leq 5L^3\#\mathfrak{N}$ (Substep 5). Each row takes values in $\mathfrak{g}$, whence a factor 3; and $3(12.2L^3+6) = 36.6L^3 + 18 \leq 38L^3$ for $L \geq 3$.

Further, $\|\mathbb{A}_{3/2}\| \le \|C^{-1}\| \le b_h + 1$. We apply Lem. 2.25 (b) with $A_1 = \mathbb{A}_{3/2}$, $A_2 = \mathbb{A}_2 \ge \kappa_1/2$, $r \le 38L^3\#\mathfrak{N}$. The resulting bound does not depend on the hypothesis $\|A_1\|\,\|A_2^{-1}\| \ge 1$ of that lemma: by the proof of Lem. 2.25 (b), all but r eigenvalues of $A_2^{-1/2}A_1A_2^{-1/2}$ equal 1, and the others are $\le \|A_1\|\,\|A_2^{-1}\| \le 2(b_h+1)/\kappa_1$, a number ≥ 1 because $b_h \ge 16$ and $\kappa_1 \le \gamma/2$. So $\det A_1/\det A_2 \le (2(b_h+1)/\kappa_1)^r$, and the square root of the third quotient is $\le (2(b_h+1)/\kappa_1)^{19L^3\#\mathfrak{N}}$.

By (U1$^\flat$), "Second" and "Third" together contribute $\le e^{(\mathrm{py}/192)\#\mathfrak{N}}$; and $\mathrm{py}/48 - \mathrm{py}/96 - \mathrm{py}/192 = \mathrm{py}/192$.

The tally for one cell of $\mathfrak{Z}_2$. To σ'', Substep 7 adds x^{-40} from the term with Ez, and $60c_\tau v\check\theta_x = 60c_\tau v x^{-1/4}$ and x^{-40} from "First", where $\#\mathfrak{Z} \le \#\mathfrak{Z}_2$. Further, $\frac{3}{2}vc_\xi^2p_0^2x^{-2} + 3v(c_q^\flat)^2x^{-1} \le 3v((c_q^\flat)^2 + c_\xi^2p_0^2)x^{-1}$ and $3x^{-40} \le x^{-30}$. Hence

$$\begin{aligned}
&\sigma'' + 60c_\tau v x^{-1/4} + 2x^{-40}\\
&\quad\le \sigma^\flat_{\mathrm{tot}} + 5K_s^\flat p_0^3 x^{-1/2}\\
&\quad + (6.3(\mathsf{N}_\nu^\flat)^2\vartheta_F^\flat + 3\mathsf{N}_Z^2(\theta_2 + L\vartheta_n^\flat))M^3p_0'^2\\
&\quad + (Q_\theta^\flat + 60c_\tau v)x^{-1/4}\\
&\quad + 3v((c_q^\flat)^2 + c_\xi^2p_0^2)x^{-1} + x^{-30}\\
&\quad\le \sigma'.
\end{aligned}$$

□

Step 9: clauses (c°), (c), (c′), (c″). Of the clauses of Thm. 4.19, as stated in item H of §4.3, each of Steps 9 to 12 proves those that its head names. Clause (c°) is not a case of (c): when $\mathfrak{X}$ is empty, (c) bounds $\hat\lambda\,1\{D(U)\cap\hat{\mathfrak{Z}}\text{ empty}\}$, which is a smaller function.

(c°) We apply Prop. 4.37 with $\mathsf{th} = 1$. There are at most $(3v)^{\#t}$ maps β. We fix $\mathfrak{N}$ and β and let n_L be the number of cells of type L, so that $\#\beta_Q + n_L = \#t$. We split $\frac{1}{192} = \frac{1}{384} + \frac{1}{768} + \frac{1}{768}$ and use the parts as follows.

(i) By the last assertion of Prop. 4.37, $e^{-(\mathrm{py}/768)\#\mathfrak{N}} \le \exp\{-\mathrm{py}\,n_L/(768(2r_1+1)^3)\}$.

(ii) If $n_L \ge 1$, then $\mathfrak{N}$ is not empty, and $e^{-(\mathrm{py}/768)\#\mathfrak{N}} \le e^{-\mathrm{py}/768}$.

(iii) We write $e^{-\rho_Q\#\beta_Q} = e^{-(\rho_Q/2)\#\beta_Q}e^{-(\rho_Q/2)\#\beta_Q}$; the second factor is $\le e^{-\rho_Q/2}$ if $\#\beta_Q \ge 1$.

If t is not empty, then $n_L \ge 1$ or $\#\beta_Q \ge 1$, and (ii), (iii) give u_1. What is left for the tested cells is $m_8^{\#t}$, with $m_8 = \max\{e^{c_\tau - \rho_Q/2}, e^{-\mathrm{py}/(768(2r_1+1)^3)}\}$, and the sum over β gives $(3vm_8)^{\#t} = u_t^{\#t}$. The sum over $\mathfrak{N}$ obeys

$$\sum_{\mathfrak{N}} e^{-(\mathrm{py}/384)\#\mathfrak{N}} \le \exp\{e^{-\mathrm{py}/384}M^3\#\mathfrak{Z}^{+1}\} \le \exp\{27M^3e^{-\mathrm{py}/384}\#\mathfrak{Z}_2\}.$$

All these numbers, each counted for one cell of $\mathfrak{Z}_2$, add up to the left side of (Z1$^\flat$), which is $\le x^{-1/8}$.

(c) On the support of 1_{ind} each X holds a p with $\phi_p > \varepsilon$; since $x\varepsilon\cdot\varepsilon = p_0^2$, sentence 4 of Thm. 4.19 (a) gives $|\lambda^o| \le \hat\lambda$ times constants. Then (c°) applies.

(c′) We apply Prop. 4.37 with $\mathsf{th} = 1_{\mathrm{ind}}$. On the support of this indicator each X holds a plaquette $p_X = (z;\mu\nu) \in D\cap\hat{\mathfrak{Z}}$; as $p_X \in W_P$ for $P = (y(z);\mu\nu)$, this P is loud, and so is $B(y(z))$. Two polymers are more than $s(x) \ge 8$ cubes apart, at fine distance $\ge 8M+1 > L-1$, so their blocks $B(y(z))$ differ; hence only the contributions with $\#\mathfrak{N} \ge \#\mathfrak{X}$ are nonzero. Now we split $\frac{1}{192} = \frac{1}{384} + \frac{3}{1152}$: the three factors $e^{-(\mathrm{py}/1152)\#\mathfrak{N}}$ give $\exp\{-\mathrm{py}\,n_L/(1152(2r_1+1)^3)\}$, then $e^{-\mathrm{py}/1152}$ if $n_L \ge 1$, and $e^{-(\mathrm{py}/1152)\#\mathfrak{X}}$. The rest of

the argument is that for (c°), with u'_t, u'_1 in place of u_t, u_1. The weak cap is the constant $Ae^{-\kappa(|X|_c-1)}$.

(c″) Of the indicator, the argument uses only that it is ≤ 1 and that on its support each X holds a plaquette of $D\cap\hat{\mathfrak{Z}}$; moreover $\hat{\mathfrak{b}}\subset\hat{\mathfrak{Z}}$.

Step 10: clauses (a), (a′), (b), (d°), (d), (d′), (e).

(a) The tests of $\hat\lambda$ lie in $\mathfrak{Z}_2$. The factor $[A]^\ell_{\mathfrak{Z}}1_{\rm ind}$ involves η on the bonds of the $p\in\hat{\mathfrak{Z}}$, of the W_c for $c\in\mathfrak{Z}$, and of $\hat{\mathfrak{Z}}$; and, for each $y\in\hat{\mathfrak{Z}}$, on the bonds of the plaquettes within $\varrho(x)+1$ of y (through the row, through ζ, and through both radii of the mask, where $\rho_0\le\varrho(x)$) and of the paths $\Gamma(y,\pi(p))$. All of these lie at coarse distance $\le\varrho_L+2$ from $\hat{\mathfrak{Z}}$. Next, $\eta_b=(G\xi)_b-a_{0,b}$ depends on ξ on Y_b, that is, within r_w, and $\varrho_L+2+r_w\le r_1M$; so everything lies under $\mathfrak{Z}^{+r_1}=\mathfrak{Z}_2$. The quantities a_0, $\boldsymbol{\nu}$, w_j do not depend on ξ.

For compatible data, cells of $\mathfrak{b}_i$ and of $\mathfrak{b}_j$ are farther than $4r_1+2$ apart; hence the $\mathfrak{b}_i^{+2r_1}$ are disjoint, and J_0 lies off them. The law of ξ and the tilt are products over the bonds of I°; the factors are Borel in ξ (by the splitting in Substep 5 of Step 8), they are nonnegative, and they depend on disjoint sets of bonds: this is sentence 3. Sentence 4: $|w|\le$ its cap, and on the support of $1_{\rm ind}$ each X holds a p with $\phi_p>\varepsilon$. The second assertion of sentence 4, that the strong cap is at most a constant where $1_{\rm ind}=1$, is needed because, as a function, the strong cap involves U on the whole of X, and X may reach beyond $\hat{\mathfrak{Z}}_2$.

(a′) The member $w(X,\cdot)$ depends on U on the bonds of $X^{+s(x)}$; U_b depends on η_b and V'; and η_b depends on ξ within r_w. Cells more than r_1 apart have their sites at coarse distance $\ge r_1M+1\ge R_2+1>2r_w+2$. The letters are Borel in ξ: Λ_ζ by the splitting in Substep 5 of Step 8, and $w(X,\cdot)$ because it is a Borel function of U (item C of §4.3) while $U=\mathsf{U}(\eta;V')$ is continuous in ξ. Borel functions of disjoint sets of the ξ_b are independent.

(b) The vector $x^{1/2}\xi_b$ is a standard Gaussian of $\mathbb{R}^3$. Hence $E_\xi\, e^{\frac{x}{2}\varsigma_1|\xi_b|^2}=(1-\varsigma_1)^{-3/2}\le e^{1.6\varsigma_1}$, and a cell has at most $3v$ bonds. Further, $E_\xi[1\{|\xi_b|>c_\xi\varepsilon\}e^{\frac{x}{2}\varsigma_1|\xi_b|^2}]=(1-\varsigma_1)^{-3/2}\Pr\{|N_3|>\tau\}$ with $\tau=c_\xi p_0(1-\varsigma_1)^{1/2}$, and $\Pr\{|N_3|>\tau\}\le 0.8(\tau+1/\tau)e^{-\tau^2/2}$. Indeed, for a standard Gaussian vector N_3 of $\mathbb{R}^3$,

$$\Pr=(2/\pi)^{1/2}\int_\tau^\infty r^2e^{-r^2/2}dr=(2/\pi)^{1/2}[\tau e^{-\tau^2/2}+\int_\tau^\infty e^{-r^2/2}dr]\le(2/\pi)^{1/2}(\tau+1/\tau)e^{-\tau^2/2},$$

and $(2/\pi)^{1/2}\le 0.798$. Moreover $e^{\varsigma_1c_\xi^2p_0^2/2}(1-\varsigma_1)^{-1/2}\le 1.01$ and $3\cdot0.8\cdot1.01\le 2.5$. Over all bonds, $e^{4.8v\varsigma_1}\le e^{x^{-1/8}}$, by the term $6vx^{-1}$ of (Z1♭).

(d°) We have $\mathbf{1}_{\mathsf{Y}}G=\mathbf{1}_{\mathsf{Y}}G\mathbf{1}_{\mathsf{Y}^{+r_1}}$, because $r_w\le r_1M$, and $\|G\|^2=\|C\|\le 2/\gamma$ at real V'; hence $|\varphi_{\mathsf{Y}}|^2\le(2/\gamma)|\xi_{\mathsf{Y}^{+r_1}}|^2$, and we take $\varsigma_1=(2/\gamma)\varsigma\le x^{-2}$. We then use sentence 3 of (a) for Y^{+r_1}, and after it (c°) and (b), with $\#\mathsf{Y}^{+r_1}\le(2r_1+1)^3\#\mathsf{Y}$. When $n=0$, the argument rests on (b) alone.

(d), (d′) On the supports of the $1_{{\rm ind},i}$ the caps are replaced by constants; then the argument for (d°) applies, with (c), respectively (c′), for each factor in place of (c°).

(e) By (W), $\mathrm{py}=d_0p_0^2/2$; so $\mathrm{py}/768=d_0p_0^2/1536$ and $\rho_Q/2=3c_\xi^2p_0^2/(256h_*^2)$. The numbers $C_\varphi^\flat$ and h_* depend on (L,a,K_*,λ_*).

Step 11: clause (f). Fix any e. A gauge transformation which is 1 at the block centres moves every U into the chart; the first paragraph of the proof of Lem. 3.9, the formula of the chart, shows this. Thus $U=\mathsf{U}(\eta;V')$ with an $\eta\in\mathfrak{L}_{\mathfrak{g}}$. Both sides of (f) are invariant, and Lem. 4.27 is stated for any element η of $\mathfrak{L}_{\mathfrak{g}}$.

Such an η exists for every U and V', also where U_b or $U_bV'(c)^{-1}$ is -1. The combs form a forest in which each tree carries one block centre (item C.I 4 of §3); on them, therefore, the transformation gives $U=1$. The map $Y\mapsto e^{iY}$ sends $\{Y\in\mathfrak{g}:|Y|\le\pi\}$ onto G: one diagonalises, and at -1 one takes π times any unit vector of $\mathfrak{g}$. We choose η_b with $e^{i\eta_b}=U_b$ at the bonds of I° inside blocks,

and with $e^{i\eta_b} = U_b V'(c)^{-1}$ at a crossing bond of c. The choice is immaterial: Lem. 4.27 holds for every η of $\mathfrak{L}_{\mathfrak{g}}$, and F, ϕ, ζ, m_y, n_c, $\mathfrak{n}_c$, de, db are functions of (U, V', e) alone.

The invariance: En_ℓ and $\sum_D \phi_p$ are invariant by the part "As to invariance" of item (3) of Step 1 of the proof of Thm. 4.5. The angle $\mathfrak{n}_c$ is invariant since $u = 1$ at the block centres and $\mathsf{M}(U^u)(c) = u(c_-)\mathsf{M}(U)(c)u(c_+)^{-1}$, by item C.I 4 of §3. The rest of (f) does not depend on U.

We apply Lem. 4.27 with $\mathfrak{Z} = \mathfrak{Z}' = \mathfrak{S} =$ all cells and the starred fields; then $\hat{\mathfrak{Z}} \setminus \hat{\mathfrak{Z}}'$ is empty and $\mathrm{Rim} = 0$. By $(\mathrm{E1}_k)$,

$$\langle \boldsymbol{\nu}, \iota(F,n)\rangle = \langle Z_*, W^V F + \mathrm{d}'^V n\rangle + \langle \Xi_*, \mathrm{d}^V F\rangle = \langle Z_*, X'\rangle - \langle Z_*, \mathrm{de}\rangle + \langle \Xi_*, \mathrm{db}\rangle,$$

and the defects are absorbed by the two moduli. Moreover $\langle Z_*, X'\rangle - \frac{1}{2}\|\boldsymbol{\nu}\|^2 = L^{-1}\mathcal{G}_{\mathrm{all}}$, since $\|\mathrm{pm}(v_*)\|^2 = L^{-1}\langle v_*, \hat{R}^V v_*\rangle$. We apply Lem. 4.31 to the cell datum $J =$ all classes, $t =$ all cells off f: then $\mathfrak{b} = \mathfrak{Z} = \mathfrak{Z}_2 =$ all cells, and (H17) holds for this datum, because J consists of all classes and hence every plaquette of D' lies in D'_j for some $j \in J$. When D' is empty, J is empty and t is the set of all cells. We drop $\mathrm{Sq} \geq 0$. This gives $L\mathcal{A}_{\mathrm{all}} \geq \mathcal{G}_{\mathrm{all}} + L\,\mathrm{Pay} - L\,\mathrm{vol}$; with $M^3\#\mathrm{cells} = N^3$ and $6N_1^2 \leq 6.3(\mathsf{N}_\nu^\flat)^2$, this is (f) with $e_2^\flat$.

Step 12: clauses (g*1), (g*2).

(g*1)
1. Each component contains a cell: every polymer is adjacent to some object that has cells, and that object belongs to the same component; so $\mathfrak{b}_m$ is not empty. The set t_m lies off f_J, which contains f_{J_m}, and the polymers keep the distance $> s(x)$; so $\check{\zeta}^{(m)}$ is a letter.
2. Different components have no adjacent objects; so they are compatible, the $\mathfrak{b}_m$ are pairwise farther than $4r_1 + 2$ apart, and hence the $\mathfrak{b}_m^{+r_1}$, and likewise the $\mathfrak{b}_m^{+2r_1}$, are disjoint.
3. Admitted: let $P \in D'$ lie in a cell $\square_P$ within $4r_1$ of $\mathfrak{b}_m$. By (H17) for (J,t), the class j of P satisfies $j \in J$. The cell $\square_P$ is a mark, hence a cell of f_j, and its distance from some cell of an object of $\check{\zeta}^{(m)}$ is at most $4r_1 + 2$; thus f_j is adjacent to that object, and $j \in J_m$.
4. The indicator: a p of $D \cap \hat{\mathfrak{Z}}_m$ is covered by an X containing its cube. Since a cell is made up of exactly L^3 fine cubes, that cube lies under $\square^{+r_1}$ for some cell $\square$ of $\mathfrak{b}_m$; so X is adjacent to an object of $\check{\zeta}^{(m)}$, and $X \in \mathfrak{X}_m$. Conversely, an X of $\mathfrak{X}_m$ that holds a plaquette of $D \cap \hat{\mathfrak{Z}}_k$ is adjacent to $\check{\zeta}^{(k)}$, so that $k = m$. Hence $1_{\mathrm{ind}} = \prod 1_{\mathrm{ind},m}$.
5. The set $\mathfrak{Z}_2 \setminus \mathfrak{b}$ is the disjoint union of the $\mathfrak{b}_m^{+2r_1} \setminus \mathfrak{b}_m$; $[A]_{\mathfrak{Z}}^{\ell}$ is a product over the rows of $\mathfrak{Z}$; $\sum_J w_j$, the sum in Res and $\#t$ split; and $u_1 \leq 1$.

(g*2) The proof is that of (g*1), with cell adjacency alone in place of adjacency and with no polymer; of its items, 1, 2, 3 and 5 enter.

Step 13: existence of $x_{\mathfrak{fl}}$, what it depends on, and its place in the order of the constants. The following claim is proved in items (1) to (4) below. Every constant of item F of §4.3 is computed from (O0) to (O10) and x, with the exception of $\mathrm{Res}_{(J,t)}$ and of the four quantities $e^{\circ}_{(J,t)}$, $e^{\circ\prime}_{(J,t)}$, $e^{o}_{\check{\zeta}}$, $e^{o\prime}_{\check{\zeta}}$ that are defined from it. These five also contain $\mathfrak{t}_\rho$, the cell datum (J,t) and the field V', and none of them occurs in a floor of item G. Every floor of item G of §4.3 is a condition on x and the letters (O0) to (O10) alone, and it holds for all large x. Hence $S_{\mathfrak{fl}}$ is not empty, and $x_{\mathfrak{fl}} = 1 + \inf S_{\mathfrak{fl}}$ exists; it is free of N, c_h, E_a, w, V' and e, and likewise of ℓ: it is one number for all rows ℓ of item C of §4.3. Of a row, only the pure numbers b_*, K_*, λ_* and the bound $\varrho(x)$ of item C on its range enter the floors of item G. The three numbers are letters of (O0), and $\varrho(x) = R = \lceil c_R \log x\rceil$ is a function of x given by c_R of (O6). Consequently, $S_{\mathfrak{fl}}$ and $x_{\mathfrak{fl}}$ are determined by the letters (O0) to (O10) alone.

(1) The constants of item F depend on K_*, λ_*, c_1^*, c_2^*, d_0, c_Ξ, κ, ρ_0, L, a, γ, b_h, $\theta_{\mathbb{C}}$, θ_D, C_D, c_R, M, A_0, $\mathfrak{p}_0$, G_0, $C_\varphi^\flat$, $\mathsf{K}_{\mathrm{c}}^0$, a_{c} and x. In item F, c_Ξ occurs only in e^o, and κ only in e^o and $e^{o\prime}$.

The constant Res contains t_ρ, on which no floor depends; it is a quantity of the pair (ℓ, ℓ') of rows in item C of §4.3, with $\mathsf{t}_\rho \le \frac{1}{8}(\frac{1}{2} - d_0 - c_\Xi)$. The quantity $\mathsf{ti}_0^\flat$ is given in item F by the explicit expression $8C_D^2 L a_\nu^2 x'^{-64}$ in the letters and x; it does not depend on the rows, and $\|\mathsf{B}'^{c}\hat{R}^{<} - 1\|_0 \le \mathsf{ti}_0^\flat$ by item (4) of Step 5. With the value $\mathsf{n} = 60$ of (O9), item D gives explicit formulas for τ_c, τ_2 and ti_c in terms of the letters and x.

(2) The radii R, ϱ_L, r_1, $r(x')$ and $Mr(x')$ are at most $\log x$ times a number that depends only on the letters (O0) to (O10). The left side of each of ($\mathrm{F}_{\mathrm{law}}$), (P1), (P3), (FA), ($\mathrm{F0}^\flat$) to ($\mathrm{F9}^\flat$), ($\mathrm{U0}^\flat$), ($\mathrm{Z0}^\flat$), ($\mathrm{Z1}^\flat$) is the product of such a number, a power of $\log x$, and one of $p_0^3 x^{-1/2}$, $p_0^2\varepsilon$, $\varepsilon^{1/2}$, $x^{-1/4}$, $p_0^2 x^{-1}$, $e^{-\mathrm{py}/384}$ or a negative power of x; it is compared with a constant, with $x^{-1/8}$, with ε' or with a positive power of x. Four kinds of clauses do not have this form:

(i) The inequality $\varepsilon \le \varepsilon'$ of (FA): both sides carry $x^{-1/2}$. Here $\varepsilon/\varepsilon' = L^{-1/2}(\log x/\log x')^{\mathfrak{p}_0}$, and $\log x/\log x' = 1 + \log L/\log x'$ decreases to 1 as x grows; so $\varepsilon \le \varepsilon'$ holds for exactly those x with $\log x' \ge \log L/(L^{1/(2\mathfrak{p}_0)} - 1)$, and these x make up an upper half line.

(ii) In ($\mathrm{U0}^\flat$) and ($\mathrm{Z0}^\flat$) the right sides $r_o = x^{-3/8}$, $0.09 r_o$, $t_0 = p_0^{1/2} x^{-1/4}$, $\breve{\theta}_x/2 = x^{-1/4}/2$ and $c_\xi\varepsilon/4$ decrease too. The left sides are a factor times ε, compared with r_o, $0.09 r_o$, $\breve{\theta}_x/2$; then r_o, compared with t_0; and a factor times $\varepsilon'^2 + \varepsilon\varepsilon'$, compared with $c_\xi\varepsilon/4$. Each factor before ε, and each factor before $\varepsilon'^2 + \varepsilon\varepsilon'$, is a number that depends only on the letters (O0) to (O10). In every case the power of x on the left is strictly lower, up to powers of $\log x$.

(iii) The conditions $R \ge \max\{M, 6\}$ and $\rho_0 \le R(x')$ of ($\mathrm{G1}^\flat$), $\mathrm{py} \ge 1$ of ($\mathrm{F9}^\flat$) and $c_\xi p_0 \ge 1$ of ($\mathrm{Z0}^\flat$): in each of them a constant is compared with a function of x that grows without bound.

(iv) The term $e^{-\mathrm{py}/384}$ in ($\mathrm{Z1}^\flat$): in the setting (W), $\mathrm{py} = d_0 A_0^2 (\log x)^{2\mathfrak{p}_0}/2$ with $2\mathfrak{p}_0 > 1$, so $e^{-\mathrm{py}/384}$ decreases faster than any power of x.

For the floors ($\mathrm{G1}^\flat$), ($\mathrm{U1}^\flat$) and (W) the comparison runs as follows.

($\mathrm{G1}^\flat$) $Mr(x') \ge 20 c_R(\log x - \log L)$ is compared with $5R + 4\varrho_L + 12 \le 6.34 c_R \log x + 23$.

($\mathrm{U1}^\flat$) $\log(1/\kappa_1) \le \log x + \mathrm{const}$, and $(\log x)^3$ is compared with py, which by (W) is $d_0 A_0^2 (\log x)^{2\mathfrak{p}_0}/2$, where $2\mathfrak{p}_0 > 4$.

(W) $x^{1/4}$ is compared with a power of $\log x$.

(3) The conditions on N reduce to (Tor), which is a datum; see the paragraph on (P0) between (P3) and (FA) in item G of §4.3.

(4) The number x_{fl} is defined after (O10) and before c_h, E_a; it enters (O13) as $L x_\flat \ge x_{\mathrm{fl}}$; and (O11) depends on δ_3, δ_3'. Every condition is one-sided, and there is no circle.

□

4.4. **Polymer expansion of the step.** We assemble the local factors into the density of the coarse field. Finite-range dependence factors the expectation over separated sets of letters. A convergent cluster expansion sums the remaining interactions. The analytic and rough terms of this expansion obey the caps at $x' = x/L$, giving closure of the real class under the step.

Theorem 4.39 ((S-asm$_k$): one step carries the class at x into the class at x/L)**.** Let the letters be chosen in the order (O0) to (O13) of item 3 of §2.1. Assume those clauses of the following statements which Step 0 of the proof lists: Thm. 3.6 with Prop. 3.10 (a), (b) and with Lem. 3.3 (e); Thm. 3.32, Lem. 3.37 and Cor. 3.38; Thms. 3.40, 3.42 and Cor. 3.43; and further Thm. 4.2 (a) to (e), Thm. 4.5 (a) to (e) and Thm. 4.19, which is made up of items A to H of §4.3 (the objects of items E1 to E12, the constants of item F, the floors of item G, the conclusions of item H). Let $x \ge L x_\flat$; we work in the setting (SD) at x, with any data that it admits; and let $\rho = e^{\mathsf{V}}\rho[u;\ell] \in C_\flat^{\mathrm{su}}(x;\ell)$. The setting (SD) names no level and no depth: the rows ℓ range over the rows of a chain at any level $k \ge 0$ and over any other rows with (SD), the empty rows $\ell = \emptyset$ among them.

Then a real V' and a real $u' = (E'^a, E'^h, w') \in M_1^{\mathrm{su}}(x')$ with the strong cap at x' are defined by (F1) to (F4), which Steps 1, 6, 9 and 12 of the proof set out, and by the mean over e of Step 13 for w', which is the last clause of (F4); with them, at every V',

$$\rho'(V') = e^{\mathsf{V}'}\rho[u';\ell'^{\mathrm{c}}](V') > 0.$$

Hence the property (S-step) holds: $\rho' \in C_\flat^{\mathrm{su}}(x/L;\ell'^{\mathrm{c}})$, where ℓ'^{c} is the cut at $\varrho(x/L)$, with the same letters d_0, c_Ξ, ρ_0, κ, c_R, M, A_0, $\mathfrak{p}_0$, $\vartheta_0^\flat$, c_h, E_a and with the caps at x/L; and ρ' is a function of V' alone. Moreover $E'^a = E'^{\mathrm{new}} + E'^{\mathrm{old}}[u]$, where E'^{new} depends on no member, so that it is a single family for all ρ. For every cell $\square$,

$$\sum_{Y \ni \square} e^{\kappa(\#Y-1)} \sup_{\mathcal{U}_Y(x')} |E'^{\mathrm{old}}(Y,\cdot)| \le 0.25\, c_a(x');$$

the same sum with E'^{new} in place of E'^{old} is $\le 0.31\, c_a(x')$; and the same sum with E'^h in place of E'^{old}, its supremum now taken over the tame real fields on $Y^{\approx}$, is $\le c_h(x')$. ■□

Proof. Step 0: the inputs, clause by clause. Below we give the clauses that are used in Steps 1 to 14 of this proof or in Step 15 of the proof of Thm. 5.7 and that are drawn from the statements that Thm. 4.39 assumes or from the base. The hypotheses of each such clause, except those of Thm. 2.57 (b) and Thm. 2.64, are met by the choice of the letters, by the setting (SD) and by the floors (A0) to (A11). Thm. 2.57 (b) and Thm. 2.64 are used in Step 15 alone, at the θ_D, C_D, c_R of (O6); at these values their use rests on the part "Conclusion" of Thm. 2.65 and on Steps H9 and H14 of its proof. The two size bounds of (Γ3) are applied only at fields that satisfy its hypothesis on the curvature: in Steps 2 and 7, and, in Step 10, within the parts of the proof of Thm. 5.15 on which this step draws.

(Γ2) Thm. 3.6 (a), (b), (c), (d), (f), (h), (i), (k) and Prop. 3.10 (a), on $\mathfrak{V}$. The matrix G of the law has $G(b,b') = 0$ unless $\mathrm{dist}(b,b') \le r_w$, and $(G\xi)_b$ depends on ξ only through its restriction to the window $Y_b = \{b' \in I^\circ : \mathrm{dist}(b,b') \le r_w\}$. The row $G(b,\cdot)$ depends only on the restriction of V' to the coarse bonds whose two ends lie within r_G of $y(b)$. The matrix $C = GG^{\mathsf{T}}$ is symmetric, $C(b,b') = 0$ unless $\mathrm{dist}(b,b') \le 2r_w \le R$, and $C(b,\cdot)$ depends on V' only within $3r_w + \varrho_L + 1$ of $y(b)$. By (A1) we have $\varrho_L + 2 \le \lceil R/2 \rceil$, and with it $r_G \le R-1$ and $3r_w + \varrho_L + 1 \le R_2 - 1$.

For G-valued u we have $G[V'^u] = \mathrm{Ad}_u\, G\, \mathrm{Ad}_u^{-1}$. Every entry is a uniform limit of polynomials in the entries of those $V'(c)$ on which it depends; it is continuous on $\mathfrak{V}$, holomorphic on $\mathfrak{V}^\circ$ and real at G-valued fields. The bounds $\|G\|_\infty,\ \|G^{\mathsf{T}}\|_\infty \le \mathsf{G}_0 \le C_\varphi^\flat$ and $\|(G\xi)_b\| \le C_\varphi^\flat \sup_{Y_b} \|\xi_{b'}\|$ hold. Under $(\mathrm{F}_{\mathrm{law}})$, and with $c_R\theta_{\mathbb{C}} \ge 128$, the matrix G is invertible and $\det G > 0$ at real fields (Thm. 3.6 (h)); moreover $S = C^{-1} - q$ is symmetric and $\|S\| \le x^{-60}$.

With these we use Lem. 3.3 (e). Its hypotheses are $L \ge 3$ odd, $aL \ge 18$, and rows of the kind fixed in item C.I 1.1 of §3. The rows of (SD) are of this kind, and the clause of item C.I 1.1 on N is treated below, among the clauses on N. The lemma gives the finite sum

$$(\mathsf{g}^V X')_{b'} = \sum_P \sum_{\mathsf{t}} g_{\mathsf{t}}\, \mathrm{Ad}_{V'(\pi_{\mathsf{t}})}\, X'_P,$$

where the numbers g_{t} depend on no field and each π_{t} is a coarse path from $y(b')$ to the base site of P, all of whose sites lie within $2\varrho_L + 1$ of $y(b')$. So g^V is a polynomial in the $\mathrm{Ad}_{V'(c)}$; and at every field of $\mathfrak{V}$ and for every X', the value $(\mathsf{g}^V X')_{b'}$ depends only on the restrictions of V' to the bonds, and of X' to the plaquettes, whose sites all lie within $2\varrho_L + 2$ of $y(b')$.

(Γ3) Thm. 3.32 (a), (c) and Cor. 3.38 (a). The data of Thm. 3.32 (a), (c) are a coarse site o, a number $0 < \hat\varepsilon \le 1/40$, and an $\mathrm{SL}(2,\mathbb{C})$ field on the box of radius R_3 about o with (e1), (e2) and $|\partial V'(P) - 1| \le 2\hat\varepsilon$. By item C.II 1.4 of §3, the conditions (e1), (e2) give (h1), (h2), two

of the conditions on the field that Thm. 3.32 assumes. For each bond b with $\mathrm{dist}(b,o)\le 1$, the theorem then gives two facts: $a_{0,b}$ is determined by the field restricted to that box, and $|a_{0,b}|\le \mathsf{a}_c\hat\varepsilon$. Cor. 3.38 (a), which is (Cen), adds

$$|(EX'-\mathrm{d}^V a_0)_p|\le \mathsf{N}_c L^{-2}\hat\varepsilon+\mathsf{K}_c\hat\varepsilon^2$$

for $y(p)=o$, with $\mathsf{K}_c=\sqrt{2}\mathsf{K}_c^0$. The hypotheses of Thm. 3.32 and Cor. 3.38 are (Fr1) to (Fr3), which (SD) provides, and $x\ge L^2$, $R\ge 6$, (Tor), (P1), (P3); further, $L\ge 3$ odd and $aL\ge 18$, rows as in item C.II 1.2, and a law with (λ0), (ρ1), (ρ2), (ρ5) as in the proposition of item C.II 2.1 of §3. The conditions $L\ge 3$ odd and $aL\ge 18$ hold by (O4); the rows of (SD) are rows as in item C.II 1.2, whose clause on N is among the clauses on N below; and the law of (Γ2) has (λ0), (ρ1), (ρ2), (ρ5): it is defined on $\mathfrak{V}$, $C(b,b')=0$ unless $\mathrm{dist}(b,b')\le 2r_w\le R$, $C(b,\cdot)$ depends on V' only within $3r_w+\varrho_L+1\le R_2-1$ of $y(b)$, and, under ($\mathrm{F_{law}}$) and $c_R\theta_{\mathbb{C}}\ge 128$, the matrix C is invertible and $\|S\|\le x^{-60}$, that is, $\mathsf{n}=60$.

The last clause of (Γ3) is the locality of a_0. It holds at every field of $\mathfrak{V}$ and for every X' whose values lie in $\mathfrak{g}$ or in its complexification; in particular it holds at every pair (V',e), rough plaquettes included, and, under the convention of item 1 of §2.1, at a complex field. It says that $a_{0,b}$ depends only on the restrictions of V' to the bonds, and of X' to the plaquettes, whose sites all lie within R_3 of $y(b)$, and on nothing else; and that, when $\mathrm{dist}(b,o)\le 1$, the same is true with o in place of $y(b)$. For the proof we write $a_{0,b}=\sum_{b'}C(b,b')(\mathsf{g}^V X')_{b'}$. By ($\Gamma$2), $C(b,b')=0$ unless $\mathrm{dist}(b,b')\le 2r_w\le R$, and $C(b,\cdot)$ depends on V' only within $3r_w+\varrho_L+1\le R_2-1$ of $y(b)$. By Lem. 3.3 (e), $(\mathsf{g}^V X')_{b'}$ depends on V' and on X' only within $2\varrho_L+2$ of $y(b')$. Altogether, $a_{0,b}$ depends on them only within $\max\{R_2-1,\ R+2\varrho_L+2\}$ of $y(b)$, and therefore, when $\mathrm{dist}(b,o)\le 1$, only within $\max\{R_2,\ R+2\varrho_L+3\}\le R_3$ of o, because $\varrho_L\le R+2$. The hypothesis $|\partial V'(P)-1|\le 2\hat\varepsilon$ of Thm. 3.32 is used for the two size bounds alone. Compare Step (0) of the proof of Thm. 3.32, as well as Step 2 (ii), (iii) of the proof of Thm. 4.5.

(Γ4) Cor. 3.43, with Thm. 3.40 (b) and Thm. 3.42 (b), (c). We put $\tilde{\mathsf{D}}:=-\mathsf{D}_k/2$. At every real V' we have $(\det C^V)^{1/2}=(\det C^1)^{1/2}\exp\sum_Y\tilde{\mathsf{D}}(Y,V')$ and $S=\sum_Y S_k(Y;V')$, where Y runs over the connected sets of cells. Each $\tilde{\mathsf{D}}(Y,\cdot)$ is real, gauge invariant and 0 at pure gauges, and $|\tilde{\mathsf{D}}|\le \mathsf{K}_1^\flat e^{-\kappa^\sharp(\#Y-1)}$ at every V'. It has a holomorphic continuation to $\mathcal{U}_Y(x')$, on which $|\tilde{\mathsf{D}}|\le 1.5\mathsf{K}_1^\flat\varepsilon' e^{-\kappa^\sharp(\#Y-1)}$. The function D_k is defined in Def. 3.41, in the display that precedes Thm. 3.42:

$$\mathsf{D}_k(Y,V')=-\mathsf{D}^q(Y,V')-[\mathsf{D}^u(Y,V')-\mathsf{D}^u(Y,1)].$$

Its continuation is given by the same formulas, taken at the member of $\mathcal{U}_Y(x')$ extended by 1, as in the proof of Cor. 3.43.

By Thm. 3.40 (b), $\mathsf{D}^q(Y,\cdot)$ is holomorphic on the open set $\{|V'(c)|<1+\vartheta_0^\flat$ on the bonds of $Y^\sim\}$; by Thm. 3.42 (b), $\mathsf{D}^u(Y,\cdot)$ is holomorphic on $\mathfrak{V}^\circ$; and each of these open sets contains every G-valued field. Hence $\tilde{\mathsf{D}}(Y,\cdot)$ is continuous on the G-valued fields. On $\mathcal{U}_Y(x')$ it is invariant under G-valued gauge transformations, by Thm. 3.40 (b) and Thm. 3.42 (b); the argument is given in Step 9, Invariance.

On $\mathfrak{V}$ the matrices $S_k(Y)$ satisfy $S_k(Y)(b,b')=0$ unless $b,b'\in\hat Y$, $S_k(Y)^{\mathsf{T}}=S_k(Y)$ and $S_k(Y;V'^u)=\mathrm{Ad}_u\,S_k\,\mathrm{Ad}_u^{-1}$; they are holomorphic on $\mathfrak{V}^\circ$ and real at real fields, and $\|S_k(Y)\|\le x^{-57}e^{-\kappa^\sharp\#Y}\le e_Y:=x^{-40}e^{-\kappa^\sharp\#Y}$. Both $\tilde{\mathsf{D}}(Y,\cdot)$ and $S_k(Y;\cdot)$ depend only on the restriction of V' to the bonds of $Y^{+\lceil R_2/M\rceil}$, and $\lceil R_2/M\rceil\le r_1-8$.

The conditions of these statements are $\kappa\ge 6$, which is (O5); ($\mathrm{M4}^\flat$); ($\mathrm{R}^\flat$); (M2); (Tor); ($\mathrm{F_{law}}$); $x\ge L^2$ and $R\ge 6$, the two floors of (P0); (P1); (S1); and (S2). In Rem. 3.44, they make up the list "The conditions, whole". Of them, ($\mathrm{M4}^\flat$) and (M2) are two of the conditions on M in (O7), ($\mathrm{R}^\flat$) is derived in (O6) through four successive steps, the floors are

among those of (A0), and (Tor) is one of the data of (SD). As under (Γ3), these statements also assume $L \geq 3$ odd, $aL \geq 18$, and rows as in item C.II 1.2, its clause on N included; these conditions are met on the grounds given there.

(Γ5) Thm. 4.2, with the following parts.

(e) At every (V', e),

$$x\,\mathsf{m}_C - \frac{x}{2}\langle a_0, Sa_0\rangle = x'\mathrm{En}'(V') + \sum_j w_j + \sum_{\square} 1_{\square\approx}(V')P(\{\square\}, V'),$$

where $w_j = \sum_{\square\in \mathrm{mk}'^{+r'}_j} T_\square$, for any partition of the marks into classes whose pairwise distance is $> 2r'$. Each mk'_j is a class of the marks $\mathrm{mk}'(V')$ of item E6 of §4.1; item E9 of §4.3 has the same set of marks, and its classes form one such partition.

(b) The function $T_\square$, given by the formula of item E7 of §4.3, is real and jointly Borel; it satisfies $T_\square(V'^u, \mathrm{Ad}_u\, e) = T_\square(V', e)$; and it depends on V' only within $4R+2\varrho_L+6 \leq R_6$ of $\square$.

(d) The function $P(\{\square\}, \cdot)$ is holomorphic on $\mathcal{U}_{\{\square\}}(x')$, invariant, real at real fields and 0 at pure gauges, with $\sup|P(\{\square\}, \cdot)| \leq K_{\mathrm{q}} p_0'^2 \varepsilon'$, the supremum taken over $\mathcal{U}_{\{\square\}}(x')$; and we put $P(Y) := 0$ for $\#Y \geq 2$.

(Γ6) Thm. 4.5, applied with the test radius R_χ. The theorem allows any radius $\geq \lfloor R/2\rfloor + \varrho_L + 2$. The test is $\chi_\square(\xi) = 1$ if and only if $|\xi_b| \leq c_\xi \varepsilon$ at all $b \in I^\circ$ within R_χ of $\hat{\square}$, and $V_\square$, $A_\square$ are those of items 3.8 and 3.9 of §4.2. In (a) to (d), $V' \in \mathcal{U}_{\{\square\}}(x')$ and $\chi_\square = 1$.

(a) The function $V_\square$ depends on ξ only within $r_w + \varrho_L + 1$ of $\hat{\square}$ and on V' only within R_v of $\square$; it depends on no member and on no e; and it is holomorphic in V'.

(b) For G-valued u, $V_\square(\mathrm{Ad}_u\,\xi, V'^u) = V_\square(\xi, V')$.

(c) The function $V_\square$ satisfies $|e^{V_\square} - 1| \leq 2K_v p_0^3 x^{-1/2}$.

(d) At G-valued $V' \in \mathcal{U}_{\{\square\}}(x')$, $V_\square$ is real and $A_\square = e^{V_\square}$.

(e) At every real ξ and every pair (V', e) the following hold.

(e.i) For $X'(V', e)$, and also for any $\mathfrak{g}$-valued X' put in its place,

$$\prod_\square A_\square = 1\{|\eta_b| < \pi \text{ all } b\} \prod_b \mathsf{j}(\eta_b)\, e^{-x\mathrm{En}_\ell(U)} e^{\frac{x}{2}\langle F^\flat, \mathsf{B}^V F^\flat\rangle} \prod_c e^{\frac{ax}{2}|n_c^\flat|^2 - ax(1-\cos \mathsf{n}_c)}.$$

(e.ii) Let $\rho = e^{\mathsf{V}}\rho[u;\ell]$ with $u \in M_A^{\mathrm{su}}(x)$, real or complex; as the step is linear in ρ, the real part and the imaginary part of ρ are treated separately, whereas in u nothing is linear. Then

$$\rho'(V') = \mathsf{c}_0 z(ax)^{-3\#\mathbb{T}'}(2\pi/x)^{\frac{3}{2}\#I^\circ} \det G^V\, e^{\mathsf{V} - x\mathcal{Q}(-a_0)} E_\xi\big[e^{-\frac{x}{2}\langle\xi, \mathsf{E}\xi\rangle + x\langle\xi, \mathsf{E}h\rangle} \prod_\square A_\square \prod_X e^{1_{X\approx}(U)E(X,U)} \Xi_w(U)\big],$$

where $\mathsf{E} = G^{\mathsf{T}} q G - 1$, $h = G^{\mathsf{T}} \mathsf{g}^V X'$ and $\mathcal{Q}(-a_0) = \mathsf{m}_C - \frac{1}{2}\langle a_0, Sa_0\rangle$.

(e.iii) The function $A_\square$ depends on (V', e) only through the restriction of the pair to the bonds and plaquettes within R_v of $\square$.

(e.iv) For G-valued u, $A_\square(\mathrm{Ad}_u\,\xi, V'^u, \mathrm{Ad}_u\, e) = A_\square(\xi, V', e)$.

(e.v) The function $A_\square$ is Borel in (ξ, V', e).

(Γ7) Thm. 4.19. The definitions that this clause uses are those of items E9 to E12 of §4.3. Item E9 gives the marks mk', the set $f = \mathrm{mk}'^{+s_0}$, the classes mk'_j under the relation "distance $\leq 2s_0 + 4r_1 + 2$", and f_j, D'_j, w_j. Item E10 gives $b_\square$, and item E11 gives $[A]^\ell_{\mathfrak{Z}} = \prod_{\square\in\mathfrak{Z}} A_\square$. Item E12 gives the cell datum (J, t), with $\mathfrak{b} = f_J \cup t$, $\mathfrak{Z} = \mathfrak{b}^{+r_1}$ and $\mathfrak{Z}_2 = \mathfrak{b}^{+2r_1}$; the function $\hat{\lambda}_{(J,t)}$; (H17); the objects and their adjacency; the letters $(J, t, \mathfrak{X})$ and the properties of being compatible and of being admitted; 1_{ind}; and λ^o. The constants that occur in this clause are those of item F of §4.3.

(a) By the first sentence of Thm. 4.19 (a), $\hat{\lambda}$ and $\hat{\lambda}1_{\mathrm{ind}}$ depend on ξ only through its restriction to the bonds of I° under $\mathfrak{b}^{+2r_1}$.

(d) We take this clause for $A = 1$ and with the strong cap. Its data are letters $\check\zeta_i$, admitted and pairwise compatible; a finite set J_0 of cells lying off every $\mathfrak{b}_i^{+2r_1}$; a further finite set Y of cells; and $0 \le \varsigma \le (\gamma/2)x^{-2}$. Then

$$E_\xi[\prod_i |\lambda^o_{\check\zeta_i}| \prod_{J_0} b_\square\, e^{\frac{x}{2}\varsigma|\varphi_{\mathsf{Y}}|^2}] \le \exp\{4.8v(2/\gamma)\varsigma(2r_1+1)^3 \#\mathsf{Y} + x^{-1/8}(\sum_i \#\mathfrak{b}_i^{+2r_1} + \#J_0)\}\, u_b^{\#J_0} \prod_i e^o_{\check\zeta_i},$$

the case of no $\check\zeta$ being included.

(e) By this clause, $u_1 = e^{-\delta_3 p_0^2}$.

(g*1) We use this clause in item 2 of Step 7.

The data of Thm. 4.19, which item C of §4.3 sets out, include a system w that is complex Borel, that is a function of U on the bonds of $X^{+s(x)}$, and that obeys the cap. The w of ρ has these properties by (cΞ), with the strong cap at $A = 1$; so they hold in (d) and in (g*1) alike.

The base consists of the following statements: Lem. 5.17 and Thm. 5.18, in Step 2 of the proof of Thm. 5.15; Thm. 5.22, in Step 8, (8.2), of the proof of Thm. 5.15; Prop. 5.8, in Step 15, item 2, of the proof of Thm. 5.7; Lems. 4.49, 4.50, 4.51, in Step 8; Lem. 5.23, in Step 10 (9) of the proof of Thm. 5.15; and Thms. 2.57 and 2.64, with (X12), in §2.2. This proof uses these passages of the proof of Thm. 5.15: in Step 2, Lem. 5.17, Thm. 5.18 and (ii); in Step 8, (8.2), and, in the part of that step headed "The clauses without a size", the extension by 1 of the G-valued gauge transformation u given on the sites of $Y^{\approx}$ or of B_ω; and in Step 10 (9), Lem. 5.23 with its proof and nothing more. These passages use statements of §3 and the identity of matrices in Step 4 of the proof of Thm. 4.5. From Sections 2.1, 4.4 and 5.2, they use the definitions in items 1 and 2 and the letters of item 3 of §2.1, and from Thms. 4.39, 5.7 and 5.3 they use nothing. That identity of matrices is an input of this proof too.

We come to the clauses on N. The clause $2D_w + 1 < N$ of (D0) and the clause $2\omega + 1 < N$ of Thm. 5.22 follow from (Tor): indeed $2D_w + 1 \le (12Mr(x) + 24)/L + 5 < 16Mr(x) \le N$, and $2\omega + 1 \le 2Mr(x) + 1 < N$, since $\omega \le Mr'$ by (A2) and $r' = r(x') \le r(x)$. From (Tor) we also obtain $\varrho/L + 2 < N/2$, a clause that item C.I 1.1 of §3 imposes on the rows; $\varrho_L + 2 < N/2$, a clause that item C.II 1.2 of that section imposes on the rows; and $2R_3 + 1 < N$, which item C.II 4 of §3, in its part on the floors, obtains from (P0). The reason is that $\omega \ge R_3 + 2 = 2R + \varrho_L + 7$, so that $\varrho/L + 2 \le \varrho_L + 2 < \omega < N/2$ for every range $\varrho \le \varrho(x) = R$, and $2R_3 + 1 < 2\omega + 1 < N$. These are the clauses on N in the hypotheses of Lem. 3.3, of Thm. 3.6 and Prop. 3.10, and of Thm. 3.32 and Cor. 3.38. At the empty rows the range is 0, while ϱ_L is still determined by the range $\varrho(x)$, not by the range 0.

Step 1: (F1), *the exact formula.* We claim that at every pair (V', e)

$$\rho'(V') = e^{\mathsf{V}_0} \exp\{-x'\mathrm{En}'(V') + \sum_Y \tilde{\mathsf{D}}(Y, V') - \sum_\square 1_{\square\approx}(V') P(\{\square\}, V')\}\, \mathcal{Z}, \tag{F1}$$

where

$$\mathcal{Z} = e^{-\sum_j w_j} E_\xi[e^{\sum_Y T_Y} \prod_\square A_\square \prod_{X\in\mathcal{D}} e^{1_{X^\approx}(U) E(X,U)} \Xi_w(U)].$$

Here $\xi \sim \mathcal{N}(0, x^{-1}1)$ on $\mathfrak{L}_\mathfrak{g}$, $\varphi = G^V\xi$, $a_0 = C^V\mathfrak{g}^V X'$, $\eta = \varphi - a_0$, $U = \mathsf{U}(\eta; V')$ is given by the chart of item 3.4 of §4.2, and $E = E^a + E^h$; further, $T_Y = \frac{x}{2}\langle\varphi, S_k(Y)\varphi\rangle - x\langle\varphi, S_k(Y)a_0\rangle$ and $e^{\mathsf{V}_0} = e^{\mathsf{V}\mathsf{c}_0} z(ax)^{-3\#\mathbb{T}'} (2\pi/x)^{\frac{3}{2}\#I^\circ} (\det C^1)^{1/2}$.

1. The function ρ is Borel, gauge invariant and bounded: $-x\mathrm{En}_\ell \le 0$, the sum of the members is $\le 1.22(c_a + c_h)$ times the number of cubes, and Ξ_w is a finite sum on a finite torus. At a real U with $1_{X^\approx}(U) = 1$, the restriction of U to $X^\approx$ lies in $\mathcal{U}_X(x)$, because

(u1) $1 < 1 + \vartheta(x)$,

(u2) $1 < 2 - 1/\log x$,

(u3) $|\partial U(p)-1| = 2\sin(\phi_p/2) \le \varepsilon < 2\varepsilon$.
So $E^a(X,U)$ is defined.

2. Thm. 4.5 (e.ii) holds at every pair (V',e). Together with (e.i) and with $\rho(U) = e^{\mathsf{V}} e^{-x\mathrm{En}_\ell(U)} \prod_X e^{1_X\approx(U) E(X,U)} \Xi_w(U)$, it gives

$$\begin{aligned}\rho'(V') = {}& \mathsf{c}_0 z(ax)^{-3\#\mathbb{T}'} (2\pi/x)^{\frac{3}{2}\#I^\circ} \det G^V \, e^{-x\mathcal{Q}(-a_0)} \\ & E_\xi\big[e^{-\frac{x}{2}\langle \xi, \mathsf{E}\xi\rangle + x\langle \xi, \mathsf{E}h\rangle} \\ & 1\{|\eta_b| < \pi\} \prod_b \mathsf{j}(\eta_b) \\ & \rho(U)\, e^{\frac{x}{2}\langle F^\flat, \mathsf{B}^V F^\flat\rangle} \\ & \prod_c e^{\frac{ax}{2}|n_c^\flat|^2 - ax(1-\cos \mathsf{n}_c)} \big],\end{aligned}$$

with $\mathsf{E} = G^{\mathsf{T}} q G - 1$ and $h = G^{\mathsf{T}} \mathsf{g}^V X'$.

3. By Prop. 3.10 (a), $\eta\cdot S\eta = -(\xi - h)\cdot \mathsf{E}(\xi - h)$, $a_0 \cdot S a_0 = -h\cdot \mathsf{E}h$ and $\mathcal{Q}(-a_0) = \mathsf{m}_C - \frac{1}{2}\langle a_0, Sa_0\rangle$. With $\eta = \varphi - a_0$ and $S = \sum_Y S_k(Y)$ this gives

$$\sum_Y T_Y = \frac{x}{2}\langle \eta, S\eta\rangle - \frac{x}{2}\langle a_0, Sa_0\rangle = -\frac{x}{2}\langle \xi, \mathsf{E}\xi\rangle + x\langle \xi, \mathsf{E}h\rangle.$$

Moreover $\det G^V = (\det C^V)^{1/2}$.

4. We insert the expression for $\rho(U)$ given in item 2 and apply Thm. 4.5 (e): the factors that depend on no member are $\prod_\square A_\square$.

5. We apply (Γ4) to $(\det C^V)^{1/2}$, and Thm. 4.2 (e) to $x\mathcal{Q}(-a_0) = x\,\mathsf{m}_C - \frac{x}{2}\langle a_0, Sa_0\rangle$, with the partition of the marks into the classes of item E9 of §4.3; their pairwise distance is $> 2s_0 + 4r_1 + 2 > 2r'$. The factor $e^{-\sum w_j}$ depends on no ξ.

Finally, the left side of (F1) contains no e; hence its right side is the same number at every e.

Step 2: a tame coarse field and passed tests make the fine field tame.

Lemma 4.40 (A tame fine field from passed tests and a tame coarse field)**.** (T1) Let p be a plaquette. If V' is real, if $\phi'_P \le \varepsilon'$ on the box of radius R_3 about $y(p)$, and if every $b' \in I^\circ$ within r_w of the four bonds of p satisfies $|\xi_{b'}| \le c_\xi \varepsilon$, then $\phi_p(U) \le 0.72\varepsilon$.

(T2) Let y be a fine site. If the hypotheses of (T1) hold at every p with $|y - \pi(p)|_\infty \le \varrho(x) + 1$, then $m_y(U) = 1$, and $\zeta_p = F_p$ on the range of ℓ_y.

(T3) Let $X \in \mathcal{D}$ with $\chi_X(\xi) := 1\{|\xi_b| \le c_\xi\varepsilon$ at the bonds of I° within R of the blocks that meet $X^\approx\} = 1$, let Y be a set of cells that contains $Y_0(X)$, and let either $V' \in \mathcal{U}_Y(x')$, or V' be real and tame on $Y^\approx$, extended by 1. In either case, the restriction of $\mathsf{U}(\eta; V')$ to the bonds of $X^\approx$ lies in $\mathcal{U}_X(x)$, and at a real V' it is real and tame on $X^\approx$. The second case is contained in the first. Indeed, a field that is real on the bonds of $Y^\approx$ and tame there belongs to $\mathcal{U}_Y(x')$: for such a field the three conditions of item 1 of Step 1, taken at x', read as follows.

(u1) $1 < 1 + \vartheta(x')$.
(u2) $1 < 2 - 1/\log x'$.
(u3) $|\partial V'(P) - 1| = 2\sin(\phi'_P/2) \le \varepsilon' < 2\varepsilon'$ at the plaquettes of $Y^\approx$.

■

Proof of Lem. 4.40. (T1) As V' is real, (e1) and (e2) hold, and $|\partial V'(P) - 1| \le \phi'_P \le \varepsilon'$ on the box of radius R_3 about $y(p)$. Taken at $\hat\varepsilon = \varepsilon'$ and $o = y(p)$, (Γ3) gives $|a_{0,b}| \le \mathsf{a}_c\varepsilon'$ and $|(EX' - \mathrm{d}^V a_0)_p| \le \mathsf{N}_c L^{-2}\varepsilon' + \mathsf{K}_c\varepsilon'^2$. On $\mathfrak{g}$ the two norms agree, $|\cdot| = \|\cdot\|$; hence $|(G\xi)_b| \le C^\flat_\varphi c_\xi \varepsilon = \varepsilon/6.5$, and since transports are isometries, $|(\mathrm{d}^V G\xi)_p| \le (4/6.5)\varepsilon \le 0.6154\varepsilon$. By (L2^{su}), $\mathsf{N}_c L^{-2}\varepsilon' \le \mathsf{N}_c L^{-3/2}\varepsilon \le 10^{-4}\varepsilon$. Moreover $|\eta_b| \le (1/6.5 + L^{1/2}\mathsf{a}_c)\varepsilon \le r_\eta$.

We now apply Lem. 5.17 at $\tau = 1$ and $r = r_\eta$; this lemma is stated in Step 2 of the proof of Thm. 5.15. Its hypotheses are met: $|X'_P| = \phi'_P \le \varepsilon'$; $4.1 r_\eta + 2.1\varepsilon' \le 1$ by (A1); and $\varepsilon' \le 1/40$, since $17T\varepsilon' \le 1/40$ by Part 3 of (A0) and $T > 1$ by (O4). It yields $|\partial U(p) - 1 - iF^\flat_p| \le 0.72(4.1 r_\eta + 2.1\varepsilon')^2$, where $F^\flat = EX' + \mathrm{d}^V \eta$. Together with (A1), the bounds obtained so far give $|\partial U(p) - 1| \le (0.6154 + 10^{-4} + 0.1)\varepsilon = 0.7155\varepsilon$. Finally, $|\partial U(p) - 1|$ is the chord $2\sin(\phi_p/2)$, and $\varepsilon \le 1/40$; therefore $\phi_p \le 0.7156\varepsilon \le 0.72\varepsilon$.

(T2) The second clause of (T2) is void at the empty rows. By (T1), $\phi_p \le 0.72\varepsilon$ at every plaquette p with $|y - \pi(p)|_\infty \le \varrho(x) + 1$. Hence no plaquette of $D(U)$ lies within $\rho_0 \le \varrho(x') \le \varrho(x) + 1$ of y, and, as $0.72\varepsilon < 1$, no plaquette with $\phi_p > 1$ lies within $\varrho(x) + 1$ of y; so $m_y = 1$ by its definition in item E2 of §4.3. Moreover $\zeta_p = F_p \min\{1, \varepsilon/\phi_p\} = F_p$.

(T3) We first locate the sets involved. By the part of item 1 of §2.1 on $Y_0(X)$, every plaquette of $X^\sim$ has a cell of $Y_0(X)$ within $M/2$ coarse units, and $X^\approx$ reaches $Mr(x)$ fine sites farther out. So the blocks that meet $X^\approx$ lie within $M + \lceil Mr(x)/L \rceil$ of the sites of $Y_0(X)$. Their D_w-neighbourhood, and the boxes of radius R_3 about them, lie within ω of these sites, where $\omega \le Mr'$ by (A2); so they lie in $Y_0(X)^{+r'}$, which is inside $Y^\approx$. There, (u1), (u2), (u3) at x' give (e1), (e2), (e3) at $\tau = 1$. At a bond b of $X^\approx$, the value $(G\xi)_b$ involves V' only on the coarse bonds within $r_G \le R_3$ of $y(b)$, and $a_{0,b}$ only on the bonds and plaquettes within R_3 of $y(b)$; neither depends on the continuation.

We apply Thm. 5.18, which is stated in Step 2 of the proof of Thm. 5.15, at $\tau = 1$, with B the set of sites within ω of $Y_0(X)$. Its hypotheses hold, as we now show.

(h1) At the bonds of $X^\approx$, $|(G\xi)_b| \le \sqrt{2}\|(G\xi)_b\| \le (\sqrt{2}/6.5)\varepsilon \le 0.2176\varepsilon$; for these bonds are based in blocks that meet $X^\approx$, or in blocks next to one, and $r_w + 1 \le R$. Moreover, by Thm. 3.32 (c) at $o = y(b)$, $|a_{0,b}| \le \mathsf{a}_\mathrm{c}\varepsilon' \le L^{1/2}\mathsf{a}_\mathrm{c}\varepsilon$; therefore $|\eta_b| \le r_\eta$.

(h2) We have $|(EX' - \mathrm{d}^V a_0)_p| \le 10^{-4}\varepsilon + \mathsf{K}_\mathrm{c}\varepsilon'^2 \le (10^{-4} + 0.1)\varepsilon$. For the transports, $|\operatorname{Ad}_g Y| \le |g|^2|Y| \le 1.01|Y|$. Here g is 1 or $V'(c)$ for a single coarse bond c, because $T_j \in \{1, \operatorname{Ad}_{V'(c)}\}$ by Step 4 of the proof of Thm. 4.5; so $|g| < 1 + \vartheta^\flat_0$ by (u1) at x', whereas (u2) alone would give only the factor 4. Thus $|(\mathrm{d}^V G\xi)_p| \le 4 \cdot 1.01 \cdot (\sqrt{2}/6.5)\varepsilon \le 0.879\varepsilon$, and the two terms together are at most $0.98\varepsilon \le 1.9\varepsilon$.

The conditions (D0), (D2), (D3) at $(1, r_\eta)$ hold by (A0) and Step 0.

At a real $V' \in \mathcal{U}_Y(x')$, the field U is real. To prove its tameness, use $|\partial V'(P) - 1| < 2\varepsilon'$ on the same boxes and apply (Γ3) with $\hat\varepsilon = \varepsilon'$. Since the transports are now isometries, the argument for (T1) gives

$$|F^\flat_p| \le (0.6154 + 10^{-4})\varepsilon + \mathsf{K}_\mathrm{c}\varepsilon'^2.$$

Use the first case of Lem. 5.17, with the principal logarithm X'_P and $(\tau, r) = (1, r_\eta)$. By (A1), the sum of its error and $\mathsf{K}_\mathrm{c}\varepsilon'^2$ is at most 0.1ε. Thus $2\sin(\phi_p/2) \le 0.7155\varepsilon$, and the last estimate in (T1) gives $\phi_p \le 0.72\varepsilon$. This uses the curvature bound supplied by the analytic domain; it does not require $\phi'_P \le \varepsilon'$.

□

Step 3: the normaliser; the numbers n_j. For a given real coarse field, we sort the connected sets of cells.

Definition 4.41. Let V' be real, and let Y be a connected set of cells. We call Y *on* if $Y^{+r'} \cap \mathrm{mk}'$ is empty, and *off* otherwise. An off set Y is *short* if it lies in f, and *long* otherwise. ■

We claim the following.

(i) Y is on if and only if $1_{Y^\approx}(V') = 1$.

(ii) A short Y lies in exactly one f_j. The number $n_j(V') := \sum_{Y \subset f_j,\ \mathrm{off}} \tilde{\mathsf{D}}(Y, V')$ is real, Borel and gauge invariant; it does not depend on e, and depends on V' only within $3r'$ cells of mk'_j; and $|n_j| \le 1.01\mathsf{K}^\flat_1(2r' + 1)^3 \#\mathrm{mk}'_j$.

(iii) A long Y has $\#Y > r'$ and a cell within r' of a mark. Moreover, if

$$\hat{w}(Y) := 2\mathsf{K}_1^\flat e^{-\kappa^\sharp(\#Y-1)+(\kappa+1)(\#Y+s')}$$

, then

$$|e^{\tilde{\mathsf{D}}(Y)} - 1|\, e^{(\kappa+1)(\#Y+s')} \le \hat{w}(Y)$$

and

$$\sum_{Y \text{ long, within } r' \text{ of } \mathrm{mk}'_j} \hat{w}(Y) \le \#\mathrm{mk}'_j$$

.

(iv) $\sum_Y \tilde{\mathsf{D}} = \sum_Y 1_{Y\approx}(V')\tilde{\mathsf{D}}(Y,V') + \sum_j n_j + \sum_{\text{long}} \tilde{\mathsf{D}}$.

We prove (i) to (iv).

(i) A mark is a cell that holds a plaquette of D', and $1_{Y\approx}(V') = 1$ means that no plaquette based in $Y^{+r'}$ is in D'; both conditions say that $Y^{+r'}$ contains no mark.

(ii) For $j \neq k$, $\mathrm{dist}(f_j, f_k) \ge \mathrm{dist}(\mathrm{mk}'_j, \mathrm{mk}'_k) - 2s_0 > 4r_1 + 2 \ge 2$, while consecutive cells of a connected set are at distance 1; so a short Y lies in a single f_j. An off Y has a cell in $\mathrm{mk}'^{+r'}$. If $Y \subset f_j$, the mark within r' of that cell lies within $3r' \le 2s_0 + 4r_1 + 2$ of mk'_j, so it belongs to the class j; and the cell is one of at most $(2r'+1)^3\#\mathrm{mk}'_j$ cells. For a fixed cell $\square$, $\sum_{Y\ni\square} \mathsf{K}_1^\flat e^{-\kappa^\sharp(\#Y-1)} \le \mathsf{K}_1^\flat/(1 - 26e\cdot e^{-100}) \le 1.01\mathsf{K}_1^\flat$, and the bound on $|n_j|$ follows. As to the dependence, $\tilde{\mathsf{D}}(Y,V')$ depends on V' in $Y^{+\lceil R_2/M\rceil}$, where $Y \subset \mathrm{mk}_j'^{+2r'}$ and $\lceil R_2/M\rceil \le r_1 \le r'$. That n_j is real and gauge invariant follows from (Γ4). It is Borel, because on each of the finitely many Borel sets $\{\mathrm{mk}'(V') = \mathsf{d}\}$ it is a finite sum of continuous functions: by (Γ4), $\tilde{\mathsf{D}}(Y,\cdot)$ is continuous on the G-valued fields, the rough ones included.

(iii) A long Y is connected and has both a cell in $\mathrm{mk}'^{+r'}$ and a cell off $\mathrm{mk}'^{+2r'}$; hence $\#Y > r'$. Then $|\tilde{\mathsf{D}}(Y)| \le \mathsf{K}_1^\flat e^{-\kappa^\sharp r'} \le 1$ by (A11), and $|e^z - 1| \le (e-1)|z| \le 2|z|$ for $|z| \le 1$. For the sum, each long Y within r' of mk'_j meets one of $(2r'+1)^3\#\mathrm{mk}'_j$ cells. For $n = \#Y > r'$ we have $s' = 8r' \le 8(n-1)$, so $(\kappa+1)(n+s') \le (\kappa+1)+9(\kappa+1)(n-1)$. Each of the $n-1$ further cells contributes to the exponent the term $\log(26e) - \kappa^\sharp + 9(\kappa+1) \le 4.2581 - 10\kappa - 40 + 9\kappa + 9 = -(\kappa + 26.7419)$, and summing gives $\sum_{n-1\ge r'} e^{-(\kappa+26.7419)(n-1)} \le 1.001e^{-(\kappa+26.74)r'}$. The sum in (iii) is therefore at most $\#\mathrm{mk}'_j(2r'+1)^3 \cdot 2.01\mathsf{K}_1^\flat e^{\kappa+1}e^{-(\kappa+26.74)r'} \le \#\mathrm{mk}'_j$, by (A11).

(iv) The on, the short and the long sets form a partition of the connected sets; so (iv) follows from (i) and (ii).

Step 4: the alphabet. We define the letters (v), (a), (h), (t), (b) and the zone letters, at a pair (V', e) or at a field of $\mathfrak{V}$, in three definitions: on letters, on zone letters, and on compatible and actual letters. These definitions also give the meaning of link, core J, set I, forced and incompatible. In their terms, we make the following four claims.

(L) Every link of a core has at most s' cells; hence $\#J_\zeta \le \sum_{o\in\zeta}(\#\mathrm{core}(o) + s')$.

(In) $I_\alpha \subset J_\alpha^{+\varrho_*}$ for every letter α other than a forced zone letter; for a forced ζ, $I_\zeta \subset J_\zeta^{+s_0+\varrho_*}$, with $\mathfrak{b}_\zeta \subset J_\zeta^{+s_0}$.

(Rd) A letter depends on (V', e) in $I_\alpha^{+r_1}$ only; w_j depends on it only within $2r'$ of mk'_j, and n_j only within $3r'$ of mk'_j.

(Inc) Two incompatible letters satisfy $\mathrm{dist}(I_\alpha, I_\beta) \le 2 \le r_1$.

Definition 4.42 (Letters)**.** The letters λ, each with its core J and its set I, are the following.

(v) At a cell $\square$: $\lambda = \chi_\square(e^{V_\square} - 1)$, with $J = I = \{\square\}$.

(a) Of $X \in \mathcal{D}$: $\chi_X(e^{E^a(X,U)} - 1)$.

(h) Of X: $\chi_X e^{E^a(X,U)}(e^{E^h(X,U)} - 1)$. For both (a) and (h), $J = Y_0(X)$ and $I = \mathrm{cl}(X^{\approx}) \cup Y_0(X)$.

(t) Of a connected set Y of cells: $e^{T_Y} - 1$, with $J = I = Y$.

(b) At a cell $\square$: $-b_\square$, with $J = I = \{\square\}$.

The letters (v), (a), (h) are used only at fields that are tame on $J_\alpha^{\approx}$ or lie in $\mathcal{U}_{J_\alpha}(x')$; there they are defined by (Γ6) and Lem. 4.40 (T3). A product with the factor $\chi_\square$ or χ_X is taken to be 0 where that factor is 0; there U may lie outside $\mathcal{U}_X(x)$ and may fail to be tame, and $V_\square$, $E^a(X,U)$, $E^h(X,U)$ may have no value. In this way every letter is defined at every ξ. For a set $\mathfrak{B}$ of pairwise compatible letters, $K(\mathfrak{B}) = E_\xi \prod_{\mathfrak{B}} \lambda_\alpha$ denotes the mean of the product of its letters. ■

Definition 4.43 (Zone letters)**.** A *zone letter* ζ is a set of objects that is connected under adjacency. The *objects* are those of item E12 of §4.3, that is, a forced class f_j, which carries the cells of f_j; a cell; and a set $X \in \mathcal{P}$, of *kind* p; to these we add the sets $X \in \mathcal{D}$, of *kind* p'. *Adjacency* is that of item E12 of §4.3: two objects with cells are adjacent if they have cells at distance $\le 4r_1 + 2$, and an object with a cell $\square$ is adjacent to an X of kind p if and only if X holds a cube of $\widehat{\square^{+r_1}}$; in addition, an object with a cell $\square$ is adjacent to an X of kind p′ if and only if $X^{\approx}$ meets $\widehat{\square^{+r_1}}$.

A zone letter is subject to three conditions: its classes J and its cells t form a cell datum, that is, t lies off f_J and $\mathfrak{b} = f_J \cup t$ is non-empty; each of its polymers is adjacent to an object with cells; and its polymers of kind p, which make up the family $\mathfrak{X}$, lie pairwise at distance $> s(x)$. Its polymers of kind p′ make up the family $\mathfrak{Y}$. The part $(J, t, \mathfrak{X})$ alone may fail to be connected. The value of ζ is

$$\lambda_\zeta := [\hat\lambda_{(J,t)} 1_{\mathrm{ind}} \prod_{\mathfrak{X}} w(X,U)] \prod_J e^{n_j} \prod_{\mathfrak{Y}} (e^{1_{X^{\approx}}(U) E(X,U)} - 1),$$

where 1_{ind} is the indicator of item E12 of §4.3, on $D(U) \cap \hat{\mathfrak{Z}}$ with $\mathfrak{Z} = \mathfrak{b}^{+r_1}$. We put $J(\zeta) := J$, $f_\zeta := f_{J(\zeta)} = f_J$ and $\mathfrak{b}_\zeta := \mathfrak{b} = f_J \cup t$. For zone letters ζ_i, ζ_k we set $J_i := J(\zeta_i)$, the classes of ζ_i, as distinct from the core J_{ζ_i}. We write t_i, $\mathfrak{b}_i$, $\mathfrak{X}_i$, $\mathfrak{Y}_i$, $\mathfrak{Z}_i$, $1_{\mathrm{ind},i}$ for the other data of ζ_i, and likewise with the index k.

A zone letter is *forced* if J is non-empty. Its *core* J_ζ is made of the marks of the classes of ζ, the cells of t and the sets $Y_0(X)$ for the polymers X of ζ, connected by diagonal paths that follow a spanning tree, chosen by a fixed order. This tree takes its edges from the adjacencies and, within each class, from the pairs of its marks at distance $\le 2s_0 + 4r_1 + 2$. The classes of item E9 of §4.3 are those of the relation generated by such pairs, so every class is connected under them. We call these diagonal paths, one for each edge of the tree, the *links* of the core. Thus $\mathrm{core}(\square) = \{\square\}$ and $\mathrm{core}(X) = Y_0(X)$, while $\mathrm{core}(f_j)$ consists of the marks of f_j, connected in this way, and has at most $(1+s')\#\mathrm{mk}'_j$ cells. The set of ζ is $I_\zeta := \mathfrak{b}^{+2r_1} \cup \mathrm{cl}(X^{\approx\approx})$ $(X \in \mathfrak{X}) \cup \mathrm{cl}(X^{\approx})$ $(X \in \mathfrak{Y}) \cup J_\zeta$.

The *root* of a letter is the first cell of J_α; for a zone letter with non-empty t, it is the first cell of t. The *universal alphabet* $\mathfrak{A}$ consists of the letters (v), (a), (h), (t), (b) and of the zone letters without a class; these do not depend on f. Finally, $d_\alpha := (\kappa+1)(\#J_\alpha + s')$, and $\alpha \sim \beta$ if and only if $\mathrm{dist}(I_\alpha, I_\beta) \le r_1$. ■

Definition 4.44 (Compatible and actual letters)**.** Two zone letters are *compatible* if every object of the first differs from every object of the second and is not adjacent to it, and if each polymer of kind p of the first lies at distance $> s(x)$ from each polymer of kind p of the second. Besides, the following pairs are *incompatible*: a letter (b) at a cell of $\mathfrak{b}_\zeta^{+2r_1}$ and ζ; a letter (v) at a cell of $\mathfrak{b}_\zeta^{+r_1}$ and ζ; a letter (a) or (h) of an X for which $\mathrm{cl}(X^{\approx}) \cap \mathfrak{b}_\zeta^{+r_1}$ is non-empty, and ζ; and the letters (a) and (h) of a single X. At V', a letter is *actual* in the following cases: it is a forced zone letter ζ, and $f \setminus f_\zeta$ has no cell within $4r_1 + 2$ of $\mathfrak{b}_\zeta$; it is a zone letter of $\mathfrak{A}$, and f has no cell within $4r_1 + 2$ of $\mathfrak{b}$; it is a letter (b) at a cell that lies off f^{+2r_1}; it is a letter (v) at a cell that lies off f^{+r_1}; it is a letter (a) or (h) of an X whose set $\mathrm{cl}(X^{\approx})$ lies off f^{+r_1}; or it is a letter (t). ■

We prove (L), (In), (Rd) and (Inc).

(L) A link bridges at most the following distances: from a cell to a cell, $4r_1 + 2$; from a cell to a mark of an adjacent class, $s_0 + 4r_1 + 2$; between marks of a single class, $2s_0 + 4r_1 + 2$. Two

classes are never adjacent, by Step 3 (ii). From a cell $\square$ to $Y_0(X)$ for an X of kind p the distance is r_1+1: the cube that X holds lies in a cell within r_1 of $\square$, and $\mathrm{cl}(X) \subset Y_0^{+1}$. For an X of kind p′, a cube of X lies within $r(x)$ cubes of $\widehat{\square^{+r_1}}$, which gives $r_1 + \lceil r(x)/L \rceil + 2$. From a mark in place of a cell, s_0 is added. By (A2), $4r_1+2 \le r'$ and $r_1+\lceil r(x)/L\rceil+2 \le r'$; so all these distances are at most $4r'+4r_1+2 \le 5r' \le s'$. A diagonal path between cells at distance d adds $d-1$ cells.

(In) We have $\mathrm{cl}(X^{\approx}) \subset Y_0^{+\lceil r(x)/L\rceil+2}$ and $\mathrm{cl}(X^{\approx\approx}) \subset Y_0^{+\lceil s(x)/L\rceil+2}$; if ζ is universal, then $\mathfrak{b} = t \subset J_\zeta$; and $f_j = \mathrm{mk}_j'^{+s_0}$.

(Rd) By Step 0, η_b depends on (V', e) only within R_3 of $y(b)$, and this holds at every pair, hence also for the forced letters and for the letters (t) near marks, where $\phi'_P > \varepsilon'$. For $(G\xi)_b$ this follows from (Γ2), as $r_G \le R-1 \le R_3$; for $a_{0,b}$ it follows from the locality of a_0, which is the last clause of (Γ3) and holds at every pair (V', e) whatever the curvature, rough plaquettes included; and e enters through X' alone. Moreover $R_3 \le r_1 M$. The letters depend on the field as follows.

(v) By (Γ6) (a), $V_\square$ does not depend on e, and depends on V' only within R_v of $\square$; moreover $R_v \le r_1 M$.

(a), (h) These letters depend on U on $X^{\approx}$, so on V' within R_3 of $\mathrm{cl}(X^{\approx})$.

(t) Such a letter depends on φ and a_0 on $\hat{Y}$ and on $S_k(Y)$. The first paragraph of the proof of (Rd) shows that φ_b and $a_{0,b}$, at $b \in \hat{Y}$, depend on the field only within R_3 of $y(b)$, and that $R_3 \le r_1 M$. By (Γ4), $S_k(Y)$ is a function of V' on the bonds of $Y^{+\lceil R_2/M\rceil}$, and $\lceil R_2/M\rceil \le r_1$.

For a zone letter, $[A]^\ell_{\mathfrak{Z}}$ depends on η within ϱ_L+2 of $\hat{\mathfrak{Z}}$, hence on V' within R_v of $\mathfrak{Z}$, that is, in $\mathfrak{b}^{+2r_1}$; and $w(X,U)$ and $E(X,U)$ depend on V' within R_3 of $\mathrm{cl}(X^{\approx\approx})$ and of $\mathrm{cl}(X^{\approx})$, respectively. Finally, w_j depends on V' within $r' + \lceil R_6/M\rceil \le 2r'$ of mk'_j, by (A2), and the claim on n_j is part of Step 3.

(Inc) We go through the incompatible pairs.

(b), (v) For a zone letter ζ and a letter (b) or (v) that is incompatible with ζ, the cell of this letter lies in $\mathfrak{b}_\zeta^{+2r_1}$, which is a part of I_ζ; the distance is 0.

(a), (h) For a letter of X and a zone letter ζ, $\mathrm{cl}(X^{\approx})$ meets $\mathfrak{b}^{+r_1}$; the distance is 0. For the two letters of a single X it is 0 as well.

For two zone letters that share an object the distance is 0; if they have cells within $4r_1+2$ of each other, their sets $\mathfrak{b}^{+2r_1}$ are within 2. If a cell $\square$ of ζ and a polymer X of ζ' are adjacent, we take a cube of $\widehat{\square^{+r_1}}$ that X holds when X is of kind p, and a cube of $X^{\approx}$ in $\widehat{\square^{+r_1}}$ when X is of kind p′. The cell of this cube lies in $\mathfrak{b}_\zeta^{+r_1}$, and also in $\mathrm{cl}(X)$ for kind p and in $\mathrm{cl}(X^{\approx})$ for kind p′; the distance is 0. If two polymers of kind p lie within $s(x)$ of each other, a cube of one lies in $X^{\approx\approx}$ of the other; again the distance is 0.

Step 5: independence of letters at a distance.

Lemma 4.45 (Letters whose sets lie far apart are independent)**.** For a set I of cells, let $\mathcal{N}(I)$ be the set of bonds b of I° for which $y(b)$ lies within the coarse distance $R_\chi+1$ of a site of I.

(i) Every letter λ_α, at every (V', e), or at every field, at which it is used, is a Borel function of (ξ_b), $b \in \mathcal{N}(I_\alpha)$.

(ii) If $\mathrm{dist}(I_\alpha, I_\beta) > r_1$, then $\mathcal{N}(I_\alpha)$ and $\mathcal{N}(I_\beta)$ are disjoint.

Consequently, families of letters whose sets I lie pairwise at distance greater than r_1 are independent. This is the hypothesis $K(\mathfrak{B} \cup \mathfrak{B}') = K(\mathfrak{B})K(\mathfrak{B}')$ at $\mathsf{r} = r_1$ of Lem. 4.50 (4), which is stated in Step 8. ∎

Proof of Lem. 4.45. (i) The field U_b depends on η_b and on V'. Here $\eta_b = (G\xi)_b - a_{0,b}$, where a_0 does not depend on ξ, and $(G\xi)_b$ depends on ξ only within r_w of b, by (Γ2). The letters depend on ξ as follows.

(b) Such a letter at a cell $\square$ depends on the ξ at the bonds of I° in $\hat{\square}$.

(v) The factor $\chi_\square$ depends on ξ within R_χ of $\hat{\square}$, and $V_\square$ within $r_w + \varrho_L + 1 \le R_\chi$, by Thm. 4.5 (a).

(a), (h) The factor χ_X depends on ξ within R of the blocks that meet $X^\approx$, and these blocks belong to cells of $\mathrm{cl}(X^\approx)$. By (ca) and (ch), $E^a(X,\cdot)$ and $E^h(X,\cdot)$ are functions of U on the bonds of $X^\approx$, hence of ξ within $r_w + 1$.

(t) Since $S_k(Y)$ is carried by $\hat{Y} \times \hat{Y}$, only φ on $\hat{Y}$ enters, and with it ξ within r_w.

For a zone letter, $\hat{\lambda} 1_{\mathrm{ind}}$ depends only on the values of ξ at the bonds of I° in $\widehat{\mathfrak{b}^{+2r_1}}$, by the first sentence of Thm. 4.19 (a); and by (cΞ), $w(X,\cdot)$ is a function of U on the bonds of $X^{\approx\approx}$, hence of ξ within $r_w + 1$ of $\mathrm{cl}(X^{\approx\approx})$. The factors $1_{X^\approx}(U) E(X,U)$ with $X \in \mathfrak{Y}$ depend on ξ within $r_w + 1$ of $\mathrm{cl}(X^\approx)$, while e^{-w_j} and e^{n_j} do not depend on ξ.

All these radii are at most $R_\chi + 1$; here a bond of a set is a bond with both ends in it, as in item 1 of §2.1 and in (ca), (ch), (cΞ). The letters are Borel, since the members are Borel and the kernels are continuous.

(ii) Cells at cell distance $\ge r_1 + 1$ have their sites at coarse distance $\ge r_1 M + 1 > 2(R_\chi + 1)$, by item 2 of §2.1; so $\mathcal{N}(I_\alpha)$ and $\mathcal{N}(I_\beta)$ are disjoint. As the law of ξ is a product over the bonds of I°, the families are independent.

□

Remark 4.46. The locality of E^a, E^h and w is a clause of the definition of the class, and so a part of the hypothesis of Thm. 4.39 (and of $M_A^{\mathrm{su}}(x)$).

Step 6: the identity (F2) *and the bijection.* We claim the following.

(F2) At every (V', e),

$$e^{\sum_j n_j} \mathcal{Z} = \sum_{\mathfrak{B}} E_\xi \prod_{\alpha \in \mathfrak{B}} \lambda_\alpha,$$

where $\mathfrak{B}$ ranges over all sets of letters that are pairwise compatible and hold every forced class.

(Act1) All letters of such a set $\mathfrak{B}$ are actual.

(Act1′) Where an actual letter (v), (a) or (h) is used, V' is tame on $J_\alpha^\approx$.

(Act2) The cell datum of an actual zone letter has (H17).

We begin with the resolution of the integrand. At every ξ,

$$1 = \sum_{\mathfrak{b} \supset f} \prod_{\mathfrak{b} \setminus f} b_\square \prod_{\square \notin \mathfrak{b}} (1 - b_\square),$$

the product, over the cells off f, of $b_\square + (1 - b_\square)$. On the support of the term $\mathfrak{b}$ the following holds.

(1) Let $\square$ be a cell off $\mathfrak{b}^{+r_1}$. The bonds within $R_\chi \le r_1 M$ of $\hat{\square}$ lie in cells within r_1 of $\square$, hence off $\mathfrak{b}$; so $\chi_\square = 1$. Moreover $\mathrm{dist}(\square, \mathrm{mk}') > s_0 + r_1 > r'$, so V' is tame on $\square^\approx$, and Thm. 4.5 (d) gives $A_\square = e^{V_\square} = 1 + \lambda^v_\square$, where $\lambda^v_\square$ is the value $\chi_\square(e^{V_\square} - 1)$ of the letter (v) at $\square$.

(2) $D(U) \subset \widehat{\mathfrak{b}^{+r_1}}$. Indeed, take a plaquette p whose base site lies in a cell $\square$ off $\mathfrak{b}^{+r_1}$. Every bond $b' \in I^\circ$ within r_w of a bond of p lies within $r_w + 1 \le R_\chi$ of $\hat{\square}$, and hence satisfies $|\xi_{b'}| \le c_\xi \varepsilon$. Each plaquette of D' has its base site in a mark, at coarse distance $> (s_0 + r_1) M \ge R_3 + 1$ from $y(p)$. Lem. 4.40 (T1) now gives $\phi_p \le 0.72\varepsilon$, and so p does not belong to $D(U)$.

(3) Let $X \in \mathcal{D}$ be such that $\mathrm{cl}(X^\approx) \cap \mathfrak{b}^{+r_1}$ is empty. Every bond within $R \le r_1 M$ of a block meeting $X^\approx$ then lies off $\mathfrak{b}$, which gives $\chi_X = 1$; moreover $1_{X^\approx}(U) = 1$ by (2). Hence

$e^{1_X\approx(U)E(X,U)} = 1+\lambda^a_X+\lambda^h_X$, where $\lambda^a_X = \chi_X(e^{E^a(X,U)}-1)$ and $\lambda^h_X = \chi_X e^{E^a(X,U)}(e^{E^h(X,U)}-1)$ are the values of the letters (a) and (h) of X.

(4) For the other $X \in \mathcal{D}$,

$$\prod e^{1_X\approx(U)E(X,U)} = \sum_{\mathfrak{Y}} \prod_{\mathfrak{Y}} (e^{1_X\approx(U)E(X,U)} - 1),$$

where $\mathfrak{Y}$ runs over the subsets of $\{X : \mathrm{cl}(X^{\approx}) \cap \mathfrak{b}^{+r_1} \text{ non-empty}\}$.

(5) By (2), $D(U) = D(U) \cap \hat{\mathfrak{Z}}$ with $\mathfrak{Z} = \mathfrak{b}^{+r_1}$. Hence $\Xi_w(U) = \sum_{\mathfrak{X}} \prod_{\mathfrak{X}} w(X,U) 1_{\mathrm{ind}}$, the sum running over the families in $\mathcal{P}$ at pairwise distance $> s(x)$, and $1_{\mathrm{ind}} = 0$ unless each X holds a cube of $\hat{\mathfrak{Z}}$.

We split three more factors. First,

$$\prod_{\square \notin \mathfrak{b}} (1 - b_\square) = \prod_{\mathfrak{b}^{+2r_1}\setminus\mathfrak{b}} (1 - b_\square) \sum_{J_b} \prod_{J_b} (-b_\square),$$

where J_b runs over the sets of cells off $\mathfrak{b}^{+2r_1}$. Second, $e^{\sum T_Y} = \prod_Y (1 + (e^{T_Y} - 1))$. Third, $\prod_\square A_\square = [A]^\ell_{\mathfrak{Z}} \prod_{\square \text{ off } \mathfrak{Z}} A_\square$. Off the support of the term $\mathfrak{b}$, its first factors vanish. So at every ξ the integrand of $\bar{\mathcal{Z}}$ is

$$\sum_{\mathfrak{b}} \prod_{\mathfrak{b}\supset f} \prod_{\mathfrak{b}\setminus f} b_\square \prod_{\mathfrak{b}^{+2r_1}\setminus\mathfrak{b}} (1 - b_\square)\, [A]^\ell_{\mathfrak{Z}} \sum_{\mathfrak{X}} \prod_{\mathfrak{X}} w(X,U) 1_{\mathrm{ind}} \sum_{\mathfrak{Y}} \prod_{\mathfrak{Y}} (e^{1_X\approx(U)E(X,U)} - 1)$$
$$\times \sum_{J_b} \prod_{J_b} (-b_\square) \prod_{\square \text{ off } \mathfrak{Z}} (1 + \lambda^v_\square) \prod_{X \text{ of (3)}} (1 + \lambda^a_X + \lambda^h_X) \prod_Y (1 + (e^{T_Y} - 1)).$$

All these sums are finite.

We turn to the bijection. Let $(\mathfrak{b}, \mathfrak{X}, \mathfrak{Y})$ be given with non-zero indicator. Then the components under adjacency of the collection made of all classes, the cells of $\mathfrak{b} \setminus f$, the polymers of $\mathfrak{X}$ and those of $\mathfrak{Y}$ are zone letters; they are pairwise compatible, and together they hold every class once. They are zone letters because for each X of $\mathfrak{X}$ there is a cell $\square$ of $\mathfrak{b}$ such that X holds a cube of $\widehat{\square^{+r_1}}$, for each X of $\mathfrak{Y}$ there is a cell $\square$ of $\mathfrak{b}$ such that $X^{\approx}$ meets $\widehat{\square^{+r_1}}$, and every cell of $\mathfrak{b}$ belongs to an object.

Conversely, suppose that zone letters ζ_i are pairwise compatible and that together they hold every class. A cell object lying inside some f_k would be at distance 0 from the object f_k; the cell datum rules this out when $k \in J(\zeta_i)$, and compatibility rules it out otherwise. So the cell objects are distinct cells off f; we put $\mathfrak{b} := \bigcup \mathfrak{b}_i \supset f$, and the ζ_i are the components.

As for the factors, $\mathrm{dist}(\mathfrak{b}_i, \mathfrak{b}_k) > 4r_1 + 2$, so the sets $\mathfrak{b}_i^{+2r_1}$ are disjoint, and the first three factors split into products. As for the indicator, $\hat{\mathfrak{Z}}$ is the disjoint union of the $\hat{\mathfrak{Z}}_i$. A plaquette p of $D(U) \cap \hat{\mathfrak{Z}}_i$ is covered by an X that holds its cube, and this cube lies in $\widehat{\square^{+r_1}}$ for some $\square \in \mathfrak{b}_i$; so X is adjacent to an object of ζ_i, and $X \in \mathfrak{X}_i$. If an X of $\mathfrak{X}_i$ holds a plaquette of $D(U) \cap \hat{\mathfrak{Z}}_k$, then it is adjacent to ζ_k, so $k = i$. Hence $1_{\mathrm{ind}} = \prod_i 1_{\mathrm{ind},i}$.

The sets J_b, the cells off $\mathfrak{Z}$, the sets X of (3), with at most one of (a), (h) to each X, and the sets Y are exactly the letters (b), (v), (a), (h), (t) that are compatible with all the ζ_i. The factors e^{-w_j} and e^{n_j} are given to the letter of the class j.

We turn to the clauses on actual letters.

(Act1) Let ζ be a zone letter. If some cell of f_k lies within $4r_1 + 2$ of a cell of $\mathfrak{b}_\zeta$, then f_k is adjacent to an object of ζ; hence either f_k belongs to ζ, or ζ and the letter ζ_k that holds f_k are incompatible. For the letters (b), (v), (a), (h), note that $\mathfrak{b}_{\zeta_k} \supset f_k$; the incompatibilities of Step 4 exclude the cells of $f_k^{+2r_1}$, the cells of $f_k^{+r_1}$, and the sets X for which $\mathrm{cl}(X^{\approx}) \cap f_k^{+r_1}$ is non-empty.

(Act1′) We take the letters in turn.

(v) $\operatorname{dist}(\square, \mathrm{mk}') > s_0 + r_1 > r'$.

(a), (h) In both cases of the definition of $Y_0(X)$ in item 1 of §2.1, $Y_0(X) \subset \mathrm{cl}(X^{\sim})$, a part of $\mathrm{cl}(X^{\approx})$ (for an X that is not small, the inclusion rests on $\mathrm{hull}(X) = X$, which holds by that item since $X \in \mathcal{D}$). Hence $\operatorname{dist}(Y_0(X), \mathrm{mk}') > s_0 + r_1 > r'$, and Lem. 4.40 (T3) applies with $Y = Y_0(X)$.

(Act2) Suppose that $P \in D'$ lies in a cell at distance at most $4r_1$ from $\mathfrak{b}_\zeta$. Such a cell is a mark and so belongs to f; actuality places it in f_ζ, and for $\zeta \in \mathfrak{A}$ no such cell exists. The marks of a class k keep a distance greater than $2s_0 + 4r_1 + 2$ from those of each class $j \neq k$; such a mark therefore lies in no f_j with $j \neq k$, and its class belongs to $J(\zeta)$.

Step 7: the product bound.

Definition 4.47. For a letter α we put $e_\alpha := e^\circ_\alpha e^{2x^{-1/8}\#I_\alpha}$, where e°_α is as follows.

(v) $2K_v p_0^3 x^{-1/2}$.
(a) $2c_a(x)e^{-\kappa(\#X-1)}$.
(h) $4c_h(x)e^{-\kappa(\#X-1)}$.
(t) $e_Y^{1/2} = x^{-20}e^{-\kappa^\sharp \#Y/2}$.
(b) u_b.

For a zone letter, e°_ζ is

$$\mathrm{Res}_{(J,t)}\, u_t^{\#t} u_1^{1[t \text{ non-empty}]} \prod_{\mathfrak{X}} \upsilon e^{-\kappa(|X|_c-1)} \prod_J e^{1.01\mathsf{K}_1^\flat (2r'+1)^3 \#\mathrm{mk}'_j} \prod_{\mathfrak{Y}} 2(c_a + c_h)(x) e^{-\kappa(\#X-1)}.$$

■

We claim that $E_\xi|\prod_{\mathfrak{B}} \lambda_\alpha| \le \prod_{\mathfrak{B}} e_\alpha$ in each of the following four situations.

(i) For pairwise compatible letters that are actual at V', at every e.
(ii) At a field that is real on the bonds of $Y^{\approx}$ and tame there, for letters of $\mathfrak{A}$ that are pairwise compatible and satisfy $J_\alpha \subset Y$.
(iii) For letters (a), (v), (t) with $J_\alpha \subset Y$, at $V' \in \mathcal{U}_Y(x')$.
(iv) For letters (b).

We now prove the claim.

1. We first bound the letters at every ξ, before any mean over ξ is taken.
   (v) The bound is (Γ6) (c).
   (a) Lem. 4.40 (T3) shows that the restriction of U to $X^{\approx}$ lies in $\mathcal{U}_X(x)$ where $\chi_X = 1$; we use $|e^z - 1| \le 2|z|$ for $|z| \le 1/2$, and (A4).
   (h) U is real and tame on $X^{\approx}$, and $e^{1/2} \cdot 2c_h(x) \le 4c_h(x)$.
   For a polymer of kind p′, where the indicator $1_{X^{\approx}}(U)$ is 1, the restriction of U to $X^{\approx}$ is real and lies in $\mathcal{U}_X(x)$, by item 1 of Step 1; so $|e^{1_{X^{\approx}}(U)E(X,U)} - 1| \le 2(c_a + c_h)(x)e^{-\kappa(\#X-1)}$ at every ξ. The factors e^{n_j} are bounded by Step 3. The hypotheses of (Γ6) and of Lem. 4.40 (T3) hold in the situation (i) by (Act1′), and in the situations (ii) and (iii) because $J_\alpha^{\approx} \subset Y^{\approx}$.
2. What is left of a zone letter ζ_i is $\hat\lambda_{(J_i,t_i)} 1_{\mathrm{ind}} \prod_{\mathfrak{X}_i} w$, and in the situation (i) the datum (J_i, t_i) has (H17) by (Act2). We apply Thm. 4.19 (g*1) to $O_i = \zeta_i$ without $\mathfrak{Y}_i$: the components $\check\zeta_i^{(m)}$ of O_i under the adjacency of item E12 of §4.3 are pairwise compatible admitted letters, and $\hat\lambda_{(J_i,t_i)} 1_{\mathrm{ind}} \prod_{\mathfrak{X}_i} w = \prod_m \lambda^o_{\check\zeta_i^{(m)}}$. Components of different ζ_i are compatible, because the ζ_i are; and the sets $\mathfrak{b}^{(m)+2r_1}$ are disjoint parts of $\mathfrak{b}_i^{+2r_1}$. Moreover $\prod_m e^o_{\check\zeta^{(m)}} \le \mathrm{Res}_{(J,t)}\, u_t^{\#t} u_1^{1[t \text{ non-empty}]} \prod_{\mathfrak{X}} \upsilon e^{-\kappa(|X|_c-1)}$, since Res and $u_t^{\#t}$ factor over the components and $u_1 \le 1$. The cells of the letters (b) of $\mathfrak{B}$ lie off all the $\mathfrak{b}_i^{+2r_1}$.
3. We turn to the letters (t) at a real V'. Transports are isometries and $|X'_P| \le \pi$; moreover $\|\ell\|_{\mathrm{r}} \le 3K_* \mathsf{g}_*$ and $\|\ell\|_{\mathrm{c}} \le K_* \mathsf{g}_*$, by the first two bounds of Def. 2.48, $\|\ell\|_{\mathrm{r}} \le 3K\mathsf{g}(\lambda)$ and

$\|\ell\|_{\mathrm{c}} \le K\mathsf{g}(\lambda)$, taken at (K_*, λ_*) with $\mathsf{g}_* = \mathsf{g}(\lambda_*)$, and by (SD). Hence $|(\mathsf{B}^V EX')_p| \le \mathsf{N}_{\mathsf{B}}\pi$. A bond lies in 4 plaquettes, so $|\mathsf{g}^V X'|_\infty \le 4\pi\mathsf{N}_{\mathsf{B}}$; and as $a_0 = GG^{\mathsf{T}}\mathsf{g}^V X'$, we get $|a_0|_\infty \le a_\infty$ at every (V', e). A cell carries at most $3v$ bonds; therefore

$$|T_Y| \le \frac{x}{2} e_Y |\varphi_{\hat{Y}}|^2 + x e_Y |\varphi_{\hat{Y}}||a_{0,\hat{Y}}| \le x e_Y \big(|\varphi_{\hat{Y}}|^2 + \frac{1}{2}|a_{0,\hat{Y}}|^2\big) \le x e_Y \big(|\varphi_{\hat{Y}}|^2 + n_a \# Y\big).$$

Put $z_Y = x e_Y^{1/2}(n_a \# Y + |\varphi_{\hat{Y}}|^2)$; then $|e^{T_Y} - 1| \le |T_Y| e^{|T_Y|} \le e_Y^{1/2} e^{2z_Y}$. Further, $\sup_\square \sum_{Y \ni \square} e_Y^{1/2} \le x^{-20} \sum_n (26e)^{n-1} e^{-50n} \le 1.01 x^{-20}$. Writing Y for the union of the sets Y, we obtain

$$\prod |e^{T_Y} - 1| \le \prod e_Y^{1/2} \cdot \exp\{2.02 x^{-19} n_a \sum \# Y\} \cdot e^{\frac{x}{2}\varsigma |\varphi_{\mathsf{Y}}|^2},$$

where $\varsigma = 4.04 x^{-20} \le (\gamma/2) x^{-2}$ by (A10).

4. For the situations (i) and (iv), we apply Thm. 4.19 (d) at $A = 1$ to the letters $\zeta_i^{\check{}(m)}$, with $J_0 = J_b$, the set of the cells of the letters (b), and to Y and ς. The harmless factor that remains is

$$\exp\big\{[2.02 x^{-19} n_a + 38.8 v \gamma^{-1} x^{-20} (2r_1 + 1)^3] \sum \# Y + x^{-1/8} \big(\sum \# \mathfrak{b}_i^{+2r_1} + \# J_b\big)\big\}$$

(here $4.8 \cdot 2 \cdot 4.04 = 38.784 \le 38.8$), and by (A10) it is at most $\prod e^{2x^{-1/8} \# I_\alpha}$, since $I_\zeta \supset \mathfrak{b}^{+2r_1}$.

5. For the situation (ii) we extend the field by 1: let $\tilde{V}' = V'$ on the bonds of $Y^{\approx}$, and 1 elsewhere. A plaquette of $D'(\tilde{V}')$ has a site off $Y^{+r'}$, and its base site is within one site of that one; so all marks of $\tilde{V}'$ are at cell distance $\ge r'$ from Y. By (A2), $2\varrho_* + r_1 \le r' - 2$. A letter of $\mathfrak{A}$ with $J_\alpha \subset Y$ depends on the field in $I_\alpha^{+r_1}$, which is a part of $Y^{+\varrho_* + r_1}$ and so of $Y^{+(r'-2)}$; hence its value does not depend on the extension, and it does not depend on e, since $\phi'_P \le \varepsilon' < \pi$ there. A zone letter has $\mathfrak{b} \subset J_\zeta \subset Y$, and $r' > 4r_1$; so no plaquette of $D'(\tilde{V}')$ lies in a cell within $4r_1$ of $\mathfrak{b}$, and (H17) holds with J empty. Items 1 to 4 therefore apply at $(\tilde{V}', e)$, for any e.
6. In the situation (iii) there is no zone letter. The extended field lies in $\mathfrak{V}^\circ$ by (u1). Since $|Y \cdot Y'| \le \|Y\| \, \|Y'\|$, item 3 holds with $\|\cdot\|$ in place of $|\cdot|$. At $b \in \hat{Y}$ we have $\|a_{0,b}\| \le \mathsf{a}_{\mathrm{c}}\varepsilon'$, because the box of radius $R_3 \le r_1 M$ about $y(b)$ lies in $Y^{\approx}$. By (A1), $(4.1\mathsf{a}_{\mathrm{c}}\varepsilon')^2 \le 0.1\varepsilon/0.72 \le 0.1/28.8$, so $\mathsf{a}_{\mathrm{c}}\varepsilon' \le 0.0144 \le 4\pi \le a_\infty$. The last inequality follows from $a_\infty = 4\pi \mathsf{N}_{\mathsf{B}} (C_\varphi^\flat)^2$, $\mathsf{N}_{\mathsf{B}} \ge 1$ and $C_\varphi^\flat \ge \mathsf{G}_0 = (2/\gamma)^{1/2} 3L^3 \mathsf{g}(\theta_{\mathbb{C}}) > 1$, by item C.I 1.2 of §3. Next, $\|\varphi_b\|^2 \le (C_\varphi^\flat)^2 \sum_{b' \in Y_b} |\xi_{b'}|^2$, and a bond b' lies in at most $3L^3(R+1)^3$ windows; so $\|\varphi_{\hat{\mathsf{Y}}}\|^2 \le 3L^3(R+1)^3 (C_\varphi^\flat)^2 |\xi_{\mathsf{Y}^{+r_1}}|^2$. Finally, $E_\xi e^{\frac{x}{2}\varsigma' |\xi_b|^2} = (1 - \varsigma')^{-3/2} \le e^{3\varsigma'}$ for $\varsigma' \le 1/2$, and there are at most $3v(2r_1+1)^3 \# \mathsf{Y}$ bonds; (A10) then gives the bound.

Step 8: sums over connected sets, and the summability condition. We attach the following weights to cells and to polymers: a cell $\square$ receives $\hat{w}(\square) = u_t e^{(\kappa+4)(1+s')}$, a polymer X of kind p receives $\hat{w}_p(X) = v e^{-\kappa(|X|_c - 1) + (\kappa+4)(\# Y_0(X) + s')}$, and a polymer X of kind p′ receives $\hat{w}_{p'}(X) = 4(c_a + c_h) e^{-\kappa(\# X - 1) + (\kappa+4)(\# Y_0(X) + s')}$.

In the setting (A3), (A6), we have

$$\sum_{\zeta \ni o} \prod_{o' \in \zeta} \hat{w}(o') \le \hat{w}(o) e^{a_o}, \qquad a_o \le \#\mathrm{core}(o)/50, \tag{An}$$

where ζ ranges over those connected sets of cells and polymers that contain o, and where the numbers a_o are the ones defined below. The first inequality holds for o a cell, or a polymer of kind p, or a polymer of kind p′; the second holds for all objects o, the classes f_j included.

Moreover, with $\mathsf{r} = r_1$, $t = x^{-1/4}$, $\varepsilon_2 = x^{-1/8}$ and $d_\beta = (\kappa+1)(\# J_\beta + s')$, the alphabet $\mathfrak{A}$ satisfies the condition (S) of Lem. 4.50 below.

We first state and prove four lemmas, and then return to the step.

Lemma 4.48 (Sums over connected sets)**.** Let a countable set carry a symmetric relation, and call two of its elements *adjacent* when they are related. Attach to each element β numbers $v_\beta \geq 0$ and $a_\beta \geq 0$, and suppose that $\sum_{\beta \text{ adjacent to } \alpha} v_\beta e^{a_\beta} \leq a_\alpha$ for all α. We call a finite set that is connected with respect to the relation an *animal*, and the hypothesis just made the *condition for animals*. Then for each α, with S running over the animals that contain α,

$$\sum_{S \ni \alpha} \prod_{\beta \in S} v_\beta \leq v_\alpha e^{a_\alpha}; \tag{An0}$$

and if only the sets S with at least two members are kept, the sum is at most $v_\alpha(e^{a_\alpha} - 1)$. ■

Proof. For $n \geq 1$ let $Z_n(\alpha)$ be the sum of $\prod_{\beta \in S} v_\beta$ over the connected sets $S \ni \alpha$ having at most n members; in particular $Z_1(\alpha) = v_\alpha$. Let $S \ni \alpha$ be connected with at most $n+1$ members, and remove α from it. What is left splits into $k \geq 0$ connected components; they are pairwise distinct, none of them has more than n members, and each contains some β adjacent to α. The components determine S, and since they are pairwise distinct, each choice of k of them occurs at least $k!$ times among the ordered k-tuples that the k-th power of $\sum_{\beta \text{ adjacent to } \alpha} Z_n(\beta)$ counts; hence

$$Z_{n+1}(\alpha) \leq v_\alpha \sum_k \frac{1}{k!} \Big(\sum_{\beta \text{ adjacent to } \alpha} Z_n(\beta) \Big)^k \leq v_\alpha \exp\Big\{ \sum_{\beta \text{ adjacent to } \alpha} v_\beta e^{a_\beta} \Big\} \leq v_\alpha e^{a_\alpha},$$

where the second inequality uses $Z_n(\beta) \leq v_\beta e^{a_\beta}$ and the third is the hypothesis. Since $v_\alpha \leq v_\alpha e^{a_\alpha}$, the bound holds for $n = 1$, and induction on n yields $Z_n(\alpha) \leq v_\alpha e^{a_\alpha}$ for every n, and the first claim follows as $n \to \infty$. In this bound the term $k = 0$ comes from the set $S = \{\alpha\}$ and equals v_α; without it, the same chain bounds the sum over the sets with at least two members by $v_\alpha(e^{a_\alpha} - 1)$. □

Lemma 4.49 (A convergence criterion for sums over polymers, due to Kotecký and Preiss)**.** Let $\mathfrak{P}$ be a finite set, whose elements we call *polymers*, and let a symmetric relation on $\mathfrak{P}$ be given; related polymers are said to *touch*, and every polymer touches itself. Attach to each P numbers $W(P) \in \mathbb{C}$ and $a(P) \geq 0$, and suppose that $\sum_{P' \text{ touches } P} |W(P')| e^{a(P')} \leq a(P)$ for all P. A *cluster* is a finite sequence $C = (P_1, \dots, P_n)$ of polymers, with $n \geq 1$, for which the graph on $\{1, \dots, n\}$ that joins $i \neq j$ whenever P_i touches P_j is connected. For a cluster we set $\varphi^T(C) := \frac{1}{n!} \sum_g (-1)^{\#g}$, where g ranges over the connected spanning subgraphs of that graph and $\#g$ denotes the number of edges of g; we write $C \ni P$ when P is one of the P_i.

Then $\sum_{C \ni P} |\varphi^T(C)| \prod_i |W(P_i)| \leq |W(P)| e^{a(P)}$ for every polymer P, and the clusters with $n \geq 2$ contribute at most $|W(P)|(e^{a(P)} - 1)$ to this sum. Moreover, $\sum_{\{P_i\}} \prod_i W(P_i) = \exp \sum_C \varphi^T(C) \prod_i W(P_i)$, where on the left $\{P_i\}$ runs over the sets of polymers no two of which touch, and the empty set contributes 1. ■

Proof. Fix a total order on the edges of the cluster graph. For a spanning tree τ, let $R(\tau)$ consist of τ and every edge whose order is greater than that of each edge of the path in τ joining its endpoints. Run the greedy tree algorithm in increasing order on a connected spanning subgraph g. If its output is τ, then every edge of $g \setminus \tau$ was rejected after its endpoints were already joined by earlier edges of τ. Hence $g \subset R(\tau)$. Conversely, if $\tau \subset g \subset R(\tau)$, induction over the edges shows that each edge of τ is accepted and each edge of $g \setminus \tau$ is rejected. The subgraphs giving τ therefore form the interval $[\tau, R(\tau)]$. The alternating sum on an interval is 0 unless $R(\tau) = \tau$, when it is $(-1)^{n-1}$. This gives the Penrose tree bound

$$\Big| \sum_g (-1)^{\#g} \Big| \leq \#\{\text{spanning trees}\}.$$

A cluster containing P can be rooted at an occurrence of P. Counting all such occurrences only increases the sum. Permutation invariance then puts the root at the first place at a cost of a factor

n. Thus the first sum of the lemma is at most $|W(P)|T(P)$, where

$$T(P) = \sum_{n\geq 1} \frac{1}{(n-1)!} \sum_{P_2,\dots,P_n} \sum_{\tau} \prod_{i=2}^{n} |W(P_i)|,$$

with $P_1 = P$ and the tree edges required to join touching polymers.

For $m \geq 0$ let $T_m(P)$ be the part of $T(P)$ that comes from the trees whose depth, measured from the root 1, is at most m; thus $T_0 = 1$. Sorting the trees according to the k subtrees that hang at the root, we find

$$T_{m+1}(P) = \exp\{ \sum_{P' \text{ touches } P} |W(P')| T_m(P')\} \leq \exp\{\sum |W(P')| e^{a(P')}\} \leq e^{a(P)},$$

where the first inequality uses $T_m(P') \leq e^{a(P')}$ and the second is the hypothesis. Induction on m gives $T_m(P) \leq e^{a(P)}$ for every m, hence $T(P) \leq e^{a(P)}$, which proves the first claim. In $T(P)$ the term $n = 1$ is the summand 1; it comes from the cluster (P), which contributes $|W(P)|$ to the sum of the first claim, and omitting it on both sides proves the second claim.

For the identity introduce a variable z and replace every activity by $zW(P)$. The coefficient of z^n in the partition function is

$$\frac{1}{n!} \sum_{P_1,\dots,P_n} \prod_{i} W(P_i) \prod_{i<j} (1 - 1\{P_i \text{ touches } P_j\}).$$

Self-touching excludes repeated polymers. Expand the product at each fixed n and partition the graph into connected components. A component contributes its connected graph sum. The factor $1/n!$ accounts for the allocation of labels to the components and their permutations, so the resulting identity of formal power series is the exponential of the connected-cluster series. The rooted estimate above, summed over the finite set of possible root polymers, gives absolute convergence of that series for $|z| \leq 1$. The formal identity thus gives equality for $|z| < 1$. Let z tend to 1 along the real interval and use absolute convergence to obtain the claimed identity. □

Lemma 4.50 (Clusters of letters under a summability condition)**.** Let a finite alphabet be given, and attach to each letter α a finite non-empty set I_α of cells and two numbers $e_\alpha \geq 0$, $d_\alpha \geq 0$. For a number $\mathsf{r} \geq 0$ we write $\alpha \sim \beta$ if and only if $\operatorname{dist}(I_\alpha, I_\beta) \leq \mathsf{r}$. Let *compatibility* be a symmetric relation between different letters, such that two letters that are not compatible are always $\sim$. Let $0 < t \leq 1$ and $\varepsilon_2 > 0$, and suppose that at every cell $\square$

$$\sum_{\beta : \operatorname{dist}(I_\beta, \square) \leq \mathsf{r}} e_\beta e^{3\varepsilon_2 \# I_\beta + d_\beta} \leq t\varepsilon_2. \tag{S}$$

A *polymer* is a non-empty set of pairwise compatible letters that is $\sim$-connected, and two polymers *touch* if some letter of the one is $\sim$ to some letter of the other. For this relation, the clusters $C = (\mathfrak{B}_1, \dots, \mathfrak{B}_n)$ of polymers and the numbers $\varphi^T(C)$ are those of Lem. 4.49, and we set $\hat\Phi(C) := |\varphi^T(C)| \prod_i \prod_{\alpha\in\mathfrak{B}_i} e_\alpha$. We denote by $n(C)$ the number of the letters of C and by $d(C)$ the sum of their numbers d_α, both counted with multiplicity. We write $C \ni \alpha$ if α belongs to some $\mathfrak{B}_i$, and $C \sim \alpha$ if some letter of C is $\sim \alpha$; in the relation $C \sim \alpha$ we allow α to be any object that carries a finite non-empty set I_α of cells. Then the following hold.

(1) For every letter α, $\sum_{C\ni\alpha} \hat\Phi(C) e^{d(C)} \leq e_\alpha e^{2\varepsilon_2 \# I_\alpha + d_\alpha}$.
(2) The clusters with $n(C) \geq 2$ contribute to the sum in (1) at most t times the right side of (1).
(3) For every such object α, $\sum_{C\sim\alpha} \hat\Phi(C) e^{d(C)} \leq t\varepsilon_2 \# I_\alpha$; and $\hat\Phi(C) e^{d(C)} \leq t\varepsilon_2$ for every cluster C.
(4) Suppose in addition that a number $K(\mathfrak{B}) \in \mathbb{C}$ is given for every set $\mathfrak{B}$ of pairwise compatible letters, with $K(\emptyset) = 1$, with $|K(\mathfrak{B})| \leq \prod_{\mathfrak{B}} e_\alpha$, and with $K(\mathfrak{B}\cup\mathfrak{B}') = K(\mathfrak{B})K(\mathfrak{B}')$ whenever

no letter of $\mathfrak{B}$ is $\sim$ to a letter of $\mathfrak{B}'$; and put $\Phi(C) := \varphi^T(C)\prod_i K(\mathfrak{B}_i)$. Then $|\Phi| \le \hat{\Phi}$, and $\sum_{\mathfrak{B}} K(\mathfrak{B}) = \exp\sum_C \Phi(C)$, where $\mathfrak{B}$ runs over the sets just named and the series converges absolutely.

If an alphabet satisfies (S), so does each of its subsets, and the claims therefore hold for each subset. Claims (1) to (3) do not involve K, and so they do not depend on any field. ■

Proof. Put $s_\alpha = \varepsilon_2 \# I_\alpha$ and $v_\alpha = e_\alpha e^{d_\alpha}$. If $\beta \sim \alpha$, then $\operatorname{dist}(I_\beta, \square) \le \mathsf{r}$ for some cell $\square$ of I_α; adding up (S) over the cells of I_α we therefore obtain

$$\sum_{\beta\sim\alpha} v_\beta e^{3s_\beta} \le t s_\alpha \tag{S$'$}$$

for every object α that carries a set I_α.

We apply Lem. 4.48 to the letters, with $\sim$ as the adjacency, with the weights $v_\beta e^{ts_\beta}$ and with $a_\beta = ts_\beta$. Its hypothesis becomes $\sum_{\beta\sim\alpha} v_\beta e^{2ts_\beta} \le t s_\alpha$, which follows from (S$'$) since $t \le 1$. The polymers that hold α are among the connected sets that hold α, so the lemma gives

$$\sum_{\mathfrak{B}\ni\alpha} \prod_{\beta\in\mathfrak{B}} v_\beta e^{ts_\beta} \le v_\alpha e^{2ts_\alpha}, \tag{$\star$}$$

the sum running over the polymers $\mathfrak{B}$ that hold α.

Next we apply Lem. 4.49 to the polymers, which form a finite set, with $W(\mathfrak{B}) = \prod_{\mathfrak{B}} v_\beta$ and $a(\mathfrak{B}) = t\sum_{\mathfrak{B}} s_\beta$. If $\mathfrak{B}'$ touches $\mathfrak{B}$, then some letter β of $\mathfrak{B}'$ is $\sim$ to some letter α of $\mathfrak{B}$; hence, by ($\star$) and (S$'$),

$$\sum_{\mathfrak{B}' \text{ touches } \mathfrak{B}} W(\mathfrak{B}') e^{a(\mathfrak{B}')} \le \sum_{\alpha\in\mathfrak{B}} \sum_{\beta\sim\alpha} \sum_{\mathfrak{B}'\ni\beta} \prod_{\mathfrak{B}'} v e^{ts} \le \sum_{\alpha\in\mathfrak{B}} \sum_{\beta\sim\alpha} v_\beta e^{2ts_\beta} \le a(\mathfrak{B}),$$

and this is the hypothesis of Lem. 4.49. For every cluster, $\hat{\Phi}(C)e^{d(C)} = |\varphi^T(C)|\prod_i W(\mathfrak{B}_i)$. We now prove the claims in turn.

(1) With the summand $\hat{\Phi}(C)e^{d(C)}$ in the first two sums, Lem. 4.49 and ($\star$) give

$$\sum_{C\ni\alpha} \le \sum_{\mathfrak{B}\ni\alpha} \sum_{C\ni\mathfrak{B}} \le \sum_{\mathfrak{B}\ni\alpha} W(\mathfrak{B}) e^{a(\mathfrak{B})} \le v_\alpha e^{2ts_\alpha} \le v_\alpha e^{2s_\alpha},$$

and $v_\alpha e^{2s_\alpha} = e_\alpha e^{2\varepsilon_2 \# I_\alpha + d_\alpha}$.

(2) Exactly one cluster with $n(C) = 1$ holds α, namely $(\{\alpha\})$, and its term is v_α. By the chain in (1), the other clusters that hold α contribute at most $v_\alpha(e^{2ts_\alpha} - 1) \le t v_\alpha(e^{2s_\alpha} - 1)$; the last inequality holds because $t \mapsto e^{2ts_\alpha}$ is convex and takes the value 1 at $t = 0$. So these clusters contribute at most t times the bound in (1).

(3) A cluster $C \sim \alpha$ holds a letter $\beta \sim \alpha$; hence, by (1) and (S$'$),

$$\sum_{C\sim\alpha} \le \sum_{\beta\sim\alpha} \sum_{C\ni\beta} \le \sum_{\beta\sim\alpha} v_\beta e^{2s_\beta} \le t s_\alpha,$$

which is the first half, as $s_\alpha = \varepsilon_2 \# I_\alpha$. For the second half, choose a letter β of C and a cell $\square$ of I_β; then (1) and (S) at $\square$ give $\hat{\Phi}(C)e^{d(C)} \le v_\beta e^{2s_\beta} \le t\varepsilon_2$.

(4) A set $\mathfrak{B}$ of pairwise compatible letters can be written in exactly one way as a union of polymers no two of which touch, namely as the union of its $\sim$-components, and $K(\mathfrak{B})$ is the product of the values of K on these components. Conversely, a union of polymers no two of which touch consists of pairwise compatible letters, since incompatible letters are $\sim$. Thus $\sum_{\mathfrak{B}} K(\mathfrak{B})$ is the left side of the identity of Lem. 4.49 for the weight $W = K$ on the polymers. Moreover $|K(\mathfrak{B})| \le \prod_{\mathfrak{B}} e_\alpha \le \prod_{\mathfrak{B}} v_\alpha$; so the hypothesis of Lem. 4.49, established above for the weight $\prod_{\mathfrak{B}} v_\beta$, holds for $W = K$ as well, the identity of that lemma becomes the identity in (4), with absolute convergence, and the same bound on K yields $|\Phi| \le \hat{\Phi}$.

□

Lemma 4.51 (Sums over letters and clusters attached to given letters)**.** Let the data of Lem. 4.50 be given and satisfy (S), and let $t \le 1/2$. Let F be a finite set of additional letters ζ; to each of them a finite non-empty set I_ζ of cells is attached, and $\sim$ is defined for them by the same rule. Let $\mathfrak{M}$ be a countable set of objects C, each of which has a relation $C \sim \alpha$ to the letters, and let numbers $\hat w_C \ge 0$ be given such that $\sum_{C\in\mathfrak{M},\, C\sim\alpha} \hat w_C \le \varepsilon_2 \# I_\alpha$ for every letter α, both of the alphabet and of F. We consider the pairs $(\mathfrak{B}_o, \mathfrak{N})$ consisting of a set $\mathfrak{B}_o$ of letters of the alphabet, every $\sim$-component of which holds a letter that is $\sim$ to a letter of F, and of a finite subset $\mathfrak{N}$ of $\mathfrak{M}$, every member of which is $\sim$ to some letter of $\mathsf{F} \cup \mathfrak{B}_o$. Then summing $\prod_{\beta\in\mathfrak{B}_o} e_\beta e^{d_\beta} \prod_{C\in\mathfrak{N}} \hat w_C$ over all these pairs gives at most $\prod_{\zeta\in\mathsf{F}} e^{2\varepsilon_2 \# I_\zeta}$. When $\mathfrak{M}$ is the set of the clusters and $\hat w_C = 2\hat\Phi(C) e^{d(C)}$, the condition on $\mathfrak{M}$ is satisfied, by Lem. 4.50 (3) and $2t \le 1$. ∎

Proof. We write again $s_\alpha = \varepsilon_2 \# I_\alpha$, now also for the letters of F, and $v_\alpha = e_\alpha e^{d_\alpha}$. Fix $\mathfrak{B}_o$. The sets $\mathfrak{N}$ for which $(\mathfrak{B}_o, \mathfrak{N})$ is one of the pairs considered in the lemma are the finite sets of members C of $\mathfrak{M}$ with $C \sim \alpha$ for some $\alpha \in \mathsf{F} \cup \mathfrak{B}_o$; therefore

$$\sum_{\mathfrak{N}} \prod \hat w_C \le \prod_{C\sim \mathsf{F}\cup\mathfrak{B}_o} (1 + \hat w_C) \le \exp \sum_{\alpha\in\mathsf{F}\cup\mathfrak{B}_o} \sum_{C\sim\alpha} \hat w_C \le \prod_{\alpha\in\mathsf{F}\cup\mathfrak{B}_o} e^{s_\alpha}.$$

It remains to sum over $\mathfrak{B}_o$. Such a set is the union of its $\sim$-components, which are pairwise distinct $\sim$-connected sets S, and each of them holds a letter $\beta \sim \zeta$ for some ζ of F. A choice of k such sets occurs at least $k!$ times among the ordered choices, so summing over k we get

$$\sum_{\mathfrak{B}_o} \prod v_\beta e^{s_\beta} \le \exp\{\sum_{\zeta\in\mathsf{F}} \sum_{\beta\sim\zeta} \sum_{S\ni\beta} \prod_S v e^s\} \le \exp\{\sum_\zeta \sum_{\beta\sim\zeta} v_\beta e^{2s_\beta}\} \le \exp\{t \sum_\zeta s_\zeta\}.$$

The second inequality is Lem. 4.48 for $\sim$ with the weights $v_\beta e^{s_\beta}$ and $a_\beta = s_\beta$, whose hypothesis $\sum_{\beta\sim\alpha} v_\beta e^{2s_\beta} \le t s_\alpha \le s_\alpha$ follows from (S′); the third inequality is (S′) at ζ. Combining the two bounds, each $\zeta \in \mathsf{F}$ receives the factor $e^{(1+t)s_\zeta} \le e^{2s_\zeta}$, and this is the claim. □

We now come to the step itself. The application of Lem. 4.48 to the objects involves the number $\varepsilon'' = x^{-1/4}$ that item 4 of §2.1 sets. It enters at two places: in the numbers $a_o := \varepsilon'' \# \operatorname{fp}(o)^{+(4r_1+2)} \le \#\operatorname{core}(o)$, where $\operatorname{fp}(o)$ is the footprint of o defined below, and in the bound $\sum_{\operatorname{fp}(o)\ni\square} \hat w(o) e^{a_o} \le \varepsilon''$ at every cell $\square$.

The *footprints* of the objects are $\operatorname{fp}(\square) = \{\square\}$ for a cell, $\operatorname{fp}(f_j) = f_j$, $\operatorname{fp}_{\mathrm p}(X) = \operatorname{cl}(X)^{+r_1}$ for a polymer of kind p, and $\operatorname{fp}_{\mathrm{p}'}(X) = \operatorname{cl}(X^{\approx})^{+r_1}$ for a polymer of kind p′. For adjacent objects o and o' we have $\operatorname{fp}(o') \cap \operatorname{fp}(o)^{+(4r_1+2)} \ne \emptyset$: if each of the two is a cell or a class f_j, then by the definition of adjacency some cell of the one lies at distance at most $4r_1 + 2$ from some cell of the other, and the footprint of each is its set of cells; if an object has a cell $\square$ through which it is adjacent to a polymer X, then $\square$ lies in the footprint of that object and in $\operatorname{fp}(X)$, whether X is of kind p or of kind p′; and two polymers are never adjacent.

We bound a_o. The set $\operatorname{fp}(o)^{+(4r_1+2)}$ is contained in $\square^{+(4r_1+2)}$ when $o = \square$ is a cell, in $Y_0^{+(5r_1+\lceil r(x)/L\rceil+4)}$ with $Y_0 = Y_0(X)$ when $o = X$ is a polymer, and in $\mathrm{mk}_j'^{+(s_0+4r_1+2)}$ when $o = f_j$. Each of these radii is at most $s_0 + \varrho_* + 3r_1 + 2$, and so (A3), together with $x^{-1/4} \le x^{-1/8}$, gives $a_o \le \#\operatorname{core}(o)/50$.

Now we fix a cell $\square$ and estimate $\sum_{\operatorname{fp}(o)\ni\square} \hat w(o) e^{a_o}$, where o ranges over the cells and over the polymers of kind p and of kind p′. The cell $\square$ itself contributes at most $u_t e^{\kappa+5} e^{(\kappa+4)s'}$.

Consider first the polymers X of kind p. If $\square$ lies in $\operatorname{fp}_{\mathrm p}(X)$, then $\square \in \operatorname{cl}(X)^{+r_1}$, and so, by the definition of cl, the fine set of some cell of $\square^{+r_1}$ meets X. As the grids fit (item 1 of §2.1), the fine set $\hat\square$ of each cell is made up of exactly L^3 fine cubes. Hence X holds one of the $L^3(2r_1+1)^3$ fine cubes that make up the fine sets of the $(2r_1+1)^3$ cells of $\square^{+r_1}$. Up to the factor $v e^{(\kappa+4)s'}$ of their weights, the polymers of kind p holding a fixed cube contribute at most $\sum_n (52e)^n e^{-\kappa(n-1)} e^{(\kappa+5)\#Y_0}$, where $n = |X|_c$. For the small X, which have $\#Y_0 = 1$, this is at most $e^{\kappa+5}\cdot 52e/(1-52e\cdot e^{-6}) \le 217.59\, e^{\kappa+5}$.

For X not small we have $n \geq (L-3)/2 \geq 6\cdot 10^{12}$ (for the second inequality, (L2^{su}) and $\mathsf{N}_{\mathrm{c}} \geq 17$ give $L \geq (1.2\cdot 10^4 \cdot 17^2)^2 \geq 1.2027\cdot 10^{13}$) and $\#Y_0 \leq 8 + 432n/L$, and the exponent of the summand of index n is at most $9\kappa+40-n[\kappa-4.952-432(\kappa+5)/L]$. For $\kappa \geq 6$ one has $(\kappa+5)/(\kappa-4.952) \leq 10.5$, and together with $L \geq 1.2\cdot 10^{13}$ this gives $432(\kappa+5)/L \leq 10^{-9}(\kappa - 4.952)$; hence the bracket is at least 1.04 and at least 0.174κ. The ratio of two consecutive terms is therefore at most $e^{-1.04}$, and the sum is at most 1.55 times its first term, whose exponent is at most $\kappa(15.7 - 0.174\cdot 6\cdot 10^{12})$ and thus below -10^{12}. Altogether, up to the same factor, the polymers of kind p holding a fixed cube contribute at most $220\, e^{\kappa+5}$.

Consider next the polymers X of kind p′. If $\square$ lies in $\mathrm{fp}_{p'}(X)$, then X has a cube in a set of $((2r_1+1)L + 2r(x))^3$ cubes that is determined by $\square$ and does not depend on X. For a fixed cube, $\sum_{X\ni\mathrm{cube}} e^{-\kappa(\#X-1)} \leq (1 - 26e\cdot e^{-6})^{-1} \leq 1.2124$; hence, up to the factor $(c_a + c_h)e^{\kappa+5}e^{(\kappa+4)s'}$, the small polymers of kind p′ that hold the cube contribute at most $4\cdot 1.2124 \leq 4.85$, and together with those that are not small at most 6. For the latter, $n = \#X \geq (L-3)/2$ and $\#Y_0 \leq 8 + 432n/L$, and, relative to $e^{\kappa+5}$, the exponent of the summand of index n is at most $4.2581(n-1) + 35 + 2160n/L - \kappa[n(1-432/L) - 8]$. The bracket is positive, so this bound decreases as κ increases; at $\kappa = 6$ it is at most $-1.74n + 79$.

Adding the contribution of the cell and those of the two kinds of polymers, we obtain the left side of (A6); hence $\sum_{\mathrm{fp}(o)\ni\square} \hat{w}(o)e^{a_o} \leq \varepsilon''/2$ at every cell $\square$.

We apply Lem. 4.48, by which $\sum_{\beta \text{ adjacent to } \alpha} v_\beta e^{a_\beta} \leq a_\alpha$ implies $\sum_{\text{connected sets}\ni\alpha} \prod v \leq v_\alpha e^{a_\alpha}$. Every object o' adjacent to o has a footprint that meets $\mathrm{fp}(o)^{+(4r_1+2)}$; applying the last bound at each cell of $\mathrm{fp}(o)^{+(4r_1+2)}$, we obtain $\sum_{o' \text{ adjacent to } o} \hat{w}(o')e^{a_{o'}} \leq \varepsilon''\#\,\mathrm{fp}(o)^{+(4r_1+2)} = a_o$. So the lemma applies to the set of all cells and of all polymers, of kind p as well as of kind p′, with the adjacency of objects, with the weights $\hat{w}$ attached to them at the beginning of the step and with the numbers a_o, and its conclusion is (An).

It remains to prove (S). Fix a cell $\square$ and a letter $\beta \in \mathfrak{A}$ with $\mathrm{dist}(I_\beta, \square) \leq r_1$. By (In), some cell $\square''$ of J_β lies within $\varrho_* + r_1 \leq s'$ of $\square$, and there are $(2s'+1)^3$ such cells $\square''$. Since J_β is connected and holds its root $\square_0$, we have $\#J_\beta > \mathrm{dist}(\square_0, \square'')$, and so, for non-negative summands,

$$\sum_{\beta : J_\beta \ni \square''} (\cdot) \leq \sum_{\square_0} e^{-\mathrm{dist}(\square_0,\square'')} \sum_{\beta \text{ rooted at } \square_0} e^{\#J_\beta}(\cdot),$$

where the sum of $e^{-\mathrm{dist}(\square_0,\square'')}$ over $\square_0$ is at most $\sum_{n\in\mathbb{Z}^3} e^{-|n|_\infty} \leq 49.98 \leq 50$. Further, $e_\beta e^{3\varepsilon_2 \#I_\beta} = e_\beta^\circ e^{5x^{-1/8}\#I_\beta} \leq e_\beta^\circ e^{\#J_\beta}$, by $\#I_\beta \leq (2\varrho_*+1)^3 \#J_\beta$ and (A3), and $e^{d_\beta} = e^{(\kappa+1)s'}e^{(\kappa+1)\#J_\beta}$. What has to be bounded for each root $\square_0$ is thus the sum of $e_\beta^\circ e^{(\kappa+3)\#J_\beta}$ over the letters β rooted at $\square_0$; we treat the kinds of letters in turn.

(a), (h) For X small, these letters give at most $L^3\cdot 1.2124\cdot(2c_a+4c_h)e^{\kappa+3} \leq 4.85L^3(c_a+c_h)e^{\kappa+3}$. For X not small, X has a cube in a set of $(L+2)^3$ cubes that is determined by the root $\square_0$ and does not depend on X. Moreover, $n \geq (L-3)/2 \geq 6\cdot 10^{12}$ and $\#Y_0 \leq 8 + 432n/L$; relative to $e^{\kappa+3}$, the exponent of the summand of index n is at most $4.2581(n-1) + 21 + 1296n/L - \kappa[n(1-432/L) - 8]$. The bracket is positive, so this bound decreases as κ increases. By (O5) we have $\kappa \geq 6$. For every such κ the bound is therefore at most its value at $\kappa = 6$, and this value is at most $-1.74n + 65$. Altogether the letters (a) and (h) give at most $5e^{\kappa+3}L^3(c_a + c_h)$.

(v) These letters give at most $2e^{\kappa+3}K_v p_0^3 x^{-1/2}$.

(t) These letters give at most $x^{-20}\sum_n (26e)^{n-1}e^{-5(\kappa+4)n+(\kappa+3)n} \leq x^{-20}$.

(b) These letters give at most $e^{\kappa+3}u_b$.

For a zone letter ζ, (L) gives $e_\zeta^\circ e^{(\kappa+3)\#J_\zeta} \leq u_1 \prod_o \hat{w}(o)$, the product running over the objects o of ζ. By (An), the sum over the zone letters $\zeta \ni \square_0$ is therefore at most $u_1\hat{w}(\square_0)e^{a_{\square_0}}$, and by the

bound at the cell $\square_0$ this is at most $u_1\varepsilon''/2 \le u_1$. With $t^{-1} = x^{1/4}$, the inequality to which these bounds reduce (S) is (A5); hence (S) holds.

Step 9: cores of clusters; the new members E'^{old}, E'^{new}, E'^{h} and the number V', as defined in (F3)*; those clauses of* (ca) *and* (ch) *that involve no size.* The polymers and clusters of the alphabet $\mathfrak{A}$, and the numbers $\varphi^T(C)$, are those of Lem. 4.50. For a set $\mathfrak{B}$ of pairwise compatible letters we put $K(\mathfrak{B}) = E_\xi \prod_{\mathfrak{B}} \lambda_\alpha$, and for a cluster C we put $\Phi(C) = \varphi^T(C) \prod_i K(\mathfrak{B}_i)$ and $\hat\Phi(C) = |\varphi^T(C)| \prod e_\alpha$; moreover, $n(C)$ is the number of the letters of C, and $d(C) = \sum d_\alpha$.

The *core* $\mathrm{co}(C)$ of a cluster C is obtained from the sets J_α of its letters by joining them with diagonal paths along a spanning tree of $\sim$, the choices being fixed by an order. We call a cluster *constant* when each of its letters is (b), *analytic* when each of its letters is (a), (v) or (t), and *hard* otherwise. Every cluster C satisfies

$$e^{\kappa(\#\,\mathrm{co}(C)-1)} \le e^{d(C)-(\kappa+1)s'-\#\,\mathrm{co}(C)-\kappa}. \tag{Clu}$$

We define

$$\begin{aligned} E'^{\mathrm{old}}(Y,\cdot) &:= \sum_{\mathrm{co}(C)=Y,\ \text{analytic, with a letter (a)}} [\Phi(C)(\cdot) - \Phi(C)(1)], \\ E'^{\mathrm{new}}(Y,\cdot) &:= \sum_{\mathrm{co}(C)=Y,\ \text{analytic, no letter (a)}} [\Phi(C)(\cdot) - \Phi(C)(1)] + \tilde{\mathsf{D}}(Y,\cdot) - P(Y,\cdot), \\ E'^{a} &= E'^{\mathrm{old}} + E'^{\mathrm{new}}, \qquad E'^{h}(Y,\cdot) := \sum_{\mathrm{co}(C)=Y,\ \text{hard}} \Phi(C)(\cdot), \\ \mathsf{V}' &:= \mathsf{V}_0 + \sum_{\text{analytic}} \Phi(C)(1) + \sum_{\text{constant}} \Phi(C). \end{aligned} \tag{F3}$$

We claim the following. For $Y \in \mathcal{D}$, the function $E'^{a}(Y,\cdot)$ is holomorphic on $\mathcal{U}_Y(x')$ and real at G-valued fields; it is invariant under G-valued gauge transformations, and it vanishes at the pure gauges. The function $E'^{h}(Y,\cdot)$ is real, Borel and gauge invariant, its domain consisting of the G-valued fields that live on the bonds of $Y^{\approx}$ and are tame there. The number V' is real, and E'^{new} does not depend on the member.

We prove (Clu). If $\alpha \sim \beta$ in $\mathfrak{A}$, then $\mathrm{dist}(J_\alpha, J_\beta) \le 2\varrho_* + r_1 \le r' - 2 \le s'$, by (In) and (A2). Each of the diagonal paths of the core therefore adds at most s' cells, so that $\#\,\mathrm{co} \le \sum(\#J_\alpha + s') - s'$, and hence $d(C) \ge (\kappa+1)(\#\,\mathrm{co} + s')$. Since $(\kappa+1)\#\,\mathrm{co} - \#\,\mathrm{co} - \kappa = \kappa(\#\,\mathrm{co} - 1)$, this gives (Clu).

Next we determine on what the values depend. By item 5 of Step 7, a letter α with $J_\alpha \subset Y = \mathrm{co}(C)$ is a function of the field in $I_\alpha^{+r_1}$, and this set is contained in $Y^{+(r'-2)}$. Consequently the value $\Phi(C)(\cdot)$ depends only on the field on the bonds of $Y^{\approx}$, that is, on the bonds with both ends in $Y^{\approx}$, the field being extended by 1. The entry e of the pair does not enter: the letters of C are functions of the field in $Y^{+(r'-2)}$, so the values $\Phi(C)(\cdot)$ with $\mathrm{co}(C) = Y$ depend on X'_P only at the plaquettes P based in $Y^{+(r'-2)}$; these plaquettes have all four sites in $Y^{+r'} = Y^{\approx}$, and at each of them $\phi'_P \le \varepsilon' < \pi$ holds on the domain of E'^{h} (the tame G-valued fields on the bonds of $Y^{\approx}$) and (u3) holds on the domain $\mathcal{U}_Y(x')$ of E'^{a}.

Convergence. By (ii), (iii) and (iv) of Step 7, $|\Phi| \le \hat\Phi$, and $\hat\Phi$ does not depend on the field. Lem. 4.50, applied with the condition (S) proved in Step 8 and with Lem. 4.45, gives $\sum_{C \ni \alpha} \hat\Phi(C) e^{d(C)} \le e_\alpha e^{2\varepsilon_2 \# I_\alpha + d_\alpha}$, and t times this bound for the clusters with $n(C) \ge 2$. The torus is finite, hence so is the alphabet, and all the sums in (F3), those of V' included, converge absolutely and uniformly in the field.

The sets $Y_0(X)$ and $\{\square\}$, as well as the sets Y of the letters (t) and of $\tilde{\mathsf{D}}$, are connected; hence $\mathrm{co}(C)$ is connected too.

Holomorphy. Fix ξ. By the locality just proved, the restriction of $\mathsf{U}(\eta; V')$ to $X^{\approx}$ is holomorphic in $V' \in \mathcal{U}_Y(x')$: the entries of G and C are holomorphic by (Γ2), g^V is a polynomial in $\mathrm{Ad}\, V'$ by Lem. 3.3 (e), and the values $X' = -i \log \partial V'$ that enter are power series by (u3). Where $\chi_X = 1$,

Lem. 4.40 (T3) places this restriction in $\mathcal{U}_X(x)$, where $E^a(X,\cdot)$ is holomorphic; where $\chi_X = 0$, the letter (a) is identically zero. Likewise, Thm. 4.5 (a) makes the letter (v) holomorphic where $\chi_\square = 1$, and it is zero elsewhere. The tests depend on ξ alone. Finally, $S_k(Y)$ is holomorphic by (Γ4). The product of the letters is dominated by the integrable majorant of (iii) of Step 7; hence $K(\mathfrak{B})$ is holomorphic, and the series converges uniformly. Finally, $\tilde{\mathsf{D}}$ is holomorphic by (Γ4), and P by Thm. 4.2 (d).

Invariance. Let u be G-valued. Then $\mathsf{U}(\mathrm{Ad}_u\,\eta; V'^u)$ arises from $\mathsf{U}(\eta; V')$ by the gauge transformation whose value at each fine site is that of u at the coarse site whose block contains that fine site; this is a property of the chart U in its two arguments η and V'. Moreover, $G[V'^u] = \mathrm{Ad}_u\, G\, \mathrm{Ad}_u^{-1}$, $X' \mapsto \mathrm{Ad}_u\, X'$, $a_0 \mapsto \mathrm{Ad}_u\, a_0$, and S_k is covariant, so that $\eta(\mathrm{Ad}_u\,\xi, V'^u) = \mathrm{Ad}_u\,\eta(\xi, V')$. The two facts together show that U at $(\mathrm{Ad}_u\,\xi, V'^u)$ is the gauge transform of U at (ξ, V'). The functions E^a, E^h and w are gauge invariant by (ca), (ch) and (cΞ), and so is $V_\square$, by Thm. 4.5 (b); the factors $b_\square$, $\chi_\square$ and χ_X are functions of the numbers $|\xi_b|$, and 1_{ind} and $1_{X^\approx}(U)$ are functions of the set $D(U)$, which a gauge transformation of U leaves unchanged; the factor $A_\square$ satisfies $A_\square(\mathrm{Ad}_u\,\xi, V'^u, \mathrm{Ad}_u\, e) = A_\square(\xi, V', e)$, by Thm. 4.5 (e.iv); and the law of ξ is invariant under Ad_u. Therefore $\lambda_\alpha(\mathrm{Ad}_u\,\xi, V'^u) = \lambda_\alpha(\xi, V')$, and K and Φ are invariant. If u is given on the sites of $Y^\approx$ alone, we extend it by 1. The extension of V'^u by 1 from the bonds of $Y^\approx$ and the field obtained from V' by extending it by 1 and then transforming it by the extended u disagree only on bonds leaving $Y^\approx$, and no quantity here depends on such bonds.

We now come to the function $\tilde{\mathsf{D}}$ on $\mathcal{U}_Y(x')$. Together with V', the field V'^u lies in $\mathcal{U}_Y(x')$, since conjugation by a unitary matrix preserves the operator norm, and so (u1), (u2) and (u3) are preserved. By (Γ4), $\tilde{\mathsf{D}} = -\mathsf{D}_k/2$ and $\mathsf{D}_k(Y,V') = -\mathsf{D}^q(Y,V') - [\mathsf{D}^u(Y,V') - \mathsf{D}^u(Y,1)]$, and the same formulas give $\tilde{\mathsf{D}}$ and D_k at complex fields. By Thm. 3.40 (b), $\mathsf{D}^q(Y,V'^u) = \mathsf{D}^q(Y,V')$ at every field of its domain $\{|V'(c)| < 1 + \vartheta_0^\flat$ on the bonds of $Y^\sim\}$: in the proof of that theorem D^q is made of traces of matrices of §3, and when V' is replaced by V'^u, each of these matrices is conjugated by Ad_u; this is an identity of matrices, so the traces agree. By Thm. 3.42 (b), $\mathsf{D}^u(Y,V'^u) = \mathsf{D}^u(Y,V')$ at every field of $\mathfrak{V}$.

An element of $\mathcal{U}_Y(x')$, extended by 1, lies in $\mathfrak{V}^\circ$ by (u1). Extend u by 1 as well, and transform the extended field by it: the result keeps the norms $|V'(c)|$ on the bonds of $Y^\approx$, is unitary on the bonds leaving $Y^\approx$, and equals 1 on the bonds with no end in $Y^\approx$; so it, too, lies in $\mathfrak{V}^\circ$. It agrees with the extension of V'^u except on the bonds leaving $Y^\approx$, whereas $\mathsf{D}_k(Y,\cdot)$ is determined by the field on the bonds of $Y^{+\lceil R_2/M\rceil}$ alone, and $\lceil R_2/M\rceil \le r_1 - 8 < r'$. Hence $\tilde{\mathsf{D}}(Y,V'^u) = \tilde{\mathsf{D}}(Y,V')$ on $\mathcal{U}_Y(x')$.

By Thm. 4.2 (d), P is invariant on $\mathcal{U}_{\{\square\}}(x')$. So $E'^a(Y,\cdot)$ is invariant on $\mathcal{U}_Y(x')$, as (ca) requires.

At G-valued fields every letter is real, by Thm. 4.5 (d), by (ca), (ch) and (cΞ), and by (Γ2) and (Γ4); hence the values $\Phi(C)(1)$ and the values $\Phi(C)$ of the constant clusters are real as well. The number V_0 is real, because Step 1 displays e^{V_0} as a product of numbers that are positive: V is real by (c0), and $\det C^1 > 0$ by (Γ2).

At the pure gauges, invariance gives $\Phi(C)(1^u) - \Phi(C)(1) = 0$; for $\tilde{\mathsf{D}}$ the vanishing at 1^u follows from (Γ4), and for P from Thm. 4.2 (d).

As to the Borel property of E'^h: the letters are jointly Borel in (ξ, V'), because the members are Borel and the kernels are continuous; Tonelli's theorem then applies to the means, and the series converges uniformly.

Finally, the letters (v) and (t), as well as $\tilde{\mathsf{D}}$ and P, do not depend on the member (Thm. 4.5 (a); T_Y depends on G, S_k and a_0).

Step 10: the clause (ca) *at* x'. We claim that for every cell $\square$

$$\sum_{Y \ni \square} e^{\kappa(\#Y-1)} \sup_{\mathcal{U}_Y(x')} |E'^{\mathrm{old}}(Y,\cdot)| \le 0.24978\, c_a(x'),$$

and that the same sum, with E'^{new} in place of E'^{old}, is $\le 0.30695\, c_a(x')$. The two bounds together give $\sup|E'^a(Y,\cdot)| \le 0.56\, c_a(x') e^{-\kappa(\#Y-1)}$.

We first compare the caps at x and at x'. By (A8),

$$c_a(x') = L^{1/2}(p_0'/p_0)^3 c_a(x) \ge 2^{-3/4} L^{1/2} c_a(x) \ge 0.59460\, L^{1/2} c_a(x).$$

The rooted bound. Let C be a cluster with $\mathrm{co}(C) \ni \square$, let α be one of its letters, and let $\square_0$ be the root of J_α. Then $\#\,\mathrm{co} \ge \mathrm{dist}(\square_0, \square) + 1$. By (Clu) and Lem. 4.50 we obtain, for the clusters that hold a letter α of a given kind,

$$\sum_{\mathrm{co}\ni\square,\ C\ni\alpha \text{ of the given kind}} \hat\Phi e^{\kappa(\#\,\mathrm{co}\,-1)} \le \sum_{\square_0} e^{-\mathrm{dist}(\square_0,\square)-1} \sum_{\alpha \text{ rooted at } \square_0} [t]\, e_\alpha^\circ e^{4x^{-1/8}\#I_\alpha + (\kappa+1)\#J_\alpha - \kappa},$$

where the bracketed factor $t = x^{-1/4}$ is present when the sum runs over the clusters with $n(C) \ge 2$ alone.

By (A3), $e^{4x^{-1/8}\#I_\alpha} \le e^{0.08\#J_\alpha}$, and at $\#J_\alpha = 1$ we have $e^{-1}e^{(\kappa+1)-\kappa} = 1$. Wherever $\Phi(C)(1)$ is subtracted, each of these bounds is doubled.

(i) Let the cluster consist of one letter (a) whose set X is small. Then $\Phi(C)(V') = E_\xi[\chi_X(e^{E^a(X,U)} - 1)] =: \mathcal{G}_X(V')$, and $\mathrm{co} = \{\square_0\}$, where $\square_0$ is the cell of the first cube of X.

Using $T > 1$, $\omega \le Mr'$ from (A2), $c_a(x) \le 1$ from (A4), and (A0), (A8), we have

$$\sup_{\mathcal{U}_{\{\square_0\}}(x')} |\mathcal{G}_X - \mathcal{G}_X(1)| \le 4T^{-2} c_a(x) e^{-\kappa(\#X-1)}.$$

The argument for this bound is that of Step 8, (8.2), of the proof of Thm. 5.15 and of the passage on u in "The clauses without a size" of the same step, together with part (ii) of Step 2 of that proof, taken at $A = 1$. These two steps rely on Lem. 5.17 and Thms. 5.18 and 5.22. The passage on u extends by 1 the G-valued gauge transformation u given on the sites of B_ω. Since the regions of radii r_G and R_3 about the blocks that meet $X^{\approx}$ lie inside B_ω, every quantity that enters $\mathcal{G}_X$ is independent of the field on the bonds that leave B_ω.

Of the test χ_X, that argument uses only the bounds $|\xi_{b'}| \le c_\xi \varepsilon$ on the windows Y_b of the bonds b of $X^{\approx}$. The test χ_X of Lem. 4.40 (T3) provides these bounds, since $r_w + 1 \le R$ and every bond of $X^{\approx}$ is based in a block that meets $X^{\approx}$.

This test is a function of the norms $|\xi_b|$ alone: it is independent of V', and it is unchanged under $\xi \mapsto \mathrm{Ad}_u\, \xi$.

Hence $\mathcal{G}_X$, defined with this test, is holomorphic and G-invariant on $U^{17T}(\mathsf{B}_\omega; x')$ and is bounded there by $2c_a(x)e^{-\kappa(\#X-1)}$, by the argument of Step 8, (8.2), of the proof of Thm. 5.15; and the statement of Thm. 5.22 covers all G-invariant holomorphic functions on that domain that are bounded.

That argument is independent of the condition $\mathsf{a}_c\varepsilon' \le 1$ of (0b). The floors on which it rests are the other conditions of (0b), then (0c) and (0d), where (0b), (0c), (0d) are clauses of (Y.0) in item 5 of §5.3, and finally $\omega \le Mr'$ and $c_a(x) \le 1/2$; all of these floors are stated in (A0), (A2), (A4), (A8).

The two hypotheses of Thm. 5.22 on ω are satisfied: $3\omega + 1 \le 3Mr' + 1 \le 6Mr' + 12 = \ell_w(x')$ by (A2), and $2\omega + 1 < N$ by the clauses on N in Step 0. Moreover, that argument is independent of Thms. 4.39 and 5.7.

Since the grids fit, there are L^3 first cubes in $\hat\square_0$; for each cube, $\sum_{X\ni\mathrm{cube}} e^{-\kappa(\#X-1)} \le 1.2124$ by Step 8; and $4 \cdot 1.2124 \le 4.85$. Summing over the small sets X whose first cube

lies in $\hat{\square}_0$, we obtain by (L2^{su})

$$\begin{aligned}&\sum_{X \text{ small, first cube in } \hat{\square}_0} \sup_{\mathcal{U}_{\{\square_0\}}(x')} |\mathcal{G}_X - \mathcal{G}_X(1)| \\ &\quad \le 4.85 L^3 T^{-2} c_a(x) = 4.85 \cdot 361 \mathsf{N}_{\mathrm{c}}^2 c_a(x) \\ &\quad \le (4.85 \cdot 361/(1.2 \cdot 10^4)) L^{1/2} c_a(x) \\ &\quad \le 0.24539\, c_a(x').\end{aligned}$$

In numbers: $4.85 \cdot 361/12000 \le 0.1459042$, and $0.1459042/0.59460 \le 0.245383$.

(ii) Let the cluster consist of one letter (a) whose set X is not small. Then $\mathrm{co} = Y_0(X) \ni \square$, so X holds one of $(L+2)^3$ cubes, and $n = \#X \ge (L-3)/2$. The term belonging to n is $\le 4 c_a (26e)^{n-1} e^{-\kappa(n-1)} e^{\kappa(7+432n/L)}$.

The ratio of two consecutive terms is $\le e^{-1.74}$; so the sum is ≤ 1.22 times its first term, and the exponent of the first term is $\le -((L-5)/2)(\kappa - 4.2581) + 223\kappa \le -\kappa L/7$. For the last inequality we note that the difference $((L-5)/2)(\kappa - 4.2581) - 223\kappa - \kappa L/7$ is $\ge L(5\kappa/14 - 2.12905) - 225.5\kappa + 10.64$; at $\kappa = 6$, $L \ge 1.2 \cdot 10^{13}$ this lower bound is $\ge 0.0138L - 1343 \ge 1.6 \cdot 10^{11} > 0$, and its derivative in κ is $5L/14 - 225.5 > 0$.

Case (ii) therefore contributes $\le 16 L^3 e^{-\kappa L/7} c_a(x) \le 10^{-6} c_a(x')$.

(iii) Let $n(C) \ge 2$, and let C hold a letter (a); we root the cluster at one of its letters (a). In the rooted sum of the sizes e°, the letters with small X give $L^3 \cdot 1.2124 \cdot 2 c_a \le 2.43 L^3 c_a$; the letters whose X is not small add a negligible amount, by the argument of case (ii); in all, the rooted sum is $\le 4.86 L^3 c_a$.

So case (iii) contributes $\le 2 \cdot 49.98 e^{0.1} x^{-1/4} \cdot 4.86 L^3 c_a(x) \le 110.48 \cdot 4.86 L^3 x^{-1/4} c_a(x) \le 0.0026 L^{1/2} c_a(x) \le 0.00438\, c_a(x')$, by (A8), since $2 \cdot 49.98 e^{0.1} \le 110.48 \le 111$ and (A8) contains the clause $111 \cdot 4.86 L^3 x^{-1/4} \le 0.0026 L^{1/2}$. The factor 49.98 comes from $\sum_{n \in \mathbb{Z}^3} e^{-|n|_\infty} \le 49.98$, and the sum on the left bounds the sum $\sum_{\square_0} e^{-\mathrm{dist}(\square_0, \square)}$ in the rooted bound, by Step 8.

Adding the three cases, we find $0.24539 + 0.000001 + 0.00438 \le 0.24978$, the bound for E'^{old}.

(iv) Let the cluster consist of one letter (v). We use the bound $e^\circ_\alpha = 2 K_v p_0^3 x^{-1/2}$ on the supremum of this letter α, from item 1 of Step 7. This bound does not carry the factor $e^{2x^{-1/8} \# I_\alpha}$ by which e_α exceeds e°_α. We obtain $2 \cdot 2 K_v p_0^3 x^{-1/2} = (4K_v/E_a) c_a(x) \le (1/8) L^{1/2} c_a(x) \le 0.21023\, c_a(x')$, as $E_a \ge 32 L^{-1/2} K_v$ by (O12).

(v) For the terms $\tilde{\mathsf{D}}$ we have $\sum_{Y \ni \square} e^{(\kappa - \kappa^\sharp)(\#Y - 1)} 1.5 \mathsf{K}_1^\flat \varepsilon' \le 1.01 \cdot 1.5 \mathsf{K}_1^\flat \varepsilon'$. Since $\varepsilon' = c_a(x')/(E_a p_0'^2)$, the bound $1.01 \cdot 1.5 \mathsf{K}_1^\flat \varepsilon'$ is $\le (1.01/17) c_a(x') \le 0.05942\, c_a(x')$, by (A8).

(vi) For the terms P we have $K_{\mathrm{q}} p_0'^2 \varepsilon' = (K_{\mathrm{q}}/E_a) c_a(x') \le (1/32) c_a(x') = 0.03125\, c_a(x')$, as $E_a \ge 32 K_{\mathrm{q}}$ by (O12).

(vii) Let $n(C) \ge 2$, and let C hold no letter (a); we root it at a letter (v) or at a letter (t). The contribution is then $\le 110.48 x^{-1/4} [2 K_v p_0^3 x^{-1/2} + x^{-20}] \le 111 x^{-1/4} (2K_v/E_a + 1) c_a(x) \le 0.003 L^{1/2} c_a(x) \le 0.00505\, c_a(x')$, by (A8).

(viii) Let the cluster consist of one letter (t). Then $2 \sum_{Y \ni \square} e^{\kappa(\#Y - 1)} e_Y^{1/2} e^{2x^{-1/8} \# Y} \le 2x^{-20} \le 0.001\, c_a(x')$, where the last inequality is the last clause of (A8).

Adding the five cases (iv) to (viii), we find $0.21023 + 0.05942 + 0.03125 + 0.00505 + 0.001 = 0.30695$, the bound for E'^{new}.

Step 11: the clause (ch) *at* x'. For every cell $\square$ we have

$$\sum_{Y \ni \square} e^{\kappa(\#Y - 1)} \sup |E'^h(Y, \cdot)| \le 10^3 L^3 c_h(x) + 50(2u_b + u_1) \le c_h(x').$$

Every hard cluster holds a letter α that is a letter (h), a letter (b) or a zone letter of $\mathfrak{A}$. We sum by means of the rooted bound of Step 10, which rests on claim (1) of Lem. 4.50.

If α is a letter (h), we obtain $49.98e^{0.08}\cdot L^3\cdot 1.2124\cdot 4c_h(x)\le 262.6L^3c_h(x)$; when the sets X that are not small are added, by the argument of case (ii) of Step 10, the total is $\le 10^3L^3c_h(x)$. If α is a letter (b), we obtain $49.98e^{0.08}u_b\le 54.2u_b\le 100u_b$.

For a zone letter ζ, (L) gives

$$e_\zeta^\circ e^{4x^{-1/8}\#I_\zeta+(\kappa+1)\#J_\zeta-\kappa}\le e_\zeta^\circ e^{(\kappa+2)\#J_\zeta}\le u_1\prod_o\hat w(o),$$

where we use that the set t of tested cells is non-empty for $\zeta\in\mathfrak{A}$. By (An), the sum over the zone letters ζ rooted at $\square_0$ is $\le u_1\hat w(\square_0)e^{a\square_0}\le u_1$, and so the zone letters give $49.98u_1\le 50u_1$.

The first clause of (A9) gives $10^3L^3c_h(x)\le(1/2)e^{-c_hp_0'^2}=(1/2)c_h(x')$.

By Thm. 4.19 (e), $u_1=e^{-\delta_3p_0^2}$; hence $50u_1\le(1/4)e^{-\delta_3p_0^2/2}$, as the second clause of (A9) is $e^{\delta_3p_0^2/2}\ge 200$. Next, $h_*=C_\varphi^\flat(4+12K_*\mathsf{g}_*+a^{1/2}L)\ge 4C_\varphi^\flat\ge 4(2/\gamma)^{1/2}>1$, since $C_\varphi^\flat\ge\mathsf{G}_0=(2/\gamma)^{1/2}3L^3\mathsf{g}(\theta_{\mathbb{C}})\ge(2/\gamma)^{1/2}$ by item C.I 1.2 of §3, where the last step rests on $3L^3\mathsf{g}(\theta_{\mathbb{C}})\ge 1$, given by the term $n=0$; therefore $\delta_3\le(3/256)c_\xi^2$ and $c_\xi^2/2-\delta_3/2\ge 0.494c_\xi^2$. With this, and with $u_b=2.5v(c_\xi p_0+(c_\xi p_0)^{-1})e^{-c_\xi^2p_0^2/2}$ as in item F of §4.3, we obtain $100u_be^{\delta_3p_0^2/2}\le 250v(c_\xi p_0+1/(c_\xi p_0))e^{-0.49c_\xi^2p_0^2}\le 1/4$, where the last inequality is the third clause of (A9).

So $50(2u_b+u_1)\le(1/2)e^{-\delta_3p_0^2/2}\le(1/2)e^{-c_hp_0'^2}$, because $c_h\le\delta_3/2$ and $p_0'\le p_0$.

Step 12: the expansion of Ξ' at a fixed e. In this step we show that at every (V',e)

$$\rho'(V')=e^{\mathsf{V}'}\exp\{-x'\mathrm{En}'(V')+\sum_Y 1_{Y^\approx}(V')[E'^a+E'^h](Y,V')\}\,\Xi',$$

where $\Xi'=\sum_{\{\mathsf{Y}_i\}}\prod_i w'(\mathsf{Y}_i,V',e)$, the sum being taken over the families of the definition of Ξ_w in item 1 of §2.1 applied at x' with cells in place of cubes (s' in place of $s(x)$, D' in place of $D(V)$), and where w' is given by the definition (F4) that follows.

Definition (F4). Let Y be a finite set of cells that is not empty. When some cell of $\mathrm{mk}'(V')$ lies in $\mathsf{Y}^{+(s'-1)}\setminus\mathsf{Y}$, we set $w'(\mathsf{Y},V',e):=0$. Otherwise $w'(\mathsf{Y},V',e)$ is the sum of

$$\prod_{\gamma\in\mathsf{G}}K(\gamma)\prod_{C\in\mathfrak{N}}(e^{-\Phi^\natural(C)}-1)\prod_{Y\in\mathfrak{N}^\flat}(e^{\tilde{\mathsf{D}}(Y,V')}-1)$$

over all triples $(\mathsf{G},\mathfrak{N},\mathfrak{N}^\flat)$ with the following properties: G is a set of rough polymers at V', no two of them $\sim$, and the marks of its forced letters are exactly $\mathrm{mk}'\cap\mathsf{Y}$, with every class met once; $\mathfrak{N}$ is a finite set of clusters $\sim\mathsf{G}$; $\mathfrak{N}^\flat$ is a set of long Y; the cores together make up Y; and the graph of links is connected. Rough polymers and $K(\gamma)$ are defined in part (a), $\Phi^\natural$ in part (b), and the elements, their cores and the links in part (c).

The definition (F4) has one more clause, its last: the third member of u' is the mean $w'(\mathsf{Y},V'):=\int_{\mathfrak{E}}w'(\mathsf{Y},V',e)\,de$, which is treated in item 5 of Step 13.

We claim that $w'(\mathsf{Y},V',e)$ depends on V' only through the bonds of $\mathsf{Y}^{+s'}$, and on e_P only at the plaquettes of cells of Y.

(a) *Rough polymers.* We call a rough polymer at V' any $\sim$-connected set γ of letters, all actual at V' and pairwise compatible, that contains a forced letter; for it we put $K(\gamma)=E_\xi\prod_\gamma\lambda_\alpha$.

We split a set $\mathfrak{B}$ of (F2) into its $\sim$-components, and we let G be the set of those components that hold a forced letter; by (Act1) these are rough polymers, and the union of the sets I_α of their letters contains f^{+2r_1}. We put $\mathfrak{A}_\mathsf{G}:=\{\alpha\in\mathfrak{A}:\alpha\text{ not }\sim\text{ any letter of }\mathsf{G}\}$.

Every letter of $\mathfrak{A}_\mathsf{G}$ is actual. For such a letter, I_α lies off f^{+3r_1}; hence the letters (v) and (b) are actual, and so are the letters (a) and (h), for which $\mathrm{cl}(X^\approx)\subset I_\alpha$; for a zone letter, $\mathfrak{b}^{+2r_1}\subset I_\alpha$, and so $\mathrm{dist}(\mathfrak{b},f)>5r_1\ge 4r_1+2$.

The remaining components of $\mathfrak{B}$ are the $\sim$-connected compatible sets of letters of $\mathfrak{A}_{\mathsf{G}}$, and all letters of such a set are compatible with all letters of G, because incompatibility occurs only between letters that are $\sim$; hence such a set and G can occur together among the components of a set of (F2).

Lems. 4.45 and 4.50, applied to the alphabet $\mathfrak{A}_{\mathsf{G}}$ (we use (i) of Step 7, and the condition (S) is weaker for a smaller alphabet), give

$$e^{\sum n_j}\mathcal{Z} = \sum_{\mathsf{G}} \prod_{\gamma} K(\gamma) \exp\{ \sum_{C \text{ not}\sim \mathsf{G}} \Phi(C)(V')\},$$

where G ranges over the sets of rough polymers that are pairwise not $\sim$ and contain every forced class once.

(b) *On and off.* A cluster C is on at V' if it is constant or if V' is tame on $\mathrm{co}(C)^{\approx}$; otherwise it is off. Every cluster C that is not $\sim \mathsf{G}$ is on: its letters have $\mathrm{dist}(J_\alpha, \mathrm{mk}') > s_0 + 3r_1$, and every cell of a joining path is within r' of one of the J_α, so that $\mathrm{dist}(\mathrm{co}, \mathrm{mk}') > s_0 - r' = r'$. The value of such a cluster at V' is its value at the restriction of V' to $\mathrm{co}^{\approx}$, by Step 9.

We put $\Phi^\natural(C) := \Phi(C)(V')$ if C is on, $\Phi(C)(1)$ if C is analytic and off, and 0 if C is hard and off. This quantity is independent of e, and $|\Phi^\natural| \le \hat{\Phi}$.

We go through the types. For an analytic cluster we have $\Phi(V') = \Phi(1) + (\Phi(V') - \Phi(1))$ if it is not $\sim \mathsf{G}$, $0 = \Phi(1) + (\Phi(V') - \Phi(1)) - \Phi(V')$ if it is $\sim \mathsf{G}$ and on, and $0 = \Phi(1) - \Phi(1)$ if it is $\sim \mathsf{G}$ and off. For a hard cluster the same three cases read $\Phi(V') = \Phi(V')$, $0 = \Phi(V') - \Phi(V')$ and $0 = 0 - 0$. For a constant cluster they read $\Phi = \Phi$ if it is not $\sim \mathsf{G}$ and $0 = \Phi - \Phi$ if it is $\sim \mathsf{G}$.

Combining these identities gives

$$\sum_{C \text{ not}\sim \mathsf{G}} \Phi(C)(V') = [\sum_{\text{analytic}} \Phi(C)(1) + \sum_{\text{constant}} \Phi(C)] + \sum_{Y} 1_{Y^{\approx}}(V')[E'^{a} - \tilde{\mathsf{D}} + P + E'^{h}](Y, V') - \sum_{C \sim \mathsf{G}} \Phi^\natural(C),$$

where all sums converge absolutely.

We insert this, together with (iv) of Step 3, into (F1). The terms $1_{Y^{\approx}}\tilde{\mathsf{D}}$ and $1_{\square^{\approx}}P$ cancel, and $\sum n_j$ is now carried by the letters. Hence the formula for $\rho'(V')$ holds with

$$\Xi' = \sum_{\mathsf{G}} \prod K(\gamma) e^{-\sum_{C\sim\mathsf{G}} \Phi^\natural(C)} e^{\sum_{\text{long}} \tilde{\mathsf{D}}} = \sum_{(\mathsf{G}, \mathfrak{N}, \mathfrak{N}^\flat)} \prod K(\gamma) \prod_{\mathfrak{N}} (e^{-\Phi^\natural} - 1) \prod_{\mathfrak{N}^\flat} (e^{\tilde{\mathsf{D}}} - 1),$$

where $\mathfrak{N}$ ranges over the finite subsets of $\{C \sim \mathsf{G}\}$ and $\mathfrak{N}^\flat$ over the sets of long Y.

(c) *Elements, links and a bijection.* The elements of a triple are its letters, each with the core J_α, the clusters C of $\mathfrak{N}$, each with the core $\mathrm{co}(C)$, and the sets Y of $\mathfrak{N}^\flat$, each with the core Y. Two elements are linked if their cores lie within s' of each other.

When no cell of $\mathrm{mk}'(V')$ lies in $\mathsf{Y}^{+(s'-1)} \setminus \mathsf{Y}$, that is, when the first clause of (F4) does not make w' vanish, every mark outside Y lies at a distance greater than $s' - 1 \ge 5r' \ge 2s_0 + 4r_1 + 2$ from Y; therefore $\mathrm{mk}' \cap \mathsf{Y}$ is a union of classes.

We now set up a bijection. If a triple as in part (b) is given, each component of its graph of links yields a set Y_i; these sets lie at pairwise distance $> s'$, and each of them carries a triple of (F4). Indeed, two letters of one γ that are $\sim$, and a cluster C together with a letter of G to which C is $\sim$, have cores within $2(s_0 + \varrho_*) + r_1 = 4r' + 5r_1 + 2\lceil s(x)/L \rceil + 4 \le 5r' - 2$ of each other, by (A2); and every long Y comes within r' of some mark, and that mark belongs to the core of a forced letter. Every Y_i therefore holds a forced letter, thus a mark and thus a plaquette of D'; since every mark lies in some Y_i, the set D' lies in their union; and no Y_i has a mark in $\mathsf{Y}_i^{+(s'-1)} \setminus \mathsf{Y}_i$.

Conversely, let sets Y_i be given, at pairwise distance $> s'$, which together hold mk', each with a triple. Letters of different indices i have $\mathrm{dist}(I_\alpha, I_\beta) > s' - 2(s_0 + \varrho_*) = 4r' - 2\varrho_* \ge r_1$, so they are not $\sim$ and hence compatible. The condition of compatibility on polymers of

kind p holds for them as well, since, by the proof of (Inc) in Step 4, two zone letters α, β with polymers of kind p within $s(x)$ of each other have $\operatorname{dist}(I_\alpha, I_\beta) = 0$. The mark that makes a set Y of $\mathfrak{N}_i^\flat$ off lies in $\mathsf{Y}_i^{+r'}$, hence in Y_i. Therefore the union of the given triples is a triple as in part (b), and its components are the Y_i.

If D' is empty, only the empty triple remains, and $\Xi' = 1$.

(d) *On what w' depends.* Measured in cells from Y, the ingredients of w' depend on V' within the following distances: a forced letter within $s_0 + \varrho_* + r_1 \le 3r'$, by (In) and by (Rd), which holds at every pair; the class terms w_j within $2r'$; the numbers n_j within $3r'$; a long Y within $\lceil R_2/M \rceil$; and $\Phi^\natural(C)$ within r'.

The marks that decide whether a letter is actual or forced, what the classes are, and whether a cluster or a set is on, off, short or long lie within $2s_0 + 4r_1 + 2 \le 5r'$ of a core, and the marks that decide whether the first clause of (F4) applies are those in $\mathsf{Y}^{+(s'-1)} \setminus \mathsf{Y}$, within distance $s' - 1$ of Y. All these distances are $\le s' - 1$, and the plaquettes based in a cell decide whether that cell is a mark; so w' depends on V' only through the bonds of $\mathsf{Y}^{+s'}$.

The component e_P of e occurs at $P \in D'$ only; hence e enters only through plaquettes that belong to marks lying within $s' - 1$ of Y, and, when $w' \neq 0$, all such marks are cells of Y; finally, the first clause of (F4) is independent of e.

Step 13: the strong cap and the mean over e. In this step we show that at every (V', e)

$$|w'(\mathsf{Y}, V', e)| e^{\kappa(|\mathsf{Y}|_c - 1)} \le \exp\{-c_\Xi x' \varepsilon' \sum_{P \in D' \cap \mathsf{Y}} \phi'_P\},$$

that the mean $w'(\mathsf{Y}, V') := \int w'(\mathsf{Y}, V', e)\, de$ satisfies (cΞ) at x' with the strong cap, and that $\Xi'(V') = \sum_{\{\mathsf{Y}_i\}} \prod w'(\mathsf{Y}_i, V')$ at every V'. Thm. 4.39 follows from this.

1. Let $w' \neq 0$. By (i) of Step 7, $|K(\gamma)| \le \prod e_\alpha$. The cores are connected and are linked within s'; hence $|\mathsf{Y}|_c \le \sum_{\text{elements}} (\#\text{core} + s') - s'$, and so
$$e^{\kappa(|\mathsf{Y}|_c - 1)} \le \prod_{\text{letters}} e^{d_\alpha} \prod_{\mathfrak{N}} e^{d(C)} \prod_{\mathfrak{N}^\flat} e^{(\kappa+1)(\#Y + s')}.$$
Since $|\Phi^\natural| \le \hat\Phi \le 1$, we have $|e^{-\Phi^\natural} - 1| \le 2\hat\Phi$. By Step 3, $|e^{\tilde{\mathsf{D}}(Y)} - 1| e^{(\kappa+1)(\#Y + s')} \le \hat w(Y)$.
2. We apply Lem. 4.51, with $t = x^{-1/4} \le 1/2$, taking for F the forced letters of G, for $\mathfrak{B}_o$ the other letters of G, and for $\mathfrak{M}$ the clusters, with $\hat w_C = 2\hat\Phi e^{d(C)}$. Each $\sim$-component of $\mathfrak{B}_o$ holds a letter that is $\sim$ to a letter of F, since γ is $\sim$-connected; and the condition of the lemma holds with $2t\varepsilon_2 \le \varepsilon_2$.

In addition to the lemma, we use a separate fact: for a further finite set $\mathfrak{M}^\flat$ with $\sum_{\mathfrak{M}^\flat} \hat w \le \sum_{\mathsf{F}} A_\zeta$ we have $\sum_{\mathfrak{N}^\flat \subset \mathfrak{M}^\flat} \prod \hat w \le \prod(1 + \hat w) \le e^{\sum A_\zeta}$. We take for $\mathfrak{M}^\flat$ the long Y within r' of the marks in Y, and $A_\zeta = \sum_{J(\zeta)} \#\mathrm{mk}'_j$, by (iii) of Step 3.

The outcome is
$$|w'| e^{\kappa(|\mathsf{Y}|_c - 1)} \le \sum_{\mathsf{F}} \prod_{\zeta \in \mathsf{F}} e_\zeta e^{d_\zeta + 2\varepsilon_2 \#I_\zeta + A_\zeta},$$
where F runs over those sets of forced letters for which each class with marks in Y occurs exactly once among their classes and no other class occurs. The right side is a majorant, since the forced letters in it need not be actual at V'.
3. Since $\#I_\zeta \le (2(s_0 + \varrho_*) + 1)^3 \#J_\zeta$, (A3) gives $e^{4x^{-1/8}\#I_\zeta} \le e^{\#J_\zeta}$. With (L) it follows that $e_\zeta e^{d_\zeta + 2\varepsilon_2 \#I_\zeta + A_\zeta} \le e^{(\kappa+1)s'} \prod_{o \in \zeta} \hat w(o)$, where $\hat w(\square)$, $\hat w_p$, $\hat w_{p'}$ are the weights of Step 8, and

$$\hat w(f_j) := \exp\{-(r_\Xi + c_\Xi) x' \varepsilon' \sum_{D'_j} \phi'_P + [1.01 \mathsf{K}_1^\flat (2r' + 1)^3 + 1] \#D'_j + (\kappa + 4)((1 + s') \#\mathrm{mk}'_j + s')\}.$$

Into $\hat w(f_j)$ enter the term Res, the factor $e^{|n_j|}$, the number A_ζ, and the inequality $\#\mathrm{mk}'_j \le \#D'_j$.

We have $r_\Xi \ge r_\Xi^\flat \ge (3/4)(1/2 - d_0 - c_\Xi) > 0$, since $\mathtt{t}_\rho \le (1/8)(1/2 - d_0 - c_\Xi)$ by (SD) and $3e_1^\flat$ obeys the same bound by (A7). Each $P \in D'$ has $x'\varepsilon'\phi'_P > p_0'^2$, so $-r_\Xi x'\varepsilon' \sum_{D'_j} \phi'_P \le -r_\Xi^\flat p_0'^2 \#D'_j$, and therefore $\hat w(f_j) e^{2\#\mathrm{mk}'_j} \le e^{-c_\Xi x'\varepsilon' \sum_{D'_j} \phi'_P} \upsilon_f^{\#D'_j}$.

4. For every class we take out the factor $e^{-c_\Xi(\cdot)} \upsilon_f^{\#D'_j/2}$. With $\hat w''(f_j) = \upsilon_f^{\#D'_j/2} e^{-2\#\mathrm{mk}'_j}$ the condition of Lem. 4.48 holds: at a cell, the cells and polymers contribute $\varepsilon''/2$, by (A6); at most one class has $f_j \ni \square$; and $a_{f_j} \le \#\mathrm{mk}'_j$.

   This last bound comes from the definition of a_o in Step 8, independently of the bound $a_o \le \#\mathrm{core}(o)/50$ obtained there, because $\mathrm{core}(f_j)$ may have $(1+s')\#\mathrm{mk}'_j$ cells: we have $f_j^{+(4r_1+2)} \subset \mathrm{mk}_j'^{+(s_0+4r_1+2)}$, so $a_{f_j} \le x^{-1/4}(2(s_0+4r_1+2)+1)^3 \#\mathrm{mk}'_j \le \#\mathrm{mk}'_j/50$ by (A3).

   Hence $\hat w''(f_j) e^{a_{f_j}} \le \upsilon_f^{1/2} \le \varepsilon''/4$, by (A7), and so $\sum_{\zeta \ni f_j} e^{(\kappa+1)s'} \prod \hat w'' \le e^{(\kappa+1)s'} \upsilon_f^{1/2} \le 1$. A set F is determined by choosing, for each class, either nothing or the letter of which that class is the first class; thus $\sum_{\mathsf{F}} \prod \le \prod_j 2$. Finally $\prod_j 2\upsilon_f^{\#D'_j/2} \le 1$.
5. We apply Lem. 5.23, which is stated and proved in (9) of Step 10 of the proof of Thm. 5.15, to the function

$$\Xi'(V') := \rho'(V') e^{-\mathsf{V}'} \exp\{x'\mathrm{En}' - \sum 1_{Y \approx} [E'^a + E'^h]\},$$

   which is independent of e. Its hypotheses (i) to (iv) hold, as follows.

   (i) The curvature X', the masks, the marks and the classes are Borel in (V', e); $T_\square$ is jointly Borel, by Thm. 4.2 (b); the kernels are continuous; $\tilde{\mathsf{D}}(Y, \cdot)$ is continuous and n_j is Borel, by (Γ4) and by (ii) of Step 3; $A_\square$ is Borel, by Thm. 4.5 (e.v); and the members are Borel.

   Hence every letter is jointly Borel in (ξ, V', e), and every $K(\gamma)$ is Borel in (V', e), by Tonelli's theorem and Step 7. The Borel sets $\{\mathrm{mk}'(V') \cap \mathsf{Y}^{+(s'-1)} = \mathsf{d}\}$ are finite in number, and on every one of them the sum of (F4) is a series of functions of this kind that converges absolutely; moreover, the bound obtained in items 1 to 4 is independent of e.

   (ii) The marks, the classes, and the properties actual, on, off, short and long are functions of the ϕ'_P. Further, $X'(V'^u, \mathrm{Ad}_u\, e) = \mathrm{Ad}_u\, X'(V', e)$; hence a_0 and η are covariant, and $T_\square$, w_j and the letters are invariant together with $\xi \mapsto \mathrm{Ad}_u\, \xi$, by Step 9; and $\Phi^\natural$, $\tilde{\mathsf{D}}$, n_j are invariant.

   (iii) Hypothesis (iii) is the content of part (d) of Step 12.

   (iv) Hypothesis (iv) is the content of Step 12.

   The lemma yields that the mean is real, Borel and gauge invariant, that it depends on V' only through the bonds of $\mathsf{Y}^{+s'}$, that it satisfies the bound, and that $\Xi' = \sum \prod w'(\mathsf{Y}_i, V')$ at *every* V'.
6. We conclude. Clause (c0) holds because V' is real, by Step 9; clause (ca) holds by Steps 9 and 10, clause (ch) by Steps 9 and 11, and clause (cΞ) by item 5 above. Moreover $\rho' > 0$, as a positive kernel integrated against a positive density, and ℓ'^c has range $\le \varrho(x')$.

   The letters of the class $C_\flat^{\mathrm{su}}$ listed in the statement of Thm. 4.39 are the same at x' as at x, and ρ' is a function of V' alone; besides the clauses just verified, these two facts are what that statement asserts of ρ'.

Step 14: the floors on half lines and the order of the constants. In this step we show that each of the floors (A0) to (A11) holds for all large x once the letters (O0) to (O12) are fixed; hence the set $S(c_h, E_a)$ of (O13) is non-empty, and $x_\flat^0 = 1 + \inf S(c_h, E_a)$ exists. We show moreover that each

entry of the table in item 3 of §2.1 depends only on entries before it, and that each condition is one-sided.

We put $\Lambda = c_R \log x/M$. Item 1 of §2.1 defines $s' = s(x')$, where $s(x) = 8r(x)$ and $r(x) = \lceil 20 c_R \log x/M \rceil$. Since $x' \le x$, it follows that $s' \le 8 + 160\Lambda$, and so $e^{(\kappa+4)s'} \le e^{8(\kappa+4)} x^{1/12}$, because $(\kappa+4)160 c_R/M \le 1/12$ by (M2).

For (A0): the inputs state that their floors hold for all large x; they do so in §3, Part C.II, items 4 and 5 (Floors), in item G (Floors) of §4.1, in item 6 (Floors) of §4.2 with Step 14 of the proof of Thm. 4.5, and in item G of §4.3 with Step 13 of the proof of Thm. 4.19. Of (P0), only the clauses $x \ge L^2$ and $R \ge 6$ are among these floors, since (Tor) is a datum by (A0).

For the conditions (D2), (D3) and (C0) to (C2) named in (A0), we use ℓ_w, D_w, $\omega = O(\log x)$, $\mathsf{g}_x \ge c(\log x)^{-2}$, and r_η, r_T, $\varepsilon' = O(x^{-1/2}(\log x)^{\mathfrak{p}_0})$. In $3D_w \le \ell_w(x')$, the leading term $360\Lambda M/L$ of the left side is smaller than the leading term $120\Lambda M$ of the right side; in $\mathsf{K}_c(17T\varepsilon')^2 \le 0.12\varepsilon$, the left side is of the order ε^2 and the right side of the larger order ε.

For (A1): $\varepsilon \to 0$; $\varrho(x') \to \infty$; $R/L + 3 \le R/2$; and in the last condition ε^2 on the left is of smaller order than ε on the right.

For (A2): $r_1 \le (2 + 3/L)\Lambda + \text{const}$ and $2\lceil s(x)/L \rceil \le 320\Lambda/L + \text{const}$, so the left side is $\le (10 + 335/L)\Lambda + \text{const}$, while $r' \ge 20\Lambda - 20 c_R \log L/M$; thus the condition holds for large x if and only if $L \ge 34$, as $10 + 335/L$, rounded to two places, is 20.15 at $L = 33$ and 19.85 at $L = 34$. Further, for large x the upper bound in $\omega \le (2 + 141/L)R + M + \text{const}$ lies below the lower bound in $Mr' \ge 20R - \text{const}$, and so does the upper bound in $R_6 + 1 \le (4 + 6/L)R + \text{const}$.

For (A3), (A4), (A8) and (A10), powers of x dominate powers of $\log x$ for large x; moreover $(p_0'/p_0) \to 1$, and n_a is independent of x.

For (A5): $1/4 + 1/12 - 1/2 = -1/6 < -1/8$, and u_b, u_1 decay faster than every power.

For (A6): in the third term, $-1/2 + 1/12 = -5/12 < -1/4$, and $\upsilon = e^{-c_\Xi p_0^2}$ decays faster than every power. Further, $-\log u_t \ge \min\{\rho_Q/2 - c_\tau,\ \mathrm{py}/(768(2r_1+1)^3)\} - \log 3v$, which is of the order $(\log x)^{2\mathfrak{p}_0 - 3}$ under the floor (W) and exceeds $(1/4 + 1/12)\log x$ for large x; it is here that we use $\mathfrak{p}_0 > 2$, which gives $2\mathfrak{p}_0 - 3 > 1$.

For (A7): $e_1^\flat = O(\varepsilon')$, and $r_\Xi^\flat p_0'^2$, with $r_\Xi^\flat \to (7/8)(1/2 - d_0 - c_\Xi) > 0$, is of the order $(\log x)^{2\mathfrak{p}_0}$ and so of larger order than $(\log x)^3$. For (A9): $p_0^2 - p_0'^2 \ge 2\mathfrak{p}_0 A_0^2 (\log x')^{2\mathfrak{p}_0 - 1} \log L \to \infty$. For (A11): $r' \to \infty$.

The order of the constants is as follows. First (O0) to (O3) fix N_Z, N_Ξ, $\mathsf{N}_\sharp$, N_ν, $\mathsf{N}_\nu^\flat$, N_c as functions of L; then come L and a; then κ; then γ, N_B, b, $A^\flat$, $\theta^\flat$, $\theta_\mathbb{C}$, $\vartheta_0^\flat$, the number $\Lambda = 2 + A^\flat/b$ of (O6), μ, G_0, $C_\varphi^\flat$, determined by (L, a, K_*, λ_*); then θ_D, C_D, c_R; then K_c^0 and a_c, which are independent of M and of x; then M, by (M2), (M4$^\flat$), (M5r); then κ_+, $\mathsf{K}_1^\flat$; then $\mathfrak{p}_0$, A_0; then c_ξ, h_*; then K_q, K_v, the list of item F of §4.3, and δ_3; then a_∞, n_a; then c_h; then E_a; and last $x_\flat$.

The conditions are lower bounds on L, C_D, M, E_a, $x_\flat$ and upper bounds on θ_D, c_h; and c_R is defined as $64/\theta_D$.

The letters E_a and $1/c_h$ enter the floors only through $c_a(x)$, $c_h(x)$ and $1/E_a$; hence Thm. 4.39 holds on half lines in c_h, E_a, $x_\flat$. In other words, for each pair (c_h, E_a) taken from the two half lines, some number $x_\flat^0(c_h, E_a)$ exists such that Thm. 4.39 holds for every $x_\flat \ge x_\flat^0(c_h, E_a)$.

Finally, $x_\mathfrak{fl}$ is independent of ℓ and of N, by item G of §4.3. So the order of the constants has no circle. □

5. Iteration and cutoff comparison

5.1. **The first Wilson step.** We start the iteration by writing Wilson's weight as a member of the class with empty rows. The free-kernel estimates at depth zero give the data for the first step. Thm. 5.3 checks that the thresholds in these two arguments can be chosen together and uniformly in the volume.

Item 0: names. We take over from items 1 to 4 of §2.1 and from Thm. 4.39 the following: the lattice, the cubes and the cells, and the way the grids fit; the families $\mathcal{D}$ and $\mathcal{P}$, the number $|X|_c$ and the sets X^{+j}; the numbers p_0, ε, $R = \varrho(x)$, $r(x)$, $s(x)$; the step $\rho \to \rho'$; the function S_ε and the energy En_ℓ; the domains $\mathcal{U}_X(x)$, with their clauses (u1) to (u3) (item 1 of §2.1); the sets $M_A^{\mathrm{su}}(x)$ of members, with the strong cap; the densities $\rho[u;\ell]$ and the sums Ξ_w; the class $C_\flat^{\mathrm{su}}(x;\ell)$; the data (SD); the letters (O0) to (O13); and the floors (A0) to (A11).

By (F1) to (F4) we mean the formulas that Thm. 4.39 names: those of Steps 1, 6, 9 and 12 of its proof, and, for w', Step 13 of that proof, which takes the mean over e.

Two conventions fixed in §2.1 are in force throughout. The first stems from item 4 and (A7): (A7) is defined with $r_\Xi^\flat = \frac{7}{8}(\frac{1}{2} - d_0 - c_\Xi) - 3e_1^\flat$. The second stems from item 1, on cubes: a bond, a path or a plaquette *of* A has all of its sites in A, whereas a plaquette p lies *in* A as soon as its base site does.

We write $\emptyset$ for the family of rows with $\mathsf{H}_y = \{0\}$; for it, $\mathrm{En}_\emptyset(U) = \sum_p S_\varepsilon(\phi_p)$. We write $\emptyset'$ for the family of rows that Prop. 2.28 builds from $\emptyset$; at the empty rows its group (5) is void. We set $\ell_1 := (\emptyset')^{\mathrm{c}}$, that is, the cut of $\emptyset'$ at $\varrho(\beta/L)$, tensored with $1_{\mathfrak{g}}$ and dressed. Finally, $m_\Xi := \frac{1}{2} - d_0 - c_\Xi$.

The hull $\mathrm{hull}(X)$ is a member of $\mathcal{D}$ that contains X and has the least number of cubes among such members, an order fixing the choice; $|X|_c = \#\mathrm{hull}(X)$. Both are those of item 1 of §2.1, on cubes.

The letters ρ, ρ', $\rho_1 = \rho'$, ρ_β are densities, and $\rho_\beta = e^{-\beta\sum_p(1-\cos\phi_p)}$ is Wilson's density at β.

Further, $\ell_w(x) = 6Mr(x) + 12$ is a number of bonds (item 1 of §2.1, at x). The number D_w and the radius ω in (D2) and (C0) to (C2) are set in item 2 of §2.1, and the number g_x in (A0).

The k in $K_{\mathrm{q},k}$ and in $(\mathrm{G1}_k)$, $(\mathrm{F8}_k)$, $(\mathrm{F9}_k)$ is a mark and not a level. The letter K has three senses. In $0 \le k \le K$, and also in the phrase 'the tower $K = 1$', it is the number of steps of a tower. It denotes the bound of the rows in $\mathcal{R}(b, K, \lambda, \tau)$, and hence in $K = K_*$ and in $K \ge 0$. In $K \otimes 1_{\mathfrak{g}}$ it is a scalar kernel. The letter U is a fine field in $\mathrm{En}_\emptyset(U)$ and in ℓ_y^U; in the clause $U = 1$ of the first sentence on covariance of Rem. 5.6, it is the orthogonal map $U_{\mathsf{t},y}$ of Def. 2.46.

Item 1.

Definition 5.1 (Wilson's system at x, for any grid of fine cubes)**.** For $|t| < 2$ we put

$$h(t) := \sum_{n\ge 2} \frac{(2t)^n}{n^2\binom{2n}{n}} = \frac{t^2}{6} + \frac{2t^3}{45} + \frac{t^4}{70} + \dots .$$

We call $X \in \mathcal{P}$ *s-connected* when no splitting of X into two non-empty parts separates them by a distance $> s(x)$.

We set $u_W := (E_W^a, 0, w_W)$. Here, for a single cube Q, $E_W^a(\{Q\}, V) := x\sum_{p\in Q} h(1 - \frac{1}{2}\operatorname{tr}\partial V(p))$, while $E_W^a(X, \cdot) := 0$ for $\#X \ge 2$; and

$$\begin{aligned} w_W(X, V) := {} & 1\{X \text{ is } s\text{-connected}\} \\ & 1\{\text{every cube of } X \text{ holds a plaquette of } D(V)\} \\ & \exp\Big\{x \sum_{Q\in X^{+r(x)}} \sum_{p\in Q,\, p\notin D(V)} h(1-\cos\phi_p) \\ & - x \sum_{p\in D(V)\cap X} \big[1 - \cos\phi_p - d_0\varepsilon\phi_p\big]\Big\}. \end{aligned}$$

■

Item 2.

Lemma 5.2 (Wilson's weight in the class at the empty rows)**.** Let $x > 1$ satisfy the following three conditions.

(w1) $\varepsilon \le \frac{1}{40}$.

(w2) $2.01M^3\varepsilon \le E_a$.
(w3)

$$(m_\Xi - \frac{\varepsilon^2}{24})p_0^2 \ge \kappa s(x) + \frac{M^3(2r(x)+1)^3 p_0^4}{8x}.$$

Conditions (w1) and (w3) force $x > 1.5\cdot 10^5$. Indeed, inserted into (w3), the bounds $\kappa \ge 6$ of (O5), $s(x) = 8r(x) \ge 8$ and $m_\Xi < \frac{1}{2}$ give $p_0^2 > 96$, and (w1) gives $x = p_0^2/\varepsilon^2 \ge 1600 p_0^2$. Consequently $\log x > 1$, $\vartheta(x) = \vartheta_0^\flat(1 - 1/\log x) > 0$ and $2 - 1/\log x > 1$, and the domains $\mathcal{U}_X(x)$ are non-empty: a G-valued field that is tame on $Q^\approx$ has $|V(b)| = |V(\pi)| = 1$, and $|\partial V(p) - 1| = 2\sin(\phi_p/2) \le \varepsilon$ at the plaquettes of $Q^\approx$, so that it belongs to $\mathcal{U}_{\{Q\}}(x)$.

Then, on every torus $\mathbb{Z}^3/n\mathbb{Z}^3$ with $M \mid n$ and for every grid of cubes, the following hold.

(a) $e^{-x\sum_p(1-\cos\phi_p)} = \rho[u_W;\emptyset](V)$ at every V.
(b) u_W is real, lies in $M_1^{\mathrm{su}}(x)$ and has the strong cap. Hence Wilson's weight at $\beta = x$ lies in $C_\flat^{\mathrm{su}}(x;\emptyset)$, with $\mathsf{V} = 0$ and $E^h = 0$.
(c)

$$\sup_{\mathcal{U}_{\{Q\}}(x)} |E_W^a(\{Q\},\cdot)| \le 2.01M^3p_0^4/x = (2.01M^3\varepsilon/E_a)\, c_a(x) \le 0.0016\, c_a(x),$$

the last inequality holding if $E_a \ge 32M^3$.
(d) The conditions (w1) to (w3) follow from (O12), from $x \ge L^2$ of (A0), from (A1), (A4), (A6), (A7), and from $K_{\mathrm{q}} \ge M^3$ (item F of §4.1); so they hold for every $x \ge Lx_\flat^0$.

Moreover, u_W is independent of each of c_h, E_a, $x_\flat$, N, c_Ξ, κ. ■

Item 3.

Theorem 5.3 (The first step from Wilson's action stays in the class)**.** Assume the clauses and base statements listed in Step 0 of the proof of Thm. 4.39, together with the conclusion of Thm. 2.65. Choose (O0) to (O10) in the order of item 3 of §2.1, and define

$$E_a^0 := \max\{32L^{-1/2}K_v, 32K_{\mathrm{q}}\}, \qquad x_\flat^{(1)} := x_\flat^0(\delta_3/2, E_a^0).$$

Here $x_\flat^0(c_h, E_a) = 1 + \inf S(c_h, E_a)$ is the threshold of (O13): $S(c_h, E_a)$ consists of those $y \ge e^2$ for which (A0) to (A11) hold at every $x \ge Ly$. Step 14 of the proof of Thm. 4.39 shows that this is a nonempty upper half line, so $1 + \inf S$ belongs to it, whether or not its infimum does. The threshold $x_\flat^{(1)}$ therefore depends on (O0) to (O10) alone.

All size conditions entering these floors are recorded in (A0). In particular, it includes the floors of §4.3 through $x \ge x_{\mathrm{fl}}$, the chart conditions (D2), (D3) at $(1, r_\eta)$ and $(17T, r_T)$, and (C0) to (C2). The volume condition (Tor) is a datum of the step and is excluded from these floors, as stipulated in (A0) and (P0). Thus $x_\flat^{(1)}$ is independent of N and of the grid.

Conclusion. In this setting, let $0 < c_h \le \delta_3/2$, $E_a \ge E_a^0$ and $x_\flat \ge x_\flat^{(1)}$ be arbitrary, let $\beta \ge Lx_\flat$, let N be any even number with $M \mid N$ and $N \ge 2Ms(\beta)$, and let any grid of cells on $\mathbb{T}'$ be given. Then the image ρ' of $e^{-\beta\sum_p(1-\cos\phi_p)}$ on $\mathbb{Z}^3/NL\mathbb{Z}^3$ under the step at $x = \beta$ lies in $C_\flat^{\mathrm{su}}(\beta/L;\ell_1)$, with the letters d_0, c_Ξ, κ, ρ_0, c_R, M, A_0, $\mathfrak{p}_0$, $\vartheta_0^\flat$, c_h, E_a and with $x' = \beta/L$ exactly. The formula of the class holds at every V', and V' is its only argument. One system covers every $(c_h, E_a, x_\flat)$ of the three half lines: (V', u') is built by (F1) to (F4) from $(\mathsf{V}, u, \ell) = (0, u_W, \emptyset)$, with the fine grid of origin $o = Lo' - \mathsf{h}_L$ mod M, and w' is the mean over e of the $w'(\Upsilon, V', e)$ of (F4) (Step 13 of the proof of Thm. 4.39). The same conclusion holds at any (c_h, E_a) of the half lines for $\beta \ge Lx_\flat^0(c_h, E_a)$.

Besides. The following also hold.

(i′) For every cell $\square$,

$$\sum_{Y \ni \square} e^{\kappa(\#Y-1)} \sup_{\mathcal{U}_Y(x')} |E'^a(Y,\cdot)| \le 0.56\,(E_a^0/E_a)\, c_a(x').$$

(ii′) For every cell $\square$,

$$\sum_{Y\ni\square} e^{\kappa(\#Y-1)} \sup|E'^h(Y,\cdot)| \le e^{-(\delta_3/2)p_0(\beta/L)^2}.$$

(iii′) The clauses that follow are those of the scalar rows, namely of the cut of $\emptyset'$ at $\varrho(\beta/L)$, not yet tensored with $1_{\mathfrak{g}}$ or dressed: indeed $K\otimes 1_{\mathfrak{g}}$ has the norms of K (Step H13 of the proof of Thm. 2.65, the group), and at G-valued fields the transports of the dressing are isometries. The rows ℓ_1 are translation covariant and have range $\le \varrho(\beta/L)$; further, $|\ell_{1,y'}(P)| \le K_* e^{-\lambda_* \mathrm{dist}(y',P)}$, $1 \le \mathsf{B}' \le \frac{3}{2}\mathsf{r}_\sharp^{-1}$, $\|\ell_1\|_{\mathrm{r}}\|\ell_1\|_{\mathrm{c}} \le 3K_*^2\mathsf{g}(\lambda_*)^2$ and $\mathsf{t}_\rho \le \frac{1}{8}m_\Xi$. These facts come from the conclusion of Thm. 2.65, with (Fr0) and (Fr4), at $k=0$, $x_0=\beta$; the estimate of the two norms from Def. 2.48, at (K_*,λ_*); the covariance of $\emptyset'$ from Prop. 2.55 (a); the covariance of the cut from the proof of Prop. 2.55 (d); and the range from the cut.

(iv′) $\theta_D \le \theta_{\mathbb{C}}/2$, $\theta_D \le \lambda_*/8$ and $c_R\theta_{\mathbb{C}} \ge 128$, by (O6) and by item C.II 2.2 of §3 (source and hypotheses).

■□

Remark 5.4 (Comparison of half lines of c_h). For every number $\delta > 0$ we have $\frac{1}{2}\min\{\delta,\delta_3'\} \le \frac{1}{2}\min\{\delta,\delta_3\} \le \delta_3/2$. Hence the half lines $c_h \le \frac{1}{2}\min\{\delta,\delta_3'\}$ and $c_h \le \frac{1}{2}\min\{\delta,\delta_3\}$ are contained in the half line $c_h \le \delta_3/2$ of Thm. 5.3, whatever $\delta > 0$ is. For instance, at (O11) of item 3 of §2.1 the table takes $c_h \le \delta_3'/2$, and $\delta_3'/2 = \frac{1}{2}\min\{\delta,\delta_3'\}$ for every $\delta \ge \delta_3'$.

Remark 5.5 (Statements for later use). For later sections we collect the following.

(R0) ρ_0 itself lies in $C_\flat^{\mathrm{su}}(\beta;\emptyset)$, at all (c_h,E_a) of the half lines and for every $\beta \ge Lx_\flat^{(1)}$; here ρ_0 is Wilson's density $e^{-\beta\sum_p(1-\cos\phi_p)}$, the ρ_0 of Thm. 5.7, and not the integer of (O3). Indeed, neither (w1) nor (w3) depends on c_h or on E_a; (w2) at E_a^0 gives (w2) at every $E_a \ge E_a^0$; $E^h = 0$ meets (ch) at every c_h; and Lem. 5.2 (b) then applies.

(R1) $\rho_1 \in C_\flat^{\mathrm{su}}(\beta/L;\ell_1)$ at every V', for every grid, where $\rho_1 = \rho'$ is the image of $e^{-\beta\sum_p(1-\cos\phi_p)}$ under the step at $x=\beta$, as in the conclusion of Thm. 5.3.

By (R0), the membership $\rho_k \in C_\flat^{\mathrm{su}}(x_k;\ell_k)$ of Thm. 5.7 is true also at $k=0$, where $x_0=\beta$ and $\ell_0=\emptyset$; it is therefore true for $0 \le k \le K$.

Proof of Lem. 5.2 and of Thm. 5.3. Step S0: the hypotheses at Wilson's weight. To apply Thm. 4.39, we need the letters and floors, the data (SD), and a real member with the strong cap representing Wilson's density. Steps S1–S4 prove the representation and its analytic and rough caps; Step S5 checks its three size conditions; and Step S6 proves (SD) for $(\emptyset,\emptyset')$. The remaining conditions hold directly: $\mathsf{V}=0$, $E^h=0$, Wilson's density is positive and gauge invariant, and the chosen fine grid fits the given coarse grid. The hypotheses on N are those in the theorem.

The empty rows are allowed in items C.I 1.1 and C.II 1.2 of §3, in Remark (4) of item I of §4.1, and in the data of Sections 4.2–4.4. They are also members of $\mathcal{R}(1,0,\lambda,0)$ by Def. 2.48. Rem. 5.6 gives an equivalent formulation using one-dimensional zero rows. The window law is constructed within the step from $q_\emptyset^V$; it is not an additional datum of the class.

Step S1: the function h. We show that h has the following four properties.

(1) The function h is holomorphic in $|t|<2$, and its coefficients are >0.
(2) For $0 \le \phi < \pi$ we have $h(1-\cos\phi) = \phi^2/2 - (1-\cos\phi)$.
(3) Moreover $0 \le h(1-\cos\phi) \le \phi^4/24$.
(4) In the same disc, $|h(t)| \le \dfrac{|t|^2/6}{1-|t|/2}$.

The proof rests on two facts: the expansion

$$\arcsin^2 z = \frac{1}{2}\sum_{n\ge1}\frac{(2z)^{2n}}{n^2\binom{2n}{n}},$$

and the relation $\arccos(1-t) = 2\arcsin\big((t/2)^{1/2}\big)$.

To prove the first, we put $f = \arcsin^2 z$. This function is even and holomorphic in $|z|<1$, and it satisfies $f(0)=0$ and the equation $(1-z^2)f'' - zf' = 2$, which we obtain by differentiating $(1-z^2)^{1/2} f' = 2\arcsin z$.

If we write $f = \sum_{n\ge1} c_n z^{2n}$, this equation says that $2c_1 = 2$ and that $(2n+2)(2n+1)c_{n+1} = 4n^2 c_n$.

On the other hand, the numbers $b_n = \frac12 4^n/(n^2\binom{2n}{n})$ satisfy $b_1 = 1$ and $b_{n+1}/b_n = 4n^2/((2n+2)(2n+1))$, since $\binom{2n+2}{n+1}/\binom{2n}{n} = (2n+2)(2n+1)/(n+1)^2$. Therefore $c_n = b_n$.

The second fact is the identity $\cos 2a = 1 - 2\sin^2 a$ at $\sin a = (t/2)^{1/2}$, where $0 \le t \le 2$.

From the two facts together we obtain

$$\frac12 \arccos(1-t)^2 = \sum_{n\ge1} \frac{(2t)^n}{n^2\binom{2n}{n}} = t + h(t).$$

In this series, the quotient of two consecutive coefficients equals $n^2/((n+1)(2n+1))$, which is $< \frac12$. Hence the radius of convergence is 2, and the tail of the series from $n=2$ on is bounded in absolute value by $|t|^2/6$, the absolute value of its first term, times $1/(1-|t|/2)$.

The bound $\phi^4/24$ comes from the inequality $\cos\phi \le 1 - \phi^2/2 + \phi^4/24$, which is valid for every real ϕ.

Step S2: clause (ca) *for E^a_W, in the proof of Lem. 5.2 (c).* We show that $E^a_W(\{Q\},\cdot)$ is holomorphic on $\mathcal{U}_{\{Q\}}(x)$ and real at G-valued fields, that it is invariant under G-valued gauge transformations and vanishes at pure gauges, and that its supremum is bounded by $2.01M^3p_0^4/x$, a number that is itself $\le c_a(x)$.

Let p be a plaquette based in Q. All its sites lie in Q^{+1}, and Q^{+1} is contained in $Q^{+r(x)}$ because $r(x) \ge 1$. Thus p is a plaquette of $Q^{\approx}$, and (u3) yields $|\partial V(p) - 1| < 2\varepsilon$.

We set $g = \partial V(p) = 1 + A \in \mathrm{SL}(2,\mathbb{C})$. This uses that $\mathcal{U}_X(x)$ is a set of $\mathrm{SL}(2,\mathbb{C})$ fields (item 1 of §2.1, on domains); for a cell, the domain of item C6 of §4.1 at x' is a set of $\mathrm{SL}(2,\mathbb{C})$ fields as well.

Because $\det g = 1$, we have $\operatorname{tr} A = -\det A$, and therefore $t_p := 1 - \frac12 \operatorname{tr} g = \frac12 \det A$. Without $\det g = 1$, t_p would be of first order in A.

Since $|\det A|$ is the product of the two singular values of A, it follows that $|t_p| \le \frac12 |A|^2 < 2\varepsilon^2 \le \frac{1}{800}$.

The number t_p is a polynomial in the entries of V, so $h(t_p)$ is holomorphic. Under gauge transformations the trace does not change. At G-valued fields, $t_p = 1 - \cos\phi_p$ is real. At $\partial V = 1$ we get $t_p = 0$, and $h(0) = 0$.

Step S1 gives $|h(t_p)| \le \dfrac{4\varepsilon^4/6}{1-\varepsilon^2} \le 0.668\,\varepsilon^4$. As a cube holds $3M^3$ plaquettes, $|E^a_W| \le x \cdot 3M^3 \cdot 0.668\,\varepsilon^4 \le 2.01 M^3 p_0^4/x$.

Since $c_a(x) = E_a p_0^3 x^{-1/2}$, dividing this bound by $c_a(x)$ gives $2.01M^3\varepsilon/E_a$, and by (w2) this quotient is ≤ 1. With $E_a \ge 32K_{\mathrm{q}} \ge 32M^3$ and $\varepsilon \le \frac{1}{40}$, it is even ≤ 0.0016.

Step S3: the identity, in the proof of Lem. 5.2 (a). We show that at every G-valued V

$$e^{-x\sum_p(1-\cos\phi_p)} = \exp\Big\{-x\sum_p S_\varepsilon(\phi_p) + \sum_Q 1_{Q^{\approx}}(V) E^a_W(\{Q\},V)\Big\}\,\Xi_{w_W}(V).$$

We denote by $d(V)$ the set of those cubes that hold a plaquette of $D(V)$. In this step and the next we abbreviate $D = D(V)$, $r = r(x)$ and $s = s(x)$, and under a sum over p we write h for $h(1-\cos\phi_p)$.

1. *The quotient.* If $p \notin D$, then $-x(1-\cos\phi_p) + x\phi_p^2/2 = x\,h(1-\cos\phi_p)$. If $p \in D$, the corresponding expression is $-x(1-\cos\phi_p) + x d_0 \varepsilon \phi_p$. Dividing the left side by $e^{-x\sum S_\varepsilon}$, we therefore obtain $\exp\{x\sum_{p\notin D} h - x\sum_{p\in D}[1-\cos\phi_p - d_0\varepsilon\phi_p]\}$.

2. *The indicator.* The equality $1_{Q\approx}(V) = 1$ holds exactly when no plaquette based in Q^{+r} belongs to D, that is, exactly when $\operatorname{dist}(Q, d(V)) > r$. Hence $\sum_Q 1_{Q\approx} E^a_W = x \sum_{Q\notin d(V)^{+r}} \sum_{p\in Q} h$, and each p occurring in this sum is tame. The factor that remains for Ξ is thus $\exp\{x \sum_{Q\in d(V)^{+r}} \sum_{p\in Q,\, p\notin D} h - x \sum_{p\in D} [\,\cdot\,]\}$, where $[\,\cdot\,]$ is the bracket of part 1.
3. *The families.* Any family $\{X_i\}$ in Ξ_w has pairwise distances $> s$, and D lies inside its union, that is, each plaquette of D is in some cube of the union. When its product is not 0, every cube of every X_i lies in $d(V)$, and hence the union is exactly $d(V)$. Every X_i is s-connected, while two different X_i are at distance $> s$; so the X_i are the classes of $d(V)$ under "distance $\le s$". There is thus exactly one such family, and when D is empty this is the empty family, whose value is 1.
4. *The product.* As $s = 8r > 2r$, the sets X_i^{+r} are pairwise disjoint, and their union is $d(V)^{+r}$; the sets $D \cap X_i$ form a partition of D. Consequently $\prod_i w_W(X_i, V)$ equals the factor that remains in part 2.

Step S4: clause (cΞ) *with the strong cap, in the proof of Lem. 5.2 (b).* We show that $w_W(X, \cdot)$ is real, Borel and gauge invariant, that it is a function of V on the bonds of $X^{+s(x)}$, and that at every V

$$|w_W(X, V)| \le \exp\{-c_\Xi x\varepsilon \sum_{D(V)\cap X} \phi_p - \kappa(|X|_c - 1)\}.$$

The function w_W depends only on the angles ϕ_p at the plaquettes based in X^{+r}. The sites of these plaquettes lie in X^{+r+1}, and this set is contained in X^{+s} because $r + 1 \le 8r$. Each ϕ_p is a continuous class function of $\partial V(p)$.

The bound. Suppose $w_W \ne 0$, and put $n = \#X$.

(1) *The rough plaquettes.* The function $(1 - \cos\phi)/\phi$ has the derivative $(\phi \sin\phi - 1 + \cos\phi)/\phi^2$. The numerator $N(\phi)$ of this derivative satisfies $N(0) = 0$ and $N' = \phi \cos\phi$; so N increases on $(0, \pi/2)$, then decreases, and $N(\pi) = -2$. Therefore the function first increases and then decreases on $(0, \pi]$, and on $[\varepsilon, \pi]$ it is bounded below by the smaller of its values at the two end points. These values satisfy $(1 - \cos\varepsilon)/\varepsilon \ge \varepsilon/2 - \varepsilon^3/24$ and $2/\pi \ge \varepsilon/2$. For $p \in D$ we thus get $1 - \cos\phi_p \ge (\frac{1}{2} - \frac{\varepsilon^2}{24})\varepsilon\phi_p$, and hence $1 - \cos\phi_p - d_0\varepsilon\phi_p \ge (c_\Xi + m_\Xi - \frac{\varepsilon^2}{24})\varepsilon\phi_p$.

(2) Each cube of X holds some $p \in D$, and for this p we have $x\varepsilon\phi_p > x\varepsilon^2 = p_0^2$. Summing, $x\varepsilon \sum_{D\cap X} \phi_p \ge n p_0^2$.

(3) We have $\#X^{+r} \le n(2r+1)^3$; a cube holds $3M^3$ plaquettes; and at tame p we have $h \le \varepsilon^4/24$. The positive part of the exponent is therefore at most $n(2r+1)^3\, 3M^3\, x\varepsilon^4/24 = nM^3(2r+1)^3 p_0^4/(8x)$.

(4) *The hull.* For $X \in \mathcal{P}$, $\operatorname{hull}(X)$ is a member of $\mathcal{D}$ that contains X and has the least number of cubes among such members, the choice being fixed by an order; and $|X|_c := \#\operatorname{hull}(X)$. Consequently, every connected set of cubes that contains X has at least $|X|_c$ cubes. The set X has a spanning tree whose edges have distance $\le s$, and for two cubes at sup distance d there are $d - 1$ further cubes that link them through corners. Taking the n cubes of X together with, for each of the $n - 1$ edges of the tree, its $d - 1 \le s - 1$ further cubes, we obtain a connected set of cubes that contains X; this is true on a torus as well. Hence $|X|_c \le n + (n-1)(s-1)$, that is, $|X|_c - 1 \le (n-1)s$.

(5) *Conclusion.* The number $m_\Xi - \varepsilon^2/24$ is > 0, since the right side of (w3) is > 0 and $p_0^2 > 0$; thus we may multiply the inequality of (2) by this number. We arrive at

$$\log w_W \le -c_\Xi x\varepsilon \sum \phi_p - n\Big[(m_\Xi - \frac{\varepsilon^2}{24})p_0^2 - \frac{M^3(2r+1)^3 p_0^4}{8x}\Big] \le -c_\Xi x\varepsilon \sum \phi_p - n\kappa s,$$

where the second inequality follows from (w3); and $n\kappa s \ge \kappa(|X|_c - 1)$.

Step S5: the floors, in the proof of Lem. 5.2 (d). We show that, for any $c_h \le \delta_3/2$ and any $E_a \ge E_a^0$, the three conditions (w1), (w2), (w3) hold at every $x \ge L x_\flat^0$.

(w1) This condition is the first clause of (A1).

(w2) By (O12) we have $E_a \geq 32K_{\mathrm{q}}$, and $K_{\mathrm{q}} = K_{\mathrm{q},k} = M^3[3.09L^2K_Y + 520LC_D(1+4C_D^2)+1] \geq M^3$ (item F of §4.1). Hence $2.01M^3\varepsilon \leq M^3 \leq E_a$.

(w3) We proceed in four parts.

(i) Condition (A7) gives $\upsilon_f^{1/2} \leq \frac{1}{4}x^{-1/4}e^{-(\kappa+1)s'}$. Taking logarithms, we find

$$r_\Xi^\flat p_0'^2 \geq 2(\kappa+1)s' + (\kappa+4)(1+2s') + 4 + 1.01\mathsf{K}_1^\flat(2r'+1)^3 + 2\log 4 \geq (4\kappa+10)s' + \kappa + 8.$$

Here $r_\Xi^\flat = \frac{7}{8}m_\Xi - 3e_1^\flat \leq m_\Xi$. The signs we use are these: $e_1^\flat > 0$ and $u_t > 0$ (item F of §4.3); and $\mathsf{K}_1^\flat > 0$ (Part C.II of §3, in the display before Thm. 3.40). The sign of $\mathsf{K}_1^\flat$ enters the displayed line of logarithms, while $u_t > 0$, $\upsilon > 0$ and $c_h(x) > 0$ are used in (iii), where two terms of (A6) are dropped. Since moreover $p_0' \leq p_0$, it follows that $m_\Xi p_0^2 \geq (4\kappa+10)s' + \kappa + 8$.

(ii) By (A0), $x \geq L^2$, and this gives $\log x \leq 2\log x'$. So $r(x) = \lceil 20c_R \log x/M\rceil \leq \lceil 2(20c_R \log x'/M)\rceil \leq 2r'$, and $s(x) \leq 2s'$. Hence $\kappa s(x) \leq 2\kappa s'$.

(iii) From the third term of (A6) we have $6((2r_1+1)L + 2r(x))^3 c_a(x) \leq \frac{1}{2}$, so that $(2r(x)+1)^3 c_a(x) \leq \frac{1}{12}$. Also $c_a(x) = E_a p_0^2 \varepsilon \geq 32M^3 p_0^2 \varepsilon$. Hence $M^3(2r(x)+1)^3 p_0^4/(8x) = M^3(2r(x)+1)^3 p_0^2 \varepsilon\,(\varepsilon/8) \leq \frac{1}{384}\cdot\frac{1}{320}$.

(iv) Condition (A4) gives $32M^3 p_0^2 \varepsilon \leq \frac{1}{2}$, and so $\varepsilon^2 p_0^2/24 \leq \frac{1}{64}\cdot\frac{1}{40}\cdot\frac{1}{24}$.

Adding these estimates, we find that the left side of (w3) minus its right side is $\geq (2\kappa+10)s' + \kappa + 8 - 10^{-4} > 0$.

Step S6: the data (SD) *for* $(\emptyset, \emptyset')$. Apply Thm. 2.65 with $x_0 = \beta$, $k = 0$ and the tower $K = 1$, whose tori have sides NL and N. Its hypotheses hold: (O4) gives odd $L \geq L_A$ and $aL \geq 3/c_L$; N is even; and $x_1 = \beta/L \geq x_\flat \geq e^2$. The condition (X12) holds for every $x_\flat \geq 2$, by Step H16 of that theorem's proof. Its conclusions give

$$\emptyset \in \mathcal{R}(b_*, K_*, \lambda_*, 1/9), \qquad \mathrm{range}(\emptyset) = 0,$$

(Dec)(C_D, θ_D) for all sixteen kernels, (N′) with eq. 2.vii at every cut and mask, and $\mathsf{t}_\rho \leq m_\Xi/8$, through (Fr0), (Fr2), (Fr3′) and (Fr4). The value of C_D in (O6) is at least the value in that theorem, so it preserves these estimates.

The remaining clauses (L1^{su}) and (L2^{su}) are imposed by (O4). They also imply the older (L1), (L2) of Def. 2.42: since $\mathsf{N}_\nu \geq \mathsf{N}_\sharp$, $2200 \geq 60$ and $40 \geq 4$,

$$\mathsf{c}_{\mathrm{loss}} \leq [\mathsf{N}_\nu/L + 2200\mathsf{N}_Z + 40\mathsf{N}_\Xi]L^{-1/2} \leq d_0/4.$$

Also $L \geq (1.2\cdot 10^4)^2 \mathsf{N}_{\mathrm{c}}^4$, with $\mathsf{N}_{\mathrm{c}} = 17(\mathsf{N}_\sharp + 1)$, implies $L \geq 115\mathsf{N}_\sharp^2$. Thus (SD) holds.

For later use, (Fr0) also gives $|\emptyset'_{y'}(P)| \leq K_* e^{-\lambda_* \mathrm{dist}(y',P)}$ and $1 \leq \mathsf{B}' \leq \frac{3}{2}\mathsf{r}_\sharp^{-1}$. Translation covariance follows from Prop. 2.55(a), and cutting by distance preserves it by the proof of Prop. 2.55(d). Tensoring a scalar kernel with $1_{\mathfrak{g}}$ preserves its norms; dressing at a real field uses isometries. These are the tensoring and dressing operations of item C.II 2.2 of §3 and item 3.5 of §4.2.

Remark 5.6 (Empty rows as zero rows). Let 0 denote the rows with $\mathsf{H}_y = \mathbb{R}$ and $\ell_y(p) = 0$. This family is translation covariant, with $U = 1$; it has $b_{\mathsf{B}} = 1$, $\tau = 0$ and range 0, and hence belongs to $\mathcal{R}(1, 0, \lambda_*, 0)$.

Adding a zero component to a row does not change its column norms or its Gram matrix. Thus 0 and $\emptyset$ give the same B, R, q_ℓ^V, g^V and En_ℓ; their row norms vanish, as do $\sigma = \ell\nu$, $|\sigma_y|$ and $\ell^{\mathsf{T}}\sigma$. The transported row $(\ell^V F)_y$ and every step of kind (R) vanish as well. The auxiliary output spaces $\bigoplus_y \mathsf{H}_y$ and Hil may differ, but the extra coordinates are zero and contribute no norm or energy.

By group (5) of Prop. 2.28, $0'$ is $\emptyset'$ with L^3 additional zero components. Their column norms and Gram matrices therefore agree, as do B', $H^\sharp$, $\mathsf{N}^\sharp$, T, X, the cut forms, t_ρ and En'. It follows that (SD) holds for $(0, 0')$ exactly when it holds for $(\emptyset, \emptyset')$, and that the densities and classes at both scales agree. Consequently either family yields the same hypothesis and conclusion in Thm. 4.39.

Step S7: the proof of Thm. 5.3, with (i′) *to* (iv′). We now prove the theorem of item 3 of §5.1.

1. We fix the letters (O0) to (O10). We set $(c_h, E_a) = (\delta_3/2, E_a^0)$, both of them computed from (O0) to (O10), and $x_\flat^{(1)} = x_\flat^0$ at these two values (Step 14 of the proof of Thm. 4.39). Let $\beta \geq L x_\flat^{(1)}$, let N be as in the statement, and let o' be the origin of an arbitrary grid of cells.
2. We choose the grid of fine cubes with origin $o = Lo' - \mathsf{h}_L \bmod M$; with this choice the grids fit.
3. Wilson's density $\rho_\beta = e^{-\beta \sum_p (1-\cos\phi_p)}$ at β is the left side of the identity of Step S3 at $x = \beta$, and Lem. 5.2 (a), taken at $x = \beta$, identifies it with $\rho[u_W; \emptyset]$. Step S5 and Lem. 5.2 applied with $x = \beta$ show that ρ_β lies in $C_\flat^{\mathrm{su}}(\beta; \emptyset)$.
4. Step S6 shows that (SD) holds at $x = \beta$ for $(\emptyset, \emptyset')$.
5. Thm. 4.39 then yields the following: the formulas (F1) to (F4) define a real V' and a real $u' \in M_1^{\mathrm{su}}(x')$ with the strong cap at $x' = \beta/L$. Moreover $\rho' = e^{\mathsf{V}'} \rho[u'; (\emptyset')^{\mathrm{c}}] > 0$ at every V', and ρ' has no argument other than V'.
6. *Half lines.* Let $c_h \leq \delta_3/2$ and $E_a \geq E_a^0$. In the definition of the class, E_a occurs only in c_a, and c_h only in $c_h(x)$. Now $c_a(x')$ increases with E_a, and $e^{-c_h p_0'^2}$ increases when c_h decreases; hence the same (V', u') satisfies (ca) and (ch) at (c_h, E_a). The clause "V real" of the class, the clause (cΞ) and the formula of the class depend on neither letter. A larger $x_\flat$ merely shrinks the set of β.
7. *One system.* The system u_W depends on none of c_h, E_a and $x_\flat$ (item 1 of §5.1). The formulas (F1) to (F4) depend on letters, clusters and cores, but not on c_h, E_a or $x_\flat$; and neither does the mean over e in Step 13 of the proof of Thm. 4.39.
8. For (i′) and (ii′): we apply the part "Moreover" of Thm. 4.39 with $(c_h, E_a) = (\delta_3/2, E_a^0)$. For every cell, this part bounds the sum on the left of (i′) by $(0.25{+}0.31)\, E_a^0 p_0'^3 x'^{-1/2}$, which equals $(0.25 + 0.31)(E_a^0/E_a) c_a(x')$. Its bound for the sum on the left of (ii′), at the same cell, is $e^{-(\delta_3/2) p_0'^2}$.
9. For (iii′) and (iv′) we use Step S6, (O6), and item C.II 2.2 of §3 (in its remark "Source and hypotheses").
10. For Rem. 5.4 we recall from (O10) that $\delta_3 = \min\{d_0/1536,\, 3c_\xi^2/(256 h_*^2)\}$ and $\delta_3' = \min\{d_0/2304,\, 3c_\xi^2/(256 h_*^2)\}$. Since $d_0/2304 < d_0/1536$, while the other entry is the same in both, $\delta_3' \leq \delta_3$. A minimum increases with each of its entries and does not exceed any of them; therefore $\min\{\delta, \delta_3'\} \leq \min\{\delta, \delta_3\} \leq \delta_3$.
11. The last clause of the theorem, where $\beta \geq L x_\flat^0(c_h, E_a)$, follows from parts 2 to 5 of this step, carried out with the values (c_h, E_a) themselves.

Step S8: order of the constants and dependencies. The numbers δ_3, K_v and K_{q} are fixed at (O10). They determine E_a^0 and $x_\flat^{(1)}$ before c_h, E_a and $x_\flat$ are chosen in (O11)–(O13).

The proof uses Thm. 2.65 and Thm. 4.39. The latter uses Sections 3–4.3 and the base statements listed in Step 0 of its proof, all independently of the present theorem. Its use of §5.3 is limited to the chart lemmas Lem. 5.17 and Thm. 5.18, Step 2(ii), Step 8(8.2) and its paragraph on u, and Lem. 5.23 in Step 10(9) of the proof of Thm. 5.15. Those passages use the definitions and constants of §2.1, the results of §3, and the chart identity from Step 4 of the proof of Thm. 4.5; they do not use Thm. 5.3 or Thm. 5.7. The references in item 2 of §5.3 to Step S6 and Rem. 5.6 concern only empty-row data, which those two passages derive from §2.2 and the definitions. Thus the dependency order is the free row estimates, the real step, the present first step, and then the iteration Thm. 5.7. □

5.2. **Iteration of the real step.** We fix the constants for the initial step and the iteration. The side condition at the last step implies the conditions at all earlier steps. The row estimates and the fitting of the grids then permit an induction from the Wilson weight through all K steps.

Theorem 5.7 (The K steps from Wilson's weight stay in the class, (S-K))**.** In the setting (O0) to (O10) of item 3 of §2.1, let $0 < c_h \le \delta_3/2$ and $E_a \ge E_a^0$, and let $x_\flat \ge \max\{x_\flat^{(1)}, x_\flat^0\}$ satisfy (X12), which is item (8) in the order of the constants of Thm. 2.65. Here $x_\flat^{(1)}$ denotes the floor of Thm. 5.3, namely $x_\flat^0$ evaluated at $(\delta_3/2, E_a^0)$, while $x_\flat^0$ itself is evaluated at the chosen pair (c_h, E_a). Let $K \ge 1$ and $\beta \ge L^K x_\flat$, and let N be even, with $M \mid N$ and $N \ge 2Ms(L^{1-K}\beta)$. For $0 \le k \le K$ let $\mathbb{T}_k = \mathbb{Z}^3/NL^{K-k}\mathbb{Z}^3$, with the grids $o_K = 0$ and $o_k = Lo_{k+1} - \mathsf{h}_L$, and put $x_k = L^{-k}\beta$. Let $\rho_0 = e^{-\beta\sum_p(1-\cos\phi_p)}$ be Wilson's density on $\mathbb{T}_0$, and for $0 \le k < K$ let $\rho_{k+1} = (\rho_k)'$ be the image of ρ_k under the step at $x = x_k$, in the sense of item 1 of §2.1, with the noise ax_k, by which we mean the index ax of the kernel of that step.

Then, given Thm. 5.3 and Thm. 4.39, we have $\rho_k \in C_\flat^{\mathrm{su}}(x_k; \ell_k)$ for $1 \le k \le K$, where the rows are $\ell_0 = \emptyset$ and $\ell_{k+1} = ((\ell_k)')^{\mathrm{c}}$, cut at $\varrho(x_{k+1})$; these are the scalar rows of §2.2, tensored with $1_{\mathfrak{g}}$ and dressed. This holds with the same letters at every k, and at every field of $\mathbb{T}_k$. No argument other than the field of level k enters ρ_k.

We make the following additional assertion: for $1 \le k \le K$ we have $\rho_k = e^{\mathsf{V}_k}\rho[u_k; \ell_k]$, where u_1 is a system of members given by Thm. 5.3 and, for $k < K$, u_{k+1} is made from u_k by (F1) to (F4). The map of members so obtained is not the map F_ℓ of Thm. 5.15: the formulas (F1) to (F4) and the formulas (Fm1) to (Fm4) of F_ℓ are defined with two different tests χ_X, so that their means $K(\mathfrak{B})$ need not agree, and neither need the members E'^a, E'^h, w' built with these means; however, both the map given by (F1) to (F4) and the map F_ℓ yield members that represent the same ρ'. The members are not unique. ■□

Proof. Step 15: the tower of K steps. Item 1: the letters. Thm. 5.3 holds for all $0 < c_h \le \delta_3/2$, all $E_a \ge E_a^0$ and all $x_\flat \ge x_\flat^{(1)}$, and the number $x_\flat^{(1)}$ depends on none of N, c_h, E_a; this is true for any $L \ge L_*^{\mathrm{su}}$ and for any M that satisfies the bounds of (O7). In turn, Thm. 4.39 is valid on the half lines of Step 14 of its proof. Intersecting the two ranges gives the choice of Thm. 5.7, and this is one choice of all the letters.

Item 2: the condition (Tor). We begin with a proposition on numbers.

Proposition 5.8 (The condition on the side passes to finer levels)**.** If $x \ge e^2$, $L \ge 3$ and $N \ge 2Ms(x)$, then $NL^j \ge 2Ms(L^j x)$ for every integer $j \ge 0$. ■

Proof of Prop. 5.8. Put $\alpha = 20c_R/M$ and $t = \alpha \log x$, so that $r(x) = \lceil t \rceil$. Then

$$\alpha \log(L^j x) = t(1 + j \log L/\log x) \le t(1 + (j/2)\log L) \le L^j t \le L^j \lceil t \rceil$$

where the first inequality uses $\log x \ge 2$ and the second uses $1 + v/2 \le e^v$ at $v = j \log L$. Since $r(L^j x) = \lceil \alpha \log(L^j x) \rceil$ and $L^j \lceil t \rceil$ is an integer, this gives $r(L^j x) \le L^j r(x)$. Consequently

$$2Ms(L^j x) = 16Mr(L^j x) \le L^j \cdot 2Ms(x) \le L^j N$$

which proves the proposition. □

We now apply the proposition. The coarse side of step k is $N_k = NL^{K-k-1}$, which is even and a multiple of M. Moreover $x_k = L^{K-1-k}x_{K-1}$, and $x_{K-1} = L^{1-K}\beta \ge Lx_\flat \ge e^2$. Prop. 5.8, applied with $x = x_{K-1}$ and $j = K - 1 - k$, therefore turns (Tor) at the last step, which is the condition $N \ge 2Ms(L^{1-K}\beta)$ of the data, into (Tor) at every step, namely $N_k \ge 2Ms(x_k)$.

Item 3: the grids. The grids of the data obey $o_k = Lo_{k+1} - \mathsf{h}_L$, and this relation is the fit of the grids that (SD) requires at every step.

Item 4: the level $k = 0$. Thm. 5.3, applied with $\beta \ge Lx_\flat$, gives $\rho_1 \in C_\flat^{\mathrm{su}}(\beta/L; \ell_1)$. Of that theorem we use here that it holds at every field of $\mathbb{T}_1$ and for every grid of cells, hence in particular for the grid o_1. The condition $N_0 \ge 2Ms(\beta)$ of that theorem is (Tor) of item 2 for $k = 0$. The rows are $\ell_1 = (\emptyset')^{\mathrm{c}}$, cut at $\varrho(\beta/L)$.

Item 5: from k to $k+1$, for $1 \le k < K$. Here $x_k \ge Lx_\flat$, and ℓ_k has range $\le \varrho(x_k)$. We use Thm. 2.65 for this k, with $x_{k+1} \ge x_\flat$, with (X12), with N_k even, with $L \ge L_A$ odd, and with

$aL \geq 3/c_L$ from (O4). The conclusion of that theorem gives (Dec)(C_D, θ_D) for all sixteen kernels, with $\theta_D = \frac{1}{2}\min\{\lambda_2^*, \theta_{\mathbb{C}}, \theta_J\}$ and with the C_D of (O6). Indeed, in that conclusion C_D stands for the expression displayed in item (6) of its order of the constants; since (Dec)(C, θ_D) holds for every C at least as large as this expression, it holds for the letter C_D of (O6). The same conclusion yields (Fr3) at every cut.

Two steps of the proof of Thm. 2.65 are used as well: Step H9, which is the membership (Fr0) of its conclusion, and Step H14, which, with the present letters, applies Thm. 2.57 (b) and proves Thm. 2.64 in the form that Thm. 2.65 uses. For the scalar rows ℓ_k of §2.2, before they are tensored with $1_{\mathfrak{g}}$, the two steps give $\ell_k \in \mathcal{R}(b_*, K_*, \lambda_*, 1/9)$, then (N′) with eq. 2.vii at every cut, and finally $\mathtt{t}_\rho \leq \frac{1}{8}(\frac{1}{2} - d_0 - c_\Xi)$. All of these are statements on scalar kernels. For a scalar kernel K, the kernel $K \otimes 1_{\mathfrak{g}}$ has the same norms as K, by item C.II 2.2, "Source and hypotheses", of §3; hence these statements hold for the rows $\ell \otimes 1_{\mathfrak{g}}$ as well. Therefore $\ell_k \in \mathcal{R}(b_*, K_*, \lambda_*, 1/9)$, and the pair $(\ell_k, (\ell_k)')$ has the four conclusions named in (SD), with (L1) and (L2) in the forms (L1^{su}) and (L2^{su}) by (O4). Together with items 2 and 3, this is (SD) at step k; it rests on the conclusion of Thm. 2.65, namely on (Fr0), (Fr2), (Fr3′) and (Fr4), at every k with $x_{k+1} \geq x_\flat$. Thm. 4.39 now carries $\rho_k \in C_\flat^{\mathrm{su}}(x_k; \ell_k)$ over to $\rho_{k+1} \in C_\flat^{\mathrm{su}}(x_{k+1}; \ell_{k+1})$.

Item 6: the members. The proof of Thm. 4.39 constructs the members of ρ_{k+1} from those of ρ_k by (F1) to (F4). □

5.3. **The complex step.** The cutoff comparison needs a step defined on complex members with only a weak cap on the rough term. We use a different test χ_X from the real step, which gives a different representation of the same coarse density. Thm. 5.15 proves holomorphic dependence and restores the strong cap at the output. Cauchy's estimate then contracts the weak distance between inputs into the strong distance between outputs.

Remark 5.9 (Item 0: signs). The signs $D(U)$ and D' are sets of rough plaquettes. We call a map G-valued when its values lie in SU(2). Throughout Step 0 (i) of the proof of Thm. 5.15, $c > 0$ denotes a number determined by the letters (O0) to (O12) and independent of x, and the value of c may change from one occurrence to the next. On $\mathfrak{g}_{\mathbb{C}}$ we write $|\cdot|$ for the operator norm and $\|\cdot\|$ for the norm of $\mathbb{C}^3$; these obey $\|X\| \leq |X| \leq \sqrt{2}\,\|X\|$, and $|X| = \|X\|$ on $\mathfrak{g}$.

Definition 5.10 (Item 1: the letters, in the order of the table)**.** The letters are those of the table of item 3 of §2.1, in its order (O0) to (O13).

(O0) The pure numbers are $b_*, K_*, \lambda_*, L_A, L_B, c_1^*, c_2^*, \lambda_2^*, \theta_J$. All of them but L_B are fixed in Thm. 2.65, in item (0) of its part "The order of the constants", and L_B is the pure number defined in item 1, (O0), of §5.4. We set $\mathtt{g}(\lambda) = \sum_{n \in \mathbb{Z}^3} e^{-\lambda |n|_\infty}$ and $\mathtt{g}_* = \mathtt{g}(\lambda_*)$.

(O1) We fix $d_0 \in (0, \frac{1}{2})$.

(O2) We fix c_Ξ with $0 < c_\Xi \leq \frac{1}{2}(\frac{1}{2} - d_0)$.

(O3) Let ρ_0 be the least integer ≥ 0 such that

$$t_\rho := 3K_*^2\, \mathtt{g}(\lambda_*)\, \mathtt{g}(\lambda_*/2)\, e^{-\lambda_* \rho_0/2} \leq \frac{1}{8}(\frac{1}{2} - d_0 - c_\Xi);$$

this is the display eq. 2.vi of §2.2.

Item E5 of §4.3, the data (SD) in item 1 of §2.1, and §2.2 denote by $\mathtt{t}_\rho$ a different object, a quantity of the rows, namely

$$\|\ell'^{\mathrm{c}}\|_{\mathrm{r}} \sup_{P_0} \sum_{y' \colon \mathrm{dist}(y', P_0) > \rho_0} |\ell'^{\mathrm{c}}_{y'}(P_0)|.$$

For the rows of (SD) that quantity is at most $3K_* \mathtt{g}(\lambda_*) \cdot K_* \sum_{|u|_\infty > \rho_0} e^{-\lambda_* |u|_\infty}$, and this in turn is at most the number t_ρ above. The comparison rests on two facts. By Thm. 2.57 (a), $|\ell'_{y'}(P)| \leq K_* e^{-\lambda_* \mathrm{dist}(y', P)}$ for every $\ell \in \mathcal{R}(b_*, K_*, \lambda_*, \frac{1}{9})$, the empty rows included, where

$K = K_*$ is allowed because $L \geq L_A$; and the sum is bounded by the estimate of t_ρ in the proof of Thm. 2.64.

We put $\mathsf{N}_Z = \mathsf{N}_\mathsf{B} = 1 + 3K_*^2\mathsf{g}_*^2$ and $\mathsf{P}_\rho = 1 + 3(2 + 3(2\rho_0 + 1)^3)K_*^2\mathsf{g}_*^2$. The number P_ρ is fixed by the entries (O0) to (O3), while the numbers $\mathsf{l}_L = 1 + \log L$, $\mathsf{N}_\sharp = c_1^* \mathsf{l}_L \mathsf{N}_Z \mathsf{P}_\rho$ and $\mathsf{N}_\mathrm{c} = 17(\mathsf{N}_\sharp + 1)$ are functions of L; since L is chosen in (O4) by inequalities in these functions, no circle arises.

(O4) L is odd and $L \geq L_*^{\mathrm{su}}$, where L_*^{su} is the least integer $\geq \max\{L_A, L_B\}$ such that ($\mathrm{L1}^{\mathrm{su}}$), ($\mathrm{L1}^\flat$) at $\frac{3}{16}d_0$ and ($\mathrm{L2}^{\mathrm{su}}$) hold for all values of L from L_*^{su} on (item 3, (O4), of §2.1). The condition ($\mathrm{L2}^{\mathrm{su}}$) is the inequality $L^{1/2} \geq 1.2 \cdot 10^4 \mathsf{N}_\mathrm{c}^2$. As $\mathsf{N}_\mathrm{c} = 17(\mathsf{N}_\sharp + 1) \geq 17$, it gives $L \geq (1.2 \cdot 10^4 \cdot 17^2)^2 > 1.2 \cdot 10^{13}$. The number $c_L = (L^2 - 1)/(6L^2)$ depends on L, and $a \geq 3/(c_L L)$; as $c_L < \frac{1}{6}$, this yields $aL \geq 18$.

(O5) $\kappa \geq 6$, and $\kappa^\sharp = 10(\kappa + 4)$.

(O6) This entry consists of $\gamma = (73L^4)^{-1}$, of $b = 16(1 + 3K_*^2\mathsf{g}_*^2) + a/L$, of the numbers $\theta_\mathbb{C}$ and $\vartheta_0^\flat$ of item C.I 1.2 of §3, taken at K_*, λ_*, of $\theta_D = \frac{1}{2}\min\{\lambda_2^*, \theta_\mathbb{C}, \theta_J\}$, of C_D, and of $c_R = 64/\theta_D$, so that $c_R\theta_\mathbb{C} \geq 128$. As $L \geq L_A$ yields $\lambda_* L \geq 12$, the condition ($\mathrm{R}^\flat$) $c_R\mu \geq 1.35$ of Thm. 3.40, Thm. 3.42 and Cor. 3.43 follows (item C.II 5 of §3, under "Conditions on the letters"; item 3, (O6), of §2.1).

(O7) $M = L^{m'}$ with $m' \geq 1$, subject to (M2) $M \geq 1920(\kappa + 4)c_R$, to ($\mathrm{M4}^\flat$) (item C.II 5 of §3, under "Conditions on the letters") and to (M5r) $M \geq \rho_0$. The condition ($\mathrm{M4}^\sharp$) is a consequence of ($\mathrm{M4}^\flat$). We put $v := (LM)^3$.

(O8) $\mathfrak{p}_0 > 2$; further $A_0 > 0$. We set $p_0(x) = A_0(\log x)^{\mathfrak{p}_0}$, $\varepsilon = p_0 x^{-1/2}$ and $x' = x/L$; a prime on a letter means that it is taken at x'; and $\varepsilon' \leq L^{1/2}\varepsilon$.

(O9) $\mathsf{G}_0 \leq C_\varphi^\flat$ (item C.I 1.2 of §3). These two numbers are already determined at (O6), since they depend on (L, a, K_*, λ_*) alone. Further, we set $\mathsf{n} = 60$ and $c_\xi = (6.5\, C_\varphi^\flat)^{-1}$.

(O10) Thm. 3.32 and Cor. 3.38 supply a_c (item C.II 4 of §3, under "Constants") and $\mathsf{K}_\mathrm{c} := \sqrt{2}\,\mathsf{K}_\mathrm{c}^0$. Here K_c is the constant of (Cen) in Cor. 3.38 (a), where a single constant suffices for every $0 < \hat{\varepsilon} \leq \frac{1}{40}$. Thm. 3.40, Thm. 3.42 and Cor. 3.43 supply $\mathsf{K}_1^\flat$; the numbers $K_1^\sharp := 2\mathsf{K}_1^\flat$ and $K_2^\sharp := 3\mathsf{K}_1^\flat$ bound D_k, and $\tilde{\mathsf{D}} = -\mathsf{D}_k/2$ obeys the halves of these bounds. Thm. 4.2 supplies $K_{\mathrm{q},k}$, Thm. 4.5 supplies $K_v^\flat$, and Thm. 4.19, through item F of §4.3, supplies $u_b, u_t, u_1, \delta_3, \delta_3', e_1^\flat, \mathrm{Res}$. Formed in this section are $T := L^{3/2}/(19\mathsf{N}_\mathrm{c})$, $a_\infty := 4\pi\mathsf{N}_\mathsf{B}(C_\varphi^\flat)^2$ and $n_a := \frac{3}{2}\, v\, a_\infty^2$.

(O11) $0 < c_h \leq \delta_3'/2$. As $\delta_3' \leq \delta_3$, it follows that $c_h \leq \delta_3/2$.

(O12) $E_a \geq \max\{32K_{\mathrm{q},k},\ 1500\, K_v^\flat L^{-1/2}\}$.

(O13) $x_9 := 1 + \inf S_9$, where S_9 is the set of the $y \geq x_{\mathrm{fl}}$ (item G of §4.3) such that (Y.0) to (Y.9) of item 5 hold for all $x \geq y$. The set S_9 is an upper half line, and it is not empty by Step 0 (i) of the proof of Thm. 5.15. The floors (Y.0) to (Y.9) involve numbers rounded up to an integer, such as the radii R, ϱ_L, $r(x)$, r' and r_1 of item 2, and $\lceil s(x)/L \rceil$ and $\lceil r(x)/L \rceil$. As these numbers jump as functions of x, S_9 need not contain its infimum. But $1 + \inf S_9 \in S_9$, and this number depends on what S_9 depends on. The floor of Thm. 5.3 and that of item 3, (O13), of §2.1 have the same form. The table requires $Lx_\flat \geq x_9$.

The caps are $c_a(x) = E_a p_0^3 x^{-1/2}$ and $c_h(x) = e^{-c_h p_0^2}$.

No constant depends on A, u, z or N, nor on more of ℓ than the pure numbers b_*, K_*, λ_* and its range. ∎

Definition 5.11 (Item 2: the data (SD) of a step, and the radii)**.** The data (SD) of a step begin with the torus: $x \geq x_9$; N is even, $M \mid N$, and (Tor) $N \geq 2Ms(x)$, with N the coarse side; the fine torus is $\mathbb{Z}^3/NL\mathbb{Z}^3$.

The lattices, the blocks, the base sites, the coarse distance, the cubes, the cells $\square$ and $\hat{\square}$ (the fine set under $\square$), and $\mathcal{D}$, $\mathcal{P}$, $|X|_c$ and X^{+j} are those of item 1 of §2.1 and of item 3.1 of §4.2. The radii in force are those defined below.

The grids of cubes are fitted as in item 1 of §2.1, under "Cubes", so that the fine set $\hat{\square}$ under each cell $\square$ consists of exactly L^3 fine cubes. An object is attached to the cube, or the cell, of its base site. We say that X *holds* p if p belongs to a cube of X.

The bonds, paths and plaquettes *of* a set A of cubes, cells or sites are those all of whose sites lie in A (item 1 of §2.1; item 3.3 of §4.2). A plaquette p is *in* A if its base site lies in A; this is the sense of "in" in 1_A and in $D(V) \cap X$, and also of "based in" and "belongs to". A field on the bonds of A is *tame there* if $\phi_p \leq \varepsilon$ at the plaquettes of A.

For the rows, (SD) requires that $\ell \in \mathcal{R}(b_*, K_*, \lambda_*, \frac{1}{9})$ have range $\leq \varrho(x)$ and that (ℓ, ℓ') satisfy the four conclusions of Thm. 2.64. These are listed in (SD) in item 1 of §2.1. Here ℓ' is the family of rows that Prop. 2.28 forms from ℓ, so that $\mathsf{B}' = 1 + \ell'^{\mathsf{T}} \ell'$, tensored with $1_{\mathfrak{g}}$; at $\ell = \emptyset$ its group (5) is void. Further, ℓ'^{c} is the cut of ℓ' at $\varrho(x')$.

Empty rows are admitted. The rows $\ell = \emptyset$, with $\mathsf{H}_y = \{0\}$ for all y, lie in $\mathcal{R}(1, 0, \lambda_*, 0) \subset \mathcal{R}(b_*, K_*, \lambda_*, \frac{1}{9})$ (by the display and the last sentence of Def. 2.48) and have range 0. For them $\mathrm{En}_\ell = \sum_p S_\varepsilon(\phi_p)$ and $\mathsf{B}^V = 1$, while the numbers and the radii remain the ones evaluated at $(K_*, \lambda_*, \varrho(x))$. Neither a level nor a depth enters these data.

For $\ell = \emptyset$ the hypotheses of Thm. 5.15, that is, the data (SD) and the clauses (Γ2) to (Γ7), are met as follows.

(1) Thm. 5.15 is stated for every choice of the data (SD), hence for $(\emptyset, \emptyset')$, which satisfies the four conclusions that (SD) requires. Of these, (1), (2) and (3) follow from Thm. 2.65 at $k = 0$ and $x_0 = x$, namely from (Fr2) for the sixteen kernels, (Fr3′) and (Fr4). At these values, $x/L \geq 2$ gives (X12). Conclusion (4) consists of (L1^{su}) and (L2^{su}) and holds by (O4). Step S6 of the proof of Thm. 5.3 treats this case.

(2) Each of the clauses (Γ2) to (Γ7) admits $\ell = \emptyset$ within the data of the section to which it belongs: in items C.I 1.1 and C.II 1.2 of §3, with $K \geq 0$; in §4.1, item B and item I, Remark (4); in §4.2, item 2; and in §4.3, item C.

(3) Besides the reason (2), a second argument runs as follows. By Rem. 5.6 in Step S6 of the proof of Thm. 5.3, the objects that (SD) and the classes attach to the rows are the same at the empty rows as at the zero rows whose H_y is the real line; and the zero rows fall under the hypothesis of those sections as a datum with rows.

(4) (Γ3) is stated in the setting (Fr1) to (Fr3). For $(\emptyset, \emptyset')$ these conditions follow from (SD), by the last sentence of (SD) in item 1 of §2.1.

At $\ell = \emptyset$, Steps 0 to 12 of the proof of Thm. 5.15 use ℓ only through (i) to (iv) below.

(i) $\mathrm{En}_\ell \geq 0$ and B^V, in items 1, 3 and 4 of Step 1 of that proof. For $\ell = \emptyset$ we have $\mathrm{En}_\ell = \sum_p S_\varepsilon(\phi_p)$, $\mathsf{B}^V = 1$ and $|\ell^V F^\flat|^2 = 0$, and the identity stays as it is.

(ii) ζ_p, ℓ_y and the mask m_y, in Step 2 (iv) of that proof. For $\ell = \emptyset$ there is no such object, and $A_\square = e^{V_\square}$ holds factor by factor.

(iii) The range $\leq \varrho(x)$, in the dependence of $[A]^\ell_3$ in Step 3 (V) of that proof. For $\ell = \emptyset$ the range is $0 \leq \varrho(x)$, so $[A]^\ell_3$ depends on no more than what Step 3 (V) states.

(iv) $\|\ell^{V*} \ell^V\|_\infty \leq 3 K_*^2 \mathsf{g}_*^2$, in Step 6, (t), of that proof. For $\ell = \emptyset$ this norm is 0.

The family $\ell' = \emptyset'$ is not empty. The steps need two things from it: the energy $\mathrm{En}' = \mathrm{En}_{\ell'^{\mathrm{c}}}$, through (5a), and the inequality $\mathsf{t}_\rho \leq t_\rho$ for $(\emptyset')^{\mathrm{c}}$, in which t_ρ is the number of (O3) and which holds by (O3), Thm. 2.57 (a) and the proof of Thm. 2.64.

For $\ell = \emptyset$ the letters are those of the table: $E_a \geq \max\{32 K_{\mathrm{q},k}, 1500 K_v^\flat L^{-1/2}\}$ of (O12), and not merely $E_a \geq E_a^0$ of Thm. 5.3; $c_h \leq \delta_3'/2$ of (O11); and $x \geq x_9$, not merely $x \geq L x_\flat^{(1)}$.

For $\ell = \emptyset$ there are two systems. Let u_W be the member of Wilson's weight, so that $e^{-x\sum_p(1-\cos\phi_p)} = \rho[u_W;\emptyset]$ by Lem. 5.2 (a). The member $F_\emptyset(u_W)$ need not coincide with the system u' of Thm. 5.3, which (F1) to (F4) of Thm. 4.39 construct; each of the two represents the image of Wilson's weight under the step (see item 8).

A pair (V', e) consists of a coarse field V' and of $e \in \mathfrak{E} = \prod_P S^2$, where S^2 is the unit sphere of $(\mathfrak{g}, \|\cdot\|)$, one copy for each coarse plaquette P. The measure de is the product, taken over all P, of the rotation invariant probability measures on the spheres. It has three properties: $\int_{\mathfrak{E}} de = 1$; the e_P are independent under de; and for every G-valued u the map $e \mapsto \mathrm{Ad}_u\, e$, given by $(\mathrm{Ad}_u\, e)_P = \mathrm{Ad}_{u(y_P)}\, e_P$ with y_P the base site of P, leaves de invariant, since every $\mathrm{Ad}_{u(y_P)}$ acts as a rotation on the three-dimensional Euclidean space $(\mathfrak{g}, \|\cdot\|)$. The mean over e is the integral against de.

For a pair we put $X'_P = \mathsf{L}(\partial V'(P)) + \pi e_P 1\{\partial V'(P) = -1\}$; then $|X'_P| = \phi'_P \le \pi$.

The step acts on a bounded Borel function ρ by

$$\rho'(V') = \int \rho(U) \prod_c k_{ax}(V'(c), \mathsf{M}(U)(c))\, dU$$

(item C.I 4 of §3, under "The step, the chart"; item 3.4 of §4.2), and on a complex ρ by linearity.

The radii in coarse units are $R = \varrho(x) = \lceil c_R \log x \rceil$, $\varrho_L = \lceil R/L \rceil$, $r_w = \lfloor R/2 \rfloor$, $r_G = r_w + \varrho_L + 1$, $R_2 = 2R + 2$, $R_3 = 2R + \varrho_L + 5$, $R_\chi = R + \varrho_L + 2$ and $R_v = 2R + 2\varrho_L + 7$. Further, we put $\ell_w(x) = 6Mr(x) + 12$, $D_w = \lceil \ell_w(x)/L \rceil + 1$ and $\omega = M + \lceil Mr(x)/L \rceil + D_w + R_3 + 2$.

The radii in units of cubes or of cells are $r(x) = \lceil 20 c_R \log x/M \rceil$ and $s(x) = 8r(x)$. We write $X^{\approx} = X^{+r(x)}$ and $X^{\approx\approx} = X^{+s(x)}$, we put $r' = r(x')$, $s' = s(x')$ and $s_0 = 2r'$, and for sets of cells we write $Y^{\approx} = Y^{+r'}$. Finally $r_l = \lceil (\varrho_L + 2)/M \rceil$, $r_1 = \lceil R_2/M \rceil + 3r_l + 5$, $\varrho_* = 2r_1 + \lceil s(x)/L \rceil + 2$, $k_t = \lceil (r_w + 1)/M \rceil$ and $k_\chi = \lceil R_\chi/M \rceil$. ∎

Definition 5.12 (Item 3: domains, members, caps, densities and distances)**.** For $X \in \mathcal{D}$, the domain $\mathcal{U}_X(x)$ (item 1 of §2.1, under "Domains") consists of the maps V that send the oriented bonds of $X^{\approx}$ into $\mathrm{SL}(2,\mathbb{C})$, with $\mathsf{V}(\bar b) = \mathsf{V}(b)^{-1}$, such that the following hold.

(u1) $|\mathsf{V}(b)| < 1 + \vartheta(x)$, where $\vartheta(x) = \vartheta_0^\flat(1 - 1/\log x)$.
(u2) $|\mathsf{V}(\pi)| < \mathsf{b}(x) = 2 - 1/\log x$ for every path π in $X^{\approx}$ of at most $\ell_w(x)$ bonds.
(u3) $|\partial\mathsf{V}(p) - 1| < 2\varepsilon$.

For a set Y of cells, the domain $\mathcal{U}_Y(x')$ is defined in the same way, with x' in the place of x. The fields in $\mathcal{U}_Y(x')$ are extended off $Y^{\approx}$ by the value 1, and X' is then defined by the convention given below for $U^\tau(B; x')$.

For $\tau \ge 1$ and a set B of coarse sites, $U^\tau(B; x')$ (item 3.3 of §4.2) is the set of the $\mathrm{SL}(2,\mathbb{C})$ fields on the bonds of B such that the following hold.

(e1) $|V'(c)| < 1 + \vartheta(x') + \mathsf{g}_x$.
(e2) $|V'(\pi)| < \mathsf{b}(x') + \mathsf{g}_x$ for every path π in B of at most $\ell_w(x')$ bonds.
(e3) $|\partial V'(P) - 1| < 2\tau\varepsilon'$.

Here $\mathsf{g}_x = \vartheta_0^\flat \log L/(4(\log x)^2)$. On such a field we put $X'_P := -i \log \partial V'(P)$ for each P with all four sites in B; the field is extended by 1, and $X'_P := 0$ for every other P, which is the convention of item 3.7 of §4.2.

The quantity $a_{0,b}$ depends on X' within R_3 of $y(b)$ ((Γ3), "The dependence at every pair"). In what follows, $a_{0,b}$ enters only at those bonds b for which the box of radius R_3 about $y(b)$ is contained in B (Step 2 (ii), Step 6 (iii) and Step 8, (8.2), of the proof of Thm. 5.15); hence every value of $a_{0,b}$ that enters is independent of the convention.

We put $1_Y(V) = 1\{\phi_p \le \varepsilon \text{ for all } p \text{ in } Y\}$, and we say that V is *tame on* Y if $1_Y(V) = 1$. The energy is

$$\mathrm{En}_\ell(U) = \sum_p S_\varepsilon(\phi_p) + \frac{1}{2}\sum_y m_y(U)\, |\ell_y^U \zeta|^2,$$

where S_ε, ζ, the dressed rows ℓ_y^U and the mask m_y, of radii ρ_0 and $\varrho(x)+1$, are those of items E1 and E2 of §4.3.

For $A \geq 1$, $M_A^{\mathrm{su}}(x)$ is the set of the triples $u = (E^a, E^h, w)$ with the three properties below.

(ca_A) For $X \in \mathcal{D}$, $E^a(X,\cdot)$ is holomorphic on $\mathcal{U}_X(x)$, invariant under G-valued gauge transformations and 0 at pure gauges, and $\sup_{\mathcal{U}_X(x)} |E^a(X,\cdot)| \leq A\, c_a(x)\, e^{-\kappa(\#X-1)}$.

(ch_A) $E^h(X,\cdot)$ is defined on those G-valued fields on the bonds of $X^{\approx}$ which are tame there; it is complex, Borel and gauge invariant, and $\sup |E^h(X,\cdot)| \leq A\, c_h(x)\, e^{-\kappa(\#X-1)}$, with the supremum over these fields.

($\mathrm{c}\Xi_A$) For $X \in \mathcal{P}$, $w(X,\cdot)$ is complex, Borel and gauge invariant, it is a function of V on the bonds of $X^{\approx\approx}$, and $|w(X,V)| \leq A\, e^{-\kappa(|X|_c-1)}$ at every V; this bound is the *weak cap*.

The *strong cap* is the bound

$$|w(X,V)| \leq \exp\{-c_\Xi x\varepsilon \sum_{p \in D(V)\cap X} \phi_p - \kappa(|X|_c - 1)\}.$$

We call u *real* if E^a is real at G-valued fields and E^h and w are real.

Consider a triple $u = (E^a, E^h, w)$ with those clauses of (ca), (ch) and (cΞ) which carry no size. For it, $\|u\|_x$ is the smallest $A \geq 0$ with which all three size bounds are true, that is, the largest of the three weighted suprema; it is a norm (item 3 of §5.4), and $M_A^{\mathrm{su}}(x) = \{\|u\|_x \leq A\}$. The pure gauges are the fields 1^u with G-valued u.

We put $1E(X)(V) := 1_{X^{\approx}}(V)\,[E^a(X,V) + E^h(X,V)]$ and

$$\rho[u;\ell](V) := \exp\{-x\mathrm{En}_\ell(V) + \sum_{X\in\mathcal{D}} 1E(X)(V)\} \cdot \Xi_w(V),$$

where $\Xi_w(V) = \sum_{\{X_i\}} \prod_i w(X_i,V)$, the sum running over the finite families of sets $X_i \in \mathcal{P}$ of pairwise distance $> s(x)$ such that every X_i holds a plaquette of $D(V)$ and $D(V)$ lies inside their union. The empty family, whose product is 1, is admitted exactly when $D(V)$ is empty. These families are called the *families of* (cΞ).

We put

$$C_\flat^{\mathrm{su}}(x;\ell) = \{e^{\mathsf{V}}\rho[u;\ell] > 0 :\ \mathsf{V} \text{ real},\ u \in M_1^{\mathrm{su}}(x) \text{ real with the strong cap}\}.$$

For two members u_1 and u_2 we put $\mathsf{D}_x^{\mathrm{in}}(u_1,u_2) := \|u_1 - u_2\|_x$, and we let $\mathsf{D}_x(u_1,u_2)$ be the largest of the three numbers

$$\sup_X \sup_{\mathcal{U}_X(x)} |E_1^a - E_2^a|\, e^{\kappa(\#X-1)}/c_a(x),$$

$\sup_X \sup |E_1^h - E_2^h|\, e^{\kappa(\#X-1)}/c_h(x)$, with the inner supremum over the tame G-valued fields, and

$$\sup_{X,V} |w_1 - w_2|(X,V) \exp\{c_\Xi x\varepsilon \sum_{D(V)\cap X} \phi_p + \kappa(|X|_c - 1)\}.$$

Hence $\mathsf{D}_x^{\mathrm{in}} \leq \mathsf{D}_x$: the first two of these numbers are the corresponding weighted suprema of $u_1 - u_2$, and in the third the weight of $|w_1 - w_2|$ is at least the weight $e^{\kappa(|X|_c-1)}$ of the weak cap. ∎

Item 4: what is used of the earlier theorems. The inputs are the following clauses of earlier theorems, each stated in full.

(Γ2) Thm. 3.6 (a), (b), (c), (d), (f), (i) and (k), together with Lem. 3.3 (e). On the set $\mathfrak{V} = \{\mathrm{SL}(2,\mathbb{C})$ fields with $|V'(c)| \leq 1 + \vartheta_0^\flat\}$ there is a matrix $G = G^{V'}$ on I°, the set of fine bonds off the combs, such that G and the matrices C and S obtained from it have the properties (2a) to (2f), while (2g) is a property of g^V.

(2a) The entry $G(b,b')$ vanishes as soon as b' lies farther than r_w from b, and $(G\xi)_b$ depends on ξ only through its values on a window Y_b of bonds within the distance r_w of b.

(2b) The row $G(b,\cdot)$ depends on V' only through the coarse bonds within the distance r_G of $y(b)$. The matrix $C := GG^{\mathsf{T}}$ is symmetric; its entry $C(b,b')$ vanishes as soon as b' lies farther than $2r_w$ from b, where $2r_w \le R$; and its row $C(b,\cdot)$ depends on V' only within the distance $3r_w + \varrho_L + 1$ of $y(b)$, where $3r_w + \varrho_L + 1 \le R_2 - 1$.

(2c) For every G-valued u, the relation $G[V'^u] = \mathrm{Ad}_u\, G[V']\, \mathrm{Ad}_u^{-1}$ holds on all of $\mathfrak{V}$.

(2d) The entries of G are holomorphic in the interior of $\mathfrak{V}$, and they are real at G-valued fields.

(2e) For every ξ and every bond b we have $\|(G\xi)_b\| \le C_\varphi^\flat \sup_{Y_b} \|\xi\|$, and the norms obey $\|G\|_\infty, \|G^{\mathsf{T}}\|_\infty \le \mathsf{G}_0 \le C_\varphi^\flat$.

(2f) The matrix G is invertible, and $S := C^{-1} - q_\ell^V$ is symmetric.

(2g) We recall Lem. 3.3 (e). It expresses $(\mathrm{g}^V X')_{b'} = \sum_P \sum_{\mathfrak{t}} g_{\mathfrak{t}}\, \mathrm{Ad}_{V'(\pi_{\mathfrak{t}})}\, X'_P$ as a finite sum in which the numbers $g_{\mathfrak{t}}$ are independent of every field and each $\pi_{\mathfrak{t}}$ is a coarse path that runs from $y(b')$ to the base site of P and keeps all its sites within the distance $2\varrho_L + 1$ of $y(b')$. Consequently, at every field of $\mathfrak{V}$ and for every $\mathfrak{g}_{\mathbb{C}}$-valued X', the value $(\mathrm{g}^V X')_{b'}$ depends on V' only through its values on bonds, and on X' only through its values on plaquettes, all of whose sites lie within the distance $2\varrho_L + 2$ of $y(b')$.

(Γ3) Thm. 3.32 (a) and (c), together with Cor. 3.38 (a), which is (Cen). We put $a_0 := C\mathrm{g}^V X'$.

The two bounds. Let p be a fine plaquette, let b be a bond of p, let $0 < \hat\varepsilon \le \frac{1}{40}$, and take an $\mathrm{SL}(2,\mathbb{C})$ field on the box of radius R_3 about $y(p)$ that satisfies (e1), (e2) and $|\partial V'(P) - 1| \le 2\hat\varepsilon$. Then

$$|a_{0,b}| \le \mathsf{a}_{\mathrm{c}}\hat\varepsilon \qquad \text{and} \qquad |(EX' - \mathrm{d}^V a_0)_p| \le \mathsf{N}_{\mathrm{c}} L^{-2}\hat\varepsilon + \mathsf{K}_{\mathrm{c}}\hat\varepsilon^2.$$

Both quantities depend on (V', e) only through its values on that box. Thm. 3.32 (a) (item C.II 4 of §3) is a clause on the dependence only and contains no bound, whereas Cor. 3.38 (a) states the two bounds alone; for $a_{0,b}$, where $\mathrm{dist}(b, y(p)) \le 1$, the dependence just stated is what that clause asserts, and for $(EX' - \mathrm{d}^V a_0)_p$ it is derived below from the values $a_{0,b}$ at the four bonds of p. The bound $|a_{0,b}| \le \mathsf{a}_{\mathrm{c}}\hat\varepsilon$ is Thm. 3.32 (c) as well as part of Cor. 3.38 (a); the bound on $|(EX' - \mathrm{d}^V a_0)_p|$ comes from that corollary alone.

Only the clause on the dependence, Thm. 3.32 (a), is free of the conditions (h2), (h3) and (Fr2); the two bounds are not. Suppose that the field is given on a set containing the box and that X' is defined on the plaquettes of that set. Then at each bond b with $\mathrm{dist}(b, y(p)) \le 1$, in particular at the four bonds of p, the value $a_{0,b}$ coincides with the value computed on the box. The same is true of $(EX' - \mathrm{d}^V a_0)_p$: either $(EX')_p = X'_P$ for a P based at $y(p)$ or $(EX')_p = 0$, and $(\mathrm{d}^V a_0)_p$ is built from these four values $a_{0,b}$ by transports along no more than one coarse bond from $y(p)$ (Step (0) of the proof of Thm. 3.32). For this reason the two bounds can be applied where the field is given on a larger set.

The dependence at every pair. At every field of $\mathfrak{V}$ and for every $\mathfrak{g}_{\mathbb{C}}$-valued X', hence at every pair (V', e), rough plaquettes included, the value $a_{0,b}$ depends on V' only through its values on bonds, and on X' only through its values on plaquettes, all of whose sites lie within the distance R_3 of $y(b)$, and it depends on nothing else.

Proof of the dependence at every pair. We have $a_{0,b} = \sum_{b'} C(b,b')(\mathrm{g}^V X')_{b'}$. By (2b), $C(b,b') = 0$ as soon as b' lies farther than $2r_w$ from b, and $2r_w \le R$; moreover the row $C(b,\cdot)$ depends on V' only within the distance $R_2 - 1$ of $y(b)$. By (2g), each value $(\mathrm{g}^V X')_{b'}$ depends on V' and on X' only within the distance $2\varrho_L + 2$ of $y(b')$. Combining the two facts, all sites that enter lie within the distance $\max\{R_2 - 1,\ R + 2\varrho_L + 2\} \le R_3$ of $y(b)$, where the last inequality holds because $\varrho_L \le R + 3$. If, in addition, $\mathrm{dist}(b, o) \le 1$ for a coarse site o, as is the case for a bond b of p and $o = y(p)$, then all these sites also lie within the distance $\max\{R_2,\ R + 2\varrho_L + 3\} \le R_3$ of o, the last inequality now holding because $\varrho_L \le R + 2$. □

(Γ4) Thm. 3.40 (b), Thm. 3.42 (a), (b) and (c), and Cor. 3.43, all taken at $\mathsf{n} = 60$. Summing over the connected sets Y of cells, we have $\log\det C^1 - \log\det C^{V'} = \sum_Y \mathsf{D}_k(Y, V')$ at every V', and $S = \sum_Y S_k(Y; V')$ on $\mathfrak{V}$.

(4a) The function $\mathsf{D}_k(Y, \cdot)$ is real and gauge invariant, it vanishes at pure gauges, and $|\mathsf{D}_k| \le K_1^\sharp e^{-\kappa^\sharp(\#Y-1)}$ at every V'; it has a holomorphic continuation to $\mathcal{U}_Y(x')$, on which $|\mathsf{D}_k| \le K_2^\sharp \varepsilon' e^{-\kappa^\sharp(\#Y-1)}$.
By definition (Def. 3.41, the display before Thm. 3.42), $\mathsf{D}_k(Y, V') = -\mathsf{D}^q(Y, V') - [\mathsf{D}^u(Y, V') - \mathsf{D}^u(Y, 1)]$; the continuation is given by the same formulas, evaluated at the element of $\mathcal{U}_Y(x')$ extended by 1 (proof of Cor. 3.43).
The function $\mathsf{D}^q(Y, \cdot)$ is holomorphic on the open set

$$\{|V'(c)| < 1 + \vartheta_0^\flat \text{ on the bonds of } Y^{+1}\}$$

(Thm. 3.40 (b)), and $\mathsf{D}^u(Y, \cdot)$ is holomorphic in the interior of $\mathfrak{V}$ (Thm. 3.42 (b)); each of these two open sets contains all G-valued fields. It follows that $\mathsf{D}_k(Y, \cdot)$ is continuous on the G-valued fields. On $\mathcal{U}_Y(x')$ the function is invariant under G-valued gauge transformations, by Thm. 3.40 (b) and Thm. 3.42 (b); the argument is given in Step 8 of the proof of Thm. 5.15, paragraph "The clauses without a size".

(4b) We have $S_k(Y)(b, b') = 0$ unless both b and b' lie in $\hat{Y}$. The matrix $S_k(Y)$ is symmetric, covariant as in (2c), holomorphic and real at G-valued fields, and $\|S_k(Y; V')\| \le x^{-57} e^{-\kappa^\sharp \# Y}$ at every field of $\mathfrak{V}$.

(4c) Both $\mathsf{D}_k(Y, \cdot)$ and $S_k(Y; \cdot)$ depend on V' only through its values on the bonds of $Y^{+\lceil R_2/M \rceil}$. In §3 the number $\lceil R_2/M \rceil$ is written r_u.

We put $\tilde{\mathsf{D}} := -\mathsf{D}_k/2$ and $e_Y := x^{-40} e^{-\kappa^\sharp \# Y}$.

(Γ5) Thm. 4.2 (b), (d) and (e): below, (5a) is its clause (e), (5b) its clause (b) and (5c) its clause (d).

(5a) At every pair (V', e),

$$x\,\mathsf{m}_C - \frac{x}{2}\langle a_0, S a_0\rangle = x' \mathrm{En}'(V') + \sum_j w_j + \sum_{\square} 1_{\square \approx}(V')\, P(\{\square\}, V').$$

Here $\mathsf{m}_C = \frac{1}{2}\langle EX', \mathsf{B}^V EX'\rangle - \frac{1}{2}\langle \mathsf{g}^V X', C\mathsf{g}^V X'\rangle$; the term $\mathrm{En}' = \mathrm{En}_{\ell^c}$ is taken at x'; and $w_j = \sum_{\square \in \mathrm{mk}_j'^{+r'}} T_\square$, where $T_\square$ is that of item E7 of §4.3 and the classes are those of item E9 of §4.3.

(5b) The function $T_\square$ is real, jointly Borel and invariant under $(V'^u, \mathrm{Ad}_u\, e)$, and it depends on (V', e) only within the distance $Mr' - 1$ of $\square$.

(5c) The function $P(\{\square\}, \cdot)$ is holomorphic on $\mathcal{U}_{\{\square\}}(x')$, invariant, real at G-valued fields and zero at pure gauges, and its supremum does not exceed $K_{\mathrm{q},k}\, p_0'^2 \varepsilon'$. For $\#Y \ge 2$ we put $P(Y, \cdot) := 0$.

(Γ6) Thm. 4.5 (a), (b), (c), (d), (e.iii), (e.iv) and (e.v). Here $V_\square(\xi, V')$ is the vertex term defined by the display of item 3.8 of §4.2, with the law of Thm. 3.6, and $\chi_\square(\xi) = 1$ if and only if $|\xi_b| \le c_\xi \varepsilon$ at all $b \in I^\circ$ within the distance R_χ of $\hat{\square}$.

Let $V' \in \mathcal{U}_{\{\square\}}(x')$, and let ξ be real with $\chi_\square = 1$. Then $V_\square$ is holomorphic in V'; it depends on ξ only within the distance $r_w + \varrho_L + 1$ of $\hat{\square}$ and on V' only within the distance R_v of $\square$; it is independent of every member and of e; it is invariant under $(\mathrm{Ad}_u\, \xi, V'^u)$; it satisfies $|V_\square| \le K_v^\flat p_0^3 x^{-1/2}$; and it is real at G-valued V'.

(6e) Thm. 4.5 (e.iii), (e.iv) and (e.v) state the following for the $A_\square$ of item 6 of this section (item E11 of §4.3; the same product appears in item 3.9 of §4.2). At every real ξ and every pair (V', e), tame or rough, whether the tests are passed or failed, $A_\square$ depends on ξ only through its values on the bonds of I° within the distance $r_w + \varrho_L + 2$ of $\square$, and on (V', e) only through its values on bonds and plaquettes within the distance R_v

of $\square$; moreover $A_\square(\mathrm{Ad}_u\,\xi, V'^u, \mathrm{Ad}_u\,e) = A_\square(\xi, V', e)$ for G-valued u, and $A_\square$ is Borel in (ξ, V', e).

(Γ7) Thm. 4.19. From §4.3 we take item E (the objects), item F (the numbers) and item G (the floor $x_{\mathfrak{fl}}$), and the clauses (7a), (7d) and (7e).

(7a) Of clause (a) of Thm. 4.19 we use one assertion alone: $\hat\lambda_{(\mathsf{J},t)}$ and $\hat\lambda_{(\mathsf{J},t)}\cdot 1_{\mathrm{ind}}$ depend on ξ only through its values on the bonds of I° under $\mathfrak{b}^{+2r_1}$. The set of classes of a cell datum, written J in Sections 2.1, 4.3 and 4.4, is written J in this section, where J is the core of a letter.

(7d) We state clause (d°) of Thm. 4.19. Its data consist of cell data (J_i, t_i) that are admitted and pairwise compatible; of a set J_b of finitely many cells (in Thm. 4.19 this set is written J_0), disjoint from every $\mathfrak{b}_i^{+2r_1}$; of a finite set Y of cells, with $|\varphi_{\mathsf{Y}}|^2$ defined as in item E1 of §4.3, that is, by summing $|(G\xi)_b|^2$ over all bonds b of I° in $\hat{\mathsf{Y}}$; of a number ς with $0 \le \varsigma \le \frac{\gamma}{2}x^{-2}$; and of an arbitrary pair (V', e). For such data,

$$E_\xi[\prod_i \hat\lambda_{(\mathsf{J}_i,t_i)}\cdot\prod_{J_b} b_\square\cdot e^{\frac{x}{2}\varsigma|\varphi_{\mathsf{Y}}|^2}] \le \exp\{4.8\,\upsilon\,\frac{2}{\gamma}\,\varsigma\,(2r_1+1)^3\#\mathsf{Y} + x^{-1/8}(\sum_i \#\mathfrak{b}_i^{+2r_1} + \#J_b)\}\cdot u_b^{\#J_b}\cdot\prod_i e^\circ_{(\mathsf{J}_i,t_i)}.$$

The sizes in this bound are $e^\circ_{(\mathsf{J},t)} = \mathrm{Res}_{(\mathsf{J},t)}\cdot u_t^{\#t}\cdot u_1^{1[t\text{ non-empty}]}$ and $\mathrm{Res} = \exp\{-(\frac12 - d_0 - t_\rho - 3e_1^\flat)\,x'\varepsilon'\sum_{\mathsf{J}}\sum_{D'_j}\phi'_P\}$, defined with the number t_ρ of (O3). The quantity Res of Thm. 4.19, defined with the t_ρ of the rows, is at most this one. Hence Thm. 4.19 (d°) implies the bound above in the form given here, and each of r_Ξ, υ_f, υ_* and x_9 depends on the rows only through their range and the numbers b_*, K_*, λ_*.

(7e) By clause (e) of Thm. 4.19, $u_1 = e^{-\delta_3 p_0^2}$.

Tools. The proof of Thm. 5.15 uses the following tools: the formula of the chart, Lem. 3.9, in item C.I 4 of §3; Lem. 5.17 and Thm. 5.18, in Step 2 of the proof of Thm. 5.15; Thm. 5.22, in Step 8, (8.2), of the proof of Thm. 5.15; Lem. 4.50, together with Lem. 4.48, and Lem. 4.51, in Step 8 of the proof of Thm. 4.39; and Lem. 5.23, in Step 10 (9) of the proof of Thm. 5.15.

From item 1 of §2.1 the proof takes the set $Y_0(X)$ and the notion "small", the grids, and two bounds: a given cube lies in at most $(26e)^{n-1}$ members of $\mathcal{D}$ with n cubes, and in at most $(52e)^n$ members of $\mathcal{P}$ with $|X|_c = n$.

Definition 5.13 (Item 5: the numbers defined here and the floor (Y*))**.** We put $t = x^{-1/4}$, $\varepsilon_2 = x^{-1/8}$, $\varepsilon'' = x^{-1/4}$, $\varsigma'' = 1.01\,x^{-20}$, $d_\alpha = (\kappa+1)(\#J_\alpha + s')$ and $r_T := \sqrt{2}\,C_\varphi^\flat c_\xi\varepsilon + 17T\mathsf{a}_{\mathrm{c}}\varepsilon'$. Further, $r_\Xi = \frac12 - d_0 - t_\rho - 3e_1^\flat - c_\Xi$,

$$\upsilon_f = \exp\{-r_\Xi p_0'^2 + 1.01\,K_1^\sharp(2r'+1)^3 + 4 + (\kappa+4)(1+2s')\},$$

and $\upsilon_* = (\max\{u_t^{1/2}, \upsilon_f^{1/4}\})^{1/L^3}$.

The floor (Y*) consists of the following conditions (Y.0) to (Y.9) on x.

(Y.0) This condition is made up of (0a) to (0e).

(0a) It contains the floors (S1) and (S2) of item C.II 5 of §3, paragraph "Floors", with $r_u = \lceil R_2/M\rceil$ as there, and the floors of Thm. 4.2 and of Thm. 4.5, as item G of §4.1 and item 6 of §4.2 list them.

The only constraint that Sections 4.1 and 4.2 put on their letter $x_\flat$ is that the floors of the section hold for all $x \ge Lx_\flat$, and §4.1 asks in addition that $x_\flat \ge e^2$. We therefore apply Thm. 4.2 and Thm. 4.5 with x_9/L in the role of their $x_\flat$; here $x_9/L \ge x_{\mathfrak{fl}}/L \ge L \ge e^2$, because $x_{\mathfrak{fl}} \ge L^2$ (item G of §4.3).

The condition $x \ge x_{\mathfrak{fl}}$ comes from (O13) of item 1 of this section: there S_9 is the set of the $y \ge x_{\mathfrak{fl}}$, with $x_{\mathfrak{fl}}$ as in item G of §4.3, for which (Y.0) to (Y.9) are satisfied at every $x \ge y$, and $x_9 = 1 + \inf S_9 \in S_9$. So, for $x \ge x_9$, the condition $x \ge x_{\mathfrak{fl}}$ holds in addition to (Y.0) to (Y.9). The condition $x \ge x_{\mathfrak{fl}}$ entails (G1$^\flat$), which gives $R \ge 6$. The same

condition also entails ($\mathrm{F_{law}}$), the clauses $x \geq L^2$ and $R \geq 6$ of (P0), the conditions (P1) and (P3) of item C.II 4 of §3, the bound $\varepsilon' \leq \frac{1}{40}$, and (W).
The third clause of (P0), as item C.II 4 of §3, paragraph "Floors", defines it, is the clause (Tor). This clause depends on N; it is a datum of (SD) and not a floor. In the other sections whose floors enter the definition of S_9, by way of (0a) or of $x_{\mathfrak{fl}}$, the clause (Tor) is likewise among the data and not among the floors: in Step 14 of the proof of Thm. 4.5, in item G of §4.3 and in item G of §4.1. In the last of these, (P0) consists of the two clauses $x \geq L^2$ and $R \geq 6$. The condition (W0) of Thm. 4.5 contains both of them and no other clause of (P0). Consequently, none of the conditions in the definition of S_9 depends on N.

(0b) $17T\varepsilon' \leq \frac{1}{40}$, $\mathsf{K_c}(17T\varepsilon')^2 \leq 0.12\,\varepsilon$, $(p_0'/p_0)^4 \geq \frac{1}{2}$ and $\mathsf{a_c}\varepsilon' \leq 1$.

(0c) $3D_w \leq \ell_w(x')$, together with the two conditions below.
(D2) $\ell_w(x)\,(7.1\,r_T + 18D_w \cdot 17T\varepsilon') \leq \mathsf{g}_x/2$.
(D3) $(4.1\,r_T + 2.1 \cdot 17T\varepsilon')^2 \leq 0.1\,\varepsilon$.
The conditions (D2) and (D3) are the conditions (D2)(τ, r_T) and (D3)(τ, r_T) of Thm. 5.18, at $\tau = 17T$. Since both left sides increase with τ, the two conditions hold at every $1 \leq \tau \leq 17T$. The condition (D0)(τ) of the same theorem consists of $2D_w + 1 < N$, which is part of Step 0 (iv) of the proof of Thm. 5.15, of $3D_w \leq \ell_w(x')$, and of $\tau\varepsilon' \leq \frac{1}{40}$, which comes from (0b).

(0d) The following three conditions.
(C0) $18\omega\varepsilon' \leq \frac{1}{10}$ and $160T\omega\varepsilon' \leq 1$.
(C1) $2640\,\ell_w(x')(T+1)\omega\varepsilon' \leq \mathsf{g}_x$.
(C2) $102400\,T\omega^2\varepsilon' \leq 1$.

(0e) $3e_1^\flat \leq \frac{1}{8}(\frac{1}{2} - d_0 - c_\Xi)$.

(Y.1) $16(c_a + c_h)(x) \leq \frac{1}{2}$ and $K_v^\flat p_0^3 x^{-1/2} \leq \frac{1}{2}$.

(Y.2) $50\,x^{-1/8}(2(s_0 + \varrho_*) + 1)^3 \leq 1$.

(Y.3) The inequality

$$x^{1/4}(2s'+1)^3 \cdot 50\,e^{(\kappa+1)s'} \cdot [\,5e^{\kappa+3}L^3 \cdot 16(c_a + c_h) + 2e^{\kappa+3}K_v^\flat p_0^3 x^{-1/2} + x^{-20} + e^{\kappa+3}u_b + u_1\,] \leq x^{-1/8}.$$

(Y.4) The inequality

$$[\,u_t^{1/2}e^{\kappa+5} + 220((2r_1+1)L)^3 e^{\kappa+5} \cdot 16\upsilon_* + 6((2r_1+1)L + 2r(x))^3 e^{\kappa+5} \cdot 16(c_a + c_h)\,] \cdot e^{(\kappa+4)s'} \leq \frac{1}{2}x^{-1/4},$$

together with $e^{(\kappa+1)s'}\upsilon_f^{1/2} \leq \frac{1}{4}x^{-1/4}$.

(Y.5) $16 \cdot 10^3 L^3 e^{-c_h(p_0^2 - p_0'^2)} \leq \frac{1}{2}$ and $50(2u_b + u_1) \leq \frac{1}{2}e^{-\delta_3 p_0^2/2}$.

(Y.6) $10^3 e^{\kappa+3}L^{5/2}x^{-1/4} \leq 10^{-3}$, $8K_2^\sharp \leq E_a p_0'^2$ and $x^{-10} \leq c_a(x')/32$.

(Y.7) $5r_1 + 2\lceil s(x)/L \rceil + 6 \leq r'$ and $R_3 + D_w + M(\lceil r(x)/L \rceil + 2) \leq Mr'$.

(Y.8) $4.04\,x^{-20} \leq \frac{\gamma}{2}x^{-2}$,

$$2x\varsigma'' n_a + 4.8 \cdot 4.04\,v\,\frac{2}{\gamma}\,x^{-20}(2r_1+1)^3 \leq x^{-1/8},$$

$$2x\varsigma'' n_a + 36\varsigma''(C_\varphi^\flat)^2 \cdot 3L^3(R+1)^3 \cdot v(2r_1+1)^3 \leq x^{-1/8},$$

and $4\varsigma''(C_\varphi^\flat)^2 \cdot 3L^3(R+1)^3 \leq \frac{1}{2}$.

(Y.9) $2.001\,K_1^\sharp e^{\kappa+1}(2r'+1)^3 e^{-(\kappa+26.7)r'} \leq 1$ and $K_1^\sharp e^{-\kappa^\sharp r'} \leq 1$.

■

Definition 5.14 (Item 6: the letters of the alphabet, the clusters and the formulas for the output)**.** All objects of this item are defined at a pair (V', e). We let $\xi \sim \mathcal{N}(0, x^{-1}1)$ on I°, and we put $\varphi = G\xi$, $\eta = \varphi - a_0$ and $U = \mathsf{U}(\eta; V')$, with the chart of item 3.4 of §4.2 and of item C.I 4 of §3.

From items E3 and E9 to E12 of §4.3 we take $F^\flat$ and $n^\flat$; the marks mk' and the set $f = \mathrm{mk}'^{+s_0}$; the classes mk'_j, f_j and D'_j; the tests $b_\square$; the factor $[A]^\ell_{\mathfrak{Z}}$; the cell data (J, t), with $\mathfrak{b} = f_{\mathsf{J}} \cup t$ and $\mathfrak{Z} = \mathfrak{b}^{+r_1}$, and the functions $\hat\lambda_{(\mathsf{J},t)} \geq 0$; the notion "admitted", which is (H17); and the notion of "compatible" cell data.

We put $A_\square := [A]^\ell_{\{\square\}}$ and $T_Y := \frac{x}{2}\langle\varphi, S_k(Y)\varphi\rangle - x\langle\varphi, S_k(Y)a_0\rangle$, and we let $\chi_X(\xi) = 1$ if and only if $|\xi_b| \leq c_\xi\varepsilon$ at all $b \in I^\circ$ within the distance $r_w + 1$ of a bond of $X^{\approx}$.

The set $Y_0(X)$ and the notion "small", which means $n \leq L/2 - 2$, with $n = \#X$ for $X \in \mathcal{D}$ and $n = |X|_c$ for $X \in \mathcal{P}$, are those of item 1 of §2.1. The cubes and the cells are ordered once and for all, and the words "first" and "root" refer to these orders.

We say that a set Y is *on* at V' if $Y^{+r'}$ misses mk', and *off* otherwise. A set Y that is off is *short* if it lies inside f, and *long* otherwise. We put $n_j(V') := \sum_{Y \subset f_j,\ \mathrm{off}} \tilde{\mathsf{D}}(Y, V')$.

The letters. Each letter α carries a function, a core J_α and a set I_α; there are the following letters.

(a) For each $X \in \mathcal{D}$, the function is $\lambda^a_X = \chi_X \cdot 1_{X^{\approx}}(U)\,(e^{E^a(X,U)} - 1)$, and at complex V' it is $\chi_X(e^{E^a(X,U)} - 1)$; the core is $J = Y_0(X)$, and I consists of J and of the cells that meet $X^{\approx}$. Here and in (h) and (v) we agree that a product containing the factor χ_X, $\chi_\square$ or $1_{X^{\approx}}(U)$ is 0 wherever that factor is 0, even where $E^a(X,U)$, $E^h(X,U)$ or $V_\square$ has no value.

(h) For each $X \in \mathcal{D}$, the function is $\lambda^h_X = \chi_X \cdot 1_{X^{\approx}}(U)\, e^{E^a(X,U)}(e^{E^h(X,U)} - 1)$, and J and I are as for (a).

(v) For each cell $\square$, the function is $\lambda^v_\square = \chi_\square(A_\square - 1)$, and at complex V' it is $\chi_\square(e^{V_\square} - 1)$; here $J = I = \{\square\}$.

(t) For each connected set Y of cells, the function is $e^{T_Y} - 1$, and $J = I = Y$.

(b) For each cell $\square$, the function is $-b_\square$, and $J = I = \{\square\}$.

Zone letters. The objects are of four sorts: a forced class f_j, whose cells are those of f_j; a cell; a polymer of kind p, that is, some $X \in \mathcal{P}$; and a polymer of kind p′, that is, some $X \in \mathcal{D}$. Two objects with cells are adjacent when they have cells at distance at most $4r_1 + 2$. An object with a cell $\square$ is adjacent to a polymer X of kind p if and only if X contains a cube of the fine set under $\square^{+r_1}$, and to a polymer X of kind p′ if and only if $X^{\approx}$ meets that fine set.

A *zone letter* $\zeta = (\mathsf{J}, t, \mathfrak{X}, \mathfrak{Y})$ is a set of objects that is connected for the adjacency and has these properties: (J, t) is a cell datum; each polymer in it is adjacent to some object with cells; and the polymers X of $\mathfrak{X}$, all of kind p, are at mutual distances $> s(x)$. The set $\mathfrak{Y}$ collects its polymers of kind p′. We call ζ *forced* when J is non-empty, and *admitted* when (J, t) is admitted. We put

$$1_{\mathrm{ind},\zeta} = 1\{D(U) \cap \hat{\mathfrak{Z}} \text{ lies in the union of } \mathfrak{X}, \text{ and each } X \text{ of } \mathfrak{X} \text{ holds a plaquette of } D(U) \cap \hat{\mathfrak{Z}}\}$$

and $\Lambda_\zeta := \hat\lambda_{(\mathsf{J},t)} \cdot 1_{\mathrm{ind},\zeta}$, and we give ζ the function

$$\lambda_\zeta := \Lambda_\zeta \cdot \prod_{\mathsf{J}} e^{n_j} \cdot \prod_{\mathfrak{X}} w(X,U) \cdot \prod_{\mathfrak{Y}} (e^{1E(X)(U)} - 1).$$

The core J_ζ is built from the marks of the classes of ζ, from its tested cells t and from the sets $Y_0(X)$, linked by diagonal paths that follow a spanning tree, fixed by the order, of the graph whose edges are the adjacencies together with, within each class, the links of the marks of that class. The set I_ζ is the union of $\mathfrak{b}^{+2r_1}$, of the cells that meet $X^{\approx\approx}$ for X in $\mathfrak{X}$ or $X^{\approx}$ for X in $\mathfrak{Y}$, and of J_ζ.

Two zone letters are *compatible* when each object of the one differs from, and is not adjacent to, every object of the other, and when the polymers of kind p of the two letters stay at distance $> s(x)$ from one another. In addition, the following pairs are incompatible: ζ and a letter (b) at a cell of $\mathfrak{b}_\zeta^{+2r_1}$; ζ and a letter (v) at a cell of $\mathfrak{Z}_\zeta$; ζ and a letter (a) or (h) of a polymer X for which $X^{\approx}$ meets $\hat{\mathfrak{Z}}_\zeta$; and the two letters (a) and (h) of the same X.

We write $\alpha \sim \beta$ if $\mathrm{dist}(I_\alpha, I_\beta) \leq r_1$. The universal alphabet $\mathfrak{A}$ consists of the letters (a), (h), (v), (t) and (b), and of the zone letters without a class.

The letters *actual* at V' are the following. A forced ζ is actual if $f \setminus f_\zeta$ has no cell within the distance $4r_1+2$ of $\mathfrak{b}_\zeta$, and a zone letter of $\mathfrak{A}$ is actual if f has no cell within the distance $4r_1+2$ of $\mathfrak{b}_\zeta$. A letter (b) is actual if it lies off f^{+2r_1}, and a letter (v) if it lies off f^{+r_1}. A letter (a) or (h) of X is actual if $X^{\approx}$ lies off the fine set under f^{+r_1}. Every letter (t) is actual.

Clusters. The clusters, the function $K(\mathfrak{B}) = E_\xi \prod_{\mathfrak{B}} \lambda_\alpha$, and $\Phi(C)$, $\hat\Phi(C)$, $n(C)$ and $d(C)$ are those of Lem. 4.50 (Step 8 of the proof of Thm. 4.39), defined for $\mathfrak{A}$ with $\mathsf{r} = r_1$. The core $\mathrm{co}(C)$ of a cluster C is obtained by linking the cores J_α of its letters with diagonal paths that follow a spanning tree of $\sim$ (fixed by an order). A cluster C is *constant* if all its letters are letters (b), and *analytic* if all its letters are letters (a), (v) or (t); an analytic cluster is *new* if it has no letter (a), and *old* otherwise. All remaining clusters are *hard*.

The formulas for the output.

(Fm1) On $\mathcal{U}_Y(x')$ we put

$$E'^{\mathrm{new}}(Y,V') := \sum_{\mathrm{co}(C)=Y,\ \mathrm{new}} [\Phi(C)(V') - \Phi(C)(1)] + \tilde{\mathsf{D}}(Y,V') - P(Y,V'),$$

$$E'^{\mathrm{old}}[u](Y,V') := \sum_{\mathrm{co}(C)=Y,\ \mathrm{old}} [\Phi(C)(V') - \Phi(C)(1)],$$

and $E'^{a} := E'^{\mathrm{new}} + E'^{\mathrm{old}}[u]$.

(Fm2) We put $\mathsf{V}' := \mathsf{V}_0 + \sum_{\mathrm{analytic}} \Phi(C)(1) + \sum_{\mathrm{constant}} \Phi(C)$ with

$$\mathsf{V}_0 := \log[\mathsf{c}_0\, z(ax)^{-3\#\mathbb{T}'} (2\pi/x)^{\frac{3}{2}\#I^\circ} (\det C^1)^{1/2}],$$

where $\mathsf{c}_0 = (2\pi^2)^{-\#I^\circ}$ (item C.I 4 of §3, paragraph "The step, the chart") and z is the normaliser of k.

(Fm3) For G-valued fields given on the bonds of $Y^{\approx}$ and tame there, where each such field enters through its extension by 1 off these bonds (Step 6 of the proof of Thm. 5.15, paragraph "The extension by $\mathbf{1}$"), we put $E'^{h}(Y,V') := \sum_{\mathrm{co}(C)=Y,\ \mathrm{hard}} \Phi(C)(V')$.

(Fm4) A *rough polymer* at V' is a $\sim$-connected set γ of letters that are pairwise compatible and all actual at V', and among which there is a forced letter. A cluster C is *on* at V' if it is constant or if V' is tame on $\mathrm{co}(C)^{\approx}$, and *off* otherwise. We put $\Phi^\natural(C) := \Phi(C)(V')$ if C is on, $\Phi^\natural(C) := \Phi(C)(1)$ if C is analytic and off, and $\Phi^\natural(C) := 0$ if C is hard and off.

Let Y be a finite set of cells. We put $w'(\mathsf{Y},V',e) := 0$ if $\mathrm{mk}'(V')$ has a cell in $\mathsf{Y}^{+(s'-1)} \setminus \mathsf{Y}$. The radius is $s'-1$ because the plaquettes based in a cell decide whether that cell is a mark, and these plaquettes involve bonds of the next layer; this is the clause of (F4) in Step 12 of the proof of Thm. 4.39.

In the remaining case, $w'(\mathsf{Y},V',e)$ is the sum of

$$\prod_{\gamma\in\mathfrak{G}} K(\gamma) \cdot \prod_{C\in\mathfrak{N}} (e^{-\Phi^\natural(C)} - 1) \cdot \prod_{Y\in\mathfrak{N}^\flat} (e^{\tilde{\mathsf{D}}(Y,V')} - 1)$$

over all triples $(\mathfrak{G},\mathfrak{N},\mathfrak{N}^\flat)$ with the following properties: $\mathfrak{G}$ is a set of rough polymers at V', no two of which are related by $\sim$, and the forced letters of $\mathfrak{G}$ have as their marks exactly $\mathrm{mk}' \cap \mathsf{Y}$, with each class appearing once; $\mathfrak{N}$ is a finite set of clusters of $\mathfrak{A}$ in which every cluster has some letter related by $\sim$ to some letter of $\mathfrak{G}$; $\mathfrak{N}^\flat$ is a set of long sets Y; the cores J_α, $\mathrm{co}(C)$ and Y have the union Y; and the graph that joins two elements whenever their cores lie within the distance s' of each other is connected.

Finally, $w'(\mathsf{Y},V') := \int_{\mathfrak{E}} w'(\mathsf{Y},V',e)\,de$.

The output is $F_\ell(u) := (E'^{a}, E'^{h}, w')$. ∎

Theorem 5.15 (Item 7: one drawn step for complex members within wider caps)**.** In the setting of the letters of item 1, taken in the order (O0) to (O13), in the setting (SD) of item 2, whatever

its data, and in the setting of the inputs (Γ2) to (Γ7) of item 4, let $x \geq x_9$, let $1 \leq A \leq 16$ and let $u \in M_A^{\mathrm{su}}(x)$. Then all series in (Fm1) to (Fm4) converge absolutely, and (a) to (e) hold.

(a) The identity $(\rho[u;\ell])' = e^{\mathsf{V}'} \cdot \rho[F_\ell(u);\ell'^{\mathrm{c}}]$ holds at every V', and V' is its only argument.

(b) By (Fm1), $E'^a = E'^{\mathrm{new}} + E'^{\mathrm{old}}[u]$, and E'^{new} is one and the same for all u. For every cell $\square$,

$$\sum_{Y \ni \square} e^{\kappa(\#Y-1)} \sup_{\mathcal{U}_Y(x')} |E'^{\mathrm{new}}(Y,\cdot)| \leq 0.35\, c_a(x'),$$

$$\sum_{Y \ni \square} e^{\kappa(\#Y-1)} \sup_{\mathcal{U}_Y(x')} |E'^{\mathrm{old}}[u](Y,\cdot)| \leq \vartheta_w\, A\, c_a(x'), \qquad \vartheta_w = 0.248 \leq \frac{1}{4},$$

and

$$\sum_{Y \ni \square} e^{\kappa(\#Y-1)} \sup |E'^h(Y,\cdot)| \leq c_h(x').$$

Moreover, at every V',

$$|w'(\mathsf{Y},V')| \leq \exp\{-c_\Xi x' \varepsilon' \sum_{P \in D' \cap \mathsf{Y}} \phi'_P - \kappa(|\mathsf{Y}|_c - 1)\},$$

which is the strong cap at x'. Of (ca), (ch) and (cΞ), the clauses without a size all hold at x'. Hence $F_\ell(u) \in M^{\mathrm{su}}_{\max\{1,\,0.35+0.248A\}}(x')$ with the strong cap, and at $A = 2$ the bound on $E'^a = E'^{\mathrm{new}} + E'^{\mathrm{old}}[u]$, in units of $c_a(x')$, is $0.35 + 0.496 = 0.846 \leq 1$.

(c) Let $z \mapsto u_z \in M_{16}^{\mathrm{su}}(x)$ be affine on an open set of $\mathbb{C}$. Along it, the number V'_z and every value $E'^a_z(Y,V')$ with $V' \in \mathcal{U}_Y(x')$, $E'^h_z(Y,V')$ and $w'_z(\mathsf{Y},V')$ depend holomorphically on z.

(d) If u is real, then $F_\ell(u)$ and V' are real; if, besides, $\rho[u;\ell] > 0$, then $(\rho[u;\ell])' > 0$.

(e) For $u_1, u_2 \in M_2^{\mathrm{su}}(x)$,

$$\mathsf{D}_{x'}(F_\ell(u_1), F_\ell(u_2)) \leq 0.4\, \mathsf{D}^{\mathrm{in}}_x(u_1,u_2) \leq 0.4\, \mathsf{D}_x(u_1,u_2);$$

the inputs are thus measured in the weak distance and the outputs in the strong one.

Conclusions (a), (b), (c) and (e) are independent of the reality, of the sign and of the positivity of the member u. ■□

Remark 5.16 (Item 8: uses of Thm. 5.15 in other theorems). In Steps 1, 6, 9 and 12 of the proof of Thm. 4.39, the formulas (F1) to (F4) of that theorem, which use a different test χ_X, define a map of members other than the map F_ℓ of (Fm1) to (Fm4). Each of the two maps represents ρ'.

Thm. 5.48, which is the second half of the change of rows, uses Thm. 5.15, applied to inputs in M_2^{su} for which the size of w is bounded by the weak cap alone.

Proof of Thm. 5.15. Step 0: the floor (Y*)*, the constants and the radii.*

(i) Each of the conditions (Y.0) to (Y.9) holds for all large x; hence S_9 is not empty, and $x_9 = 1 + \inf S_9$ exists. The number x_9 is computed from (O0) to (O12) alone; it depends on none of A, u, z and N, and it depends on ℓ only through the numbers b_*, K_*, λ_* and the range.

(ii) The inequalities (Geo): for $R \geq 6$ we have $2k_\chi \leq r_1 - 1$, $k_t \leq r_1$, $Mr_1 \geq R_v \geq R_3$, $3r_w + \varrho_L + 1 \leq R_2 - 1$ and $\lceil R_2/M \rceil \leq r_1 - 8$.

(iii) The inequalities (Rad), which follow from (Y.7): $4r_1 \leq r'$, $r_1 \leq r'/5$, $2\varrho_* + r_1 \leq r'$, $2(s_0 + \varrho_*) + r_1 \leq 5r' \leq s'$, $2s_0 + 4r_1 + 2 \leq 5r'$, $s_0 + r_1 + \lceil r(x)/L \rceil + 2 \leq 5r'$, $\omega \leq Mr'$, $3\omega + 1 \leq \ell_w(x')$ and $2(s_0 + k_t) + 1 \leq s(x) + 1$.

(iv) We have $T > 1$, $r_\Xi \geq \frac{3}{4}(\frac{1}{2} - d_0 - c_\Xi) > 0$, $u_t, \upsilon_f, \upsilon_* < 1$ and $2\upsilon_f^{1/4} \leq 1$; moreover $2D_w + 1 < N$ and $2\omega + 1 < N$, and $\varrho(x)/L + 2 \leq \varrho_L + 2 < N/2$ and $2R_3 + 1 < N$.

We prove (ii). First, $Mr_1 \geq R_2 + 3(\varrho_L + 2) + 5M = 2R + 3\varrho_L + 8 + 5M \geq R_v = R_3 + \varrho_L + 2$. Next, $2k_\chi \leq 2R_\chi/M + 2 = R_2/M + (2\varrho_L + 2)/M + 2 \leq \lceil R_2/M \rceil + 2r_l + 2 \leq r_1 - 1$, and $k_t \leq k_\chi$. As $r_l \geq 1$, we have $\lceil R_2/M \rceil = r_1 - 3r_l - 5 \leq r_1 - 8$. Finally, for $R \geq 6$, $3r_w + \varrho_L + 1 \leq 3R/2 + \lceil R/3 \rceil + 1 \leq 2R + 1$.

We prove (iii), with the help of (Y.7). We have $2\varrho_* + r_1 = 5r_1 + 2\lceil s(x)/L\rceil + 4$, $2(s_0+\varrho_*)+r_1 = 4r' + (2\varrho_* + r_1)$ and $2s_0 + 4r_1 + 2 = 4r' + 4r_1 + 2$, and moreover $s_0 + r_1 + \lceil r(x)/L\rceil + 2 \le 3r'$. Because $M\lceil r(x)/L\rceil \ge \lceil Mr(x)/L\rceil$ and $2M \ge M+2$, we obtain $\omega \le R_3 + D_w + M(\lceil r(x)/L\rceil + 2) \le Mr'$, and so $3\omega + 1 \le 3Mr' + 1 \le 6Mr' + 12$. Lastly, $2s_0 + 2k_t + 1 \le 4r' + 0.4\,r' + 1 \le 4.4\,r(x) + 1 \le 8r(x) + 1$, since $r' \le r(x)$.

We prove (iv). We have $T = L^{3/2}/(19\mathsf{N}_c) \ge (1.2 \cdot 10^4 \mathsf{N}_c^2)^3/(19\mathsf{N}_c) > 10^{17} > 1$. By (O3), $t_\rho \le \frac{1}{8}(\frac{1}{2} - d_0 - c_\Xi)$, and together with (0e) this yields $r_\Xi \ge \frac{3}{4}(\frac{1}{2} - d_0 - c_\Xi) \ge \frac{3}{8}(\frac{1}{2} - d_0)$. From (Y.4) we get $u_t^{1/2} < 1$ and $\upsilon_f^{1/2} \le \frac{1}{4}$. Moreover, $N \ge 2Ms(x) = 16Mr(x) > 2Mr' + 1 \ge 2\omega + 1 \ge 2D_w + 1$.

The same bounds give three further clauses: the clause $\varrho/L + 2 < N/2$ on the rows (item C.I 1.1 of §3), the clause $\varrho_L + 2 < N/2$ on the rows (item C.II 1.2 of §3), and the inequality $2R_3 + 1 < N$ in (P0) (item C.II 4 of §3, paragraph "Floors"). Indeed, $\omega \ge R_3 + 2 = 2R + \varrho_L + 7$; hence $\varrho/L + 2 \le \varrho_L + 2 < \omega < N/2$ for every range $\varrho \le \varrho(x) = R$, and $2R_3 + 1 < 2\omega + 1 < N$. These are the clauses on N of the setting in which §3 states (Γ2) and (Γ3). For empty rows the range is 0, while ϱ_L remains the one belonging to $\varrho(x)$.

We prove (i), condition by condition. As to the orders, R, $r(x)$, r', r_1, s', ϱ_*, ω, D_w and ℓ_w are $O(\log x)$; $\varepsilon, \varepsilon' = O((\log x)^{\mathfrak{p}_0} x^{-1/2})$; and $\mathsf{g}_x \ge c(\log x)^{-2}$. By (M2), $(\kappa+4)s' \le (\kappa+4)(160 c_R \log x/M + 8) \le \frac{1}{12}\log x + 8(\kappa+4)$.

(0a) That the floors of (0a) hold for all large x is stated in each of Sections 3, 4.1 and 4.2, each section doing so for its own floors, and item G of §4.3 states that x_{fl} exists; of (P0), the clauses $x \ge L^2$ and $R \ge 6$ alone are floors, by (0a).

(0b) We have $17T\varepsilon' \to 0$, $\mathsf{K}_c(17T\varepsilon')^2/\varepsilon = O(\varepsilon)$, $p_0'/p_0 = (1 - \log L/\log x)^{\mathfrak{p}_0} \to 1$ and $\mathsf{a}_c\varepsilon' \to 0$.

(0c) Here $3D_w \le 18Mr(x)/L + 43$, and $6Mr' + 12$ is $\ell_w(x')$. For all large x we find that $18Mr(x)/L + 43$ is at most $\ell_w(x')$, using $L > 10^{13}$ and $r(x) \le r' + 1 + 20c_R \log L/M$, and the first condition of (0c) follows.

2), (D3), (C0) to (C2) In each of these conditions the left side is a number computed from the letters times $(\log x)^{k+\mathfrak{p}_0} x^{-1/2}$ with $k \le 2$ (in (D3), after division by ε, it is $O(\varepsilon)$), and the right side is a number free of x or is at least $c(\log x)^{-2}$, since $\mathsf{g}_x \ge c(\log x)^{-2}$.

(0e) The number $e_1^\flat$ is ε' times a number free of x.

(Y.1) We have c_a, c_h, $p_0^3 x^{-1/2} \to 0$.

(Y.2) The left side is of the order of $(\log x)^3 x^{-1/8}$.

(Y.3) Before the bracket of (Y.3) there are the three factors $x^{1/4}$, $50(2s'+1)^3 = O((\log x)^3)$ and $e^{(\kappa+1)s'}$, and the bound on $(\kappa+4)s'$ given above yields $e^{(\kappa+1)s'} \le e^{8(\kappa+4)} x^{1/12}$. Inside the bracket, the power $x^{-1/2}$ occurs in the term with c_a and in the one with $K_v^\flat$, each of which also contains the factor $p_0^3 = O((\log x)^{3\mathfrak{p}_0})$. For the products of these two terms with the three factors, the exponents of x add up to $\frac{1}{4} + \frac{1}{12} - \frac{1}{2} = -\frac{1}{6} < -\frac{1}{8}$, and $x^{-1/6+1/8} = x^{-1/24}$ absorbs the powers of $\log x$. For the term x^{-20} of the bracket, the sum of the exponents is $\frac{1}{4} + \frac{1}{12} - 20$, and $c_h(x)$, u_b and u_1 tend to 0 as $x \to \infty$ at the rate $e^{-c(\log x)^{2\mathfrak{p}_0}}$.

(Y.4) In the third term, the power of x is $1/12 - 1/2 = -5/12 < -1/4$.

For the first and the second term, (W) gives $\mathrm{py} = d_0 p_0^2/2$, and therefore

$$\log(1/u_t) = \min\{\rho_Q/2 - c_\tau,\ \mathrm{py}/(768(2r_1+1)^3)\} - \log 3v \ge c(\log x)^{2\mathfrak{p}_0 - 3}$$

with a constant $c > 0$. Next,

$$\log(1/\upsilon_f) \ge (3/8)(1/2 - d_0) A_0^2 (\log x')^{2\mathfrak{p}_0} - O((\log x)^3).$$

Together these two bounds give

$$\log(1/\upsilon_*) = L^{-3} \min\{(1/2)\log(1/u_t), (1/4)\log(1/\upsilon_f)\} \ge cL^{-3}(\log x)^{2\mathfrak{p}_0 - 3},$$

again with a constant $c > 0$, where $2\mathfrak{p}_0 - 3 > 1$. This lower bound is to be compared with $(1/4 + 1/12)\log x + O(\log\log x)$, and since the power of $\log x$ in it exceeds 1, it is the larger

of the two for all large x. It is here that the strict inequality $\mathfrak{p}_0 > 2$ is used, for it is what gives $2\mathfrak{p}_0 - 3 > 1$.

The second inequality of (Y.4) follows from the bound on $\log(1/\upsilon_f)$ obtained above.

(Y.5) We have $p_0^2 - p_0'^2 \geq 2\mathfrak{p}_0 A_0^2 \log L\,(\log x')^{2\mathfrak{p}_0-1} \to \infty$. By (7e), $u_1 = e^{-\delta_3 p_0^2}$. The rate of u_b is $c_\xi^2/2 > \delta_3/2$, because $\delta_3 \leq 3c_\xi^2/(256h_*^2)$ and $h_* \geq 1$.

(Y.6) This condition follows from the factor $x^{-1/4}$, from $p_0' \to \infty$, and from the comparison of x^{-10} with $x^{-1/2}$.

(Y.7) For the first inequality, $5r_1 + 2\lceil s(x)/L\rceil + 6 \leq (10 + 335/L)c_R \log x/M + \text{const}$, which is to be compared with $r' \geq 20c_R(\log x - \log L)/M$; and $10 + 335/L < 20$ for $L \geq 34$. For the second inequality, $R_3 + D_w + M(\lceil r(x)/L\rceil + 2) \leq (2 + 141/L)c_R \log x + \text{const} \cdot M$, which is to be compared with $20c_R \log x'$.

(Y.8) This condition follows from the factors x^{-18} and $x^{-19}(\log x)^6$.

(Y.9) This condition follows from $r' \to \infty$.

As to the order of the choices: the conditions depend only on x and on the letters (O0) to (O12). The numbers c_h and E_a occur in (Y.1) and in (Y.3) to (Y.6), and they are fixed before x_9. The floor $x_{\mathfrak{fl}}$ is free of c_h, E_a, ℓ and N (item G of §4.3).

Step 1: the formula for the step at every pair. Let $u \in M_A^{\mathrm{su}}(x)$ with $A \leq 16$. Then $\rho[u;\ell]$ is a bounded, Borel and gauge invariant complex function, and at every pair (V', e)

$$(\rho[u;\ell])'(V') = e^{\mathsf{V}_0} \cdot \exp\{-x'\mathrm{En}'(V') + \sum_Y \tilde{\mathsf{D}}(Y, V') - \sum_\square 1_{\square\approx}(V')P(\{\square\}, V')\} \cdot Z_u,$$

where

$$Z_u := e^{-\sum_j w_j} \cdot E_\xi[e^{\sum_Y T_Y} \cdot \prod_\square A_\square \cdot \prod_{X\in\mathcal{D}} e^{1E(X)(U)} \cdot \Xi_w(U)].$$

Proof of Step 1. 1. A G-valued field that is tame on $X^{\approx}$ belongs to $\mathcal{U}_X(x)$: indeed $|U_b| = 1 < 1 + \vartheta(x)$, $|U(\pi)| = 1 < \mathsf{b}(x)$, and $|\partial U(p) - \mathbf{1}| = 2\sin(\phi_p/2) \leq \varepsilon < 2\varepsilon$. Therefore $1E(X)(V)$ is defined at every V, and $|1E(X)| \leq A(c_a + c_h)e^{-\kappa(\#X-1)}$. The function Ξ_w is a finite sum of bounded terms. The energy En_ℓ is nonnegative and gauge invariant, and the three kinds of members and the indicators are gauge invariant as well. This gives the first assertion.

2. We use the formula of the chart, Lem. 3.9. Since the formula is linear in ρ, it holds for the real and for the imaginary part of ρ alike, and so

$$\rho'(V') = \mathsf{c}_0 \int_{\mathfrak{D}} \rho(\mathsf{U}(\eta; V')) \prod_c k_{ax} \prod_{b\in I^\circ} \mathsf{j}(\eta_b)\, d\eta,$$

where $k_{ax} = z(ax)^{-1}e^{-ax(1-\cos \mathsf{n}_c)}$.

3. Let X' be any $\mathfrak{g}$-valued function on the coarse plaquettes; in particular, it may be the one of the pair. We put $\mathcal{Q}(\eta) := (1/2)\langle F^\flat, \mathsf{B}^V F^\flat\rangle + (a/2)|Q_1^V \eta|^2$, with $F^\flat = EX' + \mathrm{d}^V\eta$ and $\mathsf{B}^V = 1 + \ell^{V*}\ell^V$, and we insert $1 = e^{-x\mathcal{Q}(\eta)}e^{x\mathcal{Q}(\eta)}$. By the definition in item E11 of §4.3,

$$e^{x\mathcal{Q}(\eta)} \cdot e^{-x\mathrm{En}_\ell(U)} \cdot \prod_c e^{-ax(1-\cos \mathsf{n}_c)} \cdot 1_{\mathfrak{D}}(\eta) \prod_b \mathsf{j}(\eta_b) = \prod_\square A_\square,$$

because $\langle F^\flat, \mathsf{B}^V F^\flat\rangle = |F^\flat|^2 + |\ell^V F^\flat|^2$ and $n^\flat = -Q_1^V \eta$.

4. Expanded, $\mathcal{Q}(\eta) = (1/2)\langle \eta, q\eta\rangle + \langle \eta, \mathsf{g}^V X'\rangle + (1/2)\langle EX', \mathsf{B}^V EX'\rangle$. With $C^{-1} = q + S$, $\mathsf{g}^V X' = (q + S)a_0$ and $\eta = \varphi - a_0$ this becomes

$$\begin{aligned}\mathcal{Q} &= (1/2)\langle \varphi, q\varphi\rangle + \langle \varphi, Sa_0\rangle - (1/2)\langle a_0, qa_0\rangle - \langle a_0, Sa_0\rangle + (1/2)\langle EX', \mathsf{B}^V EX'\rangle \\ &= [\mathsf{m}_C - (1/2)\langle a_0, Sa_0\rangle] + (1/2)\langle \varphi, C^{-1}\varphi\rangle - (1/2)\langle \varphi, S\varphi\rangle + \langle \varphi, Sa_0\rangle.\end{aligned}$$

5. Now let $\varphi = G\xi$. As $C = GG^{\mathsf{T}}$ and G is invertible by (2f), we get $\langle \varphi, C^{-1}\varphi\rangle = |\xi|^2$ and $d\eta = |\det G|\, d\xi = (\det C)^{1/2} d\xi$; moreover $\int e^{-(x/2)|\xi|^2} d\xi = (2\pi/x)^{(3/2)\#I^\circ}$. By ($\Gamma$4), $(x/2)\langle \varphi, S\varphi\rangle - x\langle \varphi, Sa_0\rangle = \sum_Y T_Y$.

6. Again by (Γ4), $(\det C^{V'})^{1/2} = (\det C^{\mathbf{1}})^{1/2} \exp\{\sum_Y \tilde{\mathsf{D}}(Y, V')\}$, and (5a) rewrites $x\,\mathsf{m}_C - (x/2)\langle a_0, Sa_0\rangle$. Inserting all of this into the formula of the chart, we obtain the formula of Step 1.

The argument is independent of the reality of u and of its sign. □

Step 2: tameness of the fine field where the tests are passed and the coarse field is tame, and the domains. The following assertions hold.

(i) Let V' be G-valued with $\phi'_P \le \varepsilon'$ on the box of radius R_3 about $y(p)$, and let $|\xi_b| \le c_\xi \varepsilon$ at the bonds b of I° within $r_w + 1$ of p. Then $\phi_p(U) \le \varepsilon$.

(i′) If $p \in D(U)$, then some bond β of I° within $r_w + 1$ of p has $|\xi_\beta| > c_\xi\varepsilon$, or some plaquette of D' lies in the box of radius R_3 about $y(p)$.

(ii) Let $X \in \mathcal{D}$ with $\chi_X(\xi) = 1$, let $\tau \in \{1, 17T\}$, let B be a set of coarse sites that contains the D_w-neighbourhood of the sites whose blocks meet $X^{\approx}$ and the boxes of radius R_3 about them, and let $V' \in U^\tau(B; x')$. Then U, restricted to $X^{\approx}$, lies in $\mathcal{U}_X(x)$ and is holomorphic in V'. If $V' \in \mathcal{U}_Y(x')$ with $Y \supset Y_0(X)$, such a set B can be chosen in $Y^{\approx}$.

(iii) Let $\mathfrak{b} \supset f$ and $\mathfrak{Z} = \mathfrak{b}^{+r_1}$. On the event that every cell off $\mathfrak{b}$ passes its tests, $D(U)$ lies in $\hat{\mathfrak{Z}}$, $\chi_\square = 1$ for every $\square$ off $\mathfrak{Z}$, and $\chi_X = 1 = 1_{X^{\approx}}(U)$ for every X with $X^{\approx}$ off $\hat{\mathfrak{Z}}$.

(iv) If V' is G-valued and tame on $\square^{\approx}$ and $\chi_\square(\xi) = 1$, then $A_\square = e^{V_\square}$.

We first prove a lemma and a theorem.

Lemma 5.17 (The plaquette of the chart to first order)**.** Let p be a fine plaquette in the plane $\mu\nu$, let $P = (y(p); \mu\nu)$, and let $\tau \ge 1$ with $\tau\varepsilon' \le 1/40$. For the field there are two cases. In the first case, V' is an $\mathrm{SL}(2,\mathbb{C})$ field with $|V'(c)| \le 1 + \vartheta_0^\flat$ at the two bonds of P at $y(p)$ and with $|\partial V'(P) - \mathbf{1}| < 2\tau\varepsilon'$, and $X'_P = -i \log \partial V'(P)$. In the second case, (V', e) is a pair with $\phi'_P \le \tau\varepsilon'$. Let r be any number such that $|\eta_b| \le r$ at the bonds of p and $4.1r + 2.1\tau\varepsilon' \le 1$. Then $|\partial U(p) - \mathbf{1} - iF_p^\flat| \le 0.72(4.1r + 2.1\tau\varepsilon')^2$, where $F^\flat = EX' + \mathrm{d}^V\eta$. ■

Proof. We use the identity

$$\partial U(p) = e^{i\eta_1} e^{iT_2\eta_2}[e^{iX'_P}]e^{-iT_3\eta_3}e^{-i\eta_4},$$

with $T_j \in \{1, \mathrm{Ad}_{V'(c)}\}$, which is given in full in Step 4 of the proof of Thm. 4.5; the bracket is present if and only if p is an edge plaquette under P, and the exponents add up to $F_p^\flat$. Now $|\mathrm{Ad}_g Y| \le |g|^2|Y| \le 1.01|Y|$. In the first case $|X'_P| \le 2.06\tau\varepsilon'$ (item C.II 1.4 of §3); in the second, $|X'_P| = \phi'_P$. Hence the norms of the exponents add up to $S \le 4.02r + 2.06\tau\varepsilon' \le 1$, and (E) of item C.I 1.1 of §3 gives $|\partial U(p) - \mathbf{1} - iF_p^\flat| \le 0.72S^2$. □

Theorem 5.18 (The chart maps into the fine domain)**.** Let $\tau \ge 1$, $r > 0$ and $\varepsilon \le 1/40$. Let $X \in \mathcal{D}$, let B be a set of coarse sites that contains the D_w-neighbourhood of the sites whose blocks meet $X^{\approx}$, let $V' \in U^\tau(B; x')$, and let η be a $\mathfrak{g}_{\mathbb{C}}$-valued function on I° with the following two properties.

(h1) $|\eta_b| \le r$ at the bonds of $X^{\approx}$.

(h2) $|F_p^\flat| \le 1.9\varepsilon$ at the plaquettes of $X^{\approx}$.

Let, moreover, the following floors hold.

(D0)(τ) $2D_w + 1 < N$, $3D_w \le \ell_w(x')$, $\tau\varepsilon' \le 1/40$.

(D2)(τ, r) $\ell_w(x)(7.1r + 18D_w\tau\varepsilon') \le \mathfrak{g}_x/2$.

(D3)(τ, r) $(4.1r + 2.1\tau\varepsilon')^2 \le 0.1\varepsilon$.

Then the restriction of $\mathsf{U}(\eta; V')$ to the bonds of $X^{\approx}$ lies in $\mathcal{U}_X(x)$; it is polynomial in the entries of V' and entire in η. ■□

The condition $\varepsilon \le 1/40$ is the floor (A1) of item 4 of §2.1. Here $\varepsilon < \varepsilon' \le 1/40$ by the identity $\varepsilon' = L^{1/2}(p'_0/p_0)\varepsilon$ and by (0b).

Proof. First we note by how much the radii at x exceed those at x': $\vartheta(x)-\vartheta(x') = \vartheta_0^\flat \log L/(\log x \log x') > 4\mathsf{g}_x$ and $\mathsf{b}(x)-\mathsf{b}(x') = \log L/(\log x \log x') > 4\mathsf{g}_x$; moreover $\mathsf{g}_x \le 1$ and $\mathsf{b}(x')+\mathsf{g}_x < 2$ (item C.II 1.4 of §3). By (D2), as $\ell_w(x) \ge 18$, we have $r \le \mathsf{g}_x/255$.

(u3): A plaquette p of $X^{\approx}$ has $y(p)$ among the sites named in the hypothesis on B, and the sites of P lie within 1 of $y(p)$, hence in B. By (D3), $4.1r + 2.1\tau\varepsilon' \le 1$. So Lem. 5.17, together with (h2) and (D3), yields $|\partial U(p) - \mathbf{1}| \le 1.9\varepsilon + 0.072\varepsilon < 2\varepsilon$.

(u1): Each U_b is $\mathbf{1}$, $e^{i\eta_b}$ or $e^{i\eta_b}V'(c)$. Hence

$$|U_b| \le e^r(1+\vartheta(x')+\mathsf{g}_x) < 1+\vartheta(x')+\mathsf{g}_x+2.1r < 1+\vartheta(x')+2\mathsf{g}_x < 1+\vartheta(x).$$

(u2): Let π be a path in $X^{\approx}$ of $m \le \ell_w(x)$ bonds $b_1,\dots,b_m$ through the sites $z_0,\dots,z_m$, and put $y_i := y(z_i)$ and $o_1 := y_0$. Since $|z_i - z_0|_\infty \le m$, we get $\operatorname{dist}(y_i,o_1) \le \lceil m/L \rceil \le D_w - 1$. Let B_1 be the box of radius $D_w - 1$ about o_1. It is embedded, as $2D_w + 1 < N$; it lies in B; and $3(D_w-1)+1 \le \ell_w(x')$. By (e2) and (e3), the field V' has on B_1 the properties (h2) and (h3) of item C.II 1.4 of §3, with the value $\hat\varepsilon = \tau\varepsilon'$ of their parameter.

We put $\mathsf{k}(y) := V'(\Gamma(o_1,y))$ and $\mathsf{W}(c) := \mathsf{k}(c_-)V'(c)\mathsf{k}(c_+)^{-1}$. By Lem. 3.27 (a), $|\mathsf{k}(y)| = |\mathsf{k}(y)^{-1}| \le 2$ and $|\mathsf{W}(c)| = |\mathsf{W}(c)^{-1}| \le e^{18D_w\tau\varepsilon'}$. Let $U(b_i)$ be the factor of b_i in $U(\pi)$, taken in its direction, and put $g_i := \mathsf{k}(y_{i-1})U(b_i)\mathsf{k}(y_i)^{-1}$. As $\mathsf{k}(o_1) = \mathbf{1}$, this gives $U(\pi) = g_1\cdots g_m\cdot\mathsf{k}(y_m)$.

If b_i stays in a block, g_i is $\mathbf{1}$ or $\mathsf{k}e^{\pm i\eta}\mathsf{k}^{-1}$. If b_i crosses c forwards, $g_i = [\mathsf{k}(c_-)e^{i\eta}\mathsf{k}(c_-)^{-1}]\mathsf{W}(c)$; if it crosses c backwards, $g_i = \mathsf{W}(c)^{-1}[\mathsf{k}(c_-)e^{-i\eta}\mathsf{k}(c_-)^{-1}]$. Consequently $|g_i| \le (1+4(e^r-1))e^{18D_w\tau\varepsilon'} \le e^{7.1r+18D_w\tau\varepsilon'}$. Further, $|\mathsf{k}(y_m)| < \mathsf{b}(x') + \mathsf{g}_x$ by (e2). With (D2) we conclude that

$$|U(\pi)| < e^{\mathsf{g}_x/2}(\mathsf{b}(x')+\mathsf{g}_x) < \mathsf{b}(x') + 2.3\mathsf{g}_x < \mathsf{b}(x).$$

The argument is independent of the rows of ℓ. □

Proof of Step 2. (i) Since $\eta = \varphi - a_0$ and $\varphi = G\xi$, we have $F_p^\flat = (EX' - \mathrm{d}^V a_0)_p + (\mathrm{d}^V G\xi)_p$. A G-valued field has (e1) and (e2), and $|\partial V'(P) - \mathbf{1}| = 2\sin(\phi'_P/2) \le 2\varepsilon'$. We apply (Γ3) with $\hat\varepsilon = \varepsilon' \le L^{1/2}\varepsilon$: the first term is at most $\mathsf{N}_{\mathrm{c}}L^{-3/2}\varepsilon + \mathsf{K}_{\mathrm{c}}\varepsilon'^2 \le (10^{-4}+0.12)\varepsilon$, by ($\mathrm{L2^{su}}$) and by (0b) with $T > 1$. By (2a) and (2e), the second term is at most $4C_\varphi^\flat c_\xi\varepsilon \le 0.6154\varepsilon$.

At the bonds of p, $|\eta_b| \le C_\varphi^\flat c_\xi\varepsilon + \mathsf{a}_{\mathrm{c}}\varepsilon' \le r_T$, where r_T is the number of item 5 of §5.3; so Lem. 5.17, with (D3), gives $|\partial U(p) - \mathbf{1} - iF_p^\flat| \le 0.72\cdot 0.1\varepsilon$. Adding up, $2\sin(\phi_p/2) = |\partial U(p) - \mathbf{1}| \le 0.8075\varepsilon \le 2\sin(\varepsilon/2)$, and since $2\sin(\cdot/2)$ increases on $[0,\pi]$, assertion (i) follows.

(i′) This is the contrapositive of (i).

(ii) We apply Thm. 5.18, using (Γ3) with $\hat\varepsilon = \tau\varepsilon' \le 17T\varepsilon'$. As to (h1): $|(G\xi)_b| \le \sqrt{2}\|(G\xi)_b\| \le \sqrt{2}C_\varphi^\flat c_\xi\varepsilon$ and $|a_{0,b}| \le \mathsf{a}_{\mathrm{c}}\cdot 17T\varepsilon'$, so $|\eta_b| \le r_T$. As to (h2): the two terms of (Cen) give $17T\cdot\mathsf{N}_{\mathrm{c}}L^{-2}\varepsilon' \le (17/19)\varepsilon$ and $\mathsf{K}_{\mathrm{c}}(17T\varepsilon')^2 \le 0.12\varepsilon$, while $|(\mathrm{d}^V G\xi)_p| \le 4\cdot 1.01\cdot\sqrt{2}\cdot C_\varphi^\flat c_\xi\varepsilon \le 0.8790\varepsilon$; the sum is at most $1.8938\varepsilon \le 1.9\varepsilon$.

The floors hold: (D0) by Step 0 (iv), (0c) and (0b), while (D2) and (D3) are contained in (0c). Holomorphy follows from (2d), from X' being a power series by (e3), and from the chart being polynomial in V' and entire in η.

It remains to show that, for $V' \in \mathcal{U}_Y(x')$ with $Y \supset Y_0(X)$, a set B as in (ii) can be chosen in $Y^{\approx}$. The set $\operatorname{hull}(X)^{+1}$ lies within $M/2$ of $Y_0(X)$ when X is small, and under $Y_0(X)$ otherwise (item 1 of §2.1). Passing to $X^{\approx}$ adds $r(x)$ cubes, which is at most $M\lceil r(x)/L\rceil$ coarse units; then D_w and R_3 are added, and by the second inequality of (Y.7) the resulting set lies in $Y^{\approx}$. Moreover, at x' the conditions (u1) to (u3) imply (e1) to (e3) with $\tau = 1$.

(iii) Let p lie in a cell $\square_p$ off $\mathfrak{Z}$. The bonds within r_w+1 of p lie in cells within $k_t \le r_1$ of $\square_p$, hence off $\mathfrak{b}$, so they pass. The box of radius R_3 about $y(p)$ lies in cells within $\lceil R_3/M\rceil \le r_1$ of $\square_p$, hence off $\mathfrak{b} \supset \mathrm{mk}'$, so no plaquette of D' lies in it. By (i), $\phi_p \le \varepsilon$. For $\chi_\square$, the bonds within R_χ of $\hat\square$ lie in cells within $k_\chi \le r_1$ of $\square$; for χ_X, the bonds within r_w+1 of $X^{\approx}$ lie

within k_t of cells off $\mathfrak{Z}$. In both cases these bonds lie in cells off $\mathfrak{b}$ and pass, and $1_{X^{\approx}}(U) = 1$ because $D(U)$ lies in $\hat{\mathfrak{Z}}$.

(iv) Let p be a plaquette within $\varrho(x) + 1$ fine units of a site of $\hat{\square}$. Then $y(p)$ lies within $\varrho_L + 1$ of $\square$; the bonds within $r_w + 1$ of p lie within $r_w + \varrho_L + 2 \le R_\chi$ of $\hat{\square}$, and they pass; and the box of radius R_3 about $y(p)$ lies within $R_v \le M\, r_1 \le M\, r'$ of $\square$, where V' is tame. By (i), p is tame.

Hence, by the definitions (items E2 and E11 of §4.3, item 3.8 of §4.2), we have $S_\varepsilon(\phi_p) = \phi_p^2/2 = (1/2)F_p \cdot F_p$; $\zeta_p = F_p$ on the reach of ℓ_y; for each of the two radii of the mask, no rough plaquette lies within that radius of y, so that $m_y = 1$; $1 - \cos \mathsf{n}_c = 1 - (1/2)\operatorname{tr} V'(c)\mathsf{M}(U)(c)^{-1}$; and $|\eta_b| \le r_T < \pi$. Comparing factor by factor, $A_\square = e^{V_\square}$.

□

Step 3: the alphabet. The following assertions hold.

(I) $e^{\sum_j n_j} \cdot Z_u = \sum_{\mathfrak{B}} E_\xi \prod_{\alpha \in \mathfrak{B}} \lambda_\alpha$, where the sum runs over the sets $\mathfrak{B}$ of pairwise compatible letters such that $\mathfrak{B}$ holds every forced class.

(II) (Act1): All letters of such a set $\mathfrak{B}$ are actual. An actual zone letter is admitted.

(III) Any two incompatible letters α and β satisfy $\alpha \sim \beta$.

(IV) Families of letters are independent as soon as their sets I lie pairwise at distance more than r_1.

(V) Except for the factors e^{-w_j} and e^{n_j} of a forced letter, every letter α depends on (V', e) on the cells of $I_\alpha^{+r_1}$ alone. Here w_j is determined by (V', e) within $2r'$ cells of mk'_j, and n_j by (V', e) within $3r'$ cells of mk'_j. For $\alpha \in \mathfrak{A}$, the set $I_\alpha^{+r_1}$ lies in $J_\alpha^{+r'}$. The entries e_P enter at the plaquettes of D' alone.

(VI) At a G-valued V', the two formulas for the letter (a) of X agree if V' is tame on $Y_0(X)^{\approx}$, and those for the letter (v) of $\square$ agree if V' is tame on $\square^{\approx}$.

Proof of Step 3. (I) At every ξ we have

$$1 = \sum_{\mathfrak{b} \supset f} \prod_{\mathfrak{b} \setminus f} b_\square \cdot \prod_{\square \text{ off } \mathfrak{b}} (1 - b_\square),$$

the sum running over the sets $\mathfrak{b}$ of cells that contain f. On the support of the term of $\mathfrak{b}$, Step 2 (iii) applies with $\mathfrak{Z} = \mathfrak{b}^{+r_1}$.

There $D(U) = D(U) \cap \hat{\mathfrak{Z}}$, and hence $\Xi_w(U) = \sum_{\mathfrak{X}} \prod_{\mathfrak{X}} w(X, U) \cdot 1\{D(U) \cap \hat{\mathfrak{Z}}$ lies in the union of $\mathfrak{X}$, and each X of $\mathfrak{X}$ holds a plaquette of $D(U) \cap \hat{\mathfrak{Z}}\}$, where $\mathfrak{X}$ runs over the finite families in $\mathcal{P}$ at pairwise distance $> s(x)$. A family with a polymer that holds no cube under $\mathfrak{Z}$ has indicator 0, and we drop it.

Further,

$$\prod_X e^{1E(X)} = \prod_{X^{\approx} \text{ off } \hat{\mathfrak{Z}}} (1 + \lambda_X^a + \lambda_X^h) \cdot \sum_{\mathfrak{Y}} \prod_{\mathfrak{Y}} (e^{1E(X)} - 1),$$

where $\mathfrak{Y}$ runs over the subsets of $\{X : X^{\approx}$ meets $\hat{\mathfrak{Z}}\}$; this follows from $e^{E^a + E^h} = 1 + (e^{E^a} - 1) + e^{E^a}(e^{E^h} - 1)$ and from $\chi_X = 1_{X^{\approx}}(U) = 1$ for the X with $X^{\approx}$ off $\hat{\mathfrak{Z}}$. Likewise $\prod_\square A_\square = [A]_{\mathfrak{Z}}^{\ell} \cdot \prod_{\square \text{ off } \mathfrak{Z}} (1 + \lambda_\square^v)$, $\prod_{\square \text{ off } \mathfrak{b}^{+2r_1}} (1 - b_\square) = \sum_{J_b} \prod_{J_b} (-b_\square)$, and $e^{\sum T_Y} = \prod_Y (1 + (e^{T_Y} - 1))$.

Off the support of the term of $\mathfrak{b}$, the product $\prod_{\mathfrak{b} \setminus f} b_\square \cdot \prod_{\mathfrak{b}^{+2r_1} \setminus \mathfrak{b}} (1 - b_\square) \cdot \sum_{J_b} \prod (-b_\square)$ vanishes. The expanded identity therefore holds at every ξ, and every letter is defined at every ξ. All sums are finite.

We now pass to components: the set consisting of all classes, the cells of $\mathfrak{b} \setminus f$, and the polymers of $\mathfrak{X}$ and of $\mathfrak{Y}$ splits into its components ζ_i under adjacency. Every X of $\mathfrak{X}$ holds a cube under some $\square^{+r_1}$ with $\square \in \mathfrak{b}$, and every X of $\mathfrak{Y}$ has $X^{\approx}$ meeting such a set. So every polymer is adjacent to an object with cells, and each ζ_i is a zone letter.

Objects with cells in different components are farther than $4r_1 + 2$ apart. Hence the sets $\mathfrak{b}_i^{+2r_1}$ are pairwise disjoint, the sets $\mathfrak{Z}_i$ are pairwise disjoint, and $\mathfrak{Z}$ is their union; the products $\prod b_\square$ and $\prod(1 - b_\square)$ and the factor $[A]^\ell_{\mathfrak{Z}}$ factorise.

The indicator factorises as well. A plaquette p of $D(U) \cap \hat{\mathfrak{Z}}_i$ is covered by a polymer X of $\mathfrak{X}$ that holds the cube of p. By (SD) that cube lies under the cell of p, and this cell lies in $\square^{+r_1}$ for a cell $\square$ of an object of ζ_i. So X is adjacent to that object, and $X \in \mathfrak{X}_i$. Conversely, a polymer $X \in \mathfrak{X}_i$ holds a plaquette of $D(U)$ in some $\hat{\mathfrak{Z}}_k$; it is then adjacent to an object of ζ_k, so that $k = i$.

For the converse direction, let the ζ_i be pairwise compatible and hold every class. A cell object of ζ_i inside some f_k is at distance 0 from the object f_k: if $k \in \mathsf{J}(\zeta_i)$, the definition of a cell datum excludes it; if not, ζ_i is incompatible with the letter that holds f_k. Let $\mathfrak{b} := \bigcup_i \mathfrak{b}_i$. It follows that $\mathfrak{b}$ contains f, that the ζ_i are the components, under adjacency, of the set of all their objects, and that the polymers of the union of the $\mathfrak{X}_i$ are at pairwise distance $> s(x)$. The other letters that are compatible with all ζ_i are exactly the following: the letters (b) off $\mathfrak{b}^{+2r_1}$; the letters (v) off $\mathfrak{Z}$; for X with $X^{\approx}$ off $\hat{\mathfrak{Z}}$, the letter (a) or the letter (h) of X, not both; and any letter (t). The factors e^{-w_j} and e^{n_j} are absorbed into the function of the letter of the class j.

(II) A cell of f_k within $4r_1 + 2$ of a cell of $\mathfrak{b}_\zeta$ makes f_k adjacent to an object of ζ; so either f_k is in ζ, or ζ is incompatible with the letter that holds f_k. For a letter (b), a cell of $f_k^{+2r_1}$ belongs to $\mathfrak{b}^{+2r_1}$; for a letter (v), a cell of $f_k^{+r_1}$ belongs to $\mathfrak{Z}$; and for a letter (a) or (h), the fine set under $f_k^{+r_1}$ lies in $\hat{\mathfrak{Z}}$. In each of these three cases, $\mathfrak{b}^{+2r_1}$, $\mathfrak{Z}$ and $\hat{\mathfrak{Z}}$ are the sets of the letter that holds f_k.

It remains to see that an actual zone letter is admitted. Let $P \in D'$ lie in a cell within $4r_1$ of $\mathfrak{b}_\zeta$; this cell is a mark, and so a cell of f. By actuality it lies in f_ζ, and for $\zeta \in \mathfrak{A}$ there is no such P. A mark of the class k is farther than $2s_0 + 4r_1 + 2$ from the marks of every $j \neq k$, so it does not lie in f_j; hence its class is in $\mathsf{J}(\zeta)$.

(III) If two zone letters have cells within $4r_1 + 2$ of each other, then $\operatorname{dist}(\mathfrak{b}^{+2r_1}, \mathfrak{b}'^{+2r_1}) \leq 2 \leq r_1$. If a cell $\square$ of ζ is adjacent to a polymer X of ζ', then a cell of $\square^{+r_1}$, which lies in I_ζ, meets X or $X^{\approx}$, and so it lies in $I_{\zeta'}$. If two zone letters share a polymer, their sets I have a common cell. For polymers of kind p within $s(x)$ of each other, $X^{\approx\approx}$ meets X', where X' denotes the second polymer. For a letter (b), (v), (a) or (h) that is incompatible with ζ, the cell that makes them incompatible lies in both sets I. The letters (a) and (h) of one X have the same set I.

(IV) Each letter depends on ξ only on the bonds of the cells of $I_\alpha^{+k_\chi}$. For (v) this comes from the radius R_χ. For (a), (h), $w(X, U)$ and $e^{1E(X)}$: U_b depends on $(G\xi)_b$, hence on ξ within r_w, and χ_X depends on ξ within $r_w + 1$. The factor Λ_ζ depends on ξ under $\mathfrak{b}^{+2r_1}$, by (7a). A letter (t) depends on ξ within r_w of $\hat{Y}$, and a letter (b) on ξ in $\hat{\square}$. Since $2k_\chi \leq r_1 - 1$, for $\operatorname{dist}(I, I') > r_1$ the sets I^{+k_χ} and I'^{+k_χ} are disjoint; and the ξ_b are independent.

(V) The variable η_b depends on (V', e) within R_3 of $y(b)$. For $G\xi$ this follows from (2b), since $r_G \leq R_3$; for a_0 it follows from the part of (Γ3) by which, at every pair, $a_{0,b}$ depends on V' and X' within R_3 of $y(b)$ alone. That part holds at every pair, hence also for forced letters and for letters (t) near marks, where the hypothesis of (Cen) on the curvature fails. The chart depends on V' at $y(b)$.

For (a), (h), w and e^{1E}, this dependence reaches at most $R_3 \leq M r_1$ from the cells of I_α; for (v) it reaches at most $R_v \leq M r_1$; for (t) the claim follows from (4c) and R_3. The factors $[A]^\ell_{\mathfrak{Z}}$ and 1_{ind} depend on U at the plaquettes within $\varrho(x) + 1$ fine units of $\hat{\mathfrak{Z}}$, hence on (V', e) within $R_3 + \varrho_L + 2 = R_v \leq M r_1$ of $\mathfrak{Z}$.

For w_j the claim is (5b). For n_j: each Y in its sum lies in $\mathrm{mk}_j'^{+2r'}$, (4c) applies, and whether Y is off depends on the marks within r'.

For $\alpha \in \mathfrak{A}$, the set I_α lies in $J_\alpha^{+\varrho_*}$, because $\mathfrak{b}$ lies in J_ζ and the cells that meet $X^{\approx\approx}$ lie in $Y_0(X)^{+\lceil s(x)/L\rceil+2}$; moreover $\varrho_* + r_1 \le r'$. Finally, e enters through X'_P alone, at the plaquettes P with $\partial V'(P) = -\mathbf{1}$.

(VI) For (a), Step 2 (i) with (Y.7) gives $1_{X^{\approx}}(U) = 1$ where $\chi_X = 1$. For (v), we use Step 2 (iv). □

Step 4: the skeleton bound. Let (V', e) be any pair, let ζ_i be pairwise compatible admitted zone letters, let J_b be a finite set of cells off the union of the $\mathfrak{b}_i^{+2r_1}$, let Y be a finite set of cells, and let $0 \le \varsigma \le (\gamma/2)x^{-2}$. Then

$$E_\xi[\prod_i \Lambda_{\zeta_i} \cdot \prod_{J_b} b_\square \cdot e^{(x/2)\varsigma|\varphi_{\mathsf{Y}}|^2}] \le \exp\{4.8v(2/\gamma)\varsigma(2r_1+1)^3 \#\mathsf{Y} + x^{-1/8}(\sum_i \#\mathfrak{b}_i^{+2r_1} + \#J_b)\} \cdot u_b^{\#J_b} \cdot \prod_i e^\circ_{(\mathsf{J}_i,t_i)}.$$

Proof of Step 4. 1. Pointwise, $0 \le \Lambda_\zeta = \hat\lambda_{(\mathsf{J},t)} \cdot 1_{\mathrm{ind},\zeta} \le \hat\lambda_{(\mathsf{J},t)}$. Indeed, $\hat\lambda \ge 0$ by its form, since it is built of exponentials, indicators and Jacobians j (item E12 of §4.3), and the indicator is at most 1. Neither of them depends on the member u.

2. The pairs (J_i, t_i) are cell data, by the definition of a zone letter. Any two of the zone letters ζ_i are compatible, so no object with cells of the one is adjacent to an object with cells of the other; that is, no cell of $\mathfrak{b}_i$ lies within $4r_1+2$ of a cell of $\mathfrak{b}_k$. The cell data are therefore pairwise compatible in the sense of item E12 of §4.3. They are admitted by hypothesis. Item E12 admits cell data that are not connected.

3. By parts 1 and 2 we may apply (7d), which follows from Thm. 4.19 (d°).

The functions $\hat\lambda_{(\mathsf{J},t)}$ make up the skeleton, which is free of members. □

Step 5: packing of the fine polymers under the weak cap.

Lemma 5.19 (At most L^3 fine polymers for each cell of t and each mark of the classes in J)**.** Let (V', e) be any pair, and let $\zeta = (\mathsf{J}, t, \mathfrak{X}, \mathfrak{Y})$ be a zone letter that is admitted at V'. If $\Lambda_\zeta(\xi) \ne 0$ holds for at least one ξ, then $\#\mathfrak{X} \le L^3 \cdot (\#t + \sum_{j\in\mathsf{J}} \#\mathrm{mk}'_j)$. ■

Corollary 5.20 (A factor for each fine polymer)**.** In the bound of Step 4, each factor $e^\circ_{(\mathsf{J}_i,t_i)}$ may be replaced by the number e^*_ζ of the zone letter $\zeta = \zeta_i$, where, for a zone letter $\zeta = (\mathsf{J}, t, \mathfrak{X}, \mathfrak{Y})$,

$$e^*_\zeta := \mathrm{Res}_{(\mathsf{J},t)} \cdot v_f^{-(1/4)\sum_{\mathsf{J}} \#D'_j} \cdot u_t^{\#t/2} \cdot u_1^{1[t \text{ non-empty}]} \cdot v_*^{\#\mathfrak{X}}.$$

■□

Proof of Lem. 5.19. Let ξ be such that $\Lambda_\zeta(\xi) \ne 0$. Since Λ_ζ carries the factor $\prod_{\mathfrak{b}^{+2r_1}\setminus\mathfrak{b}}(1 - b_\square)$ and the indicator $1_{\mathrm{ind},\zeta}$, every cell of $\mathfrak{b}^{+2r_1} \setminus \mathfrak{b}$ passes all of its tests at ξ, and each $X \in \mathfrak{X}$ holds a plaquette p_X of $D(U) \cap \hat{\mathfrak{Z}}$. We write q_X for the cube of p_X; it is a cube of X, and by (SD) it lies under the cell $\square(p_X)$, which is a cell of $\mathfrak{Z}$.

1. By Step 2 (i′), at least one of the following two alternatives holds.
   (A) Some bond β of I° within $r_w + 1$ of p_X has $|\xi_\beta| > c_\xi\varepsilon$.
   (B) Some plaquette P of D' lies in the box of radius R_3 about $y(p_X)$.
2. Case (A). The cell $\square_\beta$ of β lies within $k_t \le r_1$ of $\square(p_X)$, and $\square(p_X)$ is a cell of $\mathfrak{Z} = \mathfrak{b}^{+r_1}$; so $\square_\beta$ belongs to $\mathfrak{b}^{+2r_1}$. As $|\xi_\beta| > c_\xi\varepsilon$, the cell $\square_\beta$ fails its test, and therefore it lies in $\mathfrak{b} = f_{\mathsf{J}} \cup t$. If $\square_\beta \in t$, we charge X to $\square_\beta$, and then q_X lies under $\square_\beta^{+k_t}$. If $\square_\beta \in f_j = \mathrm{mk}_j'^{+s_0}$, we charge X to a mark $\square_m$ of mk'_j within s_0 of $\square_\beta$, and then q_X lies under $\square_m^{+(s_0+k_t)}$.
3. Case (B). The cell that holds P is a mark; it lies within $\lceil R_3/M\rceil \le r_1$ of $\square(p_X)$, hence within $2r_1 \le 4r_1$ of $\mathfrak{b}$. Condition (H17) therefore applies and shows that P belongs to D'_j

for some $j \in \mathsf{J}$. We charge X to that mark $\square_m$; then q_X lies under $\square_m^{+r_1}$, where $r_1 \le s_0 + k_t$. If J is empty, (H17) excludes such a P, and case (B) does not occur.

4. Counting. Put $k = s_0 + k_t$. The polymers of $\mathfrak{X}$ lie at distance $> s(x)$ from one another and q_X lies in X, so the cubes q_X have pairwise sup distance $> s(x)$. By the two cases above, and as $k_t \le k$ and $r_1 \le k$, every X charged to a cell $\square$ has its cube q_X in the fine set under $\square^{+k}$. This fine set is a box of $((2k+1)L)^3$ cubes; it is embedded, and distances within it are those of the torus, because $2k+1 \le 4.4\, r(x) + 1 \le 8r(x) \le N/(2M)$. Here the first inequality is $2k+1 = 2s_0 + 2k_t + 1 \le 4r' + 0.4\, r' + 1 \le 4.4\, r(x) + 1$, since $s_0 = 2r'$, $k_t \le r_1$ by (Geo), $r_1 \le r'/5$ by (Rad) and $r' \le r(x)$; the last one, $8r(x) = s(x) \le N/(2M)$, is (Tor), that is, $N \ge 2M s(x)$.

   We divide this box into subboxes with $s(x)+1$ cubes along each edge. Any two cubes in a common subbox are at distance at most $s(x)$, and so a subbox contains at most one of the cubes q_X. Since $2k+1 \le s(x)+1$, there are at most $\lceil (2k+1)L/(s(x)+1) \rceil^3 \le L^3$ subboxes, and at most L^3 polymers are charged to any one cell.
5. Each $X \in \mathfrak{X}$ is charged either to a cell of t or to a mark of a class of J. Summing over these cells and marks, we obtain $\#\mathfrak{X} \le L^3(\#t + \sum \#\mathrm{mk}'_j)$, the bound of Lem. 5.19.

□

Proof of Cor. 5.20. If $\#\mathfrak{X}_i > L^3(\#t_i + \sum \#\mathrm{mk}'_j)$ for some i, then Λ_{ζ_i} vanishes identically by Lem. 5.19, and the mean of Step 4 equals 0.

In the remaining case we use $\upsilon_* < 1$, $\upsilon_*^{L^3} \ge u_t^{1/2}$, $\upsilon_*^{L^3} \ge \upsilon_f^{1/4}$ and $\#\mathrm{mk}'_j \le \#D'_j$, together with $\#\mathfrak{X} \le L^3(\#t + \sum \#\mathrm{mk}'_j)$ for every zone letter; they give

$$u_t^{\#t/2} \cdot \upsilon_f^{(1/4)\sum \#D'_j} \le \upsilon_*^{L^3(\#t + \sum \#D'_j)} \le \upsilon_*^{L^3(\#t + \sum \#\mathrm{mk}'_j)} \le \upsilon_*^{\#\mathfrak{X}}.$$

Consequently $u_t^{\#t} \le u_t^{\#t/2} \cdot \upsilon_f^{-(1/4)\sum \#D'_j} \cdot \upsilon_*^{\#\mathfrak{X}}$ for each zone letter separately, while the factor u_1 is left as it stands. Since $\mathrm{Res}_{(\mathsf{J},t)}$ and $u_1^{1[t \text{ non-empty}]}$ are factors of both $e^\circ_{(\mathsf{J},t)}$ and e^*_ζ, this is the replacement that the corollary asserts. □

Remark 5.21 (The idea of Step 5). A rough fine plaquette under $\mathfrak{Z}$ can arise only near a failed test or near a rough coarse plaquette, and each such place has already contributed a small factor to the bound: a tested cell with the factor u_t, a rough coarse plaquette with the factor υ_f. A single place carries at most L^3 polymers, because $s(x)$, measured in cubes, exceeds $2(s_0 + k_t)$, measured in cells.

The price of Step 5 is the exponent $1/L^3$ in υ_*, which makes the floor x_9 very large.

Step 6: the product bound, in four cases. To each letter α we attach the size $e_\alpha := e^\circ_\alpha \cdot e^{2x^{-1/8}\#I_\alpha}$, where e°_α is given according to the kind of α.

(a) For a letter (a) at X: $e^\circ(\mathrm{a}) = 4A\, c_a e^{-\kappa(\#X - 1)}$.
(h) For a letter (h) at X: $e^\circ(\mathrm{h}) = 4A\, c_h e^{-\kappa(\#X-1)}$.
(v) For a letter (v) at $\square$: $e^\circ(\mathrm{v}) = 2K_v^\flat p_0^3 x^{-1/2}$.
(t) For a letter (t) at Y: $e^\circ(\mathrm{t}) = e_Y^{1/2} = x^{-20} e^{-\kappa^\sharp \#Y/2}$.
(b) For a letter (b) at $\square$: $e^\circ(\mathrm{b}) = u_b$.

For a zone letter ζ, with e^*_ζ as in Cor. 5.20, we put

$$e^\circ_\zeta = e^*_\zeta \cdot \prod_{\mathsf{J}} e^{|n_j|} \cdot \prod_{\mathfrak{X}} A\, e^{-\kappa(|X|_c - 1)} \cdot \prod_{\mathfrak{Y}} 4A(c_a + c_h) e^{-\kappa(\#X - 1)}.$$

The step asserts that, for a set $\mathfrak{B}$ of letters,

$$E_\xi \Big| \prod_{\mathfrak{B}} \lambda_\alpha \Big| \le \prod_{\mathfrak{B}} e_\alpha$$

in each of the following four cases.

(i) The letters of $\mathfrak{B}$ are pairwise compatible and actual at V'; the bound then holds at every e.
(ii) The letters of $\mathfrak{B}$ are pairwise compatible letters of $\mathfrak{A}$ with J_α inside Y, and either V' is G-valued and tame on $Y^{\approx}$, or V' is a G-valued field on the bonds of $Y^{\approx}$, tame there, and the left side is taken at its extension $\tilde V'$ by $\mathbf{1}$.
(iii) The letters of $\mathfrak{B}$ are of the kinds (a), (v), (t), with J_α inside Y, and $V' \in \mathcal{U}_Y(x')$.
(iv) The letters of $\mathfrak{B}$ are letters (b).

Proof of Step 6. Uniform bounds on the members. By (Y.1), $|z| \le 1/2$ for $z = E^a, E^h, 1E(X)$, and therefore $|e^z - 1| \le |z|e^{|z|} \le 2|z|$.

(a) For a letter (a), $|\lambda^a_X| \le 2A\, c_a e^{-\kappa(\#X-1)}$. At real fields the factor $1_{X^{\approx}}(U)$ keeps U in $\mathcal{U}_X(x)$; at complex fields Step 2 (ii) does so.
(h) For a letter (h), $|\lambda^h_X| \le e^{1/2} \cdot 2A\, c_h e^{-\kappa(\#X-1)}$.

For the factors that a zone letter carries, $|e^{1E(X)} - 1| \le 2A(c_a + c_h)e^{-\kappa(\#X-1)}$ and $|w(X,U)| \le A\, e^{-\kappa(|X|_c-1)}$ hold at every U.

(v) For a letter (v) at $\square$, the restriction of V' to $\square^{\approx}$ lies in $\mathcal{U}_{\{\square\}}(x')$: in case (i) because $\operatorname{dist}(\square, \mathrm{mk}') > s_0 + r_1 > r'$, and in cases (ii), (iii) because $\square^{\approx}$ lies in $Y^{\approx}$. Step 2 (iv), (Γ6) and (Y.1) then give $|\lambda^v_\square| \le e^{|V_\square|} - 1 \le 2K^\flat_v p_0^3 x^{-1/2}$.
(t) *Real fields.* As $a_0 = C\mathsf{g}^V X'$ and $C = GG^{\mathsf{T}}$, we have $a_0 = G(G^{\mathsf{T}}\mathsf{g}^V X')$. A bond lies in 4 plaquettes, $|EX'|_\infty \le \pi$, and $\|\ell^{V*}\ell^V\|_\infty \le (K_*\mathsf{g}_*)(3K_*\mathsf{g}_*)$; these three facts give $|(\mathsf{g}^V X')_b| \le 4\pi \mathsf{N}_\mathsf{B}$. Together with (2e) this yields $|a_{0,b}| \le (C^\flat_\varphi)^2 \cdot 4\pi\mathsf{N}_\mathsf{B} = a_\infty$ at every pair (V', e), and since $\hat Y$ has $3v \cdot \#Y$ bonds, $(1/2)|a_{0,Y}|^2 \le (1/2) \cdot 3v \cdot \#Y \cdot a_\infty^2 = n_a \#Y$; here, as in item E1 of §4.3, $|a_{0,Y}|^2$ is the sum of the $|a_{0,b}|^2$ over the bonds b of I° in $\hat Y$, and the forms with $\|\cdot\|$ in the place of $|\cdot|$ are defined alike.

By (4b),

$$|T_Y| \le x\, e_Y((1/2)|\varphi_Y|^2 + |\varphi_Y||a_{0,Y}|) \le x\, e_Y(|\varphi_Y|^2 + n_a\#Y) = e_Y^{1/2} z_Y, \qquad z_Y := x\, e_Y^{1/2}(n_a\#Y + |\varphi_Y|^2),$$

and then $|e^T - 1| \le |T|e^{|T|} \le e_Y^{1/2} e^{2z_Y}$.

Moreover $\sum_{Y \ni \square} e_Y^{1/2} \le x^{-20} \sum_n (26e)^{n-1} e^{-50n} \le \varsigma''$, where e^{-50n} comes from $\kappa^\sharp/2 = 5\kappa + 20 \ge 50$, by (O5). Let Y denote the union of the sets Y of the letters (t). Then

$$\prod |e^{T_Y} - 1| \le \prod e_Y^{1/2} \cdot \exp\{2x\varsigma''(n_a\#\mathsf{Y} + |\varphi_\mathsf{Y}|^2)\},$$

and the exponential is the tilt of Step 4 at $\varsigma = 4\varsigma'' = 4.04\, x^{-20} \le (\gamma/2)x^{-2}$, by (Y.8).

We now treat the cases of the claim.

(i), (ii), (iv) With the members and the letters (t) bounded as above, what is left is the mean of Step 4, and we bound it by Cor. 5.20. By compatibility, the cells of the letters (b) lie off the sets $\mathfrak{b}_i^{+2r_1}$. The zone letters are admitted: in case (i) by Step 3 (II), and in case (ii) because J is empty, $\mathfrak{b}_\zeta$ lies in J_ζ, J_ζ lies in Y, and $r' \ge 4r_1$. Moreover $|\lambda_\zeta| \le \Lambda_\zeta \cdot \prod e^{|n_j|}$ times the bounds found above for the factors $w(X,U)$ and $e^{1E(X)} - 1$.

What remains is the harmless factor

$$\exp\{[2x\varsigma'' n_a + 4.8 \cdot 4.04 \cdot v(2/\gamma)x^{-20}(2r_1+1)^3]\#\mathsf{Y} + x^{-1/8}(\textstyle\sum \#\mathfrak{b}_i^{+2r_1} + \#J_b)\},$$

and it is at most $\prod e^{x^{-1/8}\#I_\alpha}$, by the second inequality of (Y.8) and because I_ζ contains $\mathfrak{b}_\zeta^{+2r_1}$.

(ii) *The extension by* $\mathbf{1}$. Let V' be a G-valued field on the bonds of $Y^{\approx}$, tame there, and let $\tilde V' = V'$ on these bonds and $\mathbf{1}$ elsewhere. Every plaquette of $D'(\tilde V')$ has a site off $Y^{+r'}$, and

its base site lies within one site of that site; hence all marks of $\tilde V'$ lie at cell distance $\geq r'$ from Y.

By (Y.7), $2\varrho_* + r_1 = 5r_1 + 2\lceil s(x)/L\rceil + 4 \leq r' - 2$. By Step 3 (V), a letter of $\mathfrak{A}$ with J_α inside Y depends on the field in $I_\alpha^{+r_1}$, which lies inside $Y^{+(\varrho_*+r_1)}$ and thus inside $Y^{+(r'-2)}$. Its value is therefore a function of the field on the bonds of $Y^{\approx}$ alone, not of the extension, and e does not enter, since $\phi'_P \leq \varepsilon' < \pi$ there.

A zone letter has $\mathfrak{b}_\zeta$ inside J_ζ, and J_ζ inside Y. As $4r_1 < r'$, no plaquette of $D'(\tilde V')$ lies in a cell within $4r_1$ of $\mathfrak{b}_\zeta$, and so (H17) holds, since J is empty and (H17) then requires nothing more.

For a letter (v), $\square^{\approx}$ lies in $Y^{\approx}$, so the restriction to $\square^{\approx}$ lies in $\mathcal{U}_{\{\square\}}(x')$; the plaquettes that Step 2 (iv) uses have all their sites within $R_v \leq M\, r_1$ of $\square$, and are therefore plaquettes of $Y^{\approx}$, on which $\tilde V'$ is tame. The letters (a) and (h) carry the factor $1_{X^{\approx}}(U)$. The bound of Step 4 holds at every pair.

Hence the bound holds at $(\tilde V', e)$ for every e, and the value of the left side is a function of V' on the bonds of $Y^{\approx}$.

(iii) *Complex fields.* Here no zone letters occur. Let $V' \in \mathcal{U}_Y(x')$. Extended by $\mathbf{1}$, it lies in $\mathfrak{V}$, by (u1).

We write Y_t for the connected set of cells at which a letter (t) of $\mathfrak{B}$ is placed. It is the set J_α of that letter, and so, in case (iii), it lies inside Y, the set with $V' \in \mathcal{U}_Y(x')$. On all of $\mathfrak{V}$, $\|S_k(Y_t; V')\| \leq e_{Y_t}$ by (4b). The pairings are bilinear, and $|\varphi \cdot S\psi| \leq \|S\|\|\varphi\|\|\psi\|$.

Let b be a bond in $\hat Y_t$. It is a bond of a plaquette p whose box of radius R_3 lies in $Y^{\approx}$, because $R_3 + 1 \leq M\, r'$. On that box, (u1) to (u3) give (e1), (e2) and $|\partial V'(P) - \mathbf{1}| < 2\varepsilon'$, so ($\Gamma$3) applies at $\hat\varepsilon = \varepsilon'$ and gives $\|a_{0,b}\| \leq |a_{0,b}| \leq \mathsf{a_c}\varepsilon' \leq 1 \leq a_\infty$, by (0b). Hence $(1/2)\|a_{0,Y_t}\|^2 \leq n_a \# Y_t$. By this inequality, the bound on $\|S_k(Y_t; V')\|$ and the estimate of the pairings, the two bounds of the real case hold here as well: $|T_{Y_t}| \leq e_{Y_t}^{1/2} z_{Y_t}$ and $|e^{T_{Y_t}} - 1| \leq e_{Y_t}^{1/2} e^{2z_{Y_t}}$, where z_{Y_t} is the z_Y defined for real fields, taken at $Y = Y_t$ and with $\|\varphi_{Y_t}\|$ in the place of $|\varphi_Y|$.

By (2a) and (2e), $\|(G\xi)_b\|^2 \leq (C_\varphi^\flat)^2 \sum_{b' \in Y_b} |\xi_{b'}|^2$. Each bond b' lies in at most $3L^3(2r_w + 1)^3 \leq 3L^3(R+1)^3$ windows, and so $\|\varphi_{\mathsf{Y}}\|^2 \leq (C_\varphi^\flat)^2 \cdot 3L^3(R+1)^3 \cdot |\xi_{\mathsf{Y}^{+r_1}}|^2$.

For a single bond, $E_\xi e^{(x/2)\tilde\varsigma|\xi_b|^2} = (1-\tilde\varsigma)^{-3/2} \leq e^{3\tilde\varsigma}$ at $\tilde\varsigma = 4\varsigma''(C_\varphi^\flat)^2 \cdot 3L^3(R+1)^3 \leq 1/2$, since $-\log(1-\varsigma) \leq 1.39\varsigma$ on $[0, 1/2]$. At most $3v(2r_1+1)^3 \# \mathsf{Y}$ bonds occur, and the resulting bound is the content of the third and fourth inequalities of (Y.8).

□

Step 7: the animals at $\varepsilon'' = x^{-1/4}$, and condition (S). The objects of zone letters receive the weights

$$\hat w(\square) = u_l^{1/2} e^{(\kappa+4)(1+s')}, \qquad \hat w_p(X) = 16 \upsilon_* e^{-\kappa(|X|_c - 1) + (\kappa+4)(\#Y_0(X) + s')}, \qquad \hat w_{p'}(X) = 64(c_a + c_h) e^{-\kappa(\#X - 1) + (\kappa+4)(\#Y_0(X)+s')}$$

for a cell $\square$, a polymer X of kind p and a polymer X of kind p′, respectively, and the cores $\mathrm{core}(\square) = \{\square\}$ and $\mathrm{core}(X) = Y_0(X)$, while $\mathrm{core}(f_j)$ consists of the marks of the class j together with their links; here $\#\mathrm{core}(f_j) \leq (1+s')\#\mathrm{mk}'_j$. With these, the following holds.

(o) Each link of a core costs at most s' further cells, so that $\#J_\zeta \leq \sum_{o \in \zeta}(\#\mathrm{core}(o) + s')$. The set I_ζ lies in $J_\zeta^{+s_0+\varrho_*}$, and in $J_\zeta^{+\varrho_*}$ if ζ is not forced.

(i) Put $\mathrm{fp}(\square) = \{\square\}$ and $\mathrm{fp}(f_j) = f_j$, and let $\mathrm{fp}_p(X)$, $\mathrm{fp}_{p'}(X)$ be the set of cells $\square$ adjacent to X. With $a_o := \varepsilon''\#\,\mathrm{fp}(o)^{+(4r_1+2)}$ we have $a_o \leq \#\mathrm{core}(o)$, indeed $a_{f_j} \leq \#\mathrm{mk}'_j$, and, at every cell $\square$,

$$\sum_{o : \mathrm{fp}(o) \ni \square} \hat w(o) e^{a_o} \leq \varepsilon''/2,$$

where the sum runs over cells and polymers. Consequently $\sum_{\zeta\ni o}\prod_{o'\in\zeta}\hat w(o')\le \hat w(o)e^{a_o}$.

(ii) Condition (S) of Lem. 4.50 holds for $\mathfrak A$ with the sizes of Step 6, at every $A\le 16$, with $t=x^{-1/4}$, $\varepsilon_2=x^{-1/8}$ and $\mathsf r=r_1$.

Proof of Step 7. (o) A link has length at most $4r_1+2$ from a cell to a cell, s_0+4r_1+2 from a cell to a mark of an adjacent class, and $2s_0+4r_1+2$ from a mark to a mark. From a cell to $Y_0(X)$ it has length at most r_1+1 when X is of kind p, since a small X lies under Y_0^{+1} and any other X under Y_0, and at most s_0+r_1+1 from a mark; when X is of kind p′, the bound is $r_1+\lceil r(x)/L\rceil+2$ from a cell, and s_0 more from a mark. All of these are $\le 5r'\le s'$, by (Rad).

The set $\mathfrak b$ lies in J^{+s_0}, so $\mathfrak b^{+2r_1}$ lies in $J^{+s_0+2r_1}$; and the cells that meet $X^{\approx\approx}$ lie in $Y_0(X)^{+\lceil s(x)/L\rceil+2}$.

(i) The set $\mathrm{fp}_p(X)$ lies in $Y_0^{+r_1+1}$, and $\mathrm{fp}_{p'}(X)$ in $Y_0^{+r_1+\lceil r(x)/L\rceil+2}$. Using $8r_1+5\le 2(s_0+\varrho_*)+1$, $5r_1+3\le s_0+\varrho_*$ and $5r_1+\lceil r(x)/L\rceil+4\le s_0+\varrho_*$, we find $\#\,\mathrm{fp}(o)^{+(4r_1+2)}\le(2(s_0+\varrho_*)+1)^3\#\mathrm{core}(o)$.

For f_j, the inequality $2(s_0+4r_1+2)+1\le 2(2(s_0+\varrho_*)+1)$ costs a factor 8. Then (Y.2) and $x^{-1/4}\le x^{-1/8}/50$ give $a_o\le\#\mathrm{core}(o)$.

If two objects o,o' are adjacent, then $\mathrm{fp}(o')$ meets $\mathrm{fp}(o)^{+(4r_1+2)}$. Therefore $\sum_{o'\text{ adjacent to }o}\hat w(o')e^{a_{o'}}\le\varepsilon''\#\,\mathrm{fp}(o)^{+(4r_1+2)}=a_o$, and (An0) of Step 8 of the proof of Thm. 4.39 applies.

The sums at a cell $\square$. A cell contributes $u_t^{1/2}e^{\kappa+5}e^{(\kappa+4)s'}$.

A polymer X of kind p holds one of $((2r_1+1)L)^3$ cubes. For small X,

$$\sum_n(52e)^n e^{-\kappa(n-1)}e^{\kappa+5}\le 217.59\,e^{\kappa+5}$$

; the number is the value at $\kappa=6$, and the sum decreases as κ grows.

For X of kind p that is not small, $n>L/2-2$ and $\#Y_0\le 8+432n/L$, and the term n is $(52e)^ne^{-\kappa(n-1)}e^{(\kappa+4)\#Y_0+a_o}$. Since $a_o\le\#Y_0$, the bound on $\#Y_0$ gives $(\kappa+4)\#Y_0+a_o\le(\kappa+5)(8+432n/L)$. The factors of the resulting bound that do not depend on n are e^κ, from $-\kappa(n-1)$, and $e^{8(\kappa+5)}$, from $(\kappa+5)\cdot 8$; their product is $e^{9\kappa+40}=e^{\kappa+5}\cdot e^{35+8\kappa}$. Hence, relative to $e^{\kappa+5}$, the exponent of the term n is at most $n(4.952+2160/L)+35-\kappa[n(1-432/L)-8]$. For $n>L/2-2$ the bracket is positive, so this bound decreases as κ grows. Since $\kappa\ge 6$ is chosen in (O5) only after L has been fixed in (O4), this bound on the exponent, with the factor $e^{35+8\kappa}$ included, is needed at that L for all $\kappa\ge 6$ and, by the monotonicity just established, does not exceed its value at $\kappa=6$. At $\kappa=6$ the bound is $\le -1.04n+83$, the geometric sum costs a factor <3, and the contribution lies below $e^{-L/4}$. Altogether, the polymers of kind p give at most $220e^{\kappa+5}$.

A polymer X of kind p′ holds one of $((2r_1+1)L+2r(x))^3$ cubes. We take $16(c_a+c_h)e^{\kappa+5}e^{(\kappa+4)s'}$ as the unit for one cube. A small X has $\#Y_0(X)=1$ and $a_o\le 1$, and $\sum_{X\ni\text{cube}}e^{-\kappa(\#X-1)}\le 1.2124$; as $64=4\cdot 16$, the small X contribute $1.2124\cdot 4\le 4.85$ in this unit. For X not small, again $n>L/2-2$ and $\#Y_0\le 8+432n/L$, and relative to $e^{\kappa+5}$ the exponent of the term n is at most $4.2581(n-1)+35+2160n/L-\kappa[n(1-432/L)-8]$. The bracket is positive for $n>L/2-2$, so this bound, too, decreases as κ grows; at $\kappa=6$ it is $\le -1.74n+79$, and the contribution lies below $e^{-L/4}$. Altogether, in this unit, the small X contribute at most 4.85 and the X that are not small at most $6-4.85$, so the polymers of kind p′ give at most 6, the coefficient of the third term in the first of the inequalities (Y.4).

The three contributions add up to the left side of the first inequality of (Y.4).

(ii) Let $\beta\in\mathfrak A$ with $\mathrm{dist}(I_\beta,\square)\le r_1$. Then some cell $\square''$ of J_β lies within $\varrho_*+r_1\le s'$ of $\square$, which leaves $(2s'+1)^3$ cells for $\square''$. The set J_β is connected and has a root $\square_0$, namely its first cell, or its first tested cell for a zone letter. Hence, for non-negative summands, $\sum_{\beta:J_\beta\ni\square''}(\cdot)\le$

$\sum_{\square_0} e^{-\mathrm{dist}(\square_0,\square'')} \sum_{\beta \text{ rooted at } \square_0} e^{\#J_\beta}(\cdot)$, where $\sum_{\square_0} e^{-\mathrm{dist}(\square_0,\square'')} \le \mathsf{g}(1) \le 49.98$. Further, $e^{5x^{-1/8}\#I_\beta} \le e^{\#J_\beta}$ by (Y.2), and $e^{d_\beta} = e^{(\kappa+1)s'} e^{(\kappa+1)\#J_\beta}$.

It remains to bound, kind by kind, the sums of $e^\circ_\beta e^{(\kappa+3)\#J_\beta}$ over the letters β rooted at a given cell $\square_0$.

(a), (h) For small X, this rooted sum is at most $L^3 \cdot 1.2124 \cdot 4 \cdot 16(c_a + c_h) e^{\kappa+3}$.
For X not small, X holds one of the $(L+2)^3$ cubes about $\hat{\square}_0$; with $n > L/2 - 2$ cubes and $\#Y_0 \le 8 + 432n/L$, the exponent of the term n, relative to $e^{\kappa+3}$, is at most $4.2581(n-1) + 21 + 1296n/L - \kappa[n(1 - 432/L) - 8]$. Since the bracket is positive for $n > L/2 - 2$, this bound decreases as κ grows, as for the polymers of kind p in (i); at $\kappa = 6$ it is $\le -1.74n + 65$, and the contribution lies below $e^{-L/4}$. Altogether, the letters (a), (h) give at most $5e^{\kappa+3} L^3 \cdot 16(c_a + c_h)$.

(v) The rooted sum is at most $2e^{\kappa+3} K_v^\flat p_0^3 x^{-1/2}$.

(t) The rooted sum is at most $x^{-20} \sum_n (26e)^{n-1} e^{-(5\kappa+20)n + (\kappa+3)n} \le x^{-20}$.

(b) The rooted sum is at most $e^{\kappa+3} u_b$.

A zone letter of $\mathfrak{A}$ has a tested cell, and Res $= 1$ for it. Hence $e^\circ_\zeta e^{(\kappa+3)\#J_\zeta} \le u_1 \prod_o \hat{w}(o)$, and by (i) at the object $\square_0$ the rooted sum is $\le u_1 \hat{w}(\square_0) e^{a_{\square_0}} \le u_1$, with $a_{\square_0}$ the number a_o of (i) at $o = \square_0$; the last inequality holds as $\hat{w}(\square_0) e^{a_{\square_0}} \le \varepsilon''/2 \le 1$, by the sum at a cell in (i).

With $t^{-1} = x^{1/4}$, this is (Y.3); the factor u_1 enters it with its full power, as no part of u_1 is spent on the sums of this step.

□

Step 8: the first two bounds of Thm. 5.15 (b), and the clauses of (ca) *at x' without a size.* The two parts of E'^a in (Fm1) are E'^{new} and $E'^{\mathrm{old}}[u]$. The functions $E'^{\mathrm{new}}(Y,\cdot)$ and $E'^{\mathrm{old}}[u](Y,\cdot)$ are holomorphic on $\mathcal{U}_Y(x')$, invariant under G-valued gauge transformations and 0 at pure gauges, and they are functions of the field on $Y^{\approx}$; moreover, E'^{new} does not depend on u. At every cell $\square$, the sums over $Y \ni \square$ of the suprema of their moduli on $\mathcal{U}_Y(x')$, weighted by $e^{\kappa(\#Y-1)}$, are $\le 0.35\, c_a(x')$ for E'^{new} and $\le 0.248\, A\, c_a(x')$ for $E'^{\mathrm{old}}[u]$.

Proof of Step 8. (Clu) For $\alpha \sim \beta$ in $\mathfrak{A}$ we have $\mathrm{dist}(J_\alpha, J_\beta) \le 2\varrho_* + r_1 \le r' \le s'$. For a cluster C, therefore, $\#\,\mathrm{co}(C) \le \sum(\#J_\alpha + s') - s'$ and $d(C) \ge (\kappa+1)(\#\,\mathrm{co} + s')$, and it follows that $e^{\kappa(\#\,\mathrm{co}\,-1)} \le e^{d(C) - (\kappa+1)s' - \#\,\mathrm{co}\,\kappa}$.

(R) If α is a letter of C with root $\square_0$ and $\mathrm{co}(C) \ni \square$, then $\#\,\mathrm{co} \ge \mathrm{dist}(\square_0, \square) + 1$. We apply Lem. 4.50 to the alphabet $\{\alpha \in \mathfrak{A} : J_\alpha \text{ inside } Y\}$, on which the hypotheses of the lemma hold: by Step 7 (ii), condition (S); by Step 6 (ii) to (iv), the bound $|K(\mathfrak{B})| \le \prod_{\mathfrak{B}} e_\alpha$; by Step 3 (III), the fact that two letters which are not compatible are $\sim$; and by Step 3 (IV), the factorisation $K(\mathfrak{B} \cup \mathfrak{B}') = K(\mathfrak{B}) K(\mathfrak{B}')$ whenever $\mathfrak{B}$ has no letter that is $\sim$ to a letter of $\mathfrak{B}'$. For each kind of letter this gives

$$\sum_{\mathrm{co}(C) \ni \square,\ C \ni \text{ a letter of that kind}} \hat{\Phi}(C) e^{\kappa(\#\,\mathrm{co}\,-1)} \le \sum_{\square_0} e^{-\mathrm{dist}(\square_0,\square)-1} \sum_{\alpha \text{ of that kind rooted at } \square_0} e_\alpha e^{2\varepsilon_2 \#I_\alpha + (\kappa+1)\#J_\alpha - \kappa}.$$

The clauses without a size. For letters (a), (v), (t), the mean $K(\mathfrak{B})(V') = E_\xi \prod \lambda_\alpha$ is holomorphic on $\mathcal{U}_Y(x')$: the letters (a) are holomorphic by Step 2 (ii), the letters (v) by (Γ6), and the letters (t) by (2d) and (4b); X' is a power series; and the integrand is dominated by Step 6 (iii). The convergence is uniform. The dependence on the field is that of Step 3 (V).

We have $\lambda_\alpha(\mathrm{Ad}_u \xi, V'^u) = \lambda_\alpha(\xi, V')$, by (2c), (4b) and (Γ6), because $\mathsf{U}(\mathrm{Ad}_u \eta; V'^u)$ is the gauge transform of $\mathsf{U}(\eta; V')$ by u, taken constant on blocks, and E^a is invariant. The indicators χ and the law of ξ are invariant under Ad_u.

Here u is given on the sites of $Y^{\approx}$ (in part (8.2) below, on those of B_ω) and is extended by $\mathbf{1}$. The transform of the extended field differs from the extension of V'^u only on the bonds that leave

$Y^{\approx}$ (respectively B_ω); on them it is unitary, so it lies in $\mathfrak{V}$. None of the quantities in use depends on the field on these bonds: G depends on the field within r_G, a_0 within R_3 of the blocks that meet $X^{\approx}$, $V_\square$ within R_v of $\square$, and $S_k(Y_t)$ within $\lceil R_2/M\rceil$ cells of Y_t, and all these regions lie inside $Y^{\approx}$ (respectively B_ω).

Together with V', also V'^u lies in $\mathcal{U}_Y(x')$, because the operator norm is invariant under multiplication by unitary matrices on the left and on the right, so that (u1), (u2) and (u3) hold for V'^u as they do for V'. By (4a), $\mathsf{D}_k(Y,V') = -\mathsf{D}^q(Y,V') - [\mathsf{D}^u(Y,V') - \mathsf{D}^u(Y,\mathbf{1})]$, with the same formulas at complex fields.

Here $\mathsf{D}^q(Y,V'^u) = \mathsf{D}^q(Y,V')$ at every field of its domain, by Thm. 3.40 (b): in its proof, the matrices of §3 whose traces make up D^q go over into their conjugates by Ad_u, and as this is an identity of matrices, the traces agree. Likewise $\mathsf{D}^u(Y,V'^u) = \mathsf{D}^u(Y,V')$ at every field of $\mathfrak{V}$, by Thm. 3.42 (b).

An element of $\mathcal{U}_Y(x')$, extended by $\mathbf{1}$, lies in the interior of $\mathfrak{V}$, by (u1). So does its transform by the extended u: on the bonds of $Y^{\approx}$ its norms are the numbers $|V'(c)|$, on the bonds that leave $Y^{\approx}$ it is unitary, and on all other bonds it equals $\mathbf{1}$. This transform differs from the extension of V'^u only on the bonds that leave $Y^{\approx}$, whereas $\mathsf{D}_k(Y,\cdot)$ depends on the bonds of $Y^{+\lceil R_2/M\rceil}$ alone, by (4c), and $\lceil R_2/M\rceil \le r_1 - 8 < r'$. Therefore $\tilde{\mathsf{D}}(Y,V'^u) = \tilde{\mathsf{D}}(Y,V')$ on $\mathcal{U}_Y(x')$.

The function P is invariant by (5c), and $\tilde{\mathsf{D}}$ and P are holomorphic on $\mathcal{U}_Y(x')$ by (4a) and (5c). Thus E'^{new} has the invariance that (ca) demands at x'.

Both $E'^{\mathrm{new}}(Y,\cdot)$ and $E'^{\mathrm{old}}[u](Y,\cdot)$ vanish at pure gauges: the cluster terms by the invariance and the subtraction, and $\tilde{\mathsf{D}}$ and P by (4a) and (5c). No letter of a new cluster depends on u, that is, on any of E^a, E^h and w, and neither $\tilde{\mathsf{D}}$ nor P does; so E'^{new} does not depend on u.

(8.1) By (R), the new clusters contribute at most

$$49.98\, e^{4x^{-1/8}}[2K_v^\flat p_0^3 x^{-1/2} + x^{-20}] \le 102 K_v^\flat p_0^3 x^{-1/2} + x^{-19}.$$

From $r_1 \ge 1+3+5 = 9$ we get $\varrho_* \ge 2r_1 + 3 \ge 21$, and $s_0 \ge 2$; so (Y.2) gives $x^{-1/8} \le (50\cdot 47^3)^{-1} < 2\cdot 10^{-7}$, whence $e^{4x^{-1/8}} < 1.000001$, $2\cdot 49.98\cdot 1.000001 \le 102$ and $49.98\cdot 1.000001\cdot x^{-20} \le x^{-19}$.

The subtraction doubles this bound. Next,

$$c_a(x') = L^{1/2}(p_0'/p_0)^3 c_a(x) \ge 0.594 L^{1/2} E_a p_0^3 x^{-1/2}$$

, by (0b). As $E_a \ge 1500 K_v^\flat L^{-1/2}$ by (O12), the first term is at most $204/(1500\cdot 0.594) \le 0.2290$ in units of $c_a(x')$; and the third inequality of (Y.6) gives

$$2x^{-19} = 2x^{-9}\cdot x^{-10} \le 2x^{-9} c_a(x')/32 \le 0.02 c_a(x')$$

. So, in units of $c_a(x')$, the clusters give at most 0.25.

For $\tilde{\mathsf{D}}$,

$$(K_2^\sharp/2)\varepsilon' \cdot \sum_n (26e)^{n-1} e^{-(9\kappa+40)(n-1)} \le 1.01 (K_2^\sharp/2)\varepsilon' \le (1.01/16) c_a(x'),$$

because $c_a(x') = E_a p_0'^2 \varepsilon'$ and, by the second inequality of (Y.6), $8K_2^\sharp \le E_a p_0'^2$. For P, $K_{\mathrm{q},k} p_0'^2 \varepsilon' \le c_a(x')/32$, as $E_a \ge 32 K_{\mathrm{q},k}$ by (O12).

Adding up, $0.25 + 0.06313 + 0.03125 = 0.34438 \le 0.35$.

(8.2) Consider a cluster C of one letter (a), at a small X whose first cube lies in $\hat{\square}_0$; let o be the centre site of $\square_0$, and B_ω the box of radius ω about o. Then $\Phi(C) = \mathcal{G}_X(V') := E_\xi[\chi_X(e^{E^a(X,U)} - 1)]$ and $\mathrm{co}(C) = \{\square_0\}$.

The blocks that meet $X^{\approx}$ lie within $M + \lceil M\, r(x)/L\rceil$ of o, so B_ω contains the set B of Step 2 (ii). We extend the field on B_ω by $\mathbf{1}$. No quantity of this item depends on the field outside B_ω. By (e1), $|V'(c)| < 1 + \vartheta(x') + \mathsf{g}_x < 1 + \vartheta_0^\flat$, because $\mathsf{g}_x = \vartheta_0^\flat \log L/(4(\log x)^2) < \vartheta_0^\flat/\log x'$; so the extended field lies in the interior of $\mathfrak{V}$, and (2a) to (2f) apply.

By Step 2 (ii) at $\tau = 17T$, the function $\mathcal{G}_X$ is holomorphic and G-invariant on $U^{17T}(\mathsf{B}_\omega; x')$, with $|\mathcal{G}_X| \le 2A\, c_a e^{-\kappa(\#X-1)}$. The restriction to B_ω of an element of $\mathcal{U}_{\{\square_0\}}(x')$ satisfies (u1) to (u3) at x', since $\omega \le M\, r'$.

Thm. 5.22 applies, its hypotheses being $T > 1$, $17T\varepsilon' \le 1/40$, the embedding of B_ω, $3\omega + 1 \le \ell_w(x')$, and (C0) to (C2); it yields

$$|\mathcal{G}_X(V') - \mathcal{G}_X(\mathbf{1})| \le 4T^{-2} A\, c_a(x) e^{-\kappa(\#X-1)}.$$

There are L^3 choices of the first cube, and $\sum_{X \ni \text{cube}} e^{-\kappa(\#X-1)} \le 1.2124$. The total is therefore $4.85 \cdot 361 \cdot \mathsf{N}_c^2 \cdot A\, c_a(x) \le (2947.6\, \mathsf{N}_c^2 / L^{1/2}) A\, c_a(x') \le 0.2457\, A\, c_a(x')$, by (L2^{su}).

Theorem 5.22 (Bounded invariant holomorphic functions vary little)**.** Let (C0), (C1), (C2) of (0d) hold, suppose that $T > 1$, $17T\varepsilon' \le 1/40$, $2\omega + 1 < N$ and $3\omega + 1 \le \ell_w(x')$, let o be a coarse site, and let B_ω be the box of radius ω about o. Let Ψ be a bounded holomorphic function on $\mathcal{U} := U^{17T}(\mathsf{B}_\omega; x')$ that is invariant under G-valued gauge transformations. Then every $\mathrm{SL}(2,\mathbb{C})$ field V' on the bonds of B_ω that satisfies (u1), (u2) and (u3) at x' obeys

$$|\Psi(V') - \Psi(\mathbf{1})| \le 2T^{-2} \sup|\Psi|.$$

■□

Proof of Thm. 5.22. Step 1: the curve. We follow Step (1) of the proof of Thm. 3.34, with ω in place of R_3. The field V' has the properties (h2) and (h3) of item C.II 1.4 of §3 with $\hat{\varepsilon} = \varepsilon'$. Along the comb paths we define the comb gauge: $\mathsf{k}(y) := V'(\Gamma(o,y))$ for every site y of B_ω, and $\mathsf{W}(c) := \mathsf{k}(c_-) V'(c) \mathsf{k}(c_+)^{-1}$ for every bond c of B_ω.

By Lem. 3.27 (a) and by (C0), $|\mathsf{k}(y)| = |\mathsf{k}(y)^{-1}| \le 2$ and $|\mathsf{W}(c) - \mathbf{1}| \le e^{18\omega\varepsilon'} - 1 \le 19\omega\varepsilon' \le 0.106$. We put $A_c := -i \log \mathsf{W}(c)$; this matrix is traceless, and $|A_c| \le \mathsf{a} := 21\omega\varepsilon'$.

For complex z let $V'_z(c) := \mathsf{k}(c_-)^{-1} e^{izA_c} \mathsf{k}(c_+)$. This field is entire in z and takes its values in $\mathrm{SL}(2,\mathbb{C})$, and $V'_1 = V'$. Moreover $V'_z(c) = V'(c)(\mathbf{1} + \delta_c)$, where $\mathbf{1} + \delta_c = \mathsf{k}(c_+)^{-1} e^{i(z-1)A_c} \mathsf{k}(c_+)$.

Now let $|z| < T$. From (C1), together with $\ell_w(x') \ge 18$ and $\mathsf{g}_x \le 1$, we obtain $(T+1)\mathsf{a} \le 10^{-3}$. Consequently $|\delta_c|$ and $|(\mathbf{1} + \delta_c)^{-1} - \mathbf{1}|$ are both $\le 4(e^{(T+1)\mathsf{a}} - 1) \le \delta := 85(T+1)\omega\varepsilon'$. We show that V'_z satisfies the three conditions that define $\mathcal{U}$.

(e1) For every bond c of B_ω, $|V'_z(c)| < (1 + \vartheta(x'))(1+\delta) \le 1 + \vartheta(x') + 1.01\delta \le 1 + \vartheta(x') + \mathsf{g}_x$, the last inequality by (C1).

(e2) Let π be a path in B_ω of $n \le \ell_w(x')$ bonds. In the product $V'_z(\pi)$ we move every factor $(\mathbf{1} + \delta_c)^{\pm 1}$ to the left, past the partial products of V'; by (u2), these partial products have norm < 2. In this way $|V'_z(\pi)| \le (1 + 4\delta)^n |V'(\pi)| < e^{4n\delta} \mathsf{b}(x') \le \mathsf{b}(x') + \mathsf{g}_x$, since (C1) gives $4n\delta \le (340/2640)\mathsf{g}_x$.

(e3) Let P be a plaquette of B_ω with base site y_P, and let $A_1, \dots, A_4$ be the matrices A_c of its four bonds. Then $\partial V'_z(P) = \mathsf{k}(y_P)^{-1} \Pi_z \mathsf{k}(y_P)$ with $\Pi_z := e^{izA_1} e^{izA_2} e^{-izA_3} e^{-izA_4}$. As $4T\mathsf{a} = 84T\omega\varepsilon' \le 1$ by (C0), the estimate (E) of item C.I 1.1 of §3 yields $|\Pi_z - \mathbf{1} - iz(\mathrm{d}'A)_P| \le (4|z|\mathsf{a})^2$, where $(\mathrm{d}'A)_P$ is the coarse coboundary. At $z = 1$ we have $|\Pi_1 - \mathbf{1}| = |\mathsf{k}(y_P)(\partial V'(P) - \mathbf{1})\mathsf{k}(y_P)^{-1}| < 8\varepsilon'$, so that $|(\mathrm{d}'A)_P| < 8\varepsilon' + 16\mathsf{a}^2$. Therefore $|\partial V'_z(P) - \mathbf{1}| \le 4|\Pi_z - \mathbf{1}| < 32T\varepsilon' + 128T^2\mathsf{a}^2 = (32 + 56448T\omega^2\varepsilon')T\varepsilon' < 34T\varepsilon'$, by (C2).

Thus $V'_z \in \mathcal{U}$ for $|z| < T$. Hence $f(z) := \Psi(V'_z)$ is a holomorphic function on $|z| < T$, with $|f| \le \mathsf{s} := \sup|\Psi|$ and $f(1) = \Psi(V')$.

Step 2: complex gauges. We follow Step (2) of the proof of Lem. 3.33. The set $\mathcal{U}$ is open, as it is given by finitely many strict inequalities. For $\mathsf{Z} \in \mathcal{U}$ and $\lambda : \mathsf{B}_\omega \to \mathfrak{g}_{\mathbb{C}}$ we define $\mathfrak{d}(\mathsf{Z}; \lambda) := (d/d\sigma)\Psi(\mathsf{Z}^{\exp(i\sigma\lambda)})$ at $\sigma = 0$. Because Ψ is holomorphic, $\mathfrak{d}$ is $\mathbb{C}$-linear in λ; because Ψ is invariant, $\mathfrak{d}$ vanishes for $\mathfrak{g}$-valued λ. An arbitrary λ has the form $\lambda_1 + i\lambda_2$ with $\mathfrak{g}$-valued λ_1 and λ_2, and so $\mathfrak{d} = 0$ for every λ.

It follows that $\Psi(\mathsf{Z}^{u_s})$ is constant along every C^1 curve $s \mapsto u_s$ of $\mathrm{SL}(2,\mathbb{C})$-valued gauges for which $\mathsf{Z}^{u_s} \in \mathcal{U}$, since its derivative in s is $\mathfrak{d}(\mathsf{Z}^{u_s}; -i(du_s/ds)u_s^{-1})$, which vanishes.

We write $\mathsf{k}(y)^{-1}$ in polar form, $\mathsf{k}(y)^{-1} = k_0(y)e^{h(y)}$, where k_0 is G-valued and h is hermitian and traceless, and we set $u_s := k_0 e^{sh}$ for $0 \le s \le 1$. Then $V_0' = \mathbf{1}^{u_1}$.

Let π be a path from y to y''. Cordes' inequality, which is proved in Step (1) of the proof of Lem. 3.33, gives $|\mathbf{1}^{u_s}(\pi)| = |e^{sh(y)}e^{-sh(y'')}| \le |e^{h(y)}e^{-h(y'')}|^s = |V_0'(\pi)|^s \le |V_0'(\pi)|$. By Step 1 of the present proof, $V_0' \in \mathcal{U}$. Hence $\mathbf{1}^{u_s}$ satisfies (e1) and (e2), and its curvature is $\mathbf{1}$; so $\mathbf{1}^{u_s} \in \mathcal{U}$ for every s. The constancy just proved, applied with $\mathsf{Z} = \mathbf{1}$, and the invariance of Ψ under the G-valued gauge $u_0 = k_0$ give $f(0) = \Psi(\mathbf{1}^{u_1}) = \Psi(\mathbf{1}^{u_0}) = \Psi(\mathbf{1})$.

Step 3: the derivative of f at 0. We show that $f'(0) = 0$. Let $\mathsf{Z}_z := (e^{izA_c})_c$, so that $V_z' = \mathsf{Z}_z^{u_1}$. The set $\{\mathbf{1}^{u_s}\}$ is compact and lies in the open set $\mathcal{U}$; hence there is $z_0 > 0$ such that $\mathsf{Z}_z^{u_s} \in \mathcal{U}$ for $|z| < z_0$ and for all s. For these z, Step 2 of the present proof and the invariance of Ψ give $f(z) = \Psi(\mathsf{Z}_z^{u_0}) = \Psi(\mathsf{Z}_z)$.

Let $\Theta(A) := (d/dz)\Psi(\mathsf{Z}_z)$ at $z = 0$; it is $\mathbb{C}$-linear in A. For a constant $v \in G$ we have $\mathsf{Z}_z^v = (e^{iz\,\mathrm{Ad}_v A_c})_c$, and therefore $\Theta(\mathrm{Ad}_v A) = \Theta(A)$. For a traceless matrix Y, the mean of $\mathrm{Ad}_v Y$ over G commutes with G and is traceless, so it is 0. Averaging over v and using linearity, we find $\Theta(A) = \Theta(\int \mathrm{Ad}_v A\, dv) = 0$; thus $f'(0) = 0$.

Step 4: the bound. Since $f'(0) = 0$, the function $(f(z) - f(0))/z^2$ is holomorphic on $|z| < T$, and $|f - f(0)| \le 2\mathsf{s}$. The maximum principle on the circles $|z| = T_1$, with $T_1 \to T$, gives at the point $z = 1$ the bound $|f(1) - f(0)| \le 2\mathsf{s}T^{-2}$; as $f(1) = \Psi(V')$ and $f(0) = \Psi(\mathbf{1})$, this is the claim of the theorem.

The theorem and its proof are independent of the rows ℓ. □

(8.3) Here the cluster consists of one letter of kind (a) at a set X that is not small, so $n = \#X > L/2 - 2$; its core is $\mathrm{co}(C) = Y_0(X) \ni \square$, and $\#Y_0 \le 8 + 432n/L$. For such a cluster, $|\mathcal{G}_X(V') - \mathcal{G}_X(\mathbf{1})| \le 4A\, c_a(x) e^{-\kappa(n-1)}$. The cell $\square$ meets X^{+1}, so X contains one of $(L+2)^3$ cubes, and, for each n, at most $(26e)^{n-1}$ sets X contain a given cube.

Summing over these clusters, we obtain at most $4A\, c_a(x)(L+2)^3 \sum_{n>L/2-2} \exp\{(4.2581 - \kappa)(n-1) + \kappa(7 + 432n/L)\}$. For $\kappa \ge 6$ we have $\kappa - 4.2581 \ge 0.29\kappa$. Since $L > 10^{13}$, the exponent is at most $-0.289\kappa(n-1) + 7.1\kappa$. Summing this geometric majorant gives

$$4A\, c_a(x)(L+2)^3 \frac{\exp\{-\kappa[0.289(L/2-3) - 7.1]\}}{1 - e^{-0.289\kappa}}.$$

Here $0.289(L/2-3) - 7.1 \ge (0.8/6)L$, $4/(1 - e^{-0.289\kappa}) < 5$, and $5(L+2)^3 < e^{0.3L}$. The contribution is thus at most $e^{-L/2} A\, c_a(x) \le 10^{-3} A\, c_a(x')$.

(8.4) Now $n(C) \ge 2$, and we root C at a letter of kind (a). The sizes are computed with the number A for which $u \in M_A^{\mathrm{su}}(x)$, and condition (S) holds for them, by Step 7 (ii). Claim (2) of Lem. 4.50 then yields the bound $t\, e_\alpha e^{2\varepsilon_2 \# I_\alpha + d_\alpha}$, in which a single size e_α occurs; it is linear in A.

Two applications of (R) give $2 \cdot 49.98 \cdot 4.85 \cdot e^{0.08} \cdot x^{-1/4} L^3 A\, c_a(x) \le 525.2\, L^3 x^{-1/4} A\, c_a(x)$, where the letters at sets X that are not small are bounded by the count of case (8.3). This is $\le 500 e^{\kappa+3} L^3 x^{-1/4} A\, c_a(x) \le 842 e^{\kappa+3} L^{5/2} x^{-1/4} A\, c_a(x') \le 10^{-3} A\, c_a(x')$, the last step by (Y.6).

The three cases (8.2), (8.3) and (8.4) cover all old clusters: those that consist of one letter of kind (a) at a small set X, those that consist of one letter of kind (a) at a set X that is not small, and those with $n(C) \ge 2$. Adding the bounds of these cases, we obtain the second bound of Step 8: in units of $A\, c_a(x')$, the sum $\sum_{Y \ni \square} e^{\kappa(\#Y - 1)} \sup_{\mathcal{U}_Y(x')} |E'^{\mathrm{old}}[u](Y, \cdot)|$ is at most $0.2457 + 0.001 + 0.001 \le 0.248$. □

Step 9: E'^h; the third bound of Thm. 5.15 (b), and the clauses of (ch) *at x' without a size.* For every set Y of cells, $E'^h(Y,\cdot)$ is a complex, Borel and gauge invariant function of the G-valued fields on the bonds of $Y^{\approx}$ that are tame there, and it is independent of e. Moreover, $\sum_{Y\ni\square} e^{\kappa(\#Y-1)} \sup|E'^h(Y,\cdot)| \le c_h(x')$ for every $A \le 16$ with $u \in M_A^{\mathrm{su}}(x)$.

Proof of Step 9. The function $E'^h(Y,\cdot)$ is well defined, and its series converges absolutely, by Lem. 4.50 for the alphabet $\{\alpha \in \mathfrak{A} : J_\alpha \text{ inside } Y\}$, together with Step 6 (ii), (iv) and Step 7 (ii). It is Borel, being the sum of a uniformly convergent series of integrals of Borel functions; that the letters are jointly Borel is shown in the verification of hypothesis (i) of Lem. 5.23 in Step 10 (9).

The function is gauge invariant: by the argument of Step 8, $b_\square$, $A_\square$, $w(X,U)$, E^a, E^h, $1_{X^\approx}(U)$ and 1_{ind} are invariant under $(\mathrm{Ad}_u\,\xi, V'^u)$. For $A_\square$ this rests on (6e), since in a hard cluster $A_\square$ appears also at failed tests, and there it is not $e^{V_\square}$.

For the dependence on the field we use Step 3 (V) and the extension by $\mathbf{1}$ in Step 6: $\Phi(C)(\cdot)$ is evaluated at the extension of the field by $\mathbf{1}$, and only the values of the field in $Y^{+(r'-2)}$ enter. On $Y^{\approx}$ every angle satisfies $\phi'_P \le \varepsilon' < \pi$, so no e enters.

We now bound the size. A hard cluster contains a letter of kind (h), a letter of kind (b) or a zone letter, and in each case we use (R) of Step 8.

(h) For a letter of kind (h), the bound is $49.98 \cdot L^3 \cdot 4.85 \cdot e^{0.08} \cdot A\, c_h(x) \le 262.6\, L^3 A\, c_h(x)$; the sets X that are not small add a negligible amount, by the count of case (8.3) of Step 8 with $c_h(x)$ in place of $c_a(x)$. Since $A \le 16$, this is $\le 10^3 L^3 \cdot 16 c_h(x) \le c_h(x')/2$, by the first inequality of (Y.5), because $c_h(x)/c_h(x') = e^{-c_h(p_0^2 - p_0'^2)}$.

(b) For a letter of kind (b), the bound is $49.98\, e^{4x^{-1/8}} u_b$.

Zone For a zone letter ζ, $e_\zeta e^{2\varepsilon_2 \#I_\zeta + (\kappa+1)\#J_\zeta} \le e_\zeta^\circ e^{(\kappa+2)\#J_\zeta} \le u_1 \prod_o \hat{w}(o)$, where o runs over the objects of ζ. Summing over the zone letters rooted at $\square_0$ by means of the bound for animals, we obtain at most $u_1 \cdot (1/2)x^{-1/4}$.

The letters of kind (b) and the zone letters together give $\le 50(2u_b + u_1) \le (1/2)e^{-\delta_3 p_0^2/2} \le c_h(x')/2$, by the second inequality of (Y.5), by $c_h \le \delta_3/2$ and by $p'_0 \le p_0$; with the bound $c_h(x')/2$ for kind (h), this gives the bound of Step 9. □

Step 10: Ξ' and w'; clause (a) *of Thm. 5.15, the strong cap of the output in clause* (b)*, and the mean over e.* The following hold.

(8a) $e^{\sum n_j} Z_u = \sum_{\mathfrak{G}} \prod_{\gamma\in\mathfrak{G}} K(\gamma) \cdot \exp\{\sum_{C \not\sim \mathfrak{G}} \Phi(C)(V')\}$, where $\mathfrak{G}$ runs over the sets of rough polymers that are pairwise not related by $\sim$ and that contain every forced class exactly once.

(8b) The formula of Thm. 5.15 (a) holds at (V', e), with $\Xi' = \sum_{(\mathfrak{G},\mathfrak{N},\mathfrak{N}^\flat)} \prod K(\gamma) \prod (e^{-\Phi^\natural(C)} - 1) \prod (e^{\tilde{\mathrm{D}}(Y)} - 1)$.

(8c) $\Xi' = \sum_{\{\mathsf{Y}_i\}} \prod_i w'(\mathsf{Y}_i, V', e)$, the sum running over the families of (cΞ) at x'. Here $w'(\mathsf{Y},\cdot,\cdot)$ depends on V' only through the bonds of $\mathsf{Y}^{+s'}$, and on e_P only at the plaquettes of cells of Y.

(8d) At every pair (V', e), $|w'(\mathsf{Y}, V', e)| e^{\kappa(|\mathsf{Y}|_c - 1)} \le \exp\{-c_\Xi x' \varepsilon' \sum_{D' \cap \mathsf{Y}} \phi'_P\}$.

(9) The mean $w'(\mathsf{Y}, V') = \int w'(\mathsf{Y}, V', e)\, de$ is a complex, Borel and gauge invariant function, which depends on V' only through the bonds of $\mathsf{Y}^{+s'}$ and obeys the same bound; and $\Xi' = \sum_{\{\mathsf{Y}_i\}} \prod w'(\mathsf{Y}_i, V')$ at every V'.

Proof of Step 10. We begin with the normaliser n_j, which we treat by (4a). A set Y is on exactly when $1_{Y^\approx}(V') = 1$. A short Y is connected and lies in f, and the parts f_j of f are farther than $4r_1 + 2$ apart; hence it lies in exactly one f_j. An off Y inside f_j has a cell within r' of a mark, and this mark lies within $r' + s_0$ of mk'_j, so it belongs to class j. Consequently $|n_j| \le (2r'+1)^3 \#\mathrm{mk}'_j \cdot 1.01 \cdot K_1^\sharp/2$.

The function n_j is real, Borel and gauge invariant. It is Borel because on each of the finitely many Borel sets $\{\mathrm{mk}'(V') = \mathsf{d}\}$ it is a finite sum of functions $\tilde{\mathsf{D}}(Y,\cdot)$, and these are continuous on G-valued fields, the rough ones included, by (4a).

A long Y has a cell within r' of a mark and a cell farther than $2r'$ from it, so $\#Y > r'$. By (Y.9) we then have $|\tilde{\mathsf{D}}(Y)| \le 1/2$, and therefore $|e^{\tilde{\mathsf{D}}(Y)} - 1| e^{(\kappa+1)(\#Y+s')} \le \hat{w}(Y) := K_1^\sharp e^{-\kappa^\sharp(\#Y-1)} e^{(\kappa+1)(\#Y+s')}$.

The exponent per cell is $4.2581 - \kappa^\sharp + \kappa + 1 \le -(9\kappa + 34.74)$, and $s' = 8r'$. Hence $\sum_{Y \text{ long, within } r' \text{ of } \mathrm{mk}'_j} \hat{w}(Y) \le (2r'+1)^3 \#\mathrm{mk}'_j \cdot 1.001 \cdot K_1^\sharp e^{\kappa+1} e^{-(\kappa+26.74)r'} \le \#\mathrm{mk}'_j$, the last step by (Y.9).

Each set Y is on, short or long, and exactly one of these holds; this partition gives $\sum_Y \tilde{\mathsf{D}} = \sum_Y 1_{Y\approx} \tilde{\mathsf{D}} + \sum_j n_j + \sum_{\text{long}} \tilde{\mathsf{D}}$.

(8a) We split each set $\mathfrak{B}$ of Step 3 (I) into its $\sim$-components and let $\mathfrak{G}$ consist of the components that contain a forced letter; by (Act1), they are rough polymers. The union of the sets I_α, where α runs over the letters of the polymers of $\mathfrak{G}$, contains f^{+2r_1}. The remaining components are $\sim$-connected compatible sets of letters of $\mathfrak{A}_{\mathfrak{G}} := \{\alpha \in \mathfrak{A} : \alpha \not\sim \text{ any letter of } \mathfrak{G}\}$.

All letters of $\mathfrak{A}_{\mathfrak{G}}$ are actual. Indeed, $\mathrm{dist}(I_\alpha, f^{+2r_1}) > r_1$ for each of them; for a zone letter, $\mathrm{dist}(\mathfrak{b}, f) > 5r_1 \ge 4r_1 + 2$, and for a letter of kind (a) or (h), $X^{\approx}$ lies off the fine set under f^{+r_1}. The independence of Step 3 (IV) and Lem. 4.50 for $\mathfrak{A}_{\mathfrak{G}}$, with complex weights and with Step 6 (i), now give (8a); condition (S) for the smaller alphabet is weaker.

(8b) Every cluster $C \not\sim \mathfrak{G}$ is on. Indeed, its letters satisfy $\mathrm{dist}(J_\alpha, \mathrm{mk}') > s_0 + 3r_1$, and each cell of the joining path lies within r' of some J_α, so $\mathrm{dist}(\mathrm{co}(C), \mathrm{mk}') > r'$.

We sort the clusters by type. For an analytic cluster C we write $\Phi(V') = \Phi(\mathbf{1}) + (\Phi(V') - \Phi(\mathbf{1}))$ if $C \not\sim \mathfrak{G}$, $0 = \Phi(\mathbf{1}) + (\Phi(V') - \Phi(\mathbf{1})) - \Phi(V')$ if $C \sim \mathfrak{G}$ is on, and $0 = \Phi(\mathbf{1}) - \Phi(\mathbf{1})$ if $C \sim \mathfrak{G}$ is off. Hard and constant clusters are sorted in the same way, with these changes: a hard or constant $C \not\sim \mathfrak{G}$ keeps its value $\Phi(C)(V')$; for a hard $C \sim \mathfrak{G}$ that is on, and for a constant $C \sim \mathfrak{G}$, we write $0 = \Phi(C)(V') - \Phi(C)(V')$; and a hard $C \sim \mathfrak{G}$ that is off contributes 0, as $\Phi^\natural(C) = 0$ there.

Summing over all clusters, we obtain

$$\sum_{C \not\sim \mathfrak{G}} \Phi(C)(V') = [\sum_{\text{analytic}} \Phi(C)(\mathbf{1}) + \sum_{\text{constant}} \Phi(C)] + \sum_Y 1_{Y\approx}(V')[E'^a - \tilde{\mathsf{D}} + P + E'^h](Y, V') - \sum_{C \sim \mathfrak{G}} \Phi^\natural(C),$$

where all sums converge absolutely and $|\Phi^\natural| \le \hat{\Phi}$. We insert this identity into the formula of Step 1, together with the partition of the sets Y into on, short and long ones, and we expand $e^{-\sum \Phi^\natural}$ and $e^{\sum_{\text{long}} \tilde{\mathsf{D}}}$; this gives (8b).

(8c) The graph of (Fm4) for a triple $(\mathfrak{G}, \mathfrak{N}, \mathfrak{N}^\flat)$ of (8b) has the letters of $\mathfrak{G}$, the clusters of $\mathfrak{N}$ and the sets of $\mathfrak{N}^\flat$ as its vertices, and it joins two of them whenever their cores lie within s' of each other. Its components give sets Y_i at pairwise distance $> s'$, each with a triple as in (Fm4). Indeed, two letters of one γ that are related by $\sim$, and also a cluster C and a letter of $\mathfrak{G}$ to which C is related by $\sim$, have cores within $2(s_0 + \varrho_*) + r_1 \le 5r' \le s'$ of each other; and a long Y has a cell within r' of a mark, which lies in the core of a forced letter.

Hence each Y_i contains a mark, D' lies in the union of the Y_i, and no Y_i has a mark in $\mathsf{Y}_i^{+s'} \setminus \mathsf{Y}_i$; in particular, none has a mark in $\mathsf{Y}_i^{+(s'-1)} \setminus \mathsf{Y}_i$.

Conversely, two letters, one of the triple of Y_i and the other of the triple of $\mathsf{Y}_{i'}$ with $i \ne i'$, satisfy $\mathrm{dist}(I_\alpha, I_\beta) > s' - 2(s_0 + \varrho_*) \ge r_1$; so they are not related by $\sim$, and hence they are compatible. A mark that makes a set Y of $\mathfrak{N}_i^\flat$ off lies in $\mathsf{Y}_i^{+r'}$, and as $r' \le s' - 1$, it lies in Y_i. In the remaining case of the definition of w' in (Fm4), when $\mathrm{mk}'(V')$ has no cell in $\mathsf{Y}^{+(s'-1)} \setminus \mathsf{Y}$, the marks outside Y are farther than $s' - 1 \ge 5r' \ge 2s_0 + 4r_1 + 2$ from Y, and so $\mathrm{mk}' \cap \mathsf{Y}$ is a union of classes.

For the dependence on the field we use Step 3 (V), which is valid at every pair, since by (Γ3) each $a_{0,b}$ depends at every pair only on V' and X' within R_3 of $y(b)$; here $I_\zeta^{+r_1}$ lies inside $J_\zeta^{+s_0+\varrho_*+r_1}$. Whether a letter is actual, the classes, and whether a set is on, off, short or long depend only on the marks within $2s_0+4r_1+2$ of the core.

Measured in cells from Y, the reach of each ingredient is as follows: for a forced letter it is $s_0+\varrho_*+r_1 \le 3r'$, for w_j it is $2r'$, for n_j it is $3r'$, for a long Y it is $\lceil R_2/M\rceil \le r_1$, for $\Phi^\natural(C)$ it is r', for the marks that decide whether a letter is actual, what the classes are, and whether a set is on, off, short or long, it is $2s_0+4r_1+2 \le 5r'$, and for the first clause of the definition of w' in (Fm4), which sets $w'(\mathsf{Y},V',e) := 0$ if $\mathrm{mk}'(V')$ has a cell in $\mathsf{Y}^{+(s'-1)} \setminus \mathsf{Y}$, it is $s'-1$. All of these are $\le s'-1$.

Whether a cell is a mark depends on the plaquettes based in it, whose sites lie in the cell or in a neighbour of it; so only bonds with both ends in $\mathsf{Y}^{+s'}$ enter. The choice e_P enters only at $P \in D'$, hence at plaquettes of marks within $s'-1$ of Y; where $w' \neq 0$, these marks are cells of Y. The first clause of the definition of w' in (Fm4) is independent of e.

(8d) By Step 6 (i), $|K(\gamma)| \le \prod e_\alpha$. As $|\mathsf{Y}|_c \le \sum_{\text{elements}}(\#\text{core}+s') - s'$, we have $e^{\kappa(|\mathsf{Y}|_c-1)} \le \prod_{\text{elements}} e^d$. Moreover $|e^{-\Phi^\natural}-1| \le 2\hat\Phi$: indeed, $\hat\Phi \le t\varepsilon_2 \le 1$ by (S), and $e^s-1 \le (e-1)s \le 2s$ on $[0,1]$; besides, $t\varepsilon_2 = x^{-3/8} \le \log 2$.

We apply Lem. 4.51, with $t \le 1/2$, taking for F the forced letters of $\mathfrak{G}$, for $\mathfrak{B}_o$ its other letters, and for $\mathfrak{M}$ the clusters, with $\hat w_C = 2\hat\Phi e^{d(C)}$; the condition of the lemma holds, with $2t\varepsilon_2 \le \varepsilon_2$. The sets $\mathfrak{N}^\flat$ contribute $\prod(1+\hat w(Y)) \le \exp\sum \hat w(Y) \le \exp\sum_{\mathsf{F}} A_\zeta$, where $A_\zeta := \sum_{\mathsf{J}(\zeta)} \#\mathrm{mk}'_j$.

Therefore $|w'|e^{\kappa(|\mathsf{Y}|_c-1)} \le \sum_{\mathsf{F}} \prod_{\zeta\in\mathsf{F}} e_\zeta e^{d_\zeta+2\varepsilon_2\#I_\zeta+A_\zeta}$, where F runs over the sets of forced letters whose classes are the classes with marks in Y, each taken once. Here the numbers e_ζ are used for actual letters and for the others alike, which gives a majorant.

By (Y.2), $e^{4x^{-1/8}\#I_\zeta} \le e^{\#J_\zeta}$, and with $u_1 \le 1$ we obtain $e_\zeta e^{d_\zeta+2\varepsilon_2\#I_\zeta+A_\zeta} \le e^{(\kappa+1)s'}\prod_{o\in\zeta}\hat w(o)$, with the weights $\hat w(\square), \hat w_p, \hat w_{p'}$ of Step 7 and

$$\hat w(f_j) := \upsilon_f^{-\#D'_j/4}\exp\{-(r_\Xi+c_\Xi)x'\varepsilon'\sum_{D'_j}\phi'_P + [1.01K_1^\sharp(2r'+1)^3+1]\#D'_j + (\kappa+4)((1+s')\#\mathrm{mk}'_j+s')\}.$$

Since $x'\varepsilon'\phi'_P > p_0'^2$ on D' and $\#\mathrm{mk}'_j \le \#D'_j$, it follows that $\hat w(f_j)e^{2\#\mathrm{mk}'_j} \le e^{-c_\Xi x'\varepsilon'}\sum\phi'\upsilon_f^{(3/4)\#D'_j}$.

We take out the factor $e^{-c_\Xi(\cdot)}\upsilon_f^{\#D'_j/4}$. With $\hat w''(f_j) = \upsilon_f^{\#D'_j/2}e^{-2\#\mathrm{mk}'_j}$, the condition for animals holds: at most one class has $f_j \ni \square$, and $\hat w'' e^{a_{f_j}} \le \upsilon_f^{1/2} \le \varepsilon''/4$, while the cells and the polymers contribute $\le \varepsilon''/2$. Hence $\sum_{\zeta\ni f_j} e^{(\kappa+1)s'}\prod\hat w'' \le e^{(\kappa+1)s'}\upsilon_f^{1/2} \le 1$, by (Y.4).

The set F is fixed by choosing, for each class, either nothing or a letter that has this class as its first class; hence $\sum_{\mathsf{F}}\prod \le \prod_j 2$.

As a result, the left side of (8d) is $\le e^{-c_\Xi x'\varepsilon'}\sum\phi'\prod_j 2\upsilon_f^{\#D'_j/4} \le e^{-c_\Xi x'\varepsilon'}\sum\phi'$. The strong cap of the output thus comes from the factor $\exp\{-(r_\Xi+c_\Xi)x'\varepsilon'\sum_{D'_j}\phi'_P\}$ of $\hat w(f_j)$, and not from the input w.

(9) Here $\Xi'(V') := (\rho[u;\ell])'(V')e^{-\mathsf{V}'}\exp\{x'\mathrm{En}' - \sum 1_{Y\approx}[E'^a + E'^h]\}$, which is independent of e. Lem. 5.23 below, applied to the complex weights w', gives (9).

Lemma 5.23 (Averaging the choice e out of a polymer sum)**.** Let Ξ be a complex function of V', and for every finite set Y of cells let $w(\mathsf{Y},\cdot,\cdot)$ be a complex function of (V',e), such that the following hold.

(i) The function w is jointly Borel, and $|w(\mathsf{Y}, V', e)| \leq B(\mathsf{Y}, V') < \infty$ with B independent of e.
(ii) $w(\mathsf{Y}, V'^u, \mathrm{Ad}_u\, e) = w(\mathsf{Y}, V', e)$ for all G-valued u.
(iii) $w(\mathsf{Y}, V', e)$ depends on V' only through the bonds of $\mathsf{Y}^{+s'}$, and on e_P only at the plaquettes that belong to cells of Y.
(iv) At every (V', e), $\Xi(V') = \sum_{\{\mathsf{Y}_i\}} \prod_i w(\mathsf{Y}_i, V', e)$, the sum running over the families of (cΞ) at x'.

Then $\bar{w}(\mathsf{Y}, V') := \int_{\mathfrak{E}} w(\mathsf{Y}, V', e)\, de$ is Borel and gauge invariant, it is real if w is real, it depends on V' only through the bonds of $\mathsf{Y}^{+s'}$, it satisfies $|\bar{w}| \leq B$, and $\Xi(V') = \sum_{\{\mathsf{Y}_i\}} \prod_i \bar{w}(\mathsf{Y}_i, V')$ at every V'. ■

Proof of Lem. 5.23. The positive and negative parts of $\mathrm{Re}\, w$ and $\mathrm{Im}\, w$ are jointly Borel and ≥ 0; by Tonelli's theorem, their means over e are Borel, and by (i) they are finite. The dependence on V' and the bound carry over to the mean.

For G-valued u we have $\bar{w}(\mathsf{Y}, V'^u) = \int w(\mathsf{Y}, V'^u, \mathrm{Ad}_u\, e)\, de = \bar{w}(\mathsf{Y}, V')$; the first equality holds because $e \mapsto \mathrm{Ad}_u\, e$ is a product of rotations of spheres and preserves de, and the second is (ii).

Fix V'. The index set in (iv) depends on $D'(V')$ and not on e, and it is finite; the left side of (iv) is independent of e, and we integrate (iv) over e. Within one family the sets Y_i are pairwise disjoint, so by (iii) the factors depend on disjoint sets of independent e_P, and the mean of the product is the product of the means. □

We now show that Ξ' and w' satisfy the hypotheses of Lem. 5.23.

(i) Every letter is jointly Borel in (ξ, V', e). For $A_\square$ this is (6e); for $T_\square$, and hence for w_j, it is (5b). The variable X', the marks and the classes are Borel in (V', e). The entries of G, C and $S_k(Y)$ are continuous at G-valued fields, which lie in the interior of $\mathfrak{V}$, by (2d) and (4b); and g^V is a polynomial in $\mathrm{Ad}_{V'(c)}$, by (2g). So φ, a_0, η, T_Y are Borel. The chart is continuous, so U is Borel. The function E^a is continuous on $\mathcal{U}_X(x)$, and E^h and w are Borel, by (ca), (ch) and (cΞ). Finally, $b_\square, \chi_\square, \chi_X, 1_{X^\approx}(U), 1_{\mathrm{ind}}$ are indicators of Borel sets.
Every $K(\gamma)$ is jointly Borel in (V', e), by Tonelli's theorem together with the bound of Step 6 (i). Further, $e^{\tilde{\mathsf{D}}(Y,\cdot)} - 1$ is continuous by (4a), and n_j is Borel, as shown for the normaliser at the start of the proof of Step 10. On each of the finitely many Borel sets $\{\mathrm{mk}' \cap \mathsf{Y}^{+(s'-1)} = \mathsf{d}\}$, the sum that defines w' is an absolutely convergent series, and its real and imaginary parts are Borel; the bound (8d) is independent of e.

(ii) The marks, the classes, whether a letter is actual, and whether a cluster or a set is on or off depend only on the angles ϕ'_P. Moreover $X'(V'^u, \mathrm{Ad}_u\, e) = \mathrm{Ad}_u\, X'(V', e)$; together with $\xi \mapsto \mathrm{Ad}_u\, \xi$, which preserves the law of ξ, this makes a_0 and η covariant, by (2c) and (2g), and it makes $\mathsf{U}(\mathrm{Ad}_u\, \eta; V'^u)$ the gauge transform of $\mathsf{U}(\eta; V')$, as shown for the clauses without a size in Step 8.
The members, $1_{X^\approx}(U)$ and 1_{ind} are invariant; $b_\square$ and χ depend only on $|\xi_b|$; $A_\square$ is invariant by (6e), $T_\square$ by (5b), T_Y by (4b) and n_j by (4a). Hence every letter satisfies $\lambda_\alpha(\mathrm{Ad}_u\, \xi, V'^u, \mathrm{Ad}_u\, e) = \lambda_\alpha(\xi, V', e)$, and $K(\gamma)(V'^u, \mathrm{Ad}_u\, e) = K(\gamma)(V', e)$; the map $e \mapsto \mathrm{Ad}_u\, e$ preserves de.

(iii) For V', this is (8c), with the dependence on the field established in the proof of (8c). The choice e_P enters at plaquettes of marks in $\mathsf{Y}^{+(s'-1)}$, and where $w' \neq 0$, these marks are cells of Y.

(iv) This is (8c). The index set depends on D' and not on e; the sets Y_i are disjoint, so the bounded factors depend on disjoint sets of e_P, and Fubini's theorem applies.

□

Step 11: clauses (c) *and* (d) *of Thm. 5.15.* Let $z \mapsto u_z$ be an affine map from an open set $\Omega \subset \mathbb{C}$ into $M_{16}^{\mathrm{su}}(x)$. Then the number V'_z and all the values $E'^a_z(Y,V')$, $E'^h_z(Y,V')$, $w'_z(\mathsf{Y},V')$ are holomorphic in z. If u is real, then $F_\ell(u)$ and V' are real; if moreover $\rho[u;\ell] > 0$, then $(\rho[u;\ell])' > 0$.

Proof of Step 11. At fixed (ξ, V', e) every letter is holomorphic in z on Ω, as a product of affine values $w_z(X,U)$, of exponentials of the affine functions E^a_z and E^h_z, and of factors that contain no member.

The alphabet, the adjacency, the compatibility, the cores, χ, U, the indicators, whether a letter is actual, whether a cluster or a set is on or off, whether a set is long, and the alternative of Lem. 5.19, which depends only on the supports of Λ_ζ, are all independent of z.

The majorant of Step 6 at $A = 16$ is integrable and independent of z. Hence $K_z(\mathfrak{B})$ is holomorphic, by the theorems of Morera and Fubini.

Each $\Phi_z(C)$ is a polynomial in the K_z, and $|\Phi_z| \le \hat{\Phi}$ at $A = 16$. Therefore (Fm1) to (Fm3) and the sum in (Fm4) converge uniformly in z, and $e^{-\Phi^{\natural}_z} - 1$ is holomorphic.

The mean over e of a bounded and jointly measurable family that is holomorphic in z is again holomorphic; here the measurability is shown in the verification of hypothesis (i) of Lem. 5.23 in Step 10 (9), together with (6e).

Reality: at real u and G-valued V', every letter is real, by (2d), (4a), (4b), (5b), (5c) and (Γ6), and $A_\square > 0$ or $A_\square = 0$. Hence $K, \Phi, \tilde{\mathsf{D}}, P, n_j, w_j, \mathsf{V}_0$ are real.

The block kernel is strictly positive. Integrating the positive fine density against its product gives a positive coarse density at every V'. □

Step 12. Contraction along a segment. Let $u_1, u_2 \in M_B^{\mathrm{su}}(x)$ for $B = 1$ or 2, and put $d = \|u_2 - u_1\|_x$. The case $d = 0$ is immediate. Otherwise $0 < d \le 2B$, and the affine family

$$u_z = u_1 + \frac{z}{d}(u_2 - u_1)$$

preserves the linear conditions on the members and satisfies $\|u_z\|_x \le B + |z|$. Its values lie in $M_{16}^{\mathrm{su}}(x)$ for $|z| < 16 - B$. At each $t \in [0,d]$ a disc of radius $16 - 3B$ lies in that domain. For any scalar output $f(z)$ with bound m on $|z| < 16 - B$, clause (c) and Cauchy's estimate give

$$|f(d) - f(0)| \le \frac{d\mathsf{m}}{16 - 3B}.$$

For the analytic part take $f = E'^{\mathrm{old}}[u_z](Y,V')$, since E'^{new} is independent of z. Its bound is $16 \cdot 0.248$ times the analytic cap in clause (b). For E'^h and w' the bounds are their caps at x', using the strong cap for w'. After dividing by the three caps and taking the suprema in the strong distance, we obtain the factors $3.968/(16-3B)$, $1/(16-3B)$ and $1/(16-3B)$. At $B = 2$ the largest is $0.3968 \le 0.4$, which proves clause (e).

At $A = 1$, clauses (a), (b) and (d) also give the class inclusion of (S-step) for $x \ge x_9$. Indeed the analytic cap is multiplied by $0.35 + 0.248 < 1$, the output has the strong cap, and positive real densities retain positivity and reality. At $A = 2$ the analytic factor is $0.846 < 1$, again with the strong output cap. The inputs from Sections 4.1 through 4.3 concern the chart, energy and tests of the step, which are fixed independently of the input members. □

5.4. **Contraction and row perturbations.** To compare cutoffs we first express two densities over one family of rows. The difference in their reference energies is assigned to the analytic term in tame cells and to a rough weight in each remaining cell. The change is small in the weak distance. The complex step then restores the strong caps and contracts the difference, so both runs can continue over the same rows.

Item 0: what is used; notation.

Remark 5.24 (What is used of other sections). From other sections we use Thm. 5.15, with its clauses (a), (b), (c) and (d), its data (SD), its floor x_9 and Step 0 (iv) of its proof, together with (Y.0)(0b), (Y.4) and (Y.7) of the floors (Y.0) to (Y.9); item 8 of §5.3; Cor. 5.81, with its chains, with (RU) and with its hypotheses, among which are (X12) and (X13); Steps 1, 2 (d) and item 1 of Step 4 of the proof of Thm. 5.63; Step S4 (4) of the proof of Thm. 5.3; and Thm. 2.65, Thm. 2.57 (a), Prop. 2.27 (a), Lem. 2.53 (a), Thm. 2.64 and Prop. 2.28. Cor. 5.81 belongs to the later §5.5, where it is stated and proved in Step 1a; every statement on which a proof in Step 1a rests is stated either in Step 1a itself or in §2.2, and the proof of Cor. 5.81 uses nothing of this section.

Definition 5.25 (Notation)**.** Throughout this section $G = \mathrm{SU}(2)$. The letter V stands for a G-valued field of the level at x, and W for an $\mathrm{SL}(2,\mathbb{C})$-valued field. The letters $\mathsf{V}_1, \mathsf{V}_2, \mathsf{V}', \mathsf{V}'_1, \mathsf{V}'_2$ are numbers. We write $D(V)$ for the set of all rough plaquettes, D_x and $\mathsf{D}_x^{\mathrm{in}}$ for distances of members, D_y for a difference of kernels of rows, and $\mathcal{D}$, $\mathcal{P}$ for families of sets of cubes. The letter b is the label of a run, a number $\geq x_\flat$, whereas $\mathsf{b}(x)$ is the number $2 - 1/\log x$ of (u2). A prime on a letter means that the letter is taken at $x' = x/L$. ■

Remark 5.26 (Letters with two senses). The letter u is a system of members (E^a, E^h, w), but in V^u, $u(y)$, $\mathbf{1}^u$ and $\mathrm{Ad}_{u(y)}$ it is a gauge transformation.

Only in Step 9 (E4), F and $\bar{F}$ each stand for a real 2-cochain, X' for a coarse 2-form (and not for X at x'), and we set $\mathsf{B} = 1 + \ell^{\mathsf{T}}\ell$.

The radius $r(x)$ is written r wherever x is fixed.

In Steps 9 and 10, $\sigma_3 = \mathrm{diag}(1, -1)$.

We write (9a) to (9d) for the clauses (a) to (d) of Thm. 5.15.

Item 1: letters.

Definition 5.27 (Letters)**.** We give the letters in the order (O0) to (O14) of the complete table.

The complete table appears in item 1 of §5.5. Its letters are defined in that item as well as in "The order of the constants" (0) and (6) in §2.2, in item C.I 1.2 of §3, in (O3), (O4), (O6), (O7), (O10), (O13) in item 3 of §2.1, and in (O13) in item 1 of §5.3.

(O0) The pure numbers (Def. 2.56) are b_*, γ_*, $\lambda_* = \gamma_*^{1/2}/44$, $K_* = 3\gamma_*^{-1/2}$ (> 1), L_A, and the others of the table. Here $\gamma_* = \mathsf{r}_\sharp^2/12296 \leq 9.05513 \cdot 10^{-25} < 1$ ("The order of the constants" (0) in §2.2), whence $K_* > 3$. We put $c_0 := 108\, b_* \gamma_*^{-1/2} \mathsf{g}(\lambda_*/4)^3$, $c_{\mathrm{fix}} := 48\, \mathsf{g}(\lambda_\flat) C_1 c_0^{1/2}$, $L_B := \max\{L_A, c_{\mathrm{fix}}^{10}\}$ and $\vartheta_1 := c_{\mathrm{fix}} L^{-1/2}$ ("Rates, constants", Step 1a of §5.5); all of these are pure numbers except ϑ_1, which depends on L. Further, $\mathsf{g}(\lambda) := \sum_{n\in\mathbb{Z}^3} e^{-\lambda|n|_\infty}$. The definition "Rates, constants" in Step 1a of §5.5 fixes $\lambda_\flat := \lambda_*/16$ and $C_1 := \sqrt{3}\, K_*\, \mathsf{g}(\lambda_*/2)^{3/2}$; hence $C_1 > 1$, because $K_* > 3$ and $\mathsf{g}(\lambda_*/2) \geq 1$.

(O1) $d_0 \in (0, \frac{1}{2})$.

(O2) $0 < c_\Xi \leq \frac{1}{2}(\frac{1}{2} - d_0)$.

(O3) The radius ρ_0 is given by eq. 2.vi of §2.2.

(O4) L is odd with $L \geq L_*^{\mathrm{su}}$ of the table; $c_L = (L^2 - 1)/(6L^2)$; and $a \geq 3/(c_L L)$. In particular, $L \geq L_B$.

(O5) $\kappa \geq 6$.

(O6) $\theta^\flat \leq 1$, $\theta_{\mathbb{C}} = \theta^\flat/6$ and $\vartheta_0^\flat = \theta^\flat/24$, where $\theta^\flat = \min\{\theta_1, \gamma/(2A^\flat)\}$ and $\theta_1 = \min\{1, \lambda_* L/12\}$, as in item C.I 1.2 of §3 at $(K, \lambda) = (K_*, \lambda_*)$ and in "The order of the constants" (6) in §2.2. Further, $\theta_D = \frac{1}{2}\min\{\lambda_2^*, \theta_{\mathbb{C}}, \theta_J\}$, C_D is as in the table, and $c_R = 64/\theta_D$. We have $c_R \geq 768$, since $\theta_D \leq \theta_{\mathbb{C}}/2 \leq 1/12$.

(O7) $M = L^{m'}$ with $m' \geq 1$, subject to (M2), (M4$^\flat$) and (M5r) $M \geq \rho_0$; the condition (M4$^\sharp$) is a consequence of (M4$^\flat$).

(O8) $\mathfrak{p}_0 > 2$ and $A_0 > 0$; $p_0(x) = A_0(\log x)^{\mathfrak{p}_0}$ and $\varepsilon = p_0 x^{-1/2}$, so that $x\varepsilon^2 = p_0^2$.
(O10) This section defines $c_Q^{\mathrm{su}} := 306\, M^3\, \mathsf{g}(\lambda_\flat)^2$.
(O11) $0 < c_h \le \delta_3'/2$.
(O12) E_a is as in the table. In particular, $E_a > 0$.
(O13) $x_\flat \ge e^2$ with $L\, x_\flat \ge x_9$ (Thm. 5.15). The number $x_9 := 1 + \inf S_9$ is defined in (O13) of item 1 of §5.5 and in (O13) of item 1 of §5.3, each of which gives the set S_9. Of x_9 this section uses that (Y.0)(0b), (Y.4), (Y.7) and Thm. 5.15 hold at every $x \ge x_9$, and that $x_9 \ge x_{\mathrm{fl}} \ge L^2 e$. Further, $x_\flat$ is chosen such that for all $x \ge x_\flat$
(X14) $p_0(x) \ge 1$;
(X15) $c_\Xi p_0(x)^2 \ge 1$.
Cor. 5.49 uses in addition $16 c_a(x) \le \frac{1}{2}$ for all $x \ge x_\flat$, together with (X12) and (X13) of Cor. 5.81, which read
(X12) $3K_* \mathsf{g}(\lambda_*/2)\, x_\flat^{-e_R}/(1 - L^{-e_R}) \le \frac{1}{9}$, where $e_R := \lambda_* c_R/2$;
(X13) $16\, x_\flat^{-e_R/4} \le 1$.
The full entry, with $x_\flat \ge x_\flat^{(1)}$, $x_\flat \ge x_\flat^0$ and (X16) to (X18), is (O13) of item 1 of §5.5.
(O14) The numbers of this entry, ϑ_w, n_w, ϑ_{10}, ϑ, c_4, $\Lambda(\mathsf{b})$, $K_0(\mathsf{b})$ and k_0, are all defined after (O13), and no floor depends on any of them. Here $\vartheta_w = 0.248$ and $n_w = 0.35$ are those of Thm. 5.15 (b). We put $\vartheta_{10} := 16\vartheta_w/10 = 0.3968$, and the ϑ of the table is $\vartheta := (16\vartheta_w + \frac{1}{2})/10 = 0.4468$. Thus $\vartheta_{10} \le 0.4 \le 0.4318 \le \vartheta \le 0.45$, so that the factor 0.4 of Thm. 5.15 (e), the factors 0.4318 and 0.45 of Thm. 5.46 (iv), and ϑ itself, each as the factor in $\mathsf{D}_{x'} \le \text{factor} \cdot \mathsf{D}_x^{\mathrm{in}}$ for pairs in $M_2^{\mathrm{su}}(x)$, are upper bounds of ϑ_{10}. Finally, $c_4 := 15\, c_Q^{\mathrm{su}}\, C_1^2/E_a$, $\Lambda(\mathsf{b}) := \log_L(c_4 \mathsf{b}^{1/2})$, $K_0(\mathsf{b}) := \frac{5}{2}\Lambda + \frac{9}{4}$ and, for the length K of a run, an integer ≥ 1 (the K of Cor. 5.49), $k_0 := \lceil (5K + 10\Lambda)/9 \rceil$.

The caps are $c_a(x) = E_a p_0^3 x^{-1/2}$ and $c_h(x) = e^{-c_h p_0^2}$. No number defined here depends on N, on a member, or on more of the rows than their range. As their formulas show, p_0, ε and the caps are functions of x; apart from these, none depends on x. ■

Item 2: the level at x.

Definition 5.28 (Lattice)**.** Let N be even, with $M \mid N$ and (Tor) $N \ge 2M\, s(x)$; this N is the coarse side of the step that starts at x. Let $\mathbb{T} = \mathbb{Z}^3/NL\mathbb{Z}^3$. Plaquettes are written $p = (z; \mu\nu)$ with $\mu < \nu$, and $\pi(p) = z$ is the base site of p; the base site of a bond (z, μ) is z. We set $|y - p| := |y - \pi(p)|_\infty$, measured periodically (Def. 2.43).

Cubes are the boxes of M^3 sites of one fixed grid of $\mathbb{T}$. Each bond and each plaquette belongs to the cube of its base site, and "p in X" or "X holds p" means that p belongs to a cube of X. The distance of cubes is the sup distance counted in cubes, and X^{+j} consists of the cubes within j of X. By $\mathcal{D}$ we denote the family of all non-empty sets of cubes that are connected through faces, edges and corners, and by $\mathcal{P}$ the family of all non-empty finite sets of cubes; further, $|X|_c = \#\mathrm{hull}(X)$. Here $\mathrm{hull}(X)$ is a member of $\mathcal{D}$ that holds X and has the least number of cubes, singled out by an order (Step S4 (4) of the proof of Thm. 5.3); thus $|X|_c$ is the least number of cubes of a member of $\mathcal{D}$ holding X, as in item 2 of §5.5.

The bonds of X are those with both ends in cubes of X. The paths and plaquettes of a set A of cubes are those all of whose sites lie in A, while "p in A" means that the base site of p lies in A. We call a field on the bonds of A tame there when every plaquette of A has $\phi_p \le \varepsilon$ (see item 1 of §2.1 and item 2 of §5.3). If $1_A(V) = 1$, the restriction of V to the bonds of A is tame there, since a plaquette of A is based in A. The suprema of E^h in $\|\cdot\|_x$, $\mathsf{D}_x^{\mathrm{in}}$ and D_x are taken over the fields that are tame there. ■

Definition 5.29 (At x)**.** At x we put $\varrho(x) = \lceil c_R \log x \rceil$, $r(x) = \lceil 20 c_R \log x/M \rceil$ (≥ 1), $s(x) = 8r(x)$ and $\ell_w(x) = 6M\, r(x) + 12$, and we write $X^{\approx} = X^{+r(x)}$ and $X^{\approx\approx} = X^{+s(x)}$. ■

Definition 5.30 (Group)**.** Let $\mathfrak{g}$ be the space of hermitian traceless 2×2 matrices, with $\langle Y,Y'\rangle = \frac12 \operatorname{tr} YY'$ and $|Y|^2=\langle Y,Y\rangle$; this $|Y|$ is the operator norm. We put $\operatorname{ang}(g)=\arccos(\frac12\operatorname{tr} g)\in[0,\pi]$. For $g\neq -1$, $\mathsf{L}(g)$ is the unique $Y\in\mathfrak{g}$ with $e^{iY}=g$ and $|Y|<\pi$, and $\mathsf{L}(-1):=0$. On complex 2×2 matrices, $|\cdot|$ denotes the operator norm and $Y\cdot Y':=\frac12\operatorname{tr} YY'$; hence $|Y\cdot Y'|\le |Y||Y'|$, and on $\mathfrak{g}$ we have $Y\cdot Y'=\langle Y,Y'\rangle$. For $g\in \mathrm{SL}(2,\mathbb{C})$ one has $|g^{-1}|=|g|$; with $\operatorname{Ad}_g Y = gYg^{-1}$ we get $|\operatorname{Ad}_g Y|\le |g|^2|Y|$ and $(\operatorname{Ad}_g Y)\cdot(\operatorname{Ad}_g Y')=Y\cdot Y'$. ■

Definition 5.31 (Fields)**.** For a field V we have $V(\bar b)=V(b)^{-1}$; $V(\pi)$ is the ordered product along a path π, and $V^u(b)=u(b_-)V(b)u(b_+)^{-1}$. The curvature is

$$\partial V(z;\mu\nu)=V(z,\mu)V(z+e_\mu,\nu)V(z+e_\nu,\mu)^{-1}V(z,\nu)^{-1}.$$

We set $\phi_p=\operatorname{ang}\partial V(p)$, $F_p=\mathsf{L}(\partial V(p))$ and $D(V)=\{p:\phi_p>\varepsilon\}$, and we put $1_Y(V):=1\{\phi_p\le\varepsilon \text{ for all } p \text{ in } Y\}$; when it equals 1 we say that V is tame on Y. ■

Definition 5.32 (Transports)**.** For sites y,z whose sup distance is less than half the side, $\Gamma(y,z)$ is the coordinate path from y to z that moves first in the direction e_1, then e_2, then e_3, each time the short way; and $T(y,z):=\operatorname{Ad}_{V(\Gamma(y,z))}$. ■

Definition 5.33 (Rows)**.** Rows, columns and range are those of Def. 2.22, with one exception, the mask: the mask there has the one radius ρ_0, whereas the mask of this section, defined below, has two radii. Rows ℓ on $\mathbb{T}$ of range $\le\varrho$ consist of a Euclidean space H_y of finite dimension for each site y and of columns $\ell_y(p)\in\mathsf{H}_y$, with $\ell_y(p)=0$ if $|y-p|>\varrho$. We put $\mathsf{E}_y(p,p'):=\langle \ell_y(p),\ell_y(p')\rangle$, so that $\mathsf{E}_y=\ell_y^{\mathsf{T}}\ell_y$ ("Site forms", Step 1a of §5.5). For a family $D=(D_y)$ of kernels,

$$\|D\|_{(\lambda)}:=\sup_{y,p,p'}|D_y(p,p')|\,e^{\lambda(|y-p|+|y-p'|)}$$

("Norms of a family of symmetric operators", Step 1a of §5.5). The empty rows $\ell=\emptyset$ are those with all $\mathsf{H}_y=0$. ■

Definition 5.34 (Clip, dressed rows, mask, energy)**.** The clip is $\zeta_p:=F_p\min\{1,\varepsilon/\phi_p\}$, with $\zeta_p=0$ if $\partial V(p)=1$ or -1; hence $|\zeta_p|\le\varepsilon$ at every V, and $\zeta_p=F_p$ if $\phi_p\le\varepsilon$. Dressing the rows with the transported clip, we put $\ell_y^V\zeta:=\sum_p \ell_y(p)\otimes T(y,\pi(p))\zeta_p\in\mathsf{H}_y\otimes\mathfrak{g}$. The mask is $m_y(V):=1$ if no plaquette of $D(V)$ has $|y-p|\le\rho_0$ and no plaquette with $\phi_p>1$ has $|y-p|\le\varrho(x)+1$, and $m_y(V):=0$ otherwise. With $S_\varepsilon(t)=t^2/2$ $(t\le\varepsilon)$, $d_0\varepsilon t$ $(t>\varepsilon)$, the energy is

$$\mathrm{En}_\ell(V):=\sum_p S_\varepsilon(\phi_p)+\frac12\sum_y m_y(V)\,|\ell_y^V\zeta|^2.$$

The mask, the clip and the paths depend on V, ε, ρ_0 and $\varrho(x)$, and on no row. ■

Proposition 5.35 (Invariance)**.** Let u be a gauge transformation, and let V be replaced by V^u. Then ϕ_p, $D(V)$, m_y and 1_Y are invariant; ζ_p becomes $\operatorname{Ad}_{u(\pi(p))}\zeta_p$; and $T(y,\pi(p))\zeta_p$ becomes $\operatorname{Ad}_{u(y)}T(y,\pi(p))\zeta_p$. ■

Proof. One has $\partial V^u(p)=u(\pi(p))\,\partial V(p)\,u(\pi(p))^{-1}$. As ang is a class function, ϕ_p is unchanged, and so are $D(V)$, m_y and 1_Y, which are defined through the angles ϕ_p alone. Next, $\mathsf{L}(ugu^{-1})=\operatorname{Ad}_u\mathsf{L}(g)$, where at $g=-1$ both sides are 0. Therefore $F_p=\mathsf{L}(\partial V(p))$ is mapped to its image under $\operatorname{Ad}_{u(\pi(p))}$, and so is ζ_p, the factor $\min\{1,\varepsilon/\phi_p\}$ being invariant. Finally, $V^u(\Gamma(y,z))=u(y)V(\Gamma(y,z))u(z)^{-1}$. For $z=\pi(p)$, in $\operatorname{Ad}_{u(y)V(\Gamma(y,\pi(p)))u(\pi(p))^{-1}}\operatorname{Ad}_{u(\pi(p))}\zeta_p$ the factor $u(\pi(p))^{-1}$ cancels $u(\pi(p))$, so $T(y,\pi(p))\zeta_p$ is mapped to its image under $\operatorname{Ad}_{u(y)}$, which is an isometry of $\mathfrak{g}$. □

Definition 5.36 (The step)**.** For bounded Borel ρ,

$$\rho'(V')=\int\rho(U)\prod_c k_{ax}(V'(c),\mathsf{M}(U)(c))\,dU,$$

and for complex ρ the definition extends by linearity. The map $\rho \mapsto \rho'$ is linear in ρ, it depends on x, a, L and on no row, and $k_{ax} > 0$. The kernel k_s, its normaliser $z(s)$ and the average M are defined in item 2 of §5.5, "The step". ■

Item 3: domains, members, distances, the class.

Remark 5.37. The definitions of this item are those of item 3 of §5.3.

Definition 5.38 (Domain)**.** For $X \in \mathcal{D}$, let $\mathcal{U}_X(x)$ be the set of maps W from the oriented bonds of $X^{\approx}$ to $\mathrm{SL}(2,\mathbb{C})$ with $\mathsf{W}(\bar b) = \mathsf{W}(b)^{-1}$ that satisfy (u1), (u2) and (u3):

(u1) $|\mathsf{W}(b)| < 1 + \vartheta_0^{\flat}(1 - 1/\log x)$;
(u2) $|\mathsf{W}(\pi)| < \mathsf{b}(x) := 2 - 1/\log x$ for every path π in $X^{\approx}$ of at most $\ell_w(x)$ bonds;
(u3) $|\partial\mathsf{W}(p) - 1| < 2\varepsilon$ for every plaquette p of $X^{\approx}$.

This set is open and invariant under G-valued gauge transformations, and it contains every G-valued field that is tame on $X^{\approx}$, because the norms of such a field are 1 and $|\partial V(p) - 1| = 2\sin(\phi_p/2) \le \varepsilon$. ■

Definition 5.39 (Members)**.** Let $\mathfrak{M}(x)$ be the complex linear space of all triples $u = (E^a, E^h, w)$ that have $\|u\|_x < \infty$ and satisfy the clauses without a size, which are the three conditions on E^a, E^h and w that follow. For $X \in \mathcal{D}$, $E^a(X,\cdot)$ is holomorphic on $\mathcal{U}_X(x)$, invariant under G-valued gauge transformations, and 0 at the pure gauges $\mathbf{1}^u$. For $X \in \mathcal{D}$, $E^h(X,\cdot)$ is a gauge invariant complex Borel function defined on those G-valued fields on the bonds of $X^{\approx}$ which are tame there. For $X \in \mathcal{P}$, $w(X,\cdot)$ is a gauge invariant complex Borel function of the restriction of V to the bonds of $X^{\approx\approx}$.

The number $\|u\|_x$ is the largest of the three numbers

$$\sup_X \sup_{\mathcal{U}_X(x)} |E^a(X,\cdot)|\, e^{\kappa(\#X-1)}/c_a(x), \qquad \sup_X \sup |E^h(X,\cdot)|\, e^{\kappa(\#X-1)}/c_h(x), \qquad \sup_{X,V} |w(X,V)|\, e^{\kappa(|X|_c-1)},$$

and $\|\cdot\|_x$ is a norm on $\mathfrak{M}(x)$. For $A \ge 1$ we put $M_A^{\mathrm{su}}(x) := \{u \in \mathfrak{M}(x) : \|u\|_x \le A\}$; these sets are given by (ca_A), (ch_A), $(\mathrm{c}\Xi_A)$ of item 3 of §5.3, and the cap of w in them is the weak cap. The strong cap is the bound

$$|w(X,V)| \le \exp\{-c_\Xi x\varepsilon \sum_{p \in D(V)\cap X} \phi_p - \kappa(|X|_c - 1)\}$$

at every V. We call u real if E^a is real at G-valued fields and E^h, w are real. ■

Definition 5.40 (Density and class)**.** The density of u over the rows ℓ is

$$\rho[u;\ell](V) := \exp\{-x\mathrm{En}_\ell(V) + \sum_{X\in\mathcal{D}} 1_{X^{\approx}}(V)[E^a(X,V) + E^h(X,V)]\} \cdot \Xi_w(V),$$

where $\Xi_w(V) := \sum_{\{X_i\}} \prod_i w(X_i,V)$ is summed over the finite families of $X_i \in \mathcal{P}$ with pairwise distance $> s(x)$, each of which holds a plaquette of $D(V)$, and whose union contains $D(V)$; these are the families of $(\mathrm{c}\Xi)$. If $D(V) = \emptyset$, the only such family is the empty one, and $\Xi_w = 1$. For rows ℓ on $\mathbb{T}$ of range $\le \varrho(x)$, the class $C_\flat^{\mathrm{su}}(x;\ell)$ is the set of all densities $e^{\mathsf{V}_1}\rho[u;\ell] > 0$ with V_1 real and $u \in M_1^{\mathrm{su}}(x)$ real with the strong cap. ■

Definition 5.41 (Distances)**.** We put $\mathsf{D}_x^{\mathrm{in}}(u_1,u_2) := \|u_1 - u_2\|_x$, the weak distance. The strong distance $\mathsf{D}_x(u_1,u_2)$ is defined in the same way, except that in the third supremum the weight $e^{\kappa(|X|_c-1)}$ is replaced by $\exp\{c_\Xi x\varepsilon \sum_{D(V)\cap X}\phi_p + \kappa(|X|_c - 1)\}$. Hence $\mathsf{D}_x^{\mathrm{in}} \le \mathsf{D}_x$, and $\mathsf{D}_x(u_1,u_2) \le 2$ when u_1, u_2 both lie in $M_1^{\mathrm{su}}(x)$ and have the strong cap. ■

Definition 5.42 (The level at x')**.** At x', the definitions of item 2, with two exceptions, and those of item 3 hold on $\mathbb{T}' = L\mathbb{Z}^3/NL\mathbb{Z}^3$, in its own units $\mathbb{Z}^3/N\mathbb{Z}^3$, with cells in the place of cubes. The exceptions are the definition "The step" and part of the definition "Lattice", namely the conditions

that N is even, that $M \mid N$ and that (Tor) holds, together with the role of N as the coarse side of the step starting at x and the torus $\mathbb{T} = \mathbb{Z}^3/NL\mathbb{Z}^3$. ■

Remark 5.43. Of the conditions on the side N in "Lattice", only the following is used at x': the side N of $\mathbb{T}'$ is even, $M \mid N$, and $N \geq 2M\, s(x) \geq 2M\, s(x') = 16M\, r(x')$, because r is non-decreasing. Nothing at x' depends on N being L times the coarse side of a further step: (SD) does not provide this, at the last level of a run it fails in general, and no step starts there. Four facts suffice instead. First, wherever x' is used (in item 4 and in Thms. 5.46 and 5.48), we have $x \geq x_9 \geq x_{\mathrm{fl}} \geq L^2 e$ ((O13) in item 1 of §5.5), and so $x' \geq Le > e$. Second, (F2) therefore holds at x', by the argument for it in Step 0 (Fl), which uses only $x \geq e$ and $c_R \geq 768$; that is, $\varrho(x') + 1 < M\, r(x') \leq N/16$. Third, the paths $\Gamma(y, z)$ of En_{ℓ^c} and the mask at x' are therefore defined, since the sup distances involved are less than half the side N. Fourth, every box $\square^{+r(x')}$ of cells, whose side is $(2r(x') + 1)M \leq 3M\, r(x') \leq 3N/16$, embeds. This is the level of item 2 of §5.5, "Lattice, cubes".

Item 4. Inputs from the complex step. The tags (9a) through (9d) refer to the corresponding clauses of Thm. 5.15. They give the exact image, the caps, holomorphy along affine families, and reality and positivity. We write $n_w = 0.35$ for the analytic contribution independent of the input and $\vartheta_w = 0.248$ for the coefficient multiplying the input cap.

Item 5: the conditions of Lem. 5.47.

Definition 5.44 (Five conditions)**.** The five conditions of the change of rows are these.

(M5r) $M \geq \rho_0$.
(X15) $c_\Xi p_0(x)^2 \geq 1$.
(F1) $\varepsilon(x) \leq 1/40$.
(F2) $\varrho(x) + 2 \leq M\, r(x)$.
(F3) $(2r(x) + 1)^3 c_a(x) \leq 34$.

■

Remark 5.45. In the setting of the table, at every $x \geq L\, x_\flat$, condition (M5r) is (O7) and condition (X15) is (O13), while (F1), (F2) and (F3) follow from $x \geq x_9$ and $c_R \geq 768$ (Step 0), with $\varepsilon < 1/680$ and $(2r(x) + 1)^3 c_a(x) \leq x^{-1/4}/(192e^{11}) < 10^{-7}$.

Item 6: the statements.

Theorem 5.46 (One step contracts from the weak to the strong distance)**.** Let $x \geq x_9$, let the data (SD) at x be given, and let $u_1, u_2 \in M_2^{\mathrm{su}}(x)$, complex members with the weak cap. Then (9b) and (9c) imply the following.

(i) $F_\ell(u_i)$ lies in $M_1^{\mathrm{su}}(x')$ and has the strong cap at x'; and $\sup_{\mathcal{U}_Y(x')} |E'^a(Y, \cdot)| \leq (n_w + 2\vartheta_w)\, c_a(x')\, e^{-\kappa(\#Y - 1)}$ for every Y, where $n_w + 2\vartheta_w = 0.846$ at the numbers of (O14).

(ii) We have

$$\mathsf{D}_{x'}(F_\ell(u_1), F_\ell(u_2)) \leq \vartheta_{10}\, \mathsf{D}_x^{\mathrm{in}}(u_1, u_2) \leq \vartheta_{10}\, \mathsf{D}_x(u_1, u_2), \qquad \vartheta_{10} = 0.3968;$$

for the three members the factors are 0.3968, 0.1 and 0.1.

(iii) For $u_1, u_2 \in M_1^{\mathrm{su}}(x)$ the factor is $\max\{16\vartheta_w/13, 1/13\} = 3.968/13 = 0.30523\cdots \leq 0.3053$.

(iv) In a robust form: if, for E'^a, the two sums of (9b) are replaced by the single requirement that $\sup_{\mathcal{U}_Y(x')} |E'^a(Y, \cdot)| \leq (\vartheta_w A + n_w)\, c_a(x')\, e^{-\kappa(\#Y - 1)}$ for every Y, with numbers $\vartheta_w, n_w \geq 0$ and $2\vartheta_w + n_w \leq 1$, while every other clause of (9b) stays in force (the sum for E'^h, the strong cap of w', and the clauses without a size), then (i) still holds and the factor in (ii) becomes $\max\{(16\vartheta_w + n_w)/10, 1/10\}$. That factor never exceeds 0.8; it is 0.4318 at $(0.248, 0.35)$ and 0.45 at $(\frac{1}{4}, \frac{1}{2})$.

■□

Lemma 5.47 (A density over other rows)**.** Let $x > e$, let the lattice of item 2 be given, and let $\ell, \tilde\ell$ be rows on $\mathbb{T}$ of range $\le \varrho(x)$. Let $\tilde{\mathsf{E}}_y(p,p') := \langle \tilde\ell_y(p), \tilde\ell_y(p')\rangle$, the kernel E_y of "Rows" with $\tilde\ell$ in the place of ℓ, and let $D_y := \tilde{\mathsf{E}}_y - \mathsf{E}_y$, $\delta := \|D\|_{(\lambda_\flat)}$ and

$$\delta_\Delta := c_Q^{\mathrm{su}} p_0^2 \delta / c_a(x) = (c_Q^{\mathrm{su}}/E_a)\, x^{1/2} \delta / p_0(x).$$

Let the five conditions of item 5 hold, and let $\delta_\Delta \le 1$. Then each real $\tilde u = (\tilde E^a, \tilde E^h, \tilde w) \in M_1^{\mathrm{su}}(x)$ having the strong cap admits a real $u^\flat = (\tilde E^a + E^\Delta, \tilde E^h, w^\flat) \in M_2^{\mathrm{su}}(x)$ such that (B1) to (B4) hold:

(B1) $\rho[\tilde u; \tilde\ell](V) = \rho[u^\flat; \ell](V)$ at every V.

(B2) $E^\Delta(X,\cdot) = 0$ unless $\#X = 1$, and $\sup_{\mathcal{U}_{\{\square\}}(x)} |E^\Delta(\{\square\},\cdot)| \le \delta_\Delta c_a(x)$; moreover E^Δ depends on $\ell, \tilde\ell, x$ and not on $\tilde u$.

(B3) $|w^\flat(X,V)| \le e^{-\kappa(|X|_c - 1)}$ and $|w^\flat - \tilde w|(X,V) \le \frac12 \delta_\Delta e^{-\kappa(|X|_c-1)}$ at every X and V. Here

$$w^\flat = \tilde w \cdot \prod \{ e^{q_\square(V)} : \square \text{ off at } V,\ p_*(\square, V) \text{ in } X\}$$

(see Steps 3 and 5 of the proof); the product involves only the values of V on the bonds of $X^{+(2r(x)+1)}$, so that $w^\flat(X,\cdot)$ is, just as $\tilde w(X,\cdot)$, a function of V on the bonds of $X^{\approx\approx} = X^{+s(x)}$.

(B4) $\mathsf{D}_x^{\mathrm{in}}(\tilde u, u^\flat) \le \delta_\Delta$.

■

Theorem 5.48 (Two families of rows in one class)**.** Let $x \ge x_9$, let the data (SD) at x be given for (ℓ, ℓ'), and suppose that the conditions of item 5 hold; let $\tilde\ell$ be rows on the same $\mathbb{T}$ with the same grid, of range $\le \varrho(x)$, and let $\delta_\Delta \le 1$. Let $\rho_1 = e^{\mathsf{V}_1}\rho[u_1;\ell] \in C_\flat^{\mathrm{su}}(x;\ell)$ and $\rho_2 = e^{\mathsf{V}_2}\rho[u_2;\tilde\ell] \in C_\flat^{\mathrm{su}}(x;\tilde\ell)$, and let $u_2^\flat$ be the system that Lem. 5.47 gives for $\tilde u = u_2$.

Then, by (9a) to (9d), both images lie in one and the same class $C_\flat^{\mathrm{su}}(x';\ell'^{\mathrm{c}})$:

$$\rho_1' = e^{\mathsf{V}_1'}\rho[F_\ell(u_1);\ell'^{\mathrm{c}}], \qquad \rho_2' = e^{\mathsf{V}_2'}\rho[F_\ell(u_2^\flat);\ell'^{\mathrm{c}}],$$

where $\mathsf{V}_1' = \mathsf{V}_1 + \mathsf{V}'[u_1]$ and $\mathsf{V}_2' = \mathsf{V}_2 + \mathsf{V}'[u_2^\flat]$ are real; and

$$\mathsf{D}_{x'}(F_\ell(u_1), F_\ell(u_2^\flat)) \le \vartheta_{10}(\mathsf{D}_x(u_1,u_2) + \delta_\Delta) \le 3\vartheta_{10} \le 3\vartheta.$$

■□

In Thm. 5.48 the hypotheses of (SD) are made on the pair (ℓ, ℓ') alone, and none of them on $\tilde\ell$.

Corollary 5.49 (The level of the switch)**.** Let $\mathsf{b} \ge x_\flat$, $K' \ge K \ge 1$, $m = K' - K$ and $x_k = L^{K-k}\mathsf{b}$. Let ℓ_k and $\tilde\ell_{m+k}$ be the two chains of Cor. 5.81, under its hypotheses, that is, with $L \ge L_B$ odd, $aL \ge 3/c_L$, $\theta_D \le \lambda_*/8$, (X12) and (X13); and let the side N of the torus on which both chains end be even. In this corollary and in its proof, N denotes this side, so that the coarse side of the step that starts at x_k is NL^{K-k-1} for $0 \le k \le K-1$.

Then for $0 \le k \le K$ the number δ_Δ at $(x_k; \ell_k, \tilde\ell_{m+k})$ is $\le c_4 \mathsf{b}^{1/2} L^{K/2 - 9k/10}$, and this bound is ≤ 1 if and only if $k \ge (5K + 10\Lambda)/9$. Moreover $\Lambda > 0$, so that $k_0 \ge 1$; and $k_0 \le K-1$ if and only if $K \ge K_0(\mathsf{b})$. Further, $K - k_0 \ge \frac49 K - \frac{10}{9}\Lambda - 1$. With the numeral $\varphi := 5/9$ of (O14) in item 1 of §5.5, one has $k_0 < \varphi K + \frac{10}{9}\Lambda + 1$, since $k_0 = \lceil (5K + 10\Lambda)/9 \rceil$; hence $\vartheta^{K-k_0} \le \vartheta^{-\frac{10}{9}\Lambda - 1}(\vartheta^{4/9})^K$, and the rate in K of this bound is $\vartheta^{4/9} \le 0.7013$ (and 0.6632 at ϑ_{10}). ■□

The reference coupling changes by $x' = x/L$ in the exact identity (9a). Matching the reference couplings of two runs therefore requires a change of rows only.

Proposition 5.50 (An unbounded strong distance after a change of rows)**.** Let $x \ge x_9$, let the five conditions of item 5 hold, and let $\delta_\Delta \in (0,1]$. Then there are rows ℓ of range 0 and, with $\tilde\ell = \emptyset$, a real $\tilde u \in M_1^{\mathrm{su}}(x)$ with the strong cap, with the following property for the system $u^\flat$ that

Lem. 5.47 forms from these data: for every n such that $\mathbb{T}$ has n sites whose pairwise sup distances are $\geq 2M(r(x)+2)$, there exist a field V and a set X such that

$$|w^\flat - \tilde{w}|(X,V) \cdot \exp\{c_\Xi x\varepsilon \sum_{D(V)\cap X} \phi_p + \kappa(|X|_c - 1)\} \;\geq\; \exp\{n\delta_\Delta c_a(x)/(8c_Q^{\mathrm{su}})\} - 1.$$

Hence, for this system $u^\flat$ of Lem. 5.47, $\mathsf{D}_x(\tilde{u}, u^\flat) > \delta_\Delta$ as soon as $n \geq 8c_Q^{\mathrm{su}}/c_a(x)$, and $\sup_N \mathsf{D}_x(\tilde{u}, u^\flat) = \infty$. ∎

Remark 5.51 (Item 7: weaker contractions implied by Thm. 5.46, the caps in Lem. 5.47, the constants, the change of rows, empty rows). *Weaker contractions implied by Thm. 5.46.* A contraction with the strong distance D_x on the right, and with a factor $\vartheta < 1$ computed from the letters, follows from Thm. 5.46 (ii), because $\mathsf{D}_x^{\mathrm{in}} \leq \mathsf{D}_x$. The factor 0.45 comes from Thm. 5.46 (iv). The stronger form is (ii), and Thm. 5.48 requires it.

The caps in Lem. 5.47. In Lem. 5.47, $\tilde{u}$ has the caps of the class, the strong cap of $\tilde{w}$ included, since every point of a class has it; and $u^\flat \in M_2^{\mathrm{su}}(x)$, with the weak cap for $w^\flat$.

The constants.

(1) The contraction factor of Thm. 5.46 (ii) is $\vartheta_{10} = 0.3968$, because E'^{new} cancels, as it does in Thm. 5.15 (e). The numbers 0.4 and 0.4318 are upper bounds of this factor. The letter ϑ of the table has the value 0.4468.
(2) One has $c_Q^{\mathrm{su}} = 306M^3\mathsf{g}(\lambda_\flat)^2$.
(3) The constant c_4 may be replaced by any larger number.
(4) The condition (F3) of item 5 gives $\delta' \leq \frac{1}{2}\delta_\Delta$ in Step 6, and the inequality $c_a(5+2r)^3 \leq \frac{1}{2}$ is (X18) of (O13) in item 1 of §5.5. At every $x \geq x_9$ both inequalities follow from (Y.4) of item 5 of §5.3, by Step 0 (Fl).
(5) The upper bound of c_h is $c_h \leq \delta_3'/2$, which is (O11). By (O10) in item 3 of §2.1 we have $\delta_3' \leq \delta_3$, and hence $c_h \leq \delta_3/2$, the bound of (O11) in that item.

The change of rows in Thm. 5.15. By item 8 of §5.3, all that Thm. 5.15 requires of the change of rows is $u^\flat \in M_2^{\mathrm{su}}(x)$, and (B2), (B3) give it.

Empty rows.

(1) Of the rows $\ell, \tilde{\ell}$ themselves, Lem. 5.47 assumes only that their range is $\leq \varrho(x)$; its one further hypothesis on the pair is $\delta_\Delta \leq 1$, and it assumes no lower bound of a row and no membership in a set $\mathcal{R}(\cdot)$, and it allows the rows to be empty. With $\ell = \emptyset$, with $\tilde{\ell} = \emptyset$, or with both, Steps 3 to 6 hold without any change. The reasons are the implication $\mathsf{H}_y = 0 \Rightarrow \mathsf{E}_y = 0$, the independence of the mask, the clip and the paths from the rows, and the fact that the argument nowhere divides by δ or calls on a lower bound of a row. When both are empty, $D = 0$, $\delta_\Delta = 0$, $E^\Delta = 0$, all $q_\square = 0$, and $u^\flat = \tilde{u}$. Prop. 5.50 is the case $\tilde{\ell} = \emptyset$.
(2) The condition (SD) admits $\ell = \emptyset$: one has $\emptyset \in \mathcal{R}(1, 0, \lambda_*, 0) \subset \mathcal{R}(b_*, K_*, \lambda_*, \frac{1}{9})$, by the last sentence and the display of Def. 2.48. At $\ell = \emptyset$ the pair (ℓ, ℓ') also has the four conclusions required by (SD); Thm. 2.65 at $k = 0$ and $x_0 = x$ yields the first three of them through (Fr2) for the sixteen kernels, (Fr3′) and (Fr4). Item (8) of that theorem asks for a number > 1 subject to (X12), and here we take x' for it, not the $x_\flat$ of (O13). Step H16 (v) of the proof of Thm. 2.65 shows that every number ≥ 2 satisfies (X12), and $x_0 = x \geq x_9$ gives $x_1 = x' \geq Le \geq 2$; so x' is admissible, and x_1 is at least x', as the theorem requires at $k = 0$. The fourth conclusion, made up of (L1^{su}) and (L2^{su}), holds by (O4). The proofs of Thms. 5.46 and 5.48 draw on no row: at $\ell = \emptyset$ they consist of the same deductions from the four clauses (9a), (9b), (9c) and (9d), which, by §5.3, hold for all data in the setting (SD); and the condition (SD) of §5.3 admits $\ell = \emptyset$ expressly, in item 2 of that section. So at $\ell = \emptyset$ Thms. 5.46 and 5.48 rest on Thm. 5.15 at $\ell = \emptyset$, for members that are complex, lie within up to sixteen caps and obey the weak cap, and for $x \geq x_9$.

Preliminaries to the proofs of this section. Step 0: geometry, the conditions of item 5, and the order of the constants. We claim the following seven statements, (G1) to (Ord), and prove them in the paragraphs after the list.

(G1) If y is a site of a cube $\square$ and z is a site with $|z-y|_\infty \le jM$, then z lies in a cube of $\square^{+j}$.
(G2) Let (F2) hold, and let $y \in \square$ and $|y-p| \le \varrho(x)$. Then all sites of $\Gamma(y,\pi(p))$ and of p lie within $\varrho(x)+1 \le Mr(x)-1$ of y. Consequently $\Gamma(y,\pi(p))$ is a path in $\square^{\approx}$ with no more than $3\varrho(x) \le \ell_w(x)$ bonds, and p is a plaquette of $\square^{\approx}$ whose four bonds are bonds of $\square^{\approx}$.
(G3) Let (F2) hold. Then every path $\Gamma(y,\pi(p))$ with $|y-p| \le \varrho(x)+1$ is defined, and $\square^{\approx}$ is an embedded box.
(G4) For each cube, at most $(2r(x)+1)^3$ cubes $\square$ have it in $\square^{\approx}$.
(G5) For every y, $\sum_{p,p'} |D_y(p,p')| \le 9\mathsf{g}(\lambda_\flat)^2\delta$.
(Fl) For $x \ge x_9$: $\varepsilon < \frac{1}{680}$; (F2) holds; $(5+2r(x))^3 c_a(x) \le x^{-1/4}/(192e^{11})$, so that (F3) holds; and $r(x) \ge 6$.
(Ord) The definitions of $c_Q^{\mathrm{su}}, c_4, \Lambda, K_0, k_0$ form no circle.

For (G1): the cubes are the boxes $o + Mn + \{0,\dots,M-1\}^3$, and moving one coordinate by no more than jM moves the index n by no more than j. For (G2): the coordinate path runs inside the box spanned by y and $\pi(p)$, so its sites lie within $\varrho(x)$ of y; the sites of p are $\pi(p)$, $\pi(p)+e_\mu$, $\pi(p)+e_\nu$, $\pi(p)+e_\mu+e_\nu$, which lie within $\varrho(x)+1$ of y. By (F2) and (G1) all these sites lie in cubes of $\square^{+r(x)}$, and the path has at most $3\varrho(x) \le 3Mr(x) \le \ell_w(x)$ bonds.

For (G3): by (Tor), the side of $\mathbb{T}$ is $NL \ge 16Mr(x)L \ge 48Mr(x)$; so $\varrho(x)+1 \le Mr(x)$ is less than half the side, and the box $\square^{\approx}$, of side $(2r(x)+1)M \le 3Mr(x)$, embeds. For (G4): $\square_* \subset \square^{+r}$ holds if and only if $\square \subset \square_*^{+r}$, so the cubes $\square$ in question are among those of $\square_*^{+r}$. For (G5): $|D_y(p,p')| \le \delta e^{-\lambda_\flat(|y-p|+|y-p'|)}$; each base site carries three plaquettes, and the periodic sum is bounded by the sum over $\mathbb{Z}^3$, so that $\sum_p e^{-\lambda_\flat|y-p|} \le 3\mathsf{g}(\lambda_\flat)$.

For (Fl), first the bound on ε: the condition (Y.0)(0b) of item 5 of §5.3 contains $17T\varepsilon' \le \frac{1}{40}$ and $(p_0'/p_0)^4 \ge \frac{1}{2}$, with the number $T = L^{3/2}/(19\mathsf{N_c}) > 1$, by (L2^{su}) (Step 0 (iv) of the proof of Thm. 5.15). Therefore $\varepsilon' < \frac{1}{680}$, and $\varepsilon = (p_0/p_0')L^{-1/2}\varepsilon' \le 2^{1/4}L^{-1/2}\varepsilon' < \varepsilon'$.

Next, (F2): $\varrho(x)+2 \le c_R \log x + 3 \le 20 c_R \log x \le Mr(x)$, because $19c_R \log x \ge 19 \cdot 768 > 3$ for $x \ge e$.

Next, (F3): each term of the bracket in (Y.4) of item 5 of §5.3 is ≥ 0; hence

$$6((2r_1+1)L+2r(x))^3 e^{\kappa+5} \cdot 16 c_a(x) \cdot e^{(\kappa+4)s'} \le \frac{1}{2}x^{-1/4}.$$

Here the third term of that bracket contains $16(c_a+c_h)(x)$, and the summand $c_h(x) > 0$ is dropped. As to the letters: $r_1 = \lceil R_2/M \rceil + 3r_l + 5 \ge 9$ and $s' = s(x')$ are radii of item 2 of §5.3, where $r_1 \ge 9$ comes from item 2 of §2.1; and $\mathsf{N_c} = 17(\mathsf{N}_\sharp + 1)$, which enters T, is the number of (O3) in item 3 of §2.1. Now $(2r_1+1)L \ge 19L \ge 5$, $e^{\kappa+5} \ge e^{11}$ and $e^{(\kappa+4)s'} \ge 1$, which gives $96e^{11}(5+2r(x))^3 c_a(x) \le \frac{1}{2}x^{-1/4}$; and $192e^{11} \ge 1.1495 \cdot 10^7$.

Last, $r(x) \ge 6$: the condition (Y.7) of item 5 of §5.3 contains $5r_1 + 2\lceil s(x)/L \rceil + 6 \le r'$, and $r(x) \ge r(x') = r'$.

For (Ord): c_Q^{su} depends on M and on the pure number $\lambda_\flat$, and it appears in (O10). No floor depends on it, since $\delta_\Delta \le 1$ is a hypothesis on the pair of rows, not on $x_\flat$. Further, c_4 depends on $c_Q^{\mathrm{su}}, C_1, E_a$, and Λ, K_0, k_0 depend on c_4, L, b, K; all of them appear in (O14). □

Proof of Thm. 5.46. A sum in (9b) bounds the term for any Y containing the cell in that sum. Hence at $A = 2$ the analytic factor is $0.35 + 2 \cdot 0.248 = 0.846$. The other caps and the conditions without a size also follow from (9b), and the strong cap implies the weak cap. This proves (i).

Apply Step 12 of the proof of Thm. 5.15 with $B = 2$ for (ii) and with $B = 1$ for (iii). The weak distance is bounded by the strong one because the strong weight for w is larger.

Under the replacement in (iv), the analytic factor at $A = 2$ is at most 1. In the segment estimate use the full analytic member in place of E'^{old} and its bound at $A = 16$. This gives the factor $\max\{16\vartheta_w + n_w, 1\}/10$. Since $16\vartheta_w + n_w = 8(2\vartheta_w + n_w) - 7n_w \le 8$, the factor is at most 0.8. Substitution gives the two stated numerical values. □

Proof of Lem. 5.47. Step 3: the identity of the energies. We claim that at every V one has $-x\mathrm{En}_{\tilde\ell}(V) = -x\mathrm{En}_\ell(V) + \sum_\square q_\square(V)$, where $q_\square := \sum_{y\in\square} q_y$ and

$$q_y(V) := -\frac{x}{2}\, m_y(V) \sum_{p,p'} D_y(p,p')\langle T(y,\pi(p))\zeta_p, T(y,\pi(p'))\zeta_{p'}\rangle;$$

further, that q_y is real, Borel and gauge invariant, that $|q_y(V)| \le \frac{9}{2}\mathsf{g}(\lambda_\flat)^2 p_0^2\delta$, and that q_y depends on V only at sites within $\max\{\rho_0, \varrho(x)+1\}+1$ of y.

In $\mathsf{H}_y \otimes \mathfrak{g}$ we have

$$|\ell_y^V\zeta|^2 = \sum_{p,p'}\langle \ell_y(p), \ell_y(p')\rangle\langle T(y,\pi(p))\zeta_p, T(y,\pi(p'))\zeta_{p'}\rangle = \sum_{p,p'} \mathsf{E}_y(p,p')\langle T(y,\pi(p))\zeta_p, T(y,\pi(p'))\zeta_{p'}\rangle,$$

so the spaces H_y enter through E_y alone. The sum runs over $|y-p|, |y-p'| \le \varrho(x)$, where the paths are defined, by (G3). The terms $S_\varepsilon(\phi_p)$, the clip ζ, the mask m_y and the paths coincide for ℓ and $\tilde\ell$, since by item 2 none of them depends on a row; in particular the second radius of the mask is $\varrho(x)+1$, which is a function of x and not the range of the rows. Subtracting the two energies and multiplying by $-x$ gives the identity.

As to the size: at a G-valued V each T is an isometry of $\mathfrak{g}$; $|\zeta_p| \le \varepsilon$ at every V; $m_y \le 1$; and $x\varepsilon^2 = p_0^2$. Together with (G5) this yields $|q_y| \le \frac{x}{2}\varepsilon^2 \cdot 9\mathsf{g}(\lambda_\flat)^2\delta$. The function q_y is Borel since L, ang and min are Borel. It is invariant by Prop. 5.35: under $V \to V^u$ each of the two entries of $\langle\cdot,\cdot\rangle$, of the form $T(y,\pi(p))\zeta_p$, becomes $\mathrm{Ad}_{u(y)}\, T(y,\pi(p))\zeta_p$, and $\mathrm{Ad}_{u(y)}$ is an isometry of $\mathfrak{g}$; and m_y is invariant. As to dependence, the mask depends on ϕ_p at $|y-p| \le \max\{\rho_0, \varrho(x)+1\}$, the sites of such a p lie at most one unit farther out, and the paths and the ζ_p stay within $\varrho(x)+1$.

Step 4: the tame part at complex fields. For $\mathsf{W} \in \mathcal{U}_{\{\square\}}(x)$ let $F_p := -i\log\partial\mathsf{W}(p)$, defined by the power series, and

$$E^\Delta(\{\square\}, \mathsf{W}) := -\frac{x}{2}\sum_{y\in\square}\sum_{p,p'} D_y(p,p')\,(\mathrm{Ad}_{\mathsf{W}(\Gamma(y,\pi(p)))}\, F_p)\cdot(\mathrm{Ad}_{\mathsf{W}(\Gamma(y,\pi(p')))}\, F_{p'}),$$

and let $E^\Delta(X,\cdot) := 0$ for $\#X \ge 2$. We claim: E^Δ has the clauses without a size of E^a; it is real at G-valued fields; $\sup_{\mathcal{U}_{\{\square\}}(x)} |E^\Delta(\{\square\},\cdot)| \le c_Q^{\mathrm{su}} p_0^2\delta = \delta_\Delta c_a(x)$; and $E^\Delta(\{\square\}, V) = q_\square(V)$ at every G-valued V that is tame on $\square^{\approx}$.

Dependence: by (G2), every path and every plaquette that occurs lies in $\square^{\approx}$ and keeps a margin of one site there. Holomorphy: by (u3) and (F1), $|\partial\mathsf{W}(p) - 1| < 2\varepsilon \le \frac{1}{20}$; hence the series $\log(1+A) = \sum_{k\ge1}(-1)^{k+1}A^k/k$ converges, and $|F_p| \le -\log(1-2\varepsilon) \le 2.0518\,\varepsilon \le 2.06\,\varepsilon$, because $t \mapsto -\log(1-t)/t$ is increasing and its value at $t = \frac{1}{20}$ is ≤ 1.02587. The entries of $\mathsf{W}(\bar b) = \mathsf{W}(b)^{-1}$ are those of the adjugate. Thus E^Δ is built, by finitely many sums and products, from polynomials in the entries and from convergent power series, so it is holomorphic on $\mathcal{U}_{\{\square\}}(x)$.

As to the size: by (G2) and (u2), $|\mathsf{W}(\Gamma)| < \mathsf{b}(x) < 2$, where the empty path gives 1; so $|\mathrm{Ad}_{\mathsf{W}(\Gamma)}\, F_p| \le |\mathsf{W}(\Gamma)|^2|F_p| \le 4\cdot 2.06\,\varepsilon = 8.24\,\varepsilon$, and $|Y\cdot Y'| \le |Y||Y'|$. A single term is therefore at most $\frac{x}{2}|D_y(p,p')|(8.24\,\varepsilon)^2 = 33.9488\, p_0^2|D_y(p,p')|$. By (G5), and since $\square$ has M^3 sites, the whole sum is at most $33.9488\cdot 9\cdot M^3\mathsf{g}(\lambda_\flat)^2p_0^2\delta \le 305.54\, M^3\mathsf{g}(\lambda_\flat)^2p_0^2\delta \le c_Q^{\mathrm{su}}p_0^2\delta$. At $\#X = 1$ the weight $e^{\kappa(\#X-1)}$ is 1.

Invariance under a G-valued u: $\partial\mathsf{W}^u(p) = u(\pi(p))\partial\mathsf{W}(p)u(\pi(p))^{-1}$, so $F_p \to \mathrm{Ad}_{u(\pi(p))}\, F_p$; also $\mathsf{W}^u(\Gamma(y,z)) = u(y)\mathsf{W}(\Gamma(y,z))u(z)^{-1}$, so $\mathrm{Ad}_{\mathsf{W}(\Gamma)}\, F_p \to \mathrm{Ad}_{u(y)}\,\mathrm{Ad}_{\mathsf{W}(\Gamma)}\, F_p$; and the form $\cdot$ is Ad-invariant. At a pure gauge, $\partial\mathsf{W} = 1$ and $F = 0$.

Reality at the G-valued fields of the domain: write $\partial V(p) = e^{iY}$ with $Y = \mathsf{L}(\partial V(p))$, so that $|Y| = \phi_p$ and $2\sin(\phi_p/2) < \frac{1}{20}$. The series gives $\log e^{iY} = iY$, hence $F_p = \mathsf{L}(\partial V(p)) \in \mathfrak{g}$. For $g \in G$ we have $\mathrm{Ad}_g Y \in \mathfrak{g}$ for $Y \in \mathfrak{g}$, and on $\mathfrak{g}$ one has $\cdot = \langle , \rangle$.

Agreement with $q_\square$: let V be tame on $\square^{\approx}$; its restriction then lies in $\mathcal{U}_{\{\square\}}(x)$, by item 3. Take $y \in \square$. A plaquette with $|y - p| \le \rho_0 \le M$, by (M5r), lies in $\square^{+1}$, which is inside $\square^{\approx}$ because $r(x) \ge 1$; so it does not belong to $D(V)$. A plaquette with $|y - p| \le \varrho(x) + 1 \le Mr(x)$ lies in $\square^{\approx}$, by (G1), and so $\phi_p \le \varepsilon \le 1$. Hence $m_y(V) = 1$. For $|y - p| \le \varrho(x)$ we have $\phi_p \le \varepsilon$, so $\zeta_p = F_p$, and the two formulas agree term by term.

Step 5: the part that is off and (B1). A cube $\square$ is called *on* at V if $1_{\square^{\approx}}(V) = 1$, and *off* otherwise. For an off $\square$, let $p_*(\square, V)$ be the first plaquette of $D(V)$ lying in $\square^{\approx}$, with respect to an ordering of the plaquettes of $\mathbb{T}$ fixed once and for all. We set

$$w^\flat(X, V) := \tilde{w}(X, V) \cdot \prod\{e^{q_\square(V)} : \ \square \text{ off at } V,\ p_*(\square, V) \text{ in } X\}.$$

We claim: $\Xi_{w^\flat}(V) = \Xi_{\tilde{w}}(V) \cdot \prod_{\square \text{ off}} e^{q_\square(V)}$ at every V; (B1) holds; and $w^\flat(X, \cdot)$ is real, Borel and gauge invariant. Moreover the product $\prod\{e^{q_\square(V)} : \square \text{ off at } V,\ p_*(\square, V) \text{ in } X\}$ depends only on the restriction of V to the bonds of $X^{+(2r(x)+1)}$, a set inside $X^{\approx\approx}$; therefore $w^\flat(X, \cdot)$ depends, as $\tilde{w}(X, \cdot)$ does, only on the restriction of V to the bonds of $X^{\approx\approx}$.

The factorisation: in a family of (cΞ), the sets X_i are pairwise more than $s(x) \ge 8$ apart, hence disjoint, and $D(V)$ is contained in their union. Each $p_*(\square, V)$, being a plaquette of $D(V)$, therefore belongs to exactly one X_i. Thus $\prod_i w^\flat(X_i, V) = \prod_i \tilde{w}(X_i, V) \cdot \prod_{\square \text{ off}} e^{q_\square(V)}$, where the last factor does not change from one family to another, and we sum over the families. If $D(V) = \emptyset$, all cubes are on and both Ξ equal 1.

For (B1): by Steps 3 and 4, $-x\mathrm{En}_{\tilde{\ell}} = -x\mathrm{En}_\ell + \sum_{\text{on}} q_\square + \sum_{\text{off}} q_\square$, and $\sum_{\text{on}} q_\square(V) = \sum_\square 1_{\square^{\approx}}(V) E^\Delta(\{\square\}, V) = \sum_{X \in \mathcal{D}} 1_{X^{\approx}}(V) E^\Delta(X, V)$. Hence

$$\rho[\tilde{u}; \tilde{\ell}] = \exp\{-x\mathrm{En}_\ell + \sum_X 1_{X^{\approx}}[\tilde{E}^a + E^\Delta + \tilde{E}^h]\} \cdot \prod_{\text{off}} e^{q_\square} \cdot \Xi_{\tilde{w}} = \rho[u^\flat; \ell].$$

The dependence of the product: if $p_*(\square, V)$ is in X, the cube of p_* lies both in X and in $\square^{\approx}$, and so $\square \subset X^{+r}$. For the cubes $\square$ of X^{+r}, whether $\square$ is off, and which plaquette is $p_*(\square, V)$, is decided by ϕ_p at the plaquettes in $\square^{\approx}$; these lie inside X^{+2r}, and the end points of their bonds lie in $X^{+(2r+1)}$. By Step 3, (M5r) and (F2), $q_\square$ depends on V at sites within $\max\{\rho_0, \varrho(x)+1\}+1 \le \max\{M+1, Mr\}$ of $\square$, hence, by (G1), in $\square^{+\max\{2,r\}}$, which lies inside $X^{+(2r+1)}$ because $\max\{2, r\} \le r+1$. Moreover $2r + 1 \le 3r < 8r = s(x)$. Finally, $\tilde{w}(X, \cdot)$ depends on V through the bonds of $X^{\approx\approx}$, by its clause without a size, and it may depend on every one of them; so $w^\flat(X, \cdot)$, like $\tilde{w}(X, \cdot)$, is a function of V on the bonds of $X^{\approx\approx}$.

For the remaining clauses: the product is finite and is taken over a finite Borel partition of the fields according to $D(V)$, so $w^\flat(X, \cdot)$ is Borel; it is real because $\tilde{w}$ and the $q_\square$ are real; and it is gauge invariant because $D(V)$, the property of being off, p_*, q_y and $\tilde{w}$ are.

Step 6: the two bounds on $w^\flat$, (B2) *to* (B4)*, and $u^\flat \in M_2^{\mathrm{su}}(x)$.* We claim that $|w^\flat| \le e^{-\kappa(|X|_c - 1)}$ and $|w^\flat - \tilde{w}| \le \frac{2}{e}\,\delta' e^{-\kappa(|X|_c - 1)}$, where

$$\delta' := (2r(x)+1)^3 M^3 \frac{9}{2}\,\mathfrak{g}(\lambda_\flat)^2 p_0^2 \delta = \big((2r(x)+1)^3 c_a(x)/68\big)\delta_\Delta \le \frac{1}{2}\delta_\Delta.$$

First, fix a plaquette p_* of $D(V)$ and let $\square_*$ be the cube that holds it. Each cube $\square$ with $p_*(\square, V) = p_*$ has $\square_*$ in $\square^{\approx}$; by (G4) there are at most $(2r+1)^3$ such cubes, each with M^3 sites. By Step 3, $\sum\{|q_\square(V)| : p_*(\square, V) = p_*\} \le \delta'$. The second form of δ' follows from $c_Q^{\mathrm{su}} p_0^2 \delta = \delta_\Delta c_a(x)$ and $4.5/306 = 1/68$; and (F3) together with $\delta_\Delta \le 1$ gives $\delta' \le \frac{1}{2}\delta_\Delta \le \frac{1}{2}$.

Let $n := \#(D(V) \cap X)$ and $Q := \sum\{q_\square(V) : \square \text{ off},\ p_*(\square, V) \text{ in } X\}$; then $|Q| \le \delta' n$. If $n = 0$, the product is empty, $w^\flat = \tilde{w}$, and $|\tilde{w}| \le e^{-\kappa(|X|_c - 1)}$. If $n \ge 1$, then $\phi_p > \varepsilon$ on $D(V)$, so $x\varepsilon\phi_p > p_0^2$ there,

and the strong cap of $\tilde{w}$ together with (X15) gives $|\tilde{w}(X,V)| \le e^{-c_\Xi p_0^2 n} e^{-\kappa(|X|_c-1)} \le e^{-n} e^{-\kappa(|X|_c-1)}$. Then $|w^\flat| = |\tilde{w}|e^Q \le e^{-n+n/2} e^{-\kappa(|X|_c-1)}$, and

$$|w^\flat - \tilde{w}| = |\tilde{w}||e^Q - 1| \le |\tilde{w}| \cdot \delta' n \cdot e^{\delta' n} \le \delta' \cdot n e^{-n/2} \cdot e^{-\kappa(|X|_c-1)} \le \frac{2}{e}\,\delta' e^{-\kappa(|X|_c-1)}.$$

Here we use $|e^Q - 1| \le |Q| e^{|Q|}$, valid for either sign of Q, and $te^{-t/2} \le 2/e$.

Hence (B3). (B2) is Step 4. For (B4): $u^\flat - \tilde{u} = (E^\Delta, 0, w^\flat - \tilde{w})$, and the three numbers of item 3 whose largest is $\|u^\flat - \tilde{u}\|_x$ are at most δ_Δ, 0 and $\frac{1}{e}\delta_\Delta$. Finally, $u^\flat \in M_2^{\mathrm{su}}(x)$: the member $\tilde{E}^a + E^\Delta$ lies within $1 + \delta_\Delta \le 2$ caps, $\tilde{E}^h$ within one cap, and $w^\flat$ within one weak cap; the clauses without a size hold by Steps 4 and 5 together with linearity; and reality holds by Steps 4 and 5. □

Remark 5.52 (On Step 6). In the setting of the table and at every $x \ge x_9$, $\delta' \le 1.3 \cdot 10^{-9}\delta_\Delta$, by Step 0.

Proof of Thm. 5.48. Step 7: the step applied to the two densities. We prove the two formulas for ρ_1' and ρ_2', that both of them lie in $C_\flat^{\mathrm{su}}(x'; \ell'^{\mathrm{c}})$, and the bound on $\mathsf{D}_{x'}$. The definition of the class makes u_2 real, in $M_1^{\mathrm{su}}(x)$, and subject to the strong cap. So Lem. 5.47 applies and gives $\rho_2 = e^{\mathsf{V}_2}\rho[u_2^\flat; \ell]$ as functions on the fields of $\mathbb{T}$, with $u_2^\flat$ real and in $M_2^{\mathrm{su}}(x)$. Since the step is linear and independent of the rows (item 2), (9a) at $A = 2$ gives

$$\rho_2' = e^{\mathsf{V}_2}(\rho[u_2^\flat; \ell])' = e^{\mathsf{V}_2 + \mathsf{V}'[u_2^\flat]}\rho[F_\ell(u_2^\flat); \ell'^{\mathrm{c}}],$$

and (9a) at $A = 1$ gives ρ_1' in the same way.

The class: by Thm. 5.46 (i), $F_\ell(u_1)$ and $F_\ell(u_2^\flat)$ lie in $M_1^{\mathrm{su}}(x')$ with the strong cap, and by (9d) they are real, as are $\mathsf{V}'[u_1]$ and $\mathsf{V}'[u_2^\flat]$. Since $\rho[u_1; \ell] = e^{-\mathsf{V}_1}\rho_1 > 0$ and $\rho[u_2^\flat; \ell] = e^{-\mathsf{V}_2}\rho_2 > 0$, the images are > 0 by (9d); and ℓ'^{c} has range $\le \varrho(x')$.

The distance: $\mathsf{D}_x^{\mathrm{in}}$ is the distance of a norm, so item 3 and (B4) give

$$\mathsf{D}_x^{\mathrm{in}}(u_1, u_2^\flat) \le \mathsf{D}_x^{\mathrm{in}}(u_1, u_2) + \mathsf{D}_x^{\mathrm{in}}(u_2, u_2^\flat) \le \mathsf{D}_x(u_1, u_2) + \delta_\Delta.$$

By item 3, $\mathsf{D}_x(u_1, u_2) \le 2$; and Thm. 5.46 (ii) is applied to $(u_1, u_2^\flat)$, both of which lie in $M_2^{\mathrm{su}}(x)$. □

Proof of Cor. 5.49. Step 8: the level of the switch. We claim that $\delta_\Delta(k) \le c_4 \mathsf{b}^{1/2} L^{K/2 - 9k/10}$, that $\Lambda > 0$, that $k_0 \ge 1$, that $k_0 \le K - 1$ if and only if $K \ge \frac{5}{2}\Lambda + \frac{9}{4}$, and that $K - k_0 \ge \frac{4}{9}K - \frac{10}{9}\Lambda - 1$.

Cor. 5.81 gives $\delta_k := \|\tilde{\mathsf{E}}_{m+k} - \mathsf{E}_k\|_{(\lambda_\flat)} \le 15 C_1^2 \vartheta_1^k \le 15 C_1^2 L^{-2k/5}$, in the norm of Lem. 5.47; by Step 1 of the proof of Thm. 5.63, the hypotheses of that corollary hold for the letters of the table. The last inequality holds because $\vartheta_1 = c_{\mathrm{fix}} L^{-1/2} \le L^{1/10} L^{-1/2}$, as $L \ge L_B \ge c_{\mathrm{fix}}^{10}$ by (O0) and (O4). Both chains, at partner levels, live on a single torus and have range $\le \varrho(x_k)$. The inequality $\theta_D \le \lambda_*/8$ follows from the definitions alone: $\theta_D \le \lambda_2^*/2$ by (O6), and $\lambda_2^* = \min\{\lambda_*/4, [6K_*^2 \mathsf{g}(\lambda_*/2)\mathsf{g}_1(\lambda_*/4)]^{-1}\} \le \lambda_*/4$, by the order of the constants, (0), in §2.2.

By (X14),

$$\delta_\Delta(k) = (c_Q^{\mathrm{su}}/E_a) x_k^{1/2} \delta_k / p_0(x_k) \le (c_Q^{\mathrm{su}}/E_a) \mathsf{b}^{1/2} L^{(K-k)/2} \cdot 15 C_1^2 L^{-2k/5} = c_4 \mathsf{b}^{1/2} L^{K/2 - 9k/10},$$

and this is ≤ 1 if and only if $\frac{9}{10}k \ge \frac{K}{2} + \Lambda$. As to $\Lambda > 0$: $16 c_a(\mathsf{b}) \le \frac{1}{2}$ and (X14) give $\mathsf{b}^{1/2} \ge 32 E_a p_0(\mathsf{b})^3 \ge 32 E_a$, hence $c_4 \mathsf{b}^{1/2} \ge 480 c_Q^{\mathrm{su}} C_1^2 > 1$, since $c_Q^{\mathrm{su}} \ge 306$ and $C_1 > 1$. Put $y := (5K + 10\Lambda)/9 > 0$. Then $k_0 = \lceil y \rceil \ge 1$; and since $K - 1$ is an integer, $\lceil y \rceil \le K - 1$ holds if and only if $y \le K - 1$, that is, if and only if $4K \ge 10\Lambda + 9$. Also $K - k_0 \ge K - y - 1$.

At $k = k_0 \le K - 1$ we have $x_{k_0} \ge L\mathsf{b} \ge L x_\flat \ge x_9$. So the floor of Thm. 5.48 is met there, and, by Step 0 (Fl), so are the conditions (F1), (F2), (F3); (M5r) and (X15) are entries of the table; and $\delta_\Delta \le 1$ is the bound just obtained. The remaining hypotheses of Thm. 5.48 are established elsewhere: (SD) at $(x_{k_0}, NL^{K-k_0-1}, \ell_{k_0})$, for N even with $M \mid N$ and $N \ge 2Ms(L\mathsf{b})$, by Step 2 (d) of the proof of Thm. 5.63; and the membership of the two densities in their classes by item 1 of Step 4

there. With these, Thm. 5.48 applies at k_0. Numerically, $0.45^{4/9} \le 0.70125$, $0.4468^{4/9} \le 0.69903$ and $0.3968^{4/9} \le 0.66312$. □

Proof of Prop. 5.50. Step 10: the lower bound. The letters and $x \ge x_9$ are fixed. By Step 0 (Fl), $r(x) \ge 6$ and $\varepsilon \le \frac{1}{40}$; and $M \ge 3$ by (M2), the condition $M \ge 1920(\kappa+4)c_R$ of (O7).

The rows: $\tilde{\ell} = \emptyset$; $\mathsf{H}_y = \mathbb{R}$, and $\ell_y(p) = t$ if $p = p_y := (y;12)$, while $\ell_y(p) = 0$ otherwise, so ℓ has range 0. Then $D_y(p,p') = -t^2$ at $p = p' = p_y$ and $D_y(p,p') = 0$ otherwise, so $\delta = t^2$; we choose $t^2 := \delta_\Delta c_a(x)/(c_Q^{\mathrm{su}} p_0^2)$. Since $T(y,y) = 1$, $q_y(V) = \frac{x}{2} t^2 m_y(V) |\zeta_{p_y}|^2 \ge 0$ at every V.

The members: $\tilde{u} = (0,0,\tilde{w})$ with $\tilde{w}(X,V) := \exp\{-c_\Xi x\varepsilon \sum_{D(V)\cap X} \phi_p - \kappa(|X|_c - 1)\}$. This function is real, Borel and gauge invariant, depends on ϕ on X only, and meets the strong cap with equality; moreover $\rho[\tilde{u};\emptyset] > 0$.

The field: take sites $z_1, \dots, z_n$ whose pairwise sup distances are $\ge 2M(r(x)+2)$; put $d := \max\{\rho_0, \varrho(x)+1\}+3$ and $y_i := z_i + de_1$; and let $V := 1$ except for $V(z_i,3) := e^{2i\varepsilon\sigma_3}$ and $V(y_i,1) := e^{i(\varepsilon/2)\sigma_3}$. The plaquettes containing the bond $(z_i,3)$ are $(z_i;13)$, $(z_i - e_1;13)$, $(z_i;23)$, $(z_i - e_2;23)$; they have $\phi = 2\varepsilon$, which is ≤ 1, and they are rough. The plaquettes containing $(y_i,1)$ are $(y_i;12)$, $(y_i - e_2;12)$, $(y_i;13)$, $(y_i - e_3;13)$; they have $\phi = \varepsilon/2$ and are tame. Because $d \ge 4$, no plaquette contains two of the changed bonds, and every other plaquette has $\phi = 0$. Moreover $d \le Mr(x)$, as $\rho_0 + 3 \le M + 3 \le 2M$ and $\varrho(x) + 4 \le c_R \log x + 5 \le 20 c_R \log x$; so, by (G1), the cube of z_i is one of the cubes of $\square^{+r(x)}$ for the cube $\square$ of y_i, and that cube $\square$ is therefore off. The four rough plaquettes that contain the bond $(z_i,3)$ satisfy $|y_i - p| \ge d > \max\{\rho_0, \varrho(x)+1\}$, and the rough plaquettes that contain a bond $(z_j,3)$ with $j \ne i$ satisfy $|y_i - p| \ge 2M(r+2) - d - 1 > Mr$; hence $m_{y_i}(V) = 1$. The plaquette p_{y_i} contains $(y_i,1)$, so $|\zeta_{p_{y_i}}| = \varepsilon/2$ and $q_{y_i}(V) = t^2 p_0^2/8$.

The set: let X consist of the cubes that hold a plaquette of $D(V)$. Each off cube has its p_* in $D(V)$ and hence in X; $q_y \ge 0$ for every y; and the y_i are distinct sites of off cubes. Therefore $w^\flat(X,V)/\tilde{w}(X,V) = e^Q$ with $Q \ge nt^2 p_0^2/8 = n\delta_\Delta c_a(x)/(8c_Q^{\mathrm{su}})$. Since $\tilde{w}$ equals the cap, the weighted difference is $e^Q - 1$. For $n \ge 8c_Q^{\mathrm{su}}/c_a(x)$ this gives $Q \ge \delta_\Delta$ and $e^Q - 1 > \delta_\Delta$. Sites $z_1, \dots, z_n$ whose pairwise sup distances are $\ge 2M(r(x)+2)$ exist on $\mathbb{T}$ as soon as $\lfloor NL/(2M(r(x)+2)) \rfloor^3 \ge n$. The conditions of item 2 on N are N even, $M \mid N$ and (Tor) $N \ge 2M\, s(x)$. None of them bounds N from above. □

5.5. **Comparison of two cutoffs.** Two runs with the same final torus and label have the same coupling at partner levels. Their site forms approach each other until one run can be expressed over the rows of the other. The contraction then applies at every subsequent step. A lower bound for the normalizing integral converts the resulting comparison into total variation at every fixed depth. We also apply the comparison to the response to a fixed abelian flux.

Item 0: setting, what is used, signs. Our group is SU(2), the dimension is $d = 3$, and the step is the drawn step. We rely on Thm. 5.7; on clauses (a), (b), (d), (e) of Thm. 5.15, together with the data (SD) of item 2 of §5.3 and with item 3 of §5.3; and on Lem. 5.47.

The label b is the coupling at the last level, and $G = \mathrm{SU}(2)$. For laws with densities f, g against Haar measure we define $\|\mu - \nu\|_{\mathrm{TV}} := \int |f - g|\, dV$, without a factor $\frac{1}{2}$. With an argument, $s(x) = 8r(x)$ is a radius; without an argument, s denotes the depth $K - k$; so in $NL^s \ge 2M\, s(L^s \mathsf{b})$ the exponents are the depth while $s(\cdot)$ is the radius; and in k_s and $z(s)$ the letter s is the noise, which is a third meaning. The number of (O11) is c_h, and the cap is $c_h(x)$; next to c_a and without an argument, as in $(c_a + c_h)$, the letter c_h denotes the cap at the coupling in question. Laws are written μ, ν, but μ in Step 9 (4) is Haar measure of G. By $K_{\mathrm{q},k}$ and $K_v^\flat$ we mean the K_{q} and K_v of (O10) in item 3 of §2.1. A system of members is written u; in V^u, $u(y)$, u', $\mathrm{Ad}_{u(y)}$ the letter stands for a gauge transformation. The coupling of the level is x, and $x' = x/L$. In item 2, in the paragraph of the definition "The step" that defines M and in the proof of the three clauses, and only there, x and $x' = x + Le_\mu$ are fine sites and $y(x)$ is a coarse site, as in item C.I 4 of §3; in

k_{ax}, and where we state on what the step depends, x is the coupling. In Cor. 5.66 and in Step 12, $\sigma_3 = \mathrm{diag}(1,-1)$. Finally, (9a), (9b), (9d), (9e) stand for clauses (a), (b), (d), (e) of Thm. 5.15.

A step that we cite without naming a statement belongs to a proof of this section: Steps 0 to 6 to the proof of Thm. 5.63, Steps 7 to 10 to that of Thm. 5.64, Step 11 to that of Cor. 5.65, and Step 12 to that of Cor. 5.66.

Item 1: letters. We go through the entries (O0) to (O13) of the one table, item 3 of §2.1, and the entry (O14), in their order.

(O0) Pure numbers. Def. 2.56 provides b_*, γ_*, $\lambda_* = \gamma_*^{1/2}/44$ (≤ 1), $K_* = 3\gamma_*^{-1/2}$, L_A and $\mathsf{t}_*(\varrho) = 3K_*\mathsf{g}(\lambda_*/2)e^{-\lambda_*\varrho/2}$. The numbers $c_1^*, c_2^*, \lambda_2^*, \theta_J$ are those of Thm. 2.65, "The order of the constants" (0); $c_1^*, c_2^*, \lambda_2^*$ also appear in Def. 2.61, and θ_J in Rem. 2.36. Here $\mathsf{g}(\lambda) = \sum_{n\in\mathbb{Z}^3} e^{-\lambda|n|_\infty}$. We set $C_1 = \sqrt{3}\,K_*\,\mathsf{g}(\lambda_*/2)^{3/2}$, $c_0 := 108\,b_*\gamma_*^{-1/2}\mathsf{g}(\lambda_*/4)^3$, $c_{\mathrm{fix}} := 48\,\mathsf{g}(\lambda_\flat)C_1c_0^{1/2}$ and $L_B = \max\{L_A, c_{\mathrm{fix}}^{10}\}$, where $\lambda_\flat = \lambda_*/16$; and $\vartheta_1 := c_{\mathrm{fix}}L^{-1/2}$, a number that depends on L.

(O1) $d_0 \in (0, \frac12)$.

(O2) $0 < c_\Xi \le \frac12(\frac12 - d_0)$.

(O3) The radius ρ_0 is given by eq. 2.vi of §2.2, and $\mathsf{N}_Z = 1 + 3K_*^2\mathsf{g}(\lambda_*)^2$.

(O4) L is odd and $L \ge L_*^{\mathrm{su}}$, where L_*^{su} is the least integer $\ge \max\{L_A, L_B\}$ such that (L1^{su}), (L1$^\flat$) with right side $\frac{3}{16}d_0$ and (L2^{su}) hold at that integer and at every larger L ((O4) in item 3 of §2.1). The number $c_L = (L^2-1)/(6L^2)$ depends on L, and $a \ge 3/(c_L L)$.

(O5) $\kappa \ge 6$.

(O6) This entry fixes $\gamma = (73L^4)^{-1}$, $A^\flat$, $\theta^\flat$, $\theta_{\mathbb{C}} = \theta^\flat/6$ and $\vartheta_0^\flat = \theta^\flat/24$. In it, $A^\flat = 2e^2(16+a) + 384e^8K_*^2\mathsf{g}_2(\lambda_*/2)^2$, $\theta^\flat = \min\{\theta_1, \gamma/(2A^\flat)\}$ and $\theta_1 = \min\{1, \lambda_*L/12\}$, taken from item C.I 1.2 of §3, at $(K,\lambda) = (K_*, \lambda_*)$, and from Thm. 2.65, "The order of the constants" (6).

We further put $\theta_D = \frac12\min\{\lambda_2^*, \theta_{\mathbb{C}}, \theta_J\}$, and we take $C_D \ge 4$ and no smaller than the number C_D of Thm. 2.65, "The order of the constants" (6), defined with this θ_D. This choice is legitimate: (Dec) of §2.2 bounds sixteen kernels from above by C_D, so it stays true for every larger C_D; and a number of this kind is what Sections 5.1, 4.4 and 5.3 mean by C_D. Then $c_R = 64/\theta_D$.

We have $\theta_D \le \lambda_*/8$, which follows from the choice of the letters (Step 0 (i)). Hence, as $c_R = 64/\theta_D$ and $\lambda_* \le 1$, we get $e_R := \lambda_* c_R/2 \ge 256$ and $c_R \ge 512$.

(O7) $M = L^{m'}$ with $m' \ge 1$, subject to three conditions.

(M2) $M \ge 1920(\kappa+4)c_R$.

(M4$^\flat$) The condition of this name in (O7) of item 3 of §2.1.

(M5r) $M \ge \rho_0$.

The condition (M4$^\sharp$) is a consequence of (M4$^\flat$).

(O8) $\mathfrak{p}_0 > 2$ and $A_0 > 0$; moreover $p_0(x) = A_0(\log x)^{\mathfrak{p}_0}$ and $\varepsilon = \varepsilon(x) = p_0(x)x^{-1/2}$.

(O9), (O10) These two entries contain the numbers of the law of windows and those of Thm. 5.3 and of Sections 3, 4.1, 4.2, 4.3. Four of these numbers are $\delta_3, \delta_3', K_{\mathrm{q},k}, K_v^\flat$. From §5.4 we take $c_Q^{\mathrm{su}} := 306\,M^3\mathsf{g}(\lambda_\flat)^2$.

(O11) $0 < c_h \le \delta_3'/2$.

(O12) $E_a \ge \max\{32K_{\mathrm{q},k},\, 1500K_v^\flat L^{-1/2}\}$. The caps are $c_a(x) = E_a p_0(x)^3 x^{-1/2}$ and $c_h(x) = e^{-c_h p_0(x)^2}$.

(O13) This entry is the last of the choices. It is a number $x_\flat \ge e^2$ such that $x_\flat \ge x_\flat^{(1)}$, $x_\flat \ge x_\flat^0$ and $L\,x_\flat \ge x_9$, such that (X12) and (X13) hold, and such that (X14) to (X18) hold for all $x \ge x_\flat$.

Here $x_\flat^{(1)}$ is the number of Thm. 5.3; it coincides with the $x_\flat^0$ of (O13) in item 3 of §2.1 evaluated at $c_h = \delta_3/2$, $E_a = \max\{32L^{-1/2}K_v^\flat, 32K_{\mathrm{q},k}\}$. Next, $x_\flat^0$ is the number of (O13)

in item 3 of §2.1, computed from the c_h, E_a of (O11), (O12); and x_9 is the number of (O13) in item 1 of §5.3.

We recall the definitions of three numbers, in this order: $x_{\mathfrak{fl}}$ (item G of §4.3), $x_\flat^0 = x_\flat^0(c_h, E_a)$ and x_9. First, $x_{\mathfrak{fl}} := 1 + \inf S_{\mathfrak{fl}}$, where $S_{\mathfrak{fl}}$ is the set of the $y \geq L^2 e$ for which every floor of item G (Floors) of §4.3 is valid at each $x \geq y$; here $e = \exp 1$. Second, $x_\flat^0 = x_\flat^0(c_h, E_a) := 1 + \inf S(c_h, E_a)$, where $S(c_h, E_a)$ consists of the $y \geq e^2$ for which (A0) to (A11) of item 4 of §2.1 are valid at each $x \geq Ly$. Third, $x_9 := 1 + \inf S_9$, where S_9 is the set of the $y \geq x_{\mathfrak{fl}}$ for which (Y.0) to (Y.9) of item 5 of §5.3 are valid at each $x \geq y$.

Each of these sets is an upper half line: together with y it contains every $\tilde{y} \geq y$. None of them is empty, because each floor holds for all large x, as Step 13 of the proof of Thm. 4.19, Step 14 of the proof of Thm. 4.39 and Step 0 (i) of the proof of Thm. 5.15 show. A set of this kind need not contain its infimum, since the inequalities of the floors involve numbers rounded up to an integer, $\lceil \cdot \rceil$, and these, as functions of x, are continuous from the left and jump upwards to the right. The number $1 + \inf$ does lie in the set, which is an upper half line, and it depends on what the set depends on.

The definitions of S_9 and $S_{\mathfrak{fl}}$ give $x_9 \geq x_{\mathfrak{fl}} \geq L^2 e$.

The conditions named in this entry read as follows.

(X12) $3K_* \mathsf{g}(\lambda_*/2)\, x_\flat^{-e_R}/(1 - L^{-e_R}) \leq \frac{1}{9}$.
(X13) $16\, x_\flat^{-e_R/4} \leq 1$.
(X14) $p_0(x) \geq 1$.
(X15) $c_\Xi p_0(x)^2 \geq 1$.
(X16) $\varepsilon(x) \leq \frac{1}{40}$.
(X17) $16(c_a + c_h)(x) \leq \frac{1}{2}$.
(X18) $c_a(x)(5 + 2r(x))^3 \leq \frac{1}{2}$.

The condition (X13) follows from the choice of the letters and imposes nothing more: $x_\flat \geq e^2$ and $e_R \geq 256$ give $16 x_\flat^{-e_R/4} \leq 16 e^{-128}$, which is less than 1. The condition (X12) is a consequence of that choice as well: $3K_* \mathsf{g}(\lambda_*/2) < 4 \cdot 10^{56}$, $x_\flat^{-e_R} \leq e^{-512} < 10^{-222}$ and $1 - L^{-e_R} \geq \frac{1}{2}$, and these three bounds give (X12).

The bound $4 \cdot 10^{56}$ rests on $\mathsf{g}(\lambda) \leq 51 \lambda^{-3}$ for $0 < \lambda \leq 1$, one of the bounds after the display of $\mathsf{g}_1, \mathsf{g}_2$ in Def. 2.44, and on $K_* \leq 3.1527 \cdot 10^{12}$, $\lambda_* \geq 2.1626 \cdot 10^{-14}$, from Thm. 2.65, "The order of the constants" (0). Indeed, these give $3K_* \mathsf{g}(\lambda_*/2) \leq 3 \cdot 3.1527 \cdot 10^{12} \cdot 51 \cdot 8/(2.1626 \cdot 10^{-14})^3 \leq 3.82 \cdot 10^{56}$.

(O14) The numbers of this entry are defined once all the choices (O0) to (O13) have been made, and no floor depends on them. We have $\vartheta_w = 0.248$ (Thm. 5.15 (b)) and $\vartheta := (16\vartheta_w + \frac{1}{2})/10 = 0.4468$.

We put $c_4 := 15\, c_Q^{\mathrm{su}} C_1^2 / E_a$, and for $\mathsf{b} \geq x_\flat$ we put $\Lambda := \log_L(c_4 \mathsf{b}^{1/2})$ and $K_0(\mathsf{b}) := \frac{5}{2}\Lambda + \frac{9}{4}$, which is the least length of a run. For an integer K the level of the switch is $k_0 := \lceil (5K + 10\Lambda)/9 \rceil$. Moreover $\varphi := 5/9$ and $\theta := \vartheta^{4/9}$, so that $0.69902 < \theta \leq 0.69903$, and we set $\varepsilon_L := \frac{10}{9} \log(1/\vartheta)/\log L$ and $C_\mathsf{b} := (3/\vartheta)(c_4 \mathsf{b}^{1/2})^{\varepsilon_L}$, where $3/\vartheta \leq 6.715$.

Finally, with $n = (N/M)^3$,

$$C_{\mathrm{vol}}^{\mathrm{su}}(N, x) := 4n\, e^{220n}\, e^{\frac{3}{2} \mathsf{N}_Z p_0(x)^2 N^3}\, (384/\varepsilon(x)^3)^{3N^3},$$

and

$$Q^{\mathrm{su}}(\mathsf{b}) := 19.52\, M^{-3} [E_a p_0(\mathsf{b}) \mathsf{b}^{1/2} + \mathsf{b}\, c_h(\mathsf{b}) p_0(\mathsf{b})^{-2}].$$

Item 2: objects at a level. We now fix the objects that live at one level.

Definition 5.53 (Lattice, cubes)**.** A level is a torus $\mathbb{Z}^3/N\mathbb{Z}^3$, with N even and $M \mid N$, that carries a grid of cubes of M^3 sites each; $n := (N/M)^3$ is the number of its cubes. It has $3N^3$ bonds b and $3N^3$ plaquettes p.

The base site $\pi(p)$ of a plaquette is its least corner, and $|y-p| := |y-\pi(p)|_\infty$. Each object is assigned to the cube that contains its base site, and we say that X holds p when p lies in a cube of X. The distance between two cubes is their sup distance, counted in cubes.

By $\mathcal{D}$ we denote the family of the non-empty sets of cubes that are connected through faces, edges and corners. *First count:* a given cube is held by at most $(26e)^{j-1}$ members of $\mathcal{D}$ with j cubes.

By $\mathcal{P}$ we denote the family of the non-empty finite sets of cubes, and $|X|_c$ is the smallest number of cubes that a member of $\mathcal{D}$ holding X can have. *Second count:* a given cube is held by at most $(52e)^j$ members X of $\mathcal{P}$ with $|X|_c = j$. ■

Proof of the two counts. We regard the cubes as the vertices of a graph of degree at most $\Delta = 26$, in which two cubes are neighbours when their sup distance is 1, and we number the neighbours of each cube by $1, \dots, \Delta$. Let X be a connected set of j cubes that holds $\square$. It has a spanning tree rooted at $\square$, and we fix one by a rule: breadth first, and least numbers first. Reading the numbers along it, we see this tree as a rooted subtree, on j vertices, of the infinite rooted tree whose vertices each have Δ numbered children. The set X consists of the cubes reached from $\square$ along this subtree, so distinct sets X give distinct subtrees. These subtrees are counted by the Fuss–Catalan number $\binom{\Delta j}{j}/((\Delta-1)j+1)$.

Now $\binom{\Delta j}{j} \le (\Delta j)^j/j! \le \Delta^j e^{j-1}$, because $j! \ge e(j/e)^j$. The last inequality is an equality at $j=1$, and when j is replaced by $j+1$ its left side gains the factor $j+1$, while its right side gains the factor $(j+1)(1+1/j)^j/e \le j+1$. Moreover $(\Delta-1)j+1 \ge \Delta$. Hence the number of the sets X is $\le (e\Delta)^{j-1}$, which is the first count. (Encoding the tree by a walk along its contour gives only 676^{j-1}, and $676e^{-\kappa} > 1$ at $\kappa = 6$, the least value that (O5) admits.)

For $\mathcal{P}$, let X hold the given cube, with $|X|_c = j$, and pick a member of $\mathcal{D}$ with j cubes that holds X. That member holds the given cube as well, so it is one of at most $(26e)^{j-1}$ sets, and X is one of its 2^j subsets. As $(26e)^{j-1}2^j \le (52e)^j$, the second count follows. □

Definition 5.54 (Lattice, cubes, continued)**.** For a set X of cubes, X^{+j} is the set of the cubes at distance $\le j$ from X. For a set A of cubes, the bonds of A are the bonds with both ends in A, while the paths and the plaquettes of A are those all of whose sites lie in A; a plaquette p is said to be in A when its base site lies in A.

A field on the bonds of A is called tame there if $\phi_p \le \varepsilon$ at the plaquettes of A (item 1 of §2.1; item 2 of §5.3). If $1_A(V) = 1$, with 1_A as in the definition "Group, fields" below, then V, restricted to the bonds of A, is tame there, because each plaquette of A is in A. In $\|\cdot\|_x$, D_x^{in} and D_x, the suprema of E^h are taken over the G-valued fields on the bonds of $X^{\approx}$ which are tame there. ■

Definition 5.55 (At x)**.** The radii at x are $\varrho(x) = \lceil c_R \log x \rceil$, $r(x) = \lceil 20 c_R \log x/M \rceil$ and $s(x) = 8r(x)$; further $\ell_w(x) = 6M\, r(x) + 12$, $X^{\approx} = X^{+r(x)}$ and $X^{\approx\approx} = X^{+s(x)}$. ■

Definition 5.56 (Group, fields)**.** Let $\mathfrak{g}$ be the space of the hermitian traceless 2×2 matrices, with $\langle Y, Y'\rangle = \frac{1}{2}\operatorname{tr} YY'$ and $|Y|^2 = \langle Y, Y\rangle$; for $g \in G$ the map $\operatorname{Ad}_g Y = gYg^{-1}$ is an isometry of $\mathfrak{g}$. The angle of g is $\operatorname{ang}(g) = \arccos(\frac{1}{2}\operatorname{tr} g) \in [0,\pi]$. For $g \ne -1$ we let $\mathsf{L}(g)$ be the unique $Y \in \mathfrak{g}$ with $e^{iY} = g$ and $|Y| < \pi$, and we set $\mathsf{L}(-1) := 0$; then $|\mathsf{L}(g)| = \operatorname{ang}(g)$ for $g \ne -1$.

A field V assigns to each oriented bond an element of G, with $V(\bar b) = V(b)^{-1}$, and dV denotes the product of the Haar probability measures. We write $\partial V(z;\mu\nu) = V(z,\mu)V(z+e_\mu,\nu)V(z+e_\nu,\mu)^{-1}V(z,\nu)^{-1}$, denote by $V(\pi)$ the ordered product along a path π, and put $V^u(b) = u(b_-)V(b)u(b_+)^{-1}$, as in the definition "Fields" in item 2 of §5.4. Finally $\phi_p = \operatorname{ang}\partial V(p)$ and $F_p = \mathsf{L}(\partial V(p))$; the rough plaquettes form the set $D(V) = \{p : \phi_p > \varepsilon\}$; and $1_Y(V) = 1\{\phi_p \le \varepsilon \text{ for all } p \text{ in } Y\}$, where we say that V is tame on Y when $1_Y(V) = 1$. ■

Definition 5.57 (Energy)**.** Rows $\ell = (\ell_y)$ take their values $\ell_y(p)$ in a real Hilbert space H_y and have range $\le \varrho(x)$, which means $\ell_y(p) = 0$ for $|y-p| > \varrho(x)$. Their site forms are $\mathsf{E}_y(p,p') :=$

$\langle \ell_y(p), \ell_y(p') \rangle$, and for two families of site forms we put

$$\|\tilde{\mathsf{E}} - \mathsf{E}\|_{(\lambda)} := \sup_{y,p,p'} |(\tilde{\mathsf{E}}_y - \mathsf{E}_y)(p,p')|\, e^{\lambda(|y-p|+|y-p'|)};$$

this is the second norm of the definition "Norms of a family of symmetric operators" in Step 1a.

The coordinate path $\Gamma(y, y'')$ goes first in the direction e_1, then e_2, then e_3, each time the short way; it is defined when the sup distance is less than half the side, and we put $T(y, y'') = \mathrm{Ad}_{V(\Gamma(y,y''))}$. The clip is $\zeta_p = F_p \min\{1, \varepsilon/\phi_p\}$, with multiplier 1 at $\phi_p = 0$; hence it is 0 at $\partial V(p) = 1$ or -1. The dressed rows are $\ell_y^V \zeta = \sum_p \ell_y(p) \otimes T(y, \pi(p))\zeta_p$.

The mask $m_y(V)$ is 1 when $D(V)$ has no plaquette within ρ_0 of y and there is no plaquette with $\phi_p > 1$ within $\varrho(x)+1$ of y; otherwise the mask is 0. With $S_\varepsilon(t) = t^2/2$ for $t \le \varepsilon$ and $d_0 \varepsilon t$ for $t > \varepsilon$, the energy is

$$\mathrm{En}_\ell(V) = \sum_p S_\varepsilon(\phi_p) + \frac{1}{2} \sum_y m_y(V) |\ell_y^V \zeta|^2.$$

The clip, the mask and the paths are functions of $V, \varepsilon, \rho_0, \varrho(x)$, and of no rows. ■

Definition 5.58 (Domain)**.** For $X \in \mathcal{D}$, the complex domain $\mathcal{U}_X(x)$ consists of the maps V that send the oriented bonds of $X^{\approx}$ to $\mathrm{SL}(2,\mathbb{C})$, satisfy $\mathsf{V}(\bar{b}) = \mathsf{V}(b)^{-1}$, and have the following three properties, $|\cdot|$ being the operator norm.

(u1) $|\mathsf{V}(b)| < 1 + \vartheta_0^\flat (1 - 1/\log x)$.
(u2) $|\mathsf{V}(\pi)| < 2 - 1/\log x$ for every path π in $X^{\approx}$ with at most $\ell_w(x)$ bonds.
(u3) $|\partial \mathsf{V}(p) - 1| < 2\varepsilon$.

For $x > e$ this set contains every G-valued field that is tame on $X^{\approx}$. Indeed, for such a field the norms in (u1), (u2) are equal to 1, and $1 - 1/\log x > 0$; further $|\partial V(p) - 1| = 2\sin(\phi_p/2) \le \varepsilon < 2\varepsilon$, since the eigenvalues of $\partial V(p)$ are $e^{\pm i\phi_p}$. ■

Definition 5.59 (Members, class, distances)**.** The members, the class and the distances below are those of item 3 of §5.3. For $A \ge 1$, $M_A^{\mathrm{su}}(x)$ is the set of the triples $u = (E^a, E^h, w)$ that have the following three properties.

(ca_A) For $X \in \mathcal{D}$, the function $E^a(X, \cdot)$ is holomorphic on $\mathcal{U}_X(x)$, invariant under G-valued gauge transformations and equal to 0 at pure gauges, and $\sup_{\mathcal{U}_X(x)} |E^a(X,\cdot)| \le A\, c_a(x) e^{-\kappa(\#X - 1)}$.

(ch_A) $E^h(X, \cdot)$ is a complex Borel gauge invariant function of those G-valued fields on the bonds of $X^{\approx}$ that are tame there, and $\sup |E^h(X,\cdot)| \le A\, c_h(x) e^{-\kappa(\#X-1)}$, where the supremum runs over these fields.

($\mathrm{c}\Xi_A$) For $X \in \mathcal{P}$, the function $w(X, \cdot)$ is complex, Borel and gauge invariant, it depends on V only through the bonds of $X^{\approx\approx}$, and $|w(X,V)| \le A\, e^{-\kappa(|X|_c - 1)}$ at every V; this bound is the weak cap.

The strong cap is the bound $|w(X,V)| \le \exp\{-c_\Xi x \varepsilon \sum_{p \in D(V) \cap X} \phi_p - \kappa(|X|_c - 1)\}$ at every V. We call u real when E^a is real at G-valued fields and E^h, w are real. For a triple that satisfies every requirement of (ca_A), (ch_A), ($\mathrm{c}\Xi_A$) except the three bounds in which A occurs, $\|u\|_x$ is the least $A \ge 0$ with which the three size bounds hold; this is a norm, as in item 3 of §5.3 and in item 3 of §5.4.

We put

$$\rho[u;\ell](V) := \exp\{-x \mathrm{En}_\ell(V) + \sum_{X \in \mathcal{D}} 1_{X^{\approx}}(V) [E^a + E^h](X,V)\} \cdot \Xi_w(V),$$

where $\Xi_w(V) = \sum_{\{X_i\}} \prod_i w(X_i, V)$, the sum running over the finite families of sets $X_i \in \mathcal{P}$ at pairwise distance $> s(x)$ such that each X_i holds a plaquette of $D(V)$ and their union contains $D(V)$; the empty family, with the product 1, is admitted exactly when $D(V)$ is empty. The class is

$$C_\flat^{\mathrm{su}}(x;\ell) := \{e^{\mathsf{V}} \rho[u;\ell] > 0 :\ \mathsf{V} \text{ real},\ u \in M_1^{\mathrm{su}}(x) \text{ real with the strong cap}\},$$

and such a u is called a system of members of the density; it is not unique.

We put $\mathsf{D}^{\mathrm{in}}_x(u_1,u_2) := \|u_1-u_2\|_x$. The distance $\mathsf{D}_x(u_1,u_2)$ is the largest of three numbers: first $\sup_X \sup_{\mathcal{U}_X(x)} |E^a_1 - E^a_2| e^{\kappa(\#X-1)}/c_a(x)$; second $\sup_X \sup |E^h_1 - E^h_2| e^{\kappa(\#X-1)}/c_h(x)$, where the inner supremum runs over the tame G-valued fields, that is, over the G-valued fields on the bonds of $X^{\approx}$ which are tame there; and third $\sup_{X,V} |w_1 - w_2|(X,V) \exp\{c_\Xi x\varepsilon \sum_{D(V)\cap X} \phi_p + \kappa(|X|_c - 1)\}$. Comparing these three numbers with the three size bounds, we find $\mathsf{D}^{\mathrm{in}}_x \le \mathsf{D}_x$. ■

Definition 5.60 (The step)**.** The drawn step leads from $\mathbb{Z}^3/N'L\mathbb{Z}^3$ to $\mathbb{Z}^3/N'\mathbb{Z}^3$; it maps ρ to

$$\rho'(V') = \int \rho(U) \prod_c k_{ax}(V'(c), \mathsf{M}(U)(c))\, dU,$$

with the kernel $k_s(g,h) = z(s)^{-1} e^{-s(1-\cos \mathrm{ang}(gh^{-1}))}$, where $z(s) = \int_G e^{-s(1-\cos \mathrm{ang}\, g)} dg$.

Here M is the average introduced in the definition "The step, the chart" in item C.I 4 of §3. For a fine site x, the coarse site $y(x)$ is the one whose centred block $B(y)$ holds x, and $\mathsf{t}_U(x) := U(\Gamma(y(x),x))$. Given a coarse bond $c=(y,y')$, with $y' = y + Le_\mu$ in fine units, and a site $x \in B(y)$, we let $x' := x + Le_\mu$, write $[x,x']$ for the straight path, and put $h_{c,x} := \mathsf{t}_U(x) U([x,x']) \mathsf{t}_U(x')^{-1}$, $\bar\tau_c := L^{-3} \sum_{x\in B(y)} \mathsf{L}(h_{c,x} h_{c,y}^{-1})$ and $\mathsf{M}(U)(c) := e^{i\bar\tau_c} h_{c,y}$. Thus M is a Borel function of U, built only from U, L, the blocks and the paths.

At every U the kernel is a probability kernel: the right invariance of Haar measure gives $\int k_s(g,h)\, dg = z(s)^{-1} \int e^{-s(1-\cos \mathrm{ang}(gh^{-1}))} dg = 1$, and hence $\int \rho' dV' = \int \rho\, dU$, by Fubini. Three clauses hold: the kernel is gauge covariant; if $\rho \ge 0$ is not zero almost everywhere, then $\rho' > 0$; and if ρ is gauge invariant, so is ρ'. The step depends on x, a and L only: on no rows, on no grid of cubes, on no members, and not on K. ■

Proof of the three clauses. Because $(ugv^{-1})(uhv^{-1})^{-1} = u(gh^{-1})u^{-1}$ and ang is a class function, $k_s(ugv^{-1}, uhv^{-1}) = k_s(g,h)$. Next, $\mathsf{t}_{U^u}(x) = u(y(x)) \mathsf{t}_U(x) u(x)^{-1}$. Therefore $h_{c,x} \mapsto u(y) h_{c,x} u(y')^{-1}$, so that $h_{c,x} h_{c,y}^{-1}$ is carried into its conjugate by $u(y)$ and $\bar\tau_c \mapsto \mathrm{Ad}_{u(y)}\, \bar\tau_c$; for the last point we use $\mathsf{L}(ugu^{-1}) = \mathrm{Ad}_u\, \mathsf{L}(g)$, where both sides are 0 at $g=-1$. This gives $\mathsf{M}(U^u)(c) = u(y) \mathsf{M}(U)(c) u(y')^{-1}$, and with the invariance of k_s just shown, the kernel is gauge covariant.

Given a coarse u', we take the fine $u(x) := u'(y(x))$ and change the variable $U \to U^u$, which leaves Haar measure invariant; when ρ is invariant, this yields $\rho'(V'^{u'}) = \rho'(V')$. Finally $k_s > 0$, so that $\rho' > 0$ as soon as $\rho \ge 0$ is not zero almost everywhere. □

Definition 5.61 (Runs)**.** Let $\mathsf{b} \ge x_\flat$, let N be even with $M \mid N$, and let $K \ge 1$. On $\mathbb{Z}^3/NL^K\mathbb{Z}^3$ we put $\rho_0^{[K]} = e^{-L^K \mathsf{b} \sum_p (1-\cos\phi_p)}$, and we define $\rho_{k+1}^{[K]} = (\rho_k^{[K]})'$ by the step with the noise $a\, x_k$, where $x_k = L^{K-k}\mathsf{b}$. Then $\rho_k^{[K]}$ lives on $\mathbb{T}_k = \mathbb{Z}^3/NL^{K-k}\mathbb{Z}^3$, with the grids of origin $o_K = 0$, $o_k = L\, o_{k+1} - (L-1)/2$. The rows $\ell_k^{[K]}$ are the chain of Def. 2.49, $\ell_0 = \emptyset$, $\ell_{k+1} = ((\ell_k)')^{\mathrm{c}}$, built from the rows of Prop. 2.28 cut at $\varrho(x_{k+1})$, then tensored with $1_{\mathfrak{g}}$ and dressed.

For $K' = K+m$, the partner of the level k of the run of length K is the level $m+k$ of the run of length K', and the depth of this level k is $s = K-k$. Finally $\mu_{[s]}^{[K]}$ is the law on the fields of $\mathbb{Z}^3/NL^s\mathbb{Z}^3$ with the density $\rho_{K-s}^{[K]} / \int \rho_{K-s}^{[K]} dV$. ■

Item 3: what is used. We write out each clause that we use.

(R8) *Thm. 5.7.* The letters are those of item 1. Let $K \ge 1$, $\beta \ge L^K x_\flat$, N even, $M \mid N$ and $N \ge 2M\, s(L^{1-K}\beta)$. Then, for $1 \le k \le K$, the density ρ_k of the run that starts from $e^{-\beta \sum (1-\cos\phi_p)}$ satisfies $\rho_k \in C^{\mathrm{su}}_\flat(L^{-k}\beta; \ell_k)$ at every field, with the rows, tori and grids of the definition "Runs" in item 2 and with the same letters at every k.

(R9) *Thm. 5.15.* The data (SD) at x (item 2 of §5.3) are as follows. First, $x \ge x_9$; N' is even, $M \mid N'$, and (Tor) $N' \ge 2M\, s(x)$ holds, where N' is the coarse side; the fine torus is $\mathbb{Z}^3/N'L\mathbb{Z}^3$; and

the grids fit, that is, $o = L\,o' - (L-1)/2 \bmod M$. Second, $\ell \in \mathcal{R}(b_*, K_*, \lambda_*, \frac{1}{9})$ (see Def. 2.48) on the fine torus, of range $\le \varrho(x)$, and the pair (ℓ, ℓ') has the four conclusions of Thm. 2.64: (Dec)(C_D, θ_D); (N′) with eq. 2.vii at every cut; $\mathsf{t}_\rho \le \frac{1}{8}(\frac{1}{2} - d_0 - c_\Xi)$; and (L1), (L2) in the forms (L1^{su}), (L2^{su}). By ℓ'^{c} we denote the cut of ℓ' at $\varrho(x/L)$.

For all data (SD) at x, all $1 \le A \le 16$ and all $u \in M_A^{\mathrm{su}}(x)$ there exist a number $\mathsf{V}' \in \mathbb{C}$ and a triple $F_\ell(u) = (E'^a, E'^h, w')$, given by the formulas (Fm1) to (Fm4) of item 6 of §5.3, such that the following clauses hold with $x' = x/L$.

(9a) $(\rho[u;\ell])' = e^{\mathsf{V}'}\rho[F_\ell(u); \ell'^{\mathrm{c}}]$ at every V'.

(9b) $F_\ell(u)$ satisfies, at x', all clauses of (ca), (ch), (cΞ) other than the size bounds; $e^{\kappa(\#Y-1)}\sup_{\mathcal{U}_Y(x')}|E'^a(Y,\cdot)| \le (0.35 + 0.248A)c_a(x')$; $e^{\kappa(\#Y-1)}\sup|E'^h(Y,\cdot)| \le c_h(x')$; and w' has the strong cap at x'. Consequently, for $A \le 2$, where $0.35 + 0.496 = 0.846$, we have $F_\ell(u) \in M_1^{\mathrm{su}}(x')$ with the strong cap.

(9d) If u is real, then $F_\ell(u)$ and V' are real; if moreover $\rho[u;\ell] > 0$, then $(\rho[u;\ell])' > 0$.

(9e) For $u_1, u_2 \in M_2^{\mathrm{su}}(x)$, $\mathsf{D}_{x'}(F_\ell(u_1), F_\ell(u_2)) \le 0.4\,\mathsf{D}_x^{\mathrm{in}}(u_1, u_2)$. Here u_1, u_2 are measured in the weak distance $\mathsf{D}_x^{\mathrm{in}}$ and $F_\ell(u_1), F_\ell(u_2)$ in the strong distance $\mathsf{D}_{x'}$.

Clause (9e) is Thm. 5.46 (ii), where the factor is 0.3968.

(R11) *Lem. 5.47.* Let $x \ge e^2$, and consider a torus $\mathbb{Z}^3/N_f\mathbb{Z}^3$ with $N_f = N'L$, where N' is even, $M \mid N'$ and $N' \ge 2M\,s(x)$; this is the lattice of item 2 of §5.4, whose N is the present N'. The torus carries a grid of cubes, and $\ell, \tilde{\ell}$ are any rows on it of range $\le \varrho(x)$, with no further condition on them. We put $\delta := \|\tilde{\mathsf{E}} - \mathsf{E}\|_{(\lambda_\flat)}$ and $\delta_\Delta := c_Q^{\mathrm{su}} p_0^2 \delta / c_a(x) = (c_Q^{\mathrm{su}}/E_a)x^{1/2}\delta/p_0(x)$. Let $\delta_\Delta \le 1$, and let (M5r), $\varepsilon(x) \le \frac{1}{40}$, $\varrho(x) + 2 \le M\,r(x)$, $c_\Xi p_0(x)^2 \ge 1$ and $(2r(x)+1)^3 c_a(x) \le 34$ hold. Then each real $\tilde{u} = (\tilde{E}^a, \tilde{E}^h, \tilde{w}) \in M_1^{\mathrm{su}}(x)$ that has the strong cap admits a real $u^\flat = (\tilde{E}^a + E^\Delta, \tilde{E}^h, w^\flat) \in M_2^{\mathrm{su}}(x)$ such that $\rho[\tilde{u};\tilde{\ell}] = \rho[u^\flat;\ell]$ at every V and $\mathsf{D}_x^{\mathrm{in}}(\tilde{u}, u^\flat) \le \delta_\Delta$. For $w^\flat$ the lemma gives the weak cap alone.

(RU) *Cor. 5.81.* Let $L \ge L_B$ be odd, $aL \ge 3/c_L$, $\theta_D \le \lambda_*/8$ and $c_R = 64/\theta_D$; let $x_\flat$ satisfy (X12), (X13); and let $\mathsf{b} \ge x_\flat$, $K \le K'$, $m = K' - K$, and N even. Let ℓ_k be the chain on the tower $\mathbb{Z}^3/NL^{K-k}\mathbb{Z}^3$ with the radii $\varrho(L^{K-j}\mathsf{b})$, and $\tilde{\ell}_k$ the chain of length K' ending on $\mathbb{Z}^3/N\mathbb{Z}^3$, with the radii $\varrho(L^{K'-j}\mathsf{b})$. For $0 \le k \le K$ we then have $\|\tilde{\mathsf{E}}_{m+k} - \mathsf{E}_k\|_{(\lambda_\flat)} \le 15C_1^2\vartheta_1^k$, where $\vartheta_1 = c_{\mathrm{fix}}L^{-1/2}$.

(RB) *Base.* We use Thm. 2.57 (b).

We use Thm. 2.64 with (C_D, θ_D) of (O6) as its pair of constants. Its conclusions hold with this pair by Thm. 2.65 (clause (Fr2); Step H13 of its proof) and by the proof of Thm. 2.64, which amounts to clauses (Fr3′), (Fr4) of the conclusion of Thm. 2.65 (Step H14 of its proof); Step 2 (d) explains this.

We use the bounds $\|\ell\|_{\mathrm{r}} \le 3K_*\mathsf{g}(\lambda_*)$ and $\|\ell\|_{\mathrm{c}} \le K_*\mathsf{g}(\lambda_*)$ on $\mathcal{R}(b_*, K_*, \lambda_*, \cdot)$, which are stated after the display of Def. 2.48.

The torus condition passes to all finer levels. This is Prop. 5.8: if $x \ge e^2$, $L \ge 3$ and $N \ge 2M\,s(x)$, then $NL^j \ge 2M\,s(L^j x)$ for all $j \ge 0$.

The angle satisfies $\mathrm{ang}(gh) \le \mathrm{ang}\,g + \mathrm{ang}\,h$ and $\mathrm{ang}\,g^{-1} = \mathrm{ang}\,g$ for $g, h \in G$; this is part of Lem. 4.20 (a).

Proposition 5.62 (Every point of the class is gauge invariant)**.** All points of $C_\flat^{\mathrm{su}}(x;\ell)$ are gauge invariant. ■

Proof. By Prop. 5.35, the objects ϕ_p, $D(V)$, m_y and 1_Y are invariant, while $\ell_y^V\zeta$ is rotated by $1 \otimes \mathrm{Ad}_{u(y)}$, which is an isometry. Therefore En_ℓ, the indicators $1_{X^\approx}$ and the families of (cΞ) are invariant, and each member is invariant by the clauses that define it. □

Item 4: the statements.

Theorem 5.63 (Two cutoffs meet)**.** In the setting (O0) to (O13) of item 1, with the letters chosen in this order, let $\mathsf{b} \geq x_\flat$; let N be even, with $M \mid N$ and (Tor) $N \geq 2M\, s(L\mathsf{b})$; let $K' \geq K \geq K_0(\mathsf{b})$ and $m := K' - K$; and write $x_k = L^{K-k}\mathsf{b}$ and $\ell_k := \ell_k^{[K]}$. Then the following hold.

(i) $\Lambda > 0$ and $1 \leq k_0 \leq K - 1$.

(ii) Fix a system u_{k_0} of members of $\rho_{k_0}^{[K]}$ (there is one by (R8)), and put $u_{k+1} := F_{\ell_k}(u_k)$ for $k_0 \leq k < K$. Then for every k with $k_0 \leq k \leq K$, u_k is a system of members of $\rho_k^{[K]}$: it is real, it lies in $M_1^{\mathrm{su}}(x_k)$, it satisfies the strong cap, and $\rho_k^{[K]} = e^{\mathsf{V}}\rho[u_k; \ell_k]$ with a real V.

(iii) For every k with $k_0 < k \leq K$ there exist a real $\mathfrak{n}_k \in M_1^{\mathrm{su}}(x_k)$ satisfying the strong cap and a real V such that $\rho_{m+k}^{[K']} = e^{\mathsf{V}}\rho[\mathfrak{n}_k; \ell_k^{[K]}]$ at every field. Thus $\rho_k^{[K]}$ and $\rho_{m+k}^{[K']}$ belong to one and the same class $C_\flat^{\mathrm{su}}(x_k; \ell_k^{[K]})$.

(iv) For $k_0 < k \leq K$ one has $\mathsf{D}_{x_k}(u_k, \mathfrak{n}_k) \leq 3\vartheta^{k-k_0}$, and at $k = K$ one has $\mathsf{D}_{\mathsf{b}}(u_K, \mathfrak{n}_K) \leq C_{\mathsf{b}}\theta^K$. In terms of the depth $s = K - k$, this says: the level $K - s$ is $> k_0$ if and only if $K - s \geq K_0(L^s\mathsf{b}) = \frac{5}{2}\Lambda + \frac{5}{4}s + \frac{9}{4}$, and in that case $\mathsf{D} \leq 3\vartheta^{\frac{4}{9}K - s - \frac{10}{9}\Lambda - 1} = C_{L^s\mathsf{b}}\theta^{K-s}$.

(v) In a cruder form, $k_0 < \varphi K + K_0(\mathsf{b})$ and $k_0 \leq \lceil \varphi K \rceil + K_0(\mathsf{b})$. Consequently, at every level k with $\varphi K + K_0(\mathsf{b}) < k \leq K$, clause (iii) holds and $\mathsf{D} \leq 3\vartheta^{k - \lceil \varphi K \rceil - K_0(\mathsf{b})}$.

Here $\varphi, \vartheta, \theta$ are numerals, and $K_0(\mathsf{b})$, C_{b} depend on L, M, E_a, b and pure numbers; none of these quantities depends on N, K', m or a row. ■□

Theorem 5.64 (From the distance of members to densities)**.** In the setting (O0) to (O12) of item 1, let $x \geq e^2$; let N be even, with $M \mid N$ and $N \geq 2M\, s(x)$; let $\kappa \geq 6$, $\varepsilon(x) \leq 1$ and $(c_a + c_h)(x) \leq \frac{1}{2}$; let ℓ be rows on $\mathbb{Z}^3/N\mathbb{Z}^3$ of range $\leq \varrho(x)$; and let $\rho_i = e^{\mathsf{V}_i}\rho[u_i; \ell] \in C_\flat^{\mathrm{su}}(x; \ell)$ for $i = 1, 2$. Put $\mathsf{D} := \mathsf{D}_x^{\mathrm{in}}(u_1, u_2)$, which is $\leq \mathsf{D}_x(u_1, u_2)$, so that each bound below remains valid with D_x in place of D; and put $n = (N/M)^3$. Then the following hold.

(a) At every V for which $D(V)$ is empty,

$$|\log \rho_1(V) - \log \rho_2(V) - (\mathsf{V}_1 - \mathsf{V}_2)| \leq 1.22\, n(c_a + c_h)(x)\, \mathsf{D}.$$

(b) $\sup_V |\rho_1 - e^{\mathsf{V}_1 - \mathsf{V}_2}\rho_2| \leq e^{\mathsf{V}_1} \cdot 2n\, e^{219n}\, \mathsf{D}$.

(c) If in addition $|\ell_y(p)| \leq K_* e^{-\lambda_* |y-p|}$, then

$$\int \rho_1 dV \geq e^{\mathsf{V}_1} e^{-0.61n} e^{-\frac{3}{2}\mathsf{N}_Z p_0^2 N^3} (\varepsilon^3/384)^{3N^3}$$

and

$$\int |\rho_1 / \textstyle\int \rho_1 - \rho_2 / \int \rho_2|\, dV \leq C_{\mathrm{vol}}^{\mathrm{su}}(N, x)\, \mathsf{D}.$$

(d) The integral $\int |\rho_1/\int\rho_1 - \rho_2/\int\rho_2|\, dV$ is $\leq 2 \max_i r_i + e^{2.44n(c_a+c_h)\mathsf{D}} - 1$, where r_i denotes the mass of $\{D(V) \text{ non-empty}\}$ under $\rho_i/\int\rho_i$.

■□

Corollary 5.65 (The laws converge in total variation)**.** Let the letters, b and N satisfy the hypotheses that Thm. 5.63 imposes on them, and let $s \geq 0$.

(a) For all $K' \geq K \geq s + K_0(L^s\mathsf{b})$,

$$\|\mu_{[s]}^{[K']} - \mu_{[s]}^{[K]}\|_{\mathrm{TV}} \leq C_{\mathrm{vol}}^{\mathrm{su}}(NL^s, L^s\mathsf{b})\, C_{L^s\mathsf{b}}\, \theta^{K-s}.$$

(b) As $K \to \infty$, the full sequence converges: $\mu_{[s]}^{[K]} \to \mu_{[s]}^{\infty}$ in total variation, at the rate of (a). The limit $\mu_{[s]}^{\infty}$ is a gauge invariant probability law, absolutely continuous with respect to Haar measure.

(c) For $s \geq 1$, the law $\mu_{[s-1]}^{\infty}$ is obtained from $\mu_{[s]}^{\infty}$ by the drawn step with the noise $aL^s\mathsf{b}$.

(d) For $K \geq s + K_0(L^s\mathsf{b})$, the distance in $L^1(dV)$ between the density of $\mu^{\infty}_{[s]}$ and the normalised density of the point $\rho^{[K]}_{K-s}$ of $C^{\mathrm{su}}_{\flat}(L^s\mathsf{b}; \ell^{[K]}_{K-s})$ is at most the bound of (a).

■□

Corollary 5.66 (The flux response)**.** Let the letters, b and N satisfy the hypotheses that Thm. 5.63 imposes on them, let moreover $N^2 \geq 4\pi/\varepsilon(\mathsf{b})$, and let m_0 be an integer with $\varepsilon(\mathsf{b})/2 \leq \varepsilon_0 := 2\pi m_0/N^2 \leq \varepsilon(\mathsf{b})$; such an integer exists. On $\mathbb{Z}^3/N\mathbb{Z}^3$, with $0 \leq z_i < N$, let V_* be given by $V_*(z, e_2) = e^{i\varepsilon_0 z_1 \sigma_3}$, by $V_*(z, e_1) = e^{-i\varepsilon_0 N z_2 \sigma_3}$ if $z_1 = N-1$, and by the value 1 at all other bonds. For $\rho > 0$ set

$$\mathfrak{x}(\rho) := \frac{2}{N^3 \varepsilon_0^2} [\log \rho(1) - \log \rho(V_*)].$$

Then for all $K' \geq K \geq K_0(\mathsf{b})$,

$$|\mathfrak{x}(\rho^{[K']}_{K'}) - \mathfrak{x}(\rho^{[K]}_{K})| \leq Q^{\mathrm{su}}(\mathsf{b})\, C_{\mathsf{b}}\, \theta^K.$$

■□

Item 5: remarks.

Remark 5.67. The system $\mathfrak{n}_k$ of Thm. 5.63 (iii) is a system of members over the rows $\ell^{[K]}_k$ of the shorter run, while the run of length K' has members of its own, over its rows $\ell^{[K']}_{m+k}$; each of the two is a system of members of one and the same function $\rho^{[K']}_{m+k}$. Members are not unique, and the proofs use no identity between the two systems.

The proofs depend on c^{su}_Q only through two facts, that (R11) holds with it and that $480 c^{\mathrm{su}}_Q C_1^2 > 1$, and on ϑ only through $0.4 \leq \vartheta < 1$. Inserting $\vartheta = 0.4$ in all formulas of (O14) gives $\theta = 0.4^{4/9} < 0.66549$.

Proof of Thm. 5.63. Step 0: the letters; three inequalities that follow from their choice; the existence of a number $x_\flat$ that satisfies all conditions of (O13)*; no circularity in the choice of the letters.* We claim the following six facts.

(i) $\theta_D \leq \lambda_*/8$, so that $c_R \geq 512$ and $e_R \geq 256$.
(ii) For $x \geq e^2$, $\varrho(x) + 2 \leq M\, r(x)$ and $2(\varrho(x) + 1) < 2M\, s(x)$.
(iii) $480\, c^{\mathrm{su}}_Q C_1^2 > 2000$.
(iv) There is a number $x_\flat$ that satisfies all conditions of (O13).
(v) (R8), (R9), (R11) and (RU) hold for one and the same choice of the letters, and no entry of this choice depends on a later one.
(vi) (X18) gives $(2r+1)^3 c_a \leq 34$.

We prove the six facts in turn.

(i) By (O6), $\theta_D \leq \theta_{\mathbb{C}}/2$. Next, $\theta_{\mathbb{C}} = \theta^\flat/6 \leq \gamma/(12A^\flat)$ and $A^\flat \geq 384 e^8 K_*^2$; these follow from (O6), from clause (6) of the order of the constants in Thm. 2.65, and from $\mathsf{g}_2 \geq 1$. As $K_*^2 = 9/\gamma_*$ and $\gamma \leq 1$, we get $\theta_{\mathbb{C}} \leq \gamma_*/(41472e^8)$, while $\lambda_*/4 = \gamma_*^{1/2}/176$. The bound $\theta_{\mathbb{C}} \leq \lambda_*/4$ therefore comes down to $\gamma_*^{1/2} \leq 41472e^8/176$, which follows from $\gamma_*^{1/2} = 44\lambda_* \leq 44 < 41472/176$. Thus $\theta_D \leq \lambda_*/8$, whence $c_R = 64/\theta_D \geq 512/\lambda_* \geq 512$ and $e_R = \lambda_* c_R/2 \geq 256$.
(ii) We have $M\, r(x) \geq 20 c_R \log x$ and $\varrho(x) + 2 \leq c_R \log x + 3$, and moreover $19 c_R \log x \geq 19 \cdot 512 \cdot 2 > 3$; this gives the first inequality. The second follows from $2M\, s(x) = 16 M\, r(x) \geq 320 c_R \log x > 2(c_R \log x + 2) \geq 2(\varrho(x) + 1)$.
(iii) From $K_* \lambda_* = 3/44$ and $\lambda_* \leq 1$ we obtain $K_* \geq 3/44$; since also $\mathsf{g}(\lambda_*/2) \geq 1$, it follows that $C_1^2 = 3K_*^2 \mathsf{g}(\lambda_*/2)^3 \geq 3(3/44)^2 > 0.0139$. We also have $c^{\mathrm{su}}_Q \geq 306$, and $480 \cdot 306 \cdot 0.0139 > 2000$.

(iv) The conditions $x_\flat \geq x_\flat^{(1)}$, $x_\flat \geq x_\flat^0$, $L\,x_\flat \geq x_9$, (X12) and (X13) of (O13) bound $x_\flat$ from below by finite numbers. Here $x_\flat^{(1)}$, $x_\flat^0$ and x_9 are numbers defined in (O13) of item 1, while (X12) and (X13) hold for all large $x_\flat$, because $x_\flat^{-e_R} \to 0$ as $x_\flat \to \infty$. For the floors (X14) to (X18), the limits of p_0, ε, c_a and c_h as $x \to \infty$ follow from the definitions of item 1. By (O8), $p_0(x) = A_0(\log x)^{\mathfrak{p}_0}$ with $A_0 > 0$ and $\mathfrak{p}_0 > 2$, and $\varepsilon = p_0 x^{-1/2}$; hence $p_0 \to \infty$ and $\varepsilon = p_0 x^{-1/2} \to 0$. Under (O12), the caps are $c_a(x) = E_a p_0(x)^3 x^{-1/2}$ and $c_h(x) = e^{-c_h p_0(x)^2}$, so that $c_a, c_h \to 0$. With $r(x) = \lceil 20 c_R \log x/M \rceil$, these definitions give

$$c_a(5+2r)^3 \leq E_a A_0^3 (\log x)^{3\mathfrak{p}_0} x^{-1/2} (7 + 40 c_R \log x/M)^3 \to 0.$$

Thus the five floors hold for all large x: the limit of p_0 yields (X14) and (X15), that of ε yields (X16), those of c_a and c_h yield (X17), and the display yields (X18). The inequality of each floor holds strictly at the limit that yields it; hence each floor is satisfied by all x beyond some number.

(v) The conditions of (O13) depend on the letters of (O0) to (O12) only. Thm. 5.3 holds for all $c_h \leq \delta_3/2$, all $E_a \geq \max\{32 L^{-1/2} K_v^\flat, 32 K_{\mathrm{q},k}\}$ and all $x_\flat \geq x_\flat^{(1)}$, where $x_\flat^{(1)}$ does not depend on c_h, E_a, N. Thm. 5.7 holds for $c_h \leq \delta_3/2$, $E_a \geq \max\{32 L^{-1/2} K_v^\flat, 32 K_{\mathrm{q},k}\}$ and $x_\flat \geq \max\{x_\flat^{(1)}, x_\flat^0\}$ with (X12). Thm. 5.15 holds for $c_h \leq \delta_3'/2$, $E_a \geq \max\{32 K_{\mathrm{q},k}, 1500 K_v^\flat L^{-1/2}\}$ and $x \geq x_9$. Since $\delta_3' \leq \delta_3$, and since the maximum in (O12) is the largest of these maxima, the choice of the table lies on all these half-lines; and the class does not depend on $x_\flat$. The condition (M5r) is a lower bound on M in terms of the radius ρ_0 that (O3) defines, and Thms. 5.3, 4.39, 5.7 and 5.15 admit every larger $M = L^{m'}$. Finally, (O14) is defined last and enters no floor.

(vi) Indeed $(2r+1)^3 \leq (5+2r)^3$, and $\frac{1}{2} \leq 34$.

Step 1: Cor. 5.81 for the letters of SU(2). We claim that, for $\mathsf{b} \geq x_\flat$, $K' \geq K$, $m = K' - K$, N even and $0 \leq k \leq K$,

$$\delta_k := \|\tilde{\mathsf{E}}_{m+k} - \mathsf{E}_k\|_{(\lambda_\flat)} \leq 15 C_1^2 L^{-2k/5},$$

where E_k and $\tilde{\mathsf{E}}_j$ denote the site forms of the rows $\ell_k^{[K]}$ and $\ell_j^{[K']}$.

By the definition "Runs" of item 2 and by (R8), the rows of both runs are the scalar chains of Def. 2.49, tensored with $1_{\mathfrak{g}}$, and $\langle \ell_y(p) \otimes 1, \ell_y(p') \otimes 1 \rangle$ enters only through the scalar site form $\mathsf{E}_y(p,p') = \langle \ell_y(p), \ell_y(p') \rangle$.

We take the hypotheses of (RU) one by one. That L is odd, $L \geq L_B$ and $aL \geq 3/c_L$ is (O4). The relations $\theta_D \leq \lambda_*/8$ and $c_R = 64/\theta_D$ come from (O6) and Step 0 (i). Conditions (X12) and (X13) hold by (O13), and $\mathsf{b} \geq x_\flat$, $K \leq K'$ and the evenness of N belong to the data. As to the radii, the chain of the run of length K is cut at $\varrho(x_j) = \varrho(L^{K-j}\mathsf{b})$, the chain of the run of length K' at $\varrho(L^{K'-j}\mathsf{b})$, and both chains end on $\mathbb{Z}^3/N\mathbb{Z}^3$.

Hence $\delta_k \leq 15 C_1^2 \vartheta_1^k$, and $\vartheta_1 = c_{\mathrm{fix}} L^{-1/2} \leq L^{1/10} L^{-1/2} = L^{-2/5}$ because $L \geq L_B \geq c_{\mathrm{fix}}^{10}$. Cor. 5.81, together with Thm. 5.80, Prop. 5.79, Lems. 5.73 and 5.74, Prop. 5.75 and Lems. 5.76, 5.77 and 5.78 on which it rests, is stated and proved in Step 1a.

Step 1a: (RU)*, that is Cor. 5.81, and its proof.* The claim of this step is (RU) of item 3, which is the last inequality of Cor. 5.81.

Objects and statements are taken from §2.2, each with the meaning given to it by the passage named in the parentheses after it: the lattices, W, Q_i and the facts on them (Def. 2.17 and Lem. 2.18); d, dist and $\mathsf{g}(\lambda)$ (Def. 2.19); own units, $F_\beta(z)$, g_1, ∂_α, ∇ and ∇' (Defs. 2.43, 2.44 and 2.45), where $|(\nabla F)_p|^2 := \sum_\alpha |(\partial_\alpha F)_p|^2$; rows, covariance, $\tau(\ell)$ and $\mathcal{R}(b, K, \lambda, \tau)$ (Def. 2.22, and Defs. 2.46 to 2.48); pairs, Ξ_Z, $\mathsf{T}_\sharp$ and $R_\sharp$ (Def. 2.26); B', $H^\sharp$, $\mathsf{N}^\sharp$ and the statement that defines them (Prop. 2.27 (a)); Φ (Def. 2.29); the five groups of ℓ' (Prop. 2.28); S_0 and $\mathfrak{R}$ (Def. 2.30); $\mathsf{r}_\sharp$, γ_1 and Prop. 2.33 (Def. 2.32 and Rem. 2.34); the cut and $\ell'^{>}$ (Def. 2.39); towers and chains (Def. 2.49); and the pure numbers and (Cut) (Def. 2.56 and Thm. 2.57).

Lems. 2.52 and 2.53, Prop. 2.55 and Thm. 2.57 are proved in §2.2. Clauses (a) and (b) of Lem. 5.73 are Lem. 2.51, and the condition (X12) of §2.2 is the (X12) of (O13).

For the statements of Step 1a and for their proofs, we fix the meaning of several letters. Wherever Step 1a treats the passage from the fine torus $\mathbb{T}$ to the coarse torus $\mathbb{T}'$ in a single step, N is the coarse side of the step, with $\mathbb{T} = \mathbb{Z}^3/NL\mathbb{Z}^3$ and $\mathbb{T}' = L\mathbb{Z}^3/NL\mathbb{Z}^3$ as in Def. 2.17; it is the N' of this section. Throughout Step 1a we allow every even $N \geq 2$ and do not impose the condition $M \mid N$. In Cor. 5.81, by contrast, N is the side of the last torus, as in the definition "Runs" of item 2. In Thm. 5.80, $k \leq k'$ are the lengths of the two chains and n is the side of the torus on which both end; Cor. 5.81 applies that theorem with $(k, k', n) = (k, m+k, NL^{K-k})$. The letters $\mathrm{d}, \mathrm{d}', \delta$ are operators of the free model of §2.2. In Lem. 5.76 (b), $\hat{\mathsf{B}}(0)$ and $\hat{\mathsf{B}}'(0)$ denote the values taken at the momentum 0 by the symbols of B, B'; in other words, they are the 3×3 matrices $1 + \mathsf{c}(\mathsf{E})$ and $1 + \mathsf{c}(\mathsf{E}')$ by which B, B' act on the constant forms. In Prop. 5.79 (b) and in Step (7) of its proof, $\varrho \in \mathbb{N} \cup \{\infty\}$ is a radius in the sense of Def. 2.49. The distance $r_P = \operatorname{dist}(y', P)$ equals $|y' - P|$ in the units of $\mathbb{T}'$ (Def. 2.19, for coarse cells); accordingly Lem. 5.77 is applied on $\mathbb{T}'$ with this distance. In the proofs of Prop. 5.75 and Lem. 5.76, $\mathsf{R} = \mathsf{B}^{-1}$, as in Def. 2.22.

The proofs of Step 1a use the exact constants and outward intervals in Thm. 2.65 (0), together with the following five numerical facts. In them, C_1, c_0, c_{fix}, L_B and ϑ_1 are as in (O0) of item 1, C_0 and μ are fixed in the definition "Rates, constants" below, and y is the number defined in Step (6) of the proof of Prop. 5.79.

(1) The 27 points with $|n|_\infty \leq 1$ give $\mathsf{g}(\lambda) \geq 27e^{-\lambda}$, hence $\mathsf{g}(\lambda) \geq 10$ for $\lambda \leq \lambda_*$; this yields the bound $\mathsf{g}(\lambda_*/4)^2 \geq 100$. Moreover $\mathsf{g}(\lambda) \geq 3$ gives $3K_*\mathsf{g}(\lambda_*/2) \leq \sqrt{3}K_*\mathsf{g}(\lambda_*/2)^{3/2} = C_1$.
(2) One has $\gamma_*^{-1/2}/(9b_*) = \sqrt{12296}/(9b_*\mathsf{r}_\sharp) = \sqrt{12296}/18 < 6.17$; consequently $C_0/c_0 = (\gamma_*^{-1/2}/(9b_*))\,\mathsf{g}(\lambda_*)/\mathsf{g}(\lambda_*/4)^3 \leq 6.17/100$.
(3) Since $2\mu = \lambda_*/4 < 10^{-14}$, we have $e^{\lambda_*} \leq 1.1$, $6e^{2\mu} + 6 \leq 13$ and $3 + 3e^{2\mu} + 1 \leq 7.1$.
(4) One has $b_* \geq 27$.
(5) One has $c_{\mathrm{fix}} \geq 48$, because $\mathsf{g}(\lambda) \geq 1$, $C_1 > 1$ and $c_0 > 1$. Hence $L \geq L_B \geq 48^{10} > 32$; further $\vartheta_1 \leq L^{-2/5} \leq 48^{-4}$, and $\mathsf{y} = \vartheta_1/48 \leq \frac{1}{100}$.

We now recall statements of §2.2, beginning with Lem. 2.52. On $\Omega'^2 \oplus \Omega^3 \oplus \mathsf{H}$ let

$$\mathbb{A}^{\mathrm{e}}(Z, \Xi, \sigma) = (W^*Z + \delta\Xi - \ell^{\mathsf{T}}\sigma,\ \sigma,\ a^{-1/2}\mathrm{d}'^*Z,\ \mathsf{S}_0^{1/2}Q_3\Xi),$$

whose rows are indexed, component by component, by p, by the components of the spaces H_y (each placed at y), by c, and by q'; let $\mathfrak{R}^{\mathrm{e}} = \mathbb{A}^{\mathrm{e}\mathsf{T}}\mathbb{A}^{\mathrm{e}}$ and $\mathsf{D}^{\mathrm{e}} = \operatorname{diag}(L, L^{-2}, 1)$. Lem. 2.52 states the following.

(a) At the source $(\delta_P, 0, 0)$, $\mathbb{A}^{\mathrm{e}}(\mathfrak{R}^{\mathrm{e}})^{-1}(\delta_P, 0, 0) = (\nu, \ell\nu, a^{1/2}\mathsf{N}^\sharp\delta_P, \cdot)$, where $\nu = H^\sharp\delta_P$.
(b) $\mathfrak{R}^{\mathrm{e}} \geq (\gamma_1/2b)\mathsf{D}^{\mathrm{e}}$ for $b \geq b_{\mathsf{B}}$.
(c) For $\ell \in \mathcal{R}(b_*, K_*, \lambda_*, \cdot)$ and $L \geq L_A$, the bound (sinh) of Lem. 5.73 is valid for $\mathbb{A}^{\mathrm{e}}, \mathsf{D}^{\mathrm{e}}, \mathsf{d}$ at λ_*, with some $\varkappa \leq \frac{1}{4}\gamma_*^{1/2}$. Here Lem. 2.52 (c) has the rate λ', a minimum defined in that lemma, and $\lambda' = \lambda_*$ for $\ell \in \mathcal{R}(b_*, K_*, \lambda_*, \cdot)$ and $L \geq L_A$, because L is at least the first member of the maximum that defines L_A.

Next we recall Lem. 2.53. A pair (F, n) is *admitted at* X' if $WF + \mathrm{d}'\mathsf{n} = X'$ and $\mathrm{d}F \perp \ker Q_3$; these pairs form the set $\mathrm{Adm}(X')$, which does not depend on ℓ. Each entry below bears the tag of the clause of Lem. 2.53 to which it belongs.

(a) $(\bar{F}, \bar{\mathsf{n}}) \in \mathrm{Adm}(X')$, and $\langle(\bar{F}, \bar{\mathsf{n}}), (F, \mathsf{n})\rangle_{\mathcal{H}} = \langle \bar{Z}, X'\rangle$ for every $(F, \mathsf{n}) \in \mathrm{Adm}(X')$; this identity is shown in the proof of Lem. 2.53 (a). Consequently $(\bar{F}, \bar{\mathsf{n}})$ minimises $\|\cdot\|_{\mathcal{H}}$ on $\mathrm{Adm}(X')$ and is $\mathcal{H}$-orthogonal to $\mathrm{Adm}(0)$. Conversely, an element of $\mathrm{Adm}(X')$ which is $\mathcal{H}$-orthogonal to $\mathrm{Adm}(0)$ is the minimiser, since its difference with the minimiser lies in $\mathrm{Adm}(0)$ and is orthogonal to both of them.
(b) Φ is monotone.

Finally we recall Prop. 2.55 and Thm. 2.57 for the starred constants. Let $L \geq L_A$ be odd, let $aL \geq 3/c_L$, let $N \geq 2$ be even, and let $\ell \in \mathcal{R}(b_*, K_*, \lambda_*, \tau)$ with $\tau \leq \frac{1}{9}$. Then ℓ' is translation covariant on $\mathbb{T}'$, $|\ell'_{y'}(P)| \leq K_* e^{-\lambda_* \mathrm{dist}(y', P)}$, $1 \leq \mathsf{B}' \leq \frac{3}{2}\mathsf{r}_\sharp^{-1}$, and

$$\sum_p e^{2\lambda_* \mathsf{d}(p,P)} (H^\sharp \delta_P)_p^2 \leq 4\gamma_*^{-1} L^{-1}.$$

If moreover $\mathsf{t}_*(\varrho) \leq \frac{1}{9}$, then $\|\ell'^{>}\|_{\mathrm{r}} := \sup_{y'} \sum_P |\ell'^{>}_{y'}(P)| \leq \mathsf{t}_*(\varrho)$ and $\ell'^{\mathrm{c}} \in \mathcal{R}(b_*, K_*, \lambda_*, \tau + \mathsf{t}_*(\varrho))$. Along a chain with (Cut), $\ell_k \in \mathcal{R}(b_*, K_*, \lambda_*, \sum_{j \leq k} \mathsf{t}_*(\varrho_j))$; this is shown in the proof of Thm. 2.57 (b). For every F, and for rows with $|\ell_y(p)| \leq K e^{-\lambda|y-p|}$, one has $|\ell F| \leq 3K \mathsf{g}_1(\lambda) |\nabla F| + \tau(\ell)|F|$, as shown within the proof of Prop. 2.55 (b).

Definition 5.68 (Rates, constants)**.** We fix the pure rates $\mu := \frac{1}{8}\lambda_*$ and $\lambda_\flat := \mu_0 := \frac{1}{16}\lambda_*$, and the pure constants

$$C_1 := \sqrt{3}\, K_* \, \mathsf{g}(\tfrac{\lambda_*}{2})^{3/2}, \quad c_0 := 108\, b_* \gamma_*^{-1/2} \mathsf{g}(\tfrac{\lambda_*}{4})^3, \quad C_0 := 12\, \gamma_*^{-1} \mathsf{g}(\lambda_*),$$

$$c_{\mathrm{fix}} := 48\, \mathsf{g}(\lambda_\flat)\, C_1\, c_0^{1/2}, \qquad L_B := \max\{L_A, c_{\mathrm{fix}}^{10}\}.$$

We also set $\vartheta_0 := c_0/L$ and $\vartheta_1 := c_{\mathrm{fix}} L^{-1/2}$, the latter being $\leq L^{-2/5}$ for $L \geq L_B$. Note that $C_0 \leq c_0$, because $\gamma_*^{-1/2}/9b_* = \sqrt{12296}/18 < 6.17$, $\mathsf{g}(\lambda_*) \leq \mathsf{g}(\frac{\lambda_*}{4})$ and $\mathsf{g}(\frac{\lambda_*}{4})^2 \geq 100$. ■

Definition 5.69 (Site forms)**.** The site form at y is $\mathsf{E}_y := \ell_y^{\mathsf{T}} \ell_y$, a positive operator on Ω^2, and $\sum_y \mathsf{E}_y = \mathsf{B} - 1$. The cut acts on site forms by $\mathsf{E}'^{\mathrm{c}}_{y'} = \chi_{y'} \mathsf{E}'_{y'} \chi_{y'}$, where $\chi_{y'} = 1\{\mathrm{dist}(y', \cdot) \leq \varrho\}$. ■

Definition 5.70 (Constant forms, charge)**.** Let the constant 2-form $e_\beta \in \Omega^2$ take the value 1 on the plaquettes whose normal is the unit vector e_β and the value 0 on the others, and write $h = \sum h_\beta e_\beta$; let e'_β and h' be the same objects on $\mathbb{T}'$, and let $h \leftrightarrow h'$ pair forms with equal components. For translation covariant rows, put $\mathsf{c}(\mathsf{E})_{\alpha\beta} := \langle e_\alpha, \mathsf{E}_y e_\beta \rangle = \langle \ell_y e_\alpha, \ell_y e_\beta \rangle$. This is a positive 3×3 matrix that does not depend on y, because $\ell_{y+\mathsf{t}} e_\alpha = U_{\mathsf{t},y} \ell_y e_\alpha$; moreover $\mathsf{c}_{\beta\beta} \leq \tau(\ell)^2$ by the definition of $\tau(\ell)$, since $|e_\beta|_\infty = 1$. ■

Definition 5.71 (Seminorms at a site)**.** In own units, for $\eta \geq 0$, let

$$n_y(F)^2 := \sum_p e^{-2\mu|y-p|} |(\nabla F)_p|^2, \qquad |F|_y^2 := \sum_p e^{-2\mu_0|y-p|} F_p^2, \qquad \hat{n}_{y,\eta}(F)^2 := n_y(F)^2 + \eta^2 |F|_y^2.$$

■

Definition 5.72 (Norms of a family of symmetric operators)**.** For a family $D = (D_y)_y$ of symmetric operators on Ω^2, let

$$\|D\|_\eta := \inf\{\epsilon : \ |\langle F, D_y G \rangle| \leq \epsilon\, \hat{n}_{y,\eta}(F)\, \hat{n}_{y,\eta}(G) \text{ for all } y, F, G\}$$

and

$$\|D\|_{(\lambda)} := \sup_{y,p,p'} |D_y(p,p')|\, e^{\lambda(|y-p|+|y-p'|)}.$$

■

These seminorms and norms satisfy two bounds.

(n1) For $\eta \leq 1$, $\hat{n}_{y,\eta}(G)^2 \leq 14 |G|_y^2$. Indeed, $|(\nabla G)_p|^2 \leq 2 \sum_\alpha (G_{p+e_\alpha}^2 + G_p^2)$, and therefore

$$\sum_p e^{-2\mu|y-p|} |(\nabla G)_p|^2 \leq (6e^{2\mu} + 6) \sum_p e^{-2\mu|y-p|} G_p^2 \leq 13 |G|_y^2.$$

(n2) $n_y(\delta_p)^2 \leq (3 + 3e^{2\mu}) e^{-2\mu|y-p|}$; hence, for $\eta \leq 1$, $\hat{n}_{y,\eta}(\delta_p)^2 \leq 7.1 e^{-2\mu_0|y-p|}$ and $\|D\|_{(\lambda_\flat)} \leq 7.1 \|D\|_\eta$.

Moreover $\|D\|_\eta$ does not increase in η, and $\|D\|_0 < \infty$ only if every D_y kills constants, since $n_y(h) = 0$ for each constant form h.

The standing conditions for the rest of Step 1a are that L is odd, that $aL \geq 3/c_L$, and that $N \geq 2$ is even.

Lemma 5.73 (Exponentially weighted estimates for $\mathfrak{R}^{-1}$ and for the range projection of $\mathbb{A}$)**.** Let I and J be finite sets, and let each point of I and of J be assigned a place, that is, a point of a pseudo-metric space with distance d. In $\mathsf{d}(r,i)$, $\mathsf{d}(i,j)$ and $\mathsf{d}(r,j)$, the distance is taken between the places of the two points. Let $\mathbb{A}$ be a real $J \times I$ matrix and $\mathfrak{R} = \mathbb{A}^{\mathsf{T}}\mathbb{A}$; let D be a positive diagonal matrix on I, with $N(\phi) = \langle \phi, \mathsf{D}\phi\rangle$; and let $\gamma > 0$ satisfy $\mathfrak{R} \geq \gamma \mathsf{D}$. Let $\lambda > 0$ and $0 \leq \varkappa \leq \frac{1}{4}\gamma^{1/2}$ be such that

$$\sum_{r\in J}\Big(\sum_{i\in I} |\mathbb{A}(r,i)| \sinh(\lambda \mathsf{d}(r,i))|\phi_i|\Big)^2 \leq \varkappa^2 N(\phi) \tag{sinh}$$

for all ϕ. Then the following hold.

(a) For $j \in I$ and $\Phi = \mathfrak{R}^{-1}\delta_j$, $\sum_i \mathsf{D}_i e^{2\lambda \mathsf{d}(i,j)}\Phi_i^2 \leq 4\gamma^{-2}\mathsf{D}_j^{-1}$.

(b) For the same j and Φ, $\sum_r e^{2\lambda \mathsf{d}(r,j)}(\mathbb{A}\Phi)_r^2 \leq 4\gamma^{-1}\mathsf{D}_j^{-1}$.

(c) Let $\Pi := \mathbb{A}\mathfrak{R}^{-1}\mathbb{A}^{\mathsf{T}}$, which is the orthogonal projection of $\mathbb{R}^J$ onto $\operatorname{ran}\mathbb{A}$. If w is any real function on $I \cup J$ such that $|w_i - w_r| \leq \lambda \mathsf{d}(r,i)$ for all $r \in J$, $i \in I$, then each $\rho \in \mathbb{R}^J$ satisfies $|e^w \Pi \rho| \leq 3|e^w\rho|$ and $|e^w(1-\Pi)\rho| \leq 4|e^w\rho|$.

■

Proof. Clauses (a) and (b) are Lem. 2.51 under the same hypotheses. To prove (c), w is a real function on $I \cup J$ as in (c), so that $|w_i - w_r| \leq \lambda \mathsf{d}(r,i)$, and we put $\mathfrak{R}_w := e^w \mathfrak{R} e^{-w}$. To a real vector ϕ on I we attach the two vectors on J with the entries $\mathsf{C}_r = \sum_i \mathbb{A}(r,i)\cosh(w_i - w_r)\phi_i$ and $\mathsf{S}_r = \sum_i \mathbb{A}(r,i)\sinh(w_i - w_r)\phi_i$.

Writing $e^{w_i - w_r} = \cosh(w_i - w_r) + \sinh(w_i - w_r)$, we find $e^{-w_r}(\mathbb{A}e^w\phi)_r = \mathsf{C}_r + \mathsf{S}_r$, and in the same way $e^{w_r}(\mathbb{A}e^{-w}\phi)_r = \mathsf{C}_r - \mathsf{S}_r$. Since $\mathfrak{R} = \mathbb{A}^{\mathsf{T}}\mathbb{A}$, it follows that $\langle \phi, \mathfrak{R}_w\phi\rangle = \langle \mathbb{A}e^w\phi, \mathbb{A}e^{-w}\phi\rangle = |\mathsf{C}|^2 - |\mathsf{S}|^2$.

Next we compare both vectors with $\mathbb{A}\phi$. As sinh increases on $[0,\infty)$ and $\cosh t - 1 \leq \sinh|t|$, the condition on w and (sinh) give $|\mathsf{S}|, |\mathsf{C} - \mathbb{A}\phi| \leq \varkappa N^{1/2} \leq \frac{1}{4}|\mathbb{A}\phi|$, where $N = N(\phi)$; the last inequality follows from $\varkappa \leq \frac{1}{4}\gamma^{1/2}$ and $|\mathbb{A}\phi| \geq \gamma^{1/2}N^{1/2}$, which is a consequence of $\mathfrak{R} \geq \gamma\mathsf{D}$.

From these bounds and the identity above we obtain

$$|\mathsf{C}| - |\mathsf{S}| \geq \frac{1}{2}|\mathbb{A}\phi|, \qquad |\mathsf{C}| + |\mathsf{S}| \leq \frac{3}{2}|\mathbb{A}\phi|, \qquad \langle \phi, \mathfrak{R}_w\phi\rangle \geq \frac{1}{2}|\mathbb{A}\phi|^2 \geq \frac{\gamma}{2}N(\phi). \tag{†}$$

For (c) we fix ρ on J and put $\Phi := \mathfrak{R}^{-1}\mathbb{A}^{\mathsf{T}}\rho$ and $\phi := e^w\Phi$; then $\Pi\rho = \mathbb{A}\Phi$ and $\mathfrak{R}_w\phi = e^w\mathbb{A}^{\mathsf{T}}\rho$. By the Cauchy–Schwarz inequality,

$$\langle \phi, \mathfrak{R}_w\phi\rangle = \langle e^w\phi, \mathbb{A}^{\mathsf{T}}\rho\rangle = \sum_r (\mathbb{A}e^w\phi)_r\rho_r = \sum_r (\mathsf{C}_r + \mathsf{S}_r)e^{w_r}\rho_r \leq (|\mathsf{C}| + |\mathsf{S}|)\,|e^w\rho|.$$

On the other hand, the identity for $\langle \phi, \mathfrak{R}_w\phi\rangle$ shown above gives $\langle \phi, \mathfrak{R}_w\phi\rangle = (|\mathsf{C}| - |\mathsf{S}|)(|\mathsf{C}| + |\mathsf{S}|)$. If $|\mathsf{C}| + |\mathsf{S}| = 0$, then $\phi = 0$ by (†), and so $|\mathsf{C}| - |\mathsf{S}| = 0$; otherwise we divide by $|\mathsf{C}| + |\mathsf{S}|$. In both cases $|\mathsf{C}| - |\mathsf{S}| \leq |e^w\rho|$.

With (†) this gives $|\mathbb{A}\phi| \leq 2|e^w\rho|$, and therefore $|e^w\Pi\rho| = |e^w\mathbb{A}\Phi| = |\mathsf{C} - \mathsf{S}| \leq \frac{3}{2}|\mathbb{A}\phi| \leq 3|e^w\rho|$, where the second equality is the identity $e^{w_r}(\mathbb{A}e^{-w}\phi)_r = \mathsf{C}_r - \mathsf{S}_r$. The bound for $1 - \Pi$ follows by the triangle inequality, $|e^w(1-\Pi)\rho| \leq |e^w\rho| + |e^w\Pi\rho|$. □

Lemma 5.74 (A Weitzenböck identity for 2-forms)**.** For $F, G \in \Omega^2(\mathbb{Z}^3/\mathcal{N}\mathbb{Z}^3)$ we have $\sum_\alpha \langle \partial_\alpha F, \partial_\alpha G\rangle = \langle \mathrm{d}F, \mathrm{d}G\rangle + \langle \mathrm{d}^*F, \mathrm{d}^*G\rangle$. In particular $|\mathrm{d}F|, |\mathrm{d}^*F| \leq |\nabla F|$. ■

Proof of Lem. 5.74. By polarisation it suffices to treat $G = F$. All operators that occur commute with the translations of the torus, so that by Parseval's identity we may consider one momentum

k at a time. At this momentum we write $v_\beta = e^{ik_\beta} - 1$, and we let $E \in \mathbb{C}^3$ be the vector of the components F_β, where the orientation signs are absorbed into E; then $\widehat{\mathrm{d}F} = \sum_\beta v_\beta E_\beta$.

On a 1-form A the coboundary is $(\mathrm{d}A)_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$, which at the momentum k is the vector product $v \times \hat{A}$. Taking adjoints, $\widehat{\mathrm{d}^* F} = (v\times)^* E = -\bar{v} \times E$.

For complex vectors the Lagrange identity reads $|a \times b|^2 = |a|^2|b|^2 - |\sum a_i \bar{b}_i|^2$. At $a = \bar{v}$ and $b = E$ it becomes $|\bar{v} \times E|^2 + |\sum v_\beta E_\beta|^2 = |v|^2|E|^2$. The left side is the symbol of $|\mathrm{d}^*F|^2 + |\mathrm{d}F|^2$, and the right side is the symbol of $|\nabla F|^2$; this proves the identity. □

Proposition 5.75 (A gradient bound for the minimiser)**.** Let the rows ℓ be translation covariant, with $\mathsf{B} \le b$ and $\mathsf{B}' \le b'$. Then every X' satisfies $|\nabla H^\sharp X'| \le (b' + \sqrt{27}(bb')^{1/4}) L^{-3/2} |\nabla' X'|$. If $b = b_*$ and $b' \le b_*$, this gives $|\nabla H^\sharp X'| \le 2 b_* L^{-3/2} |\nabla' X'|$, and for the column of a coarse plaquette P it gives $|\nabla H^\sharp \delta_P| \le 2\sqrt{6}\, b_* L^{-3/2}$. ■

Proof of Prop. 5.75. Let $\bar{F} = H^\sharp X'$. By the definition of $H^\sharp$ we have $\mathsf{B}\bar{F} = \bar{Y} = W^* \bar{Z} + \delta \Xi_{\bar{Z}}$, which defines $\bar{Y}$. As ℓ is translation covariant, $\mathsf{B} \ge 1$ commutes with each ∂_α, and Lem. 5.74 gives

$$|\nabla \bar{F}|^2 \le \sum_\alpha \langle \partial_\alpha \bar{F}, \mathsf{B} \partial_\alpha \bar{F} \rangle = \sum_\alpha \langle \partial_\alpha \bar{F}, \partial_\alpha \bar{Y} \rangle = \langle \mathrm{d}\bar{F}, \mathrm{d}\bar{Y} \rangle + \langle \mathrm{d}^* \bar{F}, \mathrm{d}^* \bar{Y} \rangle.$$

We bound the two terms on the right in turn.

For the first term, $\mathrm{d}\bar{F} \perp \ker Q_3$, because $(\bar{F}, \bar{\mathsf{n}})$ lies in $\mathrm{Adm}(X')$; so $\mathrm{d}\bar{F} = Q_3^* \theta$ for a 3-cochain θ. From $Q_3 \mathrm{d} = \mathrm{d}' W$, $W\bar{F} + \mathrm{d}'\bar{\mathsf{n}} = X'$ and $\mathrm{d}'\mathrm{d}' = 0$ we get $Q_3 \mathrm{d}\bar{F} = \mathrm{d}' W \bar{F} = \mathrm{d}'(X' - \mathrm{d}'\bar{\mathsf{n}}) = \mathrm{d}' X'$. Hence $\theta = (Q_3 Q_3^*)^{-1} \mathrm{d}' X'$, and as $Q_3 Q_3^* \ge L^3/27$ and $|\mathrm{d}' X'| \le |\nabla' X'|$ by Lem. 5.74 on $\mathbb{T}'$, we have $|\theta| \le 27 L^{-3} |\nabla' X'|$.

It follows that $\langle \mathrm{d}\bar{F}, \mathrm{d}\bar{Y} \rangle = \langle \theta, Q_3 \mathrm{d}\bar{Y} \rangle = \langle \theta, \mathrm{d}' W \bar{Y} \rangle$. The map $\mathcal{O} : X' \mapsto W\bar{Y}$ is linear, and $\mathcal{O} X' = W\bar{Y} = W \mathsf{B} H^\sharp X'$ because $\bar{F} = H^\sharp X'$ and $\mathsf{B}\bar{F} = \bar{Y}$. Since ℓ is translation covariant, B commutes with the translations of $\mathbb{T}$, and the proof of Prop. 2.55 (a) shows that W, $H^\sharp$ and B' commute with those of $\mathbb{T}'$; hence so does $\mathcal{O}$. To bound its norm, note that $|\bar{Y}|^2 \le b \langle \bar{Y}, \mathsf{R}\bar{Y} \rangle \le b L^{-1} \langle X', \mathsf{B}' X' \rangle$, where the first inequality uses $\mathsf{B} \le b$ and the second uses $\langle \bar{Y}, \mathsf{R}\bar{Y} \rangle = \langle \bar{F}, \mathsf{B}\bar{F} \rangle \le \|(\bar{F}, \bar{\mathsf{n}})\|_{\mathcal{H}}^2 = L^{-1} \langle X', \mathsf{B}' X' \rangle$; with $\|W\|^2 \le L$ and $\mathsf{B}' \le b'$ we obtain $\|\mathcal{O}\| \le (bb')^{1/2}$. By Lem. 5.74 on $\mathbb{T}'$ and the commutation with translations, $|\mathrm{d}' W \bar{Y}| \le |\nabla' \mathcal{O} X'| = |\mathcal{O} \nabla' X'| \le (bb')^{1/2} |\nabla' X'|$.

For the second term we use $\mathrm{d}^* \delta = 0$ and $\mathrm{d}^* W^* = Q_1^* \mathrm{d}'^*$; with $\bar{Z} = L^{-1} \mathsf{B}' X'$ they give $\mathrm{d}^* \bar{Y} = L^{-1} Q_1^* \mathrm{d}'^* \mathsf{B}' X'$. As we saw in bounding the first term, B' commutes with the translations of $\mathbb{T}'$; hence $\nabla' \mathsf{B}' X' = \mathsf{B}' \nabla' X'$. Lem. 5.74 on $\mathbb{T}'$, this identity and $\mathsf{B}' \le b'$ yield $|\mathrm{d}'^* \mathsf{B}' X'| \le |\mathsf{B}' \nabla' X'| \le b' |\nabla' X'|$, so that $\|Q_1\|^2 \le L^{-1}$ implies $|\mathrm{d}^* \bar{Y}| \le b' L^{-3/2} |\nabla' X'|$; moreover $|\mathrm{d}^* \bar{F}| \le |\nabla \bar{F}|$ by Lem. 5.74.

If $\nabla' X' = 0$, the bounds above give $\nabla \bar{F} = 0$, and the claim holds. If $\nabla' X' \ne 0$, then for $u = |\nabla \bar{F}| L^{3/2} / |\nabla' X'|$ the bounds for the two terms combine to $u^2 \le 27 (bb')^{1/2} + b' u$, so $u \le b' + \sqrt{27}(bb')^{1/4}$, because every $u > b' + \sqrt{27}(bb')^{1/4}$ satisfies $u(u - b') > 27 (bb')^{1/2}$. This is the first claim.

Finally, $b_* \ge 27$, so that $\sqrt{27}(b_* b')^{1/4} \le \sqrt{27}\, b_*^{1/2} \le b_*$ when $b' \le b_*$; this gives the second claim. The third is the second at $X' = \delta_P$, as $|\nabla' \delta_P|^2 = 6$. □

Lemma 5.76 (The step on constant forms)**.** Let L be odd, let the rows ℓ on $\mathbb{T}$ be translation covariant and $\tau(\ell)$ arbitrary, let h' be a constant form on $\mathbb{T}'$, and let h be the constant form on $\mathbb{T}$ with the same components as h'.

(a) $H^\sharp h' = L^{-2} h$ and $\mathsf{N}^\sharp h' = 0$.

(b) For all sites y' of $\mathbb{T}'$ and y of $\mathbb{T}$, all constant forms h_1, h_2 on $\mathbb{T}$ and the constant forms h_1', h_2' on $\mathbb{T}'$ with the same components, $\langle \ell'_{y'} h_1', \ell'_{y'} h_2' \rangle = \langle \ell_y h_1, \ell_y h_2 \rangle$; in other words, $\mathsf{c}(\mathsf{E}') = \mathsf{c}(\mathsf{E})$ as 3×3 matrices. Consequently $\tau(\ell') = \tau(\ell)$ and $\hat{\mathsf{B}}'(0) = \hat{\mathsf{B}}(0)$.

■

Proof of Lem. 5.76. We begin with (a). As B commutes with translations, $\mathsf{B}h =: g$ is a constant form. Let g' be the constant form on $\mathbb{T}'$ with the components of g, and put $Z := L^{-1}g'$.

Since W couples only plaquettes of equal orientation and $\sum_P W(P,p) = L^{-1}$, we get $W^*Z = L^{-1}\cdot L^{-1}g$; also $\mathrm{d}'^*Z = 0$, as Z is constant. Therefore $\mathrm{pair}(Z,0) = (L^{-2}\mathsf{R}g, 0) = (L^{-2}h, 0)$. This element lies in $\mathrm{Adm}(h')$, because $\sum_p W(P,p) = L^2$ gives $W(L^{-2}h) = h'$, and because $dh = 0$.

For every $(G,\mathsf{m}) \in \mathrm{Adm}(0)$ we have $\langle \mathrm{pair}(Z,0), (G,\mathsf{m})\rangle_{\mathcal{H}} = \langle Z, WG + \mathrm{d}'\mathsf{m}\rangle = 0$. Lem. 2.53 (a), as recalled in Step 1a of the proof of Thm. 5.63, states that an element of $\mathrm{Adm}(X')$ that is $\mathcal{H}$-orthogonal to $\mathrm{Adm}(0)$ is the minimiser. Hence $\mathrm{pair}(Z,0)$ is the pair $(H^\sharp h', \mathsf{N}^\sharp h')$, and (a) is proved.

For (b) we fix y' and treat the groups of $\ell'_{y'}$ from Prop. 2.28 one at a time.

(1), (2) These groups consist of differences of ζ at y', and they vanish on constants.

(3) Here part (a) and $W^*h' = L^{-1}h$ give $L^{-2}h - L^{-1}W^*h' = 0$, so that this group vanishes at h'.

(4) The two terms of this group, the one with $\mathsf{N}^\sharp$ and the one with d'^*, are 0 at h', by (a) and $\mathrm{d}'^*h' = 0$.

(5) By (a), this group contributes $L\sum_{y\in B(y')}\langle \ell_y L^{-2}h_1, \ell_y L^{-2}h_2\rangle = L\cdot L^{-4}\cdot L^3\, h_1^{\mathsf{T}}\mathsf{c}(\mathsf{E})h_2$ to the inner product, since $B(y')$ has L^3 sites and, by the covariance of ℓ, $\langle \ell_y h_1, \ell_y h_2\rangle = h_1^{\mathsf{T}}\mathsf{c}(\mathsf{E})h_2$ for every y.

Adding the contributions of the groups, we find $\langle \ell'_{y'}h'_1, \ell'_{y'}h'_2\rangle = h_1^{\mathsf{T}}\mathsf{c}(\mathsf{E})h_2$, which is (b); the last two equalities in (b) follow, since τ and $\hat{\mathsf{B}}(0)$ are determined by the matrix $\mathsf{c}(\mathsf{E})$. □

Lemma 5.77 (A form compared with its value at a site)**.** Let y be a site of $\mathbb{Z}^3/n\mathbb{Z}^3$, and let $\mathsf{a} \geq 0$ be a function on the plaquettes with $\mathsf{a}(p) \leq Ae^{-\theta|y-p|}$, where $\theta \geq 4\mu$. For $F \in \Omega^2$ let $h^{(y)}$ be the constant form with the components $F_\beta(y)$.

Claim 1. $\sum_p \mathsf{a}(p)\,|F_p - h_p^{(y)}| \leq \sqrt{3}\,A\,\mathsf{g}(\frac{\theta}{2})^{3/2}\,n_y(F)$.

Claim 2. For rows with $|\ell_y(p)| \leq K_* e^{-\lambda_*|y-p|}$, $|\ell_y F| \leq C_1 n_y(F) + \tau(\ell)|F|_y$.

Claim 3. If moreover $\tau(\ell) \leq C_1\eta$, then $|\ell_y F| \leq \sqrt{2}\,C_1\hat{n}_{y,\eta}(F)$.

■

Proof of Lem. 5.77. Each class of $\mathbb{Z}^3/n\mathbb{Z}^3$ has exactly one representative $u \in (-\frac{n}{2}, \frac{n}{2}]^3$, and the sup norm of u is the periodic norm of the class; we work with these representatives and write $|u|$ for this norm. For such u, $\mathrm{path}(u)$ is the lattice path from 0 to u that runs first along e_1, then along e_2 and then along e_3; it is simple, that is, it visits no point twice, and it stays in the box spanned by 0 and u.

Each step of the path joins some v and $v + e_\alpha$, and we call v its lower end. Along this step $F_\beta(y + \cdot)$ changes by $\pm(\partial_\alpha F)_{(y+v;\beta)}$, where $(y+v;\beta)$ denotes the plaquette based at $y+v$ with normal e_β. Let $c(u,v)$ be the number of steps of $\mathrm{path}(u)$ whose lower end is v. On a simple path every point lies on at most two steps, and the starting point on at most one; so $c \leq 2$ and $c(u,0) \leq 1$. Moreover $c(u,v) \neq 0$ only for $v \in \mathrm{path}(u)$, and then $|v|_\infty \leq |u|_\infty$.

Summing along the path and using $|(\partial_\alpha F)_p| \leq |(\nabla F)_p|$, we obtain $|F_\beta(y+u) - F_\beta(y)| \leq \sum_v c(u,v)|(\nabla F)_{(y+v;\beta)}|$. The plaquette based at $y+u$ with normal e_β has distance $|u|$ from y, so

$$\sum_p \mathsf{a}(p)\,|F_p - h_p^{(y)}| \leq \sum_{\beta,v}\kappa(v)|(\nabla F)_{(y+v;\beta)}|, \qquad \kappa(v) := A\sum_u c(u,v)e^{-\theta|u|}.$$

At $v = 0$, the bound $c(u,0) \leq 1$ gives $\kappa(0) \leq A\mathsf{g}(\theta) \leq A\mathsf{g}(\frac{\theta}{2})$.

We define $\mathrm{path}(u)$ for every $u \in \mathbb{Z}^3$ in the same way as for the representatives, and we write $|u|$ for the sup norm $|u|_\infty$ of such u. Now let $v \neq 0$, and let $U_v := \{u \in \mathbb{Z}^3 : v \in \mathrm{path}(u)\}$. In the definition of $\kappa(v)$, the summation variable u runs over the representatives, a subset of $\mathbb{Z}^3$. For $u \in U_v$ each coordinate v_i lies between 0 and u_i, and some v_i is nonzero and has the sign of u_i;

hence v lies outside the box spanned by 0 and $-u$. This shows $-U_v \cap U_v = \emptyset$, and both sets are contained in $\{|u|_\infty \geq |v|_\infty\}$.

From $c \leq 2$, the symmetry $|-u| = |u|$, this disjointness and $e^{-\theta|u|} \leq e^{-\frac{\theta}{2}|v|} e^{-\frac{\theta}{2}|u|}$ for $|u| \geq |v|$ we get

$$\sum_u c(u,v) e^{-\theta|u|} \leq \sum_{U_v} e^{-\theta|u|} + \sum_{-U_v} e^{-\theta|u|} \leq \sum_{|u|\geq|v|} e^{-\theta|u|} \leq e^{-\frac{\theta}{2}|v|} \mathsf{g}(\frac{\theta}{2}).$$

Together with the case $v = 0$, this shows $\kappa(v) \leq A e^{-\frac{\theta}{2}|v|} \mathsf{g}(\frac{\theta}{2})$ for all v.

We apply the Cauchy–Schwarz inequality to the sum over β and v, with the weights $e^{\mp\mu|v|}$. The factor with $e^{-\mu|v|}$ is at most $n_y(F)$, since $e^{-2\mu|v|}$ is the weight of the plaquette $(y+v;\beta)$ in $n_y(F)^2$; the other factor is the square root of

$$3\sum_v e^{2\mu|v|} \kappa(v)^2 \leq 3A^2 \mathsf{g}(\frac{\theta}{2})^2 \sum_v e^{-(\theta-2\mu)|v|} \leq 3A^2 \mathsf{g}(\frac{\theta}{2})^3,$$

where the last step uses $\theta - 2\mu \geq \frac{\theta}{2}$. This proves Claim 1.

For Claim 2 we write $|\ell_y F| \leq \sum_p |\ell_y(p)||F_p - h_p^{(y)}| + |\ell_y h^{(y)}|$ and apply Claim 1 with $\mathsf{a}(p) = |\ell_y(p)|$, $A = K_*$ and $\theta = \lambda_* = 8\mu$, which bounds the sum by $C_1 n_y(F)$. For the last term, $|\ell_y h^{(y)}| \leq \tau |h^{(y)}|_\infty \leq \tau |F|_y$, because each of the three plaquettes based at y enters $|F|_y$ with the weight 1. This proves Claim 2. If moreover $\tau(\ell) \leq C_1 \eta$, the right side of Claim 2 is at most $C_1(n_y(F) + \eta|F|_y) \leq \sqrt{2}\, C_1 \hat{n}_{y,\eta}(F)$, which is Claim 3. □

Lemma 5.78 (The change of the minimiser as a projection)**.** Let $\ell, \tilde{\ell}$ be rows on $\mathbb{T}$. Objects with a tilde are defined from $\tilde{\ell}$ as the corresponding objects without a tilde are defined from ℓ: $\tilde{\mathsf{B}} = 1 + \tilde{\ell}^{\mathsf{T}}\tilde{\ell}$; the inner product of $\tilde{\mathcal{H}}$ is the inner product of Def. 2.26 with B replaced by $\tilde{\mathsf{B}}$, $\langle (F,\mathsf{n}), (F',\mathsf{n}')\rangle_{\tilde{\mathcal{H}}} = \langle F, \tilde{\mathsf{B}} F'\rangle + a\langle \mathsf{n}, \mathsf{n}'\rangle$; $\tilde{H}^\sharp$ and $\tilde{\mathsf{N}}^\sharp$ are the maps $H^\sharp$ and $\mathsf{N}^\sharp$ of Prop. 2.27 (a), defined with $\tilde{\mathsf{B}}$ instead of B; and $\tilde{\mathsf{E}}_y = \tilde{\ell}_y^{\mathsf{T}} \tilde{\ell}_y$. Put $\mathsf{D} := \tilde{\mathsf{B}} - \mathsf{B}$, $\Delta H := \tilde{H}^\sharp - H^\sharp$ and $\Delta\mathsf{N} := \tilde{\mathsf{N}}^\sharp - \mathsf{N}^\sharp$. Let $\tilde{\Pi}$ be the projection of Lem. 5.73 (c) for $\tilde{\mathbb{A}}^{\mathrm{e}}$, the matrix of Lem. 2.52 with $\tilde{\ell}$ in place of ℓ.

(a) $\ker \tilde{\mathbb{A}}^{\mathrm{e}\mathsf{T}} = \{(G, \tilde{\ell}G, a^{1/2}\mathsf{m}, 0) : (G,\mathsf{m}) \in \mathrm{Adm}(0)\}$, and the Euclidean inner product of two vectors of this form is the inner product $\langle\cdot,\cdot\rangle_{\tilde{\mathcal{H}}}$ of the corresponding pairs (G,m).

(b) For every X',

$$(\Delta H X',\ \tilde{\ell}\,\Delta H X',\ a^{1/2}\Delta\mathsf{N} X',\ 0) = -(1-\tilde{\Pi})(\mathsf{D} H^\sharp X', 0, 0, 0).$$

■

Proof of Lem. 5.78. For (a), the definition of $\tilde{\mathbb{A}}^{\mathrm{e}}$ gives $\tilde{\mathbb{A}}^{\mathrm{e}\mathsf{T}}(\rho_1,\rho_2,\rho_3,\rho_4) = (W\rho_1 + a^{-1/2}\mathrm{d}'\rho_3,\ \mathrm{d}\rho_1 + \mathsf{S}_0^{1/2} Q_3^* \rho_4,\ \rho_2 - \tilde{\ell}\rho_1)$. Let $(\rho_1,\rho_2,\rho_3,\rho_4)$ lie in the kernel, and put $G = \rho_1$ and $\mathsf{m} = a^{-1/2}\rho_3$. Then $WG + \mathrm{d}'\mathsf{m} = 0$, $\rho_2 = \tilde{\ell}G$ and $\mathrm{d}G = -\mathsf{S}_0^{1/2} Q_3^* \rho_4$. Applying Q_3 to the last identity, we find on its left side $Q_3\mathrm{d}G = \mathrm{d}'WG = -\mathrm{d}'\mathrm{d}'\mathsf{m} = 0$; so $Q_3 Q_3^* \rho_4 = 0$, hence $\rho_4 = 0$ because $Q_3 Q_3^* \geq L^3/27$, and then $\mathrm{d}G = 0$. Thus $(G,\mathsf{m}) \in \mathrm{Adm}(0)$, and the vector has the stated form.

Conversely, let $(G,\mathsf{m}) \in \mathrm{Adm}(0)$. Then $\mathrm{d}G \in \operatorname{ran} Q_3^*$, since $\mathrm{d}G \perp \ker Q_3$, and $Q_3\mathrm{d}G = 0$, because $Q_3\mathrm{d} = \mathrm{d}'W$, $WG = -\mathrm{d}'\mathsf{m}$ and $\mathrm{d}'\mathrm{d}' = 0$; together these give $\mathrm{d}G = 0$, and the formula for the transpose shows that $(G, \tilde{\ell}G, a^{1/2}\mathsf{m}, 0)$ lies in the kernel. The statement on the inner products holds because $\tilde{\mathsf{B}} = 1 + \tilde{\ell}^{\mathsf{T}}\tilde{\ell}$.

For (b), note that $(\Delta H X', \Delta\mathsf{N} X') \in \mathrm{Adm}(0)$, being the difference of two elements of $\mathrm{Adm}(X')$. As Adm does not depend on the rows, it is the same for ℓ and for $\tilde{\ell}$. Lem. 2.53 (a), as recalled in Step 1a of the proof of Thm. 5.63, shows that $(H^\sharp X', \mathsf{N}^\sharp X')$ is $\mathcal{H}$-orthogonal to $\mathrm{Adm}(0)$, and, with $\tilde{\ell}$ in place of ℓ, that $(\tilde{H}^\sharp X', \tilde{\mathsf{N}}^\sharp X')$ is $\tilde{\mathcal{H}}$-orthogonal to $\mathrm{Adm}(0)$. For $(G,\mathsf{m}) \in \mathrm{Adm}(0)$ this gives

$$\langle (G,\mathsf{m}), (\Delta H X', \Delta\mathsf{N} X')\rangle_{\tilde{\mathcal{H}}} = 0 - \langle (G,\mathsf{m}), (H^\sharp X', \mathsf{N}^\sharp X')\rangle_{\tilde{\mathcal{H}}} = -\langle G, \mathsf{D} H^\sharp X'\rangle - 0,$$

where in the second step we write $\tilde{\mathsf{B}} = \mathsf{B} + \mathsf{D}$.

Hence, by (a), the left side of (b) is a vector of $\ker \tilde{\mathbb{A}}^{\mathsf{eT}} = \operatorname{ran}(1-\tilde{\Pi})$, and, by (a) again, its inner product with any vector of that space equals the inner product of $-(\mathsf{D}H^\sharp X', 0, 0, 0)$ with the same vector. It is therefore the orthogonal projection of $-(\mathsf{D}H^\sharp X', 0, 0, 0)$ onto $\operatorname{ran}(1-\tilde{\Pi})$, which is (b). □

Proposition 5.79 (One step contracts the difference of the site forms)**.** Let $L \geq L_B$ be odd, $aL \geq 3/c_L$ and $N \geq 2$ even. Let $\ell, \tilde{\ell} \in \mathcal{R}(b_*, K_*, \lambda_*, \tau)$ be rows on $\mathbb{T}$ with $\tau \leq \frac{1}{9}$, and let $0 \leq \eta \leq 1$ with $\tau \leq C_1\eta$. Let $D_y := \tilde{\mathsf{E}}_y - \mathsf{E}_y$, with $\epsilon := \|D\|_\eta \leq 2C_1^2$, and put $\eta_+ := L^{1/2}\eta$.

(a) For the uncut rows ℓ' and $\tilde{\ell}'$, $\|\tilde{\mathsf{E}}' - \mathsf{E}'\|_{\eta_+} \leq \vartheta_1\,\epsilon$.

(b) For the rows ℓ'^{c} and $\tilde{\ell}'^{\mathrm{c}}$, both cut at the same radius ϱ, $\|\tilde{\mathsf{E}}'^{\mathrm{c}} - \mathsf{E}'^{\mathrm{c}}\|_{\eta'} \leq \vartheta_1\,\epsilon$ with $\eta'^2 := L\eta^2 + 6e^{-\lambda_*\varrho/8}$.

■

Proof of Prop. 5.79. We fix $y' \in \mathbb{T}'$ and define $w := -\lambda_*\mathsf{d}(\cdot, y')$ on all cells and rows of the step; by the triangle inequality, $|w_i - w_r| \leq \lambda_*\mathsf{d}(r,i)$. We write $H = H^\sharp$ for the map defined from the rows ℓ. For $F \in \Omega^2$ we set $\beta[F]^2 := L(|e^w\nabla F|^2 + \eta^2|e^w F|^2)$, and we abbreviate $\mathsf{g}_\flat := \mathsf{g}(\lambda_\flat)$.

Step (1): sites of a block. We claim that $L\sum_{y\in B(y')} \hat{n}_{y,\eta}(F)^2 \leq 1.1\,\mathsf{g}_\flat\,\beta[F]^2$. Fix a plaquette p and let $\mathsf{a} := \min_{y\in B(y')}|y-p|$; as $\mu_0 = \lambda_\flat$, we have $\sum_{y\in B(y')} e^{-2\mu_0|y-p|} \leq e^{-\mu_0\mathsf{a}}\mathsf{g}_\flat$. On the other hand $|p-y'| \leq \mathsf{a} + \mathsf{h}_L$, so that $2\lambda_*\mathsf{d}(p,y') \leq \frac{2\lambda_*}{L}\mathsf{a} + \lambda_* \leq \mu_0\mathsf{a} + \lambda_*$, where we use $L \geq 32$; since $e^{\lambda_*} \leq 1.1$, this gives $e^{-\mu_0\mathsf{a}} \leq 1.1\,e^{2w_p}$. The same bounds hold with $\mu \geq \mu_0$ in place of μ_0 in the sum over y. Exchanging the sums over y and p in $L\sum_{y\in B(y')}\hat{n}_{y,\eta}(F)^2$ and inserting these bounds, we obtain the claim.

Step (2): the source. We claim that $|e^w\mathsf{D}F| \leq \sqrt{14}\,\mathsf{g}_\flat\,\epsilon\,L^{-1/2}\beta[F]$, where $\mathsf{D} = \sum_y D_y$. Writing $\hat{n}_y = \hat{n}_{y,\eta}$, we get from the definition of ϵ and the Cauchy–Schwarz inequality, for every G,

$$|\langle G, \mathsf{D}F\rangle| \leq \epsilon\sum_y \hat{n}_y(G)\hat{n}_y(F) \leq \epsilon\Big(\sum_y e^{-2w_y}\hat{n}_y(G)^2\Big)^{1/2}\Big(\sum_y e^{2w_y}\hat{n}_y(F)^2\Big)^{1/2}.$$

Since $|w_y - w_p| \leq \frac{\lambda_*}{L}|y-p| \leq \frac{1}{2}\mu_0|y-p|$, again by $L \geq 32$, we have $\sum_y e^{\mp 2w_y}e^{-2\mu_0|y-p|} \leq e^{\mp 2w_p}\mathsf{g}_\flat$, and the same with μ in place of μ_0. With (n1), the first sum is thus at most $14\mathsf{g}_\flat|e^{-w}G|^2$ and the second at most $\mathsf{g}_\flat L^{-1}\beta[F]^2$, so that $|\langle G, \mathsf{D}F\rangle| \leq \epsilon\sqrt{14}\,\mathsf{g}_\flat|e^{-w}G|\,L^{-1/2}\beta[F]$. Taking $G = e^{2w}\mathsf{D}F$, we obtain the claim.

Step (3): the minimiser under the weight. We claim that every X' satisfies $\beta[HX']^2 \leq \vartheta_0\,\hat{n}_{y',\eta_+}(X')^2$. For a coarse plaquette P we put $\nu_P := H\delta_P$ and $r_P := \operatorname{dist}(y', P)$. Prop. 2.55 (a) for the given rows $\ell \in \mathcal{R}(b_*, K_*, \lambda_*, \tau)$ with $\tau \leq \frac{1}{9}$, recalled in Step 1a of the proof of Thm. 5.63 in the form $\sum_p e^{2\lambda_*\mathsf{d}(p,P)}(H^\sharp\delta_P)_p^2 \leq 4\gamma_*^{-1}L^{-1}$, together with $w_p \leq -\lambda_* r_P + \lambda_*\mathsf{d}(p,P)$, gives $|e^w\nu_P| \leq 2(\gamma_* L)^{-1/2}e^{-\lambda_* r_P}$.

As $|w_p - w_{p+e_\alpha}| \leq \lambda_*/L$, we have $|e^w\partial_\alpha\nu_P| \leq (1+e^{\lambda_*/L})|e^w\nu_P|$, and with $1 + e^{\lambda_*/L} \leq 2.1$ this yields $|e^w\nabla\nu_P| \leq \sqrt{3}\cdot 2.1\cdot 2(\gamma_* L)^{-1/2}e^{-\lambda_* r_P} \leq 7.3(\gamma_* L)^{-1/2}e^{-\lambda_* r_P}$. A second bound comes from Prop. 5.75, which applies since $\mathsf{B}' \leq \frac{3}{2}\mathsf{r}_\sharp^{-1} \leq b_*$: as $w \leq 0$, it gives $|e^w\nabla\nu_P| \leq 2\sqrt{6}\,b_* L^{-3/2}$. Taking the geometric mean of the two bounds, we find $|e^w\nabla\nu_P| \leq 6(b_*\gamma_*^{-1/2})^{1/2}L^{-1}e^{-\frac{1}{2}\lambda_* r_P}$; here we use $2\sqrt{6}\cdot 7.3 < 35.8 < 36$.

By Lem. 5.76 (a), $\nabla H h' = 0$ for the constant form $h' = h'^{(y')}$ of X', that is, for the constant form on $\mathbb{T}'$ with the components of X' at y'. Hence Claim 1 of Lem. 5.77, used on $\mathbb{T}'$ with $\theta = \frac{1}{2}\lambda_* = 4\mu$, gives

$$|e^w\nabla HX'| \leq \sum_P |e^w\nabla\nu_P||X'_P - h'_P| \leq \sqrt{3}\cdot 6(b_*\gamma_*^{-1/2})^{1/2}L^{-1}\mathsf{g}\Big(\frac{\lambda_*}{4}\Big)^{3/2}n_{y'}(X'),$$

and so $L|e^w\nabla HX'|^2 \leq \vartheta_0 n_{y'}(X')^2$. For the second part of β, the bound on $|e^w\nu_P|$ and the Cauchy–Schwarz inequality give $L^{1/2}|e^w HX'| \leq 2\gamma_*^{-1/2}\sum_P e^{-\lambda_* r_P}|X'_P| \leq 2\gamma_*^{-1/2}(3\mathsf{g}(2\lambda_* - 2\mu_0))^{1/2}|X'|_{y'}$,

hence $L|e^w HX'|^2 \le C_0|X'|^2_{y'}$, as $\mathsf{g}(2\lambda_* - 2\mu_0) \le \mathsf{g}(\lambda_*)$. Since $C_0\eta^2 \le c_0\eta^2 = \vartheta_0 L\eta^2$ and $\eta_+^2 = L\eta^2$, the two parts add up to the claim.

Step (4): the algebra. Put $G := H - L^{-1}W^*$ and $\Gamma := \mathsf{N}^\sharp - (aL)^{-1}\mathrm{d}'^*$, and let $\tilde G$, $\tilde\Gamma$ and $\tilde H$ be the same operators defined from $\tilde\ell$, that is, with $\tilde H^\sharp$ and $\tilde{\mathsf{N}}^\sharp$. We set

$$\tilde u(X') := (L^{1/2}\mathbf{1}_{B(y')}\tilde G X',\ (aL)^{1/2}\mathbf{1}_{y'}\tilde\Gamma X',\ L^{1/2}(\tilde\ell_y \tilde H X')_{y\in B(y')}),$$

which is the part of $\tilde\ell'_{y'}X'$ consisting of the groups (3), (4) and (5), and

$$\Delta u(X') := (L^{1/2}\mathbf{1}_{B(y')}\Delta H X',\ (aL)^{1/2}\mathbf{1}_{y'}\Delta\mathsf{N}X',\ L^{1/2}(\tilde\ell_y \Delta H X')_{y\in B(y')}),\qquad v := \tilde u - \Delta u.$$

The groups (1) and (2) are the same for $\ell'_{y'}$ and for $\tilde\ell'_{y'}$.

The vector $v(X')$ consists of the groups (3) and (4) of $\ell'_{y'}X'$ and of $L^{1/2}(\tilde\ell_y HX')_{y\in B(y')}$. Since $\langle\tilde\ell_y F, \tilde\ell_y F'\rangle = \langle \ell_y F, \ell_y F'\rangle + \langle F, D_y F'\rangle$, we therefore find that $\langle X', \mathsf{E}'_{y'}X''\rangle$ is the sum of three terms: the contribution of the groups (1) and (2), the term $\langle v(X'), v(X'')\rangle$ and the term $-L\sum_{y\in B(y')}\langle HX', D_y HX''\rangle$. On the other hand, $\langle X', \tilde{\mathsf{E}}'_{y'}X''\rangle$ is the same contribution plus $\langle\tilde u(X'), \tilde u(X'')\rangle$. With $D'_{y'} := \tilde{\mathsf{E}}'_{y'} - \mathsf{E}'_{y'}$ and $v = \tilde u - \Delta u$ we obtain

$$\langle X', D'_{y'}X''\rangle = L\sum_{y\in B(y')}\langle HX', D_y HX''\rangle + \langle\Delta u(X'), \tilde u(X'')\rangle + \langle\tilde u(X'), \Delta u(X'')\rangle - \langle\Delta u(X'), \Delta u(X'')\rangle.$$

Step (5): the three bounds. By the definition of ϵ, the Cauchy–Schwarz inequality and Step (1), the first term on the right satisfies $\big|L\sum_{y\in B(y')}\langle HX', D_y HX''\rangle\big| \le \epsilon L\sum_{B(y')}\hat n_y(HX')\hat n_y(HX'') \le 1.1\mathsf{g}_\flat\epsilon\,\beta[HX']\beta[HX'']$.

The vector $\Delta u(X')$ is $L^{1/2}$ times the rows of the vector of Lem. 5.78 (b) that are placed in $B(y')$ or at y', and on these rows $e^w \ge e^{-\lambda_*/2}$. By Lem. 2.52 (b) and (c), the bound (sinh) of Lem. 5.73 holds for $\tilde{\mathbb{A}}^{\mathrm{e}}$ at λ_*, so that Lem. 5.73 (c) applies with our weight w; followed by Step (2), it gives $|\Delta u(X')| \le L^{1/2}e^{\lambda_*/2}\,4\,|e^w \mathsf{D}HX'| \le 15.1\,\mathsf{g}_\flat\,\epsilon\,\beta[HX']$. The factor 15.1 rests on $4\sqrt{14} < 14.97$ and on $\lambda_* < 10^{-13}$, which follows from the interval for λ_* in Thm. 2.65 (0); together they give $e^{\lambda_*/2}\cdot 4\sqrt{14} \le 14.97\cdot 1.001 \le 15.1$.

Finally $|\tilde u(X'')| \le |\tilde\ell'_{y'}X''| \le \sqrt{2}\,C_1\hat n_{y',\eta_+}(X'')$. The first inequality holds because $\tilde u(X'')$ is a part of $\tilde\ell'_{y'}X''$. The second is Claim 3 of Lem. 5.77 on $\mathbb{T}'$, which applies by Thm. 2.57 (a), recalled in Step 1a of the proof of Thm. 5.63 in the form $|\tilde\ell'_{y'}(P)| \le K_* e^{-\lambda_*\mathrm{dist}(y',P)}$, and because $\tau(\tilde\ell') = \tau(\tilde\ell) \le C_1\eta \le C_1\eta_+$, where the equality is Lem. 5.76 (b).

Step (6): the sum. Let $\mathsf{y} := \mathsf{g}_\flat C_1\vartheta_0^{1/2} = \vartheta_1/48 \le \frac{1}{100}$. We insert the three bounds of Step (5) into the identity of Step (4), and we use Step (3) and $\epsilon \le 2C_1^2$; this gives

$$|\langle X', D'_{y'}X''\rangle| \le \epsilon\,[1.1\mathsf{g}_\flat\vartheta_0 + 2\cdot 15.1\sqrt{2}\,\mathsf{y} + 15.1^2\cdot 2\,\mathsf{y}^2]\,\hat n_{y',\eta_+}(X')\hat n_{y',\eta_+}(X'').$$

The bracket is $\le (0.02 + 42.8 + 4.6)\mathsf{y} \le 48\mathsf{y} = \vartheta_1$, and (a) is proved.

Step (7): the cut. For the cut families, $D'^{\mathrm{c}}_{y'} = \chi D'_{y'}\chi$ with $\chi = \chi_{y'}$. We say that a coarse plaquette is inside if χ takes the value 1 on it, and outside otherwise. Then $(\partial_\alpha(\chi X'))_P$ equals $(\partial_\alpha X')_P$ if P and $P + e_\alpha$ are both inside, 0 if both are outside, and, in the remaining case, where exactly one of the two is inside, $-X'_P$ if this one is P and X'_{P+e_α} if it is $P + e_\alpha$.

In the remaining case, the one of the two plaquettes that is inside has distance exactly ϱ from y', since the periodic sup distance takes integer values and varies by at most 1 from a plaquette to its neighbour; moreover $|y' - P| \ge \varrho$, and each plaquette that is inside occurs in at most six such terms. Hence

$$n_{y'}(\chi X')^2 \le n_{y'}(X')^2 + 6e^{-2\mu\varrho}\sum_{|y'-P|=\varrho} X'^2_P \le n_{y'}(X')^2 + 6e^{-2(\mu-\mu_0)\varrho}|X'|^2_{y'},$$

and since $|\chi X'|_{y'} \le |X'|_{y'}$, $2(\mu-\mu_0) = \lambda_*/8$ and $\eta_+^2 = L\eta^2$, we get $\hat n_{y',\eta_+}(\chi X') \le \hat n_{y',\eta'}(X')$. Part (a), applied to $\chi X'$ and $\chi X''$, now gives (b). □

Theorem 5.80 (Contraction of the site forms along two chains)**.** Let $L \ge L_B$ be odd, $aL \ge 3/c_L$, n even, $0 \le k \le k'$ and $m := k'-k$. Let $(\ell_j)_{j\le k}$ and $(\tilde\ell_j)_{j\le k'}$ be two chains that end on $\mathbb{Z}^3/n\mathbb{Z}^3$, with radii (ϱ_j) and $(\tilde\varrho_j)$ that both satisfy the condition (Cut), that is, $\sum_{j\le k}\mathsf{t}_*(\varrho_j) \le \frac19$ and $\sum_{j\le k'}\mathsf{t}_*(\tilde\varrho_j) \le \frac19$, and with equal radii at equal distance from the end, $\tilde\varrho_{m+j} = \varrho_j$ for $1 \le j \le k$.

Put

$$\eta_0 := C_1^{-1}\sum_{j\le m}\mathsf{t}_*(\tilde\varrho_j), \qquad \eta_i^2 := L\,\eta_{i-1}^2 + 6\,e^{-\lambda_*\varrho_i/8} \quad (1\le i\le k),$$

and let the radii satisfy

$$\eta_k \le 1. \tag{Bud}$$

Then for $0 \le i \le k$, on $\mathbb{Z}^3/nL^{k-i}\mathbb{Z}^3$, $\|\tilde{\mathsf{E}}_{m+i} - \mathsf{E}_i\|_{\eta_i} \le 2C_1^2\,\vartheta_1^i$. In particular, for all y,p,p',

$$\big|(\mathsf{E}_{k,y} - \tilde{\mathsf{E}}_{k',y})(p,p')\big| \le 15\,C_1^2\,\vartheta_1^k\,e^{-\lambda_\flat(|y-p|+|y-p'|)}, \qquad \vartheta_1 = c_{\mathrm{fix}}L^{-1/2} \le L^{-2/5}, \quad \lambda_\flat = \tfrac{1}{16}\lambda_*.$$

■□

Proof of Thm. 5.80. By Thm. 2.57 (b), together with the bound on τ established in its proof, all rows of both chains lie in $\mathcal{R}(b_*,K_*,\lambda_*,\frac19)$, with $\tau(\ell_i) \le \sum_{j\le i}\mathsf{t}_*(\varrho_j)$ and $\tau(\tilde\ell_{m+i}) \le T_i := C_1\eta_0 + \sum_{j\le i}\mathsf{t}_*(\varrho_j)$; for the second chain we use $\tilde\varrho_{m+j} = \varrho_j$.

We show by induction on i the bound (I$_i$): $T_i \le C_1\eta_i$. For $i=0$ it holds by the definition of η_0. If (I$_i$) holds, then

$$(C_1\eta_i + \mathsf{t}_*(\varrho_{i+1}))^2 \le 2C_1^2\eta_i^2 + 2\mathsf{t}_*^2 \le C_1^2(L\eta_i^2 + 6e^{-\lambda_*\varrho_{i+1}/8}),$$

by $L \ge 2$ and by $2\mathsf{t}_*(\varrho)^2 = 18K_*^2\mathsf{g}(\frac{\lambda_*}{2})^2e^{-\lambda_*\varrho} \le 6C_1^2e^{-\lambda_*\varrho/8}$, where $6C_1^2 = 18K_*^2\mathsf{g}(\frac{\lambda_*}{2})^3$. As $T_{i+1} = T_i + \mathsf{t}_*(\varrho_{i+1})$, this is (I$_{i+1}$). Since the η_i increase with i, (Bud) gives $\eta_i \le \eta_k \le 1$.

Now we prove the bound of the theorem by induction on i. At the start, $\mathsf{E}_0 = 0$ because $\ell_0 = \emptyset$, and Claim 3 of Lem. 5.77, which applies as $\tau(\tilde\ell_m) \le T_0 = C_1\eta_0$, gives $|\langle F, \tilde{\mathsf{E}}_{m,y}G\rangle| \le |\tilde\ell_{m,y}F||\tilde\ell_{m,y}G| \le 2C_1^2\hat n_{y,\eta_0}(F)\hat n_{y,\eta_0}(G)$; this is the case $i=0$.

For the step from i to $i+1 \le k$ we apply Prop. 5.79 (b) to the pair $(\ell_i, \tilde\ell_{m+i})$, with $\eta = \eta_i$, $\varrho = \varrho_{i+1}$ and $\tau = \min\{C_1\eta_i, \frac19\}$; here the letter N of the proposition is nL^{k-i-1}. Its hypotheses hold by the first paragraph of the proof, by (I$_i$), by $\eta_i \le 1$ and by the bound of the theorem for the index i, which is the hypothesis of the induction and gives $\epsilon \le 2C_1^2\vartheta_1^i \le 2C_1^2$. Since $\eta' = \eta_{i+1}$ and both families are cut at $\varrho_{i+1} = \tilde\varrho_{m+i+1}$, the proposition yields this bound for the index $i+1$. Finally, as $\eta_k \le 1$, we may use (n2) with $\eta = \eta_k$; it gives the bound on the entries, because $7.1\cdot 2 \le 15$. □

Corollary 5.81 (Row families of two runs approach each other)**.** Let $L \ge L_B$ be odd, let $aL \ge 3/c_L$ and $\theta_D \le \frac18\lambda_*$, and put $c_R = 64/\theta_D$, $\varrho(x) = \lceil c_R \log x\rceil$ and $e_R = \frac12\lambda_*c_R$ (≥ 256). Let $x_\flat$ satisfy the condition (X12) of the entry (O13), together with (X13): $16x_\flat^{-e_R/4} \le 1$. Let $\mathsf{b} \ge x_\flat$, let $K \le K'$ with $m = K'-K$, and let N be even. Let ℓ_k be the chain on the tower $\mathbb{Z}^3/NL^{K-k}\mathbb{Z}^3$ with the radii $\varrho_j = \varrho(L^{K-j}\mathsf{b})$, and let $\tilde\ell_k$ be the chain of length K' ending on $\mathbb{Z}^3/N\mathbb{Z}^3$, with the radii $\tilde\varrho_j = \varrho(L^{K'-j}\mathsf{b})$. Then for $0 \le k \le K$, with $x_k = L^{K-k}\mathsf{b}$, the condition (Bud) holds with $\eta_k^2 \le 16x_k^{-e_R/4}$, we have $\|\tilde{\mathsf{E}}_{m+k} - \mathsf{E}_k\|_{\eta_k} \le 2C_1^2\vartheta_1^k$, and

$$\|\tilde{\mathsf{E}}_{m+k} - \mathsf{E}_k\|_{(\lambda_\flat)} \le 15C_1^2\vartheta_1^k. \tag{RU}$$

■□

Proof of Cor. 5.81. At partner levels the radii of the two chains agree: $\tilde\varrho_{m+j} = \varrho_j$. The condition (Cut) follows from (X12), by the argument that the proof of Thm. 2.64 gives for (Cut). Next, $\mathsf{t}_*(\varrho(x)) \le 3K_*\mathsf{g}(\frac{\lambda_*}{2})x^{-e_R}$ and $\sum_{j\le m}(L^{K'-j}\mathsf{b})^{-e_R} \le (L^K\mathsf{b})^{-e_R}/(1-L^{-e_R})$; together with

$3K_*\mathsf{g}(\frac{\lambda_*}{2}) \le C_1$, these bounds yield $\eta_0 \le 2(L^K\mathsf{b})^{-e_R}$. Further, $e^{-\lambda_*\varrho_j/8} \le x_j^{-e_R/4}$, where $x_j = L^{k-j}x_k$.

Therefore

$$\eta_k^2 = L^k\eta_0^2 + 6\sum_{j\le k} L^{k-j}e^{-\lambda_*\varrho_j/8} \le 4L^{k-2e_RK}\mathsf{b}^{-2e_R} + 6x_k^{-e_R/4}\sum_i L^{i(1-e_R/4)} \le 4x_k^{-e_R/4} + 12x_k^{-e_R/4},$$

where the last inequality rests on $k + \frac{1}{4}(K-k)e_R \le 2e_RK$ and on $\mathsf{b} > 1$. This is the bound on η_k^2 in the statement. As $x_k \ge \mathsf{b} \ge x_\flat$, we have $16x_k^{-e_R/4} \le 16x_\flat^{-e_R/4} \le 1$ by (X13), so that $\eta_k \le 1$, which is (Bud) for the two chains cut off at the level k. Thm. 5.80, applied to these two chains, now gives the two bounds on $\tilde{\mathsf{E}}_{m+k} - \mathsf{E}_k$. □

Step 2: partners, and the data of a step at each level. In this step of the proof of Thm. 5.63, under the hypotheses and in the notation of that theorem, we claim that the following clauses (a) to (e) hold.

(a) Level k of the run K and level $m+k$ of the run K' share one torus, one number x_k, one grid of cubes, one set of the numbers $\varepsilon(x_k), p_0(x_k), \varrho(x_k), r(x_k), s(x_k), \ell_w(x_k)$, the caps and the domains, one mask and one clip, and one step map.
(b) (R8) applies to both runs.
(c) For $0 \le k \le K$, $\ell_k \in \mathcal{R}(b_*, K_*, \lambda_*, \frac{1}{9})$, and ℓ_k has range $\le \varrho(x_k)$.
(d) For $1 \le k \le K-1$, (SD) holds at $(x_k,\ N' = NL^{K-k-1},\ \ell_k)$.
(e) For $0 \le k \le K$, $NL^{K-k} \ge 2M\, s(x_k)$.

Taken by themselves, and with $K \ge 1$ in place of the hypothesis $K \ge K_0(\mathsf{b})$ of Thm. 5.63, the clauses (c) and (e) are a statement in its own right. Among the letters, which are chosen in the order (O0) to (O13), its data are only these: $L \ge L_A$ odd, $aL \ge 3/c_L$, $x_\flat \ge e^2$ with (X12); $\mathsf{b} \ge x_\flat$; N even, $M \mid N$, $N \ge 2M\, s(L\mathsf{b})$; $K \ge 1$; $x_k = L^{K-k}\mathsf{b}$, and the rows $\ell_k = \ell_k^{[K]}$ of the runs (item 2 of §5.5). With these data, (c) and (e) hold for $0 \le k \le K$. Clause (a) is a list of identities between level k of the run K and level $m+k$ of the run $K' = K+m$; it holds for every $K' \ge K \ge 1$ as well. The arguments for (c) and (e) below rest on these data alone: they are independent of $K_0(\mathsf{b})$, of k_0, of Λ, of K', of the members, and of Steps 3 to 6.

For (a): the tori agree, $\mathbb{Z}^3/NL^{K'-(m+k)}\mathbb{Z}^3 = \mathbb{Z}^3/NL^{K-k}\mathbb{Z}^3$, and $L^{K'-(m+k)}\mathsf{b} = L^{K-k}\mathsf{b} = x_k$. The origins are defined recursively, downwards from the last level $j = K$ of the run, by $o_K = 0$ and $o_j = L\,o_{j+1} - (L-1)/2$; hence $o_{m+k}^{[K']} = o_k^{[K]}$. Each number of the list is a function of x and of the letters, and the mask, the clip and the paths depend on $V, \varepsilon, \rho_0, \varrho(x)$ (the definition "Energy" in item 2 of §5.5). The step depends on x_k, a, L alone and is independent of the rows, of the grid and of the members (the definition "The step" in item 2 of §5.5). So it is one map for both runs, and $(\rho_{m+k}^{[K']})' = \rho_{m+k+1}^{[K']}$.

For (b): $\beta = L^K\mathsf{b} \ge L^K x_\flat$ and $L^{1-K}\beta = L\mathsf{b}$, and the same holds with K' in place of K. The condition $N \ge 2M\, s(L^{1-K}\beta)$ is thus (Tor) of Thm. 5.63, for both runs.

For (c) we begin with (Cut). From $\varrho(x) \ge c_R \log x$ we obtain $\mathsf{t}_*(\varrho(x_j)) \le 3K_*\mathsf{g}(\lambda_*/2)x_j^{-e_R}$; as $x_j \ge L^{K-j}x_\flat$, the geometric series and (X12) give $\sum_{j=1}^K \mathsf{t}_*(\varrho(x_j)) \le \frac{1}{9}$. Thm. 2.57 (b), whose data are $L \ge L_A$ odd, $aL \ge 3/c_L$ and N even, then yields $\ell_k \in \mathcal{R}(b_*, K_*, \lambda_*, \frac{1}{9})$ for $0 \le k \le K$, and the range is $\le \varrho(x_k)$ because of the cut.

For (d), let $1 \le k \le K-1$. Then $x_k \ge L\mathsf{b} \ge L\,x_\flat \ge x_9$, and $N' = NL^{K-k-1}$ is even and a multiple of M. As to (Tor): Prop. 5.8, one of the base clauses (RB), applied with $x = L\mathsf{b} \ge e^2$ and $j = K-1-k$, turns $N \ge 2M\, s(L\mathsf{b})$ into $NL^j \ge 2M\, s(L^j \cdot L\mathsf{b}) = 2M\, s(x_k)$. The grids fit, by their definition.

The four conclusions for the pair $(\ell_k, (\ell_k)')$ are provided by Thm. 2.64 at this k, since its two hypotheses on k and on $x_\flat$, namely $x_{k+1} \ge \mathsf{b} \ge x_\flat$ and (X12), hold. Applied with (C_D, θ_D) of (O6)

as its pair of constants, the theorem gives them in the following sense. The bound (Dec)(C_D, θ_D) at $\theta_D = \frac{1}{2}\min\{\lambda_2^*, \theta_{\mathbb{C}}, \theta_J\}$ is the conclusion (Fr2) of Thm. 2.65, which covers all sixteen kernels of (Dec); the hypotheses of that theorem, $L \geq L_A$ odd, $aL \geq 3/c_L$, N even and $x_{k+1} \geq x_\flat$ with (X12), are met. The condition (N′) with eq. 2.vii at every cut, and the bound $\mathsf{t}_\rho \leq \frac{1}{8}(\frac{1}{2} - d_0 - c_\Xi)$, are (Fr3′) and (Fr4) in the conclusion of Thm. 2.65. The properties of ℓ_k and $(\ell_k)'$ that these two clauses use are the ones stated in (Fr0) of that theorem. In the proof of Thm. 2.64, the condition and the bound are derived from $\ell_k \in \mathcal{R}(b_*, K_*, \lambda_*, \frac{1}{9})$ and $|(\ell_k)'_{y'}(P)| \leq K_* e^{-\lambda_* \mathrm{dist}(y', P)}$, which Thm. 2.57 (b) gives; from the radius ρ_0 fixed by eq. 2.vi; and from Thm. 2.60 (d) with $(b, K, \lambda, \lambda^\circ) = (b_*, K_*, \lambda_*, \lambda_*)$, as in Def. 2.61, hence with the constants c_1^* and c_2^*. This derivation depends on θ_D only through (Cut), which (c) gives when c_R has the value fixed by (O6).

The hypothesis $L \geq L_*(d_0, c_\Xi)$ of Thm. 2.64 follows from (O4). With $\mathsf{N}_\nu = \mathsf{N}_\sharp + (3 + 4c_2^* L)\mathsf{N}_Z$, $\mathsf{N}_\sharp = c_1^* \mathsf{I}_L \mathsf{N}_Z \mathsf{P}_\rho$ and $\mathsf{N}_\Xi = c_2^* \mathsf{N}_Z$, the left side of (L1^{su}) equals

$$[c_1^* \mathsf{I}_L \mathsf{P}_\rho / L + 3/L + 2200 + 44 c_2^*] \mathsf{N}_Z \, L^{-1/2} \geq (c_1^* \mathsf{I}_L \mathsf{P}_\rho / L + 60 + 4c_2^*) \mathsf{N}_Z \, L^{-1/2}.$$

From (L2^{su}) we get $L \geq 1.44 \cdot 10^8 \cdot 17^4 (\mathsf{N}_\sharp + 1)^4 \geq 115 \mathsf{N}_\sharp^2$. The definition of $L_*(d_0, c_\Xi)$, in Def. 2.63, asks for its two inequalities at all real $L \geq L_*(d_0, c_\Xi)$. For each of the two, the left side divided by the right side decreases in the real variable $L > e$: I_L / L has the derivative $-\log L / L^2$, $L^{-1/2}$ decreases, and I_L^2 / L has the derivative $\mathsf{I}_L(2 - \mathsf{I}_L)/L^2 < 0$ for $\mathsf{I}_L > 2$. The two bounds deduced from (L1^{su}) and (L2^{su}) show that both inequalities hold at the L of (O4), where (L1^{su}) and (L2^{su}) hold; by this monotonicity they hold at every real number $\geq L$. As L is an integer $\geq L_A$, it follows that $L_*(d_0, c_\Xi) \leq L$. The conditions (L1) and (L2) hold in the forms (L1^{su}) and (L2^{su}), by (O4).

For (e): as $r(x) = \lceil 20 c_R \log x / M \rceil$ is the ceiling of an increasing function, the function $x \mapsto s(x) = 8r(x)$ is non-decreasing, so $N \geq 2M\, s(L\mathsf{b}) \geq 2M\, s(\mathsf{b})$, and Prop. 5.8, applied with $x = \mathsf{b} \geq e^2$ and $j = K - k$, gives $N L^{K-k} \geq 2M\, s(x_k)$.

Step 3: Thm. 5.63 (i), and the size of δ_Δ. We claim that $\Lambda > 0$, that $k_0 \geq 1$, and that $k_0 \leq K - 1$ holds if and only if $K \geq K_0(\mathsf{b})$. As $K \geq K_0(\mathsf{b})$ is a hypothesis of Thm. 5.63, this gives $1 \leq k_0 \leq K - 1$, which, together with $\Lambda > 0$, makes up clause (i) of that theorem. Moreover, at every level k with $k_0 \leq k \leq K$

$$\delta_\Delta(k) := (c_Q^{\mathrm{su}} / E_a)\, x_k^{1/2} \delta_k / p_0(x_k) \leq L^{\Lambda + K/2 - 9k/10} \leq 1.$$

First, $\Lambda > 0$. By (X17) at $x = \mathsf{b}$ we have $16 c_a(\mathsf{b}) \leq \frac{1}{2}$, that is, $\mathsf{b}^{1/2} \geq 32 E_a p_0(\mathsf{b})^3$, and the right side is $\geq 32 E_a$ by (X14). Hence $c_4 \mathsf{b}^{1/2} \geq 480 c_Q^{\mathrm{su}} C_1^2 > 1$, by Step 0 (iii). Next, k_0 is a level: as $(5K + 10\Lambda)/9 > 0$, we have $k_0 \geq 1$. As $K - 1$ is an integer, $\lceil y \rceil \leq K - 1$ holds if and only if $y \leq K - 1$; and $(5K + 10\Lambda)/9 \leq K - 1$ holds if and only if $4K \geq 10\Lambda + 9$, that is, if and only if $K \geq \frac{5}{2}\Lambda + \frac{9}{4}$.

We turn to δ_Δ. By Step 1 and by (X14) at $x_k \geq \mathsf{b} \geq x_\flat$,

$$\delta_\Delta(k) \leq (c_Q^{\mathrm{su}} / E_a) \mathsf{b}^{1/2} L^{(K-k)/2} \cdot 15 C_1^2 L^{-2k/5} = c_4 \mathsf{b}^{1/2} L^{K/2 - 9k/10} = L^{\Lambda + K/2 - 9k/10}.$$

The exponent is ≤ 0 if and only if $k \geq (5K + 10\Lambda)/9$, and hence for every $k \geq k_0$.

Step 4: the switch at the step $k_0 \to k_0 + 1$, following the argument of Thm. 5.48. We show three things. First, for some real number V and some real $\mathfrak{n}_{k_0+1} \in M_1^{\mathrm{su}}(x_{k_0+1})$ having the strong cap, the identity $\rho^{[K']}_{m+k_0+1} = e^{\mathsf{V}} \rho[\mathfrak{n}_{k_0+1}; \ell^{[K]}_{k_0+1}]$ holds at every field. Second, $u_{k_0+1} = F_{\ell_{k_0}}(u_{k_0})$ is a system of members of $\rho^{[K]}_{k_0+1}$. Third, $\mathsf{D}_{x_{k_0+1}}(u_{k_0+1}, \mathfrak{n}_{k_0+1}) \leq 0.4(2 + \delta_\Delta) \leq 1.2 \leq 3\vartheta$.

We put $x = x_{k_0}$, $\ell = \ell^{[K]}_{k_0}$ and $\tilde{\ell} = \ell^{[K']}_{m+k_0}$. Both row families live on $\mathbb{T}_{k_0}$ and have the same grid of cubes (Step 2 (a)).

1. *Members.* We have $1 \leq k_0 \leq K$ and $1 \leq m + k_0 \leq K'$. By (R8) for both runs (Step 2 (b)), $\rho^{[K]}_{k_0} = e^{\mathsf{V}_1} \rho[u_{k_0}; \ell]$ and $\rho^{[K']}_{m+k_0} = e^{\mathsf{V}_2} \rho[\tilde{u}; \tilde{\ell}]$, where the numbers V_i are real and $u_{k_0}, \tilde{u}$ are real elements of $M_1^{\mathrm{su}}(x)$ with the strong cap.

2. *The rows are changed.* We go through the conditions of (R11) at x in turn. First, $x \geq e^2$. The side is $N_f = NL^{K-k_0} = N'L$ with $N' = NL^{K-k_0-1}$, as $k_0 \leq K-1$ by Step 3; and N' is even, a multiple of M, and $\geq 2M\, s(x)$, by Step 2 (d) at $k = k_0$. The families ℓ and $\tilde{\ell}$ have range $\leq \varrho(x)$; for $\tilde{\ell}$ this holds because it is cut at $\varrho(L^{K'-m-k_0}\mathsf{b}) = \varrho(x)$. Further, $\delta_\Delta \leq 1$ by Step 3 at $k = k_0$; (M5r) holds by (O7); $\varepsilon \leq \frac{1}{40}$ is (X16); $\varrho + 2 \leq M\, r$ is Step 0 (ii); $c_\Xi p_0^2 \geq 1$ is (X15); and $(2r+1)^3 c_a \leq 34$ is Step 0 (vi). Hence (R11) provides a real $u^\flat \in M_2^{\mathrm{su}}(x)$ with $\rho[\tilde{u};\tilde{\ell}] = \rho[u^\flat;\ell]$ at every V and $\mathsf{D}_x^{\mathrm{in}}(\tilde{u}, u^\flat) \leq \delta_\Delta$. Thus $\rho^{[K']}_{m+k_0} = e^{\mathsf{V}_2}\rho[u^\flat;\ell]$ as functions.

3. *The step.* By Step 2 (d), as $k_0 \leq K-1$, (SD) holds at (x, NL^{K-k_0-1}, ℓ). The step is linear in ρ, and it is one map for both runs. By (9a) at $A = 2$,

$$\rho^{[K']}_{m+k_0+1} = e^{\mathsf{V}_2}(\rho[u^\flat;\ell])' = e^{\mathsf{V}_2+\mathsf{V}'}\rho[F_\ell(u^\flat);\ell'^{\mathrm{c}}],$$

and $\ell'^{\mathrm{c}} = ((\ell_{k_0})')^{\mathrm{c}}$, cut at $\varrho(x_{k_0+1})$, is $\ell^{[K]}_{k_0+1}$, the row family of the short run. We put $\mathfrak{n}_{k_0+1} := F_\ell(u^\flat)$. By (9b) at $A = 2$, it lies in $M_1^{\mathrm{su}}(x_{k_0+1})$ and has the strong cap; by (9d), it is real, and so is V'. Also by (9d), $\rho^{[K']}_{m+k_0+1} > 0$, because $\rho[u^\flat;\ell] = e^{-\mathsf{V}_2}\rho^{[K']}_{m+k_0} > 0$. So $\rho^{[K']}_{m+k_0+1}$ is a point of $C_\flat^{\mathrm{su}}(x_{k_0+1}; \ell^{[K]}_{k_0+1})$. The clauses (9a), (9b) and (9d), now taken at $A = 1$ and applied to the system $u_{k_0} \in M_1^{\mathrm{su}}(x)$ of $\rho^{[K]}_{k_0}$, give the claim on u_{k_0+1}.

4. *The distance.* As $\|\cdot\|_x$ is a norm,

$$\mathsf{D}_x^{\mathrm{in}}(u_{k_0}, u^\flat) \leq \|u_{k_0}\|_x + \|\tilde{u}\|_x + \mathsf{D}_x^{\mathrm{in}}(\tilde{u}, u^\flat) \leq 2 + \delta_\Delta \leq 3.$$

Both systems lie in $M_2^{\mathrm{su}}(x)$, so (9e) applies; it gives $\mathsf{D}_{x_{k_0+1}}(u_{k_0+1}, \mathfrak{n}_{k_0+1}) \leq 0.4 \cdot 3 = 1.2 \leq 3\vartheta = 1.3404$.

Remark 5.82. By (R11), the system $u^\flat$ lies in $M_2^{\mathrm{su}}(x)$, that is, it satisfies the three size bounds with $A = 2$; and $w^\flat$ has the weak cap only. The definition of the class asks for a system of members in $M_1^{\mathrm{su}}(x)$ with the strong cap. Hence (R11) does not provide $u^\flat$ as such a system, nor does it give the membership of $\rho[u^\flat;\ell]$ in $C_\flat^{\mathrm{su}}(x;\ell)$. The property (S-step) of Thm. 4.39 concerns the images of points of the class alone. We therefore use Thm. 5.15 at $A = 2$, which admits the weak cap, and we obtain reality from Thm. 5.15 (d).

Step 5: after the switch, and the last level $k = K$; Thm. 5.63 (ii), (iii), (iv) at k and at K. We claim that for $k_0 < k \leq K$ the systems u_k and $\mathfrak{n}_k$ are real, lie in $M_1^{\mathrm{su}}(x_k)$, have the strong cap, and are systems of members of $\rho^{[K]}_k$ and of $\rho^{[K']}_{m+k}$ respectively, both over the same row family $\ell^{[K]}_k$, and that $\mathsf{D}_{x_k}(u_k, \mathfrak{n}_k) \leq 3\vartheta^{k-k_0}$. At $k = K$ this distance is $\leq C_\mathsf{b}\theta^K$.

We argue by induction on k, starting from $k_0 + 1$, which is Step 4. Let the claim hold at some k with $k_0 < k < K$. By Step 2 (d), (SD) holds at $(x_k, NL^{K-k-1}, \ell_k)$. We put $\mathfrak{n}_{k+1} := F_{\ell_k}(\mathfrak{n}_k)$, and we have $u_{k+1} = F_{\ell_k}(u_k)$. By (9a) at $A = 1$, together with Step 2 (a),

$$\rho^{[K']}_{m+k+1} = (\rho^{[K']}_{m+k})' = e^{\mathsf{V}+\mathsf{V}'}\rho[\mathfrak{n}_{k+1}; (\ell_k)'^{\mathrm{c}}],$$

and $(\ell_k)'^{\mathrm{c}} = \ell^{[K]}_{k+1}$. By (9b) at $A = 1$, the sizes of $\mathfrak{n}_{k+1}$ are at most 0.598 of the cap c_a and at most the cap c_h; it has the strong cap, and it satisfies at x_{k+1} all clauses of (ca), (ch) and (cΞ) other than the size bounds. By (9d), $\mathfrak{n}_{k+1}$ and V' are real, and the image is > 0. Applied to u_k in place of $\mathfrak{n}_k$, the clauses (9a), (9b), (9d) give these properties for u_{k+1} as well. By (9e),

$$\mathsf{D}_{x_{k+1}}(u_{k+1}, \mathfrak{n}_{k+1}) \leq 0.4\, \mathsf{D}_{x_k}^{\mathrm{in}}(u_k, \mathfrak{n}_k) \leq \vartheta\, \mathsf{D}_{x_k}(u_k, \mathfrak{n}_k).$$

Starting from $\mathsf{D}_{x_{k_0+1}} \leq 3\vartheta$, this gives $\mathsf{D}_{x_k} \leq 3\vartheta^{k-k_0}$.

The last level $k = K$. Since $k_0 < (5K + 10\Lambda)/9 + 1$, we have $K - k_0 > \frac{4}{9}K - \frac{10}{9}\Lambda - 1$. As $\vartheta < 1$, the distance at the level K, where $x_K = \mathsf{b}$, satisfies

$$\mathsf{D}_\mathsf{b} \leq 3\vartheta^{\frac{4}{9}K - \frac{10}{9}\Lambda - 1} = (3/\vartheta) \cdot \vartheta^{-\frac{10}{9}\Lambda} \cdot (\vartheta^{4/9})^K.$$

Here

$$\vartheta^{-\frac{10}{9}\Lambda} = \exp\{\frac{10}{9}\Lambda \log(1/\vartheta)\} = (L^\Lambda)^{\varepsilon_L} = (c_4 \mathsf{b}^{1/2})^{\varepsilon_L},$$

so that $\mathsf{D}_\mathsf{b} \le C_\mathsf{b}\theta^K$. Numerically, $0.4468^{4/9} \le 0.69903$ and $3/0.4468 \le 6.715$.

Step 6: Thm. 5.63 (iv) at the depth s, and (v). We claim the following.

(a) $K - s > k_0$ if and only if $K - s \ge K_0(L^s\mathsf{b}) = \frac{5}{2}\Lambda + \frac{5}{4}s + \frac{9}{4}$; in that case $3\vartheta^{K-s-k_0} \le 3\vartheta^{\frac{4}{9}K - s - \frac{10}{9}\Lambda - 1} = C_{L^s\mathsf{b}}\theta^{K-s}$.

(b) $k_0 < \varphi K + K_0(\mathsf{b})$ and $k_0 \le \lceil \varphi K \rceil + K_0(\mathsf{b})$.

For (a): as $K - s - 1$ is an integer, $k_0 \le K - s - 1$ holds if and only if $(5K + 10\Lambda)/9 \le K - s - 1$, that is, if and only if $4K \ge 10\Lambda + 9s + 9$, that is, if and only if $K - s \ge \frac{5}{2}\Lambda + \frac{5}{4}s + \frac{9}{4}$. Moreover $\Lambda(L^s\mathsf{b}) = \log_L(c_4 L^{s/2}\mathsf{b}^{1/2}) = \Lambda + s/2$, so that $K_0(L^s\mathsf{b}) = \frac{5}{2}\Lambda + \frac{5}{4}s + \frac{9}{4}$. Further, $K - s - k_0 > \frac{4}{9}K - s - \frac{10}{9}\Lambda - 1$; as $\vartheta < 1$, this gives the inequality in (a). And

$$C_{L^s\mathsf{b}}\theta^{K-s} = (3/\vartheta)\vartheta^{-\frac{10}{9}(\Lambda + s/2)}\vartheta^{\frac{4}{9}(K-s)} = 3\vartheta^{\frac{4}{9}K - \frac{4}{9}s - \frac{5}{9}s - \frac{10}{9}\Lambda - 1},$$

with the same exponent.

For (b): $k_0 < \varphi K + \frac{10}{9}\Lambda + 1 \le \varphi K + \frac{5}{2}\Lambda + \frac{9}{4}$. The inequality $\lceil \alpha + \beta \rceil \le \lceil \alpha \rceil + \lceil \beta \rceil$ gives $k_0 \le \lceil \varphi K \rceil + \lceil \frac{10}{9}\Lambda \rceil \le \lceil \varphi K \rceil + \frac{10}{9}\Lambda + 1 \le \lceil \varphi K \rceil + K_0(\mathsf{b})$. So $k > \varphi K + K_0(\mathsf{b})$ implies $k > k_0$; and $k - k_0 \ge k - \lceil \varphi K \rceil - K_0(\mathsf{b})$, together with $\vartheta < 1$, gives $3\vartheta^{k-k_0} \le 3\vartheta^{k - \lceil \varphi K \rceil - K_0}$. For $K \le \frac{9}{4}K_0(\mathsf{b})$ the range of k in (v) is empty, and the claim at $k = K$ is (iv). □

Proof of Thm. 5.64. Step 7: Thm. 5.64 (a). For $i = 1, 2$ let

$$S_i(V) := \sum_{X \in \mathcal{D}} 1_{X^\approx}(V)[E_i^a + E_i^h](X, V),$$

and let $c_S := 1.22(c_a + c_h)(x) \le 0.61$. We claim that $|S_i| \le c_S n$ and $|S_1 - S_2| \le c_S n \mathsf{D}$ at every V, and that $\Xi_i = 1$ where $D(V)$ is empty; together these give (a).

We have $\rho_i = e^{\mathsf{V}_i} e^{-x\mathrm{En}_\ell} e^{S_i}\Xi_i$, with one and the same En_ℓ for $i = 1, 2$, as ℓ and x are the same. The indicators $1_{X^\approx}$ depend on V and ε only, and so they are the same for $i = 1, 2$. Let $1_{X^\approx}(V) = 1$. Then V, restricted to the bonds of $X^\approx$, takes its values in G and is tame there, so that it belongs to $\mathcal{U}_X(x)$ (the definition "Domain" in item 2 of §5.5; here $x > e$). Accordingly, at such V,

$$|E_i^a + E_i^h|(X, V) \le (c_a + c_h)e^{-\kappa(\#X - 1)}$$

and

$$|E_1^a - E_2^a| + |E_1^h - E_2^h| \le \mathsf{D}(c_a + c_h)e^{-\kappa(\#X-1)};$$

for these two parts of the members, D^{in} and D_x are the same quotients. Each X contains a cube, so for the positive terms $e^{-\kappa(\#X-1)}$ we have $\sum_X e^{-\kappa(\#X-1)} \le \sum_\square \sum_{X \ni \square} e^{-\kappa(\#X-1)}$, and

$$\sum_{X \ni \square} e^{-\kappa(\#X-1)} \le \sum_{j \ge 1}(26e)^{j-1}e^{-6(j-1)} = 1/(1 - 26e^{-5}) \le 1.2124 \le 1.22;$$

summed over the n cubes, this gives the two bounds on S_i and on $S_1 - S_2$. Each S_i is real, because u_i is real.

If $D(V)$ is empty, no non-empty family is admitted, since each X_i of a family must hold a plaquette of $D(V)$; the empty family gives $\Xi_i = 1$. So there $\log \rho_1 - \log \rho_2 - (\mathsf{V}_1 - \mathsf{V}_2) = S_1 - S_2$.

Step 8: Thm. 5.64 (b), the bound on the supremum. We show that $|\Xi_i| \le e^{218n}$, that $|\Xi_1 - \Xi_2| \le \mathsf{D}ne^{218n}$, and that $\sup_V |\rho_1 - e^{\mathsf{V}_1 - \mathsf{V}_2}\rho_2| \le e^{\mathsf{V}_1} 2ne^{219n}\mathsf{D}$.

The sets of a family as in (cΞ) lie pairwise at a distance $> s(x) \ge 8$, and so they are disjoint. We fix an order of the cubes and send each set of the family to its first cube; different sets go to different first cubes. By the strong cap, $|w_i(X, V)| \le e^{-\kappa(|X|_c - 1)}$. Hence

$$|\Xi_i| \le \prod_\square (1 + \sum_{X \ni \square} e^{-\kappa(|X|_c - 1)}) \le \exp\{n \sum_{j \ge 1}(52e)^j e^{-6(j-1)}\},$$

and the sum equals $52e/(1-52e^{-5}) \le 217.588 \le 218$.

By telescoping,

$$|\prod_i w_1(X_i) - \prod_i w_2(X_i)| \le \sum_j |w_1 - w_2|(X_j) \prod_{i\neq j} e^{-\kappa(|X_i|_c - 1)} \le \mathsf{D} \cdot \#(\text{family}) \cdot \prod_i e^{-\kappa(|X_i|_c-1)},$$

where we use the bound $|w_1 - w_2|(X, V) \le \mathsf{D} e^{-\kappa(|X|_c - 1)}$ that the weak distance provides. A family has at most n sets. So $|\Xi_1 - \Xi_2| \le \mathsf{D}\, n\, e^{218n}$.

By Step 7 and by $|e^\alpha - e^\beta| \le e^{\max\{\alpha,\beta\}}|\alpha - \beta|$,

$$|e^{S_1}\Xi_1 - e^{S_2}\Xi_2| \le |e^{S_1} - e^{S_2}||\Xi_1| + e^{S_2}|\Xi_1 - \Xi_2| \le e^{0.61n} \cdot 0.61n\mathsf{D} \cdot e^{218n} + e^{0.61n} \cdot n\mathsf{D} \cdot e^{218n} = 1.61\, n\mathsf{D}\, e^{218.61n} \le 2n\, e^{219n}\mathsf{D}.$$

Moreover $e^{-x\mathrm{En}_\ell} \le 1$, because $S_\varepsilon \ge 0$, $m_y \in \{0, 1\}$, and the other terms are squares. Finally $\rho_1 - e^{\mathsf{V}_1 - \mathsf{V}_2}\rho_2 = e^{\mathsf{V}_1} e^{-x\mathrm{En}_\ell}(e^{S_1}\Xi_1 - e^{S_2}\Xi_2)$, and the bound on the supremum follows.

Step 9: Thm. 5.64 (c), a lower bound on $\int \rho$ over a neighbourhood of 1. We claim that

$$\int \rho_1 dV \ge e^{\mathsf{V}_1} e^{-0.61n} e^{-\frac{3}{2}\mathsf{N}_Z p_0^2 N^3} (\varepsilon^3/384)^{3N^3},$$

and that this gives the bound in L^1 with $C^{\mathrm{su}}_{\mathrm{vol}}$.

Let $\mathfrak{B} := \{V : \operatorname{ang} V(b) \le \varepsilon/4 \text{ at all } 3N^3 \text{ bonds}\}$.

(1) *Tame.* At $V \in \mathfrak{B}$, $\partial V(p)$ is the product of four $V(b)^{\pm 1}$, so ϕ_p is at most the sum of four angles, which is $\le \varepsilon$; here we use the clause $\operatorname{ang}(gh) \le \operatorname{ang} g + \operatorname{ang} h$, $\operatorname{ang} g^{-1} = \operatorname{ang} g$ of (RB). Hence $D(V)$ is empty, $\Xi_1 = 1$, and $S_1 \ge -0.61n$ by Step 7.

(2) *Mask and clip.* No plaquette has $\phi_p > \varepsilon$, and $\varepsilon \le 1$. So $m_y = 1$ for every y, $\zeta_p = F_p$ with $|F_p| = \phi_p \le \varepsilon$, and $S_\varepsilon(\phi_p) = \phi_p^2/2$.

(3) *Energy.* The paths $\Gamma(y, \pi(p))$ with $|y - p| \le \varrho(x)$ are defined, because $2(\varrho(x)+1) < 2M\, s(x) \le N$ (Step 0 (ii)). Here the rows are dressed, and the form differs from the scalar B. At G-valued fields, T is an isometry of $\mathfrak{g}$, and $|h \otimes Y| = |h||Y|$; therefore $|\ell_y^V F| \le \sum_p |\ell_y(p)|\phi_p$. By Schur's test,

$$\sum_y (\sum_p |\ell_y(p)|\phi_p)^2 \le \sum_y (\sum_p |\ell_y(p)|)(\sum_p |\ell_y(p)|\phi_p^2) \le \|\ell\|_{\mathrm{r}} \|\ell\|_{\mathrm{c}} \sum_p \phi_p^2 \le 3K_*^2 \mathsf{g}(\lambda_*)^2 \sum_p \phi_p^2;$$

here we use that a base site carries three plaquettes, that the sums over the torus are at most the sums over $\mathbb{Z}^3$, and the bounds on $\|\ell\|_{\mathrm{r}}, \|\ell\|_{\mathrm{c}}$ in (RB), which Def. 2.48 states in the sentence after its display. Hence

$$x\mathrm{En}_\ell \le \frac{x}{2}\mathsf{N}_Z \sum_p \phi_p^2 \le \frac{x}{2}\mathsf{N}_Z \varepsilon^2 \cdot 3N^3 = \frac{3}{2}\mathsf{N}_Z p_0^2 N^3.$$

(4) *Haar measure.* The group G is the unit sphere S^3, and $\operatorname{ang} g$ is the polar angle from 1. So the law of $\operatorname{ang} g$ is $(2/\pi)\sin^2\tau\, d\tau$, and $\mu\{\operatorname{ang} g \le t\} = (t - \sin t \cos t)/\pi$. From $\sin u \le u - u^3/6 + u^5/120$ for $u \ge 0$, taken at $u = 2t$, we get $t - \frac{1}{2}\sin 2t \ge \frac{2}{3}t^3 - \frac{2}{15}t^5 \ge \frac{8}{15}t^3$ on $(0, 1]$; and $8/(15\pi) \ge 0.1697 \ge \frac{1}{6}$. At $t = \varepsilon/4 \le 1$ this gives $\mu \ge \varepsilon^3/384$ for each bond; since dV is the product measure, $\int_{\mathfrak{B}} dV \ge (\varepsilon^3/384)^{3N^3}$.

(5) As $\rho_1 > 0$, we have $\int \rho_1 \ge \int_{\mathfrak{B}} \rho_1$, and by (1) to (4) the right side is at least the bound claimed.

(6) For positive f, g,

$$f/\int f - g/\int g = (f - g)/\int f + g(\int g - \int f)/(\int f \int g),$$

and hence $\|f/\int f - g/\int g\|_1 \le 2\|f - g\|_1/\int f$. We take $f = \rho_1$ and $g = e^{\mathsf{V}_1 - \mathsf{V}_2}\rho_2$, so that $g/\int g = \rho_2/\int \rho_2$. As dV, the product of the Haar probability measures, has total mass 1, Step 8 gives $\|f - g\|_1 \le e^{\mathsf{V}_1} 2n\, e^{219n}\mathsf{D}$. Since $2 \cdot 2n\, e^{219n} e^{0.61n} \le 4n\, e^{220n}$, division by the lower bound on $\int \rho_1 dV$ gives the bound with $C^{\mathrm{su}}_{\mathrm{vol}}$.

Remark 5.83. The bound is exponential in N^3, with the factor ε^{-9N^3}.

Step 10: Thm. 5.64 (d). Let $\mu_i = \rho_i / \int \rho_i$, $T = \{D(V) \text{ empty}\}$, $r_i = \mu_i(\text{complement of } T)$ and $\eta = 1.22n(c_a + c_h)\mathsf{D}$. We show that

$$\int |\mu_1 - \mu_2| dV \le 2 \max r_i + e^{2\eta} - 1.$$

If some $r_i = 1$, the right side is ≥ 2, and there is nothing to show. Otherwise $\nu_i := \mu_i 1_T/(1-r_i)$ are two probability densities on T. On T, by (a), $\rho_1/\rho_2 = e^{\mathsf{V}_1 - \mathsf{V}_2} e^{S_1 - S_2}$ with $|S_1 - S_2| \le \eta$. So

$$\nu_1/\nu_2 = e^{S_1-S_2} / \int e^{S_1-S_2} d\nu_2 \in [e^{-2\eta}, e^{2\eta}],$$

and $\int |\nu_1 - \nu_2| = \int |\nu_1/\nu_2 - 1| d\nu_2 \le e^{2\eta} - 1$. Let $r_1 \ge r_2$. Then

$$\int_T |\mu_1 - \mu_2| = \int |(1-r_1)(\nu_1 - \nu_2) - (r_1 - r_2)\nu_2| \le (e^{2\eta} - 1) + (r_1 - r_2).$$

Off T the integral is $\le r_1 + r_2$. The sum of the two parts is $e^{2\eta} - 1 + 2r_1$. □

Proof of Cor. 5.65. Step 11: the laws. We prove Cor. 5.65 (a) to (d).

Relative to Haar measure dV', the step is given by a probability kernel (the definition "The step" in item 2 of §5.5); hence $\int \rho_{k+1} dV' = \int \rho_k dU$. So $\rho_k^{[K]} / \int \rho_k^{[K]}$ is the density of the law of the field obtained by k steps from the Wilson measure at $\beta = L^K \mathsf{b}$ on $\mathbb{Z}^3/NL^K\mathbb{Z}^3$.

For (a), let $K' \ge K \ge s + K_0(L^s\mathsf{b})$. Then $K \ge K_0(\mathsf{b})$, and $k := K - s$ is a level $> k_0$ (Step 6 (a)). By Thm. 5.63 (ii), (iii), (iv) at this k, the densities $\rho_{K-s}^{[K]} = e^{\mathsf{V}_1} \rho[u_k; \ell_k]$ and $\rho_{K'-s}^{[K']} = e^{\mathsf{V}_2} \rho[\mathfrak{n}_k; \ell_k]$ are points of a common class at $x = x_k = L^s\mathsf{b}$ on $\mathbb{Z}^3/NL^s\mathbb{Z}^3$, with $\mathsf{D} \le C_{L^s\mathsf{b}} \theta^{K-s}$. The hypotheses of Thm. 5.64 (c) hold there: $x \ge x_\flat \ge e^2$; NL^s is even, a multiple of M, and $\ge 2M\, s(L^s\mathsf{b})$ (Step 2 (e)); $\kappa \ge 6$; $\varepsilon \le \frac{1}{40}$ by (X16); $c_a + c_h \le \frac{1}{32}$ by (X17); and ℓ_k has range $\le \varrho(x_k)$ and satisfies $|\ell_{k,y}(p)| \le K_* e^{-\lambda_* |y-p|}$ (Step 2 (c)). Both densities live on the grid with origin $o_{K-s}^{[K]} = o_{K'-s}^{[K']}$ (Step 2 (a)), and the class is that of this grid. Thm. 5.64 (c) therefore gives (a).

For (b): by (a), and as $\theta < 1$, the densities form a Cauchy sequence in $L^1(dV)$, which is complete. The limit is ≥ 0 and has integral 1. Since $\rho_k^{[K]} \in C_\flat^{\mathrm{su}}(x_k; \ell_k)$ for $1 \le k \le K$, by (R8) (Step 2 (b)), and here $k = K - s \ge 1$, the density $\rho_k^{[K]}$ is gauge invariant by Prop. 5.62; Haar measure is invariant under gauge transformations; and invariance passes to limits in L^1. So the limit density is gauge invariant, and with it $\mu_{[s]}^\infty$, as Cor. 5.65 (b) asserts.

For (c): for every K, $\mu_{[s-1]}^{[K]}$ is the image of $\mu_{[s]}^{[K]}$ under the kernel,

$$\prod_c k_{aL^s\mathsf{b}}(V'(c), \mathsf{M}(U)(c)) dV',$$

since the noise is $a\, x_{K-s} = aL^s\mathsf{b}$; this kernel is independent of K. A probability kernel does not increase $\|\cdot\|_{\mathrm{TV}}$. We let $K \to \infty$ on both sides, using (b) at s and at $s-1$.

For (d), we let $K' \to \infty$ in (a). □

Proof of Cor. 5.66. Step 12: the flux response. We claim that

$$|\mathfrak{x}(\rho_{K'}^{[K']}) - \mathfrak{x}(\rho_K^{[K]})| \le Q^{\mathrm{su}}(\mathsf{b})\, C_\mathsf{b}\, \theta^K$$

for $K' \ge K \ge K_0(\mathsf{b})$.

The integer m_0 exists: the interval $[\varepsilon(\mathsf{b})/2, \varepsilon(\mathsf{b})]$ has the length $\varepsilon(\mathsf{b})/2 \ge 2\pi/N^2$, and $2\pi/N^2$ is the spacing of the numbers $2\pi m_0/N^2$.

The field V_* is abelian, as all its values are of the form $e^{i\alpha\sigma_3}$; so $\partial V_*(p) = e^{iF\sigma_3}$, where F is the lattice curl of the exponents. At the plaquettes with normal e_3 and $z_1 < N-1$, the curl is $\varepsilon_0(z_1+1) - \varepsilon_0 z_1 = \varepsilon_0$. At those with $z_1 = N-1$ and $z_2 < N-1$, it is $-\varepsilon_0 N z_2 + \varepsilon_0 N(z_2+1) - \varepsilon_0(N-1) = \varepsilon_0$. At those with $z_1 = z_2 = N-1$, it is $-\varepsilon_0 N(N-1) - \varepsilon_0(N-1) = \varepsilon_0 - 2\pi m_0$. At

all other plaquettes, $F = 0$. So ϕ_p is ε_0 or 0, since $0 < \varepsilon_0 \le \varepsilon(\mathsf{b}) \le \frac{1}{40} < \pi$, and $D(V_*) = D(1) = \emptyset$ at $x = \mathsf{b}$. Here $\varepsilon_0 \le \varepsilon(\mathsf{b})$ holds by the hypothesis on m_0, and $\varepsilon(\mathsf{b}) \le \frac{1}{40}$ by (X16) at $x = \mathsf{b} \ge x_\flat$.

By Thm. 5.63 at $k = K$, both densities have the form $e^{\mathsf{V}_i}\rho[\cdot;\ell_K]$ and lie in a common class, with $\mathsf{D} \le C_\mathsf{b}\theta^K$. We apply Thm. 5.64 (a) at $V = 1$ and at $V = V_*$; its hypotheses at (b, N) are those of Step 11, taken at $s = 0$. The numbers $\mathsf{V}_1 - \mathsf{V}_2$ cancel in the difference, and

$$|[\log\rho_1(1) - \log\rho_1(V_*)] - [\log\rho_2(1) - \log\rho_2(V_*)]| \le 2.44(N/M)^3(c_a + c_h)(\mathsf{b})\mathsf{D}.$$

We multiply by $2/(N^3\varepsilon_0^2) \le 8/(N^3\varepsilon(\mathsf{b})^2) = 8\mathsf{b}/(N^3 p_0^2)$. Since $c_a(\mathsf{b})\mathsf{b}/p_0^2 = E_a p_0 \mathsf{b}^{1/2}$ and $8 \cdot 2.44 = 19.52$, the left side of the claim is at most $Q^{\mathrm{su}}(\mathsf{b})\mathsf{D} \le Q^{\mathrm{su}}(\mathsf{b})\, C_\mathsf{b}\, \theta^K$. □

Remark 5.84. The bound of the factor $2/(N^3\varepsilon_0^2)$ by $8\mathsf{b}/(N^3 p_0^2)$ rests on $\varepsilon_0 \ge \varepsilon(\mathsf{b})/2$: at $m_0 = 1$ and N large, the factor exceeds $8\mathsf{b}/(N^3 p_0^2)$.

6. Source estimates for one step

6.1. **Setup for the sourced step.** The following sections analyse the RG flow in the presence of sources. The transport of the quadratic source forms is linear, and their charges obey an exact block-averaging identity. The analytic part of the RG step follows this transport up to a controlled remainder. The shared notation and the additional bounds on the starting scale are collected below.

Definition 6.1 (Item (0): the shared letters and their senses)**.** Every letter of Sections 2.1 to 5.5 has here the sense it has there, with the exceptions that entries 1 to 13 of this item name. One of them is the following: the letter κ of those sections appears here as κ_1, as entry 1 explains.

Entry 1: the rates κ. We put $\kappa_1 := \kappa + 1$, and $\tilde\kappa$ ranges over the interval $[\kappa, \kappa_1]$, whose lower end is $\kappa \ge 6$. The rate of (O5) is κ_1: the statements of Sections 3 to 5.5 are used with their letter κ replaced by κ_1, and so are the entries (O5) to (O13) of the table. Further, $\kappa^\sharp = 10(\kappa_1 + 4)$.

In this section and in Sections 6.2 to 8.1 the symbol κ_1 always stands for $\kappa + 1$. The rate $\kappa_k = \kappa + \sum_{k \le j < k_d} \check\kappa_j$ of the level k is set in item 2, (o4), of §7.3.

In §4.3, κ_1 is not the rate $\kappa + 1$ but the number $\mathrm{py}/(240\pi^2 L^3 x)$, which enters the present section only through $x_\mathfrak{fl}$.

The factor $e^{\tilde\kappa+3}$ of §6.4 is taken at κ_1 where it enters a constant, and at κ where it appears on the right side of the inequality of a floor.

Entry 2: the letters n, q_N and ν. The letter n is the number of sources. We write $q_N := (N/M)^3$ for the number of cubes in a level of side N; in §5.5 this number is called n. The symbol ν_p denotes the local coupling of Thm. 7.1.

We write ν_a for the homogeneous factor of the a-channel, measured in the units u_3. By entry 9 one has $\nu_a = \vartheta_a L^{1/4} = 0.2431$, and by the condition (L3su) the factor in the bound of Thm. 6.7 (h3) is $\le \vartheta_a$.

Entry 3: floors and levels. The floors are x_3, x_4, x_9^U, $x_\mathfrak{fl}$ and x_s, besides $x_\flat$ and x_9 of Sections 2.1 to 5.5. The floor of Thm. 6.7 is x_9^U; we write x_3 for the floor of Thm. 6.33, and x_4 is that of Thm. 7.1. The floor of Thm. 8.2 is x_s, which comes last in the order of (O13) of item (1). The index of x_s is a mark and not the depth s of Thm. 8.2 (T7+) or of Thm. 7.19 (D3). The letters x_k, x_j, $x_{\check k}$, x_{k_0}, x_{k_d}, x_K, x_{K-1}, x_0 and x_1 are levels.

Entry 4: the symbols D, $\mathbb{D}$ and sep. Any family of kernels is written $D = (D_y)$, as in §5.4. The source forms are written $\mathbb{D}_k[f]$. The separations are written sep_k and sep_*. Item 2, (o4), of §7.3 sets $\mathrm{sep}^{[i]}$, $\mathrm{sep}_k = \mathrm{sep}^{[K-k]}$ and sep_*, and from them j^* and k_d.

Entry 5: the symbols $s_x[D]$, σ_D and σ_M. For a cube $\square$ we set

$$s_x[D](\{\square\}, \mathsf{W}) := \frac{x}{2} \sum_{y \in \square} \sum_{p,p'} D_y(p,p')\, (\mathrm{Ad}_{\mathsf{W}(\Gamma(y,\pi(p)))}\, F_p) \cdot (\mathrm{Ad}_{\mathsf{W}(\Gamma(y,\pi(p')))}\, F_{p'}),$$

where $F_p = -i \log \partial\mathsf{W}(p)$, and $s_x[D](X,\cdot) = 0$ whenever $\#X \ge 2$. One has $E^\Delta[D] = -s_x[D]$. We further set $\sigma_D := c_Q^{\mathrm{su}}\, p_0^2\, \|D\|_{(\lambda_\flat)}$ and $\sigma_M := 160(\kappa_1 + 1)c_R/M$.

Entry 6: amplitudes, and the letters α and μ. The amplitude at x is $\mathrm{amp}(x) := c_Q^{\mathrm{su}} K_5 x^{1/2}/(\mathsf{b} E_a p_0(x))$, and we put $\mathrm{amp}_W := 1/(c_\nu \mathsf{b}\, \varepsilon(\beta))$. The two satisfy the inequality $1.01\,\mathrm{amp}_W + 0.01/\mathsf{b} \le \mathrm{amp}(x_0)$, for the following reasons. Since $c_Q^{\mathrm{su}} = 306 M^3 \mathsf{g}_\flat^2$, we have $\mathrm{amp}(x_0) = 51 \mathsf{g}_\flat^2 K_5\, \mathrm{amp}_W$, where $\mathsf{g}_\flat, K_5 \ge 1$. Next, (X17) and (X14) of §5.5 give $x^{1/2}/(E_a p_0) \ge 32 p_0^2 \ge 32$. Consequently $\mathrm{amp}_W = 6M^3 x_0^{1/2}/(\mathsf{b} E_a p_0(x_0)) \ge 192 M^3/\mathsf{b} \ge 0.01/\mathsf{b}$. The left side of the inequality is therefore at most $2.01\,\mathrm{amp}_W \le 51\,\mathrm{amp}_W$, and since $\mathsf{g}_\flat, K_5 \ge 1$ the identity above shows that this is at most $\mathrm{amp}(x_0)$.

The letter α is a letter of the alphabet of Thm. 5.15, as in item 6 of §5.3. Taylor indices are written μ.

Entry 7: the function u_3, the letters u with an index, and $\check\omega_3$. We set $u_3(x) := C_3 x^{1/4}$. The families of the levels are written $u_k(z)$, $\bar u_k$, u_k^{pol}, $u_0(\nu)$ and $u_1(\nu)$.

The symbol $u_k(z)$ has two senses. In the first it is the honest family of Thm. 7.1 (h) on Pd_0; it has this sense in that clause, in the data of Prop. 7.12, and at every place where the family is called honest. In the second it stands for the representatives of item 4 of §7.3, that is, for u_k^{pol} when $k \le k_d$ and for their iterates under F beyond k_d; it has this sense in (J6) to (J8) of Prop. 7.12, in Thm. 7.19 (D1), (D3), (D4), (D6). The two share the base point $\bar u_k$ and the multi-affine jets, by (J1) and (J6); beyond that, the two families can differ.

We set $\check\omega_3(x) := C_3 E_a p_0(x) x^{-1/4}/c_Q^{\mathrm{su}}$. This closed form is defined at (O12b) and is free of b, and it satisfies the identity $\check\omega_3(x) = u_3(x) K_5/(\mathsf{b}\,\mathrm{amp}(x))$.

Entry 8: radii. The radii $r(x)$, $s(x)$ and $\varrho(x)$ are those of item 2 of §5.3. The polydiscs have the radii $\mathfrak{r}(x) := 3.5/(n\,\mathrm{amp}(Lx))$, $\mathfrak{r}_k := \mathfrak{r}(x_k)$ and $\mathfrak{r}_* := \mathfrak{r}_{k_d}$; here $\mathfrak{r}_*$ is set in item 2, (o5), of §7.3, and k_d in item 2, (o4), of that section. We write $\mathrm{Pd}(r) := \{|z_i| < r_i\}$, and when r is a single number every r_i equals r. The remainder of Prop. 7.12 (J3) is $\mathrm{rem}_{k,i}$: the a-part of the jet at the level k in the source i alone, minus $s_{x_k}[\mathbb{D}_k[f_i]/\mathsf{b}]$. We put $\check r(x) := r(x) + 3$.

Entry 9: the thetas. From Sections 2.1 to 5.5 we take $\vartheta = 0.4468$, $\theta = \vartheta^{4/9} \le 0.69903$, $\vartheta_w = 0.248$, ϑ_{10}, $\vartheta_1 = c_{\mathrm{fix}} L^{-1/2}$, and $\vartheta(x)$ of (u1). The thetas introduced in Sections 6.2 to 8.1 are $\vartheta_s := c_s L^{-15/32}$, $\vartheta_5 := \max\{\vartheta_s, \vartheta_1^{15/16}\}$, $\vartheta_a := 0.2431\, L^{-1/4}$ and $\vartheta[B] := \max\{16\vartheta_w, 1\}/(16 - B)$.

Entry 10: constants. The number C_5 is that of (O0a) (item 2 of §6.3), and $C_5^{\#} := L^{15/32} C_5 + K_5$. We set $C_3 := 1686\, e^{\kappa+4} L^{11/4} c_Q^{\mathrm{su}}/E_a$. The constants C_n and C_n' are those of Thm. 8.2. The numbers K_5 and C_{ch} belong to (O0a). We set $C_g := 263 L^3$; it is bounded above by $10^3 L^3$, and at the level 0 Thm. 7.1 (e) gives the sharper value $2.43 L^3$. The numbers k_1, k_2, k_3 are those of (O10a) (item 1 of §6.4).

Entry 11: further symbols, and symbols with a local sense. The threshold of Thm. 8.2 (T2) is $K_T(\mathsf{b}, n, d)$.

We set $\tau_7 = \max\{1, B_0 C_{\mathrm{vol}}\}$. This is a function of the datum of Cor. 7.28 (F0) and not a single number; in the clauses of Thm. 8.2 the datum is as follows:

(T1) $B_0 = 0.91$ and $C_{\mathrm{vol}}(N, \mathsf{b})$;
(T2) $B_0 = 1$ and $C_{\mathrm{vol}}(N, \mathsf{b})$;
(T7+) $B_0 = 1$ and $C_{\mathrm{vol}}(NL^s, L^s\mathsf{b})$.

We write $C_{\mathrm{vol}} := C_{\mathrm{vol}}^{\mathrm{su}}$. The radius of the balls of the supports is R_f. The exponent of the side is m_l, and $m = K' - K$.

The letter l is the side, and $T_l = \mathbb{R}^3/l\mathbb{Z}^3$ is the torus. The letter d is the separation. In entry 5 of this section, $\pi(p)$ is the base site of p.

We set $m_\Xi := \frac{1}{2} - d_0 - c_\Xi$. The set of the pairs p of Thm. 6.7 is $\mathbb{P}$, and P_5 is the number of Thm. 6.21 (S3). The number $\check\kappa_k$ is that of item 2, (o4), of §7.3. It satisfies $\check\kappa_k = \kappa_k - \kappa_{k+1}$ for $k < k_d$. The numbers of Thm. 7.19 (D2) are a_k^D, h_k^D and e_k, and $\mathrm{Tail}(J)$ and Ser_2 are the series of item 1.3 of §7.4. We set $\gamma_r := 1/4$.

The symbol V' is the number of (Fm2), and V' is the coarse field.

Entry 12: tags. The following clause names are used in the estimates for the sourced flow. The clauses (U), (c+), (f1) to (f6), (g1) to (g4), (h), (h1) to (h3), (h#) and (b#) are those of Thm. 6.7; (G1) to (G4) and (J0) to (J8) are those of §7.3; (B1) to (B8) are those of Lem. 7.10; (D0) to (D7) are those of Thm. 7.19; (a^C) to (d^C), (S) and (c1) to (c4) are those of Thm. 7.27; (F0) to (F5) are those of Cor. 7.28; (R0) to (R3) are those of Cor. 7.29; and (S0) to (S6) are those of Thm. 6.21.

The floors (Y.0) to (Y.9) are stated in item 5 of §5.3, and (Y.10) in item 0, (O13) (v), of §6.2. For the floor x_9^U, (Y.0) to (Y.9) are taken in the form in which item 0, (O13) (i) to (iv), of §6.2 lists them. Further, (X5) is stated in item 2 of §6.3, (Z.0) to (Z.4) in item 1 of §6.4, and (W1) to (W5) in item 1 of §7.1. A second pair (W2), (W3) occurs in item 1.3 of §7.4. Where (W2) and (W3) are cited by their tags alone, in item (1) below and in item 1 of §8.1, they are the two of item 1.3 of §7.4, not those of item 1 of §7.1.

Entry 13: the observable. The observable is $\mathcal{O}_K(f) := a_K^{-1} \sum_p f(x(p))\, a_p(U)$, in which $x(p)$ is the barycentre (Def. 1.1; item 2 of §8.1 writes it out among its data). Moreover $s_p = a_p$. ■

Definition 6.2 (Item (1): the numbers and floors introduced in Sections 6.2 to 8.1, each at its place)**.** Each of them is listed at its place in the one table, together with what it depends on.

(O0a) For §6.4: $\mathsf{g}_\flat = \mathsf{g}(\lambda_\flat)$, g_{1b} and g_b^-, which depend on λ_*. For §6.3, the pure numbers $C_{\mathrm{ch}} = 4e^{2\lambda_*}/\gamma_*$; $K_5 = 1.35\, C_{\mathrm{ch}}$, which equals $0.60\, e^{2\lambda_*} K_*^2 \le 0.61 K_*^2$; $\mathsf{g}' = \mathsf{g}(\lambda_\flat/2)$; $\mathsf{g}_1' = \mathsf{g}_1(\lambda_\flat/2)$; a_G; c_s; $t_5 = 18\, \mathsf{g}_\flat \mathsf{g}'$; a_H; c_I; $A_1 = \max\{1, 15C_1^2\}$; $I_5 = c_I K_5 A_1$; and $C_5 = 0.3(K_5 + 1) + 1.02\, I_5$.

(O2a) The number m_Ξ (§7.1), which depends on d_0 and c_Ξ.

(O4a) Two conditions on L.

(L3su) $L^{1/4} \ge 10^4 \mathsf{N}_{\mathrm{c}}^2$. It depends on N_{c} of (O3), which grows like $\log L$. The condition can be met: one has $\mathsf{N}_{\mathrm{c}} = 17(c_1^*(1 + \log L)\mathsf{N}_Z \mathsf{P}_\rho + 1)$, where c_1^*, N_Z and P_ρ do not depend on L, and the quotient $L^{1/4}/(1 + \log L)^2$ increases as soon as $\log L \ge 7$; hence (L3su) holds for every L from some value on, as do the three conditions of (O4), and it is required in addition to them. Squaring (L3su) gives $L^{1/2} \ge 10^8 \mathsf{N}_{\mathrm{c}}^4 \ge 1.2 \cdot 10^4 \mathsf{N}_{\mathrm{c}}^2$, so that (L3su) implies ($\mathrm{L2}^{\mathrm{su}}$).

(L5*) $c_s \le L^{43/160}$, which depends on (O0). It follows from $L \ge L_B \ge c_{\mathrm{fix}}^{10}$, because $\log_{10} c_s \le 145.34 < 204.89 \le \log_{10} c_{\mathrm{fix}}$ (Rem. 6.23). It therefore adds nothing to the conditions on L.

(O5) The rates κ, κ_1 and $\kappa^\sharp$.

(O7) The conditions (M2), ($\mathrm{M4}^\flat$) and (M5r), taken at κ_1 and $\kappa^\sharp$.

(O7a) The number $\sigma_M := 160(\kappa_1 + 1)c_R/M$, which depends on κ_1, c_R and M. By (M2), $\sigma_M \le 1/12$.

(O10a) The numbers k_1, k_2, k_3 (§6.4), which depend on $\mathsf{g}_\flat$, g_{1b}, g_b^- of (O0a) and on L, a_{c}, K_{c}, C_D and θ_D; and C_g, which depends on L.

(O11) The number c_h of item 3, (O11), of §2.1; $c_h/2$ is computed from it, and no condition arises.

(O12a) The number $c_\nu := E_a/(6M^3)$ (§7.1), and C_3, which depends on κ, L and E_a, and on c_Q^{su} of (O10), hence on M and on $\mathsf{g}_\flat$ of (O0a).

(O12b) Three functions of x. They are defined before x_s, they are given in closed form, and they are built from (O2), (O4), (O8), (O10), (O11), (O12) and (O12a) and from nothing else: $\eta_h(x) = e^{-(c_h/2)(p_0(x)^2 - p_0(x/L)^2)}$, $\eta_w(x) = e^{-c_\Xi p_0(x)^2}$ and $\check{\omega}_3(x) = C_3 E_a p_0(x) x^{-1/4}/c_Q^{\mathrm{su}}$. They are free of b, n and d. The floor x_s depends on them through (W2) and (W3), and through $\check{\omega}_3$ it depends on C_3.

(O13) The one-sided floors, which come in the following order. First, $x_9^U \ge x_{\mathrm{fl}}$, subject to (Y.0) to (Y.9), in the form in which item 0, (O13) (i) to (iv), of §6.2 lists them, and to (Y.10), which reads $1800\, L^{7/2} x^{-1/4} \le 1$. Each of (Y.0) to (Y.9) is taken at the pair of $\mathbb{P} = [\kappa, \kappa_1] \times \{c_h, c_h/2\}$ at which it is strongest.

Second, $x_3 \geq x_9^U$, subject to (Z.0) to (Z.4) of item 1 of §6.4, with (Z.4) taken at κ. Third, $x_4 \geq x_9^U$, subject to (W1) to (W5) of item 1 of §7.1, taken at κ_1.

Last, $x_s \geq e^2$. It is subject to all of (O13) of item 1 of §5.5, taken at κ_1; to $Lx_s \geq \max\{x_9^U, x_3, x_4\}$; and to the three conditions below, of which the last two are required for all $x \geq Lx_s$.

(X5) $1.01\,(4/3)\,t_5 C_{\mathrm{ch}} L\, x_s^{-16} \leq 1/100$;

(W2) $\eta_h + e^{1/2}\eta_w \leq 1/L$;

(W3) $\check{\omega}_3(x),\ \check{\omega}_3(x/L) \leq 10^{-3}$.

Four remarks follow, and none of them adds a floor. First, (Y.10) and (L3su) give $[2430\mathsf{N}_{\mathrm{c}}^2 L^{-1/2} + L^{-1}]L^{1/4} \leq 0.2431$. Second, there are the identities $\vartheta_a u_3(x)/u_3(x/L) = 0.2431 \leq 0.3452$ and $\check{\omega}_3(x/L)/\check{\omega}_3(x) = L^{1/4} p_0'/p_0 \leq L^{1/4}$. Third, $\vartheta_5 \leq L^{-1/5}$ is (L5*) together with $L \geq L_B$. Fourth, at the last level only $\tilde{c} = c_h$ is used, and what is required there is (X17) of §5.5.

(O14) The numbers on which no floor depends. They are listed in an order in which no entry depends on a later one. They are the entries of (O14) of item 1 of §5.5, taken at κ_1, namely c_4, Λ, K_0, k_0, ε_L, $C_{\mathfrak{b}}$ and C_{vol}; then ϑ_s; ϑ_5; $C_5^{\#}$; ϑ_a; $\vartheta[B]$; amp; u_3; the identity $\check{\omega}_3 = u_3 K_5/(\mathfrak{b}\,\mathrm{amp})$; $\mathfrak{r}$; the functions $\varepsilon_2^{\sharp}(x)$, $a^{\sharp}(x)$ and $\tau^{\sharp}(x)$ of x from item 0 of §6.2; $\check{r}$; sep_k; $\mathrm{sep}_* = 8\log(4nL^{1/2})$; j^*; k_d; $\check{\kappa}_k$; κ_k; $\mathfrak{r}_*$; $N_1^{[j]}$; $m^{[j]}$; K_T with (K1a) and (K1b), and $\Omega^{\#}$ (item 1.3 of §7.4); Tail; Ser_1; Ser_2; P_* (item 1 of §8.1); τ_7; C_n; and C_n'.

No circle. There is no circle in the table, and the reasons are these. Every entry depends only on places that come strictly earlier. The conditions (L3su) and (L5*) tie L to (O0) and (O3). No number of (O10a) depends on c_h, on E_a or on x. The number c_ν depends on (O7) and (O12), and C_3 depends on (O0a), (O4), (O5), (O7), (O10) and (O12).

Every floor has the same form, a family of one-sided inequalities in x. Their left sides are a polylogarithmic factor times a negative power of x, or $e^{-cp_0^2}$, where the exponent $\mathfrak{p}_0$ of p_0 is > 2; or else, for η_h in (W2), the left side is $e^{-c(p_0(x)^2 - p_0(x/L)^2)}$. In the last case the mean value theorem gives $p_0(x)^2 - p_0(x/L)^2 \geq 2\mathfrak{p}_0 A_0^2 \log L\,(\log(x/L))^{2\mathfrak{p}_0 - 1}$, and this lower bound tends to infinity because $2\mathfrak{p}_0 - 1 > 3$.

No floor depends on N, K, $\mathfrak{b}$, n, d, f, on a member or on a row. Finally, $x_{\mathfrak{b}}^{(1)}$ of Thm. 5.3 is free of c_h and E_a, as that theorem states. ∎

The linear source transport and its comparison across two cutoffs are proved in §6.3. §6.4 identifies this transport as the leading part of the derivative of the analytic RG step. The initial source family is given in §7.1. The holomorphic estimates of §7.2 allow the flow to be analysed through its squarefree Taylor coefficients.

Sections 7.3 and 7.4 estimate those coefficients for one and two runs, respectively. In the two-run comparison the free source forms are compared before the two families are evolved at common rows. §7.5 controls the final integral, and §8.1 proves the finite-volume cutoff limit. §8.2 gives the estimates for arbitrarily many separated sources using the number of sources visible to each polymer.

6.2. **Estimates at two decay rates.** In this section we extend Thm. 5.15, which is stated for one pair of rates, to every pair of rates. Since the formulas (Fm1)–(Fm4) contain neither the rate $\tilde{\kappa} \in [\kappa, \kappa_1]$, nor $\tilde{c} \in \{c_h, c_h/2\}$, nor E_a, nor A, they define one map F_ℓ, one number V' and one family v, and the clauses (a)–(e) hold at every pair, with the same numerals, above a single floor x_9^U. New clauses treat holomorphic families in several variables, the members on which an image member depends together with its reach, the number V' as a sum of local numbers with bounds in the cell sum, and the derivative of E'^{old} in a direction δ, split into the part of one letter and a rest, with bounds that in turn bound E'^{old}; two of the bounds, the first bound on the rest Rst in (h1)

and the bound on r in (g3), also hold when $x^{-1/4}$ is replaced by $\tau^\sharp(x)$, but with larger numerals: $291\,L^3\tau^\sharp(x)$ in the place of $217\,L^3x^{-1/4}$ and $146\,L^3\tau^\sharp(x)$ in the place of $109\,L^3x^{-1/4}$.

Remark 6.3 (What the theorem is, and where its signs come from). Thm. 6.7 extends Thm. 5.15 from one pair of rates to every pair: it keeps the clauses (a)–(e) of that theorem and adds to them the clauses (U), (b#), (c+), (f), (g), (h) and (h#), and it is accompanied by Lem. 6.9 and Lem. 6.8.

From items 0 to 6 of §5.3 we take the signs, the lattices, the cubes and the cells $\square$, the families $\mathcal{D}$ and $\mathcal{P}$, the radii, the domains $\mathcal{U}_X(x)$ and $U^\tau(B;x')$, the members and the distances, the data (SD), the inputs (Γ2)–(Γ7) and the tools, the alphabet $\mathfrak{A}$ together with J_α, I_α, the relation $\sim$ and the relation "compatible", $\mathrm{co}(C)$, $n(C)$, Φ and $\hat\Phi$, the formulas (Fm1)–(Fm4), and the floors (Y.0)–(Y.9). From item 1 of §2.1 we take $Y_0(X)$, the word "small", $\mathrm{cl}(X)$ and the fitted grids. From Lems. 4.49 and 4.50 in Step 8 of the proof of Thm. 4.39 we take the polymers, the relation "touch", the clusters, which are finite sequences, and φ^T.

The letters and the numbers are those of items (0) and (1) of §6.1. Throughout this section, "Step n" denotes Step n of the proof of Thm. 5.15, while the steps of the proof given here are named S1 to S13.

Remark 6.4 (Letters with more than one sense). The letter t is the number $x^{-1/4}$, but as a member of a cell datum (J,t), and so in $\hat\lambda_{(\mathsf{J},t)}$ and in a zone letter $(\mathsf{J},t,\mathfrak{X},\mathfrak{Y})$, it stands for the set of the tested cells; and the families of numbers of item 1 are written v, $v(Y;u)$ and $v^0(Y)$, while in Step S7, in $v(2r_1+1)^3$, we have $v=(LM)^3$. Outside the proof of Lem. 6.13, the bare A denotes the number of caps, whereas in that proof the bare A is the family $(A_c)_c$ of the A_c, indexed by the bonds c of B_ω and with values in $\mathfrak{g}_\mathbb{C}$, and $\mathrm{Ad}_u A := (\mathrm{Ad}_u A_c)_c$. The tags (a), (b) and (h) name clauses of Thm. 5.15 and of Thm. 6.7, and also kinds of letters, as in $(\mathrm{a})_X$, $(\mathrm{h})_X$ and (b). The two senses of the bare letter G are the group and the matrix of the law, those of the bare letter P are a coarse plaquette and the function $P(Y,\cdot)$ of (Fm1), and the context shows which sense is meant.

Definition 6.5 (Item 0: the letters and their place in the table)**.** The places (O0)–(O4), (O6), (O8)–(O10) and (O12) of the table are as in item 1 of §5.3. The remaining places, and the conditions added to them, read as follows.

(O4a) The place (O4a) consists of the conditions of (O4) in item 1 of §5.3, together with one more condition,

$$L^{1/4} \geq 10^4 \mathsf{N}_\mathrm{c}^2. \tag{L3su}$$

By item 1, (O3), of §5.3, $\mathsf{N}_\mathrm{c} = 17(c_1^*\mathsf{l}_L\mathsf{N}_Z\mathsf{P}_\rho+1)$ with $\mathsf{l}_L = 1+\log L$, and the numbers c_1^*, N_Z, P_ρ are computed from (O0)–(O3) and are independent of L. For $\log L \geq 7$ we have

$$\frac{d}{d\log L}\log(L^{1/4}/\mathsf{N}_\mathrm{c}^2) \geq \frac14 - \frac{2}{\mathsf{l}_L} \geq 0,$$

and $L^{1/4}/\mathsf{N}_\mathrm{c}^2 \to \infty$. Hence (L3su) holds from some L on. Moreover (L3su) implies $(\mathrm{L2}^{\mathrm{su}})$, since it gives $L^{1/2} \geq 10^8\mathsf{N}_\mathrm{c}^4 \geq 1.2\cdot 10^4\mathsf{N}_\mathrm{c}^2$. As (L3su) depends on L and on (O0)–(O3) alone, we may impose it on L at the same time as (O4), before any of the later letters is chosen.

(O5) The number $\kappa \geq 6$ is the lower end of the range $[\kappa,\kappa_1]$ of the rate $\tilde\kappa$. We set $\kappa_1 := \kappa+1$: the table, and the chain of statements of Sections 3 to 5.5, are taken with the letter κ of §5.3 equal to κ_1. We set $\kappa^\sharp := 10(\kappa_1+4)$.

(O7) The side M is chosen by (M2), $(\mathrm{M4}^\flat)$ and (M5r) at κ_1 and $\kappa^\sharp$. Hence every number of (O6)–(O12), the floor x_fl and every radius of item 2 of §5.3 are defined once, and so are the objects G, C, a_0, $\mathsf{D}_k(Y)$, $S_k(Y)$, P, $T_\square$, $V_\square$, $\hat\lambda_{(\mathsf{J},t)}$, $b_\square$.

(O11) The number c_h is as in item 1 of §5.3, and $c_h/2$ is computed from it. As (O11) bounds c_h from above, $c_h/2$ obeys it just as c_h does.

(O13) We let x_9^U be 1 plus the infimum of the numbers $y \geq x_\mathrm{fl}$ such that, for all $x \geq y$, the following conditions hold.

(i) (Y.0), (Y.2), (Y.7), (Y.8), the second inequality of (Y.5), the second and the third inequality of (Y.6) and the second inequality of (Y.9), all as in item 5 of §5.3, with $\kappa^\sharp$ as in (O5) above.
(ii) (Y.1), (Y.3), (Y.4), together with the υ_f and υ_* that occur in it, and the first inequality of (Y.6), with κ_1 in the place of the letter κ of §5.3 and $e^{-(c_h/2)p_0^2}$ in the place of $c_h(x)$.
(iii) The first inequality of (Y.5), with $c_h/2$ in the place of c_h.
(iv) The first inequality of (Y.9), with the lower end κ in the place of the letter κ of §5.3.
(v) The inequality

$$1800\, L^{7/2} x^{-1/4} \le 1. \tag{Y.10}$$

Each inequality of (i)–(v) holds for all large x, by Step S1, and it depends on x and on (O0)–(O12) alone.

The set of these y is an upper half line, and Step S1 shows that it is not empty. It need not contain its infimum, because the radii are step functions of x and some of them occur on the side of an inequality that is asserted to be the larger, as r' does on the right side of (Y.7). The number x_9^U lies in the set and depends on what the set depends on; this is the same form of a floor as in Thm. 5.3 (item 3 of §5.1).

We have $x_9^U \ge x_9$, where $x_9 = 1 + \inf S_9$ is the floor of item 1, (O13), of §5.3, taken at κ_1: by Step S1 the conditions (i)–(iv) imply (Y.0)–(Y.9) at the pair (κ_1, c_h), so that the set of these y lies in S_9.

In the table of item (1) of §6.1, the floor x_s is taken with $Lx_s \ge x_9^U$.

Pairs. We set $\mathbb{P} := [\kappa, \kappa_1] \times \{c_h, c_h/2\}$, and we write its elements as $p = (\tilde\kappa, \tilde c)$. For $p \in \mathbb{P}$, the class $M_A(x;p)$, the norm $\|\cdot\|_{x;p}$ and the distances $\mathsf{D}_{x;p}$, $\mathsf{D}^{\mathrm{in}}_{x;p}$ are the class $M_A^{\mathrm{su}}(x)$, the norm $\|\cdot\|_x$ and the distances D_x, $\mathsf{D}^{\mathrm{in}}_x$ of item 3 of §5.3, defined with $\tilde\kappa$ in the place of the letter κ of that section and with $e^{-\tilde c p_0^2}$ in the place of $c_h(x)$; at x' they are defined with $\tilde\kappa$ and with $e^{-\tilde c p_0'^2}$. Then $M_A(x;\kappa_1,c_h) \subset M_A(x;p)$ for every p, and every $M_A(x;p)$ is contained in $M_A(x;\kappa,c_h/2)$.

Numbers.

(O7a) $\sigma_M := 160(\kappa_1+1)c_R/M$, which is $\le 1/12$ by (M2).
(O10a) $C_g := 263L^3$.
(O14) No floor depends on (O14). It sets $\vartheta_a := 0.2431\, L^{-1/4}$ and three functions of x, like t and ε_2 of item 5 of §5.3 (here s_0, ϱ_* and s' depend on x):

$$\varepsilon_2^\sharp = \varepsilon_2^\sharp(x) := [6(2(s_0+\varrho_*)+1)^3]^{-1},$$
$$a^\sharp(x) := 50(2s'+1)^3 e^{(\kappa_1+1)s'}\,[\,5e^{\kappa_1+3}L^3\cdot 16c_a(x) + 2e^{\kappa_1+3}K_v^\flat p_0^3 x^{-1/2} + x^{-20}\,],$$
$$\tau^\sharp(x) := a^\sharp(x)/\varepsilon_2^\sharp.$$

As in item 5 of §5.3, $t = x^{-1/4}$ and $\varepsilon_2 = x^{-1/8}$. ∎

Definition 6.6 (Item 1: the objects)**.** *Cell sum.* For a family $f = (f(Y,\cdot))$ indexed by the connected sets Y of cells we set

$$|f|_{\mathrm{cs}} := \sup_{\square} \sum_{Y \ni \square} e^{\tilde\kappa(\#Y-1)} \sup|f(Y,\cdot)|,$$

where the inner supremum is taken over $\mathcal{U}_Y(x')$. For a family of numbers, the modulus of each number takes the place of the inner supremum. The cell sum depends on $\tilde\kappa$.

We let $T^a(x)$ be the set of the families $\delta = (\delta(X,\cdot))$, $X \in \mathcal{D}$, of bounded functions which have the clauses of (ca) that carry no size, namely that $\delta(X,\cdot)$ is holomorphic on $\mathcal{U}_X(x)$, invariant under G-valued gauge transformations and 0 at pure gauges. For $\delta \in T^a(x)$, $B = B_p(\delta)$ is the least number with $\sup_{\mathcal{U}_X(x)}|\delta(X,\cdot)| \le B\, c_a(x) e^{-\tilde\kappa(\#X-1)}$ for all X; here B may be any number ≥ 0. Further, $A_a = A_a(u) := \sup_X \sup_{\mathcal{U}_X(x)} |E^a(X,\cdot)|\, e^{\tilde\kappa(\#X-1)}/c_a(x)$ is the a-part of $\|u\|_{x;p}$, so that $A_a \le A$ on $M_A(x;p)$.

We set

$$v(Y;u) := \sum_{\mathrm{co}(C)=Y,\ \mathrm{old}} \Phi(C)(1), \qquad v^0(Y) := \sum_{\mathrm{co}(C)=Y,\ \mathrm{new}} \Phi(C)(1) + \sum_{\mathrm{co}(C)=Y,\ \mathrm{constant}} \Phi(C),$$

$V_0^+ := \mathsf{V}_0 + \sum_Y v^0(Y)$ and $A(u) := E'^{\mathrm{old}}[u]$, and

$$G_X^\delta[u](V') := E_\xi[\chi_X(\xi)\, e^{E^a(X,U)}\, \delta(X,U)], \qquad U = \mathsf{U}(G\xi - a_0; V').$$

For an analytic cluster C and $V' \in \mathcal{U}_{\mathrm{co}(C)}(x')$, we let $D\Phi(C)[\delta](V')$ be the derivative $\frac{d}{dz}\big|_{z=0}$ of $\Phi(C)(V')$ evaluated at $u + z(\delta,0,0)$; by Step S9, $\Phi(C)(V')$ so evaluated is an entire function of z. We set

$$DA(u)[\delta](Y,V') := \sum_{\mathrm{co}(C)=Y,\ \mathrm{old}} [D\Phi(C)[\delta](V') - D\Phi(C)[\delta](1)], \qquad Dv(Y;u)[\delta] := \sum_{\mathrm{co}(C)=Y,\ \mathrm{old}} D\Phi(C)[\delta](1).$$

For a cluster $C = (\mathfrak{B}_1, \dots, \mathfrak{B}_n)$, a finite sequence of polymers as in item 3, we set $\mathrm{mult}_\alpha(C) := \#\{i : \mathfrak{B}_i \text{ holds the letter } \alpha\}$. The root of an analytic letter α is the first cell of J_α. ■

Theorem 6.7 (Item 2: the step at every pair of rates)**.** In the setting (SD) of item 2 of §5.3, take the data of a step at any $x \geq x_9^U$, the empty rows included wherever they obey (SD), and let $p \in \mathbb{P}$, $1 \leq A \leq 16$ and $u \in M_A(x;p)$. Then (U), (b#), (c+), (f), (g), (h) and (h#) hold.

(U) The formulas (Fm1)–(Fm4), with $s'-1$ in (Fm4) as in item 6 of §5.3, depend neither on $\tilde\kappa$, nor on $\tilde c$, nor on E_a, nor on A. They define one map F_ℓ, one number V' and one family v on $M_{16}(x;\kappa,c_h/2)$, and all their series converge absolutely. The clauses (a)–(e) of Thm. 5.15 hold at p, with $\tilde\kappa$ and $\tilde c$ in the classes, in the weights $e^{\tilde\kappa(\#Y-1)}$, in the caps and in the distances, and with the numerals 0.35, 0.248, 0.4 of that theorem.

(b#)
$$|E'^{\mathrm{old}}[u]|_{\mathrm{cs}} \leq [2430\mathsf{N}_c^2 L^{-1/2} + L^{-1}]\, A\, c_a(x') \leq \vartheta_a\, A\, c_a(x').$$

(c+) Let Ω be open in $\mathbb{C}^n$, and let $z \mapsto u_z \in M_{16}(x;p)$ be such that every value $E_z^a(X,V)$ with $V \in \mathcal{U}_X(x)$, every value $E_z^h(X,V)$ and every value $w_z(X,V)$ is holomorphic on Ω. Then V'_z, every $v(Y;u_z)$ and every value of $F_\ell(u_z)$ are holomorphic on Ω. Here, as in (f6) and in Step S5, the n of $\mathbb{C}^n$ is any integer ≥ 1, and not the number of sources in entry 2 of item (0) of §6.1.

(f) *Locality in the members.*

(f1) The objects E'^{new}, v^0 and V_0^+ depend on no member. Both $E'^{\mathrm{old}}[u](Y,\cdot)$ and $v(Y;u)$ are functions of $\{E^a(X,\cdot) : X \in \mathcal{D},\ Y_0(X) \subset Y\}$ alone; hence neither E'^a nor V' depends on E^h or on w.

(f2) The member $E'^h(Y,\cdot)$ is a function of $\{E^a(X), E^h(X) : X \in \mathcal{D},\ Y_0(X) \subset Y\}$ and $\{w(X) : X \in \mathcal{P},\ Y_0(X) \subset Y\}$ alone.

(f3) The member $w'(\mathsf{Y},\cdot)$ is a function of $\{E^a(X), E^h(X) : X \in \mathcal{D},\ Y_0(X) \subset \mathsf{Y}\}$ and $\{w(X) : X \in \mathcal{P},\ Y_0(X) \subset \mathsf{Y}\}$ alone.

(f4) In $F_\ell(u)$ and in V', the member $w(X,V)$ enters only at fields V such that X holds a plaquette of $D(V)$, and $E^h(X,V)$ enters only at fields V that are tame on $X^{\approx}$. Moreover, each of V', E'^h, w' and $E'^a(Y,V')$, the last for G-valued V' tame on $Y^{\approx}$, depends on $E^a(X,V)$ only at G-valued fields V that are tame on $X^{\approx}$; at the remaining V' of $\mathcal{U}_Y(x')$, the value $E'^a(Y,V')$ draws on $E^a(X,\cdot)$ on $\mathcal{U}_X(x)$.

(f5) *Reach.* Let S be a set of fine cubes, $S' := \mathrm{cl}(S)$, $\hat\rho \geq 0$ an integer and $\hat\rho' := \lceil \hat\rho/L \rceil + 1$. If $u_z(X)$, that is, each of the three kinds of member at X, is independent of z for every X with $X^{+\hat\rho}$ off S, then $F_\ell(u_z)(Y)$ and $v(Y;u_z)$ are independent of z for every Y with $Y^{+\hat\rho'}$ off S'.

(f6) Let $z \mapsto u_z$ be a family as in (c+), and let $S_1, \dots, S_n$ be sets such that $u_z(X)$ is independent of z_i whenever $X^{+\hat\rho}$ is off S_i. Then, for every set I of indices, the mixed

derivative $\prod_{i\in I}\partial_{z_i}$ of $F_\ell(u_z)(Y)$ and of $v(Y;u_z)$ vanishes, unless $Y^{+\hat\rho'}$ meets S_i' for every $i\in I$.

(g) *The number.*

(g1) $\mathsf{V}'(u)=V_0^+ +\sum_Y v(Y;u)$, where Y runs over the connected sets of cells and the sum converges absolutely; and $v(Y;u)=0$ if $E^a=0$.

(g2) $|v(\cdot;u)|_{\rm cs}\le C_g\,A\,c_a(x)$; besides, $|v^0|_{\rm cs}\le 55\,[2K_v^\flat p_0^3 x^{-1/2}+x^{-20}+u_b]$.

(g3) For $\delta\in T^a(x)$,

$$Dv(Y;u)[\delta]=\sum_{X\colon Y_0(X)=Y} G_X^\delta[u](1)+r(Y),\qquad |r|_{\rm cs}\le 109\,L^3x^{-1/4}\,B\,c_a(x).$$

If $A_a<16$, and so if $A<16$, then $Dv(Y;u)[\delta]=\frac{d}{dz}|_0\,v(Y;u+z(\delta,0,0))$, and, since the torus is finite, $\frac{d}{dz}|_0\,\mathsf{V}'(u+z(\delta,0,0))=\sum_Y Dv(Y;u)[\delta]$. If $u_\tau:=u_0+\tau(\delta,0,0)\in M_{16}(x;p)$ for $0\le\tau\le 1$, and $A_a(u_\tau)<16$ for every $\tau<1$, then $v(Y;u_1)-v(Y;u_0)=\int_0^1 Dv(Y;u_\tau)[\delta]\,d\tau$.

(g4) For $u_1,u_2\in M_2(x;p)$,

$$|v(\cdot;u_1)-v(\cdot;u_2)|_{\rm cs}\le 421\,L^3c_a(x)\,\|E_1^a-E_2^a\|,$$

where $\|E_1^a-E_2^a\|=B_p(E_1^a-E_2^a)$, with B_p as in item 1.

(h) *The one-letter formula.* Let $\delta\in T^a(x)$. The series $DA(u)[\delta](Y,V')$ converges absolutely and uniformly on $\mathcal{U}_Y(x')$; it is linear in δ; it has, at x', the clauses of (ca) without a size; of u it depends on E^a alone; along a family as in (c+) it is holomorphic in z; and if $A_a<16$, and so if $A<16$, it equals $\frac{d}{dz}|_0\,E'^{\rm old}[u+z(\delta,0,0)](Y,V')$. Moreover

$$DA(u)[\delta](Y,V')=\sum_{X\colon Y_0(X)=Y}[G_X^\delta[u](V')-G_X^\delta[u](1)]+{\rm Rst}(Y,V'),$$

where

$${\rm Rst}(Y,V'):=\sum_{{\rm co}(C)=Y,\ {\rm old},\ n(C)\ge 2}[D\Phi(C)[\delta](V')-D\Phi(C)[\delta](1)],$$

and the following bounds hold.

(h1) $|{\rm Rst}|_{\rm cs}\le 217\,L^3x^{-1/4}B\,c_a(x)\le 366\,L^{5/2}x^{-1/4}B\,c_a(x')$.

(h2) For every cell $\square_0$,

$$\sum_{X\text{ small, first cube under }\square_0}\ \sup_{\mathcal{U}_{\{\square_0\}}(x')}|G_X^\delta[u](V')-G_X^\delta[u](1)|\le 2.4248\cdot 361\,\mathsf{N}_{\rm c}^2e^{Ac_a(x)}B\,c_a(x)\le 2430\,\mathsf{N}_{\rm c}^2L^{-1/2}B\,c_a(x'),$$

and for every cell $\square$,

$$\sum_{X\text{ not small, }Y_0(X)\ni\square}e^{\tilde\kappa(\#Y_0(X)-1)}\sup|G_X^\delta[u](V')-G_X^\delta[u](1)|\le e^{-L/2}B\,c_a(x).$$

(h3) $|DA(u)[\delta]|_{\rm cs}\le[2430\mathsf{N}_{\rm c}^2L^{-1/2}+L^{-1}]\,B\,c_a(x')\le\vartheta_a\,B\,c_a(x')$, where the last inequality rests on (L3su); under (L2$^{\rm su}$) alone the bracket is ≤ 0.2026.

(h#) *The bounds at $\tau^\sharp(x)$.* We have

$$\tau^\sharp(x)\le 0.12\,x^{-1/4}$$

and

$$\tau^\sharp(x)\le 300\,(2(s_0+\varrho_*)+1)^3(2s'+1)^3e^{9\kappa_1+11}[81L^3E_a+2K_v^\flat]\,p_0^3\,x^{-1/2+\sigma_M};$$

moreover (h1) holds with $291\,L^3\tau^\sharp(x)$ in the place of $217\,L^3x^{-1/4}$, and (g3) with $146\,L^3\tau^\sharp(x)$ in the place of $109\,L^3x^{-1/4}$.

■□

Lemma 6.8 (Rooted cluster sum with multiplicities and a power of τ)**.** Assume the hypotheses (SD), $x \geq x_9^U$, $p = (\tilde{\kappa}, \tilde{c}) \in \mathbb{P}$ and $1 \leq A \leq 16$ of Thm. 6.7. Let e_α be the sizes of Step 6 at $(\tilde{\kappa}, A)$, let α be an analytic letter with root $\square_0$, let $\square$ be a cell and let $\tau \in [\tau^\sharp(x), 1]$. Then

$$\sum_{C \text{ analytic, } \mathrm{co}(C) \ni \square} \mathrm{mult}_\alpha(C)\, \hat{\Phi}(C)\, e^{\tilde{\kappa}(\#\,\mathrm{co}(C)-1)}\, \tau^{-(n(C)-1)} \leq e^{-\mathrm{dist}(\square_0,\square)-1}\, e_\alpha\, e^{2\varepsilon_2^\sharp \#I_\alpha + (\tilde{\kappa}+1)\#J_\alpha - \tilde{\kappa}}.$$

■

Lemma 6.9 (Item 3: the cluster bound with powers of τ and with multiplicities)**.** Let a finite alphabet be given, and to each letter α a non-empty finite set I_α of cells. Let $\mathsf{r} \geq 0$, and let $\alpha \sim \beta$ if and only if $\mathrm{dist}(I_\alpha, I_\beta) \leq \mathsf{r}$. Let "compatible" be a symmetric relation on pairs of distinct letters under which incompatible letters are $\sim$. Let $e_\alpha \geq 0$, $d_\alpha \geq 0$, $0 < \tau \leq 1$ and $\epsilon > 0$ be numbers such that for every cell $\square$

$$\sum_{\beta:\ \mathrm{dist}(I_\beta,\square) \leq \mathsf{r}} \tau^{-1} e_\beta\, e^{3\epsilon \#I_\beta + d_\beta} \leq \epsilon. \tag{S_τ}$$

The polymers are the non-empty sets of pairwise compatible letters that are $\sim$-connected, and two polymers touch when some letter of the first is $\sim$ to some letter of the second. The clusters, as in Step 8 of the proof of Thm. 4.39 (Lems. 4.49 and 4.50), are the finite sequences $C = (\mathfrak{B}_1, \dots, \mathfrak{B}_n)$, $n \geq 1$, of polymers such that the graph of touching on $\{1, \dots, n\}$ is connected, and

$$\varphi^T(C) := \frac{1}{n!} \sum_g (-1)^{\#g},$$

where g runs over the connected spanning subgraphs of that graph and $\#g$ is the number of the edges of g. We set $\hat{\Phi}(C) := |\varphi^T(C)| \prod e_\alpha$, we let $n(C)$ be the number of the letters of the polymers of C, and we set $d(C) := \sum d_\alpha$; here products and sums run over these letters, and both in them and in $n(C)$ every letter is counted with multiplicity.

Then for every letter α

$$\sum_C \mathrm{mult}_\alpha(C)\, \hat{\Phi}(C)\, e^{d(C)}\, \tau^{-n(C)} \leq \tau^{-1} e_\alpha\, e^{2\epsilon \#I_\alpha + d_\alpha};$$

consequently, for every $m \geq 1$, the part of $\sum_C \mathrm{mult}_\alpha(C) \hat{\Phi}(C) e^{d(C)}$ with $n(C) \geq m$ is $\leq \tau^{m-1} e_\alpha\, e^{2\epsilon \#I_\alpha + d_\alpha}$.

Suppose further that a complex number $K(\mathfrak{B})$ is attached to each set $\mathfrak{B}$ of pairwise compatible letters, with $K(\emptyset) = 1$, with $K(\mathfrak{B} \cup \mathfrak{B}') = K(\mathfrak{B}) K(\mathfrak{B}')$ whenever $\mathfrak{B}$ has no letter that is $\sim$ to a letter of $\mathfrak{B}'$, and with $|K(\mathfrak{B})| \leq \prod_{\mathfrak{B}} e_\alpha$. Then

$$\sum_{\mathfrak{B}} K(\mathfrak{B}) = \exp \sum_C \Phi(C), \qquad \Phi(C) = \varphi^T(C) \prod K(\mathfrak{B}_i),$$

and the series converge absolutely. ■

Remark 6.10 (Lem. 6.9 beside Lem. 4.50). Taken at $(\tau, \epsilon) = (t, \varepsilon_2)$ and with $m = 1, 2$, the claims of Lem. 6.9 are the two claims of Lem. 4.50 on which the proof of Thm. 4.39, in its Step 9 on convergence, and the proof of Thm. 5.15, in Step 8, rely. The definitions in Lem. 6.9 are those of Step 8 of the proof of Thm. 4.39 (Lems. 4.49 and 4.50). The paragraph "Clusters" in item 6 of §5.3 takes from Lem. 4.50 the clusters as finite sequences, $K(\mathfrak{B})$, Φ, $\hat{\Phi}$, $n(C)$ and $d(C)$, and with them the polymers as sets of letters, the relation "touch" and φ^T; these are the ones defined in Lem. 6.9, for the alphabet $\mathfrak{A}$ with $\mathsf{r} = r_1$, and the polymers of the kinds p and p′ of the zone letters are not among them. The condition (S) of Step 7 (ii) is (S_t) at $\epsilon = \varepsilon_2$.

Compared with Lem. 4.50 (1), (2), Lem. 6.9 brings two additions: the factor $\mathrm{mult}_\alpha(C)$ takes the place of $1\{C \ni \alpha\}$, and the power $\tau^{-n(C)}$ is added; indeed, if Lem. 4.50 is taken at the sizes e_α/τ, with $t = 1$ and $\varepsilon_2 = \epsilon$, its condition (S) becomes (S_τ). The last claim of Lem. 6.9 is Lem. 4.50 (4).

The display (An0) of Lem. 4.48, Lem. 4.50 (3), (4) and Lem. 4.51 are used in Step 7 (i) and in Step 10 (8a) and (8d), at p as well; Lem. 6.9 covers only Lem. 4.50 (1), (2) and (4).

Remark 6.11 (Item 4: what is used). The constants and the floors of Thm. 6.7 are free of A, u, z, δ, N, K, b and f, and of ℓ they involve only the pure numbers b_*, K_*, λ_* and the range. (maj) Each majorant behind (b#), (g2), (g3), (h1), (h2), (h3) is a function of A and B only: these bounds stay true with the supremum over $u \in M_A(x;p)$, and over all δ with $B_p(\delta) \le B$, placed inside the cell sum.

Thm. 6.7, like Thm. 5.15, is proved in the setting (Γ2)–(Γ7), with the tools of item 4 of §5.3. Its clause (U) is not obtained by applying the statement of Thm. 5.15 at each pair, since in item 1, (O5), of §5.3 the number $\kappa^\sharp$ is tied to the single letter κ of that section, since (O11) has a single c_h, and since that statement speaks of one rate at a time. Clause (U) is obtained instead in part (3) of Step S2, by repeating the proof of Thm. 5.15 step by step at p: the letter κ of §5.3 and c_h occur only in Steps 0, 1, 6 to 10 and 12 of that proof.

In the proof of Thm. 6.7 we also use, in the manner of clauses of a statement, individual steps from the proof of Thm. 5.15, namely Step 0 (iii) and (iv), Step 2 (i) and (ii), Step 3 (V) and (VI), Step 6 (iii) and (iv), Step 7 (ii), Step 8 with (Clu), with "The clauses without a size" and with the setting of (8.2), Step 10 (8d), and Step 11. The proof of Thm. 6.7 thus uses §5.3 together with the proof of Thm. 5.15, and not the statement of that theorem alone; Step S13 lists the statements that this section and the following ones use in addition to those of §5.3 and of Step 8 of the proof of Thm. 4.39.

Proof of Thm. 6.7. Step S1: the floor x_9^U, its worst corners, its existence and its place in the table. We show two things. First, each inequality of (Y.0)–(Y.9) holds, at every $p \in \mathbb{P}$ and every $x \ge x_9^U$, when it is taken with $\tilde\kappa$ in the place of the letter κ of §5.3 and with $e^{-\tilde c p_0^2}$ in the place of $c_h(x)$, while $\kappa^\sharp$ is kept. Second, x_9^U exists and is determined by (O0)–(O12) alone; each of the conditions (i)–(v) of (O13) that define it holds for all large x; and x_9^U is free of A, u, z and N, and of all of ℓ but the pure numbers b_*, K_*, λ_* and the range.

Where the two letters occur. Apart from its place inside $\kappa^\sharp$, the letter κ of §5.3 occurs in (Y.3), in $e^{(\kappa+1)s'}$ and $e^{\kappa+3}$; in (Y.4), in $e^{\kappa+5}$, $e^{(\kappa+4)s'}$ and $e^{(\kappa+1)s'}$, and in $(\kappa+4)(1+2s')$ in the exponent of υ_f, hence also in $\upsilon_* = (\max\{u_t^{1/2}, \upsilon_f^{1/4}\})^{1/L^3}$; in the first inequality of (Y.6), in $e^{\kappa+3}$; and in the first inequality of (Y.9). The number c_h occurs in (Y.1), (Y.3) and (Y.4) through $c_h(x)$, and in the first inequality of (Y.5). Neither letter occurs in any other inequality, and u_t, u_b, u_1, $K_1^\sharp$, $K_v^\flat$, δ_3 are numbers of the table, defined once.

Directions. In (Y.3), (Y.4) and the first inequality of (Y.6), each term of the left side is non-decreasing in the letter κ of §5.3, and so are υ_f and hence υ_*; the worst end is κ_1. The left side of the first inequality of (Y.9) is

$$2.001\, K_1^\sharp (2r'+1)^3 \exp\{1 - 26.7 r' - \kappa(r'-1)\},$$

where κ is the letter of §5.3; as $r' \ge 1$, it is non-increasing in that letter, and the worst end is the lower end κ. The factors $e^{-\tilde c p_0^2}$ and $e^{-\tilde c(p_0^2 - p_0'^2)}$ grow as $\tilde c$ falls, so the worst end is $c_h/2$. Since every inequality holds at every p once it holds at the worst ends, and since (i)–(iv) of (O13) take the inequalities there, these give every inequality at every p.

Existence. The argument on orders of Step 0 (i) applies word for word and shows that each inequality of (i)–(v) holds for all large x; at the new places its inputs are the following. By (M2) at κ_1,

$$(\kappa_1+4)s' \le (\kappa_1+4)(160 c_R \log x/M + 8) \le \frac{1}{12}\log x + 8(\kappa_1+4).$$

Hence the exponent of x in the terms of (Y.3) with c_a and with $K_v^\flat$ is $\frac14 + \frac{1}{12} - \frac12 = -\frac16 < -\frac18$, and in the third term of (Y.4) it is $\frac{1}{12} - \frac12 = -\frac{5}{12} < -\frac14$. The factor $e^{-(c_h/2)p_0^2} = \exp\{-(c_h/2)A_0^2 (\log x)^{2\mathfrak{p}_0}\}$

still decays faster than every power of x. As in Step 0 (i), $\log(1/\upsilon_f) \geq \frac{3}{8}(\frac{1}{2} - d_0)A_0^2(\log x')^{2\mathfrak{p}_0} - O((\log x)^3)$, because $(\kappa_1+4)(1+2s') = O(\log x)$; so $\log(1/\upsilon_*) \geq cL^{-3}(\log x)^{2\mathfrak{p}_0-3}$ for a number $c>0$, where $2\mathfrak{p}_0 - 3 > 1$. Further, $(c_h/2)(p_0^2 - p_0'^2) \geq c_h\mathfrak{p}_0 A_0^2 \log L\,(\log x')^{2\mathfrak{p}_0-1} \to \infty$. In the first inequality of (Y.9), $r' \to \infty$, and in (Y.10), $x^{-1/4} \to 0$.

The inequality (Y.10) is an additional condition: the first inequality of (Y.6) bounds $1800L^{7/2}x^{-1/4}$ only by $1.8\cdot 10^{-3}e^{-(\kappa_1+3)}L$, a bound that is larger than 1 once $L > 556\,e^{\kappa_1+3}$.

Place. The conditions depend on x and (O0)–(O12); the floor x_{fl} is free of c_h, E_a, ℓ and N (item G of §4.3, on floors, and Step 13 of the proof of Thm. 4.19); and, besides the data, only the floors x_3, x_4 and x_s of item (1) of §6.1 depend on x_9^U.

The rest of Step 0. (Geo), (Rad), $T > 1$, $r_\Xi \geq \frac{3}{4}(\frac{1}{2} - d_0 - c_\Xi)$ and the clauses on N contain neither letter. Applied at κ_1, (Y.4) gives $u_t, \upsilon_f, \upsilon_* < 1$ and $2\upsilon_f^{1/4} \leq 1$, and these hold for the υ_f at every $\tilde\kappa \leq \kappa_1$.

Step S2: one map, and Thm. 5.15 at every pair. We show that (Fm1)–(Fm4) are free of $\tilde\kappa$, $\tilde c$, E_a and A; that, for every $p \in \mathbb{P}$, the clauses (a)–(e) of Thm. 5.15 hold on $M_A(x;p)$ with the numerals 0.35, 0.248, 0.4; and that the classes are nested as stated in item 0.

(1) *The formulas.* We examine the formulas in item 6 of §5.3. The functions of the letters (a), (h), (v), (t), (b) and of the zone letters are made of u, χ, U, $A_\square$, $V_\square$, T_Y, $b_\square$, $\hat\lambda$, 1_{ind} and n_j. The sets J_α and I_α, the relations $\sim$, "adjacent", "compatible" and "actual", the words "on", "off", "short" and "long", and $\mathrm{co}(C)$, "small" and Y_0 are determined by r_1, s_0, s', r', $s(x)$, L and the fixed orders; and $\Phi = \varphi^T \prod K$. The letter κ of §5.3, as well as c_h, E_a and A, occurs only in d_α, e_α, $\hat\Phi$, $\hat w$, υ_f and υ_*, which are majorants and appear in no formula. Hence (Fm1)–(Fm4) are the same expressions at every p.

(2) *The inputs.* Thm. 5.15 is derived from the clauses that item 4 of §5.3 writes out. These contain the letter κ of that section only through $\kappa^\sharp$, namely in (4a), (4b) and in e_Y, and they contain no c_h. The objects are fixed at κ_1, and $\kappa^\sharp = 10(\kappa_1+4) \geq 10(\tilde\kappa+4)$. It follows from the clauses of item 4 of §5.3 that $\mathsf{D}_k(Y)$, $S_k(Y)$ and $\hat\lambda$ constitute a single family of objects at fixed $(M,\kappa^\sharp)$. Since $\kappa^\sharp$ is determined by κ_1, it does not depend on p.

(3) *The proof of Thm. 5.15 at p.* We repeat that proof at p, with $\tilde\kappa$ in the place of the letter κ of §5.3 in the classes (ca), (ch), (cΞ), in the weights, in $d_\alpha = (\tilde\kappa+1)(\#J_\alpha + s')$ and in e°, $\hat w$, υ_f, υ_*, with $e^{-\tilde c p_0^2}$ in the place of $c_h(x)$, and keeping $\kappa^\sharp$. We treat its steps in order.

Steps 1, 2 and 3 involve neither the letter κ of §5.3 nor c_h, except that Step 1 uses them in the statement that $\rho[u;\ell]$ is bounded, and there alone. Steps 4 and 5 involve no member, and $\upsilon_f, \upsilon_* < 1$ by Step S1. In Step 6, (Y.1) enters at $c_h/2$, and the bound $\sum_{Y\ni\square} e_Y^{1/2} \leq x^{-20}\sum_n (26e)^{n-1}e^{-50n}$ rests on $\kappa^\sharp/2 \geq 50$, which holds, since $\kappa^\sharp/2 \geq 55$.

In Step 7, and throughout the discussion of that step, κ denotes the letter of §5.3 as a running variable. We have $52e/(1-52e\cdot e^{-6}) \leq 217.59$ and $(1-26e\cdot e^{-6})^{-1} \leq 1.2124$, and the left sides are the values at $\kappa = 6$ of expressions that decrease in κ. The exponents per cube, $4.952 - \kappa + (\kappa+5)432/L \leq -1.04$ and $4.2581 - \kappa + (\kappa+3)432/L \leq -1.7$, decrease in κ for $L > 432$.

For X not small, the prefactors also contain κ, namely $e^{9\kappa+40}$ in Step 7 (i) and $e^{8(\kappa+3)}$ in Step 7 (ii), and κ is chosen after L; so we keep the whole exponent. Measured against $e^{\kappa+5}$ for the kinds p and p′, and against $e^{\kappa+3}$ for (a) and (h), the term with $n > L/2-2$ has an exponent at most

$$n(4.952 + 2160/L) + 35 - \kappa[n(1-432/L) - 8] \quad \text{for the kind p,}$$

$$4.2581(n-1) + 35 + 2160n/L - \kappa[n(1-432/L) - 8] \quad \text{for the kind p′,}$$

$$4.2581(n-1) + 21 + 1296n/L - \kappa[n(1-432/L) - 8] \quad \text{for (a) and (h).}$$

When $n > L/2 - 2$ and $L > 10^{13}$, the bracket is positive; so each exponent decreases in κ, and its largest value, at $\kappa = 6$, is $\le -1.04n + 83$. With the geometric factor, which is ≤ 1.55, each sum is therefore $\le e^{-0.52L+90} \le e^{-L/4}$, uniformly in $\tilde\kappa \ge 6$. In Step 7, the bound for the letters of kind (t) uses $\kappa^\sharp/2 = 5\kappa_1 + 20 \ge 5\tilde\kappa + 20$, and Step 7 (i) is (Y.4) and Step 7 (ii) is (Y.3), both at p.

In Step 8, (Clu) and (R) are pure algebra in the letter κ of §5.3; $\tilde{\mathsf{D}}$ rests on $\kappa^\sharp - \tilde\kappa \ge 9\tilde\kappa + 40$; (8.3) rests on $\kappa - 4.2581 \ge 0.29\kappa$ for the running letter κ, which is true for $\kappa \ge 6$, since at 6 it reads $1.7419 \ge 1.74$ and the slope of the left side is 1 against 0.29; (8.4) rests on the first inequality of (Y.6); (8.1) rests on (O12) and (Y.6); and the numerals $0.34438 \le 0.35$ and $0.2457 + 0.001 + 0.001 \le 0.248$ are unchanged. In Step 9, the number c_h is used at two places: in the first inequality of (Y.5), which enters at $\tilde c$, and in the inequality $\frac{1}{2} e^{-\delta_3 p_0^2/2} \le e^{-\tilde c p_0'^2}/2$, which requires $\tilde c \le \delta_3/2$ and only becomes easier as $\tilde c$ falls.

In Step 10, the exponent per cell of a long Y is $4.2581 - \kappa^\sharp + \tilde\kappa + 1 \le -(9\tilde\kappa + 34.74)$; with $s' = 8r'$ and $\#Y \ge r' + 1$, the sum is $\le (2r'+1)^3 \#\mathrm{mk}_j' \cdot 1.001 K_1^\sharp e^{\tilde\kappa+1} e^{-(\tilde\kappa + 26.74)r'}$, which the first inequality of (Y.9) at $\tilde\kappa$ bounds; the second inequality of (Y.4) enters at $\tilde\kappa$; and $|\tilde{\mathsf{D}}(Y)| \le 1/2$ by the second inequality of (Y.9). Steps 11 and 12 use no floor, and the Cauchy bounds from (b) and (c) at p give $16 \cdot 0.248/10 = 0.3968$, 0.1 and 0.1. Every floor that is used is an inequality of (Y.0)–(Y.9) at p, and so it holds by Step S1.

(4) *Inclusions.* For $n \ge 0$ we have $e^{-\kappa_1 n} \le e^{-\tilde\kappa n} \le e^{-\kappa n}$, and $e^{-c_h p_0^2} \le e^{-\tilde c p_0^2} \le e^{-(c_h/2) p_0^2}$; hence $M_A(x; \kappa_1, c_h) \subset M_A(x; p) \subset M_A(x; \kappa, c_h/2)$.

(5) *One map.* Part (3), taken at $(\kappa, c_h/2)$, shows that all series converge absolutely on $M_{16}(x; \kappa, c_h/2)$; by part (1), restricting them to $M_{16}(x; p)$ gives the map, the number and the family v at p.

(6) *Rows.* The rows enter the proof only through (Γ2)–(Γ7), through En_ℓ and through the upper bound $\|\ell^{V*} \ell^V\|_\infty \le (K_* \mathsf{g}_*)(3K_* \mathsf{g}_*)$ of Step 6. No quantity is divided by a norm of the rows, and the argument does not depend on any row being non-zero.

Step S3: item (VII) *of the alphabet, and* (f1) *to* (f4). We prove the following further item of the alphabet and deduce (f1)–(f4) from it. (VII) The member u at X enters a letter α only if $Y_0(X) \subset J_\alpha$: the letter $(\mathrm{a})_X$ involves $E^a(X, \cdot)$; the letter $(\mathrm{h})_X$ involves $E^a(X, \cdot)$ and $E^h(X, \cdot)$; a zone letter $(\mathsf{J}, t, \mathfrak{X}, \mathfrak{Y})$ involves $w(X, \cdot)$ with $X \in \mathfrak{X}$, where $1_{\mathrm{ind},\zeta} = 1$, and $1E(X)$ with $X \in \mathfrak{Y}$; while (v), (t), (b), Λ_ζ, n_j, $\tilde{\mathsf{D}}$ and P depend on no member.

Item (VII) is read off item 6 of §5.3, from λ_X^a, λ_X^h and $\lambda_\zeta = \Lambda_\zeta \prod e^{n_j} \prod_{\mathfrak{X}} w(X, U) \prod_{\mathfrak{Y}} (e^{1E(X)(U)} - 1)$: for (a) and (h), $J = Y_0(X)$, and J_ζ contains the sets $Y_0(X)$ of its polymers.

(f1) The set $\mathrm{co}(C)$ contains every J_α of C, and an analytic cluster contains letters (a), (v) and (t) only. Therefore an old C with $\mathrm{co}(C) = Y$ depends on the $E^a(X, \cdot)$ with $Y_0(X) \subset Y$ and on no other part of u; a new or a constant C depends on no member; and the same holds for $\tilde{\mathsf{D}}$, P and V_0. Together with (Fm1) and (Fm2), this gives (f1).

(f2) In (Fm3), a hard C with $\mathrm{co}(C) = Y$ is made of letters of $\mathfrak{A}$ with $J_\alpha \subset Y$, and so (VII) gives (f2).

(f3) In (Fm4), the union of the cores, which are the J_α of the letters of the rough polymers, the sets $\mathrm{co}(C)$ and the long Y, equals Y. Moreover $\Phi^\natural(C)$ is one of $\Phi(C)(V')$, $\Phi(C)(1)$ and 0, and which one it is depends on V' alone, as do the marks and the attributes "actual", "on", "off" and "long". The argument rests neither on the linking distance s' nor on the clause with $s' - 1$, and the mean over e depends on no member.

(f4) The member w occurs in zone letters alone, with the factor $1_{\mathrm{ind},\zeta}$, which vanishes unless each X of $\mathfrak{X}$ holds a plaquette of $D(U)$. The member E^h occurs in (h) and in $1E(X)$, each time with the factor $1_{X^\approx}(U)$. The member E^a carries the factor $1_{X^\approx}(U)$ wherever it occurs in (h), in $1E(X)$, or in the formula of (a) at a pair (V', e); it is these formulas that appear in $K(\gamma)$ of (Fm4) and in (Fm3). In $\Phi^\natural(C)$ for analytic C, and in V', the field either equals 1

or is tame on $\mathrm{co}(C)^{\approx}$; in $E'^{a}(Y,V')$ at a G-valued V' tame on $Y^{\approx}$ the field is tame as well; and there the two formulas of (a) coincide by Step 3 (VI), so that $1_{X^{\approx}}(U)=1$ on $\chi_X=1$. At the other fields of $\mathcal{U}_Y(x')$ we use the holomorphic formula $\chi_X(e^{E^a(X,U)}-1)$, with U on $X^{\approx}$ in $\mathcal{U}_X(x)$ by Step 2 (ii); and at those of them that are G-valued, the remark that follows shows that U is G-valued and tame on $X^{\approx}$.

Remark 6.12 (The G-valued fields V' of $\mathcal{U}_Y(x')$). In the proof of Step 2 (i), the bound "$\phi'_P\le\varepsilon'$" enters only through $|\partial V'(P)-1|\le 2\varepsilon'$, and this is the hypothesis of (Γ3) at $\hat\varepsilon=\varepsilon'$. A G-valued V' of $\mathcal{U}_Y(x')$ satisfies (e1) and (e2), and by (u3) it has $|\partial V'(P)-1|<2\varepsilon'$ at the plaquettes of $Y^{\approx}$; moreover, for $Y_0(X)\subset Y$, the boxes of radius R_3 about $y(p)$, with p a plaquette of $X^{\approx}$, lie in $Y^{\approx}$, by the second inequality of (Y.7), as in Step 2 (ii). For such a V' we use Lem. 5.17 in its first case, with the parameter $\tau=1$ of that lemma: $|V'(c)|=1$, $|\partial V'(P)-1|<2\varepsilon'$ by (u3), and $X'_P=-i\log\partial V'(P)$ by the convention of item 3 of §5.3. Together with (D3), this gives 0.072ε, as in the proof of Step 2 (i). On $\chi_X=1$ the numbers of that proof are therefore unchanged, namely $(10^{-4}+0.12)\varepsilon$, 0.6154ε and 0.072ε, and $2\sin(\phi_p(U)/2)\le 0.8075\varepsilon\le 2\sin(\varepsilon/2)$. So, at every G-valued V' of $\mathcal{U}_Y(x')$, the field U is G-valued and tame on $X^{\approx}$.

Step S4: the reach, (f5) *and* (f6).

We begin with (f5). Let $Y^{+\hat\rho'}$ be disjoint from S', and let $Y_0(X)\subset Y$, where $X\in\mathcal{D}$ or $X\in\mathcal{P}$. By item 1 of §2.1, $\mathrm{cl}(X)\subset Y_0(X)^{+1}$, and therefore $\mathrm{cl}(X)\subset Y^{+1}$.

Suppose that some cube q of S is within distance $\hat\rho$ of a cube q' of X. In a given coordinate let i and j be the indices of q and q', so that $|i-j|\le\hat\rho$. Because the grids fit (item 1 of §2.1), a cell is made of exactly L^3 fine cubes; so the cells of q,q' carry the indices $\lfloor i/L\rfloor,\lfloor j/L\rfloor$ in this coordinate, and $|\lfloor i/L\rfloor-\lfloor j/L\rfloor|\le\lceil|i-j|/L\rceil\le\lceil\hat\rho/L\rceil$. On the torus we choose lifts of q,q' with $|i-j|\le\hat\rho$; as the grids fit, the index of a cell is $\lfloor\cdot/L\rfloor$ of the lift, taken modulo the side.

The cell of q' lies in $\mathrm{cl}(X)$, hence in Y^{+1}; consequently the cell of q lies in $Y^{+1+\lceil\hat\rho/L\rceil}=Y^{+\hat\rho'}$. It lies in S' as well, which is impossible, since $Y^{+\hat\rho'}$ does not meet S'. Therefore $X^{+\hat\rho}$ does not meet S, and $u_z(X)$ is constant. By (f1) to (f3), $F_\ell(u_z)(Y)$ and $v(Y;u_z)$ are functions of these values $u_z(X)$ alone, which proves (f5).

For (f6) suppose that $Y^{+\hat\rho'}$ does not meet S'_i for some $i\in I$; we fix this i and all other variables. Then (f5) for S_i shows that the value does not depend on z_i, and by (c+) it is holomorphic. So its derivative ∂_{z_i} is 0, and with it the mixed derivative.

Step S5: holomorphic families in several variables, (c+). The argument is that of Step 11, with "holomorphic on Ω, open in $\mathbb{C}^n$" in place of "affine on an open set of $\mathbb{C}$". As in (c+), n is any integer ≥ 1.

(1) Fix (ξ,V',e). Each letter is then a product of values $w_z(X,U)$, of exponentials of values $E^a_z(X,U)$ and $E^h_z(X,U)$, and of factors in which no member occurs; hence it is holomorphic on Ω. Indeed, at $V'\in\mathcal{U}_Y(x')$ and where $\chi_X=1$, the field U on $X^{\approx}$ lies in $\mathcal{U}_X(x)$, and there the value is holomorphic in z by hypothesis.

(2) None of the following depends on z: the alphabet, the adjacency, the compatibility, the cores, χ, U, the indicators, the notions actual, on, off and long, and the alternative of Lem. 5.19, which is decided by the supports of the Λ_ζ.

(3) For fixed z each letter is a Borel function of ξ, and for fixed ξ it is continuous in z; being a Carathéodory function, it is jointly measurable. The majorant of Step 6, at $A=16$ and at p, is integrable and independent of z. By the theorems of Fubini and Morera, $K_z(\mathfrak{B})$ is holomorphic in each variable z_i separately; it is bounded by $\prod e_\alpha$; and Osgood's lemma says that a bounded function which is holomorphic in each variable separately is holomorphic.

(4) $\Phi_z(C)$ is a polynomial in the K_z, and $|\Phi_z|\le\hat\Phi$ at $A=16$, a bound independent of z. Hence the following converge uniformly on Ω: the series (Fm1) to (Fm3); the series of $v(Y;\cdot)$; and the sum in (Fm4), since $e^{-\Phi^\natural_z}-1$ is holomorphic and the majorant (8d) of Step 10 applies. By Weierstrass' theorem the sums are holomorphic.

(5) $w'_z(\Upsilon, V')$ is the mean over e of a bounded, jointly measurable family that is holomorphic in z; by the argument of (3), namely the theorems of Fubini and Morera and Osgood's lemma, it is holomorphic.

Step S6: polymers and clusters, the proof of Lem. 6.9.

(1) *The gas of polymers.* Every set $\mathfrak{V}$ of pairwise compatible letters has a unique splitting into $\sim$-components. These are polymers, no two of which touch, and K factorises over them. Conversely, let polymers be given, no two of which touch. A letter of one of them and a letter of another are then not $\sim$; so they are distinct, as $\sim$ is reflexive, and compatible, as incompatible letters are $\sim$. Hence the union is pairwise compatible, and its components are the given polymers. It follows that $\sum_{\mathfrak{V}} K(\mathfrak{V})$ is the partition function of the polymers with the weights K and the hard core "touch", where a polymer touches itself.

(2) *Animals.* We claim: if $v_\beta, a_\beta \geq 0$ satisfy $\sum_{\beta\sim\alpha} v_\beta e^{a_\beta} \leq a_\alpha$ for all α, then

$$\sum_{\mathfrak{V}\ni\alpha,\ \sim\text{-connected}} \prod_{\mathfrak{V}} v \leq v_\alpha e^{a_\alpha}$$

. Let $S_N(\alpha)$ be the part of this sum over the sets with at most N letters; we argue by induction on N, starting from $S_1(\alpha) = v_\alpha$. A connected set $\mathfrak{V} \ni \alpha$ with at most $N+1$ letters consists of $\{\alpha\}$ together with the components $\mathfrak{V}_1, \dots, \mathfrak{V}_k$ of $\mathfrak{V} \setminus \{\alpha\}$; these are distinct connected sets of at most N letters, and each contains a letter $\beta_i \sim \alpha$. Every unordered family of this kind arises from at least $k!$ tuples $((\beta_i, \mathfrak{V}_i))_i$. With the induction hypothesis and the assumption we obtain

$$S_{N+1}(\alpha) \leq v_\alpha \sum_k \frac{1}{k!} \Big(\sum_{\beta\sim\alpha} S_N(\beta)\Big)^k \leq v_\alpha \exp\Big\{\sum_{\beta\sim\alpha} v_\beta e^{a_\beta}\Big\} \leq v_\alpha e^{a_\alpha}.$$

(3) We put $a_\beta := \epsilon \# I_\beta$ and $v_\beta := \tau^{-1} e_\beta e^{d_\beta + \epsilon \# I_\beta}$. Summing (S_τ) over the cells of I_α we get $\sum_{\beta\sim\alpha} v_\beta e^{2a_\beta} \leq \epsilon \# I_\alpha = a_\alpha$. A polymer is a $\sim$-connected set, so (2) yields $\sum_{\text{polymers } \mathfrak{V}\ni\beta} \prod_{\mathfrak{V}} v \leq v_\beta e^{a_\beta}$.

(4) *The criterion of Kotecký and Preiss.* We use it with one marked polymer, that is, Lem. 4.49 of Step 8 of the proof of Thm. 4.39 in the form that its proof yields. The claim is: if weights $\hat{z}(\gamma) \geq 0$ and numbers $a(\gamma) \geq 0$ satisfy $\sum_{\gamma' \text{ touching } \gamma} \hat{z}(\gamma') e^{a(\gamma')} \leq a(\gamma)$ for every γ, then for every γ_1

$$1 + \sum_{k\geq 2} k \sum_{\gamma_2,\dots,\gamma_k} |\varphi^T(\gamma_1, \dots, \gamma_k)| \prod_{j\geq 2} \hat{z}(\gamma_j) \leq e^{a(\gamma_1)};$$

here φ^T is the function of Lem. 4.49, $\frac{1}{k!}\sum_g (-1)^{\#g}$. Moreover, for all complex weights ϖ with $|\varpi| \leq \hat{z}$, the partition function equals the exponential of the absolutely convergent sum of $\varphi^T \prod \varpi$.

For the proof we use Penrose's bound from the proof of Lem. 4.49: $k!\,|\varphi^T(\gamma_1, \dots, \gamma_k)| \leq N_{\mathrm{tr}}(\gamma_1, \dots, \gamma_k)$, the number of trees on $\{1, \dots, k\}$ in which every edge joins two touching polymers. It gives

$$k \sum_{\gamma_2,\dots,\gamma_k} |\varphi^T(\gamma_1, \dots, \gamma_k)| \prod_{j\geq 2} \hat{z}(\gamma_j) \leq \frac{1}{(k-1)!} \sum_{\gamma_2,\dots,\gamma_k} N_{\mathrm{tr}}(\gamma_1, \dots, \gamma_k) \prod_{j\geq 2} \hat{z}(\gamma_j).$$

Summed over $k \geq 1$, where the term for $k = 1$ is 1, the right sides give the quantity $T(\gamma_1)$ of that proof at $W = \hat{z}$. The recursion used there shows $T(\gamma) \leq e^{a(\gamma)}$: with $T_0 = 1$,

$$T_{i+1}(\gamma) = \exp\Big\{\sum_{\gamma' \text{ touching } \gamma} \hat{z}(\gamma') T_i(\gamma')\Big\} \leq \exp\Big\{\sum_{\gamma' \text{ touching } \gamma} \hat{z}(\gamma') e^{a(\gamma')}\Big\} \leq e^{a(\gamma)}.$$

The formula for the partition function is the last claim of Lem. 4.49, applied with the weights $W = \varpi$.

Let $\mathrm{mult}_\gamma(C) := \#\{j : \gamma_j = \gamma\}$. Since φ^T and $\prod \hat z$ are symmetric in the sequence, all k places contribute the same sum, and so

$$\sum_C \mathrm{mult}_\gamma(C)|\varphi^T(C)|\prod \hat z = \sum_k \sum_{(\gamma_1,\dots,\gamma_k)} \#\{j:\gamma_j=\gamma\}|\varphi^T|\prod \hat z = \hat z(\gamma)[1+\sum_{k\ge 2} k \sum_{\gamma_2,\dots,\gamma_k} |\varphi^T(\gamma,\gamma_2,\dots)|\prod_{j\ge 2}\hat z(\gamma_j)] \le \hat z(\gamma)e^{a(\gamma)}.$$

We choose $\hat z(\mathfrak B) := \prod_{\mathfrak B} \tau^{-1} e_\beta e^{d_\beta}$ and $a(\mathfrak B) := \epsilon \sum_{\mathfrak B} \#I_\beta$, so that $\hat z(\mathfrak B)e^{a(\mathfrak B)} = \prod_{\mathfrak B} v$. With this choice

$$\sum_{\mathfrak B' \text{ touching } \mathfrak B} \hat z(\mathfrak B')e^{a(\mathfrak B')} \le \sum_{\alpha\in\mathfrak B}\sum_{\beta\sim\alpha}\sum_{\mathfrak B'\ni\beta}\prod_{\mathfrak B'} v \le \sum_{\alpha\in\mathfrak B}\sum_{\beta\sim\alpha} v_\beta e^{a_\beta} \le \sum_{\alpha\in\mathfrak B} a_\alpha = a(\mathfrak B).$$

(5) We have $\mathrm{mult}_\alpha(C) = \sum_{\mathfrak B\ni\alpha}\mathrm{mult}_{\mathfrak B}(C)$ and $\hat\Phi(C)e^{d(C)}\tau^{-n(C)} = |\varphi^T(C)|\prod \hat z$. Hence the left side of the bound claimed in Lem. 6.9 is at most $\sum_{\mathfrak B\ni\alpha}\hat z(\mathfrak B)e^{a(\mathfrak B)} = \sum_{\mathfrak B\ni\alpha}\prod_{\mathfrak B} v \le v_\alpha e^{a_\alpha} = \tau^{-1}e_\alpha e^{d_\alpha+2\epsilon\#I_\alpha}$. When $n(C)\ge m$, we use $1\le \tau^m\tau^{-n(C)}$. The identity for $\sum_{\mathfrak B}K(\mathfrak B)$ follows from (1) and (4), because $|K(\mathfrak B)| \le \prod e_\beta \le \hat z(\mathfrak B)$.

Step S7: two bounds and a sum, (M), (R*) *and* (Σa).

(M) Let $\mathfrak B$ be a set of letters (a), (v), (t) with $J_\beta \subset Y$; let $V' \in \mathcal U_Y(x')$, $(\mathrm a)_X \in \mathfrak B$ and $\delta \in T^a(x)$. Then

$$E_\xi|\chi_X e^{E^a(X,U)}\delta(X,U)\cdot \prod_{\beta\in\mathfrak B,\ \beta\ne(\mathrm a)_X}\lambda_\beta| \le m_X \prod_{\beta\ne(\mathrm a)_X} e_\beta,$$

where

$$m_X := e^{Ac_a(x)}\sup_{\mathcal U_X(x)}|\delta(X,\cdot)|\, e^{2x^{-1/8}\#I_{(\mathrm a)_X}}$$

; consequently $m_X/e_{(\mathrm a)_X} \le e^{Ac_a(x)}B/(4A)$.

(R*) Take the sizes e_α of Step 6 at $(\tilde\kappa, A)$, and $d_\alpha = (\tilde\kappa+1)(\#J_\alpha + s')$. Then for every analytic letter α with root $\square_0$, every cell $\square$, and $m = 1, 2$,

$$\sum_{C \text{ analytic},\ \mathrm{co}(C)\ni\square,\ n(C)\ge m} \mathrm{mult}_\alpha(C)\hat\Phi(C)e^{\tilde\kappa(\#\,\mathrm{co}(C)-1)} \le t^{m-1}e^{-\mathrm{dist}(\square_0,\square)-1}e_\alpha e^{2\varepsilon_2\#I_\alpha+(\tilde\kappa+1)\#J_\alpha-\tilde\kappa}.$$

(Σa) For every cell $\square$,

$$\sum_{X\in\mathcal D} e^{-\mathrm{dist}(\square_0(X),\square)-1}e^{-\tilde\kappa(\#X-1)}e^{4x^{-1/8}\#I_{(\mathrm a)_X}+(\tilde\kappa+1)\#Y_0(X)-\tilde\kappa} \le 65.642L^3,$$

where $\square_0(X)$ denotes the root of $(\mathrm a)_X$.

We prove (M) by the argument of Step 6 (iii), in which each factor containing a member is bounded in modulus by a number that does not depend on ξ, and these numbers then appear as constant factors in front of E_ξ. Where $\chi_X = 1$, the field U on $X^{\approx}$ lies in $\mathcal U_X(x)$, by Step 2 (ii), since $Y \supset Y_0(X)$; on that domain $|E^a(X,\cdot)| \le Ac_a(x)$, and $\delta(X,\cdot)$ is bounded by its supremum. Each remaining letter (a) or (v) is handled in the same way: its modulus is at most $2Ac_a(x)e^{-\tilde\kappa(\#X'-1)} \le e^\circ(\mathrm a)$ or $2K_v^\flat p_0^3 x^{-1/2} = e^\circ(\mathrm v)$, respectively, a number independent of ξ. What remains is $E_\xi\prod|e^{T_{Y_t}}-1|$, the product running over the letters (t), and Step 6 (iii) bounds this by $\prod e_{Y_t}^{1/2}\cdot\prod e^{x^{-1/8}\#I}$, again over the letters (t).

In fact Step 6 (iii) leaves the factor

$$\exp\{[2x\varsigma'' n_a + 36\varsigma''(C_\varphi^\flat)^2\cdot 3L^3(R+1)^3\cdot v(2r_1+1)^3]\#\mathsf Y\} \le e^{x^{-1/8}\#\mathsf Y},$$

with $\mathsf Y$ the union of the Y_t; this inequality follows from the third and fourth inequalities of (Y.8), and $\#\mathsf Y \le \sum \#Y_t$.

The factors $e^{2x^{-1/8}\#I}$ in e_β and in m_X are at least 1. For the last assertion of (M) we use $e_{(\mathrm a)_X} = 4Ac_a(x)e^{-\tilde\kappa(\#X-1)}e^{2x^{-1/8}\#I_{(\mathrm a)_X}}$.

We turn to (R*). The letters (a), (v), (t) are pairwise compatible: item 6 of §5.3 lists no incompatibility among them. The condition (S_t) at $\epsilon = \varepsilon_2$, $\mathsf{r} = r_1$, holds for $\mathfrak{A}$ at p and every $A \le 16$, by Step 7 (ii), which applies by Step S2; hence it holds for the smaller alphabet as well. Lem. 6.9 bounds the part with $n(C) \ge m$ of $\sum_C \mathrm{mult}_\alpha \hat\Phi e^{d(C)}$ by $t^{m-1} e_\alpha e^{2\varepsilon_2 \# I_\alpha + d_\alpha}$. By (Clu) of Step 8, $\#\,\mathrm{co}(C) \le \sum(\#J_\beta + s') - s'$, the sum being taken with multiplicity; thus $e^{\tilde\kappa(\#\,\mathrm{co}(C)-1)} \le e^{d(C)-(\tilde\kappa+1)s'-\#\,\mathrm{co}(C)-\tilde\kappa}$. Since $\mathrm{co}(C)$ is connected and contains both $\square_0$ and $\square$, $\#\,\mathrm{co}(C) \ge \mathrm{dist}(\square_0,\square)+1$. Finally, $d_\alpha - (\tilde\kappa+1)s' = (\tilde\kappa+1)\#J_\alpha$.

We turn to (Σa). By Step 3 (V), $\#I_\alpha \le (2\varrho_*+1)^3 \#J_\alpha$; with (Y.2) this gives $4x^{-1/8}\#I_\alpha \le 0.08\#J_\alpha$. Consider first a small X. Then $\#J = 1$, and $e^{-1+(\tilde\kappa+1)-\tilde\kappa} = 1$. Next, $\sum_{\square_0} e^{-\mathrm{dist}(\square_0,\square)} \le \mathsf{g}(1) \le 49.9791$; indeed the distance on the torus is the minimum over the lifts, so that the sum over the torus does not exceed the sum over $\mathbb{Z}^3$. The exact value is $\mathsf{g}(1) = 49.97902\ldots$. The first cube of X is one of the L^3 fine cubes that make up $\hat\square_0$, and $\sum_{X\ni\mathrm{cube}} e^{-\tilde\kappa(\#X-1)} \le 1.2124$. Altogether the small X contribute at most $49.9791 \cdot 1.2124 \cdot e^{0.08} L^3 \le 65.6415 L^3$.

Now let X be not small, so $n = \#X > L/2-2$. Then $\#Y_0 \le 8 + 432n/L$; $\square_0$ meets X^{+1}, so X contains one of $(L+2)^3$ cubes; and a given cube lies in at most $(26e)^{n-1}$ sets X with n cubes, where $\log 26e \le 4.2581$. So these X contribute at most

$$49.9791(L+2)^3 \sum_{n>L/2-2} \exp\{(4.2581-\tilde\kappa)(n-1) + (\tilde\kappa+1.08)(8+432n/L) - \tilde\kappa - 1\}.$$

From $\tilde\kappa - 4.2581 \ge 0.29\tilde\kappa$, $1.08 \le 0.18\tilde\kappa$ and $L > 1.2\cdot 10^{13}$, each exponent is at most $-0.289\tilde\kappa(n-1) + 9.5\tilde\kappa$. Consecutive terms have ratio at most $e^{-1.73}$, and the first exponent is at most $-0.289\tilde\kappa(L/2-36) \le -0.867L + 63$. Hence this part is at most $\exp\{-0.867L + 70 + 3\log(L+2)\} \le \frac18 e^{-L/2}$, and together with the small X we arrive at (Σa).

Step S8: the cluster series, (g1) *and* (g2). (g1) is (Fm2) with the clusters sorted by $\mathrm{co}(C)$. The alphabet is finite, and, by Lem. 6.9, the sums of $\hat\Phi$ over the clusters containing a given letter are finite; so all the sums converge absolutely. At $E^a = 0$ each λ_X^a vanishes identically; therefore $K(\mathfrak{B}) = 0$ for every $\mathfrak{B}$ containing a letter (a), and $\Phi(C) = 0$ for every old C.

For the first bound of (g2), the bound on $v(\cdot;u)$, we note that the field 1 belongs to $\mathcal{U}_Y(x')$ for every Y; so, by Step 6 (iii), applied with $Y = \mathrm{co}(C)$, $|\Phi(C)(1)| \le \hat\Phi(C)$ for every analytic C. Every old C contains a letter (a), and hence

$$\sum_{Y\ni\square} e^{\tilde\kappa(\#Y-1)} |v(Y;u)| \le \sum_X \sum_{C\ni(\mathrm{a})_X,\ \mathrm{co}(C)\ni\square} \hat\Phi(C) e^{\tilde\kappa(\#\,\mathrm{co}(C)-1)}.$$

We apply (R*) for $m=1$, with $e_\alpha e^{2\varepsilon_2\#I_\alpha} = 4Ac_a(x) e^{-\tilde\kappa(\#X-1)} e^{4x^{-1/8}\#I_\alpha}$, and then (Σa): the right side is at most $4 \cdot 65.642 L^3 Ac_a(x) \le 262.57 L^3 Ac_a(x) \le C_g Ac_a(x)$.

For the second bound of (g2), the bound on v^0, we observe that a new C contains a letter (v) or (t), and a constant C contains a letter (b); $|\Phi| \le \hat\Phi$ by Step 6 (iii), (iv); and Lem. 6.9 for $\mathfrak{A}$, together with (Clu), gives (R*) for these letters too. For the letters (v) and (b) we have $\#J = \#I = 1$ and $\mathsf{g}(1) e^{4x^{-1/8}} \le 49.9791 e^{0.08} \le 54.142$, and this number is multiplied by $2K_v^\flat p_0^3 x^{-1/2}$ for the letters (v) and by u_b for the letters (b). For the letters (t), $\sum_n (26e)^{n-1} e^{-(5\tilde\kappa+20)n + (\tilde\kappa+1.08)n - \tilde\kappa - 1} \le e^{-40}$, to be multiplied by $49.9791 x^{-20}$. Here e°(t) carries $\kappa^\sharp/2 = 5\kappa_1 + 20$; as $\kappa_1 \ge \tilde\kappa$, we have $e^{-(5\kappa_1+20)n} \le e^{-(5\tilde\kappa+20)n}$.

Step S9: the derivative and the one-letter formula, (h) *up to and including* (h1)*;* (g3), (g4) *and* (maj).

(0) *Each term is entire.* Let $\mathfrak{B}$ be a set of letters (a), (v), (t) with $J \subset Y$, and let $V' \in \mathcal{U}_Y(x')$. When u is replaced by $u + z(\delta,0,0)$, the letter $(\mathrm{a})_X$ becomes $\chi_X(e^{E^a(X,U)+z\delta(X,U)} - 1)$, an entire function of z, bounded on $|z| \le \rho$ by $e^{Ac_a(x)+\rho\sup|\delta(X)|} + 1$; the other factors keep the integrable majorant of Step 6 (iii). Hence $K_z(\mathfrak{B})(V')$ is entire, the derivative may be taken under E_ξ, and at $z=0$ one has $\frac{d}{dz}\lambda_X^a = \chi_X e^{E^a(X,U)}\delta(X,U)$. So $\Phi_z(C) = \varphi^T \prod K_z(\mathfrak{B}_i)$ is entire.

(1) *The majorant.* The product rule and (M) give

$$|DK(\mathfrak{B})[\delta]| \le \sum_{(\mathrm{a})_X \in \mathfrak{B}} (m_X / e_{(\mathrm{a})_X}) \prod_{\mathfrak{B}} e_\beta$$

. Consequently, for $V' \in \mathcal{U}_{\mathrm{co}(C)}(x')$,

$$|D\Phi(C)[\delta](V')| \le \hat{\Phi}(C) \sum_X \mathrm{mult}_{(\mathrm{a})_X}(C)\, m_X / e_{(\mathrm{a})_X},$$

a bound in which no field enters. From (R*) and (Σa), with

$$m_X e^{2\varepsilon_2 \# I_{(\mathrm{a})_X}} \le e^{Ac_a(x)} B c_a(x) e^{-\tilde{\kappa}(\# X - 1)} e^{4x^{-1/8} \# I_{(\mathrm{a})_X}}$$

and with $e^{Ac_a(x)} \le e^{1/2}$ by (Y.1), we obtain for $m = 1, 2$

$$\sum_{C \text{ old},\ \mathrm{co}(C) \ni \square,\ n(C) \ge m} e^{\tilde{\kappa}(\# \mathrm{co}(C) - 1)} \sup |D\Phi(C)[\delta]| \le t^{m-1} e^{1/2} 65.642 L^3 B c_a(x) \le t^{m-1} 108.23 L^3 B c_a(x). \quad (D_m)$$

(2) Therefore the series of $DA(u)[\delta](Y, \cdot)$, of $Dv(Y; u)[\delta]$, of Rst and of r converge absolutely and uniformly. Each of them is linear in δ, as $D\Phi(C)[\delta]$ is; and, by (f1), of the components of u only E^a enters them.

(3) *The derivative.* Let $A_a < 16$. For $|z|B \le 16 - A_a$ we have $u + z(\delta, 0, 0) \in M_{16}(x; p)$, because the clauses without a size are linear, $\|\cdot\|_{x;p}$ is the largest of three weighted suprema (item 3 of §5.3), and $z(\delta, 0, 0)$ changes only the first of them. On this disc $|\Phi_z| \le \hat{\Phi}$ at $A = 16$, so the series of E'^{old} converges uniformly in z, and by Weierstrass' theorem its derivative at 0 is the sum of the derivatives of its terms, which is $DA(u)[\delta]$. For v we use, at the field 1, the bound $|\Phi_z(C)(1)| \le \hat{\Phi}(C)$ at $A = 16$; then the series of v converges uniformly in z, and Weierstrass' theorem shows that $Dv(Y; u)[\delta]$ is $\frac{d}{dz}$ at 0 of $v(Y; u + z(\delta, 0, 0))$. By (g1) and (f1), and because there are finitely many connected sets Y of cells, $\frac{d}{dz}$ at 0 of $\mathsf{V}'(u + z(\delta, 0, 0))$ equals $\sum_Y Dv(Y; u)[\delta]$.

The integral of (g3). Let u_τ be as in (g3). By (0), every $\Phi(C)(1)$ defined with u_τ is entire in τ, its derivative being $D\Phi(C)[\delta](1)$ evaluated at u_τ. On $[0, 1]$ the terms are at most $\hat{\Phi}(C)$ at $A = 16$, and their derivatives are bounded in modulus by the majorant of (1) at $A = 16$; both majorants are independent of τ and summable, by (R*) and (D_1). We may therefore differentiate term by term: $\tau \mapsto v(Y; u_\tau)$ is continuously differentiable on $[0, 1]$, with derivative $Dv(Y; u_\tau)[\delta]$, and we integrate.

(4) *Families.* Along the family u_z of (c+), $D\Phi_z(C)[\delta](V')$ is holomorphic in z. To see this, note first that, by (0) and the product rule,

$$DK_z(\mathfrak{B})[\delta](V') = \sum_{(\mathrm{a})_X \in \mathfrak{B}} E_\xi\big[\chi_X e^{E^a_z(X,U)} \delta(X, U) \prod_{\beta \in \mathfrak{B},\ \beta \ne (\mathrm{a})_X} \lambda_{\beta,z}\big].$$

For fixed ξ the integrand is holomorphic in z on Ω, by (1) and (2) of Step S5 and because δ is independent of z; it is also jointly measurable. By the estimate of the proof of (M), applied with $A = 16$, the modulus of the integrand is at most $e^{1/2} \sup |\delta(X, \cdot)|$ times the numbers e° of the other letters (a), (v) times $\prod |e^{T Y_t} - 1|$; this is an integrable majorant independent of z, also where $\|u_z\| = 16$. The theorems of Fubini and Morera and Osgood's lemma, used as in (3) of Step S5, now show that $DK_z(\mathfrak{B})[\delta]$ is holomorphic on Ω. Further, $D\Phi_z(C)[\delta] = \varphi^T \sum_i DK_z(\mathfrak{B}_i)[\delta] \prod_{j \ne i} K_z(\mathfrak{B}_j)$ is a polynomial in holomorphic functions. Since the majorant of (1) at $A = 16$ is independent of z, (D_1) makes the series of $DA(u_z)[\delta](Y, V')$ converge uniformly in z, and Weierstrass' theorem applies.

(5) *The split.* An old cluster with $n(C) = 1$ is $C = (\{(\mathrm{a})_X\})$, the sequence consisting of the single polymer $\{(\mathrm{a})_X\}$, and for it $\varphi^T = 1$, $\Phi(C) = \mathcal{G}_X = E_\xi[\chi_X(e^{E^a(X,U)} - 1)]$, $\mathrm{co}(C) = Y_0(X)$ and $D\Phi(C)[\delta] = G^\delta_X[u]$. In $DA(u)[\delta]$ the clusters with $n(C) \ge 2$ make up Rst, and in $Dv(Y; u)[\delta]$ they make up $r(Y) := \sum_{\mathrm{co}(C) = Y,\ \mathrm{old},\ n(C) \ge 2} D\Phi(C)[\delta](1)$.

(6) For (h1) we apply (D_2) twice, on account of the subtraction; the result is

$$216.46 L^3 x^{-1/4} B c_a(x) \le 217 L^3 x^{-1/4} B c_a(x)$$

. By (Y.0)(0b),

$$c_a(x') = L^{1/2}(p_0'/p_0)^3 c_a(x) \ge 0.594 L^{1/2} c_a(x)$$

, because $2^{-3/4} \ge 0.5946$; and $217/0.594 \le 365.32 \le 366$. For (g3) we apply (D_2) once: $108.23 \le 109$.

(7) For (g4) recall that, by (f1), v depends on E^a alone. The family $u_\tau := (E_1^a + \tau(E_2^a - E_1^a), E_1^h, w_1)$, $0 \le \tau \le 1$, lies in the convex set $M_2(x;p)$, and $\delta := E_2^a - E_1^a \in T^a(x)$ with $B = \|E_1^a - E_2^a\|$. By (3), $v(Y;u_2) - v(Y;u_1) = \int_0^1 Dv(Y;u_\tau)[\delta]\, d\tau$. Taking moduli and the cell sum under the integral and using (D_1), we bound this by $108.23 L^3 c_a(x) B \le 421 L^3 c_a(x) B$.

(8) *Clauses without a size.* We treat them as in Step 8. The function $DK(\mathfrak{B})[\delta](V')$ is the mean of a product which depends holomorphically on $V' \in \mathcal{U}_Y(x')$, by Step 2 (ii) and by (Γ6), (2d) and (4b) of item 4 of §5.3, and which is dominated by (M); the convergence being uniform, holomorphy in V' follows. The dependence on the field on $Y^{\approx}$ alone follows from Step 3 (V). For the invariance: like $E^a(X,\cdot)$, the function $\delta(X,\cdot)$ is invariant, so the marked letter satisfies $\lambda(\mathrm{Ad}_u\,\xi, V'^u) = \lambda(\xi, V')$, just as the other letters do; moreover the χ and the law of ξ are invariant under Ad_u. The invariance and the subtraction give the value 0 at pure gauges.

(9) For (maj) we note that the right sides of (1), and the number S_X of Step S10, depend on A and B alone; hence the supremum over u and δ may be moved inside every sum above.

Step S10: the one-letter part, (h2). Let X be small, with its first cube in $\hat{\square}_0$; let o be the centre site of $\square_0$, and B_ω the box of radius ω about o. This is the setting of Step 8, (8.2). The blocks meeting $X^{\approx}$ lie within $M + \lceil Mr(x)/L \rceil$ of o, so that B_ω contains the set of sites of Step 2 (ii). A field on B_ω is extended by 1; it then lies in the interior of $\mathfrak{V}$, and no quantity of the argument depends on the extension.

Indeed, the values of U on $X^{\approx}$ are determined by V' at $y(b)$, by G at distance at most r_G and by a_0 at distance at most R_3 from the blocks meeting $X^{\approx}$; this follows from (2b) and, for a_0, from the paragraph "The dependence at every pair" of (Γ3) of item 4 of §5.3, by which, at every field of $\mathfrak{V}$, $a_{0,b}$ depends on V' through its values on bonds and on the $\mathfrak{g}_{\mathbb{C}}$-valued variables X'_P of plaquettes P, every site of these bonds and plaquettes being within R_3 of $y(b)$. By the definition of ω, all these sites lie in B_ω. Hence, if a member of $\mathcal{U}_{\{\square\}}(x')$ is extended by 1 outside the neighbourhood of the cell, $G_X^\delta[u]$ takes at it the same value as at its restriction to B_ω, extended by 1.

(1) We apply Step 2 (ii) with $\tau = 17T$: for $V' \in U^{17T}(\mathsf{B}_\omega; x')$ and $\chi_X(\xi) = 1$, the field U on $X^{\approx}$ lies in $\mathcal{U}_X(x)$, and it depends holomorphically on V'. Therefore the integrand of $G_X^\delta[u]$ is holomorphic in V' and bounded by $S_X := e^{Ac_a(x)} B c_a(x) e^{-\tilde{\kappa}(\#X - 1)}$; so $G_X^\delta[u]$ is holomorphic on $U^{17T}(\mathsf{B}_\omega; x')$, with $|G_X^\delta[u]| \le S_X$.

(2) The function $G_X^\delta[u]$ is invariant under gauge transformations with values in G. Indeed, $E^a(X,\cdot)$ and $\delta(X,\cdot)$ are invariant; the gauge transform of $\mathsf{U}(\eta; V')$ by the blockwise constant u is $\mathsf{U}(\mathrm{Ad}_u\,\eta; V'^u)$; G is covariant, by (2c) of item 4 of §5.3, and so is a_0, by its definition in (Γ3) together with (2c) and the formula of (2g) of that item; χ_X and the law of ξ are invariant under Ad_u; and a transformation u given on the sites of B_ω is extended by 1, as in the part of Step 8 on the clauses without a size.

(3) The restriction of a member of $\mathcal{U}_{\{\square_0\}}(x')$ to B_ω has (u1) to (u3) at x', since $\omega \le Mr'$. We now apply Lem. 6.13 below with $T = L^{3/2}/(19\mathsf{N}_c)$, $\Psi = G_X^\delta[u]$ and $S = S_X$. Here $T > 1$ and $2\omega + 1 < N$ hold by Step 0 (iv), so that B_ω is embedded; $3\omega \le \ell_w(x')$ follows from $3\omega + 1 \le \ell_w(x')$, which is given by Step 0 (iii) with (Rad); and (C0) to (C2) are, for this T, the conditions (Y.0)(0d), which are satisfied since $x \ge x_9^U$ and (Y.0) is one of the conditions of (O13) (i) of item 0 of this section. Thus the hypotheses of the lemma are met; and its conclusion says that any complex Ψ which is

holomorphic, G-invariant and bounded by S on $U^{17T}(\mathsf{B}_\omega; x')$ satisfies $|\Psi(V') - \Psi(1)| \le 2ST^{-2}$ at every such V'. Hence $|G^\delta_X[u](V') - G^\delta_X[u](1)| \le 2T^{-2}S_X$.

The conclusion of Lem. 6.13 is that of Thm. 5.22; the theorem assumes in addition $17T\varepsilon' \le 1/40$ and $3\omega + 1 \le \ell_w(x')$.

Lemma 6.13 (Invariant holomorphic functions on a box)**.** Let $T > 1$ be arbitrary. Let B_ω be the box of radius ω about a coarse site o, embedded, that is, $2\omega + 1 < N$, and let $3\omega \le \ell_w(x')$. Let the following conditions hold; for $T = L^{3/2}/(19\mathsf{N_c})$ they are the conditions (Y.0)(0d).

(C0) $18\omega\varepsilon' \le 1/10$ and $160T\omega\varepsilon' \le 1$;
(C1) $2640\,\ell_w(x')(T+1)\omega\varepsilon' \le \mathsf{g}_x$;
(C2) $102400\,T\omega^2\varepsilon' \le 1$.

Let Ψ be a complex function on $U^{17T}(\mathsf{B}_\omega; x')$ that is holomorphic, bounded by S, and invariant under the gauge transformations with values in G at the sites of B_ω. Then $|\Psi(V') - \Psi(1)| \le 2ST^{-2}$ for every field V' on B_ω that satisfies (u1) to (u3) at x'. ■

Proof of Lem. 6.13. We write $|\cdot|$ for the operator norm. In $\mathrm{SL}(2,\mathbb{C})$ one has $M^{-1} = (\operatorname{tr} M)1 - M$; hence $|M^{-1}| = |M| \ge 1$ and $|M^{-1} - 1| = |M - 1|$. For k on the sites of B_ω with values in $\mathrm{SL}(2,\mathbb{C})$ we set $V'^k(c) := k(c_-)V'(c)k(c_+)^{-1}$. The domain $U^{17T}(\mathsf{B}_\omega; x')$ is open, and it is invariant under G-valued k.

(i) *A tree gauge.* Let $\Gamma(o, y')$ be the path in B_ω from o to y' that follows e_1 first, then e_2, and finally e_3; it has at most $3\omega \le \ell_w(x')$ bonds. We put $\mathsf{k}(y') := V'(\Gamma(o, y'))$; by (u2), $|\mathsf{k}|, |\mathsf{k}^{-1}| < 2$. The field $\mathsf{W} := V'^{\mathsf{k}}$ equals 1 on the bonds of the paths Γ, and $\partial\mathsf{W}(P) = \mathsf{k}\,\partial V'(P)\mathsf{k}^{-1}$ with k at the base site of P; so (u3) gives $|\partial\mathsf{W}(P)^{\pm 1} - 1| < 4 \cdot 2\varepsilon' = 8\varepsilon'$.

Each bond along e_3 lies on a path Γ. Consider a bond $c = (y'; e_2)$. The bond $(y'_1, y'_2, 0; e_2)$ lies on a path Γ, and we pass from it to c across $|y'_3|$ plaquettes forming a ladder in the plane $(2,3)$; the sides of this ladder along e_3 lie on paths Γ, and each plaquette contributes one factor $\partial\mathsf{W}(P)^{\pm 1}$ to W. For a bond $(y'; e_1)$ we start from $(y'_1, 0, 0; e_1)$ and cross $|y'_2|$ plaquettes in the plane $(1,2)$ at height 0, then $|y'_3|$ plaquettes in the plane $(1,3)$. So $\mathsf{W}(c)$ is a product of at most 2ω factors $1 + a_i$ with $|a_i| < 8\varepsilon'$, and $|\mathsf{W}(c) - 1| \le e^{16\omega\varepsilon'} - 1 \le 18\omega\varepsilon' \le 1/10$ by (C0). Hence $A_c := -i \log \mathsf{W}(c)$ is traceless, and $|A_c| \le 1.054 \cdot 18\omega\varepsilon' \le 19\omega\varepsilon'$.

(ii) *The value at* W. We show that $\Psi(V') = \Psi(\mathsf{W})$. Let $\mathsf{k} = hu$ be the polar decomposition, where h is positive of determinant 1 and u has its values in G; then $|h| = |h^{-1}| = |\mathsf{k}| < 2$. Unitary factors leave the norms unchanged, so $V'' := V'^u$ again has (u1) to (u3); moreover $\Psi(V'') = \Psi(V')$ by the invariance, and $\mathsf{W} = V''^h$. For complex s let $V_s := V''^{h^s}$, where $h^s = e^{s \log h}$. Then V_s is entire in s, $V_0 = V''$, $V_1 = \mathsf{W}$, and $V_{s+i\sigma} = (V_s)^{h^{i\sigma}}$, with $h^{i\sigma}$ taking its values in G. We claim that V_s lies in the domain for $0 \le s \le 1$. If so, V_s lies in the domain for all s in an open set containing the strip $0 \le \operatorname{Re} s \le 1$. On the strip itself this holds because $V_{s+i\sigma} = (V_s)^{h^{i\sigma}}$; the set $\{s : V_s \text{ in the domain}\}$ is open, and we take its component containing the strip, which is a connected open set. On this set $s \mapsto \Psi(V_s)$ is holomorphic, and by the invariance it is constant on the imaginary axis; so it is constant, and $\Psi(V') = \Psi(V'') = \Psi(\mathsf{W})$.

Proof of the claim. For (e3): $|\partial V_s(P) - 1| \le |h^s||h^{-s}||\partial V''(P) - 1| < 8\varepsilon' < 34T\varepsilon'$. For (e1) and (e2), let π be a path in B_ω from y' to y'' with no more than $\ell_w(x')$ bonds, a single bond in the case of (e1). From (i), from (C1) with $T + 1 > 2$, and from $\mathsf{g}_x \le 1$ we get $|\mathsf{W}(\pi)^{\pm 1} - 1| \le e^{19\omega\varepsilon'\ell_w(x')} - 1 \le \mathsf{g}_x/250$; here $\mathsf{g}_x \le 1$ because $\mathsf{g}_x < \vartheta_0^\flat / \log x'$, by Step 8, (8.2), and $\vartheta_0^\flat \le 10^{-4}$, by item C.II 1.3 of §3.

Next, $V''(\pi) = h(y')^{-1}\mathsf{W}(\pi)h(y'')$; so, with $\theta := 1 - s \in [0,1]$,

$$V_s(\pi) = h(y')^{-\theta}h(y'')^\theta + h(y')^{-\theta}(\mathsf{W}(\pi) - 1)h(y'')^\theta,$$

and the second term has norm at most $4\mathsf{g}_x/250$. For positive matrices H_1, H_2 and $0 \le \theta \le 1$, $|H_1^\theta H_2^\theta| \le |H_1 H_2|^\theta$; this bound is Cordes' inequality, which is a form of the Löwner–Heinz theorem,

and it is proved in Step (1) of the proof of Lem. 3.33. We take $H_1 = h(y')^{-1}$ and $H_2 = h(y'')$. Then $H_1H_2 = V''(\pi)[1 + h(y'')^{-1}(\mathsf{W}(\pi)^{-1} - 1)h(y'')]$, so $|H_1H_2| \le |V''(\pi)|(1 + 4\mathsf{g}_x/250)$; as the determinant equals 1, $|H_1H_2| \ge 1$ and $|H_1H_2|^\theta \le |H_1H_2|$. Using $|V''(\pi)| < 2$, we find $|V_s(\pi)| \le |V''(\pi)| + 12\mathsf{g}_x/250 < |V''(\pi)| + \mathsf{g}_x$, and this is $< \mathsf{b}(x') + \mathsf{g}_x$ by (u2) in the case of (e2), where π has at most $\ell_w(x')$ bonds, or $< 1 + \vartheta(x') + \mathsf{g}_x$ by (u1) in the case of (e1), where π is a single bond.

(iii) *A family of fields.* The family $\mathsf{W}_t := (e^{itA_c})_c$, t complex, $|t| < T$, lies in the domain. For (e1) and (e2): $|\mathsf{W}_t(\pi)| \le e^{19T\omega\varepsilon'\ell_w(x')} \le e^{\mathsf{g}_x/138} < 1 + \mathsf{g}_x$, by (C1). For (e3): $\partial\mathsf{W}_t(P)$ is a product of four exponentials $e^{\pm itA_c}$, with $\sigma_t := \sum |tA_c| \le 76T\omega\varepsilon' \le 1/2$ by (C0); therefore $|\partial\mathsf{W}_t(P) - 1 - it(\mathrm{d}A)_P| \le e^{\sigma_t} - 1 - \sigma_t \le 0.6\sigma_t^2$, where $(\mathrm{d}A)_P$ denotes the signed sum of the four A_c. Taking $t = 1$ we get $|(\mathrm{d}A)_P| \le |\partial\mathsf{W}(P) - 1| + 0.6 \cdot 76^2\omega^2\varepsilon'^2 \le 8\varepsilon' + 3466\omega^2\varepsilon'^2$. Hence $|\partial\mathsf{W}_t(P) - 1| \le T\varepsilon'[8 + 3466\omega^2\varepsilon' + 3466T\omega^2\varepsilon'] \le 8.1T\varepsilon' < 34T\varepsilon'$, by (C2).

(iv) *Schwarz' lemma.* The function $\psi(t) := \Psi(\mathsf{W}_t)$ is holomorphic on $|t| < T$, with $|\psi| \le S$, $\psi(0) = \Psi(1)$ and $\psi(1) = \Psi(\mathsf{W}) = \Psi(V')$. Its derivative at the origin is $\psi'(0) = \mathcal{L}(A)$, where $\mathcal{L}$, the derivative of Ψ at the field 1, is complex linear in A. For a constant $u \in G$ we have $(\mathsf{W}_t)^u = (e^{it\,\mathrm{Ad}_u A_c})_c$, and so $\mathcal{L}(\mathrm{Ad}_u A) = \mathcal{L}(A)$. The Haar mean of Ad_u on $\mathfrak{g}_{\mathbb{C}}$ is 0, since the adjoint representation of SU(2) is irreducible and not trivial; hence $\mathcal{L}(A) = 0$. Thus $\psi - \psi(0)$ is bounded by $2S$ and vanishes to second order at 0, and Schwarz' lemma gives $|\psi(1) - \psi(0)| \le 2ST^{-2}$.

□

Step S10, continued. (4) There are L^3 first cubes, and $\sum_{X\ni\mathrm{cube}} e^{-\tilde\kappa(\#X-1)} \le 1.2124$; since $L^3T^{-2} = 361\mathsf{N}_c^2$, the sum over the small X is at most $2.4248 \cdot 361\mathsf{N}_c^2 e^{Ac_a(x)}Bc_a(x)$. Using $e^{Ac_a(x)} \le e^{1/2}$ and $c_a(x) \le c_a(x')/(0.594L^{1/2})$, together with $2.4248 \cdot 361 \cdot 1.6488/0.594 \le 2429.8 \le 2430$, we bound it by $2430\mathsf{N}_c^2L^{-1/2}Bc_a(x')$.

Now let X be not small. On $\mathcal{U}_{Y_0(X)}(x')$, Step 2 (ii), applied with $\tau = 1$, gives $|G_X^\delta[u](V') - G_X^\delta[u](1)| \le 2S_X \le 3.3Bc_a(x)e^{-\tilde\kappa(n-1)}$. A cell $\square \in Y_0(X)$ meets X^{+1}, so the sum over these X is at most

$$3.3Bc_a(x)(L+2)^3 \sum_{n>L/2-2} \exp\{(4.2581 - \tilde\kappa)(n-1) + \tilde\kappa(7 + 432n/L)\},$$

and the counting in the proof of (Σa) bounds this by $\exp\{-0.867L + 70 + 3\log(L+2)\}Bc_a(x) \le \frac{1}{8}e^{-L/2}Bc_a(x)$.

Step S11: summing the parts, (h3) *and* (b#). For (h3) we arrange the cell sum of the one-letter part according to the sets Y: a set $Y = \{\square\}$ receives, with weight 1, the small X whose first cube lies in $\hat\square$, and an X that is not small, with $Y_0(X) = Y \ni \square$, enters with the weight $e^{\tilde\kappa(\#Y_0-1)}$. Adding (h2) and (h1), we obtain the bound $[2430\mathsf{N}_c^2L^{-1/2} + 366L^{5/2}x^{-1/4}]Bc_a(x') + e^{-L/2}Bc_a(x)$.

By (Y.10),

$$366L^{5/2}x^{-1/4} \le (366/1800)L^{-1} \le 0.2034L^{-1}$$

. Also

$$e^{-L/2}c_a(x) \le e^{-L/2}c_a(x')/(0.594L^{1/2}) \le 0.1L^{-1}c_a(x')$$

. So the bracket is at most $2430\mathsf{N}_c^2L^{-1/2} + L^{-1}$. In the setting (L2^{su}), $2430/(1.2 \cdot 10^4) = 0.2025$, and $L^{-1} < 10^{-13}$. In the setting (L3su), $2430\mathsf{N}_c^2L^{-1/2} \le 0.243L^{-1/4}$, and

$$L^{-1} = L^{-3/4}L^{-1/4} \le 1.6 \cdot 10^{-10}L^{-1/4}$$

; thus the bracket is at most $0.2431L^{-1/4} = \vartheta_a$.

For (b#) let $u_\tau := (\tau E^a, E^h, w)$, $0 \le \tau \le 1$; this family lies in $M_A(x;p)$, and $E^a \in T^a(x)$ with $B_p(E^a) \le A$. By Step S8, $E'^{\mathrm{old}}[u_0] = 0$. For $\tau < 1$ and $|z| \le 16/A - \tau$ we have $u_\tau + z(E^a, 0, 0) = u_{\tau+z} \in M_{16}(x;p)$. As $A_a(u_\tau) \le \tau A < 16$, (3) of Step S9 shows that $\tau \mapsto E'^{\mathrm{old}}[u_\tau](Y, V')$ is differentiable on $[0,1)$ with the derivative $DA(u_\tau)[E^a](Y, V')$. Continuity on $[0,1]$ holds as well: the series consists of functions continuous in τ and is dominated by the majorant $\hat\Phi$ with the sizes e_α for this A, valid for the members within A caps, that is, on $M_A(x;p)$, which contains every u_τ

with $0 \le \tau \le 1$. Hence $E'^{\mathrm{old}}[u](Y,V') = \int_0^1 DA(u_\tau)[E^a](Y,V')\,d\tau$. We take the supremum over V' and the cell sum under the integral, and we apply (h3) at $u_\tau \in M_A(x;p)$, with $B \le A$.

Step S12: the bounds at $\tau^\sharp$, Lem. 6.8 and (h#).

(1) *The condition* (S_τ) *for* {(a), (v), (t)} *at* $(\tau^\sharp, \varepsilon_2^\sharp)$, *with* $\mathsf{r} = r_1$. We follow the lines of Step 7 (ii). If $\mathrm{dist}(I_\beta, \square) \le r_1$, then some cell $\square''$ of J_β is within $\varrho_* + r_1 \le s'$ of $\square$, and there are $(2s'+1)^3$ such cells. For the sum over the letters β with $J_\beta \ni \square''$ we use $\sum_{\beta: J_\beta \ni \square''} \le \sum_{\square_0} e^{-\mathrm{dist}(\square_0, \square'')} \sum_{\beta \text{ rooted at } \square_0} e^{\#J_\beta}$, together with $\mathsf{g}(1) \le 50$. Further, $e_\beta e^{3\varepsilon_2^\sharp \# I_\beta} = e_\beta^\circ e^{(2x^{-1/8} + 3\varepsilon_2^\sharp)\# I_\beta} \le e_\beta^\circ e^{\#J_\beta}$, because $\#I_\beta \le (2(s_0+\varrho_*)+1)^3 \#J_\beta$ gives $3\varepsilon_2^\sharp \#I_\beta \le \#J_\beta/2$ and, by (Y.2), $2x^{-1/8}\#I_\beta \le \#J_\beta/25$. Also $e^{d_\beta} = e^{(\tilde\kappa+1)s'} e^{(\tilde\kappa+1)\#J_\beta}$.

The rooted sums of $e_\beta^\circ e^{(\tilde\kappa+3)\#J_\beta}$ are at most $5e^{\tilde\kappa+3}L^3 \cdot 16 c_a(x)$, $2e^{\tilde\kappa+3}K_v^\flat p_0^3 x^{-1/2}$ and x^{-20}, for the letters (a), (v) and (t) respectively. Therefore $\sum_{\beta: \mathrm{dist}(I_\beta,\square) \le r_1} e_\beta e^{3\varepsilon_2^\sharp \# I_\beta + d_\beta} \le a^\sharp(x) = \tau^\sharp \varepsilon_2^\sharp$, and ($\mathrm{S}_\tau$) holds for every $\tau \ge \tau^\sharp$.

(2) *The bound $\tau^\sharp \le 1$.* We take (Y.3) with κ_1 in the place of κ. Its bracket and that of $a^\sharp$ are multiplied by the same factor $50(2s'+1)^3 e^{(\kappa_1+1)s'}$. Each of the three terms of the bracket of $a^\sharp$ is also a term of the bracket of (Y.3), except that the first has $16c_a(x)$ where (Y.3) has $16c_a(x)$ plus a term ≥ 0; the remaining terms of the bracket of (Y.3) are ≥ 0. Therefore the bracket of $a^\sharp$ does not exceed that of (Y.3), so $x^{1/4}a^\sharp \le x^{-1/8}$. By (Y.2), $1/\varepsilon_2^\sharp = 6(2(s_0+\varrho_*)+1)^3 \le 0.12 x^{1/8}$. Consequently $\tau^\sharp \le 0.12 x^{-1/4}$.

(3) *Lem. 6.8.* By (1), the alphabet {(a), (v), (t)}, with $\mathsf{r} = r_1$, the sizes e_α at $(\tilde\kappa, A)$ and $d_\alpha = (\tilde\kappa+1)(\#J_\alpha + s')$, satisfies the hypothesis (S_τ) of Lem. 6.9 at $\epsilon = \varepsilon_2^\sharp$ for every $\tau \ge \tau^\sharp$. Let τ with $\tau^\sharp \le \tau \le 1$ be as in Lem. 6.8. Lem. 6.9, applied with this τ and with $\epsilon = \varepsilon_2^\sharp$, and multiplied by τ, gives $\sum_C \mathrm{mult}_\alpha \hat\Phi e^{d(C)} \tau^{-(n(C)-1)} \le e_\alpha e^{2\varepsilon_2^\sharp \# I_\alpha + d_\alpha}$. We conclude as in the proof of (R*), with (Clu) and with $\#\,\mathrm{co}(C) \ge \mathrm{dist}(\square_0, \square) + 1$.

(4) *The power.* Since $s' = 8r' \le 160 c_R \log x/M + 8$, we have $e^{(\kappa_1+1)s'} \le e^{8(\kappa_1+1)} x^{\sigma_M}$. The third inequality of (Y.6) gives $x^{-10} \le c_a(x')/32 \le L^{1/2} c_a(x)/32$, and (Y.10) gives $x^{-10} L^{1/2} \le 1$; hence $x^{-20} \le c_a(x)$. So the bracket of $a^\sharp$ is at most $e^{\kappa_1+3}[81L^3 E_a + 2K_v^\flat] p_0^3 x^{-1/2}$; moreover $6 \cdot 50 = 300$ and $8(\kappa_1+1) + \kappa_1 + 3 = 9\kappa_1 + 11$. Finally $\sigma_M \le 160(\kappa_1+1)/(1920(\kappa_1+4)) \le 1/12$.

(5) *The bounds* (h1) *and* (g3) *at $\tau^\sharp$.* In (1) of Step S9 we use Lem. 6.8, applied with $\tau = \tau^\sharp$, instead of (R*); the part with $n(C) \ge 2$ then has $\tau^\sharp$ in place of t, and $e^{2\varepsilon_2^\sharp \# I}$ in place of $e^{2\varepsilon_2 \# I}$. For small X, $(2\varepsilon_2^\sharp + 2x^{-1/8})\#I \le 1/3 + 1/25$, and $49.9791 \cdot 1.2124 \cdot e^{0.3734} \cdot e^{1/2} \le 145.13 \le 146$; twice this gives $290.26 \le 291$. For X not small, the factor $e^{0.3734 \# Y_0}$ replaces $e^{0.08 \# Y_0}$; we have $1.3734 \le 0.23\tilde\kappa$, the exponents are at most $-0.289\tilde\kappa(n-1) + 10\tilde\kappa$, and the counting in the proof of (Σa) remains valid. Indeed, the first exponent is at most $-0.289\tilde\kappa(L/2 - 37.7) \le -0.867L + 65.4$. Consecutive terms of the sum over n have ratio at most $e^{-0.289\tilde\kappa} \le e^{-1.734}$, as $\tilde\kappa \ge 6$, so this sum is at most its first term times $1/(1-e^{-1.734}) \le 1.215$, the sum of the geometric series with ratio $e^{-1.734}$. Since $\log 49.98 + \log 1.215 \le 4.11$, the constant term becomes at most $69.6 \le 70$.

Step S13: additional statements used from this section on. The present section and the sections after it use the statements of §5.3, the proof of Thm. 5.15 included, and the statements of Step 8 of the proof of Thm. 4.39. They also use the following statements.

Instead of the floor x_9 of (O13) of item 1 of §5.3, we use in this section and in the later ones the floor x_9^U, where x_9^U is defined in (O13) of item 0 of this section, with (Y.10) among the conditions that define it, and $x_9^U \ge x_9$. Item (VII) of the alphabet, stated and proved in Step S3, is used besides the items of Step 3. Together with the product bound of Step 6 we use the bound (M); Step S7 proves it. In addition to item (R) of Step 8, we apply the claim (R*), with mult_α, proved in Step S7. Step S10 carries out the argument of (8.2) for $G_X^\delta[u]$, and this argument is used alongside (8.2) of Step 8. Step 11 concerns affine families on an open set of $\mathbb{C}$, and together with it we apply

clause (c+) of Thm. 6.7, which is proved in Step S5 and yields the conclusion of Step 11 for families that are holomorphic on an open set of $\mathbb{C}^n$.

Lem. 4.50 is part of Step 8 of the proof of Thm. 4.39, and besides its claims (1) and (2) we use the following two claims:

(1*) if (S) of Lem. 4.50 holds with τ in place of t, $0 < \tau \le 1$, then

$$\sum_C \mathrm{mult}_\alpha(C)\hat{\Phi}(C)e^{d(C)}\tau^{-(n(C)-1)} \le e_\alpha e^{2\varepsilon_2 \# I_\alpha + d_\alpha}$$

, where $\mathrm{mult}_\alpha(C) = \#\{i : \alpha \in \mathfrak{B}_i\}$;

(2*) consequently the claims (1) and (2) of Lem. 4.50 hold with the factor $\mathrm{mult}_\alpha(C)$ under the sum.

The claims (1*) and (2*) are Lem. 6.9. That lemma is proved in Step S6, and (1*) is its first claim, taken with $\epsilon = \varepsilon_2$ and multiplied by τ. In Step 8, (8.2), Thm. 5.22 is used. In the place of Thm. 5.22, Lem. 6.13, which gives the conclusion of that theorem for every $T > 1$ for which (C0) to (C2) hold, is applied in (3) of Step S10 and, with another value of T, in Step 4 of the proof of Thm. 6.33; the lemma is stated and proved in Step S10. Besides (O4) of item 1 of §5.3, L also satisfies the condition (L3su), which is stated in (O4a) of item 0 of this section. □

6.3. **Linear flow of quadratic sources.** All objects of this section are scalar kernels on tori: no field, no group, no member and no cap enters them, apart from the letters of (S5)(iii), and no source appears in any kernel, law, centre or row. We show that one step $D \mapsto T_\ell D$ on families of kernels contracts by the factor $\vartheta_s = c_s L^{-15/32}$ away from the charges, that is, away from the values on constant forms; that the step averages the charges exactly, whatever the defect $\tau(\ell)$ of the rows may be; that a cut moves the charges by a power of x; and that the steps at two sets of rows differ by $c_I \delta^{15/16}$, with charges 0. From these facts we obtain Thm. 6.20, which is clause (S1) of Thm. 6.21, and clause (S3). This section uses §2.2; from §5.5 it uses (O0) in item 1, the definition "Runs" in item 2, (RU) in item 3, Steps 0 (i) and 1 of the proof of Thm. 5.63, and (X12), (X13); from §5.4 it uses Lem. 5.47 and Cor. 5.49.

Item 1: the setting.

Remark 6.14. The letters of this section are those listed in item (0), and its numbers those listed in item (1), of §6.1.

A "Step n" cited without the name of a statement is Step n of the proof of Thms. 6.20 and 6.21.

The letter K is the number of steps, except in $K = K_*$ (Step 1 (a)), where it is the bound of the rows of §2.2.

Letters and data. The letters and data come from the table, item 3 of §2.1, in the form given in item 1 of §5.5, which also contains (O14). Its entries read as follows.

(O0) The numbers $b_*, \gamma_*, \lambda_*, K_*, L_A, C_1, c_{\mathrm{fix}}$, together with $L_B = \max\{L_A, c_{\mathrm{fix}}^{10}\}$ and $\lambda_\flat = \lambda_*/16$; moreover $\mathsf{g}(\lambda) = \sum_{n\in\mathbb{Z}^3} e^{-\lambda|n|_\infty}$, and $\mathsf{g}_1(\lambda)$ is the same sum with the factor $|n|_\infty$ in each term.

(O4) L is odd, $L \ge L_*^{\mathrm{su}} \ge \max\{L_A, L_B\}$ and $a \ge 3/(c_L L)$; moreover $\vartheta_1 = c_{\mathrm{fix}} L^{-1/2}$.

(O6) The number c_R, with $e_R = \lambda_* c_R/2 \ge 256$ (Step 0 (i) of the proof of Thm. 5.63), and $\varrho(x) = \lceil c_R \log x \rceil$.

(O13) Of the conditions on the last floor x_s, this section rests on (X12) and (X13), so that $\mathsf{b} \ge x_s$ meets the requirement $\mathsf{b} \ge x_\flat$ of §2.2 and of (RU), and on (X5) below; in (S5)(iii) alone, through Cor. 5.49, it rests also on (X14) $p_0(x) \ge 1$.

The data are: $\mathsf{b} \ge x_s$; N even; $K' \ge K \ge 1$; $m = K' - K$; $x_k = L^{K-k}\mathsf{b}$; $a_K = L^{-K}$; and $\mathbb{T}_k = \mathbb{Z}^3/NL^{K-k}\mathbb{Z}^3$ for $0 \le k \le K$, each in the units of its own level. In step k we have $\mathbb{T} = \mathbb{T}_k$, and $\mathbb{T}' = L\mathbb{Z}^3/NL^{K-k}\mathbb{Z}^3$ is $\mathbb{T}_{k+1}$.

Definition 6.15 (Rows, partners and the kernels of the step)**.** *Rows.* We write $\ell_k = \ell_k^{[K]}$ and $\tilde{\ell}_j = \ell_j^{[K']}$ for the scalar chains of the runs K and K', as given by the definition "Runs" in item 2 of §5.5: $\ell_0 = \emptyset$, and $\ell_{k+1} = ((\ell_k)')^{\mathrm{c}}$ is obtained by cutting the rows of Prop. 2.28 at $\varrho(x_{k+1})$. This recursion is used for every even N, and the condition $M \mid N$ imposed in that definition plays no part in these chains.

Site forms. $\mathsf{E}_{k,y}(p,p') = \langle \ell_{k,y}(p), \ell_{k,y}(p') \rangle$, so that $\mathsf{B}_k = 1 + \sum_y \mathsf{E}_{k,y}$; the forms $\tilde{\mathsf{E}}_j$ are given by the same formula with the rows $\tilde{\ell}_j$ in place of the rows ℓ_k.

Partners. Level $m+k$ of the run K' lives on $\mathbb{T}_k$, and $\tilde{x}_{m+k} = x_k$; partners are therefore cut at the same radius.

Kernels of the step. We let $H_k := H^\sharp$ be the map of Prop. 2.27 (a) at the rows ℓ_k, from $\Omega^2(\mathbb{T}_{k+1})$ to $\Omega^2(\mathbb{T}_k)$; $\tilde{H}_j$ is the same map at the rows $\tilde{\ell}_j$. We write δ_P and δ_p for the 2-forms on the coarse level and on the fine level that are 1 at P and at p, respectively, and 0 at every other plaquette. The column P of $H^\sharp$ is $\nu^P := H^\sharp \delta_P$, which is denoted ν in §2.2.

Blocks. $B(y')$ is the centred block. Further, $B^0(y) = \{y\}$ and $B^{k+1}(y') = \bigcup_{y \in B(y')} B^k(y)$; thus $B^k(y)$ consists of the sites z of $\mathbb{T}_0$ with $|z - L^k y|_\infty \le (L^k - 1)/2$. ■

Definition 6.16 (Places)**.** Throughout, f is a complex bounded function on $T_N = \mathbb{R}^3/N\mathbb{Z}^3$. The site y of $\mathbb{T}_k$ is placed at $a_K L^k y$; in the run K', the site y of $\tilde{\mathbb{T}}_j$ is placed at $a_{K'} L^j y$, so partners have the same places. For a plaquette p, $\iota(p) \in \{1,2,3\}$ is the direction normal to p, and $(z;\iota)$ is the plaquette based at z with normal e_ι, spanned by e_μ, e_ν. Its barycentre is $x(p) = x^{[K]}(z;\iota) = a_K(z + \frac{1}{2}(e_\mu + e_\nu))$ (entry 13 of item (0) of §6.1). We denote by h_ι the constant 2-form whose component ι equals 1 and whose other components are 0. The Lipschitz constant Lip is taken with respect to the Euclidean distance, and $\|f\|_\sharp = \max\{\|f\|_\infty, 40M \operatorname{Lip} f\}$. ■

Definition 6.17 (Families, norm, charge, step and cut)**.** A family on a level $\mathbb{T}$ is $D = (D_y)_{y \in \mathbb{T}}$, where every D_y is a symmetric complex scalar kernel on pairs of plaquettes of $\mathbb{T}$, and we write $D_y(F,G) = \sum_{p,p'} D_y(p,p') F_p G_{p'}$. The norm of a family is

$$\|D\| = \|D\|_{(\lambda_\flat)} := \sup_{y,p,p'} |D_y(p,p')| e^{\lambda_\flat(|y-p|+|y-p'|)},$$

with $|y-p|$ and $|y-p'|$ measured in the units of the level $\mathbb{T}$ (Def. 2.43). The charge of D at y is the 3×3 matrix $c_y(D)(\iota,\iota') := D_y(h_\iota, h_{\iota'})$, and $|c(D)|$ is the supremum over y of its operator norm. The step at the rows ℓ is $(T_\ell D)_{y'} := L \sum_{y \in B(y')} H^{\mathsf{T}} D_y H$, with $H = H^\sharp$ at ℓ; entry by entry, $(T_\ell D)_{y'}(P,P') = L \sum_{y \in B(y')} D_y(\nu^P, \nu^{P'})$. The cut at radius ϱ of a family D' on the coarse level is

$$(\mathrm{cut}_\varrho D')_{y'}(P,P') := D'_{y'}(P,P') 1\{|y' - P| \le \varrho,\ |y' - P'| \le \varrho\}.$$

■

Definition 6.18 (Source forms and Riemann means)**.** The source forms are given by

$$\mathbb{D}_{0,y}[f] := \sum_{p \text{ based at } y} f(x(p))\, \delta_p \delta_p^{\mathsf{T}}$$

and, recursively, by $\mathbb{D}_{k+1}[f] := \mathrm{cut}_{\varrho(x_{k+1})}(T_{\ell_k} \mathbb{D}_k[f])$. The forms $\tilde{\mathbb{D}}_j[f]$ of the run K' are given by the same two formulas with $x^{[K']}$, $\tilde{\ell}_j$ and $\varrho(\tilde{x}_{j+1})$ in place of $x^{[K]}$, ℓ_k and $\varrho(x_{k+1})$. The Riemann means are $\mathrm{Mn}^{[K]}_{k,y,\iota}(f) := L^{-3k} \sum_{z \in B^k(y)} f(x^{[K]}(z;\iota))$, and we set

$$\mathrm{os}_i := \sup_{y \in \mathbb{T}_i, \iota} |\mathrm{Mn}^{[K']}_{m+i,y,\iota}(f) - \mathrm{Mn}^{[K]}_{i,y,\iota}(f)|,$$

together with $\mathrm{osc}_K(f) := \mathrm{os}_0$. ■

Item 2: the numbers and their places.

Definition 6.19 (Entries of the table)**.** (O0a) Pure numbers: $\mathsf{g}_\flat := \mathsf{g}(\lambda_\flat)$, $\mathsf{g}' := \mathsf{g}(\lambda_\flat/2)$ and $\mathsf{g}_1' := \mathsf{g}_1(\lambda_\flat/2)$. Next, $C_{\mathrm{ch}} := 4e^{2\lambda_*}/\gamma_*$ and $K_5 := 1.35\,C_{\mathrm{ch}}$ $(= 0.60\,e^{2\lambda_*}K_*^2 \le 0.61K_*^2)$. Then $a_G := (23b_*^2)^{15/16}(7\gamma_*^{-1/2})^{1/16}$, $c_s := 63e^{2\lambda_*}\mathsf{g}_1'\mathsf{g}'a_G\gamma_*^{-1/2}$ and $t_5 := 18\mathsf{g}_\flat\mathsf{g}'$. Further, $a_H := (4\gamma_*^{-1/2})^{1/16}(3\mathsf{g}_\flat^2 b_*^{1/2})^{15/16}$ and $c_I := 12e^{2\lambda_*}\mathsf{g}'^2 a_H\gamma_*^{-1/2}$. Finally, $A_1 := \max\{1, 15C_1^2\}$, $I_5 := c_I K_5 A_1$ and $C_5 := 0.3(K_5+1) + 1.02I_5$.

(O4a) The condition (L5*) $c_s \le L^{43/160}$. It depends on (O0) alone, and it follows from $L \ge L_B$ (Rem. 6.23).

(O13) On x_s, the condition (X5) $1.01 \cdot \frac{4}{3}t_5 C_{\mathrm{ch}} L x_s^{-16} \le \frac{1}{100}$, which depends on (O0) and (O4).

(O14) The numbers $\vartheta_s := c_s L^{-15/32}$, $\vartheta_5 := \max\{\vartheta_s, \vartheta_1^{15/16}\}$ and $C_5^{\#} := L^{15/32}C_5 + K_5$, on which no floor depends. Then $L^{-15/32} \le \vartheta_s \le \vartheta_5 \le L^{-1/5} \le \frac{1}{4}$.

Numerals. The numerals come from those of "The order of the constants", (0), §2.2, and from the bounds $\mathsf{g}(\lambda) \le 51\lambda^{-3}$ and $\mathsf{g}_1(\lambda) \le 182\lambda^{-4}$ of Def. 2.44. We have $\log_{10}\mathsf{g}_\flat \le 46.32$, $\log_{10}\mathsf{g}' \le 47.22$, $\log_{10}\mathsf{g}_1' \le 62.95$, $\log_{10}C_{\mathrm{ch}} \le 24.646$, $\log_{10}K_5 \le 24.776$, $\log_{10}a_G \le 21.352$, $\log_{10}c_s \le 145.34$, $\log_{10}t_5 \le 94.79$, $\log_{10}a_H \le 92.90$, $\log_{10}c_I \le 200.44$, $\log_{10}A_1 \le 157.47$, $\log_{10}I_5 \le 382.68$ and $\log_{10}C_5 \le 382.69$. Accordingly, (L5*) holds for $\log_{10}L \ge 540.8$, and (X5) holds once $16\log_{10}x_s \ge 121.57 + \log_{10}L$. None of the constants below depends on $\mathsf{b}, N, K, K', k, f$. ■

Item 3: the free flow of the source forms.

Theorem 6.20 (Bound, range and locality of the source forms)**.** With the letters and data of item 1, and under the conditions (L5*) and (X5), the following hold for all $\mathsf{b} \ge x_s$, all even N, all $K \ge 1$, all $0 \le k \le K$ and all complex bounded f.

(U2.1) $\|\mathbb{D}_k[f]\|_{(\lambda_\flat)} \le K_5\|f\|_\infty$; site by site, this reads $|\mathbb{D}_{k,y}[f](p,p')| \le K_5\|f\|_\infty e^{-\lambda_\flat(|y-p|+|y-p'|)}$.

(U2.2) $\mathbb{D}_{k,y}[f]$ is symmetric, linear in f, real when f is real, and of range $\le \varrho(x_k)$: it vanishes unless $|y-p|, |y-p'| \le \varrho(x_k)$. At $k=0$ its range is 0.

(U2.3) $\mathbb{D}_{k,y}[f]$ depends on f only through the values of f at the barycentres of the plaquettes based in $B^k(y)$; in particular, it is 0 if f vanishes at all of them.

■□

Item 4: one step on families of kernels, and two runs.

Theorem 6.21 (One step on families of kernels, and the source forms of two runs)**.** Under (O0) to (O13), (L5*) and (X5), the following hold for all data of item 1.

(S0) *One step.* Let $\mathbb{T} = \mathbb{Z}^3/N'L\mathbb{Z}^3$ with N' even, let $\mathbb{T}'$ be its coarse torus, let D be any family on $\mathbb{T}$, and let D' be any family on $\mathbb{T}'$.

- (a) For every translation covariant ℓ on $\mathbb{T}$, cut or not, and whatever $\tau(\ell)$ is, $H^\sharp h' = L^{-2}h$ and $\mathsf{N}^\sharp h' = 0$ for every constant form h' on $\mathbb{T}'$, where h is the constant form on $\mathbb{T}$ with the components of h'.
- (b) For $\ell \in \mathcal{R}(b_*, K_*, \lambda_*, \frac{1}{9})$, $\|T_\ell D\| \le \vartheta_s\|D\| + C_{\mathrm{ch}}|c(D)|$.
- (c) For every translation covariant ℓ, the identity $c_{y'}(T_\ell D) = L^{-3}\sum_{y\in B(y')} c_y(D)$ holds exactly.
- (d) $\|\mathrm{cut}_\varrho D'\| \le \|D'\|$ and $|c_{y'}(\mathrm{cut}_\varrho D') - c_{y'}(D')| \le t_5 e^{-\lambda_\flat\varrho/2}\|D'\|$; at $\varrho = \varrho(x)$ with $x \ge 1$, the right side is $\le t_5 x^{-e_R/16}\|D'\| \le t_5 x^{-16}\|D'\|$.
- (e) For $\ell, \tilde\ell \in \mathcal{R}(b_*, K_*, \lambda_*, \frac{1}{9})$ on $\mathbb{T}$, and with $\delta := \|\tilde{\mathsf{E}} - \mathsf{E}\|_{(\lambda_\flat)}$ of any size, $\|(T_{\tilde\ell} - T_\ell)D\| \le c_I\delta^{15/16}\|D\|$, and every charge of $(T_{\tilde\ell} - T_\ell)D$ vanishes.

(S1) The assertions of Thm. 6.20 hold. Besides, $\mathbb{D}_{k,y+\mathsf{t}}[f(\cdot - a_K L^k\mathsf{t})](p+\mathsf{t}, p'+\mathsf{t}) = \mathbb{D}_{k,y}[f](p,p')$ for every translation t of $\mathbb{T}_k$.

(S2) *Charges.* For $0 \le k \le K$,

$$|c_y(\mathbb{D}_k[f]) - \mathrm{diag}_\iota \mathrm{Mn}^{[K]}_{k,y,\iota}(f)| \le 0.0102\,L^{-1}(x_s/x_k)^{16}\|f\|_\infty.$$

(S3) *Two runs.* For $1 \le k \le K$,

$$\|\tilde{\mathbb{D}}_{m+k}[f] - \mathbb{D}_k[f]\|_{(\lambda_\flat)} \le K_5 \operatorname{osc}_K(f) + C_5 k \vartheta_5^{k-1} \|f\|_\infty.$$

Here os_i does not increase with i, and $\operatorname{osc}_K(f) \le \sqrt{3} a_K \mathrm{Lip} f \le (\sqrt{3}/40M) a_K \|f\|_\sharp$; at $k = 0$ the left side is $\le (K_5 + 1)\|f\|_\infty$. *Finer form:* if $\omega_5, P_5 \ge 0$ and $\mathrm{os}_i \le \omega_5 + P_5 \vartheta_5^i \|f\|_\infty$ for all $i < k$, then the right side may be replaced by $K_5 \omega_5 + (C_5 + 1.02 C_{\mathrm{ch}} P_5) k \vartheta_5^{k-1} \|f\|_\infty$.

(S3c) *Corollary.* For $1 \le k \le K$ and Lipschitz f, $\|\tilde{\mathbb{D}}_{m+k}[f] - \mathbb{D}_k[f]\|_{(\lambda_\flat)} \le C_5^\# k \vartheta_5^k \|f\|_\sharp$, with $\vartheta_5 \le L^{-1/5}$.

(S4) *The hybrid chain.* For $1 \le k_0 \le K$, let $\mathbb{D}^b_{k_0} := \tilde{\mathbb{D}}_{m+k_0}[f]$ and $\mathbb{D}^b_{k+1} := \mathrm{cut}_{\varrho(x_{k+1})}(T_{\ell_k} \mathbb{D}^b_k)$. Then $\|\mathbb{D}^b_k - \mathbb{D}_k[f]\| \le 2.1 K_5 \operatorname{osc}_K(f) + 1.2 C_5 k_0 \vartheta_5^{k_0 - 1} \|f\|_\infty$ for $k_0 \le k \le K$.

(S5) *Other forms.*

(i) *Totals.* Let $N_k[f] := \sum_y \mathbb{D}_{k,y}[f]$. The operator norm of $\tilde{N}_{m+k}[f] - N_k[f]$ is $\le 3\mathsf{g}_\flat^2$ times the right side of (S3), and its entries are $\le \mathsf{g}' e^{-(\lambda_\flat/2)|p-p'|}$ times that right side.

(ii) Let $f_1, \dots, f_n$ be Lipschitz with pairwise disjoint sets $\{f_i \ne 0\}$, and let $w = \sum z_i f_i$ with complex z_i. Then $\sum z_i \mathbb{D}_k[f_i] = \mathbb{D}_k[w]$, $\|w\|_\infty = \max |z_i| \|f_i\|_\infty$ and $\mathrm{Lip} w = \max |z_i| \mathrm{Lip} f_i$; so (S1) to (S4) hold for w with $\max_i$ and without any factor n.

(iii) For $1 \le k \le K$ and the family $D := (\tilde{\mathsf{E}}_{m+k} - \tilde{\mathbb{D}}_{m+k}[w/\mathsf{b}) - (\mathsf{E}_k - \mathbb{D}_k[w/\mathsf{b}),$

$$(c_Q^{\mathrm{su}}/E_a) x_k^{1/2} \|D\|_{(\lambda_\flat)} / p_0(x_k) \le c_4 \mathsf{b}^{1/2} L^{K/2 - 9k/10} + \mathrm{amp}(x_k) \big[\operatorname{osc}_K(w) + (C_5/K_5) k \vartheta_5^{k-1} \|w\|_\infty\big],$$

where c_4 comes from Cor. 5.49, and the left side equals δ_Δ, the number of Lem. 5.47, when that number is evaluated at this D.

(S6) *Bounded f.*

(i) For $K \ge 3$ and $m \ge 1$, the bound (S3) without the term $K_5 \operatorname{osc}_K(f)$ fails for a bounded Borel f; the instance is that of Step 10 (i).

(ii) If f is the indicator of a box, that is, of a product of three arcs, then $\mathrm{os}_i \le 12 L^{-i} \le 12 \vartheta_5^i$; the finer form therefore holds with $\omega_5 = 0$ and $P_5 = 12$.

(iii) If f has a modulus of continuity ω_f, then $\operatorname{osc}_K(f) \le \omega_f(\sqrt{3} a_K)$.

■□

Item 5: three remarks.

Remark 6.22 (The uncut chain). Let ℓ_k be the uncut chain, in which all radii are infinite; let $H_k = H^\sharp$ at ℓ_k and $\mathbb{H}_k := H_0 H_1 \cdots H_{k-1}$; and let $D^{\mathrm{unc}}_k[f]$ be the forms built without any cut. Then

$$D^{\mathrm{unc}}_{k,y}[f](P, P') = L^k \sum_{p \text{ based in } B^k(y)} f(x(p)) \mathbb{H}_k(p, P) \mathbb{H}_k(p, P')$$

and

$$|D^{\mathrm{unc}}_{k,y}[f](P, P')| \le 243 K_*^2 \|f\|_\infty e^{-\lambda_*(|y-P| + |y-P'|)}$$

, so that the left side is also $\le 250 K_*^2 \|f\|_\infty e^{-(\lambda_*/4)(|y-P| + |y-P'|)}$. For $f \ge 0$ the total obeys $0 \le N^{\mathrm{unc}}_k[f] \le \|f\|_\infty \mathsf{B}_k$.

Remark 6.23 ((L5*) follows from $L \ge L_B$). We have $c_s \le 10^{145.34} < 10^{204.89} \le c_{\mathrm{fix}}$; therefore $L \ge L_B \ge c_{\mathrm{fix}}^{10}$ gives $L^{43/160} \ge c_{\mathrm{fix}}^{43/16} \ge c_s$. Moreover, $L_B \ge c_{\mathrm{fix}}^{10} \ge 10^{2048.9}$, so this floor on L is larger than $10^{540.8}$, and item 2 shows that (L5*) holds for all $L \ge 10^{540.8}$.

Remark 6.24 (The kernel of the step). The step T_ℓ and the cut of this section coincide exactly with those of Thm. 6.33, namely $D'_{y'} = \mathrm{cut}_{\varrho(x')}(T_\ell D)_{y'}$ with $H^\sharp$ at the rows ℓ. Like the rows, the source forms are chosen by hand.

Proof of Thms. 6.20 and 6.21 and of Rems. 6.22 and 6.23. Step 1: the columns $\nu^P = H^\sharp \delta_P$; constant forms. This step proves (S0)(a).

Lemma 6.25 (Bounds on the columns of the minimiser, and its value on constants)**.** Let $\ell \in \mathcal{R}(b_*, K_*, \lambda_*, \frac{1}{9})$ be rows on $\mathbb{T}$, and let $L \geq L_A$. For a set A of sites, let $|F|_A$ be the ℓ^2 norm of F over the plaquettes based in A, and let $|\nabla F|_A^2 := \sum_{\alpha,\iota} \sum_{z\in A} |\partial_\alpha F_\iota(z)|^2$. Let y' be a coarse site, P a coarse plaquette and $v \in \mathbb{Z}^3$. Then the following hold.

(a) $|\nu^P|_{B(y')+v} \leq 2\gamma_*^{-1/2} L^{-1/2} e^{\lambda_*(1+|v|/L)} e^{-\lambda_* \mathrm{dist}(y',P)}$.
(b) $|\nabla \nu^P| \leq 23 b_*^2 L^{-1}$.
(c) $|\nabla \nu^P|_{B(y')+v} \leq a_G L^{-31/32} e^{\lambda_\flat(1+(|v|+1)/L)} e^{-\lambda_\flat \mathrm{dist}(y',P)}$.
(d) The assertion of (S0)(a) holds, for every translation covariant ℓ.

■

Proof of Lem. 6.25. We take the four assertions in turn.

(a) At $b = b_*$, $K = K_*$ and $\lambda = \lambda_*$, the rate λ' of Lem. 2.52 (c) is λ_*, since $L \geq L_A$ (proof of Thm. 2.57 (a)), and $\gamma_\mathrm{e} = \gamma_*$. Prop. 2.55 (a), taken at $\lambda'' = \lambda_*$, thus yields $\sum_p e^{2\lambda_* \mathsf{d}(p,P)} (\nu_p^P)^2 \leq 4/(\gamma_* L)$. For p based in $B(y') + v$, the triangle inequality for the periodic distance gives $\mathsf{d}(p, P) \geq \mathrm{dist}(y', P) - (\mathsf{h}_L + |v|)/L$, with $\mathsf{h}_L = (L-1)/2$ from Def. 2.17, and the right side is in turn $\geq \mathrm{dist}(y', P) - \frac{1}{2} - |v|/L$. Keeping in the sum only the plaquettes based in $B(y') + v$ and inserting this lower bound for $\mathsf{d}(p, P)$, we obtain (a). The empty rows also belong to the set of the lemma, as they lie in $\mathcal{R}(1, 0, \lambda_*, 0)$.

(b) We apply Lem. 2.54 with $X' = \delta_P$ and $b = b' = b_*$; the choice $b' = b_*$ is allowed, since $\|\mathsf{B}'\| \leq \frac{3}{2}/\mathsf{r}_\sharp \leq b_*$ (Thm. 2.57 (a)). Under $b_\mathsf{B} \leq b$ and $\|\mathsf{B}'\| \leq b'$, the lemma gives $|\nabla H^\sharp X'| \leq 23\, b\, b'\, L^{-1} |X'|$, whence the bound $23 b_*^2 L^{-1}$.

(c) For all α, ι and z, $|\partial_\alpha \nu_\iota(z)|^2 \leq 2\nu_\iota(z+e_\alpha)^2 + 2\nu_\iota(z)^2$. Summing over ι, over $z \in A = B(y') + v$ and over the three directions α, and then applying (a) to the shifts v and $v + e_\alpha$, each of length at most $|v| + 1$, we get $|\nabla \nu|_A^2 \leq \sum_\alpha (2|\nu|_{A+e_\alpha}^2 + 2|\nu|_A^2) \leq 12$ times the square of the bound (a) at $|v| + 1$. Hence $|\nabla \nu|_A \leq \sqrt{48} \gamma_*^{-1/2} L^{-1/2} e^{\lambda_*(1+(|v|+1)/L)} e^{-\lambda_* \mathrm{dist}}$, where $\sqrt{48} \leq 6.93 \leq 7$. Let X be the bound (b) and Y the bound just found; both bound $|\nabla \nu|_A$, and we use $\min\{X, Y\} \leq X^{15/16} Y^{1/16}$. The power of L is then $-\frac{15}{16} - \frac{1}{32} = -\frac{31}{32}$, the rate is $\lambda_*/16 = \lambda_\flat$ and the constant is a_G; this is (c).

(d) The pair $(L^{-2}h, 0)$ is admitted at h' in the sense of Lem. 2.53: indeed $Wh = L^2 h'$, because $\sum_p W(P, p) = L^2$ (Lem. 2.18), and $dh = 0$. Now let (F, n) be any pair admitted at h', and add up the relation $WF + \mathrm{d}'\mathsf{n} = h'$ over the N'^3 plaquettes P of one component ι. Since $\sum_P W(P, p) = 1/L$ (Lem. 2.18) and $\sum_P (\mathrm{d}'\mathsf{n})_P = 0$, each bond variable entering twice and with opposite signs, we find $(1/L) \sum_{p:\iota(p)=\iota} F_p = N'^3 h'_\iota$. As there are $(N'L)^3$ such p, the mean of F_ι is $L^{-2} h'_\iota$; that is, $F = L^{-2}h + G$ with G of mean 0 in each component. The form B commutes with all translations of $\mathbb{T}$ (Def. 2.46), so $\mathsf{B}h$ is a constant form and $\langle \mathsf{B}h, G\rangle = 0$. Consequently

$$\|(F, \mathsf{n})\|_\mathcal{H}^2 = L^{-4} \langle h, \mathsf{B}h\rangle + \langle G, \mathsf{B}G\rangle + a|\mathsf{n}|^2 \geq \|(L^{-2}h, 0)\|_\mathcal{H}^2.$$

Lem. 2.53 (a) states that $(H^\sharp h', \mathsf{N}^\sharp h')$ is an admitted pair and that no admitted pair has a smaller norm; the last display, applied to this pair, shows that its norm is at least that of $(L^{-2}h, 0)$, and hence the two norms coincide. The last identity of Lem. 2.53 (a), at $(F, \mathsf{n}) = (L^{-2}h, 0)$, then gives $\|(L^{-2}h, 0) - (H^\sharp h', \mathsf{N}^\sharp h')\|_\mathcal{H}^2 = 0$, and $\mathsf{B} \geq 1$ together with $a > 0$ finishes the proof of (d). This argument does not depend on $\tau(\ell)$; in this it sharpens Prop. 2.55 (c).

The statement of clause (S0)(a) is also Lem. 5.76 (a), which appears in Step 1a of the proof of Thm. 5.63, where it has a different proof.

□

Step 2: one step on a family, away from its charges. This step proves (S0)(b), (c) and (d).

Lemma 6.26 (One step contracts a family away from its charges)**.** Let $\ell \in \mathcal{R}(b_*, K_*, \lambda_*, \frac{1}{9})$, and let D be any family on $\mathbb{T}$. Then

$$|(T_\ell D)_{y'}(P, P')| \le (\vartheta_s \|D\| + C_{\mathrm{ch}}|c(D)|)e^{-\lambda_\flat(|y'-P|+|y'-P'|)}.$$

Moreover, $c_{y'}(T_\ell D)$ is the mean of $c_y(D)$ over $B(y')$. Finally, a cut at radius ϱ never increases the norm, and it changes a charge by at most $18\mathsf{g}_\flat\mathsf{g}'e^{-\lambda_\flat\varrho/2}$ times the norm. ∎

Proof of Lem. 6.26. Freezing at y. We write $\nu = \nu^P$ and $\nu' = \nu^{P'}$, and we let $\nu(y)$ be the constant form whose components are the numbers $\nu_\iota(y)$. By bilinearity,

$$D_y(\nu, \nu') = D_y(\nu(y), \nu'(y)) + D_y(\nu - \nu(y), \nu') + D_y(\nu(y), \nu' - \nu'(y)),$$

and $D_y(\nu(y), \nu'(y)) = \sum_{\iota,\iota'} c_y(D)(\iota, \iota')\nu_\iota(y)\nu'_{\iota'}(y)$.

The charge term. The first term contributes to $(T_\ell D)_{y'}(P, P')$ an amount of modulus at most

$$L \sum_{y \in B(y')} |c_y(D)||\nu(y)|_2|\nu'(y)|_2 \le L|c(D)||\nu|_{B(y')}|\nu'|_{B(y')},$$

where we use $|c_y(D)| \le |c(D)|$ and then the Cauchy–Schwarz inequality over $y \in B(y')$. By Lem. 6.25 (a) at $v = 0$, the right side is $\le 4e^{2\lambda_*}\gamma_*^{-1}|c(D)|e^{-\lambda_*(\mathrm{dist}(y',P)+\mathrm{dist}(y',P'))}$, and the factor in front of the exponential is $C_{\mathrm{ch}}|c(D)|$.

The cross terms. Let $\mathrm{Gr}_y(\nu) := \sum_p e^{-\lambda_\flat|y-p|}|\nu_p - \nu_{\iota(p)}(y)|$ and $\mathrm{Ab}_y(\nu') := \sum_p e^{-\lambda_\flat|y-p|}|\nu'_p|$. By the definition of the norm, $|D_y(\nu - \nu(y), \nu')| \le \|D\|\mathrm{Gr}_y(\nu)\mathrm{Ab}_y(\nu')$, and the sum over $y \in B(y')$ is then estimated by the Cauchy–Schwarz inequality. In the sums over p we write $p = (y + u; \iota)$, where $u \in \mathbb{Z}^3$ is a representative with $|u|_\infty = |y - p|$.

For Gr, Minkowski's inequality in $\ell^2(y \in B(y'))$ gives

$$(\sum_y \mathrm{Gr}_y^2)^{1/2} \le \sum_{u,\iota} e^{-\lambda_\flat|u|}(\sum_y |\nu_\iota(y+u) - \nu_\iota(y)|^2)^{1/2}.$$

The difference $\nu_\iota(y+u) - \nu_\iota(y)$ is a sum of $|u|_1 \le 3|u|$ differences $\pm\partial_\alpha\nu_\iota(y + v_s)$ with $|v_s| \le |u|$. So the inner root is $\le 3|u| \max_{|v|\le|u|}|\nabla\nu|_{B(y')+v}$, and the sum over ι multiplies this bound by 3. Lem. 6.25 (c), together with $\lambda_\flat(1 - 1/L) \ge \lambda_\flat/2$ and $1 + 1/L \le 2$, then gives

$$(\sum_y \mathrm{Gr}_y^2)^{1/2} \le 9\mathsf{g}'_1 a_G e^{2\lambda_\flat} L^{-31/32} e^{-\lambda_\flat \mathrm{dist}(y',P)}.$$

For Ab, Minkowski's inequality in $\ell^2(y \in B(y'))$, the Cauchy–Schwarz inequality over ι, Lem. 6.25 (a) and $\lambda_*/L \le \lambda_\flat/2$, which holds since $L \ge 32$, give

$$(\sum_y \mathrm{Ab}_y^2)^{1/2} \le \sum_u e^{-\lambda_\flat|u|}\sum_\iota(\sum_y \nu'_\iota(y+u)^2)^{1/2} \le \sqrt{3}\sum_u e^{-\lambda_\flat|u|}|\nu'|_{B(y')+u} \le 2\sqrt{3}\mathsf{g}'\gamma_*^{-1/2}e^{\lambda_*}L^{-1/2}e^{-\lambda_*\mathrm{dist}(y',P')}.$$

In the second cross term, the factor $\mathrm{Gr}_y(\nu')$ obeys the bound found for Gr with P' in place of P, and the frozen factor is $\sum_p e^{-\lambda_\flat|y-p|}|\nu_{\iota(p)}(y)| \le \mathsf{g}_\flat\sum_\iota|\nu_\iota(y)|$, whose $\ell^2(y)$ norm is $\le \sqrt{3}\mathsf{g}_\flat|\nu|_{B(y')}$; as $\mathsf{g}_\flat \le \mathsf{g}'$, the frozen factor obeys the bound found for Ab, with P in place of P'.

Conclusion. Apart from their exponential factors, the two cross terms contribute at most

$$L \cdot 2 \cdot 9\mathsf{g}'_1 a_G e^{2\lambda_\flat} L^{-31/32} \cdot 2\sqrt{3}\mathsf{g}'\gamma_*^{-1/2}e^{\lambda_*}L^{-1/2}\|D\| = 36\sqrt{3}e^{2\lambda_\flat+\lambda_*}\mathsf{g}'_1\mathsf{g}'a_G\gamma_*^{-1/2}L^{-15/32}\|D\| \le c_s L^{-15/32}\|D\|,$$

because $36\sqrt{3} \le 62.36 \le 63$, $2\lambda_\flat \le \lambda_*$ and $1 - \frac{31}{32} - \frac{1}{2} = -\frac{15}{32}$. For coarse cells, $\mathrm{dist}(y', P)$ is $|y' - P|$ measured in the units of the level $\mathbb{T}'$ (Def. 2.19), and $e^{-\lambda_*} \le e^{-\lambda_\flat}$; so every exponential factor is at most $e^{-\lambda_\flat(|y'-P|+|y'-P'|)}$, which proves the first assertion of the lemma and, with it, (S0)(b).

We turn to (S0)(c) and (S0)(d).

(c) By (S0)(a),

$$(T_\ell D)_{y'}(h'_\iota, h'_{\iota'}) = L \sum_{y \in B(y')} D_y(L^{-2}h_\iota, L^{-2}h_{\iota'}) = L^{-3} \sum_{y \in B(y')} c_y(D)(\iota, \iota').$$

(d) The cut sets entries to 0; hence it does not raise the norm. The operator norm of a 3×3 matrix is at most the sum of the moduli of its entries, and the entries that the cut removes have $|y'-P|>\varrho$ or $|y'-P'|>\varrho$; the sum of their moduli is therefore

$$\le \|D'\|\cdot 2\cdot (3\sum_{|u|>\varrho} e^{-\lambda_\flat|u|})(3\mathsf{g}_\flat) \le 18\mathsf{g}_\flat\mathsf{g}' e^{-\lambda_\flat\varrho/2}\|D'\|.$$

At $\varrho=\varrho(x)$, finally, $e^{-\lambda_\flat\varrho(x)/2}\le x^{-\lambda_* c_R/32}=x^{-e_R/16}\le x^{-16}$.

□

Step 3: two sets of rows on one torus. This step proves (S0)(e).

Lemma 6.27 (The step at two sets of rows)**.** Let $\ell,\tilde\ell\in\mathcal{R}(b_*,K_*,\lambda_*,\frac19)$ be two sets of rows on $\mathbb{T}$ for one and the same a, let the objects defined with $\tilde\ell$ carry a tilde, and let $\delta=\|\tilde{\mathsf{E}}-\mathsf{E}\|_{(\lambda_\flat)}$. Then the following hold.

(a) $\|\tilde{\mathsf{B}}-\mathsf{B}\|\le 3\mathsf{g}_\flat^2\delta$.

(b) For every X', $|(\tilde H-H)X'|\le 3\mathsf{g}_\flat^2\delta(b_*/L)^{1/2}|X'|$.

(c) For every P,

$$(\sum_p e^{2\lambda_\flat \mathsf{d}(p,P)}|(\tilde H-H)(p,P)|^2)^{1/2}\le a_H L^{-1/2}\delta^{15/16}.$$

(d) The statement (S0)(e) holds for the pair $\ell,\tilde\ell$.

■

Proof of Lem. 6.27. We take the four clauses in turn.

(a) The entries of the difference of the two forms are $(\tilde{\mathsf{B}}-\mathsf{B})(p,p')=\sum_y(\tilde{\mathsf{E}}_y-\mathsf{E}_y)(p,p')$, and by the definition of δ each of them has modulus at most $\delta\sum_y e^{-\lambda_\flat(|y-p|+|y-p'|)}$. For a fixed p, the sum of these bounds over p' does not exceed $\delta\cdot\mathsf{g}_\flat\cdot 3\mathsf{g}_\flat$. The difference is a symmetric matrix, so its column sums obey the same bound as its row sums, and Schur's test yields (a).

(b) Fix X', write $\mathsf{B}_1=\mathsf{B}$ and $\mathsf{B}_2=\tilde{\mathsf{B}}$, and for $i=1,2$ let (F_i,n_i) be the minimiser at this X' for the form B_i, and $\|\cdot\|_i$ the norm $\|\cdot\|_{\mathcal{H}}$ built with B_i. Whether a pair is admitted at X' does not depend on the rows; hence both pairs are admitted at X' for either form.

Lem. 2.53 (a), applied in the model with $\mathsf{B}=\mathsf{B}_1$ to the admitted pair (F_2,n_2), gives $\|(F_2,\mathsf{n}_2)-(F_1,\mathsf{n}_1)\|_1^2=\|(F_2,\mathsf{n}_2)\|_1^2-\|(F_1,\mathsf{n}_1)\|_1^2$. Applied in the model with $\mathsf{B}=\mathsf{B}_2$ to the admitted pair (F_1,n_1), that is with the indices 1 and 2 exchanged, it gives $\|(F_1,\mathsf{n}_1)-(F_2,\mathsf{n}_2)\|_2^2=\|(F_1,\mathsf{n}_1)\|_2^2-\|(F_2,\mathsf{n}_2)\|_2^2$.

We add these two identities and write $\mathrm{d}F=F_2-F_1$, $\mathrm{d}\mathsf{n}=\mathsf{n}_2-\mathsf{n}_1$, $\mathrm{d}\mathsf{B}=\mathsf{B}_2-\mathsf{B}_1$. On the right the terms in n_1 and n_2 cancel, and since $\mathrm{d}\mathsf{B}$ is symmetric we arrive at

$$\langle \mathrm{d}F,(\mathsf{B}_1+\mathsf{B}_2)\mathrm{d}F\rangle+2a|\mathrm{d}\mathsf{n}|^2=\langle F_1,\mathrm{d}\mathsf{B}F_1\rangle-\langle F_2,\mathrm{d}\mathsf{B}F_2\rangle=-\langle \mathrm{d}F,\mathrm{d}\mathsf{B}(F_1+F_2)\rangle.$$

As $\mathsf{B}_i\ge 1$, the left side is at least $2|\mathrm{d}F|^2$, while the modulus of the right side is at most $|\mathrm{d}F|\,\|\mathrm{d}\mathsf{B}\|(|F_1|+|F_2|)$. Thus $2|\mathrm{d}F|^2\le|\mathrm{d}F|\,\|\mathrm{d}\mathsf{B}\|(|F_1|+|F_2|)$.

Moreover $|F_i|^2\le\|(F_i,\mathsf{n}_i)\|_i^2=\langle X',\mathsf{B}_i'X'\rangle/L\le b_*|X'|^2/L$, where the equality is given by Lem. 2.53 (a), and the second inequality by the bound $\|\mathsf{B}_i'\|\le\frac32/\mathsf{r}_\sharp\le b_*$ of Thm. 2.57 (a). We insert this into the last inequality and divide by $2|\mathrm{d}F|$ where this is not 0; the result is $|\mathrm{d}F|\le\|\mathrm{d}\mathsf{B}\|(b_*/L)^{1/2}|X'|$. Since $\mathrm{d}F=(\tilde H-H)X'$, the bound (a) on $\|\mathrm{d}\mathsf{B}\|$ turns this into (b).

(c) Fix P and let $v=(\tilde H-H)\delta_P$; thus $v=\tilde\nu-\nu$ is the difference of the columns of $\tilde H$ and H at P. Write d for $\mathsf{d}(p,P)$. Since $\lambda_\flat=\lambda_*/16$, Hölder's inequality with the exponents 16 and $16/15$ gives

$$\sum e^{2(\lambda_*/16)\mathsf{d}}v^2\le(\sum e^{2\lambda_*\mathsf{d}}v^2)^{1/16}(\sum v^2)^{15/16}.$$

To bound the first sum on the right we use $v^2 \le 2\tilde\nu^2 + 2\nu^2$ together with Prop. 2.55 (a), applied to each of the two columns with $\lambda'' = \lambda_*$ and $\gamma_{\mathrm e} = \gamma_*$ as in the proof of Lem. 6.25 (a), where the bound for one column is $4/(\gamma_* L)$; this sum is then at most $16/(\gamma_* L)$. The second sum equals $|(\tilde H - H)\delta_P|^2$, and (b) at $X' = \delta_P$ bounds it by $9\mathsf g_\flat^4 \delta^2 b_*/L$. The square root of the product of the two bounds is $(4\gamma_*^{-1/2})^{1/16}(3\mathsf g_\flat^2 b_*^{1/2})^{15/16}\delta^{15/16}L^{-1/2}$, and (c) follows.

(d) Let D be a family, and let ν, ν' and $\tilde\nu, \tilde\nu'$ denote the columns of H and of $\tilde H$ at P and at P'. The definition of the step and the bilinearity of D_y give

$$((T_{\tilde\ell} - T_\ell)D)_{y'}(P, P') = L \sum_{y \in B(y')} [D_y(\tilde\nu - \nu, \tilde\nu') + D_y(\nu, \tilde\nu' - \nu')].$$

Each of the two sums is at most $L\|D\|(\sum_y \mathrm{Ab}_y(\cdot)^2)^{1/2}(\sum_y \mathrm{Ab}_y(\cdot)^2)^{1/2}$, where the two vectors of that sum take the places of the dots.

We bound these factors by the argument of Step 2, now with (c) as the input for a difference of columns and with the inequality $\mathsf d(p, P) \ge \mathrm{dist}(y', P) - \frac12 - |u|/L$. This yields

$$(\sum_y \mathrm{Ab}_y(\tilde\nu - \nu)^2)^{1/2} \le \sqrt{3}\mathsf g' a_H e^{\lambda_\flat} L^{-1/2}\delta^{15/16} e^{-\lambda_\flat \mathrm{dist}(y', P)},$$

and (c) at P' gives the same bound for $\tilde\nu' - \nu'$, with P' in the place of P. For the factors of ν and of $\tilde\nu'$, the bound for Ab of Step 2 gives

$$(\sum_y \mathrm{Ab}_y(\nu)^2)^{1/2} \le 2\sqrt{3}\mathsf g' \gamma_*^{-1/2} e^{\lambda_*} L^{-1/2} e^{-\lambda_* \mathrm{dist}(y', P)}$$

and

$$(\sum_y \mathrm{Ab}_y(\tilde\nu')^2)^{1/2} \le 2\sqrt{3}\mathsf g' \gamma_*^{-1/2} e^{\lambda_*} L^{-1/2} e^{-\lambda_* \mathrm{dist}(y', P')}.$$

Twice the product of the two factors, times L, bounds the norm of $(T_{\tilde\ell} - T_\ell)D$ by $12 e^{\lambda_\flat + \lambda_*}\mathsf g'^2 a_H \gamma_*^{-1/2}\delta^{15/16}\|D\| \le c_I \delta^{15/16}\|D\|$. The charges of $(T_{\tilde\ell} - T_\ell)D$ vanish, because $\tilde H h' = H h' = L^{-2}h$ by (S0)(a). This proves (d).

□

Step 4: bounds uniform in k and K, with the cuts. We prove (S1), which is Thm. 6.20, and (S2). That is, we show for $0 \le k \le K$ the bound $\|\mathbb D_k[f]\| \le K_5\|f\|_\infty$ and

$$|c_y(\mathbb D_k[f]) - \mathrm{diag}\,\mathrm{Mn}^{[K]}_{k,y,\iota}(f)| \le 0.0102 L^{-1}(x_s/x_k)^{16}\|f\|_\infty,$$

and further (U2.2), (U2.3) and the covariance.

We begin with the rows. Since $\mathtt t_*(\varrho(x)) \le 3K_*\mathsf g(\lambda_*/2)x^{-e_R}$ and $x_j \ge L^{K-j}x_s$, (X12) yields the condition (Cut) of Thm. 2.57 (b); this deduction is the one made in Step (Cut) of the proof of Thm. 2.64. Consequently $\ell_k \in \mathcal R(b_*, K_*, \lambda_*, \frac19)$ for $0 \le k \le K$: for $k < K$ by Thm. 2.65 (Fr0), and for $k = K$ by Thm. 2.57 (b). So (S0) applies at every step.

We normalise $\|f\|_\infty = 1$ and put $\mathrm{Nm}_k := \|\mathbb D_k[f]\|$, $\mathrm{Ch}_k := |c(\mathbb D_k[f])|$ and $\mathrm{ct}_i := x_{i+1}^{-16}$. At the start, $\mathbb D_{0,y}$ is diagonal and is carried by the p with $|y - p| = 0$; hence $\mathrm{Nm}_0 \le 1$. Its charge is $c_y(\mathbb D_0) = \mathrm{diag}_\iota f(x(y;\iota))$, so that $\mathrm{Ch}_0 \le 1$.

From (S0)(b), (S0)(c) and (S0)(d) we obtain the recursion

$$\mathrm{Nm}_{k+1} \le \vartheta_s \mathrm{Nm}_k + C_{\mathrm{ch}}\mathrm{Ch}_k,$$

$$\mathrm{Ch}_{k+1} \le \mathrm{Ch}_k + t_5 \mathrm{ct}_k(\vartheta_s \mathrm{Nm}_k + C_{\mathrm{ch}}\mathrm{Ch}_k);$$

here we use that a mean of matrices of norm at most Ch_k has norm at most Ch_k.

To solve it, let $\mathrm{Ch}^* := \max\{1, \max_k \mathrm{Ch}_k\}$. Iterating the first inequality from $\mathrm{Nm}_0 \le 1$ and summing the geometric series, we find with $\vartheta_s \le \frac14$ and $C_{\mathrm{ch}} \ge 1$ that $\mathrm{Nm}_k \le \vartheta_s^k + \frac43(1-\vartheta_s^k)C_{\mathrm{ch}}\mathrm{Ch}^* \le \frac43 C_{\mathrm{ch}}\mathrm{Ch}^*$; it follows that $\vartheta_s \mathrm{Nm}_k + C_{\mathrm{ch}}\mathrm{Ch}_k \le \frac43 C_{\mathrm{ch}}\mathrm{Ch}^*$. The cut errors form a geometric series:

$$\sum_{i<k} \mathrm{ct}_i = \sum_{j=1}^{k} (x_k L^{k-j})^{-16} \le x_k^{-16}/(1-L^{-16}) \le 1.01 x_k^{-16}.$$

Summing the second inequality of the recursion and using $x_k \ge x_s$, we get $\mathrm{Ch}^* \le 1 + 1.01 t_5 \frac43 C_{\mathrm{ch}} x_s^{-16} \mathrm{Ch}^* \le 1 + \mathrm{Ch}^*/(100L)$, the last step by (X5). Therefore $\mathrm{Ch}^* \le 1/0.99 \le 1.0102$, and $\mathrm{Nm}_k \le \frac43 \cdot 1.0102\, C_{\mathrm{ch}} \le 1.347 C_{\mathrm{ch}} \le K_5$, which is the first bound.

Next we follow the charges. The mean $\mathrm{Mn}_{k+1,y'}$ is the mean of the $\mathrm{Mn}_{k,y}$ over $y \in B(y')$, since $B^{k+1}(y')$ is the disjoint union of the $B^k(y)$. Put $\mathrm{ce}_k := \sup_y |c_y(\mathbb{D}_k) - \operatorname{diag} \mathrm{Mn}_{k,y}|$. Then $\mathrm{ce}_0 = 0$, and by (S0)(c) and (S0)(d), $\mathrm{ce}_{k+1} \le \mathrm{ce}_k + t_5 \mathrm{ct}_k \frac43 C_{\mathrm{ch}} \mathrm{Ch}^*$. Summing these increments and using (X5) once more,

$$\mathrm{ce}_k \le 1.01 x_k^{-16} t_5 \frac{4}{3} C_{\mathrm{ch}} \cdot 1.0102 \le (1.0102/100L)(x_s/x_k)^{16},$$

and since $1.0102/100 \le 0.0102$ this is the second bound.

The remaining clauses are shown by induction on k. Symmetry, linearity in f and reality pass from one level to the next, because H is real and both the step and the cut are linear. The bound on the range comes from the cut. (U2.3) holds since the step sums over $y \in B(y')$ while the cut acts on P, P' alone. The covariance follows from three facts: H_k commutes with the translations of $\mathbb{T}_{k+1}$, as the proof of Prop. 2.55 (a) shows; translations map blocks to blocks; and the cut is defined by a distance. No positivity enters this argument, so f may be complex.

Step 5: the start of the comparison, Riemann means at barycentres. On $\mathbb{T}_k$ let $\mathrm{d}\mathbb{D}_k := \tilde{\mathbb{D}}_{m+k}[f] - \mathbb{D}_k[f]$. We show that $|c(\mathrm{d}\mathbb{D}_0)| \le \mathrm{osc}_K(f) + 0.0102 L^{-16K-1}\|f\|_\infty$ and $\|\mathrm{d}\mathbb{D}_0\| \le (K_5+1)\|f\|_\infty$, that $\mathrm{osc}_K(f) \le \sqrt{3} a_K \mathrm{Lip} f$, and that $\mathrm{os}_{i+1} \le \mathrm{os}_i$.

By (S2) in the run K' at its level m, where $\tilde{x}_m = L^K \mathsf{b}$, the charge $c_y(\tilde{\mathbb{D}}_m[f])$ coincides with $\operatorname{diag} \mathrm{Mn}^{[K']}_{m,y,\iota}(f)$ up to an error of at most $0.0102 L^{-1}(x_s/(L^K\mathsf{b}))^{16} \le 0.0102 L^{-16K-1}$ times $\|f\|_\infty$. In the run K we have exactly $c_y(\mathbb{D}_0[f]) = \operatorname{diag} f(x^{[K]}(y;\iota)) = \operatorname{diag} \mathrm{Mn}^{[K]}_{0,y,\iota}(f)$. These two facts give the bound on the charge. The bound on the norm follows from (S1) in the run K' and from $\mathrm{Nm}_0 \le 1$, that is $\|\mathbb{D}_0[f]\| \le \|f\|_\infty$.

For the distances, let $z \in B^m(y)$. Then $a_{K'}|z - L^m y|_\infty \le a_{K'}(L^m - 1)/2 = (a_K - a_{K'})/2$. The offsets of the two barycentres $x^{[K']}(z;\iota)$ and $x^{[K]}(y;\iota)$ differ by $(a_K - a_{K'})/2$ in the directions μ, ν, and by 0 in the normal one. Hence $|x^{[K']}(z;\iota) - x^{[K]}(y;\iota)|_\infty \le a_K - a_{K'} < a_K$, and the Euclidean distance of the two points is $< \sqrt{3} a_K$. So each term of the mean differs from $f(x^{[K]}(y;\iota))$ by $\le \sqrt{3} a_K \mathrm{Lip} f$, which is the bound on $\mathrm{osc}_K(f)$; for f with a modulus of continuity ω_f, each term differs from it by $\le \omega_f(\sqrt{3} a_K)$, so that $\mathrm{osc}_K(f) \le \omega_f(\sqrt{3} a_K)$, which is (S6) (iii).

Finally, $B^{m+i+1}(y')$ and $B^{i+1}(y')$ are the unions over $y \in B(y')$ of the blocks $B^{m+i}(y)$ and $B^i(y)$, respectively. Hence the difference of the two means at level $i+1$ is the mean of the differences at level i, and $\mathrm{os}_{i+1} \le \mathrm{os}_i$.

Step 6: the linear system for the difference of the two runs. We prove (S3). Let $\|f\|_\infty = 1$, and now let $\mathrm{Nm}_k := \|\mathrm{d}\mathbb{D}_k\|$ and $\mathrm{Ch}_k := |c(\mathrm{d}\mathbb{D}_k)|$. We show that

$$\mathrm{Nm}_k \le 1.347 C_{\mathrm{ch}} \mathrm{Ch}_0 + \vartheta_5^{k-1}[(\vartheta_5 + 0.0102)\mathrm{Nm}_0 + 1.0102 k I_5]$$

for $1 \le k \le K$, and we derive from this (S3) and its finer form.

Partners have one and the same x, hence one and the same cut radius; and the step and the cut are linear. Therefore

$$\mathrm{d}\mathbb{D}_{k+1} = \mathrm{cut}_{\varrho(x_{k+1})}[T_{\ell_k} \mathrm{d}\mathbb{D}_k + I_k],$$

with $I_k := (T_{\tilde{\ell}_{m+k}} - T_{\ell_k})\tilde{\mathbb{D}}_{m+k}[f]$.

By Step 4, applied to both runs, the rows ℓ_k and $\tilde{\ell}_{m+k}$ lie in $\mathcal{R}(b_*, K_*, \lambda_*, \frac{1}{9})$ on $\mathbb{T}_k$. We bound I_k with (S0)(e), with (S1) in the run K', and with (RU) of item 3 of §5.5, which gives $\delta_k := \|\tilde{\mathsf{E}}_{m+k} - \mathsf{E}_k\|_{(\lambda_\flat)} \le 15C_1^2\vartheta_1^k$; the hypotheses of (RU) are verified in Step 1 of the proof of Thm. 5.63. The result is $\|I_k\| \le c_I K_5 (15C_1^2)^{15/16}\vartheta_1^{15k/16} \le I_5\vartheta_5^k$, and the charges of I_k are 0.

From (S0)(b), (S0)(c), (S0)(d) and $\vartheta_s \le \vartheta_5$ we obtain the recursion

$$\mathrm{Nm}_{k+1} \le \vartheta_5 \mathrm{Nm}_k + C_{\mathrm{ch}}\mathrm{Ch}_k + I_5\vartheta_5^k,$$

$$\mathrm{Ch}_{k+1} \le \mathrm{Ch}_k + t_5 \mathrm{ct}_k(\vartheta_5\mathrm{Nm}_k + C_{\mathrm{ch}}\mathrm{Ch}_k + I_5\vartheta_5^k).$$

We solve it as follows, with $\mathrm{Ch}_k^* := \max_{i\le k}\mathrm{Ch}_i$.

(1) Iterating the first inequality and summing the geometric series with $\vartheta_5 \le \frac{1}{4}$, we get $\mathrm{Nm}_k \le \vartheta_5^k\mathrm{Nm}_0 + kI_5\vartheta_5^{k-1} + \frac{4}{3}C_{\mathrm{ch}}\mathrm{Ch}_{k-1}^*$.

(2) Hence, as $\frac{4}{3}\vartheta_5 + 1 \le \frac{4}{3}$,

$$\vartheta_5\mathrm{Nm}_i + C_{\mathrm{ch}}\mathrm{Ch}_i + I_5\vartheta_5^i \le \vartheta_5^{i+1}\mathrm{Nm}_0 + (i+1)I_5\vartheta_5^i + \frac{4}{3}C_{\mathrm{ch}}\mathrm{Ch}_i^*.$$

(3) We have $\mathrm{ct}_i = (x_k L^{k-1-i})^{-16}$ and $L^{-16}/\vartheta_5 \le L^{-16+15/32} \le 0.0099$; summing a geometric series of ratio at most 0.0099, we get $\sum_{i<k}\mathrm{ct}_i\vartheta_5^i \le 1.01 x_k^{-16}\vartheta_5^{k-1}$.

(4) The sums over $i<k$ that bound Ch_k grow with k, so they also bound Ch_k^*. With (2), (3) and the bound $1.01x_k^{-16}$ for $\sum_{i<k}\mathrm{ct}_i$ we obtain

$$\mathrm{Ch}_k^* \le \mathrm{Ch}_0 + 1.01t_5x_k^{-16}[\vartheta_5^k\mathrm{Nm}_0 + kI_5\vartheta_5^{k-1}] + 1.01t_5x_k^{-16}\frac{4}{3}C_{\mathrm{ch}}\mathrm{Ch}_k^*.$$

(5) By (X5), $1.01t_5x_k^{-16} \le 3/(400C_{\mathrm{ch}}L)$, so the last term of (4) is at most $\mathrm{Ch}_k^*/(100L)$; moreover $1/(L\vartheta_5) \le L^{-17/32} \le 1$. We move the last term to the left and divide by $1-1/(100L)$, whose inverse is at most 1.0102; this gives

$$\mathrm{Ch}_k^* \le 1.0102\mathrm{Ch}_0 + 1.0102(0.0075/C_{\mathrm{ch}})(\mathrm{Nm}_0 + kI_5)\vartheta_5^k \le 1.0102\mathrm{Ch}_0 + (0.0076/C_{\mathrm{ch}})(\mathrm{Nm}_0 + kI_5)\vartheta_5^k.$$

(6) We insert (5), taken at $k-1$, into (1). Since $\frac{4}{3}\cdot 1.0102 \le 1.347$ and $\frac{4}{3}\cdot 0.0076 \le 0.01014 \le 0.0102$, the outcome is the claim.

We now derive (S3). By Step 5, $\mathrm{Nm}_0 \le K_5 + 1$ and $\mathrm{Ch}_0 \le \mathrm{osc}_K(f) + 0.0102L^{-16K-1}$. Although the first of these is not small, the contraction does the work, since Nm_0 appears in the claim with the coefficient $\vartheta_5^{k-1}(\vartheta_5 + 0.0102)$. Further, we have $1.347C_{\mathrm{ch}} \le K_5$ and $L^{-16K-1}\vartheta_5^{1-k} \le L^{-16K-1+15K/32} \le 1$. Therefore

$$\begin{aligned}
&\mathrm{Nm}_k \\
&\quad\le K_5\mathrm{osc}_K(f) \\
&\quad+ \vartheta_5^{k-1}[(0.25 + 0.0102)(K_5+1) + 0.0102K_5 + 1.0102kI_5] \\
&\quad\le K_5\mathrm{osc}_K(f) \\
&\quad+ \vartheta_5^{k-1}[0.2704(K_5+1) + 1.0102kI_5] \\
&\quad\le K_5\mathrm{osc}_K(f) + C_5k\vartheta_5^{k-1}.
\end{aligned}$$

For the finer form, let $d_i := \sup_y |c_y(\mathrm{d}\mathbb{D}_i) - \mathrm{diag}(\mathrm{Mn}_{m+i,y}^{[K']} - \mathrm{Mn}_{i,y}^{[K]})|$, so that $\mathrm{Ch}_i \le \mathrm{os}_i + d_i$. As the means average exactly and I_i has charges 0, we have $d_{i+1} \le d_i + t_5\mathrm{ct}_i(\vartheta_5\mathrm{Nm}_i + C_{\mathrm{ch}}\mathrm{Ch}_i + I_5\vartheta_5^i)$, and $d_0 \le 0.0102L^{-16K-1}$. If $\mathrm{os}_i \le \omega_5 + P_5\vartheta_5^i$, then $\mathrm{Ch}_i \le (\omega_5 + d_i) + P_5\vartheta_5^i$, and the pair $(\mathrm{Nm}_i, \omega_5 + d_i)$ obeys the recursion above with $I_5 + C_{\mathrm{ch}}P_5$ in place of I_5 and with $\omega_5 + d_0$ in place of Ch_0, so that the claim, applied to this pair, gives

$$\mathrm{Nm}_k \le 1.347C_{\mathrm{ch}}(\omega_5 + d_0) + \vartheta_5^{k-1}[(\vartheta_5 + 0.0102)\mathrm{Nm}_0 + 1.0102k(I_5 + C_{\mathrm{ch}}P_5)].$$

As in the derivation of (S3), we now use $1.347C_{\mathrm{ch}} \le K_5$, $d_0 \le 0.0102L^{-16K-1}$, $L^{-16K-1}\vartheta_5^{1-k} \le 1$, $\mathrm{Nm}_0 \le K_5+1$, $\vartheta_5 \le \frac14$, $1.0102 \le 1.02$, and $k \ge 1$, and obtain

$$\mathrm{Nm}_k \le K_5\omega_5 + \vartheta_5^{k-1}[0.2704(K_5+1) + 1.0102kI_5 + 1.0102kC_{\mathrm{ch}}P_5] \le K_5\omega_5 + (C_5 + 1.02C_{\mathrm{ch}}P_5)k\vartheta_5^{k-1},$$

and the last expression is the right side of the finer form of (S3).

Nothing is lost at the last levels. The errors of the cuts are of order x_k^{-16}, which is not small in K when k is near K. Since both runs are cut at the same radius, and since (S0)(a) eliminates $\tau(\ell_k)$ completely, such an error enters only as a factor of Nm_i, of Ch_i or of ϑ_5^i.

Step 7: the size of ϑ_5, and the order of the numbers. In this step we prove (S3c), the bound $\vartheta_5 \le L^{-1/5}$ and Rem. 6.23. We show that $L^{-15/32} \le \vartheta_5 \le L^{-1/5} \le \frac14$, that the right side of (S3) is at most $C_5^{\#}k\vartheta_5^k\|f\|_\sharp$, and that $c_s \le c_{\mathrm{fix}}$; and we show that the numbers are fixed in an order without a circle.

First the sizes. Since $c_s \ge 63$, we have $\vartheta_5 \ge \vartheta_s \ge L^{-15/32}$. By (L5*), $c_sL^{-15/32} \le L^{43/160-75/160} = L^{-1/5}$. Next, $\vartheta_1 = c_{\mathrm{fix}}L^{-1/2} \le L^{-2/5}$, because $L \ge L_B \ge c_{\mathrm{fix}}^{10}$ by Step 1 of the proof of Thm. 5.63; hence $\vartheta_1^{15/16} \le L^{-3/8} \le L^{-1/5}$. Finally $L \ge 1024$ gives $L^{-1/5} \le \frac14$.

For (S3c) we use $\vartheta_5^{k-1} \le L^{15/32}\vartheta_5^k$, $a_K = L^{-K} \le \vartheta_5^K \le \vartheta_5^k$, $\sqrt{3}/(40M) \le 1$, $\|f\|_\infty \le \|f\|_\sharp$ and $k \ge 1$. Combined, they give

$$K_5\mathrm{osc}_K(f) + C_5k\vartheta_5^{k-1}\|f\|_\infty \le (K_5 + L^{15/32}C_5)k\vartheta_5^k\|f\|_\sharp.$$

For Rem. 6.23, the closed forms in the proof of the bounds of Def. 2.44 give, with $t = \lambda/2 \le 0.2$,

$$\mathsf{g}(\lambda) \ge 24\cosh t/(4\sinh^3 t) \ge 24/(4(1.01t)^3) \ge 46\lambda^{-3}.$$

By (O0) in item 1 of §5.5,

$$c_0 = 108b_*\gamma_*^{-1/2}\mathsf{g}(\lambda_*/4)^3, \qquad C_1 = \sqrt{3}K_*\mathsf{g}(\lambda_*/2)^{3/2}, \qquad c_{\mathrm{fix}} = 48\mathsf{g}_\flat C_1c_0^{1/2}.$$

Use the exact definitions and outward intervals in the order of the constants, (0), of Thm. 2.65. Substitution of $\mathsf{g}(t) \ge 46t^{-3}$ into these three formulas gives $\log_{10} c_{\mathrm{fix}} \ge 204.892 > 204.89$. For the other direction, substitute $\mathsf{g}(t) \le 51t^{-3}$ and $\mathsf{g}_1(t) \le 182t^{-4}$ from Def. 2.44 into

$$a_G = (23b_*^2)^{15/16}(7\gamma_*^{-1/2})^{1/16}, \qquad c_s = 63e^{2\lambda_*}\mathsf{g}_1(\lambda_*/32)\mathsf{g}(\lambda_*/32)a_G\gamma_*^{-1/2}.$$

They give $\log_{10} c_s \le 145.332 < 145.34$. So $c_s \le c_{\mathrm{fix}} \le c_{\mathrm{fix}}^{43/16} \le L_B^{43/160} \le L^{43/160}$.

It remains to fix the order of the numbers. The numbers of (O0a) depend on $b_*, \gamma_*, \lambda_*, C_1$ alone. (L5*) bounds L from below in terms of (O0) only, and (X5) bounds x_s from below in terms of (O0) and (O4). The numbers $\vartheta_s, \vartheta_5, C_5^{\#}$ are fixed afterwards, and no floor is defined through them; so there is no circle. None of these numbers depends on $\mathsf{b}, N, K, K', k, f, M, \kappa, p_0, c_h, E_a$.

Step 8: the hybrid chain. We prove (S4): for $1 \le k_0 \le k \le K$,

$$\|\mathbb{D}_k^b - \mathbb{D}_k[f]\| \le 2.1K_5\mathrm{osc}_K(f) + 1.2C_5k_0\vartheta_5^{k_0-1}\|f\|_\infty.$$

Let $\|f\|_\infty = 1$. From k_0 on, the difference $\mathbb{D}_k^b - \mathbb{D}_k$ obeys the homogeneous system of Step 4, since both chains then use one set of rows and one cut. Let Nm_k and Ch_k be its norm and its charge, and $\mathrm{Ch}^* := \max_{k_0 \le i \le K} \mathrm{Ch}_i$. Iterating from k_0 with $\vartheta_s \le \frac14$ we find $\mathrm{Nm}_k \le \vartheta_s^{k-k_0}\mathrm{Nm}_{k_0} + \frac43 C_{\mathrm{ch}}\mathrm{Ch}^*$, and therefore $\vartheta_s\mathrm{Nm}_i + C_{\mathrm{ch}}\mathrm{Ch}_i \le \mathrm{Nm}_{k_0}/4 + \frac43 C_{\mathrm{ch}}\mathrm{Ch}^*$.

Summing the inequality for the charges from k_0 on, we get $\mathrm{Ch}^* \le \mathrm{Ch}_{k_0} + 1.01t_5x_s^{-16}[\mathrm{Nm}_{k_0}/4 + \frac43 C_{\mathrm{ch}}\mathrm{Ch}^*]$, and with (X5) this becomes $\mathrm{Ch}^* \le 1.0102[\mathrm{Ch}_{k_0} + (3/(400C_{\mathrm{ch}}L))\mathrm{Nm}_{k_0}]$. Inserted into the bound for Nm_k, it yields $\mathrm{Nm}_k \le 1.0102\mathrm{Nm}_{k_0} + 1.347C_{\mathrm{ch}}\mathrm{Ch}_{k_0}$.

Into this we insert (S3) for Nm_{k_0}, and (5) of Step 6 for Ch_{k_0}, which gives

$$\mathrm{Ch}_{k_0} \le 1.0102(\mathrm{osc}_K + 0.0102L^{-16K-1}) + (0.0076/C_{\mathrm{ch}})(K_5 + 1 + k_0I_5)\vartheta_5^{k_0}.$$

Moreover $1.347 \cdot 1.0102\,C_{\mathrm{ch}} \le 1.3608C_{\mathrm{ch}} = 1.008K_5$. So the coefficient of osc_K is $(1.0102 + 1.008)K_5 = 2.0182K_5 \le 2.1K_5$.

We measure the remaining terms in units of $C_5 k_0 \vartheta_5^{k_0-1}$, using $K_5 \le C_5/0.3$, $K_5 + 1 + k_0 I_5 \le k_0 C_5/0.3$, $L^{-16K-1} \le \vartheta_5^{k_0-1}$ and $\vartheta_5 \le \frac{1}{4}$. Their sum is at most

$$1.0102 + 1.008 \cdot 0.0102/0.3 + 1.347 \cdot 0.0076/(4 \cdot 0.3) \le 1.0102 + 0.0343 + 0.0086 = 1.0531 \le 1.2,$$

which proves the claim.

Step 9: totals, several sources, and the number δ_Δ. We prove the three parts of (S5).

(i) Let D be a family. Then

$$|\sum_y D_y(p,p')| \le \|D\| \sum_y e^{-\lambda_\flat(|y-p|+|y-p'|)} \le \|D\| \mathsf{g}' e^{-(\lambda_\flat/2)|p-p'|},$$

where the second inequality is obtained by the computation in Def. 2.48, in which the triangle inequality gives $\sum_y e^{-\lambda(|y-p|+|y-p'|)} \le e^{-\frac{\lambda}{2}|p-p'|} \sum_y e^{-\frac{\lambda}{2}|y-p|}$. For fixed p, the middle term of the chain in the display, summed over p', is $\|D\| \sum_{p'} \sum_y e^{-\lambda_\flat(|y-p|+|y-p'|)} \le \|D\| \, \mathsf{g}_\flat \cdot 3\mathsf{g}_\flat$, as in the proof of Lem. 6.27 (a). Hence the row sums of $\sum_y D_y$ are at most $3\mathsf{g}_\flat^2 \|D\|$, and as $\sum_y D_y$ is symmetric, Schur's test bounds its norm by the same number; for $D = \mathrm{d}\mathbb{D}_k$ this gives the two bounds of (S5) (i), since then $\sum_y D_y = \tilde{N}_{m+k}[f] - N_k[f]$ and $\|D\|$ is at most the right side of (S3).

(ii) The linearity is (U2.2). At each point at most one of the f_i differs from 0, which gives the claim on the sup norm.

For the upper bound on $\mathrm{Lip} w$, let $w(x) = z_i f_i(x)$ and $w(x') = z_j f_j(x')$ with $i \ne j$ and both values not 0, and take a shortest segment from x to x'. Let ξ be its first point with $f_i = 0$, and ξ' its last point with $f_j = 0$. These points exist: the f_j are Lipschitz, hence continuous, and $f_i(x') = 0$, $f_j(x) = 0$, since the sets $\{f_i \ne 0\}$ are disjoint.

On $[x, \xi)$ we have $f_i \ne 0$, so $f_j = 0$ there, whereas $f_j \ne 0$ on $(\xi', x']$. Hence ξ lies before ξ' on the segment, or is equal to it, and

$$|w(x) - w(x')| \le |z_i| \mathrm{Lip} f_i |x - \xi| + |z_j| \mathrm{Lip} f_j |\xi' - x'| \le \max_i |z_i| \mathrm{Lip} f_i \, |x - x'|.$$

The remaining cases are direct. If $i = j$, then $|w(x) - w(x')| = |z_i||f_i(x) - f_i(x')| \le |z_i| \mathrm{Lip} f_i |x - x'|$. If one value is 0, say $w(x') = 0$ while $w(x) = z_i f_i(x) \ne 0$, then $z_i \ne 0$ forces $f_i(x') = 0$, and again $|w(x) - w(x')| = |z_i||f_i(x) - f_i(x')| \le |z_i| \mathrm{Lip} f_i |x - x'|$. If both values are 0, there is nothing to prove.

For the lower bound $\mathrm{Lip} w \ge |z_i| \mathrm{Lip} f_i$, we use once more that the f_j are continuous, being Lipschitz, and take two points x, x'. If $f_i \ne 0$ at both, then $w = z_i f_i$ at both, and $|z_i||f_i(x) - f_i(x')| = |w(x) - w(x')| \le \mathrm{Lip} w |x - x'|$. If $f_i = 0$ at both, there is nothing to prove.

Up to exchanging x and x', the last case is $f_i(x) \ne 0$ and $f_i(x') = 0$. Let ξ be the first point with $f_i(\xi) = 0$ on a shortest segment from x to x'. On $[x, \xi)$ we have $f_i \ne 0$, so the other f_j vanish and $w = z_i f_i$ there; by continuity $w(\xi) = 0$. Hence

$$|z_i||f_i(x) - f_i(x')| = |w(x) - w(\xi)| \le \mathrm{Lip} w |x - \xi| \le \mathrm{Lip} w |x - x'|.$$

We remark that $\mathrm{Lip} w = \max |z_i| \mathrm{Lip} f_i$ holds as well, the value ∞ allowed, for bounded f_i with pairwise disjoint closed supports and all $z_i \ne 0$. If all f_i are Lipschitz, this is the case just treated. Let $\mathrm{Lip} f_i = \infty$ for one i. The supports are compact and finitely many, so they lie at a positive least distance from each other, and $w = z_i f_i$ at all points nearer to $\mathrm{supp} f_i$ than that distance.

Take x_n, x_n' with $|f_i(x_n) - f_i(x_n')|/|x_n - x_n'| \to \infty$. As f_i is bounded, $|x_n - x_n'| \to 0$. One of the two points lies in $\mathrm{supp} f_i$, since otherwise both values would be 0; so from some n on, both points lie nearer to $\mathrm{supp} f_i$ than the least distance between the supports, and the difference quotients of w are $|z_i|$ times those of f_i. Thus $\mathrm{Lip} w = \infty$.

(iii) For the family D of this part, $\|D\| \le \|\tilde{\mathsf{E}}_{m+k} - \mathsf{E}_k\| + \|\tilde{\mathbb{D}}_{m+k}[w] - \mathbb{D}_k[w]\|/\mathsf{b}$. We multiply by $(c_Q^{\mathrm{su}}/E_a)x_k^{1/2}/p_0(x_k) = \mathrm{amp}(x_k)\mathsf{b}/K_5$. Then the first term is bounded by Cor. 5.49, and the second by (S3) for w.

Step 10: the class of f. We prove (S6) (i) and (ii).

(i) For $K \ge 3$ and $m \ge 1$, there exists a bounded Borel f for which (S3) fails once the term $K_5\mathrm{osc}_K(f)$ is deleted.

(ii) For the indicator of a box, $\mathrm{os}_i \le 12L^{-i}$.

We first note a lower bound for the norm: for every family D, $|c_y(D)| \le \sum_{p,p'} |D_y(p,p')| \le 9\mathsf{g}_\flat^2\|D\|$.

(i) Let $m \ge 1$, and let b, L, N, K be any data of item 1 subject to (O0) to (O13), with $K \ge 3$. Thus $L \ge L_B$, and the range $3 \le k \le K$ is not empty. Let $f = 1$ at the barycentres $x^{[K]}(p)$ of all plaquettes p of level 0 in the run K, and $f = 0$ elsewhere. This f is bounded and Borel, and $\|f\|_\infty = 1$. We show that (S3) with the term $K_5\mathrm{osc}_K(f)$ deleted fails at every k with $3 \le k \le K$.

Indeed, with ι normal to e_μ, e_ν and ι' normal to $e_{\mu'}, e_{\nu'}$, the equation $x^{[K']}(z;\iota) = x^{[K]}(y;\iota')$ says $z + \frac12(e_\mu + e_\nu) = L^m(y + \frac12(e_{\mu'} + e_{\nu'}))$. As L^m is odd, the half-integer coordinates of the two sides agree only if $\iota = \iota'$, and then $z = L^m y + \frac{L^m-1}{2}(e_\mu + e_\nu)$, a single site of $B^m(y)$.

Hence $\mathrm{Mn}^{[K']}_{m+i,y,\iota}(f) = L^{-3m}$ and $\mathrm{Mn}^{[K]}_{i,y,\iota}(f) = 1$, so $\mathrm{os}_i = 1 - L^{-3m}$ for all i. By (S2) for both runs and the lower bound above, $\|\mathrm{d}\mathbb{D}_k\| \ge (1 - L^{-3m} - 0.0204/L)/(9\mathsf{g}_\flat^2)$ for all $k \le K$. Since $\vartheta_5 \le L^{-1/5} \le \frac14$ by Step 7, we have $(k+1)\vartheta_5^k \le 2\vartheta_5 \cdot k\vartheta_5^{k-1} \le \frac12 k\vartheta_5^{k-1}$ for $k \ge 1$, and hence $C_5 k\vartheta_5^{k-1} \le 3C_5\vartheta_5^2$ for $k \ge 3$. By the numerals of item 2 of §6.3, $\log_{10} C_5 \le 382.69$ and $\log_{10} \mathsf{g}_\flat \le 46.32$, and by Rem. 6.23, $L \ge L_B \ge 10^{2048.9}$. Therefore $3C_5\vartheta_5^2 \le 3\cdot 10^{382.69}L^{-2/5} \le 10^{0.48+382.69-819.56} \le 10^{-436}$ and $(1 - L^{-3m} - 0.0204/L)/(9\mathsf{g}_\flat^2) \ge 0.99/(9\cdot 10^{92.64}) \ge 10^{-94}$, so that $\|\mathrm{d}\mathbb{D}_k\| \ge 10^{-94} > 10^{-436} \ge C_5 k\vartheta_5^{k-1}$ for $3 \le k \le K$. What Thm. 8.2 (T2) has and this instance lacks is the continuity of f.

(ii) Let f be the indicator of a box. By Step 5, the points $x^{[K']}(z';\iota)$, $z' \in B^m(z)$, lie at sup distance $< a_K$ from $x^{[K]}(z;\iota)$. Therefore the difference $\mathrm{Mn}^{[K']}_{m,z,\iota}(f) - f(x^{[K]}(z;\iota))$, of modulus at most 1, vanishes unless for some μ an end c of the μ-th arc of the box lies between two μ-coordinates that differ by less than a_K, that is, unless $|x^{[K]}(z;\iota)_\mu - c| < a_K$ for one of the six pairs (μ, c).

For one pair, the coordinate z_μ then lies in an open interval of length 2, so it takes at most 2 values; this leaves at most $2L^{2i}$ sites of $B^i(y)$ for each pair, and $12L^{2i}$ in all. As os_i is a supremum of means over L^{3i} sites, $\mathrm{os}_i \le 12L^{-i}$. By Step 7, $\vartheta_5 \ge L^{-15/32}$, so that $L^{-1} \le \vartheta_5$ and therefore $\mathrm{os}_i \le 12L^{-i} \le 12\vartheta_5^i$, which is the second bound of (S6) (ii).

Step 11: the uncut chain. We prove Rem. 6.22: the formula for $D_k^{\mathrm{unc}}[f]$, the bound $|D_{k,y}^{\mathrm{unc}}[f](P,P')| \le 243K_*^2\|f\|_\infty e^{-\lambda_*(|y-P|+|y-P'|)}$, and $0 \le N_k^{\mathrm{unc}}[f] \le \|f\|_\infty \mathsf{B}_k$ for $f \ge 0$.

The formula follows by induction on k from the definition of T_ℓ, because $B^{k+1}(y')$ is the disjoint union of the $B^k(y)$.

For the bound we introduce auxiliary rows. Let $\tau_0 := 1/243$, $\hat a := (1+\tau_0)a$ and $\hat\ell_{0,y}\zeta := \tau_0^{1/2}(\zeta_p)_{p \text{ based at } y}$. These rows are translation covariant, $\hat{\mathsf{B}}_0 = (1+\tau_0)1$, $|\hat\ell_{0,y}(p)| \le \tau_0^{1/2} \le K_*$, and $\hat\ell_{0,y}(p) = 0$ unless $|y-p| = 0$; moreover $\tau(\hat\ell_0) = (3\tau_0)^{1/2} = \frac19$ and $\hat a L \ge 3/c_L$. We set $\hat\ell_{k+1} := (\hat\ell_k)'$, given by Prop. 2.28 in the model with $\hat a$, without cut.

We have $\langle F, (1+\tau_0)\mathsf{B}F\rangle + \hat a|\mathsf{n}|^2 = (1+\tau_0)[\langle F, \mathsf{B}F\rangle + a|\mathsf{n}|^2]$, and being admitted depends neither on the rows nor on a. Hence Lem. 2.53 (a) shows: if $\hat{\mathsf{B}}_k = (1+\tau_0)\mathsf{B}_k$, then the minimisers of the two models agree, $\hat H_k = H_k$, and $\hat{\mathsf{B}}_{k+1} = (1+\tau_0)\mathsf{B}_{k+1}$. By Thm. 2.57 (a), in which the constants do

not depend on a, and by $\tau(\ell') \le \tau(\ell)$ from Prop. 2.55 (c), induction on k gives $\hat\ell_k \in \mathcal{R}(b_*, K_*, \lambda_*, \frac{1}{9})$ and $|\hat\ell_{k,y}(P)| \le K_* e^{-\lambda_*|y-P|}$ for $k \ge 1$.

By group (5) of Prop. 2.28, and by induction, $\hat\ell_{k,y}\zeta$ has among its components the numbers $\tau_0^{1/2} L^{k/2}(\mathbb{H}_k\zeta)_p$, p based in $B^k(y)$. Therefore $L^k \sum_{p \text{ based in } B^k(y)} \mathbb{H}_k(p,P)^2 \le 243|\hat\ell_{k,y}(P)|^2 \le 243 K_*^2 e^{-2\lambda_*|y-P|}$, and the Cauchy–Schwarz inequality, applied in the formula for $D^{\mathrm{unc}}_{k,y}[f]$, gives the bound.

We set $N^{\mathrm{unc}}_k[f] := \sum_y D^{\mathrm{unc}}_{k,y}[f]$, the total of Rem. 6.22, and turn to the inequality $0 \le N^{\mathrm{unc}}_k[f] \le \|f\|_\infty \mathsf{B}_k$ for $f \ge 0$. Since $\mathbb{T}_0$ is the disjoint union of the blocks $B^k(y)$, $y \in \mathbb{T}_k$, the formula for $D^{\mathrm{unc}}_{k,y}[f]$ gives $N^{\mathrm{unc}}_k[f] = L^k \mathbb{H}_k^{\mathsf{T}} \mathrm{diag}(f(x(p)))\mathbb{H}_k$; so $0 \le N^{\mathrm{unc}}_k[f] \le \|f\|_\infty N^{\mathrm{unc}}_k[1]$ for $f \ge 0$. Further, we have $N^{\mathrm{unc}}_0[1] = 1 = \mathsf{B}_0$, and if $N^{\mathrm{unc}}_k[1] \le \mathsf{B}_k$, then by Prop. 2.27 (a)

$$\langle X', N^{\mathrm{unc}}_{k+1}[1]X'\rangle = L\langle H_k X', N^{\mathrm{unc}}_k[1] H_k X'\rangle \le L\langle H_k X', \mathsf{B}_k H_k X'\rangle \le L\|(H_k X', \mathsf{N}^\sharp X')\|^2_{\mathcal{H}} = \langle X', \mathsf{B}_{k+1} X'\rangle.$$

So $N^{\mathrm{unc}}_k[1] \le \mathsf{B}_k$ for every k, and the inequality $0 \le N^{\mathrm{unc}}_k[f] \le \|f\|_\infty \mathsf{B}_k$ for $f \ge 0$ follows. □

6.4. **One-step derivative.** This section studies one drawn step in the setting (SD), for SU(2) in $d = 3$, with kernels, law, centre and rows free of sources, and differentiates it in the direction of the quadratic member $s_x[D]$, where D is any complex symmetric family of kernels of range $\le \varrho(x)$. The a-part of this derivative is the quadratic member $s_{x'}[D']$, with D' one cut step of the free source flow, plus a remainder $n[u; D]$; the remainder splits as $n^{\mathrm{fix}} + n^{\mathrm{mem}}$, its cell sum is bounded by $u_3(x')\|D\|_{(\lambda_\flat)}$ caps at x', and further clauses treat analyticity, locality, the dependence on the member, linearity in D and the constants. Further results are the derivative of the number, the estimate (2τ) of a trace, and a corollary in which the direction is $s_x[D]$ + rem and the remainders rem, rem′ obey $t' \le 0.2431t + 1$.

Remark 6.28 (Comparison of the bounds on the remainder and on the leading member). By Thm. 6.33 (a), the remainder n is at most $u_3(x')\|D\|_{(\lambda_\flat)}$ caps at x', a bound that is not small by itself. The leading member $s_{x'}[D']$, however, is bounded by $c^{\mathrm{su}}_Q p_0'^2 \|D'\|_{(\lambda_\flat)}$, that is, by $(c^{\mathrm{su}}_Q/E_a) x'^{1/2} p_0'^{-1} \|D'\|_{(\lambda_\flat)}$ caps at x'; this follows from Step 1 (1) of the proof of Thm. 6.33, applied at x'. The coefficient of $\|D\|_{(\lambda_\flat)}$ in the first bound, divided by that of $\|D'\|_{(\lambda_\flat)}$ in the second, is $\check\omega_3(x') = C_3 E_a p_0' x'^{-1/4}/c^{\mathrm{su}}_Q$ (entry 7 of item (0) of §6.1), and this quotient is small by (W3) in item (1) of §6.1.

Item 1: the letters. All letters are those of items (0) and (1) of §6.1. The list below gives entries (O0a) to (O14) of the table which occur in this section, and its entry (O13) also defines x_3 and the conditions (Z.0) to (Z.4).

(O0a) $\lambda_\flat = \lambda_*/16$; $\mathsf{g}(t) = \sum_{m\in\mathbb{Z}^3} e^{-t|m|_\infty}$; $\mathsf{g}_\flat = \mathsf{g}(\lambda_\flat)$; $g_{1b} = \sum_m |m|_\infty e^{-\lambda_\flat|m|_\infty}$; $g_b^- = \mathsf{g}(8\lambda_\flat/9)$.
(O4) L is odd and satisfies (L2^{su}), that is $L^{1/2} \ge 1.2\cdot 10^4 \mathsf{N}_{\mathrm{c}}^2$. Moreover (L3su) reads $L^{1/4} \ge 10^4 \mathsf{N}_{\mathrm{c}}^2$.
(O5) $\kappa \ge 6$, $\kappa_1 = \kappa + 1$ and $\tilde\kappa \in [\kappa, \kappa_1]$; $\mathbb{P} = [\kappa, \kappa_1] \times \{c_h, c_h/2\}$, with elements $p = (\tilde\kappa, \tilde c)$.
(O6) $\gamma = (73L^4)^{-1}$; $\theta_D \le 1$; $C_D \ge 4$; $c_R = 64/\theta_D$.
(O7) The letter M.
(O8) $p_0(x) = A_0(\log x)^{\mathfrak{p}_0}$, $\varepsilon = p_0 x^{-1/2}$ and $x' = x/L$; a prime means "at x'".
(O9) $C^\flat_\varphi$, and $c_\xi = (6.5 C^\flat_\varphi)^{-1}$.
(O10) a_{c} and K_{c}; $C^\flat_\eta = 1/6.5 + \max\{\mathsf{a}_{\mathrm{c}}, 7\} L^{1/2}$ (§4.2); $c^{\mathrm{su}}_Q = 306 M^3 \mathsf{g}_\flat^2$ (§5.4).
(O10a) $k_1 := 0.9475 (C^\flat_\eta)^3 (1320 g_{1b}/\mathsf{g}_\flat + 216)$; $k_2 := 0.1359 L (g_b^-/\mathsf{g}_\flat) C_D [4\mathsf{K}_{\mathrm{c}} + 1 + 1.51\cdot 10^6 \theta_D^{-2} C_D]$; $k_3 := 6730 L^4$.
(O12) E_a, and $c_a(x) = E_a p_0^3 x^{-1/2}$.
(O12a) $C_3 := 1686 e^{\kappa+4} L^{11/4} c^{\mathrm{su}}_Q / E_a$.
(O13) $x_3 := 1 + \inf S_3$, where S_3 is the set of $y \ge x_9^U$ such that the five conditions (Z.0) to (Z.4) below hold for every $x \ge y$, where $T_s := x^{1/4}$ and $n_\chi := 3(M(2r(x)+1) + 2L(r_w+2) + 1)^3$:

(Z.0) $160T_s\omega\varepsilon' \le 1$, $2640\ell_w(x')(T_s+1)\omega\varepsilon' \le \mathfrak{g}_x$ and $102400T_s\omega^2\varepsilon' \le 1$;
(Z.1) $c_\xi p_0 \ge 1$;
(Z.2) $7L^3 c_\xi p_0 n_\chi e^{-c_\xi^2 p_0^2/2} \le x^{-10}$;
(Z.3) $17.9L^4 C_D^2 x^{-31} \le x^{-20}$;
(Z.4) $18.2L^3 c_a(x) + L^3(2k_1\varepsilon + k_2\varepsilon') + k_3 L^3 p_0^{-2} x^{-1/2} + 2x^{-10} \le e^{\kappa+3} L^3 x^{-1/4}$, with κ the lower end of the range of $\tilde\kappa$.

The table, in item (1), (O13), of §6.1, requires the last floor x_s to satisfy $Lx_s \ge x_3$. The floor x_3 imposes no condition on $d_0, c_\Xi, \rho_0, L, a, \kappa, M, \mathfrak{p}_0, c_h, E_a$.

(O14) $u_3(x) = C_3 x^{1/4}$; $\vartheta_a = 0.2431 L^{-1/4}$; $\sigma_D = c_Q^{\mathrm{su}} p_0(x)^2 \|D\|_{(\lambda_\flat)}$.

The radii and domains $\varrho(x)$, ϱ_L, r_w, R_2, R_3, R_T, $r(x)$, ℓ_w, D_w, ω, $\mathfrak{g}_x$, $\mathcal{U}_X(x)$, $\mathcal{U}_Y(x')$, $U^\tau(B;x')$, the chart $U = \mathsf{U}(\eta; V')$ with $\eta = G\xi - a_0$ and $\xi \sim \mathcal{N}(0, x^{-1}1)$, and the symbols χ_X, $Y_0(X)$, $F^\flat$ and $\bar F = EX' - \mathrm{d}^V a_0$ are taken from items 2, 3 and 6 of §5.3 and from Sections 4.1, 3 and 4.2. From item 2 of §5.4 we take $|y-p| = |y-\pi(p)|_\infty$, the norm

$$\|D\|_{(\lambda)} = \sup_{y,p,p'} |D_y(p,p')| e^{\lambda(|y-p|+|y-p'|)},$$

the paths $\Gamma(y,z)$, and the form $Y \cdot Y' = \frac{1}{2}\operatorname{tr} YY'$. The transport $T(y',y'') = \mathrm{Ad}_{V'(\Gamma(y',y''))}$ is that of item C3 of §4.1.

Definition 6.29 (The cell sum of a family)**.** Let $a = (a(Y,\cdot))$ be a family indexed by the connected sets Y of cells. Its cell sum is

$$\mathrm{CS}_{\tilde\kappa}(a) := \sup_{\square} \sum_{Y \ni \square} e^{\tilde\kappa(\#Y-1)} \sup_{\mathcal{U}_Y(x')} |a(Y,\cdot)|,$$

where the first supremum runs over the cells $\square$, and we put $|a|_{a,x'}^{\mathrm{cs},(\tilde\kappa)} := \mathrm{CS}_{\tilde\kappa}(a)/c_a(x')$. ■

The number $\mathrm{CS}_{\tilde\kappa}(a)$ is the $|a|_{\mathrm{cs}}$ of item 1 of §6.2, taken at $\tilde\kappa$. The quantity $|a|_{a,x'}^{\mathrm{cs},(\tilde\kappa)}$ dominates the size $\sup_Y e^{\tilde\kappa(\#Y-1)} \sup|a(Y,\cdot)|/c_a(x')$, which is the size in which Thm. 6.7 (h) measures a direction.

Remark 6.30 (Letters with more than one meaning). The letters p, p' denote fine plaquettes, and p is the pair $p = (\tilde\kappa, \tilde c)$ in $\mathbb{P}$ only in $M_A(x;p)$, in $p \in \mathbb{P}$, in "at p" and "all p", and in the list of clause (f) of Thm. 6.33; the remainders are $n[u;D]$, n^{fix}, n^{mem}, and $n = n[u;D]$ as Thm. 6.33 sets it, also in $n(Y,\cdot)$, whereas $n(C)$, told apart from them by its argument, a cluster C, is the number of letters in the polymers of C, each counted with multiplicity (Lem. 6.9); and outside (CD) the letter r without argument and without prime is the number $C_\eta^\flat\varepsilon$ of Step 0 (vi) of the proof of Thm. 6.33, while $r(x)$ and r' are the radii, and $r(Y)$, as well as r where Step 10 of that proof uses it for the family $(r(Y))$, is the remainder of Thm. 6.7 (g3). The symbol $T(y',y'')$ denotes the transport, $T > 1$ is the parameter of Lem. 6.13, and $T_s = x^{1/4}$ of (O13) is the value of T with which Step 4 of the proof of Thm. 6.33 applies that lemma. The letter u is a member; in $u(y')$, Ad_u and V'^u the letter u is a gauge transformation.

Item 2: the data. We fix $x \ge x_3$ and work in the setting (SD) of item 2 of §5.3, with x_9^U in place of x_9: thus N is even, $M \mid N$, (Tor) holds, the grids are fitted, and the rows $\ell \in \mathcal{R}(b_*, K_*, \lambda_*, \frac{1}{9})$ have range $\le \varrho(x)$ and the four conclusions named there. The empty rows, those with all $\mathsf{H}_y = 0$, are admitted wherever they satisfy (SD). We fix $p \in \mathbb{P}$ and a member $u \in M_8(x;p)$, which may be complex and carries the weak cap. Finally, $D = (D_y)$, with y running over the fine torus, is a family of complex kernels on the fine plaquettes such that $D_y(p,p') = D_y(p',p)$, and $D_y(p,p') = 0$ unless $|y-p|, |y-p'| \le \varrho(x)$. Nothing further is asked of D: no sign, no covariance under translations, no factorisation through the rows and no bound on its size.

Item 3: the objects.

Definition 6.31 (The quadratic member, the step on kernels, the derivative and the remainder)**.** For a cube $\square$ and a field $\mathsf{W} \in \mathcal{U}_{\{\square\}}(x)$, the quadratic member is

$$s_x[D](\{\square\},\mathsf{W}) := +\frac{x}{2}\sum_{y\in\square}\sum_{p,p'} D_y(p,p')\,(\mathrm{Ad}_{\mathsf{W}(\Gamma(y,\pi(p)))}\,F_p)\cdot(\mathrm{Ad}_{\mathsf{W}(\Gamma(y,\pi(p')))}\,F_{p'}),$$

where $F_p = -i\log\partial\mathsf{W}(p)$, and $s_x[D](X,\cdot) := 0$ for $\#X \geq 2$. This family is $-E^{\Delta}[D]$. At x' the same formula is used, with cells, coarse sites y', coarse plaquettes P and $X'_P = -i\log\partial V'(P)$.

The step on kernels is

$$(T_\ell D)_{y'}(P,P') := L\sum_{y\in B(y')}\sum_{p,p'} H^\sharp(p,P)D_y(p,p')H^\sharp(p',P'),$$

where $H^\sharp$ is the free kernel of item E1 of §4.1, taken at the rows ℓ; its cut is

$$D'_{y'}(P,P') := 1\{|y'-P|,|y'-P'|\leq\varrho(x')\}\,(T_\ell D)_{y'}(P,P'),$$

the distances being measured in coarse units. Kernel and cut are those of Thm. 6.20 (see Rem. 6.24): item E1 of §4.1 defines $H^\sharp$ by $H^\sharp = L^{-1}\mathsf{B}^{-1}\mathsf{T}_\sharp\mathsf{B}'$ in the form of Prop. 2.27 (a), and item 1 of §6.3 takes the kernel from §2.2 in the form of that proposition.

We write $DA_\ell(u)[\delta](Y,V')$ for the derivative $DA(u)[\delta](Y,V')$ that item 1 of §6.2 defines as a sum, term by term, of derivatives, here taken at the rows ℓ. For $u \in M_8(x;p)$, $\delta \in T^a(x)$ and $V' \in \mathcal{U}_Y(x')$ it equals $\frac{d}{dz}E'^{\mathrm{old}}[u+z(\delta,0,0)](Y,V')$ at $z=0$, by Thm. 6.7 (h), as $A = 8 < 16$. By (Fm1), E'^a depends on E^a alone and E'^{new} on no member; hence $DA_\ell(u)[\delta]$ is the a-part of $DF_\ell(u)$, the derivative of the step at the member u, in every direction whose a-part is δ.

The remainder is $n[u;D] := DA_\ell(u)[s_x[D]] - s_{x'}[D']$. ■

Item 4: the theorem.

Lemma 6.32 (Quadratic source members)**.** Suppose $\varepsilon(x) \leq 1/40$, $\varrho(x)+2 \leq Mr(x)$, and the torus has side at least $16Mr(x)$. Let $D = (D_y)$ be a complex family with $D_y(p,p') = 0$ unless both $|y-p|$ and $|y-p'|$ are at most $\varrho(x)$. Then $s_x[D] \in T^a(x)$ and

$$\sup_{\mathcal{U}_{\{\square\}}(x)} |s_x[D](\{\square\},\cdot)| \leq 306M^3\mathsf{g}_\flat^2 p_0(x)^2\|D\|_{(\lambda_\flat)} = \sigma_D.$$

For real D the member is real on G-valued fields. ■

Proof. For $y \in \square$ and $|y-p| \leq \varrho(x)$, the plaquette p and the path $\Gamma(y,\pi(p))$ lie in the embedded box $\square^{\approx}$. The path has length at most $3\varrho(x) \leq \ell_w(x)$, as in (G2) and (G3) of Step 0 in §5.4. On $\mathcal{U}_{\{\square\}}(x)$, (u2) bounds the norm of the path transport by 2. Also, (u3) and $\varepsilon(x) \leq 1/40$ give

$$|F_p| \leq -\log(1-2\varepsilon(x)) \leq 2.06\varepsilon(x).$$

The norm of each transported F_p is thus at most $8.24\varepsilon(x)$. Since $\sum_{p,p'}|D_y(p,p')| \leq 9\mathsf{g}_\flat^2\|D\|_{(\lambda_\flat)}$, the size bound follows from $x\varepsilon(x)^2 = p_0(x)^2$ and $(9/2)(8.24)^2 \leq 306$.

The logarithm defining F_p is holomorphic by the same plaquette bound. A gauge transformation conjugates both transported vectors at y, preserving their bilinear product. At a pure gauge every F_p vanishes. For G-valued fields the transported vectors belong to $\mathfrak{g}$ and their bilinear product is real. These properties and the fact that $s_x[D]$ vanishes on sets of two or more cubes give the asserted membership. □

Theorem 6.33 (The step differentiated in a quadratic direction)**.** In the setting of item 2, the remainder $n = n[u;D]$ of Def. 6.31 has the following properties.

(a) *Bound.* The remainder splits as $n = n^{\mathrm{fix}}[D] + n^{\mathrm{mem}}[u; D]$, the two parts being defined in Step 1 of the proof. The part n^{fix} depends on no member and vanishes unless $\#Y = 1$, and for every cell $\square$

$$\sup_{\mathcal{U}_{\{\square\}}(x')} |n^{\mathrm{fix}}(\{\square\}, \cdot)| \le \sigma_D \big[L^3(2k_1\varepsilon + k_2\varepsilon') + k_3 L^3 p_0^{-2} x^{-1/2} + 2x^{-10}\big].$$

The other part satisfies

$$\mathrm{CS}_{\tilde\kappa}(n^{\mathrm{mem}}) \le \sigma_D \big[1000 e^{\tilde\kappa+3} L^3 x^{-1/4} + 18.2 L^3 c_a(x)\big] =: B_{\mathrm{mem}}(\tilde\kappa).$$

In n^{mem}, the clusters with $n(C) \ge 2$ make up the family Rst of Thm. 6.7 (h); by (h1) applied with $\delta = s_x[D]$, whose size B satisfies $B c_a(x) \le \sigma_D$, they are bounded by $217 L^3 x^{-1/4}\sigma_D$, and $217 \le 1000 e^{\tilde\kappa+3}$. Consequently $\mathrm{CS}_{\tilde\kappa}(n) \le 1001 e^{\tilde\kappa+3} L^3 x^{-1/4}\sigma_D$ and $|n[u; D]|^{\mathrm{cs},(\tilde\kappa)}_{a,x'} \le u_3(x')\|D\|_{(\lambda_\flat)}$, for every $p = (\tilde\kappa, \tilde c) \in \mathbb{P}$ and every $u \in M_8(x; p)$, each at its own $\tilde\kappa$.

(b) The functions $n(Y, \cdot)$, $n^{\mathrm{fix}}(Y, \cdot)$ and $n^{\mathrm{mem}}(Y, \cdot)$ have the clauses of (ca) without a size at x': they are holomorphic on $\mathcal{U}_Y(x')$, invariant under G-valued gauge transformations, 0 at pure gauges, and functions of the field on $Y^{\approx}$. They are real at G-valued fields if u and D are real.

(c) *Locality.* The function $n(Y, \cdot)$ depends on D_y only at the fine sites y under Y, and it vanishes unless some D_y with y under Y is not 0. The kernel $D'_{y'}$ depends only on the D_y with $y \in B(y')$.

(d) *In the member.* The remainder n depends on the member u through E^a alone, and each value $n[u; D](Y, V')$ is holomorphic along every affine line in $M_8(x; p)$. For $u_1, u_2 \in M_2(x; p)$,

$$\mathrm{CS}_{\tilde\kappa}(n[u_1; D] - n[u_2; D]) \le \frac{1}{3} B_{\mathrm{mem}}(\tilde\kappa)\, d_a,$$

where $d_a := \sup_X \sup_{\mathcal{U}_X(x)} |E^a_1 - E^a_2| e^{\tilde\kappa(\#X-1)}/c_a(x)$ is the a-part of $\mathsf{D}^{\mathrm{in}}_x(u_1, u_2)$.

(e) The map $D \mapsto (s_x[D], D', n^{\mathrm{fix}}[D], n^{\mathrm{mem}}[u; D])$ is complex linear. Hence (a) and (d) hold for $\tilde D - D$, with $\|\tilde D - D\|_{(\lambda_\flat)}$ in place of $\|D\|_{(\lambda_\flat)}$.

(f) The constants k_1, k_2, k_3, C_3, x_3 and all numerals are independent of $N, K, k, \mathsf{b}, p, u, D$; of the rows ℓ they involve only the pure numbers b_*, K_*, λ_* and the range.

(g) *The number.* The derivative of the number in the direction $s_x[D]$ is

$$D\mathsf{V}'(u)[s_x[D]] = \sum_y \frac{3}{2} \sum_{p,p'} D_y(p, p')(\mathrm{d}C^1 \mathrm{d}^{\mathsf{T}})(p, p') + \sum_{\square} e_{\square},$$

the last sum running over the cells, with $|e_{\square}| \le \sigma_D \big[L^3(k_1\varepsilon + 9.1 c_a(x)) + 109 L^3 x^{-1/4} + x^{-10}\big]$; here C^1 is C at $V' = 1$, and its blocks are scalars.

■□

Item 5: the clause (2τ). The clause below is a corollary of Lem. 6.13, which is stated in Step S10 (3) of the proof of Thm. 6.7 and proved there by Cordes' inequality; Step 4 of the proof of Thm. 6.33 proves (2τ).

Corollary 6.34 ((2τ): the variation of a dressed trace of the covariance)**.** In the setting of item 2, let y be a fine site and $y' = y(y)$, let p, p' be fine plaquettes with $|y - p|, |y - p'| \le \varrho(x)$, let o be a coarse site within M of y', and let B_ω be the box of radius ω centred at o. Put

$$\tau(V') := \mathrm{tr}\big[T(y', y(p))\, (\mathrm{d}^V C \mathrm{d}^{V\mathsf{T}})(p, p')\, T(y', y(p'))^{\mathsf{T}}\big].$$

For every $T > 1$ at which (C0), (C1) and (C2) of (Y.0)(0d) (item 5 of §5.3) hold, the function τ is holomorphic and G-invariant on $U^{17T}(\mathsf{B}_\omega; x')$, depends on V' only within $R_2 + \varrho_L$ of y', and

satisfies $|\tau| \le 3134/\gamma$ there; moreover $\tau(1) = 3(\mathrm{d}C^{1}\mathrm{d}^{\mathsf{T}})(p,p')$. At the fields on B_{ω} with (u1) to (u3) at x',

$$|\tau(V') - \tau(1)| \le 2(3134/\gamma)T^{-2}.$$

At $T = T_s$ this bound becomes $6268\gamma^{-1}x^{-1/2}$. ■□

Item 6: the corollary. Besides Thm. 6.33, the next corollary rests on Thm. 6.7 (h3) and on (L3su).

Corollary 6.35 (A direction with a remainder, and the recursion of the remainders)**.** In the setting of item 2, let $\mathrm{rem} \in T^{a}(x)$ be of size B_r caps at p. Then

$$DA_{\ell}(u)[s_x[D] + \mathrm{rem}] = s_{x'}[D'] + \mathrm{rem}',$$

where $\mathrm{rem}' := DA_{\ell}(u)[\mathrm{rem}] + n[u;D] \in T^{a}(x')$, and $|\mathrm{rem}'|^{\mathrm{cs},(\tilde{\kappa})}_{a,x'} \le \vartheta_a B_r + u_3(x')\|D\|_{(\lambda_\flat)}$. Let $\bar{d} > 0$ with $\|D\| \le \bar{d}$; in the units $t := B_r/(u_3(x)\bar{d})$ and $t' := |\mathrm{rem}'|/(u_3(x')\bar{d})$ one has $t' \le 0.2431t + 1$. Along levels $x_{k+1} = x_k/L$ with $\|D_k\| \le \bar{d}$, it follows that $t_k \le \max\{t_0, 1.3212\}$, which is at most 2 if $t_0 \le 2$. ■□

Item 7: four remarks.

Remark 6.36 (n^{fix} in caps). In caps at x',

$$|n^{\mathrm{fix}}|_{a,x'} \le (c^{\mathrm{su}}_{Q}/E_a)(L^{5/2}/0.5946)(2k_1 + L^{1/2}k_2 + k_3 + 1)\|D\|_{(\lambda_\flat)},$$

a bound that carries no power of x and no logarithm. The power $x'^{1/4}$ of u_3 comes from Rst alone, and n^{fix} cancels in the difference of two members.

Remark 6.37 (The identity over sub-alphabets). Let X be a cube under Y, and let s_X be the member equal to $s_x[D]$ at the cube X and to 0 at all other sets of cubes. Let $\mathbb{A}$ be a finite set of letters (a), (v), (t) with cores in Y which contains the letter (a) at X, and let $V' \in \mathcal{U}_Y(x')$. With $W_{\mathbb{A}} := \sum \chi_X E^{a}(X,U) + \sum \chi_{\square} V_{\square} + \sum T_Y$, the sums running over $\mathbb{A}$, we have

$$\sum_{C \text{ in } \mathbb{A}} \frac{d}{dz}\Big|_{z=0} \Phi(C)[u + z s_X] = E_{\xi}[\chi_X s(X,U) e^{W_{\mathbb{A}}}]/E_{\xi}[e^{W_{\mathbb{A}}}].$$

The cluster of the single letter $(\mathrm{a})_X$ yields $E_{\xi}[\chi_X e^{E^{a}} s]$, and the clusters with $n(C) \ge 2$ yield the truncated part. Thus n is, polymer by polymer, the remainder of the family itself, and not a new decomposition.

Remark 6.38 (The sum rule). Let u be a real member of M_1 with the strong cap and with $\rho[u;\ell] > 0$, and let D be real. Put

$$q_D(V) := \frac{x}{2} \sum_{y} m_y(V) \sum D_y(p,p') \langle T(y,\pi(p))\zeta_p, T(y,\pi(p'))\zeta_{p'} \rangle,$$

and let $u_t := (E^{a} + t s_x[D], E^{h}, w_t)$, where

$$w_t(X,V) := w(X,V) \prod \{ e^{t q_{\square}(V)} : \ \square \text{ off at } V,\ p_*(\square, V) \text{ in } X \}.$$

Here $q_{\square}$, with the sign $+$, is the part of q_D with $y \in \square$; a cube $\square$ off at V and $p_*(\square, V)$ are as in Step 5 of the proof of Lem. 5.47; m_y and ζ_p are the mask and the clip of item 2 of §5.4; and $q'_{D'}$ is q_D taken at x', with D' in place of D. Then $\rho[u_t;\ell] = \rho[u;\ell] e^{t q_D}$ at every V, and $\frac{d}{dt}\log(\rho[u_t;\ell])'(V')$ at 0 is the mean of $q_D(U)$ given V'. At a G-valued V' without a rough plaquette, this derivative equals

$$D\mathsf{V}'(u)[s_x[D]] + q'_{D'}(V') + \sum_{Y} (n[u;D] + DE'^{h}(u)[du_t/dt])(Y, V').$$

Remark 6.39 (The size of D'). A bound on $\|D'\|_{(\lambda_\flat)}$ is the subject of Thm. 6.20.

Item 8: what this section rests on.

Remark 6.40 (The results on which this section rests). This section rests on §6.1, on Thm. 6.7 (U) and on (c+), (h), (h1), (g1), (g3) and (h3) of that theorem, on Lems. 6.13 and 6.9, on Rem. 5.6, and on Sections 3, 4.1, 4.2, 2.1, 4.4, 5.3 and 5.4.

Proof of Thm. 6.33. Step 0: the floor x_3, the order of the constants, and the inequalities that all later steps use. The floor x_3 exists, and each new letter depends only on letters introduced before it. For $x \geq x_3$ we have the following.

(i) $p_0 \geq p_0' \geq 1$.
(ii) $\varepsilon < \varepsilon' \leq L^{1/2}\varepsilon$, and $\varepsilon' \leq 1/40$.
(iii) $c_a(x) \leq c_a(x')/(0.5946L^{1/2})$.
(iv) $\theta_D/L \leq \lambda_\flat/9$.
(v) $8c_a(x) \leq 1/4$.
(vi) With $r := C_\eta^\flat\varepsilon$, one has $r \geq 7\varepsilon'$, $r \leq 1/(22L)$ and $17.1r\varrho(x) \leq 0.1$.
(vii) $\omega \leq Mr'$, $3\omega \leq \ell_w(x')$ and $\omega \geq M + R_2 + \varrho_L$, and B_ω is embedded.

Order. The numbers $\mathsf{g}_\flat, g_{1b}, g_b^-$ depend on λ_* of (O0). The number k_1 depends on $g_{1b}/\mathsf{g}_\flat$ of (O0a) and on $C_\eta^\flat$, that is, on a_c of (O10) and on L. The number k_2 depends on $g_b^-/\mathsf{g}_\flat$ of (O0a), on L, on C_D and θ_D of (O6), and on K_c of (O10); and $k_3 = 6730L^4$ depends on L. None of these three numbers depends on M, κ, c_h, E_a, x; they are introduced at (O10a). The constant C_3 depends on κ, L and M, on $\mathsf{g}_\flat$ of (O0a) (through $c_Q^{\mathrm{su}} = 306M^3\mathsf{g}_\flat^2$ of (O10)), and on E_a. The floors of this section are free of C_3; the last floor x_s depends on it, by way of $\check{\omega}_3$ in (W3) of item (1) of §6.1, and since C_3 is fixed at (O12a), ahead of (O13), the order contains no circle. Finally, x_3 depends on x_9^U and on (O0) to (O12a).

Existence. Each of the conditions (Z.0) to (Z.4) is one-sided and holds for all large x. In (Z.0), $T_s\varepsilon' = L^{1/2}p_0'x^{-1/4}$, while ω and $\ell_w(x')$ are $O(\log x)$ and $1/\mathsf{g}_x = 4(\log x)^2/(\vartheta_0^\flat \log L)$; so each of its three inequalities, divided by its right side, has on the left at most a constant multiple of a fixed power of $\log x$, times $x^{-1/4}$. (Z.1) holds since p_0 grows. In (Z.2), $n_\chi = O((\log x)^3)$ and $p_0^2 = A_0^2(\log x)^{2\mathfrak{p}_0}$ with $2\mathfrak{p}_0 > 4$, so that, as x grows, $e^{-c_\xi^2 p_0^2/2}$ tends to 0 more quickly than any negative power of x. (Z.3) amounts to $x^{-11} \leq (17.9L^4C_D^2)^{-1}$. In (Z.4) the left side is at most a constant multiple of a fixed power of $\log x$, times $x^{-1/2}$. Thus S_3 is a non-empty upper half line and $x_3 = 1 + \inf S_3$ belongs to it. No condition depends on N, K, b, a member, a row or D.

The least floor. The factors ω, $\ell_w(x')$ and n_χ are integer valued, do not decrease in x, and appear as factors on the left sides; all other factors, and g_x, are continuous and positive. Suppose that (Z.0) to (Z.4) hold for all $x > t$. If, at such x, we give these three factors their values at t, the left sides become no larger; letting x tend to t from above, we obtain the conditions at t. Hence the set of the thresholds is a closed half line, and it has a minimum.

The inequalities. Since $x \geq x_3 \geq x_9^U$, the floors (Y.0) to (Y.9) of item 5 of §5.3 hold, and with them, by (Y.0)(0a), the floors of Sections 4.1 and 4.2. In (Y.0)(0b), and in (0d) as §5.3 states it, T is the number $L^{3/2}/(19\mathsf{N}_\mathrm{c}) > 1$ of Step 0 (iv) of the proof of Thm. 5.15. (0b) contains $\mathsf{K}_\mathrm{c}(17T\varepsilon')^2 \leq 0.12\varepsilon$, and since $17T \geq 1$ this gives $\mathsf{K}_\mathrm{c}\varepsilon'^2 \leq 0.12\varepsilon$.

(i) By (F8$_k$) of item G of §4.1, $p_0' \geq 1$; and p_0 increases.
(ii) $\varepsilon'/\varepsilon = L^{1/2}p_0'/p_0 \leq L^{1/2}$. By (0b), $(p_0'/p_0)^4 \geq 1/2$, hence $\varepsilon/\varepsilon' \leq 2^{1/4}L^{-1/2} < 1$; and $\varepsilon' \leq 1/40$ by (F8$_k$).
(iii) $c_a(x')/c_a(x) = L^{1/2}(p_0'/p_0)^3 \geq 2^{-3/4}L^{1/2}$, where $2^{-3/4} \geq 0.5946$.
(iv) By (O6) in item A of §4.1, $\theta_D \leq \theta_\mathbb{C}/2 = \theta^\flat/12 \leq \theta_1/12 \leq \lambda_*L/144$, and $\lambda_*/144 = \lambda_\flat/9$.
(v) This is (Y.1).
(vi) The bound $r \geq 7\varepsilon'$ follows from $C_\eta^\flat \geq 7L^{1/2}$ and (ii); the other two bounds are (W1) and (W2) of item 6 of §4.2 at $\tau = 1$.

(vii) The bound $\omega \le Mr'$ is the second inequality of (Y.7), together with $\lceil Mr(x)/L\rceil \le M\lceil r(x)/L\rceil$ and $M+2 \le 2M$; then $3\omega \le \ell_w(x')$, because $3Mr' \le \ell_w(x')$. Next, $\omega \ge M + R_3 + 2 = M + 2R + \varrho_L + 7$. Finally, $2\omega + 1 < N$ by (Tor) (the clauses on N in Step 0 of the proof of Thm. 4.39), so B_ω is embedded.

Step 1: the member $s_x[D]$, the derivative, and the split into five terms. We prove the following three facts.

(1) $s_x[D] \in T^a(x)$ and $\sup_{\mathcal{U}_{\{\boxdot\}}(x)} |s_x[D](\{\boxdot\},\cdot)| \le \sigma_D$; moreover $s_{x'}[D'] \in T^a(x')$.
(2) For $V' \in \mathcal{U}_Y(x')$,

$$DA_\ell(u)[s_x[D]](Y,V') = 1\{Y = \{\square_0\}\} \sum_{X \text{ cube under } \square_0} [g_X(V') - g_X(1)] + \mathrm{Rst}(Y,V'),$$

where $g_X := E_\xi[\chi_X e^{E^a(X,U)} s(X,U)]$ and $s := s_x[D]$.
(3) With the objects defined below,

$$g_X = M_X + \frac{1}{2}\mathrm{Tr}_X + E_\xi[\chi_X R_X] - E_\xi[(1-\chi_X)Q_X] + E_\xi[\chi_X(e^{E^a(X,U)} - 1)s(X,U)].$$

Proof of (1). We apply Lem. 6.32 to D at x and to D' at x'. The hypotheses (F1) and (F2) hold at x by the claim (Fl) of Step 0 in §5.4, and at x' by $\varepsilon' \le 1/40$ and $\varrho(x') + 2 \le c_R \log x' + 3 \le 20 c_R \log x' \le Mr'$. (Tor) gives $N \ge 16Mr(x)$ and hence the required lower bounds on the fine side NL and on the coarse side $N \ge 16Mr(x')$. On the coarse torus the coordinate paths are defined and $\square^\approx$ embeds, by (0.3) of Step 0 of the proof of Thm. 4.2. The family D' is complex, symmetric because D is, and of range $\le \varrho(x')$; so the same conclusions hold at x'.

Proof of (2). If $\sigma_D = 0$, everything vanishes. Otherwise $u + z(s,0,0) \in M_{16}(x;p)$ for $|z| \le z_0 := 8c_a(x)/\sigma_D$, so by Thm. 6.7 (U) and (c+) the derivative exists and is one and the same object for all p. Thm. 6.7 (h), applied with $\delta = s$, gives the formula with the sum over the X with $Y_0(X) = Y$, and $s(X) = 0$ for $\#X \ge 2$. Every single cube is small, as $1 \le L/2 - 2$; by Step 8, (8.2), of the proof of Thm. 5.15, for such a cube X the equality $Y_0(X) = \{\square_0\}$ then holds, where $\square_0$ is the cell that contains the cube in its fine set. As the grids are fitted ("The grids fit" in item 1 of §2.1, with the proof given there), the fine set of $\square_0$ is the union of L^3 cubes, and it is also the union of the blocks $B(y')$, $y' \in \square_0$.

Proof of (3). Fix a cube X under $\square_0$ and $V' \in \mathcal{U}_{\{\square_0\}}(x')$, extended by 1. Put

$$Q_X(\xi,V') := \frac{x}{2} \sum_{y\in X} \sum_{p,p'} D_y(p,p')\, (T(y(y),y(p))F_p^\flat) \cdot (T(y(y),y(p'))F_{p'}^\flat),$$

with $F^\flat = \bar F + \mathrm{d}^V G\xi$. This is a polynomial of degree 2 in the real variable ξ, defined for all ξ, and its coefficients are holomorphic in V' by (2d) of (Γ2) in item 4 of §5.3, since $X'_P = -i \log \partial V'(P)$, the logarithm being given by its power series (item C6 of §4.1). On $\chi_X = 1$ put $R_X := s(X,U) - Q_X$. Then $\chi e^{E^a} s = \chi s + \chi(e^{E^a} - 1)s$ and $\chi s = Q - (1-\chi)Q + \chi R$. As $E_\xi \xi = 0$, the cross term has mean 0; moreover $E_\xi(G\xi)(G\xi)^{\mathsf T} = x^{-1} GG^{\mathsf T} = C/x$, the transpose being that of the bilinear form, also at complex V'; and, written in an orthonormal basis of $\mathfrak{g}$, the form $Y \cdot Y'$ becomes the standard bilinear form of $\mathbb{C}^3$. Therefore $E_\xi Q_X = M_X + \frac{1}{2}\mathrm{Tr}_X$, with

$$M_X := \frac{x}{2} \sum_{y\in X} \sum D_y(p,p')\, (T\bar F_p) \cdot (T\bar F_{p'})$$

and

$$\mathrm{Tr}_X := \sum_{y\in X} \sum D_y(p,p')\ \mathrm{tr}[T(y(y),y(p))\, (\mathrm{d}^V C \mathrm{d}^{V\mathsf T})(p,p')\, T(y(y),y(p'))^{\mathsf T}].$$

At $V' = 1$ we have $X' = 0$, $a_0 = 0$ and $\bar F = 0$, so that $M_X(1) = 0$.

The two parts of the remainder. We define

$$n^{\mathrm{fix}}(\{\square_0\},V') := \sum_X \{E_\xi[\chi R_X] - E_\xi[(1-\chi)Q_X] + \frac{1}{2}\mathrm{Tr}_X\}(V') - \text{the same at } 1 + [\sum_X M_X(V') - s_{x'}[D'](\{\square_0\},V')],$$

and $n^{\mathrm{fix}}(Y) := 0$ for $\#Y \geq 2$; further

$$n^{\mathrm{mem}}(Y,V') := 1\{Y = \{\square_0\}\} \sum_X \{E_\xi[\chi_X(e^{E^a} - 1)s](V') - \text{the same at } 1\} + \mathrm{Rst}(Y,V').$$

The part n^{fix} contains no member.

Step 2: the member $s(X,U)$ against Q_X where $\chi_X = 1$, at complex V'.

Lemma 6.41 (Comparison of the member at the chart with its quadratic form at the centre)**.** In the setting of item 2 and with the objects of Step 1, let $V' \in \mathcal{U}_{\{\square_0\}}(x')$, let X be a cube under $\square_0$, and let ξ be real with $\chi_X(\xi) = 1$. Then

$$|R_X| \leq 289.92 M^3 \mathsf{g}_\flat (1320 g_{1b} + 216 \mathsf{g}_\flat) \|D\| x r^3 \leq k_1 \varepsilon \sigma_D.$$

The part of n^{fix} consisting of the terms $E_\xi[\chi R_X]$ is at most $2L^3 k_1 \varepsilon \sigma_D$ in modulus. ■

Proof of Lem. 6.41. (i) *The size of η.* Let $y \in X$ and $|y-p| \leq \varrho(x)$. Then, by (G2) of Step 0 in §5.4, the four bonds of p, as well as every bond of $\Gamma(y,\pi(p))$, are bonds of $X^{\approx}$. For each such bond b, its window Y_b lies within r_w of b by (2a); hence χ_X tests that window, and (2e) gives $\|(G\xi)_b\| \leq C^\flat_\varphi c_\xi \varepsilon = \varepsilon/6.5$. Such a bond $b = (z,\mu)$ is a bond of a plaquette based at z, and the block of z meets $X^{\approx}$. By the last sentence of Step 2 (ii) of the proof of Thm. 5.15, the boxes of radius R_3 centred at the sites whose blocks meet $X^{\approx}$ lie in $\square_0^{\approx\approx}$, where V' satisfies (e1), (e2) and $|\partial V'(P) - 1| < 2\varepsilon'$. So (Γ3) of item 4 of §5.3, applied with $\hat\varepsilon = \varepsilon'$, gives $\|a_{0,b}\| \leq |a_{0,b}| \leq \mathsf{a_c}\varepsilon' \leq \mathsf{a_c} L^{1/2}\varepsilon$. Since $\eta = 0$ on the combs and $\varepsilon/6.5 + \mathsf{a_c}L^{1/2}\varepsilon \leq C^\flat_\eta \varepsilon$, we obtain $\|\eta_b\| \leq r$ on all bonds that occur.

(ii) *Steps 4 and 5 of the proof of Thm. 4.5.* In clause (CD) below, (C) is Step 4 and (D) is Step 5 of that proof, each stated under the hypotheses on which its proof rests.

(CD) Let $0 < r \leq 1/(22L)$, let V' be an $\mathrm{SL}(2,\mathbb{C})$ field with $|V'(c)| \leq 1 + 10^{-4}$, let η be a $\mathfrak{g}_{\mathbb{C}}$-valued function on I°, with $\eta = 0$ on the combs, and let $U = \mathsf{U}(\eta; V')$.

(C) If p is a fine plaquette with $\|\eta_b\| \leq r$ on its four bonds and, in case p is an edge plaquette under P, $\partial V'(P) = e^{iX'_P}$ with $|X'_P| \leq (2.06/7)r$, then $|F^\flat_p| \leq 6r$, $|F_p - F^\flat_p| \leq 54r^2$ and $|F_p| \leq 8.25r$.

(D) If y is a fine site with $n := |y-p| \leq \varrho(x)$ and $n + L < NL/2$, if $17.1 r \varrho(x) \leq 0.1$, if $\|\eta_b\| \leq r$ on the bonds of $\Gamma(y,\pi(p))$, and if $|V'(\pi'_j)| < 2$ on the first pieces π'_j of $\Gamma(y(y),y(p))$, then $\|\mathrm{Ad}_{U(\Gamma(y,\pi(p)))} - T(y(y),y(p))\| \leq 160 r n$, and the norm of $\mathrm{Ad}_{U(\Gamma(y,\pi(p)))}$ is at most 4.9, that of $T(y(y),y(p))$ less than 4.

Proof of (CD). The proof is that of Steps 4 and 5 of the proof of Thm. 4.5, word for word, together with Lem. 4.8 (item 3.2 of §4.2), which these two steps cite. The hypotheses of (CD) take the place of what the two steps invoke, namely Steps 2 and 3 of that proof, $r \geq 7\hat\varepsilon$, $(\mathrm{W1}_\tau)$ and $(\mathrm{W2}_\tau)$; in particular $r \leq 1/(22L)$ takes the place of $(\mathrm{W1}_\tau)$, and $17.1 r\varrho(x) \leq 0.1$ that of $(\mathrm{W2}_\tau)$. Beyond this, the two steps depend neither on $\chi_\square = 1$ nor on y being under $\square$, and they depend on no row. □

The two steps are stated under $\chi_\square = 1$ for y under $\square$; the hypotheses of (CD) hold here. To begin with, $|V'(c)| < 1 + \vartheta(x') < 1 + \vartheta^\flat_0$ by (u1) at x', and $\vartheta^\flat_0 = \theta^\flat/24 \leq 10^{-4}$ by (O6) in item A of §4.1. If p is an edge plaquette under P, then $y(p) = y_P$; the block of $y(p)$ meets $X^{\approx}$, because it contains $\pi(p)$ by (G2); and P is a plaquette of the box of radius R_3 centred at $y(p)$, which lies in $\square_0^{\approx\approx}$ by (i). Hence $|\partial V'(P) - 1| < 2\varepsilon' \leq 1/20$ by (u3), the logarithm in $X'_P = -i \log \partial V'(P)$ is given by its power series, $e^{iX'_P} = \partial V'(P)$, and $|X'_P| \leq 2.06\varepsilon'$ by item C6 of §4.1. As for the paths, $3\varrho_L \leq \ell_w(x')$ by (0.2) of Step 0 of the proof of Thm. 4.2, and $\varrho_L \leq Mr'$ by $(\mathrm{G1}_k)$ of item G of §4.1. Thus $|V'(\pi'_j)| < 2$ on the first pieces of $\Gamma(y(y),y(p))$: these are coarse paths of at most

$3\varrho_L \le \ell_w(x')$ bonds within ϱ_L of $y(y)$ in $\square_0$, so they lie in $\square_0^{\approx}$, where (u2) holds. The bound $\|\eta_b\| \le r$ is (i); the inequalities $7\varepsilon' \le r$, $r \le 1/(22L)$ and $17.1 r\varrho(x) \le 0.1$ are Step 0 (vi), so that $|X'_P| \le 2.06\varepsilon' \le (2.06/7)r$; and $|y-p|+L < NL/2$ follows from (Tor) with (F2) of Step 1 (1). The two steps, that is (C) and (D) of (CD), therefore give $|F_p^\flat| \le 6r$, $|F_p - F_p^\flat| \le 54r^2$, $|F_p| \le 8.25r$ and $\| \mathrm{Ad}_{U(\Gamma(y,\pi(p)))} - T(y(y), y(p))\| \le 160 r|y-p|$, the norm of the first of these two operators being at most 4.9 and that of the second less than 4.

(iii) *The sum.* Put $A_p := \mathrm{Ad}_{U(\Gamma(y,\pi(p)))} F_p$ and $B_p := T(y(y), y(p)) F_p^\flat$. Then $A_p - B_p = (\mathrm{Ad}_U - T)F_p + T(F_p - F_p^\flat)$, so that $\|A_p - B_p\| \le (160 \cdot 8.25|y-p| + 4\cdot 54)r^2 = (1320|y-p| + 216)r^2$, while $\|A_p\| + \|B_p\| \le (4.9 \cdot 8.25 + 4\cdot 6)r = 64.425r$. Since D_y and the form are symmetric, the cross terms cancel, and

$$\sum D_y(p,p')(A_p \cdot A_{p'} - B_p \cdot B_{p'}) = \sum D_y(p,p')(A_p - B_p)\cdot(A_{p'} + B_{p'}).$$

Each base site carries three plaquettes, so $\sum_p e^{-\lambda_\flat |y-p|}|y-p| \le 3 g_{1b}$ and $\sum_p e^{-\lambda_\flat |y-p|} \le 3\mathsf{g}_\flat$. With $|Y \cdot Y'| \le \|Y\|\|Y'\|$, the modulus of the sum is at most $\|D\| \cdot 3(1320 g_{1b} + 216 \mathsf{g}_\flat) r^2 \cdot 3\mathsf{g}_\flat \cdot 64.425 r$. Multiplication by $\frac{x}{2}M^3$ brings in the factor $\frac{1}{2}\cdot 9 \cdot 64.425 = 289.9125$. Now $xr^3 = (C_\eta^\flat)^3 p_0^2 \varepsilon$, and division by $\sigma_D = 306 M^3 \mathsf{g}_\flat^2 p_0^2 \|D\|$ gives $289.9125/306 \le 0.94743 \le 0.9475$, which is the bound $k_1 \varepsilon \sigma_D$. The weight $|y-p|$ is absorbed by $e^{-\lambda_\flat |y-p|}$, so that no logarithm appears. Finally, $|E_\xi[\chi_X R_X]|$ is at most the supremum; the subtraction at 1, a field that lies in $\mathcal{U}_{\{\square_0\}}(x')$, doubles it; and there are L^3 cubes. This gives the bound $2L^3 k_1 \varepsilon \sigma_D$ in n^{fix}. □

Step 3: the tree part. In this step we compare the form D, taken at the centre, with $s_{x'}[D']$.

Lemma 6.42 (The tree part at the centre)**.** Let $\square_0$ and M_X be as in Step 1. Then every $V' \in \mathcal{U}_{\{\square_0\}}(x')$ satisfies

$$\Big| \sum_{X \text{ under } \square_0} M_X(V') - s_{x'}[D'](\{\square_0\}, V') \Big| \le \sigma_D \, [L^3 k_2 \varepsilon' + x^{-20}].$$

■

Proof. We fix a site $y' \in \square_0$ and a site $y \in B(y')$, and we write $o := y' = y(y)$. We also put $\tilde{X}_P := T(y', y_P) X'_P$; this is the dressing at y' with which $s_{x'}[D']$ is defined.

(1) *The field.* Let Bx be the box of radius R_T about o. By (0.4) in Step 0 of the proof of Thm. 4.2, the box Bx lies inside $W_{\square_0}$ and inside $\square_0^{\approx}$. On Bx the field V' has (u), and hence, by (0.6) of the same Step 0, it has (h) at $\hat{\varepsilon} = \varepsilon'$. For every plaquette p with $|y-p| \le \varrho(x)$ we have $\mathrm{dist}(y,p) \le \varrho_L$, and the box of radius R_3 about $y(p)$ is contained in Bx, since $R_3 + 2\varrho_L \le R_T$ by (0.1). Now (C3.1) and (C3.2) of item D of §4.1, together with (0.5), give

$$\bar{F}_p = \sum_P H^\natural(p,P)\, T(y(p), y_P) X'_P + \Lambda_p$$

with a remainder that satisfies $\|\Lambda_p\| \le \mathsf{K}_c \varepsilon'^2$.

(2) We put $dH := H^\natural - H^{\sharp <}$. This kernel vanishes beyond R_2 by (C3.4), and its row sums are at most ti_c by (C3.5). In $T(y', y(p))\bar{F}_p$ the kernel dH gives a sum over P in which X'_P is carried by two transports, each of norm at most 4 by (0.2) and (0.7), while $|X'_P| \le 2.06\varepsilon'$. So this contribution is at most $4 \cdot 4 \cdot 2.06\, \mathsf{ti}_c \varepsilon' = 32.96\, \mathsf{ti}_c \varepsilon' \le \varepsilon'^2$, the last inequality being (F9$_k$) of item G of §4.1.

(3) We apply Lem. 4.3, which is stated in Step 3 of the proof of Thm. 4.2, with $n = 2$ and $i_0 = y$. As K_1 we take the point kernel at (y,p); it is scalar, its coarse range does not exceed ϱ_L, and $\|K_1\|_\theta = e^{\theta\, \mathrm{dist}(y,p)}$. As the second kernel we take $K_2 := H^{\sharp <}$; its range is R_2, and its norm obeys $\|\cdot\|_{\theta_D} \le C_D$ by (Dec). The remaining choices are $\theta = \theta_D \le 1$, $\rho_B = R_T$ and $\hat{\varepsilon} = \varepsilon'$. The hypotheses of the lemma are met: $\varrho_L + R_2 \le R_T - 1$ holds by (0.1), $3R_T + 1 \le \ell_w(x')$ by (0.2), $4R_T + 2 \le N$ by (0.3), and $35100 R_T^2 \varepsilon' \le 1$ by (F8$_k$).

Let $\tilde a_p := \sum_P H^{\sharp<}(p,P)\tilde X_P$. Since $\sup\|\tilde X_P\| \le 8.24\varepsilon'$, the lemma yields

$$\|T(y',y(p))\sum_P H^{\sharp<}(p,P)T(y(p),y_P)X'_P - \tilde a_p\| \le 7020\cdot 8\cdot 1.2\cdot 8.24\,\theta_D^{-2}e^{\theta_D \mathrm{dist}(y,p)}C_D\varepsilon'^2,$$

and the numerical factor is $7020\cdot 8\cdot 1.2\cdot 8.24 \le 555310.1$. From $\mathrm{dist}(y,p)\le (|y-p|+L-1)/L$, from $\theta_D\le 1$ and from Step 0 (iv) we get $e^{\theta_D \mathrm{dist}} \le e\cdot e^{\lambda_\flat|y-p|/9}$; moreover $555310.1\,e \le 1.50949\cdot 10^6$.

(4) Collecting (1) to (3), we obtain

$$b_p := T(y',y(p))\bar F_p = \tilde a_p + e_p$$

with

$$\|e_p\| \le [4\mathsf{K}_\mathrm{c} + 1 + 1.51\cdot 10^6\theta_D^{-2}C_D e^{\lambda_\flat|y-p|/9}]\varepsilon'^2;$$

here the term $4\mathsf{K}_\mathrm{c}$ stems from Λ_p and one transport, the term 1 from (2), and the last term from (3). By (Γ3), together with (L2^{su}) and with (0b) of (Y.0) in item 5 of §5.3, we have $|\bar F_p| \le \mathsf{N}_\mathrm{c}L^{-2}\varepsilon' + \mathsf{K}_\mathrm{c}\varepsilon'^2 \le \varepsilon'$, so that $\|b_p\|\le 4\varepsilon'$. Further, we have $\|\tilde a_p\|\le 8.24C_D\varepsilon'$, and since $C_D\ge 4$, also $\|b_p+\tilde a_p\|\le 9.24C_D\varepsilon'$.

(5) As $b_p = \tilde a_p + e_p$, the difference $b_p\cdot b_{p'} - \tilde a_p\cdot\tilde a_{p'}$ equals $e_p\cdot b_{p'} + \tilde a_p\cdot e_{p'}$. Since D_y and the form $Y\cdot Y'$ are symmetric, exchanging p and p' in the sum $\sum D_y(p,p')\,\tilde a_p\cdot e_{p'}$ turns it into the sum of $D_y(p,p')\,e_p\cdot\tilde a_{p'}$; with this argument, the same as in Step 2 (iii), we get

$$\sum D_y(p,p')(b_p\cdot b_{p'} - \tilde a_p\cdot\tilde a_{p'}) = \sum D_y(p,p')\,e_p\cdot(b_{p'}+\tilde a_{p'}).$$

We use $\sum_p e^{-(8/9)\lambda_\flat|y-p|}\le 3g_b^-$ and $\mathsf{g}_\flat \le g_b^-$. By the definition of $\|D\|$, the decay of $D_y(p,p')$ in p absorbs the factor $e^{\lambda_\flat|y-p|/9}$ in the upper bound for $\|e_p\|$ obtained in (4), and the contribution of a single site y to the right side of (5) is at most

$$\|D\|\cdot 3g_b^-[4\mathsf{K}_\mathrm{c}+1+1.51\cdot 10^6\theta_D^{-2}C_D]\varepsilon'^2\cdot 3\mathsf{g}_\flat\cdot 9.24C_D\varepsilon'.$$

For a cube, this is multiplied by $(x/2)M^3$. Now $x\varepsilon'^3 = Lp_0'^2\varepsilon' \le Lp_0^2\varepsilon'$ and $9\cdot 9.24/612\le 0.1359$, so that each cube contributes at most $k_2\varepsilon'\sigma_D$; there are L^3 cubes.

(6) *The free identity.* Let $D^< := L\sum_{y\in B(y')} H^{\sharp<\mathsf{T}}D_yH^{\sharp<}$. Then

$$(x/2)\sum_{y\in B(y')}\sum_{p,p'} D_y(p,p')\,\tilde a_p\cdot\tilde a_{p'} = (x'/2)\sum_{P,P'} D^<_{y'}(P,P')\,\tilde X_P\cdot\tilde X_{P'},$$

because $x = Lx'$. By Step 1, the sites under $\square_0$ are the sites of the blocks of the sites of $\square_0$.

(7) *Tails.* Let ct denote the cut at $\varrho(x')$. Then

$$D^< - D' = \{D^< - \mathrm{ct}\,D^<\mathrm{ct}\} - \mathrm{ct}\,(T_\ell D - D^<)\,\mathrm{ct}.$$

By (0.8), the row sums of $H^{\sharp>}$ are at most C_Dx^{-128}. If $|y'-P|>\varrho(x')$, then $\mathrm{dist}(p,P)>\varrho(x')-\varrho_L$, and therefore the part of a row sum of $H^{\sharp<}$ that comes from such P is at most $C_Dx'^{-64}e^{\theta_D\varrho_L}$. Here $\theta_D\varrho_L\le 64\log x/L+2$; next, $x'^{-64}\le x^{-32}$, because $x\ge L^2$ by (P0) of item G of §4.1; and $x^{64/L}\le x$, because $L\ge 64$, which holds since (L2^{su}) with $\mathsf{N}_\mathrm{c}\ge 1$ gives $L\ge 1.44\cdot 10^8$. Altogether, that part of a row sum is at most $e^2C_Dx^{-31}$.

Hence

$$\sum_{P,P'}|D^< - D'| \le L^4\cdot 9\mathsf{g}_\flat^2\|D\|C_D^2(2x^{-128}+2e^2x^{-31}) \le 161L^4\mathsf{g}_\flat^2C_D^2\|D\|x^{-31},$$

because $9(2+2e^2)\le 151.01\le 161$. The transports in $\tilde X_P$ run along paths of at most $3R_T$ bonds, whence $|\tilde X_P\cdot\tilde X_{P'}|\le (8.24\varepsilon')^2$. We multiply by $(x'/2)(8.24\varepsilon')^2M^3 = 33.9488\,p_0'^2M^3$ and divide by σ_D; since $33.9488\cdot 161/306\le 17.862$, the tails contribute at most $17.9L^4C_D^2x^{-31}\sigma_D \le x^{-20}\sigma_D$, the last inequality being (Z.3). □

Step 4: the loop part and clause (2τ). We show that for $V' \in \mathcal{U}_{\{\square_0\}}(x')$ and for every cube X under $\square_0$

$$\frac{1}{2}|\mathrm{Tr}_X(V') - \mathrm{Tr}_X(1)| \le S_{\mathrm{tr}} T_s^{-2},$$

where $S_{\mathrm{tr}} := 28204\, M^3 \mathsf{g}_\flat^2 \|D\|/\gamma \le k_3 p_0^{-2} \sigma_D$. Summing over the L^3 cubes, we get $k_3 L^3 p_0^{-2} x^{-1/2} \sigma_D$. The step also yields clause (2τ) of item 5 of §6.4.

The gauge-box estimate. We apply Lem. 6.13 with $T = T_s$. Then (C0) to (C2) are (Z.0), completed by $18\omega\varepsilon' \le 1/10$, which follows from (0d) of (Y.0) in item 5 of §5.3; the requirements on ω and on B_ω hold by Step 0 (vii).

The domain. Let o be the centre site of $\square_0$, and extend a field of $U^{17T_s}(\mathsf{B}_\omega; x')$ by the value 1 to the bonds outside B_ω. By (e1) and $\mathsf{g}_x < \vartheta_0^\flat / \log x'$, the field so extended lies in the interior of $\mathfrak{V}$; this is the inequality $|V'(c)| < 1 + \vartheta(x') + \mathsf{g}_x < 1 + \vartheta_0^\flat$ of Step 8, (8.2), of the proof of Thm. 5.15. Moreover (e2) gives $|V'(\pi)| < 2$ for every path of at most $\ell_w(x')$ bonds.

The part of V' on which Tr_X depends. The coarse site $y' = y(y)$ lies within M of o. The transport $T(y', y(p))$ depends on V' within ϱ_L of y'; the factor $\mathrm{d}^V(p, b)$ depends on the bond between $y(p)$ and $y(b)$; by (2b) of (Γ2) in item 4 of §5.3, $C(b, \cdot)$ depends on V' within $R_2 - 1$ of $y(b)$; and $y(b)$ lies within $\varrho_L + 1$ of y'. Altogether, Tr_X depends on V' only within $R_2 + \varrho_L$ of y', a region inside B_ω by Step 0 (vii).

Holomorphy and invariance. The function Tr_X is holomorphic by (2d), its other factors being polynomials in V'. Under a gauge transformation u we have $T(y', y'') \to \mathrm{Ad}_{u(y')}\, T\, \mathrm{Ad}_{u(y'')}^{-1}$, and $\mathrm{d}^V(p, b)$ transforms in the same manner; for $C = GG^{\mathsf{T}}$ we use (2c) together with $\mathrm{Ad}_u^{\mathsf{T}} = \mathrm{Ad}_u^{-1}$. Thus the matrix under the trace is conjugated by $\mathrm{Ad}_{u(y')}$, and the trace is unchanged. A gauge transformation u on B_ω is extended by 1, in the manner of Step 8 of the proof of Thm. 5.15.

Size. On $\mathfrak{V}$ we have $|V'(c)| \le 1 + \vartheta_0^\flat \le 1 + 10^{-4}$, by (O6) of item A of §4.1; hence $\|\mathrm{d}^V(p, b)\| \le |V'(c)|^2 \le 1.01$. A plaquette has four bonds, so $\|(\mathrm{d}^V C \mathrm{d}^{V\mathsf{T}})(p, p')\| \le 4.04^2 \sup \|C(b, b')\|$. On $\mathfrak{V}$ we also have $\|C(b, b')\| \le \|C\| \le 4/\gamma$ by (C2.5) of item D of §4.1; this is Lem. 3.26 (a), which is stated on all of $\mathfrak{V}$. With $\|T\|, \|T^{\mathsf{T}}\| < 4$ and $|\operatorname{tr}| \le 3\|\cdot\|$ we find $|\tau| \le 3 \cdot 16 \cdot 16.3216 \cdot 4/\gamma \le 3133.75/\gamma$. As $\sum_{p,p'} |D_y| \le 9\mathsf{g}_\flat^2 \|D\|$ and there are M^3 sites, $\sup|\mathrm{Tr}_X| \le 28203.8\, M^3 \mathsf{g}_\flat^2 \|D\|/\gamma \le S_{\mathrm{tr}}$. Finally, $S_{\mathrm{tr}}/\sigma_D \le 92.170/(\gamma p_0^2) = 6728.41\, L^4/p_0^2 \le k_3/p_0^2$.

The bound. Because $\omega \le Mr'$, the restriction to B_ω of a field in $\mathcal{U}_{\{\square_0\}}(x')$ has (u1) to (u3). The lemma therefore gives $|\mathrm{Tr}_X(V') - \mathrm{Tr}_X(1)| \le 2 S_{\mathrm{tr}} x^{-1/2}$.

Clause (2τ). Here the lemma is applied to the single function τ in the place of Tr_X, with any value of T for which (C0) to (C2) hold, and about any coarse site o within M of y'. The fields are extended by 1 and lie in the interior of $\mathfrak{V}$ by (e1); the function τ depends on V' only within $R_2 + \varrho_L$ of y', hence inside B_ω by Step 0 (vii); it is holomorphic and invariant by the two arguments just given; and $|\tau| \le 3133.75/\gamma$. For the value $\tau(1)$ let $V' = 1$ and let u be constant: then (2c) gives $C^1(b, b')\, \mathrm{Ad}_u = \mathrm{Ad}_u\, C^1(b, b')$; since Ad is irreducible on $\mathbb{C}^3$, each block is a scalar by Schur's lemma, and $\tau(1) = 3(\mathrm{d}C^1\mathrm{d}^{\mathsf{T}})(p, p')$.

Step 5: the cut-off and the member factor. We show that for every $V' \in \mathcal{U}_{\{\square_0\}}(x')$ and every cube X under $\square_0$

$$|E_\xi[(1 - \chi_X) Q_X]| \le 3.47\, c_\xi p_0 n_\chi e^{-c_\xi^2 p_0^2/2} \sigma_D$$

and

$$|E_\xi[\chi_X (e^{E^a(X,U)} - 1) s(X, U)]| \le 9.09\, c_a(x) \sigma_D.$$

When these bounds are doubled and summed over the L^3 cubes, they become at most $x^{-10}\sigma_D$ and $18.2 L^3 c_a(x) \sigma_D$ respectively.

The cut-off. Let $\Xi := \sup \|\xi_b\|$, the supremum running over the bonds tested by χ_X. For a tested bond b' there is a bond b of $X^{\approx}$ with $|y(b') - y(b)|_\infty \le L(r_w + 1)$; hence the base site of b' lies within $L(r_w + 1) + L - 1 < L(r_w + 2)$ of the box $X^{\approx}$, which has side $M(2r(x) + 1)$, and the number of tested bonds is at most n_χ. Recall that $\|T\| < 4$. By (Γ3), (L2^{su}) and (0b) we have

$\|\bar F_p\| \le \mathsf{N}_c L^{-3/2}\varepsilon + \mathsf{K}_c\varepsilon'^2 \le (10^{-4}+0.12)\varepsilon \le \varepsilon$. By (2a) and (2e), $\|(\mathrm{d}^V G\xi)_p\| \le 4\cdot 1.01\, C_\varphi^\flat \Xi$; the factor 1.01 is the bound $\|\mathrm{d}^V(p,b)\| \le |V'(c)|^2 \le 1.01$ of Step 4, obtained here from (u1) at x'.

On the event $\chi_X = 0$ we have $\varepsilon < \Xi/c_\xi = 6.5\, C_\varphi^\flat \Xi$, and therefore $\|F_p^\flat\| \le 10.54\, C_\varphi^\flat \Xi$ and

$$|Q_X| \le (x/2)\, 16\cdot 9 M^3 \mathsf{g}_\flat^2 \|D\| (10.54\, C_\varphi^\flat)^2 \Xi^2.$$

Moreover $\Xi^2 1\{\Xi > t\} \le \sum_b \|\xi_b\|^2 1\{\|\xi_b\| > t\}$, the left side being the largest term on the right. Each ξ_b has the form $\xi_b = x^{-1/2} g$ with g a standard Gaussian vector in $\mathbb{R}^3$, and for $t_g \ge 1$

$$E_\xi |g|^2 1\{|g| > t_g\} = (2/\pi)^{1/2}\{(t_g^3+3t_g)e^{-t_g^2/2} + 3\int_{t_g}^\infty e^{-v^2/2}dv\} \le 0.79789\,(t_g^3+3t_g+3/t_g)e^{-t_g^2/2} \le 5.6\, t_g^3 e^{-t_g^2/2},$$

since $0.79789\cdot 7 = 5.58523$.

We take $t_g = c_\xi p_0$, which corresponds to $t = c_\xi \varepsilon$ because $\varepsilon = p_0 x^{-1/2}$, and which satisfies $t_g \ge 1$ by (Z.1). Using $C_\varphi^\flat = (6.5 c_\xi)^{-1}$ we arrive at

$$|E_\xi(1-\chi)Q| \le 72(10.54/6.5)^2 \cdot 5.6\, M^3 \mathsf{g}_\flat^2 \|D\| n_\chi c_\xi p_0^3 e^{-c_\xi^2 p_0^2/2}.$$

After division by σ_D the constant is $(72/306)(10.54/6.5)^2\cdot 5.6 \le 3.4647$. Doubling and summing over the L^3 cubes gives the factor $6.93 \le 7$, so that (Z.2) yields the bound $x^{-10}\sigma_D$.

The member factor. On the event $\chi_X = 1$, Step 2 (ii) of the proof of Thm. 5.15, taken at $\tau = 1$, shows that the restriction of U to $X^{\approx}$ lies in $\mathcal{U}_X(x)$. Hence $|s(X,U)| \le \sigma_D$ and $|E^a(X,U)| \le 8c_a(x) \le 1/4$. For $|w| \le 1/4$ we have $|e^w - 1| \le e^{|w|} - 1 \le 1.13611|w|$, and $8\cdot 1.13611 \le 9.0889$. Doubling and summing over the L^3 cubes yields $18.178 \le 18.2$. The term $E_\xi[(1-\chi_X)Q_X]$ contains no member, and it belongs to n^{fix} by the definitions of Step 1; the term $E_\xi[\chi_X(e^{E^a}-1)s]$ contains E^a and belongs to n^{mem}.

Step 6: the clusters with $n(C) \ge 2$. We show that

$$\mathrm{CS}_{\tilde\kappa}(\mathrm{Rst}) \le 217 L^3 x^{-1/4}\sigma_D \le 1000\, e^{\tilde\kappa+3} L^3 x^{-1/4}\sigma_D.$$

To this end we apply Thm. 6.7 (h1) to the member $u \in M_8(x;p)$, with $\delta = s_x[D]$. Since δ lives on single cubes, Step 1 (1) shows that at every $\tilde\kappa$ its size in caps at p is $B = \sup|s|/c_a(x) \le \sigma_D/c_a(x)$. Then (h1) gives the bound $217 L^3 x^{-1/4} B c_a(x)$, whence the first inequality. The second follows from $1000\, e^{\tilde\kappa+3} \ge 1000\, e^9 \ge 8.1\cdot 10^6$.

Step 7: assembly, clause (a), and the first of the four remarks of item 7. We prove the two bounds of Thm. 6.33 (a),

$$\sup_{\mathcal{U}_{\{\square\}}(x')} |n^{\mathrm{fix}}(\{\square\},\cdot)| \le \sigma_D\big[L^3(2k_1\varepsilon + k_2\varepsilon') + k_3 L^3 p_0^{-2} x^{-1/2} + 2x^{-10}\big]$$

and $\mathrm{CS}_{\tilde\kappa}(n^{\mathrm{mem}}) \le B_{\mathrm{mem}}(\tilde\kappa)$, and their two consequences

$$\mathrm{CS}_{\tilde\kappa}(n) \le 1001\, e^{\tilde\kappa+3} L^3 x^{-1/4}\sigma_D$$

and

$$|n|_{a,x'}^{\mathrm{cs},(\tilde\kappa)} \le u_3(x')\|D\|_{(\lambda_\flat)},$$

together with the first of the four remarks of item 7 of §6.4.

By Step 1 we have $n = DA_\ell(u)[s] - s_{x'}[D'] = n^{\mathrm{fix}} + n^{\mathrm{mem}}$, since $M_X(1) = 0$ and $s_{x'}[D'](\cdot,1) = 0$. For n^{fix}, Steps 2, 5, 4 and 3 give the bound

$$\sigma_D[2L^3 k_1\varepsilon + x^{-10} + k_3 L^3 p_0^{-2} x^{-1/2} + L^3 k_2\varepsilon' + x^{-20}].$$

For n^{mem} we use Steps 5 and 6; a term on a single cell has weight 1 in the cell sum. Applying (Z.4), which is taken at the lower end $\kappa \le \tilde\kappa$, we get

$$\mathrm{CS}_{\tilde\kappa}(n) \le (1000\, e^{\tilde\kappa+3} + e^{\kappa+3}) L^3 x^{-1/4}\sigma_D \le 1001\, e^{\tilde\kappa+3} L^3 x^{-1/4}\sigma_D.$$

In caps at x'. As $x = Lx'$ and $c_a(x') = E_a p_0'^3 x'^{-1/2}$, the quotient of this bound by $c_a(x')$ equals

$$1001\, e^{\tilde\kappa+3} L^{3-1/4} (c_Q^{\mathrm{su}}/E_a)\, x'^{1/4} (p_0^2/p_0'^3) \|D\|.$$

By Step 0 (i) and (iii), $p_0^2/p_0'^3 = (p_0/p_0')^3/p_0 \le 1/0.5946$; furthermore $1001/0.5946 \le 1683.5 \le 1686$ and $e^{\tilde\kappa+3} \le e^{\kappa+4}$. The quotient is therefore at most $C_3 x'^{1/4}\|D\|$, which is the second of the two consequences.

Proof of the first of the four remarks of item 7. We use three relations. First, $\sigma_D \varepsilon = (c_Q^{\mathrm{su}}/E_a) c_a(x) \|D\|$, and $\varepsilon' \le L^{1/2}\varepsilon$. Second,

$$\sigma_D p_0^{-2} x^{-1/2} = (c_Q^{\mathrm{su}}/E_a) c_a(x) p_0^{-3} \|D\| \le (c_Q^{\mathrm{su}}/E_a) c_a(x) \|D\|.$$

Third, $2x^{-10}\sigma_D \le L^3 (c_Q^{\mathrm{su}}/E_a) c_a(x) \|D\|$. We apply these three relations to the bound on n^{fix} and find that each of its terms is at most a multiple of $(c_Q^{\mathrm{su}}/E_a)\, c_a(x)\|D\|$. By Step 0 (iii), that is, by $c_a(x) \le c_a(x')/(0.5946 L^{1/2})$, we then pass from $c_a(x)$ to $c_a(x')$, and $L^3/(0.5946 L^{1/2}) = L^{5/2}/0.5946$ is the factor that the remark carries. This proves the bound of that remark, in caps at x'.

Step 8: clauses (b), (c), (e) and (f). We prove Thm. 6.33 (b), (c), (e) and (f).

Clause (b). For $0 < \rho_z < z_0$, Cauchy's formula gives

$$DA_\ell(u)[s](Y, V') = (2\pi i)^{-1} \int_{|z|=\rho_z} E'^{\mathrm{old}}[u + zs](Y, V')\, z^{-2} dz.$$

At every z the integrand has the clauses without a size, by Thm. 5.15 (b), which applies at p by Thm. 6.7 (U). That clause is a statement on $E'^a = E'^{\mathrm{new}} + E'^{\mathrm{old}}$; since E'^{new} does not depend on z and $\int_{|z|=\rho_z} z^{-2} dz = 0$, the integral of $E'^{\mathrm{old}}[u+zs] z^{-2}$ coincides with that of $E'^a[u+zs] z^{-2}$. The integrand is bounded, and it is holomorphic in z and in V' separately, hence jointly by Osgood's lemma. Consequently the integral is holomorphic on $\mathcal{U}_Y(x')$, invariant, equal to 0 at pure gauges, and a function of the field on $Y^{\approx}$. By Step 1, $s_{x'}[D']$ has these clauses; hence n has them too.

The functions g_X and $E_\xi[\chi_X(e^{E^a} - 1)s]$ are holomorphic on $\mathcal{U}_{\{\square_0\}}(x')$, by Step 2 (ii) of the proof of Thm. 5.15, with domination by $\sigma_D e^{1/4}$. They are invariant: the transformation $(\xi, V') \to (\mathrm{Ad}_u \xi, V'^u)$ leaves χ_X, $E^a(X,U)$ and $s(X,U)$ unchanged, and Ad_u preserves the law of ξ; see the part on the clauses without a size in Step 8 of the proof of Thm. 5.15. They depend on the field in $\square_0^{\approx}$, and their differences with their values at 1 vanish at pure gauges. Hence $\mathrm{Rst} = DA_\ell(u)[s] - \sum[g_X - g_X(1)]$ has the clauses; so does n^{mem}, and then also $n^{\mathrm{fix}} = n - n^{\mathrm{mem}}$.

Reality. Let u and D be real, and let z be real. Then $u + zs$ is real, and E'^a is real at G-valued fields by Thm. 5.15 (d). Since E'^{new} does not depend on z, the z-derivative of E'^a, which equals that of E'^{old}, is real. As $H^\sharp$ is real, so is D', and $s_{x'}[D']$ is real at G-valued fields by Step 4 of the proof of Lem. 5.47; the same holds for g_X.

Clause (c). By (Fm1), $E'^{\mathrm{old}}[u](Y)$ is a sum over the clusters C with $\mathrm{co}(C) = Y$, and $\mathrm{co}(C)$ contains the cores J_α of the letters of C. A letter (a) at X has the core $J = Y_0(X)$, and $\Phi(C)$ depends on E^a only through its letters (a). So $E'^{\mathrm{old}}[u+zs](Y)$ depends on $s(X)$ only at the cubes X with $Y_0(X) \subset Y$, which are the cubes under Y; and $s(\{X\})$ depends on the D_y with $y \in X$. The second term of n, namely $s_{x'}[D'](\{\square\})$, depends on the $D'_{y'}$ with $y' \in \square$, and $D'_{y'}$ depends on the D_y with $y \in B(y')$, which lie under $\square$. Therefore both terms are 0 when all D_y under Y vanish.

Clause (e). The following are complex linear in D: $s_x[D]$, $T_\ell D$, the cut, the derivative $DA_\ell(u)[\cdot]$, and every one of the five terms of Step 1.

Clause (f). As their definitions show, the numbers k_1, k_2, k_3, C_3 and the conditions (Z.0) to (Z.4) are functions of the letters of Step 0 only. Among these letters, the numbers that come from the rows are C_D, θ_D (for every $\ell \in \mathcal{R}(b_*, K_*, \lambda_*, 1/9)$), K_{c}, a_{c} and ti_{c}.

Empty rows. We show that the steps above hold also for $\ell = \emptyset$, a case admitted by item 2 of §6.4.

By Rem. 5.6 in Step S6 of the proof of Thm. 5.3, the zero rows ($\mathsf{H}_y = \mathbb{R}$, $\ell_y(p) = 0$) yield the same objects as the empty rows; moreover, the zero rows form a member of $\mathcal{R}(b_*, K_*, \lambda_*, 1/9)$, and this member satisfies (SD) exactly when the empty rows satisfy it. Hence every statement made for all ℓ with (SD) holds for the zero rows. The objects that Rem. 5.6 lists are those of Sections 2.2 and 3 to 4.4. From the rows, Sections 5.3 and 5.4, Thm. 6.7 and the present section build only objects of three kinds: objects on that list; the numbers $|\ell_y(p)|$, $|\ell'_{y'}(P)|$ and $\mathsf{E}_y(p,p') = \langle \ell_y(p), \ell_y(p')\rangle$; and T_ℓ, which depends on $H^\sharp$. Each of them is expressed through the columns $\ell_y(p)$, and so takes the same value at 0 as at $\emptyset$.

Step 9: clause (d), the member. We show three facts: n depends on the member u through E^a alone; n is holomorphic along affine lines in $M_8(x;p)$; and, for $u_1, u_2 \in M_2(x;p)$,

$$\mathrm{CS}_{\tilde\kappa}(n[u_1;D] - n[u_2;D]) \le \frac{1}{3} B_{\mathrm{mem}}(\tilde\kappa)\, d_a.$$

Neither n^{fix} nor $s_{x'}[D']$ depends on the member, and E'^{old} depends on E^a alone, by the proof of clause (c) in Step 8.

Holomorphy. Let $\zeta \to u_\zeta$ be an affine map with values in $M_8(x;p)$ on an open set. Then the map $(\zeta, z) \to u_\zeta + zs$ is holomorphic value by value and, for $|z| < z_0$, takes its values in $M_{16}(x;p)$. By Thm. 6.7 (c+), $E'^{\mathrm{old}}[u_\zeta + zs](Y, V')$ is therefore holomorphic in (ζ, z); consequently its z-derivative at 0 is holomorphic in ζ.

The bound. As n depends on E^a alone, we may take $E_2^h = E_1^h$ and $w_2 = w_1$. If $d_a = 0$, both sides of the bound vanish; in any case $d_a \le 4$. For $|\zeta| < \zeta_0 := 6/d_a$, a radius with $\zeta_0 \ge 3/2$, the member $u_\zeta := u_1 + \zeta(u_2 - u_1)$ has an a-part within $2 + |\zeta| d_a < 8$ caps. Let $f(\zeta) := n^{\mathrm{mem}}[u_\zeta; D] - n^{\mathrm{mem}}[u_1; D]$; by Thm. 6.33 (a), $\mathrm{CS}_{\tilde\kappa}(f(\zeta)) \le 2B_{\mathrm{mem}}$ on the disc, and $f(0) = 0$. Since the torus is finite, only finitely many Y contain a given cell $\square$. For a choice of points $V'_Y \in \mathcal{U}_Y(x')$ and of numbers $|c_Y| \le 1$, the function

$$\phi(\zeta) := \sum_{Y \ni \square} e^{\tilde\kappa(\#Y - 1)} c_Y f(\zeta)(Y, V'_Y)$$

is a finite sum of holomorphic functions, with $|\phi| \le 2B_{\mathrm{mem}}$ and $\phi(0) = 0$. By Schwarz's lemma, $|\phi(1)| \le 2B_{\mathrm{mem}}/\zeta_0 = \frac{1}{3} B_{\mathrm{mem}} d_a$. The supremum over all such choices is the sum at $\square$. We bound this sum as a whole, and not term by term in Y followed by a summation, because Thm. 6.7 (h1) bounds the sum at each member.

Step 10: clause (g), the number V'. We show that

$$D\mathsf{V}'(u)[s_x[D]] = \sum_y \frac{3}{2} \sum_{p,p'} D_y(p,p')(\mathrm{d}C^1 \mathrm{d}^{\mathsf{T}})(p,p') + \sum_{\square} e_\square,$$

with the bound of Thm. 6.33 (g).

By Thm. 6.7 (g1), $\mathsf{V}'(u) = V_0^+ + \sum_Y v(Y;u)$, a finite sum, and V_0^+ does not depend on the member. Let $D\mathsf{V}'(u)[s]$ be the derivative at $z = 0$ of $\mathsf{V}'(u + z(s,0,0))$; as the sum over Y is finite, it equals $\sum_Y Dv(Y;u)[s]$. Because here $A = 8 < 16$, the quantity $Dv(Y;u)[s]$ of item 1 of §6.2, which is a sum of derivatives of terms, equals the derivative d/dz at 0 of $v(Y; u + z(s,0,0))$. This equality is stated in Thm. 6.7 (g3), for the case $A_a < 16$.

Thm. 6.7 (g3), applied with $\delta = s$, gives

$$Dv(Y;u)[s] = \sum_{Y_0(X) = Y} g_X(1) + r(Y),$$

where the weighted cell sum of r is at most $109 L^3 x^{-1/4} \sigma_D$. By Step 1 (3) at $V' = 1$,

$$g_X(1) = \frac{1}{2}\mathrm{Tr}_X(1) + \{E_\xi[\chi R_X] - E_\xi[(1-\chi)Q_X] + E_\xi[\chi(e^{E^a} - 1)s]\}(1),$$

and by Step 4,

$$\frac{1}{2}\mathrm{Tr}_X(1) = \sum_{y\in X}\frac{3}{2}\sum_{p,p'} D_y(p,p')(\mathrm{d}C^1\mathrm{d}^{\mathsf{T}})(p,p').$$

We define

$$e_\square := \sum_{X \text{ under } \square} \{E_\xi[\chi R_X] - E_\xi[(1-\chi)Q_X] + E_\xi[\chi(e^{E^a}-1)s]\}(1) + \sum\{r(Y) : \square \text{ the first cell of } Y\}.$$

By Steps 2 and 5 without the doubling, the first sum in the definition of $e_\square$ has modulus $\le \sigma_D[L^3 k_1\varepsilon + \frac{1}{2}x^{-10} + 9.09L^3 c_a(x)]$. For the second sum we have $|\sum\{r(Y) : \square \text{ the first cell of } Y\}| \le \sum_{Y\ni\square}|r(Y)| \le 109L^3x^{-1/4}\sigma_D$, by the bound on the weighted cell sum of r stated above, whose weights $e^{\tilde\kappa(\#Y-1)}$ are ≥ 1. Since $9.09 \le 9.1$ and $\frac{1}{2}x^{-10} \le x^{-10}$, adding the two estimates, we obtain $|e_\square| \le \sigma_D[L^3(k_1\varepsilon + 9.1c_a(x)) + 109L^3x^{-1/4} + x^{-10}]$, which is the bound stated in Thm. 6.33 (g).

Step 11: proof of Cor. 6.35. We prove the identity for rem′, its bound, the inequality $t' \le 0.2431\,t + 1$, and $t_k \le \max\{t_0, 1.3212\}$.

Since $DA_\ell(u)[\cdot]$ is linear,

$$DA_\ell(u)[s_x[D] + \mathrm{rem}] = s_{x'}[D'] + n[u;D] + DA_\ell(u)[\mathrm{rem}].$$

Thm. 6.7 (h3), in the setting (L3su), gives

$$\mathrm{CS}_{\tilde\kappa}(DA_\ell(u)[\mathrm{rem}]) \le [2430\mathsf{N}_{\mathrm{c}}^2 L^{-1/2} + L^{-1}]B_r c_a(x') \le \vartheta_a B_r c_a(x'),$$

by $[0.2430 + L^{-3/4}]L^{-1/4} \le 0.2431L^{-1/4}$. Together with Thm. 6.33 (a), this yields the upper bound for rem′ that Cor. 6.35 asserts. The clauses without a size hold for rem′: for $DA_\ell(u)[\mathrm{rem}]$ by the argument given for clause (b) in Step 8, with rem in the place of s, and for n by Thm. 6.33 (b). From $u_3(x)/u_3(x') = L^{1/4}$ we get $\vartheta_a u_3(x)/u_3(x') = 0.2431$. The map $t \to 0.2431\,t + 1$ has the fixed point $1/0.7569 \le 1.32118$ and maps $[0, \max\{t_0, 1.3212\}]$ into itself.

Step 12: proofs of the second and the third of the four remarks of item 7. The second of these remarks is the identity over sub-alphabets, and the third is the sum rule; both are stated in item 7 of §6.4.

Proof of the second of the four remarks of item 7. Item 6 of §5.3 calls a pair of letters incompatible only if the pair contains a zone letter or a letter (h): namely two zone letters, a letter (b), (v), (a) or (h) together with a zone letter, or the letters (a) and (h) of one X. Hence any two of the letters (a), (v), (t) are compatible, and

$$Z_{\mathbb{A}} := \sum_{\mathfrak{B}\subset\mathbb{A}} E_\xi \prod_{\mathfrak{B}} \lambda_\alpha = E_\xi \prod_{\mathbb{A}}(1+\lambda_\alpha).$$

Since χ takes only the values 0 and 1, we have $1 + \lambda_X^a = e^{\chi_X E^a(X,U)}$ and $1 + \lambda_\square^v = e^{\chi_\square V_\square}$; for the function $\lambda_Y^t := e^{T_Y} - 1$ of the letter (t) at Y in item 6 of §5.3 we have $1 + \lambda_Y^t = e^{T_Y}$; so $Z_{\mathbb{A}} = E_\xi e^{W_{\mathbb{A}}}$.

We apply the last claim of Lem. 6.9 to the alphabet $\mathbb{A}$, with $K(\mathfrak{B}) := E_\xi \prod_{\mathfrak{B}} \lambda_\alpha$. It gives $Z_{\mathbb{A}} = \exp\sum_{C \text{ in } \mathbb{A}} \Phi(C)$, with an absolutely convergent series; in particular $Z_{\mathbb{A}} \ne 0$. The hypotheses of the lemma are satisfied. First, $K(\mathfrak{B}\cup\mathfrak{B}') = K(\mathfrak{B})K(\mathfrak{B}')$ whenever no letter of $\mathfrak{B}$ is $\sim$ to a letter of $\mathfrak{B}'$, by the independence shown in Step 3 (IV) of the proof of Thm. 5.15. Second, $|K(\mathfrak{B})| \le \prod e_\alpha$, by Step 6 (iii) of the proof of Thm. 5.15, which concerns the letters (a), (v), (t) whose cores lie in Y, at $V' \in \mathcal{U}_Y(x')$. Third, (S_τ) holds a fortiori on a part of the alphabet.

We differentiate at $E^a + zs_X$. When $|z| < z_0$, the member stays within 16 caps and the sizes e_α to be used are the ones at $A = 16$; hence the convergence of the series is uniform in z, and summing the $(d/dz)\Phi(C)$ gives $(dZ_{\mathbb{A}}/dz)/Z_{\mathbb{A}}$. The letter (a) at X belongs to $\mathbb{A}$; were it absent, the left side would be 0 while the right side would not. The cluster that consists of this one letter is $E_\xi[\chi_X(e^{E^a}-1)]$, and its derivative is $E_\xi[\chi_X e^{E^a}s]$. Finally, the split by $\mathrm{co}(C)$ is the one of (Fm1).

Proof of the third of the four remarks of item 7. The proof is that of Steps 3 to 6 of the proof of Lem. 5.47, with changes that we give below. Those steps depend on D only through its range,

its reality and (G5) of Step 0 of §5.4. In all of them, D is an arbitrary real symmetric family of range $\le \varrho(x)$, which takes the place of $\tilde{\mathsf{E}} - \mathsf{E}$, and in q_y the factor $+(x/2)$ replaces $-(x/2)$. In the present section $\square$ denotes a cell, and we write $\boxdot$ for a cube; in §5.4 a cube is written $\square$, as in $q_\square$, $e^{q\square}$ and $p_*(\square, V)$ of that proof. From Step 3 of that proof we take the definitions of q_y and of $q_\boxdot$, with the properties that q_y is real, Borel and gauge invariant and that $|q_y| \le \frac{9}{2}\mathsf{g}_\flat^2 p_0^2 \|D\|_{(\lambda_\flat)}$, but not the identity of two energies of that step. Of Step 4 of that proof we use two parts: its claims, in the form of Lem. 6.32, and its agreement, which here reads $s_x[D](\{\boxdot\}, V) = q_\boxdot(V)$ at every G-valued V that is tame on $\boxdot^\approx$. That step proves, in §5.4, the agreement $E^\Delta(\{\square\}, V) = q_\square(V)$, in which both sides carry the factor $-(x/2)$; since the two sides change sign together, its proof, which proceeds term by term with $m_y = 1$ and $\zeta_p = F_p$ and rests on (M5r), applies unchanged. In Step 5 of that proof, w replaces $\tilde{w}$, w_t replaces $w^\flat$, and $e^{tq\boxdot}$ replaces $e^{q\boxdot}$. In Step 6 of that proof, the condition $|t|\delta' \le c_\Xi p_0^2$ replaces (X15), (F3) and $\delta_\Delta \le 1$.

With $q_\boxdot := \sum_{y\in\boxdot} q_y$ and $q_D = \sum q_\boxdot$, we have $q_\boxdot(V) = s_x[D](\{\boxdot\}, V)$ at every cube that is on, by the agreement proved in Step 4 of that proof. By Step 5 of that proof, each off cube is assigned through $p_*(\boxdot, V)$, and these cubes factor out of Ξ_w; hence $\rho[u;\ell]e^{tq_D} = \rho[u_t;\ell]$ at every V.

Let t be complex with $|t|\delta' \le c_\Xi p_0^2$, where $\delta' = (2r(x)+1)^3 M^3 \frac{9}{2}\mathsf{g}_\flat^2 p_0^2 \|D\|$. The counting of Step 6 of that proof, combined with the strong cap of w, gives $|w_t| \le e^{-\tilde{\kappa}(|X|_c - 1)}$; we give the details. Let $m := \#(D(V) \cap X)$, and let Q denote the sum, over the off cubes with $p_*(\boxdot, V) \in X$, of the $q_\boxdot(V)$. Here $D(V)$ is the set of the rough plaquettes of V, and $D'(V')$ that of V'; written without a field, $D = (D_y)$ and D' are the kernels. The counting of Step 6, which depends on Step 3 of that proof and on (G4) alone, gives $|Q| \le \delta' m$. If $m = 0$, then $w_t = w$. If $m \ge 1$, then $x\varepsilon\phi_p > p_0^2$ on $D(V)$, so that the strong cap gives $|w| \le e^{-c_\Xi p_0^2 m} e^{-\tilde{\kappa}(|X|_c - 1)}$, and

$$|w_t| = |w||e^{tQ}| \le |w| e^{|t|\delta' m} \le e^{-\tilde{\kappa}(|X|_c - 1)}.$$

Finally, (M5r) is (O7), and (F1), (F2) hold at x by Step 1 (1) of the present proof.

Hence $u_t \in M_2$ for small complex t, and the map $t \to u_t$ is holomorphic value by value. As q_D is bounded, we may differentiate under the integral of the step $\rho \to \rho'$; the result is the mean of q_D given V'. By Thm. 5.15 (a), $(\rho[u_t;\ell])' = e^{\mathsf{V}'[u_t]}\rho[F_\ell(u_t);\ell'^c]$, and all values are holomorphic in t by Thm. 6.7 (c+). At a G-valued V' with $D'(V')$ empty, that is without a rough plaquette, we have $\Xi' = 1$, since the only family is the empty one, and $1_{Y^\approx}(V') = 1$ for every Y. Hence, for t close to 0, the left side lies close to $\rho'(V') > 0$, and there the logarithm of the right side equals

$$\mathsf{V}'[u_t] - x'\mathrm{En}_{\ell'^c}(V') + \sum_Y (E'^a + E'^h)(Y, V').$$

By (Fm2), the number V' depends on the member through E^a alone. The t-derivative of the logarithm is

$$D\mathsf{V}'(u)[s] + \sum_Y DA_\ell(u)[s](Y, V') + \sum_Y DE'^h,$$

where $DA_\ell(u)[s] = s_{x'}[D'] + n$, and $\sum_\square s_{x'}[D'](\{\square\}, V') = q'_{D'}(V')$ by Step 4 of the proof of Lem. 5.47, taken at x'. $\square$

7. Iteration with sources

7.1. **The Wilson measure with sources.** In this section Wilson's weight with a complex coupling ν, which assigns to each plaquette p of $\mathbb{T}_0$ a value ν_p of its own and satisfies $|\nu|_\infty \le 15.99 c_\nu \varepsilon(\beta)$, is expressed through a system of members at the empty rows: at every field, $\exp\{-\beta\sum_p(1-\nu_p)s_p\} = \rho[u_0(\nu);\emptyset]$, where $u_0(\nu)$ stays within 16 caps and carries the strong cap at κ_1 together with a reserve. We verify (SD) at $(x_k;\ell_k)$ for $0 \le k < K$, that is, at the levels from which a step is made, among them $(x_0;\ell_0) = (\beta;\emptyset)$ with the empty rows, and we trace the first step $F_\emptyset$ along the family, with bounds on the members of $u_1(\nu)$, a Lipschitz bound, holomorphy, locality in ν, and the vacuum number estimated cell by cell. We then compute the jets in the sources z at level 0 and at level

1, follow the family $u_k(z)$ over all K steps on a polydisc, and deduce in Cor. 7.2 that each ρ_k, $0 \le k \le K$, lies in the class $C_\flat^{\mathrm{su}}(x_k;\ell_k)$ with the system $\bar u_k$.

Item 0: names. We work with the group SU(2) in dimension $d = 3$. The letters are fixed in item (0) of §6.1, and the numbers in item (1) of §6.1.

We take over from item 1 of §2.1 the following notions and notation: the lattice, the cubes and the cells; the shift h_L of the grids; the operations $\hat{}$ and cl; the convention that an object is *of* A when all its sites lie in A and *in* A when its base site lies in A; the families $\mathcal{D}$ and $\mathcal{P}$; hull, $|X|_c$, X^{+j} and $Y_0(X)$; the functions p_0, ε, $\varrho(x)$, $r(x)$ and $s(x) = 8r(x)$; the caps $c_a(x) = E_a p_0^3 x^{-1/2} = E_a p_0^2 \varepsilon$ and $c_h(x)$; S_ε and En_ℓ; the sets $\mathcal{U}_X(x)$ with (u1) to (u3); the members, with the weak and the strong cap; the densities $\rho[u;\ell]$; Ξ_w; the class $C_\flat^{\mathrm{su}}(x;\ell)$; and the data (SD).

Throughout this section, every notion and every statement that comes from Sections 2.1 to 5.5 is used with its letter κ replaced by κ_1.

The sets $M_A(x;p)$, $p \in \mathbb{P}$, are those defined in the part "Pairs" of item 0 of §6.2. We write $M_A(x) := M_A(x;\kappa_1,c_h)$; it is a subset of every $M_A(x;p)$.

The map F_ℓ, the number V', and (Fm1) to (Fm4) come from item 6 of §5.3, within the range that Thm. 6.7 (U) allows.

By $\emptyset$, the empty rows, we mean the rows with $\mathsf{H}_y = \{0\}$. Hence $\mathrm{En}_\emptyset(U) = \sum_p S_\varepsilon(\phi_p)$, where $S_\varepsilon(t) = t^2/2$ for $t \le \varepsilon$ and $S_\varepsilon(t) = d_0\varepsilon t$ for $t > \varepsilon$.

When Y is a set of cells, an object is said to lie under Y if it is based in $\hat Y$.

The hull and the number $|X|_c = \#\mathrm{hull}(X)$ are those of item 0 of §5.1: $\mathrm{hull}(X)$ is an element of $\mathcal{D}$ that contains X and has the least number of cubes among such elements, the choice being fixed by an order. By item 1 of §2.1, $\mathrm{hull}(X) = X$ for $X \in \mathcal{D}$.

Letters with more than one sense. The letter r is the radius $r(\beta)$ (item 2), $r_\tau(Y)$ in Step 8 of the proof of Thm. 7.1 is the remainder of Thm. 6.7 (g3) at u^τ, and the r_i of Thm. 7.1 (g) are radii; n is the number of sources and, in the series for Arc, the index of summation; and $n[u;D]$ is the remainder of Thm. 6.33. For u, the argument decides the sense: $u_0(\nu)$, $u_1(\nu)$, $u_k(z)$, $\bar u_k$ and u^τ, whose argument is ν, z or absent, are members indexed by a level or a parameter, and $u_3(x_1)$, whose argument is the level x_1, is the number $C_3 x_1^{1/4}$ of §6.4. The symbols x_3, x_4, x_9^U and x_s are floors, whereas the $x_k = L^{-k}\beta$ of item 2, among them x_0 and x_1, are levels.

Item 1: new entries of the table.

(O2a) We set $m_\Xi := \frac12 - d_0 - c_\Xi$. By (O2), $\frac12(\frac12 - d_0) \le m_\Xi < \frac12$.

(O12a) We set $c_\nu := E_a/(6M^3)$.

(O13) We put $x_4 := 1 + \inf S_4$, where S_4 is the set of $y \ge x_9^U$ for which (W1) to (W5) below hold at every $x \ge y$:

- (W1) $\varepsilon(x) \le 1/40$;
- (W2) $201M^3\varepsilon(x) \le E_a$;
- (W3) $\varepsilon^2/24 + 8c_\nu\varepsilon \le m_\Xi/4$;
- (W4) $(m_\Xi/4)p_0^2 \ge \kappa_1 s(x) + (2r(x)+1)^3[M^3p_0^4/(8x) + 4c_a(x)]$;
- (W5) $e^{-(m_\Xi/2)p_0^2} \le x^{-10}$.

The table moreover gives $Lx_s \ge \max\{x_9^U, x_3, x_4\}$.

Item 2: data. We fix an integer $K \ge 1$ and a number $\mathsf{b} \ge x_s$, and we put $\beta = x_0 = L^K\mathsf{b}$ and $x_k = L^{-k}\beta$.

The integer N is even, $M \mid N$, and the condition (Tor), namely $N \ge 2Ms(L\mathsf{b})$, holds.

The tori are $\mathbb{T}_k = \mathbb{Z}^3/NL^{K-k}\mathbb{Z}^3$, with the grids $o_K = 0$ and $o_k = Lo_{k+1} - \mathsf{h}_L \bmod M$ for $0 \le k < K$.

The rows are $\ell_0 = \emptyset$ and $\ell_{k+1} = ((\ell_k)')^{\mathrm{c}}$ cut at $\varrho(x_{k+1})$, and the steps $\rho \to \rho'$ are made with the noise ax_k. Wilson's density on $\mathbb{T}_0$ is $\rho_0 = e^{-\beta\sum_p(1-\cos\phi_p)}$, and $\rho_{k+1} = (\rho_k)'$ for $0 \le k < K$.

The rows, the steps, the noise and these densities are those of Thm. 5.7. The kernels, the law, the centre and the rows do not depend on any source.

Without an argument, ε, p_0, r, s and c_a are taken at β.

A coupling ν is a map from the plaquettes of $\mathbb{T}_0$ to $\mathbb{C}$, and $|\nu|_\infty$ is the supremum of its modulus.

Item 3: objects. For a field V and a plaquette p we set $s_p(V) := 1 - \frac{1}{2}\operatorname{tr}\partial V(p)$; this is a polynomial on the fields with values in $\mathrm{SL}(2,\mathbb{C})$, and at G-valued fields $s_p = 1 - \cos\phi_p = a_p$.

For $|s| < 2$ we write

$$\mathrm{Arc}(s) := \arccos(1-s)^2 = 2\sum_{n\geq 1}(2s)^n/(n^2\binom{2n}{n}) = 2s + s^2/3 + 4s^3/45 + \dots,$$

and $h(s) := \mathrm{Arc}(s)/2 - s$, which is the function h of §5.1.

We put $q_p(V) := \beta h(s_p(V))$. The plaquettes in $D(V) = \{\phi_p > \varepsilon\}$ are called rough, and all other plaquettes are called tame.

A set $X \in \mathcal{P}$ is called *s-connected* if the graph on its cubes in which two cubes are joined whenever they lie at distance $\leq s$ is connected.

For a coupling ν we define the triple $u_0(\nu) := (E^a_\nu, 0, w_\nu)$ as follows. For a single cube Q,

$$E^a_\nu(\{Q\}, V) := \sum_{p\in Q}[q_p(V) + \beta\nu_p s_p(V)],$$

and $E^a_\nu(X,\cdot) := 0$ whenever $\#X \geq 2$.

For $X \in \mathcal{P}$,

$$\begin{aligned} w_\nu(X,V) := {} & 1\{X \text{ } s\text{-connected}\} \\ & 1\{\text{every cube of } X \text{ holds a plaquette of } D(V)\} \\ & \exp\{-\beta \sum_{p\in X,\ \text{rough}} [(1-\nu_p)s_p - d_0\varepsilon\phi_p] \\ & + \sum_{p\in X^{+r},\ \text{tame}} [q_p + \beta\nu_p s_p]\}. \end{aligned}$$

At $\nu = 0$, the triple $u_0(0)$ is Wilson's system u_W of §5.1, taken at $x = \beta$.

Sources. Let $f_1, \dots, f_n$ be bounded complex functions for which the sets $\mathrm{supp}_0 f_i := \{p : f_i(x(p)) \neq 0\}$ are pairwise disjoint, where $x(p)$ denotes the barycentre of p.

We denote by $S(f_i)$ the set of cubes of $\mathbb{T}_0$ that hold a plaquette of $\mathrm{supp}_0 f_i$; it coincides with $S_i(0)$ of item 2, (o3), of §7.3.

For complex numbers z_i we put $\nu(z)_p := \mathsf{b}^{-1}\sum_i z_i f_i(x(p))$, so that $|\nu(z)|_\infty \leq \max_i |z_i|\,\|f_i\|_\infty/\mathsf{b}$. Since $\beta/\mathsf{b} = L^K = a_K^{-1}$ and $\mathcal{O}_K(f) = a_K^{-1}\sum_p f(x(p))a_p$ by Def. 1.6,

$$\exp\{-\beta\sum_p(1-\nu(z)_p)s_p\} = \rho_0\exp\{\sum_i z_i\mathcal{O}_K(f_i)\}.$$

We put $\mathrm{amp}_W := 1/(c_\nu \mathsf{b}\varepsilon(\beta)) = 6M^3 x_0^{1/2}/(\mathsf{b}E_a p_0(x_0))$, and we consider the set $\mathrm{Pd}_0 := \{|z_i|\,\|f_i\|_\infty \leq 1.99/\mathrm{amp}_W\}$ together with the open set $\mathrm{Pd}_0^o := \{|z_i|\,\|f_i\|_\infty < 15.99/\mathrm{amp}_W\}$.

For a non-empty $S \subset \{1,\dots,n\}$ we write $d_S := \prod_{i\in S} d/dz_i$ at $z = 0$.

Item 4: the theorem and the corollary.

Theorem 7.1 (Wilson's weight with sources through all levels)**.** In the setting of items 0 to 3, the following hold.

(a) *Level 0.* If $|\nu|_\infty \leq 15.99 c_\nu\varepsilon$, then

$$\exp\{-\beta\sum_p(1-\nu_p)s_p(U)\} = \rho[u_0(\nu);\emptyset](U)$$

at every G-valued field U; that is, the identity holds with the factor e^{V} at $\mathsf{V}=0$, where V is the exponent of this factor in the form $e^{\mathsf{V}}\rho[u;\ell]$ of a point of $C^{\mathrm{su}}_\flat(x;\ell)$ (item 1 of §2.1). This value of V is the V_0 of (h). Moreover (a1) to (a4) hold.

(a1) For every cube Q,

$$\sup_{\mathcal{U}_{\{Q\}}(\beta)} |E^a_\nu(\{Q\},\cdot)| \le (0.00996+|\nu|_\infty/(c_\nu\varepsilon))c_a \le (0.01+|\nu|_\infty/(c_\nu\varepsilon))c_a \le 16c_a,$$

with equality in the last inequality exactly when $|\nu|_\infty = 15.99c_\nu\varepsilon$. In addition, $(0.00996+|\nu|_\infty/(c_\nu\varepsilon))c_a \le 15.99996c_a < 16c_a$.

(a2) For every field V and every $X\in\mathcal{P}$,

$$|w_\nu(X,V)| \le \exp\{-(c_\Xi + m_\Xi/2)\beta\varepsilon \sum_{p\in D(V)\cap X}\phi_p - \kappa_1(|X|_c-1)\},$$

and wherever $w_\nu(X,V)\ne 0$, also $\beta\varepsilon\sum_{D(V)\cap X}\phi_p \ge p_0^2\#X$. In other words, w_ν has the strong cap at κ_1 with the reserve $e^{-(m_\Xi/2)p_0^2\#X} \le x_0^{-10\#X}$. Consequently $u_0(\nu)\in M_A(\beta)$, where $A:=\max\{1, 0.01+|\nu|_\infty/(c_\nu\varepsilon)\}\le 16$, and $u_0(\nu)$ has the strong cap.

(a3) In each of (ca), (ch) and (cΞ), the clauses without a size hold for $u_0(\nu)$. For real ν, the triple $u_0(\nu)$ is real and $\rho[u_0(\nu);\emptyset]>0$; if ν is real with $|\nu|_\infty \le 0.99c_\nu\varepsilon$, the weight of (a) is a point of $C^{\mathrm{su}}_\flat(\beta;\emptyset)$.

(a4) The function $E^a_\nu(\{Q\},V)$ is affine in ν and depends only on the ν_p with $p\in Q$; the function $w_\nu(X,V)$ is entire in the ν_p and depends only on those with $p\in X^{+r}$.

Clause (a) and (a1) to (a4) hold for every $\beta\ge x_4$, on every torus whose side is divisible by M, and for every grid.

(b) (SD) holds at $(x_k;\ell_k)$ for every $0\le k<K$. For $k=0$ this is $(\beta;\emptyset)$, and there $\ell'^{\mathrm{c}} = (\emptyset')^{\mathrm{c}}$ cut at $\varrho(\beta/L)$, so that $\ell'^{\mathrm{c}}=\ell_1$.

(c) *The first step.* For $|\nu|_\infty \le 15.99c_\nu\varepsilon$, let $u_1(\nu):=F_\emptyset(u_0(\nu))$, and let $V_1(\nu)$ be the number V' of (Fm2). Then

$$(\exp\{-\beta\sum(1-\nu_p)s_p\})' = e^{V_1(\nu)}\rho[u_1(\nu);\ell_1]$$

at every coarse field V', and $E'^a = E'^{\mathrm{new}} + E'^{\mathrm{old}}[u_0(\nu)]$, where E'^{new} does not depend on ν.

Moreover, for each cell and each $\tilde\kappa\in[\kappa,\kappa_1]$, the three sums, over the Y that hold the cell and with the weight $e^{\tilde\kappa(\#Y-1)}$, of $\sup|E'^{\mathrm{new}}(Y,\cdot)|$, of $\sup|E'^{\mathrm{old}}[u_0(\nu)](Y,\cdot)|$ and of $\sup|E'^h(Y,\cdot)|$ are $\le 0.35c_a(x_1)$, $\le 0.248Ac_a(x_1)$ and $\le c_h(x_1)$ respectively, and w' has the strong cap at x_1. Hence $u_1(\nu)\in M_1(x_1)$ for $|\nu|_\infty\le 1.99c_\nu\varepsilon$, and $u_1(\nu)\in M_2(x_1)$ for $|\nu|_\infty\le 6.6c_\nu\varepsilon$.

If $|\nu|_\infty,|\tilde\nu|_\infty\le 1.99c_\nu\varepsilon$, then $\mathsf{D}_{x_1}(u_1(\nu),u_1(\tilde\nu))\le 0.4|\nu-\tilde\nu|_\infty/(c_\nu\varepsilon)$. For real ν, both $u_1(\nu)$ and $V_1(\nu)$ are real. Finally, the number $V_1(\nu)$ and all values of $u_1(\nu)$ depend holomorphically on (ν_p) in the region $|\nu|_\infty<15.99c_\nu\varepsilon$; this is Thm. 6.7 (c+).

(d) *Locality.* The function $E'^{\mathrm{old}}[u_0(\nu)](Y,\cdot)$ and the number $v_0(Y;\nu)$ of (e) depend only on the ν_p with p under Y. The functions $E'^h(Y,\cdot)$ and $w'(Y,\cdot)$ depend only on the ν_p with p under $Y^{+\lceil r(\beta)/L\rceil+1}$.

(e) *Vacuum.* One has $V_1(\nu)=V_1^o+\sum_{Y\in\mathcal{D}}v_0(Y;\nu)$, where $v_0(Y;\nu):=v(Y;u_0(\nu))$ is the number of Thm. 6.7 (g1) and V_1^o does not depend on ν. For every cell and every $\tilde\kappa\in[\kappa,\kappa_1]$,

$$\sum_{Y \text{ holding the cell}} e^{\tilde\kappa(\#Y-1)}|v_0(Y;\nu)| \le 1.30L^3(0.01+|\nu|_\infty/(c_\nu\varepsilon))c_a(\beta) \le 2.43L^3Ac_a(\beta) \le C_gAc_a(\beta).$$

(f) *Jets at level 0.* Let $\nu=\nu(z)$ and $\delta_S(0):=d_Su_0(\nu(z))$. For the a-part, $\delta^a_i(0) = s_{x_0}[\mathbb{D}_0[f_i]/\mathsf{b}] + \mathrm{rem}_{0,i}$, where

$$s_{x_0}[\mathbb{D}_0[f]/\mathsf{b}](\{Q\},V) = (x_0/2\mathsf{b})\sum_{p\in Q} f(x(p))F_p\cdot F_p$$

and

$$\mathrm{rem}_{0,i}(\{Q\},V) = -(1/\mathsf{b})\sum_{p\in Q} f_i(x(p))q_p(V).$$

Both terms are carried by single cubes of $S(f_i)$ and satisfy the clauses of (ca) without a size. Measured in caps $c_a(\beta)$, $|\delta_i^a(0)| \le \mathrm{amp}_W\|f_i\|_\infty$, $|s_{x_0}[\mathbb{D}_0[f_i]/\mathsf{b}]| \le 1.01\,\mathrm{amp}_W\|f_i\|_\infty$ and $|\mathrm{rem}_{0,i}| \le 0.01\|f_i\|_\infty/\mathsf{b}$. For $|S| \ge 2$ the a-part of $\delta_S(0)$ is 0.

The h-part of $\delta_S(0)$ is 0. Its w-part is

$$\delta_S^w(0)(X,V) = w_0(X,V)\prod_{i\in S}(x_0/\mathsf{b})\sum_{p\in \mathrm{Pl}(X,V)} f_i(x(p))s_p,$$

where $\mathrm{Pl}(X,V)$ is the set consisting of the rough plaquettes in X together with the tame plaquettes in X^{+r}. The w-part is 0 unless X^{+r} meets $S(f_i)$ for each $i \in S$, and $|\delta_S^w(0)(X,V)| \le e^{-(m_\Xi/2)p_0^2\#X}\prod_{i\in S}(\mathrm{amp}_W\|f_i\|_\infty/15.99)$ times the strong cap at κ_1. The leading term of $\delta_i(0)$ is the triple $(s_{x_0}[\mathbb{D}_0[f_i]/\mathsf{b}],0,0)$, and the whole w-jet is counted in the remainder.

(g) *Jets at level 1.* Let $\delta_S(1) := d_S u_1(\nu(z))$. *First half.* Then the functions $\delta_S^a(1)(Y,\cdot)$ and the numbers $d_S v_0(Y)$ vanish except when Y meets $\mathrm{cl}\,S(f_i)$ for every $i \in S$, while $\delta_S^h(1)(Y,\cdot)$ and $\delta_S^w(1)(Y,\cdot)$ vanish except when $Y^{+\lceil r(\beta)/L\rceil+1}$ meets $\mathrm{cl}\,S(f_i)$ for every $i \in S$.

The bounds below are stated for $0 < t \le 15.98$, with $r_i := t/(\mathrm{amp}_W\|f_i\|_\infty)$ and $A_t := \max\{1, t+0.01\}$, and they hold for every cell and every $\tilde\kappa \in [\kappa,\kappa_1]$. If $\|f_i\|_\infty = 0$, then the coupling $\nu(z)$ is independent of z_i, each $\delta_S(1)$ and each $d_S v_0$ with $i \in S$ is 0, and we take r_i^{-1} to be 0. The sums over the Y that hold the cell, with the weight $e^{\tilde\kappa(\#Y-1)}$, of $\sup|\delta_S^a(1)(Y,\cdot)|$, of $\sup|\delta_S^h(1)(Y,\cdot)|$ and of $|d_S v_0(Y)|$ are $\le 0.248 A_t c_a(x_1)\prod_S r_i^{-1}$, $\le c_h(x_1)\prod_S r_i^{-1}$ and $\le 2.43 L^3 A_t c_a(\beta)\prod_S r_i^{-1}$ respectively, and $|\delta_S^w(1)(Y,V')| \le \prod_S r_i^{-1}$ times the strong cap at x_1.

Second half. Moreover, $\delta_i^a(1) = s_{x_1}[\mathbb{D}_1[f_i]/\mathsf{b}] + \mathrm{rem}_{1,i}$ with $\mathrm{rem}_{1,i} = n[u_0(0);\mathbb{D}_0[f_i]/\mathsf{b}] + DA_\emptyset(u_0(0))[\mathrm{rem}_{0,i}]$, and for all $\tilde\kappa \in [\kappa,\kappa_1]$,

$$|\mathrm{rem}_{1,i}|_{a,x_1}^{\mathrm{cs},(\tilde\kappa)} \le (u_3(x_1)K_5 + 0.01\,\vartheta_a)\|f_i\|_\infty/\mathsf{b}.$$

(h) *The honest family.* Put $u_0(z) := u_0(\nu(z))$ and $V_0 := 0$, and for $0 \le k < K$ put $u_{k+1}(z) := F_{\ell_k}(u_k(z))$ and let $V_{k+1}(z)$ be the number V' at $u_k(z)$; we call $u_k(z)$, $V_k(z)$, $0 \le k \le K$, the honest family. On Pd_0^o all of these are defined; there $u_k(z) \in M_{A^{(k)}}(x_k)$ with $A^{(0)} = 16$, $A^{(1)} = 4.318$, $A^{(2)} = 1.421$ and $A^{(k)} = 1$ for $k \ge 3$, each $u_k(z)$ has the strong cap, and $V_k(z)$ and every value of $u_k(z)$ are holomorphic. On Pd_0, $u_0(z) \in M_2(x_0)$ and $u_k(z) \in M_1(x_k)$ for $1 \le k \le K$.

Let $\rho_0^z := \rho_0\exp\{\sum z_i\mathcal{O}_K(f_i)\}$ and $\rho_{k+1}^z := (\rho_k^z)'$. Then $\rho_k^z = \exp\{\sum_{j\le k} V_j(z)\}\rho[u_k(z);\ell_k]$ at every field of $\mathbb{T}_k$, and

$$Z_K(z) := \int \rho_0^z = \exp\{\sum_{j\le K} V_j(z)\}\int \rho[u_K(z);\ell_K].$$

When all f_i are real, all these objects are real at each real point z of Pd_0^o, and $\rho[u_k(z);\ell_k] > 0$ there; for arbitrary f_i the same is true at $z = 0$.

Hence, for $0 \le k \le K$, $\bar u_k := u_k(0)$ is a system of members of ρ_k: it is real, it lies in $M_1(x_k)$, where $M_1(x_k) = M_1(x_k;\kappa_1,c_h)$ by item 0, and it has the strong cap. The radius $1.99/\mathrm{amp}_W = 1.99 c_\nu \mathsf{b}^{1/2}L^{-K/2}p_0(L^K\mathsf{b})$ depends on K.

(i) The numbers c_ν, m_Ξ, x_4, and every numeral above, do not depend on K, N, b, ν, f or n.

■□

Corollary 7.2 (Wilson's density and its images lie in the classes)**.** With the letters as fixed in the table, at every field, $\rho_0 = e^{-\beta\sum(1-\cos\phi_p)} \in C_\flat^{\mathrm{su}}(\beta;\emptyset)$ and $\rho_k \in C_\flat^{\mathrm{su}}(x_k;\ell_k)$ for $1 \le k \le K$, with the systems $\bar u_k$. For $k = 0$ and $k = 1$ the two memberships are (R0) and (R1) of §5.1, in the case of the letters as fixed in the table and of the grids o_k of item 2 of §7.1; at $k = 1$ the system is

$\bar u_1 = F_\emptyset(u_W)$, defined with $F_\emptyset$ of §5.3 in the place of the map of (F1) to (F4) of §4.4, and the two maps give systems of the same ρ_1.

In Thm. 5.7, for the data of item 2 of §7.1, that is, with the values of the table, for $\mathsf b \ge x_s$ and at the rate κ_1, Thm. 7.1 (a), (b) can be used instead of the hypothesis that Thm. 5.3 is given; the induction then begins at $k = 0$. In Thm. 5.63 (ii), the system that is fixed may be $\bar u_{k_0}$. ■□

Proof of Thm. 7.1 and of Cor. 7.2. Step 1: the floor x_4 and its place in the table. We first show that x_4 exists. Only strictly earlier places of the table enter m_Ξ, c_ν and x_4; none of N, K, $\mathsf b$, ν, f, n enters them, and neither does a member or a row. Once $x \ge x_4$, we have $c_\nu\varepsilon \le 1/64$, and therefore $\mathrm{amp}_W \ge 64/\mathsf b$.

By (O8), $p_0^2 = A_0^2(\log x)^{2\mathfrak p_0}$ with $\mathfrak p_0 > 2$, while both $\varepsilon = p_0 x^{-1/2}$ and $c_a = E_a p_0^3 x^{-1/2}$ tend to 0 as x grows; hence (W1), (W2) and (W3) are true from some x on. In (W4) the left side grows like $(\log x)^{2\mathfrak p_0}$, where $2\mathfrak p_0 > 4$; on the right side, $\kappa_1 s(x) \le 8\kappa_1(20 c_R \log x/M + 1)$, while $(2r(x)+1)^3$ is of order $(\log x)^3$ and multiplies a factor tending to 0. Condition (W5) states that $(m_\Xi/2)A_0^2(\log x)^{2\mathfrak p_0} \ge 10\log x$. So S_4 is a non-empty upper half line, and $x_4 = 1 + \inf S_4$ belongs to it.

The least threshold. Apart from r and s, which are non-decreasing and enter only the right side of (W4), all functions involved are continuous in x. Assume that the five conditions hold for every $x > t$. Replacing $r(x)$, $s(x)$ by $r(t)$, $s(t)$ does not increase the right side of (W4); letting x decrease to t, we conclude that the five conditions hold at t. Hence the set of thresholds is a closed half line, and it has a least point.

Order. The number m_Ξ depends on (O1) and (O2), the number c_ν on (O7) and (O12), and x_4 on these, on κ_1 (O5), c_R (O6), M (O7), A_0 and $\mathfrak p_0$ (O8), E_a (O12), and on x_9^U. The floor x_s is chosen after x_4. *The last claim.* From (W3) and $m_\Xi < 1/2$ we obtain $8c_\nu\varepsilon \le 1/8$.

Step 2: five elementary facts. We need the following five facts.

(E1) Let $g \in \mathrm{SL}(2,\mathbb C)$. Then $1 - \frac12 \operatorname{tr} g = \frac12 \det(g-1)$, and consequently $|1 - \frac12 \operatorname{tr} g| \le \frac12 |g-1|^2$.

(E2) The coefficients of the series h are > 0, and its radius of convergence is 2. Moreover $|h(t)| \le h(|t|) \le (|t|^2/6)/(1-|t|/2)$, and for $0 \le \phi < \pi$ the number $h(1-\cos\phi) = \phi^2/2 - (1-\cos\phi)$ lies in $[0, \phi^4/24]$.

(E3) Let $g \in \mathrm{SL}(2,\mathbb C)$ with $|g-1| < 1/20$, and let $F = -i\log g$. Then $F\cdot F = \mathrm{Arc}(1 - \frac12 \operatorname{tr} g)$.

(E4) If $0 < \varepsilon \le 1$ and $\phi \in [\varepsilon, \pi]$, then $1 - \cos\phi \ge (\frac12 - \varepsilon^2/24)\varepsilon\phi$.

(E5) If $X \in \mathcal P$ is s-connected, then $|X|_c - 1 \le s(\#X - 1)$.

Proof of (E1). Write $g = 1 + y$. From $1 = \det(1+y) = 1 + \operatorname{tr} y + \det y$ we get $\operatorname{tr} y = -\det y$, so that $1 - \frac12 \operatorname{tr} g = -\frac12 \operatorname{tr} y = \frac12 \det y$. Finally $|\det y|$ is the product of the two singular values of y, and is therefore $\le |y|^2$.

Proof of (E2). It rests on the classical series

$$\arcsin^2 z = \frac12 \sum_{n\ge1} (2z)^{2n}/(n^2 \binom{2n}{n}),$$

which we derive first. The function $\mathrm{fa}(z) := \arcsin^2 z$ is even and holomorphic on $|z| < 1$; it satisfies $\mathrm{fa}(0) = 0$ and $(1-z^2)\mathrm{fa}'^2 = 4\,\mathrm{fa}$. Differentiating this relation and dividing by $2\,\mathrm{fa}'$, we get $(1-z^2)\mathrm{fa}'' - z\,\mathrm{fa}' = 2$. Written for $\mathrm{fa} = \sum_{n\ge1} \breve\gamma_n z^{2n}$, it means that $\breve\gamma_1 = 1$ and $(2n+2)(2n+1)\breve\gamma_{n+1} = 4n^2\breve\gamma_n$; both are satisfied by $\breve\gamma_n = \frac12 4^n/(n^2\binom{2n}{n})$, which proves the displayed formula.

Since $\arccos(1-t) = 2\arcsin((t/2)^{1/2})$, the series gives

$$\frac12 \arccos(1-t)^2 = \sum_{n\ge1} (2t)^n/(n^2\binom{2n}{n}) = t + h(t).$$

Two consecutive coefficients have the quotient $n^2/((n+1)(2n+1)) < 1/2$. Hence the radius is 2, and the part of the series from $n = 2$ on is bounded by its first term $|t|^2/6$ times $1/(1-|t|/2)$. At

$t = 1 - \cos\phi < 2$ we have $\arccos(\cos\phi) = \phi$. Finally $\cos\phi \le 1 - \phi^2/2 + \phi^4/24$ for real ϕ; together with the positivity of the coefficients, this places $h(1-\cos\phi)$ in $[0, \phi^4/24]$.

Proof of (E3). We have $\det g = e^{i\,\mathrm{tr}\,F} = 1$ and $|\,\mathrm{tr}\,F| \le 2|F| < 2\pi$, so $\mathrm{tr}\,F = 0$. For a traceless 2×2 matrix, the theorem of Cayley–Hamilton gives $F^2 = -\det F\,1 = (F\cdot F)1$, as $F\cdot F = \frac{1}{2}\,\mathrm{tr}\,F^2$. Therefore, with $w = F\cdot F$,

$$g = e^{iF} = \cos(w^{1/2})1 + iF\sin(w^{1/2})/w^{1/2},$$

and $1 - \frac{1}{2}\,\mathrm{tr}\,g = c(w) := 1 - \cos w^{1/2}$, which is an entire function of w. On $|w| < 1$ we have $|c(w)| \le \cosh 1 - 1 \le 0.5431 < 2$, so that $\mathrm{Arc}(c(w))$ is holomorphic there; since it coincides with w on $[0,1)$, it coincides with w on the whole disc. It remains to note that $|w| \le |F|^2 \le (\log(20/19))^2 \le 0.0513^2 < 1$.

Proofs of (E4) *and* (E5). The monotonicity argument in Step S4 (1) of the proof of Thm. 5.3 proves (E4) for $0 < \varepsilon \le 1$: its endpoint comparison only requires $\varepsilon/2 \le 2/\pi$. For (E5) apply the spanning-tree construction of Step S4 (4) there. It uses only s-connectivity and the definition of $|X|_c$ as the least size of a connected set containing X, and applies on a torus as well.

Step 3: (a1), (a3) *and* (a4)*, the analytic member and the clauses without a size.* We prove (a1); that $E^a_\nu(\{Q\},\cdot)$ is holomorphic on $\mathcal{U}_{\{Q\}}(\beta)$, does not change under G-valued gauge transformations, vanishes at pure gauges, and is real at G-valued fields when ν is real; that $w_\nu(X,\cdot)$ is Borel, gauge invariant, and a function of V on the bonds of X^{+s}; and (a4).

The sites of a plaquette in Q lie in Q^{+1}, which is a subset of Q^{+r} ($r \ge 1$ since $\beta > 1$); thus the plaquette is a plaquette *of* Q^{+r}, and (u3) gives $|\partial V(p) - 1| < 2\varepsilon$. Now (E1) yields $|s_p| < 2\varepsilon^2 \le 1/800$, and (E2) yields

$$|q_p| \le \beta(4\varepsilon^4/6)/(1-\varepsilon^2) \le 0.66709\beta\varepsilon^4 = 0.66709 p_0^4/\beta.$$

As a cube holds $3M^3$ plaquettes,

$$|E^a_\nu| \le 2.00127 M^3 p_0^4/\beta + 6M^3|\nu|_\infty p_0^2.$$

In the first term, $p_0^4/\beta = \varepsilon p_0^3 \beta^{-1/2}$, and (W2) gives $M^3\varepsilon \le E_a/201$; so the first term is $\le (2.00127/201)c_a \le 0.00996 c_a$. In the second term we use $c_a = E_a p_0^2 \varepsilon$ and $6M^3 c_\nu = E_a$, which give $6M^3|\nu|_\infty p_0^2 = (|\nu|_\infty/(c_\nu\varepsilon))c_a$. For $|\nu|_\infty \le 15.99 c_\nu\varepsilon$ the sum is therefore $\le 15.99996 c_a$.

The clauses. Since s_p is a polynomial in the entries of V and $|s_p| < 2$, the function $h(s_p)$ is holomorphic. A gauge transformation takes $\partial V(p)$ to $u\,\partial V(p) u^{-1}$, which leaves the trace unchanged. At a pure gauge, $\partial V(p) = 1$, so $s_p = 0$ and $h(0) = 0$. At G-valued fields, $s_p = 1 - \cos\phi_p$ is real. Moreover $E^h = 0$. The weight $w_\nu(X,\cdot)$ is built from ϕ_p and s_p at plaquettes in X^{+r}, whose sites lie in X^{+r+1}, a subset of X^{+s} ($r+1 \le 8r$); and, ϕ_p being a continuous class function of $\partial V(p)$, the indicators are Borel. For real ν the weight $w_\nu \ge 0$ is real, and by (a) (Step 4) $\rho[u_0(\nu);\emptyset]$ equals $\exp\{-\beta\sum_p(1-\nu_p)s_p\}$, the left side of (a), so that $\rho[u_0(\nu);\emptyset] > 0$. If moreover $|\nu|_\infty \le 0.99 c_\nu\varepsilon$, then $0.01 + |\nu|_\infty/(c_\nu\varepsilon) \le 0.01 + 0.99 = 1$, so $A = 1$ in (a2), and (a2) (Step 5) places $u_0(\nu)$ in $M_1(\beta)$, with the strong cap; as $u_0(\nu)$ is real and $\rho[u_0(\nu);\emptyset] > 0$ (both just shown), and as the exponent of the factor in front of $\rho[u_0(\nu);\emptyset]$ is 0, a real number ((a), Step 4), the weight of (a) is then a point of $C_\flat^{\mathrm{su}}(\beta;\emptyset)$. Clause (a4) follows from the formulas of item 3 of §7.1, since $\mathbb{T}_0$ has finitely many plaquettes.

Step 4: (a)*, the identity at every field.* We show that, for every complex ν and every G-valued V,

$$\exp\{-\beta\sum_p(1-\nu_p)s_p\} = \exp\{-\beta\sum_p S_\varepsilon(\phi_p) + \sum_Q 1_{Q^{+r}}(V) E^a_\nu(\{Q\},V)\}\,\Xi_{w_\nu}(V).$$

Denote by $d(V)$ the set of those cubes that hold a rough plaquette, and by $X_1,\dots,X_j$ its classes under the links at distance $\le s$.

1. *One family.* Take a family $\{X_i'\}$ of Ξ_w; its members are at pairwise distance $> s$, and $D(V)$ lies inside their union. Suppose that its product does not vanish. Then each cube of each

X_i' holds a rough plaquette, so that the union equals $d(V)$; since every X_i' is s-connected and no link connects two distinct members, the family must be $\{X_1, \dots, X_j\}$. When $D(V)$ is empty, this is the empty family, whose value is 1.

2. *On and off.* We have $1_{Q^{+r}}(V) = 1$ exactly when no plaquette in Q^{+r} is rough, that is, exactly when Q is not in $d(V)^{+r}$. For such V the field on the bonds of Q^{+r} lies in $\mathcal{U}_{\{Q\}}(\beta)$, because $|V(b)| = |V(\pi)| = 1$ and $|\partial V(p) - 1| = 2\sin(\phi_p/2) \le \varepsilon$. We say that a cube Q is on at V if $1_{Q^{+r}}(V) = 1$, and off at V otherwise. Two distinct classes are at distance $\ge s + 1 = 8r + 1 > 2r$; hence the sets X_i^{+r} are pairwise disjoint, their union is $d(V)^{+r}$, and a rough plaquette in X_i^{+r} lies in X_i.
3. *Plaquette by plaquette.* At a tame p, (E2) gives $q_p = \beta\phi_p^2/2 - \beta s_p$. A tame p in an on cube carries the factors $e^{-\beta\phi_p^2/2} e^{q_p + \beta\nu_p s_p}$. A tame p in an off cube lies in exactly one X_i^{+r} and carries the same two factors, the second of which now appears inside $w_\nu(X_i, V)$. A rough p lies in exactly one X_i and carries $e^{-\beta d_0 \varepsilon\phi_p} e^{-\beta[(1-\nu_p)s_p - d_0\varepsilon\phi_p]}$. In all three cases the product of the factors is $e^{-\beta(1-\nu_p)s_p}$, and the identity follows on multiplying over all plaquettes.

Step 5: (a2)*, the strong cap at κ_1 with a reserve.* We prove (a2) for $|\nu|_\infty \le 15.99 c_\nu \varepsilon$, at every V and for every $X \in \mathcal{P}$. Assume that $w_\nu(X, V) \ne 0$, and set $\check{\Sigma} := \beta\varepsilon \sum_{p \in D(V) \cap X} \phi_p$.

(1) *Rough plaquettes.* Here $\mathrm{Re}[(1 - \nu_p)s_p] \ge (1 - |\nu|_\infty)s_p$, and (E4) gives $s_p \ge (\frac{1}{2} - \varepsilon^2/24)\varepsilon\phi_p$. By (W3),

$$(1 - |\nu|_\infty)(\frac{1}{2} - \varepsilon^2/24) \ge \frac{1}{2} - \varepsilon^2/24 - |\nu|_\infty/2 \ge \frac{1}{2} - (\varepsilon^2/24 + 8c_\nu\varepsilon) \ge \frac{1}{2} - m_\Xi/4.$$

Hence the rough plaquettes contribute to the real part of the exponent at most $-(\frac{1}{2} - d_0 - m_\Xi/4)\check{\Sigma} = -(c_\Xi + 3m_\Xi/4)\check{\Sigma}$.

(2) Each cube of X holds a rough p, and for it $\beta\varepsilon\phi_p > \beta\varepsilon^2 = p_0^2$; therefore $\check{\Sigma} \ge p_0^2 \# X$.

(3) *Tame plaquettes.* We have $\#X^{+r} \le (2r+1)^3 \#X$; each cube holds $3M^3$ plaquettes; $0 \le q_p \le \beta\varepsilon^4/24 = p_0^4/(24\beta)$ by (E2); and $|\beta\nu_p s_p| \le \beta|\nu|_\infty \varepsilon^2/2 \le 8c_\nu \varepsilon p_0^2$. The tame plaquettes therefore contribute $\le \#X(2r+1)^3[M^3 p_0^4/(8\beta) + 24M^3 c_\nu \varepsilon p_0^2]$, where $24M^3 c_\nu \varepsilon p_0^2 = 4E_a p_0^2 \varepsilon = 4c_a$.

(4) From (2) and (W4), $(m_\Xi/4)\check{\Sigma} \ge \#X(m_\Xi/4)p_0^2$, and the right side is at least $\kappa_1 s \#X$ plus the bound found in (3). By (E5), $\kappa_1 s \#X \ge \kappa_1(|X|_c - 1)$.

(5) *Conclusion.* Adding up, $\log|w_\nu(X, V)| \le -(c_\Xi + m_\Xi/2)\check{\Sigma} - \kappa_1(|X|_c - 1)$. For the reserve, (2) and (W5) give $e^{-(m_\Xi/2)\check{\Sigma}} \le e^{-(m_\Xi/2)p_0^2 \#X} \le x_0^{-10\#X}$.

Rough plaquettes in X^{+r} that lie off X occur in no sum that defines w_ν. Therefore (1) to (5) are valid at *every* V, and not only at those V for which X is a class of $d(V)$.

Step 6: (b)*, the data* (SD) *at every level, the empty rows included.* Let $0 \le k < K$ and $x = x_k$. We show: the coarse side $N_k := NL^{K-k-1}$ is even, divisible by M, and satisfies $N_k \ge 2Ms(x_k)$; the grids fit; $\ell_k \in \mathcal{R}(b_*, K_*, \lambda_*, 1/9)$ has range $\le \varrho(x_k)$, and the pair $(\ell_k, (\ell_k)')$ has the four conclusions named in (SD); and $x_k \ge \max\{x_9^U, x_3, x_4\}$.

Floors. We have $x_k \ge x_{K-1} = L\mathsf{b} \ge Lx_s$.

The side of the torus, (Tor). We use the following lemma.

Lemma 7.3 (Growth of s under scaling)**.** If $x \ge e^2$ and $j \ge 0$ is an integer, then $L^j s(x) \ge s(L^j x)$. ∎

To prove it, put $\alpha = 20c_R/M$. As $e^t - 1 \ge t$, we have $(L^j - 1)\log x \ge 2(L^j - 1) \ge 2j \log L$, which gives $L^j \alpha \log x \ge \alpha \log(L^j x)$. The number $L^j \lceil \alpha \log x \rceil$ is an integer and is $\ge L^j \alpha \log x$; hence it is $\ge \lceil \alpha \log(L^j x) \rceil$. Since $r(x) = \lceil 20c_R \log x / M \rceil$ and $s(x) = 8r(x)$, this is the lemma.

At $x = L\mathsf{b} \ge e^2$ and $j = K - 1 - k$, the lemma gives $N_k = NL^j \ge 2ML^j s(L\mathsf{b}) \ge 2Ms(x_k)$. Moreover N_k is even, and $M \mid N_k$ holds because it holds for N.

Grids. The relation $o_k = Lo_{k+1} - \mathsf{h}_L \bmod M$ is the fit of item 1 of §2.1, under which the hat of each cell consists of exactly L^3 fine cubes; and M divides the side NL^{K-k} of $\mathbb{T}_k$. At $k = 0$ this

fixes the grid of $\mathbb{T}_0$. The weight $\exp\{-\beta\sum_p(1-\nu_p)s_p\}$ does not depend on that grid, and (a) holds for every grid.

Rows. Thm. 2.65, with its clauses (Fr0), (Fr2), (Fr3′) and (Fr4), is valid for every x_0, for every $k \geq 0$ with $x_{k+1} \geq x_\flat$, for every tower with $N \geq 2$ even, and for the pair $(\ell_k, (\ell_k)')$. Here $x_{k+1} \geq \mathsf{b} \geq x_s$, and x_s meets every demand that (O13) makes on $x_\flat$, (X12) included (item (1), (O13), of §6.1). The statement therefore provides the following.

(Fr0) $\ell_k \in \mathcal{R}(b_*, K_*, \lambda_*, 1/9)$, of range $\leq \varrho(x_k)$; for ℓ_0, which is empty, the range is 0.
(Fr2) This is (Dec)(C_D, θ_D) for the sixteen kernels, with (C_D, θ_D) as in (O6).
(Fr3′) This is (N′) with the numbers eq. 2.vii of §2.2, at every cut and every mask.
(Fr4) $\mathsf{t}_\rho \leq \frac{1}{8} m_\Xi$.

The conditions (L1^{su}) and (L2^{su}) are (O4). These statements concern scalar kernels; the kernel $K \otimes 1_{\mathfrak{g}}$ has the same norms as K, by item 5 of Step 15 of the proof of Thm. 5.7; and at $\ell = \emptyset$ tensoring and dressing act on nothing. What we have obtained are the four conclusions of (SD) (item 1 of §2.1, the last clause of (SD)). At $k = 0$, ℓ'^{c} is $(\emptyset')^{\mathrm{c}}$ cut at $\varrho(\beta/L)$, and this is the definition of ℓ_1.

The empty rows. The data (SD) of §3, of item B of §4.1, of item 2 of §4.2, of item C of §4.3 and of item 1 of §2.1 admit the empty rows; this is shown in Steps S0 and S6 of the proof of Thm. 5.3. In §3, items C.I 1.1 and C.II 1.2 require $|\ell_y(p)| \leq K e^{-\lambda|y-p|}$, $K \geq 0$; the inequalities $\gamma|\eta|^2 \leq q_\emptyset \leq q_\ell \leq b|\eta|^2$ are not demanded by these items but follow from Lem. 3.24 (a) and Lem. 3.3 (a), in the case $K = 0$ as in the case $K > 0$; and Remark (4) of item I of §4.1 treats the case $\ell = \emptyset$. The data (SD) of item 2 of §5.3 admit $\ell = \emptyset$ as well.

For item 2 of §5.3 this means that Thm. 5.15 itself, with its clauses (a) to (e), applies at $\ell = \emptyset$, at every complex u within 16 caps that has the weak cap.

Step 7: (c)*, the first step along the family.* By (a), $u_0(\nu) \in M_A(\beta)$, a subset of $M_A(\beta; p)$ for every $p \in \mathbb{P}$; by (b), (SD) holds at $(\beta; \emptyset)$; and $\beta \geq x_9^U$. Hence Thm. 6.7 (U) yields clauses (a) to (e) of Thm. 5.15 at every p, with a single map $F_\emptyset$ and a single number V'. Combined with (a) of Thm. 7.1, Thm. 5.15 (a) gives the formula of (c) at every V'. Thm. 5.15 (b) shows that E'^{new} is the same for all members, hence free of ν, and it supplies the three sums, the strong cap, and $F_\emptyset(u_0(\nu)) \in M_{\max\{1, 0.35+0.248A\}}(x_1)$.

Numerals. If $|\nu|_\infty \leq 1.99 c_\nu \varepsilon$, then $A \leq 2$ and $0.35 + 0.496 = 0.846 \leq 1$. If $|\nu|_\infty \leq 6.6 c_\nu \varepsilon$, then $A \leq 6.61$ and $0.35 + 1.63928 = 1.98928 \leq 2$. Reality comes from Thm. 5.15 (d).

The Lipschitz bound. Let $\check\delta = \tilde\nu - \nu$. Applied on M_2, Thm. 5.15 (e) gives $\mathsf{D}_{x_1}(u_1(\nu), u_1(\tilde\nu)) \leq 0.4 \mathsf{D}^{\mathrm{in}}_\beta(u_0(\nu), u_0(\tilde\nu))$. In the a-part, $E^a_\nu - E^a_{\tilde\nu} = -\beta\sum_{p\in Q}\check\delta_p s_p$, whose modulus is $\leq 6M^3|\check\delta|_\infty p_0^2 = (|\check\delta|_\infty/(c_\nu\varepsilon))c_a$; the h-part is 0. In the w-part, for $\tau \in [0,1]$ and complex ζ with $|\zeta|\,|\check\delta|_\infty \leq 14 c_\nu\varepsilon$ we have $|\nu + (\tau+\zeta)\check\delta|_\infty \leq 15.99 c_\nu\varepsilon$. On that disc, Cauchy's estimate together with (a2) gives

$$|(d/d\tau) w_{\nu+\tau\check\delta}(X, V)| \leq (|\check\delta|_\infty/(14 c_\nu\varepsilon)) x_0^{-10} e^{-\kappa_1(|X|_{\mathrm{c}}-1)}.$$

Thus $\mathsf{D}^{\mathrm{in}} \leq |\check\delta|_\infty/(c_\nu\varepsilon)$.

Holomorphy. On the open set $|\nu|_\infty < 15.99 c_\nu\varepsilon$ of $\mathbb{C}^{\#\mathrm{plaquettes}}$, the family $\nu \mapsto u_0(\nu)$ takes its values in $M_{16}(\beta; p)$, and by (a4) each of its values is affine or entire. So Thm. 6.7 (c+) applies.

Step 8: (d) *and* (e)*, locality in ν and the vacuum number cell by cell.* For (d), Thm. 6.7 (f1) states that $E'^{\mathrm{old}}[u](Y)$ and $v(Y; u)$ depend on $\{E^a(X) : Y_0(X) \subset Y\}$ alone. The function E^a_ν is carried by single cubes Q. Single cubes are small ($1 \leq L/2 - 2$), and the cell holding Q is $Y_0(\{Q\})$; by (a4), $E^a_\nu(\{Q\})$ depends on ν in Q. So both quantities depend only on ν under Y. By Thm. 6.7 (f2), (f3), the quantities $E'^h(Y)$ and $w'(Y)$ depend on the members at those X with $Y_0(X) \subset Y$. Here $\mathrm{cl}(X) \subset Y_0(X)^{+1}$ (item 1 of §2.1); since a cell is L cubes wide, X^{+r} lies under $Y_0(X)^{+1+\lceil r/L\rceil}$; and $w_\nu(X)$ depends on ν in X^{+r}. This is Thm. 6.7 (f5) at the radius $r(\beta)$.

For (e), Thm. 6.7 (g1) gives $V_1(\nu) = V_0^+ + \sum_Y v(Y; u_0(\nu))$, and $V_1^o := V_0^+$ depends on no member, by Thm. 6.7 (f1).

The bound. Put $a_\nu := 0.00996 + |\nu|_\infty/(c_\nu\varepsilon) < 16$ and $u^\tau := (\tau E^a_\nu, 0, w_\nu)$. For $|\tau| < 16/a_\nu$, an open disc containing $[0,1]$, we have $u^\tau \in M_{16}(\beta;p)$ with affine values; so by Thm. 6.7 (c+) the map $\tau \mapsto v(Y;u^\tau)$ is holomorphic, and its derivative is $Dv(Y;u^\tau)[E^a_\nu]$. For $0 \le \tau \le 1$ we have $u^\tau \in M_A(\beta;p)$ with $A = \max\{1, \tau a_\nu\} < 16$; on this range v may be differentiated term by term, so that its derivative is the Dv defined in item 1 of §6.2 as the sum of the derivatives of the terms. This is part of Thm. 6.7 (g3).

By the closing clause of Thm. 6.7 (g1), $v(Y;u) = 0$ if $E^a = 0$, so that $v(Y;u^0) = 0$. Hence, by Thm. 6.7 (g3) applied with $\delta = E^a_\nu$, which has size $B \le a_\nu$,

$$v_0(Y;\nu) = \int_0^1 \{ \sum_{Q: Y_0(\{Q\})=Y} G^\delta_{\{Q\}}[u^\tau](1) + r_\tau(Y)\} d\tau.$$

Clause (g3) holds on all of $M_{16}(\beta;p)$, since item 2 of §6.2 has $1 \le A \le 16$.

The first sum vanishes when $\#Y \ge 2$, and for $Y = \{\text{cell}\}$ it has L^3 terms. The field $V' = 1$ lies in $\mathcal{U}_{\{\text{cell}\}}(x_1)$. Item (ii) of Step 2 of the proof of Thm. 5.15, applied at $V' = 1$ with $X = \{Q\}$, with $Y = Y_0(\{Q\})$ and with the upper index τ of its domain U^τ of coarse fields equal to 1, shows that the restriction of U to Q^{+r} lies in $\mathcal{U}_{\{Q\}}(\beta)$ on the set where $\chi_Q = 1$; and there $|E^a_\nu| \le a_\nu c_a$ by (a1). Each term, being a mean of $\chi e^{\tau E^a} E^a$, therefore has modulus $\le e^{\tau a_\nu c_a} a_\nu c_a$. As $a_\nu c_a \le 16 c_a \le 1/2$ according to (Y.1) of item 5 of §5.3 (we take this item at $x = \beta$; it is in force there since $\beta \ge x_9^U$, by item 0, (O13) (ii), of §6.2), and $(e^{1/2} - 1)/(1/2) \le 1.2975$,

$$\int_0^1 e^{\tau a_\nu c_a} a_\nu c_a \, d\tau = e^{a_\nu c_a} - 1 \le 1.2975 a_\nu c_a.$$

The remainder, with the sum over Y taken under the integral, is $\le 109 L^3 \beta^{-1/4} a_\nu c_a \le 0.00011 L^3 a_\nu c_a$, because the first inequality of (Y.6), in the same item, gives $\beta^{-1/4} \le 10^{-6}$. Both parts together are $\le 1.2977 L^3 a_\nu c_a \le 1.30 L^3 a_\nu c_a$. Finally $a_\nu \le 0.01 + |\nu|_\infty/(c_\nu\varepsilon) \le A$, and $C_g = 263 L^3$.

Step 9: (f), *the jets at level 0. The a-part.* The expression

$$E^a_{\nu(z)}(\{Q\}, V) = \sum_{p\in Q} q_p + (x_0/\mathsf{b}) \sum_i z_i \sum_{p\in Q} f_i(x(p)) s_p$$

is affine in z. Therefore $\delta^a_i(0)(\{Q\}, V) = (x_0/\mathsf{b}) \sum_{p\in Q} f_i(x(p)) s_p$, while the a-part of $\delta_S(0)$ vanishes for $|S| \ge 2$. Using $|s_p| < 2\varepsilon^2$ we find

$$|\delta^a_i(0)| \le 3M^3 (x_0/\mathsf{b}) 2\varepsilon^2 \|f_i\|_\infty = 6M^3 p_0^2 \|f_i\|_\infty/\mathsf{b} = \mathrm{amp}_W \|f_i\|_\infty c_a.$$

Next, $s_p = \mathrm{Arc}(s_p)/2 - q_p/x_0$, and on $\mathcal{U}_{\{Q\}}(\beta)$ we have $\mathrm{Arc}(s_p) = F_p \cdot F_p$ by (E3), since $2\varepsilon \le 1/20$. Consequently

$$\delta^a_i(0) = (x_0/2\mathsf{b}) \sum f_i(x(p)) F_p \cdot F_p - (1/\mathsf{b}) \sum f_i(x(p)) q_p.$$

The first term equals $s_{x_0}[\mathbb{D}_0[f_i]/\mathsf{b}]$. Indeed, $\mathbb{D}_{0,y}[f](p,p') = f(x(p))$ if $p = p'$ is based at y, and 0 otherwise; the path $\Gamma(y, \pi(p))$ is trivial; and the formula of entry 5 of item (0) of §6.1 gives $(x_0/2) \sum_{y\in Q} \sum_{p \text{ based at } y} (f(x(p))/\mathsf{b}) F_p \cdot F_p$.

Sizes. For $|s| < 2\varepsilon^2$ and $\varepsilon \le 1/40$ we have $|\mathrm{Arc}(s)| \le \mathrm{Arc}(|s|) = 2|s| + 2h(|s|) \le 4.00084\varepsilon^2$; hence the first term is $\le 3M^3 (x_0/2\mathsf{b}) 4.00084 \varepsilon^2 \|f_i\|_\infty = 1.00021 \,\mathrm{amp}_W \|f_i\|_\infty c_a$. By Step 3, the second term is $\le (\|f_i\|_\infty/\mathsf{b}) 3M^3 \, 0.66709 p_0^4/x_0 \le 0.00996 c_a \|f_i\|_\infty/\mathsf{b}$. Just as in Step 3, both terms are holomorphic on $\mathcal{U}_{\{Q\}}(\beta)$, gauge invariant and 0 at pure gauges; and they vanish unless Q lies in $S(f_i)$.

The h-part. Here $E^h = 0$. *The w-part.* We have $w_\nu(X,V) = w_0(X,V) \exp\{x_0 \sum_{p\in \mathrm{Pl}(X,V)} \nu_p s_p\}$, where $\mathrm{Pl}(X,V)$ does not depend on ν. One differentiation in each z_i, $i \in S$, at $z = 0$ produces the product formula. One of its factors is 0 when no plaquette in X^{+r} lies in $\mathrm{supp}_0 f_i$. For the size: if $\|f_i\|_\infty = 0$ for some $i \in S$, the jet is 0. Otherwise consider the torus $|z_i| = 15.99/(\mathrm{amp}_W \|f_i\|_\infty)$,

$i \in S$, with the other $z_j = 0$. As the supports are disjoint, $|\nu(z)|_\infty \le 15.99 c_\nu \varepsilon$ there, so (a2) bounds $|w_{\nu(z)}(X,V)|$ there; and the Cauchy estimate, applied to a derivative of first order in every variable z_i, $i \in S$, produces the factor $\prod_S(\mathrm{amp}_W \|f_i\|_\infty/15.99)$. When the supports overlap, (a) to (e) remain valid as stated in ν, whereas in z the radii shrink by the factor n.

Step 10: (g)*, the jets at level 1. First half.* Clause (c) and Thm. 6.7 (c+) show that each value of $u_1(\nu(z))$, and each number $v_0(Y;\nu(z))$, is holomorphic on Pd_0^o. *Vanishing.* By (d), any value in which ν enters only off $\mathrm{supp}_0 f_i$ is independent of z_i. *Bounds.* If $\|f_i\|_\infty = 0$ for some $i \in S$, everything vanishes, by the vanishing just shown. Otherwise, on $|z_i| = r_i$ ($i \in S$), with $z_j = 0$ for the remaining j, we have $|\nu(z)|_\infty \le t c_\nu \varepsilon$, so that $u_0(\nu(z)) \in M_{A_t}(\beta)$; this torus lies in Pd_0^o because $t \le 15.98$. As E'^{new} is free of z, $\delta_S^a(1) = d_S E'^{\mathrm{old}}$. We write d_S as the Cauchy integral over the torus $|z_i| = r_i$, and bring the supremum over V', together with the weighted sum over Y, under the integral sign; at each z the integrand is then bounded by (c) or by (e), whichever applies to it. For w', the strong cap at x_1 does not depend on z.

Second half. We proceed in four parts.

(1) By Thm. 6.7 (f1), E'^{old} depends on E^a alone, and $E^a_{\nu(z)} = E^a_0 + \sum z_i \delta_i^a(0)$ is affine. Hence $\delta_i^a(1) = (d/dt)E'^{\mathrm{old}}[(E^a_0 + t\delta_i^a(0), 0, w_0)]$ at $t = 0$, and this is $DA_\emptyset(u_0(0))[\delta_i^a(0)]$ in the sense of Thm. 6.7 (h). The map DA is additive in its argument, since by Thm. 6.7 (c+) the two-parameter family is holomorphic.

(2) By (f),

$$\delta_i^a(1) = DA_\emptyset(u_0(0))[s_{x_0}[D]] + DA_\emptyset(u_0(0))[\mathrm{rem}_{0,i}], \qquad D := \mathbb{D}_0[f_i]/\mathsf{b}.$$

(3) We apply Thm. 6.33 at $x = \beta \ge x_3$, with (SD) $= (\beta;\emptyset)$ (its data include the empty rows), with $u = u_0(0) \in M_1(\beta)$, a subset of $M_8(\beta;p)$, and with D complex, symmetric (diagonal) and of range 0. It gives $DA_\emptyset(u_0(0))[s_{x_0}[D]] = s_{x_1}[D'] + n[u_0(0);D]$, where $D' = \mathrm{cut}_{\varrho(x_1)}(T_\emptyset D) = \mathbb{D}_1[f_i]/\mathsf{b}$ by the definition in item 1 of §6.3 (with $\ell_0 = \emptyset$) and by linearity. Thm. 6.33 (a), with Thm. 6.20 (U2.1) at $k = 0$, gives $|n|^{\mathrm{cs},(\bar\kappa)}_{a,x_1} \le u_3(x_1)\|D\|_{(\lambda_\flat)} \le u_3(x_1) K_5 \|f_i\|_\infty/\mathsf{b}$.

(4) The remainder $\mathrm{rem}_{0,i}$ belongs to $T^a(\beta)$ and has size $B \le 0.01\|f_i\|_\infty/\mathsf{b}$ at every p, as it lives on single cubes. By Thm. 6.7 (h3), the weighted cell sum of $DA_\emptyset(u_0(0))[\mathrm{rem}_{0,i}]$ is $\le \vartheta_a B c_a(x_1)$, and a cell sum bounds the weighted supremum. Since $\vartheta_a = 0.2431 L^{-1/4}$, we get $0.01\,\vartheta_a \le 0.00244$.

Step 11: (h), (i) *and the corollary. Induction on* k *over* Pd_0^o. For $k = 0$, (a) gives $u_0(z) \in M_{16}(x_0)$ with the strong cap, and its values are affine or entire in z. For the passage $k \to k+1$: by (b), (SD) holds at $(x_k;\ell_k)$, $x_k \ge x_9^U$, and $u_k(z) \in M_{A^{(k)}}(x_k)$ with $A^{(k)} \le 16$. Thm. 6.7 (U), together with Thm. 5.15 (b), then gives $u_{k+1}(z) \in M_{A^{(k+1)}}(x_{k+1})$ with $A^{(k+1)} = \max\{1, 0.35 + 0.248 A^{(k)}\}$, and the strong cap. This yields 4.318, then $1.420864 \le 1.421$, then at most $0.702408 < 1$, so that the maximum is 1. By Thm. 6.7 (c+), $V_{k+1}(z)$ and all values are holomorphic on the open set Pd_0^o. On Pd_0 one has $A^{(0)} = 2$, then $0.846 \le 1$, then $0.598 \le 1$.

The densities. By (a), $\rho_0^z = \rho[u_0(z);\emptyset]$. Suppose that $\rho_k^z = e^c \rho[u_k(z);\ell_k]$ for a complex number c. As the step is linear in ρ (item 2 of §5.3), Thm. 5.15 (a) gives, at every field,

$$\rho_{k+1}^z = e^c(\rho[u_k(z);\ell_k])' = e^{c+V_{k+1}(z)}\rho[u_{k+1}(z);\ell_{k+1}].$$

Normalisation. The kernel $k_{ax}(\cdot,h)$ is a probability density in $V'(c)$, by item 2 of §5.5, which describes the step: there $k_s(g,h) = z(s)^{-1} e^{-s(1-\cos \mathrm{ang}(gh^{-1}))}$ with $z(s) = \int_G e^{-s(1-\cos \mathrm{ang}\, g)} dg$, and the right invariance of Haar measure gives $\int k_s(g,h)dg = 1$. The function ρ_k^z is bounded and Borel on a compact group carrying its Haar probability, so Fubini's theorem yields $\int \rho_{k+1}^z = \int \rho_k^z$. Hence $Z_K(z) = \int \rho_K^z$.

Real z and real f_i, or $z = 0$. In this case $\nu(z)$ is real; so $u_0(z)$ is real, and $\rho_0^z > 0$ by (a3). Applying Thm. 5.15 (d) step by step, we find that $u_k(z)$ and $V_k(z)$ are real. From $\rho_0^z > 0$ we get $\rho_k^z > 0$, and then $\rho[u_k(z); \ell_k] > 0$.

Clause (i). It follows from Steps 1 to 10: among the constants of Thm. 6.7, Thm. 6.20 and Thm. 6.33 that enter, none depends on K, N, b, on a member, or on any part of the rows other than the numbers b_*, K_*, λ_* and the range.

The corollary. At $z = 0$, $\bar{u}_0 = u_0(0)$ is real, lies in $M_1(\beta)$ ($A = 1$) and has the strong cap, and $\rho[\bar{u}_0; \emptyset] > 0$; so ρ_0 is a point of $C_\flat^{\mathrm{su}}(\beta; \emptyset)$. For $k \geq 1$, $\bar{u}_k$ is real, lies in $M_1(x_k)$ and has the strong cap, and $\rho_k = e^{\sum V_j(0)} \rho[\bar{u}_k; \ell_k] > 0$, where the exponent $\sum V_j(0)$ is real. Further, $u_0(0) = u_W$, since the two formulas agree term by term ($q_p = \beta h(s_p)$). Finally, Thm. 5.63 (ii) asks that a system be fixed, and any system will do. □

7.2. **Holomorphic contraction.** In this section we work on the complex space $\mathfrak{M}(x)$ of members. For bounded maps that are holomorphic along lines we give Cauchy bounds on their derivatives, a Lipschitz bound and holomorphy along families, in every seminorm that is a supremum of finite linear forms, as the weighted suprema and the cell sums are; for families we give the Cauchy bound on every Taylor coefficient at the origin and the Schwarz bound on the values of a family that vanishes at the origin, we show that the Taylor coefficients of a bounded family of members are again members, and we write the coefficient of z^S of a composed map as a sum over the partitions of S. We then follow the step map F_ℓ on families within $B \leq 2.6$ caps: their images lie within one cap and obey the strong cap, the factor $\vartheta[B]$ leads from the weak distance of the inputs to the strong distance of the images, and we give the channels of the first and second derivatives, the local vacuum numbers and the reflection $u \mapsto u^*$. Finally, a lemma of eight clauses (B1) to (B8) gives the properties of an affine map $\mathbb{B}_D$ that absorbs into the members a complex symmetric kernel D of the density, added to the kernels E_y of the rows, and we carry out one step for two families, one of them over a kernel $D(z)$.

Remark 7.4. The statements of this section follow from §5.4, from standard facts, and from Thm. 6.7, which asserts Thm. 5.15 at every pair of rates and adds further clauses.

Item 0: the letters.

(1) *Letters and data kept.* From items 2 and 3 of §5.4 we keep the lattice $\mathbb{Z}^3/NL\mathbb{Z}^3$ with (Tor), the cubes and the cells, the neighbourhoods $X^{\approx} = X^{+r(x)}$ and $X^{\approx\approx} = X^{+s(x)}$, and the numbers $\varrho(x)$, $r(x)$ and $s(x) = 8r(x)$.

From the same items we take the group $G = \mathrm{SU}(2)$ with the operator norm $|\cdot|$ and the form $Y \cdot Y' = \frac{1}{2} \operatorname{tr} YY'$, the fields V with values in G, and the fields W with values in $\mathrm{SL}(2, \mathbb{C})$.

These items also give us ϕ_p, F_p, ζ_p, the rough set $D(V)$, the indicator $1_Y(V)$ of the property that V is tame on Y, the path $\Gamma(y, z)$, the transport $T(y, z)$ and the mask $m_y(V)$.

We keep as well the rows ℓ with their kernels $\mathsf{E}_y = \ell_y^{\mathsf{T}} \ell_y$, the norm $\|D\|_{(\lambda)}$ of kernels, the energy $\mathrm{En}_\ell(V)$ and the step $\rho \to \rho'$.

We keep the domains $\mathcal{U}_X(x)$ with (u1), (u2) and (u3), and the complex linear space $\mathfrak{M}(x)$ of those triples $u = (E^a, E^h, w)$ that obey the clauses without a size and have $\|u\| < \infty$ (item 3 of §5.4).

We keep the density $\rho[u; \ell]$, the sum Ξ_w and the families of (cΞ), the class $C_\flat^{\mathrm{su}}(x; \ell)$, and the property of a triple u of being real.

From item 1 of §5.4 we keep $p_0(x)$ and $\varepsilon = \varepsilon(x) = p_0(x) x^{-1/2}$, for which $x\varepsilon^2 = p_0^2$ is (O8); the caps $c_a(x) = E_a p_0(x)^3 x^{-1/2}$ and $c_h(x) = e^{-c_h p_0(x)^2}$; and the numbers c_h of (O11) and E_a of (O12). None of these depends on a rate.

From §5.3 we keep (SD), the map $F_\ell(u) = (E'^a, E'^h, w')$ with $E'^a = E'^{\mathrm{new}} + E'^{\mathrm{old}}[u]$, and $\mathsf{V}'[u]$.

From §6.1 we keep κ and κ_1, the set $\mathbb{P} = [\kappa, \kappa_1] \times \{c_h, c_h/2\}$ of pairs, and the numbers $\vartheta_w = 0.248$, $\vartheta_a = 0.2431 L^{-1/4}$, $\vartheta[B] = \max\{16\vartheta_w, 1\}/(16 - B) = 3.968/(16 - B)$, $C_g = 263 L^3$, $\mathsf{g}_\flat = \mathsf{g}(\lambda_\flat)$ and $c_Q^{\mathrm{su}} = 306 M^3 \mathsf{g}_\flat^2$.

From that section we keep, too, $\eta_h(x) = e^{-(c_h/2)(p_0(x)^2 - p_0(x/L)^2)}$, $\eta_w(x) = e^{-c_\Xi p_0(x)^2}$, the floor x_9^U, the member $s_x[D]$ and the polydisc $\mathrm{Pd}(r) = \{z \in \mathbb{C}^n : |z_i| < r_i\}$.

From §6.2 we keep $v(Y; u)$, V_0^+, $T^a(x)$ and $DA(u)[\delta]$, where $A = E'^{\mathrm{old}}$.

In x', p_0', V', W', E', w' and ℓ' the prime denotes the object at the level $x' = x/L$.

At the level x' the definitions of items 2 and 3 of §5.4 are taken on the coarse torus $\mathbb{T}' = \mathbb{Z}^3/N\mathbb{Z}^3$, with cells in the place of cubes; the definition "The level at x'" in item 3 of §5.4 carries them over, and the remark directly after it shows that this rests on three conditions alone, namely that N is even, that $M \mid N$ and that $N \geq 16 M r(x')$.

Lems. 7.5, 7.7 and 7.8 depend on three features of a level, that is, of x with its torus, and on nothing else of it: the index sets are finite, each $\mathcal{U}_X$ is an open set of a complex manifold, and the weights of items 0 (2) and 0 (3) are free of z.

Hence the three lemmas hold at x' on $\mathbb{T}'$ just as at x; in the same way they hold for $\mathfrak{M}(\mathsf{b})$ on the last torus $\mathbb{T}_K$ of §7.3, which is the torus $\mathbb{T}'$ of the step at $x = L\mathsf{b}$.

(2) *Norms at a pair.* Let $p = (\tilde\kappa, \tilde c) \in \mathbb{P}$ and $\delta = (\delta^a, \delta^h, \delta^w) \in \mathfrak{M}(x)$. We set

$$|\delta|_a := \sup_X \sup_{\mathcal{U}_X(x)} |\delta^a(X, \cdot)|\, e^{\tilde\kappa(\#X - 1)}/c_a(x),$$

$$|\delta|_h := \sup_X \sup |\delta^h(X, \cdot)|\, e^{\tilde\kappa(\#X-1)} e^{\tilde c p_0(x)^2},$$

$$|\delta|_w^{\mathrm{wk}} := \sup_{X,V} |\delta^w(X, V)|\, e^{\tilde\kappa(|X|_c - 1)},$$

and we let $|\delta|_w^{\mathrm{st}}$ be the same supremum as $|\delta|_w^{\mathrm{wk}}$, with the weight

$$\exp\{c_\Xi x\varepsilon \sum_{p \in D(V) \cap X} \phi_p + \tilde\kappa(|X|_c - 1)\}$$

in the place of $e^{\tilde\kappa(|X|_c - 1)}$. We also set $\|\delta\| := \max\{|\delta|_a, |\delta|_h, |\delta|_w^{\mathrm{wk}}\}$ and $\|\delta\|^{\mathrm{st}} := \max\{|\delta|_a, |\delta|_h, |\delta|_w^{\mathrm{st}}\}$.

We write $M_A(x; p) = \{\|u\| \leq A\}$, $\mathsf{D}_x^{\mathrm{in}}(u_1, u_2) = \|u_1 - u_2\|$ and $\mathsf{D}_x(u_1, u_2) = \|u_1 - u_2\|^{\mathrm{st}}$. A member obeys the *strong cap* if $|w|_w^{\mathrm{st}} \leq 1$. On the torus the index sets are finite, so that $\mathfrak{M}(x)$ is one and the same set for all p. Where the context does not make clear which pair is meant, we write an upper index (p) or $(\tilde\kappa)$.

(3) *Cell sums at x'.* For a family $e(Y, \cdot)$ we set

$$|e|_a^\Sigma := \sup_\square \sum_{Y \ni \square} e^{\tilde\kappa(\#Y - 1)} \sup_{\mathcal{U}_Y(x')} |e(Y, \cdot)|/c_a(x'),$$

where $\square$ runs over the cells. We define $|e|_h^\Sigma$ in the same way, with the supremum over those G-valued fields on the bonds of $Y^{\approx}$ that are tame there, and with $e^{-\tilde c p_0(x')^2}$ in the place of $c_a(x')$. For numbers $v = (v(Y))$ we set $|v|^\Sigma := \sup_\square \sum_{Y \ni \square} e^{\tilde\kappa(\#Y-1)} |v(Y)|$. Since every Y holds a cell, $|e|_a \leq |e|_a^\Sigma$ and $|e|_h \leq |e|_h^\Sigma$.

The quantity $\sup_\square \sum_{Y \ni \square} e^{\tilde\kappa(\#Y-1)} \sup |e(Y, \cdot)|$ is the $|e|_{\mathrm{cs}}$ of item 1 of §6.2 and the $\mathrm{CS}_{\tilde\kappa}(e)$ of item 1 of §6.4, and neither of the two is divided by a cap. Thus $|e|_a^\Sigma = |e|_{\mathrm{cs}}/c_a(x')$, and this is the $|e|_{a,x'}^{\mathrm{cs},(\tilde\kappa)}$ of Sections 6.4 and 7.1.

(4) *Values, admissible seminorms, domination of coordinates.* A *value* of u is one of the numbers $E^a(X, \mathsf{W})$, $E^h(X, V)$, $w(X, V)$. Let I be an index set. A seminorm N on a linear subspace of $\mathbb{C}^I$ is *admissible* if $N(a) = \sup_{\mathfrak{l} \in \mathfrak{L}} |\mathfrak{l}(a)|$ for a set $\mathfrak{L}$ of linear forms $\mathfrak{l}(a) = \sum_i c_i a_i$, each with finitely many $c_i \neq 0$. Every weighted supremum norm of item 0 (2) is admissible: each of its forms $\mathfrak{l}$ is the weight times one value. Every cell sum of item 0 (3) is admissible, with

the forms $\sum_{Y\ni\square} c_Y e^{\tilde\kappa(\#Y-1)} e(Y,\mathsf{W}_Y)/c_a(x')$ over the cells $\square$, the arguments W_Y and the phases $|c_Y|=1$; each such sum is finite.

Each of these seminorms N *dominates each coordinate*: for every i there is a $c_i>0$ with $c_i|a_i|\le N(a)$. Indeed, a weighted value is itself a form of $\mathfrak{L}$; and in a cell sum every Y holds a cell and the phases are free, so that the sum of the moduli, and each of its terms, is at most $N(a)$. Fix such a representing family $\mathfrak{L}$ and use the same supremum to define N on all of $\mathbb{C}^I$, allowing the value $+\infty$. Its finite domain is a linear subspace, and N is a seminorm there. A bound on a coordinatewise derivative or Taylor coefficient also asserts finiteness of this extended seminorm. Membership of Taylor coefficients in $\mathfrak{M}(x)$ is proved separately in Lem. 7.7 (S3). An admissible seminorm can fail to dominate: $N(a)=|a_1|$ on $\mathbb{C}^2$ is admissible and does not dominate the coordinate a_2.

(5) *Families, Taylor coefficients, u^{pol}.* Let $n\ge 1$. For $\mu\in\mathbb{N}^n$, where $\mathbb{N}^n$ is the set of the n-tuples of integers ≥ 0, we write $r^{-\mu}=\prod r_i^{-\mu_i}$, and for a non-empty subset S of $\{1,\dots,n\}$ we write $z^S=\prod_{i\in S} z_i$. A *family* at x on an open set $\Omega\subset\mathbb{C}^n$ within B caps at p is a map $z\mapsto u(z)\in M_B(x;p)$ all of whose values are holomorphic in z. For families, $\mathsf{D}_{x;\Omega}$ and $\mathsf{D}^{\mathrm{in}}_{x;\Omega}$ denote the suprema over $z\in\Omega$. We write u_μ for the Taylor coefficients at 0, defined value by value, and u_S for the coefficient of z^S, and we set $u^{\mathrm{pol}}(z):=u(0)+\sum_S z^S u_S$.

(6) *Holomorphy along lines, derivatives.* A function Φ on a norm-open set O of $\mathfrak{M}(x)$ is *holomorphic along lines* if for all $u,g\in\mathfrak{M}(x)$ the map $s\mapsto\Phi(u+sg)$ is holomorphic on $\{s\in\mathbb{C}: u+sg\in O\}$. For such a Φ we set $D^j\Phi(u)[g_1,\dots,g_j]:=\partial_{s_1}\cdots\partial_{s_j}\Phi(u+\sum s_ig_i)$ at $s=0$.

(7) *The reflection.* For a group element g we put $g^{\#}:=(g^\dagger)^{-1}$, for a field W we put $\mathsf{W}^{\#}(b):=\mathsf{W}(b)^{\#}$, and for a member u we put

$$u^*:=(\overline{E^a(X,\mathsf{W}^{\#})},\overline{E^h(X,V)},\overline{w(X,V)}).$$

(8) *Letters with several senses.*

The letter p is the pair $(\tilde\kappa,\tilde c)$ when a norm, a ball $M_A(x;p)$ or a family is taken at it, and also in p° and in the upper index (p); it is a plaquette when it indexes a quantity of the field or of a kernel, is an argument of such a quantity or is the variable of a sum, as in ϕ_p, $D_y(p,p')$ and $\sum_{p\in D(V)\cap X}$, and this role tells the two senses apart.

The letter A is a number of caps; in DA and D^2A it is the map E'^{old}; and in $\log(1+A)$ (Step 7 of this section) it is a 2×2 matrix.

The letter r denotes the radii in $\mathrm{Pd}(r)$ and in $r^{-\mu}$, and it is $r(x)$ in $(2r+1)^3$ and in X^{+r}.

Lemma 7.5 (Cauchy bounds on a ball of members)**.** Let $\|\cdot\|$ be the norm of item 0 (2) at one pair p, and let $0\le B<A_{\#}\le 16$. Let $\Phi=(\Phi_i)_{i\in I}$ be functions on $\{u\in\mathfrak{M}(x):\|u\|<A_{\#}\}$, each of them holomorphic along lines and bounded on that ball, and let N be an admissible seminorm with $N(\Phi(u))\le\mathsf{m}$ for all $\|u\|<A_{\#}$.

(a) For $\|u\|\le B$ and $j\ge 1$, every $D^j\Phi_i(u)$ is symmetric and complex multilinear, and

$$N(D^j\Phi(u)[g_1,\dots,g_j])\le \mathsf{m}\,(j/(A_{\#}-B))^j\prod\|g_i\|.$$

(b) For $\|u_1\|,\|u_2\|\le B$ we have $N(\Phi(u_1)-\Phi(u_2))\le\mathsf{m}\,\|u_1-u_2\|/(A_{\#}-B)$.

(c) For every family u within B caps at p, every $\Phi_i(u(z))$ is holomorphic in z.

(a°) Off the centre: let $u_0\in\mathfrak{M}(x)$, let $t>0$ be any number, and let Φ be a function on $\{u\in\mathfrak{M}(x):\|u-u_0\|<t\}$ that is holomorphic along lines and obeys $|\Phi|\le\mathsf{m}$ there. Then for every $j\ge 1$ the form $D^j\Phi(u_0)$ is symmetric and complex multilinear, and

$$|D^j\Phi(u_0)[g_1,\dots,g_j]|\le\mathsf{m}\,(j/t)^j\prod\|g_i\|.$$

■

Remark 7.6. The boundedness of each Φ_i follows from $N(\Phi(u)) \le \mathsf{m}$ wherever N dominates each coordinate (item 0 (4)); so it follows for every norm of item 0 (2) and for every cell sum of item 0 (3).

For Φ a value of F_ℓ, of V' or of $v(Y;\cdot)$, Lem. 7.5 (c) is Thm. 6.7 (c+).

Lemma 7.7 (Cauchy and Schwarz for families; coefficients are members)**.** Let $a(z) = (a_i(z))_{i\in I}$, with every a_i holomorphic on $\mathrm{Pd}(r)$, and let N be an admissible seminorm with $N(a(z)) \le m$ on $\mathrm{Pd}(r)$.

(S1) Every Taylor coefficient a_μ at 0 obeys $N(a_\mu) \le m\, r^{-\mu}$.
(S2) If $a(0) = 0$, then $N(a(z)) \le m \max_i |z_i|/r_i$.
(S3) If $g : \mathrm{Pd}(r) \to \mathfrak{M}(x)$ has holomorphic values and $\sup_z \|g(z)\| < \infty$, then every g_μ lies in $\mathfrak{M}(x)$.

Hence (S1) and (S2) hold for such a g in $\|\cdot\|$, in $\|\cdot\|^{\mathrm{st}}$, in each of the suprema of item 0 (2), and in the cell sums of item 0 (3). ■

Lemma 7.8 (The coefficient of z^S of a composed map)**.** Let $u_0 \in \mathfrak{M}(x)$ and $t > 0$, and let Φ be bounded and holomorphic along lines on $\{\|u - u_0\| < t\}$. Let u be a map from a neighbourhood of 0 in $\mathbb{C}^n$ to $\mathfrak{M}(x)$ that is bounded in norm, has holomorphic values and satisfies $u(0) = u_0$. Then $\Phi(u(z))$ is holomorphic near 0, and its coefficient of z^S is

$$\sum_{\pi\in\mathrm{Part}(S)} D^{|\pi|}\Phi(u_0)[u_{S_1},\dots,u_{S_{|\pi|}}],$$

where $\pi = \{S_1,\dots,S_{|\pi|}\}$ runs over the partitions of S into non-empty blocks.

So this coefficient depends on $u(0)$ and on the $u_{S'}$, with S' a non-empty subset of S, alone, and $\Phi(u(z))$ and $\Phi(u^{\mathrm{pol}}(z))$ have the same coefficient of z^S for every S.

In particular, in the setting of Thm. 7.9 ($x \ge x_9^U$, (SD) at x, $p \in \mathbb{P}$), and for a family u within $B < 16$ caps at p near 0, the assertions above hold, with $t = 16 - B$, for every value of $F_\ell(u(z))$, for $\mathsf{V}'[u(z)]$ and for $v(Y;u(z))$.

They hold also for $I[u] := \int \rho[u;\ell](V)\,dV$, at any $x > e$, for any rows ℓ of range $\le \varrho(x)$, any u_0 and any t. The lattice for $I[u]$ is that of item 2 of §5.4 at x, or a torus $\mathbb{Z}^3/N\mathbb{Z}^3$ with N even, $M \mid N$ and $N \ge 16Mr(x)$, on which the definitions of items 2 and 3 of §5.4 are set up by the definition "The level at x'" in item 3 of that section; this torus is the $\mathbb{T}'$ of that definition, with x in the place of its x'.

Thus they hold for $I[u]$ on $\mathbb{T}_K$ at $x = \mathsf{b}$, where $N \ge 2Ms(L\mathsf{b}) \ge 16Mr(\mathsf{b})$.

The claim for $I[u]$ rests on three facts. First, at each V the density $\rho[u;\ell](V)$ is $e^{-x\mathrm{En}_\ell(V)}$ times the exponential of a finite sum of values of E^a and E^h, times a polynomial in finitely many values of w, and so it is entire along lines; the factor $e^{-x\mathrm{En}_\ell(V)}$ is free of the members and is at most 1, since $\mathrm{En}_\ell \ge 0$, the energy of item 2 of §5.4 being a sum of numbers $S_\varepsilon(\phi_p) \ge 0$ and of squares. Second, on $\{\|u\| < R\}$ the density is bounded, uniformly in V, by

$$\exp\{R(c_a(x) + e^{-\tilde{c}p_0^2})\#\mathcal{D}\}(1+R)^{\#\mathcal{P}},$$

where, on the finite torus, $\#\mathcal{D}$ and $\#\mathcal{P}$ are the numbers of elements of the families $\mathcal{D}$ and $\mathcal{P}$ of item 2 of §5.4. Third, by the theorems of Fubini and Morera the integral is holomorphic along lines.

Thus the two hypotheses of the lemma on Φ, boundedness on the ball and holomorphy along lines on it, hold for four kinds of Φ: for a value of F_ℓ, for $\mathsf{V}'[\cdot]$, for $v(Y;\cdot)$ and for $I[\cdot]$. ■

Theorem 7.9 (The contraction for pairs, for families; family form of Thm. 5.46)**.** Let $x \ge x_9^U$, in the setting (SD) at x; let $p = (\tilde{\kappa},\tilde{c}) \in \mathbb{P}$ and $1 \le B \le 2.6$; and let u_1 and u_2 be families on Ω within B caps at p, with the weak cap. All norms are taken at p.

(i) Each $F_\ell(u_i(\cdot))$ is a family at x' in $M_1(x';p)$ that obeys the strong cap, and $|E'^a|_a^\Sigma \le 0.35 + 0.248B \le 0.9948$; moreover $\mathsf{V}'[u_i(\cdot)]$ and every $v(Y;u_i(\cdot))$ are holomorphic.

(ii) We have $\mathsf{D}_{x';\Omega}(F_\ell u_1, F_\ell u_2) \le \vartheta[B]\,\mathsf{D}^{\mathrm{in}}_{x;\Omega}(u_1,u_2)$. Here $\vartheta[1] \le 0.26454$, $\vartheta[2] \le 0.28343$, $\vartheta[2.455] \le 0.29295$ and $\vartheta[2.6] \le 0.29612$, by Step 4 of this section. For the three members the factors are $16\vartheta_w/(16-B)$, $1/(16-B)$ and $1/(16-B)$, and the first two of them hold in $|\cdot|^\Sigma_a$ and $|\cdot|^\Sigma_h$. Besides, $|E'^a[u_1] - E'^a[u_2]|^\Sigma_a \le \vartheta_a |E^a_1 - E^a_2|_a$ at each z.

(iii) On $\Omega = \mathrm{Pd}(r)$, for every μ,

$$\|(F_\ell u_1 - F_\ell u_2)_\mu\|^{\mathrm{st}} \le \vartheta[B]\, r^{-\mu} \mathsf{D}^{\mathrm{in}}_{x;\mathrm{Pd}(r)}(u_1,u_2).$$

(iv) *Channels.* Let $\|u\| \le B$ and $g, g_1, g_2 \in \mathfrak{M}(x)$, and put $g^{w\circ}(X,V) := g^w(X,V)\,1\{D(V)\cap X \ne \emptyset\}$. The inputs are measured in caps at x and the outputs in caps at x', the a- and h-members in the cell sums and w' in $|\cdot|^{\mathrm{st}}_w$. For the first derivatives, the channel from a to a obeys $|DA(u)[g^a]|^\Sigma_a \le \vartheta_a |g^a|_a$, and the channels from h and from w to a are 0. The channels from a to h and w are at most $|g^a|_a/(16-B)$. The channels from w to h and w are at most $|g^{w\circ}|^{\mathrm{wk}}_w/(16-B)$, and $|g^{w\circ}|^{\mathrm{wk}}_w \le \eta_w |g^w|^{\mathrm{st}}_w$, the input being measured in the strong norm. At $\tilde c = c_h$, the channels from h to h and w are at most $\eta_h |g^h|_h/(16-B)$.

For the second derivatives with output a, $|D^2A(u)[g^a_1, g^a_2]|^\Sigma_a \le 16\vartheta_a\,(4/(16-B)^2)\,|g^a_1|_a |g^a_2|_a$, and the second derivative is 0 as soon as an entry is an h- or a w-member. Hence on M_B

$$|(DA(u_1) - DA(u_2))[g^a]|^\Sigma_a \le (64\vartheta_a/(16-B)^2)|E^a_1 - E^a_2|_a |g^a|_a,$$

and also $|(DA(u_1) - DA(u_2))[g^a]|^\Sigma_a \le (\vartheta_a/(16-B))|E^a_1 - E^a_2|_a |g^a|_a$; of these two bounds the second is the smaller. For the second derivatives with outputs h and w, the bound is $4/(16-B)^2$ times the product of the sizes of the two entries, where an a-entry is measured by $|\cdot|_a$, a w-entry by $|g^{w\circ}|^{\mathrm{wk}}_w \le \eta_w |g^w|^{\mathrm{st}}_w$, and an h-entry, at $\tilde c = c_h$, by $\eta_h |g^h|_h$. An entry can also be left whole, as a mixed triple g, and is then measured by $\|g\|$ at p; a mixed entry that is split is measured by the sum $|g^a|_a + \eta_h |g^h|_h + |g^{w\circ}|^{\mathrm{wk}}_w$. The factor $4/(16-B)^2$ is $1/56.25$ at $B=1$ and $1/49$ at $B=2$.

(v) At each z,

$$|v(\cdot;u_1) - v(\cdot;u_2)|^\Sigma \le 16 C_g c_a(x) |E^a_1 - E^a_2|_a/(16-B) \le 16 C_g c_a(x) \mathsf{D}^{\mathrm{in}}_x(u_1,u_2)/(16-B),$$

and at $B=2$ the factor $16C_g c_a(x)/(16-B)$ is at most $300.6 L^3 c_a$. On $\mathrm{Pd}(r)$,

$$|(v(\cdot;u_1(z)))_\mu|^\Sigma \le r^{-\mu} C_g B c_a(x)$$

and

$$|(v(\cdot;u_1(z)) - v(\cdot;u_2(z)))_\mu|^\Sigma \le r^{-\mu} 16 C_g c_a(x) \mathsf{D}^{\mathrm{in}}_{x;\mathrm{Pd}(r)}/(16-B).$$

(vi) *Reflection.* The map $u \mapsto u^*$ is an antilinear involution of $\mathfrak{M}(x)$, isometric in every norm of item 0 (2), and its fixed points are real. For $\|u\| < 16$ we have $\mathsf{V}'[u^*] = \overline{\mathsf{V}'[u]}$, and $E'^h[u^*] = \overline{E'^h[u]}$ and $w'[u^*] = \overline{w'[u]}$ at every argument; further, $E'^a[u^*](Y,\mathsf{W}') = \overline{E'^a[u](Y,\mathsf{W}'^\#)}$ for each W' in $\mathcal{U}_Y(x')$ with values in G, and also on each connected component of $\mathcal{U}_Y(x')$ that contains a field with values in G. Hence $\rho[F_\ell(u^*);\ell'^c] = \overline{\rho[F_\ell(u);\ell'^c]}$.

Clauses (ii) to (v) hold for every $1 \le B < 16$; among the hypotheses of the theorem, the bound $B \le 2.6$ is used for (i) alone. ■□

Lemma 7.10 (Absorbing into the members a complex kernel added to those of the rows; complex form of Lem. 5.47)**.** Let $x > e$. We work on the lattice of item 2 of §5.4 and in the setting of the five conditions of item 5 of §5.4: (M5r) $M \ge \rho_0$, (X15) $c_\Xi p_0(x)^2 \ge 1$, (F1) $\varepsilon(x) \le 1/40$, (F2) $\varrho(x) + 2 \le Mr(x)$ and (F3) $(2r(x)+1)^3 c_a(x) \le 34$. In the setting of the table of item (1) of §6.1 these five conditions hold at every $x \ge Lx_s$, by (O7), by (O13) and by Step 0 (Fl) of §5.4, which is stated for $x \ge x_9$, the floor of Thm. 5.15, evaluated at κ_1; every $x \ge Lx_s$ is at least x^U_9, since $Lx_s \ge x^U_9$ is a clause of (O13). As $x^U_9 \ge x_9$ by item 0 (O13) of §6.2, we have $x \ge Lx_s \ge x^U_9 \ge x_9$. Let ℓ be rows of range $\le \varrho(x)$. Let $D = (D_y)$ be complex kernels with $D_y(p,p') = D_y(p',p)$ and

with $D_y(p,p')=0$ unless $|y-p|,|y-p'|\le\varrho(x)$, whatever their sign, whether or not they are Gram forms and whether or not they are covariant. Put $\delta:=\|D\|_{(\lambda_\flat)}$ and $\delta_\Delta=\delta_\Delta(D):=c_Q^{\mathrm{su}}p_0^2\delta/c_a(x)$, and let $\delta_\Delta\le1$ and $A>0$. The norms of item 0 (2) are taken at any rate $\tilde\kappa\ge0$ and any $\tilde c>0$.

We set

$$q_y^D(V):=-(x/2)\,m_y(V)\sum_{p,p'}D_y(p,p')\langle T(y,\pi(p))\zeta_p,T(y,\pi(p'))\zeta_{p'}\rangle$$

and, for a cube $\square$, $q_\square^D:=\sum_{y\in\square}q_y^D$. We set $E^\Delta[D]:=-s_x[D]$, that is,

$$E^\Delta[D](\{\square\},\mathsf W)=-(x/2)\sum_{y\in\square}\sum_{p,p'}D_y(p,p')\,(\mathrm{Ad}_{\mathsf W(\Gamma(y,\pi(p)))}F_p)\cdot(\mathrm{Ad}_{\mathsf W(\Gamma(y,\pi(p')))}F_{p'}),$$

with $F_p=-i\log\partial\mathsf W(p)$, and $E^\Delta[D]$ is 0 at $\#X\ge2$. A cube $\square$ is *on* at V if V is tame on $\square^{\approx}$, and *off* otherwise; $p_*(\square,V)$ is the first plaquette of $D(V)$ in $\square^{\approx}$, in a fixed order. We set $Q_D(X,V):=\sum\{q_\square^D(V):\square\text{ off},\ p_*(\square,V)\in X\}$ and $\rho[u;\ell;D]:=\rho[u;\ell]\exp\{\sum_\square q_\square^D\}$; if $D=\tilde{\mathsf E}-\mathsf E$ for rows $\tilde\ell$ of range $\le\varrho(x)$, then $\rho[u;\ell;D]=\rho[u;\tilde\ell]$.

Then for every complex $\tilde u=(\tilde E^a,\tilde E^h,\tilde w)\in\mathfrak M(x)$ with $|\tilde E^a|_a,|\tilde E^h|_h,|\tilde w|_w^{\mathrm{st}}\le A$, the triple $\mathbb B_D(\tilde u):=(\tilde E^a+E^\Delta[D],\tilde E^h,w^\flat)$, where $w^\flat:=\tilde we^{Q_D}$, lies in $M_{A+\delta_\Delta}(x;p)$, and (B1) to (B8) hold. In them $\mathfrak n:=\#(D(V)\cap X)$, a number determined by (X,V).

(B1) $\rho[\tilde u;\ell;D]=\rho[\mathbb B_D\tilde u;\ell]$ at every V.

(B2) The member $E^\Delta[D]$ is complex linear in D, lives on $\#X=1$ and obeys those of the clauses without a size that concern E^a; it obeys $\sup_{\mathcal U_{\{\square\}}(x)}|E^\Delta[D](\{\square\},\cdot)|\le\delta_\Delta c_a(x)$; it depends neither on ℓ nor on $\tilde u$; and $E^\Delta[D](\{\square\},V)=q_\square^D(V)$ at every V tame on $\square^{\approx}$. Clause (B2) holds for every D that satisfies the hypotheses that the lemma makes on D other than $\delta_\Delta\le1$, whatever $\delta_\Delta(D)$ is: both sides of its bound are homogeneous of degree 1 in D. Likewise, whatever $\delta_\Delta(D)$ is,

$$|Q_D(X,V)|\le((2r(x)+1)^3c_a(x)/68)\,\delta_\Delta(D)\,\mathfrak n\le(\delta_\Delta(D)/2)\,\mathfrak n,$$

by Step 8 of this section together with (F3).

(B3) $|w^\flat(X,V)|\le Ae^{-(c_\Xi p_0^2-1/2)\mathfrak n}e^{-\tilde\kappa(|X|_c-1)}$ and $|w^\flat-\tilde w|(X,V)\le(A/e)\,\delta_\Delta e^{-\tilde\kappa(|X|_c-1)}$; and $e^{Q_D(X,\cdot)}$ depends on V through its values on the bonds of $X^{+(2r(x)+1)}$, which lie inside $X^{\approx\approx}$.

(B4) $\mathsf D_x^{\mathrm{in}}(\tilde u,\mathbb B_D\tilde u)\le\max\{1,A/e\}\,\delta_\Delta$.

(B5) The map $\mathbb B_D$ is affine in $\tilde u$. For a family $\tilde u(z)$ whose values are holomorphic and whose three sizes are $\le A$ at each z, and for a constant D, the a- and h-coefficients of z^μ, $\mu\ne0$, are unchanged, and $w_\mu^\flat=\tilde w_\mu e^{Q_D}$, which is 0 wherever $\tilde w_\mu$ is 0; moreover $|w_\mu^\flat|_w^{\mathrm{wk}}\le|\tilde w_\mu|_w^{\mathrm{st}}$, and $|(w_\mu^\flat)^\circ|_w^{\mathrm{wk}}\le e^{1/2}\eta_w|\tilde w_\mu|_w^{\mathrm{st}}$ for the part on $D(V)\cap X\ne\emptyset$. If $z\mapsto D(z)$ has holomorphic entries and at each z the kernel $D(z)$ satisfies the hypotheses that the lemma makes on D, with $\sup_z\delta_\Delta(D(z))\le1$, then $\mathbb B_{D(z)}\tilde u(z)$ is a family within $A+1$ caps.

(B6) $\mathsf D_x^{\mathrm{in}}(\mathbb B_{D_1}\tilde u_1,\mathbb B_{D_2}\tilde u_2)\le\mathsf D_x(\tilde u_1,\tilde u_2)+\max\{1,A/e\}\,\delta_\Delta(D_1-D_2)$.

(B7) The derivative $(d/dt)\mathbb B_{tD}\tilde u$ at $t=0$ is $(E^\Delta[D],0,\tilde wQ_D)$, and

$$|\tilde wQ_D|(X,V)\exp\{(c_\Xi/2)x\varepsilon\sum_{p\in D(V)\cap X}\phi_p+\tilde\kappa(|X|_c-1)\}\le(A/e)\,\delta_\Delta,$$

with half of the strong weight. With the whole strong weight instead of its half, the left-hand side is bounded by $A(\delta_\Delta/2)\mathfrak n$. For a lower bound let $x\ge x_9$, the floor of Thm. 5.15, evaluated at κ_1 (so every $x\ge x_9^U$ qualifies, by item 0 (O13) of §6.2), and let $A\ge1$; take the rows, the field and the set of Step 10 of §5.4, the proof of Prop. 5.50, with the kernel $D=\tilde{\mathsf E}-\mathsf E$ of that step, for which $D_y(p,p')=-t^2$ at $p=p'=p_y$ and 0 otherwise, and the

member $\tilde{u} = (0, 0, \tilde{w})$ with

$$\tilde{w}(X, V) := \exp\{-c_{\Xi} x \varepsilon \sum_{p \in D(V) \cap X} \phi_p - \tilde{\kappa}(|X|_c - 1)\},$$

which is the member of that proposition with the rate $\tilde{\kappa}$ of the norm in the place of its κ. In $D_y(p, p') = -t^2$ the letter t denotes not the parameter of $\mathbb{B}_{tD}$ but the entry $\ell_y(p_y) = t$ of the rows of that step. For this kernel, $\delta = t^2$, so that $t^2 = \delta_\Delta c_a(x)/(c_Q^{\mathrm{su}} p_0^2)$. Then, with the whole strong weight,

$$|\tilde{w} Q_D|(X, V) \exp\{c_{\Xi} x \varepsilon \sum_{p \in D(V) \cap X} \phi_p + \tilde{\kappa}(|X|_c - 1)\}$$

equals $Q_D(X, V)$ and is $\geq \mathfrak{n}' \delta_\Delta c_a(x)/(8 c_Q^{\mathrm{su}})$, where $\mathfrak{n}'$ is the number n of the sites $z_1, \dots, z_n$ of that step, which grows with N.

(B8) $E^\Delta[\overline{D}] = (E^\Delta[D])^*$ on all of $\mathcal{U}_{\{\square\}}(x)$, $\mathbb{B}_{\overline{D}}(\tilde{u}^*) = (\mathbb{B}_D \tilde{u})^*$, and $\mathbb{B}_D \tilde{u}$ is real if D and $\tilde{u}$ are.

At $A = 1$, with $D = \tilde{\mathsf{E}} - \mathsf{E}$, with $\tilde{u}$ real and with the norms taken at the pair $p = (\kappa_1, c_h)$, the membership of $\mathbb{B}_D(\tilde{u})$ in $M_{A+\delta_\Delta}(x; p)$, (B1) to (B4) and (B8) give Lem. 5.47, since $1/e \leq \frac{1}{2}$ in (B3). ■

Theorem 7.11 (One step for two families, one of them over a kernel $D(z)$; family form of Thm. 5.48)**.** Let $x \geq x_9^U$, in the setting (SD) at x for (ℓ, ℓ') and in the setting of the conditions of item 5 of §5.4, and let $p \in \mathbb{P}$. Let $D(z)$, $z \in \Omega$, be kernels as in Lem. 7.10 (B5), with $\delta_\Delta := \sup_z \delta_\Delta(D(z)) \leq 1$. Let u_1 be a family on Ω within B_1 caps, and let $\tilde{u}_2$ be a family with $|\tilde{E}_2^a|_a, |\tilde{E}_2^h|_h, |\tilde{w}_2|_w^{\mathrm{st}} \leq B_2$; let $B := \max\{B_1, B_2 + 1\} \leq 2.6$, and put $u_2^\flat(z) := \mathbb{B}_{D(z)} \tilde{u}_2(z)$.

(C1) At every z and V', $(\rho[\tilde{u}_2; \ell; D])' = e^{\mathsf{V}'[u_2^\flat]} \rho[F_\ell(u_2^\flat); \ell'^{\mathrm{c}}]$.

(C2) $F_\ell(u_1(\cdot))$ and $F_\ell(u_2^\flat(\cdot))$ are families in $M_1(x'; p)$ that obey the strong cap, and $\mathsf{V}'[u_2^\flat(\cdot)]$ is holomorphic.

(C3) $\mathsf{D}_{x';\Omega}(F_\ell u_1, F_\ell u_2^\flat) \leq \vartheta[B]\,(\mathsf{D}^{\mathrm{in}}_{x;\Omega}(u_1, \tilde{u}_2) + \delta_\Delta)$.

(C4) At one point z, with $D = \tilde{\mathsf{E}} - \mathsf{E}$, with u_1 and $\tilde{u}_2$ real, in M_1 and obeying the strong cap, with $\rho > 0$ and with $p = (\kappa_1, c_h)$, the theorem becomes Thm. 5.48, with $\vartheta[2] \leq 0.28343 \leq \vartheta_{10}$.

■□

The steps of the proofs in this section are numbered consecutively from Step 1 to Step 9.

Proof of Lem. 7.5 (a), (b). Step 1: Cauchy's estimate on the ball, in an admissible seminorm.

(a) We begin with the scalar case, in which I is a single point and $N = |\cdot|$. Fix u with $\|u\| \leq B$ and directions $g_1, \dots, g_j$. If $g_i = 0$ for some i, both sides of the bound are 0; we may therefore take every g_i to be non-zero.

The set $O := \{s \in \mathbb{C}^j : \|u + \sum s_i g_i\| < A_\#\}$ is open and convex. On O the function $\Psi(s) := \Phi(u + \sum s_i g_i)$ is bounded by m, and it is holomorphic in each s_i when the other variables are held fixed, because $s_i \mapsto u + \sum s_i g_i$ then runs along an affine line. By Osgood's lemma, a bounded function that is holomorphic in each variable separately is holomorphic; hence Ψ is holomorphic on O. Consequently the mixed derivative of Ψ at 0, which is $D^j \Phi(u)[g_1, \dots, g_j]$, exists and is symmetric in the directions.

This derivative is linear in g_1. Indeed, for $g_1 = cg + c'g'$ we apply the same argument to the $j + 1$ directions $g, g', g_2, \dots, g_j$, and the chain rule gives $\partial_{s_1} = c\,\partial_s + c'\partial_{s'}$.

For the bound we put $R_i := (A_\# - B)/(j\|g_i\|)$. Where $|s_i| < R_i$ for every i, we have $\|u + \sum s_i g_i\| < B + j(A_\# - B)/j = A_\#$, so this polydisc lies in O. Cauchy's estimate on the polydisc with radii $R_i' < R_i$, followed by the limit $R_i' \to R_i$, gives

$$|\partial_1 \cdots \partial_j \Psi(0)| \leq \mathsf{m} \prod R_i^{-1} = \mathsf{m}\,(j/(A_\# - B))^j \prod \|g_i\|.$$

We now let N be general. Each Φ_i is bounded on the ball, so the scalar case applies to Φ_i, with the bound of Φ_i in the place of m: the derivative $D^j\Phi_i(u)$ exists and is symmetric and complex multilinear. Let $\mathfrak{l} \in \mathfrak{L}$. The function $\mathfrak{l}(\Phi(\cdot)) = \sum c_i\Phi_i$ is a finite sum, hence holomorphic along lines, and $|\mathfrak{l}(\Phi(u))| \le N(\Phi(u)) \le \mathsf{m}$ on the ball; moreover D^j commutes with the finite sum. The scalar case, applied to $\mathfrak{l}(\Phi(\cdot))$, therefore gives

$$|\mathfrak{l}(D^j\Phi(u)[g_1,\dots,g_j])| \le \mathsf{m}\,(j/(A_\# - B))^j \prod \|g_i\|,$$

and we take the supremum over $\mathfrak{L}$.

For a cell sum this argument amounts to the following: each derivative is written as a Cauchy integral, and the sum over Y is taken inside the integral, together with the suprema over the fields. Applying Cauchy's estimate to each Y separately and summing afterwards would instead give $\sum_Y \sup_s$, and this can be larger than $\sup_s \sum_Y$. For example, for $v_Y(z) = ((1 + ze^{-2\pi iY/8})/2)^{12}$, $Y = 1,\dots,8$, on $|z| = 1$ the first quantity is 8, and the second is the value of the sum $\sum_Y |v_Y(z)|$, which is constant on $|z| = 1$ and equals $231/128 \le 1.805$.

(b) The closed ball is convex, so $u_t := u_2 + t(u_1 - u_2)$ satisfies $\|u_t\| \le B$ for $0 \le t \le 1$. Let $\mathfrak{l} \in \mathfrak{L}$. The function $t \mapsto \mathfrak{l}(\Phi(u_t))$ is C^1 with derivative $\mathfrak{l}(D\Phi(u_t)[u_1 - u_2])$, and by (a) at $j = 1$ this derivative is bounded by $\mathsf{m}\|u_1 - u_2\|/(A_\# - B)$. We integrate over $[0,1]$ and take the supremum over $\mathfrak{L}$.

Step 2 of the proof of Thm. 5.46 uses the bound $\|u_z\| \le B + |z|$ on the segment as well, which leads to its constant $16 - 3B = 10$ at $B = 2$. Here the norm on the segment is $\le B$, and this gives $16 - B$. □

Proof of Lem. 7.7. Step 2: the estimates of Cauchy and of Schwarz; the coefficients are members.

(S1) Let $\mathfrak{l} \in \mathfrak{L}$ and $h(z) := \mathfrak{l}(a(z))$. Then h is a finite sum of holomorphic functions, $|h| \le m$ on $\mathrm{Pd}(r)$, and the Taylor coefficient of h of index μ is $\mathfrak{l}(a_\mu)$. Cauchy's estimate on the polydisc gives $|\mathfrak{l}(a_\mu)| \le m\,r^{-\mu}$, and we take the supremum over $\mathfrak{L}$.

(S2) Now $h(0) = 0$. For $z \ne 0$ let $t := \max |z_i|/r_i < 1$. The function $c \mapsto h(cz/t)$ is holomorphic on $|c| < 1$, bounded by m and equal to 0 at 0, so the Schwarz lemma at $c = t$ gives $|h(z)| \le mt$.

(S3) Let $0 < r_i' < r_i$. The coefficients are given, at each value, by

$$g_\mu(\text{value}) = (2\pi i)^{-n} \oint_{|z_i|=r_i'} g(z)(\text{value}) \prod dz_i/z_i^{\mu_i+1}.$$

We first treat E^a. The set $\mathcal{U}_X(x)$ is open in the complex manifold $\mathrm{SL}(2,\mathbb{C})^{\text{bonds of } X^{\approx}}$. In a chart, the map $(z, \mathsf{W}) \mapsto E^a_z(X, \mathsf{W})$ is bounded and separately holomorphic, so it is holomorphic by Osgood's lemma. Integrating a jointly holomorphic function over a compact torus in z gives a function that is holomorphic in W.

Invariance under G-valued gauge transformations, the value 0 at the pure gauges, the dependence sets $X^{\approx}$ and $X^{\approx\approx}$, and the domains of E^h and w are linear conditions that hold at each z; hence they hold for the integral.

The coefficients are Borel. Indeed, the integrand is continuous in z, so the integral is a pointwise limit of Riemann sums, and these are finite combinations of Borel functions of V.

Finally $\|g_\mu\| < \infty$ by (S1), and the norms and cell sums that the lemma names are admissible by item 0 (4) of §7.2.

□

Proof of Lem. 7.5 (c), (a°) and of Lem. 7.8. Step 3: one power series.

(a°) Let $|\Phi| \le \mathsf{m}$ on $\{\|u - u_0\| < t\}$. We apply Step 1 with the centre u_0 in place of 0, with $B = 0$ and with $A_\# = t$; this is possible because the proof of Step 1 uses only that the ball

is open and convex. It gives

$$|D^m\Phi(u_0)[g_1,\dots,g_m]| \le \mathsf{m}\,(m/t)^m \prod \|g_i\|,$$

so $D^m\Phi(u_0)$ is multilinear and continuous for the norm.

(1) Let $\|h\| < t$. The function $s \mapsto \Phi(u_0 + sh)$ is holomorphic on $|s| < t/\|h\|$, and its Taylor series at 0, evaluated at $s = 1$, is

$$\Phi(u_0 + h) = \sum_{m\ge 0} (1/m!) D^m\Phi(u_0)[h,\dots,h].$$

Here the m-th derivative at 0 equals $D^m\Phi(u_0)[h,\dots,h]$, by the chain rule applied to the function of m variables from Step 1 with all $g_i = h$.

(2) Let $\|u(z)\| \le B'$ on $\mathrm{Pd}(q)$. Lem. 7.7 gives $u_\mu \in \mathfrak{M}(x)$ and $\|u_\mu\| \le B' q^{-\mu}$.

Hence

$$\varsigma(z) := \sum_{\mu\ne 0} |z^\mu|\,\|u_\mu\| \le B'[\prod (1 - |z_i|/q_i)^{-1} - 1],$$

which tends to 0 with z.

Value by value we have $h(z) := u(z) - u_0 = \sum_{\mu\ne 0} z^\mu u_\mu$. The norm is a weighted supremum of values, so the norm of a tail of this series is at most the sum of the norms of its terms; thus the series converges in norm.

(3) On the set $\{\varsigma(z) < t/e\}$, multilinearity and continuity give

$$D^m\Phi(u_0)[h(z),\dots,h(z)] = \sum_{(\mu^1,\dots,\mu^m),\ \text{all}\ne 0} z^{\mu^1+\dots+\mu^m} D^m\Phi(u_0)[u_{\mu^1},\dots,u_{\mu^m}],$$

and the sum of the moduli on the right is at most $\mathsf{m}\,(m/t)^m\varsigma^m$; moreover $m^m/m! \le e^m$.

Therefore $\Phi(u(z))$ is the sum of a multiple power series whose sum of moduli is at most $\mathsf{m}\sum_m (e\varsigma/t)^m < \infty$. So $\Phi(u(z))$ is holomorphic near 0, and its coefficients are obtained by collecting terms.

(4) The coefficient of z^S collects the tuples with $\mu^1 + \dots + \mu^m = 1_S$ and all $\mu^i \ne 0$. For such a tuple $\mu^i = 1_{S_i}$, where $(S_1,\dots,S_m)$ is an ordered partition of S into non-empty blocks.

By symmetry the $m!$ orderings of one partition give equal terms, which cancels the factor $1/m!$; this is the formula. It involves only u_0 and the $u_{S'}$ with $S' \subset S$, and u and u^{pol} have these in common.

(c) We prove Lem. 7.5 (c). At $z_0 \in \Omega$ we take $\Phi = \Phi_i$ and let m be the bound of Φ_i on $\{\|u\| < A_\#\}$. This ball contains $\{\|u - u_0\| < t\}$ for $u_0 = u(z_0)$ and $t = A_\# - B$. We apply (3) to the map $z \mapsto u(z_0 + z)$ with $B' = B$, and obtain holomorphy near z_0.

It remains to prove the sentence of Lem. 7.8 that is introduced by the words *in particular*. By Thm. 6.7 (U), Thm. 5.15 (b) holds at p, and so a value of F_ℓ is bounded on $M_{16}(x;p)$. By Thm. 6.7 (g2) we have $|v(Y;u)| \le 16 C_g c_a(x)$, and by Thm. 6.7 (g1), (g2) we have $|\mathsf{V}'[u]| \le |V_0^+| + 16 C_g c_a(x)$ times the number of cells. All these maps are holomorphic along lines, by Thm. 6.7 (c+) applied to affine families.

For $I[u]$ the two hypotheses are shown in the statement of Lem. 7.8 itself, by the three facts named there. □

Proof of Thm. 7.9. Step 4: the image, the contraction and the coefficients. In this step we prove clauses (i) to (iii) of Thm. 7.9.

(i) By Thm. 6.7 (U), Thm. 5.15 (b) holds at p, and we use it at $A = B$. It gives $|E'^{\mathrm{new}}|_a^\Sigma \le 0.35$ and $|E'^{\mathrm{old}}[u]|_a^\Sigma \le 0.248B$, so that

$$|E'^a|_a^\Sigma \le 0.35 + 0.248 \cdot 2.6 = 0.9948 \le 1.$$

Moreover $|E'^h|_h^\Sigma \le 1$; the member w' carries the strong cap at x' with the rate $\tilde\kappa$, and hence also the weak cap; and the clauses without a size hold at x'.

Since a sum of non-negative terms bounds each of its terms, $F_\ell(u(z)) \in M_1(x';p)$. All values, V' and $v(Y;\cdot)$ are holomorphic in z by Thm. 6.7 (c+).

(ii) Fix z; then $\|u_i(z)\| \le B$. We apply Lem. 7.5 (b), with $A_\# = 16$, to three maps Φ on $\{\|u\| < 16\}$, each of which is holomorphic along lines by Thm. 6.7 (c+): the map $\Phi_a = (E'^{\mathrm{old}}[u](Y,\mathsf{W}'))_{Y,\mathsf{W}'}$, with $N = |\cdot|_a^\Sigma$ and $\mathsf{m} = 16\vartheta_w = 3.968$ by Thm. 5.15 (b) at $A = 16$; the map $\Phi_h = (E'^h(Y,V'))$, with $N = |\cdot|_h^\Sigma$ and $\mathsf{m} = 1$; and the map $\Phi_w = (w'(Y,V'))$, with $N = |\cdot|_w^{\mathrm{st}}$ at x' and $\mathsf{m} = 1$.

The member E'^{new} is the same for all u, so it cancels in $E'^a[u_1] - E'^a[u_2]$.

Hence the three members of $F_\ell(u_1) - F_\ell(u_2)$ are at most $3.968/(16-B)$, $1/(16-B)$ and $1/(16-B)$ times $\|u_1 - u_2\|$, and the largest of these factors is $\vartheta[B]$. Here we use $|\cdot|_a \le |\cdot|_a^\Sigma$ and $|\cdot|_h \le |\cdot|_h^\Sigma$. Finally we take the supremum over z.

We turn to the further bound in (ii). By Thm. 6.7 (f1), E'^{old} depends on E^a alone. Along $u_t = u_2 + t(u_1 - u_2)$ we have $(d/dt)E'^{\mathrm{old}}[u_t] = DA(u_t)[E_1^a - E_2^a]$ by Thm. 6.7 (h): the series DA, defined in §6.2 termwise as a sum of derivatives, coincides with this derivative wherever $\|u_t\| < 16$.

The difference $E_1^a - E_2^a$ lies in $T^a(x)$, since its norm is finite and so each of its values is bounded. At each t, Thm. 6.7 (h3) gives the bound $\vartheta_a |E_1^a - E_2^a|_a$ in the cell sum. As in the proof of (b) in Step 1, we apply each linear form of the cell sum, integrate over $0 \le t \le 1$, and take the supremum over these forms.

(iii) Let $g(z) := F_\ell(u_1(z)) - F_\ell(u_2(z))$. This map takes $\mathrm{Pd}(r)$ to $\mathfrak{M}(x')$: by Thm. 6.7 (U), Thm. 5.15 (b) holds at p, and at $A = B < 16$ it yields finite sizes together with all clauses without a size at x', namely $F_\ell(u) \in M_{\max\{1,0.35+0.248B\}}(x';p)$.

The values of g are holomorphic by Thm. 6.7 (c+). Neither of these two facts depends on $B \le 2.6$; for $B \le 2.6$ they are (i). By (ii), $\|g(z)\|^{\mathrm{st}} \le \vartheta[B]\mathsf{D}^{\mathrm{in}}_{x;\mathrm{Pd}(r)}$, and we apply Lem. 7.7 (S1), (S3) in the norm $\|\cdot\|^{\mathrm{st}}$ at x'.

The contraction factors satisfy $\vartheta[1] \le 0.26454$, $\vartheta[2] \le 0.28343$, $\vartheta[2.455] \le 0.29295$ and $\vartheta[2.6] \le 0.29612$. Also $(1 - 0.35)/0.248 > 2.6$, so the radius 2.6 is admitted.

The numbers 0.3968 of Thm. 5.46 (ii) and 0.4 of Thm. 5.15 (e) are upper bounds of $\vartheta[2]$.

Step 5: the channels. We prove the bounds on the first and second derivatives in Thm. 7.9 (iv).

By Step 1, $DF_\ell(u)[g]$ is linear in g, so we may split $g = (g^a,0,0) + (0,g^h,0) + (0,0,g^w)$; likewise D^2 is bilinear.

For the channel $a \to a$ we use Thm. 6.7 (h3) with $\delta = g^a \in T^a(x)$, whose size is $|g^a|_a$ caps at p, in the setting (L3su).

For the channels $h, w \to a$ we use Thm. 6.7 (f1): E'^{new} depends on no member and $E'^{\mathrm{old}}[u]$ on E^a alone, so $E'^a[u + s(0,g^h,g^w)]$ is constant in s.

For the channels $a \to h, w$ we use Lem. 7.5 (a) with $j = 1$ and $A_\# = 16$, for the maps Φ_h and Φ_w of Step 4, with $\mathsf{m} = 1$, in the direction $g = (g^a,0,0)$, for which $\|g\| = |g^a|_a$.

For the channels $w \to h, w$ we use Thm. 6.7 (f4): F_ℓ and V' use $w(X,V)$ only at those V for which X holds a plaquette of $D(V)$. Hence $F_\ell(u) = F_\ell(u^\circ)$, where $w^\circ(X,V) := w(X,V)\,1\{D(V) \cap X \ne \emptyset\}$ and $u^\circ := (E^a, E^h, w^\circ)$.

Moreover $g^\circ := (0,0,g^{w\circ}) \in \mathfrak{M}(x)$, because the indicator is Borel and gauge invariant, being a function of ϕ_p at the plaquettes of X.

Since $(u + \sum s_i g_i)^\circ = (u + \sum s_i g_i^\circ)^\circ$, we have $F_\ell(u + \sum s_i g_i) = F_\ell(u + \sum s_i g_i^\circ)$ for small s. Every derivative may therefore be taken in the directions g_i°, and we apply Lem. 7.5 (a) with $\|g^\circ\| = |g^{w\circ}|_w^{\mathrm{wk}}$.

Where $D(V) \cap X \ne \emptyset$, some p has $\phi_p > \varepsilon$, and then $x\varepsilon\phi_p > x\varepsilon^2 = p_0^2$. There the strong weight is at least $e^{c\Xi p_0^2}$ times the weak one, and so $|g^{w\circ}|_w^{\mathrm{wk}} \le \eta_w |g^w|_w^{\mathrm{st}}$.

For the channels $h \to h, w$ let $p = (\tilde\kappa, c_h)$, and let $p^\circ := (\tilde\kappa, c_h/2)$, which is a pair of $\mathbb{P}$. The norms at p and at p° differ only in the h-unit: $|\cdot|_h^{(p^\circ)} = e^{-(c_h/2)p_0(x)^2} |\cdot|_h^{(p)}$.

Hence $\|u\|^{(p^\circ)} \le \|u\|^{(p)} \le B$ and $\|(0,g^h,0)\|^{(p^\circ)} = e^{-(c_h/2)p_0^2}|g^h|_h^{(p)}$.

By Thm. 6.7 (U) the same map F_ℓ satisfies Thm. 5.15 (b) at p° on $M_{16}(x;p^\circ)$: the cell sum of E'^h is at most $e^{-(c_h/2)p_0'^2}$, and w' has the strong cap.

We apply Lem. 7.5 (a) in the ball of the norm at p°. The cell sum of $DE'^h(u)[g^h]$ is then at most $e^{-(c_h/2)p_0'^2}e^{-(c_h/2)p_0^2}|g^h|_h/(16-B) = c_h(x')\,e^{(c_h/2)p_0'^2-(c_h/2)p_0^2}|g^h|_h/(16-B) = c_h(x')\,\eta_h|g^h|_h/(16-B)$, and $|Dw'(u)[g^h]|_w^{\mathrm{st}} \le e^{-(c_h/2)p_0^2}|g^h|_h/(16-B) \le \eta_h|g^h|_h/(16-B)$.

We turn to the second derivatives and begin with the output a. Every point u of the open ball $\{\|u\|<16\}$ belongs to $M_A(x;p)$ for $A=\max\{1,\|u\|\}<16$; hence Thm. 6.7 (b#), applied at values $A<16$ alone, gives $|E'^{\mathrm{old}}[u]|_a^\Sigma \le 16\vartheta_a$ on $\{\|u\|<16\}$.

Lem. 7.5 (a) with $j=2$ then gives the bound $16\vartheta_a(2/(16-B))^2\|g_1\|\|g_2\|$. By Thm. 6.7 (f1) each g_i may be replaced by $(g_i^a,0,0)$, and the derivative is 0 if one of the entries has no a-part.

For the Lipschitz bound we write $(DA(u_1)-DA(u_2))[g^a] = \int_0^1 D^2A(u_t)[u_1-u_2,g^a]\,dt$; with the bound just obtained, the constant is $16\cdot 4 = 64$.

We turn to the second Lipschitz bound of Thm. 7.9 (iv), namely $|(DA(u_1)-DA(u_2))[g^a]|_a^\Sigma \le (\vartheta_a/(16-B))|E_1^a-E_2^a|_a|g^a|_a$. The map $\Phi(u) := DA(u)[g^a]$, with $N = |\cdot|_a^\Sigma$, is bounded by $\mathsf{m} = \vartheta_a|g^a|_a$ on $\{\|u\|<16\}$ by Thm. 6.7 (h3); by Thm. 6.7 (h) it is holomorphic along lines, and it depends on u through E^a alone.

We apply Lem. 7.5 (b) to the two points u_1 and (E_2^a,E_1^h,w_1), whose distance is $|E_1^a-E_2^a|_a$; it gives $\vartheta_a|g^a|_a|E_1^a-E_2^a|_a/(16-B)$. Since $1/(16-B)\le 64/(16-B)^2$, this second bound is the smaller of the two.

For the second derivatives in the outputs h and w we use Lem. 7.5 (a) with $j=2$ and $\mathsf{m}=1$. If no entry is an h-member, we work at p, with g° in place of each w-entry.

If $j_h \ge 1$ entries are h-members, we work at p°. The entries then give the factor $e^{-j_h(c_h/2)p_0^2}$, and the return of E'^h to the unit $c_h(x')$ gives the factor $e^{(c_h/2)p_0'^2}$.

Since $\eta_h^{j_h} = e^{-j_h(c_h/2)p_0^2+j_h(c_h/2)p_0'^2}$ and $j_h\ge 1$, we have $e^{(c_h/2)p_0'^2-j_h(c_h/2)p_0^2} \le \eta_h^{j_h}$. The bound for w' carries no factor $e^{(c_h/2)p_0'^2}$, and $e^{-(c_h/2)p_0^2}\le\eta_h$. The a-unit and the w-unit are the same at p and at p°.

An entry may also be left whole. Lem. 7.5 (a) is applied in the max norm, so g_1 may be kept unsplit. At p° we have $\|g_1\|^{(p^\circ)} \le \|g_1\|^{(p)}$, because the h-unit at p° is the larger one.

If g_2 is a pure entry, the bound is $(4/(16-B)^2)\|g_1\|^{(p)}$ times $|g_2^a|_a$, or times $|g_2^{w\circ}|_w^{\mathrm{wk}}$ (at p, with g_2° in place of g_2), or times $\eta_h|g_2^h|_h$ (at p°, with $j_h=1$ and with the factor $e^{(c_h/2)p_0'^2}$ occurring once), according to the kind of g_2. Since D^2 is bilinear, a mixed g_2 gives the sum of the three.

The numerals are $4/225 = 1/56.25$, $4/196 = 1/49$ and $64/225 = 16/56.25$, which is 0.28444 to five decimals.

Step 6: the local vacuum numbers and the reflection. We prove clauses (v) and (vi) of Thm. 7.9.

(v) Let $\Phi = (v(Y;u))_Y$ on $\{\|u\|<16\}$, with $N = |\cdot|^\Sigma$. It is holomorphic along lines by Thm. 6.7 (c+), and $N(\Phi(u)) \le 16C_gc_a(x)$ by Thm. 6.7 (g2) at $A=16$.

By Thm. 6.7 (f1), $v(Y;u)$ depends on E^a alone. Hence $v(Y;u_2) = v(Y;u_2^\circ)$ with $u_2^\circ := (E_2^a,E_1^h,w_1)$, where $\|u_2^\circ\|\le B$ and $\|u_1-u_2^\circ\| = |E_1^a-E_2^a|_a$, and we apply Lem. 7.5 (b). The numeral is $16\cdot 263/14 = 300.57$ to two decimals.

For the coefficients we use Lem. 7.7 (S1) in $|\cdot|^\Sigma$: for $a(z) = (v(Y;u_1(z)))_Y$, which is bounded by $C_gBc_a(x)$ by Thm. 6.7 (g2) at $A=B$, and for the difference, which is bounded by the first sentence of (v).

(vi) We begin with the involution. The map $g\mapsto g^\#$ is an antiholomorphic automorphism of $\mathrm{SL}(2,\mathbb{C})$ that fixes G pointwise; thus $(gh)^\# = g^\#h^\#$ and $(g^{-1})^\# = (g^\#)^{-1}$.

Hence $\mathsf{W}^\#$ is a field, with $\mathsf{W}^\#(\mathrm{path}) = \mathsf{W}(\mathrm{path})^\#$ and $\partial\mathsf{W}^\#(p) = (\partial\mathsf{W}(p))^\#$. Since $|g^\#| = |g^{-1}| = |g|$ (the definition "Group" in item 2 of §5.4), the conditions (u1) and (u2) are preserved under $\mathsf{W}\mapsto\mathsf{W}^\#$.

For $h \in \mathrm{SL}(2,\mathbb{C})$ we have $h^{-1} = (\operatorname{tr} h)1 - h = \operatorname{adj} h$. The map adj is linear on 2×2 matrices, with $\operatorname{adj} 1 = 1$ and $\operatorname{adj} A = JA^{\mathsf{T}}J^{-1}$ for a unitary J. Therefore $|h^{\#} - 1| = |(h^{-1}-1)^{\dagger}| = |\operatorname{adj}(h-1)| = |h-1|$, and (u3) is preserved as well.

So $\mathcal{U}_X(x)$ is mapped onto itself. The map $\mathsf{W} \mapsto \overline{E^a(X,\mathsf{W}^{\#})}$ is holomorphic, $(\mathsf{W}^u)^{\#} = (\mathsf{W}^{\#})^u$ for G-valued u, and the pure gauges are fixed.

Hence $u^* \in \mathfrak{M}(x)$, $u^{**} = u$ and $(cu)^* = \bar{c}\,u^*$, and every supremum of item 0 (2) of §7.2 is unchanged under $u \mapsto u^*$.

If $u = u^*$, then $E^a(X,V) = \overline{E^a(X,V)}$ at G-valued V, and E^h and w are real; thus u is real.

Next we prove the identity. Let Φ be either V' or, at a G-valued argument, a value of F_ℓ, and put $\psi(u) := \overline{\Phi(u^*)} - \Phi(u)$ on $\{\|u\| < 16\}$. This function is bounded, and it is holomorphic along lines, since $(u+sg)^* = u^* + \bar{s}\,g^*$.

If $u = u^*$, then u is real, so $\Phi(u)$ is real by Thm. 5.15 (d) at p, which holds by Thm. 6.7 (U); hence $\psi(u) = 0$.

For an arbitrary u we put $u_1 := (u+u^*)/2$ and $u_2 := (u-u^*)/(2i)$. Both are fixed by $*$, and $u = u_1 + iu_2$.

The set $O := \{(s,t) \in \mathbb{C}^2 : \|su_1 + tu_2\| < 16\}$ is open and convex and contains $(0,0)$ and $(1,i)$. The function $\chi(s,t) := \psi(su_1 + tu_2)$ is holomorphic on O by Osgood's lemma, and it vanishes at the real points of O.

Hence all derivatives of χ at $(0,0)$ vanish, so $\chi = 0$ near $(0,0)$ and therefore on the connected set O. In particular $\chi(1,i) = \psi(u) = 0$.

It remains to treat complex arguments. The map $\mathsf{W}' \mapsto E'^a[u^*](Y,\mathsf{W}') - \overline{E'^a[u](Y,\mathsf{W}'^{\#})}$ is holomorphic on $\mathcal{U}_Y(x')$ and vanishes at its G-valued points.

Now G^{bonds} is a totally real submanifold of $\mathrm{SL}(2,\mathbb{C})^{\mathrm{bonds}}$ of full real dimension. A holomorphic function that is zero on an open piece of such a submanifold is zero near that piece, and so on the whole component.

The reason is that in the chart $\mathsf{W}(b) = V(b)\exp\{i\sum_k c_{b,k}\sigma_k\}$, with complex c, the G-valued fields are those with real c, and the derivatives of the function in all real directions are zero there.

Finally, the density $\rho[\cdot;\ell'^{\mathrm{c}}]$ uses the members at G-valued fields alone.

□

Proof of Lem. 7.10. Step 7: the objects at a complex kernel. In this step we prove the bound $|q_y^D| \le (9/2)\mathsf{g}_\flat^2 p_0^2 \delta$, clause (B2), the factorisation $\Xi_{w^\flat} = \Xi_{\tilde{w}} \prod_{\mathrm{off}} e^{q_\square^D}$, clause (B1), the statement on the dependence set of e^{Q_D}, and the identity $\rho[u;\ell;\tilde{\mathsf{E}} - \mathsf{E}] = \rho[u;\tilde{\ell}]$.

We use the facts (G1) to (G5) on the geometry, from Step 0 of §5.4. The fact (G5), namely $\sum_{p,p'} |D_y(p,p')| \le 9\mathsf{g}_\flat^2\delta$, involves $|D_y(p,p')|$ alone; by its own proof it holds for every family $D = (D_y)$ of complex kernels.

That proof runs as follows. The norm $\|D\|_{(\lambda)}$ is defined in item 2 of §5.4 for a family $D = (D_y)$ of kernels, and its definition gives $|D_y(p,p')| \le \delta e^{-\lambda_\flat(|y-p|+|y-p'|)}$. Three plaquettes share a base site, and the periodic sum is at most the sum over $\mathbb{Z}^3$; hence $\sum_p e^{-\lambda_\flat|y-p|} \le 3\mathsf{g}_\flat$.

We turn to q_y^D. At a G-valued V each $T(y,\cdot)$ is an isometry of the real space $\mathfrak{g}$, $|\zeta_p| \le \varepsilon$, $m_y \le 1$ and $x\varepsilon^2 = p_0^2$; therefore

$$|q_y^D| \le (x/2)\varepsilon^2 \cdot 9\mathsf{g}_\flat^2\delta = (9/2)\mathsf{g}_\flat^2 p_0^2\delta.$$

The brackets $\langle\cdot,\cdot\rangle$ are real, Borel and gauge invariant (Prop. 5.35), and they are functions of V at the sites within $\max\{\rho_0, \varrho(x)+1\}+1$ of y; the number q_y^D is a complex linear combination of them.

If $D = \tilde{\mathsf{E}} - \mathsf{E}$, Step 3 of the proof of Lem. 5.47 gives $-x\mathrm{En}_{\tilde{\ell}} = -x\mathrm{En}_\ell + \sum_\square q_\square^D$, and so $\rho[u;\ell;D] = \rho[u;\tilde{\ell}]$. For every family $D = (D_y)$ of kernels that satisfies the hypotheses of Lem. 7.10 on D, the

lemma defines $\rho[u;\ell;D] := \rho[u;\ell]\exp\{\sum_\square q^D_\square\}$. Only for $D = \tilde{\mathsf{E}} - \mathsf{E}$ does the identity of the energies make it the density over the rows $\tilde\ell$.

(B2) By (G2), every path and every plaquette that occurs lies in $\square^\approx$.

By (u3) and (F1), $|\partial\mathsf{W}(p) - 1| < 2\varepsilon \le 1/20$. Hence the series $\log(1+A)$ converges, and $|F_p| \le -\log(1-2\varepsilon) \le 1.02587\cdot 2\varepsilon \le 2.06\varepsilon$.

By (u2), $|\mathsf{W}(\Gamma)| < 2$, so $|\operatorname{Ad}_{\mathsf{W}(\Gamma)} F_p| \le 4\cdot 2.06\varepsilon = 8.24\varepsilon$; moreover $|Y\cdot Y'| \le |Y||Y'|$.

Consequently one term is at most $(x/2)|D_y(p,p')|(8.24\varepsilon)^2 = 33.9488\, p_0^2|D_y(p,p')|$. Together with (G5) and the M^3 sites, the sum is at most

$$305.5392\, M^3 \mathfrak{g}_\flat^2 p_0^2\delta \le c_Q^{\mathrm{su}} p_0^2\delta = \delta_\Delta c_a(x).$$

Each term is holomorphic on $\mathcal{U}_{\{\square\}}(x)$; it is invariant under G-valued gauge transformations, because such a transformation u takes each of the two entries Y to $\operatorname{Ad}_{u(y)} Y$ and the form is Ad-invariant; and it is 0 at the pure gauges, where $F = 0$. Hence every complex combination of the terms has these properties.

The formula shows the linearity in D and the independence of ℓ and of $\tilde u$.

Let V be tame on $\square^\approx$ and $y \in \square$. By (M5r), (F2) and (G1), no plaquette within $\rho_0 \le M$ of y is rough, and none within $\varrho(x)+1 \le Mr(x)$ has $\phi_p > 1$; so $m_y = 1$. Further, $\zeta_p = F_p = \mathsf{L}(\partial V(p))$ for $|y-p| \le \varrho(x)$, and on $\mathfrak{g}$ we have $\operatorname{Ad} = T$ and $\cdot = \langle\,,\rangle$. Hence, for any coefficients $D_y(p,p')$, the two formulas coincide term by term.

(B1) We first prove the factorisation. In a family of (cΞ) the sets X_i are pairwise farther apart than $s(x) \ge 8$, so they are disjoint, and $D(V)$ lies in their union; therefore each $p_*(\square, V)$ belongs to exactly one X_i.

So $\prod_i w^\flat(X_i,V) = \prod_i \tilde w(X_i,V)\cdot\prod_{\mathrm{off}} e^{q^D_\square(V)}$, where the last factor is the same for all families, and we sum over the families. In the case $D(V) = \emptyset$, all cubes are on.

Now we prove (B1) itself. We split $\sum_\square q^D_\square = \sum_{\mathrm{on}} + \sum_{\mathrm{off}}$, and

$$\sum_{\mathrm{on}} q^D_\square(V) = \sum_\square 1_{\square^\approx}(V) E^\Delta[D](\{\square\},V) = \sum_X 1_{X^\approx}(V) E^\Delta[D](X,V).$$

Hence

$$\rho[\tilde u;\ell]\exp\{\sum q^D\} = \exp\{-x\mathrm{En}_\ell + \sum_X 1_{X^\approx}[\tilde E^a + E^\Delta[D] + \tilde E^h]\}\cdot\prod_{\mathrm{off}} e^{q^D}\cdot\Xi_{\tilde w} = \rho[\mathbb{B}_D\tilde u;\ell].$$

We turn to the dependence set. If $p_*(\square,V) \in X$, then $\square \subset X^{+r}$. Whether a cube is off, and the plaquette p_*, depend on ϕ_p on $\square^\approx$, which lies inside X^{+2r}, and hence on bonds with ends in $X^{+(2r+1)}$.

The number $q^D_\square$ depends on V within $\max\{M+1, Mr\}$ of the cube, hence inside $X^{+(2r+1)}$; measured in cubes, this is within $\max\{2,r\} \le r+1$ of the cube, so the paragraph on the dependence of the product in Step 5 of the proof of Lem. 5.47 applies. Moreover $2r+1 < 8r = s$.

The member $w^\flat$ is Borel, being a finite product taken over a finite Borel partition by $D(V)$, and it is gauge invariant, because $D(V)$, the set of cubes that are off, p_*, q^D_y and $\tilde w$ are.

The numerals are $8.24^2/2 = 33.9488$, $33.9488\cdot 9 = 305.5392$ and $-\log(0.95)/0.05 \le 1.025866$.

Step 8: the clauses (B3) *to* (B8) *and the membership.*

Fix $p_* \in D(V)$, in the cube c_*. The cubes $\square$ with $p_*(\square,V) = p_*$ have $c_* \subset \square^\approx$; by (G4) there are at most $(2r+1)^3$ of them, each with M^3 sites.

Hence

$$\sum\{|q^D_\square| : p_*(\square,V) = p_*\} \le \delta' := (2r+1)^3 M^3 (9/2)\mathfrak{g}_\flat^2 p_0^2\delta = ((2r+1)^3 c_a(x)/68)\,\delta_\Delta,$$

since $4.5/306 = 1/68$, and by (F3) this is $\le \delta_\Delta/2 \le 1/2$. Consequently $|Q_D(X,V)| \le \delta'\mathfrak{n}$.

On $D(V)$ we have $x\varepsilon\phi_p > p_0^2$, so $|\tilde w(X,V)| \le Ae^{-c\Xi p_0^2\mathfrak{n}} e^{-\tilde\kappa(|X|_c-1)}$, and by (X15) this is $\le Ae^{-\mathfrak{n}} e^{-\tilde\kappa(|X|_c-1)}$.

(B3) Since $|e^{Q}| \le e^{|Q|}$, we obtain $|w^\flat| \le Ae^{-c_\Xi p_0^2 \mathfrak{n} + \mathfrak{n}/2} e^{-\tilde\kappa(|X|_c - 1)}$.

Since $|e^{Q} - 1| \le |Q| e^{|Q|}$ for complex Q,

$$|w^\flat - \tilde w| \le Ae^{-\mathfrak{n}} \delta' \mathfrak{n} e^{\mathfrak{n}/2} e^{-\tilde\kappa(|X|_c - 1)} \le (2/e) A\delta' e^{-\tilde\kappa(|X|_c - 1)} \le (A/e)\delta_\Delta e^{-\tilde\kappa(|X|_c - 1)},$$

because $te^{-t/2} \le 2/e$. At $\mathfrak{n} = 0$ we have $Q = 0$. The dependence is treated in Step 7.

We now prove the membership $\mathbb{B}_D \tilde u \in M_{A+\delta_\Delta}(x; p)$. We have $|\tilde E^a + E^\Delta[D]|_a \le A + \delta_\Delta$, since at $\#X = 1$ the weight is 1; further $|\tilde E^h|_h \le A$, and $|w^\flat|_w^{\mathrm{wk}} \le A$ since $c_\Xi p_0^2 \ge 1 > 1/2$. The clauses without a size follow from Step 7 and linearity.

(B4) We have $\mathbb{B}_D \tilde u - \tilde u = (E^\Delta[D], 0, w^\flat - \tilde w)$, whose members have the sizes δ_Δ, 0 and $(A/e)\delta_\Delta$.

(B5) The objects $E^\Delta[D]$ and Q_D, the set of cubes that are off, and p_* depend on no $\tilde u$. Hence $\mathbb{B}_D(\tilde u) = (E^\Delta[D], 0, 0) + (\tilde E^a, \tilde E^h, \tilde w e^{Q_D})$ is a constant plus a linear map.

For constant D, the coefficient of z^μ with $\mu \ne 0$ is $(\tilde E^a_\mu, \tilde E^h_\mu, \tilde w_\mu e^{Q_D})$, and $\tilde w_\mu$ is a member with finite $|\cdot|_w^{\mathrm{st}}$ by Lem. 7.7.

Moreover $|\tilde w_\mu e^{Q_D}| e^{\tilde\kappa(|X|_c - 1)} \le |\tilde w_\mu|_w^{\mathrm{st}} e^{-(c_\Xi p_0^2 - 1/2)\mathfrak{n}}$. This is always $\le |\tilde w_\mu|_w^{\mathrm{st}}$, and for $\mathfrak{n} \ge 1$ it is at most $e^{-(c_\Xi p_0^2 - 1/2)} = e^{1/2}\eta_w$ times $|\tilde w_\mu|_w^{\mathrm{st}}$.

For holomorphic $D(z)$, the objects $q_y^{D(z)}$ and $E^\Delta[D(z)]$ are linear in the entries and so holomorphic in z; products and the exponential keep this property, and the caps hold at each z.

(B6) In the a-part the bound is $|\tilde E^a_1 - \tilde E^a_2|_a + \delta_\Delta(D_1 - D_2)$, by linearity and (B2), whose bound on the size holds without the condition $\delta_\Delta \le 1$. In the h-part it is $|\tilde E^h_1 - \tilde E^h_2|_h$.

In the w-part we put $w_i := \tilde w_i$ and $Q_i := Q_{D_i}(X, V)$ for $i = 1, 2$, and D_x stands for $\mathsf{D}_x(\tilde u_1, \tilde u_2)$. We write $w_1 e^{Q_1} - w_2 e^{Q_2} = (w_1 - w_2)e^{Q_1} + w_2(e^{Q_1} - e^{Q_2})$. Here $|Q_i| \le \mathfrak{n}/2$, by the bound on $|Q_D(X, V)|$ at the start of this step together with $\delta_\Delta(D_i) \le 1$ from the data.

The first term is at most $\mathsf{D}_x e^{-c_\Xi p_0^2 \mathfrak{n} + \mathfrak{n}/2} e^{-\tilde\kappa(|X|_c - 1)} \le \mathsf{D}_x e^{-\tilde\kappa(|X|_c - 1)}$; this is where the strong distance is used.

For the second term, $e^{Q_1} - e^{Q_2} = (Q_1 - Q_2)\int_0^1 e^{Q_2 + t(Q_1 - Q_2)} dt$, so its modulus is at most $|Q_1 - Q_2| e^{\max|Q_i|}$. Moreover $Q_1 - Q_2 = Q_{D_1 - D_2}$. We put $\delta'_{12} := ((2r+1)^3 c_a(x)/68)\,\delta_\Delta(D_1 - D_2)$. The count at the start of this step, which does not need $\delta_\Delta \le 1$, shows that the modulus of $Q_{D_1 - D_2}$ is at most $\delta'_{12}\mathfrak{n}$. By (F3), $\delta'_{12} \le \delta_\Delta(D_1 - D_2)/2$. So the second term is at most $Ae^{-\mathfrak{n}}\delta'_{12}\mathfrak{n}e^{\mathfrak{n}/2} \le (A/e)\delta_\Delta(D_1 - D_2)$.

(B7) We have $\mathbb{B}_{tD}\tilde u = (\tilde E^a + tE^\Delta[D], \tilde E^h, \tilde w e^{tQ_D})$.

Then

$$|\tilde w Q_D| \exp\{(c_\Xi/2)x\varepsilon \sum_{p \in D(V) \cap X} \phi_p + \tilde\kappa(|X|_c - 1)\} \le A \exp\{-(c_\Xi/2)x\varepsilon \sum_{p \in D(V)\cap X} \phi_p\} \delta' \mathfrak{n} \le A\delta' \mathfrak{n} e^{-\mathfrak{n}/2} \le (2/e) A\delta'.$$

With the whole weight instead of half of it, the quantity is at most $A\delta'\mathfrak{n}$.

In Step 10 of the proof of Prop. 5.50 we use the rate $\tilde\kappa$ of the norm in place of κ in its $\tilde w$. Then $\tilde w$ coincides with the strong cap taken at the rate $\tilde\kappa$; the kernel is $D_y = -t^2$ at a single diagonal entry; and the number Q of that step, at its X and V, is $Q_D(X, V)$. Since $q_y \ge 0$ at every y for this kernel, $Q_D(X, V) \ge 0$; moreover $Q_D(X, V) \ge \mathfrak{n}' t^2 p_0^2/8 = \mathfrak{n}' \delta_\Delta c_a(x)/(8c_Q^{\mathrm{su}})$. So $|\tilde w Q_D|(X, V)$, multiplied by the whole strong weight $\exp\{c_\Xi x\varepsilon \sum_{p \in D(V)\cap X} \phi_p + \tilde\kappa(|X|_c - 1)\}$, equals $Q_D(X, V) \ge \mathfrak{n}'\delta_\Delta c_a(x)/(8c_Q^{\mathrm{su}})$.

Step 10 of the proof of Prop. 5.50 uses κ only in $\tilde w$: its rows, its field, its set and the cubes that are off do not depend on the rate, and the identities $m_{y_i} = 1$ and $q_{y_i} = t^2 p_0^2/8$ hold for every rate. It uses $x \ge x_9$ to obtain $r(x) \ge 2$, which it needs for the inequality $d \le Mr(x)$. In that step the member is $\tilde u = (0, 0, \tilde w)$, and its sizes are 0, 0 and 1. Hence $\tilde u$ satisfies the hypothesis $|\tilde E^a|_a, |\tilde E^h|_h, |\tilde w|_w^{\mathrm{st}} \le A$ of Lem. 7.10 if and only if $A \ge 1$, which is why (B7) assumes $A \ge 1$.

(B8) We have

$$F_p(\mathsf{W}^{\#}) = -i\log((\partial\mathsf{W}(p))^{\#}) = -i(-(\log\partial\mathsf{W}(p))^{\dagger}) = (F_p(\mathsf{W}))^{\dagger},$$

and also $\mathrm{Ad}_{g^{\#}} Y^{\dagger} = (\mathrm{Ad}_g Y)^{\dagger}$ and $Y^{\dagger}\cdot Y'^{\dagger} = \overline{Y\cdot Y'}$.

Hence $\overline{E^{\Delta}[D](\{\square\},\mathsf{W}^{\#})} = E^{\Delta}[\overline{D}](\{\square\},\mathsf{W})$ holds term by term on all of $\mathcal{U}_{\{\square\}}(x)$.

At G-valued V the brackets in q_y^D are real, so $\overline{q_y^D} = q_y^{\overline{D}}$ and $\overline{e^{Q_D}} = e^{Q_{\overline{D}}}$.

The numeral is $\max_n ne^{-n/2} = 2/e \le 0.73576$, the maximum being attained at $n = 2$. □

Proof of Thm. 7.11. Step 9: one step for two families, one of them over $D(z)$.

By Lem. 7.10 (B5), $u_2^{\flat}$ is a family on Ω within $B_2 + \delta_\Delta \le B_2 + 1 \le B$ caps at p, with the weak cap.

(C1) By Lem. 7.10 (B1), $\rho[\tilde{u}_2(z);\ell;D(z)] = \rho[u_2^{\flat}(z);\ell]$ as bounded Borel functions of V. The step is linear in ρ and free of the rows (the definition "The step" in item 2 of §5.4). Thm. 5.15 (a) at p, which holds by Thm. 6.7 (U), applied at $A = B \le 2.6$, gives the formula at every V'.

(C2) This is Thm. 7.9 (i) for u_1 and for $u_2^{\flat}$.

(C3) At each z, by Lem. 7.10 (B4),

$$\mathsf{D}_x^{\mathrm{in}}(u_1,u_2^{\flat}) \le \mathsf{D}_x^{\mathrm{in}}(u_1,\tilde{u}_2) + \mathsf{D}_x^{\mathrm{in}}(\tilde{u}_2,u_2^{\flat}) \le \mathsf{D}_x^{\mathrm{in}}(u_1,\tilde{u}_2) + \max\{1,B_2/e\}\delta_\Delta,$$

and $B_2 \le 1.6 < e$. Then we apply Thm. 7.9 (ii).

(C4) Here $B_1 = B_2 = 1$ and $B = 2$. By Step 7, $\rho[\tilde{u}_2;\ell;\tilde{\mathsf{E}} - \mathsf{E}] = \rho[\tilde{u}_2;\tilde{\ell}]$, and $u_2^{\flat}$ is real by Lem. 7.10 (B8). The images and the numbers V' are real by Thm. 5.15 (d), and the images are > 0; the rows ℓ'^{c} have range $\le \varrho(x')$; and $\mathsf{D}_x^{\mathrm{in}} \le \mathsf{D}_x$.

□

7.3. **Source derivatives for one run.** This section treats a single run of K drawn steps from Wilson's weight, in which n sources with pairwise separated supports enter through the bare weight alone. Starting at the level 0, we carry the multi-affine jets of the members at $z = 0$ from level to level; the families of members having these jets are each defined for z in a polydisc: at every level at which the distance between the supports, counted in cubes of the level, is large, the radius increases from that level to the next, and all later levels share one and the same polydisc. At the end we write S_n^T as a sum of local vacuum numbers and one last integral, in Prop. 7.12 (J7), and we bound the local vacuum numbers in (J8).

Item 0: what the run with sources takes from the one without sources, and what the proof of the proposition uses. Kernels, law, centre and rows are taken from the run without sources. In the sum of local vacuum numbers and the last integral that together give S_n^T, nothing is subtracted. The proof of Prop. 7.12 uses Thms. 6.7, 6.20, 6.33 and 7.1, Lems. 7.5, 7.7 and 7.8, and Thm. 7.9 (i), (ii), (iv).

Item 1: the data. The data are those of Thm. 8.2 (T1). The group is $G = \mathrm{SU}(2)$ and the dimension is 3; the letters are taken from item (0) of §6.1, the numbers and the floors from item (1) of §6.1, and the chain runs at $\kappa_1 = \kappa + 1$ with $\kappa \ge 6$. We take $\mathsf{b} \ge x_s$, and $N = 2L^{m_l}$ with $m_l \ge m'$; since $M = L^{m'}$, this gives $M \mid N$. The data include (Tor), which here reads $N \ge 2M\, s(L\mathsf{b})$.

The number of steps is $K \ge 1$. We set $\beta = x_0 = L^K\mathsf{b}$ and $x_k = L^{K-k}\mathsf{b}$, so that $x_{k+1} = x_k/L$ and $x_K = \mathsf{b}$, and the spacing is $a_K = L^{-K}$. The lattices are $\mathbb{T}_k = \mathbb{Z}^3/NL^{K-k}\mathbb{Z}^3$, with the fitted grids fixed by $o_K = 0$ and $o_k = Lo_{k+1} - \mathsf{h}_L$. The rows start from $\ell_0 = \emptyset$, and $\ell_{k+1} = (\ell'_k)^{\mathrm{c}}$ cut at $\varrho(x_{k+1})$, following Thm. 5.7 and Thm. 7.1 (b).

We place $\mathbb{T}_0$ in $T_l = \mathbb{R}^3/N\mathbb{Z}^3$ with the spacing a_K, the site z being the point $a_K z$. In this way $\mathbb{T}_0$ becomes the lattice $a_K\mathbb{Z}^3/N\mathbb{Z}^3$ in T_l, which is the $\Lambda_{K,\ell}$ of Def. 1.1 for $\ell = N$.

We fix $n \ge 2$, $d > 0$, $R_f > 0$ and real bounded functions $f_1,\dots,f_n$ on T_l with $\|f_i\|_\infty \le 1$, in such a way that any two of the sets $\{f_i \ne 0\}$ are at sup distance $\ge d$ from each other and each of these sets lies within sup distance R_f of some point.

For a plaquette p we write $s_p = a_p = 1 - \frac{1}{2}\operatorname{tr}\partial U(p)$, and $x(p)$ for its barycentre: if $p = (z;\mu\nu)$ is the plaquette based at the site z, then $x(p) = a_K(z + (e_\mu + e_\nu)/2)$. For a function f on T_l let $\mathcal{O}_K(f) = a_K^{-1}\sum_p f(x(p))s_p$. We define

$$Z_K(z) = \int \exp\{-\beta\sum_p s_p + \sum_i z_i\mathcal{O}_K(f_i)\}\,dU$$

and we let

$$S_n^T = S_{n;K,l,\beta}^T(f_1,\dots,f_n) = \partial_1\cdots\partial_n \log Z_K(z)$$

at $z = 0$.

So defined, S_n^T is the cumulant $\langle\mathcal{O}_K(f_1);\dots;\mathcal{O}_K(f_n)\rangle = \partial_1\cdots\partial_n \mathrm{Log}\,\Gamma(0)$, with $\Gamma(z) = Z_K(z)/Z_K(0)$, of Def. 1.1; here Log denotes the principal branch of the complex logarithm, taken on $|w-1| \le 1/2$. That definition introduces $x(p)$, a_p, the measure, $\mathcal{O}(f)$ and S_n^T on a torus, and they agree with the objects defined above; the model is the one of Def. 1.1. In this item, L and M are those of the table, which is item 3 of §2.1, with L subject to (O4) and M to (O7), and they also satisfy the entries (O4a) and (O7) of item (1) of §6.1.

Item 2: the objects (o1) *to* (o9). We introduce the following objects.

(o1) *The honest family.* For $z \in \mathbb{C}^n$ we put $\nu_p(z) := \mathsf{b}^{-1}\sum_i z_i f_i(x(p))$. The sets $\{f_i \ne 0\}$ are pairwise disjoint, so at any point at most one f_i differs from 0; therefore $|\nu(z)|_\infty \le \mathsf{b}^{-1}\max_i|z_i|\,\|f_i\|_\infty \le \mathsf{b}^{-1}\max_i|z_i|$, where the second inequality uses $\|f_i\|_\infty \le 1$.

The set $\mathrm{Pd}_0 = \{|z_i|\,\|f_i\|_\infty \le 1.99c_\nu\mathsf{b}\,\varepsilon(\beta)\}$ is the one of item 3, "Sources", of §7.1, and here $1.99c_\nu\mathsf{b}\,\varepsilon(\beta) = 1.99/\mathrm{amp}_W$. On Pd_0 we define $u_0(z) := u_0(\nu(z))$, $u_{k+1}(z) := F_{\ell_k}(u_k(z))$ and $V_{k+1}(z) := \mathsf{V}'_{\ell_k}[u_k(z)]$, as in Thm. 7.1 (h). All these values are holomorphic on the open set $\mathrm{Pd}_0^o = \{|z_i|\,\|f_i\|_\infty < 15.99/\mathrm{amp}_W\}$ of Thm. 7.1 (h), and this open set contains the closed set Pd_0. Whenever the honest family is regarded as a family in the sense of (o6), we take it on the interior of Pd_0. Finally, $\bar u_k := u_k(0)$.

(o2) *Jets.* For a non-empty subset S of $\{1,\dots,n\}$ we write $z^S := \prod_{i\in S} z_i$ and $\partial_S := \prod_{i\in S}\partial/\partial z_i$. Value by value we define $\delta_S(k) := \partial_S u_k(z)$ at $z = 0$; its parts are $\delta_S(k) = (\delta_S^a, \delta_S^h, \delta_S^w)(k)$, and for a single index we abbreviate $\delta_i := \delta_{\{i\}}$. With the sum over all such S, we put $u_k^{\mathrm{pol}}(z) := \bar u_k + \sum_S z^S\delta_S(k)$.

(o3) *Blocks.* We denote by $S_i(k)$ the set of those cubes of $\mathbb{T}_k$ for which some plaquette p of $\mathbb{T}_0$ with $f_i(x(p)) \ne 0$ has its base site in the bare box of the cube, and we call these cubes the k-blocks of $\operatorname{supp} f_i$. The bare box of a cube consists of the sites of $\mathbb{T}_0$ lying in the k-fold blocks of its M^3 sites, and it is a box of side ML^k. As the grids fit, $S_i(k+1) = \mathrm{cl}(S_i(k))$ is the set of cells containing a cube of $S_i(k)$. For $k = 0$, $S_i(0)$ is the set $S(f_i)$ of Thm. 7.1.

(o4) *Reach, separation, rates.* The reach is $\check r(x) = r(x) + 3$. For a depth $i \ge 0$ we put $\mathrm{sep}^{[i]} := \lfloor dL^i/M\rfloor - 2\check r(L^i\mathsf{b}) - 2$, and at the level k we write $\mathrm{sep}_k := \mathrm{sep}^{[K-k]}$. With $\mathrm{sep}_* := 8\log(4nL^{1/2})$, we let j^* be the least $j \ge 1$ such that $\mathrm{sep}^{[i]} \ge \mathrm{sep}_*$ holds for all $i \ge j-1$, and we let $k_d := \max\{K - j^*, 0\}$. For $k < k_d$ we define $\check\kappa_k := 2\log(4nL^{1/2})/\mathrm{sep}_{k+1}$, and the rate of the level k is $\kappa_k := \kappa + \sum_{k\le j<k_d}\check\kappa_j$.

(o5) *Amplitudes, radii.* The amplitude is $\mathrm{amp}(x) = c_Q^{\mathrm{su}}K_5 x^{1/2}/(\mathsf{b}E_a p_0(x))$, where $c_Q^{\mathrm{su}} = 306M^3\mathsf{g}_\flat^2$ as in (O10) of item 1 of §5.4, and we also use $\mathrm{amp}_W = 1/(c_\nu\mathsf{b}\,\varepsilon(\beta)) = 6M^3x_0^{1/2}/(\mathsf{b}E_a p_0(x_0))$. Further, $u_3(x) = C_3x^{1/4}$. The function $\check\omega_3(x) = C_3E_a p_0(x)x^{-1/4}/c_Q^{\mathrm{su}}$ is defined in (O12b) of item (1) of §6.1; it is independent of b, n and d, which is why the condition (W3) on x_s, given in (o9) below, may involve it.

These definitions yield the identity

$$\check\omega_3(x) = u_3(x)K_5/(\mathsf{b}\,\mathrm{amp}(x))$$

for, once u_3 and amp are inserted, the right side reads $C_3 x^{1/4} K_5\, \mathsf{b} E_a p_0(x)/(\mathsf{b}\, c_Q^{\mathrm{su}} K_5 x^{1/2})$, which is $\breve\omega_3(x)$ after b and K_5 cancel.

The radii are $\mathfrak{r}(x) = 3.5/(n\,\mathrm{amp}(Lx))$, $\mathfrak{r}_k := \mathfrak{r}(x_k)$ and $\mathfrak{r}_* := \mathfrak{r}_{k_d}$, and for $r > 0$ we write $\mathrm{Pd}(r) = \{z \in \mathbb{C}^n : |z_i| < r \text{ for all } i\}$.

(o6) *Norms.* Each norm below is taken at a coupling x and at a rate $\tilde\kappa \in [\kappa, \kappa_1]$; it is defined on the triples $\delta = (\delta^a, \delta^h, \delta^w)$ obeying those clauses of (ca), (ch), (cΞ) which carry no size and satisfying $\|\delta\|_x < \infty$, where $\|\cdot\|_x$ is the norm of item 3 of §5.4; these triples form the space $\mathfrak{M}(x)$ of that item. We set

$$|\delta^a|_a^{(\tilde\kappa)} := \sup_X \sup_{\mathcal{U}_X(x)} |\delta^a(X,\cdot)|\, e^{\tilde\kappa(\#X-1)}/c_a(x)$$

and

$$|\delta^h|_h^{(\tilde\kappa)} := \sup_X \sup |\delta^h(X,\cdot)|\, e^{\tilde\kappa(\#X-1)}/c_h(x)$$

where the inner supremum runs over the tame G-valued fields, and

$$|\delta^w|_w^{\mathrm{st},(\tilde\kappa)} := \sup_{X,V} |\delta^w(X,V)| \exp\{c_\Xi x\varepsilon \sum_{p\in D(V),\, p\in X} \phi_p + \tilde\kappa(|X|_c - 1)\}.$$

The strong norm $\|\delta\|^{\mathrm{st},(\tilde\kappa)}$ is the largest of these three numbers. The weak norm $\|\delta\|^{(\tilde\kappa)}$ is built in the same way, except that the term with c_Ξ is dropped from the exponential; it never exceeds the strong one. A norm written without a rate is taken at κ_1.

The space $T^a(x)$ consists of the a-members that obey the clauses of (ca) without a size and are each bounded, as in item 1 of §6.2. In Sections 6.4 and 7.1 the symbol $|\cdot|_{a,x'}^{\mathrm{cs},(\tilde\kappa)}$ stands for the cell sum divided by $c_a(x')$. As Step 0 (c) of the proof of Prop. 7.12 shows, it dominates the supremum $|\cdot|_a^{(\tilde\kappa)}$ used in this section. A family on an open set of $\mathbb{C}^n$ is a map $z \mapsto u(z)$ into some $M_A(x;p)$ with $A \le 16$ all of whose values are holomorphic.

(o7) We take $v_k(Y;u)$ and V_0^+ from Thm. 6.7 (g1), applied with the coupling and the rows $(x_k;\ell_k)$; there Y runs over the connected sets of cells, that is, of cubes of $\mathbb{T}_{k+1}$. We put $I[u] := \int \rho[u;\ell_K](V)\, dV$, the integral being over the fields of $\mathbb{T}_K$.

(o8) The source forms $\mathbb{D}_k[f]$, the map T_ℓ and cut are those of Thm. 6.20, in item 1 of §6.3; $s_x[D]$ is that of entry 5 of item (0) of §6.1, and $n[u;D]$ that of Thm. 6.33. By $DA_\ell(u)[\delta]$ we mean the series $DA(u)[\delta]$ of item 1 of §6.2 taken at the rows ℓ, as in item 3 of §6.4. Thm. 6.7 (h) shows that it is linear in $\delta \in T^a(x)$ and that, wherever $A_a(u) < 16$, it equals $\frac{d}{dz}|_0\, E'^{\mathrm{old}}[u + z(\delta,0,0)]$; here $A_a(u)$ is the a-part of the norm of u, as in item 1 of §6.2. We recall $\vartheta_a = 0.2431 L^{-1/4}$ and $C_g = 263 L^3$. The functions η_h, η_w are those of item (1) of §6.1.

(o9) *One symbol, two senses.* The symbol κ_1 always means $\kappa+1$, as in entry 1 of item (0) of §6.1, while κ_k, κ_{k+1}, κ_{k_d} and κ_0 are the rates of (o4) at the levels; $u_3(x)$ is always the number $C_3 x^{1/4}$, whereas $u_k(z)$, $\bar u_k$ and u_k^{pol}, whose index is a letter, 0 or K, are the families of the levels. The symbols x_3, x_4, x_9^U, x_s are floors of (O13) of item (1) of §6.1, the index in x_s being a mark, which is neither a level nor a depth; x_k with a letter index, x_0 and x_K are the couplings of the levels. The letter p is a plaquette, except where it denotes an element of $\mathbb{P}$, namely a pair $(\tilde\kappa, \tilde c)$ consisting of a rate and a constant; this is the case for the last argument of $M_A(x;p)$, for the p of "at p" and wherever p is written as such a pair.

The tags (W2) and (W3) denote the two final conditions that (O13) of item (1) of §6.1 imposes on x_s; for all $x \ge L x_s$ they read

(W2) $\eta_h(x) + e^{1/2}\eta_w(x) \le 1/L$,

(W3) $\breve\omega_3(x)$, $\breve\omega_3(x/L) \le 10^{-3}$.

We apply (W2) at the couplings x_k with $k < K$ and (W3) at x_k with $k \le K$; at $k = K$, (W3) enters through its second member $\breve\omega_3(x/L)$, taken at $x = L\mathsf{b}$.

Item 3: the numbers of (o4) *and* (o5). These numbers have the following properties.

(G1) For every $i \geq 0$ we have $\mathrm{sep}^{[i+1]} \geq L\,\mathrm{sep}^{[i]}$. Consequently j^* is the least $j \geq 1$ with $\mathrm{sep}^{[j-1]} \geq \mathrm{sep}_*$; it is finite, it depends on b, n, d, L, M, c_R alone, and it does not decrease when b increases, so that it is locally bounded in b.

(G2) We have $k_d \leq K-1$. For $k < k_d$, $\mathrm{sep}_{k+1} \geq L\,\mathrm{sep}_*$ and $\check{\kappa}_k \leq 1/(4L) \leq 1/4$. The rate κ_k is non-increasing in k, $\kappa_{k_d} = \kappa$, and $\kappa \leq \kappa_k \leq \kappa + 1/(4(L-1))$; hence κ_k lies in $[\kappa, \kappa + 0.26]$ and therefore in $[\kappa, \kappa_1]$.

(G3) For $x \geq x_s$ we have

$$5.01\,\mathrm{amp}(x) \leq 2^{-1/4}L^{1/2}\mathrm{amp}(x) \leq \mathrm{amp}(Lx) \leq L^{1/2}\mathrm{amp}(x)$$

and so $\mathfrak{r}_k \leq \mathfrak{r}_{k+1} \leq L^{1/2}\mathfrak{r}_k$; moreover $\mathfrak{r}_* \geq \mathfrak{r}(L^{j^*}\mathsf{b})$ for every $K \geq 1$, with equality if $K \geq j^*$.

(G4) The sets $S_i(k)^{+\check{r}(x_k)}$, $i = 1, \dots, n$, are pairwise at a distance $\geq \mathrm{sep}_k + 2$ in cubes.

Item 4: the step for jets of one run.

Proposition 7.12 (The step for jets of one run)**.** In the setting of item 1, (J0) to (J5) hold for $0 \leq k \leq k_d$, and (J6) to (J8) hold as well.

(J0) Thm. 6.20, applied with $f = f_i$, shows that $\mathbb{D}_k[f_i]$ is real, symmetric and linear in f_i, that its range is $\leq \varrho(x_k)$, that $\|\mathbb{D}_k[f_i]\|_{(\lambda_\flat)} \leq K_5\|f_i\|_\infty$ site by site, and that $\mathbb{D}_{k,y}[f_i]$ depends only on the values of f_i at the barycentres of the plaquettes based in $B^k(y)$. In addition, $s_{x_k}[\mathbb{D}_k[f_i]/\mathsf{b}]$ lies in $T^a(x_k)$ and lives on single cubes of $S_i(k)$, and $|s_{x_k}[\mathbb{D}_k[f_i]/\mathsf{b}]|_a \leq \mathrm{amp}(x_k)$ holds in the norm $|\cdot|_a^{(\tilde{\kappa})}$ for every rate $\tilde{\kappa}$ of (o6).

(J1) Let $k < K$. For every S, the coefficient of z^S in $F_{\ell_k}(u_k^{\mathrm{pol}}(z))$ is $\delta_S(k+1)$, and the coefficient of z^S in $\mathsf{V}'_{\ell_k}[u_k^{\mathrm{pol}}(z)]$ is $\partial_S V_{k+1}(0)$.

(J2) We have $\delta_S(k)(X) = 0$ unless $X^{+\check{r}(x_k)}$ meets $S_i(k)$ for every $i \in S$. Consequently, if $|S| \geq 2$ and $\mathrm{sep}_k \geq 1$, then $\delta_S(k)(X) \neq 0$ forces $\#X \geq |S|\,\mathrm{sep}_k/2$ for the a- and h-parts, X being connected, and $|X|_c \geq |S|\,\mathrm{sep}_k/2$ for the w-part.

(J3) We have $\delta_i^a(k) = s_{x_k}[\mathbb{D}_k[f_i]/\mathsf{b}] + \mathrm{rem}_{k,i}$ with $\mathrm{rem}_{k,i} \in T^a(x_k)$ and

$$|\mathrm{rem}_{k,i}|_a^{(\kappa_1)} \leq 2u_3(x_k)K_5/\mathsf{b} = 2\breve{\omega}_3(x_k)\mathrm{amp}(x_k) \leq 0.002\,\mathrm{amp}(x_k).$$

(J4) We have $|\delta_i^h(k)|_h^{(\kappa_1)} \leq 0.12\,\mathrm{amp}(Lx_k)$ and $|\delta_i^w(k)|_w^{\mathrm{st},(\kappa_1)} \leq 0.12\,\mathrm{amp}(Lx_k)$. The leading term of a single jet $\delta_i(k)$ is $(s_{x_k}[\mathbb{D}_k[f_i]/\mathsf{b}], 0, 0)$, so the second bound covers the whole w-jet.

(J5) We have $n\,\mathfrak{r}_k \max_i \|\delta_i(k)\|^{\mathrm{st},(\kappa_1)} \leq 0.7$ and

$$\sum_{|S|\geq 2} \mathfrak{r}_k^{|S|}\|\delta_S(k)\|^{\mathrm{st},(\kappa_k)} \leq 0.21$$

so that u_k^{pol} is a family on $\mathrm{Pd}(\mathfrak{r}_k)$ in $M_{1.91}(x_k; \kappa_k, c_h)$ and $\sup_{\mathrm{Pd}(\mathfrak{r}_k)} \|u_k^{\mathrm{pol}}(z) - \bar{u}_k\|^{\mathrm{st},(\kappa_k)} \leq 0.91$.

(J6) From here on we set $u_k(z) := u_k^{\mathrm{pol}}(z)$ on $\mathrm{Pd}(\mathfrak{r}_k)$ for $k \leq k_d$. By the representatives we mean these $u_k(z)$ and the $u_k(z)$, $k_d < k \leq K$, on $\mathrm{Pd}(\mathfrak{r}_*)$ defined below. Where the family of (o1) itself is meant, and not a representative, we call it the honest family. For $k_d \leq k < K$ we put $u_{k+1}(z) := F_{\ell_k}(u_k(z))$ on $\mathrm{Pd}(\mathfrak{r}_*)$, beginning with $u_{k_d}(z) = u_{k_d}^{\mathrm{pol}}(z)$. When $k_d < k \leq K$, the $u_k(z)$ so obtained are families in $M_1(x_k; \kappa, c_h)$, and they carry the strong cap. For $k_d \leq k \leq K$ we have

$$\sup_{\mathrm{Pd}(\mathfrak{r}_*)} \|u_k(z) - \bar{u}_k\|^{\mathrm{st},(\kappa)} \leq 0.91 \cdot 0.4^{k-k_d}$$

and $u_k(0) = \bar{u}_k$, and for every S the derivative $\partial_S u_k(z)$ at 0 is $\delta_S(k)$, the true multi-affine jet. For $0 \leq k \leq K$, $\bar{u}_k$ is real, it lies in $M_1(x_k; \kappa_1, c_h)$ with the strong cap, and $\rho[\bar{u}_k; \ell_k] > 0$, by Thm. 7.1 (h); thus $\bar{u}_k$ is a system of members of a point of $C_\flat^{\mathrm{su}}(x_k; \ell_k)$, and in particular,

for every $K \geq 1$, $\bar{u}_K$ is a system of members of a point of $C^{\mathrm{su}}_\flat(\mathsf{b}; \ell_K)$. As $k_d \leq K-1$, the family $u_K(z)$ is an output of $F_{\ell_{K-1}}$.

(J7) With the representatives,

$$S_n^T = \sum_{0 \leq k < K} \sum_Y \partial_1 \cdots \partial_n v_k(Y; u_k(z))(0) + \partial_1 \cdots \partial_n \log I[u_K(z)](0)$$

and $I[\bar{u}_K] > 0$. A set Y contributes 0 unless $Y^{+\check{r}(x_{k+1})}$ meets $S_i(k+1)$ for every $i = 1, \dots, n$.

(J8) For $0 \leq k < K$,

$$|\sum_Y \partial_1 \cdots \partial_n v_k(Y; u_k(z))(0)| \leq \sum_Y |\partial_1 \cdots \partial_n v_k(Y; u_k(z))(0)| \leq \mathfrak{r}^{-n}_{\min(k,k_d)} N_1(k)\, 2C_g c_a(x_k) e^{-\kappa(m_k - 1)}$$

where $N_1(k) := (2R_f L^{K-k-1}/M + 2\check{r}(x_{k+1}) + 3)^3$, and where $m_k := \lceil n \operatorname{sep}_{k+1}/2 \rceil$ for $k < k_d$ and $m_k := 1$ for $k \geq k_d$.

■

Remark 7.13 (Item 5: constants and the range of the clauses). Every numerical constant that appears above is independent of K, k, N, b, n, d, R_f and f. As for the data, amp and $\mathfrak{r}$ are functions of (b, n) and of the depth $K-k$; sep, j^*, $\check{\kappa}$, κ_k and m_k are functions of (b, n, d) and of the depth; N_1 is a function of (b, R_f) and of the depth; and none of them depends on N.

The proof given below establishes (J1) to (J4) of Prop. 7.12 for every $k \leq K-1$. Its arguments also yield (J0) for every $k \leq K$, and (J2), (J3), (J4) at $k = K$ as well. Indeed, Step 2 shows, for $0 \leq k \leq K$, that every $\delta_S(k)$ belongs to the space of members and has a finite norm $\|\delta_S(k)\|^{\mathrm{st},(\kappa_1)}$; Steps 3, 5 and 6 pass from a level $k < K$ to the level $k+1 \leq K$; Step 0 (a), together with the part of its proof that concerns the lattice, and Thm. 6.20 apply at $k = K$; and at $k = K$ the condition (W3) enters through the second of its two inequalities. At $k = K$ the clause (J1) has no meaning, because no step starts at level K.

For complex f_i with $\|f_i\|_\infty \leq 1$ the arguments of Steps 2 to 6 go through without any change, with one exception: the reality asserted in (J0) refers to real f_i ((U2.2) of Thm. 6.20). The reason is that Thm. 6.20 and item 3 of §7.1, on sources, treat complex bounded f; Thm. 6.33 treats complex D; Thm. 6.7 (h), (h3) and Thm. 7.9 (iv) admit complex directions; and the only reality that enters is that of $\bar{u}_k$.

Proof of Prop. 7.12 and of (G1) *to* (G4) *of item 3. Step 0: tools.* This step provides the following four tools.

(a) *The size of the leading term.* Let $x \geq x_s$, and let $D = (D_y)$ be complex, symmetric in the sense that $D_y(p,p') = D_y(p',p)$, and of range $\leq \varrho(x)$. Then $s_x[D] \in T^a(x)$; it vanishes on every X with $\#X \geq 2$; and $\sup_{\mathcal{U}_{\{\square\}}(x)} |s_x[D](\{\square\}, \cdot)| \leq \sigma_D = c^{\mathrm{su}}_Q p_0^2 \|D\|_{(\lambda_\flat)}$. Measured in caps, and at every rate, $|s_x[D]|_a \leq c^{\mathrm{su}}_Q x^{1/2} \|D\|_{(\lambda_\flat)} / (E_a p_0(x))$.

(b) *Three numbers.* (G3) holds; $\mathrm{amp}_W \leq \mathrm{amp}(x_0)/51$; and $u_3(x) K_5 \geq 1$ for $x \geq L x_s$.

(c) *Norms.* We have $M_A(x; \kappa_1, c_h) \subset M_A(x; \tilde{\kappa}, c_h)$; any norm taken at the rate $\tilde{\kappa}$ is at most the same norm taken at the rate κ_1; and a bound for the cell sums $\sum_{Y \ni \mathrm{a\ cell}} e^{\tilde{\kappa}(\#Y - 1)} \sup |\cdot(Y)|$ is also a bound for the weighted supremum.

(d) *Representatives.* Let Φ be a value of $F_\ell(u)$, the number $\mathsf{V}'_\ell[u]$, or a number $v(Y; u)$, each taken for some data (SD) at an $x \geq x_9^U$ and some $p \in \mathbb{P}$; or else let Φ be the number $I[u]$. Suppose that $u(z)$ and $\tilde{u}(z)$ are families near $z = 0$, of norm < 16, such that $\partial_{S'} u(0) = \partial_{S'} \tilde{u}(0)$ holds for each subset S' of S, including the empty one. Then $\partial_S \Phi(u(z))(0) = \partial_S \Phi(\tilde{u}(z))(0)$.

Proof of (a). Condition (X16) gives $\varepsilon(x) \leq 1/40$ for $x \geq x_s$. Since $x \geq e^2$ and $c_R \geq 768$,

$$\varrho(x) + 2 \leq c_R \log x + 3 \leq 20 c_R \log x \leq M r(x).$$

The side condition of Lem. 6.32 follows from (Tor) for $k < K$, when the coarse side NL^{K-k-1} is at least $16Mr(x_k)$. At $k = K$, the initial condition $N \geq 2Ms(L\mathsf{b})$ gives $N \geq 16Mr(\mathsf{b})$. Thus the lemma applies on every $\mathbb{T}_k$, including the last torus. The bound in caps follows by division by $c_a(x) = E_a p_0(x)^3 x^{-1/2}$, since a single cube has weight 1 at every rate.

Proof of (b). We begin with (G3). We have $\mathrm{amp}(Lx)/\mathrm{amp}(x) = L^{1/2} p_0(x)/p_0(Lx)$, and since p_0 is increasing, this ratio does not exceed $L^{1/2}$. On the other hand, (Y.0)(0b) of item 5 of §5.3, applied at $Lx \geq Lx_s \geq x_9^U$, gives $(p_0(x)/p_0(Lx))^4 \geq 1/2$; it is one of the conditions defining the floor x_9^U ((O13) (i) of item 0 of §6.2). Hence the ratio is at least $2^{-1/4} L^{1/2} \geq 0.8408 L^{1/2} \geq 5.01$, because $L > 1.2 \cdot 10^{13}$ ((O4) of item 1 of §5.3). Moreover $\mathfrak{r}_{k+1}/\mathfrak{r}_k = \mathrm{amp}(Lx_k)/\mathrm{amp}(x_k)$. If $K \geq j^*$, then $x_{k_d} = L^{j^*}\mathsf{b}$. If $K < j^*$, then $k_d = 0$ and $\mathfrak{r}_* = \mathfrak{r}(L^K \mathsf{b}) \geq \mathfrak{r}(L^{j^*}\mathsf{b})$, as $i \mapsto \mathfrak{r}(L^i \mathsf{b})$ is decreasing.

Next, $\mathrm{amp}_W/\mathrm{amp}(x_0) = 6M^3/(306 M^3 \mathsf{g}_\flat^2 K_5) = 1/(51 \mathsf{g}_\flat^2 K_5) \leq 1/51$, because $\mathsf{g}_\flat \geq 1$ and $K_5 = 1.35 C_{\mathrm{ch}}$, where $C_{\mathrm{ch}} = 4e^{2\lambda_*}/\gamma_* > 4$ as $\gamma_* < 1$ ((O0) of item 1 of §5.4).

Finally, we turn to (Y.3) of item 5 of §5.3. Each term inside its bracket is nonnegative, and each factor in front of the bracket is at least 1. In the definition of x_9^U, (Y.3) is imposed with κ_1 in place of κ and with $e^{-(c_h/2)p_0^2}$ in place of $c_h(x)$ ((O13) (ii) of item 0 of §6.2); our argument involves (Y.3) only through its term in c_a. In both forms this term is $16 c_a$ times 5 times factors ≥ 1, so that the inequality $x^{1/4} \cdot 50 \cdot 5 \cdot 16 c_a(x) \leq x^{-1/8}$ follows from either form. Therefore $E_a p_0^3 \leq x^{1/8}/4000$, and $x^{1/4}/E_a \geq 4000 p_0^3 x^{1/8} \geq 1$. The last inequality uses $p_0 \geq 1$; this is (X14) in (O13) of item 1 of §5.4, a floor of x_s. Since (Y.3) in turn is a floor of $x_9^U \leq Lx_s$, we obtain the range $x \geq Lx_s$. It follows that $u_3(x) K_5 = 1686 e^{\kappa+4} L^{11/4} c_Q^{\mathrm{su}} K_5 x^{1/4}/E_a \geq 1$.

Proof of (c). As $\tilde{\kappa} \leq \kappa_1$, we have $e^{\tilde{\kappa}(\#X-1)} \leq e^{\kappa_1(\#X-1)}$, which gives the inclusion and the comparison of norms. For the last assertion, observe that every Y contains a cell, and that in a sum of nonnegative terms each term is bounded by the sum.

Proof of (d). Let $B < 16$ be a common bound of $\|u(z)\|$ and $\|\tilde{u}(z)\|$ for z near 0. Then (d) follows from Lem. 7.8, applied at $u_0 = u(0) = \tilde{u}(0)$ with $t = 16 - B$: the two coefficients are given by one and the same sum over the partitions of S. The lemma covers all four kinds of Φ: the first three at $x \geq x_9^U$, and $\Phi = I$ for any u_0 and any t.

Step 1: geometry. We establish the following.

- (r) $r(Lx) \leq r(x) + \log L/960 + 1$.
- (G1) The clause (G1) of item 3 holds.
- (G2) The clause (G2) of that item holds.
- (G4) The clause (G4) of that item holds.
- (R) $\lceil \check{r}(x_k)/L \rceil + 1 \leq \check{r}(x_{k+1})$.
- (C) Let X be a connected set of cubes, and suppose that X meets $j \geq 2$ sets which lie pairwise at distance $\geq \sigma \geq 1$. Then $\#X \geq j\lceil \sigma/2 \rceil$. For $X \in \mathcal{P}$ meeting $j \geq 2$ such sets, the same bound holds for $|X|_c$.
- (A) The set $S_1(k+1)^{+\check{r}(x_{k+1})}$ has at most $N_1(k)$ cells.

All distances are periodic sup distances.

Proof of (r). We have $r(x) = \lceil 20 c_R \log x/M \rceil$ and $\lceil a + a' \rceil \leq \lceil a \rceil + a' + 1$, which we apply with $a = 20 c_R \log x/M$ and $a' = 20 c_R \log L/M$; moreover $20 c_R/M \leq 1/(96(\kappa_1 + 4)) \leq 1/960$, by (M2) $M \geq 1920(\kappa+4) c_R$ ((O7) of item 1 of §5.3), taken at κ_1 as in (O7) of item (1) of §6.1.

Proof of (G1). We use $\lfloor La \rfloor \geq L \lfloor a \rfloor$ together with (r); they give

$$\mathrm{sep}^{[i+1]} \geq L\lfloor dL^i/M \rfloor - 2(\check{r}(L^i\mathsf{b}) + \log L/960 + 1) - 2 = L\,\mathrm{sep}^{[i]} + (2L-2)\check{r}(L^i\mathsf{b}) + 2L - 4 - \log L/480 \geq L\,\mathrm{sep}^{[i]},$$

where the last step uses $\check{r} \geq 3$. Hence, at every depth greater than the first depth at which $\mathrm{sep}^{[i]} \geq \mathrm{sep}_* > 0$ holds, this inequality is true as well. Such a depth exists, since $\mathrm{sep}^{[i]} \geq dL^i/M - 40 c_R \log(L^i\mathsf{b})/M - 11$, and the right side tends to ∞ with i. As r does not decrease when x grows, $\mathrm{sep}^{[i]}$ does not increase with b, and therefore j^* does not decrease as b grows.

Proof of (G2). As $j^* \geq 1$, we get $k_d \leq K-1$. For $k < k_d$, the depth of $k+1$ is $K-k-1 \geq j^*$; hence $\mathrm{sep}_{k+1} \geq L^{K-k-j^*}\mathrm{sep}^{[j^*-1]} \geq L\,\mathrm{sep}_*$, and $\check\kappa_k = (\mathrm{sep}_*/4)/\mathrm{sep}_{k+1} \leq 1/(4L)$. Summing over the levels, $\sum_{k\leq j<k_d}\check\kappa_j \leq (1/4)\sum_{t\geq 1}L^{-t} = 1/(4(L-1))$.

Proof of (G4). Let $i \neq j$, and let p, p' be plaquettes with $f_i(x(p)) \neq 0$ and $f_j(x(p')) \neq 0$. Write z, z' for their base sites in bare units; these lie in the bare boxes of two cubes Q, Q' of $\mathbb{T}_k$, at distance D from each other. Since $x(p) = a_K(z+(e_\mu+e_\nu)/2)$, we have $|x(p)-x(p')| \leq a_K(|z-z'|+1/2)$, and therefore $|z-z'| \geq dL^K - 1/2$. On the other hand, two sites lying in boxes of side ML^k at distance D satisfy $|z-z'| \leq (D+1)ML^k - 1$. It follows that $D+1 > dL^{K-k}/M$, that is, $D \geq \lfloor dL^{K-k}/M\rfloor$. Fattening both sets $S_i(k)$ and $S_j(k)$ by $\check r(x_k)$ lowers the distance by at most $2\check r(x_k)$, so the fattened sets lie at distance $\geq \mathrm{sep}_k + 2$.

Proof of (R). Using (r) in the second inequality, we get

$$\lceil (r(x_k)+3)/L\rceil + 1 \leq (r(x_k)+3)/L + 2 \leq (r(x_{k+1}) + \log L/960 + 4)/L + 2 \leq r(x_{k+1}) + 3,$$

because $(\log L/960+4)/L \leq 1$.

Proof of (C). For each i we pick a cube c_i lying both in X and in the i-th set, and we put $h := \lceil \sigma/2\rceil - 1$. Then $2h < \sigma$, so the balls $\{c : \mathrm{dist}(c,c_i) \leq h\}$ are pairwise disjoint. Since X is connected, it links c_i to some cube whose distance from c_i is $\geq \sigma > h$; along a chain of neighbouring cubes, the distance from c_i changes by at most 1 from one link to the next. Hence, for each of the values $0, 1, \dots, h$, the set X contains a cube at that distance from c_i. Counting these cubes in the disjoint balls, we find $\#X \geq j(h+1) = j\lceil\sigma/2\rceil \geq j\sigma/2$.

Now let $X \in \mathcal{P}$. Here $\mathrm{hull}(X)$ is a member of $\mathcal{D}$ that contains X and has the least number of cubes among such members, and $|X|_c = \#\mathrm{hull}(X)$; for these notions see item 0 of §5.1, on names, and S4 (4) of the proof of Thm. 5.3, on the hull, as well as item 1 of §2.1, and, for $|X|_c$, item 2 of §5.5. Thus $\mathrm{hull}(X)$ is connected and meets every set that is met by X, and the first half of (C), applied to $\mathrm{hull}(X)$, gives the second half.

Proof of (A). The barycentres at which $f_1 \neq 0$ lie within R_f of some point, and so the base sites lie within $R_f + a_K/2$ of it; in bare units, each coordinate of these sites ranges over an interval of length $\lambda = 2R_fL^K + 1$. The integer points of an interval of length λ meet no more than $\lambda/s + 2$ intervals of a grid of intervals of length s. For $s = ML^{k+1}$ this number is at most $2R_fL^{K-k-1}/M + 1/(ML^{k+1}) + 2 \leq 2R_fL^{K-k-1}/M + 3$. The fattening by $\check r(x_{k+1})$ adds $2\check r(x_{k+1})$ to it.

Step 2: the run, (J0) *and* (J1). We show the following. For $0 \leq k < K$, (SD) holds at $(x_k;\ell_k)$ with $\ell_k'^{\mathrm c} = \ell_{k+1}$, and $x_k \geq Lx_s \geq \max\{x_9^U, x_3, x_4\}$. For $0 \leq k \leq K$, $\bar u_k$ is real and lies in $M_1(x_k;\kappa_1,c_h)$ with the strong cap, and $\rho[\bar u_k;\ell_k] > 0$. For $0 \leq k \leq K$, every $\delta_S(k)$ lies in the space of members $\mathfrak{M}(x_k)$ of (o6) of item 2, and $\|\delta_S(k)\|^{\mathrm{st},(\kappa_1)}$ is finite. Finally, (J0) and (J1) hold.

By Thm. 7.1 (a), for $|\nu|_\infty \leq 15.99c_\nu\varepsilon(\beta)$ we have $\exp\{-\beta\sum_p(1-\nu_p)s_p\} = \rho[u_0(\nu);\emptyset]$ at every field. By Thm. 7.1 (b), (SD) holds at $(x_k;\ell_k)$ for every $0 \leq k < K$, with $\ell_k'^{\mathrm c} = \ell_{k+1}$; for $k = 0$ this is $(\beta;\emptyset)$, with $\ell'^{\mathrm c} = \ell_1$. The empty rows are admitted by (SD) of item 2 of §5.3, and also by the setting at the opening of item 2 of §6.2; and Thm. 5.15 itself holds at $\ell = \emptyset$, for every complex u within 16 caps that has the weak cap. At $k = 0$, Steps 2 to 10 apply Thms. 6.7, 6.33 and 7.9 at $(\beta;\emptyset)$.

By Thm. 7.1 (h), on Pd_0 the honest family $u_k(z)$ takes values in $M_1(x_k;\kappa_1,c_h)$ with the strong cap when $k \geq 1$, and in M_2 with the strong cap when $k = 0$ (by (a1), under which the a-part E_ν^a is at most $0.01 + 1.99$ caps $c_a(\beta)$ on Pd_0, and by (a2), both of Thm. 7.1); and it is holomorphic. At $z = 0$ it is real ((a3) of Thm. 7.1; Thm. 5.15 (d)) and lies within one cap (by (a1), which bounds the same a-part by 0.01 caps at $z = 0$); and $\rho[\bar u_k;\ell_k] > 0$ for $0 \leq k \leq K$, by the part of Thm. 7.1 (h) that concerns $z = 0$. Further, $x_k \geq L\mathsf{b} \geq Lx_s$ for $k < K$, and $Lx_s \geq \max\{x_9^U, x_3, x_4\}$ by (O13) of item (1) of §6.1.

We apply Lem. 7.7 on the open polydisc $\mathrm{Pd}(1.99/\mathrm{amp}_W)$, with the strong weighted sup norm at κ_1. Because $\|f_i\|_\infty \leq 1$, this polydisc is contained in the interior of Pd_0, inside the open set

Pd_0^o of Thm. 7.1 (h), on which every value is holomorphic. The lemma shows that each coefficient $\delta_S(k)$ lies in the space of members and has a finite norm. Consequently $u_k^{\mathrm{pol}}(z)$ is a family of norm < 16 near $z = 0$; by its form, $\partial_{S'} u_k^{\mathrm{pol}}(0) = \delta_{S'}(k) = \partial_{S'} u_k(0)$ for every S', and $u_k^{\mathrm{pol}}(0) = \bar{u}_k$. Now Step 0 (d), applied with Φ a value of F_{ℓ_k}, yields the first half of (J1); applied with $\Phi = \mathsf{V}'_{\ell_k}$, it yields the second half.

For (J0) we apply Thm. 6.20 with $f = f_i$, in the setting (L5*), (X5) of the table (item (1) of §6.1). Step 0 (a), applied with $x = x_k$ and $D = \mathbb{D}_k[f_i]/\mathsf{b}$, gives the bound $|\cdot|_a \le c_Q^{\mathrm{su}} x_k^{1/2} K_5/(\mathsf{b} E_a p_0(x_k)) = \mathrm{amp}(x_k)$. If a cube does not belong to $S_i(k)$, then for no y in that cube does $B^k(y)$ contain the base site of a plaquette p with $f_i(x(p)) \neq 0$; since $\mathbb{D}_{k,y}[f_i]$ is linear and depends only on the values of f_i there, it vanishes.

Step 3: the reach and (J2). A family $u(z)$ at level k is said to *have* (P_k) if, for every i, the value $u(z)(X)$ is independent of z_i for all X such that $X^{+\check{r}(x_k)}$ misses $S_i(k)$. We claim: if $u(z)$ has (P_k), where $k < K$, then $F_{\ell_k}(u(z))$ has (P_{k+1}), and moreover $v_k(Y; u(z))$ is independent of z_i for all Y such that $Y^{+\check{r}(x_{k+1})}$ misses $S_i(k+1)$. From this claim we shall obtain (J2).

Proof of the claim. We fix z_j for $j \neq i$. Thm. 6.7 (f5), applied with $S = S_i(k)$ and $\hat{\rho} = \check{r}(x_k)$, shows that $F_{\ell_k}(u(z))(Y)$ and $v_k(Y; u(z))$ are constant in z_i for every Y such that $Y^{+\hat{\rho}'}$ does not meet $\mathrm{cl}(S_i(k)) = S_i(k+1)$, where $\hat{\rho}' = \lceil \check{r}(x_k)/L \rceil + 1 \le \check{r}(x_{k+1})$ by Step 1 (R). If $Y^{+\check{r}(x_{k+1})}$ does not meet $S_i(k+1)$, then neither does $Y^{+\hat{\rho}'}$.

Proof of (J2) *at* $k = 0$. The a-part $E_\nu^a(\{\square\}) = \sum_{p \in \square} [q_p + \beta \nu_p s_p]$ (item 3 of §7.1) is affine in z. Hence $\delta_i^a(0)(\{\square\}) = (\beta/\mathsf{b}) \sum_{p \in \square} f_i(x(p)) s_p$, which vanishes unless the cube lies in $S_i(0)$, and $\delta_S^a(0) = 0$ for $|S| \ge 2$. The h-part is 0. As for the w-part, Thm. 7.1 (f) shows that it is 0 unless $X^{+r(x_0)}$ meets $S_i(0)$ for each $i \in S$; and $r \le \check{r}$.

Proof of (J2) *from* k *to* $k{+}1$. By (J2) at level k, the family u_k^{pol} has (P_k): whenever $X^{+\check{r}(x_k)}$ misses $S_i(k)$, each term $z^S \delta_S(k)(X)$ with $i \in S$ is 0. By the claim, $F_{\ell_k}(u_k^{\mathrm{pol}}(z))(Y)$ is then independent of z_i whenever $Y^{+\check{r}(x_{k+1})}$ misses $S_i(k+1)$, so that for such Y its coefficient of z^S with $i \in S$ vanishes. By (J1), this coefficient is $\delta_S(k+1)(Y)$.

The count. Suppose that $\delta_S(k)(X) \neq 0$. Then, for the a- and h-parts, X meets each of the $|S|$ sets $S_i(k)^{+\check{r}(x_k)}$, and $\mathrm{hull}(X)$ does so for the w-part; by (G4) these sets lie pairwise at distance $\ge \mathrm{sep}_k + 2$. Step 1 (C) therefore bounds $\#X$ in the first case, and $|X|_c$ in the second, from below by $|S| \lceil (\mathrm{sep}_k + 2)/2 \rceil \ge |S| \, \mathrm{sep}_k/2$.

Step 4: level 0. For every $K \ge 1$ we treat (J3), (J4) and (J5) at $k = 0$ by proving the following: $t_0 := |\mathrm{rem}_{0,i}|_a \mathsf{b}/(u_3(x_0) K_5) \le 0.01$; $\delta_i^h(0) = 0$; $|\delta_i^w(0)|_w^{\mathrm{st}} \le \mathrm{amp}(x_0)/815$; $\|\delta_i(0)\|^{\mathrm{st}} \le 0.2 \, \mathrm{amp}(L x_0)$; and $\sum_{|S| \ge 2} \mathfrak{r}_0^{|S|} \|\delta_S(0)\|^{\mathrm{st},(\kappa_1)} \le 4 \cdot 10^{-7}$.

By Thm. 7.1 (f), $\delta_i^a(0) = s_{x_0}[\mathbb{D}_0[f_i]/\mathsf{b}] + \mathrm{rem}_{0,i}$, where $|\mathrm{rem}_{0,i}|_a \le 0.01/\mathsf{b}$; and $\mathrm{rem}_{0,i} \in T^a(x_0)$, being the difference of two elements of that space. As $u_3(x_0) K_5 \ge 1$ by Step 0 (b), we get $t_0 \le 0.01$. The h-part is 0. Thm. 7.1 (f) for $S = \{i\}$ gives $|\delta_i^w(0)|_w^{\mathrm{st}} \le \mathrm{amp}_W/15.99 \le \mathrm{amp}(x_0)/(51 \cdot 15.99) \le \mathrm{amp}(x_0)/815$. Combining these bounds with (J0) and (W3), we obtain $\|\delta_i(0)\|^{\mathrm{st}} \le 1.002 \, \mathrm{amp}(x_0) \le (1.002/5.01) \mathrm{amp}(L x_0) = 0.2 \, \mathrm{amp}(L x_0)$, and $n \, \mathfrak{r}_0 \cdot 0.2 \, \mathrm{amp}(L x_0) = 3.5 \cdot 0.2 = 0.7$.

For the mixed jets, the a- and h-parts vanish, and $\|\delta_S(0)\|^{\mathrm{st},(\kappa_1)} \le (\mathrm{amp}_W/15.99)^{|S|}$, since the factor $e^{-(m\Xi/2)p_0^2 \# X}$ does not exceed 1. Now $\mathfrak{r}_0 \, \mathrm{amp}_W/15.99 \le 3.5/(n \cdot 5.01 \cdot 815) \le 0.00086/n =: q_0$, and so $\sum_{|S| \ge 2} q_0^{|S|} = (1+q_0)^n - 1 - n q_0 \le e^{0.00086} - 1.00086 \le 4 \cdot 10^{-7}$. Since $\kappa_0 \le \kappa_1$, the norm at κ_0 is not larger than the norm at κ_1. The bounds of this step are independent of the separation of the supports.

Step 5: the a*-part of the single jets,* (J3). For $k < K$ we show the identity $\mathrm{rem}_{k+1,i} = D A_{\ell_k}(\bar{u}_k)[\mathrm{rem}_{k,i}] + n[\bar{u}_k; \mathbb{D}_k[f_i]/\mathsf{b}]$, and we show that $t_k := |\mathrm{rem}_{k,i}|_a^{(\kappa_1)} \mathsf{b}/(u_3(x_k) K_5)$ satisfies $t_{k+1} \le 0.2431 t_k + 1$; it follows that $t_k \le 2$ for every k.

By Thm. 6.7 (f1), E'^{new} does not depend on any member, and $A_\ell(u) := E'^{\mathrm{old}}[u]$ depends on E^a only; thus the a-channel is autonomous. From (J1) for $S = \{i\}$ we derive $\delta_i^a(k+1) = DA_{\ell_k}(\bar u_k)[\delta_i^a(k)]$, as follows. By (J1) and Lem. 7.8, applied with $S = \{i\}$, where a single partition occurs, the coefficient of z_i in $E'^a[u_k^{\mathrm{pol}}(z)](Y,\mathsf{W}')$ equals $\frac{d}{ds}|_0\, E'^a[\bar u_k + s\delta_i(k)](Y,\mathsf{W}')$. By Thm. 6.7 (f1), this is $\frac{d}{ds}|_0\, E'^{\mathrm{old}}[\bar u_k + s(\delta_i^a(k),0,0)](Y,\mathsf{W}')$. Here $\delta_i^a(k) \in T^a(x_k)$: being the a-part of a member whose norm is finite (Step 2), it satisfies the clauses of (ca) without a size, and all its values are bounded. Also $A_a(\bar u_k) \le 1 < 16$ at $p = (\kappa_1, c_h)$. Hence, by Thm. 6.7 (h), at $x_k \ge x_9^U$ and with the data (SD) that Step 2 provides at $(x_k;\ell_k)$, this derivative is given by the series $DA(\bar u_k)[\delta_i^a(k)](Y,\mathsf{W}')$ of item 1 of §6.2, taken at the rows ℓ_k; this series is the DA_{ℓ_k} of (o8) of item 2 of the present section, the object of Thm. 6.7 (h3) and of Thm. 6.33.

By Thm. 6.7 (h) this series is linear in δ; we split it according to $\delta_i^a(k) = s_{x_k}[\mathbb{D}_k[f_i]/\mathsf{b}] + \mathrm{rem}_{k,i}$, where both summands lie in $T^a(x_k)$ (by (J0), and by (J3) at k). Here (J3) at level k serves as the induction hypothesis. To the first summand we apply Thm. 6.33 at $x = x_k$, with the data (SD) that Step 2 provides at $(x_k;\ell_k)$ (the empty rows at $k=0$), with $u = \bar u_k \in M_1(x_k;\kappa_1,c_h)$, which lies inside M_8, and with $D = \mathbb{D}_k[f_i]/\mathsf{b}$, which is symmetric, of range $\le \varrho(x_k)$ and of norm $\le K_5/\mathsf{b}$ by (J0). Its $D' = \mathrm{cut}_{\varrho(x_{k+1})}(T_{\ell_k} D)$ is $\mathbb{D}_{k+1}[f_i]/\mathsf{b}$, by the definition of the source forms in item 1 of §6.3. Therefore $DA_{\ell_k}(\bar u_k)[s_{x_k}[D]] = s_{x_{k+1}}[\mathbb{D}_{k+1}[f_i]/\mathsf{b}] + n[\bar u_k; D]$, and together with the splitting this gives the identity for $\mathrm{rem}_{k+1,i}$ stated at the beginning of the step.

Moreover $\mathrm{rem}_{k+1,i} \in T^a(x_{k+1})$, since it is $\delta_i^a(k+1)$ (Step 2) minus $s_{x_{k+1}}[\cdot]$ (Step 0 (a), which is valid at $x_{k+1} \ge \mathsf{b} \ge x_s$, and hence also when $k+1 = K$); Cor. 6.35 gives the same, $\mathrm{rem}' \in T^a(x')$. Thm. 6.33 (a) at $\tilde\kappa = \kappa_1$ gives $|n|_a \le u_3(x_{k+1})K_5/\mathsf{b}$ at x_{k+1}. Thm. 6.7 (h3), applied with $u = \bar u_k$, $p = (\kappa_1, c_h)$, $\delta = \mathrm{rem}_{k,i}$ and $B = |\mathrm{rem}_{k,i}|_a^{(\kappa_1)}$, in the setting (L3su), bounds the cell sum of $DA(\bar u_k)[\mathrm{rem}_{k,i}]$ by $\vartheta_a Bc_a(x_{k+1})$; by Step 0 (c), the norm obeys the same bound. Since $u_3(x_k) = L^{1/4}u_3(x_{k+1})$ and $\vartheta_a L^{1/4} = 0.2431$, we conclude that $t_{k+1} \le 0.2431 t_k + 1$. Now $t_0 \le 0.01$ (Step 4), and $t \le 2$ implies $0.2431t + 1 \le 1.4862 \le 2$; the fixed point is $1/0.7569 \le 1.3212$. Finally, $2u_3(x_k)K_5/\mathsf{b} = 2\breve\omega_3(x_k)\mathrm{amp}(x_k)$ by the identity of (o5) of item 2, and $\breve\omega_3(x_k) \le 10^{-3}$ by (W3). This condition is imposed for all $x \ge Lx_s$ and consists of the two inequalities $\breve\omega_3(x)$, $\breve\omega_3(x/L) \le 10^{-3}$; between them they give $\breve\omega_3(x) \le 10^{-3}$ at every $x \ge x_s$, through the first inequality from Lx_s on and through the second on $[x_s, Lx_s)$. At $x_K = \mathsf{b}$ the bound comes from the second inequality at $x = L\mathsf{b}$.

Step 6: the h- and w-parts of the single jets, (J4). Let $H_k := \max_i \max\{|\delta_i^h(k)|_h^{(\kappa_1)}, |\delta_i^w(k)|_w^{\mathrm{st},(\kappa_1)}\}$. We show that $H_{k+1} \le (1/15)\,1.002\,\mathrm{amp}(x_k) + ((\eta_h+\eta_w)(x_k)/15)H_k$, and hence that $H_k \le 0.12\,\mathrm{amp}(Lx_k)$.

For $k=0$ the bound $H_0 \le \mathrm{amp}(x_0)/815$ follows from the second and third claims of Step 4, namely $\delta_i^h(0) = 0$ and $|\delta_i^w(0)|_w^{\mathrm{st}} \le \mathrm{amp}(x_0)/815$. Since $\mathrm{amp}(x_0)/815 \le \mathrm{amp}(Lx_0)/(815\cdot 5.01) \le 0.12\,\mathrm{amp}(Lx_0)$ by (G3), we have $H_0 \le 0.12\,\mathrm{amp}(Lx_0)$, which is the estimate of this step at $k=0$. For the passage from k to $k+1$, (J1) for $S=\{i\}$ gives $(\delta_i^h,\delta_i^w)(k+1) = D(F^h,F^w)_{\ell_k}(\bar u_k)[\delta_i(k)]$, which is linear in $\delta_i(k) = (\delta^a,0,0) + (0,\delta^h,0) + (0,0,\delta^w)$. Here we set $(F^h,F^w)_\ell(u) := (E'^h, w')$, the h- and w-members of $F_\ell(u)$, and we write D for the derivative of Lem. 7.5 (a). Thm. 7.9 (iv), at the base $\bar u_k$, with $B=1$ and the rate κ_1, bounds each of the three contributions to the h- and w-parts by a factor times the size of its input: the contribution of a by $|\delta_i^a(k)|_a/15$, that of h by $\eta_h(x_k)\,|\delta_i^h(k)|_h/15$, and that of w, with the input in strong units, by $\eta_w(x_k)\,|\delta_i^w(k)|_w^{\mathrm{st}}/15$; the h-output is measured in the cell sum, in caps $c_h(x_{k+1})$, and the w-output in strong units at x_{k+1}. By Step 0 (c), this cell sum bounds $|\delta_i^h(k+1)|_h^{(\kappa_1)}$.

By (J0) and (J3), $|\delta_i^a(k)|_a \le 1.002\,\mathrm{amp}(x_k)$. By (W2) at $x_k \ge Lx_s$, $\eta_h + \eta_w \le 1/L$; and by (G3), $\mathrm{amp}(Lx_k) \le L^{1/2}\mathrm{amp}(x_k)$. So, if $H_k \le 0.12\,\mathrm{amp}(Lx_k)$, then

$$H_{k+1} \le 0.0668\,\mathrm{amp}(x_k) + (0.12/15)L^{-1}L^{1/2}\mathrm{amp}(x_k) = (0.0668 + 0.008L^{-1/2})\mathrm{amp}(x_k) \le 0.12\,\mathrm{amp}(x_k) = 0.12\,\mathrm{amp}(Lx_{k+1}).$$

A single bounded complex source. The single-jet arguments of Steps 2 and 4–6 also apply to a single complex bounded source f, with no condition on its support. With $\|f\|_\infty \le 1$, Thm. 7.1 (h) gives the honest family for $|z| < 1.99/\mathrm{amp}_W$, so Step 2 applies to its first jet. Thm. 7.1 (f) gives the level-zero bounds for complex f, Thm. 6.20 gives its complex symmetric source form, and all derivatives in Steps 5 and 6 are complex linear. The base $\bar u_k$ remains the sourceless run. The arguments therefore give the splitting and estimates of (J0), (J3) and (J4) for this single jet on every $\mathbb{T}_k$. In particular,

$$\|\delta[f](k)\|^{\mathrm{st},(\kappa_1)} \le \max\{1.002\,\mathrm{amp}(x_k), 0.12\,\mathrm{amp}(Lx_k)\} \le 0.2\,\mathrm{amp}(Lx_k).$$

Linearity gives the factor $\|f\|_\infty$ for a general f.

Step 7: the early levels, (J5). We show: if (J5) holds at some $k < k_d$, then it holds at $k+1$; in fact $\sum_{|S|\ge 2} \mathfrak{r}_{k+1}^{|S|}\|\delta_S(k+1)\|^{\mathrm{st},(\kappa_{k+1})} \le 0.0160$.

At $k = 0$, (J5) is given by Step 4.

Single jets at level $k+1$. By (J0), (J3), (J4) and (G3),

$$\|\delta_i(k+1)\|^{\mathrm{st},(\kappa_1)} \le \max\{1.002\,\mathrm{amp}(x_{k+1}), 0.12\,\mathrm{amp}(x_k)\} \le 0.2\,\mathrm{amp}(x_k) = 0.2\,\mathrm{amp}(Lx_{k+1}),$$

where we used $1.002/5.01 = 0.2$; and $n\,\mathfrak{r}_{k+1}\cdot 0.2\,\mathrm{amp}(Lx_{k+1}) = 0.7$.

The family at level k. We have $\|\bar u_k\| \le 1$. On $\mathrm{Pd}(\mathfrak{r}_k)$, using Step 0 (c) for the single jets and for $\bar u_k$, and taking 0.7 from the first clause of (J5),

$$\|u_k^{\mathrm{pol}}(z) - \bar u_k\|^{\mathrm{st},(\kappa_k)} \le n\,\mathfrak{r}_k \max_i \|\delta_i(k)\|^{\mathrm{st},(\kappa_1)} + \sum_{|S|\ge 2} \mathfrak{r}_k^{|S|}\|\delta_S(k)\|^{\mathrm{st},(\kappa_k)} \le 0.7 + 0.21 = 0.91.$$

Hence $u_k^{\mathrm{pol}}(z)$ and $\bar u_k$ lie in $M_2(x_k;\kappa_k,c_h)$; and $(\kappa_k,c_h)\in\mathbb{P}$, by (G2).

Mixed jets. Thm. 5.15 (e) at (κ_k,c_h) (Thm. 6.7 (U)) gives, at x_{k+1},

$$\sup_{\mathrm{Pd}(\mathfrak{r}_k)} \|F_{\ell_k}(u_k^{\mathrm{pol}}(z)) - F_{\ell_k}(\bar u_k)\|^{\mathrm{st},(\kappa_k)} \le 0.4\cdot 0.91 = 0.364.$$

By Thm. 6.7 (c+), the difference on the left, as a function of z, is a family, and by (J1) its coefficient of z^S is $\delta_S(k+1)$. Lem. 7.7, that is, Cauchy's estimate in the variables of S on circles of radius $r < \mathfrak{r}_k$, with the remaining variables put equal to 0, followed by the limit $r \to \mathfrak{r}_k$, yields $\|\delta_S(k+1)\|^{\mathrm{st},(\kappa_k)} \le 0.364\,\mathfrak{r}_k^{-|S|}$. By (J2) at $k+1$, $\delta_S(k+1)$ is carried by sets whose $\#X$ (for the a- and h-parts) or $|X|_c$ (for the w-part) is at least $|S|\,\mathrm{sep}_{k+1}/2$; since $\kappa_{k+1} = \kappa_k - \check\kappa_k$, it follows that

$$\|\delta_S(k+1)\|^{\mathrm{st},(\kappa_{k+1})} \le e^{-\check\kappa_k(|S|\mathrm{sep}_{k+1}/2-1)}\cdot 0.364\,\mathfrak{r}_k^{-|S|}.$$

By (G3), $\mathfrak{r}_{k+1} \le L^{1/2}\mathfrak{r}_k$. Let $q := L^{1/2}e^{-\check\kappa_k \mathrm{sep}_{k+1}/2} = L^{1/2}/(4nL^{1/2}) = 1/(4n)$. Then $\mathfrak{r}_{k+1}^{|S|}\|\delta_S(k+1)\|^{\mathrm{st},(\kappa_{k+1})} \le 0.364\,e^{\check\kappa_k}q^{|S|}$, and therefore

$$\begin{aligned}
&\sum_{|S|\ge 2} \mathfrak{r}_{k+1}^{|S|}\|\delta_S(k+1)\|^{\mathrm{st},(\kappa_{k+1})}\\
&\quad\le 0.364\,e^{1/4}[(1+q)^n - 1 - nq]\\
&\quad\le 0.364\cdot 1.28403\cdot (e^{1/4} - 5/4)\\
&\quad\le 0.364\cdot 1.28403\cdot 0.034026\\
&\quad\le 0.0160 \le 0.21.
\end{aligned}$$

Here we used that $(1+1/(4n))^n$ increases to $e^{1/4}$: the supremum of the bracket is $e^{1/4} - 5/4 \le 0.034026$, and no n attains it.

Step 8: the late levels and the short runs, (J6). We prove (J6) for every $K \ge 1$. At $k = k_d$ it is (J5), with $\kappa_{k_d} = \kappa$.

From k to $k+1$, for $k_d \le k < K$. We apply Thm. 7.9 (i), (ii) at $p = (\kappa, c_h)$ to the pair of families consisting of $u_k(z)$ and the constant family $\bar u_k$ on $\mathrm{Pd}(\mathfrak{r}_*)$; they lie within $B = 1.91$ caps at $k = k_d$, and within $B = 1$ at the later levels. The image is a family in $M_1(x_{k+1}; \kappa, c_h)$ with the strong cap, because $0.35 + 0.248 \cdot 1.91 \le 0.824 \le 1$. The strong distance of the images is at most $\vartheta[B]$ times the weak distance of the inputs, where $\vartheta[1.91] = 3.968/14.09 \le 0.2817$ and $\vartheta[1] \le 0.2646$; both values are ≤ 0.4. The weak distance does not exceed the strong one. Moreover $F_{\ell_k}(\bar u_k) = \bar u_{k+1}$.

Jets. If $u_k(z)$ and the honest family have the same $\partial_{S'}$ at 0 for every S', then Step 0 (d), applied to each value, shows that their images have the same property; at $k = k_d$ this hypothesis holds by the form of u^{pol}.

Short runs. If $K < j^*$, then $k_d = 0$. In that case (J5) at $k = 0$ is given by Step 4, whose bounds are independent of the separation; and the families start from u_0^{pol} on $\mathrm{Pd}(\mathfrak{r}_0)$, where $\mathfrak{r}_0 = \mathfrak{r}_* \ge \mathfrak{r}(L^{j^*}\mathfrak{b})$ by (G3). There is then no early level. In every case $k_d \le K - 1$, by (G2).

The base points. For $0 \le k \le K$ they are provided by Step 2 (Thm. 7.1 (h)).

Step 9: the identity (J7). Since $\beta\nu_p(z)s_p = L^K \sum_i z_i f_i(x(p)) s_p$, we have

$$\exp\{-\beta \sum_p s_p + \sum_i z_i \mathcal{O}_K(f_i)\} = \exp\{-\beta \sum_p (1 - \nu_p(z)) s_p\},$$

which is the weight of Thm. 7.1. With the honest family, Thm. 7.1 (h) gives $Z_K(z) = \exp\{\sum_{1 \le j \le K} V_j(z)\} I[u_K(z)]$ on Pd_0; here the kernels are normalised, and Fubini's theorem is used. Also $\rho[\bar u_K; \ell_K] > 0$ (Thm. 7.1 (a3) at $\nu = 0$; Thm. 5.15 (d), applied step by step), so that $I[\bar u_K] > 0$. Therefore, near $z = 0$, $\log Z_K(z) = \sum_j V_j(z) + \log I[u_K(z)]$ up to an additive constant. By Thm. 6.7 (g1), $V_{k+1}(z) = V_0^+ + \sum_Y v_k(Y; u_k(z))$, a finite sum because $\mathbb{T}_{k+1}$ is finite; and V_0^+ does not depend on any member (Thm. 6.7 (f1)), so ∂_1 annihilates it. This proves (J7) with the honest family.

Representatives. These have the same $\partial_{S'}$ at 0 as the honest family, for all S' ((o2) of item 2; (J6)). Applied with $S = \{1, \dots, n\}$ and $\Phi = v_k(Y; \cdot)$, Step 0 (d) shows that the derivative is unchanged. As to the last term, $\partial_1 \cdots \partial_n \log I[u(z)](0)$ is a polynomial in the numbers $\partial_{S'} I[u(z)](0)/I[u(0)]$, $S' \subset \{1, \dots, n\}$, and we apply Step 0 (d) with $\Phi = I$ for every S'.

Vanishing. The representatives have the property (P_k) of Step 3: at the levels $k \le k_d$ by (J2), and at the levels $k > k_d$ by the claim of Step 3, applied from one level to the next. By that claim, $v_k(Y; u_k(z))$ is independent of z_i when $Y^{+\check r(x_{k+1})}$ misses $S_i(k+1)$, and then $\partial_1 \cdots \partial_n$ annihilates it.

Step 10: the local vacuum numbers, (J8). Fix $k < K$; set $\hat{\mathfrak{r}}_k := \mathfrak{r}_{\min(k, k_d)}$, and let $\hat\kappa := \kappa_k$ for $k < k_d$ and $\hat\kappa := \kappa$ for $k \ge k_d$. On $\mathrm{Pd}(\hat{\mathfrak{r}}_k)$ the representative takes values in $M_2(x_k; \hat\kappa, c_h)$ ((J5), (J6)), and $z \mapsto v_k(Y; u_k(z))$ is holomorphic on this polydisc (Thm. 6.7 (c+)). For $r < \hat{\mathfrak{r}}_k$ we have

$$\partial_1 \cdots \partial_n v_k(Y; u_k(z))(0) = (2\pi i)^{-n} \oint_{|z_1| = \cdots = |z_n| = r} v_k(Y; u_k(z)) \prod_i dz_i / z_i^2.$$

Let $\check Y$ be the family of those Y that remain in (J7). We take the sum over Y inside the integral:

$$\sum_{Y \in \check Y} |\partial_1 \cdots \partial_n v_k(Y; \cdot)(0)| \le r^{-n} \sup_{|z_i| = r} \sum_{Y \in \check Y} |v_k(Y; u_k(z))|.$$

Every $Y \in \check Y$ contains a cell of $S_1(k+1)^{+\check r(x_{k+1})}$, a set with at most $N_1(k)$ cells (Step 1 (A)). Every $Y \in \check Y$ is connected and satisfies $\#Y \ge m_k$: for $k < k_d$, Y meets each of the n sets $S_i(k+1)^{+\check r(x_{k+1})}$, which lie pairwise at distance $\ge \mathrm{sep}_{k+1} + 2$ by (G4), so that Step 1 (C) gives $\#Y \ge n\lceil \mathrm{sep}_{k+1}/2 \rceil \ge \lceil n \, \mathrm{sep}_{k+1}/2 \rceil$, the number $n\lceil \mathrm{sep}_{k+1}/2 \rceil$ being an integer $\ge n \, \mathrm{sep}_{k+1}/2$; for $k \ge k_d$ we have $m_k = 1$ by the definition of m_k in (J8), and $\#Y \ge 1$ because Y contains a cell.

At each z, Thm. 6.7 (g2) at $p = (\hat\kappa, c_h)$ and $A = 2$ gives

$$\sum_{Y \in \check Y} |v_k(Y; u)| \le N_1(k) e^{-\hat\kappa(m_k - 1)} \max_{\text{cells}} \sum_{Y \ni \text{the cell}} e^{\hat\kappa(\#Y - 1)} |v_k(Y; u)| \le N_1(k) e^{-\kappa(m_k - 1)} 2 C_g c_a(x_k).$$

Letting $r \to \hat{\mathfrak{r}}_k$, we obtain (J8). No entropy factor appears, because the factor $e^{-\kappa(m_k-1)}$ is drawn out of the weight of Thm. 6.7 (g2). At $k=0$ the step is the one at $(\beta;\emptyset)$, and Thm. 6.7 (g1), (g2) apply there. □

7.4. **Comparison of sourced runs.** We compare two runs of the drawn step, of lengths $K \le K'$, which start from Wilson's weight with n separated sources and end on one and the same torus; along each run we follow the base points of the members and their multi-affine jets at $z=0$, on polydiscs whose radii grow with the level k as long as $k \le k_d$, and on one fixed polydisc for $k_d \le k \le K$, namely $\mathrm{Pd}(\mathfrak{r}_*/2)$ where the two runs are compared and $\mathrm{Pd}(\mathfrak{r}_*)$ for one run alone. At the level k_0 we rewrite the long run over the rows of the short one by the map $\mathbb{B}_D$, $D = \tilde{\mathsf{E}}_{m+k_0} - \mathsf{E}_{k_0}$; from there on a single map F_{ℓ_k} carries both runs, and the distances between their base points, between their jets and between their families are bounded by a numeral times ϑ^{k-k_0} plus a numeral times $\Omega^{\#}$. From this we derive the bound (D3) at the end of the runs, the bounds (D4) and (D5) on the vacuum numbers, the identity (D6) for $S^T_{n;K'} - S^T_{n;K}$, and in (D7) the local boundedness in b of all constants.

Remark 7.14 (Content, place and setting). The tags (D0) to (D7) denote the clauses of Thm. 7.19. Thm. 7.19 is the fourth of four statements on holomorphic families of complex members; the other three are Thm. 7.9, Lem. 7.10 and Thm. 7.11, all of them in §7.2.

Our setting consists of the group $G = \mathrm{SU}(2)$, dimension three, the drawn step, and one fixed torus.

Item 1: notation, norms and the table.

Definition 7.15 (Item 1.1: what is kept)**.** From §5.5 we keep the following objects, taken at the rate $\kappa_1 = \kappa + 1$ of (O5), item (1) of §6.1: the lattice, the cubes, the fields, the rows, the site forms, the norm $\|\cdot\|_{(\lambda_\flat)}$, the energy En_ℓ, the domain $\mathcal{U}_X(x)$, the clauses (ca), (ch), (cΞ), the weak and the strong cap, the density $\rho[u;\ell]$, the class, the step, the runs, the partners and the depth. Here $\kappa \ge 6$, and κ is the lower end of the interval $[\kappa,\kappa_1]$ of the rates $\tilde{\kappa}$.

We write $x_k = L^{K-k}\mathsf{b}$, $\ell_k = \ell_k^{[K]}$ and $\tilde{\ell}_j = \ell_j^{[K']}$, and we denote by E_k and $\tilde{\mathsf{E}}_j$ the site forms of these rows; moreover $m = K' - K$.

For the run K let $S_i(k) := S_i^{[K]}(k)$ be the k-blocks of $\operatorname{supp} f_i$ as in (o3), item 2 of §7.3; for the run K' the corresponding sets are $\tilde{S}_i(k) := S_i^{[K']}(m+k)$. Both are sets of cubes of one torus $\mathbb{T}_k = \mathbb{Z}^3/NL^{K-k}\mathbb{Z}^3$, and the two may differ, since the bare lattice of K' has barycentres that are missing from the bare lattice of K.

The numbers $\Lambda = \log_L(c_4\mathsf{b}^{1/2})$, $K_0(\mathsf{b}) = (5/2)\Lambda + 9/4$, $k_0 = \lceil (5K+10\Lambda)/9 \rceil$, $\theta = \vartheta^{4/9} \le 0.69903$, ε_L and $C_\mathsf{b} = (3/\vartheta)(c_4\mathsf{b}^{1/2})^{\varepsilon_L}$ are entries of (O14), item 1 of §5.5, taken at the rate κ_1, and $\vartheta = 0.4468$ is also an entry of this (O14).

The symbols F_ℓ, $\mathsf{V}'[\cdot]$, $v(Y;\cdot)$ and $A_\ell := E'^{\mathrm{old}}$ refer to the one map of Thm. 6.7 (U). Finally, $I[u] := \int \rho[u;\ell_K]\, dV$.

■

Definition 7.16 (Item 1.2: norms, families and local letters)**.** The norms are those of §7.2. Let $g = (g^a, g^h, g^w)$ be a triple that satisfies at x the clauses without a size, and let $\tilde{\kappa} \in [\kappa,\kappa_1]$ be a rate. We set

$$|g|_a := \sup_X \sup_{\mathcal{U}_X(x)} |g^a(X,\cdot)|\, e^{\tilde{\kappa}(\#X-1)}/c_a(x)$$

and

$$|g|_h := \sup_X \sup |g^h(X,\cdot)|\, e^{\tilde{\kappa}(\#X-1)}/c_h(x),$$

where in the second formula the inner supremum is taken over the tame G-valued fields. For the third component we put $|g|_w^{\mathrm{wk}} := \sup_{X,V} |g^w(X,V)|\, e^{\tilde{\kappa}(|X|_c-1)}$, and $|g|_w^{\mathrm{st}}$ is defined in the same way

with the additional factor $\exp\{c_\Xi x\varepsilon \sum_{p\in D(V)\cap X}\phi_p\}$. Then $\|g\| := \max\{|g|_a, |g|_h, |g|_w^{\mathrm{wk}}\}$, and $\|g\|^{\mathrm{st}}$ is obtained likewise with $|g|_w^{\mathrm{st}}$ in place of $|g|_w^{\mathrm{wk}}$.

The two distances are $\mathsf{D}_x^{\mathrm{in}}(u_1,u_2) = \|u_1-u_2\|$ and $\mathsf{D}_x(u_1,u_2) = \|u_1-u_2\|^{\mathrm{st}}$. An upper index $(\tilde\kappa)$ names the rate; where no index is written, the rate is κ_1. The norm at a higher rate dominates the norm at a lower rate. We write $M_B(x;\tilde\kappa,c_h) = \{\|u\|^{(\tilde\kappa)} \le B\}$ and $\mathrm{Pd}(r) = \{z\in\mathbb{C}^n : \ |z_i|<r\}$; for a non-empty subset S of $\{1,\dots,n\}$ we put $z^S = \prod_{i\in S} z_i$.

A family on an open set Ω within B caps at $p=(\tilde\kappa,\tilde c)$ is a map $z\mapsto u(z)\in M_B(x;p)$ such that all values of $u(z)$ depend holomorphically on z; its coefficient of z^S at 0 is denoted by u_S, and these coefficients are its multi-affine jets.

A step cited with no proof named beside it is a step of the proof of Thm. 7.19.

The symbols $r(x)$ and $s(x)$ denote two of the radii at x of item 2 of §5.5, and $\partial_i = \partial/\partial z_i$.

$s_x[D]$ is the map of entry 5 of item (0) of §6.1.

The letter D stands for $\tilde{\mathsf{E}}_{m+k_0} - \mathsf{E}_{k_0}$ in (D1) and in Step 2, and for any family of kernels in $s_x[D]$ and $n[u;D]$; in Step 3 these families are $\mathbb{D}_k/\mathsf{b}$ and $(\tilde{\mathbb{D}}_{m+k_0} - \mathbb{D}_{k_0})/\mathsf{b}$. Further, $D(V)$ is the set of the rough plaquettes, and D placed before A or F denotes a derivative.

Without an argument, c_h is the number of (O11), item 1 of §5.5. The caps are $c_h(x) = e^{-c_h p_0(x)^2}$ and $c_a(x) = E_a p_0(x)^3 x^{-1/2}$.

m_l and m' are the exponents in $N = 2L^{m_l}$ and $M = L^{m'}$.

The symbol $u_k(z)$ has two senses, and in the first it is the given or the honest family, iterated by $u_{k+1}(z) = F_{\ell_k}(u_k(z))$ from its starting level. In (D0) and in the part on (D0) of Step 2, that level is k_0, and the family $u_{k_0}(z)$ is given on Pd. In the part on (D1) (c) of Step 2 and in Step 8 up to its part "Representatives", $u_k(z)$ is, for $0\le k\le K$, the honest family of (o1), item 2 of §7.3, and on the interior of Pd_0 of the run K, that family is an iterate of this kind with starting level 0. For each run, Pd_0 and Pd_0^o denote the sets that the part "Sources" of item 3 of §7.1 defines for that run. With its argument, $\tilde u_j(z)$ always denotes the given or the honest family of the run K'. Everywhere else $u_k(z)$ has the second sense: like the symbol without an argument, it is the representative of Prop. 7.12 (J6), namely u_k^{pol} on $\mathrm{Pd}(\mathfrak{r}_k)$ for $k\le k_d$ and, for $k_d\le k\le K$, the family on $\mathrm{Pd}(\mathfrak{r}_*)$ iterated from $u_{k_d}^{\mathrm{pol}}$. The families in the two senses share the base point $\bar u_k$ and the multi-affine jets $\delta_S(k)$, by (o2) of item 2 of §7.3 and Prop. 7.12 (J6); beyond these they may differ. Without an argument, as in $\mathsf{V}'_{\ell_k}[u_k]$, $\mathsf{V}'_{\tilde\ell_j}[\tilde u_j]$ and $\mathsf{V}'_{\ell_k}[\mathfrak{n}_k]$ in (D4) to (D6), the symbol means the representative of (D1): the polynomial up to k_d (up to k'_d for the run K'), and the family after it.

■

Definition 7.17 (Item 1.3: the table)**.** Of item (1) of §6.1 we use the following entries, in the order (O0) to (O14) of the table. The entry (O0a) gives the number K_5. From (O4a) we take the conditions (L3su), of which this section uses $L^{1/4}\ge 10^4$, and (L5*), which enters through $\vartheta_5\le L^{-1/5}$. The entry (O10a) gives the number $C_g = 263L^3$, and $c_Q^{\mathrm{su}} = 306M^3\mathsf{g}(\lambda_\flat)^2$ is the number of (O10), item 1 of §5.4. From (O12a) we take the number C_3.

After (O12a) and before (O13) appear three functions of x alone: $\eta_h(x) := e^{-(c_h/2)(p_0(x)^2 - p_0(x/L)^2)}$, $\eta_w(x) := e^{-c_\Xi p_0(x)^2}$ and $\breve\omega_3(x) := C_3 E_a p_0(x) x^{-1/4}/c_Q^{\mathrm{su}}$. They contain x, the function p_0 and the letters c_h, c_Ξ, C_3, E_a, c_Q^{su}, L, all of which are defined before (O13), and none of b, n, d.

By (O13), $\mathsf{b}\ge x_s$, so that $x_k \ge L\mathsf{b} \ge Lx_s \ge \max\{x_9^U, x_3\}$ for $k<K$. Also by (O13), the following two conditions hold for all $x\ge Lx_s$:

(W2) $\eta_h(x) + e^{1/2}\eta_w(x) \le 1/L$;
(W3) $\breve\omega_3(x),\ \breve\omega_3(x/L) \le 10^{-3}$.

Under (O13) there is moreover, from §5.3, the condition (Y.0)(0b): $p_0(x)^4 \le 2p_0(x/L)^4$ for $x\ge x_9^U$.

From (O14) we take first the following functions of x and numbers. We set $\mathrm{amp}(x) = c_Q^{\mathrm{su}} K_5 x^{1/2}/(\mathsf{b} E_a p_0(x))$ and $u_3(x) = C_3 x^{1/4}$; they obey the identity $u_3(x)K_5/(\mathsf{b}\,\mathrm{amp}(x)) = \breve{\omega}_3(x)$, in which b and K_5 cancel. Further, $\mathfrak{r}(x) = 3.5/(n\,\mathrm{amp}(Lx))$ and $\mathfrak{r}_k = \mathfrak{r}(x_k)$; $\vartheta_a = 0.2431 L^{-1/4}$ and $\nu_a := 0.2431$; $\vartheta[B] = 3.968/(16-B)$, so that $\vartheta[1] \le 0.26454$, $\vartheta[2] \le 0.28343$ and $\vartheta[2.455] \le 0.29295$; finally $\check{r}(x) = r(x) + 3$ and $C_5^{\#} = L^{15/32} C_5 + K_5$.

The further entries of (O14) are functions of the depth; they depend on b, n, d, R_f and on letters, but on none of K, K', N and on no row. For $i \ge 0$ let $\mathrm{sep}^{[i]} = \lfloor dL^i/M \rfloor - 2\,\check{r}(L^i\mathsf{b}) - 2$, and let $\mathrm{sep}_* = 8\log(4nL^{1/2})$. We let j^* be the least $j \ge 1$ such that $\mathrm{sep}^{[i]} \ge \mathrm{sep}_*$ for all $i \ge j-1$. For every $j \ge 1$ we set $N_1^{[j]} = (2R_f L^{j-1}/M + 2\,\check{r}(L^{j-1}\mathsf{b}) + 3)^3$; moreover $m^{[j]} = \lceil n\,\mathrm{sep}^{[j-1]}/2 \rceil$ for $j > j^*$ and $m^{[j]} = 1$ for $1 \le j \le j^*$.

At the level k of a run of length K we put $\mathrm{sep}_k = \mathrm{sep}^{[K-k]}$ and $k_d = \max\{K - j^*, 0\}$; for $k < k_d$ we put $\check{\kappa}_k = 2\log(4nL^{1/2})/\mathrm{sep}_{k+1}$, as in (o4), item 2 of §7.3. The rates are $\kappa_k = \kappa + \sum_{k \le j < k_d} \check{\kappa}_j \in [\kappa, \kappa + 0.26]$, and $\kappa_k = \kappa$ for $k \ge k_d$. Finally $\mathfrak{r}_* = \mathfrak{r}_{k_d}$, $N_1(k) = N_1^{[K-k]}$ and $m_k = m^{[K-k]}$.

The two series of the table are

$$\mathrm{Tail}(J) = \sum_{j > J} \mathfrak{r}(L^{\max(j,j^*)}\mathsf{b})^{-n} N_1^{[j]}\, 2C_g c_a (L^j \mathsf{b})\, e^{-\kappa(m^{[j]}-1)}$$

and

$$\mathrm{Ser}_2 = \sum_{j \ge 1} \vartheta^{-j} \big(2/\mathfrak{r}(L^{\max(j,j^*)}\mathsf{b})\big)^n N_1^{[j]}\, 1.2 C_g c_a (L^j \mathsf{b})\, e^{-\kappa(m^{[j]}-1)}.$$

The threshold $K_T(\mathsf{b}, n, d)$ is the least integer $K \ge K_0(\mathsf{b})$ that satisfies the two conditions

(K1a) $(4/9)K - (10/9)\Lambda - 1 \ge j^* + 1$,
(K1b) $C_5^{\#} K_5^{-1} K\, 0.3683^{(4/9)K - (10/9)\Lambda - 1} \le 0.1$.

Both conditions then hold for every larger K, by Step 1 (c). We put $\Omega^{\#} := C_5^{\#} K_5^{-1} K\, 0.3683^{K - k_0}$.

The conditions (W2) and (W3) use η_h, η_w and $\breve{\omega}_3$ in the closed forms above, which are defined before (O13). No floor depends on an entry of (O14), and no entry depends on a later one. ∎

Definition 7.18 (Item 3: the data)**.** The data are those of Thm. 8.2 (T2). The letters are those of item (1) of §6.1, and $\mathsf{b} \ge x_s$. We take $N = 2L^{m_l}$ with $m_l \ge m'$, so that N is even and $M \mid N$, and we require (Tor) $N \ge 2M\, s(L\mathsf{b})$. Further, let $n \ge 2$ and $d, R_f > 0$. The functions $f_1, \dots, f_n$ are real and Lipschitz on the torus of side N, with $\|f_i\|_\sharp = \max\{\|f_i\|_\infty, 40M\,\mathrm{Lip} f_i\} \le 1$, where Lip refers to the Euclidean distance. The sets $\{f_i \ne 0\}$ lie pairwise at sup distance at least d, and each of them lies within sup distance R_f of one point, as in item 1 of §7.3; these are the distances of item 1 of §8.1. Finally $K' \ge K \ge K_T(\mathsf{b}, n, d)$. ∎

Theorem 7.19 (Item 4: two runs of jets meet on growing radii)**.** In the setting of item 3, the following hold.

(D0) *A fixed polydisc.* For this clause it suffices that $K \ge K_0(\mathsf{b})$. Let Pd be a polydisc about 0 on which the two sourced runs have families $u_{k_0}(z)$ and $\tilde{u}_{m+k_0}(z)$ in $M_1(x_{k_0}; \kappa_1, c_h)$ with the strong cap, such that $\rho_{k_0}^{[K],z} = e^{\mathsf{V}_0(z)} \rho[u_{k_0}(z); \ell_{k_0}]$ and $\rho_{m+k_0}^{[K'],z} = e^{\tilde{\mathsf{V}}_0(z)} \rho[\tilde{u}_{m+k_0}(z); \tilde{\ell}_{m+k_0}]$, where $\mathsf{V}_0(z)$ and $\tilde{\mathsf{V}}_0(z)$ are numbers that depend holomorphically on z; by Thm. 7.1 (h), this is so on the interior of Pd_0 of the run K'. Here $\rho_k^{[K],z}$ is the ρ_k^z of Thm. 7.1 (h) taken for the run K, that is, $\rho_0^{[K],z} = \rho_0^{[K]} \exp\{\sum_i z_i \mathcal{O}_K(f_i)\}$ and $\rho_{k+1}^{[K],z} = (\rho_k^{[K],z})'$, and $\rho_j^{[K'],z}$ is defined in the same way for the run K'. Put $\mathfrak{n}_{k_0}^\circ(z) := \mathbb{B}_{\tilde{\mathsf{E}}_{m+k_0} - \mathsf{E}_{k_0}}(\tilde{u}_{m+k_0}(z))$, and recursively $\mathfrak{n}_{k+1}^\circ(z) := F_{\ell_k}(\mathfrak{n}_k^\circ(z))$ and $u_{k+1}(z) := F_{\ell_k}(u_k(z))$. Then, for $k_0 < k \le K$ and every z of Pd,

three assertions hold. First, both $u_k(z)$ and $\mathfrak{n}^\circ_k(z)$ are families on Pd in $M_1(x_k;\kappa_1,c_h)$ with the strong cap. Second, at every field,

$$\rho^{[K'],z}_{m+k} = \exp\{\tilde{\mathsf{V}}_0(z) + \sum_{k_0\le j<k} \mathsf{V}'_{\ell_j}[\mathfrak{n}^\circ_j(z)]\}\,\rho[\mathfrak{n}^\circ_k(z);\ell_k],$$

where the numbers in the exponent are holomorphic. Third, $\sup_{\mathrm{Pd}} \mathsf{D}_{x_k}(u_k(z),\mathfrak{n}^\circ_k(z)) \le 3\vartheta^{k-k_0}$. If moreover, at the real z of Pd, the members $u_{k_0}(z)$ and $\tilde{u}_{m+k_0}(z)$ are real and $\rho[u_{k_0}(z);\ell_{k_0}]$, $\rho[\tilde{u}_{m+k_0}(z);\tilde{\ell}_{m+k_0}] > 0$ (the honest families have this property by Thm. 7.1 (h), since the f_i are real by item 3), then at these z all $u_k(z)$ and $\mathfrak{n}^\circ_k(z)$ and all numbers $\mathsf{V}'_{\ell_j}[u_j(z)]$, $\mathsf{V}'_{\ell_j}[\mathfrak{n}^\circ_j(z)]$ are real, and $\rho[u_k(z);\ell_k]$, $\rho[\mathfrak{n}^\circ_k(z);\ell_k] > 0$.

(D1) *Objects.* The objects $\bar{u}_k = u_k(0)$, $\delta_S(k)$, u^{pol}_k for $0 \le k \le k_d$, and $u_k(z)$ for $k_d \le k \le K$ on $\mathrm{Pd}(\mathfrak{r}_*)$, are those of Prop. 7.12 for the run K; the objects $\bar{\tilde{u}}_j$, $\tilde{\delta}^{\mathrm{own}}_S(j)$, $\tilde{u}^{\mathrm{pol}}_j$ are the corresponding ones for the run K', defined over its own rows. Let $Q := Q_D$ be as in Lem. 7.10, taken at $D = \tilde{\mathsf{E}}_{m+k_0} - \mathsf{E}_{k_0}$ and $x = x_{k_0}$. On $\mathrm{Pd}(\mathfrak{r}_{k_0}/2)$ put $\mathfrak{n}_{k_0}(z) := \mathbb{B}_D(\tilde{u}^{\mathrm{pol}}_{m+k_0}(z))$. For $k_0 \le k \le k_d$ put $\mathfrak{n}^{\mathrm{pol}}_k(z) := \bar{\mathfrak{n}}_k + \sum_S z^S \tilde{\delta}_S(k)$, with $\mathfrak{n}^{\mathrm{pol}}_{k_0} = \mathfrak{n}_{k_0}$. For $k_0 \le k < k_d$ let $\bar{\mathfrak{n}}_{k+1} := F_{\ell_k}(\bar{\mathfrak{n}}_k)$, and let $\tilde{\delta}_S(k+1)$ be the coefficient of z^S of $F_{\ell_k}(\mathfrak{n}^{\mathrm{pol}}_k(z))$. For $k_d \le k < K$ put $\mathfrak{n}_{k_d}(z) := \mathfrak{n}^{\mathrm{pol}}_{k_d}(z)$ and $\mathfrak{n}_{k+1}(z) := F_{\ell_k}(\mathfrak{n}_k(z))$ on $\mathrm{Pd}(\mathfrak{r}_*/2)$, and $\bar{\mathfrak{n}}_{k+1} := \mathfrak{n}_{k+1}(0) = F_{\ell_k}(\bar{\mathfrak{n}}_k)$. Then:

(a) $k_0 + 2 \le k_d = K - j^*$, and the run K' has $k'_d = m + k_d$; partners share x_k, $\mathfrak{r}_k$, sep_k, $\check{\kappa}_k$, κ_k, $\check{r}$, the cubes and the L-blocks, while the sets $S_i(k)$, $\tilde{S}_i(k)$ of item 1.1 are not among the shared objects.

(b) $\mathfrak{n}_{k_0}(z)$ is a multi-affine family in $M_{2.455}(x_{k_0};\kappa_{k_0},c_h)$, with the weak cap, and $\tilde{\delta}_S(k_0) = (\tilde{\delta}^{\mathrm{own},a}_S, \tilde{\delta}^{\mathrm{own},h}_S, \tilde{\delta}^{\mathrm{own},w}_S e^Q)(m+k_0)$.

(c) *True jets.* $\bar{\mathfrak{n}}_k$ is the base point, and the $\tilde{\delta}_S(k)$ are the multi-affine jets, of the family $\mathfrak{n}^\circ_k(z)$ that (D0) provides on the interior of Pd_0 of the run K', that is, of a system of members of the sourced long run, taken over the rows ℓ_k of the short run. For $k > k_0$, $\bar{\mathfrak{n}}_k$ is an $\mathfrak{n}_k$ as in Thm. 5.63 (iii); that clause asserts existence, and $\bar{\mathfrak{n}}_k$ is the system that Steps 4 and 5 of the proof of Thm. 5.63 construct from the systems $u_{k_0} = \bar{u}_{k_0}$ and $\tilde{u} = \bar{\tilde{u}}_{m+k_0}$. Of this, only (D0) at $z = 0$ enters: $\bar{\mathfrak{n}}_k$ is real, lies in $M_1(x_k;\kappa_1,c_h)$ and carries the strong cap; for some real number V one has $\rho^{[K']}_{m+k} = e^{\mathsf{V}}\rho[\bar{\mathfrak{n}}_k;\ell_k] > 0$ at every field; and $\mathsf{D}_{x_k}(\bar{u}_k,\bar{\mathfrak{n}}_k) \le 3\vartheta^{k-k_0}$.

(d) *Reach.* $\tilde{\delta}_S(k)(X) = 0$ unless $X^{+\check{r}(x_k)}$ meets $\tilde{S}_i(k)$ for every $i \in S$; so, where $\tilde{\delta}_S(k)(X) \ne 0$ and $|S| \ge 2$, we have $\#X \ge |S|\,\mathrm{sep}_k/2$ for the components a and h, and $|X|_c \ge |S|\,\mathrm{sep}_k/2$ for the component w.

(e) For $k_0 < k \le k_d$, u^{pol}_k and $\mathfrak{n}^{\mathrm{pol}}_k$ are families on $\mathrm{Pd}(\mathfrak{r}_k/2)$ in $M_2(x_k;\kappa_k,c_h)$, with $\|u^{\mathrm{pol}}_k(z) - \bar{u}_k\|^{\mathrm{st},(\kappa_k)} \le 0.455$ and $\|\mathfrak{n}^{\mathrm{pol}}_k(z) - \bar{\mathfrak{n}}_k\|^{\mathrm{st},(\kappa_k)} \le 0.455 + 0.34$.

(f) For $k_d < k \le K$, $u_k(z)$ and $\mathfrak{n}_k(z)$ are families on $\mathrm{Pd}(\mathfrak{r}_*/2)$ in $M_1(x_k;\kappa,c_h)$ with the strong cap; they are outputs of F and carry the true multi-affine jets, and $\|u_k(z) - \bar{u}_k\|^{\mathrm{st},(\kappa)} \le 0.455$ and $\|\mathfrak{n}_k(z) - \bar{\mathfrak{n}}_k\|^{\mathrm{st},(\kappa)} \le 0.795$, each right side being multiplied by $\vartheta[2]^{k-k_d}$.

(D2) *Pair bound.* Put $e_k := \mathsf{D}_{x_k}(\bar{u}_k,\bar{\mathfrak{n}}_k)$ for $k > k_0$ and $e_{k_0} := 3$. Put $a^D_k := \max_i |\delta^a_i(k) - \tilde{\delta}^a_i(k)|_a$ and, for $k > k_0$, $h^D_k := \max_i \max\{|\delta^h_i - \tilde{\delta}^h_i|_h, |\delta^w_i - \tilde{\delta}^w_i|^{\mathrm{st}}_w\}(k)$. For $k_0 < k \le k_d$ let

$$\Delta_k := e_k + (n\mathfrak{r}_k/2)\max\{a^D_k, h^D_k\} + \sum_{|S|\ge 2} (\mathfrak{r}_k/2)^{|S|}\|\delta_S(k) - \tilde{\delta}_S(k)\|^{\mathrm{st},(\kappa_k)},$$

and let $\Delta_{k_0} := 3.91$. Then $\Delta_k \geq \sup_{\mathrm{Pd}(\mathfrak{r}_k/2)} \mathsf{D}^{(\kappa_k)}_{x_k}(u^{\mathrm{pol}}_k, \mathfrak{n}^{\mathrm{pol}}_k)$, where at k_0 the distance is D^{in}. Moreover, for $k_0 < k \leq k_d$, with

$$\Omega_0 := \max_i \|\tilde{\mathbb{D}}_{m+k_0}[f_i] - \mathbb{D}_{k_0}[f_i]\|_{(\lambda_\flat)}/(K_5 \check{\omega}_3(x_{k_0})),$$

the bounds (i) to (iii) hold:

(i) $a^D_k \leq (u_3(x_k)K_5/\mathsf{b})\,[7.6\vartheta^{k-k_0} + \nu_a^{k-k_0}\Omega_0]$;
(ii) $n\mathfrak{r}_k a^D_k \leq 0.0054\vartheta^{k-k_0} + 0.7\Omega^{\#}$ and $n\mathfrak{r}_k h^D_k \leq 0.49\vartheta^{k-k_0} + 0.25\Omega^{\#}$;
(iii) $\Delta_k \leq 3.36\vartheta^{k-k_0} + 0.36\Omega^{\#} \leq 3.45\vartheta^{k-k_0} + 1.36\Omega^{\#}$, where $\Omega^{\#} \leq 0.1$, and $\Delta_k - e_k \leq 0.195 \leq 0.34$.

(D3) *End.*

$$\sup_{\mathrm{Pd}(\mathfrak{r}_*/2)} \mathsf{D}^{(\kappa)}_{\mathsf{b}}(u_K(z), \mathfrak{n}_K(z)) \leq (1.15 + 0.46C_5^{\#}K_5^{-1}K)\, C_{\mathsf{b}}\theta^K.$$

At a depth $s \geq 0$ with $K - s > k_0$, that is, with $K - s \geq K_0(L^s\mathsf{b})$, we have $\sup \mathsf{D}^{(\kappa_{K-s})}_{L^s\mathsf{b}} \leq (1.15 + 0.46C_5^{\#}K_5^{-1}K)\, C_{L^s\mathsf{b}}\theta^{K-s}$; there the supremum is that of the distance between the two families over $\mathrm{Pd}(\mathfrak{r}_*/2)$ if $s < j^*$, and that of the distance between the two polynomials over $\mathrm{Pd}(\mathfrak{r}(L^s\mathsf{b})/2)$ if $s \geq j^*$.

(D4) *Vacuum, pairs.* For $k_d < k \leq K$ put $\Delta_k := \sup_{\mathrm{Pd}(\mathfrak{r}_*/2)} \mathsf{D}^{(\kappa)}_{x_k}(u_k(z), \mathfrak{n}_k(z))$. For $k_0 \leq k < K$,

$$|\partial_1 \cdots \partial_n(\mathsf{V}'_{\ell_k}[\mathfrak{n}_k] - \mathsf{V}'_{\ell_k}[u_k])(0)| \leq (2/\mathfrak{r}_{\min(k,k_d)})^n N_1(k)\, 1.2C_g c_a(x_k)\, e^{-\kappa(m_k-1)}\Delta_k,$$

with the polynomials for $k \leq k_d$ and the families after; the sum of the left sides over k is at most $\mathrm{Ser}_2\,(4 + 1.36C_5^{\#}K_5^{-1}K)\,\vartheta^{K-k_0}$.

(D5) *Vacuum, early levels.*

$$\sum_{0\leq k<k_0} |\partial_1 \cdots \partial_n \mathsf{V}'_{\ell_k}[u_k](0)| \leq \mathrm{Tail}(K - k_0)$$

and

$$\sum_{0\leq j<m+k_0} |\partial_1 \cdots \partial_n \mathsf{V}'_{\tilde{\ell}_j}[\tilde{u}_j](0)| \leq \mathrm{Tail}(K - k_0).$$

For every $J \geq j^*$,

$$\mathrm{Tail}(J) \leq 2\,\mathfrak{r}(L^{J+1}\mathsf{b})^{-n} N_1^{[J+1]}\, 2C_g c_a(L^{J+1}\mathsf{b})\, e^{\kappa}(4nL^{1/2})^{-4\kappa n L^{J+1-j^*}}.$$

(D6) *Identity.* Put $S^T_{n;K} := S^T_{n;K,l,L^K\mathsf{b}}(f_1, \ldots, f_n)$ and $S^T_{n;K'} := S^T_{n;K',l,L^{K'}\mathsf{b}}(f_1, \ldots, f_n)$, where the right sides are those of item 1 of §7.3, taken with $l = N$ and with the f_i of item 3. Then

$$\begin{aligned} S^T_{n;K'} - S^T_{n;K} = &\sum_{0\leq j<m+k_0} \partial_1 \cdots \partial_n \mathsf{V}'_{\tilde{\ell}_j}[\tilde{u}_j](0) \\ &- \sum_{0\leq k<k_0} \partial_1 \cdots \partial_n \mathsf{V}'_{\ell_k}[u_k](0) \\ &+ \sum_{k_0\leq k<K} \partial_1 \cdots \partial_n (\mathsf{V}'_{\ell_k}[\mathfrak{n}_k] - \mathsf{V}'_{\ell_k}[u_k])(0) \\ &+ \partial_1 \cdots \partial_n (\log I[\mathfrak{n}_K(z)] - \log I[u_K(z)])(0). \end{aligned}$$

Here the step $0 \to 1$ of each run is the term 0 of its early sum.

(D7) *What the constants depend on.* The numbers j^*, K_T, $\mathfrak{r}_* = \mathfrak{r}(L^{j^*}\mathsf{b})$, Ser_2, $\mathrm{Tail}(J)$, C_{b} and Λ depend only on L, M, E_a, κ, c_R, on the letters A_0, $\mathfrak{p}_0$ of (O8) (through $p_0(x) = A_0(\log x)^{\mathfrak{p}_0}$, which enters amp and c_a), on pure numbers, and on b, n, d, R_f. In detail, j^* depends on L, M, c_R and on b, n, d; Λ and C_{b} depend on L, M, E_a, pure numbers and b; K_T depends on all of these; $\mathfrak{r}_*$ depends in addition on A_0, $\mathfrak{p}_0$, through amp; and Ser_2, $\mathrm{Tail}(J)$ depend in addition on κ, R_f, and once more on p_0, through c_a. None of them depends on N, K',

m, a row or an f_i. Each of them, as a function of b, is bounded on every compact subset of the range $[x_s, \infty)$ of b, and $\mathfrak{r}_*$ is bounded there from below. All numerals of (D1) to (D4) are free of every letter.

■□

Proof of Thm. 7.19. Step 1: the arithmetic of the table. We first collect the following facts on the entries of item 1.3.

(a) Five relations hold:
  (t1) $\vartheta_a u_3(x)/u_3(x/L) = \nu_a = 0.2431$;
  (t2) $u_3(x)K_5/\mathsf{b} = \check{\omega}_3(x)\,\mathrm{amp}(x)$;
  (t3) $5 \le 2^{-1/4}L^{1/2} \le \mathrm{amp}(Lx)/\mathrm{amp}(x) \le L^{1/2}$ for $Lx \ge x_9^U$, and $\mathfrak{r}_{k+1}/\mathfrak{r}_k = \mathrm{amp}(Lx_k)/\mathrm{amp}(x_k)$, $n\mathfrak{r}_{k+1} = 3.5/\mathrm{amp}(x_k)$;
  (t4) $\check{\omega}_3(x_k)/\check{\omega}_3(x_{k_0}) \le L^{(k-k_0)/4}$ for $k \ge k_0$;
  (t5) $c_a(Lx) \le c_a(x)$ for $Lx \ge x_9^U$.

(b) $1 \le k_0 \le K-1$; $K - k_0 > \mathrm{Ex}_K := (4/9)K - (10/9)\Lambda - 1$; $k_0 + 2 \le k_d = K - j^*$; $k'_d = m + k_d$; and partners share every number of the table.

(c) (K1a) and (K1b) hold at every $K \ge K_T$, and $\Omega^{\#} \le 0.1$.

(d) $(K-k_0)/4 - k_0/5 \le -\Lambda/2 < 0$.

(e) $\mathrm{sep}^{[i+1]} \ge L\,\mathrm{sep}^{[i]}$; hence j^* is the least $j \ge 1$ with $\mathrm{sep}^{[j-1]} \ge \mathrm{sep}_*$, it is finite, and $\mathrm{sep}^{[i]} \ge L^{i+1-j^*}\mathrm{sep}_*$ for $i \ge j^* - 1$; moreover $\check{\kappa}_k \le 1/4$ for $k < k_d$.

(f) $N_1^{[j+1]} \le L^3 N_1^{[j]}$.

We prove these facts in turn.

(a) We take the five relations one by one.
  (t1) Since $u_3(x) = C_3 x^{1/4}$, the quotient $u_3(x)/u_3(x/L)$ equals $L^{1/4}$, and this factor is cancelled by $\vartheta_a = 0.2431 L^{-1/4}$.
  (t2) We insert the formulas for amp and for u_3; then b and K_5 cancel.
  (t3) We have $\mathrm{amp}(Lx)/\mathrm{amp}(x) = L^{1/2}p_0(x)/p_0(Lx)$. Since p_0 is increasing, $p_0(x) \le p_0(Lx)$; hence $p_0(x)/p_0(Lx) \le 1$, and the upper bound $L^{1/2}$ follows. For the lower bound we apply (Y.0)(0b) at the point Lx, which is admissible as $Lx \ge x_9^U$; it gives $p_0(x)/p_0(Lx) \ge 2^{-1/4}$. Since $L^{1/2} \ge 10^8$, also $2^{-1/4}L^{1/2} \ge 5$. The two formulas for $\mathfrak{r}$ follow from $\mathfrak{r}(x) = 3.5/(n\,\mathrm{amp}(Lx))$ and $Lx_{k+1} = x_k$.
  (t4) Here $\check{\omega}_3(x) = C_3 E_a p_0(x) x^{-1/4}/c_Q^{\mathrm{su}}$ and $x_k = L^{-(k-k_0)}x_{k_0}$; so the quotient $\check{\omega}_3(x_k)/\check{\omega}_3(x_{k_0})$ is $L^{(k-k_0)/4}$ times $p_0(x_k)/p_0(x_{k_0})$, and the latter factor is at most 1 because $p_0(x_k) \le p_0(x_{k_0})$.
  (t5) By the form of c_a, $c_a(Lx)/c_a(x) = (p_0(Lx)/p_0(x))^3 L^{-1/2} \le 2^{3/4}10^{-8}$, with the two bounds used for (t3); this number is below 1.

(b) Since $K \ge K_0(\mathsf{b})$, Thm. 5.63 (i) gives $1 \le k_0 \le K-1$. From $k_0 < (5K + 10\Lambda)/9 + 1$ we obtain $K - k_0 > \mathrm{Ex}_K$. Together with (K1a) this yields $K - k_0 > j^* + 1$, and as both sides are integers, $k_0 \le K - j^* - 2$. In particular $K > j^*$, so that $k_d = K - j^*$; and $K' \ge K$ gives $k'_d = K' - j^* = m + k_d$. Every entry of item 1.3 that is taken at the level k depends on K and k only through the depth $K - k$, and the level $m + k$ of K' has the same depth $K - k$. The tori, grids, cubes and L-blocks, the x_k, the caps and the domains are the same for partners, by Step 2 (a) of the proof of Thm. 5.63, which involves neither a source nor a support. So partners share every number of the table.

(c) The condition (K1a) is monotone in K. The left side of (K1b) is a constant times Kc^K, where $c = 0.3683^{4/9} \le 0.64151$; passing from K to $K+1$ multiplies it by $(1 + 1/K)c \le (4/3)0.64151 \le 0.8554 < 1$ when $K \ge 3$, and every integer $K \ge K_0(\mathsf{b}) > 9/4$ satisfies $K \ge 3$. Hence each of the two conditions, once it holds, keeps holding for all larger K.

Since $0.3683 < 1$ and $K - k_0 > \mathrm{Ex}_K$, we get $\Omega^{\#} \le C_5^{\#} K_5^{-1} K\, 0.3683^{\mathrm{Ex}_K} \le 0.1$, the last inequality being (K1b).

(d) Since $k_0 \ge (5K + 10\Lambda)/9$,

$$(K - k_0)/4 - k_0/5 = K/4 - (9/20)k_0 \le K/4 - (9/20)(5K + 10\Lambda)/9 = -\Lambda/2,$$

and $\Lambda > 0$ by Thm. 5.63 (i).

(e) By (G1) of item 3 of §7.3, established in Step 1 of the proof of Prop. 7.12, we have $\mathrm{sep}^{[i+1]} \ge L\,\mathrm{sep}^{[i]}$ at each $i \ge 0$, and j^* is a finite number. Consequently the inequality $\mathrm{sep}^{[i]} \ge \mathrm{sep}_* > 0$, once it holds at i, holds at $i+1$ as well, with a gain of the factor L; this gives both the description of j^* and the lower bound $\mathrm{sep}^{[i]} \ge L^{i+1-j^*}\mathrm{sep}_*$ for $i \ge j^* - 1$. By (G2) of item 3 of §7.3, $\check\kappa_k \le 1/(4L) \le 1/4$ for $k < k_d$.

(f) By Step 1 (r) of the proof of Prop. 7.12, $\check r(Lx) \le \check r(x) + \log L/960 + 1 \le L\,\check r(x)$, the second inequality because $\check r \ge 3$. Therefore $2R_f L^j/M + 2\,\check r(L^j \mathsf{b}) + 3 \le L\,(2R_f L^{j-1}/M + 2\,\check r(L^{j-1}\mathsf{b}) + 3)$, and since both sides are positive, raising them to the third power gives (f).

Step 2: the switch for families, the true jets, the reach and the level k_0. In this step we prove (D0) and the parts (a), (b), (c), (d) of (D1), together with (Sep): the sets $(S_i(k) \cup \tilde S_i(k))^{+\check r(x_k)}$, $i = 1, \dots, n$, lie pairwise at distance at least $\mathrm{sep}_k + 2$, counted in cubes, and $(S_1(k+1) \cup \tilde S_1(k+1))^{+\check r(x_{k+1})}$ has at most $N_1(k)$ cells. We also show that $\Delta_{k_0} = 3.91$ bounds $\sup_{\mathrm{Pd}(\mathfrak r_{k_0}/2)} \mathsf{D}^{\mathrm{in},(\kappa_{k_0})}_{x_{k_0}}(u^{\mathrm{pol}}_{k_0}, \mathfrak n_{k_0})$.

Notation. We put $x = x_{k_0}$, $\ell = \ell_{k_0}$, $\tilde\ell = \tilde\ell_{m+k_0}$ and $D = \tilde{\mathsf E}_{m+k_0} - \mathsf E_{k_0}$.

The conditions of Lem. 7.10 at (x, D). Both the requirements of Lem. 7.10 on the lattice and the five conditions of item 5 of §5.4 are satisfied, as item 2 of Step 4 of the proof of Thm. 5.63 shows. The family D is real and symmetric, and its range is at most $\varrho(x)$, because both families of rows are cut at $\varrho(x)$. Finally $\delta_\Delta \le 1$, by Step 3 of the proof of Thm. 5.63 taken at $k = k_0$.

The rate. The map $\mathbb B_D(\tilde u) = (\tilde E^a + E^\Delta[D], \tilde E^h, \tilde w e^Q)$ changes no index set: the term of $\mathbb B_D(\tilde u)$ at X is obtained from the term of $\tilde u$ at the same X; E^Δ lives on $\#X = 1$, and Q depends on V and D alone. In (B3) and (B5) the weight $e^{\tilde\kappa(|X|_c - 1)}$ is a common factor of the two sides. Hence Lem. 7.10 holds, with the same numerals, at every rate $\tilde\kappa$ of $[\kappa, \kappa_1]$; indeed its data admit any rate $\tilde\kappa \ge 0$.

Proof of (D0). At each z of Pd we apply Lem. 7.10 at $A = 1$: the member $\mathfrak n^\circ_{k_0}(z)$ lies in $M_2(x; \kappa_1, c_h)$, by (B1) we have $\rho[\tilde u_{m+k_0}(z); \tilde\ell] = \rho[\mathfrak n^\circ_{k_0}(z); \ell]$ at every field, and by (B5) $\mathfrak n^\circ_{k_0}(z)$ is a family. For $k_0 \le k \le K - 1$, (SD) holds at $(x_k, NL^{K-k-1}, \ell_k)$, by Step 2 (d) of the proof of Thm. 5.63. By Step 2 (a) of the same proof, the step acts linearly on ρ, involves no row, and is the same map for two partners. Applied first at $A = 2$ and then at $A = 1$, clause (a) of Thm. 5.15, which is available through Thm. 6.7 (U) at $p = (\kappa_1, c_h)$, gives

$$\rho^{[K'],z}_{m+k+1} = (\rho^{[K'],z}_{m+k})' = e^{\cdots + \mathsf V'_{\ell_k}[\mathfrak n^\circ_k(z)]}\rho[\mathfrak n^\circ_{k+1}(z); \ell_{k+1}],$$

since $(\ell_k)'^{\mathrm c} = \ell_{k+1}$. By Thm. 7.9 (i) the images are families in M_1 with the strong cap, and the numbers are holomorphic. As for the distance, $\|u_{k_0}(z)\| \le 1$ and $\|\mathfrak n^\circ_{k_0}(z)\| \le 2$, so that the two inputs are at distance at most 3; clause (e) of Thm. 5.15, used through Thm. 6.7 (U) at $p = (\kappa_1, c_h)$ and at each z, then gives $0.4 \cdot 3 = 1.2 \le 3\vartheta$ at the first level and a further factor $0.4 \le \vartheta$ at each later level.

Real z. Now let the further hypothesis of (D0) hold, and let z be real. Then $D = \tilde{\mathsf E} - \mathsf E$ is real, since the rows lie in real Hilbert spaces; by (B8) the member $\mathfrak n^\circ_{k_0}(z)$ is real, and by (B1) $\rho[\mathfrak n^\circ_{k_0}(z); \ell] = \rho[\tilde u_{m+k_0}(z); \tilde\ell] > 0$. From there on, the reality of the members and of the numbers $\mathsf V'$ passes from each level to the next by clause (d) of Thm. 5.15, which is available through Thm. 6.7 (U); the same clause, together with (a), passes on the positivity of the density; this holds for both runs. For the honest families the further hypothesis follows from Thm. 7.1 (a3), (h), the f_i being real.

These arguments are those of Steps 4 and 5 of the proof of Thm. 5.63, carried out at each z, and none of the estimates in those steps depends on the sign of a quantity. The items of §5.5 on members are stated for real members, by way of (R8) and (R11) of its item 3; at a complex z the argument is independent of them. At a complex z we draw on Lem. 7.10, on Thm. 6.7 (U), on Thm. 7.9 (i), and on those items of the proof of Thm. 5.63 that involve no member, namely Step 2 (a) and (d), Step 3, and item 2 of Step 4.

Clause (D1) (a) is proved in Step 1 (b).

For (D1) (b) we apply Prop. 7.12 (J5) to the run K' at its level $m+k_0 \le k'_d$; it gives $n\,\mathfrak{r}_{k_0}\max_i \|\tilde\delta_i^{\mathrm{own}}\|^{\mathrm{st}} \le 0.7$ and $\sum_{|S|\ge 2}\mathfrak{r}_{k_0}^{|S|}\|\tilde\delta_S^{\mathrm{own}}\|^{\mathrm{st},(\kappa_{k_0})} \le 0.21$. Let $|z_i| < \mathfrak{r}_{k_0}/2$ for every i. Then the singles contribute at most $0.7/2$ and the sets with $|S|\ge 2$ at most $0.21/4$, so that $\|\sum_S z^S\tilde\delta_S^{\mathrm{own}}\|^{\mathrm{st},(\kappa_{k_0})} \le 0.7/2 + 0.21/4 = 0.4025 \le 0.455$. Since moreover $\|\bar{\tilde u}_{m+k_0}\|^{\mathrm{st}} \le 1$ by Thm. 7.1 (h), each of the three members of $\tilde u^{\mathrm{pol}}_{m+k_0}(z)$ has strong size at most 1.455. Lem. 7.10, taken with $A = 1.455$ and at the rate κ_{k_0}, now yields $\mathfrak{n}_{k_0}(z) \in M_{2.455}(x;\kappa_{k_0},c_h)$. The map $\mathbb{B}_D$ is affine in $\tilde u$, and its shift $E^\Delta[D]$ as well as its factor e^Q are free of z; consequently $\mathfrak{n}_{k_0}(z)$ is multi-affine in z, and its coefficients are those named in (D1) (b). Its base point is $\bar{\mathfrak{n}}_{k_0} = \mathbb{B}_D(\bar{\tilde u}_{m+k_0})$, an element of $M_2(x;\kappa_1,c_h)$.

For (D1) (c) we start from the honest family $\tilde u_{m+k_0}(z)$ of Thm. 7.1 (h): restricted to the interior of Pd_0 of the run K', it satisfies the hypothesis of (D0). The interior is taken because Pd_0 is closed, while a family is defined on an open set. Thm. 7.1 (h) provides membership in M_1 with the strong cap on Pd_0 at the levels ≥ 1, and holomorphy on the larger open set Pd_0^o; from then on only germs at 0 are used. The family $u_{k_0}(z)$ satisfies the hypothesis on the same set, because Pd_0 of the run K' is contained in Pd_0 of the run K. To see this, note that $\varepsilon(x) = p_0(x)x^{-1/2} = c_Q^{\mathrm{su}}K_5/(\mathsf{b}E_a\,\mathrm{amp}(x))$, so that (t3) gives $\varepsilon(Lx) \le \varepsilon(x)/5$ whenever $Lx \ge x_9^U$. Applying this m times, starting from $x = L^K\mathsf{b}$, we obtain $\varepsilon(L^{K'}\mathsf{b}) \le 5^{-m}\varepsilon(L^K\mathsf{b})$; in this iteration only the points $L^j\mathsf{b}$ occur. The radii of Pd_0 are $1.99c_\nu\mathsf{b}\,\varepsilon(L^K\mathsf{b})/\|f_i\|_\infty$ (item 3 of §7.1).

By definition, the base point and the multi-affine jets of $\tilde u_{m+k_0}(z)$ are $\bar{\tilde u}_{m+k_0}$ and the $\tilde\delta_S^{\mathrm{own}}$ ((o1), (o2) in item 2 of §7.3, for the run K'). Since $\mathbb{B}_D$ is affine, $\mathfrak{n}^\circ_{k_0}(z)$ and $\mathfrak{n}_{k_0}(z)$ have the same base point and the same multi-affine jets. By Lem. 7.8, the coefficient of z^S in $F_\ell(u(z))$ depends only on $u(0)$ and on the $u_{S'}$ with $S'\subset S$. An induction on k therefore shows that $\mathfrak{n}^\circ_k(z)$, $\mathfrak{n}^{\mathrm{pol}}_k(z)$ for $k \le k_d$, and $\mathfrak{n}_k(z)$ for $k \ge k_d$ share the same base point and the same multi-affine jets.

We now identify the base points. The base points $\bar u_{k_0}$ and $\bar{\tilde u}_{m+k_0}$, that is, the values at $z=0$, are real systems of members of the two densities without sources; they lie in M_1 and obey the strong cap (Thm. 7.1 (a3), (h); Thm. 5.15 (d), through Thm. 6.7 (U)). For the short run, Thm. 5.63 (ii) admits every such system. For the long run, Thm. 5.63 (iii) asserts that a system $\mathfrak{n}_k$ exists, without naming one: item 1 of Step 4 of its proof takes $\tilde u$ from (R8) in item 3 of §5.5, which is Thm. 5.7. Yet Steps 4 and 5 of that proof use only three properties of $\tilde u$: it is a real system of members of $\rho^{[K']}_{m+k_0}$, it lies in M_1, and it obeys the strong cap; and (R11) of the same item, which is Lem. 5.47, is stated for every such $\tilde u$. We choose $\tilde u := \bar{\tilde u}_{m+k_0}$. The $u^\flat$ of Lem. 5.47 is then $(\tilde E^a + E^\Delta, \tilde E^h, \tilde w e^{Q_D}) = \mathbb{B}_D(\bar{\tilde u}_{m+k_0})$, by the formula in Lem. 5.47 (B3) that expresses $w^\flat$ in terms of $\tilde w$, by the definition $\mathbb{B}_D(\tilde u) = (\tilde E^a + E^\Delta[D], \tilde E^h, w^\flat)$ with $w^\flat := \tilde w e^{Q_D}$ in Lem. 7.10, and by the closing sentence of Lem. 7.10; it is real by Lem. 5.47 and Lem. 7.10 (B8). As Steps 4 and 5 of the proof of Thm. 5.63 put $\mathfrak{n}_{k_0+1} = F_{\ell_{k_0}}(u^\flat)$ and $\mathfrak{n}_{k+1} = F_{\ell_k}(\mathfrak{n}_k)$, they hold word for word, and $\mathfrak{n}_k = \bar{\mathfrak{n}}_k$. In this paragraph $\mathfrak{n}_{k_0+1}$, $\mathfrak{n}_{k+1}$ and $\mathfrak{n}_k$, written without an argument, denote the real system of Thm. 5.63 (iii) that Steps 4 and 5 of its proof build, not the family $\mathfrak{n}_k(z)$ of (D1) and not its representative. Hence $e_k \le 3\vartheta^{k-k_0}$ for $k_0 < k \le K$, by Thm. 5.63 (iv), at the rate κ_1; Step 5 of the proof of that theorem proves this bound. The properties obtained in this way (reality; membership in M_1; the strong cap; and, for each of these k, the existence of a real number V with $\rho^{[K']}_{m+k} = e^{\mathsf{V}}\rho[\bar{\mathfrak{n}}_k;\ell_k]$ at every field, the positivity of this density at every field, and the bound $\mathsf{D}_{x_k}(\bar u_k,\bar{\mathfrak{n}}_k) \le 3\vartheta^{k-k_0}$) are (D0) at $z=0$.

We prove (D1)(d). At k_0 the clause comes from Prop. 7.12 (J2), applied to the run K' at its level $m+k_0$, where the blocks are the $\tilde S_i(k_0)$: the members of $\mathbb{B}_D(\tilde u) = (\tilde E^a + E^\Delta[D], \tilde E^h, \tilde w e^{Q_D})$ at an index set X are computed from those of $\tilde u$ at the same X, the partner levels have the same cubes, the same $\check r$ and the same sep, and (Sep), proved below, holds. For the passage from k to $k+1$, assume that $\mathfrak{n}_k^{\mathrm{pol}}(z)$ can depend on z_i at X only when $X^{+\check r(x_k)}$ meets $\tilde S_i(k)$. Thm. 6.7 (f6) then makes the mixed derivative over $i \in S$ of $F_{\ell_k}(\mathfrak{n}_k^{\mathrm{pol}})(Y)$ equal to zero unless $Y^{+\hat\rho'}$ meets each set $\mathrm{cl}(\tilde S_i(k)) = \tilde S_i(k+1)$, see (o3) in item 2 of §7.3 for the run K'; and $\hat\rho' = \lceil \check r(x_k)/L\rceil + 1 \le \check r(x_{k+1})$ by Step 1 (R) of the proof of Prop. 7.12.

We prove (Sep) and measure lengths in the units of T_l. In these units the site y of $\mathbb{T}_k$ lies at $L^{k-K}y$, and every bare site of either run that belongs to the block of y is at distance at most $(L^{k-K} - \mathrm{sp})/2$ from it in each coordinate; here $\mathrm{sp} = a_K$ or $a_{K'}$ is the spacing of the lattice of that run, and we use that the blocks are centred (item 1 of §6.3). The origins of the two grids agree (Step 2 (a) of the proof of Thm. 5.63). Consequently a cube of $\mathbb{T}_k$ is, for both runs, one and the same box of side $\mathrm{sd} := ML^{k-K}$, and its bare sites are at distance at least $\mathrm{sp}/2$ from each of its walls.

Now let cb be a cube in $S_i(k)\cup\tilde S_i(k)$ and cb′ a cube in $S_{i'}(k)\cup\tilde S_{i'}(k)$, where $i \ne i'$, at the distance dc in cubes; all distances are periodic sup distances. Let ξ, ξ' be base sites in cb and cb′ of plaquettes p, p' with $f_i(x(p)) \ne 0$ and $f_{i'}(x(p')) \ne 0$, and let $\mathrm{sp}, \mathrm{sp}'$ be their spacings. In each coordinate $|\xi - \xi'| \le (\mathrm{dc}+1)\mathrm{sd} - \mathrm{sp}/2 - \mathrm{sp}'/2$, while $x(p) - \xi$ is 0 or $\mathrm{sp}/2$ and $x(p') - \xi'$ is 0 or $\mathrm{sp}'/2$. Since the sources are separated by d, we get $d \le |x(p) - x(p')| \le (\mathrm{dc}+1)\mathrm{sd} - \min\{\mathrm{sp}, \mathrm{sp}'\}/2 < (\mathrm{dc}+1)\mathrm{sd}$, that is, $\mathrm{dc} \ge \lfloor dL^{K-k}/M \rfloor$. Passing to the neighbourhoods costs $2\check r(x_k)$, so the enlarged sets are at distance at least $\mathrm{sep}_k + 2$, as in (G4) in item 3 of §7.3.

Count. In each coordinate, the base sites belonging to f_1, in the two runs taken together, are contained in a single interval of length $2R_f + a_K$. Such an interval meets at most $(2R_f + a_K)L^{K-k-1}/M + 2 \le 2R_f L^{K-k-1}/M + 3$ cells of $\mathbb{T}_{k+1}$, and the neighbourhood adds $2\check r(x_{k+1})$; this gives the bound $N_1(k)$. For the lower bound in (D1)(d) on $\#X$ or $|X|_c$ of a set X that meets $|S| \ge 2$ of the enlarged sets, Step 1 (C) of the proof of Prop. 7.12, applied with $\sigma = \mathrm{sep}_k + 2$, shows that any connected set of cubes meeting $|S| \ge 2$ of these sets contains at least $|S|\,\mathrm{sep}_k/2$ cubes. If $X \in \mathcal{P}$, then $|X|_c = \#Z$ with Z a connected set of cubes containing X (item 2 of §5.5), and Z meets the sets that X meets.

Finally we bound the majorant at the level k_0. As $\bar u_{k_0} \in M_1$ and $\bar{\mathfrak{n}}_{k_0} \in M_2$, the base points satisfy $\|\bar u_{k_0} - \bar{\mathfrak{n}}_{k_0}\| \le 1 + 2 = 3$. On $\mathrm{Pd}(\mathfrak{r}_{k_0}/2)$, Prop. 7.12 (J5) for the run K gives $\|\sum z^S \delta_S(k_0)\|^{\mathrm{st},(\kappa_{k_0})} \le 0.455$. For the jets of $\mathfrak{n}_{k_0}$ we have $\|\sum z^S \tilde\delta_S(k_0)\|^{(\kappa_{k_0})} \le 0.455$ in the weak norm, because $|\tilde w_S e^Q|_w^{\mathrm{wk}} \le |\tilde w_S|_w^{\mathrm{st}}$ by Lem. 7.10 (B5). The three bounds add up to 3.91, which is the number Δ_{k_0}.

Step 3: the singles in the a-channel. The singles are not carried by the contraction of §5.5: on the radii $\mathfrak{r}_k$ (item 1.3; (o5) in item 2 of §7.3), for which (t3) allows $\mathfrak{r}_{k+1}/\mathfrak{r}_k$ up to $L^{1/2}$, the factor $\vartheta[2]$ of Thm. 7.9 (ii) times this growth is $\vartheta[2]L^{1/2} > 1$, and the factor ϑ_a of Thm. 6.7 (h3) times it is $\vartheta_a L^{1/2} = 0.2431 L^{1/4} > 1$.

We put $\hat a_k := a_k^D\,\mathsf{b}/(u_3(x_k)K_5)$ and show that $\hat a_{k_0} \le 4 + \Omega_0$ and that $\hat a_{k+1} \le \nu_a \hat a_k + c_e e_k$ for $k_0 \le k < k_d$, where $c_e = 2\nu_a/14 + 1/3 \le 0.36807$ at $k = k_0$ and $c_e = 2\nu_a/15 + 1/3 \le 0.36575$ for the later k. These bounds yield (D2)(i). Writing $\Omega_k := \nu_a^{k-k_0}\check\omega_3(x_k)\Omega_0$, we show further that $\Omega_k \le C_5^{\#} K_5^{-1} K\, \nu_a^{k-k_0} L^{-(K-k)/4} \le \Omega^{\#}$ and that $n\,\mathfrak{r}_k a_k^D \le 0.7\,\check\omega_3(x_k)\hat a_k \le 0.0054\,\vartheta^{k-k_0} + 0.7\,\Omega^{\#}$.

We fix i and write $\mathbb{D}_k = \mathbb{D}_k^{[K]}[f_i]$ for the forms of the short run; both runs are decomposed along these forms. Let $\mathrm{rem}_k := \delta_i^a(k) - s_{x_k}[\mathbb{D}_k/\mathsf{b}]$, which is the $\mathrm{rem}_{k,i}$ of Prop. 7.12 (J3), and $\widetilde{\mathrm{rem}}_k := \tilde\delta_i^a(k) - s_{x_k}[\mathbb{D}_k/\mathsf{b}]$. Then $a_k^D = \max_i |\mathrm{rem}_k - \widetilde{\mathrm{rem}}_k|_a$. All these elements lie in $T^a(x_k)$: the jets by Lem. 7.7, and $s_x[D] = -E^\Delta[D]$ by Lem. 7.10.

We derive the recursion. By Thm. 6.7 (f1), E'^{new} depends on no member, and $A = E'^{\mathrm{old}}$ depends on E^a alone. Taking the coefficient of z_i we find $\delta_i^a(k+1) = DA_{\ell_k}(\bar u_k)[\delta_i^a(k)]$ and $\tilde\delta_i^a(k+1) =$

$DA_{\ell_k}(\bar{\mathfrak{n}}_k)[\tilde{\delta}_i^a(k)]$. Thm. 6.33 applies with (SD) $= (x_k, \ell_k)$, as $x_k \geq x_3$, at $u = \bar{u}_k$ and at $u = \bar{\mathfrak{n}}_k$, which both lie in M_2 and so inside M_8, and with $D = \mathbb{D}_k/\mathsf{b}$, which is symmetric and of range $\leq \varrho(x_k)$ by Thm. 6.20. It yields

$$DA_{\ell_k}(u)[s_{x_k}[\mathbb{D}_k/\mathsf{b}]] = s_{x_{k+1}}[\mathbb{D}_{k+1}/\mathsf{b}] + n[u; \mathbb{D}_k/\mathsf{b}];$$

here $D' = \mathrm{cut}_{\varrho(x_{k+1})}(T_{\ell_k}\mathbb{D}_k)/\mathsf{b} = \mathbb{D}_{k+1}/\mathsf{b}$ is determined by the rows ℓ_k alone and is independent of u. After the switch the two runs use one row family, so the same $s_{x_{k+1}}[\mathbb{D}_{k+1}/\mathsf{b}]$ is subtracted from both. Since $DA_{\ell_k}(u)$ is linear, it follows that $\mathrm{rem}_{k+1} = DA_{\ell_k}(\bar{u}_k)[\mathrm{rem}_k] + n[\bar{u}_k; \mathbb{D}_k/\mathsf{b}]$ and $\widetilde{\mathrm{rem}}_{k+1} = DA_{\ell_k}(\bar{\mathfrak{n}}_k)[\widetilde{\mathrm{rem}}_k] + n[\bar{\mathfrak{n}}_k; \mathbb{D}_k/\mathsf{b}]$, and subtraction gives

$$\mathrm{rem}_{k+1} - \widetilde{\mathrm{rem}}_{k+1} = DA_{\ell_k}(\bar{\mathfrak{n}}_k)[\mathrm{rem}_k - \widetilde{\mathrm{rem}}_k] + (DA_{\ell_k}(\bar{u}_k) - DA_{\ell_k}(\bar{\mathfrak{n}}_k))[\mathrm{rem}_k] + (n[\bar{u}_k; \mathbb{D}_k/\mathsf{b}] - n[\bar{\mathfrak{n}}_k; \mathbb{D}_k/\mathsf{b}]).$$

We now bound the three terms at the rate κ_1, with the caps at x_k on the input side and the caps at x_{k+1} on the output side; here $B = 2$ at $k = k_0$ and $B = 1$ for the later k. A sum over cells dominates the supremum over Y. By Thm. 6.7 (h3) at $u = \bar{\mathfrak{n}}_k$, the first term is at most $\vartheta_a a_k^D$.

For the second term we fix $g \in T^a(x_k)$ and consider $\Phi(u) := DA_{\ell_k}(u)[g]$, with values in the space that carries the cell-sum seminorm. By (h3), Φ is bounded there by $\vartheta_a|g|_a$ caps on $\{\|u\| < 16\}$. It is holomorphic along affine lines because, by Thm. 6.7 (h), $DA(u)[g]$ is holomorphic along families. On the open ball this can also be seen directly, since there $DA_{\ell_k}(u)[g]$ is, by (h), the t-derivative of $A(u + tg)$ at 0: the map $(s, t) \mapsto A(u + sg' + tg)$ is holomorphic by Thm. 6.7 (c+), hence the t-derivative at 0 depends holomorphically on s. Lem. 7.5 (b) and Prop. 7.12 (J3) now bound the second term by

$$\vartheta_a|\mathrm{rem}_k|_a\|\bar{u}_k - \bar{\mathfrak{n}}_k\|/(16 - B) \leq \vartheta_a\,(2u_3(x_k)K_5/\mathsf{b})\,e_k/(16 - B).$$

In the third term the part n^{fix} depends on no member, so it cancels. By Thm. 6.33 (d) on M_2, the difference of the parts n^{mem} is at most 1/3 of their bound in (a) times the a-part of D^{in}, which does not exceed e_k. Call that bound B_{mem}; it is a part of the sum that (a) bounds by $u_3(x_{k+1})\|D\|$. Indeed, (Z.4), in item 1 of §6.4, implies $18.2L^3c_a(x) \leq e^{\kappa+3}L^3x^{-1/4}$, hence B_{mem} is at most the term $1001e^{\tilde{\kappa}+3}L^3x^{-1/4}\sigma_D$ of the last sentence of Thm. 6.33 (a). As $\|\mathbb{D}_k/\mathsf{b}\| \leq K_5/\mathsf{b}$, the third term is at most $(1/3)u_3(x_{k+1})(K_5/\mathsf{b})e_k$. Dividing the three bounds by $u_3(x_{k+1})K_5/\mathsf{b}$ and using (t1), we arrive at $\hat{a}_{k+1} \leq \nu_a\hat{a}_k + (2\nu_a/(16 - B) + 1/3)e_k$, which is the recursion of the claim.

Start. At the start, the a-jets pass through $\mathbb{B}_D$ unchanged, by (D1) (b). Prop. 7.12 (J3) for the run K' gives $\tilde{\delta}_i^a(k_0) = s_x[\tilde{\mathbb{D}}_{m+k_0}/\mathsf{b}] + \widetilde{\mathrm{rem}}^{\mathrm{own}}$ with $|\widetilde{\mathrm{rem}}^{\mathrm{own}}|_a \leq 2u_3K_5/\mathsf{b}$, where u_3 and amp are taken at $x = x_{k_0}$. Therefore $\mathrm{rem}_{k_0} - \widetilde{\mathrm{rem}}_{k_0} = \mathrm{rem}_{k_0} - \widetilde{\mathrm{rem}}^{\mathrm{own}} - s_x[(\tilde{\mathbb{D}}_{m+k_0} - \mathbb{D}_{k_0})/\mathsf{b}]$. Lem. 7.10 (B2) gives $|s_x[D]|_a \leq c_Q^{\mathrm{su}}p_0^2\|D\|/c_a(x) = (\mathrm{amp}(x)\mathsf{b}/K_5)\|D\|$; since this bound is homogeneous and s_x is linear in D, it holds also when $\delta_\Delta > 1$. The terms rem_{k_0} and $\widetilde{\mathrm{rem}}^{\mathrm{own}}$ contribute $2u_3K_5/\mathsf{b}$ each, by (J3) for the two runs, and we obtain $a_{k_0}^D \leq (\mathrm{amp}/K_5)\|\tilde{\mathbb{D}}_{m+k_0} - \mathbb{D}_{k_0}\| + 4u_3K_5/\mathsf{b}$; by (t2) this is $\hat{a}_{k_0} \leq \Omega_0 + 4$.

Solution. Recall that $e_{k_0+i} \leq 3\vartheta^i$ for $i \geq 0$. As both values of c_e are at most 0.36807, for $j \geq 1$ we have

$$\hat{a}_{k_0+j} \leq \nu_a^j(4 + \Omega_0) + 3 \cdot 0.36807 \sum_{i<j} \nu_a^{j-1-i}\vartheta^i \leq \nu_a^j\Omega_0 + [4(\nu_a/\vartheta)^j + 1.10421\,\vartheta^{-1}/(1 - \nu_a/\vartheta)]\vartheta^j.$$

Here $\nu_a/\vartheta \leq 0.54410$, $1/(1 - \nu_a/\vartheta) \leq 2.19343$ and $1.10421 \cdot 2.19343/0.4468 \leq 5.4208$, while $4 \cdot 0.54410 = 2.1764$; the sum $7.5972 \leq 7.6$ bounds the bracket.

In units of the radius, (t2) and (t3) give $n\,\mathfrak{r}_k a_k^D = 3.5\,(\mathrm{amp}(x_k)/\mathrm{amp}(Lx_k))\,\breve{\omega}_3(x_k)\hat{a}_k \leq 0.7\,\breve{\omega}_3(x_k)\hat{a}_k$. By (W3), the part with ϑ is at most $0.7 \cdot 10^{-3} \cdot 7.6 = 0.00532$ times ϑ^{k-k_0}, and the form part is $0.7\,\Omega_k$. By (t4), by Thm. 6.21 (S3c) at k_0 (with $\|f_i\|_\sharp \leq 1$, $k_0 \leq K$ and $\vartheta_5 \leq L^{-1/5}$) and by Step 1 (d),

$$\Omega_k = \nu_a^{k-k_0}(\breve{\omega}_3(x_k)/\breve{\omega}_3(x_{k_0}))\|\tilde{\mathbb{D}} - \mathbb{D}\|/K_5 \leq C_5^{\#}K_5^{-1}K\,\nu_a^{k-k_0}L^{(k-k_0)/4-k_0/5},$$

and $(k-k_0)/4 - k_0/5 = (K-k_0)/4 - k_0/5 - (K-k)/4 \le -(K-k)/4$. Since $\nu_a \le 0.3683$ and $L^{-1/4} \le 10^{-4} \le 0.3683$, it follows that $\Omega_k \le \Omega^{\#}$, also at $k = k_0$.

Step 4: the singles in the h- and w-channels. Let $\hat h_k := n\,\mathfrak{r}_k h_k^D$. We show that $\hat h_{k_0+1} \le 0.2158 + 0.25\,\Omega^{\#}$ and that, for $k_0 < k < k_d$,

$$\hat h_{k+1} \le (1/15)\big(3.5\,\breve\omega_3(x_k)\hat a_k + L^{-1/2}\hat h_k\big) + (e_k/56.25)\big(3.5(1+2\breve\omega_3(x_k)) + 0.42L^{-1/2}\big);$$

consequently $\hat h_k \le 0.49\,\vartheta^{k-k_0} + 0.25\,\Omega^{\#}$ for $k_0 < k \le k_d$.

Taking the coefficient of z_i in the h- and w-members, we get $\delta_i^{hw}(k+1) = DF_{\ell_k}^{hw}(\bar u_k)[\delta_i(k)]$ and $\tilde\delta_i^{hw}(k+1) = DF_{\ell_k}^{hw}(\bar{\mathfrak{n}}_k)[\tilde\delta_i(k)]$. Their difference splits as

$$DF_{\ell_k}^{hw}(\bar{\mathfrak{n}}_k)[\delta_i - \tilde\delta_i] + \int_0^1 D^2F_{\ell_k}^{hw}(u_\tau)[\bar u_k - \bar{\mathfrak{n}}_k, \delta_i]\,d\tau, \qquad u_\tau = \bar{\mathfrak{n}}_k + \tau(\bar u_k - \bar{\mathfrak{n}}_k),$$

and u_τ lies in the convex set M_B, with B as in Step 3.

In the mixed entry $D^2F_{\ell_k}^{hw}(u_\tau)[g_1,g_2]$ the first argument $g_1 = \bar u_k - \bar{\mathfrak{n}}_k$ is a whole triple. Thm. 7.9 (iv) covers this case. In that clause an argument g that is a mixed triple is measured by $\|g\|$ at p if it is left whole, and by the sum $|g^a|_a + \eta_h|g^h|_h + |g^{w\circ}|_w^{\mathrm{wk}}$ if it is split, where $g^{w\circ} := g^w 1\{X \text{ holds a plaquette of } D(V)\}$. We keep g_1 whole, with $\|g_1\| \le e_k$, and split only $g_2 = \delta_i$. If $g_2 = (g^a,0,0)$, we use Lem. 7.5 (a) with $j=2$ and $A_{\#} = 16$ in the norm with the pair $p = (\kappa_1, c_h)$, where m is one cap of E'^h for the output E'^h and the strong cap of w' for the output w' (Thm. 6.7 (U); Thm. 5.15 (b)); this gives the bound $4\|g_1\||g^a|_a/(16-B)^2$. If $g_2 = (0,0,g^w)$, then Thm. 6.7 (f4) allows us to replace g^w by $g^{w\circ}$, and the same lemma applies, now with $|g^{w\circ}|_w^{\mathrm{wk}} \le \eta_w|g^w|_w^{\mathrm{st}}$ by Thm. 7.9 (iv). If $g_2 = (0,g^h,0)$, we use the same lemma in the norm with $(\kappa_1, c_h/2)$, which is a pair of $\mathbb{P}$. At this pair neither the norm of g_1 nor that of u_τ grows, since the unit of h is larger; the size of g^h is $e^{-(c_h/2)p_0(x)^2}|g^h|_h$; and to bring E'^h back to the unit $c_h(x')$ costs $e^{(c_h/2)p_0(x')^2}$. These factors together give η_h, while for w' the bound holds as it stands: nothing needs to be brought back to that unit, in which only E'^h is measured. Summing the three cases,

$$|D^2F_{\ell_k}^{hw}(u_\tau)[g_1,g_2]| \le (4/(16-B)^2)\|g_1\|(|g_2^a|_a + \eta_h|g_2^h|_h + \eta_w|g_2^w|_w^{\mathrm{st}}).$$

First let $k > k_0$. We use the channels of Thm. 7.9 (iv) with $B=1$ and the rate κ_1, with strong units on the output side. The first term is then at most $(a_k^D + (\eta_h+\eta_w)h_k^D)/15$, and the second term at most $(4/225)e_k(|\delta_i^a|_a + (\eta_h+\eta_w)\max\{|\delta_i^h|_h, |\delta_i^w|_w^{\mathrm{st}}\})$. Here $|\delta_i^a(k)|_a \le \mathrm{amp}(x_k)(1+2\breve\omega_3(x_k))$, by Prop. 7.12 (J3), Lem. 7.10 (B2), $\|\mathbb{D}_k\| \le K_5$ and (t2); and the maximum is at most $0.12\,\mathrm{amp}(Lx_k)$ by Prop. 7.12 (J4). We multiply by $n\,\mathfrak{r}_{k+1} = 3.5/\mathrm{amp}(x_k)$. Then $n\,\mathfrak{r}_{k+1}a_k^D = 3.5\,\breve\omega_3(x_k)\hat a_k$, $n\,\mathfrak{r}_{k+1}h_k^D = (\mathfrak{r}_{k+1}/\mathfrak{r}_k)\hat h_k \le L^{1/2}\hat h_k$, and $n\,\mathfrak{r}_{k+1}\,0.12\,\mathrm{amp}(Lx_k) \le 0.42L^{1/2}$. Finally (W2) at x_k turns $L^{1/2}(\eta_h+\eta_w)$ into $L^{-1/2}$, and $225/4 = 56.25$; this is the recursion of the claim.

Now let $k = k_0$, so that $B = 2$ and $e_{k_0} = 3$; we write $x = x_{k_0}$. For the w-jets of $\mathfrak{n}_{k_0}$ our bound is in the weak norm, so we use the channel $w \to h,w$ of Thm. 7.9 (iv) in its first form, $\le |g^{w\circ}|_w^{\mathrm{wk}}/(16-B)$. The same clause gives $|\delta_i^{w\circ}|_w^{\mathrm{wk}} \le \eta_w|\delta_i^w|_w^{\mathrm{st}}$, and Lem. 7.10 (B5) gives $|(\tilde\delta_i^{\mathrm{own},w}e^Q)^\circ|_w^{\mathrm{wk}} \le e^{1/2}\eta_w|\tilde\delta_i^{\mathrm{own},w}|_w^{\mathrm{st}}$. Hence the w-input contributes at most $(1+e^{1/2})\eta_w\,0.12\,\mathrm{amp}(Lx)/14$, and the h-input at most $\eta_h\,0.24\,\mathrm{amp}(Lx)/14$ (Thm. 7.9 (iv)). Altogether this is at most $0.24\,\mathrm{amp}(Lx)(\eta_h + e^{1/2}\eta_w)/14 \le 0.24\,\mathrm{amp}(Lx)/(14L)$. The a-input contributes $a_{k_0}^D/14$, and $3.5\,\breve\omega_3(x_{k_0})\hat a_{k_0} \le 3.5(4\cdot10^{-3} + \Omega_{k_0}) \le 0.014 + 3.5\,\Omega^{\#}$. The second term is $(4/196)\cdot 3$ times the same bracket as for $k > k_0$. Therefore

$$\hat h_{k_0+1} \le (1/14)(0.014 + 3.5\,\Omega^{\#} + 0.84L^{-1/2}) + (12/196)(3.5\cdot 1.002 + 0.42L^{-1/2}) \le 0.001 + 0.25\,\Omega^{\#} + 0.21472 \le 0.2158 + 0.25\,\Omega^{\#},$$

and $0.2158 \le 0.48300\,\vartheta \le 0.49\,\vartheta$.

For the induction, let $\hat h_k \le 0.49\,\vartheta^j + 0.25\,\Omega^{\#}$ with $j = k - k_0 \ge 1$. By Step 3, $\breve\omega_3(x_k)\hat a_k \le 7.6\cdot10^{-3}\vartheta^j + \Omega_k$. In the recursion the ϑ-part is at most

$$(3.5/15)\,7.6\cdot10^{-3} + 3\cdot 3.507/56.25 + 10^{-8} \le 0.00178 + 0.18704 + 10^{-8} \le 0.18883 \le 0.42263\,\vartheta$$

times ϑ^j, and the Ω-part is at most $(3.5/15)\Omega^{\#} + 10^{-8}\Omega^{\#} \le 0.2334\,\Omega^{\#} \le 0.25\,\Omega^{\#}$; so the bound passes from k to $k+1$.

The recursion closes because the system is triangular. A single has an a-part smaller than $1/\mathfrak{r}_k$ by the factor $L^{-1/2}$, because the radius $\mathfrak{r}_k$ is defined in terms of $\mathrm{amp}(Lx_k)$; the factor $1/L$ is carried by the channels $h, w \to h, w$; and by Thm. 6.7 (f1) the channels $h, w \to a$ vanish.

Step 5: the sets with $|S| \ge 2$ on the growing radii. Fix k with $k_0 \le k < k_d$, and suppose that u_k^{pol} and $\mathfrak{n}_k^{\mathrm{pol}}$ are families in $M_B(x_k; \kappa_k, c_h)$, where $B \le 2.6$, on the polydisc $\mathrm{Pd}(\mathfrak{r}_k/2)$. We claim that

$$\sum_{|S|\ge 2} (\mathfrak{r}_{k+1}/2)^{|S|} \|\delta_S(k+1) - \tilde\delta_S(k+1)\|^{\mathrm{st},(\kappa_{k+1})} \le c_B \Delta_k, \qquad c_B = 0.0436895\,\vartheta[B];$$

in particular $c_2 \le 0.012383$ and $c_{2.455} \le 0.012799$.

By the triangle inequality,

$$\sup_{\mathrm{Pd}(\mathfrak{r}_k/2)} \mathsf{D}^{\mathrm{in},(\kappa_k)}(u_k^{\mathrm{pol}}, \mathfrak{n}_k^{\mathrm{pol}}) \le e_k + \sum_S (\mathfrak{r}_k/2)^{|S|} \|\delta_S - \tilde\delta_S\|^{(\kappa_k)} \le \Delta_k;$$

here the n singles contribute $(n\,\mathfrak{r}_k/2)\max\{a_k^D, h_k^D\}$, norms at the rate κ_1 dominate those at κ_k, and at k_0 the bound is that of Step 2. For $k > k_0$, the same bound with strong norms is the first claim of (D2). Thm. 7.9 (ii) with the pair $p = (\kappa_k, c_h) \in \mathbb{P}$ gives $\sup \mathsf{D}^{(\kappa_k)}_{x_{k+1}}(F_{\ell_k} u_k^{\mathrm{pol}}, F_{\ell_k}\mathfrak{n}_k^{\mathrm{pol}}) \le \vartheta[B]\Delta_k$. Lem. 7.7 in the norm $\|\cdot\|^{\mathrm{st},(\kappa_k)}$ at x_{k+1}, together with Prop. 7.12 (J1) and the definition of $\tilde\delta_S(k+1)$, then yields $\|\delta_S(k+1) - \tilde\delta_S(k+1)\|^{\mathrm{st},(\kappa_k)} \le \vartheta[B]\Delta_k(\mathfrak{r}_k/2)^{-|S|}$.

Prop. 7.12 (J2) and (D1) (d) show that the a- and h-parts of both jets are zero at Y unless $\#Y$ is at least $|S|\,\mathrm{sep}_{k+1}/2$, and their w-parts unless $|Y|_c$ is at least this number; therefore the passage to the rate $\kappa_{k+1} = \kappa_k - \check\kappa_k$ gains the factor $e^{-\check\kappa_k(|S|\mathrm{sep}_{k+1}/2-1)}$. With (t3),

$$q := (\mathfrak{r}_{k+1}/\mathfrak{r}_k) e^{-\check\kappa_k \mathrm{sep}_{k+1}/2} \le L^{1/2}/(4nL^{1/2}) = 1/(4n),$$

the factor $1/2$ of the radius cancelling in the quotient; and $e^{\check\kappa_k} \le e^{1/4}$ by Step 1 (e). Multiplying the above bound on $\|\delta_S(k+1) - \tilde\delta_S(k+1)\|^{\mathrm{st},(\kappa_k)}$ by $(\mathfrak{r}_{k+1}/2)^{|S|}$ and by the gain, and summing over the sets S with $|S| \ge 2$, we find that the sum of the claim is at most $\vartheta[B]\Delta_k e^{1/4}[(1+q)^n - 1 - nq]$.

Since $q \le 1/(4n)$, the bracket is at most $(1 + 1/(4n))^n - 5/4$, which increases with n towards $e^{1/4} - 5/4 \le 0.0340255$. The product of the latter, $e^{1/4}(e^{1/4} - 5/4)$, has the decimal expansion $0.0436894998\ldots$ and is therefore a number ≤ 0.0436895. Finally, $0.0436895\,\vartheta[2] \le 0.012383$ and $0.0436895\,\vartheta[2.455] \le 0.012799$.

Step 6: the induction closes; (D1) *(e) and* (D2). We claim: when $k_0 < k \le k_d$, the two polynomials belong to $M_2(x_k; \kappa_k, c_h)$ on $\mathrm{Pd}(\mathfrak{r}_k/2)$, $\Delta_k \le 3.36\,\vartheta^{k-k_0} + 0.36\,\Omega^{\#}$, and $\Delta_k - e_k \le 0.195$.

Let $\check\Sigma_k := (1/2)\max\{n\,\mathfrak{r}_k a_k^D, \hat h_k\}$. Neither Step 3 nor Step 4 uses the membership of a polynomial u_k^{pol} or $\mathfrak{n}_k^{\mathrm{pol}}$, so the bounds of these two steps can be used in the induction below. From Steps 3 and 4 we obtain $\check\Sigma_k \le \max\{0.0027\,\vartheta^j + 0.35\,\Omega^{\#},\ 0.245\,\vartheta^j + 0.125\,\Omega^{\#}\} \le 0.245\,\vartheta^j + 0.35\,\Omega^{\#}$, where $j = k - k_0 \ge 1$; and $e_k \le 3\vartheta^j$ by Step 2.

For the first step, Step 2 gives $u_{k_0}^{\mathrm{pol}} \in M_{1.455}$, $\mathfrak{n}_{k_0} \in M_{2.455}$ and $\Delta_{k_0} = 3.91$. Step 5 with $B = 2.455$ bounds the sets with $|S| \ge 2$ at $k_0 + 1$ by $0.012799 \cdot 3.91 = 0.05004409$; this is at most $0.112006\,\vartheta = 0.05004428\ldots$, and at most 0.050045. Hence $\Delta_{k_0+1} \le (3 + 0.245 + 0.112006)\vartheta + 0.35\,\Omega^{\#} = 3.357006\,\vartheta + 0.35\,\Omega^{\#}$.

Membership at a level $k > k_0$ follows from the bound at k:

$$\|\mathfrak{n}_k^{\mathrm{pol}}(z)\|^{\mathrm{st},(\kappa_k)} \le \|\bar{\mathfrak{n}}_k\|^{\mathrm{st}} + \|\sum z^S \delta_S(k)\|^{\mathrm{st},(\kappa_k)} + \|\sum z^S(\tilde\delta_S - \delta_S)(k)\|^{\mathrm{st},(\kappa_k)} \le 1 + 0.455 + (\Delta_k - e_k);$$

the three bounds come from (D1) (c) (Thm. 5.63 (iii)), from Prop. 7.12 (J5) and from the definition of Δ_k. Here $\Delta_k - e_k = \check\Sigma_k + (\text{the sets } |S| \ge 2) \le 0.245 \cdot 0.4468 + 0.35 \cdot 0.1 + 0.050045 \le 0.19452$, for the

last summand is largest at k_0+1: at the later levels it is at most $0.012383(3.36\,\vartheta+0.036)\le 0.01904$. Hence the norm is at most $1.65\le 2$; for u_k^{pol} it is at most 1.455.

For the step from k to $k+1$, with $k_0<k<k_d$, Step 5 with $B=2$ gives

$$\begin{aligned}\Delta_{k+1}&\le e_{k+1}+\check\Sigma_{k+1}+0.012383\,\Delta_k\\&\le[3.245+0.012383\cdot 3.36/0.4468]\vartheta^{j+1}\\&\quad+[0.35+0.012383\cdot 0.36]\Omega^{\#}\\&\le 3.338122\,\vartheta^{j+1}+0.354458\,\Omega^{\#}.\end{aligned}$$

Both coefficients are below 3.36 and 0.36, respectively. The first bound of (D2) (iii), $\Delta_k\le 3.36\,\vartheta^{k-k_0}+0.36\,\Omega^{\#}$, therefore holds; it implies the second bound of (D2) (iii), because $3.36\le 3.45$ and $0.36\le 1.36$. The last clause of (D2) (iii) follows from $\Delta_k-e_k\le 0.195\le 0.34$. For (D1) (e), let $k_0<k\le k_d$. On $\mathrm{Pd}(\mathfrak r_k/2)$ the difference $\mathfrak n_k^{\mathrm{pol}}(z)-\bar{\mathfrak n}_k$ is the sum of the two quantities whose norms are the second and third summands in the display above that gives membership, so that $\|\mathfrak n_k^{\mathrm{pol}}(z)-\bar{\mathfrak n}_k\|^{\mathrm{st},(\kappa_k)}\le 0.455+(\Delta_k-e_k)\le 0.455+0.195\le 0.455+0.34$; there, moreover, $\|u_k^{\mathrm{pol}}(z)-\bar u_k\|^{\mathrm{st},(\kappa_k)}\le 0.7/2+0.21/4\le 0.455$ by Prop. 7.12 (J5). Both polynomials belong to $M_2(x_k;\kappa_k,c_h)$, as shown after that display.

Step 7: the late levels; (D1) *(f),* (D3) *and the depth s.* We prove (D1) (f), the bound $\Delta_k\le\vartheta^{k-k_d}\Delta_{k_d}$ for $k_d\le k\le K$, and (D3) both at $s=0$ and at each depth s for which $K-s>k_0$. The rate at k_d equals $\kappa_{k_d}=\kappa$, and by Step 6, as $k_d>k_0$, the two polynomials are families in $M_2(x_{k_d};\kappa,c_h)$ on $\mathrm{Pd}(\mathfrak r_*/2)$. The hypothesis (SD) holds at every level $k<K$, and $k_d\le K-1$. Thm. 7.9 (i) shows that the images belong to M_1, as families with the strong cap, at every level up to K, each of them being an output of F. Thm. 7.9 (ii), first with $B=2$ and then with $B=1$, gives at each level a factor $\vartheta[2]\le 0.28343\le\vartheta$, with the weak distance on the input side and the strong distance on the output side; hence $\Delta_k\le\vartheta^{k-k_d}\Delta_{k_d}$. The same clause, applied to the pairs $(u_k(z),\bar u_k)$ and $(\mathfrak n_k(z),\bar{\mathfrak n}_k)$, whose second entries are constant families, and started at k_d from 0.455 and from $0.455+0.34$, gives (D1) (f). The jets are the true ones: for $\mathfrak n_k(z)$ by Step 2 and (D1) (c), and for $u_k(z)$ by Prop. 7.12 (J6).

At the last level this gives

$$\sup\mathsf D_{\mathsf b}^{(\kappa)}\le\vartheta^{K-k_d}(3.45\,\vartheta^{k_d-k_0}+1.36\,\Omega^{\#})\le(3.45+1.36\,C_5^{\#}K_5^{-1}K)\vartheta^{K-k_0},$$

because $0.3683\le\vartheta$. By Step 5 of the proof of Thm. 5.63, $3\vartheta^{K-k_0}\le C_{\mathsf b}\theta^K$; and $3.45/3=1.15$ and $1.36/3\le 0.46$.

Now let $k=K-s>k_0$. In both cases, namely for $k\le k_d$ by (D2) and for $k>k_d$ by the contraction $\Delta_k\le\vartheta^{k-k_d}\Delta_{k_d}$ together with (D2) at k_d, as at the last level, we have $\Delta_k\le 3.45\,\vartheta^{k-k_0}+1.36\,\Omega^{\#}$, and $\Omega^{\#}\le C_5^{\#}K_5^{-1}K\,\vartheta^{K-k_0}\le C_5^{\#}K_5^{-1}K\,\vartheta^{k-k_0}$. Step 6 (a) of the proof of Thm. 5.63 shows that $K-s>k_0$ is equivalent to $K-s\ge K_0(L^s\mathsf b)$, in which case $3\vartheta^{K-s-k_0}\le C_{L^s\mathsf b}\theta^{K-s}$. Finally $k\le k_d$ if and only if $s\ge j^*$, and in that case $\mathfrak r_k=\mathfrak r(L^s\mathsf b)$.

Step 8: the identity (D6). All derivatives in (D6) are taken on the representatives of (D1). *Normalisation.* Since the kernels of the step are probability kernels (item 2 of §5.5), Fubini's theorem and linearity give $\int\rho'\,dV'=\int\rho\,dU$ for every bounded complex Borel function ρ. It follows that $\int\rho_k^z\,dV=Z_K(z)$ at every level.

We work on the interior of Pd_0 of the run K'. For the run K, Thm. 7.1 (h) gives

$$Z_K(z)=\exp\{\sum_{0\le k<K}\mathsf V'_{\ell_k}[u_k(z)]\}\,I[u_K(z)].$$

For the run K' we apply Thm. 7.1 (h) as far as the level $m+k_0$, next Lem. 7.10 (B1), and finally (D0). Since (B1) is the identity $\rho[\tilde u;\ell;D]=\rho[\mathbb B_D\tilde u;\ell]$ at every field, without a prefactor $e^{\mathsf V}$, the

switch adds nothing to the exponent of $Z_{K'}(z)$, and we obtain

$$Z_{K'}(z) = \exp\{ \sum_{0\le j<m+k_0} \mathsf{V}'_{\tilde\ell_j}[\tilde u_j(z)] + \sum_{k_0\le k<K} \mathsf{V}'_{\ell_k}[\mathfrak{n}^\circ_k(z)]\}\, I[\mathfrak{n}^\circ_K(z)].$$

At $z=0$ all factors are >0; we take the logarithm near 0 and apply $\partial_1\cdots\partial_n$ at 0. For the run K we have $\rho_0^z = \rho_0 \exp\{\sum z_i \mathcal{O}_K(f_i)\}$, and for the run K' we have $\rho_0^z = \rho_0 \exp\{\sum z_i \mathcal{O}_{K'}(f_i)\}$ with the bare weight ρ_0 of that run; in both cases ρ_0^z is the weight of Thm. 7.1 taken at $\nu_p = \mathsf{b}^{-1}\sum z_i f_i(x(p))$, where p is a plaquette of the bare lattice of the run. Hence the truncated function S_n^T of item 1 of §7.3, at the same label b and on the same torus, is $S^T_{n;K} = \partial_1\cdots\partial_n \log Z_K(0)$ for the run K and $S^T_{n;K'} = \partial_1\cdots\partial_n \log Z_{K'}(0)$ for the run K'; the left side of (D6) consists of these two terms.

Representatives. Consider a function Φ that is holomorphic along affine lines in a ball about $u(0)$ and bounded on that ball. Since Lem. 7.8 holds for every such Φ, it gives

$$\partial_1\cdots\partial_n(\Phi\circ u)(0) = \sum_\pi D^{|\pi|}\Phi(u(0))[u_{S_1},\dots,u_{S_{|\pi|}}],$$

where π runs over the partitions of $\{1,\dots,n\}$ and $S_1,\dots,S_{|\pi|}$ are the blocks of π, as in Lem. 7.8; since every variable is differentiated exactly once, the right side involves only $u(0)$ and the multi-affine jets. We apply this to $\mathsf{V}'_\ell[\cdot]$, to every $v(Y;\cdot)$, and, near a base point at which $I>0$, to $\log I$. For I the reason is the following. For every V, $\rho[u;\ell_K](V)$ equals a polynomial in finitely many values of w times the exponential of a linear form in finitely many values of E^a and E^h, and it is bounded on M_{16} uniformly in V, the torus being finite; therefore I is bounded and holomorphic along affine lines, by the theorems of Fubini and Morera. Moreover $I[\bar u_K]>0$ and $I[\bar{\mathfrak{n}}_K]>0$, by Thm. 5.63 (ii), (iii) taken as in (D1) (c): both densities are >0 at every field.

The ball. We take the norm at $p=(\kappa,c_h)$, the pair of the last level. The reason given for I holds at every p (Lem. 7.8, its part on $I[u]$); in this norm both the honest families and the representatives at the level K remain bounded (Thm. 7.1 (h), (D0), (D1) (f)); and each of the two base points u_0 has $\|u_0\|\le 1$, since the norm at the rate κ_1 dominates. Let m be a bound of $|I|$ on $\{\|u\|<16\}$. For $\|g\|\le 1$, the map $s\mapsto I[u_0+sg]$ is holomorphic and bounded by m on $|s|<15$. Applying Cauchy's formula on circles of radius $R<14$ centred at the points with $|s|\le 1$ and letting $R\to 14$, we see that at these points the derivative has modulus at most $\mathsf{m}/14$. Taking $s=\|u-u_0\|>0$ and $g=(u-u_0)/s$, we get $|I[u]-I[u_0]|\le \mathsf{m}\|u-u_0\|/14$ for $\|u-u_0\|\le 1$; this is Lem. 7.5 (b) with $A_\#=16$, $B=2$ for the single function I, carried out on a line.

Put $t:=\min\{1, 7I[u_0]/\mathsf{m}\}$. For $\|u-u_0\|<t$ the quotient $I[u]/I[u_0]$ then lies in $\{|w-1|\le 1/2\}$. So $\mathrm{Log}(I[u]/I[u_0])$, where Log is the principal branch of the complex logarithm on $|w-1|\le 1/2$, is defined on the ball $\{\|u-u_0\|<t\}$, bounded there by $\log 2$ and holomorphic along lines, and $\Phi[u]:=\log I[u_0]+\mathrm{Log}(I[u]/I[u_0])$ satisfies the hypothesis of Lem. 7.8.

To conclude, Prop. 7.12 (J1), (J6) and (D1) (c) allow us to replace the honest families by u_k^{pol}, $\mathfrak{n}_k^{\mathrm{pol}}$ for $k\le k_d$ and by $u_k(z)$, $\mathfrak{n}_k(z)$ for $k\ge k_d$. Subtracting the two identities, we obtain (D6).

Step 9: (D4) *and* (D5)*; the numbers* V' *and the two series.* We prove (D4) and (D5), and that, once $j>j^*$, consecutive terms of Tail, and likewise of Ser_2, have quotient at most $1/2$.

For (D4), fix k with $k_0\le k<K$ and a radius $r<\mathfrak{r}_{\min(k,k_d)}/2$, and let $u(z)$, $\mathfrak{n}(z)$ be the two representatives; they are families in $M_B(x_k;\kappa_k,c_h)$ with $B\le 2.455$. By Thm. 6.7 (g1), (f1), $\mathsf{V}'[\mathfrak{n}(z)]-\mathsf{V}'[u(z)] = \sum_Y [v(Y;\mathfrak{n}(z))-v(Y;u(z))]$, since V_0^+ depends on no member. We determine which Y contribute. For u, the dependence on z_i at X requires that $X^{+\check r(x_k)}$ meet $S_i(k)$, and for $\mathfrak{n}$ that it meet $\tilde S_i(k)$; for the polynomials this is Prop. 7.12 (J2) and (D1) (d), and after k_d it follows level by level from Thm. 6.7 (f5). By Thm. 6.7 (f6), $\partial_1\cdots\partial_n v(Y;\cdot)(0)=0$ unless $Y^{+\check r(x_{k+1})}$ meets, for every i, the set $S_i(k+1)$ for $v(Y;u(z))$ and the set $\tilde S_i(k+1)$ for $v(Y;\mathfrak{n}(z))$. Every such Y contains one of the $N_1(k)$ cells that lie within $\check r(x_{k+1})$ of $S_1(k+1)\cup\tilde S_1(k+1)$, by the part "Count" of (Sep) in Step 2; and, for $k<k_d$, $\#Y\ge m_k$ follows from (Sep) at $k+1$ and Step 1 (C) of the proof of Prop. 7.12.

For these Y we write the mixed derivative at 0 of $v(Y;\mathfrak{n}(z))-v(Y;u(z))$ by Cauchy's formula:

$$\partial_1\cdots\partial_n[v(Y;\mathfrak{n}(z))-v(Y;u(z))](0)=(2\pi i)^{-n}\int_{|z_i|=r}[v(Y;\mathfrak{n}(z))-v(Y;u(z))]\prod dz_i/z_i^2.$$

Moving the moduli inside the integral and summing over Y, we apply at each z and at each anchor cell Thm. 7.9 (v) with the rate $\kappa_k\ge\kappa$; it gives $\sum_{Y\text{ holding the cell},\,\#Y\ge m_k}|v(Y;\mathfrak{n}(z))-v(Y;u(z))|\le e^{-\kappa(m_k-1)}16C_gc_a(x_k)\mathsf{D}^{\mathrm{in}}/(16-B)$, where $16/13.545\le1.18125\le1.2$ and $\mathsf{D}^{\mathrm{in}}\le\Delta_k$. Then we let r tend to the radius.

For the sum over the levels, $\Delta_{k_0}=3.91\le4$, and for $k>k_0$ we have $\Delta_k\le3.45\,\vartheta^{k-k_0}+1.36\,\Omega^{\#}$ by Steps 6 and 7, with $\Omega^{\#}\le C_5^{\#}K_5^{-1}K\,\vartheta^{K-k_0}$. Consequently, for every k, $\Delta_k\le(4+1.36\,C_5^{\#}K_5^{-1}K)\vartheta^{K-k_0}\vartheta^{-(K-k)}$. With $j=K-k\ge1$, $x_k=L^j\mathsf{b}$ and $\mathfrak{r}_{\min(k,k_d)}=\mathfrak{r}(L^{\max(j,j^*)}\mathsf{b})$, the sum is at most Ser_2 times the bracket times ϑ^{K-k_0}.

For (D5), Prop. 7.12 (J7), (J8) for either run gives

$$|\partial_1\cdots\partial_n\mathsf{V}'_{\ell_k}[u_k](0)|=|\sum_Y\partial_1\cdots\partial_nv_k(Y;u_k(z))(0)|\le\mathfrak{r}_{\min(k,k_d)}^{-n}N_1(k)\,2C_gc_a(x_k)e^{-\kappa(m_k-1)},$$

and the right side is the term $j=K-k$ of Tail. For the run K, $0\le k<k_0$ means $K\ge j>K-k_0$; for the run K', $0\le k'<m+k_0$ means $K'\ge K'-k'>K-k_0$. The level 0 is covered by (J8).

Let $j>j^*$. By (t3), Step 1 (f) and (t5), $\mathfrak{r}(L^j\mathsf{b})/\mathfrak{r}(L^{j+1}\mathsf{b})\le L^{1/2}$, $N_1^{[j+1]}\le L^3N_1^{[j]}$, and $c_a(L^{j+1}\mathsf{b})\le c_a(L^j\mathsf{b})$. Moreover $m^{[j+1]}-m^{[j]}\ge n(\mathrm{sep}^{[j]}-\mathrm{sep}^{[j-1]})/2-1\ge n(L-1)\mathrm{sep}_*/2-1$ by Step 1 (e), and $\mathrm{sep}_*\ge4\log L$. The quotient is therefore at most $\vartheta^{-1}L^{n/2+3}e^{\kappa}L^{-2\kappa n(L-1)}\le1/2$. Both series converge; $\mathrm{Ser}_2\le\sum_{j\le j^*}(\text{its terms})+2\,(\text{its term }j^*+1)$, and $\mathrm{Tail}(J)\le2\,(\text{its term }J+1)$ for $J\ge j^*$. In the latter term $m^{[J+1]}\ge n\,\mathrm{sep}^{[J]}/2\ge4nL^{J+1-j^*}\log(4nL^{1/2})$.

Step 10: (D7)*; on what the numbers depend, and local boundedness in* b. The dependence follows from the definitions in item 1.3. In $c_Q^{\mathrm{su}}=306M^3\mathsf{g}(\lambda_\flat)^2$ only M enters, besides a pure number. The constants K_5, C_5 and C_1 are pure numbers, by (O0a) in item (1) of §6.1 and by (O0) in item 1 of §5.5; $C_5^{\#}$, C_g and ε_L involve L; and $c_4=15c_Q^{\mathrm{su}}C_1^2/E_a$. The functions amp and c_a contain $p_0(x)=A_0(\log x)^{\mathfrak{p}_0}$ ((O8) in item 1 of §5.5), and $\check r(x)=r(x)+3$, with $r(x)=\lceil20c_R\log x/M\rceil$ (item 2 of §5.5), contains c_R and M. In the remainder of this step p_0 enters only through being continuous and >0 on $[x_s,\infty)$, and through (t3). Finally $\mathfrak{r}_*=\mathfrak{r}_{k_d}=\mathfrak{r}(L^{j^*}\mathsf{b})$, because $K>j^*$.

We turn to monotonicity in b. As $\check r(L^i\mathsf{b})$ does not decrease in b, $\mathrm{sep}^{[i]}$ does not increase and j^* does not decrease; on $[\mathsf{b}_1,\mathsf{b}_2]$, therefore, $j^*(\mathsf{b})\le j^*(\mathsf{b}_2)<\infty$ (Step 1 (e)). The numbers Λ and K_0 are continuous and increasing in b. The left side of (K1a) decreases in Λ and its right side increases in j^*, and the left side of (K1b) increases in Λ; hence $K_T(\mathsf{b})$ is at most the K_T obtained with $\Lambda(\mathsf{b}_2)$ and $j^*(\mathsf{b}_2)$. For every j the function $\mathsf{b}\mapsto\mathfrak{r}(L^j\mathsf{b})$ is >0 and continuous; since j^* has only finitely many values on $[\mathsf{b}_1,\mathsf{b}_2]$, $\mathfrak{r}_*$ is bounded below there. The function C_{b} is continuous as well.

On $[\mathsf{b}_1,\mathsf{b}_2]$, each term of the series is bounded by the term defined as follows: amp and c_a at their largest values, which exist since they are continuous functions on a compact set (in $\mathfrak{r}(L^{\max(j,j^*)}\mathsf{b})$ we put $j^*(\mathsf{b}_2)$ for $j^*(\mathsf{b})$, which is possible because $\mathrm{amp}(L^{i+1}\mathsf{b})\ge\mathrm{amp}(L^i\mathsf{b})$ by (t3)); $N_1^{[j]}$ at b_2; and $e^{-\kappa(m^{[j]}-1)}$ replaced by 1 for $j\le j^*(\mathsf{b}_2)$ and by its value at b_2 for $j>j^*(\mathsf{b}_2)$, since $m^{[j]}$ does not increase in b there. This majorant converges by Step 9.

Concerning the numerals: Steps 3 to 7 rest on $L^{1/4}\ge10^4$ and on (W2), (W3), (Y.0)(0b). □

7.5. **Normalization at the last scale.** We compare the densities $\rho[u_2;\ell]$ and $\rho[u_3;\ell]$, whose members may be complex, with one real positive first member: we bound their difference in the supremum and in L^1, and, when the condition (S) holds, we bound quotients and logarithms. Along a pointwise holomorphic family $u(z)$ in the sources the integral $I[u(z)]$ is holomorphic, and Cauchy bounds control $\partial_S\log I$ and, in L^1, the jets of the laws; only the multi-affine jets count. For the sourced runs from two cutoffs the datum (J) implies a formula for $S^T_{n;K}$, namely $\partial_1\cdots\partial_n\mathfrak{V}(0)$ plus

a term of modulus at most $\log 2\,\tau_7^n/\prod r_i$; a bound for what remains of the difference $S^T_{n;K'} - S^T_{n;K}$ when $\partial_1\cdots\partial_n\mathfrak{V}'(0) - \partial_1\cdots\partial_n\mathfrak{V}(0)$ is subtracted from it; and bounds in L^1 for the jets $\pi^{[K]}_{S,[s]}$ of the laws that the drawn fields follow.

Item 0: what the section does.

Remark 7.20 (What passes from two real members to complex ones). Thm. 5.64 concerns two points of one class, where both systems of members are real and both densities are positive. For the jets in the sources the further members are complex. Thm. 5.64 (a) and (b) carry over to this case, (b) with an exponent that depends on the sizes. The first inequality of Thm. 5.64 (c) depends on the first member alone and holds here unchanged. For a complex second member the argument for the second inequality of Thm. 5.64 (c), an L^1 bound with C_{vol}, does not carry over, since Step 9 (6) of the proof of Thm. 5.64 uses $\|g\|_1 = \int g$, which requires $g \geq 0$; instead, when the condition (S) holds, (c2) gives the bound with $2C_{\mathrm{vol}}$ in the place of C_{vol}.

Item 1: letters, signs and definitions. We use the letters of §6.1 and of items 1 and 2 of §5.5. The chain is run at the rate $\kappa_1 = \kappa + 1$, with $\kappa \geq 6$: the sets $M^{\mathrm{su}}_A(x)$, the strong cap, the class $C^{\mathrm{su}}_\flat(x;\ell)$, the norm $\|\cdot\|_x$ and the distances $\mathsf{D}^{\mathrm{in}}_x$ and D_x are all defined with κ_1. The number of sources is n, and $q_N = (N/M)^3$ is the number of cubes.

For a subset S of $\{1,\dots,n\}$ we write $z^S = \prod_{i\in S} z_i$, $\partial_S = \prod_{i\in S}\partial/\partial z_i$ and $r^S = \prod_{i\in S} r_i$. We write $\mathrm{Pd}(r) = \{z \in \mathbb{C}^n : |z_i| < r_i\}$, and $\mathrm{Pd}(r/t)$ for the polydisc with the radii r_i/t.

The letter s denotes the depth when it has no argument and the radius $s(x) = 8r(x)$ of item 2 of §5.5 when it has one, and both senses occur in $2M\,s(L^s\mathsf{b})$, whereas in x_s the index is a mark and does not denote the depth s. The symbol $z(ax)$ refers not to the sources z but to the normalisation of the kernel k_{ax}, as defined in item 2 of §5.5. The rate κ_1 is always $\kappa+1$, but κ_K, κ_{K-s} and κ_k, whose indices are letters that denote levels, are the rates of (o4) in item 2 of §7.3, and for $K = 1$ we have $\kappa_K = \kappa$, not $\kappa + 1$. The caps are $c_h(x)$ and $c_a(x)$, while in the phrase "with the cap c_h" and in $M_1(\mathsf{b};\kappa_1,c_h)$ the letter c_h, without an argument, is the number of (O11) in item 1 of §5.5, and such a phrase says that the cap is $c_h(x) = e^{-c_h p_0(x)^2}$ itself.

Definition 7.21 (Triple without a size)**.** A *triple without a size* at x is a triple $u = (E^a, E^h, w)$ that satisfies all clauses of (ca), (ch) and (cΞ) of item 2 of §5.5 except the inequality. These triples form a complex vector space. The proofs use of such a triple only that $1_{X\approx}(V)E^a(X,V)$, $1_{X\approx}(V)E^h(X,V)$ and $w(X,V)$ are complex Borel functions of the G-valued field V. ■

Definition 7.22 (Rate and norms at a rate)**.** A *rate* is a number $\tilde\kappa \in [\kappa,\kappa_1]$, and the cap of E^h is $c_h(x)$ itself. The norms below take their values in $[0,\infty]$. We let $\|u\|^\sim_E$ be the least A such that $\sup_{\mathcal{U}_X(x)}|E^a(X,\cdot)| \leq A\,c_a(x)e^{-\tilde\kappa(\#X-1)}$ and $\sup|E^h(X,\cdot)| \leq A\,c_h(x)e^{-\tilde\kappa(\#X-1)}$ for all $X \in \mathcal{D}$. We let $\|u\|^\sim_w$ be the least A such that $|w(X,V)| \leq A\,e^{-\tilde\kappa(|X|_c-1)}$ for all $X \in \mathcal{P}$ and all V, that is, at the weak cap. We let $\|u\|^\sim$ be the larger of the two. Since $\tilde\kappa \leq \kappa_1$, we have $\|u\|^\sim \leq \|u\|_x$; hence $\|u_1 - u_2\|^\sim \leq \mathsf{D}^{\mathrm{in}}_x(u_1,u_2) \leq \mathsf{D}_x(u_1,u_2)$, and $\|u\|^\sim \leq 1$ on $M^{\mathrm{su}}_1(x)$. If a norm measures w against the strong cap at $\tilde\kappa$, it dominates $\|\cdot\|^\sim$. An inequality whose right side is infinite says nothing. ■

Definition 7.23 (The last integral and the sum of the members)**.** For $\|u\|^\sim$ finite we put $I[u] := \int \rho[u;\ell](V)\,dV$. For the member u_i we put $\mathrm{Sm}_i(V) := \sum_{X\in\mathcal{D}} 1_{X\approx}(V)\,[E^a_i + E^h_i](X,V)$. ■

Definition 7.24 (The volume constant)**.** We put

$$C_{\mathrm{vol}} = C_{\mathrm{vol}}(N,x) := C^{\mathrm{su}}_{\mathrm{vol}}(N,x) = 4q_N e^{220q_N} e^{(3/2)\mathsf{N}_Z p_0(x)^2 N^3}(384/\varepsilon(x)^3)^{3N^3},$$

which is (O14) in item 1 of §5.5. If $\varepsilon(x) \leq 1$, then $C_{\mathrm{vol}} \geq 4q_N e^{220q_N} \geq 400q_N$. At fixed N, C_{vol} is a continuous function of x on $[e^2,\infty)$. ■

Definition 7.25 (Pointwise holomorphic family)**.** Let $\Omega \subset \mathbb{C}^n$ be open. A *pointwise holomorphic family* on Ω, in the sense of Thm. 6.7 (c+), assigns to every $z \in \Omega$ a triple without a size $u(z)$ such that every value $E^a(X, \mathsf{W}; z)$ with $\mathsf{W} \in \mathcal{U}_X(x)$, every value $E^h(X, V; z)$ and every value $w(X, V; z)$ is holomorphic in z. ■

Definition 7.26 (Congruence modulo the squares of the sources)**.** For germs f and g of holomorphic functions at $0 \in \mathbb{C}^n$ we write $f \equiv g$ if $\partial_S f(0) = \partial_S g(0)$ for all S, the empty set included, and we call f and g congruent modulo z_i^2. Two families are congruent if all their values are. ■

Item 2: the theorem.

Theorem 7.27 (Two complex members against one positive member)**.** Let the following data (B) be given: $x \geq e^2$; N even, $M \mid N$ and $N \geq 2M\, s(x)$; $\varepsilon(x) \leq 1$; $(c_a + c_h)(x) \leq 1/2$; rows ℓ on $\mathbb{Z}^3/N\mathbb{Z}^3$ of range $\leq \varrho(x)$; and $\tilde{\kappa} \in [\kappa, \kappa_1]$. As the first member, let $\rho_1 = e^{\mathsf{V}_1}\rho[u_1; \ell]$ be a point of $C_\flat^{\mathrm{su}}(x; \ell)$ and u_1 a system of members of it; thus u_1 is real, lies in $M_1^{\mathrm{su}}(x)$ and has the strong cap, V_1 is real, and $\rho_1 > 0$. As further members, let u_2, u_3 be triples without a size and $\mathsf{V}_2, \mathsf{V}_3 \in \mathbb{C}$, and put $\rho_i := e^{\mathsf{V}_i}\rho[u_i; \ell]$. Put $\mathsf{D}_{ij} := \|u_i - u_j\|^\sim$. Whenever $\|u_i\|^\sim < \infty$, write $I_i := I[u_i]$. Then the following hold.

(a^C) At every V for which $D(V)$ is empty, $\rho_i(V) = \exp\{\mathsf{V}_i - x\mathrm{En}_\ell(V) + \mathrm{Sm}_i(V)\}$, which is not 0, and $|\mathrm{Sm}_i - \mathrm{Sm}_j|(V) \leq 1.22\, q_N (c_a + c_h)(x)\, \mathsf{D}_{ij}$.

(b^C) Let $a_E \geq \|u_2\|_E^\sim, \|u_3\|_E^\sim$ and $a_w \geq \max\{1, \|u_2\|_w^\sim, \|u_3\|_w^\sim\}$. Then

$$\sup_V |e^{-\mathsf{V}_2}\rho_2 - e^{-\mathsf{V}_3}\rho_3| \leq 1.61\, q_N e^{(0.61 a_E + 218 a_w) q_N}\, \mathsf{D}_{23}.$$

In three cases the bound reads as follows.

(i) If $400 q_N \max\{\mathsf{D}_{12}, \mathsf{D}_{13}\} \leq 1$, the bound is $\leq 2 q_N e^{219 q_N}\mathsf{D}_{23}$; for $u_3 = u_1$ it is the bound of Thm. 5.64 (b).

(ii) If $a_w = 1$ and $a_E \leq 2$, the bound is $\leq 1.61\, q_N e^{219.22 q_N}\mathsf{D}_{23}$. This case covers every output $F_\ell(u)$ of Thm. 5.15 with $A \leq 6$ and with $p = (\tilde{\kappa}, c_h)$ as the pair of parameters, because Thm. 5.15 (b), which holds for this p by Thm. 6.7 (U), bounds the E^a of such an output by $0.35 + 0.248 \cdot 6 = 1.838$ times the cap c_a and its E^h by the cap c_h, and gives its w the strong cap; hence $a_E \leq 2$ and $a_w = 1$.

(iii) If $\|u_2\|^\sim, \|u_3\|^\sim \leq A$ with $A \geq 1$, the bound is $\leq 2 q_N e^{219 A q_N}\mathsf{D}_{23}$.

(c^C) Let moreover the rows obey $|\ell_y(p)| \leq K_* e^{-\lambda_* |y-p|}$. Then

$$I_1 \geq e^{-0.61 q_N} e^{-(3/2)\mathsf{N}_Z p_0^2 N^3} (\varepsilon^3/384)^{3N^3};$$

this is the first inequality of Thm. 5.64 (c), and it depends on the first member alone. In the cases (i) and (ii), $\int |\rho[u_2; \ell] - \rho[u_3; \ell]|\, dV \leq \frac{1}{2} C_{\mathrm{vol}} \mathsf{D}_{23} I_1$; in the case (iii) the same holds with $C_{\mathrm{vol}} e^{219(A-1)q_N}$ in the place of C_{vol}. If, in addition, $C_{\mathrm{vol}} \max\{\mathsf{D}_{12}, \mathsf{D}_{13}\} \leq 1$, which is the condition (S) and implies the case (i), then we have the following.

(c1) We have $|I_2 - I_3| \leq \frac{1}{2} C_{\mathrm{vol}} \mathsf{D}_{23} I_1$ and $|I_i/I_1 - 1| \leq \frac{1}{2} C_{\mathrm{vol}} \mathsf{D}_{1i} \leq \frac{1}{2}$; moreover $I_i \neq 0$ and $\int |\rho[u_i; \ell]|\, dV \leq \frac{3}{2} I_1 \leq 3|I_i|$.

(c2) We have

$$\int |\rho_1 / \int \rho_1 - \rho_2 / \int \rho_2|\, dV \leq C_{\mathrm{vol}} \mathsf{D}_{12} / (1 - C_{\mathrm{vol}} \mathsf{D}_{12}/2) \leq 2 C_{\mathrm{vol}} \mathsf{D}_{12}.$$

(c3) We have $\int |\rho_2/\int \rho_2 - \rho_3/\int \rho_3|\, dV \leq 4 C_{\mathrm{vol}} \mathsf{D}_{23}$.

(c4) We have $|\mathrm{Log}(I_2/I_1) - \mathrm{Log}(I_3/I_1)| \leq C_{\mathrm{vol}} \mathsf{D}_{23}$, where Log is the principal branch on $|w - 1| \leq 1/2$.

(d^C) Let $\eta_d := 1.22\, q_N (c_a + c_h) \mathsf{D}_{12} < \log 2$. Since $c_a + c_h > 0$ and $\|u_1\|^\sim \leq 1$, this implies $\|u_2\|^\sim \leq 1 + \mathsf{D}_{12} < \infty$. Let $I_2 \neq 0$, let $\mathrm{Tm} := \{D(V) \text{ empty}\}$, let ms_1 be the mass of the

complement of Tm under $\rho_1/\int\rho_1$, and let $\mathrm{ms}_2 := \int_{\mathrm{off\,Tm}} |\rho_2|\,dV/|\int\rho_2|$. Then

$$\int |\rho_1/\int\rho_1 - \rho_2/\int\rho_2|\,dV \le 2\,\mathrm{ms}_1 + \mathrm{ms}_2(1+e^{\eta_d}/(2-e^{\eta_d})) + 2(e^{\eta_d}-1)/(2-e^{\eta_d}).$$

■□

Item 3: the first corollary.

Corollary 7.28 (Families in the sources and their jets)**.** Let the data (B), the bound on the rows of (c^C) and the datum (F0) be given; then (F1) to (F5) hold.

(F0) The datum is a pointwise holomorphic family $u(z)$ on $\mathrm{Pd}(r)$ with $u(0) = u_1$, the first member, together with a finite number $B_0 \ge \sup_{\mathrm{Pd}(r)} \|u(z) - u_1\|^\sim$. We put $\tau_7 := \max\{1, B_0 C_{\mathrm{vol}}\}$.

(F1) The map $z \mapsto I[u(z)]$ is holomorphic on $\mathrm{Pd}(r)$, with $\partial_S I[u(z)] = \int \partial_S \rho[u(z);\ell](V)\,dV$, and $\|u(z) - u_1\|^\sim \le B_0 \max_i |z_i|/r_i$.

(F2) On $\mathrm{Pd}(r/\tau_7)$ we have $|I[u(z)]/I_1 - 1| \le 1/2$, and for every non-empty S we have $|\partial_S \log I[u(z)]|_{z=0} \le \log 2\ \tau_7^{|S|}/r^S$.

(F3) Let $\mathfrak{n}(z)$ be a pointwise holomorphic family on $\mathrm{Pd}(r/2\tau_7)$, with $\mathfrak{n}(0)$ arbitrary, let $\Delta := \sup_{\mathrm{Pd}(r/2\tau_7)} \|u(z) - \mathfrak{n}(z)\|^\sim$, and let S be non-empty; if $C_{\mathrm{vol}}\Delta \le 1/2$, then $I[\mathfrak{n}(0)] \ne 0$ and

$$|\partial_S \log I[u(z)] - \partial_S \log I[\mathfrak{n}(z)]|_{z=0} \le C_{\mathrm{vol}}\Delta\,(2\tau_7)^{|S|}/r^S.$$

If $\mathfrak{n}(z)$ is a pointwise holomorphic family on all of $\mathrm{Pd}(r)$, if $\mathfrak{n}(0)$ is a system of members of a point of $C^{\mathrm{su}}_\flat(x;\ell)$, hence real, in $M_1^{\mathrm{su}}(x)$, with the strong cap and with $\rho[\mathfrak{n}(0);\ell] > 0$, and if $\sup_{\mathrm{Pd}(r)} \|\mathfrak{n}(z) - \mathfrak{n}(0)\|^\sim \le B_0$, then the left side is $\le 3C_{\mathrm{vol}}\Delta\,(2\tau_7)^{|S|}/r^S$ for every value of Δ.

(F4) *A lemma on jets.*

(i) If δ_S, for non-empty S, are triples without a size with $\sum_S r^S \|\delta_S\|^\sim \le B_0$, then $u^{\mathrm{pol}}(z) := u_1 + \sum_S z^S \delta_S$ obeys (F0).

(ii) If $u(z)$ and $\hat u(z)$ are congruent pointwise holomorphic families near 0, then $\rho[u(z);\ell](V) \equiv \rho[\hat u(z);\ell](V)$ at every V. If, moreover, the norms $\|\cdot\|^\sim$ of both families stay bounded near 0 and $I[u(0)] \ne 0$, then $\partial_S \log I[u(z)] = \partial_S \log I[\hat u(z)]$ at 0 for every non-empty S.

(iii) Let $u(z)$ be a pointwise holomorphic family whose norm $\|u(z)\|^\sim$ stays bounded near 0, with $I[u(0)] \ne 0$, and let $\mathfrak{V}(z)$ be holomorphic near 0. Let $F(z,V)$ be a function that is Borel in V, holomorphic in z near 0 for each fixed V, and bounded uniformly in (z,V), such that $F(\cdot,V) \equiv e^{\mathfrak{V}(z)}\rho[u(z);\ell](V)$ holds for each V. Then $\partial_S \log \int F\,dV = \partial_S \mathfrak{V} + \partial_S \log I[u(z)]$ at $z = 0$ for each non-empty S, and for each V we have $F(z,V)/\int F\,dV \equiv \rho[u(z);\ell](V)/I[u(z)]$.

Hence, under the hypotheses of the second sentence of (ii), the congruence $\rho[u(z);\ell](V)/I[u(z)] \equiv \rho[\hat u(z);\ell](V)/I[\hat u(z)]$ holds at every V.

(F5) For the laws $\mu_u^z := \rho[u(z);\ell]/I[u(z)]$ and for non-empty S, the jets obey $\int |\partial_S \mu_u^z|_{z=0}\,dV \le 3\tau_7^{|S|}/r^S$, and, with $\mathfrak{n}$ and Δ as in (F3), if $C_{\mathrm{vol}}\Delta \le 1/2$, then

$$\int |\partial_S \mu_u^z - \partial_S \mu_{\mathfrak{n}}^z|_{z=0}\,dV \le 4C_{\mathrm{vol}}\Delta\,(2\tau_7)^{|S|}/r^S.$$

Under the hypotheses of the second sentence of (F3), this bound holds for every value of Δ.

■□

Item 4: the second corollary.

Corollary 7.29 (Two sourced runs and the jets of their laws)**.** *Setting.* The letters are those of §6.1; $\mathsf{b} \geq x_s$; $N = 2L^{m_l}$ with $m_l \geq m'$; the condition (Tor) $N \geq 2M\, s(L\mathsf{b})$ holds; $K' \geq K \geq 1$ and $m = K' - K$; and the runs, tori, grids, rows $\ell_k = \ell_k^{[K]}$ and numbers $x_k = L^{K-k}\mathsf{b}$ are those of the definition "Runs" in item 2 of §5.5. Let $f_1, \dots, f_n$ be complex bounded functions on T_l, and let $\mathcal{O}_K(f) = a_K^{-1} \sum_p f(x(p))\, a_p(U)$, where $a_K = L^{-K}$, $x(p)$ is the barycentre of p and $a_p = 1 - \cos\phi_p$. We put $\rho_0^{[K],z} := \rho_0^{[K]} \exp\{\sum_i z_i \mathcal{O}_K(f_i)\}$ and $\rho_{k+1}^{[K],z} := (\rho_k^{[K],z})'$, where the prime denotes the unsourced step with the noise ax_k. We put $Z_K(z) := \int \rho_0^{[K],z}$ and $S_{n;K}^T = \partial_1 \cdots \partial_n \log Z_K(0)$.

Datum. We call the clauses (J.a) to (J.c) below the datum (J) at the depth $s \geq 0$; it is taken at the level $k = K - s \geq 0$, at $x = x_k = L^s\mathsf{b}$, on the torus $\mathbb{Z}^3/NL^s\mathbb{Z}^3$ and with the rows ℓ_k, and it is the hypothesis of each clause below that names it. At $k = 0$ the rows ℓ_0 are empty, and the data (B), the bound on the rows of (c^C), with $\ell_y = 0$ for empty rows, and Step 9 (1) to (5) of the proof of Thm. 5.64 admit empty rows as well.

(J.a) $\bar{u}_k$ is a system of members of the point $\rho_k^{[K]}$ of $C_\flat^{\mathrm{su}}(x_k; \ell_k)$.

(J.b) $u(z)$ is a pointwise holomorphic family on $\mathrm{Pd}(r)$ with $u(0) = \bar{u}_k$ and $\sup_{\mathrm{Pd}(r)} \|u(z) - \bar{u}_k\|^\sim \leq B_0$, $\mathfrak{V}(z)$ is holomorphic near 0, and $\rho_k^{[K],z}(V) \equiv e^{\mathfrak{V}(z)} \rho[u(z); \ell_k](V)$ at every V.

(J.c) $\mathfrak{n}(z)$ is a pointwise holomorphic family on $\mathrm{Pd}(r/2\tau_7)$, bounded in $\|\cdot\|^\sim$, $\mathfrak{V}'(z)$ is holomorphic near 0, and $\rho_{m+k}^{[K'],z}(V) \equiv e^{\mathfrak{V}'(z)} \rho[\mathfrak{n}(z); \ell_k](V)$ at every V, with the rows ℓ_k of the short run; Δ is as in (F3).

Then, with $C_{\mathrm{vol}} = C_{\mathrm{vol}}(NL^s, L^s\mathsf{b})$ and $\tau_7 = \max\{1, B_0 C_{\mathrm{vol}}\}$, the following hold.

(R0) For every level k and every $z \in \mathbb{C}^n$ we have $\int \rho_k^{[K],z}\, dV = Z_K(z)$.

(R1) If (J.a) and (J.b) hold, then $S_{n;K}^T = \partial_1 \cdots \partial_n \mathfrak{V}(0) + \partial_1 \cdots \partial_n \log I[u(z)](0)$, and the last term has modulus at most $\log 2\, \tau_7^n / \prod r_i$.

(R2) If (J.a) to (J.c) hold and $C_{\mathrm{vol}}\Delta \leq 1/2$, then

$$|S_{n;K'}^T - S_{n;K}^T - (\partial_1 \cdots \partial_n \mathfrak{V}'(0) - \partial_1 \cdots \partial_n \mathfrak{V}(0))| \leq C_{\mathrm{vol}}\Delta\, (2\tau_7)^n / \prod r_i,$$

and under the hypotheses of the second sentence of (F3) the same left side is $\leq 3C_{\mathrm{vol}}\Delta\, (2\tau_7)^n / \prod r_i$ for every Δ. For polynomials $u = \bar{u}_k + \sum z^S \delta_S$ and $\mathfrak{n} = \bar{\mathfrak{n}}_k + \sum z^S \tilde{\delta}_S$ we have

$$\Delta \leq \|\bar{u}_k - \bar{\mathfrak{n}}_k\|^\sim + \sum_S (r/2\tau_7)^S \|\delta_S - \tilde{\delta}_S\|^\sim.$$

(R3) For non-empty S and $0 \leq s \leq K$ we put

$$\pi_{S,[s]}^{[K]} := \partial_S (\rho_{K-s}^{[K],z} / Z_K(z))\big|_{z=0},$$

written π_S for short; it is a bounded Borel function on the fields of $\mathbb{Z}^3/NL^s\mathbb{Z}^3$.

(i) For every bounded Borel function obs, the integral $\int \mathrm{obs}\, \pi_{S,[s]}^{[K]}\, dV$ is the joint truncated function of the $\mathcal{O}_K(f_i)$, $i \in S$, and of $\mathrm{obs}(V_{K-s})$ under the law of the run. Moreover $\int \pi_{S,[s]}^{[K]}\, dV = 0$, and π_S is gauge invariant, and real for real f_i.

(ii) For $1 \leq s \leq K$ the drawn step with the noise $aL^s\mathsf{b}$ maps $\pi_{S,[s]}^{[K]}$ to $\pi_{S,[s-1]}^{[K]}$; this step does not increase the L^1 norm of a complex function, and it does not depend on K.

(iii) If (J) holds, then $\int |\pi_{S,[s]}^{[K]}|\, dV \leq 3\tau_7^{|S|}/r^S$, and if moreover $C_{\mathrm{vol}}\Delta \leq 1/2$, then

$$\int |\pi_{S,[s]}^{[K']} - \pi_{S,[s]}^{[K]}|\, dV \leq 4C_{\mathrm{vol}}\Delta\, (2\tau_7)^{|S|}/r^S.$$

Under the hypotheses of the second sentence of (F3), that is, if $\mathfrak{n}$ is a family on all of $\mathrm{Pd}(r)$, $\mathfrak{n}(0)$ is a system of members of a point of the class and $\sup_{\mathrm{Pd}(r)} \|\mathfrak{n}(z) - \mathfrak{n}(0)\|^\sim \leq$

B_0, the same bound holds for every value of Δ; the condition $C_{\mathrm{vol}}(NL^s, L^s\mathsf{b})\Delta \le 1/2$ depends on N and s. With the datum of item 5, K has to satisfy $K \ge K_T(\mathsf{b}, n, d)$ and $K - s \ge K_0(L^s\mathsf{b})$, and these conditions depend on b, n, d, s and on no N.

(iv) Let the datum hold at the depths s and $s-1$ for all $K' \ge K \ge K_\#$, with r and B_0 free of K, K' and with $\Delta \le \breve{\Delta}_K$ for some numbers $\breve{\Delta}_K \to 0$; with the datum of item 5, $\breve{\Delta}_K$ is the bound of Thm. 7.19 (D3) at the depth s. Then, as $K \to \infty$, each of the sequences $\pi^{[K]}_{S,[s]}$ and $\pi^{[K]}_{S,[s-1]}$ converges in $L^1(dV)$, without passing to a subsequence. The $L^1(dV)$ distance of $\pi^{[K]}_{S,[s]}$ from its limit satisfies the second bound of (iii) with that limit in the place of $\pi^{[K']}_{S,[s]}$ and with $\breve{\Delta}_K$ in the place of Δ, for the $K \ge K_\#$ with $C_{\mathrm{vol}}\breve{\Delta}_K \le 1/2$, and for all $K \ge K_\#$ under the hypotheses of the second sentence of (F3). At the depth $s-1$ the same is true of $\pi^{[K]}_{S,[s-1]}$, with the constants of that depth. The limits have the properties of (i), namely integral 0, invariance and reality, and they obey the first bound of (iii); the step of (ii) maps the limit at s to the limit at $s-1$. Since their integral is 0, none of them is ≥ 0 unless it vanishes almost everywhere.

(v) Let $K - s \ge 1$, and let $\hat{f}_i(y) := f_i(L^{-s}y)$ on T_{NL^s}. Then $\pi^{[K]}_{S,[s]}$, defined with (K, b, N, f), equals $L^{s|S|}$ times $\pi^{[K-s]}_{S,[0]}$, defined with $(K-s, L^s\mathsf{b}, NL^s, \hat{f})$. Hence, for the run in which $(K-s, L^s\mathsf{b}, NL^s, \hat{f})$ replaces (K, b, N, f), the datum (J) may be taken at the depth 0. The triple $(L^s\mathsf{b}, NL^s, \hat{f})$ meets the conditions of the "Setting" of this corollary. If, moreover, the f_i are in the setting of item 3 of §7.4, then any two of the sets $\{\hat{f}_i \ne 0\}$ have sup distance $\ge L^s d$, each of them lies within sup distance $L^s R_f$ of a point, $\|\hat{f}_i\|_\infty = \|f_i\|_\infty$, $\mathrm{Lip}\,\hat{f}_i = L^{-s}\mathrm{Lip}\,f_i$ and $\|\hat{f}_i\|_\sharp = \max\{\|f_i\|_\infty, 40M\,L^{-s}\mathrm{Lip}\,f_i\} \le 1$, so that the $\hat{f}_i$ lie in that setting as well, with $(L^s\mathsf{b}, NL^s, L^s d, L^s R_f)$ instead of (b, N, d, R_f). In that case the datum (J) holds at the depth 0 for this run whenever $K' - s \ge K - s \ge K_T(L^s\mathsf{b}, n, L^s d)$, as follows from the parts "First" and "Second" of item 5 when $(K-s, K'-s, L^s\mathsf{b}, NL^s, \hat{f}_i, L^s d, L^s R_f)$ replaces $(K, K', \mathsf{b}, N, f_i, d, R_f)$ in them; there $r_i = \mathfrak{r}_*/2$ and $B_0 = 1$, by Thm. 7.19 (D1) (f), and Δ is at most the bound of Thm. 7.19 (D3). Alternatively the datum is taken at the depth s itself, as in the part "Datum at the depth $s \ge 0$" of item 5, for $K \ge K_T(\mathsf{b}, n, d)$ and $K - s \ge K_0(L^s\mathsf{b})$; this threshold is the lower one, since $\max\{K_T(\mathsf{b}, n, d),\ s + K_0(L^s\mathsf{b})\} \le s + K_T(L^s\mathsf{b}, n, L^s d)$ by the part "The two thresholds" of Step P10 (v).

■□

Item 5: the datum.

Remark 7.30 (Which statements supply the datum). The setting of item 4 admits any n and any complex bounded f_i. The clauses supplying the datum are stated in narrower settings, and the parts "First" and "Second" below assume these settings. Prop. 7.12 (J6), (J7) hold in the setting of item 1 of §7.3: $n \ge 2$; $d, R_f > 0$; the f_i are real and bounded with $\|f_i\|_\infty \le 1$; any two of the sets $\{f_i \ne 0\}$ have sup distance $\ge d$, and each of them lies within sup distance R_f of a point; and $K \ge 1$. The clauses of Thm. 7.19 are formulated in the setting of item 3 of §7.4, which asks besides that the f_i be Lipschitz with $\|f_i\|_\sharp \le 1$ and that $K' \ge K \ge K_T(\mathsf{b}, n, d)$. Since S_n^T and π_S are linear in each f_i, the conditions $\|f_i\|_\infty \le 1$ and $\|f_i\|_\sharp \le 1$ are no restriction: the general case follows by scaling each f_i.

First. We supply (F0), (J.a) and (J.b) at $s = 0$. The last sentences of Thm. 7.1 (h) give (J.a) for every $K \ge 1$. Prop. 7.12 (J6), (J7) give the family $u_K(z)$ on $\mathrm{Pd}(\mathfrak{r}_*)$, with $r_i = \mathfrak{r}_*$, $B_0 = 0.91$ and $\tilde{\kappa} = \kappa_K = \kappa$. This family is an output of Thm. 5.15 at the rate $\kappa < \kappa_1$, taken for a pair p for which the clauses of that theorem hold by Thm. 6.7 (U), and it falls under the case (ii) of Thm. 7.27 (b^C); on $\mathrm{Pd}(r/\tau_7)$, where (F2) applies, the condition (S) holds and implies the case (i), which suffices

there. The family is pointwise holomorphic by Thm. 6.7 (c+), and it has the true multi-affine jets. Then $\tau_7 = \max\{1, 0.91\, C_{\mathrm{vol}}(N, \mathsf{b})\}$.

Second. We supply the hypotheses of the second sentence of (F3), and (J.c), at $s = 0$. Thm. 7.19 (D1), (D3) give $\mathfrak{n}_K(z)$ on $\mathrm{Pd}(\mathfrak{r}_*/2)$, with $r_i = \mathfrak{r}_*/2$, with $B_0 = 1$, and with Δ at most the bound of (D3). That $\mathfrak{n}_K(0)$ is a system of members of a point of the class follows from Thm. 5.63 (iii) together with Thm. 7.19 (D1) (c). The value $B_0 = 1$ comes from Thm. 7.19 (D1) (f).

Datum at the depth $s \geq 0$. We work in the setting of item 3 of §7.4, that is, with $n \geq 2$, with real Lipschitz f_i such that $\|f_i\|_\sharp \leq 1$, and with any two of the sets $\{f_i \neq 0\}$ at sup distance $\geq d$, each within sup distance R_f of a point; we take $K \geq K_T(\mathsf{b}, n, d)$ and $k = K - s > k_0$, that is, $K - s \geq K_0(L^s\mathsf{b})$, and $\tilde{\kappa} = \kappa_{K-s}$. The clauses of the datum are then supplied as follows.

(J.a) Thm. 7.1 (h) gives this clause.

(J.b) We take $u = u_k^{\mathrm{pol}}$ if $s \geq j^*$ and $u = u_k(z)$ if $s < j^*$, on $\mathrm{Pd}(r)$ with $r_i = \mathfrak{r}(L^{\max(s,j^*)}\mathsf{b})/2$, and $B_0 = 1$, since by Thm. 7.19 (D1) (e), (f) the family $u(z)$ lies within 0.455 of $\bar{u}_k$ on $\mathrm{Pd}(r)$. The congruence follows from Thm. 7.1 (h), which holds at every field, from the definition of the jets in (o2) of item 2 of §7.3, from Prop. 7.12 (J6) and from (F4) (ii).

(J.c) For this clause and for the hypotheses of the second sentence of (F3) we take $\mathfrak{n} = \mathfrak{n}_k^{\mathrm{pol}}$ if $s \geq j^*$ and $\mathfrak{n} = \mathfrak{n}_k(z)$ if $s < j^*$, on the same $\mathrm{Pd}(r)$; it lies within 0.795 of $\bar{\mathfrak{n}}_k$, a system of members of a point of the class, by Thm. 7.19 (D1) (c), (e), (f) and Thm. 5.63 (iii). The congruence follows from Thm. 7.19 (D0), (D1) (c) and (F4) (ii), as in Step P11, and Δ is at most the bound of Thm. 7.19 (D3) at the depth s.

Here r and B_0 depend on no K, K', N, and $\tau_7 = C_{\mathrm{vol}}(NL^s, L^s\mathsf{b})$.

Item 6: the place of this section in the order of the letters.

Remark 7.31 (The letters and the floors of this section). In this section we impose no further condition on the letters (O0) to (O12), and the floors in force here are those of (O13). Indeed, at $x = L^s\mathsf{b}$, where $\mathsf{b} \geq x_s$, the data (B) follow from $x_s \geq e^2$, (X16), (X17) and Step 2 (c), (e) of the proof of Thm. 5.63. Here we use that, with the rate κ_1, the number x_s satisfies all of (O13) of item 1 of §5.5 ((O13) in item (1) of §6.1), so that the inequality $\mathsf{b} \geq x_s$ supplies what the floors (X16), (X17), (X12) and the arguments for these two items of Step 2 require of $x_\flat$.

Step 2 belongs to the proof of Thm. 5.63, among whose data is the condition $K \geq K_0(\mathsf{b})$, while the setting of item 4 of this section admits every $K \geq 1$. The arguments for the items (c) and (e) of that step are free of K_0:

(c) The argument for (c) rests on $\mathsf{b} \geq x_\flat$, on (X12), on the geometric series over $1 \leq j \leq K$, and on Thm. 2.57 (b), applied under its hypotheses $L \geq L_A$ odd, $aL \geq 3/c_L$ and N even; this clause yields $\ell_k \in \mathcal{R}(b_*, K_*, \lambda_*, \frac{1}{9})$ for $0 \leq k \leq K$, for every tower and all radii subject to (Cut), and its statement is free of K_0.

(e) The argument for (e) rests on $N \geq 2M\, s(L\mathsf{b}) \geq 2M\, s(\mathsf{b})$ and on a clause of (RB) in item 3 of §5.5, namely Prop. 5.8: if $x \geq e^2$, $L \geq 3$ and $N \geq 2M\, s(x)$, then $NL^j \geq 2M\, s(L^j x)$ for all $j \geq 0$. We apply this clause with $x = \mathsf{b} \geq e^2$.

Hence both items hold for every $K \geq 1$.

The item (a) of Step 2 enters Step P9 (R2) for all $K' \geq K \geq 1$. It asserts that the level $m + k$ of the run of length K' has the same torus, the same value x_k, the same grid, the same numbers listed in item (a) and the same step as the level k of the run of length K. The reasons are that $K' - (m + k) = K - k$ is the exponent of L in x_k and in the side of the torus, that the origins of the grids are fixed backwards from the last level of a run, that each of these numbers is a function of x_k and of the letters, and that the step depends on x_k, a and L and is independent of the rows, of the grid and of the members. Its argument, too, is free of K_0.

The bound $|\ell_{k,y}(p)| \le K_* e^{-\lambda_*|y-p|}$ is the clause on decay in the definition of $\mathcal{R}(b_*, K_*, \lambda_*, \cdot)$ (Def. 2.48). The number τ_7 is an entry of (O14), in item (1) of §6.1, and every floor is free of it. Finally, (S) and $C_{\mathrm{vol}}\Delta \le 1/2$ are conditions on distances, and no letter depends on them.

The proofs, in Steps P1 to P11.

Proof of Thm. 7.27. Step P1: the bound in the supremum norm. We show that

$$\sup_V |\rho[u_2;\ell] - \rho[u_3;\ell]| \le 1.61\, q_N e^{(0.61 a_E + 218 a_w) q_N}\, \mathsf{D}_{23},$$

together with the cases (i), (ii), (iii) and with (a^C).

We write $\rho[u_i;\ell] = e^{-x\mathrm{En}_\ell} e^{\mathrm{Sm}_i} \Xi_i$, where $\Xi_i := \Xi_{w_i}$. The energy En_ℓ, the indicators $1_{X^\approx}$ and the set of admitted families are the same for every i: they are determined by V, ε, $s(x)$ and the rows, and by nothing else. Moreover $0 < e^{-x\mathrm{En}_\ell} \le 1$, because $S_\varepsilon \ge 0$, $m_y \in \{0,1\}$ and all remaining terms are squares.

(1) *The sums.* Let $1_{X^\approx}(V) = 1$. Then the restriction of V to the bonds of $X^\approx$ takes values in G and is tame there; since $x > e$, it lies in $\mathcal{U}_X(x)$, by the definition "Domain" of item 2 of §5.5. For such V we therefore have

$$|E_i^a + E_i^h|(X,V) \le \|u_i\|_E^\sim (c_a + c_h) e^{-\tilde\kappa(\#X - 1)}$$

and

$$|(E_i^a + E_i^h) - (E_j^a + E_j^h)|(X,V) \le \mathsf{D}_{ij}(c_a + c_h) e^{-\tilde\kappa(\#X-1)}.$$

Every X holds a cube, so that $\sum_X \le \sum_{\text{cube}} \sum_{X \text{ holding it}}$. By the count $(26e)^{j-1}$ of item 2 of §5.5, and since $\tilde\kappa \ge 6$, the inner sum is at most $\sum_{j\ge1}(26e^{-5})^{j-1} = 1/(1-26e^{-5}) \le 1.22$.

Summing over the q_N cubes, we get $|\mathrm{Sm}_i - \mathrm{Sm}_j| \le 1.22\, q_N (c_a + c_h)\mathsf{D}_{ij}$; as $c_a + c_h \le 1/2$, it follows that $|\mathrm{Sm}_i| \le 0.61\, q_N a_E$ for $i = 2, 3$ and that $|\mathrm{Sm}_2 - \mathrm{Sm}_3| \le 0.61\, q_N \mathsf{D}_{23}$.

When $D(V)$ is empty, the only admitted family is the empty one, because each member of an admitted family has to hold a plaquette of $D(V)$; the empty family gives $\Xi_i = 1$. This is (a^C).

(2) *The exponentials.* For complex σ, σ' we have $e^\sigma - e^{\sigma'} = (\sigma - \sigma') \int_0^1 e^{\sigma' + t(\sigma - \sigma')} dt$, and therefore $|e^\sigma - e^{\sigma'}| \le e^{\max(\mathrm{Re}\,\sigma, \mathrm{Re}\,\sigma')} |\sigma - \sigma'|$. Together with (1) this gives $|e^{\mathrm{Sm}_2} - e^{\mathrm{Sm}_3}| \le e^{0.61 a_E q_N}\, 0.61\, q_N \mathsf{D}_{23}$ and $|e^{\mathrm{Sm}_3}| \le e^{0.61 a_E q_N}$.

(3) *A majorant for Ξ.* Two members of one family lie at a distance $> s(x) \ge 8$ from each other, hence they are disjoint. We fix an order of the cubes and assign to each member its first cube in that order; distinct members receive distinct first cubes, and so no family has more than q_N members.

Let $\mathrm{wt}(X) := e^{-\tilde\kappa(|X|_c - 1)}$. Then

$$\sum_{\text{families}} \prod_j a_w \mathrm{wt}(X_j) \le \prod_{\text{cube}} \Big(1 + a_w \sum_{X \text{ holding it}} \mathrm{wt}(X)\Big) \le \exp\{a_w q_N \sum_{j\ge1} (52e)^j e^{-6(j-1)}\},$$

and the sum in the exponent equals $52e/(1 - 52e^{-5}) \le 217.588 \le 218$. Consequently $|\Xi_2|, |\Xi_3| \le e^{218 a_w q_N}$.

(4) *Telescoping.* For finite products,

$$\prod_j y_j - \prod_j y_j' = \sum_j \Big(\prod_{i<j} y_i\Big)(y_j - y_j')\Big(\prod_{i>j} y_i'\Big).$$

We combine this with $|w_2 - w_3|(X,V) \le \mathsf{D}_{23}\mathrm{wt}(X)$ and $|w_i|(X,V) \le a_w \mathrm{wt}(X)$. For a family with #fam members we find

$$|\prod w_2 - \prod w_3| \le \mathsf{D}_{23}\, \#\mathrm{fam}\; a_w^{\#\mathrm{fam}-1} \prod \mathrm{wt}(X_j) \le \mathsf{D}_{23}\, q_N \prod a_w \mathrm{wt}(X_j),$$

where the second inequality holds as $\#\mathrm{fam} \le q_N$ and $a_w \ge 1$; for the empty family the difference is 0. Summing over the families and using (3), we arrive at $|\Xi_2 - \Xi_3| \le \mathsf{D}_{23}\, q_N e^{218 a_w q_N}$.

(5) *The sum.* From (2), (3) and (4),

$$|e^{\mathrm{Sm}_2}\Xi_2 - e^{\mathrm{Sm}_3}\Xi_3| \le |e^{\mathrm{Sm}_2} - e^{\mathrm{Sm}_3}||\Xi_2| + |e^{\mathrm{Sm}_3}||\Xi_2 - \Xi_3| \le (0.61+1)\, q_N \mathsf{D}_{23}\, e^{(0.61 a_E + 218 a_w) q_N}.$$

We now turn to the three cases.

(i) Here $\|u_1\|^\sim \le 1$, so that $a_E = a_w = 1 + \mathsf{D}_{\max}$, with $\mathsf{D}_{\max} := \max\{\mathsf{D}_{12}, \mathsf{D}_{13}\}$, are admitted. The exponent is then $218.61\, q_N + 218.61\, q_N \mathsf{D}_{\max} \le 218.61\, q_N + 218.61/400$, and $1.61\, e^{0.546525} \le 2.78086 < 2e^{0.39} \le 2e^{0.39 q_N}$.

(ii) In this case the coefficient of q_N in the exponent is $0.61 \cdot 2 + 218 = 219.22$.

(iii) Here $a_E = a_w = A$, and we use $218.61 A \le 219 A$ and $1.61 \le 2$.

Step P2: quotients and logarithms. We first prove an inequality. Let $h > 0$, and let f, g be complex integrable functions with $\eta_f := \|f - h\|_1 / \int h < 1$ and $\eta_g := \|g - h\|_1 / \int h < 1$. Then

$$\|f / \int f - g / \int g\|_1 \le 2\|f - g\|_1 / (\int h\, (1 - \eta_f)(1 - \eta_g)),$$

and when $f = h$ the left side is $\le 2\eta_g/(1 - \eta_g)$. From this inequality we then derive (c^C), with its clauses (c1) to (c4).

For complex f, g whose integrals are different from 0 one has the identity

$$f / \int f - g / \int g = (f - g) / \int f + g\, (\int g - \int f) / (\int f \int g).$$

Taking the L^1 norm of both sides, we see that the left side has L^1 norm $\le (\|f - g\|_1 / |\int f|)(1 + \|g\|_1 / |\int g|)$. At this point Step 9 (6) of the proof of Thm. 5.64 uses $\|g\|_1 / \int g = 1$. Here we use instead $\|g\|_1 \le (1 + \eta_g) \int h$, $|\int g| \ge (1 - \eta_g) \int h$ and $|\int f| \ge (1 - \eta_f) \int h$, together with $1 + (1 + \eta_g)/(1 - \eta_g) = 2/(1 - \eta_g)$.

The bound in L^1. Since dV is the product of the Haar probability measures and so has total mass 1, the integral $\int |\rho[u_2; \ell] - \rho[u_3; \ell]|\, dV$ is at most the supremum bound of Step P1. For the lower bound on I_1 we take the first inequality of Thm. 5.64 (c), applied with $\rho_2 := \rho_1$ and with the rate $\kappa_1 \ge 6$. Its proof, which is Step 9 (1) to (5) of the proof of Thm. 5.64, rests on $\mathrm{Sm}_1 \ge -0.61\, q_N$, on $\Xi_1 = 1$ on $\{\mathrm{ang}\ V(b) \le \varepsilon/4$ at all bonds $b\}$, on the bound on the rows and on $\rho_1 > 0$. Dividing the bound in L^1 by this lower bound, we obtain in the three cases:

(i) $2 q_N e^{219 q_N} e^{0.61 q_N} = 2 q_N e^{219.61 q_N} \le \frac{1}{2}\, 4 q_N e^{220 q_N}$;

(ii) $1.61\, q_N e^{219.83 q_N} \le 2 q_N e^{220 q_N}$;

(iii) $2 q_N e^{219 A q_N + 0.61 q_N} \le 2 q_N e^{220 q_N} e^{219 (A-1) q_N}$.

The other two factors of C_{vol} are those of the lower bound.

When (S) holds, the inequality $C_{\mathrm{vol}} \ge 400 q_N$ places us in case (i), and it does so for the pairs (u_i, u_1) as well, where $\mathsf{D}_{11} = 0$. With $h = \rho[u_1; \ell]$ we then have $\eta_i := \|\rho[u_i; \ell] - h\|_1 / I_1 \le \frac{1}{2} C_{\mathrm{vol}} \mathsf{D}_{1i} \le \frac{1}{2}$.

(c1) We have $|I_2 - I_3| \le \|\rho[u_2; \ell] - \rho[u_3; \ell]\|_1$ and $|I_i / I_1 - 1| \le \eta_i$; moreover $\int |\rho[u_i; \ell]| \le (1 + \eta_i) I_1 \le \frac{3}{2} I_1$, and $I_1 \le 2 |I_i|$.

(c2) Here $\rho_i / \int \rho_i = \rho[u_i; \ell] / I_i$, and we use the case $f = h$ of the inequality. The function $t \mapsto 2t/(1 - t)$ is increasing, and its value at $t = C_{\mathrm{vol}} \mathsf{D}_{12}/2$ is $C_{\mathrm{vol}} \mathsf{D}_{12} / (1 - C_{\mathrm{vol}} \mathsf{D}_{12}/2) \le 2 C_{\mathrm{vol}} \mathsf{D}_{12}$.

(c3) The inequality, with the bounds just obtained, gives $2 \cdot (C_{\mathrm{vol}} \mathsf{D}_{23}/2) \cdot 2 \cdot 2 = 4 C_{\mathrm{vol}} \mathsf{D}_{23}$.

(c4) Both quotients lie in the convex set $\{|w - 1| \le 1/2\}$, on which $|d \operatorname{Log} w / dw| = 1/|w| \le 2$; hence the difference of their logarithms is $\le 2 |I_2 - I_3| / I_1 \le C_{\mathrm{vol}} \mathsf{D}_{23}$.

Step P3: the laws, with a complex second member. With $\mu_i := \rho_i / \int \rho_i$, we prove that for $\eta_d < \log 2$

$$\int |\mu_1 - \mu_2|\, dV \le 2\, \mathrm{ms}_1 + \mathrm{ms}_2 (1 + e^{\eta_d}/(2 - e^{\eta_d})) + 2(e^{\eta_d} - 1)/(2 - e^{\eta_d}).$$

First, $\mathrm{ms}_1 < 1$. Indeed, the set Tm contains $\{\mathrm{ang}\ V(b) \le \varepsilon/4$ at all bonds $b\}$, whose Haar mass is positive, as Step 9 (1), (4) of the proof of Thm. 5.64 shows; moreover $\rho_1 > 0$.

On Tm, (a^C) gives $\mu_2 = c' e^{\sigma} \mu_1$, where $\sigma := \mathrm{Sm}_2 - \mathrm{Sm}_1$ satisfies $|\sigma| \le \eta_d$ and where $c' := e^{\mathsf{V}_2 - \mathsf{V}_1} \int \rho_1 / \int \rho_2$. Splitting the integral of μ_2 into its parts on and off Tm, we get $1 = \int \mu_2 = c'(1 - \mathrm{ms}_1)(1 + t) + \mathrm{mo}$; here $t := \int_{\mathrm{Tm}} (e^{\sigma} - 1) \mu_1 \, dV / (1 - \mathrm{ms}_1)$ satisfies $|t| \le e^{\eta_d} - 1 < 1$, and $\mathrm{mo} := \int_{\text{off Tm}} \mu_2$ satisfies $|\mathrm{mo}| \le \mathrm{ms}_2$. It follows that $(1 - \mathrm{ms}_1) c' = (1 - \mathrm{mo})/(1 + t)$, and that

$$\int_{\mathrm{Tm}} |\mu_2 - \mu_1| \le (1 - \mathrm{ms}_1) \sup |c' e^{\sigma} - 1| = \sup |(1 - \mathrm{mo}) e^{\sigma}/(1 + t) - (1 - \mathrm{ms}_1)| \le \sup |e^{\sigma}/(1 + t) - 1| + \mathrm{ms}_2 e^{\eta_d}/(2 - e^{\eta_d}) + \mathrm{ms}_1.$$

Further, $|e^{\sigma}/(1 + t) - 1| \le (|e^{\sigma} - 1| + |t|)/(1 - |t|) \le 2(e^{\eta_d} - 1)/(2 - e^{\eta_d})$.

Off Tm, the integral of $|\mu_1 - \mu_2|$ is at most $\mathrm{ms}_1 + \mathrm{ms}_2$. □

Proof of Cor. 7.28. Step P4: the integral along a family. We prove (F1): $I[u(z)]$ is holomorphic on $\mathrm{Pd}(r)$, the derivatives ∂_S may be taken under the integral, and $\|u(z) - u_1\|^{\sim} \le B_0 \max_i |z_i|/r_i$.

(1) *At a fixed field.* As the torus is finite, the sets $\mathcal{D}$ and $\mathcal{P}$ are finite, and so is the set of families. Thus $\rho[u(z); \ell](V)$ is the product of $e^{-x \mathrm{En}_\ell(V)}$, of the exponential of a finite sum, and of a polynomial in finitely many values, each of these values being holomorphic in z; hence it is holomorphic on $\mathrm{Pd}(r)$. For fixed z it is a Borel function of V. By Step P1 (1) and (3), applied with $a_E = a_w = 1 + B_0$, its modulus on $\mathrm{Pd}(r)$ is at most $\mathrm{Bd} := e^{218.61(1 + B_0) q_N}$.

(2) *The integral.* Fix radii $0 < r' < r'' < r$, the inequalities holding entry by entry. Take z in the closure of $\mathrm{Pd}(r')$ and change z_i by an amount of modulus $\le r_i'' - r_i'$. The difference quotient in z_i is then at most the supremum of $|\partial_i \rho|$ over the segment joining the two points, and Cauchy's estimate bounds that supremum by $\mathrm{Bd}/(r_i - r_i'')$. By dominated convergence, I is therefore continuous and complex differentiable in each z_i, with $\partial_i I = \int \partial_i \rho \, dV$. Here $\partial_i \rho$ is a Borel function of V, being a limit of Borel functions, and it is holomorphic in z and bounded on $\mathrm{Pd}(r'')$. Osgood's lemma states that a continuous function which is holomorphic in each variable separately is holomorphic; so I is holomorphic, and by iteration $\partial_S I = \int \partial_S \rho \, dV$.

Only three properties of the integrand enter (2): it is holomorphic in z at each V, Borel in V at each z, and bounded. For $\rho[u(z); \ell]$ they are supplied by (1). If u is any pointwise holomorphic family with $\sup \|u(z)\|^{\sim} \le A$ on a polydisc, then Step P1 (1) and (3), applied with $a_E = A$ and $a_w = \max\{1, A\}$, bound the modulus of $\rho[u(z); \ell]$ there by $e^{(0.61A + 218 \max\{1, A\}) q_N}$, which gives the third property. The other two properties follow for such a family from the first three sentences of (1), which use only that the family is pointwise holomorphic. The argument of (2) therefore applies to every pointwise holomorphic family that is bounded in $\| \cdot \|^{\sim}$, and to every function $F(z, V)$ that is holomorphic in z, Borel in V and bounded.

(3) *Schwarz's lemma for each value.* Fix X and W, and let $\mathrm{val}(z) := (E^a(X, \mathsf{W}; z) - E^a(X, \mathsf{W}; 0)) \, e^{\tilde{\kappa}(\#X - 1)}/c_a(x)$. This function is holomorphic on $\mathrm{Pd}(r)$, with $|\mathrm{val}| \le B_0$ and $\mathrm{val}(0) = 0$. For $z \ne 0$ let $t := \max_i |z_i|/r_i < 1$. The map $\zeta \mapsto \mathrm{val}(\zeta z/t)$ is holomorphic on the unit disc, bounded by B_0 and equal to 0 at 0, so Schwarz's lemma at $\zeta = t$ gives $|\mathrm{val}(z)| \le B_0 t$. The values of E^h and of w are treated in the same way, each multiplied by its own factor in $\| \cdot \|^{\sim}$ in place of $e^{\tilde{\kappa}(\#X - 1)}/c_a(x)$. Taking suprema over all values, we obtain the bound on $\|u(z) - u_1\|^{\sim}$.

Step P5: the logarithm of the integral. We prove (F2): on $\mathrm{Pd}(r/\tau_7)$ we have $|I[u(z)]/I_1 - 1| \le 1/2$, and $|\partial_S \log I[u(z)]|_0 \le \log 2 \, \tau_7^{|S|}/r^S$.

On $\mathrm{Pd}(r/\tau_7)$, (F1) gives $\|u(z) - u_1\|^{\sim} \le B_0/\tau_7 \le 1/C_{\mathrm{vol}}$. Hence (S) holds for $u_2 = u_3 = u(z)$, and (c1) yields $|\breve{\rho}_u(z) - 1| \le 1/2$, where $\breve{\rho}_u(z) := I[u(z)]/I_1$. So $\mathrm{Log}\, \breve{\rho}_u$ is holomorphic on this polydisc; and $|\mathrm{Log}(1 + v)| \le \sum_j |v|^j/j = -\log(1 - |v|) \le \log 2$ whenever $|v| \le 1/2$. As $I_1 > 0$ is a constant, for non-empty S we have $\partial_S \log I[u(z)] = \partial_S \mathrm{Log}\, \breve{\rho}_u(z)$ near 0, whichever branch of the logarithm is taken.

We use Cauchy's estimate in the variables of S, on the circles $|z_i| = t\, r_i/\tau_7$, with the other variables set to 0 and one derivative in each variable of S; letting $t \to 1$, we get $|\partial_S \mathrm{Log}\, \breve{\rho}_u(0)| \le \log 2 \prod_{i\in S} \tau_7/r_i$.

Step P6: two families against one positive reference. We prove the two sentences of (F3).

The first sentence. On $\mathrm{Pd}(r/2\tau_7)$, (F1) gives $\|u(z) - u_1\|^\sim \le B_0/(2\tau_7) \le 1/(2C_{\mathrm{vol}})$, and then $\|\mathfrak{n}(z) - u_1\|^\sim \le \Delta + 1/(2C_{\mathrm{vol}}) \le 1/C_{\mathrm{vol}}$. So (S) holds for $u_2 = u(z)$ and $u_3 = \mathfrak{n}(z)$, with u_1 as first member. Since $\|\mathfrak{n}(z)\|^\sim \le 2$, Step P4 (2) shows that $I[\mathfrak{n}(z)]$ is holomorphic. By (c1), both $\breve{\rho}_u(z)$ and $\breve{\rho}_{\mathfrak{n}}(z) := I[\mathfrak{n}(z)]/I_1$ lie in $\{|w-1| \le 1/2\}$; in particular $I[\mathfrak{n}(0)] \ne 0$.

By (c4), the function $\breve{\delta}(z) := \mathrm{Log}\, \breve{\rho}_u(z) - \mathrm{Log}\, \breve{\rho}_{\mathfrak{n}}(z)$ is holomorphic, with $|\breve{\delta}| \le C_{\mathrm{vol}}\Delta$. We apply Cauchy's estimate to it on $\mathrm{Pd}(r/2\tau_7)$, in the manner of Step P5: in the variables of S, on circles whose radii are t times those of $\mathrm{Pd}(r/2\tau_7)$, with the other variables at 0 and one derivative in each variable of S, and then we let $t \to 1$. For non-empty S the constant $\log I_1$ is annihilated by ∂_S.

The second sentence. If $C_{\mathrm{vol}}\Delta \le 1/2$, the first sentence gives a stronger bound. If $C_{\mathrm{vol}}\Delta > 1/2$, we apply (F2) to u, and (F2) to $\mathfrak{n}$ with $\mathfrak{n}(0)$ as its own first member. Here C_{vol} depends on N and x alone, and the number $\max\{1, C_{\mathrm{vol}} \sup_{\mathrm{Pd}(r)} \|\mathfrak{n}(z) - \mathfrak{n}(0)\|^\sim\}$, which (F0) assigns to $\mathfrak{n}$ with the first member $\mathfrak{n}(0)$, is $\le \tau_7$, since $\sup_{\mathrm{Pd}(r)} \|\mathfrak{n}(z) - \mathfrak{n}(0)\|^\sim \le B_0$ by a hypothesis of the second sentence of (F3). The left side is therefore

$$\le 2\log 2\, \tau_7^{|S|}/r^S < 2\log 2 \cdot 2C_{\mathrm{vol}}\Delta\, \tau_7^{|S|}/r^S \le 3C_{\mathrm{vol}}\Delta\, (2\tau_7)^{|S|}/r^S.$$

Step P7: the lemma on jets. We prove the clauses (i), (ii), (iii) of (F4).

(0) *The ideal.* Let $\mathfrak{a}$ be the set of the germs c with $c \equiv 0$; we show that $\mathfrak{a}$ is an ideal in the ring of germs. When two monomials are multiplied, the result is z^S only if they are $z^{S'}$ and $z^{S''}$ for disjoint S', S'' with $S' \cup S'' = S$. Hence, if $c \in \mathfrak{a}$ and f is a germ, the coefficient of z^S in cf equals $\sum_{S'\subset S} c_{S'} f_{S\setminus S'} = 0$. Consequently $\equiv$ is compatible with finite sums and finite products, and every $c \in \mathfrak{a}$ satisfies $c(0) = 0$.

Moreover $e^c - 1 = c \cdot \mathrm{pw}_1(c)$ and $\log(1+c) = c \cdot \mathrm{pw}_2(c)$, where pw_1 is entire and pw_2 is holomorphic near 0; so both lie in $\mathfrak{a}$. It follows that $f \equiv g$ implies $e^f - e^g = e^g(e^{f-g} - 1) \in \mathfrak{a}$; and if $f \equiv g$ and $g(0) \ne 0$, then $1/f - 1/g = (g-f)/(fg) \in \mathfrak{a}$, and $\log f - \log g = \log(1 + (f-g)/g) \in \mathfrak{a}$ for branches that agree at 0.

(i) Since the triples without a size form a complex vector space (Def. 7.21), $u^{\mathrm{pol}}(z)$ is such a triple for every z. Its values are polynomials, $u^{\mathrm{pol}}(0) = u_1$, and on $\mathrm{Pd}(r)$ we have $\|u^{\mathrm{pol}}(z) - u_1\|^\sim \le \sum_S |z^S|\, \|\delta_S\|^\sim \le B_0$.

(ii) At a fixed V, $\rho[\cdot;\ell](V)$ is computed from finitely many values by sums, products and a single exponential, so (0) applies to it. Step P4 (2) lets us take ∂_S under the integral for each of the two families; hence $I[u(z)] \equiv I[\hat{u}(z)]$. As $I[u(0)] \ne 0$, (0) applies to the logarithm, and ∂_S at 0 picks out the coefficient of z^S.

(iii) By Step P4 (2), applied to F and to $\rho[u(z);\ell]$, we have for all S

$$\partial_S \int F\, dV|_0 = \int \partial_S F(0,V)\, dV = \int \partial_S (e^{\mathfrak{V}(z)} \rho[u(z);\ell](V))|_0\, dV = \partial_S (e^{\mathfrak{V}(z)} I[u(z)])|_0.$$

So $\int F\, dV \equiv e^{\mathfrak{V}} I[u(z)]$, whose value at 0 is $e^{\mathfrak{V}(0)} I[u(0)] \ne 0$. By (0), $\log \int F\, dV \equiv \mathfrak{V} + \log I[u(z)]$ up to an additive constant in $2\pi i\mathbb{Z}$, and $F/\int F\, dV \equiv e^{\mathfrak{V}} \rho[u(z);\ell](V)/(e^{\mathfrak{V}} I[u(z)])$.

Step P8: the jets of the laws in L^1. We prove (F5):

$$\int |\partial_S \mu_u^z|_0\, dV \le 3\tau_7^{|S|}/r^S,$$

and

$$\int |\partial_S \mu_u^z - \partial_S \mu_{\mathfrak{n}}^z|_0\, dV \le 4C_{\mathrm{vol}}\Delta\, (2\tau_7)^{|S|}/r^S$$

if $C_{\mathrm{vol}}\Delta \le 1/2$, and also for every Δ under the hypotheses of the second sentence of (F3).

On $\mathrm{Pd}(r/\tau_7)$ the condition (S) holds, by Step P5, and (c1) gives $\|\mu_u^z\|_1 = \int |\rho[u(z);\ell]|/|I[u(z)]| \le 3$. On $\mathrm{Pd}(r/2\tau_7)$ the condition (S) holds for the pair $u(z)$, $\mathfrak{n}(z)$, by Step P6, and (c3) gives $\|\mu_u^z - \mu_{\mathfrak{n}}^z\|_1 \le 4C_{\mathrm{vol}}\Delta$.

Let $m^z(V)$ be holomorphic in z at each fixed V. Then $\partial_S m^z(V)|_0$ equals $(2\pi i)^{-|S|}$ times the integral of $m^z(V)\prod_{i\in S} dz_i/z_i^2$ over the circles $|z_i| = \mathrm{rd}_i$, $i \in S$, the other variables being 0. Since $m^z(V)$ is continuous in z and Borel in V, it is jointly Borel, and Tonelli's theorem gives $\int |\partial_S m^z|_0\, dV \le \sup_z \|m^z\|_1 / \prod_{i\in S} \mathrm{rd}_i$. We let rd_i tend to r_i/τ_7 for the first bound and to $r_i/2\tau_7$ for the second.

The second sentence. If $C_{\mathrm{vol}}\Delta \le 1/2$, it is contained in the first. So let $C_{\mathrm{vol}}\Delta > 1/2$. On $\mathrm{Pd}(r/\tau_7)$, (S) holds for $u_2 = u(z)$, with u_1 as first member, by Step P5, and (c2) gives $\|\mu_u^z - \mu_u^0\|_1 \le 2C_{\mathrm{vol}}\|u(z) - u_1\|^\sim \le 2$. As μ_u^0 is free of z, $\partial_S \mu_u^z = \partial_S(\mu_u^z - \mu_u^0)$ for non-empty S. The Cauchy formula and Tonelli's theorem, now applied to $\mu_u^z - \mu_u^0$, whose L^1 norm is at most 2, with rd_i tending to r_i/τ_7, give $\int |\partial_S \mu_u^z|_0\, dV \le 2\tau_7^{|S|}/r^S$.

The same bound holds for $\mathfrak{n}$, with $\mathfrak{n}(0)$ as its own first member; here the number that (F0) assigns to $\mathfrak{n}$ is $\max\{1, C_{\mathrm{vol}} \sup_{\mathrm{Pd}(r)} \|\mathfrak{n}(z) - \mathfrak{n}(0)\|^\sim\} \le \tau_7$, because $\sup_{\mathrm{Pd}(r)} \|\mathfrak{n}(z) - \mathfrak{n}(0)\|^\sim \le B_0$ is one of the hypotheses of the second sentence of (F3). Adding the two bounds, we get $4\tau_7^{|S|}/r^S < 8C_{\mathrm{vol}}\Delta\, \tau_7^{|S|}/r^S \le 4C_{\mathrm{vol}}\Delta\, (2\tau_7)^{|S|}/r^S$, because $2^{|S|} \ge 2$. □

Proof of Cor. 7.29. Step P9: the integrals of the runs and the two formulas. We prove $\int \rho_k^{[K],z} = Z_K(z)$ and the two formulas for S^T.

Bounds at a fixed K. We have $0 \le a_p \le 2$, and the bare lattice has $3(NL^K)^3$ plaquettes, so that $|\mathcal{O}_K(f)| \le 6L^K(NL^K)^3\|f\|_\infty$. It follows that $\rho_0^{[K],z}$ is continuous in U and entire in z, and that it is bounded on compact sets of z, uniformly in U. The kernel $\prod_c k_{ax}(V'(c), \mathsf{M}(U)(c))$, where M is the average in the definition "The step" of item 2 of §5.5, is a Borel function of (V', U) with values > 0 and $\le z(ax)^{-\#c}$, $\#c$ being the number of coarse bonds; at every U it is a probability density in V'. By induction on k, with the help of Step P4 (2), every $\rho_k^{[K],z}$ is Borel in V and entire in z, and it is bounded on compact sets of z, uniformly in V.

(R0) Fubini's theorem for a bounded Borel complex function gives $\int \rho'(V')\, dV' = \int \rho(U)\, [\int \prod_c k\, dV']\, dU = \int \rho(U)\, dU$.

(R1) At $x = L^s\mathsf{b}$, on $\mathbb{Z}^3/NL^s\mathbb{Z}^3$, both the data (B) and the bound on the rows are satisfied. Indeed, $\mathsf{b} \ge x_s \ge e^2$; NL^s is even, a multiple of M, and $\ge 2M\, s(L^s\mathsf{b})$, by Step 2 (e) of the proof of Thm. 5.63; $\varepsilon \le 1/40$ by (X16); $c_a + c_h \le 1/32$ by (X17); and ℓ_k has range $\le \varrho(x_k)$ and satisfies $|\ell_{k,y}(p)| \le K_* e^{-\lambda_*|y-p|}$, by Step 2 (c) of the proof of Thm. 5.63 and by Step 11 (a) of the proof of Cor. 5.65. We apply (F4) (iii) with $F = \rho_k^{[K],z}$ and $S = \{1, \dots, n\}$; together with (R0) it gives $S^T_{n;K} = \partial_1 \cdots \partial_n \log \int F\, dV|_0$. We then apply (F2).

(R2) We apply (F4) (iii) to both runs. At the level $m + k$ of the run of length K' and at the level k of the run of length K, the two runs share the torus, the value x_k, the grid and the step, by Step 2 (a) of the proof of Thm. 5.63, and both densities are defined with the rows ℓ_k. Here $I[\mathfrak{n}(0)] \ne 0$, by the first sentence of (F3), or since the density of which $\mathfrak{n}(0)$ is a system of members is positive. We subtract the two formulas and bound the difference by (F3).

For polynomial families, $\|u(z) - \mathfrak{n}(z)\|^\sim \le \|\bar{u}_k - \bar{\mathfrak{n}}_k\|^\sim + \sum_S |z^S|\, \|\delta_S - \tilde{\delta}_S\|^\sim$.

Step P10: the jets of the laws of the drawn fields. We prove the clauses (i) to (v) of (R3).

As $Z_K(0) > 0$, the quotient $\rho_{K-s}^{[K],z}(V)/Z_K(z)$ is holomorphic near 0 and bounded uniformly in V, by Step P9. Therefore π_S is bounded, by Cauchy's estimate, and Borel, as a limit of Borel functions.

(i) By Step P4 (2), $\int \mathrm{obs}\,\pi_S\,dV = \partial_S\big[\int \mathrm{obs}\,\rho_{K-s}^{[K],z}\,dV/Z_K(z)\big]_0$. Under the joint law of the bare field U and of the drawn fields we have $E[\mathrm{obs}(V_{K-s})\,e^{\sum z_i\mathcal{O}_K(f_i)}] = \int \mathrm{obs}\,\rho_{K-s}^{[K],z}\,dV/Z_K(0)$, since no kernel depends on a source. Hence the bracket equals $\frac{d}{dt}\log E[e^{\sum z_i\mathcal{O}_K(f_i)+t\,\mathrm{obs}}]$ at $t=0$, and its derivative ∂_S at $z=0$ is the joint cumulant, that is, the truncated function. For $\mathrm{obs}=1$ we get $\partial_S 1 = 0$.

The function $\mathcal{O}_K(f)$ is gauge invariant, and the step, which is linear in ρ, carries invariant functions into invariant functions (the definition "The step" of item 2 of §5.5). Hence $\rho_k^{[K],z}$ and π_S are gauge invariant as well. When the f_i are real, the quotient takes real values at real z.

(ii) Let $\mathrm{Kn}\,g(V') := \int g(U)\prod_c k_{aL^s\mathsf{b}}(V'(c),\mathsf{M}(U)(c))\,dU$. Since $|\mathrm{Kn}\,g| \le \mathrm{Kn}|g|$ and $\int \mathrm{Kn}|g| = \int |g|$, we have $\|\mathrm{Kn}\,g\|_1 \le \|g\|_1$. Further, $\mathrm{Kn}\,\rho_{K-s}^{[K],z} = \rho_{K-s+1}^{[K],z}$, the noise being $ax_{K-s} = aL^s\mathsf{b}$; $Z_K(z)$ does not depend on the field and is the same number in $\pi_{S,[s]}^{[K]}$ as in $\pi_{S,[s-1]}^{[K]}$; and ∂_S passes under the integral over U, by Step P4 (2).

(iii) By (R0), $\int F\,dV = Z_K(z)$ for $F = \rho_{K-s}^{[K],z}$, and $\int F\,dV = Z_{K'}(z)$ for $F = \rho_{K'-s}^{[K'],z}$. The second claim of (F4) (iii) yields $\pi_{S,[s]}^{[K]} = \partial_S\mu_u^z|_0$ for the first F and $\pi_{S,[s]}^{[K']} = \partial_S\mu_{\mathfrak{n}}^z|_0$ for the second; here $I[\mathfrak{n}(0)] \ne 0$, either by the first sentence of (F3) or because $\mathfrak{n}(0)$ is a system of members of a density that is positive. We then apply (F5), together with its second sentence.

(iv) Clause (iii) shows that the sequence is a Cauchy sequence in the complete space $L^1(dV)$; we let $K' \to \infty$ in (iii). The integral, the invariance (the Haar measure being invariant), the reality and the first bound are preserved under limits in L^1. We let $K \to \infty$ on both sides of (ii), using that Kn is continuous in L^1 and independent of K.

(v) The run $(K-s, L^s\mathsf{b}, NL^s)$ starts from $e^{-L^{K-s}L^s\mathsf{b}\sum(1-\cos\phi_p)}$ on $\mathbb{Z}^3/(NL^s)L^{K-s}\mathbb{Z}^3$, and its noises are $aL^{K-s-k}L^s\mathsf{b} = ax_k$. Hence at the levels $0 \le k \le K-s$ it has the same densities as the run (K,b,N), and π_S is independent of the grid, of the rows and of the members.

With $\mathrm{bar}(p)$ the barycentre of p in lattice units,

$$\mathcal{O}_{K-s}(\hat f) = L^{K-s}\sum_p f(L^{-s}L^{-(K-s)}\mathrm{bar}(p))\,a_p = L^{-s}\mathcal{O}_K(f)$$

. Thus $\sum z_i\mathcal{O}_K(f_i) = \sum(L^s z_i)\mathcal{O}_{K-s}(\hat f_i)$, and every derivative produces a factor L^s.

With $(L^s\mathsf{b}, NL^s)$ in the place of (b,N), all conditions of the setting of item 4 of this section are met: $L^s\mathsf{b} \ge x_s$; $NL^s = 2L^{m_l+s}$; and (Tor) follows from (RB) in item 3 of §5.5, by which $N \ge 2M\,s(x)$ implies $NL^j \ge 2M\,s(L^j x)$, applied with $x = L\mathsf{b}$.

The two thresholds. Let $\mathrm{Kt} := K_T(L^s\mathsf{b}, n, L^s d)$. We claim that $\mathrm{Kt}+s \ge K_T(\mathsf{b},n,d)$ and $\mathrm{Kt} \ge K_0(L^s\mathsf{b})$; then every K with $K-s \ge \mathrm{Kt}$ satisfies $K \ge K_T(\mathsf{b},n,d)$ and $K-s \ge K_0(L^s\mathsf{b})$. The second claim is part of the definition of K_T.

For the first claim, note that $\Lambda(L^s\mathsf{b}) = \Lambda + s/2$, whence $K_0(L^s\mathsf{b}) = K_0(\mathsf{b}) + (5/4)s$; that the $\mathrm{sep}^{[i]}$ computed from $(L^s\mathsf{b}, L^s d)$ is the $\mathrm{sep}^{[i+s]}$ computed from (b,d), while sep_* depends on n and L alone, so that the definition of j^*, with $(L^s\mathsf{b}, L^s d)$ in the place of (b,d), gives $\max\{j^*-s,1\}$; and that $C_5^{\#}$ and K_5 are independent of b. Let $\mathrm{Ex}_t := (4/9)\mathrm{Kt} - (10/9)(\Lambda+s/2) - 1$. This is the left side of (K1a) and the exponent of 0.3683 in (K1b) when Kt replaces K and $L^s\mathsf{b}$ replaces b, and hence $\Lambda + s/2$ replaces Λ. Let $K := \mathrm{Kt} + s$. Then $(4/9)K - (10/9)\Lambda - 1 = \mathrm{Ex}_t + s$.

(K1a) This condition holds for the data (K,b,d), since $\mathrm{Ex}_t + s \ge \max\{j^*-s,1\}+1+s \ge j^*+1$.

(K1b) This condition holds there as well, since $C_5^{\#}K_5^{-1}K\,0.3683^{\mathrm{Ex}_t+s} \le 0.1\,(1 + s/\mathrm{Kt})\,0.3683^s \le 0.1\,(1+s/3)\,0.3683^s \le 0.1$. Here an integer $\mathrm{Kt} \ge K_0(L^s\mathsf{b}) > 9/4$ is ≥ 3, and $(1+s/3)\,0.3683^s \le 0.4911$ for $s \ge 1$: the value at $s=1$ is 0.49107, and the quotient of the value at $s+1$ by the value at s is $\le (5/4)\,0.3683 < 1$.

Finally $K \geq K_0(L^s\mathsf{b}) + s \geq K_0(\mathsf{b})$. So K is an integer $\geq K_0(\mathsf{b})$ with (K1a) and (K1b), and $K_T(\mathsf{b}, n, d)$ is the least such integer; this proves the first claim.

□

Proof of Rem. 7.30. Step P11: the origin of the families and of the datum. We show that the families of Prop. 7.12 (J6) and of Thm. 7.19 (D1) at the last level have these properties: they are pointwise holomorphic; they are measured in a norm which dominates $\|\cdot\|^\sim$ at a rate in $[\kappa, \kappa_1]$, with the cap c_h; they are defined on polydiscs, the family of (J6) on one containing $\mathrm{Pd}(\mathfrak{r}(L^{j^*}\mathsf{b}))$ and the family of (D1) on one containing $\mathrm{Pd}(\mathfrak{r}(L^{j^*}\mathsf{b})/2)$, with the radius $\mathfrak{r}(L^{j^*}\mathsf{b})$ independent of K; and they start at systems of members of points of the class.

The assertion on the radius of the polydiscs follows from (G3) in item 3 of §7.3, by which $\mathfrak{r}_* \geq \mathfrak{r}(L^{j^*}\mathsf{b})$ for every $K \geq 1$, with equality if $K \geq j^*$, while for $K < j^*$ the radius $\mathfrak{r}_* = \mathfrak{r}(L^K\mathsf{b})$ depends on K. The restriction of a family to a smaller polydisc is a family as well, with the same B_0.

Prop. 7.12 (J6) provides, for $k_d < k \leq K$, the family $u_k(z)$ on $\mathrm{Pd}(\mathfrak{r}_*)$, in M_1 with the strong cap, and for $k_d \leq k \leq K$ the bound $\|u_k(z) - \bar{u}_k\|^{\mathrm{st},(\kappa)} \leq 0.91 \cdot 0.4^{k-k_d} \leq 0.91$, in the norm with the rate κ_k, where $\kappa_k = \kappa$ for $k \geq k_d$. At $k = k_d$ the family is $u^{\mathrm{pol}}_{k_d}$, which lies in $M_{1.91}$ by (J5); we use the case $k = K > k_d$ alone. The norm $\|\cdot\|^{\mathrm{st},(\kappa)}$ measures w in strong units, so it dominates $\|\cdot\|^\sim$ at $\tilde{\kappa} = \kappa$. Since $k_d \leq K - 1$, we have $u_K(z) = F_{\ell_{K-1}}(u_{K-1}(z))$; for this output the clauses of Thm. 5.15 hold for the pair (κ, c_h), whose rate is $\kappa < \kappa_1$, by Thm. 6.7 (U), and Thm. 6.7 (c+) shows that it is pointwise holomorphic.

As to the true multi-affine jets: $u_K(z) \equiv \hat{u}_K(z)$, where $\hat{u}_K(z)$ is the family of Thm. 7.1 (h), for which $\rho_K^{[K],z} = e^{\mathfrak{V}(z)}\rho[\hat{u}_K(z); \ell_K]$ at every field. This congruence is shown at the end of (3) below, and with (F4) (ii) it gives (J.b). Moreover Prop. 7.12 (J7) states the formula of (R1) at depth 0.

We now supply what (F3) requires for $s = 0$: the second family, the bound on Δ, the polydisc and the base point. On $\mathrm{Pd}(\mathfrak{r}_*/2)$, Thm. 7.19 (D1) provides $\mathfrak{n}_K(z)$, and (D3) bounds $\sup \mathsf{D}^{(\kappa)}_{\mathsf{b}}(u_K(z), \mathfrak{n}_K(z))$, the rate of this distance being $\kappa_K = \kappa$; the distance $\mathsf{D}^{(\kappa)}_{\mathsf{b}}$ dominates $\|\cdot\|^\sim$; and at $r_i = \mathfrak{r}_*/2$ the polydisc $\mathrm{Pd}(r/2\tau_7)$ lies in $\mathrm{Pd}(\mathfrak{r}_*/2)$. Further, $\mathfrak{n}_K(0) = \bar{\mathfrak{n}}_K$ is a real system as in Thm. 5.63 (iii). That clause asserts that such a system exists, without singling one out, and the system in question is the one of Thm. 7.19 (D1) (c). What we use of it is Thm. 7.19 (D0) at $z = 0$: the system is real, in $M_1(\mathsf{b}; \kappa_1, c_h)$, with the strong cap, and $\rho[\bar{\mathfrak{n}}_K; \ell_K] > 0$, this density being a positive multiple of $\rho_{K'}^{[K']}$.

We use three further facts: (1) for (J.a), for every $K \geq 1$; (2) for the value $B_0 = 1$ in the hypotheses of the second sentence of (F3); and (3) for (J.c) and for the congruence $u_K(z) \equiv \hat{u}_K(z)$ of the short run.

(1) For every $K \geq 1$, $\bar{u}_K$ is a system of members of a point of the class, its reality and positivity coming from Thm. 5.15 (d) from level 0 on. This is what the last sentences of Thm. 7.1 (h) say: at $z = 0$, where the sources disappear from the weight, whatever the f_i are, all members are real and $\rho[u_k(z); \ell_k] > 0$; the f_i of the setting may be complex, and for real f_i the same holds at real z. Hence $\bar{u}_k$ is a system of members of ρ_k for $0 \leq k \leq K$: it is real, in $M_1^{\mathrm{su}}(x_k)$, with the strong cap.

(2) We have $\sup_{\mathrm{Pd}(\mathfrak{r}_*/2)} \|\mathfrak{n}_K(z) - \mathfrak{n}_K(0)\|^\sim \leq 1$, the supremum being at most $0.455 + 0.34$. This is Thm. 7.19 (D1) (f), which gives $\leq 0.795\,\vartheta[2]^{K-k_d}$ in the strong norm with the rate κ, and that norm dominates $\|\cdot\|^\sim$; at a depth $s \geq j^*$ it is (D1) (e).

(3) The pointwise congruence $\rho_{K'}^{[K'],z}(V) \equiv e^{\mathfrak{V}'(z)}\rho[\mathfrak{n}_K(z); \ell_K](V)$ is (J.c) at $s = 0$. The families of (D1) are constructed so that this congruence holds level by level, by Lem. 7.10 (B1) at the switch and by Thm. 5.15 (a) and Lem. 7.8 at every later step. Two clauses of Thm. 7.19 yield it.

By (D0), on the interior of the polydisc Pd_0 named there for the run of length K' we have, at every field,

$$\rho_{m+k}^{[K'],z} = \exp\{\tilde{\mathsf{V}}_0(z) + \sum_{k_0 \le j < k} \mathsf{V}'_{\ell_j}[\mathfrak{n}_j^\circ(z)]\}\, \rho[\mathfrak{n}_k^\circ(z); \ell_k],$$

where the numbers in the exponent are holomorphic; we write $\mathfrak{V}'(z)$ for the exponent. By (D1) (c) for $k \le k_d$, and by (D1) (c) with (f) for $k_d < k \le K$, the family $\mathfrak{n}_k(z)$, which for $k \le k_d$ is $\mathfrak{n}_k^{\mathrm{pol}}(z)$, and the family $\mathfrak{n}_k^\circ(z)$ have the same base point and the same multi-affine jets; so they are congruent. The first claim of (F4) (ii) then gives $\rho[\mathfrak{n}_k(z); \ell_k](V) \equiv \rho[\mathfrak{n}_k^\circ(z); \ell_k](V)$ at every V, and multiplication by $e^{\mathfrak{V}'(z)}$ preserves $\equiv$, by Step P7 (0). This proves (J.c).

For the short run the argument is the same, with Thm. 7.1 (h), (o2) in item 2 of §7.3 and Prop. 7.12 (J6) in place of the two clauses of Thm. 7.19; this proves the congruence $u_K(z) \equiv \hat{u}_K(z)$ used above for (J.b).

□

8. Correlation bounds and limits

8.1. **Fixed-torus convergence.** On a fixed torus we bound the truncated functions of the action density uniformly in the cutoff, show that they converge, with the explicit bound $C'_n K \theta^K$ of (T2) and (T3) on the speed of the convergence in K, and derive the properties of their limit. In Steps 0 to 3 of the proof of Thm. 8.2, whose clauses are (T1) to (T7) and (T7+), we introduce the generating function, prove Lem. 8.4 on the numbers of (O14), deduce (T1) from Prop. 7.12 and Thm. 7.27, and deduce (T2) from Thm. 7.19 and Thm. 7.27. Steps 4 to 11 construct the limit, prove the dilation identities, the continuity in b, the invariance, the reflection positivity and the regularity, pass to every $\mathsf{b} > 0$ and every side, and treat the laws together with their jets.

Item 1: letters and numbers. We retain the parameter table and floors of §6.1, with $\kappa_1 = \kappa + 1$. In particular $M = L^{m'}$ with $m' \ge 1$, $p_0(x) = A_0(\log x)^{\mathfrak{p}_0}$, $\mathfrak{p}_0 > 2$, $\varepsilon(x) = p_0(x) x^{-1/2}$, $c_a(x) = E_a p_0(x)^3 x^{-1/2}$ and $c_h(x) = e^{-c_h p_0(x)^2}$. The floor x_s satisfies (O13) there, including all the unsourced floors at κ_1, $Lx_s \ge \max\{x_9^U, x_3, x_4\}$, (X5), and the final conditions (W2), (W3). Thus $p_0 \ge 1$, $\varepsilon \le 1/40$ and $16(c_a + c_h) \le 1/2$ for $x \ge x_s$.

Radii. We use $r(x) = \lceil 20 c_R \log x/M \rceil$, $s(x) = 8r(x)$, $\varrho(x) = \lceil c_R \log x \rceil$ and $\check{r}(x) = r(x) + 3$.

The quantities Λ, K_0, k_0, ϑ, θ, C_{b}, amp, $\mathfrak{r}$, $\mathrm{sep}^{[i]}$, sep_*, j^*, $N_1^{[j]}$, $m^{[j]}$, K_T, Tail and Ser_2 are those of items 1.1 and 1.3 of §7.4, for the data (b, n, d, R_f). The volume constant is Def. 7.24. No new condition is imposed on these quantities. We additionally put

$$\mathrm{Ser}_1 := \mathrm{Tail}(0), \qquad P_* := \mathfrak{r}(L^{j^*}\mathsf{b})^{-n} N_1^{[j^*]} (C_g/16) e^{\kappa}.$$

The definitions give the identity

$$C_{L^s\mathsf{b}} \theta^{K-s} = 3\vartheta^{(4/9)K - s - (10/9)\Lambda - 1}.$$

In a run of length K, $x_k = L^{K-k}\mathsf{b}$, the depth is $K - k$, $k_d = \max\{K - j^*, 0\}$, $\mathfrak{r}_k = \mathfrak{r}(x_k)$ and $\mathfrak{r}_* = \mathfrak{r}_{k_d} \ge \mathfrak{r}(L^{j^*}\mathsf{b})$, by (G3) of item 3 of §7.3. We write $\mathrm{Pd}(r) = \{z \in \mathbb{C}^n : |z_i| < r\}$.

Distances. The separation d, the radius R_f and the balls refer to the sup distance of T_l; a tuple that is of the type $(d; R)$ in Def. 8.1, where the distance is Euclidean, is of the type $(d/\sqrt{3}; R)$ here. The Lipschitz constant $\mathrm{Lip} f$, by contrast, refers to the Euclidean distance, and $\|f\|_\sharp = \max\{\|f\|_\infty, 40M \,\mathrm{Lip} f\}$, with M the letter of (O7).

We use the following conventions for test functions and isometries.

Definition 8.1 (Separated tuples and functionals on them)**.** Let $\mathbb{E} = T_l$. For O open in $\mathbb{E}$ and $n \ge 0$, the set $\mathfrak{S}_n(O)$ consists of the n-tuples $X = (f_1, \dots, f_n)$ of real C^1 functions whose supports are compact, contained in O and pairwise disjoint; the entry 0 is allowed. The set $\mathfrak{S}_0(O)$ consists

of the empty tuple alone, $\mathfrak{S}(O) = \bigcup_{n\geq 0} \mathfrak{S}_n(O)$, and $\mathfrak{S}_n^\infty(\mathbb{E})$ is the set of the tuples in $\mathfrak{S}_n(\mathbb{E})$ all of whose entries are C^∞.

For a set J of indices we write $X_J = (f_i)_{i\in J}$, and $X \cdot Y$ is the tuple X followed by the tuple Y; for an isometry r we put $rf = f \circ r^{-1}$ and $rX = (rf_1, \dots, rf_n)$; and $\tau_v : x \mapsto x + v$. The cubic group W_3 consists of the 48 signed permutation matrices of $\mathbb{R}^3$. Each of them maps $l\mathbb{Z}^3$ and $a_K\mathbb{Z}^3$ onto themselves, so that W_3 acts on T_l and on every lattice of spacing a_K; the reflection $y_\mu \mapsto -y_\mu$ is an element of W_3.

A tuple is of the *type* $(d; R)$ if the supports of any two different entries are at distance $\geq d$ from each other and each support lies in a ball of radius R. We write $\mathrm{Part}_{\geq 2}(n)$ for the set of the partitions of $\{1, \dots, n\}$ into blocks of size ≥ 2; the set $\mathrm{Part}_{\geq 2}(0)$ consists of the empty partition, and $\mathrm{Part}_{\geq 2}(1)$ is empty.

A map on $\mathfrak{S}_n(O)$ is *multilinear* if it is n-linear on $C_c^1(O_1) \times \cdots \times C_c^1(O_n)$ for all open sets $O_i \subset O$ with pairwise disjoint closures, and *symmetric* if every permutation of the entries leaves it unchanged. A *functional on separated tuples* is a multilinear map $\sigma : \mathfrak{S}(\mathbb{E}) \to \mathbb{R}$ that takes the value 1 at the empty tuple. A *family of connected functions* is a sequence $S = (S_n)_{n\geq 2}$ of multilinear maps $S_n : \mathfrak{S}_n(\mathbb{E}) \to \mathbb{R}$; its *centred functional* eq. 1.i is

$$\sigma_S(X) = \sum_{\pi\in \mathrm{Part}_{\geq 2}(n)} \prod_{J\in\pi} S_{|J|}(X_J)$$

for tuples X of length $n \geq 0$. A family S is said to have a property of functionals when σ_S has it, with one exception: S is called symmetric when every S_n is symmetric. A map is *invariant* under r if its value at rX equals its value at X.

A *mirror* of T_l is a pair $\Sigma = (\mu, c)$ with $\mu \in \{1, 2, 3\}$ and $c \in \mathbb{R}/l\mathbb{Z}$; its reflection θ_Σ is induced by $y \mapsto y - 2(y_\mu - c)e_\mu$, and its open side is $\mathbb{E}_+(\Sigma) = \{c < y_\mu < c + l/2\}$. A functional σ is *reflection positive at* Σ, in the sense of eq. C.viii, if

$$\sum_{i,k} \bar\zeta_i \zeta_k \, \sigma(\theta_\Sigma X_i \cdot X_k) \geq 0$$

for every r, all $X_1, \dots, X_r \in \mathfrak{S}(\mathbb{E}_+(\Sigma))$ and all complex numbers ζ_i.

Let $\mathbb{E}^n_{\neq}$ be the set of the points of $\mathbb{E}^n$ whose entries are pairwise different. A *distribution* on $\mathbb{E}^n_{\neq}$ is a complex linear map D on $C_c^\infty(\mathbb{E}^n_{\neq}; \mathbb{C})$ such that for each compact set there are k and C with $|D(F)| \leq C \max_{|\alpha|\leq k} \sup |\partial^\alpha F|$ for all F carried by that set; it is *of order* $\leq k$ *on compact sets* if a single k suffices for all compact sets. A functional σ is *regular* if for every n there is a distribution σ_n on $\mathbb{E}^n_{\neq}$ with $\sigma_n(f_1 \otimes \cdots \otimes f_n) = \sigma(f_1, \dots, f_n)$ on $\mathfrak{S}_n^\infty(\mathbb{E})$. ■

On the lattice (8.1) the functional is $\sigma_{K,l,\beta}(X) = \langle \prod_i (\mathcal{O}_K(f_i) - \langle \mathcal{O}_K(f_i)\rangle)\rangle$, and $\sigma_{K,l,\beta} = \sigma_S$ at $S = (S^T_{n;K,l,\beta})_{n\geq 2}$; this is the classical expression of centred moments through cumulants.

In (T6) and in Steps 5, 6 and 10, which are free of sources, $z \in \mathbb{Z}$ is the integer of the dilation; everywhere else z stands for the sources in $\mathbb{C}^n$. The letter s is the depth in (T7+), in $C_{L^s\mathfrak{b}}$, in $C^{(s)}$ and in Step 11; $s(\cdot)$ denotes the radius function $s(x) = 8r(x)$, also where s without an argument is said to be non-decreasing; in the kernel $k_s(g, h)$ and in its formula, s is the noise; and in x_s it is part of the name of the floor, not the depth, whereas x_k, $x_{\check k}$ and x_{K-1}, with a level as index, are the couplings of these levels. In Steps 4 and 6, e is a tolerance, and the upper index e of f^e is the same tolerance. With a plaquette as argument, $\partial U(p)$ is the product of U along the boundary of p (1.1); in Step 8, $\theta_0 := \theta_{(\mu,0)}$ and $\theta_c := \theta_{(\mu,c)}$ are the reflections of the mirrors $(\mu, 0)$ and (μ, c) of T_l; and in Step 10, n_i and n_k are the lengths of X_i and X_k.

In this section the tags (a^C) to (d^C), (S) and (c1) to (c4) are those of Thm. 7.27; (F0) to (F5) are those of Cor. 7.28; (R0) to (R3) those of Cor. 7.29; (D0) to (D7) those of Thm. 7.19; (J0) to (J8) those of Prop. 7.12; and (G1) to (G4) those of item 3 of §7.3.

Item 2: the statement.

Theorem 8.2 (Truncated functions of the action density: bounds, rate and limit)**.** *Data.* The group is $G = \mathrm{SU}(2)$, the dimension is 3, and the letters are those of item 1. The variables are $\mathsf{b} \geq x_s$, the side $l = N = 2L^{m_l}$ with $m_l \geq m'$, so that N is even and $M \mid N$, subject to (Tor) $N \geq 2Ms(L\mathsf{b})$, and $K \geq 1$; we put $\beta = L^K\mathsf{b}$ and $a_K = L^{-K}$. The torus is $T_l = \mathbb{R}^3/l\mathbb{Z}^3$, and the lattice $a_K\mathbb{Z}^3/l\mathbb{Z}^3 \subset T_l$ of 1.1 has spacing a_K; as a graph it is $\mathbb{Z}^3/NL^K\mathbb{Z}^3$, with the site y of the graph placed at the point $a_K y$. So 0 is a site, $a_K\mathbb{Z}^3 \subset a_{K'}\mathbb{Z}^3$ for $K' \geq K$, and $l/2 = L^{K+m_l}a_K$. The plaquette based at y and spanned by $e_\mu, e_{\mu'}$ has the barycentre $x(p) = a_K(y + (e_\mu + e_{\mu'})/2)$, as in item 1 of §6.3.

Let $a_p(U) = 1 - \frac{1}{2}\operatorname{tr}\partial U(p)$, $\mu_{K,l,\beta} = Z^{-1}e^{-\beta\sum_p a_p}dU$ and $\mathcal{O}_K(f) = a_K^{-1}\sum_p f(x(p))a_p(U)$, where $x(p)$ is the barycentre, and let $S^T_{n;K,l,\beta}(f_1,\dots,f_n) = \langle\mathcal{O}_K(f_1);\dots;\mathcal{O}_K(f_n)\rangle$ denote the joint cumulant. These coincide with the objects of Def. 1.1, whose formulas are taken at the L and M of the table ((O4), (O7)) rather than at the L and M in the base Wilson model. Beyond the table and the data, the theorem has no hypothesis. In this setting the following hold.

(T1) For $n \geq 2$, $d > 0$ and $R_f > 0$, the bound

$$|S^T_{n;K,l,L^K\mathsf{b}}(f_1,\dots,f_n)| \leq C_n \prod \|f_i\|_\infty$$

holds for every $K \geq 1$ and for all real bounded functions f_i on T_l whose supports are pairwise at distance $\geq d$, each support lying in a ball of radius R_f. Here $C_n := \mathrm{Ser}_1(\mathsf{b}, n, d, R_f) + \log 2\,(\tau_7/\mathfrak{r}(L^{j^*}\mathsf{b}))^n$ with $\tau_7 = \max\{1, 0.91C_{\mathrm{vol}}(N,\mathsf{b})\}$; this constant is free of K and locally bounded in b.

(T2) For real Lipschitz functions f_i as in (T1) and for $K' \geq K \geq K_T(\mathsf{b}, n, d)$,

$$|S^T_{n;K',l,L^{K'}\mathsf{b}}(f) - S^T_{n;K,l,L^K\mathsf{b}}(f)| \leq C'_n K\theta^K \prod \|f_i\|_\sharp,$$

where $\theta = \vartheta^{4/9} \leq 0.69903$ and

$$C'_n := 4P_*\vartheta^{-j^*-(10/9)\Lambda} + (C_{\mathsf{b}}/3)\mathrm{Ser}_2(4 + 1.36C_5^\#/K_5) + 3C_{\mathrm{vol}}(N,\mathsf{b})C_{\mathsf{b}}(1.15 + 0.46C_5^\#/K_5)(4C_{\mathrm{vol}}(N,\mathsf{b})/\mathfrak{r}(L^{j^*}\mathsf{b}))^n.$$

Since K_T and C'_n are locally bounded in b, the bound is uniform on $I_N := \{\mathsf{b} \geq x_s : (\mathrm{Tor})\}$, which is either empty or one compact interval $[x_s, \mathsf{b}_N]$. The threshold K_T does not depend on N.

(T3) For real Lipschitz tuples with pairwise disjoint supports, the limit $S_n^{(l,\mathsf{b})}(f) := \lim_K S^T_{n;K,l,L^K\mathsf{b}}(f)$ exists as $K \to \infty$ through all integers K, and $|S_n^{(l,\mathsf{b})} - S^T_{n;K}| \leq C'_n K\theta^K \prod \|f_i\|_\sharp$ for $K \geq K_T$. The limit is real and symmetric, it is n-linear on every product of spaces of functions supported in pairwise disjoint closed sets, one set for each factor, and it satisfies (T1). For continuous entries with pairwise disjoint supports the limit exists as well, and it coincides there with the extension given by (T1).

(T4) For tuples f as in (T3), Lipschitz or continuous, the map $\mathsf{b} \mapsto S_n^{(l,\mathsf{b})}(f)$ is continuous. If $K_j \to \infty$ and $a_{K_j}\beta_j \to \mathsf{b}$, then $S^T_{n;K_j,l,\beta_j}(f) \to S_n^{(l,\mathsf{b})}(f)$. Together with (T6), both statements hold for every $\mathsf{b} > 0$; and if (K_j, β_j) is a sequence of data with $K_j \to \infty$ whose labels $a_{K_j}\beta_j$ remain in a compact subset of $(0,\infty)$, one can pass to a subsequence along which $S^T_{n;K_j,l,\beta_j}(X)$ converges simultaneously for all $n \geq 2$ and all X.

(T5) The family $S = (S_n^{(l,\mathsf{b})})_{n\geq 2}$, taken on the separated tuples $\mathfrak{S}_n(T_l)$ of Def. 8.1, whose entries are C^1 with pairwise disjoint supports, forms a family of connected functions as defined in 8.1 and written out in item 1 of this section; moreover:

(i) S is invariant under all translations of T_l and under W_3;

(ii) the centred functional $\sigma_S(X) = \sum_{\pi\in\mathrm{Part}_{\geq 2}(n)}\prod_{J\in\pi} S_{|J|}(X_J)$ of eq. 1.i is, at every mirror (μ, c) of T_l, reflection positive as in eq. C.viii;

(iii) each S_n is symmetric;

(iv) there is exactly one distribution $\tilde{S}_n$ on $(T_l)^n$ off the diagonals such that $\tilde{S}_n(f_1 \otimes \cdots \otimes f_n) = S_n(f_1, \dots, f_n)$ on $\mathfrak{S}_n^\infty(T_l)$; its order on compact sets is $\le 3n+1$, and σ_S is regular.

(T6) By eq. 1.ii and eq. 1.iii, the clauses (T3) to (T5) hold for every $\mathsf{b} > 0$ and every side $l = 2L^m$, $m \in \mathbb{Z}$. Moreover, for every z with $L^z\mathsf{b} \ge x_s$, $m + z \ge m'$ and $L^z l \ge 2Ms(L^{z+1}\mathsf{b})$,

$$|S_n^{(l,\mathsf{b})}(f)| \le L^{zn} C_n(L^z\mathsf{b}, n, L^z d, L^z R_f, L^z l) \prod \|f_i\|_\infty,$$

and, for real Lipschitz f_i and $K - z \ge K_T(L^z\mathsf{b}, n, L^z d)$,

$$|S_n^{(l,\mathsf{b})}(f) - S^T_{n;K,l,L^K\mathsf{b}}(f)| \le L^{zn} C'_n(L^z\mathsf{b}, n, L^z d, L^z R_f, L^z l)(K-z)\theta^{K-z} \prod_i \max\{\|f_i\|_\infty, 40ML^{-z}\mathrm{Lip} f_i\};$$

this is the rate of (T3), with its constants taken at the dilated data.

(T7) Cor. 5.65 holds with the rate κ_1 for the b and N of the data.

(T7+) *The jets of the laws.* In the setting of (T2), let $\|f_i\|_\sharp \le 1$, let S be a non-empty subset of $\{1, \dots, n\}$, and let $0 \le s \le K$. Let $\rho_k^{[K],z}$ be the run that starts from $\rho_0^{[K]} \exp\{\sum z_i \mathcal{O}_K(f_i)\}$, and $\pi^{[K]}_{S,[s]} := \partial_S[\rho^{[K],z}_{K-s}/Z_K(z)]_{z=0}$. Then $\pi^{[K]}_{S,[s]}$ is a real function; it lies in $L^1(dV)$ on the fields of $\mathbb{Z}^3/NL^s\mathbb{Z}^3$; and, for every bounded Borel F, $\int F\pi\, dV$ is the joint cumulant of F, evaluated at the drawn field of depth s, with the $\mathcal{O}_K(f_i)$, $i \in S$. For $K' \ge K \ge K_T$ with $K - s - 1 \ge K_0(L^{s+1}\mathsf{b})$,

$$\|\pi^{[K']}_{S,[s]} - \pi^{[K]}_{S,[s]}\|_{L^1} \le 8C^{(s)}(1.15 + 0.46C_5^\# K_5^{-1}K) C_{L^s\mathsf{b}} \theta^{K-s} (4C^{(s)}/\mathfrak{r}(L^{\max(s+1,j^*)}\mathsf{b}))^{|S|},$$

where $C^{(s)} = C_{\mathrm{vol}}(NL^s, L^s\mathsf{b})$. Hence $\pi^{[K]}_{S,[s]}$ converges in L^1 as $K \to \infty$ through all integers K, and, for $s \ge 1$, the kernel of the drawn step with the noise $aL^s\mathsf{b}$ carries the limit at depth s to the limit at depth $s-1$.

■□

Remark 8.3. The proof uses the blocking factor and cell scale of the data. Equation 1.ii transports the estimates to other sides and labels.

Proof of Thm. 8.2. Step 0: normalisation and the generating function. We claim three things. First, it suffices to prove (T1) for $\|f_i\|_\infty \le 1$ and (T2) for $\|f_i\|_\sharp \le 1$. Second, for such f and for the function Z_K defined below, we have $S^T_{n;K,l,\beta}(f) = \partial_1 \cdots \partial_n \log Z_K(0)$, and $Z_K(z)$ is the integral of the weight of Thm. 7.1: $Z_K(z) = \int \exp\{-\beta \sum_p (1-\nu_p) s_p\} dU$, with $s_p = a_p$ and $\nu_p := \mathsf{b}^{-1} \sum_i z_i f_i(x(p))$. Third, the data of Prop. 7.12 are those of (T1), and the data of Thm. 7.19 are those of (T2).

For fixed K the joint cumulant is n-linear in $(f_1, \dots, f_n)$, and the right sides of (T1) and (T2) change in the same way when one entry is multiplied by a number. If the norm of an entry vanishes, that entry vanishes at all barycentres, and both sides are 0. We may therefore divide each f_i by its norm.

The function $Z_K(z) := \int \exp\{-\beta \sum_p a_p + \sum_i z_i \mathcal{O}_K(f_i)\} dU$ is entire in $z \in \mathbb{C}^n$, because its integrand is continuous and the group is compact; moreover $Z_K(0) > 0$, and $\partial_1 \cdots \partial_n \log Z_K(0)$ is the joint cumulant, by the classical link between cumulants and the logarithm of the generating function. As $\beta = L^K\mathsf{b}$ and $a_K^{-1} = L^K$, we get $\beta\nu_p a_p = L^K \sum_i z_i f_i(x(p)) a_p$, and the sum over p gives $\sum_p \beta\nu_p a_p = \sum_i z_i \mathcal{O}_K(f_i)$; so the weight is $\exp\{-\beta \sum_p (1-\nu_p) s_p\}$, the one of Thm. 7.1 (a). Since the supports are pairwise disjoint, at each p at most one $f_i(x(p)) \ne 0$, and hence $|\nu|_\infty \le \max_i |z_i|/\mathsf{b}$.

The variables $\mathsf{b} \ge x_s$, $N = 2L^{m_l}$ with $m_l \ge m'$, (Tor) and $K \ge 1$ make up the data of Thm. 7.1, of Prop. 7.12 and of Thm. 7.19. The kernels of the step are probability kernels (item 2 of §5.5, in its part "The step"), and the step is linear in the density; hence, by Fubini's theorem, $\int \rho_k^z dV = Z_K(z)$ at every level, for complex z as well as for real z.

Step 1: the numbers of (O14). We first prove a lemma on these numbers.

Lemma 8.4 (Growth of the separations and convergence of the tail series)**.** In the setting of item 1, the following hold for all data (b, n, d, R_f) with $\mathsf{b} \geq x_s$, $n \geq 2$, $d > 0$ and $R_f > 0$.

(N1) $r(Lx) \leq r(x) + \log L/960 + 1$ for $x > 1$.
(N2) $\mathrm{sep}^{[i+1]} \geq L\,\mathrm{sep}^{[i]}$ for all $i \geq 0$; the number j^* exists and is the least $j \geq 1$ with $\mathrm{sep}^{[j-1]} \geq \mathrm{sep}_*$; $\mathrm{sep}^{[j^*-1+t]} \geq L^t \mathrm{sep}_*$ for $t \geq 0$; and $\mathsf{b} \mapsto j^*$ is non-decreasing.
(N3) For $t \geq 1$, the term $j = j^* + t$ of Tail is $\leq P_* \exp\{-(\kappa n\,\mathrm{sep}_*/4)L^t\}$.
(N4) $\mathrm{Tail}(J) \leq 2P_* \exp\{-(\kappa n\,\mathrm{sep}_*/4)L^{J+1-j^*}\}$ for $J \geq j^*$, and $\mathrm{Ser}_1 < \infty$; the numbers Ser_1, Tail and P_* depend neither on K nor on N.
(N5) Ser_1, P_* and j^* are bounded on every compact set of $\mathsf{b} \in [x_s, \infty)$.

■

Proof of Lem. 8.4. For (N1) we apply $\lceil u + v \rceil \leq \lceil u \rceil + v + 1$ to $r(Lx)$, with $v = 20c_R \log L/M$; by (M2) and $\kappa_1 + 4 \geq 11$, this v satisfies $v \leq \log L/(96 \cdot 11) \leq \log L/960$.

For (N2) we put $t_i = dL^i/M$ and $r_i = r(L^i\mathsf{b})$, so that $\mathrm{sep}^{[i]} = \lfloor t_i \rfloor - 2r_i - 8$. Then

$$\mathrm{sep}^{[i+1]} - L\,\mathrm{sep}^{[i]} = [\lfloor Lt_i \rfloor - L\lfloor t_i \rfloor] + 2(Lr_i - r_{i+1}) + 8(L-1).$$

The bracket is ≥ 0, since L is an integer, and (N1) gives $r_{i+1} - Lr_i \leq \log L/960 + 1 \leq 4(L-1)$; so the difference is ≥ 0. As t_i grows geometrically while r_i grows at most linearly in i, we have $\mathrm{sep}^{[i]} \to \infty$; hence some $\mathrm{sep}^{[j-1]} \geq \mathrm{sep}_* > 0$, and from that index on the sequence grows by the factor L. Finally, $\check{r}(L^i\mathsf{b})$ is non-decreasing in b, so that $\mathrm{sep}^{[i]}$ is non-increasing and j^* is non-decreasing in b.

For (N3) let $j = j^* + t$ with $t \geq 1$. We bound the factors of the term j one by one.

(i) Since p_0 is increasing, $\mathrm{amp}(L^{j+1}\mathsf{b})/\mathrm{amp}(L^{j^*+1}\mathsf{b}) = L^{t/2} p_0(L^{j^*+1}\mathsf{b})/p_0(L^{j+1}\mathsf{b}) \leq L^{t/2}$, and therefore $\mathfrak{r}(L^j\mathsf{b})^{-n} \leq L^{nt/2}\mathfrak{r}(L^{j^*}\mathsf{b})^{-n}$.
(ii) By (N1), $\check{r}(L^{j-1}\mathsf{b}) \leq \check{r}(L^{j^*-1}\mathsf{b}) + t(\log L/960 + 1)$. Combined with $2\check{r} + 3 \geq 9$, $L^t - 1 \geq t(L-1)$ and $9(L-1) \geq 2(\log L/960 + 1)$, this yields $2\check{r}(L^{j-1}\mathsf{b}) + 3 \leq L^t(2\check{r}(L^{j^*-1}\mathsf{b}) + 3)$, and so $N_1^{[j]} \leq L^{3t} N_1^{[j^*]}$.
(iii) By (X17), $2C_g c_a \leq C_g/16$.
(iv) $m^{[j]} \geq n\,\mathrm{sep}^{[j^*-1+t]}/2 \geq (n/2)L^t \mathrm{sep}_*$.

Consequently the term is $\leq P_* L^{(n/2+3)t} \exp\{-(\kappa n\,\mathrm{sep}_*/2)L^t\}$. One half of the exponent absorbs the power, because

$$(\kappa n\,\mathrm{sep}_*/4)L^t = 2\kappa n \log(4nL^{1/2})L^t \geq 6nL^t \log L \geq 2nt \log L \geq (n/2+3)t \log L,$$

where the last inequality uses $n \geq 2$.

For (N4) put $c := \kappa n\,\mathrm{sep}_*/4 \geq 24 \log(8L^{1/2}) > 49$. The quotient of consecutive bounds is $\exp\{-cL^t(L-1)\} \leq 1/2$, whence $\sum_{t \geq T} \exp\{-cL^t\} \leq 2\exp\{-cL^T\}$; here we take $T = J + 1 - j^*$. Moreover, Ser_1 consists of the j^* terms with $j \leq j^*$ together with $\mathrm{Tail}(j^*)$, and none of the entries of the terms involves K or N.

For (N5) we put $\hat{\jmath} := \max j^*$ on a given compact set, see (N2). The terms with $j \leq \hat{\jmath}$ are products of the following factors: $c_a(L^j\mathsf{b})$, continuous in b; $\mathfrak{r}(L^i\mathsf{b})^{-n}$ at $i = \max(j, j^*(\mathsf{b})) \leq \hat{\jmath}$, which is one of finitely many continuous functions of b (and is $\leq \mathfrak{r}(L^j\mathsf{b})^{-n}$ by (G3) of item 3 of §7.3); the non-decreasing $N_1^{[j]}$; and a factor ≤ 1. So these terms are bounded. The number P_* is the value at $j' = j^* \leq \hat{\jmath}$ of a product of this kind, and it is bounded as well. For $j > \hat{\jmath}$ the term is $\leq P_* \exp\{-cL^{j-j^*}\} \leq P_* \exp\{-cL^{j-\hat{\jmath}}\}$. □

Step 2: the bound (T1). We claim that, for $\|f_i\|_\infty \leq 1$ and all $K \geq 1$,

$$|S^T_{n;K,l,L^K\mathsf{b}}(f)| \leq \mathrm{Ser}_1 + \log 2\,(\tau_7/\mathfrak{r}(L^{j^*}\mathsf{b}))^n,$$

with $\tau_7 = \max\{1, 0.91 C_{\mathrm{vol}}(N, \mathsf{b})\}$, and that this bound is free of K and locally bounded in b. By Step 0 and Prop. 7.12 (J7),

$$S_n^T = \sum_{0 \le k < K} \sum_{Y} \partial_1 \cdots \partial_n v_k(Y; u_k(z))(0) + \partial_1 \cdots \partial_n \log I[u_K(z)](0),$$

where $I[u] := \int \rho[u; \ell_K] dV$. We treat the levels first and the last integral afterwards.

The levels. By Prop. 7.12 (J8), the term k is at most $\mathfrak{r}_{\min(k,k_d)}^{-n} N_1(k)\, 2C_g c_a(x_k) e^{-\kappa(m_k - 1)}$; in terms of the depth $j = K - k$ of the level k, we have $x_k = L^j \mathsf{b}$ and $N_1(k) = N_1^{[j]}$. If $K \ge j^*$, then $k_d = K - j^*$ and $\mathfrak{r}_{\min(k,k_d)} = \mathfrak{r}(L^{\max(j,j^*)}\mathsf{b})$; moreover $k < k_d$ exactly when $j > j^*$, so that $m_k = m^{[j]}$. If $K < j^*$, then $k_d = 0$, every $m_k = 1 = m^{[j]}$ because $j \le K < j^*$, and $\mathfrak{r}_{\min(k,0)} = \mathfrak{r}_0 = \mathfrak{r}_* \ge \mathfrak{r}(L^{j^*}\mathsf{b}) = \mathfrak{r}(L^{\max(j,j^*)}\mathsf{b})$ by (G3) of item 3 of §7.3. In both cases the term k is bounded by the term j of Tail, where j runs over $1, \dots, K$; therefore the sum over the levels is at most $\mathrm{Tail}(0) = \mathrm{Ser}_1$, which is finite by (N4).

The last integral. For the last integral we use Thm. 7.27 on $\mathbb{Z}^3/N\mathbb{Z}^3$, applied with $x = \mathsf{b}$ and with the pair (κ, c_h), that is, with the rate κ and the cap $c_h(x)$. Its data are met: $\mathsf{b} \ge x_s \ge e^2$; N is even, $M \mid N$, and $N \ge 2Ms(L\mathsf{b}) \ge 2Ms(\mathsf{b})$ because s is non-decreasing; $\varepsilon(\mathsf{b}) \le 1$ and $(c_a + c_h)(\mathsf{b}) \le 1/2$ by (X16), (X17); and ℓ_K has range $\le \varrho(\mathsf{b})$, with $|\ell_{K,y}(p)| \le K_* e^{-\lambda_* |y-p|}$, by Step 2 (c) of the proof of Thm. 5.63.

These data are available at every $K \ge 1$. Indeed, Step 2 of the proof of Thm. 5.63 is carried out under the data of that theorem, which include $K \ge K_0(\mathsf{b})$, but the reasoning in its items (c) and (e) makes no use of K_0:

- (c) uses $\mathsf{b} \ge x_\flat$ and (X12), the geometric series over $1 \le j \le K$ and Thm. 2.57 (b), with $L \ge L_A$ odd, $aL \ge 3/c_L$ and N even, and it gives $\ell_k \in \mathcal{R}(b_*, K_*, \lambda_*, 1/9)$ for $0 \le k \le K$, with range $\le \varrho(x_k)$, since the definition of the rows ℓ_k cuts them at the radius $\varrho(x_k)$; here, by (O13) of item (1) of §6.1, x_s serves as the $x_\flat$ of §5.5, with (X12), so that $\mathsf{b} \ge x_s$ gives $\mathsf{b} \ge x_\flat$;
- (e) uses $N \ge 2Ms(L\mathsf{b}) \ge 2Ms(\mathsf{b})$ and Prop. 5.8, applied with $x = \mathsf{b} \ge e^2$; this proposition is stated with its proof in item 2 of Step 15 of the proof of Thm. 5.7.

So both items hold for every $K \ge 1$. The pointwise bound above is the decay clause in the definition of $\mathcal{R}$ (the second condition in the display of Def. 2.48).

The family is $u_K(z)$ on $\mathrm{Pd}(\mathfrak{r}_*)$, as in (J6). Since $j^* \ge 1$, we have $k_d \le K - 1$, so that $u_K(z) = F_{\ell_{K-1}}(u_{K-1}(z))$ is an output of Thm. 5.15 as extended by Thm. 6.7 (U); here the rate is $\kappa_{K-1} = \kappa$, while Thm. 5.15 itself has a single rate, namely κ_1. This is case (ii) of (b^C). The argument does not depend on this case: (F2) of Cor. 7.28 is asserted and derived on the polydisc $\mathrm{Pd}(\mathfrak{r}_*/\tau_7)$, where (S) holds, and (S) implies case (i) by Thm. 7.27 (c^C).

We have $u_K(0) = \bar{u}_K$, a system of members of the point $\rho_K^{[K]}$ of the class, by Thm. 7.1 (h) at $z = 0$ and Thm. 5.15 (d). The norm $\|u_K(z) - \bar{u}_K\|^\sim$ is at most the norm $\|\cdot\|^{\mathrm{st}}$, with the rate $\kappa_K = \kappa$, of the same difference, and the latter is $\le 0.91 \cdot 0.4^{K - k_d} \le 0.91 =: B_0$. Now (F2), applied with $r_i = \mathfrak{r}_*$ and $S = \{1, \dots, n\}$, gives

$$|\partial_1 \cdots \partial_n \log I[u_K(z)](0)| \le \log 2\, (\tau_7/\mathfrak{r}_*)^n \le \log 2\, (\tau_7/\mathfrak{r}(L^{j^*}\mathsf{b}))^n.$$

Local boundedness. Finally, the right side of the claim is locally bounded in b, by (N5), by the continuity of $\mathsf{b} \mapsto \mathfrak{r}(L^j\mathsf{b})$ and of $\mathsf{b} \mapsto C_{\mathrm{vol}}(N, \mathsf{b})$, and since j^* is bounded on compact sets.

Step 3: the clause (T2). We show that, for $\|f_i\|_\sharp \le 1$ and $K' \ge K \ge K_T(\mathsf{b}, n, d)$, one has $|S^T_{n;K'} - S^T_{n;K}| \le C'_n K \theta^K$, where C'_n is the constant of Thm. 8.2; that K_T and C'_n are locally bounded in b; and that I_N is empty or a compact interval.

Since $K \ge K_T \ge K_0(\mathsf{b})$, Thm. 5.63 (i) gives $\Lambda > 0$ and $1 \le k_0 \le K - 1$. From $k_0 < (5K + 10\Lambda)/9 + 1$ we obtain $J := K - k_0 > \frac{4}{9}K - \frac{10}{9}\Lambda - 1 \ge j^* + 1$, the second inequality being (K1a). As J is an integer, it is therefore at least $j^* + 2$, and Thm. 7.19 (D1) (a) then yields $k_0 + 2 \le k_d$.

In (D6), the early sum of each run has the step $0 \to 1$ of that run as its term 0. With $m = K' - K$, the identity (D6), written in its four groups of terms, reads

$$S^T_{n;K'} - S^T_{n;K} = [\text{early levels } 0 \le k < m + k_0 \text{ of } K'] - [\text{early levels } 0 \le k < k_0 \text{ of } K] + [\text{pairs } k_0 \le k < K] + [\text{last}].$$

The early levels. By (D5), each of the two brackets of early levels is at most $\mathrm{Tail}(J)$, because a level and its partner have the same depth and j^* depends on (b, n, d) alone. We put $T := J+1-j^* \ge 3$ and $c := \kappa n \,\mathrm{sep}_*/4 > 49$; then (N4) gives $\mathrm{Tail}(J) \le 2P_* e^{-cL^T}$. Since $\log(1/\vartheta) \le 0.8057$, we have $cL^T \ge T \log(1/\vartheta)$, and hence $e^{-cL^T} \le \vartheta^T$. On the other hand $\frac{4}{9}K < J + 1 + \frac{10}{9}\Lambda = T + j^* + \frac{10}{9}\Lambda$, whence $\theta^K = \vartheta^{(4/9)K} \ge \vartheta^{T+j^*+(10/9)\Lambda}$. The two brackets together are thus at most $4P_*\vartheta^{-j^*-(10/9)\Lambda}\theta^K$.

The pairs. By (D4), the bracket of the pairs is at most $\mathrm{Ser}_2(4 + 1.36C_5^{\#}K_5^{-1}K)\vartheta^{K-k_0}$. By Thm. 5.63 (iv), $3\vartheta^{K-k_0} \le 3\vartheta^{(4/9)K-(10/9)\Lambda-1} = C_{\mathsf{b}}\theta^K$. Since $K \ge 1$, the bracket of the pairs is therefore bounded by $(C_{\mathsf{b}}/3)\mathrm{Ser}_2(4 + 1.36C_5^{\#}/K_5)K\theta^K$.

The last bracket. Here we apply the second sentence of Cor. 7.28 (F3) at $x = \mathsf{b}$, with the rate κ, the cap $c_h(\mathsf{b})$ and $r_i = \mathfrak{r}_*/2$, and with the data of Step 2. The first family is $u_K(z)$; by Prop. 7.12 (J6) it stays within 0.364 of $\bar{u}_K$. The second family is the family $\mathfrak{n}_K(z)$ of (D1) on $\mathrm{Pd}(\mathfrak{r}_*/2)$. As $K - 1 \ge k_d > k_0$, we have $\mathfrak{n}_{K-1}(z) \in M_2$, since by (D1) the member at the level k_d is the polynomial and every later member is an output of Thm. 5.15 lying in M_1; and $\mathfrak{n}_{K-1}(0) \in M_1$, by (D0) at $z = 0$ together with (D1) (c), and by Thm. 5.63 (iii).

We use Thm. 7.9 (ii) with $B = 2$ and $p = (\kappa, c_h)$ at $x = x_{K-1} = L\mathsf{b} \ge Lx_s \ge x_9^U$, where (SD) holds at $(x_{K-1}; \ell_{K-1})$ by Thm. 7.1 (b), and we take the second family constant. This gives

$$\|\mathfrak{n}_K(z) - \mathfrak{n}_K(0)\|^{\sim} \le \vartheta[2](2+1) \le 0.28343 \cdot 3 \le 0.8503 \le 1,$$

and Thm. 7.19 (D1) (f) provides the bound 0.795 besides. Hence $B_0 = 1$ and $\tau_7 = \max\{1, C_{\mathrm{vol}}\} = C_{\mathrm{vol}}$.

The member $\mathfrak{n}_K(0)$ is real and lies in M_1 with the strong cap, and $\rho[\mathfrak{n}_K(0); \ell_K]$ is a positive multiple of $\rho^{[K']}_{K'}$; so $\rho^{[K']}_{K'}$ is a point of $C^{\mathrm{su}}_{\flat}(\mathsf{b}; \ell_K)$, and $\mathfrak{n}_K(0)$ is a system of members of it, as the second sentence of (F3) requires. This follows from (D0) at $z = 0$ with (D1) (c) and from Thm. 5.63 (iii). With the rate κ and the cap $c_h(\mathsf{b})$, the norm $\|\cdot\|^{\sim}$ is bounded by the distance D_{b} with the rate $\kappa_K = \kappa$, and therefore (D3) yields $\Delta \le (1.15 + 0.46C_5^{\#}K_5^{-1}K)C_{\mathsf{b}}\theta^K$, where $3.45/3 = 1.15$ and $1.36/3 \le 0.4534$. Now (F3) gives

$$|\text{last}| \le 3C_{\mathrm{vol}}\Delta(2\tau_7)^n(2/\mathfrak{r}_*)^n \le 3C_{\mathrm{vol}}C_{\mathsf{b}}(1.15 + 0.46C_5^{\#}/K_5)(4C_{\mathrm{vol}}/\mathfrak{r}(L^{j^*}\mathsf{b}))^n K\theta^K.$$

No threshold is imposed on Δ; consequently K_T does not depend on N.

The sum. Adding the bounds for the early levels, for the pairs and for the last bracket, we arrive at the bound $C'_n K\theta^K$.

Uniformity. The quantities Λ, C_{b}, $\mathfrak{r}(L^j\mathsf{b})$ and $C_{\mathrm{vol}}(N, \mathsf{b})$ depend continuously on b; j^* and P_* are bounded on compact sets by (N5); and Ser_2 and K_T are bounded on compact sets by (D7). The function $s(x) = 8\lceil 20c_R \log x/M\rceil$ is a non-decreasing step function and is continuous from the left. Hence $\{\mathsf{b} : 2Ms(L\mathsf{b}) \le N\}$ is a half line closed on the right, and either $I_N = [x_s, \mathsf{b}_N]$ or I_N is empty.

Step 4: the clause (T3). On real Lipschitz tuples with pairwise disjoint supports, the limit exists as $K \to \infty$ through all integers K, and the convergence has the rate of (T2). This limit is real and symmetric, it is n-linear on every product of spaces of functions whose carriers are closed and pairwise disjoint, and it obeys (T1). For continuous entries the limit exists as well.

Supports are closed and T_l is compact, so the least distance between two of the pairwise disjoint supports is a number $d > 0$; each support lies in the ball of radius $R_f = N/2$, which is T_l itself. By (T2), the sequence $(S^T_{n;K})_K$ is Cauchy, because $K\theta^K \to 0$; and letting $K' \to \infty$ in (T2) gives the rate.

At every K the cumulant is real, symmetric and n-linear. Let $A_1, \dots, A_n$ be closed sets that are pairwise at distance $\geq d$. On the product of the spaces of Lipschitz functions carried by these sets, all tuples are of the same type $(d; N/2)$; therefore these properties, and (T1) as well, pass to the limit.

Continuous entries. Let the f_i be continuous and carried by A_i. We choose Lipschitz functions f_i^e with $\|f_i - f_i^e\|_\infty \leq e$ that are carried by the closed $(d/4)$-neighbourhood of A_i; such functions are obtained by mollification, and these neighbourhoods are $\geq d/2$ apart. A telescoping sum in the entries, n-linearity at finite K and (T1) for tuples of the type $(d/2; N/2)$ give, uniformly in K,

$$|S^T_{n;K}(f) - S^T_{n;K}(f^e)| \leq nC_n(d/2)\, e \max_i (\|f_i\|_\infty + e)^{n-1}.$$

Consequently $(S^T_{n;K}(f))_K$ is a Cauchy sequence. Its limit is the extension of S_n by continuity with respect to the sup norms, and it retains all the properties above.

Step 5: dilation. Lem. 1.3 gives the finite-cutoff identity eq. 1.ii for every $\beta > 0$ and $z \in \mathbb{Z}$. With $\beta = L^K \mathsf{b} = L^{K-z}(L^z \mathsf{b})$ it also gives eq. 1.iii and the equivalence of existence of the two limits.

Step 6: the clause (T4). We show that $\mathsf{b} \mapsto S_n^{(l,\mathsf{b})}(f)$ is continuous at every $\mathsf{b} > 0$; that $S^T_{n;K_j,l,\beta_j}(f)$ converges whenever $K_j \to \infty$ and $a_{K_j}\beta_j \to \mathsf{b}$; and that every sequence (K_j, β_j) with $K_j \to \infty$ and with labels $a_{K_j}\beta_j$ in a compact subset of $(0, \infty)$ has a subsequence along which $S^T_{n;K_j,l,\beta_j}(f)$ converges.

(i) *On I_N.* For fixed K the function $\mathsf{b} \mapsto S^T_{n;K,l,L^K\mathsf{b}}(f)$ is continuous: it is a polynomial in the moments, and every moment is the quotient of two integrals, over a compact group, of integrands that depend continuously on b, the denominator being positive. By (T2) and (T3) we have, for $K \geq \sup_{I_N} K_T$,

$$\sup_{I_N} |S_n^{(l,\mathsf{b})} - S^T_{n;K}| \leq \sup_{I_N} C'_n K \theta^K \prod \|f_i\|_\sharp \to 0.$$

Being a uniform limit of continuous functions, $\mathsf{b} \mapsto S_n^{(l,\mathsf{b})}(f)$ is continuous on I_N.

(ii) *A compact interval about any $\mathsf{b} > 0$, for any side $l = 2L^m$.* We choose z with $L^z\mathsf{b}/2 \geq x_s$, $m + z \geq m'$ and $L^z l \geq 2Ms(2L^{z+1}\mathsf{b})$. Such a z exists, because $s(2L^{z+1}\mathsf{b}) = 8\lceil 20 c_R((z+1)\log L + \log 2\mathsf{b})/M\rceil$ grows linearly in z. Then $[L^z\mathsf{b}/2, 2L^z\mathsf{b}] \subset I_{L^z l}$.

By Step 5, $S_n^{(l,\mathsf{b}')}(f) = L^{zn} S_n^{(L^z l, L^z \mathsf{b}')}(f^{[z]})$ for every b' in the interval $[\mathsf{b}/2, 2\mathsf{b}]$, and $f^{[z]}$ is Lipschitz, of the type $(L^z d; L^z R_f)$. Applied on the torus of side $L^z l$, part (i) now gives the continuity at b. In particular, the end points of I_N and all $\mathsf{b} < x_s$ are covered.

(iii) Let $K_j \to \infty$ and $\mathsf{b}_j := a_{K_j}\beta_j \to \mathsf{b}$. For all large j we have $\mathsf{b}_j \in [\mathsf{b}/2, 2\mathsf{b}]$ and $K_j - z \geq \sup K_T$. By Step 5,

$$S^T_{n;K_j,l,\beta_j}(f) = L^{zn} S^T_{n;K_j - z, L^z l, \beta_j}(f^{[z]}),$$

where the label on the right is $L^z \mathsf{b}_j$. Therefore

$$|S^T_{n;K_j,l,\beta_j}(f) - S_n^{(l,\mathsf{b})}(f)| \leq L^{zn} \sup C'_n (K_j - z)\theta^{K_j - z} \prod \|f_i^{[z]}\|_\sharp + |S_n^{(l,\mathsf{b}_j)}(f) - S_n^{(l,\mathsf{b})}(f)| \to 0,$$

by (ii) and by (T3), applied on the torus of side $L^z l$ uniformly for the labels in $[L^z\mathsf{b}/2, 2L^z\mathsf{b}]$, a subset of $I_{L^z l}$, that is, for $\mathsf{b}_j \in [\mathsf{b}/2, 2\mathsf{b}]$. The suprema of K_T and of C'_n are taken over this dilated interval.

(iv) A sequence of labels in a compact subset of $(0, \infty)$ has a convergent subsequence $\mathsf{b}_j \to \mathsf{b} > 0$. Along this one subsequence, (iii) holds for all n and all tuples at once: the subsequence does not depend on n or on the tuple.

(v) *Continuous entries.* Let the f_i be continuous with pairwise disjoint supports, let f^e be as in Step 4, and take the interval $[\mathsf{b}/2, 2\mathsf{b}]$ and the z of (ii). Combining Step 5 with the

n-linearity at finite K and with (T1) on the torus of side $L^z l$, we obtain

$$|S^T_{n;K,l,L^K\mathsf{b}'}(f)-S^T_{n;K,l,L^K\mathsf{b}'}(f^e)| \le n \sup_{\mathsf{b}'\in[\mathsf{b}/2,2\mathsf{b}]} L^{zn}C_n(L^z\mathsf{b}',n,L^zd/2,L^zl/2,L^zl)\, e \max_i(\|f_i\|_\infty+e)^{n-1}$$

for every $K \ge z+1$ and every $\mathsf{b}' \in [\mathsf{b}/2, 2\mathsf{b}]$. The supremum is finite, since C_n is locally bounded in b, and the same bound holds in the limit $K\to\infty$. Hence (ii), (iii) and (iv) pass from f^e to f by a uniform limit.

Step 7: the clauses (T5) (i) *and* (iii)*: a family of connected functions; translations, W_3 and symmetry.* The family $S = (S_n^{(l,\mathsf{b})})_{n\ge2}$ on $\mathfrak{S}_n(T_l)$ is symmetric and, in the sense of 8.1, multilinear. Further, every translation r of T_l and every $r \in W_3$ satisfy $S_n(rX) = S_n(X)$, and consequently σ_S is invariant.

A C^1 function on T_l is Lipschitz, so (T3) applies on $\mathfrak{S}_n(T_l)$. The open sets $O_1,\dots,O_n \subset T_l$ in the definition of multilinearity have pairwise disjoint closures, and every entry in $C^1_c(O_i)$ is supported in the closure of O_i. Applied to these n closed sets, one for each factor, (T3) shows that S_n is n-linear on $C^1_c(O_1)\times\cdots\times C^1_c(O_n)$; hence S_n is multilinear as 8.1 requires, and it is symmetric by (T3).

At finite K. Let r be the map $x \mapsto r^\circ x + v$ with $r^\circ \in W_3$ and $v \in a_K\mathbb{Z}^3$. It maps the lattice onto itself, each plaquette onto a plaquette and each barycentre to a barycentre. It does not, however, map base sites to base sites, and this is why the barycentre is required here. With $(rU)(b) = U(r^{-1}b)$ we have $a_p(rU) = a_{r^{-1}p}(U)$, since $\operatorname{tr} g = \operatorname{tr} g^{-1}$ on SU(2) and the trace around a plaquette depends neither on the corner at which it starts nor on its orientation. As the Haar measure and the action are invariant, $\mathcal{O}_K(f)(rU) = \mathcal{O}_K(f\circ r)(U)$, and $S^T_{n;K}(rX) = S^T_{n;K}(X)$.

The limit. For $K' \ge K$ we have $a_K\mathbb{Z}^3 \subset a_{K'}\mathbb{Z}^3$. So the identity holds for all large K whenever $r^\circ \in W_3$ and v lies in the union A of the lattices $a_K\mathbb{Z}^3$; by (T3) it then holds for S_n. The set A is dense in $\mathbb{R}^3$.

All v. Let X be of the type $(d; R_f)$ and let v be any vector of $\mathbb{R}^3$, in A or not; let $v_j \in A$ with $|v - v_j| \le d/4$ and $v_j \to v$. We write $\tau_v X - \tau_{v_j}X$ as a telescoping sum. The entries of its i-th term are $\tau_{v_j}f_1,\dots,\tau_v f_i - \tau_{v_j}f_i,\dots,\tau_v f_n$; the entry in the k-th place is carried by the closed $(d/4)$-neighbourhood of $\tau_v \operatorname{supp} f_k$, and these neighbourhoods are $\ge d/2$ apart. By n-linearity and (T1) for tuples of the type $(d/2; R_f + d/4)$,

$$|S_n(\tau_v X) - S_n(\tau_{v_j}X)| \le nC_n\sqrt{3}|v-v_j| \max_i[\operatorname{Lip} f_i \prod_{k\ne i}\|f_k\|_\infty] \to 0,$$

while $S_n(\tau_{v_j}X) = S_n(X)$. Finally, σ_S is a polynomial in the $S_{|J|}(X_J)$, and $(rX)_J = r(X_J)$.

Step 8: the clause (T5) (ii)*, reflection positivity of the centred functional.* For every mirror $\Sigma = (\mu, c)$ of T_l, with $\mathbb{E}_+(\Sigma) = \{c < y_\mu < c + l/2\}$, for all $X_1,\dots,X_r \in \mathfrak{S}(\mathbb{E}_+(\Sigma))$ and for all $\zeta \in \mathbb{C}^r$, we show that

$$\sum_{i,k} \bar\zeta_i\zeta_k\, \sigma_S(\theta_\Sigma X_i \cdot X_k) \ge 0.$$

(a) *Finite K and $c = 0$.* Both planes $y_\mu = 0$ and $y_\mu = l/2 = L^{K+m_l}a_K$ are planes of sites, and the reflection $\theta : y_\mu \mapsto -y_\mu$ maps the lattice onto itself and leaves each of the two planes pointwise fixed. We divide the bonds into three groups: U_0 consists of the bonds lying in the two planes, U_+ of the remaining bonds in $\{0 \le y_\mu \le l/2\}$, and U_- of the mirror images of the latter. Every plaquette lies either in one of the two planes or in exactly one of the two closed halves, so that

$$\sum_p a_p = A_0(U_0) + A_+(U_0, U_+) + A_+(U_0, \theta U_-).$$

If the barycentre of a plaquette lies in the open side, then the plaquette lies in the closed half. Indeed, the μ-coordinate of the barycentre is then either a multiple of a_K strictly

between the two planes, for a plaquette spanned by the two directions other than μ and lying in a plane $y_\mu = \text{const}$ of sites, or an odd multiple of $a_K/2$, for a plaquette lying between two neighbouring planes of sites. Hence, for f carried by $\mathbb{E}_+$, $\mathcal{O}_K(f)$ is a function of (U_0, U_+).

We put $G := \sum_k \zeta_k \prod_{f\in X_k}(\mathcal{O}_K(f) - \langle \mathcal{O}_K(f)\rangle)$. By Step 7, $\mathcal{O}_K(f)(\theta U) = \mathcal{O}_K(\theta f)(U)$ and $\langle \mathcal{O}_K(\theta f)\rangle = \langle \mathcal{O}_K(f)\rangle$, both being real. Fubini's theorem therefore gives

$$\sum \bar\zeta_i \zeta_k \, \sigma_{K,l,\beta}(\theta X_i \cdot X_k) = \langle \overline{G\circ\theta}\, G\rangle = Z^{-1} \int dU_0\, e^{-\beta A_0} \Big| \int dU_+ \, e^{-\beta A_+} G \Big|^2 \geq 0.$$

In this computation θ reverses every bond of direction μ: at every y,

$$(\theta U)(\langle y, y + a_K e_\mu\rangle) = U(\langle \theta y - a_K e_\mu, \theta y\rangle)^{-1};$$

since the Haar measure is invariant under inversion, the substitution $U_- \mapsto \theta U_-$ carries dU_- to dU_+; and A_0, A_+ and the $\mathcal{O}_K(f)$ are real. Here $\sigma_{K,l,\beta}(X) = \langle \prod(\mathcal{O}_K(f_i) - \langle \mathcal{O}_K(f_i)\rangle)\rangle$.

(b) The tuple $\theta X_i \cdot X_k$ is separated: the entries of X_k are pairwise disjoint in $\mathbb{E}_+$, the entries of θX_i are pairwise disjoint in $\theta\mathbb{E}_+$, and $\theta\mathbb{E}_+$ does not meet $\mathbb{E}_+$. By 8.1, $\sigma_{K,l,\beta} = \sigma_S$ at $S = (S^T_{n;K,l,\beta})_{n\geq 2}$: centering makes the singleton cumulants zero, and all other cumulants are evaluated at separated tuples.

(c) *The limit.* The number $\sigma_S(Y)$ is a finite polynomial in the $S_{|J|}(Y_J)$, and each factor converges by (T3). A limit of non-negative reals is non-negative.

(d) *All c.* Let $\tau_v y = y + v$ with $v = ce_\mu$. Then $\theta_{(\mu,c)} = \tau_v \theta_{(\mu,0)} \tau_{-v}$, since $\tau_v\theta_0(y - v) = y - 2(y_\mu - c)e_\mu$, and τ_{-v} maps $\mathbb{E}_+(\mu, c)$ onto $\mathbb{E}_+(\mu, 0)$. By Step 7, $\sigma_S(\theta_c X_i \cdot X_k) = \sigma_S(\theta_0 X_i' \cdot X_k')$ with $X' = \tau_{-v}X \in \mathfrak{S}(\mathbb{E}_+(\mu, 0))$, and so the claim follows from (c).

Step 9: the clause (T5) (iv)*, regularity.* Let $n \geq 2$ and $\mathbb{E} = T_l$, and let $\mathbb{E}^n_{\neq}$ be the set of points of $\mathbb{E}^n$ whose entries are pairwise different. We show that $\mathbb{E}^n_{\neq}$ carries exactly one distribution $\tilde S_n$ with $\tilde S_n(f_1 \otimes \cdots \otimes f_n) = S_n(f_1, \ldots, f_n)$ on the C^∞ separated tuples, that this distribution is of order $\leq 3n + 1$ on compact sets, and that σ_S is regular in the sense of 8.1.

The local construction. Let $\xi \in \mathbb{E}^n_{\neq}$, let $\delta := \min_{i\neq k} |\xi_i - \xi_k|/4$, the distance being the sup distance, let B_i be the open ball with centre ξ_i and radius δ, and let $Q = \prod B_i$. We fix $\chi_i \in C^\infty(T_l)$ with $0 \leq \chi_i \leq 1$, equal to 1 on B_i and carried by the closed ball of radius $3\delta/2$ about ξ_i. These closed balls are $\geq \delta$ apart. Let $S^{\mathbb{C}}_n$ be the extension of S_n that is n-linear over $\mathbb{C}$ and is defined on entries that are continuous, complex valued and carried by these balls. Splitting every entry into its real and imaginary parts, we get from (T1) and (T3)

$$|S^{\mathbb{C}}_n(g_1, \ldots, g_n)| \leq 2^n C_n(\delta) \prod \|g_i\|_\infty.$$

For $F \in C^\infty_c(Q; \mathbb{C})$ let $\hat F(k)$, $k \in (\mathbb{Z}^3)^n$, be its Fourier coefficients on $(T_l)^n$, where $e_k(x) = e^{2\pi i k\cdot x/l}$. For $k \neq 0$ we integrate by parts $3n + 1$ times in a coordinate with $|k_a| = |k|_\infty$ and obtain $|\hat F(k)| \leq (l/2\pi|k|_\infty)^{3n+1} \sup|\partial_a^{3n+1} F|$; for $k = 0$ we have $|\hat F(0)| \leq \sup|F|$, and this term is covered by the maximum below. The number of indices with $|k|_\infty = t$ is at most $6n(2t + 1)^{3n-1}$, and $\sum_t t^{-3n-1}(2t + 1)^{3n-1} \leq 3^{3n-1} \sum_t t^{-2}$. Hence

$$\sum_k |\hat F(k)| \leq c(n, l) \max_{|\alpha|\leq 3n+1} \sup|\partial^\alpha F|.$$

As $F = (\chi_1 \otimes \cdots \otimes \chi_n)F$, we define

$$D_Q(F) := \sum_k \hat F(k) S^{\mathbb{C}}_n(\chi_1 e_{k_1}, \ldots, \chi_n e_{k_n}).$$

Then D_Q is linear, and $|D_Q(F)| \leq 2^n C_n(\delta) c(n, l) \max_{|\alpha|\leq 3n+1} \sup|\partial^\alpha F|$.

On products. Let $f_i \in C^\infty_c(B_i)$, and let F be their tensor product. Then $\hat F(k) = \prod \hat f_i(k_i)$, and the series $\sum_k \hat f_i(k)\chi_i e_k$ converges uniformly to $\chi_i f_i = f_i$. Using n-linearity and the continuity in the sup norms, we conclude that $D_Q(f_1 \otimes \cdots \otimes f_n) = S_n(f_1, \dots, f_n)$.

Uniqueness on Q. Given $F \in C^\infty_c(Q)$, we take $\chi'_i \in C^\infty_c(B_i)$ equal to 1 on the i-th projection of $\mathrm{supp} F$. The sums $(\chi'_1 \otimes \cdots \otimes \chi'_n) \sum_{|k|\le R} \hat F(k) e_k$ are finite linear combinations of products of functions in $C^\infty_c(B_i; \mathbb{C})$; they are all carried by a single compact subset of Q, and they converge to F in every C^m. On that compact set every distribution has finite order, and so it is determined by its values on products.

Gluing. The sets Q cover $\mathbb{E}^n_{\neq}$. Each intersection $Q \cap Q'$ is again a product of open sets whose closures are pairwise disjoint; the argument for uniqueness works on it, so that $D_Q = D_{Q'}$ there. We now use the gluing theorem (L. Hörmander, *The Analysis of Linear Partial Differential Operators I*, Theorem 2.2.4): let open sets X_i cover X, and for each i let u_i be a distribution on X_i, such that $u_i = u_j$ on $X_i \cap X_j$ for all i, j; then X carries a unique distribution u that restricts to u_i on each X_i. Applied in the flat charts of $(T_l)^n$, it yields exactly one distribution $\tilde S_n$ with $\tilde S_n = D_Q$ on every Q.

Any compact set is covered by finitely many sets Q, and with a partition of unity we get the order $\le 3n + 1$, with a constant that depends on the compact set. Now let a C^∞ separated tuple be given, with separation d. A partition of unity splits each f_i into pieces carried by small balls, namely open balls of sup radius $d/4$ about points ξ_i of $\mathrm{supp} f_i$. At such a point ξ we have $\delta = \min |\xi_i - \xi_k|/4 \ge d/4$, so that every product of pieces is carried by a single Q. Since both sides are n-linear, $\tilde S_n(f_1 \otimes \cdots \otimes f_n) = S_n(f)$. If another distribution has this property, then on every Q it coincides with D_Q on products; so it is $\tilde S_n$.

The functional σ_S. We put $\sigma_n := \sum_{\pi \in \mathrm{Part}_{\ge 2}(n)} \bigotimes_{J\in\pi} \tilde S_{|J|}$, each tensor product being restricted to the open subset $\mathbb{E}^n_{\neq}$ of $\prod_{J\in\pi} \mathbb{E}^J_{\neq}$; and $\sigma_1 = 0$, because $\mathrm{Part}_{\ge 2}(1)$ is empty. Each σ_n is thus a distribution on $\mathbb{E}^n_{\neq}$. For $X = (f_1, \dots, f_n) \in \mathfrak{S}^\infty_n(\mathbb{E})$ and each π, the tensor product over $J \in \pi$ takes at the product function $f_1 \otimes \cdots \otimes f_n$ the value $\prod_{J\in\pi} \tilde S_{|J|}(\otimes_{i\in J} f_i) = \prod_{J\in\pi} S_{|J|}(X_J)$; hence

$$\sigma_n(f_1 \otimes \cdots \otimes f_n) = \sum_{\pi \in \mathrm{Part}_{\ge 2}(n)} \prod_{J\in\pi} S_{|J|}(X_J) = \sigma_S(X),$$

and σ_S is regular in the sense of 8.1.

Step 10: the clause (T6). For every $\mathsf{b} > 0$ and every side $l = 2L^m$, $m \in \mathbb{Z}$, the clauses (T3), (T4) and (T5) are true, with the bound and the rate that Thm. 8.2 (T6) states.

We take z with $L^z \mathsf{b} \ge x_s$, $m + z \ge m'$ and $L^z l \ge 2Ms(L^{z+1}\mathsf{b}) = 16M\lceil 20 c_R \log(L^{z+1}\mathsf{b})/M\rceil$; every large z will do, as the right side is linear in z. Then $(L^z l, L^z \mathsf{b})$ satisfies the conditions imposed on the side and on b in the paragraph "Data" of Thm. 8.2, and by Step 5 every clause is carried back; the map $f \mapsto f^{[z]}$ is a bijection of the tuples.

(T3) The map $f \mapsto f^{[z]}$ preserves Lipschitz, C^1 and continuous entries as well as disjoint supports; $\|f^{[z]}\|_\infty = \|f\|_\infty$; and the type $(d; R_f)$ turns into $(L^z d; L^z R_f)$, which gives the bound. As to the rate, Step 5 at $\beta = L^K \mathsf{b} = L^{K-z}(L^z \mathsf{b})$ shows that $S^T_{n;K,l,L^K\mathsf{b}}(f)$ and $S^{(l,\mathsf{b})}_n(f)$ equal L^{zn} times the corresponding terms at $(K - z, L^z l, L^z \mathsf{b}, f^{[z]})$; there (T3) holds for $K - z \ge K_T(L^z \mathsf{b}, n, L^z d)$, and $\mathrm{Lip} f^{[z]} = L^{-z} \mathrm{Lip} f$. Two admissible values of z lead to the same number, as eq. 1.iii is an identity.

(T4) This is the content of parts (ii) to (v) of Step 6, which are formulated there for every $\mathsf{b} > 0$ and every side.

(T5) We have $(\tau_v f)^{[z]} = \tau_{L^z v} f^{[z]}$, and W_3 commutes with the dilation; under the dilation, the mirror (μ, c) is mapped onto the mirror $(\mu, L^z c)$ of the torus of side $L^z l$, and its open side onto the open side of the image, whose extent in the direction μ is $L^z l/2$, half the side of that torus. In σ_S every partition of n contributes $\prod_J L^{z|J|} = L^{zn}$; hence $\sigma_{S^{(l,\mathsf{b})}}(X) =$

$L^{zn}\sigma_{S(L^{z}l,L^{z}\mathsf{b})}(X^{[z]})$, and for the form of Step 8 we absorb the factor $L^{z(n_i+n_k)}$ into $\zeta_i L^{zn_i}$, a positive multiple. For the distributions, let $\Psi_z : (T_l)^n_{\neq} \to (T_{L^z l})^n_{\neq}$ be the componentwise dilation. With D' the distribution constructed in Step 9 at the dilated data, define

$$D(F) := L^{zn} D'(F \circ \Psi_z^{-1}), \qquad F \in C_c^\infty((T_l)^n_{\neq}; \mathbb{C}).$$

Composition with Ψ_z^{-1} sends compact off-diagonal supports to compact off-diagonal supports and satisfies $\partial^\alpha(F \circ \Psi_z^{-1}) = L^{-z|\alpha|}(\partial^\alpha F) \circ \Psi_z^{-1}$. Hence D has order at most $3n+1$ on compact sets. On tensor products this formula is exactly the dilation identity for S_n; the uniqueness argument of Step 9 therefore identifies D with $\tilde{S}_n$ at (l, b).

Step 11: the clauses (T7) *and* (T7+). Cor. 5.65 holds for the b and the N of the data of Thm. 8.2, and (T7+) holds.

(T7) The number x_s satisfies all of (O13) of item 1 of §5.5, evaluated at κ_1; and the conditions $\mathsf{b} \geq x_s$, N even, $M \mid N$ and (Tor) are the hypotheses of Thm. 5.63 on b and N. To the conditions just named, the table of item (1) of §6.1 only adds further ones. Therefore Cor. 5.65 holds.

(T7+) We put $k = K - s$ and $\check{k} = k - 1$. By Thm. 5.63 (iv), the condition $K - (s+1) \geq K_0(L^{s+1}\mathsf{b})$ means that $\check{k} > k_0$.

(0) *The type of each object.* This part holds for every $K \geq 1$, every $0 \leq s \leq K$ and any finite number of real bounded f_i. Here k ranges over all the levels $0 \leq k \leq K$, whereas from part (1) on we take $k = K - s$ again. We have $\rho_0^{[K]} = e^{-\beta \sum_p a_p} \leq 1$ and $|\mathcal{O}_K(f_i)| \leq 6L^K(NL^K)^3 \|f_i\|_\infty$, because $0 \leq a_p \leq 2$ and there are $3(NL^K)^3$ plaquettes. Hence $\rho_0^z = \rho_0 \exp\{\sum z_i \mathcal{O}_K(f_i)\}$ is continuous in U, entire in z at every U, and bounded for z in a compact set uniformly in U.
Next, the step from the level k has the kernel $\prod_c k_{ax_k}(V'(c), \mathsf{M}(U)(c))$. It is Borel in (V', U), since k_s is continuous on $G \times G$ and M is a Borel function of U (item 2 of §5.5, in its part "The step"). Its values lie between 0 and a bound that depends only on ax_k and on the number of coarse bonds, because $k_s(g,h) \leq 1/\int_G e^{-s(1-\cos \mathrm{ang}\, g)} dg$.
By induction on k it follows that $\rho_k^z(V)$ is Borel in V, entire in z at every V, and bounded for z in a compact set uniformly in V; and, by Cauchy's formula and Fubini's theorem, ∂_S may be taken under the integrals of the steps and under $\int F \cdot dV$. The function Z_K is entire, and $Z_K(0) > 0$ by Step 0; hence $\pi^{[K]}_{S,[s]}$ is a bounded Borel function on a compact group, so it lies in $L^1(dV)$. For this we use Cauchy's estimate on a polydisc on which $|Z_K(z)| \geq Z_K(0)/2$, and the fact that a derivative is a limit of Borel functions. At real z the functions ρ_k^z and the number $Z_K(z)$ are real, as the f_i are real; so are their derivatives at 0.
The kernels do not depend on the sources. Hence, for every bounded Borel function F,

$$\int F \rho^z_{K-s} dV / Z_K(z) = \langle F(V_{K-s}) e^{\sum z_i \mathcal{O}_K(f_i)} \rangle / \langle e^{\sum z_i \mathcal{O}_K(f_i)} \rangle,$$

where $\langle\ \rangle$ denotes the expectation under the joint law of the bare field $V_0 = U$, whose law is $\mu_{K,l,\beta}$, and of the fields $V_1, \dots, V_K$ drawn in the run at $z = 0$. This is the derivative d/dt at $t = 0$ of $\log \langle e^{tF(V_{K-s}) + \sum z_i \mathcal{O}_K(f_i)} \rangle$, and differentiating it by ∂_S at $z = 0$ yields the joint cumulant of $F(V_{K-s})$ and the $\mathcal{O}_K(f_i)$, $i \in S$.

(1) *Families at the level k.* If $\check{k} \leq k_d$, let $u^*_{\check{k}}$ and $\mathfrak{n}^*_{\check{k}}$ be the polynomials $u^{\mathrm{pol}}_{\check{k}}$ and $\mathfrak{n}^{\mathrm{pol}}_{\check{k}}$; if $\check{k} > k_d$, let them be the families $u_{\check{k}}(z)$ and $\mathfrak{n}_{\check{k}}(z)$ of Prop. 7.12 (J6) and Thm. 7.19 (D1). We consider them on $\mathrm{Pd}(r)$, where $r := \mathfrak{r}_{\min(\check{k},k_d)}/2 = \mathfrak{r}(L^{\max(s+1,j^*)}\mathsf{b})/2$; note that $K \geq K_T$ gives $K > j^*$. By (J5), (J6) and (D1), both lie in $M_2(x_{\check{k}}; \kappa_{\check{k}}, c_h)$. We put $\mathbb{U}(z) := F_{\ell_{\check{k}}}(u^*_{\check{k}}(z))$ and $\check{N}(z) := F_{\ell_{\check{k}}}(\mathfrak{n}^*_{\check{k}}(z))$. Since $x_{\check{k}} = L^{s+1}\mathsf{b} \geq L x_s \geq x_9^U$, and since

(SD) holds at $(x_{\check{k}};\ell_{\check{k}})$ by Thm. 7.1 (b), Thm. 7.9 (i) shows that these are families in $M_1(x_k)$ with the strong cap, and outputs of Thm. 5.15.

(2) *The distance.* For $\check{k} \le k_d$, (D2) gives $\sup \mathsf{D}^{\mathrm{in}}(u^*,\mathfrak{n}^*) \le \Delta_{\check{k}} \le 3.45\vartheta^{\check{k}-k_0} + 1.36\Omega^{\#}$. For $\check{k} > k_d$, Thm. 7.9 (ii), applied level by level with $B = 2$, gives the bound $\vartheta[2]^{\check{k}-k_d}\Delta_{k_d}$, where $\vartheta[2] \le \vartheta$. Moreover $\Omega^{\#} = C_5^{\#}K_5^{-1}K\,0.3683^{K-k_0} \le C_5^{\#}K_5^{-1}K\vartheta^{K-k_0}$. In both cases, therefore, $\sup \mathsf{D}^{\mathrm{in}} \le (3.45 + 1.36C_5^{\#}K_5^{-1}K)\vartheta^{\check{k}-k_0}$. A further application of Thm. 7.9 (ii) gives

$$\Delta := \sup_{\mathrm{Pd}(r)} \mathsf{D}_{x_k}(\mathbb{U},\check{N}) \le (3.45 + 1.36C_5^{\#}K_5^{-1}K)\vartheta^{K-s-k_0} \le (1.15 + 0.46C_5^{\#}K_5^{-1}K)C_{L^s\mathsf{b}}\theta^{K-s},$$

since $3\vartheta^{K-k_0-s} \le 3\vartheta^{(4/9)K-s-(10/9)\Lambda-1}$.

(3) *The choice $B_0 = 1$.* By (J5), where $0.7/2 + 0.21/4 = 0.4025$, and by (J6), we have $\sup\|\mathbb{U}(z) - \mathbb{U}(0)\| \le \vartheta[2]\cdot 0.455$; and $\sup\|\check{N}(z) - \check{N}(0)\| \le \vartheta[2](2+1) \le 0.8503$.

(4) *The true jets.* On Pd_0 we have $\rho_k^{[K],z} = \exp\{\sum_{j\le k} V_j(z)\}\rho[u_k(z);\ell_k]$ with the honest family, by Thm. 7.1 (h). In this part and in the following ones, $I[u] = \int \rho[u;\ell_k]\,dV$, with the rows ℓ_k of the level k. By Step 0, $\int \rho_k^z\,dV = Z_K(z)$ at every level; hence, near $z = 0$, where $Z_K(z) \ne 0$ because $Z_K(0) > 0$, we have $Z_K(z) = \exp\{\sum_{j\le k} V_j(z)\}I[u_k(z)]$, so that $I[u_k(z)] \ne 0$ as well, and $\rho_k^z/Z_K(z) = \rho[u_k(z);\ell_k]/I[u_k(z)]$. By (J1) if $\check{k} \le k_d$, by (J6) if $\check{k} > k_d$, and by Lem. 7.8, the values of $\mathbb{U}(z)$ and those of the honest $u_k(z)$ agree modulo the ideal (z_i^2). For fixed V, $\rho[u;\ell](V)$ is an entire function of a finite number of values of u; so the two densities agree modulo the ideal. By bounded convergence, each coefficient of I is the integral of the corresponding coefficient of ρ; a quotient whose denominator does not vanish at $z = 0$ preserves congruences; and ∂_S at 0 annihilates the ideal. Therefore

$$\pi_{S,[s]}^{[K]} = \partial_S[\rho[\mathbb{U}(z);\ell_k]/I[\mathbb{U}(z)]](0).$$

For K' the same identity holds with $\check{N}$ in place of $\mathbb{U}$, by (D1): the coefficients of $\check{N}$ are jets of a system of members of the long run with sources, taken over the rows $\ell_k^{[K]}$.

(5) *Thm. 7.27 at $x = L^s\mathsf{b}$ on $\mathbb{Z}^3/NL^s\mathbb{Z}^3$, with the rate $\kappa_{\check{k}}$ and the cap $c_h(L^s\mathsf{b})$.* The inequality $NL^s \ge 2Ms(L^s\mathsf{b})$ and the bound of the rows hold by Step 2 (e), (c) of the proof of Thm. 5.63, and the floors hold by (X16) and (X17). The first members $\mathbb{U}(0) = \bar{u}_k$ and $\check{N}(0) = \mathfrak{n}_k(0)$ are systems of members of points of one and the same class: for $\bar{u}_k$ by Thm. 7.1 (h), and for $\mathfrak{n}_k(0)$ by (D0) at $z = 0$ and (D1) (c) together with Thm. 5.63 (ii), (iii). Here $\tau_7 = C^{(s)}$. We denote the two normalised densities by p_2 and p_3.

By parts (1), (3) and the first paragraph of (5), the two families satisfy (F0) and the hypotheses of the second sentence of (F3), with $B_0 = 1$ and $\tau_7 = C^{(s)}$. Their integrals and normalized densities are therefore holomorphic on the smaller polydiscs specified in Cor. 7.28.

(6) *The estimate for the jets.* Part (4) identifies the true jets with those of the normalized densities. Apply Cor. 7.28 (F5), including its assertion for every Δ, to obtain

$$\|\pi_{S,[s]}^{[K']} - \pi_{S,[s]}^{[K]}\|_1 \le 4C^{(s)}\Delta(2\tau_7/r)^{|S|}.$$

Since $2\tau_7/r = 4C^{(s)}/\mathfrak{r}(L^{\max(s+1,j^*)}\mathsf{b})$, part (2) gives the claimed bound of (T7+) (with 4 in place of its 8).

(7) *The limit and consistency.* The bound tends to zero with K, uniformly for $K' \ge K$, so completeness of L^1 gives the limit. By Cor. 7.29 (R3)(ii), the step at noise $aL^s\mathsf{b}$ maps

$\pi^{[K]}_{S,[s]}$ to $\pi^{[K]}_{S,[s-1]}$, is independent of K and is a contraction on L^1. Passing to the limits at the two fixed depths proves consistency, as in (R3)(iv) there.

□

8.2. **Estimates for separated sources.** For Q separated sources, the radii and decay rates can be chosen independently of Q. Linearity bounds the combined single jets by the sup norm of the combined source. A mixed jet can contribute to a polymer only through the sources within its reach, and a polymer meeting many separated source blocks must be large. These two facts give the estimates in Prop. 8.11. Its vacuum bounds and the last-field integral estimate use the volume factor only once in Cor. 8.14.

Remark 8.5 (Item B: the range of depths). Each clause of Prop. 8.11 that is applied in §11 holds without an upper bound on the depth $K-k$, on x_k or on ϱ. Prop. 8.11 (J5°) (d) and (J8°) (a′) assume $1 \le k \le K$, $\varrho \ge 1$ and a single lower bound, $\mathrm{sep}_k \ge 8\log(4\varrho L^{1/2})$. Prop. 8.11 (lin), (J2°) to (J4°), (J5°) (a′) and (J7°) (iii) hold for all $0 \le k \le K$, while (J1°) and (J7°) (ii), which concern the step from k to $k+1$, hold for all $0 \le k < K$, and Prop. 8.11 (L0) holds for every $K \ge 1$. So they can be used at every depth $j \ge j_*(\mathsf{b})$, at the level $k = K - j \ge 1$, where $j_*(\mathsf{b})$ is the depth of §11, set among its definitions: $j_*(\mathsf{b})$ is the least $j \ge 1$ with $L^j\mathsf{b} \ge \max\{x^\#, y(\mathsf{b})^{4/3}\}$, where $x^\#$ is that of §11 and $y(\mathsf{b}) = \mathsf{b}\max\{c_{\mathrm{vol}}(\mathsf{b})^{1/3}, L^{1-m'}\}$.

Item C: the data and the strong units.

Definition 8.6 (One run with Q separated sources)**.** We take the data of item 1 of §7.3: the group $G = \mathrm{SU}(2)$ in dimension 3; $\mathsf{b} \ge x_s$; $N = 2L^{m_l}$ with $m_l \ge m'$; the condition (Tor), $N \ge 2Ms(L\mathsf{b})$; and $K \ge 1$. As there, $\beta = x_0 = L^K\mathsf{b}$, $x_k = L^{K-k}\mathsf{b}$ and $a_K = L^{-K}$, with $\mathbb{T}_k$, the fitted grids and the rows ℓ_k. In the letters of §1, $Q \ge 1$ is any number, $\check\delta > 0$ and $\check\rho > 0$. The sources $g_1, \dots, g_Q$ are bounded functions on $T_l = \mathbb{R}^3/N\mathbb{Z}^3$, real or complex, with $\|g_t\|_\infty \le 1$. The sets $\{g_t \ne 0\}$ lie pairwise at sup distance $\ge \check\delta$, and each of them lies within sup distance $\check\rho$ of one point.

The letters n, f_i, d, R_f of §7.3 become Q, g_t, $\check\delta$, $\check\rho$ here. We let $z \in \mathbb{C}^Q$, and S stands for a non-empty subset of $\{1, \dots, Q\}$. If no g_t is 0 and $\check\delta \le N$, then $Q \le (N/\check\delta)^3$; a number of sources proportional to the volume is thus the general case. The objects $\mathcal{O}_K$, $Z_K(z)$, $\nu(z)$, the honest family $u^{\mathrm{hon}}_k(z)$ of (o1) at the level k, $\delta_S(k)$, u^{pol}_k, $S_t(k)$, $\check r$, amp, amp_W, the norms, v_k, V_0^+ and $I[u]$ are those of item 1 and of (o1) to (o8) in item 2 of §7.3. ■

Definition 8.7 (Strong units at a rate)**.** Let $\varkappa$ be a real number. The strong unit at $\varkappa$ at (X, V) is $c_a(x_k)e^{-\varkappa(\#X-1)}$ for the a-member and $c_h(x_k)e^{-\varkappa(\#X-1)}$ for the h-member, and for the w-member it is

$$\exp\{-c_\Xi x_k \varepsilon \sum_{p \in D(V)\cap X} \phi_p - \varkappa(|X|_c - 1)\}.$$

■

Item D: numbers.

Definition 8.8 (Numbers that depend on b, $\check\delta$ and $\check\rho$ alone)**.** (D1) The numbers below belong to the line (O14) and are defined after the table and after b. No floor depends on them, and none of them depends on Q, N, K or the places of the sources.

We set $\mathrm{sep}^\circ := 8\log(8L^{1/2})$; this is the one numeral among them, and it is sep_* of (o4) at $n = 2$. Next, $\mathrm{sep}^{[i]} := \lfloor \check\delta L^i/M \rfloor - 2\check r(L^i\mathsf{b}) - 2$, and $j^\circ = j^\circ(\mathsf{b}, \check\delta)$ is the least $j \ge 1$ with $\mathrm{sep}^{[j-1]} \ge \mathrm{sep}^\circ$, which is j^* of (o4) at $n = 2$ and $d = \check\delta$. We put $\mathfrak{r}^\circ(x) := 3.5/\mathrm{amp}(Lx)$ and $r^\circ := \mathfrak{r}^\circ(L^{j^\circ}\mathsf{b})$, and

$$n_{\mathrm{loc}} := (\lfloor (2\check r(\mathsf{b}) + 1)M/\check\delta \rfloor + 1)^3, \qquad N_1^{[j]} := (2\check\rho\, L^{j-1}/M + 2\check r(L^{j-1}\mathsf{b}) + 3)^3.$$

For $j > j^\circ$ we put

$$E^{[j]} := N_1^{[j]}\, 2C_g c_a(L^j\mathsf{b})\, e^{1/(4L)} \exp\{-(\kappa - 1/(4L))(\mathrm{sep}^{[j-1]} + 1)\}.$$

With these,

$$C_W^\circ := \tfrac{1}{2}\sum_{j>j^\circ} E^{[j]} + 6 \sum_{1\le j\le j^\circ} N_1^{[j]} C_g c_a(L^j\mathsf{b}),$$

and the letter of §1 is

$$C_W(\mathsf{b},\check\delta,\check\rho) := \sum_{j>j^\circ} 2e\, N_1^{[j]} C_g c_a(L^j\mathsf{b}) e^{-(\kappa-1)(\mathrm{sep}^{[j-1]}-1)} + \sum_{1\le j\le j^\circ} 6N_1^{[j]} C_g c_a(L^j\mathsf{b}).$$

Term by term, $C_W^\circ \le C_W$. Finally, $e_h := 0.45\, N_1^{[1]} c_h(\mathsf{b})$ and $G_1(\mathsf{b},\check\delta,\check\rho) := \exp\{C_W + e_h\}/r^\circ$, and

$$c_{\mathrm{vol}}(\mathsf{b}) := 225/M^3 + (3/2)\mathsf{N}_Z p_0(\mathsf{b})^2 + 3\log(384/\varepsilon(\mathsf{b})^3).$$

■

Definition 8.9 (Quantities of a level)**.** (D2) The quantities of a level, defined below, depend on K only through the depth $K-k$, with two exceptions: k° and $\mathfrak{r}_*^\circ$ depend on K through $\min(K, j^\circ)$, as $k^\circ = K - \min(K, j^\circ)$ and $\mathfrak{r}_*^\circ = \mathfrak{r}^\circ(L^{\min(K,j^\circ)}\mathsf{b})$. The sets $I_k(X)$ and the numbers $j_k(X)$ depend on the places of the sources, being defined on $\mathbb{T}_k$; the other quantities depend neither on Q nor on N.

We set $\mathrm{sep}_k := \mathrm{sep}^{[K-k]}$ and $k^\circ := \max\{K - j^\circ, 0\}$. We set $\check\kappa_k^\circ := 2\log(8L^{1/2})/\mathrm{sep}_{k+1}$ for $k < k^\circ$, and $\kappa_k^\circ := \kappa + \sum_{k\le j<k^\circ} \check\kappa_j^\circ$. Next, $\mathfrak{r}_k^\circ := \mathfrak{r}^\circ(x_k)$ and $\mathfrak{r}_*^\circ := \mathfrak{r}_{k^\circ}^\circ$. Further, $I_k(X) := \{t : X^{+\check r(x_k)}$ meets $S_t(k)\}$, $j_k(X) := \#I_k(X)$ and $\mathfrak{h}_k := \lceil(\mathrm{sep}_k + 2)/2\rceil$. Finally, $N_1(k) := N_1^{[K-k]}$. ■

Item E: the statements.

Lemma 8.10 (The multi-affine part of a bounded holomorphic function)**.** Let $r = (r_1, \dots, r_n)$ be a polyradius, and let g be holomorphic on $\mathrm{Pd}(r) \subset \mathbb{C}^n$ with $|g| \le m$. Put $Mg(z) := \sum_S z^S \partial_S g(0)$, where the sum runs over all subsets S, the empty one included. Then $|Mg| \le m$ on $\{|z_i| \le r_i/2\}$, and no constant in this bound depends on n. ■

Proposition 8.11 (Jets of one run for any number of sources)**.** In the setting of Defs. 8.6, 8.7, 8.8 and 8.9, the following hold.

(G°) The clauses (G1) to (G4) of item 3 of §7.3 hold in the following form.

(G1), (G2) These two hold with $(\mathrm{sep}^\circ, j^\circ, k^\circ, \check\kappa^\circ, \kappa^\circ)$ in place of $(\mathrm{sep}_*, j^*, k_d, \check\kappa, \kappa_k)$: we have $\mathrm{sep}^{[i+1]} \ge L\,\mathrm{sep}^{[i]}$ and $k^\circ \le K-1$; for $k < k^\circ$ we have $\mathrm{sep}_{k+1} \ge L\,\mathrm{sep}^\circ$ and $\check\kappa_k^\circ \le 1/(4L)$; further, $\kappa_{k^\circ}^\circ = \kappa$ and $\kappa \le \kappa_k^\circ \le \kappa + 1/(4(L-1))$; and j° is finite, depends on b, $\check\delta$, L, M and c_R alone, and does not decrease as b grows.

(G3) We have $\mathfrak{r}_k^\circ \le \mathfrak{r}_{k+1}^\circ \le L^{1/2}\mathfrak{r}_k^\circ$, and $\mathfrak{r}_*^\circ \ge r^\circ$, with equality if $K \ge j^\circ$.

(G4) The sets $S_t(k)^{+\check r(x_k)}$ lie pairwise at distance $\ge \mathrm{sep}_k + 2$ in cubes.

(G5°) (a) If $\mathrm{sep}_k \ge -1$ and $j_k(X) \ge 2$, then $\#X$ (for connected X), or $|X|_c$ (for $X \in \mathcal{P}$), is $\ge j_k(X)\mathfrak{h}_k \ge j_k(X)(\mathrm{sep}_k/2 + 1)$.

(b) In every case, $j_k(X) \le n_{\mathrm{loc}}\#X$.

(L0) *Level 0 in terms of ν.* Let $z \mapsto \nu(z)$ be complex linear from $\mathbb{C}^m$ to the maps from the plaquettes of $\mathbb{T}_0$ to $\mathbb{C}$; let R be a polyradius with $\sup_{\mathrm{Pd}(R)} |\nu(z)|_\infty \le \mathfrak{r}_0^\circ/\mathsf{b}$; and let $u_0^{\mathrm{pol}}(z)$ be the multi-affine part at 0, taken value by value, of $u_0(\nu(z))$. Then $\sup_{\mathrm{Pd}(R)} \|u_0^{\mathrm{pol}}(z) - \bar u_0\|^{\mathrm{st},(\kappa_1)} \le 0.014$ for every $K \ge 1$ and every m, and this bound depends on no separation and on no disjointness of supports. The value of $u_0^{\mathrm{pol}}(z)$ at X does not depend on z_i unless some $\nu_p(\cdot)$ with $p \in X^{+r(x_0)}$ depends on z_i. If, besides, each $\nu_p(\cdot)$ depends on at most one variable, as it does in the setting of Def. 8.6, then the value at X depends on z only through

the $\nu_p(z)$ with $p \in X^{+r(x_0)}$. In the setting of Def. 8.6 we have $\nu(z) = \mathsf{b}^{-1}\sum_t z_t g_t(x(\cdot))$, and the polyradius with $R_t = \mathfrak{r}_0^\circ$ is one such R.

(lin) This clause, (J0°), (J3°), (J4°) and the bound stated after (J4°) concern the single jets, for one complex bounded source of any support and for $0 \le k \le K$; (J1°) and (J2°) hold in the setting of Def. 8.6, for its Q sources and every S. For a complex bounded f on T_l and $0 \le k \le K$, let $\delta[f](k)$ be the derivative $\frac{d}{dz}\big|_{z=0}$, at the level k, of the honest family of (o1) for the run with the single source f, in the variable $z \in \mathbb{C}$. Then $f \mapsto \delta[f](k)$ is complex linear, and $\delta_t(k) = \delta[g_t](k)$. Moreover, $\sum_t z_t\delta_t(k) = \delta[g_z](k)$ with $g_z := \sum_t z_t g_t$, and $\|g_z\|_\infty \le \max_t |z_t|$.

(J0°) The quadratic a-member $s_{x_k}[\mathbb{D}_k[f]/\mathsf{b}]$ lies in $T^a(x_k)$, lives on single cubes, and has $|\cdot|_a \le \mathrm{amp}(x_k)\|f\|_\infty$ at every rate.

(J3°) We have $\delta^a[f](k) = s_{x_k}[\mathbb{D}_k[f]/\mathsf{b}] + \mathrm{rem}_k[f]$ with $\mathrm{rem}_k[f] \in T^a(x_k)$, and

$$|\mathrm{rem}_k[f]|_a^{(\kappa_1)} \le 2u_3(x_k)K_5\|f\|_\infty/\mathsf{b} \le 0.002\,\mathrm{amp}(x_k)\|f\|_\infty.$$

(J4°) We have $|\delta^h[f](k)|_h^{(\kappa_1)}$, $|\delta^w[f](k)|_w^{\mathrm{st},(\kappa_1)} \le 0.12\,\mathrm{amp}(Lx_k)\|f\|_\infty$.
Hence $\|\delta[f](k)\|^{\mathrm{st},(\kappa_1)} \le 0.2\,\mathrm{amp}(Lx_k)\|f\|_\infty$ for $0 \le k \le K$.

(J1°) For $k < K$ and every S, the quantities $\delta_S(k+1)$, $\partial_S V_{k+1}(0)$ and $\partial_S v_k(Y; u_k^{\mathrm{hon}}(z))(0)$ are the coefficients of z^S of $F_{\ell_k}(u_k^{\mathrm{pol}}(z))$, $\mathsf{V}'_{\ell_k}[u_k^{\mathrm{pol}}(z)]$ and $v_k(Y; u_k^{\mathrm{pol}}(z))$, respectively.

(J2°) For $0 \le k \le K$, $\delta_S(k)(X) = 0$ unless $S \subset I_k(X)$.

(AD)

Definition 8.12 (Admissible radii and rates)**.** Admissible data consist of an integer $0 \le k_1 \le K$; radii $R_0, \dots, R_{k_1} > 0$; and rates $\kappa_1 \ge \varkappa_0 \ge \cdots \ge \varkappa_{k_1-1} \ge \kappa$ and a real $\varkappa_{k_1} \le \varkappa_{k_1-1}$ (for $k_1 = 0$: $\varkappa_0 \le \kappa_1$), where we write $\check\Delta_k := \varkappa_k - \varkappa_{k+1}$; they are subject to (i) and (ii).

(i) $R_k \le \mathfrak{r}_k^\circ$ for $k < k_1$; no condition is placed on R_{k_1}.

(ii) For $k < k_1$: $\mathrm{sep}_{k+1} \ge -1$; $q_k := (1 + R_{k+1}/R_k)e^{-\check\Delta_k \mathfrak{h}_{k+1}} \le 1$; and $0.364\, e^{\check\Delta_k} q_k^2 \le 0.03$.

The standard data are $k_1 = k^\circ$, $R_k = \mathfrak{r}_k^\circ$ and $\varkappa_k = \kappa_k^\circ$. ■

The standard data are admissible, with $0.364\, e^{\check\Delta_k} q_k^2 \le 0.0057$.

(J5°) (a) For $0 \le k \le K$ and $R > 0$,

$$\sup_{\mathrm{Pd}(R)} \|\sum_t z_t\delta_t(k)\|^{\mathrm{st},(\kappa_1)} \le 0.2\,\mathrm{amp}(Lx_k)R = 0.7R/\mathfrak{r}_k^\circ.$$

(a′) *Value by value.* At every (X, V) and for each member, $\sum_t |\delta_t(k)(X, V)| \le 0.2\,\mathrm{amp}(Lx_k)$ strong units at κ_1.

(b) Let admissible data be given and let $1 \le k \le k_1$. Then $\sup_{(X,V)} \sum_{|S|\ge 2} R_k^{|S|}|\delta_S(k)(X, V)| \le 0.03$ strong units at $\varkappa_k$, separately for every one of the three members.

(c) For $k < k_1$, and for $k = k_1$ if $R_{k_1} \le \mathfrak{r}_{k_1}^\circ$ and $\varkappa_{k_1} \ge \kappa$, the map u_k^{pol} is a family on $\mathrm{Pd}(R_k) \subset \mathbb{C}^Q$ in $M_{1.91}(x_k; \varkappa_k, c_h)$, and $\sup \|u_k^{\mathrm{pol}}(z) - \bar u_k\|^{\mathrm{st},(\varkappa_k)} \le 0.73$; for $k = 0$ the bound is ≤ 0.014, at κ_1.

(d) *Free last ratio.* Let $1 \le k \le K$ and $\varrho \ge 1$, with $\mathrm{sep}_k \ge 8\log(4\varrho L^{1/2})$. Put $\check\kappa_i := 2\log(4L^{1/2})/\mathrm{sep}_{i+1}$ for $i < k-1$, and $\check\kappa_{\mathrm{last}} := 2\log(4\varrho L^{1/2})/\mathrm{sep}_k \le 1/4$. Then the data $k_1 = k$; $R_{k'} = \mathfrak{r}_{k'}^\circ$ for $k' < k$; $R_k = \varrho\,\mathfrak{r}_k^\circ$; $\varkappa_{k'} = \kappa + 1/4 + \check\kappa_{\mathrm{last}} + \sum_{k'\le i<k-1}\check\kappa_i$ for $k' < k$; and $\varkappa_k = \kappa + 1/4$ are admissible, and all their rates lie in $[\kappa+1/4,\ \kappa+1/2+1/(4(L-1))]$. Consequently,

$$\sup_{(X,V)} \sum_{|S|\ge 2} (\varrho\,\mathfrak{r}_k^\circ)^{|S|}|\delta_S(k)(X, V)| \le 0.03$$

strong units at $\kappa + 1/4$. This clause depends neither on j° nor on k°.

(J6°) *The representatives.* Put $u_k(z) := u_k^{\mathrm{pol}}(z)$ on $\mathrm{Pd}(\mathfrak{r}_k^\circ)$ for $k \le k^\circ$, and $u_{k+1}(z) := F_{\ell_k}(u_k(z))$ on $\mathrm{Pd}(\mathfrak{r}_*^\circ)$ for $k^\circ \le k < K$. The following holds for every $K \ge 1$; if $K < j^\circ$, then $k^\circ = 0$.

(a) For $k^\circ < k \le K$, the u_k are families in $M_1(x_k; \kappa, c_h)$ with the strong cap.

(b) For $k^\circ \le k \le K$, $\sup_{\mathrm{Pd}(\mathfrak{r}_*^\circ)} \|u_k(z) - \bar u_k\|^{\mathrm{st},(\kappa)} \le 0.91 \cdot 0.4^{k-k^\circ}$.

(c) We have $u_k(0) = \bar u_k$ and $\partial_S u_k(0) = \delta_S(k)$ for every S.

(d) Every representative has the property (P_k) of Step 3 of the proof of Prop. 7.12, and $u_K(z)$ is an output of $F_{\ell_{K-1}}$.

(e) We have $|\delta_t^h(K)|_h^\Sigma \le 0.0646/\mathfrak{r}_*^\circ$ at (κ, c_h); so $\sum_X \sup|\delta_t^h(K)(X,\cdot)| \le 0.0646\, N_1^{[1]} c_h(\mathsf{b})/r^\circ$, and $e_t := \sum_X \delta_t^h(K)(X,1)$ satisfies $r^\circ |e_t| \le 0.065\, N_1^{[1]} c_h(\mathsf{b}) \le e_h$.

(J7°) Let $\equiv$ be the congruence of germs at 0 in $\mathbb{C}^Q$ modulo z_t^2, as in Def. 7.26. By any representatives we mean a choice, at every level k', of a map $z \mapsto u_{k'}(z)$ defined on some neighbourhood of 0 and mapping into $\mathfrak{M}(x_{k'})$, of norm < 16, all of whose values, in the sense of item 0 (4) of §7.2, are holomorphic in z, with $u_{k'}(0) = \bar u_{k'}$ and $\partial_S u_{k'}(0) = \delta_S(k')$ for all S. Among them are the honest family, $u_{k'}^{\mathrm{pol}}$ at every $k' \le K$, and those of (J6°). We put $\gamma_t^{(k)} := \sum_{k'<k} \partial_t V_{k'+1}(0)$ and $\gamma_t := \gamma_t^{(K)}$, and

$$W_{k'}(z) := \sum_Y \sum_{|S|\ge 2} z^S \partial_S v_{k'}(Y; u_{k'})(0);$$

this is a polynomial, and it does not depend on the representatives.

(i) For $0 \le k \le K$, at every field V of $\mathbb{T}_k$ and with any representatives,

$$\rho_k^z(V) \equiv \exp\{\sum_{j\le k} V_j(0) + \sum_{k'<k} \sum_Y [v_{k'}(Y; u_{k'}(z)) - v_{k'}(Y; \bar u_{k'})]\}\, \rho[u_k(z); \ell_k](V).$$

(ii) We have $\partial_S v_k(Y; u_k)(0) = 0$ unless $S \subset I_{k+1}(Y)$.

(iii) Moreover,

$$\rho_k^z(V) \equiv \exp\{\sum_{j\le k} V_j(0) + \sum_t z_t \gamma_t^{(k)} + \sum_{k'<k} W_{k'}(z)\}\, \rho[u_k^{\mathrm{pol}}(z); \ell_k](V).$$

(iv) With the representatives of (J6°), the congruence (iii) holds with $W^{(k)}$ in place of $\sum W_{k'}$ and $u_k(z)$ in place of $u_k^{\mathrm{pol}}(z)$, where

$$W^{(k)} := \sum_{k'<\min(k,k^\circ)} W_{k'} + \sum_{k^\circ \le k' < k} [w_{k'} - \sum_t z_t \partial_t w_{k'}(0)]$$

and $w_{k'}(z) := \sum_Y [v_{k'}(Y; u_{k'}(z)) - v_{k'}(Y; \bar u_{k'})]$, which is holomorphic on $\mathrm{Pd}(\mathfrak{r}_*^\circ)$. We put $W := W^{(K)}$.

(v) We have $Z_K(z)/Z_K(0) \equiv \exp\{\sum_t z_t \gamma_t + W(z)\}\, I[u_K(z)]/I[\bar u_K]$, and $I[\bar u_K] > 0$.

(vi) For every S, $\partial_S \log Z_K(0) = \sum_{k<K} \sum_Y \partial_S v_k(Y; u_k)(0) + \partial_S \log I[u_K(z)](0)$.

(vii) The number $\hat c_K(g_t) := \partial/\partial z_t \log \rho_K^z(1)$ at $z = 0$ equals $\gamma_t + e_t$.

(J8°) (a) *Free outer radius.* For admissible data, for $k < k_1$, for every t and for every $R > 0$ with $\lambda := \log(1 + R/R_k)/\mathfrak{h}_{k+1} \le \varkappa_k$,

$$\sum_{S \ni t,\, |S|\ge 2} R^{|S|} \sum_Y |\partial_S v_k(Y; u_k^{\mathrm{pol}})(0)| \le N_1(k)\, 2C_g c_a(x_k)\, e^\lambda \exp\{-(\varkappa_k - \lambda)(\mathrm{sep}_{k+1} + 1)\}.$$

(a°) For the standard data, with $R = \mathfrak{r}_*^\circ$, $k < k^\circ$ and the depth $j = K - k > j^\circ$, we have $\lambda \le 1/(4L)$, the right side of the bound in (a) is $\le E^{[j]}$, and $\sup_{\mathrm{Pd}(\mathfrak{r}_*^\circ)} |W_k| \le (Q/2) E^{[j]}$.

(a′) In the setting of (J5°) (d), for every t,

$$\begin{aligned}\sum_{S\ni t,\,|S|\ge 2} (\varrho\,\mathfrak{r}_k^\circ)^{|S|} \sum_{k'<k} \sum_{Y} |\partial_S v_{k'}(Y; u_{k'}^{\mathrm{pol}})(0)| \\ \le \sum_{1\le m\le k} N_1(k-m)\, 2C_g c_a(x_{k-m})\, e^{1/4} \\ \exp\{-(\kappa-1/4)(L^{m-1}\mathrm{sep}_k+1)\} \\ \le E_{\mathrm{vac}},\end{aligned}$$

where

$$E_{\mathrm{vac}} := \sum_{m\ge 1} N_1^{[K-k+m]}\, 2C_g c_a(L^{K-k+m}\mathsf{b})\, e^{1/4} \exp\{-(\kappa-1/4)(L^{m-1}\mathrm{sep}_k+1)\}.$$

Of K and k, E_{vac} depends on the depth $K-k$ alone.

(b) *Late levels.* For $k^\circ \le k < K$, on $\mathrm{Pd}(\mathfrak{r}_*^\circ)$,

$$\sum_Y |v_k(Y; u_k(z)) - v_k(Y; \bar{u}_k)| \le 3QN_1(k)C_g c_a(x_k), \qquad |\partial_t w_k(0)| \le 3N_1(k)C_g c_a(x_k)/\mathfrak{r}_*^\circ.$$

(c) *Late levels, anchored.* At these levels, $\sum_{S\ni t}(\mathfrak{r}_*^\circ/n_{\mathrm{loc}})^{|S|}\sum_Y |\partial_S v_k(Y; u_k)(0)| \le 2e\, N_1(k)C_g c_a(x_k)$.

(d) We have $\sup_{\mathrm{Pd}(\mathfrak{r}_*^\circ)} |W(z)| \le QC_W^\circ \le QC_W$.

■

Lemma 8.13 (The tail of the vacuum numbers, and finiteness)**.** For $t \ge 1$,

$$E^{[j^\circ+t]} \le N_1^{[j^\circ]}(C_g/16)\, e^{1/(4L)} \exp\{-20L^t \log L\}.$$

The numbers C_W° and C_W are finite, depend neither on K nor on N nor on Q, and are bounded on compact sets of $\mathsf{b} \in [x_s, \infty)$. ■

Corollary 8.14 (The moment of Q sources, with the volume constant to the first power)**.** The following holds for all $K \ge 1$, all $Q \ge 1$ and all complex e_t with $r^\circ|e_t| \le e_h$, hence in particular for the e_t of Prop. 8.11 (J6°) (e).

(a) We have

$$|\langle \prod_{t\le Q} (\mathcal{O}_K(g_t) - \gamma_t - e_t)\rangle| \le G_1^Q(1 + 0.182\, C_{\mathrm{vol}}(N, \mathsf{b})) \le G_1^Q e^{c_{\mathrm{vol}}(\mathsf{b})N^3}.$$

Together with Prop. 8.11 (J7°) (vii) this gives

$$|\langle \prod_t (\mathcal{O}_K(g_t) - \hat{c}_K(g_t))\rangle| \le G_1^Q(1 + 0.182\, C_{\mathrm{vol}}(N, \mathsf{b})) \prod_t \|g_t\|_\infty$$

for bounded g_t of any norms. Here C_{vol} appears to the first power, and every sub-collection of the sources also falls under the setting of Def. 8.6.

(b) Let F be a bounded Borel function on the fields of $\mathbb{T}_K$. In

$$e^{-\sum_t z_t(\gamma_t+e_t)} \int F(V)\rho_K^z(V)\, dV/Z_K(0)$$

the coefficient of $z_1 \cdots z_Q$ is then, in modulus, $\le G_1^Q(1 + 0.182\, C_{\mathrm{vol}}(N, \mathsf{b})) \sup |F|$.

■□

Remark 8.15 (Constants). Every numeral of this section is free of K, k, N, b, Q, $\check{\delta}$, $\check{\rho}$ and g. The numerals 0.03 and 0.0057 depend on $L > 1.2 \cdot 10^{13}$ of the table only through $(1 + L^{-1/2})^2 \le 1.001$. The number $\mathfrak{r}^\circ$ depends on b and the depth; sep, j°, $\check{\kappa}^\circ$, κ° and n_{loc} depend on $(\mathsf{b}, \check{\delta})$ and the depth; and N_1, $E^{[j]}$, C_W°, C_W, e_h and G_1 depend on $(\mathsf{b}, \check{\delta}, \check{\rho})$. The number Q occurs only as the dimension

of Pd, as the factor $Q/2$ in Prop. 8.11 (J8°) (a°), as the factor Q in (J8°) (b) and (d), and as the exponent in Cor. 8.14.

Remark 8.16 (Short distances, and the volume once). The proofs of all clauses of Prop. 8.11, and that of Lem. 8.13, work at short distances only. Lem. 8.10, on the multi-affine part on polyradii, is proved by function theory in one variable (Schwarz–Pick), one variable after the other.

The proof of Cor. 8.14 uses the volume once, through the L^1 bound of Thm. 7.27 (c^C), case (ii), and it does not depend on any control at large distances; the step to a bound free of N, by the chessboard estimate, is taken in §8.3.

Proof. Proof 1: Lem. 8.10 on the multi-affine part. We show: if r is a polyradius and g is holomorphic on $\mathrm{Pd}(r)$ with $|g| \le m$, then $|Mg| \le m$ on $\{|z_i| \le r_i/2\}$, where $Mg = \sum_S z^S \partial_S g(0)$.

One variable. Let ϕ be holomorphic on $|z| < r$ with $|\phi| \le 1$, and put $a = |\phi(0)|$. By the Schwarz–Pick lemma, $|\phi'(0)| \le (1-a^2)/r$. Consequently, whenever $|z| \le r/2$,

$$|\phi(0) + z\phi'(0)| \le a + (1-a^2)/2 = 1 - (1-a)^2/2 \le 1.$$

Many variables. Define $(M_i g)(z) := g|_{z_i=0} + z_i(\partial_i g)|_{z_i=0}$. On power series, M_i deletes exactly the monomials z^μ with $\mu_i \ge 2$ and leaves all others in place; therefore the operators M_i commute with one another, and $M = M_n \cdots M_1$. The result is immediate for $m = 0$. For $m > 0$ put $g_0 = g/m$ and $g_i = M_i g_{i-1}$.

We claim that g_i is holomorphic, of modulus at most 1, on

$$A_i := \{|z_j| \le r_j/2,\ j \le i\} \times \{|z_j| < r_j,\ j > i\};$$

since it is affine in each of the first i variables, it is holomorphic on a neighbourhood in these variables. For $i = 0$ the claim is the hypothesis on g, as $A_0 = \mathrm{Pd}(r)$.

To pass from $i-1$ to i, freeze all variables except z_i at a point of A_{i-1}. Then $z_i \mapsto g_{i-1}$ is holomorphic on $|z_i| < r_i$ with modulus at most 1, and g_i arises from it as $\phi(0) + z_i\phi'(0)$ arises from ϕ; the bound in one variable gives $|g_i| \le 1$ for $|z_i| \le r_i/2$. At the end, $g_n = Mg/m$ on $A_n = \{|z_i| \le r_i/2\}$, which proves the lemma.

Proof 2: the clauses (G°) *and* (G5°). We show that (G1) to (G4) of item 3 of §7.3 hold with the numbers of (G°), and that $\#X \ge j_k(X)\mathfrak{h}_k$ and $j_k(X) \le n_{\mathrm{loc}}\#X$.

(G°) The numbers $\mathrm{sep}^\circ, j^\circ, k^\circ, \check\kappa_k^\circ, \kappa_k^\circ$ are the numbers $\mathrm{sep}_*, j^*, k_d, \check\kappa_k, \kappa_k$ of (o4) in item 2 of §7.3, taken at $n = 2$ and $d = \check\delta$.

(G1), (G2) These two conditions are proved in Step 1 (G1), (G2) of the proof of Prop. 7.12. That argument involves n only through $\mathrm{sep}_* > 0$ and through $\check\kappa = (\mathrm{sep}_*/4)/\mathrm{sep}_{k+1}$, so it holds with the numbers of (G°).

(G3) By (G3) of item 3 of §7.3 and Step 0 (b) of the proof of Prop. 7.12, both taken at $x = x_k$, we have $\mathfrak{r}^\circ_{k+1}/\mathfrak{r}^\circ_k = \mathrm{amp}(Lx_k)/\mathrm{amp}(x_k) \in [5.01, L^{1/2}]$. The condition (G3) is stated for $x \ge x_s$, and $x_k \ge \mathfrak{b} \ge x_s$. If $K \ge j^\circ$, then $x_{k^\circ} = L^{j^\circ}\mathfrak{b}$, and hence $\mathfrak{r}^\circ_* = \mathfrak{r}^\circ(x_{k^\circ}) = r^\circ$, which is the case of equality in (G3). If $K < j^\circ$, then $k^\circ = 0$ and $\mathfrak{r}^\circ_* = \mathfrak{r}^\circ(L^K\mathfrak{b}) \ge r^\circ$, because the map $i \mapsto \mathfrak{r}^\circ(L^i\mathfrak{b})$ is decreasing.

(G4) The argument of Step 1 (G4) of the proof of Prop. 7.12 concerns one pair, and so it gives (G4) pair by pair.

(G5°) (a) The set $X^{+\check r(x_k)}$ meets $S_t(k)$ if and only if X meets $S_t(k)^{+\check r(x_k)}$. Let $j = j_k(X) \ge 2$. Then X, and likewise $\mathrm{hull}(X)$, meets the j sets $S_t(k)^{+\check r(x_k)}$ with $t \in I_k(X)$; here $\mathrm{hull}(X)$ is connected, contains X and has $\#\mathrm{hull}(X) = |X|_c$, by Step 1 (C) of the proof of Prop. 7.12. By (G4), these j sets lie pairwise at distance at least $\sigma := \mathrm{sep}_k + 2 \ge 1$. Step 1 (C) of the proof of Prop. 7.12 then bounds $\#X$ (for connected X) or $|X|_c$ (by way of $\mathrm{hull}(X)$) from below by $j\lceil\sigma/2\rceil = j\mathfrak{h}_k$.

(b) Since $I_k(X) = \bigcup_c I_k(\{c\})$, the union running over the cubes c of X, it suffices to treat $X = \{c\}$. For each $t \in I_k(\{c\})$ there is a plaquette p_t with $g_t(x(p_t)) \ne 0$ that is based

in the bare box of a cube of $c^{+\check r}$. The lifts of these base sites lie in a box whose side has $(2\check r(x_k)+1)ML^k$ sites. As $x(p)=a_K(z+(e_\mu+e_\nu)/2)$, the lifted barycentres spread, in each coordinate, over at most $a_K((2\check r+1)ML^k-1/2)<s_k:=(2\check r(x_k)+1)ML^{-(K-k)}$, so after a translation they lie in $[0,s_k)^3$. The sup distance of two lifts is at least their periodic distance, hence at least $\check\delta$. We cut each coordinate interval $[0,s_k)$ into $\lfloor s_k/\check\delta\rfloor+1$ half-open intervals of length $\check\delta$; two points in one cell of the resulting grid are closer than $\check\delta$, so each cell contains at most one of the lifted barycentres $x(p_t)$, and $\#I_k(\{c\})\le(\lfloor s_k/\check\delta\rfloor+1)^3$. Finally, Step 1 (r) of the proof of Prop. 7.12 gives

$$(2\check r(Lx)+1)/L\le(2\check r(x)+3+\log L/480)/L\le 2\check r(x)+1,$$

and therefore s_k is largest at depth 0.

Proof 3: the run in terms of ν, and the clauses (lin), (J1°) *and* (J2°). We show that $U_k(\nu)$ is holomorphic on $|\nu|_\infty<15.99c_\nu\varepsilon(\beta)$, that $\delta[f](k)$ is linear in f with $\sum z_t\delta_t(k)=\delta[g_z](k)$, and that (J1°) and (J2°) hold for every $Q\ge1$.

1. *Plaquette sources.* For a plaquette p of $\mathbb{T}_0$, let 1_p be the function on T_l equal to 1 at $x(p)$ and to 0 elsewhere. The point $x(p)=a_K(z+(e_\mu+e_\nu)/2)$ determines $(z;\mu\nu)$, so $\mathrm{supp}_0 1_p=\{p\}$. Thus the 1_p are complex bounded functions with pairwise disjoint supp_0, and the part "Sources" of item 3 of §7.1 admits them. The variables of these sources are ζ_p, and we write $\nu_p=\zeta_p/\mathsf b$; then

$$\mathrm{Pd}_0^o=\{|\zeta_p|<15.99/\mathrm{amp}_W\}=\{|\nu|_\infty<15.99c_\nu\varepsilon(\beta)\}=:B_\nu.$$

   By Thm. 7.1 (h), the maps $U_0(\nu)=u_0(\nu)$, $U_{k+1}=F_{\ell_k}(U_k)$ and $\mathsf V_{k+1}=\mathsf V'[U_k]$ are defined on B_ν, of norm at most 16 in the strong cap, with all their values holomorphic in the finitely many ν_p; moreover, writing ρ_k^ν for the ρ_k^z of that clause for the plaquette sources 1_p, with their variables ζ_p in the place of z, we have $\rho_k^\nu=\exp\{\sum_{j\le k}\mathsf V_j(\nu)\}\rho[U_k(\nu);\ell_k]$ at every field of $\mathbb{T}_k$.
2. For a complex linear map $z\mapsto\nu(z)$ we put $u_k(z):=U_k(\nu(z))$ wherever $\nu(z)\in B_\nu$. In the setting of Def. 8.6, this is the honest family of (o1) in item 2 of §7.3: both arise by one and the same recursion from the same $u_0(\nu(z))$.
3. Value by value, the chain rule gives

$$\delta[f](k)=\mathsf b^{-1}\sum_p f(x(p))\,(\partial U_k/\partial\nu_p)(0),$$

   which is complex linear in f. The same rule gives $\delta_t(k)=\delta[g_t](k)$, and linearity then gives $\sum z_t\delta_t(k)=\delta[g_z](k)$. Since at each point at most one g_t differs from 0, and $\|g_t\|_\infty\le1$, we have $\|g_z\|_\infty\le\max|z_t|$.
4. We prove (J1°). By Step 2 of the proof of Prop. 7.12, for $k<K$ the condition (SD) of item 2 of §5.3 holds at $(x_k;\ell_k)$, through Thm. 7.1 (b), and $x_k\ge Lx_s\ge\max\{x_9^U,x_3,x_4\}$; for $0\le k\le K$, moreover, $\bar u_k$ is real and lies in $M_1(x_k;\kappa_1,c_h)$ in the strong cap. Lem. 7.7 (S1) and (S3), applied on $\mathrm{Pd}(1.99/\mathrm{amp}_W)$, put every $\delta_S(k)$, $0\le k\le K$, in $\mathfrak M(x_k)$, with finite strong norm at κ_1. Hence u_k^{pol} is a map defined near 0, of norm less than 16, that has the jets of the honest family. Lem. 7.8, with Φ a value of F_{ℓ_k}, of $\mathsf V'_{\ell_k}$ or of $v_k(Y;\cdot)$, now yields (J1°); here Q enters only as the dimension.
5. We prove (J2°). Let $k=0$. By Thm. 7.1 (a4), $u_0(\nu)(X)$ depends only on the ν_p with $p\in X^{+r(x_0)}$. Now $\nu_p(z)$ depends on z_t only if $g_t(x(p))\ne0$, that is, only if the cube that holds p lies in $S_t(0)$; and $r\le\check r$. Hence $u_0(z)$, and with it u_0^{pol}, has the property (P_0) of the proof of Prop. 7.12. For the passage from k to $k+1$ with $k<K$ we use the claim of Step 3 of the proof of Prop. 7.12, which comes from Thm. 6.7 (f5) at $S=S_t(k)$ and $\hat\rho=\check r(x_k)$, applied one variable at a time, together with Step 1 (R) of the same proof: $F_{\ell_k}(u_k^{\mathrm{pol}}(z))(Y)$ does

not depend on z_t if $Y^{+\check{r}(x_{k+1})}$ misses $S_t(k+1)$. By (J1°) its coefficient of z^S is $\delta_S(k+1)(Y)$, which therefore vanishes whenever $t \in S$.

Proof 4: the single jets. Apply the single complex source estimate at the end of Step 6 of the proof of Prop. 7.12. Restriction to one source variable gives the same first jet as the run with that source alone. The splitting and bounds on all levels give (J0°), (J3°) and (J4°).

Proof 5: the clause (L0) *at level* 0, *for every* $K \geq 1$ *and any number of variables.* We show that

$$\sup_{\mathrm{Pd}(R)} \|u_0^{\mathrm{pol}}(z) - \bar{u}_0\|^{\mathrm{st},(\kappa_1)} \leq 0.014$$

whenever $\nu(\cdot)$ is linear with $|\nu(z)|_\infty \leq \mathfrak{r}_0^\circ/\mathsf{b}$ on $\mathrm{Pd}(R)$, and we prove the two sentences of (L0) that say on what the value at X depends.

Let $\theta := \mathrm{amp}_W \mathfrak{r}_0^\circ = 3.5\,\mathrm{amp}_W/\mathrm{amp}(Lx_0)$; by Step 0 (b) of the proof of Prop. 7.12 and (G3) of item 3 of §7.3, $\theta \leq 3.5/(51 \cdot 5.01) \leq 0.0136981$. Let $c := 15.99/\theta$; then $c \geq 1167.3$. By linearity, $|\nu(z)|_\infty \leq c\,\mathfrak{r}_0^\circ/\mathsf{b} = 15.99/(\mathsf{b}\,\mathrm{amp}_W) = 15.99 c_\nu \varepsilon(\beta)$ on $\mathrm{Pd}(cR)$. There Thm. 7.1 (a) to (a4) hold, and $u_0(\nu(z))$ has norm at most 16.

The a-part. By Thm. 7.1 (a4), $E^a_\nu(\{c'\}, V) = \sum_{p \in c'} [q_p + \beta \nu_p s_p]$ is affine in ν, hence of degree at most 1 in z, and so it coincides with its own multi-affine part. The a-part of $u_0^{\mathrm{pol}}(z) - \bar{u}_0$ is therefore $\beta \sum_{p \in c'} \nu_p(z) s_p$. Fix z and let f take the value $\mathsf{b}\nu_p(z)$ at $x(p)$ and the value 0 off the barycentres. This is a single complex bounded source, with $\|f\|_\infty = \mathsf{b}|\nu(z)|_\infty$, and $\delta^a[f](0)(\{c'\}) = (\beta/\mathsf{b}) \sum f(x(p)) s_p$ equals the expression above. The a-part of Thm. 7.1 (f) bounds it by $\mathrm{amp}_W \|f\|_\infty$ caps, and on $\mathrm{Pd}(R)$ we have $\mathrm{amp}_W \|f\|_\infty \leq \mathrm{amp}_W \mathfrak{r}_0^\circ = \theta \leq 0.0137$.

The h-part of $u_0^{\mathrm{pol}}(z) - \bar{u}_0$ is 0.

The w-part. Fix (X, V), and let cap be the strong unit at κ_1 at (X, V). The function $g(z) := w_{\nu(z)}(X, V)$ is entire by Thm. 7.1 (a4), and $|g| \leq \mathrm{cap}$ on $\mathrm{Pd}(cR)$ by Thm. 7.1 (a2). The value of u_0^{pol} at (X, V) is Mg, that of $\bar{u}_0$ is $g(0)$, and Lem. 8.10 gives $|Mg| \leq \mathrm{cap}$ on $\mathrm{Pd}(cR/2)$. Thus $Mg - g(0)$ vanishes at 0 and is at most $2\,\mathrm{cap}$ on $\mathrm{Pd}(cR/2)$, and Lem. 7.7 (S2) gives on $\mathrm{Pd}(R)$

$$|Mg - g(0)| \leq 2\,\mathrm{cap} \cdot \max |z_i|/(cR_i/2) \leq (4/c)\,\mathrm{cap} \leq 0.003427\,\mathrm{cap}.$$

The strong norm at κ_1 of $u_0^{\mathrm{pol}}(z) - \bar{u}_0$ is the largest of the norms of its three parts, hence at most $0.0137 \leq 0.014$. The coefficients of u_0^{pol} lie in $\mathfrak{M}(x_0)$, by Lem. 7.7 (S3).

On what the value depends. Fix (X, V) and one of the members, and let $g(z)$ be the value of that member of $u_0(\nu(z))$ at (X, V); the corresponding value of u_0^{pol} is Mg. By Thm. 7.1 (a4), g is a function of the $\nu_p(z)$ with $p \in X^{+r(x_0)}$. If none of these $\nu_p(\cdot)$ depends on z_i, then g does not depend on z_i, so $\partial_S g(0) = 0$ for every $S \ni i$, and Mg does not depend on z_i: this is the first sentence of (L0). Suppose besides that each $\nu_p(\cdot)$ depends on at most one variable, and let T be the set of those i on which some $\nu_p(\cdot)$ with $p \in X^{+r(x_0)}$ depends. For $i \in T$ there is such a p with $\nu_p(z) = a_p z_i$, $a_p \neq 0$, so that $z_i = \nu_p(z)/a_p$. By the first sentence, Mg depends on $(z_i)_{i \in T}$ alone, and so it is a function of the $\nu_p(z)$ with $p \in X^{+r(x_0)}$: this is the second sentence. In the setting of Def. 8.6, at most one t has $g_t(x(p)) \neq 0$; for that t we have $\nu_p(z) = \mathsf{b}^{-1} z_t g_t(x(p))$, and if there is none, $\nu_p(z) = 0$.

Proof 6: the estimate (J5°) (a) *to* (d). We show: for admissible data the mixed jets are at most 0.03, as a supremum over (X, V) of a sum of moduli at $\varkappa_k$, and u_k^{pol} lies within 0.73 of $\bar{u}_k$ on $\mathrm{Pd}(R_k)$; moreover, the standard data and the data of (d) are admissible.

(a) We have $\sum z_t \delta_t(k) = \delta[g_z](k)$ with $\|g_z\|_\infty \leq \max |z_t| < R$, so Proof 4, applied to the source g_z, gives (a).

(a′) Fix (X, V). By a choice of the phases of the z_t, $\sup_{\mathrm{Pd}(R)} |\sum z_t \delta_t(k)(X, V)| = R \sum_t |\delta_t(k)(X, V)|$; dividing by R gives (a′).

(b), (c) We argue by induction on k. For $k = 0$ we have $R_0 \leq \mathfrak{r}_0^\circ$ (if $k_1 = 0$, this is a hypothesis) and $|\nu(z)|_\infty \leq \max |z_t|/\mathsf{b}$, so (L0) yields the bound 0.014 for the norm at the rate κ_1, and

therefore, as $\kappa_1 \geq \varkappa_0$, for the norm at the rate $\varkappa_0$ as well. Now let $k < k_1$; we pass from k to $k+1$.

1. By (c) at k and by $\|\bar{u}_k\| \leq 1$ at κ_1 (Step 0 (c) of the proof of Prop. 7.12), $u_k^{\mathrm{pol}}(z)$ and $\bar{u}_k$ lie in $M_2(x_k; \varkappa_k, c_h)$ for $z \in \mathrm{Pd}(R_k)$. Moreover $(\varkappa_k, c_h) \in \mathbb{P}$, and $k < K$, so (SD) of item 2 of §5.3 holds and $x_k \geq x_9^U$.
2. Thm. 5.15 (e) at $p = (\varkappa_k, c_h)$ (Thm. 6.7 (U)) gives
$$\|F_{\ell_k}(u_k^{\mathrm{pol}}(z)) - \bar{u}_{k+1}\|^{\mathrm{st},(\varkappa_k)} \leq 0.4 \cdot 0.91 = 0.364$$
on $\mathrm{Pd}(R_k)$.
3. Fix (Y, V') and one of the members. The value of that member of $F_{\ell_k}(u_k^{\mathrm{pol}}(z))$ at (Y, V') is holomorphic on $\mathrm{Pd}(R_k)$ by Thm. 6.7 (c+) at $\Omega = \mathrm{Pd}(R_k)$; the hypothesis of that clause, $u_z \in M_{16}(x_k; p)$ on all of Ω with holomorphic values, is supplied by item 1, the values being polynomials. By (J1°), the coefficient of z^S in this value is $\delta_S(k+1)(Y, V')$. Lem. 7.7 (S1), applied to the seminorm given by this one weighted value, yields $|\delta_S(k+1)(Y, V')| \leq 0.364\, R_k^{-|S|}$ units at $\varkappa_k$.
4. By (J2°), $\delta_S(k+1)(Y, V') = 0$ unless $S \subset I_{k+1}(Y)$. Since a mixed jet has $|S| \geq 2$, only those Y with $j := j_{k+1}(Y) \geq 2$ matter, and with $I = I_{k+1}(Y)$
$$\sum_{S \subset I, |S| \geq 2} (R_{k+1}/R_k)^{|S|} \leq (1 + R_{k+1}/R_k)^j.$$
5. A unit at $\varkappa_k$ equals $e^{-\check{\Delta}_k(\#Y - 1)}$ units at $\varkappa_{k+1}$, with $|Y|_c$ in the place of $\#Y$ for the w-member. By (G5°) (a), $\#Y$ for the a- and h-members, or $|Y|_c$ in the case of the w-member, is at least $j\mathfrak{h}_{k+1}$.
6. Collecting items 3 to 5, the sum over $|S| \geq 2$ of $|\delta_S(k+1)(Y, V')| R_{k+1}^{|S|}$, in units at $\varkappa_{k+1}$, is at most
$$0.364\, e^{\check{\Delta}_k}[(1 + R_{k+1}/R_k)e^{-\check{\Delta}_k \mathfrak{h}_{k+1}}]^j = 0.364\, e^{\check{\Delta}_k} q_k^j \leq 0.364\, e^{\check{\Delta}_k} q_k^2 \leq 0.03,$$
which is (b) at $k+1$. Here R_{k+1} enters only through the ratio R_{k+1}/R_k, and no estimate is applied at the rate $\varkappa_{k+1}$; so at $k+1 = k_1$ no bound on R_{k_1} enters, and $\varkappa_{k_1}$ may lie outside $[\kappa, \kappa_1]$.
7. If $R_{k+1} \leq \mathfrak{r}_{k+1}^\circ$ and $\varkappa_{k+1} \geq \kappa$, then (c) holds at $k+1$: by (a) the single jets contribute at most $0.2 \operatorname{amp}(Lx_{k+1})\mathfrak{r}_{k+1}^\circ = 0.7$ at κ_1, and the mixed jets at most 0.03, together 0.73. The coefficients lie in $\mathfrak{M}$ by Lem. 7.7 (S3), the values are polynomials in z, and $\|\bar{u}_{k+1}\| \leq 1$; so the family lies in $M_{1.73}$, hence in $M_{1.91}$.

The decay factor is raised to the power $j(Y)$, one factor for each of the sources near Y, and not to the power $|S|$, one for each element of S: with the power $|S|$, it would leave the sum $(1+q')^j - 1 - jq'$, which grows with j. Here $q' = (R_{k+1}/R_k)e^{-\check{\Delta}_k \mathfrak{h}_{k+1}}$.

Standard data. For these, condition (i) of (AD) holds with equality. Let $k < k^\circ$. Then $\mathrm{sep}_{k+1} \geq L \,\mathrm{sep}^\circ$, $R_{k+1}/R_k \leq L^{1/2}$ and $\mathfrak{h}_{k+1} \geq \mathrm{sep}_{k+1}/2 + 1$; as $\check{\kappa}_k^\circ = (\mathrm{sep}^\circ/4)/\mathrm{sep}_{k+1}$ and $\mathrm{sep}^\circ = 8 \log(8L^{1/2})$, this gives $\check{\kappa}_k^\circ \mathfrak{h}_{k+1} \geq \log(8L^{1/2}) + \check{\kappa}_k^\circ$, that is, $e^{-\check{\kappa}_k^\circ \mathfrak{h}_{k+1}} \leq e^{-\check{\kappa}_k^\circ}/(8L^{1/2})$. Hence $q_k \leq e^{-\check{\kappa}_k^\circ}(1 + L^{-1/2})/8 < 1$, and
$$0.364\, e^{\check{\Delta}} q^2 \leq 0.364((1 + L^{-1/2})/8)^2 \leq 0.0057.$$

(d) By (G1) of item 3 of §7.3, $\mathrm{sep}_{i+1} \geq L^{k-i-1}\mathrm{sep}_k > 0$ for $i < k$. Since $\mathrm{sep}_k \geq 8 \log(4\varrho L^{1/2}) \geq 8 \log(4L^{1/2})$, we have $\check{\kappa}_{\mathrm{last}} \leq 1/4$ and $\sum_{i<k-1} \check{\kappa}_i \leq \frac{1}{4} \sum_{m \geq 1} L^{-m} = 1/(4(L-1))$. So the rates lie in $[\kappa + 1/4, \kappa + 1/2 + 1/(4(L-1))]$, inside $[\kappa, \kappa_1]$. For $i < k-1$, $q_i \leq e^{-\check{\kappa}_i}(1 + L^{-1/2})/4$ and $0.364\, e^{\check{\Delta}} q^2 \leq 0.364((1 + L^{-1/2})/4)^2 \leq 0.02276$. At the last step, $R_k/R_{k-1} \leq \varrho L^{1/2}$ and

$e^{-\check\kappa_{\mathrm{last}}\mathfrak{h}_k} \le e^{-\check\kappa_{\mathrm{last}}}/(4\varrho L^{1/2})$, so $q \le e^{-\check\kappa_{\mathrm{last}}}(1+1/(\varrho L^{1/2}))/4$, and there too $0.364\,e^{\check\Delta}q^2 \le 0.02276$.

Remark. The standard data of (AD) remain admissible when sep° is replaced by an arbitrary numeral $\widehat{\mathrm{sep}} \ge 8\log(4L^{1/2})$. Moreover, if two such numerals lie within a factor L of one another, their values of j° differ by at most 1. We put any numeral $\widehat{\mathrm{sep}} \ge 8\log(4L^{1/2})$ in the place of sep°, and we define with it the number j° of (D1) as well as the number k° and the rates κ_k° of (D2): j° is the least $j \ge 1$ with $\mathrm{sep}^{[j-1]} \ge \widehat{\mathrm{sep}}$, $k^\circ = \max\{K - j^\circ, 0\}$, $\check\kappa_k := (\widehat{\mathrm{sep}}/4)/\mathrm{sep}_{k+1}$ for $k < k^\circ$, and the rates are $\kappa + \sum_{k\le j<k^\circ}\check\kappa_j$. In the two estimates below, $\check\kappa$ without an index denotes $\check\kappa_k$. The data so obtained, namely the standard data of (AD) changed in this one respect, with $k_1 = k^\circ$, $R_k = \mathfrak{r}_k^\circ$ and these rates as $\varkappa_k$, are the data of the first assertion. For them, the two estimates given above for the standard data go through with these changes: for $k < k^\circ$, with this k°, we have $\mathrm{sep}_{k+1} \ge L\,\widehat{\mathrm{sep}}$, and $\check\kappa\,\mathfrak{h}_{k+1} \ge \widehat{\mathrm{sep}}/8 + \check\kappa \ge \log(4L^{1/2}) + \check\kappa$, so that $e^{-\check\kappa\,\mathfrak{h}_{k+1}} \le e^{-\check\kappa}/(4L^{1/2})$; hence $q_k \le e^{-\check\kappa}(1+L^{-1/2})/4 < 1$ and $0.364\,e^{\check\Delta}q^2 \le 0.364((1+L^{-1/2})/4)^2 \le 0.02276 \le 0.03$, while condition (i) of (AD) again holds with equality. For $k < k^\circ$ we have $\mathrm{sep}_{k+1} \ge L^{K-k-j^\circ}\,\widehat{\mathrm{sep}}$ with $K - k - j^\circ \ge 1$: this follows from (G1) of item 3 of §7.3, applied from the depth $j^\circ - 1$ on, by the reasoning of Step 1 (G2) of the proof of Prop. 7.12, where $\widehat{\mathrm{sep}}$ takes the place of sep_*. We therefore get $\sum_{k<k^\circ}\check\kappa_k \le \frac14\sum_{m\ge1}L^{-m} = 1/(4(L-1))$. So the rates $\varkappa_k = \kappa + \sum_{k\le j<k^\circ}\check\kappa_j$, $0 \le k \le k^\circ$, are non-increasing in k, since each $\check\kappa_j > 0$, and they lie in $[\kappa, \kappa + 1/(4(L-1))]$, inside $[\kappa, \kappa_1]$; they thus form the chain of rates that (AD) requires. By (G1) of item 3 of §7.3, the second assertion of the remark follows.

These estimates use the L of the table, $L > 1.2\cdot10^{13}$ ((O4) in item 1 of §5.3, as cited in Step 0 (b) of the proof of Prop. 7.12); the size of L matters for the numeral $0.364((1+L^{-1/2})/4)^2$, which arises when $4L^{1/2}$ is under the logarithm, rather than the $8L^{1/2}$ of the standard data: at $L = 13$, for instance, it would be $0.0371 > 0.03$.

Proof 7: the clause (J6°)*, on the representatives.* We prove (a) to (e) of (J6°) for every $K \ge 1$.

The induction starts at $k = k^\circ$. There (J5°) (c), taken for the standard data, for which $\kappa_{k^\circ}^\circ = \kappa$, shows that $u_{k^\circ}^{\mathrm{pol}}$ is a family on $\mathrm{Pd}(\mathfrak{r}_*^\circ)$ within $B = 1.91$ caps at $p = (\kappa, c_h)$, whose distance to $\bar u_{k^\circ}$ is $\le 0.73 \le 0.91$. If $K < j^\circ$, then $k^\circ = 0$, and the start is (L0), in which no separation enters.

The passage from k to $k+1$, for $k^\circ \le k < K$, is Step 8 of the proof of Prop. 7.12, carried out with the following data. We apply Thm. 7.9 (i), (ii) with $p = (\kappa, c_h)$, on $\Omega = \mathrm{Pd}(\mathfrak{r}_*^\circ) \subset \mathbb{C}^Q$ (item 0 (5) of §7.2 allows any $n \ge 1$), to the pair consisting of $u_k(z)$ and the constant family $\bar u_k$, with $B = 1.91$ at $k = k^\circ$ and $B = 1$ at the later levels. By Thm. 7.9 (i), the image is a family in $M_1(x_{k+1}; \kappa, c_h)$ that obeys the strong cap, and its a-member satisfies $|E'^a|_a^\Sigma \le 0.35 + 0.248\cdot1.91 = 0.82368 \le 1$. The strong distance of the images is at most $\vartheta[B]$ times the weak distance of the pair, where $\vartheta[1.91] = 3.968/14.09 \le 0.2817$ and $\vartheta[1] \le 0.2646$, both ≤ 0.4. Since $F_{\ell_k}(\bar u_k) = \bar u_{k+1}$, this gives (a) and (b).

(c) If at level k the family $u_k(z)$ and the honest family have the same $\partial_{S'}$ at 0 for every S', then Lem. 7.8, applied value by value, shows that so do their images. At k° this holds by the form of u^{pol}.

(d) The property (P_k) of the proof of Prop. 7.12 holds for $k \le k^\circ$ by (J2°), and it passes from each level to the next by the claim of Step 3 of that proof. Here $k^\circ \le K - 1$, by (G2) of item 3 of §7.3, as restated in (G°).

(e) The last step is $k = K - 1 \ge k^\circ$. There Thm. 7.9 (ii), in its part on the factors of the three members, bounds the h-member in the cell sum $|\cdot|_h^\Sigma$, with the factor $1/(16 - B)$: $|E'^h[u_1] - E'^h[u_2]|_h^\Sigma \le \mathsf{D}^{\mathrm{in}}/(16-B) \le 0.91/14.09 \le 0.0646$ on $\mathrm{Pd}(\mathfrak{r}_*^\circ)$, in caps $c_h(\mathfrak{b})$, at (κ, c_h). The cell sum is an admissible seminorm (item 0 (4) of §7.2), so Lem. 7.7 (S1), applied with $\mu = 1_{\{t\}}$, yields $|\delta_t^h(K)|_h^\Sigma \le 0.0646/\mathfrak{r}_*^\circ$.

By (J2°) for $k = K$ and $S = \{t\}$, $\delta_t^h(K)(X) = 0$ unless X contains a cell of $S_t(K)^{+\check{r}(\mathsf{b})}$, and there are at most $N_1(K-1) = N_1^{[1]}$ such cells (Step 1 (A) of the proof of Prop. 7.12). We add up the cell sums over these cells and use $\mathfrak{r}_*^\circ \geq r^\circ$. The field 1, at which e_t is evaluated, is G-valued and tame.

Proof 8: the clause (J7°)*, on congruences and vacuum functions.* We prove (i) to (vii) of (J7°). We use the squarefree congruence of Def. 7.26. Its compatibility with sums, products and the exponential was proved in Step P7 of the proof of Cor. 7.28.

(i) Thm. 7.1 (h), applied on Pd_0^o to the honest family, gives $\rho_k^z = \exp\{\sum_{j\leq k} V_j(z)\}\rho[u_k^{\mathrm{hon}}(z);\ell_k]$ at every field of $\mathbb{T}_k$. By Thm. 6.7 (g1), (f1), $V_{k'+1}(z) - V_{k'+1}(0) = \sum_Y [v_{k'}(Y;u_{k'}^{\mathrm{hon}}(z)) - v_{k'}(Y;\bar{u}_{k'})]$, a finite sum; the term V_0^+ involves no member. Lem. 7.8, applied with $\Phi = v_{k'}(Y;\cdot)$, shows that the coefficient of z^S depends only on $u(0)$ and the $u_{S'}$; therefore $v_{k'}(Y;u_{k'}^{\mathrm{hon}}(z)) \equiv v_{k'}(Y;u_{k'}(z))$.

Lem. 7.8 applies as well with $\Phi = \rho[\cdot;\ell_k](V)$, with V fixed, by its last part, the one on $I[u]$: at each V this function is $e^{-x\mathrm{En}_\ell(V)}$, in which no member enters, times the exponential of a finite sum of values of E^a, E^h, times a polynomial in finitely many values of w; it is entire along lines and bounded on $\{\|u\| < R\}$. The lattices $\mathbb{T}_k$ with $k < K$ are those of item 2 of §5.4 at x_k, and $\mathbb{T}_K$ is the one named in Lem. 7.8. Thus $\rho[u_k^{\mathrm{hon}}(z);\ell_k](V) \equiv \rho[u_k(z);\ell_k](V)$, and $\equiv$ then passes through the sum, the exponential and the product.

(ii) By (J2°), u_k^{pol} has the property (P_k). The claim of Step 3 of the proof of Prop. 7.12 then shows that $v_k(Y;u_k^{\mathrm{pol}}(z))$ does not depend on z_t if $Y^{+\check{r}(x_{k+1})}$ misses $S_t(k+1)$, so that its coefficient of z^S vanishes whenever $t \in S$. By Lem. 7.8, this coefficient is the same for all representatives.

(iii) We take u^{pol} at all levels and replace each $v_{k'}(Y;u_{k'}^{\mathrm{pol}}(z))$ by its multi-affine part $v_{k'}(Y;\bar{u}_{k'}) + \sum_S z^S \partial_S v_{k'}(Y;u_{k'}^{\mathrm{pol}})(0)$. The terms with $|S| = 1$ add up to $\sum z_t \gamma_t^{(k)}$, because $\partial_t V_{k'+1}(0) = \sum_Y \partial_t v_{k'}(Y;u_{k'}^{\mathrm{pol}})(0)$.

(iv) For $k' < k^\circ$ we replace, exactly as in (iii), each $v_{k'}(Y;u_{k'}^{\mathrm{pol}}(z))$ by its multi-affine part. For $k' \geq k^\circ$ we write $w_{k'} = \sum z_t \partial_t w_{k'}(0) + [w_{k'} - \sum z_t \partial_t w_{k'}(0)]$, where $\partial_t w_{k'}(0) = \partial_t V_{k'+1}(0)$. Each $w_{k'}$ is a finite sum of functions holomorphic on $\mathrm{Pd}(\mathfrak{r}_*^\circ)$, by Thm. 6.7 (c+).

(v) By Thm. 7.1 (h), $Z_K(z) = \exp\{\sum V_j(z)\} I[u_K^{\mathrm{hon}}(z)]$, and Lem. 7.8 applies with $\Phi = I$. As $\rho[\bar{u}_K;\ell_K] > 0$ (Thm. 7.1 (h)), we have $I[\bar{u}_K] > 0$, and we may divide by $Z_K(0) = \exp\{\sum V_j(0)\} I[\bar{u}_K]$.

(vi) Near 0 we have $\log Z_K = \sum V_j(z) + \log I[u_K^{\mathrm{hon}}(z)]$. The argument of Step 9 of the proof of Prop. 7.12, with S in place of $\{1,\dots,n\}$, shows that $\partial_S \log I$ is a polynomial in the $\partial_{S'} I/I(0)$, $S' \subset S$.

(vii) Since $\rho_K^0(1) = \exp\{\sum V_j(0)\}\rho[\bar{u}_K;\ell_K](1) > 0$, the logarithm of $\rho_K^z(1)$ exists near 0. Thm. 7.1 (h) at $V = 1$ gives $\partial_t \log \rho_K^z(1)(0) = \gamma_t + \partial_t \log \rho[u_K^{\mathrm{hon}}(z);\ell_K](1)(0)$.

The set $D(1)$ is empty, since $\phi_p = 0$ at the field 1. Every polymer in a family contributing to $\Xi_w(V)$ contains a plaquette of $D(V)$, by item 3 of §5.3. At $V = 1$ only the empty family contributes, so $\Xi_w(1) = 1$. Thus $\rho[u;\ell_K](1) = \exp\{-\mathsf{b}\mathrm{En}_\ell(1) + \sum_X [E^a + E^h](X,1)\}$, with En_ℓ independent of the members. Moreover $E^a(X,1) = 0$, by the clause "0 at pure gauges", a clause without a size which every member of the family satisfies. As E^a is 0 at the field 1 for every z, we obtain $\partial_t \log \rho[u_K^{\mathrm{hon}}(z);\ell_K](1)(0) = \sum_X \partial_t [E^a + E^h](X,1)(0) = \sum_X \delta_t^h(K)(X,1) = e_t$.

Setting $z_s = 0$ for $s \neq t$ turns the run with Q sources into the run with the one source g_t. Hence $\partial_t \log \rho_K^z(1)(0)$ is $\hat{c}_K(g_t)$, which is $c_K^{(0)}(g_t)$ of §8.3, the subtraction of its item 2 and of Thm. 8.17 (R0).

Proof 9: the clauses (J8°) (a), (a°), (a′), *on the early vacuum numbers per source.* For (a) we show that

$$\sum_{S\ni t,|S|\ge 2} R^{|S|}\sum_{Y}|\partial_S v_k(Y;u_k^{\mathrm{pol}})(0)|\le N_1(k)2C_gc_a(x_k)e^{\lambda}\exp\{-(\varkappa_k-\lambda)(\mathrm{sep}_{k+1}+1)\},$$

where $\lambda=\log(1+R/R_k)/\mathfrak{h}_{k+1}\le\varkappa_k$.

Fix $k<k_1$ and $r<R_k$, and put $\phi_Y(z):=v_k(Y;u_k^{\mathrm{pol}}(z))$; by Thm. 6.7 (c+), this function is holomorphic on $\mathrm{Pd}(R_k)$.

1. One mean suffices for every S: $\partial_S\phi_Y(0)=r^{-|S|}\mathrm{Mean}[\phi_Y(re^{i\theta})e^{-i\sum_{s\in S}\theta_s}]$, where the mean is taken over the whole Q-torus, because the mean of $z^\mu e^{-i\sum_S\theta}$ vanishes unless $\mu=1_S$. Hence $R^{|S|}|\partial_S\phi_Y(0)|\le(R/r)^{|S|}\mathrm{Mean}|\phi_Y|$, with the same mean for all S.
2. By (J7°) (ii), only sets $S\subset I_{k+1}(Y)$ occur. Over the $S\subset I_{k+1}(Y)$ with $t\in S$ and $|S|\ge2$, the factors $(R/r)^{|S|}$ add up to at most $(1+R/r)^{j(Y)}$. This sum is empty unless $Y\in\check{Y}_t:=\{Y$ connected: $t\in I_{k+1}(Y),\ j(Y)\ge2\}$, where $j(Y)=j_{k+1}(Y)$.
3. The sums over S and over Y are finite, so we may take them inside the mean. Thus the left side of the bound is $\le\sup_{|z_s|=r}\sum_{Y\in\check{Y}_t}(1+R/r)^{j(Y)}|v_k(Y;u_k^{\mathrm{pol}}(z))|$.
4. By (G5°) (a) for $k+1$, $\#Y\ge j(Y)\mathfrak{h}$ with $\mathfrak{h}=\mathfrak{h}_{k+1}$; therefore $(1+R/r)^{j(Y)}\le e^{\lambda_r\#Y}$, where $\lambda_r=\log(1+R/r)/\mathfrak{h}$. Also $\#Y\ge2\mathfrak{h}\ge\mathrm{sep}_{k+1}+2$.
5. Each $Y\in\check{Y}_t$ contains a cell of $S_t(k+1)^{+\check{r}(x_{k+1})}$, and for any t there are $\le N_1(k)$ such cells (Step 1 (A) of the proof of Prop. 7.12).
6. We use Thm. 6.7 (g2) with $A=2$ and $p=(\varkappa_k,c_h)$; it applies because $u_k^{\mathrm{pol}}(z)\in M_2$ by (J5°) (c). For a cell, provided that $\lambda_r\le\varkappa_k$,

$$\sum_{Y\ni\mathrm{cell},\#Y\ge2\mathfrak{h}}e^{\lambda_r\#Y}|v|=\sum e^{\lambda_r-(\varkappa_k-\lambda_r)(\#Y-1)}e^{\varkappa_k(\#Y-1)}|v|\le e^{\lambda_r}e^{-(\varkappa_k-\lambda_r)(2\mathfrak{h}-1)}2C_gc_a(x_k).$$

Letting $r\to R_k$ gives the bound of (a); in the case $\lambda=\varkappa_k$ it follows by continuity of both sides in R.

(a°) Let $t':=k^\circ-k\ge1$. By (G3) of item 3 of §7.3, as restated in (G°), $\mathfrak{r}_*^\circ/\mathfrak{r}_k^\circ\le L^{t'/2}$, and $\log(1+L^{t'/2})\le t'\log(2L^{1/2})\le t'\mathrm{sep}^\circ/8$. The depth of $k+1$ is $j^\circ-1+t'$, so by (G1) of the same item $\mathfrak{h}>\mathrm{sep}_{k+1}/2\ge L^{t'}\mathrm{sep}^\circ/2$, and hence $\lambda\le t'/(4L^{t'})\le1/(4L)$. Also $\varkappa_k\ge\kappa$. The sum over t counts each S exactly $|S|$ times, and $|S|\ge2$; therefore $\sup_{\mathrm{Pd}(\mathfrak{r}_*^\circ)}|W_k|\le(Q/2)E^{[j]}$.

(a′) Put $k'=k-m$, $R=\varrho\,\mathfrak{r}_k^\circ$ and $R_{k'}=\mathfrak{r}_{k'}^\circ$. Then $R/R_{k'}\le\varrho L^{m/2}$, and $\log(1+\varrho L^{m/2})\le m\log(2\varrho L^{1/2})\le m\,\mathrm{sep}_k/8$. Since $\mathfrak{h}_{k'+1}>\mathrm{sep}_{k'+1}/2\ge L^{m-1}\mathrm{sep}_k/2$, we obtain $\lambda\le m/(4L^{m-1})\le1/4$. Moreover $\varkappa_{k'}-\lambda\ge\kappa\ge\kappa-1/4$ and $\mathrm{sep}_{k'+1}+1\ge L^{m-1}\mathrm{sep}_k+1$, and we sum over $1\le m\le k$. The numbers $N_1(k-m)=N_1^{[K-k+m]}$ and $x_{k-m}=L^{K-k+m}\mathfrak{b}$ are defined for every $m\ge1$, and all terms are ≥0; so the finite sum is $\le E_{\mathrm{vac}}$.

The series E_{vac} is finite. Item (ii) of the proof of Lem. 8.4 (N3) uses (N1) alone among the clauses of that lemma; carried out with $K-k+1$ in place of j^*, with $m-1$ in place of the offset t of that item and with $\check{\rho}$ in place of R_f, it gives $N_1^{[K-k+m]}\le L^{3(m-1)}N_1^{[K-k+1]}$, with equality at $m=1$; moreover $2C_gc_a\le C_g/16$ by item (iii) of that proof. Since $\mathrm{sep}_k\ge8\log(4L^{1/2})\ge4\log L$ and $\kappa-1/4\ge5.75$,

$$(\kappa-1/4)L^{m-1}\mathrm{sep}_k\ge23L^{m-1}\log L\ge3(m-1)\log L+20L^{m-1}\log L.$$

So $E_{\mathrm{vac}}\le N_1^{[K-k+1]}(C_g/8)e^{1/4}L^{-20}$.

Proof 10: the clauses (J8°) (b), (c), (d), *and Lem. 8.13.* We bound the late levels without shrinking the radius, prove $|W|\le QC_W^\circ\le QC_W$, and show that the series are finite and locally bounded.

(b) Let $k^\circ \le k < K$ and $z \in \mathrm{Pd}(\mathfrak{r}_*^\circ)$. The representative has the property (P_k), by (J6°) (d). If $I_{k+1}(Y)$ is empty, then by the claim of Step 3 of the proof of Prop. 7.12, $v_k(Y; u_k(z))$ does not depend on any z_t, so it equals its value at 0, and the term of Y is 0. Every other Y contains one of the $\le QN_1(k)$ cells of $\bigcup_t S_t(k+1)^{+\check r(x_{k+1})}$. Thm. 6.7 (g2), applied with $p = (\kappa, c_h)$, with $A = 1.91$ (at $k = k^\circ$) or 1 for $u_k(z)$ and with $A = 1$ for $\bar u_k$, bounds the contribution of each cell by $2.91\, C_g c_a(x_k)$.

For $\partial_t w_k(0)$ we set $z_s = 0$ for $s \ne t$. Then every Y with $t \notin I_{k+1}(Y)$ gives 0, and each of the others contains one of $N_1(k)$ cells; hence $|w_k| \le 3N_1(k) C_g c_a$ on $|z_t| < \mathfrak{r}_*^\circ$, and Cauchy's estimate applies. So $|w_k - \sum z_t \partial_t w_k(0)| \le 6QN_1(k) C_g c_a(x_k)$ on $\mathrm{Pd}(\mathfrak{r}_*^\circ)$.

(c) We repeat items 1 to 3 and 5 of Proof 9 with these changes: throughout, $R = \mathfrak{r}_*^\circ / n_{\mathrm{loc}}$; in item 1 the function is $v_k(Y; u_k(z))$ of (b), and the radius is any $r < \mathfrak{r}_*^\circ$; in items 2 and 3 the sums run over all $S \ni t$; in item 5 each connected Y with $t \in I_{k+1}(Y)$, without the condition $j(Y) \ge 2$, contains one of the $\le N_1(k)$ cells of $S_t(k+1)^{+\check r(x_{k+1})}$. Letting $r \to \mathfrak{r}_*^\circ$, we get $(1 + R/r)^{j(Y)} \to (1 + 1/n_{\mathrm{loc}})^{j(Y)} \le (1 + 1/n_{\mathrm{loc}})^{n_{\mathrm{loc}} \# Y} \le e^{\# Y}$, by (G5°) (b) for $k+1$. For a cell, $\sum_{Y \ni \mathrm{cell}} e^{\# Y} |v| \le e \sum e^{\kappa(\# Y - 1)} |v| \le e \cdot 1.91\, C_g c_a$; the first inequality holds because $\kappa \ge 1$, and the second follows from Thm. 6.7 (g2), applied with $p = (\kappa, c_h)$, with $A = 1.91$ (at $k = k^\circ$) or 1. Summing over the $\le N_1(k)$ cells and using $1.91e \le 2e$, we obtain the bound $2e\, N_1(k) C_g c_a(x_k)$ of (J8°) (c).

(d) We pass to the depth $j = K - k$, so that $x_k = L^j \mathsf{b}$ and $N_1(k) = N_1^{[j]}$. Then $k < k^\circ$ holds if and only if $j > j^\circ$, in which case $K > j^\circ$; so the late levels are the depths $1, \dots, \min(K, j^\circ)$. Combining (a°) and (b) gives

$$|W| \le Q[\frac{1}{2} \sum_{j > j^\circ} E^{[j]} + 6 \sum_{j \le j^\circ} N_1^{[j]} C_g c_a(L^j \mathsf{b})].$$

For $C_W^\circ \le C_W$ we put $s = \mathrm{sep}^{[j-1]} > 0$; then $e^{1/(4L)} \le 2e$ and $(\kappa - 1/(4L))(s+1) \ge (\kappa - 1)(s - 1)$.

We turn to Lem. 8.13. Let $j = j^\circ + t$ with $t \ge 1$. Item (ii) of the proof of Lem. 8.4 (N3) uses (N1) alone among the clauses of that lemma; carried out with j° in place of j^* and with $\check\rho$ in place of R_f, it gives $N_1^{[j]} \le L^{3t} N_1^{[j^\circ]}$. By item (iii) of the same proof, $2C_g c_a \le C_g/16$, and by (G1) of item 3 of §7.3, as restated in (G°), $\mathrm{sep}^{[j-1]} \ge L^t \mathrm{sep}^\circ$. Moreover $(\kappa - 1/(4L))\mathrm{sep}^\circ \ge 5.9 \cdot 8 \log(8L^{1/2}) \ge 23.6 \log L$ and $L^{3t} \le e^{3L^t \log L}$. Therefore $E^{[j]} \le N_1^{[j^\circ]} (C_g/16) e^{1/(4L)} \exp\{-20 L^t \log L\}$, and these terms are summable.

For the early terms of C_W, again with $j = j^\circ + t$, we have $\mathrm{sep}^\circ = 8 \log 8 + 4 \log L$, $\kappa - 1 \ge 5$ and $L^t \mathrm{sep}^\circ - 1 > 0$. Hence

$$(\kappa - 1)(\mathrm{sep}^{[j-1]} - 1) \ge 5(L^t \mathrm{sep}^\circ - 1) = 20 L^t \log L + 40 \log 8 \cdot L^t - 5 \ge 20 L^t \log L + 83 L^t - 5.$$

The term $83L^t$ absorbs the factor L^{3t} in $N_1^{[j]} \le L^{3t} N_1^{[j^\circ]}$, as $L^{3t} = e^{3t \log L}$ and $L^t \ge tL \ge t \log L$. Next, $2ec_a \le 2e/32 < 1$ by (X17) of (O13), item 1 of §5.5, which is the inequality $16(c_a + c_h)(x) \le \frac{1}{2}$, that is, $c_a + c_h \le 1/32$. So

$$2e N_1^{[j]} C_g c_a(L^j \mathsf{b}) e^{-(\kappa-1)(\mathrm{sep}^{[j-1]} - 1)} \le N_1^{[j^\circ]} C_g \exp\{-20 L^t \log L + 5\}.$$

Local boundedness follows the proof of Lem. 8.4 (N5). Let b range over a compact set, and put $\hat\jmath := \max j^\circ$, where the maximum is taken over this set. Here j° is non-decreasing in b. The terms with $j \le \hat\jmath$, in either form, are products of the continuous $c_a(L^j \mathsf{b})$, of the non-decreasing $N_1^{[j]}$ and of factors ≤ 6, and so they are bounded: in the late form the factor is 6, and in the early form it is $2e$ times an exponential ≤ 1, because $\mathrm{sep}^{[j-1]} \ge L \mathrm{sep}^\circ \ge 1$ there. For $j > \hat\jmath$ the term is $\le \hat P \exp\{-20 L^{j - \hat\jmath} \log L + 5\}$, where $\hat P$ is an upper bound for $N_1^{[j^\circ]} C_g$ on the compact set. No term depends on K, N or Q.

Proof 11: Cor. 8.14. We prove (a), the bound on the moment of Q sources with C_{vol} to the first power, for all $K \geq 1$, and (b), the same bound with a bounded Borel function of the last field.

The data (B) of Thm. 7.27 with $x = \mathsf{b}$ on $\mathbb{Z}^3/N\mathbb{Z}^3$, with $\tilde{\kappa} = \kappa$, are met for every $K \geq 1$: this is shown in Step 2 of the proof of Thm. 8.2, in its part on the last integral, by an argument into which no source enters. These data are: $\mathsf{b} \geq x_s \geq e^2$; N even, $M|N$, $N \geq 2Ms(\mathsf{b})$; $\varepsilon(\mathsf{b}) \leq 1$; $(c_a + c_h)(\mathsf{b}) \leq 1/2$; and ℓ_K of range $\leq \rho(\mathsf{b})$ with $|\ell_{K,y}(p)| \leq K_* e^{-\lambda_* |y-p|}$.

The first member is $u_1 = \bar{u}_K$, which by Thm. 7.1 (h) is a system of members of a point of the class; we put $u_3 := u_1$. For the second we put $u_2 := u_K(z)$ with $z \in \mathrm{Pd}(\mathfrak{r}_*^\circ)$. By (J6°) (a) it lies in $M_1(\mathsf{b}; \kappa, c_h)$ with the strong cap, so $\|u_2\|_E^\sim \leq 1$ and $\|u_2\|_w^\sim \leq 1$; thus $a_w = 1$ and $a_E \leq 2$, which is case (ii) of Thm. 7.27 (b^C). The pair is (κ, c_h), since at $c_h/2$ the cap of E^h would not give $a_E \leq 2$. Then $\mathsf{D}_{23} = \|u_K(z) - \bar{u}_K\|^\sim \leq \|\cdot\|^{\mathrm{st},(\kappa)} \leq 0.91 \cdot 0.4^{K-k^\circ} \leq 0.364$, since $k^\circ \leq K - 1$.

In its part before the condition (S), Thm. 7.27 (c^C) gives the L^1 bound $\int |\rho[u_2; \ell] - \rho[u_3; \ell]|\, dV \leq \frac{1}{2} C_{\mathrm{vol}} \mathsf{D}_{23} I_1$ in the cases (i), (ii). In case (ii), with $\mathsf{D}_{23} \leq 0.364$, this gives $\int |\rho[u_K(z); \ell_K] - \rho[\bar{u}_K; \ell_K]| dV \leq 0.182\, C_{\mathrm{vol}} I_1$. Hence, on $\mathrm{Pd}(\mathfrak{r}_*^\circ)$,

$$\int |\rho[u_K(z); \ell_K]| dV \leq (1 + 0.182\, C_{\mathrm{vol}}) I_1. \tag{$*$}$$

(a) The moment $\langle \prod_t (\mathcal{O}_K(g_t) - a_t) \rangle$ is the coefficient of $z_1 \cdots z_Q$ in $e^{-\sum a_t z_t} Z_K(z)/Z_K(0)$, a squarefree coefficient. With $a_t = \gamma_t + e_t$ and (J7°) (v), it equals the same coefficient of $G(z) := \exp\{W(z) - \sum e_t z_t\} I[u_K(z)]/I_1$. The function G is holomorphic on $\mathrm{Pd}(\mathfrak{r}_*^\circ) \supset \mathrm{Pd}(r^\circ)$, by (J7°) (iv) and by Cor. 7.28 (F0), (F1) with its datum $B_0 = 0.91$. On $\mathrm{Pd}(r^\circ)$ we have $|W| \leq Q C_W$ by (J8°) (d), $|\sum e_t z_t| \leq Q e_h$, and $|I/I_1| \leq 1 + 0.182\, C_{\mathrm{vol}}$ by $(*)$. By Cauchy's estimate the coefficient is $\leq (r^\circ)^{-Q} e^{Q(C_W + e_h)} (1 + 0.182\, C_{\mathrm{vol}}) = G_1^Q (1 + 0.182\, C_{\mathrm{vol}})$.

With $q = (N/M)^3 \geq 1$,

$$1 + 0.182\, C_{\mathrm{vol}} \leq [1 + 0.728\, q e^{220q}] e^{(c_{\mathrm{vol}}(\mathsf{b}) - 225/M^3) N^3} \leq e^{220.728q} e^{(c_{\mathrm{vol}}(\mathsf{b}) - 225/M^3) N^3} \leq e^{c_{\mathrm{vol}}(\mathsf{b}) N^3},$$

where $c_{\mathrm{vol}}(\mathsf{b}) - 225/M^3 = \frac{3}{2} \mathsf{N}_Z p_0(\mathsf{b})^2 + 3 \log(384/\varepsilon(\mathsf{b})^3)$ and $225q = 225N^3/M^3$; here we use the formulas that define C_{vol} in item 1 of §7.5 and c_{vol} in (D1) of item D of §8.2. These steps rest on $1 + aqe^{220q} \leq e^{220q}(1 + aq) \leq e^{(220+a)q}$ at $a = 0.728$, and on $220.728 \leq 225$.

The moment is linear in each g_t because $\gamma_t, e_t, \hat{c}_K$ are linear in g_t. Scaling each g_t gives the factor $\prod_t \|g_t\|_\infty$. The moment is zero if any g_t vanishes at all barycentres. A sub-collection of the sources has the same $\check{\delta}$ and $\check{\rho}$.

(b) By (J7°) (iv) for $k = K$, $\rho_K^z(V)/Z_K(0) \equiv \exp\{\sum z_t \gamma_t + W(z)\} \rho[u_K(z); \ell_K](V)/I_1$ at every V. Both sides are holomorphic near 0, bounded uniformly in V (by the last part of Lem. 7.8, $\mathbb{T}_K$ being finite), and Borel in V; so ∂_S at 0 is a Cauchy integral, which commutes with $\int F(V) dV$ by Fubini's theorem. The coefficient is therefore that of $\exp\{W - \sum e_t z_t\} \int F \rho[u_K(z); \ell_K] dV/I_1$. This function is holomorphic on $\mathrm{Pd}(\mathfrak{r}_*^\circ)$, by the theorems of Fubini and Morera, and through $(*)$ it is bounded on $\mathrm{Pd}(r^\circ)$ by $e^{Q(C_W + e_h)}(1 + 0.182\, C_{\mathrm{vol}}) \sup |F|$; Cauchy's estimate then gives the bound in (b). Here $\int \rho_K^z dV = Z_K(z)$, by Step 0 of the proof of Thm. 8.2.

□

8.3. **Uniform bounds and limit procedures.** This section proves one theorem, Thm. 8.17, with the clauses (R0) to (R8). They provide a subtraction $c_K^{(s)}$; a bound on moments per unit of volume, valid for every number Q of sources; estimates in each cell; estimates of σ and of S_n^T that do not depend on K, on N or on where the sources lie on the torus; and the extension to every $\mathsf{b} > 0$ through a dilation. They also give the limits taken continuum first, volume first and jointly; the properties of every limit, namely invariance, reflection positivity, the distribution and the dilation identity; and the Hilbert space with Ω, $H \geq 0$ and the momenta. Finally they treat the block fields,

with locality, reflection positivity without a collar, the laws in the limit and mixed bounds. The proofs occupy Lems. 8.18 to 8.39.

Item 0.1: what is used of Sections 6.2 to 8.1. Of Thm. 6.7 we use the clause (g1). Of item 1 of §8.1 we use the table of letters, (O0) to (O13), with the entries (O4a) to (O13) in the form in which that item lists them; the radii $r(x)$, $s(x)$, $\varrho(x)$, $\check{r}(x)$; the entry (O14); the numbers $\mathrm{sep}^{[i]}$ and $N_1^{[j]}$, which the item sets at the data (b, n, d, R_f); and x_k in a run of length K. From the same item we take the paragraph on the sup distance of T_l, with the separation d, the radius R_f and the balls, and the definition in that item of separated tuples and functionals on them, together with the letters and names of Def. 8.1 that this definition states: $\mathfrak{S}_n(O)$, the type $(d; R)$, $\mathrm{Part}_{\geq 2}$, σ_S, the mirror, reflection positivity, distributions and regularity. Of Thm. 8.2 we use the data and the clauses (T2), (T3), (T6), (T7). Further, we use the following steps of the proof of Thm. 8.2.

- Step 0, both its statement and its argument.
- Step 1, which is Lem. 8.4; we use Lem. 8.4 (N2), and item (ii) of the proof of Lem. 8.4 (N3).
- Step 2, its part "The last integral": the data of Thm. 7.27, and the sentences there stating that items (c) and (e) of Step 2 of the proof of Thm. 5.63 hold for every $K \geq 1$.
- Step 5, its statement and its argument.
- Step 7, its parts "At finite K" and "All v".
- Step 8 (a), (b) and (d).
- Step 9, with its parts "The local construction", "On products", "Uniqueness on Q", "Gluing" and "The functional σ_S".
- Step 11 (0), and the first sentence of Step 11 (5).

Of §7.1 we use the part "Sources" of item 3 and Thm. 7.1 (a), (h), (i). We use Lem. 7.8 together with its sentences on $I[u]$. Of §7.3 we use (o3), (o6) and (o7) of item 2; the sentence on $\bar{u}_k$ in Prop. 7.12 (J6); the number $N_1(k)$ of (J8); and, from the proof of that proposition, Step 1 (A) and the property (P_k) of Step 3. Of item 1 of §7.5 we use q_N, Def. 7.22, the number C_{vol} and Def. 7.26, and of Thm. 7.27 its data (B). We use Thm. 7.27 (b^C) with its cases (ii) and (iii). Of Thm. 7.27 (c^C) we use the two assertions that come before (S): the lower bound on I_1, and the L^1 bound in the cases (i), (ii) and (iii). We use Cor. 7.28 (F0) and (F1). We use the entries (X16) and (X17) of (O13), item 1 of §5.5, in the form in which they enter Step P9 of the proof of Cor. 7.29 (R1) and the part "The last integral" of Step 2 of the proof of Thm. 8.2.

Item 0.2: what is used of §5.5, its items 2 and 3. Of item 2 we use the part "Lattice, cubes", with $\mathcal{D}$ and the count $(26e)^{j-1}$ that is proved there; the set $X^{\approx}$ of the part "At x"; the part "Group, fields" and the part "Energy"; and the clauses (ca), (ch), (cΞ). From the same item we use $\rho[u;\ell]$ and Ξ_w, the part "The step" and the part "Runs". Of item 3 we use (9a), (9b) and (9d), which are clauses of Thm. 5.15. We use Cor. 5.65 (b) and (c).

Item 0.3. Wilson-measure estimates. We use the model of §1.1, the board and its reflected copies from Def. A.4, coordinate reflection positivity from Cor. A.3, and the chessboard seminorm and product estimate from Def. A.10 and Prop. A.11. Lem. D.1 gives the reflection Hilbert space, and Def. B.1 describes the Gibbs laws and the torus-limit laws.

Item 0.4: what is used of §8.2, namely of Prop. 8.11. We use Prop. 8.11 with the single numeral $\mathrm{sep}^\circ := 8\log(8L^{1/2})$, for any number $Q \geq 1$ of sources and for every $K \geq 1$. Its data are those of Def. 8.6: the functions g_t are real and bounded with $\|g_t\|_\infty \leq 1$, and the sets $\{g_t \neq 0\}$ lie pairwise at sup distance $\geq \check{\delta}$, each of them within $\check{\rho}$ of a point. We use four of its clauses, under the names (P1.0) to (P1.3).

(P1.0) This is (G°): the number $j^\circ(\mathsf{b}, \check{\delta})$ exists, and $k^\circ := \max\{K - j^\circ, 0\} \leq K - 1$; further $\mathfrak{r}_k^\circ \leq \mathfrak{r}_{k+1}^\circ \leq L^{1/2}\mathfrak{r}_k^\circ$, and $\mathfrak{r}_*^\circ := \mathfrak{r}_{k^\circ}^\circ \geq r^\circ$; finally $\kappa_k^\circ \in [\kappa, \kappa + 1/(4(L-1))]$.

(P1.1) This is (J5°) (c) at the standard data together with (J6°) (a) to (d), on the representatives $u_k(z)$, $0 \leq k \leq K$:

- for $k \le k^\circ$, $u_k = u_k^{\mathrm{pol}}$ is a family on $\mathrm{Pd}(\mathfrak{r}_k^\circ)$ in $M_{1.91}(x_k;\kappa_k^\circ,c_h)$, at distance at most 0.91 from $\bar u_k$ in $\|\cdot\|^{\mathrm{st},(\kappa_k^\circ)}$;
- for $k^\circ < k \le K$, $u_k(z) = F_{\ell_{k-1}}(u_{k-1}(z))$ on $\mathrm{Pd}(\mathfrak{r}_*^\circ)$; this family lies in $M_1(x_k;\kappa,c_h)$, satisfies the strong cap, and stays within $0.91\cdot 0.4^{k-k^\circ}$ of $\bar u_k$ in $\|\cdot\|^{\mathrm{st},(\kappa)}$;
- for all k, $u_k(0) = \bar u_k$, and $\partial_S u_k(0) = \delta_S(k)$ for every S, where $\delta_S(k)$ are the jets of the honest family of Thm. 7.1 (h); and (P_k) holds.

(P1.2) This clause collects (J7°) (i), (iv) and (v); it is the identity at a level k, which holds pointwise in V. Thus, for $0 \le k \le K$, at each field V of $\mathbb{T}_k$ and as germs modulo z_t^2:

(i)
$$\rho_k^z(V) \equiv \exp\{\sum_{j\le k} V_j(0) + \sum_{k'<k}\sum_Y [v_{k'}(Y;u_{k'}(z)) - v_{k'}(Y;\bar u_{k'})]\}\,\rho[u_k(z);\ell_k](V);$$

(iv)
$$\rho_k^z(V) \equiv \exp\{\sum_{j\le k} V_j(0) + \sum_t z_t\gamma_t^{(k)} + W^{(k)}(z)\}\,\rho[u_k(z);\ell_k](V),$$
where $\gamma_t^{(k)} := \sum_{k'<k}\partial_t V_{k'+1}(0)$ and
$$W^{(k)} := \sum_{k'<\min(k,k^\circ)} W_{k'} + \sum_{k^\circ\le k'<k}[w_{k'} - \sum_t z_t\partial_t w_{k'}(0)],$$
with $W_{k'}$ and $w_{k'}$ as in (P1.3);

(v)
$$Z_K(z)/Z_K(0) \equiv \exp\{\sum_t z_t\gamma_t^{(K)} + W^{(K)}(z)\}\,I[u_K(z)]/I[\bar u_K].$$

Here $V_0 := 0$, as in Thm. 7.1 (h); hence $\gamma_t^{(k)}$ is the number $\partial_t\sum_{j\le k}V_j(0)$ of Lem. 8.21 (c).

(P1.3) This is (J8°) (a°), (b) and (d), level by level, on $\mathrm{Pd}(\mathfrak{r}_*^\circ)$, where $j = K-k$ is the depth:

- for $k < k^\circ$, the polynomial $W_k(z) = \sum_Y\sum_{|S|\ge 2} z^S\partial_S v_k(Y;u_k)(0)$ obeys
$$|W_k| \le Q\,N_1^{[j]}C_g c_a(L^j\mathsf{b})\,e^{1/(4L)}\exp\{-(\kappa - 1/(4L))(\mathrm{sep}^{[j-1]}+1)\};$$
- for $k^\circ \le k < K$, the function $w_k(z) = \sum_Y[v_k(Y;u_k(z)) - v_k(Y;\bar u_k)]$ obeys $|w_k| \le 3Q\,N_1^{[j]}C_g c_a(L^j\mathsf{b})$ and $|\partial_t w_k(0)| \le 3N_1^{[j]}C_g c_a(L^j\mathsf{b})/\mathfrak{r}_*^\circ$;

(d) $|W^{(K)}| \le Q\,C_W$.

We use Cor. 8.14 (a) and (b). We use Prop. 8.11 (J7°) (vii) and (J6°) (e). For $K < j^\circ$ we use (L0), (J5°) (c) at $k = k_1 = 0$, and (J6°) at $k^\circ = 0$. Prop. 8.11 (lin) enters as a definition only: for $0 \le k \le K$, $\delta[f](k)$ is the derivative d/dz at 0 of the honest run with the single source f, and its h-member $\delta^h[f](k)$ is the letter that enters the display of (R0). The numbers of (D1), item D of §8.2, that this section uses are sep°, $\mathrm{sep}^{[i]}$, j°, $\mathfrak{r}^\circ$, r°, $N_1^{[j]}$, C_W, e_h, G_1 and c_{vol}, and item 2 below defines each of them exactly as (D1) does.

Item 0.5: local letters. The letter s is the depth, and $s(x) = 8r(x)$; in k_s it stands for the noise of the kernel, in the sense of "The step", item 2 of §5.5. The letter θ is a reflection; with an exponent, it is the numeral of Thm. 8.2 (T2), (T3). The letter z denotes the variables of the sources; it is also the integer of a dilation, a bound variable, with $z(\mathsf{b})$ as in the setting (S7) of §1; and $z(t)$ normalises the kernel.

Item 1: data. The group is $G = \mathrm{SU}(2)$, and the dimension is 3. The table (O0) to (O13) and its floors are exactly those of item (1) of §6.1, which is item 1 of §8.1. The blocking factor L is odd, and $M = L^{m'}$.

We recall the data of the setting (S2) of §1: a label $\mathsf{b} \ge x_s$; a side $l = N = 2L^{m_l}$ with $m_l \ge m'$, subject to (Tor), that is, $N \ge 2M\,s(L\mathsf{b})$; an integer $K \ge 1$; and $\beta = L^K\mathsf{b}$, $a_K = L^{-K}$. We write $\mu_{K,N,\beta}$ for Wilson's measure on $a_K\mathbb{Z}^3/N\mathbb{Z}^3$ and set $\mathcal{O}_K(f) = a_K^{-1}\sum_p f(x(p))\,a_p(U)$, where

$a_p = 1 - \frac{1}{2}\operatorname{tr}\partial U(p)$ and $x(p)$ is the barycentre of p. The joint cumulant is S_n^T, and $\sigma_{K,N,\beta}(X) = \langle\langle \prod_i (\mathcal{O}_K f_i - \langle \mathcal{O}_K f_i\rangle)\rangle\rangle$.

The level k of the run is the lattice $L^{-(K-k)}\mathbb{Z}^3/N\mathbb{Z}^3$, whose depth is $K-k$. The point 0 is a site of every level, and the last lattice has spacing 1. The fields $(V_0 = U, V_1, \dots, V_K)$ carry the joint law of Step 11 (0) of the proof of Thm. 8.2. Without an argument, V_k is the field drawn at the level k, with $V_0 = U$, whereas $V_j(z)$, $V_j(0)$ and the value $V_0 := 0$ are numbers of Thm. 7.1 (h), not these fields; and V_0^+ is the number of Thm. 6.7 (g1). All distances are sup distances, and the support of a bounded function g means the set $\{g \neq 0\}$.

Boards. For $-m_l \le i \le K-1$ and $o \in a_K\mathbb{Z}^3$, the board $B_i(o)$ has its walls at $o_\mu + L^{-i}(\mathbb{Z} + \frac{1}{2})$ and its cells of side L^{-i}; we put $B_i := B_i(0)$. In the letters with which §A.2 writes a board, $B_i(o) = (a_K,\ \lambda = L^{K-i},\ \mathsf{n}_\mu = 2L^{m_l+i},\ o)$.

No hypothesis enters beyond the data and the table.

Item 2: numbers. The numbers are fixed in this order: first the table, then b, then (O14). No floor depends on the numbers below, and all of these numbers are locally bounded in b. None of them depends on N, on K, on the number of the sources or on their places, except as follows: $C_n^{(2)}$ and its $\check\delta_n$ depend on n, the number of the entries, but on none of N, K and the places; $\mathfrak{r}_k^\circ = \mathfrak{r}^\circ(x_k)$ depends on K and on k, but only through x_k; and the subtractions $c_K^{(s)}$ and $\hat c_K$ at the end of the item depend on K and on N.

We set $\mathrm{sep}^\circ = 8\log(8L^{1/2})$ and $\mathrm{sep}^{[i]} = \lfloor \check\delta\, L^i/M\rfloor - 2\,\check r(L^i\mathsf{b}) - 2$, and we let $j^\circ(\mathsf{b},\check\delta)$ be the least $j \ge 1$ with $\mathrm{sep}^{[j-1]} \ge \mathrm{sep}^\circ$. Next, $\mathfrak{r}^\circ(x) = 3.5/\mathrm{amp}(Lx)$, $\mathfrak{r}_k^\circ = \mathfrak{r}^\circ(x_k)$, $r^\circ = \mathfrak{r}^\circ(L^{j^\circ}\mathsf{b})$ and $r_s^\circ = \mathfrak{r}^\circ(L^{\max(s,j^\circ)}\mathsf{b})$, so that $r_0^\circ = r^\circ$. We put $N_1^{[j]} = (2\,\check\rho\, L^{j-1}/M + 2\,\check r(L^{j-1}\mathsf{b}) + 3)^3$ and

$$C_W(\mathsf{b},\check\delta,\check\rho) = \sum_{j>j^\circ} 2e\, N_1^{[j]} C_g c_a(L^j\mathsf{b})\, e^{-(\kappa-1)(\mathrm{sep}^{[j-1]}-1)} + \sum_{1\le j\le j^\circ} 6N_1^{[j]} C_g c_a(L^j\mathsf{b});$$

since, for the terms with $j > j^\circ$, we have $\mathrm{sep}^{[j-1]} \ge 0$ and the inequalities $e^{1/(4L)} \le 2e$ and $(\kappa - 1/(4L))(\mathrm{sep}^{[j-1]}+1) \ge (\kappa-1)(\mathrm{sep}^{[j-1]}-1)$, this number dominates the bounds of (P1.3) term by term.

Further, $e_h = e_h^{(0)} = 0.45\, N_1^{[1]} c_h(\mathsf{b})$, and $e_h^{(s)} = 1.11\, N_1^{[s+1]} c_h(L^s\mathsf{b})$ for $s \ge 1$. We set $G_1(\mathsf{b},\check\delta,\check\rho) = e^{C_W + e_h}/r^\circ$ and $G_{1,s} = e^{C_W + e_h^{(s)}}/r_s^\circ$. The letter c_{vol} stands for $c_{\mathrm{vol}}(\mathsf{b}) = 225/M^3 + \frac{3}{2}\mathsf{N}_Z p_0(\mathsf{b})^2 + 3\log(384/\varepsilon(\mathsf{b})^3)$; hence $1 + 0.182\, C_{\mathrm{vol}} \le 1 + C_{\mathrm{vol}} \le 2C_{\mathrm{vol}} \le e^{c_{\mathrm{vol}}N^3}$. For $s \ge 1$ we set $c_{\mathrm{vol}}^{(s)}(\mathsf{b}) = L^{3s}(c_{\mathrm{vol}}(L^s\mathsf{b}) + 200/M^3)$, and $c_{\mathrm{vol}}^{(0)} = c_{\mathrm{vol}}$.

We set $B_0(\mathsf{b},\check\delta) = 6L^{j^\circ}(\check\delta\, L^{j^\circ}/2 + 2)^3$; this is a number, whereas the board B_0 carries no argument. Then $G(\mathsf{b},\check\delta) = \max\{G_1(\mathsf{b},\check\delta,\check\delta/4)\, e^{c_{\mathrm{vol}}(\mathsf{b})},\, B_0(\mathsf{b},\check\delta)\}$. With $\check\delta_n := \min\{d, 1/(4n)\}$ we define $C_n^{(2)}(\mathsf{b}; d, R_f) := 1.45^{n+1}\, n!\, (4R_f/\check\delta_n + 2)^{3n}\, G(\mathsf{b},\check\delta_n)^n$.

Subtractions. For $0 \le s \le K-1$ we set $c_K^{(s)}(f) := \frac{d}{dz}\log \rho_{K-s}^{[zf]}(1)\big|_{z=0}$, where $\rho_k^{[zf]}$ is the density drawn k times from Wilson's weight multiplied by $e^{z\mathcal{O}_K(f)}$, taken at the trivial field of $\mathbb{T}_k$. We write $\hat c_K := c_K^{(0)}$.

Item 3: the statement.

Theorem 8.17 (Uniform bounds, limits in three orders, reconstruction and block fields)**.** In every clause the objects, and the letters that denote them, are those of item 1, and the numbers are those of item 2. The clauses (R0) to (R3) and (R8) assume the data of item 1, with $\mathsf{b} \ge x_s$, $m_l \ge m'$ and (Tor); the clauses (R4) to (R7) carry their own data, every $\mathsf{b} > 0$ and, in (R4), every side $l = 2L^m$. Under these assumptions the following clauses hold.

(R0) *Subtraction.*

- The number $c_K^{(s)}(f)$ is the mean of $\mathcal{O}_K(f)$ under Wilson's measure weighted by the kernel density at $V_{K-s} = 1$; it is real for real f, and it is linear in f.

- $c_K^{(s)}(f\circ R)=c_K^{(s)}(f)$ for every map $R: y_\mu\mapsto c-y_\mu$ with $c\in L^{-s}\mathbb{Z}$, and for every translation by a vector of $L^{-s}\mathbb{Z}^3$. For $\hat c_K$ the invariance holds at every integer c, even or odd; the mirrors with odd c are those in the walls of B_0.
- For the data of the setting (S2) and $0\le s\le K-1$,

$$c_K^{(s)}(f)=\frac{d}{dz}\Big|_{z=0}\sum_{1\le j\le K-s}V_j(z)+\sum_X \delta^h[f](K-s)(X,1),$$

where X runs over $\mathcal{D}$ of the level $K-s$, and $V_j(z)$ are the numbers of Thm. 7.1 (h) along the run with the single source f and the variable z. If f is of the type of one source of (R1), then, at any $\check\delta>0$, the modulus of the last sum, multiplied by r_s°, is $\le e_h^{(s)}\|f\|_\infty$.

(R1) *Per unit of volume.* Let $K\ge 1$, let $Q\ge 1$ be arbitrary, and let g_t be real bounded functions on $T_N=\mathbb{R}^3/N\mathbb{Z}^3$ whose supports lie pairwise at distance $\ge\check\delta$, each of them inside a ball of radius $\check\rho$. Then

$$|\langle\prod_t(\mathcal{O}_Kg_t-\hat c_Kg_t)\rangle|\le G_1^Q\,(1+0.182\,C_{\rm vol}(N,\mathsf{b}))\prod_t\|g_t\|_\infty.$$

Here $C_{\rm vol}$ enters with the exponent 1, whatever Q is.

(R2) *Cells.* Let $B_i(o)$ be a board with $-m_l\le i\le 0$ and $o\in\mathbb{Z}^3$, let $h=L^{-i}$, and take the data of (R1); suppose in addition that every support lies inside one open cell, at distance $\ge\check\delta/2$ from the walls, and let q cells be occupied. Then the moment of (R1) is $\le G_1^Q\,e^{q\,c_{\rm vol}(\mathsf{b})h^3}\prod_t\|g_t\|_\infty$.

(i) On B_0, whose walls are $\mathbb{Z}+\frac12$ and where $h=1$, the factor is $e^{q\,c_{\rm vol}(\mathsf{b})}$. At $Q=1$ this gives $|\langle\mathcal{O}_Kg\rangle-\hat c_Kg|\le G_1e^{c_{\rm vol}}\|g\|_\infty$.

(ii) *Placed anywhere.* Let $n\ge1$ sources of the type $(d;R_f)$ be placed anywhere, and let $h=L^{-i}>n(2R_f+d+1)$ and $N\ge 2h$. Then some $o\in\mathbb{Z}^3$ meets the condition above at $\check\delta=d$, $\check\rho=R_f$; hence the moment is $\le G_1(\mathsf{b},d,R_f)^n\,e^{n\,c_{\rm vol}h^3}\prod_i\|f_i\|_\infty$.

(R3) *Uniform bounds.* Let $n\ge2$, $d>0$ and $R_f>0$, and let f_i be real bounded functions whose supports lie pairwise at distance $\ge d$, each of them inside a ball of radius R_f, wherever these balls lie. Recall from item 2 that $\check\delta_n=\min\{d,1/(4n)\}$. Then, for all $K\ge1$ and all N of the data,

$$|\sigma_{K,N,\beta}(f)|\le(4R_f/\check\delta_n+2)^{3n}\,(2G(\mathsf{b},\check\delta_n))^n\prod_i\|f_i\|_\infty,$$

and $|S_n^T(f)|\le C_n^{(2)}\prod_i\|f_i\|_\infty$, where $C_n^{(2)}(\mathsf{b};d,R_f)=1.45^{n+1}\,n!\,(4R_f/\check\delta_n+2)^{3n}\,G(\mathsf{b},\check\delta_n)^n$ is the number of item 2. The limits of Thm. 8.2 (T3) obey the same bounds.

(R4) *Every $\mathsf{b}>0$, every side $l=2L^m$.* Let $n\ge2$, $d>0$ and $R_f>0$, and let f_i be real bounded functions on $T_l=\mathbb{R}^3/l\mathbb{Z}^3$ of the type $(d;R_f)$, as in (R3): their supports lie pairwise at distance $\ge d$, each of them inside a ball of radius R_f, wherever these balls lie. Let $z\in\mathbb{Z}$ be such that $L^z\mathsf{b}\ge x_s$, $K-z\ge1$, $m+z\ge m'$ and $L^zl\ge 2M\,s(L^{z+1}\mathsf{b})$. Then

$$|S^T_{n;K,l,L^K\mathsf{b}}(f)|\le L^{zn}\,C_n^{(2)}(L^z\mathsf{b};L^zd,L^zR_f)\prod_i\|f_i\|_\infty,$$

and the same bound holds for σ.

(R5) *Limits, for every $\mathsf{b}>0$.* Let the sides $l_j=2L^{m_j}$ tend to ∞. In (a) to (d), the cumulants and their limits are evaluated at tuples f whose entries $f_1,\dots,f_n$ are continuous functions on $\mathbb{R}^3$ with compact and pairwise disjoint supports, periodised to T_l as soon as l exceeds twice the diameter of the union of the supports.

(a) *Continuum first.* At each side, $\lim_K$ exists as $K\to\infty$ through all integers K, by Thm. 8.2 (T3), (T6). From every sequence of sides one can extract a subsequence on

which $S_n^{(l_j,\mathsf{b})}(f)$ converges simultaneously for all $n \geq 2$ and all such tuples. The same is true for labels $\mathsf{b}_j \to \mathsf{b} > 0$.

(b) *Volume first.*
- One subsequence of sides suffices for all K, and along it $\mu_{K,l_j,L^K\mathsf{b}}$ converges weakly to a state ν_K on $G^{\text{bonds of } a_K\mathbb{Z}^3}$.
- ν_K is a Gibbs state of Wilson's action, and it is invariant under $a_K\mathbb{Z}^3$ and under W_3; so it lies in $\mathbb{G}_K(\mathsf{b})$, the set of the setting (S6) of §1.
- ν_K is reflection positive both with respect to planes of sites and with respect to the planes that lie midway between sites.
- The cumulants of ν_K are the limits as $j \to \infty$, along the subsequence of the first bullet, of the cumulants $S^T_{n;K,l_j,L^K\mathsf{b}}(f)$ of $\mu_{K,l_j,L^K\mathsf{b}}$, and they obey (R3) and (R4); these cumulants of ν_K then converge along a subsequence of the cutoffs K.

(c) *Jointly.* Every sequence (K_j, l_j) in which both entries tend to ∞ has a convergent subsequence; the same is true for labels $\mathsf{b}_j \to \mathsf{b} > 0$.

(d) If $K_j/l_j^3 \to \infty$, the limit points of (c) coincide with those of (a) along the same sides l_j and the same fixed or convergent labels.

(R6) *Properties of every limit.* In (R6) and (R7), the definition in item 1 of §8.1 is taken with $\mathbb{R}^3$ in the place of T_l: a mirror is (μ, c) with c real, and its open side is $\{y_\mu > c\}$. Every limit $S = (S_n)_{n\geq 2}$ of (R5) has the following properties.

(i) S_n is real and symmetric, and it is n-linear on every product of n spaces in which each factor is the space of the continuous functions that vanish off one compact set, the n sets being pairwise disjoint.

(ii) S obeys (R3) and (R4), uniformly with respect to where the supports of the entries lie. For labels $\mathsf{b}_j \to \mathsf{b}$ the constant is the supremum over $U(\mathsf{b})$, at $z = z(\mathsf{b})$, with $U(\mathsf{b})$ and $z(\mathsf{b})$ as in the setting (S7) of §1.

(iii) S is invariant under all translations of $\mathbb{R}^3$ and under W_3.

(iv) σ_S is reflection positive in the sense of eq. C.viii at every coordinate mirror (μ, c).

(v) On the open set of the points of $(\mathbb{R}^3)^n$ whose entries are pairwise different there is exactly one distribution whose value at $f_1 \otimes \cdots \otimes f_n$ equals $S_n(f_1, \ldots, f_n)$ whenever $f_1, \ldots, f_n$ are smooth functions with pairwise disjoint compact supports, and on every compact subset of that open set this distribution has order $\leq 3n + 1$. Moreover σ_S is regular.

(vi) The identity $S_n^{\mathsf{b}}(f) = L^{zn} S_n^{L^z\mathsf{b}}(f^{[z]})$ holds along matched subsequences, that is, if the limit for the label b is taken through the sides l_j and the cutoffs K, and the limit for the label $L^z\mathsf{b}$ through the sides $L^z l_j$ and the cutoffs $K - z$, with the same order of the limits in both cases; when the labels move, it holds for each member of the sequence, by Step 5 of the proof of Thm. 8.2.

(vii) The group $\{R \in \mathrm{O}(3) : \sigma_S(RX) = \sigma_S(X) \text{ for all } X\}$ is closed, and it is W_3 or $\mathrm{O}(3)$.

(R7) *Reconstruction, in each coordinate direction μ.* Let D consist of the finite linear combinations, with complex coefficients, of the tuples of $\mathfrak{S}(\mathbb{E}_+)$, where $\mathbb{E}_+ = \{y_\mu > 0\}$, and equip it with $\langle F, F'\rangle = \sum \bar{\zeta}_i \zeta'_k \, \sigma_S(\theta X_i \cdot X'_k)$. The completion $\mathcal{H}(S)$ of D/null is separable, and $\Omega =$ [empty tuple] has $\|\Omega\| = 1$. The maps $\mathcal{T}_t[F] = [\tau_{te_\mu} F]$ form a strongly continuous semigroup of self-adjoint contractions e^{-tH}, with $H \geq 0$ and $H\Omega = 0$. The translations of the mirror plane act by strongly continuous unitary operators $U(v)$, with $U(v)\Omega = \Omega$ for all v in that plane; their momenta P_1, P_2 satisfy $P_j\Omega = 0$, and H, P_1, P_2 commute strongly; the joint spectrum lies in $[0, \infty) \times \mathbb{R}^2$, and W_2 acts by a unitary representation, where $W_2 := \{w \in W_3 : \ we_\mu = e_\mu\}$. The class $[X]$ is defined also for tuples whose entries are continuous with compact disjoint supports in $\mathbb{E}_+$, with $\langle [X], [Y]\rangle = \sigma_S(\theta X \cdot Y)$, and $\langle \Omega, [f]\rangle = 0$.

(R8) *Block fields.* In the setting of item 1, with $\mathsf{b} \geq x_s$, let $s \geq 0$ be the depth and let $K \geq s+1$.

(a) *Locality.* Let $O_1, \dots, O_m$ be pairwise disjoint open sets of T_N bounded by planes $y_\mu \in L^{-s}(\mathbb{Z}+\frac{1}{2})$, and let F_i be a bounded Borel function of V_{K-s} that depends only on the bonds with both ends in O_i. Then $E[F_i|U]$ is bounded and Borel, and it depends on U only through the bonds with both ends in O_i. Moreover $E[\prod_i F_i|U] = \prod_i E[F_i|U]$, and $E[RF_i|U] = R\,E[F_i|U]$ for the maps R of (R0) at depth s.

(b) *Reflection positivity without a collar.* Let $c \in L^{-s}(\mathbb{Z}+\frac{1}{2})$ and $T_+ = \{c < y_\mu < c+N/2\}$, and let F be a bounded Borel function of the values of V_{K-s} on the bonds that have both ends in T_+. Then $E[\overline{\theta F}\, F] \geq 0$.

(c) *Limits.* Along a further subsequence of sides, and for every $s \geq 0$, the laws $\mu^{\infty}_{[s]}$ of Cor. 5.65 (b) converge, locally in total variation, to a law of the fields on $L^{-s}\mathbb{Z}^3$. This limit has the following properties.

- It is gauge invariant.
- It is invariant under $L^{-s}\mathbb{Z}^3$ and under $y_\mu \mapsto c - y_\mu$, $c \in L^{-s}\mathbb{Z}$.
- For every finite set B of bonds of $L^{-s}\mathbb{Z}^3$, its marginal on B has a density, and that density lies between $(z^{-1}e^{-2t})^{\#B}$ and $z^{-\#B}$, where $t = aL^{s+1}\mathsf{b}$ and $z = z(t)$.
- It is consistent under the drawn step with the noise $aL^s\mathsf{b}$.
- It is reflection positive as in (b), with $T_+ = \{y_\mu > c\}$.

From every sequence (K_j, N_j) with $N_j \to \infty$ one can extract a subsequence along which the same holds for all $s \leq \liminf K_j - 1$.

For $\mathsf{b} < x_s$, the dilation by L^z gives the lattices $L^{-(s+z)}\mathbb{Z}^3$ only.

(d) *Mixed bounds.*

(i) *Without cells.* Let F be a bounded Borel function of V_{K-s}, and take the data of (R1). Then

$$|\langle F(V_{K-s}) \prod_t (\mathcal{O}_K g_t - c_K^{(s)} g_t)\rangle| \leq \sup|F|\; G_{1,s}^{Q}\, e^{c_{\mathrm{vol}}^{(s)} N^3} \prod_t \|g_t\|_\infty .$$

At $s = 0$ the factor $1 + 0.182\, C_{\mathrm{vol}}$ replaces the exponential.

(ii) *With cells.* Let $B_i(o)$ be a board with $-m_l \leq i \leq s$ and $o \in L^{-s}\mathbb{Z}^3$, let $h = L^{-i}$, and take sources as in (R2). Let the factors F_{cell} be functions of V_{K-s} on those bonds that have both ends in the open cell, and let q cells be occupied, each by a source or by a factor $\neq 1$. Then the moment with these factors is bounded by $\prod \sup|F_{\mathrm{cell}}|\; G_{1,s}^{Q}\, e^{q\, c_{\mathrm{vol}}^{(s)} h^3} \prod_t \|g_t\|_\infty$. At $s = 0$ an inner coarse bond exists only for $h \geq L$, while at $s \geq 1$ there is one already in unit cells.

■□

The proof first estimates the subtracted moments at fixed volume. Chessboard bounds then replace the volume factor by a product over occupied cells. The resulting uniform bounds give compactness of the correlations and allow reflection positivity to pass to the limits.

Statements and proofs of other sections, and how they are used. The statements named below are used as stated, and any change in their use is given in the paragraph that names them. The data of Thm. 8.2 and Step 0 of its proof, for Q sources: $Z_K(z)$ is entire; the weight is that of Thm. 7.1 at $\nu_p = \mathsf{b}^{-1}\sum_t z_t g_t(x(p))$; since the supports are disjoint, $|\nu|_\infty \leq \max_t |z_t|\, \|g_t\|_\infty/\mathsf{b}$; and $\int \rho_k^z dV = Z_K(z)$ at every level. The first three of these assertions are Lem. 8.18; the fourth is the last paragraph of Step 0 of the proof of Thm. 8.2.

The bound (R3) replaces Thm. 8.2 (T1): where (T1) has $C_n = \mathrm{Ser}_1(\mathsf{b}, n, d, R_f) + \log 2\,(\tau_7/\mathfrak{r}(L^{j^*}\mathsf{b}))^n$, $\tau_7 = \max\{1, 0.91\, C_{\mathrm{vol}}(N, \mathsf{b})\}$, the bound (R3) has the constant $C_n^{(2)}$, for all N and all $K \geq 1$. Of Step 2 of the proof of Thm. 8.2 we use only the verification of the data (B) of Thm. 7.27 at $x = \mathsf{b}$ for every $K \geq 1$, namely the part "The last integral" with its sentences on every $K \geq 1$; no line of this verification depends on the sources.

Thm. 8.2 (T2). In the constant C'_n of (T2), only $3C_{\mathrm{vol}}(N,\mathsf{b})C_{\mathsf{b}}(1.15+0.46C_5^{\#}/K_5)(4C_{\mathrm{vol}}(N,\mathsf{b})/\mathfrak{r}(L^{j^*}\mathsf{b}))^n$ depends on N, so that $\log C'_n \le (n+1)c_{\mathrm{vol}}N^3+O(1)$; and, as (T2) states, K_T does not depend on N.

Thm. 8.2 (T3) and (T6). At a fixed side, the limit as $K\to\infty$ through all integers K exists as those clauses state, also for continuous entries. Moreover this limit obeys (R3) and (R4), being a limit of numbers that obey these bounds.

Step 7 of the proof of Thm. 8.2. Its part "At finite K" gives $\mathcal{O}_K(f)(rU)=\mathcal{O}_K(f\circ r)(U)$ for every bounded f. Its part "All v" holds with $C_n^{(2)}$ at $(d/2;R_f+d/4)$. Here $C_n^{(2)}$ at $(d;R)$ denotes the constant of (R6) (ii), namely $C_n^{(2)}(\mathsf{b};d,R)$ at a fixed $\mathsf{b}\ge x_s$, $L^{zn}C_n^{(2)}(L^z\mathsf{b};L^zd,L^zR)$ at a fixed $0<\mathsf{b}<x_s$, and the supremum over $U(\mathsf{b})$ at $z(\mathsf{b})$ for moving labels. This constant is free of K and of l, and so the part "All v" applies to all three orders.

Step 8 (a), (b), (d) of the proof of Thm. 8.2. Part (a) is taken at finite (K,N) and $c=0$; it enters the proof of (R6) (iv) in each of the three orders, since every compact support in $\{y_\mu>0\}$ is contained in $\{0<y_\mu<l/2\}$ once l is large; and from (R6) (iii) we then obtain (d).

Step 9 of the proof of Thm. 8.2. In its part "The local construction" the Fourier series is taken with the period 4δ about ξ_i, in the place of the period l. The nearest copy of $\operatorname{supp}F$ begins at the distance 3δ, and χ_i ends at $3\delta/2$; hence $F=(\chi\times\cdots\times\chi)F^{\mathrm{per}}$. The constant $c(n,l)$ becomes $c(n,4\delta)$, and $C_n(\delta)$ becomes $C_n^{(2)}$ at $(\delta;3\delta/2)$, the constant of (R6) (ii). The parts "On products", "Uniqueness on Q" and "Gluing" are taken word for word.

Step 5 of the proof of Thm. 8.2: the identity $S^T_{n;K,l,\beta}(f)=L^{zn}S^T_{n;K-z,L^zl,\beta}(f^{[z]})$ carries (R3) to every $\mathsf{b}>0$. The side drops out, since $C_n^{(2)}$ depends on no side, and the conditions on z are those of Thm. 8.2 (T6).

Thm. 8.2 (T7), Cor. 5.65 (b), (c), and Step 11 (0), (5) of the proof of Thm. 8.2. The clauses (b) and (c) of that corollary apply at each N. The rate $C_{\mathrm{vol}}(NL^s,\dots)\theta^{K-s}$ is lost as N grows, and local densities that are equi-Lipschitz take its place; they come from the kernel alone (Lem. 8.39). Step 11 (0) gives the joint law and the identity $\int F\rho^z_{K-s}dV/Z_K(z)=\langle Fe^{\sum_t z_t\mathcal{O}_K(g_t)}\rangle/\langle e^{\sum_t z_t\mathcal{O}_K(g_t)}\rangle$, and the first sentence of Step 11 (5) gives the data (B) at $(NL^s,L^s\mathsf{b})$.

The part "Sources" of item 3 of §7.1 and Thm. 7.1 (a), (h), (i). Clause (h) states that $\rho^z_k=\exp\{\sum_{j\le k}V_j(z)\}\rho[u_k(z);\ell_k]$ at every field of $\mathbb{T}_k$; it states also that $\bar u_k$ is a system of members and that $\rho[\bar u_k]>0$. In (i) the numerals are free of n.

Thm. 6.7 (g1), the convention $V_0:=0$ of Thm. 7.1 (h), and (X16), (X17). By (g1), $\mathsf{V}'(u)=V_0^+ +\sum_Y v(Y;u)$; so $V_{k'+1}(0)=V_0^+ +\sum_Y v_{k'}(Y;\bar u_{k'})$, and together with $V_0:=0$ this is how the exponent of (P1.2) (i) is computed. By (X16) and (X17), Step P9 of the proof of Cor. 7.29 (R1) has $\varepsilon\le 1/40$ and $c_a+c_h\le 1/32$ at $x=L^s\mathsf{b}$, $\mathsf{b}\ge x_s$.

Of §7.3, (o3), (o6), (o7), *the sentence of Prop. 7.12 (J6) on $\bar u_k$, the number $N_1(k)$, Step 1 (A) and the property (P_k) of Step 3.* By (o6), the h-norm is a supremum over X and the field, in units $c_h(x)e^{-\check\kappa(\#X-1)}$. By (A), at most $N_1(k-1)$ cubes of $\mathbb{T}_k$ lie in $S_t(k)^{+\check r(x_k)}$. By (P_k), $u(z)(X)$ does not depend on z_t if $X^{+\check r(x_k)}$ misses $S_t(k)$.

In the place of Prop. 7.12 (J5) to (J8) we use Prop. 8.11, for any number $Q\ge 1$ of sources and clause by clause, as (P1.0) to (P1.3), at $\mathrm{sep}^\circ=8\log(8L^{1/2})$. Its radii are free of Q; it provides representatives at all levels; it gives the congruence at a level k, pointwise in V; and it treats the vacuum sums per source, level by level.

Thm. 7.27 with its data (B), (b^C) *(ii), (iii), and the two assertions of* (c^C) *that come before* (S). The clause that carries the volume is this: in the cases (i) and (ii), $\int|\rho[u_2;\ell]-\rho[u_3;\ell]|dV\le\frac12 C_{\mathrm{vol}}\mathsf{D}_{23}I_1$, and in the case (iii) the same holds with $C_{\mathrm{vol}}e^{219(A-1)q_N}$. It is used at $u_3=u_1=\bar u_k$, $u_2=u_k(z)$, and at $x=\mathsf{b}$ as well as at $x=L^s\mathsf{b}$ on $\mathbb{Z}^3/NL^s\mathbb{Z}^3$. The case (ii) of ($\mathrm{b}^C$) is applied with the rate κ and the cap c_h, the case (iii) with the rate κ°_k and the cap c_h.

Defs. 7.22 and 7.26, Cor. 7.28 (F0), (F1) and Lem. 7.8. Besides $I[u]$, of which these statements speak, we use the functional $\Phi_F[u] = \int F\rho[u;\ell]dV$, with F bounded Borel. This functional is holomorphic along a family by the proof of Cor. 7.28 (F1), which is Step P4 of the proof of that corollary: pointwise holomorphy, together with the sup bound of Thm. 7.27 (b^C).

The part "Energy" of item 2 of §5.5, the clauses (ca), (ch), (cΞ), *and* $\rho[u;\ell]$, Ξ_w, *at the field* 1. There $\phi_p = 0$, so that $S_\varepsilon(0) = 0$, $\zeta = 0$ and $\mathrm{En}_\ell(1) = 0$. The set $D(1)$ is empty, so that only the empty family is admitted and $\Xi_w(1) = 1$. Further, $1_{X\approx}(1) = 1$, and the clause (ca) includes the condition that the value at pure gauges is 0.

The parts "The step" and "Runs" of item 2 of §5.5. This section proves two further properties of the drawn step. The first, Lem. 8.20, gives $\mathsf{M}(RU) = R\,\mathsf{M}(U)$ and the invariance of the kernel, for integer mirrors and coarse translations. The second, Lem. 8.27 A, states that $\mathsf{M}(U)(c)$ depends on the bonds of $B(y) \cup B(y')$ alone. The proof given in "The step" yields gauge covariance only. "The step" states also that the step depends on x, a and L, and on no rows, no grid of cubes, no members and no K; hence the covariance iterates. By "Runs", the noise of the step $k \to k+1$ is $a\,x_k$.

The clauses (9a), (9b), (9d) *of Thm. 5.15, in item 3 of §5.5.* Clause (9a) holds at every V'. By (9b), the outputs keep all the clauses that carry no size, so that $E^a(X,1) = 0$; and an input in M_2 gives an output in M_1 with the strong cap. Clause (9d) gives positivity.

Def. A.4 (2) and the paragraphs of §A.2 that follow it and that item 0.3 names. $B_i(o)$ is a board: $\lambda = L^{K-i}$ is odd and ≥ 3 if and only if $i \leq K-1$; $\mathsf{n}_\mu = 2L^{m_l+i}$ is an even integer if and only if $i \geq -m_l$; and $o \in a_K\mathbb{Z}^3$. The measure, a_p, $x(p)$, $\mathcal{O}(f)$, $(rU)(b) = U(r^{-1}b)$ for a bond b, $(rF)(U) = F(r^{-1}U)$ and $rf = f \circ r^{-1}$ coincide with the objects so named in the data of Thm. 8.2 and in Step 7 of its proof.

Cor. A.3 is stated for all bounded Borel functions of the bonds that have both ends in T_+, with $c \in a(\mathbb{Z} + \frac{1}{2})$.

Def. A.10, Prop. A.11 and Prop. A.12 (f) are applied on $B_i(o)$ and over the occupied cells. With $N_c = (N/h)^3$, the number of the cells of $B_i(o)$, the root $1/N_c$ turns $e^{c_{\mathrm{vol}}N^3}$ into $e^{c_{\mathrm{vol}}h^3}$.

Lem. D.1. Its hypotheses are a semigroup, commuting maps, the fixing of 1, and (1) to (3). The conclusions (a) to (d) of Lem. D.1 are (R7).

Lemma 8.18 (The field and the generating identity)**.** Let $Q \geq 1$, and let g_t be real bounded functions with pairwise disjoint supports. Then $Z_K(z) := \int \exp\{-\beta\sum_p a_p + \sum_t z_t\mathcal{O}_K(g_t)\}dU$ is entire, and $Z_K(0) > 0$. For any numbers m_t, and with $m \cdot z = \sum_t m_t z_t$, the coefficient of $z_1 \cdots z_Q$ in $e^{-m\cdot z}Z_K(z)/Z_K(0)$ is $\langle\prod_t(\mathcal{O}_K g_t - m_t)\rangle$. Finally, the weight is that of Thm. 7.1 at $\nu(z)_p = \mathsf{b}^{-1}\sum_t z_t g_t(x(p))$, and $|\nu(z)|_\infty \leq \max_t|z_t|\,\|g_t\|_\infty/\mathsf{b}$. ■

Proof. The argument of Step 0 of the proof of Thm. 8.2 applies word for word, with the one change that Q replaces n: the integrand is continuous on a compact group, and it is entire in z. Next, $e^{-m\cdot z}Z_K(z)/Z_K(0) = \langle\exp\sum_t z_t(\mathcal{O}_K g_t - m_t)\rangle$, and the coefficient of $z_1\cdots z_Q$ in it is the mean of the product $\prod_t(\mathcal{O}_K g_t - m_t)$. Since $\beta = L^K\mathsf{b}$, we have $\beta\nu_p a_p = L^K\sum_t z_t g_t(x(p))a_p$. At each p at most one of the numbers $g_t(x(p))$ is $\neq 0$, by the part "Sources" of item 3 of §7.1. Therefore the domain Pd_0 of Thm. 7.1 (h) depends on $\max_t|z_t|$ alone, and not on Q. □

Remark 8.19 (The field). The factor $\exp\{\sum_t z_t\mathcal{O}_K(g_t)\}$ turns Wilson's weight into the weight at the local label $\mathsf{b} - \sum_t z_t g_t(x)$. Hence S_n is the n-th derivative of $\log Z$ with respect to minus the local label, taken at different points. These are the truncated functions of the action density. They concern one real gauge invariant scalar, which is given modulo a constant, and they are taken for $n \geq 2$, off the diagonals.

Lemma 8.20 (Covariance of the drawn step under integer mirrors and coarse translations)**.** On the fine lattice $\mathbb{Z}^3/N'L\mathbb{Z}^3$, consider the drawn step that is defined in "The step", item 2 of §5.5. Let R be either the map $x_\mu \mapsto Lc - x_\mu$, in one direction μ and with $c \in \mathbb{Z}$, or a translation by a

vector of $L\mathbb{Z}^3$, and let $(RU)(\text{bond}) = U(R^{-1}\text{bond})$, where $U(\text{reversed bond}) = U(\text{bond})^{-1}$. Then $\mathsf{M}(RU)(c) = \mathsf{M}(U)(R^{-1}c)$ for every coarse bond c, and

$$\prod_c k_s((RV')(c), \mathsf{M}(RU)(c)) = \prod_c k_s(V'(c), \mathsf{M}(U)(c)).$$

Consequently, consider, in the units of the last lattice, a reflection $R : y_\mu \mapsto c - y_\mu$ with $c \in L^{-s}\mathbb{Z}$, or a translation R by a vector of $L^{-s}\mathbb{Z}^3$. For every $k \le K - s$, the kernel density $\mathbb{K}_k(U; V)$, which is the density of V_k conditional on U, satisfies $\mathbb{K}_k(RU; RV) = \mathbb{K}_k(U; V)$. ■

Proof. For translations the claim is evident. If R is a reflection, then R is an involution.

Step (1): geometry. The map R carries $L\mathbb{Z}^3$ onto itself, and it carries the centred block $B(y) = y + \{-(L-1)/2, \dots, (L-1)/2\}^3$ onto $B(Ry)$. Hence $y(Rx) = R\,y(x)$.

The map R carries the coordinate path $\Gamma(y, x)$, which runs first along e_1, then along e_2, then along e_3, each time the short way, onto the coordinate path $\Gamma(Ry, Rx)$: only the steps in the direction μ change their sign, and the order of the axes is kept. For a path π we have $(RU)(\pi) = U(R\pi)$; applied to the path $\Gamma(y(x), x)$, together with $y(Rx) = Ry(x)$, this gives $\mathsf{t}_{RU}(x) = \mathsf{t}_U(Rx)$.

Step (2): the bond $c = (y, y + Le_\nu)$ with $\nu \neq \mu$. For the straight path $[x, x']$ along e_ν we have $R[x, x'] = [Rx, Rx + Le_\nu]$, and therefore $h_{c,x}(RU) = h_{Rc,Rx}(U)$. The map $x \mapsto Rx$ is a bijection of $B(y)$ onto $B(Ry)$, and it sends y to Ry. Hence $\bar\tau_c(RU) = \bar\tau_{Rc}(U)$ and $\mathsf{M}(RU)(c) = \mathsf{M}(U)(Rc)$.

Step (3): the bond $c = (y, y')$ with $y' = y + Le_\mu$. Now $Rc = (Ry, Ry - Le_\mu)$ is the reverse of the bond $\tilde c := (Ry', Ry)$. For $x \in B(y)$ we put $\tilde x := Rx' = Rx - Le_\mu$; then $\tilde x \in B(Ry')$ and $\tilde x + Le_\mu = Rx$. As $[Rx, Rx']$ is the path $[\tilde x, Rx]$ run backwards,

$$h_{c,x}(RU) = \mathsf{t}_U(Rx)U([Rx, Rx'])\mathsf{t}_U(Rx')^{-1} = [\mathsf{t}_U(\tilde x)U([\tilde x, Rx])\mathsf{t}_U(Rx)^{-1}]^{-1} = g_x^{-1}, \qquad g_x := h_{\tilde c,\tilde x}(U).$$

Next, $g_x^{-1}g_y = g_y^{-1}(g_x g_y^{-1})^{-1}g_y$. The logarithm satisfies $\mathsf{L}(ugu^{-1}) = \mathrm{Ad}_u\,\mathsf{L}(g)$ and $\mathsf{L}(g^{-1}) = -\mathsf{L}(g)$; at $g = -1$ both sides of each identity are 0. Hence $\mathsf{L}(g_x^{-1}g_y) = -\,\mathrm{Ad}_{g_y^{-1}}\,\mathsf{L}(g_x g_y^{-1})$. The map $x \mapsto \tilde x$ is a bijection of $B(y)$ onto $B(Ry')$ which sends y to the base site Ry' of $\tilde c$; summing over $x \in B(y)$ we obtain $\bar\tau_c(RU) = -\,\mathrm{Ad}_{g_y^{-1}}\,\bar\tau_{\tilde c}(U)$. Finally,

$$\mathsf{M}(RU)(c) = \exp\{i\bar\tau_c(RU)\}g_y^{-1} = g_y^{-1}e^{-i\bar\tau_{\tilde c}(U)} = (e^{i\bar\tau_{\tilde c}(U)}g_y)^{-1} = \mathsf{M}(U)(\tilde c)^{-1} = \mathsf{M}(U)(Rc);$$

the second equality holds because the exponential turns $\mathrm{Ad}_{g_y^{-1}}$ into conjugation by g_y^{-1}, and the last because Rc is $\tilde c$ reversed.

Step (4) Kernel: both entries inverted. We have $(g^{-1})(h^{-1})^{-1} = g^{-1}h$, and this is conjugate to $hg^{-1} = (gh^{-1})^{-1}$. Since ang is a class function and $\mathrm{ang}(u^{-1}) = \mathrm{ang}(u)$, it follows that $k_s(g^{-1}, h^{-1}) = k_s(g, h)$. The map R permutes the unoriented coarse bonds; at a bond whose orientation it reverses, both entries of the kernel are inverted, and the factor of that bond keeps its value.

Step (5) The last assertion: the levels down to depth s. In the step from $\mathbb{T}_k$ to $\mathbb{T}_{k+1}$ the coarse site y is located at the fine site Ly; so all levels are the lattices $L^{-j}\mathbb{Z}^3$ in one torus $\mathbb{R}^3/N\mathbb{Z}^3$, and 0 is a site of each. In the units of depth j the reflection $y_\mu \mapsto c - y_\mu$ of the last assertion of Lem. 8.20 becomes the map $x \mapsto L^j c - x$, where $L^j c = L(L^{j-1}c)$ and $L^{j-1}c \in \mathbb{Z}$ for $j - 1 \ge s$. In the same units a translation by a vector $v \in L^{-s}\mathbb{Z}^3$ becomes the translation by $L^j v = L(L^{j-1}v)$, where $L^{j-1}v \in \mathbb{Z}^3$ for $j - 1 \ge s$; so it is a translation by a vector of $L\mathbb{Z}^3$, which is one of the maps R of the first assertion of the lemma. Hence (1) to (4) hold at every step down to depth s.

The step depends on x, a and L alone: it depends on no rows, on no grid of cubes, on no members, and not on K; this is the last sentence of item 2 ("The step") of §5.5. The density $\mathbb{K}_k$ is the integral over $V_1, \dots, V_{k-1}$ of the product of the kernels of the steps. In this integral we change the variables of integration, $V_j \to RV_j$; the Haar measure is invariant under permutations of the bonds and under inversion, and (1) to (4) hold at each of these steps; so $\mathbb{K}_k(RU; RV) = \mathbb{K}_k(U; V)$ for $k \le K - s$. □

Lemma 8.21 (The subtraction as a mirror invariant weighted mean)**.** Let $0 \le s \le K-1$ and $k = K - s$.

(a) We have

$$c_K^{(s)}(f) = \int \mathcal{O}_K(f) e^{-\beta A} \mathbb{K}_k(U;1)\, dU \Big/ \int e^{-\beta A} \mathbb{K}_k(U;1)\, dU;$$

this number is real for real f, and it is linear in f. Here $A = \sum_p a_p$, with $a_p = 1 - \frac{1}{2} \operatorname{tr} \partial U(p)$ (item 1 of §8.3).

(b) For the maps R of the last assertion of Lem. 8.20 (in the units of the last lattice, the reflections $y_\mu \mapsto c - y_\mu$ with $c \in L^{-s}\mathbb{Z}$ and the translations by vectors of $L^{-s}\mathbb{Z}^3$), $c_K^{(s)}(f \circ R) = c_K^{(s)}(f)$.

(c) Let the g_t be the $Q \ge 1$ sources of the data of (R1) of Thm. 8.17: real bounded functions whose supports are pairwise at distance $\ge \check{\delta}$, each in a ball of radius $\check{\rho}$. Then $c_K^{(s)}(g_t) = \gamma_t^{(k)} + e_t^{(s)}$, where $\gamma_t^{(k)} := \partial_t \sum_{j \le k} V_j(0)$ and $e_t^{(s)} := \sum_X \delta_t^h(k)(X,1)$. If $\|g_t\|_\infty \le 1$, then $r_s^\circ |e_t^{(s)}| \le e_h^{(s)}$.

■

At $s = 0$, clause (c) is Prop. 8.11 (J7°) (vii) together with (J6°) (e), where the constant 0.065 appears in the place of 0.45.

Proof. Step (a): the weighted mean. By the definition of the run, $\rho_k^{[zf]}(1) = \int e^{-\beta A + z\mathcal{O}_K(f)} \mathbb{K}_k(U;1)\, dU$. By Step 11 (0) of the proof of Thm. 8.2, the integrand is Borel, the integral is entire in z, and d/dz may be taken under the integrals. As $\mathbb{K}_k > 0$, we have $\rho_k(1) > 0$. Differentiating the logarithm at $z = 0$ we obtain the formula of (a); it shows that $c_K^{(s)}(f)$ is linear in f, and real for real f.

Step (b): mirror invariance. In both integrals we change the variable of integration, $U \to RU$. The Haar measure and A are invariant under this change. By the last assertion of Lem. 8.20, $\mathbb{K}_k(RU;RV) = \mathbb{K}_k(U;V)$ for $k \le K - s$, and so $\mathbb{K}_k(RU;1) = \mathbb{K}_k(RU;R1) = \mathbb{K}_k(U;1)$. By Step 7 ("At finite K") of the proof of Thm. 8.2, $\mathcal{O}_K(f)(RU) = \mathcal{O}_K(f \circ R)(U)$. So the numerator for f equals the numerator for $f \circ R$, the denominator stays as it is, and (b) follows.

Step (c) The formula: the value at the trivial field. By Thm. 7.1 (h), on Pd_0 we have

$$\rho_k^z(V) = \exp\{\sum_{j \le k} V_j(z)\}\, \rho[u_k^{\mathrm{hon}}(z); \ell_k](V)$$

at *every* field V of $\mathbb{T}_k$. Here $u_k^{\mathrm{hon}}(z)$ is the honest family, the family $u_k(z)$ of Thm. 7.1 (h). We write $u_k(z)$, without the superscript hon, for the representative of (P1.1). By (P1.1) the jets at 0 of both families are the $\delta_S(k)$. Thus $E_k^h(X,1;z)$ below belongs to $u_k^{\mathrm{hon}}(z)$, while $E_k^h(X,\cdot\,;z)$ in the next step belongs to the representative. We evaluate the right side at the trivial field $V = 1$, in four parts.

First, $\phi_p = 0$, so that $S_\varepsilon(0) = 0$, $\zeta_p = 0$ and $\mathrm{En}_\ell(1) = 0$ (item 2, "Energy", of §5.5). Second, $D(1)$ is empty; as every X_i of a family must hold a plaquette of $D(V)$, while the empty family is admitted, we get $\Xi_w(1) = 1$ (item 2 of §5.5, the definition of $\rho[u;\ell]$ and of Ξ_w). Third, the field 1 is tame, so $1_{X^\approx}(1) = 1$. Fourth, $E^a(X,1) = 0$ by the clause "0 at pure gauges" of (ca) in item 2 of §5.5; this clause is not one of the size bounds, and the outputs of Thm. 5.15 obey every clause of (ca), (ch) and (cΞ) other than the size bounds ((9b) of item 3 of §5.5; here $k \ge 1$), as well as the size bounds of Thm. 5.15 (b).

Hence

$$\log \rho_k^z(1) = \sum_{j \le k} V_j(z) + \sum_X E_k^h(X,1;z),$$

a finite sum of functions holomorphic near 0. The derivative ∂_t at 0, with the other variables set to 0, is the derivative in the run with the single source g_t; this gives the formula of (c).

Step (c) The size: the bound on $e_t^{(s)}$. By (P1.1) in item 0.4 of §8.3, $\delta_t^h(k)$ is $\partial_t E^h$ of the representative. First let $k > k^\circ$. Then $s < j^\circ$, and $r_s^\circ = r^\circ \le \mathfrak{r}_*^\circ$. On the disc $|z_t| < r_s^\circ$, with the other variables 0, the part of (P1.1) for $k^\circ < k \le K$, taken in the norm $|\cdot|_h^{(\kappa)}$ of (o6) (item 2 of §7.3), gives

$$|E_k^h(X,\cdot\,;z) - \bar{E}_k^h(X,\cdot)| \le 0.91 \cdot 0.4\, c_h(x_k) e^{-\kappa(\#X-1)},$$

and Cauchy's estimate gives $|\delta_t^h(k)(X,1)| \le 0.364\, c_h(L^s\mathsf{b}) e^{-\kappa(\#X-1)}/r_s^\circ$.

Now let $k \le k^\circ$. Then $s \ge j^\circ$, and $\mathfrak{r}_k^\circ = \mathfrak{r}^\circ(L^s\mathsf{b}) = r_s^\circ$. On the same disc the part of (P1.1) for $k \le k^\circ$, taken in the norm $|\cdot|_h^{(\kappa_k^\circ)}$ of (o6), gives the bound of the previous case with 0.91 in the place of $0.91 \cdot 0.4$ and with the rate κ_k° in the place of κ; since $\kappa_k^\circ \ge \kappa$, Cauchy's estimate now gives $|\delta_t^h(k)(X,1)| \le 0.91\, c_h(L^s\mathsf{b}) e^{-\kappa(\#X-1)}/r_s^\circ$.

In both cases only the X that hold a cube of $S_t(k)^{+\check{r}(x_k)}$ count: for the others $\delta_t(k)(X) = 0$, by (P_k) (Step 3 of the proof of Prop. 7.12; (P1.1)). The number of these cubes is at most $N_1(k-1) = N_1^{[s+1]}$ (Step 1 (A) of the proof of Prop. 7.12). The connected X of m cubes that hold a given cube are at most $(26e)^{m-1}$ in number (item 2, "Lattice, cubes", of §5.5). As $\kappa \ge 6$, the geometric series gives $\sum_m (26e\, e^{-6})^{m-1} \le 1.2124$. So $r_s^\circ |e_t^{(s)}|$ is at most $N_1^{[s+1]} c_h(L^s\mathsf{b})$ times $0.364 \cdot 1.2124$ if $k > k^\circ$, and times $0.91 \cdot 1.2124$ if $k \le k^\circ$. Now $0.364 \cdot 1.2124 \le 0.45$; this settles the case $s = 0$, which falls under $k > k^\circ$ because $j^\circ \ge 1$. For $s \ge 1$ the numerical factor is at most 1.11 in either case: if $k > k^\circ$, that is, if $1 \le s < j^\circ$, it is at most 0.45, and if $k \le k^\circ$, it is $0.91 \cdot 1.2124 \le 1.11$. As $e_h^{(0)} = 0.45\, N_1^{[1]} c_h(\mathsf{b})$ and $e_h^{(s)} = 1.11\, N_1^{[s+1]} c_h(L^s\mathsf{b})$ for $s \ge 1$ (item 2 of §8.3), we obtain $r_s^\circ |e_t^{(s)}| \le e_h^{(s)}$ for every $0 \le s \le K-1$, which is the bound of (c). □

Remark 8.22 (The constant of the subtraction). The constant of the subtraction is $c_K^{(s)}$ rather than the vacuum numbers γ_t alone, because the grids of cubes and the base sites are not mirror covariant.

Lemma 8.23 (The last integral bounded as a moment)**.** Assume the data of item 0.4 of §8.3, let $u_k(z)$ be the representative of (P1.1) for $k = K - s$ with $0 \le s \le K-1$, let $z \in \mathrm{Pd}(r_s^\circ)$, and put $I_1 := I[\bar{u}_k] > 0$. Then $\int |\rho[u_k(z);\ell_k]|\, dV \le \mathfrak{V}_s I_1$, where

$$\mathfrak{V}_0 = 1 + 0.182\, C_{\mathrm{vol}}(N,\mathsf{b}) \le e^{c_{\mathrm{vol}}(\mathsf{b}) N^3}$$

and, for $s \ge 1$,

$$\mathfrak{V}_s = 1 + 0.455\, C_{\mathrm{vol}}(NL^s, L^s\mathsf{b}) e^{200 q_N} \le e^{c_{\mathrm{vol}}^{(s)}(\mathsf{b}) N^3}, \qquad q_N = (NL^s/M)^3.$$

Moreover, for every bounded Borel function F on the fields of $\mathbb{T}_k$, the map $z \mapsto \int F \rho[u_k(z);\ell_k]\, dV$ is holomorphic on $\mathrm{Pd}(r_s^\circ)$. ■

Proof. Step (Data (B)*): at $x = L^s\mathsf{b}$ the data of Thm. 7.27 are met.* We take the data (B) of Thm. 7.27 at $x = L^s\mathsf{b}$ on $\mathbb{Z}^3/NL^s\mathbb{Z}^3$, with the rate $\tilde{\kappa} = \kappa$ if $k > k^\circ$ and $\tilde{\kappa} = \kappa_k^\circ$ if $k \le k^\circ$; both rates lie in $[\kappa, \kappa_1]$. These data are met. First, $x \ge x_s \ge e^2$. Next, $NL^s = 2L^{m_l+s}$ is even, and M divides it. Further, $NL^s \ge 2Ms(L^s\mathsf{b})$: for $s = 0$ by (Tor), since $s(\cdot)$ is non-decreasing, and for $s \ge 1$ by Step 11 (5) of the proof of Thm. 8.2.

Moreover, $\varepsilon \le 1$ and $c_a + c_h \le 1/2$ at $x \ge x_s$, by (X16) and (X17) (entry (O13) of item 1 of §5.5). The rows ℓ_k have range $\le \varrho(x_k)$ and obey the decay clause, for $0 \le k \le K$ and every $K \ge 1$ (this is Step 2, "The last integral", of the proof of Thm. 8.2). As the first member we take $u_1 = \bar{u}_k$; by Prop. 7.12 (J6) and Thm. 7.1 (h), it is a system of members of a point of $C_\flat^{\mathrm{su}}(x_k;\ell_k)$. We put $u_3 := u_1$ and $u_2 := u_k(z)$.

Step (Case $k > k^\circ$): case (ii) *of* (b^C). Here $u_2 = F_{\ell_{k-1}}(u_{k-1}(z))$, with its input in $M_{1.91} \subset M_2$. So u_2 is an output of Thm. 5.15 applied with $A \le 6$. For this output, and with $u_3 \in M_1$, we may

take $a_w = 1$ and $a_E \le 2$ in Thm. 7.27 (b^C): this is case (ii) of that clause, used with the pair (κ, c_h). Further, $\mathsf{D}_{23} = \|u_2 - u_1\|^{\sim} \le \|u_2 - u_1\|^{\mathrm{st},(\kappa)} \le 0.364$, since the strong norm of the same difference dominates (Def. 7.22). The L^1 bound of Thm. 7.27 (c^C) for the cases (i), (ii) and (iii) then gives in the case (ii)

$$\int |\rho[u_2] - \rho[u_1]|\, dV \le \frac{1}{2} C_{\mathrm{vol}} \cdot 0.364\, I_1.$$

Step (Case $k \le k^\circ$): case (iii) *of* (b^C). Here $\|u_2\|^{\sim} \le 1.91$ and $\|u_3\|^{\sim} \le 1$: this is case (iii) of (b^C), applied with $A = 1.91$ and $\mathsf{D}_{23} \le 0.91$. The same integral is now at most $\frac{1}{2} C_{\mathrm{vol}} e^{219 \cdot 0.91 q_N}\, 0.91\, I_1$, and $219 \cdot 0.91 = 199.29 \le 200$.

Step (Add): the first member. We add $\int \rho[u_1]\, dV = I_1$ to the bounds of the two cases; this gives the first claim.

Step (Arithmetic): the two exponentials. Let X be the part of C_{vol} that is a power with exponent N^3. For $\mathfrak{V}_0$ we use $1 + 0.728\, q_N e^{220 q_N} X \le e^{220.728 q_N} X$, and for $\mathfrak{V}_s$ we use $1 + 1.82\, q_N e^{420 q_N} X \le e^{421.82 q_N} X \le e^{425 q_N} X$; here $q_N = L^{3s} N^3 / M^3$.

Step (Holomorphy): the integrals against F. For each field V, $\rho[u; \ell](V)$ is an entire function of finitely many values of u (the clause of Lem. 7.8 on $I[u]$); hence $\rho[u_k(z); \ell_k](V)$ is holomorphic in z. By (b^C) of Thm. 7.27, $\sup_V |\rho[u_2] - \rho[u_1]|$ is bounded on the polydisc, and $\rho[u_1]$ is bounded. The function $F(V)\rho[u_k(z); \ell_k](V)$ is holomorphic in z, Borel in V and bounded on $\mathrm{Pd}(r_s^\circ)$; so Step P4 (2) of the proof of Cor. 7.28 applies to it, and it gives both the second claim and the identity $\partial_S \int F \rho[u_k(z); \ell_k]\, dV = \int F\, \partial_S \rho[u_k(z); \ell_k]\, dV$. □

Remark 8.24 (On the last integral). The lower bound on I_1 is crude, and this crudeness costs only the size of c_{vol}.

Lemma 8.25 (The master bound for centred sources)**.** Assume the data of (R1) of Thm. 8.17, let $0 \le s \le K - 1$ and $k = K - s$, and let F be a bounded Borel function of the fields of $\mathbb{T}_k$. Then

$$\big|\big\langle F(V_k) \prod_t (\mathcal{O}_K g_t - c_K^{(s)} g_t) \big\rangle\big| \le \sup |F|\; G_{1,s}^Q\, \mathfrak{V}_s \prod \|g_t\|_\infty.$$

■

Clause (R1) of Thm. 8.17 is the case $s = 0$, $F = 1$. At $s = 0$ the lemma is Cor. 8.14 (a), (b) of §8.2.

Proof. By homogeneity we may take $\|g_t\|_\infty = 1$ for every t.

Step (1) The ideal: congruence of germs. The germs at 0 in $\mathbb{C}^Q$ whose squarefree coefficients all vanish form the ideal (z_t^2): every monomial of such a germ is divisible by some z_t^2, and we group the monomials by the least such t. Hence $f \equiv g$ implies $fh \equiv gh$. It also implies $e^f \equiv e^g$, since $e^f - e^g = e^g(e^{f-g} - 1)$.

Step (2) The exponent: its higher part. By (P1.2), in the form (J7°) (iv) of Prop. 8.11, the exponent at the level k is $\sum_{j \le k} V_j(0) + \sum_t z_t \gamma_t^{(k)} + W^{(k)}(z)$, with

$$W^{(k)} := \sum_{k' < \min(k, k^\circ)} W_{k'} + \sum_{k^\circ \le k' < k} [w_{k'} - \text{linear part}].$$

Bounding term by term with (P1.3) on $\mathrm{Pd}(\mathfrak{r}_*^\circ) \supset \mathrm{Pd}(r_s^\circ)$, we get $|W^{(k)}| \le Q C_W$. For a late term, of depth $j = K - k'$, we have $|w_{k'}| \le 3Q N_1^{[j]} C_g c_a (L^j \mathsf{b})$ and $|\text{linear part}| \le Q\, \mathfrak{r}_*^\circ \cdot 3 N_1^{[j]} C_g c_a (L^j \mathsf{b}) / \mathfrak{r}_*^\circ$, which together give $6 Q N_1^{[j]} C_g c_a (L^j \mathsf{b})$.

Step (3) The identity: the generating function. By Step 11 (0) of the proof of Thm. 8.2, $\langle F(V_k) e^{\sum z_t \mathcal{O}_K g_t} \rangle = \int F \rho_k^z\, dV / Z_K(0)$. We take (P1.2) at the level k, pointwise in V; the derivatives ∂_S may be taken under $\int F \cdot dV$ on both sides, on the left by Step 11 (0) and on the right by Step

(Holomorphy) of the proof of Lem. 8.23. Moreover $Z_K(0) = \exp\{\sum_{j\le k} V_j(0)\} I[\bar u_k]$ (by Thm. 7.1 (h) and Step 0 of the proof of Thm. 8.2). With Steps (1) and (2) and Lem. 8.21 (c) we arrive at

$$e^{-\sum z_t c_K^{(s)} g_t} \langle F(V_k) e^{\sum z_t \mathcal{O}_K g_t}\rangle \equiv G_F(z) := \exp\{W^{(k)}(z) - \sum_t z_t e_t^{(s)}\} \int F \rho[u_k(z); \ell_k]\, dV / I[\bar u_k].$$

Step (4) Cauchy: the coefficient of $z_1 \cdots z_Q$. The function G_F is holomorphic on $\mathrm{Pd}(r_s^\circ)$, and there $|G_F| \le e^{QC_W} e^{Qe_h^{(s)}} \sup|F|\, \mathfrak{V}_s$, by Lems. 8.21 and 8.23. The coefficient of $z_1 \cdots z_Q$ on the left is the moment, as in Lem. 8.18; by Step (1) it is also the coefficient of G_F, and Cauchy's estimate bounds it by $(r_s^\circ)^{-Q} \sup|G_F|$. As $G_{1,s} = e^{C_W + e_h^{(s)}}/r_s^\circ$, this is the claim. □

Remark 8.26 (On the master bound). In Lem. 8.25 the volume factor enters the bound with the exponent *one*, whatever Q is. All the ultraviolet input of the lemma lies in (P1.1) to (P1.3), in Thm. 5.15, in Thm. 6.7 (g2) and in Thm. 7.27 (b^C).

Lemma 8.27 (The joint law: locality, conditional independence, covariance)**.** Consider the joint law

$$d\mu(U) \prod_{k<K} \prod_c k_{ax_k}(V_{k+1}(c), \mathsf{M}(V_k)(c))\, dV_{k+1};$$

it is the law of Step 11 (0) of the proof of Thm. 8.2. Let $0 \le s \le K-1$. The following assertions hold for this law.

- A For a coarse bond $c = (y, y')$, $\mathsf{M}(U)(c)$ is a function of U on the bonds with both ends in $B(y) \cup B(y')$.
- B Let O be an open set bounded by planes $y_\mu \in L^{-s}(\mathbb{Z} + \frac12)$, and let F be a bounded Borel function of the values of V_{K-s} at finitely many bonds, each with both ends in O. Then the function $E[F(V_{K-s})|U]$ of U is bounded and Borel, and it depends only on the bonds with both ends in O.
- C Let the sets O_i be pairwise disjoint, and let each F_i be as in B for O_i. Then $E[\prod F_i|U] = \prod E[F_i|U]$.
- D For a map R of the last assertion of Lem. 8.20 and a bounded Borel function F of V_{K-s} we put $(RF)(V) = F(R^{-1}V)$, $\Phi = E[F|U]$ and $(R\Phi)(U) := \Phi(R^{-1}U)$. Then $E[(RF)(V_{K-s})|U] = (R\Phi)(U)$.

■

Proof. Step A. Reach of M*: the bonds on which the average depends.* Let $c = (y, y')$ be as in A. The path $\Gamma(y, x)$ leads from the centre of the box $B(y)$ to $x \in B(y)$ and stays in this box; the same is true of $\Gamma(y', x')$; and the straight path $[x, x + Le_\mu]$ stays in the two boxes. These three facts give claim A.

Step B. Locality: functions of an open set. Let O be as in B. As L is odd, $L^{-s}(\mathbb{Z} + \frac12) \subset L^{-j}(\mathbb{Z} + \frac12)$ for all $j \ge s$. The block of a site y of depth $j-1 \ge s$ consists of the sites of depth j in $y + L^{-(j-1)}(-\frac12, \frac12)^3$; so at every step down to depth s the planes are faces of blocks, and no site lies on them. By A, the average at a bond of depth $j-1$ with both ends in O depends only on bonds of depth j with both ends in O.

Let F be a bounded Borel function of V_{k+1} on a finite set B_F of such bonds. Then

$$E[F(V_{k+1})|V_k] = \int F(V') \prod_{c \in B_F} k_{ax_k}(V'(c), \mathsf{M}(V_k)(c))\, dV',$$

because the kernels at the other bonds integrate to 1. This function is bounded; it is Borel, as k_s is continuous and M is Borel; and it is a function of V_k on the bonds with both ends in O. Iterating along the Markov chain we obtain claim B.

Step C. Conditional independence: disjoint open sets. Let O_i and F_i be as in C. Then the sets B_{F_i} are disjoint, and since the kernel is a *product* over the coarse bonds, $E[\prod F_i(V_{k+1})|V_k] = \prod_i \Phi_i(V_k)$,

where $\Phi_i(V_k) := E[F_i(V_{k+1})|V_k]$ is the function of Step B for F_i. Induction on the level gives claim C.

Step D. Covariance: the maps R. In the integral against $\mathbb{K}$ we change the variable of integration, $V \to RV$; as the last assertion of Lem. 8.20 gives $\mathbb{K}_k(RU; RV) = \mathbb{K}_k(U; V)$ for $k \le K - s$, we obtain claim D. □

Lemma 8.28 (Boards, functions of a cell, mirror maps, folds)**.** Let $0 \le s \le K-1$, $-m_l \le i \le \min(s, K-1)$, $o \in L^{-s}\mathbb{Z}^3$, $h = L^{-i}$ and $\mathsf{n}_\mu = N/h$. The cells of $B_i(o)$ are indexed by $\varkappa, \varkappa' \in \prod_\mu \mathbb{Z}/\mathsf{n}_\mu\mathbb{Z}$, and the cell $\varkappa$ is centred at $o + h\varkappa$.

(a) $B_i(o)$ is a board in the sense of Def. A.4 (2), with $N_c = (N/h)^3$ cells, and its walls lie in $L^{-s}(\mathbb{Z} + \frac{1}{2}) \subset a_K(\mathbb{Z} + \frac{1}{2})$.
(b) If $\{g \ne 0\}$ lies in an open cell, then $\mathcal{O}_K(g)$ is a function of that cell.
(c) Every mirror map $r_{\varkappa'\varkappa}$ is, coordinate by coordinate, a map of (R0) of Thm. 8.17 at depth s.
(d) Let $Y = \prod_{t\in I}(\mathcal{O}_K g_t - c_K^{(s)} g_t)$, with supports in the open cell $\varkappa$, and let $\Phi = E[F(V_{K-s})|U]$, where F depends only on the bonds that have both ends in that open cell. Then

$$\langle \mathrm{Fold}(\Phi Y)\rangle = \langle \prod_{\varkappa'} F^{[\varkappa']}(V_{K-s}) \prod_{\varkappa'}\prod_{t\in I}(\mathcal{O}_K(rg_t) - c_K^{(s)}(rg_t))\rangle,$$

where $r = r_{\varkappa'\varkappa}$.

■

Proof. Step (a): the board. For the board $B_i(o)$, $\lambda = L^{K-i}$ is odd, and $\lambda \ge 3$ because $i \le K-1$; $\mathsf{n}_\mu = N/h = 2L^{m_l+i}$ is an even integer ≥ 2; and $o \in a_K\mathbb{Z}^3$ because $s \le K$. For the walls, $L^{-i}(m + \frac{1}{2}) = L^{-s}L^{s-i}(2m+1)/2$, where $L^{s-i}(2m+1)$ is odd; adding $o_\mu \in L^{-s}\mathbb{Z}$ keeps the wall in $L^{-s}(\mathbb{Z} + \frac{1}{2})$.

Step (b): functions of a cell. A plaquette p enters $\mathcal{O}_K(g)$ only if its barycentre $x(p)$ lies in $\{g \ne 0\}$, hence in the open cell. In the two directions of p, the coordinate of $x(p)$ lies in $a_K(\mathbb{Z} + \frac{1}{2})$, as the walls do; so it is at distance $\ge a_K$ from them, and the corners, at $\pm a_K/2$, lie inside. In the third direction the coordinate is that of a site, which lies off the walls. Hence the four corners of p lie in the cell, and its four bonds are inner bonds; so $\mathcal{O}_K(g)$ is a function of the cell. Here The inner bonds are those of Def. A.4. Cell functions are described in eq. A.xi and Lem. A.5(d).

Step (c): mirror maps. By eq. A.xiii, $r_{\varkappa'\varkappa}$ is the map $x \mapsto o + h\varkappa' + \sigma(x - o - h\varkappa)$, where $\sigma = \sigma(\varkappa', \varkappa) = (\sigma_\mu)_\mu$, with $\sigma_\mu = (-1)^{\varkappa'_\mu - \varkappa_\mu}$, acts coordinate by coordinate: $\sigma u = (\sigma_\mu u_\mu)_\mu$. In a coordinate μ with $\varkappa'_\mu - \varkappa_\mu$ even, it is the translation by $h(\varkappa'_\mu - \varkappa_\mu) \in L^{-s}\mathbb{Z}$. In a coordinate with $\varkappa'_\mu - \varkappa_\mu$ odd, it is $y \mapsto 2o_\mu + h(\varkappa'_\mu + \varkappa_\mu) - y$, with the constant in $L^{-s}\mathbb{Z}$. Thus $r_{\varkappa'\varkappa}$ is a composition of maps in the different coordinates, and it is covered by (R0) applied to one coordinate after the other.

Step (d): the fold. By Lem. 8.27 B, Φ is a function of the cell. By eq. A.xiv, $(\Phi Y)^{[\varkappa']}$ is $r(\Phi Y)$ or its complex conjugate, and Y is real. By Step 7 of the proof of Thm. 8.2, $r\mathcal{O}_K(g) = \mathcal{O}_K(g)(r^{-1}\cdot) = \mathcal{O}_K(g \circ r^{-1}) = \mathcal{O}_K(rg)$, and by Lem. 8.21 (b), $c_K^{(s)}(g) = c_K^{(s)}(rg)$; so the copy of Y is $\prod_{t\in I}(\mathcal{O}_K(rg_t) - c_K^{(s)}(rg_t))$. By Lem. 8.27 D, $r\Phi = E[rF|U]$; complex conjugation passes through E, and so $\Phi^{[\varkappa']} = E[F^{[\varkappa']}(V_{K-s})|U]$. As the open cells are disjoint, Lem. 8.27 C gives $\prod_{\varkappa'} E[F^{[\varkappa']}|U] = E[\prod F^{[\varkappa']}|U]$. The factors that depend on U are measurable in U; taking the mean we obtain (d). □

Lemma 8.29 (Pieces, placing and mirror copies)**.** (a) Let $f_1, \dots, f_n$ be of the type $(d; R_f)$, and let $0 < \check{\delta} \le d$. Each f_i is a sum of at most $(4R_f/\check{\delta} + 2)^3$ pieces $g = f_i 1_{\mathrm{cube}}$, over half-open cubes of side $\check{\delta}/2$. In a tuple with one piece of each f_i, the supports are pairwise at distance $\ge \check{\delta}$, and each of them lies in a ball of radius $\check{\delta}/4$.

(b) Let, besides, $\check\delta \le 1/(4n)$ and $a_K < 5/(8n)$. Then there is $v \in a_K\mathbb{Z}^3$ such that every cube of such a tuple, translated by v, is at distance $\ge \check\delta/2$ from all the planes $y_\mu \in \mathbb{Z} + \frac12$, and so lies in one open cell of B_0. The condition on a_K holds for $K \ge j^\circ(\mathsf{b}, \check\delta)$.

(b′) Let $f_1, \dots, f_n$ be of the type $(d; R_f)$, let $h = L^{-i} > n(2R_f + d + 1)$, and let $N \ge 2h$. Then there is $o \in \mathbb{Z}^3$ such that each of the n balls of radius R_f that contain the supports of $f_1, \dots, f_n$ in the definition of the type is at distance $\ge d/2$ from the walls of $B_i(o)$.

(c) On any board, let the sources of one cell be pairwise at distance $\ge \check\delta$, each in a ball of radius $\check\rho$ and at distance $\ge \check\delta/2$ from the walls. Then their mirror copies in all the cells are pairwise at distance $\ge \check\delta$, each lies in a ball of radius $\check\rho$, and they have the same sup norms. ■

Proof. Step (a): the pieces. If $2R_f < N$, the ball lifts to a cube of side $2R_f$ in $\mathbb{R}^3$, which we cut by the grid $(\check\delta/2)\mathbb{Z}^3$; an interval of length $2R_f$ meets at most $4R_f/\check\delta + 2$ of the grid intervals. If $2R_f \ge N$, we cut the fundamental cube $[0, N)^3$, the last row being smaller; per axis there are at most $2N/\check\delta + 1 \le 4R_f/\check\delta + 2$ intervals. Pieces of different f_i are at distance $\ge d \ge \check\delta$, and a half-open cube of side $\check\delta/2$ lies in a closed sup ball of radius $\check\delta/4$.

Step (b): the translation. We fix μ; the i-th cube projects to $[\alpha_i, \alpha_i + \check\delta/2]$. The values of v_μ modulo 1 for which $[\alpha_i + v_\mu, \alpha_i + \check\delta/2 + v_\mu]$ comes closer than $\check\delta/2$ to $\mathbb{Z} + \frac12$ form an *open* arc of length $\check\delta/2 + \check\delta = 1.5\,\check\delta$. Altogether these arcs have length $\le 1.5\, n\check\delta \le 3/8$. Their complement in $\mathbb{R}/\mathbb{Z}$ consists of at most n *closed* arcs of total length $\ge 5/8$, so one of them has length $\ge 5/(8n) > a_K$. That arc contains a point of $a_K\mathbb{Z}/\mathbb{Z}$, which is a grid of spacing a_K because $1/a_K \in \mathbb{Z}$; we take v_μ there. An interval at positive distance from all the walls contains none of them; hence each translated cube lies in one open cell. Finally, if $K \ge j^\circ$, then $\mathrm{sep}^{[j^\circ - 1]} \ge \mathrm{sep}^\circ$ forces $\check\delta L^{j^\circ - 1}/M \ge \mathrm{sep}^\circ$, and so $a_K \le L^{-j^\circ} \le \check\delta/(LM\,\mathrm{sep}^\circ) < 5/(8n)$.

Step (b′)*: the point o.* We fix μ and call $o_\mu \in \{0, \dots, h-1\}$ bad if some wall $o_\mu + h(\mathbb{Z} + \frac12)$ lies at distance less than $d/2$ from the projection of a ball, which is an interval of length $\le 2R_f$. For one ball the bad o_μ are points of spacing one in an open interval of length $2R_f + d$, so there are at most $2R_f + d + 1$ of them. As $n(2R_f + d + 1) < h$, a good o_μ remains. The condition $N \ge 2h$ is the condition $N/h = 2L^{m_l + i} \ge 2$.

Step (c): the mirror copies. Inside one cell the copies are the image of the sources under one isometry, which preserves distances and sup norms and maps each ball to a ball of the same radius. If two cells are different, there is a coordinate μ in which they differ. Then, on the circle $\mathbb{R}/N\mathbb{Z}$, each arc that joins the two projections meets a wall; as each projection keeps a distance $\ge \check\delta/2$ from that wall, the distance is $\ge \check\delta$. □

Lemma 8.30 (The bound per occupied cell)**.** In the setting of Lem. 8.28, let $B_i(o)$ be its board, assume the data of (R1) of Thm. 8.17, and let each source lie in one open cell, at distance $\ge \check\delta/2$ from the walls; let the factors F_{cell} be as in Lem. 8.28 (d). We call a cell occupied if a source lies in it or if $F_{\mathrm{cell}} \ne 1$, and we write q for the number of occupied cells. Then

$$\Big|\Big\langle \prod_{\text{cells}} \Big[F_{\mathrm{cell}}(V_{K-s}) \prod_{t \in I_{\mathrm{cell}}} (\mathcal{O}_K g_t - c_K^{(s)} g_t)\Big]\Big\rangle\Big| \le \prod \sup|F_{\mathrm{cell}}|\; G_{1,s}^Q\, e^{q\, c_{\mathrm{vol}}^{(s)} h^3} \prod \|g_t\|_\infty .$$

■

Clause (R2) of Thm. 8.17 is the case $s = 0$, $F = 1$, $o \in \mathbb{Z}^3$.

Proof. By Lem. 8.27 C and B, the mean is $\langle \prod_{\mathrm{cells}} \Phi_{\mathrm{cell}} Y_{\mathrm{cell}} \rangle$, where $\Phi_{\mathrm{cell}} = E[F_{\mathrm{cell}}|U]$ and Y_{cell} is the product of the centred sources of the cell; each $\Phi_{\mathrm{cell}} Y_{\mathrm{cell}}$ is a function of its cell (Lem. 8.28 (b) and Lem. 8.27 B). We apply Prop. A.12 (f) over the q occupied cells: the mean is at most $\prod_{\mathrm{cells}} \|\Phi_{\mathrm{cell}} Y_{\mathrm{cell}}\|_{\mathrm{ch}}$, where $\|\cdot\|_{\mathrm{ch}}^{N_c} = \langle \mathrm{Fold}(\cdot) \rangle$ (Def. A.10).

We fix an occupied cell and denote by n_{cell} the number of its sources. First let $n_{\mathrm{cell}} \geq 1$. By Lem. 8.28 (d), $\langle \mathrm{Fold}(\Phi Y)\rangle$ is a mean of the form of Lem. 8.25: its observable is $\prod_{\varkappa'} F^{[\varkappa']}(V_{K-s})$, whose sup is at most $(\sup|F|)^{N_c}$, and it has $n_{\mathrm{cell}}N_c$ sources, which by Lem. 8.29 (c) satisfy the data $(\check\delta;\check\rho)$ and have the sup norms of the sources of the cell. Lem. 8.25 bounds it by $(\sup|F|)^{N_c} G_{1,s}^{n_{\mathrm{cell}}N_c} e^{c_{\mathrm{vol}}^{(s)}N^3} \prod \|g_t\|^{N_c}$. As $N^3/N_c = h^3$, the N_c-th root is $\sup|F|\, G_{1,s}^{n_{\mathrm{cell}}} e^{c_{\mathrm{vol}}^{(s)}h^3} \prod_{t\in I_{\mathrm{cell}}} \|g_t\|$. If $n_{\mathrm{cell}} = 0$, then $Y = 1$, and Lem. 8.28 (d) gives $\langle \mathrm{Fold}\,\Phi\rangle = \langle \prod_{\varkappa'} F^{[\varkappa']}(V_{K-s})\rangle \leq (\sup|F|)^{N_c}$, which is below the bound claimed: its N_c-th root is at most $\sup|F| \leq \sup|F|\, e^{c_{\mathrm{vol}}^{(s)}h^3}$. The product of these bounds over the occupied cells gives the claim. For (R2) (ii) we use Lem. 8.29 (b′). □

Remark 8.31 (Why one cell). If the support of a source meets $r > 1$ cells, then $\mathcal{O}_K(g)$ is a *sum* over cells, and the moment is not a product of functions of cells.

Remark 8.32 (The placing of the sources). For the cumulants and for σ we use unit cells, we translate the *sources* by $a_K\mathbb{Z}^3$, and we cut into pieces of side $\check\delta_n/2$. This is allowed, since S_n^T and σ are invariant under $a_K\mathbb{Z}^3$. It is not allowed for moments centred at $\hat c$, which are invariant under $\mathbb{Z}^3$ alone. For moments centred at $\hat c$, with the sources placed anywhere and $n = 1$ included, we move the board instead, by $o \in \mathbb{Z}^3$, with $h > n(2R_f + d + 1)$. Both forms hold: they are (R2) (i) and (R2) (ii) of Thm. 8.17.

Lemma 8.33 (Uniform bounds, short runs and local boundedness in b)**.** Thm. 8.17 (R3) holds: the bounds on $\sigma_{K,N,\beta}$ and on S_n^T hold for all $K \geq 1$, for all N of the data and wherever the balls lie, and they hold for the limits of (T3) of Thm. 8.2 too. Moreover, $\sum_{\pi\in\mathrm{Part}(n)}(|\pi|-1)! \leq n!\,1.45^{n+1}$ for every $n \geq 2$, and $C_n^{(2)}$ is locally bounded in b. ■

Proof. Step (Case $K \geq j^\circ(\mathsf{b},\check\delta_n)$): pieces on the unit board. Let K be finite. Since S_n^T and σ are n-linear in bounded entries, each of them is a sum of its values at the tuples of pieces of Lem. 8.29 (a) applied with $\check\delta = \check\delta_n$, and there are at most $(4R_f/\check\delta_n + 2)^{3n}$ such tuples. We fix one tuple and take v as in Lem. 8.29 (b). By Step 7 ("At finite K") of the proof of Thm. 8.2, which holds for bounded entries, S_n^T and σ are unchanged under τ_v: both are computed from means, and these are invariant under $a_K\mathbb{Z}^3$.

For the translated pieces we put $Y_i = \mathcal{O}_K(g_i) - \hat c_K(g_i)$. Cumulants of order ≥ 2 do not depend on shifts by constants; hence

$$S_n^T = \sum_{\pi}(-1)^{|\pi|-1}(|\pi|-1)! \prod_{J\in\pi}\langle\prod_{i\in J} Y_i\rangle.$$

We bound each factor by Thm. 8.17 (R2) (i), on the *same* board B_0 for all factors, applied with the data $(\check\delta_n;\check\delta_n/4)$ and with $q_J \leq |J|$: each factor is at most $(G_1 e^{c_{\mathrm{vol}}})^{|J|}\prod_J\|g_i\|$. For σ we write, for each of the translated pieces $g = g_i$ and with $Y = Y_i$, $\mathcal{O}_K(g) - \langle\mathcal{O}_K(g)\rangle = Y - \eta$, where $|\eta| = |\langle Y\rangle| \leq G_1 e^{c_{\mathrm{vol}}}\|g\|$ by (R2) applied with $Q = 1$; expanding the product gives 2^n terms, each at most $(G_1 e^{c_{\mathrm{vol}}})^n\prod\|g_i\|$.

Step (Case $K < j^\circ$): short runs. The barycentres of the plaquettes of one orientation form a translate of $a_K\mathbb{Z}^3$; a half-open cube of side $\check\delta/2$ contains at most $(\check\delta L^K/2+1)^3$ of them; there are three orientations, and $0 \leq a_p \leq 2$. Hence $|\mathcal{O}_K(g)| \leq 6L^K(\check\delta L^K/2+1)^3\|g\| \leq B_0(\mathsf{b},\check\delta)\|g\|$. The expansion of S_n^T over the partitions, now with $Y = \mathcal{O}_K(g)$, gives the bound on S_n^T. For σ we bound the product of the n factors directly, each factor by $|\mathcal{O}_K(g) - \langle\mathcal{O}_K(g)\rangle| \leq 2B_0(\mathsf{b},\check\delta)\|g\|$; this gives $|\sigma| \leq \prod_i \sup|\mathcal{O}_K(g_i) - \langle\mathcal{O}_K(g_i)\rangle| \leq (2B_0(\mathsf{b},\check\delta))^n\prod\|g_i\|$.

In neither case does N enter; if supports are far apart, the one effect is that they lie in different cells, and no factor depends on their distance. So the bounds of the two cases hold for all N of the data and wherever the balls lie.

Step (The sum): the combinatorial bound. Let $a_n := \sum_\pi (|\pi|-1)!$. Its exponential generating function is $-\log(2-e^x)$, with derivative $2/(2-e^x)-1$; hence $a_{n+1} = 2f_n$ for $n \geq 1$, where $f_n = \sum_{m\geq 0} m^n 2^{-m-1}$. As $t^n e^{-t\log 2}$ is unimodal, the sum is at most the integral plus the maximum:

$$f_n \leq \frac{1}{2}[n!/(\log 2)^{n+1} + (n/(e\log 2))^n] \leq 1.2214\, n!\,(1/\log 2)^n.$$

Therefore $a_n \leq 2.443\,(n-1)!\,1.4427^{n-1} \leq n!\,1.45^{n+1}$ for $n \geq 2$.

Step (Local boundedness in b*): the constants.* The integer $j^\circ(\mathsf{b}, \check\delta)$ is non-decreasing in b and finite (Lem. 8.4 (N2); (P1.0), that is, Prop. 8.11 (G°)), hence bounded on compact sets. The functions amp, c_a, c_h, p_0 and ε are continuous, and $\check r$ is a bounded step function. The tail of C_W is bounded uniformly: $N_1^{[j^\circ+t]} \leq L^{3t} N_1^{[j^\circ]}$ (item (ii) of the proof of Lem. 8.4 (N3)) is compensated by $e^{-(\kappa-1)(L^t \mathrm{sep}^\circ - 1)}$ (Lem. 8.4 (N2)), with $c_a \leq 1/32$. In this step (N2) of Lem. 8.4 and item (ii) of the proof of its (N3) are taken with $(\check\delta, \check\rho, \mathrm{sep}^\circ, j^\circ)$ in the place of $(d, R_f, \mathrm{sep}_*, j^*)$, and their proofs hold as they stand: $\mathrm{sep}^{[i]}$ and $N_1^{[j]}$ are the same functions, the proof of (N2) uses no property of sep_* other than $\mathrm{sep}_* > 0$, and item (ii) uses no property of j^*.

Step (The limits of (T3)*): the bounds pass to the limits.* The limits of (T3) are limits of numbers bounded in this way, and so they obey the same bounds; for continuous entries the bounds hold for the extension given by (T1). □

Remark 8.34 (How sep° enters the closed form of §9). Let $j^\circ \geq 2$, and put $t = \check\delta L^{j^\circ-2}/M$. By the choice of j°, $\mathrm{sep}^{[j^\circ-2]} < \mathrm{sep}^\circ$; with $\check r(x) \leq \log x/1056 + 4$ this gives $t < \mathrm{sep}^\circ + 11 + \log(L^{j^\circ-2}\mathsf{b})/528$. Hence $t(1-1/528) < \mathrm{sep}^\circ + 11 + [\log \mathsf{b} + \log(M/\check\delta)]/528$. Thus sep° enters L^{j°, r°, $N_1^{[j]}$ for $j \leq j^\circ$, and $B_0(\mathsf{b}, \check\delta)$ as a letter.

Lemma 8.35 (Every positive label, by dilation)**.** Thm. 8.17 (R4) holds, with the integer z subject to the conditions $L^z\mathsf{b} \geq x_s$, $K - z \geq 1$, $m + z \geq m'$ and $L^z l \geq 2Ms(L^{z+1}\mathsf{b})$. ■

Proof. We take the dilation $x \mapsto L^z x$ of Step 5 of the proof of Thm. 8.2. It maps the lattice of (K, l) onto the lattice of $(K-z, L^z l)$, sending plaquettes to plaquettes and barycentres to barycentres. The two graphs thus correspond to each other, and β is the same for both; as Wilson's measure is determined by the graph and by β, the dilation carries the one measure onto the other. Under this correspondence $\mathcal{O}_{K-z}(f^{[z]}) = L^{-z}\mathcal{O}_K(f)$, and a cumulant of order n, like a centred moment of order n, is homogeneous of degree n in its entries. It follows that

$$S^T_{n;K,l,\beta}(f) = L^{zn} S^T_{n;K-z,L^z l,\beta}(f^{[z]}),$$

where $\beta = L^{K-z}(L^z\mathsf{b})$. In the same way the centred moment of order n obeys $\sigma_{K,l,\beta}(f) = L^{zn}\sigma_{K-z,L^z l,\beta}(f^{[z]})$. Under the dilation the type changes as $(d; R_f) \to (L^z d; L^z R_f)$, while the sup norms stay the same.

By the four conditions, $(L^z\mathsf{b}, L^z l, K-z)$ are data: $L^z\mathsf{b} \geq x_s$ is the condition on the label, $K - z \geq 1$ the condition on the number of steps, $m + z \geq m'$ the condition on the exponent of the side $L^z l$, and $L^z l \geq 2Ms(L^{z+1}\mathsf{b})$ is (Tor) for the label $L^z\mathsf{b}$. So both bounds of (R3) apply to the dilated entries for these data. With $\check\delta_n = \min\{L^z d, 1/(4n)\}$, the bound for σ reads

$$|\sigma_{K-z,L^z l,\beta}(f^{[z]})| \leq (4L^z R_f/\check\delta_n + 2)^{3n}(2G(L^z\mathsf{b}, \check\delta_n))^n \prod_i \|f_i\|_\infty,$$

and, since $2^n \leq 1.45^{n+1} n!$ for $n \geq 2$, the factor in front of the product obeys

$$(4L^z R_f/\check\delta_n + 2)^{3n}(2G(L^z\mathsf{b}, \check\delta_n))^n \leq C_n^{(2)}(L^z\mathsf{b}; L^z d, L^z R_f).$$

Together with the two identities above, the two bounds of (R3) thus give $|S^T_{n;K,l,\beta}(f)| \leq L^{zn} C_n^{(2)}(L^z\mathsf{b}; L^z d, L^z R_f)\prod_i \|f_i\|_\infty$ and the same bound for $|\sigma_{K,l,\beta}(f)|$. One integer suffices for all

labels $\mathsf{b}' \in U(\mathsf{b})$, namely $z = z(\mathsf{b})$ of Def. 1.10, once the sides are large enough. The constant does not depend on the side. □

Lemma 8.36 (Compactness; the orders of the limits)**.** Thm. 8.17 (R5) holds in each of its parts:

(a) the continuum limit taken first;
(b) the volume limit taken first;
(c) the two limits taken jointly;
(d) the joint limits with $K_j/l_j^3 \to \infty$.

■

Proof. Step (Carrying): entries on the torus. For a function f on $\mathbb{R}^3$ with compact support, $f^{\langle l\rangle}(x)$ denotes the sum of the values of f at all the lifts of x. If l is larger than twice the diameter of the union of the supports, then on that union the covering map is an isometry for the sup distance, and so the periodised tuple has the type of the tuple.

Step (Frames): countably many frames. A frame is an n-tuple $(A_1, \dots, A_n)$ of sets, each a finite union of closed cubes with rational corners, the sets being pairwise disjoint; there are countably many frames. Every tuple of compact supports with the separation d is carried by a frame: for A_i we take the union of those closed cubes of a rational grid of side $< d/4$ that meet $\operatorname{supp} f_i$, and then any two of the A_i are $\geq d/2$ apart. For each i, the space $C(A_i)$ of the continuous functions vanishing off A_i is separable.

Let a frame be of the type $(d_A; R_A)$. On the entries that it carries, all the forms $S^T_{n;K,l,L^K\mathsf{b}'}$ and $S_n^{(l,\mathsf{b}')}$ are n-linear and bounded by $L^{zn}C_n^{(2)}(L^z\mathsf{b}'; L^z d_A, L^z R_A)\prod\|\cdot\|_\infty$. At $z = z(\mathsf{b})$ this bound is uniform in (K, l) with $K \geq z+1$ and l large, and in $\mathsf{b}' \in U(\mathsf{b})$, by (R3), (R4), Def. 1.10 and Lem. 8.33: the last assertion of the lemma, the local boundedness of $C_n^{(2)}$ in the label, is used on the set $L^{z(\mathsf{b})}U(\mathsf{b})$, which by the definition is compact and contained in $[x_s, \infty)$. So the forms are equicontinuous on each frame. Taking a diagonal sequence over the frames and over countable dense subsets of the spaces $C(A_i)$, we obtain convergence on these subsets, and the equicontinuity extends it to all entries.

Step (a)*: the continuum limit first.* We apply Step (Frames) to the forms $S_n^{(l_j,\mathsf{b}_j)}$, which exist by Thm. 8.2 (T3), (T6), for continuous entries as well.

Step (c)*: the joint limits.* We apply Step (Frames) to the forms $S^T_{n;K_j,l_j,L^{K_j}\mathsf{b}_j}$.

Step (b)*. Volume limit first.* At fixed K and β, periodize the torus configurations on $a_K\mathbb{Z}^3$. Prop. B.3(c) gives a weak limit along a subsequence of sides, and the limit ν_K is a translation-invariant Gibbs law. Choose the same subsequence for all K by diagonal extraction. Prop. B.28 gives its cubic invariance and reflection positivity at coordinate site and bond mirrors. Every fixed lattice observable $\mathcal{O}_K(f)$ with compactly supported f is continuous and local. Thus its moments and cumulants converge on this subsequence, and the uniform bounds pass to ν_K. The frame argument then extracts the limit in K on all separated tuples.

Step (d)*: comparison along the same sides and labels.* Suppose $\mathsf{b}_j \to \mathsf{b} > 0$ and put $z = z(\mathsf{b})$ and $N_j = L^z l_j$. For large j the dilated labels $L^z\mathsf{b}_j$ belong to the compact set $L^zU(\mathsf{b}) \subset [x_s, \infty)$ and the sides satisfy (Tor) uniformly on this set. For Lipschitz entries, (T2), (T3) and (T6) of Thm. 8.2 give

$$|S^T_{n;K_j,l_j,L^{K_j}\mathsf{b}_j}(f) - S_n^{(l_j,\mathsf{b}_j)}(f)| \leq L^{zn}C'_{n,j}(K_j - z)\theta^{K_j - z}\prod_i \max\{\|f_i\|_\infty, 40ML^{-z}\operatorname{Lip} f_i\}.$$

The cutoff threshold is independent of the side and locally bounded in the label. The displayed expression for C'_n in (T2) gives $\log C'_{n,j} \leq CN_j^3 + C'$ with C, C' uniform on the compact set of labels. Since $N_j = L^z l_j$ with fixed z, the hypothesis $K_j/l_j^3 \to \infty$ implies $K_j/N_j^3 \to \infty$, and the right side tends to zero. The two sequences thus have the same limit points along the given sides and labels.

For continuous entries we approximate uniformly inside a frame and invoke the equicontinuity. □

Lemma 8.37 (What every limit inherits)**.** Thm. 8.17 (R6) holds in its parts (i) to (vii). ■

Proof. Step (i), (ii)*: reality, symmetry and the bound.* The forms S_n are limits of real, symmetric, n-linear forms that all obey one bound; hence they are real, symmetric and n-linear themselves, and they obey the same bound.

We fix a convention for the rest of this proof and for Lem. 8.38: $C_n^{(2)}$ at $(d;R)$ denotes the constant of (ii). For a fixed label $\mathsf{b} \geq x_s$ this is the number $C_n^{(2)}(\mathsf{b};d,R)$ itself. For a fixed label $0 < \mathsf{b} < x_s$ it is $L^{zn}C_n^{(2)}(L^z\mathsf{b};L^z d,L^z R)$, with z as (R4) admits. For moving labels it is the supremum of the latter over $\mathsf{b}' \in U(\mathsf{b})$, taken at $z = z(\mathsf{b})$. The arguments below depend on this constant only through the fact that it is independent of K, of l and of the places of the entries.

Step (iii)*: invariance.* At finite (K,l) the identity $S_n^T(rX) = S_n^T(X)$ is exact for $r^\circ \in W_3$, $v \in a_K\mathbb{Z}^3$ (r: $x \mapsto r^\circ x + v$), and these sets of maps grow with K. In the order (a), the identity at finite K passes to the limit in K of Thm. 8.2 (T3), (T6). Thus $S_n(rX) = S_n(X)$ for $r^\circ \in W_3$ and for every v in the union A of the lattices $a_K\mathbb{Z}^3$, a dense set.

For all v and for C^1 entries, the argument is that of the part "All v" of Step 7 of the proof of Thm. 8.2, changed in one point: where that step uses C_n, we use $C_n^{(2)}$ at $(d/2;R_f + d/4)$. For a sequence $v_j \in A$ with $v_j \to v$ the outcome is

$$|S_n(\tau_v X) - S_n(\tau_{v_j} X)| \leq nC_n^{(2)}\sqrt{3}\,|v - v_j| \max_i [\mathrm{Lip} f_i \prod_{k\neq i} \|f_k\|_\infty] \to 0,$$

and since $S_n(\tau_{v_j}X) = S_n(X)$, the identity holds at v. For continuous entries we conclude by the uniform continuity of the entries together with (ii).

Step (iv)*: reflection positivity.* Let $X_1,\dots,X_r \in \mathfrak{S}(\{y_\mu > 0\})$; when l is large, their supports lie in $\{0 < y_\mu < l/2\}$. At finite (K,l), Step 8 (a), (b) of the proof of Thm. 8.2 yields $\sum \bar\zeta_i\zeta_k\,\sigma_{K,l,\beta}(\theta X_i \cdot X_k) \geq 0$, where $\sigma_{K,l,\beta}$ is σ_S taken at S^T. Now σ is a polynomial in the $S_{|J|}$, each factor of which converges, and a limit of non-negative numbers is non-negative, whether the limit is single, iterated or joint. For an arbitrary c we write $\theta_c = \tau_v\theta_0\tau_{-v}$ with $v = ce_\mu$, and (iii) carries the inequality from θ_0 over to θ_c.

Step (v)*: Step 9 of the proof of Thm. 8.2 on* $\mathbb{R}^3$. With $\mathbb{R}^3$ in place of T_l, that step changes as follows. Let ξ be a point whose entries are pairwise different, let $\delta = \min|\xi_i - \xi_k|/4$, let B_i be the open ball with centre ξ_i and radius δ, and let $Q = \prod B_i$. We choose χ_i smooth, with $0 \leq \chi_i \leq 1$, equal to 1 on B_i and carried by the closed ball with centre ξ_i and radius $3\delta/2$; any two of these closed balls are at least δ apart. For $F \in C_c^\infty(Q)$ let F^{per} be the function that repeats F with the period 4δ in every one of the $3n$ coordinates, let $\hat F(k)$ be its Fourier coefficients, and let $e_k(x) = e^{2\pi ik\cdot x/(4\delta)}$.

In the i-th variable the nearest copy of $\operatorname{supp} F$ in F^{per} other than $\operatorname{supp} F$ itself, namely $\operatorname{supp} F$ shifted by 4δ in one coordinate of that variable, begins at the distance 3δ from ξ_i, while the support of χ_i ends at $3\delta/2$; therefore $F = (\chi_1 \otimes \cdots \otimes \chi_n)F^{\mathrm{per}}$. We define

$$D_Q(F) := \sum_k \hat F(k)\, S_n^{\mathbb{C}}(\chi_1 e_{k_1},\dots,\chi_n e_{k_n}).$$

The bounds of the part "The local construction" of Step 9, which hold there with the period l, the constant $c(n,l)$ and the bound $C_n(\delta)$, give here, with $C_n^{(2)}$ at $(\delta;3\delta/2)$,

$$|D_Q(F)| \leq 2^n C_n^{(2)}\, c(n,4\delta) \max_{|\alpha|\leq 3n+1} \sup|\partial^\alpha F|.$$

The parts "On products", "Uniqueness on Q", "Gluing" and "The functional σ_S" of Step 9 then carry over word for word.

Step (vi)*: dilation.* At finite K we use Step 5 of the proof of Thm. 8.2, matching the subsequence (m_j) belonging to the label b with the subsequence $(m_j + z)$ belonging to the label $L^z\mathsf{b}$.

Step (vii)*: the group of symmetries.* The group in (vii) is closed: if $R_j \to R$, then $\|f \circ R_j^{-1} - f \circ R^{-1}\|_\infty \to 0$ on carriers that stay disjoint, and (ii) carries the invariance to the limit. The group contains W_3, by (iii). Write H for its rotation subgroup and $O = W_3 \cap \mathrm{SO}(3)$, so $O \subset H$ and $|O| = 24$. We give the subgroup argument explicitly.

If H is infinite, compactness gives nonidentity rotations $h_j \to 1$. Write $h_j = \exp(\theta_j J_{v_j})$, where $\theta_j \downarrow 0$ after taking a subsequence, $|v_j| = 1$ and $J_v x = v \times x$. Passing to a further subsequence gives $v_j \to v$. For each real t, choose integers m_j with $m_j\theta_j \to t$. Then $h_j^{m_j} \to \exp(tJ_v)$, which belongs to H by closedness. Conjugating by O gives the circles with axes wv, $w \in O$. Their axes span $\mathbb{R}^3$, because the action of O is irreducible: averaging a vector with its half-turn about a coordinate axis projects onto that axis, and cyclically permuting the axes then generates all three from any nonzero invariant subspace. Choose three independent axes v_1, v_2, v_3 from this orbit. The map

$$(t_1, t_2, t_3) \longmapsto e^{t_1 J_{v_1}} e^{t_2 J_{v_2}} e^{t_3 J_{v_3}}$$

has invertible derivative at zero. The inverse function theorem therefore puts a neighborhood of 1 in H. As SO(3) is connected, $H = \mathrm{SO}(3)$.

Suppose instead that H is finite, of order $q \geq 24$. Let $P \subset S^2$ be the set of poles of its nonidentity rotations, and let $n_1, \ldots, n_r \geq 2$ be the stabilizer orders for its orbits in P. Each stabilizer is a finite group of rotations about one axis, hence cyclic. Counting pairs (h, x) with $h \neq 1$ and $hx = x$ gives

$$\sum_{i=1}^{r} \frac{q}{n_i}(n_i - 1) = 2(q-1), \qquad \sum_{i=1}^{r} \left(1 - \frac{1}{n_i}\right) = 2 - \frac{2}{q}.$$

Thus $2 \leq r \leq 3$. If $r = 2$, then $1/n_1 + 1/n_2 = 2/q$ and $n_i \leq q$ force $n_1 = n_2 = q$, giving an invariant axis, contrary to the irreducibility of O. For $r = 3$, order the n_i. The inequality $\sum_i 1/n_i > 1$ gives $n_1 = 2$ and $n_2 \leq 3$. The case $(2, 2, n_3)$ gives $n_3 = q/2 \geq 12$. Its cyclic stabilizer has index two, hence is normal, so its axis is invariant under H, again impossible. The remaining triples are $(2, 3, 3)$, $(2, 3, 4)$ and $(2, 3, 5)$, giving $q = 12, 24, 60$. Since 24 divides q, only $q = 24$ remains, and $H = O$.

Finally, $-1 \in W_3$, so the whole symmetry group is $H \cup (-1)H$. It is therefore W_3 or O(3). □

Lemma 8.38 (Hilbert space, vacuum, energy and momenta)**.** The data $(D, \langle\cdot,\cdot\rangle, 1 = \text{empty tuple}, \mathcal{T}_t^\circ = \tau_{te_\mu}, U^\circ(v) = \tau_v)$, with D and the form of Thm. 8.17 (R7) and with v in the mirror plane, satisfy the hypotheses of Lem. D.1 applied with $N = 2$, the dimension of the mirror plane. The Hilbert space that they define is separable, and the construction extends to the class of continuous entries. Consequently, by Lem. D.1 (a) to (d), Thm. 8.17 (R7) holds. ■

Proof. Step (The form): Hermitian, positive, normalised. The form is Hermitian: σ_S is real, and $\sigma_S(\theta X \cdot X') = \sigma_S(X \cdot \theta X') = \sigma_S(\theta X' \cdot X)$, the first equality by the invariance under $\theta \in W_3$ and the second by the symmetry. The form is positive by (R6) (iv). Moreover $\sigma_S(\emptyset) = 1$.

Step (The maps): translations of tuples. The maps τ_{te_μ}, for $t \geq 0$, and τ_v send $\mathfrak{S}(\mathbb{E}_+)$ into itself and act linearly on D; the first family is a semigroup and the second a group; the two commute, and both fix the empty tuple.

Step (1)*: symmetry and isometry.* From $\theta\tau_{te_\mu} = \tau_{-te_\mu}\theta$ and (R6) (iii) we get $\sigma_S(\theta X \cdot \tau_t X') = \sigma_S(\tau_{-t}\theta X \cdot X') = \sigma_S(\theta\tau_t X \cdot X')$. From $\theta\tau_v = \tau_v\theta$ and (R6) (iii) again, $\sigma_S(\theta\tau_v X \cdot \tau_v X') = \sigma_S(\theta X \cdot X')$, so that τ_v is an isometry.

Step (2)*: bounded matrix elements.* Let $F = \sum \zeta_i X_i$, where X_i has n_i entries and the separation d_i. For all $t \geq 0$ the tuple $\theta X_i \cdot \tau_t X_k$ has $n_i + n_k$ entries and is of the type $(\min\{d_i, d_k, 2\,\mathrm{dist}(\text{supports}, \text{mirror})\}; R)$, where R is a radius such that each entry of X_i and of X_k has its support in a ball of radius R. The bound that (R3), (R4) give for σ does not depend

on the places. Hence the number $c_F := \sup_{t\geq 0} |\langle F, \mathcal{T}_t^\circ F\rangle|$ of hypothesis (2) of Lem. D.1 obeys $c_F \leq \sum |\zeta_i||\zeta_k| C(i,k) < \infty$, where $C(i,k)$ is that bound for the pair (i,k). What matters here is that the walls of the unit board B_0 in the proof of (R3) enclose each piece of Lem. 8.29 (a), and that no cell has to contain the whole tuple. Indeed, Lem. 8.29 (b) places each piece in one open cell of B_0.

Step (3)*: continuity.* By the definition of $\mathfrak{S}$, the entries of the tuples of $\mathfrak{S}(\mathbb{E}_+)$ are C^1. We pass from $\tau_t X_k$ to X_k by changing one entry at a time, which is the telescoping of Step 7 of the proof of Thm. 8.2; each term then has one entry of the form $\tau_{te_\mu} f - f$, and for small t the carriers remain $d'/2$ apart, d' being the separation of $\theta X_i \cdot X_k$. This yields

$$|\sigma_S(\theta X_i \cdot \tau_t X_k) - \sigma_S(\theta X_i \cdot X_k)| \leq C\sqrt{3}\, t \max \mathrm{Lip} \to 0.$$

For τ_v, the telescoping in the entries gives $|\sigma_S(\theta X_i \cdot \tau_v X_k) - \sigma_S(\theta X_i \cdot X_k)| \leq C\sqrt{3}\,|v| \max \mathrm{Lip}$, which tends to 0 with v.

Step (Conclusions): the operators. Lem. D.1 (a) to (c) now provides $\|\mathcal{T}_t^\circ F\| \leq \|F\|$, the operators $\mathcal{T}_t$ and $U(v)$, the identity $\mathcal{T}_t = e^{-tH}$ with $H \geq 0$ and $H\Omega = 0$, the momenta P_1 and P_2, and a joint spectral measure carried by $[0,\infty) \times \mathbb{R}^2$. For Lem. D.1 (d), we use the group W_2 of (R7), whose elements, acting on the mirror plane, are the 8 signed permutation matrices of that plane. Let W° be the action of an element w of W_2. It is a linear bijection of D that fixes the empty tuple 1; it commutes with θ and with τ_{te_μ}, it maps $\mathbb{E}_+$ onto itself, and $W^\circ \tau_v = \tau_{wv} W^\circ$. By $W^\circ \theta = \theta W^\circ$ and (R6) (iii) it leaves the form unchanged, $\langle W^\circ F, W^\circ F'\rangle = \langle F, F'\rangle$; so we obtain a unitary representation of W_2. Finally, Lem. D.1 (a) gives $|\langle F, \mathcal{T}_t F\rangle| \leq \|F\|^2$, and therefore $c_F = \langle F, F\rangle$.

Step (Separable): a countable dense set. By (R6) (ii), for tuples inside one frame (Step (Frames) of the proof of Lem. 8.36), $\|[X] - [X']\|^2$ is bounded in terms of the sup norms of the differences of the entries. There are countably many frames, and every $C(A_i)$ is separable; hence the Hilbert space is separable.

Step (Continuous entries): extension by continuity. Let f be continuous with compact support in $\mathbb{E}_+$. We choose C^1 functions $f^e \to f$ uniformly, all carried by one compact subset of $\mathbb{E}_+$. Then

$$\|[f^e] - [f^{e'}]\|^2 = S_2(\theta(f^e - f^{e'}), f^e - f^{e'}) \leq C_2^{(2)} \|f^e - f^{e'}\|_\infty^2,$$

and for tuples we change the entries one at a time. The classes therefore form a Cauchy sequence, the limit does not depend on the approximating functions, and the form extends by continuity. Moreover $\langle \Omega, [f]\rangle = \sigma_S(f) = 0$, because $\mathrm{Part}_{\geq 2}(1)$ is empty.

Every mirror c can be used equally well, by (R6) (iii). An element of W_3 that carries e_μ to e_ν intertwines the three constructions, one for each direction. □

Lemma 8.39 (Reflection positivity without a collar; the limit laws on $\mathbb{R}^3$)**.** Thm. 8.17 (R8) holds in its parts (b) and (c). ■

Proof. Step (b)*: positivity without a collar.* Let $c \in L^{-s}(\mathbb{Z} + \frac{1}{2}) \subset a_K(\mathbb{Z} + \frac{1}{2})$. Since $N/2 = L^{m_l}$ is a multiple of L^{-s}, we also have $c + N/2 \in L^{-s}(\mathbb{Z} + \frac{1}{2})$. Hence T_+ and θT_+ are disjoint open sets whose boundaries are planes $y_\mu \in L^{-s}(\mathbb{Z} + \frac{1}{2})$. For F as in (b) let $\Phi := E[F(V_{K-s})|U]$. By Lem. 8.27 B, Φ is bounded and Borel and depends only on the bonds whose two ends lie in T_+; it therefore lies in the class $\mathfrak{F}_+$ of Cor. A.3. The reflection θ is $y_\mu \mapsto 2c - y_\mu$, and $2c \in L^{-s}\mathbb{Z}$; hence $E[\theta F|U] = \theta\Phi$, by Lem. 8.27 D, and $E[\overline{\theta F} F|U] = \overline{\theta\Phi}\Phi$, by Lem. 8.27 C. Consequently $E[\overline{\theta F} F] = \langle \overline{\theta\Phi}\Phi\rangle \geq 0$, by Cor. A.3 (a).

Step (c) *Densities: bounds on the marginals.* Let $K \geq s+1$, and let B be a finite set of bonds of depth s. The marginal of V_{K-s} on B has the density

$$p_B(V) = E[\prod_{c\in B} k_t(V(c), \mathsf{M}(V_{K-s-1})(c))],$$

where $t = ax_{K-s-1} = aL^{s+1}\mathsf{b}$ (the definition "Runs" in item 2 of §5.5). Because $1 - \cos \operatorname{ang}(g) = 1 - \frac{1}{2}\operatorname{tr} g \in [0,2]$, we have $z(t)^{-1}e^{-2t} \le k_t \le z(t)^{-1}$. Moreover $|k_t(g,h) - k_t(g',h)| \le z^{-1}t\,|\frac{1}{2}\operatorname{tr}((g-g')h^{-1})| \le z^{-1}t\,\|g-g'\|$, uniformly in h. Hence p_B is bounded from above and from below and is equi-Lipschitz, uniformly in K and in N.

Step (c) $K \to \infty$ *at fixed* N. The law $\mu_{[s]}^{[K]}$ of V_{K-s} converges in total variation by Cor. 5.65 (b), which holds for b and N of the data by Thm. 8.2 (T7). So the marginal densities converge in L^1; by Arzelà–Ascoli they converge uniformly along subsequences, and the limit has a density that obeys the same bounds.

Step (c) $N_j \to \infty$. The pairs (s,B) form a countable set. Arzelà–Ascoli and a diagonal sequence give the uniform convergence of all the densities $p_B^{(N_j)}$, which is the same as convergence in total variation on every finite set of bonds. Marginalisation passes to the limit, and Kolmogorov's theorem yields the law on $G^{\text{bonds of } L^{-s}\mathbb{Z}^3}$.

Step (c) *Consistency: from depth* s *to depth* $s-1$. Let F be a bounded Borel function on a finite set B' of bonds of depth $s-1$. For each side N, Cor. 5.65 (c) gives $\int F\,d\mu_{[s-1]}^{\infty} = \int (KF)\,d\mu_{[s]}^{\infty}$, with

$$KF(V) = \int F(V') \prod_{c \in B'} k_{aL^s\mathsf{b}}(V'(c), \mathsf{M}(V)(c))\,dV';$$

by Lem. 8.27 A, this function is bounded, Borel and local. The map M is discontinuous, so that weak convergence would not carry the identity to the limit. Convergence in total variation on finite sets of bonds, by contrast, implies that both sides of the identity converge.

Step (c) *Invariances: identities in the limit.* Each of the following is an identity or an inequality between integrals of bounded local Borel functions: the invariance under gauge transformations of finite support; the invariance at depth s under the maps of (R0), which comes from Lem. 8.20 and the invariance of $\mu_{K,N,\beta}$; and (b) for local F. All of them are valid for (K,N) with N large, and they remain valid for the limits in total variation, taken first in K and then in N.

Step (Other sequences). For $N_j \to \infty$ the argument requires of (K_j) only that $K_j \ge s+1$ from some j on, that is $s \le \liminf K_j - 1$. It provides a subsequence of (K_j, N_j), and along this subsequence the conclusions of (c) hold for these depths s. At finite K the consistency is the definition of the run (the definition "Runs" in item 2 of §5.5), which then takes the place of Cor. 5.65 (c).

Step ($\mathsf{b} < x_s$)*: small labels.* By Lem. 8.35, the run with the data (K,l,b) is the run with the data $(K-z, L^z l, L^z\mathsf{b})$. In the units of the data (K,l,b), which are the units before the dilation, the depth s of this run is the lattice $L^{-(s+z)}\mathbb{Z}^3$. □

9. Growth on flat test functions

In this section we bound the numbers $j^\circ, r^\circ, N_1^{[j]}, C_W, e_h, B_0(\mathsf{b}, \check\delta)$ of §8.3 in closed form by powers of $1/\check\delta$ on $0 < \check\delta \le 1$; this is clause (G1) of Thm. 9.4, whose clauses are (G1) to (G6+). A chessboard estimate, carried out cell by cell on the unit board together with Thm. 8.17 (R0) and (R1), gives bounds for moments that contain no factorial, (G3); from these we obtain the product form $A_5^n (n!)^5 \max\{1, d^{-5n}\}(1+R_f)^{3n}$, which is (G2). This bound holds uniformly in the cutoff K, in the positions of the supports and in the length N of the sides of the torus; by a dilation it extends from $\mathsf{b} \ge x_s$ to all $\mathsf{b} > 0$, as stated in (G2) (iv). We then pass to $\mathcal{S}_{\neq}(\mathbb{R}^{3n})$, where the growth condition of Def. C.17 holds with $(13, 1, C_E(\mathsf{b}), 13)$, (G4), together with reality, symmetry, invariance and positivity, (G5); and a single field set between two vectors obeys a bound free of n, (G6) and (G6+).

0.1. Inputs and conventions. (A) We use the table, radii and numbers of item 1 of §8.1, and the lattice data of Thm. 8.2. In particular, $\kappa \ge 6$, $\kappa_1 = \kappa + 1$, $M = L^{m'}$ with $m' \ge 1$, $M \ge 1920(\kappa_1+4)c_R$, and $L^{1/4} \ge 10^4 \mathsf{N}_{\mathrm{c}}^2$. The floors give $x_s \ge e^2$, $Lx_s \ge x_9^U$, $p_0 \ge 1$ and $\varepsilon \le 1/40$ on

$x \geq x_s$. We take $\mathrm{sep}^{[i]}$ and $N_1^{[j]}$ at $d = \check{\delta}$ and $R_f = \check{\rho}$; item 1 below recalls the numbers needed here.

The separated tuples, their centred functionals and reflection positivity are those of item 1 of §8.1. We use its convention $rf = f \circ r^{-1}$, with $\tau_v : x \mapsto x + v$, and $\|f\|_\sharp = \max\{\|f\|_\infty, 40M \operatorname{Lip} f\}$. A Euclidean type $(d; R)$ is a sup type $(d/\sqrt{3}; R)$, as in (S3) of §1.

By Lem. 8.4 (N2), $\mathrm{sep}^{[i+1]} \geq L \,\mathrm{sep}^{[i]}$ for $i \geq 0$ and $d > 0$. Its proof also gives $\mathrm{sep}^{[i]} \to \infty$: dL^i/M grows geometrically, whereas $r(L^i \mathsf{b})$ grows at most linearly in i.

We use the bound $c_a(x) \leq x^{-3/8}/4000$ on the whole range $x \geq L x_s$. Indeed, (O13) (ii) of item 0 of §6.2 imposes (Y.3) with κ_1 and $e^{-(c_h/2)p_0^2}$ for every $x \geq x_9^U$. Its non-negative c_a term and the factors at least 1 give $x^{1/4} \cdot 50 \cdot 5 \cdot 16 c_a(x) \leq x^{-1/8}$; now use $L x_s \geq x_9^U$.

Step 5 of the proof of Thm. 8.2 supplies dilation for $l = 2L^m$, $m \in \mathbb{Z}$, $K \geq 1 - m$, $\beta > 0$ and $z \in \mathbb{Z}$: with $f^{[z]}(x) = f(L^{-z}x)$, the data (K, l, β) and $(K - z, L^z l, \beta)$ describe the same graph and measure, and $\mathcal{O}_{K-z}(f^{[z]}) = L^{-z}\mathcal{O}_K(f)$. Thus the cumulants scale by L^{zn}; the corresponding limits exist together and obey the same identity along matched subsequences, also at moving labels. Step 7 of that proof supplies the finite-cutoff covariance $\mathcal{O}_K(f)(rU) = \mathcal{O}_K(f \circ r)(U)$ and invariance of the measure for $r : x \mapsto r^\circ x + v$, $r^\circ \in W_3$, $v \in a_K \mathbb{Z}^3$.

(B) We use the boards, mirrors, copies and folds of Def. A.4 (2), with the action $(rF)(U) = F(r^{-1}U)$, and the chessboard estimate of Def. A.10 and Prop. A.11. For flat test functions we use Def. C.11, the norm eq. C.xx, $\mathsf{C}_{\neq} = 10^{53}$ of eq. C.xxiv, Cor. C.16, Prop. C.13, Def. C.17 and Prop. C.15 (a)–(c). The table is admissible for these results: Rem. C.8 admits odd $L \geq 13$ and $M = L^{m'}$ with $m' \geq 1$.

The conventions for $d(X)$, $R(X)$ and $\mathbb{C}\mathfrak{S}(O)$ are those of Def. C.1, with Euclidean distance. Defs. C.9 (4), C.4 (2), (3) and C.6 (1), (3) are used as definitions only, at $\mathbb{E} = \mathbb{R}^3$; item 1 of §8.1 gives the corresponding notions at T_l.

(C) From §8.3 we take the locally bounded numbers of item 2 and Thm. 8.17 (R0), (R1), (R5)–(R7). We use (R0) at $s = 0$, and (R1) only at $K \geq j^\circ(\mathsf{b}, \check{\delta})$, for real bounded g_t whose nonzero sets are pairwise at sup distance $\geq \check{\delta}$ and lie each in a sup-ball of radius $\check{\rho} > 0$. The limits of (R5), including moving labels in parts (a), (c), are taken on continuous entries with compact, pairwise disjoint supports. Their reality, symmetry, invariance, reflection positivity and dilation follow from (R6) (i), (iii), (iv), (vi); (R7) supplies $\mathcal{H}(S)$, its unit vacuum and the contraction semigroup $e^{-tH}[F] = [\tau_{te_3} F]$.

When all cells carry the same pair $(\check{\delta}, \check{\rho})$, (G3) (a) is Thm. 8.17 (R2) (i) restricted to $K \geq j^\circ(\mathsf{b}, \check{\delta})$. The range $K < j^\circ$ of (R2) lies outside (G3).

0.2. Local letters and tags.

Remark 9.1 (Letters set here). The letter θ is the map $x_3 \mapsto -x_3$.

The letter T_N is the torus.

The labels (E0′) to (E4) denote the Euclidean axioms of Def. F.17.

In the first sentence of (G2) (iv) and in (G4) (e), $z = z_0(\mathsf{b})$, in the setting (S7) of §1.

1. Data and numbers.

Definition 9.2 (Lattice data, centred moments and the unit board)**.** We recall the data of Thm. 8.2. Let $G = \mathrm{SU}(2)$ and $d = 3$. Let $\mathsf{b} \geq x_s$, and let $N = 2L^{m_l}$ with $m_l \geq m'$ and (Tor) $N \geq 2Ms(L\mathsf{b})$. Let $K \geq 1$, $\beta = L^K \mathsf{b}$ and $a_K = L^{-K}$; we write $\mu_{K,N,\beta}$ for the measure, and $\mathcal{O}_K(f) = a_K^{-1} \sum_p f(x(p)) a_p$.

We put $\sigma_{K,N}(f_1, \dots, f_n) = \langle \prod_i (\mathcal{O}_K f_i - \langle \mathcal{O}_K f_i \rangle) \rangle$. Distances are sup distances, except where they are called Euclidean.

The board B_0, written without arguments, is $(a_K, L^K, (N, N, N), 0)$ in the notation $(a, \lambda, \underline{\mathsf{n}}, o)$ of Def. A.4 (2) for a board. Its cells are $\varkappa + [-1/2, 1/2]^3$, and its walls are $y_\mu \in \mathbb{Z} + 1/2$, through the

midpoints of bonds. The number $B_0(\mathsf{b},\check\delta)$ always carries its arguments. By S we denote any limit of (R5). ■

Definition 9.3 (The numbers of the section)**.** In the order in which the constants are fixed, the table (O0) to (O13) comes first, then b, and then (O14), to which all numbers below belong. No floor depends on them, and none of them depends on n, N, K or on where the sets $\{f_i\neq 0\}$ lie on T_N; all are locally bounded in b, by (Lab1).

From item 2 of §8.3 we recall the following numbers: $\mathrm{sep}^\circ := 8\log(8L^{1/2})$, the one numeral among them; $\mathrm{sep}^{[i]} = \lfloor\check\delta L^i/M\rfloor - 2\check r(L^i\mathsf{b}) - 2$; $j^\circ(\mathsf{b},\check\delta)$, the least $j\geq 1$ with $\mathrm{sep}^{[j-1]}\geq\mathrm{sep}^\circ$; $\mathfrak{r}^\circ(x) = 3.5/\mathrm{amp}(Lx)$ and $r^\circ = \mathfrak{r}^\circ(L^{j^\circ}\mathsf{b})$; $N_1^{[j]} = (2\check\rho L^{j-1}/M + 2\check r(L^{j-1}\mathsf{b}) + 3)^3$;

$$C_W(\mathsf{b},\check\delta,\check\rho) = \sum_{j>j^\circ} 2eN_1^{[j]}C_gc_a(L^j\mathsf{b})e^{-(\kappa-1)(\mathrm{sep}^{[j-1]}-1)} + \sum_{1\leq j\leq j^\circ} 6N_1^{[j]}C_gc_a(L^j\mathsf{b});$$

$e_h = 0.45N_1^{[1]}c_h(\mathsf{b})$; $G_1 = e^{C_W+e_h}/r^\circ$; $B_0(\mathsf{b},\check\delta) = 6L^{j^\circ}(\check\delta L^{j^\circ}/2+2)^3$; and $c_{\mathrm{vol}}(\mathsf{b}) := 225/M^3 + (3/2)\mathsf{N}_Zp_0(\mathsf{b})^2 + 3\log(384/\varepsilon(\mathsf{b})^3)$.

We also set $\mathfrak{l}(\mathsf{b},\check\delta) := \mathrm{sep}^\circ + 11 + [\log\mathsf{b} + \log^+(M/(264\check\delta))]/528$, $\mathfrak{l}_1(\mathsf{b}) := \mathrm{sep}^\circ + 11 + \log(\mathsf{b}M)/528$ and $\check\lambda(x) := \mathrm{sep}^\circ + 11 + \log x/528$.

We set $A_2 := c_Q^{\mathrm{su}}K_5L^{3/2}(2M)^{1/2}/(3.5E_a)$ and $T_L := \sum_{t\geq 1}L^{3t}\exp\{-(\kappa-1)(L^t\mathrm{sep}^\circ - 1)\}$.

We set

$$C_W^*(\mathsf{b}) := C_gL^3\{6\sum_{j\geq 1}\check\lambda(L^j\mathsf{b})^3c_a(L^j\mathsf{b}) + 2eT_L\sup_{j\geq 1}\check\lambda(L^j\mathsf{b})^3c_a(L^j\mathsf{b})\}$$

and $e_h^*(\mathsf{b}) := 0.45(2\check r(\mathsf{b})+4)^3c_h(\mathsf{b})$.

We set $\check\Gamma(\mathsf{b},\check\delta,\check\rho) := G_1(\mathsf{b},\check\delta,\check\rho)(1+e^{c_{\mathrm{vol}}(\mathsf{b})})$ and $\check\Gamma(\mathsf{b},\check\delta) := \check\Gamma(\mathsf{b},\check\delta,\check\delta/4)$.

We set $A_4(\mathsf{b}) := \mathfrak{l}_1^4\max\{A_2\mathsf{b}^{-1/2}e^{C_W^*+e_h^*}(1+e^{c_{\mathrm{vol}}})e^{c_{\mathrm{vol}}},\ 24L^2M(L^2M+2)^3\}$, $A_5 := 64(4e)^5A_4$, $A_6 := 3^{5/2}A_5$ and $C_E(\mathsf{b}) := \mathsf{C}_{\neq}^6M8A_6$.

We set $\check\Gamma_*(\mathsf{b},\eta) := 2A_2e^{C_W^*+e_h^*}(1+e^{c_{\mathrm{vol}}})e^{c_{\mathrm{vol}}}(\mathfrak{l}(\mathsf{b},\eta/4)/(\mathsf{b}\eta))^{1/2}$.

For $\eta\in(0,1]$ let $p' := \lceil 1/(2\eta)\rceil$ and $\eta' := 1/(2p')\in[\eta/2,\eta]$; equality on the left holds at $\eta=1$ alone. For $y\in\mathbb{R}^3$ let $\|\!|h|\!\|_{\eta,y} := \sum_{c\in y+(\eta'/8)\mathbb{Z}^3}\sup\{|h(x)| : |x-c|_\infty\leq\eta'/8\}$, and $\|\!|h|\!\|_\eta := \|\!|h|\!\|_{\eta,0}$. ■

Labels. As functions of the label b, the numbers obey the following.

(Lab1) Every number above is locally bounded in b on $[x_s,\infty)$, and so is $j^\circ(\mathsf{b},\check\delta)$ at fixed $\check\delta$. The numbers $A_5(\mathsf{b})$, $A_6(\mathsf{b})$, $C_E(\mathsf{b})$ of (G2) (iv) and (G4) (e) are locally bounded on $(0,\infty)$.

(Lab2) Let $e_h^{*c}(\mathsf{b}) := 0.45(\log\mathsf{b}/528+12)^3c_h(\mathsf{b})$; then $e_h^*\leq e_h^{*c}$. For X one of A_4, A_5, A_6, C_E, $\check\Gamma_*(\cdot,\eta)$ and C_n^∞ of (G2) (iii), let X^c be X with e_h^{*c} in the place of e_h^*. Then $X\leq X^c$, and X^c is continuous in b on $[x_s,\infty)$.

(Lab3) Let S be a limit of (R5) (a) or (c) along labels $\mathsf{b}_j\to\mathsf{b}$. Then every number $X(\mathsf{b})$ in (G2) (iii), (G4) (a) to (c), (G6) and (G6+) means $\limsup_j X(\mathsf{b}_j)$; this limsup may be taken along any subsequence, the limit S staying the same.

That number is at most $\sup X$ over any set that contains all b_j of the subsequence from some j on, for instance over $U(\mathsf{b})$ of the setting (S7) of §1 when $\mathsf{b}>x_s$. If $\mathsf{b}_j\geq x_s$ for infinitely many j, then along these j it is at most $X^c(\mathsf{b})$, a number of b alone.

If $\mathsf{b}_j<x_s$ for infinitely many j, we take a single integer $z\geq 1$ with $L^z\mathsf{b}_j\geq x_s$ from some j on, and we use (G2) (iv) at that z: the number is then at most $L^{4z}A_5^c(L^z\mathsf{b})$, and likewise for A_6 and C_E.

As to (G6), its main clause and its first three bullets are asserted only when $\mathsf{b}_j\geq x_s$ for infinitely many j, and along these j. If $\mathsf{b}_j<x_s$ from some j on, (G6) is asserted in its last bullet alone, at such a z, with a number at most $L^z\check\Gamma_*^c(L^z\mathsf{b},L^z\eta_z)$.

2. The theorem.

Theorem 9.4 (Growth condition and bounds in closed form)**.** In every clause the objects, letters and numbers are those of item 1. The data as recalled in item 1 are $\mathsf{b} \geq x_s$, $N = 2L^{m_l}$ with $m_l \geq m'$, (Tor) and $K \geq 1$. They are assumed in full in the clauses at finite (K, N): (G1), (G2) (i) and (ii), (G3) and (G4) (d). The clauses on a limit S with $\mathsf{b} \geq x_s$ are (G2) (iii), (G4) (a) to (c), (G5), the main clause and the first three bullets of (G6), and (G6+) (m), which states its own condition on the b_j, the same as that of the restriction in (G6). What these clauses assume of the data is $\mathsf{b} \geq x_s$ for the label b of the limit, while N, (Tor) and K concern the clauses at finite (K, N). The labels b_j of the members are subject only to the conditions of (Lab3) and of the restriction in (G6): $\mathsf{b}_j < x_s$ is admitted in (G2) (iii), (G4) (a) to (c) and (G5), with the numbers taken in the sense of (Lab3). In (G2) (iv) and (G4) (e), the label, the sides and K are those that the clause itself states, in the place of those of item 1. The last bullet of (G6) and (G6+) (n) restate the label alone. The following hold.

(G1) *The numbers.* For $\mathsf{b} \geq x_s$ and $0 < \check{\delta} \leq 1$ the following lines hold.

0. $1 + 0.182\, C_{\mathrm{vol}}(N, \mathsf{b}) \leq e^{c_{\mathrm{vol}}(\mathsf{b})N^3}$ for every N of the data; this line involves no $\check{\delta}$.
1. j° exists and is non-increasing in $\check{\delta}$; $L^{j^\circ} \leq 2L^2(M/\check{\delta})\mathfrak{l}$; and $a_K \leq \check{\delta}/(LM(\mathrm{sep}^\circ + 8))$ for $K \geq j^\circ$.
2. $1/r^\circ \leq A_2(\mathfrak{l}/(\mathsf{b}\check{\delta}))^{1/2}$.
3. At $\check{\rho} = \check{\delta}/4$, $N_1^{[j]} \leq (L\,\check{\lambda}(L^j\mathsf{b}))^3$ for $1 \leq j \leq j^\circ$, and $N_1^{[j^\circ+t]} \leq (L^{t+1}\check{\lambda}(L^{j^\circ+t}\mathsf{b}))^3$ for $t \geq 1$.
4. $C_W(\mathsf{b}, \check{\delta}, \check{\rho}) \leq C_W(\mathsf{b}, \check{\delta}, \check{\delta}/4) \leq C_W^*(\mathsf{b})$ for $\check{\rho} \leq \check{\delta}/4$, and $C_W^*(\mathsf{b}) \leq 0.27L^6((\mathrm{sep}^\circ + 11)^3 + 1)(12 + 2eT_L)(L\mathsf{b})^{-1/4}$; so $C_W^*(\mathsf{b})$ is finite, free of $\check{\delta}$, and bounded on $[x_s, \infty)$.
5. $e_h \leq e_h^*(\mathsf{b})$ for $\check{\rho} \leq \check{\delta}/4$.
6. $|\mathcal{O}_K(g)| \leq B_0(\mathsf{b}, \check{\delta})\|g\|_\infty$ for $K \leq j^\circ$ whenever $\{g \neq 0\}$ lies in a sup-ball of radius $\check{\delta}/4$; and $B_0(\mathsf{b}, \check{\delta}) \leq 12L^2M(L^2M + 2)^3\mathfrak{l}^4/\check{\delta}$.
7. $\mathfrak{l} \leq \mathfrak{l}_1(1 + \log(1/\check{\delta})/528)$, and $\mathfrak{l}^4 \leq \mathfrak{l}_1^4/\check{\delta}$.
8. $\max\{\check{\Gamma}(\mathsf{b}, \check{\delta})e^{c_{\mathrm{vol}}}, 2B_0(\mathsf{b}, \check{\delta})\} \leq A_4(\mathsf{b})\check{\delta}^{-2}$.

(G2) *Product form.* Let $K \geq 1$ be arbitrary, let N be any side of the data, $n \geq 2$, $d > 0$ and $R_f > 0$, and let the f_i be real bounded functions on T_N whose sets $\{f_i \neq 0\}$ are pairwise at distance $\geq d$ and lie each in a sup-ball of radius R_f, wherever these balls lie.

(i) $|\sigma_{K,N}(f)| \leq A_5(\mathsf{b})^n(n!)^5 \max\{1, d^{-5n}\}(1 + R_f)^{3n} \prod \|f_i\|_\infty$.

(ii) $|S^T_{n;K,N,L^K\mathsf{b}}(f)|$ is at most $1.45^{n+1}n!$ times the right side of (i).

(iii) Every limit S of (R5) obeys (i) and (ii) on continuous entries, at moving labels in the sense of (Lab3); for tuples of the Euclidean type $(d; R)$ it obeys them with A_6 in the place of A_5. The number $C_n^\infty(d; R) := 1.45^{n+1}n!\, A_6^n(n!)^5 \max\{1, d^{-5n}\}(1 + R)^{3n}$ is non-increasing in d and non-decreasing in R.

(iv) For $0 < \mathsf{b} < x_s$ and $z = z_0(\mathsf{b})$, in the setting (S7) of §1, (i) to (iii) hold with $A_5(\mathsf{b}) := L^{4z}A_5(L^z\mathsf{b})$, for $K \geq z + 1$ and for the sides $l = 2L^m$ with $m + z \geq m'$ and $L^z l \geq 2Ms(L^{z+1}\mathsf{b})$. The same holds for every $\mathsf{b} > 0$ with any integer $z \geq 0$ such that $L^z\mathsf{b} \geq x_s$, and with the number $L^{4z}A_5(L^z\mathsf{b})$.

(G3) *Cell by cell on B_0, free of factorials.* Let $g_1, \ldots, g_Q$ be real bounded functions on T_N, each set $\{g_i \neq 0\}$ lying in one cell. For each occupied cell t let $\check{\delta}_t \in (0, 1]$ and $\check{\rho}_t > 0$ be such that the n_t sources of t are pairwise at distance $\geq \check{\delta}_t$, at distance $\geq \check{\delta}_t/2$ from all walls, and each in a sup-ball of radius $\check{\rho}_t$. Let $K \geq \max_t j^\circ(\mathsf{b}, \check{\delta}_t)$.

(a) $|\langle \prod_i(\mathcal{O}_K g_i - \hat{c}_K g_i)\rangle| \leq \prod_t [G_1(\mathsf{b}, \check{\delta}_t, \check{\rho}_t)^{n_t} e^{c_{\mathrm{vol}}(\mathsf{b})}] \prod \|g_i\|_\infty$; at $Q = 1$ this reads $|\langle \mathcal{O}_K g\rangle - \hat{c}_K g| \leq G_1 e^{c_{\mathrm{vol}}}\|g\|_\infty$.

(b) $|\sigma_{K,N}(g_1, \ldots, g_Q)| \leq \prod_t [\check{\Gamma}(\mathsf{b}, \check{\delta}_t, \check{\rho}_t)^{n_t} e^{c_{\mathrm{vol}}(\mathsf{b})}] \prod \|g_i\|_\infty$.

If $\check{\rho}_t \leq \check{\delta}_t/4$, the condition that a source lies in one cell follows from its distance to the walls.

(G4) *The growth condition.*

(a) σ_S satisfies the growth condition of Def. C.17 with $(k_{\mathrm{E}}, \mathsf{c}_{\mathrm{E}}, \mathsf{C}_{\mathrm{E}}, \varkappa_{\mathrm{E}}) = (13, 1, C_E(\mathsf{b}), 13)$: for every n there is exactly one complex linear map $S_n^{\sigma_S}$ on $\mathcal{S}_{\neq}(\mathbb{R}^{3n})$ which is bounded for some $|\cdot|_k$ and takes the value $\sigma_S(X)$ at the tensors of $\mathfrak{S}_n^{\infty}(\mathbb{R}^3)$, and $|S_n^{\sigma_S}(\varphi)| \le C_E(\mathsf{b})^n (n!)^{13} |\varphi|_{13n-1}$. Moreover $S_1 = 0$.

(b) Independently of Prop. C.13, the sharper bound

$$|S_n^{\sigma_S}(\varphi)| \le (10^{36} A_5(\mathsf{b}))^n (n!)^5 |\varphi|_{12n+1}$$

holds. Since $10^{36} A_5 \le C_E$ and $12n+1 \le 13n$, (b) gives the quadruple of (a) in the form eq. C.xxxii.

(c) The extension $\tilde{S}_n$ of the truncated S_n obeys (a) and (b) with the further factor $1.45^{n+1} n!$.

(d) At finite (K, N), for φ smooth on $\mathbb{R}^{3n}$, flat on the diagonals and carried by $(-N/4+1, N/4-1)^{3n}$,

$$|a_K^{-n} \sum_{p_1,\dots,p_n} \varphi(x(p_1),\dots,x(p_n)) \langle \prod_i (a_{p_i} - \langle a_{p_i} \rangle) \rangle| \le (10^{36} A_5)^n (n!)^5 |\varphi|_{12n+1},$$

a bound free of K and N.

(e) For $0 < \mathsf{b} < x_s$, (a) to (d) and (G5) hold with $A_5(\mathsf{b})$ of (G2) (iv), with $A_6(\mathsf{b}) := 3^{5/2} A_5(\mathsf{b})$ and with $C_E(\mathsf{b}) := \mathsf{C}_{\neq}^6 M 8 A_6(\mathsf{b})$; in (d) they hold for $K \ge z+1$, with the sides l of (G2) (iv) in the place of N.

(G5) On $\mathcal{S}_{\neq}(\mathbb{R}^{3n})$, with the axioms named as in Def. F.17, the functionals $S_n^{\sigma_S}$ of (G4) are real, have the symmetry (E3), and are invariant under all translations and under W_3, which is (E1) for that group; and (E2) holds in the form eq. C.xxx at every coordinate mirror, for Schwartz functions flat outside $V_+^{(n)}$.

(G6) *A field between two vectors, free of n.* For $\mathsf{b} \ge x_s$, $\eta \in (0,1]$, $F, F' \in \mathbb{C}\mathfrak{S}(\{x_3 > \eta\})$, h real in $C_c^1(\{|x_3| < \eta/4\})$ and every $y \in \mathbb{R}^3$,

$$|\sigma_S(\theta \overline{F} \cdot (h) \cdot F')| \le \check{\Gamma}_*(\mathsf{b}, \eta)\, \|\!|h|\!\|_{\eta,y}\, \|[\tau_{-\eta e_3} F]\|\, \|[\tau_{-\eta e_3} F']\|.$$

At $y = 0$ the clause holds with $\|\!|h|\!\|_\eta$.

Restriction. At moving labels, the main clause and the first three bullets below hold when $\mathsf{b}_j \ge x_s$ for infinitely many j, with $\check{\Gamma}_*$ in the sense of (Lab3) along these j. When $\mathsf{b} > x_s$, every sequence $\mathsf{b}_j \to \mathsf{b}$ is covered. When $\mathsf{b} = x_s$ and $\mathsf{b}_j < x_s$ from some j on, the last bullet holds, and so does (G6+) (n).

- $\check{\Gamma}_*(\mathsf{b}, \eta) \le 4 A_2 e^{C_W^* + e_h^*} (1 + e^{c_{\mathrm{vol}}}) e^{c_{\mathrm{vol}}} (\mathfrak{l}_1/\mathsf{b})^{1/2}/\eta$. For $h \in C_c^3$ and every y, $\|\!|h|\!\|_{\eta,y} \le 4096\, \eta^{-3} \sum_{\alpha \in \{0,1\}^3} \eta^{|\alpha|} \|\partial^\alpha h\|_{L^1}$.
- In time order: let the h_i be real in C_c^1 with $\operatorname{supp} h_i \subset \{|x_3 - t_i| < \eta_i/4\}$ and $t_{i+1} - t_i \ge \eta_i + \eta_{i+1}$; then $|\sigma_S(h_1,\dots,h_n)| \le \prod_i \check{\Gamma}_*(\mathsf{b}, \eta_i) \|\!|h_i|\!\|_{\eta_i}$.
- The same holds in each coordinate direction.
- Let $\mathsf{b} > 0$, and let $z \ge 1$ be an integer with $L^z \mathsf{b}_j \ge x_s$ from some j on (at a fixed label: $L^z \mathsf{b} \ge x_s$). Let $\eta_z := \min\{\eta, L^{-z}\}$, let h be real in $C_c^1(\{|x_3| < \eta_z/4\})$, and let F, F' be as above. Then the left side is at most $L^z \check{\Gamma}_*(L^z \mathsf{b}, L^z \eta_z)\, \|\!|h^{[z]}|\!\|_{L^z \eta_z, y}\, \|[\tau_{-\eta e_3} F]\|\, \|[\tau_{-\eta e_3} F']\|$, with $\check{\Gamma}_*(L^z \mathsf{b}, \cdot)$ in the sense of (Lab3) along the labels $L^z \mathsf{b}_j$. Such a z exists for every sequence $\mathsf{b}_j \to \mathsf{b} > 0$.

(G6+) *The whole slab.*

(m) Let the label be fixed with $\mathsf{b} \ge x_s$, or let $\mathsf{b}_j \ge x_s$ for infinitely many j, $\check{\Gamma}_*$ then being taken in the sense of (Lab3) along these j. For any $\eta > 0$ put $\hat{\eta} := \min\{1, 3\eta/4\}$. Then for $F, F' \in \mathbb{C}\mathfrak{S}(\{x_3 > \eta\})$, h real in $C_c^1(\{|x_3| < \eta/4\})$ and every y,

$$|\sigma_S(\theta \overline{F} \cdot (h) \cdot F')| \le \check{\Gamma}_*(\mathsf{b}, \hat{\eta})\, \|\!|h|\!\|_{\hat{\eta},y}\, \|[\tau_{-\eta e_3} F]\|\, \|[\tau_{-\eta e_3} F']\|.$$

(n) Let $\mathsf{b} > 0$, let $z \geq 1$ be an integer with $L^z \mathsf{b}_j \geq x_s$ from some j on, let $\eta \in (0,1]$, and put $e_z := \min\{1, 3L^z\eta/4\}$. Then for $F, F' \in \mathbb{C}\mathfrak{S}(\{x_3 > \eta\})$, h real in $C^1_c(\{|x_3| < \eta/4\})$ and every $y \in \mathbb{R}^3$, $|\sigma_S(\theta\overline{F}\cdot(h)\cdot F')|$ is at most $L^z \check{\Gamma}_*(L^z\mathsf{b}, e_z)\, \|\!|h^{[z]}|\!\|_{e_z,y}\, \|[\tau_{-\eta e_3}F]\|\, \|[\tau_{-\eta e_3}F']\|$. Here the slab $\{|x_3| < \eta/4\}$ is the whole slab, not the thin slab $\{|x_3| < \eta_z/4\}$ of the last bullet of (G6). In the case $\mathsf{b} = x_s$, $\mathsf{b}_j < x_s$ from some j on, of the restriction in (G6), this holds with $z = 1$.

■□

The useful dependence on separation comes from summing the decaying analytic bounds over the late scales before exponentiating. The resulting powers of the separation are combined with a dyadic partition off the diagonals. Flatness of the tests makes the resulting Schwartz bound summable over that partition.

Step 1: clause (G1)*, with* (Lab1) *and* (Lab2)*.*

Lemma 9.5 (Closed forms of the numbers, and their dependence on the label)**.** In the setting of item 1, for $\mathsf{b} \geq x_s$ and $0 < \check{\delta} \leq 1$, the nine lines 0 to 8 of (G1) hold, and so do (Lab1) and (Lab2). ■

Proof. Step (0): radii. By (M2) and $\kappa_1 + 4 \geq 11$ we have $20c_R/M \leq 1/1056$. Hence $r(x) \leq \log x/1056 + 1$, and as $\check{r} = r + 3$ we get $2\check{r}(x) + 2 \leq \log x/528 + 10$, then $2\check{r}(x) + 3 \leq \log x/528 + 11 \leq \check{\lambda}(x)$, and $2\check{r}(x) + 4 \leq \log x/528 + 12$.

Step (i): line 1. The argument for (N2) in the proof of Lem. 8.4, as recalled in item 0.1 (A), gives $\mathrm{sep}^{[i]} \to \infty$ at $d = \check{\delta}$, and by (N2) the sequence $\mathrm{sep}^{[i]}$ grows by the factor L from its first positive value on. Hence j° exists, and $\mathrm{sep}^{[j-1]} < \mathrm{sep}^\circ$ for $1 \leq j < j^\circ$. The number $\mathrm{sep}^{[i]}$ is non-decreasing in $\check{\delta}$; so j° is non-increasing in $\check{\delta}$.

Let $j^\circ \geq 2$, and put $Y = L^{j^\circ - 2}$ and $t = \check{\delta}Y/M$. The inequality $\mathrm{sep}^{[j^\circ-2]} < \mathrm{sep}^\circ$ reads $\lfloor t \rfloor < \mathrm{sep}^\circ + 2\check{r}(Y\mathsf{b}) + 2$, and with step (0) it gives $t < \lfloor t \rfloor + 1 < \mathrm{sep}^\circ + 11 + \log(Y\mathsf{b})/528$. Here $\log(Y\mathsf{b}) = \log \mathsf{b} + \log(M/(264\check{\delta})) + \log(264t)$, and $\log(264t) < 264t$. Since $\log \leq \log^+$, we obtain $t < \mathfrak{l} + t/2$, hence $t < 2\mathfrak{l}$ and $L^{j^\circ} = L^2 tM/\check{\delta} < 2L^2(M/\check{\delta})\mathfrak{l}$. If $j^\circ = 1$, then $L \leq 2L^2(M/\check{\delta})\mathfrak{l}$ holds as $\check{\delta} \leq 1$ and $\mathfrak{l} \geq 11$.

For the last clause of line 1, $\mathrm{sep}^{[j^\circ-1]} \geq \mathrm{sep}^\circ$ gives $\check{\delta}L^{j^\circ-1}/M \geq \mathrm{sep}^\circ + 2\check{r} + 2 \geq \mathrm{sep}^\circ + 8$, so $a_K \leq L^{-j^\circ} \leq \check{\delta}/(LM(\mathrm{sep}^\circ + 8))$ for $K \geq j^\circ$.

Step (ii): line 2. By the definitions and (X14),

$$1/r^\circ = \mathrm{amp}(L^{j^\circ+1}\mathsf{b})/3.5 = c_Q^{\mathrm{su}} K_5 (L^{j^\circ+1}\mathsf{b})^{1/2}/(3.5\,\mathsf{b}E_a p_0) \leq [c_Q^{\mathrm{su}} K_5 L^{1/2}/(3.5E_a)](L^{j^\circ}/\mathsf{b})^{1/2}.$$

We insert the bound of step (i) for L^{j°; by the definition of A_2 this is line 2.

Step (iii): line 3, at $\check{\rho} = \check{\delta}/4$. At $\check{\rho} = \check{\delta}/4$ we have $N_1^{[j]} = (\check{\delta}L^{j-1}/(2M) + 2\check{r}(L^{j-1}\mathsf{b}) + 3)^3$. Let $2 \leq j \leq j^\circ$. Then $\mathrm{sep}^{[j-2]} < \mathrm{sep}^\circ$ gives $\check{\delta}L^{j-2}/M < \check{\lambda}(L^{j-2}\mathsf{b}) \leq \check{\lambda}(L^j\mathsf{b})$, hence $\check{\delta}L^{j-1}/(2M) < (L/2)\check{\lambda}(L^j\mathsf{b})$, and with step (0) we find $N_1^{1/3} \leq (L/2+1)\check{\lambda} \leq L\,\check{\lambda}$. For $j = 1$ the bound requires $1/2 + \log \mathsf{b}/528 + 11 \leq \check{\lambda}(L\mathsf{b})$, which says $1/2 \leq \mathrm{sep}^\circ + \log L/528$ and holds by the definition of sep°. For $j = j^\circ + t$ we have $\check{\delta}L^{j-1}/(2M) = L^t\check{\delta}L^{j^\circ-1}/(2M) \leq L^t(L/2)\check{\lambda}(L^j\mathsf{b})$, where for $j^\circ = 1$ we use $\check{\delta}/(2M) \leq 1/2$; so $N_1^{1/3} \leq (L^{t+1}/2+1)\check{\lambda} \leq L^{t+1}\check{\lambda}(L^j\mathsf{b})$.

Step (iv): line 4. Since N_1 is non-decreasing in $\check{\rho}$, it suffices to treat $\check{\rho} = \check{\delta}/4$, where line 3 applies. For the late levels,

$$\sum_{j \leq j^\circ} 6N_1 C_g c_a \leq 6C_g L^3 \sum_{j \geq 1} \check{\lambda}(L^j\mathsf{b})^3 c_a(L^j\mathsf{b});$$

this is the whole series, and it does not depend on j°. For the early levels $j = j^\circ + t$, (N2) gives $\mathrm{sep}^{[j-1]} \geq L^t\mathrm{sep}^\circ$, so the term is at most $2eC_gL^3L^{3t}e^{-(\kappa-1)(L^t\mathrm{sep}^\circ-1)}\check{\lambda}^3 c_a$, and the sum of these terms is at most $2eC_gL^3T_L \sup \check{\lambda}^3 c_a$. Hence $C_W(\mathsf{b}, \check{\delta}, \check{\rho}) \leq C_W(\mathsf{b}, \check{\delta}, \check{\delta}/4) \leq C_W^*(\mathsf{b})$ for $\check{\rho} \leq \check{\delta}/4$.

The number $C_W^*(\mathsf{b})$ is finite. For $x = L^j\mathsf{b} \geq Lx_s$, the bound recalled from Step 0 (b) of the proof of Prop. 7.12 gives $c_a \leq x^{-3/8}/4000$. We also have $\check{\lambda}^3 \leq 4((\mathrm{sep}^\circ + 11)^3 + (\log x/528)^3)$, and $(\log x)^3 \leq (24/e)^3 x^{1/8} \leq 688.3\, x^{1/8}$ with $688.3/528^3 < 1$. Hence $\check{\lambda}^3 c_a \leq ((\mathrm{sep}^\circ+11)^3+1)x^{-1/4}/1000$. By (L3su), $\sum_j (L^j\mathsf{b})^{-1/4} \leq 2(L\mathsf{b})^{-1/4}$. With $C_g = 263L^3$ this yields $C_W^* \leq 0.263L^6((\mathrm{sep}^\circ + 11)^3 + 1)(12 + 2eT_L)(L\mathsf{b})^{-1/4}$, which gives line 4, since $T_L < \infty$ by the double exponential decay of its terms.

We sum the whole series for the following reason. With $c_a \leq 1/32$ alone, each of the $j^\circ \sim \log(1/\check{\delta})$ late terms would be of the size $\check{\lambda}^3 \sim \log^3$, and G_1 would contain $\exp\{c\log^4(1/\check{\delta})\}$, which is not a power of $\check{\delta}$.

Step (v): line 5. For $\check{\rho} \leq \check{\delta}/4$ we have $N_1^{[1]} \leq (\check{\delta}/(2M) + 2\check{r}(\mathsf{b}) + 3)^3 \leq (2\check{r}(\mathsf{b}) + 4)^3$, so $e_h \leq e_h^*(\mathsf{b})$.

Step (vi): line 6. We have $0 \leq a_p \leq 2$. The barycentres of the plaquettes of one orientation form a translate of $a_K\mathbb{Z}^3$, so a closed cube of side $\check{\delta}/2$ contains at most $(\check{\delta}L^K/2+1)^3$ of them; and there are three orientations. A sup-ball of radius $\check{\delta}/4$ is such a cube, hence $|\mathcal{O}_K g| \leq L^K \cdot 2 \cdot 3(\check{\delta}L^K/2+1)^3\|g\| \leq B_0(\mathsf{b}, \check{\delta})\|g\|$ for $K \leq j^\circ$. With step (i), $B_0(\mathsf{b}, \check{\delta}) \leq 12L^2M(\mathfrak{l}/\check{\delta})(L^2M\mathfrak{l}+2)^3 \leq 12L^2M(L^2M+2)^3\mathfrak{l}^4/\check{\delta}$, as $\mathfrak{l} \geq 1$.

Step (vii): line 7. For $\check{\delta} \leq 1$, $\log^+(M/(264\check{\delta})) \leq \log M + \log(1/\check{\delta})$. Hence, with $u = \log(1/\check{\delta})$, $\mathfrak{l} \leq \mathfrak{l}_1 + u/528 \leq \mathfrak{l}_1(1 + u/528)$; and $(1 + u/528)^4 \leq e^u$, which gives $\mathfrak{l}^4 \leq \mathfrak{l}_1^4/\check{\delta}$.

Step (viii): line 8. By lines 6 and 7, $2B_0(\mathsf{b}, \check{\delta}) \leq 24L^2M(L^2M + 2)^3\mathfrak{l}_1^4\check{\delta}^{-2}$. By lines 2, 4 and 5, $\check{\Gamma}\, e^{c_{\mathrm{vol}}} \leq A_2\mathsf{b}^{-1/2}e^{C_W^* + e_h^*}(1+e^{c_{\mathrm{vol}}})e^{c_{\mathrm{vol}}}(\mathfrak{l}/\check{\delta})^{1/2}$, and by step (vii), $(\mathfrak{l}/\check{\delta})^{1/2} \leq \mathfrak{l}_1^{1/2}\check{\delta}^{-5/8} \leq \mathfrak{l}_1^4\check{\delta}^{-2}$. Line 8 follows by the definition of A_4.

Step (ix): line 0, the volume factor. We write $C_{\mathrm{vol}} = 4qe^{220q}E$, with $q = (N/M)^3 \geq 8$, since $N/M = 2L^{m_l - m'} \geq 2$, and with $E = e^{(3/2)\mathsf{N}_Z p_0^2 N^3}(384/\varepsilon^3)^{3N^3} \geq 1$, since $\varepsilon \leq 1/40$. Then $1 + 0.182\, C_{\mathrm{vol}} \leq (1 + 0.728\, qe^{220q})E$. On $q \geq 8$ we have $\log(1 + 0.728\, qe^{220q}) \leq 220.23\, q$: the quotient is largest at $q = 8$, where its value is at most 220.2203. So this logarithm is at most $225N^3/M^3$, and line 0 follows from the definition of $c_{\mathrm{vol}}(\mathsf{b})$.

Step (x): the labels. For (Lab1): by (O8) of the table, p_0, ε, c_a, c_h are explicit positive functions, continuous on $x > 1$; so c_{vol}, $\mathfrak{l}$, $\mathfrak{l}_1$ and $\check{\lambda}$ are continuous in b. The radius $\check{r}$ is a step function of $\log \mathsf{b}$, bounded on compact sets; C_W^* is bounded by line 4; and A_2, T_L do not depend on b. The numbers A_4, A_5, A_6, C_E, $\check{\Gamma}_*$ are continuous expressions in these numbers, and $j^\circ(\mathsf{b}, \check{\delta})$ is locally bounded by line 1, as $\mathfrak{l}$ is continuous. For G_1, $\check{\Gamma}(\mathsf{b}, \check{\delta}, \check{\rho})$ and $B_0(\mathsf{b}, \check{\delta})$ we use item 2 of §8.3. On $(0, x_s)$ the least z is a step function of b, bounded on compact sets.

For (Lab2): by step (0), $(2\check{r}(\mathsf{b}) + 4)^3 \leq (\log \mathsf{b}/528 + 12)^3$, so $e_h^* \leq e_h^{*c}$, and e_h^{*c} is continuous. As to C_W^*, the terms $u_j(\mathsf{b}) = \check{\lambda}(L^j\mathsf{b})^3 c_a(L^j\mathsf{b})$ are continuous and, by step (iv), at most $((\mathrm{sep}^\circ + 11)^3 + 1)(L^j x_s)^{-1/4}/1000$ on $[x_s, \infty)$. So the series converges uniformly, and $\sup_j u_j$ is the uniform limit of $\max_{j \leq J} u_j$; both are continuous. Each X of (Lab2) is a continuous expression which does not decrease as e_h^* grows; hence $X \leq X^c$, and X^c is continuous. □

Step 2: the cut, the placing on B_0, and the copies.

Lemma 9.6 (Cutting, placing and copying sources)**.** In the setting of item 1 the following hold.

(a) Let f be real and bounded on T_N, with $\{f \neq 0\}$ in a sup-ball of radius R_f, and let $\check{\delta} > 0$. Then f is a sum of at most $(4R_f/\check{\delta} + 2)^3$ pieces g with $\|g\| \leq \|f\|$ and $\{g \neq 0\} \subset \{f \neq 0\}$, each set $\{g \neq 0\}$ lying in a sup-ball of radius $\check{\delta}/4$.

(b) Let n sets lie each in a sup-ball of radius $\check{\delta}/4$, where $\check{\delta} \leq 1/(4n)$, and let $a_K \leq 5/(8n)$. Then there is $v \in a_K\mathbb{Z}^3$ such that, after the translation by v, all of them keep a distance $\geq \check{\delta}/2$ from all walls $y_\mu \in \mathbb{Z} + 1/2$.

(c) Let the sources of one cell be pairwise at distance $\geq \check{\delta}$, and at distance $\geq \check{\delta}/2$ from the walls. Then their N^3 families of mirror copies are pairwise at distance $\geq \check{\delta}$.

■

Proof. Step (a): cutting. The sup-ball is the image under the projection pr of a cube C of $\mathbb{R}^3$ of side $\min\{2R_f, N\}$, half-open if the side is N, and pr is injective on C. We cut $\mathbb{R}^3$ into the half-open cubes $(\check\delta/2)(k+[0,1)^3)$. An interval of length $2R_f$ meets at most $4R_f/\check\delta+2$ of the intervals of length $\check\delta/2$, so at most $(4R_f/\check\delta+2)^3$ of these cubes meet C. For each of them we put $g := f\,1_{\mathrm{pr}(C\cap\mathrm{cube})}$; these pieces add up to f and have the properties stated in (a).

Step (b): placing. We fix μ. The projection of the set i to the coordinate μ lies in an interval of length $\check\delta/2$. The values of v_μ mod 1 for which this interval, shifted by v_μ, lies at distance $< \check\delta/2$ from $\mathbb{Z}+1/2$ form an open arc of length $\check\delta/2+\check\delta$. All these arcs together have length at most $1.5\,n\check\delta \le 3/8$. Their complement is closed, consists of at most n arcs, and has length $\ge 5/8$; so one of its arcs is closed of length $\ge 5/(8n) \ge a_K$, and it contains a point of $a_K\mathbb{Z}$ mod 1, as $1/a_K$ is an integer. The same holds for $\mu = 1,2,3$, and the three choices give v.

Step (c): copies. Inside one cell the copies form an isometric image of the sources, so they have the same pairwise distances as the sources. Copies in two different cells differ in some coordinate μ; in that coordinate the two supports lie in different unit intervals of $\mathbb{R}/N\mathbb{Z}$ cut out by $\mathbb{Z}+1/2$, each at distance $\ge \check\delta/2$ from every wall, and every path between them in that coordinate crosses a wall. So their distance is $\ge \check\delta$.

Step "in one cell": the last sentence of (G3). Two points of a set which lie on the two sides of a wall, each at distance $\ge \check\delta/2$ from it, are at distance $\ge \check\delta$ from each other; but a sup-ball of radius $\check\delta/4$ has diameter $\check\delta/2$. So a source in such a ball, at distance $\ge \check\delta/2$ from the walls, lies in one cell. □

Step 3: clause (G3)*, the chessboard estimate cell by cell, and centring at the true mean.*

Lemma 9.7 (The chessboard estimate cell by cell, and centring at the true mean)**.** In the setting of item 1, (G3) (a) and (b) hold. ■

Proof. Step (1): cell functions. For constants k_i we put $F_t := \prod_{i\in I_t}(\mathcal{O}_K g_i - k_i)$, where I_t indexes the n_t sources of the cell t. Let p be a plaquette with $g_i(x(p)) \neq 0$. Its four sites lie at sup distance $a_K/2$ from $x(p)$, the point $x(p)$ lies at distance $\ge \check\delta_t/2$ from the walls, and $a_K < \check\delta_t$ by line 1 of (G1). So the sites lie in the cell and off the walls, which contain no site, and the four bonds of p are inner bonds. Hence F_t is a real bounded continuous function of the cell t.

Step (2): the chessboard estimate. We apply Prop. A.11 on the board B_0, together with Def. A.10. A cell of the board that holds no source receives the constant function 1, and its fold is $\mathrm{Fold}\,1 = 1$. The chessboard estimate then reads

$$|\langle \prod_t F_t\rangle| \le \prod_t \langle \mathrm{Fold}\, F_t\rangle^{1/N^3},$$

where both products run over all cells t of the board.

Step (3): the fold. Let $\varkappa$ and $\varkappa'$ be cell indices, and let $\sigma = \sigma(\varkappa',\varkappa)$ be their signs. The mirror map $r = r_{\varkappa'\varkappa} : x \mapsto \varkappa' + \sigma(x-\varkappa)$ is of the form $r^\circ x + v$, where the linear part r° is a diagonal element of W_3 and $v \in \mathbb{Z}^3$. By Step 7 of the proof of Thm. 8.2, in its paragraph on finite K, the copy of $\mathcal{O}_K g_i - k_i$ under r is $\mathcal{O}_K(g_i \circ r^{-1}) - k_i$. Coordinate by coordinate, r is either a translation by an even integer or a reflection $y \mapsto (\varkappa'_\mu + \varkappa_\mu) - y$, in which $\varkappa'_\mu + \varkappa_\mu$ is an odd integer. The parity of $\varkappa' - \varkappa$ is well defined, because N is even.

Step (4): proof of (a). In case (a) of clause (G3) of Thm. 9.4, the constants are $k_i = \hat c_K g_i$. By (R0) of Thm. 8.17, also $k_i = \hat c_K(g_i \circ r^{-1})$ for every mirror map r. Hence $\mathrm{Fold}\, F_t$ is a product of factors $(\mathcal{O}_K g' - \hat c_K g')$, one for each of $n_t N^3$ sources g'. Any two of these sources lie at distance at least $\check\delta_t$ from each other, by Lem. 9.6 (c); each of them lies in a ball of radius $\check\rho_t$; and each has the sup norm of the source of which it is a copy. We apply (R1) with $Q = n_t N^3$ and $K \ge j^\circ(\mathsf{b}, \check\delta_t)$ and

obtain

$$\langle \mathrm{Fold}\, F_t\rangle \le G_1^{n_t N^3}(1+0.182 C_{\mathrm{vol}}) \prod \|g_i\|_\infty^{N^3},$$

where the product runs over $i \in I_t$. We take the N^3-th root and use line 0 of (G1); this bounds $\langle \mathrm{Fold}\, F_t\rangle^{1/N^3}$ by $G_1^{n_t} e^{c_{\mathrm{vol}}} \prod \|g_i\|_\infty$.

Step (5): proof of (b). In case (b) the constants are $k_i = \langle \mathcal{O}_K g_i\rangle$; as the measure is invariant under the mirror maps, $k_i = \langle \mathcal{O}_K g_i\rangle = \langle \mathcal{O}_K (g_i \circ r^{-1})\rangle$. For each copy g' we split

$$\mathcal{O}_K g' - \langle \mathcal{O}_K g'\rangle = (\mathcal{O}_K g' - \hat{c}_K g') - \mathfrak{d}(g'), \qquad \mathfrak{d}(g') := \langle \mathcal{O}_K g'\rangle - \hat{c}_K g'.$$

Case (a), applied with $Q = 1$ to g' as the only source of one cell with the same $\check{\delta}_t$ and $\check{\rho}_t$, gives $|\mathfrak{d}(g')| \le G_1(\mathsf{b}, \check{\delta}_t, \check{\rho}_t) e^{c_{\mathrm{vol}}} \|g'\|_\infty$. We expand the product over the $n_t N^3$ copies into a sum over the subsets A of the copies: each copy in A contributes the factor $\mathcal{O}_K g' - \hat{c}_K g'$, and each copy outside A contributes the number $-\mathfrak{d}(g')$. To every sub-family A we apply (R1), which admits every $Q \ge 1$; for empty A the moment is 1. Summing over A, we find

$$\langle \mathrm{Fold}\, F_t\rangle \le (1+0.182 C_{\mathrm{vol}}) \prod_{\text{copies}} (G_1 + G_1 e^{c_{\mathrm{vol}}}) \|g'\|_\infty.$$

The N^3-th root of this bound, taken with line 0 of (G1), is at most $\check{\Gamma}^{n_t} e^{c_{\mathrm{vol}}} \prod \|g_i\|_\infty$, with $\check{\Gamma} = \check{\Gamma}(\mathsf{b}, \check{\delta}_t, \check{\rho}_t)$. □

Remark 9.8 (On the proof of Lem. 9.7). The proof of Lem. 9.7 involves no sum over partitions, and so its bounds carry no factorial. The fold $\mathrm{Fold}\, F_t$ of the function F_t of a cell t is made of copies of the sources of t and of nothing else; for this reason the bounds on $\langle \mathrm{Fold}\, F_t\rangle^{1/N^3}$ at the end of Steps (4) and (5) involve, besides b, n_t and the sup norms of the sources of t, the pair $(\check{\delta}_t, \check{\rho}_t)$ of t itself, and no pair of another cell. The copy $r_{\varkappa'\varkappa} F$ of eq. A.xiv is $(r_{\varkappa'\varkappa} F)(U) = F(r_{\varkappa'\varkappa}^{-1} U)$, that is, F composed with the action on bond fields of the inverse of the mirror map $r_{\varkappa'\varkappa}$ of Step (3). This follows from the notation recalled at the opening of §A.2. In that opening, the paragraph "The lattice, the action and the integrals" provides the action of a diagonal isometry on functions and on bond fields, while the paragraph "Mirror maps, copies and folds" provides eq. A.xiv, the fact that a mirror map is a diagonal isometry, and the definition eq. A.xvi of the bond field u^σ.

Step 4: clause (G2)*, with* (Lab3).

Lemma 9.9 (The bounds of clause (G2))**.** In the setting of Thm. 9.4, the bounds (i) to (iv) of its clause (G2) hold, and (Lab3) of item 1 of §9 holds for (G2) (iii). ■

Proof. Throughout the proof we put $\check{\delta} := \check{\delta}_n := \min\{d, 1/(4n)\} \le 1/8$.

Step 4 (a): the cut. The functional $\sigma_{K,N}$ is n-linear on real bounded functions. By Lem. 9.6 (a) and this n-linearity, $\sigma_{K,N}(f)$ is the sum of the numbers $\sigma_{K,N}(g_1, \dots, g_n)$, one for each way of picking a single piece g_i of every f_i; this sum has at most $(4R_f/\check{\delta} + 2)^{3n}$ terms. Pieces of different f_i are at distance at least $d \ge \check{\delta}$ from each other, so every tuple of picked pieces is of type $(\check{\delta}; \check{\delta}/4)$.

Step 4 (b): the case $K \ge j^\circ(\mathsf{b}, \check{\delta})$. Line 1 of (G1) of Thm. 9.4 gives $a_K < \check{\delta} \le 1/(4n) < 5/(8n)$. Let v be the vector of Lem. 9.6 (b). By Step 7 of the proof of Thm. 8.2, in its paragraph on finite K, the tuple X of picked pieces satisfies $\sigma_{K,N}(\tau_v X) = \sigma_{K,N}(X)$. We use (G3) (b) with $\check{\delta}_t = \check{\delta}$ and $\check{\rho}_t = \check{\delta}/4$; at most n cells are occupied, and we obtain

$$|\sigma_{K,N}(g_1, \dots, g_n)| \le (\check{\Gamma}(\mathsf{b}, \check{\delta}) e^{c_{\mathrm{vol}}})^n \prod \|g_i\|_\infty.$$

Step 4 (c): the case $K < j^\circ$. Here line 6 of (G1) gives $|\mathcal{O}_K g - \langle \mathcal{O}_K g\rangle| \le 2B_0(\mathsf{b}, \check{\delta}) \|g\|_\infty$ for each piece g.

Step 4 (d): proof of (i). In both cases, line 8 of (G1) bounds each term of the sum of Step 4 (a) by $(A_4 \check{\delta}^{-2})^n \prod \|f_i\|_\infty$. As $4R_f/\check{\delta} + 2 \le 4(1+R_f)/\check{\delta}$, the number of terms is at most $(4(1+R_f)/\check{\delta})^{3n}$.

So $|\sigma_{K,N}(f)| \le [64A_4(1+R_f)^3\check{\delta}^{-5}]^n \prod \|f_i\|_\infty$. Finally, $1/\check{\delta} = \max\{1/d, 4n\} \le 4n\max\{1, 1/d\}$ and $n^n \le e^n n!$; inserted into the last bound, these two inequalities give (i).

Step 4 (e): cumulants, and proof of (ii). For $n \ge 2$ the joint cumulant does not change when constants are added to its entries, and it is expressed through the centred moments:

$$S_n^T = \sum_\pi (-1)^{|\pi|-1}(|\pi|-1)! \prod_{J\in\pi} \sigma_{K,N}(f_J),$$

where π runs over the partitions of $\{1,\dots,n\}$. A partition with a block of size 1 contributes 0, since the centred moment of a single function vanishes. We apply (i) to each block with $|J| \ge 2$ and use $\prod |J|! \le n!$; then every product in the sum is at most the right side of (i). Hence $|S_n^T|$ is at most a_n times the right side of (i), where $a_n := \sum_\pi (|\pi|-1)!$. The exponential generating function of the numbers a_n is $-\log(2-e^x)$, whose derivative is $2/(2-e^x) - 1$. Expanding this derivative into a series, one finds $a_n = 2f_{n-1}$ for $n \ge 2$, with $f_m = \sum_{j\ge 0} j^m 2^{-j-1}$. Since $t \mapsto t^m 2^{-t}$ is unimodal on $[0,\infty)$, the sum $\sum_j j^m 2^{-j}$ is at most the integral of this function plus its maximum; these are $m!(\log 2)^{-m-1}$ and $(m/(e\log 2))^m$, and $m!(\log 2)^{-m-1} + (m/(e\log 2))^m \le m!(\log 2)^{-m-1}(1+\log 2)$. It follows that $a_n \le 1.694\,(n-1)!\,1.4427^n \le 1.45^{n+1} n!$, which gives (ii).

Step 4 (f): proof of (iii). The bounds obtained so far hold, for every (K,N), for the entries carried to the torus T_N, and the carried entries keep their type as soon as $N \ge 2(\mathrm{diam} + d)$, where diam is the diameter, for the sup distance, of the union of the supports of the entries. A limit of numbers obeys every bound that the numbers obey, and σ_S is a polynomial in the $S_{|J|}$. By item 1 of §8.1, in its part on distances, a tuple of Euclidean type $(d; R)$ is of sup type $(d/\sqrt{3}; R)$; this costs the factor $3^{5n/2}$.

Step 4 (f′): proof of (Lab3). Let the labels move, $\mathsf{b}_j \to \mathsf{b}$. The sides tend to ∞, while $s = 8r$ is bounded on compact sets by item 1 of §8.1, in its part on radii; hence, for every large j, (Tor) holds for the label b_j. The j-th member of the sequence obeys (i) and (ii) with $A_5(\mathsf{b}_j)$. The limit of the values is at most the lim sup of the bounds, and at fixed n, d, R_f this is the bound taken at $\limsup_j A_5(\mathsf{b}_j)$. Passing to a subsequence does not change the limit of the values; so the lim sup may be computed along a subsequence. The value of a lim sup is not affected by finitely many members; hence it is at most the supremum over any set that contains a tail of the sequence. If $\mathsf{b}_j \ge x_s$ for infinitely many j, then along these j we have $A_5(\mathsf{b}_j) \le A_5^c(\mathsf{b}_j) \to A_5^c(\mathsf{b})$, by (Lab2) of item 1 of §9. Otherwise the set $\{\mathsf{b}_j\} \cup \{\mathsf{b}\}$ is compact in $(0,\infty)$, and one integer $z \ge 1$ satisfies $L^z\mathsf{b}_j \ge x_s$ for all j. For every large j the ranges of (G2) (iv) hold for this z, because the sides tend to ∞, the cutoff tends to ∞ (at each side in (R5) (a) and along the sequence in (R5) (c)), and s is bounded on the compact set $\{L^{z+1}\mathsf{b}_j\} \cup \{L^{z+1}\mathsf{b}\}$. Then (iv), applied with that z, gives the bound with $L^{4z}A_5(L^z\mathsf{b}_j)$, whose lim sup is at most $L^{4z}A_5^c(L^z\mathsf{b})$.

Step 4 (g): proof of (iv). The argument of Step 5 of the proof of Thm. 8.2 gives

$$\sigma_{K,l,L^K\mathsf{b}}(f) = L^{zn}\sigma_{K-z,L^z l,L^{K-z}(L^z\mathsf{b})}(f^{[z]}).$$

In addition, $(1+L^zR_f)^3 \le L^{3z}(1+R_f)^3$, and $\max\{1,(L^z d)^{-5n}\} \le \max\{1, d^{-5n}\}$ for $z \ge 0$. The argument applies to every integer $z \ge 0$ with $L^z\mathsf{b} \ge x_s$, and it does not use that z is the least such integer. □

Remark 9.10 (Where n enters). The integer n enters the bounds of (G2) at the following places. The factor $(4n)^{3n}$ comes from the number of the pieces; this number remains as a factor because the bound of Step 4 (d) uses the sup norm of each piece and not the volume of its support. The factor $(4n)^{2n}$ comes from $A_4\check{\delta}^{-2}$. The factor $n!$ comes from Step 4 (e). Everything else is of the form C^n, because the exponent in (R1) is linear in Q and C_{vol} occurs only once.

Step 5: the extension lemma, and (G4) (b), (d).

Lemma 9.11 (Extension to test functions flat on the diagonals)**.** Let $n \ge 1$, $a \in \mathbb{N}_0$ and $\Gamma \ge 0$. Let T be a real n-linear map on the n-tuples of continuous functions on $\mathbb{R}^3$ with compact and pairwise

disjoint supports. Suppose that $|T(f)| \le \mathit{\Gamma} \max\{1, \check{\delta}^{-a}\} \prod \|f_i\|_\infty$ for every $\check{\delta} > 0$ and every such tuple whose supports are pairwise at sup distance at least $\check{\delta}$ and each lie in a sup-ball of radius 1, wherever in $\mathbb{R}^3$ these balls are placed. Then there is a complex linear functional $\tilde{T}$ on $\mathcal{S}_{\neq}(\mathbb{R}^{3n})$ such that $\tilde{T}(\bigotimes X) = T(X)$ for every X in $\mathfrak{S}_n^\infty(\mathbb{R}^3)$ and $|\tilde{T}(\varphi)| \le \mathit{\Gamma}\, C_4^n\, 39^{a+3n+1} |\varphi|_{a+7n+1}$, where $C_4 \le 2.3 \cdot 10^{20}$. ∎

Steps (h) and (i) of the proof of Lem. 9.11 yield (G4) (b) and (d) of Thm. 9.4.

Proof. Step (a): a partition of unity off the diagonals. Let $\rho(t) = c_\rho(1 - 16t^2)^5$ for $|t| \le \frac{1}{4}$ and $\rho(t) = 0$ for $|t| > \frac{1}{4}$; with $c_\rho = 693/128$, its integral is 1. The function $\phi_1 = 1_{[-1/2,1/2]} * \rho$ is of class C^5, is carried by $[-\frac{3}{4}, \frac{3}{4}]$, satisfies $0 \le \phi_1 \le 1$, and has $\sum_k \phi_1(t-k) = 1$, where k runs over $\mathbb{Z}$. Put $\chi(y) = \prod_\mu \phi_1(y_\mu)$ for $y \in \mathbb{R}^3$. Then $X_4 := \max_{|\alpha| \le 4} \sup |\partial^\alpha \chi| \le 15131.7$: the derivatives of ϕ_1 are pieced together from explicit polynomials, and $\sup |\phi_1^{(j)}| \le 1,\ 5.42,\ 45.1,\ 866.3,\ 15131.7$ for $j = 0, \dots, 4$.

For $j \ge 1$, $\phi_1^{(j)}(t) = \rho^{(j-1)}(t + \frac{1}{2}) - \rho^{(j-1)}(t - \frac{1}{2})$, and the two terms have disjoint carriers; hence $\sup |\phi_1^{(j)}| = \sup |\rho^{(j-1)}|$. In the variable $u = 4t$ one has $\rho = c_\rho(1-u^2)^5$. The first derivative $\rho' = -40 c_\rho u (1-u^2)^4$ has its largest absolute value at $u = \frac{1}{3}$, namely $40 c_\rho \cdot 4096/19683 \le 45.07$. The second derivative $\rho'' = 16 c_\rho (1-u^2)^3 (90u^2 - 10)$ has its largest absolute value at $u = 0$, namely $160 c_\rho = 866.25$. The third derivative is $\rho''' = 64 c_\rho \cdot 240 u (1-u^2)^2 (1 - 3u^2)$; the polynomial $u(1-u^2)^2(1-3u^2)$ has its critical points at $u^2 = 1$, at $u^2 = (14 - \sqrt{112})/42$ and at $u^2 = (14 + \sqrt{112})/42$, where its absolute values are 0, ≤ 0.1819584 and ≤ 0.0995 respectively, and so $|\rho'''|$ is at most $15360 c_\rho \cdot 0.1819584 \le 15131.7$. For mixed derivatives of χ of total order at most 4, the products of these suprema are smaller ($5.42 \cdot 866.3$, 45.1^2, $5.42^2 \cdot 45.1$).

For $m \ge 0$ let $h_m = 2^{-m-1}$, and for $c \in (\mathbb{Z}^3)^n$ let $\Phi_{m,c}(\xi) = \prod_i \chi(\xi_i/h_m - c_i)$, where $\xi = (\xi_1, \dots, \xi_n)$ with $\xi_i \in \mathbb{R}^3$. We call c admissible when $|c_i - c_k|_\infty \ge 6$ for $i \ne k$. Let $\mathsf{A}_m = \sum_{c \text{ admissible}} \Phi_{m,c}$, and $\mathsf{A}_{-1} = 0$. Suppose that $\mathsf{A}_{m-1}(\xi) > 0$. Then $|\xi_i - \xi_k| \ge 4.5 h_{m-1} = 9 h_m$ for $i \ne k$, so that every c with $\Phi_{m,c}(\xi) > 0$ has $|c_i - c_k| \ge 7.5$; all these c are admissible, and $\mathsf{A}_m(\xi) = 1$. It follows that $\mathsf{A}_m(1 - \mathsf{A}_{m-1}) = \mathsf{A}_m - \mathsf{A}_{m-1} =: \Upsilon_m$, and that $\sum_{m \le m_1} \Upsilon_m = \mathsf{A}_{m_1}$, which equals 1 wherever $\min |\xi_i - \xi_k| \ge 7.5 h_{m_1}$. Thus $1 = \sum_m \Upsilon_m$ off the diagonals, and the sum is locally finite; the Υ_m are obtained by sums and products alone, without any quotient. Finally, $1 - \mathsf{A}_{m-1}$ is the sum of the $\Phi_{m-1,c'}$ over the non-admissible c'; at every point of its support, some pair satisfies $|\xi_i - \xi_k|_\infty \le 6.5 h_{m-1} = 13 h_m$.

Step (b): one piece. Let $\varphi \in \mathcal{S}_{\neq}(\mathbb{R}^{3n})$. Fix m and an admissible c, put $h = h_m$, and let $G = \Phi_{m,c}(1 - \mathsf{A}_{m-1})\varphi$. Then G is carried by $\prod_i \{|x - h c_i| \le 3h/4\}$. It has continuous derivatives ∂^γ for each γ of order at most 4 in the three coordinates of every entry, and so of total order at most $4n$: the cut-offs are finite sums of products of C^5 functions, each of a single coordinate, and φ is smooth. Only these derivatives of G enter the argument. For each i we choose a continuous $\tilde{\chi}_i$ with $0 \le \tilde{\chi}_i \le 1$, equal to 1 on $|x - h c_i| \le 3h/4$ and carried by $|x - h c_i| \le 5h/4$; the supports of the $\tilde{\chi}_i$ are at least $3.5h$ apart, and each lies in a ball of radius $1.25h < 1$. A Fourier series of period $4h$ in each coordinate gives

$$G = \Big(\prod_i \tilde{\chi}_i\Big) \sum_k \hat{G}(k) \prod_i e_{k_i}, \qquad e_k(x) = \exp\{2\pi i k \cdot (x - h c_i)/(4h)\}.$$

With $\lambda = 4h/(2\pi)$ we have $(1 - \lambda^2 \Delta_i) e_{k_i} = (1 + |k_i|^2) e_{k_i}$, and integration by parts yields

$$|\hat{G}(k)| \le \prod_i (1 + |k_i|^2)^{-2} \sup \Big| \prod_i (1 - \lambda^2 \Delta_i)^2 G \Big|.$$

We set

$$\tilde{T}(G) := \sum_k \hat{G}(k)\, T^{\mathbb{C}}(\tilde{\chi}_1 e_{k_1}, \dots, \tilde{\chi}_n e_{k_n}),$$

where $T^{\mathbb{C}}$ denotes the complex n-linear extension of T. The complex entries cost a factor 2^n, and $\max\{1,(3.5h)^{-a}\} \le h^{-a}$. Further, $\sum_{\mathbb{Z}^3}(1+|k|^2)^{-2} \le 10.1$, by Poisson's formula, which gives the value $\pi^2 \sum_m e^{-2\pi|m|} \le 9.9986$ for this sum.

Step (c): derivatives. Written out, $(1-\lambda^2\Delta)^2$ consists of $1+6+9=16$ monomials $\lambda^{|\gamma|}\partial^\gamma$, $|\gamma| \le 4$. The Leibniz rule over the three factors of G produces at most 3^{4n} terms. The first factor obeys $|\partial^\gamma \Phi_{m,c}| \le X_4^n h^{-|\gamma|}$. At any point at most 2^{3n} of the c' contribute to $1-\mathsf{A}_{m-1}$, so $|\partial^\gamma(1-\mathsf{A}_{m-1})| \le 8^n X_4^n h^{-|\gamma|}$. Since $\lambda/h = 2/\pi < 0.64 < 1$ and $\lambda < 1$, no power of λ or of λ/h exceeds 1. Collecting these factors and those of Step (b), we obtain

$$|\tilde T(G)| \le \Gamma h^{-a}(2\cdot 10.1\cdot 16\cdot 81\cdot 8\cdot X_4^2)^n \max_{|\gamma|\le 4n} \sup_{\operatorname{supp} G} |\partial^\gamma \varphi|.$$

Step (d): flatness, for $m \ge 1$. Let $m \ge 1$ and $\xi \in \operatorname{supp} G$, and take the pair $i \ne k$ of Step (a), with $|\xi_i - \xi_k|_\infty \le 13h$. Let ξ^* be ξ with ξ_i replaced by ξ_k; this is a point of a diagonal, where every derivative of φ vanishes. Put $v = \xi_i - \xi_k$; then $|v|_1 \le 39h$. Taylor's formula with remainder on the segment from ξ^* to ξ gives, for each order r,

$$|\partial^\gamma \varphi(\xi)| \le \frac{(39h)^r}{r!} \max_{|\beta|=r} \sup_{\text{seg}} |\partial^{\gamma+\beta}\varphi|.$$

Step (e): weights. Because $(1+|y|^2)^n \ge \prod_i (1+|y_i|^2)$, we have, for $k \ge 4n$, the bound $|\partial^\alpha \varphi(y)| \le |\varphi|_k \prod_i (1+|y_i|^2)^{-2}$ whenever $|\alpha| \le k$. On the segment of Step (d), $|y_i - hc_i|_\infty \le 13.75h \le 6.875$. The inequality $1+|u|^2 \le 2(1+|u-y|^2)(1+|y|^2)$ then gives $(1+|y_i|^2)^{-2} \le 8.2\cdot 10^4 (1+|hc_i|^2)^{-2}$, as $4(1+3\cdot 6.875^2)^2 \le 81{,}564$. The same inequality, used on the cubes of side h, together with the value π^2 of the integral of $(1+|x|^2)^{-2}$ over $\mathbb{R}^3$, gives $\sum_{c\in\mathbb{Z}^3}(1+h^2|c|^2)^{-2} \le 5.641\pi^2 h^{-3} \le 57h^{-3}$. It is at this point that the hypothesis of the lemma is used for supports placed anywhere in $\mathbb{R}^3$.

Step (f): the sum. We put $C_4 := 2\cdot 10.1\cdot 16\cdot 81\cdot 8\cdot X_4^2 \cdot 8.2\cdot 10^4 \cdot 57 \le 2.3\cdot 10^{20}$. Let $G_{m,c}$ be the function G of Step (b) for the scale m and the admissible c. By Steps (c) to (e), the contributions $|\tilde T(G_{m,c})|$ of a scale $m \ge 1$ add up to at most $\Gamma C_4^n h^{-a-3n}\frac{(39h)^r}{r!}|\varphi|_{4n+r}$; the choice $r = a+3n+1$ makes this at most $\Gamma C_4^n 39^r h_m |\varphi|_{a+7n+1}$, and $\sum_{m\ge 1} h_m = \frac12$. At the scale $m=0$, without the flatness of Step (d), the contributions add up to at most $\Gamma C_4^n 2^{a+3n}|\varphi|_{4n}$. Altogether, the contributions of all scales add up to at most $\Gamma C_4^n 39^{a+3n+1}|\varphi|_{a+7n+1}$. Hence the series $\tilde T(\varphi) := \sum_{m,c}\tilde T(G_{m,c})$ converges absolutely, and $\tilde T(\varphi)$ is linear in φ.

Step (g): values at tensors. Let $X = (f_1,\dots,f_n) \in \mathfrak{S}_n^\infty$ and $K_i = \operatorname{supp} f_i$. Restricted to the product of the spaces of continuous functions carried by K_i, the map T induces a linear map Λ on their algebraic tensor product, which embeds in the space of functions on $\prod K_i$. Let $G = \bigotimes g_i$ be a tensor with $g_i \in C^5$ carried by $|x - hc_i| \le 3h/4$. Then $g_i = \tilde\chi_i \sum_k \hat g_i(k) e_k$ with uniform convergence; as T is continuous for the sup norm in each entry on fixed separated carriers, $\tilde T(G) = T(g)$. At $m=0$, $G_{0,c}$ is the tensor of the functions $\chi(\cdot/h_0 - c_i)f_i$, because $\mathsf{A}_{-1} = 0$. For $m \ge 1$ use the complementary coarse-grid sum. Then $G_{m,c}$ is a finite sum over c' of tensors of $\chi(\cdot/h_m - c_i)\chi(\cdot/h_{m-1} - c_i')f_i$. Only finitely many (m,c) contribute, since $\Upsilon_m = 0$ on the support $\prod K_i$ of $\bigotimes X$ once the sets K_i are pairwise at sup distance at least $7.5h_{m-1}$. As functions, these finitely many tensors add up to $\bigotimes X$. Therefore $\tilde T(\bigotimes X) = \Lambda(\bigotimes X) = T(X)$.

Step (h): proof of (G4) (b). We take $T = \sigma_S$. By (G2) (iii) of Thm. 9.4, applied with $R_f = 1$, the hypothesis of the lemma holds with $a = 5n$ and $\Gamma = (8A_5)^n (n!)^5$. As $39^{8n+1} \le 39^{9n}$ and $8C_4 39^9 \le 3.9\cdot 10^{35}$, the conclusion of the lemma becomes the bound of (G4) (b), with a norm of order $12n+1$.

Step (i): proof of (G4) (d). Now let $T = \sigma_{K,N}$ on bounded functions carried by $(-N/4, N/4)^3$; there the distance on the torus is that of $\mathbb{R}^3$. The hypothesis of the lemma is (G2) (i) for every $d > 0$; the scales $h_m < a_K$ are covered by Step 4 (c). We define the lattice sum

$$\tilde\sigma(G) := a_K^{-n} \sum G(x(p_1),\dots)\langle \prod (a_p - \langle a_p\rangle)\rangle.$$

It is linear in G, it depends on finitely many values of G, and $\tilde\sigma(\bigotimes f) = T(f)$. Distinct plaquettes have distinct barycentres, and $\varphi = 0$ at the points where two of the barycentres coincide. At each point the series of Step (b) converges absolutely, so $\tilde\sigma(G_{m,c}) = \tilde T(G_{m,c})$; and at the finitely many points that enter the lattice sum, only finitely many (m, c) contribute. The carrier of $\tilde\chi_i$ stays within $1.25h + 0.75h = 2h \le 2h_0 = 1$ of the i-th projection of $\operatorname{supp}\varphi$: it lies within $1.25h$ of hc_i, and hc_i lies within $0.75h$ of a point of that projection, since otherwise $G_{m,c} = 0$. The margin 1 in the statement of (G4) (d) is therefore exactly what is needed, and the box in that statement is open. □

Step 6: (G4) (a), (c), (e) *by Prop. C.13, uniqueness, and* (G5).

Lemma 9.12 (Bound, uniqueness and symmetries of the extension)**.** In the setting of Thm. 9.4, for every $n \ge 1$ and every $\varphi \in \mathcal{S}_{\neq}(\mathbb{R}^{3n})$,

$$|S_n^{\sigma_S}(\varphi)| \le C_E(\mathsf{b})^n (n!)^{13} |\varphi|_{13n-1};$$

for every n there is exactly one complex linear map $S_n^{\sigma_S}$ on $\mathcal{S}_{\neq}(\mathbb{R}^{3n})$ that is bounded with respect to some norm $|\cdot|_k$ and whose value at the tensor of each X in $\mathfrak{S}_n^\infty(\mathbb{R}^3)$ is $\sigma_S(X)$; and these maps satisfy (E2) of Def. F.17 in the form eq. C.xxx at every coordinate mirror, are real and have the symmetry (E3), and are invariant under all translations and under W_3, which is (E1) for $\mathbb{R}^3 \rtimes W_3$. The same is true for labels $0 < \mathsf{b} < x_s$, with the numbers $A_5(\mathsf{b})$, $A_6(\mathsf{b})$ and $C_E(\mathsf{b})$ of (G4) (e) of Thm. 9.4, and for a limit S of (R5) (a) or (c) along labels $\mathsf{b}_j \to \mathsf{b}$, with $C_E(\mathsf{b})$ taken as in (Lab3) of item 1 of §9, namely as $\limsup_j C_E(\mathsf{b}_j)$ along any subsequence. Uniqueness and the symmetries hold in both cases, whatever the label. ■

Proof. Step 6 (a): the bound. Fix $n \ge 2$. By (R6) (i) of Thm. 8.17, the functional $T := \sigma_S$ on $\mathfrak{S}_n^\infty(\mathbb{R}^3)$ is multilinear in the sense of Def. C.9. Take any real number $d > 0$. Let X satisfy $R(X) \le 1$ and $d(X) \ge d$, both in the Euclidean sense. Since every Euclidean ball is contained in the sup-ball of equal radius, (G2) (iii) of Thm. 9.4 gives

$$|T(X)| \le A_6^n (n!)^5 \max\{1, d^{-5n}\} 2^{3n} \prod \|f_i\|_\infty \le \mathsf{c}_\mathsf{T} \max\{1, d^{-5n}\} \prod \|f_i\|_\sharp, \qquad \mathsf{c}_\mathsf{T} = (8A_6)^n (n!)^5.$$

This is eq. C.xxiii with $\mathsf{a} = 5n$, and it holds for every such X, since the bound of (G2) does not depend on where the supports lie. Then eq. C.xxv gives

$$|\tilde T(\varphi)| \le \mathsf{C}_{\neq}^{6n} M^n (n!)^8 \mathsf{c}_\mathsf{T} |\varphi|_{13n-1} = C_E^n (n!)^{13} |\varphi|_{13n-1} \le C_E^n (n!)^{13} |\varphi|_{13n}.$$

For $n = 1$ the set $\mathrm{Part}_{\ge 2}(1)$ is empty, so $\sigma_S(f) = 0$, and $S_1^{\sigma_S} = 0$: the zero functional is bounded, its value at the tensors is $\sigma_S(f)$, and by Cor. C.16 it is the only such functional. Hence σ_S satisfies the growth condition of Def. C.17, with the quadruple of constants $(13, 1, C_E, 13)$.

Step 6 (b): one functional. Cor. C.16 states that two extensions which are bounded for some norms and coincide on the tensors are equal. Hence Lem. 9.11 and Prop. C.13 give one and the same functional, and (G4) (b) bounds that same object. Since $10^{36} A_5 \le 10^{318} M\, 8 \cdot 3^{5/2} A_5$ and $(n!)^5 \le (n!)^{13}$, Lem. 9.11 and Cor. C.16 suffice by themselves for the growth condition of Def. C.17 with the quadruple $(13, 1, C_E, 13)$.

Step 6 (c): the cumulants. By (R6) (i), the truncated function S_n of the limit S is n-linear. For $T = S_n$ we carry out Step 6 (a), with A_6, and Step (h) of the proof of Lem. 9.11, with A_5 and $R_f = 1$; in both we use the assertion of (G2) (iii) on bound (ii), for S_n, in place of its assertion on bound (i), for σ_S. By Cor. C.16 the two steps give the same functional, and the factor $1.45^{n+1} n!$ enters both bounds.

Step 6 (d): proof of (G5). The functional σ_S is a symmetric functional on separated tuples over $\mathbb{E} = \mathbb{R}^3$, as in Def. C.4 (3), and it satisfies the growth condition. We apply the parts of Prop. C.15 one after the other.

(a) Prop. C.15 (a) yields reality and symmetry.

(b) Prop. C.15 (b) is applied isometry by isometry, to every translation and to every element of W_3; its hypothesis, the invariance of σ_S under each of these isometries, is given by (R6) (iii).
(c) Prop. C.15 (c) yields eq. C.xxx at $(e_3, 0)$. The hypothesis of this part is (R6) (iv): σ_S is reflection positive at this mirror in the sense of Def. C.6 (3). For any other coordinate mirror (μ, c), we take the element of $\mathbb{R}^3 \rtimes W_3$ that maps it to $(e_3, 0)$, conjugate by it, and use the invariance obtained in item (b) of this list.

Step 6 (e): labels $0 < \mathsf{b} < x_s$. Steps 6 (a) to 6 (d), and Steps (h) and (i) of the proof of Lem. 9.11, depend on the label only through (G2) (i) to (iii). By (G2) (iv), the clauses (i) to (iii) hold for these labels, with the number $A_5(\mathsf{b}) = L^{4z} A_5(L^z \mathsf{b})$. Moreover, (R5) holds for every $\mathsf{b} > 0$, and every limit of (R5) satisfies (R6) (i), (iii), (iv). Hence these steps carry over to the labels $0 < \mathsf{b} < x_s$, with the numbers of (G4) (e).

Step 6 (f): moving labels. Steps 6 (a) to 6 (c) depend on the label only through the single number A_5 (or A_6) in (G2) (iii). This number is taken as (Lab3) of item 1 of §9 prescribes, and C_E is linear in it. □

Remark 9.13 (The exponent of d and the order of the norm). Flatness of order $a + 3n + 1$ pays for the factor h^{-a} and for the factor h^{-3n} that comes from the sum over the grid points. Hence the exponent a is allowed to be linear in n, provided the norm has an order that is linear in n too; this is the form of the growth condition of Def. C.17, with its norm of order $k_{\mathrm{E}} n$. A factor d^{-cn^2} or $e^{c/d}$ would not do, and neither would a prefactor e^{cn^4}.

Step 7: the field between two vectors, and (G6), (G6+).

Lemma 9.14 (Norm of the operator of one small piece)**.** In the setting of item 1 of §9, let S be any limit of (R5), whose label is any $\mathsf{b} > 0$, either fixed or approached by labels $\mathsf{b}_j \to \mathsf{b}$; let $\eta \in (0, 1]$, and let g be a real function of class C^1_c that is carried by a cube of side $\eta'/4$ lying inside $\{|x_3| < \eta'/2\}$. Put $\mathbb{E}_+ = \{x_3 > 0\}$. Then the operator given on representatives $F \in \mathbb{C}\mathfrak{S}(\mathbb{E}_+)$ by $\hat{A}_g[F] := [(\tau_{\eta' e_3} g) \cdot \tau_{2\eta' e_3} F]$ is well defined on $D = [\mathbb{C}\mathfrak{S}(\mathbb{E}_+)]$. If the label is fixed and $\mathsf{b} \geq x_s$, then

$$\|\hat{A}_g\| \leq \check{\Gamma}(\mathsf{b}, \eta'/2) e^{2\eta' c_{\mathrm{vol}}} \|g\|_\infty.$$

If $\mathsf{b}_j \to \mathsf{b}$, in the order (a) or (c) of (R5), and $\mathsf{b}_j \geq x_s$ for infinitely many j, then

$$\|\hat{A}_g\| \leq \limsup_j \check{\Gamma}(\mathsf{b}_j, \eta'/2) e^{2\eta' c_{\mathrm{vol}}(\mathsf{b}_j)} \|g\|_\infty,$$

the upper limit being taken along these j. ■

From Lem. 9.14 we obtain the main clause of Thm. 9.4 (G6), every bullet of (G6), and also (G6+).

Proof of Lem. 9.14, of Thm. 9.4 (G6) *and of* (G6+). *Step (a): reduction to one piece.* As $\eta' \geq \eta/2$, we have $\eta/4 \leq \eta'/2$, so the test function h of Thm. 9.4 (G6) has $\operatorname{supp} h \subset \{|x_3| < \eta'/2\}$; we define $\hat{A}_h$ by the formula of the lemma. For $G, G' \in \mathbb{C}\mathfrak{S}(\mathbb{E}_+)$, the scalar product of $\mathcal{H}(S)$ gives

$$\langle [G], \hat{A}_h [G'] \rangle = \sigma_S(\theta \overline{G} \cdot \tau_{\eta' e_3} h \cdot \tau_{2\eta' e_3} G').$$

After a translation of every entry by $-\eta' e_3$, and with $\tau_{-\eta' e_3} \theta = \theta \tau_{\eta' e_3}$, the right side equals $\sigma_S(\theta \overline{(\tau_{\eta' e_3} G)} \cdot h \cdot \tau_{\eta' e_3} G')$. Now let F, F' be as in (G6), and choose $G = \tau_{-\eta' e_3} F$ and $G' = \tau_{-\eta' e_3} F'$. Then

$$\sigma_S(\theta \overline{F} \cdot (h) \cdot F') = \langle \Psi, \hat{A}_h \Psi' \rangle$$

with $\Psi' = [\tau_{-\eta' e_3} F']$ and

$$\Psi = [\tau_{-\eta' e_3} F] = e^{-(\eta - \eta')H} [\tau_{-\eta e_3} F];$$

as $e^{-(\eta-\eta')H}$ is a contraction, $\|\Psi\| \leq \|[\tau_{-\eta e_3} F]\|$.

We now split h. Let $s = \eta'/8$, let $y \in \mathbb{R}^3$ be arbitrary, and for $c \in y + s\mathbb{Z}^3$ put $\chi_c(x) = \prod_\mu w((x_\mu - c_\mu)/s)$, where $w \in C^5$ is carried by $[-1,1]$ and satisfies $0 \le w \le 1$ and $\sum_k w(t-k) = 1$; we take $w = 1_{[-1/2,1/2]} * \rho_2$, with ρ_2 carried by $[-\frac12, \frac12]$. Then $h = \sum_c h\chi_c$ is a finite sum, and each piece $h\chi_c$ is of class C^1 and is carried by the intersection of its cube of side $\eta'/4$ with $\operatorname{supp} h$, hence by a cube of side $\eta'/4$ inside $\{|x_3| < \eta'/2\}$. Moreover $\|h\chi_c\|_\infty \le \sup_{|x-c|\le\eta'/8} |h|$. Since $\hat A$ is linear in h,

$$\|\hat A_h\| \le \sum_c \|\hat A_{h\chi_c}\|,$$

and the lemma bounds the term of c by $\check\Gamma(\mathsf{b}, \eta'/2) e^{2\eta' c_{\mathrm{vol}}} \sup_{|x-c|\le\eta'/8} |h|$; summed over c, these suprema give $\|\!|h|\!\|_{\eta,y}$. The steps (b) to (g) concern one piece g, and they are free of the grid.

Step (b): the adjoint. The reflection $\theta_{\eta'} : x_3 \mapsto 2\eta' - x_3$ leaves σ_S invariant. We have $\theta_{\eta'}\theta = \tau_{2\eta' e_3}$, then $\theta_{\eta'}\tau_{\eta' e_3} g = \tau_{\eta' e_3} g^r$ with $g^r(x) = g(x_1, x_2, -x_3)$, and $\theta_{\eta'}\tau_{2\eta' e_3} F = \theta F$. Applying $\theta_{\eta'}$ to every entry of $\langle [F'], \hat A_g [F]\rangle = \sigma_S(\theta\overline{F'} \cdot \tau_{\eta' e_3} g \cdot \tau_{2\eta' e_3} F)$, we obtain $\sigma_S(\tau_{2\eta' e_3}\overline{F'} \cdot \tau_{\eta' e_3} g^r \cdot \theta F)$. Next we apply θ to every entry, as Thm. 8.17 (R6) (iii) permits; with $\theta^2 = 1$ and the symmetry of σ_S, this gives $\sigma_S(\theta\tau_{\eta' e_3} g^r \cdot \theta\tau_{2\eta' e_3}\overline{F'} \cdot F)$, which is $\langle \hat A_{g^r}[F'], [F]\rangle$ by the definition, since $\overline{g^r} = g^r$ for real g. Thus

$$\langle [F'], \hat A_g [F]\rangle = \langle \hat A_{g^r}[F'], [F]\rangle$$

for all $F, F' \in \mathbb{C}\mathfrak{S}(\mathbb{E}_+)$.

Step (c): multiple reflections. Put $B = \hat A_{g^r}\hat A_g$ and, for $\Psi \in D$ and $k \ge 0$, $a_k = \langle \Psi, B^{2^k}\Psi\rangle$. By step (b), $a_0 = \|\hat A_g \Psi\|^2$ and B is symmetric, so the Cauchy–Schwarz inequality yields $a_k \le \|\Psi\| a_{k+1}^{1/2}$. Applying this for $0, 1, \dots, k-1$ in turn, we get

$$a_0 \le \|\Psi\|^{2-2^{1-k}} a_k^{2^{-k}}.$$

These inequalities hold as well for the seminorm on $\mathbb{C}\mathfrak{S}(\mathbb{E}_+)$, with $\hat A_g$ acting on representatives; taken there, they show that $\hat A_g$ maps null vectors to null vectors, so that $\hat A_g$ is well defined on D.

Step (d): the moment a_k. Let $m = 2^k$ and $\Psi = [F_0]$ with $F_0 \in \mathbb{C}\mathfrak{S}(\mathbb{E}_+)$. Unfolding the $2m$ factors of B^m, we find

$$a_k = \sigma_S(\theta\overline{F_0} \cdot g_1 \cdots g_{2m} \cdot \tau_{4m\eta' e_3} F_0),$$

where $g_j = \tau_{(2j-1)\eta' e_3}(g^r \text{ or } g)$. The copy g_j lies in the slab $((2j-2)\eta', 2j\eta')$, at distance at least $\eta'/2$ from its faces. So the g_j are pairwise at least η' apart, each is at least $\eta'/2$ away from the entries of θF_0 and of $\tau_{4m\eta' e_3} F_0$, and each lies in a cube of side $\eta'/4$.

Step (e): placing. We put $\check\delta := \eta'/2$, so that the copies g_j are of the type $(\check\delta; \check\delta/4)$. We write $F_0 = \sum \zeta_a X_a$ with tuples X_a of lengths $\le n_F$, and we fix a term $(\theta X_a, X_{a'})$ of the resulting expansion of a_k. Let

$$\check\delta_F := \min\{\check\delta,\ 1/(32 n_F),\ \text{the least distance between two different entries of the same tuple, } X_a \text{ or } X_{a'}\};$$

this number depends on F_0 and η, but not on m. With the partition of step (a), taken at $s = \check\delta_F/4$, we cut each entry of X_a and of $X_{a'}$ into pieces carried by cubes of side $\check\delta_F/2$. A term of the resulting sum contains one piece of each entry, so that its pieces are pairwise at least $\check\delta_F$ apart; and the number of terms does not depend on m.

Next we translate the whole tuple by a real vector v. In coordinate 3 we have $1 = p' \cdot 2\eta'$; hence, if one wall lies within $\eta'/4$ of a face of a slab, every wall does, and then the copies are at distance at least $\eta'/4 = \check\delta/2$ from the walls. The v_3 for which this happens form a set of measure $p' \cdot \eta'/2 = \frac14$ modulo 1. Each of the at most $2n_F$ pieces forbids an arc of length $1.5\check\delta_F$, so together they forbid at most $3/32$. In the coordinates 1 and 2, the stack of copies forbids $\eta'/4 + \eta'/2 \le 3/8$, and the pieces forbid at most $3/32$. Hence a translation v that avoids every forbidden set exists.

Step (f): the bound on a_k. To the translated term we apply Thm. 9.4 (G3) (b) at finite (K, N), with N large and with $K \ge j^\circ(\mathsf{b}, \check\delta_F) \ge j^\circ(\mathsf{b}, \check\delta)$. By line 1 of (G1), both numbers $j^\circ(\mathsf{b}, \check\delta_F)$ and

$j^\circ(\mathsf{b},\check\delta)$ exist, and j° is non-increasing in $\check\delta$; since $\check\delta_F \le \check\delta$, this gives the second inequality. A pure cell, that is, one that holds copies of g alone, holds at most p' copies, and for it $(\check\delta_t,\check\rho_t) = (\check\delta,\check\delta/4)$; since the stack has length $2m/p'$, there are at most $2m/p'+2$ pure cells. The mixed cells, which hold pieces of entries, number at most $2n_F$; for them $(\check\delta_t,\check\rho_t) = (\check\delta_F,\check\delta/4)$ and $n_t \le 2n_F + p'$. Each source lies in one cell, at distance at least $\check\delta_F/2$ from the walls. We conclude that

$$|a_k| \le C[\check\Gamma(\mathsf{b},\check\delta)e^{c_{\mathrm{vol}}/p'}\|g\|_\infty]^{2m},$$

where $C = C(F_0,\eta,g)$ does not depend on m: it is made of the number of terms, $|\zeta|^2$, the sup norms of the entries, $e^{(2+2n_F)c_{\mathrm{vol}}}$, $\max\{1,\check\Gamma(\mathsf{b},\check\delta_F,\check\delta/4)\}^{2n_F(2n_F+p')}$, $\max\{1,\|g\|_\infty\}^{2n_Fp'}$ and, for $g\neq 0$, $\max\{1,(\check\Gamma(\mathsf{b},\check\delta)\|g\|_\infty)^{-2n_Fp'}\}$. This bound passes to S. It is here that the cell-wise form of (G3) is needed: with one global $\check\delta$ we would get $\check\Gamma(\mathsf{b},\check\delta_F)^{2m}$, which depends on F_0.

Step (g): the norm of $\hat A_g$. As C is free of m, the bound of step (f) on $a_k^{1/m}$ tends to $[\check\Gamma e^{c_{\mathrm{vol}}/p'}\|g\|_\infty]^2$ as $k\to\infty$, where $\check\Gamma = \check\Gamma(\mathsf{b},\check\delta)$. Together with step (c), and with $\check\delta = \eta'/2$ and $1/p' = 2\eta'$, this gives

$$\|\hat A_g\| \le \check\Gamma(\mathsf{b},\eta'/2)e^{2\eta' c_{\mathrm{vol}}}\|g\|_\infty,$$

which is the bound of the lemma. Next we use lines 2, 4 and 5 of Thm. 9.4 (G1) at $\check\delta = \eta'/2 \in [\eta/4,\tfrac14]$: since $\mathfrak{l}$ is non-increasing in $\check\delta$, $1/\check\delta \le 4/\eta$ and $2\eta' \le 1$, the right side is at most $\check\Gamma_*(\mathsf{b},\eta)\|g\|_\infty$. Combined with step (a), this bound gives the main clause of Thm. 9.4 (G6), for every $y\in\mathbb{R}^3$. Indeed, by step (a), $|\sigma_S(\theta\overline{F}\cdot(h)\cdot F')| = |\langle\Psi,\hat A_h\Psi'\rangle|$, where $\|\Psi\| \le \|[\tau_{-\eta e_3}F]\|$ and Ψ' obeys the same bound with F' in place of F; and by step (a), with the bound just proved for each piece, $\|\hat A_h\| \le \sum_c \|\hat A_{h\chi_c}\| \le \check\Gamma_*(\mathsf{b},\eta)\,|\!|\!|h|\!|\!|_{\eta,y}$. Hence

$$|\sigma_S(\theta\overline{F}\cdot(h)\cdot F')| \le \check\Gamma_*(\mathsf{b},\eta)\,|\!|\!|h|\!|\!|_{\eta,y}\,\|[\tau_{-\eta e_3}F]\|\,\|[\tau_{-\eta e_3}F']\|.$$

Line 7 of (G1) gives $\mathfrak{l}(\mathsf{b},\eta/4) \le \mathfrak{l}_1\cdot 4/\eta$, whence

$$\check\Gamma_* \le 4A_2e^{C_W^*+e_h^*}(1+e^{c_{\mathrm{vol}}})e^{c_{\mathrm{vol}}}(\mathfrak{l}_1/\mathsf{b})^{1/2}/\eta.$$

Step (h): time-ordered products. Let $g_i = \tau_{-t_ie_3}h_i \in C_c^1(\{|x_3|<\eta_i/4\})$, and let $A_i = \hat A_{g_i}$ be the operator defined with η_i'. Once translated into $\mathbb{E}_+$, the class of $(h_1,\dots,h_n)$ is

$$e^{-s_0H}A_1e^{-s_1H}A_2\cdots A_n\Omega,$$

where $s_i = t_{i+1}-t_i-\eta_i'-\eta_{i+1}' \ge 0$, and $\sigma_S(h)$ is the scalar product of Ω with this vector. Step (a) with the grid at $y=-t_ie_3$, together with step (g), gives $\|A_i\| \le \check\Gamma_*(\mathsf{b},\eta_i)|\!|\!|g_i|\!|\!|_{\eta_i,-t_ie_3}$. Moreover

$$|\!|\!|\tau_{-te_3}h|\!|\!|_{\eta,-te_3} = \sum_{c_0\in(\eta'/8)\mathbb{Z}^3} \sup\{|h(x+te_3)| : |x-(c_0-te_3)| \le \eta'/8\} = |\!|\!|h|\!|\!|_{\eta,0}.$$

Therefore the bound holds with the fixed grid $y=0$, that is, with the norms $|\!|\!|h_i|\!|\!|_{\eta_i}$, as stated in Thm. 9.4 (G6).

Step (i): the norm $|\!|\!|\cdot|\!|\!|$. On a cube Q of side s,

$$\sup_Q|u| \le \sum_{\alpha\in\{0,1\}^3} s^{|\alpha|-3}\int_Q|\partial^\alpha u|;$$

indeed, in one variable the supremum is at most the mean plus $\int|u'|$, and we iterate this over the three coordinates. We apply it on the cubes that enter $|\!|\!|h|\!|\!|_{\eta,y}$, whose side is $s=\eta'/4\ge\eta/8$, so that $s^{|\alpha|-3}\le 512\,\eta^{|\alpha|-3}$; almost every point lies in 8 of these cubes, on every translate of the grid. The two factors give the constant $8\cdot512 = 4096$.

Step (j): other directions. For the other coordinate directions we pass to e_3 by an element of the group W_3.

Step (k): the last bullet of Thm. 9.4 (G6). Let $\mathsf{b}>0$, let $\mathsf{b}_j\to\mathsf{b}$ (or $\mathsf{b}_j=\mathsf{b}$ for all j), and let $z\ge1$ be an integer with $L^z\mathsf{b}_j \ge x_s$ from some j on; such a z exists because $\{\mathsf{b}_j\}\cup\{\mathsf{b}\}$ is compact

in $(0,\infty)$. If the label is fixed, Thm. 8.17 (R6) (vi) gives $S_n(f) = L^{zn} S'_n(f^{[z]})$; here S' is taken along the matched subsequence, at the label $L^z\mathsf{b} \geq x_s$, as a limit in the same order. For moving labels we rely on Step 5 of the proof of Thm. 8.2; its statement is valid for all $\beta > 0$ and all z, and by it the member with index j of the sequence forming S is L^{zn} times the value at $f^{[z]}$ of a member with label $L^z\mathsf{b}_j$. In the order (c) of (R5) this reads

$$S^T_{n;K_j,l_j,L^{K_j}\mathsf{b}_j}(f) = L^{zn} S^T_{n;K_j-z,L^z l_j,L^{K_j-z}(L^z\mathsf{b}_j)}(f^{[z]}),$$

and in the order (a), by the last sentence of that statement,

$$S_n^{(l_j,\mathsf{b}_j)}(f) = L^{zn} S_n^{(L^z l_j, L^z\mathsf{b}_j)}(f^{[z]}).$$

The dilation $f \mapsto f^{[z]}$ maps the set of tuples whose entries are continuous, with compact and pairwise disjoint supports, one to one onto itself; hence the dilated members converge on all such tuples. Their limit S' is a limit of (R5), in the same order of limits, taken along sides $L^z l_j = 2L^{m_j+z} \to \infty$ and labels $L^z\mathsf{b}_j \to L^z\mathsf{b}$; these labels are $\geq x_s$ for all large j, and $S_n(f) = L^{zn} S'_n(f^{[z]})$. Condition (Tor) of item 1 of §9 at $L^z\mathsf{b}_j$ holds for large j. Since σ_S is a sum, over partitions, of products, we get $\sigma_S(X) = L^{zn}\sigma_{S'}(X^{[z]})$. Let D_z be the linear map $X \mapsto L^{zn}X^{[z]}$. Dilation commutes with θ, and $(\tau_v f)^{[z]} = \tau_{L^z v} f^{[z]}$. Consequently $\langle [F],[F']\rangle_S = \langle [D_zF],[D_zF']\rangle_{S'}$, and

$$\sigma_S(\theta\overline{F}\cdot(h)\cdot F') = \sigma_{S'}(\theta\overline{D_zF}\cdot(L^z h^{[z]})\cdot D_zF').$$

Here every entry of every tuple of D_zF has its support in $\{x_3 > L^z\eta\} \subset \{x_3 > L^z\eta_z\}$, $\operatorname{supp} h^{[z]}$ lies in $\{|x_3| < L^z\eta_z/4\}$, and $L^z\eta_z \leq 1$. Clause (G6) for S' at $(L^z\mathsf{b}, L^z\eta_z)$, which holds for moving labels by step (l), bounds the modulus of this quantity by

$$L^z\check\Gamma_*(L^z\mathsf{b}, L^z\eta_z)\|\!|h^{[z]}|\!\|_{L^z\eta_z,y}\|[\tau_{-L^z\eta_z e_3}D_zF]\|_{S'}\|[\tau_{-L^z\eta_z e_3}D_zF']\|_{S'}.$$

Finally

$$\|[\tau_{-L^z\eta_z e_3}D_zF]\|_{S'} = \|[\tau_{-\eta_z e_3}F]\|_S = \|e^{-(\eta-\eta_z)H}[\tau_{-\eta e_3}F]\| \leq \|[\tau_{-\eta e_3}F]\|,$$

and the same chain holds with F' in place of F.

Step (l): moving labels. Suppose that $\mathsf{b}_j \geq x_s$ for infinitely many j; we pass to these j, and the subsequence has the same limit S. In step (f), the j-th member of the sequence carries the label b_j. By (Lab1) of item 1 of §9, $j^\circ(\mathsf{b}_j, \check\delta_F)$ is bounded in j, while $K \to \infty$ in Thm. 8.17 (R5) (a), (c); so the condition on K is met from some j on, and, as in Step 4 (f′) of the proof of Lem. 9.9, condition (Tor) at b_j holds for large j, because the sides go to infinity while $s(x) = 8r(x)$ is bounded on compact sets. The numbers that make up C stay bounded in j: c_{vol} is continuous; by item 2 of §8.3 and Thm. 9.4 (G1), $\check\Gamma(\cdot, \check\delta_F, \check\delta/4)$ is locally bounded; and since $1/r^\circ = \mathrm{amp}(L^{j^\circ+1}\mathsf{b})/3.5$, with $j^\circ(\cdot,\check\delta)$ locally bounded on $[x_s,\infty)$ by (Lab1) and with amp continuous and positive jointly in x and the label, we have $\check\Gamma(\cdot,\check\delta) \geq 1/r^\circ > 0$, where $1/r^\circ$ has a positive lower bound on every compact set of labels in $[x_s,\infty)$. Hence

$$|a_k| \leq C'[\limsup_j \check\Gamma(\mathsf{b}_j,\check\delta)e^{c_{\mathrm{vol}}(\mathsf{b}_j)/p'}\|g\|_\infty]^{2m},$$

with C' free of m. From this and step (c), with $\check\delta = \eta'/2$ and $1/p' = 2\eta'$, we argue as in step (g): the limit $k \to \infty$ yields the first inequality in

$$\|\hat A_g\| \leq \limsup_j \check\Gamma(\mathsf{b}_j,\eta'/2)e^{2\eta' c_{\mathrm{vol}}(\mathsf{b}_j)}\|g\|_\infty \leq \limsup_j \check\Gamma_*(\mathsf{b}_j,\eta)\|g\|_\infty,$$

and lines 2, 4 and 5 of (G1) at $\check\delta = \eta'/2$, applied at each $\mathsf{b}_j \geq x_s$, yield the second, as in (Lab3). In the remaining case, $\mathsf{b}_j < x_s$ from some j on (so that $\mathsf{b} \leq x_s$), and steps (e) to (g) do not apply to these members. Indeed, $\check\Gamma_*(\mathsf{b}_j,\cdot)$ and $j^\circ(\mathsf{b}_j,\cdot)$ are defined under the data, and (G1) and (R1) are stated under these data, whose label is $\geq x_s$. This case is treated by step (k), and by step (n).

Step (m): Thm. 9.4 (G6+) (m). Let $\eta > 0$ be arbitrary, let $\hat\eta = \min\{1, 3\eta/4\}$ and $s = (\hat\eta)'/8$, and write $h = \sum_c h\chi_c$ on the grid $y + s\mathbb{Z}^3$, with the partition of step (a). We fix a piece different from 0 and let t be the point of $[-\eta/4, \eta/4]$ nearest to c_3. This piece lies in $\{|x_3 - t| \le s\}$. Indeed, $t = c_3$ when $|c_3| \le \eta/4$; when $c_3 > \eta/4$, the points of the piece satisfy $c_3 - s \le x_3 < \eta/4 = t$, whence $0 < t - x_3 < s$; and when $c_3 < -\eta/4$, they satisfy $-\eta/4 = t < x_3 \le c_3 + s$, whence $0 < x_3 - t < s$. The function $g := \tau_{-te_3}(h\chi_c)$ is real and of class C^1, and it is carried by a cube of side $2s = (\hat\eta)'/4$ inside $\{|x_3| \le s\} \subset \{|x_3| < (\hat\eta)'/2\}$.

We translate all entries by $-te_3$, as Thm. 8.17 (R6) (iii) permits, and use $\tau_{-te_3}\theta = \theta\tau_{te_3}$; this gives

$$\sigma_S(\theta\overline{F}\cdot(h\chi_c)\cdot F') = \sigma_S(\theta\overline{(\tau_{te_3}F)}\cdot(g)\cdot\tau_{-te_3}F').$$

The tuples $\tau_{te_3}F$ and $\tau_{-te_3}F'$ are carried by $\{x_3 > \eta - |t|\} \subset \{x_3 > 3\eta/4\} \subset \{x_3 > \hat\eta\}$. Steps (a) and (g), for the single piece g at $\hat\eta \in (0, 1]$ (and step (l) at moving labels), bound the term by

$$\check\Gamma_*(\mathsf{b}, \hat\eta)\|g\|_\infty \|[\tau_{(t-(\hat\eta)')e_3}F]\|\,\|[\tau_{(-t-(\hat\eta)')e_3}F']\|.$$

Now $[\tau_{(t-(\hat\eta)')e_3}F] = e^{-(\eta+t-(\hat\eta)')H}[\tau_{-\eta e_3}F]$ with $\eta + t - (\hat\eta)' \ge \eta - \eta/4 - 3\eta/4 = 0$, so its norm is at most $\|[\tau_{-\eta e_3}F]\|$, the semigroup being a contraction by (R7); the same holds with $-t$ in place of t and F' in place of F. Finally $\|g\|_\infty \le \sup\{|h(x)| : |x - c| \le s\}$, and summing over c we obtain $\|\!|h|\!\|_{\hat\eta,y}$.

Step (n): Thm. 9.4 (G6+) (n). We take the argument of step (k) without change, from the choice of b_j and z up to the identity

$$\sigma_S(\theta\overline{F}\cdot(h)\cdot F') = \sigma_{S'}(\theta\overline{D_zF}\cdot(L^z h^{[z]})\cdot D_zF').$$

Now D_zF, D_zF' are carried by $\{x_3 > L^z\eta\}$, and $h^{[z]}$ by $\{|x_3| < L^z\eta/4\}$. Clause (G6+) (m), which admits any positive number, applied to S' with $L^z\eta$ in the place of η, gives the bound

$$L^z\check\Gamma_*(L^z\mathsf{b}, e_z)\|\!|h^{[z]}|\!\|_{e_z,y}\|[\tau_{-L^z\eta e_3}D_zF]\|_{S'}\|[\tau_{-L^z\eta e_3}D_zF']\|_{S'},$$

and $\|[\tau_{-L^z\eta e_3}D_zF]\|_{S'} = \|[\tau_{-\eta e_3}F]\|_S$, and the same identity holds with F' in place of F. □

Remark 9.15 (The lower bound on $\check\Gamma(\mathsf{b}_j, \check\delta)$ in step (l)). In step (l) of the proof of Lem. 9.14, the lower bound of $\check\Gamma(\mathsf{b}_j, \check\delta)$ enters only through the factor $\max\{1, (\check\Gamma\|g\|_\infty)^{-2n_F p'}\}$. By the definitions of item 1 of §9, we have $\check\Gamma(\mathsf{b}_j, \check\delta) = G_1(\mathsf{b}_j, \check\delta, \check\delta/4)(1 + e^{c_{\mathrm{vol}}(\mathsf{b}_j)}) \ge G_1$ and $G_1 = e^{C_W + e_h}/r^\circ \ge 1/r^\circ$.

Remark 9.16 (Def. A.4 and Prop. A.11). The notions recalled here are defined in Def. A.4 (2) and in the paragraphs "Boards" and "Mirror maps, copies and folds", which follow that definition in §A.2. The quadruple $(a_K, L^K, (N, N, N), 0)$ is a board: L^K is odd and ≥ 3, $N = 2L^{m_l}$ is even and ≥ 2, a cell has side 1, and there are N^3 cells. The mirrors $o_\mu + \mathsf{l}(t + \frac12)$ pass through midpoints of bonds and hold no site eq. A.xii. A function of a cell is a bounded Borel function of its inner bonds. A mirror map has the form $x \mapsto o + \mathsf{l}\varkappa' + \sigma(\varkappa', \varkappa)(x - o - \mathsf{l}\varkappa)$ eq. A.xiii. The fold Fold F is the product of the N^3 copies, conjugated for odd pairs, which changes nothing for real F eq. A.xiv, eq. A.xv. With $\|F\|_{\mathrm{ch}} = \langle \mathrm{Fold}\, F\rangle^{1/N^3}$ as in Def. A.10, eq. A.xxiii, Prop. A.11 states $|\langle\prod F_\varkappa\rangle| \le \prod\langle \mathrm{Fold}\, F_\varkappa\rangle^{1/N^3}$.

Remark 9.17 (§12). Let $z(\mathsf{b})$ be the least $z \ge 0$ with $L^z\mathsf{b} > x_s$, as in (S7) of §1. In Thm. 12.4 (E5), the growth bound with its quadruple of constants holds with $\mathsf{C}^6_{\neq} M L^z(1 + L^z)^3 A_6(L^z\mathsf{b})$, $z = z(\mathsf{b})$, in the place of C_E; the factor $L^z(1 + L^z)^3$ there is to be compared with the factor $8L^{4z}$ here. When $z(\mathsf{b}) = z_0(\mathsf{b})$, the number $\bar c_*(\mathsf{b}) = (3/(4\pi))L^{4z}A_6(L^z\mathsf{b})$ and the number $(3/(4\pi))A_6(\mathsf{b})$, with $A_6(\mathsf{b}) = L^{4z}A_6(L^z\mathsf{b})$ of (G4) (e), are given by one formula, $(3/(4\pi))L^{4z}A_6(L^z\mathsf{b})$. In (E5), however, $A_6(L^z\mathsf{b})$ is the supremum over $L^zU(\mathsf{b})$, whereas here it is the value at the label, or the number of (Lab3); hence $\bar c_*(\mathsf{b}) \ge (3/(4\pi))A_6(\mathsf{b})$.

Remark 9.18 (What (G1) uses of sep°). Clause (G1) of Thm. 9.4 uses $\mathrm{sep}^\circ \ge 0$ in Step (0) of the proof of Lem. 9.5, for $2\check r + 3 \le \check\lambda$; it uses $\mathrm{sep}^\circ > 0$ in line 1 and in $T_L < \infty$; and it uses

$\frac{1}{2} \le \mathrm{sep}^\circ + \log L/528$ in line 3 at $j = 1$. At the value $\mathrm{sep}^\circ = 8\log(8L^{1/2})$ of item 2 of §1, we have $\mathrm{sep}^\circ \ge 8\log 8 > 16.6$, and all three conditions hold.

10. Symmetries of continuum limits

On one torus and at one label, we bound the chessboard seminorm of a product of centred sources that lie in one cell, with free centres, first on the unit board and then on boards with small cells. Domination at a fixed cutoff on the infinite lattice carries these bounds over to every Gibbs state that is invariant under $a_K\mathbb{Z}^3$; this yields one table of moments for all $K \ge 1$, and compactness. For every limit $S \in \mathrm{Lim}_G(\mathsf{b})$ we then show that σ_S has the properties (R0) to (R7) of Def. D.10: an insertion bound with the power η^{-1}, a growth of the plates with exponent $7/2 < 4$, and positivity at the coordinate and at the diagonal mirrors. We then obtain invariance under $\mathbb{R}^3 \rtimes \mathrm{O}(3)$, under $\mathrm{O}(3)$ from Thm. D.31 and under translations from (R1); positivity at every mirror from Prop. D.7 (c), or from Prop. C.7 (b), (c); the circular cone and the bound $\|\mathcal{T}(z)\| \le 1$ from Cor. D.8; and the group $\mathrm{O}(2)$ from Prop. D.6. The dilation $x \mapsto L^z x$ carries all of this over to every label $\mathsf{b} > 0$. Besides, we treat joint limits with large sides, and we show that for every limit, reflection positivity at one diagonal mirror is equivalent to invariance under $\mathrm{O}(3)$.

Of Thm. A, this section proves clause (Y5), the third line of (Y2), the last two sentences of (Y3) and the last paragraph of (Y4). The letters, the board, the number c_{vol} and the ranges are those of §1.

Remark 10.1 (The letters of this section). Without a reference, (R0) to (R7) always denote the properties of Def. D.10, while a clause of Thm. 8.17 always appears together with that reference. The letter N always stands for the side of the torus, and l always for the side of a cell; where the data of Thm. 8.2 write l for the side of the torus, we write N, and m_l is the exponent of the side in those data. The threshold in K is $j^\circ_U(\mathsf{b}, \check\delta)$; neither the number $K_0(\mathsf{b})$ of entry (O14) in item 1 of §8.1 nor the index set of eq. B.xli carries a letter here. Without arguments, B_0 is the board; with its arguments, $B_0(\mathsf{b}, \check\delta)$ is the number of item 2 of §8.3. The number c_{I} is the one of Def. D.10; no number of this section depends on the letter c_I that item (1) of §6.1 adds to entry (O0).

0.1. Data and constants. We take the data of Thm. 8.2 and of item 1 of §8.1: the group $G = \mathrm{SU}(2)$ in dimension 3, a label $\mathsf{b} \ge x_s$, the side $N = 2L^{m_l}$ with $m_l \ge m'$ and the condition (Tor) $N \ge 2Ms(L\mathsf{b})$, a number of steps $K \ge 1$, and $\beta = L^K\mathsf{b}$. From the same place come Wilson's measure $\mu_{K,N,\beta}$ on $a_K\mathbb{Z}^3/N\mathbb{Z}^3$, where $a_K = L^{-K}$, and the source $\mathcal{O}_K(f)$. There, too, the separation d, the radii and the balls refer to the sup distance of the torus, and a Euclidean type $(d; R)$ counts as a sup type $(d/\sqrt{3}; R)$. The entries (O4a) and (O7) in item (1) of §6.1, the table of item 1 of §8.1 and the remark after Thm. 8.2 in item 2 of §8.1 fix L and $M = L^{m'}$: by that remark L is odd with $L \ge L^{\mathrm{su}}_* \ge 10^{2048.9}$, and by the table $m' \ge 1$. These values lie in the ranges in which the sections A.4, B.10, B.5 and B.31 are stated, namely L odd with $L \ge 13$ and $m' \ge 0$ (Def. B.31 (1)), and Thm. 10.4 is asserted for the L and M of the table alone. From item 1 of §7.5 we take $q_N = (N/M)^3 \ge 1$ and

$$C_{\mathrm{vol}}(N, x) = 4q_N e^{220q_N} e^{(3/2)\mathsf{N}_Z p_0(x)^2 N^3} (384/\varepsilon(x)^3)^{3N^3}.$$

The table and the radii of item 1 of §8.1 provide $p_0(x) = A_0(\log x)^{\mathfrak{p}_0}$, $\varepsilon(x) = p_0(x)x^{-1/2}$ and $c_h(x) = e^{-c_h p_0(x)^2}$, the floor $x_s \ge e^2$, the bound $p_0 \ge 1$ on $[x_s, \infty)$, and $\check r(x) = \lceil 20c_R \log x/M\rceil + 3$.

0.2. Results of earlier sections. Thm. 8.17 (R0) gives the lattice symmetries of the subtraction. It says that the subtraction $\hat c_K(f) = \frac{d}{dz}\log \rho_K^{[zf]}(1)\big|_{z=0}$ is real, linear in f, and dependent on K and N (we write it $\hat c_{K,N}$ where the dependence on N matters), and that $\hat c_K(f \circ r) = \hat c_K(f)$ for $r : y_\mu \mapsto c - y_\mu$ at every integer c; we use it at odd c. The subtracted moment estimate is Thm. 8.17 (R1) and we use it for $K \ge j^\circ$. Its data are a label $\mathsf{b} \ge x_s$, a side N allowed by the data of Thm. 8.2, a number $K \ge j^\circ(\mathsf{b}, \check\delta)$, an arbitrary $Q \ge 1$, and real bounded functions g_t on

T_N whose sets $\{g_t \neq 0\}$ are pairwise at sup distance at least $\check\delta$ and lie each in a sup-ball of radius $\check\rho$. For these it gives the bound

$$|\langle \prod_t (\mathcal{O}_K g_t - \hat c_K g_t)\rangle_{K,N}| \le G_1(\mathsf{b},\check\delta,\check\rho)^Q (1 + 0.182\, C_{\mathrm{vol}}(N,\mathsf{b})) \prod \|g_t\|_\infty .$$

On the board B_0, Thm. 8.17 (R2) (i), taken at $K \ge j^\circ(\mathsf{b},\check\delta)$, is Thm. 10.4 (4.1) (b), (c). In this section these two clauses also follow from Thm. 10.4 (4.1) (a) (Step 1 of the proof of Thm. 10.4). Of Thm. 9.4 we use clause (G1), on $0 < \check\delta \le 1$, with the following lines:

Line 1: $L^{j^\circ(\mathsf{b},\check\delta)} \le 2L^2(M/\check\delta)\, \mathfrak{l}(\mathsf{b},\check\delta)$;
Line 2: $1/r^\circ \le A_2(\mathfrak{l}/(\mathsf{b}\check\delta))^{1/2}$;
Line 4: $C_W(\mathsf{b},\check\delta,\check\delta/4) \le C_W^*(\mathsf{b})$;
Line 5: $e_h \le e_h^*(\mathsf{b})$;
Line 7: $\mathfrak{l}(\mathsf{b},\check\delta) \le \mathfrak{l}_1(\mathsf{b})(1+\log(1/\check\delta)/528)$;
Line 8: $\max\{\check\Gamma(\mathsf{b},\check\delta)e^{c_{\mathrm{vol}}}, 2B_0(\mathsf{b},\check\delta)\} \le A_4(\mathsf{b})\check\delta^{-2}$.

We use Thm. 8.17 (R5), Thm. 8.17 (R6) (i) to (iv) and (vi), Thm. 9.4 (G2) (i) to (iii), of which (i) and (ii) are lines 1 and 2 of Thm. A (Y2), and the main clause and the first bullet of Thm. 9.4 (G6).

Definition 10.2 (Item 1: labels, the class of states, limit sets, boards, normal form)**.** *Labels.* A number $\mathsf{b} > 0$ is the label of a limit, and b' is the label of an approximant; always $\beta = L^K\mathsf{b}' > 0$.

The class. We denote by $\mathbb{G}_K(\mathsf{b}')$ the set of the Gibbs states ν at $(a_K, L^K\mathsf{b}')$, in the sense of Def. B.1 (2), with $(\tau_v)_*\nu = \nu$ for all $v \in a_K\mathbb{Z}^3$. By Def. B.1 (3), with period number 1, we have $\mathbb{G}_K(\mathsf{b}') \subset \mathcal{G}^{\mathrm{per}}_{K,\beta}$; by Prop. B.3 (c), $\mathbb{G}_K(\mathsf{b}') \supset \mathcal{G}^{\mathrm{tor}}_{K,\beta}$, so that $\mathbb{G}_K(\mathsf{b}')$ is not empty. For $\rho,\nu \in \mathbb{G}_K(\mathsf{b}')$ and a tuple $X = (f_1,\dots,f_n)$ we put $X^\nu := \prod_i(\mathcal{O}_K f_i - \nu(\mathcal{O}_K f_i))$, give the empty tuple the value 1, and extend linearly; then

$$\sigma_\rho(X) := \rho(X^\rho) = \sum_{\pi\in \mathrm{Part}_{\ge 2}(n)} \prod_{J\in\pi} S^T_{|J|}[\rho](X_J).$$

Limit sets. We let $\mathrm{Lim}_G(\mathsf{b})$ be the set of the families $S = (S_n)_{n\ge 2}$ with $S_n(X) = \lim_j S_n^T[\rho_j](X)$ for all $n \ge 2$ and all separated C_c^1 tuples X over $\mathbb{R}^3$, where $K_j \to \infty$, $\rho_j \in \mathbb{G}_{K_j}(\mathsf{b}_j)$ and $\mathsf{b}_j \to \mathsf{b}$. The set $\mathrm{Lim}_{\mathrm{tor}}(\mathsf{b})$ consists of those families for which $\rho_j \in \mathcal{G}^{\mathrm{tor}}$. The limits of Thm. 8.17 (R5) (b), in which first $N \to \infty$ along a subsequence at fixed K and then $K \to \infty$, lie in $\mathrm{Lim}_{\mathrm{tor}}(\mathsf{b})$: by Thm. 8.17 (R5) (b) the measures converge weakly along the subsequence of sides, so that their limit lies in $\mathcal{G}^{\mathrm{tor}}$ by Def. B.1 (4), and the cumulants are polynomials in means of continuous local functions. For every family $S = (S_n)_{n\ge2}$, σ_S denotes the centred functional eq. 1.i, with $\sigma_S(\emptyset) = 1$.

Boards. We use one family B_i, the board $(i,0)$ of $\mathbb{L}_K$ for $-m_l \le i \le K-1$, whose cells are $\mathsf{l}k + [-\mathsf{l}/2,\mathsf{l}/2]^3$ with $\mathsf{l} = L^{-i}$ and whose walls are $\mathsf{l}(\mathbb{Z}+\frac12)$; besides, we use its translates (i,o) with $o \in a_K\mathbb{Z}^3$. In this section only $i \ge 0$ occurs.

Normal form. The set $U(\mathsf{b})$ is that of item (S7) of §1; for $\mathsf{b} > x_s$, $U(\mathsf{b}) = [\mathsf{b}/2, 2\mathsf{b}] \cap [x_s,\infty)$. The labels are in normal form if $\mathsf{b} > x_s$; then $\mathsf{b}_j \in U(\mathsf{b})$ from some j on.

Remark. At $\mathsf{b} = x_s$, the clauses are given by Thm. 10.4 (4.8) at $z = 1$, for every sequence. For sequences with $\mathsf{b}_j \ge x_s$, the numbers defined by the supremum over $[x_s, 2x_s]$ may be used in the clauses as well, because throughout each proof the label is a single fixed $\mathsf{b}' \ge x_s$.

Every number $X(\mathsf{b})$ below means $\sup_{\mathsf{b}'\in U(\mathsf{b})} X(\mathsf{b}')$. Each of them is thus one function of b, which depends neither on the sequence (b_j) nor on the state. These suprema are finite, on the following grounds. The number c_{vol} is continuous on $[x_s,\infty)$, by the formulas for p_0 and ε in the table of item 1 of §8.1; there $p_0 \ge 1$, so that $\varepsilon > 0$. The number C_W^* is bounded on $[x_s,\infty)$, by line 4 of Thm. 9.4 (G1). On $U(\mathsf{b})$ we have $e_h^*(\mathsf{b}') = 0.45(2\check r(\mathsf{b}')+4)^3 c_h(\mathsf{b}') \le 0.45(2\check r(2\mathsf{b})+4)^3$, since $c_h(\cdot) \le 1$ and $\check r$ does not decrease (the table and the radii of item 1 of §8.1). The function $\mathfrak{l}_1$ is continuous.

The numbers A_4, $\hat G$, A_6, $C_{\mathrm I}$ and C_φ are continuous expressions in these four numbers and in $1/\mathsf b'$. Finally, $\check\Gamma(\mathsf b',\check\delta)\le \hat G(\mathsf b')/\check\delta$ and $\Gamma(\mathsf b',\mathsf l)\le 3\hat G(\mathsf b')e^{c_{\mathrm{vol}}(\mathsf b')}/(2\mathsf l)$.

We put $j^\circ_U(\mathsf b,\check\delta):=\sup_{\mathsf b'\in U(\mathsf b)} j^\circ(\mathsf b',\check\delta)$, which is finite by line 1 of Thm. 9.4 (G1), as $\mathfrak l$ is continuous in $\mathsf b'$. Clauses (4.1) and (4.2) of Thm. 10.4 are statements on one torus at one label, and no supremum is taken in them. In these two clauses, and throughout the proofs of clauses (4.3) to (4.5), a single label $\mathsf b'$ is kept fixed, and the supremum is taken at the end. Every other label $\mathsf b>0$ is treated in clause (4.8). ∎

Definition 10.3 (Item 2: the numbers of the theorem)**.** All numbers of this item are placed at (O14). No floor depends on them, and none of them depends on N, K, n, the places of the sources or the state, with one exception: $\check\delta_n(d)$, which depends on n and on d. We set $\mathrm{sep}^\circ:=8\log(8L^{1/2})$; it enters here only as a letter, inside j°, $\mathfrak l$, $\mathfrak l_1$, $\check\lambda$ and T_L. We set

$$c_{\mathrm{vol}}(\mathsf b):=225/M^3+\frac{3}{2}\mathsf N_Z p_0(\mathsf b)^2+3\log(384/\varepsilon(\mathsf b)^3),$$

so that $1+0.182\,C_{\mathrm{vol}}(N,\mathsf b)\le e^{c_{\mathrm{vol}}(\mathsf b)N^3}$. We define $\check\Gamma(\mathsf b,\check\delta):=G_1(\mathsf b,\check\delta,\check\delta/4)(1+e^{c_{\mathrm{vol}}(\mathsf b)})$ and

$$\hat G(\mathsf b):=A_2e^{C^*_W(\mathsf b)+e^*_h(\mathsf b)}(1+e^{c_{\mathrm{vol}}(\mathsf b)})(\mathfrak l_1(\mathsf b)/\mathsf b)^{1/2}.$$

For $0<\check\delta\le 1$,

$$\check\Gamma(\mathsf b,\check\delta)\le \hat G(\mathsf b)\check\delta^{-1/2}(1+\log(1/\check\delta))^{1/2}\le \hat G(\mathsf b)/\check\delta;$$

the first inequality comes from lines 2, 4, 5 and 7 of Thm. 9.4 (G1), and the second from $x(1+\log(1/x))\le 1$ on $(0,1]$, taken at $x=\check\delta$. We put $\Gamma(\mathsf b,\mathsf l):=\check\Gamma(\mathsf b,2\mathsf l/3)e^{c_{\mathrm{vol}}(\mathsf b)\mathsf l^3}$, $\check\delta_n(d):=\min\{d/\sqrt3,1/(4n)\}$, where $d=\infty$ is allowed, and $A_6(\mathsf b):=3^{5/2}\cdot 64(4e)^5A_4(\mathsf b)$. In the order of Def. D.10, $(c_{\mathrm I},C_{\mathrm I},\mathsf p_{\mathrm I}):=(1/(48L),\ 24Le^{c_{\mathrm{vol}}(\mathsf b)}\hat G(\mathsf b),\ 1)$. Moreover $\mathsf a_\Gamma:=7/2$ and

$$C_\varphi:=288\cdot 512^2(16\lambda_\varphi+1)^2\|\varphi\|_\infty^2e^{2c_{\mathrm{vol}}(\mathsf b)}\hat G(\mathsf b)^2.$$

For an integer $z\ge 1$ with $L^z\mathsf b>x_s$, we put $(c_{\mathrm I}^{(z)},C_{\mathrm I}^{(z)},\mathsf p_{\mathrm I}):=(c_{\mathrm I}L^{-z}/2,\ 2L^zC_{\mathrm I}(L^z\mathsf b),\ 1)$. We also put

$$C_\varphi^{(z)}:=L^{2z}\max\{L^{-7z/2}C_{\varphi^{[z]}}(L^z\mathsf b),\ 9\|\varphi\|_\infty^2C_2^\sigma(2;(2+\lambda_\varphi)L^z)\},$$

where the table C^σ of Thm. 10.4 (4.6) is taken at the label $L^z\mathsf b$, and where $\varphi^{[z]}(x)=\varphi(L^{-z}x)$ has $\lambda_{\varphi^{[z]}}=L^z\lambda_\varphi$ and the sup norm of φ. ∎

Theorem 10.4 (Invariant Gibbs states; rotations and mirrors of the limits taken volume first)**.** In clauses (4.3) to (4.7) the labels are in normal form; clauses (4.1) and (4.2) hold at a label $\mathsf b\ge x_s$, in the setting of item (S2) of §1.

(4.1) *The unit board B_0, with walls $\mathbb Z+\frac12$.* Let $\mathsf b\ge x_s$, let $N=2L^{m_l}$ with $m_l\ge m'$ satisfy (Tor), and let $K\ge j^\circ(\mathsf b,\check\delta)$.

(a) *One cell, free centres.* Let $g_1,\dots,g_r$ be real bounded functions that vanish outside one cell, such that the sets $\{g_i\ne 0\}$ are pairwise at distance at least $\check\delta$, lie at distance at least $\check\delta/2$ from the faces, and lie each in a sup-ball of radius $\check\rho$, and let the c_i be real numbers. Then

$$\|\prod_i(\mathcal O_Kg_i-c_i)\|^{(0,0)}_{\mathrm{ch},N}\le e^{c_{\mathrm{vol}}(\mathsf b)}\prod_i(G_1(\mathsf b,\check\delta,\check\rho)\|g_i\|_\infty+|c_i-\hat c_{K,N}(g_i)|).$$

(b) This clause is the first sentence of Thm. 8.17 (R2) (i), taken at $K\ge j^\circ(\mathsf b,\check\delta)$. Let $Q\ge 1$ sources with these data be given, each in one cell, and let q cells be occupied. Then

$$|\langle\prod_t(\mathcal O_Kg_t-\hat c_Kg_t)\rangle_{K,N}|\le G_1^Qe^{qc_{\mathrm{vol}}(\mathsf b)}\prod\|g_t\|_\infty.$$

(c) For one source g with the data of clause (a) and $m_N(g) := \langle \mathcal{O}_K g\rangle_{K,N}$ we have $|m_N(g) - \hat{c}_{K,N}(g)| \le G_1 e^{c_{\mathrm{vol}}(\mathsf{b})}\|g\|_\infty$. (This clause is the second sentence of Thm. 8.17 (R2) (i), taken at $K \ge j^\circ(\mathsf{b}, \check{\delta})$.)

(4.2) *Small cells, centred at the torus mean.* Let $i \ge 0$, $\mathsf{l} = L^{-i}$ and $K \ge \max\{i+1, j^\circ(\mathsf{b}, \check{\delta})\}$, and let (i, o) be any board of $\mathbb{L}_K$. Let $g^1, \dots, g^r$ be real bounded functions that vanish outside one cell, such that the sets where they do not vanish are pairwise at distance at least $\check{\delta}$, lie at distance at least $\check{\delta}/2$ from the faces, and lie each in a sup-ball of radius $\check{\delta}/4$, and let the c^s be real numbers. Then

$$\|\prod_s (\mathcal{O}_K g^s - c^s)\|_{\mathrm{ch},N}^{(i,o)} \le e^{c_{\mathrm{vol}}(\mathsf{b})\mathsf{l}^3} \prod_s (\check{\Gamma}(\mathsf{b}, \check{\delta})\|g^s\|_\infty + |c^s - m_N(g^s)|).$$

(4.3) *All of $\mathbb{G}_K(\mathsf{b}')$.* Let $\mathsf{b}' \in U(\mathsf{b})$ and $\rho, \nu \in \mathbb{G}_K(\mathsf{b}')$.

(a) *Cell by cell.* Let sources on B_0 be given, with the data of clause (4.1) (a) at $(\check{\delta}_t; \check{\delta}_t/4)$ in the cell t, and let $K \ge \max_t j^\circ(\mathsf{b}', \check{\delta}_t)$, for which $K \ge \max_t j_U^\circ(\mathsf{b}, \check{\delta}_t)$ suffices. Then

$$|\rho(\prod_i (\mathcal{O}_K g_i - \nu(\mathcal{O}_K g_i)))| \le \prod_{t \text{ occupied}} [e^{c_{\mathrm{vol}}} \check{\Gamma}(\mathsf{b}, \check{\delta}_t)^{n_t}] \cdot \prod \|g_i\|_\infty.$$

On a board (i, o) with $i \ge 0$, with the data of clause (4.2) at the label b', the same holds with the factor $e^{c_{\mathrm{vol}}\mathsf{l}^3}[\check{\Gamma}(\mathsf{b}, \check{\delta}_t)(1 + e^{c_{\mathrm{vol}}\mathsf{l}^3})]^{n_t}$ for each occupied cell.

(b) *The table.* This clause is the third line of Thm. A (Y2). For all $K \ge 1$, all $n \ge 1$ and all real bounded f_i of Euclidean type $(d; R)$, wherever they lie,

$$|\rho(\prod (\mathcal{O}_K f_i - \nu(\mathcal{O}_K f_i)))| \le A_6(\mathsf{b}')^n (n!)^5 \max\{1, d^{-5n}\}(1+R)^{3n} \prod \|f_i\|_\infty.$$

Here the plain value $A_6(\mathsf{b}')$ is obtained by evaluating the formula of A_6 in item 2 at b' itself, without a supremum over a set $U(\cdot)$ (item (S8) (n1) of §1; the paragraph "The label" of Step 3 of the proof of Thm. 10.4), and $A_6(\mathsf{b}') \le A_6(\mathsf{b})$. Since every $\mathsf{b}' \ge x_s$ lies in $U(\mathsf{b})$ for some $\mathsf{b} > x_s$, this bound holds at every $\mathsf{b}' \ge x_s$. For $n \ge 2$, moreover, $|S_n^T[\rho](f)| \le 1.45^{n+1} n!$ times the right side of the display of this clause.

(c) Every sequence $(K_j, \rho_j, \mathsf{b}_j)$ with $K_j \to \infty$ and $\mathsf{b}_j \to \mathsf{b}$ has a subsequence that converges on all separated tuples. Hence $\mathrm{Lim}_{\mathrm{tor}}(\mathsf{b})$ and $\mathrm{Lim}_G(\mathsf{b})$ are not empty.

(4.4) *Insertion on the lattice.* Let (i, o) be a board with $i \ge 0$, let $\{x_3 = c\}$ be one of its walls, and let $\square$ be a cell in $\{c \le x_3 \le c + \mathsf{l}\}$. Let f be real and bounded and vanish outside $\square$, with $\{f \ne 0\}$ at distance at least $\mathsf{l}/3$ from the faces and in a sup-ball of radius $\mathsf{l}/6$. Let $\mathsf{b}' \in U(\mathsf{b})$, let $K \ge \max\{i+1, j^\circ(\mathsf{b}', 2\mathsf{l}/3)\}$, for which $K \ge \max\{i+1, j_U^\circ(\mathsf{b}, 2\mathsf{l}/3)\}$ suffices, let $\rho \in \mathbb{G}_K(\mathsf{b}')$, and let $F, F' \in \mathcal{C}_{\mathrm{loc}}(\mathbb{H}_c^+)$. Then

$$|\rho(\overline{\theta_c F'}\,[\mathcal{O}_K f - \rho(\mathcal{O}_K f)]\,\tau_{\mathsf{l}e_3} F)| \le 2\Gamma(\mathsf{b}', \mathsf{l})\|f\|_\infty \|F'\|_{\rho,c}\|F\|_{\rho,c},$$

and $\Gamma(\mathsf{b}', \mathsf{l}) \le \Gamma(\mathsf{b}, \mathsf{l})$.

(4.5) *Plates.* Let $\mathsf{b}' \in U(\mathsf{b})$, $\rho \in \mathbb{G}_K(\mathsf{b}')$, $0 < t \le 1$ and $\alpha \in \mathsf{Prof}_t$, let φ be a plate label and $f = \alpha \otimes \varphi$, and let $K \ge j_U^\circ(\mathsf{b}, t/8)$. Then $|S_2^T[\rho](\theta f, f)| \le C_\varphi t^{-7/2}$. More precisely, the left side is at most of the order $t^{-3}(1 + \log(8/t))$, as we show in Step 6 of the proof by summing over the pairs of pieces.

(4.6) *The table of hypotheses.* For every $S \in \mathrm{Lim}_G(\mathsf{b})$, σ_S is a functional on separated tuples over $\mathbb{R}^3$ with the properties (R0) to (R7) of Def. D.10:

(R0) holds by eq. 1.i;

(R1) holds by Proposition B.28 (b) and clause (4.3) (b);

(R2) holds with $C_n^\sigma(d; R) := A_6^n (n!)^5 \max\{1, d^{-5n}\}(1+R)^{3n}$, a table that is non-increasing in d by its form and whose bound is in the sup norm, where $\|\cdot\|_\infty \le \|\cdot\|_\sharp$;

(R3) holds by Proposition B.28 (a) at $(e_3, 0)$;

(R4) holds by Prop. D.5 with Remark D.13 (1), (2);

(R5) holds with $(c_{\mathrm{I}}, C_{\mathrm{I}}, \mathsf{p}_{\mathrm{I}})$, by clause (4.4);
(R6) holds by Proposition B.28 (a) at $(e_1+e_3, 0)$, and that clause gives the full positivity at Σ;
(R7) holds with $\mathsf{a}_\Gamma = 7/2 < 4$, by clause (4.5).

(4.7) *Rotations and mirrors.* For every $S \in \mathrm{Lim}_G(\mathsf{b})$ we have $\sigma_S(rX) = \sigma_S(X)$ and $S_n(rX) = S_n(X)$ for all $r \in \mathbb{R}^3 \rtimes \mathrm{O}(3)$, and σ_S is reflection positive at every mirror (u, c) of $\mathbb{R}^3$. Moreover $H \geq |P_k|$ for $k = 1, 2$, the joint spectrum lies in $\{E \geq |q|\}$, and $\|\mathcal{T}(z)\| \leq 1$ for $\mathrm{Re}\, z_3 \geq |\mathrm{Im}(z_1, z_2)|$. The group $\mathrm{O}(2)$ acts on $\mathcal{H}(\sigma_S)$ by a strongly continuous unitary representation, which fixes Ω, commutes with H and extends the representation of W_2. Even without Thm. D.31, the reflection positivity of σ_S holds at every coordinate plane, and also at each plane that belongs to one of the six diagonal families.

(4.8) *Every label* $\mathsf{b} > 0$. Take an integer $z \geq 0$ with $L^z\mathsf{b} > x_s$. For $K \geq z+1$, the relabelling $x \mapsto L^z x$ carries $\mathbb{G}_K(\mathsf{b}')$ onto $\mathbb{G}_{K-z}(L^z\mathsf{b}')$, and $\mathcal{O}_K(f) = L^z\mathcal{O}_{K-z}(f^{[z]})$ with $f^{[z]}(x) = f(L^{-z}x)$. Hence every $S \in \mathrm{Lim}_G(\mathsf{b})$ is of the form $S_n(X) = L^{zn}S'_n(X^{[z]})$ with $S' \in \mathrm{Lim}_G(L^z\mathsf{b})$. The map $D_z : x \mapsto L^z x$ carries $\tilde{\mu}_{K,2L^m,\beta}$ to $\tilde{\mu}_{K-z,2L^{m+z},\beta}$; since limit points depend on tails only (Definition B.1 (4)), D_z carries $\mathcal{G}^{\mathrm{tor}}_{K,\beta}$ onto $\mathcal{G}^{\mathrm{tor}}_{K-z,\beta}$, and $\mathrm{Lim}_{\mathrm{tor}}(\mathsf{b})$ corresponds to $\mathrm{Lim}_{\mathrm{tor}}(L^z\mathsf{b})$. Clause (4.3) (c) holds at b, as one sees by applying it to $(K_j - z, D_z\rho_j, L^z\mathsf{b}_j)$; so $\mathrm{Lim}_{\mathrm{tor}}(\mathsf{b})$ and $\mathrm{Lim}_G(\mathsf{b})$ are not empty. Clause (4.7) holds for S word for word. The properties (R0) to (R4) and (R6) hold for S, with the table $L^{zn}C^\sigma_n(L^z d; L^z R)$ taken at the label $L^z\mathsf{b}$. For $z = 0$, properties (R5) and (R7) have the constants of (4.6). For $z \geq 1$, they hold with $(c_{\mathrm{I}}^{(z)}, C_{\mathrm{I}}^{(z)}, 1)$ and $(7/2, C_\varphi^{(z)})$, by Lem. 10.5.

(4.9) *Further limits.* Let $\mathsf{b} > 0$ be any label, and let $\mathsf{b}_j \to \mathsf{b}$.
(i) There is a threshold $N^*(K, \mathsf{b}')$, which is not explicit, with the following property: whenever a joint sequence σ_{K_j,N_j}, taken at labels $\mathsf{b}_j \to \mathsf{b}$ and with $N_j \geq N^*(K_j, \mathsf{b}_j)$, has a limit point, that limit point equals the functional of some member of $\mathrm{Lim}_{\mathrm{tor}}(\mathsf{b})$.
(ii) This clause covers every limit of Thm. 8.17 (R5): those of (a) and (c) also at labels $\mathsf{b}_j \to \mathsf{b}$, and those of (b) at a fixed label; a limit in the order of (b) at moving labels is a member of $\mathrm{Lim}_{\mathrm{tor}}(\mathsf{b})$, and for it clauses (4.6) to (4.8) apply. Every such limit has the properties (R0) to (R3), (R5) and (R7), with these numbers: in normal form, the table C^σ_n of clause (4.6), the triple $(c_{\mathrm{I}}, C_{\mathrm{I}}, \mathsf{p}_{\mathrm{I}})$ and the pair $(\mathsf{a}_\Gamma, C_\varphi)$; otherwise, with $z \geq 1$ and $L^z\mathsf{b} > x_s$, the table $L^{zn}C^\sigma_n(L^z d; L^z R)$ taken at $L^z\mathsf{b}$, the triple $(c_{\mathrm{I}}^{(z)}, C_{\mathrm{I}}^{(z)}, 1)$ and the pair $(7/2, C_\varphi^{(z)})$. For such a limit the following are equivalent: invariance under $\mathrm{O}(3)$; reflection positivity at the one mirror $\Sigma = ((e_1+e_3)/\sqrt{2}, 0)$; and (R4) together with (R6).

■□

The per-cell estimate transfers to periodic Gibbs states by chessboard domination. We use the transferred estimates in two ways. A one-cell insertion bounds the local form in the reflection Hilbert space. A sum over pairs of small pieces bounds the plate covariance by $O(t^{-3}(1+\log(8/t)))$. This is the input for the rotation argument in Appendix D.

The volume factor used below satisfies

$$1 + 0.182\, C_{\mathrm{vol}}(N, \mathsf{b}) \leq e^{c_{\mathrm{vol}}(\mathsf{b})N^3}.$$

Indeed, write $q = (N/M)^3 \geq 1$ and $C_{\mathrm{vol}} = 4qe^{220q}Y$, where $Y \geq 1$ is the remaining factor in the formula of paragraph 0.1. Then

$$1 + 0.182\, C_{\mathrm{vol}} \leq e^{220q}(1+q)Y \leq e^{221q}Y \leq e^{225q}Y = e^{c_{\mathrm{vol}}N^3}.$$

Proof of Thm. 10.4. Step 1: the unit board B_0 with walls $\mathbb{Z}+\frac{1}{2}$, with free centres. We prove (4.1) (a), (b) and (c).

Board. For $K \geq 1$ the pair $B_0 = (0,0)$ is a board of $\mathbb{L}_K$, as $\mathsf{l}/a_K = L^K$ is odd and at least 3. Its cells are the cubes $k + [-\frac{1}{2}, \frac{1}{2}]^3$, and its walls $\mathbb{Z} + \frac{1}{2}$ lie in $a_K(\mathbb{Z} + \frac{1}{2})$. The side of the torus satisfies $N = 2L^{m_\mathsf{l}} \geq 2\mathsf{l}$.

Functions of a cell. By Lem. B.11 (b), each $\mathcal{O}_K(g_i)$ is a real continuous element of $\mathcal{A}(\square)$; so is $F := \prod_i (\mathcal{O}_K(g_i) - c_i)$.

Fold. By eq. B.xviii in Lem. B.13,

$$\|F\|_{\mathrm{ch},N}^{N^3} = \tilde{\mu}_{K,N,\beta}\Big(\prod_{k'} F^{[k']}\Big),$$

the product running over one period box of N^3 cells; we call this mean the fold. As F is real, the conjugations drop out of the formula (Lem. B.11 (e)). The map $\hat{r}_{k' \leftarrow k}$ is a composite of wall reflections $y_\mu \mapsto (2t+1) - y_\mu$, and it carries plaquettes to plaquettes and barycentres to barycentres. The action a_p of a plaquette does not depend on its orientation, because $\operatorname{tr} g^{-1} = \operatorname{tr} g$ in SU(2). Hence $\hat{r}\mathcal{O}_K(g) = \mathcal{O}_K(g \circ \hat{r}^{-1}) =: \mathcal{O}_K(g^{k'})$, and $F^{[k']} = \prod_i (\mathcal{O}_K(g_i^{k'}) - c_i)$. After composition with per_N, the fold is the mean under $\mu_{K,N,\beta}$ of a product over the sources $g^{k'}$ on T_N.

Data of the rN^3 copies on T_N. Inside one cell the copies are isometric images of the given sources; so they are pairwise at least $\check{\delta}$ apart, and each of them lies in a sup-ball of radius $\check{\rho}$. Now take two points of copies that lie in different cells of the board on the torus. In some coordinate μ their cell indices differ modulo N, and on the circle $\mathbb{R}/N\mathbb{Z}$ both arcs joining their μ-coordinates leave the interval of one cell and enter that of the other; hence each arc has length at least $\check{\delta}/2 + \check{\delta}/2$. Since N is even, the pattern of the copies closes up on the torus.

Subtraction. We apply (Y0) with $c = 2t + 1$ to each factor of $\hat{r}$ and obtain $\hat{c}_{K,N}(g_i^{k'}) = \hat{c}_{K,N}(g_i)$. We set $\varepsilon_i := \hat{c}_{K,N}(g_i) - c_i$.

Expansion. Then

$$\prod_{k',i} [(\mathcal{O}_K(g_i^{k'}) - \hat{c}_{K,N}(g_i^{k'})) + \varepsilon_i] = \sum_T \prod_T (\mathcal{O}_K - \hat{c}_{K,N}) \cdot \prod_{T^c} \varepsilon,$$

where T runs over the subsets of the rN^3 copies. To each sub-collection T we apply (Y1), with the same data; here $|T| \geq 1$ is arbitrary, while $T = \emptyset$ contributes 1. This gives

$$\big|\big\langle \prod_T (\mathcal{O}_K - \hat{c}_{K,N}) \big\rangle_{K,N}\big| \leq (1 + 0.182 C_{\mathrm{vol}}) \prod_T G_1 \|g\|_\infty,$$

and the sum over T yields

$$|\mathrm{fold}| \leq (1 + 0.182 C_{\mathrm{vol}}(N, \mathsf{b})) \prod_i (G_1 \|g_i\|_\infty + |\varepsilon_i|)^{N^3}.$$

Root. Since $(1 + 0.182 C_{\mathrm{vol}})^{1/N^3} \leq e^{c_{\mathrm{vol}}(\mathsf{b})}$, the N^3-th root of this bound is (a).

Proof of (b). Clause (b) is Thm. 8.17 (R2) (i); we derive it from (a). Let $F_t := \prod_{i \in I_t} (\mathcal{O}_K(g_i) - \hat{c}_{K,N}(g_i))$, a function of the cell t, since every source lies in one cell. Cor. B.14 (b), applied to the set of the occupied cells, gives $|\langle \prod_t F_t \rangle_{K,N}| \leq \prod_t \|F_t\|_{\mathrm{ch},N}$, and (a), applied with $c_i = \hat{c}_{K,N}(g_i)$, bounds each factor.

Proof of (c). Clause (c) is Thm. 8.17 (R2) (i) in the case $Q = 1$, and it too follows from (a): by Cor. B.14 (c), $|\tilde{\mu}(F)| \leq \|F\|_{\mathrm{ch},N}$, and we apply (a) with $r = 1$ and $c = \hat{c}_{K,N}(g)$.

Step 2: the chessboard seminorm of sources on a board with small cells. We prove (4.2).

(i) *Offset.* By eq. B.xix applied with $r = \tau_{-o}$, $\|F\|_{\mathrm{ch},N}^{(i,o)} = \|\tau_{-o} F\|_{\mathrm{ch},N}^{(i,0)}$. Moreover $\tau_{-o}\mathcal{O}_K(g) = \mathcal{O}_K(\tau_{-o} g)$, and $m_N(\tau_{-o} g) = m_N(g)$, because $\mu_{K,N,\beta}$ is invariant under $a_K\mathbb{Z}^3$. We may therefore take $o = 0$; then the walls are $\mathsf{l}(\mathbb{Z} + \frac{1}{2})$. As L^i is odd, $k + \frac{1}{2} = \mathsf{l}(L^i k + (L^i - 1)/2 + \frac{1}{2})$, so every plane of $\mathbb{Z} + \frac{1}{2}$ is a wall of B_i.

(ii) *Fold.* We have

$$\|\cdot\|_{\mathrm{ch},N}^{Q_c} = \Big\langle \prod_\kappa \prod_s (\mathcal{O}_K(g_\kappa^s) - c^s) \Big\rangle_{K,N},$$

with $Q_c = (N/\mathsf{l})^3$. The reasoning of Step 1 carries over: the functions are real, so that no conjugation occurs, and the wall reflections carry plaquettes to plaquettes and barycentres to barycentres, so that the mirror copies are again sources. On T_N the rQ_c copies are pairwise at least $\check\delta$ apart, and each lies in a sup-ball of radius $\check\delta/4$; here $N/\mathsf{l} = 2L^{m_l+i}$ is even.

(iii) *One mean.* A wall reflection $y_\mu \mapsto 2w - y_\mu$ has $2w \in a_K\mathbb{Z}$. It is therefore an isometry of $a_K\mathbb{Z}^3/N\mathbb{Z}^3$, and $\mu_{K,N,\beta}$ is invariant under it. Hence $\langle \mathcal{O}_K(g^s_\kappa)\rangle_{K,N} = m_N(g^s) =: m^s$ for every κ.

(iv) *The subtraction.* For $i \geq 1$, (Y0) yields no invariance of $\hat c_{K,N}$ under the small mirrors. We set $\delta^s_\kappa := \hat c_{K,N}(g^s_\kappa) - m^s$. By (i), each copy stays at least $\check\delta/2$ away from the planes $\mathbb{Z}+\frac12$ and lies in a sup-ball of radius $\check\delta/4$; it therefore lies in one unit cell, and (4.1) (c) gives $|\delta^s_\kappa| \leq G_1(\mathsf{b}, \check\delta, \check\delta/4)e^{c_{\mathrm{vol}}}\|g^s\|_\infty$.

(v) *Expansion.* We write

$$\mathcal{O}_K(g^s_\kappa) - c^s = (\mathcal{O}_K(g^s_\kappa) - \hat c_{K,N}(g^s_\kappa)) + (\delta^s_\kappa + m^s - c^s)$$

and expand the product of these sums over the subsets T of the copies: the term of T is the mean of $\prod_T(\mathcal{O}_K - \hat c_{K,N})$, multiplied by the constants $\delta^s_\kappa + m^s - c^s$ of the copies outside T. For each T, (Y1) bounds that mean by $(1 + 0.182C_{\mathrm{vol}})\prod_T G_1\|g^s\|_\infty$, and $T = \emptyset$ contributes 1. With the bound of (iv), the sum over T gives

$$|\mathrm{fold}| \leq (1 + 0.182C_{\mathrm{vol}})\prod_{\kappa,s}[G_1\|g^s\|_\infty + G_1e^{c_{\mathrm{vol}}}\|g^s\|_\infty + |c^s - m^s|].$$

(vi) *Root.* Finally $(1 + 0.182C_{\mathrm{vol}}(N,\mathsf{b}))^{(\mathsf{l}/N)^3} \leq e^{c_{\mathrm{vol}}(\mathsf{b})\mathsf{l}^3}$, and the Q_c-th root of the last bound gives (4.2).

Remark. We use eq. B.xix and Lem. B.11 in the form recalled in Def. B.31 (2)(e).

Step 3: all of $\mathbb{G}_K(\mathsf{b}')$, a single table of moment bounds for all $K \geq 1$, compactness. We prove (4.3) (a), (b) and (c).

The label. We prove (a) and (b) at one label $\mathsf{b}' \in U(\mathsf{b})$. The data, (Y1), (4.1), (4.2) and (G1) are taken at b': each number is the plain value $X(\mathsf{b}')$, and each threshold is $j^\circ(\mathsf{b}', \cdot)$. Only at the end do we use $X(\mathsf{b}') \leq X(\mathsf{b})$, where $X(\mathsf{b})$ is the supremum over $U(\mathsf{b})$.

(a) *on B_0.* At fixed K we have $|m_N(g)| \leq 2a_K^{-1}\#\{p : x(p) \in \operatorname{supp} g\}\|g\|_\infty$; together with (4.1) (c), it follows that $\hat c_{K,N}(g)$ is bounded in N as well. There is therefore a sequence $N_\bullet = (N_k) \to \infty$ in $\{2L^m\}$, all of whose members are sides of the data, along which $\hat c_{K,N_k}(g_i) \to c^\infty_i$ for each of the finitely many i. By Cor. B.33 (a), ν has (Dom; $N_\bullet$, $\mathcal{C}_{\mathrm{loc}}$). First take the one cell of g_i and $F = \mathcal{O}_K(g_i) - c^\infty_i$, which does not depend on N_k; by eq. B.xli and (4.1) (a),

$$|\nu(\mathcal{O}_K(g_i)) - c^\infty_i| \leq \liminf e^{c_{\mathrm{vol}}}(G_1\|g_i\|_\infty + |c^\infty_i - \hat c_{K,N_k}(g_i)|) = e^{c_{\mathrm{vol}}}G_1\|g_i\|_\infty.$$

Next let $F_t := \prod_{I_t}(\mathcal{O}_K(g_i) - \nu(\mathcal{O}_K(g_i)))$; it belongs to $\mathcal{A}(\square_t) \cap \mathcal{C}_{\mathrm{loc}}$ and does not depend on N_k. By eq. B.xli for ρ and (4.1) (a) applied with $c_i = \nu(\mathcal{O}_K(g_i))$,

$$|\rho(\prod_t F_t)| \leq \liminf \prod_t e^{c_{\mathrm{vol}}}\prod_{I_t}(G_1\|g_i\|_\infty + |\nu(\mathcal{O}_K(g_i)) - \hat c_{K,N_k}(g_i)|).$$

This limit exists, and by the bound for one cell it is at most $\prod_t e^{c_{\mathrm{vol}}}\prod_{I_t} G_1(1 + e^{c_{\mathrm{vol}}})\|g_i\|_\infty$, where $G_1 = G_1(\mathsf{b}', \check\delta_t, \check\delta_t/4)$. Only the copies of the cell t occur in its fold; so the data may differ from cell to cell.

(a) *on a board (i,o).* We repeat the argument for B_0, with the following changes. The means are bounded in N by the first inequality above, and we take a sequence $N_\bullet$ of sides of the data with $m_{N_k}(g_i) \to m^*_i$. Each use of (4.1) (a) is replaced by a use of (4.2), so that the factor $e^{c_{\mathrm{vol}}}$ of a cell becomes $e^{c_{\mathrm{vol}}\mathsf{l}^3}$. For one cell, with $F = \mathcal{O}_K(g_i) - m^*_i$, the bound becomes $|\nu(\mathcal{O}_K(g_i)) - m^*_i| \leq e^{c_{\mathrm{vol}}\mathsf{l}^3}\check\Gamma\|g_i\|_\infty$. For the product over the cells of the board (i,o), each factor of the final bound becomes $\check\Gamma\|g_i\|_\infty + e^{c_{\mathrm{vol}}\mathsf{l}^3}\check\Gamma\|g_i\|_\infty$.

(b)*: the table.* Let $\check\delta := \check\delta_n(d) \leq 1/(4n)$. A Euclidean distance of at least d gives a sup distance of at least $d/\sqrt{3} \geq \check\delta$.

(b)*: cut.* We cut each f_i sharply along the half-open cubes of side $\check\delta/2$ of one grid. A Euclidean ball of radius R lies in a cube of side $2R$; so f_i falls into at most $(4R/\check\delta+2)^3$ pieces, each lying in a sup-ball of radius $\check\delta/4$. Then $\mathcal{O}_K(f_i)$ is exactly the sum of the $\mathcal{O}_K$ of its pieces, and by multilinearity the moment is a sum over the choices of one piece from each f_i. Pieces of different f_i are at least $\check\delta$ apart, and two pieces of one and the same f_i never occur in one term.

The argument also applies to tuples of sup type $(d;R)$: their separation is at least $d\ge\check\delta$, and each sup-ball of radius R is a cube of side $2R$. Thus the number of pieces and every subsequent factor are unchanged.

(b)*, case $K\ge j^\circ(\mathsf{b}',\check\delta)$: placing.* In each coordinate, a piece forbids an arc of length $\check\delta/2+\check\delta$ on the circle $\mathbb{R}/\mathbb{Z}$ of shifts. The n pieces together forbid at most $3n\,\check\delta/2\le 3/8$, and the complement consists of at most n arcs of total length at least $5/8$; so one of these arcs has length at least $5/(8n)\ge(5/2)\check\delta$. From $\mathrm{sep}^{[j^\circ-1]}\ge\mathrm{sep}^\circ$ we obtain $\check\delta L^{j^\circ-1}/M\ge\mathrm{sep}^\circ+2$; hence $a_K\le L^{-j^\circ}<\check\delta$, and the arc contains a point of $a_K\mathbb{Z}$. Thus some $v\in a_K\mathbb{Z}^3$ moves all n pieces to distance at least $\check\delta/2$ from $\mathbb{Z}+\frac12$, each into one cell. As ρ and ν are both invariant under τ_v, the moment of the translated pieces is the same, and (a), with $q\le n$ cells, bounds it by $(e^{c_{\mathrm{vol}}}\check\Gamma(\mathsf{b}',\check\delta))^n$ times the product of the sup norms of the n pieces.

(b)*, case $K<j^\circ(\mathsf{b}',\check\delta)$.* Here $0\le a_p\le 2$, and a half-open cube of side $\check\delta/2$ contains at most $3(\check\delta L^K/2+1)^3$ barycentres. So, for a piece g, $|\mathcal{O}_K(g)|\le 6L^K(\check\delta L^K/2+2)^3\|g\|_\infty\le B_0(\mathsf{b}',\check\delta)\|g\|_\infty$, and $|\mathcal{O}_K(g)-\nu(\mathcal{O}_K(g))|\le 2B_0(\mathsf{b}',\check\delta)\|g\|_\infty$ under any law.

(b)*: sum.* By line 8 of (G1), in both cases each choice of pieces contributes at most $(A_4\check\delta^{-2})^n$ times the product of the sup norms. Moreover $(4R/\check\delta+2)^3\le 64(1+R)^3\check\delta^{-3}$, $\check\delta^{-5n}\le 3^{5n/2}(4n)^{5n}\max\{1,d^{-5n}\}$ and $n^n\le e^n n!$. Combining these, we bound the moment by $A_6^n(n!)^5\max\{1,d^{-5n}\}(1+R)^{3n}$ times the product of the sup norms of the f_i.

(b)*: cumulants, $n\ge2$.* By eq. C.vi with $\sigma_\rho((f))=0$,

$$|S_n^T|\le\sum_\pi(|\pi|-1)!\prod_J\mathrm{bound}(|J|),$$

where $\mathrm{bound}(|J|)$ denotes the bound just found, for the $|J|$ entries of a block J. Here $\prod_J(|J|!)^5\le(n!)^5$, and

$$\sum_\pi(|\pi|-1)!\le F_n:=\sum_k k!\,S(n,k)\le 0.52\cdot1.45^{n+1}n!,$$

with $S(n,k)$ the Stirling numbers of the second kind. The last inequality follows from $F_n=(n!/2)\sum_{k\in\mathbb{Z}}(\log2+2\pi ik)^{-n-1}$. In $F_n/n!$ the term $k=0$ is $\frac12(1/\log2)^{n+1}\le\frac12\cdot1.45^{n+1}$, as $1/\log2\le1.4427$. For $n\ge1$ the moduli of the terms $k\ne0$ add up to

$$\frac12\sum_{k\ne0}|\log2+2\pi ik|^{-n-1}\le\sum_{k\ge1}(2\pi k)^{-n-1}\le(\pi^2/6)(2\pi)^{-n-1}\le0.02\cdot1.45^{n+1},$$

because $\sum_{k\ge1}k^{-n-1}\le\pi^2/6$. Hence $F_n/n!\le0.52\cdot1.45^{n+1}$.

(c)*: compactness.* Let $\mathcal{U}$ denote the countable set of all tuples $(O_1,\dots,O_n)$ whose members are finite unions of open rational cubes and have pairwise disjoint closures. Each $C_c^1(O_i)$ is separable in the sup norm, and we fix countable dense sets $D(O_i)$. The values $\sigma_{\rho_j}(X)$ with $X\in\prod D(O_i)$ are bounded by (b), the numbers being suprema over $U(\mathsf{b})$; hence a diagonal subsequence converges at all of them. A separated tuple X has $\mathrm{supp}\,f_i\subset O_i$ for some member of $\mathcal{U}$. On $\prod C_c^1(O_i)$ the type $(d;R)$ is fixed, and multilinearity together with (b) gives, uniformly in j,

$$|\sigma_{\rho_j}(X)-\sigma_{\rho_j}(X')|\le C\sum_i\|f_i-f_i'\|_\infty\prod_{k\ne i}\max\{\|f_k\|_\infty,\|f_k'\|_\infty\}.$$

Approximating X by tuples X' from $\prod D(O_i)$, we see that $\sigma_{\rho_j}(X)$ is a Cauchy sequence. By eq. C.vi and eq. 1.i, the convergence of all σ amounts to the convergence of all S_n^T with $n \geq 2$. Finally, $\mathbb{G}_K(\mathsf{b}_j) \supset \mathcal{G}^{\mathrm{tor}}$ is not empty (Prop. B.3 (c)).

Remark. Let $S \in \mathrm{Lim}_G(\mathsf{b})$, with labels b_j in normal form, and let X be a separated C_c^1 tuple of type $(d; R)$ with n entries f_i. For all j from some j on, namely those with $\mathsf{b}_j \in U(\mathsf{b})$, (4.3) (b) applied with $\nu = \rho = \rho_j$ bounds $|\sigma_{\rho_j}(X)|$ by the table $C_n^\sigma(d; R) = A_6^n (n!)^5 \max\{1, d^{-5n}\}(1+R)^{3n}$ times the product of the sup norms of the f_i, because $A_6(\mathsf{b}_j) \leq A_6(\mathsf{b})$ for $\mathsf{b}_j \in U(\mathsf{b})$. As $\sigma_S(X) = \lim_j \sigma_{\rho_j}(X)$ on separated C_c^1 tuples (Step 4), the bound passes to $\sigma_S(X)$ in the limit, and since $\|\cdot\|_\infty \leq \|\cdot\|_\sharp$ it holds also with the norm $\|\cdot\|_\sharp$. By its form the table does not increase in d; and for $n = 1$ we have $\sigma_S((f)) = 0$, which is (R0). Altogether, these facts give (R2) of (4.6).

Step 4: symmetries and mirrors. We prove (R0), (R1), (R3), (R6) and (R4) of (4.6). Thus, for $S \in \mathrm{Lim}_G(\mathsf{b})$, we show: σ_S is a functional, centred and symmetric; it is invariant under $\mathbb{R}^3 \rtimes W_3$; it is reflection positive (Def. C.6 (3)) in each coordinate plane and in each plane of the six diagonal families; and $H \geq |P_1|$.

Functional; (R0). Each S_n is a limit of real, symmetric and multilinear maps; hence σ_S is a functional (Def. C.4 (3), (4)), and on every separated tuple $\sigma_S = \lim \sigma_{\rho_j}$. As $\mathrm{Part}_{\geq 2}$ of a set with one element is empty, $\sigma_S((f)) = 0$.

(R1)*: W_3.* Each ρ_j is invariant under $\mathsf{Iso}(\mathbb{L}_{K_j}) = a_{K_j}\mathbb{Z}^3 \rtimes W_3$ (Prop. B.28 (b)), and $r\mathcal{O}_K(f) = \mathcal{O}_K(f \circ r^{-1})$ (Step 1). Hence $r(X^\rho) = (rX)^\rho$, and $\sigma_{\rho_j}(rX) = \sigma_{\rho_j}(X)$ holds exactly, for $r \in \mathsf{Iso}(\mathbb{L}_{K_j})$.

(R1)*: translations (equicontinuity).* For real v we pick $v_j \in a_{K_j}\mathbb{Z}^3$ with $v_j \to v$, and we replace $f_i(\cdot - v)$ by $f_i(\cdot - v_j)$, one entry at a time. The difference has sup norm at most $\mathrm{Lip}\, f_i\, |v - v_j|$, and for large j the tuples are of the type $(d/2; 2R)$. So (4.3) (b) gives

$$|\sigma_{\rho_j}(\tau_v X) - \sigma_{\rho_j}(\tau_{v_j} X)| \leq C|v - v_j| \sum_i \mathrm{Lip}\, f_i \prod_{k \neq i} \|f_k\|_\infty \to 0,$$

while $\sigma_{\rho_j}(\tau_{v_j} X) = \sigma_{\rho_j}(X)$; in the limit $\sigma_S(\tau_v X) = \sigma_S(X)$. The same estimate holds if only some of the entries are moved, provided the tuples stay separated by a fixed amount.

Mirrors. Let $n = e_3$ or $n = e_1 + e_3$, and $c = 0 \in \mathrm{Adm}(n)$; here $\mathbb{E}_+$ is the open side of the mirror $(n, 0)$ at hand. Let $X_1, \dots, X_r \in \mathfrak{S}(\mathbb{E}_+)$, let ζ_a be complex numbers, and put $F_j := \sum_a \zeta_a X_a^{\rho_j}$. The supports are compact and lie in the open side, so $n \cdot x \geq \delta > 0$ on them. A point of a plaquette differs from the barycentre by at most $a_K/2$ in each of two coordinates; hence, for large j, $n \cdot x \geq \delta - a_K \geq 0$ on its bonds, and $F_j \in \mathcal{F}_{\mathrm{loc}}(\mathbb{H}^+(n, 0))$. Prop. B.28 (a) gives $\rho_j(\overline{\theta_{n,0} F_j}\, F_j) \geq 0$. Here $\theta_{e_1+e_3,0} x = x - (x_1 + x_3)(e_1 + e_3) = (-x_3, x_2, -x_1) = \theta_\Sigma x$ (Def. D.10 (R6)), and as $\theta_{n,0} \in \mathsf{Iso}(\mathbb{L}_K)$, we have $\theta(X^\rho) = (\theta X)^\rho$. The left side is therefore

$$\sum \bar{\zeta}_a \zeta_b\, \sigma_{\rho_j}(\theta X_a \cdot X_b) \to \sum \bar{\zeta}_a \zeta_b\, \sigma_S(\theta X_a \cdot X_b),$$

so that the limit is ≥ 0. This is (R3) and the full positivity at Σ; (R6) is the case of tuples with one entry. With (R1) and Prop. C.7 (d), (a), we reach every coordinate mirror and every diagonal mirror, on both sides.

(R4). By Rem. D.13 (1), (R1) and (R2) place σ_S in some $\mathsf{Fct}(\underline{C}, \bar{\mathfrak{c}})$. Prop. D.5 applied with $k = 1$, $\varepsilon = 1$, with $e_+ = (e_3 + e_1)/\sqrt{2}$, gives $H_\sigma \geq |P_{1,\sigma}|$, and by Rem. D.13 (2) this is (R4).

Step 5: the insertion bound. On the lattice we prove (4.4). On the continuum we prove (R5) in the following form. Let $\eta \in (0, 1]$; let h be a real C_c^1 function with $\mathrm{supp}\, h \subset B(y, \eta/(48L))$, where $|y_3| \leq \eta/8$; and let $F, G \in \mathbb{C}\mathfrak{S}(\{x_3 > \eta/4\})$. Then

$$|\sigma_S(\theta \bar{F} \cdot (h) \cdot G)| \leq C_{\mathrm{I}} \eta^{-1} \|h\|_\sharp\, \|[\tau_{-(\eta/4)e_3} F]\|\, \|[\tau_{-(\eta/4)e_3} G]\|.$$

Lattice. We fix a label $\mathsf{b}' \in U(\mathsf{b})$, and we take (4.2) and the data at b'; at the end we use $X(\mathsf{b}') \leq X(\mathsf{b})$, the supremum over $U(\mathsf{b})$. The state ρ has (Inv) and (Pos) by Prop. B.28(a), (b), since $c \in a_K(\mathbb{Z} + \frac{1}{2}) \subset \mathrm{Adm}(e_3)$, and it has (Dom) by Cor. B.33(a). We choose a sequence $N_\bullet$ of sides of the data with $m_{N_k}(f) \to m^*$, and we put $D := \mathcal{O}_K(f) - \rho(\mathcal{O}_K(f))$, which lies in $\mathcal{A}(\square) \cap \mathcal{C}_{\mathrm{loc}}$

(Lem. B.11 (b)). The single source f has the data of (4.2) with $r = 1$ and $\check\delta = 2\mathsf{l}/3$: the set $\{f \neq 0\}$ is at distance at least $\mathsf{l}/3 = \check\delta/2$ from the faces, and it lies in a sup-ball of radius $\mathsf{l}/6 = \check\delta/4$. By Cor. B.14 (c),

$$\|D\|_{\mathrm{ch},N_k} \le \|\mathcal{O}_K(f) - m^*\|_{\mathrm{ch},N_k} + |m^* - \rho(\mathcal{O}_K(f))|,$$

and by (4.2),

$$\|\mathcal{O}_K(f) - m^*\|_{\mathrm{ch},N_k} \le e^{c_{\mathrm{vol}}(\mathsf{b}')\mathsf{l}^3}\big(\check\Gamma(\mathsf{b}', 2\mathsf{l}/3)\|f\|_\infty + |m^* - m_{N_k}(f)|\big) \to \Gamma(\mathsf{b}', \mathsf{l})\|f\|_\infty.$$

Also, eq. B.xli for the one cell gives $|\rho(\mathcal{O}_K(f)) - m^*| \le \Gamma(\mathsf{b}', \mathsf{l})\|f\|_\infty$. So $\|D\|_{\mathrm{ch},N_\bullet} \le 2\,\Gamma(\mathsf{b}', \mathsf{l})\,\|f\|_\infty$, and (4.4) follows by Prop. B.35, with the number $2\Gamma(\mathsf{b}', \mathsf{l})$. At the end of the step we pass to $\Gamma(\mathsf{b}, \mathsf{l})$, since $\Gamma(\mathsf{b}', \mathsf{l}) \le \Gamma(\mathsf{b}, \mathsf{l})$ by $X(\mathsf{b}') \le X(\mathsf{b})$.

Geometry. The half-open intervals $(x/L, x]$ with $x = L^k\eta/8$, $k \in \mathbb{Z}$, tile $(0, \infty)$, and 1 lies in exactly one of them; so exactly one i has $\mathsf{l} = L^{-i} \in (\eta/(8L), \eta/8]$, and $i \ge 1$, as $\mathsf{l} \le 1/8$. Then $c_{\mathrm{I}}\eta = \eta/(48L) < \mathsf{l}/6$, strictly. We take j so large that $K = K_j \ge \max\{i+1, j_U^\circ(\mathsf{b}, 2\mathsf{l}/3)\}$, $a_K/2 \le \mathsf{l}/6 - c_{\mathrm{I}}\eta$ and $a_K < \eta/8$. We pick $o \in a_K\mathbb{Z}^3$ with $|o_\mu - y_\mu| \le a_K/2$, and we let $\square = o + [-\mathsf{l}/2, \mathsf{l}/2]^3$ and $c := o_3 - \mathsf{l}/2$; then $c \in a_K(\mathbb{Z} + \frac12)$, because L^{K-i} is odd. As a Euclidean ball lies in the sup-ball of the same radius, $\operatorname{supp} h$ is at distance at least $\mathsf{l}/2 - a_K/2 - c_{\mathrm{I}}\eta \ge \mathsf{l}/3$ from the faces and lies in a sup-ball of radius $c_{\mathrm{I}}\eta \le \mathsf{l}/6$. Finally $c \ge -3\eta/16 - a_K/2 > -\eta/4$ and $c + \mathsf{l} \le 3\eta/16 + a_K/2 < \eta/4$.

The two sides, on the lattice. By the rules of Def. B.31 (2)(d), $\theta_c\theta_0 = \tau_{2ce_3}$, and $\theta_c\tau_v = \tau_{-v}\theta_c$ for v parallel to e_3; moreover $2c$, l and $2c + 2\mathsf{l}$ lie in $a_K\mathbb{Z}$. Let F^ρ be the linear extension of $X \mapsto X^\rho$. It commutes with $\mathsf{Iso}(\mathbb{L}_K)$, and $\overline{F^\rho} = (\bar F)^\rho$, the X^ρ being real. With D defined as above from the source h,

$$\sigma_\rho(\theta\bar F \cdot (h) \cdot G) = \rho(\overline{\theta_0 F^\rho}\, D\, G^\rho).$$

Put $F' := \tau_{2ce_3}F^\rho$ and $F'' := \tau_{-\mathsf{l}e_3}G^\rho$; then $\theta_c F' = \theta_0 F^\rho$ and $\tau_{\mathsf{l}e_3}F'' = G^\rho$. The entries of F' lie in $x_3 > \eta/4 + 2c > c$, and those of F'' in $x_3 > \eta/4 - \mathsf{l} > c$. All bonds of a plaquette lie in $\mathbb{H}_c^+$ as soon as its barycentre lies strictly above the bond mirror c: the height x_3 of the barycentre either lies in $a_K\mathbb{Z}$, and is then at least $c + a_K/2$, the plaquette being horizontal; or it lies in $a_K(\mathbb{Z} + \frac12)$, and is then at least $c + a_K$. Hence $F', F'' \in \mathcal{C}_{\mathrm{loc}}(\mathbb{H}_c^+)$, and (4.4) applies. Its two norms are $\|F'\|_{\rho,c}^2 = \rho(\overline{\theta_0 F^\rho}\,\tau_{2ce_3}F^\rho) = \sigma_\rho(\theta\bar F \cdot \tau_{2ce_3}F)$ and, since $\theta_c\tau_{-\mathsf{l}e_3} = \tau_{(2c+\mathsf{l})e_3}\theta_0$, $\|F''\|_{\rho,c}^2 = \sigma_\rho(\tau_{(2c+\mathsf{l})e_3}\theta\bar G \cdot \tau_{-\mathsf{l}e_3}G)$. A translation by the lattice vector $-(2c+\mathsf{l})e_3$ turns the latter into $\sigma_\rho(\theta\bar G \cdot \tau_{-2(c+\mathsf{l})e_3}G)$. The gaps of these two tuples are $\eta/2 + 2c$ and $\eta/2 - 2(c+\mathsf{l})$, and both are at least $\eta/8 - a_K > 0$. We stress that no translation by $(\eta/4)e_3$ is made on the lattice: the vector $(\eta/4)e_3$ need not lie in $a_K\mathbb{Z}^3$, and σ_ρ is invariant under $a_K\mathbb{Z}^3$ alone.

Limit. The index i is fixed by η, and $c_j = o_{3,j} - \mathsf{l}/2 \to c_\infty := y_3 - \mathsf{l}/2$, because $|o_{3,j} - y_3| \le a_{K_j}/2$. From $|y_3| \le \eta/8$ and $\mathsf{l} \le \eta/8$ we get $-3\eta/16 \le c_\infty$ and $c_\infty + \mathsf{l} \le 3\eta/16$. The left side tends to $\sigma_S(\theta\bar F \cdot (h) \cdot G)$, since the tuples are fixed. By the equicontinuity of Step 4, with a common positive separation given by the minimum of $\eta/16$ and the internal separations of the finitely many tuples occurring in F and G, we have $\sigma_{\rho_j}(\theta\bar F \cdot \tau_{2c_je_3}F) \to \sigma_S(\theta\bar F \cdot \tau_{2c_\infty e_3}F)$; in the same way $\sigma_{\rho_j}(\theta\bar G \cdot \tau_{-2(c_j+\mathsf{l})e_3}G) \to \sigma_S(\theta\bar G \cdot \tau_{-2(c_\infty+\mathsf{l})e_3}G)$.

After the limit, by (R1) *for* σ_S. Let $F_0 := \tau_{-(\eta/4)e_3}F \in \mathbb{C}\mathfrak{S}(\mathbb{E}_+)$; then $\theta\bar F = \tau_{-(\eta/4)e_3}\theta\bar F_0$. A translation by $+(\eta/4)e_3$ gives

$$\sigma_S(\theta\bar F \cdot \tau_{2c_\infty e_3}F) = \sigma_S(\theta\bar F_0 \cdot \tau_{(2c_\infty+\eta/2)e_3}F_0) = \langle [F_0], \mathcal{T}_{2c_\infty+\eta/2}[F_0]\rangle \le \|[F_0]\|^2.$$

In the last step we use $2c_\infty + \eta/2 \ge \eta/8 \ge 0$ and $0 \le \mathcal{T}_s \le 1$ (Lem. D.11 (b), which applies as σ_S has (R1) to (R3)). In the same manner, with $G_0 := \tau_{-(\eta/4)e_3}G$, the translation by $+(\eta/4)e_3$ gives

$$\sigma_S(\theta\bar G \cdot \tau_{-2(c_\infty+\mathsf{l})e_3}G) = \langle [G_0], \mathcal{T}_{\eta/2-2(c_\infty+\mathsf{l})}[G_0]\rangle \le \|[G_0]\|^2,$$

where now $\eta/2 - 2(c_\infty + \mathsf{l}) \ge \eta/8 \ge 0$.

Power. Keep the plain label $\mathsf{b}' \in U(\mathsf{b})$ in the lattice estimate until the complete coefficient has been bounded. Pointwise in that label,

$$2\Gamma(\mathsf{b}', \mathsf{I}) \le \frac{3e^{c_{\mathrm{vol}}(\mathsf{b}')}\hat{G}(\mathsf{b}')}{\mathsf{I}} < \frac{24Le^{c_{\mathrm{vol}}(\mathsf{b}')}\hat{G}(\mathsf{b}')}{\eta} \le \frac{C_{\mathrm{I}}(\mathsf{b})}{\eta},$$

where $C_{\mathrm{I}}(\mathsf{b})$ is the supremum over $U(\mathsf{b})$ of the whole plain-label expression $24Le^{c_{\mathrm{vol}}(\mathsf{b}')}\hat{G}(\mathsf{b}')$. Thus this last coefficient bounds every member of the sequence before taking the two limits above. Since $\|h\|_\infty \le \|h\|_\sharp$, (4.4) and those limits give the bound stated at the beginning of the step.

Step 6: plates, their growth exponent $7/2 < 4$. We prove (4.5) and (R7). Here b', ρ, t, α, φ, f and K are as in clause (4.5). From (4.5) it follows that $\sigma_S(\theta f, f) = S_2(\theta f, f) \le C_\varphi t^{-7/2}$, which is (R7).

One choice for all t. We take $\check{\delta} := t/8 \le 1/8$, on B_0.

Pieces. The function f vanishes outside the box $\{x_3 \in [t, 2t],\ x_1 \in [-t, t],\ x_2 \in$ the interval of $\varphi\}$, and since it is continuous it also vanishes on the boundary of the box. We cut f sharply by half-open cubes of side $\check{\delta}/2 = t/16$, aligned with the box. This yields $16 \cdot 32 \cdot \lceil 16\lambda_\varphi/t \rceil \le 512(16\lambda_\varphi + 1)/t$ pieces g_p, with $\|g_p\|_\infty \le 3\|\varphi\|_\infty$, each in a sup-ball of radius $\check{\delta}/4$. Bilinearity, at the lattice level, gives $S_2^T[\rho](\theta f, f) = \sum_{p,q} \sigma_\rho(\theta g_p, g_q)$. Along x_3, θg_p and g_q are at least $2t \ge \check{\delta}$ apart; between two pieces on the same side no condition is required.

Placing. In each coordinate two pieces forbid at most $3\,\check{\delta} \le 3/8$, so one free arc has length at least $5/16 > a_K$. A point of $a_K\mathbb{Z}$ on the free arc of each coordinate gives, as in Step 3, a common $v \in a_K\mathbb{Z}^3$ which moves both pieces to distance at least $\check{\delta}/2$ from $\mathbb{Z} + \frac{1}{2}$, each into one cell; and σ_ρ is invariant under τ_v.

Bound. Retain the plain label b' throughout the estimate. The fixed-label proof of (4.3) (a), applied with $\nu = \rho$, covers both cases, that of one cell with two separated sources and that of two cells:

$$|\sigma_\rho(\theta g_p, g_q)| \le e^{2c_{\mathrm{vol}}(\mathsf{b}')}\check{\Gamma}(\mathsf{b}', t/8)^2 \|g_p\|_\infty \|g_q\|_\infty.$$

Here the numbers at b' are their plain values. By (G1), $\check{\Gamma}(\mathsf{b}', t/8)^2 \le \hat{G}(\mathsf{b}')^2(8/t)(1 + \log(8/t))$.

Sum. Summing over the pairs of pieces gives

$$72 \cdot 512^2(16\lambda_\varphi + 1)^2 \|\varphi\|_\infty^2 e^{2c_{\mathrm{vol}}(\mathsf{b}')}\hat{G}(\mathsf{b}')^2 t^{-3}(1 + \log(8/t)).$$

The function $t^{1/2}(1 + \log(8/t))$ increases on $(0, 1]$, its maximum lying at $t = 8/e > 1$; so it is at most $1 + \log 8 \le 3.0795 \le 4$. Writing $t^{-3} = t^{-7/2}t^{1/2}$, this is at most $C_\varphi(\mathsf{b}')t^{-7/2}$, where $C_\varphi(\mathsf{b}')$ is the plain expression in item 2 with the factor $288 = 72 \cdot 4$. Now take the supremum of this whole expression over $\mathsf{b}' \in U(\mathsf{b})$ to obtain $C_\varphi(\mathsf{b})t^{-7/2}$.

Limit. The pair $(\theta f, f)$ is a separated C_c^1 tuple (Rem. D.12); hence $S_2^T[\rho_j](\theta f, f) \to S_2(\theta f, f) = \sigma_S(\theta f, f)$.

Remark: the margin. The argument needs $G_1 \le C\,\check{\delta}^{-p}$ with $2 + 2p < 4$, that is, $p < 1$. Line 2 of (G1) yields the exponent $p = 1/2$, with a logarithm.

Remark: why (4.5) *has a threshold in* K. The square $B_0(\mathsf{b}, \check{\delta})^2$ is of the order of $\check{\delta}^{-2}$, and the closed form d^{-5n} gives t^{-10}. These bounds are of use for $K < j^\circ$ alone, and not for (R7); hence the threshold in K in (4.5).

Step 7: Thm. D.31 applied; every mirror; the cone; O(2)*; every* $\mathsf{b} > 0$. We prove (4.7) and (4.8).

(4.7)*: labels in normal form.* By Steps 3 to 6, σ_S is a functional with (R0) to (R7), where $c_{\mathrm{I}} = 1/(48L) \in (0, 1/16]$, $C_{\mathrm{I}} > 0$, $\mathsf{p}_{\mathrm{I}} = 1 \ge 0$ and $\mathsf{a}_\Gamma = 7/2 < 4$. Thm. D.31 gives $\sigma_S(rX) = \sigma_S(X)$ for $r \in \mathrm{O}(3)$, and with (R1) we obtain invariance under all of $\mathbb{R}^3 \rtimes \mathrm{O}(3)$. By eq. C.vi with $\sigma_S((f)) = 0$, $S_n = S_n^{\sigma_S}$, and that formula commutes with r.

Every mirror. This is Prop. D.7 (c). By Rem. D.13 (1), σ_S lies in some $\mathsf{Fct}(\underline{C}, \bar{\mathfrak{c}})$, and Cor. D.8 then gives $H \ge |P_k|$ for $k = 1, 2$, eq. D.xi, $\|\mathcal{T}(z)\| \le 1$, and all of Prop. D.6.

(4.8). The dilation D_z, $x \mapsto L^z x$, maps $a_K\mathbb{Z}^3$ onto $a_{K-z}\mathbb{Z}^3$ for $K \geq z+1$, and it maps bonds to bonds and plaquettes to plaquettes. As $H_B = -\beta \sum a_p$ does not depend on the spacing (eq. B.v), D_z carries the Gibbs states at (a_K, β) onto those at (a_{K-z}, β), with $\beta = L^{K-z}(L^z \mathsf{b}')$, and it carries $a_K\mathbb{Z}^3$-invariance to $a_{K-z}\mathbb{Z}^3$-invariance. Further,

$$\mathcal{O}_K(f) = a_K^{-1} \sum f(x(p))a_p = L^z a_{K-z}^{-1} \sum f^{[z]}(L^z x(p))a_p,$$

and $X \mapsto X^{[z]}$ is a bijection of the separated C_c^1 tuples. Therefore $S'_n(Y) := \lim S_n^T[D_z\rho_j](Y)$ exists on all tuples, $S' \in \mathrm{Lim}_G(L^z\mathsf{b})$, and $S_n(X) = L^{zn} S'_n(X^{[z]})$. The dilation D_z carries $\tilde{\mu}_{K,2L^m,\beta}$ to $\tilde{\mu}_{K-z,2L^{m+z},\beta}$; since limit points depend on the tails only (Def. B.1 (4)), it carries $\mathcal{G}^{\mathrm{tor}}_{K,\beta}$ onto $\mathcal{G}^{\mathrm{tor}}_{K-z,\beta}$, and $\mathrm{Lim}_{\mathrm{tor}}(\mathsf{b})$ corresponds to $\mathrm{Lim}_{\mathrm{tor}}(L^z\mathsf{b})$. Since $L^z\mathsf{b}_j \to L^z\mathsf{b} > x_s$, we have $L^z\mathsf{b}_j \in U(L^z\mathsf{b})$ from some j on, so the labels are in normal form. The invariance follows from $(rX)^{[z]} = r(X^{[z]})$ for $r \in \mathrm{O}(3)$ and $(\tau_v X)^{[z]} = \tau_{L^z v} X^{[z]}$.

Mirrors. We have $(\theta_{(u,c)}X)^{[z]} = \theta_{(u,L^z c)}X^{[z]}$. Hence, with $\theta' = \theta_{(u,L^z c)}$ and with n_a the number of entries of X_a,

$$\sum \bar{\zeta}_a \zeta_b \, \sigma_S(\theta X_a \cdot X_b) = \sum \overline{\zeta_a L^{zn_a}} \, (\zeta_b L^{zn_b}) \, \sigma_{S'}(\theta' X_a^{[z]} \cdot X_b^{[z]}) \geq 0.$$

For (R2) of σ_S, the type $(d; R)$ becomes $(L^z d; L^z R)$ and the sup norm of each entry does not change under $f \mapsto f^{[z]}$; the table $L^{zn} C_n^\sigma(L^z d; L^z R)$ does not increase in d. Hence Rem. D.13 (1) and Cor. D.8 are applicable to σ_S itself; (R4) is contained in this, and (R0), (R1), (R3), (R6) are contained in the invariance and the mirrors. For $z = 0$, (R5) and (R7) are already part of (4.6). For $z \geq 1$, apply Lem. 10.5 (Step 8) to $\sigma' = \sigma_{S'}$, which has (R0) to (R7) by (4.6).

Step 8: joint limits along large sides, the diagonal mirror Σ at which reflection positivity is equivalent to invariance under $\mathrm{O}(3)$*, and the dilation lemma, for all* $\mathsf{b} > 0$. We prove (4.9) (i) and (ii), at every $\mathsf{b} > 0$. First we state the lemma on dilations.

Lemma 10.5 (Dilation by L^z carries the hypotheses)**.** Let $z \geq 1$ be an integer, and put $D_z X := L^{z|X|} X^{[z]}$. Let σ' be a functional with (R0) to (R3), for a table $C'_n(d; R)$ that gives the bound of (R2) in terms of the sup norms $\|f_i\|_\infty$ of the entries and does not decrease in R; with (R5), for the triple $(c_{\mathrm{I}}, C'_{\mathrm{I}}, 1)$; and with (R7), for the exponent $7/2$ and numbers C'_φ. Then $\sigma := \sigma' \circ D_z$ is a functional with (R0) to (R3), for the table $L^{zn} C'_n(L^z d; L^z R)$; with (R5), for the triple $(c_{\mathrm{I}} L^{-z}/2,\ 2L^z C'_{\mathrm{I}},\ 1)$; and with (R7), for the exponent $7/2$ and the numbers $L^{2z} \max\{L^{-7z/2} C'_{\varphi^{[z]}},\ 9\|\varphi\|_\infty^2 C'_2(2; (2+\lambda_\varphi)L^z)\}$. ∎

Proof of (4.9) (i)*, at fixed* (K, b')*, for any* $\mathsf{b}' > 0$. Let $X^1, X^2, \ldots$ be the countable family of Step 3 (c), and fix K and b'. We claim that there is an N^* such that for every side $N \geq N^*$ in $\{2L^m\}$ some $\nu \in \mathcal{G}^{\mathrm{tor}}_{K,L^K\mathsf{b}'}$ satisfies $|\sigma_{K,N}(X^q) - \sigma_\nu(X^q)| \leq 1/K$ for all $q \leq K$. Otherwise some $N_k \to \infty$ violates this for every ν. Since $\mathcal{V}(\mathbb{L}_K)$ is compact, a subsequence of $\tilde{\mu}_{K,N_k,\beta}$ converges weakly to some $\nu \in \mathcal{G}^{\mathrm{tor}}$ (Def. B.1 (4)). The finitely many $\sigma(X^q)$ are polynomials in means of continuous local functions, so they converge along this subsequence, a contradiction. Besides, let $z_0(\mathsf{b}')$ be the least $z \geq 0$ with $L^z\mathsf{b}' \geq x_s$ (item (S7) of §1). We choose $N^*(K, \mathsf{b}')$ large enough that, whenever $N \geq N^*$, the side $L^z N$ belongs to the data at the label $L^z\mathsf{b}'$, both for $z = z_0(\mathsf{b}')$ and for $z = z_0(\mathsf{b}') + 1$. The conditions are $m + z \geq m'$ and $L^z N \geq 2Ms(L^{z+1}\mathsf{b}')$, and they hold for all large N.

(i)*: the limit, in normal form.* Along sides $N_j \geq N^*(K_j, \mathsf{b}_j)$ we choose such states ν_j; then $\sigma_{K_j,N_j}(X^q) - \sigma_{\nu_j}(X^q) \to 0$ for every q. Both families obey sup-norm bounds uniform in j: on the tori by Thm. 9.4 (G2)(i), and for the ν_j by (4.3) (b). The argument of Step 3 (c) therefore applies to both: on each product $\prod C_c^1(O_i)$ with $(O_1, \ldots, O_n) \in \mathcal{U}$ the type is fixed, and multilinearity with these bounds yields the estimate of Step 3 (c) for both families, uniformly in j. As the differences tend to 0 on the countable family, they tend to 0 on every separated tuple; so the two families have the same limit points on all tuples.

(i)*: the limit, outside the normal form.* Let $z = z(\mathsf{b})$ (item (S7) of §1); for $\mathsf{b} \le x_s$ it is the least $z \ge 1$ with $L^z\mathsf{b} > x_s$. For large j we have $L^z\mathsf{b}_j > x_s$ and $L^{z-2}\mathsf{b}_j < x_s$; thus z is $z_0(\mathsf{b}_j)$ or $z_0(\mathsf{b}_j)+1$, and L^zN_j is a side of the data at $L^z\mathsf{b}_j$. The map $x \mapsto L^z x$ carries $a_K\mathbb{Z}^3/N\mathbb{Z}^3$ onto $a_{K-z}\mathbb{Z}^3/L^zN\mathbb{Z}^3$, and Wilson's weight does not depend on the spacing. Hence, by the formula for $\mathcal{O}_K(f)$ in the proof of (4.8), $\sigma_{K,N}(X)$ at the label b' equals $L^{z|X|}\sigma_{K-z,L^zN}(X^{[z]})$ at the label $L^z\mathsf{b}'$. So Thm. 9.4 (G2)(i), applied to the torus with cutoff, side and label $(K_j - z, L^zN_j, L^z\mathsf{b}_j)$, bounds the torus family by L^{zn} times the table at $(L^zd; L^zR)$, with the numbers as suprema over $U(L^z\mathsf{b})$; and (4.8) bounds σ_{ν_j} in the same way. We now follow the argument of the normal form, with one change: these bounds take the place of Thm. 9.4 (G2)(i) and of (4.3) (b). The choice of the ν_j along $N_j \ge N^*(K_j, \mathsf{b}_j)$ stays as it is, the estimate of Step 3 (c) holds for both families uniformly in j, and the two families have the same limit points on all tuples.

Proof of Lem. 10.5. Functional. The map $X \mapsto X^{[z]}$ is a bijection of the separated C^1_c tuples, and it is linear in each entry; hence σ is a functional, and $\sigma(\emptyset) = 1$.

Intertwining. We have $(rX)^{[z]} = r(X^{[z]})$ for $r \in \mathrm{O}(3)$, $(\tau_v X)^{[z]} = \tau_{L^zv}X^{[z]}$, $(\theta X)^{[z]} = \theta X^{[z]}$ and $D_z(X \cdot Y) = D_zX \cdot D_zY$; and we extend D_z linearly to $\mathbb{C}\mathfrak{S}$. Properties (R0), (R1) and (R3) follow from these relations as in the proof of (4.8) in Step 7: $\sigma((f)) = L^z\sigma'((f^{[z]})) = 0$, and σ is symmetric, since D_z acts entry by entry; for $r \in W_3$ and real v, $\sigma(rX) = L^{z|X|}\sigma'(rX^{[z]}) = \sigma(X)$ and $\sigma(\tau_v X) = L^{z|X|}\sigma'(\tau_{L^zv}X^{[z]}) = \sigma(X)$; and, with n_a the number of entries of X_a, $\sum \bar\zeta_a\zeta_b\,\sigma(\theta X_a \cdot X_b) = \sum \overline{\zeta_a L^{zn_a}}\,(\zeta_b L^{zn_b})\,\sigma'(\theta X_a^{[z]} \cdot X_b^{[z]}) \ge 0$. For (R2), the type $(d; R)$ becomes $(L^zd; L^zR)$, while the sup norm of each entry does not change under $f \mapsto f^{[z]}$.

Norms. For $F \in \mathbb{C}\mathfrak{S}(\mathbb{E}_+)$ we have $\|[F]\|^2_\sigma = \sigma'(\theta\overline{D_zF} \cdot D_zF) = \|[D_zF]\|^2_{\sigma'}$, and $D_z\tau_{-(\eta/4)e_3}F = \tau_{-(L^z\eta/4)e_3}D_zF$.

(R5) *at $\eta \le \eta_1 := L^{-z}$, for* $\operatorname{supp} h \subset B(y, c_{\mathrm{I}}\eta)$. Put $\tilde\eta := L^z\eta \le 1$ and $\tilde h := L^zh^{[z]}$. Then $\operatorname{supp}\tilde h \subset B(L^zy, c_{\mathrm{I}}\tilde\eta)$, $|L^zy_3| \le \tilde\eta/8$, and $D_zF, D_zG \in \mathbb{C}\mathfrak{S}(\{x_3 > \tilde\eta/4\})$. Moreover $\sigma(\theta\bar F \cdot (h) \cdot G) = \sigma'(\theta\overline{D_zF} \cdot (\tilde h) \cdot D_zG)$, and

$$\|\tilde h\|_\sharp = L^z \max\{\|h\|_\infty, 40ML^{-z}\operatorname{Lip} h\} \le L^z\|h\|_\sharp.$$

Property (R5) of σ' at $\tilde\eta$ bounds the modulus by $C'_{\mathrm{I}}L^{-z}\eta^{-1}L^z\|h\|_\sharp$ times the two norms; by the relations under Norms, this equals

$$C'_{\mathrm{I}}\eta^{-1}\|h\|_\sharp\,\|[\tau_{-(\eta/4)e_3}F]\|\,\|[\tau_{-(\eta/4)e_3}G]\|.$$

(R5) *at $\eta \in (\eta_1, 1]$, for* $\operatorname{supp} h \subset B(y, c_{\mathrm{I}}\eta_1\eta/2)$. Put $\eta_0 := \min\{\eta/2, \eta_1\}$; then $c_{\mathrm{I}}\eta_1\eta/2 \le c_{\mathrm{I}}\eta_0$. We translate by $-y_3e_3$, using $\tau_{-y_3e_3}\theta = \theta\tau_{y_3e_3}$: the functions $F_1 := \tau_{y_3e_3}F$ and $G_1 := \tau_{-y_3e_3}G$ live in $\{x_3 > \eta/8\} \subset \{x_3 > \eta_0/4\}$, and $h_1 := \tau_{-y_3e_3}h$ has its centre at height 0. The case above, at η_0, gives the bound

$$C'_{\mathrm{I}}\eta_0^{-1}\|h\|_\sharp\,\|[\tau_{-(\eta_0/4)e_3}F_1]\|\,\|[\tau_{-(\eta_0/4)e_3}G_1]\|.$$

Here $[\tau_{-(\eta_0/4)e_3}F_1] = \mathcal{T}_s[\tau_{-(\eta/4)e_3}F]$ with $s = \eta/4 + y_3 - \eta_0/4 \ge \eta/8 - \eta_0/4 \ge 0$, and $\mathcal{T}_s$ is a contraction (Lem. D.11 (b), which applies as σ has (R1) to (R3)); the same holds for G_1, with $-y_3$ in the place of y_3. Moreover $\eta_0^{-1} \le 2L^z/\eta$. At $\eta \le \eta_1$ the smaller ball lies in $B(y, c_{\mathrm{I}}\eta)$, and $C'_{\mathrm{I}} \le 2L^zC'_{\mathrm{I}}$; so one triple covers all $\eta \in (0, 1]$, and $c_{\mathrm{I}}L^{-z}/2 \in (0, 1/16]$.

(R7)*; $f = \alpha \otimes \varphi$, $\alpha \in \mathsf{Prof}_t$, $0 < t \le 1$.* We have $f^{[z]} = \alpha^{[z]} \otimes \varphi^{[z]}$. The function $\alpha^{[z]}$ has its support in $[L^zt, 2L^zt] \times [-L^zt, L^zt]$, satisfies $|\alpha^{[z]}| \le 3$, and has $\operatorname{Lip}\alpha^{[z]} \le 12/(L^zt)$; so $\alpha^{[z]} \in \mathsf{Prof}_{L^zt}$, and $\varphi^{[z]}$ is a plate label. Moreover $\sigma(\theta f, f) = L^{2z}\sigma'(\theta f^{[z]}, f^{[z]})$. For $L^zt \le 1$, (R7) of σ' bounds this by $L^{2z}C'_{\varphi^{[z]}}(L^zt)^{-7/2}$. For $1 < L^zt \le L^z$, the pair is of the type $(2L^zt; R)$, where $2L^zt > 2$ and R is half the diagonal of the box: $R = (L^z/2)(5t^2 + \lambda_\varphi^2)^{1/2} \le (2 + \lambda_\varphi)L^z$. Then (R2) of σ' for $n = 2$, whose table does not rise with d, gives the bound $L^{2z}C'_2(2; (2+\lambda_\varphi)L^z)\,9\|\varphi\|^2_\infty$, which is a constant; and $1 \le t^{-7/2}$. □

Proof of (4.9) (ii)*, in normal form.* Properties (R0) to (R3) come from Thm. 8.17 (R6) (i) to (iv). The table comes from Thm. 9.4 (G2) (i), (iii) for $n \geq 2$; for the Euclidean type, this is the C_n^σ of (4.6), which is monotone in d by its form. For $n = 1$, $\sigma_S((f)) = 0$.

For (R7) we follow the proof of (4.5) in Step 6, now on the tori, with the following changes. In Placing, the invariance of $\mu_{K,N,\beta}$ under $a_K\mathbb{Z}^3$ is used. In Bound, Cor. B.14 (b) takes the place of eq. B.xli, on which (4.3) (a) rests; and (4.1) (a) applied with $c_i = m_N(g_i)$, together with (4.1) (c), gives $|\sigma_{K,N}(\theta g_p, g_q)| \leq e^{2c_{\mathrm{vol}}}\check{\Gamma}^2\|g_p\|_\infty\|g_q\|_\infty$ for $K \geq j^\circ(\mathsf{b}', t/8)$ at the label b' of the torus, hence for $K \geq j_U^\circ(\mathsf{b}, t/8)$. In Limit, the bound passes to every limit.

(ii)*, in normal form:* (R5) *from Thm. 9.4 (G6).* Let η_0, h, y, F, G be as in (R5), with $c_{\mathrm{I}} = 1/(48L) \leq 1/32$. We translate by $-y_3e_3$, using $\tau_{-y_3e_3}\theta = \theta\tau_{y_3e_3}$, and obtain $\sigma(\theta\bar{F}\cdot(h)\cdot G) = \sigma(\theta\bar{F}_1\cdot(h_1)\cdot G_1)$, where $F_1 = \tau_{y_3e_3}F$ and $G_1 = \tau_{-y_3e_3}G$ live in $\{x_3 > \eta_0/8\}$, and $h_1 = \tau_{-y_3e_3}h$ lives in $\{|x_3| < \eta_0/32\}$. Now (G6) at $\eta = \eta_0/8$ gives the bound

$$\check{\Gamma}_*(\mathsf{b}, \eta)\,\|\!|h_1|\!\|_\eta\,\|[\tau_{-\eta e_3}F_1]\|\,\|[\tau_{-\eta e_3}G_1]\|.$$

Here $[\tau_{-\eta e_3}F_1] = \mathcal{T}_{y_3+\eta_0/8}[\tau_{-(\eta_0/4)e_3}F]$ with $y_3+\eta_0/8 \geq 0$, and the operator $\mathcal{T}_{y_3+\eta_0/8}$ is a contraction; the same holds for G, with $-y_3$. Let $\eta' \in [\eta/2, \eta]$ be as in item 1 of §9, taken at $\eta = \eta_0/8$. In each coordinate, at most $256c_{\mathrm{I}} + 3 < 4$ points of the grid $(\eta'/8)\mathbb{Z}$ lie at distance at most $\eta'/8$ from the image of the support of h_1 under the projection onto that coordinate axis. The sum that defines $\|\!|h_1|\!\|_\eta$ runs over $(\eta'/8)\mathbb{Z}^3$. Its term at a point vanishes unless each coordinate of that point is one of these grid points; hence $\|\!|h_1|\!\|_\eta \leq 4^3\|h\|_\infty$. At each plain label $\mathsf{b}' \in U(\mathsf{b})$, the first bullet of (G6) gives

$$\check{\Gamma}_*(\mathsf{b}', \eta) \leq \frac{32e^{c_{\mathrm{vol}}(\mathsf{b}')}\hat{G}(\mathsf{b}')}{\eta_0}.$$

For moving labels use the upper limit prescribed by (Lab3), and then bound it by the supremum over $U(\mathsf{b})$ of this whole expression. Combining with the factor 4^3 just obtained gives

$$\frac{2048}{24L}\,C_{\mathrm{I}}(\mathsf{b})\eta_0^{-1}\|h\|_\infty$$

times the two norms. Since $2048 \leq 24L$, the triple $(c_{\mathrm{I}}, C_{\mathrm{I}}, \mathsf{p}_{\mathrm{I}})$ of item 2 of §10 suffices.

(ii)*, outside the normal form.* Let $z \geq 1$ with $L^z\mathsf{b} > x_s$. Thm. 8.17 (R6) (vi), at moving labels, gives $S_n(X) = L^{zn}S'_n(X^{[z]})$. At finite (K, N) the identity is the one found in (i) above. Here S' is a limit of Thm. 8.17 (R5) along the matched subsequence, and its two limits are taken in the same order as for S, namely in one of the orders (a), (b) and (c) of that clause. The labels of S' are $L^z\mathsf{b}_j \to L^z\mathsf{b}$, and they are in normal form. Hence $\sigma_S = \sigma_{S'} \circ D_z$, by eq. 1.i. With the numbers of the normal form at $L^z\mathsf{b}$, Lem. 10.5 now gives (R0) to (R3), (R5) and (R7) for σ_S, with the numbers stated in (4.9) (ii).

Equivalence, at every $\mathsf{b} > 0$. This part uses only that σ_S has (R0) to (R3), (R5) and (R7). Positivity at Σ gives (R6), which is this positivity for tuples with one entry. It also gives (R4), by Prop. D.5 with Rem. D.13. Then (R4) and (R6), together with (R0) to (R3), (R5) and (R7), give invariance of σ_S under O(3) by Thm. D.31. Conversely, invariance under O(3) gives positivity at Σ (Prop. D.7 (c)). □

11. A reflected test for nontriviality

In this section we prove clause (Y7) of Thm. A, the bound $\mathsf{b}^2S_2^T(\theta f_*, f_*) \geq c_* > 0$, in which θ is a reflection and c_* a pure number: first on every torus of the data and for every cutoff $K \geq j + 1$, then for the limits and for every label $\mathsf{b} > 0$. The witness is a function G_- of the drawn block field $V^{(k)}$, and it lives on one side of a mirror; by reflection positivity, $\mathrm{Cov}^2 \leq \mathrm{Var}\cdot S_2^T$. We bound the covariance from below by integration by parts in ν_k and by the free Green function, and the variance from above. All conditions on the depth j are collected in one list, and we give a depth $j_*(\mathsf{b})$ in closed form that is bounded on compact sets.

Item 0: what is used, letters.

Item 0.1: what is used of Sections 2.1 to 8.1. From item 1 of §2.1, "Cubes", we take the counts $(26e)^{n-1}$ and $(52e)^n$, together with the meaning that is given there to "plaquettes of A", to "p in A" and to "tame there". From the same item, "At x", we take p_0, ε, $\rho(x)$ (see item 0.4), $r(x)$, $s(x) = 8r(x)$, $\ell_w(x)$, c_a and c_h, and from its parts "Fields" and "Domains" the conditions (u1) to (u3). The members of its part "Members" are objects of the same kinds as those of item 3 of §5.3, to which item 0 of §6.2, "Pairs", refers: they are the same triples, with the same functions, domains and invariance. From "Members" we take (ca), (ch) and (cΞ), the strong cap, the formula for $\rho[u;\ell]$, and Ξ_w together with its covering clause, by which, in the sum that defines Ξ_w, each set X_i holds a plaquette of $D(V)$ and $D(V)$ lies in the union of the X_i. Item 1 of §2.1 also gives (SD), and item 3 of §2.1, (O3), gives $\mathsf{N}_Z = \mathsf{N}_{\mathsf{B}} = 1 + 3K_*^2\mathsf{g}_*^2$.

From item 2 of §5.5 we take "Energy", "The step" and "Runs". From item 3 of §5.5, (R9), we take (9a) and (9e); they are Thm. 5.15 (a), (e).

Of the floor (W0) of Thm. 4.5 we use one clause alone, $\rho_0 \le \rho(x) + 1$. Here it enters as the entry (j7) (vii) of the list in Thm. 11.1, rather than as a floor.

From §2.2 we take the following: the bounds $\mathsf{g} \le 51\lambda^{-3}$ and $\mathsf{g}_1 \le 182\lambda^{-4}$, and the count of $24r^2 + 2$ points at distance r (Def. 2.44, after the display of $\mathsf{g}_1, \mathsf{g}_2$); the covariant rows (Def. 2.46); the defect $\tau(\ell)$ (Def. 2.47); the sets $\mathcal{R}(b, K, \lambda, \tau)$ with their bounds (Def. 2.48); the numbers b_*, λ_*, K_*, $\mathsf{t}_*(\rho)$ and L_A (Def. 2.56); the statements of Thm. 2.57 (a), (b); the number $b_{\mathsf{B}} = \|\mathsf{B}\|$ (Def. 2.22); and the numerals (the order of the constants in Thm. 2.65, (0)).

We keep $\kappa \ge 6$ as a letter, with $\kappa_1 = \kappa + 1$ (entry 1 of item (0) of §6.1; item 0 (O5) of §6.2; item 1 (O5) of §6.4). The quadratic member $s_x[D]$ is that of entry 5 of item (0) of §6.1. By (M2), $\sigma_M = 160(\kappa_1 + 1)c_R/M \le 1/12$ (item (1) (O7a) of §6.1), and (M2) itself is stated in the table of item 1 of §8.1. The function η_w is that of item (1) (O12b) of §6.1, and the floors and $\mathbb{P}$ are those of item (1) (O13) there. By item (1), "No circle", of §6.1, the exponent of p_0 is > 2, and each floor is a family of one-sided inequalities in x.

Of Thm. 8.2 we use the paragraph "Data", (T3) and (T6). We use the part "Pairs" of item 0 of §6.2, and Thm. 6.7 (U), (c+), (f1), (f4), (g1), (g2).

From §6.3 we take $e_R \ge 256$ (item 1, "Letters and data"); "Rows", "Blocks", "Places", the norm of a family and the charge (the definition "Families, norm, charge, step and cut"), the source forms (the definition "Source forms and Riemann means") and Mn (item 1); the numerals (item 2), together with the letter g_{1b} (item 1 (O0a) of §6.4); Thm. 6.20 (U2.1) to (U2.3); and Thm. 6.21 (S2). We use Thm. 7.1 (b), (h); $C_g = 263L^3$ (item 0 (1) of §7.2); Lem. 7.7; and Thm. 7.9 (i).

From §7.3 we take the data (item 1) and (o2) to (o7) (item 2); (G1) to (G4) (item 3); Prop. 7.12 (J0) to (J4); the statement of item 5 that (J1) to (J4) hold for all $k \le K - 1$, by the same proof; and (C) and (A) of Step 1 of the proof of Prop. 7.12.

From §7.5 we take q_N, the norms, C_{vol} and the congruence (item 1); and from Thm. 7.27 we take the data (B), (b^C) (ii), and (c^C), the sentence before (S).

From Step 2 of the proof of Thm. 8.2, "The last integral", we take $\ell_k \in \mathcal{R}(b_*, K_*, \lambda_*, \frac{1}{9})$ for $0 \le k \le K$, for every $K \ge 1$. This inclusion is proved in Step 2 (c) of the proof of Thm. 5.63, where its data include $K \ge 1$.

Item 0.2. The reflection and chessboard estimates. We use the board, copies and fold of Def. A.4, the reflected weight identities of Cor. A.3, and the seminorm and product estimate of Def. A.10 and Prop. A.11. The two special cases for a constant and for one cell follow from Prop. A.12.

Item 0.3: what is used of earlier sections. Of Prop. 8.11 we use the setting of Def. 8.6, with any number $Q \ge 1$ of sources, and then the following, cited by the tags of the proposition: (G°) and (G5°) (a); (lin), together with the line "Hence $\|\delta[f](k)\|^{\mathrm{st},(\kappa_1)} \le 0.2\,\mathrm{amp}(Lx_k)\|f\|_\infty$" that follows (J4°); (J1°), (J2°), (J3°) and (J4°); of (L0), the bound 0.014 in the norm with the rate κ_1, this bound being also (J5°) (c) at $k = 0$; (J5°) (a′), (c) at $k = 0$, and (d); (J7°) (ii), (iii); (J8°) (a′); and $\mathrm{sep}^\circ := 8\log(8L^{1/2})$, as it is set in item 2 of §1.

Of Thm. 8.17 we use, in its own letters, (R4), (R5) (a), (b), (c), (R6) (i), (iii), (vi), and (R7); they are (Y3), (Y4) and (Y8) of Thm. A.

Of §1 we use (S9), the boards B_i for $-m_l \le i \le K-1$, with walls $L^{-i}(\mathbb{Z}+\frac12)$; item 2, with $c_{\mathrm{vol}}(\mathsf{b}) = 225/M^3 + \frac32 \mathsf{N}_Z p_0(\mathsf{b})^2 + 3\log(384/\varepsilon(\mathsf{b})^3)$; and (S5), by which $Q \ge 1$ is arbitrary. Further, $\beta = L^K \mathsf{b} > 0$ by the data.

The boards carry no offset. Every other place is reached by a translation as in Lem. 11.3 (K), namely by a site of level k, in $L^{-j}\mathbb{Z}^3$; and if a board is shifted by an offset in $L^{-j}\mathbb{Z}^3$ and carries the cell function translated accordingly, its fold is the same.

Item 0.4: letters. We put $a_u := \langle u, C_B u\rangle$, where C_B is that of Lem. 11.15. The sign $\mathfrak{h}_k$ has the meaning given in (D2) of §8.2, and $\mathrm{ex} := x^{-8L}$.

Without an argument, N_1 is $(D_0/4M + 2\check r(x) + 3)^3$ (Lem. 11.4), whereas $N_1(k-m)$ is that of (D2) of §8.2 (Lem. 11.8). For $s \ge 0$, $\tau_s f$ is the translate of f by the distance s along the normal to the mirror of θ, away from that mirror.

The bare tags (f1), (f4), (g1), (g2), (c+), (U), (S2), (U2.1) to (U2.3), (SD), (9a), (9e), (b^C), (c^C), (M2), (W0), (J0) to (J4) and (G1) to (G4) refer to the statements and items named in item 0.1. With an index, as in $R_{k'}$ and R_k, the letter R denotes the radii of (AD) of Prop. 8.11. In Sections 2.1 to 8.1, the radius $\rho(x) = \lceil c_R \log x \rceil$ of (S1) of §1 is denoted by $\varrho(x)$; in this section and in §12, ϱ without an argument is $x^{1/2}p_0^6$.

Theorem 11.1 (A lower bound on the two-point function, uniform in cutoff, side, depth and label)**.** The gauge group is SU(2), and $d = 3$.

Data. We work in the setting of the paragraph "Data" of Thm. 8.2: $\mathsf{b} \ge x_s$; $N = 2L^{m_l}$ with $m_l \ge m'$; (Tor) $N \ge 2Ms(L\mathsf{b})$; $K \ge 1$; and $\beta = L^K\mathsf{b}$. The observable is $\mathcal{O}_K(f) = a_K^{-1}\sum_p f(x(p))a_p$. Let $j \ge 1$, $k := K - j \ge 1$ and $x := x_k = L^j\mathsf{b}$; we write $\varepsilon = p_0(x)x^{-1/2}$, where $p_0 = A_0(\log x)^{\mathfrak{p}_0}$ and $\mathfrak{p}_0 > 2$, and $\rho = \rho(x)$. Let ν_k be the law of $V = V^{(k)}$ on $\mathbb{T}_k = \mathbb{Z}^3/NL^j\mathbb{Z}^3$; its density is $\rho_k/\int\rho_k$, where $\rho_k = e^{\mathsf{V}}\rho[\bar u_k;\ell_k] > 0$ at every field. The member $\bar u_k$ is real and lies in $M_1(x_k;\kappa_1,c_h)$, with the strong cap (Thm. 7.1 (h)). The table (O0) to (O13) and all floors are those of Thm. 8.2; none of them is chosen here, and the new numbers come after them, at (O14). Beyond these data, the table and the floors, we make no hypothesis.

Geometry. We measure in units of level k, that is in units of L^{-j}, and $r = |y|$ is the sup distance. We centre the construction at the site 0, which is the base site of P_0 and the centre of the boxes and of the cell 0; other places are reached by the translations of Lem. 11.3 (K). The mirror is $y_1 = \frac12$, and it is a wall of B_j. Let P_0 be the 23-plaquette based at 0, and let $u := e_{P_0} - e_{P_0+e_2}$. Let $T_y := \mathrm{Ad}_{V(\Gamma(0,y))}$, $\tilde F_p := T_{\pi(p)}F_p$ and $u^\alpha(F) := \sum_p u_p \tilde F_p^\alpha$. The boxes are $B_R = \{r \le R\}$, $B' = \{r \le R + 2\rho + 3\}$ and $\mathrm{Box} = \{r \le r_B\}$, with $r_B := R + h/8 + 1$.

Let χ be the product of $\chi_0(\phi_p/\varepsilon)$ over the plaquettes of B', where χ_0 is smooth, equal to 1 on $[0,\frac12]$ and to 0 on $[\frac34,\infty)$, with $|\chi_0'| \le 8$ and $|\chi_0''| \le 250$; such a χ_0 exists, since for $-\chi_0'$ we can take a smooth trapezium of height 16/3 with two ramps of width 1/16. Let χ_- be the same product over those plaquettes all of whose sites lie in $y_1 < \frac12$. We put $G_- := \frac{x}{2}\sum_\alpha u^\alpha(F)^2\chi_-$; its values lie in $[0,\frac98 p_0^2]$, and it depends only on the bonds with $\frac12 - NL^j/2 < y_1 < \frac12$. By G we denote the same function with χ in place of χ_-.

Let $\chi_R(y) := \prod_\mu \chi_1(|y_\mu|/R)$, where $\chi_1 = 1$ on $[0,\frac12]$, $\chi_1 = 0$ on $[1,\infty)$, and $\chi_1(t) = 1 - 3s^2 + 2s^3$ for $\frac12 \le t \le 1$, with $s = 2t - 1$; thus $|\chi_1'| \le 3$ and χ_1' is 24-Lipschitz. Let $y_c := \lceil D_0/\sqrt{2}\rceil(e_1 + e_2)$. Let $f_0 \in C^\infty$ satisfy $0 \le f_0 \le 1$, $f_0 = 1$ on $|y - y_c| \le D_0/16$ and $f_0 = 0$ off $|y - y_c| \le D_0/8$, and let $f_*(\xi) := f_0(L^j\xi)$. The boards are B_{j-1}, whose cells have side L (Lem. 11.5), and B_{j-i_h}, whose cells have side h (Lems. 11.10 and 11.13); in both, the cell 0 is centred at 0, and there is no offset.

New numbers. All new numbers are placed at (O14), after $\mathfrak{r}^\circ$ and sep°. No floor depends on them; none of them depends on N or on K; and each depends only on those that precede it, in the

order below. The pure numbers are $D_0 := 10^{140}$ and

$$c_F := \frac{3}{4} \sum_{|y-y_c| \le D_0/16-1} |(\Pi u)(y)|^2 \in [5.4 \cdot 10^{-5}, 4.0 \cdot 10^{-4}] D_0^{-5},$$

where Π is the lattice projection on exact 2-forms; then $c_* := c_F^2/7 \ge 10^{-1410}$, $r_1 := 10^{130}$ and $W_* := 10^3 r_1^4$; the numbers C_e, C_1, C_2, C_S, $C_R := 60 C_S C_1$, C_6 and C_{lat} of item 17 of §11, the table of the numerals; and $C_h := 64$ and $C_{\mathrm{ext}} := 32$.

The following depend on x alone: $R := \lceil C_R p_0^5 / c_F \rceil$; $h = L^{i_h}$, the least power of L that is $\ge \max\{16R, 64LM \log x\}$; $n_0 := \lfloor h/16M \rfloor$, $t := x^{-2}$ and $r_\eta := x^4$; $\varrho := x^{1/2} p_0^6$ and $r_z/\mathsf{b} = 3.5 E_a p_0^6 p_0(Lx)/(c_Q^{\mathrm{su}} K_5 L^{1/2})$; $\tau_x := 3.82 \cdot 10^{56} x^{-256}/(1-L^{-256})$; $W_0(x) := 78 r_1^3 (1+\log R(x))$; $\Theta(x) := \varepsilon(x)(2\ell_w(x) + 4/\vartheta_0^\flat) W_0(x)$, in which $\vartheta_0^\flat$ is the number of (u1); and $C_a(x) := 2.5(2r(x)+1)^3 E_a W_*$. After these comes $r_z := \varrho\, \mathfrak{r}^\circ(x)$, which depends on b besides x.

The list of the conditions on the depth. The list below contains every condition on the depth. Its entries fall into two families:

(X) the entries that depend on x alone:
- (j1) each of (E1) to (E6) of Lem. 11.17 is $\le c_F/12$; it follows that (E7) $:= 4[(\mathrm{E2}) + (\mathrm{E3}) + (\mathrm{E4}) + (\mathrm{E6})] \le 4c_F/3 < 1$, and the bound on (E2) holds of itself;
- (j5) $C_{\mathrm{ext}} r_B^2 \varepsilon \le 1$;
- (j6) $x^{10} \eta_w(x) \le 1$, $p_0^{10} \le x$ and $(r_z/\mathsf{b})(4C_S p_0^2 + 5) \le x$;
- (j7) the conditions
  - (i) $(c_P/2) p_0^2 \ge L^3 (24 \mathsf{N}_Z + 4.5 \log x + 5 + 220 M^{-3})$, where $c_P := \min\{c_\Xi, \frac{1}{8}\}$;
  - (ii) $R/2 - 1 \ge 2D_0 + M(r(x)+2)$;
  - (iii) $8\rho(x) + 8 \le R$;
  - (iv) $8M(r(x)+1) \le R$;
  - (vi) $p_0(x) \ge 4$, $(c_a + c_h)(x) \le \frac{1}{2}$, $\log x \ge 3$ and $\Theta(x) \le 1$;
  - (vii) $\rho_0 \le \rho(x) + 1$;

(B) the entries that depend on b besides x:
- (j2) $c_{\mathrm{vol}}(\mathsf{b}) h^3 L^{-3j} \le 1$;
- (j3) $i_h \le j + m' - 1$.

Once (X) holds, the following hold of themselves (Lem. 11.22 (b)). First, (E2) $\le c_F/(12 p_0)$ and (E7) < 1. Second, the two conditions

(j4) in both of its lines, $\lfloor h/2M \rfloor - 2\check{r}(x) - 2 \ge 8 \log(4\varrho L^{1/2})$ and $\lfloor h/2LM \rfloor - 2\check{r}(x/L) - 2 \ge \mathrm{sep}^\circ$;
(j7) (v) $\varrho L^{1/2} \ge 75$.

Third, $R \ge 6 r_1^3$, because $R \ge 10^{1410} p_0^5$; moreover $R \ge 4 r_1$, $4t \sup|w| < \varepsilon/4$ for the 1-form $w := \mathrm{d}^*(-\Delta)^{-1}\omega$ of Lem. 11.15, $x \ge L$ and $h < 0.71 x^{1/4}$.

Definitions. Let $J(\mathsf{b}) := \{j \ge 1$: (X) holds at $x = L^j \mathsf{b}$, and (B) holds$\}$. Let $x^\# := 1 + \inf\{y \ge L$: (X) holds at every $x \ge y\}$; this number depends on the table and on the pure numbers alone. Let $y(\mathsf{b}) := \mathsf{b} \max\{c_{\mathrm{vol}}(\mathsf{b})^{1/3}, L^{1-m'}\}$. For $\mathsf{b} \ge x_s$, let $j_*(\mathsf{b})$ be the least integer $j \ge 1$ with $L^j \mathsf{b} \ge \max\{x^\#, y(\mathsf{b})^{4/3}\}$; for $0 < \mathsf{b} < x_s$, let $j_*(\mathsf{b}) := z + j_*(L^z \mathsf{b})$, where $z = z_0(\mathsf{b})$ is that of (S7) of §1. For $\mathsf{b} \ge x_s$, let $j^+(\mathsf{b})$ be the supremum of j_* over $[\mathsf{b}/2, 2\mathsf{b}] \cap [x_s, \infty)$. Finally, let $x^{\#\#} := \max\{x^\#,\ L^3 x_s,\ \sup_{\mathsf{b}' \in [x_s, L^2 x_s]} y(\mathsf{b}')^{4/3}\}$.

Claims. Let θ be the reflection $y_1 \to 1 - y_1$.

(D) The number $x^\#$ is finite; $[j_*(\mathsf{b}), \infty) \subset J(\mathsf{b})$; and j_* is bounded on compact sets of $[x_s, \infty)$, so that $j^+(\mathsf{b})$ is finite.

For every $j \in J(\mathsf{b})$, hence for all $j \ge j_*(\mathsf{b})$, and for all $K \ge j+1$ and all N of the data, the following clauses hold.

(N0) $\mathrm{Cov}(G_-, \mathcal{O}_K(f))^2 \le \mathrm{Var}(G_-)\, S_2^T(\theta f, f)$ for every real bounded f such that $f(x(p)) = 0$ unless all sites of p lie in $\frac{1}{2} < y_1 < \frac{1}{2} + NL^j/2$. This clause does not depend on any entry

of the list: it holds for every $j \geq 1$ with $k \geq 1$, once the coordinate y_1 is taken in the fundamental domain $\frac{1}{2} - NL^j/2 < y_1 \leq \frac{1}{2} + NL^j/2$.

(N1) $\mathsf{b}\,\mathrm{Cov}(G_-(V^{(k)}), \mathcal{O}_K(f_*)) \geq c_F$.

(N2) $\mathrm{Var}(G_-) \leq \frac{3}{2}a_u^2 + 1 \leq 7$.

(N3) $\mathsf{b}^2 S^T_{2;K,N,L^K\mathsf{b}}(\theta f_*, f_*) \geq c_*$, where c_* is pure, that is free of the cutoff, of the side, of j and of b.

(N4) The same bound holds for $S_2^{(N,\mathsf{b})}$ of (T3); for each limit of (R5) (a), (b), (c) at the label b; and, at every $j \geq j^+(\mathsf{b})$, for each limit of (R5) (a), (c) with labels $\mathsf{b}_i \to \mathsf{b}$, $\mathsf{b}_i \geq x_s$.

(N5) For every limit S of (R5), $S_2(\theta\tau_s f_*, \tau_s f_*) > 0$ for every $s \geq 0$, and $\dim \mathcal{H}(S) \geq 2$; at moving labels this holds at the j of (N4) or at the i of (N6′).

(N6) The same holds for every $\mathsf{b} > 0$ and every $j \geq j_*(\mathsf{b})$, because $\mathsf{b}^2 S_2$ is invariant under the dilation of (T6), (R4), (R6) (vi). This clause does not include the clause of (N4) on moving labels; for labels below x_s, that clause is carried by (N6′) with $Lx^{\#\#}$.

(N6′) One inequality holds for all labels: for every $\mathsf{b} > 0$ and every integer i with $L^i\mathsf{b} \geq x^{\#\#}$, $\mathsf{b}^2 S_2(\theta_i f_0(L^i\cdot), f_0(L^i\cdot)) \geq c_*$, where θ_i is the mirror at $\xi_1 = L^{-i}/2$. It is valid for each limit of (R5) at the fixed label b and, provided $K \geq i+1$, on every torus whose side satisfies the conditions that (R4) sets for it at some integer z with $L^z\mathsf{b} \in [x_s, L^2 x_s]$; with $Lx^{\#\#}$ in place of $x^{\#\#}$, it is valid also for labels $\mathsf{b}_n \to \mathsf{b}$.

■□

Remark 11.2. The function f_* depends on b through the depth $j \geq j_*(\mathsf{b})$. The number c_* is pure but tiny, and the depth j_* is astronomically large; the numerals are generous, and none of them is sharp. For $x^{\#}$, $x^{\#\#}$ and j_* we give closed forms only, since the table gives L, M, A_0, $\mathfrak{p}_0$, c_Ξ, c_h, E_a, C_3, $\vartheta_0^\flat$, ρ_0, c_R and κ as letters. The depth $j_*(\mathsf{b})$ is defined by its closed form, and not as the least depth at which the list holds; for large b this closed form grows like $\frac{1}{3}\log_L \mathsf{b}$.

The parts of the proof. The proof of Thm. 11.1 falls into the parts below.

The first part consists of two statements. One is the clause (N0), which is Lem. 11.3 (W), the witness inequality; it follows from (L), (I) and (K) of that lemma (the joint law: locality, independence and the covariance of the kernel) and from A.3 (c), (d). The other, Lem. 11.4, is the split $E[\mathcal{O}_K(f)|V] = c_f + A_f + W_f$; it is used for (N1). The inputs of this part are "The step" and "Runs" of item 2 of §5.5, the clauses (c) and (d) of A.3, Thm. 7.1 (h), Prop. 7.12 (J0) to (J4), and item 5 of §7.3.

Lem. 11.5 is the Peierls bound at $\varepsilon/2$; it is free of N, K and b, and it is stated in the norm form (P°). In its proof, the numerator and the denominator are treated cell by cell, and then the root of order equal to the number of cells is taken. In this way the volume drops out by soft means: by A.11 on B_{j-1} for the block law, by the covering clause together with the strong cap, and by a lower bound for the denominator, the integral of the density, which we obtain on a small ball of fields about $V = 1$ (Step 7 of the proof of Lem. 11.5).

Lem. 11.6 extends a field that is tame on the box $\{r \leq r_B\}$ to a globally tame field on $\mathbb{T}_k$. Lem. 11.7 compares two sums, the sum Λ of the lemma and the sum $c_f + A_f$ of the split, with their images under the maps of the fold of B_{j-i_h}, that is under the mirrors and the translations of its cells; only these sums enter the comparison, not the single members, so that no covariance of a member is used. In both lemmas no member is reflected.

Lem. 11.8 is Prop. 8.11 with a free last ratio. It is (J5°) (d) and (J8°) (a′) of Prop. 8.11; its proof is an induction and uses (9e), (g2), (J1°) and (J2°).

For any $\hat{w} \geq 0$ that takes the place of the w-member of $\bar{u}_k$ and lies within the cap $2.6e^{-\kappa(|X|_c - 1)}$, Lem. 11.9 bounds the integral of the density by $(1 + C_{\mathrm{vol}}) \int \rho[\bar{u}_k; \ell_k]$, and it gives the root bound. Lems. 11.10 and 11.13 build their majorants and use this lemma. Lem. 11.10 bounds the part W_f of the split, and Lem. 11.13 the defect of the integration by parts caused by Ξ_w; both the part and the defect come from the weights, the w-member of $\bar{u}_k$. The last two lemmas are stated in the

norm forms (W°) and (X°). Their bounds are independent of N and of K, whereas the first bound of Lem. 11.9 contains the factor $1 + C_{\mathrm{vol}}$, which depends on N. The proofs of Lems. 11.10 and 11.13 take the fold over the cells of B_{j-i_h} and then the root of order equal to the number of cells. The volume factors that arise in these proofs are removed by A.11 on B_{j-i_h} and by both bounds of Lem. 11.9. The proof of the first of these bounds, with its factor $1 + C_{\mathrm{vol}}$, uses (9a) with (f1), that is the exact cancellation of the vacuum factors along $k \le i < K$; Thm. 7.9 (i); and Thm. 7.27 (c^C). The last bound of Lem. 11.9 follows from (j2): the root of $1 + C_{\mathrm{vol}}$ of order equal to the number of cells of B_{j-i_h} is $\le e$.

Lems. 11.14, 11.16 and 11.17 carry out the integration by parts in ν_k, and give the key line, four identities and the list of errors. Every power in their estimates is explicit.

Lems. 11.15 and 11.19 give the Green function of $\mathrm{d}^*\mathsf{B}_k\mathrm{d}$ uniformly in the rows, and $q \ge c_F$. Both concern the free model, and both have explicit constants.

The clauses (N1) to (N6) and (N6′) are derived in Lem. 11.21. The derivation rests on (T3) and (T6), and on (R4), (R5), (R6) (i), (iii), (vi) and (R7) of Thm. 8.17.

Lem. 11.22 treats the list of the conditions on j and the depth $j_*(\mathsf{b})$. It gives each entry together with its use and the reason why it holds; $x^{\#}$ is finite, $[j_*(\mathsf{b}), \infty) \subset J(\mathsf{b})$, and j_* is locally bounded.

The numerals are collected in one table. For C_2, C_6 and the numeral of (E4), the table gives generous closed forms, with the rows from which they come.

Statements and items of other sections, as used here.

Item 1 of §2.1 ("Cubes", "At x", "Fields" and "Domains", "Members", (SD)*) and item 3 there,* (O3). These two items supply the formula for the score of ν_k. On fields that are tame on B', $S_\varepsilon(t) = t^2/2$, $\zeta = F$, and $m_y = 1$ for every row that the flow moves.

Item 2 of §5.5, "Energy", with (W0) *of Thm. 4.5.* The mask depends on $D(V)$ within ρ_0 of y, and on the plaquettes with $\phi_p > 1$ within $\rho(x) + 1$ of y; the rows have range $\le \rho(x)$; and the paths go along e_1, then e_2, then e_3. Once $\rho_0 \le \rho(x) + 1$, which is (j7) (vii), this fixes the radius $R + 2\rho + 3$ of B'.

Item 2 of §5.5, "The step", "Runs". The average $\mathsf{M}(U)(c)$ depends on $B(y)$, $B(y')$ and straight paths. Since the kernel k_{ax} of the step is a probability kernel, $\int \rho' = \int \rho$. The step is independent of the rows, of the grid and of the members. The site y of $\mathbb{T}_k$ lies at $L^{-j}y$ ("Places", item 1 of §6.3, which sets it at $a_K L^k y = L^{-j} y$).

Item 3 of §5.5, (R9)*, with* (9a), (9e). We apply (9a), which holds at every V', to a single real member $\hat{u}$ and to its iterates (Lem. 11.9); and in the range over which Thm. 6.7 (U) extends the clauses (a) to (e) of Thm. 5.15, that is at every pair $p \in \mathbb{P}$, we apply (9e) to $u_1, u_2 \in M_2$ (Lem. 11.8).

§2.2, with the definitions and Thm. 2.57 named in item 0.1. The statement of Thm. 2.57 (a) gives $\ell'^{\mathrm{c}} \in \mathcal{R}(b_*, K_*, \lambda_*, \tau + \mathsf{t}_*(\rho))$, and induction from $\tau(\emptyset) = 0$ gives $\tau(\ell_k) \le \sum_{i \le k} \mathsf{t}_*(\rho(x_i))$. The definition of t_* (Def. 2.56), together with $\rho(x) \ge c_R \log x$ and $e_R \ge 256$, then yields $\tau(\ell_k) \le \tau_x$. If only the stated bound $\tau \le \frac{1}{9}$ were at hand, the term $C_\tau \tau^2 r^{-4}$ of Lem. 11.15 (b) would swamp the continuum form π of Lem. 11.19 (A), which is of size r^{-4}; the sharper bound is therefore needed. The bound $b_{\mathsf{B}} \le b_*$ holds at the momenta of the torus, and Step 0 of the proof of Lem. 11.15 carries it to all p.

Item 0 of §6.2, "Pairs" (ℙ), and Thm. 6.7 (U), (c+), (f1), (f4), (g1), (g2). By (f1), neither E'^a nor V' depends on E^h or on w, so the vacuum factors of $\hat{u}$ and $\bar{u}$ cancel exactly. The function $w(X, V)$ of (f4) enters only for sets X that hold a plaquette of $D(V)$. We apply (g2) with $A = 2$ and with a pair $p \in \mathbb{P}$ whose first entry is a rate $\ge \kappa$, one of the rates of Lem. 11.8, and whose second entry is c_h.

§6.3, with "Rows", "Blocks", "Places", the charge (the definition "Families, norm, charge, step and cut" of item 1), Mn, (U2.1) *to* (U2.3), (S2) *and the numerals.* The barycentres of $B^k(y)$ lie within $\frac{1}{2}$ of y, in units of level k. Hence $\mathrm{Mn} = 1$ for $|y - y_c| \le D_0/16 - \frac{1}{2}$, $\mathrm{Mn} \ge 0$ always, and

$\mathbb{D}_{k,y}[f_*] = 0$ for $|y - y_c| > D_0/8 + \frac{1}{2}$. For each site y, the error of (S2) is bounded in operator norm, and these bounds are summed over the sites y of the outer ball $|y - y_c| \le D_0/8 + \frac{1}{2}$.

Thm. 7.1 (b), (h). The condition (SD) holds at $(x_i; \ell_i)$ for $0 \le i < K$. At every field, $\rho_k^z = \exp\{\sum V_i(z)\}\rho[u_k(z); \ell_k]$.

Lem. 7.7 and Thm. 7.9 (i). We apply Thm. 7.9 (i) with $B = 2.6$ and $p = (\kappa, c_h)$: the theorem assumes families within B caps at p, with the weak cap, and its clause (i) yields $F_\ell(u_i)$ in $M_1(x'; p)$, with the strong cap $(0.35 + 0.248 \cdot 2.6 = 0.9948)$. Lem. 7.7 (S1) is applied value by value.

Prop. 7.12, with items 1, 2, 3, 5 of §7.3 and with (C), (A) *of Step 1 of its proof.* We use (J2), (J3) and (J4) for one source. Prop. 8.11 contains them as (J2°), (J3°) and (J4°), for one complex source of any support and for $0 \le k \le K$. Prop. 7.12 by itself yields them for $n = 2$ as well, if the second source is a translate, shifted by the distance $(h/2)L^{-j}$, and is taken at $z_2 = 0$.

Thm. 7.27, with item 1 of §7.5. The sentence $\int |\rho[u_2; \ell] - \rho[u_3; \ell]| \le \frac{1}{2} C_{\mathrm{vol}} \mathsf{D}_{23} I_1$ of (c^C), case (ii), is applied in Lem. 11.9, with $x = \mathsf{b}$, with $u_1 = u_3 = \bar{u}_K$ and with $u_2 = \hat{u}_K$, an output of Thm. 5.15 with $A \le 2.6 \le 6$; and $\mathsf{D}_{23} \le 2$.

A.4 (2), A.6 (1), A.3 (c), (d), (e), A.10 with eq. A.xxiii, A.11, A.12 *(b), (f).* The boards have λ odd with $\lambda \ge 3$, n_μ even with $\mathsf{n}_\mu \ge 2$, and walls at the midpoints of bonds. The functions are bounded Borel functions of those bonds whose two ends lie in the cell. Clause (c) of A.3 is the Schwarz inequality. For $\mathfrak{F}_+$ we take a half torus, and clause (d) of A.3 gives the other half. The copies are $F(u^\sigma)$; for real F, no conjugation enters.

Prop. 8.11, with (G°), (G5°) *(a),* (lin), (J1°) *to* (J4°), (L0), (J5°) *(a′), (c), (d),* (J7°) *(ii), (iii) and* (J8°) *(a′).* Lem. 11.8 (A) is (J5°) (d), and Lem. 11.8 (B) is (J8°) (a′). We take $R_{k'} = \mathfrak{r}^\circ_{k'}$ for $k' < k$, and $R_k = \varrho\, \mathfrak{r}^\circ_k$. The rates satisfy $\varkappa_{k'} \in [\kappa + \frac{1}{4}, \kappa + 0.51]$, and the last drop satisfies $2\log(4\varrho L^{1/2})/\mathrm{sep}_k \le \frac{1}{4}$. The hypothesis (AD) (ii) holds: for its number $q_{k'}$ we have $q_{k'} \le (1 + L^{1/2})/(4\, L^{1/2})$ for $k' < k - 1$ and $q_{k-1} \le (1 + \varrho L^{1/2})/(4\, \varrho L^{1/2})$. Both right-hand sides are $\le (1 + 1/75)/4$, hence ≤ 1, and since every drop $\varkappa_{k'} - \varkappa_{k'+1}$ is $\le \frac{1}{4}$, it follows that $0.364 e^{1/4} q_{k'}^2 \le 0.029996$. Here $L^{1/2}$ and $\varrho L^{1/2}$ are the bounds on $R_{k'+1}/R_{k'}$ that (G3) gives.

Thm. 8.17, with (R4), (R5) *(a) to (c),* (R6) *(i), (iii), (vi) and* (R7). The dilation covariance is (R6) (vi). Moving labels are treated in Step 4 of the proof of Lem. 11.21 and in (N6′). The space $\mathcal{H}(S)$ and all real translations are at hand for the limits of (R5) alone.

(S9) *of §1, the boards.* The mirror $y_1 = \frac{1}{2}$ is a wall of B_j $(\lambda = L^k \ge 3)$. The board of Lem. 11.5 is B_{j-1} $(\lambda = L^{k+1}$, $\mathsf{n}_\mu = 2L^{m_l + j - 1})$. The board of Lems. 11.10 and 11.13 is $B_{j - i_h}$ $(\lambda = L^{k+i_h}$, $\mathsf{n}_\mu = 2L^{m_l + j - i_h} \ge 2L$ by (j3)); its cells are centred at $h\mathbb{Z}^3$, and $h(\mathbb{Z} + \frac{1}{2}) \subset \mathbb{Z} + \frac{1}{2}$ because h is odd.

Item 2 of §1, with $c_{\mathrm{vol}}(\mathsf{b})$ and the subtraction $c_f + A_f$. We have

$$\log(1 + C_{\mathrm{vol}}) \le \log(8q) + 220q + [\frac{3}{2}\mathsf{N}_Z p_0(\mathsf{b})^2 + 3\log(384/\varepsilon(\mathsf{b})^3)]N^3 \le c_{\mathrm{vol}}(\mathsf{b}) N^3,$$

with the numeral 225 in $c_{\mathrm{vol}}(\mathsf{b})$, because $q = q_N = (N/M)^3 \ge 1$ (item 1 of §7.5) and $\log(8q) \le 2.08q$ for $q \ge 1$. The subtraction $c_f + A_f$ is neither $\hat{c}_K$ nor the torus mean; its mirror defect is treated in Lem. 11.7.

Lemma 11.3 (The joint law and the witness inequality)**.** Let P be the joint law of $(U = V^{(0)}, V^{(1)}, \dots, V^{(K)})$, that is, Wilson's measure multiplied by the kernels $\prod_c k_{ax_i}(V^{(i+1)}(c), \mathsf{M}(V^{(i)})(c))$, $0 \le i < K$, as they are defined in item 2 of §5.5, "The step" and "Runs". As M is Borel and $k_s(\cdot, h)dg$ is a probability measure, P is a probability measure; we denote by ν_k the law of the component $V^{(k)}$ under P. For a set A of sites of $\mathbb{T}_k$ we write $A^{(i)}$ for the union of the $(k - i)$-fold centred blocks of the sites of A.

(L) The kernels from which the values of $V^{(i+1)}$ on the bonds of $A^{(i+1)}$ are drawn depend on $V^{(i)}$ only through its values on the bonds of $A^{(i)}$.

(I) Let $A_1, \dots, A_m$ be pairwise disjoint, and for every s let Φ_s be a bounded Borel function of the values of U on the bonds of $A_s^{(0)}$ together with the values of $V^{(k)}$ on the bonds of A_s.

Then $E[\prod \Phi_s|U] = \prod E[\Phi_s|U]$, and each $E[\Phi_s|U]$ is bounded, Borel, and determined by the values of U on the bonds of $A_s^{(0)}$ alone.

(K) Let $\check\theta$ act in each coordinate either as $y_\mu \to y_\mu + n_\mu$ or as $y_\mu \to n_\mu - y_\mu$, with $n_\mu \in \mathbb{Z}$ measured in units of level k, and define $(\check\Theta V)(b) := V(\check\theta\, b)$ at every level $i \le k$, under the convention $V(\text{reversed } b) = V(b)^{-1}$. Then $\mathsf{M}(\check\Theta U)(c) = \mathsf{M}(U)(\check\theta\, c)$; acting on the levels $0, \dots, k$ simultaneously, $\check\Theta$ leaves the marginal of P on these levels invariant; $E[\Phi \circ \check\Theta|U] = E[\Phi|\cdot](\check\Theta U)$; $\rho_k \circ \check\Theta = \rho_k$; and $\rho_k^{[zf]}(\check\Theta V) = \rho_k^{[zf\circ\check\theta^{-1}]}(V)$, where $\rho_k^{[zf]}(V)$ is the integral of Wilson's weight, times $e^{z\mathcal{O}_K(f)}$, times the k kernels.

(W) Let G_- be a bounded real Borel function of those bonds of $V^{(k)}$ whose two ends lie in the region $\frac12 - NL^j/2 < y_1 < \frac12$, and let f be as in (N0). Then $\operatorname{Cov}(G_-, \mathcal{O}_K(f))^2 \le \operatorname{Var}(G_-) S_2^T(\theta f, f)$.

■

Proof. Step (L)*: locality.*

1. Let $c = (y, y')$ with $y' = y + Le_\mu$. The formula for M (item 2 of §5.5, "The step") depends on U only through $\mathsf{t}_U(x) = U(\Gamma(y(x), x))$, through the straight paths $[x, x + Le_\mu]$ with $x \in B(y)$, and through $\mathsf{t}_U(x')$.
2. A coordinate path running from the centre of a box to one of its points changes a single coordinate at a time, each of them monotonically; therefore it never leaves the box.
3. The straight path is contained in $B(y) \cup B(y')$.
4. Consequently $\mathsf{M}(U)(c)$ depends only on bonds with both ends in $B(y) \cup B(y')$, and (L) follows by induction on the levels.

Step (I)*: independence.*

1. Conditionally on U, the law of $(V^{(1)}, \dots, V^{(K)})$ has a density, namely a product over all bonds of the levels $1, \dots, K$ that contains one kernel factor for every bond.
2. The factors of the levels above k integrate to 1, level by level, beginning with K and descending.
3. We call a bond of level i a bond of s if both of its ends lie in $A_s^{(i)}$, and free if it is a bond of no s.
4. By (L), the factor of a bond of s depends only on bonds of s one level lower.
5. We integrate over the free bonds, beginning at level k and descending. A free bond of level k occurs neither in any Φ_s nor in any factor besides its own, so its factor integrates to 1.
6. Suppose that the free bonds of all levels above i are already integrated out. The factors that remain at level $i+1$ belong to bonds of some s, and by item 4 they depend only on bonds of s at level i; hence a free bond of level i occurs in no remaining factor except its own, and its factor integrates to 1. We apply this for $i = k-1, k-2, \dots, 1$ in turn.
7. What is left is a product over s of integrals, and the s-th integral depends only on the values of U on the bonds of $A_s^{(0)}$.

Step (K)*: covariance.*

1. At the transition from level i to level $i+1$, $i < k$, the offsets of $\check\theta$ in fine units are $L^{k-i}n_\mu$, hence multiples of L. Apply Lem. 8.20 to the coordinate reflections and translations composing $\check\theta^{-1}$; its convention $(RV)(b) = V(R^{-1}b)$ then agrees with $\check\Theta$. This gives the covariance of M and the invariance of each transition kernel, including reversed bonds and the value $\mathsf{L}(-1) = 0$.
2. Wilson's weight and the Haar measure are invariant, as $\operatorname{tr} g^{-1} = \operatorname{tr} g$.
3. Barycentres are mapped to barycentres; hence $\mathcal{O}_K(f)(\check\Theta U) = a_K^{-1} \sum_p f(x(p)) a_{\check\theta p}(U) = \mathcal{O}_K(f \circ \check\theta^{-1})(U)$.

4. The levels above k integrate to 1 (item 2 of Step (I)). The change of variables $V^{(i)} \to \check\Theta V^{(i)}$ at the levels $0 \le i \le k$ then yields the five assertions of (K).

Step (W)*: the witness inequality.*

1. At level k the planes $y_1 = \frac12$ and $y_1 = \frac12 + NL^j/2$ lie in $\mathbb{Z} + \frac12$, because $NL^j/2 = L^{m_l+j}$.
2. At level i, the plane $m + \frac12$ of level $i+1$ becomes the plane $L(m+\frac12)$. Since L is odd, this plane lies in $\mathbb{Z}+\frac12$; moreover it is a face of the centred blocks. In bare units, therefore, both planes lie in $\mathbb{Z} + \frac12$.
3. Let $G_+ := G_- \circ \check\Theta$ with $\check\theta = \theta$ ($n_1 = 1$). Applied with A the right half, (I) shows that $\check\Gamma := E[G_+|U] - EG_+$ belongs to $\mathfrak{F}_+$ of A.3; and (K) gives $E[G_-|U] - EG_- = \theta\check\Gamma$.
4. By the hypothesis on f, also $F := \mathcal{O}_K(f) - \langle \mathcal{O}_K(f)\rangle$ belongs to $\mathfrak{F}_+$.
5. All functions here are real, and $\mathrm{Cov}(G_-, \mathcal{O}_K(f)) = E[(E[G_-|U] - EG_-)F] = \langle \theta\,\check\Gamma\, F\rangle$.
6. By A.3 (c), $\langle \theta\check\Gamma F\rangle^2 \le \langle \theta\check\Gamma\check\Gamma\rangle\langle \theta F F\rangle$.
7. By the Schwarz inequality and the invariance of the measure, $\langle \theta\check\Gamma\check\Gamma\rangle \le \langle \check\Gamma^2\rangle$; furthermore $\langle\check\Gamma^2\rangle = \mathrm{Var}\, E[G_+|U] \le \mathrm{Var}\, G_+ = \mathrm{Var}\, G_-$.
8. Item 3 of Step (K) gives $\langle \theta F F\rangle = S_2^T(\theta f, f)$. Combining items 5 to 8 we arrive at $\mathrm{Cov}^2 \le \mathrm{Var}(G_-) S_2^T(\theta f, f)$.

□

Lemma 11.4 (A decomposition of the conditional mean)**.** At every V of $\mathbb{T}_k$ put $T_f(V) := \frac{d}{dz}\log \rho_k^{[zf]}(V)$ at $z=0$. Then $T_f(V)$ is a version of $E[\mathcal{O}_K(f)|V^{(k)} = V]$ that is defined at every V, and $T_f = c_f + A_f(V) + W_f(V)$. Here $c_f = \sum_{i\le k} \frac{d}{dz}\big|_{z=0} V_i(z)$; next, $A_f = \sum_{X\in\mathcal{D}} 1_{X\approx}(\delta^a + \delta^h)(X,V) = S_f/\mathsf{b} + \mathrm{Rem}$, where S_f denotes the sum over all cubes of $1_{\mathrm{cube}\approx} s_x[\mathbb{D}_k[f]]$; and $W_f(V) = [\frac{d}{dz}\big|_{z=0}\Xi_{w(z)}(V)]/\Xi_{\bar w}(V)$. In these formulas $V_i(z)$ and $u_k(z) = (\cdot,\cdot,w(z))$ are those of Thm. 7.1 (h) along the run with the single source f and the variable z, and $\bar w = w(0)$. For $f = f_*$ we have $\sup|S_f| \le C_S p_0^2$ and

$$\mathsf{b}\sup|\mathrm{Rem}| \le 1.2124 N_1 [2C_3K_5E_a p_0^3 x^{-1/4} + 0.12 c_Q^{\mathrm{su}} K_5 (Lx)^{1/2} c_h(x)/(E_a p_0(Lx))],$$

with $N_1 = (D_0/4M + 2\check r(x) + 3)^3$; hence $\mathsf{b}\sup|A_f| \le C_S p_0^2 + 1$ under the condition (j1). ■

Proof. Step 1: the conditional mean as a logarithmic derivative. The density $\rho_k^{[zf]}(V)$ is the integral of Wilson's weight, times $e^{z\mathcal{O}_K(f)}$, times the k kernels. Since $\mathcal{O}_K(f)$ is bounded and $\rho_k > 0$ at every V (Thm. 7.1 (h)), we have $E[e^{z\mathcal{O}_K(f)}|V] = \rho_k^{[zf]}(V)/\rho_k(V)$, and differentiation at 0 is legitimate.

Step 2: the density in terms of members. According to Thm. 7.1 (h), whose density ρ_k^z is $\rho_k^{[zf]}$ for the single source f, we have $\rho_k^{[zf]} = \exp\{\sum V_i(z)\}\rho[u_k(z);\ell_k]$ at every field, and the values are holomorphic.

Step 3: the three parts. The formula for $\rho[u;\ell]$ (item 1 of §2.1, "Members") produces the three parts of T_f. Here $\Xi_{\bar w}(V) > 0$, as $\rho[\bar u_k] > 0$.

Step 4: the a-part. From Prop. 7.12 (J3) we have $\delta^a = s_x[\mathbb{D}_k[f]/\mathsf{b}] + \mathrm{rem}$, where

$$|\mathrm{rem}(X,\cdot)| \le 2u_3(x)K_5\mathsf{b}^{-1}c_a(x)e^{-\kappa_1(\#X-1)} = 2C_3K_5E_a p_0^3 x^{-1/4}\mathsf{b}^{-1}e^{-\kappa_1(\#X-1)}.$$

Step 5: the h-part. From Prop. 7.12 (J4) we have

$$|\delta^h(X,\cdot)| \le 0.12\,\mathrm{amp}(Lx)c_h(x)e^{-\kappa_1(\#X-1)}$$

, and

$$\mathsf{b}\,\mathrm{amp}(Lx) = c_Q^{\mathrm{su}}K_5(Lx)^{1/2}/(E_a p_0(Lx))$$

by item 2 of §7.3, (o5).

Step 6: the sets that occur. Prop. 7.12 (J2) shows that an X occurs only when $X^{+\check r}$ meets $S(k)$.

Step 7: the anchor cubes. The support of f_* is a sup-ball of radius $D_0/8$. It meets no more than $(D_0/4M+2)^3$ cubes, and there are therefore at most N_1 anchor cubes.

Step 8: connected sets through a cube. For a given cube, the sum of $e^{-\kappa(\#X-1)}$ over the connected X that contain it is at most $1/(1-26e\,e^{-6}) < 1.2124$ (item 1 of §2.1, "Cubes"; $\kappa \geq 6$).

Step 9: the quadratic member. The member $s_x[D]$ is given by the formula of entry 5 of item (0) of §6.1. On fields tame on cube$^{\approx}$, each F_p entering it satisfies $|F_p| \leq \varepsilon$, because the range does not exceed $\rho(x) \leq Mr(x) - 2$.

Step 10: the bound on S_f. We use (U2.1), the three orientations at a site, and (U2.3), by which only sites $y \in Y_{\mathrm{out}} = \{|y - y_c| \leq D_0/8 + \frac{1}{2}\}$ occur; this gives $|S_f| \leq \frac{x}{2}\#Y_{\mathrm{out}} K_5(3\mathsf{g}_\flat)^2\varepsilon^2 = C_S p_0^2$.

Step 11: a single source. Steps 4 to 6 use (J3), (J4) and (J2) for the single source f at the level k. Prop. 8.11 contains them, with the same bounds, as (J3°), (J4°) and (J2°) for one complex source of any support and for $0 \leq k \leq K$. □

Lemma 11.5 (A Peierls bound at the threshold $\varepsilon/2$)**.** (P°) In norm form: at $j \in J(\mathsf{b})$, on the board B_{j-1}, and for p based at 0,

$$\|E[1\{\phi_p \geq \varepsilon/2\}|U]\|_{\mathrm{ch}} \leq \exp\{-c_P p_0^2 + L^3(24\mathsf{N}_Z + 4.5\log x + 5 + 220M^{-3})\} \leq e^{-(c_P/2)p_0^2}.$$

Consequently, for every plaquette p of $\mathbb{T}_k$,

$$\nu_k(\phi_p \geq \varepsilon/2) \leq \exp\{-c_P p_0^2 + L^3(24\mathsf{N}_Z + 4.5\log x + 5 + 220M^{-3})\} \leq e^{-(c_P/2)p_0^2},$$

where $c_P = \min\{c_\Xi, \frac{1}{8}\}$. This bound does not depend on N, K or b. Hence

$$\nu_k(\text{some } \phi_p \geq \varepsilon/2 \text{ in } B') \leq 3(2R+4\rho+7)^3 e^{-(c_P/2)p_0^2} =: \nu_P.$$

■

Proof. Step 1: translation to the origin. Using Lem. 11.3 (K) for translations, we may take p based at 0.

Step 2: the board. The board B_{j-1} has cells of side L in units of level k, with $\lambda = L^{k+1}$ and $\mathsf{n}_\mu = 2L^{m_l+j-1}$. The cell 0 is centred at 0; as $L \geq 13$, the plaquette p is therefore inner.

Step 3: a function of one cell. According to Lem. 11.3 (I), $F := E[1\{\phi_p \geq \varepsilon/2\}|U]$ is a function of the cell 0.

Step 4: the chessboard estimate. By A.11, with the function 1 in all other cells (A.12 (b), (f)), we get $\nu_k(\phi_p \geq \varepsilon/2) = \langle F\rangle \leq \langle \mathrm{Fold}\, F\rangle^{1/\#\mathrm{cells}} = \|F\|_{\mathrm{ch}}$. Steps 5 to 8 bound the fold and thereby prove (P°).

Step 5: the copies. The copies are $F \circ \check{\Theta}_c$, with $\check{\Theta}_c$ as in the data of Lem. 11.7, but with the side L in place of h; the map $\check{\theta}_c^{-1}$ is of the kind allowed in (K). From (K) and (I) we get $\langle \mathrm{Fold}\, F\rangle = \nu_k(\phi_{\check{\theta}_c^{-1}p} \geq \varepsilon/2 \text{ in every cell } c)$. The event is the same for either orientation of the mirrored plaquette $\check{\theta}_c^{-1}p$, since $\mathrm{ang}\, g^{-1} = \mathrm{ang}\, g$.

Step 6: the numerator, pointwise. To begin with, $\mathrm{En}_\ell \geq \sum_p S_\varepsilon(\phi_p)$ (item 2 of §5.5, "Energy"). Next, writing q for the number of cubes, $|\sum 1(E^a + E^h)| \leq 1.2124q(c_a + c_h) \leq 0.61q$. Further, since $\bar{w}$ has the strong cap, and since the members of any family are pairwise disjoint and together cover $D(V)$, we obtain

$$|\Xi_w| \leq \exp\{-c_\Xi x\varepsilon \sum_{D(V)} \phi_p\} \prod_{\text{cubes}} (1 + \sum_{X \text{ holding the cube}} e^{-\kappa(|X|_c - 1)}) \leq \exp\{-c_\Xi x\varepsilon \sum_{D(V)} \phi_p\} e^{217.6q},$$

where $\sum_{n\geq 1}(52e)^n e^{-6(n-1)} = 52e/(1-52e\,e^{-6}) < 217.6$. Lastly, the marked plaquette of a cell contributes the factor $e^{-x(\varepsilon/2)^2/2} = e^{-p_0^2/8}$ if $\phi \leq \varepsilon$, and the factor $e^{-c_\Xi x\varepsilon\phi} \leq e^{-c_\Xi p_0^2}$ if $\phi > \varepsilon$.

Step 7: the denominator. We restrict the integral to the ball $\{\mathrm{ang}\, V(b) \leq x^{-1/2} \text{ for all } b\}$. On this ball $\phi_p \leq 4x^{-1/2} \leq \varepsilon$ by (j7) (vi), so that $D(V)$ is empty and $\Xi = 1$. The total energy $x\mathrm{En}_\ell = \frac{x}{2}\langle F, \mathsf{B}^V F\rangle$ is at most $\frac{x}{2}\mathsf{N}_Z \cdot 3 \cdot 16/x = 24\mathsf{N}_Z$ per site (Def. 2.48, second paragraph; item 3 of §2.1,

(O3)). The members are bounded below by $-0.61q$. Finally, $\mathrm{Haar}\{\mathrm{ang} \le \check\delta\} = \frac{2}{\pi}\int_0^{\check\delta} \sin^2 \ge \check\delta^3/5$ for $\check\delta^2 \le 0.28$; since a site carries three bonds, the restriction costs $3(1.5\log x + \log 5) \le 4.5\log x + 4.83$ per site.

Step 8: conclusion. The factor e^{V} cancels. A cell contains L^3 sites, and $218.9q$ does not exceed $220M^{-3}$ per site. This proves the first inequality of (P°); the second one is (j7) (i). □

Lemma 11.6 (Extension of a field that is tame on a box)**.** Let V be a field with values in G, defined on the bonds of the box $\{r \le r_B\}$ and tame there. Suppose that $r_B \ge 7$, $C_{\mathrm{ext}} r_B^2 \varepsilon \le 1$, and the side of $\mathbb{T}_k$ is at least $10r_B + 2$. Then V is the restriction of a field $\tilde V$ on $\mathbb{T}_k$ that is globally tame. ■

Proof. Step 1: the comb gauge. Let $g(y) = V(\Gamma(0,y))$ and $V^g(b) = g(y)V(b)g(y')^{-1} =: e^{iA_b}$. Then V_b^g is the holonomy of the loop $\Gamma(0,y).b.\Gamma(0,y')^{-1}$. When b points along e_3, this loop is trivial; when b points along e_2, the loop bounds a strip of $|y_3|$ plaquettes; and when b points along e_1, the loop encloses two strips, one of $|y_2|$ and one of $|y_3|$ plaquettes. In each case no more than $2r_B$ plaquettes of the box are involved. This holonomy is a product of conjugates of the holonomies of plaquettes; as $\mathrm{ang}(gh) \le \mathrm{ang}\, g + \mathrm{ang}\, h$ and ang is a class function, it follows that $|A_b| \le a := 2r_B\varepsilon$. Since $C_{\mathrm{ext}} = 32$ and $r_B \ge 7$, the assumed bound gives $a \le 1/(16r_B) < 0.01$. The prescribed box radius satisfies $r_B \ge 7$.

Step 2: the continuation. Let pr be the map that replaces each coordinate of a site by the nearest value that this coordinate takes in the box; for a site y let $n(y)$ be its sup distance from the box; and let $s(n) := \max(0, 1 - n/(4r_B))$. If b is a bond outside the box, its image $\mathrm{pr}\, b$ is a bond or a single point; the former happens when pr leaves the coordinate along b unchanged, and then b has the same n at both ends. We put $\tilde V_b := \exp(is(n)A_{\mathrm{pr}\, b})$ in the first case and $\tilde V_b := 1$ in the second. All bonds with $n \ge 4r_B$ then carry the value 1, and the side, being at least $2(5r_B) + 2$, leaves room for this on the torus.

Step 3: plaquettes outside the box. The image of such a plaquette under pr is a plaquette, a bond or a point.

(i) A plaquette. Here n is constant. Write its four oriented link logarithms as $A_1, \dots, A_4$, with $|A_i| \le a$, and put

$$F(sA) = -i\log(e^{isA_1}\cdots e^{isA_4}) = \sum_{q\ge 1} s^q F_q.$$

We verify $|d^2F(sA)/ds^2| \le 14a^2$ on $0 \le s \le 1$. The six commutators in degree two give $|F_2| \le 6a^2$. For completeness, after m factors write the first three coefficients as S_1, S_2, S_3. Multiplication by e^{isA}, followed by expansion of the logarithm through degree three, gives the bounds

$$|S_2^{\mathrm{new}}| \le |S_2| + ma^2, \qquad |S_3^{\mathrm{new}}| \le |S_3| + a|S_2| + \tfrac{1}{3}(m^2+m)a^3,$$

using $|S_1| \le ma$ and the bracket constant 2. The successive bounds for $|S_2|/a^2$ are $1, 3, 6$, and those for $|S_3|/a^3$ are $2/3, 11/3, 32/3$. Thus $|F_3| \le (32/3)a^3$. More generally, put $S = \sum_i |A_i| \le 4a$. The coefficients of the logarithm are bounded by those of $H(Ss)$, where $H(u) = -\log(2 - e^u)$: the product minus 1 is majorized coefficient by coefficient by $e^{Ss} - 1$, and then the logarithm series is absolutely majorized by $H(Ss)$. In particular, the remaining coefficients are bounded by those of $H(4as)$. Since all coefficients of H are nonnegative,

$$\left|\frac{d^2}{ds^2}F(sA)\right| \le 12a^2 + 64a^3 + 16a^2\big[H''(4a) - 2 - 24a\big] < 13.01a^2 < 14a^2.$$

Here $H''(u) = 2e^u/(2-e^u)^2$, and for $a \le 0.01$ the coefficient on the right is at most $12 + 0.64 + 16[H''(0.04) - 2 - 0.24] < 13.01$. Now let $f(s) = F(sA) - sF(A)$. It has

$f(0) = f(1) = 0$; the Green function for the second derivative on $[0,1]$ therefore gives $|f(s)| \le 7s(1-s)a^2$. Consequently

$$|F(sA)| \le s\varepsilon + (1-s)28r_B^2\varepsilon^2 \le \varepsilon.$$

(ii) A bond b'. Here the holonomy is $\exp(i(s-s')A_{b'})$ with $|n-n'| \le 1$, and its angle is at most $a/(4r_B) = \varepsilon/2$.

(iii) A point. Here the holonomy is 1.

Step 4: back to the original gauge. We undo the comb gauge: the transformation is g^{-1} at the sites of the box and 1 at all other sites. Tameness is gauge invariant, and on the bonds of the box the resulting field restricts to V. □

Lemma 11.7 (Mirrors and a comparison of sums)**.** At $j \in J(\mathsf{b})$, on the board B_{j-i_h}, let $\check\theta_c$ be the map of the fold that takes the cell c onto the cell 0; in each coordinate it is given by $y_\mu \to y_\mu - c_\mu h$ when c_μ is even and by $y_\mu \to c_\mu h - y_\mu$ when c_μ is odd. Let $\check\Theta_c V := V \circ \check\theta_c^{-1}$, which is the map $\check\Theta$ of Lem. 11.3 (K) applied with $\check\theta = \check\theta_c^{-1}$. For a bounded function f let $f_c := f \circ \check\theta_c$, and let $\mathrm{Box}_c := \check\theta_c^{-1}\mathrm{Box}$, with B'_c and $B_{R,c}$ defined in the same manner. Accordingly, the copy $\Phi \circ \check\Theta_c$ of a function Φ of the cell 0 is a function of the cell c, and Box_c is contained in the cell c. If the support of f lies in the cell 0, as it does for $f = f_*$, then the support of f_c lies in the cell c. Let $\Lambda := -x\mathrm{En}_\ell + \sum 1(\bar E^a + \bar E^h)$ and $\Lambda_c := \Lambda \circ \check\Theta_c$. With an upper index tr, a sum over X is restricted to the X with $\#X \le n_0$, and the index long refers to all other X. Finally, let τ_t be the flow of Lem. 11.14 and $\tau^c := \check\Theta_c^{-1}\tau\,\check\Theta_c$.

(a) $\rho_k \circ \check\Theta_c = \rho_k$, and $T_f(\check\Theta_c V) = T_{f_c}(V)$.

(b) At globally tame V, $\Lambda_c = \Lambda$ and $c_f + A_f(\check\Theta_c V) = c_{f_c} + A_{f_c}(V)$.

(c) Let $f = f_*$. Put $\Delta_c(V) := c_{f_c} + A^{\mathrm{tr}}_{f_c}(V) - c_f - A^{\mathrm{tr}}_f(\check\Theta_c V)$ and $d_c(V) := (\Lambda^{\mathrm{tr}} \circ \tau_t^c - \Lambda^{\mathrm{tr}}) - (\Lambda_c^{\mathrm{tr}} \circ \tau_t^c - \Lambda_c^{\mathrm{tr}})$. Each of them depends on the bonds of Box_c alone. On fields that are tame on Box_c, and in the case of d_c satisfy moreover $\phi_p \le 3\varepsilon/4 + 4t$ on B'_c, we have $\mathsf{b}|\Delta_c|$, $|d_c| \le \mathrm{ex}$. At every field, $\mathsf{b}|\Delta_c| \le 4(C_S p_0^2 + 1)$. On fields with $\phi_p \le 3\varepsilon/4 + 4t$ on B'_c, $|d_c| \le 1$.

■

Proof. Step (a)*: the mirrors as maps of the kind* (K). The inverse map acts by $y_\mu \to y_\mu + c_\mu h$ if c_μ is even and by $y_\mu \to c_\mu h - y_\mu$ if c_μ is odd; so $\check\theta_c^{-1}$ is of the kind (K), with $n_\mu = c_\mu h$. Lem. 11.3 (K), applied with $\check\theta = \check\theta_c^{-1}$, gives (a), because $f \circ \check\theta^{-1} = f \circ \check\theta_c = f_c$.

Step (b)*: globally tame fields.*

1. If $D(V)$ is empty, then the empty family is the only family in $\Xi_{w(z)}$ (item 1 of §2.1, "Members"). Hence $\Xi_{w(z)}(V) = 1$ for all z, and $W_f(V) = 0$.
2. It follows that $\Lambda = \log \rho_k - \mathsf{V}$ and $c_f + A_f = T_f$ at such fields.
3. The field $\check\Theta_c V$ is globally tame as well, and (a) now gives (b).

Step (c)*, dependence: the bonds that enter.*

1. When a short X contributes to A^{tr}_f, the set $X^{+\check r}$ meets $\mathrm{supp} f$; and this support lies in $r \le 0.84D_0 + 2 \le R/2$.
2. When it contributes to $\Lambda^{\mathrm{tr}} \circ \tau - \Lambda^{\mathrm{tr}}$, the set $X^{\approx}$ meets B_R^{+1}.
3. The term depends on $X^{\approx}$, and every point of $X^{\approx}$ has r at most $R+1+M(n_0+2r(x)+4) \le R+1+h/16+R \le r_B$; the three inequalities used are $Mn_0 \le h/16$, $M(2r+4) \le R$ from (j7) (ii), and $R \le h/16$.
4. The moved terms of En_ℓ depend on B', which lies in Box since $2\rho + 3 \le R$.
5. Under a mirror, the least corner of a cube or a plaquette moves by one unit; this is where the $+1$ in r_B comes from.

Step (c)*, smallness: fields tame on the box.*

1. By Lem. 11.6 the restriction of V to Box_c extends to a field $\tilde V$; the side obeys $NL^j \ge 2Lh \ge 10r_B + 2$ by (j3).
2. Then $\Delta_c(V) = \Delta_c(\tilde V)$, and by (b) this equals minus the long parts.
3. Because s_x lives on single cubes, the long parts consist of rem and δ^h only.
4. They are bounded by
$$2N_1 \sum_{n>n_0} (26e)^{n-1} e^{-\kappa_1(n-1)} [2C_3 K_5 E_a p_0^3 x^{-1/4} + 0.12 c_Q^{\mathrm{su}} K_5 (Lx)^{1/2} c_h(x)/(E_a p_0(Lx))]/\mathsf{b}$$
, where the bracket is that of the display of Lem. 11.4.
5. As $\kappa_1 - \log(26e) \ge 2.7419$ and $n_0 \ge 4L\log x - 1$, the bound is $x^{-10.9L} e^{2.75}$ times $2N_1[\,\cdot\,]/\mathsf{b}$, with the same bracket, and this is at most ex/b: the number $2N_1[\,\cdot\,]$ is at most 1 by (E1) $\le c_F/12$.
6. Turning to d_c: the flow τ_t^c moves only plaquettes touching $B_{R,c}$, each by at most $4t \sup|w| \le 4t$, where w is the form fixed for Lem. 11.14 and not the third member. All sites of a moved plaquette lie in B_c', where $\phi_p(\tilde V) = \phi_p(V) \le 3\varepsilon/4 + 4t$ by the hypothesis of (c), since $B_c' \subset \mathrm{Box}_c$; at $\tau_t^c \tilde V$ such a plaquette therefore obeys $\phi_p \le 3\varepsilon/4 + 8t \le \varepsilon$. The last inequality holds because $32x^{-2} \le p_0 x^{-1/2}$, by $t = x^{-2}$ and (j7) (vi), which gives $p_0 \ge 4$ and $\log x \ge 3$. A plaquette that is not moved keeps $\phi_p(\tilde V) \le \varepsilon$, so $\tau_t^c \tilde V$ is globally tame.
7. Applying (b) both at $\tilde V$ and at $\tau_t^c \tilde V$, we see that the untruncated differences cancel. There remain four long sums, each bounded by $(2R/M + 2r + 3)^3 (c_a + c_h) x^{-10.9L} e^{2.75}$. Since $2R/M + 2r + 3 \le h < x^{1/4}$ (Lem. 11.22 (b)), the four together are at most ex.

Step (c)*, rough bounds: all fields.*

1. At $V = 1$, (b) gives $|c_{f_c} - c_f| \le 2\sup|A|$, and so $\mathsf{b}|\Delta_c| \le 4\mathsf{b}\sup|A|$.
2. For d_c, let $\Lambda^a := -x\mathrm{En}_\ell + \sum 1\bar E^a$, so that $\Lambda = \Lambda^a + \sum 1\bar E^h$. We split both differences, each into the a-part coming from Λ^a and the h-part coming from $\sum 1\bar E^h$. The index tr acts on the sums over X alone.
3. We estimate the two a-parts along the flow segments. Their moving plaquettes have angles at most $3\varepsilon/4 + 8t < \varepsilon$: indeed $32x^{-2} < p_0 x^{-1/2}$ follows from $p_0 \ge 4$ and $\log x \ge 3$. The masks of every varying energy row are therefore 1, as in Step 2 of the proof of Lem. 11.14.

   Write $D = D_w^\alpha$ in the cell's translated coordinates, and use the sums W_0, W_1, W_2 of Lem. 11.14. Steps 1–3 of the proof of Lem. 11.16, which do not use the Green-function identity, give
$$|xD\mathrm{En}_\ell| \le x\varepsilon N_{B0} W_1 + C_2 p_0^2 R.$$
   The leading term is bounded by $\varepsilon N_{B0} W_1$ because the absolute row sums of B are at most N_{B0}. The remainder includes both the curvature error and the derivatives of the row transports.

   For the other difference, use the reflected oriented bonds and plaquettes, with the root translated to the origin. Reversing a bond turns a left-invariant derivative into a right-invariant one; it preserves its speed $|w_b|$ and the signed plaquette sum $\mathrm{d}w$. We bound the resulting conjugations directly. In the comb gauge each link of a plaquette based at radius r has angle at most $2(r+2)\varepsilon$. A plaquette prefix uses at most four links, and conversion of a right derivative to a left derivative adds at most one endpoint link. Thus each derivative axis differs from e_α by at most $20(r+2)\varepsilon$. The logarithm differential of Step 1 of the proof of Lem. 11.16, with $\varepsilon \le 0.01$, gives the safe bound
$$|DF_p - (\mathrm{d}w)_p e_\alpha| \le \varepsilon\{40(r+2)|w|_p + |\mathrm{d}w|_p\}.$$
   The scalar row matrix is a signed reindexing of B, so its absolute row-sum bound remains N_{B0}. The displacement sums use the pulled-back oriented plaquettes and their distinguished endpoints, so their path lengths and displacement counts are unchanged. A varying curvature plaquette lies within radius $R+1$, so a retained row has center at radius at most

$R+\rho+1$; its paths, plaquettes and mask lie within $R+2\rho+3$, hence in B'. If the least base site changes under reflection, its sup radius changes by at most one. The added unit in $r+2$ and this possible change are both covered by

$$W_2 \text{ replaced by } W_2+8W_0, \qquad V_1 \text{ replaced by } V_1+W_1, \qquad V_1=\sum_p (r_p+1)|(\mathrm{d}w_R)_p|.$$

Here $\sum_p |w|_p = 4W_0$. In the notation of Step 3 there, the row remainder is thus at most

$$\begin{aligned} &p_0^2\big[N_{B0}\{40(W_2+8W_0)+W_1\}+18N_{B1}W_0 \\ &\quad +24N_{B1}(V_1+W_1)+72N_{B2}W_1\big] \le 400(N_{B0}+N_{B1}+N_{B2})r_1^4p_0^2R. \end{aligned}$$

The bare plaquette remainder is at most $12000r_1^3p_0^2R$. Indeed $W_0 \le r_1^3R$ for $R \ge 10^6$, $W_2 \le 289r_1^3R$, and $W_1 \le 400r_1^4 \le r_1^3R$. The two remainders together have coefficient at most

$$12000r_1^3+400(N_{B0}+N_{B1}+N_{B2})r_1^4 < 10^{665} < C_2 = 10^{670}.$$

Thus the same energy bound holds for the reflected orientation.

The first Cauchy estimate of Lem. 11.14 also holds on these slightly longer segments. In Step 7.1 of its proof, replace the starting plaquette bound by ε; the domain estimate becomes $\varepsilon + e^{1/2}\varepsilon/4 < 2\varepsilon$. Its other domain estimates are unchanged. For the reflected orientation use $\delta_p = |\mathrm{d}w|_p + 40(r_p+2)\varepsilon|w|_p$ and the same definition of s_X. The frames remain constant along the conjugated flow, as follows by conjugating Step 7.0. The enlarged weighted correction is less than 9, since

$$W_0 \le 78r_1^3(1+\log R) \le r_1^3R, \qquad \varepsilon \le (288R^2)^{-1}, \qquad R \ge 6r_1^3.$$

Indeed $5\varepsilon W_2 \le 1$ gives $40\varepsilon W_2 \le 8$, while $320\varepsilon W_0 \le 320/1728 < 1$. Hence $W_1 + 40\varepsilon(W_2 + 8W_0) \le 400r_1^4 + 9 \le 401r_1^4$, retaining the constants $1605r_1^4$ and $C_a(x)$ of that proof. Restricting the sum to short X only decreases its positive majorant.

For either a-part, we therefore have

$$\sup|D\Lambda^{a,\mathrm{tr}}| \le x\varepsilon N_{B0}W_1 + (C_2R+C_a(x))p_0^2 \le 2N_{B0}W_*p_0x^{1/2}.$$

Indeed $W_1 \le 0.4W_*$, while (E3)$\le c_F/12$ gives $(C_2R+C_a)p_0^2 \le c_Fx^{1/2}/(60C_Sp_0^3)$. Multiplying by t bounds the corresponding difference.

4. The h-part is at most $2.5(2R/M+2r+3)^3c_h$.
5. By (E6) $\le c_F/12$, both parts are at most $\frac14$, because $C_6 \ge N_{B0}W_*$.

□

Lemma 11.8 (Jets of many sources with a free last ratio)**.** In the setting of Def. 8.6 with Q sources, denoted there by g_t and here by f_i, each with sup norm $\|f_i\|_\infty \le 1$, let $1 \le k \le K-1$, let ϱ be any number ≥ 1 (the lemma is used with $\varrho = x^{1/2}p_0^6$), and let $\mathrm{sep}_k \ge 8\log(4\varrho L^{1/2})$. Here $\varrho L^{1/2} \ge 75$, because $L > 1.2\cdot 10^{13}$.

(A) In the strong unit of each of the three members at the rate $\kappa+\frac14$,

$$\sup_{(X,V)} \sum_{|S|\ge 2} (\varrho\,\mathfrak{r}_k^\circ)^{|S|}|\delta_S(k)(X,V)| \le 0.03.$$

(B) For each i,

$$\sum_{S\ni i,|S|\ge 2} (\varrho\,\mathfrak{r}_k^\circ)^{|S|} \sum_{k'<k}\sum_Y |\partial_S v_{k'}(Y;u_{k'}^{\mathrm{pol}})(0)| \le E_{\mathrm{vac}},$$

where $E_{\mathrm{vac}} := \sum_{m\ge 1} N_1(k-m)2C_gc_a(x_{k-m})e^{1/4}\exp\{-(\kappa-\frac14)(L^{m-1}\mathrm{sep}_k+1)\}$.

Neither of the two bounds depends on Q, N or K. ■

Proof. Apply Prop. 8.11 (J5°) (d) with $R_{k'} = \mathfrak{r}^\circ_{k'}$ for $k' < k$ and $R_k = \varrho\mathfrak{r}^\circ_k$. Its rates are

$$\varkappa_{k'} = \kappa + \tfrac{1}{4} + \frac{2\log(4\varrho L^{1/2})}{\mathrm{sep}_k} + \sum_{k'\le i<k-1} \frac{2\log(4L^{1/2})}{\mathrm{sep}_{i+1}} \quad (k'<k), \qquad \varkappa_k = \kappa + \tfrac{1}{4}.$$

The assumed separation is precisely the hypothesis of that clause, which proves admissibility of these data and gives (A). The additional $1/4$ in the rates is retained: in (J8°) (a′), the loss $\lambda \le 1/4$ leaves $\varkappa_{k'} - \lambda \ge \kappa$. That clause therefore gives (B) with the stated E_{vac}, since $N_1(k-m) = N_1^{[K-k+m]}$ and $x_{k-m} = L^m x_k$. Neither result imposes an upper bound on the depth or on ϱ.

Step: the size of E_{vac}. Throughout this step we take the sources f_c of Lem. 11.10, $\check\delta = (h/2)L^{-j}$, $\check\rho = (D_0/8)L^{-j}$, $k = K-j$ and $x = x_k$, and (X) is in force. We have $N_1(k-m) \le ((D_0/4M)L^{m-1} + 3 + 2\check r(L^{m-1}x))^3$ and $\mathrm{sep}_k \ge \lfloor h/2M\rfloor - 2\check r(x) - 2 \ge 31L\log x$. The latter holds because (M2) gives $r(x) \le 1 + \log x/768$, hence $2\check r(x) + 2 \le 10 + \log x/384$, while $h/2M \ge 32L\log x$ (Lem. 11.22 (b)). Moreover $C_g = 263L^3$ (item 0 (1) of §7.2), and $x^{8L} \ge L^{8L}$. As (M2) is a condition on the letters alone, it gives $r(y) \le 1 + \log y/768$ at every $y > 1$; at $y = L^{m-1}x$ this yields $3 + 2\check r(L^{m-1}x) \le 11 + (\log x + (m-1)\log L)/384$. We have $(\kappa - \frac14)\cdot 31 \ge 178.25$, and $c_a(x_{k-m}) = c_a(L^m x) \le 1/32$ by (X17) (an entry of (O13), item 1 of §5.5). Hence the m-th term of E_{vac} is at most

$$((D_0/4M)L^{m-1} + 11 + (\log x + (m-1)\log L)/384)^3 \cdot 526L^3 e^{1/4} x^{-178.25L^m}.$$

By $x^{8L} \ge L^{8L}$, the term with $m = 1$ is at most $\frac12 x^{-170L}$, and the terms with $m \ge 2$ form a smaller geometric tail. Summing over $m \ge 1$, we obtain $E_{\mathrm{vac}} \le x^{-170L} \le \mathrm{ex}$. □

Lemma 11.9 (A positive majorant and its integral)**.** Let $\hat w(X,V) \ge 0$ be Borel and gauge invariant, depending on V only through its values on $X^{\approx\approx}$. Suppose that $\hat w = 0$ unless X holds a plaquette of $D(V)$, and that $\hat w \le 2.6e^{-\kappa(|X|_c - 1)}$. Let $\hat u := (\bar E^a_k, \bar E^h_k, \hat w)$. Then

$$0 \le \int \rho[\hat u; \ell_k]dV \le (1 + C_{\mathrm{vol}}(N, \mathsf b)) \int \rho[\bar u_k; \ell_k]dV,$$

and $(1 + C_{\mathrm{vol}})^{1/Q} \le \exp\{c_{\mathrm{vol}}(\mathsf b)h^3L^{-3j}\} \le e$ for $Q = (NL^j/h)^3$, at $j \in J(\mathsf b)$. ■

Proof. Step 1: the class of $\hat u$. The triple $\hat u$ is real and belongs to $M_{2.6}(x;\kappa,c_h)$: each of its a- and h-members has a single cap at κ_1, and therefore one at κ as well (item 0 of §6.2, "Pairs"). As $\hat w \ge 0$, also $\rho[\hat u] \ge 0$. Membership in M_A does not depend on any translation covariance of w (item 1 of §2.1, "Members").

Step 2: the data along the run. For $k \le i < K$, Thm. 7.1 (b) gives (SD) at $(x_i;\ell_i)$; moreover $x_i \ge L\mathsf b \ge Lx_s \ge x_9^U$ (item (1) (O13) of §6.1).

Step 3: the chain of members. Let $\hat u_k := \hat u$, and define $\hat u_{i+1} := F_{\ell_i}(\hat u_i)$ recursively. From (9a), applied at every V', together with $\int \rho' = \int \rho$ (item 3 of §5.5, (R9), and item 2 of the same section, "The step"), we get $\int \rho[\hat u_i;\ell_i] = e^{\mathsf V'(\hat u_i)} \int \rho[\hat u_{i+1};\ell_{i+1}]$.

Step 4: the classes along the chain. Thm. 7.9 (i), applied with $B = 2.6$ and $p = (\kappa, c_h)$, gives $\hat u_{k+1} \in M_1(x_{k+1};p)$ with the strong cap. At each later level i the same theorem, applied to $\hat u_i$, gives $\hat u_{i+1} \in M_1(x_{i+1};p)$ with the strong cap.

Step 5: the vacuum numbers. According to (f1) (Thm. 6.7), neither E'^a nor $\mathsf V'$ depends on E^h or on w. The a-members of $\hat u_k$ and $\bar u_k$ coincide; by induction they coincide at every i, and so $\mathsf V'(\hat u_i) = \mathsf V'(\bar u_i)$.

Step 6: exact cancellation. Since $\bar u_{i+1} = F_{\ell_i}(\bar u_i)$ (Thm. 7.1 (h)), the vacuum factors cancel exactly, and

$$\int \rho[\hat u;\ell_k] / \int \rho[\bar u_k;\ell_k] = \int \rho[\hat u_K;\ell_K] / \int \rho[\bar u_K;\ell_K].$$

Step 7: the last level. For every $K \ge 1$ the data (B) of Thm. 7.27 are satisfied for $x = \mathsf b$ and the pair (κ, c_h), by Step 2 of the proof of Thm. 8.2, and $|\ell_{K,y}(p)| \le K_* e^{-\lambda_*|y-p|}$ by the same step, "The last integral". Here $u_1 = u_3 = \bar u_K$, while $u_2 = \hat u_K$ arises as an output of Thm. 5.15 at

$A \le 2.6 \le 6$: since $j \ge 1$, we have $k = K - j < K$, so the chain of Step 3 has at least one step, and $\hat{u}_K = F_{\ell_{K-1}}(\hat{u}_{K-1})$ with $\hat{u}_{K-1} \in M_A$ and $A \le 2.6$ (Steps 3 and 4). We are in case (ii), with $a_w = 1$ and $a_E \le 2$ (Thm. 7.27 (b^C) (ii)). Moreover $\mathsf{D}_{23} \le 2$. The sentence of (c^C) before (S) in Thm. 7.27 then yields $\int |\rho[\hat{u}_K] - \rho[\bar{u}_K]| \le \frac{1}{2} C_{\mathrm{vol}} \cdot 2 \cdot I_1$.

Step 8: the root. We have

$$\log(1 + C_{\mathrm{vol}}) \le \log(2C_{\mathrm{vol}}) = \log(8q_N) + 220 q_N + [\frac{3}{2} \mathsf{N}_Z p_0(\mathsf{b})^2 + 3 \log(384/\varepsilon(\mathsf{b})^3)] N^3 \le c_{\mathrm{vol}}(\mathsf{b}) N^3,$$

since $q_N = (N/M)^3$ (item 1 of §7.5: q_N; C_{vol}) and $\log(8q) \le 2.08q \le 5q$ for $q \ge 1$. It remains to divide by Q and to apply (j2). □

Lemma 11.10 (The part of the source carried by the weights)**.** In the setting of Thm. 11.1, at $j \in J(\mathsf{b})$, the following hold. (W°) In norm form: on the board B_{j-i_h} and for every constant $g \in [0, (9/8)p_0^2]$,

$$\|E[(G_- - g)(\mathcal{O}_K(f_*) - c_f - A_f^{\mathrm{tr}}(V)) \mid U]\|_{\mathrm{ch}} \le (9/8)p_0^2 \cdot 2e^{2.2}/r_z.$$

As a consequence, for every such g, and so in particular for $g = EG_-$,

$$\mathsf{b}\, |E[(G_- - g)W_{f_*}]| \le (9/8)p_0^2 \cdot 2e^{2.2}\mathsf{b}/r_z + (9/4)p_0^2\, \mathrm{ex} \le (\mathrm{E5}),$$

where we define

$$(\mathrm{E5}) := 5.81\, c_Q^{\mathrm{su}} K_5 L^{1/2}/(E_a p_0^4 p_0(Lx)) + (9/4)p_0^2\, \mathrm{ex}.$$

Here $\mathsf{b}/r_z = c_Q^{\mathrm{su}} K_5 (Lx)^{1/2}/(3.5 \varrho E_a p_0(Lx))$. The bound is independent of N and of K. ■

Proof. We write f for f_*.

Step 1: the cell function. We work on the board B_{j-i_h}, which has $Q = (NL^j/h)^3$ cells. The box Box lies in the cell 0 and is centred in it ($r_B \le (h-1)/2$). We set $\Phi := (G_- - g)(\mathcal{O}_K(f) - c_f - A_f^{\mathrm{tr}}(V))$; it is a bounded Borel function of the bonds of U and of $V^{(k)}$ that belong to this cell. Lem. 11.4 gives $E\Phi = E[(G_- - g)(W_f + A_f^{\mathrm{long}})]$, because by that lemma $T_f = c_f + A_f + W_f$ is a version of $E[\mathcal{O}_K(f)|V]$, and because G_- is a function of V; the indices tr and long are those of the statement of Lem. 11.7. The bound $\mathsf{b} \sup |A^{\mathrm{long}}| \le \mathrm{ex}$ follows from the sizes in Lem. 11.4 and from items 3 to 5 of Step (c), smallness, of the proof of Lem. 11.7. Indeed, A^{long} is the long part of Rem; as s_x lives on single cubes, only rem and δ^h have long parts, and

$$\mathsf{b} \sup |A^{\mathrm{long}}| \le N_1 \sum_{n > n_0} (26e)^{n-1} e^{-\kappa_1 (n-1)}\, \mathsf{b}\, (\text{sizes of Lem. 11.4}) \le \mathrm{ex},$$

where N_1 counts the anchors, $n_0 \ge 4L \log x - 1$, and the prefactor is at most 1 by (E1) $\le c_F/12$.

Step 2: the chessboard estimate. By Lem. 11.3 (I), $E[\Phi|U]$ is a function of the cell. Prop. A.11, with the function 1 in all other cells, yields

$$|E\Phi|^Q \le \langle \mathrm{Fold}\, E[\Phi|U] \rangle = \|E[\Phi|U]\|_{\mathrm{ch}}^Q.$$

Steps 3 to 8 bound the fold, and in doing so they prove (W°). By (K) and (I) of Lem. 11.3, the fold equals $E \prod_c \Phi \circ \check{\Theta}_c$.

Step 3: the copies. For every cell c we have $\Phi \circ \check{\Theta}_c = (G_c - g)(\mathcal{O}_K(f_c) - a_c(V))$, where $G_c = G_- \circ \check{\Theta}_c$ and

$$a_c = c_f + A_f^{\mathrm{tr}}(\check{\Theta}_c V) = c_{f_c} + A_{f_c}^{\mathrm{tr}}(V) - \Delta_c(V).$$

The functions f_c are themselves Wilson sources, with $\sup |f_c| \le 1$, and they enter as $\mathcal{O}_K(f_c)$; each of them is supported within D_0 of a point, and the supports of two distinct ones are at least $h - 2D_0 \ge h/2$ apart. Hence the sources f_c satisfy the conditions of the setting of Def. 8.6, with $\check{\delta} = (h/2)L^{-j}$ and $\check{\rho} = (D_0/8)L^{-j}$, and $\mathrm{sep}_k \ge \lfloor h/2M \rfloor - 2\check{r}(x) - 2$. The first line of (j4) then gives the hypothesis of Lem. 11.8.

Step 4: conditioning. Each G_c is a function of V. Moreover

$$E[\prod_c (\mathcal{O}_K(f_c) - a_c) \mid V]\, \rho_k(V) = \partial_1 \cdots \partial_Q [e^{-\sum z_c a_c(V)} \rho_k^z(V)](0).$$

We apply Prop. 8.11 (J7°) (iii) at the level k and at this V, with $\gamma_c^{(k)} = c_{f_c}$, modulo z_c^2 (the congruence of item 1 of §7.5); it shows that the bracket equals

$$e^{\mathsf{V}+\Lambda(V)} \exp\{W_2(z) + \mathrm{Mix}(z,V)\} \prod_c (1 + z_c(\Delta_c + A_{f_c}^{\mathrm{long}})(V))\, \Xi_{w^{\mathrm{pol}}(z)}(V).$$

Here W_2 gathers the vacuum jets with $|S| \geq 2$ of the levels $k' < k$, and $\mathrm{Mix} = \sum_X 1_{X^{\approx}} \sum_{|S|\geq 2} z^S (\delta_S^a + \delta_S^h)$. The terms of a single source, $c_{f_c} + A_{f_c}^{\mathrm{tr}}$, cancel identically against a_c.

Step 5: the majorant. The function Ξ_w is a polynomial in the values of w, and its coefficients are non-negative. Hence the modulus of the coefficient of $z_1 \cdots z_Q$ is at most r_z^{-Q} times the value at $z_c = r_z$ of the majorant series, in which every coefficient is replaced by its modulus. For the majorants in the exponent, we have $\hat{W}_2 \leq (Q/2) E_{\mathrm{vac}} \leq Q\,\mathrm{ex}$: the first inequality is Lem. 11.8 (B), and the second follows from the step on the size of E_{vac} in the proof of Lem. 11.8, which gives $E_{\mathrm{vac}} \leq x^{-170L} \leq \mathrm{ex}$. Next, Lem. 11.8 (A) with Prop. 8.11 (J2°) gives

$$\widehat{\mathrm{Mix}} \leq Q N_1\, 0.03 (c_a + c_h) \sum_{n \geq \mathrm{sep}_k} (26e)^{n-1} e^{-(\kappa+1/4)(n-1)} \leq Q\,\mathrm{ex}.$$

We put

$$\hat{w} := [\,|\bar{w}| + \sum_S r_z^{|S|} |\delta_S^w|\,]\, 1\{X \text{ holds a plaquette of } D(V)\}.$$

Prop. 8.11 (J5°) (a′), taken value by value, gives $\sum_c r_z |\delta_c^w| \leq 0.2\,\mathrm{amp}(Lx) r_z = 0.7\varrho$ strong units at κ_1. So $\hat{w} \leq (1.03 + 0.7\varrho)$ strong units at $\kappa + 1/4$.

Step 6: the factors. Let $\mathsf{y} := (r_z/\mathsf{b})(4C_S p_0^2 + 5)$; then $\mathsf{y} \leq x$, by (j6). By Lem. 11.7 (c), on fields tame on Box_c we have $1 + r_z|\Delta_c + A^{\mathrm{long}}| \leq 1 + (r_z/\mathsf{b}) 2\,\mathrm{ex} \leq 1 + 2x^{1-8L}$, the last step by (j6), while on the other fields this quantity is at most $1 + \mathsf{y}\, 1\{D(V) \text{ meets } \mathrm{Box}_c\}$. A field fails to be tame on Box_c exactly when some plaquette of Box_c lies in $D(V)$ (item 1 of §2.1, on cubes). Let S be a set of cells. On the event that $D(V)$ meets Box_c for all $c \in S$, the covering clause shows that in every family there is, for each $c \in S$, a member holding a site of Box_c. Consequently $\prod_{c\in S} 1\{\ldots\}\, \Xi_{\hat{w}} \leq x^{-|S|} \Xi_{\hat{w}'}$, where $\hat{w}' := \hat{w}\, x^{m(X)}$ and $m(X) := \#\{c : X \text{ meets } \mathrm{Box}_c\}$. Finally, $\sum_S (\mathsf{y}/x)^{|S|} \leq 2^Q$.

Step 7: the cap of $\hat{w}'$. At a set X holding a rough plaquette, a strong unit at a rate is at most $\eta_w(x) e^{-\mathrm{rate}\,(|X|_c - 1)}$. If $m \leq 1$, then $(1.03 + 0.7\varrho) x\, \eta_w \leq x^{10} \eta_w \leq 1$, by (j6). If $m \geq 2$, the boxes are more than $h - 2r_B > h/2$ apart, so that their $h/4$-neighbourhoods are disjoint; the connected hull of X leaves each of these neighbourhoods, and therefore $|X|_c \geq mh/(4M)$. The margin $1/4$ in the rate then gives $e^{1/4}(x e^{-h/(16M)})^m \leq 1$, because $h \geq 64LM \log x$. Thus $\hat{w}' \leq 2.6 e^{-\kappa(|X|_c - 1)}$; moreover $\hat{w}'$ is Borel, gauge invariant and non-negative, and it is a function of V on $X^{\approx\approx}$.

Step 8: use of Lem. 11.9. Lem. 11.9 now gives

$$\langle \mathrm{Fold}\, E[\Phi|U] \rangle = \|E[\Phi|U]\|_{\mathrm{ch}}^Q \leq ((9/8)p_0^2)^Q r_z^{-Q} 2^Q e^{Q(2\mathrm{ex} + 2x^{1-8L})}(1 + C_{\mathrm{vol}}),$$

and by Step 2 the right side also bounds $|E\Phi|^Q$. Taking the Q-th root we get $\|E[\Phi|U]\|_{\mathrm{ch}} \leq (9/8) p_0^2 r_z^{-1} 2 e^{1.01}$, and the same bound for $|E\Phi|$. As $e^{1.01} \leq e^{2.2}$, the bound on $\|E[\Phi|U]\|_{\mathrm{ch}}$ implies (W°) of the statement; and as $(9/8) \cdot 2e^{2.2}/3.5 < 5.802 \leq 5.81$, the value of b/r_z given in the statement yields the first term of (E5). □

Remark 11.11 (One kind of source in each cell). The witness G_c carries no source. It is taken out as a supremum after the conditioning, and this is legitimate because the bound of Step 5 of the proof of Lem. 11.10 holds pointwise in V and is positive.

Remark 11.12 (The role of the subtracted constant). The proof of Lem. 11.10 depends on g only through the bound $\sup|G_c - g| \le (9/8)p_0^2$, which enters in Step 8. The identity of Step 1 holds for every constant g.

For Lem. 11.13 we fix the following data. Let $w := w_R$ be the form of Lem. 11.15, and let τ_t, $0 < t \le x^{-2}$, be the flow of D_w^α of Lem. 11.14. In Lems. 11.13 and 11.14 and their proofs, w denotes the w-member $\bar w$ of $\bar u_k$ where it appears as an index of Ξ or carries the arguments (X, V). The lemma depends on w only through these properties: w is real, it is carried by the bonds of B_R, $|w| \le 1$, and the two bounds on d_c in Lem. 11.7 (c) hold. Let Φ be a bounded Borel function that depends on the bonds of B' and is carried by $\{\phi_p \le 3\varepsilon/4 \text{ on } B'\}$. We split $\Lambda = \Lambda^a + H$, with $H = \sum 1\bar E^h$, and we put $\Psi := (\Phi e^{\Lambda^a}) \circ \tau_t\, e^{-\Lambda^a}$.

Lemma 11.13 (The defect of the integration by parts through the weights)**.** In the setting of Thm. 11.1, at $j \in J(\mathsf{b})$, and with the data fixed before the lemma, the following hold. (X°) In norm form: on the board B_{j-i_h},

$$\|E[\Phi_{\mathrm{cell}}^{\mathrm{tr}} \mid U]\|_{\mathrm{ch}} \le \sup|\Psi| \cdot 2e^{2.2}/(r_\eta - 1),$$

where $\Phi_{\mathrm{cell}}^{\mathrm{tr}} := \Psi^{\mathrm{tr}} \circ \tau_{-t}\, e^{\Lambda^{\mathrm{tr}} \circ \tau_{-t} - \Lambda^{\mathrm{tr}}} - \Psi^{\mathrm{tr}}$, and Ψ^{tr} is defined as Ψ is, but with $\Lambda^{a,\mathrm{tr}}$. Claim. With $r_\eta = x^4$, we have

$$\Big|\int \Psi e^{\Lambda}[\Xi_w \circ \tau_t - \Xi_w]\, dV\Big| \le \sup|\Psi|\,(2e^{2.2}/(r_\eta - 1) + \mathrm{ex}) \int e^{\Lambda}\Xi_w,$$

and the bound is independent of N and of K. ∎

Proof. Step 1: a mean. The flow τ_t preserves Haar measure (Step 1 of the proof of Lem. 11.14). Hence the integral on the left side of the "Claim" equals $E_\nu[\Phi_{\mathrm{cell}}] \int e^{\Lambda}\Xi_w$, with $\Phi_{\mathrm{cell}} = \Psi \circ \tau_{-t}\, e^{\Lambda \circ \tau_{-t} - \Lambda} - \Psi$. The function Ψ is carried by $\{\phi_p \le 3\varepsilon/4 + 4t \text{ on } B'\}$. We replace Λ and Λ^a by Λ^{tr} and $\Lambda^{a,\mathrm{tr}}$ once, before the chessboard estimate. We justify its pointwise cost and the comparison of the two suprema. Put $H = \sum 1\bar E^h$ and let $\delta_s A = A \circ \tau_s - A$. The long-polymer estimate in Step (c) of the proof of Lem. 11.7 gives, for $s = \pm t$ on the supports at issue,

$$|\delta_s \Lambda^{a,\mathrm{long}}| + |\delta_s H^{\mathrm{long}}| \le T := 2(2R/M + 2r + 3)^3 (c_a + c_h) e^{2.75} x^{-10.9L} \le \mathrm{ex}/100.$$

For the last inequality use $2R/M + 2r + 3 \le h < x^{1/4}$, $c_a + c_h \le 1/2$, $L \ge 13$, $\log x \ge 3$, and $\mathrm{ex} = x^{-8L}$. In fact $T/\mathrm{ex} \le e^{2.75} x^{3/4 - 2.9L} \le e^{2.75 + 3(3/4 - 2.9\cdot 13)} = e^{-108.1} < 1/100$. The energy itself has no long part. The definitions give the exact identities

$$\Phi_{\mathrm{cell}} = \Phi e^{\delta_{-t} H} - \Psi, \qquad \Phi_{\mathrm{cell}}^{\mathrm{tr}} = \Phi e^{\delta_{-t} H^{\mathrm{tr}}} - \Psi^{\mathrm{tr}}, \qquad \Psi^{\mathrm{tr}} = \Psi e^{-\delta_t \Lambda^{a,\mathrm{long}}}.$$

The flow bound in the data of Lem. 11.14 gives $\sup|\Phi| \le e^{1/2} \sup|\Psi|$, and the rough h-part bound of Lem. 11.7 gives $|\delta_{-t} H^{\mathrm{tr}}| \le 1/4$ on $\operatorname{supp}\Phi$. Consequently

$$|\Phi_{\mathrm{cell}} - \Phi_{\mathrm{cell}}^{\mathrm{tr}}| \le (e^{3/4} + 1)(e^T - 1) \sup|\Psi| \le \mathrm{ex} \sup|\Psi|, \qquad \sup|\Psi^{\mathrm{tr}}| \le (1 + \mathrm{ex}) \sup|\Psi|.$$

Thus there is one pointwise truncation error, and the latter comparison supplies the factor used in Step 7. The function $\Phi_{\mathrm{cell}}^{\mathrm{tr}}$ is then a function of the field on Box alone, by items 2 to 4 of Step (c), dependence, of the proof of Lem. 11.7: there $\Lambda^{\mathrm{tr}} \circ \tau - \Lambda^{\mathrm{tr}}$ and $\Lambda^{a,\mathrm{tr}} \circ \tau - \Lambda^{a,\mathrm{tr}}$ depend on bonds with $r \le r_B$, and the moved terms of En_ℓ depend on B', which lies in Box.

Step 2: the chessboard estimate on B_{j-i_h}. By (I) and (K) of Lem. 11.3 and by Prop. A.11,

$$|E\Phi_{\mathrm{cell}}^{\mathrm{tr}}|^Q \le \|E[\Phi_{\mathrm{cell}}^{\mathrm{tr}}|U]\|_{\mathrm{ch}}^Q = E \prod_c [\Psi_c \circ \tau_{-t}^c e^{a_c} - \Psi_c].$$

Steps 3 to 7 bound this fold, and so they prove (X°). Here $\Psi_c = \Psi^{\mathrm{tr}} \circ \breve\Theta_c$ and $a_c = \Lambda_c^{\mathrm{tr}} \circ \tau_{-t}^c - \Lambda_c^{\mathrm{tr}}$. The function Ψ_c is bounded, and no regularity of it is at hand.

Step 3: the expansion. We expand the product over the subsets S' of the set of cells, and in the term of S' we change the variable, $V \to \tau_t^{S'} V$. The flows commute, since each of them depends

only on the bonds of its own $B_{R,c}$ and moves only these. The function Ψ_c is independent of the bonds moved by $\tau^{c'}$ for $c' \neq c$. Hence

$$E \prod_c [\Psi_c \circ \tau^c_{-t} e^{a_c} - \Psi_c] = \int \prod \Psi_c \, e^{\Lambda} \sum_{S'} (-1)^{Q-|S'|} e^{\mathsf{E}_{S'}} \, \Xi_w \circ \tau^{S'}_t \Big/ \int e^{\Lambda} \Xi_w,$$

with $\mathsf{E}_{S'} = \Lambda \circ \tau^{S'}_t - \Lambda - \sum_{c \in S'} (\Lambda^{\mathrm{tr}}_c \circ \tau^c_t - \Lambda^{\mathrm{tr}}_c)$.

Step 4: the switch. On the support of $\prod \Psi_c$, the plaquettes of B'_c have angles at most $3\varepsilon/4 + 4t$. Along $\tau^{S'}_s$, $0 \le s \le t$, their angles are at most $3\varepsilon/4 + 8t < \varepsilon$, by (j7) (vi). Thus $D(V)$ and the angles of its plaquettes stay fixed, as do all the indicators and masks. The clip equals the varying curvature F_p on each moved plaquette and is unchanged on the others. The difference $\mathrm{En}_\ell \circ \tau^{S'}_t - \mathrm{En}_\ell$ is additive over c, because each energy term that moves under τ^c depends on B'_c, and the other flows leave those bonds fixed. A short X has $X^{\approx}$ of diameter at most $h/8$, so it touches at most one of the sets $B_{R,c}$. For a function g of V we put

$$g(\eta) := \sum_{S''} \prod_{c \in S''} \eta_c \prod_{c \notin S''} (1 - \eta_c) \, g(\tau^{S''}_t V),$$

with one variable η_c for each cell c for which g depends on the bonds of $B_{R,c}$. Then $\mathsf{E}_{S'}$ is the value at $\eta = 1_{S'}$ of $\sum_c \eta_c d_c + \mathsf{L}(\eta)$, where $\mathsf{L}(\eta) := \sum_{X \,\mathrm{long}} ([1E(X)](\eta) - 1E(X))$. We write L for $\mathsf{L}(\eta)$. We put

$$\mathsf{G}(\eta) := \int \prod \Psi_c \, e^{\Lambda} \prod_c (1 + \eta_c (e^{d_c} - 1)) \, e^{\mathsf{L}(\eta)} \, \Xi_{w(\eta)} \, dV.$$

This function is entire, it takes the right values at the vertices, and the families of Ξ are the same at all vertices. Therefore $E \prod_c [\Psi_c \circ \tau^c_{-t} e^{a_c} - \Psi_c] = [\prod_c (\mathrm{ev}_1 - \mathrm{ev}_0)\mathsf{G}] / \int e^{\Lambda} \Xi_w$.

Step 5: bounds for $|\eta_c| \le r_\eta$. In this range, $|g(\eta)| \le (2r_\eta + 1)^{\text{number of variables}} \sup |g|$. Let $\tau(X)$ be the set of cells c such that $X^{\approx\approx}$ meets $B_{R,c}$. Then (j7) (iv) gives $\tau(X) \subset \{c : X \text{ meets } \mathrm{Box}_c\}$, hence $|\tau(X)| \le m(X)$. Next,

$$|\mathsf{L}| \le Q(2R/M + 2r + 3)^3 \sum_{n > n_0} (26e)^{n-1} 2(2r_\eta + 1)^{\max(1, 4Mn/h)} (c_a + c_h) e^{-\kappa_1 (n-1)} \le Q \, \mathrm{ex}.$$

Further, $|w(\eta)(X, V)|$ is at most $(2r_\eta + 1)^{|\tau(X)|}$ times the strong cap at κ_1 at V. By Lem. 11.7 (c), $|1 + \eta_c(e^{d_c} - 1)| \le 1 + 2r_\eta \mathrm{ex}$ on fields tame on Box_c, and $|1 + \eta_c(e^{d_c} - 1)| \le 1 + 2r_\eta 1\{D(V) \text{ meets } \mathrm{Box}_c\}$ on the others.

We now bound the product of these factors with $\Xi_{w(\eta)}$ by the argument of Step 6 of the proof of Lem. 11.10, with the following changes. The factor $2r_\eta$ takes the place of y, and on tame fields the error $\mathrm{ex}' := 2r_\eta \mathrm{ex} = 2x^{4-8L}$ takes the place of $2x^{1-8L}$. The covering clause enters unchanged: on the event that $D(V)$ meets Box_c for all c in a set S of cells, every family has, for each $c \in S$, a member holding a site of Box_c. The weight receives the factor $(2r_\eta)^{m(X)}$ in place of $x^{m(X)}$; this factor absorbs $(2r_\eta)^{|S|} \prod_{c \in S} 1\{\ldots\}$, so that the sum over S gives 2^Q in place of $\sum_S (\mathsf{y}/x)^{|S|}$. In this way the product is at most $e^{Q\mathrm{ex}'} 2^Q \Xi_{\hat{w}'}$, with

$$\hat{w}' := (2r_\eta + 1)^{|\tau(X)|} (2r_\eta)^{m(X)} \exp\{-c_\Xi x \varepsilon \sum_{D \cap X} \phi_p - \kappa_1 (|X|_c - 1)\} \, 1\{X \text{ holds a plaquette of } D(V)\}.$$

The weight $\hat{w}'$ is an explicit function of V on X; so the bound is unaffected by the dependence of $w(X, \tau V)$ on comb paths outside $X^{\approx\approx}$. For the cap of $\hat{w}'$ there are two cases. If $m \le 1$, then $(2x^4 + 1)^2 \eta_w \le x^{10} \eta_w \le 1$. If $m \ge 2$, then $|X|_c \ge mh/(4M)$, and the margin $\kappa_1 - \kappa = 1$ absorbs $(3x^4)^{2m}$, because $h/(4M) \ge 16L \log x$.

Step 6: a contour integral. In each variable,

$$g(1) - g(0) = (1/2\pi i) \oint_{|\zeta| = r_\eta} g(\zeta)/(\zeta(\zeta - 1)) \, d\zeta,$$

and the modulus of this is at most $\sup |g| / (r_\eta - 1)$.

Step 7: the conclusion. Lem. 11.9, followed by the Q-th root, gives $\|E[\Phi^{\mathrm{tr}}_{\mathrm{cell}}|U]\|_{\mathrm{ch}} \le \sup|\Psi|(1+\mathrm{ex})2e^{1.01}/(r_\eta-1)$, and by Step 2 the same estimate holds for $|E\Phi^{\mathrm{tr}}_{\mathrm{cell}}|$. As $(1+\mathrm{ex})e^{1.01} \le e^{2.2}$, the bound on the norm implies (X°); and the bound on $|E\Phi^{\mathrm{tr}}_{\mathrm{cell}}|$, together with the single pointwise error $\sup|\Psi|\,\mathrm{ex}$ of Step 1, yields the "Claim". □

For Lem. 11.14 we fix the following data. Let w be real on the bonds of B_R, with $|w| \le 1$, and let $t\sup|D^\alpha_w \Lambda^a| \le 1/2$ hold on $\{\phi_p \le 3\varepsilon/4 + 4t \text{ on } B'\}$. The integration-by-parts lemma is applied to $w = w_R$ of Lem. 11.15. For this form, Step 0 of the proof of Lem. 11.17 gives $\sup|D\Lambda^a| \le 2p_0 x^{1/2}$. Its proof also gives the same bound for $\Lambda^{a,\mathrm{tr}}$, since the estimate on the analytic corrections is a sum of absolute values. As (j6) gives $p_0^{10} \le x$ and (j7) (vi) gives $\log x \ge 3$, we have

$$t\sup|D\Lambda^a| \le 2p_0 x^{-3/2} \le 2x^{-1.4} < \tfrac{1}{2}.$$

In the reflected copies used in Lems. 11.7 and 11.13, the flow and the weight are conjugated together:

$$\tau^c_s = \check\Theta_c^{-1}\tau_s\check\Theta_c, \qquad \Lambda^a_c = \Lambda^a \circ \check\Theta_c, \qquad \frac{d}{ds}\Lambda^a_c(\tau^c_s V)\Big|_{s=0} = (D\Lambda^a)(\check\Theta_c V).$$

Thus the same $2p_0x^{1/2}$ bound applies to these conjugated weights, and to their truncations, on the corresponding reflected tame sets.

The comparison of $\Lambda^{a,\mathrm{tr}}$ itself along τ^c_s in Lem. 11.7 uses the separate bound $2N_{B0}W_*p_0x^{1/2}$ proved in its rough-bounds step. Its product with t is also less than $1/2$: from (E6)$\le c_F/12$ and $C_6 \ge N_{B0}W_*$,

$$2N_{B0}W_*p_0x^{-3/2} \le \frac{c_F}{36p_0^4} < \tfrac{1}{2}.$$

This comparison does not apply the Green-function identity to a reflected form against the original row matrix. There is no circularity: Lems. 11.15 and 11.16, and the part "Besides" of Lem. 11.14, use nothing of the "Claim" of Lem. 11.14 or of Lem. 11.13. We put $D^\alpha_w := \sum_b w_b \cdot$ (derivative of $V_b \to e^{isX_b}V_b$ at $s=0$), where $X_b := T^{-1}_{\pi(b)}e_\alpha$. Here the e_α, $\alpha = 1,2,3$, form an orthonormal basis of $(\mathfrak{g}, \langle Y, Y'\rangle = \frac{1}{2}\operatorname{tr} YY')$, and we write $Y^\alpha = \langle Y, e_\alpha\rangle$; further, $\pi(b)$ is the base site of the bond b, namely its least end, so that $\pi(b) = z$ for $b = (z, z+e_\mu)$. Finally, Φ is C^2 along the flow, depends on the bonds of B', and is carried by $\{\phi_p \le 3\varepsilon/4 \text{ on } B'\}$.

Lemma 11.14 (Integration by parts for members that are only Borel)**.** In the setting of Thm. 11.1, at $j \in J(\mathsf{b})$, the following hold. Claim. With the data fixed before the lemma and at $t = x^{-2}$, the bound

$$|E_\nu[D^\alpha_w\Phi + \Phi D^\alpha_w\Lambda^a]| \le tB_2 + t^{-1}e\sup|\Phi|\,(2.5(2R/M + 2r(x) + 3)^3 c_h(x) + 6e^{2.2}x^{-4} + 2\mathrm{ex})$$

holds, where $B_2 = \sup|e^{-\Lambda^a}D^2_w(\Phi e^{\Lambda^a})|$. Besides. At $w = w_R$ and on the set $\{\phi_p \le 3\varepsilon/4 + 4t \text{ on } B'\}$ that occurs in the data fixed before the lemma, two bounds of Cauchy type hold: first, $|D_w\sum 1\bar E^a| \le C_a(x)p_0^2$, with $C_a(x) = 2.5(2r(x)+1)^3E_aW_*$; second, $|D^2_w\sum 1\bar E^a| \le 12.5E_ap_0x^{1/2}(2r(x)+1)^3W_*^2$. The sums. Let $W_0 = \sum_b|w_b|$, $W_{00} = \sum_b w_b^2$, $W_1 = \sum_p|\mathrm{d}w|_p$ and $W_2 = \sum_p(r_p+1)|w|_p$, where $|w|_p$ denotes the sum over the four bonds of p. At $w = w_R$ these sums satisfy $W_0 \le W_0(x)$, $W_{00} \le 60r_1^3$, $W_1 \le 400r_1^4$, $W_2 \le 289r_1^3R$, and $5\varepsilon W_2 \le 1$. ■

The available bound on W_0 grows logarithmically with R; the argument uses it only through the product $\Theta(x)$.

Proof. Step 1: the divergence. The simple path $\Gamma(0,\pi(b))$ can contain the bond b only as its last step, run backwards. In that case $V(\Gamma) = V(\Gamma')V_b^{-1}$, hence $X_bV_b = V_bX'_b$ with X'_b free of V_b; in the other case X_b itself is free of V_b. In both cases the b-component is an invariant vector field in the variable V_b. So its divergence is 0, and the flow τ_t, smooth on a compact manifold, preserves Haar measure.

Step 2. Regularity along the relevant flow segments. For a configuration V, consider $\tau_s V$ for $0 \le s \le t$. If the segment misses $\operatorname{supp}\Phi$, the product $\Phi(\tau_s V)e^{\Lambda^a(\tau_s V)-\Lambda^a(V)}$ vanishes there with its derivatives. Otherwise some point of the segment lies in $\operatorname{supp}\Phi$. The displacement bound $4t$ then puts the whole segment in $\{\phi_p \le 3\varepsilon/4 + 4t \text{ on } B'\}$. Since $4t < \varepsilon/4$, all these plaquettes remain tame. Every $1_{X^{\approx}}$ is constant, because the plaquettes off B' stay fixed. A row depending on a plaquette that moves, or having a path $\Gamma(y,\pi p)$ that crosses a moving bond, has $r(y) \le R+\rho+1$. By (j7) (vii), the mask of such a row depends on the plaquettes within $\max\{\rho_0, \rho(x)+1\} = \rho(x)+1$ of y; recall that the mask has two radii, ρ_0 and $\rho(x)+1$ (item 2 of §5.5, on the energy). These plaquettes belong to B', so $m_y = 1$ and $\zeta = F$. There En_ℓ is a polynomial in F and in the $T(y,\cdot)$, and the E^a are holomorphic.

Step 3. Taylor's formula. On each segment meeting the support, apply Taylor's formula to the product from Step 2. Its second τ-derivative has modulus at most $B_2 e^{t\sup|D_w\Lambda^a|}$ by the bounds on the enlarged tame set. On the remaining segments the product vanishes. Integrating this pointwise estimate for $J(t) = E_\nu[\Phi\circ\tau_t\, e^{\Lambda^a\circ\tau_t-\Lambda^a}]$ gives

$$|J(t) - J(0) - tJ'(0)| \le (t^2/2)B_2 e^{t\sup|D\Lambda^a|} \le t^2 B_2,$$

by the data, as $e^{1/2} < 2$.

Step 4: a split. We put $Z := \int e^{\Lambda}\Xi_w\, dV$. By Step 1, $J(0) = \int(\Phi e^{\Lambda^a}e^H\Xi_w)\circ\tau_t/Z$. So

$$J(t) - J(0) = \int (\Phi e^{\Lambda^a})\circ\tau_t\,[e^H\Xi_w - (e^H\Xi_w)\circ\tau_t]/Z,$$

and this is

$$-\int \Phi\circ\tau_t(1-e^{H-H\circ\tau_t})\,(e^{\Lambda}\Xi_w)\circ\tau_t/Z - \int \Psi e^{\Lambda}[\Xi_w\circ\tau_t - \Xi_w]/Z.$$

Step 5: the first term. Here $|H - H\circ\tau_t| \le 2\cdot 1.2124(2R/M + 2r(x) + 3)^3 c_h$. This error acts multiplicatively on $(e^{\Lambda}\Xi_w)\circ\tau_t > 0$, whose integral is Z; and $|1-e^y| \le e|y|$.

Step 6: the second term. Here we apply Lem. 11.13, with $\sup|\Psi| \le \sup|\Phi|e^{t\sup|D\Lambda^a|} \le e^{1/2}\sup|\Phi|$ by the data.

Step 7: the Cauchy estimates for E^a. Step 7.0: the frames do not move. We have $D_w^\alpha X_b = 0$ exactly, and so $\tau_s V_b = e^{is w_b X_b(V)}V_b$. Let $g_z := V(\Gamma(0,z))$. We claim that $D_w g_z = i\,\check\sigma(z)e_\alpha g_z$, where $\check\sigma(z)$ is the signed sum of the w_b along $\Gamma(0,z)$. This follows by induction along the comb path, whose initial segments are the comb paths of the sites it passes through (e_1, then e_2, then e_3: item 2 of §5.5, on the energy). At a forward step $b = (z, z+e_\mu)$, with $\pi(b) = z$, we have $DV_b = iw_b X_b V_b$ and $X_b = g_z^{-1}e_\alpha g_z$, so that $g_z DV_b = iw_b e_\alpha g_z V_b$. At a backward step $b = (z-e_\mu, z)$, with $\pi(b) = z - e_\mu$, we set $g := g_z V_b^{-1}$; then $X_b = g^{-1}e_\alpha g$ and $D(V_b^{-1}) = -iw_b V_b^{-1}X_b$, so that $g_z D(V_b^{-1}) = -iw_b e_\alpha g$. Hence $DX_b = g^{-1}[e_\alpha, i\,\check\sigma\, e_\alpha]g = 0$: the field is constant along its own flow.

Step 7.1: the curve and its domain. For complex s let $V_b(s) = e^{isw_b X_b}V_b$. By Step 7.0 this curve is the flow itself for real s, so that d^n/ds^n at 0 is D_w^n, for every n. Carrying all four factors of a plaquette to its base site, we find $\partial V(s)(p) = \prod_b e^{\pm isw_b Y_b}\,\partial V(p)$, where $Y_b = T_{\pi p}^{-1}\operatorname{Ad}_{\mathrm{hol}_b} e_\alpha$ does not depend on s and, with $Y := T_{\pi p}^{-1}e_\alpha$, satisfies $|Y_b - Y| \le 5(r_p+1)\varepsilon$ (Step 1 of the proof of Lem. 11.16). By telescoping, $|\prod_b e^{\pm isw_b Y_b} - 1| \le |s|(|\mathrm{d}w|_p + 5(r_p+1)\varepsilon|w|_p)e^{|s||w|_p}$. We put $\delta_p := |\mathrm{d}w|_p + 5(r_p+1)\varepsilon|w|_p$, we let δ_X be the supremum of δ_p over the plaquettes of $X^{\approx}$, and w_X the supremum of $|w_b|$ over $X^{\approx}$. We define s_X by $1/s_X := 4\delta_X/\varepsilon + (2\ell_w(x) + 4/\vartheta_0^\flat)w_X$. For $|s| \le s_X$ the following hold.

(u3) $|\partial V(s)(p) - 1| \le 3\varepsilon/4 + 4t + e^{1/2}\varepsilon/4 < 2\varepsilon$.
(u2) $|V(s)(\text{path})| \le e^{|s|\ell_w w_X} \le e^{1/2} < 2 - 1/\log x$.
(u1) $|V_b(s)| \le e^{\vartheta_0^\flat/4} \le 1 + \vartheta_0^\flat/2 < 1 + \vartheta_0^\flat(1 - 1/\log x)$.

Here (u1), (u2) and (u3) are the three conditions of the domain $\mathcal{U}_X(x)$ (item 1 of §2.1, on the domains). In (u3) the last inequality follows from the bound $4t < \varepsilon/4$ of Step 2, and in (u1) and (u2) it rests on $\log x \geq 3$, which is (j7) (vi) of the list of Thm. 11.1.

Step 7.2: the first order. The Cauchy estimate gives $|D_w E^a(X,V)| \leq c_a e^{-\kappa_1(\#X-1)}/s_X$, and $1/s_X \leq \sum_{\text{cubes in } X} 1/s_{\text{cube}}$. We have $\sum_{\text{cubes}} \sup_{p\in\text{cube}\approx} \delta_p \leq (2r+1)^3 \sum_p \delta_p$, where $\sum_p$ is the sum over all plaquettes. Likewise, $\sum_{\text{cubes}} \sup_{b\in\text{cube}\approx} |w_b| \leq (2r+1)^3 \sum_b |w_b|$, with $\sum_b$ running over all bonds. We bound the sums with $|w| \leq \min\{1, r_1^3 r^{-3}\}$ and $|\mathrm{d}w_R| \leq 3r_1^4 r^{-4}$ (Lem. 11.15 (a0), (a)), and with the count of $3(24r^2+2) \leq 78r^2$ bonds, and of as many plaquettes, at distance $r \geq 1$ (Def. 2.44). This gives $W_0 \leq 3 + 78\sum_{1\leq r\leq r_1} r^2 + 78 r_1^3 \log(R/r_1) \leq W_0(x)$ and $W_1 \leq 234 r_1^4 \pi^2/6 + (\text{terms at } 0) \leq 400 r_1^4$. For W_2, a plaquette based at radius r has its bond bases at radius at least $r-1$, and it cannot meet a moved bond if $r > R+1$. For $r \leq r_1+1$ use $|w|_p \leq 4$. The sum of $4(72r^2+6)(r+1)$ over those radii, including the origin, is at most $73 r_1^4$. For $r \geq r_1+2$ use $|w|_p \leq 4r_1^3/(r-1)^3$. The corresponding summand is at most $288.5 r_1^3$, since $4(72r^2+6)(r+1)/(r-1)^3 \leq 288.5$ for $r \geq r_1+2$. There are at most $R-r_1$ such radii. Hence $W_2 \leq 289 r_1^3 R$. As $r_B \geq R + h/8 \geq 3R$, (j5) gives $\varepsilon \leq 1/(288R^2)$; hence $5\varepsilon W_2 \leq 5.02 r_1^3/R \leq 1$, by $R \geq 6r_1^3$, one of the entries of the list of Thm. 11.1 that hold of themselves. Also $W_{00} \leq 3 + \sum 78 r^2 \min\{1, r_1^6 r^{-6}\} \leq 60 r_1^3$. So

$$\sum_{\text{cubes}} 1/s_{\text{cube}} \leq (2r+1)^3 [4(W_1 + 5\varepsilon W_2) + \Theta(x)]/\varepsilon \leq (2r+1)^3 \cdot 1605 r_1^4/\varepsilon,$$

by $\Theta \leq 1$. Since $c_a/\varepsilon = E_a p_0^2$ and $1.2124 \cdot 1605 r_1^4 \leq 2.5 W_*$, this yields $C_a(x)$.

Step 7.3: the second order. The Cauchy estimate gives $|D_w^2 E^a(X,V)| \leq 2c_a e^{-\kappa_1(\#X-1)}/s_X^2$. As δ_X and w_X are maxima over the cubes of X,

$$1/s_X^2 \leq \sum_{\text{cubes in } X} [32\delta_{\text{cube}}^2/\varepsilon^2 + 2(2\ell_w + 4/\vartheta_0^\flat)^2 w_{\text{cube}}^2].$$

We sum over the cubes in the manner of Step 7.2, that is, with the factor $(2r+1)^3$ times the sum over all plaquettes or over all bonds; the result is at most

$$2 \cdot 1.2124 c_a (2r+1)^3 \varepsilon^{-2} [32 \sum_p \delta_p^2 + 2(\varepsilon(2\ell_w + 4/\vartheta_0^\flat))^2 W_{00}].$$

Here $\sum \delta_p^2 \leq (W_1 + 5\varepsilon W_2)^2 \leq (401 r_1^4)^2 \leq 0.161 W_*^2$, and $\varepsilon(2\ell_w + 4/\vartheta_0^\flat) \leq 1$, by $\Theta \leq 1$ and $W_0(x) \geq 1$. With $c_a \varepsilon^{-2} = E_a p_0 x^{1/2}$ the total is $2.425(5.146 W_*^2 + 120 r_1^3) \leq 12.5 W_*^2$ (the coefficient is 12.48). □

Lemma 11.15 (The Green function, uniformly in the rows)**.** Let $\ell = \ell_k^{[K]}$, $1 \leq k \leq K$, be unfolded to $\mathbb{Z}^3$. We put $B = 1 + \ell^{\mathsf{T}}\ell$, $M = \mathrm{d}^* B \mathrm{d}$, $C_B = \mathrm{d}M^+\mathrm{d}^*$, $\omega := C_B u$, $w := \mathrm{d}^*(-\Delta)^{-1}\omega$, $w_R := \chi_R w$ and $e := \mathrm{d}^* u - M w_R$. Then the following hold.

(a0) At every y we have $|w| \leq 1$, $|\omega| \leq \sqrt{2}$, $\mathrm{d}w = \omega$ and $Mw = \mathrm{d}^* u$.
(a) For $r \geq r_1 = 10^{130}$: $|w| \leq C_w r^{-3}$, $|\nabla w| \leq C_{gw} r^{-4}$, $|\omega| \leq C_\omega r^{-4}$ and $|\nabla\omega| \leq C_{g\omega} r^{-5}$. For all $r \geq 1$: $|w| \leq r_1^3 r^{-3}$ and $|\mathrm{d}w_R| \leq 3 r_1^4 r^{-4}$.
(b) For $r \geq r_1$: $|\omega - \Pi u| \leq C_G r^{-6} + C_\tau \tau_x^2 r^{-4}$; and, for the gradient, $|\nabla(\omega - \Pi u)| \leq C'_G r^{-7} + 1.9 \cdot 10^{36} \tau_x^2 r^{-5} + A_{\text{out}} r^{-16}$.
(c) $0 \leq a_u = \langle u, \omega\rangle \leq 2$.
(d) If $8\rho + 8 \leq R$ and $R \geq 4r_1$, then e is carried by $R/2 - 2\rho - 2 \leq r \leq R + 2\rho + 2$, $|e| \leq C_e R^{-5}$, and $\|e\|_1 \leq 81 C_e R^{-2}$.

None of the constants depends on $\rho(x)$, k, K or N; the values of C_w, C_{gw}, C_ω, $C_{g\omega}$, C_G, C'_G, C_τ, A_{out} and C_e are given in item 17 of §11. ■

Proof. Step 0: unfolding. The rows have range at most $\rho(x)$, and the side is $S = NL^j > 4\rho + 2$ (by (Tor) in (SD), and Thm. 7.1 (b)). Hence, for $|y - p| \leq \rho$, the formula $\ell_y(p) := \ell_{[y]}([p])$ gives

covariant rows on $\mathbb{Z}^3$; their forms E_y are unchanged, and so are the bound $K_* e^{-\lambda_*|y-p|}$ (Step 2 of the proof of Thm. 8.2, in the part headed "The last integral") and the defect τ. Identify the target spaces H_y orthogonally with H_0 using translation covariance. The unfolded row has finite support, so its symbol $\hat\ell(q)$ is a matrix of finite Laurent polynomials and $\hat B(q) = 1 + \hat\ell(-q)^{\mathsf T}\hat\ell(q)$. For real p, $\hat B(p) = 1 + \hat\ell(p)^*\hat\ell(p) \geq 1$. The inequality $S > 4\rho + 2$ ensures that periodizing the unfolded kernel, whose range is at most 2ρ, recovers the torus kernel. At $p \in (2\pi/S)\mathbb{Z}^3$ the symbol $\hat B(p)$ is therefore that of the torus, and it is at most b_*. At any other p we choose $p' \in (2\pi/S)\mathbb{Z}^3$ with $|p - p'|_\infty \leq \pi/S$; then

$$\|\hat B(p) - \hat B(p')\| \leq 9K_*^2 \mathsf{g}(\lambda_*/2)\mathsf{g}_1(\lambda_*/2)|p - p'|_\infty \leq 4.81 \cdot 10^{127}|p - p'|_\infty < 1,$$

because $S \geq 2L \geq 2L_A \geq 10^{149}$ (by Def. 2.56, with $\mathsf{g}_1(\lambda) \geq 2.2\lambda^{-4}$). So $1 \leq \hat B \leq b^\circ := b_* + 1$ at all p. In the application to Lem. 11.16, $j \in J(\mathsf{b})$. Condition (j3) gives $S \geq 2Lh$, while $h \geq 16R$ and (j7)(iii) gives $8\rho + 8 \leq R$. Thus B' and every range-2ρ interaction entering Mw_R fit inside one fundamental domain. The torus and $\mathbb{Z}^3$ therefore give the same Mw_R there; no torus Green function is identified with the infinite-lattice Green function.

We now prove the bound $\tau(\ell_k) \leq \tau_x$. Thm. 2.57 (a) says that $\ell \in \mathcal{R}(b_*, K_*, \lambda_*, \tau)$, $\tau \leq 1/9$ and $\mathsf{t}_*(\rho) \leq 1/9$ imply $\ell'^{\mathrm c} \in \mathcal{R}(b_*, K_*, \lambda_*, \tau + \mathsf{t}_*(\rho))$. Along the runs (item 2 of §5.5; the rows of item 1 of §6.3) we have $\ell_i = (\ell'_{i-1})^{\mathrm c}$, with the cut at $\rho(x_i)$, and $\tau(\emptyset) = 0$. All ℓ_i lie in $\mathcal{R}(b_*, K_*, \lambda_*, 1/9)$, by Step 2 of the proof of Thm. 8.2 (for $0 \leq k \leq K$ and every $K \geq 1$). Therefore $\tau(\ell_k) \leq \sum_{i \leq k} \mathsf{t}_*(\rho(x_i))$. The definition of t_* (Def. 2.56), together with $\rho(x) \geq c_R \log x$ and $e_R = \lambda_* c_R/2 \geq 256$ (item 1 of §6.3), gives $\mathsf{t}_*(\rho(x)) \leq 3K_*\mathsf{g}(\lambda_*/2)x^{-e_R}$. As $x_i = L^{k-i}x$, the sum is a geometric series and is at most $3K_*\mathsf{g}(\lambda_*/2)x^{-e_R}/(1 - L^{-e_R}) \leq \tau_x$.

Step 1: symbols. In Steps 1–5, Fourier-vector norms are Euclidean; the spatial radius remains $r = |y|_\infty$. We write $k_\mu = e^{ip_\mu} - 1$ and $k^\#_\mu = e^{-ip_\mu} - 1$; then $(2/\pi)|p| \leq |k| \leq |p|$. The range of $\hat{\mathrm d}$ is $\{F : k^{\mathsf T}F = 0\}$. With $\mathsf R = B^{-1}$ and $s = k^{\mathsf T}\mathsf R k^\#$ we have $\hat C_B = \mathsf R - \mathsf R k^\# k^{\mathsf T}\mathsf R/s$: both sides annihilate $k^\#$, map into $\{k^{\mathsf T}F = 0\}$, and invert $\hat B$ on that space modulo $k^\#$. Hence $\hat B\hat C_B = 1 - k^\# k^{\mathsf T}\mathsf R/s$, and $\widehat{\mathrm d^*}k^\# = 0$; so $\mathrm d^*BC_Bu = \mathrm d^*u$, that is, $Mw = \mathrm d^*u$. On 2-forms $\mathrm{dd}^* + \mathrm d_2^*\mathrm d_2 = -\Delta$, and $k^{\mathsf T}\hat\omega = 0$; so $\mathrm dw = \omega$. Further, $\hat u = e_1(1 - e^{-ip_2})$. Let $P := \hat\Pi = 1 - k^\# k^{\mathsf T}/(k^{\mathsf T}k^\#)$ be the symbol of the projection Π of Thm. 11.1, continued to the discs. We have $\hat C_B\hat B P = P$, hence $\hat C_B - P = -\hat C_B(\hat B - 1)P$. The symbol $(\hat B - 1)(q)$ is holomorphic for $|\operatorname{Im} q_\mu| \leq \lambda_*/6$. Finally, $|\hat\ell(q)e_i - \hat\ell(q')e_i| \leq (\mu_1/3)|q - q'|_\infty$, where $\mu_1 := 9K_*\mathsf{g}_1(\lambda_*/2) \leq 3.78 \cdot 10^{71}$, and $|\hat\ell(0)e_i| \leq \tau_x$.

Step 2: inner discs. Let $|p| \leq p_1 := 1/(20\mu_1)$, and let q differ from p in one coordinate only, with $|q_\mu - p_\mu| \leq |p|/20$. Then $|\sum q^2| \geq (1 - 0.1 - 0.0025)|p|^2 = 0.8975|p|^2$ and $\sum |q|^2 \leq 1.05^2|p|^2 = 1.1025|p|^2$, and $\|\hat B(q) - 1\| \leq 3(\tau_x + 1.05\mu_1|p|)^2 \leq 0.0086$. Hence $|s| \geq 0.80|p|^2$, $\|\hat C_B\| \leq 2.5107$, $\|P\| \leq 2.442$ and $|\hat u| \leq 1.051|p|$. The gradients below are full tensors, with their Euclidean norms: their symbols are $k \otimes \hat w$, $k \otimes \hat\omega$ and $k \otimes ((C_B - \Pi)u)^\wedge$. The pure numbers give $p_1 < 10^{-12}$; thus $|q| \leq 1.05|p|$ and the exponential series imply $|k(q)|, |k^\#(q)| \leq 1.0501|p|$. Six symbols get majorants of the form $A'|p|^a$: for $\hat w$ it is 3.502, with $a = 0$; for $k \otimes \hat w$, $3.68|p|$; for $\hat\omega$, $2.64|p|$; for $k \otimes \hat\omega$, $2.78|p|^2$; for $((C_B - \Pi)u)^\wedge$, $42.7\mu_1^2|p|^3 + 38.7\tau_x^2|p|$; and for its full gradient, $1.051|p|$ times that. Indeed $3.502 \cdot 1.0501 < 3.68$ and $2.64 \cdot 1.0501 < 2.78$. The Cauchy estimate for these vector- or tensor-valued symbols gives $|\partial_\mu^i m| \leq i!(20/|p|)^i A'|p|^a$. The symbol P is discontinuous at 0, and we take no derivative of it there.

Step 3: outer discs. Now let $|p| > p_1$, and take discs of radius $d_{\mathrm{out}} := \min\{p_1/(200b^\circ), 1/(80b^\circ\breve\Lambda)\} \geq 4.2 \cdot 10^{-90}$, where $\breve\Lambda = 3\mu_1 b^{\circ 1/2}$. Write $b = b^\circ$, $d = d_{\mathrm{out}}$ and $K = |k(p)|$. The row-symbol bound of Step 1 gives $\|\hat\ell(q) - \hat\ell(p)\| \leq \mu_1 d/\sqrt{3}$ and the same bound at $-q, -p$. Since $\|\hat\ell(p)\| \leq \sqrt{b}$, it follows that

$$\|\hat B(q) - \hat B(p)\| \leq 2\sqrt{b/3}\,\mu_1 d + \mu_1^2 d^2/3 \leq \breve\Lambda d \leq \frac{1}{80b}.$$

Thus $\|\mathsf{R}(q)\| \le 80/79 < 1.013$ and $\|\mathsf{R}(q)-\mathsf{R}(p)\| \le 1/(79b) < 0.013/b$. Also $K \ge 2|p|/\pi$ and $|k(q)-k(p)|, |k^{\#}(q)-k^{\#}(p)| \le 2d \le 0.016K/b$. The real symbol has $s(p) \ge K^2/b$. Expanding its three factors gives

$$|s(q)-s(p)| \le \frac{K^2}{b}[0.016\cdot 1.013\cdot 1.016 + 0.013\cdot 1.016 + 0.016] < 0.046K^2/b,$$

so $|s(q)| > 0.95K^2/b$. Likewise $|k(q)\cdot k^{\#}(q)| \ge (1-2\cdot 0.016-0.016^2)K^2 > 0.967K^2$. The formulas of Step 1 now give $\|\hat{C}_B(q)\| < 3b$ and $\|P(q)\| < 3$. Moreover $|k(q)|, |k^{\#}(q)| < 3.5$, $|\hat{u}(q)| < 2.1$ and

$$|\hat{w}(q)| \le \frac{|k^{\#}(q)|^2}{|k(q)\cdot k^{\#}(q)|}\,\|\hat{C}_B(q)\| < 4b.$$

The six full symbols of Step 2 are therefore bounded respectively by $4b$, $14b$, $6.3b$, $22.05b$, $12.6b$ and $44.1b$, all less than $70b$. Consequently $|\partial_\mu^{16} m| \le 16!\, d_{\mathrm{out}}^{-16} 70 b^{\circ} =: A_{\mathrm{out}} \le 10^{1455.5}$.

Step 4: a bound for multipliers. We first state it. Let m be periodic and C^n off 0, with $a \ge 0$ and $n \ge a+4$; let $|\partial_\mu^i m| \le A|p|^{a-i}$ for $|p| \le p_1$ and $i \le n$, and let $|\partial_\mu^n m| \le A_{\mathrm{out}}$ for $|p| > p_1$. Then, for $r = |y|_\infty = |y_\mu| \ge \sqrt{3}/p_1$,

$$|m^{\vee}(y)| \le c_{a,n} A r^{-3-a} + A_{\mathrm{out}} r^{-n},$$

with $c_{a,n} = (8\cdot 3^{a/2}(n+1) + 4\pi/(n-a-3))/(2\pi)^3$.

We now prove this bound. On $Q = \{|p|_\infty \le 1/r\}$ we have $\int |m| \le 8r^{-3}A(\sqrt{3}/r)^a$. Off Q we integrate by parts n times in p_μ, and the boundary terms at the ends $p_\mu = \pm\pi$ cancel by periodicity. Only the lines with $|p_\perp|_\infty \le 1/r$ have further ends, namely at $\pm 1/r$, and the boundary terms there are at most $\sum_{i<n} r^{-i-1}\cdot 2\cdot (4/r^2)A\cdot 3^{a/2} r^{i-a}$. What remains is at most $r^{-n}[A\cdot 4\pi \int_{1/r}^{\infty} s^{a-n+2} ds + (2\pi)^3 A_{\mathrm{out}}]$. This proof is linear in the majorant.

Step 5: clauses (a) *and* (b). Apply Step 4 with $n = 16$ and $A = 16!\,20^{16}A'$ separately to the powers in each majorant of Step 2. The outer remainder is $A_{\mathrm{out}} r^{-16}$, with $A_{\mathrm{out}} \le 10^{1455.5}$. At $r_1 = 10^{130}$ its ratio to $c_{a,16} A r^{-3-a}$ is at most $10^{-9.29}$ for the four symbols $\hat{w}, k\otimes\hat{w}, \hat{\omega}, k\otimes\hat{\omega}$ and for the term $42.7\mu_1^2|p|^3$ of $((C_B-\Pi)u)^{\wedge}$. These ratios decrease for $r \ge r_1$. The numerical constants $C_w, C_{gw}, C_\omega, C_{g\omega}, C_G$ in item 17 exceed $(1+10^{-9.29})c_{a,16}A$ for the corresponding majorants. This proves (a) and absorbs the outer remainder in $C_G r^{-6}$ in the first estimate of (b).

The terms containing τ_x^2 are handled by linearity of the integral estimate in Step 4. Their coefficients are bounded by C_τ and $1.9\cdot 10^{36}$. For the gradient of the $\mu_1^2|p|^3$ term, use $a = 4$ and the constant C_G'. At $r = r_1$ the outer remainder cannot be absorbed into this r^{-7} bound, so it remains as the explicit $A_{\mathrm{out}} r^{-16}$ term in (b).

For (a0): $|\hat{w}| \le |k||\hat{\omega}|/|k|^2 \le |k_2|/|k| \le 1$, and $|\omega(y)| \le \|C_B u\|_2 \le \sqrt{2}$. For $r < r_1$ we have $1 \le r_1^3 r^{-3}$. Finally, $\mathrm{d}w_R = \chi_R \omega + \eta_1$. Summing the squares of the three antisymmetric components in the formula for η_1 below gives

$$|\eta_1(y)|^2 \le \frac{54}{R^2}\max_\alpha |w(y+e_\alpha)|^2, \qquad |\mathrm{d}w_R(y)| \le \sqrt{2} + 8/R.$$

Here and in this display the norms include all form components. The prescribed R satisfies $R \ge 4r_1$. Thus $\mathrm{d}w_R = \omega$ for $r < r_1$. If $\eta_1(y) \ne 0$, then $R/2 - 1 \le r \le R$, and the bound on w gives

$$|\eta_1(y)| \le \frac{8r_1^3}{R(r-1)^3} \le 9r_1^3 r^{-4}.$$

For $r \ge r_1$, therefore, $|\mathrm{d}w_R| \le (C_\omega + 9r_1^3)r^{-4} < 3r_1^4 r^{-4}$; for smaller r the same conclusion follows from $|\omega| \le \sqrt{2}$.

Step 6: clause (c). The operator $B^{1/2} C_B B^{1/2}$ is an orthogonal projection. So $0 \le C_B \le B^{-1} \le 1$; and $|u|^2 = 2$.

Step 7: clause (d)*, an exact identity and the bounds.* At a cell, χ is evaluated at the base site of the cell. Then $\mathrm{d}(\chi w) = \chi\omega + \eta_1$, where

$$\eta_1(y;\mu\nu) = (\partial_\mu\chi)(y)w_\nu(y+e_\mu) - (\partial_\nu\chi)(y)w_\mu(y+e_\nu);$$

next, $B(\chi\omega) = \chi B\omega + \eta_3$, where

$$\eta_3(p) = \sum_{p'}(B-1)(p,p')(\chi(p') - \chi(p))\omega_{p'};$$

and $\mathrm{d}^*(\chi G) = \chi \mathrm{d}^* G + \eta_2$. We take $G = B\omega$, for which $\mathrm{d}^*G = \mathrm{d}^*u$; since $\chi = 1$ on $\operatorname{supp}\mathrm{d}^*u$, we obtain the exact identity $e = -\eta_2 - \mathrm{d}^*\eta_3 - \mathrm{d}^*B\eta_1$.

We now turn to the bounds of clause (d). For the product cut-off, $|\partial_\alpha\chi| \le 3/R$ in each direction, $|\chi(y') - \chi(y)| \le (9/R)|y - y'|_\infty$, and the second differences are at most $24/R^2$. We write $S_\cdot$ for a supremum over $r \ge R/2 - 2\rho - 2 \ge R/4 \ge r_1$. Then $|\eta_2| \le (6/R)N_{B0}S_\omega$ and $|\mathrm{d}^*\eta_3| \le 4(9/R)N_{B1}S_\omega$. The form η_1 is only of order $R^{-1}R^{-3}$. However, d^*B annihilates constant forms for any covariant rows, because Bc is then constant. Let $D_1(\eta_1)$ be the supremum of $|\partial_\alpha\eta_1(p)|$ over directions and plaquettes of all orientations in this region. Subtract the constant form $c_i = \eta_1(y_b;i)$. Each component difference is a sum along a coordinate path of length at most $3(|z|_\infty + 1)$; the four terms in d^* therefore give $|\mathrm{d}^*B\eta_1(b)| \le 12(N_{B0} + N_{B1})D_1(\eta_1)$. The product rule in each direction gives $D_1(\eta_1) \le 2[(24/R^2)S_w + (3/R)S_{gw}]$. Here $N_{B0} = 1 + 3K_*^2\mathsf{g}(\lambda_*/2)^2 \le 4.9\cdot 10^{112}$ and $N_{B1} = 3K_*^2\mathsf{g}(\lambda_*/2)\mathsf{g}_1(\lambda_*/2) \le 1.61\cdot 10^{127}$; moreover $S_w \le 64C_wR^{-3}$, $S_\omega \le 256C_\omega R^{-4}$ and $S_{gw} \le 256C_{gw}R^{-4}$. Altogether, this gives

$$C_e = 256(6N_{B0} + 36N_{B1})C_\omega + 24(N_{B0}+N_{B1})(1536C_w + 768C_{gw}) \le 7.1\cdot 10^{166}.$$

The number of bonds that carry e is at most $3(2R + 4\rho + 5)^3 \le 81R^3$. □

Lemma 11.16 (The key line with its non-abelian remainders)**.** In the setting of Thm. 11.1, at $j \in J(\mathsf{b})$, on fields with $\phi_p < \varepsilon$ for every plaquette of B' and for $\alpha = 1,2,3$,

$$|x\,u^\alpha(F) - x\,D^\alpha_{w_R}\mathrm{En}_\ell(V)| \le C_1 x\varepsilon/R + C_2 p_0^2 R,$$

where $C_1 = 203C_e$. In its applications the fields satisfy the stronger bound $\phi_p \le 3\varepsilon/4 + 4t < \varepsilon$ on B'. ■

Proof. Step 0: the gauge. Under a gauge transformation g with $g(0) = 1$, the quantities $\tilde F$ and En_ℓ are invariant, and the vector field is covariant ($X_b \to \mathrm{Ad}_{g(\pi b)}X_b$). We work in the comb gauge on B', as in Step 1 of the proof of Lem. 11.6. There, at the point V, $T_y = 1$ and $X_b = e_\alpha$; also $\tilde F = F$, and $V = e^{iA}$, where $|A_b| \le 2r\varepsilon$ by the count of plaquettes in that step: for a bond at the site y, it gives at most $|y_2| + |y_3| \le 2r$ plaquettes.

Step 1: the variation of a plaquette, in any gauge. The product $\delta(\partial V(p))\,\partial V(p)^{-1}$ is i times the sum, over the four bonds, of $\pm w_b \mathrm{Ad}_{\mathrm{path}}X_b$. Each term equals $T_{\pi p}^{-1}\mathrm{Ad}_{\mathrm{hol}_b}e_\alpha$, where hol_b is the holonomy of a loop composed of the comb paths to two neighbouring sites and closed along p; its area is at most $2r+2$. Since Ad_h is a rotation by 2 ang h, we get $|\mathrm{Ad}_{\mathrm{hol}_b}e_\alpha - e_\alpha| \le 2(2r+2)\varepsilon$. For the left-trivialized variation $\delta g\, g^{-1} = iY$, write $g = \cos\phi\,1 + i\sin\phi\, n\cdot\boldsymbol{\sigma}$ in an orthonormal Pauli basis, with $\phi = \mathrm{ang}\,g$ and $|n| = 1$, and decompose $Y = Y_\parallel + Y_\perp$ along n. Differentiating this formula gives

$$d\mathsf{L}_g(Y) = Y_\parallel + \phi\cot\phi\, Y_\perp + \phi\, n\times Y_\perp.$$

For $0 < \phi \le 0.01$, the inequalities $\tan\phi \ge \phi$, $\sin\phi \le \phi$ and $\cos\phi \ge 1 - \phi^2/2$ give $0 \le 1 - \phi\cot\phi \le \phi^2/2$. Orthogonality of $Y_\perp$ and $n\times Y_\perp$ yields $|d\mathsf{L}_g(Y) - Y| \le \phi\sqrt{1+\phi^2/4}|Y| \le 1.01\phi|Y|$; the value at $\phi = 0$ follows by continuity. Here $\phi \le \varepsilon$, as the plaquette p is tame, and $\varepsilon \le 2r_B\varepsilon \le 0.01$ by Step 1 of the proof of Lem. 11.6. Since $|\mathrm{d}w|_p \le |w|_p$, the excess over $\phi|Y|$ fits into the room between $4(r+1)$ and $5(r+1)$. Hence $\delta F_p = T_{\pi p}^{-1}[(\mathrm{d}w)_p e_\alpha] + \mathrm{rem}_p$, with $|\mathrm{rem}_p| \le \varepsilon(5(r+1)|w|_p + |\mathrm{d}w|_p)$.

Step 2: the plaquette part. We have

$$x\sum_p F_p\cdot\delta F_p = x\langle\tilde F^\alpha,\mathrm{d}w_R\rangle + O(x\varepsilon^2\sum_p(5(r+1)|w|_p+|\mathrm{d}w|_p)).$$

With $|w|\le r_1^3 r^{-3}$ and $24r^2+2$ sites at distance r, the sum is at most $1600r_1^3R$. So the error is $1600r_1^3p_0^2R$.

Step 3: the row part. Write $\mathrm{En}_{\mathrm{row}}=\frac12\sum_y m_y(V)|\ell_y^V\zeta|^2$. Every tame indicator is locally constant along the flow: each plaquette that can move lies in B' and has angle strictly below ε. Retain the rows whose curvature arguments or transport paths meet a bond moved by w_R. Their centers have radius at most $R+\rho+1$, and their plaquettes, transport paths and masks lie in B', using $\rho_0\le\rho+1$. On these rows the tame indicators equal 1 and the clipped curvature equals F. The other rows contribute neither to the derivative nor to its proposed leading term. We may therefore use $|F_p|\le\varepsilon$ on every retained row and extend all ensuing positive bounds to the full lattice.

Put $r_p=|\pi(p)|_\infty$, $d=|y-\pi(p)|_\infty$, $d'=|y-\pi(p')|_\infty$ and

$$A_{y,p}=\mathrm{Ad}_{V(\Gamma(y,\pi(p)))},\qquad a_{y,p}=\sum_{b\in\Gamma(y,\pi(p))}|w_b|.$$

In the gauge of Step 0, differentiation and symmetry give the row derivative, before multiplication by x, as

$$\sum_{y,p,p'}\langle\ell_y(p),\ell_y(p')\rangle\langle D(A_{y,p}F_p),A_{y,p'}F_{p'}\rangle.$$

The leading term is $\langle F^\alpha,(B-1)\mathrm{d}w_R\rangle$. We bound its three errors separately.

The curvature remainder of Step 1 contributes at most $\varepsilon^2N_{B0}(5W_2+W_1)$, with the sums from Lem. 11.14. Differentiating the transport gives $\|DA_{y,p}\|\le 2a_{y,p}$. For a fixed displacement $v=y-\pi(p)$ and fixed plaquette orientation ι, summing over translates of its coordinate path gives

$$\sum_{z\in\mathbb{Z}^3}a_{z+v,(z;\iota)}\le 3|v|_\infty W_0.$$

Indeed each of its at most $3d$ bond positions contributes at most W_0. Summing both plaquette orientations and using the exponential row bound therefore makes this error at most

$$54\varepsilon^2K_*^2\mathsf{g}(\lambda_*)\mathsf{g}_1(\lambda_*)W_0\le 18\varepsilon^2N_{B1}W_0.$$

Finally, each bond on $\Gamma(y,\pi(p))$ has radius at most $|y|_\infty+d$ and angle at most $2(|y|_\infty+d)\varepsilon$. The adjoint bound and telescoping along the path yield

$$\|A_{y,p}-1\|\le 12d(|y|_\infty+d)\varepsilon.$$

Since $|y|_\infty\le r_p+d$, the discrepancy between the two transports in the leading term is bounded by

$$\|A_{y,p'}^{-1}A_{y,p}-1\|\le 12\varepsilon\big[(d+d')(r_p+1)+3d^2+3d'^2\big].$$

For each fixed p, the row weights summed over y,p' have zeroth, first and second displacement moments bounded by N_{B0}, N_{B1} and N_{B2}, respectively, for either d or d'. The first two constants are those of Step 7 of the proof of Lem. 11.15; the last is

$$N_{B2}=3K_*^2\mathsf{g}(\lambda_*/2)\,600(\lambda_*/2)^{-5}\le 4.9\cdot10^{141}.$$

These bounds follow by summing $K_*^2e^{-\lambda_*(d+d')}$ over the three orientations of p' and then using the shell sums of Def. 2.44. Thus the transport discrepancy contributes at most

$$\varepsilon^2(24N_{B1}V_1+72N_{B2}W_1),\qquad V_1:=\sum_p(r_p+1)|(\mathrm{d}w_R)_p|.$$

To bound the new sum, use $|\mathrm{d}w_R| \le \min\{2, 3r_1^4 r^{-4}\}$ and $72r^2+6$ plaquettes at radius $r \ge 1$. The radii $r \le r_1$ contribute less than $40r_1^4$; at $r > r_1 \ge 1000$ each shell contributes at most $217r_1^4/r$. Since $\mathrm{d}w_R$ is supported at radii at most $R+1$,

$$V_1 \le 500 r_1^4 (1+\log(R+1)) \le r_1^4 R.$$

The last inequality holds for $R \ge 10^6$, as is the case here. Likewise the existing bounds give $W_0, W_1, W_2 \le r_1^4 R$, since $r_1 \ge 289$ and $R \ge 6r_1^3$. Combining the three errors gives

$$\begin{aligned} &|xD\mathrm{En}_{\mathrm{row}} - x\langle F^\alpha, (B-1)\mathrm{d}w_R\rangle| \\ &\quad \le p_0^2 [N_{B0}(5W_2+W_1) + 18N_{B1}W_0 + 24N_{B1}V_1 + 72N_{B2}W_1] \\ &\quad \le 200(N_{B0}+N_{B1}+N_{B2}) r_1^4 p_0^2 R. \end{aligned}$$

Step 4: the two parts together. So $xD\mathrm{En}_\ell = x\langle \tilde{F}^\alpha, B\,\mathrm{d}w_R\rangle + O(p_0^2 R)$, with the undressed scalar B.

Step 5: from F to A. We have $F_p = (\mathrm{d}A)_p + O(28r^2\varepsilon^2)$. Here r is the distance of p. At second order, the four factors $e^{\pm iA_b}$ of $\partial V(p)$ give the term $\frac{1}{2}\sum_{\mathrm{pairs}}[A_b, A_{b'}]$, a sum over six pairs. Of the bonds of p, the one at the neighbouring site has $|A_b| \le (2r+1)\varepsilon$, two have $|A_b|$ at most $r\varepsilon$ or 0, and the fourth has $|A_b| \le 2r\varepsilon$. With the bracket constant 2, that is, with $|[Y,Y']| \le 2|Y||Y'|$, the term of second order is therefore at most $\frac{1}{2}\cdot 2\sum_{\mathrm{pairs}}|A_b||A_{b'}| \le (13r^2+4r)\varepsilon^2 \le 24r^2\varepsilon^2$ for $r \ge 1$. For the higher terms put $S = \sum_b |A_b| \le (6r+1)\varepsilon \le 7r\varepsilon \le 0.035$, since $2r\varepsilon \le 2r_B\varepsilon \le 0.01$. The logarithm coefficient bound proved in Step 3 (i) of Lem. 11.6, with $H(s) = -\log(2-e^s)$, bounds the terms of degree at least three by $H(S) - S - S^2$. The series converges absolutely since $e^S - 1 < 1$. Since

$$H'''(s) = \frac{2e^s(2+e^s)}{(2-e^s)^3}, \qquad H'''(0.035)/6 < 1.18,$$

Taylor's theorem gives $H(S) - S - S^2 \le 1.18 S^3 \le 2.024 r^2\varepsilon^2$, using $r\varepsilon \le 0.005$. For real A_b, the logarithm after multiplication by $-i$, its homogeneous terms and their remainder are traceless Hermitian matrices; their operator norm is the norm $|\cdot|$ used here. The total error is consequently less than $(24+2.024)r^2\varepsilon^2 < 28r^2\varepsilon^2$. At $r=0$ three of the four A_b vanish, so that $F_p = (\mathrm{d}A)_p$ exactly. Since $r_p^2 \le 2r_{p'}^2 + 2|p-p'|^2$, and with $|\mathrm{d}w_R| \le \min\{\sqrt{2}+8/R,\, 3r_1^4 r^{-4}\}$ for the factor 370,

$$\sum_p r^2 |B\,\mathrm{d}w_R|_p \le 2N_{B0}\cdot 225 r_1^4 R + 2N_{B2}\cdot 370 r_1^4.$$

The factor 370 is unchanged: $R \ge 4r_1$ gives $\sqrt{2}+8/R < 1.415$, and the three plaquette types give

$$\sum_p |\mathrm{d}w_R|_p \le r_1^4 \left[81\cdot 1.415 + 216\sum_{m\ge 2} m^{-2} + 18\sum_{m\ge 2} m^{-4}\right] < 337 r_1^4 < 370 r_1^4,$$

using the integral bounds $\sum_{m\ge 2} m^{-2} \le 1$ and $\sum_{m\ge 2} m^{-4} \le 1/3$. The error is 28 times the preceding weighted sum, times p_0^2.

Step 6: summation by parts inside B'. The form $B\,\mathrm{d}w_R$ lives in $r \le R+2\rho+1$, and

$$\langle \mathrm{d}A^\alpha, B\,\mathrm{d}w_R\rangle = \langle A^\alpha, M w_R\rangle = \langle A^\alpha, \mathrm{d}^* u\rangle - e(A^\alpha) = u^\alpha(F) + O(100\varepsilon^2) - e(A^\alpha).$$

Step 7: the shell. By Lem. 11.15 (d) and (j7) (iii),

$$|e(A)| \le \|e\|_1 \cdot 2(R+2\rho+2)\varepsilon \le 81 C_e R^{-2}\cdot 2.5 R\varepsilon \le 203 C_e \varepsilon/R.$$

The unknown infrared part of the field enters the proof only in this step; as u annihilates constants, it is bounded by a constant times ε/R, where ε is the bound of a tame field.

Step 8: the constant C_2. The constant $C_2 := 10^{670}$ covers Steps 2, 3, 5 and 6; the largest of their contributions is $28\cdot 740 N_{B2} r_1^4 \le 1.1\cdot 10^{666}$. The contributions are collected in the rows for C_2 in item 17 of §11. □

Lemma 11.17 (Three bounds for the covariance and the variance; the list of errors)**.** In the setting of Thm. 11.1, put $q := \langle \mathrm{d}w, \mathbb{D}_k[f_*]\mathrm{d}w\rangle = \sum_y \mathbb{D}_{k,y}[f_*](\omega,\omega)$. Then the following three bounds hold at $j \in J(\mathsf{b})$.

(i) $|\mathrm{Cov}_\nu(G, S_f) - (3/2)q| \le$ (E2)+(E3)+(E4)+(E6).
(ii) $|\mathsf{b}\,\mathrm{Cov}(G_-, \mathcal{O}_K(f_*)) - (3/2)q| \le$ (E1)+…+(E6).
(iii) $|\mathrm{Var}(G_-) - (3/2)a_u^2| \le$ (E7).

Here the errors are the following quantities.

(E1) is the number

$$(9/4)p_0^2 \cdot 1.2124 N_1[2C_3K_5E_a p_0^3 x^{-1/4} + 0.12 c_Q^{\mathrm{su}} K_5 (Lx)^{1/2} c_h(x)/(E_a p_0(Lx))];$$

(E2) is the number $5C_S C_1 p_0^4/R$, and this number is at most $c_F/(12p_0)$;
(E3) is the number $5C_S(C_2R + C_a(x))p_0^5 x^{-1/2}$;
(E4) is the number $10^5 W_* C_S p_0^4 \nu_P$, where $\nu_P = 3(2R+4\rho+7)^3 e^{-(c_P/2)p_0^2}$;
(E5) is the bound given by Lem. 11.10;
(E6) is the number

$$6[C_6 p_0^5 x^{-3/2} + 1.5 e C_S p_0^3 x^{3/2}(2.5(2R/M + 2r(x) + 3)^3 c_h(x) + 6e^{2.2}x^{-4} + 2\mathrm{ex})];$$

(E7) is the number 4[(E2)+(E3)+(E4)+(E6)].

■

Proof. Throughout the proof we write $D = D^\alpha_{w_R}$.

Step 0: the tool. On $\mathrm{supp}\,\chi$ we have $D\Lambda^a = -xu^\alpha + \mathsf{k}$, where $|\mathsf{k}| \le C_1 x\varepsilon/R + C_2 p_0^2 R + C_a(x)p_0^2$: Lem. 11.16 is used for $-xD\mathrm{En}_\ell$, and the first bound of the part "Besides" of Lem. 11.14, namely $|D_w \sum 1\bar{E}^a| \le C_a(x)p_0^2$, accounts for the term $C_a(x)p_0^2$ of k. Neither of the two depends on the "Claim" of Lem. 11.14, and that first bound does not depend on Lem. 11.13 either. This remainder satisfies $|\mathsf{k}| \le 0.5 p_0 x^{1/2}$: the choice of R yields $C_1/R \le c_F/(60C_S p_0^5)$, and the inequality (E3) $\le c_F/12$ yields $(C_2R + C_a)p_0^2 \le c_F x^{1/2}/(60C_S p_0^3)$. The identity and this bound remain valid on the set $\{\phi_p \le 3\varepsilon/4 + 4t$ on $B'\}$, on which $|xu^\alpha| \le 1.5p_0x^{1/2} + 8/x$. The two bounds on C_1/R and on $(C_2R+C_a)p_0^2$ give $|\mathsf{k}| \le c_F x^{1/2}/(30C_S p_0^3) \le x^{1/2}$, where the second inequality follows from $c_F < 1 \le C_S$. Since $8/x \le x^{1/2}$ by $x \ge L$ and $0.5p_0x^{1/2} \ge 2x^{1/2}$ by $p_0 \ge 4$, we obtain $|\mathsf{k}| + 8/x \le 0.5p_0x^{1/2}$. Hence we have $\sup|D\Lambda^a| \le 2p_0x^{1/2}$ on that set, and this is the datum that Lems. 11.13 and 11.14 take as input. With it, Lem. 11.14 states that $E[xu^\alpha\Phi^\alpha] = E[D\Phi^\alpha] + O(\sup|\Phi|\sup|\mathsf{k}|) + \mathrm{def}(\Phi)$, where the defect $\mathrm{def}(\Phi)$ is bounded by the right side of the "Claim" of Lem. 11.14 at $t = x^{-2}$:

$$|\mathrm{def}(\Phi)| \le tB_2 + t^{-1}e\sup|\Phi|\,(2.5(2R/M + 2r(x) + 3)^3 c_h(x) + 6e^{2.2}x^{-4} + 2\mathrm{ex}),$$

with $B_2 = \sup|e^{-\Lambda^a} D^2(\Phi e^{\Lambda^a})|$.

On $\mathrm{supp}\,\chi$ one has $|u(F)| \le 2(3\varepsilon/4)$, and so $G \le (9/8)p_0^2$. For the transported field strength we have

$$D^\alpha \tilde{F}_p^\beta = (\mathrm{d}w_R)_p \delta_{\alpha\beta} + O(\varepsilon(5(r+1)|w|_p + |\mathrm{d}w|_p + 2\sum_{\Gamma(0,\pi p)} |w|)).$$

For the two plaquettes defining u, the numerical error can be bounded independently of their radii. Work in the comb gauge. In each of the 23-plaquettes based at 0 and e_2, three edges belong to the comb, so their link values are 1. If g is its plaquette holonomy, Step 7.0 of the proof of Lem. 11.14 gives

$$g(s) = e^{isae_\alpha} g e^{-isbe_\alpha}, \qquad |a|, |b| \le 2, \qquad a - b = (\mathrm{d}w_R)_p.$$

Here a and b are the sums of the weights on the two pairs of edges. The logarithm differential from Step 1 of the proof of Lem. 11.16 obeys $\|d\mathsf{L}_g\| \le 1.01$: its perpendicular norm is $\phi/\sin\phi \le$

$(1-\phi^2/6)^{-1} < 1.01$ for $\phi \le 0.01$. Since $\|\operatorname{Ad}_g - 1\| \le 2\phi$, the left-trivialized generator $ae_\alpha - b\operatorname{Ad}_g e_\alpha$ gives

$$|D^\alpha F_p - (\mathrm{d}w_R)_p e_\alpha| \le 4 \cdot 1.01\varepsilon + 2 \cdot 1.01 \cdot 2\varepsilon = 8.08\varepsilon.$$

The derivative of the transport to the origin vanishes at the first base. At the base e_2 its signed comb sum has modulus at most 1, and its contribution is at most 2ε. Thus the error for their difference is at most $18.16\varepsilon < 50\varepsilon$. As $\chi_R = 1$ near these plaquettes and $\langle u, \mathrm{d}w\rangle = a_u$, we obtain $D^\alpha u^\beta = a_u\delta_{\alpha\beta} + O(50\varepsilon)$.

Step 1: the first identity. Let $\Phi^\alpha = u^\alpha\chi(S_f - \bar{s})/2$, with $\bar{s} = ES_f$; then $\sup|\Phi| \le 1.5C_S\varepsilon p_0^2$. On the left, $\sum_\alpha xu^\alpha\Phi^\alpha = G(S_f - \bar{s})$, and the mean of this product is $\operatorname{Cov}_\nu(G, S_f)$. On the right, $\sum_\alpha D\Phi^\alpha$ is the sum of the three parts (a), (b), (c) below.

(a) The first part equals $(3/2)a_u\chi(S_f - \bar{s}) + O(\varepsilon C_S p_0^2)$. In the mean of its main term the factor $E[\chi(S_f - \bar{s})] = E[(\chi - 1)(S_f - \bar{s})] \le 2C_S p_0^2\nu_P$ appears.

(b) The second part is the one in which $D\chi$ occurs. Here $|D\chi| \le 8(W_1 + 5\varepsilon W_2)/\varepsilon$, and $D\chi$ vanishes outside the event {some $\phi_p \ge \varepsilon/2$ in B'}; the mean of this part is therefore at most $(3/2)(3\varepsilon/2)(16W_*/\varepsilon)(2C_S p_0^2)\nu_P$.

(c) The third part is $(1/2)\sum_\alpha u^\alpha\chi DS_f$.

The remainder k produces two further errors. Its shell term $C_1 x\varepsilon/R$ costs $3 \cdot 1.5C_S\varepsilon p_0^2 \cdot C_1 x\varepsilon/R = 4.5C_S C_1 p_0^4/R$, and the remaining part of k costs $4.5C_S p_0^3 x^{-1/2}(C_2 p_0^2 R + C_a p_0^2)$.

Step 2: the derivative of S_f. We prove

$$DS_f = x\langle \mathbb{D}_k \mathrm{d}w, \tilde{F}^\alpha\rangle + O(C_D p_0^2), \qquad C_D := 10^{200}C_S,$$

where C_D is a constant of this step. By (U2.3), only the centers $y \in Y_{\mathrm{out}}$ occur, and $|y| \le D_0$. A plaquette in the range of such a center obeys $|\pi(p)| \le D_0 + \rho(x)$, which may exceed $2D_0$. Nevertheless, (j7) (ii) and $\rho(x) \le Mr(x) - 2$ put all these plaquettes in the region where $\chi_R = 1$. Thus $\mathrm{d}w_R = \mathrm{d}w$ there. The expanded source cubes lie in B', so every indicator in S_f is identically 1 in a neighborhood of $\operatorname{supp}\chi$.

Work at the field in the root comb gauge and set

$$d = |y - \pi(p)|_\infty, \qquad d' = |y - \pi(p')|_\infty, \qquad A_{y,p} = \operatorname{Ad}_{V(\Gamma(y,\pi(p)))}.$$

There are at most $3d$ bonds on this path, each based at radius at most $D_0 + d$. The comb-gauge bound of Step 0 of the proof of Lem. 11.16 and the adjoint bound of Lem. 4.20 (b) therefore give

$$\|A_{y,p} - 1\| \le 12d(D_0 + d)\varepsilon, \qquad \|DA_{y,p}\| \le 6d.$$

The second estimate follows by differentiating the path product: each bond has speed at most $|w_b| \le 1$, and the adjoint action has differential norm at most twice that speed. All these paths and their combs lie in B'. By Lem. 11.15 (a0), $|\mathrm{d}w_R| \le \sqrt{2} + 8/R < 2$, while $|w|_p \le 4$. Step 1 of the proof of Lem. 11.16 consequently yields

$$|DF_p - (\mathrm{d}w)_p e_\alpha| \le \varepsilon[20(D_0 + d) + 22].$$

Differentiate the quadratic expression of entry 5 of item (0) of §6.1. Symmetry of $\mathbb{D}_{k,y}$ combines the two differentiated factors. Since $|F_p|, |F_{p'}| \le \varepsilon$, the preceding three estimates bound the error for a pair (p, p'), after removing $x|\mathbb{D}_{k,y}(p, p')|\varepsilon^2$, by

$$20D_0 + 26d + 22 + 24D_0(d + d') + 24(d^2 + d'^2) \le 100(D_0 + 1)(1 + d + d')^2.$$

Here the last two terms account for replacing both $A_{y,p}$ and $A_{y,p'}$ by 1 in the leading term; at the field in this gauge, $\tilde{F} = F$.

The displacement factors are summable uniformly in the range. Put $\lambda = \lambda_\flat$. For $s \ge 0$,

$$(1 + s)^2 e^{-\lambda s} \le (1 + 4/(e\lambda))^2 e^{-\lambda s/2}.$$

By (U2.1) and the three plaquette orientations, the error is at most

$$100(D_0+1)p_0^2\#Y_{\mathrm{out}}K_5(1+4/(e\lambda))^2[3\mathsf{g}(\lambda/2)]^2.$$

For completeness, writing $z=2v+e$ with $e\in\{0,1\}^3$ gives $\mathsf{g}(\lambda/2)\le 8e^{\lambda/2}\mathsf{g}(\lambda)$. Also $\lambda=\lambda_*/16>10^{-15}$ by the pure-number intervals in Thm. 2.65, and $\lambda\le 1$. Comparing with $C_S=\frac12\#Y_{\mathrm{out}}K_5[3\mathsf{g}(\lambda)]^2$, the coefficient is at most

$$200(D_0+1)(1+4/(e\lambda))^2\left[\frac{\mathsf{g}(\lambda/2)}{\mathsf{g}(\lambda)}\right]^2 C_S<10^{176}C_S\le C_D.$$

This proves the derivative estimate without an upper bound on $\rho(x)$ or on the depth.

Step 3: the second identity. Let $\Phi'^{\alpha}=\chi\langle\mathbb{D}_k\mathrm{d}w,\tilde F^{\alpha}\rangle/2$; its supremum is at most $(\varepsilon/2)\cdot 2\sqrt{2}C_S$. The sum $\sum_\alpha E[xu^\alpha\Phi'^\alpha]$ is the mean of the leading term in part (c) of Step 1. By Step 2, the error in that substitution is at most $\frac94 C_D\varepsilon p_0^2$, since $|u^\alpha|\le 3\varepsilon/2$ and there are three values of α.

The derivative of the root transport at p contributes at most $6|\pi(p)|\varepsilon$. Together with Step 1 of the proof of Lem. 11.16, this gives

$$|D^\alpha\tilde F_p-(\mathrm{d}w)_p e_\alpha|\le\varepsilon(26|\pi(p)|+22).$$

The same displacement-moment bound as in Step 2, using $|\pi(p)|\le D_0+|y-\pi(p)|_\infty$ and $|\mathrm{d}w|\le 2$, therefore shows that

$$\sum_\alpha D^\alpha\Phi'^\alpha=\tfrac32 q\chi+O(3C_D\varepsilon)+\tfrac12\sum_\alpha(D^\alpha\chi)\langle\mathbb{D}_k\mathrm{d}w,\tilde F^\alpha\rangle.$$

The last term and the replacement of $q\chi$ by q are included in (E4), as are the cutoff terms of Step 1. The derivative error in part (a) of Step 1 is at most $150C_S\varepsilon p_0^2$, by the 50ε estimate of Step 0. Thus all three non-cutoff derivative errors just identified total at most

$$150C_S\varepsilon p_0^2+\tfrac94 C_D\varepsilon p_0^2+3C_D\varepsilon\le 10^{202}C_S p_0^3x^{-1/2}.$$

We record the budget in (E2) and (E3). The two shell costs are at most $C_SC_1p_0^4R^{-1}(4.5+4.3/p_0^2)<5C_SC_1p_0^4/R$. The two remaining costs from k are at most

$$C_S(C_2R+C_a(x))p_0^5x^{-1/2}(4.5+4.3/p_0^2).$$

Since $p_0\ge 4$, $R\ge 1$ and $C_2=10^{670}$, $10^{202}\le 0.01C_2Rp_0^2$. The displayed derivative errors therefore add at most 0.01 to this last coefficient, and $4.5+4.3/16+0.01<5$. All these errors fit within (E2) and (E3); the integration-by-parts defects are those of (E6). This proves (i).

Step 4: the bound (ii). Lems. 11.3 and 11.4 give

$$\mathsf{b}\,\mathrm{Cov}(G_-,\mathcal{O}_K(f))=\mathrm{Cov}_\nu(G_-,S_f)+\mathsf{b}\,\mathrm{Cov}_\nu(G_-,\mathrm{Rem})+\mathsf{b}\,E[(G_--g)W_f].$$

Here $f=f_*$ and $g:=EG_-$, so that $E(G_--g)=0$. The second term is at most $2(9/8)p_0^2\mathsf{b}\sup|\mathrm{Rem}|\le$ (E1), and the third is bounded by (E5). In the first term we compare G_- with G: one has $G_--G\neq 0$ only where some $\phi_p>\varepsilon/2$ in B', so $|\mathrm{Cov}(G_--G,S_f)|\le(9/2)C_Sp_0^4\nu_P$, an amount that is included in (E4).

Step 5: the bound (iii). The third identity uses $\Phi=u^\alpha\chi/2$ and gives $EG=(3/2)a_u+\mathrm{err}_1$, the shell part of err_1 being $(9/4)C_1p_0^2/R$. The fourth identity uses $\Phi=(x/4)u^\alpha|u|^2\chi^2$, whose supremum is at most $(x/4)(3\varepsilon/2)^3=(27/32)\varepsilon p_0^2$, smaller than $1.5C_S\varepsilon p_0^2$. For this choice $\sum_\alpha xu^\alpha\Phi=G^2$, and $\sum_\alpha D(u^\alpha|u|^2)=(3+2)a_u|u|^2+O(\varepsilon^3)$, so that $\sum_\alpha D\Phi=(5/2)a_uG\chi+\ldots$, up to terms that enter err_2. Hence $EG^2=(15/4)a_u^2+\mathrm{err}_2$, and so $\mathrm{Var}(G)=(3/2)a_u^2+\mathrm{err}_2-3a_u\mathrm{err}_1-\mathrm{err}_1^2$. Here err_2 contains, besides the errors of the fourth identity itself, the term $(5/2)a_u\,\mathrm{err}_1$. Each entry of err_1 and of err_2 is at most C_S^{-1} times the error that arises in the same way, namely (E2) for the shell term, (E3) for the remaining part of k and, as in Step 3, for the terms $O(\varepsilon)$, (E4) for the rows with ν_P and (E6) for the defect: the suprema $3\varepsilon/4$ and $(27/32)\varepsilon p_0^2$ take the place of $1.5C_S\varepsilon p_0^2$, the rows with ν_P have no factor C_S, and for both functions $B_2\le 10^3N_{B0}W_*^2p_0^5x^{1/2}$. Since [(E2)+(E3)+(E4)+(E6)] <1,

we obtain $|\mathrm{err}_2| + 6|\mathrm{err}_1| + \mathrm{err}_1^2 \le 8C_S^{-1}[(\mathrm{E2})+(\mathrm{E3})+(\mathrm{E4})+(\mathrm{E6})]$, which lies far below (E7). The passage from G to G_- costs $(81/16)p_0^4\nu_P$.

Step 6: the defect. Work on the tame segments of Step 2 of the proof of Lem. 11.14. The moving plaquettes and the plaquettes entering a varying row or its mask lie in B'. On these segments the corresponding energy terms are bare curvature squares and dressed row squares.

For the plaquette product in Step 7.1 of that proof, set $\delta_p = |\mathrm{d}w|_p + 5(r_p+1)\varepsilon|w|_p$. Its first derivative has norm at most δ_p. Its second derivative is the square of the first-order generator plus the pairwise commutators, followed by right multiplication by $\partial V(p)$. Each commutator of unit generators has norm at most $20(r_p+1)\varepsilon$, by comparison with their common generator. The absolute pair weights sum to at most $|w|_p^2/2$. On $\|g-1\| \le \varepsilon \le 1/2$ the first and second derivatives of the matrix logarithm have norms at most 2 and 4, by its power series. It follows that

$$|DF_p| \le 2\delta_p, \qquad |D^2F_p| \le 6\delta_p^2 + 20(r_p+1)\varepsilon|w|_p^2.$$

Put $A = \sum_p \delta_p$ and $B = \sum_p |D^2F_p|$. By the sums in Lem. 11.14 and the weighted square sum $\sum_p (r_p+1)|w|_p^2 \le 10^3 r_1^4$ in the numerical bounds below, $A \le 0.401 W_*$ and $B \le W_*^2$. The bare curvature squares contribute at most $4A^2 + \varepsilon B < 2W_*^2$.

For a path Γ put $a_\Gamma = \sum_{b\in\Gamma} |w_b|$. Differentiation of its adjoint product gives $\|DT\| \le 2a_\Gamma$ and $\|D^2T\| \le 4a_\Gamma^2$. At displacement v, the path has length at most $3|v|_\infty$, so $a_\Gamma \le 3|v|_\infty$ and its ℓ^2 norm over path translates is at most $3|v|_\infty W_{00}^{1/2}$. For $G_y = \ell_y^V F$, apply the row bound $K_* e^{-\lambda_*|v|_\infty}$ and sum over the three plaquette types. Schur's bound on the term with DF and the triangle inequality on the transport terms give

$$\|DG\|_2 \le 2\sqrt{N_{B0}-1}\, A + 18K_*\mathtt{g}_1(\lambda_*)\varepsilon W_{00}^{1/2},$$
$$\sum_y |D^2G_y| \le K_*\mathtt{g}(\lambda_*)B + 24K_*\mathtt{g}_1(\lambda_*)A + 108K_*\mathtt{g}_2(\lambda_*)\varepsilon W_{00}.$$

In the second line, sum the terms differentiating F first over the row centers. For the term with D^2T, sum a_Γ^2 over the translates before summing the displacements. On the varying rows, $|G_y| \le 3K_*\mathtt{g}(\lambda_*)\varepsilon$. The bounds in Def. 2.44, together with $\mathtt{g}_2(\lambda) \le (1+4/(e\lambda))^2\mathtt{g}(\lambda/2)$, give $K_*\mathtt{g}_1(\lambda_*) < 10^{72}$ and $K_*\mathtt{g}_2(\lambda_*) < 10^{88}$. With $W_{00} \le 60 r_1^3$ and $W_* = 10^3 r_1^4$ we get $\|DG\|_2 < 0.81\sqrt{N_{B0}}W_*$ and $\sum_y |G_y||D^2G_y| < 0.51 N_{B0}W_*^2$. Including the bare part gives $|D^2\mathrm{En}_\ell| \le 5N_{B0}W_*^2$.

Where a derivative of χ_0 is nonzero, $\phi_p \ge \varepsilon/2$. The group distance gives $|D\phi_p| \le \delta_p$. Differentiating $|F_p|$ then gives $|D^2\phi_p| \le |D^2F_p| + 8\delta_p^2/\varepsilon$. The product rule yields

$$|D\chi| \le 8A/\varepsilon, \qquad |D^2\chi| \le [(64+250+64)A^2 + 8\varepsilon B]/\varepsilon^2 \le 100 W_*^2/\varepsilon^2.$$

The transports in S_f, u and the linear form of Step 3 have displacement at most R, with $\varepsilon R^2 \le 1/288$. Thus $|D(TF)| \le 2A + 6R\varepsilon \le W_*$ and

$$\varepsilon|D^2(TF)| \le \varepsilon B + 24R\varepsilon A + 36R^2\varepsilon^2 \le W_*^2.$$

For $\widehat{D} = \varepsilon D/W_*$, the bounds on TF and its first two scaled derivatives are therefore all ε. The sum of absolute quadratic coefficients of $2S_f/x$ is at most $2C_S$, by the definition of C_S. The sum of absolute coefficients of $\langle \mathbb{D}_k \mathrm{d}w, \tilde{F}^\alpha\rangle$ is at most $8C_S$, since $|\mathrm{d}w|_p \le 4$. The product rule gives $|D^i\Phi| \le 10^3 C_S \varepsilon p_0^2 (W_*/\varepsilon)^i$ for $i = 1, 2$ in the four tests of Steps 1, 3 and 5. For the last two tests one can replace C_S by 1. For example, the cubic $u^\alpha|u|^2$ and its first two scaled derivatives are bounded by $8\varepsilon^3$, $24\varepsilon^3$ and $72\varepsilon^3$. Those of χ^2 are bounded by 1, 8 and 232. The factor $x/4$ in the last test gives the stated bound.

Finally,

$$e^{-\Lambda^a} D^2(\Phi e^{\Lambda^a}) = D^2\Phi + 2D\Phi D\Lambda^a + \Phi\big(D^2\Lambda^a + (D\Lambda^a)^2\big).$$

Use $|\Phi| \le 1.5 C_S\varepsilon p_0^2$, $|D\Lambda^a| \le 2p_0 x^{1/2}$ and the energy estimate above. The Cauchy term in $|\Phi D^2\Lambda^a|$ is at most $48 C_S E_a (2r+1)^3 W_*^2 p_0^4$ by Step 7.3 of the proof of Lem. 11.14. This is

$19.2 C_S W_* p_0^4 C_a(x)$ and hence at most $0.32 W_* c_F x^{1/2}/p_0$ by (E3) $\le c_F/12$. Adding the estimates gives $B_2 \le 10^3 C_S N_{B0} W_*^2 p_0^5 x^{1/2} \le C_6 p_0^5 x^{1/2}$. For the two tests of Step 5 the same bound holds with $10^3 N_{B0} W_*^2$ in place of C_6. Apply Lem. 11.14 six times for (i) and six times for (iii). The resulting defect in (i) is bounded by (E6), and the defect in (iii) is at most C_S^{-1} times that bound. □

Remark 11.18 (The factor 3/2). For $Y \cdot Y' = (1/2)\operatorname{tr} YY'$ there are three colours of unit length. For Gaussian colours with covariance C/x one finds

$$\operatorname{Cov}((x/2)|u\cdot\phi|^2, (x/2)\langle\phi, D\phi\rangle) = 3\cdot(1/4)\cdot 2\langle Cu, DCu\rangle.$$

Lemma 11.19 (The lower bound for q)**.** In the setting of Thm. 11.1 the following hold.

(A) For y off $\operatorname{supp} u$ with $r = |y|_\infty \ge 1.2\cdot 10^5$, one has $|(\Pi u)(y) - \pi(y)| \le C_{\mathrm{lat}} r^{-5}$, where $C_{\mathrm{lat}} = 10^{11}$ and $\pi_i := \partial_i(3y_1y_2/(4\pi|y|_2^5))$.

(B) At $D_0 = 10^{140}$, for all rows of the runs, all $1 \le k \le K$ and all $x \ge L$, one has $q \ge c_F$, and $5.4\cdot 10^{-5} D_0^{-5} \le c_F \le 4.0\cdot 10^{-4} D_0^{-5}$.

■

Proof of (A). *Step 1: the multiplier.* We pass to the dual picture, in which a 2-form becomes a vector field through the normal directions of the plaquettes. Then $u = e_1(\delta_0 - \delta_{e_2})$ and $\Pi u = u - \nabla\Delta^{-1}\operatorname{div} u$. Off $\operatorname{supp} u$ this gives $(\Pi u)_i = [m_i]^\vee$ with $m_i = k_i^{\#} k_1 k_2^{\#}/(k\cdot k^{\#})$. On $\mathbb{R}^3$ the multiplier $m_{0,i} = -ip_ip_1p_2/|p|^2$ has the transform π_i, because $\partial_1\partial_2(1/r) = 3y_1y_2/r^5$.

Step 2: bounds on small discs. All Fourier-vector norms in this proof are Euclidean. For each coordinate μ, take $q = p + ze_\mu$ with $|z| \le P/20$, where $0 < P = |p|_2 \le 1/2$. Then $|q|_2 \le 1.05P$ and $|\sum_\nu q_\nu^2| \ge 0.8975P^2$. Write $k_\nu = iq_\nu(1+e_\nu)$ and $k_\nu^{\#} = -iq_\nu(1+e_\nu^{\#})$, using the continuous values at $q_\nu = 0$. The exponential series gives $|e_\nu|, |e_\nu^{\#}| \le 0.6304P$. The cosine series gives

$$\Big|k\cdot k^{\#} - \sum_\nu q_\nu^2\Big| \le 0.0842 \sum_\nu |q_\nu|^4 \le 0.0842(1.05P)^4.$$

Thus $k\cdot k^{\#} = (\sum_\nu q_\nu^2)(1-\eta)$ with $|\eta| \le 0.1141P^2 < 0.0793P$. Moreover,

$$|m_0(q)|_2 \le \frac{|q|_2^3}{2|\sum_\nu q_\nu^2|} \le \frac{1.05^3}{2\cdot 0.8975} P < 0.757P.$$

The three numerator factors and the denominator therefore give

$$|m(q) - m_0(q)|_2 \le 0.757P\left(\frac{(1+0.6304P)^3}{1-0.0793P} - 1\right) \le 2.08P^2.$$

The quotient of the parenthesis by P increases on $(0, 1/2]$; its endpoint gives the last bound. These one-coordinate discs suffice for every Cauchy derivative estimate used below.

Step 3: the difference as an integral. We have

$$\Pi u - \pi = (2\pi)^{-3}\Big[\int_{T^3} (m - m_0)e^{ipy} - \int_{\mathbb{R}^3\setminus T^3} m_0 e^{ipy}\Big].$$

To bound the right side we follow Step 4 of the proof of Lem. 11.15, where the inverse transform of a multiplier is estimated, and we take $n = 6$. The homogeneous multiplier m_0 has degree one. Interpret its inverse transform as a tempered distribution. To justify the integration by parts in the high-frequency contribution, insert a smooth cutoff that equals one on $|p| \le R_\infty$ and vanishes for $|p| \ge 2R_\infty$. Six integrations by parts produce terms with derivatives of this cutoff. A term with $j \ge 1$ cutoff derivatives is $O(R_\infty^{-j} R_\infty^{1-(6-j)})$ on the annulus. Its integral is $O(R_\infty^{-2})$ and tends to zero. The sixth derivative of m_0 is integrable at infinity. This gives the inverse transform away from the origin with the bounds below. The boundary terms of m at opposite faces of the torus cancel by periodicity and $y \in \mathbb{Z}^3$. For m_0 the inner and outer face terms cancel before the cutoff is removed.

Step 4: the numbers. With the bound of Step 2, the argument yields the constant $c_{2,6} \cdot 6! \cdot 20^6 \cdot 2.08 < 7.0 \cdot 10^{10}$. The multiplier m_0 is homogeneous of degree 1, so the bound $|m_0| \le 0.757|p|$ of that step holds on the discs $|q-p| \le |p|/20$ for every p; hence m_0 on $|p| > 1/2$, on both sides of ∂T^3, contributes $3.6 \cdot 10^9$, a coefficient of r^{-6}. For m on $|p|_2 > 1/2$, use the one-coordinate discs $q = p + ze_\mu$, $|z| \le 1/100$, and put $K = |k(p)|_2 \ge 0.318$. Writing $D(q) = k(q) \cdot k^{\#}(q)$, the cosine formula and $|\sin p_\mu| \le |k_\mu(p)|$ give

$$|D(q) - D(p)| \le 2K \sinh(0.01) + 2(\cosh(0.01) - 1) < 0.064K^2.$$

Thus $|D(q)| \ge 0.935K^2$. Also $|k_\nu(q) - k_\nu(p)|, |k_\nu^{\#}(q) - k_\nu^{\#}(p)| \le 0.0102$, and only one component changes. Hence

$$|k_1(q)k_2^{\#}(q)| \le \tfrac{1}{2}K^2 + 0.0102K < 0.547K^2, \qquad |k^{\#}(q)|_2 \le \sqrt{12} + 0.0102.$$

Consequently $|m(q)|_2 \le (\sqrt{12} + 0.0102)\, 0.547/0.935 < 2.1$, and Cauchy's estimate gives $|\partial_\mu^6 m|_2 \le 6! \cdot 100^6 \cdot 2.1 < 3.3 \cdot 10^{15}$. In sum, $7.0 \cdot 10^{10} r^{-5} + 3.4 \cdot 10^{15} r^{-6} \le 10^{11} r^{-5}$. □

Proof of (B). We use $r_c := |y_c|_2 \in [D_0, D_0 + \sqrt{2}]$ and the two sets $Y_{\text{out}} = \{|y - y_c| \le D_0/8 + 1/2\}$ and $Y_{\text{in}} = \{|y - y_c| \le D_0/16 - 1\}$.

Step 1: localisation. We write $\mathbb{D}_y := \mathbb{D}_{k,y}[f_*]$ and $Q_y := \mathbb{D}_{k,y}[f_*](\omega, \omega)$. In units of level k, the barycentres of the plaquettes based in $B^k(y)$ lie within $1/2$ of y; the places of the sites and the barycentres $x(p)$ are those of the definition "Places" in item 1 of §6.3. With Thm. 6.20 (U2.3) and $\operatorname{supp} f_0 \subset \{|y - y_c| \le D_0/8\}$, this gives $q = \sum_{Y_{\text{out}}} Q_y$. By Thm. 6.20 (U2.1), $|\mathbb{D}_y(p, p')| \le K_5 e^{-\lambda_\flat(|y-p| + |y-p'|)}$.

Step 2: the region. The set $Z = \{|p - y_c| \le 0.1875 D_0 + 1/2\}$ contains every p with $|y - p| \le D_0/16$ for some $y \in Y_{\text{out}}$, and on Z we have $|p|_\infty \ge 0.5195 r_c \ge r_1$ and $|p|_2 \ge 0.7347 r_c$. A closed form is available: with $Y = 3y_1y_2$,

$$|\nabla(Y/r^5)|^2 = |\nabla Y|^2/r^{10} + (25 - 20)Y^2/r^{12} = 9(y_1^2 + y_2^2)/r^{10} + 45 y_1^2 y_2^2/r^{12} \le 20.25/r^8.$$

Therefore $|\pi| \le (20.25/157.9)^{1/2}/0.7347^4\, r_c^{-4} < 1.23 r_c^{-4}$ on Z. To bound the Jacobian, put $n = y/r$, where $r = |y|_2$. Direct differentiation gives

$$\begin{aligned} r^5 \partial_i\partial_j\partial_1\partial_2(1/r) = {} & 105 n_i n_j n_1 n_2 \\ & - 15(\delta_{ij} n_1 n_2 + \delta_{i1} n_j n_2 + \delta_{i2} n_j n_1 + \delta_{j1} n_i n_2 + \delta_{j2} n_i n_1) \\ & + 3(\delta_{i1}\delta_{j2} + \delta_{i2}\delta_{j1}). \end{aligned}$$

For $i = j \in \{1, 2\}$ the right side has modulus at most 30; for $\{i, j\} = \{1, 2\}$ it is at most $177/4$; the remaining cases are at most 45. These bounds use $|n_a n_b| \le 1/2$ for $a \ne b$ and $\sum_a n_a^2 = 1$. In particular every component is bounded by 105. Since $(\operatorname{Jac} \pi)_{ij} = (4\pi)^{-1} \partial_i\partial_j\partial_1\partial_2(1/r)$, the nine entries and the lower bound $r \ge 0.7347 r_c$ give $|\operatorname{Jac} \pi|_F \le G_4 r_c^{-5}$ with $G_4 = 3 \cdot 105/(4\pi \cdot 0.7347^5) < 117.1$. On Y_{in} one has $y_1, y_2 \ge 0.6446 r_c$ and $r \le 1.0902 r_c$; it follows that $|\pi|^2 r_c^8 \ge 0.0374$ there, and also $|\pi|^2 r_c^8 \le 0.2689$.

Step 3: ω against π on Z. Lem. 11.15 (b) gives

$$|\omega - \Pi u| \le [C_G \cdot 50.9/r_c^2 + 13.8 C_\tau \tau_x^2] r_c^{-4} \le (10^{-98.6} + 3 \cdot 10^{150} x^{-512}) r_c^{-4},$$

and (A) gives $|\Pi u - \pi| \le 26.5 C_{\text{lat}} r_c^{-5}$.

Step 4: the split. For fixed y let $\bar\omega_y$ be the constant form that has the components of ω at y, and set $d_p := \omega_p - \bar\omega_{y,\iota(p)}$. For $|y - p| \le D_0/16$,

$$|d_p| \le \sqrt{3}|y - p| G_4 r_c^{-5} + 2 C_{\text{lat}} (0.5195 r_c)^{-5} + 3|y - p|[C_G'(0.5195 r_c)^{-7} + \ldots];$$

here the gradient is taken only of the smooth function π and of $\omega - \Pi u$. Put $\mathtt{t} := r_c^4 \sum_p e^{-\lambda_\flat |y-p|} |d_p|$. Then $\mathtt{t} \le 3\sqrt{3} g_{1\flat} G_4/D_0 + 159 \mathsf{g}_\flat C_{\text{lat}}/D_0 + \ldots +$ the far part of the sum, and this bound is at most $3.3178 \cdot 10^{-76} + 3.3 \cdot 10^{-81} + 6 \cdot 10^{-158} + 8.5\, r_c^4\, \mathsf{g}' e^{-\lambda_\flat D_0/32}$, where $\lambda_\flat D_0/32 \ge 4 \cdot 10^{123}$. Here

$6 \cdot 10^{-158}$ bounds the contribution of all three terms of the bracket in the bound on $|d_p|$. The term $3|y-p|C_G'(0.5195r_c)^{-7}$ contributes $9g_{1b}C_G'\,0.5195^{-7}D_0^{-3} \le 4.2 \cdot 10^{-176}$, summed over the three orientations; the other two terms, denoted by the dots in the bracket, are those of Lem. 11.15 (b) at $0.5195r_c$, namely $1.9 \cdot 10^{36}\tau_x^2(0.5195r_c)^{-5}$ and $A_{\mathrm{out}}(0.5195r_c)^{-16}$, and the largest of the three contributions is $9g_{1b}A_{\mathrm{out}}0.5195^{-16}D_0^{-12} \le 5.6 \cdot 10^{-158}$.

Step 5: the error from replacing ω by $\bar\omega$. With $\log_{10} K_5 \le 24.776$, $\log_{10} \mathsf{g}_\flat \le 46.315$, the factor 1.23 and $\mathsf{t} \le 3.318 \cdot 10^{-76}$,

$$|\mathbb{D}_y(\omega,\omega) - \mathbb{D}_y(\bar\omega,\bar\omega)| \le K_5(6\mathsf{g}_\flat \cdot 1.23\mathsf{t} + \mathsf{t}^2)r_c^{-8} \le 3.02 \cdot 10^{-4} r_c^{-8}.$$

Step 6: the charge. For the constant form, $\mathbb{D}_y(\bar\omega,\bar\omega) = c_y(\bar\omega,\bar\omega)$. By Thm. 6.21 (S2), c_y coincides with $\operatorname{diag}\mathrm{Mn}$ up to $0.0102/L$ in operator norm. Moreover $\mathrm{Mn} \ge 0$, and $\mathrm{Mn} = 1$ on Y_{in}.

Step 7: the number of points. We have $\#Y_{\mathrm{out}}/\#Y_{\mathrm{in}} \le ((D_0/4+2)/(D_0/8-3))^3 \le 8.01$. Hence

$$q \ge \sum_{Y_{\mathrm{in}}} |\bar\omega|^2 - \#Y_{\mathrm{out}}(3.02 \cdot 10^{-4} + 0.0155/L)r_c^{-8} \ge \sum_{Y_{\mathrm{in}}} |\Pi u|^2[1 - 10^{-97} - 8.01(3.02 \cdot 10^{-4} + 0.0155/L)/0.037],$$

where the term with $1/L$ costs $3.4/L < 10^{-148}$. The bracket is at least $0.93 \ge 3/4$, and so $q \ge c_F$.

Step 8. Bounds on c_F. On Y_{in}, Step 2 gives $0.0374 \le r_c^8|\pi(y)|^2 \le 0.2689$. By (A) the difference $r_c^4|\Pi u - \pi|$ is at most $10^{11}(0.6446)^{-5}/r_c < 10^{-127}$ there. In particular, $0.037 \le r_c^8|\Pi u|^2 \le 0.2690$. Counting sites in the sup cube gives $(D_0/8-3)^3 \le \#Y_{\mathrm{in}} \le (D_0/8+1)^3$. With $D_0 \le r_c \le D_0 + \sqrt{2}$, we obtain

$$5.4 \cdot 10^{-5} D_0^{-5} \le \tfrac{3}{4} \sum_{y \in Y_{\mathrm{in}}} |\Pi u(y)|^2 \le 4.0 \cdot 10^{-4} D_0^{-5}.$$

□

Remark 11.20 (The inner numerals of Step 2 of the proof of (A)). In Step 2 of the proof of (A), the quotient $(e^t - 1 - t)/t$ at $t = 0.525$ is less than $0.6304 \cdot 0.5$; the sharper bound $|\eta| \le 0.069|p|$ lies below $0.0793|p|$; and $0.757[(1+0.6304|p|)^3/(1-0.0793|p|) - 1] \le 2.073|p|$ for $|p| \le 1/2$, which lies below 2.08.

Lemma 11.21 (Assembly of the clauses)**.** In the setting of Thm. 11.1, the clauses (N1) to (N6) and (N6′) of that theorem hold at each $j \in J(\mathsf{b})$. ■

Proof. Step 1: the clause (N1). By Lem. 11.17 (ii), Lem. 11.19 and (j1),

$$\mathsf{b}\,\mathrm{Cov} \ge (3/2)q - 6c_F/12 \ge (3/2)c_F - c_F/2 = c_F.$$

Step 2: the clause (N2). Lem. 11.17 (iii), together with $a_u \le 2$ and (E7) < 1, gives $\mathrm{Var} \le (3/2) \cdot 4 + 1 = 7$.

Step 3: the clause (N3). Clause (W) of Lem. 11.3 applies. Indeed, the support of f_* lies in $y_1 \ge (0.707 - 0.125)D_0 \ge 0.58D_0$ and in $y_1 \le D_0 < NL^j/2$, while G_- depends on P_0, on $P_0 + e_2$ and on the one bond between them, all of which have $y_1 = 0$, and on the plaquettes of B' on the left. By (N1), $\mathrm{Cov} \ge c_F/\mathsf{b} > 0$. Therefore (N0) forces $\mathrm{Var}(G_-) > 0$, and $\mathsf{b}^2 S_2^T(\theta f_*, f_*) \ge c_F^2/\mathrm{Var}(G_-) \ge c_F^2/7$.

Step 4: the clause (N4). The function f_* is smooth, and θf_* and f_* have disjoint supports. Hence Thm. 8.2 (T3) applies: it provides the limit through all cutoffs $K \ge j+1$ at a fixed side, and the closed inequality persists in that limit. For the limits (a) and (c) of Thm. 8.17 (R5) at the fixed label b, one has $K_i \ge j+1$ for every i from some index on, and once more the closed inequality persists.

Consider next the limits (a) and (c) of (R5) along labels $\mathsf{b}_i \to \mathsf{b}$ with $\mathsf{b}_i \ge x_s$. We have $\mathsf{b}_i \in [\mathsf{b}/2, 2\mathsf{b}]$ for all but finitely many i. We take $j \ge j^+(\mathsf{b})$, which is finite by Lem. 11.22 (d); then $j \ge j_*(\mathsf{b}_i)$, and so (N3) holds at (j, b_i). The function f_* and the mirror depend on j alone, whereas G_-, R and h may change with i. Further, $\mathsf{b}_i^2 \to \mathsf{b}^2$, and for all sufficiently large i the sides satisfy (Tor) at $\mathsf{b}_i \le 2\mathsf{b}$. In the case (c) we have in addition $K_i \ge j+1$.

Labels $\mathsf{b}_i < x_s$ can occur only when $\mathsf{b} = x_s$, and they are handled by (N6′). For the limit (b) of (R5) we note that S_2^T is a polynomial in means of bounded continuous local functions; it therefore passes to the weak limit in the side at each $K \geq j+1$, and then along K.

Step 5: the clause (N5). Let S be a limit of (R5). For such limits, and only for them, Thm. 8.17 (R6) (iii) and (R7) supply all real translations, the space $\mathcal{H}(S)$ and the operators e^{-sH}. By (R6) (iii) we move the mirror to that of (R7). In $\mathcal{H}(S)$, let v be the class of the tuple with the single entry f_*. Then $\langle \Omega, v\rangle = \sigma_S(f_*) = 0$, because the centred functional is a sum over partitions into blocks of size ≥ 2. Moreover $\|v\|^2 = S_2(\theta f_*, f_*) \geq c_*/\mathsf{b}^2 > 0$, so that $\dim \mathcal{H}(S) \geq 2$. Finally $S_2(\theta\tau_s f_*, \tau_s f_*) = \|e^{-sH}v\|^2 > 0$, since e^{-sH} is injective, H being self-adjoint and ≥ 0. Thus, for $s > 0$, the argument yields $S_2(\theta\tau_s f_*, \tau_s f_*) > 0$, but no lower bound for this quantity.

Step 6: the clause (N6). Let $\mathsf{b} > 0$, and choose z with $L^z\mathsf{b} \geq x_s$. We regard the lattice $\mathbb{Z}^3/NL^K\mathbb{Z}^3$ at $\beta = L^K\mathsf{b}$ as the lattice with the data $(K-z, NL^z, L^z\mathsf{b})$; then $a_{K-z} = L^z a_K$. With $f^{[z]}(\xi) := f(L^{-z}\xi)$ we have $\mathcal{O}_{K-z}(f^{[z]}) = L^{-z}\mathcal{O}_K(f)$, and hence $\mathsf{b}^2 S_2^{(\mathsf{b})}(f, f') = (L^z\mathsf{b})^2 S_2^{(L^z\mathsf{b})}(f^{[z]}, f'^{[z]})$. We apply (N3) to (N5) at the dilated data, for $K \geq z + j + 1$ and for the sides of (R4), with the test function $f_*^{[-z]}$; for this we cite Thm. 8.2 (T6) and Thm. 8.17 (R4), (R6) (vi). In the label b the depth is $j + z$: indeed $f_*^{[-z]}(\xi) = f_0(L^{j+z}\xi)$, and $x = L^{j+z}\mathsf{b}$ is the same number. With $z = z_0(\mathsf{b})$ this is the rule $j_*(\mathsf{b}) = z + j_*(L^z\mathsf{b})$ of the definitions of Thm. 11.1.

Step 7: the clause (N6′). Let $\mathsf{b} > 0$ with $L^i\mathsf{b} \geq x^{\#\#}$. Choose $z \in \mathbb{Z}$ with $\mathsf{b}' := L^z\mathsf{b} \in [x_s, L^2 x_s]$, and put $j := i - z$. Then $x = L^j\mathsf{b}' = L^i\mathsf{b} \geq L^3 x_s \geq L\mathsf{b}'$, so $j \geq 1$; moreover $x \geq \max\{x^{\#}, y(\mathsf{b}')^{4/3}\}$, so $j \geq j_*(\mathsf{b}')$. By Step 6, $\mathsf{b}^2 S_2^{(\mathsf{b})}(f, f') = \mathsf{b}'^2 S_2^{(\mathsf{b}')}(f^{[z]}, f'^{[z]})$. Here $f^{[z]} = f_0(L^j\cdot)$ if and only if $f = f_0(L^i\cdot)$, and the mirror $y_1 = 1/2$ of depth j lies at $\xi_1 = L^{-z}L^{-j}/2 = L^{-i}/2$. We apply (N3) to (N5) at $(K-z, NL^z, \mathsf{b}')$. This requires $K - z \geq j + 1$, that is $K \geq i + 1$, and the sides of (R4), namely $m + z \geq m'$ and $L^z l \geq 2Ms(L^{z+1}\mathsf{b})$. For $\mathsf{b} > L^2 x_s$ the integer z is negative, and only large sides are reached; for the limits of (R5) this loses nothing. For moving labels $\mathsf{b}_n \to \mathsf{b}$ we choose z with $L^z\mathsf{b} \in [L^{1/2}x_s, L^{3/2}x_s]$ and require $L^i\mathsf{b} \geq Lx^{\#\#}$; then the same z and the same i can be used for all large n. □

Lemma 11.22 (The conditions on the depth)**.** In the setting of Thm. 11.1, every condition on $x = L^j\mathsf{b}$, and on b and m', that Lems. 11.3 to 11.21 use is an entry of the list of Thm. 11.1, or follows from that list. These conditions affect no floor. Moreover the following hold.

(a) Each entry of (X) holds for all sufficiently large x. The number $x^{\#}$ is finite and depends on the table alone, and (X) holds at every $x \geq x^{\#}$.
(b) In the setting (X), $h < 0.71x^{1/4}$, and the entries that Thm. 11.1 lists as holding of themselves do hold.
(c) $[j_*(\mathsf{b}), \infty) \subset J(\mathsf{b})$.
(d) $\sup_{[\mathsf{b}_1, \mathsf{b}_2]} j_* \leq \max\{1, \lceil \log_L(\max\{x^{\#}, \sup y^{4/3}\}/\mathsf{b}_1)\rceil\}$.
(e) The clauses (N1) to (N3) at the depth j follow from (X) and (B) taken at the single value $x = L^j\mathsf{b}$, that is, from $j \in J(\mathsf{b})$ in the sense of the definitions of Thm. 11.1; the list enters at this one value of x and at no other.

■

Proof. The growth of the quantities. By (O8) of item 1 of §8.1, $p_0 = A_0(\log x)^{\mathfrak{p}_0}$ with $\mathfrak{p}_0 > 2$, and $\varepsilon = p_0 x^{-1/2}$. The quantities $\eta_w = e^{-c_\Xi p_0^2}$ and $c_h(x) = e^{-c_h p_0^2}$ both decay faster than every power of x, since $p_0^2 \geq A_0^2(\log x)^4$. By the definition "At x" in item 1 of §2.1, and by item 2 of §7.3, (o4), we have $\rho = \lceil c_R \log x\rceil$, $r = \lceil 20 c_R \log x/M\rceil$, $\ell_w = 6Mr + 12$ and $\check{r} = r + 3$, so that ρ, Mr and ℓ_w are linear in $\log x$. From $\sigma_M \leq 1/12$ (item (1) of §6.1, by (M2)) and $\kappa_1 \geq 7$ we obtain $20c_R/M \leq 1/768$, and hence $r(x) \leq 1 + \log x/768$. Finally R is of the order $(\log x)^{5\mathfrak{p}_0}$, where $5\mathfrak{p}_0 > 10$, and $h \in [m, Lm)$ with $m = \max\{16R, 64LM\log x\}$.

The entries of (X). Every entry is a lower bound on x: the quotient of its left side by its right side tends to 0 as x grows. We take the entries in turn.

(j1) The errors (E1) to (E6) are each at most $c_F/12$. This entry is used in Steps 1 and 2 of the proof of Lem. 11.21, in Lem. 11.7 (c) for the smallness and for the rough bound, in Lem. 11.4 for $\mathsf{b}\sup|\mathrm{Rem}| \le 1$. As to the growth: (E1), (E3) and the first term of (E6) are at most a constant times a power of $\log x$ times $x^{-1/4}$, $x^{-1/2}$ and $x^{-3/2}$, respectively; (E2) is at most $c_F/(12p_0)$ by the choice of R, so it imposes no condition; (E4) and the c_h-term of (E6) are a power of x times $e^{-cp_0^2}$; and (E5), of the order p_0^{-5}, decays the slowest.

(j5) $C_{\mathrm{ext}} r_B^2 \varepsilon \le 1$. It is used in Lem. 11.6, for $a \le 0.01$, for $5\varepsilon W_2 \le 1$ and for $h < 0.71x^{1/4}$. It holds for large x because its left side is at most a constant times a power of $\log x$ times $x^{-1/2}$.

(j6) This entry consists of three inequalities. The first, $x^{10}\eta_w \le 1$, is used for the caps in Lems. 11.10 and 11.13. The second, $p_0^{10} \le x$, is used for $\varrho x \le x^{2.1}$ and in (j4). The third, $(r_z/\mathsf{b})(4C_S p_0^2 + 5) \le x$, is used in Step 6 of the proof of Lem. 11.10; here $r_z/\mathsf{b} = 3.5E_a p_0^6 p_0 (Lx)/(c_Q^{\mathrm{su}} K_5 L^{1/2})$ is at most a constant times a power of $\log x$ and is free of b (item 2 of §7.3, (o5)).

(j7) (i) $(c_P/2)p_0^2 \ge L^3(24\mathsf{N}_Z + 4.5\log x + 5 + 220M^{-3})$. This entry is used for the second inequality of Lem. 11.5. It holds for large x because p_0^2 outgrows $L^3 \log x$, as $2\mathfrak{p}_0 > 4$.

(j7) (ii) $R/2 - 1 \ge 2D_0 + M(r(x) + 2)$. It is used to make the indicators of S_f equal to 1 and to have $\mathrm{d}w_R = \mathrm{d}w$ (Step 2 of the proof of Lem. 11.17), and for the dependence on Box alone (Lem. 11.7 (c)).

(j7) (iii) $8\rho + 8 \le R$. It is used in Lem. 11.15 (d), in Step 7 of the proof of Lem. 11.16, and for the inclusion of B' in Box.

(j7) (iv) $8M(r(x) + 1) \le R$. It is used to compare $\tau(X)$ with $m(X)$ (Lem. 11.13).

The entries (ii) to (iv) hold for large x for one common reason: $R \ge 10^{1410} A_0^5 (\log x)^{5\mathfrak{p}_0}$, to be compared with D_0, with $Mr(x) \le 20c_R \log x + M$ and with $\rho \le c_R \log x + 1$. They are genuine conditions, since $M = L^{m'}$ may exceed 10^{1411}; what carries them is the growth of p_0^5 against $\log x$, and not the numeral.

(j7) (vi) $p_0 \ge 4$, $(c_a + c_h)(x) \le 1/2$ and $\log x \ge 3$. They are used in Steps 6 and 7 of the proof of Lem. 11.5, in (E2), and in (u1), (u2).

The fourth entry of (j7) (vi) is $\Theta(x) \le 1$. It is used for $C_a(x)$ and for the second-order Cauchy estimate, in Steps 7.2 and 7.3 of the proof of Lem. 11.14. It holds for large x because

$$\Theta \le p_0 x^{-1/2}(12M(1 + \log x/768) + 24 + 4/\vartheta_0^\flat) \cdot 78 r_1^3 (3252 + 5\log p_0),$$

which is at most a constant times a power of $\log x$ times $\log\log x$ times $x^{-1/2}$; here $\log R \le 3251 + 5\log p_0$.

(j7) (vii) $\rho_0 \le \rho(x) + 1$. It is used for the mask in Step 2 of the proof of Lem. 11.14 and for the radius $R + 2\rho + 3$ of B'. It holds for large x because $\rho(x)$ is nondecreasing and tends to infinity.

The entries of (B). They are used as follows.

(j2) $c_{\mathrm{vol}}(\mathsf{b}) h^3 L^{-3j} \le 1$. It is used for the root in Lem. 11.9.

(j3) $i_h \le j + m' - 1$. It is used for the fact that $\mathsf{n}_\mu = 2L^{m_l + j - i_h}$ is an even integer $\ge 2L$, and for the side $\ge 10r_B + 2$ (Lems. 11.6 and 11.7).

Step (a)*: $x^\#$ is finite.* The list (X) has finitely many entries. By the growth of the quantities, each entry has the form $A(x) \le 1$ with $A(x) \to 0$, so that each holds for all sufficiently large x. Among the definitions of Thm. 11.1, that of $x^\#$ requires of a number y that (X) hold at every $x \ge y$, and hence no monotonicity of the $A(x)$ in x is needed. The quantities R, h, r, ρ are ceilings, and an entry such as (j7) (ii) may become true just after a jump of R. The set of such y is an upper half-line, whose endpoint may be excluded. With $x^\# := 1 + \inf$, (X) holds at every $x \ge x^\#$. The

number $x^{\#}$ depends on the letters of the table and on pure numbers alone; it is free of b, j, K and N.

Step (b)*: in the setting* (X). First, $h < 0.71x^{1/4}$: as $r_B > h/8$, (j5) gives $32(h^2/64)p_0x^{-1/2} < 1$, hence $h^2 < 2x^{1/2}/p_0 \le x^{1/2}/2$, by $p_0 \ge 4$. The first inequality of (j4) is used for $\mathrm{sep}_k \ge 8\log(4\varrho L^{1/2})$ in Lem. 11.8. Its left side is at least $32L\log x - 1 - 2(\log x/768+4) - 2 \ge 31L\log x$, and its right side is $8\log 4 + 4\log x + 48\log p_0 + 4\log L \le 11.1 + 12.8\log x$, by $p_0^{10} \le x$ and $x \ge L$. The second inequality of (j4) has its left side at least $31.9\log x - 11$ and its right side $\mathrm{sep}^\circ = 8\log 8 + 4\log L \le 16.64 + 4\log x$.

The entry (j7) (v) holds since $\varrho L^{1/2} \ge L^{1/2} \ge 75$, because $L > 1.2\cdot 10^{13}$; it is used for the numeral 0.029996 of Lem. 11.8. Next, $R \ge 6r_1^3 \ge 4r_1$, because $R \ge 10^{1410}p_0^5$; the bound $R \ge 6r_1^3$ is used for $5\varepsilon W_2 \le 1$ (Step 7.2 of the proof of Lem. 11.14). Further, $4t\sup|w| < \varepsilon/4$, because $|w| \le 1$ and $16x^{-2} < p_0x^{-1/2}$; and $x \ge L$, because $j \ge 1$ and $\mathsf{b} \ge x_s \ge 1$. Finally, (E7) < 1.

Step (c)*:* $[j_*(\mathsf{b}), \infty) \subset J(\mathsf{b})$. As $L^j = x/\mathsf{b}$, the entry (j2) holds if and only if $x/h \ge \mathsf{b}\, c_{\mathrm{vol}}(\mathsf{b})^{1/3}$. The entry (j3) says that $h \le (x/\mathsf{b})L^{m'-1}$, that is, $x/h \ge \mathsf{b}L^{1-m'}$. By $h^2 < x^{1/2}/2$ of Step (b), $x/h > 1.41x^{3/4} \ge y(\mathsf{b})$ as soon as $x \ge y(\mathsf{b})^{4/3}$. Every $j \ge j_*(\mathsf{b})$ has $L^j\mathsf{b} \ge \max\{x^{\#}, y(\mathsf{b})^{4/3}\}$, so (X) and (B) hold at j.

Step (d)*: local boundedness.* The function

$$c_{\mathrm{vol}}(\mathsf{b}) = 225/M^3 + (3/2)\mathsf{N}_Z p_0(\mathsf{b})^2 + 3\log(384\mathsf{b}^{3/2}/p_0(\mathsf{b})^3)$$

is continuous on $[x_s, \infty)$, and so is y. Hence $\sup_{[\mathsf{b}_1,\mathsf{b}_2]} j_* \le \max\{1, \lceil \log_L(\max\{x^{\#}, \sup y^{4/3}\}/\mathsf{b}_1)\rceil\}$. On the whole of $[x_s, \infty)$, however, j_* is unbounded.

Step (e)*: one x at a time.* We show that, at the depth j, the clauses (N1) to (N3) require the list only at the one value $x = L^j\mathsf{b}$. Finer levels $k-m$, with $x_{k-m} = L^m x$, appear in two places in which a bound is needed at $L^m x$, $m \ge 1$. In Lem. 11.8 they occur through (G1) (item 3 of §7.3) and through $N_1(k-m)c_a(L^m x)$; this product is multiplied by the factor $\exp\{-5.75L^{m-1}\mathrm{sep}_k\}$, where $\mathrm{sep}_k \ge 31L\log x$. Step 0 of the proof of Lem. 11.15 uses them through $\mathsf{t}_*(\rho(L^m x))$. In both places the argument depends on no entry of the list at $L^m x$. Coarser levels $i > k$, with $x_i \ge L\mathsf{b}$, occur in Lem. 11.9 alone, through clauses of Sections 2.2 to 8.1 under their own floors. Thus (N1) to (N3) hold at every $j \in J(\mathsf{b})$.

Every condition is a condition on j; none of them is a condition on x_s, L, M, κ or c_R. Hence these conditions affect no floor. □

Item 17: the numerals in one table. Each constant of the section is stated below by its numerical value or, where letters of the table (O0) to (O13) occur, by a closed expression in those letters.

Input. Use the exact definitions in Def. 2.56 and the sums of Def. 2.44. For $\lambda > 0$, writing $z = e^{-\lambda}$ gives the useful closed forms

$$\mathsf{g}(\lambda) = 1 + \frac{24z(1+z)}{(1-z)^3} + \frac{2z}{1-z}, \qquad \mathsf{g}_1(\lambda) = \frac{24z(1+4z+z^2)}{(1-z)^4} + \frac{2z}{(1-z)^2}.$$

They follow by differentiating the geometric series and counting the $24r^2 + 2$ sites in the shell of radius $r \ge 1$. The bounds used below are $\log_{10}\mathsf{g}_\flat \le 46.315$, $\log_{10} g_{1\flat} \le 61.7366$ and $\log_{10} K_5 \le 24.776$. Here $g_{1\flat} = \mathsf{g}_1(\lambda_\flat)$, and $e_R \ge 256$.

Since $\kappa \ge 6$, the connected-set sums are bounded by their values at $\kappa = 6$. In particular, $(1 - 26e^{-5})^{-1} < 1.2124$, $52e/(1-52e^{-5}) < 217.59$ and $\kappa_1 - \log(26e) > 2.7419$.

The numerals of Lem. 11.15. The defining formulas and the sums in Def. 2.44 give

$$\mu_1 \le 3.78\cdot 10^{71}, \quad \check{\Lambda} \le 1.56\cdot 10^{77}, \quad p_1 \ge 1.32\cdot 10^{-73}, \quad d_{\mathrm{out}} \ge 4.2\cdot 10^{-90}, \quad A_{\mathrm{out}} \le 10^{1455.5}.$$

We choose

$$C_w = 5.3\cdot 10^{34}, \quad C_{gw} = 9.7\cdot 10^{34}, \quad C_\omega = 7.0\cdot 10^{34}, \quad C_{g\omega} = 1.3\cdot 10^{35},$$

and

$$C_G = 4.8\cdot 10^{179}, \quad C'_G = 8.7\cdot 10^{179}, \quad C_\tau = 1.1\cdot 10^{36}.$$

For example, $16!\,20^{16} = 1.371195958099968 \cdot 10^{34}$. The coefficients $c_{a,16}16!\,20^{16}A'$ for the first four majorants in Step 2 are respectively below $2.66 \cdot 10^{34}$, $4.82 \cdot 10^{34}$, $3.46 \cdot 10^{34}$ and $6.29 \cdot 10^{34}$. The remaining coefficients are below $2.39{\cdot}10^{179}$, $4.35{\cdot}10^{179}$, $5.07{\cdot}10^{35}$ and $9.21{\cdot}10^{35}$. These bounds verify the choices above. The exact formulas for N_{B0}, N_{B1}, N_{B2} give the bounds $4.9 \cdot 10^{112}$, $1.61 \cdot 10^{127}$ and $4.9 \cdot 10^{141}$. The formula in Step 7 then gives $C_e \le 7.1 \cdot 10^{166}$.

The numerals of Lem. 11.16. Here $C_1 = 203C_e \le 1.5 \cdot 10^{169}$, and $C_2 := 10^{670}$; its largest contribution is at most $1.1 \cdot 10^{666}$.

The numerals of Lem. 11.19. Here $C_{\mathrm{lat}} = 10^{11}$, because $7.0{\cdot}10^{10}+(3.6{\cdot}10^{9}+3.3{\cdot}10^{15})/r \le 9.8{\cdot}10^{10}$ for $r \ge 1.2 \cdot 10^5$, and $D_0 = 10^{140}$.

The numerals of Lems. 11.4 and 11.17. We have $C_S = (9/2)K_5\mathfrak{g}_\flat^2 \#Y_{\mathrm{out}}$ and $C_R = 60C_SC_1$. The closed forms above and $\#Y_{\mathrm{out}} = (D_0/4+1)^3$ give $1.58{\cdot}10^{536} < C_S < 1.59{\cdot}10^{536}$, $10^{707} < C_R < 1.02{\cdot}10^{707}$ and $\log_{10}(C_R/c_F) < 1411.3$. The constant $W_* = 10^3 r_1^4 = 10^{523}$ bounds $W_1 + 5\varepsilon W_2 + 1 \le 402 r_1^4$ alone and is no bound for W_0. The sums of w_R (Step 7.2 of the proof of Lem. 11.14) are as follows: $W_0 \le W_0(x) = 78r_1^3(1+\log R(x))$; $W_{00} \le 60r_1^3$; $W_1 \le 400r_1^4$; $W_2 \le 289 r_1^3 R$; the sum of $|w|$ along a comb path is at most $5r_1$; and $\sum_p (r_p+1)|w|_p^2 \le 10^3 r_1^4$. For the last sum, a shell of radius r has at most $72r^2+6$ plaquettes, and $|w|_p \le 4$ on each plaquette. Summing the shells through r_1+1 gives at most $300r_1^4$. For $r \ge r_1+2$ we have $|w|_p \le 4r_1^3/(r-1)^3$, so the shell contributes at most $1156 r_1^6 r^{-3}$. The tail is bounded by $578r_1^4$. The numerical bounds used above include $1.2124 \cdot 1605 < 2500$ and $2 \cdot 1.2124 \cdot 32 \cdot 0.401^2 < 12.5$. We set $C_6 := 10^{1700}$, which covers $10^3 C_S N_{B0} W_*^2 < 10^{1699}$. The numeral of (E4) gives $10^5 W_* C_S < 1.59 \cdot 10^{1064}$.

Input for the rows behind C_2, the 10^5 of (E4) *and C_6.* The rows use $|w| \le \min\{1, r_1^3 r^{-3}\}$ and $|\mathrm{d}w_R| \le 3r_1^4 r^{-4}$; there are $72r^2+6$ plaquettes at distance r, and as many bonds; $N_{B0} \le 4.9 \cdot 10^{112}$, $N_{B1} \le 1.61 \cdot 10^{127}$, $N_{B2} \le 4.9 \cdot 10^{141}$; and $W_* = 10^{523}$ and $C_S < 1.59 \cdot 10^{536}$.

The rows for C_2. The steps named in these rows are those of the proof of Lem. 11.16, and every row is a multiple of $p_0^2 = x\varepsilon^2$.

1. Step 2: $5W_2 + W_1 \le 1445 r_1^3 R + 400 r_1^4 \le 1600 r_1^3 R$, for $R \ge 2.6 r_1$.
2. Step 3: the curvature remainder, the derivative of the transports and their discrepancy contribute, together, at most

$$N_{B0}(5W_2 + W_1) + 18 N_{B1} W_0 + 24 N_{B1} V_1 + 72 N_{B2} W_1 \le 200(N_{B0} + N_{B1} + N_{B2}) r_1^4 R.$$

   The displacement moments and the estimate $V_1 \le r_1^4 R$ are proved in that step.
3. Step 5: $\sum r^2 |\mathrm{d}w_R| \le 216 r_1^4 R + 30 r_1^4 \le 225 r_1^4 R$, which gives $28[2N_{B0} \cdot 225 r_1^4 R + 2N_{B2} \cdot 370 r_1^4]$; the largest piece is $28 \cdot 740 N_{B2} r_1^4 \le 1.1 \cdot 10^{666}$.
4. Step 6: 100, from $100\varepsilon^2$ times x.

Summing the rows, we obtain a total $\le 4 \cdot 10^{665} R + 1.1 \cdot 10^{666}$ + smaller terms $\le 10^{670} R$.

The rows for the 10^5 of (E4). Every row is a multiple of $C_S \nu_P$.

(a) The part (a) of Step 1 of the proof of Lem. 11.17 gives $6p_0^2$.
(b) The part (b) gives $72 W_* p_0^2$, from $|D\chi| \le 8 \cdot 401 r_1^4/\varepsilon = 3.208 W_*/\varepsilon \le 16 W_*/\varepsilon$.

The part with $D\chi$ of the second identity gives $3{\cdot}3.208 W_* \cdot \sqrt{2} < 13.7 W_*$; the term $(3/2)q\nu_P$ enters with $q \le 4C_S$; and the passage from G to G_- gives $(9/2)p_0^4$. In (iii) the rows are $(81/16)p_0^4/C_S$, and the parts with $D\chi$ are at most $40 W_* p_0^2/C_S$. Altogether the sum is at most $10^3 W_* C_S p_0^4 \nu_P$, which does not exceed (E4).

Remark 11.23 (The size of the constants). The choice of the constants C_2, C_6 and the numeral of (E4) does not affect the clauses: no floor depends on (O14), and a larger constant only raises $x^\#$.

12. Components and reconstruction

In this section we show that the Gibbs states at cutoff K invariant under $a_K\mathbb{Z}^3$ make up one class, with bounds that depend neither on the cutoff nor on the state nor on the places; from this

we obtain the frame and the canonical decomposition of σ_S for $S \in \mathrm{Lim}_G(\mathsf{b})$, and each component has full Euclidean invariance, reflection positivity at every mirror, a unique vacuum, clustering, the axioms of Def. F.17, point functions and a Wightman theory (Thm. 12.4). We then carry the lower bound of §11 over from the tori to all periodic Gibbs states (Thm. 12.5), and further to each component at each ξ (Thm. 12.6). In consequence (Cor. 12.7), for every $\mathsf{b} > 0$ each component of each $S \in \mathrm{Lim}_G(\mathsf{b})$ is, up to a shift, the functional of some member of $\mathrm{Lim}_G(\mathsf{b})$ whose canonical component measure is concentrated on that component, and it satisfies all the axioms together with $S_2 \neq 0$.

Remark 12.1 (The premises of the results). Every result below is a consequence of the clauses named in items A to D.

Item A: the model, the step and the runs. From the data of Thm. 8.2, in item 2 of §8.1, we take the model, together with $\mu_{K,l,\beta}$, $\mathcal{O}_K(f)$ and S^T, which are those of Def. 1.1. The same data give the sides $N = 2L^{m_l}$ with $m_l \geq m'$, and the condition (Tor). By the part of item 1 of §8.1 on distances, a tuple of type $(d; R)$ in the sense of Def. 8.1, where distances are Euclidean, is of type $(d/\sqrt{3}; R)$ in the sup distance. From the step of item 2 of §5.5 we take the formula of $\mathsf{M}(U)(c)$, which is a Borel function of U built from U, L, the blocks and the paths alone. By the same item, the step, that is, the kernel $\prod_c k_{ax}(V'(c), \mathsf{M}(U)(c))$, is a probability kernel at every U. The step depends on x, a and L, but on no rows, on no grid of cubes, on no members, and not on K. The runs of item 2 of §5.5 supply the following: ρ_0 is Wilson's weight at $L^K\mathsf{b}$; the noises are $a\,x_k$ with $x_k = L^{K-k}\mathsf{b}$; and the rows form the chain described in Def. 2.49.

Item B. Gibbs and continuum inputs. For the Gibbs-state estimates we use the chessboard domination B.18 and its finite-side form B.19. The symmetry and conditioning statements are B.28 and B.29. For the limit we use the frame and component construction E.8–E.13 and the conclusions of F.22. The point-insertion bound supplies the reconstruction hypotheses through F.21.

Item C: the clauses of other sections that this section uses. From §8.3 we take Thm. 8.17 (R0) for every integer c and for all translations by $\mathbb{Z}^3$. We use Thm. 8.17 (R1) for every $Q \geq 1$, with C_{vol} to the first power, and the inequality $1 + 0.182\, C_{\mathrm{vol}}(N, \mathsf{b}) \leq e^{c_{\mathrm{vol}}(\mathsf{b})N^3}$. The numbers sep°, j°, G_1, $B_0(\mathsf{b}, \check{\delta})$, c_{vol} and $\hat{c}_K$ of item 2 of §8.3 enter as well; each of them is locally bounded in b.

§9 contributes a single inequality of Thm. 9.4 (G1), namely $\max\{\check{\Gamma}(\mathsf{b}, \check{\delta})e^{c_{\mathrm{vol}}}, 2B_0(\mathsf{b}, \check{\delta})\} \leq A_4(\mathsf{b})\check{\delta}^{-2}$ on $0 < \check{\delta} \leq 1$. Besides this inequality, we take the letters of item 1 of §9 and (Lab1).

Thm. 10.4 (4.3) (a), (b) is the moment bound of (E0). By Thm. 10.4 (4.6), every $S \in \mathrm{Lim}_G(\mathsf{b})$ has (R0) to (R7) of Def. D.10, and (R5) holds with $(c_{\mathrm{I}}, C_{\mathrm{I}}, \mathsf{p}_{\mathrm{I}}) = (1/(48L), 24Le^{c_{\mathrm{vol}}}\hat{G}(\mathsf{b}), 1)$. We use what Thm. 10.4 (4.7) states for all isometries, for every mirror, for the circular cone and for $\mathrm{O}(2)$. From Thm. 10.4 (4.8) we take the dilation, and from Thm. 10.4 (4.9) (i) the joint limits.

From §11 we take the following. We use its geometry and, of its numbers, D_0, c_F and $c_* = c_F^2/7$, and also R, $h = L^{i_h}$, n_0, t, r_η, ϱ, r_z and a_u. The depth $j_*(\mathsf{b})$ of the data enters with Thm. 11.1 (D) and Lem. 11.22 (a), (c): the number $x^\#$ is finite and depends on the table alone, and every entry of (j1) to (j7) and of (N0) to (N6) is valid for all $j \geq j_*(\mathsf{b})$. We also use Lem. 11.22 (d), by which j_* is bounded on compact sets. We need Lem. 11.3 (L), (I), (K), that is, locality, independence, and the covariance of the k-fold kernel under $y_\mu \to n_\mu - y_\mu$ and under $y_\mu \to y_\mu + n_\mu$, for $n_\mu \in \mathbb{Z}$ of level k. The statement of Lem. 11.4, together with Steps 4 to 8 of its proof, where the remainder is bounded term by term, is used as well, and so are two parts of the proof of Lem. 11.7: item 1 of Step (b), which gives $W_f = 0$ at globally tame fields, and Step (c), on dependence, that is, on the bonds that enter. Lems. 11.5, 11.10 and 11.13 are used in their norm forms (P°), (W°) and (X°), with (W°) for every constant $g \in [0, \frac{9}{8}p_0^2]$. We rely, moreover, on Steps 2, 5, 6 and 7 of the proof of Lem. 11.14 together with the part "Besides" of that lemma, the data of Lem. 11.15 and its clause (c), Lem. 11.16, Steps 0 to 6 of the proof of Lem. 11.17 with the list of errors (E1) to (E7)

of Lem. 11.17, which in this section always appear together with that lemma, and Lem. 11.19 (B), which gives $q \geq c_F$. Finally we take Thm. 11.1 (N3), (N6).

Item D: classical inputs. We use the following textbook facts, each by its name: the spectral theorem, dominated convergence, normal families, the identity theorem, the reflection principle, and the mean ergodic theorem. The boundary-value estimate is Lem. F.12.

Definition 12.2 (Data)**.** We recall the settings (S1), (S2), (S3), (S6), (S7), (S8) and (S9) of §1. The group is $G = \mathrm{SU}(2)$ and the dimension is 3. The table (O0) to (O13) and the floors are those of item (1) of §6.1 and of item 1 of §8.1; L and $M = L^{m'}$ are the L and the M of the table, and L is odd. We write $a_K = L^{-K}$; the number $\mathsf{b} > 0$ is the label of the limit, b' is the label of an approximant, and always $\beta = L^K \mathsf{b}' > 0$. We set

$$\mathcal{O}_K(f) = a_K^{-1} \sum_p f(x(p))\, a_p(U),$$

where $a_p = 1 - \frac{1}{2} \operatorname{tr} \partial U(p) \in [0,2]$ and $x(p)$ is the barycentre of p; and $\|f\|_\sharp = \max\{\|f\|_\infty, 40M \operatorname{Lip} f\}$.

The class. The class $\mathbb{G}_K(\mathsf{b}')$ is the set of the $\nu \in \mathcal{G}_{a_K, L^K \mathsf{b}'}$, the Gibbs states of Def. B.1 (2), with $(\tau_v)_* \nu = \nu$ for all $v \in a_K \mathbb{Z}^3$. We have $\mathbb{G}_K(\mathsf{b}') \subset \mathcal{G}^{\mathrm{per}}$, with period number 1 in the sense of Def. B.1 (3), and $\mathbb{G}_K(\mathsf{b}') \supset \mathcal{G}^{\mathrm{tor}}$ by Prop. B.3 (c). We remark that $\mathbb{G}_K(\mathsf{b}')$ is in fact all of $\mathcal{G}^{\mathrm{per}}$, by Prop. B.28 (c).

The boards. For $i \leq K-1$ the board B_i has its walls at $L^{-i}(\mathbb{Z} + \frac{1}{2})$ and its cells of side L^{-i}, the cell 0 being centred at 0; these are the boards $(i,0)$ of Def. B.10, and B_0 is the unit board. Written without arguments, B_0 is this board; written with them, $B_0(\mathsf{b}, \check{\delta})$ is the number. The boards have no offset; other places are reached by translating the functions by vectors of $a_K \mathbb{Z}^3$, as in Lem. 12.9, or of $L^{-j}\mathbb{Z}^3$, as in Thm. 12.5.

We put $X_K^\nu = \prod_i (\mathcal{O}_K f_i - \nu(\mathcal{O}_K f_i))$, and $S_n^T[\rho]$ stands for the joint cumulants under ρ. The set $\mathrm{Lim}_G(\mathsf{b})$ consists of the $S = (S_n)_{n \geq 2}$ such that $S_n(X) = \lim_j S_n^T[\nu_j](X)$ for all separated C_c^1 tuples, for some $K_j \to \infty$, $\nu_j \in \mathbb{G}_{K_j}(\mathsf{b}_j)$ and $\mathsf{b}_j \to \mathsf{b}$; and $\mathrm{Lim}_{\mathrm{tor}}(\mathsf{b})$ consists of those obtained with $\nu_j \in \mathcal{G}^{\mathrm{tor}}$. The functional σ_S is the centred functional of eq. 1.i. The components of S are the σ_ξ of a canonical decomposition $(\mathsf{X}, \lambda, (\sigma_\xi))$ of σ_S, that is, of one with the properties of Def. E.12. Further, θ is the reflection at $(e_3, 0)$, $\mathbb{E}_+ = \{x_3 > 0\}$, and sh_c is the algebra map $(f) \mapsto (f) + c \int f$.

Labels. We let $z(\mathsf{b})$ be the least integer $z \geq 0$ with $L^z \mathsf{b} > x_s$; it is a z as in Thm. 10.4 (4.8). The set $U(\mathsf{b}) := [\mathsf{b}/2, 2\mathsf{b}] \cap [L^{-z(\mathsf{b})} x_s, \infty)$ is a compact neighbourhood of b in $(0,\infty)$ that is determined by b alone and satisfies $L^{z(\mathsf{b})} U(\mathsf{b}) \subset [x_s, \infty)$; for $\mathsf{b} > x_s$ we have $z(\mathsf{b}) = 0$ and $U(\mathsf{b}) = [\mathsf{b}/2, 2\mathsf{b}] \cap [x_s, \infty)$. Of a sequence $\mathsf{b}_j \to \mathsf{b}$ only finitely many labels lie outside $U(\mathsf{b})$, and we drop these j: by Def. E.8 (5), a subsequence of a class system is again a class system, with the same bound system, and the limit does not change.

Normal form. The normal form is the case $\mathsf{b} > x_s$. There every number "at b" is defined as its supremum over $U(\mathsf{b})$; it is thus one function of b, which depends neither on the sequence (b_j) nor on the state. Numbers at the label b' of an approximant are plain values, whereas numbers at the limit label b are suprema over $U(\mathsf{b})$. All the numbers are locally bounded in b: see item 2 of §8.3 and (Lab1) of §9; further, c_{vol}, p_0 and ε are continuous on $[x_s, \infty)$, and for j_* we refer to the paragraph "The depth" below. The labels outside the normal form, those with $\mathsf{b} \leq x_s$, are the subject of (E5).

The depth. We recall the depth from the definitions of Thm. 11.1, from Thm. 11.1 (D) and from Lem. 11.22 (a), (c). The number $x^\#$ is the one of those definitions, namely $1 + \inf\{y \geq L :$ each entry of the list of §11 depending on x alone is valid for all $x \geq y\}$. Next, $y(\mathsf{b}) := \mathsf{b} \max\{c_{\mathrm{vol}}(\mathsf{b})^{1/3}, L^{1-m'}\}$, and for $\mathsf{b} \geq x_s$ the number $j_*(\mathsf{b})$ is the least $j \geq 1$ with $L^j \mathsf{b} \geq \max\{x^\#, y(\mathsf{b})^{4/3}\}$. If $j \geq j_*(\mathsf{b})$, then also $L^{j'} \mathsf{b} \geq \max\{x^\#, y(\mathsf{b})^{4/3}\}$ for every $j' \geq j$. Moreover, as in Lem. 11.22 (d), $j_*(\mathsf{b}') \leq \max\{1, \lceil \log_L(\max\{x^\#, y(\mathsf{b}')^{4/3}\}/\mathsf{b}') \rceil\}$, and since y is continuous on $[x_s, \infty)$, j_* is bounded on compact sets.

The geometry of §11, recalled. We measure in units of level $k = K - j$, that is, in units of L^{-j}; $r = |y|$ is the sup distance; and $x = L^j \mathsf{b}'$, $\varepsilon = p_0(x)x^{-1/2}$, $\rho = \rho(x)$. The mirror is the plane $y_1 = \frac{1}{2}$, and u, T_y, $\tilde{F}_p$ and $u^\alpha(F)$ are those of the geometry of Thm. 11.1. We set $B_R = \{r \le R\}$, $B' = \{r \le R + 2\rho + 3\}$ and $\mathrm{Box} = \{r \le r_B\}$, where $r_B := R + h/8 + 1 \le (h-1)/2$. Let χ be the product over the plaquettes of B' of $\chi_0(\phi_p/\varepsilon)$, where χ_0 is smooth, equal to 1 on $[0, \frac{1}{2}]$ and to 0 on $[\frac{3}{4}, \infty)$, and let χ_- be the analogous product taken over those plaquettes all of whose sites lie in $y_1 < \frac{1}{2}$. We set $G_- := \frac{x}{2}\sum_\alpha u^\alpha(F)^2\chi_- \in [0, \frac{9}{8}p_0^2]$; it is a function of those drawn bonds of B' that lie in $y_1 < \frac{1}{2}$, and G is defined in the same way with χ. Let $y_c := \lceil D_0/\sqrt{2}\rceil(e_1 + e_2)$, and let $f_0 \in C^\infty$ with $0 \le f_0 \le 1$, equal to 1 on $|y - y_c| \le D_0/16$ and to 0 off $|y - y_c| \le D_0/8$; we set $f_* := f_0(L^j\,\cdot\,)$, and it keeps a distance $\ge 0.58\, D_0 L^{-j}$ from the mirror.

The pure numbers are $D_0 = 10^{140}$, $c_F \in [5.4 \cdot 10^{-5}, 4.0 \cdot 10^{-4}]\, D_0^{-5}$ and $c_* = c_F^2/7$. The following are functions of x alone: $R = \lceil C_R p_0^5/c_F\rceil$; $h = L^{i_h}$, the least power of L that is $\ge \max\{16R, 64LM\log x\}$; $n_0 = \lfloor h/16M\rfloor$; $t = x^{-2}$; $r_\eta = x^4$; $\varrho = x^{1/2}p_0^6$; and r_z/b'. Here $r_z := \varrho\,\mathfrak{r}^\circ(x)$, which is not itself a member of the list, depends on b' as well as on x. The boards are B_{j-1}, with cells of side L, and B_{j-i_h}, with cells of side h.

■

Definition 12.3 (New numbers)**.** All the numbers below are defined at (O14), after the table and b. No floor depends on them, and they are independent of K, N, the state, the sequence of labels, ξ and the places. From item 2 of §8.3 we recall $\mathrm{sep}^\circ := 8\log(8L^{1/2})$, the only numeral among the numbers of that item, and, from the same item,

$$c_{\mathrm{vol}}(\mathsf{b}) := 225/M^3 + \frac{3}{2}\mathsf{N}_Z p_0(\mathsf{b})^2 + 3\log(384/\varepsilon(\mathsf{b})^3).$$

We recall $\check{\Gamma}(\mathsf{b}, \check{\delta}) = G_1(\mathsf{b}, \check{\delta}, \check{\delta}/4)(1 + e^{c_{\mathrm{vol}}})$; and A_4 and $\hat{G}(\mathsf{b})$ are those of item 1 of §9 and of item 2 of §10. From item 1 of §9 we recall $A_6(\mathsf{b}) := 3^{5/2} \cdot 64(4e)^5 A_4(\mathsf{b})$, and we set $C_1(R) := A_6(1 + R)^3$. From item 2 of §1 we recall

$$C_n^\infty(d; R) := 1.45^{n+1} n!\, A_6^n (n!)^5 \max\{1, d^{-5n}\}(1 + R)^{3n},$$

the value $d = \infty$ included, and we set $C^{\mathsf{b}} := (C_1, (C_n^\infty))$. We let $\bar{c}(\mathsf{b})$ be the value of the formula eq. E.xliv at C^{b}. We set $\bar{c}_*(\mathsf{b}) := \frac{3}{4\pi}A_6(\mathsf{b})$.

From Thm. 9.4 (G4) (a) we recall the quadruple $(13, 1, C_E, 13)$, with $C_E = \mathsf{C}_{\neq}^6\, M\, 8A_6(\mathsf{b})$ and $\mathsf{C}_{\neq} = 10^{53}$. From item 2 of §10 we recall $(c_{\mathrm{I}}, C_{\mathrm{I}}, \mathsf{p}_{\mathrm{I}}) = (1/(48L),\ 24Le^{c_{\mathrm{vol}}}\hat{G}(\mathsf{b}),\ 1)$. The formulas eq. D.xxxvii at $(c_{\mathrm{I}}, C_{\mathrm{I}}, \mathsf{p}_{\mathrm{I}})$ give $\wp = \mathsf{p}_{\mathrm{I}} + 4 = 5$ and

$$C_{\mathrm{pt}} = 320 \cdot 8^4\, \mathsf{mv}_3\, c_{\mathrm{I}}^{-4} C_{\mathrm{I}} M,$$

where $\mathsf{mv}_3 = m_3$ is defined in Def. D.21; we set $C_\wp = 32C_{\mathrm{pt}}$, $c_0 = \frac{1}{2}$ and $C_* = C_\wp + \bar{c}_*$. For $\mathsf{b} \ge x_s$ we recall from the definitions of Thm. 11.1 that $j^+(\mathsf{b})$ is the supremum of j_* over $[\mathsf{b}/2, 2\mathsf{b}] \cap [x_s, \infty)$; for $\mathsf{b} > x_s$ this set is $U(\mathsf{b})$, and the supremum is finite, since j_* is bounded on compact sets by the last sentence of the paragraph on the depth in the data. We set $c_{**} := (4/L)^6 c_*$ and, in the normal form, $r_*(\mathsf{b}) := D_0 L^{-j^+(\mathsf{b})} > 0$; in general $r_*(\mathsf{b})$ is given in (C4). We write D_W for the domain $\mathcal{D}_0$ of Thm. F.16.

■

Theorem 12.4 (The class, the limits and their components)**.** The following hold.

(E0) *Class and bounds.* Let $\mathsf{b}' \ge x_s$ and $K \ge 1$. Then $\mathbb{G}_K(\mathsf{b}')$ is a class, in the sense of (Cls1), (Cls2) of Def. E.8. For $\rho, \nu \in \mathbb{G}_K(\mathsf{b}')$, for $n \ge 1$ and for real bounded functions f_i of the Euclidean type $(d; R)$, wherever the balls lie,

$$|\rho(X_K^\nu)| \le A_6(\mathsf{b}')^n (n!)^5 \max\{1, d^{-5n}\}(1 + R)^{3n} \prod_i \|f_i\|_\infty,$$

and for $n \geq 2$ also $|S_n^T[\rho](X)| \leq C_n^\infty(d;R)\prod_i \|f_i\|_\infty$. These two bounds are Thm. 10.4 (4.3) (b). For $K_j \to \infty$, $\mathsf{b} > x_s$ and $\mathsf{b}_j \in U(\mathsf{b})$, the sequence $(K_j, \mathbb{G}_{K_j}(\mathsf{b}_j))$ is a class system that obeys C^{b} in the sense of (Bd1), (Bd2).

(E1) *The limit.* Let $K_j \to \infty$, $\mathsf{b}_j \to \mathsf{b} > x_s$ and $\nu_j \in \mathbb{G}_{K_j}(\mathsf{b}_j)$. After discarding finitely many indices we have $\mathsf{b}_j \in U(\mathsf{b})$. Every such frame has a convergent subsequence. By Def. E.9, the sequence (K_j, ν_j) is a frame in the class system $(K_j, \mathbb{G}_{K_j}(\mathsf{b}_j))$ of (E0). Along (K_j, ν_j) an element $S \in \mathrm{Lim}_G(\mathsf{b})$ is obtained if and only if the frame converges in the sense of eq. E.xxv, and then its limit σ_{frame}, the functional of eq. E.xxv, equals σ_S. The functional σ_S has the properties of Thm. F.22 (1)(a) to (e); it is invariant under $\mathbb{R}^3 \rtimes \mathrm{O}(3)$ and reflection positive at every plane; it satisfies the growth condition eq. C.xxxii with the quadruple; and it has data (Rec1) to (Rec4) with $(C_\wp, 5, \frac{1}{2})$, where (Rec1) to (Rec5) are as recalled in F.20 before Prop. F.21.

(E2) *The decomposition.* A canonical decomposition of σ_S exists, and it is unique up to exactly one isomorphism. The map $P_0[F] \mapsto (\xi \mapsto \sigma_\xi(F))$ is a unitary map of $\ker H$ onto $L^2(\mathsf{X}, \lambda)$ that carries Ω to 1. Moreover $\#\mathsf{X} = \dim\ker H$ when X is finite, and $\ker H$ is infinite-dimensional when X is infinite, and every isometry acts on X as the identity.

(E3) *Every limit in the frame.* Let σ' be any limit in the frame; in particular, σ' may be any component σ_ξ.

- (i) $(S_n^{\sigma'})_{n\geq 2} \in \mathrm{Lim}_G(\mathsf{b})$, and we write σ'^{c} for its centred functional; moreover $\sigma' = \sigma'^{\mathrm{c}} \circ \mathrm{sh}_c$ with $c = c_{\sigma'}$. The functional σ' is invariant under $\mathbb{R}^3 \rtimes \mathrm{O}(3)$, reflection positive at every plane, and regular of order $\leq 4n+1$. When σ' is a component σ_ξ, it has approximants $\rho_j = \nu_j(\,\cdot \mid E_j)$ with E_j invariant and $\nu_j(E_j) > 0$.
- (ii) $S_1 = c_{\sigma'} \int f$ with $|c_{\sigma'}| \leq \bar{c}_*$; the number c_ξ is continuous in ξ, and $\int c_\xi \, d\lambda = 0$. The moment bound of (E0) holds for σ', and so does the growth condition eq. C.xxxii of (E1), with the same quadruple.
- (iii) $\mathcal{H}(\sigma')$ is separable and $H \geq 0$; the joint spectrum lies in $E \geq |q|$, and $\mathrm{O}(2)$ acts. When $\sigma' = \sigma_\xi$, we have $\ker H_\xi = \mathbb{C}\Omega$.
- (iv) $\sigma_\xi(X \cdot \tau_v Y) \to \sigma_\xi(X)\sigma_\xi(Y)$ as $|v| \to \infty$, uniformly in ξ; Prop. E.20 gives, in addition, the error $C(X,Y)/|v|$.

(E4) *Reconstruction.* The following hold.

- (a) The Schwinger functions of every σ_ξ satisfy (E0′), (E1) for $\mathbb{R}^3 \rtimes \mathrm{O}(3)$, (E2), (E3) and (E4) of Def. F.17.
- (b1) Every limit σ' in the frame has (PtB) of Def. F.20 with $(C_\wp, 5, \frac{1}{2})$.
- (b2) Every σ_ξ has data (Rec1) to (Rec5) with $(C_*, 5, \frac{1}{2})$.
- (b3) *Point functions.* For each n there is one real-analytic, symmetric and isometry-invariant function S_n^ξ on $\{x_i \neq x_k\}$ such that $\sigma_\xi(f_1, \dots, f_n) = \int \prod_i f_i(x_i)\, \mathsf{S}_n^\xi(x)\, dx$ for all separated tuples; and

$$|\mathsf{S}_n^\xi(x)| \leq [C_* \max\{1, (2n^2/d(x))^5\}]^n,$$

  where $d(x)$ is the least distance between distinct entries, with $d(x) = +\infty$ for $n = 1$.
- (b4) *Wightman.* For each n there is a function W_n, holomorphic on T^{n-1}, where $T = \{(z^0, \mathbf{z}) \in \mathbb{C} \times \mathbb{C}^2 : \mathrm{Re}\, z^0 > |\mathrm{Im}\, \mathbf{z}|\}$, such that $|\mathsf{W}_n| \leq (C_* \varepsilon_T^{-5})^n$ with $\varepsilon_T = \min\{1, \frac{1}{2}\min_k(\mathrm{Re}\, z_k^0 - |\mathrm{Im}\, \mathbf{z}_k|)\}$, with $\varepsilon_T = 1$ when $n = 1$. At points of increasing height, $\mathsf{S}_n(x) = \mathsf{W}_n(x_2 - x_1, \dots, x_n - x_{n-1})$. The quintuple $(\mathcal{H}(\sigma_\xi), U, \Omega, D_W, \Phi)$ is a Wightman theory of one Hermitian scalar field on $\mathbb{R}^{1+2}$, with the domain, covariance, locality, spectrum and continuation properties of Thm. F.16. For σ_S all of this holds too, except the uniqueness of the vacuum.

(E5) *Every* $\mathsf{b} > 0$. For $\mathsf{b} > x_s$, use (E0) to (E4) with their stated constants. For $0 < \mathsf{b} \leq x_s$, put $z = z(\mathsf{b}) \geq 1$, take $K \geq z + 1$, and let every number "at $L^z\mathsf{b}$" be its supremum over

$L^zU(\mathsf{b})$. Then (E0) to (E4) hold with the changes listed below. In both cases (E7) holds as well, since its proof uses only (E2) to (E4). For a label $\mathsf{b}' < x_s$ of $U(\mathsf{b})$, Thm. 12.5 is to be understood with $(K-z, L^z\mathsf{b}')$ in place of (K, b'). The changes are these: $A_6(\mathsf{b})(1+R)^3$ becomes $L^zA_6(L^z\mathsf{b})(1+L^zR)^3$; the growth condition eq. C.xxxii of (E1) holds with the quadruple in which C_E is replaced by $\mathsf{C}_{\neq}^6 M L^z(1+L^z)^3A_6(L^z\mathsf{b})$; $\bar{c}_*(\mathsf{b}) = \frac{3}{4\pi}L^{4z}A_6(L^z\mathsf{b})$; $(c_{\mathrm{I}}, C_{\mathrm{I}})$ becomes $(c_{\mathrm{I}}L^{-z}/2,\ 2L^zC_{\mathrm{I}}(L^z\mathsf{b}))$; and C_{pt} and C_* are computed at these numbers, so that $C_{\mathrm{pt}}(\mathsf{b}) = 32L^{5z}C_{\mathrm{pt}}(L^z\mathsf{b})$.

(E6) *Joint limits.* For any $\mathsf{b} > 0$, the limit points of joint sequences at labels $\mathsf{b}_j \to \mathsf{b}$ with $N_j \geq N^*(K_j, \mathsf{b}_j)$ are functionals of members of $\mathrm{Lim}_{\mathrm{tor}}(\mathsf{b})$; this is Thm. 10.4 (4.9) (i).

(E7) *The relation between S_2 and the components.* Let $g_f(\xi) := S_2^{\sigma_\xi}(\theta f, f)$ and $V := \int c_\xi^2\, d\lambda$.

(a) For $f \in C_c^1(\mathbb{E}_+)$ we have $S_2(\theta f, f) = \int g_f\, d\lambda + V(\int f)^2$; here $g_f \geq 0$ is continuous, $\int g_f\, d\lambda = \|(1-P_0)[(f)]\|^2$, and $V(\int f)^2 = \|P_0[(f)]\|^2 \leq \bar{c}_*^2(\int f)^2$.

(b) For a single ξ the following are equivalent: $\dim \mathcal{H}(\sigma_\xi) = 1$; $S_2^{\sigma_\xi} = 0$; $g_f(\xi) = 0$ for one $f \geq 0$ with $f \neq 0$; and $\sigma_\xi(f_1, \dots, f_n) = \prod_i c_\xi \int f_i$. The centred two-point function of σ_ξ is $s_\xi(|x-y|)$, where $s_\xi \geq 0$ is non-increasing.

(c) The conditions $S \neq 0$ and $\dim \mathcal{H}(S) \geq 2$ by themselves decide nothing about the components; Rem. E.24 gives an instance.

■□

Theorem 12.5 (A lower bound in every periodic Gibbs state)**.** Let $\mathsf{b}' \geq x_s$, $j \geq j_*(\mathsf{b}')$ and $K \geq j+1$ be arbitrary, put $k = K - j$ and $x = L^j\mathsf{b}'$, and let $\rho \in \mathbb{G}_K(\mathsf{b}')$ be arbitrary. Let P_ρ be ρ times the k-fold kernel of the step of item 2 of §5.5, on the infinite lattices, and write $V = V^{(k)}$. Let θ_* be the reflection at $x_1 = cL^{-j}$, where $c \in \mathbb{Z} + \frac{1}{2}$; for $c = \frac{1}{2}$, G_- and f_* are those of the geometry in the data, and for other c they are their translates by $(c - \frac{1}{2})L^{-j}e_1$. Then the following hold.

(P0) $\mathrm{Cov}_{P_\rho}(G_-, \mathcal{O}_K f)^2 \leq \mathrm{Var}_{P_\rho}(G_-)\, S_2^T[\rho](\theta_* f, f)$ for every real bounded local f carried by the open right half.

(P1) $\mathsf{b}'\,\mathrm{Cov}_{P_\rho}(G_-, \mathcal{O}_K f_*) \geq c_F$.

(P2) $\mathrm{Var}_{P_\rho}(G_-) \leq 7$.

(P3) $\mathsf{b}'^2 S_2^T[\rho](\theta_* f_*, f_*) \geq c_*$.

(P4) For $\nu \in \mathbb{G}_K(\mathsf{b}')$, ν-almost surely, $\mathsf{b}'^2\,\mathrm{Cov}_\nu(\mathcal{O}_K\theta_* f_*, \mathcal{O}_K f_* \mid \mathcal{I}) \geq c_*$, where $\mathcal{I}$ is the one of Def. B.1 (5) at $a = a_K$.

■□

Thm. 12.5 involves no torus and no side, and each of its clauses is a statement at one j.

Theorem 12.6 (A lower bound for every component)**.** Let $S \in \mathrm{Lim}_G(\mathsf{b})$ and $\xi \in \mathsf{X}$, where every point of X is admitted as ξ. Clauses (C1) to (C3) below are stated in normal form, $\mathsf{b} > x_s$. Clause (C4) gives the corresponding bounds for every $\mathsf{b} > 0$, including $\mathsf{b} = x_s$.

(C1) For every $j \geq j^+(\mathsf{b})$, with the mirror carried to $(e_3, 0)$, we have $\mathsf{b}^2 g_{f_*}(\xi) \geq c_*$; hence every component is non-trivial in the sense of (E7) (b).

(C2) $s_\xi(r) \geq c_{**}\mathsf{b}^{-2}r^{-6}$ for $0 < r \leq r_*(\mathsf{b})$. Moreover, H_ξ is unbounded and $\dim \mathcal{H}(\sigma_\xi)$ is infinite.

(C3) $\|(1-P_0)[(f_*)]\|^2 \geq c_*/\mathsf{b}^2$; hence $H \neq 0$ on $\mathcal{H}(\sigma_S)$.

(C4) The same holds for every $\mathsf{b} > 0$, with $z = z(\mathsf{b})$, $f_* = f_0(L^{j+z}\,\cdot\,)$ and $j \geq j^+(L^z\mathsf{b})$, where $j^+(L^z\mathsf{b})$ is the supremum of j_* over $L^zU(\mathsf{b}) = U(L^z\mathsf{b})$. The quantities b^2S_2 and $\mathsf{b}^2r^6s(r)$ are invariant under the dilation, and $r_*(\mathsf{b}) = L^{-z}D_0L^{-j^+(L^z\mathsf{b})}$.

■□

The clauses of Thm. 12.6 differ in the values of j they require. (C1), (C3) and the non-triviality of the components require one $j \geq j^+(\mathsf{b})$; the lower bound on s_ξ in (C2) requires all $j \geq j^+(\mathsf{b})$; and the unboundedness of H_ξ together with the infinite dimension in (C2) requires infinitely many. Thm. 11.1 (D) and Lem. 11.22 (a), (c) provide the whole half line of such j.

Corollary 12.7 (Each component is a theory in its own right, satisfying all axioms, with $S_2^{\sigma_\xi} \neq 0$)**.** Let $\mathsf{b} > 0$. Then $\mathrm{Lim}_G(\mathsf{b}) \neq \emptyset$. For every $S \in \mathrm{Lim}_G(\mathsf{b})$, every component σ_ξ of S is, up to the shift, the functional of a member of $\mathrm{Lim}_G(\mathsf{b})$ whose only component is itself: $\sigma_\xi = \sigma_\xi^{\mathrm{c}} \circ \mathrm{sh}_{c_\xi}$, where σ_ξ^{c} is the centred functional of $S^\xi = (S_n^{\sigma_\xi})_{n\geq 2} \in \mathrm{Lim}_G(\mathsf{b})$. Each such component has the following properties. It satisfies (E0′) to (E4) of Def. F.17, with full Euclidean invariance. It is reflection positive at every plane. It has a unique vacuum, and it clusters. It carries the point functions of clause (E4)(b3) of Thm. 12.4 and the Wightman theory of its clause (E4)(b4). Finally, $S_2^{\sigma_\xi} \neq 0$, with the lower bound (C2).

■□

Remark 12.8 (Two words of this section). In this section, an *ergodic component* is understood as in Def. E.12. The word *non-trivial* is meant in the sense of (E7)(b).

The moment estimates give the class and bound system needed for canonical decomposition. The centering law is held fixed while the integrating law is conditioned. The point-insertion bound then applies to all limits of the resulting frame. To prove nontriviality in each component, we first transfer the reflected test estimate to all periodic Gibbs states and then pass to the conditioned limits.

The letter P_0 denotes the projection on $\ker H$. We also write U for the unitary representation $v \mapsto U(v)$ of $\mathbb{R}^2$ on $\mathcal{H}(\sigma')$ with $U(v)[F] = [\tau_{(v,0)}F]$, which is the $\mathcal{U}$ of (Rec2) and of Prop. D.2.

Lemma 12.9 (Step 1: the class, the moment bound and the bound system)**.** Let $\mathsf{b}' \geq x_s$, which in this step we write b; let $K \geq 1$, and let $\rho, \nu \in \mathbb{G}_K(\mathsf{b})$. Every number of this step, that is, j°, G_1, $\check{\Gamma}$, $B_0(\cdot,\cdot)$, c_{vol} and A_4, is evaluated at the single label b itself, not as a supremum over $U(\cdot)$.

(a) *One cell.* Let $g_1, \dots, g_r$ be real bounded functions carried by one cell of B_0, at sup distance $\geq \check{\delta}$ from one another and at distance $\geq \check{\delta}/2$ from the walls $\mathbb{Z} + \frac{1}{2}$, each in a sup-ball of radius $\check{\delta}/4$, where $0 < \check{\delta} \leq 1$ and $K \geq j^\circ(\mathsf{b}, \check{\delta})$; let the c_i be real. Then, for every N of the data,

$$\|\prod_i (\mathcal{O}_K g_i - c_i)\|_{\mathrm{ch},N} \leq e^{c_{\mathrm{vol}}} \prod_i (G_1 \|g_i\|_\infty + |c_i - \hat{c}_{K,N}(g_i)|),$$

where $G_1 = G_1(\mathsf{b}, \check{\delta}, \check{\delta}/4)$.

(b) *Cell by cell.* Let finitely many pieces be placed so that, in each cell t, the n_t pieces lying in t satisfy the conditions of (a) with a number $\check{\delta}_t$ in the place of $\check{\delta}$. Then

$$|\rho(\prod_i (\mathcal{O}_K g_i - \nu(\mathcal{O}_K g_i)))| \leq \prod_{t \text{ occupied}} [\check{\Gamma}(\mathsf{b}, \check{\delta}_t)^{n_t} e^{c_{\mathrm{vol}}}] \cdot \prod_i \|g_i\|_\infty.$$

(c) The moment bound of (E0), the bound on $|\rho(X_K^\nu)|$, holds for every $n \geq 1$, for real bounded f_i of the Euclidean type $(d; R)$ and for all $K \geq 1$.

(d) The bound of (E0) on $|S_n^T[\rho](X)|$ holds for $n \geq 2$, and so do (Cls1), (Cls2), (Bd1) and (Bd2).

■

Proof. Clause (a) is Thm. 10.4(4.1)(a) at $\check{\rho} = \check{\delta}/4$. The fixed-label estimate in the proof of (4.3)(a) gives (b) before the supremum over a label neighborhood is taken. Its application to each occupied cell uses the same sequence of sides for the integrating law and the centering law. This gives (c) by (4.3)(b), which also gives the cumulant bound in (d). The one-point bound (Bd2) is (c) at $n = 1$, and (Bd1) follows from the cumulant estimate since $\|f\|_\infty \leq \|f\|_\sharp$. Prop. B.28(a), (b) gives (Cls1). For (Cls2), Prop. B.29(a) gives the Gibbs equations after invariant conditioning, and the conditioned law remains invariant under $a_K\mathbb{Z}^3$. □

Proof of Thm. 12.4. Step 3: from (E0) *to* (E1), (E2), (E3) *and* (E4)*(a).* In this step we derive (E1) to (E4)(a) from three inputs: (E0); Thm. 10.4 (4.6), (4.7); and the statements proved in the supporting sections that item B of §12 cites.

(1) *The data of Def. F.19.* Once the j whose labels lie outside $U(\mathsf{b})$ are dropped, $(K_j, \mathbb{G}_{K_j}(\mathsf{b}_j))$ is a class system obeying C^{b}, and (K_j, ν_j) is a frame. We have $\sigma_{\nu_j}(X) = \sum_{\mathrm{Part}_{\geq 2}} \prod_J S^T_{|J|}[\nu_j](X_J)$; conversely, the cumulants are fixed polynomials in the moments (eq. C.vi). Hence $S_n = \lim S^T_n[\nu_j]$ exists on all tuples if and only if the frame converges, and then $\sigma_{\mathrm{frame}} = \sigma_S$. Converging subsequences exist by Lem. E.14, and a decomposition exists by Thm. E.13.

(2) *The hypotheses of Thm. F.22.* We substitute the data of Step 3 (1) for the letters of Thm. F.22; then the hypotheses of that theorem are met. Its clauses (1)(a) to (e) give the assertion of (E1) that σ_S has the properties of these clauses, while the other three assertions of (E1) on σ_S come from Step 3 (3), which gives the invariance under $\mathbb{R}^3 \rtimes \mathrm{O}(3)$ and the reflection positivity at every mirror, from Step 3 (4), which gives the growth condition eq. C.xxxii with the quadruple stated there, and from Step 4 in the case $c = 0$, which gives the data (Rec1) to (Rec4) with $(C_\varphi, 5, \frac{1}{2})$. Its clauses (2)(a) and (b), which are the content of Prop. E.15 (a) and of Thm. E.22 by way of Cor. E.23, give (E2) except for the assertion that every isometry acts on X as the identity; this assertion is given by clause (2)(c) with $h_r = \mathrm{id}$, which the condition (Rot) of Step 3 (3) yields. Its clauses (3)(i), (ii)(a), the first half of (ii)(b), (iii) and (iv) give (E3) for the components $\sigma' = \sigma_\xi$, apart from its part on rotations. For a general limit σ' in the frame, Prop. E.18 (a), (b), (c), (e) shows that σ' is symmetric, invariant under translations and W_3, and reflection positive at the coordinate and the diagonal mirrors, that $\sigma' \in \mathsf{Fct}$, and that σ' is regular of order $\leq 4n+1$; Prop. E.19 (a) gives $S_1 = c_{\sigma'} \int f$. The datum (Rec2) of Prop. 12.10 (b2) shows that $\mathcal{H}(\sigma')$ is separable and that $H \geq 0$, since, by Step 3 (3), every limit in the frame is invariant under $\mathrm{O}(3)$, and this invariance is the hypothesis of clause (b2). As $\sigma' \in \mathsf{Fct}$ by Prop. E.18 (e), and as Step 3 (3) gives the invariance of σ' under $\mathbb{R}^3 \rtimes \mathrm{O}(3)$ and its reflection positivity at $(e_3, 0)$, Cor. D.8 applies to σ' itself: it gives the representation of $\mathrm{O}(2)$ of Prop. D.6, and it shows that the joint spectrum lies in the cone $E \geq |q|$, which is the datum (Rec3) of Prop. 12.10 (b2).

(3) *The condition* (Rot) *for every limit in the frame.* Let σ' be a limit in the frame, with approximants $\rho_j \in \mathbb{G}_{K_j}(\mathsf{b}_j)$. Prop. E.18 (d) gives $S^{\sigma'}_n = \lim_j S^T_n[\rho_j]$ for $n \geq 2$, and $\sigma'^{\mathrm{c}} = \lim \sigma_{\rho_j}$. By the definition of $\mathrm{Lim}_G(\mathsf{b})$, the sequence $(S^{\sigma'}_n)_{n\geq 2}$ is therefore a member of $\mathrm{Lim}_G(\mathsf{b})$, and its centred functional is σ'^{c}. Thm. 10.4 (4.7) then shows that σ'^{c} is invariant under $\mathbb{R}^3 \rtimes \mathrm{O}(3)$ and reflection positive at every mirror. Next, Prop. E.19 (a) gives $\sigma'((f)) = c \int f$, and from eq. C.vii we get $\sigma' = \sigma'^{\mathrm{c}} \circ \mathrm{sh}_c$. The map sh_c commutes with every isometry r, because $\int f \circ r = \int f$; it commutes with conjugation, because c is real; and it commutes with concatenation. Hence $\sigma'(rX) = \sigma'(X)$, and for every mirror m

$$\sigma'(\theta_m \bar{F} \cdot F) = \sigma'^{\mathrm{c}}(\theta_m \overline{\mathrm{sh}_c F} \cdot \mathrm{sh}_c F) \geq 0.$$

This is the condition (Rot) of Def. F.20. It yields the clauses (1)(g) and (2)(c) of Thm. F.22, with $h_r = \mathrm{id}$, and those halves of (3)(i) and (iii) that are conditional on (Rot): the circular cone, by Cor. D.8, and the group $\mathrm{O}(2)$, by Prop. D.6.

(4) *Growth.* By (E0), $\rho(X^{\nu_j}) \in [-B, B]$ for all $\rho \in \mathbb{G}_{K_j}(\mathsf{b}_j)$, where the interval is closed and B is the bound of (E0). Prop. E.18 (f), applied with $k = 1$, therefore gives $|\sigma'(X)| \leq B$. For $R \leq 1$ this becomes

$$|\sigma'(X)| \leq (8A_6)^n (n!)^5 \max\{1, d^{-5n}\} \prod \|f_i\|_\sharp.$$

We now apply Prop. C.13 to $\mathsf{T} = \sigma'$, with $\mathsf{a} = 5n$ and $\mathsf{c}_\mathsf{T} = (8A_6)^n (n!)^5$. It yields the complex linear functional $\mathsf{S}^{\sigma'}_n$ of Def. C.17 on $\mathcal{S}_{\neq}(\mathbb{R}^{3n})$, which satisfies $\mathsf{S}^{\sigma'}_n(\bigotimes X) = \sigma'(X)$ on tensors, together with the bound

$$|\mathsf{S}^{\sigma'}_n(\varphi)| \leq \mathsf{C}^{6n}_{\neq} M^n (n!)^{13} (8A_6)^n |\varphi|_{13n-1},$$

the case $n = 1$ being included. This is the bound eq. C.xxxii with the quadruple $(13, 1, C_E, 13)$.

(5) *The axioms.* We verify the hypotheses of Prop. F.18 for the components σ_ξ. Membership in $\mathsf{Fct}(C,\bar c)$ is supplied by Prop. E.18 (e), the growth condition by Step 3 (4), the invariance by Step 3 (3), reflection positivity at $(e_3,0)$ by Prop. E.18 (c), and clustering by Prop. E.20.

(6) *The constant $\bar c_*$.* For $n=1$ the bound of Step 3 (4) reads $|c_{\sigma'}|\,|\int f| \le A_6(1+R)^3\|f\|_\infty$. We take $f\in C^1_c$ with $0\le f\le 1$ which equals 1 on a ball of radius $R-1$ and is carried by the ball of radius R, and we let $R\to\infty$. Since $(1+R)^3/(\frac{4\pi}{3}(R-1)^3)\to\frac{3}{4\pi}$, this yields $|c_{\sigma'}|\le\frac{3}{4\pi}A_6$.

Step 4: the clauses (E4)(b1) *and* (b2). They make up the following proposition.

Proposition 12.10 (Point bounds and reconstruction data)**.** (b1) The point bound eq. F.lxx holds in the form

$$|\sigma'^{\mathrm c}(\theta\bar F\cdot(h)\cdot F')| \le 32 C_{\mathrm{pt}}\,\eta^{-5}\|h\|_{L^1}\,\|[\tau_{-\eta}F]\|\,\|[\tau_{-\eta}F']\|$$

for every limit σ' in the frame, every $\eta\in(0,1]$, all $F,F'\in\mathbb{CS}(\{x_3>\eta\})$ and every $h\in C^1_c(\{|x_3|<\eta/2\})$.

(b2) Every limit σ' in the frame that is invariant under O(3) carries the data (Rec1) to (Rec4) with $(C_\wp+|c_{\sigma'}|,5,\frac12)$, and the components σ_ξ satisfy (Rec5) as well. ∎

Proof of Prop. 12.10. The clause (b1). By Step 3 (3), $\sigma'^{\mathrm c}$ is the centred functional of a member of $\mathrm{Lim}_G(\mathsf b)$. Thm. 10.4 (4.6) therefore gives it the properties (R1) to (R4), and (R5) with the triple $(c_{\mathrm I},C_{\mathrm I},\mathsf p_{\mathrm I})$ of item 2 of §10, where $C_{\mathrm I}$ is the supremum of the whole plain-label expression; these numbers do not depend on σ'. They are exactly the hypotheses of Cor. D.24, with $\wp=\mathsf p_{\mathrm I}+4=5$ and with C_{pt} as in eq. D.xxxvii. Clause (b) of that corollary, applied with $d=\eta$, gives $\sigma'^{\mathrm c}(\theta\bar F\cdot(h)\cdot F')=\int h(x)\mathsf v_\eta(x)\,dx$ with a function $\mathsf v_\eta$ that obeys, by eq. D.lii,

$$|\mathsf v_\eta(x)| \le C_{\mathrm{pt}}\min\{1,\eta-|x_3|\}^{-5}\|[\tau_{-\eta}F]\|\,\|[\tau_{-\eta}F']\|.$$

On the support of h we have $\eta-|x_3|>\eta/2$; as also $\eta\le1$, the factor with the minimum is $\le 2^5\eta^{-5}$ there, and (b1) follows.

For (b2), Euclidean invariance of the frame limits was proved in Step 3(3). Use (b1) as the point-bound hypothesis of Prop. F.21. Its scalar shift gives $C_\wp+|c_{\sigma'}|$, and the one-dimensional zero-energy space for a component gives (Rec5). □

Lemma 12.11 (One Hilbert space for a limit and its centred functional; the point bound carries over)**.** Let σ' be a limit in the frame, and let $c=c_{\sigma'}$, so that $\sigma'=\sigma'^{\mathrm c}\circ\mathrm{sh}_c$. Then

$$\langle[F],[F']\rangle_{\sigma'}=\sigma'^{\mathrm c}(\theta\overline{\mathrm{sh}_cF}\cdot\mathrm{sh}_cF')=\langle[\mathrm{sh}_cF],[\mathrm{sh}_cF']\rangle_{\sigma'^{\mathrm c}}.$$

The map $[F]\mapsto[\mathrm{sh}_cF]$ is isometric and onto, it intertwines e^{-tH} and $U(v)$, and it fixes Ω. Moreover, σ' satisfies (Rec4) with the constant $C_\wp+|c|$, and $|c|\le\bar c_*$. ∎

Proof. Prop. D.3 gives the unitary and its intertwining properties, with $\sigma=\sigma'^{\mathrm c}$ and shift c. Prop. 12.10(b1) supplies the centered bound required in Lem. F.4. That lemma gives (Rec4) with constant $C_\wp+|c|$. The one-point estimate in Step 3(6) bounds $|c|$ by $\bar c_*$. For σ_S the constant c is zero. □

The combined constant. We work in normal form, $\mathsf b>x_s$. The convention of (S8) takes the supremum of the whole plain-label expression $C_\wp+\bar c_*$, so we keep these two terms at one label. Let σ' be a limit in the frame, represented by $\rho_j\in\mathbb G_{K_j}(\mathsf b_j)$ with the original centring laws ν_j. By local boundedness we may pass to a subsequence, without changing σ', on which the plain values satisfy

$$A_6(\mathsf b_j)\longrightarrow A,\qquad C_{\mathrm I}(\mathsf b_j)\longrightarrow I.$$

The fixed-label lattice estimate in Step 5 of the proof of Thm. 10.4 has coefficient $C_{\mathrm I}(\mathsf b_j)\eta^{-1}$. Its limiting argument therefore gives (R5) for $\sigma'^{\mathrm c}$ with $I+\epsilon$ in place of $C_{\mathrm I}$, for every $\epsilon>0$. Applying

Cor. D.24 and letting $\epsilon \downarrow 0$ yields the centered point bound with coefficient

$$P := 32 \cdot 320 \cdot 8^4 \mathsf{mv}_3 c_{\mathrm{I}}^{-4} MI = \lim_j C_\wp(\mathsf{b}_j).$$

The plain-label estimate of Thm. 10.4 (4.3)(b), used with $n = 1$ and the centring law ν_j and passed to the same subsequence, gives

$$|c_{\sigma'}| \left| \int f \right| \le A(1+R)^3 \|f\|_\infty.$$

The ball tests of Step 3(6) imply $|c_{\sigma'}| \le 3A/(4\pi)$. The shift lemma F.4 now gives (Rec4) for σ' with coefficient at most

$$P + \frac{3A}{4\pi} = \lim_j (C_\wp(\mathsf{b}_j) + \bar{c}_*(\mathsf{b}_j)) \le \sup_{\mathsf{b}' \in U(\mathsf{b})} (C_\wp(\mathsf{b}') + \bar{c}_*(\mathsf{b}')) = C_*(\mathsf{b}).$$

All quantities inside this supremum are plain values. The same argument applies after the fixed dilation in (E5), using its modified plain coefficients before taking their sum.

Point kernels and reconstruction. The preceding argument supplies the insertion bound for the component functionals with constant at most C_*. Apply the ordered kernel construction eq. F.xxiv– eq. F.xxx and the extension eq. F.xxxv to each component. If $n \ge 2$, choose a direction with successive projection gaps at least $d(x)/n^2$ by eq. F.xxxiv. The bound eq. F.xxxvi with $\eta = \min\{1, d(x)/(2n^2)\}$ is the estimate in (E4)(b3). For $n = 1$ use $\mathbf{W}_1 = \langle \Omega, B(1,1)\Omega \rangle$.

Thm. F.16 gives (E4)(b4), including locality and tempered matrix elements. The complex time variable in that statement is the third Euclidean coordinate. The norm estimate eq. F.xxvii at $\eta = \varepsilon_T$ gives the claimed bound on W_n. The same construction applies to σ_S with its vacuum space $L^2(\mathsf{X}, \lambda)$. The one-dimensional vacuum conclusion uses (Rec5) for a component.

Proposition 12.12 (Every label $\mathsf{b} > 0$ and labels $\mathsf{b}_j \to \mathsf{b}$; joint limits at every b)**.** The clauses (E0) to (E4) hold at labels $\mathsf{b} \le x_s$, with the constants of (E5) and with $z = z(\mathsf{b})$. In particular (R5) holds for every $\eta \in (0,1]$, with $(c_{\mathrm{I}} L^{-z}/2,\ 2L^z C_{\mathrm{I}}(L^z \mathsf{b}),\ 1)$. The clause (E6) holds at every $\mathsf{b} > 0$. ∎

Proof. (1) *Dilation.* The map $x \mapsto L^z x$ carries $a_K \mathbb{Z}^3$ to $a_{K-z}\mathbb{Z}^3$, and its pushforward is a bijection from $\mathbb{G}_K(\mathsf{b}')$ to $\mathbb{G}_{K-z}(L^z \mathsf{b}')$, since $\beta = L^{K-z}(L^z \mathsf{b}')$. Moreover $\mathcal{O}_K(f) = L^z \mathcal{O}_{K-z}(f^{[z]})$ with $f^{[z]}(y) = f(L^{-z}y)$. The type $(d; R)$ of a tuple becomes $(L^z d; L^z R)$, and $f^{[z]}$ has the same sup norm as f. Hence (E0) holds for $K \ge z + 1$ with $L^z A_6(L^z \mathsf{b})(1 + L^z R)^3$ for each entry, and $\max\{1, (L^z d)^{-5n}\} \le \max\{1, d^{-5n}\}$. The one number $z = z(\mathsf{b})$ suffices for all labels of $U(\mathsf{b})$, because $L^z U(\mathsf{b}) \subset [x_s, \infty)$; we discard the finitely many j with $K_j \le z$ or $\mathsf{b}_j \notin U(\mathsf{b})$. At $\mathsf{b} = x_s$ the definition of $z(\mathsf{b})$ gives $z = 1$ and $U(\mathsf{b}) = [x_s/2, 2x_s]$, so that the same numbers also cover the sequences that tend to x_s from below. With this bound system, the frame and point estimates of Steps 3 and 4, scalar reconstruction in Appendix F, and the argument of Step 9 apply at label b. Parts (2) to (5) below verify the corresponding constants and hypotheses.

(2) *The numbers.* For $R \le 1$ we have $\mathsf{c}_\mathsf{T} = (L^z(1+L^z)^3 A_6(L^z \mathsf{b}))^n (n!)^5$; with this c_T in place of $(8A_6)^n (n!)^5$, Prop. C.13 gives, as in Step 3 (4), the number that takes the place of C_E. From $|c|\,|\int f| \le L^z A_6(L^z \mathsf{b})(1 + L^z R)^3 \|f\|_\infty$, with the functions f of Step 3 (6) and $R \to \infty$, we obtain $\bar{c}_*(\mathsf{b}) = \frac{3}{4\pi} L^{4z} A_6(L^z \mathsf{b})$.

(3) *The properties* (R1) *to* (R4). For σ'^{c} the properties (R1) to (R4) do not depend on the label. The properties (R1), (R2), (R3) come from Prop. E.18 (a), (b), (c), (e); the property (R4) comes from Prop. D.5, by way of the diagonal mirrors; and the invariance under O(3) comes from Thm. 10.4 (4.8).

(4) *The property* (R5). Put $D_z X = L^{z|X|} X^{[z]}$. The centered functionals satisfy $\sigma'^{\mathrm{c}} = \sigma''^{\mathrm{c}} \circ D_z$, where σ'' is the corresponding frame limit at $(K_j - z, L^z \mathsf{b}_j)$. By (E3)(i) in normal form, σ''^{c} is the centered functional of a member of $\mathrm{Lim}_G(L^z \mathsf{b})$. It therefore satisfies the hypotheses of Lem. 10.5, by Thm. 10.4 (4.6). That lemma gives (R5) for every $\eta \in (0,1]$, with $(c_{\mathrm{I}} L^{-z}/2,\ 2L^z C_{\mathrm{I}}(L^z \mathsf{b}),\ 1)$.

(5) *The conditions* (PtB) *and* (Rec). They follow by the proof of Prop. 12.10 in Step 4, with these changes: (R1) to (R4) are taken from Step 8 (3), and (R5) from Step 8 (4); where that proof applies Cor. D.24, C_{pt} is taken at $(c_{\mathrm{I}}L^{-z}/2,\, 2L^{z}C_{\mathrm{I}}(L^{z}\mathsf{b}))$, which is $2^{4}L^{4z}\cdot 2L^{z} = 32L^{5z}$ times $C_{\mathrm{pt}}(L^{z}\mathsf{b})$; and in Lem. 12.11 the bound on $|c|$ is the one of Step 8 (2).

(6) *The clause* (E6). This clause is Thm. 10.4 (4.9) (i). □

Step 9: the exact relation between S_2 and the components. In this step we prove the clauses (E7)(a), (b) and (c) of Thm. 12.4.

(a) *The clause* (E7)(a). By eq. E.xxxvii, applied with $q=1$, $\sigma_S(\theta f, f) = \int \sigma_\xi(\theta f, f)\, d\lambda$; moreover $\sigma_S(\theta f, f) = S_2(\theta f, f)$, because $S_1[\sigma_S] = 0$. As $\int \theta f = \int f$, we have $\sigma_\xi(\theta f, f) = S_2^\xi(\theta f, f) + c_\xi^2(\int f)^2$. In $\mathcal{H}(\sigma_\xi)$ we have $g_f(\xi) = \|[(f)] - c_\xi \int f\,\Omega\|^2 \geq 0$, and g_f is continuous in ξ by Def. E.12. Prop. E.21 (a) and eq. E.xxxvii, applied with $q=2$, give

$$\|P_0[(f)]\|^2 = \lim_t \sigma_S(\theta f \cdot \tau_t f) = \int \sigma_\xi((\theta f))\sigma_\xi((f))\, d\lambda = V(\int f)^2.$$

Subtracting, we obtain $\int g_f\, d\lambda = \|(1-P_0)[(f)]\|^2$. Finally $V \leq \bar{c}_*^2$ by (E3)(ii).

(b) *The clause* (E7)(b). By (b3), $\mathsf{S}_2^\xi(x,y) - c_\xi^2 = s(|x-y|)$. Write $B_{\mathsf{e}} = B(\mathsf{e},\mathsf{e})$ as in eq. F.xiii. It is self-adjoint, and $c_\xi = \langle \Omega, B_{\mathsf{e}}\Omega\rangle$ by the one-point case of eq. F.xxiv. We place two points at the distance t along e_3 and put $\psi_{\mathsf{e}} := B_{\mathsf{e}}\Omega - c_\xi\Omega$, a vector orthogonal to Ω. Then

$$s(t) = \langle \psi_{\mathsf{e}}, e^{-(t-2\mathsf{e})H}\psi_{\mathsf{e}}\rangle = \int e^{-(t-2\mathsf{e})E}\, d\mu_{\mathsf{e}}(E)$$

for $0 < \mathsf{e} \leq t/2$, with $\mu_{\mathsf{e}} \geq 0$. Hence $s \geq 0$, and s is non-increasing. If $s(t_0) = 0$, then $e^{-(t_0-2\mathsf{e})H/2}\psi_{\mathsf{e}} = 0$, and so $\psi_{\mathsf{e}} = 0$, since e^{-sH} is injective. As $\psi_{\mathsf{e}'} = e^{-(\mathsf{e}'-\mathsf{e})H}\psi_{\mathsf{e}}$ for $\mathsf{e}' \geq \mathsf{e}$, it follows that $\psi_{\mathsf{e}} = 0$ for every e. Suppose now that $g_f(\xi) = \iint f(\theta x) f(y) s(|x-y|)\, dx\, dy = 0$ for a continuous $f \geq 0$ with $f \neq 0$. Then s vanishes somewhere, and hence $B_{\mathsf{e}}\Omega = c_\xi\Omega$ for all e. In eq. F.xxvi the last insertion then gives $c_\xi\Omega$. The translations fix Ω, so induction gives $\Psi_n(x) = c_\xi^n\Omega$. The ordered tuple classes are total by the approximation in §F.3.4, and therefore $\dim\mathcal{H} = 1$; moreover $\mathsf{S}_n = c_\xi^n$, which is the product formula. Conversely, if $\dim\mathcal{H} = 1$, then $[(f)] = \sigma_\xi((f))\Omega$, so that $g_f = 0$ for all f, and polarisation together with translations and O(3) gives $S_2^\xi = 0$.

(c) *The clause* (E7)(c). As an instance we take $\frac{1}{2}(\sigma_0 + \sigma_0 \circ \mathrm{sh}_{c'})$, centred, where σ_0 is the functional equal to 1 at the empty tuple and to 0 elsewhere. It has $S_2 = (c'^2/4)\int f \int f'$, two trivial components and $\dim\mathcal{H}(S) = 2$, by Rem. E.24. Hence Thm. 11.1 (N5) by itself decides nothing about the components; non-triviality of every component follows from Thm. 12.6. □

Proof of Thm. 12.5. Step 10: domination at one side; cell functions of the pair $(U, V^{(k)})$. The first of the three steps towards Thm. 12.5 consists of two lemmas: Lem. 12.13 gives domination at one side for a cell function that is allowed to depend on the side, and Lem. 12.14 treats the cell functions of the pair $(U, V^{(k)})$, with the three bounds of §11 as seminorms.

Lemma 12.13 (Domination at one side for a cell function allowed to depend on the side)**.** Let $\rho \in \mathbb{G}_K(\mathsf{b}')$, and let B_i be a board of $\mathbb{L}_K$ with cells of side $l_c = a_i$. Let $N \in \{2L^m\}$ with $N \geq 2l_c$, and let $F \in \mathcal{A}(\square_0)$ with $\sup|F| \leq \mathsf{m}$. Put $\epsilon_N := 36\beta(N/a_K)^2\lfloor N/2l_c\rfloor^{-3}$. If $\epsilon_N \leq \frac{1}{4}$, then $|\rho(F)| \leq \|F\|_{\mathrm{ch},N} + 2\mathsf{m}\,\epsilon_N^{1/2}$. Moreover $\epsilon_N \leq 2304\,\beta a_K^{-2} l_c^3/N \to 0$ as $N\to\infty$. As a consequence, if $F_N \in \mathcal{A}(\square_0)$, $\sup_N \sup|F_N| < \infty$ and $\|F_N\|_{\mathrm{ch},N} \leq A$ for all large N, then $\limsup_N |\rho(F_N)| \leq A$. ■

Lemma 12.14 (Cell functions of the pair $(U, V^{(k)})$, and three bounds as seminorms)**.** Let Φ be a bounded real Borel function of the inner bonds of a cell and the drawn bonds of level k whose two k-blocks both lie in that cell, where the walls of the board lie in $\mathbb{Z} + \frac{1}{2}$ at level k.

(a) $F := E[\Phi \mid U] \in \mathcal{A}(\square)$, and $P_\rho(\Phi) = \rho(F)$.

(b) $\|F\|^{Q}_{\mathrm{ch},N} = E_{\mathrm{torus}\,N} \prod_c \Phi \circ \check\Theta_c$, where Q is the number of cells and $\check\theta_c$, $\check\Theta_c$ are the maps of the fold, those of the data of Lem. 11.7 with h replaced by the side of the cells of the board, counted in units of level k.

(c) For every N of the data, that is, every $N = 2L^{m_l}$ with $m_l \geq m'$ and with (Tor) holding at b', for all $K \geq j+1$ and all $j \geq j_*(\mathsf{b}')$, and on boards without offset, the three bounds below hold.

(P°) (Lem. 11.5) On B_{j-1} and for p based at 0, $\|E[1\{\phi_p \geq \varepsilon/2\} \mid U]\|_{\mathrm{ch},N} \leq e^{-(c_P/2)p_0^2}$.

(W°) (Lem. 11.10) On B_{j-i_h} and for every constant $g \in [0, \frac{9}{8}p_0^2]$,

$$\|E[(G_- - g)(\mathcal{O}_K f_* - c_f^N - A_f^{\mathrm{tr},N}(V)) \mid U]\|_{\mathrm{ch},N} \leq \frac{9}{8}p_0^2 \cdot 2e^{2.2}/r_z.$$

(X°) (Lem. 11.13) On the same board and at $w = w_R$,

$$\|E[\Psi \circ \tau_{-t}\, e^{\Lambda^{\mathrm{tr}} \circ \tau_{-t} - \Lambda^{\mathrm{tr}}} - \Psi \mid U]\|_{\mathrm{ch},N} \leq \sup|\Psi^\circ| \cdot 2e^{2.2}/(r_\eta - 1).$$

Here Ψ is the function of Step 11, that is, the Ψ^{tr} of Lem. 11.13, and $\Psi^\circ := (\Phi e^{\Lambda^a}) \circ \tau_t\, e^{-\Lambda^a}$, with $t = x^{-2}$ the number t of §11, is the Ψ of Lem. 11.13. In (X°), Φ is not the function of the opening sentence of this lemma but, as in the data of Lem. 11.13, a bounded Borel function of the bonds of B', carried by $\{\phi_p \leq 3\varepsilon/4 \text{ on } B'\}$.

■

Proof of Lem. 12.13. 1. Since ρ is invariant under $a_K\mathbb{Z}^3$, the number $P = 1$ is a period number in the sense of Def. B.1 (3).

2. As data of Def. B.17 we take $\mathcal{K}_0 = \{0\}$, $F_0 = F$, $\mathsf{q} = 2$, which is even and a multiple of P, and $\mathbf{k}_0 = 0$. With these data the letter P of Def. B.17 equals $2l_c$, and the number ϵ_ℓ of eq. B.xxii is ϵ_N.

3. Cor. B.19, applied at this one side, is the claim. In that corollary m is $\sup|F|$, and a larger choice of m would only make the bound weaker.

4. Because $\lfloor y \rfloor \geq y/2$ for $y \geq 1$, we have $\lfloor N/2l_c \rfloor^{-3} \leq (4l_c/N)^3$, and hence $\epsilon_N \leq 36 \cdot 64\,\beta a_K^{-2} l_c^3/N$.

5. *A cell function that may depend on the side.* Thm. B.18 (a) and Cor. B.19 assume that the data of a domination are given and that ℓ belongs to the set of sides with $\ell \geq \mathsf{P}$; the corollary assumes in addition that $\epsilon_\ell \leq \frac{1}{4}$. For each N_0 we give the data with $F = F_{N_0}$, and we use the conclusion at $\ell = N_0$ alone. No hypothesis asks that F be the same at two sides.

□

Proof of Lem. 12.14. (a) This clause follows as well from Lem. 11.3 (L) and (I); we give the argument. By item 2 of §5.5, which describes the step, the average $\mathsf{M}(U)(c)$ is determined by t_U on $B(y)$ and on $B(y')$ together with the straight paths $[x, x + Le_\mu]$, all of which lie in $B(y) \cup B(y')$. This average is a Borel function of U, and k_s is a probability kernel. We follow the dependence level by level, from level k to level 0: a drawn bond of level k depends on the drawn bonds of level $k-1$ in its two blocks, and so in the end on the bare bonds in the union of its two centred k-blocks. Since L is odd, every plane in $\mathbb{Z} + \frac{1}{2}$ of level k is a face of the centred blocks of each finer level, and lies in $a_K(\mathbb{Z} + \frac{1}{2})$.

Therefore F is a bounded Borel function of the inner bonds. The law P_ρ is ρ times a product kernel, and Φ depends on finitely many of its draws; it follows that $P_\rho(\Phi) = \rho(F)$. The formula of the kernel is local and does not depend on the side, so $F \circ \mathsf{per}_N$ is the function that the same formula gives on the torus.

(b) The identity $\check\Theta_c E[\Phi \mid U] = E[\Phi \circ \check\Theta_c \mid U]$ is the third claim of Lem. 11.3 (K). Its proof uses two pointwise facts, in which no torus enters: $\mathsf{M}(\check\Theta U)(c) = \mathsf{M}(U)(\check\theta\, c)$ and $k_s(g^{-1}, h^{-1}) = k_s(g, h)$. Here $\check\theta$ and $\check\Theta$ are the maps of Lem. 11.3 (K) for any choice, in each coordinate, of a translation $y_\mu \to y_\mu + n_\mu$ or a reflection $y_\mu \to n_\mu - y_\mu$. The draws used in different cells,

at all levels $\le k$, form disjoint families, and by Lem. 11.3 (I) these families are independent given U. All functions here are real.

By Def. B.12 (2), $\|F\|_{\mathrm{ch},N} = \|F \circ \mathsf{per}_N\|_{\mathrm{ch}}$, the seminorm on the right being that of eq. A.xxiii, whose Q-th power is the mean of the product of the copies. By the identity and the independence above, this mean equals $E_{\mathrm{torus}\,N} \prod_c \Phi \circ \breve{\Theta}_c$.

(c) The three clauses come from §11. There Lems. 11.5, 11.10 and 11.13 are formulated as bounds on $\|E[\Phi \mid U]\|_{\mathrm{ch}}$: they give (P°), (W°) for every constant $g \in [0, \frac{9}{8}p_0^2]$, and (X°). By (b), the fold $E\prod_c \Phi \circ \breve{\Theta}_c$ is $\|F\|^Q_{\mathrm{ch},N}$. In (X°) the truncation is part of the definition of Λ and of Λ^{tr} in Step 11, and these two are the $\Lambda^{a,\mathrm{tr}}$ and Λ^{tr} of §11.

□

The boards. The cells of the two boards have the sides a_{j-1} and a_{j-i_h}, where $h = L^{i_h}$; both indices are $\le j \le K-1$. There is no offset: as in §11, the cell 0 is centred at 0. All sites of a plaquette p based at 0 lie in the cell of side L, and all sites of Box lie in the cell of side h, because $r_B \le (h-1)/2$. Hence every bond of p or of Box is a drawn bond of level k with both of its k-blocks in the respective cell, as in the opening sentence of Lem. 12.14. Every other place is reached by a translation in $L^{-j}\mathbb{Z}^3$, the set of sites of level k, which lies in $a_K\mathbb{Z}^3$. Since ρ is invariant under these translations and the kernel is covariant under them by Lem. 11.3 (K), the law P_ρ is invariant under them. In each direction there are $2L^{m_l+j-1}$ and $2L^{m_l+j-i_h} \ge 2L$ cells respectively, by entry (j3) of the list of conditions on the depth in §11.

Sup bounds free of N, for Lem. 12.13. Keep K, j and b' fixed. Only plaquettes whose barycentres meet $\operatorname{supp} f_*$ contribute to $\mathcal{O}_K f_*$. This support lies in a cube of side $(D_0/4)L^{-j}$. For each orientation its barycentres form a translate of $a_K\mathbb{Z}^3$, so

$$\sup|\mathcal{O}_K f_*| \le 6L^K\big((D_0/4)L^{K-j}+2\big)^3 =: C_{K,j}.$$

This number is independent of N. The termwise estimates of Lem. 11.4 give

$$\max\{\sup|A_f|, \sup|A_f^{\mathrm{tr},N}|\} \le (C_S p_0^2+1)/\mathsf{b}' =: A_*.$$

At $V=1$, the field is globally tame, so $W_f(1)=0$ by Step (b) of the proof of Lem. 11.7. Consequently Lem. 11.4 gives $|c_f^N| \le C_{K,j}+A_*$. For every $g \in [0, 9p_0^2/8]$, the cell function in (W°) therefore has supremum at most $(9/4)p_0^2(C_{K,j}+A_*)$. The cell function in (X°) has supremum at most $2e\sup|\Phi|$; for the four tests used below, $\sup|\Phi| \le 1.5C_S\varepsilon p_0^2$. Conditional expectation preserves these supremum bounds. Thus each application of Lem. 12.13 has an error bounded by a constant independent of N times $\epsilon_N^{1/2}$.

Step 11: two lemmas at one side. We fix a side N of the data, that is, $N = 2L^{m_l}$ with $m_l \ge m'$ and with (Tor) holding at b', such that $\epsilon_N \le \frac14$ on B_{j-i_h}; both lemmas below are stated at this side. The rows, the members, w_R, S_f and $\mathbb{D}_k$ are those of the torus of side N, and what follows holds however they depend on N. For the function Φ of Lem. 12.15 below we put $\delta_N := 4e\sup|\Phi|\,\epsilon_N^{1/2}$.

Lemma 12.15 (Approximate integration by parts under every periodic Gibbs state)**.** Let $\rho \in \mathbb{G}_K(\mathsf{b}')$, and put $\Lambda := \Lambda^{a,\mathrm{tr}}$. Let $t = x^{-2}$ and $w = w_R$; here w_R is a real function on the bonds of B_R with $|w_R| \le 1$. Suppose that the datum $t\sup|D_w^\alpha \Lambda| \le \frac12$ holds on $\{\phi_p \le 3\varepsilon/4 + 4t$ on $B'\}$. Let Φ be as in the data of Lem. 11.14, that is, of class C^2 along the flow, a function of B', and carried by $\{\phi_p \le 3\varepsilon/4$ on $B'\}$, and on the support of Φ put $\varkappa := D_w\Lambda + xu^\alpha$. Then

$$|P_\rho[D_w^\alpha\Phi - xu^\alpha\Phi]| \le tB_2 + t^{-1}e\sup|\Phi|\,\big(2.5(2R/M+2r(x)+3)^3c_h(x) + 6e^{2.2}x^{-4}\big) + \sup|\Phi|\sup|\varkappa| + \delta_N/t.$$

■

At this w the datum is satisfied. Indeed, Step 0 of the proof of Lem. 11.17 proves $\sup|D\Lambda^a| \le 2p_0x^{1/2}$ by bounding the terms one by one, so the same bound holds for $\Lambda = \Lambda^{a,\mathrm{tr}}$, a truncated sum being a part of the sum, and therefore $t\sup|D_w\Lambda| \le 2p_0x^{-3/2}$. Moreover the data of Lem. 11.14,

taken at the same x, include the inequality $2p_0x^{-3/2} < \frac{1}{2}$ between the numbers x and p_0 alone; so the datum holds for the truncated sum Λ. In the claim, B_2 is defined with Λ.

Lemma 12.16 (Covariance and variance, centred at the means of the state)**.** Let $\rho \in \mathbb{G}_K(\mathsf{b}')$, and let N be the side fixed in Step 11. Let $a_{u,N} = \langle u, C_B u\rangle$ and $q_N = \langle \mathrm{d}w, \mathbb{D}_k[f_*]\mathrm{d}w\rangle$. Then there is a multiple Δ_N of $\epsilon_N^{1/2}$, with a factor that depends on K, j and b' but not on N, such that the following hold.

(i) $|\mathrm{Cov}_{P_\rho}(G, S_f) - \frac{3}{2}q_N| \le$ (E2) + (E3) + (E4) + (E6) $+\Delta_N$, where the four errors are those of Lem. 11.17.

(ii) $|\mathsf{b}'\mathrm{Cov}_{P_\rho}(G_-, \mathcal{O}_K f_*) - \frac{3}{2}q_N| \le$ (E1) $+\cdots+$ (E6) $+\Delta_N$, where the errors are those of Lem. 11.17.

(iii) $|\mathrm{Var}_{P_\rho}(G_-) - \frac{3}{2}a_{u,N}^2| \le$ (E7) $+\Delta_N$, where the error is that of Lem. 11.17. ∎

By Lem. 11.19 (B), $q_N \ge c_F$ for all rows of the runs, and by Lem. 11.15 (c), $0 \le a_{u,N} \le 2$.

Proof of Lem. 12.15. Throughout, all members are truncated at $\#X \le n_0$ and taken on Box.

1. With Λ as in the statement, $\Lambda^{\mathrm{tr}} = \Lambda + H^{\mathrm{tr}}$, where H^{tr} is the truncated sum of the E^h. We define $\Psi := (\Phi e^{\Lambda}) \circ \tau_t\, e^{-\Lambda}$ and
$$\Phi_{\mathrm{cell}} := \Psi \circ \tau_{-t}\, e^{\Lambda^{\mathrm{tr}}\circ\tau_{-t} - \Lambda^{\mathrm{tr}}} - \Psi.$$
The argument depends on Λ and H^{tr} only through the differences $\Lambda\circ\tau - \Lambda$ and $H^{\mathrm{tr}}\circ\tau - H^{\mathrm{tr}}$. These are finite sums over the terms that depend on a moved bond, so they are functions of Box, as Step (c), dependence, of the proof of Lem. 11.7 shows.
2. *Pointwise algebra.* We have $\Psi \circ \tau_{-t} = \Phi e^{\Lambda - \Lambda\circ\tau_{-t}}$. Hence the first term of Φ_{cell} is
$$\Phi \exp\{\Lambda - \Lambda\circ\tau_{-t} + \Lambda\circ\tau_{-t} + H^{\mathrm{tr}}\circ\tau_{-t} - \Lambda - H^{\mathrm{tr}}\} = \Phi e^{dH},$$
with $dH = H^{\mathrm{tr}}\circ\tau_{-t} - H^{\mathrm{tr}}$. The second term of Φ_{cell} is $\Phi \circ \tau_t\, e^{\Lambda\circ\tau_t - \Lambda}$.
3. *Taylor's formula along the flow.* Use the support argument in Step 2 of the proof of Lem. 11.14. A segment meeting the support stays in the enlarged tame set, with fixed indicators and smooth Λ. On the other segments all terms involving Φ and its derivatives vanish. Thus the Taylor identity holds on every configuration:
$$\Phi \circ \tau_t\, e^{\Lambda\circ\tau_t - \Lambda} = \Phi + t(D_w\Phi + \Phi D_w\Lambda) + \mathrm{rem},$$
where, by the datum, $|\mathrm{rem}| \le (t^2/2)B_2 e^{t\sup|D_w\Lambda|} \le t^2 B_2$.
4. Pointwise, $|dH| \le 2 \cdot 1.2124(2R/M + 2r(x) + 3)^3 c_h \le 1$. The first inequality follows from Step 5 of the proof of Lem. 11.14, since a truncated sum is a part of the sum. For the second, entry (j1) of the list of §11 bounds the error (E6) of Lem. 11.17 by $c_F/12$, and this gives $2.5(2R/M + 2r(x) + 3)^3 c_h \le \frac{1}{4}$.
5. It follows that, pointwise,
$$t(D_w\Phi + \Phi D_w\Lambda) = -\Phi_{\mathrm{cell}} + \Phi(e^{dH} - 1) - \mathrm{rem}.$$
6. Applying P_ρ, we use Lem. 12.14 (a), then Lem. 12.13 at the side N, then (X°), and obtain
$$|P_\rho(\Phi_{\mathrm{cell}})| \le \sup|\Psi^\circ|\, 2e^{2.2}/(r_\eta - 1) + \delta_N.$$
Here $\sup|\Psi^\circ| \le e^{1/2}\sup|\Phi|$ by Step 6 of the proof of Lem. 11.14, and the datum of Lem. 11.14 is met at w_R because of Step 0 in the proof of Lem. 11.17; further, $\sup|\Psi| \le e^{1/2}\sup|\Phi|$ by the datum, and $\sup|\Phi_{\mathrm{cell}}| \le 2e\sup|\Phi|$.
7. On the support of Φ we have $D_w\Lambda = -xu^\alpha + \varkappa$ with $|\varkappa| \le C_1 x\varepsilon/R + C_2 p_0^2 R + C_a(x)p_0^2$, where C_1 and C_2 are the numbers of Lem. 11.16. This rests on Lem. 11.16, which is pointwise and deterministic, and on the Cauchy bound for the truncated sum of the E^a, given by the part "Besides" of Lem. 11.14, by Step 7 of its proof and by Step 0 of the proof of Lem. 11.17;

the caps in that bound do not depend on N. The number $C_a(x)$ is the function of x alone from §11, a fixed multiple of $(2r(x)+1)^3$.

8. Items 5 and 7 give

$$P_\rho[D_w^\alpha \Phi - x u^\alpha \Phi] = t^{-1} P_\rho[-\Phi_{\mathrm{cell}} + \Phi(e^{dH}-1) - \mathrm{rem}] - P_\rho[\Phi \varkappa].$$

To bound the right side we use item 4 together with $2 \cdot 1.2124 \le 2.5$ and $|e^y - 1| \le e|y|$ for $|y| \le 1$, then item 3 in the form $|\mathrm{rem}|/t \le t B_2$, and item 6 with $r_\eta = x^4$ and $e^{1/2} \cdot 2/(x^4 - 1) \le 6e\, x^{-4}$. This gives the claim, an inequality at the one side N.

□

Remark 12.17. On the torus, $E_\nu \Phi_{\mathrm{cell}}$ becomes an integral against $e^\Lambda[\Xi_w \circ \tau_t - \Xi_w]$ by changing variables in the density: the proof of Lem. 11.14 reaches this integral in its Step 4, after the divergence, the regularity and Taylor's formula, and Step 1 of the proof of Lem. 11.13 identifies it with $E_\nu[\Phi_{\mathrm{cell}}] \int e^\Lambda \Xi_w$. A periodic Gibbs state has no density of this kind. Under ρ the argument does not depend on the change of variables in the density, because Φ_{cell} is a bounded cell function given explicitly, whose fold is a quantity on the torus. No term $2\,\mathrm{ex}$ arises, since the truncation is part of the definition; consequently (E6), in the form in which Lem. 11.17 states it, is the larger bound.

Proof of Lem. 12.16. This is the proof of Lem. 11.17 with the changes in items 1 to 8 below.

1. *The split, without a conditional expectation.* Let $g := P_\rho(G_-) \in [0, \frac{9}{8}p_0^2]$. As g is the mean, plain algebra gives

$$\mathrm{Cov}_{P_\rho}(G_-, \mathcal{O}_K f_*) = P_\rho[(G_- - g)(\mathcal{O}_K f_* - c_f^N - A_f^{\mathrm{tr},N}(V))] + \mathrm{Cov}_{P_\rho}(G_-, A_f^{\mathrm{tr},N}).$$

By (W°), Lem. 12.14 and Lem. 12.13, the first term is at most $\frac{9}{8}p_0^2 \cdot 2e^{2.2}/r_z$ plus the term of Lem. 12.13, which goes into Δ_N. For the error (E5) of Lem. 11.17 we have $\frac{9}{8}p_0^2 \cdot 2e^{2.2}/r_z \le (\mathrm{E5})/\mathsf{b}'$. Since single cubes are short, $A^{\mathrm{tr}} = S_f/\mathsf{b}' + \mathrm{Rem}^{\mathrm{tr}}$, and $|\mathrm{Cov}_{P_\rho}(G_-, \mathrm{Rem}^{\mathrm{tr}})| \le 2 \cdot \frac{9}{8}p_0^2 \sup|\mathrm{Rem}^{\mathrm{tr}}| \le (\mathrm{E1})/\mathsf{b}'$. For the second inequality we use that the bound of Lem. 11.4 on $\mathsf{b}' \sup|\mathrm{Rem}|$ comes, in its proof, from estimating each term separately, with the sizes of the terms from Steps 4 and 5 of that proof, the number of anchors from Steps 6 and 7 of that proof and the constant 1.2124 from Step 8 of that proof. As a truncated sum is a part of the sum, the same bound holds for $\mathsf{b}' \sup|\mathrm{Rem}^{\mathrm{tr}}|$, and $2 \cdot \frac{9}{8}p_0^2$ times this bound equals the error (E1) of Lem. 11.17.
2. *Peierls bound.* A translation in $L^{-j}\mathbb{Z}^3$ carries p to a plaquette based at 0, and P_ρ is invariant under it by the paragraph on the boards in Step 10. By Lem. 12.14(a), this probability is $\rho(E[1\{\phi_p \ge \varepsilon/2\} \mid U])$. The conditional expectation is a cell function independent of the torus side. Apply Thm. B.18(b) to this function and use (P°) to obtain $P_\rho(\phi_p \ge \varepsilon/2) \le e^{-(c_P/2)p_0^2}$. So $P_\rho(\chi < 1) \le 3(2R + 4\rho(x) + 7)^3 e^{-(c_P/2)p_0^2} = \nu_P$. Since $G_- - G \ne 0$ only where some $\phi_p > \varepsilon/2$ in B', we get $|\mathrm{Cov}_{P_\rho}(G_- - G, S_f)| \le \frac{9}{2} C_S p_0^4 \nu_P$.
3. *First identity.* We apply Lem. 12.15 with $\Phi^\alpha = u^\alpha \chi (S_f - \bar{s})/2$ as its function Φ, where $\bar{s} := P_\rho(S_f)$. The sup bound of this function is the one of Lem. 11.17, because $|S_f - \bar{s}| \le 2 \sup|S_f|$. Now $\sum_\alpha x u^\alpha \Phi^\alpha = G(S_f - \bar{s})$, and its P_ρ-mean is $\mathrm{Cov}_{P_\rho}(G, S_f)$. From $D u^\alpha = a_u + O(\varepsilon)$, and as $\bar{s}$ is the mean of S_f, we get

$$\frac{3}{2} a_u P_\rho[\chi(S_f - \bar{s})] = -\frac{3}{2} a_u P_\rho[(1-\chi)(S_f - \bar{s})] = O(\sup|S_f|\, P_\rho(\chi < 1)).$$

The term with $D\chi$ is carried by $\{\text{some } \phi_p \ge \varepsilon/2 \text{ in } B'\}$. There remains $\frac{1}{2}\sum_\alpha P_\rho[u^\alpha \chi D S_f]$.
4. Pointwise, $DS_f = x\langle \mathbb{D}_k \mathrm{d}w, \tilde{F}^\alpha \rangle + O(p_0^2 C_D)$, by Step 2 of the proof of Lem. 11.17.
5. *Second identity.* The choice $\Phi'^\alpha = \chi \langle \mathbb{D}_k \mathrm{d}w, \tilde{F}^\alpha \rangle/2$ yields $\frac{3}{2} q_N P_\rho(\chi)$ plus errors.

6. *Variance.* The choice $\Phi = u^\alpha\chi/2$ yields $P_\rho G = \frac{3}{2}a_u + \text{err}$. The function $\Phi = \frac{x}{4}u^\alpha|u|^2\chi^2$ satisfies $\sup \le 0.844\, p_0^2\varepsilon$ and yields $P_\rho G^2 = \frac{15}{4}a_u^2 + \text{err}$. Recall that in Lem. 11.17 the error (E7) is 4[(E2) + (E3) + (E4) + (E6)], with the errors of that lemma on the right.
7. All the other lines of the proof of Lem. 11.17 are pointwise. The list of errors is the one of Lem. 11.17, with Δ_N added. In Δ_N we collect the contribution of the single application of (W°) and of the twelve applications of (X°), which arise from three colours and four functions; each enters with a factor that does not depend on N, namely b', $1/t$ and the sup bounds of Step 10.
8. *What does not depend on the side.* Neither G_-, nor f_*, nor ρ, nor the kernel depends on the side; hence neither $\mathrm{Cov}_{P_\rho}(G_-, \mathcal{O}_K f_*)$ nor $\mathrm{Var}_{P_\rho}(G_-)$ does. The argument holds whether or not q, a_u, S_f, $\mathbb{D}_k$ and w_R depend on the side.

□

Step 12: the witness inequality (P0) *under* ρ*, and the clauses* (P1) *to* (P4). It is enough to take $c = \frac{1}{2}$, since the remaining mirrors are reached by the translations in $L^{-j}\mathbb{Z}^3$ from the paragraph on the boards in Step 10.

(P0)
1. The plane $x_1 = cL^{-j}$ lies in $a_K(\mathbb{Z} + \frac{1}{2})$, and so in $(a_K/2)\mathbb{Z} = \mathrm{Adm}(e_1)$; it is therefore a lattice mirror.
2. Let $G_+ := G_- \circ \theta_*$, $\breve{\Gamma} := E[G_+ \mid U] - \rho(E[G_+ \mid U])$, $F := \mathcal{O}_K f - \rho(\mathcal{O}_K f)$. Unlike G_+, which is a function of the drawn field $V^{(k)}$, the two functions $\breve{\Gamma}$ and F are bounded Borel functions of finitely many bonds of U in $\mathbb{H}^+$, by the argument of Lem. 12.14 (a) with a half space instead of a cell: a drawn bond whose two ends lie on one side depends on that side only. If the barycentre of a plaquette lies strictly to the right of a mid-bond plane, then every site of the plaquette lies to the right of that plane.
3. The identity $E[G_- \mid U] = \theta_* E[G_+ \mid U]$ is the third claim of Lem. 11.3 (K) for the map $\breve{\theta}$ given by the reflection $y_1 \mapsto n_1 - y_1$, with $n_1 = 2c$ an odd integer. The state ρ is θ_*-invariant by Prop. B.28 (b); so the two means agree, and $\rho(\mathcal{O}_K\theta_* f) = \rho(\mathcal{O}_K f)$.
4. Since $\mathcal{O}_K f$ is a function of U, $\mathrm{Cov}_{P_\rho}(G_-, \mathcal{O}_K f) = \rho(\theta_*\breve{\Gamma}\, F)$.
5. By Prop. B.28 (a), the form $(A, B) \mapsto \rho(\theta_* A\, B)$ is positive when restricted to real functions in $\mathcal{F}_{\mathrm{loc}}(\mathbb{H}^+)$, and it is symmetric by (b) of that theorem and $\theta_*^2 = 1$. The Schwarz inequality then gives $\rho(\theta_*\breve{\Gamma}\, F)^2 \le \rho(\theta_*\breve{\Gamma}\,\breve{\Gamma})\,\rho(\theta_* F\, F)$.
6. By Cauchy's inequality and invariance, $\rho(\theta_*\breve{\Gamma}\,\breve{\Gamma}) \le \rho(\breve{\Gamma}^2)$. Moreover $\rho(\breve{\Gamma}^2) = \mathrm{Var}_{P_\rho} E[G_+ \mid U] \le \mathrm{Var}_{P_\rho} G_+ = \mathrm{Var}_{P_\rho} G_-$.
7. Finally, $\rho(\theta_* F\, F) = S_2^T[\rho](\theta_* f, f)$.

Items 4 to 7 together give (P0), in product form, so that no quotient occurs.

(P1) We combine Lem. 12.16 (ii), taken at the side N, with $q_N \ge c_F$ and with entry (j1) of the list of §11; this gives

$$\mathsf{b}'\mathrm{Cov}_{P_\rho}(G_-, \mathcal{O}_K f_*) \ge \frac{3}{2}c_F - 6c_F/12 - \Delta_N = c_F - \Delta_N.$$

The left side does not depend on N, and letting $N \to \infty$ we obtain (P1).

(P2) Clause (iii), together with $a_{u,N} \le 2$ and entry (j1), yields Var $\le \frac{3}{2}\cdot 4 +$ (E7) $+\Delta_N \le 6 + 4c_F/3 + \Delta_N$, where the error is that of Lem. 11.17. As $N \to \infty$, this becomes Var ≤ 7.

(P3) (P1) forces $\mathrm{Var}_{P_\rho}(G_-) > 0$. Then (P0), (P1) and (P2) give $\mathsf{b}'^2 S_2^T \ge c_F^2/7 = c_*$.

(P4) The last sentence of Prop. B.30 (2) gives $\nu^U \in \mathcal{G}^{\mathrm{per}}$ for ν-almost every U. This state is ergodic, hence invariant under $a_K\mathbb{Z}^3$, and therefore $\nu^U \in \mathbb{G}_K(\mathsf{b}')$. Moreover $\mathrm{Cov}_\nu(\cdot, \cdot \mid \mathcal{I})(U)$ is the covariance under ν^U. We apply (P3) to ν^U.

□

Remark 12.18. For non-triviality alone, (P2) can be replaced by the bound $\mathrm{Var} \le (\frac{9}{8})^2 p_0^4$, which holds because G_- takes its values in $[0, \frac{9}{8}p_0^2]$.

Proof of Thm. 12.6 and of Cor. 12.7. Step 13: the depth, the clauses (C1) *to* (C4)*, and the corollary.* We first work in normal form, $\mathsf{b} > x_s$. Item (C0) shows that $j^+(\mathsf{b}) < \infty$ and $r_*(\mathsf{b}) > 0$, so that the ranges $j \ge j^+(\mathsf{b})$ and $0 < r \le r_*(\mathsf{b})$ are not empty. Item (C4) treats every $\mathsf{b} > 0$.

(C0) *The depth.* By Thm. 11.1 (D) and Lem. 11.22 (a), (c), the conditions on the depth listed in §11 and the clauses (N0) to (N6) of Thm. 11.1 are satisfied for every $j \ge j_*(\mathsf{b}')$. By Lem. 11.22 (d), j_* is bounded on the compact set $U(\mathsf{b}) \subset [x_s, \infty)$. Hence $j^+(\mathsf{b})$ is an integer, $r_*(\mathsf{b}) = D_0 L^{-j^+(\mathsf{b})} > 0$, and every $j \ge j^+(\mathsf{b})$ is $\ge j_*(\mathsf{b}')$ for every $\mathsf{b}' \in U(\mathsf{b})$.

(C1) We take b in normal form. Let $S \in \mathrm{Lim}_G(\mathsf{b})$ with the frame (K_i, ν_i), $\nu_i \in \mathbb{G}_{K_i}(\mathsf{b}_i)$, $\mathsf{b}_i \to \mathsf{b}$, all $\mathsf{b}_i \in U(\mathsf{b})$, and let $\xi \in \mathsf{X}$. By Thm. E.13 (iii) there are $\rho_i = \nu_i(\cdot \mid E_i) \in \mathbb{G}_{K_i}(\mathsf{b}_i)$ which are approximants of σ_ξ, and this for every $\xi \in \mathsf{X}$. By Prop. E.18 (d), $S_2^{\sigma_\xi} = \lim_i S_2^T[\rho_i]$ on $\mathfrak{S}_2(\mathbb{R}^3)$.

Fix $j \ge j^+(\mathsf{b})$. For every i we then have $j \ge j_*(\mathsf{b}_i)$, and once i is large, $K_i \ge j+1$. Only D_0 and j enter $f_* = f_0(L^j\,\cdot)$ and the mirror, whereas G_-, R and h depend on $x_i = L^j \mathsf{b}_i$; these are allowed to change with i, since they are used only as witnesses. As f_0 is C^∞ and the supports of $\theta_* f_*$ and f_* are $\ge D_0 L^{-j}$ apart, $(\theta_* f_*, f_*)$ is a separated C_c^1 tuple, so it lies in $\mathfrak{S}_2(\mathbb{R}^3)$, the set on which the limit above holds. By (P3), $S_2^T[\rho_i](\theta_* f_*, f_*) \ge c_*/\mathsf{b}_i^2 \to c_*/\mathsf{b}^2$. By (E3)(i) the mirror can be carried to $(e_3, 0)$. By (E7)(b), non-triviality amounts to $g_f(\xi) > 0$ for one $f \ge 0$.

(C2) We have $g_{f_*}(\xi) = \iint f_*(\theta_* x) f_*(y) s_\xi(|x-y|)$, where $s_\xi \ge 0$ is non-increasing. The set $\operatorname{supp} f_*$ keeps a distance $\ge (\lceil D_0/\sqrt{2}\rceil - \frac{1}{2} - D_0/8)L^{-j} \ge 0.58 D_0 L^{-j}$ from the mirror, as $1/\sqrt{2} - 1/8 = 0.5821\ldots$; hence $|\theta_* x - y| \ge D_0 L^{-j}$ for x and y in that set. Since $0 \le f_* \le 1$ and f_* is carried by a sup-ball of radius $(D_0/8)L^{-j}$, we have $\int f_* \le (D_0/4)^3 L^{-3j}$. As s_ξ is non-increasing and $|\theta_* x - y| \ge D_0 L^{-j}$, it follows that $g_{f_*}(\xi) \le s_\xi(D_0 L^{-j})(\int f_*)^2 \le s_\xi(D_0 L^{-j})(D_0/4)^6 L^{-6j}$. By (C1), $c_*/\mathsf{b}^2 \le g_{f_*}(\xi)$, and so $c_*/\mathsf{b}^2 \le s_\xi(D_0 L^{-j})(D_0/4)^6 L^{-6j}$. For $D_0 L^{-j-1} < r \le D_0 L^{-j}$ this gives

$$s_\xi(r) \ge s_\xi(D_0 L^{-j}) \ge c_* \mathsf{b}^{-2} (4/D_0)^6 L^{6j} > c_* (4/L)^6 \mathsf{b}^{-2} r^{-6},$$

because $L^j > D_0/(Lr)$. Taken together, all $j \ge j^+$ cover $(0, r_*]$.

Suppose that H_ξ were bounded, $H_\xi \le \check{\Lambda}$. For $t < t_0$ and one $\mathsf{e} \le t/2$ we would then have $s(t_0) = \int e^{-(t_0 - 2\mathsf{e})E}\, d\mu_{\mathsf{e}} \ge e^{-(t_0-t)\check{\Lambda}} s(t)$, so that s would stay bounded as $t \to 0$, which contradicts the lower bound with r^{-6}. In finite dimension H is necessarily bounded; hence $\dim \mathcal{H}(\sigma_\xi)$ is infinite.

(C3) This follows from (E7)(a) and (C1).

(C4) Put $z = z(\mathsf{b})$. Step 8 (1) gives $S_2^T[\rho; K](f, f') = L^{2z} S_2^T[\rho; K - z](f^{[z]}, f'^{[z]})$; moreover Thm. 12.5, taken at $(K - z, L^z \mathsf{b}')$, concerns the same measures. We take $f = f_0(L^{j+z}\,\cdot)$, so that $f^{[z]} = f_0(L^j\,\cdot)$, with $j \ge j^+(L^z \mathsf{b})$. Then

$$\mathsf{b}_i^2 S_2^T[\rho_i; K_i](\theta_* f, f) = (L^z \mathsf{b}_i)^2 S_2^T[\rho_i; K_i - z](\theta_* f^{[z]}, f^{[z]}) \ge c_*.$$

The steps of (C1) and (C2) go through with L^{-j-z} in the place of L^{-j} in each of their formulas. In the limit, $S_n^{(\mathsf{b})}(X) = L^{zn} S_n^{(L^z \mathsf{b})}(X^{[z]})$, and hence $\mathsf{S}_n^{(\mathsf{b})}(x) = L^{4zn} \mathsf{S}_n^{(L^z \mathsf{b})}(L^z x)$ and $s^{(\mathsf{b})}(r) = L^{8z} s^{(L^z \mathsf{b})}(L^z r)$. The quantities $\mathsf{b}^2 S_2$ and $\mathsf{b}^2 r^6 s(r)$ are therefore invariant, as $2 + 6 = 8$.

The corollary. By (E1), $\mathrm{Lim}_G(\mathsf{b})$ is not empty, since $\mathcal{G}^{\mathrm{tor}}$ is not empty by compactness. For $\xi \in \mathsf{X}$ we have $S^\xi := (S_n^{\sigma_\xi})_{n \ge 2} \in \mathrm{Lim}_G(\mathsf{b})$, with centred functional σ_ξ^{c}, and $\sigma_\xi = \sigma_\xi^{\mathrm{c}} \circ \mathrm{sh}_{c_\xi}$ by (E3)(i). Thus, up to the shift, σ_ξ is the functional of a member, and it coincides with that functional exactly when $c_\xi = 0$. By Prop. E.20, σ_ξ clusters, and clustering passes through sh_{-c}; so the one-point space

is a canonical decomposition of σ_ξ^{c}, and by Prop. E.15 (a) it is the only one. The list of the corollary follows from (E3), (E4) and Thm. 12.6. This list is stated for σ_ξ; Lem. 12.11 carries it over to σ_ξ^{c}, and the two have the same S_2.

□

Appendix A. Reflection positivity and chessboard estimates

A.1. **Wilson reflection positivity.** To estimate the source derivatives uniformly in the volume, we first prove the reflection inequality for the Wilson measure. We use rectangular tori because the chessboard proof treats one coordinate direction at a time.

Let $a > 0$, let $N_\mu \geq 3$ be integers, and put

$$T = \prod_{\mu=1}^{3} \mathbb{R}/(aN_\mu\mathbb{Z}), \qquad \Lambda = \prod_{\mu=1}^{3} a\mathbb{Z}/(aN_\mu\mathbb{Z}).$$

Positive bonds and plaquettes have the usual coordinate orientations. For reversed bonds set $U(\bar{b}) = U(b)^{-1}$. Let $\partial U(p)$ denote the ordered product around p and put $a_p(U) = 1 - \frac{1}{2}\operatorname{tr}\partial U(p)$. For $\beta \geq 0$ define

$$A(U) = \sum_p a_p(U), \qquad \mathcal{O}_a(f) = a^{-1}\sum_p f(x(p))a_p(U), \qquad d\mu_{a,T,\beta} = Z_{a,T,\beta}^{-1} e^{-\beta A(U)}\, dU. \tag{A.i}$$

Here dU is product Haar probability measure and $Z_{a,T,\beta}$ is the integral of $e^{-\beta A}$. Since $0 \leq a_p \leq 2$, this normalization is strictly positive, and $\mathcal{O}_a(f)$ is bounded for every function f on the finite set of plaquette centres. For a bounded Borel function F of the configuration, write

$$\mathcal{Z}_{a,T,\beta}(F) = \int F(U) e^{-\beta A(U)}\, dU, \qquad \langle F\rangle_{a,T,\beta} = Z_{a,T,\beta}^{-1}\mathcal{Z}_{a,T,\beta}(F).$$

Thus, on the cubic lattice with $a = L^{-K}$, $Z_{K,\ell,\beta}[w] = \mathcal{Z}_{a,T_\ell,\beta}(e^{\mathcal{O}_a(w)})$.

For $n \geq 1$ the joint cumulant is

$$S^T_{n;a,T,\beta}(f_1,\dots,f_n) = \partial_{z_1}\cdots\partial_{z_n} \log\left\langle e^{\sum_i z_i\mathcal{O}_a(f_i)}\right\rangle_{a,T,\beta}\Big|_{z=0}. \tag{A.ii}$$

The expectation is entire and equals one at zero, so the logarithm in this formula exists near zero. If $\mathcal{P}_n$ denotes the set of partitions of $\{1,\dots,n\}$, differentiating the logarithm gives

$$S^T_n(f_1,\dots,f_n) = \sum_{\pi\in\mathcal{P}_n} (-1)^{|\pi|-1}(|\pi|-1)! \prod_{B\in\pi}\left\langle \prod_{i\in B}\mathcal{O}_a(f_i)\right\rangle. \tag{A.iii}$$

Conversely, the joint moment is the sum over $\pi \in \mathcal{P}_n$ of the products of the cumulants indexed by its blocks. Both identities follow by comparing the coefficient of $z_1\cdots z_n$ in the exponential identity relating the expectation to its logarithm. In particular, cumulants are symmetric and multilinear. For $n \geq 2$ the cumulant of the random variables $\mathcal{O}_a(f_i)$ is unchanged if a scalar constant is added to any of these random variables: such a change adds a linear function of z to the logarithm.

Translations and signed coordinate permutations transport oriented bonds and preserve product Haar measure, with inversion at reversed bonds. They carry a plaquette holonomy to a conjugate of the original holonomy or its inverse, both with the same trace in SU(2). The Wilson law is therefore invariant under translations and under signed permutations that preserve the rectangular period lattice. Other signed permutations carry it to the Wilson law on the permuted torus. In either case $\mathcal{O}_a(f)$ transforms by pullback of f.

A.1.1. *The operator across one layer.* Fix a coordinate direction, write N for the number of sites in that direction, and let Σ be the two-dimensional cross-section. Let $\mathsf{B}(\Sigma)$ be its positive bonds and set $\mathcal{X} = \mathrm{SU}(2)^{\mathsf{B}(\Sigma)}$. A configuration consists of the transverse fields $V^t \in \mathcal{X}$ and the bonds $S^t \in \mathrm{SU}(2)^{\Sigma}$ joining layers t and $t+1$, with t taken modulo N. Write $A_\Sigma(V)$ for the sum of the plaquette actions within a transverse layer and put

$$A_\times(V,S,W) = \sum_{b \in \mathsf{B}(\Sigma)} \left(1 - \tfrac{1}{2}\operatorname{tr} V_b S_{b_+} W_b^{-1} S_{b_-}^{-1}\right). \tag{A.iv}$$

The two types of plaquettes partition all plaquettes, giving

$$A(U) = \sum_{t \bmod N} \left[A_\Sigma(V^t) + A_\times(V^t, S^t, V^{t+1})\right].$$

On $\mathcal{H} = L^2(\mathcal{X}, dV)$ define the integral operator

$$\mathsf{K}(W,V) = \int e^{-\beta A_\times(V,S,W)}\, dS.$$

The kernel is continuous, symmetric and bounded by one. Symmetry follows by replacing S by S^{-1} and using invariance of the trace under cyclic permutations and inversion. Thus K is self-adjoint and Hilbert–Schmidt. Its positivity needs a separate argument.

Proposition A.1. The crossing operator satisfies $\mathsf{K} \geq 0$. ∎

Proof. Put $\mathsf{k}(u) = e^{-\beta(1-\operatorname{tr} u/2)}$ and $\mathsf{K}_0(V,W) = \prod_b \mathsf{k}(V_b W_b^{-1})$. For $R \in \mathrm{SU}(2)^\Sigma$ let $(V^R)_b = R_{b_-} V_b R_{b_+}^{-1}$. The crossing weight is $\mathsf{K}_0(V, W^S)$, and $\mathsf{K}_0(V^R, W^R) = \mathsf{K}_0(V,W)$. Haar invariance therefore gives

$$\mathsf{K}(W,V) = \int dR\, dS\, \mathsf{K}_0(V^R, W^S). \tag{A.v}$$

For $v, w \in \mathrm{SU}(2)$, $\operatorname{tr}(vw^{-1}) = \sum_{i,j} v_{ij}\overline{w_{ij}}$. Expanding the exponential at every bond consequently yields

$$\mathsf{K}_0(V,W) = \sum_\alpha c_\alpha \phi_\alpha(V)\overline{\phi_\alpha(W)}, \qquad c_\alpha \geq 0.$$

Here α specifies a nonnegative integer n_b and n_b pairs of matrix indices at each bond,

$$c_\alpha = e^{-\beta|\mathsf{B}(\Sigma)|} \prod_b \frac{(\beta/2)^{n_b}}{n_b!}, \qquad \phi_\alpha(V) = \prod_b \prod_{s=1}^{n_b} (V_b)_{i_{b,s} j_{b,s}}.$$

Every matrix entry has modulus at most one. There are 4^{n_b} choices of indices at bond b, so

$$\sum_\alpha c_\alpha = e^{-\beta|\mathsf{B}(\Sigma)|} \prod_b \sum_{n \geq 0} \frac{(2\beta)^n}{n!} = e^{\beta|\mathsf{B}(\Sigma)|} < \infty.$$

This proves absolute uniform convergence. Put $\widehat{\phi}_\alpha(V) = \int \phi_\alpha(V^R)\, dR$. Substituting the series in eq. A.v gives

$$\mathsf{K}(W,V) = \sum_\alpha c_\alpha \widehat{\phi}_\alpha(V)\overline{\widehat{\phi}_\alpha(W)}.$$

Since dV is a probability measure, every $\psi \in \mathcal{H}$ is integrable. Absolute convergence permits termwise integration, giving

$$\langle \psi, \mathsf{K}\psi \rangle = \sum_\alpha c_\alpha \left| \int \widehat{\phi}_\alpha(V)\psi(V)\, dV \right|^2 \geq 0.$$

□

A.1.2. *Slabs and the trace formula.* For an integer $q \geq 2$, a slab of q layers has variables $(V^1, \dots, V^q; S^1, \dots, S^{q-1})$ and weight

$$w_q = \exp\left[-\beta \sum_{t=1}^{q} A_\Sigma(V^t) - \beta \sum_{t=1}^{q-1} A_\times(V^t, S^t, V^{t+1})\right].$$

For a bounded Borel function Φ of these variables define

$$\mathfrak{S}[\Phi](W, V) = \int \prod_{1<t<q} dV^t \prod_{1\leq t<q} dS^t \, (\Phi w_q)\big|_{V^1=V, V^q=W}, \qquad \mathfrak{Y}[\Phi] = \mathsf{K}^{1/2}\mathfrak{S}[\Phi]\mathsf{K}^{1/2}. \tag{A.vi}$$

These are Hilbert–Schmidt operators, since the first kernel is bounded by $\|\Phi\|_\infty$ and $\mathsf{K}^{1/2}$ is bounded. Reverse the order of the layers, invert the joining bonds, and conjugate the value of Φ; denote the resulting function by Φ^Θ. This change of variables preserves Haar measure and w_q, and exchanges the two endpoint fields. Consequently

$$\mathfrak{S}[\Phi^\Theta] = \mathfrak{S}[\Phi]^*, \qquad \mathfrak{Y}[\Phi^\Theta] = \mathfrak{Y}[\Phi]^*. \tag{A.vii}$$

Proposition A.2 (Trace formula)**.** Partition the N layers cyclically into $r \geq 2$ consecutive slabs of lengths $q_s \geq 2$. Let Φ_s be a bounded Borel function of the inner bonds of slab s. Then

$$\mathcal{Z}_{a,T,\beta}\left(\prod_{s=1}^{r} \Phi_s\right) = \operatorname{Tr}(\mathfrak{S}[\Phi_r]\mathsf{K} \cdots \mathfrak{S}[\Phi_1]\mathsf{K}) = \operatorname{Tr}(\mathfrak{Y}[\Phi_r] \cdots \mathfrak{Y}[\Phi_1]). \tag{A.viii}$$

All the operators attached to a slab depend on its length, the cross-section and β, but not on the total length N. ■

Proof. Keep the first and last transverse fields V_s, W_s of each slab. Every plaquette either lies inside one slab or crosses from its last layer to the first layer of the next slab. Integrating all other variables gives

$$\int \prod_{s=1}^{r} dV_s \, dW_s \prod_{s=1}^{r} \mathfrak{S}[\Phi_s](W_s, V_s)\mathsf{K}(V_{s+1}, W_s), \qquad V_{r+1} = V_1.$$

The integrands are bounded and all measures are finite, so Fubini applies. The kernel formula for a product, followed by the Hilbert–Schmidt identity $\operatorname{Tr}(CD) = \iint C(y,x)D(x,y)\,dx\,dy$, gives the first trace. There are at least two Hilbert–Schmidt factors, so the product is trace class. Splitting each K into two square roots and moving the last square root around the trace gives the second identity. The remaining factors form a trace-class operator, which justifies this cyclic permutation. The definitions contain no total layer number N, proving the last assertion. □

Corollary A.3 (Reflection at bond midpoints)**.** Suppose that N is even. Let θ be reflection in a coordinate plane at a half-integer multiple of a, and let T_+ be one open half-torus bounded by this plane and its opposite parallel plane. Write $\mathfrak{F}_+$ for the bounded Borel functions depending only on bonds whose two endpoints lie in T_+.

(a) The form $\langle \overline{\theta F} H\rangle_{a,T,\beta}$ on $\mathfrak{F}_+$ is positive semidefinite.
(b) It is Hermitian and sesquilinear.
(c) For $F, H \in \mathfrak{F}_+$,

$$|\langle \overline{\theta F} H\rangle|^2 \leq \langle \overline{\theta F} F\rangle \langle \overline{\theta H} H\rangle. \tag{A.ix}$$

(d) The assertions hold on the other open half-torus as well.
(e) When the plane is a wall of a board in Def. A.4, the product of functions of cells on one side is a function of that open half-torus.

■□

Proof. The two halves contain $q = N/2 \geq 2$ layers each. Their inner bond variables identify F, H with slab functions Φ, Ψ. Reflection sends the first slab to the second and reverses its layer order, so $\overline{\theta F}$ is the placed function Φ^{Θ}. By eq. A.vii and A.2,

$$\mathcal{Z}_{a,T,\beta}(\overline{\theta F} H) = \mathrm{Tr}(\mathfrak{Y}[\Phi]^* \mathfrak{Y}[\Psi]).$$

This is the Hilbert–Schmidt inner product. Its positivity, Hermitian symmetry and Cauchy–Schwarz inequality survive division by $Z_{a,T,\beta} > 0$. The same trace proof starts with either half-torus. For (e), the two walls bounding a half-torus have half-integer coordinates in units of a. They contain no sites. The cells of that half-torus have their sites strictly between the two walls, so all their inner bonds have both endpoints on that side. □

A.2. **The chessboard estimate on a torus.** The trace formula reduces the chessboard estimate in one coordinate direction to an inequality for the slab operators. Applying this bound in all three directions gives the estimate on the full board.

A.2.1. *Cells and reflected copies.*

Definition A.4 (A board on a torus)**.** (1) We use the Wilson measure of eq. A.i and its action on functions by lattice isometries.

(2) Fix $a > 0$, an odd integer $\lambda \geq 3$, even integers $\mathsf{n}_\mu \geq 2$, and an offset $o \in a\mathbb{Z}^3$. We call these data a board. Its torus has sides $\ell_\mu = \mathsf{n}_\mu \mathsf{l}$, where $\mathsf{l} = \lambda a$. Write

$$\mathsf{Z} = \prod_{\mu=1}^{3} \mathbb{Z}/\mathsf{n}_\mu\mathbb{Z}, \qquad \mathsf{N} = \prod_{\mu} \mathsf{n}_\mu, \qquad h_\lambda = (\lambda - 1)/2. \tag{A.x}$$

The cell $\square_\kappa$ is the image in the torus of $o + \mathsf{l}\tilde{\kappa} + [-\mathsf{l}/2, \mathsf{l}/2]^3$, with $\tilde{\kappa}$ any integer representative of $\kappa \in \mathsf{Z}$. Its sites are

$$o + \mathsf{l}\tilde{\kappa} + an, \qquad n \in \mathsf{C}_\lambda := \{-h_\lambda, \dots, h_\lambda\}^3. \tag{A.xi}$$

The walls normal to e_μ have coordinates

$$x_\mu = o_\mu + \mathsf{l}(t + \tfrac{1}{2}), \qquad t \in \mathbb{Z}/\mathsf{n}_\mu\mathbb{Z}. \tag{A.xii}$$

The cell faces contain no sites since their varying coordinate is a half-integer multiple of a. A plaquette whose centre lies in a cell interior has all its corners in that cell. Otherwise its centre lies on a cell face.

■

Lemma A.5 (Cell coordinates)**.** (a) The sites of each cell have the representatives eq. A.xi.

(b) The number of cells is N.

(c) Every site belongs to exactly one cell.

(d) A function of a cell is a bounded complex Borel function of its inner bond variables, where an inner bond has both endpoints in the cell. Transporting the positive bonds of C_λ to $\square_\kappa$ identifies such a function with a unique model function F_0 on the finite product $\mathrm{SU}(2)^{\mathsf{B}^+(\mathsf{C}_\lambda)}$; its placement in $\square_\kappa$ is denoted by $F_0^{(\kappa)}$.

■

Proof. The site formula gives (a). The cardinality in (b) follows by multiplying the three index ranges. In each coordinate the sets of λ consecutive sites of the cells are disjoint and exhaust the sites in that coordinate. This gives (c). For (d), arbitrary values of the inner bond variables extend to a torus configuration, so placement preserves the supremum norm and determines the model function uniquely. □

Definition A.6 (Copies and folds)**.** (1) For $\kappa, \kappa' \in \mathsf{Z}$ set $\epsilon_\mu = (-1)^{\tilde{\kappa}'_\mu - \tilde{\kappa}_\mu}$ and define the diagonal isometry

$$\mathsf{r}_{\kappa'\kappa}(x) = o + \mathsf{l}\tilde{\kappa}' + \epsilon(x - o - \mathsf{l}\tilde{\kappa}). \tag{A.xiii}$$

It carries $\square_\kappa$ to $\square_{\kappa'}$. The evenness of each n_μ makes the signs independent of the representatives; changing a representative changes the formula only by a torus period. Transport configurations and observables by the action used in A.1, including inversion when a bond orientation is reversed. For a function F of $\square_\kappa$, define

$$F^{[\kappa']} = \begin{cases} \mathsf{r}_{\kappa'\kappa} F, & \prod_\mu \epsilon_\mu = 1, \\ \overline{\mathsf{r}_{\kappa'\kappa} F}, & \prod_\mu \epsilon_\mu = -1, \end{cases} \tag{A.xiv}$$

(2) The fold is

$$\operatorname{Fold} F = \prod_{\kappa' \in \mathsf{Z}} F^{[\kappa']}. \tag{A.xv}$$

(3) For a sign vector ϵ and a model bond variable u, put

$$u^\epsilon(\langle n, n + e_\mu \rangle) = \begin{cases} u(\langle \epsilon n, \epsilon n + e_\mu \rangle), & \epsilon_\mu = 1, \\ u(\langle \epsilon n - e_\mu, \epsilon n \rangle)^{-1}, & \epsilon_\mu = -1. \end{cases} \tag{A.xvi}$$

The sign map preserves the model cube, so both positive bond arguments lie in the model. Formula eq. A.xiv is therefore the placement of $F_0(u^\epsilon)$, with conjugation for a negative product of the signs.

■

Lemma A.7 (Operations on copies)**.** Copies preserve products and sums and commute with conjugation. For all cell indices,

$$(F^{[\kappa']})^{[\kappa'']} = F^{[\kappa'']}, \qquad \operatorname{Fold}(F^{[\kappa']}) = \operatorname{Fold} F. \tag{A.xvii}$$

■

Proof. Pullback and conjugation preserve products and sums and commute with conjugation. The mirror maps compose since their sign matrices multiply to the sign matrix for the composite map. The determinants multiply as well, so the composite copy conjugates exactly when the composite mirror map has negative determinant. This proves the first identity. Multiplying over κ'' gives the second. □

Exactly half the cells have each parity, since changing one cell index by one interchanges the parities. Consequently

$$\operatorname{Fold}(cF) = |c|^{\mathsf{N}} \operatorname{Fold} F, \qquad \operatorname{Fold} \bar{F} = \overline{\operatorname{Fold} F}, \qquad |\operatorname{Fold} F| \le \|F\|_\infty^{\mathsf{N}}. \tag{A.xviii}$$

A.2.2. *Multiple reflections.* We first give the trace inequality needed for an arbitrary even number of slabs. For a Hilbert–Schmidt operator A put $\|A\|_{2k} = (\operatorname{Tr}(A^* A)^k)^{1/(2k)}$. This is finite. Cyclicity of the trace gives $\|A^*\|_{2k} = \|A\|_{2k}$. The spectral theorem for $A^* A$ shows that this number vanishes only when $A = 0$.

Lemma A.8 (Trace Hölder inequality)**.** For $k \ge 1$ and Hilbert–Schmidt operators $A_1, \dots, A_{2k}$, the product is trace class and

$$|\operatorname{Tr}(A_1 \cdots A_{2k})| \le \prod_{s=1}^{2k} \|A_s\|_{2k}. \tag{A.xix}$$

■

Proof. The trace-class property follows from the presence of two Hilbert–Schmidt factors. Unless one factor is zero, normalize each A_s to have $\|A_s\|_{2k} = 1$. Let $\mathcal{A}$ be the finite set consisting of these normalized operators and their adjoints, and maximize $|\operatorname{Tr}(B_1 \cdots B_{2k})|$ over $\mathcal{A}^{2k}$; denote the maximum by m. An alternating tuple $(C, C^*, \dots, C, C^*)$ has trace one, so $m \geq 1$.

For $X = B_1 \cdots B_k$ and $Y = B_{k+1} \cdots B_{2k}$, Hilbert–Schmidt Cauchy–Schwarz gives

$$|\operatorname{Tr}(XY)|^2 \leq \operatorname{Tr}(XX^*) \operatorname{Tr}(Y^*Y).$$

Both traces on the right are nonnegative and correspond to tuples in $\mathcal{A}^{2k}$. If the original tuple maximizes the modulus, both reflected tuples also maximize it. Among all maximizing tuples choose one with a longest string of consecutive entries alternating between an operator and its adjoint, counting positions cyclically. Let its length be j. If $j \leq k$, cyclically shift it to end at position k; if $k < j < 2k$, shift it to start at position one. Reflect the first k entries to obtain $(B_1, \dots, B_k, B_k^*, \dots, B_1^*)$. This is again a maximizer and has an alternating string of length $2\min\{j, k\} > j$, a contradiction. Thus $j = 2k$ and the maximizing trace is one. Rescaling proves the inequality. □

Let $\mathbf{F} = (F_\kappa)_{\kappa \in \mathsf{Z}}$ be a family with F_κ a function of $\square_\kappa$, and abbreviate

$$Z[\mathbf{F}] = \mathcal{Z}_{a,T,\beta}\left(\prod_\kappa F_\kappa\right).$$

For a direction μ and an index $t \bmod \mathsf{n}_\mu$, write $\kappa|t$ for κ with its μth component replaced by t. Define the reflected family by

$$(R_{\mu,t}\mathbf{F})_\kappa = (F_{\kappa|t})^{[\kappa]}. \tag{A.xx}$$

The cells with $\kappa_\mu = t$ lie in a slab of λ layers, starting just after layer $\tau_0(t) = o_\mu/a + \lambda t - h_\lambda - 1$. Their product is a bounded Borel function $\Phi^{(t)}$ of the inner slab bonds. Here $2 \leq \lambda \leq \lambda\mathsf{n}_\mu - 1$, so these are slabs to which A.2 applies.

Proposition A.9. For every board, every $\beta \geq 0$, and every such family,

$$Z[R_{\mu,t}\mathbf{F}] = \operatorname{Tr}(\mathfrak{Y}[\Phi^{(t)}]^*\mathfrak{Y}[\Phi^{(t)}])^{\mathsf{n}_\mu/2} = \|\mathfrak{Y}[\Phi^{(t)}]\|_{\mathsf{n}_\mu}^{\mathsf{n}_\mu} \geq 0, \tag{A.xxi}$$

$$|Z[\mathbf{F}]| \leq \prod_{t \bmod \mathsf{n}_\mu} Z[R_{\mu,t}\mathbf{F}]^{1/\mathsf{n}_\mu}. \tag{A.xxii}$$

■

Proof. The trace formula expresses $Z[\mathbf{F}]$ as the trace of the product of the n_μ slab operators, in reverse cyclic order. In $R_{\mu,t}\mathbf{F}$, an even displacement from slab t is a translation of its slab function; an odd displacement is a translation of the reflected, conjugated function $(\Phi^{(t)})^\Theta$. Indeed, in the varying coordinate the map between two slabs is respectively a translation by $(t'-t)\mathsf{l}$ or reflection followed by translation, with reflection formula $x_\mu \mapsto 2o_\mu + (t+t')\mathsf{l} - x_\mu$. The other coordinates are unchanged. Thus the operators in this trace alternate between $Y = \mathfrak{Y}[\Phi^{(t)}]$ and Y^*, by eq. A.vii. Since n_μ is even, their trace is $\operatorname{Tr}(Y^*Y)^{\mathsf{n}_\mu/2}$, independently of the initial parity. This proves eq. A.xxi. Apply A.8 to the original slab product and use eq. A.xxi for each factor to obtain eq. A.xxii. □

Composition of copies in eq. A.xvii implies that $R_{3,t_3}R_{2,t_2}R_{1,t_1}\mathbf{F}$ is the family of all copies of $F_{(t_1,t_2,t_3)}$. In particular its product is $\operatorname{Fold} F_{(t_1,t_2,t_3)}$. For the family of all copies of one F, applying $R_{1,t}$ does not change the family. Equation eq. A.xxi therefore shows that $\mathcal{Z}_{a,T,\beta}(\operatorname{Fold} F) \geq 0$.

Definition A.10 (Chessboard seminorm)**.** Define

$$\|F\|_{\mathrm{ch}} = \langle \operatorname{Fold} F \rangle_{a,T,\beta}^{1/\mathsf{N}}. \tag{A.xxiii}$$

■

Proposition A.11 (Chessboard estimate)**.** For every board, every $\beta \geq 0$, and every family of bounded Borel functions F_κ of their respective cells,

$$\left| \left\langle \prod_{\kappa \in \mathsf{Z}} F_\kappa \right\rangle_{a,T,\beta} \right| \leq \prod_{\kappa \in \mathsf{Z}} \|F_\kappa\|_{\mathrm{ch}}. \tag{A.xxiv}$$

■

Proof. Apply eq. A.xxii in directions 1, 2, and 3, in that order. The first reflected integrals are nonnegative, so the inequality can be applied successively under their positive roots. The composition of copies just proved gives

$$|Z[\mathbf{F}]| \leq \prod_{t_1,t_2,t_3} \mathcal{Z}_{a,T,\beta}(\mathrm{Fold}\, F_{(t_1,t_2,t_3)})^{1/\mathsf{N}}.$$

Divide by $Z_{a,T,\beta} > 0$, distributing its Nth root among the N factors. □

Proposition A.12 (Properties of the chessboard seminorm)**.** (a) $\|cF\|_{\mathrm{ch}} = |c|\|F\|_{\mathrm{ch}}$.

(b) $\|1\|_{\mathrm{ch}} = 1$.
(c) $\|\bar{F}\|_{\mathrm{ch}} = \|F\|_{\mathrm{ch}}$.
(d) A mirror copy has the same fold and the same seminorm.
(e) $\|F\|_{\mathrm{ch}} \leq \|F\|_\infty$.
(f) $|\langle F \rangle_{a,T,\beta}| \leq \|F\|_{\mathrm{ch}}$. More generally, for functions F_k of distinct cells, $|\langle \prod_k F_k \rangle_{a,T,\beta}| \leq \prod_k \|F_k\|_{\mathrm{ch}}$.
(g) $\|F + H\|_{\mathrm{ch}} \leq \|F\|_{\mathrm{ch}} + \|H\|_{\mathrm{ch}}$ for functions of the same cell.
(h) For a fixed model function the seminorm is independent of the chosen cell and of the offset of the board.

■

Proof. The homogeneity, conjugation, and supremum assertions follow from eq. A.xvii–eq. A.xviii and the nonnegativity of the folded integral. Set the other cell functions equal to one in Prop. A.11 to get (f). A constant-one function has fold one, which gives (b). For the triangle inequality, expand

$$\mathrm{Fold}(F + H) = \sum_{J \subset \mathsf{Z}} \prod_{\kappa \in J} F^{[\kappa]} \prod_{\kappa \notin J} H^{[\kappa]}.$$

The chessboard estimate and invariance under copies bound the modulus of the mean of the Jth term by $\|F\|_{\mathrm{ch}}^{|J|} \|H\|_{\mathrm{ch}}^{\mathsf{N}-|J|}$. Summing and taking the Nth root gives the triangle inequality. Finally, changing (o, κ) to (o', κ') for the same model function translates its entire fold by $o' - o + \mathsf{l}(\tilde{\kappa}' - \tilde{\kappa}) \in a\mathbb{Z}^3$. The Wilson integral is translation invariant. □

Appendix B. Periodic Gibbs states

B.1. **Infinite-volume lattice states.** We first take the volume limit at a fixed spacing. The configuration space is compact, but the finite periodic measures do not themselves satisfy the infinite-lattice Gibbs equations. We therefore check those equations on each fixed set of bonds before passing to a limit.

Let $a > 0$, let $\mathbb{L}_a = a\mathbb{Z}^3$, and let Bd be its positively oriented nearest-neighbour bonds. A configuration is an element of

$$\mathcal{V}(\mathbb{L}_a) = \mathrm{SU}(2)^{\mathsf{Bd}}, \qquad U(\bar{b}) = U(b)^{-1}.$$

We use the product topology. This is a compact metrizable space. For $B \subset \mathsf{Bd}$, write $\mathcal{F}(B)$ for the bounded Borel functions depending only on U_B, and $\mathcal{C}_{\mathrm{loc}}$ for the continuous functions depending on finitely many bonds. For a spatial set Q, $\mathsf{Bd}(Q)$ consists of bonds whose closed segments lie in Q. The notation $\mathcal{C}_{\mathrm{loc}}(Q)$ requires a finite bond support in $\mathsf{Bd}(Q)$.

Enumerating the bonds gives a product metric with summable coordinate weights. Replacing all but the first N coordinates by the identity approximates the identity map uniformly in this metric. Uniform continuity therefore shows that $\mathcal{C}_{\mathrm{loc}}$ is uniformly dense in $C(\mathcal{V}(\mathbb{L}_a))$. In particular, its integrals determine a Borel probability measure. We call such a measure a law. The laws form a compact metrizable space in the weak topology.

For a plaquette p, retain the notation $a_p(U) = 1 - \frac{1}{2}\operatorname{tr}\partial U(p) \in [0,2]$. For bounded compactly supported f on $\mathbb{R}^3$, put

$$\mathcal{O}_a(f) = a^{-1}\sum_{p\in\mathsf{Pl}} f(x(p))a_p. \tag{B.i}$$

The sum is finite, so $\mathcal{O}_a(f) \in \mathcal{C}_{\mathrm{loc}}$ even when f is merely bounded. For a law ν define

$$S_n^T[\nu](f_1,\dots,f_n) = \langle \mathcal{O}_a(f_1);\dots;\mathcal{O}_a(f_n)\rangle_\nu. \tag{B.ii}$$

If $X = (f_1,\dots,f_n)$ and ρ,ν are two laws, write

$$X_a^\nu = \prod_{q=1}^{n}(\mathcal{O}_a(f_q) - \nu(\mathcal{O}_a(f_q))), \qquad \sigma_{\rho,\nu}(X) = \rho(X_a^\nu), \qquad \sigma_\nu = \sigma_{\nu,\nu}. \tag{B.iii}$$

The empty product is one. The first index specifies the integrating law, the second the centring law. These expressions are symmetric and multilinear in the entries. Lattice isometries $\mathsf{r} \in a\mathbb{Z}^3 \rtimes W_3$ act by pullback on configurations and functions, with inversion when a bond orientation is reversed. Plaquette traces are unchanged by cyclic permutation and inversion, hence

$$\mathsf{r}\mathcal{O}_a(f) = \mathcal{O}_a(\mathsf{r}f), \qquad \sigma_{\mathsf{r}_*\rho,\mathsf{r}_*\nu}(\mathsf{r}X) = \sigma_{\rho,\nu}(X).$$

The same covariance holds for the cumulants.

B.1.1. *Gibbs equations and periodic limits.* A period lattice is a finite-index subgroup $\Gamma \subset a\mathbb{Z}^3$ with $\operatorname{sys}(\Gamma) := \min_{0\neq v\in\Gamma}|v|_\infty \geq 3a$. Choose one representative of every positive bond class modulo Γ. Independent Haar variables at these representatives, extended periodically, define dU_Γ on the closed space $\mathcal{V}_\Gamma$ of periodic configurations. Changing representatives only permutes the variables. If D contains one site of each class, then the plaquettes based at sites of D represent the plaquette classes once. Thus

$$S_\Gamma = \sum_{p\in\mathsf{Pl}/\Gamma} a_p, \quad Z_\Gamma[\Phi] = \int e^{-\beta S_\Gamma + \Phi}\,dU_\Gamma, \quad d\mu_\Gamma^\Phi = Z_\Gamma[\Phi]^{-1}e^{-\beta S_\Gamma+\Phi}\,dU_\Gamma \tag{B.iv}$$

are independent of representatives; here $\beta \geq 0$ and $\Phi \in C(\mathcal{V}_\Gamma)$ is real. Write $\mu_{\Gamma,\beta}$ when $\Phi = 0$. The action and Haar measure show that $\mathsf{r}_*\mu_\Gamma^\Phi = \mu_\Gamma^{\mathsf{r}\Phi}$ whenever $\mathsf{r}^\circ\Gamma = \Gamma$.

For finite $B \subset \mathsf{Bd}$, let $\mathsf{Pl}[B]$ be the plaquettes containing a bond of B, and set

$$H_B = -\beta\sum_{p\in\mathsf{Pl}[B]} a_p. \tag{B.v}$$

Define the kernel by

$$(\gamma_B F)(U) = \frac{\int F(V\vee U_{B^c})e^{H_B(V\vee U_{B^c})}\,dV_B}{\int e^{H_B(V\vee U_{B^c})}\,dV_B}. \tag{B.vi}$$

Every bond is in four plaquettes, so $\#\mathsf{Pl}[B] \leq 4\#B$. The denominator lies in $[e^{-2\beta\#\mathsf{Pl}[B]},1]$. The kernel is positive, preserves one and contracts the supremum norm. Integration over the finite set B preserves continuity; if F is local, the remaining support lies in the union of its support and the bonds of $\mathsf{Pl}[B]$, with B removed. Thus $\gamma_B\mathcal{C}_{\mathrm{loc}} \subset \mathcal{C}_{\mathrm{loc}}$.

Definition B.1 (Gibbs states and invariance)**.** (1) The Gibbs equation on a finite bond set B is

$$\nu(\gamma_B F) = \nu(F) \qquad (F \in \mathcal{C}_{\mathrm{loc}}). \tag{B.vii}$$

It then holds for every bounded Borel F, since the two probability measures agree on continuous local tests.

(2) A Gibbs state satisfies this equation for every finite B. Write $\mathcal{G}_{a,\beta}$ for all such states.

(3) A state is periodic if it is invariant under a finite-index subgroup of $a\mathbb{Z}^3$. Equivalently it is invariant under $Pa\mathbb{Z}^3$ for some period number $P \in \mathbb{N}$, because the orders of the coordinate classes in the finite quotient have a common multiple. The class of periodic Gibbs states is $\mathcal{G}^{\mathrm{per}}_{a,\beta}$.

(4) For $a = a_K$, $\mathcal{G}^{\mathrm{tor}}_{K,\beta}$ consists of the weak limits of periodically extended cubic torus laws along subsequences of sides $2L^m \to \infty$.

(5) For a translation-invariant state let $\mathcal{I}$ be the sigma field of translation-invariant events modulo null sets. Such a state is ergodic when $\mathcal{I}$ is trivial.

■

We also need locally finite perturbations. Let $\mathfrak{A} = (A_i)_{i\in I}$ be real local continuous functions, each with a nonempty finite support B_i, such that each bond lies in only finitely many B_i, counting indices. Replace H_B in eq. B.v by

$$H^{\mathfrak{A}}_B = H_B + \sum_{i:B_i\cap B\neq\emptyset} A_i$$

to define $\gamma^{\mathfrak{A}}_B$ and the corresponding Gibbs equations.

Proposition B.2. The Gibbs states form a convex weakly compact set. The subset invariant under any specified subgroup of lattice translations is also convex and weakly compact. Lattice isometries carry Gibbs states to Gibbs states and preserve period numbers. If $t_j \to t$ and $\nu_j \to \nu$ weakly, with ν_j Gibbs for $t_j\mathfrak{A}$, then ν is Gibbs for $t\mathfrak{A}$. ■

Proof. Each Gibbs equation is a closed affine condition on laws, since $\gamma_B F - F$ is continuous. Invariance is likewise expressed by $\nu(F \circ \mathsf{r} - F) = 0$. Intersect these conditions in the compact space of laws. Relabelling bonds, and inverting the reversed ones, preserves product Haar measure and gives $\gamma_{\mathsf{r}B}(\mathsf{r}F) = \mathsf{r}(\gamma_B F)$.

For the last assertion put $A_B = \sum_{i:B_i\cap B\neq\emptyset} A_i$ and $\delta_j = |t_j - t|\|A_B\|_\infty$. The ratio of the two normalized kernel densities lies between $e^{-2\delta_j}$ and $e^{2\delta_j}$. Consequently

$$\|\gamma^{t_j\mathfrak{A}}_B F - \gamma^{t\mathfrak{A}}_B F\|_\infty \le (e^{2\delta_j} - 1)\|F\|_\infty \longrightarrow 0.$$

The fixed kernel applied to $F \in \mathcal{C}_{\mathrm{loc}}$ is continuous. Pass to the limit in the equations for ν_j. □

Proposition B.3 (Periodic approximations)**.** (a) Let Γ_j be period lattices with $\mathrm{sys}(\Gamma_j) \to \infty$. Suppose that Γ_j acts on the indices of $\mathfrak{A}$, with $A_{v\cdot i} = \tau_v A_i$ and $B_{v\cdot i} = B_i + v$. Put $\Phi_j = \sum_{[i]\in I/\Gamma_j} A_i$ on $\mathcal{V}_{\Gamma_j}$. Every weak limit of $\mu^{\Phi_j}_{\Gamma_j}$ is Gibbs for $\mathfrak{A}$.

(b) Limits of $\mu_{\Gamma_j,\beta}$ are translation-invariant Gibbs states.

(c) At $a = a_K$, let $\widetilde{\mu}_{K,\ell,\beta}$ be the periodic lift of the cubic torus measure and define

$$\mathcal{G}^{\mathrm{tor}}_{K,\beta} = \bigcap_{m_0\ge 1} \overline{\{\widetilde{\mu}_{K,2L^m,\beta} : m \ge m_0\}}.$$

This is a nonempty compact subset of $\mathcal{G}^{\mathrm{per}}_{K,\beta}$. Every member is invariant under $a_K\mathbb{Z}^3 \rtimes W_3$ and has period number one.

■

Proof. The index action is free: $B_i + v = B_i$ with B_i finite nonempty forces $v = 0$. Every orbit has an index whose support meets a fixed bond transversal, so local finiteness makes I/Γ_j finite. Periodicity makes Φ_j independent of the representatives.

Fix a finite B and a local continuous F. Choose a closed cube Q containing its bond support, all plaquettes of $\mathsf{Pl}[B]$, and all B_i meeting B. For large j, its side is smaller than $\mathrm{sys}(\Gamma_j)$. Distinct

bonds and plaquettes in Q are then incongruent modulo Γ_j. Choose a bond transversal containing $\mathsf{Bd}(Q)$ and write its variables as (V_B, W). In these variables,

$$-\beta S_{\Gamma_j} + \Phi_j = H_B^{\mathfrak{A}} + H',$$

where H' is independent of V_B. Indeed, a remaining plaquette class, or perturbation orbit, depending on a bond of B would have a representative already included in $H_B^{\mathfrak{A}}$. For functions supported in Q, replacing V_B in a periodic configuration agrees on Q with the nonperiodic replacement in the definition of $\gamma_B^{\mathfrak{A}}$. Integrating first over V_B therefore gives

$$\mu_{\Gamma_j}^{\Phi_j}(\gamma_B^{\mathfrak{A}} F) = \mu_{\Gamma_j}^{\Phi_j}(F).$$

Both functions are continuous and independent of j. The equality passes to every weak limit.

For the empty perturbation family, periodic Haar measure and the action are invariant under every lattice translation. Invariance passes to weak limits. For the cubic sequence, the same is true of every signed coordinate permutation. Its nested tail closures are nonempty compact sets in the space of laws, so their intersection is nonempty and compact. Metrizability identifies this intersection with the subsequential limits. These limits are Gibbs by the first part, proving all assertions. □

Lemma B.4 (Gauge invariance)**.** Every Gibbs state is invariant under every gauge transformation. Two Gibbs states agreeing on gauge-invariant local continuous functions are equal. ■

Proof. First let a gauge transformation differ from the identity only at one site x. In the kernel for the six incident bonds, it is a left or right translation of each Haar variable and preserves every plaquette trace. Thus $\gamma_{B_x}(F \circ u) = \gamma_{B_x} F$, and the Gibbs equation gives invariance. Compose these changes for finitely many sites. On a local function, an arbitrary gauge transformation uses only its values at the endpoints of a finite bond support; hence invariance holds there and therefore for the law.

Averaging a local continuous F over the gauge variables at those endpoints gives a gauge-invariant local continuous F^{inv} with the same integral in every Gibbs state. Agreement on such averages implies agreement on $\mathcal{C}_{\mathrm{loc}}$, hence equality of the laws. □

B.1.2. *Reflection positivity for adapted periods.* The diagonal reflections require period lattices that split in the normal direction.

Definition B.5 (Lattice mirrors)**.** (1) Fix $a > 0$ and $\beta \geq 0$.

(2) Let

$$\mathfrak{N} = \{e_\mu : 1 \leq \mu \leq 3\} \cup \{e_\mu \pm e_\nu : \mu < \nu\}, \qquad \Sigma(n, c) = \{x : n \cdot x = c\}.$$

The admissible heights are $\mathrm{Adm}(n) = \frac{a}{2}\mathbb{Z}$ for coordinate normals and $\mathrm{Adm}(n) = a\mathbb{Z}$ for diagonal normals. These are exactly the heights in the respective families at which $\theta_{n,c} x = x - 2(n \cdot x - c)n/|n|^2$ preserves the lattice: the linear reflection is a signed permutation and its translation part is $2cn/|n|^2$. Opposite normals give the same mirrors upon negating the height. The coordinate mirrors are site mirrors when $c \in a\mathbb{Z}$ and bond mirrors otherwise. ■

A period lattice is adapted to n if, for some $h > 0$,

$$\Gamma = \Gamma_\perp \oplus \mathbb{Z} v_\Gamma, \qquad \Gamma_\perp = \Gamma \cap n^\perp, \qquad v_\Gamma = hn/|n|^2. \tag{B.viii}$$

Then $n \cdot \Gamma = h\mathbb{Z}$, and the linear reflection preserves Γ. If n is coordinate, $h \in a\mathbb{N}$ and $h \geq 3a$; if n is diagonal, $h \in 2a\mathbb{N}$ and $h/2 \geq 3a$. Thus $c + h/2$ is also admissible. The reflections in c and $c + h/2$ differ by translation through v_Γ and act identically on $\mathcal{V}_\Gamma$.

Proposition B.6 (Two-mirror positivity)**.** Let $a > 0$, $\beta \geq 0$, $n \in \mathfrak{N}$, and let Γ be a period lattice adapted to n, with height period h. For admissible c, put $\theta = \theta_{n,c}$ and $\mathbb{S}^+ = \{c \leq n \cdot x \leq c + h/2\}$. Then

$$\mu_{\Gamma,\beta}(\overline{\theta F} F) \geq 0 \qquad (F \in \mathcal{F}(\mathsf{Bd}(\mathbb{S}^+))). \tag{B.ix}$$

The law is θ-invariant. Both assertions hold also for μ_Γ^Φ if $\Phi = (\Phi_+ + \theta\Phi_+)|_{\mathcal{V}_\Gamma}$ with Φ_+ real continuous and dependent only on the bonds of $\mathbb{S}^+$. No gauge-invariance assumption on F is imposed; bonds lying in either mirror are included. ■

Proof. We retain the boundary variables instead of integrating them into either half. Write $t(x) = n \cdot x - c$. Modulo $\Gamma_\perp$, choose representatives R^0 of the bonds lying in $t = 0$ or $t = h/2$, R^+ of the other bonds contained in $0 \leq t \leq h/2$, and $R^\times$ of the bonds crossing either boundary. The remaining representatives are the reflected bonds $(\theta b)^+$, $b \in R^+$. For a bond or plaquette, use its midpoint or barycentre to measure height. Translation by jv_Γ changes this height by jh. Every object has a representative in the two half-slabs, their boundaries, or the crossing sets, with height in $(-h/2, h/2]$. Two such representatives congruent modulo Γ differ by an element of $\Gamma_\perp$, since their height difference has modulus less than h. This proves that the chosen representatives are disjoint and exhaustive. There are finitely many of them.

Consequently the coordinates

$$U_0 = U_{R^0}, \quad U_+ = U_{R^+}, \quad U'_+ = (\theta U)_{R^+}, \quad U_\times = U_{R^\times}$$

are independent Haar variables. The necessary inversions in U'_+ preserve Haar measure. On the boundary bonds θ fixes the coordinates modulo Γ, and it interchanges U_+ and U'_+. Thus $F(U) = \widehat{F}(U_0, U_+)$ and $(\theta F)(U) = \widehat{F}(U_0, U'_+)$.

Choose plaquette representatives in the same way. Those lying in the boundaries contribute a weight $W_0(U_0) > 0$. Those in the positive half but not in its boundaries contribute $W_+(U_0, U_+) > 0$, and their reflected copies contribute $W_+(U_0, U'_+)$. The remaining plaquettes meet both sides of a mirror. Their form follows by listing the corner heights

$$t(x), \quad t(x) + an_\mu, \quad t(x) + a(n_\mu + n_\nu), \quad t(x) + an_\nu.$$

A site mirror has none. At a diagonal mirror the only possibility has two opposite corners z, z' in the plane and two reflected corners off it, at heights $\pm a$ relative to the plane. With y the corner in $\mathbb{S}^+$, set $v_p(U) = U(\langle z, y\rangle)U(\langle y, z'\rangle)$. These two bonds belong to the positive half and not to its boundary; the inequality $a < h/2$ ensures that they do not reach the other mirror. Tracing the plaquette boundary gives

$$e^{-\beta a_p(U)} = k_\beta(v_p(U_+)v_p(U'_+)^{-1}), \qquad k_\beta(g) = e^{-\beta} e^{(\beta/2)\,\mathrm{tr}\, g}.$$

At a bond mirror, let x, y be neighbouring sites in the adjacent layer of $\mathbb{S}^+$. Put $u_b = U(\langle x, y\rangle)$, $u'_b = (\theta U)(\langle x, y\rangle)$ and $w_x = U(\langle x, \theta^{(k)} x\rangle)$, where $\theta^{(k)}$ is reflection in the boundary under consideration. The four-bond loop gives

$$e^{-\beta a_p(U)} = k_\beta(u_b w_y (u'_b)^{-1} w_x^{-1}).$$

The crossing variables at the two mirrors are disjoint: their midpoint heights differ by $h/2$, not by a multiple of h. Their Haar integral at each boundary is therefore

$$K(u, u') = \int \prod_{b=(x,y)} k_\beta(u_b (w \cdot u')_b^{-1})\, dw, \qquad (w \cdot u')_b = w_x u'_b w_y^{-1}.$$

Each resulting kernel has a uniformly absolutely convergent expansion $\sum_\alpha c_\alpha \Phi_\alpha(V)\overline{\Phi_\alpha(V')}$ with $c_\alpha \geq 0$ and $\sum_\alpha c_\alpha \|\Phi_\alpha\|_\infty^2 < \infty$. Indeed, the matrix-entry expansion used in A.1 gives

$$k_\beta(gg'^{-1}) = e^{-\beta} \sum_{N \geq 0} \frac{(\beta/2)^N}{N!} \sum_{I,J \in \{1,2\}^N} g_{IJ}\overline{g'_{IJ}}, \qquad \sum_{N,I,J} e^{-\beta} \frac{(\beta/2)^N}{N!} = e^\beta.$$

Products and pullbacks preserve this property. For a bond boundary, simultaneous gauge invariance and Haar invariance replace the single average in K by independent averages on its two entries. The expansion then has features $\Phi_\alpha^{\mathrm{av}}(u) = \int \Phi_\alpha(w \cdot u)\, dw$; their supremum norms do not increase. This proves the assertion also for the product $\mathcal{K}$ of the two boundary kernels.

After integrating $U_\times$, the full Wilson weight is

$$W_0(U_0)W_+(U_0, U_+)W_+(U_0, U'_+)\mathcal{K}(U_+, U'_+).$$

Termwise integration is justified by the summable supremum bounds and gives the unnormalized quadratic form

$$\int dU_0\, W_0(U_0) \sum_\alpha c_\alpha \left| \int \widehat{F}(U_0, U_+)W_+(U_0, U_+)\Phi_\alpha(U_+)\, dU_+ \right|^2 \geq 0.$$

Division by $Z_\Gamma[0] > 0$ proves eq. B.ix. For the perturbed measure apply this calculation to Fe^{Φ_+}. Finally θ preserves Haar measure and the action, and the stated perturbation is θ-invariant, proving invariance. □

B.1.3. *The torus-limit states.* Periodization $f^{\langle \ell \rangle}(x) = \sum_{v \in \ell\mathbb{Z}^3} f(\widetilde{x} + v)$ is well defined for compactly supported f. Grouping the plaquettes by period class gives the exact identity

$$\mathcal{O}_K(f)(\mathrm{per}_\ell\, U) = \mathcal{O}_{a_K}(f^{\langle \ell \rangle})(U). \tag{B.x}$$

If the supports lie in $(-\ell/4, \ell/4)^3$, periodization preserves their stated separation and per-entry radius bounds, and their $\| \cdot \|_\sharp$ norms. To check the distance assertion, each coordinate difference of two points in this cube has modulus less than $\ell/2$, so it is already the shortest representative of its class modulo ℓ. The same argument and local lifting give the supremum and Lipschitz norm identities. Periodization commutes with the lattice isometries whenever the source and its image are considered on the same quotient.

Proposition B.7. Let $K \geq 1$, $\beta \geq 0$ and $\nu \in \mathcal{G}^{\mathrm{tor}}_{K,\beta}$. There are sides $\ell_j \to \infty$ with $\widetilde{\mu}_{K,\ell_j,\beta} \to \nu$ weakly. Along every such sequence, for bounded compactly supported test functions and $w = \sum_q z_q f_q$,

$$\nu(e^{\mathcal{O}_K(w)}) = \lim_j \frac{Z_{K,\ell_j,\beta}[w^{\langle \ell_j \rangle}]}{Z_{K,\ell_j,\beta}[0]}, \tag{B.xi}$$

$$S_n^T[\nu](X) = \lim_j S^T_{n;K,\ell_j,\beta}(X^{\langle \ell_j \rangle}), \qquad \sigma_\nu(X) = \lim_j \sigma_{K,\ell_j,\beta}(X^{\langle \ell_j \rangle}). \tag{B.xii}$$

Any bound on the right uniform for tuples in $(-\ell_j/4, \ell_j/4)^3$ with prescribed separation and per-entry radius passes to the left for tuples anywhere in $\mathbb{R}^3$. The state ν is reflection positive at every coordinate site and bond mirror, for bounded local functions of the closed positive half-space. ■

Proof. The sequence exists by the tail-closure description in B.3. Every product of the observables and their exponentials is local continuous. Weak convergence and eq. B.x therefore give convergence of all moments and the first formula. Cumulants and centred moments are fixed finite polynomials in these moments, proving eq. B.xii. A fixed compact union of supports eventually lies in the central cube, so each uniform bound passes to the limit.

The cubic period lattice is adapted to every coordinate normal, with height period ℓ_j. A fixed finite bond support in $\{x_\mu \geq c\}$ eventually lies in $\{c \leq x_\mu \leq c + \ell_j/2\}$. Apply B.6 and pass to the weak limit for continuous local F. To extend the resulting inequality to bounded Borel F with the same finite support, approximate its function of the finite bond coordinates in L^2 of the corresponding marginal of ν by continuous functions. Reflection invariance gives the same L^2 convergence for their reflected functions. Cauchy–Schwarz then passes the quadratic form to the limit. Reversing the side follows by reflection invariance. □

The last assertion does not supply diagonal positivity. For $n = e_\mu \pm e_\nu$, the heights of $\ell\mathbb{Z}^3$ are $\ell\mathbb{Z}$, whereas its shortest nonzero pure-normal vector is ℓn, of height 2ℓ. Thus the cubic period lattice does not satisfy eq. B.viii for a diagonal normal.

B.1.4. *Periodic arrays and their reflection.* The comparison of Gibbs states below uses two finite-volume inequalities. The first compares a periodic array with its expectation in a Gibbs state; the second increases its partition function by reflecting the array across successive slabs.

For a site transversal D of Γ, write Bd_D for the positive bonds starting in D, and let

$$\partial_r D = \{x \in D : x + w \notin D \text{ for some } w \in a\mathbb{Z}^3, |w|_\infty \le ra\}.$$

If $\Gamma \subset \Gamma' \subset a\mathbb{Z}^3$ have finite index, the array of a local function A is $\mathbb{A} = \sum_{v\in\Gamma'/\Gamma} \tau_v A$ on $\mathcal{V}_\Gamma$, with $|\mathbb{A}| = [\Gamma' : \Gamma]$.

Lemma B.8 (Lower tangent)**.** Let Γ be a period lattice, $\Gamma' \supset \Gamma$ a finite-index subgroup, D a site transversal of Γ, and $\nu \in \mathcal{G}_{a,\beta}$ invariant under Γ'. Let $A \in \mathcal{C}_{\mathrm{loc}}$ be real and $t \in \mathbb{R}$. If N_{in} is the number of classes in Γ'/Γ having a representative v for which $\tau_v A$ has a support in Bd_D, then

$$\log \frac{Z_\Gamma[t\mathbb{A}]}{Z_\Gamma[0]} \ge t|\mathbb{A}|\nu(A) - 24\beta \#\partial_1 D - 2|t|\|A\|_\infty(|\mathbb{A}| - N_{\mathrm{in}}). \tag{B.xiii}$$

If A has a bond support in $x_0 + [0, ra]^3$, with $x_0 \in a\mathbb{Z}^3$ and integer $r \ge 1$, then

$$|\mathbb{A}| - N_{\mathrm{in}} \le \#\partial_r D \le (2r-1)^3 \#\partial_1 D.$$

■

Proof. Put $B = \mathsf{Bd}_D$. Plaquettes whose four bonds lie in B give identical action terms for $V \vee U_{B^c}$ and the periodic extension V^{per}. They represent distinct period classes. Let n_1 count the remaining plaquettes meeting B, and n_2 the period classes not represented by an interior plaquette. Then

$$\sum_{p\in\mathsf{Pl}[B]} a_p(V \vee U_{B^c}) - S_\Gamma(V^{\mathrm{per}}) = \Sigma_1 - \Sigma_2, \qquad 0 \le \Sigma_i \le 2n_i.$$

If a positive bond starts at $x \notin \partial_1 D$, all plaquettes containing it have their bonds in B. There are at most nine plaquettes meeting the three positive bonds starting at a specified x, and three plaquettes based at x. Hence $n_1 \le 9\#\partial_1 D$ and $n_2 \le 3\#\partial_1 D$.

Choose one good representative for each class counted by N_{in}, and set $\Phi_B = t\sum_{\mathrm{good}} \tau_v A$ and $e = |t|\|A\|_\infty(|\mathbb{A}| - N_{\mathrm{in}})$. The preceding action comparison, applied to numerator and denominator, gives for every exterior configuration

$$\frac{Z_\Gamma[\Phi_B]}{Z_\Gamma[0]} \ge e^{-2\beta(n_1+n_2)} \gamma_B(e^{\Phi_B}).$$

Jensen's inequality for the normalized kernel, followed by the Gibbs equation and Γ'-invariance, yields

$$\log \frac{Z_\Gamma[\Phi_B]}{Z_\Gamma[0]} \ge tN_{\mathrm{in}}\nu(A) - 2\beta(n_1+n_2).$$

On periodic configurations $|t\mathbb{A} - \Phi_B| \le e$; also $|t(|\mathbb{A}| - N_{\mathrm{in}})\nu(A)| \le e$. The two errors give eq. B.xiii.

For the last estimate, assign each array class the unique point of $x_0 + v + \Gamma$ in D. If that point is outside $\partial_r D$, the translated cube and all its bonds lie in Bd_D. Different bad classes therefore give different points of $\partial_r D$. Finally, from $x \in \partial_r D$ follow a lattice path to a point outside D, moving at most a in each coordinate at each step and using at most r steps. The point immediately before its first exit from D lies in $\partial_1 D$ and differs from x by a vector of sup norm at most $(r-1)a$. Thus $\partial_r D$ is contained in the union of $(2r-1)^3$ translates of $\partial_1 D$. □

Fix admissible c_0, $h>0$ with $v_h = hn/|n|^2 \in a\mathbb{Z}^3$, and put

$$\Sigma_j = \Sigma(n, c_0 + jh), \quad \theta_j = \theta_{n,c_0+jh}, \quad \mathbb{S}_j = \{(j-1)h \le n\cdot x - c_0 \le jh\}.$$

The maps $\mathsf{w}_{2k+1} = \tau_{2kv_h}$ and $\mathsf{w}_{2k} = \theta_k$ carry $\mathbb{S}_1$ to $\mathbb{S}_j$. Define $f^{[j]} = \mathsf{w}_j f$ for odd j and $f^{[j]} = \overline{\mathsf{w}_j f}$ for even j. Writing $\vartheta = \theta_{n,c_0+h/2}$, the reflection formulas give

$$\theta_m \mathsf{w}_j = \mathsf{w}_{2m+1-j}, \quad \mathsf{w}_{j+2N} = \tau_{2Nv_h}\mathsf{w}_j, \quad \tau_{v_h}\mathsf{w}_j = \mathsf{w}_{j+1}\vartheta. \tag{B.xiv}$$

All these maps preserve the lattice. The group generated by the θ_j consists of them and the translations τ_{2kv_h}.

Proposition B.9 (Reflection of arrays)**.** Suppose Γ is adapted to n with height period $2Nh$, $N \ge 1$. If a law μ on $\mathcal{V}_\Gamma$ is reflection positive at every pair Σ_j, Σ_{j+N}, then

$$\left|\mu\left(\prod_{j=1}^{2N} f_j^{[j]}\right)\right| \le \prod_{j=1}^{2N} \mu\left(\prod_{k=1}^{2N} f_j^{[k]}\right)^{1/(2N)} \tag{B.xv}$$

for bounded Borel functions of the bonds in $\mathbb{S}_1$; each quantity under a root is nonnegative.

Suppose additionally that $\Gamma' \supset \Gamma$ has finite index and the splitting eq. B.viii with height period h. No lower bound on its systole is required. For real $A \in \mathcal{C}_{\mathrm{loc}}(\mathbb{S}_1)$, let T represent $(\Gamma' \cap n^\perp)/(\Gamma \cap n^\perp)$, and set

$$A_T = \sum_{v\in T} \tau_v A, \qquad \mathbb{A} = \sum_{j=1}^{2N} \tau_{(j-1)v_h} A_T, \qquad \mathbb{A}^{\mathrm{f}} = \sum_{j=1}^{2N} \mathsf{w}_j A_T.$$

These define functions on $\mathcal{V}_\Gamma$ independent of representatives, and $|\mathbb{A}| = 2N\#T$. For every real t,

$$Z_\Gamma[t\mathbb{A}] \le Z_\Gamma[t\mathbb{A}^{\mathrm{f}}]. \tag{B.xvi}$$

The law $\mu_\Gamma^{t\mathbb{A}^{\mathrm{f}}}$ is reflection positive at each pair Σ_j, Σ_{j+N}, invariant under the wall group and under $\Gamma' \cap n^\perp$, and satisfies

$$\mu_\Gamma^{t\mathbb{A}^{\mathrm{f}}}(\mathbb{A}^{\mathrm{f}}) = |\mathbb{A}|\mu_\Gamma^{t\mathbb{A}^{\mathrm{f}}}(A). \tag{B.xvii}$$

■

Proof. Let $\mathrm{cy}(f_1, \dots, f_{2N})$ denote the integral on the left of eq. B.xv. The copies are cyclic modulo $2N$ on $\mathcal{V}_\Gamma$. Reflection in Σ_m changes index j to $2m+1-j$ and conjugates the copy. Cauchy–Schwarz for the positive form at that pair therefore bounds $|\mathrm{cy}|^2$ by the product of the two integrals obtained by reflecting either half of the cyclic array into the other half. Those two integrals are nonnegative. In particular constant arrays have nonnegative integrals.

Assume first that the constant-array roots $\|f_j\|$ are positive. Among the finitely many arrays with entries chosen from $f_1, \dots, f_{2N}$ maximize $|\mathrm{cy}(\underline{f})|/\prod_j \|f_j\|$. The maximum is at least one and is unchanged by either half-reflection: the product bound and maximality force both reflected arrays to attain it. Choose a maximizing array with a longest constant cyclic string, of length k. Reflect across its terminal boundary. If $k \le N$, this doubles the string; if $N < k < 2N$, it makes the whole array constant. Maximality of k therefore forces $k = 2N$, so the maximum is one. If a root vanishes, replace every f_j by $f_j + \varepsilon$ with real ε. Each constant-array integral is a nonnegative polynomial of degree $2N$ with leading coefficient one. Choose $\varepsilon \to 0$ avoiding their finitely many zeros and pass to the limit. This proves eq. B.xv.

For the array comparison take $f = e^{tA_T}$. The splitting of both period groups gives the displayed expression for $\mathbb{A}$. For odd j, $\tau_{(j-1)v_h} = \mathsf{w}_j$; for even j it is $\mathsf{w}_j\vartheta$. Thus

$$Z_\Gamma[t\mathbb{A}]/Z_\Gamma[0] = \mathrm{cy}(f, \vartheta f, \dots, f, \vartheta f), \quad Z_\Gamma[t\mathbb{A}^{\mathrm{f}}]/Z_\Gamma[0] = \mathrm{cy}(f, \dots, f).$$

The unperturbed measure has the required positivity by B.6. Translation by v_h takes the fold of f to the fold of ϑf, by eq. B.xiv; translation invariance therefore gives $\|\vartheta f\| = \|f\|$. Inequality eq. B.xv proves eq. B.xvi.

For any m, the folded perturbation is $\Phi_+ + \theta_m \Phi_+$, where $\Phi_+ = t \sum_{j=m+1}^{m+N} \mathsf{w}_j A_T$. Apply the perturbed assertion of B.6. This gives positivity and invariance under every wall reflection. Translations in $\Gamma' \cap n^\perp$ permute the transverse classes and commute with the w_j, so they preserve the perturbation and the law. All $2N\#T$ summands in $\mathbb{A}^{\mathrm{f}}$ consequently have the same expectation, which proves eq. B.xvii. □

B.1.5. *Domination of periodic Gibbs states.*

Definition B.10 (Infinite-lattice boards)**.** Fix $K \geq 1$, $a = a_K = L^{-K}$ and $\beta > 0$. For each integer $i \leq K - 1$ and offset $o \in a\mathbb{Z}^3$, the board (i, o) has cells

$$\square_k = o + lk + [-l/2, l/2]^3, \qquad l = a_i = L^{-i}, \qquad k \in \mathbb{Z}^3.$$

The space $\mathcal{A}(\square_k)$ consists of the bounded Borel functions of the inner bonds of the cell. Mirror copies are defined by eq. A.xiii and eq. A.xiv with indices in $\mathbb{Z}^3$. The reflections in the walls generate the wall group of the board. ■

Lemma B.11 (Cell properties)**.**

(a) The walls of (i, o) are bond-midpoint mirrors. Each site belongs to exactly one cell and lies in its interior. The inner bond sets of distinct cells are disjoint.

(b) A plaquette whose barycentre is in the interior of $\square_k$ has all its bonds in $\square_k$. If f vanishes outside the interior of $\square_k$, the observable $\mathcal{O}_K(f)$ is continuous and belongs to $\mathcal{A}(\square_k)$.

(c) The site chart identifies the inner bond coordinates with those of the integer cube $\mathsf{C}_{l/a}$ from eq. A.xi. It is $n \mapsto o + lk + an$.

(d) The mirror maps eq. A.xiii are lattice isometries and carry the bond coordinates of one cell to those of the other, using inverse variables at reversed bonds.

(e) The copies preserve products, sums and sup norms and commute with conjugation. Copies compose as in eq. A.xvii. An even copy is the pullback by its mirror map, while an odd copy also conjugates the function.

■

Proof. The side $l/a = L^{K-i}$ is odd, so $o_\mu + l(t + 1/2)$ is a half-integer multiple of a. Consecutive wall positions in each coordinate enclose exactly l/a sites. A plaquette can cross a wall only along one of its varying coordinates, in which case its barycentre in that coordinate is the wall midpoint. These facts give (a)–(c). The signs in eq. A.xiii preserve $a\mathbb{Z}^3$ and map the corresponding site charts onto one another. In each coordinate an odd index displacement gives reflection in the wall halfway between the two centres. An even index displacement gives translation by a multiple of $2l$, which is a product of two wall reflections. Thus the full map lies in the wall group and (d) holds. The composition and parity calculation in Lem. A.7 applies also with integer indices and proves (e). □

Definition B.12 (Period boxes and seminorms)**.** Write $\mathfrak{L} = \{2L^m : m \in \mathbb{Z}\}$ for the sides. Let (i, o) be a board and $\ell \in \mathfrak{L}$ with $\ell \geq 2l$.

(1) A period box of cell indices is $k_0 + \{0, \dots, \ell/l - 1\}^3$, where $k_0 \in \mathbb{Z}^3$.

(2) For a function F of one cell, $\|F\|_{\mathrm{ch},\ell}^{(i,o)}$ is the chessboard seminorm of its restriction to periodically lifted configurations, as a function of the corresponding torus cell. Write $\|F\|_{\mathrm{ch},\ell}$ when the board is understood.

■

Lemma B.13 (Period-box formula)**.** For a period box $\mathcal{K}_\ell$ and a cell function F,

$$\|F\|_{\mathrm{ch},\ell}^{(\ell/l)^3} = \widetilde{\mu}_{K,\ell,\beta} \left(\prod_{k' \in \mathcal{K}_\ell} F^{[k']} \right). \tag{B.xviii}$$

■

Proof. Since ℓ/l is an even integer, taking a copy commutes with periodic projection and its parity is unchanged by changing the lift of an index. Every torus cell has exactly one index in the box. The restriction of the product to periodic configurations is therefore the torus fold. Apply eq. A.xxiii. □

Corollary B.14 (Chessboard estimates for the lifted laws)**.** (a) The lifted measure $\widetilde{\mu}_{K,\ell,\beta}$ has reflection positivity on local functions in each open half-period strip at a coordinate bond mirror.

(b) For cell functions F_k in any subset of a single period box,

$$\left|\widetilde{\mu}_{K,\ell,\beta}\left(\prod_k F_k\right)\right| \le \prod_k \|F_k\|_{\mathrm{ch},\ell}.$$

(c) The properties of Prop. A.12 hold for the lifted seminorms, with the same normalization, bound by the sup norm, and invariance under mirror copies.

■□

Proof. Periodic projection carries the functions in (a) into the half-torus algebra. Apply Cor. A.3. In (b) distinct indices in a period box project to distinct cells. Apply Prop. A.11, with all unused factors equal to one. Since restriction to the periodic configurations preserves products, sums and conjugation and commutes with copies, Prop. A.12 gives (c). □

Lemma B.15 (Moving a board)**.** Let r be a lattice isometry with linear part r° and (i,o) a board.

(a) It carries cell k of (i,o) to cell $r^\circ k$ of (i,ro) and carries their inner bond algebras onto one another.

(b) Conjugation by r carries the mirror map from k to k' to the mirror map from $r^\circ k$ to $r^\circ k'$ and preserves its parity.

(c) For each $\ell \in \mathfrak{L}$ with $\ell \ge 2l$,

$$\|rF\|_{\mathrm{ch},\ell}^{(i,ro)} = \|F\|_{\mathrm{ch},\ell}^{(i,o)}. \tag{B.xix}$$

■

Proof. A signed permutation preserves the cube $[-l/2, l/2]^3$. Substitution in the mirror formula eq. A.xiii gives the conjugate map, with the diagonal signs permuted and determinant unchanged. This proves (a) and (b). The image of a period box is a period box, and the lifted cubic Wilson measure is invariant under r. Formula eq. B.xviii therefore proves (c). □

The following surface estimate removes the boundary condition.

Proposition B.16. Let $\ell \in \mathfrak{L}$ with ℓ/a an integer at least 3. For $y \in a(\mathbb{Z}+1/2)^3$, put $D = (y+[0,\ell]^3) \cap a\mathbb{Z}^3$ and let W be the set of positive bonds whose initial point lies in D. For every nonnegative bounded Borel function Ψ of W and every exterior configuration ξ,

$$e^{-36\beta(\ell/a)^2}\widetilde{\mu}_{K,\ell,\beta}(\Psi) \le \gamma_W\Psi(\xi) \le e^{36\beta(\ell/a)^2}\widetilde{\mu}_{K,\ell,\beta}(\Psi). \tag{B.xx}$$

The same inequalities hold with $\gamma_W\Psi(\xi)$ replaced by $\nu(\Psi)$ for any Gibbs state ν. For every real bounded function H of W,

$$\nu(H) \le \log \widetilde{\mu}_{K,\ell,\beta}(e^H) + 36\beta(\ell/a)^2. \tag{B.xxi}$$

■

Proof. Use the bonds of W as Haar coordinates for both the Gibbs kernel and the torus measure, and put $n = \ell/a$. In each of the three plaquette orientations, $n(n-1)^2$ plaquettes have all their bonds in W and occur with the same action in both measures. There are n^3 torus plaquettes and n^3+2n^2 plaquettes touching W in that orientation. Thus, after cancellation, the Gibbs action has $N_1 = 12n^2-3n$ terms and the torus action has $N_2 = 6n^2-3n$ terms. Since $0 \le a_p \le 2$, their difference is

between $-2N_2$ and $2N_1$. Comparing both numerators and both normalizing denominators gives a density ratio between $e^{-2\beta(N_1+N_2)}$ and $e^{2\beta(N_1+N_2)}$; $2(N_1+N_2) = 36n^2 - 12n \le 36n^2$ proves eq. B.xx. Integrate the kernel inequality against ν and use its Gibbs equation. Finally Jensen's inequality in the kernel gives $\gamma_W H \le \log \gamma_W e^H$; apply eq. B.xx and integrate once more to obtain eq. B.xxi. □

Definition B.17 (Domination data)**.** Let ν be a periodic Gibbs state with period number P, let (i, o) be a board with cell side l, and let $F_k \in \mathcal{A}(\square_k)$ for k in a nonempty finite set $\mathcal{K}_0$. Choose an even positive multiple q of P and k_0 with $\mathcal{K}_0 \subset k_0 + \{0, \dots, q-1\}^3$. Such a choice exists by choosing coordinatewise minima for k_0 and increasing q. Set $\mathsf{P} = ql$ and $m = \prod_k \|F_k\|_\infty$. For $\ell \in \mathfrak{L}$, $\ell \ge \mathsf{P}$, set

$$N_\ell = \lfloor \ell/\mathsf{P} \rfloor^3, \qquad \epsilon_\ell = 36\beta(\ell/a)^2/N_\ell. \tag{B.xxii}$$

■

Theorem B.18 (Chessboard domination)**.** (a) For data of Def. B.17, $\ell \ge \mathsf{P}$ and $t > 0$ with $2tm \le 1$,

$$\left| \nu\left(\prod_k F_k \right) \right| \le \prod_k \|F_k\|_{\mathrm{ch},\ell} + tm^2 + \epsilon_\ell/t. \tag{B.xxiii}$$

(b) For every such state and family and every sequence $\ell_j \in \mathfrak{L}$ tending to infinity,

$$\left| \nu\left(\prod_{k\in\mathcal{K}_0} F_k \right) \right| \le \liminf_{j\to\infty} \prod_{k\in\mathcal{K}_0} \|F_k\|_{\mathrm{ch},\ell_j} \le \prod_{k\in\mathcal{K}_0} \limsup_{j\to\infty} \|F_k\|_{\mathrm{ch},\ell_j}. \tag{B.xxiv}$$

(c) For a real bounded function H of one cell, the same packing, with $\ell \ge \mathsf{P}$, gives

$$\nu(H) \le \log \|e^H\|_{\mathrm{ch},\ell} + \epsilon_\ell. \tag{B.xxv}$$

(d) If ν is invariant under every reflection in the walls of the board, then for every $\ell \in \mathfrak{L}$ with $\ell \ge 2l$,

$$\nu(H) \le \log \|e^H\|_{\mathrm{ch},\ell} + 36\beta l^3 a^{-2} \ell^{-1}. \tag{B.xxvi}$$

■□

Proof. The vectors

$$V_\ell = \{v \in \mathsf{P}\mathbb{Z}^3 : 0 \le v_\mu \le \ell - \mathsf{P}\}$$

are periods of ν, since l/a is an integer and $P \mid q$. They belong to $2l\mathbb{Z}^3$, so $\tau_v F_k$ is the mirror copy in cell $k + v/l$. All these cell indices are distinct and fit in one ℓ-period box: division with remainder by q separates the translated blocks. Their inner bonds lie in the set W of B.16, with corner $y = o + l(k_0 - (1/2, 1/2, 1/2))$. Applying eq. B.xxi to the sum of the N_ℓ translates of a real local function Φ gives

$$N_\ell \nu(\Phi) \le 36\beta(\ell/a)^2 + \log \widetilde{\mu}_{K,\ell,\beta}\left(\prod_{v\in V_\ell} e^{\tau_v \Phi} \right). \tag{B.xxvii}$$

Here $\#V_\ell = N_\ell$, and $\lfloor \ell/\mathsf{P} \rfloor \ge \ell/(2\mathsf{P})$ proves the bound $\epsilon_\ell \le 288\beta \mathsf{P}^3 a^{-2} \ell^{-1}$.

Write $F = \prod_k F_k$, and choose $|\zeta| = 1$ so that $\zeta\nu(F) = |\nu(F)|$. If $m = 0$ there is nothing to prove. Otherwise put $G = \mathrm{Re}(\zeta F)$ and $\Phi = \log(1 + tG)$. Since $|tG| \le 1/2$ and $\log(1+x) \ge x - x^2$ on $[-1/2, 1/2]$,

$$\nu(\Phi) \ge t|\nu(F)| - t^2 m^2.$$

Expand $\prod_{v\in V_\ell}(1 + t\tau_v G)$ over subsets $Y \subset V_\ell$. Each product $\prod_{v\in Y} \tau_v G$ is $2^{-|Y|}$ times a sum of $2^{|Y|}$ products of mirror copies of the F_k or $\overline{F_k}$, with coefficients of modulus one. The finite chessboard

estimate and invariance of its seminorm under copying and conjugation bound each such mean by $\Pi_\ell^{|Y|}$, where $\Pi_\ell = \prod_k \|F_k\|_{\mathrm{ch},\ell}$. Consequently

$$\widetilde{\mu}_{K,\ell,\beta}\left(\prod_{v\in V_\ell} e^{\tau_v \Phi}\right) \le (1+t\Pi_\ell)^{N_\ell} \le e^{N_\ell t \Pi_\ell}.$$

Insert this in eq. B.xxvii and divide by $N_\ell t$ to obtain eq. B.xxiii. First let $\ell = \ell_j \to \infty$ at fixed t, and then let $t \downarrow 0$, proving the first inequality of eq. B.xxiv. The second follows by bounding each factor eventually by its upper limit plus an arbitrary positive error, since the family is finite and all the factors are bounded. For H, choose the corner k_0 so that the packing box contains its cell, retaining the same q. Apply eq. B.xxvii and bound the torus mean by $\|e^H\|_{\mathrm{ch},\ell}^{N_\ell}$ to obtain eq. B.xxv. The seminorm is positive because its defining integrand is bounded below by a positive number.

For eq. B.xxvi, sum all the mirror copies of H in an ℓ-period box. The mean of this sum is $(\ell/l)^3\nu(H)$, while the torus exponential mean is exactly $\|e^H\|_{\mathrm{ch},\ell}^{(\ell/l)^3}$. Use eq. B.xxi and divide by $(\ell/l)^3$. □

Corollary B.19 (Finite-side error)**.** With the data of Def. B.17, if $\ell \ge \mathsf{P}$ and $\epsilon_\ell \le 1/4$, then

$$\left|\nu\left(\prod_k F_k\right)\right| \le \prod_k \|F_k\|_{\mathrm{ch},\ell} + 2m\sqrt{\epsilon_\ell}.$$

For $\ell \ge \mathsf{P}$, the error condition follows from $\ell \ge 1152\beta\mathsf{P}^3 a^{-2}$. ■□

Proof. The case $m = 0$ is immediate. If $\epsilon_\ell = 0$, let $t \downarrow 0$ in eq. B.xxiii. Otherwise take $t = \sqrt{\epsilon_\ell}/m$ there. The estimate for ϵ_ℓ in the proof of Thm. B.18 gives the threshold. □

B.2. **Two-layer exchange and comparison states.** The comparison argument for diagonal reflections requires more than positivity of a finite-volume measure. We first prove an exchange identity for two neighbouring layers of every periodic Gibbs state. The proof inserts a unitary operator in the transfer matrix and compares the resulting partition function with a Gibbs expectation. Strict positivity of the Fourier coefficients of the plaquette weight then turns this operator identity into an identity for arbitrary continuous functions of the two layers.

B.2.1. *The plaquette kernel and its square root.* Throughout this subsection $G = \mathrm{SU}(2)$ has normalized Haar measure and $\beta > 0$. Set

$$k_\beta(g) = \exp\{-\beta(1 - \tfrac{1}{2}\operatorname{tr} g)\}, \qquad k_S(u,u') = \prod_{b\in S} k_\beta(u(b)u'(b)^{-1})$$

for a finite set S. We write K_S for the integral operator with kernel k_S on $L^2(G^S)$. Inner products are antilinear in the first argument.

For $l \in \frac{1}{2}\mathbb{N}_0$, let D^l be the representation on the homogeneous polynomials of degree $2l$ in two variables, acting by $\phi(z) \mapsto \phi(zg)$, with an invariant inner product obtained by averaging any positive inner product over G. Choose an orthonormal basis and write D^l_{mn} for its matrix coefficients and $\chi_l = \operatorname{tr} D^l$ for its character. Put $\chi_{-1/2} = 0$. The normalized products

$$e_\alpha(u) = \prod_{b\in S} (2l_b+1)^{1/2} D^{l_b}_{m_b n_b}(u(b))$$

are indexed by $\alpha = (\underline{l}, \underline{m}, \underline{n})$. Their finite linear combinations form $\mathcal{T}(G^S)$.

Lemma B.20. The coefficients satisfy

$$\int_G D^l_{mn}(g)\overline{D^{l'}_{m'n'}(g)}\,dg = \frac{\delta_{ll'}\delta_{mm'}\delta_{nn'}}{2l+1}. \tag{B.xxviii}$$

The functions e_α are an orthonormal basis of $L^2(G^S)$. The algebra $\mathcal{T}(G^S)$ is invariant under conjugation and uniformly dense in $C(G^S)$. Each fixed-spin coefficient space is invariant under conjugation. Moreover, χ_l is real, $\chi_l(g^{-1}) = \chi_l(g)$, $|\chi_l| \le 2l+1$, and

$$(\operatorname{tr} g)^N = \sum_l m_{N,l}\chi_l(g), \quad m_{N,l} \ge 0, \quad m_{2l,l} = 1, \quad \sum_l (2l+1)m_{N,l} = 2^N, \tag{B.xxix}$$

where $m_{N,l} = 0$ for $2l > N$. ∎

Proof. In the monomial basis $z_1^j z_2^{2l-j}$, a diagonal group element with entries $e^{i\omega}, e^{-i\omega}$ acts with weights $2j - 2l$. Taking $\omega = \pi/(2l+1)$ makes these eigenvalues distinct. An invariant subspace therefore contains every nonzero monomial component of each of its elements, by polynomial spectral projections. The matrix $2^{-1/2}\left(\begin{smallmatrix} 1 & 1 \\ -1 & 1 \end{smallmatrix}\right)$ sends any monomial to a polynomial with a nonzero z_1^{2l} coefficient, and sends z_1^{2l} to a polynomial with every monomial coefficient nonzero. Thus the representation is irreducible. An intertwiner has invariant kernel and image; it is zero between different dimensions, and is scalar in one irreducible representation by subtracting an eigenvalue. Averaging $D^l(g)BD^{l'}(g)^{-1}$ and taking B to be a matrix unit proves eq. B.xxviii.

Let $\mathcal{P}_N$ be the restrictions to G of polynomials of degree at most N in its four matrix entries. The coefficients with $2l \le N$ span a subspace of dimension $\sum_{j=0}^N (j+1)^2 = (N+1)(N+2)(2N+3)/6$. The identity $g_{11}g_{22} - g_{12}g_{21} = 1$ expresses every monomial of degree at most N as a sum of monomials of degrees N and $N-1$. The dimensions of these two homogeneous polynomial spaces add to the same number. Consequently $\mathcal{P}_N$ is exactly the orthogonal sum of these coefficient spaces. Conjugating an entry of an SU(2) matrix gives another entry, up to sign. Hence $\mathcal{P}_N$ is conjugation invariant, as is its orthogonal complement to $\mathcal{P}_{N-1}$, the spin-$N/2$ space. Products over S now give a self-adjoint algebra which contains the constants and separates points. Stone–Weierstrass and eq. B.xxviii prove the asserted density and completeness.

Every $g \in G$ is conjugate in G to $\operatorname{diag}(e^{i\omega}, e^{-i\omega})$, so $\chi_l(g) = \sum_{j=0}^{2l} e^{i(2j-2l)\omega}$. This proves the reality and bounds and gives $\chi_{1/2}\chi_l = \chi_{l+1/2} + \chi_{l-1/2}$. Starting with $m_{0,l} = \delta_{0l}$, the recursion $m_{N+1,l} = m_{N,l-1/2} + m_{N,l+1/2}$ proves eq. B.xxix; evaluation at the identity gives its last equality. □

Lemma B.21. Define

$$\lambda_l = \frac{1}{2l+1}\int_G k_\beta(g)\chi_l(g)\,dg, \qquad \lambda_\alpha = \prod_{b\in S}\lambda_{l_b}.$$

Then

$$\frac{e^{-\beta}(\beta/2)^{2l}}{(2l)!(2l+1)} \le \lambda_l \le \frac{\beta^{2l}}{(2l+1)^2(2l)!}, \qquad \sum_l (2l+1)^2\lambda_l = 1. \tag{B.xxx}$$

The operator K_S is positive, injective and trace class, with $K_S e_\alpha = \lambda_\alpha e_\alpha$ and $\operatorname{Tr} K_S = 1$. Its positive square root has the continuous real symmetric kernel

$$q_S(u,u') = \prod_{b\in S} q_\beta(u(b)u'(b)^{-1}), \qquad q_\beta = \sum_l (2l+1)\sqrt{\lambda_l}\chi_l.$$

Writing $q_u = q_S(u,\cdot)$, we have

$$\langle q_u, q_{u'}\rangle = k_S(u,u'), \qquad \|q_u\| = 1. \tag{B.xxxi}$$

Both K_S and its square root map each fixed-spin space to itself by a strictly positive scalar. In particular, K_S maps $\mathcal{T}(G^S)$ onto itself. If a finite complex measure σ on $G^S \times Y$, with Y compact metrizable, annihilates $(K_S\varphi)(u)\psi(y)$ for every $\varphi \in \mathcal{T}(G^S)$ and a uniformly dense collection of $\psi \in C(Y)$, then $\sigma = 0$. ∎

Proof. Expanding the exponential and using eq. B.xxix gives

$$(2l+1)\lambda_l = e^{-\beta}\sum_{N\ge 2l}\frac{(\beta/2)^N}{N!}m_{N,l}.$$

The term $N = 2l$ gives the lower bound. For the upper bound use $m_{N,l} \le 2^N/(2l+1)$ and $\sum_{N \ge j} \beta^N/N! \le \beta^j e^\beta/j!$. The character expansion $k_\beta = \sum_l (2l+1)\lambda_l \chi_l$ converges absolutely uniformly, since the exponential expansion has a summable majorant. At the identity it gives the sum rule in eq. B.xxx. Schur orthogonality gives the stated eigenvalues of K_S; their sum with multiplicities is 1. Positivity, injectivity and the trace assertions follow.

The upper bound in eq. B.xxx gives $\sum_l (2l+1)^2 \sqrt{\lambda_l} < \infty$, by the ratio test. Thus the series defining q_β converges uniformly. Applying orthogonality once more shows that its product kernel has eigenvalues $\sqrt{\lambda_\alpha}$, and convolution of two copies gives k_S. This proves eq. B.xxxi. Finally, division by the nonzero eigenvalues is legitimate on every finite coefficient sum. The assertion about measures follows from this surjectivity, density of products in $C(G^S \times Y)$, and uniqueness of representing measures. □

For a finite bond scheme with vertex set X and bond set S, put $(w \cdot u)(b) = w(b_-)u(b)w(b_+)^{-1}$ and

$$(R_w \varphi)(u) = \varphi(w^{-1} \cdot u), \qquad P\varphi = \int R_w \varphi \, dw.$$

Haar invariance shows that R_w is unitary and P is the orthogonal projection onto gauge invariant functions. Conjugation invariance of q_β gives $q_S(w \cdot u, w \cdot u') = q_S(u, u')$. Consequently K_S and $K_S^{1/2}$ commute with P and every R_w. If C has continuous kernel c, then CP has kernel $\int c(u, w \cdot u') \, dw$.

B.2.2. *Operator insertions and the finite transfer formula.* Use the third direction as time. A configuration is specified by horizontal layers u_t on the positive bonds of $a\mathbb{Z}^2$ and vertical variables $w_t(y)$ between heights ta and $(t+1)a$. The vertical plaquette over a horizontal bond b has weight

$$k_\beta(u_t(b)[(w_t \cdot u_{t+1})(b)]^{-1}). \tag{B.xxxii}$$

Indeed, its holonomy is $u_t(b) w_t(b_+) u_{t+1}(b)^{-1} w_t(b_-)^{-1}$. Horizontal plaquette actions are unchanged by a planar gauge transformation.

For a bounded operator X on $L^2(G^S)$ define

$$\kappa_X(u, u') = \langle q_u, X q_{u'} \rangle, \qquad A_X(u, u') = \frac{\kappa_X(u, u')}{k_S(u, u')}, \qquad A_X^{(t)}(U) = A_X(u_t|_S, (w_t \cdot u_{t+1})|_S). \tag{B.xxxiii}$$

The last expression uses the horizontal copies of S at heights t and $t+1$ and the vertical bonds at their endpoints. In particular, it is a continuous local function supported in that closed two-layer slab.

Proposition B.22. The map $X \mapsto A_X$ is linear, $A_1 = 1$, and

$$\|A_X\|_\infty \le e^{2\beta \# S} \|X\|.$$

It preserves reality, tensor products on disjoint bond sets, and relabeling of bonds. The function κ_X is a continuous kernel of $K_S^{1/2} X K_S^{1/2}$.

Let $n_1, n_2, s \ge 3$ be integers and $\Gamma = a(n_1\mathbb{Z} \times n_2\mathbb{Z} \times s\mathbb{Z})$. Let S_0 and X_0 be the horizontal bonds and vertices modulo the first two periods. On $L^2(G^{S_0})$, let E multiply by $e^{-\beta V}$, where V is the sum of the horizontal plaquette actions in one layer, and set

$$Q = K_{S_0}^{1/2}, \qquad M = QPEQ.$$

Then $M \ge 0$, M is trace class and $Z_\Gamma[0] = \operatorname{Tr} M^s$. At each height t, insert operators $X_{t,j}$ on finitely many horizontal bond sets $S_{t,j}$. Require that the map $(j, b) \mapsto [b]$ from the disjoint union of the $S_{t,j}$ to S_0 is injective. Relabel each operator by the resulting bijection onto its image in S_0, and let X_t be their tensor product with the identity on the remaining bonds. Then

$$\mu_{\Gamma,\beta}\left(\prod_{t=1}^{s} \prod_j A_{X_{t,j}}^{(t)}\right) = \frac{\operatorname{Tr}(X_1 M \cdots X_s M)}{\operatorname{Tr} M^s}, \qquad \left|\mu_{\Gamma,\beta}\left(\prod_{t,j} A_{X_{t,j}}^{(t)}\right)\right| \le \prod_{t,j} \|X_{t,j}\|. \tag{B.xxxiv}$$

Heights in this formula are understood modulo s. ∎

Proof. The map $u \mapsto q_u$ is continuous in L^2, by uniform continuity of the kernel. Hence κ_X is continuous and $|\kappa_X| \le \|X\|$. Since $k_S \ge e^{-2\beta \# S}$, the norm bound follows. Applying the root kernel on both sides of X gives the kernel assertion, first against continuous test functions and then in L^2. Formula eq. B.xxxi gives $A_1 = 1$. Product Haar measure and the product formula for q_S prove the tensor and relabeling assertions. They also prove preservation of reality.

The torus Haar measure is the product of the measures $du_t\, dw_t$, and eq. B.xxxii gives

$$e^{-\beta S_\Gamma(U)} = \prod_{t=1}^{s} e^{-\beta V(u_t)} k_{S_0}(u_t, w_t \cdot u_{t+1}), \qquad u_{s+1} = u_1.$$

Multiplication by the insertions replaces the tth kernel by κ_{X_t}. Integrating w_t gives the continuous kernel of $F_t = EQX_tQP$. Integrating the remaining cyclic product of kernels therefore gives $\operatorname{Tr}(F_1 \cdots F_s)$. Each factorization needed for cyclicity contains two Hilbert–Schmidt factors Q; moving EQ from the beginning to the end yields

$$Z_\Gamma[0]\mu_{\Gamma,\beta}\left(\prod_{t,j} A^{(t)}_{X_{t,j}}\right) = \operatorname{Tr}(X_1 M \cdots X_s M).$$

Since V is gauge invariant, P commutes with $E^{1/2}$ and $M = (E^{1/2}PQ)^*(E^{1/2}PQ)$ is positive trace class. Empty insertions give $Z_\Gamma[0] = \operatorname{Tr} M^s > 0$.

For the norm bound, let $L \ge 0$ be trace class and let $B_1, \dots, B_s$ be bounded. The trace Hölder inequality gives

$$|\operatorname{Tr}(B_1 L \cdots B_s L)| \le \prod_t \|B_t\| \operatorname{Tr} L^s. \tag{B.xxxv}$$

Indeed, apply Lem. A.8 to the $2s$ factors $B_t L^{1/2}, L^{1/2}$. For positive trace class $A \le B$, $\operatorname{Tr} A^s \le \operatorname{Tr} B^s$: in an eigenbasis of A, bound each eigenvalue by $\langle e, Be\rangle$, and then use scalar Jensen in an eigenbasis of B. Applying this to $L^{1/2}B_t^*B_tL^{1/2} \le \|B_t\|^2 L$ bounds the Schatten norm of $B_tL^{1/2}$ by $\|B_t\|(\operatorname{Tr} L^s)^{1/(2s)}$. This proves eq. B.xxxv, and hence the second part of eq. B.xxxiv. □

B.2.3. *Exchange of two neighbouring layers.*

Proposition B.23. Let $a > 0$, $\beta > 0$, and let ν be a periodic Gibbs state. For a finite horizontal bond set S, an integer t, and every $H \in C(G^S \times G^S)$,

$$\nu H(u_t|_S, (w_t \cdot u_{t+1})|_S) = \nu H((w_t \cdot u_{t+1})|_S, u_t|_S).$$

More precisely, if D is a bounded real operator on $L^2(G^S)$ with transpose $D^{\mathsf T} = -D$, then $\nu(A^{(t)}_D) = 0$. ∎

Proof. For a real operator, $D^{\mathsf T} = D^*$, so $X_\varepsilon = e^{\varepsilon D}$ is real unitary for real ε. Its remainder after $1 + \varepsilon D$ has norm at most $\frac{1}{2}\varepsilon^2\|D\|^2 e^{|\varepsilon|\|D\|}$. By Prop. B.22, for sufficiently small $|\varepsilon|$ the continuous local insertion $A^{(t)}_{X_\varepsilon}$ lies in $[1/2, 3/2]$. Set $A = \log A^{(t)}_{X_\varepsilon}$. Then

$$\|A\|_\infty \le \log 2, \qquad \|A - \varepsilon A^{(t)}_D\|_\infty \le C_D\varepsilon^2. \tag{B.xxxvi}$$

Choose a period number N_0 of ν and a multiple $N' \ge 3$ of N_0 large enough that distinct translates of S by $N'a\mathbb{Z}^2$ are disjoint. For each $i \ge 1$, take $\Gamma' = N'a\mathbb{Z}^3$, $\Gamma = iN'a\mathbb{Z}^3$ and the array $\mathbb{A}$ of A over Γ'/Γ. On the torus, $e^{\mathbb{A}}$ is a product of unitary insertions: at heights $t + N'k_3$ use the i^2 translates by $N'a(k_1, k_2)$, $0 \le k_1, k_2 < i$. Their bond sets are pairwise incongruent modulo the horizontal periods: a congruence would give a nonzero $N'a\mathbb{Z}^2$ translation intersecting S, or equal representatives. Thus eq. B.xxxiv gives

$$\log \frac{Z_\Gamma[\mathbb{A}]}{Z_\Gamma[0]} \le 0.$$

Enclose the support of A in a lattice cube of side ra. The lower tangent inequality B.8, applied on the coordinate domain of side $iN'a$, gives

$$0 \geq i^3\nu(A) - (24\beta + 2\log 2(2r-1)^3)\#\partial_1 D_i.$$

Since $\#\partial_1 D_i \leq 6i^2N'^2$, division by i^3 and $i \to \infty$ yield $\nu(A) \leq 0$. Using eq. B.xxxvi for both signs of ε and then letting $\varepsilon \to 0$ proves $\nu(A_D^{(t)}) = 0$.

It remains to recover arbitrary functions of the layers. Let Φ be a real antisymmetric trigonometric polynomial on $G^S \times G^S$, and write its finite expansion as

$$\Phi(u,u') = \sum_{\alpha,\alpha'} c_{\alpha\alpha'} e_\alpha(u)\overline{e_{\alpha'}(u')}.$$

Divide its coefficients by $\sqrt{\lambda_\alpha \lambda_{\alpha'}}$ to obtain Φ'. This operation preserves reality and antisymmetry: each fixed pair of spin spaces is conjugation invariant, and swapping the two spaces does not change the positive divisor. The finite-rank operator D with kernel Φ' is therefore real and skew-adjoint. Applying $K_S^{1/2}$ in both variables gives $\kappa_D = \Phi$, and hence

$$\nu\left[\frac{\Phi}{k_S}(u_t|_S, (w_t \cdot u_{t+1})|_S)\right] = 0.$$

For real continuous H, the function $k_S(u,u')[H(u,u') - H(u',u)]$ is real and antisymmetric. Approximate it uniformly by trigonometric polynomials, then take real parts and antisymmetrize. Division by $k_S \geq e^{-2\beta\#S}$ preserves uniform convergence, proving the desired exchange identity for real H. Complex linearity finishes the proof. □

B.2.4. *Comparison with a state positive at a prescribed mirror.*

Proposition B.24. Let $a > 0$, $\beta \geq 0$, $n \in \mathfrak{N}$ and $c \in \mathrm{Adm}(n)$. For every periodic Gibbs state ν and every real $A \in \mathcal{C}_{\mathrm{loc}}(\mathbb{H}^+(n,c))$ there exists a periodic Gibbs state ν_0, reflection positive at $\Sigma(n,c)$, such that $\nu(A) \leq \nu_0(A)$. ■

Proof. Choose a period number N_0 of ν, a nonempty finite bond support $B_A \subset \mathsf{Bd}(\mathbb{H}^+(n,c))$, and a lattice cube $Q = x_0 + [0, ra]^3$ containing its bonds. Choose $h \in 2N_0a\mathbb{N}$ so that B_A lies in $\mathbb{S}_1 = \{c \leq n\cdot x \leq c+h\}$, and put $v_h = hn/|n|^2$. If $n = e_\mu$, let $\mathfrak{T} = N_0a(\mathbb{Z}e_{\mu'} \oplus \mathbb{Z}e_{\mu''})$. If $n = e_\mu + se_\nu$, $s = \pm 1$, let $\mathfrak{T} = N_0a\mathbb{Z}(e_\mu - se_\nu) \oplus N_0a\mathbb{Z}e_\kappa$, where κ is the remaining direction. Set

$$\Gamma' = \mathfrak{T} \oplus \mathbb{Z}v_h, \qquad \iota = [a\mathbb{Z}^3 : \Gamma'], \qquad \Gamma_i = i\mathfrak{T} \oplus \mathbb{Z}(2iv_h), \quad i \geq 3.$$

These groups are adapted to n, with respective height periods h and $2ih$. Moreover $\Gamma' \subset N_0a\mathbb{Z}^3$, so ν is Γ' invariant, and $\mathrm{sys}(\Gamma_i) \geq ia$.

In the coordinate case choose the domain

$$D_i = \{x \in a\mathbb{Z}^3 : 0 \leq x_\mu < 2ih,\ 0 \leq x_{\mu'}, x_{\mu''} < iN_0a\}.$$

In the diagonal case choose

$$D_i = \{x \in a\mathbb{Z}^3 : 0 \leq x_\mu - sx_\nu < 2iN_0a, \quad 0 \leq x_\kappa < iN_0a, \quad 0 \leq x_\mu + sx_\nu < 2ih\}.$$

The two diagonal coordinates divided by a have the same parity. Reducing them modulo $2iN_0$ and $2ih/a$ preserves this parity and gives one representative per class. Thus these sets are domains of Γ_i. In the definition of $\partial_1 D_i$, the displacements w satisfy $|w|_\infty \leq a$. Each ordinary coordinate changes by at most a, while $x_\mu - sx_\nu$ and $x_\mu + sx_\nu$ change by at most $2a$. Count one lattice layer at each end of an ordinary coordinate interval and two at each end of a diagonal coordinate interval. The parity count above gives the same number of sites in each layer. The union bound therefore gives

$$\frac{\#\partial_1 D_i}{\#D_i} \leq \frac{2}{i}\left(\frac{2}{N_0} + \frac{a}{h}\right).$$

Let $\mathbb{A}$ be the array of A over Γ'/Γ_i, and $\mathbb{A}^{\mathrm{f}}$ its folded array from §B.1. Its number of terms is $|\mathbb{A}| = \#D_i/\iota$. For $t > 0$, the lower tangent inequality, folding inequality, and Jensen's inequality in the folded measure give, in that order,

$$\begin{aligned} t|\mathbb{A}|\nu(A) - (24\beta + 2t\|A\|_\infty(2r-1)^3)\#\partial_1 D_i &\le \log \frac{Z_{\Gamma_i}[t\mathbb{A}]}{Z_{\Gamma_i}[0]} \\ &\le \log \frac{Z_{\Gamma_i}[t\mathbb{A}^{\mathrm{f}}]}{Z_{\Gamma_i}[0]} \\ &\le t\mu_{\Gamma_i}^{t\mathbb{A}^{\mathrm{f}}}(\mathbb{A}^{\mathrm{f}}) = t|\mathbb{A}|\mu_{\Gamma_i}^{t\mathbb{A}^{\mathrm{f}}}(A). \end{aligned}$$

The last equality is the wall symmetry of the folded measure. Hence

$$\nu(A) \le \mu_{\Gamma_i}^{t\mathbb{A}^{\mathrm{f}}}(A) + \iota\left(\frac{24\beta}{t} + 2\|A\|_\infty(2r-1)^3\right)\frac{\#\partial_1 D_i}{\#D_i}. \tag{B.xxxvii}$$

We keep t fixed while taking the volume limit. To identify its Gibbs specification, write the infinite folded family with its indices:

$$I = \mathbb{Z} \times \mathfrak{T}, \qquad A_{(j,v)} = tw_j\tau_v A, \qquad B_{(j,v)} = w_j(B_A + v). \tag{B.xxxviii}$$

Here w_j are the alternating translations and reflections which send the first slab to the jth slab. Every bond belongs to at most $2\#B_A$ of these supports. For $g = g_\perp + 2ikv_h \in \Gamma_i$, define $g\cdot(j,v) = (j+2ik, v+g_\perp)$. The relations $w_{j+2ik} = \tau_{2ikv_h}w_j$ and $\tau_{g_\perp}w_j = w_j\tau_{g_\perp}$ show that this free index action translates both the functions and their supports. Representatives of I/Γ_i are $\{1,\dots,2i\}\times R_i$, where R_i represents $\mathfrak{T}/i\mathfrak{T}$, and their sum on the torus is $t\mathbb{A}^{\mathrm{f}}$. Thus the finite-volume DLR limit applies to one fixed locally finite perturbation family.

By compactness, a subsequence of the folded torus measures converges to a Gibbs state ν_t for this family. The finite measures, and hence their limit, are invariant under $\mathfrak{T} \oplus \mathbb{Z}(2v_h)$. A fixed continuous local F in $\mathbb{H}^+(n,c)$ lies in $\{c \le n\cdot x \le c+ih\}$ for all sufficiently large i. The two-mirror positivity of the folded torus measure then gives $\mu_{\Gamma_i}^{t\mathbb{A}^{\mathrm{f}}}(\overline{\theta F}F) \ge 0$, and this inequality passes to ν_t. The error in eq. B.xxxvii tends to zero, so $\nu(A) \le \nu_t(A)$.

Finally take a weak limit of a subsequence of $\nu_{1/k}$. Continuity of the local DLR kernels as the strength of the fixed family tends to zero makes its limit ν_0 an unperturbed Gibbs state. The common period group, reflection positivity, and the expectation inequality all pass to this limit. The period group has finite index, so ν_0 is periodic, as required. □

B.3. **Symmetries of periodic Gibbs states.** We first use the two-layer identity to obtain a common translation group for all periodic Gibbs states. This makes their class compact and convex. The comparison states can then test membership in its reflection-positive subclass by separation.

B.3.1. *Temporal gauge and determination by two layers.* Fix an integer t_0. Given the vertical variables w_t, define

$$\hat{w}_{t_0} = 1, \qquad \hat{w}_{t+1} = \hat{w}_t w_t, \qquad \tilde{u}_t = \hat{w}_t \cdot u_t.$$

This defines a continuous map $\mathsf{tg} : \mathcal{V}(\mathbb{L}_a) \to (\mathcal{V}^{(2)})^{\mathbb{Z}}$: each of its coordinates uses finitely many vertical variables and one horizontal bond. The gauge transformation $\hat{w}$ makes all vertical bonds equal to 1. Write $U[\tilde{u}]$ for the resulting configuration, with horizontal layers $\tilde{u}$ and vertical variables 1. For a gauge invariant local continuous function F,

$$F(U) = \tilde{F}(\mathsf{tg}(U)), \qquad \tilde{F}(\tilde{u}) = F(U[\tilde{u}]).$$

For finite horizontal S, let $V_S(u)$ be the sum of the planar plaquette actions which meet S. It contains at most $2\#S$ terms, so $0 \le V_S \le 4\#S$. Define the conditional layer kernel

$$\tilde{k}_{S,t}(v;\tilde{u}) = k_S(\tilde{u}_{t-1}|_S, v)e^{-\beta V_S(v\vee\tilde{u}_t|_{S^c})}k_S(v,\tilde{u}_{t+1}|_S). \tag{B.xxxix}$$

The kernel $\tilde\gamma_{S,t}$ replaces $\tilde u_t|_S$ by v and integrates with the normalized density $\tilde k_{S,t}(v;\tilde u)\,dv$. This density and its normalizer are bounded below by $e^{-8\beta\#S}$ before normalization. They depend continuously on finitely many coordinates. A layer law obeys the chain equations if it is fixed by every $\tilde\gamma_{S,t}$ on continuous local functions.

Lemma B.25. The temporal-gauge image of every Gibbs state obeys the chain equations. The equations are invariant under each index reflection $(\tilde\theta_m\tilde u)_t = \tilde u_{m-t}$. If $\mathsf{tg}_*\nu$ is invariant under $\tilde\theta_{2t_0+1}$, then ν is invariant under the bond reflection $\theta_{e_3,(t_0+1/2)a}$. ■

Proof. The vertical plaquette formula becomes $1-\frac12\operatorname{tr}(\tilde u_t(b)\tilde u_{t+1}(b)^{-1})$, and the horizontal plaquette action is unchanged by $\hat w_t$. If only the horizontal bonds $B=\{b^{(t)} : b\in S\}$ are integrated in the DLR kernel, the variables $\hat w_s$ do not change. The substitution $v=\hat w_t\cdot U_B$ preserves product Haar measure. The plaquettes meeting B consist of the planar plaquettes meeting S, and one vertical plaquette above and one below each bond of S. Their weight is precisely eq. B.xxxix. Thus

$$\gamma_B(\psi\circ\mathsf{tg}) = (\tilde\gamma_{S,t}\psi)\circ\mathsf{tg},$$

which proves the chain equations.

Symmetry of k_S exchanges the two neighbouring layers in eq. B.xxxix; hence $(\tilde\gamma_{S,m-t}\psi)\circ\tilde\theta_m = \tilde\gamma_{S,t}(\psi\circ\tilde\theta_m)$. This proves covariance of the equations. With $\theta=\theta_{e_3,(t_0+1/2)a}$ one has $\theta U[\tilde u] = U[\tilde\theta_{2t_0+1}\tilde u]$. Consequently invariance of the layer law implies that ν and $\theta_*\nu$ agree on all gauge invariant local continuous functions. Both measures are Gibbs states, so the gauge averaging lemma B.4 makes them equal. □

Lemma B.26. For $\beta>0$, two laws obeying the chain equations and agreeing on two or more consecutive entire layers agree on all layers. ■

Proof. Suppose that the two laws agree on layers $m,\dots,n$, with $m<n$. We extend agreement to $n+1$. Fix a nonempty finite horizontal set S, and use the coordinates

$$u'=\tilde u_{n+1}|_S,\qquad v=\tilde u_n|_S,\qquad c=(\tilde u_n|_{S^c},\tilde u_{n-1},\dots,\tilde u_m)\in Y.$$

Here Y is a compact countable product of copies of G. Put

$$\Phi(v;c)=k_S(\tilde u_{n-1}|_S,v)e^{-\beta V_S(v\vee\tilde u_n|_{S^c})},\qquad N(u',c)=\int k_S(v,u')\Phi(v;c)\,dv.$$

The assumption $m<n$ ensures that $\tilde u_{n-1}$ is among the coordinates of c. Both functions are continuous and local, with $\Phi\ge e^{-6\beta\#S}$ and $N\ge e^{-8\beta\#S}$. Let ρ_i' be the image of the ith law on (u',c) and set $\sigma_i=N^{-1}\rho_i'$. The chain equation at (S,n) reads

$$\rho_i(\psi)=\int\sigma_i(du',dc)\int k_S(v,u')\Phi(v;c)\psi(u',v,c)\,dv \tag{B.xl}$$

for every continuous local ψ. Take $\psi=\varphi(v)\chi(c)/\Phi(v;c)$, with $\varphi\in\mathcal T(G^S)$ and χ continuous local. This is a function of the already equal layers. Subtracting the two identities gives

$$\int(K_S\varphi)(u')\chi(c)\,d(\sigma_1-\sigma_2)=0.$$

The measure injectivity in Lem. B.21 yields $\sigma_1=\sigma_2$. Equation eq. B.xl then gives agreement for all continuous local functions of (u',v,c). Enlarging S to contain the finitely many coordinates of any test on layer $n+1$ proves agreement on layers $m,\dots,n+1$.

Apply the same argument after replacing t by $m+n-t$, using the covariance in Lem. B.25, to extend agreement to layer $m-1$. Iterating both extensions gives agreement on every finite block of layers. Continuous local functions determine laws on this compact product, so the two laws are equal. □

Proposition B.27. Every periodic Gibbs state is invariant under every coordinate bond reflection, and consequently under translations by $2a\mathbb{Z}^3$. The periodic Gibbs states are exactly the Gibbs states invariant under $2a\mathbb{Z}^3$; their set is nonempty, compact and convex. ■

Proof. For $\beta > 0$, take the temporal gauge based at t_0. Its layers $t_0, t_0 + 1$ are u_{t_0} and $w_{t_0} \cdot u_{t_0+1}$. Prop. B.23 says that their joint law is invariant under exchange. The layer law and its $\tilde{\theta}_{2t_0+1}$ reflection both obey the chain equations and agree on these two layers. Lem. B.26 makes them equal, and Lem. B.25 gives the bond reflection invariance of ν. The same reasoning in each coordinate direction follows by applying it to the coordinate permutation of ν, which is again a periodic Gibbs state.

At $\beta = 0$, the DLR equation on the support of a local function sets its expectation equal to its product Haar integral. This determines a unique Gibbs state, which is invariant under all lattice isometries.

The product of the reflections at $x_\mu = a/2$ and $x_\mu = 3a/2$ is translation by $2ae_\mu$. Thus every periodic state is invariant under $2a\mathbb{Z}^3$. Conversely, this group has finite index, so a Gibbs state invariant under it is periodic. Compactness and convexity follow from the DLR equations and the fixed common period group; nonemptiness follows from the periodic volume construction. □

B.3.2. *All lattice reflections.*

Proposition B.28. For every $a > 0$ and $\beta \geq 0$ the following hold.

(a) Every periodic Gibbs state is reflection positive at all coordinate site, coordinate bond and diagonal lattice mirrors, on bounded local Borel functions of either closed half-space.
(b) Every periodic Gibbs state is invariant under $a\mathbb{Z}^3 \rtimes W_3$.
(c) The periodic Gibbs states are exactly the translation-invariant Gibbs states. Their class is nonempty, compact and convex and contains every cubic torus-limit state.

■

Proof. Fix a mirror $\Sigma(n, c)$, $n \in \mathfrak{N}$, and write $\theta = \theta_{n,c}$. Let $\mathcal{R}$ be the periodic Gibbs states reflection positive at this mirror. It is nonempty by Prop. B.24 with $A = 0$. It is compact and convex: Prop. B.27 gives those properties for the ambient class, and each condition $\rho(\overline{\theta F} F) \in [0, \infty)$, with F continuous local in the positive half-space, is closed and affine.

Suppose $\nu \notin \mathcal{R}$. There is a real continuous f and $\delta > 0$ such that $\nu(f) \geq \rho(f) + 3\delta$ for every $\rho \in \mathcal{R}$. Here separation can be obtained in a finite-dimensional space. Indeed, if every finite family of continuous test integrals of ν were attained by an element of $\mathcal{R}$, compactness and the finite intersection property would give an element with all the same integrals, hence $\nu \in \mathcal{R}$. Otherwise the finite vector of integrals of ν lies outside the corresponding compact convex set. Its nearest point separates it by the vector joining that point to the given vector, which gives f and a positive gap.

Choose real local A_0 with $\|f - A_0\|_\infty \leq \delta$. Then $\nu(A_0) \geq \rho(A_0) + \delta$ for every $\rho \in \mathcal{R}$. Translate A_0 by $2kan$, with k so large that its support lies in $\mathbb{H}^+(n, c)$, and call the result A. All the states in question are invariant under this translation by Prop. B.27, so the same strict inequality holds for A. Prop. B.24 now supplies $\nu_0 \in \mathcal{R}$ with $\nu(A) \leq \nu_0(A)$, a contradiction. This proves positivity for continuous local functions.

The same positivity holds for bounded Borel local functions. On their finite compact coordinate space, approximate such a function in L^2 for the sum of the marginal laws of ν and $\theta_*\nu$ by continuous functions. Cauchy–Schwarz bounds the change in $\nu(\overline{\theta F} F)$, so positivity passes to the limit. Replacing F by the conjugate of its reflection also gives positivity on the negative half-space and for the opposite normal.

To obtain reflection invariance, positivity makes the form $\nu(\overline{\theta F} F')$ Hermitian. Taking one entry equal to 1 gives $\nu(F) = \nu(\theta F)$ for local F in the positive half-space. Any local F_0 can be translated into that half-space by $v = 2kan$. Since both v and $-v$ are periods of ν and $\theta\tau_v = \tau_{-v}\theta$, the same

equality holds for F_0. Thus $\theta_*\nu = \nu$ on all continuous local tests and hence on all Borel sets. The coordinate and diagonal lattice reflections generate $a\mathbb{Z}^3 \rtimes W_3$: adjacent site and bond mirrors give unit translations, coordinate mirrors give sign changes, and $x_\mu = x_\nu$ gives coordinate transpositions. This proves the symmetry assertion. The description and compactness of the state class then follow from the translation-invariant DLR equations, and the torus-limit inclusion is already known. □

B.3.3. *Invariant conditioning.*

Proposition B.29 (Invariant reweighting)**.** (a) Let ν be a periodic Gibbs state. Suppose $h \geq 0$ is bounded, is $\mathcal{I}$-measurable and satisfies $\nu(h) = 1$. Then $h\nu$ is a periodic Gibbs state.
(b) The extreme points of the periodic Gibbs class are exactly its ergodic states.

■

Proof. By Prop. B.28 the state ν is invariant under all lattice translations. Fix a finite bond set B. Approximate h in $L^1(\nu)$ by uniformly bounded continuous local functions, and translate each approximant so that its support is outside B. The error is unchanged because both h and ν are translation invariant. The DLR equation on B implies $\nu(h_j\gamma_B F) = \nu(h_j F)$ for each translated approximant h_j. Pass to the limit to get the same equality for h. The law $h\nu$ is translation invariant by the invariance of h and ν, so this proves (a).

If an invariant event has probability strictly between zero and one, (a) decomposes ν into the two conditioned Gibbs laws. Conversely, in any decomposition into invariant states with positive weights, each summand has a bounded Radon–Nikodym density with respect to ν. Uniqueness of that density makes it invariant. Ergodicity makes each density constant, so the decomposition is trivial. □

Proposition B.30 (Ergodic conditional Gibbs laws)**.** (1) In the periodic Gibbs class, ergodicity is equivalent to extremality.
(2) Let ν be a periodic Gibbs state and let ν^U denote its regular conditional law given $\mathcal{I}$. Then for ν-almost every U, the law ν^U belongs to $\mathcal{G}^{\mathrm{per}}_{a,\beta}$ and is ergodic.

■

Proof. Part (1) is Prop. B.29(b). For (2), the DLR and translation-invariance equalities hold after integration over any invariant event by Prop. B.29(a). Thus their conditional expectations given $\mathcal{I}$ vanish. Use a countable uniformly dense family of continuous local tests, and the countability of finite bond sets and translations, to obtain all the equalities for ν^U simultaneously outside one null set. The equality on the dense family extends to continuous tests by contraction of the kernels in supremum norm. This proves the Gibbs and invariance assertions.

For a continuous local F, the cubic averages $A_n F$ of its translates converge in $L^2(\nu)$ to $\nu(F \mid \mathcal{I})$. Indeed the commuting coordinate translations are unitary. Their one-direction Cesàro averages converge strongly to their invariant projections, by the spectral theorem, and the product converges to the common invariant projection. Choose a common subsequence with summable squared errors for every member of the countable dense family. By disintegration, the same subsequence converges in $L^2(\nu^U)$ for almost every U, with constant limit $\nu^U(F)$. Here the invariant random variable $\nu(F \mid \mathcal{I})$ equals its value at U, ν^U-almost surely. Applying the same Hilbert-space mean argument under ν^U identifies its invariant projection of each test as that constant. Density proves that the invariant subspace consists of constants, so ν^U is ergodic. □

B.4. **Insertion estimates for periodic Gibbs states.**

Definition B.31 (Insertion setting)**.** (1) Fix the odd integer L and the cube parameter M of the renormalisation argument. The insertion estimates only require $L \geq 3$ odd and do not restrict M.

(2) At fixed cutoff K and coupling $\beta > 0$, use (a) the lattice $a_K\mathbb{Z}^3$, (b) the Wilson laws and their periodic lifts, (c) the Gibbs equations of Def. B.1, (d) the mirrors and admissible heights of Def. B.5, and (e) the boards and their seminorms from Defs. B.10 and B.12.

■

Definition B.32 (Invariance, positivity and domination)**.** For a law ρ on $a_K\mathbb{Z}^3$ and a sequence $\ell_\bullet$ of admissible sides tending to infinity, use the following conditions.

(1) (Inv) is invariance under all lattice isometries.
(2) (Pos) is reflection positivity at every bond mirror $x_3 = c$ for bounded local functions in the positive half-space.
(3) (Dom) for a class of functions $\mathfrak{C}$ means that on every board and for every finite family of cell functions $F_k \in \mathfrak{C}$,

$$\left|\rho\left(\prod_k F_k\right)\right| \le \liminf_{j\to\infty} \prod_k \|F_k\|_{\mathrm{ch},\ell_j}. \tag{B.xli}$$

Here $\mathfrak{C}$ is the bounded local Borel class or its continuous subclass.
(4) For a cell function D, write $\|D\|_{\mathrm{ch},\ell_\bullet} = \liminf_j \|D\|_{\mathrm{ch},\ell_j}$.
(5) At a bond mirror c, put $\langle F', F\rangle_{\rho,c} = \rho(\overline{\theta_c F'} F)$ and, under (Pos), $\|F\|_{\rho,c} = \langle F, F\rangle_{\rho,c}^{1/2}$.

■

Corollary B.33 (Domination and positivity of Gibbs laws)**.** (a) Every periodic Gibbs state satisfies (Dom) for every sequence of sides tending to infinity and for both function classes.
(b) Each limit of cubic torus measures belongs to the periodic Gibbs class and satisfies (Inv) and (Pos) as well.

■□

Proof. Thm. B.18(b) gives (a). Prop. B.3(c) gives the Gibbs inclusion in (b), and Prop. B.28 gives the invariance and positivity. In fact the same two properties hold for every periodic Gibbs state. □

B.4.1. *Multiple reflections with an insertion.* We need the following consequence of the finite chessboard estimate. Call (i', o') coarser than (i, o) if $i' \le i$ and $o' - o \in l\mathbb{Z}^3$. Then $d = a_{i'}/a_i = L^{i-i'}$ is odd, and each coarse cell is the union of d^3 fine cells.

Lemma B.34. For a coarse cell C' and functions D_k of its fine cells,

$$\left\| \prod_{C_k \subset C'} D_k \right\|_{\mathrm{ch},\ell}^{\mathrm{coarse}} \le \prod_{C_k \subset C'} \|D_k\|_{\mathrm{ch},\ell}^{\mathrm{fine}}, \qquad \ell \ge 2a_{i'}. \tag{B.xlii}$$

■

Proof. Writing $z = (o' - o)/l$, the fine cells of coarse cell j have indices

$$z + dj + \{-(d-1)/2, \dots, (d-1)/2\}^3.$$

Every coarse wall is a fine wall. Thus a coarse mirror copy sends each fine cell to a fine cell by the corresponding fine mirror map; its parity is unchanged, since parity is the determinant of that map. The disseminated coarse product is consequently a product over all fine cells in a torus period box. Apply the fine chessboard inequality to this product. Every original factor has one copy for each coarse cell, and its fine seminorm is unchanged by copying. Taking the root whose order is the number of coarse cells proves eq. B.xlii. □

Proposition B.35. Let ρ satisfy (Inv), (Pos) and (Dom) along $\ell_\bullet$ for the class $\mathfrak{C}$ of Def. B.32. Suppose a cell C of side l lies between consecutive board walls $x_3 = c$ and $x_3 = c+l$. For a function $D \in \mathfrak{C}$ of this cell and functions $F, F' \in \mathfrak{C}$ in $\{x_3 \geq c\}$,

$$\left|\rho(\overline{\theta_c F'} D \tau_{le_3} F)\right| \leq \|D\|_{\mathrm{ch},\ell_\bullet} \|F'\|_{\rho,c} \|F\|_{\rho,c}. \tag{B.xliii}$$

■

Proof. Use the reflected form of Def. B.32(5), and write $\|\cdot\|$ for its seminorm. Put $\widehat{D}F = D\tau_{le_3}F$ and $D^\sharp = \overline{\theta_{c+l/2} D}$, where the middle reflection preserves the cell. Translation invariance and $\theta_c \theta_{c+l/2} = \tau_{-le_3}$ give

$$\langle F', \widehat{D} F\rangle_{\rho,c} = \langle \widehat{D^\sharp} F', F\rangle_{\rho,c}.$$

Let $F_n = (\widehat{D^\sharp}\widehat{D})^n F$ and $s_n = \langle F, F_n\rangle_{\rho,c}$. Moving adjoints gives $s_{2p} = \|F_p\|^2$ and $s_{2p+1} = \|\widehat{D}F_p\|^2$. The odd-indexed moment is also $\langle F_p, F_{p+1}\rangle_{\rho,c}$, and the even-indexed moment s_{2p}, for $p \geq 1$, is $\langle \widehat{D}F_{p-1}, \widehat{D}F_p\rangle_{\rho,c}$. Cauchy–Schwarz in these two identities gives $s_n^2 \leq s_{n-1}s_{n+1}$ for $n \geq 1$. Therefore, for $n \geq 2$,

$$s_1^n \leq s_0^{n-1} s_n. \tag{B.xliv}$$

If $s_1 > 0$, log convexity first gives $s_n > 0$ for all $n \geq 0$ and then makes the ratios s_{n+1}/s_n increasing. If $s_1 = 0$, the inequality holds directly.

Expanding the iterates yields

$$s_n = \rho\left(\overline{\theta_c F} \prod_{r=0}^{2n-1} D_r\, \tau_{2nle_3} F\right), \qquad D_r = \begin{cases} \tau_{rle_3} D^\sharp, & r \text{ even}, \\ \tau_{rle_3} D, & r \text{ odd}. \end{cases}$$

These are the successive mirror copies of $D^\sharp$. Reflection in the cell centre preserves the board, so Lem. B.15(c) and invariance under conjugation give $\|D^\sharp\|_{\mathrm{ch},\ell} = \|D\|_{\mathrm{ch},\ell}$. Choose an odd coarsening factor $d = L^{r_0}$ large enough that F is supported in a coarse cell directly above the wall c. Such a board has side $l' = dl$ and center obtained from that of C by adding $(d-1)le_3/2$; its bottom wall remains c. For $n = n'd$, group the $2n$ functions D_r into $2n'$ consecutive coarse cells, with F and its reflected copy in the two end cells. At each side $\ell_j \geq 2l'$, the product of coarse seminorms is at most $\|F\|_\infty^2 \|D\|_{\mathrm{ch},\ell_j}^{2n}$ by eq. B.xlii and the supremum bound on the end factors. Chessboard domination and the lower limit give

$$s_n \leq \|F\|_\infty^2 \|D\|_{\mathrm{ch},\ell_\bullet}^{2n}.$$

Combine this estimate with eq. B.xliv, take nth roots and let $n' \to \infty$. This gives $\|\widehat{D}F\| \leq \|D\|_{\mathrm{ch},\ell_\bullet}\|F\|$ when $\|F\| > 0$. If $\|F\| = 0$, eq. B.xliv gives $s_1 = 0$ directly. Cauchy–Schwarz now proves eq. B.xliii. □

Appendix C. Separated tests and growth

C.1. **Separated functions and compactness.** Uniform estimates at separated supports give one subsequence for all orders and all test tuples. The support restriction matters in this argument: approximating the entries must preserve their separation. We give the compactness statement on both a fixed torus and $\mathbb{R}^3$.

Definition C.1 (Separated tuples)**.** Let $\mathbb{E} = T_\ell$ or $\mathbb{R}^3$. For an open set $O \subset \mathbb{E}$, let $\mathfrak{S}_n(O)$ be the tuples $(f_1, \ldots, f_n)$ of real functions in $C_c^1(O)$ with pairwise disjoint supports. Zero entries are allowed. Put $\mathfrak{S}_0(O) = \{\varnothing\}$ and $\mathfrak{S}(O) = \bigcup_{n\geq 0} \mathfrak{S}_n(O)$. A tuple has type $(d; R)$ if distinct supports are at distance at least $d > 0$ and each support lies in a ball of radius $R > 0$; the centres of these balls may differ. Every separated tuple has such a type, since its supports are compact. For a tuple $X = (f_1, \ldots, f_n)$, let $d(X)$ be the least distance between distinct supports, with value ∞ when fewer than two supports are nonempty. Let $R(X)$ be the largest of the least radii of closed balls

containing the individual supports, with $R(\varnothing) = 0$. These radii are attained by compactness of the support and of the set of candidate centres within any fixed radius of a support point. We write $|X| = n$, $\operatorname{supp} X = \bigcup_i \operatorname{supp} f_i$, and $\|X\|_\sharp = \prod_i \|f_i\|_\sharp$. For $J \subseteq \{1, \dots, n\}$ the tuple X_J retains the entries indexed by J in order. The tuples with smooth entries form $\mathfrak{S}_n^\infty(O)$.

An isometry r acts on tests by $rf = f \circ r^{-1}$ and entrywise on tuples. The notation $X \cdot Y$ denotes concatenation when the combined tuple is separated. We write $\mathbb{C}\mathfrak{S}(O)$ for the free complex vector space with basis $\mathfrak{S}(O)$. Actions on tuples extend linearly, and concatenation extends bilinearly whenever every pair of basis tuples with nonzero coefficients can be concatenated. Conjugation $F \mapsto \bar{F}$ conjugates the coefficients of a formal sum. A tuple of complex tests is evaluated by expansion in real and imaginary parts. This expansion and the usual linear relations in each entry hold after applying a multilinear functional. ■

A map on $\mathfrak{S}_n(O)$ is called multilinear if its restriction to $\prod_i C_c^1(O_i)$ is multilinear whenever the open sets $O_i \subset O$ have pairwise disjoint closures. We use

$$\|f\|_\sharp = \max(\|f\|_\infty, 40M \operatorname{Lip} f) \tag{C.i}$$

on both spaces. With $\|f\|_{C^1} = \max(\|f\|_\infty, \|\nabla f\|_\infty)$, integration along a shortest line segment, lifted on the torus, gives $\|f\|_\sharp \le 40M\|f\|_{C^1}$.

Fix $n_0 \ge 1$ and finite constants $C_n(d; R) \ge 0$ for $n \ge n_0$ and $d, R > 0$. A family $\Phi = (\Phi_n)_{n \ge n_0}$ obeys these bounds if

$$|\Phi_n(f_1, \dots, f_n)| \le C_n(d; R) \prod_i \|f_i\|_\sharp \tag{C.ii}$$

on every tuple of type $(d; R)$. We impose no bound as d tends to zero.

Proposition C.2 (Simultaneous subsequence extraction)**.** Let $O_j \subset \mathbb{E}$ be open sets such that every compact subset of $\mathbb{E}$ is contained in O_j for all sufficiently large j. Suppose that the real multilinear families $\Phi^{(j)}$ on $\mathfrak{S}(O_j)$ satisfy eq. C.ii for $n \ge n_0$, with the same constants, once $j \ge j_0(n, R)$. Then there is a subsequence along which $\Phi_n^{(j)}(X)$ converges for every $n \ge n_0$ and every $X \in \mathfrak{S}_n(\mathbb{E})$. The subsequence can also be chosen so that any prescribed countable collection of bounded real sequences converges along it. The limiting family is multilinear and satisfies the same bounds. Symmetry is preserved. ■

Proof. A countable family with controlled supports. Let $\mathfrak{O}$ consist of finite unions of rational open boxes, using their projections on the torus. These sets have compact closures and form a countable basis under finite unions. For each $W \in \mathfrak{O}$ choose a countable C^1-dense subset $\mathfrak{f}(W) \subset C_c^1(W)$.

Here is the separability argument with the support fixed. For $f \in C_c^1(W)$ consider $g_f = (f, \partial_1 f, \partial_2 f, \partial_3 f)$ on $\overline{W}$. The C^1 distance of two functions is the uniform distance of these maps for the norm $|(v_0, v')|_* = \max(|v_0|, |v'|)$. For each integer $p \ge 1$ choose a finite $1/p$-net in $\overline{W}$. Classify the functions by a common modulus bound $|g_f(x) - g_f(y)|_* \le 1/q$ for $|x - y| \le 1/p$, and by rational approximations within $1/q$ to their values at the points of the net. There are countably many classes. Each has diameter at most $4/q$, by comparison through a nearest net point. Uniform continuity of g_f on $\overline{W}$ places every f in such a class for each q, with p sufficiently large. Choosing one member of each nonempty class therefore gives the required dense subset. For $W = \varnothing$ take $\mathfrak{f}(W) = \{0\}$.

Take all tuples of all lengths with entries in $\mathfrak{f}(W_i)$, where the W_i have pairwise disjoint closures. This is a countable family; enumerate it by $(X^k)_{k \ge 1}$. Given a tuple $X = (f_1, \dots, f_n)$ of type $(d; R)$, compactness of each support permits a choice of such sets with

$$\operatorname{supp} f_i \subset W_i \subset \{x : \operatorname{dist}(x, \operatorname{supp} f_i) < d/3\}. \tag{C.iii}$$

Their mutual distances are at least $d/3$, and each lies in a ball of radius $R + d/3$. The union of their closures is compact. For a zero entry choose $W_i = \varnothing$.

Uniform approximation of evaluations. If f_i, h_i have supports in W_i, multilinearity gives

$$\Phi_n(f_1,\dots,f_n)-\Phi_n(h_1,\dots,h_n)=\sum_{i=1}^{n}\Phi_n(h_1,\dots,h_{i-1},f_i-h_i,f_{i+1},\dots,f_n).$$

Every tuple on the right retains separation $d/3$. Choose $h_i\in\mathfrak{f}(W_i)$ with $\|f_i-h_i\|_{C^1}\le\epsilon\le 1$. For every family obeying eq. C.ii at $(n,R+d/3)$, the preceding identity gives

$$|\Phi_n(X)-\Phi_n(h_1,\dots,h_n)|\le C_n(d/3;R+d/3)\,n(40M)^n\epsilon(1+\max_i\|f_i\|_{C^1})^{n-1}. \tag{C.iv}$$

The constant does not depend on the approximating family or its domain, provided the latter contains all W_i.

Extraction and extension. For each X^k of length at least n_0, its evaluation by $\Phi^{(j)}$ is defined and bounded for all sufficiently large j. Assign arbitrary values to the finitely many undefined entries. Apply the diagonal subsequence argument to these countably many bounded real sequences and to the additional sequences in the statement. Each selected evaluation then converges. For a fixed arbitrary X, use eq. C.iv to approximate it by one X^k, with an error uniform in all sufficiently large j. The triangle inequality shows that the evaluations at X are Cauchy. This proves convergence on every separated tuple along the same subsequence.

Each multilinearity or symmetry identity involves only finitely many evaluations, all eventually defined, so it passes to the limit. The same is true of eq. C.ii for every fixed tuple and type. □

Proposition C.3. The set of symmetric multilinear families on $\mathfrak{S}(\mathbb{E})$ satisfying eq. C.ii is compact and metrizable for pointwise convergence on separated tuples. ■

Proof. Use the determining family X^k in the preceding proof and let n_k be the length of X^k. A metric is

$$d(\Phi,\Psi)=\sum_{k:n_k\ge n_0}2^{-k}\min\{1,|\Phi_{n_k}(X^k)-\Psi_{n_k}(X^k)|\}. \tag{C.v}$$

The triangle inequality follows term by term. If this expression vanishes, eq. C.iv makes every evaluation of the two families equal, so it is a metric. That same estimate shows that every evaluation is continuous for the metric: approximate its tuple by one X^k, uniformly over all the families. Conversely, the tail of the series is uniformly bounded by $\sum_{k>N}2^{-k}$, and its first N terms are continuous for pointwise convergence. The two topologies thus agree. By Prop. C.2, every sequence has a convergent subsequence whose limit still belongs to the set. Sequential compactness in this metric proves compactness. □

Definition C.4 (Functionals and cumulants)**.** In (1), for a finite set J, let $\mathrm{Part}(J)$ be its set of partitions and $\mathrm{Part}_{\ge 2}(J)$ the partitions with all blocks of size at least two. For $J=\{1,\dots,n\}$ write $\mathrm{Part}(n)$ and $\mathrm{Part}_{\ge 2}(n)$. The empty set has one partition, with no blocks. In (2), multilinearity is on products of test spaces with separated closures and symmetry means permutation invariance. In (3), a functional σ is real and multilinear on separated tuples, with $\sigma(\varnothing)=1$. For $n\ge 1$, its connected functions are

$$S_n^{\sigma}(X)=\sum_{\pi\in\mathrm{Part}(n)}(-1)^{|\pi|-1}(|\pi|-1)!\prod_{J\in\pi}\sigma(X_J). \tag{C.vi}$$

In (4), for a family $S=(S_n)_{n\ge 2}$, the centred functional σ_S is defined by eq. 1.i. For a functional σ, we write $\sigma^{\mathrm{c}}=\sigma_{(S_n^{\sigma})_{n\ge 2}}$. The notation $\mathsf{Fct}(\underline{C},\bar{c})$ denotes symmetric functionals whose one-point function is $c\int f$ with $|c|\le\bar{c}$ and whose connected functions obey the table $\underline{C}=(C_n(d;R))_{n\ge 2}$ of eq. C.ii. The corresponding symmetric connected families form $\mathcal{K}_{\mathbb{E}}(\underline{C})$. ■

Lemma C.5 (Moment inversion and centring)**.** Every symmetric functional obeys

$$\sigma(X)=\sum_{A\subseteq\{1,\dots,n\}}\prod_{i\in A}\sigma((f_i))\,\sigma^{\mathrm{c}}(X_{A^{\mathrm{c}}}). \tag{C.vii}$$

In particular, if $\sigma((f)) = c\int f$, the map sh_c replaces the entry (f) by $(f)+c(\int f)\varnothing$ in each position and $\sigma = \sigma^{\mathrm{c}} \circ \mathrm{sh}_c$. It is invertible with inverse sh_{-c}. The shift commutes with isometries and with concatenation on separated arguments. ■

Proof. In the commutative algebra on variables $z_1, \dots, z_n$ subject to $z_i^2 = 0$, let $z_J = \prod_{i\in J} z_i$ and $M = \sum_J \sigma(X_J) z_J$. The coefficient of z_J in $\log M$ is the cumulant of the subtuple X_J, since an unordered partition with k blocks appears $k!$ times in $(M-1)^k$. Exponentiating gives the partition expansion for moments. Grouping the partitions by their singleton blocks gives eq. C.vii. The shift identities follow by expanding the product of the replacements. Isometries preserve $\int f$. □

Definition C.6 (Mirrors and reflection positivity)**.** In (1), a mirror in $\mathbb{R}^3$ (u, c), $|u| = 1$, has reflection $\theta x = x - 2(u \cdot x - c)u$ and open side $\{x : u \cdot x > c\}$. Its opposite mirror is $(-u, -c)$. The normals of coordinate and diagonal mirrors are $\pm e_\mu$ and $(\pm e_\mu \pm e_\nu)/\sqrt{2}$, $\mu \neq \nu$, respectively. In (2), a mirror on T_ℓ (μ, c) is the projection of the two parallel planes with $x_\mu = c$ and $x_\mu = c + \ell/2$. Its reflection is induced by the coordinate reflection about $x_\mu = c$ and its open side is the projection of $c < x_\mu < c + \ell/2$. The opposite mirror is $(\mu, c + \ell/2)$. All torus formulas descend since a change of lift shifts coordinates by periods.

In (3), a functional is reflection positive at the mirror if for all $F \in \mathbb{C}\mathfrak{S}(\mathbb{E}_+)$ on its open side,

$$\sigma(\theta\bar{F} \cdot F) \geq 0. \tag{C.viii}$$

The inequality means a real nonnegative number. Each concatenation is separated since the reflection maps the open side to the opposite open side. ■

Proposition C.7 (Transport of reflection positivity)**.** For a real symmetric functional, reflection positivity passes to (a) the opposite side, (b) every translate of the mirror by a translation preserving the functional, and (c) every image under a linear isometry preserving the functional. On the torus the linear isometry must preserve the period lattice. Consequently, (d) invariance under translations and W_3 gives positivity at every mirror of the same kind as a positive mirror. Full Euclidean invariance and positivity at one mirror give positivity at every plane. ■

Proof. In the matrix of the reflected form on the opposite side substitute $X_i = \theta Y_i$ with Y_i on the original side. The entries become $\sigma(Y_i \cdot \theta Y_j) = \sigma(\theta Y_j \cdot Y_i)$ by symmetry. The resulting matrix is the transpose of the original positive semidefinite matrix and is positive semidefinite. For an isometry r carrying one mirror to another, substitute $X_i = rY_i$ and use $\theta_{r\Sigma} r = r\theta_\Sigma$. The transports in (b) and (c) follow by invariance. The cubic group acts transitively on the coordinate normals and on the diagonal normals, and O(3) acts transitively on all unit normals. This proves the last assertions. □

C.2. **Distributional extensions and growth.**

Remark C.8 (Admissible geometric constants). The estimates on separated tests and their flat Schwartz extension below hold for $M \geq 1$. In particular, they hold for every odd $L \geq 13$ and $M = L^{m'}$ with $m' \geq 1$.

Definition C.9 (Regularity on separated tests)**.** Write $E^n_{\neq}$ for the configurations of n distinct points in E. A functional is regular of order at most k on compact subsets of this set if its values on separated smooth tensors have a distributional extension there of order at most k on each compact set. Multilinearity in this definition is multilinearity on products of spaces of tests with mutually separated supports as above. ■

The bounds on separated tests define distributions away from coincident points. Their power dependence on the separation also controls Schwartz functions flat on the diagonals. We prove both facts by one local Fourier construction. The dyadic partition used below keeps the supports of the individual Fourier factors separated; flatness then makes its fine-scale sum converge.

Write $E^n_{\neq} = \{(x_1, \dots, x_n) : x_i \neq x_j \ (i \neq j)\}$, where $E = \mathbb{R}^3$ or T_ℓ, and set $m(x) = \min_{i \neq j} |x_i - x_j|_\infty$, with $m = \infty$ for $n = 1$. We complexify multilinear forms by complex linearity in each entry. Thus a bound by $c \prod_i \|f_i\|_\sharp$ on real entries gives the bound $c \prod_i (\|\operatorname{Re} f_i\|_\sharp + \|\operatorname{Im} f_i\|_\sharp)$ on complex entries.

C.2.1. *A partition adapted to the diagonals.* Put $v(t) = e^{-1/t}$ for $t > 0$ and $v(t) = 0$ for $t \leq 0$, $\vartheta(t) = v(1 - 16t^2)$, and $c = \int \vartheta$. The two cutoffs

$$\chi(t) = c^{-1} \int_{t-1/2}^{t+1/2} \vartheta(s)\, ds, \qquad \widetilde{\chi}(t) = c^{-1} \int_{t-1}^{t+1} \vartheta(s)\, ds \tag{C.ix}$$

have values in $[0, 1]$. The support of χ is $[-3/4, 3/4]$, $\sum_{z \in \mathbb{Z}} \chi(t - z) = 1$, and at most two summands are nonzero. The second cutoff equals one on $[-3/4, 3/4]$, vanishes outside $[-5/4, 5/4]$, and satisfies $|\widetilde{\chi}'| \leq 16$. We shall use

$$|v^{(k)}| \leq 8^k (k!)^2, \qquad |\vartheta^{(k)}| \leq 200^k (k!)^2, \qquad |\chi^{(k)}| \leq 200^k (k!)^2. \tag{C.x}$$

Here is the derivative estimate. On $t > 0$, $v^{(k)}(t) = P_k(1/t) e^{-1/t}$, where $P_{k+1}(s) = s^2 (P_k(s) - P_k'(s))$ and $P_0 = 1$. The weighted coefficient sum $A_k = \sum_j |[s^j] P_k| j!$ satisfies $A_{k+1} \leq 2(2k+1)^2 A_k \leq 8(k+1)^2 A_k$; also $s^j e^{-s} \leq j!$. All derivatives tend to zero at $t = 0$, so this proves the first bound globally. For the second, the chain rule gives

$$\vartheta^{(k)}(t) = \sum_{0 \leq 2j \leq k} \frac{k!}{j!(k-2j)!} (-32t)^{k-2j} (-16)^j v^{(k-j)}(1 - 16t^2).$$

On $|t| < 1/4$, use $|32t| \leq 8$, $k!/(j!(k-2j)!) \leq 3^k j!$, and $j!((k-j)!)^2 \leq (k!)^2$. The resulting sum is bounded by $(32/31) 192^k (k!)^2 \leq 200^k (k!)^2$ for $k \geq 1$. Outside this interval it vanishes. Finally $c \geq \frac{1}{4} e^{-4/3} > 1/16$, and differentiating eq. C.ix proves the last bound and the stated properties of the cutoffs.

Let $h_q = 2^{-q}/16$, $q \geq 0$. On $\mathbb{R}^3$ use the grid $h_q \mathbb{Z}^3$; on T_ℓ use its quotient and periodize the cutoffs. Here $\ell = 2L^m$, $m \geq 0$, so ℓ / h_q is an integer and the cutoff supports embed in the torus. Define

$$\chi_{q,z}(y) = \prod_{\mu=1}^{3} \chi(y_\mu / h_q - z_\mu), \qquad \widetilde{\chi}_{q,z}(y) = \prod_{\mu=1}^{3} \widetilde{\chi}(y_\mu / h_q - z_\mu),$$

with the periodized interpretation on the torus. Let $Q_{q,z}$ and $\widetilde{Q}_{q,z}$ be the closed cubes of sup radii $3h_q/4$ and $5h_q/4$ about $h_q z$. An index tuple $\mathbf{z} = (z_1, \dots, z_n)$ is good when $D(\mathbf{z}) := \min_{i \neq j} |z_i - z_j|_\infty \geq 6$. Write $\chi_{q,\mathbf{z}} = \bigotimes_i \chi_{q,z_i}$ and

$$G_q = \sum_{\mathbf{z} \text{ good}} \chi_{q,\mathbf{z}}, \qquad G_{-1} = 0, \qquad \Upsilon_q = G_q (1 - G_{q-1}). \tag{C.xi}$$

The enlarged cubes in a good tuple have mutual distance at least $(D(\mathbf{z}) - 5/2) h_q \geq 7h_q/2$ and each lies in a ball of radius less than one. The grid identities give

$$G_q = 0 \quad (m \leq 9h_q/2), \qquad G_q = 1 \quad (m \geq 15 h_q/2), \tag{C.xii}$$

$$G_q G_{q-1} = G_{q-1}, \qquad \Upsilon_q = G_q - G_{q-1} \geq 0, \qquad \sum_{q \geq 0} \Upsilon_q = 1 \quad \text{on } E^n_{\neq}. \tag{C.xiii}$$

Indeed, a nonzero grid factor has $|x_i - h_q z_i|_\infty < 3h_q/4$, so the distance of two centres differs from that of their points by less than $3h_q/2$. This proves the two plateaux. If $G_{q-1} \neq 0$, then $m > 9h_q$, which proves the product identity. In particular, for $q \geq 1$,

$$\operatorname{supp} \Upsilon_q \subset \{9h_q/2 \leq m \leq 15 h_q\}.$$

The partition is locally finite on $E^n_{\neq}$.

For a pure derivative in any coordinate, the product rule and eq. C.x yield

$$|\partial_{i\mu}^k(\chi_{q,\mathbf{z}}(1-G_{q-1})F)| \le 2\cdot 600^k (k!)^2 \max_{0\le j\le k} h_q^{j-k}|\partial_{i\mu}^j F|. \tag{C.xiv}$$

For the derivatives of G_q, only two terms in the differentiated coordinate can contribute; the sums in every other coordinate are one. Thus $|\partial_{i\mu}^k G_q| \le 2h_q^{-k}200^k(k!)^2$. For mixed derivatives of a sum of any subfamily of the grid tensors, the corresponding bound is $2^{3n}h_q^{-|\alpha|}200^{|\alpha|}(|\alpha|!)^2$.

One count will supply summability over positions. For $q \ge 1$ and each unit box $\mathbf{w}+[0,1)^{3n}$, there are at most

$$\binom{n}{2} 33^3 h_q^{-(3n-3)} \tag{C.xv}$$

good $\mathbf{z}$ with $h_q\mathbf{z}$ in that box and $Q_{q,\mathbf{z}}$ meeting $\operatorname{supp}\Upsilon_q$. At an intersection point some pair has distance at most $15h_q$; the two centres therefore differ by at most $16h_q$ in each coordinate. Choose the pair, the other $n-1$ centres, and one of 33^3 differences. At $q=0$ the unrestricted count is 16^{3n}.

C.2.2. *The local Fourier construction.*

Proposition C.10 (Kernel on separated points)**.** Let T be a real multilinear form on separated smooth compactly supported n-tuples over E. Suppose that for every $d>0$ there is $c_T(d)<\infty$ such that

$$|T(f_1,\dots,f_n)| \le c_T(d)\prod_i \|f_i\|_\sharp \tag{C.xvi}$$

whenever the supports have mutual distance at least d and each lies in a ball of radius one. There is a unique distribution $\widetilde{T}$ on $E^n_{\neq}$ such that $\widetilde{T}(\bigotimes_i f_i) = T(f_1,\dots,f_n)$. It has order at most $4n+1$ on compact sets, with a bound depending only on that compact set, M, and the function c_T. Symmetries of T under permutations or isometries are inherited by $\widetilde{T}$. ∎

Proof. Fix a good box at scale q and put $h=h_q$, $\lambda = 5h/2$. Lift its enlarged cubes to cubes $C_i\subset\mathbb{R}^3$ of side λ. For $\omega\in\mathbb{Z}^3$ let $e_\omega(t)=e^{2\pi i\omega\cdot t/\lambda}$ and let $E_{i,\omega}$ be $\widetilde{\chi}_{q,z_i}e_\omega$, interpreted through the lift and periodized if necessary. For a smooth F supported in $Q_{q,\mathbf{z}}$, extend its lift by zero and set

$$\widehat{F}(\omega) = \lambda^{-3n}\int_{\prod_i C_i} F(t)\prod_i \overline{e_{\omega_i}(t_i)}\,dt, \qquad [F]_{q,k} = \max\left\{\|F\|_\infty, (\lambda/2\pi)^k \max_{i,\mu}\|\partial_{i\mu}^k F\|_\infty\right\}.$$

Integration by parts in a coordinate attaining $|\omega|_\infty$ gives

$$|\widehat{F}(\omega)| \le 2^k(1+|\omega|_\infty)^{-k}[F]_{q,k}.$$

The real and imaginary parts of $E_{i,\omega}$ have sharp norm at most $1109Mh^{-1}(1+|\omega|_\infty)$: the cutoff gradient is at most $16\sqrt{3}/h$, and differentiating the exponential adds at most $16\sqrt{3}|\omega|_\infty/h$. Also, grouping by $|\omega|_\infty$ gives $\sum_{\omega\in\mathbb{Z}^{3n}}(1+|\omega|_\infty)^{-3n-1} \le 3n8^n$. With $k=4n+1$ these bounds prove absolute convergence of

$$T_{q,\mathbf{z}}(F) = \sum_{\omega\in(\mathbb{Z}^3)^n} \widehat{F}(\omega)T(E_{1,\omega_1},\dots,E_{n,\omega_n}) \tag{C.xvii}$$

and the estimate

$$|T_{q,\mathbf{z}}(F)| \le (2\cdot 10^6)^n M^n c_\square h^{-n}[F]_{q,4n+1}. \tag{C.xviii}$$

Here $c_\square$ can be either $c_T(7h/2)$ or $c_T((D(\mathbf{z})-5/2)h)$, whenever the argument is finite; for $n=1$ any valid uniform one-entry bound serves. The numerical factor before $M^n c_\square h^{-n}[F]$ is $6n\,283904^n \le (2\cdot 10^6)^n$. Changing a lift multiplies the entries by phases and the Fourier coefficient by their inverse product; hence eq. C.xvii is independent of the lifts.

The Fourier partial sums, multiplied by $\bigotimes_i \widetilde{\chi}_{q,z_i}$, converge to F with all derivatives and have support in $\widetilde{Q}_{q,\mathbf{z}}$. For completeness, integration by parts makes the Fourier series of every derivative

absolutely summable. Its sum is that derivative: the product Fejér kernels are nonnegative approximate identities, so their convolution converges uniformly to a continuous periodic function and also to the sum of an absolutely convergent Fourier series. The cutoff equals one on the support of F, which proves the claimed convergence after multiplication. If $F = \bigotimes_i f_i$ with $\operatorname{supp} f_i \subset Q_{q,z_i}$, its coefficients factor. The partial sums therefore give T of the individual Fourier approximations; their convergence in the sharp norm and the telescoping estimate prove $T_{q,\mathbf{z}}(F) = T(f_1, \dots, f_n)$.

Define

$$\widetilde{T}(F) = \sum_{q \geq 0} \sum_{\mathbf{z} \text{ good}} T_{q,\mathbf{z}}(\chi_{q,\mathbf{z}}(1 - G_{q-1})F). \tag{C.xix}$$

For F in a fixed compact subset of $E^n_{\neq}$, only finitely many scales and positions occur. Combining eq. C.xiv with eq. C.xviii gives the asserted distribution bound of order $4n + 1$. For a tensor $F = \bigotimes_i f_i$, each piece in eq. C.xix is the finite sum

$$\bigotimes_i (\chi_{q,z_i} f_i) - \sum_{\mathbf{z}' \text{ good at } q-1} \bigotimes_i (\chi_{q,z_i} \chi_{q-1,z'_i} f_i),$$

where the second sum is absent at $q = 0$. All entries stay inside the original separated supports. Multilinearity and eq. C.xiii therefore give $\widetilde{T}(F) = T(f_1, \dots, f_n)$. In making this computation, the linear form induced on finite sums of tensors is well defined: on the finite-dimensional spans of their factors, point evaluations span the duals, so a tensor representing the zero function has all its coefficients zero.

Finally, the same partition and the local Fourier approximation approximate every test in $C^\infty_c(E^n_{\neq})$ by finite sums of separated tensors, in all derivatives with supports in one compact subset of $E^n_{\neq}$. This proves uniqueness. Applying uniqueness to the pullback by a symmetry proves the last assertion. □

If the form is originally defined on separated C^1_c tuples with the same bounds, these values are determined by its kernel. Indeed, cut each entry by the $q = 0$ partition, mollify the finitely many pieces, and let the mollifier radius tend to zero. Their supports retain positive separation, their sharp norms converge, and the telescoping bound recovers the original value.

C.2.3. *Flat tests and the global bound.* For $\varphi \in \mathcal{S}(\mathbb{R}^{3n})$ put

$$|\varphi|_k = \sup_{x,\, |\alpha| \leq k} (1 + |x|^2)^{k/2} |\partial^\alpha \varphi(x)|. \tag{C.xx}$$

Definition C.11 (Flat Schwartz test spaces)**.** Let $\mathcal{S}_{\neq}(\mathbb{R}^{3n})$ consist of Schwartz functions all of whose derivatives vanish whenever two entries coincide. Let $\mathcal{S}_+$ be the subspace additionally flat outside $V^{(n)}_+ = \{x \in (\mathbb{R}^3)^n_{\neq} : x_{i3} > 0 \text{ for all } i\}$. In degree zero both spaces are $\mathbb{C}$. ■

Lemma C.12. Finite sums of tensors of separated smooth compactly supported tests are dense in $\mathcal{S}_{\neq}$ for every norm $|\cdot|_k$. In $\mathcal{S}_+$ the approximating entries may all be supported in the open positive half-space. For $k \geq 4$ and $A \subset \{1, \dots, n\}$, partial integration $\psi_A(y) = \int \varphi(y, x_A)\, dx_A$ maps $\mathcal{S}_{\neq}(\mathbb{R}^{3n})$ into $\mathcal{S}_{\neq}(\mathbb{R}^{3(n-|A|)})$ and satisfies

$$|\psi_A|_{k(n-|A|)} \leq C_{\int}(k)^{|A|} |\varphi|_{kn}, \qquad C_{\int}(k) = \int_{\mathbb{R}^3} (1 + |x|^2)^{-k/2}\, dx. \tag{C.xxi}$$

In particular $C_{\int}(4) = \pi^2$ and $C_{\int}(10) = 5\pi^2/64$. Also $|\varphi \otimes \psi|_k \leq |\varphi|_k |\psi|_k$. If both factors belong to $\mathcal{S}_{\neq}$ and are flat outside the negative and positive half-spaces respectively, their tensor is flat on every diagonal. ■

Proof. Flatness gives the following Taylor estimate. If φ is flat at x^*, $|x - x^*| \leq r$ with $r \geq 1$, and $N \geq |\alpha| + s$, then

$$|\partial^\alpha \varphi(x)| \leq \frac{|x - x^*|_1^s}{s!} (1 + 2r^2)^{N/2} |\varphi|_N (1 + |x|^2)^{-N/2}. \tag{C.xxii}$$

Apply Taylor's formula with integral remainder to $t \mapsto \partial^\alpha \varphi(x^* + t(x - x^*))$; along this segment $1+|x|^2 \le (1+2r^2)(1+|x^*+t(x-x^*)|^2)$. When $m(x)$ is small, replace a closest pair by its midpoint. This gives a diagonal point x^* with $|x-x^*|_1 \le 3m(x)$ and $|x-x^*| \le \sqrt{3/2}m(x)$. In the half-space case one may instead replace a smallest third coordinate by its minimum with zero.

For the half-space partition restrict the good indices further by $z_{i3} \ge 6$, and denote the resulting sum by G_q^+. The plateau and derivative estimates above hold with $m^+(x) = \min\{m(x), x_{13}, \dots, x_{n3}\}$. Taylor's estimate with $s = k+1$, followed by the mixed derivative bound for the cutoffs, gives

$$|(1 - G_Q)\varphi|_k \le C(n,k) h_Q |\varphi|_{2k+1},$$

and the same bound with G_Q^+ for $\varphi \in \mathcal{S}_+$. Indeed, the derivatives of the cutoff cost at most h_Q^{-k}, whereas Taylor's remainder contributes h_Q^{k+1}. Fix Q to make this error small. Among the remaining finitely many scales, discard grid centres outside a large box. Their derivatives are bounded with Q fixed, while one extra Schwartz weight makes their $|\cdot|_k$ contribution tend to zero. Only finitely many boxes remain. Approximate each by its local Fourier tensors as in the preceding proof. Their enlarged supports are separated, and in the half-space case have third coordinate at least $19h_q/4$. This proves both density assertions.

For partial integrals, use

$$(1+|x|^2)^{kn/2} \ge (1+|y|^2)^{k(n-|A|)/2} \prod_{i \in A} (1+|x_i|^2)^{k/2}.$$

This permits differentiation under the integral, proves eq. C.xxi, and preserves flatness on the remaining diagonals. Polar coordinates give the two stated constants. The tensor bound follows from $1+|x|^2+|y|^2 \le (1+|x|^2)(1+|y|^2)$. At a diagonal joining the negative and positive groups, one factor and all its derivatives vanish; diagonals inside a group are already covered by its flatness. □

Proposition C.13. Suppose that T satisfies eq. C.xvi on $\mathbb{R}^3$ with $a \in \mathbb{N}_0$ and the bound

$$|T(f_1, \dots, f_n)| \le c_T \max\{1, d^{-a}\} \prod_i \|f_i\|_\sharp. \tag{C.xxiii}$$

Its kernel has a unique extension to $\mathcal{S}_{\ne}$ bounded for a Schwartz norm. Use the constant

$$C_{\ne} = 10^{53}. \tag{C.xxiv}$$

Then

$$|\widetilde{T}(\varphi)| \le C_{\ne}^{n+a} M^n (n!)^8 c_T |\varphi|_{8n+a-1}. \tag{C.xxv}$$

■

Proof. Use the series eq. C.xix, now summed over all scales and positions. Set $k = 4n+1$, $s = a + 4n - 2$, and $N = k + s$. For $F_{q,\mathbf{z}} = \chi_{q,\mathbf{z}}(1 - G_{q-1})\varphi$, eq. C.xiv and $\lambda/(2\pi) < 2h_q/5$ give

$$[F_{q,\mathbf{z}}]_{q,k} \le 2 \cdot 240^k (k!)^2 \sup_{\operatorname{supp} F_{q,\mathbf{z}}} \max_{i,\mu,\, 0 \le j \le k} h_q^j |\partial_{i\mu}^j \varphi|. \tag{C.xxvi}$$

For $q \ge 1$, this support has $m \le 15h_q$. The midpoint choice in eq. C.xxii has $|x - x^*|_1 \le 45h_q$ and $|x - x^*| < 1$. Since $N \ge 4n$, it follows that

$$|\partial_{i\mu}^j \varphi(x)| \le \frac{(45h_q)^s}{s!} 3^{N/2} |\varphi|_N \prod_{l=1}^n (1+|x_l|^2)^{-2}, \qquad j \le k.$$

To sum the positions, assign $h_q\mathbf{z}$ to its unit box $\mathbf{w} + [0,1)^{3n}$. Its support lies in $\mathbf{w} + [-1/20, 21/20]^{3n}$, and

$$\sum_{\mathbf{w}} \sup_{x \in \mathbf{w} + [-1/20, 21/20]^{3n}} \prod_l (1+|x_l|^2)^{-2} \le (81\pi^2)^n.$$

For one factor, compare a point in the enlarged unit box with a point in its unit cube through the common centre: the squared weights differ by at most 81. Integrate over that cube and sum; then

take the product in the n entries. Combining this estimate, eq. C.xv, eq. C.xviii, and eq. C.xxvi bounds the sum at scale $q \geq 1$ by

$$\Sigma_q \leq (2 \cdot 10^6)^n M^n c_T (2/7)^a 2 \cdot 240^k (k!)^2 \frac{45^s}{s!} 3^{N/2} \binom{n}{2} 33^3 (81\pi^2)^n h_q^{s-a-n-(3n-3)} |\varphi|_N. \tag{C.xxvii}$$

The exponent of h_q is one. Thus the fine-scale sum converges.

At $q = 0$, split the centres into $D(\mathbf{z}) \geq 19$ and $6 \leq D(\mathbf{z}) \leq 18$. In the first group the enlarged supports are separated by more than one, so $c_\square = c_T$; use the Schwartz bound without Taylor's formula and the count 16^{3n}. In the second group a pair in the box has distance at most $39/32$ in the sup norm. Its midpoint has $|x - x^*|_1 \leq 117/32$ and $|x - x^*| \leq 3/2$; hence eq. C.xxii supplies the factor $(117/32)^s (11/2)^{N/2}/s!$, and $c_\square = c_T (32/7)^a$. These give, respectively,

$$\Sigma_0^{\mathrm{A}} \leq (2 \cdot 10^6)^n M^n c_T 16^{4n} 2 \cdot 240^k (k!)^2 (81\pi^2)^n |\varphi|_N,$$

$$\Sigma_0^{\mathrm{B}} \leq (2 \cdot 10^6)^n M^n c_T 16^{4n} 2 \cdot 240^k (k!)^2 (81\pi^2)^n (32/7)^a \frac{(117/32)^s}{s!} (11/2)^{N/2} |\varphi|_N.$$

For $n = 1$ the second group and every $q \geq 1$ contribution are empty. In general use $(k!)^2 \leq 25^n 4^{8n} (n!)^8$, $(2/7)^a 3^{N/2} \leq 81^n$, and $45^s/s! \leq 3^{45}$ in the fine-scale bound, with $\sum_{q \geq 1} h_q = 1/16$. For the second coarse group use $(32/7)^a (11/2)^{a/2} \leq 11^a \leq 11^s$ and $(11 \cdot 117/32)^s/s! \leq 3^{41}$. The three sums, divided by $M^n (n!)^8 c_T |\varphi|_N$, are at most $10^{28}(10^{28})^n$, $10^3 (10^{30})^n$, and $10^{23}(10^{33})^n$. For $n \geq 2$ their sum is at most $3(10^{45})^n$; for $n = 1$ the sole coarse sum is at most 10^{33}. Both are bounded by $C_{\neq}^n$.

The series is therefore absolutely convergent, linear, and satisfies the stronger bound with $C_{\neq}^n$ in place of $C_{\neq}^{n+a}$. On compactly supported tests away from the diagonals it is the previous finite sum. Uniqueness follows from the density in Lem. C.12. □

Proposition C.14 (Growth of the Schwinger functions)**.** Suppose a symmetric family $S = (S_n)_{n \geq 2}$ on separated C_c^1 tuples satisfies

$$|S_n(f_1, \dots, f_n)| \leq A \max\{1, d^{-a}\} B^n (n!)^{\varkappa} \prod_i \|f_i\|_\sharp$$

when the separation is at least d and each support lies in a ball of radius one, where $a \in \mathbb{N}$, $A, B \geq 1$, and $\varkappa \geq 0$. Its centred moments σ_S extend uniquely to functionals $\mathsf{S}_n^{\sigma_S}$ on $\mathcal{S}_{\neq}$, with $\mathsf{S}_0^{\sigma_S} = 1$, $\mathsf{S}_1^{\sigma_S} = 0$, and

$$|\mathsf{S}_n^{\sigma_S}(\varphi)| \leq C_c^n (n!)^{\varkappa+9} |\varphi|_{kn}, \qquad k = 8 + \lceil a/2 \rceil, \qquad C_c = C_{\neq}^{1+a/2} M B \sqrt{A}. \tag{C.xxviii}$$

If a functional σ has these connected functions and one-point function $c \int f$, with $|c| \leq \bar{c}$, the same conclusion holds with C_c replaced by $C = C_c + \bar{c} C_{\int}(k)$. ■

Proof. In the moment partition sum, there are at most $\lfloor n/2 \rfloor$ blocks and no singleton blocks. Also $\prod_{J \in \pi} |J|! \leq n!$, and the number of partitions is at most $n!$: each partition is the cycle partition of a permutation, choosing increasing cycles. Thus

$$|\sigma_S(f_1, \dots, f_n)| \leq A^{\lfloor n/2 \rfloor} \max\{1, d^{-a\lfloor n/2 \rfloor}\} B^n (n!)^{\varkappa+1} \prod_i \|f_i\|_\sharp.$$

Apply Prop. C.13 with singularity order $a\lfloor n/2 \rfloor$. The derivative order is $8n + a\lfloor n/2 \rfloor - 1 \leq kn$, and its bound is eq. C.xxviii. For $n = 1$ the functional is zero. For an uncentred functional use the finite expansion

$$\mathsf{S}_n^{\sigma}(\varphi) = \sum_{A_0 \subset \{1, \dots, n\}} c^{|A_0|} \mathsf{S}_{n-|A_0|}^{\sigma_S}(\psi_{A_0}). \tag{C.xxix}$$

It agrees with the original moments on separated tensors. By eq. C.xxi, each integrated entry costs at most $C_{\int}(k)$, so the binomial formula gives C^n. Uniqueness again follows from density. □

Proposition C.15. Suppose the moments of a real symmetric functional σ extend to continuous linear functionals S_n on $\mathcal{S}_{\neq}(\mathbb{R}^{3n})$, agreeing with σ on separated smooth tensors and satisfying

$$|\mathsf{S}_n(\varphi)| \le c_E C_E^n (n!)^{\kappa} |\varphi|_{k_E n}$$

for fixed $k_E \in \mathbb{N}$ and $c_E, C_E, \kappa \ge 0$. In (a) these extensions preserve reality and symmetry. In (b), every Euclidean isometry under which σ is invariant also leaves them invariant. In (c), if σ is reflection positive, then for $\varphi_n \in \mathcal{S}_+(\mathbb{R}^{3n})$, $0 \le n \le n_0$, with $\mathcal{S}_+(\mathbb{R}^0) = \mathbb{C}$,

$$\sum_{n,n'=0}^{n_0} \mathsf{S}_{n+n'}(\Theta \bar{\varphi}_n \otimes \varphi_{n'}) \ge 0. \tag{C.xxx}$$

Here Θ reflects every time coordinate. ■

Proof. Conjugation and permutation preserve the Schwartz seminorms and flatness on diagonals. The respective identities hold on separated smooth tensors and extend by Lem. C.12. For an isometry $rx = \rho x + v$, its pullback preserves the flatness condition and satisfies

$$|\varphi \circ (r^{-1})^{\times n}|_k \le [6(1 + n|v|^2)]^{k/2} |\varphi|_k. \tag{C.xxxi}$$

Indeed each row of ρ has sum of absolute values at most $\sqrt{3}$, while $1 + |x|^2 \le 2(1 + n|v|^2)(1 + |(r^{-1})^{\times n} x|^2)$. The chain rule gives the estimate and shows that the pullback is Schwartz and flat on the diagonals. Density now transfers the isometry identity.

For positivity, approximate each φ_n by finite sums $\varphi_n^{(j)}$ of separated positive-time smooth tensors in the seminorm of order $k = 2k_{\mathrm{E}} n_0$; leave φ_0 unchanged. Reflection and conjugation preserve this seminorm. The tensor estimate and the flatness assertion in Lem. C.12 give

$$|\Theta \bar{\varphi}_n \otimes \varphi_{n'} - \Theta \bar{\varphi}_n^{(j)} \otimes \varphi_{n'}^{(j)}|_k \le |\varphi_n - \varphi_n^{(j)}|_k |\varphi_{n'}|_k + |\varphi_n^{(j)}|_k |\varphi_{n'} - \varphi_{n'}^{(j)}|_k \longrightarrow 0.$$

Since $k_{\mathrm{E}}(n+n') \le k$, continuity permits this limit in every term of eq. C.xxx. For each j, expansion of the finite tensor sums expresses the result as one reflection-positive quadratic form for σ, including the empty tuple. Its limit is nonnegative. □

Corollary C.16 (Uniqueness on the flat test space)**.** Two linear functionals on $\mathcal{S}_{\neq}(\mathbb{R}^{3n})$, each bounded for some Schwartz seminorm, agree if their values on separated smooth compactly supported tensors agree. ■□

Proof. Take the larger of the two orders of the seminorms and use the density statement in Lem. C.12. □

Definition C.17 (Growth condition)**.** The Schwinger functionals S_n of a symmetric functional satisfy the growth condition with $(k_{\mathrm{E}}, \mathsf{c}_{\mathrm{E}}, \mathsf{C}_{\mathrm{E}}, \varkappa_{\mathrm{E}})$ if their restrictions to separated smooth tensors are its moments and, for each $n \ge 1$ and $\varphi \in \mathcal{S}_{\neq}(\mathbb{R}^{3n})$,

$$|\mathsf{S}_n(\varphi)| \le \mathsf{c}_{\mathrm{E}}\, \mathsf{C}_{\mathrm{E}}^n (n!)^{\varkappa_{\mathrm{E}}} |\varphi|_{k_{\mathrm{E}} n}. \tag{C.xxxii}$$

Here $k_{\mathrm{E}} \ge 1$ is an integer and the other entries are finite nonnegative numbers. In degree zero $\mathsf{S}_0(z) = z$. Cor. C.16 gives uniqueness for each degree. ■

Appendix D. Continuum reflection and rotations

D.1. Reflection positivity and the translation spectrum.

Lemma D.1 (Reconstruction from a positive form)**.** Let $\mathcal{D}$ carry a positive semidefinite Hermitian form, antilinear in its first argument, and a unit vector $\mathsf{1}$. Suppose T_t° and $U^{\circ}(v)$ are commuting linear representations of $[0, \infty)$ and $\mathbb{R}^N$ on $\mathcal{D}$, with both identities represented by the identity and both representations fixing $\mathsf{1}$. Assume

(1) T_t° is symmetric for the form and $U^{\circ}(v)$ preserves it.
(2) For every F, $c_F = \sup_{t \ge 0} |\langle F, T_t^{\circ} F \rangle| < \infty$.

(3) $\langle F, T_t^\circ F\rangle$ and $\langle F, U^\circ(v)F\rangle$ tend to $\|F\|^2$ as $t \downarrow 0$ and $v \to 0$, respectively.

In the completion $\mathcal{H}$ of the null-space quotient, (a) the induced T_t are a strongly continuous semigroup of positive self-adjoint contractions, and $U(v)$ a strongly continuous unitary group. (b) $T_t = e^{-tH}$ for a unique $H \geq 0$, with $H[1] = 0$. (c) There is a joint projection-valued measure of H and the momentum generators, supported in $[0,\infty) \times \mathbb{R}^N$. (d) Any group of form-preserving maps on $\mathcal{D}$ fixing 1 induces a unitary action. When the original maps commute with time or intertwine spatial translations, the induced action has the same property. ■

Proof. Cauchy–Schwarz gives

$$\|T_t^\circ F\|^2 = \langle F, T_{2t}^\circ F\rangle \leq \|F\|\,\|T_{2t}^\circ F\|. \tag{D.i}$$

Iterating j times gives $\|T_t^\circ F\| \leq \|F\|^{1-2^{-j}} c_F^{2^{-j-1}}$. For non-null F send j to infinity, and for null F the first inequality already shows the image is null. The U° maps preserve the null space and have isometric inverses. The identity $T_t = T_{t/2}^2$ gives positivity. At zero, the norm-square of $(T_t - 1)[F]$ is bounded by $2\|F\|^2 - 2\,\mathrm{Re}\langle F, T_t^\circ F\rangle$, and the analogous expression for $U(v)$ is exact. This proves strong continuity on a dense subspace and hence on the completion. The spectral theorem for a strongly continuous self-adjoint contraction semigroup gives $H = -\log T_1 \geq 0$, and the joint spectral theorem with the commuting unitary translation group gives (c). The fixed-vector assertion follows by the spectral theorem from $T_t[1] = [1]$. The final assertion follows by passing each form-preserving invertible map and each intertwining identity to the quotient and then to the completion. □

The bounds on separated correlations suffice to construct the Hilbert space of a reflection. The additional positivity at a diagonal plane then restricts the joint energy–momentum spectrum. This construction uses neither coincident-point moments nor a probability measure on continuum fields.

Let σ be a real, symmetric, normalized functional on $\mathfrak{S}(\mathbb{R}^3)$, multilinear on products of test spaces with disjoint closures. Assume that its one-point function is $\sigma(f) = c\int f$ and that its connected functions satisfy a bound table $C_n(d;R)$ as in section C.1. Thus, on tuples of separation at least d whose individual supports lie in balls of radius R,

$$\sigma(X) = \sum_{A\subset\{1,\dots,n\}} \prod_{i\in A} \Big(c\int f_i\Big) \sum_{\pi\in \mathrm{Part}_{\geq 2}(A^c)} \prod_{J\in\pi} S^\sigma_{|J|}(X_J). \tag{D.ii}$$

In particular the absolute value is bounded by the same finite sum with $|c|\|f_i\|_1$ in place of $c\int f_i$ and $C_{|J|}(d;R)\prod_{i\in J}\|f_i\|_\sharp$ in place of each connected function. Multilinear telescoping shows that σ is continuous when the entries converge in the sharp norm and their supports remain in fixed separated compact sets.

For a unit vector u, put $E_+(u) = \{x : u\cdot x > 0\}$, let $\theta_u x = x - 2(u\cdot x)u$, and let $\mathcal{D}_u = \mathbb{C}\mathfrak{S}(E_+(u))$ be the free complex vector space on the separated tuples in this half-space. Concatenation is used only when the resulting tuple is separated. The product $\theta_u\bar{F}\cdot G$ is always defined for $F, G \in \mathcal{D}_u$. Reflection positivity means

$$\sigma(\theta_u\bar{F}\cdot F) \geq 0 \qquad (F \in \mathcal{D}_u).$$

It makes

$$\langle F, G\rangle_{\sigma,u} = \sigma(\theta_u\bar{F}\cdot G) \tag{D.iii}$$

a positive semidefinite Hermitian form, antilinear in its first argument. Quotient by its null space and complete to obtain $\mathcal{H}(\sigma,u)$. Write $[F]$ for a class and $\Omega = [\varnothing]$, so that $\|\Omega\| = 1$.

Proposition D.2 (The translation operators)**.** Suppose that σ is translation invariant and reflection positive at $u\cdot x = 0$, with the bound table above. In (a), the reflection form and the uncompleted translation maps satisfy the hypotheses of Lem. D.1, with $N = 2$. In (b), $\mathcal{H}(\sigma,u)$ is separable. In (c), the maps

$$T_t[F] = [\tau_{tu}F] \quad (t \geq 0), \qquad U(v)[F] = [\tau_v F] \quad (v \in u^\perp)$$

extend to a strongly continuous semigroup of positive self-adjoint contractions and a strongly continuous unitary group. They commute and fix Ω. There are strongly commuting self-adjoint operators $H \geq 0, P_1, P_2$ such that

$$\begin{aligned} T_t &= e^{-tH}, & U(v) &= e^{i(v_1 P_1 + v_2 P_2)}, \\ e^{i\tau H} U(v) &= \int e^{i\tau E + iv\cdot q}\, d\mathcal{P}(E,q), & T_t U(v) &= \int e^{-tE + iv\cdot q}\, d\mathcal{P}(E,q), \end{aligned} \tag{D.iv}$$

where coordinates are taken in any fixed orthonormal basis of $u^\perp$, and $\mathcal{P}$ is carried by $[0,\infty)\times\mathbb{R}^2$. Moreover $\ker T_t = 0$, $H\Omega = P_1\Omega = P_2\Omega = 0$, and $T_t \to P_{\ker H}$ strongly as $t\to\infty$.

In (d), if an orthogonal transformation r fixes u and leaves σ invariant, $[F]\mapsto[rF]$ extends to a unitary $U(r)$ fixing Ω. It commutes with H and satisfies $U(r)U(v)U(r)^* = U(rv)$. The joint spectrum is invariant under $(E,q)\mapsto(E,rq)$. ■

Proof. On $\mathcal{D}_u$ write $T_t^0 = \tau_{tu}$ and $U^0(v) = \tau_v$. The identities $\theta_u\tau_{tu} = \tau_{-tu}\theta_u$ and $\theta_u\tau_v = \tau_v\theta_u$, together with translation invariance, give

$$\langle F, T_t^0 G\rangle = \langle T_t^0 F, G\rangle, \qquad \langle U^0(v)F, U^0(v)G\rangle = \langle F, G\rangle.$$

For each $F = \sum_i z_i X_i$ the number

$$c_F = \sup_{t\geq 0} |\langle F, T_t^0 F\rangle|$$

is finite. Choose $\delta > 0$ below the Euclidean separations of the finitely many tuples $\theta_u X_i \cdot X_j$, and choose $R > 0$ bounding the radii of their individual supports. Put $d = \delta/\sqrt{3}$. Within each half-space the Euclidean separations remain at least δ: the negative entries stay fixed and all positive entries receive the same translation. Across the mirror,

$$|y + tu - x|^2 = |y-x|^2 + 2t\, u\cdot(y-x) + t^2 \geq |y-x|^2 \quad (u\cdot x < 0 < u\cdot y).$$

Thus $|y+tu-x|_\infty \geq |y+tu-x|/\sqrt{3} \geq d$ for all $t\geq 0$. Every translated tuple has sup separation at least d. The sharp and L^1 norms and the radii of individual supports are unchanged. The bound following eq. D.ii, summed with coefficients $|z_i z_j|$, therefore bounds c_F.

For small translations, enlarge the finitely many supports by less than $d/4$. They remain separated by $d/2$. Translations of a compactly supported C^1 function converge in the sharp norm, by uniform continuity of the function and its first derivatives. Hence the continuity following eq. D.ii gives $\langle F, T_t^0 F\rangle \to \|F\|^2$ and $\langle F, U^0(v)F\rangle \to \|F\|^2$ at the origin. Thus all three hypotheses of Lem. D.1 hold. That lemma gives the strongly continuous translations and their generators. The joint representation $e^{isH}U(v)$ has spectral measure eq. D.iv. The fixed vector Ω has spectral measure at $(0,0)$. The spectral theorem gives $\ker T_t = 0$ for finite t and $T_t \to P_{\ker H}$ strongly as $t\to\infty$.

For separability use the countable family of separated test tuples constructed in section C.1. Given $X \in \mathfrak{S}(E_+(u))$, choose $d > 0$ smaller than all sup separations in $\theta_u X\cdot X$. Rational neighborhoods of its positive supports of sup width less than $d/(4\sqrt{3})$ still lie in $E_+(u)$: on those supports $u\cdot x \geq d/2$, and $|u\cdot v| \leq \sqrt{3}\,|v|_\infty$. Approximate each entry in C^1 inside these neighborhoods. The four terms in

$$\|[X]-[Y]\|^2 = \sigma(\theta_u X\cdot X) - \sigma(\theta_u X\cdot Y) - \sigma(\theta_u Y\cdot X) + \sigma(\theta_u Y\cdot Y)$$

then converge to the same value. Rational complex linear combinations of the resulting countable set of classes are dense. Finally r commutes with θ_u and preserves the form. Its unitary extension has the asserted covariance on the dense set of classes, hence everywhere. Spectral uniqueness gives the covariance of $\mathcal{P}$. □

D.1.1. *Changing the one-point constant.* For real b define, on a separated tuple $X = (f_1, \dots, f_n)$,

$$\mathsf{sh}_b X = \sum_{A \subset \{1,\dots,n\}} \prod_{i \in A} \left(b \int f_i \right) X_{A^c}. \tag{D.v}$$

The map extends linearly, preserves every open-set tuple space, commutes with isometries and conjugation, and respects every defined concatenation. Expanding the subsets gives $\mathsf{sh}_b \mathsf{sh}_{b'} = \mathsf{sh}_{b+b'}$. Thus it is invertible with inverse sh_{-b}. The moment–cumulant partition formula shows that $\sigma' = \sigma \circ \mathsf{sh}_b$ has one-point constant $c + b$ and the same connected functions of order at least two.

Proposition D.3 (Recentering and reflection)**.** The functionals σ and σ' have the same isometry invariances and the same reflection positivities. They cluster simultaneously. In the setting of Prop. D.2, the map

$$[F]_{\sigma',u} \longmapsto [\mathsf{sh}_b F]_{\sigma,u}$$

extends to a unitary which fixes the vacuum and intertwines the translation operators and their joint spectral measures. ■

Proof. The algebraic identities just established imply

$$\sigma'(\theta_u \bar{F} \cdot G) = \sigma(\theta_u \overline{\mathsf{sh}_b F} \cdot \mathsf{sh}_b G). \tag{D.vi}$$

They prove reflection positivity in both directions, using sh_{-b} for the converse, and give a surjective isometry on the quotients. The same commutation identities intertwine translations and isometries. Uniqueness of the spectral measure proves the last assertion. Invariance follows also directly from the subset expansion. Each shifted clustering difference is a finite sum of unshifted clustering differences of subtuples, with constant integral factors; the converse uses the inverse shift. □

D.1.2. *Positivity at a diagonal plane.* The following measure estimate is the analytic step in comparing the Hilbert spaces of two reflections.

Lemma D.4. Let μ be a finite positive measure on $[0,\infty) \times \mathbb{R}$, and set $w = (E - iq)/\sqrt{2}$. Suppose that, on a Hilbert space $\mathcal{K}$, there are vectors a, b and self-adjoint operators $H_+ \geq 0, Q$ with

$$\int e^{-sw - s'\bar{w}} \, d\mu(E,q) = \langle a, e^{-sH_+} e^{-is'Q} b \rangle \qquad (s, s' \geq 0). \tag{D.vii}$$

Then $\mu(\{|q| > E\}) = 0$. No commutation of H_+ and Q is required. ■

Proof. For $c \in L^1(0,\infty)$ put $p(z) = \int_0^\infty c(s) e^{-sz} \, ds$ on $\operatorname{Re} z \geq 0$. Multiply eq. D.vii by $c(s)\overline{c(s')}$ and integrate. Fubini is justified by the bound $|e^{-sw-s'\bar{w}}| \leq 1$, the finiteness of μ, and the contraction and unitary bounds on the right. Spectral calculus in the indicated order gives

$$\int |p(w)|^2 \, d\mu = \langle a, p(H_+) p^\sharp(Q) b \rangle \leq \|a\| \|b\| \sup_{x \geq 0} |p(x)| \sup_{y \in \mathbb{R}} |p(iy)|, \qquad p^\sharp(y) = \overline{p(-iy)}. \tag{D.viii}$$

Take $z_0 = \varepsilon + i\kappa$, where $\varepsilon > 0$ and $\kappa \in \mathbb{R}$, and $c(s) = \Gamma(z_0)^{-1} s^{z_0 - 1} e^{-\eta s}$ with $\eta > 0$. The Gamma integral gives $p(z) = (z + \eta)^{-z_0}$, with the principal logarithm. Consequently

$$\int |w + \eta|^{-2\varepsilon} e^{2\kappa \arg(w+\eta)} \, d\mu \leq \|a\| \|b\| \eta^{-2\varepsilon} e^{|\kappa|\pi/2}.$$

Fatou's lemma first as $\varepsilon \downarrow 0$ and then as $\eta \downarrow 0$ yields

$$\int_{w \neq 0} e^{2\kappa \arg w} \, d\mu \leq \|a\| \|b\| e^{|\kappa|\pi/2}. \tag{D.ix}$$

For every $\delta > 0$, the measure of $\{\arg w \geq \pi/4 + \delta\}$ is at most $\|a\| \|b\| e^{-2\kappa\delta}$ for all $\kappa > 0$, and hence is zero. Taking $\kappa < 0$ proves the corresponding assertion for $\arg w \leq -\pi/4 - \delta$. Their countable union is exactly $\{|q| > E\}$. □

Proposition D.5 (The spectral bound from two reflections)**.** Let σ satisfy the bounds and translation invariance of Prop. D.2. Fix $k \in \{1,2\}$ and $\epsilon \in \{1,-1\}$. If σ is reflection positive for the normals e_3 and $e_+ = (e_3 + \epsilon e_k)/\sqrt{2}$, then its spectral measure in $\mathcal{H}(\sigma, e_3)$ satisfies

$$\mathcal{P}\{(E,q) : E < |q_k|\} = 0.$$

We denote this spectral-support assertion by $H \geq |P_k|$. ■

Proof. Put $e_- = (e_3 - \epsilon e_k)/\sqrt{2}$ and $W = \{x : x_3 > |x_k|\}$. For $F \in \mathbb{C}\mathfrak{S}(W)$, both F and $\theta_{e_+}\theta_{e_3}F$ lie in the positive half-space for e_+. Let μ be the image of the positive spectral measure $\langle [F], \mathcal{P}(\cdot)[F]\rangle$ under $(E,q) \mapsto (E, \epsilon q_k)$. The same correlation has two representations:

$$\begin{aligned} \int e^{-sw - s'\bar{w}}\, d\mu &= \sigma(\theta_{e_3}\bar{F} \cdot \tau_{se_+ + s'e_-} F) \\ &= \big\langle [\theta_{e_+}\theta_{e_3}F]_+, T_s^+ U^+(s'e_-)[F]_+ \big\rangle_+ . \end{aligned} \tag{D.x}$$

Indeed $se_+ + s'e_- = (s+s')e_3/\sqrt{2} + \epsilon(s-s')e_k/\sqrt{2}$, which gives the first equality by eq. D.iv; the second is the defining reflection form in $\mathcal{H}(\sigma, e_+)$. There $T_s^+ = e^{-sH_+}$ and $U^+(s'e_-) = e^{-is'Q}$ for the negative of the momentum generator along e_-. Hence Lem. D.4 shows that the spectral projection of $\{E < |q_k|\}$ annihilates $[F]$.

For arbitrary $F \in \mathcal{D}_{e_3}$, a sufficiently large positive time translation puts all its finitely many supports in W. The same projection therefore annihilates $T_r[F]$. It commutes with T_r, whose kernel is zero by Prop. D.2, and so annihilates $[F]$. Density proves that the projection is zero. □

Proposition D.6 (Rotations fixing the time axis)**.** If a functional as in Prop. D.2 is invariant under the isometries $\widehat{R}$ acting by $R \in \mathrm{O}(2)$ on $e_3^\perp$ and fixing e_3, then $\mathcal{U}(R)[F] = [\widehat{R}F]$ defines a strongly continuous unitary representation of $\mathrm{O}(2)$. It fixes Ω, commutes with time translations, and satisfies $\mathcal{U}(R)U(v)\mathcal{U}(R)^{-1} = U(Rv)$. ■

Proof. The reflection commutes with $\widehat{R}$, so invariance of the functional preserves the reflection form and the null space. The inverse action is that of R^{-1}. The commutation and covariance identities hold on tuple classes by the same identities for their supports and tests. For $R_j \to R$, the moved tests converge in the C^1 norm and for large j have supports in fixed separated compact sets on the positive side. The reflection form therefore gives norm convergence on tuple classes, hence on their span and, by unitarity, the full Hilbert space. □

Proposition D.7 (The circular spectral cone)**.** Suppose the hypotheses of Prop. D.6 hold. If $H \geq |P_1|$, then the joint spectral measure satisfies

$$\mathcal{P}(\{(E,q) : E < |q|\}) = 0. \tag{D.xi}$$

For $z \in \mathbb{C}^3$ with $\operatorname{Re} z_3 \geq |\operatorname{Im}(z_1, z_2)|$, the integral

$$\mathcal{T}(z) = \int e^{-z_3 E + i(z_1 q_1 + z_2 q_2)}\, d\mathcal{P}(E,q)$$

defines a contraction. In addition, (c) a Euclidean-invariant functional that is reflection positive at $x_3 = 0$ is reflection positive at every mirror. ■

Proof. Conjugating the spectral support inequality by $\mathcal{U}(R)$ gives $E \geq |q \cdot Re_1|$ for each $R \in \mathrm{O}(2)$. Use a countable dense set of Re_1 and take the supremum over it to obtain eq. D.xi. The norm bound follows from $-E \operatorname{Re} z_3 - q \cdot \operatorname{Im}(z_1, z_2) \leq 0$ on this cone. Clause (c) is Prop. C.7. □

Corollary D.8 (Euclidean invariance and the circular cone)**.** A Euclidean-invariant functional as in Prop. D.2 which is reflection positive at $x_3 = 0$ is reflection positive at every mirror and has the circular spectral cone, the complex translation contractions, and the unitary $\mathrm{O}(2)$ action just described. ■□

Proof. Transport reflection positivity to the diagonal plane with normal $(e_1+e_3)/\sqrt{2}$. Prop. D.5 gives $H \geq |P_1|$, so Prop. D.7 applies. □

D.2. **Slab operators and analytic translations.** We now isolate the hypotheses used to recover rotations from the cubic symmetries. The insertion estimate constructs bounded operators across a slab. Together with the spectral bound, these operators will give holomorphic correlation kernels. A separate two-point growth estimate will exclude the nonconstant angular harmonics.

Throughout this section, complex test functions are expanded multilinearly into real and imaginary parts. On them we use $\|h\|_\sharp = \max\{\|h\|_\infty, 40M \operatorname{Lip} h\}$; each of the real and imaginary parts has norm at most $\|h\|_\sharp$. Such multilinear identities hold also for Hilbert-space classes: pairing their difference with every $[G]$ reduces the assertion to the multilinearity of σ, and density makes the difference zero.

Definition D.9 (Plate geometry)**.** Write $y = (x_3, x_1)$, leaving x_2 as the third coordinate. Let R_t rotate this plane by $(y^0, y^1) \mapsto (y^0\cos t - y^1 \sin t, y^0 \sin t + y^1 \cos t)$. Put $n_\upsilon = (\cos\upsilon, \sin\upsilon)$, so $n_0 = (1,0)$ corresponds to e_3. A plate is $f(x) = \alpha(y)\varphi(x_2)$, where $\varphi \in C_c^1(\mathbb{R})$ is real and $\alpha \in C_c^1(\mathbb{R}^2;\mathbb{C})$. Its label is φ; denote the length of the smallest interval containing $\operatorname{supp}\varphi$ by λ_φ, with $\lambda_0 = 0$. Set

$$\operatorname{Prof}_t = \{\alpha \in C_c^1(\mathbb{R}^2) : \operatorname{supp}\alpha \subset [t,2t]\times[-t,t],\ |\alpha| \leq 3,\ \operatorname{Lip}\alpha \leq 12/t\}. \tag{D.xii}$$

■

Definition D.10 (Hypotheses for the rotation argument)**.** For a real normalized functional σ on separated tuples, the hypotheses are as follows.

(R0) $\sigma(f) = 0$ for every $f \in C_c^1(\mathbb{R}^3)$.

(R1) σ is symmetric and invariant under translations and the cubic group W_3 of signed coordinate permutations.

(R2) For each $n \geq 1$ and $d, R > 0$ there is a finite $C_n^\sigma(d;R)$, non-increasing in d, such that

$$|\sigma(f_1,\dots,f_n)| \leq C_n^\sigma(d;R)\prod_i \|f_i\|_\sharp$$

for every tuple with separation at least d and each support in some ball of radius R; the centers are arbitrary.

(R3) σ is reflection positive at $x_3 = 0$.

(R4) In the corresponding Hilbert space, $H \geq |P_1|$ in the sense of spectral support.

(R5) There are $0 < c_{\mathrm{I}} \leq 1/16$, $C_{\mathrm{I}} > 0$ and $p_{\mathrm{I}} \geq 0$ such that, for $0 < \eta \leq 1$, every real h supported in $B(z, c_{\mathrm{I}}\eta)$ with $|z_3| \leq \eta/8$, and $F, G \in \mathbb{C}\mathfrak{S}(\{x_3 > \eta/4\})$,

$$|\sigma(\theta\bar{F}\cdot(h)\cdot G)| \leq C_{\mathrm{I}}\eta^{-p_{\mathrm{I}}}\|h\|_\sharp \|[\tau_{-(\eta/4)e_3}F]\|\,\|[\tau_{-(\eta/4)e_3}G]\|. \tag{D.xiii}$$

(R6) For the diagonal reflection with normal $(e_1+e_3)/\sqrt{2}$, the two-point form is positive:

$$\sum_{a,b}\bar{z}_a z_b \sigma(\theta_\Sigma f_a, f_b) \geq 0 \quad (f_a \in C_c^1(\{x_1+x_3>0\})).$$

(R7) There is an exponent $a_\Gamma < 4$ such that for every label φ there is $C_\varphi < \infty$ with

$$\sigma(\theta f, f) \leq C_\varphi t^{-a_\Gamma} \quad (0 < t \leq 1,\ f = \alpha\otimes\varphi,\ \alpha \in \operatorname{Prof}_t). \tag{D.xiv}$$

■

The Hilbert construction of Prop. D.2 applies under (R1)–(R3). Its proof uses only the uniform moment bound in (R2); alternatively the partition formula bounds each connected function by a finite sum of products of those bounds.

Lemma D.11 (Coordinate reflection pictures)**.** Under (R1)–(R3), (a) the three coordinate reflection Hilbert spaces are unitarily identified by the signed coordinate permutations that send one positive normal to another. (b) Their time translations satisfy $0 \le \mathcal{T}_t \le 1$ for $t \ge 0$. ∎

Proof. The substitution $F \mapsto wF$ in the reflection form for $w \in W_3$ preserves the form by (R1), since $w\theta_u w^{-1} = \theta_{wu}$. It also intertwines translations. The contraction and positivity are the conclusions of Prop. D.2, whose bounds follow from (R2) by the partition formula for cumulants. □

The coordinate identifications transport (R4) to every ordered pair $\mu \neq \nu$:

$$H^{(\mu)} \ge |P_\nu^{(\mu)}|. \tag{D.xv}$$

Remark D.12 (Conventions in the rotation hypotheses). In (R5) the ball for the insertion test is open. In (R7) the supports of the plate and its reflection are separated by at least $2t$ in the x_3 coordinate when $\alpha \in \mathrm{Prof}_t$.

Remark D.13 (Moment bounds and the energy–momentum inequality). (1) If (R1) and (R2) hold, translation invariance and the one-point bound give $\sigma((f)) = c\int f$. In fact a continuous translation-invariant linear functional on compactly supported tests is a multiple of Lebesgue measure: it vanishes on each derivative of a smooth compactly supported test by differentiating translation invariance, and then agrees with $c\int f$ by convolution with a fixed unit-integral test. The cumulant partition formula converts the moment table to a connected-function table, taking the sum of products of the finitely many bounds. Thus $\sigma \in \mathsf{Fct}(\underline{C}, |c|)$ for some table $\underline{C}$.

(2) The spectral measure of the e_3 picture is exactly the joint measure of the reflected time and spatial translations of Prop. D.2. In particular, the spectral assertion of Prop. D.5 for $k = 1$ is (R4). Under Euclidean invariance its square-cone integral agrees with $\mathcal{T}(z)$ on the smaller tube where both are defined, by eq. D.xi.

D.2.1. *The holomorphic translation semigroup.* For $z \in \mathbb{C}^3$ and $\xi \in \mathbb{C}^2$ put

$$d_3(z) = \mathrm{Re}\, z_3 - |\,\mathrm{Im}\, z_1| - |\,\mathrm{Im}\, z_2|, \qquad \mathsf{Tb}_3 = \{d_3 > 0\}, \qquad z^* = (-\bar z_1, -\bar z_2, \bar z_3), \tag{D.xvi}$$

and

$$d_{\mathsf{Tb}}(\xi) = \mathrm{Re}\,\xi^0 - |\,\mathrm{Im}\,\xi^1|, \quad \mathsf{Tb} = \{d_{\mathsf{Tb}} > 0\}, \quad \xi^* = (\bar\xi^0, -\bar\xi^1), \quad \xi^\pm = \xi^0 \pm i\xi^1. \tag{D.xvii}$$

Complex rotations are defined by $(R_\zeta \xi)^\pm = e^{\pm i\zeta}\xi^\pm$. Both tubes are convex cones, stable under their respective involutions. The map $\xi \mapsto (\xi^1, 0, \xi^0)$ embeds Tb in Tb_3.

Lemma D.14. Under (R1)–(R4),

$$\mathsf{T}_3(z) = \int e^{-z_3 E + i(z_1 q_1 + z_2 q_2)}\, d\mathcal{P}(E, q) \quad (z \in \overline{\mathsf{Tb}_3}) \tag{D.xviii}$$

is a contraction semigroup, strongly continuous on the closed tube and holomorphic in operator norm on its interior. It satisfies $\mathsf{T}_3(z)^* = \mathsf{T}_3(z^*)$, $\mathsf{T}_3(z)\Omega = \Omega$, and $\mathsf{T}_3(x)[F] = [\tau_x F]$ for real x with $x_3 \ge 0$. Its restriction is denoted $\mathsf{T}(\xi) = \mathsf{T}_3(\xi^1, 0, \xi^0)$. ∎

Proof. On the support of $\mathcal{P}$, eq. D.xv gives $|q_1|, |q_2| \le E$, so the integrand has modulus at most $e^{-d_3(z)E} \le 1$. Spectral calculus gives the semigroup, adjoint, vacuum and real-translation identities. Dominated convergence in each scalar spectral measure proves strong continuity. On a compact subset of the open tube there is $\delta > 0$ with $d_3 \ge \delta$. Every derivative of degree m in the integrand is bounded by $E^m e^{-\delta E}$. Taylor's formula on a smaller neighborhood, with this uniform bound, permits differentiation in operator norm. This proves holomorphy. □

D.2.2. *Insertion across a slab.*

Proposition D.15 (Slab operators)**.** Under (R1)–(R3) and (R5), for every $l > 0$ and $h \in C_c^1(\{0 < x_3 < l\}; \mathbb{C})$ there is a unique bounded operator $B_l(h)$ such that

$$B_l(h)[F] = [(h) \cdot \tau_{le_3} F]. \tag{D.xix}$$

It is complex linear in h. For every compact subset K of the slab there is $C(K, l) < \infty$ with $\|B_l(h)\| \le C(K, l)\|h\|_\sharp$ when $\operatorname{supp} h \subset K$. It has the identities

$$B_l(h) = T_c B_{c'-c}(\tau_{-ce_3} h) T_{l-c'}, \qquad \operatorname{supp} h \subset \{c < x_3 < c'\}, \tag{D.xx}$$

$$B_l(h)^* = B_l(\overline{\theta_{l/2} h}), \qquad U(v) B_l(h) U(-v) = B_l(\tau_v h), \tag{D.xxi}$$

where $0 \le c < c' \le l$ and $v \perp e_3$. For real h supported in $B(z, c_{\mathrm{I}}\eta)$ with $\eta/8 \le z_3 \le 3\eta/8$, $0 < \eta \le 1$,

$$\|B_{\eta/2}(h)\| \le C_{\mathrm{I}} \eta^{-p_{\mathrm{I}}} \|h\|_\sharp. \tag{D.xxii}$$

If $0 = c_0 < \cdots < c_n$ and $\operatorname{supp} h_j \subset \{c_{j-1} < x_3 < c_j\}$, then

$$[(h_1, \dots, h_n)] = \prod_{j=1}^{n} B_{c_j - c_{j-1}}(\tau_{-c_{j-1}e_3} h_j)\Omega, \tag{D.xxiii}$$

with increasing j from left to right. ■

Proof. For a small piece as in eq. D.xxii, put $h_0 = \tau_{-(\eta/4)e_3} h$. Apply (R5) to h_0 and $\tau_{(\eta/4)e_3} F, \tau_{(\eta/4)e_3} G$. Translation invariance changes the resulting form into

$$\sigma(\theta \bar{F} \cdot (h) \cdot \tau_{(\eta/2)e_3} G),$$

whose modulus is at most $C_{\mathrm{I}} \eta^{-p_{\mathrm{I}}} \|h\|_\sharp \|[F]\| \|[G]\|$. It descends to the quotient and extends continuously to the Hilbert space. The bounded-form representation theorem gives an operator with this bound. Pairing with every class identifies its value on $[G]$ with $[(h) \cdot \tau_{(\eta/2)e_3} G]$.

For general compact K in the slab, set $\delta = \min_{x \in K} \min(x_3, l - x_3) > 0$, $\eta = \min(1, 4\delta)$ and $b = c_{\mathrm{I}}\eta/2$. Use the C^1 partition

$$\chi_{b,k}(x) = \prod_{\mu=1}^{3} \chi(x_\mu/b - k_\mu), \quad \chi(s) = \begin{cases} \cos^2(\pi s/2), & |s| \le 1, \\ 0, & |s| > 1. \end{cases}$$

The identity $\chi(s) + \chi(s-1) = 1$ for $0 \le s \le 1$ gives $\sum_k \chi_{b,k} = 1$. Each support is a cube of radius b, and $\operatorname{Lip} \chi_{b,k} \le \sqrt{3}\pi/(2b)$. Only finitely many of these cubes meet K. Write $h_k = \chi_{b,k} h$ and let c_k be the projection of $bk_3 - \eta/4$ onto $[0, l - \eta/2]$. Because $b \le \eta/32$ and $\eta/4 \le \delta$, the center of $\tau_{-c_k e_3} h_k$ has third coordinate between $7\eta/32$ and $9\eta/32$. Its support lies in a ball of radius $\sqrt{3}b < c_{\mathrm{I}}\eta$. The small-piece construction therefore applies, and

$$B_l(h) = \sum_k T_{c_k} B_{\eta/2}(\tau_{-c_k e_3} h_k) T_{l - c_k - \eta/2}$$

has the required value on classes. The product estimate

$$\|h_k\|_\sharp \le \left(1 + \frac{40\sqrt{3}\pi M}{c_{\mathrm{I}}\eta}\right) \|h\|_\sharp$$

gives the compact-support bound, with an extra factor two for the real and imaginary parts. The construction is independent of the partition because its values on the dense set of classes are prescribed.

Translation of the tuple in eq. D.xix proves eq. D.xx and tangential covariance. Reflection in $x_3 = l/2$ interchanges its left and right entries; symmetry and conjugation give the adjoint formula. Finally, iterating eq. D.xix through the ordered slabs gives eq. D.xxiii. □

D.2.3. *Removing coincident planar projections.*

Proposition D.16. Assume (R1)–(R4). Fix $n \geq 2$ and a real angle t. If $\sigma(R_t X) = \sigma(X)$ for every n-tuple whose supports have pairwise disjoint projections onto the (x_3, x_1) plane, the identity holds for all separated n-tuples. ■

Proof. First suppose that two groups of entries lie respectively in $x_2 < c - \delta$ and $x_2 > c + \delta$, where $\delta > 0$. In the Hilbert space with normal e_2, their correlation after a relative translation su, $u \perp e_2$, $|u| = 1$, is a matrix element of $T^{(2)}_{2\delta} U^{(2)}(su)$. For its finite complex spectral measure m this is

$$\Phi(s) = \int e^{-2\delta E + is(u_1 q_1 + u_3 q_3)}\, dm(E, q).$$

Since $|q_1|, |q_3| \leq E$ and $|u_1| + |u_3| \leq \sqrt{2}$, this is holomorphic for $|\operatorname{Im} s| < \sqrt{2}\delta$. Thus the relative planar translation is analytic whenever there is such an x_2 gap.

Let A_j be the planar projections of the supports and I_j their x_2 interval hulls. Suppose first that every pair of intersecting A_j has a level separating its two intervals and lying outside all the I_j. Induct on the number of intersecting pairs. Choose a separating level for one pair, and translate all entries above it by se_1. For all sufficiently large s, the planar projections on opposite sides no longer meet, and the number of intersecting pairs strictly decreases. The interval hulls and their separating levels are unchanged. The induction hypothesis gives rotation invariance for these large s. Both correlations are holomorphic on a common strip: after rotation, the relative translation is $sR_t e_1$, still perpendicular to e_2. The identity theorem gives equality also at $s = 0$.

For a general tuple of separation $d > 0$, choose $0 < \varepsilon < d/n$ and partition each entry in the x_2 coordinate with $\chi(2x_2/\varepsilon - k)$. Multilinearity gives a finite sum of tuples whose interval hulls have length at most ε. If two planar projections of a nonzero summand meet, points over their common projection are at x_2 distance at least d. The gap between their two interval hulls has length at least $d - 2\varepsilon > (n-2)\varepsilon$. The other $n-2$ interval hulls cannot cover this gap, so it contains a level outside all hulls. The preceding induction applies to each summand. The partition commutes with R_t, and summing proves the assertion. Tuples with a zero entry vanish by multilinearity. □

D.3. **The two-point rotation argument.** For a plate label φ, reflection positivity gives a positive spectral measure on the energy–momentum half-plane. Its transform represents the two-point function. The plate growth bound leaves only the angular frequencies $0, 4, -4$. Positivity at the coordinate and diagonal planes gives opposite signs for the coefficient at frequency four.

D.3.1. *The spectral measure of a label.* Put $Y = [0, \infty) \times \mathbb{R}$, $V = \{(E, q) : E \geq |q|\}$, and, for a profile supported in $y^0 > 0$, set

$$\widetilde{\alpha}(E, q) = \int \alpha(y) e^{-y^0 E + i y^1 q}\, dy. \tag{D.xxiv}$$

This transform is continuous and bounded by $\|\alpha\|_1 e^{-\varepsilon E}$ if the support has $y^0 \geq \varepsilon > 0$. Convolution of profiles multiplies their transforms.

We first work with any symmetric bilinear two-point form s on separated supports which is translation invariant, satisfies the order-two bound in (R2), and is reflection positive at $y^0 = 0$. The form $s(\theta \bar{f}, h)$ on $C^1_c(\{x_3 > 0\}; \mathbb{C})$ defines a one-field Hilbert space. The proof of Prop. D.2, with single test functions instead of tuples and with no distinguished vacuum, gives positive time translations, tangential unitaries and a joint spectral measure $\mathcal{P}^s$ on Y.

Proposition D.17. For every real label φ there is a unique positive Borel measure $\rho[s, \varphi]$ on Y, finite on compact sets, such that all profile transforms are in $L^2(\rho)$ and

$$s\big(\theta(\bar{\alpha} \otimes \varphi), \alpha' \otimes \varphi\big) = \int \overline{\widetilde{\alpha}}\, \widetilde{\alpha}'\, d\rho. \tag{D.xxv}$$

More precisely, for every Borel set $B \subset Y$,

$$\langle [\alpha \otimes \varphi], \mathcal{P}^s(B)[\alpha' \otimes \varphi] \rangle = \int_B \overline{\widetilde{\alpha}}\, \widetilde{\alpha}'\, d\rho. \tag{D.xxvi}$$

■

Proof. Write $\Psi_\alpha = [\alpha \otimes \varphi]$ and $\mu_{\alpha,\alpha'}(B) = \langle \Psi_\alpha, \mathcal{P}^s(B)\Psi_{\alpha'} \rangle$. Riemann sums for convolution converge in C^1, with supports in a fixed compact subset of the positive half-plane. The two-point bound therefore gives

$$\Psi_{\alpha' * \alpha} = \widetilde{\alpha}'(\mathcal{P}^s)\Psi_\alpha. \tag{D.xxvii}$$

On the right, the same Riemann sums converge weakly by the spectral representation of translations and dominated convergence; this identifies their norm limit on the left. Commutativity of convolution implies

$$|\widetilde{\alpha}'|^2 \mu_{\alpha,\alpha} = |\widetilde{\alpha}|^2 \mu_{\alpha',\alpha'}. \tag{D.xxviii}$$

On $O_\alpha = \{\widetilde{\alpha} \neq 0\}$ define $d\rho_\alpha = |\widetilde{\alpha}|^{-2} d\mu_{\alpha,\alpha}$. Equation eq. D.xxviii makes these local measures agree on overlaps. The sets O_α cover Y: at (E_0, q_0) take $\alpha(y) = \chi(y^0 - 2)\chi(y^1)e^{-iy^1 q_0}$, whose transform there is strictly positive. Compact exhaustion of Y gives a countable subcover. Partitioning it into disjoint Borel pieces defines a positive measure ρ with the prescribed restrictions. It is locally finite, since near a point one transform is bounded away from zero.

Equation eq. D.xxviii also shows that $\mu_{\alpha,\alpha}$ vanishes on the zeros of $\widetilde{\alpha}$: intersect that zero set with each member of the countable cover and use the nonzero transform there. Hence $d\mu_{\alpha,\alpha} = |\widetilde{\alpha}|^2 d\rho$ on all of Y. Polarization proves eq. D.xxvi, and taking $B = Y$ gives eq. D.xxv.

For uniqueness, a finite complex measure m on Y is determined by its Laplace–Fourier transform. Indeed, vanishing of $\int e^{-tE+ivq} dm$ at all integers $t \geq 0$ and all $v \in \mathbb{R}$ first gives vanishing against $p(e^{-E})e^{ivq}$ for polynomials p. Uniform polynomial approximation on $[0,1]$, followed by Fourier uniqueness in q, gives vanishing against $g(e^{-E})\mathbf{1}_B(q)$ for continuous g and Borel B. Approximation of indicators of closed subsets of $[0,\infty)$ and the monotone class theorem then give $m = 0$. Thus vanishing against every $\widetilde{\alpha}$ also implies $m = 0$, by testing the continuous Laplace–Fourier transform against profiles. If two proposed measures satisfy eq. D.xxv, replace α' by $\alpha'' * \alpha$. The preceding uniqueness applies to the finite measure $|\widetilde{\alpha}|^2(\rho' - \rho'')$ and makes it zero. Division on O_α and the countable cover give $\rho' = \rho''$. □

Apply this to $s = \sigma|_{\mathfrak{S}_2}$ under (R1)–(R4) and write $\rho_\varphi = \rho[s, \varphi]$. The one-field space embeds isometrically into $\mathcal{H}(\sigma)$ through $[f] \mapsto [(f)]$. It intertwines translations, so $|\widetilde{\alpha}|^2 \rho_\varphi$ is the image of the full spectral measure of $[(\alpha \otimes \varphi)]$ under $(E, q_1, q_2) \mapsto (E, q_1)$. Consequently ρ_φ is carried by V. Reflection of x_1 and uniqueness in Prop. D.17 make it invariant under $q \mapsto -q$.

D.3.2. *From plate bounds to a kernel bound.* Let $a_0(s) = 1 + \cos(2\pi(s - 3/2))$ on $[1,2]$, zero elsewhere, and let $b_1 = (3/2)\mathbf{1}_{[-1/3,1/3]}$, $b = b_1 * b_1 * b_1$. For $1/2 \leq \lambda \leq 1$ define $\alpha_{t,\lambda}(y) = a_0(y^0/t)b(y^1/(\lambda t))$. These profiles belong to Prof_t: $\int a_0 = \int b = 1$, $0 \leq a_0 \leq 2$, $|a_0'| \leq 2\pi$, $0 \leq b \leq 9/8$, and $|b'| \leq 9/4$. The last bounds follow by writing $b = (3/2)\int_{u-1/3}^{u+1/3} T(v)dv$ with $T(v) = (9/4)(2/3 - |v|)_+$.

There is an absolute $C_b < \infty$ such that every positive locally finite measure ρ carried by V satisfies

$$\Gamma[\rho](5t) := \int e^{-5tE}\, d\rho \leq C_b t^{-4} \sup_{1/2 \leq \lambda \leq 1} \int |\widetilde{\alpha}_{t,\lambda}|^2 d\rho. \tag{D.xxix}$$

Here the inequality is valid also for infinite integrals. To see it, the transform of b is $\widehat{b}(q) = (\sin(q/3)/(q/3))^3$, and $\widehat{a}_0(s) \geq e^{-2s}$ for $s \geq 0$. Thus

$$|\widetilde{\alpha}_{t,\lambda}(E,q)|^2 \geq (t^4/4)e^{-4tE}\widehat{b}(\lambda t q)^2.$$

Let $c_6 > 0$ be the infimum of the means of $\sin^6$ over intervals of length at least $1/2$. It is positive by periodicity, compactness for lengths between $1/2$ and π, and counting full periods for larger

lengths. Set $c_b = \min(\sin^6 1, 3^6 c_6)$. For $|s| \le 3$, concavity of sine gives $\widehat{b}(\lambda s)^2 \ge \sin^6 1$; for $|s| \ge 3$, integration over λ gives the mean of $\sin^6$ on an interval of length $|s|/6$. Therefore

$$\int_{1/2}^{1} 2\widehat{b}(\lambda s)^2 d\lambda \ge c_b(1+|s|)^{-6}.$$

On V this is at least $c_b 6^{-6} e^5 e^{-tE}$ when $s = tq$, since $(1+x)^{-6} \ge 6^{-6} e^5 e^{-x}$. Tonelli proves eq. D.xxix with $C_b = 4 \cdot 6^6 e^{-5}/c_b$.

For ρ_φ, the supremum in eq. D.xxix is finite at each fixed t by (R2), so $\Gamma_\varphi(t)$ is finite for all $t > 0$. Under (R7), eq. D.xxix gives $\Gamma_\varphi(5t) \le C_b C_\varphi t^{-4-a_\Gamma}$ for $0 < t \le 1$. If $a_\Gamma < -4$, letting $t \downarrow 0$ and using monotonicity forces $\rho_\varphi = 0$. If $a_\Gamma \ge -4$, apply this bound at $\min\{t/5, 1\}$ and use monotonicity again. In both cases,

$$\Gamma_\varphi(t) \le C'_\varphi \max(1, t^{-1})^{4+a_\Gamma}, \qquad C'_\varphi = 5^{4+a_\Gamma} C_b C_\varphi. \tag{D.xxx}$$

Define

$$k_\varphi(\xi) = \int_V e^{-\xi^0 E + i\xi^1 q}\, d\rho_\varphi. \tag{D.xxxi}$$

It is holomorphic on Tb and bounded by $\Gamma_\varphi(d_{\mathsf{Tb}}(\xi))$. The integrable majorant follows from $|q| \le E$; on smaller tubes it also bounds all derivatives. It is even in ξ^1 and obeys $\overline{k_\varphi(\xi)} = k_\varphi(\xi^*)$.

Proposition D.18 (The two-point kernels)**.** Under (R1)–(R4), k_φ extends by quarter turns to a real analytic function $\widetilde{k}_\varphi$ on $\mathbb{R}^2 \setminus \{0\}$. For disjointly supported profiles,

$$\sigma(\alpha \otimes \varphi, \alpha' \otimes \varphi) = \iint \alpha(y)\alpha'(y')\widetilde{k}_\varphi(y'-y)\, dy\, dy'. \tag{D.xxxii}$$

For two labels set $k_{\varphi,\psi} = (k_{\varphi+\psi} - k_{\varphi-\psi})/4$, and likewise for $\widetilde{k}$. Then

$$\sigma(\alpha \otimes \varphi, \alpha' \otimes \psi) = \sigma(\alpha \otimes \psi, \alpha' \otimes \varphi) = \iint \alpha(y)\alpha'(y')\widetilde{k}_{\varphi,\psi}(y'-y)\, dy\, dy'. \tag{D.xxxiii}$$

■

Proof. If the profiles lie on opposite sides of $y^0 = c$, translate to $c = 0$ and apply eq. D.xxv. Fubini gives eq. D.xxxii with k_φ on the positive half-plane. For $v^0, v^1 > 0$, take profiles in small neighborhoods of $0, v$ and rotate both through $-\pi/2$. Cubic invariance equates their correlations, and the integral representation gives $k_\varphi(v) = k_\varphi(R_{-\pi/2}v)$ by shrinking the neighborhoods. This equality makes the quarter-turned definitions agree on overlaps. They cover the punctured plane and give $\widetilde{k}_\varphi$. Reality follows from the evenness in q.

For arbitrary disjoint profile supports of distance $d > 0$, partition each into squares of diameter less than $d/4$ using the functions χ above. Any pair of nonzero pieces is ordered along one of the signed coordinate axes, since one coordinate of a vector between them has modulus at least $d/\sqrt{2}$. The ordered formula applies after a quarter turn, and its sum is eq. D.xxxii.

Polarization in the labels gives the sum of the two mixed correlations in eq. D.xxxiii. To see that they are equal, use an even approximate identity j_δ on the plane. Translation invariance and Riemann sums give

$$\sigma((\alpha * j_\delta) \otimes \varphi, (\alpha' * j_\delta) \otimes \psi) = \iint \alpha(u)\alpha'(u')h_\delta(u'-u)\, du\, du',$$

where $h_\delta(v) = \sigma(j_\delta \otimes \varphi, (\tau_v j_\delta) \otimes \psi)$ and δ is smaller than a quarter of the separation. Apply the same identity to $\alpha'(-u), \alpha(-u)$ and change both integration variables to their negatives. Its right side is unchanged. Rotation by π and permutation of the two entries turn its left side into the expression with the labels exchanged. As $\delta \downarrow 0$, convolution converges in C^1 on fixed separated neighborhoods. The order-two bound permits the limit and proves the equality of the mixed correlations. □

D.3.3. *Eliminating the fourth harmonic.*

Proposition D.19. Under (R1)–(R4) and (R7), for each $r > 0$ the function $k_{\varphi,r}(\zeta) = k_\varphi(R_\zeta(r,0))$, initially defined for $|\operatorname{Re}\zeta| < \pi/2$, extends to an entire function with period $\pi/2$. There are real $c_0(r), c_1(r)$ such that

$$k_{\varphi,r}(\zeta) = c_0(r) + 2c_1(r)\cos 4\zeta, \qquad \widetilde{k}_\varphi(r\cos t, r\sin t) = c_0(r) + 2c_1(r)\cos 4t. \tag{D.xxxiv}$$

■

Proof. For $\zeta = t + is$ in the initial strip, $d_{\mathsf{Tb}}(R_\zeta(r,0)) = re^{-|s|}\cos t$. The quarter-turn identity of Prop. D.18 agrees on overlapping strips by holomorphic uniqueness, so translates by $j\pi/2$ give one entire periodic function. Move t modulo $\pi/2$ into $[-\pi/4, \pi/4]$. Then

$$|k_{\varphi,r}(t+is)| \le \Gamma_\varphi(re^{-|s|}/\sqrt{2}) \le C_r e^{A|s|}, \qquad A = \max(0, 4 + a_\Gamma) < 8.$$

For an entire $\pi/2$-periodic function with this bound, shifting the contour of its Fourier coefficient from the real axis to height s shows that the coefficient of $e^{4im\zeta}$ vanishes if $4|m| > A$. Only $m = 0, 1, -1$ remain. Evenness in ξ^1 equates the last two coefficients, and reality on real points makes both remaining coefficients real. This proves the formulas. □

Proposition D.20 (Rotation invariance at two plates)**.** Under (R1)–(R4), (R6) and (R7), $c_1(r) = 0$ for all labels and all $r > 0$. Thus $\widetilde{k}_{\varphi,\psi}(v)$ depends only on $|v|$, and correlations of two plates with disjoint planar supports are invariant under every R_t. Moreover, for $v^0 > 0$ and real s,

$$k_{\varphi,\psi}(R_{is}v) = k_{\varphi,\psi}(v). \tag{D.xxxv}$$

■

Proof. For imaginary angles the integral in eq. D.xxxi is

$$k_{\varphi,r}(is) = \int_V e^{-r(E\cosh s + q\sinh s)}\, d\rho_\varphi \ge 0.$$

Dividing $c_0(r) + 2c_1(r)\cosh 4s$ by $\cosh 4s$ and letting $s \to \infty$ gives $c_1(r) \ge 0$.

For the opposite sign, turn the two-point form through $\pi/4$: $s^{\mathrm{d}}(f,h) = \sigma(R_{\pi/4}f, R_{\pi/4}h)$. It is translation invariant. The rotation preserves the sharp and L^1 norms. Since it and its inverse have sup-norm operator norm $\sqrt{2}$, it sends a tuple of sup type $(d;R)$ to one of type $(d/\sqrt{2}; \sqrt{2}R)$. Thus its order-two moment bound has table $C_2^\sigma(d/\sqrt{2}; \sqrt{2}R)$, and (R6) gives its coordinate reflection positivity. Hence Prop. D.17 constructs a positive measure ρ_{d} on Y. At this point no cone support for ρ_{d} is assumed.

Let $b_s(E,q) = (E\cosh s + q\sinh s, q\cosh s + E\sinh s)$. It preserves V, has inverse b_{-s}, and its first coordinate is at least $e^{-|s|}E$ on V. Thus $(b_s)_*\rho_\varphi$ has finite Laplace transforms, and its kernel at real $v^0 > 0$ is $k_\varphi(R_{is}v)$. The three harmonics in eq. D.xxxiv give

$$k_\varphi(R_i v) + k_\varphi(R_{-i}v) = (\cosh 4 + 1)k_\varphi(v) - (\cosh 4 - 1)\widetilde{k}_\varphi(R_{\pi/4}v).$$

Multiply by $\overline{\alpha(u)}\alpha'(u')$ at $v = u' - \theta u$ and integrate. The last term represents the turned form by eq. D.xxxii; the other terms are the spectral representations for the boosted measures. Uniqueness in Prop. D.17 therefore gives the identity of positive measures

$$(\cosh 4 - 1)\rho_{\mathrm{d}} + (b_1)_*\rho_\varphi + (b_{-1})_*\rho_\varphi = (\cosh 4 + 1)\rho_\varphi. \tag{D.xxxvi}$$

In particular $0 \le \rho_{\mathrm{d}} \le (\cosh 4+1)(\cosh 4-1)^{-1}\rho_\varphi$. This proves both its cone support and finiteness of its Laplace transforms. Its kernel equals $\widetilde{k}_\varphi(R_{\pi/4}v)$ on real $v^0 > 0$, by the integral identity against profiles. Analytic continuation in the angle thus gives

$$k[\rho_{\mathrm{d}}](R_{is}(r,0)) = c_0(r) - 2c_1(r)\cosh 4s \ge 0.$$

The same limit proves $c_1(r) \le 0$. Hence it vanishes. Polarization in $\varphi \pm \psi$ gives the radiality of the mixed kernel. Formula eq. D.xxxiii proves invariance of the two-plate correlations. Holomorphic uniqueness in the initial strip gives eq. D.xxxv. □

D.4. **Point fields and plate kernels.** The insertion estimate first gives operators smeared inside a slab. Holomorphy of the translation semigroup improves this to a pointwise operator bound. The essential step is a norm Cauchy estimate for mollified insertions; weak compactness alone would not give the products needed below. Throughout this section we assume (R1)–(R5).

Write $B_d^{\mathrm{c}}(h) = B_{2d}(\tau_{de_3} h)$.

Definition D.21 (Point-bound constants)**.** Put

$$\wp = p_{\mathrm{I}} + 4, \qquad C_{\mathrm{plt}} = 480 \cdot 8^3 m_2 c_{\mathrm{I}}^{-4} C_{\mathrm{I}} M, \qquad C_{\mathrm{pt}} = 320 \cdot 8^4 m_3 c_{\mathrm{I}}^{-4} C_{\mathrm{I}} M. \tag{D.xxxvii}$$

The absolute constants are obtained from the radial weight

$$m(s) = \begin{cases} c_m, & 0 \le s \le 1/3, \\ c_m \cos^2(3\pi(s - 1/3)/4), & 1/3 \le s \le 1, \\ 0, & s \ge 1, \end{cases} \qquad c_m = \frac{9\pi}{5\pi^2 - 8},$$

by setting $m_0 = c_m$, $m_1 = 3\pi c_m/4$, $m_2 = \sqrt{2} m_0 m_1 + 2m_0^2$ and $m_3 = m_0^3 + \sqrt{3} m_0^2 m_1$. The function m is C^1, nonnegative and constant near zero; direct integration gives $2\pi \int_0^1 m(s) s \, ds = 1$. Let $j_\delta^{(N)}$, $N = 2, 3$, be $m(2|x|/\delta)$ divided by its integral. These are nonnegative radial C^1 mollifiers supported in $|x| \le \delta/2$. We also write $\mathsf{mv}_3 = m_3$. ■

For $r > 0$, the product weight

$$\alpha_{r,t}^{(N)}(x) = \prod_{i=1}^{N} r^{-2} m(r^{-1} \sqrt{x_i^2 + t_i^2})$$

has supremum at most $m_0^N r^{-2N}$ and Lipschitz constant at most $\sqrt{N} m_0^{N-1} m_1 r^{-2N-1}$. Iterating the one-variable mean-value formula in polar coordinates gives

$$u(0) = \int_{(-r,r)^N} dt \int dx \, \alpha_{r,t}^{(N)}(x) u(x + it) \tag{D.xxxviii}$$

whenever u is holomorphic on a neighborhood of the closed polydisc of radius r. All operator integrals below converge in norm; their integrands are continuous with compact support.

D.4.1. *Removing the smearing.*

Proposition D.22 (Point plates)**.** For every real label $\varphi \in C_c^1(\mathbb{R})$ there is a norm-holomorphic family $B_\varphi(\xi, \xi')$, linear in φ, on $\mathsf{Tb} \times \mathsf{Tb}$ such that

$$\|B_\varphi(\xi, \xi')\| \le C_{\mathrm{plt}} (\lambda_\varphi + 1) \|\varphi\|_{C^1} \min\{1, d_{\mathsf{Tb}}(\xi), d_{\mathsf{Tb}}(\xi')\}^{-\wp}. \tag{D.xxxix}$$

For $\kappa, \kappa' \in \overline{\mathsf{Tb}}$ it satisfies

$$\mathsf{T}(\kappa) B_\varphi(\xi, \xi') \mathsf{T}(\kappa') = B_\varphi(\kappa + \xi, \xi' + \kappa'), \tag{D.xl}$$

$$B_\varphi(\xi, \xi')^* = B_\varphi(\xi'^*, \xi^*). \tag{D.xli}$$

If $\operatorname{supp} \alpha \subset \{|y^0| < d\}$, then

$$B_d^{\mathrm{c}}(\alpha \otimes \varphi) = \int \alpha(y) B_\varphi(dn_0 + y, dn_0 - y) \, dy. \tag{D.xlii}$$

These properties determine the family already among weakly continuous families satisfying the shift law and the integral identity for real profiles. ■

Proof. Put $j_\delta = j_\delta^{(2)}$ and

$$A_\delta = B_\delta^{\mathrm{c}}(j_\delta \otimes \varphi), \qquad B^\delta(\xi,\xi') = \mathsf{T}(\xi - \delta n_0) A_\delta \mathsf{T}(\xi' - \delta n_0).$$

This is defined when both tube distances are at least δ, and holomorphic when they are larger. Radiality of the mollifier and the slab adjoint formula give $A_\delta^* = A_\delta$. The shift and adjoint laws follow at once. Slab localization and tangential covariance give

$$B^\delta(dn_0 + y, dn_0 - y) = B_d^{\mathrm{c}}((\tau_y j_\delta) \otimes \varphi).$$

Consequently, for continuous α supported in $|y^0| < d - \delta$,

$$\int \alpha(y) B^\delta(dn_0 + y, dn_0 - y)\, dy = B_d^{\mathrm{c}}((\alpha * j_\delta) \otimes \varphi). \tag{D.xliii}$$

Indeed, convolution Riemann sums converge in C^1 with one compact support, and the slab bound permits passage to the limit.

We first bound B^δ independently of δ. Set $\eta = \min(1, d_{\mathsf{Tb}}(\xi), d_{\mathsf{Tb}}(\xi'))$, $r = c_{\mathrm{I}}\eta/8$ and $0 < \delta \le r$. For unit vectors v, v', apply eq. D.xxxviii to $u(z) = \langle v, B^\delta(\xi + z, \xi' - z)v'\rangle$. The polydisc of radius $2r$ lies in its domain because $\eta \ge 128r$. Move the imaginary shift into the vectors by the two contractions with arguments $\xi - \eta n_0/4 + it$ and $\xi' - \eta n_0/4 - it$. The remaining inner integral is a matrix element of

$$B_{\eta/4}^{\mathrm{c}}(\gamma_t \otimes \varphi), \qquad \gamma_t = \alpha_{r,t}^{(2)} * j_\delta.$$

Its profile is supported in $[-2r, 2r]^2$, with supremum $m_0^2 r^{-4}$ and Lipschitz constant $\sqrt{2} m_0 m_1 r^{-5}$.

Partition the label as $\varphi_k(s) = \chi(s/(4r) - k)\varphi(s)$, using the partition in the proof of Prop. D.15. At most $3(\lambda_\varphi + 1)/(c_{\mathrm{I}}\eta)$ terms are nonzero. Each $\gamma_t \otimes \varphi_k$ is supported in $B(4rke_2, c_{\mathrm{I}}\eta)$ and has sharp norm at most $40 M m_2 r^{-5}\|\varphi\|_{C^1}$. The centered version of eq. D.xxii, followed by integration over $(-r, r)^2$, therefore gives

$$\|B^\delta(\xi,\xi')\| \le (2r)^2 \frac{3(\lambda_\varphi + 1)}{c_{\mathrm{I}}\eta} C_{\mathrm{I}} \eta^{-p_{\mathrm{I}}} 40 M m_2 r^{-5} \|\varphi\|_{C^1}.$$

This is eq. D.xxxix with B^δ in place of B_φ.

To obtain norm convergence, fix $\epsilon > 0$ and set

$$\delta_\epsilon = \frac{c_{\mathrm{I}}}{16} \min(1, \epsilon), \qquad b_\epsilon = C_{\mathrm{plt}}(\lambda_\varphi + 1)\|\varphi\|_{C^1} (\tfrac{1}{2}\min(1,\epsilon))^{-\wp}.$$

For $\delta \le \delta_\epsilon$, the holomorphic map $z \mapsto B^\delta(\epsilon n_0 + z, \epsilon n_0 - z)$ is bounded by b_ϵ on the polydisc of radius $\epsilon/4$. Cauchy's derivative estimate gives the bound $8b_\epsilon|y|/\epsilon$ for its change from $z = 0$ to a real y with $|y| \le \epsilon/8$. Integrate against $j_{\delta'}$ and use eq. D.xliii; interchange δ, δ' and use commutativity of convolution. The result is

$$\|B^\delta(\epsilon n_0, \epsilon n_0) - B^{\delta'}(\epsilon n_0, \epsilon n_0)\| \le \frac{4 b_\epsilon}{\epsilon}(\delta + \delta').$$

Thus $B_{[\epsilon]} = \lim_{\delta \downarrow 0} B^\delta(\epsilon n_0, \epsilon n_0)$ exists in norm. For $0 < \epsilon \le \min(d_{\mathsf{Tb}}(\xi), d_{\mathsf{Tb}}(\xi'))$, define

$$B_\varphi(\xi,\xi') = \mathsf{T}(\xi - \epsilon n_0) B_{[\epsilon]} \mathsf{T}(\xi' - \epsilon n_0). \tag{D.xliv}$$

This is the limit of $B^\delta(\xi,\xi')$, uniformly when both tube distances exceed ϵ, so it does not depend on the chosen ϵ. Choosing a smaller ϵ locally proves norm holomorphy directly from this formula. The bound, shifts, adjoints and linearity pass to the limit. In eq. D.xliii, uniform norm convergence on the support and $\alpha * j_\delta \to \alpha$ in C^1 prove eq. D.xlii.

For uniqueness, the integral identity against all real profiles and weak continuity identify the matrix elements at $(dn_0 + y, dn_0 - y)$, in particular at (dn_0, dn_0). The shift law then identifies every pair of tube points. □

Proposition D.23 (Point fields)**.** There is a unique weakly continuous family $B(z,z')$, $(z,z') \in \mathsf{Tb}_3^2$, satisfying the three-dimensional shift law and the following slab integral formula. It is norm holomorphic and obeys

$$\|B(z,z')\| \le C_{\mathrm{pt}} \min\{1, d_3(z), d_3(z')\}^{-\wp}, \tag{D.xlv}$$

$$\mathsf{T}_3(\kappa) B(z,z') \mathsf{T}_3(\kappa') = B(\kappa + z, z' + \kappa'), \tag{D.xlvi}$$

$$B(z,z')^* = B(z'^*, z^*), \tag{D.xlvii}$$

$$B_d^{\mathrm{c}}(h) = \int h(x) B(de_3 + x, de_3 - x)\, dx. \tag{D.xlviii}$$

Here $\kappa, \kappa' \in \overline{\mathsf{Tb}_3}$, and in the last identity $h \in C_c^1(\{|x_3| < d\}; \mathbb{C})$. ■

Proof. Use $j_\delta^{(3)}$ in the preceding construction, with $A_\delta = B_\delta^{\mathrm{c}}(j_\delta^{(3)})$ and T_3 in place of T. The convolution identity and the shift and adjoint laws have the same proofs. For completeness, the change in the uniform estimate is as follows. Put $\eta = \min(1, d_3(z), d_3(z'))$ and $r = c_{\mathrm{I}}\eta/8$. The polydisc of radius $2r$ loses at most $6r$ in tube distance. The three-variable mean-value formula, after moving the imaginary shifts into contractions, gives the centered insertion of $\gamma_t = \alpha_{r,t}^{(3)} * j_\delta^{(3)}$. Its support lies in $[-2r, 2r]^3 \subset B(0, c_{\mathrm{I}}\eta)$, and $\|\gamma_t\|_\sharp \le 40 M m_3 r^{-7}$. No label partition is needed. Hence the bound is

$$(2r)^3 C_{\mathrm{I}} \eta^{-p_{\mathrm{I}}} 40 M m_3 r^{-7} = C_{\mathrm{pt}} \eta^{-\wp}.$$

At $(\epsilon e_3, \epsilon e_3)$ use a polydisc of radius $\epsilon/6$. With $b_\epsilon = C_{\mathrm{pt}}(\frac{1}{2}\min(1,\epsilon))^{-\wp}$, Cauchy's estimate and the symmetric convolution argument give

$$\|B^\delta(\epsilon e_3, \epsilon e_3) - B^{\delta'}(\epsilon e_3, \epsilon e_3)\| \le 6 b_\epsilon (\delta + \delta')/\epsilon.$$

Norm convergence, extension to arbitrary tube pairs, holomorphy, the integral formula and uniqueness now follow exactly by the limit and shift arguments above. □

Corollary D.24 (Slab operators in L^1)**.** Let σ satisfy (R1)–(R5), with $\wp$ and C_{pt} as in eq. D.xxxvii. In (a), for $0 \le d' < d$ and $\operatorname{supp} h \subset \{|x_3| \le d'\}$,

$$\|B_d^{\mathrm{c}}(h)\| \le C_{\mathrm{pt}} \min(1, d - d')^{-\wp} \|h\|_{L^1}. \tag{D.xlix}$$

In (b), if $F, F' \in \mathbb{C}\mathfrak{S}(\{x_3 > d\})$, the function

$$v_d(x) = \langle [\tau_{-de_3} F], B(de_3 + x, de_3 - x)[\tau_{-de_3} F'] \rangle \tag{D.l}$$

is real analytic for $|x_3| < d$ and satisfies

$$\sigma(\theta \bar{F} \cdot (h) \cdot F') = \int h(x) v_d(x)\, dx, \tag{D.li}$$

$$|v_d(x)| \le C_{\mathrm{pt}} \min(1, d - |x_3|)^{-\wp} \|[\tau_{-de_3} F]\| \|[\tau_{-de_3} F']\|. \tag{D.lii}$$

In (c), identifying $\xi = (\xi^0, \xi^1)$ with $(\xi^1, 0, \xi^0)$,

$$B_\varphi(\xi, \xi') = \int \varphi(s) B(\xi + se_2, \xi' - se_2)\, ds. \tag{D.liii}$$

■□

Proof. Integrate eq. D.xlv to obtain eq. D.xlix. For every $h \in C_c^1(\{|x_3| < d\})$, the slab form gives eq. D.li; its density is the restriction of a holomorphic matrix element on $|\operatorname{Re} z_3| + |\operatorname{Im} z_1| + |\operatorname{Im} z_2| < d$, and the same point bound gives eq. D.lii. The right side of eq. D.liii is weakly continuous, satisfies the plate shift law, and, by Fubini and eq. D.xlviii, represents the slab operator of $\alpha \otimes \varphi$. Uniqueness in Prop. D.22 proves the identity. □

D.4.2. *Products and their analytic continuation.* At an interior junction of two point plates, only the sum of the two adjoining arguments matters. More precisely, for $\kappa_1, \kappa_2 \in \mathsf{Tb}$, $\kappa_0 \in \overline{\mathsf{Tb}}$ and $0 < \epsilon \le \min(d_{\mathsf{Tb}}(\kappa_1), d_{\mathsf{Tb}}(\kappa_2))$,

$$B_\varphi(\xi, \kappa_1)\mathsf{T}(\kappa_0)B_\psi(\kappa_2, \xi') = B_\varphi(\xi, \epsilon n_0)\mathsf{T}(\kappa_1 + \kappa_0 + \kappa_2 - 2\epsilon n_0)B_\psi(\epsilon n_0, \xi'). \tag{D.liv}$$

This follows by extracting the two semigroup factors with eq. D.xl. Thus one may replace the adjoining arguments by half their sum. An outer argument next to Ω may be changed arbitrarily, since $\mathsf{T}(\kappa)\Omega = \Omega$. The same statements hold for point fields and T_3.

For $\xi_1, \dots, \xi_{n-1} \in \mathsf{Tb}$, define

$$K_n(\xi; \underline{\varphi}) = \langle \Omega, B_{\varphi_1}(\kappa, \xi_1/2)B_{\varphi_2}(\xi_1/2, \xi_2/2) \cdots B_{\varphi_n}(\xi_{n-1}/2, \kappa)\Omega \rangle \,. \tag{D.lv}$$

Here $\kappa \in \mathsf{Tb}$ is arbitrary; the preceding outer argument identity makes the answer independent of it. Set $K_0 = 1$ and $K_1(\varphi) = \langle \Omega, B_\varphi(\kappa, \kappa)\Omega \rangle$. For

$$\mathsf{Tb}^{(n)} = \{x : x_1 \in \mathsf{Tb},\ x_{j+1} - x_j \in \mathsf{Tb}\},$$

let $\Psi(x; \underline{\varphi})$ be the operator product in eq. D.lv, applied to Ω, with successive differences $\xi_j = x_{j+1} - x_j$ and first outer argument x_1 in place of κ; set $\Psi(\varnothing) = \Omega$.

Proposition D.25. The kernels K_n are holomorphic, and Ψ is norm holomorphic on $\mathsf{Tb}^{(n)}$. They are multilinear in the labels, and for $n \ge 2$,

$$|K_n(\xi; \underline{\varphi})| \le \prod_{a=1}^{n} (2^\wp C_{\mathrm{plt}}(\lambda_{\varphi_a} + 1)\|\varphi_a\|_{C^1}) \min\{1, \min_a d_{\mathsf{Tb}}(\xi_a)\}^{-\wp n}. \tag{D.lvi}$$

For profiles whose supports are strictly ordered in time,

$$\sigma(\alpha_1 \otimes \varphi_1, \dots, \alpha_n \otimes \varphi_n) = \int \prod_a \alpha_a(y_a) K_n(y_2 - y_1, \dots, y_n - y_{n-1}; \underline{\varphi})\, dY. \tag{D.lvii}$$

If the first support also lies in $y^0 > 0$, the class of this tuple is

$$[(\alpha_1 \otimes \varphi_1, \dots, \alpha_n \otimes \varphi_n)] = \int \prod_a \alpha_a(y_a) \Psi(Y; \underline{\varphi})\, dY. \tag{D.lviii}$$

The vectors satisfy

$$\begin{aligned} \langle \Psi(x; \underline{\varphi}), \Psi(z; \underline{\psi}) \rangle = K_{n+n'}\big((x_n - x_{n-1})^*, \dots, (x_2 - x_1)^*, \\ x_1^* + z_1, z_2 - z_1, \dots, z_{n'} - z_{n'-1}; \varphi_n, \dots, \varphi_1, \psi_1, \dots, \psi_{n'}\big). \end{aligned} \tag{D.lix}$$

When one side is empty, its formula is obtained by removing the first outer argument of the other vector; when both sides are empty, the value is one. In addition,

$$\overline{K_n(\xi; \underline{\varphi})} = K_n(\bar{\xi}; \underline{\varphi}), \qquad K_2(\xi; \varphi, \psi) = k_{\varphi,\psi}(\xi). \tag{D.lx}$$

Under (R0), $K_1 = 0$. ∎

Proof. Holomorphy follows by taking products and matrix elements of the norm-holomorphic point plates. Choose $\kappa = n_0$ and apply eq. D.xxxix to each factor to get eq. D.lvi. Taking adjoints reverses the first product; at its junction with the second, eq. D.liv replaces x_1^*, z_1 by half their sum. This gives the Gram formula, including its empty-vector variants.

For the integral formulas choose separating times $0 = c_0 < c_1 < \cdots < c_n$ for positive-time profiles. The slab product eq. D.xxiii and eq. D.xlii express the class as the integral of

$$\prod_a \alpha_a(y_a) \prod_a B_{\varphi_a}(y_a - c_{a-1}n_0, c_a n_0 - y_a)\Omega.$$

At each junction the arguments sum to $y_{a+1} - y_a$. Replacing them by half this sum gives eq. D.lviii. Taking the inner product with Ω gives eq. D.lvii. A common translation reduces arbitrary time-ordered supports to this case.

Reality of σ on real tests shows that the imaginary part of the continuous kernel vanishes on real time-ordered configurations: test it against arbitrary products of real profiles in small disjoint neighborhoods. The holomorphic identity theorem gives the first identity in eq. D.lx. Applying the same argument to the two integral representations of the two-point function identifies K_2 with $k_{\varphi,\psi}$ first on $\mathbb{R}^2_+$ and then on Tb. Finally a profile of integral one identifies $K_1(\varphi)$ with the one-point value, which vanishes under (R0). □

For a permutation $\mathfrak{s}$ and a frame $\upsilon \in \frac{\pi}{2}\mathbb{Z}$, set

$$\begin{aligned} \mathcal{Z}(\mathfrak{s},\upsilon) &= \{Z : z_{\mathfrak{s}(a+1)} - z_{\mathfrak{s}(a)} \in R_\upsilon \mathsf{Tb},\ a<n\},\\ K_\upsilon^{\mathfrak{s}}(Z;\underline{\varphi}) &= K_n\big(R_{-\upsilon}\partial(\mathfrak{s}Z);\mathfrak{s}\underline{\varphi}\big). \end{aligned} \tag{D.lxi}$$

Here $\partial Z = (z_{a+1}-z_a)_{a<n}$, and permutations act on points and labels in the same order. Let $\mathcal{Z}_n$ be the union of these tubes and $\mathcal{O}_n = \mathcal{Z}_n \cap (\mathbb{R}^2)^n$.

Lemma D.26. The functions $K_\upsilon^{\mathfrak{s}}$ agree on overlaps and define one holomorphic function $\mathcal{K}_n(Z;\underline{\varphi})$ on the connected open set $\mathcal{Z}_n$. This set is stable under common complex translations, permutations, quarter turns, conjugation, and R_{is} for real s. The kernel is invariant under the first three operations, with labels permuted together with points, and obeys $\overline{\mathcal{K}_n(Z;\underline{\varphi})} = \mathcal{K}_n(\bar{Z};\underline{\varphi})$. Its real domain is

$$\mathcal{O}_n = \{Y : \text{all } y_a^0 \text{ are distinct, or all } y_a^1 \text{ are distinct}\}.$$

For profiles whose product support lies in one tube, integration of $K_\upsilon^{\mathfrak{s}}$ against their product gives $\sigma(\alpha_1\otimes\varphi_1,\dots,\alpha_n\otimes\varphi_n)$. ■

Proof. Each tube is a nonempty open convex cone, the preimage of $(R_\upsilon\mathsf{Tb})^{n-1}$ under a surjective linear difference map. A real configuration belongs to it precisely when $n_\upsilon \cdot y_{\mathfrak{s}(a)}$ is strictly increasing. Permuting the entries and making the quarter turn therefore reduces its integral formula to eq. D.lvii, by cubic invariance of σ.

Every nonempty intersection of two tubes is convex and contains the real part of each of its points. On its real open subset, the two kernels have equal integrals against all products of small supported profiles. Their difference therefore vanishes there. The holomorphic identity theorem on the connected intersection proves agreement everywhere. This proves that the kernels glue without a choice of continuation path.

The stated stability follows from the difference inequalities defining the tubes; in light-cone coordinates R_{is} multiplies the two coordinates by positive real numbers. The kernel symmetries follow from its definition and eq. D.lx. There are at most $4n!$ tubes. The complement of $\mathcal{O}_n$ is the finite union

$$\{y_a^0 = y_b^0,\ y_k^1 = y_l^1\}, \qquad a\neq b,\ k\neq l.$$

Each subspace has codimension two, so its finite complement is connected. Every tube is connected and meets this real set. Their union $\mathcal{Z}_n$ is therefore connected. □

D.5. **Rotation invariance of all correlations.** We now assume (R0)–(R5). The polynomial growth of the point plates implies that their kernels have only finitely many angular frequencies. Positivity imposes an arithmetic restriction on the highest frequency; this reduces the rotation problem to the two-point function.

D.5.1. *Angular frequencies and their phases.*

Lemma D.27. Put $\mathcal{M}_n = \{m\in\mathbb{Z} : 4|m| \le \wp n\}$. For each tuple of $n\ge 2$ labels there are unique holomorphic functions c_m on $\mathcal{Z}_n$, zero for $m\notin\mathcal{M}_n$, such that

$$\mathcal{K}_n(R_{is}Z;\underline{\varphi}) = \sum_{m\in\mathcal{M}_n} c_m(Z;\underline{\varphi})e^{-4ms}. \tag{D.lxii}$$

They inherit permutation and translation invariance and satisfy

$$c_m(R_{is}Z) = e^{-4ms}c_m(Z), \qquad \overline{c_m(X)} = c_{-m}(X) \quad (X \in \mathcal{O}_n).$$

They extend uniquely to real-analytic functions $\widetilde{c}_m$ on configurations of distinct planar points, with

$$\widetilde{c}_m(v + R_t(X - v); \underline{\varphi}) = e^{4imt}\widetilde{c}_m(X; \underline{\varphi}). \tag{D.lxiii}$$

Here a common vector v is subtracted from every point. ■

Proof. First let the consecutive differences of a real configuration X make angles of modulus less than $\pi/8$ with n_0. For every complex $\zeta = t + is$, choose a quarter-turn frame υ with $|t - \upsilon| \le \pi/4$. All the differences of $R_\zeta X$ lie in $R_\upsilon \mathsf{Tb}$, with tube distances at least

$$\min_a |x_{a+1} - x_a| \cos(3\pi/8) e^{-|s|}.$$

Thus $h_X(\zeta) = \mathcal{K}_n(R_\zeta X)$ is entire, $\pi/2$-periodic, and bounded by $C_X e^{\wp n|s|}$. Shifting the Fourier contour vertically shows that its coefficient at $e^{4im\zeta}$ vanishes if $4|m| > \wp n$. Fourier uniqueness on the real circle then gives a finite expansion, valid throughout the complex plane.

To define the coefficients at every Z, let $\mu = \#\mathcal{M}_n$ and invert the Vandermonde matrix $V_{lm} = e^{-4ml}$, $0 \le l < \mu$. If $W = V^{-1}$, put

$$c_m(Z) = \sum_{l=0}^{\mu-1} W_{ml}\mathcal{K}_n(R_{il}Z).$$

These functions are holomorphic. On each tube, permuted quarter-turned configurations of the preceding narrow angular type form a nonempty real open set. On that set the formula recovers the Fourier coefficients. The holomorphic identity theorem on the tube proves eq. D.lxii everywhere. The same Vandermonde comparison proves uniqueness and all the stated symmetry identities.

We give the phase argument explicitly because $\mathcal{O}_n$ need not contain an entire real rotation orbit. If X is a real point of a fixed tube, write its successive differences in that frame as $r_a n_{\vartheta_a}$, $|\vartheta_a| < \pi/2$. For

$$t \in I_X = \bigcap_a (-\pi/2 - \vartheta_a, \pi/2 - \vartheta_a),$$

the full vertical strip $R_{t+is}X$ remains in this tube. Analytically continuing eq. D.lxii in the parameter and comparing coefficients gives $c_m(R_tX) = e^{4imt}c_m(X)$ on I_X.

Consider next configurations with no coincident points and no perpendicular pair of distinct difference vectors. They form a rotation-invariant dense open set contained in $\mathcal{O}_n$. For density, the product of all squared differences and all such scalar products is a polynomial nonzero at $((a, a^2))_{a=1}^n$; on the line segment from any configuration to this one it has only finitely many zeros. For these configurations the preceding local phase identity holds along the entire orbit, so $t \mapsto e^{-4imt}c_m(R_tX)$ is locally constant on $\mathbb{R}$, hence constant. Continuity extends the identity to every $X, R_tX \in \mathcal{O}_n$.

For any configuration of distinct points, all but finitely many angles modulo π give distinct time coordinates. Choose such a t and define

$$\widetilde{c}_m(X) = e^{-4imt}c_m(R_tX).$$

The preceding identity makes this independent of t. Near each configuration one may keep t fixed, giving a holomorphic local extension and hence real analyticity. Translation invariance and the same definition give eq. D.lxiii, including rotations about arbitrary centers. □

Set $\bar{m}(0) = \bar{m}(1) = 0$. For $n \ge 2$, let $\bar{m}(n)$ be the largest $|m|$ for which c_m is nonzero for some labels, or zero if all kernels vanish. Then $0 \le \bar{m}(n) \le \wp n/4$. A nonzero harmonic is nonzero at some strictly time-ordered real configuration, after permuting labels: otherwise the identity theorem on a tube would make it zero. Conjugation shows that, if $\bar{m}(n) > 0$, the coefficient $c_{-\bar{m}(n)}$ is nonzero at such a configuration. Write $\widetilde{K}_n = \sum_m \widetilde{c}_m$.

D.5.2. *Positivity of the highest harmonic.*

Lemma D.28. For all $a, b \geq 0$,

$$2\bar{m}(a+b) \leq \bar{m}(2a) + \bar{m}(2b). \tag{D.lxiv}$$

For $a \geq 1$, let $t_a = \widetilde{c}^{(2a)}_{-\bar{m}(2a)}$ on labelled configurations, where a labelled configuration is the unordered set of its distinct points and their labels. On configurations of a points in $\mathbb{R}^2_+$,

$$G_a(S,T) = t_a(\theta S \cup T)$$

is a positive Hermitian kernel. If L is any line, and S, T lie strictly on one side, then

$$|t_a(\theta_{\mathsf{L}} S \cup T)|^2 \leq |t_a(\theta_{\mathsf{L}} S \cup S)||t_a(\theta_{\mathsf{L}} T \cup T)|. \tag{D.lxv}$$

■

Proof. For strictly time-ordered positive configurations x, z, the Gram formula and $(R_{is}x)^* = R_{is}x^*$ give

$$\langle \Psi(R_{is}x), \Psi(R_{is}z)\rangle = \mathcal{K}_{a+b}(R_{is}(\theta x_a, \dots, \theta x_1, z_1, \dots, z_b)).$$

An empty configuration gives Ω. If $\bar{m}(a+b) = 0$, eq. D.lxiv follows from nonnegativity. Otherwise, by symmetry, take $b \geq 1$. Choose a time-ordered configuration where $c_{-\bar{m}(a+b)} \neq 0$ and translate a separating time between its first a and last b points to zero; when $a = 0$, choose this time below the first point. Cauchy–Schwarz for the two resulting vectors bounds the squared expansion with leading exponent $8\bar{m}(a+b)s$ by a product whose maximal exponent is $4(\bar{m}(2a)+\bar{m}(2b))s$. Letting $s \to +\infty$ proves eq. D.lxiv.

For a finite family of a-point positive configurations, multiply its Gram matrix at R_{is} by $e^{-4\bar{m}(2a)s}$ and let $s \to +\infty$. Its limit is the matrix of G_a, hence positive Hermitian. If some times agree, perturb the kth time by $k\epsilon$, avoiding the finitely many exceptional ϵ, and use continuity of $\widetilde{c}_m$.

If the positive normal to L has angle υ and $v \in \mathsf{L}$, the map $\iota y = R_{-\upsilon}(y - v)$ takes its selected side to $\mathbb{R}^2_+$. The phase law gives

$$t_a(\theta_{\mathsf{L}} S \cup T) = e^{-4i\bar{m}(2a)\upsilon} G_a(\iota S, \iota T).$$

Cauchy–Schwarz for this positive kernel proves eq. D.lxv; the common phase has modulus one. □

Proposition D.29. One has

$$\bar{m}(2a) = a\bar{m}(2), \qquad \bar{m}(n) \leq \frac{n}{2}\bar{m}(2). \tag{D.lxvi}$$

■

Proof. We first prove $a \mid 2\bar{m}(2a)$. Let $\mathcal{N}_a(S)$ mean that t_a vanishes on every $2a$-point labelled configuration containing S.

We need a halving observation. Suppose S' has $j' \leq a$ points on one side of a line L, and $S \subset \theta_{\mathsf{L}} S' \cup S'$ satisfies $\mathcal{N}_a(S)$. Fix the points and labels of $\theta_{\mathsf{L}} S'$ and arbitrary labels on another $2a - j'$ points. The resulting value of t_a is real analytic in these free points on the complement of their coincidence sets. This domain is connected, since the excluded affine subspaces have codimension two. On the nonempty open subset where $a - j'$ free points lie on the negative side and a on the positive side, write the configuration as $\theta_{\mathsf{L}} T \cup T'$ with $S' \subset T$. The diagonal value $t_a(\theta_{\mathsf{L}} T \cup T)$ vanishes by $\mathcal{N}_a(S)$; eq. D.lxv makes the mixed value zero. The real-analytic identity theorem therefore gives $\mathcal{N}_a(\theta_{\mathsf{L}} S')$ on all configurations. Exchanging the two sides proves $\mathcal{N}_a(S')$ as well.

Suppose now $a \nmid 2\bar{m}(2a)$, and fix any label φ. Give all vertices of a regular $2a$-gon this label. Rotation through π/a preserves the labelled configuration, whereas the phase law multiplies its top harmonic by $e^{-4i\bar{m}(2a)\pi/a} \neq 1$. Thus $\mathcal{N}_a$ holds for every full polygon. We descend from arcs of length $j \geq 2$ to arcs of length $j' = \lceil j/2 \rceil$. Write the vertices as $P_k = p + \rho n_{\omega + k\pi/a}$. Reflection in the

line through p of direction $\omega+(k_0+1/2)\pi/a$ sends P_{k_0+k} to P_{k_0+1-k}. The arc $P_{k_0+1},\dots,P_{k_0+j'}$ lies strictly on one side, and its reflected union is an arc of length $2j'$ containing an arc of length j. The halving observation proves $\mathcal{N}_a$ for the shorter arc. Iteration reaches a singleton at any prescribed point. Since the label was arbitrary, t_a then vanishes on every configuration. This contradicts the definition of the nonzero top harmonic. Hence the divisibility holds.

Put $\nu(a)=\bar{m}(2a)$. By eq. D.lxiv, ν is a convex integer sequence with $\nu(0)=0$ and $\nu(a)\le \wp a/2$. Its increments are increasing bounded integers, hence eventually equal to an integer d. Thus $\nu(a)=da-c$ eventually, where $c=\sum_k(d-(\nu(k)-\nu(k-1)))\ge 0$ is a finite integer. Since $a\mid 2\nu(a)$, every sufficiently large a divides $2c$, forcing $c=0$. All increments equal $d=\nu(1)$. This proves the even identity. Apply eq. D.lxiv with $a=\lfloor n/2\rfloor$ and $b=n-a$ to obtain the second one. □

D.5.3. *From kernels to separated tests.*

Proposition D.30. For arbitrary profiles with pairwise disjoint supports,

$$\sigma(\alpha_1\otimes\varphi_1,\dots,\alpha_n\otimes\varphi_n)=\int\prod_a \alpha_a(y_a)\widetilde{K}_n(Y;\underline{\varphi})\,dY. \tag{D.lxvii}$$

If $\bar{m}(n)=0$, the functional is invariant under R_t on all n-tuples with pairwise disjoint planar projections. ■

Proof. Subtract the right side of eq. D.lxvii from the left and call the difference $D(\underline{\alpha})$. It is multilinear when each profile is restricted to its own separated compact support. We first use real profiles; complexification then gives the general statement.

Let A_a^0, A_a^1 be the time and space interval hulls of the supports. Suppose each pair whose A_a^1 intersect has a separating time level outside every A_e^0. Induct on the number of such pairs. When there are none, all supports lie in one permuted quarter-turned tube, so Lem. D.26 gives $D=0$. For the induction step choose a level c and split the profiles below and above it. Translate the upper group by $sn_{\pi/2}$. Both terms of D are real analytic in s. For the correlation this follows from the matrix element of $\mathsf{T}(2\delta,s)$, where $2\delta>0$ is a time gap about c; it is holomorphic on $|\operatorname{Im} s|<2\delta$. For the kernel integral, the compact product support remains in the distinct-point domain for every real s. Local holomorphic extensions on a finite cover of this compact set justify integration and give real analyticity. For large s all cross-group spatial intervals are disjoint, so the number of intersecting pairs decreases. The induction hypothesis and the identity theorem imply $D=0$ also at $s=0$.

For general profiles with separation δ, partition the plane into the C^1 squares obtained from $\chi(2y^0/\epsilon-j^0)\chi(2y^1/\epsilon-j^1)$, with $0<\epsilon<\delta/(n+1)$. Each nonzero tuple of pieces has interval lengths at most ϵ. If two spatial intervals overlap, their time intervals have a gap at least $\sqrt{\delta^2-4\epsilon^2}>(n-2)\epsilon$. The other $n-2$ time intervals cannot cover this gap, so there is a level outside all intervals. The preceding argument applies to every tuple of pieces; multilinearity proves eq. D.lxvii.

If $\bar{m}(n)=0$, the phase law makes $\widetilde{K}_n=\tilde{c}_0$ invariant under all real rotations. Changing variables in eq. D.lxvii proves invariance for plates. For a general tuple with disjoint planar projections, convolve each entry with the product mollifier

$$J_\epsilon(x)=\prod_{\mu=1}^{3}\frac{2}{\epsilon}\chi(2x_\mu/\epsilon).$$

For fixed sufficiently small ϵ, the convolution Riemann sums are finite sums of translates of J_ϵ, each of which is a plate. Their planar supports stay in fixed mutually disjoint neighborhoods. Multilinearity proves rotation invariance for these sums. They converge in C^1 to the convolutions, which converge in C^1 to the original entries. The telescoping consequence of (R2),

$$|\sigma(g_1,\dots,g_n)-\sigma(g_1',\dots,g_n')|\le C_n^\sigma(d;R)\sum_a\|g_a-g_a'\|_\sharp\prod_{b\ne a}\max(\|g_b\|_\sharp,\|g_b'\|_\sharp),$$

on fixed separated compact supports permits both limits. □

Theorem D.31 (Rotation invariance)**.** Every functional satisfying (R0)–(R7) is invariant under O(3). ■□

Proof. The two-point result Prop. D.20 and the identification $K_2 = k$ show that $K_2(R_{is}\xi)$ is independent of s for real $\xi \in \mathbb{R}^2_+$. Comparing the coefficients in eq. D.lxii, and then applying the identity theorem, gives $\bar{m}(2) = 0$. By eq. D.lxvi, all $\bar{m}(n)$ vanish. Prop. D.30 gives invariance under R_t on tuples with disjoint planar projections, and Prop. D.16 gives it on every separated tuple. The empty and one-point values are invariant by normalization and (R0).

Cubic conjugation gives rotations about the other coordinate axes. These generate SO(3): rotate an arbitrary unit vector to e_3 by two coordinate-plane rotations, then the remaining transformation fixes e_3 and is a rotation in its perpendicular plane. A coordinate reflection already belongs to the cubic group, so adjoining it gives all of O(3). □

Appendix E. Clustering and decomposition

E.1. **Uniform clustering on a lattice.** The estimate needed for decomposition into continuum components must be uniform as the spacing tends to zero. A decay estimate involving the supremum of a lattice polynomial would not suffice. We instead bound correlations by the reflection norms of their local factors. The argument uses coordinate transfer operators, the spectral restriction imposed by a diagonal reflection, and a Fourier cutoff which removes zero energy.

Throughout this section, ν is a translation-invariant probability law on the configurations over $a\mathbb{Z}^3$, $a > 0$. We assume RP at the six coordinate mirrors $\Sigma(e_\mu, 0)$ and $\Sigma(e_\mu, a/2)$, and at one diagonal mirror $\Sigma(e_\mu + \epsilon e_\lambda, c)$ in each coordinate plane, where $\epsilon \in \{1, -1\}$ and $c \in a\mathbb{Z}$. Translations move these diagonal mirrors to height zero and move the coordinate mirrors to every height in $(a/2)\mathbb{Z}$. In particular these hypotheses hold for every periodic Gibbs state by Prop. B.28.

E.1.1. *Transfer operators and the lattice spectral restriction.* For a mirror $\Sigma(n, c)$ and a positive normal lattice vector v, suppose RP also holds at $\Sigma(n, c + n \cdot v/2)$. Complete the bounded local functions supported in the upper half-space in the form $\langle [F], [G] \rangle = \nu(\overline{\theta_{n,c}F}\, G)$, after quotienting its null space. This is the reflection Hilbert space at the mirror. Coordinate pictures use $n = e_\mu$ and $v = ae_\mu$; diagonal pictures use $n = e_\mu + \epsilon e_\lambda$ and $v = an$.

Lemma E.1. In these pictures the normal translation and the tangential translations induce commuting operators

$$T[F] = [\tau_v F], \qquad V(w)[F] = [\tau_w F] \quad (n \cdot w = 0), \qquad 0 \le T \le 1,$$

where the $V(w)$ are unitary. Moreover

$$\langle [F], T^k V(w)[G] \rangle = \nu(\overline{\theta_{n,c}F}\, \tau_{kv+w} G), \qquad k \ge 0, \tag{E.i}$$

and $\|[F]\|^2 \le \nu(|F|^2) \le \|F\|^2_\infty$. In a coordinate picture there is a joint spectral measure $\mathfrak{E}$ on $[0, 1] \times (S^1)^2$ such that

$$T^k V(ay) = \int t^k \zeta^y \, d\mathfrak{E}(t, \zeta), \qquad y_\mu = 0. \tag{E.ii}$$

■

Proof. Polarization of RP, followed by a translation of an arbitrary local function into a half-space, gives invariance under the mirror. Ordinary Cauchy–Schwarz therefore bounds the reflection norm by the $L^2(\nu)$ norm. Normal translation is symmetric in the reflection form. Reflection Cauchy–Schwarz gives

$$\|[\tau_v F]\|^2 \le \|[F]\| \, \|[\tau_{2v} F]\|.$$

Iteration and the uniform bound by $\|F\|_\infty$ show $\|[\tau_v F]\| \le \|[F]\|$. Thus translation descends through the null space to a self-adjoint contraction. The identity $\theta_{n,c} = \theta_{n,c+n\cdot v/2}\tau_v$ and RP

at the second mirror show $\langle [F], T[F]\rangle \ge 0$. Tangential translations preserve the form and have inverse translations, hence induce unitaries. Their commutation and eq. E.i follow on local classes and extend by continuity. The joint spectral theorem for the commuting normal operators $T, V(ae_\lambda), V(ae_{\lambda'})$ gives eq. E.ii. □

The following elementary complex-variable estimate isolates the spectral consequence of comparing a coordinate picture with a diagonal picture.

Lemma E.2. Suppose a finite positive measure η on the closed unit disc satisfies, for every polynomial q,

$$\int |q|^2 \, d\eta \le C \sup_{[0,1]} |q| \sup_{S^1} |q|. \tag{E.iii}$$

Then $\eta\{|1+z|^2 < 4|z|\} = 0$. ■

Proof. Uniform Taylor approximation extends eq. E.iii to functions holomorphic on a disc of radius greater than one. On the unit disc put

$$\Lambda(\xi) = \log \frac{1+\xi}{1-\xi}, \qquad g_s(z) = \cos\bigl(s\Lambda(\sqrt{z})\bigr), \qquad s > 0.$$

The principal logarithm is analytic because its argument is in the right half-plane. Since Λ is odd and cosine is even, g_s is a single-valued holomorphic function of z. Write $A(z) = |\operatorname{Im} \Lambda(\sqrt{z})|$. Then

$$\sinh^2(sA(z)) \le |g_s(z)|^2 \le \cosh^2(sA(z)), \qquad 0 \le A(z) < \pi/2,$$

and $A = 0$ on $[0,1)$. Applying eq. E.iii to $g_s(rz)$, $0 < r < 1$, gives

$$\int \sinh^2(sA(rz)) \, d\eta(z) \le C \cosh(s\pi/2).$$

Extend A to the unit circle by $A = \pi/2$ away from 1 and $A(1) = 0$; these are its radial limits. Bounded convergence allows $r \uparrow 1$. For every $\delta > 0$ the last inequality implies

$$\eta\{A \ge \pi/4 + \delta\} \le 16 C e^{-2s\delta}$$

for all sufficiently large s, so this set is null. Inside the disc,

$$\tan A(z) = \frac{\sqrt{2(|z| - \operatorname{Re} z)}}{1 - |z|}.$$

Consequently $A(z) > \pi/4$ is equivalent to $|1+z|^2 < 4|z|$, with the same equivalence at the boundary by the radial values. A countable union over $\delta > 0$ proves the claim. □

Proposition E.3. In every coordinate picture and for each tangential direction $\lambda \ne \mu$,

$$(1-T)^2 \ge T\bigl(2 - V(ae_\lambda) - V(ae_\lambda)^*\bigr). \tag{E.iv}$$

Equivalently, its joint spectral measure is supported where $(1-t)^2 \ge t|1-\zeta_\lambda|^2$. In particular, with

$$\mathsf{k} = \frac{2}{\pi} \log(1+\sqrt{2}) > 0, \tag{E.v}$$

the spectral measure is supported where

$$t \le \exp\{-\mathsf{k} |\arg \zeta_\lambda|\}. \tag{E.vi}$$

■

Proof. Use the available diagonal normal $n = e_\mu + \epsilon e_\lambda$ and write $n^\perp = e_\mu - \epsilon e_\lambda$. For a local upper-half-space function F, let ς be the image of its spectral measure $\langle [F], \mathfrak{E}(\cdot)[F]\rangle$ under $(t,\zeta) \mapsto z = t\zeta_\lambda^\epsilon$. Expansion of a polynomial $q(z) = \sum q_k z^k$ gives

$$\int |q(z)|^2 |z|^{2N} \, d\varsigma = \sum_{k,l} q_k \overline{q_l} \, \nu\bigl(\overline{\theta F} \, \tau_{kan + lan^\perp} \tau_{2Nae_\mu} F\bigr). \tag{E.vii}$$

Choose a diagonal mirror $\Sigma(n,c)$ beyond every endpoint of the bonds supporting θF, then choose N so that every endpoint supporting $\tau_{2Nae_\mu}F$ lies above that mirror. Both choices depend only on F. Set $F_1 = \theta_{n,c}\theta F$ and $F_2 = \tau_{2Nae_\mu}F$. Both functions lie in its upper half-space. In the diagonal picture, the right side of eq. E.vii equals

$$\langle [F_1], q(T_\parallel)\overline{q}(V_\perp)[F_2]\rangle,$$

where $\overline{q}$ denotes coefficient conjugation. Its modulus is at most $\|F\|_\infty^2 \sup_{[0,1]}|q| \sup_{S^1}|q|$. Lem. E.2 applied to $|z|^{2N}d\varsigma$ shows that ς gives no mass to $|1+z|^2 < 4|z|$: the weight is strictly positive there, since $z=0$ is outside this set. The identity

$$|1+t\zeta_\lambda^\epsilon|^2 - 4t = (1-t)^2 - t|1-\zeta_\lambda|^2$$

proves the spectral restriction for every local vector, hence for the whole spectral measure by density. Functional calculus gives eq. E.iv.

For $t = e^{-x} > 0$ and $\zeta_\lambda = e^{iy}$, the restriction is $\sinh(x/2) \geq |\sin(y/2)|$. The function $2\operatorname{arcsinh}(\sin(y/2))$ is concave on $[0,\pi]$: its derivative is $\cos(y/2)(1+\sin^2(y/2))^{-1/2}$, a product of nonnegative decreasing functions. It therefore lies above the chord $\mathsf{k}y$. This proves eq. E.vi; the case $t=0$ is immediate. □

E.1.2. *The invariant part of a correlation.* Let $\mathcal{I}$ be the sigma-algebra of translation-invariant Borel events, completed modulo ν-null sets. In a coordinate picture let Π_0 be the orthogonal projection onto $\ker(1-T)$. It describes the nondecaying part of a correlation.

Proposition E.4. One has $T^n \to \Pi_0$ strongly and $\Pi_0 V(w) = \Pi_0$ for tangential translations. Moreover

$$\langle [F], \Pi_0[G]\rangle = \nu(\overline{E_\nu[F \mid \mathcal{I}]}E_\nu[G \mid \mathcal{I}]). \tag{E.viii}$$

Conditional expectation onto $\mathcal{I}$ is unchanged by translations or coordinate reflections. In particular, ν is ergodic precisely when $\ker(1-T) = \mathbb{C}[1]$. ■

Proof. By the spectral theorem, $\Pi_0 = \mathfrak{E}(\{t=1\})$ and $T^n \to \Pi_0$ strongly. Prop. E.3 in both tangential directions says that $t=1$ forces $\zeta = (1,1)$, proving $\Pi_0 V(w) = \Pi_0$. Let P_μ be the $L^2(\nu)$ projection onto the functions fixed by τ_{ae_μ}. The mean ergodic theorem and eq. E.i give, for upper-half-space functions,

$$\langle [F], \Pi_0[G]\rangle = \langle \theta F, P_\mu G\rangle_{L^2(\nu)}. \tag{E.ix}$$

Since $\Pi_0 V(w) = \Pi_0$, the right side is unchanged on replacing G by $\tau_w G$. For arbitrary local F_1, F_2, choose $u = kae_\mu$ large enough that $F = \theta\tau_{-u}F_1$ and $G = \tau_u F_2$ are in the upper half-space. Substitution in this identity, using $P_\mu \tau_{\pm u} = P_\mu$, gives

$$\langle F_1, P_\mu \tau_w F_2\rangle_{L^2(\nu)} = \langle F_1, P_\mu F_2\rangle_{L^2(\nu)}.$$

Local functions are dense in $L^2(\nu)$, so $P_\mu\tau_w = P_\mu$ and, by adjoints, $\tau_w P_\mu = P_\mu$. Thus a function fixed by the one normal translation is fixed by every lattice translation. The invariant subspace is $L^2(\mathcal{I},\nu)$, so $P_\mu = E_\nu[\,\cdot \mid \mathcal{I}]$.

Reflection preserves that subspace and its orthogonal complement, hence commutes with P_μ. Positivity of the left side of eq. E.ix at $G=F$ implies $\langle \theta P_\mu F, P_\mu F\rangle_{L^2(\nu)} \geq 0$. The vectors $P_\mu F$ with F in the upper half-space are dense in the invariant subspace: translate each local function there without changing its projection. Reflection is consequently a positive involution on that subspace and must be the identity; its negative eigenspace is zero. Formula eq. E.ix now becomes eq. E.viii. If ν is ergodic, the right side is $\overline{\nu(F)}\nu(G)$, so Π_0 has range $\mathbb{C}[1]$. Conversely that range identity implies $\|E_\nu[F \mid \mathcal{I}]\|_2 = |\nu(F)|$, first for local F and then for all L^2 functions. For $F = 1_E$, $E \in \mathcal{I}$, it says $\nu(E)^{1/2} = \nu(E)$, proving ergodicity. □

E.1.3. *A bounded density after removing zero energy.* Write $Q_r = [-ra, ra]^3$, where $r \geq 1$ is an integer, and $\theta_\mu = \theta_{e_\mu,0}$. For a bounded Borel function A of the bonds in Q_r, use the site picture to set

$$\psi_A^\mu = [\tau_{rae_\mu} A]_\mu.$$

If S_8 is the group of coordinate sign changes, define

$$\mathcal{N}_r(A) = \max_{\mu,\, s\in\mathsf{S}_8} \|\psi_{sA}^\mu\|^2, \tag{E.x}$$

$$\mathcal{N}'_{r,\mu}(A) = \max\left\{\|\psi_A^\mu\|^2, \max_{\lambda\neq\mu} \|\psi_{\theta_\lambda A}^\lambda\|\|\psi_A^\lambda\|\right\}. \tag{E.xi}$$

Then $\mathcal{N}'_{r,\mu}(A) \leq \mathcal{N}_r(A) \leq \nu(|A|^2) \leq \|A\|_\infty^2$. The square root of $\mathcal{N}_r$ is a seminorm, invariant under S_8. Consequently the parity parts $A_\pm = (A \pm \theta_\mu A)/2$ satisfy $\mathcal{N}_r(A_\pm) \leq \mathcal{N}_r(A)$. Translation in the reflection form also gives

$$\nu(\overline{A}\tau_x B) = \langle \psi_{\theta_\mu A}^\mu, T_\mu^{x_\mu/a-2r} V_\mu(x_\perp)\psi_B^\mu\rangle, \qquad x_\mu \geq 2ra. \tag{E.xii}$$

The correlation at $-x$ is the conjugate of $\nu(\overline{B}\tau_x A)$; coordinate reflection changes x_μ to $-x_\mu$ and reflects both functions. These identities allow the same correlation to be estimated in any coordinate picture.

We require a scalar weight which vanishes at $t = 1$ but controls $1 - t$ from below. Let z_m be the even, period-four sequence $(z_0, z_1, z_2, z_3) = (0, 1, -2, 1)$, and let

$$B(u) = 8(1 - 4|u|/\pi)_+ \quad (-\pi \leq u \leq \pi)$$

be extended with period 2π. Its Fourier coefficients are

$$b_0 = 1, \qquad b_m = \left(\frac{8\sin(m\pi/8)}{m\pi}\right)^2 \quad (m \neq 0).$$

Integration by parts gives this formula; in particular the Fourier series converges absolutely. Put

$$B_z(u) = \sum_{m\in\mathbb{Z}} z_m b_m e^{imu}, \qquad \Psi(t) = 2\sum_{m\geq 1} z_m b_m t^{m-1}, \quad 0 \leq t \leq 1. \tag{E.xiii}$$

The convention $0^0 = 1$ is used here.

Lemma E.5. The above series converge uniformly. One has

$$\sum_m |z_m| b_m = 8, \qquad |B_z| \leq 8, \qquad B_z(u) = 0 \quad \text{if } \operatorname{dist}(u, 2\pi\mathbb{Z}) \leq \pi/4.$$

Moreover $0 \leq \Psi \leq 8$, $\Psi(1) = 0$, $\Psi(0) = 2b_1 > 0$, and

$$\Psi(e^{-x}) \geq c_\Psi \tanh x \quad (x > 0), \qquad c_\Psi = (1 + \sqrt{2})^{-1}. \tag{E.xiv}$$

■

Proof. The identities $|z_m| = 1 - \cos(m\pi/2)$ and $z_m = \cos(m\pi/2) - \cos(m\pi)$ give

$$\sum_m |z_m| b_m = B(0) - B(\pi/2) = 8, \qquad B_z(u) = \tfrac{1}{2}B(u + \pi/2) + \tfrac{1}{2}B(u - \pi/2) - B(u + \pi).$$

The three translated triangles have disjoint interiors of their supports and vanish for $\operatorname{dist}(u, 2\pi\mathbb{Z}) \leq \pi/4$. This proves the stated bounds and, at $u = 0$, $\Psi(1) = 0$. Absolute summability gives the uniform convergence. For $0 < t < 1$, the Poisson kernel $P_t(u) = (1 - t^2)/(1 - 2t\cos u + t^2)$ yields

$$t\Psi(t) = \frac{1}{2\pi}\int_{-\pi/4}^{\pi/4} B(v)\left[\tfrac{1}{2}P_t(v - \pi/2) + \tfrac{1}{2}P_t(v + \pi/2) - P_t(v - \pi)\right] dv.$$

Writing $s = 1 + t^2$, the bracket is at least

$$(1-t^2)\left[\frac{1}{s} - \frac{1}{s+\sqrt{2}t}\right] = \frac{(1-t^2)\sqrt{2}t}{s(s+\sqrt{2}t)} :$$

the average of $(s \mp 2t\sin v)^{-1}$ is at least $1/s$, whereas $\cos v \geq 1/\sqrt{2}$. Since $(2\pi)^{-1}\int B = 1$, division by t gives

$$\Psi(t) \geq \frac{1-t^2}{1+t^2}\frac{\sqrt{2}}{1+t^2+\sqrt{2}t} \geq c_\Psi \frac{1-t^2}{1+t^2}.$$

This is eq. E.xiv. Together with $\Psi(0) > 0$ and $\Psi(1) = 0$ it proves positivity on $[0, 1)$. □

Set

$$C_{\mathrm{dn}} = \frac{1}{4\pi^2}\left[128 + 64\left(\frac{16}{\pi\mathsf{k}} + 2 + \frac{64}{(\pi\mathsf{k})^2}\right)\right]. \tag{E.xv}$$

Lemma E.6. If Φ is supported in Q_r and $\theta_\mu\Phi = \epsilon\Phi$, $\epsilon \in \{1, -1\}$, let $\varrho_\psi = \langle\psi, \mathfrak{E}_\mu(\cdot)\psi\rangle$, where $\psi = \psi^\mu_\Phi$. The image of $\Psi(t^{2r})d\varrho_\psi(t,\zeta)$ on $(S^1)^2$ has a continuous density h relative to unnormalized angular Lebesgue measure $d\alpha$ on $(-\pi,\pi]^2$, and

$$0 \leq h \leq C_{\mathrm{dn}} r^2 \mathcal{N}'_{r,\mu}(\Phi). \tag{E.xvi}$$

■

Proof. Write $u(v) = \nu(\overline{\theta_\mu\Phi}\tau_v\Phi) = \epsilon\nu(\overline{\Phi}\tau_v\Phi)$ and

$$K(y) = \sum_{m\in\mathbb{Z}} z_m b_m\, u(a(2rme_\mu + y)), \qquad y_\mu = 0.$$

The series is absolutely convergent. Reflection parity makes $u(a(-ke_\mu + y)) = u(a(ke_\mu + y))$. By eq. E.xii, for $k \geq 2r$,

$$u(a(ke_\mu + y)) = \langle\psi, T_\mu^{k-2r} V_\mu(ay)\psi\rangle.$$

Pairing m with $-m$, and observing $z_0 = 0$, therefore gives

$$K(y) = \int \Psi(t^{2r})\zeta^y\, d\varrho_\psi(t,\zeta), \qquad |K(y)| \leq 8\|\psi\|^2. \tag{E.xvii}$$

Thus K is the sequence of Fourier coefficients of the weighted momentum measure.

To bound large y, choose a tangential coordinate λ with $|y_\lambda| = |y|_\infty$. Since $K(-y) = \overline{K(y)}$, assume $y_\lambda \geq 2r$. Applying eq. E.xii in the λ picture and summing its absolutely convergent series gives

$$K(y) = \epsilon\langle\psi^\lambda_{\theta_\lambda\Phi}, X\psi^\lambda_\Phi\rangle, \quad X = \int t^{y_\lambda - 2r} B_z(2r\arg\zeta_\mu)\zeta_{\lambda'}^{y_{\lambda'}}\, d\mathfrak{E}_\lambda.$$

Where $B_z(2r\arg\zeta_\mu)$ is nonzero, $|\arg\zeta_\mu| > \pi/(8r)$. The lattice spectral restriction eq. E.vi then gives $t \leq e^{-\pi\mathsf{k}/(8r)}$. Consequently

$$|K(y)| \leq 8e^{-\pi\mathsf{k}(|y|_\infty - 2r)/(8r)}\mathcal{N}'_{r,\mu}(\Phi), \qquad |y|_\infty \geq 2r. \tag{E.xviii}$$

There are at most $16r^2$ points with $|y|_\infty < 2r$ and $8j$ points with $|y|_\infty = j$. For $c = \pi\mathsf{k}/(8r)$,

$$\sum_{j\geq 2r} 64je^{-c(j-2r)} = 64\left[\frac{2r}{1-e^{-c}} + \frac{e^{-c}}{(1-e^{-c})^2}\right] \leq 64\left[2r + \frac{2r}{c} + \frac{1}{c^2}\right].$$

Here $(1-e^{-c})^{-1} \leq 1 + c^{-1}$ and $e^{-c}(1-e^{-c})^{-2} \leq c^{-2}$. Together with eq. E.xvii, this proves

$$\sum_y |K(y)| \leq 4\pi^2 C_{\mathrm{dn}} r^2 \mathcal{N}'_{r,\mu}(\Phi).$$

The Fourier series $h(\zeta) = (2\pi)^{-2}\sum_y K(y)\zeta^{-y}$ is therefore uniformly convergent and bounded by the right side of eq. E.xvi. Its angular measure has the same Fourier coefficients as the weighted

momentum measure by eq. E.xvii. Density of trigonometric polynomials gives equality of the measures. Positivity of the weighted measure and continuity of h imply $h \geq 0$ everywhere. □

E.1.4. *The clustering estimate.*

Proposition E.7. Under the nine-mirror hypotheses of this section, put

$$C_{\mathrm{dc}} = 16 c_\Psi^{-1} \mathsf{k}^{-2} C_{\mathrm{dn}}. \tag{E.xix}$$

For every A, B supported in Q_r and every integer $n \geq 1$,

$$0 \leq \langle \psi_A^\mu, T_\mu^n (1 - \Pi_0) \psi_A^\mu \rangle \leq C_{\mathrm{dc}} \mathcal{N}_r(A) \left(\frac{r}{n} + 2\frac{r^2}{n^2} \right). \tag{E.xx}$$

If $x \in a\mathbb{Z}^3$ and $n = |x|_\infty / a - 2r \geq 1$, then

$$\left| \nu(\overline{A} \tau_x B) - \nu(\overline{E_\nu[A \mid \mathcal{I}]} E_\nu[B \mid \mathcal{I}]) \right| \leq C_{\mathrm{dc}} \mathcal{N}_r(A)^{1/2} \mathcal{N}_r(B)^{1/2} \left(\frac{r}{n} + 2\frac{r^2}{n^2} \right). \tag{E.xxi}$$

In particular, for $|x| \geq 4\sqrt{3}\, ra$,

$$\left| \nu(\overline{A} \tau_x B) - \nu(\overline{E_\nu[A \mid \mathcal{I}]} E_\nu[B \mid \mathcal{I}]) \right| \leq 4\sqrt{3} C_{\mathrm{dc}} \mathcal{N}_r(A)^{1/2} \mathcal{N}_r(B)^{1/2} \frac{ra}{|x|}. \tag{E.xxii}$$

The constants are independent of the spacing, the law, and the observables. For an ergodic law the subtracted term is $\overline{\nu(A)} \nu(B)$. ■

Proof. Set $D_n = T_\mu^n (1 - \Pi_0) = \int t^n 1_{t<1} \, d\mathfrak{E}_\mu$. It is positive and commutes with tangential translations. First suppose $\theta_\mu \Phi = \pm \Phi$ and let $\psi = \psi_\Phi^\mu$. By Lem. E.6, the weighted momentum measure has a bounded density. It gives no mass to $\zeta = (1,1)$; since $\Psi(t^{2r}) > 0$ for $t < 1$, the same is true of ϱ_ψ on $\{t < 1, \zeta = (1,1)\}$. For $0 < t < 1$ write $s = -\log t$. The two tangential versions of eq. E.vi imply $s \geq \mathsf{k}|\alpha|_\infty$, where $\zeta = e^{i\alpha}$. Since $s \mapsto e^{-ns} / \tanh(2rs)$ is decreasing and $\Psi(t^{2r}) \geq c_\Psi \tanh(2rs)$,

$$t^n \leq c_\Psi^{-1} \frac{e^{-n\mathsf{k}|\alpha|_\infty}}{\tanh(2r\mathsf{k}|\alpha|_\infty)} \Psi(t^{2r})$$

away from the null point $\alpha = 0$. At $t = 0$, the integrand t^n is zero. Integrating against the bounded density and using $1/\tanh u \leq 1 + 1/u$ gives

$$\begin{aligned} \langle \psi, D_n \psi \rangle &\leq c_\Psi^{-1} C_{\mathrm{dn}} r^2 \mathcal{N}'_{r,\mu}(\Phi) \int_{(-\pi,\pi]^2} \frac{e^{-n\mathsf{k}|\alpha|_\infty}}{\tanh(2r\mathsf{k}|\alpha|_\infty)} \, d\alpha \\ &\leq c_\Psi^{-1} C_{\mathrm{dn}} \mathsf{k}^{-2} \mathcal{N}'_{r,\mu}(\Phi) \left(\frac{8r^2}{n^2} + \frac{4r}{n} \right). \end{aligned}$$

Indeed the square integral equals $8 \int_0^\pi u e^{-n\mathsf{k}u} / \tanh(2r\mathsf{k}u) \, du$; extend the upper limit to infinity to obtain the displayed bound. For arbitrary $A = A_+ + A_-$, positivity yields $\langle \psi_A, D_n \psi_A \rangle \leq 2\langle \psi_{A_+}, D_n \psi_{A_+} \rangle + 2\langle \psi_{A_-}, D_n \psi_{A_-} \rangle$. Use $\mathcal{N}'_{r,\mu}(A_\pm) \leq \mathcal{N}_r(A)$ to obtain eq. E.xx with eq. E.xix.

For eq. E.xxi, choose μ with $|x_\mu| = |x|_\infty$ and first suppose $x_\mu > 0$. Insert $T_\mu^n = \Pi_0 + D_n$ into eq. E.xii. Prop. E.4 identifies the Π_0 term with the subtracted conditional expectation. Cauchy–Schwarz for D_n, its commutation with $V_\mu(x_\perp)$, and eq. E.xx bound the remainder as claimed. For $x_\mu < 0$, conjugate and interchange A, B. If $|x|_\infty \geq 4ra$, then $n \geq |x|_\infty/(2a) \geq 2r$, and $r/n + 2r^2/n^2 \leq 4ra/|x|_\infty$. Finally $|x|_\infty \geq |x|/\sqrt{3}$ proves eq. E.xxii. The ergodic assertion follows from Prop. E.4. □

E.2. **Decomposition by invariant conditional expectations.** A continuum limit of invariant states need not cluster. We construct its components from conditional expectations on the lattice. The essential estimate is uniform in the cutoff: two local products approach the product of their invariant conditional expectations at rate inverse distance. This identifies all moments of the component measure before a component is selected.

Definition E.8 (Classes, class systems and bound systems)**.** At cutoff K, let $\mathcal{I}$ be the Borel sigma-field of events invariant under lattice translations.

(1) A class $\mathcal{G}$ at cutoff K is a nonempty set of lattice laws satisfying the following conditions.
  (Cls1) The laws are invariant under lattice isometries and reflection positive at every lattice mirror.
  (Cls2) If $\nu \in \mathcal{G}$ and h is bounded, nonnegative and $\mathcal{I}$-measurable with $\nu(h) > 0$, then $\nu_h = h\nu/\nu(h)$ belongs to $\mathcal{G}$.

(2) A class system is a sequence $(K_j, \mathcal{G}_j)$ with $K_j \to \infty$ and $\mathcal{G}_j$ a class at cutoff K_j.

(3) A bound system consists of finite positive numbers $C_1(R)$ and $C_n^\infty(d;R)$ for $n \geq 2$, $d \in (0,\infty]$ and $R > 0$. All are nondecreasing in R, and the latter are nonincreasing in d.

(4) The class system obeys this bound system if, in every class,

$$|S_n^T[\rho](X)| \leq C_n^\infty(d;R)\|X\|_\sharp \qquad \text{(Bd1)} \quad (n \geq 2,\ X \text{ of type } (d;R)), \tag{E.xxiii}$$

$$|\rho(\mathcal{O}_K(f)) - \nu(\mathcal{O}_K(f))| \leq C_1(R)\|f\|_\sharp \qquad \text{(Bd2)} \quad (\rho, \nu \in \mathcal{G}). \tag{E.xxiv}$$

  In the second line $\operatorname{supp} f$ lies in a ball of radius R.

(5) The same table applies to every class in the system and to every subsequence of the system, by restriction. We write $\underline{C} = (C_n^\infty)_{n\geq 2}$ for its connected-function table.

■

For the remainder of this section, fix a class system obeying one bound system in the sense of (Bd1)–(Bd2).

Definition E.9 (Frames and limits in a frame)**.** A *frame* is a choice $\nu_j \in \mathcal{G}_j$. It converges if

$$\sigma(X) = \lim_j \nu_j(A_j(X)), \qquad A_j(X) := X_{K_j}^{\nu_j}, \tag{E.xxv}$$

exists for every separated tuple. A limit in this frame is, more generally, $\lim_j \rho_j(A_j(X))$ with $\rho_j \in \mathcal{G}_j$. The centring law remains ν_j when the integrating law changes. Every such limit is normalized, real, symmetric and multilinear on separated supports. Its connected functions are the cumulant polynomials of eq. C.vi. ■

Write $d(X)$ for the separation and $R(X)$ for the largest of the minimal radii of balls containing individual supports. Set $d(X) = \infty$ for a tuple of length at most one. Define

$$c_n(d;R) = \sum_{\pi \in \mathrm{Part}(n)} \prod_{J\in\pi} b_{|J|}, \qquad b_1 = C_1(R), \quad b_k = C_k^\infty(d;R)\ (k \geq 2), \quad c_0 = 1. \tag{E.xxvi}$$

The moment bound and the bound used in reflections are

$$C_{\mathrm{mom}}(X) = c_{|X|}(d(X);R(X))\|X\|_\sharp, \tag{E.xxvii}$$

$$\delta(X;\bar{R}) = \operatorname{dist}_\infty(\operatorname{supp} X, \mathbb{R}^3 \setminus (-\bar{R},\bar{R})^3), \tag{E.xxviii}$$

$$N(X;\bar{R}) = c_{2|X|}(\min(d(X), 2\delta(X;\bar{R}));R(X))\|X\|_\sharp^2. \tag{E.xxix}$$

Here the support is contained in the indicated open cube. Both bounds are zero if a tuple has a zero entry, and both are one on the empty tuple. These conventions avoid evaluating c_n at $R = 0$. Extend C_{mom} to finite formal sums by the triangle inequality. Put

$$k_1 = 4\sqrt{3}, \qquad k_0 = 4\sqrt{3}\, C_{\mathrm{dc}}, \tag{E.xxx}$$

where C_{dc} is the absolute constant in the lattice clustering estimate of Prop. E.7.

Proposition E.10 (Uniform conditional bounds and clustering)**.** For $\rho, \nu \in \mathcal{G}$ one has $|\rho(X_K^\nu)| \le C_{\mathrm{mom}}(X)$. If $\operatorname{supp} f_i \cup \operatorname{supp} f_i' \subset Q_i$, the sets Q_i lie in balls of radius R and are mutually at distance at least d, then

$$|\rho(X_K^\nu - X_K'^\nu)| \le c_n(d;R) \sum_i \|f_i - f_i'\|_\sharp \prod_{l \ne i} \max(\|f_l\|_\sharp, \|f_l'\|_\sharp). \tag{E.xxxi}$$

The same bounds hold almost surely for conditional expectations given $\mathcal{I}$.

Suppose $\bar R \in a_K\mathbb{N}$, both tuples are supported in $(-\bar R, \bar R)^3$, and $a_K \le 2\min(\delta(X;\bar R), \delta(Y;\bar R))$. For $v \in a_K\mathbb{Z}^3$, $|v| \ge k_1 \bar R$,

$$|\rho((X \cdot \tau_v Y)_K^\nu) - \rho(E_\rho[X_K^\nu \mid \mathcal{I}]\, E_\rho[Y_K^\nu \mid \mathcal{I}])| \le k_0 \sqrt{N(X;\bar R) N(Y;\bar R)}\, \frac{\bar R}{|v|}. \tag{E.xxxii}$$

All constants are independent of the cutoff and of the two laws. ∎

Proof. The moment–cumulant identity for two centring laws is

$$\rho(X_K^\nu) = \sum_{\pi \in \mathrm{Part}(n)} \prod_{J \in \pi} \kappa_J, \qquad \kappa_{\{i\}} = \rho(\mathcal{O}_K(f_i)) - \nu(\mathcal{O}_K(f_i)), \quad \kappa_J = S_{|J|}^T[\rho](X_J)\ (|J| \ge 2). \tag{E.xxxiii}$$

Applying eq. E.xxiii–eq. E.xxiv block by block gives $c_n(d;R)\|X\|_\sharp$ on any tuple of type $(d;R)$. Telescope the product defining $X_K^\nu - X_K'^\nu$ one factor at a time. Each resulting tuple remains supported in the same sets Q_i; the preceding bound gives eq. E.xxxi.

For a bounded Borel function A and an invariant density h, every version $e = E_\rho[A \mid \mathcal{I}]$ remains a version under ρ_h, since

$$\rho_h(kA) = \rho(hkA)/\rho(h) = \rho(hke)/\rho(h)$$

for every bounded invariant k. If $\rho'(A) \le c$ for every $\rho' \in \mathcal{G}$, condition on $\{E_\rho[A \mid \mathcal{I}] > c + \varepsilon\}$. A positive probability for this event would contradict that bound, by Def. E.8. Applying this observation to both signs of X_K^ν and of its difference from $X_K'^\nu$ proves the conditional estimates. A version can be changed to zero on its exceptional null set to make the absolute bound hold everywhere.

Put $r = \bar R/a_K$. A plaquette lies within sup-norm distance $a_K/2$ of its barycentre. The margin assumption therefore makes $A = X_K^\nu$ and $B = Y_K^\nu$ functions of the bonds in $[-ra_K, ra_K]^3$. Lattice covariance of $\mathcal{O}_K$ and invariance of ν give

$$(\mathsf{r}X)_K^\nu = \mathsf{r}(X_K^\nu), \qquad (X \cdot \tau_v Y)_K^\nu = A\, \tau_v B.$$

The concatenation is separated because $|v|_\infty \ge 4\bar R$.

The reflection size of the real function A is the maximum, over coordinate directions μ and coordinate sign changes g, of

$$\rho((\theta_\mu \mathsf{g} A)(\tau_{2\bar R e_\mu} \mathsf{g} A)).$$

Within either reflected tuple the separation is $d(X)$. Between the two tuples it is at least $2\delta(X;\bar R)$, since their μ-coordinates lie respectively below $\bar R - \delta$ and above $\bar R + \delta$. The moment bound thus makes this reflection size at most $N(X;\bar R)$, and the same argument applies to Y. The coordinate and diagonal mirror hypotheses of Prop. E.7 hold for ρ. Its estimate, with $ra_K = \bar R$, is precisely eq. E.xxxii. □

The estimates also give continuity under small translations of individual entries, uniformly in K, ρ, ν. Indeed, $\|\tau_u f - f\|_\sharp \to 0$ as $u \to 0$, because ∇f is uniformly continuous. The analogous fact holds for $\|\mathsf{R}f - f\|_\sharp$ as $\mathsf{R} \to 1$ in $\mathrm{O}(3)$: on a fixed ball the chain rule gives $\nabla(\mathsf{R}f)(x) = \mathsf{R}\nabla f(\mathsf{R}^{-1}x)$. Enlarge each support by one quarter of $\min(1, d(X))$ and apply eq. E.xxxi; the enlarged supports have at least half the original separation. This supplies a common local modulus of continuity for all the translations used below.

E.2.1. *Limits at spatial infinity.* For a converging frame put

$$e_j(X) = E_{\nu_j}[A_j(X) \mid \mathcal{I}], \qquad m_j(X^1, \dots, X^q) = \nu_j\left(\prod_{i=1}^{q} e_j(X^i)\right).$$

Choose real versions with $|e_j(X)| \le C_{\mathrm{mom}}(X)$. Versions do not affect m_j. Limits below are iterated: the shift v_2 tends to infinity first, with all later shifts fixed.

Proposition E.11 (Iterated limits)**.** For any separated tuples $X^1, \dots, X^q$, not necessarily separated from one another, $m_j(X^1, \dots, X^q)$ converges and

$$\lim_j m_j(X^1, \dots, X^q) = \lim_{|v_q|\to\infty} \cdots \lim_{|v_2|\to\infty} \sigma(X^1 \cdot \tau_{v_2} X^2 \cdots \tau_{v_q} X^q). \tag{E.xxxiv}$$

Every intermediate limit is continuous in the remaining shifts on the open set where the remaining concatenation is separated.

More precisely, put $X'_p = X^1 \cdot \tau_{v_{p+1}} X^{p+1} \cdots \tau_{v_q} X^q$ and denote the intermediate limits by $G^{(p)}$, so that $G^{(1)}$ is the unmodified correlation. If $\bar{R}' > 0$ belongs to $\bigcup_K a_K\mathbb{Z}$ and $\operatorname{supp} X'_p \cup \operatorname{supp} X^p \subset (-\bar{R}', \bar{R}')^3$, then

$$|G^{(p-1)}(v_p, \dots, v_q) - G^{(p)}(v_{p+1}, \dots, v_q)| \le k_0 \prod_{i=2}^{p-1} C_{\mathrm{mom}}(X^i) \sqrt{N(X'_p; \bar{R}') N(X^p; \bar{R}')}\, \frac{\bar{R}'}{|v_p|} \tag{E.xxxv}$$

for $|v_p| \ge k_1 \bar{R}'$. ∎

Proof. Define

$$G_j^{(p)} = \nu_j\left(\prod_{i=2}^{p} e_j(X^i)\, A_j(X'_p)\right).$$

For $p = 1$ these converge by the definition of the frame. For $p = q$ conditional expectation gives $G_j^{(q)} = m_j(X^1, \dots, X^q)$.

A real invariant weight w with $|w| \le \bar{w}$ gives

$$\nu_j(wF) = c_+\rho_+(F) - c_-\rho_-(F), \qquad c_\pm = \nu_j(w_\pm), \quad c_+ + c_- \le \bar{w}. \tag{E.xxxvi}$$

Terms with $c_\pm = 0$ are omitted; otherwise $\rho_\pm = (\nu_j)_{w_\pm} \in \mathcal{G}_j$. The conditional expectations $e_j(X)$ remain versions for $\rho_\pm$. Apply eq. E.xxxii to these two laws, centred at ν_j, with $w = \prod_{i=2}^{p-1} e_j(X^i)$. It bounds $G_j^{(p-1)} - G_j^{(p)}$ by the right side of eq. E.xxxv, whenever the shifts and the cube side belong to the lattice and j is sufficiently large. Using eq. E.xxxi in the same signed decomposition also gives a local modulus of continuity of $G_j^{(p)}$ in the remaining shifts, independent of j.

Assume $G_j^{(p-1)}$ converges. First fix lattice-compatible later shifts and a lattice-compatible $\bar{R}'$. These shifts belong to every sufficiently fine lattice. For each sufficiently large lattice-compatible v_p, the difference between the upper and lower limits of $G_j^{(p)}$ is at most twice the right side of eq. E.xxxv, since $G_j^{(p-1)}$ converges there. Letting $v_p = ze_1$, $z \in \mathbb{N}$, tend to infinity proves that $G_j^{(p)}$ converges at the chosen later shifts.

At arbitrary later shifts with X'_p separated, approximate the shifts by lattice-compatible ones in the same separated neighborhood. The common local modulus bounds the difference between the values of $G_j^{(p)}$ at the original and approximating shifts uniformly in j. The Cauchy criterion now gives convergence at the original shifts, and the same modulus gives continuity of the limit.

For the bound at arbitrary shifts, first take $|v_p| > k_1\bar{R}'$. Round all remaining shifts to the lattice at cutoff K_j. For large j, the rounded shifts still satisfy the support and distance hypotheses. Apply eq. E.xxii in the two laws of eq. E.xxxvi before bounding the reflection norms. Each term in the maximum eq. E.x for the rounded X'_p is a reflected moment. The original and rounded reflected tuples have fixed separated neighborhoods for their entries. The uniform bound eq. E.xxxi

therefore compares their moments with error $o(1)$ in both laws. The original moment is bounded by $N(X'_p;\bar{R}')$ by eq. E.xxix, since the moment bound applies to tests at arbitrary shifts. Thus the reflection norm squared for the rounded tuple is at most $N(X'_p;\bar{R}')+o(1)$. The norm for X^p is bounded by $N(X^p;\bar{R}')$ as before. The common local modulus also allows the rounded shifts in $G_j^{(p-1)}, G_j^{(p)}$ to be replaced by the original shifts with error $o(1)$. Passing to the limit gives eq. E.xxxv. The equality case $|v_p| = k_1\bar{R}'$ follows by continuity in v_p with the later shifts fixed. The right side tends to zero as $|v_p|\to\infty$, identifying the limit as $G^{(p)}$. Induction gives eq. E.xxxiv at $p=q$. □

The constants in eq. E.xxxv depend on the later shifts through X'_p and $\bar{R}'$. No joint limit with all shifts tending to infinity is asserted.

E.2.2. *Construction of the component measure.*

Definition E.12 (Canonical decomposition)**.** A decomposition of the following kind is determined by the values at spatial infinity. Its space X is compact and metrizable, its probability measure λ has full support, and each component σ_ξ is a normalized, real, symmetric, multilinear functional on separated tuples. Require that each evaluation $\xi\mapsto\sigma_\xi(X)$ be continuous, that distinct points give distinct functionals, and that

$$\int_{\mathsf{X}}\prod_{i=1}^{q}\sigma_\xi(X^i)\,\lambda(d\xi)=\lim_{|v_q|\to\infty}\cdots\lim_{|v_2|\to\infty}\sigma(X^1\cdot\tau_{v_2}X^2\cdots\tau_{v_q}X^q). \tag{E.xxxvii}$$

For $q=1$ this is the decomposition of σ itself. ■

Theorem E.13 (Existence of the component measure)**.** Every converging frame in classes satisfying Def. E.8 has a canonical decomposition as above, with the following properties.

(i) Each evaluation $\xi\mapsto\sigma_\xi(X)$ is continuous, and the components separate the points of X.
(ii) The mixed moments of the components are the iterated limits in eq. E.xxxvii.
(iii) Each component is a limit in the same frame. More precisely, for every $\xi\in\mathsf{X}$ there are invariant events E_j with $\nu_j(E_j)>0$ such that $\rho_j=(\nu_j)_{1_{E_j}}$ obeys

$$\|E_{\rho_j}[A_j(X)\mid\mathcal{I}]-\sigma_\xi(X)\|_{L^\infty(\rho_j)}\longrightarrow 0\quad\text{for every separated } X. \tag{E.xxxviii}$$

(iv) Every component satisfies $|\sigma_\xi(X)|\le C_{\mathrm{mom}}(X)$ and the difference estimate eq. E.xxxi.
(v) $\#\mathsf{X}\le\liminf_j\dim L^2(\mathcal{V}(\mathbb{L}_{K_j}),\mathcal{I},\nu_j)$, with infinity allowed on either side.

■□

Proof. Let $\mathfrak{F}=\{X_1,X_2,\dots\}$ be the countable family with controlled supports constructed in section C.1. For every nonempty tuple $X=(f_1,\dots,f_n)$ and $0<\varepsilon\le 1$, choose $X^{[\varepsilon]}\in\mathfrak{F}$ whose entries differ from those of X by at most ε in $\|\cdot\|_\sharp$. The construction permits both supports of each entry to lie in a set contained in a ball of radius $R(X)+1/4$, with different sets at distance at least $\frac12\min(1,d(X))$. Indeed, take the open neighbourhoods within distance $\frac14\min(1,d(X))$ of the original supports and use their countable dense test sets. The estimate eq. E.xxxi, also for conditional expectations, then gives the error

$$\varepsilon_X(\varepsilon)=c_n\Big(\tfrac12\min(1,d(X));R(X)+\tfrac14\Big)\,n\varepsilon\prod_i(\|f_i\|_\sharp+1)\longrightarrow 0. \tag{E.xxxix}$$

Choose the approximation using the fixed enumeration, so that it is independent of the law and of ξ.

Consider the compact metrizable product

$$\Pi=\prod_{X\in\mathfrak{F}}[-C_{\mathrm{mom}}(X),C_{\mathrm{mom}}(X)].$$

The map $y_j = (e_j(X))_{X\in\mathfrak{F}}$ is $\mathcal{I}$-measurable, and $\lambda_j = (y_j)_*\nu_j$ is a probability measure on Π. By Prop. E.11, the integral against λ_j of every monomial in the coordinates converges. Coordinate polynomials contain the constants and separate points; the real Stone–Weierstrass theorem makes them uniformly dense in $C(\Pi)$. Consequently $\lambda_j(\phi)$ converges for every $\phi \in C(\Pi)$: approximate ϕ by a polynomial and use $|\lambda_j(\phi)| \le \|\phi\|_\infty$. The limiting positive functional, equal to one on constants, is represented by a probability measure λ. Let $\mathsf{X} = \operatorname{supp}\lambda$. The countable base of Π shows $\lambda(\mathsf{X}) = 1$, so this is a compact metrizable space with full support.

For every open $V \subset \Pi$, $\liminf_j \lambda_j(V) \ge \lambda(V)$. One may verify this directly by increasing the continuous functions $\min(1, s\,\mathrm{dist}(\,\cdot\,, \Pi\setminus V))$ to 1_V. Fix $\xi \in \mathsf{X}$ and set

$$V_s = \{\zeta : |\zeta_{X_k} - \xi_{X_k}| < 1/s, \quad 1 \le k \le s\}.$$

Then $\lambda(V_s) > 0$, and hence $\lambda_j(V_s) > 0$ for all sufficiently large j. Choose increasing j_s with this property for every $j \ge j_s$. For $j_s \le j < j_{s+1}$ put $E_j = y_j^{-1}(V_s)$; for the finitely many earlier indices use the whole space. These invariant events have positive measure, and $\rho_j = (\nu_j)_{1_{E_j}} \in \mathcal{G}_j$. The unchanged versions of conditional expectation give

$$\|E_{\rho_j}[A_j(X_k) \mid \mathcal{I}] - \xi_{X_k}\|_\infty \le 1/s \qquad (k \le s,\ j_s \le j < j_{s+1}).$$

For an arbitrary nonempty X, comparison with $X^{[\varepsilon]}$ now yields

$$\limsup_j \|E_{\rho_j}[A_j(X) \mid \mathcal{I}] - \xi_{X^{[\varepsilon]}}\|_\infty \le \varepsilon_X(\varepsilon).$$

In particular the numbers $\xi_{X^{[\varepsilon]}}$ are Cauchy as $\varepsilon \downarrow 0$, with the difference between two of them at most the sum of the two errors. Define their limit to be $\sigma_\xi(X)$ and set $\sigma_\xi(\varnothing) = 1$. Then

$$|\sigma_\xi(X) - \xi_{X^{[\varepsilon]}}| \le \varepsilon_X(\varepsilon).$$

This bound and the preceding limsup prove eq. E.xxxviii. Integration shows $\rho_j(A_j(X)) \to \sigma_\xi(X)$. It follows that σ_ξ is a functional of the asserted kind and inherits the moment and difference estimates. For $X \in \mathfrak{F}$ it equals the coordinate ξ_X.

The last approximation is uniform in ξ. Thus every evaluation $\sigma_\xi(X)$ is continuous, as a uniform limit of coordinates. Equality of all evaluations implies equality of all coordinates, so the component map is injective.

For tuples in $\mathfrak{F}$, convergence of coordinate monomials proves eq. E.xxxvii. For general tuples replace each X^i by $(X^i)^{[\varepsilon]}$. Telescoping the product bounds both the component integral error and the error in m_j by

$$\sum_i \varepsilon_{X^i}(\varepsilon) \prod_{l\ne i} (C_{\mathrm{mom}}(X^l) + \varepsilon_{X^l}(\varepsilon)).$$

This tends to zero independently of j. Empty tuples contribute a factor one. Passing first in j and then in ε proves the required identity.

Finally, if the displayed liminf dimension is a finite integer N, it equals N on infinitely many indices. If X had $N+1$ distinct points, choose disjoint open neighbourhoods of them in Π. Each has positive λ-measure and therefore positive λ_j-measure for all sufficiently large j. Their inverse images under y_j give $N+1$ nonzero orthogonal invariant indicator functions, contradicting dimension N on those indices. □

Lemma E.14 (Subsequences of frames)**.** Every frame in classes obeying the moment and cumulant bounds in Def. E.8 has a convergent subsequence. ■

Proof. The bounds in eq. E.xxvii apply to all orders of centred moments. They give the common separated-support estimates of Prop. C.2 on $\mathbb{R}^3$. The same subsequence then works for cumulants, by the finite polynomial eq. C.vi, and the moments of the limit are recovered by eq. C.vii. □

E.3. **Clustering components and the vacuum space.** We first show that the preceding decomposition is intrinsic to the limit functional. Reflection positivity will then force the symmetries of each component, and the uniform estimate eq. E.xxxii will give clustering for every component.

Proposition E.15 (Uniqueness of the decomposition)**.** In (a), two decompositions satisfying eq. E.xxxvii and the stated compactness, continuity, injectivity and full-support conditions are related by a unique measure-preserving homeomorphism which identifies their component functionals.

In (b), if an isometry r preserves σ, it induces a unique measure-preserving homeomorphism h_{r} of X with

$$\sigma_{h_{\mathsf{r}}(\xi)} = \sigma_\xi \circ \mathsf{r}^{-1}, \qquad h_{\mathsf{r}\mathsf{r}'} = h_{\mathsf{r}} h_{\mathsf{r}'}. \tag{E.xl}$$

In (c), h_{r} is the identity if and only if every component is invariant under r, by the identity in (b) and injectivity. For $s > 0$ define

$$(\mathrm{dil}_s\, \sigma)(f_1, \dots, f_n) = s^{-n} \sigma(f_1(s\,\cdot), \dots, f_n(s\,\cdot)). \tag{E.xli}$$

The dilated components on the same probability space decompose $\mathrm{dil}_s\, \sigma$. ■

Proof. For any finite collection of tuples, the two component measures have the same moments in their evaluation coordinates, by eq. E.xxxvii, allowing repeated tuples. Both coordinate images lie in a compact box. Polynomial approximation and uniqueness of the representing measure show that their finite-dimensional distributions agree.

Embed the two component spaces in $\mathbb{R}^{\mathfrak{S}(\mathbb{R}^3)}$ by all evaluations. They are compact subsets of this Hausdorff product, and their embedding maps are homeomorphisms onto their images. Every basic neighbourhood of a point in the first image is determined by finitely many evaluations. Its inverse image is a nonempty open set and hence has positive measure. Equality of the finite-dimensional distributions makes it meet the second image. Since that image is closed, it contains the point. Interchanging the two spaces proves equality of the images. Their identification is the required homeomorphism; it is unique by injectivity of the evaluations. Polynomials in these evaluations are uniformly dense on either component space, so equality of their integrals also proves preservation of the measures.

For an isometry, the functions $\sigma_\xi(\mathsf{r}^{-1}X)$ satisfy the decomposition identity: conjugating translations changes v by the orthogonal linear part of r, and therefore preserves the iterated limits $|v| \to \infty$. Apply uniqueness to obtain h_{r}. Composing its defining identities proves the group law and $h_{\mathsf{r}^{-1}} = h_{\mathsf{r}}^{-1}$. For dilation, $f(s\,\cdot)$ is translated by v/s when f is translated by v. Each iterated limit is thus rescaled by $s^{-\sum_i |X^i|}$. This is exactly the product of the factors in eq. E.xli. Continuity and injectivity are preserved because dilation of test tuples is invertible. □

Proposition E.16 (Reflection invariance of the components)**.** Suppose a real symmetric functional σ is translation invariant, is reflection positive at a mirror $\Sigma = (u, c)$, and has a decomposition as above. Then σ and every component are invariant under θ_Σ. Each component is also invariant under $\mathsf{g}\theta_\Sigma\mathsf{g}^{-1}$ whenever the isometry g preserves σ.

Consequently a Euclidean-invariant σ which is reflection positive at one mirror has Euclidean-invariant components. ■

Proof. On the positive half-space, positivity for the two formal vectors $\varnothing, X$ implies $\sigma(\theta_\Sigma X) = \sigma(X)$: use coefficients $1, i$ and equate imaginary parts. Move any tuple into this half-space by a translation parallel to u and use $\theta_\Sigma \tau_{tu} = \tau_{-tu}\theta_\Sigma$. Translation invariance proves the same identity for all tuples.

Let F be a finite formal sum of tuples and put $F_t = \tau_{tu/2} F$. For large t, F_t lies in the positive half-space. Translation invariance and the case $q = 2$ of the decomposition identity give

$$0 \le \lim_{t\to\infty} \sigma(\theta_\Sigma \overline{F_t} \cdot F_t) = \int_{\mathsf{X}} \overline{\sigma_\xi(\theta_\Sigma F)}\, \sigma_\xi(F)\, \lambda(d\xi). \tag{E.xlii}$$

Indeed, term by term the expression before the limit is $\sigma(\theta_\Sigma \bar{F} \cdot \tau_{tu} F)$, whose large translation limit is prescribed by eq. E.xxxvii. Replace F by $F - \theta_\Sigma F$. The last integral is then

$$-\int_{\mathsf{X}} |\sigma_\xi(F - \theta_\Sigma F)|^2 \, \lambda(d\xi).$$

It must vanish. The integrand is continuous, and λ has full support; hence every component is fixed by θ_Σ.

Equivalently h_{θ_Σ} in eq. E.xl is the identity. Its conjugate by h_{g} is also the identity, proving the assertion for the conjugated mirror. If σ is Euclidean invariant, every mirror is such a conjugate. Products of two parallel reflections give translations. Every orthogonal map is a product of at most three reflections: reflect its image of a basis vector back to that vector, preserving the previously fixed vectors, and repeat. Thus every component is Euclidean invariant. □

Corollary E.17 (Symmetries of every component)**.** For a translation-invariant functional with a canonical decomposition, (i) cubic invariance and reflection positivity at one coordinate and one diagonal mirror imply that every component is translation and cubic invariant. In (ii), full Euclidean invariance and positivity at one mirror imply that every component is Euclidean invariant. ■□

Proof. Prop. C.7 transports the reflection positivity to all mirrors of the relevant kinds. Prop. E.16 then gives invariance of each component under their reflections. Coordinate mirrors generate translations by products of parallel reflections, and their reflections through the origin include the coordinate sign flips. Reflections in $x_i = x_j$ are the transpositions of the coordinate axes, giving (i). In (ii), all reflections are available and generate the Euclidean group. □

E.3.1. *Properties inherited from the frame.* The following conclusions apply to every component, including points of measure zero in the component space. Their lattice approximants are the conditioned laws in eq. E.xxxviii.

Proposition E.18 (Properties of a limit in a frame)**.** Let σ' be a limit in a frame satisfying Def. E.8, with integrating laws ρ_j and centring laws ν_j.

(a) It is a symmetric functional, has the moment bound eq. E.xxvii and the continuity bound eq. E.xxxi, and is continuous under small translations of individual entries.
(b) It is invariant under translations and cubic isometries.
(c) It is reflection positive at every coordinate and diagonal mirror.
(d) Its cumulants of order at least two are the limits of those of ρ_j and obey eq. E.xxiii. Its one-point function is the limit of $\rho_j(\mathcal{O}_{K_j}(f)) - \nu_j(\mathcal{O}_{K_j}(f))$. The frame with integrating and centring laws both ρ_j converges to σ'^{c}.
(e) Its one-point function is $c_{\sigma'} \int f$ with $|c_{\sigma'}| \le \bar{c}$ of eq. E.xliv. It lies in $\mathsf{Fct}(\underline{C}, \bar{c})$, its off-diagonal distributions have order at most $4n + 1$ on compact sets, and the coordinate reflection Hilbert spaces and their translation semigroups are defined.
(f) If a finite vector of tuple moments $\big(\rho(A_j(X^i))\big)_{i=1}^k$ belongs to a fixed closed subset of $\mathbb{R}^k$ for all sufficiently large j and every $\rho \in \mathcal{G}_j$, then $\big(\sigma'(X^i)\big)_{i=1}^k$ belongs to that set.

■

Proof. The bounds pass to the limit by Prop. E.10. The continuity estimate applies to translated entries since C_c^1 tests vary continuously in the sharp norm under translations. Translations on the lattice-compatible dense subgroup and cubic symmetries pass directly to the limit. Continuity extends translation invariance to every translation. This proves (a) and (b).

For (c), the laws ρ_j are reflection positive on the lattice in the coordinate and diagonal planes through the origin. The compact supports of finitely many test tuples in an open half-space are a positive distance from its plane. At sufficiently small spacing all plaquettes entering the associated smeared observables lie on the same side. Reflection covariance of the lattice observable

and invariance of the centring laws ν_j give entrywise convergence of the positive lattice reflection matrices to the continuum reflection matrices. Transport to all heights by Prop. C.7.

Cumulants are polynomials in moments, and centring shifts only the first cumulant. Inverting the cumulants gives the last assertion of (d). Lattice symmetry makes $\rho_j(a_p) - \nu_j(a_p)$ independent of the plaquette. The barycentre Riemann sum therefore gives

$$\rho_j(\mathcal{O}_{K_j}(f)) - \nu_j(\mathcal{O}_{K_j}(f)) = 3a_{K_j}^{-4}(\rho_j(a_p) - \nu_j(a_p))r_j(f), \qquad r_j(f) = \frac{a_{K_j}^3}{3}\sum_p f(x(p)) \longrightarrow \int f.$$

A unit-integral test gives convergence of the coefficient, and the bound of eq. E.xxiv gives (e) by taking the infimum over those tests in the unit ball. The distribution and Hilbert-space conclusions follow from Props. C.10 and D.2. For (f), take $\rho = \rho_j$ in the hypothesis and pass to the limit in the closed set. □

Proposition E.19 (One-point functions of frame limits)**.** (a) For a limit σ' with integrating laws ρ_j,

$$\sigma'(f) = c_{\sigma'}\int f, \qquad c_{\sigma'} = \lim_j 3a_{K_j}^{-4}(\rho_j(a_p) - \nu_j(a_p)), \qquad |c_{\sigma'}| \le \bar{c}, \tag{E.xliii}$$

where

$$\bar{c} = C_1(1)\inf\{\|f\|_\sharp : f \in C_c^1(B(0,1)),\ \int f = 1\}. \tag{E.xliv}$$

(b) In the decomposition of a converging frame, each component has the approximants of eq. E.xxxviii, $\xi \mapsto c_\xi$ is continuous, $|c_\xi| \le \bar{c}$, and

$$\sigma(f) = 0, \qquad \int_{\mathsf{X}} c_\xi\,\lambda(d\xi) = 0. \tag{E.xlv}$$

If $\kappa = \int c_\xi^2\,d\lambda$, then for disjoint f, f',

$$S_2^\sigma(f,f') = \int_{\mathsf{X}} S_2^{\sigma_\xi}(f,f')\,\lambda(d\xi) + \kappa\int f\int f'. \tag{E.xlvi}$$

■

Proof. The Riemann-sum argument in Prop. E.18 proves (a). For (b), Thm. E.13 supplies the approximants. For a unit-integral test f_0, $c_\xi = \sigma_\xi(f_0)$ is a continuous evaluation. Every one-point moment of the frame is zero by its centring, and the decomposition identity gives eq. E.xlv. Integrate $S_2^{\sigma_\xi}(f,f') = \sigma_\xi(f,f') - c_\xi^2\int f\int f'$ to obtain eq. E.xlvi. □

Proposition E.20 (Uniform clustering of the components)**.** For the decomposition of a converging frame, let $\bar{R} > 0$ belong to $\bigcup_K a_K\mathbb{Z}$ and contain $\operatorname{supp} X \cup \operatorname{supp} Y$ in its open cube. Then, for every $\xi \in \mathsf{X}$ and $|v| \ge k_1\bar{R}$,

$$|\sigma_\xi(X\cdot\tau_v Y) - \sigma_\xi(X)\sigma_\xi(Y)| \le k_0\sqrt{N(X;\bar{R})N(Y;\bar{R})}\,\frac{\bar{R}}{|v|}. \tag{E.xlvii}$$

The bound is uniform over all components. ■

Proof. Use the conditioned approximants ρ_j for the given component. For a lattice-compatible v, all the hypotheses of eq. E.xxxii hold eventually. Its first term converges to $\sigma_\xi(X\cdot\tau_v Y)$. By eq. E.xxxviii, the conditional expectations of $A_j(X)$ and $A_j(Y)$ converge in their respective $L^\infty(\rho_j)$ norms to the constants $\sigma_\xi(X)$ and $\sigma_\xi(Y)$. If their errors are $\varepsilon_j(X)$ and $\varepsilon_j(Y)$, the error in the integral of their product is at most

$$\varepsilon_j(X)(|\sigma_\xi(Y)| + \varepsilon_j(Y)) + |\sigma_\xi(X)|\varepsilon_j(Y) \longrightarrow 0.$$

Thus eq. E.xlvii follows on the dense subgroup. Approximate any remaining v by such vectors with $|v_j| \ge k_1\bar{R}$; the continuity under translations already proved gives the assertion. The right side contains no component-dependent quantity. □

E.3.2. *Zero energy and the component space.*

Proposition E.21 (Zero energy)**.** Let σ be a translation-invariant functional with the bounds of §D.1, reflection positive at the mirror with unit normal u. In its reflection Hilbert space put $P_0 = 1_{\{0\}}(H)$.

(a) For positive-half-space vectors F, F',

$$\lim_{t\to\infty} \sigma(\theta\bar{F} \cdot \tau_{tu} F') = \langle [F], P_0[F'] \rangle. \tag{E.xlviii}$$

(b) The equality $\ker H = \mathbb{C}\Omega$ is equivalent to

$$\lim_{t\to\infty} \sigma(\theta\bar{F} \cdot \tau_{tu} F') = \overline{\sigma(F)}\sigma(F') \tag{E.xlix}$$

for all such F, F'.

(c) For each component of a converging frame obeying Def. E.8, the coordinate reflection Hamiltonians have one-dimensional kernels.

■

Proof. By the spectral theorem $T_t = e^{-tH} \to P_0$ strongly. This is (a). Since $\langle \Omega, [F] \rangle = \sigma(F)$, the identity in (b) is exactly the equality of matrix elements of P_0 and $|\Omega\rangle\langle\Omega|$ on a dense subspace. Each component satisfies the limit in (b) by the bound eq. E.xlvii and reflection invariance, which proves (c).

□

Theorem E.22 (The vacuum space of the mixture)**.** Let σ satisfy the hypotheses of Prop. E.21 and have a canonical decomposition as in Def. E.12. Assume its components are translation invariant and invariant under the reflection θ defining $\mathcal{H}(\sigma, u)$, and that

$$\sup_{\xi} |\sigma_\xi(X \cdot \tau_{tu} Y) - \sigma_\xi(X)\sigma_\xi(Y)| \longrightarrow 0 \quad (t \to \infty)$$

for positive-half-space tuples X, Y. Then

$$\langle P_0[F], P_0[F'] \rangle = \int_{\mathsf{X}} \overline{\sigma_\xi(F)}\sigma_\xi(F')\, \lambda(d\xi). \tag{E.l}$$

The assignment $P_0[F] \mapsto (\xi \mapsto \sigma_\xi(F))$ extends to a unitary map $\ker H \to L^2(\mathsf{X}, \lambda)$, carrying Ω to 1. The number of components equals $\dim \ker H$ when either is finite; if X is infinite, the kernel is a separable infinite-dimensional Hilbert space.

These conclusions apply to every coordinate reflection Hilbert space of the limit of a converging frame. ■□

Proof. For positive-half-space tuples X, Y, apply eq. E.xxxvii with first tuple θX, second tuple Y and shift tu. Reflection invariance of each component gives

$$\lim_{t\to\infty} \sigma(\theta X \cdot \tau_{tu} Y) = \int_{\mathsf{X}} \sigma_\xi(X)\sigma_\xi(Y)\, d\lambda.$$

Combine this with eq. E.xlviii and extend sesquilinearly to obtain eq. E.l. It makes the proposed map well defined and isometric. The vectors $P_0[F]$ are dense in $\ker H$, so it extends to an isometry with closed range, and sends Ω to 1.

To prove surjectivity, put $\psi_F(\xi) = \sigma_\xi(F)$ and take the uniform closure of their span in $C(\mathsf{X})$, initially for positive-half-space F. It contains constants and is closed under conjugation. The uniform clustering assumption gives

$$\psi_{X \cdot \tau_{tu} Y} \longrightarrow \psi_X \psi_Y \quad \text{uniformly on } \mathsf{X}.$$

Thus the closure is an algebra. It separates points: any tuple distinguishing two components may be translated into the positive half-space without changing its values. Stone–Weierstrass makes

the closure $C(\mathsf{X})$. Since continuous functions are dense in $L^2(\lambda)$, the isometry has dense as well as closed range.

Full support implies that disjoint nonempty open sets give nonzero orthogonal indicator functions in $L^2(\lambda)$. Any n distinct points have n such neighbourhoods. Hence a finite component space has exactly that dimension, and an infinite component space has infinite dimension. Separability follows from that of the reflection Hilbert space. For frame limits, Props. E.18 and E.20 supply both invariance and uniform clustering required above. □

Corollary E.23 (Vacuum space of a frame limit)**.** Thm. E.22 applies to every canonical decomposition of a converging frame satisfying the stated moment and class bounds. ■□

Proof. For the canonical decomposition constructed by Thm. E.13, reflection invariance and translation invariance follow from Prop. E.18, and the uniform limit follows from Prop. E.20. Every other canonical decomposition agrees through the unique measure-preserving homeomorphism of Prop. E.15. □

Remark E.24 (A two-component constant field). Let $c_1 \neq c_2$ and $0 < p < 1$. The functional given by the mixture of the two deterministic constant fields with values c_1, c_2 has a two-dimensional reflection Hilbert space and $H = 0$. Indeed, the reflection form is the Euclidean inner product on the evaluation vectors at c_1, c_2 with weights $p, 1-p$. The vacuum and a one-point test of integral one already span both coordinates. Every translation acts trivially. The connected two-point function is $p(1-p)(c_1 - c_2)^2 \int f \int h$, whereas each component's connected two-point function vanishes.

Appendix F. Scalar reconstruction

F.1. **Point insertions and energy bounds for reconstruction.** The reconstruction uses more than reflection positivity and bounds on separated moments. Its additional input is an L^1 estimate for one insertion between two positive-time vectors. We first construct point forms with polynomial energy bounds, then use them to define smeared fields on a common domain.

Let σ be a real, normalized, symmetric multilinear functional on separated tuples. Throughout this section assume:

(Rec1) σ is invariant under all Euclidean isometries.

(Rec2) It is reflection positive at $x_3 = 0$. Its reflection Hilbert space $\mathcal{H}$ is separable, and its translations are represented by e^{-tH} and $\mathcal{U}(v)$, with $H \geq 0$, the stated strong continuity, and joint spectral measure $\mathcal{P}$:

$$e^{-tH}[F] = [\tau_{te_3}F], \qquad \mathcal{U}(v)[F] = [\tau_{(v,0)}F], \qquad e^{i\tau H}\mathcal{U}(v) = \int e^{i(\tau E + v\cdot q)}\, d\mathcal{P}(E,q). \tag{F.i}$$

(Rec3) $\mathcal{P}$ is carried by $\{(E,q) : E \geq |q|\}$.

(Rec4) For some $C_\wp, \wp \geq 0$ and $0 < c_0 \leq 1/2$, put $\mathrm{Sl}_\eta = \{|x_3| < c_0\eta\}$. For $0 < \eta \leq 1$, $h \in C_c^1(\mathrm{Sl}_\eta)$ and $F, F' \in \mathbb{C}\mathfrak{S}(\{x_3 > \eta\})$, assume

$$|\sigma(\theta\bar{F} \cdot (h) \cdot F')| \leq C_\wp \eta^{-\wp} \|h\|_1 \|[\tau_{-\eta e_3}F]\| \|[\tau_{-\eta e_3}F']\|. \tag{F.ii}$$

We refer to these four conditions as (Rec). Vacuum uniqueness, $\ker H = \mathbb{C}\Omega$, will be a separate assumption when needed. The numerical triple in (Rec4) is not part of the definition of the resulting field.

Write $\vec{z} = (z_1, z_2)$ and define

$$d_\circ(z) = \operatorname{Re} z_3 - |\operatorname{Im} \vec{z}|, \qquad \mathrm{Tb}_\circ = \{d_\circ > 0\}, \qquad \mathcal{T}(z) = \int e^{-z_3 E + i\vec{z}\cdot q}\, d\mathcal{P}. \tag{F.iii}$$

The last integral is defined on the closed tube. The cone bound gives $|e^{-z_3 E + i\vec{z}\cdot q}| \leq e^{-d_\circ(z)E}$. Consequently $\mathcal{T}$ consists of contractions, is strongly continuous on the closed tube and norm holomorphic

in its interior, and satisfies

$$\mathcal{T}(z)\mathcal{T}(w) = \mathcal{T}(z+w), \qquad \mathcal{T}(z)^* = \mathcal{T}(z^*), \qquad z^* = (-\bar{z}_1, -\bar{z}_2, \bar{z}_3).$$

For norm holomorphy, expand the exponential around a point with positive margin; the factor $e^{-d_\circ(z)E}$ bounds every coefficient and the quadratic remainder uniformly on a smaller ball. Dominated convergence against scalar spectral measures proves strong continuity at the boundary. In the interior, $E^k e^{-d_\circ(z)E}$ is bounded for every k, so the range of $\mathcal{T}(z)$ lies in

$$\mathcal{D}_\infty = \bigcap_{k\geq 1} \operatorname{dom} H^k.$$

The vacuum is fixed by $\mathcal{T}(z)$ and its adjoint. For Minkowski coordinates $x = (t, \vec{x})$ put

$$\iota(t, \vec{x}) = (\vec{x}, -it). \tag{F.iv}$$

This identifies the forward tube $\mathbb{R}^{1+2} + i\{y : y^0 > |\vec{y}|\}$ with $\mathrm{Tb}_\circ$, and $\mathcal{T}(\iota x) = e^{itH}\mathcal{U}(\vec{x})$ is unitary.

F.1.1. *Densities of bounded insertions.*

Lemma F.1. For $0 < \eta \leq 1$ there is a unique complex linear map $h \mapsto B^{\mathrm{in}}_\eta(h)$ from $L^1(\mathrm{Sl}_\eta)$ to bounded operators such that

$$\|B^{\mathrm{in}}_\eta(h)\| \leq C_\wp \eta^{-\wp} \|h\|_1, \tag{F.v}$$

$$B^{\mathrm{in}}_\eta(h)[F] = [(\tau_{\eta e_3} h) \cdot \tau_{2\eta e_3} F] \quad (h \in C^1_c(\mathrm{Sl}_\eta)). \tag{F.vi}$$

For real h, its adjoint is $B^{\mathrm{in}}_\eta(\theta h)$. Each scalar matrix element has an L^∞ density of norm at most $C_\wp \eta^{-\wp}\|\psi\|\|\psi'\|$. ∎

Proof. Apply eq. F.ii after translating both formal vectors by ηe_3. The bounded sesquilinear form

$$\langle [F], B^{\mathrm{in}}_\eta(h)[F'] \rangle = \sigma(\theta \overline{\tau_{\eta e_3} F} \cdot (h) \cdot \tau_{\eta e_3} F') \tag{F.vii}$$

vanishes on null classes and therefore defines a bounded operator by Riesz representation. Translate the whole tuple by ηe_3 to obtain eq. F.vi. The support inequalities $c_0 \leq 1/2$ make that tuple separated. Real linearity and the norm bound extend this map from real C^1_c functions to real L^1 functions.

For fixed vectors the real and imaginary parts of the matrix element are bounded real functionals on L^1, hence have a complex density b. The same constant bounds $|b|$: apply the norm estimate to normalized indicators of small balls and use Lebesgue differentiation. Thus $\|b\|_\infty \leq C_\wp \eta^{-\wp}\|\psi\|\|\psi'\|$. Complex linear extension now preserves this constant, since $|\int hb| \leq \|h\|_1 \|b\|_\infty$. Reflection invariance, symmetry and reality in eq. F.vii give the adjoint relation on C^1_c, then on L^1 by approximation. The class identity, density of classes, and L^1 continuity prove uniqueness. □

Proposition F.2 (Point insertions)**.** There is a unique family of bounded operators $B(s, s')$, $s, s' > 0$, with the shift law

$$T_t B(s, s') T_{t'} = B(s + t, s' + t') \qquad (t, t' \geq 0) \tag{F.viii}$$

and the density identity

$$\langle \psi, B^{\mathrm{in}}_\eta(h)\psi' \rangle = \int h(x) \langle \psi, \mathcal{U}(\vec{x}) B(\eta + x_3, \eta - x_3) \mathcal{U}(-\vec{x})\psi' \rangle \, dx. \tag{F.ix}$$

It satisfies

$$\|B(s, s')\| \leq C_\wp \min(s, s', 1)^{-\wp}, \qquad B(s, s')^* = B(s', s). \tag{F.x}$$

It is norm continuous and commutes with spatial orthogonal transformations acting on $\mathcal{H}$. The operator density

$$b_\eta(x) = \mathcal{U}(\vec{x}) B(\eta + x_3, \eta - x_3) \mathcal{U}(-\vec{x})$$

is norm continuous on Sl_η, has norm at most $C_\wp \eta^{-\wp}$ there, and has real-analytic scalar matrix elements. ∎

Proof. For $0 < \eta' < \eta \le 1$, $h \in L^1(\mathrm{Sl}_{\eta'})$ and real w with $|w_3| < c_0(\eta - \eta')$, translation covariance of eq. F.vii gives

$$B^{\mathrm{in}}_{\eta}(\tau_w h) = \mathcal{T}((\eta - \eta')e_3 + w)\, B^{\mathrm{in}}_{\eta'}(h)\, \mathcal{T}((\eta - \eta')e_3 - w). \tag{F.xi}$$

The two semigroup arguments have positive third coordinate, and the translated support stays in Sl_η. It suffices to prove the identity on C^1_c tests and dense classes; L^1 continuity then proves the assertion as written.

Fix ψ, ψ' and let b be their scalar density at margin η. Take nonnegative continuous mollifiers j_ε of integral one, supported in $B(0,\varepsilon)$, with $\varepsilon < c_0\eta'$. The functions

$$b_\varepsilon(w) = \langle \psi, \mathcal{T}((\eta - \eta')e_3 + w) B^{\mathrm{in}}_{\eta'}(j_\varepsilon) \mathcal{T}((\eta - \eta')e_3 - w)\psi' \rangle$$

are holomorphic and bounded by $C_\wp \eta'^{-\wp} \|\psi\| \|\psi'\|$ on

$$\{w \in \mathbb{C}^3 : |\operatorname{Re} w_3| + |\operatorname{Im} \vec{w}| < \eta - \eta'\}.$$

On the smaller real slab $|w_3| < c_0(\eta - \eta')$, eq. F.xi identifies them with $\int j_\varepsilon(x - w) b(x)\, dx$. Montel's theorem gives a locally uniformly convergent subsequence with holomorphic limit. Mollification also converges to b in L^1 on compact real subsets, so this limit agrees with b almost everywhere. As $\eta' \downarrow 0$ the smaller real slabs exhaust Sl_η. Their continuous representatives agree on overlaps and give one real-analytic representative of b on the whole slab.

Linearity in ψ' and antilinearity in ψ hold everywhere for these representatives: they hold almost everywhere for the original densities, and both sides are continuous. The original L^∞ bound likewise holds at every point. Riesz representation thus defines bounded operators $b_\eta(x)$ with $\|b_\eta(x)\| \le C_\wp \eta^{-\wp}$. This constructs the operator density from scalar densities.

Integrate eq. F.xi against arbitrary tests. Uniqueness of the continuous scalar densities gives

$$b_\eta(x + w) = \mathcal{T}((\eta - \eta')e_3 + w)\, b_{\eta'}(x)\, \mathcal{T}((\eta - \eta')e_3 - w). \tag{F.xii}$$

For $0 < \eta \le \min(s, s', 1)$ define

$$B(s, s') = T_{s-\eta}\, b_\eta(0)\, T_{s'-\eta}. \tag{F.xiii}$$

The case $x = w = 0$ of eq. F.xii proves independence of η. It also proves the shift law; choosing $\eta = \min(s, s', 1)$ gives the norm bound. For $x \in \mathrm{Sl}_\eta$, choose $\eta' > 0$ with $|x_3| < c_0(\eta - \eta')$ and apply eq. F.xii at $(0, x)$. Factoring spatial translations from the semigroup gives the stated formula for $b_\eta(x)$ and hence eq. F.ix.

In eq. F.xii choose a fixed smaller margin locally. Its semigroup factors have positive margins and are norm holomorphic, proving norm continuity of b_η. The same reasoning in eq. F.xiii proves norm continuity of B. The insertion adjoint identity makes $b_\eta(x)^* = b_\eta(\theta x)$, so $b_\eta(0)$ is self-adjoint. This proves the adjoint relation for B. Spatial rotations in eq. F.vii give covariance of b_η and invariance of $b_\eta(0)$; they commute with T_t, proving the remaining assertion.

For uniqueness, any bounded-operator family with the shift law has continuous scalar densities in eq. F.ix: factor it through $B(\eta/2, \eta/2)$. The density identity then determines its value at $x = 0$, hence $B(\eta, \eta)$. The shift law determines all other values. The same argument shows that changing the admissible triple $C_\wp, \wp, c_0$ does not change the operators, by first restricting insertions to the common smaller slab. □

F.1.2. *Resolvent regularization of the point form.* Put $R_H = (1 + H)^{-1}$. For real $l > \wp$ define

$$A_l = \Gamma(l)^{-2} \int_0^\infty \int_0^\infty (ss')^{l-1} e^{-s-s'} B(s, s')\, ds\, ds'. \tag{F.xiv}$$

Lemma F.3. The integral is norm integrable and defines a bounded self-adjoint operator with

$$\|A_l\| \le C_\wp \left(1 + \frac{2\Gamma(l - \wp)}{\Gamma(l)}\right). \tag{F.xv}$$

For $a > 0$ and $s, s' > 0$,

$$R_H^a A_l R_H^a = A_{l+a}, \qquad T_s A_l T_{s'} = R_H^l B(s, s') R_H^l. \tag{F.xvi}$$

Consequently

$$q_0(\psi, \psi') = \langle (1+H)^l \psi, A_l (1+H)^l \psi' \rangle \tag{F.xvii}$$

defines a Hermitian form on $\bigcup_{l>\wp} \operatorname{dom} H^l$, independent of the admissible exponent used for a given pair of vectors. It satisfies the energy bound

$$|q_0(\psi, \psi')| \le \|A_l\| \, \|(1+H)^l \psi\| \, \|(1+H)^l \psi'\|$$

on $\operatorname{dom} H^l$, and

$$q_0(T_s \psi, T_{s'} \psi') = \langle \psi, B(s, s') \psi' \rangle. \tag{F.xviii}$$

■

Proof. By eq. F.x, $\|B(s, s')\| \le C_\wp (1 + s^{-\wp} + s'^{-\wp})$. The two Gamma integrals therefore converge for $l > \wp$ and give eq. F.xv. Norm continuity makes the operator integrand measurable. Self-adjointness follows by interchanging s, s' and using $B(s, s')^* = B(s', s)$.

Functional calculus and the scalar Gamma integral give

$$R_H^a = \Gamma(a)^{-1} \int_0^\infty t^{a-1} e^{-t} T_t \, dt$$

weakly for $a > 0$. Insert this on both sides of A_l and use the shift law. Fubini is justified by the integrable bound just proved and contractivity. Set $s = r + t$ and $s' = r' + t'$. The two inner integrals are

$$\int_0^s t^{a-1} (s-t)^{l-1} \, dt = s^{a+l-1} \frac{\Gamma(a)\Gamma(l)}{\Gamma(a+l)}.$$

This proves the first identity in eq. F.xvi. For the second, shift $B(r, r')$ by (s, s'), then write $B(r + s, r' + s') = T_r B(s, s') T_{r'}$ and use the same resolvent integral.

If $l' > l > \wp$ and both vectors lie in $\operatorname{dom} H^{l'}$, the first identity and $R_H^{l'-l}(1+H)^{l'} = (1+H)^l$ prove equality of the two definitions of q_0. The domains are nested, so a common admissible exponent exists for every pair in the stated union. Hermiticity and the bound follow from those of A_l. Finally $(1+H)^l T_s$ is bounded for $s > 0$. Apply the second identity at $(s/2, s'/2)$ and cancel the resolvent factors against these bounded regularized powers. The shift law gives eq. F.xviii. □

F.1.3. *Recovering an insertion bound after centering.* The hypotheses (Rec) are also available for a functional with nonzero one-point function.

Lemma F.4. Suppose σ satisfies (Rec1)–(Rec2), its spectrum obeys $E \ge |q_1|$, and $\sigma(f) = c \int f$. If its centered functional satisfies

$$|\sigma^{\mathrm{c}}(\theta \bar{F} \cdot (h) \cdot F')| \le C_\wp \eta^{-\wp} \|h\|_1 \|[\tau_{-\eta e_3} F]\|_{\sigma^{\mathrm{c}}} \|[\tau_{-\eta e_3} F']\|_{\sigma^{\mathrm{c}}}, \tag{F.xix}$$

with the supports in (Rec4), then σ satisfies (Rec) with constants $(C_\wp + |c|, \wp, c_0)$. ■

Proof. Spatial rotations conjugate the joint spectral measure, so $E \ge |n \cdot q|$ for every unit spatial vector n. A countable dense set of such vectors gives $E \ge |q|$. The shift identities of §D.1 give $\sigma = \sigma^{\mathrm{c}} \circ \mathsf{sh}_c$ and an isometry $[F]_\sigma \mapsto [\mathsf{sh}_c F]_{\sigma^{\mathrm{c}}}$. The shift commutes with translations, conjugation and reflection, preserves supports, and is multiplicative. Writing $F_c = \mathsf{sh}_c F$, therefore,

$$\sigma(\theta \bar{F} \cdot (h) \cdot F') = \sigma^{\mathrm{c}}(\theta \bar{F}_c \cdot (h) \cdot F'_c) + c \int h \, \sigma^{\mathrm{c}}(\theta \bar{F}_c \cdot F'_c). \tag{F.xx}$$

The first term has the bound eq. F.xix with the σ norms. Cauchy–Schwarz and contractivity of T_η bound the second by $|c| \|h\|_1 \|[\tau_{-\eta e_3} F]\| \|[\tau_{-\eta e_3} F']\|$. Since $\eta \le 1$ and $\wp \ge 0$, their sum has the required bound. Centering preserves the other stated symmetries and reflection positivity by the same shift identities. □

F.2. **Analytic continuation of the Euclidean kernels.** The point-insertion bound gives a product estimate without a factorial. Rotation of the ordering direction then covers all distinct Euclidean points, including configurations with equal third coordinates.

Call $(h_1, \dots, h_n)$ *η-ordered along u* at heights $t_1, \dots, t_n$ if $u \in S^2$, $0 < \eta \le 1$, $t_{k+1} - t_k \ge 2\eta$, and

$$\operatorname{supp} h_k \subset \{|u \cdot x - t_k| < c_0\eta\}.$$

Such tuples are separated by at least η. For $u = e_3$, set $g_k = \tau_{-t_k e_3} h_k$ and $s_k = t_{k+1} - t_k - 2\eta \ge 0$.

Proposition F.5. For an η-ordered tuple along e_3,

$$\sigma(h_1, \dots, h_n) = \langle \Omega, B^{\mathrm{in}}_\eta(g_1) T_{s_1} B^{\mathrm{in}}_\eta(g_2) \cdots T_{s_{n-1}} B^{\mathrm{in}}_\eta(g_n) \Omega \rangle. \tag{F.xxi}$$

If $t_1 \ge \eta$, then

$$[(h_1, \dots, h_n)] = T_{t_1 - \eta} B^{\mathrm{in}}_\eta(g_1) T_{s_1} \cdots B^{\mathrm{in}}_\eta(g_n) \Omega, \qquad \|[(h_1, \dots, h_n)]\| \le \prod_k C_\wp \eta^{-\wp} \|h_k\|_1. \tag{F.xxii}$$

■

Proof. Starting at the right, eq. F.vi proves by downward induction that the product beginning with $B^{\mathrm{in}}_\eta(g_k)$ represents $[\tau_{(\eta - t_k)e_3}(h_k, \dots, h_n)]$. Indeed the intervening translation has height $s_k + \eta - t_{k+1} = -t_k - \eta$, and the insertion adds 2η to the remaining tuple. All these classes have supports in $\{x_3 > (1 - c_0)\eta\}$. Taking the vacuum component and using translation invariance proves eq. F.xxi. If $t_1 \ge \eta$, translate the class forward by $t_1 - \eta$ to obtain eq. F.xxii; contractivity and eq. F.v give its norm bound. □

For later use, the scalar density formula can be iterated: for $g_k \in L^1(\mathrm{Sl}_\eta)$ and bounded $A_0, \dots, A_n$,

$$\begin{aligned} &\langle \psi, A_0 B^{\mathrm{in}}_\eta(g_1) A_1 \cdots B^{\mathrm{in}}_\eta(g_n) A_n \psi' \rangle \\ &\qquad = \int \prod_k g_k(y_k) \langle \psi, A_0 b_\eta(y_1) A_1 \cdots b_\eta(y_n) A_n \psi' \rangle \, dy. \end{aligned} \tag{F.xxiii}$$

The second factor is continuous and bounded by $(C_\wp \eta^{-\wp})^n \|\psi\| \|\psi'\| \prod_j \|A_j\|$. Thus the integrand is absolutely integrable. Apply eq. F.ix to the first insertion, then to the remaining ones; Fubini proves the formula.

F.2.1. *Ordered complex configurations.* For $\zeta_1, \dots, \zeta_{n-1} \in \mathrm{Tb}_\circ$ define

$$\mathbf{W}_n(\zeta_1, \dots, \zeta_{n-1}) = \langle \Omega, B(\eta, \eta) \mathcal{T}(\zeta_1 - 2\eta e_3) B(\eta, \eta) \cdots \mathcal{T}(\zeta_{n-1} - 2\eta e_3) B(\eta, \eta) \Omega \rangle, \tag{F.xxiv}$$

where $0 < \eta \le \min(1, \frac{1}{2} \min_k d_\circ(\zeta_k))$. For $n = 1$ this is the real number $\mathbf{W}_1 = \langle \Omega, B(1, 1)\Omega \rangle$. On

$$\mathrm{Tb}_\circ^{(n)} = \{ z : z_1 \in \mathrm{Tb}_\circ, \ z_{k+1} - z_k \in \mathrm{Tb}_\circ \}$$

put

$$\eta(z) = \min \left(1, d_\circ(z_1), \tfrac{1}{2} \min_{k<n} d_\circ(z_{k+1} - z_k) \right), \tag{F.xxv}$$

$$\Psi_n(z) = \mathcal{T}(z_1 - \eta e_3) B(\eta, \eta) \mathcal{T}(z_2 - z_1 - 2\eta e_3) B(\eta, \eta) \cdots \mathcal{T}(z_n - z_{n-1} - 2\eta e_3) B(\eta, \eta) \Omega, \tag{F.xxvi}$$

using any $0 < \eta \le \eta(z)$; the inner minimum is omitted when $n = 1$.

These definitions are independent of η. To compare $\eta' < \eta$, substitute

$$B(\eta, \eta) = T_{\eta - \eta'} B(\eta', \eta') T_{\eta - \eta'}$$

at every insertion. The semigroup law absorbs the adjacent factors into the differences, and $T_t \Omega = \Omega$ removes the final factor. For the scalar product it removes the initial factor as well.

Proposition F.6. The function $\mathbf{W}_n$ is holomorphic and Ψ_n is holomorphic in norm. They satisfy

$$|\mathbf{W}_n(\zeta)| \le (C_\wp \eta^{-\wp})^n \qquad \text{if } d_\circ(\zeta_k) \ge 2\eta,\ 0 < \eta \le 1, \tag{F.xxvii}$$

$$\|\Psi_n(z)\| \le (C_\wp \eta(z)^{-\wp})^n, \tag{F.xxviii}$$

$$\mathcal{T}(w)\Psi_n(z) = \Psi_n(z_1 + w, \dots, z_n + w),$$

$$\langle \Omega, \Psi_n(z)\rangle = \mathbf{W}_n(z_2 - z_1, \dots, z_n - z_{n-1}) \tag{F.xxix}$$

for $w \in \overline{\mathrm{Tb}_\circ}$. For ordered tuples,

$$\sigma(h_1, \dots, h_n) = \int \mathbf{W}_n(x_2 - x_1, \dots, x_n - x_{n-1}) \prod_k h_k(x_k)\, dx. \tag{F.xxx}$$

If also $t_1 \ge \eta$, then

$$\langle \psi, [(h_1, \dots, h_n)]\rangle = \int \langle \psi, \Psi_n(x)\rangle \prod_k h_k(x_k)\, dx. \tag{F.xxxi}$$

■

Proof. Contractivity and eq. F.x give the bounds. Locally choose a fixed η strictly smaller than every allowed margin. All semigroup factors are then norm holomorphic; the independence of η proves holomorphy of the defined functions. The semigroup law and the vacuum identities give eq. F.xxix.

The density can be written with half the margin:

$$b_\eta(y) = \mathcal{T}(y + \tfrac{\eta}{2}e_3) B(\tfrac{\eta}{2}, \tfrac{\eta}{2}) \mathcal{T}(-y + \tfrac{\eta}{2}e_3). \tag{F.xxxii}$$

Both arguments have positive margin, since $|y_3| < c_0\eta \le \eta/2$. This follows directly from the shift law and the formula for b_η. Insert eq. F.xxxii in eq. F.xxiii, applied to the products eq. F.xxi–eq. F.xxii. With $x_k = y_k + t_k e_3$, consecutive semigroup factors combine into $\mathcal{T}(x_{k+1} - x_k - \eta e_3)$. At the right the factor fixes Ω. At the left it either disappears in the vacuum component or becomes $\mathcal{T}(x_1 - \eta e_3/2)$. These are precisely eq. F.xxiv and eq. F.xxvi with margin $\eta/2$, proving the two integral formulas. The supports are compact and stay inside the respective open tubes, so their continuous integrands are integrable. □

F.2.2. *Changing the ordering direction.* For $u \in S^2$ put

$$d^u_\circ(z) = \mathrm{Re}(u \cdot z) - |\,\mathrm{Im}(z - (u \cdot z)u)|, \qquad \mathrm{Tb}^{u,\pi}_n = \{z : d^u_\circ(z_{\pi(k+1)} - z_{\pi(k)}) > 0,\ k < n\}.$$

If $R \in \mathrm{SO}(3)$ satisfies $Re_3 = u$, then $d^u_\circ(z) = d_\circ(R^{-1}z)$. Each tube is open and convex; it contains $\mathrm{Re}\, z$ whenever it contains z. For $n = 1$ it is all of $\mathbb{C}^3$. Define on it

$$\mathbf{S}^{R,\pi}_n(z) = \mathbf{W}_n(R^{-1}(z_{\pi(2)} - z_{\pi(1)}), \dots, R^{-1}(z_{\pi(n)} - z_{\pi(n-1)})). \tag{F.xxxiii}$$

Any n distinct real points admit a separating direction. More quantitatively, if their mutual distances are at least $\delta > 0$, there is $u \in S^2$ with

$$|u \cdot (x_i - x_j)| \ge \delta/n^2 \quad (i \ne j). \tag{F.xxxiv}$$

Indeed the spherical zone $|u \cdot (x_i - x_j)/|x_i - x_j|| < n^{-2}$ has area $4\pi/n^2$. The union over the $\binom{n}{2}$ pairs has area less than 2π, so cannot cover S^2. Ordering the projections shows that the real parts of the tubes cover $(\mathbb{R}^3)^n_{\ne}$.

Proposition F.7. There is a unique continuous kernel s_n on $(\mathbb{R}^3)^n_{\ne}$ such that

$$\sigma(f_1, \dots, f_n) = \int \mathsf{s}_n(x) \prod_i f_i(x_i)\, dx \tag{F.xxxv}$$

for all separated tuples. It is real analytic, symmetric and invariant under Euclidean isometries. It extends uniquely to a holomorphic function $\mathbf{S}_n$ on $\mathrm{Tb}_n^{\mathrm{all}} = \bigcup_{u,\pi} \mathrm{Tb}_n^{u,\pi}$, whose restriction to each tube is eq. F.xxxiii, independently of the choice of R. This extension is symmetric and obeys

$$|\mathbf{S}_n(z)| \le (C_\wp \eta^{-\wp})^n \quad \text{if } d_\circ^u(z_{\pi(k+1)} - z_{\pi(k)}) \ge 2\eta, \quad 0 < \eta \le 1. \tag{F.xxxvi}$$

In particular,

$$\mathsf{s}_n(x) = \mathbf{W}_n(x_2 - x_1, \dots, x_n - x_{n-1}) \quad (x_{13} < \cdots < x_{n3}), \qquad \mathsf{s}_1 = \mathbf{W}_1. \tag{F.xxxvii}$$

■

Proof. For a tuple ordered along u, rotate it by R^{-1}, permute its entries, and apply eq. F.xxx. Euclidean invariance and symmetry give the kernel formula with $\mathbf{S}_n^{R,\pi}$. If a real configuration belongs to two ordering regions, choose sufficiently small balls about its entries so that every product of tests in those balls is ordered both ways, with one common positive margin. The two kernel formulas have the same value on every such product. Continuous functions are determined by these product tests, for example by taking products of approximate identities. Thus the kernels agree on each real overlap.

The real ordering regions cover all distinct points by eq. F.xxxiv. They therefore define one real-analytic function s_n. To obtain eq. F.xxxv for arbitrary separated supports, let their least distance be $d > 0$ and choose $\eta = \min(d/(4n^2), 1)$. Partition each test into finitely many pieces supported in open Euclidean balls of radius $c_0\eta/2$. Choose a point of the original support in every nonzero piece. Equation eq. F.xxxiv orders these chosen points with gaps at least 4η; each piece is supported in a $c_0\eta$ slab about its chosen point. Each resulting product tuple is therefore ordered. Apply the formula to each product and sum by multilinearity. For $n = 1$ the same partition uses $\eta = 1$. This proves the formula on every separated tuple. Products of arbitrarily small supported tests also prove uniqueness. Applying uniqueness to permuted or isometrically transported tests gives the claimed real symmetries.

A nonempty complex overlap of two tubes is convex, and contains the real part of each of its points. It consequently has a nonempty open real part. The two holomorphic pieces agree there by the preceding argument. The identity theorem on the connected overlap makes them equal throughout it. They therefore glue to one holomorphic function on the union. The same identity theorem on each tube proves uniqueness and independence of R; it also extends the permutation symmetry. The bound is eq. F.xxvii after rotation. All constructions depend on σ, not on the chosen admissible numerical triple. □

F.2.3. *Growth on tests flat at the diagonals.* The extension to the full space of tests used for Schwinger functions requires a bound near every diagonal, not only for products with fixed separated supports.

Proposition F.8. Put $k_{\mathrm{E}} = \lceil \wp \rceil + 4$ and $C_{\mathrm{E}} = C_\wp 2^\wp e^{2\wp} 3^{k_{\mathrm{E}}/2} \pi^2$. Then $|\mathsf{s}_1| \le C_\wp$, and for $n \ge 2$,

$$|\mathsf{s}_n(x)| \le C_\wp^n (2n^2)^{\wp n} \min(1, \min_{i \ne j} |x_i - x_j|)^{-\wp n}. \tag{F.xxxviii}$$

For every $\varphi \in \mathcal{S}_{\ne}(\mathbb{R}^{3n})$,

$$\int |\mathsf{s}_n(x)\varphi(x)|\, dx \le 2C_{\mathrm{E}}^n (n!)^{2\wp} |\varphi|_{k_{\mathrm{E}} n}. \tag{F.xxxix}$$

Consequently the Schwinger functions on this space are

$$\mathsf{S}_n^\sigma(\varphi) = \int \mathsf{s}_n(x)\varphi(x)\, dx. \tag{F.xl}$$

■

Proof. For a configuration with least Euclidean distance ρ, choose the direction in eq. F.xxxiv and use eq. F.xxxvi with $\eta = \min(\rho/(2n^2), 1)$. This proves eq. F.xxxviii; the one-point bound follows from $B(1,1)$.

Let $k = k_{\mathrm{E}} n$, $N = \lceil \wp n \rceil$. Then $N \le k$ and $k \ge 4n$. If $\rho \le 1$ and $N \ge 1$, move a closest pair to its midpoint, obtaining a diagonal point x^* with $|x - x^*| \le 1$ and $|x - x^*|_1 \le \sqrt{3}\rho$. Flatness at x^* and Taylor's remainder, with the weighted seminorm of §C.2, give

$$|\varphi(x)| \le \frac{(\sqrt{3}\rho)^N}{N!}\, 3^{k/2} |\varphi|_k (1 + |x|^2)^{-k/2} \le 2\, 3^{k/2} \rho^N |\varphi|_k (1 + |x|^2)^{-k/2}.$$

The factor $3^{k/2}$ accounts for moving at most unit distance along the Taylor segment. Since $N \ge \wp n$, this absorbs the singular factor. When $\rho > 1$ or $\wp = 0$, the unmodified seminorm bound suffices. Finally

$$n^n \le e^n n!, \qquad \int_{\mathbb{R}^{3n}} (1 + |x|^2)^{-k/2}\, dx \le \prod_{i=1}^{n} \int_{\mathbb{R}^3} (1 + |x_i|^2)^{-2}\, dx_i = \pi^{2n}.$$

Combining these estimates proves eq. F.xxxix. The integral functional agrees with σ on separated smooth tensors by eq. F.xxxv. The tensor-density and uniqueness result of §C.2 identifies it with the Schwinger function. □

F.3. **Smeared fields and their boundary distributions.** We continue under (Rec). Minkowski space has coordinates $x = (t, \vec{x})$, product $x \cdot y = ts - \vec{x} \cdot \vec{y}$, and forward cone $V_+ = \{t > |\vec{x}|\}$. Write $U(x) = \mathcal{T}(\iota x)$ and fix an integer $l > \wp$. For $f \in \mathcal{S}(\mathbb{R}^{1+2})$ put

$$c_N(f) = \|A_l\| \sum_{k=0}^{N} \binom{N}{k} \|\partial_t^k f\|_1. \tag{F.xli}$$

The Schwartz seminorms used above give $\|\partial^\alpha F\|_1 \le \pi^{2n} |F|_{|\alpha|+4n}$ on $\mathbb{R}^{3n}$, by comparison with $\prod_i (1 + |x_i|^2)^{-2}$. In particular

$$c_N(f) \le \|A_l\| 2^N \pi^2 |f|_{N+4}. \tag{F.xlii}$$

For integers $j \ge 0$, we use the graph seminorms $\|(1 + H)^j \psi\|$ on $\mathcal{D}_\infty$. The spectral projections $P_N = \mathcal{P}(\{E \le N\})$ have ranges in $\mathcal{D}_\infty$ and converge strongly to the identity. Bounded joint spectral multipliers preserve every $\operatorname{dom} H^j$ and commute there with $(1 + H)^j$. We will also use the following domain criterion:

$$|\langle (1 + H)^j \psi, \chi \rangle| \le C \|\psi\| \quad (\psi \in \mathcal{D}_\infty) \implies \chi \in \operatorname{dom} H^j, \quad \|(1 + H)^j \chi\| \le C. \tag{F.xliii}$$

Indeed, insert $\psi = (1 + H)^j P_N \chi$ to bound $\int_{E \le N} (1 + E)^{2j}\, dm_\chi$ by C^2, then let $N \to \infty$.

F.3.1. *Construction on a common domain.*

Proposition F.9. There is a unique complex linear map $f \mapsto \phi(f)$ from $\mathcal{S}(\mathbb{R}^{1+2})$ to linear operators $\mathcal{D}_\infty \to \mathcal{D}_\infty$ with

$$\langle \psi, \phi(f) \psi' \rangle = \int f(x)\, q_0(U(-x)\psi, U(-x)\psi')\, dx. \tag{F.xliv}$$

It is independent of l and of the numerical triple in (Rec). For every integer $j \ge 0$,

$$\|(1 + H)^j \phi(f) \psi\| \le c_{l+j}(f) \|(1 + H)^{2l+j} \psi\|. \tag{F.xlv}$$

On $\mathcal{D}_\infty$ it satisfies

$$\begin{aligned} \langle \psi, \phi(f) \psi' \rangle &= \langle \phi(\bar{f}) \psi, \psi' \rangle, \\ U(a) \phi(f) U(-a) &= \phi(f(\cdot - a)), \qquad [H, \phi(f)] = i \phi(\partial_t f). \end{aligned} \tag{F.xlvi}$$

■

Proof. The integrand in eq. F.xliv equals

$$f(x)\langle U(-x)(1+H)^l\psi, A_l U(-x)(1+H)^l\psi'\rangle.$$

Its second factor is bounded and continuous. Denote the resulting sesquilinear form by q_f. It obeys

$$|q_f(\psi,\psi')| \le \|f\|_1 \|A_l\| \|(1+H)^l\psi\| \|(1+H)^l\psi'\|. \tag{F.xlvii}$$

To turn this form into an operator, move energy powers from the first vector to the second. The map $t \mapsto e^{-itH}\xi$ is norm differentiable for $\xi \in \operatorname{dom} H$, with derivative $-ie^{-itH}H\xi$, by dominated convergence in the spectral calculus. Differentiating both vectors in the preceding matrix element, then integrating by parts in t, gives

$$q_f(H\psi,\psi') - q_f(\psi,H\psi') = iq_{\partial_t f}(\psi,\psi').$$

The boundary terms vanish because f is Schwartz and the matrix element and its derivative are bounded. Iteration yields

$$q_f((1+H)^N\chi,\psi') = \sum_{k=0}^{N} \binom{N}{k} i^k q_{\partial_t^k f}(\chi,(1+H)^{N-k}\psi'). \tag{F.xlviii}$$

Choose $\chi = R_H^l\psi$ and $N = l+j$. Using eq. F.xlvii and monotonicity of graph norms,

$$|q_f((1+H)^j\psi,\psi')| \le c_{l+j}(f)\|\psi\| \|(1+H)^{2l+j}\psi'\|. \tag{F.xlix}$$

At $j=0$, Riesz representation gives a unique vector $\phi(f)\psi'$. For every j, the domain criterion eq. F.xliii gives its membership in $\operatorname{dom} H^j$ and the bound eq. F.xlv. Thus it belongs to $\mathcal{D}_\infty$.

Linearity, uniqueness and independence of the auxiliary choices follow from the defining form. Hermiticity follows from that of q_0. Changing variables in eq. F.xliv proves translation covariance. The identity obtained by integration by parts, tested against dense vectors in $\mathcal{D}_\infty$, proves the commutator formula. □

F.3.2. *Regularized products and their continuity.*

Lemma F.10. Let $w, w' \in \mathrm{Tb}_\circ$ and $0 < s \le \min(1, d_\circ(w), d_\circ(w'))$. Then

$$\mathcal{T}(w)\phi(f)\mathcal{T}(w') = \int f(x)\mathcal{T}(w - se_3 + \iota x)B(s,s)\mathcal{T}(w' - se_3 - \iota x)\,dx \tag{F.l}$$

in matrix elements on $\mathcal{H}$. The integrand after removal of f is strongly continuous and bounded by $C_\wp s^{-\wp}$. For $w_1,\dots,w_n \in \mathrm{Tb}_\circ$, set $z_k(x) = \sum_{r\le k} w_r + \iota x_k$. Then

$$\begin{aligned}&\langle\psi, \mathcal{T}(w_1)\phi(f_1)\cdots\mathcal{T}(w_n)\phi(f_n)\Omega\rangle \\ &\quad = \int \prod_k f_k(x_k)\langle\psi, \Psi_n(z(x))\rangle\,dx.\end{aligned} \tag{F.li}$$

■

Proof. The positive margins put $\mathcal{T}(w)^*\psi, \mathcal{T}(w')\psi'$ in $\mathcal{D}_\infty$. Apply eq. F.xliv and split T_s from each semigroup. Equation eq. F.xviii changes the form between these two factors into $B(s,s)$; commuting the remaining spectral multipliers gives eq. F.l. The margin is unchanged by addition of ιx, so all remaining factors are contractions. Strong continuity and the point bound give the stated properties.

For the product choose $s = \min(1, d_\circ(w_1), \frac{1}{2}\min_{k\ge 2} d_\circ(w_k))$. The inner minimum is omitted when $n=1$. Split each intermediate semigroup into equal halves, and append T_s at the right, where it fixes Ω. This writes the product as a product of bounded regularized fields, each covered by eq. F.l. Successively insert those formulas. The uniform point bounds and $\prod_k |f_k| \in L^1$ justify Fubini. Consecutive semigroup arguments combine to $z_{k+1}(x) - z_k(x) - 2se_3$; the first is $z_1(x) - se_3$, and the final spatial-time translation fixes Ω. The resulting product is $\Psi_n(z(x))$ with margin s, proving eq. F.li. □

Lemma F.11. Suppose $c_N(f_k^\epsilon - f_k^0) \to 0$ for every N, and C_k^ϵ are contractive joint spectral multipliers converging strongly to C_k^0. Then

$$C_1^\epsilon \phi(f_1^\epsilon) \cdots C_n^\epsilon \phi(f_n^\epsilon)\Omega \longrightarrow C_1^0 \phi(f_1^0) \cdots C_n^0 \phi(f_n^0)\Omega$$

in every graph norm of H. ■

Proof. Work from right to left. If $\chi^\epsilon \to \chi^0$ in all graph norms, then

$$\begin{aligned}&\|(1+H)^j(\phi(f^\epsilon)\chi^\epsilon - \phi(f^0)\chi^0)\| \\ &\quad \le c_{l+j}(f^\epsilon - f^0)\|(1+H)^{2l+j}\chi^\epsilon\| + c_{l+j}(f^0)\|(1+H)^{2l+j}(\chi^\epsilon - \chi^0)\| \longrightarrow 0.\end{aligned}$$

The spectral multipliers commute with the graph powers. Their strong convergence and uniform norm bound preserve the just-proved convergence. Induction starts at $\chi^\epsilon = \Omega$. □

F.3.3. *Tempered boundary values.* Here we give the boundary argument explicitly; pointwise polynomial growth near the tube boundary alone does not justify passing a limit through a test integral.

Lemma F.12. Let $\mathbf{F}$ be holomorphic on an open set containing

$$z(x;\eta) = (\iota x_j + \eta h_j e_3)_{j=1}^n, \qquad x \in K, \quad 0 < \eta \le \eta_0 \le 1,$$

where K is compact, and suppose that for real $b, p \ge 0$, $|\mathbf{F}(z(x;\eta))| \le b\eta^{-p}$. For $g \in C_c^\infty$ supported in K, put $I_g(\eta) = \int g(x)\mathbf{F}(z(x;\eta))\,dx$ and $D_h = \sum_j h_j \partial_{t_j}$. If the integer $N \ge p + 1/2$, then

$$\sup_{0<\eta\le\eta_0} |I_g(\eta)| \le 2^N \eta_0^{-(N-1/2)} b \sum_{r=0}^{N} \|D_h^r g\|_1. \tag{F.lii}$$

The limit at $\eta = 0$ exists. ■

Proof. On each compact interval of positive heights, holomorphy permits differentiation under the integral. The Cauchy–Riemann equations and integration by parts give $I_g' = I_{-iD_h g}$. Set $J_r = I_{(-iD_h)^r g}$. For $0 \le r \le N$, these functions obey $J_r' = J_{r+1}$ when $r < N$, and $|J_r(\eta)| \le b\|D_h^r g\|_1 \eta^{-(N-1/2)}$. Integrate successively from η_0 toward η. Every integration of η^{-a} with $a \ge 3/2$ lowers the singular exponent by one at a cost at most 2; after $N-1$ integrations the exponent is $1/2$. Its final integral is bounded by 2. The values at η_0 contribute the same bound with the corresponding $\|D_h^r g\|_1$, giving eq. F.lii. Apply that estimate once more to $-iD_h g$. Then I_g' is bounded uniformly down to zero, so I_g is Lipschitz there and has a limit. □

Proposition F.13. There is a unique tempered distribution $\mathcal{W}_n^\sigma$ on $\mathbb{R}^{3n}$ satisfying

$$\mathcal{W}_n^\sigma(f_1 \otimes \cdots \otimes f_n) = \langle \Omega, \phi(f_1) \cdots \phi(f_n)\Omega \rangle. \tag{F.liii}$$

For arbitrary $\alpha_1, \ldots, \alpha_{n-1} \in V_+$,

$$\begin{aligned}&\mathcal{W}_n^\sigma(f_1 \otimes \cdots \otimes f_n) \\ &\quad = \lim_{\epsilon \downarrow 0} \int \prod_k f_k(x_k) \mathbf{W}_n\big(\iota(x_2 - x_1 + i\epsilon\alpha_1), \ldots, \iota(x_n - x_{n-1} + i\epsilon\alpha_{n-1})\big)\,dx.\end{aligned} \tag{F.liv}$$

For $n = 1$, $\mathcal{W}_1^\sigma(f) = \mathbf{W}_1 \int f$. ■

Proof. First take the vacuum component of eq. F.li, with $w_1 = e_3$ and $w_{k+1} = \iota(i\epsilon\alpha_k)$. It identifies the integral on the right of eq. F.liv with

$$\langle \Omega, \phi(f_1)\mathcal{T}(w_2)\phi(f_2) \cdots \mathcal{T}(w_n)\phi(f_n)\Omega \rangle.$$

Its integrand is bounded by $(C_\wp \eta^{-\wp})^n \prod_k |f_k(x_k)|$, with $2\eta = \min(2, \epsilon \min_k(\alpha_k^0 - |\vec{\alpha}_k|))$. The spectral contractions tend strongly to the identity. Lem. F.11 therefore proves the boundary formula on tensors of Schwartz functions.

To obtain a distribution on general tests, apply Lem. F.12 to $\mathbf{F}(z) = \mathbf{W}_n(z_2 - z_1, \dots, z_n - z_{n-1})$ with $h_j = 2j$, $\eta_0 = 1$, $b = C_\wp^n$, $p = \wp n$, and $N = \lceil \wp n \rceil + 1$. Every consecutive margin is 2η, so the hypothesis follows from eq. F.xxvii. The resulting limit functional on C_c^∞ has bound

$$|\mathcal{L}(F)| \le 2^N (N+1) C_\wp^n (n(n+1))^N \pi^{2n} |F|_{N+4n}.$$

Here $\sum_j h_j = n(n+1)$ bounds the coefficients in D_h^r, and the L^1 derivative estimate gives the Schwartz norm.

For a smooth cutoff χ equal to one near the origin, $\chi(\cdot/R)F \to F$ in every Schwartz seminorm. Thus the bound extends $\mathcal{L}$ continuously to $\mathcal{S}(\mathbb{R}^{3n})$. It agrees with eq. F.liii first on compactly supported tensors by the boundary identity, then on Schwartz tensors by Lem. F.11. For uniqueness, finite sums of compactly supported smooth tensors are dense in each fixed Schwartz seminorm: cut off the test, convolve with a product mollifier, and approximate the resulting compact integral by Riemann sums of translates of that product mollifier. All derivatives up to the fixed order converge uniformly on a common compact set. A tempered functional vanishing on those tensors must therefore vanish. □

F.3.4. *Cyclicity.* Let $\mathcal{D}_0$ be the span of Ω and all $\phi(f_1)\cdots\phi(f_n)\Omega$. It lies in $\mathcal{D}_\infty$, is invariant under the fields and translations, and is dense.

To prove the last assertion, let $\psi \perp \mathcal{D}_0$. Translation covariance first gives

$$\langle \psi, U(a_1)\phi(f_1)\cdots U(a_n)\phi(f_n)\Omega\rangle = 0.$$

Replace the translations successively by $\mathcal{T}(w_j)$, $w_j \in \mathrm{Tb}_\circ$. For one replacement, move all preceding regularized factors to the first vector using Hermiticity. The first positive-margin semigroup puts that vector in $\mathcal{D}_\infty$, so every move is legitimate. The resulting expression is $\langle \chi, U(a)\chi'\rangle$. Its vanishing for all a says that the Fourier transform of the finite joint spectral measure $m_{\chi,\chi'}$ vanishes. Uniqueness of Fourier transforms of finite measures gives $m_{\chi,\chi'} = 0$, and hence $\langle \chi, \mathcal{T}(w)\chi'\rangle = 0$. Induction proves vanishing for every fully regularized product.

By eq. F.li and product approximate identities, $\langle \psi, \Psi_n(z)\rangle = 0$ throughout $\mathrm{Tb}_\circ^{(n)}$. Equation eq. F.xxxi then makes ψ orthogonal to classes of time-ordered positive-time tuples. We must still approximate arbitrary positive-time tuples; their supports need not be time ordered.

Fix such a tuple $X = (f_1, \dots, f_n)$. Partition its entries with nonnegative functions $\chi_{h,k}$ supported in cubes of side $2h$ about hk, summing to one. Call a product of pieces good when all their center heights differ by at least $4h/c_0$. For small h, every good product, after permutation, is ordered with $\eta = 2h/c_0$ and first height at least η. Its class is orthogonal to ψ.

Let ξ_h be the sum of the remaining classes. On the compact set consisting of the original positive supports and their reflected negative supports, the continuous kernel s_{2n} is bounded by some M_X. The kernel formula gives

$$\|\xi_h\|^2 \le M_X \left(\int \prod_i |f_i(x_i)| \sum_{\mathbf{k}\ \mathrm{bad}} \prod_i \chi_{h,k_i}(x_i)\, dx \right)^2. \tag{F.lv}$$

The sum is at most one and vanishes unless some two heights differ by less than $(4/c_0 + 2)h$. Its integral is $O_X(h)$: integrate one height first over an interval of that length, bound that test by its supremum norm on its fixed spatial support, then integrate the remaining tests in L^1. Thus $\|\xi_h\| \to 0$ and $\langle \psi, [X]\rangle = \langle \psi, \xi_h\rangle = 0$. All tuple classes span a dense subspace of the reflection Hilbert space, so $\psi = 0$ and $\mathcal{D}_0$ is dense.

The translation representation already has joint spectrum $E \ge |q|$. In the convention $U(x) = \int e^{ip\cdot x}\, d\mathsf{E}(p)$ its momentum measure is the image of $\mathcal{P}$ under $(E, q) \mapsto (E, -q)$, and is carried by $\overline{V_+}$. If $\ker H = \mathbb{C}\Omega$, the only translation-invariant vectors are multiples of Ω: invariance under e^{itH} forces spectral support at $E = 0$. No vacuum-uniqueness assertion is used in the preceding construction.

F.4. **Lorentz covariance and locality.** Let $\mathcal{L} = \mathrm{SO}^+(1,2)$. Every element is a product of spatial rotations and the boosts

$$B(s)(t, x_1, x_2) = (t\cosh s + x_1 \sinh s,\ x_1 \cosh s + t \sinh s,\ x_2).$$

For completeness, Λe_0 is a future unit timelike vector. A spatial rotation and a boost send e_0 to it. Their inverse followed by Λ fixes e_0 and hence acts by an element of $\mathrm{SO}(2)$ on its orthogonal spatial plane. This proves the factorization. These matrices have determinant one and preserve V_+.

Proposition F.14. There is a unique strongly continuous unitary representation $U(\Lambda)$ of $\mathcal{L}$ fixing Ω and satisfying

$$U(\Lambda)\mathcal{D}_0 = \mathcal{D}_0, \qquad U(\Lambda)\phi(f)U(\Lambda)^{-1}\psi = \phi(f \circ \Lambda^{-1})\psi \quad (\psi \in \mathcal{D}_0). \tag{F.lvi}$$

Together with translations it defines the strongly continuous representation

$$U(a, \Lambda) = U(a)U(\Lambda), \qquad U(a, \Lambda)U(a', \Lambda') = U(a + \Lambda a', \Lambda\Lambda'). \tag{F.lvii}$$

Its spatial-rotation subgroup agrees with the representation on the reflection Hilbert space. ∎

Proof. We first prove invariance of the analytic functions:

$$\mathbf{W}_n(\iota\Lambda\zeta_1, \dots, \iota\Lambda\zeta_{n-1}) = \mathbf{W}_n(\iota\zeta_1, \dots, \iota\zeta_{n-1}) \quad (\zeta_k \in \mathbb{R}^{1+2} + iV_+). \tag{F.lviii}$$

The complex linear map $\widehat{\Lambda} = \iota\Lambda\iota^{-1}$ preserves $\mathrm{Tb}_\circ$. Both sides, viewed as functions of $\iota\zeta_k$, are holomorphic on its connected product. It suffices to compare them at real differences ξ_k of positive third coordinate. Choose $x_1 = 0$, $x_{k+1} = x_k + \xi_k$. Then $\mathbf{W}_n(\xi) = \mathsf{s}_n(x)$. Spatial-rotation invariance follows from that of the Euclidean kernel.

For a boost, use the complex rotation

$$r_z(y_1, y_2, y_3) = (y_1 \cos z + y_3 \sin z,\ y_2,\ y_3 \cos z - y_1 \sin z). \tag{F.lix}$$

It satisfies $r_{is} = \widehat{B(s)}$. Fix real positive-time differences ξ_k. The set of z for which every $r_z\xi_k$ lies in $\mathrm{Tb}_\circ$ is open and contains the imaginary axis, since real boosts preserve the forward tube. A sufficiently small neighborhood of the compact segment from 0 to is lies in this set and is connected. For small real z, r_z is an ordinary Euclidean rotation preserving the ordering of these points. Thus $\mathbf{W}_n(r_z\xi_1, \dots, r_z\xi_{n-1})$ is constant there by Euclidean invariance. The identity theorem continues this equality to $z = is$. The identity theorem in all difference variables, then the rotation–boost factorization, proves eq. F.lviii.

In eq. F.liv change variables $x_j = \Lambda y_j$ and use eq. F.lviii. The approach vectors become $\Lambda^{-1}\alpha_k \in V_+$, which are allowed in that boundary formula. Hence

$$\mathcal{W}_n^\sigma(F \circ (\Lambda^{-1})^{\times n}) = \mathcal{W}_n^\sigma(F). \tag{F.lx}$$

This first holds on tensors; continuity under invertible linear changes of Schwartz variables and tensor density extend it to all tests.

On a polynomial vector define

$$U(\Lambda)\phi(f_1)\cdots\phi(f_n)\Omega = \phi(f_1 \circ \Lambda^{-1})\cdots\phi(f_n \circ \Lambda^{-1})\Omega.$$

Inner products of two such vectors are Wightman values: move the first string across using Hermiticity, reversing its order and conjugating its tests. Equation eq. F.lx preserves every such inner product. The rule therefore respects all linear relations, is isometric, and has inverse given by Λ^{-1}. Density of $\mathcal{D}_0$ gives a unitary extension; the group law and covariance hold on its spanning vectors and hence as stated. This also proves uniqueness.

As $\Lambda \to 1$, the time derivatives of $f \circ \Lambda^{-1} - f$ tend to zero in L^1. The chain rule expresses them as finite sums of derivatives of f after the linear change; dominated convergence, using Schwartz decay uniformly for Λ near 1, proves the assertion. Lem. F.11 gives strong continuity first on polynomial vectors, then on all of $\mathcal{H}$ by unitarity and density. On polynomial vectors the

conjugation $U(\Lambda)U(a)U(\Lambda)^{-1}$ translates every test by Λa. This proves eq. F.lvii and continuity of the combined representation.

Finally, the original spatial rotations commute with H and $B(s,s')$, hence with A_l, and preserve q_0. They conjugate $U(x)$ to $U((1\oplus R)x)$. Changing variables in eq. F.xliv therefore gives the same field transformation as $U(1\oplus R)$. Both fix the vacuum, so they agree on $\mathcal{D}_0$ and on its closure. □

F.4.1. *Interchanging spacelike-separated insertions.*

Proposition F.15. If $f,h\in\mathcal{S}(\mathbb{R}^{1+2})$ have spacelike separated supports, then

$$\phi(f)\phi(h)\psi=\phi(h)\phi(f)\psi\qquad(\psi\in\mathcal{D}_0).\tag{F.lxi}$$

■

Proof. First suppose all tests have compact support. Testing against polynomial vectors and using Hermiticity reduces the assertion to invariance of $\mathcal{W}_n^{\sigma}(f_1\otimes\cdots\otimes f_n)$ under interchange of adjacent entries $k,k+1$ whose supports are spacelike separated. Set $K=\prod_j \operatorname{supp} f_j$; zero entries are trivial. On K, write $\xi=\vec{x}_{k+1}-\vec{x}_k$, $\tau=t_{k+1}-t_k$, and put

$$\delta=\min_K(|\xi|-|\tau|)>0,\quad A=4/\delta,\quad \eta_0=\min((2A)^{-1},1/2),$$
$$h_*=2A\max_{x\in K,\ j<n}(|\vec{x}_{j+1}-\vec{x}_j|+|t_{j+1}-t_j|)+6.$$

We deform the two adjacent heights past each other:

$$h_k^\lambda=-\lambda/2,\quad h_{k+1}^\lambda=\lambda/2,$$
$$h_j^\lambda=\tfrac12+(j-k-1)h_*\quad(j>k+1),\qquad h_j^\lambda=-\tfrac12-(k-j)h_*\quad(j<k),\tag{F.lxii}$$
$$z_j(x;\lambda,\eta)=\iota x_j+\eta h_j^\lambda e_3,\qquad -1\le\lambda\le1,\quad 0<\eta\le\eta_0.$$

Although the vertical ordering fails during this deformation, the points remain in the union of tilted tubes. Indeed choose $\sin\theta=A\eta$ and

$$u_\theta=\cos\theta\,e_3+\sin\theta\,(\xi/|\xi|,0).$$

Then $\cos\theta\ge\sqrt{3}/2$, and direct calculation gives

$$\begin{aligned}d_\circ^{u_\theta}(z_{j+1}-z_j)&=\eta(h_{j+1}^\lambda-h_j^\lambda)\cos\theta\\&\quad+A\eta\,\frac{\xi}{|\xi|}\cdot(\vec{x}_{j+1}-\vec{x}_j)-A\eta|t_{j+1}-t_j|.\end{aligned}\tag{F.lxiii}$$

For $j=k$ this is at least $-\eta+A\eta\delta=3\eta$. For $j\neq k$ the height gap is at least h_*; the definition of h_* makes the same lower bound hold. Consequently

$$z(x;\lambda,\eta)\in\mathrm{Tb}_n^{\mathrm{all}},\qquad|\mathbf{S}_n(z(x;\lambda,\eta))|\le(C_\wp\eta^{-\wp})^n\tag{F.lxiv}$$

uniformly in x,λ,η in these ranges. The witnessing direction may depend on x and η; the holomorphic function $\mathbf{S}_n$ is the same on all of these tubes.

For a smooth g supported in K, let $I_g(\lambda,\eta)=\int g(x)\mathbf{S}_n(z(x;\lambda,\eta))\,dx$. The Cauchy–Riemann equations and integration by parts give

$$\partial_\eta I_g=I_{-iD_\lambda g},\qquad\partial_\lambda I_g=\eta I_{-iD' g},\qquad D_\lambda=\sum_j h_j^\lambda\partial_{t_j},\quad D'=\tfrac12(\partial_{t_{k+1}}-\partial_{t_k}).\tag{F.lxv}$$

Compactness at every positive height justifies differentiation. Lem. F.12, with $N=\lceil\wp n\rceil+1$, gives the uniform bound

$$|I_g(\lambda,\eta)|\le Q(g):=2^N\eta_0^{-(N-1/2)}C_\wp^n\sum_{r=0}^{N}\sup_{|\lambda|\le1}\|D_\lambda^r g\|_1.\tag{F.lxvi}$$

The coefficients of D_λ are uniformly bounded, so $Q(g)<\infty$.

Take $g = \bigotimes_j f_j$. At $\lambda = 1$ the heights increase in the original order, and eq. F.liv identifies the limit $I_g(1,\eta)$ with the original Wightman value. At $\lambda = -1$ they increase in the order with $k, k+1$ interchanged. The symmetry of the single function $\mathbf{S}_n$ and the same boundary formula identify $I_g(-1,\eta)$ with the interchanged value. But eq. F.lxv–eq. F.lxvi imply

$$|I_g(1,\eta) - I_g(-1,\eta)| \le 2\eta Q(D'g) \longrightarrow 0.$$

This proves the adjacent interchange for compact tests.

For polynomial vectors with other Schwartz entries, cut off those entries and use Lem. F.11 to pass to the limit in both products. Density of polynomial test vectors gives eq. F.lxi for compactly supported f, h. Finally cut off f, h themselves inside their original supports. Their spacelike separation is preserved, and graph-norm continuity passes the equality to the original Schwartz functions. □

Theorem F.16 (Scalar reconstruction)**.** Under (Rec), the reflection Hilbert space carries a Hermitian scalar field ϕ with the following properties:

(1) The proper orthochronous Poincaré group acts by a strongly continuous unitary representation U fixing the unit vector Ω. Its translation spectrum lies in $\overline{V_+}$.
(2) The dense polynomial domain $\mathcal{D}_0$ contains Ω and is invariant under U and every $\phi(f)$. For $\psi, \psi' \in \mathcal{D}_0$, $f \mapsto \langle \psi, \phi(f)\psi' \rangle$ is tempered, and the Hermiticity relation in eq. F.xlvi holds.
(3) The field is covariant:
$$U(a,\Lambda)\phi(f)U(a,\Lambda)^{-1} = \phi(f(\Lambda^{-1}(\cdot - a))) \quad \text{on } \mathcal{D}_0,$$
and fields with spacelike separated test supports commute there.
(4) The vacuum is cyclic. It is the unique translation-invariant vector up to scalar multiples if $\ker H = \mathbb{C}\Omega$.

The vacuum distributions are $\mathcal{W}_n^\sigma$. Their analytic functions are $\mathbf{W}_n$, with boundary values eq. F.liv, and their Euclidean continuations recover σ on every separated C_c^1 tuple by eq. F.xxxv. On $\mathcal{S}_{\neq}(\mathbb{R}^{3n})$ the Schwinger functions are the absolutely convergent integrals eq. F.xl. Thus, with the additional vacuum-uniqueness hypothesis, these objects form a scalar Wightman theory. ■□

Proof. The domain, spectrum, cyclicity and Hermiticity were proved in §F.3. Equation eq. F.xlv, followed by eq. F.xlii, gives tempered matrix elements. Props. F.14 and F.15 give covariance and locality on $\mathcal{D}_0$. The boundary and Euclidean identities follow from eq. F.liv, eq. F.xxxv and eq. F.xxxvii. On differences with third coordinate at least $\epsilon > 0$ and real spatial coordinates, eq. F.xxvii bounds $\mathbf{W}_n$ by $(C_\wp \min(1, \epsilon/2)^{-\wp})^n$; in particular the Euclidean continuation has the required polynomial growth. Finally eq. F.xxxix proves the asserted absolute convergence. □

Definition F.17 (Euclidean axioms on flat tests)**.** The test spaces are those in Def. C.11. Isometries act by simultaneous pullback by their inverses and permutations by rearranging arguments. The axioms are growth (E0′) as in eq. C.xxxii, Euclidean invariance (E1), positivity (E2) in eq. C.xxx, permutation symmetry (E3), and clustering (E4) in eq. F.lxvii on positive-time flat tests. ■

The passage from clustering of separated tuples to clustering of Schwartz tests uses the reflection Hilbert space. Directly translating a Schwartz approximation would give bounds depending on the translation; the Hilbert-space argument keeps them uniform.

Proposition F.18. Suppose a symmetric, normalized separated-tuple functional σ has the bounds of §D.1, is Euclidean invariant and reflection positive, and has Schwinger functions satisfying

$$|\mathsf{S}_n(\varphi)| \le c_E C_E^n (n!)^\kappa |\varphi|_{kn}, \qquad \varphi \in \mathcal{S}_{\neq}(\mathbb{R}^{3n}),$$

where $k \in \mathbb{N}$ and $c_E, C_E, \kappa \ge 0$. Assume also that

$$\sigma(X \cdot \tau_v Y) \longrightarrow \sigma(X)\sigma(Y) \quad (|v| \to \infty)$$

for all separated tuples X, Y, with the concatenation defined for sufficiently large $|v|$. Then the Schwinger functions satisfy Euclidean invariance, permutation symmetry, reflection positivity eq. C.xxx, the displayed growth bound, and

$$\lim_{s\to\infty} \mathsf{S}_{n+n'}(\Theta\bar{\varphi}\otimes\tau_{sv}\varphi') = \mathsf{S}_n(\Theta\bar{\varphi})\mathsf{S}_{n'}(\varphi') \tag{F.lxvii}$$

for $\varphi\in\mathcal{S}_+(\mathbb{R}^{3n})$, $\varphi'\in\mathcal{S}_+(\mathbb{R}^{3n'})$ and $0\neq v\in\mathbb{R}^2\times\{0\}$. We refer to these five properties as (E0′)–(E4), respectively growth, Euclidean invariance, reflection positivity, symmetry and clustering. The test spaces are the flat Schwartz spaces defined in §C.2. ∎

Proof. Prop. C.15 gives the first four properties. For the last, form the algebraic direct sum $\underline{\mathcal{S}}_+ = \bigoplus_{n\geq 0}\mathcal{S}_+(\mathbb{R}^{3n})$, with the positive form

$$\langle\underline{\varphi},\underline{\psi}\rangle_{\mathcal{S}} = \sum_{n,n'} \mathsf{S}_{n+n'}(\Theta\bar{\varphi}_n\otimes\psi_{n'}).$$

Let $\mathcal{H}^{\mathcal{S}}$ be its completion modulo the null space. If both sequences vanish above degree n_0, their form is bounded in absolute value by

$$\max(1,c_E)\max(1,C_E)^{2n_0}((2n_0)!)^{\kappa}\left(\sum_{n\leq n_0}|\varphi_n|_{2kn_0}\right)\left(\sum_{n\leq n_0}|\psi_n|_{2kn_0}\right). \tag{F.lxviii}$$

This follows term by term from the tensor-product seminorm bound; the degree-zero term is ordinary multiplication.

Send a smooth separated positive-time tuple X to the sequence ιX with $\bigotimes X$ in its degree and zero elsewhere. For finite sums of such tuples,

$$\langle\iota F,\iota F'\rangle_{\mathcal{S}} = \sigma(\theta\bar{F}\cdot F'). \tag{F.lxix}$$

Lem. C.12 and eq. F.lxviii show that their classes are dense in $\mathcal{H}^{\mathcal{S}}$. They are also dense in $\mathcal{H}(\sigma)$: mollify each entry of a C^1_c tuple with a common radius tending to zero, smaller than its distance to the reflection plane and half its least support separation. The resulting tuples remain separated in the positive half-space. The moving-entry continuity estimate gives convergence of the three terms in

$$\|[X]-[X_j]\|^2 = \sigma(\theta X\cdot X) - 2\operatorname{Re}\sigma(\theta X\cdot X_j) + \sigma(\theta X_j\cdot X_j).$$

Consequently eq. F.lxix extends to a unitary $V:\mathcal{H}(\sigma)\to\mathcal{H}^{\mathcal{S}}$ carrying Ω to the constant sequence. For a fixed tangential translation, the seminorm estimate eq. C.xxxi shows that the same approximations converge after translation. Thus

$$V\mathcal{U}(v')V^{-1}[\underline{\varphi}] = [\tau_{(v',0)}\underline{\varphi}].$$

Clustering of tuples gives, first for tuple classes,

$$\langle\chi,\mathcal{U}(sv')\chi'\rangle \longrightarrow \langle\chi,\Omega\rangle\langle\Omega,\chi'\rangle, \qquad v'\neq 0.$$

The difference between the two sides has modulus at most $2\|\chi\|\|\chi'\|$, uniformly in s, since $\mathcal{U}(sv')$ is unitary. Density extends this limit to all vectors. Apply it to the inverse images under V of the classes with φ and φ' in degrees n, n'. The form identity gives exactly eq. F.lxvii. □

F.5. **Assembly of the continuum limits.**

Definition F.19 (Frame data)**.** The data are a converging frame (K_j,ν_j) in classes $\mathcal{G}_j$ obeying the hypotheses of §E.2. Write σ for its limit and σ' for an arbitrary limit in that frame, keeping the centring by ν_j when changing the integrating laws. The reconstruction hypotheses (Rec1) to (Rec4) are those of §F.1 for the functional itself. We write (Rec5) for the condition $\ker H = \mathbb{C}\Omega$. ∎

Definition F.20 (Optional frame hypotheses)**.** The rotation condition (Rot) is Euclidean invariance of all limits in the frame. The point-bound condition (PtB) asks for common constants $C_\wp, \wp, c_0$ with $0 < c_0 \le 1/2$ for which every limit satisfies

$$|\sigma'^{c}(\theta\bar{F}\cdot(h)\cdot G)| \le C_\wp \eta^{-\wp}\|h\|_1 \|[\tau_{-\eta e_3}F]\|_{\sigma'^{c}}\|[\tau_{-\eta e_3}G]\|_{\sigma'^{c}} \tag{F.lxx}$$

for $0 < \eta \le 1$, h supported in $\{|x_3| < c_0\eta\}$ and F, G supported in $\{x_3 > \eta\}$. ■

Proposition F.21 (Point bound and reconstruction hypotheses)**.** Let σ' be a limit in a frame as above. Under (Rot) and (PtB), it has (Rec1) to (Rec4). For a component it has (Rec5) as well. ■

Proof. The first condition is (Rot). The limit properties of Prop. E.18 give its one-point function, bounds and reflection positivity, so Prop. D.2 applies and gives (Rec2). Cor. D.8 gives (Rec3). By (PtB), the centered functional obeys eq. F.xix. Apply Lem. F.4 with $c = c_{\sigma'}$ to obtain (Rec4) with constant $C_\wp + |c_{\sigma'}|$. For components the clustering estimate of Prop. E.20 and the spectral limit in eq. E.xlix give (Rec5). □

Theorem F.22 (The frame limit and its components)**.** For a convergent frame, the following hold.

(1) The frame limit σ (a) is symmetric and equals the centred functional of its connected functions $S_n = \lim_j S_n^T[\nu_j]$, $n \ge 2$, with table $\underline{C}$ given by the frame bounds. It is (b) translation and cubic invariant and (c) reflection positive at coordinate and diagonal mirrors. In (d), its moments and connected functions have unique off-diagonal distributional extensions of order at most $4n+1$ on compact sets. In (e), its coordinate Hilbert space is separable, has the translation semigroups of Prop. D.2 and a unitary W_2 action, and satisfies $H \ge |P_k|$ for $k = 1, 2$. Under (Rot), (g) the invariance is Euclidean, all mirrors are positive, the spectral cone is circular and O(2) acts unitarily on the coordinate Hilbert space as in Cor. D.8.

(2) (a) The decomposition of σ is canonical and unique up to the unique measure-preserving homeomorphism matching the components. (b) The map $P_0[F] \mapsto (\xi \mapsto \sigma_\xi(F))$ gives the vacuum-space unitary of Thm. E.22, including the dimension statement. (c) An isometry preserving σ acts by a measure-preserving homeomorphism on the component space. This homeomorphism is the identity for translations and cubic isometries, and for every Euclidean isometry under (Rot).

(3) Each component σ_ξ has the following properties. (i) It is a limit in the original frame. It lies in $\mathsf{Fct}(\underline{C}, \bar{c})$ with $\bar{c}$ given by eq. E.xliv, is invariant under translations and cubic isometries, is reflection positive at coordinate and diagonal mirrors, and is regular of order at most $4n+1$ on compact off-diagonal sets. Under (Rot) it is Euclidean invariant and reflection positive at every mirror. (ii)(a) Its mean is $c_\xi \int f$, with $|c_\xi| \le \bar{c}$, c_ξ continuous in ξ, and $\int c_\xi \, d\lambda = 0$. (b) Its connected functions obey the original bound table. (iii) Its reflection Hilbert space obeys (1)(e), and its zero-energy subspace is one-dimensional. Under (Rot), the circular spectral cone and the O(2) action also hold. (iv) For each pair of tuples X, Y, $\sigma_\xi(X \cdot \tau_v Y) - \sigma_\xi(X)\sigma_\xi(Y)$ is bounded by $C(X,Y)/|v|$ for sufficiently large $|v|$, uniformly in ξ.

■□

Proof. Apply Prop. E.18 to the frame limit and to every limit in the frame, using Thm. E.13 for the conditioned approximants to each component. The frame limit has zero mean because the integrating and centring laws coincide, and eq. C.vii then gives (1)(a). The same proposition gives (b) and (c). Prop. C.10 gives (d) for both the moments and the connected functions. In (e), Prop. D.2 gives the representation and separability, the signed permutations fixing e_3 preserve its reflection form, and Prop. D.5 gives each spectral inequality by (c). Under (Rot), Cor. D.8 gives (g).

Existence in (2)(a) is Thm. E.13; uniqueness and the action on components in (c) follow from Prop. E.15. The action is the identity whenever each component is invariant, since different points

give different functionals. The vacuum-space statement (b) is Thm. E.22 using the uniform clustering estimate of Prop. E.20. For (3)(i) and (ii) apply Prop. E.18, eq. E.xlv, and reflection transport under (Rot). The spectral and Hilbert-space statements follow by the same argument as (1)(e), (g) and the equivalence eq. E.xlix. The uniform estimate (iv) is Prop. E.20. □

Corollary F.23 (Reconstruction of the components)**.** Under (Rot) and (PtB), every component has a scalar Wightman reconstruction with unique invariant vacuum. The frame limit has the scalar reconstruction with vacuum space given by Thm. E.22. If the component Schwinger functionals have the growth bound eq. C.xxxii, they satisfy the Euclidean axioms of Prop. F.18. ■□

Proof. Prop. F.21 verifies the hypotheses of Thm. F.16 and establishes the vacuum condition for the components. Prop. F.18 applies to their Schwinger functionals by the bound, reflection positivity, Euclidean invariance and uniform component clustering. □

Appendix G. Index of notation and conventions

Notation	Meaning	Definition
G, U, $a_p(U)$	Gauge group, link field and plaquette action	§1.1
L, K, a_K	Blocking factor, cutoff and lattice spacing	§1
ℓ, b, β	Physical side, label and inverse lattice coupling	§1
$\mathcal{O}_K(f)$, S_n^T	Smeared action density and connected functions	Def. 1.1
V_k, x_k	Block field and reference coupling at step k	Def. 1.7
k_{ax}, M	Coarse-link transition density and its center	§5.5
ℓ_k, ϱ	Gaussian rows and their truncation radius	§2.2
E^a, E^h, w	Local analytic term, remainder and large-field weight	§5.3
$\rho[u;\ell]$	Effective density with interaction data u and rows ℓ	§5.5
$c_K^{(s)}$, $\hat{c}_K$	Source subtractions at depth s and at the last scale	Def. 1.7
$\mathbb{G}_K(\mathsf{b})$	Translation-invariant Wilson Gibbs states	Def. 1.9
σ_S	Centered functional of the connected family S	Def. C.4
$\mathcal{S}_{\neq}$	Schwartz tests flat at coincidence diagonals	§C.2
$\mathcal{H}(\sigma), \Omega, H$	Reflection Hilbert space, vacuum and Hamiltonian	§D.1
$\mathsf{X}, \lambda, \sigma_\xi$	Component space, measure and functionals	Def. E.12
$B(s,s')$, $\mathbf{W}_n$	Point insertion operator and holomorphic vacuum functions	§F

The cutoff K increases toward the continuum, the step number k increases toward the coarser lattice, and depth is $K-k$. Unless a Euclidean distance is specified, separation of supports is measured with the sup distance. The Lipschitz constant in $\|f\|_\sharp$ uses the Euclidean distance. On the periodic lattices, distances are quotient distances. Haar measure on G has mass one.

References

[1] OpenAI (2026). *The canonical massive continuum limit of the two-dimensional* O(3) *model*. Preprint, 4 October 2026. Paper. OpenAI mathematics result family 215.